

Moving the Time Variable from a Premise of Physical Law to a Consequence

A General Theory of the Complete Description of Time Establishment
through Local Comparison and Failure Classification

Naoaki Tsuda

B.A., Faculty of Environment and Information Studies,
Keio University (Shonan Fujisawa Campus), Japan
Independent Researcher
`tsuda.naoaki@gmail.com`

2026

Theory Name

Official Japanese name (romanized): *Jikan Sanshō no Kyokusho Hikaku Riron*

Official English name: *the local-comparison theory of time reference*

English abbreviation: LCTR

The Original Text

The Japanese version of this paper is the original text, written in Japanese, the author's principal language of thought. The corresponding English version is a translation prepared for international academic use. The theoretical structure is identical in both versions; with respect to conceptual precision, nuance, and semantic placement, however, the Japanese version, grounded in the author's mode of thought, is the most faithful.

Abstract

The local-comparison theory of time reference (LCTR) transfers the time variable appearing in physical laws from a primitive premise of law description to a representation that can be used after its physical grounds have been generated within the theory. What this paper changes is the order of presupposition governing how theoretical physics states its laws: before a time variable is entered into a law, its grounds and qualification for use must be constructed.

Its theoretical structure integrates the design principles of computer science concerning types, dependencies, states, and change effects as a mathematical structure that specifies the generation of physical time, qualification for use, invalidation localization, and downstream regeneration.

The construction begins by retaining, as records, local differences in indications and states arising in apparatuses and objects, and by determining, without temporal quantities, where the information required for comparison is joined. It then proceeds through the integration and ordering of comparison results and through dynamical

description and time representation to the construction of a time variable usable within laws. It also specifies the respective establishment conditions for the validity domain of continuous approximation and for differential form. A time variable may be used only when the establishment conditions along the generation dependencies leading to that use are satisfied. Through this generation, LCTR successively gives the establishment conditions for the identity of time content, before-and-after relations, comparability and incomparability, linear order, and real-valued representation, and separates the establishment of time structure from the choice of time coordinates. This construction gives a constructive definition of physical time in terms of its establishment conditions rather than its verbal meaning.

If a condition of the generation process is not satisfied, the audit identifies the conditions that first fail along the generation dependencies and localizes nonestablishment to the corresponding positions. It then returns the validity domain of the time variable, the set of first invalidation positions, boundary information for the approximation scale, the causes of invalidation, and the structural burden. For continuous approximation and differential forms, it thereby presents, along a single dependence order, the range in which they may be used, where and why they cease to be valid, and the objects to be repaired and regenerated or recomputed. The audit thus supplies continuous approximation and differentiation with a conditional operational warranty that states their conditions of use and causes of invalidation. LCTR changes the status of classical continuous time from a primitive premise of law description to a generated approximate representation with physical establishment conditions and a maximal domain of validity.

The audit covers both formal theoretical conditions and conditions of physical realization. It keeps theoretical and engineering conditions distinct in type while connecting physical carriers, records, communication, finite resolution, and evidence provenance within a single audit dependence, and includes presentation of physical realization and the available range of use among the obligations for establishing a theory. Qualification to describe multiple objects on a common time representation and to use simultaneity judgments, synchronization, and a common time representation is obtained on the domain where each object's physical carriers, records, information coupling, communication, and multi-object relations are established; remote or global constructions additionally require consistency conditions for transport and gluing.

Primitive-time input is defined here as entering time content into law operations or result derivations while lacking a complete generation path that establishes that content from nontemporal physical inputs. Mathematical consistency, predictive performance, and empirical fit do not supply the generation path missing from primitive-time input. The evidence-effect domain is defined as the range directly supported by successful measurement, prediction, model fit, or computation. A claim that transfers a result to another object, predicate, evidence type, condition of establishment, or application domain exceeds that evidence-effect domain.

Under this criterion, 72 representative sources spanning theories, experiments, metrological standards, and engineering from antiquity to the present were audited. Every source contained primitive-time input and results dependent on it. No source generated from nontemporal physical inputs the entirety of the time content it used. Even sources possessing physical relations corresponding to part of the generation process did not reach generation of all the time content they used; primitive-time input and results dependent on it remained. This unanimity shows that primitive-time input is not confined to a particular period or field but has recurred throughout the history of physics.

Current theoretical descriptions containing primitive-time input do not satisfy physical-time qualification. Modern physics, with its present use of time variables unchanged, is therefore not established. When primitive-time input backgrounds a condition required for use, and independent evidence establishes both actual invalidation of that same condition and downstream dependence reaching a result, the result itself loses qualification for use in the evaluation domain and requires regeneration or recomputation after repair. Existing mathematical, experimental, and engineering results and physical relations, by contrast, are retained within their evidence-effect domains and can be reused safely where time content and its grounds have been generated and their validity domains and invalidation positions audited. For quantities that contain a time dimension or are defined using time, retention of physical relations is separated from qualification for use of time-dependent expressions, and the latter is determined within the post-generation validity domain.

Result-preserving re-grounding, which integrates invalidation of qualification under primitive-time premises, retention of existing results, and their safe reuse, distinguishes removal of external time from generation of physical time as separate obligations. It generates time usable within laws from records, comparisons, differences in object states, and observer-side information coupling. Generation of time order and construction of causal order are separate obligations: establishing causal order does not entail the establishment of a clock scale, time intervals, or law time. This generation order places periods, frequencies, and Hz among post-generation metrological results, identifies the objects on which Lorentz transformations act as generated time representations and spatial displays, and separates local proper time and global law time into distinct stages of establishment. It reformulates the problem of time in quantum gravity as a problem of qualification for the use of law time, specifies the arrow of time as a subsequent orientation condition on generated time representations, and imposes on claims of the creation, emergence, recovery, or nonexistence of time the obligation to prove a generation path or alternative structure and the exclusion of primitive-time dependence from all time content and subsequent results. Along that generation dependence, it places problems conventionally discussed as separate problems of time at distinct invalidation positions. Classical physics, relativity, metrology and engineering, quantum gravity, and cosmology can thereby be compared along the common axes of the burdens required to establish time, validity domains, and invalidation positions.

When formal evidence and statement evidence are collated along these common axes, the time-stretching account in relativity exhibits a two-stage misrecognition. The first stage places time whose physical grounds have not been generated in advance of laws as “time itself,” independent of physical carriers such as clocks, oscillators, signals, and records. The second attributes comparison differences on physical carriers responding to gravity or motion to stretching of that “time itself.” If statements asserting the nonexistence of time and statements attributing properties to time itself are placed within the same explanatory system, the two are incompatible when they target the same time-content type and application domain; when they target different objects or domains, the transition must be stated. A transition to clock-indication differences, clock-rate differences, proper-time differences, coordinate differences, or other relational quantities does not entail a change in an attribute of “time itself” independent of physical carriers. A further chain that transfers the time attribute to living processes and extends the force of experimental results concerning physical carriers to direct demonstration of living processes constitutes a three-stage misrecognition involving biological application. More generally, the same kind of transgression recurs across multiple fields when the

success of a limited computation, prediction, model fit, or experiment is transferred to the physical establishment of an entire mathematical expression, the qualification of ungenerated time, application to an unmeasured target, or an ontological claim about time. This paper identifies that pattern as the unbounded inheritance of limited results. The force of a result is restricted to the object, predicate, evidence type, condition of establishment, and application domain that the result directly supports.

Consider those problems treated by modern physics as independent mysteries concerning time whose constitutive conditions depend on ungenerated time. Once generated physical relations are distinguished from the attribution of properties to an ungenerated “time itself,” and time is physically generated, that constitutive condition is removed. Every such mystery consequently loses its grounds as an independent problem and disappears. What remains is to determine the conditions of establishment, validity domain, first invalidation positions, failure classification, repair order, and next evidence to be acquired for generated time representations. To use a time variable in a physical law, its time content and physical grounds must be generated before its use in the law. A time variable in a physical law is not established as a primitive premise.

1 Introduction

1.1 Problem Setting

Physics has described motion, change, measurement, and physical law by means of a conventional time variable. Physics has not, however, decomposed the requirements for using that variable into a construction problem for the establishment of time itself. Clocks, records, synchronization, coordinates, spacetime structures, and observational procedures have been introduced, but the conditions under which they jointly establish a single time reference have remained among the premises of physical-law formulation.

Background time in classical mechanics, synchronization and inertial coordinate time in special relativity, Minkowski spacetime, and dynamical spacetime in general relativity have each supported the use of a time variable in a different form[1, 2, 3, 4]. This paper places these uses of time on a single generation pathway as conditions for establishing a time reference—conditions generated from local displays, records, object-state differences, and observer-side information-coupling positions.

The theory constructed along this pathway is called the *local-comparison theory of time reference*, abbreviated below as LCTR.

The logical architecture of LCTR integrates principles systematized in computer science (CS): type separation; boundaries between inputs, operations, and outputs; dependency order; audit states; failure localization; evidence provenance; and change-impact analysis. It gives these principles mathematical form as a structure governing the establishment of physical time. For each time content, LCTR places its input type, generation conditions, downstream conditions of use, validity domain, minimal failure locations, repair target, and scope of downstream regeneration within a single dependency structure.

In classical descriptions of the world, an object’s mass distribution, center of mass, shape, moments, and other properties are introduced as physical quantities when required. The availability of a common time variable for those objects is instead treated as an already

established background. The time values appearing in the equations are therefore used as quantities already present in the world described by the theory, apart from the processes by which something physical is time-stamped, retained in a record, and made comparable.

Changes in the treatment of time between classical and relativistic descriptions have often been understood in terms of differences in scale or theoretical domain. The more basic issue is the stage at which a physical theory takes the availability of a time variable as already established within the world it describes. Classical macroscopic descriptions embed this condition in their idealized descriptions of the world. The availability of a time variable therefore does not appear as a problem and recedes into the background of law formulation.

A theory that treats the availability of a time variable as established states laws and predictions under idealized premises and tests their agreement with experiment. Such a theory does not provide, as one of its own outputs, the extent to which those premises persist in a physical realization or the point at which they fail. The causes of disagreement are consequently distributed among several explanations, leaving the gap between idealized law and physical realization outside the theory.

This paper moves that presupposed part into the theory and formulates the conditions under which time comparison, dynamical description, time representation, and continuous-description approximation are established.

1.2 Constructed Objects and Generation Targets

The objects constructed in this paper are the conditions for establishing a time reference that a physical theory requires in order to state laws using a time variable. From locally received clock-side material, retained record-side material, object-state-side material, and observer-side information-coupling positions, the construction derives the conditions governing time comparison, time representation, dynamical description, time representations used in laws, and the validity domain of continuous-description approximation.

The dynamical content specific to each physical theory is retained as theory-specific input to the generation and audit construction of LCTR. The generation targets of this paper are the conditions under which that theory-specific input is established for use in time comparison, time representation, dynamical description, and time representations used in laws.

1.3 Scope and Generation Targets

The general theory developed in this paper uses locally received material, record-side material, object-state-side material, and observer-side information-coupling positions as top-level input categories. It addresses local constructions that include the comparison rules and realization materials supporting those categories. A construction lies within the scope of this paper when the conditions for establishing a time reference can be constructed from these materials.

The scope of this paper is to describe, along a single generation pathway, the conditions for establishing a time reference, the validity domain in which a time representation can be used, and the location at which a condition fails.

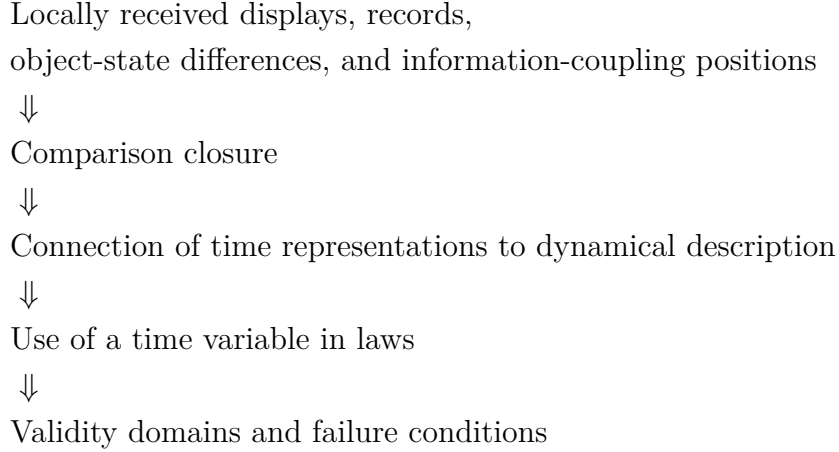
1.4 Central Thesis and Worldview Contrast

The central thesis of this paper is to shift the conventional time variable used in physical law from a primitive premise of law formulation to a law-admissible representation generated from locally received displays, retained records, object-state differences, and information-coupling positions.

This paper proposes four functional responsibilities for generating time without presupposing a time variable, and generates the time variable used in laws by a construction satisfying those responsibilities.

Conventional physical descriptions place the time variable, continuous time, coordinate time, and frequency among the premises used to formulate laws of motion and descriptions of measurement. This paper decomposes, along the generation pathway, the conditions of establishment absorbed into those premises and connects them to comparison closure, the canonical comparison quotient, the observer-time quotient, dynamical description, law-admissible time representations, validity domains, failure locations, and the conditions under which primitive premises fail.

For purposes of exposition, the construction proceeds in the following broad order.



This central thesis is developed through five claims. First, the task of establishing a time reference is formulated from local displays, records, object-state differences, and observer-side information-coupling positions, without taking external time, a total order over the external world, continuous time, or primitive frequency values as starting points. Second, the paper gives the conditions under which comparison-stage closure generates the canonical comparison quotient, exact operational time structure, representation coordinates, and comparison atlas. Third, it gives the conditions under which dynamical-description operability yields the observer-time quotient, object-state quotient, trajectory relation, canonical observable family, and law-admissible time representation. It further shows that the observer-time law core is generated when the selected law datum satisfies law-description operability. Fourth, it localizes the failure conditions for a time reference, a law-admissible time representation, and the use of a time variable in equations expressing physical laws as failure locations and audit branches associated with the failure of conditions governing comparison, representation, continuous-description approximation, dynamical description, use in laws, and differential formulation. Fifth, it classifies representative physical theories, quantum-gravitational reconstructions, cosmological initial boundaries, extreme-gravity regimes, the problem of observables, the arrow of time, quantum measurement records, microscopic physical processes, spacetime emergence, and cosmological expansion histories

by differences in time-establishment burdens, validity domains, failure locations, and audit branches along the same generation pathway.

Figure 1 shows this difference in the order of introduction. The left panel depicts a scheme in which the time variable, continuous time, coordinate time, and frequency are placed among the premises of physical-law formulation. The right panel depicts a scheme in which the conditions for establishing a time reference are constructed from local displays, records, object-state differences, and information-coupling positions.

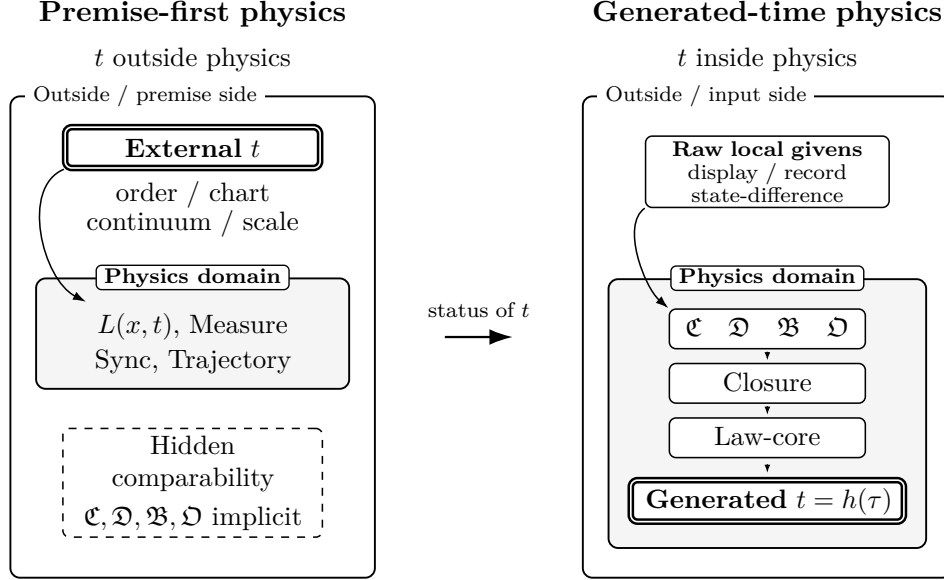


Figure 1: Contrast between the worldview of conventional physics and that developed in this paper. In the conventional scheme, the time variable, continuous time, coordinate time, and frequency are placed among the premises of physical-law formulation. In the scheme developed here, the conditions for establishing a time reference are constructed from local displays, records, object-state differences, and information-coupling positions, and the conventional time variable appears as a representation of the constructed time reference rather than as a premise of law formulation.

The generation pathway shown on the right side of Figure 1 is decomposed in Figure 2.

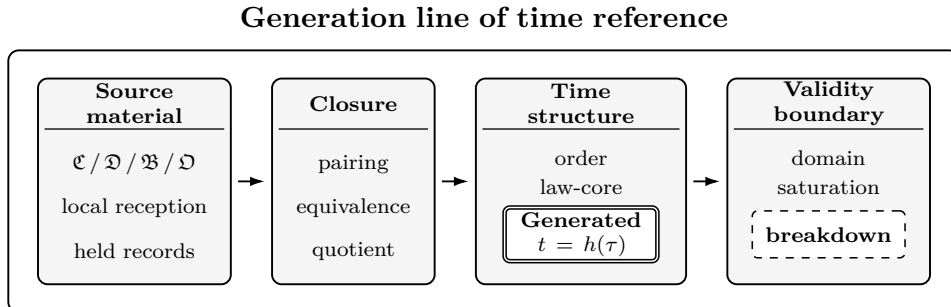


Figure 2: Generation pathway for time reference. The diagram shows the order of construction from local displays, records, object-state differences, and information-coupling positions through comparison closure and time representation to boundary and failure determination.

1.5 Non-Circularity and Input Materials

The construction developed in this paper does not use a time variable, external time, a total order over the external world, continuous time, or primitive frequency values as inputs. The input layer contains displays received locally, retained materials, object-state differences, reception positions grouped on the observer side, source-side realization materials, and local comparison rules. These are the operational materials from which a time reference is established at a later stage.

During input formation, these materials are treated as materials for difference, retention, arrival, and comparison that are available before a time variable has been established. Cycle indices and cycle-count differences are counting materials that are connected to frequency, period, and Hz after a time representation has been established at a later stage. Retained materials, object-state differences, and arrival maps are likewise incorporated into a time reference through the subsequent construction.

This paper separates the operational materials in the input layer from the time representations generated at later stages. This separation permits counting, retention, differences, arrivals, and comparisons to be used as inputs while avoiding the circularity of treating them from the outset as time values, periods, frequencies, time evolution, or time maps.

1.6 Why This Problem Was Difficult to See

This problem was difficult to see because, in formulations that placed the time variable first, the fact that time was being used as a premise had receded into the background in the fields concerned with clocks, records, synchronization, coordinates, observational procedures, and spacetime structures. In formulations that place time first, the establishment, validity, and failure of time are absorbed into the premises required to formulate laws of motion.

In classical mechanics, the conventional time variable is used as a parameter in equations of motion. Relativity brings coordinate time and proper time, comparisons between worldlines, and synchronization procedures to the foreground. Quantum theory foregrounds observational coupling and state identification. Time metrology and GPS foreground the adjustment of multiple clocks, correction records, and the gluing of comparison results. These formulations have addressed the representation, measurement, synchronization, coordinatization, and geometrization of time, together with its relation to observation, but have not formulated, as a single construction condition, the conditions under which a time reference itself is established.

The unconstructed part also remains in classical definitions of time. In Book IV, Chapter 11 of the *Physics*, Aristotle neither identified time with change nor separated it from change. He connected the continuity of magnitude, change, and time to before and after and, through distinguishing and counting two nows, defined time as the number of change with respect to before and after[5]. This was a historical achievement that connected the distinction from change, before and after, the now, perception, demarcation, counting, and continuity to the definition of time. It nevertheless used before and after, the now, change, perception, and number before the generation of time; the rules for generating them from local operational materials, record retention, comparison closure, information-coupling positions, law-admissible time representations, validity domains, and failure locations remained unconstructed.

The factors that have obscured the need to establish a time reference take different forms in the background time of Newtonian mechanics, Einstein synchronization in special relativity, Minkowski spacetime, and the spacetime structure of general relativity[1, 2, 3, 4]. This paper moves time from a premise of these descriptions to a consequence of the generation conditions at the comparison, representation, dynamical-description, and law-representation stages.

The high-precision success of relativistic calculations established clock readings, synchronization procedures, coordinate time, proper time, and comparisons between worldlines as correspondence structures that can be handled stably within physical law. This establishment obscured the fact that time was being used as a premise in spacetime structure, synchronization conventions, proper-time representations, and clock readings. This paper places the construction conditions for establishing the time reference presupposed in these correspondence structures along a single generation pathway.

In Chapter 5 of Volume I of *The Feynman Lectures on Physics*, Feynman discussed the problem of giving a satisfactory definition of time and called on his readers to acknowledge that time is probably among the things that “we,” including himself, cannot define in dictionary terms. He then shifted the discussion to how time is measured in physics[6]. This passage is direct textbook testimony that, even while a time variable was used in physical laws, the conditions for establishing a time reference itself remained an object of definition. Even after this frank statement, physics did not treat the undefinedness of time itself as a construction problem, and the fact that time was being used as a premise remained in the background across clocks, synchronization, coordinates, spacetime structure, metrology, and observational procedures. This paper treats the conditions for establishing the time presupposed in those treatments as a construction problem prior to measurement procedures and reconstructs them as the conditions under which a time reference and a law-admissible time representation are generated from local displays, records, object-state differences, and observer-side information-coupling positions.

Definition 1.1 (Difficulty of Defining Time). The difficulty of defining time is the state in which physical laws use a time variable while no satisfactory definition of time itself is supplied as the ground for that use. Feynman’s statement is direct textbook testimony to this state. Under the difficulty of defining time, the use of time as a premise recedes into the background in descriptions of clocks, measurement, synchronization, coordinate time, proper time, spacetime structure, metrology, and observational procedures, while the conditions for establishing a time reference remain unconstructed. This paper takes these conditions of establishment as objects of construction: the conditions for using the time variable that appears in physical laws.

LCTR answers this difficulty of defining time not by adding a lexical meaning of time, but by constructing the conditions of establishment under which physical time can be used.

Putting the conventional time variable first pushes the conditions for establishing a time reference outside law formulation. In that order, the properties of laws of motion are examined with respect to that time variable, while the conditions under which the time variable itself becomes available remain among the premises. This paper treats those remaining conditions as the construction problem for the establishment of time.

The historical significance of this paper lies in shifting the conventional time variable from a premise of physical law to a consequence. Physics has described motion, change, measurement, and laws in terms of time, but has not decomposed the conditions required to

establish a time reference itself into the construction materials for establishing time. This paper brings the part left outside into the theory and places the conditions for generating a time variable, the audit conditions for the use of time, and the primitive-premise failure conditions along the same theoretical pathway. Because the conditions for establishing a time variable are common premises prior to the formulation of physical laws, shifting them from premises to objects of construction revises the starting point of physical description. Its scope extends to the natural sciences built on physics and to the descriptive foundations of science and engineering that use physical laws. Problems conventionally treated as distinct mysteries of time stood as mysteries insofar as they presupposed an ungenerated time variable. Moving the time variable from a premise of physical law to an outcome of generation removes the basis on which these problems stood as mysteries and reclassifies them as conditions for establishing a time reference, validity domains, failure locations, and subjects of audit reports. This paper examines time use in the 72 audited sources spanning ancient definitions of time, modern theories and experiments, metrological standards, and engineering materials against a common criterion: whether time content and the structure grounding its establishment are generated before their use in laws.

1.7 Distinction from Existing Theories

In existing theories, clock synchronization, proper time, relational time, quantum-theoretical reconstruction of time, time metrology, and operationalism have, in their respective ways, addressed the measurement, synchronization, geometrization, relationalization, and standardization of time, together with its connection to observation. By contrast, LCTR formulates them as a single construction problem in which the conditions for establishing a time reference are determined from local displays, records, object-state differences, and information-coupling positions, thereby shifting the time variable from a premise of physical law to an outcome of generation.

The existing theories referred to here include operationalism, the problem of time in quantum gravity, external-time-free quantum-theoretical descriptions of time, relational observables, and time-metrology standards[7, 8, 9, 10, 11, 12]. This paper places the partial objects addressed by these approaches along a single generation pathway proceeding through time-reference formation, law-admissible time representations, validity domains, and failure locations.

The reduction meant here maps the use of time addressed by existing theories to the construction stages through which the time variable is shifted from a premise of physical law to a consequence, while retaining the dynamical content specific to each theory as theory-specific input to the generation and audit construction of LCTR. For each representative scheme, the correspondence among the dynamical content of the existing theory, the use of time, the conditions of establishment, and the conditions of failure is fixed.

1.8 Main Results

This paper has five main results.

First, the task of establishing a time reference is formulated from local displays, records, object-state differences, and observer-side information-coupling positions, without taking

external time, a total order over the external world, continuous time, or primitive frequency values as starting points.

Second, the paper shows that comparison-stage closure is the condition under which the canonical comparison quotient, exact operational time structure, local chart system, representation coordinates, and comparison atlas are generated. On this basis, a conventional time parameter is fixed as an order-preserving representation on the exact operational time structure thus generated.

This generation determines, prior to the use of a time variable, what may be treated as the same time content, what may be compared as before and after, and under what conditions a single order and numerical representation may be used. It also separates the establishment of the generated temporal structure from the choice of its coordinate representation.

Third, the paper shows that dynamical-description operability is the condition under which the observer-time quotient, object-state quotient, trajectory relation, graph trajectory, canonical observable family, and law-admissible time representation are generated. The observer-time law core is further generated when the selected law datum satisfies law-description operability. The generated time representation is used in the law on the common valid time domain of this law core.

Fourth, for each failure of a time reference or law-admissible time representation, the paper identifies the failure location as the point at which an establishment condition governing comparison, representation, continuous-description approximation, dynamical description, use in laws, or differential formulation ceases to hold. It further treats the use of a time variable in equations expressing physical laws as an audit branch connecting that use to establishment conditions and primitive-premise failure conditions along the generation pathway. On this basis, the range in which use of the time variable is established, its failure boundary, and its reason for failure are classified by determining which establishment conditions along the generation pathway remain satisfied, how far they remain so, and where they cease to hold. This classification turns the divergence between an idealized law and its physical realization into a theoretical output determined by the establishment conditions and failure locations governing use of the time variable.

This classification gives the conditions under which continuity and differentiation may be used. It distinguishes cases in which description by differential equations is valid from those in which it loses descriptive adequacy by the range over which those conditions are satisfied and by their failure locations. Applying the same conditions and failure locations to both theoretical description and engineering implementation connects them along the same audit pathway.

LCTR changes the stage of establishment of classical continuous time from a primitive premise of law description to a generated approximation representation that may be used within the range in which its physical conditions of establishment hold.

For each minimal failure location in each series, the paper identifies the associated condition or input. When repair of that condition succeeds, downstream results are regenerated or recomputed from the repaired input in dependency order. When the condition does not hold over the entire domain, the domain of use of the results is restricted to the region in which the condition holds. For existing forms that include continuity and differentiation, qualification for use, domains of establishment, causes of failure, and repair order are thereby returned as a single audit output.

Accepting a primitive time input as an already established time input can move the generation or establishment conditions required for that use into the background, beyond

the generation and evaluation boundary preceding use in laws. If independent evidence establishes actual failure of the same condition and a dependency path from its failure location reaches a downstream result, that result loses its qualification for use in the evaluation domain and is regenerated or recomputed from repaired input.

Physical carriers, records, communication, finite resolution, and evidence provenance constitute audit inputs that determine the domains of establishment and failure locations of a time variable. The responsibility for establishing a theory that uses a time variable therefore includes presenting its physical realization and its usable range.

Fifth, the paper reduces Newtonian mechanics, classical mechanics, special relativity, general relativity, time metrology, GPS, quantum-gravitational reconstructions, cosmological initial boundaries, extreme-gravity regimes, the problem of observables, the arrow of time, quantum measurement records, microscopic physical processes, spacetime emergence, and cosmological expansion histories to realization or connection domains along the same generation pathway. In this reduction, the use of time in each theory is connected to the conditions for establishing a time reference, validity domains, failure locations, and audit branches.

For physical quantities that include a time dimension or are defined using time, the paper retains their physical relations as theory-specific input to the generation and audit construction of LCTR and uses time-dependent representations within domains of establishment only after the required time content has been generated. This order connects the quantification domains of claims about the constancy, invariance, rate, lifetime, linewidth, or frequency of each quantity to the generation conditions for time content. The construction that retains existing physical relations as such theory-specific input, localizes failure conditions and their downstream effects, and performs repair or domain-of-use restriction provides a result-preserving re-grounding: it moves the physical grounds of retainable existing results from primitive time premises to a generated grounding structure.

The standard formulations of special and general relativity directly yield differences in clock readings, differences in coordinate time, differences in proper time along worldlines, metric structures, and the correspondences among them[2, 3, 4]. Standard textbooks and public expositions, by contrast, connect these differences to expressions according to which time itself slows, time expands or contracts, or physical processes as a whole slow together with clocks[6, 13, 14, 15, 16, 17, 18]. These sources do not define the type, state space, or deformation law of an independent time entity, yet they identify what counts as time, proper time, a clock, a rate of time, or spacetime before the construction of a time reference.

This paper does not presuppose time, proper time, or a rate of time as a primitive object. It generates a time reference and law-admissible time representations from local operational materials and determines relativistic clock differences as comparison differences among generated law-admissible time representations corresponding to multiple observers or worldlines. The fundamental reason existing theories do not give the conditions for establishing a time reference and the rules for generating a time representation is therefore not limited to an explicit assumption of an independent time entity; it also lies in identifying time, proper time, or a rate of time before constructing the time reference. A claim that expansion or contraction of an independent time entity is a literal physical mechanism additionally requires the type and identity conditions of that entity, its state space, deformation map, deformation magnitude, evolution law, and observational correspondence to clock differences. Even when no independent time entity is claimed, calling a generated comparison difference an expansion or contraction of time does not

generate time itself.

Statements that time does not exist and statements that attribute expansion, contraction, slowing, acceleration, flow, or deformation to time itself are assessed by matching the type and domain of application of the time content to which each statement refers. If they refer to the same type of time content in the same domain of application, a statement denying the existence of an object and a statement attributing physical properties to that object are incompatible. If they refer to different objects or domains of application, the transition must be stated; a transition to differences in clock indications, differences in clock rates, differences in proper time, coordinate differences, or other relational quantities does not entail a change in a property of “time itself” independent of physical carriers.

Comparison differences established on clocks, oscillators, signals, records, or other physical carriers, and explanations attributing those differences to properties of time itself independent of the carriers, have different objects and evidence. Results of measurement, prediction, model fit, or mathematical consistency hold for the objects, predicates, conditions, and domains of application directly supported by their evidence. A claim transferring such results to a different object, predicate, evidence type, establishment condition, or domain of application requires independent evidence or a valid derivation supporting that transfer.

For quantum-gravitational reconstructions, cosmological initial boundaries, extreme-gravity regimes, the problem of observables, the arrow of time, quantum measurement records, spacetime emergence, and cosmological expansion histories, the paper gives the conditions under which a time representation capable of bearing laws is generated from a construction without external time and localizes the point at which those conditions fail along the generation pathway.

Through this unified generation pathway, LCTR distinguishes local proper time from a time representation that bears physical laws as a whole, and treats Lorentz transformations as relations between time representations and spatial representations. It further distinguishes the arrow of time, statements of the nonexistence or emergence of time, and the problem of time in quantum gravity according to which problem of establishment each bears among the generation, representation, and orientation of time content and qualification for its use within laws.

A construction that presupposes primitive time identifies the time variable used in laws before generating it and places its physical establishment outside the object of explanation. It thereby relinquishes the responsibility to generate, from physical materials and operations without dependence on ungenerated temporal quantities, a time representation usable in law formulation. This paper subjects the use of time across the 72 audited sources to a unified audit under this generation responsibility and shows that the existing theories covered by the audit do not fulfill it. The unified audit yields cross-source results on the use of time across the 72 audited sources. Under the complete assessment, generated time content occurs in none of the 72 sources, and all source-specific time content remains as primitive time roots. Under the local-reach assessment augmented by formal-evidence environments, local reach is recognized in only ten sources, and primitive time roots remain in all ten. The remaining 62 sources contain no time content reached under the local-reach assessment. The construction developed in this paper generates a time reference and law-admissible time representations from local physical materials and fulfills the responsibility to generate the time variable used in physical laws.

The accompanying formal-verification assets contain Lean and Isabelle formalizations of the mathematical definitions and all proof obligations in Sections 2–6, a componentwise

correspondence table for types, quantification, premises, domains, all conclusions, and dependencies in the text and both formalizations, verification results from both systems, and results from rechecking Lean proofs with nanoda and con-leche. They also include machine-readable representations of the partial output map of a law-candidate relation in MMT, OpenMath, and Content MathML; results of applicable SMT, symbolic-computation, unit-and-dimension, counterexample, and boundary-case checks; a pinned verification environment; and rerun instructions. For each proof obligation, the assets include the results of comparing the text with the proof bodies, definitions used, theorem dependencies, and local assumptions in both formalizations.

1.9 Paper Organization

This paper proceeds as follows. First, it decomposes the conditions under which a time variable used in physical laws is established into time-value carriers, retention objects, comparison operations, law-connection targets, and prior-use dependency locations. Next, it specifies, from locally available physical materials, the local time-establishment task, four requirements, the base role-type set, local operational components, and families of object-specified role-instance bundles. It then uses these roles to form the inputs required for the comparison construction. The paper next specifies the comparison-closure conditions for local comparison and gives a construction that proceeds from that closure to the canonical comparison quotient, exact operational time structure, representation coordinates, and comparison atlas. It then connects the objects obtained by comparison to dynamical description and formulates the conditions under which the observer-time quotient, object-state quotient, trajectory relation, canonical observable family, law-admissible time representation, and observer-time law core are generated. The paper further classifies how far the construction remains valid and where it fails, organizing validity domains, failure boundaries, and structural burdens as auditable objects. Finally, it places representative physical theories, classical multi-object descriptions, quantum-gravitational reconstructions, cosmological initial boundaries, extreme-gravity regimes, the problem of observables, the arrow of time, quantum measurement records, spacetime emergence, and cosmological expansion histories along the same generation pathway and determines the establishment and failure conditions governing the shift of the time variable from a premise of physical law to a generated law-admissible time representation.

2 Achievement Conditions for Time Description and the Local Time-Generation Task

The problem presented in Chapter 1 concerns the conditions under which a time variable used in physical law is established. Physical laws have used a time variable to describe motion, change, measurement, and boundary conditions. Descriptions that use a time variable as a premise span the domains concerned with clocks, records, synchronization, coordinates, spacetime structures, and observational procedures.

The construction of time in this paper determines the time placed as a primitive premise in conventional physical laws as a relation among time-value carriers, retention objects, comparison operations, and law-connection targets.

Section 2.1 specifies four functional responsibilities required for generating time without presupposing a time variable, and Sections 2.2 and 2.3 construct a common audit form.

Sections 2.4 through 2.6 successively construct local operational inputs and families of role-instance bundles from physical materials available before the time variable has been established. Section 2.7 synthesizes this order of construction with the families of role-instance bundles that become domain-side inputs to physical-carrier realization specifications in Chapter 3.

2.1 Decomposition of Time-Description Achievement Conditions

This paper proposes four functional responsibilities for generating time without presupposing a time variable. As auditable fields for checking these four responsibilities, it expands them into four items: the time-value carrier, retention object, comparison operation, and law-connection target. A time-value carrier is an object that carries time values within a description. A retention object is a record, display, object state, or boundary condition linked to a time value. A comparison operation is an operation that determines the order, difference, or simultaneity of multiple time values or records corresponding to time values. A law-connection target is an equation of motion, field equation, measurement equation, or boundary condition into which the comparison result enters.

Clock models, coordinate time, proper time, spacetime metrics, synchronization conventions, and evolution parameters correspond to one or more of these four fields. In comparing time descriptions, the representation name is separated from the four fields, and the object corresponding to each field is identified.

In a construction in which any of the four fields is determined by dependence on prior use of the time variable, the usability of that time variable remains among the premises. This paper adopts as an achievement condition for a time description that all four fields be determined from materials available before the time variable is established.

This decomposition places the time descriptions of existing physics, primitive-time assumptions, and the construction for generating time developed in this paper on the same four-field basis for comparison. The construction developed here specifies the object, operation, and law-connection target corresponding to each field.

2.2 Assessment Form for Time-Description Achievement Conditions

The four fields in Section 2.1 provide the check fields for assessing the achievement conditions of time descriptions in general, including time terms, time symbols, clock models, coordinate time, proper time, spacetime metrics, synchronization conventions, and evolution parameters.

The assessment specifies the time-value carrier, retention object, comparison operation, and law-connection target in that order and identifies every point at which a specification depends on prior use of the time variable. The result is recorded in six fields: representation name, time-value carrier, retention object, comparison operation, law-connection target, and prior-use dependency location.

The achievement condition for a time description is that all four fields be specified by materials available before the time variable is established. A construction whose specification of the four fields is grounded in prior use of the time variable retains that variable as a primitive premise. The location at which it is retained as a primitive premise is assessed by the audit of time use in laws in Chapter 7.

This assessment form applies in common to time descriptions in existing physics, primitive-time assumptions, dependencies on prior use of a time variable, and the construction for generating time developed in this paper.

2.3 Reference Form for General Audit Conditions

The four fields in Section 2.1 and the six check fields in Section 2.2 form the general reference form for comparing time descriptions.

For a description that uses time values, time intervals, simultaneity, or an order of evolution, the representation name, time-value carrier, retention object, comparison operation, law-connection target, and prior-use dependency location are recorded. The same reference form applies to clock models, coordinate time, proper time, spacetime metrics, synchronization conventions, and evolution parameters.

The local time-generation construction developed in this paper uses this reference form to specify the objects, operations, and law-connection targets corresponding to the four functional responsibilities. The local time-generation task is to specify, from physical materials available before the time variable is established, a construction satisfying this reference form.

2.4 Local Time-Generation Task and the Local Scene

The local time-generation task is to construct the local operational input entered into the reference form of Section 2.3 without presupposing a time variable, time interval, period, frequency, or second.

Definition 2.1 (Local Time-Generation Task). The local time-generation task is the task of constructing, without placing primitive time in the input, a locally distinguishable clock-side input, an object-state-side input, a record-side input that retains and can re-present the correspondence between those two inputs, and an observer-side information-coupling input that receives the three inputs under a common operational specification and combines them into a single local operational input.

The four inputs in Definition 2.1 belong to one local scene under a common operational specification. Time quantities belong to the representational outputs generated from a realized operational construction and are excluded from the construction inputs in Definition 2.1.

2.4.1 Clock-Side Source with No Time-Quantity Input and the Local Scene

Definition 2.2 (Cycle-Count Source). A cycle-count source is a clock-side source that, before a time variable is established, supplies internal counter positions labeled by natural-number values

$$k \in \mathbb{N}$$

and their update

$$k \mapsto k + 1$$

Each internal counter position provides an internal distinction on the clock side. Phase labels and display states correspond to the respective internal counter positions, while display updates and counter increments are supplied as clock-side inputs corresponding

to the internal update. Assignments of frequency values, periods, Hz, seconds, and time intervals are representational quantities defined on a generated time representation.

Remark 2.3 (Clock-Side Repeatability). Clock-side repeatability at the primitive input layer is provided by repetition of the internal counter update in Definition 2.2 and of the corresponding phase and display updates. Equal time intervals, periods, and frequencies are representational quantities defined after this clock-side input has been connected to a generated time representation.

Definition 2.4 (Local Scene). A local scene $\mathbf{Sc}$ is a finite or locally bounded typed input domain to which construction inputs containing no time quantities belong under a common operational specification. Membership in the same $\mathbf{Sc}$ is determined by the combination of an object specification, record identifier, construction boundary, and input–output connections. Temporal simultaneity, time variables, time intervals, periods, frequencies, and seconds are not used to determine that membership.

2.5 Base Role Types and Local Four-Role Binding

The four construction inputs in Definition 2.1 are assigned role-type positions for the clock side, record side, object-state side, and observer-side information coupling. Each role type specifies an input, an action, and an output.

Definition 2.5 (Base Role-Type Set). The base role-type set for an LCTR time-generation construction that does not place primitive time in the input is defined by

$$\mathbf{Rol} = \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}, \mathfrak{O}\}$$

The input, action, and output of each role type are as follows.

- $\mathfrak{C}$ is the clock-side role type. It takes the cycle-count source of Definition 2.2 as input and supplies a locally distinguishable clock-side input.
- $\mathfrak{D}$ is the record-side role type. It takes as input the correspondence between a clock-side input and an object-state-side input, retains that correspondence, and supplies a record-side input that can be re-presented locally.
- $\mathfrak{B}$ is the object-state-side role type. It takes as input object states that have not been ordered in advance by a time variable and supplies a locally distinguishable object-state difference as an object-state-side input.
- $\mathfrak{O}$ is the observer-side information-coupling role type. It receives the clock-side, record-side, and object-state-side inputs belonging to the same local scene and combines them into a single local operational input.

Definition 2.6 (Local Operational Component Sets and Local Operational Components). For a local scene $\mathbf{Sc}$, define, for each base role type $R \in \mathbf{Rol}$, the role-indexed family of typed subsets consisting of construction components that satisfy its input, action, and output specifications within $\mathbf{Sc}$ by

$$\begin{aligned} \mathbf{LC}(\mathbf{Sc}, -) : \mathbf{Rol} &\longrightarrow \mathbf{Type}, \\ R &\longmapsto \mathbf{LC}_R(\mathbf{Sc}), \quad \mathbf{LC}_R(\mathbf{Sc}) \subseteq \mathbf{Sc} \end{aligned}$$

An arbitrary role-indexed family of local operational components corresponding to the fixed order

$$(\mathfrak{C}, \mathfrak{D}, \mathfrak{B}, \mathfrak{O})$$

of the base role types is denoted by

$$\ell := (\mathfrak{c}, \mathfrak{d}, \mathfrak{b}, \mathfrak{o}) \in \prod_{R \in \text{Rol}} \text{LC}_R(\text{Sc})$$

The components $\mathfrak{c}, \mathfrak{d}, \mathfrak{b}, \mathfrak{o}$ are called, in order, the clock-side local operational component, record-side local operational component, object-state-side local operational component, and observer-side information-coupling local operational component.

Definition 2.7 (Local Four-Role Binding Relation). For a local scene Sc , define

$$\text{Bind}_{\text{Rol}}(\text{Sc}) \subseteq \prod_{R \in \text{Rol}} \text{LC}_R(\text{Sc})$$

as the set of all role-indexed families of local operational components on the right-hand side that satisfy all four of the following conditions.

1. The correspondence between the clock-side input supplied by $\mathfrak{c}$ and the object-state-side input supplied by $\mathfrak{b}$ is entered into $\mathfrak{d}$.
2. The component $\mathfrak{d}$ retains that correspondence and supplies a record-side input that can be re-presented locally.
3. The inputs supplied by $\mathfrak{c}$, $\mathfrak{d}$, and $\mathfrak{b}$ are received by $\mathfrak{o}$.
4. The component $\mathfrak{o}$ combines the three received inputs into a single local operational input.

A role-indexed family of local operational components satisfying

$$\ell \in \text{Bind}_{\text{Rol}}(\text{Sc})$$

is called a local four-role binding witness.

Remark 2.8 (Origin of the Base Role-Type Symbols). The symbols in the base role-type set $\text{Rol} = \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}, \mathfrak{O}\}$ are the initials of *Clock*, *Detector*, *Body*, and *Observer*, respectively. The record-side role type represented by $\mathfrak{D}$ integrates detection, recording, and re-presentation, with *Detector* serving as the representative term.

Base role types, local operational components, and physical carriers belong to mutually distinct types. A base role type specifies the role of an input, action, and output; a local operational component is a construction component that satisfies that specification within the local scene Sc ; and a physical carrier is a physical object on the implementation side.

Base role types and local operational components

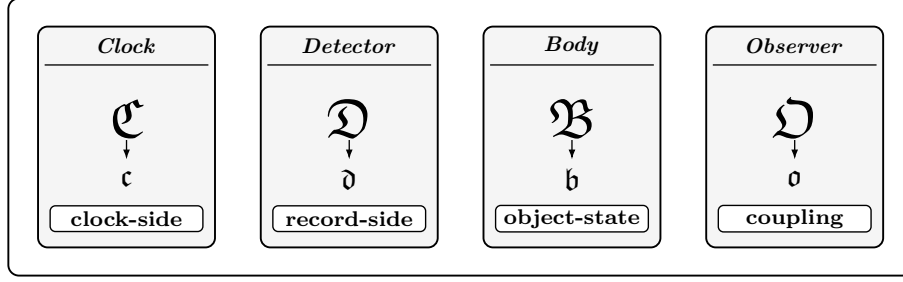


Figure 3: Construction of the base role-type set and the local four-role binding. The four base role types for the clock side, record side, object-state side, and observer-side information coupling are denoted by $\mathfrak{C}$, $\mathfrak{D}$, $\mathfrak{B}$, $\mathfrak{O}$. A quadruple of local operational components satisfying these role specifications within the same local scene forms a local four-role binding witness.

2.6 Family of Object-Specified Role-Instance Bundles and Construction Order

Take the base role-type set of Definition 2.5, the local scene of Definition 2.4, and the local four-role binding relation of Definition 2.7. An object-specification index uniquely specifies three items: a physical object being tracked, a local scene, and a local four-role binding witness.

Definition 2.9 (Object-Specification Index and Object-Specified Role Instances). An object-specification index ζ is an index that uniquely specifies the following three items.

1. One physical object being tracked.
2. One local scene Sc_ζ .
3. A local four-role binding witness

$$\ell_\zeta \in \text{Bind}_{\text{Rol}}(\text{Sc}_\zeta).$$

The component

$$\ell_\zeta(\mathfrak{B})$$

in the third item is the object-state-side local operational component abstracted from the object states of the physical object being tracked in the first item.

Define the role-instance bundle corresponding to ζ , and its role-wise evaluation at each

$$R \in \text{Rol}$$

by

$$\begin{aligned} \text{RI}(\zeta) &:= ((R, \zeta, \ell_\zeta(R)))_{R \in \text{Rol}}, \\ \text{RI}_R(\zeta) &:= \text{RI}(\zeta)(R) \end{aligned}$$

The component order of this role-indexed family is the fixed order of the base role types in Definition 2.6.

Each role instance retains the object-specification index as its second component, and the local operational component is specified by the same object-specification index. Hence, for any object-specification indices ζ_1, ζ_2 ,

$$\begin{aligned} \forall R \in \text{Rol}, \\ \text{RI}_R(\zeta_1) = \text{RI}_R(\zeta_2) \iff \zeta_1 = \zeta_2 \end{aligned}$$

holds.

Definition 2.10 (Family of Role-Instance Bundles over Object-Specification Indices). Let Z_0 be a set of object-specification indices from Definition 2.9. Define the family of role-instance bundles over Z_0 by

$$(\text{RI}[Z_0]) := \text{RI}|_{Z_0}$$

Here $\text{RI}|_{Z_0}$ is the restriction of the map $\zeta \mapsto \text{RI}(\zeta)$ on object-specification indices to Z_0 . Its role-indexed component family is

$$\left((\text{RI}_R(\zeta))_{\zeta \in Z_0} \right)_{R \in \text{Rol}}$$

Define the set of role types on the domain side of the realization map by

$$\text{Rol}^{\text{rlz}} := \text{Rol} \setminus \{\mathfrak{B}\} = \{\mathfrak{C}, \mathfrak{D}, \mathfrak{O}\}$$

Each component of the $\mathfrak{B}$ -derived component family

$$(\text{RI}_{\mathfrak{B}}(\zeta))_{\zeta \in Z_0}$$

has as its source of abstraction the physical object being tracked that ζ specifies. The domain-side component family of the realization map

$$\left((\text{RI}_R(\zeta))_{\zeta \in Z_0} \right)_{R \in \text{Rol}^{\text{rlz}}}$$

forms the domain-side input to the physical-carrier realization specification.

Proposition 2.11 (Prior Determination of the Family of Role-Instance Bundles). The family of object-specification indices Z_0 , together with the data specified by each $\zeta \in Z_0$ —namely, the physical object being tracked, local scene Sc_{ζ} , and local four-role binding witness—determines the family of role-instance bundles $\text{RI}[Z_0]$. This specification includes the abstraction-source correspondence between the $\mathfrak{B}$ -derived component family and the physical objects being tracked. The physical-carrier realization specification for the Rol^{rlz} -derived component family is additional structure that takes $\text{RI}[Z_0]$ as its domain-side input.

Proof. By Definition 2.9, each $\zeta \in Z_0$ uniquely specifies a physical object being tracked, a local scene Sc_{ζ} , and a local four-role binding witness. This specification determines the four object-specified role instances and the role-instance bundle $\text{RI}(\zeta)$. Definition 2.10 indexes these by Z_0 , retains both the object-specification positions and the four $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$ component positions, and defines $\text{RI}[Z_0]$.

The source of abstraction of each $\mathfrak{B}$ -derived component is specified as the first item of the object-specification index. The Rol^{rlz} -derived component family, as a component family of the already defined $\text{RI}[Z_0]$, forms the domain-side input to the physical-carrier realization specification. Hence the data specified in the proposition determine the family of role-instance bundles, and the physical-carrier realization specification is determined as additional structure on that component family. $\square$

2.7 Synthesis of the Construction with No Time-Quantity Input

From Section 2.1 through Section 2.3, the time-value carrier, retention object, comparison operation, law-connection target, and prior-use dependency locations are defined as an audit form common to time descriptions.

By Definitions 2.1 through 2.4, the clock-side, record-side, object-state-side, and observer-side information-coupling inputs available before a time variable is established are specified as local operational inputs belonging to the same local scene. By Definitions 2.5 through 2.7, the four inputs are assigned $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$ role types, local operational components, and a local four-role binding witness.

By Definitions 2.9 and 2.10, each object-specification index ζ determines a role-instance bundle $\text{RI}(\zeta)$, and over a family of object-specification indices Z_0 ,

$$\text{RI}[Z_0] = (\text{RI}(\zeta))_{\zeta \in Z_0}$$

is determined. The $\mathfrak{B}$ -derived component family retains the abstraction-source correspondence with each physical object being tracked, while the three $\mathfrak{C}/\mathfrak{D}/\mathfrak{O}$ -derived component families form the domain-side inputs to the physical-carrier realization specification.

Proposition 2.12 (Synthesis of the Object-Specified Role Construction). The family of object-specification indices Z_0 , together with, for each $\zeta \in Z_0$, the physical object being tracked, local scene Sc_ζ , and local four-role binding witness that it specifies, determines, while retaining both the object-specification positions and the four $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$ component positions, the family of role-instance bundles $\text{RI}[Z_0]$. This specification includes the abstraction-source correspondence between the $\mathfrak{B}$ -derived component family and the physical objects being tracked, while the three $\mathfrak{C}/\mathfrak{D}/\mathfrak{O}$ -derived component families form the domain-side inputs to the physical-carrier realization specification. The input data of this construction contain no primitive time.

Proof. Definitions 2.1 and 2.4 determine the local time-generation task and the local scene without placing primitive time in the input. Definitions 2.5 through 2.7 determine the $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$ role types, local operational components, and local four-role binding witness on the local scene.

Definition 2.9 determines, from the physical object being tracked, local scene, and local four-role binding witness specified by each $\zeta \in Z_0$, the role-instance bundle $\text{RI}(\zeta)$. Definition 2.10 indexes these by Z_0 , retains both the object-specification positions and the four $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{O}$ component positions, and defines $\text{RI}[Z_0]$. By Definition 2.10, the $\mathfrak{B}$ -derived component family retains the abstraction-source correspondence, while the three $\mathfrak{C}/\mathfrak{D}/\mathfrak{O}$ -derived component families form the domain-side inputs to the physical-carrier realization specification. Because the construction inputs from Definition 2.1 through Definition 2.10 belong to the input boundary of that task, the construction inputs in the proposition contain no primitive time. $\square$

3 Physical-Carrier Realization and Operational-Input Formation

By Proposition 2.12, the family of object-specification indices Z_0 , together with, for each $\zeta \in Z_0$, the physical object being tracked, local scene Sc_ζ , and local four-role binding witness that it specifies, determines the family of role-instance bundles $\text{RI}[Z_0]$ of Definition 2.10.

Each bundle retains both its object-specification position and the four component positions of the base role-type set Rol . The object specification includes the abstraction-source correspondence between the $\mathfrak{B}$ -derived component family and the physical objects being tracked, while the three Rol^{rlz} -derived component families form the domain-side inputs to the physical-carrier realization specification.

Define the source-side role-type set and comparison role-type set by

$$\begin{aligned}\text{Rol}^{\text{src}} &:= \text{Rol} \setminus \{\mathfrak{D}\} = \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}\}, \\ \text{Rol}^{\text{cmp}} &:= \{\mathfrak{C}, \mathfrak{D}\}\end{aligned}$$

3.1 Physical-Carrier Realization of Role Instances

Definition 3.1 (Family of Physical-Layer Constructions). For the family of object-specification indices Z_0 in Definition 2.10, the family of role-instance bundles

$$\text{RI}[Z_0]$$

over those indices, and the domain-side role-type set of realization maps

$$\text{Rol}^{\text{rlz}}$$

in the same definition, define the family of physical-layer constructions by

$$\text{Car}_{\text{phys}}[Z_0] := ((\text{Car}_R)_{R \in \text{Rol}^{\text{rlz}}}, \text{Car}_{\mathfrak{B}}[Z_0], \text{Car}_{\text{comm}})$$

The family $(\text{Car}_R)_{R \in \text{Rol}^{\text{rlz}}}$ indexes by role type R the families of physical carriers assigned to the clock-side, record-side, and observer-side information-coupling role-instance families. Define the family of physical objects tracked on the object-state side by

$$\text{Car}_{\mathfrak{B}}[Z_0] := (\text{Car}_{\mathfrak{B}}(\zeta))_{\zeta \in Z_0}$$

Each $\text{Car}_{\mathfrak{B}}(\zeta)$ is a physical object being tracked that bears the object-specification index ζ and is the source of abstraction of $\text{RI}_{\mathfrak{B}}(\zeta)$. For distinct $\zeta_1, \zeta_2 \in Z_0$, the equality

$$\text{Car}_{\mathfrak{B}}(\zeta_1) = \text{Car}_{\mathfrak{B}}(\zeta_2)$$

is allowed. The family Car_{comm} consists of physical connection carriers that establish joint information coupling among multiple physical carriers.

Expanding the role-indexed families in $\text{Car}_{\text{phys}}[Z_0]$ yields the five-component presentation

$$(\text{Car}_{\mathfrak{C}}, \text{Car}_{\mathfrak{D}}, \text{Car}_{\mathfrak{B}}[Z_0], \text{Car}_{\mathfrak{D}}, \text{Car}_{\text{comm}})$$

Definition 3.2 (Sequence of Role-Typed Correspondences between the Abstract and Physical Layers). For the family of object-specification indices Z_0 in Definition 2.10, the family of role-instance bundles $\text{RI}[Z_0]$ over those indices, and the family of physical-layer constructions $\text{Car}_{\text{phys}}[Z_0]$ in Definition 3.1, define, for each

$$R \in \text{Rol}^{\text{rlz}}$$

the realization map by

$$\text{RI}_R : \{\text{RI}_R(\zeta) \mid \zeta \in Z_0\} \longrightarrow \text{Car}_R$$

Define the object-specified physical-carrier value map by

$$\begin{aligned} \text{CarrierAt}_R &: Z_0 \longrightarrow \text{Car}_R, \\ \text{CarrierAt}_R(\zeta) &:= \text{Rlz}_R(\text{Rl}_R(\zeta)) \quad (\zeta \in Z_0) \end{aligned}$$

For each $\zeta \in Z_0$, the value $\text{CarrierAt}_R(\zeta)$ is the implementation-side counterpart assigned to the R -derived role instance.

For each

$$\zeta \in Z_0$$

define the object-state-side abstraction-source correspondence by

$$\text{Rlz}_{\mathfrak{B}}(\zeta) := (\text{Car}_{\mathfrak{B}}(\zeta), \text{Rl}_{\mathfrak{B}}(\zeta))$$

For each

$$Z \subseteq Z_0$$

define its restriction by

$$\text{Rlz}_{\mathfrak{B}}[Z] := (\text{Rlz}_{\mathfrak{B}}(\zeta))_{\zeta \in Z}$$

The first component of $\text{Rlz}_{\mathfrak{B}}(\zeta)$ is the physical object being tracked that bears the object-specification index ζ , and the second is the object-state-side role instance that retains the local operational component $\mathfrak{b}$ abstracted from the state of that object. Even when, for distinct $\zeta_1, \zeta_2 \in Z_0$,

$$\text{Car}_{\mathfrak{B}}(\zeta_1) = \text{Car}_{\mathfrak{B}}(\zeta_2)$$

holds,

$$\begin{aligned} \text{Rl}_{\mathfrak{B}}(\zeta_1) &\neq \text{Rl}_{\mathfrak{B}}(\zeta_2), \\ \text{Rlz}_{\mathfrak{B}}(\zeta_1) &\neq \text{Rlz}_{\mathfrak{B}}(\zeta_2) \end{aligned}$$

also holds, and the abstraction-source correspondence at each object-specification-index position is retained.

$$(\text{Rlz}_R)_{R \in \text{Rol}^{\text{rlz}}}$$

is the family of realization maps, while $\text{Rlz}_{\mathfrak{B}}$ is the family of abstraction-source correspondences from physical objects being tracked to object-state-side role instances.

Remark 3.3 (Role-Instance Identity and Implementation-Side Identity). The identity of an object-specified role instance is determined by the object-specification index of Definition 2.9. The implementation-side identity of a Rol^{rlz} -derived component is determined by the value of the corresponding realization map in Definition 3.2. The identity of the abstraction source of a $\mathfrak{B}$ -derived component is determined by the physical object being tracked that its object-specification index specifies.

Definition 3.4 (Role-Typed Physical-Carrier Correspondence Pattern). For two distinct object-specification indices $\zeta_1, \zeta_2 \in Z_0$ and each

$$R \in \text{Rol}^{\text{rlz}}$$

define the physical-carrier identity condition and physical-carrier difference condition, respectively, by

$$\text{CarrierAt}_R(\zeta_1) = \text{CarrierAt}_R(\zeta_2)$$

and

$$\text{CarrierAt}_R(\zeta_1) \neq \text{CarrierAt}_R(\zeta_2)$$

On the clock and record sides, the physical-carrier identity condition is called the shared type, and the physical-carrier difference condition is called the separated type.

On the observer side, the physical-carrier identity condition is called the shared type. The observer-side distributed type is the case in which the physical-carrier difference condition holds and joint information coupling is constructed between the two implementation-side counterparts through a physical connection carrier belonging to Car_{comm} .

A role-typed physical-carrier correspondence pattern is a three-component condition consisting of the shared or separated type on the $\mathfrak{C}$ side, the shared or separated type on the $\mathfrak{D}$ side, and the shared or distributed type on the $\mathfrak{V}$ side. The component on the $\mathfrak{C}$ side and the component on the $\mathfrak{D}$ side each take only the realization map of the corresponding role type as their condition input. The component on the $\mathfrak{V}$ side takes the $\mathfrak{V}$ -derived realization map as its condition input and, in the $\mathfrak{V}$ -distributed type, additionally takes the interposition of a physical connection carrier as a condition input. The $\mathfrak{B}$ -derived components are retained as the family of abstraction-source correspondences equipped with object-specification positions in Definition 3.2.

Role instances and physical-layer relations

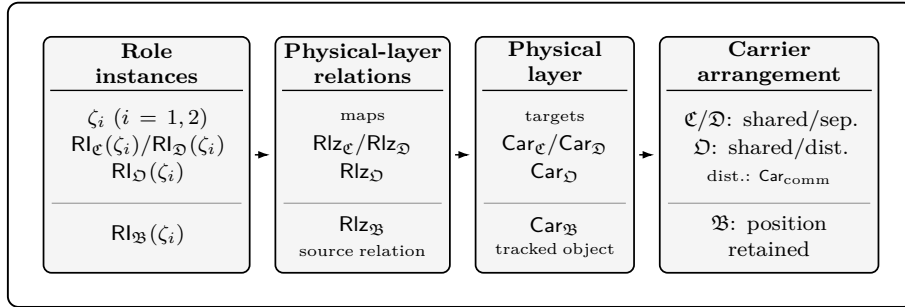


Figure 4: Role instances and physical-carrier realization. Role instances are distinguished for two object-specification indices. The physical-carrier realization maps then assign the $\mathfrak{C}$ -derived, $\mathfrak{D}$ -derived, and $\mathfrak{V}$ -derived role instances to physical carriers, while the $\mathfrak{B}$ -derived object-state-side instances are associated with the object-state-side entities corresponding to their object specifications. Sharing by the same physical carrier, separation by physical carriers distinct for each object, and distribution across multiple information-coupling carriers are distinguished on the input-formation side by equality relations among the values of the $\mathfrak{C}/\mathfrak{D}/\mathfrak{V}$ -derived physical-carrier realization maps and by physical connection carriers belonging to Car_{comm} .

Table 1: Role instances and physical-carrier correspondence patterns

Physical-carrier correspondence pattern	Physical-carrier realization condition
R -shared type ($R \in \{\mathfrak{C}, \mathfrak{D}\}$)	$\text{Rlz}_R(\text{RI}_R(\zeta_1)) = \text{Rlz}_R(\text{RI}_R(\zeta_2))$
R -separated type ($R \in \{\mathfrak{C}, \mathfrak{D}\}$)	$\text{Rlz}_R(\text{RI}_R(\zeta_1)) \neq \text{Rlz}_R(\text{RI}_R(\zeta_2))$
$\mathfrak{D}$ -shared type	$\text{Rlz}_{\mathfrak{D}}(\text{RI}_{\mathfrak{D}}(\zeta_1)) = \text{Rlz}_{\mathfrak{D}}(\text{RI}_{\mathfrak{D}}(\zeta_2))$
$\mathfrak{D}$ -distributed type	$\text{Rlz}_{\mathfrak{D}}(\text{RI}_{\mathfrak{D}}(\zeta_1)) \neq \text{Rlz}_{\mathfrak{D}}(\text{RI}_{\mathfrak{D}}(\zeta_2))$, and joint information coupling through a physical connection carrier belonging to Car_{comm}

The constructional meanings of the physical-carrier correspondence patterns in Table 1 are as follows.

- The $\mathfrak{C}$ -shared and separated types represent realization of the clock-side role by, respectively, the same physical clock carrier or physical clock carriers distinct for each object.
- The $\mathfrak{D}$ -shared and separated types represent realization of the record-side role by, respectively, the same physical record carrier or physical record carriers distinct for each object.
- In the $\mathfrak{D}$ -shared type, one physical information-coupling carrier realizes the two observer-side information-coupling roles.
- In the $\mathfrak{D}$ -distributed type, physical information-coupling carriers distinct for each object realize the observer-side information-coupling roles, and joint information coupling between them is constructed through a physical connection carrier belonging to Car_{comm} .

Proposition 3.5 (Combinability of Realization Conditions by Role Type). Take two distinct object-specification indices

$$\zeta_1, \zeta_2 \in Z_0, \quad \zeta_1 \neq \zeta_2$$

For each

$$R \in \text{Rol}^{\text{cmp}}$$

select either the shared type or the separated type. Suppose that, for each selected type, there exist physical carriers that may be specified as implementation-side counterparts,

$$p_{R,k} \in \text{Car}_R, \quad k \in \{1, 2\}$$

and that they satisfy

$$\text{CarrierAt}_R(\zeta_k) = p_{R,k} \quad (R \in \text{Rol}^{\text{cmp}}, k \in \{1, 2\})$$

If the R -shared type is selected, let

$$p_{R,1} = p_{R,2},$$

and if the R -separated type is selected, let

$$p_{R,1} \neq p_{R,2}$$

Suppose further that implementation-side counterparts satisfying either the shared or distributed type on the $\mathfrak{D}$ side are given and that, in the distributed type, joint information coupling through a physical connection carrier belonging to $\mathbf{Car}_{\text{comm}}$ is given.

Then the condition on the $\mathfrak{C}$ side and the condition on the $\mathfrak{D}$ side can each be combined, while retaining its role-type position, with the given condition on the $\mathfrak{D}$ side to construct the role-typed three-component condition of Definition 3.4. This combination retains

$$\mathbf{Rlz}_{\mathfrak{B}} [\{\zeta_1, \zeta_2\}]$$

as a family of abstraction-source correspondences equipped with object-specification positions.

Proof. The family of physical-layer constructions $\mathbf{Car}_{\text{phys}}[Z_0]$ in Definition 3.1 retains the physical-carrier families on the codomain side of the realization maps, the family of physical objects tracked on the object-state side, and the family of physical connection carriers with their role-type positions. By Definition 3.2, for each

$$R \in \mathbf{Rol}^{\text{cmp}}, \quad k \in \{1, 2\}$$

it follows that

$$p_{R,k} \in \mathbf{Car}_R$$

Hence specifying $p_{R,1}, p_{R,2}$ determines only the position of the role type R .

By Definition 3.4, the condition for each $R \in \mathbf{Rol}^{\text{cmp}}$ is determined solely by the identity or difference of $p_{R,1}, p_{R,2}$. Because each condition takes no realization-map value of another role type as a condition input, the clock-side and record-side specifications can be combined at their respective role-type positions. The condition on the $\mathfrak{D}$ side is determined by the value of the $\mathfrak{D}$ -derived realization map and, in the distributed type, by the interposition of a physical connection carrier; it does not take the clock-side or record-side specification as a condition input. The three-component condition is thereby constructed.

The object-state-side correspondence

$$\mathbf{Rlz}_{\mathfrak{B}} [\{\zeta_1, \zeta_2\}] = (\mathbf{Rlz}_{\mathfrak{B}}(\zeta))_{\zeta \in \{\zeta_1, \zeta_2\}}$$

retains, at each object-specification position, the abstraction-source correspondence between the physical object being tracked and the object-state-side role instance. Because this correspondence family takes no value of the realization-map family in Definition 3.2 as a condition input, it is unchanged by combination with the three-component condition. $\square$

3.2 Object-Specification Positions and Role-Type Positions

The base role-type set $\mathbf{Rol}$ of Definition 2.5 supplies the four component positions of a role-instance bundle. The pairing of an object-specification-index position from Definition 2.9 with a role-type position determines the object-specification membership and role-type membership of each role instance.

3.3 Role-Instance Identity and Implementation-Side Identity

By Definition 2.9, the identity of an object-specified role instance is determined by its object-specification index. The implementation-side identity of a Rol^{rlz} -derived component is determined by the value of the corresponding realization map in Definition 3.2. The identity of the abstraction source of a $\mathfrak{B}$ -derived component is determined by the physical object being tracked that its object-specification index specifies.

For a construction involving multiple object specifications, physical carriers, source sequences, or observer-side reception phases, an indexed family is used that retains the object-specification, role-type, sequence, and reception-phase positions.

3.4 Multiplicity and Physical-Placement Conditions

The role-typed physical-carrier correspondence pattern of Definition 3.4 specifies three component conditions for two distinct object-specification indices: the $\mathfrak{C}$ -shared or $\mathfrak{C}$ -separated type, the $\mathfrak{D}$ -shared or $\mathfrak{D}$ -separated type, and the $\mathfrak{D}$ -shared or $\mathfrak{D}$ -distributed type. The $\mathfrak{B}$ -derived components are retained as the family of abstraction-source correspondences equipped with object-specification positions in Definition 3.2.

For a construction with multiple clock-side, record-side, or object-state-side sequences, or multiple observer-side reception phases, the sequence and object-specification positions are retained, and the realization specifications and reception conditions are distinguished for each role type.

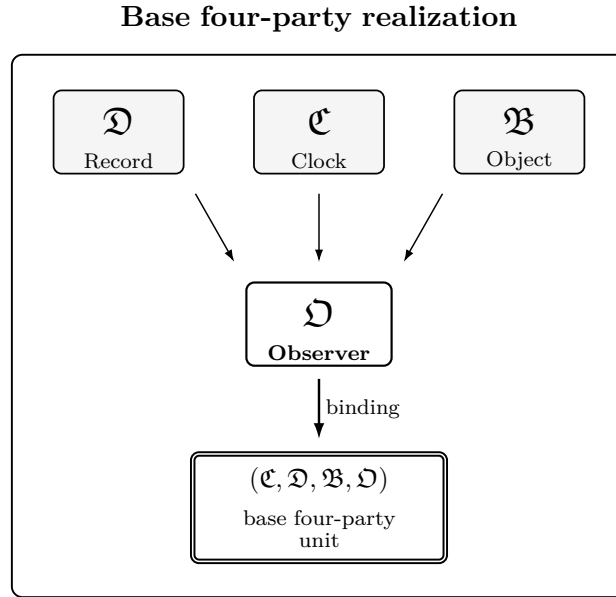


Figure 5: Base four-party realization. When the clock side $\mathfrak{C}$, record side $\mathfrak{D}$, object-state side $\mathfrak{B}$, and observer-side information-coupling position $\mathfrak{D}$ are given in the same scene, they constitute the base four-party unit of the local time-generation task.

3.5 Coupling Physical-Carrier Realization to the Reception Network

The family of physical-layer constructions in Definition 3.1 and the sequence of role-typed correspondences between the abstract and physical layers in Definition 3.2 supply physically

realized inputs that retain the object-specification and role-type positions. Definition 3.4 specifies conditions on the values of the realization maps on the $\mathfrak{C}$ side, $\mathfrak{D}$ side, and $\mathfrak{O}$ side, together with the interposition condition for a physical connection carrier in the $\mathfrak{O}$ -distributed type.

3.6 Operational-Input Formation

3.6.1 Raw Operational Input

Throughout this section, each symbol root is first written in full form with both a role type and a reception phase. Thereafter, only the reception-phase subscript may be omitted, and only when the reception phase is fixed within the same definition, proposition, or proof and the full form can be uniquely recovered. The role type is always retained; when the reception phase varies or different reception phases are treated simultaneously, the full form is retained.

For the family of object-specification indices Z_0 in Definition 2.10, take the family of role-instance bundles

$$\text{RI}[Z_0],$$

over those indices, the family of physical-layer constructions

$$\text{Car}_{\text{phys}}[Z_0],$$

in Definition 3.1, and the sequence of role-typed correspondences in Definition 3.2.

Let

$$U$$

be a nonempty index set of observer-side reception phases and comparison phases. For each $u \in U$, let an observer-side reception carrier

$$\text{RcvCar}_u \in \text{Im}(\text{RI}_{\mathfrak{D}})$$

and a family of reception sets indexed by source-side role type

$$(\text{Rcv}_{R,u})_{R \in \text{Rol}^{\text{src}}}$$

be given. Define the role-type-tagged reception set Rcv_u by

$$\text{Rcv}_u := \bigsqcup_{R \in \text{Rol}^{\text{src}}} (\{R\} \times \text{Rcv}_{R,u})$$

For each $u \in U$, let

$$(\preceq_u^{\text{rcv}}) \subseteq \text{Rcv}_u \times \text{Rcv}_u$$

be a partial order on Rcv_u . For each $R \in \text{Rol}^{\text{cmp}}$, define the partial order on the R -derived reception set, for arbitrary

$$x_R, y_R \in \text{Rcv}_{R,u}$$

by

$$x_R \preceq_u^R y_R \iff (R, x_R) \preceq_u^{\text{rcv}} (R, y_R)$$

For each $R \in \text{Rol}^{\text{src}}$, define the total local reception space derived from that source-side role type by

$$\mathcal{R}_R := \bigsqcup_{u \in U} (\{u\} \times \text{Rcv}_{R,u})$$

Definition 3.6 (Raw Operational Input). Define the reception network by

$$\text{Rcv}_U := \left(U, (\text{RcvCar}_u, (\text{Rcv}_{R,u})_{R \in \text{Rol}^{\text{src}}}, \preceq_u^{\text{rcv}})_{u \in U} \right)$$

Define the raw operational input by

$$\mathcal{I}^{\text{raw}} := (\text{Rl}[Z_0], \text{Car}_{\text{phys}}[Z_0], (\text{Rlz}_R)_{R \in \text{Rol}^{\text{rlz}}}, \text{Rlz}_{\mathfrak{B}}[Z_0], \text{Rcv}_U)$$

3.6.2 Source-Side Realization Datum

Source Structures on the Clock, Record, and Object-State Sides For the $\mathfrak{C}$ -derived source indices, let the nonempty set be

$$\text{Idx}_{\mathfrak{C}}^{\text{src}}$$

and denote the $\mathfrak{C}$ -derived source family by

$$(\text{Src}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}))_{j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}}$$

Each $\text{Src}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$ is supported by a physical carrier belonging to $\text{Car}_{\mathfrak{C}}$, supplies the internal counter positions and updates of Definition 2.2, and is a $\mathfrak{C}$ -derived physical source. The index $j_{\mathfrak{C}}^{\text{src}}$ marks a difference in physical carrier, internal update rule, initial internal position, connection path, or operational sequence.

Each $\text{Src}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$ realizes the cycle-count source of Definition 2.2 as a $\mathfrak{C}$ -derived source.

Definition 3.7 ($\mathfrak{C}$ -Derived Source-Token Family). For each

$$j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}, \quad k \in \mathbb{N}$$

the internal counter position k has a corresponding $\mathfrak{C}$ -derived individual token, denoted by

$$\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k)$$

For each $j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$, define

$$\text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}) := \{\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) \mid k \in \mathbb{N}\}$$

Within the same source, the indexed token family satisfies

$$\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, \ell) \iff k = \ell$$

Define the aggregate $\mathfrak{C}$ -derived source-token set by

$$\text{Tok}_{\mathfrak{C}}^* := \bigsqcup_{j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}} \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$$

Then

$$\begin{aligned} \forall x_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*, \\ \exists! (j_{\mathfrak{C}}^{\text{src}}, k) \in \text{Idx}_{\mathfrak{C}}^{\text{src}} \times \mathbb{N} \\ x_{\mathfrak{C}} = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) \end{aligned}$$

holds. Let the source-index map that returns the first component of this unique pair be

$$\text{idx}_{\mathfrak{C}}^{\text{src}}(\cdot) : \text{Tok}_{\mathfrak{C}}^* \rightarrow \text{Idx}_{\mathfrak{C}}^{\text{src}}$$

defined by

$$\text{idx}_{\mathfrak{C}}^{\text{src}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k)) := j_{\mathfrak{C}}^{\text{src}}$$

Definition 3.8 ($\mathfrak{C}$ -Derived Source-Internal Order). For each

$$j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$$

define the binary relation

$$\left(\preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} \right) \subseteq \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}) \times \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$$

for all $k, \ell \in \mathbb{N}$ by

$$\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, \ell) \iff k \leq \ell$$

Proposition 3.9 (Partial-Order Property of the $\mathfrak{C}$ -Derived Source-Internal Order). For each

$$j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$$

the relation

$$\preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}}$$

is a partial order on

$$\text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$$

Proof. Fix $j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$. For arbitrary

$$x_{\mathfrak{C}}, y_{\mathfrak{C}}, z_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$$

Definition 3.7 gives unique $k, \ell, m \in \mathbb{N}$ such that

$$x_{\mathfrak{C}} = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k), \quad y_{\mathfrak{C}} = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, \ell), \quad z_{\mathfrak{C}} = \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, m)$$

Reflexivity $k \leq k$ on the natural numbers and Definition 3.8 give $x_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} x_{\mathfrak{C}}$. If the two comparisons $x_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} y_{\mathfrak{C}}$ and $y_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} z_{\mathfrak{C}}$ hold, the same definition gives $k \leq \ell \leq m$. Transitivity on the natural numbers gives $k \leq m$, and hence $x_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} z_{\mathfrak{C}}$.

If $x_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} y_{\mathfrak{C}}$ and $y_{\mathfrak{C}} \preceq_{*, j_{\mathfrak{C}}^{\text{src}}}^{\mathfrak{C}} x_{\mathfrak{C}}$ hold, then $k \leq \ell$ and $\ell \leq k$. Antisymmetry on the natural numbers gives $k = \ell$, and token identity in the same definition gives $x_{\mathfrak{C}} = y_{\mathfrak{C}}$. The relation is therefore a partial order. $\square$

Definition 3.10 (Displays of $\mathfrak{C}$ -Derived Sources). Let the set of $\mathfrak{C}$ -derived display labels be

$$\text{DispLbl}_{\mathfrak{C}}$$

For each $j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$, define the display map

$$\phi_{\mathfrak{C}, j_{\mathfrak{C}}^{\text{src}}} : \text{Tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}) \rightarrow \text{DispLbl}_{\mathfrak{C}}$$

Define the aggregate display map

$$\phi_{\mathfrak{C}} : \text{Tok}_{\mathfrak{C}}^* \rightarrow \text{DispLbl}_{\mathfrak{C}}$$

by

$$\phi_{\mathfrak{C}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k)) := \phi_{\mathfrak{C}, j_{\mathfrak{C}}^{\text{src}}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k))$$

For arbitrary

$$j_{\mathfrak{C}}^{\text{src}}, \tilde{j}_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}, \\ k, \ell \in \mathbb{N}$$

if

$$\phi_{\mathfrak{C}, j_{\mathfrak{C}}^{\text{src}}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k)) = \phi_{\mathfrak{C}, \tilde{j}_{\mathfrak{C}}^{\text{src}}}(\text{tok}_{\mathfrak{C}}(\tilde{j}_{\mathfrak{C}}^{\text{src}}, \ell))$$

and

$$(j_{\mathfrak{C}}^{\text{src}}, k) \neq (\tilde{j}_{\mathfrak{C}}^{\text{src}}, \ell)$$

then

$$\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k) \neq \text{tok}_{\mathfrak{C}}(\tilde{j}_{\mathfrak{C}}^{\text{src}}, \ell)$$

The display label gives the displayed content of an individual $\mathfrak{C}$ -derived token, while source membership is given by the source-index map $\text{idx}_{\mathfrak{C}}^{\text{src}}(\cdot)$ in Definition 3.7. The display-label set and display maps assign no time variable, time interval, period, frequency, or second to a token.

Definition 3.11 (Internal Counter of $\mathfrak{D}$ -Derived Tokens). Let the nonempty set of $\mathfrak{D}$ -derived source tokens generated and retained by physical $\mathfrak{D}$ -derived record carriers be

$$\text{Tok}_{\mathfrak{D}}^*$$

Each $\mathfrak{D}$ -derived token is an individual record on a physical record carrier belonging to $\text{Car}_{\mathfrak{D}}$. Let the set of local record contents consisting of $\mathfrak{C}$ -derived display content, $\mathfrak{B}$ -derived object-specification content, and the correspondence between them on the $\mathfrak{D}$ -derived side be

$$\text{Cont}_{\mathfrak{D}}$$

and define the record-content map by

$$\phi_{\mathfrak{D}} : \text{Tok}_{\mathfrak{D}}^* \rightarrow \text{Cont}_{\mathfrak{D}}$$

Define the internal record-index map by

$$\text{seq}_{\mathfrak{D}} : \text{Tok}_{\mathfrak{D}}^* \rightarrow \mathbb{N}$$

Each $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$ is uniquely specified by the pair comprising the record content $\phi_{\mathfrak{D}}(x_{\mathfrak{D}})$ and the internal record index $\text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}})$, and for arbitrary $x_{\mathfrak{D}}, y_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$,

$$x_{\mathfrak{D}} = y_{\mathfrak{D}} \iff \phi_{\mathfrak{D}}(x_{\mathfrak{D}}) = \phi_{\mathfrak{D}}(y_{\mathfrak{D}}) \quad \text{and} \quad \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) = \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}})$$

holds.

Definition 3.12 ($\mathfrak{D}$ -Derived Record-Index Order). On the aggregate $\mathfrak{D}$ -derived source-token set

$$\text{Tok}_{\mathfrak{D}}^*$$

of Definition 3.11, define the binary relation

$$(\preceq_*^{\mathfrak{D}}) \subseteq \text{Tok}_{\mathfrak{D}}^* \times \text{Tok}_{\mathfrak{D}}^*$$

for arbitrary

$$x_{\mathfrak{D}}, y_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$$

by

$$x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} y_{\mathfrak{D}} \iff x_{\mathfrak{D}} = y_{\mathfrak{D}} \quad \text{or} \quad \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}})$$

Proposition 3.13 (Partial-Order Property of the $\mathfrak{D}$ -Derived Record-Index Order). The relation

$$\preceq_*^{\mathfrak{D}}$$

is a partial order on

$$\mathbf{Tok}_{\mathfrak{D}}^*$$

Proof. Take arbitrary $x_{\mathfrak{D}}, y_{\mathfrak{D}}, z_{\mathfrak{D}} \in \mathbf{Tok}_{\mathfrak{D}}^*$. Since $x_{\mathfrak{D}} = x_{\mathfrak{D}}$, Definition 3.12 gives $x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} x_{\mathfrak{D}}$.

Suppose that $x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} y_{\mathfrak{D}}$ and $y_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} z_{\mathfrak{D}}$. If either of the two disjunctions supplied by Definition 3.12 takes the equality branch, substituting that equality into the other disjunction gives $x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} z_{\mathfrak{D}}$. If both take the strict-order branch, then

$$\text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(z_{\mathfrak{D}})$$

and transitivity of the strict order on the natural numbers gives the same conclusion. Hence $\preceq_*^{\mathfrak{D}}$ is transitive.

Suppose that comparison holds in both directions. If $x_{\mathfrak{D}} \neq y_{\mathfrak{D}}$, Definition 3.12 gives simultaneously

$$\begin{aligned} \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) &< \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}}), \\ \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}}) &< \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) \end{aligned}$$

contrary to asymmetry of the strict order on the natural numbers. Thus $x_{\mathfrak{D}} = y_{\mathfrak{D}}$, and $\preceq_*^{\mathfrak{D}}$ is antisymmetric. $\square$

Corollary 3.14 (Incomparability of Distinct $\mathfrak{D}$ -Derived Tokens with the Same Record Index). For arbitrary

$$x_{\mathfrak{D}}, y_{\mathfrak{D}} \in \mathbf{Tok}_{\mathfrak{D}}^*$$

if

$$x_{\mathfrak{D}} \neq y_{\mathfrak{D}}, \quad \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) = \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}})$$

then

$$\begin{aligned} \neg(x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} y_{\mathfrak{D}}), \\ \neg(y_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} x_{\mathfrak{D}}) \end{aligned}$$

Proof. If $x_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} y_{\mathfrak{D}}$, then $x_{\mathfrak{D}} \neq y_{\mathfrak{D}}$, and Definition 3.12 gives

$$\text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}})$$

contrary to equality of the record indices. If $y_{\mathfrak{D}} \preceq_*^{\mathfrak{D}} x_{\mathfrak{D}}$, the same definition gives $\text{seq}_{\mathfrak{D}}(y_{\mathfrak{D}}) < \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}})$, again contrary to equality of the record indices. $\square$

Definition 3.15 ($\mathfrak{D}$ -Derived Record Cell). For each $x_{\mathfrak{D}} \in \mathbf{Tok}_{\mathfrak{D}}^*$, let the set of $\mathfrak{D}$ -derived tokens retained by a physical $\mathfrak{D}$ -derived record carrier as the same finite record unit be

$$\begin{aligned} \emptyset \neq W_{\mathfrak{D}}(x_{\mathfrak{D}}) &\subseteq \mathbf{Tok}_{\mathfrak{D}}^*, \\ |W_{\mathfrak{D}}(x_{\mathfrak{D}})| &< \infty \end{aligned}$$

and require

$$x_{\mathfrak{D}} \in W_{\mathfrak{D}}(x_{\mathfrak{D}})$$

Further require, for arbitrary $y_{\mathfrak{D}} \in \mathbf{Tok}_{\mathfrak{D}}^*$,

$$y_{\mathfrak{D}} \in W_{\mathfrak{D}}(x_{\mathfrak{D}}) \iff W_{\mathfrak{D}}(y_{\mathfrak{D}}) = W_{\mathfrak{D}}(x_{\mathfrak{D}})$$

Call $W_{\mathfrak{D}}(x_{\mathfrak{D}})$ the record cell containing $x_{\mathfrak{D}}$ on the $\mathfrak{D}$ -derived side.

This condition partitions the aggregate $\mathfrak{D}$ -derived source-token set into pairwise disjoint record cells. Each record cell is a group of $\mathfrak{D}$ -derived tokens retained as the same finite record unit.

Definition 3.16 ($\mathfrak{B}$ -Derived Source Tokens and Object-Specification Positions). Let the nonempty set of $\mathfrak{B}$ -derived source tokens supplied by physical $\mathfrak{B}$ -derived tracking objects be

$$\text{Tok}_{\mathfrak{B}}^*$$

Define the object-specification position map by

$$\zeta_{\mathfrak{B}} : \text{Tok}_{\mathfrak{B}}^* \rightarrow Z_0$$

Each $x_{\mathfrak{B}} \in \text{Tok}_{\mathfrak{B}}^*$ is supplied by the physical tracking object that is the first component of the abstraction-source correspondence retained by

$$\text{Rl}_{\mathfrak{B}}(\zeta_{\mathfrak{B}}(x_{\mathfrak{B}})) = (\text{Car}_{\mathfrak{B}}(\zeta_{\mathfrak{B}}(x_{\mathfrak{B}})), \text{Rl}_{\mathfrak{B}}(\zeta_{\mathfrak{B}}(x_{\mathfrak{B}})))$$

and is a source-side token representing the state of that object. Multiple $\mathfrak{B}$ -derived source tokens may be supplied from the same object-specification position.

Definition 3.17 (Direct-Sum Order on the $\mathfrak{C}$ -Derived Source Family). On the aggregate $\mathfrak{C}$ -derived source-token set

$$\text{Tok}_{\mathfrak{C}}^*$$

of Definition 3.7, define the binary relation

$$(\preceq_{*}^{\mathfrak{C}}) \subseteq \text{Tok}_{\mathfrak{C}}^* \times \text{Tok}_{\mathfrak{C}}^*$$

for arbitrary

$$x_{\mathfrak{C}}, y_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*$$

by

$$x_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} y_{\mathfrak{C}} \iff \text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}}) = \text{idx}_{\mathfrak{C}}^{\text{src}}(y_{\mathfrak{C}}) \quad \text{and} \quad x_{\mathfrak{C}} \preceq_{*, \text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}})}^{\mathfrak{C}} y_{\mathfrak{C}}$$

Proposition 3.18 (Partial-Order Property of the Direct-Sum Order on the $\mathfrak{C}$ -Derived Source Family). The relation

$$\preceq_{*}^{\mathfrak{C}}$$

is a partial order on

$$\text{Tok}_{\mathfrak{C}}^*$$

Proof. Take arbitrary $x_{\mathfrak{C}}, y_{\mathfrak{C}}, z_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*$. By Definition 3.7, $x_{\mathfrak{C}}$ belongs to a unique $\mathfrak{C}$ -derived source component. Reflexivity in Proposition 3.9 and Definition 3.17 give $x_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} x_{\mathfrak{C}}$.

Suppose $x_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} y_{\mathfrak{C}}$ and $y_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} z_{\mathfrak{C}}$. By Definition 3.17, the three tokens belong to the same $\mathfrak{C}$ -derived source component, and both comparisons hold in the same within-component order. Transitivity in Proposition 3.9 gives

$$x_{\mathfrak{C}} \preceq_{*, \text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}})}^{\mathfrak{C}} z_{\mathfrak{C}}$$

and Definition 3.17 therefore gives $x_{\mathfrak{C}} \preceq_{*}^{\mathfrak{C}} z_{\mathfrak{C}}$.

If comparisons in both directions hold, both tokens belong to the same $\mathfrak{C}$ -derived source component, and comparisons in both directions also hold within that component. Antisymmetry in Proposition 3.9 makes the two tokens equal. Thus $\preceq_{*}^{\mathfrak{C}}$ is a partial order. $\square$

Corollary 3.19 (Incomparability between Distinct $\mathfrak{C}$ -Derived Sources). For arbitrary

$$x_{\mathfrak{C}}, y_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*$$

if

$$\text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}}) \neq \text{idx}_{\mathfrak{C}}^{\text{src}}(y_{\mathfrak{C}})$$

then

$$\begin{aligned} &\neg(x_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} y_{\mathfrak{C}}), \\ &\neg(y_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} x_{\mathfrak{C}}) \end{aligned}$$

Proof. By Definition 3.17, each of $x_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} y_{\mathfrak{C}}$ and $y_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} x_{\mathfrak{C}}$ requires equality of the source-membership indices. The assumed inequality between the indices prevents either comparison from holding. $\square$

Definition 3.20 (Comparison-Side Arrival Map). For each $u \in U$ and $R \in \text{Rol}^{\text{src}}$, define a comparison-side arrival map with domain

$$\text{Dom}(\text{Arr}_{R,u}^{\text{cmp}}) \subseteq \text{Tok}_R^*$$

by

$$\text{Arr}_{R,u}^{\text{cmp}} : \text{Dom}(\text{Arr}_{R,u}^{\text{cmp}}) \longrightarrow \text{Rcv}_R$$

Each map sends R -derived source-side tokens to the observer-side reception set. If distinct source-side tokens map to the same reception, the cardinality of the inverse image of that reception is at least two.

Setting $R = \mathfrak{B}$ in Definition 3.20 makes $\mathfrak{B}$ -derived source tokens correspond to observer-side receptions.

Definition 3.21 (Comparison-Side Recovery Map). For each $u \in U$ and $R \in \text{Rol}^{\text{src}}$, define the uniquely recoverable reception set by

$$\text{Rcv}_{R,u}^{\text{uniq}} := \left\{ \alpha \in \text{Im}(\text{Arr}_R^{\text{cmp}}) \mid \left| (\text{Arr}_R^{\text{cmp}})^{-1}(\{\alpha\}) \right| = 1 \right\}$$

Let the comparison-side recovery map be

$$\text{Rec}_{R,u}^{\text{cmp}} : \text{Rcv}_R^{\text{uniq}} \longrightarrow \text{Tok}_R^*$$

and define it, for each $\alpha \in \text{Rcv}_R^{\text{uniq}}$, by

$$(\text{Arr}_R^{\text{cmp}})^{-1}(\{\alpha\}) = \{\text{Rec}_R^{\text{cmp}}(\alpha)\}$$

When the arrival value of a $\mathfrak{C}$ -derived token $x_{\mathfrak{C}}$ is uniquely recoverable,

$$\text{idx}_{\mathfrak{C}}^{\text{src}}(\text{Rec}_{\mathfrak{C},u}^{\text{cmp}}(\text{Arr}_{\mathfrak{C},u}^{\text{cmp}}(x_{\mathfrak{C}}))) = \text{idx}_{\mathfrak{C}}^{\text{src}}(x_{\mathfrak{C}})$$

holds. When the arrival value of a $\mathfrak{B}$ -derived token $x_{\mathfrak{B}}$ is uniquely recoverable,

$$\zeta_{\mathfrak{B}}(\text{Rec}_{\mathfrak{B},u}^{\text{cmp}}(\text{Arr}_{\mathfrak{B},u}^{\text{cmp}}(x_{\mathfrak{B}}))) = \zeta_{\mathfrak{B}}(x_{\mathfrak{B}})$$

holds.

Definition 3.22 (Comparison-Stage Two-Role Relation). For each

$$u \in U$$

define the comparison-stage relation derived from the two comparison roles by

$$\text{Rel}_u^{\text{cmp}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Rcv}_{R,u}$$

Its element

$$(\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}$$

indexes by role type the clock-side reception and record-side reception that constitute the same comparison material at observer-side reception phase u .

Definition 3.23 (Source-Side Two-Role Relation). Define the source-side relation derived from the two comparison roles by

$$\text{Rel}^{\text{src}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Tok}_R^*$$

For each

$$u \in U$$

and

$$(x_R)_{R \in \text{Rol}^{\text{cmp}}} \in \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Dom}(\text{Arr}_{R,u}^{\text{cmp}})$$

require the source-side and comparison-side relations to satisfy the consistency condition

$$(x_R)_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{src}} \iff (\text{Arr}_{R,u}^{\text{cmp}}(x_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}_u^{\text{cmp}}$$

Definition 3.24 (Source-Side Realization Datum). Define the source-side token families, display and record data, source-side orders, families of comparison-side arrival maps, families of comparison-side recovery maps, and comparison-side and source-side relations, respectively, by

$$\text{Src}_* := (\text{Tok}_R^*)_{R \in \text{Rol}^{\text{src}}},$$

$$\text{DispRec}_* := (\phi_{\mathfrak{C}}, \phi_{\mathfrak{D}}, \text{seq}_{\mathfrak{D}}, \zeta_{\mathfrak{B}}, \{W_{\mathfrak{D}}(x_{\mathfrak{D}})\}_{x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*}),$$

$$\text{SOrd}_* := (\preceq_{\mathfrak{C}}^*, \preceq_{\mathfrak{D}}^*),$$

$$\text{Arr}_U^{\text{cmp}} := (\text{Arr}_{R,u}^{\text{cmp}})_{(R,u) \in \text{Rol}^{\text{src}} \times U},$$

$$\text{Rec}_U^{\text{cmp}} := (\text{Rec}_{R,u}^{\text{cmp}})_{(R,u) \in \text{Rol}^{\text{src}} \times U},$$

and define the source-side realization datum by

$$\text{Dat}^{\text{src}} := (\text{Src}_*, \text{DispRec}_*, \text{SOrd}_*, \text{Arr}_U^{\text{cmp}}, \text{Rec}_U^{\text{cmp}}, ((\text{Rel}_u^{\text{cmp}})_{u \in U}, \text{Rel}^{\text{src}}))$$

3.6.3 Realized Operational Configuration

Definition 3.25 (Realized Operational Configuration). Define the realized operational configuration as the ordered pair comprising the raw operational input of Definition 3.6 and the source-side realization datum of Definition 3.24:

$$\mathcal{X} := (\mathcal{I}^{\text{raw}}, \text{Dat}^{\text{src}})$$

Let the realized operational configuration $\mathcal{X}$ be the domain-side input to the comparison construction.

Proposition 3.26 (Component Specification of the Realized Operational Configuration). The realized operational configuration $\mathcal{X}$ of Definition 3.25 specifies as typed components the family of role-instance bundles that retains object-specification and role-type positions, the physical-carrier realization, the observer-side reception network, the Rol^{src} -derived source-token families, display and record data, source-side orders, comparison-side arrival maps, comparison-side recovery maps, and comparison-side/source-side relations. The input data of this construction contain no time variable, time interval, period, frequency, or second.

Proof. Definition 3.6 defines as the raw operational input the family of role-instance bundles over the family of object-specification indices, the family of physical-layer constructions, the Rol^{rlz} -derived realization maps, the family of $\mathfrak{B}$ -derived abstraction-source correspondences, and the observer-side reception network.

Definition 3.24 defines as the source-side realization datum the Rol^{src} -derived source-token families, display and record data, source-side orders, the Rol^{src} -derived families of comparison-side arrival maps, the Rol^{src} -derived families of comparison-side recovery maps, and the Rol^{cmp} -derived comparison-side/source-side relations. Definition 3.25 retains these two objects as the ordered pair $\mathcal{X}$.

The object-specification positions, role-type positions, physical realizations, receptions, source tokens, display and record data, orders, arrivals, recoveries, and correspondences are therefore specified as typed components of $\mathcal{X}$. Each component is constructed from internal positions, natural-number indices, maps, relations, and reception orders, without placing a time quantity in the input. $\square$

The raw operational input of Definition 3.6 couples the observer-side reception network to the physically realized input supplied by the family of physical-layer constructions in Definition 3.1 and the sequence of role-typed correspondences between the abstract and physical layers in Definition 3.2. The source-side realization datum of Definition 3.24 retains the Rol^{src} -derived source-token families, display and record data, source-side orders, comparison-side arrival maps, comparison-side recovery maps, and comparison-side/source-side relations.

4 Comparison-Stage Closure and Canonical Representation

4.1 Generation of the Canonical Comparison Quotient

4.1.1 Definition of the Comparison-Stage Construction

From the realized operational configuration, extract the reception network used at the comparison stage, the source-side token families, the comparison-arrival maps, the comparison-

side recovery maps, and the comparison-side and source-side relations; combine these with the typed family of comparison rules to construct the comparison-stage material datum. From the comparison-admissible inputs specified by the family of comparison rules, define source-identification partial maps, comparison-transport components, typed finite words and their induced partial actions, realizations of local closed-loop specifications, comparison-reachability equivalence relations, component quotients, canonical projections, and comparison two-role relations on the quotients. When the seven comparison-stage conditions and the condition on all pure-transport closed loops hold, construct the comparison-gluing datum and the canonical comparison quotient from the generated comparison-stage objects.

In this chapter, the compound superscript (cmp, σ) on an object specific to the comparison stage is abbreviated to σ only when the symbol root together with σ uniquely determines its comparison-stage membership. To distinguish it from the source-side relation Rel^{src} , however, $\text{Rel}^{\text{cmp}, \text{src}}$ retains its complete form.

For each symbol root, one pair consisting of a role type and a reception phase first appears in complete form. Thereafter, only the reception-phase subscript is omitted within the same definition, proposition, or proof when the reception phase is fixed and the complete form is uniquely recoverable. The complete forms are retained for the role type, a varying family index, distinct reception phases, and residual modifiers that select genuine branches.

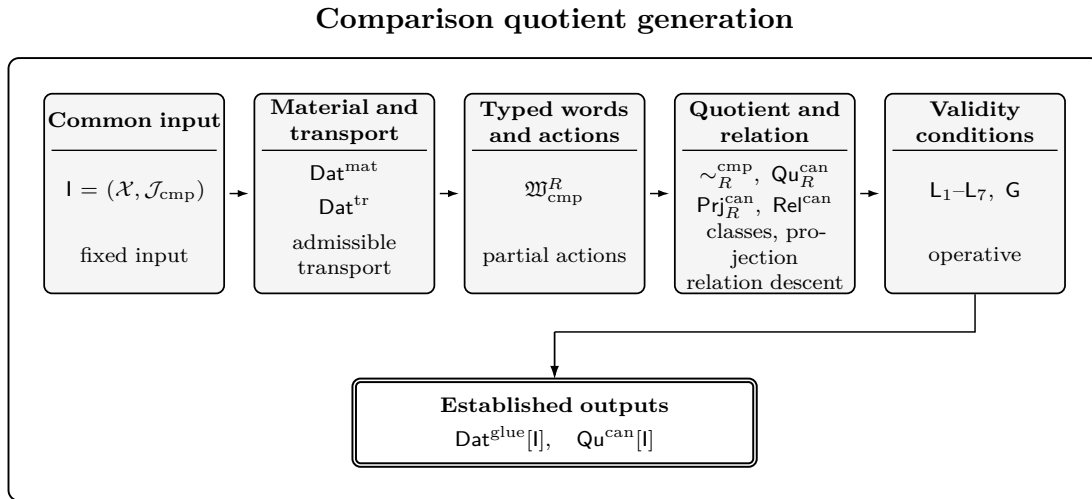


Figure 6: Comparison-stage dependency structure from the comparison-rule-dependent common input to the comparison-gluing datum and the canonical comparison quotient. From the comparison-stage input, comparison-admissible inputs, comparison-transport components, typed finite words, comparison-reachability equivalence relations, component quotients, canonical projections, and comparison two-role relations on the quotients are determined in dependency order. The seven comparison-stage conditions and the condition on all pure-transport closed loops determine whether each construction stage is established.

Definition 4.1 (Comparison-Rule-Dependent Common Input). For the realized operational configuration

$$\mathcal{X} = (\mathcal{I}^{\text{raw}}, \text{Dat}^{\text{src}})$$

of Definition 3.25 and the typed family of comparison rules

$$\mathcal{J}_{\text{cmp}}$$

define the comparison-rule-dependent common input by

$$\mathbf{l} := (\mathcal{X}, \mathcal{J}_{\text{cmp}})$$

For any object F dependent on this input, define the evaluation notation by

$$F[\mathbf{l}] := F(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

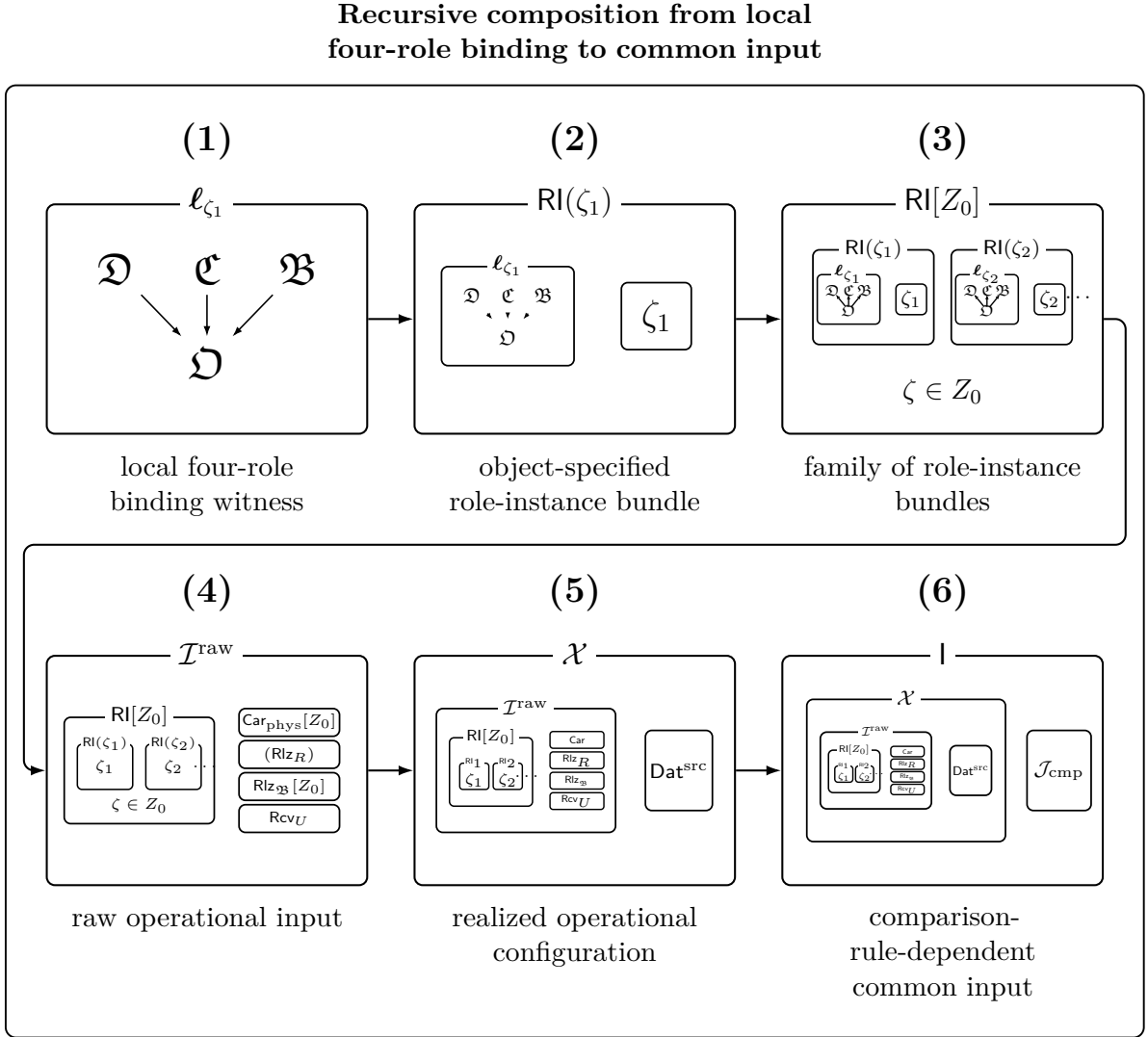


Figure 7: Recursive composition from the local four-role binding to the comparison-rule-dependent common input. An object defined as a whole at one stage is retained as a structured component at the next stage. The construction proceeds from the local four-role binding witness ℓ_{ζ_1} , through the object-specified role-instance bundle $\text{RI}(\zeta_1)$, the family $\text{RI}[Z_0]$, the raw operational input $\mathcal{I}^{\text{raw}}$, and the realized operational configuration $\mathcal{X}$, to the comparison-rule-dependent common input $\mathbf{l} = (\mathcal{X}, \mathcal{J}_{\text{cmp}})$.

Definition 4.2 (Comparison-Stage Material Datum). Let the comparison-rule-dependent common input of Definition 4.1 be

$$\mathbf{l} = (\mathcal{X}, \mathcal{J}_{\text{cmp}})$$

Define the comparison-stage material datum by

$$\begin{aligned} \text{Dat}^{\text{mat}}[\mathbb{I}] \\ := (\text{Rcv}_U, \text{Src}_*, \text{Arr}_U^{\text{cmp}}, \text{Rec}_U^{\text{cmp}}, ((\text{Rel}_u^{\text{cmp}})_{u \in U}, \text{Rel}^{\text{src}}), \mathcal{J}_{\text{cmp}}) \end{aligned}$$

The first component is the observer-side reception network, the second is the source-side token family, the third is the family of comparison-arrival maps, the fourth is the family of comparison-side recovery maps, and the fifth is the ordered pair comprising the family of comparison-side relations and the source-side relation. The sixth component is the typed family of comparison rules that specifies comparison-admissible position pairs, input synchronized-pair joint-transport relations, and local pure-transport closed-loop specifications.

4.1.2 Comparison-Stage Decision Conditions

Definition 4.3 (Comparison-Admissible Position Pairs and Input Synchronized-Pair Joint-Transport Relations). Let the comparison-stage material datum of Definition 4.2 be

$$\text{Dat}^{\text{mat}}[\mathbb{I}]$$

Let the set of comparison-admissible position pairs specified by the typed family of comparison rules

$$\mathcal{J}_{\text{cmp}}$$

be

$$\text{Adm}_{\text{cmp}}[\mathbb{I}] \subseteq U \times U$$

For each

$$(u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}]$$

let the input synchronized-pair joint-transport relation specified by the typed family of comparison rules be

$$\begin{aligned} \text{Tr}_{uv}^{\text{pair}}[\mathbb{I}] \\ \subseteq \left(\prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}) \right) \times \left(\prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,v}^{\text{cmp}}) \right) \end{aligned}$$

The flattening of an element of this product space is given by

$$((\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}, (\beta_R)_{R \in \text{Rol}^{\text{cmp}}}) \longleftrightarrow (\alpha_{\mathfrak{C}}, \alpha_{\mathfrak{D}}, \beta_{\mathfrak{C}}, \beta_{\mathfrak{D}})$$

Definition 4.4 (Family of Local Pure-Transport Closed-Loop Specifications). Let the comparison-stage material datum of Definition 4.2 and the set of comparison-admissible position pairs of Definition 4.3 be given. For each

$$R \in \text{Rol}^{\text{cmp}}$$

let the family of local pure-transport closed-loop specifications specified by the typed family of comparison rules

$$\mathcal{J}_{\text{cmp}}$$

be

$$\text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$$

The family of local pure-transport closed-loop specifications satisfies

$$\begin{aligned}
& \text{LoopSpec}_{\text{loc}}^R [I] \\
& \subseteq \left\{ (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m, \text{Dom}_{\text{Spec}}) \mid \right. \\
& m \in \mathbb{N}, \quad m \geq 1, \\
& (u_j)_{j=0}^m \in U^{m+1}, \quad u_m = u_0, \\
& (\varepsilon_j)_{j=1}^m \in \{+, -\}^m, \\
& \forall j \in \{1, \dots, m\}, \\
& \quad (\varepsilon_j = + \wedge (u_{j-1}, u_j) \in \text{Adm}_{\text{cmp}} [I]) \\
& \quad \vee (\varepsilon_j = - \wedge (u_j, u_{j-1}) \in \text{Adm}_{\text{cmp}} [I]), \\
& \left. \text{Dom}_{\text{Spec}} \subseteq \{u_0\} \times \text{Im}(\text{Arr}_{R, u_0}^{\text{cmp}}) \right\}.
\end{aligned}$$

For each

$$\text{Spec} = (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m, \text{Dom}_{\text{Spec}}) \in \text{LoopSpec}_{\text{loc}}^R [I]$$

define the base point of the local closed-loop specification by

$$\text{Base}(\text{Spec}) := u_0$$

Definition 4.5 (Comparison-Stage Decision Condition L_1 : Local Source Identification). Define the local source-identification condition by

$$L_1(\mathcal{X}) \iff \forall R \in \text{Rol}^{\text{cmp}} \quad \forall u \in U, \quad \text{Im}(\text{Arr}_{R, u}^{\text{cmp}}) \subseteq \text{Rcv}_{R, u}^{\text{uniq}}.$$

Lemma 4.6 (Local source identification and mutually inverse arrival and recovery maps). The condition $L_1(\mathcal{X})$ is equivalent to injectivity of the comparison-side arrival map on its domain for every $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$. Under this condition, for every R, u and every

$$x_R \in \text{Dom}(\text{Arr}_{R, u}^{\text{cmp}}), \quad \alpha_{R, u} \in \text{Im}(\text{Arr}_{R, u}^{\text{cmp}}),$$

the following relations hold.

$$\text{Rec}_{R, u}^{\text{cmp}} (\text{Im}(\text{Arr}_{R, u}^{\text{cmp}})) \subseteq \text{Dom}(\text{Arr}_{R, u}^{\text{cmp}}),$$

$$\text{Rec}_{R, u}^{\text{cmp}} (\text{Arr}_{R, u}^{\text{cmp}}(x_R)) = x_R,$$

$$\text{Arr}_{R, u}^{\text{cmp}} (\text{Rec}_{R, u}^{\text{cmp}}(\alpha_{R, u})) = \alpha_{R, u}.$$

Proof. By Definition 3.21, $\text{Rcv}_{R, u}^{\text{uniq}}$ consists of those points in the image of the comparison-side arrival map whose inverse images are singletons. Containment of the entire arrival image in this set is equivalent to injectivity on the domain. Under $L_1(\mathcal{X})$, the recovery map is defined on the entire arrival image and returns the unique preimage of each arrival point. Its value therefore belongs to the arrival-map domain, and the two composition identities follow. $\square$

Definition 4.7 (Comparison-Stage Condition-Family Evaluation Specification). Define the comparison-stage condition-family evaluation specification by

$$\text{Ev}^{\text{cmp}} := (\text{Ev}_i^{\text{cmp}})_{i=1}^7$$

Define each component by

$$\text{Ev}_i^{\text{cmp}} := \begin{cases} \mathcal{X}, & i = 1, \\ 1, & i \in \{2, \dots, 7\} \end{cases}$$

For each

$$i \in \{1, \dots, 7\}$$

define the bracket evaluation of a comparison-stage condition by

$$\text{L}_i[\text{Ev}_i^{\text{cmp}}] := \begin{cases} \text{L}_1(\mathcal{X}), & i = 1, \\ \text{L}_i[1], & i \in \{2, \dots, 7\} \end{cases}$$

Define the first-stage established-condition bundle by

$$\text{L}_{\leq 1}[\text{Ev}^{\text{cmp}}] := \text{L}_1[\text{Ev}_1^{\text{cmp}}]$$

Definition 4.8 (Source-Matching Partial Maps). Assume $\text{L}_1(\mathcal{X})$. For arbitrary

$$R \in \text{Rol}^{\text{cmp}}, \quad u, v \in U$$

define

$$\text{Dom}(\text{SrcMatch}_{uv}^R) := \left\{ \alpha_{R,u} \in \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}) \mid \text{Rec}_{R,u}^{\text{cmp}}(\alpha_{R,u}) \in \text{Dom}(\text{Arr}_{R,v}^{\text{cmp}}) \right\}$$

Define the source-matching partial map

$$\text{SrcMatch}_{uv}^R : \text{Dom}(\text{SrcMatch}_{uv}^R) \longrightarrow \text{Im}(\text{Arr}_{R,v}^{\text{cmp}})$$

by

$$\text{SrcMatch}_{uv}^R(\alpha_{R,u}) := \text{Arr}_{R,v}^{\text{cmp}}(\text{Rec}_{R,u}^{\text{cmp}}(\alpha_{R,u}))$$

Lemma 4.9 (Mutual Inverseness and Partial Injectivity of the Source-Matching Partial Maps). Under $\text{L}_1(\mathcal{X})$, for arbitrary

$$R \in \text{Rol}^{\text{cmp}}, \quad u, v \in U$$

and

$$(u_0, u_1) \in \{(u, v), (v, u)\}$$

one has

$$\text{Im}(\text{SrcMatch}_{u_0 u_1}^R) = \text{Dom}(\text{SrcMatch}_{u_1 u_0}^R)$$

and

$$\text{SrcMatch}_{u_1 u_0}^R \circ \text{SrcMatch}_{u_0 u_1}^R = \text{id}_{\text{Dom}(\text{SrcMatch}_{u_0 u_1}^R)}.$$

Consequently, SrcMatch_{uv}^R and SrcMatch_{vu}^R are partially injective and are mutually inverse partial maps.

Proof. Take $R \in \mathbf{Rol}^{\text{cmp}}$, $u, v \in U$, and

$$(u_0, u_1) \in \{(u, v), (v, u)\}$$

Take arbitrary

$$\alpha_{R, u_0} \in \text{Dom}(\text{SrcMatch}_{u_0 u_1}^R)$$

and set

$$x_R := \text{Rec}_{R, u_0}^{\text{cmp}}(\alpha_{R, u_0})$$

By Definition 4.8,

$$x_R \in \text{Dom}(\text{Arr}_{R, u_1}^{\text{cmp}})$$

Lemma 4.6 also gives

$$x_R \in \text{Dom}(\text{Arr}_{R, u_0}^{\text{cmp}}).$$

Set

$$\alpha_{R, u_1} := \text{SrcMatch}_{u_0 u_1}^R(\alpha_{R, u_0}) = \text{Arr}_{R, u_1}^{\text{cmp}}(x_R)$$

Lemma 4.6 gives

$$\text{Rec}_{R, u_1}^{\text{cmp}}(\alpha_{R, u_1}) = x_R$$

and therefore

$$\alpha_{R, u_1} \in \text{Dom}(\text{SrcMatch}_{u_1 u_0}^R).$$

Furthermore,

$$\begin{aligned} \text{SrcMatch}_{u_1 u_0}^R(\alpha_{R, u_1}) &= \text{Arr}_{R, u_0}^{\text{cmp}}(\text{Rec}_{R, u_1}^{\text{cmp}}(\alpha_{R, u_1})) \\ &= \text{Arr}_{R, u_0}^{\text{cmp}}(x_R) \\ &= \text{Arr}_{R, u_0}^{\text{cmp}}(\text{Rec}_{R, u_0}^{\text{cmp}}(\alpha_{R, u_0})) \\ &= \alpha_{R, u_0}. \end{aligned}$$

This proves one direction of the asserted equality between the image and the domain of the reverse map, together with the composition identity.

The ordered pair (u_0, u_1) may be either (u, v) or (v, u) . Thus the preceding inclusion and composition identity hold in both directions. The two inclusions give equality between the image and the domain of the reverse map; the two composition identities show that the maps are mutually inverse partial maps and that each is partially injective. $\square$

Definition 4.10 (Set of Component-Factorization Tuples for Synchronized-Pair Joint Transport). For each

$$(u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}]$$

define the set of component-factorization tuples for synchronized-pair joint transport by

$$\begin{aligned}
& \text{TrFac}_{uv}^{\text{cmp}}[\mathbb{I}] \\
& := \left\{ \left(P_{uv}^{\mathcal{CD}}, (\text{Fac}_{R,uv}^{\text{tr}})_{R \in \text{Rol}^{\text{cmp}}} \right) \mid \right. \\
& \quad P_{uv}^{\mathcal{CD}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}), \\
& \quad \forall R \in \text{Rol}^{\text{cmp}}, \quad \text{Fac}_{R,uv}^{\text{tr}} : \text{pr}_R(P_{uv}^{\mathcal{CD}}) \hookrightarrow \text{Im}(\text{Arr}_{R,v}^{\text{cmp}}), \\
& \quad \left. \text{Tr}_{uv}^{\text{pair}}[\mathbb{I}] \right. \\
& \quad = \left\{ \left((\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}, (\text{Fac}_{R,uv}^{\text{tr}}(\alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \right) \mid \right. \\
& \quad \quad \left. (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \in P_{uv}^{\mathcal{CD}} \right\} \Bigg\}
\end{aligned}$$

Definition 4.11 (Comparison-Stage Decision Condition L_2 : Component Factorization of Synchronized-Pair Joint Transport). Define the synchronized-pair joint-transport component-factorization condition by

$$\begin{aligned}
& L_2[\mathbb{I}] \\
& \iff \left[L_{\leq 1}[\text{Ev}^{\text{cmp}}] \implies \forall (u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}], \right. \\
& \quad \left. \# \text{TrFac}_{uv}^{\text{cmp}}[\mathbb{I}] = 1 \right]
\end{aligned}$$

Define the second-stage established-condition bundle by

$$L_{\leq 2}[\text{Ev}^{\text{cmp}}] := L_{\leq 1}[\text{Ev}^{\text{cmp}}] \wedge L_2[\text{Ev}_2^{\text{cmp}}]$$

Suppose that

$$L_1(\mathcal{X})$$

and

$$L_2[\mathbb{I}]$$

hold. For each

$$(u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}]$$

define the unique role-indexed transport datum by

$$\begin{aligned}
& \text{Dat}_{uv}^{\text{tr}}[\mathbb{I}] \\
& := \left(\text{PairDom}_{uv}[\mathbb{I}], (\text{Tr}_{uv}^R[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}} \right)
\end{aligned}$$

The first component satisfies

$$\text{PairDom}_{uv}[\mathbb{I}] \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}})$$

For each

$$R \in \mathbf{Rol}^{\text{cmp}}$$

the second component satisfies

$$\begin{aligned} & \text{Tr}_{uv}^R [\mathbb{I}] \\ & : \text{pr}_R(\text{PairDom}_{uv} [\mathbb{I}]) \hookrightarrow \text{Im}(\text{Arr}_{R,v}^{\text{cmp}}) \end{aligned}$$

Moreover,

$$\begin{aligned} & \text{Tr}_{uv}^{\text{pair}} [\mathbb{I}] \\ & = \left\{ \left((\alpha_R)_{R \in \mathbf{Rol}^{\text{cmp}}}, (\text{Tr}_{uv}^R [\mathbb{I}] (\alpha_R))_{R \in \mathbf{Rol}^{\text{cmp}}} \right) \mid \right. \\ & \quad \left. (\alpha_R)_{R \in \mathbf{Rol}^{\text{cmp}}} \in \text{PairDom}_{uv} [\mathbb{I}] \right\}. \end{aligned}$$

Definition 4.12 (Comparison-Transport Datum). Assume

$$\mathbf{L}_1(\mathcal{X})$$

and

$$\mathbf{L}_2 [\mathbb{I}]$$

Define the comparison-transport datum by

$$\begin{aligned} & \text{Dat}^{\text{tr}} [\mathbb{I}] \\ & := (\text{Dat}_{uv}^{\text{tr}} [\mathbb{I}])_{(u,v) \in \text{Adm}_{\text{cmp}} [\mathbb{I}]} \end{aligned}$$

Definition 4.13 (Tagged Comparison-Arrival Reception Spaces). For each

$$R \in \mathbf{Rol}^{\text{cmp}}$$

define the tagged comparison-arrival reception space by

$$\mathcal{R}_R^{\text{arr}} := \bigsqcup_{u \in U} (\{u\} \times \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}))$$

Definition 4.14 (Tagged Lift of a Partial Injection). For arbitrary

$$u, v \in U$$

and a partial injection f , define the tagged lift

$$\begin{aligned} & \text{TagLift}_{uv}[f] \\ & : \{u\} \times \text{Dom}(f) \hookrightarrow \{v\} \times \text{Im}(f) \end{aligned}$$

by

$$\text{TagLift}_{uv}[f](u, x) := (v, f(x))$$

Lemma 4.15 (Compatibility of Tagged Partial-Injection Lifts with Inverses). For any partial injection f and its inverse f^{-1} on its image,

$$(\text{TagLift}_{uv}[f])^{-1} = \text{TagLift}_{vu}[f^{-1}].$$

Consequently,

$$\begin{aligned} & \text{TagLift}_{vu}[f^{-1}] \circ \text{TagLift}_{uv}[f] \\ &= \text{id}_{\text{Dom}(\text{TagLift}_{uv}[f])}, \\ & \text{TagLift}_{uv}[f] \circ \text{TagLift}_{vu}[f^{-1}] \\ &= \text{id}_{\text{Im}(\text{TagLift}_{uv}[f])}. \end{aligned}$$

Proof. The domains of the two sides coincide with

$$\{v\} \times \text{Im}(f)$$

For arbitrary

$$y \in \text{Im}(f)$$

Definition 4.14 gives

$$\begin{aligned} & (\text{TagLift}_{uv}[f])^{-1}(v, y) \\ &= (u, f^{-1}(y)) \\ &= \text{TagLift}_{vu}[f^{-1}](v, y). \end{aligned}$$

The two maps therefore coincide. The two composition identities follow from the definition of an inverse map. $\square$

Definition 4.16 (Pure-Transport Terms and Formal Inverse Terms). Assume

$$\mathbf{L}_1(\mathcal{X})$$

and

$$\mathbf{L}_2[\mathbb{I}]$$

For arbitrary

$$\begin{aligned} & R \in \text{Rol}^{\text{cmp}}, \\ & (u, v) \in \text{Adm}_{\text{cmp}}[\mathbb{I}] \end{aligned}$$

define the forward pure-transport term by

$$\begin{aligned} & \text{EdgeTerm}_{uv}^{\text{tr}, R, +}[\mathbb{I}] \\ &:= \text{TagLift}_{uv}[\text{Tr}_{uv}^R[\mathbb{I}]] \end{aligned}$$

Define the formal inverse pure-transport term by

$$\begin{aligned} & \text{EdgeTerm}_{uv}^{\text{tr}, R, -}[\mathbb{I}] \\ &:= (\text{EdgeTerm}_{uv}^{\text{tr}, R, +}[\mathbb{I}])^{-1} \\ &= \text{TagLift}_{vu}[(\text{Tr}_{uv}^R[\mathbb{I}])^{-1}] \end{aligned}$$

Lemma 4.17 (Mutual Inverseness of Pure-Transport Terms and Formal Inverse Terms). For each term in Definition 4.16, for the two choices of $\pm$ and the corresponding opposite sign $\mp$,

$$\begin{aligned} & \text{EdgeTerm}_{uv}^{\text{tr}, R, \mp}[\mathbb{I}] \circ \text{EdgeTerm}_{uv}^{\text{tr}, R, \pm}[\mathbb{I}] \\ &= \text{id}_{\text{Dom}(\text{EdgeTerm}_{uv}^{\text{tr}, R, \pm}[\mathbb{I}])}, \end{aligned}$$

Proof. Apply Lemma 4.15 to

$$f = \text{Tr}_{uv}^R [\text{I}]$$

□

Definition 4.18 (Source-Matching Terms). Assume

$$\mathbf{L}_1(\mathcal{X})$$

For arbitrary

$$R \in \text{Rol}^{\text{cmp}}, \quad u, v \in U$$

define the source-matching term by

$$\text{EdgeTerm}_{uv}^{\text{src}, R} := \text{TagLift}_{uv} [\text{SrcMatch}_{uv}^R]$$

Lemma 4.19 (Mutual Inverseness of Source-Matching Terms). For each term in Definition 4.18 and arbitrary

$$R \in \text{Rol}^{\text{cmp}}, \quad u, v \in U, \quad (u_0, u_1) \in \{(u, v), (v, u)\}$$

one has

$$\begin{aligned} \text{EdgeTerm}_{u_1 u_0}^{\text{src}, R} &= (\text{EdgeTerm}_{u_0 u_1}^{\text{src}, R})^{-1}, \\ \text{EdgeTerm}_{u_1 u_0}^{\text{src}, R} \circ \text{EdgeTerm}_{u_0 u_1}^{\text{src}, R} &= \text{id}_{\text{Dom}(\text{EdgeTerm}_{u_0 u_1}^{\text{src}, R})}. \end{aligned}$$

Because the domain of the inverse map is the image of the original map, the two ordered pairs in the second identity give the two composition identities.

Proof. Applying Lemma 4.9 to this ordered pair gives

$$\text{SrcMatch}_{u_1 u_0}^R = (\text{SrcMatch}_{u_0 u_1}^R)^{-1}.$$

Apply Lemma 4.15 to

$$f = \text{SrcMatch}_{u_0 u_1}^R$$

□

Definition 4.20 (Typed Finite Partial-Injection Sequence Action). For sets Y_0, Y_1, Y_2 and partial maps

$$\begin{aligned} f &: \text{Dom}(f) \subseteq Y_0 \longrightarrow Y_1, \\ g &: \text{Dom}(g) \subseteq Y_1 \longrightarrow Y_2 \end{aligned}$$

define their composition by

$$\begin{aligned} g \circ f &: \{x \in \text{Dom}(f) \mid f(x) \in \text{Dom}(g)\} \longrightarrow Y_2, \\ x &\longmapsto g(f(x)) \end{aligned}$$

For each $m \in \mathbb{Z}_{\geq 0}$, each sequence of sets $(Y_j)_{j=0}^m$, and each sequence of partial injections

$$f_j : \text{Dom}(f_j) \subseteq Y_{j-1} \longrightarrow Y_j, \quad j \in \{1, \dots, m\}$$

define the typed finite partial-injection sequence action by

$$\begin{aligned} \text{SeqAct}[(\cdot)_{Y_0}] &:= \text{id}_{Y_0}, \\ \text{SeqAct}[(f_j)_{j=1}^m] &:= f_m \circ \dots \circ f_1 \quad (m \geq 1) \end{aligned}$$

Its maximal domain and value satisfy

$$\begin{aligned} & \text{Dom}(\text{SeqAct}[(f_j)_{j=1}^m]) \\ &= \left\{ x_0 \in Y_0 \mid \exists (x_j)_{j=1}^m \in \prod_{j=1}^m Y_j, \right. \\ & \quad \left. \forall j \in \{1, \dots, m\}, \quad x_{j-1} \in \text{Dom}(f_j), \quad x_j = f_j(x_{j-1}) \right\}, \\ & \text{SeqAct}[(f_j)_{j=1}^m](x_0) = x_m \end{aligned}$$

Lemma 4.21 (Action Laws for Typed Finite Partial-Injection Sequences). For each $m, n \in \mathbb{Z}_{\geq 0}$ and two type-compatible sequences $(f_j)_{j=1}^m$ and $(g_k)_{k=1}^n$, define the concatenated sequence and the inverse sequence by

$$\begin{aligned} h_\ell &:= \begin{cases} f_\ell, & 1 \leq \ell \leq m, \\ g_{\ell-m}, & m+1 \leq \ell \leq m+n, \end{cases} \\ f_j^{\text{rev}} &:= f_{m+1-j}^{-1}, \quad j \in \{1, \dots, m\} \end{aligned}$$

Then

$$\begin{aligned} & \text{Dom}(\text{SeqAct}[(h_\ell)_{\ell=1}^{m+n}]) \\ &= \{x \in \text{Dom}(\text{SeqAct}[(f_j)_{j=1}^m]) \mid \text{SeqAct}[(f_j)_{j=1}^m](x) \in \text{Dom}(\text{SeqAct}[(g_k)_{k=1}^n])\}, \\ & \text{SeqAct}[(h_\ell)_{\ell=1}^{m+n}] = \text{SeqAct}[(g_k)_{k=1}^n] \circ \text{SeqAct}[(f_j)_{j=1}^m], \\ & \text{SeqAct}[(f_j^{\text{rev}})_{j=1}^m] = (\text{SeqAct}[(f_j)_{j=1}^m])^{-1} \end{aligned}$$

hold. In addition, $\text{SeqAct}[(f_j)_{j=1}^m]$ is a partial injection.

Proof. The partial-map composition and associativity in Definition 4.20 give the domain equality and the composition law. When $m = 0$, the empty-sequence action is the identity map, so the inverse-sequence law holds. When $m \geq 1$,

$$(f_m \circ \dots \circ f_1)^{-1} = f_1^{-1} \circ \dots \circ f_m^{-1}$$

gives the inverse-sequence law. The existence of an inverse partial map gives partial injectivity. $\square$

Definition 4.22 (Abstract Typed Atomic-Action System). In this subsection, an atomic action is a one-stage partial action forming one component of a typed finite word. An atomic kind represents its action kind, and an admissible atom represents an admissible triple of atomic kind, initial position, and terminal position. Define an abstract typed atomic-action system by

$$\mathfrak{W} = (U, (Y_u)_{u \in U}, \text{Kind}, \text{AdmAtom}, \text{AtomAct}, \dagger)$$

Here, U is the position set, $(Y_u)_{u \in U}$ is a family of underlying sets, Kind is the set of atomic kinds, and

$$\text{AdmAtom} \subseteq \text{Kind} \times U \times U$$

is the set of admissible atoms.

For each

$$(\kappa, u, v) \in \text{AdmAtom}$$

the map

$$\text{AtomAct}(\kappa; u, v) : \text{Dom}(\text{AtomAct}(\kappa; u, v)) \subseteq Y_u \longrightarrow Y_v$$

is a partial injection.

The inverse correspondence

$$\begin{aligned} \dagger : \text{AdmAtom} &\longrightarrow \text{AdmAtom}, \\ (\kappa, u, v) &\longmapsto (\kappa^\dagger, v, u) \end{aligned}$$

satisfies

$$\begin{aligned} (\kappa^\dagger)^\dagger &= \kappa, \\ \text{AtomAct}(\kappa^\dagger; v, u) &= \text{AtomAct}(\kappa; u, v)^{-1} \end{aligned}$$

Definition 4.23 (Abstract Typed Finite Words and Induced Partial Actions). Take the abstract typed atomic-action system $\mathfrak{W}$ of Definition 4.22. Define the family of typed finite words by

$$\begin{aligned} \text{Word}(\mathfrak{W}) := \left\{ \mathbf{w} = (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m) \mid \right. \\ m \in \mathbb{Z}_{\geq 0}, \quad (u_j)_{j=0}^m \in U^{m+1}, \quad (\kappa_j)_{j=1}^m \in \text{Kind}^m, \\ \left. \forall j \in \{1, \dots, m\}, \quad (\kappa_j, u_{j-1}, u_j) \in \text{AdmAtom} \right\} \end{aligned}$$

Two words are equal when their respective components—length, position sequence, and atomic-kind sequence—are equal.

For each

$$\mathbf{w} = (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m) \in \text{Word}(\mathfrak{W})$$

define the word length, initial type, terminal type, and atomic-kind support by

$$\begin{aligned} \text{len}(\mathbf{w}) &:= m, \\ \partial_-(\mathbf{w}) &:= u_0, \\ \partial_+(\mathbf{w}) &:= u_m, \\ \text{supp}_{\text{Kind}}(\mathbf{w}) &:= \{\kappa_j \mid 1 \leq j \leq m\} \end{aligned}$$

For each

$$\text{WordFam} \subseteq \text{Word}(\mathfrak{W})$$

define the closed-loop subfamily constructor by

$$\text{Loop}(\text{WordFam}) := \{\mathbf{w} \in \text{WordFam} \mid \partial_-(\mathbf{w}) = \partial_+(\mathbf{w})\}$$

For each $\mathbf{w} = (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m) \in \text{Word}(\mathfrak{W})$ and each $j \in \{1, \dots, m\}$, define

$$\text{AtomAct}_j[\mathbf{w}] := \text{AtomAct}(\kappa_j; u_{j-1}, u_j)$$

and define the induced partial action by

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}] := \text{SeqAct}[(\text{AtomAct}_j[\mathbf{w}])_{j=1}^m]$$

The type of this partial action is

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}] : \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]) \subseteq Y_{u_0} \longrightarrow Y_{u_m}$$

For an arbitrary partial self-map on a set $\mathcal{Y}$,

$$f : \mathcal{Y} \rightharpoonup \mathcal{Y}$$

define the fixed-point set by

$$\text{Fix}(f) := \{y \in \text{Dom}(f) \mid f(y) = y\}$$

For each $u \in U$, define the empty word by

$$\mathbf{1}_u := (0, (u), ())$$

Write two words $\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}(\mathfrak{W})$ as

$$\begin{aligned} \mathbf{w}_0 &= \left(m, (u_j)_{j=0}^m, (\kappa_j^{(0)})_{j=1}^m \right), \\ \mathbf{w}_1 &= \left(n, (v_k)_{k=0}^n, (\kappa_k^{(1)})_{k=1}^n \right) \end{aligned}$$

When $u_m = v_0$, define the concatenated word that acts first by $\mathbf{w}_0$ and then by $\mathbf{w}_1$ by

$$\mathbf{w}_1 \star \mathbf{w}_0 := \left(m+n, (u_\ell^{(01)})_{\ell=0}^{m+n}, (\kappa_\ell^{(01)})_{\ell=1}^{m+n} \right)$$

where

$$\begin{aligned} u_\ell^{(01)} &:= \begin{cases} u_\ell, & 0 \leq \ell \leq m, \\ v_{\ell-m}, & m+1 \leq \ell \leq m+n, \end{cases} \\ \kappa_\ell^{(01)} &:= \begin{cases} \kappa_\ell^{(0)}, & 1 \leq \ell \leq m, \\ \kappa_{\ell-m}^{(1)}, & m+1 \leq \ell \leq m+n. \end{cases} \end{aligned}$$

For each $\mathbf{w} = (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m) \in \text{Word}(\mathfrak{W})$, define the inverse word by

$$\mathbf{w}^\dagger := \left(m, (u_{m-j})_{j=0}^m, (\kappa_{m+1-j}^\dagger)_{j=1}^m \right)$$

Lemma 4.24 (Action Laws for Abstract Typed Finite Words). For each $\mathbf{w} \in \text{Word}(\mathfrak{W})$,

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}]$$

is a partial injection, and

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}^\dagger] = (\text{Act}_{\mathfrak{W}}[\mathbf{w}])^{-1}$$

holds. For each $u \in U$,

$$\text{Act}_{\mathfrak{W}}[\mathbf{1}_u] = \text{id}_{Y_u}$$

holds.

Let $\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}(\mathfrak{W})$, and suppose that the terminal position of $\mathbf{w}_0$ equals the initial position of $\mathbf{w}_1$. Then

$$\begin{aligned} &\text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}_1 \star \mathbf{w}_0]) \\ &= \{x \in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}_0]) \mid \text{Act}_{\mathfrak{W}}[\mathbf{w}_0](x) \in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}_1])\}, \\ &\text{Act}_{\mathfrak{W}}[\mathbf{w}_1 \star \mathbf{w}_0] = \text{Act}_{\mathfrak{W}}[\mathbf{w}_1] \circ \text{Act}_{\mathfrak{W}}[\mathbf{w}_0] \end{aligned}$$

hold.

Proof. To the sequence of atomic partial injections in Definition 4.23, apply Lemma 4.21. $\square$

Definition 4.25 (Orbit Relation of Abstract Typed Word Actions). For the abstract typed atomic-action system $\mathfrak{W}$ of Definition 4.22, define the orbit relation

$$\sim_{\mathfrak{W}}^{\text{orb}}$$

for arbitrary

$$x, y \in \bigcup_{u \in U} Y_u$$

by

$$x \sim_{\mathfrak{W}}^{\text{orb}} y \iff \exists \mathbf{w} \in \text{Word}(\mathfrak{W}), \quad x \in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]), \\ \text{Act}_{\mathfrak{W}}[\mathbf{w}](x) = y$$

Lemma 4.26 (Equivalence-Relation Property of the Orbit Relation of Abstract Typed Word Actions). If

$$u \neq v \implies Y_u \cap Y_v = \emptyset$$

holds for arbitrary $u, v \in U$, then

$$\sim_{\mathfrak{W}}^{\text{orb}}$$

is an equivalence relation on

$$\bigcup_{u \in U} Y_u$$

Proof. Take arbitrary

$$x \in \bigcup_{u \in U} Y_u$$

and take $x \in Y_u$ for some $u \in U$. The empty-word action law in Lemma 4.24 and Definition 4.25 give

$$x \sim_{\mathfrak{W}}^{\text{orb}} x.$$

Next, for

$$x \sim_{\mathfrak{W}}^{\text{orb}} y$$

let the witness be $\mathbf{w}$. The inverse-word action law in the same lemma makes $\mathbf{w}^\dagger$ a witness of

$$y \sim_{\mathfrak{W}}^{\text{orb}} x$$

Finally, for

$$x \sim_{\mathfrak{W}}^{\text{orb}} y, \quad y \sim_{\mathfrak{W}}^{\text{orb}} z$$

let the respective witnesses be $\mathbf{w}_0, \mathbf{w}_1$. The action types give

$$y \in Y_{\partial_+(\mathbf{w}_0)} \cap Y_{\partial_-(\mathbf{w}_1)}.$$

Disjointness of the type domains therefore gives

$$\partial_+(\mathbf{w}_0) = \partial_-(\mathbf{w}_1).$$

The concatenated word $\mathbf{w}_1 \star \mathbf{w}_0$ is defined, and the concatenated-action law in the same lemma gives

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}_1 \star \mathbf{w}_0](x) = z.$$

Hence

$$x \sim_{\mathfrak{W}}^{\text{orb}} z.$$

The orbit relation is thus reflexive, symmetric, and transitive. $\square$

Definition 4.27 (Saturation of a Subset along an Equivalence Relation). For a set S , an equivalence relation on S , denoted by $\sim$, and a subset $S_0 \subseteq S$, define the saturation predicate by

$$\text{Sat}(S_0; \sim) \iff \left[\forall p, q \in S, \quad p \sim q \implies (p \in S_0 \iff q \in S_0) \right]$$

Lemma 4.28 (Complete Pullback and Minimality of the Image of a Saturated Set). Let S, T be sets, let an equivalence relation on S be denoted by $\sim$, and suppose that a map

$$\pi : S \longrightarrow T$$

satisfies, for arbitrary $x, y \in S$,

$$\pi(x) = \pi(y) \iff x \sim y$$

If a subset $S_0 \subseteq S$ satisfies

$$\text{Sat}(S_0; \sim)$$

then

$$\pi[S_0] \subseteq T$$

and, for arbitrary $x \in S$,

$$\pi(x) \in \pi[S_0] \iff x \in S_0$$

hold.

Furthermore, for arbitrary $A \subseteq T$,

$$[\forall x \in S_0, \quad \pi(x) \in A] \implies \pi[S_0] \subseteq A.$$

Thus $\pi[S_0]$ is the unique minimal subset containing the entire image of S_0 .

Proof. The definition of an image gives

$$\pi[S_0] \subseteq T.$$

If $x \in S_0$, the same definition gives

$$\pi(x) \in \pi[S_0].$$

Conversely, suppose

$$\pi(x) \in \pi[S_0]$$

By the definition of an image, some $y \in S_0$ satisfies

$$\pi(x) = \pi(y).$$

The fiber condition gives

$$x \sim y.$$

From

$$\text{Sat}(S_0; \sim)$$

and $y \in S_0$, it follows that $x \in S_0$.

Finally, suppose that $A \subseteq T$ satisfies

$$\forall x \in S_0, \quad \pi(x) \in A$$

For arbitrary $z \in \pi[S_0]$, some $x \in S_0$ satisfies $z = \pi(x)$, and hence $z \in A$. Therefore

$$\pi[S_0] \subseteq A.$$

If there are two minimal subsets, mutual inclusion makes them equal. □

Definition 4.29 (Identity-on-Domain Predicate). For a partial self-map on a set $\mathcal{Y}$,

$$f : \mathcal{Y} \rightharpoonup \mathcal{Y}$$

define the identity-on-domain predicate by

$$\text{IdOnDom}(f) \iff \text{Fix}(f) = \text{Dom}(f)$$

Definition 4.30 (Equivalence Classes and the Canonical Quotient Projection). For a set S and an equivalence relation $\sim$ on it, define

$$\begin{aligned} [x]_{\sim} &:= \{y \in S \mid x \sim y\}, \\ S/\sim &:= \{[x]_{\sim} \mid x \in S\} \end{aligned}$$

Define the canonical quotient projection by

$$q_{\sim} : S \longrightarrow S/\sim, \quad q_{\sim}(x) := [x]_{\sim}$$

Lemma 4.31 (Surjectivity and Equivalence-Class Criterion for the Canonical Quotient Projection). The canonical quotient projection $q_{\sim}$ of Definition 4.30 is surjective and, for arbitrary $x, y \in S$,

$$q_{\sim}(x) = q_{\sim}(y) \iff x \sim y.$$

Proof. Every equivalence class has the form $[x]_{\sim}$ for some $x \in S$; hence $q_{\sim}$ is surjective.

Suppose $[x]_{\sim} = [y]_{\sim}$. Reflexivity gives $x \in [x]_{\sim}$, and therefore $x \in [y]_{\sim}$. Thus $y \sim x$, and symmetry gives $x \sim y$.

Conversely, suppose $x \sim y$. Symmetry and transitivity give, for arbitrary $z \in S$,

$$x \sim z \iff y \sim z.$$

Extensionality gives $[x]_{\sim} = [y]_{\sim}$. □

Definition 4.32 (Closed-Word Identity-Action Predicate). For an abstract typed atomic-action system $\mathfrak{W}$, define the closed-word identity-action predicate by

$$\begin{aligned} &\text{LoopId}(\mathfrak{W}) \\ &\iff \left[\forall \mathbf{w} \in \text{Loop}(\text{Word}(\mathfrak{W})), \quad \text{IdOnDom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]) \right] \end{aligned}$$

Theorem 4.33 (Closed-Word Identity Action and Injectivity of the Orbit Quotient Projection on Type Domains). Suppose that, for arbitrary $u, v \in U$,

$$u \neq v \implies Y_u \cap Y_v = \emptyset$$

Then

$$\begin{aligned} &\text{LoopId}(\mathfrak{W}) \\ &\iff \left[\forall u \in U, \quad q_{\sim_{\mathfrak{W}}^{\text{orb}}} \upharpoonright_{Y_u} : Y_u \longrightarrow \left(\bigcup_{v \in U} Y_v \right) / \sim_{\mathfrak{W}}^{\text{orb}} \text{ is injective} \right]. \end{aligned}$$

Proof. Assume

$$\text{LoopId}(\mathfrak{W})$$

and let $u \in U$ and $x, y \in Y_u$ satisfy

$$q_{\sim_{\mathfrak{W}}^{\text{orb}}}(x) = q_{\sim_{\mathfrak{W}}^{\text{orb}}}(y)$$

Lemma 4.31 gives

$$x \sim_{\mathfrak{W}}^{\text{orb}} y.$$

By Definition 4.25, there exists

$$\mathbf{w} \in \text{Word}(\mathfrak{W})$$

such that

$$\begin{aligned} x &\in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]), \\ \text{Act}_{\mathfrak{W}}[\mathbf{w}](x) &= y. \end{aligned}$$

The action type and disjointness of the type domains give

$$\partial_-(\mathbf{w}) = u = \partial_+(\mathbf{w}).$$

Therefore

$$\mathbf{w} \in \text{Loop}(\text{Word}(\mathfrak{W})).$$

Definitions 4.32 and 4.29 give

$$\text{Act}_{\mathfrak{W}}[\mathbf{w}](x) = x.$$

Hence $x = y$, so the restriction on each type domain is injective.

Conversely, suppose that for each $u \in U$,

$$q_{\sim_{\mathfrak{W}}^{\text{orb}}} \big|_{Y_u}$$

is injective. Take arbitrary

$$\mathbf{w} \in \text{Loop}(\text{Word}(\mathfrak{W}))$$

and set

$$u := \partial_-(\mathbf{w}) = \partial_+(\mathbf{w})$$

For arbitrary

$$x \in \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}])$$

set

$$y := \text{Act}_{\mathfrak{W}}[\mathbf{w}](x)$$

The action type gives $x, y \in Y_u$, and Definition 4.25 gives

$$x \sim_{\mathfrak{W}}^{\text{orb}} y.$$

Lemma 4.31 gives

$$q_{\sim_{\mathfrak{W}}^{\text{orb}}}(x) = q_{\sim_{\mathfrak{W}}^{\text{orb}}}(y).$$

Injectivity on Y_u gives $x = y$. Therefore

$$\text{Fix}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]) = \text{Dom}(\text{Act}_{\mathfrak{W}}[\mathbf{w}]).$$

Definitions 4.29 and 4.32 give

$$\text{LoopId}(\mathfrak{W})$$

□

Definition 4.34 (Atomic-Kind Restrictions of an Abstract Typed Atomic-Action System). Take the abstract typed atomic-action system $\mathfrak{W}$ of Definition 4.22. For an inverse-correspondence-invariant subset

$$\text{Kind}_0 \subseteq \text{Kind}$$

define the atomic-kind restriction by

$$\begin{aligned} \mathfrak{W}|_{\text{Kind}_0} &:= (U, (Y_u)_{u \in U}, \text{Kind}_0, \text{AdmAtom}_0, \text{AtomAct}|_{\text{AdmAtom}_0}, \dagger|_{\text{AdmAtom}_0}), \\ \text{AdmAtom}_0 &:= \text{AdmAtom} \cap (\text{Kind}_0 \times U \times U) \end{aligned}$$

Definition 4.35 (Comparison-Stage Typed Atomic-Action System). Assume

$$\text{L}_1(\mathcal{X})$$

and

$$\text{L}_2(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

For each $R \in \text{Rol}^{\text{cmp}}$, define the underlying set at each $u \in U$ by

$$Y_u^R := \{u\} \times \text{Im}(\text{Arr}_R^{\text{cmp}})$$

Define the atomic-kind set and the atomic-kind sets used for restrictions by

$$\begin{aligned} \text{Kind}^{\text{cmp}} &:= \{\text{tr}^+, \text{tr}^-, \text{src}\}, \\ \text{Kind}^{\text{tr}} &:= \{\text{tr}^+, \text{tr}^-\}, \quad \text{Kind}^{\text{src}} := \{\text{src}\} \end{aligned}$$

Define the set of admissible atoms by

$$\begin{aligned} \text{AdmAtom}_{\text{cmp}}^R &:= \left\{ (\kappa, u, v) \in \text{Kind}^{\text{cmp}} \times U \times U \mid \right. \\ &\quad [\kappa = \text{tr}^+ \wedge (u, v) \in \text{Adm}_{\text{cmp}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})] \\ &\quad \vee [\kappa = \text{tr}^- \wedge (v, u) \in \text{Adm}_{\text{cmp}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})] \\ &\quad \left. \vee [\kappa = \text{src}] \right\} \end{aligned}$$

For each $(\kappa, u, v) \in \text{AdmAtom}_{\text{cmp}}^R$, define the atomic partial injection by

$$\text{AtomAct}_{\text{cmp}}^R(\kappa; u, v) := \begin{cases} \text{EdgeTerm}_{uv}^{\text{tr}, R, +}(\mathcal{X}; \mathcal{J}_{\text{cmp}}), & \kappa = \text{tr}^+, \\ \text{EdgeTerm}_{vu}^{\text{tr}, R, -}(\mathcal{X}; \mathcal{J}_{\text{cmp}}), & \kappa = \text{tr}^-, \\ \text{EdgeTerm}_{uv}^{\text{src}, R}, & \kappa = \text{src} \end{cases}$$

Define the inverse correspondence by

$$(\text{tr}^+)^\dagger := \text{tr}^-, \quad (\text{tr}^-)^\dagger := \text{tr}^+, \quad (\text{src})^\dagger := \text{src}$$

Define the comparison-stage typed atomic-action system by

$$\begin{aligned} \mathfrak{W}_{\text{cmp}}^R(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ := (U, (Y_u^R)_{u \in U}, \text{Kind}^{\text{cmp}}, \text{AdmAtom}_{\text{cmp}}^R, \text{AtomAct}_{\text{cmp}}^R, \dagger) \end{aligned}$$

Lemmas 4.17 and 4.19 establish the conditions of Definition 4.22.

Lemma 4.36 (Partition of the Comparison-Arrival Type-Domain Family). Under $L_{\leq 2}[\mathbf{Ev}^{\text{cmp}}]$, for each $R \in \mathbf{Rol}^{\text{cmp}}$,

$$\bigcup_{u \in U} Y_u^R = \mathcal{R}_R^{\text{arr}},$$

$$u \neq v \implies Y_u^R \cap Y_v^R = \emptyset \quad (u, v \in U)$$

hold.

Proof. The result follows from Definitions 4.13 and 4.35. $\square$

Definition 4.37 (Displayed Typed Finite Pure-Transport Words). For each $R \in \mathbf{Rol}^{\text{cmp}}$, define the following objects. For a transport-direction-encoded word

$$\mathbf{w} = (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m)$$

define

$$\text{Enc}^{\text{tr}}[\mathbf{w}] := (m, (u_j)_{j=0}^m, (\kappa_j)_{j=1}^m), \quad \kappa_j := \begin{cases} \text{tr}^+, & \varepsilon_j = +, \\ \text{tr}^-, & \varepsilon_j = - \end{cases}$$

Define the family of all pure-transport words by

$$\begin{aligned} & \text{Word}_R^{\text{tr}}[\mathbf{l}] \\ & := \left\{ \mathbf{w} \mid \text{Enc}^{\text{tr}}[\mathbf{w}] \in \text{Word} \left(\mathfrak{W}_{\text{cmp}}^R[\mathbf{l}]|_{\text{Kind}^{\text{tr}}} \right) \right\} \end{aligned}$$

The transport-direction encoding is a bijection from the family of words on the left to the family on the right.

Definition 4.38 (Induced Partial Action of a Pure-Transport Word). For each $R \in \mathbf{Rol}^{\text{cmp}}$ and each $\mathbf{w} \in \text{Word}_R^{\text{tr}}[\mathbf{l}]$, define

$$\begin{aligned} & \text{Act}_R^{\text{tr}}[\mathbf{l}][\mathbf{w}] \\ & := \text{Act}_{\mathfrak{W}_{\text{cmp}}^R[\mathbf{l}]|_{\text{Kind}^{\text{tr}}}}[\text{Enc}^{\text{tr}}[\mathbf{w}]] \end{aligned}$$

Definition 4.39 (Empty, Concatenated, and Inverse Pure-Transport Words). For each $R \in \mathbf{Rol}^{\text{cmp}}$, define the following operations. For each $u \in U$, define the empty word by

$$\text{Enc}^{\text{tr}}[\mathbf{1}_{R,u}^{\text{tr}}] := \mathbf{1}_u$$

For two words $\mathbf{w}_0$ and $\mathbf{w}_1$ whose terminal and initial positions agree,

$$\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}_R^{\text{tr}}[\mathbf{l}]$$

define the concatenated word by

$$\text{Enc}^{\text{tr}}[\mathbf{w}_1 \star \mathbf{w}_0] := \text{Enc}^{\text{tr}}[\mathbf{w}_1] \star \text{Enc}^{\text{tr}}[\mathbf{w}_0]$$

For an arbitrary pure-transport word $\mathbf{w}$, define the inverse word by

$$\text{Enc}^{\text{tr}}[\mathbf{w}^\dagger] := (\text{Enc}^{\text{tr}}[\mathbf{w}])^\dagger$$

Lemma 4.40 (Compatibility of Pure-Transport Word Operations with Induced Partial Actions). For each $R \in \text{Rol}^{\text{cmp}}$, the action system for pure-transport words defined by Definitions 4.38 and 4.39 is a special case of the abstract typed finite-word action system of Lemma 4.24. Consequently, the conclusions of that lemma for inverse-word actions, empty-word actions, concatenated-word domains, and composed actions hold.

Proof. Through Definitions 4.38 and 4.39, apply Lemma 4.24. $\square$

Lemma 4.41 (Partial Injectivity of the Induced Partial Action of a Pure-Transport Word).

For each $R \in \text{Rol}^{\text{cmp}}$ and each $\mathbf{w} \in \text{Word}_R^{\text{tr}}[\mathbf{l}]$,

$$\text{Act}_R^{\text{tr}}[\mathbf{l}][\mathbf{w}]$$

is a partial injection.

Proof. Apply Lemma 4.24. $\square$

Definition 4.42 (Family of All Pure-Transport Closed-Loop Words). For each $R \in \text{Rol}^{\text{cmp}}$, define

$$\begin{aligned} \text{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbf{l}] \\ := \{ \mathbf{w} = (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m) \in \text{Word}_R^{\text{tr}}[\mathbf{l}] \mid u_m = u_0 \} \end{aligned}$$

Definition 4.43 (Displayed Typed Finite Source-Matching Words). For each $R \in \text{Rol}^{\text{cmp}}$, define the source-matching atomic-action system by

$$\mathfrak{W}_{\text{src}}^R := \mathfrak{W}_{\text{cmp}}^R[\mathbf{l}]|_{\text{Kind}^{\text{src}}}.$$

For a source-encoded word

$$\mathbf{v} = (m, (u_j)_{j=0}^m)$$

define

$$\text{Enc}^{\text{src}}[\mathbf{v}] := (m, (u_j)_{j=0}^m, (\text{src})_{j=1}^m)$$

Define the family of all source-matching words by

$$\begin{aligned} \text{Word}_R^{\text{src}}(\mathcal{X}) \\ := \{ \mathbf{v} \mid \text{Enc}^{\text{src}}[\mathbf{v}] \in \text{Word}(\mathfrak{W}_{\text{src}}^R) \} \end{aligned}$$

The source encoding is a bijection from the family of words on the left to the family on the right.

Definition 4.44 (Induced Partial Action of a Source-Matching Word). For each $R \in \text{Rol}^{\text{cmp}}$ and each $\mathbf{v} \in \text{Word}_R^{\text{src}}(\mathcal{X})$, define

$$\text{Act}_R^{\text{src}}[\mathbf{v}] := \text{Act}_{\mathfrak{W}_{\text{src}}^R}[\text{Enc}^{\text{src}}[\mathbf{v}]]$$

Definition 4.45 (Empty, Concatenated, and Inverse Source-Matching Words). For each $R \in \text{Rol}^{\text{cmp}}$, define the following operations. For each $u \in U$, define the empty word by

$$\text{Enc}^{\text{src}}[\mathbf{1}_{R,u}^{\text{src}}] := \mathbf{1}_u$$

For two words $\mathbf{v}_0$ and $\mathbf{v}_1$ whose terminal and initial positions agree,

$$\mathbf{v}_0, \mathbf{v}_1 \in \text{Word}_R^{\text{src}}(\mathcal{X})$$

define the concatenated word by

$$\text{Enc}^{\text{src}}[\mathbf{v}_1 \star \mathbf{v}_0] := \text{Enc}^{\text{src}}[\mathbf{v}_1] \star \text{Enc}^{\text{src}}[\mathbf{v}_0]$$

For an arbitrary source-matching word $\mathbf{v}$, define the inverse word by

$$\text{Enc}^{\text{src}}[\mathbf{v}^\dagger] := (\text{Enc}^{\text{src}}[\mathbf{v}])^\dagger$$

Lemma 4.46 (Compatibility of Source-Matching Word Operations with Induced Partial Actions). For each $R \in \text{Rol}^{\text{cmp}}$, the action system of source-matching words determined by Definition 4.44 and Definition 4.45 is a special case of the abstract typed finite-word action system in Lemma 4.24. Consequently, the conclusions of that lemma concerning inverse-word actions, empty-word actions, concatenated-word domains, and composite actions hold.

Proof. Apply Definition 4.44 and Definition 4.45, and then Lemma 4.24. $\square$

Lemma 4.47 (Partial Injectivity of the Induced Partial Action of a Source-Matching Word).

For each $R \in \text{Rol}^{\text{cmp}}$ and each $\mathbf{v} \in \text{Word}_R^{\text{src}}(\mathcal{X})$,

$$\text{Act}_R^{\text{src}}[\mathbf{v}]$$

is a partial injection.

Proof. Apply Lemma 4.24. $\square$

Lemma 4.48 (Identity of a Source-Matching Word When Its Initial and Terminal Positions Coincide). For each $R \in \text{Rol}^{\text{cmp}}$ and each

$$\mathbf{v} = (m, (u_j)_{j=0}^m) \in \text{Word}_R^{\text{src}}(\mathcal{X})$$

if $u_m = u_0$, then

$$\text{Act}_R^{\text{src}}[\mathbf{v}] = \text{id}_{\text{Dom}(\text{Act}_R^{\text{src}}[\mathbf{v}])}$$

holds.

Proof. Take any

$$x_0 = (u_0, \alpha_0) \in \text{Dom}(\text{Act}_R^{\text{src}}[\mathbf{v}])$$

and set

$$\text{SrcTok}_R := \text{Rec}_{R, u_0}^{\text{cmp}}(\alpha_0)$$

By Definition 4.5 and finite induction, the corresponding stage sequence satisfies

$$x_j = \left(u_j, \text{Arr}_{R, u_j}^{\text{cmp}}(\text{SrcTok}_R) \right), \quad j \in \{0, \dots, m\}$$

The equality $u_m = u_0$ yields $x_m = x_0$. $\square$

Definition 4.49 (Displayed Typed Finite Extended Comparison Words). For each $R \in \text{Rol}^{\text{cmp}}$, define

$$\text{Word}_R^{\text{ext}}[\mathbb{I}] := \text{Word}(\mathfrak{W}_{\text{cmp}}^R[\mathbb{I}])$$

Each of its elements is displayed as

$$\mathbf{e} = (m, (u_j)_{j=0}^m, (\kappa_j(\mathbf{e}))_{j=1}^m)$$

Definition 4.50 (Induced Partial Action of an Extended Comparison Word). For each $R \in \text{Rol}^{\text{cmp}}$ and each $\mathbf{e} \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$, define

$$\begin{aligned} \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}] \\ := \text{Act}_{\mathcal{W}_{\text{cmp}}^R[\mathbb{I}]}[\mathbf{e}] \end{aligned}$$

Definition 4.51 (Empty, Concatenated, and Inverse Extended Comparison Words). For each $R \in \text{Rol}^{\text{cmp}}$, define the following objects. For each $u \in U$, define the empty word by

$$\mathbf{1}_{R,u}^{\text{ext}} := \mathbf{1}_u$$

For each $\mathbf{e}_0$ whose terminal position coincides with the initial position of $\mathbf{e}_1$, where

$$\mathbf{e}_0, \mathbf{e}_1 \in \text{Word}_R^{\text{ext}}[\mathbb{I}],$$

define the concatenated word $\mathbf{e}_1 \star \mathbf{e}_0$ by the concatenation in Definition 4.23. For each extended comparison word $\mathbf{e}$, define the inverse word $\mathbf{e}^\dagger$ by the inverse-word construction in the same definition.

Lemma 4.52 (Compatibility of Extended Comparison Word Operations with Induced Partial Actions). For each $R \in \text{Rol}^{\text{cmp}}$, the action system of extended comparison words determined by Definition 4.50 and Definition 4.51 is a special case of the abstract typed finite-word action system in Lemma 4.24. Consequently, the conclusions of that lemma concerning inverse-word actions, empty-word actions, concatenated-word domains, and composite actions hold.

Proof. Apply Definition 4.50 and Definition 4.51, and then Lemma 4.24. $\square$

Lemma 4.53 (Partial Injectivity of the Induced Partial Action of an Extended Comparison Word).

For each $R \in \text{Rol}^{\text{cmp}}$ and each $\mathbf{e} \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$,

$$\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}]$$

is a partial injection.

Proof. Apply Lemma 4.24. $\square$

Definition 4.54 (Standard Embeddings of Pure-Transport Words and Source-Matching Words). For each $R \in \text{Rol}^{\text{cmp}}$, the encoding maps of Definition 4.37 and Definition 4.43 give the standard embeddings

$$\begin{aligned} \text{Enc}^{\text{tr}} : \text{Word}_R^{\text{tr}}[\mathbb{I}] &\hookrightarrow \text{Word}_R^{\text{ext}}[\mathbb{I}], \\ \text{Enc}^{\text{src}} : \text{Word}_R^{\text{src}}(\mathcal{X}) &\hookrightarrow \text{Word}_R^{\text{ext}}[\mathbb{I}] \end{aligned}$$

Lemma 4.55 (Injectivity and Action Compatibility of the Standard Embeddings). For each $R \in \text{Rol}^{\text{cmp}}$, the two standard embeddings are respectively injections

$$\begin{aligned} \text{Word}_R^{\text{tr}}[\mathbb{I}] &\longrightarrow \text{Word}_R^{\text{ext}}[\mathbb{I}], \\ \text{Word}_R^{\text{src}}(\mathcal{X}) &\longrightarrow \text{Word}_R^{\text{ext}}[\mathbb{I}] \end{aligned}$$

For any $\mathbf{w}$ and $\mathbf{v}$ in Definition 4.54,

$$\begin{aligned} \text{Act}_R^{\text{ext}}[\mathbb{I}][\text{Enc}^{\text{tr}}[\mathbf{w}]] &= \text{Act}_R^{\text{tr}}[\mathbb{I}][\mathbf{w}], \\ \text{Act}_R^{\text{ext}}[\mathbb{I}][\text{Enc}^{\text{src}}[\mathbf{v}]] &= \text{Act}_R^{\text{src}}[\mathbb{I}][\mathbf{v}] \end{aligned}$$

hold as equalities of partial maps. For each $u \in U$,

$$\text{Enc}^{\text{tr}}[\mathbf{1}_R^{\text{tr}}] = \mathbf{1}_R^{\text{ext}} = \text{Enc}^{\text{src}}[\mathbf{1}_R^{\text{src}}]$$

holds.

Proof. The claim follows from the bijectivity in Definition 4.37 and Definition 4.43, together with Definition 4.38 and Definition 4.44. $\square$

Definition 4.56 (Family of Mixed Words and Family of All Mixed Closed-Loop Words). For each $R \in \text{Rol}^{\text{cmp}}$, define the family of mixed words by

$$\begin{aligned} & \text{Word}_R^{\text{mix}}[\mathbb{I}] \\ & := \left\{ \mathbf{e} \in \text{Word}_R^{\text{ext}}[\mathbb{I}] \mid \begin{array}{l} \text{supp}_{\text{Kind}^{\text{cmp}}}(\mathbf{e}) \cap \text{Kind}^{\text{tr}} \neq \emptyset, \\ \text{supp}_{\text{Kind}^{\text{cmp}}}(\mathbf{e}) \cap \text{Kind}^{\text{src}} \neq \emptyset \end{array} \right\} \end{aligned}$$

Define the family of all mixed closed-loop words by

$$\text{LoopWord}_{\text{all}}^{\text{mix}, R}[\mathbb{I}] := \text{Loop}(\text{Word}_R^{\text{mix}}[\mathbb{I}])$$

Definition 4.57 (One-Step Deletion of a Source-Matching Closed-Loop Subword). For each $R \in \text{Rol}^{\text{cmp}}$ and each

$$\mathbf{e} = (m, (u_j)_{j=0}^m, (\kappa_j(\mathbf{e}))_{j=1}^m) \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$$

define the following operation. When the indices $0 \leq r_0 < r_1 \leq m$ satisfy

$$u_{r_0} = u_{r_1}, \quad \forall j \in \{r_0 + 1, \dots, r_1\}, \quad \kappa_j(\mathbf{e}) = \text{src}$$

define

$$\begin{aligned} & \text{Del}_{r_0, r_1}^{\text{src}}[\mathbf{e}] \\ & := (m - (r_1 - r_0), (u_0, \dots, u_{r_0}, u_{r_1+1}, \dots, u_m), \\ & \quad (\kappa_1(\mathbf{e}), \dots, \kappa_{r_0}(\mathbf{e}), \kappa_{r_1+1}(\mathbf{e}), \dots, \kappa_m(\mathbf{e}))) \end{aligned}$$

An initial or terminal sequence that becomes empty is taken to be the empty sequence. For each $\mathbf{e}' \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$, define the one-step deletion relation by

$$\begin{aligned} \mathbf{e} \xrightarrow{\text{Del}^{\text{src}}} \mathbf{e}' & \iff \left[\begin{array}{l} \exists r_0, r_1, \quad 0 \leq r_0 < r_1 \leq m, \quad u_{r_0} = u_{r_1}, \\ \forall j \in \{r_0 + 1, \dots, r_1\}, \quad \kappa_j(\mathbf{e}) = \text{src}, \\ \mathbf{e}' = \text{Del}_{r_0, r_1}^{\text{src}}[\mathbf{e}] \end{array} \right] \end{aligned}$$

Definition 4.58 (Family of Source-Irreducible Mixed Closed-Loop Words). For each $R \in \text{Rol}^{\text{cmp}}$, define

$$\begin{aligned} & \text{LoopWord}_{\text{src-irr}}^{\text{mix}, R}[\mathbb{I}] \\ & := \left\{ \mathbf{e} \in \text{LoopWord}_{\text{all}}^{\text{mix}, R}[\mathbb{I}] \mid \nexists \mathbf{e}' \in \text{Word}_R^{\text{ext}}[\mathbb{I}], \quad \mathbf{e} \xrightarrow{\text{Del}^{\text{src}}} \mathbf{e}' \right\} \end{aligned}$$

Lemma 4.59 (Exclusive Four-Case Decomposition of Extended Comparison Words). For each $R \in \text{Rol}^{\text{cmp}}$ and each

$$\mathbf{e} \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$$

exactly one of the following four cases holds:

- (i) $\text{len}(\mathbf{e}) = 0$,
- (ii) $\text{len}(\mathbf{e}) \geq 1$, $\text{supp}_{\text{Kind}^{\text{cmp}}}(\mathbf{e}) \subseteq \text{Kind}^{\text{tr}}$,
- (iii) $\text{len}(\mathbf{e}) \geq 1$, $\text{supp}_{\text{Kind}^{\text{cmp}}}(\mathbf{e}) = \text{Kind}^{\text{src}}$,
- (iv) $\mathbf{e} \in \text{Word}_R^{\text{mix}}[\mathbb{I}]$.

In the second case, there exists a unique $\mathbf{w} \in \text{Word}_R^{\text{tr}}[\mathbb{I}]$ such that

$$\mathbf{e} = \text{Enc}^{\text{tr}}[\mathbf{w}]$$

In the third case, there exists a unique $\mathbf{v} \in \text{Word}_R^{\text{src}}(\mathcal{X})$ such that

$$\mathbf{e} = \text{Enc}^{\text{src}}[\mathbf{v}]$$

Proof. The identity

$$\text{Kind}^{\text{cmp}} = \text{Kind}^{\text{tr}} \sqcup \text{Kind}^{\text{src}}$$

makes the four cases exclusive and exhaustive. Existence and uniqueness in the second and third cases follow from Definition 4.37, Definition 4.43, and Lemma 4.55. $\square$

Lemma 4.60 (Extension of the Action under Deletion of a Source-Matching Closed-Loop Subword). For each $R \in \text{Rol}^{\text{cmp}}$ and each $\mathbf{e}, \mathbf{e}' \in \text{Word}_R^{\text{ext}}[\mathbb{I}]$, suppose that

$$\mathbf{e} \xrightarrow{\text{Del}^{\text{src}}} \mathbf{e}'$$

Then

$$\begin{aligned} \text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}]) &\subseteq \text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}']), \\ \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}']|_{\text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}])} &= \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}] \end{aligned}$$

Proof. For indices r_0, r_1 that establish the deletion relation, define the source-matching word on the deleted interval by

$$\mathbf{v}_{r_0, r_1} := (r_1 - r_0, (u_{r_0+k})_{k=0}^{r_1-r_0}) \in \text{Word}_R^{\text{src}}(\mathcal{X})$$

By Lemma 4.55 and Lemma 4.48, the stage sequence of the original word satisfies $x_{r_1} = x_{r_0}$. Removing stages $r_0 + 1, \dots, r_1$ yields the stage sequence of the word after deletion; hence the domain inclusion and equality of action values hold. $\square$

Lemma 4.61 (Finite Source Reduction of a Mixed Closed-Loop Word). For each $R \in \text{Rol}^{\text{cmp}}$ and every

$$\mathbf{e} \in \text{LoopWord}_{\text{all}}^{\text{mix}, R}[\mathbb{I}]$$

there exists a finite sequence

$$\mathbf{e}_0 = \mathbf{e}, \quad \mathbf{e}_0 \xrightarrow{\text{Del}^{\text{src}}} \dots \xrightarrow{\text{Del}^{\text{src}}} \mathbf{e}_N$$

The terminal word satisfies

$$\begin{aligned} \mathbf{e}_N &\in \text{LoopWord}_{\text{src-irr}}^{\text{mix}, R}[\mathbb{I}] \\ \text{or } \exists \mathbf{w} &\in \text{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbb{I}], \quad \mathbf{e}_N = \text{Enc}^{\text{tr}}[\mathbf{w}] \end{aligned}$$

For each $k \in \{0, \dots, N\}$,

$$\begin{aligned} \text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_0]) &\subseteq \text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_k]), \\ \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_k]|_{\text{Dom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_0])} &= \text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}_0] \end{aligned}$$

Proof. Use strong induction on word length. For a source-irreducible word, set $N = 0$. When a one-step deletion is possible, the word length decreases strictly, while the transport atoms and the closed-loop condition are preserved. If the word after deletion consists only of transport atoms, apply Lemma 4.59; otherwise apply the induction hypothesis. The two formulas concerning the action follow by applying Lemma 4.60 to each one-step deletion. $\square$

Definition 4.62 (Candidate Pure-Transport Word and Candidate Pure-Transport Action of a Local Closed-Loop Specification). Assume $L_{\leq 2}[\text{Ev}^{\text{cmp}}]$. For any $R \in \text{Rol}^{\text{cmp}}$ and

$$\text{Spec} = (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m, \text{Dom}_{\text{Spec}}) \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$$

define the candidate pure-transport word by

$$\text{Word}_R^{\text{tr}, \text{cand}}[\text{Spec}] := (m, (u_j)_{j=0}^m, (\varepsilon_j)_{j=1}^m)$$

Define the candidate pure-transport action by

$$\begin{aligned} \text{Act}_R^{\text{tr}, \text{cand}}[\text{Spec}] \\ := \text{Act}_R^{\text{tr}}[\mathbb{I}] \left[\text{Word}_R^{\text{tr}, \text{cand}}[\text{Spec}] \right] \end{aligned}$$

Lemma 4.63 (Admissibility of the Candidate Pure-Transport Word and Action from a Local Closed-Loop Specification). Under $L_{\leq 2}[\text{Ev}^{\text{cmp}}]$, for each $R \in \text{Rol}^{\text{cmp}}$ and each $\text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$,

$$\text{Word}_R^{\text{tr}, \text{cand}}[\text{Spec}] \in \text{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbb{I}]$$

and

$$\text{Act}_R^{\text{tr}, \text{cand}}[\text{Spec}] : \text{Dom} \left(\text{Act}_R^{\text{tr}, \text{cand}}[\text{Spec}] \right) \longrightarrow Y_{u_0}^R$$

is a partial injection.

Proof. The type conditions in Definition 4.4 and the equality $u_m = u_0$ place the candidate word in the family of all pure-transport closed-loop words. Partial injectivity follows from Lemma 4.41. $\square$

Definition 4.64 (Comparison-Stage Judgment Condition L_3 : Realization of Local Closed-Loop Specifications). Define the local closed-loop specification realization condition by

$$\begin{aligned} L_3[\text{Ev}_3^{\text{cmp}}] &\iff \left[L_{\leq 2}[\text{Ev}^{\text{cmp}}] \right. \\ &\implies \forall R \in \text{Rol}^{\text{cmp}}, \quad \forall \text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}], \\ &\quad \left. \text{Dom}_{\text{Spec}} \subseteq \text{Dom} \left(\text{Act}_R^{\text{tr}, \text{cand}}[\text{Spec}] \right) \right] \end{aligned}$$

Define the third-stage establishment bundle by

$$L_{\leq 3}[\text{Ev}^{\text{cmp}}] := L_{\leq 2}[\text{Ev}^{\text{cmp}}] \wedge L_3[\text{Ev}_3^{\text{cmp}}]$$

Definition 4.65 (Realized Local-Specification Closed-Loop Data). Assume $L_{\leq 3}[\text{Ev}^{\text{cmp}}]$. For any

$$R \in \text{Rol}^{\text{cmp}}$$

and

$$\text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$$

define the realized restricted action by

$$\text{Act}_R^{\text{tr},\text{real}}[\text{Spec}] := \text{Act}_R^{\text{tr},\text{cand}}[\text{Spec}] \Big|_{\text{Dom}_{\text{Spec}}}$$

Define the realized local-specification closed-loop datum by

$$\text{LoopDat}_R^{\text{real}}[\text{Spec}] := \left(\text{Spec}, \text{Word}_R^{\text{tr},\text{cand}}[\text{Spec}], \text{Dom}_{\text{Spec}}, \text{Act}_R^{\text{tr},\text{real}}[\text{Spec}] \right)$$

Define the indexed family of data corresponding to all local specifications by

$$\text{LoopDat}_R^{\text{real},\text{loc}}[\mathbb{I}] := \left(\text{LoopDat}_R^{\text{real}}[\text{Spec}] \right)_{\text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]}$$

Lemma 4.66 (Injectivity of a Realized Local Transport Action). Under $\text{L}_{\leq 3}[\text{Ev}^{\text{cmp}}]$, for any

$$R \in \text{Rol}^{\text{cmp}}$$

and

$$\text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}]$$

the map

$$\text{Act}_R^{\text{tr},\text{real}}[\text{Spec}] : \text{Dom}_{\text{Spec}} \longrightarrow Y_{\text{Base}(\text{Spec})}^R$$

is injective.

Proof. By Definition 4.64,

$$\text{Dom}_{\text{Spec}} \subseteq \text{Dom} \left(\text{Act}_R^{\text{tr},\text{cand}}[\text{Spec}] \right).$$

By Lemma 4.63, the candidate pure-transport action is a partial injection on its domain. The domain restriction in Definition 4.65 therefore makes the restricted action stated in the lemma injective. $\square$

Definition 4.67 (Comparison-Stage Judgment Condition L_4 : Identity Action of Realized Local-Specification Closed Loops). Define the identity-action condition for realized local-specification closed loops by

$$\begin{aligned} & \text{L}_4[\text{Ev}_4^{\text{cmp}}] \\ & \iff \left[\text{L}_{\leq 3}[\text{Ev}^{\text{cmp}}] \right. \\ & \quad \implies \forall R \in \text{Rol}^{\text{cmp}}, \quad \forall \text{Spec} \in \text{LoopSpec}_{\text{loc}}^R[\mathbb{I}], \\ & \quad \left. \text{IdOnDom} \left(\text{Act}_R^{\text{tr},\text{real}}[\text{Spec}] \right) \right] \end{aligned}$$

Define the fourth-stage establishment bundle by

$$\text{L}_{\leq 4}[\text{Ev}^{\text{cmp}}] := \text{L}_{\leq 3}[\text{Ev}^{\text{cmp}}] \wedge \text{L}_4[\text{Ev}_4^{\text{cmp}}]$$

Definition 4.68 (Extended Comparison-Reachability Equivalence Relation). Assume

$$\text{L}_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

For each

$$R \in \text{Rol}^{\text{cmp}}$$

define the extended comparison-reachability equivalence relation by

$$\sim_R^{\text{cmp}}[\mathbb{I}] := \sim_{\mathfrak{M}_{\text{cmp}}^R[\mathbb{I}]}^{\text{orb}}$$

Lemma 4.69 (Equivalence-Relation Property of the Extended Comparison-Reachability Relation). Under

$$L_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

for each

$$R \in \text{Rol}^{\text{cmp}}$$

the relation

$$\sim_R^{\text{cmp}} [I]$$

is an equivalence relation on

$$\mathcal{R}_R^{\text{arr}}$$

Proof. Lemma 4.36 establishes the union and pairwise-disjointness conditions on the underlying family required by Lemma 4.26. Apply that lemma to the comparison-stage typed atomic-action system $\mathfrak{M}_{\text{cmp}}^R[I]$ of Definition 4.35, and then use Definition 4.68. $\square$

Definition 4.70 (Comparison Quotient Sets and Canonical Projections). Assume

$$L_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

For each

$$R \in \text{Rol}^{\text{cmp}}$$

define the comparison quotient set by

$$\text{Qu}_R^{\text{can}}[I] := \mathcal{R}_R^{\text{arr}} / \sim_R^{\text{cmp}} [I]$$

Define the canonical projection by

$$\text{Prj}_R^{\text{can}}[I] : \mathcal{R}_R^{\text{arr}} \longrightarrow \text{Qu}_R^{\text{can}}[I]$$

and

$$\text{Prj}_R^{\text{can}}[I](x) := [x]_{\sim_R^{\text{cmp}} [I]}$$

Lemma 4.71 (Surjectivity and Equivalence-Class Criterion for the Canonical Projection). For each

$$R \in \text{Rol}^{\text{cmp}}$$

the map

$$\text{Prj}_R^{\text{can}}[I] : \mathcal{R}_R^{\text{arr}} \twoheadrightarrow \text{Qu}_R^{\text{can}}[I]$$

is surjective. Moreover, for any

$$x, y \in \mathcal{R}_R^{\text{arr}}$$

the equivalence

$$\text{Prj}_R^{\text{can}}[I](x) = \text{Prj}_R^{\text{can}}[I](y) \iff x \sim_R^{\text{cmp}} [I] y.$$

holds.

Proof. Apply Lemma 4.69 and Lemma 4.31 to

$$S = \mathcal{R}_R^{\text{arr}}, \quad \sim = \sim_R^{\text{cmp}} [I]$$

$\square$

Definition 4.72 (Local Source-Compatible Two-Role Comparison Relation). For each $u \in U$, define the local source-compatible two-role comparison relation by

$$\begin{aligned} & \text{Rel}_u^{\text{cmp}, \text{src}}[\mathbb{I}] \\ & := \left\{ (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \mid \right. \\ & \quad (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \in \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}), \\ & \quad \left. (\text{Rec}_{R,u}^{\text{cmp}}(\alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{src}} \right\} \end{aligned}$$

Definition 4.73 (Comparison-Stage Judgment Condition L_5 : Source Compatibility of the Local Two-Role Comparison Relation). Define the source-compatibility condition for the local two-role comparison relation by

$$\begin{aligned} & \text{L}_5[\text{Ev}_5^{\text{cmp}}] \\ & \iff \left[\text{L}_{\leq 4}[\text{Ev}^{\text{cmp}}] \implies \forall u \in U, \right. \\ & \quad \left. \text{Rel}_u^{\text{cmp}} = \text{Rel}_u^{\text{cmp}, \text{src}}[\mathbb{I}] \right] \end{aligned}$$

Define the fifth-stage establishment bundle by

$$\text{L}_{\leq 5}[\text{Ev}^{\text{cmp}}] := \text{L}_{\leq 4}[\text{Ev}^{\text{cmp}}] \wedge \text{L}_5[\text{Ev}_5^{\text{cmp}}]$$

Definition 4.74 (Synchronized Two-Role Comparison-Arrival Space and Local Relation Bundle). Define the synchronized two-role comparison-arrival space by

$$\mathcal{R}^{\text{pair}} := \left\{ (u, (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}) \mid \begin{array}{l} u \in U, \\ (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \in \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}) \end{array} \right\}$$

Define the local two-role comparison-relation bundle by

$$\text{Rel}^{\text{pair}} := \{(u, \alpha) \in \mathcal{R}^{\text{pair}} \mid \alpha \in \text{Rel}_u^{\text{cmp}}\}$$

where

$$\alpha = (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}$$

Definition 4.75 (Synchronized Two-Role Comparison-Pair Equivalence Relation). Assume

$$\text{L}_{\leq 2}[\text{Ev}^{\text{cmp}}]$$

Define the synchronized two-role comparison-pair equivalence relation

$$\sim^{\text{pair}}[\mathbb{I}]$$

on

$$\mathcal{R}^{\text{pair}}$$

as follows. For any

$$p, q \in \mathcal{R}^{\text{pair}}$$

written as

$$\begin{aligned} p &= (u, (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}), \\ q &= (v, (\beta_R)_{R \in \text{Rol}^{\text{cmp}}}) \end{aligned}$$

set

$$\begin{aligned} p &\sim^{\text{pair}} [\mathbb{I}] q \\ &\iff [\forall R \in \text{Rol}^{\text{cmp}}, \quad (u, \alpha_R) \sim_R^{\text{cmp}} [\mathbb{I}] (v, \beta_R)]. \end{aligned}$$

Lemma 4.76 (Equivalence-Relation Property of the Synchronized Two-Role Comparison-Pair Relation). Under

$$\mathbf{L}_{\leq 2}[\mathbf{Ev}^{\text{cmp}}]$$

the relation

$$\sim^{\text{pair}} [\mathbb{I}]$$

is an equivalence relation on

$$\mathcal{R}^{\text{pair}}$$

Proof. By Lemma 4.69, for each

$$R \in \text{Rol}^{\text{cmp}}$$

the relation

$$\sim_R^{\text{cmp}} [\mathbb{I}]$$

is an equivalence relation on the corresponding comparison-arrival space. The right-hand side of Definition 4.75 is the universal conjunction of the equivalence relations for the respective role types; it therefore satisfies reflexivity, symmetry, and transitivity componentwise. $\square$

Definition 4.77 (Comparison-Stage Judgment Condition $\mathbf{L}_6$: Consistency of Local Two-Role Comparison-Relation Values across Equivalent Representatives). Define the consistency condition for local two-role comparison-relation values across equivalent representatives by

$$\begin{aligned} &\mathbf{L}_6[\mathbf{Ev}_6^{\text{cmp}}] \\ &\iff \left[\mathbf{L}_{\leq 5}[\mathbf{Ev}^{\text{cmp}}] \implies \text{Sat}(\text{Rel}^{\text{pair}}, \sim^{\text{pair}} [\mathbb{I}]) \right] \end{aligned}$$

Define the sixth-stage establishment bundle by

$$\mathbf{L}_{\leq 6}[\mathbf{Ev}^{\text{cmp}}] := \mathbf{L}_{\leq 5}[\mathbf{Ev}^{\text{cmp}}] \wedge \mathbf{L}_6[\mathbf{Ev}_6^{\text{cmp}}]$$

Definition 4.78 (Comparison-Stage Judgment Condition $\mathbf{L}_7$: Identity Action of Mixed Closed Loops). Define the mixed closed-loop identity-action condition by

$$\begin{aligned} &\mathbf{L}_7[\mathbf{Ev}_7^{\text{cmp}}] \\ &\iff \left[\mathbf{L}_{\leq 6}[\mathbf{Ev}^{\text{cmp}}] \implies \forall R \in \text{Rol}^{\text{cmp}}, \right. \\ &\quad \forall \mathbf{e} \in \text{LoopWord}_{\text{src-irr}}^{\text{mix}, R}[\mathbb{I}], \\ &\quad \left. \text{IdOnDom}(\text{Act}_R^{\text{ext}}[\mathbb{I}][\mathbf{e}]) \right]. \end{aligned}$$

Define the seventh-stage establishment bundle by

$$\mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}] := \mathbf{L}_{\leq 6}[\mathbf{Ev}^{\text{cmp}}] \wedge \mathbf{L}_7[\mathbf{Ev}_7^{\text{cmp}}]$$

Definition 4.79 (Comparison-Stage Judgment Condition G: Identity Action of All Pure-Transport Closed Loops). Define the identity-action condition for all pure-transport closed loops by

$$\begin{aligned} & \mathbf{G}[\mathbf{Ev}^{\text{cmp}}] \\ & \iff \left[\mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}] \implies \forall R \in \mathbf{Rol}^{\text{cmp}}, \right. \\ & \quad \forall \mathbf{w} \in \mathbf{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbf{l}], \\ & \quad \left. \text{IdOnDom}(\text{Act}_R^{\text{tr}}[\mathbf{l}][\mathbf{w}]) \right]. \end{aligned}$$

Theorem 4.80 (Global Gluing Condition and Identity Action of Extended Comparison Closed Words).

$$\begin{aligned} & \mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}] \\ & \implies \left[\mathbf{G}[\mathbf{Ev}^{\text{cmp}}] \iff \forall R \in \mathbf{Rol}^{\text{cmp}}, \quad \text{LoopId}(\mathfrak{W}_{\text{cmp}}^R[\mathbf{l}]) \right]. \end{aligned}$$

Proof. Assume $\mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}]$.

Assume $\mathbf{G}[\mathbf{Ev}^{\text{cmp}}]$. For each $R \in \mathbf{Rol}^{\text{cmp}}$, take any

$$\mathbf{e} \in \text{Loop}(\mathbf{Word}_R^{\text{ext}}[\mathbf{l}])$$

Apply Lemma 4.59. For the empty-word case, use Lemma 4.52. For the standard embedding of a pure-transport word, use Definition 4.79 and Lemma 4.55. For the standard embedding of a source-matching word, use endpoint preservation by the standard embedding, Lemma 4.48, and the same embedding compatibility. For the mixed-word case, apply Lemma 4.61; use Definition 4.78 for a source-irreducible terminal word and Definition 4.79 for a pure-transport terminal word. The domain inclusion and equality of action restrictions in that lemma yield, in every case,

$$\text{IdOnDom}(\text{Act}_R^{\text{ext}}[\mathbf{l}][\mathbf{e}])$$

Definition 4.32 then gives

$$\text{LoopId}(\mathfrak{W}_{\text{cmp}}^R[\mathbf{l}])$$

Conversely, assume the same closed-word identity-action condition for each $R \in \mathbf{Rol}^{\text{cmp}}$. Map any pure-transport closed-loop word to an extended comparison closed word by the standard embedding. Definition 4.32 and Lemma 4.55 show that its pure-transport action is the identity on its domain. Definition 4.79 yields $\mathbf{G}[\mathbf{Ev}^{\text{cmp}}]$. $\square$

Lemma 4.81 (Identity Action of All Pure-Transport Closed Loops and Local Injectivity of the Canonical Projection). Assume

$$\mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}]$$

Then

$$\begin{aligned} & \mathbf{G}[\mathbf{Ev}^{\text{cmp}}] \\ & \iff \left[\forall R \in \mathbf{Rol}^{\text{cmp}}, \quad \forall u \in U, \right. \\ & \quad \left. \text{Prj}_R^{\text{can}}[\mathbf{l}]|_{Y_u^R} : Y_u^R \longrightarrow \text{Qu}_R^{\text{can}}[\mathbf{l}] \text{ is injective} \right]. \end{aligned}$$

Proof. For each $R \in \mathbf{Rol}^{\text{cmp}}$, Lemma 4.36 establishes the union and pairwise-disjointness conditions on the underlying family required by Theorem 4.33. Apply that theorem to the comparison-stage typed atomic-action system $\mathfrak{W}_{\text{cmp}}^R[\mathbb{I}]$ of Definition 4.35. By Definition 4.68 and Definition 4.70, the orbit relation, orbit quotient, and canonical quotient projection in that theorem coincide respectively with

$$\sim_R^{\text{cmp}}[\mathbb{I}], \quad \text{Qu}_R^{\text{can}}[\mathbb{I}], \quad \text{Prj}_R^{\text{can}}[\mathbb{I}]$$

Thus, for each $R \in \mathbf{Rol}^{\text{cmp}}$, that theorem identifies the closed-word identity-action condition with the local-injectivity condition on the canonical quotient projection stated in this lemma. Composing this equivalence with the equivalence in Theorem 4.80 identifies

$$\mathbf{G}[\mathbf{Ev}^{\text{cmp}}]$$

with the injectivity of every local restriction of a canonical projection. $\square$

Definition 4.82 (Canonical Projection of Synchronized Two-Role Comparison Pairs). Assume

$$\mathbf{L}_{\leq 2}[\mathbf{Ev}^{\text{cmp}}]$$

Define the canonical projection of synchronized two-role comparison pairs by

$$\begin{aligned} \text{Prj}_{\mathfrak{C}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}] : \mathcal{R}^{\text{pair}} \\ \longrightarrow \prod_{R \in \mathbf{Rol}^{\text{cmp}}} \text{Qu}_R^{\text{can}}[\mathbb{I}] \end{aligned}$$

and

$$\begin{aligned} \text{Prj}_{\mathfrak{C}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}](u, (\alpha_R)_{R \in \mathbf{Rol}^{\text{cmp}}}) \\ := (\text{Prj}_R^{\text{can}}[\mathbb{I}](u, \alpha_R))_{R \in \mathbf{Rol}^{\text{cmp}}} \end{aligned}$$

Lemma 4.83 (Fiber Criterion for the Canonical Projection of Synchronized Two-Role Comparison Pairs). Under

$$\mathbf{L}_{\leq 2}[\mathbf{Ev}^{\text{cmp}}]$$

for any

$$p, q \in \mathcal{R}^{\text{pair}}$$

the equivalence

$$\text{Prj}_{\mathfrak{C}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}](p) = \text{Prj}_{\mathfrak{C}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}](q) \iff p \sim^{\text{pair}}[\mathbb{I}] q.$$

holds.

Proof. Write

$$\begin{aligned} p &= (u, (\alpha_R)_{R \in \mathbf{Rol}^{\text{cmp}}}), \\ q &= (v, (\beta_R)_{R \in \mathbf{Rol}^{\text{cmp}}}) \end{aligned}$$

By Definition 4.82, equality of the two families of canonical images is equivalent to equality of the corresponding canonical images for each

$$R \in \mathbf{Rol}^{\text{cmp}}$$

Apply Lemma 4.71 to each component and combine the resulting equivalences with Definition 4.75. Equality of the families of canonical images is then equivalent to comparison equivalence of the synchronized pairs. $\square$

Lemma 4.84 (Descent, Local Pullback, and Minimality of the Two-Role Comparison Relation on the Quotient). Assume

$$L_{\leq 6}[\text{Ev}^{\text{cmp}}]$$

Define the two-role comparison relation on the quotient by

$$\text{Rel}^{\text{can}}[\mathbb{I}] := \text{Prj}_{\mathcal{CD}}^{\text{can,pair}}[\mathbb{I}] [\text{Rel}^{\text{pair}}]$$

Then

$$\text{Rel}^{\text{can}}[\mathbb{I}] \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Qu}_R^{\text{can}}[\mathbb{I}].$$

Moreover, for any

$$(u, \alpha) \in \mathcal{R}^{\text{pair}}, \\ \alpha = (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}$$

the equivalences

$$\begin{aligned} \text{Prj}_{\mathcal{CD}}^{\text{can,pair}}[\mathbb{I}](u, \alpha) \in \text{Rel}^{\text{can}}[\mathbb{I}] \\ \iff \alpha \in \text{Rel}_u^{\text{cmp}} \\ \iff (\text{Rec}_{R,u}^{\text{cmp}}(\alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{src}}. \end{aligned}$$

hold.

For any relation

$$\text{Rel}_*^{\text{can}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{Qu}_R^{\text{can}}[\mathbb{I}]$$

the implication

$$\text{Prj}_{\mathcal{CD}}^{\text{can,pair}}[\mathbb{I}] [\text{Rel}^{\text{pair}}] \subseteq \text{Rel}_*^{\text{can}} \implies \text{Rel}^{\text{can}}[\mathbb{I}] \subseteq \text{Rel}_*^{\text{can}}.$$

holds. Consequently,

$$\text{Rel}^{\text{can}}[\mathbb{I}]$$

is the unique minimal relation on the quotient that contains the canonical images of all local two-role comparison relations.

Proof. By Lemma 4.83, the fibers of

$$\text{Prj}_{\mathcal{CD}}^{\text{can,pair}}[\mathbb{I}]$$

are the equivalence classes of

$$\sim^{\text{pair}}[\mathbb{I}]$$

By Definition 4.77 and the assumption

$$L_{\leq 6}[\text{Ev}^{\text{cmp}}]$$

the relation

$$\text{Sat}(\text{Rel}^{\text{pair}}; \sim^{\text{pair}}[\mathbb{I}]).$$

holds. Applying Lemma 4.28 gives the type of the relation on the quotient, the local-pullback equivalence, and minimality. By Definition 4.74, the local pullback is equivalent to

$$\alpha \in \text{Rel}_u^{\text{cmp}}$$

By Definition 4.73, this is equivalent to

$$(\text{Rec}_{R,u}^{\text{cmp}}(\alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{src}}$$

□

Definition 4.85 (Comparison-Stage Preceding-Condition Establishment Bundles). Define the establishment bundle preceding the first condition by

$$L_{<1} [\mathbf{Ev}^{\text{cmp}}] := \top$$

For each

$$i \in \{2, \dots, 7\}$$

define the establishment bundle preceding condition i by

$$L_{<i} [\mathbf{Ev}^{\text{cmp}}] := L_{\leq i-1} [\mathbf{Ev}^{\text{cmp}}]$$

For each

$$i \in \{1, \dots, 7\}$$

the identity

$$L_{\leq i} [\mathbf{Ev}^{\text{cmp}}] = L_{<i} [\mathbf{Ev}^{\text{cmp}}] \wedge L_i [\mathbf{Ev}_i^{\text{cmp}}]$$

holds.

Definition 4.86 (Comparison-Stage Operability). Let

$$\text{Operative}(\mathcal{X})$$

denote comparison-stage operability, defined by

$$\text{Operative}(\mathcal{X}) \iff L_{\leq 7} [\mathbf{Ev}^{\text{cmp}}] \wedge G [\mathbf{Ev}^{\text{cmp}}]$$

Definition 4.87 (Family of Comparison Index Domains). Define the comparison index domains for the family of local objects and the family of local transitions by

$$\text{Idx}_m^{\text{cmp}} := \text{Rol}^{\text{cmp}} \times U^m, \quad m \in \{1, 2\}$$

Definition 4.88 (Comparison Gluing Datum). Assume $\text{Operative}(\mathcal{X})$. Define the comparison gluing datum by

$$\begin{aligned} \text{Dat}^{\text{glue}}[\mathbb{I}] &:= \left(\text{Dat}^{\text{tr}}[\mathbb{I}], \right. \\ &\quad (\text{LoopSpec}_{\text{loc}}^R[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad (\text{LoopDat}_R^{\text{real, loc}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad (\sim_R^{\text{cmp}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad (\text{Qu}_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad (\text{Prj}_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ &\quad \left(\text{Prj}_R^{\text{can}}[\mathbb{I}]|_{Y_u^R} \right)_{(R, u) \in \text{Idx}_1^{\text{cmp}}}, \\ &\quad \left. \text{Rel}^{\text{can}}[\mathbb{I}] \right) \end{aligned}$$

By Lemma 4.81, for each

$$(R, u) \in \text{Idx}_1^{\text{cmp}}$$

the restriction

$$\text{Prj}_R^{\text{can}}[\mathbb{I}]|_{Y_u^R}$$

is injective.

Definition 4.89 (Canonical Comparison Quotient). Assume $\text{Operative}(\mathcal{X})$. Define the canonical comparison quotient by

$$\begin{aligned} \text{Qu}^{\text{can}}[\mathbb{I}] \\ := ((\text{Qu}_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \\ (\text{Prj}_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \text{Rel}^{\text{can}}[\mathbb{I}]) \end{aligned}$$

Theorem 4.90 (Comparison-Stage Operability and Generation of the Canonical Comparison Quotient). For any realized operational configuration $\mathcal{X}$ and any family of typed comparison rules $\mathcal{J}_{\text{cmp}}$,

$$\text{Operative}(\mathcal{X}) \iff \text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}] \wedge \text{G}[\text{Ev}^{\text{cmp}}].$$

If $\text{Operative}(\mathcal{X})$ holds, then the following statements hold.

1. The comparison gluing datum

$$\text{Dat}^{\text{glue}}[\mathbb{I}]$$

and the canonical comparison quotient

$$\text{Qu}^{\text{can}}[\mathbb{I}]$$

are uniquely determined in dependency order.

2. For any

$$(R, u) \in \text{Idx}_1^{\text{cmp}}$$

the restriction

$$\text{Prj}_R^{\text{can}}[\mathbb{I}]|_{Y_u^R}$$

is injective.

3. The local-pullback equivalence in Lemma 4.84 holds.

4. By the minimality in Lemma 4.84,

$$\text{Rel}^{\text{can}}[\mathbb{I}]$$

is the unique minimal relation on the quotient that contains the canonical images of all local two-role comparison relations.

Proof. Definition 4.86 gives the first equivalence.

Assume $\text{Operative}(\mathcal{X})$. Definition 4.88 and Definition 4.89 give the first item.

Lemma 4.81 gives the second item.

Lemma 4.84 gives the third and fourth items and the uniqueness of the minimal relation on the quotient. $\square$

Comparison reachability and equivalence classes

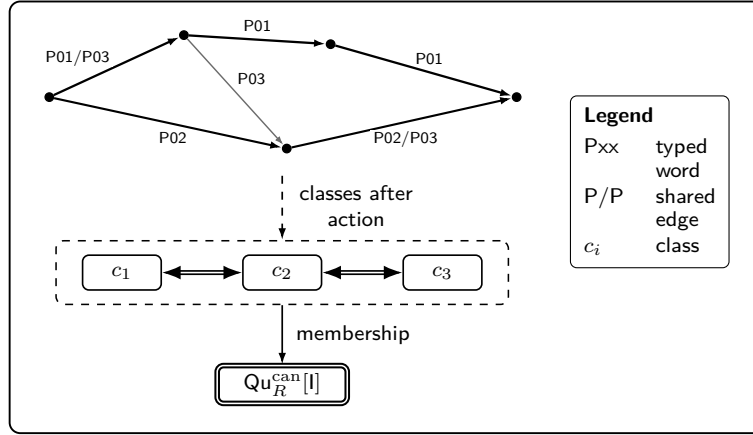


Figure 8: Actions of extended comparison words and comparison-reachability classes. Assume $L_{\leq 2}[\mathbf{Ev}^{cmp}]$ and fix $R \in \mathbf{Rol}^{cmp}$. The dots represent reception-phase types and the edges represent admissible atomic actions. P01, P02, and P03 denote three extended comparison words $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ with the same source and target types. Take a point x in the common domain of the three actions and set $y_i := \mathbf{Act}_R^{ext}[l][\mathbf{e}_i](x)$ and $c_i := \mathbf{Prj}_R^{can}[l](y_i)$. Definition 4.25 and Definition 4.68 give $x \sim_R^{cmp}[l]y_i$. Lemma 4.71 then gives $c_1 = c_2 = c_3 = \mathbf{Prj}_R^{can}[l](x)$. The lower part shows that this common class belongs to $Qu_R^{can}[l]$.

Figure 6 shows the dependency structure of the comparison stage from the comparison-rule-dependent common input to the canonical comparison quotient. Figure 8 shows that the images of a common domain point under three extended comparison words project to the same comparison-reachability class.

Theorem 4.80 states the equivalence between the judgment of closed-loop actions, including finite source-word deletion, and the identity-action condition for the family of extended comparison closed words. Theorem 4.90 connects the comparison-stage establishment conditions to the eight-component datum of Definition 4.88 and the three-component canonical comparison quotient of Definition 4.89. The injectivity of local restrictions of the canonical projections, the equivalence between local two-role comparison relations and source-side two-role comparison relations, and the minimality of the two-role comparison relation on the quotient are contained in items two through four of that theorem.

The canonical comparison quotient is an input to the local quotient charts, representational coordinates, and canonical orders of the subsequent representation stage.

4.1.3 Operational Skeleton

The operational skeleton of the comparison stage consists of four layers: input, action, quotient, and canonical construction.

$$\begin{aligned}
I &\xrightarrow{\text{gen}} \text{Dat}^{\text{mat}}[I] \\
&\xrightarrow{\text{gen}} \begin{array}{c} \text{Dat}^{\text{tr}}[I], \\ (\text{LoopSpec}_{\text{loc}}^R[I])_{R \in \text{Rol}^{\text{cmp}}} \end{array} \\
&\xrightarrow{\text{gen}} \left(\mathcal{R}_R^{\text{arr}}, \mathfrak{W}_{\text{cmp}}^R[I], \text{LoopDat}_R^{\text{real,loc}}[I] \right)_{R \in \text{Rol}^{\text{cmp}}} \\
&\xrightarrow{\text{gen}} (\sim_R^{\text{cmp}}[I], \text{Qu}_R^{\text{can}}[I], \text{Prj}_R^{\text{can}}[I])_{R \in \text{Rol}^{\text{cmp}}} .
\end{aligned}$$

Here, Definition 4.68 gives

$$\sim_R^{\text{cmp}}[I] = \sim_{\mathfrak{W}_{\text{cmp}}^R[I]}^{\text{orb}}$$

The establishment conditions connect to the respective layers as follows:

$$\begin{aligned}
&\left\{ L_{\leq 3}[\text{Ev}^{\text{cmp}}], (\text{LoopSpec}_{\text{loc}}^R[I])_{R \in \text{Rol}^{\text{cmp}}} \right\} \\
&\quad \xrightarrow{\text{gen}} \left(\text{LoopDat}_R^{\text{real,loc}}[I] \right)_{R \in \text{Rol}^{\text{cmp}}} , \\
&\left\{ L_{\leq 6}[\text{Ev}^{\text{cmp}}], (\text{Rel}_u^{\text{cmp}})_{u \in U}, (\text{Qu}_R^{\text{can}}[I], \text{Prj}_R^{\text{can}}[I])_{R \in \text{Rol}^{\text{cmp}}} \right\} \\
&\quad \xrightarrow{\text{gen}} \text{Rel}^{\text{can}}[I], \\
&L_{\leq 7}[\text{Ev}^{\text{cmp}}] \wedge G[\text{Ev}^{\text{cmp}}] \iff \text{Operative}(\mathcal{X}).
\end{aligned}$$

By Definition 4.67, the fourth condition $L_4[\text{Ev}_4^{\text{cmp}}]$ requires

$$\text{IdOnDom} \left(\text{Act}_R^{\text{tr,real}}[\text{Spec}] \right)$$

Apply Theorem 4.80 to the judgment of finite source-word deletion, source-irreducible terminal words, and pure-transport terminal words.

By Theorem 4.90, the canonical-construction layer consists of the uniquely determined objects

$$\text{Dat}^{\text{glue}}[I], \quad \text{Qu}^{\text{can}}[I]$$

The eight components of the former are given in Definition 4.88; the three components of the latter are given in Definition 4.89.

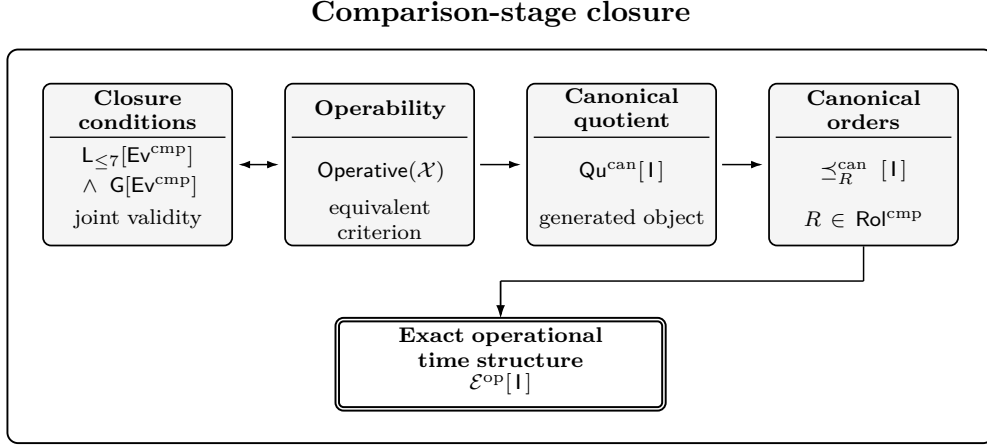


Figure 9: Separation between comparison-stage closure and the subsequent object sequence. Comparison-stage closure yields the operability judgment; the canonical comparison quotient derived from the C and D roles, the descent of canonical order data, and the exact operational time structure are determined as subsequent objects.

4.2 Exact Operational Time Structure and Representational Atlas

Definition 4.91 (Representation-Stage Input Interface). Every object at the representation stage takes as input the comparison-rule-dependent common input

$$I = (\mathcal{X}, \mathcal{J}_{\text{cmp}})$$

of Definition 4.1. By the expansion rule in that definition, evaluation of an object F that depends on this input satisfies

$$F[I] = F(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

Throughout this subsection, each symbol root is first introduced in its full form with both a role type and a reception phase. Thereafter, only the reception-phase subscript is omitted, and only when the reception phase is fixed within the same definition, proposition, or proof and the full form is uniquely recoverable. The reception-phase index u of the local incomparability relation $\asymp_{R,u}[I]$ is retained to distinguish it from the global relation $\asymp_R[I]$. For a binary relation $r \subseteq S \times S$ and a subset $A \subseteq S$, its restriction is $r|_A := r \cap (A \times A)$. The role type, any varying family index, distinct reception phases, and any residual modifier that selects a genuine branch remain explicit in the full form.

Definition 4.92 (Representation-Stage Partial Establishment Bundles). Recursively define the representation-stage partial establishment bundles by

$$\begin{aligned} \text{Repr}_{\leq 1}[I] &\iff K_{\text{CanOrdDesc}}^{\text{repr}}[I], \\ \text{Repr}_{\leq 2}[I] &\iff \text{Repr}_{\leq 1}[I] \wedge K_{\text{LocIncTrans}}^{\text{repr}}[I], \\ \text{Repr}_{\leq 3}[I] &\iff \text{Repr}_{\leq 2}[I] \wedge K_{\text{GlobIncTrans}}^{\text{repr}}[I], \\ \text{ReprCtx}_{\leq k}[I] &\iff \text{Operative}(\mathcal{X}) \wedge \text{Repr}_{\leq k}[I], \quad k \in \{1, 2, 3\}. \end{aligned}$$

The representation stage takes as input $\text{Operative}(\mathcal{X})$ and, from Definition 4.89, the $\mathfrak{C}/\mathfrak{D}$ -derived canonical comparison quotient $\text{Qu}^{\text{can}}[I]$. In dependency order, its structural generation sequence constructs the $\mathfrak{C}/\mathfrak{D}$ -source-induced arrival preorders and the exact

operational time structure; the local order-coordinate structures and the comparison-order atlas; and the global order-coordinate structures. The comparison representational coordinates, quantitative representations, and continuum-approximation evaluation input are constructed by additionally specifying a selected ordered value range and embedding, a quantitative affine realization, a finite operational granularity datum, and componentwise evaluation inputs. Definition 4.92 uses $\text{Repr}_{\leq k}[\mathbb{I}]$, ($k \in \{1, 2, 3\}$), for the cumulative establishment conditions of the three representation stages.

4.2.1 Source-Induced Arrival Preorders

Assume $\text{Operative}(\mathcal{X})$. The representation stage takes as input the $\mathfrak{C}$ -derived direct-sum order on source families from Definition 3.17, the $\mathfrak{D}$ -derived record-index order from Definition 3.12, the comparison-side recovery maps from Definition 3.21, and the tagged comparison-arrival reception spaces from Definition 4.13.

Definition 4.93 (Pullback of a Binary Relation along a Map). Let A, T be sets. On T , let $\rho \subseteq T^2$ be a binary relation, and let $r : A \rightarrow T$ be a map. Define the pullback relation by

$$r^*(\rho) := \{(a_i, a_j) \in A^2 \mid (r(a_i), r(a_j)) \in \rho\}$$

Lemma 4.94 (Preorder Property under Pullback of a Binary Relation). If ρ in Definition 4.93 is a preorder, then $r^*(\rho)$ is a preorder on A .

Proof. Reflexivity of ρ gives $(r(a_i), r(a_i)) \in \rho$. Applying transitivity to $(r(a_i), r(a_j)) \in \rho$ and $(r(a_j), r(a_k)) \in \rho$ gives $(r(a_i), r(a_k)) \in \rho$. $\square$

Definition 4.95 (Source-Induced Arrival Preorders). For each $R \in \text{Rol}^{\text{cmp}}$, define the tagged comparison-arrival recovery map and the source-induced arrival preorder by

$$\begin{aligned} \text{Rec}_R^{\text{arr}} : \mathcal{R}_R^{\text{arr}} &\longrightarrow \text{Tok}_R^*, \\ \text{Rec}_R^{\text{arr}}(u, \text{ArrVal}_{R,u}) &:= \text{Rec}_{R,u}^{\text{cmp}}(\text{ArrVal}_R), \\ \preceq_R^{\text{arr}} &:= (\text{Rec}_R^{\text{arr}})^*(\preceq_*^R) \end{aligned}$$

Expanding Lemma 4.94 yields the source-side order judgment on the recovered images of two arrival points.

Proposition 4.96 (Preorder Property of Source-Induced Arrival Preorders). For each $R \in \text{Rol}^{\text{cmp}}$,

$$(\mathcal{R}_R^{\text{arr}}, \preceq_R^{\text{arr}})$$

is a preordered set.

Proof. By Proposition 3.18 and Proposition 3.13, for each $R \in \text{Rol}^{\text{cmp}}$, the relation $\preceq_*^R$ is a preorder on Tok_R^* . By Definition 4.95, the relation $\preceq_R^{\text{arr}}$ is the pullback of $\preceq_*^R$ along $\text{Rec}_R^{\text{arr}}$. Apply Lemma 4.94. $\square$

Definition 4.97 (Descent of a Preorder to a Quotient). For a set A , an equivalence relation $\sim$ on that set, and a preorder $\preceq$ on that set, define the order-descent condition

and the order-separation condition by

$$\begin{aligned} \text{OrdDesc}(A, \sim, \preceq) &\iff \left[\begin{array}{l} \forall a_0, a'_0, a_1, a'_1 \in A, \\ (a_0 \sim a'_0) \wedge (a_1 \sim a'_1) \\ \implies (a_0 \preceq a_1 \iff a'_0 \preceq a'_1) \end{array} \right], \\ \text{OrdSep}(A, \sim, \preceq) &\iff \left[\begin{array}{l} \forall a_0, a_1 \in A, \\ (a_0 \preceq a_1) \wedge (a_1 \preceq a_0) \\ \implies a_0 \sim a_1 \end{array} \right] \end{aligned}$$

When $\text{OrdDesc}(A, \sim, \preceq)$ holds, define the quotient relation by

$$[a_0]_{\sim} \preceq / \sim [a_1]_{\sim} \iff a_0 \preceq a_1$$

Lemma 4.98 (Partial Order on a Preorder Quotient). If $\text{OrdDesc}(A, \sim, \preceq)$ and $\text{OrdSep}(A, \sim, \preceq)$ in Definition 4.97 hold, then $\preceq / \sim$ is a partial order on $A/\sim$.

Proof. The condition **OrdDesc** makes the quotient relation independent of representatives. Reflexivity and transitivity of $\preceq$ descend to the quotient relation. If the relation holds in both directions on the quotient, the representatives satisfy $\preceq$ in both directions. By **OrdSep**, the representatives are $\sim$ -equivalent, and the corresponding quotient elements are equal. $\square$

Definition 4.99 (Canonical Order-Descent Condition at the Representation Stage). Assume **Operative**($\mathcal{X}$). Define the representative-invariant descent condition for the source-induced arrival preorders by

$$K_{\text{CanOrdDesc}}^{\text{repr}}[\mathbb{I}] \iff [\forall R \in \text{Rol}^{\text{cmp}}, \quad \text{OrdDesc}(\mathcal{R}_R^{\text{arr}}, \sim_R^{\text{cmp}}[\mathbb{I}], \preceq_R^{\text{arr}})].$$

Expanding Definition 4.97 gives the representative-invariance condition on four arrival points.

Lemma 4.100 (Comparison Equivalence Obtained from Mutual Arrival Order). Assume **Operative**($\mathcal{X}$). For each $R \in \text{Rol}^{\text{cmp}}$,

$$\text{OrdSep}(\mathcal{R}_R^{\text{arr}}, \sim_R^{\text{cmp}}[\mathbb{I}], \preceq_R^{\text{arr}})$$

holds.

Proof. To prove **OrdSep** in Definition 4.97, take arbitrary

$$\text{ArrPt}_{R,0}, \text{ArrPt}_{R,1} \in \mathcal{R}_R^{\text{arr}}$$

and assume

$$\text{ArrPt}_{R,0} \preceq_R^{\text{arr}} \text{ArrPt}_{R,1}, \quad \text{ArrPt}_{R,1} \preceq_R^{\text{arr}} \text{ArrPt}_{R,0}$$

By Definition 4.13, there exist $u, v \in U$ and

$$\text{ArrVal}_{R,u}^{(0)} \in \text{Im}(\text{Arr}_{R,u}^{\text{cmp}}), \quad \text{ArrVal}_{R,v}^{(1)} \in \text{Im}(\text{Arr}_{R,v}^{\text{cmp}})$$

such that

$$\begin{aligned} \text{ArrPt}_{R,0} &= \left(u, \text{ArrVal}_{R,u}^{(0)} \right), \\ \text{ArrPt}_{R,1} &= \left(v, \text{ArrVal}_{R,v}^{(1)} \right), \\ \text{SrcTok}_{R,i} &:= \text{Rec}_R^{\text{arr}}(\text{ArrPt}_{R,i}), \quad i \in \{0, 1\}. \end{aligned}$$

Expand the preceding two assumptions with Definition 4.95, and use antisymmetry from Proposition 3.18 and Proposition 3.13 to obtain

$$\begin{aligned}\text{SrcTok}_{R,0} &\preceq_*^R \text{SrcTok}_{R,1}, \\ \text{SrcTok}_{R,1} &\preceq_*^R \text{SrcTok}_{R,0} \\ &\implies \text{SrcTok}_{R,0} = \text{SrcTok}_{R,1}.\end{aligned}$$

Definition 4.86, Definition 4.85, Definition 4.5, and Definition 4.8 give

$$\begin{aligned}\text{SrcTok}_{R,0} &\in \text{Dom}(\text{Arr}_{R,v}^{\text{cmp}}), \\ \text{Arr}_{R,v}^{\text{cmp}}(\text{SrcTok}_{R,1}) &= \text{ArrVal}_{R,v}^{(1)}, \\ \text{ArrVal}_{R,u}^{(0)} &\in \text{Dom}(\text{SrcMatch}_{uv}^R), \\ \text{SrcMatch}_{uv}^R(\text{ArrVal}_{R,u}^{(0)}) &= \text{Arr}_{R,v}^{\text{cmp}}(\text{SrcTok}_{R,0}) \\ &= \text{Arr}_{R,v}^{\text{cmp}}(\text{SrcTok}_{R,1}) \\ &= \text{ArrVal}_{R,v}^{(1)}.\end{aligned}$$

By Definition 4.49, Definition 4.18, and Definition 4.50, the length-one source-matching word and its action satisfy

$$\begin{aligned}\text{SrcldWord}_{R,01}^{(1)} &:= (1, (u, v), (\text{src})) \in \text{Word}_R^{\text{ext}}[\mathbb{I}], \\ \text{ArrPt}_{R,0} &\in \text{Dom}\left(\text{Act}_R^{\text{ext}}[\mathbb{I}] \left[\text{SrcldWord}_{R,01}^{(1)} \right]\right), \\ \text{Act}_R^{\text{ext}}[\mathbb{I}] \left[\text{SrcldWord}_{R,01}^{(1)} \right] (\text{ArrPt}_{R,0}) &= \text{ArrPt}_{R,1}\end{aligned}$$

Definition 4.68 then makes the two arrival points comparison-equivalent. This proves **OrdSep** in Definition 4.97. $\square$

Definition 4.101 (Canonical Partial Orders for the $\mathfrak{C}/\mathfrak{D}$ Roles). Assume $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$, use the comparison quotient set and canonical projection of Definition 4.70 to define the canonical partial order for the $\mathfrak{C}/\mathfrak{D}$ roles by

$$\preceq_R^{\text{can}}[\mathbb{I}] := \preceq_R^{\text{arr}} / \sim_R^{\text{cmp}}[\mathbb{I}]$$

Expanding Definition 4.97 gives the representative order judgment on canonical-projection images.

Lemma 4.102 (Partial-Order Property of the Canonical Orders for the $\mathfrak{C}/\mathfrak{D}$ Roles). Under $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$, for each $R \in \text{Rol}^{\text{cmp}}$,

$$(\text{Qu}_R^{\text{can}}[\mathbb{I}], \preceq_R^{\text{can}}[\mathbb{I}])$$

is a partially ordered set.

Proof. The preorder property of $\preceq_R^{\text{arr}}$ follows from Proposition 4.96, and the equivalence-relation property of comparison equivalence follows from Lemma 4.69. The order-descent condition follows from Definition 4.99, and the order-separation condition follows from Lemma 4.100. For the quotient relation of Definition 4.101, apply Lemma 4.98. $\square$

Definition 4.103 (Exact Operational Time Structure). Assume $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$. Define the exact operational time structure by

$$\begin{aligned} \mathcal{E}^{\text{op}}[\mathbb{I}] &:= (\text{Qu}^{\text{can}}[\mathbb{I}], (\preceq_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}), \\ (\preceq_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}} &\longleftrightarrow (\preceq_{\mathcal{C}}^{\text{can}}[\mathbb{I}], \preceq_{\mathcal{D}}^{\text{can}}[\mathbb{I}]). \end{aligned}$$

The first component is the canonical comparison quotient whose components are the family of comparison quotient sets, the family of canonical projections, and the two-role comparison relation on the quotient. The second component is the role-indexed family of canonical partial orders on the comparison quotient sets of the respective comparison roles.

Definition 4.104 (Local Canonical Quotient Regions). Assume $\text{Operative}(\mathcal{X})$. For each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$, define the local canonical quotient region by

$$\begin{aligned} \mathcal{U}_{R,u}[\mathbb{I}] &:= \text{Prj}_R^{\text{can}}[\mathbb{I}] (\{u\} \times \text{Im}(\text{Arr}_R^{\text{cmp}})) \\ &\subseteq \text{Qu}_R^{\text{can}}[\mathbb{I}]. \end{aligned}$$

Lemma 4.105 (Strict Part of a Partial Order). For a set S and a partial order $\preceq$ on that set, define a binary relation $\prec$ by

$$x \prec y \iff (x \preceq y) \wedge (x \neq y) \quad (x, y \in S)$$

Then $\prec$ is a strict partial order on S .

Proof. For any $x \in S$, the relation $x \prec x$ does not hold because it contains $x \neq x$. Suppose $x \prec y$ and $y \prec z$. Transitivity of the partial order $\preceq$ gives $x \preceq z$. If $x = z$, antisymmetry applied to $y \preceq x$ and $x \preceq y$ gives $x = y$. This is incompatible with $x \prec y$, which contains $x \neq y$. Hence $x \neq z$, and $x \prec z$ holds. $\square$

Definition 4.106 (Linearization of Incomparability in a Strict Partial Order). For a strict-partial-order datum $\mathbf{P} = (S, \prec)$, define the incomparability relation and its transitivity condition by

$$\begin{aligned} \text{Inc}[\mathbf{P}] &:= \{(x, y) \in S^2 \mid \neg(x \prec y) \wedge \neg(y \prec x)\}, \\ \text{IncTrans}[\mathbf{P}] &\iff \left[\begin{array}{c} \forall x_0, x_1, x_2 \in S, \\ (x_0 \text{ Inc}[\mathbf{P}] x_1) \wedge (x_1 \text{ Inc}[\mathbf{P}] x_2) \\ \implies x_0 \text{ Inc}[\mathbf{P}] x_2 \end{array} \right] \end{aligned}$$

When $\text{IncTrans}[\mathbf{P}]$ holds, define the incomparability linearization by

$$\begin{aligned} \text{LinQuot}[\mathbf{P}] &:= (\text{Qu}^{\text{inc}}[\mathbf{P}], \text{Prj}_{\mathbf{P}}^{\text{inc}}, <_{\mathbf{P}}^{\text{inc}}), \\ \text{Qu}^{\text{inc}}[\mathbf{P}] &:= S / \text{Inc}[\mathbf{P}], \\ \text{Prj}_{\mathbf{P}}^{\text{inc}} : S &\twoheadrightarrow \text{Qu}^{\text{inc}}[\mathbf{P}], \quad \text{Prj}_{\mathbf{P}}^{\text{inc}}(x) := [x]_{\text{Inc}[\mathbf{P}]}, \\ \text{Prj}_{\mathbf{P}}^{\text{inc}}(x) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(y) &\iff x \prec y \end{aligned}$$

Definition 4.107 (Equality Kernel of a Map). For a map

$$\Phi : S \longrightarrow T$$

define the equality kernel by

$$\text{EqKer}(\Phi) := \{(x_0, x_1) \in S^2 \mid \Phi(x_0) = \Phi(x_1)\}$$

Definition 4.108 (Ordered-Representation Judgment Condition). For a set S , binary relations $\sim$ and $\prec$ on that set, a linearly ordered set $(T, <)$, and a map $\Phi : S \rightarrow T$, define the ordered-representation judgment condition using the relation pullback of Definition 4.93 by

$$\begin{aligned} \text{OrdRepCond}(\Phi; \sim, \prec, <) \\ \iff [\text{EqKer}(\Phi) = \sim] \wedge [\Phi^*(<) = \prec] \end{aligned}$$

Lemma 4.109 (Strict Linear-Order Quotient by Transitive Incomparability). Suppose that the strict-partial-order datum $\mathbf{P} = (S, \prec)$ satisfies $\text{IncTrans}[\mathbf{P}]$. Then $\text{Inc}[\mathbf{P}]$ is an equivalence relation, and $\text{LinQuot}[\mathbf{P}]$ has a third component $<_{\mathbf{P}}^{\text{inc}}$ that is independent of representatives and is a strict linear order on $\text{Qu}^{\text{inc}}[\mathbf{P}]$. Moreover, for any $x, x', y, y' \in S$,

$$(x \text{ Inc}[\mathbf{P}] x') \wedge (y \text{ Inc}[\mathbf{P}] y') \implies (x \prec y \iff x' \prec y').$$

In addition,

$$\text{OrdRepCond}(\text{Prj}_{\mathbf{P}}^{\text{inc}}; \text{Inc}[\mathbf{P}], \prec, <_{\mathbf{P}}^{\text{inc}})$$

holds.

Proof. Irreflexivity of $\prec$ gives $\neg(x \prec x)$, and the definition then gives $x \text{ Inc}[\mathbf{P}] x$. Exchanging the two negated components in the definition shows that $x \text{ Inc}[\mathbf{P}] y$ implies $y \text{ Inc}[\mathbf{P}] x$. Together with $\text{IncTrans}[\mathbf{P}]$, these properties make $\text{Inc}[\mathbf{P}]$ an equivalence relation.

Assume $x \text{ Inc}[\mathbf{P}] x'$, $y \text{ Inc}[\mathbf{P}] y'$, and $x \prec y$. Suppose that $x' \prec y$ does not hold. If $y \prec x'$, then $x \prec y \prec x'$ gives $x \prec x'$, which is incompatible with $x \text{ Inc}[\mathbf{P}] x'$. If $y \prec x'$ also does not hold, then $x' \text{ Inc}[\mathbf{P}] y$; transitivity with $x \text{ Inc}[\mathbf{P}] x'$ gives $x \text{ Inc}[\mathbf{P}] y$, which is incompatible with $x \prec y$. Therefore $x' \prec y$.

Next suppose that $x' \prec y'$ does not hold. If $y' \prec x'$, then $y' \prec x' \prec y$ gives $y' \prec y$, which is incompatible with $y \text{ Inc}[\mathbf{P}] y'$. If $y' \prec x'$ also does not hold, then $x' \text{ Inc}[\mathbf{P}] y'$; transitivity with $y' \text{ Inc}[\mathbf{P}] y$ gives $x' \text{ Inc}[\mathbf{P}] y$, which is incompatible with $x' \prec y$. Hence $x' \prec y'$. Symmetry gives $x' \text{ Inc}[\mathbf{P}] x$ and $y' \text{ Inc}[\mathbf{P}] y$. Applying the implication just proved to (x', x, y', y) gives the reverse implication and establishes representative invariance.

The equivalence-class criterion follows from the standard equivalence-class criterion for the quotient projection of the equivalence relation $\text{Inc}[\mathbf{P}]$. Every $A \in \text{Qu}^{\text{inc}}[\mathbf{P}]$ has some $x \in S$ for which $A = \text{Prj}_{\mathbf{P}}^{\text{inc}}(x)$; hence $\text{Prj}_{\mathbf{P}}^{\text{inc}}$ is surjective.

By representative invariance, the truth of $\text{Prj}_{\mathbf{P}}^{\text{inc}}(x) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(y)$ is independent of the representatives. The relation with the same element on both sides is equivalent to $x \prec x$, and therefore does not hold.

Suppose that $\text{Prj}_{\mathbf{P}}^{\text{inc}}(x) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(y)$ and $\text{Prj}_{\mathbf{P}}^{\text{inc}}(y) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(z)$. Let y_0, y_1 be the respective intermediate representatives used in the two relations; then $y_0 \text{ Inc}[\mathbf{P}] y_1$. Representative invariance turns $x \prec y_0$ into $x \prec y_1$, and transitivity of $x \prec y_1 \prec z$ gives $x \prec z$. Therefore $\text{Prj}_{\mathbf{P}}^{\text{inc}}(x) <_{\mathbf{P}}^{\text{inc}} \text{Prj}_{\mathbf{P}}^{\text{inc}}(z)$.

If $\text{Prj}_{\mathbf{P}}^{\text{inc}}(x) \neq \text{Prj}_{\mathbf{P}}^{\text{inc}}(y)$, the equivalence-class criterion shows that $x \text{ Inc}[\mathbf{P}] y$ does not hold. By the definition of incomparability, either $x \prec y$ or $y \prec x$ holds. If both held, they would give $x \prec x$; hence exactly one holds. Thus $<_{\mathbf{P}}^{\text{inc}}$ is a strict linear order that compares any two distinct elements in a unique direction. The class criterion and order criterion in Definition 4.106 coincide with the two components in Definition 4.107 and 4.108. $\square$

Definition 4.110 (Canonical strict orders on $\mathfrak{C}/\mathfrak{D}$). Assume $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$ and any

$$(x_0, x_1) \in (\text{Qu}_R^{\text{can}}[\mathbb{I}])^2$$

define the canonical strict order by

$$\begin{aligned} \prec_R^{\text{can}} [\mathbb{I}] &\subseteq (\text{Qu}_R^{\text{can}} [\mathbb{I}])^2, \\ x_0 \prec_R^{\text{can}} [\mathbb{I}] x_1 &\iff (x_0 \preceq_R^{\text{can}} [\mathbb{I}] x_1) \\ &\quad \wedge (x_0 \neq x_1) \end{aligned}$$

Definition 4.111 (Local incomparability relations). Assume $\text{ReprCtx}_{\leq 1} [\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$, define the local strict-order datum and the local incomparability relation by

$$\begin{aligned} \text{OrdDat}_{R,u}^{\text{loc}} [\mathbb{I}] &:= (\mathcal{U}_R [\mathbb{I}], \prec_R^{\text{can}} [\mathbb{I}]|_{\mathcal{U}_R [\mathbb{I}]}) , \\ \succ_{R,u} [\mathbb{I}] &:= \text{Inc} [\text{OrdDat}_{R,u}^{\text{loc}} [\mathbb{I}]] \end{aligned}$$

Expanding Definition 4.106 gives the bidirectional incomparability condition for any two local canonical comparison quotient classes.

Definition 4.112 (Representation-stage local incomparability-transitivity condition). Assume $\text{ReprCtx}_{\leq 1} [\mathbb{I}]$. Define the representation-stage local incomparability-transitivity condition by

$$K_{\text{LocIncTrans}}^{\text{repr}} [\mathbb{I}] \iff \left[\begin{array}{c} \forall (R, u) \in \text{Idx}_1^{\text{cmp}}, \\ \text{IncTrans} [\text{OrdDat}_{R,u}^{\text{loc}} [\mathbb{I}]] \end{array} \right].$$

Expanding Definition 4.106 gives the transitivity condition quantified over three points.

Lemma 4.113 (Strict-partial-order property of the canonical strict orders on $\mathfrak{C}/\mathfrak{D}$). Under $\text{ReprCtx}_{\leq 1} [\mathbb{I}]$, for each $R \in \text{Rol}^{\text{cmp}}$,

$$(\text{Qu}_R^{\text{can}} [\mathbb{I}], \prec_R^{\text{can}} [\mathbb{I}])$$

is a strictly partially ordered set.

Proof. By Lemma 4.102, the canonical partial order is a partial order. Definition 4.110 is its strictification, so Lemma 4.105 applies. $\square$

Lemma 4.114 (Equivalence-relation property of the local incomparability relations). Under $\text{ReprCtx}_{\leq 2} [\mathbb{I}]$, for each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$,

$$(\mathcal{U}_R [\mathbb{I}], \succ_{R,u} [\mathbb{I}])$$

is a set equipped with an equivalence relation.

Proof. By Lemma 4.113, the restriction of the canonical strict order to the local canonical quotient region is a strict partial order. Definition 4.111 gives its incomparability relation, and Definition 4.112 gives transitivity. Apply Lemma 4.109. $\square$

Lemma 4.115 (Invariance of the canonical strict order along local incomparability classes).

Assume $\text{ReprCtx}_{\leq 2} [\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$, the representative invariance of Lemma 4.109 holds under

$$P = \text{OrdDat}_R^{\text{loc}} [\mathbb{I}]$$

Proof. Lemma 4.113, Definition 4.111, and Definition 4.112 permit application of the representative invariance in Lemma 4.109. $\square$

Definition 4.116 (Local order-coordinate quotients). Assume $\text{ReprCtx}_{\leq 2}[I]$. For each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$, from this section onward the superscript (ord, loc) on quotients, projections, orders, atlases, and chart changes belonging to the local order-coordinate family is abbreviated to (loc) . The full form (ord, can) is retained for the global order-coordinate family. Under this abbreviation, define the local order-coordinate quotient, the local order-coordinate projection, and the strict order on the quotient by

$$\begin{aligned}\text{LinQuot}_{R,u}^{\text{loc}}[I] &:= \text{LinQuot} [\text{OrdDat}_R^{\text{loc}}[I]] , \\ \text{LinQuot}_R^{\text{loc}}[I] &= (\text{Qu}_{R,u}^{\text{loc}}[I], \text{Prj}_{R,u}^{\text{loc}}[I], <_{R,u}^{\text{loc}}[I]) , \\ \text{OrdCls}_{R,i}^{\text{loc}} &:= \text{Prj}_R^{\text{loc}}[I](\text{CmpCls}_{R,i}), \quad i \in \{0, 1\}.\end{aligned}$$

Expanding Definition 4.106 gives the local incomparability classes, quotient set, quotient projection, and representative-based strict-order judgment.

Lemma 4.117 (Strict linearity of the local order-coordinate quotients). Under $\text{ReprCtx}_{\leq 2}[I]$, for each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$, $<_R^{\text{loc}}[I]$ is a strict linear order on $\text{Qu}_R^{\text{loc}}[I]$.

Proof. Lemma 4.113, Definition 4.111, and Definition 4.112 permit application of Lemma 4.109. The three components of Definition 4.116 coincide with LinQuot in that lemma. $\square$

Definition 4.118 (Surjective quotient-atlas datum). Let the datum whose components are a set S , a strict partial order $\prec$ on that set, an index set I , and, for each $i \in I$, sets A_i, S_i and a map ϑ_i be

$$\mathfrak{A} := (S, \prec, I, (A_i, S_i, \vartheta_i)_{i \in I})$$

and define each local strict-order datum by

$$\text{OrdDat}_i[\mathfrak{A}] := (S_i, \prec|_{S_i}), \quad i \in I$$

Define the surjective quotient-atlas condition by

$$\text{QuotAtlasDat}(\mathfrak{A}) \iff \left[\forall i \in I, \quad S_i \subseteq S, \quad \vartheta_i : A_i \twoheadrightarrow S_i, \quad \text{IncTrans}[\text{OrdDat}_i[\mathfrak{A}]] \right]$$

Definition 4.119 (Local order-coordinate projection system and local quotient-atlas datum). Assume $\text{ReprCtx}_{\leq 2}[I]$. For each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$, define the surjection from the arrival-value image to the local canonical quotient region by

$$\text{ArrCan}_{R,u}[I] : \text{Im}(\text{Arr}_R^{\text{cmp}}) \twoheadrightarrow \mathcal{U}_R[I], \quad \text{ArrVal}_R^{(0)} \longmapsto \text{Prj}_R^{\text{can}}[I](u, \text{ArrVal}_R^{(0)})$$

Using the local order-coordinate projection system of Definition 4.116 as its components, define the local quotient-atlas datum by

$$\mathfrak{A}_R^{\text{loc}}[I] := \left(\text{Qu}_R^{\text{can}}[I], <_R^{\text{can}}[I], U, \left(\text{Im}(\text{Arr}_{R,u}^{\text{cmp}}), \mathcal{U}_{R,u}[I], \text{ArrCan}_{R,u}[I] \right)_{u \in U} \right)$$

Lemma 4.120 (Local quotient-atlas datum and local order-coordinate projections). Under $\text{ReprCtx}_{\leq 2}[I]$, for each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$, the local order-coordinate projection $\text{Prj}_R^{\text{loc}}[I] : \mathcal{U}_R[I] \twoheadrightarrow \text{Qu}_R^{\text{loc}}[I]$ is surjective and satisfies, on the local canonical quotient region, the class judgment and order judgment of Lemma 4.109. Moreover,

$$\text{QuotAtlasDat}(\mathfrak{A}_R^{\text{loc}}[I])$$

holds, and its linearized-quotient component coincides with

$$\text{LinQuot}_u[\mathfrak{A}_R^{\text{loc}}[I]] = \text{LinQuot}_R^{\text{loc}}[I]$$

Proof. By Definition 4.116, the local order-coordinate quotient, the local order-coordinate projection, and the strict order on the quotient are the three components of LinQuot . Apply the surjectivity, class judgment, and order judgment of Lemma 4.109. By Definition 4.104, $\text{ArrCan}_R[\mathbb{I}]$ is surjective. By Definition 4.111,

$$\asymp_{R,u} [\mathbb{I}] = \text{Inc} [\text{OrdDat}_R^{\text{loc}}[\mathbb{I}]]$$

holds, and Definition 4.112 supplies IncTrans . Hence every condition in Definition 4.118 holds. $\square$

Definition 4.121 (Surjective quotient-atlas operations). Under $\text{QuotAtlasDat}(\mathfrak{A})$ of Definition 4.118, define the composite charts, linearized quotients, double and triple overlaps, and quotient-chart changes by

$$\begin{aligned} \text{LinQuot}_i[\mathfrak{A}] &:= \text{LinQuot}[\text{OrdDat}_i[\mathfrak{A}]] \\ &= (Q_i[\mathfrak{A}], \pi_i[\mathfrak{A}], <_i[\mathfrak{A}]), \\ \text{QuotChart}_i[\mathfrak{A}] &:= \pi_i[\mathfrak{A}] \circ \vartheta_i : A_i \twoheadrightarrow Q_i[\mathfrak{A}], \\ S_{ij}^\cap[\mathfrak{A}] &:= S_i \cap S_j, \\ S_{ijk}^\cap[\mathfrak{A}] &:= S_i \cap S_j \cap S_k, \\ \text{QuotChg}_{ij}[\mathfrak{A}] &: \pi_i[\mathfrak{A}] (S_{ij}^\cap[\mathfrak{A}]) \\ &\longrightarrow \pi_j[\mathfrak{A}] (S_{ij}^\cap[\mathfrak{A}]), \\ \text{QuotChg}_{ij}[\mathfrak{A}] (\pi_i[\mathfrak{A}](x)) &:= \pi_j[\mathfrak{A}](x) \\ &\quad (x \in S_{ij}^\cap[\mathfrak{A}]) \end{aligned}$$

If $S_{ij}^\cap[\mathfrak{A}] = \emptyset$, then $\text{QuotChg}_{ij}[\mathfrak{A}]$ is the empty map $\emptyset \rightarrow \emptyset$.

Lemma 4.122 (Composition of a surjection and a quotient projection). Under $\text{QuotAtlasDat}(\mathfrak{A})$, for every $i \in I$, $\text{QuotChart}_i[\mathfrak{A}]$ is surjective, and

$$\text{OrdRepCond} (\text{QuotChart}_i[\mathfrak{A}]; \vartheta_i^* (\text{Inc}[\text{OrdDat}_i[\mathfrak{A}]]), \vartheta_i^* (<_{|S_i}), <_i[\mathfrak{A}])$$

holds.

Proof. By Lemma 4.109, $\pi_i[\mathfrak{A}]$ is surjective and satisfies the class judgment and order judgment. Its composition with the surjection ϑ_i is surjective. By Definitions 4.107 and 4.93, precomposition with ϑ_i pulls the class-judgment relation and the order-judgment relation back to A_i , respectively. Apply Definition 4.108. $\square$

Lemma 4.123 (Overlap chart changes between subregion quotients). Under $\text{QuotAtlasDat}(\mathfrak{A})$, for every $i, j \in I$, $\text{QuotChg}_{ij}[\mathfrak{A}]$ is a representative-independent bijection and a strict linear-order isomorphism, with inverse $\text{QuotChg}_{ji}[\mathfrak{A}]$. Moreover,

$$\begin{aligned} \text{QuotChg}_{ii}[\mathfrak{A}] &= \text{id}_{Q_i[\mathfrak{A}]}, \\ \text{QuotChg}_{ji}[\mathfrak{A}] \circ \text{QuotChg}_{ij}[\mathfrak{A}] &= \text{id}_{\pi_i[\mathfrak{A}](S_{ij}^\cap[\mathfrak{A}])}, \\ \text{QuotChg}_{jk}[\mathfrak{A}] (\text{QuotChg}_{ij}[\mathfrak{A}] (\pi_i[\mathfrak{A}](x))) &= \text{QuotChg}_{ik}[\mathfrak{A}] (\pi_i[\mathfrak{A}](x)) \\ &\quad (x \in S_{ijk}^\cap[\mathfrak{A}]) \end{aligned}$$

hold.

Proof. Let $x, y \in S_{ij}^\cap[\mathfrak{A}]$. The class judgment for $\pi_i[\mathfrak{A}](x) = \pi_i[\mathfrak{A}](y)$, together with the two restricted orders that agree on the overlap, gives $\pi_j[\mathfrak{A}](x) = \pi_j[\mathfrak{A}](y)$. Using this implication after exchanging the indices gives injectivity, and the pointwise definition gives surjectivity. Since the two order judgments are both equivalent to $x \prec y$, the map preserves and reflects order. Composing the pointwise definition twice gives the inverse condition and the triple-overlap condition. The identities $S_{ii}^\cap[\mathfrak{A}] = S_i$ and $\pi_i[\mathfrak{A}](S_i) = Q_i[\mathfrak{A}]$ give the identity condition. $\square$

Definition 4.124 (Local comparison charts). Assume $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$ and $u \in U$, define the local comparison chart by

$$\begin{aligned} \text{Chart}_{R,u}^{\text{loc}}[\mathbb{I}] &:= \text{QuotChart}_u[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] \\ &= \text{Prj}_R^{\text{loc}}[\mathbb{I}] \circ \text{ArrCan}_R[\mathbb{I}] \end{aligned}$$

This family of surjections is the local comparison-chart system.

Proposition 4.125 (Canonical compositionality of local comparison charts). Under $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$, the local comparison chart is the surjection uniquely determined by $\text{ArrCan}_{R,u}[\mathbb{I}]$ and $\text{Prj}_{R,u}^{\text{loc}}[\mathbb{I}]$, and it satisfies the class judgment and order judgment of Lemma 4.122.

Proof. By Lemma 4.120, $\text{QuotAtlasDat}(\mathfrak{A}_R^{\text{loc}}[\mathbb{I}])$ holds. Apply Lemma 4.122 to the local datum of Definition 4.124. $\square$

Definition 4.126 (Local order-coordinate chart-change maps). Assume $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$ and $u, v, w \in U$, define the double and triple overlaps and the local order-coordinate chart-change maps by

$$\begin{aligned} \mathcal{U}_{R,uv}^\cap[\mathbb{I}] &:= S_{uv}^\cap[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] , \\ \mathcal{U}_{R,uvw}^\cap[\mathbb{I}] &:= S_{uvw}^\cap[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] , \\ \text{ChartChg}_{R,uv}^{\text{loc}}[\mathbb{I}] &:= \text{QuotChg}_{uv}[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] . \end{aligned}$$

On an empty overlap, $\text{ChartChg}_{R,uv}^{\text{loc}}[\mathbb{I}]$ is the empty map.

Lemma 4.127 (Well-definedness and order-isomorphism property of local order-coordinate chart-change maps). Under $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$, for each $R \in \text{Rol}^{\text{cmp}}$ and $u, v \in U$, the local order-coordinate chart-change map is a representative-independent bijection and a strict linear-order isomorphism. Its inverse is

$$\text{ChartChg}_{R,vu}^{\text{loc}}[\mathbb{I}]$$

and it satisfies the order-preservation and order-reflection formulas of Lemma 4.123.

Proof. By Lemma 4.120, $\text{QuotAtlasDat}(\mathfrak{A}_R^{\text{loc}}[\mathbb{I}])$ holds. Apply Lemma 4.123 to the local datum of Definition 4.126. $\square$

Definition 4.128 (Comparison-order atlas). Assume $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$. Define the comparison-order atlas by

$$\begin{aligned} \text{Atlas}^{\text{ord}}[\mathbb{I}] &:= \left(\left(Q_u[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] , <_u[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] , \right. \right. \\ &\quad \left. \left. \text{QuotChart}_u[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] \right)_{(R,u) \in \text{Id}_{\mathbf{x}_1}^{\text{cmp}}} , \right. \\ &\quad \left. \left(\text{QuotChg}_{uv}[\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]] \right)_{(R,u,v) \in \text{Id}_{\mathbf{x}_2}^{\text{cmp}}} \right) \end{aligned}$$

The first component is the family of quotients, orders, and composite charts on the respective $\mathfrak{C}/\mathfrak{D}$ branches; the second is the family of chart-change maps on the respective $\mathfrak{C}/\mathfrak{D}$ branches.

Proposition 4.129 (Cocycle condition for local order-coordinate chart-change maps). Under $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$, for each $R \in \text{Rol}^{\text{cmp}}$ and $u, v, w \in U$, the local order-coordinate chart-change maps satisfy the identity condition, the inverse condition, and the triple-overlap cocycle condition. The domains, codomains, and pointwise equalities of these three conditions are given by the formulas of Lemma 4.123 for $\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]$.

Proof. Apply Lemma 4.120 and Lemma 4.123 to $\mathfrak{A}_R^{\text{loc}}[\mathbb{I}]$. □

Definition 4.130 (Global incomparability relations). Assume $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$, define the global strict-order datum and the global incomparability relation by

$$\begin{aligned} \text{OrdDat}_R^{\text{can}}[\mathbb{I}] &:= (\text{Qu}_R^{\text{can}}[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]), \\ \succsim_R[\mathbb{I}] &:= \text{Inc}[\text{OrdDat}_R^{\text{can}}[\mathbb{I}]] \end{aligned}$$

Expanding Definition 4.106 gives the bidirectional incomparability condition for any two canonical comparison quotient classes.

Lemma 4.131 (Agreement under restriction between the local and global incomparability relations). Under $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$, for each $R \in \text{Rol}^{\text{cmp}}$, each $u \in U$, and any $(x_0, x_1) \in (\mathcal{U}_R[\mathbb{I}])^2$,

$$x_0 \succsim_{R,u}[\mathbb{I}] x_1 \iff x_0 \succsim_R[\mathbb{I}] x_1.$$

Proof. Definitions 4.111 and 4.130 are determined on the local canonical quotient region by the same bidirectional incomparability condition for the canonical strict order. □

Definition 4.132 (Representation-stage global incomparability-transitivity condition). Assume $\text{ReprCtx}_{\leq 2}[\mathbb{I}]$. Define the representation-stage global incomparability-transitivity condition by

$$K_{\text{GlobIncTrans}}^{\text{repr}}[\mathbb{I}] \iff [\forall R \in \text{Rol}^{\text{cmp}}, \quad \text{IncTrans}[\text{OrdDat}_R^{\text{can}}[\mathbb{I}]]].$$

Expanding Definition 4.106 gives the transitivity condition quantified over three points.

Lemma 4.133 (Equivalence-relation property of the global incomparability relations). Under $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$, for each $R \in \text{Rol}^{\text{cmp}}$, $\succsim_R[\mathbb{I}]$ is an equivalence relation on $\text{Qu}_R^{\text{can}}[\mathbb{I}]$.

Proof. Lemma 4.113, Definition 4.130, and Definition 4.132 permit application of Lemma 4.109 to the entire canonical comparison quotient. □

Lemma 4.134 (Invariance of the canonical strict order along global incomparability classes). Assume

$$\text{ReprCtx}_{\leq 3}[\mathbb{I}]$$

For each

$$R \in \text{Rol}^{\text{cmp}}$$

the representative invariance of Lemma 4.109 holds under

$$P = \text{OrdDat}_R^{\text{can}}[\mathbb{I}]$$

Proof. Lemma 4.113, Definition 4.130, and Definition 4.132 permit application of the representative invariance in Lemma 4.109 to the entire canonical comparison quotient. $\square$

Definition 4.135 (Global order-coordinate quotients on $\mathfrak{C}/\mathfrak{D}$ and canonical global order-coordinate projections).

Assume $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$, define the global order-coordinate quotient on $\mathfrak{C}/\mathfrak{D}$, the canonical global order-coordinate projection, and the strict order on the quotient by

$$\begin{aligned}\text{LinQuot}_R^{\text{can}}[\mathbb{I}] &:= \text{LinQuot}[\text{OrdDat}_R^{\text{can}}[\mathbb{I}]], \\ \text{LinQuot}_R^{\text{can}}[\mathbb{I}] &= \left(\text{Qu}_R^{\text{ord,can}}[\mathbb{I}], \text{Prj}_R^{\text{ord,can}}[\mathbb{I}], <_R^{\text{ord,can}}[\mathbb{I}] \right), \\ \text{OrdCls}_{R,i}^{\text{can}} &:= \text{Prj}_R^{\text{ord,can}}[\mathbb{I}](\text{CmpCls}_{R,i}), \quad i \in \{0, 1\}.\end{aligned}$$

Expanding Definition 4.106 gives the global incomparability classes, quotient set, quotient projection, and representative-based strict-order judgment.

Lemma 4.136 (Strict linearity of global order-coordinate quotients and canonical projections). Under

$$\text{ReprCtx}_{\leq 3}[\mathbb{I}]$$

for each

$$\begin{aligned}R &\in \text{Rol}^{\text{cmp}} \\ &<_R^{\text{ord,can}}[\mathbb{I}]\end{aligned}$$

is a strict linear order on

$$\text{Qu}_R^{\text{ord,can}}[\mathbb{I}]$$

The canonical projection

$$\text{Prj}_R^{\text{ord,can}}[\mathbb{I}] : \text{Qu}_R^{\text{can}}[\mathbb{I}] \twoheadrightarrow \text{Qu}_R^{\text{ord,can}}[\mathbb{I}]$$

is surjective and satisfies the class judgment and order judgment of Lemma 4.109 on the entire canonical comparison quotient.

Proof. Lemma 4.113, Definition 4.130, and Definition 4.132 permit application of Lemma 4.109 to the entire canonical comparison quotient. The three components of Definition 4.135 coincide with LinQuot in that lemma. $\square$

4.2.2 Representational coordinates obtained from global order-coordinate quotients

The global order-coordinate quotient sends each incomparability class in the entire canonical comparison quotient to a single order-coordinate class. The comparison representational coordinate derived from $\mathfrak{C}/\mathfrak{D}$ is determined by composing the canonical global order-coordinate projection with an order embedding into the selected linearly ordered codomain.

Definition 4.137 (Strict order embedding). For strictly ordered sets $(S, <_S)$ and $(T, <_T)$, and an injection $f : S \hookrightarrow T$, define the strict-order-embedding condition by

$$\text{OrdEmb}(f; S, <_S; T, <_T) \iff f^*(<_T) = <_S$$

Lemma 4.138 (Composition of a quotient projection and an order embedding). Take sets S, Q, V ; on S , take relations $\succsim, \prec$; on Q, V , take strict orders $<_Q, <_V$, a map π , and an injection $\iota : Q \hookrightarrow V$. Then the following implication holds:

$$\begin{aligned} \pi : S \twoheadrightarrow Q, \quad & \text{OrdRepCond}(\pi; \succsim, \prec, <_Q), \\ & \text{OrdEmb}(\iota; Q, <_Q; V, <_V), \\ & \implies \\ f := \iota \circ \pi : S \longrightarrow V, \\ & \text{OrdRepCond}(f; \succsim, \prec, <_V) \end{aligned}$$

Proof. Injectivity of ι gives $\text{EqKer}(f) = \text{EqKer}(\pi)$. Preservation and reflection of strict order give $f^*(<_V) = \pi^*(<_Q)$. Apply the assumed $\text{OrdRepCond}(\pi; \succsim, \prec, <_Q)$ and Definition 4.108. $\square$

Lemma 4.139 (Quotient-order embedding for nested domains). Let $A \subseteq S$, let Q_A, Q_S carry strict orders $<_A, <_S$, and let $\pi_A : A \twoheadrightarrow Q_A$ and $\pi_S : S \twoheadrightarrow Q_S$ be surjections. Assume that both of the following equivalences hold for every $x, y \in A$:

$$\begin{aligned} \pi_A(x) = \pi_A(y) &\iff \pi_S(x) = \pi_S(y), \\ \pi_A(x) <_A \pi_A(y) &\iff \pi_S(x) <_S \pi_S(y). \end{aligned}$$

Then the rule

$$\iota_{A,S} : Q_A \hookrightarrow Q_S, \quad \iota_{A,S}(\pi_A(x)) := \pi_S(x) \quad (x \in A)$$

defines an injection satisfying

$$\text{OrdEmb}(\iota_{A,S}; Q_A, <_A; Q_S, <_S), \quad \iota_{A,S} \circ \pi_A = \pi_S|_A.$$

Proof. Since π_A is surjective, every element of Q_A has the form $\pi_A(x)$ for some $x \in A$. By the class-judgment equivalence, $\pi_A(x) = \pi_A(y)$ implies $\pi_S(x) = \pi_S(y)$, so the value of the map is independent of the representative. By the converse class judgment, two elements with the same image satisfy $\pi_A(x) = \pi_A(y)$; hence $\iota_{A,S}$ is injective. The order-judgment equivalence and the definition give

$$\begin{aligned} \pi_A(x) <_A \pi_A(y) &\iff \iota_{A,S}(\pi_A(x)) <_S \iota_{A,S}(\pi_A(y)), \\ \iota_{A,S}(\pi_A(x)) &= \pi_S(x) \quad (x \in A) \end{aligned}$$

and thus give preservation and reflection of strict order and the commutative identity. $\square$

Definition 4.140 (Selected ordered codomains and representational coordinates). Assume $\text{ReprCtx}_{\leq 3}[I]$. For each $R \in \text{Rol}^{\text{cmp}}$, specify a selected representational codomain derived from $\mathfrak{C}/\mathfrak{D}$, $(\text{RepVal}_R, \leq_R^{\text{repr}})$, and its strict-order part $<_R^{\text{repr}}$. Define the order embedding into the representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ and the comparison representational coordinate derived from $\mathfrak{C}/\mathfrak{D}$ by

$$\begin{aligned} \text{RepEmb}_R^{\text{ord}}[I] : \text{Qu}_R^{\text{ord}, \text{can}}[I] &\hookrightarrow \text{RepVal}_R, \\ \text{OrdEmb}(\text{RepEmb}_R^{\text{ord}}[I]; \text{Qu}_R^{\text{ord}, \text{can}}[I], <_R^{\text{ord}, \text{can}}[I]; \text{RepVal}_R, <_R^{\text{repr}}), \\ \text{Rep}_R^{\text{cmp}}[I] : \text{Qu}_R^{\text{can}}[I] &\longrightarrow \text{RepVal}_R, \\ \text{Rep}_R^{\text{cmp}}[I] &:= \text{RepEmb}_R^{\text{ord}}[I] \circ \text{Prj}_R^{\text{ord}, \text{can}}[I] \end{aligned}$$

Proposition 4.141 (Canonical factorization and order judgment of representational coordinates). Under the assumptions of Definition 4.140, for each $R \in \text{Rol}^{\text{cmp}}$,

$$\text{OrdRepCond}(\text{Rep}_R^{\text{cmp}}[\mathbb{I}]; \asymp_R[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}], \prec_R^{\text{repr}})$$

holds.

Proof. In Lemma 4.138, substitute

$$\begin{aligned} \pi &= \text{Prj}_R^{\text{ord,can}}[\mathbb{I}], & \iota &= \text{RepEmb}_R^{\text{ord}}[\mathbb{I}], \\ f &= \text{Rep}_R^{\text{cmp}}[\mathbb{I}], & \asymp &= \asymp_R[\mathbb{I}], \\ \prec &= \prec_R^{\text{can}}[\mathbb{I}] \end{aligned}$$

The class judgment and order judgment of the quotient projection follow from Lemma 4.136; the order-embedding assumption and the composition identity follow from Definition 4.140. $\square$

Definition 4.142 (Local-to-global order-coordinate embeddings). Assume $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$, each $u \in U$, and any $x \in \mathcal{U}_R[\mathbb{I}]$, define the local-to-global order-coordinate embedding by

$$\begin{aligned} \text{LocEmb}_{R,u}^{\text{ord}}[\mathbb{I}] : \text{Qu}_R^{\text{loc}}[\mathbb{I}] &\hookrightarrow \text{Qu}_R^{\text{ord,can}}[\mathbb{I}], \\ \text{LocEmb}_R^{\text{ord}}[\mathbb{I}] (\text{Prj}_R^{\text{loc}}[\mathbb{I}](x)) &:= \text{Prj}_R^{\text{ord,can}}[\mathbb{I}](x) \end{aligned}$$

Proposition 4.143 (Local-to-global order-coordinate compatibility). Under the assumptions of Definition 4.140, for each

$$R \in \text{Rol}^{\text{cmp}}, \quad u \in U$$

$$\text{LocEmb}_R^{\text{ord}}[\mathbb{I}]$$

is a well-defined injection and strict order embedding. For any

$$(x_0, x_1) \in (\text{Qu}_R^{\text{loc}}[\mathbb{I}])^2$$

and

$$\begin{aligned} a &\in \text{Im}(\text{Arr}_R^{\text{cmp}}) \\ x_0 &\prec_R^{\text{loc}}[\mathbb{I}] x_1 \iff \text{LocEmb}_R^{\text{ord}}[\mathbb{I}](x_0) <_R^{\text{ord,can}}[\mathbb{I}] \text{LocEmb}_R^{\text{ord}}[\mathbb{I}](x_1), \\ y &:= \text{Prj}_R^{\text{can}}[\mathbb{I}](u, a), \\ \text{Rep}_R^{\text{cmp}}[\mathbb{I}](y) &= \text{RepEmb}_R^{\text{ord}}[\mathbb{I}] \left(\text{LocEmb}_R^{\text{ord}}[\mathbb{I}] (\text{Chart}_R^{\text{loc}}[\mathbb{I}](a)) \right) \end{aligned}$$

hold.

Proof. In Lemma 4.139, substitute

$$\begin{aligned} A &= \mathcal{U}_R[\mathbb{I}], & S &= \text{Qu}_R^{\text{can}}[\mathbb{I}], \\ \pi_A &= \text{Prj}_R^{\text{loc}}[\mathbb{I}], & \pi_S &= \text{Prj}_R^{\text{ord,can}}[\mathbb{I}]. \end{aligned}$$

The lemma gives the commutative identity

$$\begin{aligned} \text{LocEmb}_R^{\text{ord}}[\mathbb{I}] \circ \text{Prj}_R^{\text{loc}}[\mathbb{I}] \\ = \text{Prj}_R^{\text{ord,can}}[\mathbb{I}] \Big|_{\mathcal{U}_R[\mathbb{I}]} \end{aligned}$$

On the local comparison chart,

$$\begin{aligned} \text{LocEmb}_R^{\text{ord}}[\mathbb{I}] (\text{Chart}_R^{\text{loc}}[\mathbb{I}](a)) \\ = \text{Prj}_R^{\text{ord,can}}[\mathbb{I}](y) \end{aligned}$$

holds. Lemma 4.131, together with the class judgments and order judgments of Lemma 4.120 and Lemma 4.136, establishes the two equivalences required by the lemma. The commutative identity follows from its conclusion. Definition 4.124 and the commutative identity give the identity concerning the local comparison chart. Compose both sides of that identity with $\text{RepEmb}_R^{\text{ord}}[\mathbb{I}]$ and apply Definition 4.140. $\square$

4.2.3 Partial-establishment bundles and structural completion at the representation stage

Definition 4.144 (Generated-structure bundles at the representation stage). For any object a and finite tuple $(b_1, \dots, b_n)$, define the front-preserving extension by

$$\text{TupExt}(a; (b_1, \dots, b_n)) := (a, b_1, \dots, b_n)$$

Evaluate each component at $\mathbb{I}$, and define the generated-structure bundle at each stage by

$$\begin{aligned} \text{ReprStr}_{=1}[\mathbb{I}] &:= \left((\preceq_R^{\text{can}})_{R \in \text{Rol}^{\text{cmp}}}, \mathcal{E}^{\text{op}}, (\mathcal{U}_{R,u})_{(R,u) \in \text{IdX}_1^{\text{cmp}}}, \right. \\ &\quad \left. (\prec_R^{\text{can}})_{R \in \text{Rol}^{\text{cmp}}}, (\succ_{R,u})_{(R,u) \in \text{IdX}_1^{\text{cmp}}}, (\asymp_R)_{R \in \text{Rol}^{\text{cmp}}} \right), \\ \text{ReprStr}_{=2}[\mathbb{I}] &:= \left((\text{LinQuot}_{R,u}^{\text{loc}}, \text{Chart}_{R,u}^{\text{loc}})_{(R,u) \in \text{IdX}_1^{\text{cmp}}}, \right. \\ &\quad \left. (\text{ChartChg}_{R,uv}^{\text{loc}})_{(R,u,v) \in \text{IdX}_2^{\text{cmp}}}, \text{Atlas}^{\text{ord}} \right), \\ \text{ReprStr}_{=3}[\mathbb{I}] &:= \left((\text{LinQuot}_R^{\text{can}})_{R \in \text{Rol}^{\text{cmp}}}, \right. \\ &\quad \left. (\text{LocEmb}_{R,u}^{\text{ord}})_{(R,u) \in \text{IdX}_1^{\text{cmp}}} \right), \\ \text{ReprStr}_{\leq 1}[\mathbb{I}] &:= \text{ReprStr}_{=1}[\mathbb{I}], \\ \text{ReprStr}_{\leq k}[\mathbb{I}] &:= \text{TupExt}(\text{ReprStr}_{\leq k-1}[\mathbb{I}], ; \text{ReprStr}_{=k}[\mathbb{I}]), \quad k \in \{2, 3\}. \end{aligned}$$

Proposition 4.145 (Implication chain for the partial-establishment bundles at the representation stage).

$$\text{Repr}_{\leq 3}[\mathbb{I}] \implies \text{Repr}_{\leq 2}[\mathbb{I}] \implies \text{Repr}_{\leq 1}[\mathbb{I}].$$

Proof. In the recursive formula of Definition 4.92, $\text{Repr}_{\leq k}$ contains $\text{Repr}_{\leq k-1}$ as a conjunctive component. $\square$

Proposition 4.146 (Structural distinction among exact operational time structures, comparison-order atlases, global order-coordinate quotients, and representational coordinates). Assume $\text{Operative}(\mathcal{X})$. For each $k \in \{1, 2, 3\}$, under $\text{Repr}_{\leq k}[\mathbb{I}]$, $\text{ReprStr}_{\leq k}[\mathbb{I}]$ is generated in dependency order.

The exact operational time structure is a component of $\text{ReprStr}_{=1}[\mathbb{I}]$; the comparison-order atlas is a component of $\text{ReprStr}_{=2}[\mathbb{I}]$; and the global order-coordinate quotient on $\mathfrak{C}/\mathfrak{D}$ and the canonical global order-coordinate projection are components of $\text{ReprStr}_{=3}[\mathbb{I}]$.

For each $R \in \text{Rol}^{\text{cmp}}$, once the selected representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ and the order embedding into the representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ in Definition 4.140 are given, the composition formula of Definition 4.140 determines the comparison representational coordinate derived from $\mathfrak{C}/\mathfrak{D}$. Thus the exact operational time structure, comparison-order atlas, global order-coordinate quotient, and representational coordinate have distinct types and generation formulas.

Proof. For $k = 1$, Definition 4.101, Lemma 4.102, Definition 4.103, Definition 4.104, Definition 4.110, Definition 4.111, and Definition 4.130 generate $\text{ReprStr}_{=1}[\mathbb{I}]$ in dependency order.

For $k = 2$, Proposition 4.145, Lemma 4.114, Definition 4.116, Lemma 4.117, Lemma 4.120, Definition 4.124, Definition 4.126, Lemma 4.127, and Definition 4.128 generate $\text{ReprStr}_{=2}[\mathbb{I}]$ in dependency order. Proposition 4.129 gives the cocycle condition for the local order-coordinate chart-change maps.

For $k = 3$, the implication chain of Proposition 4.145, Lemma 4.133, Definition 4.135, Lemma 4.136, Definition 4.142, and Lemma 4.139 generate $\text{ReprStr}_{=3}[\mathbb{I}]$ in dependency order.

The recursive formula of Definition 4.144 gives each cumulative bundle. Definition 4.140 and Proposition 4.141 give the composition formula, equivalence-class judgment, and strict-order judgment for the representational coordinate. Proposition 4.143 gives the local-to-global compatibility identity with the local comparison chart. $\square$

Theorem 4.147 (Representation-stage generation theorem). Assume

$$\text{Operative}(\mathcal{X})$$

For each

$$k \in \{1, 2, 3\}$$

whenever

$$\text{Repr}_{\leq k}[\mathbb{I}]$$

holds,

$$\text{ReprStr}_{\leq k}[\mathbb{I}]$$

is generated in dependency order.

The local order-coordinate chart-change maps in

$$\text{ReprStr}_{=2}[\mathbb{I}]$$

satisfy the cocycle condition. Under

$$\text{Repr}_{\leq 3}[\mathbb{I}]$$

once the selected representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ and the order embedding into the representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ in Proposition 4.146 are given, the representational-coordinate factorization and the local-to-global compatibility identity with the local comparison chart hold.

Proof. Proposition 4.146 generates the three generated-structure bundles in dependency order, and Proposition 4.129 gives the cocycle condition. Definition 4.140, Proposition 4.141, and Proposition 4.143 give the factorization, equivalence-class judgment, and strict-order judgment of the representational coordinate and the local-to-global compatibility identity with the local comparison chart. $\square$

4.2.4 Conditional quantitative affine realization and frequency representation

Definition 4.148 (Conditional quantitative affine realization derived from $\mathfrak{C}/\mathfrak{D}$). Assume $\text{ReprCtx}_{\leq 3}[I]$ and the selected representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ in Definition 4.140. For each $R \in \text{Rol}^{\text{cmp}}$, let $\text{AffAct}_R^{\text{qty}}$ be, on RepVal_R , a free and transitive action of QtyVal_R . For arbitrary

$$(a_0, a_1) \in (\text{RepVal}_R)^2, \quad \delta \in \text{QtyVal}_R$$

and

$$(x_0, x_1) \in (\text{Qu}_R^{\text{can}}[I])^2$$

define the quantitatively linearly ordered field derived from $\mathfrak{C}/\mathfrak{D}$, the quantitative affine action, the quantitative affine difference, the conditional quantitative affine realization, and the quantitative difference of representational coordinates by

$$\begin{aligned} \text{QtyFld}_R &:= (\text{QtyVal}_R, +_R^{\text{qty}}, \cdot_R^{\text{qty}}, 0_R^{\text{qty}}, 1_R^{\text{qty}}, \leq_R^{\text{qty}}), \\ \text{AffAct}_R^{\text{qty}} &: \text{RepVal}_R \times \text{QtyVal}_R \longrightarrow \text{RepVal}_R, \\ \text{AffAct}_R^{\text{qty}}(a_0, \delta) = a_1 &\iff \delta = \text{AffDiff}_R^{\text{qty}}(a_1, a_0), \\ a_0 <_R^{\text{repr}} a_1 &\iff 0_R^{\text{qty}} <_R^{\text{qty}} \text{AffDiff}_R^{\text{qty}}(a_1, a_0), \\ \text{QtyAffReal}_R[I] &:= (\text{QtyFld}_R, \text{RepVal}_R, \text{AffAct}_R^{\text{qty}}, \text{AffDiff}_R^{\text{qty}}, \text{RepEmb}_R^{\text{ord}}[I]). \end{aligned}$$

Define the quantitative difference of representational coordinates derived from $\mathfrak{C}/\mathfrak{D}$ by

$$\begin{aligned} &\text{CoordDiff}_R^{\text{qty}}[I](x_1, x_0) \\ &:= \text{AffDiff}_R^{\text{qty}} \left(\text{Rep}_R^{\text{cmp}}[I](x_1), \text{Rep}_R^{\text{cmp}}[I](x_0) \right) \in \text{QtyVal}_R \end{aligned}$$

Lemma 4.149 (Nonnegativity judgment for ordered affine differences). Under Definition 4.148, for each $R \in \text{Rol}^{\text{cmp}}$ and arbitrary $a_0, a_1 \in \text{RepVal}_R$,

$$a_0 \leq_R^{\text{repr}} a_1 \iff 0_R^{\text{qty}} \leq_R^{\text{qty}} \text{AffDiff}_R^{\text{qty}}(a_1, a_0).$$

Proof. If $a_0 = a_1$, the zero-element action and uniqueness of the quantitative affine difference give $\text{AffDiff}_R^{\text{qty}}(a_1, a_0) = 0_R^{\text{qty}}$. By linearity of the order on the quantitative codomain, a nonnegative quantitative affine difference is either zero or positive. In the zero case, $a_1 = \text{AffAct}_R^{\text{qty}}(a_0, 0_R^{\text{qty}}) = a_0$; in the positive case, order compatibility gives $a_0 <_R^{\text{repr}} a_1$. $\square$

Definition 4.150 (Frequency representation). Assume the conditional quantitative affine realization derived from $\mathfrak{C}$ in Definition 4.148,

$$\text{QtyAffReal}_{\mathfrak{C}}[I]$$

Define the counting-window space, the comparison quotient classes at the counting-window endpoints, the set of admissible counting windows, the quantitative counting difference, the quantitative difference of representational coordinates, the counting-window frequency

representation derived from $\mathfrak{C}$, and the period representation by

$$\begin{aligned}
\text{CntWin}_{\mathfrak{C}} &:= U \times \text{Idx}_{\mathfrak{C}}^{\text{src}} \times \mathbb{N}^2, \\
\text{CntCls}_{\mathfrak{C},i}[\mathbf{l}; \text{win}] &:= \text{Prj}_{\mathfrak{C}}^{\text{can}}[\mathbf{l}] \left(u, \text{Arr}_{\mathfrak{C},u}^{\text{cmp}}(\text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k_i)) \right), \quad i \in \{0, 1\}, \\
\text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\mathbf{l}] &:= \left\{ \text{win} = (u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1) \in \text{CntWin}_{\mathfrak{C}} \mid \begin{array}{l} k_0 < k_1, \\ \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}}, k_i) \in \text{Dom}(\text{Arr}_{\mathfrak{C},u}^{\text{cmp}}) \ (i = 0, 1), \\ \text{CntCls}_{\mathfrak{C},0}[\mathbf{l}; \text{win}] \prec_{\mathfrak{C}}^{\text{can}}[\mathbf{l}] \text{CntCls}_{\mathfrak{C},1}[\mathbf{l}; \text{win}] \end{array} \right\}, \\
\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}] &:= (k_1 - k_0) \cdot_{\mathfrak{C}}^{\text{qty}} 1_{\mathfrak{C}}^{\text{qty}}, \\
\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[\mathbf{l}; \text{win}] &:= \text{CoordDiff}_{\mathfrak{C}}^{\text{qty}}[\mathbf{l}] (\text{CntCls}_{\mathfrak{C},1}[\mathbf{l}; \text{win}], \text{CntCls}_{\mathfrak{C},0}[\mathbf{l}; \text{win}]), \\
\nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] &:= \frac{\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}]}{\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[\mathbf{l}; \text{win}]}, \\
\text{Per}_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] &:= (\nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}])^{-1}.
\end{aligned}$$

$\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}$, $\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}$, $\nu_{\mathfrak{C}}^{\text{win}}$, and $\text{Per}_{\mathfrak{C}}^{\text{win}}$ are defined for every $\text{win} \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\mathbf{l}]$ and belong to $\text{QtyVal}_{\mathfrak{C}}$.

For each $j_{\mathfrak{C}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$, define a nonempty family of admissible counting windows belonging to the same source by

$$\begin{aligned}
\emptyset \neq \mathcal{W}_{j_{\mathfrak{C}}^{\text{src}}} &\subseteq \{(u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1) \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\mathbf{l}]\}, \\
\nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] &= \nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}'], \\
\text{Per}_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] &= \text{Per}_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}'] \quad (\text{win}, \text{win}' \in \mathcal{W}_{j_{\mathfrak{C}}^{\text{src}}}), \\
\nu_{\mathfrak{C}}^{\text{src}}[\mathbf{l}; j_{\mathfrak{C}}^{\text{src}}] &:= \nu_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}], \\
\text{Per}_{\mathfrak{C}}^{\text{src}}[\mathbf{l}; j_{\mathfrak{C}}^{\text{src}}] &:= \text{Per}_{\mathfrak{C}}^{\text{win}}[\mathbf{l}; \text{win}] \quad (\text{win} \in \mathcal{W}_{j_{\mathfrak{C}}^{\text{src}}})
\end{aligned}$$

For $\text{Src}_{\mathfrak{C}}(j_{\mathfrak{C}}^{\text{src}})$, these are the source-relative frequency representation and period representation derived from $\mathfrak{C}$.

Definition 4.151 (Frequency, period, and Hz representations). Specify a frequency-unit scalar derived from $\mathfrak{C}$, $\nu_{\mathfrak{C}}^{\text{unit}} \in \text{QtyVal}_{\mathfrak{C}}$, such that $0_{\mathfrak{C}}^{\text{qty}} <_{\mathfrak{C}}^{\text{qty}} \nu_{\mathfrak{C}}^{\text{unit}}$. For a positive frequency representation $\nu \in \text{QtyVal}_{\mathfrak{C}}$, define its Hz numerical value relative to the frequency unit by

$$\text{HzNum}_{\nu_{\mathfrak{C}}^{\text{unit}}}(\nu) := \frac{\nu}{\nu_{\mathfrak{C}}^{\text{unit}}}$$

If the unit name Hz is assigned to $\nu_{\mathfrak{C}}^{\text{unit}}$, this value is the Hz numerical value of the frequency representation.

Definition 4.152 (Standard frequency value as a representation). For the reference source derived from $\mathfrak{C}$, $\text{Src}_{\mathfrak{C}}(j_{\text{std}}^{\text{src}})$, assume its source-relative frequency representation derived from $\mathfrak{C}$. For a normalization convention $N_{\text{std}} \in \mathbb{N}_{>0}$, define the standard frequency representation derived from $\mathfrak{C}$ and the Hz unit scalar derived from $\mathfrak{C}$ by

$$\begin{aligned}
\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{l}] &:= \nu_{\mathfrak{C}}^{\text{src}}[\mathbf{l}; j_{\text{std}}^{\text{src}}], \\
\nu_{\mathfrak{C}}^{\text{Hz}}[\mathbf{l}] &:= \frac{\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{l}]}{N_{\text{std}} \cdot_{\mathfrak{C}}^{\text{qty}} 1_{\mathfrak{C}}^{\text{qty}}}, \\
\text{HzNum}_{\nu_{\mathfrak{C}}^{\text{Hz}}[\mathbf{l}]}(\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{l}]) &= N_{\text{std}} \cdot_{\mathfrak{C}}^{\text{qty}} 1_{\mathfrak{C}}^{\text{qty}}
\end{aligned}$$

Proposition 4.153 (Well-definedness and generative position of frequency and period representations).

Under $\text{ReprCtx}_{\leq 3}[I]$ and $\text{QtyAffReal}_{\mathfrak{C}}[I]$, for any $\text{win} \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[I]$,

$$\begin{aligned} 0_{\mathfrak{C}}^{\text{qty}} &<_{\mathfrak{C}}^{\text{qty}} \text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[I; \text{win}], \\ \nu_{\mathfrak{C}}^{\text{win}}[I; \text{win}] \cdot_{\mathfrak{C}}^{\text{qty}} \text{Per}_{\mathfrak{C}}^{\text{win}}[I; \text{win}] &= 1_{\mathfrak{C}}^{\text{qty}}. \end{aligned}$$

Frequency, period, Hz numerical values, and the standard frequency representation are determined from the structural representation stage, the quantitative affine realization, admissible counting windows, window invariance, and a unit-normalization convention as inputs.

Proof. Admissibility of

$$\text{win} = (u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1)$$

and

$$\text{CntCls}_{\mathfrak{C},0}[I; \text{win}] \prec_{\mathfrak{C}}^{\text{can}} [I] \text{CntCls}_{\mathfrak{C},1}[I; \text{win}]$$

hold. By the strict-order judgment of Proposition 4.141 and the order-compatibility condition of Definition 4.148, the quantitative difference of representational coordinates is positive; and $k_0 < k_1$ gives $\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}] \neq 0_{\mathfrak{C}}^{\text{qty}}$. Thus the frequency quotient and its multiplicative inverse in Definition 4.150 are defined, and the multiplication and inverse axioms of the quantitatively linearly ordered field make their product $1_{\mathfrak{C}}^{\text{qty}}$.

The domains and dependency order of the source-relative frequency representation derived from $\mathfrak{C}$, the Hz numerical value, and the standard frequency representation derived from $\mathfrak{C}$ follow from Definition 4.150, Definition 4.151, and Definition 4.152. $\square$

Remark 4.154 (Generation order of the frequency standard). For a source derived from $\mathfrak{C}$, the internal counter position $k \in \mathbb{N}$ and update $k \mapsto k + 1$ belong to the primitive input layer. Once the image of a token derived from $\mathfrak{C}$ on the canonical comparison quotient, the representational coordinate, the quantitative affine difference, and an admissible counting window have been determined in dependency order, the ratio of the cycle-count difference to the representational-coordinate difference gives the frequency representation. The Hz unit is determined from the frequency representation of the reference source derived from $\mathfrak{C}$ and the normalization convention. The cycle-count input, representational coordinate, frequency representation, and unit normalization belong to distinct generation stages.

4.2.5 Finite operational granularity input

Definition 4.155 (Quantitative coordinate width of a counting step derived from $\mathfrak{C}$). Assume

$$\begin{aligned} &\text{Operative}(\mathcal{X}), \\ &\text{Repr}_{\leq 3}[I], \\ &\text{QtyAffReal}_{\mathfrak{C}}[I] \end{aligned}$$

Define the set of admissible single-step counting windows derived from $\mathfrak{C}$ and its quantitative coordinate width by

$$\begin{aligned} \text{CntWin}_{\mathfrak{C}}^{\text{step}}[I] &:= \{ \text{win} = (u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1) \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[I] \mid k_1 = k_0 + 1 \}, \\ \text{CntStepWid}_{\mathfrak{C}}^{\text{qty}}[I; \text{win}] &:= \text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[I; \text{win}] \in \text{QtyVal}_{\mathfrak{C}}, \\ 0_{\mathfrak{C}}^{\text{qty}} &<_{\mathfrak{C}}^{\text{qty}} \text{CntStepWid}_{\mathfrak{C}}^{\text{qty}}[I; \text{win}], \quad \text{win} \in \text{CntWin}_{\mathfrak{C}}^{\text{step}}[I] \end{aligned}$$

The final inequality follows from the strict-order condition for admissible counting windows and the order-compatibility condition of Definition 4.148.

Definition 4.156 (Finite operational granularity input). Assume $\text{ReprCtx}_{\leq 3}[I]$ and $\text{QtyAffReal}_R[I]$ ($R \in \text{Rol}^{\text{cmp}}$). Specify the three nonempty auxiliary component codomains and the five component values of finite operational granularity. Define the component codomains and the product-type datum consisting of the specified values by

$$\begin{aligned} (\text{QtyVal}_R)_{\geq 0} &:= \{\delta \in \text{QtyVal}_R \mid 0_R^{\text{qty}} \leq_R^{\text{qty}} \delta\}, & R \in \text{Rol}^{\text{cmp}}, \\ \text{StabVal}_{\mathfrak{D}} &\neq \emptyset, & \text{DiscVal}_{\mathfrak{B}} \neq \emptyset, & \text{CouplVal}_{\mathfrak{D}} \neq \emptyset, \\ \text{GrnVal}^{\text{op}} &:= (\text{QtyVal}_{\mathfrak{C}})_{\geq 0} \times \text{StabVal}_{\mathfrak{D}} \times (\text{QtyVal}_{\mathfrak{D}})_{\geq 0} \times \text{DiscVal}_{\mathfrak{B}} \times \text{CouplVal}_{\mathfrak{D}}, \\ \text{Grn}^{\text{op}}(\mathcal{X}) &:= \left(\text{Grn}_{\mathfrak{C}, \text{cntstep}}, \text{Grn}_{\mathfrak{D}, \text{stab}}, \text{Grn}_{\mathfrak{D}, \text{cellwid}}, \text{Grn}_{\mathfrak{B}, \text{disc}}, \text{Grn}_{\mathfrak{D}, \text{coupl}} \right) \in \text{GrnVal}^{\text{op}} \end{aligned}$$

In order, the five components are a finite admissible scale for the quantitative coordinate width of an admissible single-step counting window derived from $\mathfrak{C}$; a finite stability value of a record value derived from $\mathfrak{D}$; a record-cell-width scale on quantitative coordinates derived from $\mathfrak{D}$; a finite discrimination scale for an object-state difference derived from $\mathfrak{B}$; and a finite limit on receptive coupling derived from $\mathfrak{D}$.

Proposition 4.157 (Structural separation of finite operational granularity input). Under $\text{ReprCtx}_{\leq 3}[I]$, Theorem 4.147 generates $\text{ReprStr}_{\leq 3}[I]$ in dependency order. Once the selected representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ and the order embedding into the representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ are given, the comparison representational coordinate derived from $\mathfrak{C}/\mathfrak{D}$ is also determined.

The finite operational granularity datum has the five-component product type

$$\text{Grn}^{\text{op}}(\mathcal{X}) \in \text{GrnVal}^{\text{op}}$$

and supplies inputs for evaluating the counting-step width derived from $\mathfrak{C}$, judging the admissibility of the record-cell width derived from $\mathfrak{D}$, and evaluating the continuum approximation.

Proof. Theorem 4.147 generates the components of $\text{ReprStr}_{\leq 3}[I]$, and Definition 4.140 generates the comparison representational coordinate derived from $\mathfrak{C}/\mathfrak{D}$. Definition 4.148 equips the selected representational codomain derived from $\mathfrak{C}/\mathfrak{D}$ with a quantitative affine structure, and Definition 4.156 makes the five components a product-type datum on $\text{GrnVal}^{\text{op}}$. These results establish the generation dependencies of the representation-stage structure, the type of the finite operational granularity datum, and the three constructions that use that datum. $\square$

4.2.6 Record-cell representational images derived from $\mathfrak{D}$ and $\mathfrak{C}$ -token record separation

Definition 4.158 (Record-cell representational image derived from $\mathfrak{D}$). Assume

$$\text{ReprCtx}_{\leq 3}[I]$$

and

$$\text{QtyAffReal}_{\mathfrak{D}}[I]$$

For each $u \in U$, each $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$, and each $y_{\mathfrak{D}} \in \text{Dom}(\text{Arr}_{\mathfrak{D},u}^{\text{cmp}})$, define the set of arrival-capable $\mathfrak{D}$ -record-cell tokens, the map from $\mathfrak{D}$ -record tokens to representational values, and the record-cell representational image derived from $\mathfrak{D}$ by

$$\begin{aligned}\text{CellTok}_{\mathfrak{D},u}^{\text{arr}}(x_{\mathfrak{D}}) &:= W_{\mathfrak{D}}(x_{\mathfrak{D}}) \cap \text{Dom}(\text{Arr}_{\mathfrak{D},u}^{\text{cmp}}), \\ \text{RecordVal}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](y_{\mathfrak{D}}) &:= \text{Rep}_{\mathfrak{D}}^{\text{cmp}}[\mathbb{I}]\left(\text{Prj}_{\mathfrak{D}}^{\text{can}}[\mathbb{I}](u, \text{Arr}_{\mathfrak{D},u}^{\text{cmp}}(y_{\mathfrak{D}}))\right) \in \text{RepVal}_{\mathfrak{D}}, \\ \text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) &:= \text{RecordVal}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}][\text{CellTok}_{\mathfrak{D},u}^{\text{arr}}(x_{\mathfrak{D}})] \subseteq \text{RepVal}_{\mathfrak{D}}\end{aligned}$$

Definition 4.159 (Extrema and quantitative coordinate width of a record-cell representational image derived from $\mathfrak{D}$). Let $\text{Bool} := \{\perp, \top\}$. For each $S \subseteq \text{RepVal}_{\mathfrak{D}}$, $u \in U$, and $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$, define the extremum-existence predicate and ordered affine width on a set of $\mathfrak{D}$ -representational values, together with the extrema and quantitative coordinate width of the record-cell representational image derived from $\mathfrak{D}$, as follows:

$$\begin{aligned}\text{HasExt}_{\mathfrak{D}}^{\text{repr}}(S) &\iff S \neq \emptyset \\ &\quad \wedge [\exists a_-, a_+ \in S \forall a \in S, \quad a_- \leq_{\mathfrak{D}}^{\text{repr}} a \leq_{\mathfrak{D}}^{\text{repr}} a_+].\end{aligned}$$

The value of this predicate satisfies

$$\text{HasExt}_{\mathfrak{D}}^{\text{repr}}(S) \in \text{Bool}$$

Further, set

$$\begin{aligned}\text{AffSpan}_{\mathfrak{D}}^{\text{qty}}(S) &:= \text{AffDiff}_{\mathfrak{D}}^{\text{qty}}(\max S, \min S), \\ S_{u,x_{\mathfrak{D}}} &:= \text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}).\end{aligned}$$

Define the existence of extrema for the record-cell representational image derived from $\mathfrak{D}$ by

$$\text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \iff \text{HasExt}_{\mathfrak{D}}^{\text{repr}}(S_{u,x_{\mathfrak{D}}})$$

and let

$$\text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \in \text{Bool}$$

Define the extrema and quantitative coordinate width by

$$\begin{aligned}\text{CellMin}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) &:= \min S_{u,x_{\mathfrak{D}}}, \\ \text{CellMax}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) &:= \max S_{u,x_{\mathfrak{D}}}, \\ \text{CellWid}_{\mathfrak{D},u}^{\text{qty}}[\mathbb{I}](x_{\mathfrak{D}}) &:= \text{AffSpan}_{\mathfrak{D}}^{\text{qty}}(S_{u,x_{\mathfrak{D}}})\end{aligned}$$

The minimum, maximum, and ordered affine width are defined on the domain where $\text{HasExt}_{\mathfrak{D}}^{\text{repr}}$ holds.

Definition 4.160 (Admissibility of record-cell width derived from $\mathfrak{D}$). Define the predicate for admissibility of record-cell width derived from $\mathfrak{D}$ by

$$\begin{aligned}\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}] \\ \iff \left[\begin{array}{l} \forall (u, x_{\mathfrak{D}}) \in U \times \text{Tok}_{\mathfrak{D}}^*, \\ (\text{CellTok}_{\mathfrak{D},u}^{\text{arr}}(x_{\mathfrak{D}}) \neq \emptyset \implies \text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}})) \\ \wedge (\text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \implies \text{CellWid}_{\mathfrak{D},u}^{\text{qty}}[\mathbb{I}](x_{\mathfrak{D}}) \leq_{\mathfrak{D}}^{\text{qty}} \text{Grn}_{\mathfrak{D},\text{cellwid}}) \end{array} \right].\end{aligned}$$

The value of this predicate satisfies

$$\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}] \in \text{Bool}$$

Definition 4.161 ($\mathfrak{C}$ -token-associated representational image of $\mathfrak{D}$ -records). For each $u \in U$ and $x_{\mathfrak{C}} \in \text{Dom}(\text{Arr}_{\mathfrak{C},u}^{\text{cmp}})$, define the $\mathfrak{C}$ -token-associated representational image of $\mathfrak{D}$ -records and the global order-coordinate class of a $\mathfrak{C}$ -token by

$$\begin{aligned} \text{RecordImg}_{\mathfrak{C}\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{C}}) &:= \bigcup_{\substack{x_{\mathfrak{D}} \in \text{Dom}(\text{Arr}_{\mathfrak{D},u}^{\text{cmp}}) \\ (x_{\mathfrak{C}}, x_{\mathfrak{D}}) \in \text{Rel}^{\text{src}}}} \text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \subseteq \text{RepVal}_{\mathfrak{D}}, \\ \text{OrdCls}_{\mathfrak{C},u}^{\text{can}}[\mathbb{I}](x_{\mathfrak{C}}) &:= \text{Prj}_{\mathfrak{C}}^{\text{ord,can}}[\mathbb{I}]\left(\text{Prj}_{\mathfrak{C}}^{\text{can}}[\mathbb{I}](u, \text{Arr}_{\mathfrak{C},u}^{\text{cmp}}(x_{\mathfrak{C}}))\right) \end{aligned}$$

The latter satisfies

$$\text{OrdCls}_{\mathfrak{C},u}^{\text{can}}[\mathbb{I}](x_{\mathfrak{C}}) \in \text{Qu}_{\mathfrak{C}}^{\text{ord,can}}[\mathbb{I}]$$

Definition 4.162 (Record separation of $\mathfrak{C}$ -tokens). Define the predicate for record separation of $\mathfrak{C}$ -tokens by

$$\begin{aligned} \text{TokSep}_{\mathfrak{C}\mathfrak{D}}^{\text{rec}}[\mathbb{I}] &\iff \forall u \in U \forall (x_0, x_1) \in (\text{Dom}(\text{Arr}_{\mathfrak{C},u}^{\text{cmp}}))^2, \\ &\quad \text{OrdCls}_{\mathfrak{C},u}^{\text{can}}[\mathbb{I}](x_0) \neq \text{OrdCls}_{\mathfrak{C},u}^{\text{can}}[\mathbb{I}](x_1) \\ &\implies \left\{ \left[\bigwedge_{i \in \{0,1\}} \text{RecordImg}_{\mathfrak{C}\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_i) \neq \emptyset \right] \right. \\ &\quad \left. \wedge [\text{RecordImg}_{\mathfrak{C}\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_0) \cap \text{RecordImg}_{\mathfrak{C}\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_1) = \emptyset] \right\}, \\ \text{TokSep}_{\mathfrak{C}\mathfrak{D}}^{\text{rec}}[\mathbb{I}] &\in \text{Bool} \end{aligned}$$

Proposition 4.163 (Generation dependencies of $\mathfrak{D}$ -record-cell width and $\mathfrak{C}$ -token record separation). Assume $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$, $\text{QtyAffReal}_{\mathfrak{D}}[\mathbb{I}]$, and $\text{Grn}^{\text{op}}(\mathcal{X})$. For arbitrary $u \in U$ and $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$,

$$\text{CellExtrema}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \implies \text{CellWid}_{\mathfrak{D},u}^{\text{qty}}[\mathbb{I}](x_{\mathfrak{D}}) \in (\text{QtyVal}_{\mathfrak{D}})_{\geq 0}.$$

$\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}]$ requires the existence of extrema of the representational image in each record cell derived from $\mathfrak{D}$ whose arrival-capable part is nonempty, and, when extrema exist, comparison of its quantitative coordinate width with $\text{Grn}_{\mathfrak{D},\text{cellwid}}$. $\text{TokSep}_{\mathfrak{C}\mathfrak{D}}^{\text{rec}}[\mathbb{I}]$ requires that, for distinct global order-coordinate classes of $\mathfrak{C}$ -tokens, the corresponding two $\mathfrak{D}$ -record representational images are both nonempty and have empty intersection.

The record-cell representational image derived from $\mathfrak{D}$, its extrema and quantitative coordinate width, record-cell-width admissibility, the $\mathfrak{C}$ -token-associated representational image of $\mathfrak{D}$ -records, the global order-coordinate class of a $\mathfrak{C}$ -token, and record separation of $\mathfrak{C}$ -tokens are determined in dependency order from record cells derived from $\mathfrak{D}$, the source-side relation derived from $\mathfrak{C}\mathfrak{D}$, the canonical comparison quotient, the global order-coordinate quotient, the representational coordinate, the quantitative affine realization, and finite operational granularity.

Proof. Assume the antecedent of the proposition's implication. Then

$$\text{CellMin}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) \leq_{\mathfrak{D}}^{\text{repr}} \text{CellMax}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}})$$

holds. By Definition 4.159 and Lemma 4.149, the record-cell width, which is the quantitative affine difference between the two extrema, is nonnegative.

The two implications in Definition 4.160 give nonemptiness of the arrival-capable part, existence of extrema, and comparison with the granularity component. Definition 4.162 gives nonemptiness and empty intersection for the two $\mathfrak{D}$ -record representational images corresponding to distinct global order-coordinate classes. The defining formulas of the respective objects and the record-cell-width granularity component derived from $\mathfrak{D}$ in Definition 4.156 determine, in dependency order from the inputs listed in the proposition, admissibility of the record-cell width derived from $\mathfrak{D}$, the $\mathfrak{C}$ -token-associated representational image of $\mathfrak{D}$ -records, and record separation of $\mathfrak{C}$ -tokens. $\square$

4.2.7 Continuum-approximation evaluation input

Definition 4.164 (Componentwise evaluation input for continuum approximation). Define the index set and component types for continuum-approximation evaluation, and the evaluation source for the counting-step-width component derived from $\mathfrak{C}$, by

$$\begin{aligned} \text{EvalIdx}^{\text{cont}} &:= \left\{ (\mathfrak{C}, \text{cntstep}), (\mathfrak{D}, \text{stab}), (\mathfrak{D}, \text{cellwid}), \right. \\ &\quad \left. (\mathfrak{B}, \text{disc}), (\mathfrak{D}, \text{coupl}), \text{OrdStab} \right\}, \\ \text{EvalVal}_{\kappa}^{\text{cont}} &\neq \emptyset, \quad \preceq_{\kappa}^{\text{cont,eval}} \subseteq (\text{EvalVal}_{\kappa}^{\text{cont}})^2, \\ 0_{\kappa}^{\text{cont,eval}} &\in \text{EvalVal}_{\kappa}^{\text{cont}}, \\ \forall z \in \text{EvalVal}_{\kappa}^{\text{cont}}, \quad 0_{\kappa}^{\text{cont,eval}} &\preceq_{\kappa}^{\text{cont,eval}} z, \\ \text{Eval}_{\kappa}^{\text{cont}}[\mathbb{I}], \text{Tol}_{\kappa}^{\text{cont}} &\in \text{EvalVal}_{\kappa}^{\text{cont}}, \\ \text{EvalComp}_{\kappa}^{\text{cont}}[\mathbb{I}] &:= \left(\begin{array}{c} \text{EvalVal}_{\kappa}^{\text{cont}}, \\ \preceq_{\kappa}^{\text{cont,eval}}, \\ 0_{\kappa}^{\text{cont,eval}}, \text{Eval}_{\kappa}^{\text{cont}}[\mathbb{I}], \\ \text{Tol}_{\kappa}^{\text{cont}} \end{array} \right), \\ \kappa &\in \text{EvalIdx}^{\text{cont}}. \end{aligned}$$

Define the evaluation source for the counting-step-width component derived from $\mathfrak{C}$ by

$$\begin{aligned} \text{EvalSrc}_{(\mathfrak{C}, \text{cntstep})}^{\text{cont}}[\mathbb{I}] &:= \left((\text{CntStepWid}_{\mathfrak{C}}^{\text{qty}}[\mathbb{I}; \text{win}])_{\text{win} \in \text{CntWin}_{\mathfrak{C}}^{\text{step}}[\mathbb{I}]}, \right. \\ &\quad \left. \text{Grn}_{\mathfrak{C}, \text{cntstep}} \right) \end{aligned}$$

The relation $\preceq_{\kappa}^{\text{cont,eval}}$ is a partial order, and $0_{\kappa}^{\text{cont,eval}}$ is its least element. The counting-step-width component derived from $\mathfrak{C}$, $\text{Eval}_{(\mathfrak{C}, \text{cntstep})}^{\text{cont}}[\mathbb{I}]$, takes $\text{EvalSrc}_{(\mathfrak{C}, \text{cntstep})}^{\text{cont}}[\mathbb{I}]$ as input.

Definition 4.165 (Order-preserving representational-stability evaluation input). The order-preserving representational-stability evaluation input is

$$\text{EvalComp}_{\text{OrdStab}}^{\text{cont}}[\mathbb{I}]$$

and its componentwise evaluation formula is defined by

$$\text{Eval}_{\text{OrdStab}}^{\text{cont}}[\mathbb{I}] \preceq_{\text{OrdStab}}^{\text{cont,eval}} \text{Tol}_{\text{OrdStab}}^{\text{cont}}$$

Definition 4.166 (Boolean truth-value evaluation inputs for record-cell width and record separation). Let the predicate for admissibility of record-cell width derived from $\mathfrak{D}$ and the predicate for record separation of $\mathfrak{C}$ -tokens be the Boolean truth-value components of the continuum-approximation evaluation-input datum by

$$(\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}], \text{TokSep}_{\mathfrak{C}\mathfrak{D}}^{\text{rec}}[\mathbb{I}]) \in \text{Bool}^2$$

Definition 4.167 (Order-topological coordinate candidates). For a subbasis $\mathcal{S}$, write $\text{TopSub}(\mathcal{S})$ for the topology it generates. For each $R \in \text{Rol}^{\text{cmp}}$ and $a_0 \in \text{RepVal}_R$, define the open rays, order topology, and candidate datum for a continuous coordinate representation by

$$\begin{aligned} \text{Ray}_R^\sigma(a_0) &:= \begin{cases} \{a \in \text{RepVal}_R \mid a <_R^{\text{repr}} a_0\}, & \sigma = -, \\ \{a \in \text{RepVal}_R \mid a_0 <_R^{\text{repr}} a\}, & \sigma = +. \end{cases} \\ \text{OrdTop}_R^{\text{repr}} &:= \text{TopSub}(\{\text{Ray}_R^\sigma(a_0) \mid \sigma \in \{-, +\}, a_0 \in \text{RepVal}_R\}), \\ \text{CoordCand}_R^{\text{cont}}[\mathbb{I}] &:= (\text{RepVal}_R, \leq_R^{\text{repr}}, \text{OrdTop}_R^{\text{repr}}, \text{Rep}_R^{\text{cmp}}[\mathbb{I}]) \end{aligned}$$

Definition 4.168 (Continuum-approximation evaluation-input datum). Assume

$$\text{ReprCtx}_{\leq 3}[\mathbb{I}]$$

and

$$\text{QtyAffReal}_R[\mathbb{I}], \quad R \in \text{Rol}^{\text{cmp}}$$

Specify the finite operational granularity datum of Definition 4.156 and the six componentwise evaluation inputs of Definition 4.164. Construct record-cell-width admissibility, record separation, and the order-topological coordinate candidates for these specifications and the generated representation-stage structures according to their defining clauses. Define the continuum-approximation evaluation-input datum by

$$\begin{aligned} \text{EvalInput}^{\text{cont}}[\mathbb{I}] &:= \left((\text{QtyAffReal}_R[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}, \text{Grn}^{\text{op}}(\mathcal{X}), \right. \\ &\quad (\text{EvalComp}_\kappa^{\text{cont}}[\mathbb{I}])_{\kappa \in \text{EvalIdx}^{\text{cont}}}, \\ &\quad \text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}], \text{TokSep}_{\mathfrak{C}/\mathfrak{D}}^{\text{rec}}[\mathbb{I}], \\ &\quad \left. (\text{CoordCand}_R^{\text{cont}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}} \right) \end{aligned}$$

Proposition 4.169 (Component types of continuum-approximation evaluation input). The componentwise establishment-judgment formulas and the types of the Boolean truth-value components of the continuum-approximation evaluation-input datum are

$$\begin{aligned} \text{Eval}_\kappa^{\text{cont}}[\mathbb{I}] &\preceq_{\kappa}^{\text{cont, eval}} \text{Tok}_\kappa^{\text{cont}}, \quad \kappa \in \text{EvalIdx}^{\text{cont}}, \\ (\text{CellWidAdm}_{\mathfrak{D}}[\mathbb{I}], \text{TokSep}_{\mathfrak{C}/\mathfrak{D}}^{\text{rec}}[\mathbb{I}]) &\in \text{Bool}^2 \end{aligned}$$

The candidate datum for a continuous coordinate representation derived from $\mathfrak{C}/\mathfrak{D}$ has as components a coordinate carrier, a linear order, an order topology, and a comparison representational coordinate.

Proof. Definition 4.164 gives the six $\text{EvalComp}_\kappa^{\text{cont}}$ an evaluation-value set, a partial order, a least element, an evaluation value, and an admissible threshold. Definition 4.165 gives order-preserving representational stability as the $\kappa = \text{OrdStab}$ component of the same component family. Definition 4.166 gives the two truth-value components, and Definition 4.167 gives the candidate for a continuous coordinate representation derived from $\mathfrak{C}/\mathfrak{D}$.

Thus the tuple of Definition 4.168 has as components conditional quantitative affine realizations, finite operational granularity, six componentwise evaluation types, two Boolean truth values, and candidates for continuous coordinate representations derived from $\mathfrak{C}/\mathfrak{D}$. The componentwise establishment judgment and continuous-representation evaluation apply the evaluation relation to the corresponding components, and the complete evaluation specification determines the establishment value of the continuum approximation. $\square$

4.2.8 Consequences of the Representation Stage

Proposition 4.170 (Separation of representation-stage structures from quantitative-evaluation inputs). Assume $\text{ReprCtx}_{\leq 3}[\mathbb{I}]$. For each $R \in \text{Rol}^{\text{cmp}}$, specify the selected $\mathfrak{C}/\mathfrak{D}$ -derived representational value range and its $\mathfrak{C}/\mathfrak{D}$ -derived representational-value-range order embedding from Definition 4.140. Then

$$\begin{aligned} \text{Repr}_{\leq 3}[\mathbb{I}] &\implies \text{ReprStr}_{\leq 3}[\mathbb{I}], \\ \text{Rep}_R^{\text{cmp}}[\mathbb{I}] &= \text{RepEmb}_R^{\text{ord}}[\mathbb{I}] \circ \text{Prj}_R^{\text{ord,can}}[\mathbb{I}] \end{aligned}$$

are determined in dependency order. The components of $\text{ReprStr}_{\leq 3}[\mathbb{I}]$ are given in Definition 4.144.

Further specify the $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations of Definition 4.148, the finite operational granularity datum of Definition 4.156, and the six componentwise evaluation inputs of Definition 4.164. Under these specifications, the $\mathfrak{D}$ -derived record-cell-width admissibility predicate, the $\mathfrak{C}$ -token record-separation predicate, and the order-topological coordinate candidates are determined and form the continuum-approximation evaluation-input datum of Definition 4.168. The bundles of structures generated at the representation stage and the quantitative-evaluation inputs have distinct types and dependency orders.

Proof. Theorem 4.147 generates $\text{ReprStr}_{\leq 3}[\mathbb{I}]$, and Definition 4.140 gives the factorization of the $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates. The specified quantitative affine realizations satisfy the compatibility conditions of Definition 4.148. Definition 4.156 forms a product-type datum from the five specified granularity components. Definitions 4.160 and 4.162 determine record-cell-width admissibility and record separation from the generated record images and the specified inputs. Definition 4.167 determines the order-topological coordinate candidates from the order on the selected representational value ranges and the comparison representational coordinates. Definition 4.168 collects these objects, together with the six specified componentwise evaluation inputs, into one tuple. Hence, the bundles of structures generated at the representation stage, the $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates, and the quantitative-evaluation inputs satisfy the types and dependency orders stated in the proposition. $\square$

Definition 4.171 (Typed generation closure). Let

Expr

denote the collection of all typed formal expressions whose components are object expressions or truth-valued expressions. For a finite tuple T , let $\text{Idx}(T)$ denote its component-index set, and define the component relation by

$$A \in_{\text{comp}} T \iff \exists i \in \text{Idx}(T), \text{pr}_i(T) = A$$

For an object set In and a family of typed generative-dependency rules

$$\text{Dep} \subseteq \mathcal{P}(\text{Expr}) \times \text{Expr}$$

define the generation closure by

$$\begin{aligned} \text{Cl}_{\text{Dep}}(\text{In}) := \bigcap \left\{ S \subseteq \text{Expr} \mid \text{In} \subseteq S, \right. \\ \left. \forall (A, Y) \in \text{Dep}, \quad A \subseteq S \implies Y \in S \right\} \end{aligned}$$

and define generability by

$$\text{GenDef}_{\text{Dep}}(Y \mid \text{In}) \iff Y \in \text{Cl}_{\text{Dep}}(\text{In})$$

Proposition 4.172 (Generation closure of continuum-approximation evaluation inputs). Define the set of direct inputs to continuum-approximation evaluation by

$$\begin{aligned} \text{In}^{\text{cont}} &:= \{A \in \text{Expr} \mid A \in_{\text{comp}} \text{EvalInput}^{\text{cont}}[\mathbb{I}]\}, \\ \text{Dep}^{\text{cont}} &:= \{(\text{In}^{\text{cont}}, \text{EvalInput}^{\text{cont}}[\mathbb{I}])\} \end{aligned}$$

Then

$$\text{GenDef}_{\text{Dep}^{\text{cont}}}(\text{EvalInput}^{\text{cont}}[\mathbb{I}] \mid \text{In}^{\text{cont}})$$

holds.

Proof. The right-hand side of Definition 4.168 is a tuple containing every element of In^{cont} as a component. Therefore,

$$(\text{In}^{\text{cont}}, \text{EvalInput}^{\text{cont}}[\mathbb{I}]) \in \text{Dep}^{\text{cont}}.$$

Apply, in order, the generation closure and the generability criterion of Definition 4.171. $\square$

By Definition 4.144, Theorem 4.147, Proposition 4.170, and Proposition 4.172, the structural generation sequence of the representation stage is

$$\begin{aligned} \text{Operative}(\mathcal{X}) &\longmapsto (\preceq_R^{\text{arr}})_{R \in \text{Rol}^{\text{cmp}}}, \\ \text{Repr}_{\leq k}[\mathbb{I}] &\longmapsto \text{ReprStr}_{\leq k}[\mathbb{I}], \quad k \in \{1, 2, 3\}, \\ (\text{ReprStr}_{\leq 3}[\mathbb{I}], (\text{RepEmb}_R^{\text{ord}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}) &\longmapsto (\text{Rep}_R^{\text{cmp}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}. \end{aligned}$$

The object $\text{Repr}_{\leq 3}[\mathbb{I}]$ of Definition 4.92 expresses the conjunction of the three representation-stage conditions. Specifying the $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations determines quantitative affine differences and frequency and period indexed by admissible counting windows. Source-relative frequency requires window invariance on a nonempty family of windows; a hertz numerical value requires a positive unit scalar. Specifying a unit through a standard frequency uses a reference source and a normalization convention. Further specifying finite operational granularity and the six componentwise evaluation inputs determines the continuum-approximation evaluation-input datum, together with record-cell-width admissibility, record separation, and coordinate candidates. Proposition 4.172 expresses tuple construction when these direct inputs are given.

Table 2: Hierarchy of generated representation-stage objects

Stage	Condition or generated object
Comparison-input layer	$\text{Operative}(\mathcal{X}) \mapsto (\preceq_R^{\text{arr}})_{R \in \text{Rol}^{\text{cmp}}}$
Canonical-order layer	$\text{Repr}_{\leq 1}[I] \mapsto \text{ReprStr}_{\leq 1}[I]$
Local order-coordinate layer	$\text{Repr}_{\leq 2}[I] \mapsto \text{ReprStr}_{\leq 2}[I]$ $\text{Repr}_{\leq 3}[I] \mapsto \text{ReprStr}_{\leq 3}[I]$
Global order-coordinate layer	$\xrightarrow{\text{RepEmb}_R^{\text{ord}}} \text{Rep}_R^{\text{cmp}}[I]$
Quantitative-representation layer	$\text{Repr}_{\leq 3}[I], \text{QtyAffReal}_R[I] \ (R \in \text{Rol}^{\text{cmp}})$ $\mapsto$ quantitative affine differences, frequency and period for each admissible counting window A hertz numerical value additionally requires a positive unit scalar.
Continuum-approximation evaluation-input layer	$\text{In}^{\text{cont}} \mapsto \text{EvalInput}^{\text{cont}}[I]$

The structures generated at each stage of Table 2 are as follows.

Comparison-input layer The $\mathfrak{C}/\mathfrak{D}$ -source-induced arrival preorders are determined from the $\mathfrak{C}/\mathfrak{D}$ source-side partial orders, the comparison-side recovery maps, and the tagged comparison-arrival reception spaces.

Canonical-order layer When $K_{\text{CanOrdDesc}}^{\text{repr}}[I]$ holds, the source-induced arrival preorders descend to the comparison equivalence classes and determine the canonical $\mathfrak{C}/\mathfrak{D}$ partial orders and the exact operational time structures.

Local order-coordinate layer When $K_{\text{LocIncTrans}}^{\text{repr}}[I]$ holds, it determines the quotients by local incomparability classes, the local strict linear orders, the local order-coordinate projections, and the local comparison charts. The local order-coordinate chart-transition maps yield the comparison-order atlas.

Global order-coordinate layer When $K_{\text{GlobIncTrans}}^{\text{repr}}[I]$ holds, it determines the quotients by global incomparability classes, the global strict linear orders, and the canonical global order-coordinate projections. Their composition with the $\mathfrak{C}/\mathfrak{D}$ -derived representational-value-range order embeddings yields the $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates.

Quantitative-representation layer Specifying the $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations determines quantitative affine differences and frequency and period indexed by admissible counting windows. Source-relative frequency and hertz numerical values use the window-family, window-invariance, and unit-normalization inputs specified in their respective definitions.

Continuum-approximation evaluation-input layer Further specify the finite operational granularity datum and the six componentwise evaluation inputs. Each evaluation input includes an evaluation-value carrier, a partial order, a least element, an evaluation value, and an admissible threshold. Construct record-cell-width admissibility, record separation, and the order-topological coordinate candidates, and collect them into the tuple of Definition 4.168.

The $\mathfrak{C}/\mathfrak{D}$ source-side partial orders, the comparison-side recovery maps, and the tagged comparison-arrival reception spaces are the generative inputs to $(\preceq_R^{\text{arr}})_{R \in \text{Rol}^{\text{cmp}}}$.

Definition 4.173 (Representation-stage dependency layers). Fix the input $\mathbb{I}$. Define the outputs of the canonical quotient and order constructions by dependency layer as

$$(\text{Out}_k^{\text{repr}})_{0 \leq k \leq 3} := \left(\begin{array}{c} \{\text{Qu}^{\text{can}}[\mathbb{I}]\}, \\ \{\mathcal{E}^{\text{op}}[\mathbb{I}]\} \cup \bigcup_{R \in \text{Rol}^{\text{cmp}}} \{\preceq_R^{\text{can}}[\mathbb{I}]\}, \\ \{\text{Atlas}^{\text{ord}}[\mathbb{I}]\} \cup \bigcup_{\substack{R \in \text{Rol}^{\text{cmp}} \\ u \in U}} \{\text{Qu}_{R,u}^{\text{loc}}[\mathbb{I}], \text{Prj}_{R,u}^{\text{loc}}[\mathbb{I}], \text{Chart}_{R,u}^{\text{loc}}[\mathbb{I}]\}, \\ \bigcup_{R \in \text{Rol}^{\text{cmp}}} \{\text{Qu}_R^{\text{ord,can}}[\mathbb{I}], \text{Prj}_R^{\text{ord,can}}[\mathbb{I}]\} \end{array} \right).$$

The selected representational value ranges and embeddings, and the quantitative-evaluation inputs, are not members of this canonical structural output family. Define the premise set for each layer's generation rules and the family of generative-dependency rules for canonical structures by

$$\begin{aligned} \text{Req}_0^{\text{repr}} &:= \{A \in \text{Expr} \mid A \in_{\text{comp}} \mathbb{I}\} \cup \{\text{Operative}(\mathcal{X})\}, \\ \text{Req}_k^{\text{repr}} &:= \text{Req}_0^{\text{repr}} \cup \{\text{Repr}_{\leq k}[\mathbb{I}]\} \cup \bigcup_{0 \leq j < k} \text{Out}_j^{\text{repr}}, \quad k \in \{1, 2, 3\}, \\ \text{Dep}^{\text{repr}} &:= \bigcup_{k=0}^3 \{(\text{Req}_k^{\text{repr}}, Y) \mid Y \in \text{Out}_k^{\text{repr}}\}, \\ \text{Out}_{\neg G}^{\text{repr}} &:= \bigcup_{k=0}^3 \text{Out}_k^{\text{repr}} \end{aligned}$$

and let the context in which global gluing fails be

$$\begin{aligned} \text{Ctx}_{\neg G}^{\text{repr}} &:= \{A \in \text{Expr} \mid A \in_{\text{comp}} \mathbb{I}\} \\ &\cup \{\text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}], \neg G[\text{Ev}^{\text{cmp}}], \neg \text{Operative}(\mathcal{X})\} \end{aligned}$$

Proposition 4.174 (Failure of global gluing prevents generation of canonical representation structures).

$$\begin{aligned} &\text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}] \wedge \neg G[\text{Ev}^{\text{cmp}}] \\ &\implies \neg \text{Operative}(\mathcal{X}) \wedge \bigwedge_{Y \in \text{Out}_{\neg G}^{\text{repr}}} \neg \text{GenDef}_{\text{Dep}^{\text{repr}}}(Y \mid \text{Ctx}_{\neg G}^{\text{repr}}). \end{aligned}$$

Proof. The equivalence in Definition 4.86 yields $\neg \text{Operative}(\mathcal{X})$ from the antecedent. Take the set

$$S_{\neg G} := \text{Expr} \setminus (\text{Out}_{\neg G}^{\text{repr}} \cup \{\text{Operative}(\mathcal{X})\})$$

By type, $\text{Ctx}_{\neg G}^{\text{repr}} \subseteq S_{\neg G}$. For each $k \in \{0, 1, 2, 3\}$,

$$\begin{aligned} \text{Operative}(\mathcal{X}) &\in \text{Req}_k^{\text{repr}}, \\ \text{Operative}(\mathcal{X}) &\notin S_{\neg G} \end{aligned}$$

and hence $\text{Req}_k^{\text{repr}} \not\subseteq S_{-G}$. Thus, S_{-G} is closed under every rule in Dep^{repr} . By the minimality of the generation closure,

$$\text{Cl}_{\text{Dep}^{\text{repr}}}(\text{Ctx}_{-G}^{\text{repr}}) \subseteq S_{-G}.$$

Therefore, no $Y \in \text{Out}_{-G}^{\text{repr}}$ belongs to the generation closure. The second conjunct of the proposition follows from the definition of generability. $\square$

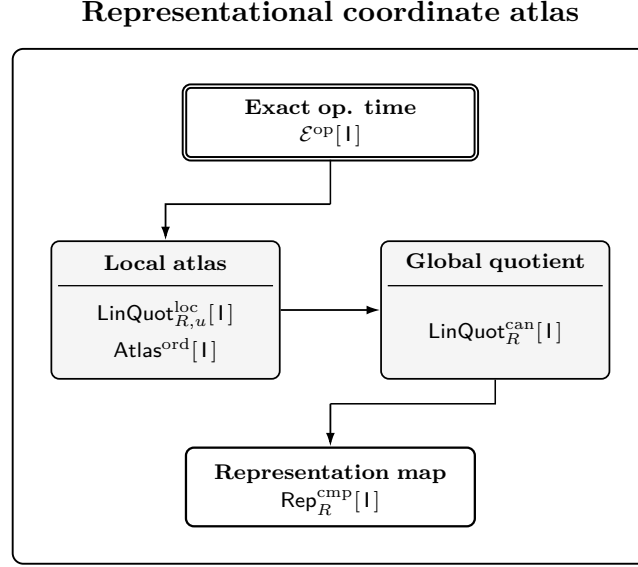


Figure 10: Generation order under $\text{ReprCtx}_{\leq 3}[l]$, from exact operational time structures through local order coordinates and the comparison-order atlas to global order-coordinate quotients. The final comparison representational coordinate is the composition of the order embedding into the selected representational value range with the canonical global order-coordinate projection.

4.3 Exact Operability, Canonical Uniqueness, and Universal Factorization

For each symbol root, one pair consisting of a role type and a reception phase first appears in complete form. Thereafter, only the reception-phase subscript is omitted within the same definition, proposition, or proof when the reception phase is fixed and the complete form is uniquely recoverable. The complete forms are retained for the role type, a varying family index, distinct reception phases, and residual modifiers that select genuine branches.

When the structural completion of the comparison stage $\text{Operative}(\mathcal{X})$ and the structural completion of the representation stage $\text{Repr}_{\leq 3}[l]$ from Definition 4.92 both hold, the dependency sequence from comparison-stage material to the global order-coordinate structure is generated. The selected $\mathfrak{C}/\mathfrak{D}$ -derived representational value ranges and the $\mathfrak{C}/\mathfrak{D}$ -derived representational-value-range order embeddings determine the $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates through composition with the canonical global order-coordinate projections. Specifying the finite operational granularity datum and the six componentwise evaluation inputs in addition to the $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations determines the continuum-approximation evaluation-input datum.

Definition 4.175 (Exact-operability object sequence and exact operability). For a realized operational configuration $\mathcal{X}$ and a typed family of comparison rules $\mathcal{J}_{\text{cmp}}$, define exact operability by

$$\text{ExactOperable}[\mathbb{I}] \iff \text{ReprCtx}_{\leq 3}[\mathbb{I}]$$

When $\text{ExactOperable}[\mathbb{I}]$ holds, define the exact-operability object sequence by

$$\text{EOSeq}[\mathbb{I}] := (\text{Dat}^{\text{mat}}[\mathbb{I}], \text{Qu}^{\text{can}}[\mathbb{I}], (\preceq_R^{\text{arr}})_{R \in \text{Rol}^{\text{cmp}}}, \text{ReprStr}_{\leq 3}[\mathbb{I}])$$

An object in the transitive closure of the component relation of a finite tuple T is called a recursive component of T . The ordering of the recursive components is determined by the dependencies among the comparison-stage material datum, the $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient, the source-induced arrival preorders, the canonical $\mathfrak{C}/\mathfrak{D}$ partial orders, the exact operational time structures, the local order-coordinate structures, the comparison-order atlas, and the global order-coordinate structures.

Proposition 4.176 (Expansion of the exact-operability conditions). For a realized operational configuration $\mathcal{X}$ and a typed family of comparison rules $\mathcal{J}_{\text{cmp}}$,

$$\begin{aligned} &\text{ExactOperable}[\mathbb{I}] \\ &\iff \text{Operative}(\mathcal{X}) \\ &\quad \wedge K_{\text{CanOrdDesc}}^{\text{repr}}[\mathbb{I}] \\ &\quad \wedge K_{\text{LocIncTrans}}^{\text{repr}}[\mathbb{I}] \\ &\quad \wedge K_{\text{GlobIncTrans}}^{\text{repr}}[\mathbb{I}]. \end{aligned}$$

holds.

When every condition in this equivalence holds, every recursive component of

$$\text{EOSeq}[\mathbb{I}]$$

from Definition 4.175 is generated according to the dependencies among those components.

Proof. Definitions 4.175 and 4.92 give

$$\begin{aligned} &\text{ExactOperable}[\mathbb{I}] \\ &\iff \text{Operative}(\mathcal{X}) \\ &\quad \wedge \text{Repr}_{\leq 3}[\mathbb{I}]. \end{aligned}$$

Substituting the equivalence from Definition 4.92 into the preceding equivalence yields the equivalence in the proposition.

By Definition 4.2, the realized operational configuration and the typed family of comparison rules determine the comparison-stage material datum

$$\text{Dat}^{\text{mat}}[\mathbb{I}]$$

By Theorem 4.90, under $\text{Operative}(\mathcal{X})$, the $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient

$$\text{Qu}^{\text{can}}[\mathbb{I}]$$

is generated.

By Theorem 4.147, under

$$\text{Operative}(\mathcal{X}), \quad \text{Repr}_{\leq 3}[\mathbb{I}]$$

the $\mathfrak{C}/\mathfrak{D}$ source-induced arrival preorders, the canonical $\mathfrak{C}/\mathfrak{D}$ partial orders, the exact operational time structures, the local order-coordinate structures, the comparison-order atlas, the global order-coordinate structures, and the local-to-global order-coordinate embeddings are generated in dependency order.

Therefore, every recursive component of Equation (4.176) is generated according to the dependencies among those components. $\square$

Let a realized operational configuration $\mathcal{X}$ from Definition 3.25 and a typed family of comparison rules $\mathcal{J}_{\text{cmp}}$ be given. The structural completion of the comparison stage is expressed by $\text{Operative}(\mathcal{X})$.

The structural completion of the representation stage is expressed by $\text{Repr}_{\leq 3}[\mathbb{I}]$ from Definition 4.92 and is equivalent to the simultaneous satisfaction of the three conditions in that definition.

Structural completion of the comparison and representation stages determines, in dependency order, the $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient, the canonical $\mathfrak{C}/\mathfrak{D}$ partial orders, the exact operational time structures, the local order-coordinate structures, the comparison-order atlas, and the global order-coordinate structures. This structural sequence is expressed by the exact-operability object sequence $\text{EOSeq}[\mathbb{I}]$ from Definition 4.175.

Specifying the selected $\mathfrak{C}/\mathfrak{D}$ -derived representational value ranges and the $\mathfrak{C}/\mathfrak{D}$ -derived representational-value-range order embeddings determines the factorization of the $\mathfrak{C}/\mathfrak{D}$ -derived comparison representational coordinates in Proposition 4.170.

Specifying the $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realizations

$$(\text{QtyAffReal}_R[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}}$$

together with the finite operational granularity datum of Definition 4.156 and the six componentwise evaluation inputs of Definition 4.164 determines the continuum-approximation evaluation-input datum of Definition 4.168,

$$\text{EvalInput}^{\text{cont}}[\mathbb{I}].$$

4.3.1 Operability Theorem

Comparison-stage operability is equivalent to the conjunction of the stage-seven satisfaction bundle and the identity-action condition for every pure transport loop.

Theorem 4.177 (Operability theorem). For every realized operational configuration $\mathcal{X}$ and typed family of comparison rules $\mathcal{J}_{\text{cmp}}$,

$$\begin{aligned} & \text{Operative}(\mathcal{X}) \\ & \iff \text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}] \wedge \text{G}[\text{Ev}^{\text{cmp}}]. \end{aligned}$$

Proof. By Definition 4.78, $\text{L}_{\leq 7}[\text{Ev}^{\text{cmp}}]$ is the stage-seven satisfaction bundle.

By Definition 4.86, comparison-stage operability is equivalent to the conjunction of the stage-seven satisfaction bundle and the identity-action condition for every pure transport loop. The claim follows. $\square$

Proposition 4.178 (Exact operability implies comparison-stage operability). For every realized operational configuration $\mathcal{X}$ and typed family of comparison rules $\mathcal{J}_{\text{cmp}}$,

$$\text{ExactOperable}[\mathbb{I}] \implies \text{Operative}(\mathcal{X})$$

holds.

Proof. The first component on the right-hand side of the equivalence in Proposition 4.176 yields $\text{Operative}(\mathcal{X})$. $\square$

4.3.2 Canonical Uniqueness and Universality

Under exact operability, the canonical objects of the comparison and representation stages are uniquely determined by the realized operational configuration and the typed family of comparison rules. Admissible comparison representations, admissible local order-atlas representations, and admissible global order representations factor, respectively, as compositions with the canonical projection, the local order-coordinate projections, and the canonical global order-coordinate projections.

Theorem 4.179 (Canonical-structure uniqueness theorem). Assume $\text{ExactOperable}[\mathbb{I}]$. Then

$$(\text{Qu}^{\text{can}}[\mathbb{I}], \text{ReprStr}_{\leq 3}[\mathbb{I}])$$

is uniquely determined by $\mathbb{I}$.

Proof. By Proposition 4.176, comparison-stage operability, the canonical-order descent condition, the local incomparability transitivity condition, and the global incomparability transitivity condition that constitute the right-hand side of its equivalence all hold.

By Theorem 4.90, the extended comparison-reachability equivalence relation, the $\mathfrak{C}/\mathfrak{D}$ comparison quotient set, the canonical projection, the quotient relation derived from $\mathfrak{C}\mathfrak{D}$, and the canonical comparison quotient derived from $\mathfrak{C}\mathfrak{D}$ are uniquely determined in dependency order.

By the representative invariance in Definition 4.99, the right-hand side of Definition 4.101 is invariant under the choice of representative of a comparison equivalence class. Hence, the canonical $\mathfrak{C}/\mathfrak{D}$ partial orders are uniquely determined as binary relations on the $\mathfrak{C}/\mathfrak{D}$ comparison quotient sets. By Definition 4.103,

$$\mathcal{E}^{\text{op}}[\mathbb{I}] = (\text{Qu}^{\text{can}}[\mathbb{I}], (\preceq_R^{\text{can}}[\mathbb{I}])_{R \in \text{Rol}^{\text{cmp}}})$$

is uniquely determined.

Definition 4.111 determines the local incomparability relations on each local canonical quotient domain from the canonical strict $\mathfrak{C}/\mathfrak{D}$ orders. By Lemma 4.114, the local incomparability relations are equivalence relations. Definition 4.116 uniquely determines the local order-coordinate quotients and the local strict linear orders. Definition 4.119 uniquely determines the local order-coordinate projections, and Definition 4.124 uniquely determines the local comparison charts as compositions of the canonical projection with the local order-coordinate projections.

By Lemma 4.127, the local order-coordinate chart-transition maps of Definition 4.126 are representative-invariant and uniquely determined. By Definition 4.128, the family of local order-coordinate projections, local comparison charts, and local order-coordinate chart-transition maps uniquely determines the comparison-order atlas.

Definition 4.130 determines the global incomparability relations from the canonical strict $\mathfrak{C}/\mathfrak{D}$ orders. By Lemma 4.133, the global incomparability relations are equivalence relations. Definition 4.135 uniquely determines the $\mathfrak{C}/\mathfrak{D}$ global order-coordinate quotients, the global strict linear orders, and the canonical global order-coordinate projections.

By Definition 4.142, each local-to-global order-coordinate embedding is uniquely determined as the map that sends a local incomparability class to the global incomparability class of the same representative. $\square$

Lemma 4.180 (Unique factorization through a surjection under equality-kernel inclusion).
Let sets

$$S, \quad Q, \quad T$$

be given, together with a surjection

$$\pi : S \twoheadrightarrow Q$$

and a map

$$\Phi : S \longrightarrow T$$

If

$$\text{EqKer}(\pi) \subseteq \text{EqKer}(\Phi)$$

holds, there exists a unique map

$$\overline{\Phi} : Q \longrightarrow T$$

such that

$$\Phi = \overline{\Phi} \circ \pi$$

Proof. For each

$$x \in S$$

define

$$\overline{\Phi}(\pi(x)) := \Phi(x)$$

By the equality-kernel inclusion, any two elements with equal values under π have equal images under Φ ; hence this map is well-defined. Surjectivity of π makes its domain equal to Q , and the defining equation gives the factorization equation. For each $x \in S$, the value of any map satisfying the factorization equation is fixed by $\pi(x) \mapsto \Phi(x)$. Since π is surjective, this map is unique. $\square$

Lemma 4.181 (Cancellation of precomposition by a surjection). If a surjection

$$\pi : S \twoheadrightarrow Q$$

and two maps

$$F, G : Q \longrightarrow T$$

satisfy

$$F \circ \pi = G \circ \pi$$

then

$$F = G$$

holds.

Proof. Take any $q \in Q$. By surjectivity of π , the equation $\pi(x) = q$ holds for some $x \in S$. Evaluating the equality after precomposition at x gives $F(q) = G(q)$. $\square$

Lemma 4.182 (Common core of the order isomorphism between representational images).
Let a set S and binary relations

$$\sim_S, \quad \prec_S \subseteq S \times S$$

be given. For each

$$a \in \{0, 1\}$$

let a linearly ordered set

$$\left(\text{RepVal}^{(a)}, <^{(a)} \right)$$

and a map

$$\text{Rep}^{(a)} : S \longrightarrow \text{RepVal}^{(a)}$$

be given. Suppose that the map satisfies

$$\text{OrdRepCond} \left(\text{Rep}^{(a)} ; \sim_S, \prec_S, <^{(a)} \right)$$

Then there exists a unique strict-order isomorphism

$$\text{Replso}^{01} : \text{Im} \left(\text{Rep}^{(0)} \right) \longrightarrow \text{Im} \left(\text{Rep}^{(1)} \right)$$

such that

$$\text{Rep}^{(1)} = \text{Replso}^{01} \circ \text{Rep}^{(0)}$$

Proof. Define

$$\text{Replso}^{01} \left(\text{Rep}^{(0)}(x) \right) := \text{Rep}^{(1)}(x)$$

By Definition 4.107 and the equality-kernel condition for $a = 0$, any two elements having equal values under $\text{Rep}^{(0)}$ are related by $\sim_S$. Applying the equality-kernel condition for $a = 1$, their images under $\text{Rep}^{(1)}$ are equal. Thus, Replso^{01} is well-defined as a map. Its definition with codomain $\text{Im} \left(\text{Rep}^{(1)} \right)$ gives surjectivity.

Suppose that two values of Replso^{01} are equal. By definition, the corresponding values of $\text{Rep}^{(1)}$ are equal. Applying the equality-kernel condition for $a = 1$, followed by that for $a = 0$, shows that the corresponding values of $\text{Rep}^{(0)}$ are equal. Thus, injectivity holds.

For any

$$x_0, x_1 \in S$$

the two order-pullback conditions give

$$\begin{aligned} & \text{Rep}^{(0)}(x_0) <^{(0)} \text{Rep}^{(0)}(x_1) \\ & \iff x_0 \prec_S x_1 \\ & \iff \text{Rep}^{(1)}(x_0) <^{(1)} \text{Rep}^{(1)}(x_1) \end{aligned}$$

Therefore, Replso^{01} preserves and reflects the strict order.

For any

$$x \in S$$

the value of a map satisfying the commutative equation is fixed by

$$\text{Rep}^{(0)}(x) \longmapsto \text{Rep}^{(1)}(x)$$

Every element of the domain is in the image of $\text{Rep}^{(0)}$; hence Replso^{01} is unique. $\square$

Theorem 4.183 (Uniqueness of the order type of representational images). Assume $\text{ExactOperable}[\mathbb{I}]$ and fix

$$R \in \text{Rol}^{\text{cmp}}$$

For each

$$a \in \{0, 1\}$$

suppose that a map into the linearly ordered set

$$\left(\text{RepVal}_R^{(a)}, <_R^{(a)} \right)$$

is given by

$$\text{Rep}_R^{(a)} : \text{Qu}_R^{\text{can}}[\mathbb{I}] \longrightarrow \text{RepVal}_R^{(a)}$$

and satisfies

$$\text{OrdRepCond} \left(\text{Rep}_R^{(a)} ; \asymp_R[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}], <_R^{(a)} \right)$$

Then there exists a unique strict-order isomorphism

$$\text{Replso}_R^{\text{ord},01} : \text{Im} \left(\text{Rep}_R^{(0)} \right) \longrightarrow \text{Im} \left(\text{Rep}_R^{(1)} \right)$$

such that

$$\text{Rep}_R^{(1)} = \text{Replso}_R^{\text{ord},01} \circ \text{Rep}_R^{(0)}$$

holds.

Further, fix

$$u \in U$$

For each

$$a \in \{0, 1\}$$

suppose that a map into the linearly ordered set

$$\left(\text{LocRepVal}_{R,u}^{(a)}, <_{R,u}^{\text{loc},(a)} \right)$$

is given by

$$\text{LocRep}_{R,u}^{(a)} : \mathcal{U}_{R,u}[\mathbb{I}] \longrightarrow \text{LocRepVal}_{R,u}^{(a)}$$

and satisfies

$$\text{OrdRepCond} \left(\text{LocRep}_{R,u}^{(a)} ; \asymp_{R,u}[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]|_{\mathcal{U}_{R,u}[\mathbb{I}]}, <_{R,u}^{\text{loc},(a)} \right)$$

Then there exists a unique strict-order isomorphism

$$\text{Replso}_{R,u}^{\text{loc},01} : \text{Im} \left(\text{LocRep}_{R,u}^{(0)} \right) \longrightarrow \text{Im} \left(\text{LocRep}_{R,u}^{(1)} \right)$$

such that

$$\text{LocRep}_{R,u}^{(1)} = \text{Replso}_{R,u}^{\text{loc},01} \circ \text{LocRep}_{R,u}^{(0)}$$

holds.

Proof. For the global representations, apply Lemma 4.182 to

$$\begin{aligned} & \left(S, \sim_S, \prec_S; \text{Rep}^{(a)}, \text{RepVal}^{(a)}, <^{(a)} \right) \\ &= \left(\text{Qu}_R^{\text{can}}[\mathbb{I}], \asymp_R[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]; \text{Rep}_R^{(a)}, \text{RepVal}_R^{(a)}, <_R^{(a)} \right) \end{aligned}$$

The assumed equality criterion and strict-order criterion yield the unique strict-order isomorphism $\text{Replso}_R^{\text{ord},01}$ and the global commutative equation.

For the local representations, apply the same lemma to

$$\begin{aligned} & \left(S, \sim_S, \prec_S; \text{Rep}^{(a)}, \text{RepVal}^{(a)}, <^{(a)} \right) \\ &= \left(\mathcal{U}_{R,u}[\mathbb{I}], \asymp_{R,u}[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]|_{\mathcal{U}_{R,u}[\mathbb{I}]}; \text{LocRep}_R^{(a)}, \text{LocRepVal}_R^{(a)}, <_R^{\text{loc},(a)} \right) \end{aligned}$$

The assumed local equality criterion and local strict-order criterion yield the unique strict-order isomorphism $\text{Replso}_R^{\text{loc},01}$ and the local commutative equation. $\square$

Definition 4.184 (Admissible comparison representation). Let a realized operational configuration $\mathcal{X}$ and a typed family of comparison rules $\mathcal{J}_{\text{cmp}}$ be given. Assume $\mathbf{L}_{\leq 2}[\text{Ev}^{\text{cmp}}]$. Let the family of comparison-representation target sets be

$$(\text{CmpTgt}_R)_{R \in \text{Rol}^{\text{cmp}}}$$

and let a relation on their product be

$$\text{Rel}^{\text{tgt}} \subseteq \prod_{R \in \text{Rol}^{\text{cmp}}} \text{CmpTgt}_R$$

For each $R \in \text{Rol}^{\text{cmp}}$, take a map

$$\Phi_R^{\text{cmp}} : \mathcal{R}_R^{\text{arr}} \longrightarrow \text{CmpTgt}_R$$

First, suppose that for every $R \in \text{Rol}^{\text{cmp}}$,

$$\text{EqKer}(\text{Prj}_R^{\text{can}}[\mathbb{I}]) \subseteq \text{EqKer}(\Phi_R^{\text{cmp}})$$

holds.

Second, suppose that for every

$$\begin{aligned} (u, \alpha) &\in \mathcal{R}^{\text{pair}}, \\ \alpha &= (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}} \end{aligned}$$

the equivalence

$$(\Phi_R^{\text{cmp}}(u, \alpha_R))_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{tgt}} \iff \alpha \in \text{Rel}_u^{\text{cmp}}$$

holds.

A pair

$$((\Phi_R^{\text{cmp}})_{R \in \text{Rol}^{\text{cmp}}}, \text{Rel}^{\text{tgt}})$$

satisfying these two conditions is called an admissible comparison representation.

Definition 4.185 (Admissible global order representation). Let a realized operational configuration $\mathcal{X}$ and a typed family of comparison rules $\mathcal{J}_{\text{cmp}}$ be given. Assume $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$. For each

$$R \in \text{Rol}^{\text{cmp}}$$

take a map into the linearly ordered set

$$\left(\text{OrdTgt}_R, <_R^{\text{ord,tgt}} \right)$$

given by

$$\Phi_R^{\text{ord}} : \text{Qu}_R^{\text{can}}[\mathbb{I}] \longrightarrow \text{OrdTgt}_R$$

When

$$\text{OrdRepCond} \left(\Phi_R^{\text{ord}} ; \asymp_R[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}], <_R^{\text{ord,tgt}} \right)$$

holds, Φ_R^{ord} is called an admissible global order representation.

Definition 4.186 (Admissible local order-atlas representation). Let a realized operational configuration $\mathcal{X}$ and a typed family of comparison rules $\mathcal{J}_{\text{cmp}}$ be given. Assume $\text{ReprCtx}_{\leq 1}[\mathbb{I}]$. For each

$$R \in \text{Rol}^{\text{cmp}}, \quad u \in U$$

take a map into the linearly ordered set

$$\left(\text{LocOrdTgt}_{R,u}, <_{R,u}^{\text{loc,tgt}} \right)$$

given by

$$\Phi_{R,u}^{\text{loc}} : \mathcal{U}_{R,u}[\mathbb{I}] \longrightarrow \text{LocOrdTgt}_R$$

Suppose that

$$\text{OrdRepCond} \left(\Phi_R^{\text{loc}} ; \asymp_{R,u}[\mathbb{I}], \prec_R^{\text{can}}[\mathbb{I}]|_{\mathcal{U}_{R,u}[\mathbb{I}]}, <_R^{\text{loc,tgt}} \right)$$

holds.

For each

$$u, v \in U$$

let the strict-order isomorphism between the overlap images be

$$\begin{aligned} \text{TgtChg}_{R,uv}^{\text{loc}} : \Phi_{R,u}^{\text{loc}}(\mathcal{U}_{R,uv}^{\cap}[\mathbb{I}]) \\ \longrightarrow \Phi_{R,v}^{\text{loc}}(\mathcal{U}_{R,uv}^{\cap}[\mathbb{I}]) \end{aligned}$$

Suppose that, for every

$$\text{CmpCls}_{R,0} \in \mathcal{U}_{R,uv}^{\cap}[\mathbb{I}]$$

the equality

$$\begin{aligned} \text{TgtChg}_{R,uv}^{\text{loc}}(\Phi_{R,u}^{\text{loc}}(\text{CmpCls}_{R,0})) \\ = \Phi_{R,v}^{\text{loc}}(\text{CmpCls}_{R,0}) \end{aligned}$$

holds.

Further, suppose that for every

$$u, v, w \in U$$

the equality

$$\mathsf{TgtChg}_{R,uu}^{\text{loc}} = \text{id}$$

holds on

$$\Phi_{R,u}^{\text{loc}}(\mathcal{U}_{R,u}[\mathbb{I}])$$

and that

$$\mathsf{TgtChg}_{R,vw}^{\text{loc}} \circ \mathsf{TgtChg}_{R,uv}^{\text{loc}} = \mathsf{TgtChg}_{R,uw}^{\text{loc}}$$

holds on

$$\Phi_{R,u}^{\text{loc}}(\mathcal{U}_{R,uvw}^{\cap}[\mathbb{I}])$$

The family of maps

$$\left((\Phi_{R,u}^{\text{loc}})_{(R,u) \in \text{Id}\mathbf{x}_1^{\text{cmp}}}, (\mathsf{TgtChg}_{R,uv}^{\text{loc}})_{(R,u,v) \in \text{Id}\mathbf{x}_2^{\text{cmp}}} \right)$$

is called an admissible local order-atlas representation.

Theorem 4.187 (Three-layer universal factorization theorem). Assume `ExactOperable` $[\mathbb{I}]$.

First, for an admissible comparison representation

$$((\Phi_R^{\text{cmp}})_{R \in \text{Rol}^{\text{cmp}}}, \text{Rel}^{\text{tgt}})$$

from Definition 4.184, there exists, for each $R \in \text{Rol}^{\text{cmp}}$, a unique map

$$\overline{\Phi}_R^{\text{cmp}} : \text{Qu}_R^{\text{can}}[\mathbb{I}] \longrightarrow \text{CmpTgt}_R$$

such that

$$\Phi_R^{\text{cmp}} = \overline{\Phi}_R^{\text{cmp}} \circ \text{Prj}_R^{\text{can}}[\mathbb{I}]$$

holds.

Moreover, for every

$$(u, \alpha) \in \mathcal{R}^{\text{pair}}, \\ \alpha = (\alpha_R)_{R \in \text{Rol}^{\text{cmp}}}$$

the equivalence

$$\begin{aligned} & \left(\overline{\Phi}_R^{\text{cmp}}(\text{Prj}_R^{\text{can}}[\mathbb{I}](u, \alpha_R)) \right)_{R \in \text{Rol}^{\text{cmp}}} \in \text{Rel}^{\text{tgt}} \\ & \iff \text{Prj}_{\mathfrak{E}\mathfrak{D}}^{\text{can,pair}}[\mathbb{I}](u, \alpha) \in \text{Rel}^{\text{can}}[\mathbb{I}] \end{aligned}$$

holds.

Second, for an admissible local order-atlas representation from Definition 4.186, there exists, for each

$$R \in \text{Rol}^{\text{cmp}}, \quad u \in U$$

a unique strict-order isomorphism

$$\overline{\Phi}_{R,u}^{\text{loc}} : \text{Qu}_R^{\text{loc}}[\mathbb{I}] \longrightarrow \text{Im}(\Phi_R^{\text{loc}})$$

such that

$$\Phi_R^{\text{loc}} = \overline{\Phi}_R^{\text{loc}} \circ \text{Prj}_R^{\text{loc}}[\mathbb{I}]$$

holds.

For every

$$u, v \in U$$

the following equality holds on the overlap image:

$$\begin{aligned} & \overline{\Phi}_{R,v}^{\text{loc}} \circ \text{ChartChg}_{R,uv}^{\text{loc}} [I] \\ &= \text{TgtChg}_{R,uv}^{\text{loc}} \circ \overline{\Phi}_{R,u}^{\text{loc}} \end{aligned}$$

Third, for an admissible global order representation from Definition 4.185, there exists, for each

$$R \in \text{Rol}^{\text{cmp}}$$

a unique strict-order isomorphism

$$\overline{\Phi}_R^{\text{ord}} : \text{Qu}_R^{\text{ord},\text{can}} [I] \longrightarrow \text{Im} (\Phi_R^{\text{ord}})$$

such that

$$\Phi_R^{\text{ord}} = \overline{\Phi}_R^{\text{ord}} \circ \text{Prj}_R^{\text{ord},\text{can}} [I]$$

holds.

Proof. Take an admissible comparison representation. For each $R \in \text{Rol}^{\text{cmp}}$, apply Lemma 4.180 to

$$\begin{aligned} & (S, Q, T; \pi, \Phi) \\ &= (\mathcal{R}_R^{\text{arr}}, \text{Qu}_R^{\text{can}} [I], \text{CmpTgt}_R; \text{Prj}_R^{\text{can}} [I], \Phi_R^{\text{cmp}}) \end{aligned}$$

The surjectivity of the canonical projection from Definition 4.70 and the equality-kernel inclusion condition in Definition 4.184 yield the unique comparison factor map and the comparison factorization equation.

Take any (u, α) quantified in the first claim. Substituting the comparison factorization equation into the relation-preservation condition of the admissible comparison representation makes the left-hand side of the theorem equivalent to $\alpha \in \text{Rel}_u^{\text{cmp}}$. By the local pullback agreement in Lemma 4.84, this is equivalent to the right-hand side of the theorem.

Take an admissible local order-atlas representation. Apply Lemma 4.182 to

$$\begin{aligned} & \left(S, \sim_S, \prec_S; \left(\text{Rep}^{(a)}, \text{RepVal}^{(a)}, <^{(a)} \right)_{a \in \{0,1\}} \right) \\ &= \left(\mathcal{U}_{R,u} [I], \asymp_{R,u} [I], \prec_R^{\text{can}} [I]|_{\mathcal{U}_{R,u} [I]}; \left(\text{Prj}_R^{\text{loc}} [I], \text{Qu}_R^{\text{loc}} [I], <_R^{\text{loc}} [I] \right), \left(\Phi_R^{\text{loc}}, \text{LocOrdTgt}_R, <_R^{\text{loc},\text{tgt}} \right) \right) \end{aligned}$$

The class criterion, order criterion, and surjectivity in Lemma 4.120, together with the order-representation criterion in Definition 4.186, yield the unique local strict-order isomorphism and factorization equation stated in the theorem.

Apply Lemma 4.181 to the local order-coordinate projection restricted to $\mathcal{U}_{R,uv}^\cap [I]$, and to the composite maps on the left- and right-hand sides of the local chart-transition commutative equation in the theorem. The restricted projection is surjective onto its image. By the local chart-transition equation in Definition 4.126, the two local factorization equations, and the overlap-compatibility equation in Definition 4.186, the two composite maps agree after precomposition with the same projection. Therefore, the local chart-transition commutative equation holds.

Take an admissible global order representation. Apply Lemma 4.182 to

$$\begin{aligned} & \left(S, \sim_S, \prec_S; \left(\text{Rep}^{(a)}, \text{RepVal}^{(a)}, <^{(a)} \right)_{a \in \{0,1\}} \right) \\ &= \left(\text{Qu}_R^{\text{can}} [I], \asymp_R [I], \prec_R^{\text{can}} [I]; \left(\text{Prj}_R^{\text{ord},\text{can}} [I], \text{Qu}_R^{\text{ord},\text{can}} [I], <_R^{\text{ord},\text{can}} [I] \right), \left(\Phi_R^{\text{ord}}, \text{OrdTgt}_R, <_R^{\text{ord},\text{tgt}} \right) \right) \end{aligned}$$

The class criterion, order criterion, and surjectivity in Lemma 4.136, together with the order-representation criterion in Definition 4.185, yield the unique global strict-order isomorphism and factorization equation stated in the theorem. $\square$

Corollary 4.188 (Uniqueness of reparameterization on representational images). Assume `ExactOperable` [I] and fix

$$R \in \mathbf{Rol}^{\text{cmp}}$$

For each

$$a \in \{0, 1\}$$

let

$$\Phi_R^{\text{ord},(a)} : \mathbf{Qu}_R^{\text{can}}[\mathbf{I}] \longrightarrow \mathbf{OrdTgt}_R^{(a)}$$

be an admissible global order representation. Then there exists a unique strict-order isomorphism

$$\mathbf{Replso}_R^{\text{ord},01} : \mathbf{Im} \left(\Phi_R^{\text{ord},(0)} \right) \longrightarrow \mathbf{Im} \left(\Phi_R^{\text{ord},(1)} \right)$$

such that

$$\Phi_R^{\text{ord},(1)} = \mathbf{Replso}_R^{\text{ord},01} \circ \Phi_R^{\text{ord},(0)}$$

holds.

Further, for each

$$a \in \{0, 1\}$$

let

$$\left(\left(\Phi_{R,u}^{\text{loc},(a)} \right)_{u \in U}, \left(\mathbf{TgtChg}_{R,uv}^{\text{loc},(a)} \right)_{(u,v) \in U \times U} \right)$$

be an admissible local order-atlas representation. Then, for each

$$u \in U$$

there exists a unique strict-order isomorphism

$$\mathbf{Replso}_R^{\text{loc},01} : \mathbf{Im} \left(\Phi_R^{\text{loc},(0)} \right) \longrightarrow \mathbf{Im} \left(\Phi_R^{\text{loc},(1)} \right)$$

such that

$$\Phi_R^{\text{loc},(1)} = \mathbf{Replso}_R^{\text{loc},01} \circ \Phi_R^{\text{loc},(0)}$$

holds.

For every

$$u, v \in U$$

the following equality holds on the overlap image:

$$\begin{aligned} & \mathbf{Replso}_{R,v}^{\text{loc},01} \circ \mathbf{TgtChg}_{R,uv}^{\text{loc},(0)} \\ &= \mathbf{TgtChg}_{R,uv}^{\text{loc},(1)} \circ \mathbf{Replso}_{R,u}^{\text{loc},01} \end{aligned}$$

Proof. For the global representations, apply Theorem 4.183 to

$$\mathbf{Rep}_R^{(a)} = \Phi_R^{\text{ord},(a)}$$

The equality criterion and strict-order criterion of an admissible global order representation satisfy the assumptions of that theorem. Hence, a unique strict-order isomorphism $\mathbf{Replso}_R^{\text{ord},01}$ exists and the global commutative equation holds.

For the local representations, at each

$$u \in U$$

apply the same theorem to

$$\text{LocRep}_R^{(a)} = \Phi_R^{\text{loc},(a)}$$

The equality criterion and strict-order criterion of an admissible local order-atlas representation satisfy the local assumptions of that theorem. Hence, a unique strict-order isomorphism $\text{Replso}_R^{\text{loc},01}$ exists and the local commutative equation holds.

For commutativity with the chart-transition maps, apply Lemma 4.181 to the restriction to $\mathcal{U}_{R,uv}^\cap[\mathbb{I}]$ of $\Phi_{R,u}^{\text{loc},(0)}$, and to the composite maps on the left- and right-hand sides of the chart-transition commutative equation in the corollary. The restricted map is surjective onto its image. By the overlap-compatibility equations of the two admissible local order-atlas representations and the two local commutative equations, the two composite maps agree after precomposition with the same map. Therefore, the chart-transition commutative equation holds. Uniqueness of the strict-order isomorphism at each phase also makes the reparameterization family satisfying this commutative equation unique. $\square$

Corollary 4.189 (Comparison-stage conditions imply operability). If

$$\mathbf{L}_{\leq 7} [\mathbf{Ev}^{\text{cmp}}] \wedge \mathbf{G} [\mathbf{Ev}^{\text{cmp}}]$$

holds, then $\text{Operative}(\mathcal{X})$ holds.

Proof. The conclusion follows from the $\Leftarrow$ direction of the equivalence in Theorem 4.177. $\square$

Corollary 4.190 (Dependence of the exact-operability object sequence on generative inputs). When $\text{ExactOperable}[\mathbb{I}]$ holds, every recursive component of $\text{EOSeq}[\mathbb{I}]$ is generated in dependency order from

$$\mathcal{I}^{\text{raw}}, \quad \text{Dat}^{\text{src}}, \quad \mathcal{J}_{\text{cmp}}$$

Proof. By Definition 3.25,

$$\mathcal{X} = (\mathcal{I}^{\text{raw}}, \text{Dat}^{\text{src}}).$$

By Definition 4.175, satisfaction of $\text{ExactOperable}[\mathbb{I}]$ gives satisfaction of $\text{Operative}(\mathcal{X})$ and $\text{Repr}_{\leq 3}[\mathbb{I}]$, and types every recursive component of $\text{EOSeq}[\mathbb{I}]$.

The comparison-stage material datum

$$\text{Dat}^{\text{mat}}[\mathbb{I}]$$

is determined from

$$\mathcal{X}, \quad \mathcal{J}_{\text{cmp}}$$

The canonical comparison quotient derived from $\mathfrak{C}\mathfrak{D}$,

$$\text{Qu}^{\text{can}}[\mathbb{I}]$$

is determined from the comparison-stage material datum, the comparison-stage seven-condition satisfaction bundle, and the global gluing condition.

Definitions 4.95 through 4.103 determine, in dependency order, the source-induced arrival preorders, the canonical $\mathfrak{C}/\mathfrak{D}$ partial orders, and the exact operational time structures.

Definitions 4.104 through 4.128 determine, in dependency order, the local order-coordinate structures, the local order-coordinate chart-transition maps, and the comparison-order atlas.

Definitions 4.130 through 4.142 determine, in dependency order, the global order-coordinate structures and the local-to-global order-coordinate embeddings.

Each generative input to the comparison-stage material datum, the canonical comparison quotient derived from $\mathfrak{C}\mathfrak{D}$, the source-induced arrival preorders, the canonical $\mathfrak{C}/\mathfrak{D}$ partial orders, the exact operational time structures, the local order-coordinate structures, the comparison-order atlas, the global order-coordinate structures, and the local-to-global order-coordinate embeddings is either

$$\mathcal{X}, \quad \mathcal{J}_{\text{cmp}}$$

or a previously generated recursive component. Substituting the construction equation for $\mathcal{X}$ from Definition 3.25 shows that every recursive component of $\text{EOSeq}[\mathbb{I}]$ is generated in dependency order from the three inputs in Equation (4.190). $\square$

Corollary 4.191 (Factorization of a conventional time parameter through the canonical global order coordinate). Assume **ExactOperable** $[\mathbb{I}]$ and fix

$$R \in \text{Rol}^{\text{cmp}}$$

Suppose that a map into the linearly ordered set

$$(\text{TimeVal}_R, <_R^{\text{time}})$$

given by

$$t_R : \text{Qu}_R^{\text{can}}[\mathbb{I}] \longrightarrow \text{TimeVal}_R$$

satisfies

$$\text{OrdRepCond} (t_R ; \asymp_R [\mathbb{I}], \prec_R^{\text{can}} [\mathbb{I}], <_R^{\text{time}})$$

Then there exists a unique strict-order isomorphism

$$\bar{t}_R : \text{Qu}_R^{\text{ord,can}}[\mathbb{I}] \longrightarrow \text{Im} (t_R)$$

such that

$$t_R = \bar{t}_R \circ \text{Prj}_R^{\text{ord,can}}[\mathbb{I}]$$

holds.

Proof. By Definition 4.108, the order-representation criterion for t_R is the condition for an admissible global order representation in Definition 4.185.

Apply the global order factorization in Theorem 4.187 to

$$(\Phi_R^{\text{ord}}, \text{OrdTgt}_R, <_R^{\text{ord,tgt}}) = (t_R, \text{TimeVal}_R, <_R^{\text{time}})$$

This yields the unique strict-order isomorphism and factorization equation stated in the corollary.

Therefore, a conventional time parameter satisfying the equality criterion and the strict-order criterion is determined as the composition of the canonical global order-coordinate projection with a strict-order isomorphism into a linearly ordered image. $\square$

5 Generation of Observer Time at the Dynamical-Description Stage

Take as input the exact operability

$$\text{ExactOperable}[\mathbb{I}]$$

of Definition 4.175. For an observer-side information-coupling position and an object-specification index, the dynamical-description stage specifies the conditions that generate the $\mathfrak{C}$ -derived observer-time quotient, the $\mathfrak{B}$ -derived object-state quotient, the $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation, graph trajectories, and the family of canonical observable functions. In dependency order, these conditions provide the construction conditions for dynamical-description operability, law-description operability, differential-stage operability, and multi-object relations.

5.1 Generation of Dynamical Descriptions from the Canonical Comparison Quotient

Definition 5.1 (Dynamical-description quotient-role family). Within the previously defined source-role family Rol^{src} , define the family of roles that carry quotients at the dynamical-description stage by

$$\text{Rol}^{\text{dyn,quot}} := \{\mathfrak{C}, \mathfrak{B}\}$$

Throughout this chapter, the compound superscript (dyn, σ) on an object specific to the dynamical-description stage is abbreviated to σ only when the symbol root and σ uniquely determine that the object belongs to the dynamical-description stage; a lone (dyn) is omitted. To distinguish it from the source-side relation Rel^{src} , the complete form $\text{Rel}^{\text{dyn,src}}$ is retained.

Definition 5.2 (Complete evaluation specification). For the comparison-rule-dependent common input $\mathbb{I}$ of Definition 4.1, an observer-side information-coupling position $o \in U$, and an object-specification index $\zeta \in Z_0$ from Definition 2.10, define the complete-evaluation-specification set by

$$\text{EvSpec} := \{\mathbb{I}\} \times U \times Z_0$$

Define a complete evaluation specification by

$$\text{Ev} := (\mathbb{I}, o, \zeta) \in \text{EvSpec}$$

Define the exactly operable complete-evaluation-specification set by

$$\text{EvSpec}^{\text{exact}} := \{(\mathbb{I}, o, \zeta) \in \text{EvSpec} \mid \text{ExactOperable}[\mathbb{I}]\}$$

Definition 5.3 (Common evaluation rule for dynamical-description conditions). The eight dynamical-description conditions below use the following common rule. For each condition, consider the subset of $\text{EvSpec}^{\text{exact}}$ on which all predecessor conditions explicitly assumed in its definition hold. Denote this subset by D , and let P be the test stated in that definition on this domain. Define the condition K , for every $\text{Ev} \in \text{EvSpec}$, by

$$K(\text{Ev}) \iff \exists e \in D, \quad \text{Ev} = e \wedge P(e).$$

The test formula in each individual definition is its expression on D ; outside D , K is false. This rule makes the conditions total without extending the domains on which their quotients or maps are constructed.

Proposition 5.4 (Foundational structure for dynamical description). For a complete evaluation specification $\mathbf{Ev} \in \mathbf{EvSpec}^{\text{exact}}$, the family of source-token sets

$$(\mathbf{Tok}_R^*)_{R \in \mathbf{Rol}^{\text{src}}}$$

is determined, and the exact-operability object sequence

$$\mathbf{EOSeq}[\mathbf{I}]$$

is generated.

Consequently, the canonical comparison quotient derived from $\mathfrak{CD}$, $\mathbf{Qu}^{\text{cmp}, \text{can}}[\mathbf{I}]$, the exact operational time structure $\mathcal{E}^{\text{op}}[\mathbf{I}]$, the local order-coordinate structures

$$(\mathbf{Qu}_{R,u}^{\text{loc}}[\mathbf{I}], <_{R,u}^{\text{loc}}[\mathbf{I}], \mathbf{Prj}_{R,u}^{\text{loc}}[\mathbf{I}])_{(R,u) \in \mathbf{Rol}^{\text{cmp}} \times U}$$

and the global order-coordinate structures

$$(\mathbf{Qu}_R^{\text{ord}, \text{can}}[\mathbf{I}], <_R^{\text{ord}, \text{can}}[\mathbf{I}], \mathbf{Prj}_R^{\text{ord}, \text{can}}[\mathbf{I}])_{R \in \mathbf{Rol}^{\text{cmp}}}$$

are determined.

Proof. By Definition 3.25, $\mathcal{X} = (\mathcal{I}^{\text{raw}}, \mathbf{Dat}^{\text{src}})$. By Definition 3.24, $(\mathbf{Tok}_R^*)_{R \in \mathbf{Rol}^{\text{src}}}$ is a component of $\mathbf{Dat}^{\text{src}}$.

By Definition 4.175, under $\mathbf{ExactOperable}[\mathbf{I}]$, $\mathbf{EOSeq}[\mathbf{I}]$ is generated. The structures listed in the proposition are recursive components of $\mathbf{EOSeq}[\mathbf{I}]$. $\square$

Definition 5.5 (Direct arrival maps at the dynamical-description stage). For a complete evaluation specification $\mathbf{Ev} = (\mathbf{I}, o, \zeta)$, define the direct arrival map at the dynamical-description stage for each $R \in \mathbf{Rol}^{\text{src}}$ by

$$\mathbf{Arr}_{R,o} : \text{Dom}(\mathbf{Arr}_{R,o}) \subseteq \mathbf{Tok}_R^* \longrightarrow \mathbf{Rcv}_{R,o}$$

Define the corresponding arrival set by

$$\mathcal{R}_{R,o}^{\text{arr}} := \text{Im}(\mathbf{Arr}_{R,o}) \quad (R \in \mathbf{Rol}^{\text{src}})$$

Definition 5.6 (Source-role relation at the dynamical-description stage). For a complete evaluation specification $\mathbf{Ev} = (\mathbf{I}, o, \zeta)$, define the source-role relation at the dynamical-description stage by

$$\mathbf{Rel}_o \subseteq \prod_{R \in \mathbf{Rol}^{\text{src}}} \mathcal{R}_{R,o}^{\text{arr}}$$

Definition 5.7 (Dynamical-description source-role relation). Define the dynamical-description source-role relation by

$$\mathbf{Rel}^{\text{dyn}, \text{src}} \subseteq \prod_{R \in \mathbf{Rol}^{\text{src}}} \mathbf{Tok}_R^*$$

Definition 5.8 (Exact four-party descent). Define exact four-party descent at an observer-side information-coupling position $o \in U$ by satisfaction of

$$\mathbf{Rel}_o = \left\{ \left(\mathbf{Arr}_{R,o}(x_R) \right)_{R \in \mathbf{Rol}^{\text{src}}} \mid (x_R)_{R \in \mathbf{Rol}^{\text{src}}} \in \mathbf{Rel}^{\text{dyn}, \text{src}}, \right. \\ \left. \forall R \in \mathbf{Rol}^{\text{src}}, \quad x_R \in \text{Dom}(\mathbf{Arr}_{R,o}) \right\}$$

Definition 5.9 (Four-party reception structure at the dynamical-description stage). Define the four-party reception structure at the dynamical-description stage at an observer-side information-coupling position $o \in U$ by

$$\text{RcvStr}_o := \left((\mathcal{R}_{R,o}^{\text{arr}})_{R \in \text{Rol}^{\text{src}}}, (\text{Arr}_{R,o})_{R \in \text{Rol}^{\text{src}}}, \text{Rel}_o, \text{Rel}^{\text{dyn,src}} \right)$$

Definition 5.10 (Recovery maps at the dynamical-description stage). For an observer-side information-coupling position $o \in U$ and each $R \in \text{Rol}^{\text{src}}$, define the recovery map at the dynamical-description stage by

$$\text{Rec}_{R,o} : \mathcal{R}_{R,o}^{\text{arr}} \longrightarrow \text{Tok}_R^*$$

For every $a_R \in \mathcal{R}_{R,o}^{\text{arr}}$,

$$\begin{aligned} \text{Rec}_{R,o}(a_R) &\in \text{Dom}(\text{Arr}_{R,o}), \\ \text{Arr}_{R,o}(\text{Rec}_{R,o}(a_R)) &= a_R \end{aligned}$$

holds.

Definition 5.11 (Four-party binding relation at the dynamical-description stage). Define the four-party binding relation at the dynamical-description stage at an observer-side information-coupling position $o \in U$ by

$$\text{Bind}_o \subseteq (\text{Rel}_o)^2$$

Definition 5.12 (Source-side four-party binding relation). For an observer-side information-coupling position $o \in U$, define the source-side four-party binding relation

$$\text{Bind}_o^{\text{src}} \subseteq (\text{Rel}^{\text{dyn,src}})^2$$

For every $\mathbf{x}_i \in \text{Rel}^{\text{dyn,src}}$, ($i \in \{0, 1\}$),

$$\begin{aligned} &\mathbf{x}_0 \text{ Bind}_o^{\text{src}} \mathbf{x}_1 \\ &\iff \left[\exists \mathbf{a}_0, \mathbf{a}_1 \in \text{Rel}_o, \quad (\mathbf{a}_0, \mathbf{a}_1) \in \text{Bind}_o, \quad \forall (i, R) \in \{0, 1\} \times \text{Rol}^{\text{src}}, \right. \\ &\quad \left. \text{Rec}_{R,o}((\mathbf{a}_i)_R) = (\mathbf{x}_i)_R \right]. \end{aligned}$$

Definition 5.13 (Generated relations and least equivalence closures). For each $R \in \text{Rol}^{\text{quot}}$, define the generated relation

$$\rightsquigarrow_{R,o} \subseteq (\text{Tok}_R^*)^2$$

by, for every $x_{R,0}, x_{R,1} \in \text{Tok}_R^*$,

$$\begin{aligned} &x_{R,0} \rightsquigarrow_{R,o} x_{R,1} \\ &\iff \left[\exists \mathbf{x}_0, \mathbf{x}_1 \in \text{Rel}^{\text{dyn,src}}, \quad \forall i \in \{0, 1\}, \quad (\mathbf{x}_i)_R = x_{R,i}, \quad \mathbf{x}_0 \text{ Bind}_o^{\text{src}} \mathbf{x}_1 \right] \end{aligned}$$

For each $R \in \text{Rol}^{\text{quot}}$, let $\sim_{R,o}$ be the least equivalence relation that contains $\rightsquigarrow_{R,o}$ and is defined on Tok_R^* . That is,

$$\rightsquigarrow_{R,o} \subseteq \sim_{R,o},$$

and, for every equivalence relation $E_R \subseteq (\text{Tok}_R^*)^2$,

$$\rightsquigarrow_{R,o} \subseteq E_R \implies \sim_{R,o} \subseteq E_R$$

holds.

Definition 5.14 (Base datum for the dynamical-description extension). For a complete evaluation specification $\text{Ev} \in \text{EvSpec}^{\text{exact}}$, define the base datum for the dynamical-description extension by

$$\text{Dat}_{o,\zeta}^0 := \left(\text{RcvStr}_o, (\text{Rec}_{R,o})_{R \in \text{Rol}^{\text{src}}}, \right. \\ \left. \text{Bind}_o, \text{Bind}_o^{\text{src}}, (\rightsquigarrow_{R,o}, \sim_{R,o})_{R \in \text{Rol}^{\text{quot}}} \right)$$

Definition 5.15 (Specified local-observable datum). For a complete evaluation specification $\text{Ev} = (\text{I}, o, \zeta)$, let $\text{ObsIdx}^{\text{loc}}$ be the local-observable index set, let $\text{ObsIdx}_{o,\zeta}^{\text{quot}}$ be the quotient-stage observable index set, and let the local-index correspondence map be

$$\text{ObsIdxMap}_{o,\zeta} : \text{ObsIdx}^{\text{loc}} \longrightarrow \text{ObsIdx}^{\text{quot}}$$

For each $u, v \in U$ and $\ell \in \text{ObsIdx}^{\text{loc}}$, let the local-observable value range, local-observable map, local-observable-value transition partial map, common comparison value range, and common comparison-value map, respectively, be

$$\begin{aligned} & \text{ObsVal}_{u,\ell}^{\text{loc}}, \\ & \text{Obs}_{u,\ell}^{\text{loc}} : \mathcal{R}_{\mathfrak{B},u}^{\text{arr}} \longrightarrow \text{ObsVal}_{u,\ell}^{\text{loc}}, \\ & \text{ObsValChg}_{uv,\ell}^{\text{loc}} : \text{ObsVal}_{u,\ell}^{\text{loc}} \rightharpoonup \text{ObsVal}_{v,\ell}^{\text{loc}}, \\ & \text{ObsVal}_{\ell}^{\text{cmp}}, \\ & \text{ObsValMap}_{u,\ell}^{\text{cmp}} : \text{ObsVal}_{u,\ell}^{\text{loc}} \longrightarrow \text{ObsVal}_{\ell}^{\text{cmp}} \end{aligned}$$

For each $\ell \in \text{ObsIdx}^{\text{loc}}$, define the local-observable family by

$$\text{ObsFam}_{\ell}^{\text{loc}} := \left(\text{ObsVal}_{\ell}^{\text{cmp}}, \right. \\ \left. (\text{ObsVal}_{u,\ell}^{\text{loc}}, \text{Obs}_{u,\ell}^{\text{loc}}, \text{ObsValMap}_{u,\ell}^{\text{cmp}})_{u \in U}, \right. \\ \left. (\text{ObsValChg}_{uv,\ell}^{\text{loc}})_{(u,v) \in U^2} \right)$$

Define the specified local-observable datum by

$$\text{Dat}^{\text{obs,loc}} := \left(\text{ObsIdx}^{\text{loc}}, \text{ObsIdx}^{\text{quot}}, \right. \\ \left. \text{ObsIdxMap}, (\text{ObsFam}_{\ell}^{\text{loc}})_{\ell \in \text{ObsIdx}^{\text{loc}}} \right)$$

5.2 Construction of the Observer-Time Quotient

Take a complete evaluation specification

$$\text{Ev} \in \text{EvSpec}^{\text{exact}}$$

from Definition 5.2.

The base datum for the dynamical-description extension

$$\text{Dat}^0,$$

from Definition 5.14 and the specified local-observable datum

$$\text{Dat}^{\text{obs,loc}}$$

from Definition 5.15 are determined.

From the $\mathfrak{C}/\mathfrak{B}$ -derived generated equivalence relations

$$\sim_{\mathfrak{C},o}, \quad \sim_{\mathfrak{B},o},$$

of Definition 5.13, the $\mathfrak{C}$ -derived source-family direct-sum order

$$\preceq_{*}^{\mathfrak{C}},$$

of Definition 3.17, and the source-side $\mathfrak{C}\mathfrak{O}\mathfrak{B}$ -derived relation

$$\text{Rel}^{\text{dyn,src}}$$

of Definition 5.7, the $\mathfrak{C}$ -derived observer-time quotient, the $\mathfrak{B}$ -derived object-state quotient, the canonical order on the $\mathfrak{C}$ -derived observer-time quotient, the $\mathfrak{C}\mathfrak{O}\mathfrak{B}$ -derived canonical trajectory relation, and the family of canonical observable functions are generated in sequence under the stage-specific establishment conditions defined below.

Definition 5.16 (The $\mathfrak{C}$ -derived observer-time quotient and the $\mathfrak{B}$ -derived object-state quotient). For a complete evaluation specification

$$\text{Ev} \in \text{EvSpec}^{\text{exact}}$$

define, for each

$$R \in \text{Rol}^{\text{quot}} = \{\mathfrak{C}, \mathfrak{B}\}$$

the canonical dynamical-description quotient and natural projection by

$$\text{Qu}_{R,o}^{\text{can}} := \text{Tok}_R^* / \sim_{R,o},$$

$$\Theta_{R,o}^{\text{can}} : \text{Tok}_R^* \twoheadrightarrow \text{Qu}_{R,o}^{\text{can}},$$

$$\Theta_{R,o}^{\text{can}}(x_R) := [x_R]_{\sim_{R,o}}$$

The component with $R = \mathfrak{C}$ is called the $\mathfrak{C}$ -derived observer-time quotient, and the component with $R = \mathfrak{B}$ is called the $\mathfrak{B}$ -derived object-state quotient.

Proposition 5.17 (Minimality of the generated closure). For each

$$R \in \text{Rol}^{\text{quot}}$$

the relation $\sim_{R,o}$ is the least equivalence relation that contains $\rightsquigarrow_{R,o}$ and is defined on Tok_R^* .

Proof. The conclusion follows from the inclusion condition and minimality condition in Definition 5.13. $\square$

Definition 5.18 (Generated time preorder and its antisymmetry condition). Fix a complete evaluation specification $\text{Ev} \in \text{EvSpec}^{\text{exact}}$. Define the image of the source order in the canonical time quotient and the preorder it generates by

$$\begin{aligned} \text{TimeOrdEdge}_o &:= \left\{ (\Theta_{\mathfrak{C},o}^{\text{can}}(x), \Theta_{\mathfrak{C},o}^{\text{can}}(y)) \mid \right. \\ &\quad \left. x, y \in \text{Tok}_{\mathfrak{C}}^*, \quad x \preceq_*^{\mathfrak{C}} y \right\}, \\ \preceq_o^{\text{time,gen}} &:= (\text{TimeOrdEdge}_o)^*. \end{aligned}$$

Here the superscript $*$ on a relation denotes its reflexive transitive closure, defined by finite paths, including paths of length zero. Define the antisymmetry condition for the generated time preorder by

$$\begin{aligned} K_{\text{GenTimeOrdAntiSym}}(\text{Ev}) &\iff \left[\forall T_0, T_1 \in \text{Qu}_{\mathfrak{C},o}^{\text{can}}, \right. \\ &\quad \left. (T_0 \preceq_o^{\text{time,gen}} T_1 \wedge T_1 \preceq_o^{\text{time,gen}} T_0) \implies T_0 = T_1 \right]. \end{aligned}$$

Definition 5.19 (Canonical observer-time order). Assume $K_{\text{GenTimeOrdAntiSym}}(\text{Ev})$. Define the canonical observer-time order by

$$\preceq_o^{\text{time,can}} := \preceq_o^{\text{time,gen}} \subseteq (\text{Qu}_{\mathfrak{C},o}^{\text{can}})^2.$$

Throughout this subsection, the time-type superscript is omitted from the symbol for the canonical observer-time order.

Proposition 5.20 (Partial-order and generation properties of the canonical observer-time order). Under

$$\text{ExactOperable}[\mathbb{I}], \quad K_{\text{GenTimeOrdAntiSym}}(\text{Ev}),$$

the relation $\preceq_o^{\text{can}}$ is a partial order on $\text{Qu}_{\mathfrak{C},o}^{\text{can}}$. The natural projection $\Theta_{\mathfrak{C},o}^{\text{can}}$ preserves the source order. Moreover, $\preceq_o^{\text{can}}$ is the least preorder on the canonical time quotient for which this natural projection preserves the source order.

Proof. Paths of length zero give reflexivity of the generated preorder, and concatenation of finite paths gives transitivity. Antisymmetry is the condition $K_{\text{GenTimeOrdAntiSym}}(\text{Ev})$. The canonical order is therefore a partial order.

If $x \preceq_*^{\mathfrak{C}} y$, the pair of natural-projection images belongs to TimeOrdEdge_o . The resulting path of length one shows that the natural projection preserves order.

Take any preorder on the canonical time quotient for which the natural projection preserves the source order. Its relation contains TimeOrdEdge_o . By reflexivity and transitivity, it also contains the reflexive transitive closure of that relation. This proves the stated minimality. $\square$

Definition 5.21 ($\mathfrak{C}/\mathfrak{B}$ source trajectory relation). Define the $\mathfrak{C}/\mathfrak{B}$ source trajectory relation by

$$\begin{aligned} \text{TrjRel}^{\text{src}} &:= \left\{ (x_{\mathfrak{C}}, x_{\mathfrak{B}}) \in \text{Tok}_{\mathfrak{C}}^* \times \text{Tok}_{\mathfrak{B}}^* \mid \right. \\ &\quad \left. \exists x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*, \quad (x_{\mathfrak{C}}, x_{\mathfrak{D}}, x_{\mathfrak{B}}) \in \text{Rel}^{\text{dyn,src}} \right\} \end{aligned}$$

Definition 5.22 (Canonical trajectory-relation descent condition at the dynamical-description stage). For a complete evaluation specification $\text{Ev} \in \text{EvSpec}^{\text{exact}}$, define representative invariance of the $\mathfrak{CDB}$ -derived trajectory relation by

$$K_{\text{CanTrjDesc}}(\text{Ev}) \iff \left[\forall (x_R, x'_R)_{R \in \text{Rol}^{\text{quot}}} \in \prod_{R \in \text{Rol}^{\text{quot}}} (\text{Tok}_R^*)^2, \right. \\ \left. \left(\bigwedge_{R \in \text{Rol}^{\text{quot}}} x_R \sim_{R,o} x'_R \right) \implies \right. \\ \left. \left((x_{\mathfrak{C}}, x_{\mathfrak{B}}) \in \text{TrjRel}^{\text{src}} \right. \right. \\ \left. \left. \iff (x'_{\mathfrak{C}}, x'_{\mathfrak{B}}) \in \text{TrjRel}^{\text{src}} \right) \right]$$

Definition 5.23 (Canonical trajectory relation derived from $\mathfrak{CDB}$). Assume

$$K_{\text{CanTrjDesc}}(\text{Ev})$$

Define the canonical trajectory relation derived from $\mathfrak{CDB}$ by

$$\text{TrjRel}_o^{\text{can}} := (\Theta_{\mathfrak{C},o}^{\text{can}} \times \Theta_{\mathfrak{B},o}^{\text{can}}) [\text{TrjRel}^{\text{src}}]$$

This relation satisfies

$$\text{TrjRel}_o^{\text{can}} \subseteq \text{Qu}_{\mathfrak{C},o}^{\text{can}} \\ \times \text{Qu}_{\mathfrak{B},o}^{\text{can}}$$

Lemma 5.24 (Descent of the $\mathfrak{CDB}$ -derived trajectory relation along the generated closure). Suppose that $K_{\text{CanTrjDesc}}(\text{Ev})$ holds. For each $R \in \text{Rol}^{\text{quot}}$, if $x_R \sim_{R,o} x'_R$, then

$$(\Theta_{\mathfrak{C},o}^{\text{can}}(x_{\mathfrak{C}}), \Theta_{\mathfrak{B},o}^{\text{can}}(x_{\mathfrak{B}})) \in \text{TrjRel}_o^{\text{can}} \\ \iff (\Theta_{\mathfrak{C},o}^{\text{can}}(x'_{\mathfrak{C}}), \Theta_{\mathfrak{B},o}^{\text{can}}(x'_{\mathfrak{B}})) \in \text{TrjRel}_o^{\text{can}}.$$

Proof. This follows from the equivalence of the source-trajectory conditions supplied by Definition 5.22 and the membership condition for the canonical trajectory relation in Definition 5.23. $\square$

Definition 5.25 (Canonical trajectory domain). Assume $K_{\text{CanTrjDesc}}(\text{Ev})$. Define the canonical trajectory domain by

$$\text{TrjDom}_o^{\text{can}} := \text{pr}_{\mathfrak{C}}(\text{TrjRel}_o^{\text{can}}) \\ = \left\{ t \in \text{Qu}_{\mathfrak{C},o}^{\text{can}} \mid \exists s \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}, (t, s) \in \text{TrjRel}_o^{\text{can}} \right\}.$$

Definition 5.26 (Canonical trajectory-function condition at the dynamical-description stage). Assume $K_{\text{CanTrjDesc}}(\text{Ev})$. Define the single-valuedness condition for the canonical trajectory relation derived from $\mathfrak{CDB}$ by

$$K_{\text{CanTrjFunc}}(\text{Ev}) \iff \left[\forall t \in \text{TrjDom}_o^{\text{can}}, \forall s_0, s_1 \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}, \right. \\ \left. \left(\bigwedge_{k=0}^1 (t, s_k) \in \text{TrjRel}_o^{\text{can}} \right) \implies s_0 = s_1 \right]$$

Proposition 5.27 (Existence and uniqueness of the graph trajectory). When

$$K_{\text{CanTrjDesc}}(\text{Ev}), \quad K_{\text{CanTrjFunc}}(\text{Ev})$$

hold, there exists a unique map

$$\text{TrjMap}_o^{\text{can}} : \text{TrjDom}_o^{\text{can}} \longrightarrow \text{Qu}_{\mathfrak{B},o}^{\text{can}}$$

satisfying

$$\text{TrjRel}_o^{\text{can}} = \{(t, \text{TrjMap}_o^{\text{can}}(t)) \mid t \in \text{TrjDom}_o^{\text{can}}\}$$

Proof. For any $t \in \text{TrjDom}_o^{\text{can}}$, Definition 5.25 gives

$$(t, s) \in \text{TrjRel}_o^{\text{can}}$$

for some $s \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}$. Definition 5.26 makes this s unique, so one may define

$$\text{TrjMap}_o^{\text{can}}(t) := s$$

By this definition, for each t , $(t, \text{TrjMap}_o^{\text{can}}(t))$ belongs to the trajectory relation. Conversely, if $(t, s) \in \text{TrjRel}_o^{\text{can}}$, then $t \in \text{TrjDom}_o^{\text{can}}$, and single-valuedness gives $s = \text{TrjMap}_o^{\text{can}}(t)$. Therefore, the stated graph equality holds.

At each t , any map satisfying the same graph equality has the same unique s as its value, so the map is unique. $\square$

Definition 5.28 (Graph trajectory). The unique map

$$\begin{aligned} &\text{TrjMap}_o^{\text{can}} : \\ &\text{TrjDom}_o^{\text{can}} \longrightarrow \\ &\text{Qu}_{\mathfrak{B},o}^{\text{can}} \end{aligned}$$

of Proposition 5.27 is called the graph trajectory.

Take as input the specified local-observable datum $\text{Dat}^{\text{obs,loc}}$ from Definition 5.15.

Definition 5.29 (Common-comparison-value composites of local observables). For each $u, v \in U$ and $\ell \in \text{ObsIdx}^{\text{loc}}$, define the common-comparison-value map and the post-transition common-comparison-value partial map of the local observable by

$$\text{ObsCmp}_{u,\ell} := \text{ObsValMap}_{u,\ell}^{\text{cmp}} \circ \text{Obs}_{u,\ell}^{\text{loc}}$$

$$\text{ObsCmpChg}_{uv,\ell} := \text{ObsValMap}_{v,\ell}^{\text{cmp}} \circ \text{ObsValChg}_{uv,\ell}^{\text{loc}} \circ \text{Obs}_{u,\ell}^{\text{loc}}$$

Definition 5.30 ($\mathfrak{B}$ -derived local-reception canonical-state maps). For each $u \in U$, define the $\mathfrak{B}$ -derived local-reception canonical-state map by

$$\begin{aligned} \Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}} &:= \Theta_{\mathfrak{B},o}^{\text{can}} \circ \text{Rec}_{\mathfrak{B},u} \\ &: \mathcal{R}_{\mathfrak{B},u}^{\text{arr}} \longrightarrow \text{Qu}_{\mathfrak{B},o}^{\text{can}} \end{aligned}$$

Definition 5.31 (Descent condition for the canonical observable-function family at the dynamical-description stage). For a complete evaluation specification $\text{Ev} \in \text{EvSpec}^{\text{exact}}$, define $K_{\text{CanObsDesc}}(\text{Ev})$ as the conjunction of the following four conditions.

First, ObsIdxMap is a surjection onto $\text{ObsIdx}^{\text{quot}}$. Within this definition, write this map as π_{obs} .

Second, for any $\ell_0, \ell_1 \in \text{ObsIdx}^{\text{loc}}$,

$$\pi_{\text{obs}}(\ell_0) = \pi_{\text{obs}}(\ell_1) \implies \text{ObsVal}_{\ell_0}^{\text{cmp}} = \text{ObsVal}_{\ell_1}^{\text{cmp}}.$$

Third, for any

$$q \in \text{ObsIdx}^{\text{quot}}, \quad s \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}$$

there exist $u \in U$, $\ell \in \text{ObsIdx}^{\text{loc}}$, and $a \in \mathcal{R}_{\mathfrak{B},u}^{\text{arr}}$ such that

$$\begin{aligned} \pi_{\text{obs}}(\ell) &= q, \\ \Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}}(a) &= s. \end{aligned}$$

Fourth, for $k \in \{0, 1\}$, let

$$u_k \in U, \quad \ell_k \in \text{ObsIdx}^{\text{loc}}, \quad a_k \in \mathcal{R}_{\mathfrak{B},u_k}^{\text{arr}}$$

If

$$\begin{aligned} \pi_{\text{obs}}(\ell_0) &= \pi_{\text{obs}}(\ell_1), \\ \Theta_{\mathfrak{B},o \leftarrow u_0}^{\text{can}}(a_0) &= \Theta_{\mathfrak{B},o \leftarrow u_1}^{\text{can}}(a_1) \end{aligned}$$

then

$$\text{ObsCmp}_{u_0,\ell_0}(a_0) = \text{ObsCmp}_{u_1,\ell_1}(a_1).$$

Definition 5.32 (Canonical observable-function family). Assume $K_{\text{CanObsDesc}}(\text{Ev})$. For each $q \in \text{ObsIdx}^{\text{quot}}$, take a solution of $\text{ObsIdxMap}(\ell) = q$ with $\ell \in \text{ObsIdx}^{\text{loc}}$, and define

$$\text{ObsVal}_{o,\zeta,q}^{\text{can}} := \text{ObsVal}_{\ell}^{\text{cmp}}$$

For each $s \in \text{Qu}_{\mathfrak{B},o}^{\text{can}}$, take a solution of

$$\Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}}(a) = s$$

with $u \in U$ and $a \in \mathcal{R}_{\mathfrak{B},u}^{\text{arr}}$, and define

$$\text{Obs}_{o,\zeta,q}^{\text{can}}(s) := \text{ObsCmp}_{u,\ell}(a)$$

Define the canonical observable-function family by

$$\mathcal{O}_{o,\zeta}^{\text{can}} := \left(\text{ObsIdx}^{\text{quot}}, \left(\text{ObsVal}_q^{\text{can}}, \text{Obs}_q^{\text{can}} \right)_{q \in \text{ObsIdx}^{\text{quot}}} \right)$$

Proposition 5.33 (Representative independence of the canonical observable-function family). The canonical observable value ranges and canonical observable functions do not depend on the choice of local-observable index or local $\mathfrak{B}$ -derived reception.

Proof. The first condition in Definition 5.31 gives the existence of a local-observable index; the second gives representative independence of the value range; and the third gives the existence of a local $\mathfrak{B}$ -derived reception. For the same quotient-stage observable index and the same $\mathfrak{B}$ -derived object-state quotient class, the fourth condition gives, for any two representing triples (u_0, ℓ_0, a_0) and (u_1, ℓ_1, a_1) ,

$$\text{ObsCmp}_{u_0,\ell_0}(a_0) = \text{ObsCmp}_{u_1,\ell_1}(a_1)$$

Therefore, the value ranges and function values in Definition 5.32 do not depend on any of these choices. $\square$

Definition 5.34 (Common-comparison-value commutativity condition for local-observable value changes at the dynamical-description stage). For a complete evaluation specification $\text{Ev} \in \text{EvSpec}^{\text{exact}}$, define $K_{\text{LocObsChgComm}}(\text{Ev})$ by requiring, for every

$$\begin{aligned} u, v &\in U, \\ \ell &\in \text{ObsIdx}^{\text{loc}}, \\ a &\in \mathcal{R}_{\mathfrak{B},u}^{\text{arr}} \end{aligned}$$

that

$$\text{Obs}_{u,\ell}^{\text{loc}}(a) \in \text{Dom}(\text{ObsValChg}_{uv,\ell}^{\text{loc}}) \implies \text{ObsCmpChg}_{uv,\ell}(a) = \text{ObsCmp}_{u,\ell}(a)$$

holds.

Proposition 5.35 (Compatibility of the canonical observable-function family with local-observable value changes). Suppose that

$$K_{\text{CanObsDesc}}(\text{Ev}), \quad K_{\text{LocObsChgComm}}(\text{Ev})$$

hold. For any $u, v \in U$, $\ell \in \text{ObsIdx}^{\text{loc}}$, and $a \in \mathcal{R}_{\mathfrak{B},u}^{\text{arr}}$, if $\text{Obs}_{u,\ell}^{\text{loc}}(a)$ belongs to the domain of $\text{ObsValChg}_{uv,\ell}^{\text{loc}}$, then

$$\text{Obs}_{\text{ObsIdxMap}(\ell)}^{\text{can}}(\Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}}(a)) = \text{ObsCmpChg}_{uv,\ell}(a).$$

Proof. By Definition 5.32,

$$\text{Obs}_{\text{ObsIdxMap}(\ell)}^{\text{can}}(\Theta_{\mathfrak{B},o \leftarrow u}^{\text{can}}(a)) = \text{ObsCmp}_{u,\ell}(a).$$

By Definition 5.34, the right-hand side equals $\text{ObsCmpChg}_{uv,\ell}(a)$. $\square$

Definition 5.36 ($\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived dynamical-description quotient). Assume

$$K_{\text{GenTimeOrdAntiSym}}(\text{Ev}), \quad K_{\text{CanTrjDesc}}(\text{Ev})$$

Define the $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived dynamical-description quotient by

$$\text{Qu}_{o,\zeta}^{\text{dyn,can}} := \left((\text{Qu}_{R,o}^{\text{can}}, \Theta_{R,o}^{\text{can}})_{R \in \text{Rel}^{\text{quot}}}, \preceq_o^{\text{can}}, \text{TrjRel}_o^{\text{can}} \right).$$

When, in addition,

$$K_{\text{CanTrjFunc}}(\text{Ev})$$

holds, the graph trajectory of Definition 5.28 satisfies

$$\text{TrjRel}_o^{\text{dyn,can}} = \left\{ \begin{aligned} &(\text{DynCls}_{\mathfrak{C},o,0}, \text{TrjMap}_o^{\text{dyn,can}}(\text{DynCls}_{\mathfrak{C},o,0})) \\ &\left| \text{DynCls}_{\mathfrak{C},o,0} \in \text{TrjDom}_o^{\text{dyn,can}} \right. \end{aligned} \right\}.$$

Definition 5.37 ($\mathfrak{D}$ -derived record code). Define the $\mathfrak{D}$ -derived record-code map, for each $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$, by

$$\text{RecordCode}_{\mathfrak{D}}^{\text{dyn}}(x_{\mathfrak{D}}) := (W_{\mathfrak{D}}(x_{\mathfrak{D}}), \phi_{\mathfrak{D}}(x_{\mathfrak{D}}), \text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}})).$$

Define the $\mathfrak{D}$ -derived record-code value range and the type of the map by

$$\begin{aligned} \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}} &:= \text{RecordCode}_{\mathfrak{D}}[\text{Tok}_{\mathfrak{D}}^*], \\ \text{RecordCode}_{\mathfrak{D}} &: \text{Tok}_{\mathfrak{D}}^* \longrightarrow \text{RecordCodeVal}_{\mathfrak{D}} \end{aligned}$$

Definition 5.38 (Typed $\mathfrak{D}$ -record-code relations). Assume

$$K_{\text{CanTrjDesc}}(\mathbf{Ev})$$

Define the $\mathfrak{C}/\mathfrak{B}$ -correspondence-typed $\mathfrak{D}$ -record-code relation by

$$\text{RecordCodeRel}_o^{\text{dyn,pair}} := \left\{ \left(\left(\Theta_{\mathfrak{C},o}^{\text{can}}(x_{\mathfrak{C}}), \Theta_{\mathfrak{B},o}^{\text{can}}(x_{\mathfrak{B}}) \right), \text{RecordCode}_{\mathfrak{D}}^{\text{dyn}}(x_{\mathfrak{D}}) \right) \mid \right. \\ \left. (x_{\mathfrak{C}}, x_{\mathfrak{D}}, x_{\mathfrak{B}}) \in \text{Rel}^{\text{dyn,src}} \right\}.$$

This relation satisfies

$$\text{RecordCodeRel}_o^{\text{pair}} \subseteq \text{TrjRel}_o^{\text{can}} \times \text{RecordCodeVal}_{\mathfrak{D}}$$

For each $R \in \text{Rol}^{\text{quot}}$, define the typed $\mathfrak{D}$ -record-code relation by

$$\text{RecordCodeRel}_{R,o}^{\text{dyn}} := (\text{pr}_R \times \text{id}) [\text{RecordCodeRel}_o^{\text{pair}}] \\ \subseteq \text{Qu}_{R,o}^{\text{can}} \times \text{RecordCodeVal}_{\mathfrak{D}}.$$

Definition 5.39 ($\mathfrak{D}$ -derived record-code datum). Define the record-code datum by

$$\text{RecordDat}_o^{\text{dyn}} := \left(\text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}}, \text{RecordCode}_{\mathfrak{D}}^{\text{dyn}}, \right. \\ \left. \left(\text{RecordCodeRel}_{R,o}^{\text{dyn}} \right)_{R \in \text{Rol}^{\text{dyn,quot}}}, \text{RecordCodeRel}_o^{\text{dyn,pair}} \right).$$

Proposition 5.40 (Projection compatibility of the typed $\mathfrak{D}$ -record-code relations). For each

$$R \in \text{Rol}^{\text{quot}}$$

the equality

$$\text{RecordCodeRel}_{R,o} = (\text{pr}_R \times \text{id}) [\text{RecordCodeRel}_o^{\text{pair}}]$$

holds.

Proof. Apply Definition 5.38 to each $R \in \text{Rol}^{\text{quot}}$. □

Definition 5.41 (Dynamical-description admissible quotients). Write the dynamical-description admissible quotients as

$$\text{AdmQuPrs}_{o,\zeta,i}^{\text{dyn}} := \left(\left(\text{Qu}_{R,o,i}^{\text{dyn,adm}}, \Theta_{R,o,i}^{\text{dyn,adm}} \right)_{R \in \text{Rol}^{\text{quot}}}, \preceq_{o,i}^{\text{dyn,time,adm}}, \text{TrjRel}_{o,i}^{\text{dyn,adm}} \right)$$

and define them by the following conditions.

(DQ1) For each $R \in \text{Rol}^{\text{quot}}$,

$$\Theta_{R,o,i}^{\text{adm}} : \text{Tok}_R^* \rightarrow \text{Qu}_{R,o,i}^{\text{adm}}$$

is surjective.

(DQ2) For each $R \in \text{Rol}^{\text{quot}}$ and all $x_R, x'_R \in \text{Tok}_R^*$,

$$x_R \sim_{R,o}^{\text{dyn}} x'_R \implies \Theta_{R,o,i}^{\text{adm}}(x_R) = \Theta_{R,o,i}^{\text{adm}}(x'_R)$$

holds.

(DQ3) $\preceq_{o,i}^{\text{time,adm}}$ is a partial order on $\text{Qu}_{\mathfrak{C},o,i}^{\text{adm}}$, and, for all $x_{\mathfrak{C}}, y_{\mathfrak{C}} \in \text{Tok}_{\mathfrak{C}}^*$,

$$x_{\mathfrak{C}} \preceq_*^{\mathfrak{C}} y_{\mathfrak{C}} \implies \Theta_{\mathfrak{C},o,i}^{\text{adm}}(x_{\mathfrak{C}}) \preceq_{o,i}^{\text{time,adm}} \Theta_{\mathfrak{C},o,i}^{\text{adm}}(y_{\mathfrak{C}})$$

holds.

(DQ4)

$$\text{TrjRel}_{o,i}^{\text{adm}} = (\Theta_{\mathfrak{C},o,i}^{\text{adm}} \times \Theta_{\mathfrak{B},o,i}^{\text{adm}}) [\text{TrjRel}^{\text{dyn,src}}].$$

Definition 5.42 (Refinement relation for dynamical-description admissible quotients).
For two dynamical-description admissible quotients

$$\text{AdmQuPrs}_i, \quad \text{AdmQuPrs}_j$$

define

$$\text{AdmQuPrs}_i \sqsubseteq_{\text{adm}} \text{AdmQuPrs}_j$$

by the existence, for each $R \in \text{Rol}^{\text{quot}}$, of a surjection

$$\text{RefMap}_{R,o;j \rightarrow i} : \text{Qu}_{R,o,j}^{\text{adm}} \twoheadrightarrow \text{Qu}_{R,o,i}^{\text{adm}}$$

satisfying

$$\Theta_{R,o,i}^{\text{adm}} = \text{RefMap}_{R,o;j \rightarrow i} \circ \Theta_{R,o,j}^{\text{adm}} \quad (R \in \text{Rol}^{\text{quot}}),$$

$$\text{TrjRel}_{o,i}^{\text{adm}} = (\text{RefMap}_{\mathfrak{C},o;j \rightarrow i} \times \text{RefMap}_{\mathfrak{B},o;j \rightarrow i}) [\text{TrjRel}_{o,j}^{\text{adm}}]$$

and by requiring $\text{RefMap}_{\mathfrak{C},o;j \rightarrow i}$ to be order-preserving from $\preceq_{o,j}^{\text{time,adm}}$ to $\preceq_{o,i}^{\text{time,adm}}$.

When there exists, between the two presentations, a family of bijections that commutes with the natural projections and the trajectory relations and whose $\mathfrak{C}$ -component is an order isomorphism between the observer-time partial orders,

$$(\text{RefMap}_{R,o;j \rightarrow i})_{R \in \text{Rol}^{\text{quot}}}$$

define

$$\text{AdmQuPrs}_i \cong_{\text{adm}} \text{AdmQuPrs}_j$$

Proposition 5.43 (Preorder property of the refinement relation). The relation $\sqsubseteq_{\text{adm}}$ is a preorder on presentations of dynamical-description admissible quotients. Two mutually refining presentations are isomorphic under $\cong_{\text{adm}}$, and $\sqsubseteq_{\text{adm}}$ induces a partial order on the isomorphism classes.

Proof. The family of identity maps satisfies every condition in Definition 5.42; hence reflexivity holds.

Suppose that

$$\text{AdmQuPrs}_0 \sqsubseteq_{\text{adm}} \text{AdmQuPrs}_1 \sqsubseteq_{\text{adm}} \text{AdmQuPrs}_2$$

For each $R \in \text{Rol}^{\text{quot}}$, define

$$\text{RefMap}_{R,o;2 \rightarrow 0} := \text{RefMap}_{R,o;1 \rightarrow 0} \circ \text{RefMap}_{R,o;2 \rightarrow 1}$$

Surjectivity, factorization through the natural projection, equality of the images of the trajectory relations, and order preservation by the $\mathfrak{C}$ -component are all preserved under composition; hence transitivity holds.

Suppose that two presentations mutually refine one another. For each $R \in \mathbf{Rol}^{\text{quot}}$, composing the two factorization identities shows that each of the two composites of the oppositely directed refinement maps agrees with the corresponding natural projection after precomposition by that projection. Since the natural projections are surjective, Lemma 4.181 implies that both composites are identity maps. The two families of maps are therefore mutually inverse families of bijections preserving the order and the trajectory relation. Thus mutual refinement gives an isomorphism, and antisymmetry holds on the isomorphism classes. $\square$

Theorem 5.44 (Maximal admissibility of the $\mathfrak{CDB}$ -derived dynamical-description quotient). Assume

$$\begin{aligned} &\text{ExactOperable}[\mathbb{I}], \\ &K_{\text{GenTimeOrdAntiSym}}(\text{Ev}), \\ &K_{\text{CanTrjDesc}}(\text{Ev}) \end{aligned}$$

The $\mathfrak{CDB}$ -derived dynamical-description quotient $\mathbf{Qu}^{\text{can}}$ is a dynamical-description admissible quotient.

For every dynamical-description admissible quotient $\mathbf{AdmQuPrs}_i$, there exists a unique family of surjections

$$\text{FactorMap}_{R,o,i} : \mathbf{Qu}_{R,o}^{\text{can}} \twoheadrightarrow \mathbf{Qu}_{R,o,i}^{\text{adm}} \quad (R \in \mathbf{Rol}^{\text{quot}})$$

such that

$$\begin{aligned} \Theta_{R,o,i}^{\text{adm}} &= \text{FactorMap}_{R,o,i} \circ \Theta_{R,o}^{\text{can}} \quad (R \in \mathbf{Rol}^{\text{quot}}), \\ \text{TrjRel}_{o,i}^{\text{adm}} &= (\text{FactorMap}_{\mathfrak{C},o,i} \times \text{FactorMap}_{\mathfrak{B},o,i}) [\text{TrjRel}_o^{\text{can}}] \end{aligned}$$

hold. Moreover, $\text{FactorMap}_{\mathfrak{C},o,i}$ is order-preserving from the canonical observer-time order to the admissible time order.

Therefore,

$$[\mathbf{Qu}^{\text{can}}]_{\cong_{\text{adm}}}$$

is the greatest element in the refinement order on the isomorphism classes under $\cong_{\text{adm}}$.

Proof. By Definition 5.16, for each $R \in \mathbf{Rol}^{\text{quot}}$, the map $\Theta_{R,o}^{\text{can}}$ is surjective and takes generated equivalence classes as its values, so it satisfies (DQ1) and (DQ2). Proposition 5.20 and Definition 5.19 give (DQ3), while Definition 5.23 gives (DQ4). Hence $\mathbf{Qu}^{\text{can}}$ is a dynamical-description admissible quotient.

For each $R \in \mathbf{Rol}^{\text{quot}}$ and $\text{DynCls}_{R,o,0} \in \mathbf{Qu}_{R,o}^{\text{can}}$, choose a representative through

$$\text{DynCls}_{R,o,0} = \Theta_{R,o}^{\text{can}}(x_R)$$

with $x_R \in \mathbf{Tok}_R^*$, and define

$$\text{FactorMap}_{R,o,i}(\text{DynCls}_{R,o,0}) := \Theta_{R,o,i}^{\text{adm}}(x_R)$$

Replacing the representative gives $x_R \sim_{R,o} x'_R$, and (DQ2) leaves the right-hand side unchanged. The family of maps is therefore well-defined. By (DQ1), every map is surjective, and the natural-projection factorization identity follows from the definition.

Take any edge in TimeOrdEdge_o . It has source representatives satisfying $x \preceq_*^{\mathfrak{C}} y$. By (DQ3), their images under the admissible natural projection are ordered in the same direction. The natural-projection factorization identity therefore shows that $\text{FactorMap}_{\mathfrak{C},o,i}$

preserves each generating edge. Reflexivity and transitivity of the admissible time order extend this preservation to all finite paths, including paths of length zero. By Definition 5.19, this finite-path relation is the canonical observer-time order. Thus the $\mathfrak{C}$ -component preserves order. Equality of the images of the trajectory relations follows from Definition 5.23 and (DQ4).

Finally, take any family of maps satisfying the natural-projection factorization identity. For each $R \in \text{Rol}^{\text{quot}}$, if

$$\text{DynCls}_{R,o,0} = \Theta_{R,o}^{\text{can}}(x_R)$$

then the factorization identity makes the value of that map equal to $\Theta_{R,o,i}^{\text{adm}}(x_R)$. Because the canonical natural projection is surjective, the family of maps is unique. $\square$

Definition 5.45 (Canonical time-and-trajectory condition bundle). Define the canonical time-and-trajectory condition bundle by

$$\begin{aligned} \text{DynCanTimeTrjCondAll}[\text{Ev}] \iff & K_{\text{GenTimeOrdAntiSym}}(\text{Ev}) \\ & \wedge K_{\text{CanTrjDesc}}(\text{Ev}) \\ & \wedge K_{\text{CanTrjFunc}}(\text{Ev}). \end{aligned}$$

Definition 5.46 (Observer-time law-input condition bundle).

Define the observer-time law-input condition bundle by

$$\begin{aligned} \text{DynLawInputCondAll}[\text{Ev}] \iff & \text{DynCanTimeTrjCondAll}[\text{Ev}] \\ & \wedge K_{\text{CanObsDesc}}(\text{Ev}) \\ & \wedge K_{\text{LocObsChgComm}}(\text{Ev}). \end{aligned}$$

Definition 5.47 (Observer-time law-input datum). Assume

$$\text{ExactOperable}[\text{I}], \quad \text{DynLawInputCondAll}[\text{Ev}]$$

Define the observer-time law-input datum by

$$\text{LawInput}_{o,\zeta}^{\text{dyn}} := (\text{Qu}^{\text{can}}, \text{TrjDom}_o^{\text{can}}, \text{TrjMap}_o^{\text{can}}, \mathcal{O}^{\text{can}}, \text{RecordDat}_o).$$

The type of each component is given, respectively, by Definition 5.36, Definition 5.25, Definition 5.28, Definition 5.32, and Definition 5.39.

Proposition 5.48 (Existence and uniqueness of the observer-time law-input datum). Under

$$\text{ExactOperable}[\text{I}], \quad \text{DynLawInputCondAll}[\text{Ev}]$$

LawInput exists uniquely.

Proof. Definition 5.36, Definition 5.25, Proposition 5.27, Definition 5.32, and Definition 5.39 uniquely determine every component and the component order in Definition 5.47. $\square$

5.3 Observer-Time Law Core and Dynamical-Description Operability

Fix a complete evaluation specification $\text{Ev} = (\text{I}, o, \zeta)$ and assume

$$\text{ExactOperable}[\text{I}], \quad K_{\text{GenTimeOrdAntiSym}}(\text{Ev})$$

Use the observer-time quotient and canonical order from Definition 5.19:

$$\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}, \quad \preceq_o^{\text{time,can}}$$

The constructions involving trajectories and observables below carry their respective existence conditions.

From here through the end of Chapter 6, each time object is first introduced in full form. Its time-type superscript is then omitted only when that type is uniquely determined by the symbol root and the remaining modifiers.

By Proposition 5.20,

$$\preceq_o^{\text{can}}$$

is a partial order on

$$\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$$

Definition 5.49 (Observer-time strict order). For any $T_0, T_1 \in \mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$, define the observer-time strict order by

$$T_0 \prec_o^{\text{time,can}} T_1 \iff (T_0 \preceq_o^{\text{can}} T_1) \wedge (T_0 \neq T_1)$$

Let the resulting strict-order datum be

$$\mathbf{P}_o^{\text{dyn,time}} := (\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}, \prec_o^{\text{can}})$$

Lemma 5.50 (Strict-partial-order property of the observer-time strict order). The relation $\prec_o^{\text{can}}$ is a strict partial order on $\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$.

Proof. Apply Proposition 5.20 and Lemma 4.105 to

$$S = \mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}, \quad \preceq = \preceq_o^{\text{can}}$$

□

Definition 5.51 (Observer-time incomparability relation). Define the observer-time incomparability relation by

$$\asymp_o^{\text{dyn,time,can}} := \text{Inc}[\mathbf{P}_o] \subseteq (\mathbf{Qu}_{\mathfrak{E},o}^{\text{can}})^2$$

Consequently, for any $T_0, T_1 \in \mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$,

$$T_0 \asymp_o^{\text{can}} T_1 \iff \neg (T_0 \prec_o^{\text{can}} T_1) \wedge \neg (T_1 \prec_o^{\text{can}} T_0).$$

Proposition 5.52 (Reflexivity and symmetry of the observer-time incomparability relation). The relation $\asymp_o^{\text{can}}$ is reflexive and symmetric.

Proof. By the irreflexivity in Lemma 5.50 and Definition 5.51, for every

$$T \in \mathbf{Qu}_{\mathfrak{E},o}^{\text{can}}$$

one has $T \asymp_o^{\text{can}} T$. The two negated components in the same definition interchange when the arguments are exchanged, so the relation is symmetric. □

Definition 5.53 (Observer-time incomparability transitivity condition). Assume $K_{\text{GenTimeOrdAntiSym}}(\text{Ev})$. Define the observer-time incomparability transitivity condition by

$$K_{\text{ObsTimeIncTrans}}^{\text{dyn}}(\text{Ev}) \iff \text{IncTrans}[\mathbf{P}_o]$$

Lemma 5.54 (Equivalence-relation property of the observer-time incomparability relation). Under $K_{\text{ObsTimeIncTrans}}(\text{Ev})$, $\succsim_o^{\text{can}}$ is an equivalence relation.

Proof. Apply Definition 5.53 and Lemma 4.109 to P_o . $\square$

Lemma 5.55 (Invariance of the strict order along observer-time incomparability classes). Under $K_{\text{ObsTimeIncTrans}}(\text{Ev})$, for any $T_0, T'_0, T_1, T'_1 \in \text{Qu}_{\mathfrak{E},o}^{\text{can}}$,

$$\begin{aligned} (T_0 \succsim_o^{\text{can}} T'_0) \wedge (T_1 \succsim_o^{\text{can}} T'_1) \\ \implies (T_0 \prec_o^{\text{can}} T_1 \iff T'_0 \prec_o^{\text{can}} T'_1). \end{aligned}$$

Proof. Apply the strict-order invariance of Lemma 4.109 to P_o . $\square$

Definition 5.56 (Observer-time order-coordinate quotient). Assume $K_{\text{ObsTimeIncTrans}}(\text{Ev})$. Define the observer-time order-coordinate quotient by

$$\begin{aligned} (\text{Qu}_o^{\text{dyn,time,ord}}, \text{Prj}_o^{\text{dyn,time,ord}}, \prec_o^{\text{dyn,time,ord}}) \\ := \text{LinQuot}[P_o] \end{aligned}$$

Consequently,

$$\begin{aligned} \text{Qu}_o^{\text{ord}} &= \text{Qu}_{\mathfrak{E},o}^{\text{can}} / \succsim_o^{\text{can}}, \\ \text{Prj}_o^{\text{ord}} : \text{Qu}_{\mathfrak{E},o}^{\text{can}} &\rightarrow \text{Qu}_o^{\text{ord}}, \quad \text{Prj}_o^{\text{ord}}(T) = [T]_{\succsim_o^{\text{can}}}, \\ \text{Prj}_o^{\text{ord}}(T_0) &\prec_o^{\text{ord}} \text{Prj}_o^{\text{ord}}(T_1) \iff T_0 \prec_o^{\text{can}} T_1. \end{aligned}$$

Lemma 5.57 (Strict-linear-order property of the observer-time order-coordinate quotient). Under $K_{\text{ObsTimeIncTrans}}(\text{Ev})$, $\prec_o^{\text{ord}}$ is a strict linear order on Qu_o^{ord} .

Proof. Apply Lemma 4.109 to P_o . $\square$

Proposition 5.58 (Surjectivity, class determination, and order determination of the canonical observer-time order-coordinate projection). Under $K_{\text{ObsTimeIncTrans}}(\text{Ev})$, $\text{Prj}_o^{\text{ord}}$ is surjective, and

$$\text{OrdRepCond}(\text{Prj}_o^{\text{ord}}; \succsim_o^{\text{can}}, \prec_o^{\text{can}}, \prec_o^{\text{ord}})$$

holds.

Proof. Apply Definition 5.56 and Lemma 4.109 to P_o , and use Definition 4.108. $\square$

Definition 5.59 (Real-valued time-order embedding condition).

Assume $K_{\text{ObsTimeIncTrans}}(\text{Ev})$. Let the set of all order embeddings from the observer-time order-coordinate quotient into the real numbers be

$$\text{OrdEmbSet}_o^{\text{time}}(\mathbb{R}) := \left\{ \iota_o \left| \begin{array}{l} \iota_o : \text{Qu}_o^{\text{ord}} \rightarrow \mathbb{R}, \\ \forall \bar{T}_0, \bar{T}_1, \\ \bar{T}_0 \prec_o^{\text{ord}} \bar{T}_1 \iff \iota_o(\bar{T}_0) < \iota_o(\bar{T}_1) \end{array} \right. \right\}.$$

Define the real-valued time-order embedding condition by

$$K_{\text{ObsTimeRealOrdEmb}}(\text{Ev}) \iff \text{OrdEmbSet}_o^{\text{time}}(\mathbb{R}) \neq \emptyset$$

Under this condition, let

$$\text{RepEmb}_o^{\text{time}} \in \text{OrdEmbSet}_o^{\text{time}}(\mathbb{R})$$

be a real-valued time-order embedding.

Proposition 5.60 (Injectivity of a real-valued time-order embedding). Under

$$K_{\text{ObsTimeIncTrans}}(\mathbf{Ev}), \quad K_{\text{ObsTimeRealOrdEmb}}(\mathbf{Ev})$$

every $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$ is injective.

Proof. If $\bar{T}_0 \neq \bar{T}_1$, Lemma 5.57 gives $\bar{T}_0 <_o^{\text{ord}} \bar{T}_1$ or its converse. Definition 5.59 gives the corresponding strict inequality between their images, so the images are unequal. $\square$

Definition 5.61 (Inverse on the image of a real-valued time-order embedding). Under Proposition 5.60, for any $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$, define its inverse on its image by

$$\begin{aligned} \rho_o^{-1} : \text{Im}(\rho_o) &\longrightarrow \text{Qu}_o^{\text{ord}}, \\ \rho_o^{-1}(t) = \bar{T} &\iff \rho_o(\bar{T}) = t \end{aligned}$$

Here $t \in \text{Im}(\rho_o)$ and $\bar{T} \in \text{Qu}_o^{\text{ord}}$.

Proposition 5.62 (Composition of a real-valued time-order embedding with its inverse on the image). Assume $K_{\text{ObsTimeIncTrans}}(\mathbf{Ev})$. For every

$$\begin{aligned} \rho_o &\in \text{OrdEmbSet}_o(\mathbb{R}), \\ \bar{T} &\in \text{Qu}_o^{\text{ord}}, \\ t &\in \text{Im}(\rho_o) \end{aligned}$$

one has

$$\rho_o^{-1}(\rho_o(\bar{T})) = \bar{T}, \quad \rho_o(\rho_o^{-1}(t)) = t.$$

Proof. This follows from Definition 5.61 and Proposition 5.60. $\square$

Definition 5.63 (Law-admissible time representation). Assume $K_{\text{ObsTimeIncTrans}}(\mathbf{Ev})$. For any $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$, define the law-admissible time representation by

$$\text{TimeRep}_o[\rho_o] := \rho_o \circ \text{Prj}_o^{\text{ord}} : \text{Qu}_{\mathfrak{E},o}^{\text{can}} \longrightarrow \mathbb{R}$$

When $K_{\text{ObsTimeRealOrdEmb}}(\mathbf{Ev})$ holds, let the time representation corresponding to RepEmb_o be

$$\text{TimeRep}_o := \text{TimeRep}_o[\text{RepEmb}_o]$$

Proposition 5.64 (Class determination and order determination by a law-admissible time representation). Assume $K_{\text{ObsTimeIncTrans}}(\mathbf{Ev})$. For every $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$,

$$\text{OrdRepCond}(\text{TimeRep}_o[\rho_o]; \prec_o^{\text{can}}, \prec_o^{\text{can}}, <)$$

holds.

Proof. Proposition 5.60, Definition 5.59, and Proposition 5.58 give the two components of Definition 4.108. $\square$

Definition 5.65 (Scalar-time trajectory factorization condition).

Assume $\text{DynCanTimeTrjCondAll}[\mathbf{Ev}]$ and $K_{\text{ObsTimeIncTrans}}(\mathbf{Ev})$. Define the projected trajectory-time domain by

$$\text{TrjDom}_o^{\text{time,ord}} := \text{Prj}_o^{\text{ord}}(\text{TrjDom}_o^{\text{can}}) \subseteq \text{Qu}_o^{\text{ord}}$$

Let the restricted projection be

$$\pi_o := \text{Prj}_o^{\text{ord}}|_{\text{TrjDom}_o^{\text{can}}}$$

Define the scalar-time trajectory factorization condition by

$$K_{\text{ObsTimeTrjFact}}(\mathbf{Ev}) \iff \text{EqKer}(\pi_o) \subseteq \text{EqKer}(\text{TrjMap}_o^{\text{can}}).$$

Proposition 5.66 (Existence and uniqueness of the scalar-time trajectory). Assume $\text{DynCanTimeTrjCondAll}[\text{Ev}]$ and $K_{\text{ObsTimeIncTrans}}(\text{Ev})$. Under $K_{\text{ObsTimeTrjFact}}(\text{Ev})$, there exists a unique map

$$\text{TrjMap}_o^{\text{time,ord}} : \text{TrjDom}_o^{\text{ord}} \longrightarrow \text{Qu}_{\mathfrak{B},o}^{\text{can}}$$

satisfying

$$\text{TrjMap}_o^{\text{can}} = \text{TrjMap}_o^{\text{ord}} \circ \pi_o$$

Proof. The map π_o is surjective onto the image in Definition 5.65. Using the equality-kernel inclusion in that definition, apply Lemma 4.180 to

$$\pi = \pi_o, \quad \Phi = \text{TrjMap}_o^{\text{can}}$$

□

Definition 5.67 (Trajectory-factorization condition bundle and all-dynamical-description-conditions bundle).

$$\begin{aligned} \text{DynTrjFactCondAll}[\text{Ev}] &\iff \text{DynCanTimeTrjCondAll}[\text{Ev}] \\ &\quad \wedge K_{\text{ObsTimeIncTrans}}(\text{Ev}) \\ &\quad \wedge K_{\text{ObsTimeTrjFact}}(\text{Ev}), \\ \text{DynCondAll}[\text{Ev}] &\iff \text{DynLawInputCondAll}[\text{Ev}] \\ &\quad \wedge K_{\text{ObsTimeIncTrans}}(\text{Ev}) \\ &\quad \wedge K_{\text{ObsTimeRealOrdEmb}}(\text{Ev}) \\ &\quad \wedge K_{\text{ObsTimeTrjFact}}(\text{Ev}). \end{aligned}$$

Definition 5.68 (Canonical observable time curves). Assume $\text{DynTrjFactCondAll}[\text{Ev}]$ and $K_{\text{CanObsDesc}}(\text{Ev})$. For any $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$, define the real-valued trajectory-time domain by

$$\begin{aligned} \text{TrjDom}_o^{\text{time,real}}[\rho_o] &:= \rho_o[\text{TrjDom}_o^{\text{ord}}] \\ &= \text{TimeRep}_o[\rho_o][\text{TrjDom}_o^{\text{can}}] \subseteq \mathbb{R}. \end{aligned}$$

For each $\ell \in \text{ObsIdx}^{\text{quot}}$, define

$$\begin{aligned} \text{ObsCurve}_{o,\zeta,\ell}^{\text{time,ord}} &:= \text{Obs}_\ell^{\text{can}} \circ \text{TrjMap}_o^{\text{ord}}, \\ \text{ObsCurve}_{o,\zeta,\ell}^{\text{time,real}}[\rho_o] &:= \text{ObsCurve}_{o,\zeta,\ell}^{\text{ord}} \circ \rho_o^{-1} \big|_{\text{TrjDom}_o^{\text{real}}[\rho_o]}. \end{aligned}$$

When $K_{\text{ObsTimeRealOrdEmb}}(\text{Ev})$ holds, let the real trajectory-time domain and the real-valued observable curve corresponding to RepEmb_o be

$$\begin{aligned} \text{TrjDom}_o^{\text{real}} &:= \text{TrjDom}_o^{\text{real}}[\text{RepEmb}_o], \\ \text{ObsCurve}_\ell^{\text{real}} &:= \text{ObsCurve}_\ell^{\text{real}}[\text{RepEmb}_o]. \end{aligned}$$

Proposition 5.69 (Canonical-time factorization of trajectories and observables). Assume $\text{DynTrjFactCondAll}[\text{Ev}]$ and $K_{\text{CanObsDesc}}(\text{Ev})$. For every $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$, $T \in \text{TrjDom}_o^{\text{can}}$, and $\ell \in \text{ObsIdx}^{\text{quot}}$,

$$\begin{aligned} \text{TrjMap}_o^{\text{can}}(T) &= \text{TrjMap}_o^{\text{ord}}(\text{Prj}_o^{\text{ord}}(T)), \\ \text{Obs}_\ell^{\text{can}}(\text{TrjMap}_o^{\text{can}}(T)) &= \text{ObsCurve}_\ell^{\text{ord}}(\text{Prj}_o^{\text{ord}}(T)) \\ &= \text{ObsCurve}_\ell^{\text{real}}[\rho_o](\text{TimeRep}_o[\rho_o](T)). \end{aligned}$$

Moreover, the real-valued time curve is the unique map on $\text{TrjDom}_o^{\text{real}}[\rho_o]$ satisfying this factorization identity.

Proof. The first identity follows from Proposition 5.66, the second from Definition 5.68, and the third from Proposition 5.62. The real-valued time domain is the image of the order-time domain under ρ_o , and injectivity gives each element a unique preimage. Since the factorization identity determines the value at that preimage, the real-valued time curve is unique. $\square$

Definition 5.70 (Dynamical-description operability). Define dynamical-description operability by

$$\text{DynOperative}[\text{Ev}] \iff \text{ExactOperable}[\text{I}] \wedge \text{DynCondAll}[\text{Ev}].$$

Definition 5.71 (Dynamical-description object). Assume $\text{DynOperative}[\text{Ev}]$, and let $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$. Define the dynamical-description time structure and the trajectory-and-observation structure relative to this embedding by

$$\begin{aligned} \text{DynTimeStr}_o[\rho_o] &:= \left(\text{Qu}_o^{\text{ord}}, <_o^{\text{ord}}, \text{Prj}_o^{\text{ord}}, \right. \\ &\quad \left. \rho_o, \text{TimeRep}_o[\rho_o] \right), \\ \text{DynTrjObsStr}_{o,\zeta}[\rho_o] &:= \left(\begin{array}{c} \text{TrjDom}_o^{\text{ord}}, \text{TrjMap}_o^{\text{ord}}, \\ \text{TrjDom}_o^{\text{real}}[\rho_o], \\ \left(\begin{array}{c} \text{ObsCurve}_\ell^{\text{ord}}, \\ \text{ObsCurve}_\ell^{\text{real}}[\rho_o] \end{array} \right)_{\ell \in \text{ObsIdx}^{\text{quot}}} \end{array} \right). \end{aligned}$$

Define the dynamical-description object by

$$\text{DynDesc}_{o,\zeta}[\rho_o] := (\text{LawInput}, \text{DynTimeStr}_o[\rho_o], \text{DynTrjObsStr}_{o,\zeta}[\rho_o]).$$

Definition 5.72 (Dynamical-description witness object). For any $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$, define the dynamical-description witness object by

$$\text{DynWit}_{o,\zeta}[\rho_o] := (\text{LawInput}, \text{DynDesc}[\rho_o]).$$

Proposition 5.73 (Existence and uniqueness of the dynamical-description object and witness object). Under $\text{DynOperative}[\text{Ev}]$, for every $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$, $\text{DynDesc}[\rho_o]$ and $\text{DynWit}[\rho_o]$ exist uniquely.

Proof. Proposition 5.48, Definition 5.56, Definition 5.63, Proposition 5.66, and Definition 5.68 uniquely determine every component of the two inner structures depending on the same ρ_o . The conclusion follows from the two outer ordered tuples. $\square$

Definition 5.74 ($\mathfrak{D}$ -record-cell code image). For any $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$, define the $\mathfrak{D}$ -record-cell code image by

$$\text{RecordImg}_{\mathfrak{D}}^{\text{code}}(x_{\mathfrak{D}}) := \text{RecordCode}_{\mathfrak{D}}[W_{\mathfrak{D}}(x_{\mathfrak{D}})] \subseteq \text{RecordCodeVal}_{\mathfrak{D}}.$$

Definition 5.75 (Observer-time-relative $\mathfrak{D}$ -record images). For a relation $R \subseteq X \times Y$ and $A \subseteq Y$, let the relational inverse image be

$$R^{-1}[A] := \{x \in X \mid \exists y \in A, (x, y) \in R\}$$

For any $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$, define the canonical-time image, order-time image, and real-time image of the $\mathfrak{D}$ -record cell by

$$\begin{aligned} \text{RecordImg}_{\mathfrak{D},o}^{\text{time,can}}(x_{\mathfrak{D}}) &:= (\text{RecordCodeRel}_{\mathfrak{C},o})^{-1}[\text{RecordImg}_{\mathfrak{D}}^{\text{code}}(x_{\mathfrak{D}})], \\ \text{RecordImg}_{\mathfrak{D},o}^{\text{time,ord}}(x_{\mathfrak{D}}) &:= \text{Prj}_o^{\text{ord}}[\text{RecordImg}_{\mathfrak{D},o}^{\text{can}}(x_{\mathfrak{D}})], \\ \text{RecordImg}_{\mathfrak{D},o}^{\text{time,real}}(x_{\mathfrak{D}}) &:= \text{TimeRep}_o[\text{RecordImg}_{\mathfrak{D},o}^{\text{can}}(x_{\mathfrak{D}})]. \end{aligned}$$

These images are subsets of $\text{Qu}_{\mathfrak{C},o}^{\text{can}}$, Qu_o^{ord} , and $\mathbb{R}$, respectively.

Proposition 5.76 (Factorization of observer-time-relative $\mathfrak{D}$ -record images). For every $x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^*$,

$$\text{RecordImg}_{\mathfrak{D},o}^{\text{real}}(x_{\mathfrak{D}}) = \text{RepEmb}_o [\text{RecordImg}_{\mathfrak{D},o}^{\text{ord}}(x_{\mathfrak{D}})]$$

holds, and the four images from the code image through the real-time image exist uniquely.

Proof. The uniqueness of the first three images follows from the map images and relational inverse images in Definition 5.74 and Definition 5.75. Applying the factorization identity in Definition 5.63 to the canonical-time image gives the displayed factorization identity and uniqueness of the real-time image. $\square$

Definition 5.77 (Disjoint union of single-object and joint-object specifications). Define the joint-object specification set and the general evaluation-specification space by

$$\begin{aligned} \text{JointEvSpec} &:= \{(l, o', J) \mid o' \in U, J \subseteq Z_0, 2 \leq |J| < \infty\}, \\ \text{EvalSpecSp} &:= \text{EvSpec} \sqcup \text{JointEvSpec}. \end{aligned}$$

Write the two injections into the disjoint union as

$$\begin{aligned} \iota_{\text{single}} &: \text{EvSpec} \longrightarrow \text{EvalSpecSp}, \\ \iota_{\text{joint}} &: \text{JointEvSpec} \longrightarrow \text{EvalSpecSp}. \end{aligned}$$

The disjoint-union tags distinguish the two branches.

Definition 5.78 (Agreement of the joint $\mathfrak{C}$ -time bases). Define the set of exactly operable joint specifications by

$$\text{JointEvSpec}^{\text{exact}} := \{(l, o, J) \in \text{JointEvSpec} \mid \text{ExactOperable}[l]\}.$$

Fix an element (l, o, J) of this set. For each $\zeta \in J$, write the generated equivalence relation relative to $\text{Ev}_{\zeta} = (l, o, \zeta)$ as $\sim_{\mathfrak{C},o;\zeta}$. The source-token set and source order are fixed by the common input l . Define agreement of the joint-time bases by

$$\begin{aligned} K_{\text{JointCTime}}^{\text{dyn}}(\iota_{\text{joint}}(l, o, J)) \\ \iff \forall \zeta, \eta \in J, \quad \sim_{\mathfrak{C},o;\zeta} = \sim_{\mathfrak{C},o;\eta}. \end{aligned}$$

The domain of this predicate is $\iota_{\text{joint}}[\text{JointEvSpec}^{\text{exact}}]$.

Definition 5.79 (Dynamical readiness of each evaluation-specification branch). Define the set of ready evaluation specifications by

$$\begin{aligned} \text{EvalSpecReady} \\ := \iota_{\text{single}}[\{\text{Ev} \in \text{EvSpec} \mid \text{DynOperative}[\text{Ev}]\}] \\ \cup \iota_{\text{joint}} \left[\left\{ (l, o, J) \in \text{JointEvSpec}^{\text{exact}} \mid \begin{array}{l} \forall \zeta \in J, \text{DynOperative}[(l, o, \zeta)], \\ K_{\text{JointCTime}}^{\text{dyn}}(\iota_{\text{joint}}(l, o, J)) \end{array} \right\} \right]. \end{aligned}$$

For every $e \in \text{EvalSpecSp}$, set

$$\text{DynReady}[e] \iff e \in \text{EvalSpecReady}.$$

Definition 5.80 (Generated time base of a ready evaluation specification). Take $e \in \text{EvalSpecReady}$. In the single-object branch $e = \iota_{\text{single}}(\mathbf{l}, o, \zeta)$, use the order-time quotient, strict linear order and trajectory domain for that complete evaluation specification:

$$Q_e := \text{Qu}_o^{\text{ord}}, \quad <_e := <_o^{\text{ord}}, \quad D_e := \text{TrjDom}_o^{\text{ord}}.$$

In the joint branch $e = \iota_{\text{joint}}(\mathbf{l}, o, J)$, the object-specific order-time quotients and orders agree. Write their common quotient and order as $Q_e, <_e$, and set

$$D_e := \bigcap_{\zeta \in J} \text{TrjDom}_{o;\zeta}^{\text{ord}} \subseteq Q_e.$$

Define the generated time base by

$$\text{EvalTimeBase}[e] := (Q_e, <_e, D_e).$$

This definition does not require the joint domain to be nonempty.

Proposition 5.81 (Uniqueness of the generated time base and existence of a real representation). For each $e \in \text{EvalSpecReady}$, the datum $\text{EvalTimeBase}[e]$ is uniquely determined, and $<_e$ is a strict linear order on Q_e . Moreover,

$$\Omega[e] := \{\rho : Q_e \rightarrow \mathbb{R} \mid \forall q, r \in Q_e, q <_e r \iff \rho(q) < \rho(r)\}$$

is nonempty.

Proof. For the single-object branch, use dynamical-description operability and Definition 5.56, Definition 5.65, and Definition 5.59. For the joint branch, the same source set, order and generated equivalence relation determine the same canonical time quotient, natural projection, generating edges and reflexive transitive closure. The antisymmetry and incomparability-transitivity conditions contained in each object's dynamical-description operability therefore give the same order-time quotient and strict linear order. Each individual trajectory domain is a subset of this common quotient, and their intersection is unique. Since the object set is nonempty, a real order embedding for one object gives an embedding of the common quotient. Agreement of the quotient and order makes this existence condition independent of the object used. $\square$

Definition 5.82 (General law-candidate relation-structure space). For a set $\mathbb{T}$, define the law-component space by

$$\begin{aligned} \text{LawCompSp}[\mathbb{T}] := \Big\{ \mathbf{L} = (X, Y, f, g, E, R, H, V, h) \mid \\ H \subseteq \mathbb{T}, \quad f : H \rightarrow X, \quad g : H \rightarrow Y, \quad E \subseteq \mathbb{T} \times X, \\ R \subseteq E \times Y, \\ V = (\mathbb{T} \times X) \times Y, \quad h : H \rightarrow V, \quad h(t) = ((t, f(t)), g(t)) \Big\}. \end{aligned}$$

For each $\mathbf{L} \in \text{LawCompSp}[\mathbb{T}]$, let its component accessors satisfy

$$\begin{aligned} \Big(\text{InValSp}, \text{OutValSp}, \text{InVal}, \text{OutVal}, \\ \text{EvalDom}, \text{CandRel}, \text{EvalTimeDom}, \\ \text{EvalTupleSp}, \text{EvalTuple} \Big)[\mathbf{L}] := \mathbf{L} \end{aligned}$$

Also let

$$\text{Bij}(S) := \{f : S \rightarrow S \mid f \text{ is a bijection}\}$$

Define the general law-candidate relation-structure space by

$$\begin{aligned} \text{CandRelStrSp}^{\text{law}} := & \left\{ (\mathbb{T}, A, (L_a)_{a \in A}, \mathcal{U}, \mathcal{F}) \mid A \neq \emptyset, \right. \\ & \forall a \in A, L_a \in \text{LawCompSp}[\mathbb{T}], \quad \mathcal{U} \subseteq \mathcal{P}(\mathbb{T}), \\ & \left. \mathcal{F} \subseteq \prod_{a \in A} \text{Bij}(\text{EvalTupleSp}[L_a]) \right\}. \end{aligned}$$

Definition 5.83 (Single-object law-evaluation components). Let LawIdx be a nonempty law-index set, and, for each $a \in \text{LawIdx}$, let

$$\text{InIdx}_a^{\text{law}}, \text{OutIdx}_a^{\text{law}} \subseteq \text{ObsIdx}^{\text{quot}}$$

Let the side-index set be $\text{IOSide} := \{\text{in}, \text{out}\}$, and set

$$\begin{aligned} \text{ObsIdx}_{a,\text{in}}^{\text{law}} &:= \text{InIdx}_a^{\text{law}}, \\ \text{ObsIdx}_{a,\text{out}}^{\text{law}} &:= \text{OutIdx}_a^{\text{law}} \end{aligned}$$

Let the individual law-evaluation time domain satisfy

$$\text{EvalTimeDom}_a^{\text{law}} \subseteq \text{TrjDom}_o^{\text{real}}$$

For each $s \in \text{IOSide}$, define

$$\begin{aligned} \text{ValSp}_{a,s}^{\text{law}} &:= \prod_{\ell \in \text{ObsIdx}_{a,s}^{\text{law}}} \text{ObsVal}_\ell^{\text{can}}, \\ \text{Val}_{a,s}^{\text{law}} : \text{EvalTimeDom}_a^{\text{law}} &\longrightarrow \text{ValSp}_{a,s}^{\text{law}}, \\ \text{Val}_{a,s}^{\text{law}}(t) &:= (\text{ObsCurve}_\ell^{\text{real}}(t))_{\ell \in \text{ObsIdx}_{a,s}^{\text{law}}} \end{aligned}$$

and let the component projections be

$$\begin{aligned} (\text{InValSp}_a^{\text{law}}, \text{OutValSp}_a^{\text{law}}) &:= (\text{ValSp}_{a,\text{in}}^{\text{law}}, \text{ValSp}_{a,\text{out}}^{\text{law}}), \\ (\text{InVal}_a^{\text{law}}, \text{OutVal}_a^{\text{law}}) &:= (\text{Val}_{a,\text{in}}^{\text{law}}, \text{Val}_{a,\text{out}}^{\text{law}}). \end{aligned}$$

Let the individual law-application domain and candidate relation satisfy

$$\begin{aligned} \text{EvalDom}_a^{\text{law}} &\subseteq \text{TrjDom}_o^{\text{real}} \times \text{InValSp}_a^{\text{law}}, \\ \text{CandRel}_a^{\text{law}} &\subseteq \text{EvalDom}_a^{\text{law}} \times \text{OutValSp}_a^{\text{law}}. \end{aligned}$$

Define the law-evaluation-tuple space and evaluation tuple by

$$\begin{aligned} \text{EvalTupleSp}_a^{\text{law}} &:= (\text{TrjDom}_o^{\text{real}} \times \text{InValSp}_a^{\text{law}}) \times \text{OutValSp}_a^{\text{law}}, \\ \text{EvalTuple}_a^{\text{law}} : \text{EvalTimeDom}_a^{\text{law}} &\longrightarrow \text{EvalTupleSp}_a^{\text{law}}, \\ \text{EvalTuple}_a^{\text{law}}(t) &:= ((t, \text{InVal}_a^{\text{law}}(t)), \text{OutVal}_a^{\text{law}}(t)). \end{aligned}$$

Let these nine components be

$$\begin{aligned} \text{LawEvalComp}_a &:= (\text{InValSp}_a^{\text{law}}, \text{OutValSp}_a^{\text{law}}, \text{InVal}_a^{\text{law}}, \text{OutVal}_a^{\text{law}}, \text{EvalDom}_a^{\text{law}}, \\ &\quad \text{CandRel}_a^{\text{law}}, \text{EvalTimeDom}_a^{\text{law}}, \text{EvalTupleSp}_a^{\text{law}}, \text{EvalTuple}_a^{\text{law}}). \end{aligned}$$

The family of admissible common law-time domains satisfies

$$\text{AdmCommonTimeDomSet}^{\text{law}} \subseteq \mathcal{P}(\text{TrjDom}_o^{\text{real}})$$

Definition 5.84 (Family of admissible faithful law-representation transformations). For each $a \in \text{LawIdx}$, let

$$\begin{aligned} \text{EvalTupleBijSet}_a^{\text{law}} &:= \text{Bij}(\text{EvalTupleSp}_a^{\text{law}}), \\ \text{FaithTransSet}^{\text{law}} &\subseteq \prod_{a \in \text{LawIdx}} \text{EvalTupleBijSet}_a^{\text{law}}. \end{aligned}$$

Definition 5.85 (Law-evaluation datum). Define the single-object law-candidate relation structure by

$$\begin{aligned} &\text{CandRelStr}_{o, \zeta}^{\text{law}}, \\ &\text{TrjDom}_o^{\text{real}}, \\ &:= \left(\begin{array}{c} \text{LawIdx}, (\text{LawEvalComp}_a)_{a \in \text{LawIdx}}, \\ \text{AdmCommonTimeDomSet}^{\text{law}}, \text{FaithTransSet}^{\text{law}} \end{array} \right) \\ &\in \text{CandRelStrSp}^{\text{law}}. \end{aligned}$$

Define the law-evaluation-datum space and the single-object law-evaluation datum by

$$\begin{aligned} \text{LawEvalDatSp} &:= \text{EvalSpecSp} \times \text{CandRelStrSp}^{\text{law}}, \\ \text{LawEvalDat} &:= (\iota_{\text{single}}(\text{Ev}), \text{CandRelStr}^{\text{law}}) \in \text{LawEvalDatSp}. \end{aligned}$$

For any $\text{LawDat} \in \text{LawEvalDatSp}$, let

$$\begin{aligned} \text{EvalSpec}[\text{LawDat}] &:= \text{pr}_1(\text{LawDat}), \\ \text{CandRelStr}[\text{LawDat}] &:= \text{pr}_2(\text{LawDat}), \\ \text{EvalCouplLoc}[\text{LawDat}] &:= \begin{cases} o', & \text{EvalSpec}[\text{LawDat}] = \iota_{\text{single}}(\text{I}, o', \zeta'), \\ o', & \text{EvalSpec}[\text{LawDat}] = \iota_{\text{joint}}(\text{I}, o', J) \end{cases} \in U. \end{aligned}$$

Write the second component as

$$\begin{aligned} \text{CandRelStr}[\text{LawDat}] &= \\ &\left(\begin{array}{c} \text{LawTimeDom}[\text{LawDat}], \text{LawIdx}[\text{LawDat}], \\ (\text{LawEvalComp}_a[\text{LawDat}])_{a \in \text{LawIdx}[\text{LawDat}]}, \\ \text{AdmCommonTimeDomSet}^{\text{law}}[\text{LawDat}], \text{FaithTransSet}^{\text{law}}[\text{LawDat}] \end{array} \right). \end{aligned}$$

For any $\text{LawDat} \in \text{LawEvalDatSp}$ and $a \in \text{LawIdx}[\text{LawDat}]$, define dot notation for an outer-component accessor F and a law-component accessor G by

$$\begin{aligned} \text{LawDat}.F &:= F[\text{LawDat}], \\ \text{LawDat}[a] &:= \text{LawEvalComp}_a[\text{LawDat}], \\ \text{LawDat}[a].G &:= G[\text{LawDat}[a]]. \end{aligned}$$

Here F ranges over the outer-component accessors of this definition, and G ranges over the law-component accessors of Definition 5.82.

Proposition 5.86 (Type consistency of the single-object law-evaluation datum). Hereafter set $\text{LawDat} := \text{LawEvalDat}$. Then

$$\begin{aligned} \iota_{\text{single}}(\text{Ev}) &\in \text{EvalSpecSp}, \\ \text{CandRelStr}^{\text{law}} &\in \text{CandRelStrSp}^{\text{law}}, \\ \text{LawDat} &\in \text{LawEvalDatSp}. \end{aligned}$$

Moreover,

$$\begin{aligned}\text{LawDat.EvalSpec} &= \iota_{\text{single}}(\text{Ev}), \\ \text{LawDat.EvalCouplLoc} &= o, \\ \text{LawDat.CandRelStr} &= \text{CandRelStr}^{\text{law}}, \\ \text{LawDat}[a] &= \text{LawEvalComp}_a \quad (a \in \text{LawIdx}).\end{aligned}$$

All projected values of the outer and law components are obtained by applying the corresponding component accessors to these record equalities.

Proof. Definition 5.2 and Definition 5.77 give $\iota_{\text{single}}(\text{Ev}) \in \text{EvalSpecSp}$. By Definition 5.83 and Definition 5.84, every law component and the two family components have the types in Definition 5.82. Thus the ordered pair of the evaluation specification and the candidate relation structure belongs to the displayed space. The displayed equalities follow by applying the dot projections of Definition 5.85 to **LawDat**. $\square$

Definition 5.87 (Individual law-application-domain condition). For any $\text{LawDat} \in \text{LawEvalDatSp}$, define the individual law-application-domain condition by

$$\begin{aligned}K_{\text{IndivEvalDomAdm}}^{\text{law}}[\text{LawDat}] &\iff \forall a \in \text{LawDat.LawIdx}, \\ &\quad \left[\text{LawDat}[a].\text{EvalTimeDom} \neq \emptyset \right. \\ &\quad \wedge \forall t \in \text{LawDat}[a].\text{EvalTimeDom}, \\ &\quad \left. (t, \text{LawDat}[a].\text{InVal}(t)) \in \text{LawDat}[a].\text{EvalDom} \right].\end{aligned}$$

Proposition 5.88 (Type consistency of the individual law-application-domain condition). For any $\text{LawDat} \in \text{LawEvalDatSp}$, $a \in \text{LawDat.LawIdx}$, and $t \in \text{LawDat}[a].\text{EvalTimeDom}$,

$$(t, \text{LawDat}[a].\text{InVal}(t)) \in \text{LawDat.LawTimeDom} \times \text{LawDat}[a].\text{InValSp}.$$

Proof. The law-component type in Definition 5.82 and the dot projections in Definition 5.85 give

$$\begin{aligned}\text{LawDat}[a].\text{EvalTimeDom} &\subseteq \text{LawDat.LawTimeDom}, \\ \text{LawDat}[a].\text{InVal} : \text{LawDat}[a].\text{EvalTimeDom} &\rightarrow \text{LawDat}[a].\text{InValSp}.\end{aligned}$$

Hence the product membership holds. $\square$

Definition 5.89 (Law-output determinacy condition). For sets X, Y and a relation $R \subseteq X \times Y$, define right uniqueness by

$$\text{RightUnique}(R) \iff \forall x \in X \forall y_0, y_1 \in Y, \quad [(x, y_0) \in R \wedge (x, y_1) \in R] \implies y_0 = y_1.$$

For any $\text{LawDat} \in \text{LawEvalDatSp}$, define the law-output determinacy condition by

$$\begin{aligned}K_{\text{CandRelRightUnique}}^{\text{law}}[\text{LawDat}] &\iff \forall a \in \text{LawDat.LawIdx}, \\ &\quad \text{RightUnique}(\text{LawDat}[a].\text{CandRel}).\end{aligned}$$

Definition 5.90 (Domains of law-candidate relations). For any $\text{LawDat} \in \text{LawEvalDatSp}$ and $a \in \text{LawDat.LawIdx}$, define the domain of the law-candidate relation by

$$\text{CandRelDom}_a^{\text{law}}[\text{LawDat}] := \text{Dom}(\text{LawDat}[a].\text{CandRel}).$$

Proposition 5.91 (Partial-output-map representation of a law-candidate relation). For any $\text{LawDat} \in \text{LawEvalDatSp}$, assume

$$K_{\text{CandRelRightUnique}}^{\text{law}}[\text{LawDat}]$$

For each $a \in \text{LawDat.LawIdx}$, there exists a unique map

$$\text{CandRelOutMap}_a^{\text{law}}[\text{LawDat}] : \text{CandRelDom}_a^{\text{law}}[\text{LawDat}] \longrightarrow \text{LawDat}[a].\text{OutValSp}$$

satisfying

$$\text{LawDat}[a].\text{CandRel} = \left\{ \begin{array}{l} (x, \text{CandRelOutMap}_a^{\text{law}}[\text{LawDat}](x)) \\ \mid x \in \text{CandRelDom}_a^{\text{law}}[\text{LawDat}] \end{array} \right\}.$$

Proof. By Definition 5.90, at least one output value corresponds to each element of the relation domain. Right uniqueness in Definition 5.89 makes that output value unique. Hence there exists a map taking each domain element to its unique corresponding output value, and its graph equals the law-candidate relation. A map with the same graph takes the same value at every domain element and is therefore unique. $\square$

Definition 5.92 (Generated law-evaluation relation-membership condition). For any $\text{LawDat} \in \text{LawEvalDatSp}$, define the generated law-evaluation relation-membership condition by

$$\begin{aligned} & K_{\text{GenEvalCandRelMem}}^{\text{law}}[\text{LawDat}] \\ & \iff K_{\text{IndivEvalDomAdm}}^{\text{law}}[\text{LawDat}] \\ & \wedge \forall a \in \text{LawDat.LawIdx}, \quad \forall t \in \text{LawDat}[a].\text{EvalTimeDom}, \\ & \quad \text{LawDat}[a].\text{EvalTuple}(t) \in \text{LawDat}[a].\text{CandRel}. \end{aligned}$$

Proposition 5.93 (Type consistency of the generated law-evaluation relation-membership condition). For any $\text{LawDat} \in \text{LawEvalDatSp}$, assume

$$K_{\text{IndivEvalDomAdm}}^{\text{law}}[\text{LawDat}].$$

For every

$$a \in \text{LawDat.LawIdx}, \quad t \in \text{LawDat}[a].\text{EvalTimeDom}$$

one has

$$\text{LawDat}[a].\text{EvalTuple}(t) \in \text{LawDat}[a].\text{EvalDom} \times \text{LawDat}[a].\text{OutValSp}.$$

Proof. Definition 5.87 gives $(t, \text{LawDat}[a].\text{InVal}(t)) \in \text{LawDat}[a].\text{EvalDom}$. The law-component type in Definition 5.82 gives $\text{LawDat}[a].\text{OutVal}(t) \in \text{LawDat}[a].\text{OutValSp}$, and expanding the definition of the evaluation-tuple projection yields the product membership. $\square$

Definition 5.94 (Faithful-representation-transformation invariance condition for law-candidate relations). For any $\text{LawDat} \in \text{LawEvalDatSp}$, define the faithful-representation-transformation invariance condition for law-candidate relations by

$$\begin{aligned} & K_{\text{CandRelFaithTransInv}}^{\text{law}}[\text{LawDat}] \\ & \iff \left[\forall (\phi_a)_{a \in \text{LawDat.LawIdx}} \in \text{LawDat.FaithTransSet}, \right. \\ & \quad \left. \forall a \in \text{LawDat.LawIdx}, \quad \phi_a[\text{LawDat}[a].\text{CandRel}] = \text{LawDat}[a].\text{CandRel} \right]. \end{aligned}$$

Proposition 5.95 (Law-candidate-relation invariance under inverses of faithful representation transformations).

For any $\text{LawDat} \in \text{LawEvalDatSp}$, assume

$$K_{\text{CandRelFaithTransInv}}^{\text{law}}[\text{LawDat}].$$

For every

$$(\phi_a)_{a \in \text{LawDat.LawIdx}} \in \text{LawDat.FaithTransSet}$$

and $a \in \text{LawDat.LawIdx}$, the inverse map

$$\phi_a^{-1} : \text{LawDat}[a].\text{EvalTupleSp} \longrightarrow \text{LawDat}[a].\text{EvalTupleSp}$$

exists and satisfies

$$\phi_a^{-1}[\text{LawDat}[a].\text{CandRel}] = \text{LawDat}[a].\text{CandRel}.$$

Consequently, for every $z \in \text{LawDat}[a].\text{EvalTupleSp}$,

$$z \in \text{LawDat}[a].\text{CandRel} \iff \phi_a^{-1}(z) \in \text{LawDat}[a].\text{CandRel}$$

holds.

Proof. By Definition 5.84, every ϕ_a is a bijection on the law-evaluation-tuple space. Apply ϕ_a^{-1} to both sides of the image-invariance identity in Definition 5.94, and use $\phi_a^{-1} \circ \phi_a = \text{id}$ to obtain the set-image identity under the inverse map. Expanding this identity as element membership gives the stated equivalence. $\square$

Definition 5.96 (Individual and common valid time domains for laws). For any $\text{LawDat} \in \text{LawEvalDatSp}$ and $a \in \text{LawDat.LawIdx}$, define the individual valid time domain for the law by

$$\begin{aligned} \text{ValidTimeDom}_a^{\text{law}}[\text{LawDat}] &:= \text{LawDat}[a].\text{EvalTimeDom} \\ &\quad \cap (\text{LawDat}[a].\text{EvalTuple})^{-1}[\text{LawDat}[a].\text{CandRel}]. \end{aligned}$$

Define the common valid time domain by

$$\text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] := \bigcap_{a \in \text{LawDat.LawIdx}} \text{ValidTimeDom}_a^{\text{law}}[\text{LawDat}].$$

Definition 5.97 (Common valid time-domain condition for laws). For any $\text{LawDat} \in \text{LawEvalDatSp}$, define the common valid time-domain condition for laws by

$$\begin{aligned} K_{\text{CommonValidTimeDomAdm}}^{\text{law}}[\text{LawDat}] &\iff \left[K_{\text{GenEvalCandRelMem}}^{\text{law}}[\text{LawDat}] \right. \\ &\quad \wedge \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] \neq \emptyset \\ &\quad \wedge \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] \\ &\quad \left. \in \text{LawDat.AdmCommonTimeDomSet} \right]. \end{aligned}$$

Proposition 5.98 (Type of the common valid time domain and simultaneous satisfaction of all laws). For any $\text{LawDat} \in \text{LawEvalDatSp}$, assume $K_{\text{CommonValidTimeDomAdm}}^{\text{law}}[\text{LawDat}]$. Then

$$\emptyset \neq \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] \subseteq \text{LawDat.LawTimeDom}.$$

Moreover, for every $t \in \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}]$ and $a \in \text{LawDat.LawIdx}$,

$$\begin{aligned} t &\in \text{LawDat}[a].\text{EvalTimeDom}, \\ \text{LawDat}[a].\text{EvalTuple}(t) &\in \text{LawDat}[a].\text{CandRel} \end{aligned}$$

holds.

Proof. Definition 5.97 and

$$\text{LawDat.AdmCommonTimeDomSet} \subseteq \mathcal{P}(\text{LawDat.LawTimeDom})$$

give nonemptiness and the subset relation. By the definition of the intersection, for every a , $t \in \text{ValidTimeDom}_a^{\text{law}}[\text{LawDat}]$. Expanding the inverse-image expression in Definition 5.96 gives the two memberships. $\square$

Definition 5.99 (Directed acyclic graph of law-condition dependencies). Define the law-condition node set and the five-condition vector by

$$\text{LawCondNodeSet} := \{1, 2, 3, 4, 5\},$$

$$\begin{aligned} \mathbf{K}^{\text{law}}[\text{LawDat}] := & \left(K_{\text{IndivEvalDomAdm}}^{\text{law}}[\text{LawDat}], K_{\text{CandRelRightUnique}}^{\text{law}}[\text{LawDat}], \right. \\ & K_{\text{GenEvalCandRelMem}}^{\text{law}}[\text{LawDat}], \\ & \left. K_{\text{CandRelFaithTransInv}}^{\text{law}}[\text{LawDat}], K_{\text{CommonValidTimeDomAdm}}^{\text{law}}[\text{LawDat}] \right). \end{aligned}$$

For every $i \in \text{LawCondNodeSet}$, let

$$\text{LawCondSat}[\text{LawDat}, i] \iff \mathbf{K}_i^{\text{law}}[\text{LawDat}].$$

Define the direct-dependency edge set and the law-condition dependency graph by

$$\begin{aligned} \text{LawCondEdgeSet} &:= \{(1, 3), (3, 5)\}, \\ \text{LawCondDAG} &:= (\text{LawCondNodeSet}, \text{LawCondEdgeSet}). \end{aligned}$$

Proposition 5.100 (Acyclicity of the law-condition dependency graph). The graph LawCondDAG is a directed acyclic graph. Its root-node set is $\{1, 2, 4\}$, and the strict-ancestor sets of its nodes are

$$\begin{aligned} \text{Anc}(1) &= \text{Anc}(2) = \text{Anc}(4) = \emptyset, \\ \text{Anc}(3) &= \{1\}, \quad \text{Anc}(5) = \{1, 3\}. \end{aligned}$$

Proof. Take the rank map

$$\begin{aligned} \text{LawCondRank} : \text{LawCondNodeSet} &\longrightarrow \{0, 1, 2\}, \\ (\text{LawCondRank}(i))_{i=1}^5 &:= (0, 0, 1, 0, 2). \end{aligned}$$

For every $(i, j) \in \text{LawCondEdgeSet}$, one has $\text{LawCondRank}(i) < \text{LawCondRank}(j)$; hence no directed cycle exists. Expanding the edge set gives the root-node set and strict-ancestor sets. $\square$

Definition 5.101 (All-law-conditions bundle). For any $\text{LawDat} \in \text{LawEvalDatSp}$, define the all-law-conditions bundle by

$$\text{LawCondAll}[\text{LawDat}] \iff \forall i \in \text{LawCondNodeSet}, \quad \mathbf{K}_i^{\text{law}}[\text{LawDat}].$$

Definition 5.102 (Law-description operability). For any $\text{LawDat} \in \text{LawEvalDatSp}$, define law-description operability by

$$\text{LawOperative}[\text{LawDat}] \iff \text{DynReady}[\text{LawDat.EvalSpec}] \wedge \text{LawCondAll}[\text{LawDat}].$$

Definition 5.103 (Transport of a law datum along a bijection of time domains). Take a law-evaluation datum and a bijection

$$D \in \text{LawEvalDatSp}, \quad \tau : D.\text{LawTimeDom} \longrightarrow \mathbb{T}'.$$

Use the following abbreviations.

$$\mathbb{T} := D.\text{LawTimeDom}, \quad A := D.\text{LawIdx}, \quad (X_a, Y_a, f_a, g_a, E_a, R_a, H_a, V_a, h_a) := D[a].$$

For each $a \in A$, define the change of time component on evaluation tuples by

$$\begin{aligned} \tau_a^{\text{eval}} : V_a &\longrightarrow (\mathbb{T}' \times X_a) \times Y_a, \\ \tau_a^{\text{eval}}((t, x), y) &:= ((\tau(t), x), y). \end{aligned}$$

Define the nine transported components by

$$\begin{aligned} X'_a &:= X_a, & Y'_a &:= Y_a, \\ H'_a &:= \tau[H_a], \\ f'_a &:= f_a \circ (\tau^{-1}|_{H'_a}), & g'_a &:= g_a \circ (\tau^{-1}|_{H'_a}), \\ E'_a &:= (\tau \times \text{id}_{X_a})[E_a], \\ R'_a &:= \tau_a^{\text{eval}}[R_a], \\ V'_a &:= (\mathbb{T}' \times X_a) \times Y_a, \\ h'_a &:= \tau_a^{\text{eval}} \circ h_a \circ (\tau^{-1}|_{H'_a}), \end{aligned}$$

and transport the families of admissible common time domains and faithful representation changes by

$$\begin{aligned} \mathcal{U}' &:= \{\tau[U] \mid U \in D.\text{AdmCommonTimeDomSet}\}, \\ \mathcal{F}' &:= \left\{ (\tau_a^{\text{eval}} \circ \phi_a \circ (\tau_a^{\text{eval}})^{-1})_{a \in A} \mid (\phi_a)_{a \in A} \in D.\text{FaithTransSet} \right\}. \end{aligned}$$

Define transport of the law datum along the bijection of time domains by

$$\begin{aligned} L'_a &:= (X'_a, Y'_a, f'_a, g'_a, E'_a, R'_a, H'_a, V'_a, h'_a), \\ \text{LawTimeTransport}[D; \tau] &:= (D.\text{EvalSpec}, (\mathbb{T}', A, (L'_a)_{a \in A}, \mathcal{U}', \mathcal{F}')). \end{aligned}$$

Proposition 5.104 (Typing and five-condition preservation under law-datum transport). Under the preceding definition,

$$D' := \text{LawTimeTransport}[D; \tau] \in \text{LawEvalDatSp}.$$

Moreover, the five law conditions and the common valid-time domain satisfy

$$\begin{aligned} \forall i \in \text{LawCondNodeSet}, \quad \mathbf{K}_i^{\text{law}}[D'] &\iff \mathbf{K}_i^{\text{law}}[D], \\ \text{CommonValidTimeDom}^{\text{law}}[D'] &= \tau[\text{CommonValidTimeDom}^{\text{law}}[D]]. \end{aligned}$$

Proof. Images under the bijection and restrictions of its inverse give the nine components the types required by Definition 5.82. The admissible domains form a family of subsets of $\mathbb{T}'$, and each conjugated change is a bijection. Thus the transported outer datum belongs to the same law-evaluation space.

For the first condition, bijective images preserve nonemptiness and membership in the applicability domain. For the second condition, replacing the time component by a bijection while retaining the input and output values preserves and reflects right uniqueness of the output. For the third condition, evaluation tuples are transported by τ_a^{eval} , which preserves and reflects membership in the candidate relation. For the fourth condition, the definitions of the relation image and conjugated changes make image invariance equivalent before and after transport. Each individual valid-time domain is the image of its original domain under τ . Since bijective images commute with intersections, the displayed equality of common valid-time domains follows. Nonemptiness and membership in the admissible-domain family are also preserved, giving equivalence of the fifth condition. $\square$

Definition 5.105 (Law-evaluation data compatible with a generated time domain). For a map ρ , use the time base of Definition 5.80 to define

$$\begin{aligned} & \text{LawEvalDatSp}^{\text{time, gen}}[\rho] \\ & := \left\{ \text{LawDat} \in \text{LawEvalDatSp} \mid \begin{array}{l} \exists e \in \text{EvalSpecReady}, \\ \text{LawDat.EvalSpec} = e, \quad \rho \in \Omega[e], \\ \text{LawDat.LawTimeDom} \subseteq \rho[D_e] \end{array} \right\}. \end{aligned}$$

Here e is uniquely determined by LawDat.EvalSpec . Below, the superscript time, gen is abbreviated to gen .

Definition 5.106 (Law-time admissibility). For any $\text{LawDat} \in \text{LawEvalDatSp}$ and map ρ , define law-time admissibility by

$$\begin{aligned} & \text{LawTimeAdm}[\text{LawDat}; \rho] \\ & \iff \text{LawDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho] \wedge \text{LawCondAll}[\text{LawDat}]. \end{aligned}$$

Definition 5.107 (Law data relative to a time representation). Fix a base representation ρ^0 and a law datum satisfying

$$\text{LawDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho^0].$$

Set

$$\begin{aligned} e &:= \text{LawDat.EvalSpec}, \\ D^0 &:= (\rho^0)^{-1}[\text{LawDat.LawTimeDom}] \subseteq D_e, \\ \tau_\rho &:= \rho \circ (\rho^0|_{D^0})^{-1} : \text{LawDat.LawTimeDom} \longrightarrow \rho[D^0], \\ \text{LawDat}[\rho] &:= \text{LawTimeTransport}[\text{LawDat}; \tau_\rho] \quad (\rho \in \Omega[e]). \end{aligned}$$

The map τ_ρ is the bijection between two representations of the same generated time subdomain D^0 . The family is uniquely determined by the base representation and base law datum, and satisfies

$$\begin{aligned} & \text{LawDat}[\rho^0] = \text{LawDat}, \\ & \text{LawDat}[\rho] \in \text{LawEvalDatSp}^{\text{gen}}[\rho]. \end{aligned}$$

For the single-object datum LawEvalDat of Definitions 5.83 and 5.85, ρ^0 is the representation used in its construction, $e = \iota_{\text{single}}(\text{Ev})$, and $D^0 = D_e$. Hereafter $\text{LawEvalDat}[\rho]$ denotes this specialization.

Definition 5.108 (Observer-time law core). Assume

$$\text{LawTimeAdm}[\text{LawEvalDat}[\text{RepEmb}_o]; \text{RepEmb}_o].$$

Define the observer-time law core relative to this real-valued time-order embedding by

$$\text{LawCore}_{o,\zeta}^{\text{obs,time}}(\text{RepEmb}_o) := \left(\text{DynDesc}[\text{RepEmb}_o], \right. \\ \left. \text{LawEvalDat}[\text{RepEmb}_o].\text{CandRelStr} \right).$$

Proposition 5.109 (Existence and uniqueness of the observer-time law core). Under

$$\text{LawTimeAdm}[\text{LawEvalDat}[\text{RepEmb}_o]; \text{RepEmb}_o],$$

$\text{LawCore}^{\text{obs}}(\text{RepEmb}_o)$ exists uniquely. Moreover,

$$\text{LawCondAll}[\text{LawEvalDat}[\text{RepEmb}_o]], \\ \text{CommonValidTimeDom}^{\text{law}}[\text{LawEvalDat}[\text{RepEmb}_o]] \neq \emptyset$$

hold.

Proof. Definition 5.106 gives law operability. The equality

$$\text{LawEvalDat}[\text{RepEmb}_o].\text{EvalSpec} \\ = \iota_{\text{single}}(\text{Ev})$$

from Definition 5.103 and Proposition 5.86, together with Proposition 5.73, gives the unique existence of the first component. The second component is unique as a record projection in Definition 5.85; hence the law core also exists uniquely.

The same law operability gives the all-law-conditions bundle. Its fifth component and Definition 5.97 imply that the common valid time domain is nonempty. $\square$

Definition 5.110 (Generated representability of law-evaluation time). For a map ρ and $\text{LawDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho]$, put $e := \text{LawDat}.\text{EvalSpec}$. Define generated representability of law-evaluation time by

$$K_{\text{CommonEvalTimeGenRep}}^{\text{law}}[\text{LawDat}; \rho] \\ \iff \emptyset \neq \text{CommonValidTimeDom}^{\text{law}}[\text{LawDat}] \\ \subseteq \rho[D_e] \cap \bigcap_{a \in \text{LawDat}.\text{LawIdx}} (\text{LawDat}[a].\text{EvalTuple})^{-1}[\text{LawDat}[a].\text{CandRel}].$$

Theorem 5.111 (Generated representability of law-evaluation time). For a map ρ and $\text{LawDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho]$, assume $\text{LawOperative}[\text{LawDat}]$. Then

$$K_{\text{CommonEvalTimeGenRep}}^{\text{law}}[\text{LawDat}; \rho].$$

Proof. Law operativity gives the fifth component of the all-law-conditions bundle. By Proposition 5.98, the common valid time domain is nonempty and lies in the inverse image of each candidate relation under its law-evaluation tuple map. Put $e := \text{LawDat.EvalSpec}$. Definition 5.105 also gives $\text{LawDat.LawTimeDom} \subseteq \rho[D_e]$. The same proposition places the common valid time domain inside this law time domain, proving the required inclusion. $\square$

Theorem 5.112 (Generation theorem for the dynamical-description stage). Under the dynamical-description operability of Definition 5.70,

$$\text{DynOperative}[\text{Ev}]$$

one has

$$\text{OrdEmbSet}_o(\mathbb{R}) \neq \emptyset,$$

and the conclusion of Proposition 5.73 holds for every element of this set.

Proof. The real-valued time-order embedding condition in Definition 5.67, together with Definition 5.59, gives the nonemptiness of the displayed set. Proposition 5.73 gives the conclusion for each element. $\square$

Theorem 5.113 (Generation at the law-description stage).

Assume $\text{LawOperative}[\text{LawEvalDat}]$ for the base law datum. For every $\rho_o \in \text{OrdEmbSet}_o(\mathbb{R})$, the observer-time law core

$$\text{LawCore}^{\text{obs}}(\rho_o)$$

exists uniquely, and

$$K_{\text{CommonEvalTimeGenRep}}^{\text{law}}[\text{LawEvalDat}[\rho_o]; \rho_o]$$

holds.

Proof. Law operativity gives dynamical readiness and the five law conditions for the base datum. The transport in Definition 5.107 preserves the evaluation specification and maps the time domain to the image of the same trajectory domain under ρ_o . Proposition 5.104 also preserves the five law conditions. Thus $\text{LawTimeAdm}[\text{LawEvalDat}[\rho_o]; \rho_o]$ holds. Proposition 5.109 gives existence and uniqueness of the law core. Applying Theorem 5.111 to the transported datum, which is law operative and compatible with the generated time domain, proves generated representability. $\square$

Definition 5.114 (Concatenation of finite tuples). For finite tuples $A = (a_1, \dots, a_m)$ and $B = (b_1, \dots, b_n)$, define their concatenation by

$$A \frown B := (a_1, \dots, a_m, b_1, \dots, b_n).$$

Definition 5.115 (Foundational datum for the differential stage). Assume

$$\text{LawTimeAdm}[\text{LawEvalDat}[\text{RepEmb}_o]; \text{RepEmb}_o].$$

Define the time foundation and the trajectory-and-observation foundation for the differential stage by

$$\begin{aligned} \text{DiffTimeBase}_o[\text{RepEmb}_o] &:= \left(\text{Qu}_o^{\text{ord}}, \right. \\ &\quad \text{RepEmb}_o, \\ &\quad \left. \text{TimeRep}_o[\text{RepEmb}_o] \right), \\ \text{DiffTrjObsBase}_{o,\zeta} &:= \left(\text{TrjMap}_o^{\text{ord}}, \right. \\ &\quad \left. \left(\text{ObsCurve}_\ell^{\text{real}}[\text{RepEmb}_o] \right)_{\ell \in \text{ObsIdx}^{\text{quot}}} \right). \end{aligned}$$

Define the foundational datum for the differential stage by

$$\begin{aligned} & \text{Dat}^{\text{diff}}(\text{RepEmb}_o) \\ & := \left(\text{DiffTimeBase}_o[\text{RepEmb}_o] \curvearrowright \text{DiffTrjObsBase} \right) \\ & \quad \curvearrowright \left(\text{LawEvalDat}[\text{RepEmb}_o].\text{CandRelStr}, \right. \\ & \quad \left. \text{LawEvalDat}[\text{RepEmb}_o].\text{FaithTransSet} \right). \end{aligned}$$

This is a seven-component ordered tuple preserving the definitional order of the tuples above.

Proposition 5.116 (Existence and uniqueness of the foundational datum for the differential stage). Under

$$\text{LawTimeAdm}[\text{LawEvalDat}[\text{RepEmb}_o]; \text{RepEmb}_o],$$

$\text{Dat}^{\text{diff}}(\text{RepEmb}_o)$ exists uniquely.

Proof. Definition 5.106 and Definition 5.102 give dynamical-description operability. Proposition 5.73 uniquely determines every component of the two inner foundations. Definition 5.107 uniquely determines the candidate relation structure and the family of faithful representation transformations transported to the same representation. The conclusion follows from the outer ordered tuple in Definition 5.115. $\square$

5.4 Trajectory Charts and Covariance of Differential Laws

Fix a base representation ρ^0 and $\text{LawEvalDat} \in \text{LawEvalDatSp}^{\text{gen}}[\rho^0]$. This section treats both single-object and joint-object specifications; put $e := \text{LawEvalDat.EvalSpec}$. For each $\rho_o \in \Omega[e]$, use $\text{LawEvalDat}[\rho_o]$ from Definition 5.107. The unindexed law indices, evaluation time domains, side-specific value spaces and value maps, and candidate relations denote components of this datum at the fixed ρ_o . When two representations are used together, their components are distinguished by $[i]$. For the generated single-object law datum, law-time admissibility also gives the unique foundational datum of Proposition 5.116.

The differential-evaluation structure below is an additional input; its existence does not follow from law operativity alone. A nonempty countable evaluation time domain cannot be covered by time charts with open images. Indexing each source by the natural numbers does not by itself imply that the index set of the entire source family is countable.

Definition 5.117 (Differential-evaluation structure). The differential-law index set satisfies

$$\emptyset \neq \text{LawIdx}^{\text{diff}} \subseteq \text{LawIdx}.$$

For each

$$a \in \text{LawIdx}^{\text{diff}},$$

use the side-index set IOSide of Definition 5.83. The required differentiation order and the coordinate dimension on each input-output side satisfy

$$k_a \in \mathbb{N}_{\geq 1}, \quad m_a^s \in \mathbb{N}_0 \quad (s \in \text{IOSide}).$$

Define the total dependent-variable coordinate dimension by

$$m_a := m_a^{\text{in}} + m_a^{\text{out}}.$$

The index set of local differential charts for time and the index set of local differential charts on each input-output side satisfy

$$\text{TimeChartIdx}_a^{\text{diff}} \neq \emptyset,$$

$$\text{ValChartIdx}_{a,s}^{\text{diff}} \neq \emptyset \quad (s \in \text{IOSide}).$$

Define the value-chart index set and the full differential-chart index set by

$$\text{ValChartIdx}_a^{\text{diff}} := \prod_{s \in \text{IOSide}} \text{ValChartIdx}_{a,s}^{\text{diff}},$$

$$\text{ChartIdx}_a^{\text{diff}} := \text{TimeChartIdx}_a^{\text{diff}} \times \text{ValChartIdx}_a^{\text{diff}}.$$

For each

$$\alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}},$$

the time-chart domain satisfies

$$\text{TimeChartDom}_{a,\alpha_0}^{\text{diff}} \subseteq \text{EvalTimeDom}_a^{\text{law}}.$$

The local numerical time chart is a bijection

$$\text{TimeChart}_{a,\alpha_0}^{\text{diff}} : \text{TimeChartDom}_{a,\alpha_0}^{\text{diff}} \longrightarrow \widehat{\text{TimeChartDom}}_{a,\alpha_0}^{\text{diff}} \subseteq \mathbb{R},$$

and its image is open.

For each

$$s \in \text{IOSide}, \quad \beta_{s,0} \in \text{ValChartIdx}_{a,s}^{\text{diff}},$$

the side-specific value-chart domain satisfies

$$\text{ValChartDom}_{a,s,\beta_{s,0}}^{\text{diff}} \subseteq \text{ValSp}_{a,s}^{\text{law}}.$$

The side-specific local numerical value chart is a bijection

$$\text{ValChart}_{a,s,\beta_{s,0}}^{\text{diff}} : \text{ValChartDom}_{a,s,\beta_{s,0}}^{\text{diff}} \longrightarrow \widehat{\text{ValChartDom}}_{a,s,\beta_{s,0}}^{\text{diff}} \subseteq \mathbb{R}^{m_s},$$

and its image is open.

For each

$$\beta_0 = (\beta_{s,0})_{s \in \text{IOSide}} \in \text{ValChartIdx}_a^{\text{diff}},$$

define the value-coordinate chart and its image by

$$\text{ValChart}_{a,\beta_0}^{\text{diff}} := \prod_{s \in \text{IOSide}} \text{ValChart}_{a,s,\beta_{s,0}}^{\text{diff}},$$

$$\widehat{\text{ValChartDom}}_{a,\beta_0}^{\text{diff}} := \prod_{s \in \text{IOSide}} \widehat{\text{ValChartDom}}_{a,s,\beta_{s,0}}^{\text{diff}} \subseteq \mathbb{R}^{m_a}.$$

Define the pair of a time-chart reference and a value-chart reference by

$$\chi_0 := (\alpha_0, \beta_0) \in \text{ChartIdx}_a^{\text{diff}}.$$

For each

$$\chi_0 \in \text{ChartIdx}_a^{\text{diff}},$$

define the local jet space by

$$\text{JetSpace}_{a;\chi_0}^{\text{diff},(k_a)} := \widehat{\text{TimeChartDom}}_{a,\alpha_0}^{\text{diff}} \times \widehat{\text{ValChartDom}}_{a,\beta_0}^{\text{diff}} \\ \times \prod_{r=1}^{k_a} \mathbb{R}^{m_a}.$$

The local differential relation satisfies

$$\text{JetRel}_{a;\chi_0}^{\text{diff}} \subseteq \text{JetSpace}_{a;\chi_0}^{\text{diff},(k_a)}.$$

For each

$$\alpha_0, \alpha_1 \in \text{TimeChartIdx}_a^{\text{diff}},$$

define the candidate time-chart change by

$$\text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}} := \text{TimeChart}_{a,\alpha_1}^{\text{diff}} \circ (\text{TimeChart}_{a,\alpha_0}^{\text{diff}})^{-1}$$

on the image of the overlap of the two time-chart domains.

For each

$$\beta_0, \beta_1 \in \text{ValChartIdx}_a^{\text{diff}},$$

define the candidate value-chart change by

$$\text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} := \text{ValChart}_{a,\beta_1}^{\text{diff}} \circ (\text{ValChart}_{a,\beta_0}^{\text{diff}})^{-1}$$

on the image of the overlap of the two value-chart domains.

Define the family of candidate time reparametrizations and the family of candidate value-chart changes by

$$\text{TimeChartChgSet}_a^{\text{diff}} := \left\{ \text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}} \mid (\alpha_0, \alpha_1) \in (\text{TimeChartIdx}_a^{\text{diff}})^2 \right\},$$

$$\text{ValChartChgSet}_a^{\text{diff}} := \left\{ \text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} \mid (\beta_0, \beta_1) \in (\text{ValChartIdx}_a^{\text{diff}})^2 \right\}.$$

For each

$$t \in \text{EvalTimeDom}_a^{\text{law}},$$

assume that there exists

$$\chi_0 \in \text{ChartIdx}_a^{\text{diff}}, \\ \beta_0 = (\beta_{s,0})_{s \in \text{IOside}}$$

such that

$$t \in \text{TimeChartDom}_{a,\alpha_0}^{\text{diff}},$$

and, for every

$$s \in \text{IOside},$$

$$\text{Val}_{a,s}^{\text{law}}(t) \in \text{ValChartDom}_{a,s,\beta_{s,0}}^{\text{diff}}$$

holds.

Define the differential-evaluation structure by

$$\begin{aligned} & \text{EvalStr}_e^{\text{diff}}[\rho_o] \\ & := \left(\text{LawIdx}^{\text{diff}}, \right. \\ & \quad \left(\begin{array}{l} k_a, (m_a^s)_{s \in \text{IOSide}}, \\ \text{TimeChartIdx}_a^{\text{diff}}, (\text{ValChartIdx}_{a,s}^{\text{diff}})_{s \in \text{IOSide}}, \\ (\text{TimeChart}_{a,\alpha_0}^{\text{diff}})_{\alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}}}, \\ (\text{ValChart}_{a,s,\beta_{s,0}}^{\text{diff}})_{\substack{s \in \text{IOSide}, \\ \beta_{s,0} \in \text{ValChartIdx}_{a,s}^{\text{diff}}}}, \\ (\text{JetRel}_{a;\chi_0}^{\text{diff}})_{\substack{\alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}}, \\ \beta_0 \in \text{ValChartIdx}_a^{\text{diff}}}}, \\ \text{TimeChartChgSet}_a^{\text{diff}}, \text{ValChartChgSet}_a^{\text{diff}} \end{array} \right)_{a \in \text{LawIdx}^{\text{diff}}} \end{aligned}$$

Below, the fixed evaluation specification e is omitted from the subscript of the differential-evaluation structure.

Definition 5.118 (Local numerical chart for observer time and trajectory). Assume

$$\text{LawEvalDat}[\rho_o] \in \text{LawEvalDatSp}^{\text{gen}}[\rho_o].$$

Suppose that the differential-evaluation structure

$$\text{EvalStr}^{\text{diff}}[\rho_o]$$

of Definition 5.117 is given.

For each

$$\begin{aligned} & a \in \text{LawIdx}^{\text{diff}}, \\ & \chi_0 \in \text{ChartIdx}_a^{\text{diff}}, \\ & \beta_0 = (\beta_{s,0})_{s \in \text{IOSide}} \end{aligned}$$

define the simultaneous three-chart evaluation-time domain by

$$\begin{aligned} & \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}} \\ & := \left\{ t \in \text{TimeChartDom}_{a,\alpha_0}^{\text{diff}} \mid \right. \\ & \quad \left. \forall s \in \text{IOSide}, \quad \text{Val}_{a,s}^{\text{law}}(t) \in \text{ValChartDom}_{a,s,\beta_{s,0}}^{\text{diff}} \right\}. \end{aligned}$$

Define its numerical time image by

$$\begin{aligned} & \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}} \\ & := \text{TimeChart}_{a,\alpha_0}^{\text{diff}}(\text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}}). \end{aligned}$$

Define the local numerical time coordinate by

$$\begin{aligned} \text{TimeCoord}_{a;\chi_0}^{\text{diff}} : \\ & \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}} \\ & \longrightarrow \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}}, \\ t & \longmapsto \text{TimeChart}_{a,\alpha_0}^{\text{diff}}(t). \end{aligned}$$

Define the local numerical value curve by

$$\begin{aligned} \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}} : \\ & \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}} \\ & \longrightarrow \widehat{\text{ValChartDom}}_{a,\beta_0}^{\text{diff}}, \\ \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}}(\theta) \\ & := \text{ValChart}_{a,\beta_0}^{\text{diff}} \left(\left(\text{Val}_{a,s}^{\text{law}} \left((\text{TimeChart}_{a,\alpha_0}^{\text{diff}})^{-1}(\theta) \right) \right)_{s \in \text{IOSide}} \right). \end{aligned}$$

For each input-output side, define the local numerical curve as the corresponding component of the local numerical value curve,

$$\text{ValCoordCurve}_{a,s;\chi_0}^{\text{diff}} := \text{pr}_s \circ \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}} \quad (s \in \text{IOSide}).$$

Define the local numerical chart for observer time and trajectory by

$$\begin{aligned} \text{TrjCoordChart}_{a;\chi_0}^{\text{diff}} \\ & := \left(\text{TimeCoord}_{a;\chi_0}^{\text{diff}}, \right. \\ & \quad \left. (\text{ValCoordCurve}_{a,s;\chi_0}^{\text{diff}})_{s \in \text{IOSide}} \right). \end{aligned}$$

Proposition 5.119 (Generation and coverage of local numerical curves). For every

$$a \in \text{LawIdx}^{\text{diff}},$$

$$\chi_0 \in \text{ChartIdx}_a^{\text{diff}}$$

the map

$$\begin{aligned} \text{TimeCoord}_{a;\chi_0}^{\text{diff}} : \\ & \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}} \\ & \longrightarrow \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}} \end{aligned}$$

is a bijection.

Moreover, for every

$$t \in \text{EvalTimeDom}_a^{\text{law}}$$

there exist

$$\chi_0 \in \text{ChartIdx}_a^{\text{diff}},$$

$$\beta_0 = (\beta_{s,0})_{s \in \text{IOSide}}$$

such that

$$t \in \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}}.$$

For this chart index and

$$\theta := \text{TimeCoord}_{a;\chi_0}^{\text{diff}}(t)$$

for every

$$s \in \text{IOSide}$$

one has

$$\begin{aligned} & \text{ValCoordCurve}_{a,s;\chi_0}^{\text{diff}}(\theta) \\ &= \text{ValChart}_{a,s,\beta_{s,0}}^{\text{diff}}(\text{Val}_{a,s}^{\text{law}}(t)), \end{aligned}$$

and, moreover,

$$\begin{aligned} & \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}}(\theta) \\ &= \text{ValChart}_{a,\beta_0}^{\text{diff}}\left(\left(\text{Val}_{a,s}^{\text{law}}(t)\right)_{s \in \text{IOSide}}\right). \end{aligned}$$

Proof. The local numerical time chart is a bijection. Its restriction to the simultaneous three-chart evaluation-time domain is therefore a bijection from the domain displayed in the proposition onto its image.

Take any

$$t \in \text{EvalTimeDom}_a^{\text{law}}.$$

By the chart-coverage condition in Definition 5.117, there exists a chart index carrying the input and output components, and the time t belongs to its simultaneous three-chart evaluation-time domain.

From

$$\theta = \text{TimeChart}_{a,\alpha_0}^{\text{diff}}(t)$$

and bijectivity,

$$\left(\text{TimeChart}_{a,\alpha_0}^{\text{diff}}\right)^{-1}(\theta) = t.$$

Substitution into the definition of the side-specific local numerical curve gives the side-specific equality. The definitions of the local numerical value curve and the value-coordinate chart give the value-coordinate equality. $\square$

Definition 5.120 (Common evaluation rule for the differential conditions). The five conditions below are predicates on the typed differential-evaluation structures of this section. An input to the fifth condition is a pair of structures belonging to the same base law datum and sharing the components specified in that definition. Let D be the subset of inputs on which all assumptions in a condition's definition hold, and let P be its test on that domain. Define the condition by

$$K[x] \iff \exists y \in D, \quad x = y \wedge P[y].$$

The test displayed in each definition is its restriction to D . The domain of the third condition is the truth domain of the second; that of the fourth is the truth domain of the first; and that of the fifth requires the first condition for both structures. In the first and second conditions, requirements that type subsequent expressions, such as openness and differentiability, are tested first. If a required type or preceding condition is absent, the condition is false; jets and change maps are not evaluated outside their domains.

Definition 5.121 (Differential-atlas regularity condition (Diff1)). Suppose that the differential-evaluation structure of Definition 5.117 and the local numerical curves of Definition 5.118 are given. For every

$$a \in \text{LawIdx}^{\text{diff}}, \quad \alpha_0, \alpha_1 \in \text{TimeChartIdx}_a^{\text{diff}}, \\ \beta_0, \beta_1 \in \text{ValChartIdx}_a^{\text{diff}}$$

define the coordinate domains of the time-chart overlap and the value-chart overlap by

$$\widehat{\text{TimeChartOverlapDom}}_{a;\alpha_0,\alpha_1}^{\text{diff}} := \text{TimeChart}_{a,\alpha_0}^{\text{diff}} \left(\text{TimeChartDom}_{a,\alpha_0}^{\text{diff}} \cap \text{TimeChartDom}_{a,\alpha_1}^{\text{diff}} \right),$$

$$\widehat{\text{ValChartOverlapDom}}_{a;\beta_0,\beta_1}^{\text{diff}} := \text{ValChart}_{a,\beta_0}^{\text{diff}} \left(\text{Dom ValChart}_{a,\beta_0}^{\text{diff}} \cap \text{Dom ValChart}_{a,\beta_1}^{\text{diff}} \right).$$

Define the value-jet-change domain and the time-jet-change domain by

$$\begin{aligned} \text{ValJetChgDom}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} &:= \widehat{\text{TimeChartDom}}_{a,\alpha_0}^{\text{diff}} \times \\ &\quad \widehat{\text{ValChartOverlapDom}}_{a;\beta_0,\beta_1}^{\text{diff}} \times \prod_{r=1}^{k_a} \mathbb{R}^{m_a}, \\ \text{TimeJetChgDom}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} &:= \widehat{\text{TimeChartOverlapDom}}_{a;\alpha_0,\alpha_1}^{\text{diff}} \times \\ &\quad \widehat{\text{ValChartDom}}_{a,\beta_0}^{\text{diff}} \times \prod_{r=1}^{k_a} \mathbb{R}^{m_a}. \end{aligned}$$

At a local numerical time θ , for a C^{k_a} -class value-curve germ γ , define the local jet by

$$\text{jet}_{\theta}^{k_a}(\gamma) := \left(\theta, \left(\frac{d^r \gamma}{d\theta^r}(\theta) \right)_{r=0}^{k_a} \right).$$

The differential-atlas regularity condition

$$K_{\text{AtlasReg}}^{\text{diff}} \left[\text{EvalStr}^{\text{diff}}[\rho_o] \right]$$

means that the following five conditions hold for every choice of the quantified variables in this definition.

First,

$$\begin{aligned} \widehat{\text{TimeChartOverlapDom}}_{a;\alpha_0,\alpha_1}^{\text{diff}}, \\ \widehat{\text{ValChartOverlapDom}}_{a;\beta_0,\beta_1}^{\text{diff}} \end{aligned}$$

are open sets.

Second,

$$\begin{aligned} \text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}} &\in C^{k_a} \left(\widehat{\text{TimeChartOverlapDom}}_{a;\alpha_0,\alpha_1}^{\text{diff}}, \widehat{\text{TimeChartOverlapDom}}_{a;\alpha_1,\alpha_0}^{\text{diff}} \right), \\ \text{TimeChartChg}_{a;\alpha_1,\alpha_0}^{\text{diff}} &= \left(\text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}} \right)^{-1} \end{aligned}$$

hold.

Third,

$$\begin{aligned} \text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} &\in C^{k_a} \left(\widehat{\text{ValChartOverlapDom}_{a;\beta_0,\beta_1}^{\text{diff}}}, \widehat{\text{ValChartOverlapDom}_{a;\beta_1,\beta_0}^{\text{diff}}} \right), \\ \text{ValChartChg}_{a;\beta_1,\beta_0}^{\text{diff}} &= \left(\text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} \right)^{-1} \end{aligned}$$

hold.

Fourth, there exists a unique value-jet-change map

$$\text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} : \text{ValJetChgDom}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \longrightarrow \text{ValJetChgDom}_{a;\alpha_0;\beta_1,\beta_0}^{\text{diff},(k_a)}$$

satisfying, for every admissible θ and γ ,

$$\text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} (\text{jet}_{\theta}^{k_a}(\gamma)) = \text{jet}_{\theta}^{k_a} (\text{ValChartChg}_{a;\beta_0,\beta_1}^{\text{diff}} \circ \gamma).$$

Fifth, there exists a unique time-jet-change map

$$\text{TimeJetChg}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} : \text{TimeJetChgDom}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} \longrightarrow \text{TimeJetChgDom}_{a;\alpha_1,\alpha_0;\beta_0}^{\text{diff},(k_a)}$$

satisfying, for every admissible θ and γ ,

$$\begin{aligned} &\text{TimeJetChg}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} (\text{jet}_{\theta}^{k_a}(\gamma)) \\ &= \text{jet}_{\text{TimeChartChg}_{a;\alpha_0,\alpha_1}^{\text{diff}}(\theta)}^{k_a} (\gamma \circ \text{TimeChartChg}_{a;\alpha_1,\alpha_0}^{\text{diff}}). \end{aligned}$$

In the value-jet-change identity, γ ranges over C^{k_a} -class curve germs into the coordinate domain of the value-chart overlap. In the time-jet-change identity, γ ranges over C^{k_a} -class curve germs on the coordinate domain of the time-chart overlap.

Definition 5.122 (Generated-trajectory-jet existence condition (Diff2)). Define the generated-trajectory-jet existence condition by

$$\begin{aligned} &K_{\text{GeneratedTrjJetExist}}^{\text{diff}} \left[\text{EvalStr}^{\text{diff}}[\rho_o] \right] \\ &\iff \left[\forall a \in \text{LawIdx}^{\text{diff}}, \right. \\ &\quad \forall \chi_0 \in \text{ChartIdx}_a^{\text{diff}}, \\ &\quad \text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}} \text{ is open,} \\ &\quad \text{ValCoordCurve}_{a;\chi_0}^{\text{diff}} \in C^{k_a} \left(\widehat{\text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}}}, \right. \\ &\quad \left. \left. \widehat{\text{ValChartDom}_{a,\beta_0}^{\text{diff}}} \right) \right]. \end{aligned}$$

Under this condition, for each

$$\theta \in \widehat{\text{ChartEvalTimeDom}_{a;\chi_0}^{\text{diff}}},$$

define the generated-trajectory jet by

$$\begin{aligned} &\text{GeneratedTrjJet}_{a;\chi_0}^{\text{diff},(k_a)}(\theta) \\ &:= \text{jet}_{\theta}^{k_a} (\text{ValCoordCurve}_{a;\chi_0}^{\text{diff}}) \\ &\in \text{JetSpace}_{a;\chi_0}^{\text{diff},(k_a)}. \end{aligned}$$

Definition 5.123 (Generated-trajectory-jet local-relation-membership condition (Diff3)).

Assume

$$K_{\text{GeneratedTrjJetExist}}^{\text{diff}} \left[\text{EvalStr}^{\text{diff}}[\rho_o] \right].$$

Define the generated-trajectory-jet local-relation-membership condition by

$$\begin{aligned} & K_{\text{GeneratedTrjJetRelMem}}^{\text{diff}} \left[\text{EvalStr}^{\text{diff}}[\rho_o] \right] \\ & \iff \left[\begin{aligned} & \forall a \in \text{LawIdx}^{\text{diff}}, \\ & \forall \chi_0 \in \text{ChartIdx}_a^{\text{diff}}, \\ & \forall \theta \in \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}}, \\ & \text{GeneratedTrjJet}_{a;\chi_0}^{\text{diff},(k_a)}(\theta) \\ & \in \text{JetRel}_{a;\chi_0}^{\text{diff}} \end{aligned} \right]. \end{aligned}$$

Proposition 5.124 (Local-jet consequences of Diff1–Diff3). Suppose that the differential-atlas regularity condition, the generated-trajectory-jet existence condition, and the generated-trajectory-jet local-relation-membership condition hold.

The value-jet-change map preserves the time-chart index and is uniquely determined by the value-chart-change map. The time-jet-change map preserves the value-chart index and is uniquely determined by the time-chart-change map.

On their common overlaps,

$$\text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_0}^{\text{diff},(k_a)} = \text{id},$$

$$\text{TimeJetChg}_{a;\alpha_0,\alpha_0;\beta_0}^{\text{diff},(k_a)} = \text{id}$$

hold. On a common overlap of three value charts,

$$\begin{aligned} & \text{ValJetChg}_{a;\alpha_0;\beta_1,\beta_2}^{\text{diff},(k_a)} \circ \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \\ & = \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_2}^{\text{diff},(k_a)}, \end{aligned}$$

and on a common overlap of three time charts,

$$\begin{aligned} & \text{TimeJetChg}_{a;\alpha_1,\alpha_2;\beta_0}^{\text{diff},(k_a)} \circ \text{TimeJetChg}_{a;\alpha_0,\alpha_1;\beta_0}^{\text{diff},(k_a)} \\ & = \text{TimeJetChg}_{a;\alpha_0,\alpha_2;\beta_0}^{\text{diff},(k_a)} \end{aligned}$$

hold.

For every

$$\begin{aligned} & \theta \in \widehat{\text{ChartEvalTimeDom}}_{a;\chi_0}^{\text{diff}}, \\ & \text{GeneratedTrjJet}_{a;\chi_0}^{\text{diff},(k_a)}(\theta) \in \text{JetRel}_{a;\chi_0}^{\text{diff}}. \end{aligned}$$

Proof. By the differential-atlas regularity condition, the time-chart-change and value-chart-change maps are bijections of class

$$C^{k_a},$$

and the reverse chart-change maps are also of class

$$C^{k_a}.$$

The

$$r$$

th derivative after a value-chart change, for

$$0 \leq r \leq k_a,$$

is determined by the chain rule from the derivatives through order

$$r$$

of the value-chart-change map and the representative curve germ. The

$$r$$

th derivative after a time-chart change is determined by the chain rule from the derivatives through order

$$r$$

of the inverse time-chart-change map and the representative curve germ. Hence the values of the two jet-change maps are independent of the choice of representative curve germ.

Every element of a jet-change domain can be represented by restricting a local polynomial curve through its zeroth-order component to a sufficiently small neighborhood, as the jet of a value-curve germ of class

$$C^{k_a}$$

of order

$$k_a.$$

Hence the two jet-change maps are defined over their entire respective jet-change domains. Two maps satisfying the same representative formula agree at every jet, so uniqueness holds.

A chart change from a chart to itself is the identity map, which gives the two identity formulas. On triple overlaps, the value-chart-change maps satisfy the value-chart composition condition, and the time-chart-change maps satisfy the time-chart composition condition. Applying two jet-change maps to a representative curve germ and using the uniqueness of each jet-change map gives the two composition formulas.

By the membership formula in Definition 5.122, each generated trajectory jet is an element of the local jet space indexed by the same law index, time-chart reference, and value-chart reference. By the membership formula in Definition 5.123, each generated trajectory jet belongs to the local jet relation indexed by the same law index, time-chart reference, and value-chart reference. $\square$

Lemma 5.125 (Transport of a relation restriction by a bijection). Suppose that a bijection

$$F : X \longrightarrow Y$$

and subsets

$$A \subseteq X, \quad B \subseteq Y$$

satisfy

$$\forall x \in X, \quad x \in A \iff F(x) \in B.$$

Then

$$F(A) = B$$

holds.

Proof. Let $y \in F(A)$. There exists $x \in A \subseteq X$ such that $y = F(x)$. Since the assumption gives $F(x) \in B$, one has $y \in B$. Conversely, let $y \in B$. Surjectivity gives $y = F(x)$ for some $x \in X$, and the assumption then gives $x \in A$. Hence $y \in F(A)$. The two inclusions give the equality. $\square$

Definition 5.126 (Value-jet-relation covariance condition (Diff4)). Assume

$$K_{\text{AtlasReg}}^{\text{diff}} \left[\text{EvalStr}^{\text{diff}}[\rho_o] \right].$$

Define the value-jet-relation covariance condition by

$$\begin{aligned} & K_{\text{ValJetRelCov}}^{\text{diff}} \left[\text{EvalStr}^{\text{diff}}[\rho_o] \right] \\ & \iff \left[\forall a \in \text{LawIdx}^{\text{diff}}, \quad \forall \alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}}, \right. \\ & \quad \forall \beta_0, \beta_1 \in \text{ValChartIdx}_a^{\text{diff}}, \quad \forall e \in \text{ValJetChgDom}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)}, \\ & \quad \left. e \in \text{JetRel}_{a;\alpha_0,\beta_0}^{\text{diff}} \iff \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)}(e) \in \text{JetRel}_{a;\alpha_0,\beta_1}^{\text{diff}} \right]. \end{aligned}$$

Proposition 5.127 (Chart-change equality for value-jet relations). When the differential-atlas regularity condition and the value-jet-relation covariance condition hold,

$$\begin{aligned} & \text{ValJetChg}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \left(\text{JetRel}_{a;\alpha_0,\beta_0}^{\text{diff}} \cap \text{ValJetChgDom}_{a;\alpha_0;\beta_0,\beta_1}^{\text{diff},(k_a)} \right) \\ & = \text{JetRel}_{a;\alpha_0,\beta_1}^{\text{diff}} \cap \text{ValJetChgDom}_{a;\alpha_0;\beta_1,\beta_0}^{\text{diff},(k_a)} \end{aligned}$$

holds.

Proof. By the differential-atlas regularity condition, the value-jet-change map is a bijection between the two displayed change domains. The value-jet-relation covariance condition gives the equivalence of relation membership along this bijection. Applying Lemma 5.125 gives the equality. $\square$

Definition 5.128 (Time-representation-change jet-relation covariance condition (Diff5)). Take two real-valued time-order embeddings

$$\rho_{o,i} \in \Omega[e], \quad i \in \{0, 1\},$$

and the set of directed pairs

$$\text{Dir}_2 := \{(0, 1), (1, 0)\}.$$

For each $(i, j) \in \text{Dir}_2$, define the order isomorphism between the time-representation images by

$$\text{TimeImIso}_{o;i \rightarrow j}^{\text{dyn}} := \rho_{o,j} \circ \rho_{o,i}^{-1} : \text{Im}(\rho_{o,i}) \longrightarrow \text{Im}(\rho_{o,j}).$$

Write the differential-evaluation structures as

$$\text{ES}_i := \text{EvalStr}^{\text{diff}}[\rho_{o,i}], \quad i \in \{0, 1\}.$$

Suppose that both satisfy the differential-atlas regularity condition and share the differential-law indices, the required differential orders, the value-chart indices, and the value-chart families. For every

$$a \in \text{LawIdx}^{\text{diff}}, \quad \alpha_i \in \text{TimeChartIdx}_a^{\text{diff},[i]} \quad (i \in \{0, 1\}),$$

$$\beta \in \text{ValChartIdx}_a^{\text{diff}}, \quad \chi_i := (\alpha_i, \beta)$$

and $(i, j) \in \text{Dir}_2$, define the inter-embedding time-chart-overlap coordinate domain by

$$\widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} := \text{TimeChart}_{a,\alpha_i}^{\text{diff},[i]} \left(\text{TimeChartDom}_{a,\alpha_i}^{\text{diff},[i]} \right. \\ \left. \cap (\text{TimeImglso}_{o;i \rightarrow j})^{-1} (\text{TimeChartDom}_{a,\alpha_j}^{\text{diff},[j]}) \right).$$

Define the inter-embedding local time-chart change and its jet-change domain by

$$\text{TimeRepChartChg}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} := \text{TimeChart}_{a,\alpha_j}^{\text{diff},[j]} \circ \text{TimeImglso}_{o;i \rightarrow j} \circ (\text{TimeChart}_{a,\alpha_i}^{\text{diff},[i]})^{-1},$$

$$\text{TimeRepJetChgDom}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j} := \widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} \\ \times \widehat{\text{ValChartDom}}_{a,\beta}^{\text{diff}} \times \prod_{r=1}^{k_a} \mathbb{R}^{m_a}.$$

The time-representation-change jet-relation covariance condition

$$K_{\text{TimeRepJetRelCov}}^{\text{diff}} [\text{ES}_0, \text{ES}_1]$$

means that the following four conditions hold for every choice of the quantified variables in this definition.

First,

$$\widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j}$$

is an open set.

Second,

$$\text{TimeRepChartChg}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} \in C^{k_a} \left(\frac{\widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j}}{\widehat{\text{TimeRepChartOverlapDom}}_{a;\alpha_j,\alpha_i}^{\text{diff},j \rightarrow i}} \right),$$

$$\text{TimeRepChartChg}_{a;\alpha_j,\alpha_i}^{\text{diff},j \rightarrow i} = \left(\text{TimeRepChartChg}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j} \right)^{-1}$$

hold.

Third, there exists a unique map

$$\text{TimeRepJetChg}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j} \\ : \text{TimeRepJetChgDom}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j} \\ \longrightarrow \text{TimeRepJetChgDom}_{a;\alpha_j,\alpha_i;\beta}^{\text{diff},(k_a),j \rightarrow i}$$

satisfying, for every admissible θ and every value-curve germ of class C^{k_a} , denoted by γ ,

$$\text{TimeRepJetChg}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j} (\text{jet}_{\theta}^{k_a}(\gamma)) \\ = \text{jet}_{\text{TimeRepChartChg}_{a;\alpha_i,\alpha_j}^{\text{diff},i \rightarrow j}(\theta)}^{k_a} \left(\gamma \circ \text{TimeRepChartChg}_{a;\alpha_j,\alpha_i}^{\text{diff},j \rightarrow i} \right).$$

Fourth, for every

$$e \in \text{TimeRepJetChgDom}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j},$$

$$e \in \text{JetRel}_{a;\chi_i}^{\text{diff},[i]} \iff \text{TimeRepJetChg}_{a;\alpha_i,\alpha_j;\beta}^{\text{diff},(k_a),i \rightarrow j}(e) \in \text{JetRel}_{a;\chi_j}^{\text{diff},[j]}$$

holds.

Proposition 5.129 (Order isomorphism between images and time-jet-relation equality).

$$\text{TimeImgIso}_{o;0 \rightarrow 1}$$

is a strictly monotone order isomorphism. When the time-representation-change jet-relation covariance condition holds,

$$\begin{aligned} & \text{TimeRepJetChg}_{a;\alpha_0,\alpha_1;\beta}^{\text{diff},(k_a),0 \rightarrow 1} \left(\text{JetRel}_{a;\chi_0}^{\text{diff},[0]} \cap \text{TimeRepJetChgDom}_{a;\alpha_0,\alpha_1;\beta}^{\text{diff},(k_a),0 \rightarrow 1} \right) \\ &= \text{JetRel}_{a;\chi_1}^{\text{diff},[1]} \cap \text{TimeRepJetChgDom}_{a;\alpha_1,\alpha_0;\beta}^{\text{diff},(k_a),1 \rightarrow 0} \end{aligned}$$

holds.

Proof. $\rho_{o,0}$ and $\rho_{o,1}$ are order isomorphisms from the observer-time order-coordinate quotient to their respective images. By Proposition 5.62, the maps between the images are mutual inverses, and each direction preserves order. Hence $\text{TimeImgIso}_{o;0 \rightarrow 1}$ is a strictly monotone order isomorphism.

By the time-representation-change jet-relation covariance condition, the time-jet-change map is a bijection between the two displayed change domains and satisfies the equivalence of relation membership. Applying Lemma 5.125 gives the image-set equality. $\square$

Definition 5.130 (Differential-stage condition DAG). Keep the base representation and law datum fixed as at the start of this section. Each component structure is relative to the law datum transported to its representation. Define the family of differential-evaluation structures over all real-valued time-order embeddings by

$$\begin{aligned} & \text{EvalStrFam}^{\text{diff}} \\ &:= \left(\text{EvalStr}^{\text{diff}}[\rho_o] \right)_{\rho_o \in \Omega[e]}. \end{aligned}$$

Each component shares

$$\text{LawIdx}^{\text{diff}}, \quad (k_a)_{a \in \text{LawIdx}^{\text{diff}}}, \quad (\text{ValChartIdx}_a^{\text{diff}})_{a \in \text{LawIdx}^{\text{diff}}}$$

and the value-chart family.

Define the differential-condition index set and the condition-specific arity by

$$\begin{aligned} I_{\text{diff}}^{\text{cond}} &:= \{1, \dots, 5\}, \\ \text{arity}_{\text{diff}}(i) &:= \begin{cases} 1, & i \in \{1, 2, 3, 4\}, \\ 2, & i = 5 \end{cases}. \end{aligned}$$

For each

$$i \in I_{\text{diff}}^{\text{cond}}$$

and

$$\mathbf{E} = (E_r)_{0 \leq r < \text{arity}_{\text{diff}}(i)},$$

define the differential-condition selection predicate by

$$\text{DiffPred}_i[\mathbf{E}] := \begin{cases} K_{\text{AtlasReg}}^{\text{diff}}[E_0], & i = 1, \\ K_{\text{GeneratedTrjJetExist}}^{\text{diff}}[E_0], & i = 2, \\ K_{\text{GeneratedTrjJetRelMem}}^{\text{diff}}[E_0], & i = 3, \\ K_{\text{ValJetRelCov}}^{\text{diff}}[E_0], & i = 4, \\ K_{\text{TimeRepJetRelCov}}^{\text{diff}}[E_0, E_1], & i = 5 \end{cases}.$$

For each

$$i \in I_{\text{diff}}^{\text{cond}},$$

define the five universal differential conditions by

$$\begin{aligned} & \text{DiffCond}_i^{\text{univ}} \\ & \iff \left[\forall \mathbf{T} = (T_r)_{0 \leq r < \text{arity}_{\text{diff}}(i)} \right. \\ & \quad \left. \in (\Omega[e])^{\text{arity}_{\text{diff}}(i)}, \right. \\ & \quad \left. \text{DiffPred}_i \left[\left(\text{EvalStr}^{\text{diff}}[T_r] \right)_{0 \leq r < \text{arity}_{\text{diff}}(i)} \right] \right]. \end{aligned}$$

Define the set of direct differential-condition-index edges by

$$E_{\text{diff}}^{\text{cond}} := \{(1, 2), (2, 3), (1, 4), (1, 5)\}.$$

Define the differential-condition node set, the set of direct differential-condition dependency edges, and the differential-stage condition DAG by

$$\begin{aligned} \text{DiffCondNodeSet} &:= I_{\text{diff}}^{\text{cond}}, \\ \text{DiffCondEdgeSet} &:= E_{\text{diff}}^{\text{cond}}, \\ \text{DiffCondDAG} &:= (\text{DiffCondNodeSet}, \text{DiffCondEdgeSet}). \end{aligned}$$

Associate node i with the condition $\text{DiffCond}_i^{\text{univ}}$. Every edge $(i, j) \in \text{DiffCondEdgeSet}$ satisfies $i < j$, so this graph is a directed acyclic graph.

Definition 5.131 (Differential-evaluation datum). Fix the base representation, base law-evaluation datum, and its differential-evaluation-structure family from this section. Call the triple

$$\text{DiffEvalDat} := (\rho^0, \text{LawEvalDat}, \text{EvalStrFam}^{\text{diff}}).$$

the differential-evaluation datum. Including the base representation makes the law data transported to each representation uniquely recoverable from this triple.

Definition 5.132 (Differential-stage completion condition). Define the differential-stage completion condition by

$$\text{DiffCondAll} \left[\text{EvalStrFam}^{\text{diff}} \right] \iff \bigwedge_{i \in I_{\text{diff}}^{\text{cond}}} \text{DiffCond}_i^{\text{univ}}.$$

Define differential-stage operability by

$$\begin{aligned} & \text{DiffOperative}[\text{DiffEvalDat}] \\ & \iff \text{LawOperative}[\text{LawEvalDat}] \\ & \quad \wedge \text{DiffCondAll} \left[\text{EvalStrFam}^{\text{diff}} \right]. \end{aligned}$$

Definition 5.133 (Chart-compatible differential law). In a context where the fixed DiffEvalDat uniquely determines e , the index e on the objects below is omitted. Assume the differential-stage completion condition

$$\text{DiffCondAll} \left[\text{EvalStrFam}^{\text{diff}} \right]$$

of Definition 5.132. For each

$$\rho_o \in \Omega[e],$$

define the chart-compatible differential law by

$$\begin{aligned} & \text{ChartCompatDiffLaw}_e(\rho_o) \\ &:= \left(\left(\text{JetRel}_{a;\chi_0}^{\text{diff}} \right)_{\substack{a \in \text{LawIdx}^{\text{diff}}, \\ \chi_0 \in \text{ChartIdx}_a^{\text{diff}}}}, \right. \\ & \quad \left. \left(\text{ValJetChg}_{a;\alpha_0;\beta_0;\beta_1}^{\text{diff},(k_a)} \right)_{\substack{a \in \text{LawIdx}^{\text{diff}}, \\ \alpha_0 \in \text{TimeChartIdx}_a^{\text{diff}}, \\ \beta_0, \beta_1 \in \text{ValChartIdx}_a^{\text{diff}}}} \right). \end{aligned}$$

The first component is the family of local jet relations to which the generated trajectory jets belong. The second component is the family of value-jet-change maps that preserve the time-chart index and satisfy the image-set equality of Proposition 5.127 .

Proposition 5.134 (Separation of order covariance from the two kinds of jet-relation covariance). The map

$$\text{TimeImglso}_{o;0 \rightarrow 1}$$

determined by two real-valued time-order embeddings is a strictly monotone order isomorphism. Value-jet-relation covariance is given by $\text{DiffCond}_4^{\text{univ}}$, and the image-set equality of Proposition 5.127 holds for the value-jet-change maps that preserve the time-chart index. Time-representation-change jet-relation covariance is given by $\text{DiffCond}_5^{\text{univ}}$, and the image-set equality of Proposition 5.129 holds for the time-jet-change maps that preserve the value-chart index.

Proof. The order isomorphism between the images follows from Proposition 5.129 . Value-jet-relation covariance follows from Definition 5.126 and Proposition 5.127 . Time-representation-change jet-relation covariance follows from Definition 5.128 and Proposition 5.129 . A value-jet change preserves the time-chart index, whereas a time-jet change preserves the value-chart index. The order isomorphism between the images assumes neither jet-relation covariance condition. The conditions establishing the three conclusions are therefore distinct. $\square$

Theorem 5.135 (Differential covariance of law-bearing dynamical descriptions). Assume

$$\text{DiffOperative}[\text{DiffEvalDat}].$$

For arbitrary

$$\rho_{o,0}, \rho_{o,1} \in \Omega[e],$$

$$\text{TimeImglso}_{o;0 \rightarrow 1} : \text{Im}(\rho_{o,0}) \longrightarrow \text{Im}(\rho_{o,1})$$

is the unique strictly monotone order isomorphism satisfying

$$\rho_{o,1} = \text{TimeImglso}_{o;0 \rightarrow 1} \circ \rho_{o,0}.$$

For the differential-evaluation structure relative to each real-valued time-order embedding, the value-jet-relation image-set equality of Proposition 5.127 holds. For the pair of differential-evaluation structures relative to the two embeddings, the time-jet-relation image-set equality of Proposition 5.129 holds. The first equality preserves the time-chart index, and the second preserves the value-chart index.

Proof. Differential-stage operability gives the five universal differential conditions. By Proposition 5.129, the map between the images is a strictly monotone order isomorphism. The displayed composition identity follows from Definition 5.128.

Take a map ψ satisfying the same composition identity. The map $\rho_{o,0}$ is surjective onto its image, and both maps, after precomposition with $\rho_{o,0}$, agree with $\rho_{o,1}$. Lemma 4.181 therefore gives $\psi = \text{TimeImglso}_{o;0 \rightarrow 1}$.

The value-jet-relation image-set equality follows from $\text{DiffCond}_4^{\text{univ}}$ and Proposition 5.127. The time-jet-relation image-set equality follows from $\text{DiffCond}_5^{\text{univ}}$ and Proposition 5.129. $\square$

Definition 5.136 (Differential-failure evaluation input). For each

$$i \in I_{\text{diff}}^{\text{cond}},$$

define the strict-ancestor index set by

$$\text{Anc}_{\text{diff}}(i) := \left\{ j \in I_{\text{diff}}^{\text{cond}} \mid (j, i) \in (E_{\text{diff}}^{\text{cond}})^+ \right\}.$$

Here the superscript $+$ denotes the transitive closure of the direct-edge set.

Define the all-ancestors-satisfied bundle for the five universal differential conditions by

$$\begin{aligned} & \text{DiffCondAncAll}_i \\ & := \text{LawOperative} [\text{LawEvalDat}] \wedge \bigwedge_{\substack{j \in \text{Anc}_{\text{diff}}(i) \\ i \in I_{\text{diff}}^{\text{cond}}}} \text{DiffCond}_j^{\text{univ}}, \end{aligned}$$

Define the differential-failure evaluation input by

$$\begin{aligned} & \text{EvalInput}_e^{\text{diff}} \\ & := \left(\text{DiffCondDAG}, (\text{DiffCond}_i^{\text{univ}})_{i \in I_{\text{diff}}^{\text{cond}}}, \right. \\ & \quad \left. (\text{DiffCondAncAll}_i)_{i \in I_{\text{diff}}^{\text{cond}}}, \text{DiffEvalDat} \right). \end{aligned}$$

Proposition 5.137 (Ancestor consistency of the differential-failure evaluation input).

$$\text{EvalInput}^{\text{diff}}$$

has the components

$$\begin{aligned} & \text{DiffCondDAG}, \quad (\text{DiffCond}_i^{\text{univ}})_{i \in I_{\text{diff}}^{\text{cond}}}, \\ & (\text{DiffCondAncAll}_i)_{i \in I_{\text{diff}}^{\text{cond}}}, \quad \text{DiffEvalDat}. \end{aligned}$$

For each

$$i \in I_{\text{diff}}^{\text{cond}},$$

the truth value of the universal differential condition

$$\text{DiffCond}_i^{\text{univ}}$$

is evaluated on the satisfaction domain of

$$\text{DiffCondAncAll}_i.$$

Proof. From the set of direct index edges in Definition 5.130, the strict-ancestor index sets are

$$(\text{Anc}_{\text{diff}}(i))_{1 \leq i \leq 5} = (\emptyset, \{1\}, \{1, 2\}, \{1\}, \{1\}).$$

Substitution into the ancestor conjunction of Definition 5.136 gives the all-ancestors-satisfied bundle for each condition. The tuple displayed in the same definition gives the four listed components. $\square$

5.5 Three-layer synthesis of dynamical, law-bearing, and differential descriptions

Dynamical-description operability, law-description operability, and differential-stage operability have a cumulative dependency relation.

Theorem 5.138 (Three-layer synthesis theorem for dynamical, law-bearing, and differential descriptions). For the single-object specification Ev , use the constructed base law datum LawEvalDat , together with its base representation ρ^0 . For the differential stage, fix a differential-evaluation-structure family belonging to this same pair. Thus $\text{LawEvalDat.EvalSpec} = \iota_{\text{single}}(\text{Ev})$. Law assertions at each representation concern $\text{LawEvalDat}[\rho_o]$ from Definition 5.107.

In the display of this theorem, the complete evaluation specification $\text{Ev} = (l, o, \zeta)$ uniquely determines the subscript pair (o, ζ) , which is omitted. The establishment conditions for the three stages satisfy

$$\begin{aligned} & \text{DynOperative}[\text{Ev}] \\ \iff & \text{ExactOperable}[l] \wedge \text{DynCondAll}[\text{Ev}], \\ & \text{LawOperative}[\text{LawEvalDat}] \\ \iff & \text{DynOperative}[\text{Ev}] \wedge \text{LawCondAll}[\text{LawEvalDat}], \\ & \text{DiffOperative}[\text{DiffEvalDat}] \\ \iff & \text{LawOperative}[\text{LawEvalDat}] \wedge \text{DiffCondAll}[\text{EvalStrFam}^{\text{diff}}]. \end{aligned}$$

Consequently,

$$\begin{aligned} \text{DiffOperative}[\text{DiffEvalDat}] & \implies \text{LawOperative}[\text{LawEvalDat}] \\ & \implies \text{DynOperative}[\text{Ev}] \\ & \implies \text{ExactOperable}[l]. \end{aligned}$$

Under dynamical-description operability, the set of real-valued time-order embeddings is nonempty, and the dynamical-description object and dynamical-description witness object relative to each embedding exist uniquely. Under law-description operability, the observer-time law core relative to each embedding exists uniquely, and generated representability of law-evaluation time holds. Under differential-stage operability, for every

$$\rho_o \in \text{OrdEmbSet}_o(\mathbb{R}),$$

$$\begin{array}{ccc} \text{DynDesc}[\rho_o], & \text{DynWit}[\rho_o], & \\ \text{LawCore}^{\text{obs,time}}(\rho_o), & \text{ChartCompatDiffLaw}(\rho_o), & \\ & \text{EvalInput}^{\text{diff}} & \end{array}$$

are determined. The dynamical-description object, dynamical-description witness object, and observer-time law core exist uniquely; moreover, the order isomorphism between time-representation embedding images, value-jet-relation covariance, and time-representation-change jet-relation covariance hold.

Proof. The first equivalence follows from Definition 5.70, the second from Definition 5.102, and the third from Definition 5.132. The first step of the implication chain follows from Definition 5.132, the second from Definition 5.102 and Proposition 5.86, and the third from Definition 5.70.

Existence and uniqueness of the dynamical-description object and dynamical-description witness object follow from Theorem 5.112; existence and uniqueness of the observer-time law core and generated representability of law-evaluation time follow from Theorem 5.113. Differential-stage operability gives the five universal differential conditions. The chart-compatible differential law and the differential-failure evaluation input are determined by Definition 5.133 and Definition 5.136, respectively. The three covariance conclusions follow from Theorem 5.135. $\square$

5.6 Multi-object relational quantities and joint trajectory relations

Suppose that the finite object-specification index family

$$Z_0$$

of Definition 2.10 satisfies

$$|Z_0| \geq 2.$$

Assign the same realized operational configuration

$$\mathcal{X}$$

and observer-side information-coupling position

$$o \in U$$

to every component of the object-specification index family.

In what follows, each object-specific construction is evaluated relative to $\text{Ev}_\zeta := (\mathbf{l}, o, \zeta)$. When constructions for several objects are used together, the generated equivalence relations, state quotients and natural projections carry the object index ζ explicitly.

Definition 5.139 (Multi-object relation input data). Define the joint dynamical-description evaluation specification by

$$\text{Ev} := \iota_{\text{joint}}(\mathbf{l}, o, Z_0) \in \text{EvalSpecSp}.$$

Let the index set of multi-object relation types be

$$\text{JRelIdx}_{Z_0},$$

and assume

$$\mathbf{JRelIdx}_{Z_0} \neq \emptyset.$$

For each $\mathbf{r} \in \mathbf{JRelIdx}_{Z_0}$, the fixed $\mathfrak{B}$ -derived symbol $\mathfrak{B}$, the section input Z_0 , and the quantified $\mathbf{r}$ uniquely determine the subscript strings $(\mathfrak{B}, \mathbf{r}, Z_0)$ and $(\mathbf{r}, Z_0)$, which are omitted within the following definition. Let the source-side domain, relation-value set, and source-side multi-object relation satisfy

$$\begin{aligned} \mathbf{JointRelDom}^{\text{src}} &\subseteq \prod_{\zeta \in Z_0} \text{Tok}_{\mathfrak{B}}^*, \\ \mathbf{JointRel}^{\text{src}} &\subseteq \mathbf{JointRelDom}^{\text{src}} \times \mathbf{JointRelVal}. \end{aligned}$$

Define the source-side multi-object relation datum by

$$\mathbf{J}^{\text{src}} := (\mathbf{JointRelDom}^{\text{src}}, \mathbf{JointRelVal}, \mathbf{JointRel}^{\text{src}}),$$

and name its three components, in order, Dom , Val , Rel . Thus

$$\begin{aligned} \mathbf{x}_{\mathfrak{B}} &= (x_{\mathfrak{B}, \zeta})_{\zeta \in Z_0} \in \mathbf{J}^{\text{src}}.\text{Dom}, \\ (\mathbf{x}_{\mathfrak{B}}, v) &\in \mathbf{J}^{\text{src}}.\text{Rel} \end{aligned}$$

represents a correspondence between the family of source tokens associated with the $\mathfrak{B}$ -derived component of each member of the object-specification index family and $v \in \mathbf{J}^{\text{src}}.\text{Val}$. For the predicate type, use

$$\mathbf{J}^{\text{src}}.\text{Val} = \{*\}.$$

Define the multi-object relation input data by

$$\mathbf{JRelDat}_{o, Z_0} := (\text{Ev}, \text{RI}[Z_0], \mathbf{JRelIdx}_{Z_0}, (\mathbf{J}^{\text{src}}_{\mathbf{r}, Z_0})_{\mathbf{r} \in \mathbf{JRelIdx}_{Z_0}}).$$

Definition 5.140 (Relation-quotient descent module). For a set X , an equivalence relation $\equiv$ on it, a subset $A \subseteq X$, a set V , and a relation $S \subseteq A \times V$, define the relation-saturation condition by

$$\begin{aligned} \text{RelSat}[\equiv; A, S] &\iff \left[\forall x, x' \in X, x \equiv x' \right. \\ &\implies (x \in A \iff x' \in A) \\ &\quad \left. \wedge \forall v \in V, ((x, v) \in S \iff (x', v) \in S) \right]. \end{aligned}$$

Further suppose that a surjection $q : X \twoheadrightarrow Q$ satisfies

$$x \equiv x' \iff q(x) = q(x').$$

Define the relation-quotient descent module by

$$\text{RelQuot}[q; A, S] := (q[A], \{(q(x), v) \mid (x, v) \in S\}).$$

Lemma 5.141 (Representative criterion for the relation-quotient descent module). Suppose that a surjection $q : X \twoheadrightarrow Q$ satisfies

$$x \equiv x' \iff q(x) = q(x')$$

and that $\text{RelSat}[\equiv; A, S]$ holds. If

$$\text{RelQuot}[q; A, S] = (\overline{A}, \overline{S}),$$

then, for every $x \in X$ and $v \in V$,

$$q(x) \in \overline{A} \iff x \in A, \quad (q(x), v) \in \overline{S} \iff (x, v) \in S.$$

Moreover, $\overline{A} \subseteq Q$ and $\overline{S} \subseteq \overline{A} \times V$ satisfying these two representative criteria are unique.

Proof. In the forward direction, take the image representation of $q(x)$ given by a representative x' , and use $q(x) = q(x')$ together with the relation-saturation condition. The reverse direction follows from the definition of RelQuot . For uniqueness, surjectivity of q represents every element of Q as $q(x)$, and the two representative criteria determine pointwise membership in $\overline{A}$ and relation membership in $\overline{S}$. $\square$

Definition 5.142 (Multi-object relation representative-invariance condition). Take the multi-object relation input data

$$\text{JRelDat}_{o, Z_0}$$

of Definition 5.139 .

For two families of $\mathfrak{B}$ -derived source tokens

$$\begin{aligned} \mathbf{x}_{\mathfrak{B}} &= (x_{\mathfrak{B}, \zeta})_{\zeta \in Z_0}, \\ \mathbf{x}'_{\mathfrak{B}} &= (x'_{\mathfrak{B}, \zeta})_{\zeta \in Z_0}, \end{aligned}$$

define componentwise $\mathfrak{B}$ -representative equivalence by

$$\begin{aligned} \mathbf{x}_{\mathfrak{B}} &\sim_{\mathfrak{B}, o, Z_0}^{\text{dyn, joint}} \mathbf{x}'_{\mathfrak{B}} \\ &\iff \left[\forall \zeta \in Z_0, \quad x_{\mathfrak{B}, \zeta} \sim_{\mathfrak{B}, o; \zeta} x'_{\mathfrak{B}, \zeta} \right]. \end{aligned}$$

Define the multi-object relation representative-invariance condition by

$$\begin{aligned} &K_{\text{CanJointRelDesc}}^{\text{dyn}}(\text{JRelDat}_{o, Z_0}) \\ &\iff \left[\forall \mathbf{t} \in \text{JRelIdx}_{Z_0}, \quad \text{RelSat} \left[\sim_{\mathfrak{B}, o, Z_0}^{\text{joint}}; \text{J}_{\mathbf{t}, Z_0}^{\text{src}}.\text{Dom}, \text{J}_{\mathbf{t}, Z_0}^{\text{src}}.\text{Rel} \right] \right]. \end{aligned}$$

Here two multi-object state families have the same quotient image exactly when they are componentwise equivalent under $\sim_{\mathfrak{B}, o; \zeta}$. Hence this condition simultaneously expresses invariance of domain membership and relation membership under the componentwise representative equivalence defined immediately above.

Definition 5.143 (Multi-object state quotient projection). Take the multi-object relation input data

$$\text{JRelDat}_{o, Z_0}$$

of Definition 5.139 .

Define the multi-object state quotient projection along the object-specification index family by

$$\Theta_{\mathfrak{B}, o; Z_0}^{\text{can}} : \prod_{\zeta \in Z_0} \text{Tok}_{\mathfrak{B}}^* \rightarrow \prod_{\zeta \in Z_0} \text{Qu}_{\mathfrak{B}, o; \zeta}^{\text{can}}$$

and

$$\begin{aligned} &\Theta_{\mathfrak{B}, o; Z_0}^{\text{can}} \left((x_{\mathfrak{B}, \zeta})_{\zeta \in Z_0} \right) \\ &:= \left(\Theta_{\mathfrak{B}, o; \zeta}^{\text{can}}(x_{\mathfrak{B}, \zeta}) \right)_{\zeta \in Z_0}. \end{aligned}$$

Lemma 5.144 (Quotient descent of multi-object relations).

Assume $K_{\text{CanJointRelDesc}}(\text{JRelDat}_{o,Z_0})$. For each $\mathfrak{r} \in \text{JRelIdx}_{Z_0}$, set

$$\begin{aligned} & \text{J}_{o,\mathfrak{r},Z_0}^{\text{can}} \\ & := \text{RelQuot} [\Theta_{\mathfrak{B},o;Z_0}^{\text{can}}; \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Dom}, \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Rel}] , \end{aligned}$$

and name its two components, in order, Dom, Rel . Then, for arbitrary

$$\begin{aligned} \mathbf{x}_{\mathfrak{B}} & \in \prod_{\zeta \in Z_0} \text{Tok}_{\mathfrak{B}}^*, \\ v & \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Val}, \\ \Theta_{\mathfrak{B},o;Z_0}^{\text{can}}(\mathbf{x}_{\mathfrak{B}}) & \in \text{J}_{o,\mathfrak{r},Z_0}^{\text{can}}.\text{Dom} \\ \iff \mathbf{x}_{\mathfrak{B}} & \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Dom}, \\ (\Theta_{\mathfrak{B},o;Z_0}^{\text{can}}(\mathbf{x}_{\mathfrak{B}}), v) & \in \text{J}_{o,\mathfrak{r},Z_0}^{\text{can}}.\text{Rel} \\ \iff (\mathbf{x}_{\mathfrak{B}}, v) & \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Rel}. \end{aligned}$$

The quotient-level domain and quotient-level relation satisfying these two criteria are unique.

Proof. The multi-object state quotient projection is surjective, and equality of the images of two of its arguments is equivalent to

$$\mathbf{x}_{\mathfrak{B}} \sim_{\mathfrak{B},o,Z_0}^{\text{joint}} \mathbf{x}'_{\mathfrak{B}}.$$

Since the assumption is the relation-saturation condition for each $\mathfrak{r}$, applying Lemma 5.141 gives the two representative criteria and uniqueness. $\square$

Definition 5.145 (Quotient-level multi-object relation). For each $\mathfrak{r} \in \text{JRelIdx}_{Z_0}$, call the relation

$$\text{J}_{o,\mathfrak{r},Z_0}^{\text{can}}.\text{Rel}$$

uniquely determined by Lemma 5.144 the quotient-level multi-object relation corresponding to the relation type $\mathfrak{r}$.

Lemma 5.146 (Common construction of the joint-time quotients). Assume $\text{ExactOperable}[I]$. Under $K_{\text{JointCTime}}(\text{Ev})$, the canonical time quotients, natural projections and generated time preorders agree across all objects. If

$$K_{\text{GenTimeOrdAntiSym}}(\text{Ev}_{\zeta}), \quad K_{\text{ObsTimeIncTrans}}(\text{Ev}_{\zeta})$$

hold for every object, the order-time quotients, order-time projections and strict linear orders also agree. The object index may therefore be omitted from these common time objects. Agreement of the individual state quotients or individual trajectory domains is not required.

Proof. The same equivalence relation on the same set defines the same quotient and natural projection. The same source order is mapped by the same projection, so the generating edges and their reflexive transitive closures agree. Under the antisymmetry condition, the canonical time partial orders agree, as do their strict parts, incomparability relations and quotients by incomparability. The definitions of the order-time projection and strict linear order give the conclusion. $\square$

Definition 5.147 (Joint scalar-time trajectory). Assume $\text{ExactOperable}[\mathbb{I}]$ and $K_{\text{JointCTime}}(\text{Ev})$. For each $\zeta \in Z_0$, define the complete evaluation specification by

$$\text{Ev}_\zeta := (\mathbb{I}, o, \zeta) \in \text{EvSpec},$$

and assume

$$\forall \zeta \in Z_0, \quad \text{DynTrjFactCondAll}[\text{Ev}_\zeta].$$

Write the individual trajectory obtained from Proposition 5.66 as

$$\text{TrjMap}_{o;\zeta}^{\text{dyn,time,ord}} : \text{TrjDom}_{o;\zeta}^{\text{dyn,time,ord}} \longrightarrow \text{Qu}_{\mathfrak{B},o;\zeta}^{\text{can}}.$$

Define the domain component and map component of the order-time joint-trajectory datum by

$$\begin{aligned} \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom} &:= \bigcap_{\zeta \in Z_0} \text{TrjDom}_{o;\zeta}^{\text{time,ord}}, \\ \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Map} : \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom} &\longrightarrow \prod_{\zeta \in Z_0} \text{Qu}_{\mathfrak{B},o;\zeta}^{\text{can}}, \\ \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Map}(s) &:= \left(\text{TrjMap}_{o;\zeta}^{\text{time,ord}}(s) \right)_{\zeta \in Z_0}. \end{aligned}$$

Definition 5.148 (Joint trajectory relation). For the multi-object relation input data of Definition 5.139, assume

$$K_{\text{CanJointRelDesc}}(\text{JRelDat}_{o,Z_0}).$$

Use the quotient-level multi-object relation of Definition 5.145 and the joint scalar-time trajectory of Definition 5.147.

For each

$$\mathfrak{r} \in \text{JRelIdx}_{Z_0},$$

define the relation component of the order-time joint-trajectory datum by

$$\begin{aligned} &\text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_\mathfrak{r} \\ &:= \left\{ (s, v) \mid \begin{array}{l} s \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom}, \\ v \in \text{J}^{\text{src}}_{\mathfrak{r},Z_0}.\text{Val}, \\ (\text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Map}(s), v) \\ \in \text{J}^{\text{can}}_{o,\mathfrak{r},Z_0}.\text{Rel} \end{array} \right\}. \end{aligned}$$

Consequently,

$$\begin{aligned} &\text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_\mathfrak{r} \\ &\subseteq \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom} \times \text{J}^{\text{src}}_{\mathfrak{r},Z_0}.\text{Val}. \end{aligned}$$

For a predicate-valued multi-object relation, use

$$\text{J}^{\text{src}}_{\mathfrak{r},Z_0}.\text{Val} = \{*\};$$

then

$$(s, *) \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_\mathfrak{r}$$

represents satisfaction of the predicate on the joint state trajectory.

Proposition 5.149 (Quotient descent of multi-object relations and restriction to a joint trajectory). Assume

$$\begin{aligned} & \text{ExactOperable}[\mathbb{I}], \\ & K_{\text{JointCTime}}(\text{Ev}), \\ & \forall \zeta \in Z_0, \text{DynTrjFactCondAll}[\text{Ev}_\zeta], \\ & K_{\text{CanJointRelDesc}}(\text{JRelDat}_{o,Z_0}). \end{aligned}$$

For each $\mathfrak{r} \in \text{JRelIdx}_{Z_0}$, the quotient-level multi-object relation, joint scalar-time trajectory, and joint trajectory relation exist uniquely.

Furthermore, for every

$$s \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom},$$

the equality

$$\Theta_{\mathfrak{B},o,Z_0}^{\text{can}}(\mathbf{x}_{\mathfrak{B}}) = \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Map}(s)$$

holds for some $\mathbf{x}_{\mathfrak{B}} \in \prod_{\zeta \in Z_0} \text{Tok}_{\mathfrak{B}}^*$, and for every such element and arbitrary $v \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Val}$,

$$(s, v) \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_{\mathfrak{r}} \iff (\mathbf{x}_{\mathfrak{B}}, v) \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Rel}.$$

Proof. Each individual trajectory is unique by Proposition 5.66, and its product map on the common domain is also unique. Since the quotient-level multi-object relation is unique by Lemma 5.144, the inverse-image relation of Definition 5.148 is also unique.

Surjectivity of the multi-object state quotient projection gives the displayed $\mathbf{x}_{\mathfrak{B}}$. Composing the defining equivalence of the joint trajectory relation with the representative criterion in Lemma 5.144 gives the final equivalence. $\square$

Definition 5.150 (Joint trajectory relation over real-valued time). Suppose that, for each $\zeta \in Z_0$,

$$K_{\text{ObsTimeRealOrdEmb}}(\text{Ev}_\zeta)$$

holds, and take

$$\rho_o \in \text{OrdEmbSet}_o(\mathbb{R}).$$

Define the domain component and relation component of the real-valued joint-trajectory datum by

$$\begin{aligned} \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Dom} &:= \rho_o [\text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Dom}] \subseteq \mathbb{R}, \\ \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Rel}_{\mathfrak{r}} &:= \{(\rho_o(s), v) \mid (s, v) \in \text{JTrj}_{o,Z_0}^{\text{ord}}.\text{Rel}_{\mathfrak{r}}\}. \end{aligned}$$

Define the joint law-time domain by

$$\mathbb{T}_{o,Z_0}^{\text{joint}}[\rho_o] := \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Dom}.$$

Definition 5.151 (Observational realization of a joint relation value). For each $\mathfrak{r} \in \text{JRelIdx}_{Z_0}$, the section inputs o, Z_0 and the quantified $\mathfrak{r}$ uniquely determine the subscript string $(o, \mathfrak{r}, Z_0)$, which is omitted within the following definition. Define the domain of observational realization of a joint relation value by

$$\begin{aligned} \text{JRelObs}[\rho_o].\text{Dom} &:= \left\{ t \in \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Dom} \mid \right. \\ & \quad \left. \exists v \in \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Val}, (t, v) \in \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Rel}_{\mathfrak{r}} \right\}. \end{aligned}$$

An observational realization of a joint relation value is a map satisfying

$$\text{JRelObs}[\rho_o].\text{Curve} : \text{JRelObs}[\rho_o].\text{Dom} \longrightarrow \text{J}_{\mathfrak{r},Z_0}^{\text{src}}.\text{Val}$$

such that, for every domain element t ,

$$(t, \text{JRelObs}[\rho_o].\text{Curve}(t)) \in \text{JTrj}_{o,Z_0}^{\text{real}}[\rho_o].\text{Rel}.$$

The family of observational realizations of joint relation values is supplied as law-evaluation input. Define the two-component datum for observational realization of a joint relation value by

$$\text{JRelObs}[\rho_o] := (\text{JRelObs}[\rho_o].\text{Dom}, \text{JRelObs}[\rho_o].\text{Curve}),$$

and name its two components, in order, Dom , Curve .

Definition 5.152 (Candidate relation structure for laws referring to joint relations). Fix the common time representation from Definition 5.150 and the family of observation realizations from Definition 5.151. For every $\zeta \in Z_0$, assume in addition that $\text{DynOperative}[\text{Ev}_\zeta]$ holds. The individual canonical observable-time curves used below are then defined. Define the multi-object canonical-observable index set by

$$\text{ObsIdx}_{o,Z_0}^{\text{dyn,quot,joint}} := \bigsqcup_{\zeta \in Z_0} (\{\zeta\} \times \text{ObsIdx}_{o,\zeta}^{\text{quot}}).$$

For a nonempty LawIdx , $a \in \text{LawIdx}$, and $s \in \text{IOSide}$, let the observable indices and joint-relation indices be

$$\begin{aligned} \text{ObsIdx}_{a,s}^{\text{joint}} &\subseteq \text{ObsIdx}_{o,Z_0}^{\text{quot,joint}}, \\ \text{RelIdx}_{a,s}^{\text{joint}} &\subseteq \text{JRelIdx}_{Z_0}. \end{aligned}$$

The input and output indices are obtained by taking $s = \text{in}, \text{out}$.

Let the individual law-evaluation time domain satisfy

$$\begin{aligned} \text{H}_a^{\text{joint}}[\rho_o] &\subseteq \text{T}_{o,Z_0}^{\text{joint}}[\rho_o] \\ &\cap \bigcap_{\substack{s \in \text{IOSide} \\ \tau \in \text{RelIdx}_{a,s}^{\text{joint}}}} \text{JRelObs}_{o,\tau,Z_0}[\rho_o].\text{Dom}. \end{aligned}$$

Define the value space and generated value family on each side by

$$\begin{aligned} \text{ValSp}_{a,s}^{\text{joint}} &:= \left(\prod_{(\zeta,\ell) \in \text{ObsIdx}_{a,s}^{\text{joint}}} \text{ObsVal}_{o,\zeta,\ell}^{\text{can}} \right) \times \left(\prod_{\tau \in \text{RelIdx}_{a,s}^{\text{joint}}} \text{J}^{\text{src}}_{\tau,Z_0}.\text{Val} \right), \\ \text{Val}_{a,s}^{\text{joint}}[\rho_o] : \text{H}_a^{\text{joint}}[\rho_o] &\longrightarrow \text{ValSp}_{a,s}^{\text{joint}}, \\ \text{Val}_{a,s}^{\text{joint}}[\rho_o](t) &:= \left((\text{ObsCurve}_{o,\zeta,\ell}^{\text{real}}[\rho_o](t))_{(\zeta,\ell) \in \text{ObsIdx}_{a,s}^{\text{joint}}}, \right. \\ &\quad \left. (\text{JRelObs}_{o,\tau,Z_0}[\rho_o].\text{Curve}(t))_{\tau \in \text{RelIdx}_{a,s}^{\text{joint}}} \right). \end{aligned}$$

Set the input and output components to

$$\begin{aligned} (\text{InValSp}_a^{\text{joint}}, \text{OutValSp}_a^{\text{joint}}) &:= (\text{ValSp}_{a,\text{in}}^{\text{joint}}, \text{ValSp}_{a,\text{out}}^{\text{joint}}), \\ (\text{InVal}_a^{\text{joint}}, \text{OutVal}_a^{\text{joint}})[\rho_o] &:= (\text{Val}_{a,\text{in}}^{\text{joint}}[\rho_o], \text{Val}_{a,\text{out}}^{\text{joint}}[\rho_o]). \end{aligned}$$

Let the individual law-application domain and law-candidate relation satisfy

$$\begin{aligned} \text{E}_a^{\text{joint}}[\rho_o] &\subseteq \text{T}_{o,Z_0}^{\text{joint}}[\rho_o] \times \text{InValSp}_a^{\text{joint}}, \\ \text{R}_a^{\text{joint}}[\rho_o] &\subseteq \text{E}_a^{\text{joint}}[\rho_o] \times \text{OutValSp}_a^{\text{joint}}. \end{aligned}$$

In the same order as the components of a general law, set

$$\begin{aligned} L_a^{\text{joint}}[\rho_o] := & \left(\text{InValSp}_a^{\text{joint}}, \text{OutValSp}_a^{\text{joint}}, \text{InVal}_a^{\text{joint}}[\rho_o], \text{OutVal}_a^{\text{joint}}[\rho_o], \right. \\ & E_a^{\text{joint}}[\rho_o], R_a^{\text{joint}}[\rho_o], H_a^{\text{joint}}[\rho_o], \left(\mathbb{T}_{o, Z_0}^{\text{joint}}[\rho_o] \times \text{InValSp}_a^{\text{joint}} \right) \times \text{OutValSp}_a^{\text{joint}}, \\ & \left. t \longmapsto \left((t, \text{InVal}_a^{\text{joint}}[\rho_o](t)), \text{OutVal}_a^{\text{joint}}[\rho_o](t) \right) \right). \end{aligned}$$

Let

$$\begin{aligned} \mathcal{U}^{\text{joint}}[\rho_o] &\subseteq \mathcal{P}\left(\mathbb{T}_{o, Z_0}^{\text{joint}}[\rho_o]\right), \\ \mathcal{F}^{\text{joint}}[\rho_o] &\subseteq \prod_{a \in \text{LawIdx}} \text{Bij}(\text{EvalTupleSp}[L_a^{\text{joint}}[\rho_o]]). \end{aligned}$$

Define the candidate relation structure for laws referring to joint relations and the law-evaluation datum by

$$\begin{aligned} \text{CandRelStr}_{o, Z_0}^{\text{law, joint}}[\rho_o] &:= \left(\mathbb{T}_{o, Z_0}^{\text{joint}}[\rho_o], \text{LawIdx}, \right. \\ &\quad \left. (L_a^{\text{joint}}[\rho_o])_{a \in \text{LawIdx}}, \mathcal{U}^{\text{joint}}[\rho_o], \mathcal{F}^{\text{joint}}[\rho_o] \right) \\ &\in \text{CandRelStrSp}^{\text{law}}, \end{aligned}$$

and, further, by

$$\begin{aligned} \text{LawEvalDat}_{o, Z_0}^{\text{joint}}[\rho_o] &:= \left(\text{Ev}, \text{CandRelStr}_{o, Z_0}^{\text{law, joint}}[\rho_o] \right) \\ &\in \text{LawEvalDatSp}. \end{aligned}$$

Proposition 5.153 (Connection of the candidate relation structure for laws referring to joint relations to the five conditions). The law-evaluation datum

$$\text{LawEvalDat}_{o, Z_0}^{\text{joint}}[\rho_o]$$

belongs to

$$\text{LawEvalDatSp}.$$

Consequently, the five-condition vector

$$\mathbf{K}^{\text{law}} \left[\text{LawEvalDat}_{o, Z_0}^{\text{joint}}[\rho_o] \right]$$

of Definition 5.99 is defined.

Proof. Each law component in Definition 5.152 belongs to

$$\text{LawCompSp} \left[\text{JTr}_{o, Z_0}^{\text{real}}[\rho_o].\text{Dom} \right],$$

and the outer five components satisfy the type of $\text{CandRelStrSp}^{\text{law}}$. Hence this law-evaluation datum belongs to LawEvalDatSp . Applying Definition 5.99 defines the displayed five-condition vector. $\square$

Proposition 5.154 (Connection of the multi-object relation condition series). The multi-object relation condition series is connected as follows.

1.

$$K_{\text{CanJointRelDesc}}(\text{JRelDat}_{o,Z_0})$$

is evaluated as the representative-invariant relation-descent condition at the dynamical-description stage.

2.

$$K_{\text{JointCTime}}(\text{Ev})$$

is the condition for a common base of the object-specific time constructions. For each

$$\zeta \in Z_0,$$

$$\text{DynTrjFactCondAll}[\text{Ev}_\zeta], \quad K_{\text{ObsTimeRealOrdEmb}}(\text{Ev}_\zeta)$$

are evaluated as the generation conditions for the corresponding individual trajectory and common time representation.

3. Separation of object correspondences by records derived from $\mathfrak{D}$ is evaluated as the record-time resolution condition derived from $\mathfrak{D}$ and the object-state distinguishability condition derived from $\mathfrak{B}$ at the continuum-approximation stage.
4. Failure of the candidate relation structure for laws referring to joint relations is evaluated through failure of the five conditions in Proposition 5.153 .
5. Use the joint law datum and the representation used in its construction as the base pair for the transported family in Definition 5.107. Given a differential-evaluation-structure family belonging to this same base pair, failure of a differential law containing joint relation values is evaluated through the differential-stage condition group.

Thus the respective failure positions of multi-object relation descent, individual-trajectory generation, record separation, laws referring to joint relations, and differential laws are identified by their corresponding condition symbols.

Give the realized operational configuration

$$\mathcal{X}$$

and the family of typed comparison rules

$$\mathcal{J}_{\text{cmp}}.$$

For the finite object-specification index family

$$Z_0$$

of Definition 2.10 , take an observer-side information-coupling position

$$o \in U$$

and an object-specification index

$$\zeta \in Z_0.$$

Take an exactly operable complete evaluation specification

$$\mathbf{Ev} = (\mathbf{l}, o, \zeta) \in \mathbf{EvSpec}^{\text{exact}}$$

of Definition 5.2 . In what follows, only within a local context in which a single complete evaluation specification is fixed, each symbol root is first displayed in full in its first formula, after which the subscript pair formed by the second and third components uniquely recoverable from that complete evaluation specification is omitted. Independent identifiers are not omitted, and the component pair is not omitted when multiple complete evaluation specifications are treated simultaneously.

By Theorem 5.138 , the respective establishment conditions for dynamical, law-bearing, and differential descriptions are given by

$$\begin{aligned} & \mathbf{ExactOperable} [\mathbf{l}] \\ & \quad \wedge \mathbf{DynCondAll} [\mathbf{Ev}] \\ & \quad \Longleftrightarrow \mathbf{DynOperative} [\mathbf{Ev}] , \end{aligned}$$

$$\begin{aligned} & \mathbf{DynOperative} [\mathbf{Ev}] \\ & \quad \wedge \mathbf{LawCondAll} [\mathbf{LawEvalDat}_{o,\zeta}] \\ & \quad \Longleftrightarrow \mathbf{LawOperative} [\mathbf{LawEvalDat}_{o,\zeta}] , \end{aligned}$$

and

$$\begin{aligned} & \mathbf{LawOperative} [\mathbf{LawEvalDat}] \\ & \quad \wedge \mathbf{DiffCondAll} [\mathbf{EvalStrFam}_{o,\zeta}^{\text{diff}}] \\ & \quad \Longleftrightarrow \mathbf{DiffOperative} [\mathbf{DiffEvalDat}_{o,\zeta}] . \end{aligned}$$

Under dynamical-description operability, the observer-time quotient derived from $\mathfrak{C}$, object-state quotient derived from $\mathfrak{B}$, canonical trajectory relation derived from $\mathfrak{CD}\mathfrak{B}$, graph trajectory, family of canonical observable functions, law-admissible time representation, dynamical-description object, and dynamical-description witness object are determined.

Under law-description operability, the observer-time law core exists uniquely, and generated representability of law-evaluation time holds.

Under differential-stage operability, the chart-compatible differential law and differential-failure evaluation input are determined, and the order isomorphism between time-representation embedding images, value-jet-relation covariance, and time-representation-change jet-relation covariance hold.

Suppose that the finite object-specification index family

$$Z_0$$

of Definition 2.10 satisfies

$$|Z_0| \geq 2.$$

Also assume agreement of the joint-time bases,

$$K_{\text{JointCTime}}(\iota_{\text{joint}}(\mathbf{l}, o, Z_0)) .$$

For each

$$\zeta \in Z_0,$$

set the complete evaluation specification to

$$\text{Ev} = (\mathbf{l}, o, \zeta),$$

and suppose that the trajectory-factorization condition bundle

$$\text{DynTrjFactCondAll} [\text{Ev}]$$

of Definition 5.67 holds.

If, in addition,

$$K_{\text{CanJointRelDesc}} (\text{JRelDat}_{o, Z_0})$$

holds, Proposition 5.149 gives the unique existence of the quotient-level multi-object relation, joint scalar-time trajectory, and joint trajectory relation.

For each

$$\zeta \in Z_0,$$

set the complete evaluation specification to

$$\text{Ev} = (\mathbf{l}, o, \zeta)$$

and suppose that

$$K_{\text{ObsTimeRealOrdEmb}} (\text{Ev})$$

holds and that

$$\text{RepEmb}_o^{\text{time}} \in \text{OrdEmbSet}_o^{\text{time}} (\mathbb{R}).$$

Definition 5.150 then determines the joint trajectory relation over real-valued time.

When $\text{DynOperative}[(\mathbf{l}, o, \zeta)]$ holds for every $\zeta \in Z_0$ and the family of observational realizations of joint relation values from Definition 5.151 is supplied as law-evaluation input, Definition 5.152 determines the candidate relation structure for laws referring to joint relations.

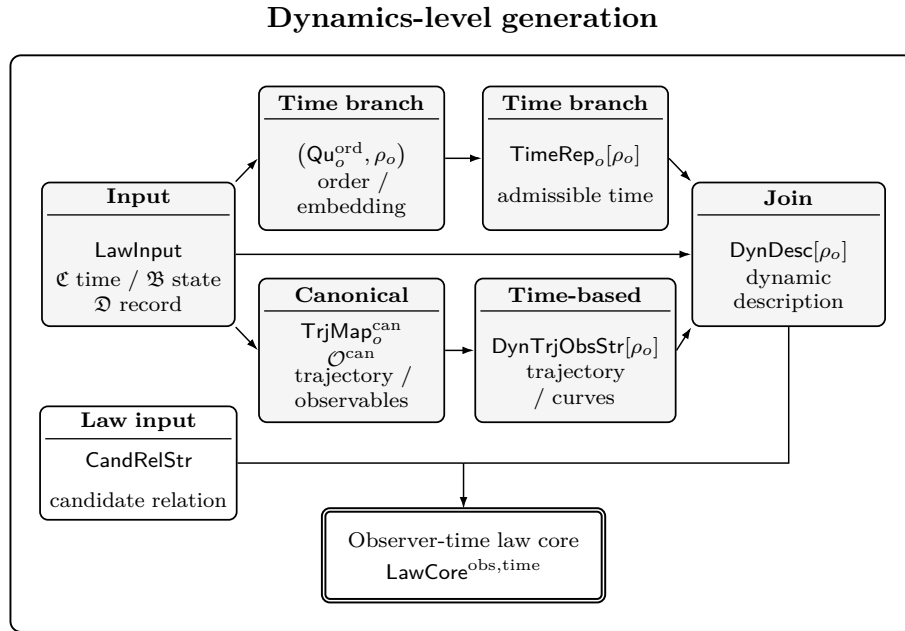


Figure 11: Generation at the dynamical-description stage. Under the conditions for each stage, the observer-time law-input datum yields the time structure and the trajectory-and-observation structure, which form the dynamical-description object. This object and the law-candidate relation structure determine the observer-time law core.

6 Localization of Validity, Failure, and Approximation

Take as input the complete evaluation specification of Definition 5.2. Validity and failure localization specifies the invalidating conditions and their dependencies for the comparison, representation, continuum-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation series, and generates a six-series failure signature, scale boundaries, and operational audit data. Engineering audit parameters provide the formal evidence used to evaluate failure in each series.

In what follows, only within a local context in which a single complete evaluation specification is fixed, each symbol root is first displayed in full in its first formula, after which the subscript pair formed by the second and third components uniquely recoverable from that complete evaluation specification is omitted. Independent identifiers are not omitted, and the component pair is not omitted when multiple complete evaluation specifications are treated simultaneously.

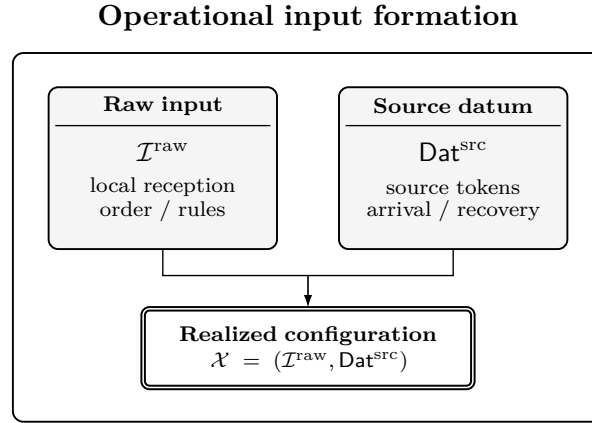


Figure 12: Formation of operational inputs. Local displays, records, object-state differences, and observer-side coupling materials are assembled as a realized operational configuration passed to the comparison and dynamical-description stages.

6.1 Basic Definitions of Validity Domains and Localization

Definition 6.1 (Validity localization). For a set

$$A$$

and a binary predicate

$$P : A \longrightarrow \{\perp, \top\}$$

define its validity localization by

$$\text{Val}_A(P) := \{a \in A \mid P[a] = \top\}$$

Further, for a set

$$B$$

and a binary predicate

$$Q : B \times A \longrightarrow \{\perp, \top\}$$

define the validity localization obtained by substituting

$$b \in B$$

into its first argument by

$$\text{Val}_A(Q; b) := \{a \in A \mid Q[b; a] = \top\}$$

Suppose that a set of complete evaluation specifications

$$\text{EvSpec},$$

a condition-index set

$$I$$

and a family of binary condition predicates

$$K_i : \text{EvSpec} \longrightarrow \{\perp, \top\}, \quad i \in I$$

are given. For any

$$J \subseteq I$$

define

$$\begin{aligned} \text{Val}_{\text{EvSpec}}((K_i)_{i \in J}) &:= \{\text{Ev} \in \text{EvSpec} \mid \forall i \in J, K_i[\text{Ev}] = \top\} \\ &= \bigcap_{i \in J} \text{Val}_{\text{EvSpec}}(K_i) \end{aligned}$$

Definition 6.2 (Approximation localization). For a set of complete evaluation specifications

$$\text{EvSpec}$$

and a totally ordered set

$$(\Lambda, \leq_\Lambda)$$

define the set of scale-indexed complete evaluation specifications by

$$\text{EvSpec}_\Lambda := \text{EvSpec} \times \Lambda$$

Suppose that an approximation-constraint index set

$$I^{\text{approx}}$$

and a family of binary condition predicates

$$K_\alpha^{\text{approx}} : \text{EvSpec}_\Lambda \longrightarrow \{\perp, \top\}, \quad \alpha \in I^{\text{approx}}$$

are given. For each

$$\text{Ev} \in \text{EvSpec}, \quad \alpha \in I^{\text{approx}}$$

the approximation localization is, by the two-argument form of Definition 6.1,

$$\text{Val}_\Lambda(K_\alpha^{\text{approx}}; \text{Ev}) = \{\lambda \in \Lambda \mid K_\alpha^{\text{approx}}[\text{Ev}; \lambda] = \top\}$$

In the remainder of this chapter, a complete evaluation specification Ev is fixed with its type in the formal environment. After the full form $X[\text{Ev}; \nu]$ has been displayed once, each subsequent display for that same Ev is abbreviated as $X[\nu]$. The subscripted form

$X_{\text{Ev},\nu}$ is likewise abbreviated as X_ν under the same condition. This abbreviation is not used on the left-hand side of a definition.

Define the scale-monotonicity condition by

$$\forall \lambda, \mu \in \Lambda, \quad \lambda \leq_\Lambda \mu \wedge K_\alpha^{\text{approx}}[\mu] = \top \implies K_\alpha^{\text{approx}}[\lambda] = \top$$

Under this condition, if

$$\mu \in \text{Val}_\Lambda(K_\alpha^{\text{approx}}; \text{Ev}), \quad \lambda \leq_\Lambda \mu$$

then

$$\lambda \in \text{Val}_\Lambda(K_\alpha^{\text{approx}}; \text{Ev})$$

Consequently, this approximation localization is an initial segment of

$$(\Lambda, \leq_\Lambda)$$

Definition 6.3 (Localization of failure positions). Let

$$(\text{FailTok}, \preceq^{\text{fail,dep}}),$$

be a finite partially ordered set, let

$$\text{FailTok}^{\text{strict}}, \quad \text{FailTok}^{\text{approx}},$$

be its subsets, and define the evaluation-state set by

$$\text{FailState} := \{\text{sat}, \text{failed}, \text{unformed}, \text{unevaluable}\}$$

where the four elements are pairwise distinct. Suppose further that an evaluation-state map

$$\text{FailStateOf} : (\text{FailTok}^{\text{strict}} \times \text{EvSpec}) \sqcup (\text{FailTok}^{\text{approx}} \times \text{EvSpec}_\Lambda) \longrightarrow \text{FailState}$$

is given. Define the failure-position localizations on the strict-series side and the continuum-approximation-series side by

$$\begin{aligned} \text{FailLoc}^{\text{strict}}(\text{Ev}) &:= \text{Min}_{\preceq^{\text{fail,dep}}} \{t \in \text{FailTok}^{\text{strict}} \mid \text{FailStateOf}(t; \text{Ev}) = \text{failed}\}, \\ \text{FailLoc}^{\text{approx}}(\text{Ev}, \lambda) &:= \text{Min}_{\preceq^{\text{fail,dep}}} \{t \in \text{FailTok}^{\text{approx}} \mid \text{FailStateOf}(t; \text{Ev}, \lambda) = \text{failed}\} \end{aligned}$$

Each value is the set of all minimal failure-position tokens whose evaluation state is

failed

in the corresponding series. The series type of each token is exactly one of comparison, representation, dynamical description, continuum approximation, law representation, or differential consistency and law-form preservation.

Definition 6.4 (Validity domain). Let the strict establishment condition and the approximation establishment condition be

$$\begin{aligned} K_{\text{strict}}^{\text{fail}} : \text{EvSpec} &\longrightarrow \{\perp, \top\}, \\ K_{\text{approx}}^{\text{fail}} : \text{EvSpec}_\Lambda &\longrightarrow \{\perp, \top\} \end{aligned}$$

respectively. By Definition 6.1, define

$$\begin{aligned}\mathcal{V}^{\text{strict}} &:= \text{Val}_{\text{EvSpec}}(K_{\text{strict}}^{\text{fail}}), \\ \mathcal{V}^{\text{approx}} &:= \text{Val}_{\text{EvSpec}_\Lambda}(K_{\text{approx}}^{\text{fail}}), \\ \mathcal{V}^{\text{full}} &:= (\mathcal{V}^{\text{strict}} \times \Lambda) \cap \mathcal{V}^{\text{approx}}\end{aligned}$$

and call these the strict validity domain, approximation validity domain, and full-evaluation validity domain, respectively.

Lemma 6.5 (Inclusion maximum of a family of sets). For a set

$$A$$

and a family of sets

$$\mathfrak{A} \subseteq \mathcal{P}(A)$$

a necessary and sufficient condition for

$$\text{Max}_\subseteq \mathfrak{A}$$

to exist is

$$\bigcup_{B \in \mathfrak{A}} B \in \mathfrak{A}$$

In this case,

$$\text{Max}_\subseteq \mathfrak{A} = \bigcup_{B \in \mathfrak{A}} B$$

and the maximum element is unique.

Proof. If a maximum element

$$B^{\text{max}} \in \mathfrak{A}$$

exists, every

$$B \in \mathfrak{A}$$

satisfies

$$B \subseteq B^{\text{max}}$$

and

$$\bigcup_{B \in \mathfrak{A}} B = B^{\text{max}} \in \mathfrak{A}.$$

Conversely, if this union belongs to

$$\mathfrak{A}$$

then it is a maximum element containing every member. Uniqueness follows from anti-symmetry of set inclusion. $\square$

Definition 6.6 (Maximum validity domain). Let the nonempty family of admissible subdomains of a validity domain

$$\mathcal{V}$$

be

$$\mathfrak{V}_{\text{adm}}(\mathcal{V}) \subseteq \mathcal{P}(\mathcal{V})$$

When

$$\mathcal{V}^{\max} \in \mathfrak{V}_{\text{adm}}(\mathcal{V})$$

satisfies

$$\forall \mathcal{V}' \in \mathfrak{V}_{\text{adm}}(\mathcal{V}), \quad \mathcal{V}' \subseteq \mathcal{V}^{\max}$$

the object

$$\mathcal{V}^{\max} = \text{Max}_{\subseteq} \mathfrak{V}_{\text{adm}}(\mathcal{V})$$

is called the maximum validity domain.

Proposition 6.7 (Existence condition for the maximum validity domain). A necessary and sufficient condition for the maximum validity domain to exist is

$$\bigcup_{\mathcal{V}' \in \mathfrak{V}_{\text{adm}}(\mathcal{V})} \mathcal{V}' \in \mathfrak{V}_{\text{adm}}(\mathcal{V})$$

In this case,

$$\mathcal{V}^{\max} = \bigcup_{\mathcal{V}' \in \mathfrak{V}_{\text{adm}}(\mathcal{V})} \mathcal{V}'$$

and it is unique.

Proof. Apply Lemma 6.5 to

$$A = \mathcal{V}, \quad \mathfrak{A} = \mathfrak{V}_{\text{adm}}(\mathcal{V})$$

□

Definition 6.8 (Maximal effective interval). For a totally ordered set

$$(\Lambda, \leq_{\Lambda})$$

and a subset

$$B \subseteq \Lambda$$

define the family of initial subsets contained in that subset by

$$\text{Init}_{\Lambda}(B) := \{\mathcal{I} \subseteq B \mid \forall \mu \in \mathcal{I}, \forall \lambda \in \Lambda, \lambda \leq_{\Lambda} \mu \implies \lambda \in \mathcal{I}\}$$

For each

$$\text{Ev} \in \text{EvSpec}$$

define the family of admissible initial intervals by

$$\mathfrak{I}_{\text{init}}[\text{Ev}] := \text{Init}_{\Lambda}(\text{Val}_{\Lambda}(K_{\text{approx}}^{\text{fail}}; \text{Ev}))$$

The object

$$\mathcal{I}^{\max}[\text{Ev}] = \text{Max}_{\subseteq} \mathfrak{I}_{\text{init}}[\text{Ev}]$$

is called the maximal effective interval at

$$\text{Ev}$$

Proposition 6.9 (Existence of the maximal effective interval). For any complete evaluation specification $\text{Ev} \in \text{EvSpec}$, the maximal effective interval exists uniquely, and

$$\bigcup_{\mathcal{I} \in \mathcal{I}_{\text{init}}[\text{Ev}]} \mathcal{I} \in \mathcal{I}_{\text{init}}[\text{Ev}]$$

Further, if the scale-monotonicity condition

$$\forall \lambda, \mu \in \Lambda, \quad \lambda \leq_{\Lambda} \mu \wedge K_{\text{approx}}^{\text{fail}}[\text{Ev}; \mu] = \top \implies K_{\text{approx}}^{\text{fail}}[\lambda] = \top$$

holds, then

$$\mathcal{I}^{\text{max}}[\text{Ev}] = \text{Val}_{\Lambda} (K_{\text{approx}}^{\text{fail}}; \text{Ev})$$

and it is unique. Its maximum endpoint exists if and only if

$$\text{Max}(\mathcal{I}^{\text{max}}[\text{Ev}])$$

exists.

Proof. Because the empty set belongs to

$$\mathcal{I}_{\text{init}}[\text{Ev}]$$

this family is nonempty. Its union is also an initial subset contained in the approximation-validity domain and therefore belongs to the same family. Applying Lemma 6.5 to

$$A = \Lambda, \quad \mathfrak{A} = \mathcal{I}_{\text{init}}[\text{Ev}]$$

yields existence and uniqueness of the maximal effective interval.

Under the scale-monotonicity condition, the approximation localization of Definition 6.2 is itself an admissible initial interval and contains all of its members. It therefore coincides with the maximal effective interval. The assertion concerning the maximum endpoint follows from the definition. $\square$

Definition 6.10 (Semantic-series-tagged failure-position token system). Define the sets of scale-invariant series, continuum-approximation series, and all failure series by

$$\begin{aligned} \text{FailSeries}^{\text{strict}} &:= \{\text{cmp}, \text{repr}, \text{dyn}, \text{law}, \text{diff}\}, \\ \text{FailSeries}^{\text{approx}} &:= \{\text{approx}\}, \\ \text{FailSeries} &:= \text{FailSeries}^{\text{strict}} \sqcup \text{FailSeries}^{\text{approx}} \end{aligned}$$

respectively. For a series

$$\tau \in \text{FailSeries}$$

define the number of failure positions, the index set, and the evaluation-specification type by

$$\begin{aligned} n_{\tau}^{\text{fail}} &:= \begin{cases} 8, & \tau \in \{\text{cmp}, \text{dyn}\}, \\ 3, & \tau = \text{repr}, \\ 9, & \tau = \text{approx}, \\ 5, & \tau \in \{\text{law}, \text{diff}\}, \end{cases} \\ I_{\tau}^{\text{fail}} &:= \{1, \dots, n_{\tau}^{\text{fail}}\}, \\ \text{FailSpec}_{\tau} &:= \begin{cases} \text{EvSpec}, & \tau \in \text{FailSeries}^{\text{strict}}, \\ \text{EvSpec}_{\Lambda}, & \tau = \text{approx} \end{cases} \end{aligned}$$

respectively.

Define the map that obtains the series-specific evaluation specification from a scale-indexed complete evaluation specification by

$$\begin{aligned} \text{FailSpecAt}_\tau : \text{EvSpec}_\Lambda &\longrightarrow \text{FailSpec}_\tau, \\ \text{FailSpecAt}_\tau[\text{Ev}; \lambda] &:= \begin{cases} \text{Ev}, & \tau \in \text{FailSeries}^{\text{strict}}, \\ (\text{Ev}, \lambda), & \tau = \text{approx} \end{cases} \end{aligned}$$

and define the six-series family of evaluation specifications and its component notation by

$$\begin{aligned} \text{Ev}^{\text{fail}} &:= \left((\text{Ev})_{\tau \in \text{FailSeries}^{\text{strict}}}, ((\text{Ev}, \lambda))_{\lambda \in \Lambda} \right), \\ \text{Ev}_\tau^{\text{fail}} &:= \text{FailSpecAt}_\tau[\text{Ev}; \lambda], \quad \tau \in \text{FailSeries}^{\text{strict}}, \\ \text{Ev}_{\text{approx}, \lambda}^{\text{fail}} &:= \text{FailSpecAt}_{\text{approx}}[\text{Ev}; \lambda] \end{aligned}$$

respectively. A scale-invariant-series component does not depend on the choice of scale.

Define the tagged index sets by

$$\begin{aligned} \text{FailIdx}^{\text{strict}} &:= \bigsqcup_{\tau \in \text{FailSeries}^{\text{strict}}} \{\tau\} \times I_\tau^{\text{fail}}, \\ \text{FailIdx}^{\text{approx}} &:= \{\text{approx}\} \times I_{\text{approx}}^{\text{fail}}, \\ \text{FailIdx} &:= \text{FailIdx}^{\text{strict}} \sqcup \text{FailIdx}^{\text{approx}} \end{aligned}$$

respectively. For each

$$(\tau, i) \in \text{FailIdx}$$

let the series-specific failure condition be

$$K_{\tau, i}^{\text{fail}} : \text{FailSpec}_\tau \longrightarrow \{\perp, \top\}.$$

For a native evaluation domain $D_{\tau, i} \subseteq \text{FailSpec}_\tau$ and a predicate $P_{\tau, i}$ supplied in a later section, define the total predicate above by

$$K_{\tau, i}^{\text{fail}}[x] \iff \exists y \in D_{\tau, i}, \quad x = y \wedge P_{\tau, i}[y].$$

For a condition already defined everywhere, use its full domain. This extension does not evaluate partial expressions outside their native domain. Conditions, regions and signatures are defined relative to fixed selected input families for each series. Judgments based on different selected inputs are not identified merely because they have the same complete evaluation specification. Define the failure-condition classification sets by

$$\begin{aligned} \text{FailCond}^{\text{strict}} &:= \{(\tau, i, K_{\tau, i}^{\text{fail}}) \mid (\tau, i) \in \text{FailIdx}^{\text{strict}}\}, \\ \text{FailCond}^{\text{approx}} &:= \{(\tau, i, K_{\tau, i}^{\text{fail}}) \mid (\tau, i) \in \text{FailIdx}^{\text{approx}}\}, \\ \text{FailCond} &:= \text{FailCond}^{\text{strict}} \sqcup \text{FailCond}^{\text{approx}} \end{aligned}$$

respectively.

Failure-position tokens are formal tagged tokens that retain the series and local index. For any $(\tau, i), (\sigma, j) \in \text{FailIdx}$, their equality is specified by

$$\text{FailTok}_{\tau, i} = \text{FailTok}_{\sigma, j} \iff \tau = \sigma \wedge i = j.$$

Define the failure-position token sets for each series, for the scale-invariant series, for the continuum-approximation series, and for all series by

$$\begin{aligned}\text{FailTok}_\tau &:= \{\text{FailTok}_{\tau,i} \mid i \in I_\tau^{\text{fail}}\}, \\ \text{FailTok}^{\text{strict}} &:= \bigsqcup_{\tau \in \text{FailSeries}^{\text{strict}}} \text{FailTok}_\tau, \\ \text{FailTok}^{\text{approx}} &:= \text{FailTok}_{\text{approx}}, \\ \text{FailTok} &:= \text{FailTok}^{\text{strict}} \sqcup \text{FailTok}^{\text{approx}}\end{aligned}$$

respectively. Define the series-type map and the failure-condition classification map by

$$\begin{aligned}\text{FailSeriesOf} : \text{FailTok} &\longrightarrow \text{FailSeries}, \quad \text{FailSeriesOf}(\text{FailTok}_{\tau,i}) := \tau, \\ \text{FailCondOf} : \text{FailTok} &\longrightarrow \text{FailCond}, \quad \text{FailCondOf}(\text{FailTok}_{\tau,i}) := (\tau, i, K_{\tau,i}^{\text{fail}})\end{aligned}$$

respectively.

Define the failure-evaluation argument set and the failure-condition evaluation map by

$$\begin{aligned}\text{FailEvalArg} &:= \bigsqcup_{\tau \in \text{FailSeries}} (\text{FailTok}_\tau \times \text{FailSpec}_\tau), \\ \text{FailEval} : \text{FailEvalArg} &\longrightarrow \{\perp, \top\}, \\ \text{FailEval}(\text{FailTok}_{\tau,i}; \text{EvalSpec}) &:= K_{\tau,i}^{\text{fail}}[\text{EvalSpec}]\end{aligned}$$

respectively.

Lemma 6.11 (Finite tagged disjoint union and classification correspondence). For a finite tag set

$$T$$

and a family of finite nonempty index sets

$$(I_t)_{t \in T}$$

let

$$X_t := \{x_{t,i} \mid i \in I_t\}, \quad X := \bigsqcup_{t \in T} X_t$$

For every $t \in T$, assume that the map $I_t \rightarrow X_t$, $i \mapsto x_{t,i}$ is injective. In what follows, identify each X_t with its image in X under the canonical inclusion. Then

$$(X_t)_{t \in T}$$

is a finite partition of

$$X$$

and

$$|X| = \sum_{t \in T} |I_t|.$$

Moreover, for a family of objects

$$(P_{t,i})_{t \in T, i \in I_t}$$

the map

$$x_{t,i} \longmapsto (t, i, P_{t,i})$$

is a bijection from

$$X$$

to

$$\{(t, i, P_{t,i}) \mid t \in T, i \in I_t\}$$

Proof. The components of the tagged disjoint union are pairwise disjoint, and each map

$$I_t \longrightarrow X_t, \quad i \longmapsto x_{t,i}$$

is surjective by the definition of X_t and injective by assumption, hence bijective. Each component is nonempty because I_t is nonempty. The partition, cardinality formula, and final bijection follow. $\square$

Lemma 6.12 (First-failure partition of a finite sequence of binary conditions). For a family of binary conditions

$$K_i : \mathcal{Y} \longrightarrow \{\perp, \top\}, \quad i \in \{1, \dots, n\},$$

and the empty conjunction

$$\bigwedge_{j < 1} K_j(y) := \top$$

one has

$$\left\{ y \in \mathcal{Y} \mid \neg \bigwedge_{i=1}^n K_i(y) \right\} = \bigsqcup_{i=1}^n \left\{ y \in \mathcal{Y} \mid \left(\bigwedge_{j < i} K_j(y) \right) \wedge \neg K_i(y) \right\}.$$

Consequently, for every element of the left-hand side,

$$\min\{i \in \{1, \dots, n\} \mid \neg K_i(y)\}$$

is uniquely determined.

Proof. The set of failed indices is finite and nonempty, and therefore has a least element

$$i_0$$

Its minimality gives membership in the

$$i_0$$

component on the right-hand side. The reverse inclusion follows from

$$\neg K_{i_0}(y)$$

in each component. Two distinct components are disjoint because the condition at the lower index is false in one and true in the other. $\square$

Definition 6.13 (Three-point nontransitivity witness predicate for a relation). For a binary relation on a set $\mathcal{Y}$,

$$\mathcal{R} \subseteq \mathcal{Y} \times \mathcal{Y}$$

define the three-point nontransitivity witness predicate by

$$\begin{aligned} & \text{NonTrans}[\mathcal{Y}; \mathcal{R}] \\ & \iff \left[\exists y_0, y_1, y_2 \in \mathcal{Y} : \bigwedge_{j \in \{0,1\}} (y_j, y_{j+1}) \in \mathcal{R}, \quad (y_0, y_2) \notin \mathcal{R} \right] \end{aligned}$$

Definition 6.14 (Maximum number of failure positions). Define the maximum number of failure positions by

$$\text{FailCount}^{\max} := |\text{FailTok}|$$

Proposition 6.15 (Finite partition of the failure-position token system and correspondence with condition classifications). The family

$$(\text{FailTok}_\tau)_{\tau \in \text{FailSeries}}$$

is a finite partition of

$$\text{FailTok}$$

and

$$\begin{aligned} & \left(|\text{FailTok}_{\text{cmp}}|, |\text{FailTok}_{\text{repr}}|, |\text{FailTok}_{\text{dyn}}|, \right. \\ & \quad \left. |\text{FailTok}_{\text{approx}}|, |\text{FailTok}_{\text{law}}|, |\text{FailTok}_{\text{diff}}| \right) \\ &= (8, 3, 8, 9, 5, 5). \end{aligned}$$

Consequently,

$$\text{FailCount}^{\max} = 38.$$

Furthermore,

$$\text{FailCondOf} : \text{FailTok} \longrightarrow \text{FailCond}$$

is a bijection.

Proof. Apply Lemma 6.11 to

$$(T, I_t, X_t, P_{t,i}) = (\text{FailSeries}, I_\tau^{\text{fail}}, \text{FailTok}_\tau, K_{\tau,i}^{\text{fail}})$$

The number of elements in each series from Definition 6.10 gives

$$|\text{FailTok}| = 8 + 3 + 8 + 9 + 5 + 5 = 38$$

□

Lemma 6.16 (Reachability order of rank-increasing edges). Suppose that a set

$$V,$$

an edge set

$$E \subseteq V \times V,$$

a strict partially ordered set

$$(R, <_R)$$

and a map

$$r : V \longrightarrow R$$

satisfy

$$(v, w) \in E \implies r(v) <_R r(w)$$

Write the relation given by equality or by the existence of a positive-length finite directed path on

$$E$$

as

$$\preceq_E$$

Then

$$E$$

has no directed cycle, and

$$\preceq_E$$

is a partial order on

$$V$$

Proof. Equality gives reflexivity, and concatenation of directed paths gives transitivity. Because

$$r$$

increases strictly along every directed path, a directed cycle would imply

$$r(v) <_R r(v)$$

contrary to the irreflexivity of a strict partial order. Two distinct points reachable from one another in both directions would yield a directed cycle, so antisymmetry also holds. $\square$

Definition 6.17 (Failure-generation dependency graph and dependency order). For each

$$\tau \in \text{FailSeries}$$

define the within-series direct dependency edges by

$$E_\tau^{\text{fail}} := \begin{cases} \{(i, i+1) \mid 1 \leq i < 8\}, & \tau = \text{cmp}, \\ \{(1, 2), (2, 3)\}, & \tau = \text{repr}, \\ \{(1, 6), (6, 7), (2, 3), (3, 8), (6, 8)\}, & \tau = \text{dyn}, \\ \{(i, 7) \mid 1 \leq i \leq 6\} \cup \{(7, 8), (7, 9)\}, & \tau = \text{approx}, \\ \{(1, 3), (1, 5), (3, 5)\}, & \tau = \text{law}, \\ \{(1, 2), (2, 3), (1, 4), (1, 5)\}, & \tau = \text{diff} \end{cases}$$

and define the within-series graphs, within-series token edges, between-series graph, and complete dependency-edge set by

$$\begin{aligned} \text{FailDAG}_\tau &:= (I_\tau^{\text{fail}}, E_\tau^{\text{fail}}), \\ \text{FailEdge}_\tau &:= \{(\text{FailTok}_{\tau,i}, \text{FailTok}_{\tau,j}) \mid (i, j) \in E_\tau^{\text{fail}}\}, \\ \text{FailEdge}^{\text{intra}} &:= \bigcup_{\tau \in \text{FailSeries}} \text{FailEdge}_\tau, \\ E_{\text{series}}^{\text{fail}} &:= \{(\text{cmp}, \text{repr}), (\text{repr}, \text{dyn}), (\text{dyn}, \text{approx}), \\ &\quad (\text{dyn}, \text{law}), (\text{law}, \text{diff})\}, \\ \text{FailDAG}^{\text{series}} &:= (\text{FailSeries}, E_{\text{series}}^{\text{fail}}), \\ \text{FailEdge}^{\text{inter}} &:= \bigcup_{(\tau_0, \tau_1) \in E_{\text{series}}^{\text{fail}}} (\text{FailTok}_{\tau_0} \times \text{FailTok}_{\tau_1}), \\ \text{FailEdge} &:= \text{FailEdge}^{\text{intra}} \cup \text{FailEdge}^{\text{inter}} \end{aligned}$$

respectively. In the reachability notation of Lemma 6.16, the failure-generation dependency relation is

$$\preceq^{\text{fail,dep}} := \preceq_{\text{FailEdge}} \subseteq \text{FailTok} \times \text{FailTok}$$

Define incomparability with respect to this order by

$$s \parallel_{\text{fail,dep}} t \iff \neg (s \preceq^{\text{fail,dep}} t) \wedge \neg (t \preceq^{\text{fail,dep}} s)$$

Proposition 6.18 (Partial-order property of the failure-generation dependency order).
The relation

$$\preceq^{\text{fail,dep}}$$

is a partial order on

$$\text{FailTok}$$

Proof. For each failure series, define the series-layer map and the family of within-series ranks by

$$\begin{aligned} \text{SeriesRank}(\tau) &:= \begin{cases} 0, & \tau = \text{cmp}, \\ 1, & \tau = \text{repr}, \\ 2, & \tau = \text{dyn}, \\ 3, & \tau \in \{\text{approx}, \text{law}\}, \\ 4, & \tau = \text{diff}, \end{cases} \\ (\text{IntraRank}_\tau(i))_{i \in I_\tau^{\text{fail}}} &:= \begin{cases} (0, 1, 2, 3, 4, 5, 6, 7), & \tau = \text{cmp}, \\ (0, 1, 2), & \tau = \text{repr}, \\ (0, 0, 1, 0, 0, 1, 2, 2), & \tau = \text{dyn}, \\ (0, 0, 0, 0, 0, 0, 1, 2, 2), & \tau = \text{approx}, \\ (0, 0, 1, 0, 2), & \tau = \text{law}, \\ (0, 1, 2, 1, 1), & \tau = \text{diff} \end{cases} \end{aligned}$$

respectively. Define the rank map on all tokens into the product equipped with the lexicographic order by

$$\text{FailRank}(\text{FailTok}_{\tau,i}) := (\text{SeriesRank}(\tau), \text{IntraRank}_\tau(i))$$

For every edge of Definition 6.17,

$$\text{FailRank}(s) <_{\text{lex}} \text{FailRank}(t)$$

holds. Apply Lemma 6.16 to

$$(V, E, R, r) = (\text{FailTok}, \text{FailEdge}, \{0, 1, 2, 3, 4\} \times \mathbb{N}_0, \text{FailRank})$$

□

Proposition 6.19 (Parallel branches of the failure-generation dependency order). With respect to the failure-generation dependency order,

$$\begin{aligned}
& s \parallel_{\text{fail,dep}} t, \\
& (s, t) \in \text{FailTok}_{\text{approx}} \times (\text{FailTok}_{\text{law}} \cup \text{FailTok}_{\text{diff}}), \\
& \text{FailTok}_{\text{dyn},4} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{dyn},5}, \\
& \text{FailTok}_{\text{approx},i} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{approx},j}, \\
& 1 \leq i < j \leq 6, \\
& \text{FailTok}_{\text{approx},8} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{approx},9}, \\
& \text{FailTok}_{\text{law},i} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{law},j}, \\
& i < j, \quad i, j \in \{1, 2, 4\}, \\
& \text{FailTok}_{\text{diff},4} \parallel_{\text{fail,dep}} \text{FailTok}_{\text{diff},5}
\end{aligned}$$

holds.

Proof. The between-series graph is

$$\text{cmp} \longrightarrow \text{repr} \longrightarrow \text{dyn} \longrightarrow \begin{cases} \text{approx}, \\ \text{law} \longrightarrow \text{diff} \end{cases}$$

and therefore has no directed path connecting the continuum-approximation series to the law or differential-consistency series. Within a series, the edge sets of Definition 6.17 contain no directed path joining any of the listed pairs. Because a between-series edge strictly increases the series layer, no directed path can join two points of the same series by passing through another series. $\square$

Lemma 6.20 (Deterministic state recursion on a finite DAG). Let

$$(V, E)$$

be a finite directed acyclic graph. Suppose that, for every vertex

$$v \in V$$

the states of its direct predecessors determine exactly one element of a state set

$$S$$

through an update map

$$\Phi_v : S^{\text{Pred}_E(v)} \longrightarrow S$$

Then the equations

$$s(v) = \Phi_v \left((s(u))_{u \in \text{Pred}_E(v)} \right), \quad v \in V$$

are satisfied by exactly one map

$$s : V \longrightarrow S$$

Proof. Choose a topological ordering

$$v_1, \dots, v_{|V|}$$

of the graph. Because every direct predecessor of each

$$v_k$$

belongs to

$$v_1, \dots, v_{k-1}$$

the map

$$\Phi_{v_k}$$

uniquely determines

$$s(v_k)$$

in sequence. Finite induction gives existence and uniqueness. $\square$

Definition 6.21 (Evaluation states and failed-token sets). Define the set of direct-predecessor tokens by

$$\text{FailPredOf}(t) := \{u \in \text{FailTok} \mid (u, t) \in \text{FailEdge}\}$$

Let the non-ancestor formation-input satisfaction map and the evaluation-input satisfaction map be

$$\text{FailInputSat}^{\text{form}}, \text{FailInputSat}^{\text{eval}} : \text{FailEvalArg} \longrightarrow \{\perp, \top\}$$

Let the evaluation-state map, the direct-predecessor satisfaction map, and the three evaluation gates be

$$\begin{aligned} \text{FailStateOf} &: \text{FailEvalArg} \longrightarrow \text{FailState}, \\ \text{FailPredSat}, \text{FailFormed}, \text{FailInputSat}, \text{FailReady} &: \text{FailEvalArg} \longrightarrow \{\perp, \top\} \end{aligned}$$

respectively. For

$$(\sigma, \tau) \in E_{\text{series}}^{\text{fail}} \cup \{(\rho, \rho) \mid \rho \in \text{FailSeries}\}$$

which forms either a direct between-series dependency edge or a same-series pair, define the evaluation-specification reindexing map by

$$\text{FailSpecRes}_{\sigma \leftarrow \tau} : \text{FailSpec}_{\tau} \longrightarrow \text{FailSpec}_{\sigma}$$

and

$$\text{FailSpecRes}_{\sigma \leftarrow \tau}(\text{FailSpecAt}_{\tau}[\text{Ev}; \lambda]) = \text{FailSpecAt}_{\sigma}[\text{Ev}; \lambda]$$

Use the identity map between scale-invariant series and within each series, and the first projection $(\text{Ev}, \lambda) \mapsto \text{Ev}$ for $(\sigma, \tau) = (\text{dyn}, \text{approx})$.

For an evaluation argument

$$a = (t; \text{EvalSpec}) \in \text{FailTok}_{\tau} \times \text{FailSpec}_{\tau}$$

and

$$u \in \text{FailPredOf}(t) \cap \text{FailTok}_{\sigma}$$

define the direct-predecessor evaluation argument by

$$\text{FailPredArgOf}(u; a) := (u; \text{FailSpecRes}_{\sigma \leftarrow \tau}(\text{EvalSpec})) \in \text{FailEvalArg}$$

The direct-predecessor satisfaction judgment is

$$\begin{aligned} \text{FailPredSat}(a) &= \top \\ \iff \forall u \in \text{FailPredOf}(t), \\ &\quad \text{FailStateOf}(\text{FailPredArgOf}(u; a)) = \text{sat}. \end{aligned}$$

For each

$$a \in \text{FailEvalArg}$$

define the three gates of input satisfaction, formation, and evaluation readiness by

$$\begin{aligned} \text{FailInputSat}(a) &= \text{FailInputSat}^{\text{form}}(a) \wedge \text{FailInputSat}^{\text{eval}}(a), \\ \text{FailFormed}(a) &= \text{FailInputSat}^{\text{form}}(a) \wedge \text{FailPredSat}(a), \\ \text{FailReady}(a) &= \text{FailPredSat}(a) \wedge \text{FailInputSat}(a) \end{aligned}$$

respectively. Here the conjunctions on the right-hand side are conjunctions on

$$\{\perp, \top\}$$

Define the evaluation state recursively by

$$\text{FailStateOf}(a) := \begin{cases} \text{unformed}, & \text{FailFormed}(a) = \perp, \\ \text{unevaluable}, & \text{FailFormed}(a) = \top \wedge \text{FailInputSat}^{\text{eval}}(a) = \perp, \\ \text{sat}, & \text{FailReady}(a) = \top \wedge \text{FailEval}(a) = \top, \\ \text{failed}, & \text{FailReady}(a) = \top \wedge \text{FailEval}(a) = \perp \end{cases}.$$

Require the input-satisfaction maps to be compatible with the native domains of Definition 6.10:

$$\text{FailReady}(t; x) = \top \implies x \in D_{\tau, i} \quad (t = \text{FailTok}_{\tau, i}).$$

If a selected input has not been supplied, its formation-input satisfaction is false. A false extended condition outside its native domain does not by itself constitute a failed state.

Define the token-specific evaluation argument and state section at a scale-indexed complete evaluation specification by

$$\begin{aligned} \text{FailArgAt}_{\text{Ev}, \lambda}(t) &:= (t; \text{FailSpecAt}_{\text{FailSeriesOf}(t)}[\text{Ev}; \lambda]), \\ \text{FailStateAt}_{\text{Ev}, \lambda}(t) &:= \text{FailStateOf}(\text{FailArgAt}_{\text{Ev}, \lambda}(t)) \end{aligned}$$

respectively.

For each

$$q \in \text{FailState}$$

define the state fiber and its failed-state components by

$$\begin{aligned} \text{FailFiber}_q[\text{Ev}; \lambda] &:= \{\text{FailTok}_{\tau, i} \in \text{FailTok} \mid \text{FailStateAt}_{\text{Ev}, \lambda}(\text{FailTok}_{\tau, i}) = q\}, \\ \text{FailedTok}[\text{Ev}; \lambda] &:= \text{FailFiber}_{\text{failed}}[\text{Ev}; \lambda], \\ \text{FailedTok}^{\text{strict}}[\text{Ev}] &:= \text{FailedTok}[\text{Ev}; \lambda] \cap \text{FailTok}^{\text{strict}}, \\ \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] &:= \text{FailedTok}[\text{Ev}; \lambda] \cap \text{FailTok}^{\text{approx}} \end{aligned}$$

respectively. The value on the scale-invariant-series side does not depend on the choice of scale. Consequently,

$$\begin{aligned} \text{FailedTok}^{\text{strict}} &: \text{EvSpec} \longrightarrow \mathcal{P}(\text{FailTok}^{\text{strict}}), \\ \text{FailedTok}^{\text{approx}} &: \text{EvSpec}_\Lambda \longrightarrow \mathcal{P}(\text{FailTok}^{\text{approx}}), \\ \text{FailedTok} &: \text{EvSpec}_\Lambda \longrightarrow \mathcal{P}(\text{FailTok}). \end{aligned}$$

Proposition 6.22 (Existence and uniqueness of the evaluation-state assignment and decomposition into state fibers). There exists exactly one evaluation-state map

$$\text{FailStateOf} : \text{FailEvalArg} \longrightarrow \text{FailState}$$

that satisfies the recursive conditions of Definition 6.21. For each

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$$

the family

$$\{\text{FailFiber}_q[\text{Ev}; \lambda] \mid q \in \text{FailState}\}$$

gives a decomposition into pairwise disjoint state fibers of

$$\text{FailTok}.$$

Empty state fibers are allowed. Its restrictions to

$$\text{FailTok}^{\text{strict}}, \quad \text{FailTok}^{\text{approx}}$$

give decompositions of the respective subsets into pairwise disjoint state fibers.

Proof. By Propositions 6.15 and 6.18,

$$(\text{FailTok}, \text{FailEdge})$$

is a finite directed acyclic graph. At each token, the four branches of Definition 6.21 return exactly one state from the family of direct-predecessor states, the two input-satisfaction values, and the failure-condition evaluation value. Lemma 6.20 therefore gives the existence and uniqueness of the evaluation-state map.

Because each token has exactly one of the four states, the four state fibers are pairwise disjoint. Their union equals

$$\text{FailTok}$$

and their restrictions to either of the two series-side subsets remain pairwise disjoint and exhaustive. $\square$

Proposition 6.23 (State-equivalence conditions and finiteness of the failed-token sets). For any

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda, \quad \text{FailTok}_{\tau,i} \in \text{FailTok}$$

one has

$$\text{FailTok}_{\tau,i} \in \text{FailedTok}[\text{Ev}; \lambda] \iff \text{FailStateAt}_{\text{Ev}, \lambda}(\text{FailTok}_{\tau,i}) = \text{failed}.$$

Furthermore,

$$\begin{aligned} \text{FailedTok}[\lambda] &= \text{FailedTok}^{\text{strict}}[\text{Ev}] \sqcup \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda], \\ \text{FailedTok}^{\text{strict}}[\text{Ev}] &= \text{FailedTok}[\lambda] \cap \text{FailTok}^{\text{strict}}, \\ \text{FailedTok}^{\text{approx}}[\lambda] &= \text{FailedTok}[\lambda] \cap \text{FailTok}^{\text{approx}} \end{aligned}$$

and all three sets are finite.

Proof. The equivalence and the three displays follow from Definition 6.21. By Proposition 6.15,

$$\text{FailTok}$$

is finite, and the three sets are its subsets. $\square$

Definition 6.24 (Strict failure-generation dependency order and three cases of minimal failure). Define the strict dependency order by

$$s \prec^{\text{fail,dep}} t \iff s \preceq^{\text{fail,dep}} t \wedge s \neq t$$

and define the set of minimal-failure cases, the number of failed tokens, and the case map by

$$\begin{aligned} \text{FailCase} &:= \{\text{FailAbsent}, \text{FailUnique}, \text{FailParallel}\}, \\ \text{FailCount}[\text{Ev}; \lambda] &:= |\text{FailedTok}[\text{Ev}; \lambda]|, \\ \text{FailCaseOf}(\text{Ev}; \lambda) &:= \begin{cases} \text{FailAbsent}, & \text{FailCount}[\text{Ev}; \lambda] = 0, \\ \text{FailUnique}, & \text{FailCount}[\text{Ev}; \lambda] = 1, \\ \text{FailParallel}, & \text{FailCount}[\text{Ev}; \lambda] \geq 2 \end{cases} \end{aligned}$$

respectively. The three cases are pairwise distinct.

Lemma 6.25 (Forward propagation of a nonsatisfied state into unformed successors). Take an evaluation specification

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$$

and a directed path

$$t_0 \longrightarrow t_1 \longrightarrow \cdots \longrightarrow t_m, \quad m \geq 1,$$

Using the notation

$$a_r := \text{FailArgAt}_{\text{Ev}, \lambda}(t_r)$$

one has

$$\text{FailStateAt}_{\text{Ev}, \lambda}(t_0) \neq \text{sat} \implies \forall r \in \{1, \dots, m\}, \quad \text{FailStateAt}_\lambda(t_r) = \text{unformed}.$$

Proof. For every

$$r \geq 1$$

one has

$$t_{r-1} \in \text{FailPredOf}(t_r).$$

The initial state is not satisfied, and at each inductive step the immediately preceding state satisfies

$$\text{unformed} \neq \text{sat}$$

so Definition 6.21 gives

$$\begin{aligned} \text{FailPredSat}(a_r) &= \text{FailFormed}(a_r) = \perp, \\ \text{FailStateOf}(a_r) &= \text{unformed} \end{aligned}$$

in succession. $\square$

Proposition 6.26 (Incomparability of failed tokens, equality with the minimal elements, and the three cases). For any

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$$

and distinct

$$s, t \in \text{FailedTok}[\text{Ev}; \lambda]$$

one has

$$s \parallel_{\text{fail,dep}} t.$$

Consequently,

$$\text{Min}_{\preceq_{\text{fail,dep}}} \text{FailedTok}[\lambda] = \text{FailedTok}[\lambda].$$

Moreover,

$$\text{FailCaseOf}(\text{Ev}; \lambda)$$

is uniquely determined as one of the following three cases:

$$\text{FailAbsent}, \quad \text{FailUnique}, \quad \text{FailParallel}$$

Proof. Suppose first that the relation

$$s \prec^{\text{fail,dep}} t$$

holds. There then exists a positive-length directed path described by

$$s = t_0 \longrightarrow \cdots \longrightarrow t_m = t$$

Proposition 6.23 then implies that the state at its initial point is

$$\text{failed} \neq \text{sat}.$$

By Lemma 6.25, the state at the endpoint is

$$\text{unformed}$$

whereas, by the same proposition, the state of

$$t \in \text{FailedTok}[\text{Ev}; \lambda]$$

is

$$\text{failed}$$

and this contradicts uniqueness of the state.

Conversely, suppose that $t \prec^{\text{fail,dep}} s$. The state at the initial point t is **failed**, and Lemma 6.25 implies that the state at the endpoint s is **unformed**. On the other hand, $s \in \text{FailedTok}[\lambda]$ implies that the state of s is **failed**, again contradicting uniqueness of the state. Distinct elements of the failed set are therefore incomparable in both directions. Hence the failed set is an antichain and every element is minimal.

Because the failed set is finite, its cardinality falls into exactly one of the following three classes:

$$0, \quad 1, \quad 2 \text{ or more}$$

and Definition 6.24 gives the three cases. □

Definition 6.27 (Series-specific completion conditions). For each

$$\tau \in \text{FailSeries}$$

define the series-specific completion condition and the six-series completion-state vector by

$$\begin{aligned} K_\tau^{\text{fail}}[\text{EvalSpec}] = \top &\iff \forall t \in \text{FailTok}_\tau, \quad \text{FailStateOf}(t; \text{EvalSpec}) = \text{sat}, \\ \mathbf{K}^{\text{fail,series}}[\text{Ev}; \lambda] &:= (K_\tau^{\text{fail}}[\text{FailSpecAt}_\tau[\text{Ev}; \lambda]])_{\tau \in \text{FailSeries}} \end{aligned}$$

Define the strict completion condition and the full-evaluation completion condition by

$$\begin{aligned} K_{\text{strict}}^{\text{fail}}[\text{Ev}] &:= \bigwedge_{\tau \in \text{FailSeries}^{\text{strict}}} K_\tau^{\text{fail}}[\text{Ev}], \\ K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda] &:= \bigwedge_{\tau \in \text{FailSeries}} K_\tau^{\text{fail}}[\text{FailSpecAt}_\tau[\text{Ev}; \lambda]] \\ &= K_{\text{strict}}^{\text{fail}}[\text{Ev}] \wedge K_{\text{approx}}^{\text{fail}}[\text{Ev}; \lambda] \end{aligned}$$

The conjunction on the right-hand side is taken in

$$\{\perp, \top\}$$

throughout.

Proposition 6.28 (Equivalence between series-specific completion conditions and evaluation states). For every

$$(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$$

one has

$$K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda] = \top \iff \forall \text{FailTok}_{\tau,i} \in \text{FailTok}, \quad \text{FailStateAt}_{\text{Ev},\lambda}(\text{FailTok}_{\tau,i}) = \text{sat}.$$

Whenever this condition holds,

$$\text{FailedTok}^{\text{strict}}[\text{Ev}] = \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] = \text{FailedTok}[\text{Ev}; \lambda] = \emptyset.$$

Proof. The series-specific conjunction in Definition 6.27 and the partition

$$\text{FailTok} = \bigsqcup_{\tau \in \text{FailSeries}} \text{FailTok}_\tau$$

give the equivalence. If every token is in the satisfied state, the failed-state fiber is empty, and so are its two restrictions. $\square$

Definition 6.29 (Approximation-series actual-failure scales and approximation-validity boundary). Define the set of scales at which actual failure occurs in the approximation series and the set of scales at which the approximation-series completion condition is false by

$$\begin{aligned} \Lambda_{\text{failed}}^{\text{approx}}[\text{Ev}] &:= \{\lambda \in \Lambda \mid \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \neq \emptyset\}, \\ \Lambda_{\text{unsat}}^{\text{approx}}[\text{Ev}] &:= \Lambda \setminus \text{Val}_\Lambda(K_{\text{approx}}^{\text{fail}}; \text{Ev}) \\ &= \{\lambda \in \Lambda \mid K_{\text{approx}}^{\text{fail}}[\text{Ev}; \lambda] = \perp\} \end{aligned}$$

respectively.

For a set-valued map

$$S : X \longrightarrow \mathcal{P}(\Lambda)$$

let

$$\begin{aligned} \text{Dom}(\text{First}_\Lambda[S]) &:= \{x \in X \mid \exists \lambda_0 \in S[x], \forall \mu \in S[x], \lambda_0 \leq_\Lambda \mu\}, \\ \text{First}_\Lambda[S](x) &:= \min_{\leq_\Lambda} S[x], \quad x \in \text{Dom}(\text{First}_\Lambda[S]) \end{aligned}$$

define its least-scale lift. The first approximation-series actual-failure scale and the approximation-validity boundary are

$$\begin{aligned} \lambda_{\text{failed}}^{\text{approx}} &:= \text{First}_\Lambda[\Lambda_{\text{failed}}^{\text{approx}}], \\ \lambda_{\text{bdry}}^{\text{approx}} &:= \text{First}_\Lambda[\Lambda_{\text{unsat}}^{\text{approx}}] \end{aligned}$$

and both are partial maps with the common type

$$\text{EvSpec} \rightarrow \Lambda$$

Lemma 6.30 (Inclusion relation for least elements and the boundary of an initial segment).
In the totally ordered set

$$(\Lambda, \leq_\Lambda)$$

if the subsets

$$S_0 \subseteq S_1$$

both have least elements, then

$$\min_{\leq_\Lambda} S_1 \leq_\Lambda \min_{\leq_\Lambda} S_0.$$

If the sets are equal, their least elements are also equal. Furthermore, if the complement of an initial segment

$$I \subseteq \Lambda$$

has the least element

$$b := \min_{\leq_\Lambda} (\Lambda \setminus I)$$

then

$$I = \{\lambda \in \Lambda \mid \lambda <_\Lambda b\}.$$

Proof. The first inequality follows from

$$\min_{\leq_\Lambda} S_0 \in S_1$$

and the least-element property of

$$S_1$$

which together yield the inequality. If the sets are equal, antisymmetry applies. For the second assertion, if

$$\lambda <_\Lambda b$$

did not belong to

$$I$$

the least-element property of

$$b$$

would be contradicted; if

$$\lambda \in I, \quad b \leq_{\Lambda} \lambda$$

then the initial-segment property would imply

$$b \in I$$

contrary to its membership in the complement. $\square$

Proposition 6.31 (Relations between approximation-series actual-failure scales and the approximation-validity boundary). For every

$$\text{Ev} \in \text{EvSpec}$$

one has

$$\Lambda_{\text{failed}}^{\text{approx}}[\text{Ev}] \subseteq \Lambda_{\text{unsat}}^{\text{approx}}[\text{Ev}].$$

If both least scales are defined, then

$$\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}] \leq_{\Lambda} \lambda_{\text{failed}}^{\text{approx}}[\text{Ev}].$$

Furthermore, if

$$\forall \lambda \in \Lambda, \quad K_{\text{approx}}^{\text{fail}}[\text{Ev}; \lambda] = \perp \implies \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \neq \emptyset$$

holds, the two scale sets are equal. Consequently, if both least scales are defined, then

$$\lambda_{\text{failed}}^{\text{approx}}[\text{Ev}] = \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}].$$

If the approximation-validity domain is an initial segment and the approximation-validity boundary is defined, then

$$\mathcal{I}^{\text{max}}[\text{Ev}] = \{ \lambda \in \Lambda \mid \lambda <_{\Lambda} \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}] \}.$$

If the first approximation-series actual-failure scale is defined, then it is unique and

$$\begin{aligned} \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda_{\text{failed}}^{\text{approx}}[\text{Ev}]] &\neq \emptyset, \\ \text{FailCaseOf}(\text{Ev}; \lambda_{\text{failed}}^{\text{approx}}[\text{Ev}]) &\in \{\text{FailUnique}, \text{FailParallel}\}. \end{aligned}$$

Proof. Because the failed state differs from the satisfied state, Definition 6.27 gives

$$\text{FailedTok}^{\text{approx}}[\lambda] \neq \emptyset \implies K_{\text{approx}}^{\text{fail}}[\lambda] = \perp.$$

This yields the first inclusion. Lemma 6.30 gives the inequality between the least scales. The reverse implication assumed above yields the reverse inclusion, so the sets and their least scales coincide.

Applying the same lemma to the initial-segment property of the approximation-validity domain and to the least element of its complement, and then applying Proposition 6.9, gives the displayed form of the maximal effective interval. The final two relations follow because the least scale belongs to the set of failure scales and because Definition 6.24 selects one of the nonempty cases. $\square$

Definition 6.32 (Covered–unrefined refinement pair). For a parent region

$$\text{ParentReg}$$

and a family of subregions

$$(\text{SubReg}_\lambda)_{\lambda \in \text{SubIdx}}$$

call

$$\begin{aligned} & \text{RefSplit}(\text{ParentReg}; (\text{SubReg}_\lambda)_{\lambda \in \text{SubIdx}}) \\ &:= \left(\bigcup_{\lambda \in \text{SubIdx}} \text{SubReg}_\lambda, \right. \\ & \quad \left. \text{ParentReg} \setminus \bigcup_{\lambda \in \text{SubIdx}} \text{SubReg}_\lambda \right) \end{aligned}$$

the covered–unrefined refinement pair.

Lemma 6.33 (Decomposition of covered and unrefined regions and refinement monotonicity). Suppose that two families of subregions satisfy

$$\begin{aligned} & \text{SubIdx}^{(0)} \subseteq \text{SubIdx}^{(1)}, \\ & \text{SubReg}_\lambda^{(\rho)} \subseteq \text{ParentReg}, \quad \rho \in \{0, 1\}, \quad \lambda \in \text{SubIdx}^{(\rho)}, \\ & \text{SubReg}_\lambda^{(0)} = \text{SubReg}_\lambda^{(1)}, \quad \lambda \in \text{SubIdx}^{(0)} \end{aligned}$$

and set

$$(\text{CoveredReg}^{(\rho)}, \text{UnrefinedReg}^{(\rho)}) := \text{RefSplit}(\text{ParentReg}; (\text{SubReg}_\lambda^{(\rho)})_{\lambda \in \text{SubIdx}^{(\rho)}})$$

Then

$$\text{ParentReg} = \text{CoveredReg}^{(\rho)} \sqcup \text{UnrefinedReg}^{(\rho)}, \quad \rho \in \{0, 1\}.$$

If each family of subregions is pairwise disjoint, then

$$\text{ParentReg} = \left(\bigsqcup_{\lambda \in \text{SubIdx}^{(\rho)}} \text{SubReg}_\lambda^{(\rho)} \right) \sqcup \text{UnrefinedReg}^{(\rho)}.$$

Furthermore,

$$\begin{aligned} & \text{CoveredReg}^{(0)} \subseteq \text{CoveredReg}^{(1)}, \\ & \text{UnrefinedReg}^{(1)} \subseteq \text{UnrefinedReg}^{(0)}. \end{aligned}$$

Proof. Each covered region is a subset of the parent region, and the unrefined region is the set difference obtained by removing the covered region from the parent region. The two regions are therefore disjoint, and their union equals the parent region. If the family of subregions is pairwise disjoint, replacing the union in the covered region by a disjoint union gives the second decomposition.

Every subregion indexed by the first index set is contained in the second covering union because the two families agree on their common indices. This proves the inclusion between the covered regions. Removing the covered regions from the same parent region reverses that inclusion, which proves the inclusion between the unrefined regions. $\square$

Definition 6.34 (Typed failure-refinement forest). Take an underlying set $\mathbf{Amb}$, a root-index set I , and, for each path ν , a child-index set $J(\nu)$. Define the refinement-path sets by

$$\begin{aligned} \text{RefPath}_0[I; J] &:= \{(i) \mid i \in I\}, \\ \text{RefPath}_{n+1}[I; J] &:= \{\nu \star \lambda \mid \nu \in \text{RefPath}_n[I; J], \lambda \in J(\nu)\}, \\ \text{RefPath}_{\leq m}[I; J] &:= \bigsqcup_{n=0}^m \text{RefPath}_n[I; J], \quad \text{RefNode}[I; J] := \bigcup_{n \geq 0} \text{RefPath}_n[I; J] \end{aligned}$$

The symbol $\star$ denotes appending a child index to the end of a path, and $J(\nu) = \emptyset$ is allowed at a leaf.

Suppose that the node-region map $R : \text{RefNode}[I; J] \rightarrow \mathcal{P}(\mathbf{Amb})$ satisfies, for every $\lambda \in J(\nu)$, the inclusion $R(\nu \star \lambda) \subseteq R(\nu)$. Define the forest, its node decomposition, its child relation, and its ancestor relation by

$$\begin{aligned} \text{RefForest}[I; J; R] &:= (\text{RefNode}[I; J], J, \star, R), \\ (R^{\text{cov}}(\nu), R^{\text{unref}}(\nu)) &:= \text{RefSplit}(R(\nu); (R(\nu \star \lambda))_{\lambda \in J(\nu)}), \\ \nu \prec_{\text{RefForest}}^{\text{child}} \omega &\iff \exists \lambda \in J(\nu), \omega = \nu \star \lambda, \quad \preceq_{\text{RefForest}}^{\text{anc}} := (\prec_{\text{RefForest}}^{\text{child}})^* \end{aligned}$$

The superscript $*$ denotes the reflexive-transitive closure of the relation.

Theorem 6.35 (Node decomposition and ancestor inclusion in a typed failure-refinement forest). Take arbitrary $\mathbf{Amb}$, I , J , and R as in Definition 6.34. For every nonleaf node $\nu \in \text{RefNode}[I; J]$ one has

$$R(\nu) = R^{\text{cov}}(\nu) \sqcup R^{\text{unref}}(\nu).$$

If the family of child-node regions is pairwise disjoint, then

$$R^{\text{cov}}(\nu) = \bigsqcup_{\lambda \in J(\nu)} R(\nu \star \lambda).$$

Furthermore, for every $\nu, \omega \in \text{RefNode}[I; J]$ one has

$$\nu \preceq_{\text{RefForest}[I; J; R]}^{\text{anc}} \omega \implies R(\omega) \subseteq R(\nu).$$

Proof. The node decomposition and the conditional disjoint union follow by substituting the node data of Definition 6.34 into Lemma 6.33. The ancestor inclusion follows by induction on the path length in the ancestor relation, using inclusion of every child-node region in its parent-node region. $\square$

6.2 Failure Positions at the Comparison Stage

The comparison-stage condition-family evaluation specification corresponding to the complete evaluation specification of Definition 5.2,

$$\text{Ev} = (\mathbf{l}, o, \zeta) \in \text{EvSpec}$$

is the family specified in Definition 4.7,

$$\text{Ev}^{\text{cmp}} = (\text{Ev}_i^{\text{cmp}})_{i=1}^7,$$

with components

$$\text{Ev}_i^{\text{cmp}} = \begin{cases} \mathcal{X}, & i = 1, \\ \text{I}, & i \in \{2, \dots, 7\} \end{cases}$$

at the seven indexed positions.

Define the bracket evaluations of the comparison-stage conditions for a complete evaluation specification by

$$\begin{aligned} \text{L}_i[\text{Ev}] &:= \text{L}_i[\text{Ev}_i^{\text{cmp}}], & i \in \{1, \dots, 7\}, \\ \text{L}_{\leq i}[\text{Ev}] &:= \text{L}_{\leq i}[\text{Ev}^{\text{cmp}}], & i \in \{1, \dots, 7\}, \\ \text{L}_{< i}[\text{Ev}] &:= \text{L}_{< i}[\text{Ev}^{\text{cmp}}], & i \in \{1, \dots, 7\}, \end{aligned}$$

and define the global-condition and operability evaluations by

$$\begin{aligned} \text{G}[\text{Ev}] &:= \text{G}[\text{Ev}^{\text{cmp}}], \\ \text{Operative}[\text{Ev}] &:= \text{Operative}(\mathcal{X}) \end{aligned}$$

respectively.

6.2.1 Comparison-Stage First-Failure Classes

Classify the first failure position in the comparison-stage condition sequence

$$(\text{L}_i)_{i=1}^7 ; \text{G}$$

into eight comparison-stage first-failure classes. Regard each first-failure class as the parent region of the corresponding first failure position. Under internal refinement, decompose each parent region into the union of a family of subregions and its complement within the parent region.

Definition 6.36 (Comparison-stage failure). Comparison-stage failure is said to occur at a complete evaluation specification

$$\text{Ev} \in \text{EvSpec}$$

if

$$\neg \text{Operative}[\text{Ev}]$$

holds.

Definition 6.37 (Comparison-stage failure domain). Define the comparison-stage failure domain in the set of complete evaluation specifications

$$\text{EvSpec}$$

by

$$\begin{aligned} \text{FailDom}_{\text{cmp}}(\text{EvSpec}) \\ := \left\{ \text{Ev} \in \text{EvSpec} \mid \neg \text{Operative}[\text{Ev}] \right\} \end{aligned}$$

Proposition 6.38 (Comparison-stage failure and the comparison-stage condition sequence). For every

$$\text{Ev} \in \text{EvSpec}$$

the following equivalence holds:

$$\begin{aligned} & \neg \text{Operative}[\text{Ev}] \\ & \iff \neg (\text{L}_{\leq 7}[\text{Ev}] \wedge \text{G}[\text{Ev}]). \end{aligned}$$

Proof. By Definition 4.86,

$$\text{Operative}[\text{Ev}] \iff \text{L}_{\leq 7}[\text{Ev}] \wedge \text{G}[\text{Ev}].$$

Negating both sides yields the asserted equivalence. $\square$

In the comparison-series raw-condition family, comparison-series preceding-completion condition, comparison-series evaluation-input satisfaction, comparison-stage first-failure-class family, and the recursive expansion of that family below, fix, unless otherwise stated, the following objects:

$$i \in I_{\text{cmp}}^{\text{fail}}, \quad \text{Ev} \in \text{EvSpec}$$

Definition 6.39 (Comparison-series raw-condition family). Define the comparison-series raw-condition family by

$$K_{\text{cmp},i}^{\text{fail}}[\text{Ev}] := \begin{cases} \text{L}_i[\text{Ev}], & i \in \{1, \dots, 7\}, \\ \text{G}[\text{Ev}], & i = 8 \end{cases}$$

Definition 6.40 (Comparison-series preceding-completion condition). Define the conjunction of the raw conditions preceding position i in the comparison series by

$$K_{\text{cmp},<i}^{\text{fail}}[\text{Ev}] := \bigwedge_{j=1}^{i-1} K_{\text{cmp},j}^{\text{fail}}[\text{Ev}]$$

In particular,

$$K_{\text{cmp},<1}^{\text{fail}}[\text{Ev}] = \top.$$

Definition 6.41 (Comparison-series evaluation-input satisfaction). Define the input-satisfaction value of a comparison-series failure-position token by

$$\text{FailInputSat}(\text{FailTok}_{\text{cmp},i}; \text{Ev}_{\text{cmp}}^{\text{fail}}) = \top$$

Definition 6.42 (Comparison-stage first-failure-class family). Define the comparison-stage first-failure-class family by

$$\begin{aligned} & \text{FC}_i[\text{Ev}] \\ & \iff \text{FailTok}_{\text{cmp},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}] \end{aligned}$$

Proposition 6.43 (Recursive expansion of comparison-series first-failure classes). The equivalence

$$\begin{aligned} & \text{FC}_i[\text{Ev}] \\ & \iff K_{\text{cmp},<i}^{\text{fail}}[\text{Ev}] \wedge [K_{\text{cmp},i}^{\text{fail}}[\text{Ev}] = \perp] \end{aligned}$$

holds. Consequently, for $i \in \{1, \dots, 7\}$,

$$(K_{\text{cmp},<i}^{\text{fail}}, K_{\text{cmp},i}^{\text{fail}})[\text{Ev}] = (\text{L}_{<i}, \text{L}_i)[\text{Ev}],$$

whereas for $i = 8$,

$$(K_{\text{cmp},<8}^{\text{fail}}, K_{\text{cmp},8}^{\text{fail}})[\text{Ev}] = (\text{L}_{\leq 7}, \text{G})[\text{Ev}].$$

Proof. By Definition 6.17, the first token has no direct predecessor token, and the sole direct predecessor of token $i \in \{2, \dots, 8\}$ is token $i - 1$. Definition 6.41 establishes the formation inputs and evaluation inputs of all tokens. Substituting these into Definition 6.21, using Proposition 6.23, and inducting along the finite chain yields, for every $i \in I_{\text{cmp}}^{\text{fail}}$,

$$\begin{aligned} \text{FailTok}_{\text{cmp},i} &\in \text{FailedTok}^{\text{strict}}[\text{Ev}] \\ \iff &\left[\bigwedge_{j=1}^{i-1} K_{\text{cmp},j}^{\text{fail}}[\text{Ev}] = \top \right] \wedge [K_{\text{cmp},i}^{\text{fail}}[\text{Ev}] = \perp]. \end{aligned}$$

At the base, the preceding conjunction is the empty conjunction; at the inductive step, the establishment state of token $i - 1$ coincides with the preceding conjunction.

Substituting Definition 6.40, Definition 6.39, and Definition 6.42 yields the asserted single-index formula and the two boundary displays. $\square$

By Definition 6.39, the explicit form of the failure condition at each position is obtained, for $i \in \{1, \dots, 7\}$, by negating the corresponding component from Definition 4.5 through Definition 4.78, and, for $i = 8$, by negating Definition 4.79.

The negation of the cardinality-one condition in component $i = 2$ jointly represents the case in which no factorization tuple exists and the case in which two or more distinct tuples exist. The negation of the domain inclusion in component $i = 3$ is equivalent to the existence, in the source of the inclusion, of an element that does not belong to the target domain. The negation of the set equality in component $i = 5$ is equivalent to the existence of a comparison two-role arrival-value family that belongs to at least one of the two directional set differences.

The comparison-stage first-failure parent regions relative to the comparison condition sequence

$$(\mathbf{L}_i)_{i=1}^7 ; \mathbf{G}$$

are represented by a single index row using the comparison-series preceding-completion condition.

Table 3: Comparison-stage first-failure parent regions relative to the comparison condition sequence

Position	First-failure parent region
$i \in I_{\text{cmp}}^{\text{fail}}$	$\text{FC}_i \iff K_{\text{cmp},<i}^{\text{fail}} \wedge \neg K_{\text{cmp},i}^{\text{fail}}$

The preceding condition at the first position is the empty conjunction $\top$; the preceding condition and the failure condition at the eighth position are

$$\mathbf{L}_{\leq 7}, \quad \neg \mathbf{G}$$

respectively. At position $i \in \{2, \dots, 7\}$, they are

$$\mathbf{L}_{<i}, \quad \neg \mathbf{L}_i$$

respectively.

For every

$$i \in I_{\text{cmp}}^{\text{fail}}$$

the class

$$\mathbf{FC}_i$$

is a parent region for internal refinement. The witness type and the object whose generation fails in each parent region correspond as follows.

$\mathbf{FC}_1$. A witness consists of distinct source-side tokens with the same arrival value for some comparison role and reception position. That is, it consists of R, u, x_R, y_R satisfying

$$\begin{aligned} R &\in \mathbf{Rol}^{\text{cmp}}, & u &\in U, \\ x_R, y_R &\in \text{Dom}(\mathbf{Arr}_{R,u}^{\text{cmp}}), \\ x_R &\neq y_R, & \mathbf{Arr}_{R,u}^{\text{cmp}}(x_R) &= \mathbf{Arr}_{R,u}^{\text{cmp}}(y_R). \end{aligned}$$

By Lemma 4.6, the existence of such a tuple is equivalent to $\neg \mathbf{L}_1(\mathcal{X})$. In this case, the generation condition fails for the family of source-matching partial maps in Definition 4.8,

$$(\mathbf{SrcMatch}_{uv}^R)_{R,u,v},$$

whose construction requires unique recovery from the entire arrival image.

$\mathbf{FC}_2$. The witness type is a comparison-admissible position pair for which the set of component-factorization tuples for synchronized-pair joint transport has cardinality

$$0$$

or at least

$$2$$

The objects whose generation fails are the unique component-factorization family

$$\left(\begin{aligned} &\mathbf{PairDom}_{uv}[\mathbf{I}], \\ &(\mathbf{Tr}_{uv}^R[\mathbf{I}])_{R \in \mathbf{Rol}^{\text{cmp}}} \end{aligned} \right)_{(u,v)}$$

and the comparison-transport datum

$$\mathbf{Dat}^{\text{cmp}, \text{tr}}[\mathbf{I}]$$

$\mathbf{FC}_3$. The witness type is a point that belongs to the declared initial domain and does not belong to the domain of the candidate transport-word action. The objects whose generation fails are the realized restricted action

$$\mathbf{Act}_R^{\text{tr}, \text{real}}[\mathbf{Spec}],$$

the realized local-specification loop datum

$$\mathbf{LoopDat}_R^{\text{real}}[\mathbf{Spec}]$$

and the indexed data family

$$\mathbf{LoopDat}_R^{\text{real}, \text{loc}}[\mathbf{I}]$$

FC₄. The witness type is a nonfixed point in the declared initial domain of a realized locally specified action. The object whose generation fails is a family of realized restricted actions that are identity maps on their declared initial domains.

FC₅. The witness type is a comparison two-role arrival-value family belonging to the set difference between a local comparison two-role relation and the full pullback of a source-side two-role relation. The object whose generation fails is agreement between the two sets.

FC₆. The witness type is a pair of synchronized pairs that are comparison-equivalent in each role component and whose membership values in the local comparison two-role relation bundle differ. The object whose generation fails is saturation of the local comparison two-role relation bundle along the synchronized two-role comparison-pair equivalence relation.

FC₇. The witness type is a nonfixed point of the induced partial action of a source-irreducible mixed closed-loop word that contains both the transport atom kind and the source-matching atom kind. The object whose generation fails is the condition that the induced partial action of every source-irreducible mixed closed-loop word

$$\mathbf{e} \in \text{LoopWord}_{\text{src-irr}}^{\text{mix}, R}[\mathbf{l}]$$

be the identity map on its domain.

FC₈. The witness type is a nonfixed point of the induced partial action of a closed-loop word composed of the pure-transport atom kind. The object whose generation fails is the condition that the induced partial action of every pure-transport closed-loop word

$$\mathbf{w} \in \text{LoopWord}_{\text{all}}^{\text{tr}, R}[\mathbf{l}]$$

be the identity map on its domain.

Definition 6.44 (Comparison-stage first-failure parent regions). For the set of complete evaluation specifications

$$\text{EvSpec}$$

and each

$$i \in I_{\text{cmp}}^{\text{fail}}$$

define the comparison-stage first-failure parent region by

$$\begin{aligned} & \text{FailReg}_{\text{cmp}, i}(\text{EvSpec}) \\ & := \left\{ \text{Ev} \in \text{EvSpec} \mid \text{FC}_i[\text{Ev}] \right\} \end{aligned}$$

Theorem 6.45 (Partition of the comparison-stage failure domain into first-failure parent regions). For every set of complete evaluation specifications

$$\text{EvSpec}$$

one has

$$\begin{aligned} & \text{FailDom}_{\text{cmp}}(\text{EvSpec}) \\ &= \bigsqcup_{i=1}^8 \text{FailReg}_{\text{cmp},i}(\text{EvSpec}). \end{aligned}$$

Equivalently, for every

$$\text{Ev} \in \text{EvSpec}$$

one has

$$\begin{aligned} & \neg \text{Operative}[\text{Ev}] \\ & \iff \exists! i \in I_{\text{cmp}}^{\text{fail}} : \text{FC}_i[\text{Ev}]. \end{aligned}$$

Proof. Apply Lemma 6.12 to

$$\begin{aligned} \mathcal{Y} &= \text{EvSpec}, \quad n = 8, \\ (K_1, \dots, K_8) &= (\text{L}_1, \dots, \text{L}_7, \text{G}) \end{aligned}$$

By Proposition 6.38 and Definition 6.37, the left-hand side of that lemma is

$$\text{FailDom}_{\text{cmp}}(\text{EvSpec})$$

By Proposition 6.43 and Definition 6.44, its i -th component is

$$\text{FailReg}_{\text{cmp},i}(\text{EvSpec})$$

The displayed disjoint union therefore holds, and the uniqueness of the first failure position in that lemma yields the pointwise necessary-and-sufficient condition. $\square$

Definition 6.46 (Comparison-stage failure-refinement forest). Given EvSpec , for each $i \in I_{\text{cmp}}^{\text{fail}}$, take a first-layer child-index set

$$\text{SubIdx}_{\text{cmp},i}$$

For each $\lambda_1 \in \text{SubIdx}_{\text{cmp},i}$, take a second-layer child-index set

$$\text{SubIdx}_{\text{cmp},i;\lambda_1}$$

Suppose that the corresponding families of regions satisfy

$$\begin{aligned} & \text{FailReg}_{\text{cmp},i;\lambda_1}(\text{EvSpec}) \subseteq \text{FailReg}_{\text{cmp},i}(\text{EvSpec}), \\ & \text{FailReg}_{\text{cmp},i;\lambda_1,\lambda_2}(\text{EvSpec}) \subseteq \text{FailReg}_{\text{cmp},i;\lambda_1}(\text{EvSpec}), \quad \lambda_2 \in \text{SubIdx}_{\text{cmp},i;\lambda_1} \end{aligned}$$

Define the child-index set at each path by

$$\text{ChildIdx}_{\text{cmp}}(i; \lambda_1, \dots, \lambda_k) := \begin{cases} \text{SubIdx}_{\text{cmp},i}, & k = 0, \\ \text{SubIdx}_{\text{cmp},i;\lambda_1}, & k = 1, \\ \emptyset, & k = 2 \end{cases}$$

Define the node-region map by

$$\begin{aligned} \text{NodeReg}_{\text{cmp}}((i); \text{EvSpec}) &:= \text{FailReg}_{\text{cmp},i}(\text{EvSpec}), \\ \text{NodeReg}_{\text{cmp}}((i; \lambda_1); \text{EvSpec}) &:= \text{FailReg}_{\text{cmp},i;\lambda_1}(\text{EvSpec}), \\ \text{NodeReg}_{\text{cmp}}((i; \lambda_1, \lambda_2); \text{EvSpec}) &:= \text{FailReg}_{\text{cmp},i;\lambda_1,\lambda_2}(\text{EvSpec}) \end{aligned}$$

Define the comparison-stage failure-refinement forest by

$$\text{RefForest}_{\text{cmp}}(\text{EvSpec}) := \text{RefForest} [I_{\text{cmp}}^{\text{fail}}; \text{ChildIdx}_{\text{cmp}}; \text{NodeReg}_{\text{cmp}}(-; \text{EvSpec})]$$

For $k \in \{0, 1\}$ and $\varsigma \in \{\text{cov}, \text{unref}\}$, obtain the layer-specific components by

$$\text{FailReg}_{\text{cmp}, i; \lambda_1, \dots, \lambda_k}^{\varsigma}(\text{EvSpec}) := \text{NodeReg}_{\text{cmp}}^{\varsigma}((i; \lambda_1, \dots, \lambda_k); \text{EvSpec})$$

When $k = 0$, omit the semicolon and everything that follows it.

For each $i \in I_{\text{cmp}}^{\text{fail}}$, take two refinement data that share the parent region $\text{FailReg}_{\text{cmp}, i}(\text{EvSpec})$ and satisfy the assumptions of Lemma 6.33:

$$\left(\text{SubIdx}_{\text{cmp}, i}^{(\rho)}, \left(\text{FailReg}_{\text{cmp}, i; \lambda}^{(\rho)}(\text{EvSpec}) \right)_{\lambda \in \text{SubIdx}_{\text{cmp}, i}^{(\rho)}} \right), \quad \rho \in \{0, 1\}$$

The two refinement snapshots at the root node are, by RefSplit in Definition 6.32,

$$\left(\text{RefSplit} \left(\text{FailReg}_{\text{cmp}, i}(\text{EvSpec}); \left(\text{FailReg}_{\text{cmp}, i; \lambda}^{(\rho)}(\text{EvSpec}) \right)_{\lambda \in \text{SubIdx}_{\text{cmp}, i}^{(\rho)}} \right) \right)_{\rho \in \{0, 1\}}$$

Proposition 6.47 (Expansion of the comparison-stage failure-refinement forest). The comparison-stage failure-refinement forest of Definition 6.46 satisfies the node decomposition, conditional disjoint union, and ancestor inclusion of Theorem 6.35. The two refinement snapshots at the root node satisfy the covered-region expansion and the unrefined-region contraction of Lemma 6.33.

Proof. The first assertion follows by substituting

$$I_{\text{cmp}}^{\text{fail}}, \quad \text{ChildIdx}_{\text{cmp}}, \quad \text{NodeReg}_{\text{cmp}}(-; \text{EvSpec})$$

from Definition 6.46 into Theorem 6.35. The second follows by substituting the two root snapshots from the same definition into Lemma 6.33. $\square$

6.2.2 Series boundary of the comparison-stage first-failure classes

Proposition 6.48 (Series boundary of the comparison-stage first-failure classes). For every Ev in the comparison-stage failure domain, $\text{FC}_i[\text{Ev}]$ holds for a unique $i \in I_{\text{cmp}}^{\text{fail}}$. The failure-position token system satisfies

$$\begin{aligned} \text{FailTok} &= \text{FailTok}_{\text{cmp}} \sqcup \bigsqcup_{\tau \in \{\text{repr}, \text{dyn}, \text{approx}, \text{law}, \text{diff}\}} \text{FailTok}_{\tau}, \\ \text{FailTok}_{\text{cmp}} &= \{ \text{FailTok}_{\text{cmp}, i} \mid i \in I_{\text{cmp}}^{\text{fail}} \}. \end{aligned}$$

The correspondence between comparison-stage first-failure classes and membership of comparison-series failure-position tokens in the failed-token set is given by Definition 6.42. For every $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$, the failed-token set is an antichain with respect to $\preceq^{\text{fail}, \text{dep}}$ and coincides with the set of all its minimal elements.

Proof. Unique existence follows from Theorem 6.45, and the disjoint-union representation follows from Definition 6.10. The antichain property and the coincidence with the set of minimal elements follow from Proposition 6.26. $\square$

6.3 Top-level signature of the comparison-stage first-failure classes

Definition 6.49 (Truth-value indicator and membership indicator function). Define the truth-value indicator of a proposition P by

$$\mathbf{1}_{\{P\}} := \begin{cases} 1, & P, \\ 0, & \neg P \end{cases}$$

and define the membership indicator function for a set A by

$$\mathbf{1}_A(x) := \mathbf{1}_{\{x \in A\}}$$

Definition 6.50 (Top-level signature of the comparison-stage first-failure classes). Take a set of complete evaluation specifications

$$\text{EvSpec}$$

For each $i \in I_{\text{cmp}}^{\text{fail}}$, define

$$\chi_{\text{FC}_i}[\text{Ev}] := \mathbf{1}_{\text{FailReg}_{\text{cmp},i}(\text{EvSpec})}(\text{Ev})$$

and define the top-level signature of the comparison-stage first-failure classes by

$$\begin{aligned} \Sigma_{\text{cmp}} : \text{FailDom}_{\text{cmp}}(\text{EvSpec}) &\longrightarrow \{0, 1\}^8, \\ \Sigma_{\text{cmp}}[\text{Ev}] &:= (\chi_{\text{FC}_i}[\text{Ev}])_{i=1}^8 \end{aligned}$$

Proposition 6.51 (Single-component property of the top-level signature of the comparison-stage first-failure classes). Let $\mathbf{e}_i^{(8)}$ denote, in $\{0, 1\}^8$, the standard basis vector with index i . For every

$$\text{Ev} \in \text{FailDom}_{\text{cmp}}(\text{EvSpec})$$

there exists a unique

$$i \in I_{\text{cmp}}^{\text{fail}}$$

such that

$$\Sigma_{\text{cmp}}[\text{Ev}] = \mathbf{e}_i^{(8)}.$$

Consequently,

$$\sum_{j=1}^8 \chi_{\text{FC}_j}[\text{Ev}] = 1.$$

Equivalently, the i -th component equals 1 and every remaining component equals 0.

Proof. By Theorem 6.45, the parent region containing Ev has a uniquely determined index $i \in I_{\text{cmp}}^{\text{fail}}$. The membership-indicator representation in Definition 6.50 then shows that the signature equals $\mathbf{e}_i^{(8)}$ and that the sum of its components is 1. $\square$

Proposition 6.52 (Top-level signature of the comparison-stage first-failure classes on unrefined regions). For every

$$i \in I_{\text{cmp}}^{\text{fail}}, \quad \text{Ev} \in \text{FailReg}_{\text{cmp},i}^{\text{unref}}(\text{EvSpec})$$

one has

$$\Sigma_{\text{cmp}}[\text{Ev}] = \mathbf{e}_i^{(8)}.$$

That is, the i -th component equals 1; for every

$$j \in I_{\text{cmp}}^{\text{fail}} \setminus \{i\}$$

the corresponding j -th component equals 0.

Proof. By Definition 6.46, the unrefined region is contained in the i -th parent region. Applying Proposition 6.51 gives the displayed identity. $\square$

6.4 Failure positions at the representation stage

The three representation-stage conditions in Definition 4.92 are placed in correspondence with the three representation failure-position tokens in Definition 6.10. Representation-stage failure is classified by the first failure position among the three conditions on the domain in which comparison-stage operativeness holds.

Definition 6.53 (Comparison-stage-operative set of complete evaluation specifications). Define the comparison-stage-operative subset of the set of complete evaluation specifications by

$$\begin{aligned} \text{EvSpec}^{\text{op}} \\ &:= \{\text{Ev} \in \text{EvSpec} \mid \text{Operative}[\text{Ev}]\} \end{aligned}$$

Definition 6.54 (Evaluation of representation-stage conditions on complete evaluation specifications). Extend the following native-domain formulas by false to EvSpec , using the rule in Definition 6.10. Outside $\text{EvSpec}^{\text{op}}$, formation-input satisfaction for the representation series is false.

Let a typed family of comparison rules $\mathcal{J}_{\text{cmp}}$ be given. For each

$$\text{Ev} \in \text{EvSpec}^{\text{op}}, \quad i \in I_{\text{repr}}^{\text{fail}} = \{1, 2, 3\}$$

define the representation-series condition family by

$$K_{\text{repr},i}^{\text{fail}}[\text{Ev}] := \begin{cases} K_{\text{CanOrdDesc}}^{\text{repr}}[\mathbb{I}], & i = 1, \\ K_{\text{LocIncTrans}}^{\text{repr}}[\mathbb{I}], & i = 2, \\ K_{\text{GlobIncTrans}}^{\text{repr}}[\mathbb{I}], & i = 3 \end{cases}$$

and set

$$\begin{aligned} \text{Repr}_{\leq 0}[\mathbb{I}] &:= \top, \\ \text{Repr}_{\leq i}[\mathbb{I}] &\iff \bigwedge_{m=1}^i K_{\text{repr},m}^{\text{fail}}[\text{Ev}] \quad (i \in I_{\text{repr}}^{\text{fail}}) \end{aligned}$$

Definition 6.55 (Representation-failure evaluation context). Define the pair consisting of the set of complete evaluation specifications and the typed family of comparison rules by

$$\Gamma_{\text{repr}} := (\text{EvSpec}; \mathcal{J}_{\text{cmp}})$$

Definition 6.56 (Representation-series prefix-satisfaction condition). For each $i \in I_{\text{repr}}^{\text{fail}}$ and $\text{Ev} \in \text{EvSpec}^{\text{op}}$, define

$$K_{\text{repr}, < i}^{\text{fail}}[\text{Ev}] := \bigwedge_{m=1}^{i-1} K_{\text{repr}, m}^{\text{fail}}[\text{Ev}],$$

$$K_{\text{repr}, < 1}^{\text{fail}}[\text{Ev}] = \top$$

Definition 6.57 (Representation-series evaluation-input satisfaction). For each $i \in I_{\text{repr}}^{\text{fail}}$ and $\text{Ev} \in \text{EvSpec}^{\text{op}}$, define the input-satisfaction value of the representation-series failure-position token by

$$\text{FailInputSat}(\text{FailTok}_{\text{repr}, i}; \text{Ev}_{\text{repr}}^{\text{fail}}) = \top$$

Definition 6.58 (Representation-stage failure domain). Define the representation-stage failure domain by

$$\begin{aligned} \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] &:= \{ \text{Ev} \in \text{EvSpec}^{\text{op}} \mid \exists i \in I_{\text{repr}}^{\text{fail}}, \neg K_{\text{repr}, i}^{\text{fail}}[\text{Ev}] \} \\ &= \{ \text{Ev} \in \text{EvSpec}^{\text{op}} \mid \neg \text{Repr}_{\leq 3}[\mathbb{I}] \}. \end{aligned}$$

Definition 6.59 (Representation-stage first-failure parent regions). For each

$$i \in I_{\text{repr}}^{\text{fail}}, \quad \text{Ev} \in \text{EvSpec}$$

define

$$\begin{aligned} \text{FailReg}_{\text{repr}, i}[\Gamma_{\text{repr}}] &:= \{ \text{Ev} \in \text{EvSpec}^{\text{op}} \mid \text{FailTok}_{\text{repr}, i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}] \}, \\ \text{FR}_i[\text{Ev}] &\iff \text{Ev} \in \text{FailReg}_{\text{repr}, i}[\Gamma_{\text{repr}}]. \end{aligned}$$

Proposition 6.60 (Recursive expansion of the representation-stage first-failure predicates). For each $i \in I_{\text{repr}}^{\text{fail}}$ and $\text{Ev} \in \text{EvSpec}^{\text{op}}$,

$$\text{FR}_i[\text{Ev}] \iff K_{\text{repr}, < i}^{\text{fail}}[\text{Ev}] \wedge [K_{\text{repr}, i}^{\text{fail}}[\text{Ev}] = \perp]$$

holds.

Proof. By Definition 6.53 and Proposition 6.38, all tokens in the comparison series are satisfied. Applying Definition 6.17, Definition 6.57, and Definition 6.21 along the three nodes of the representation series shows that failure of the i -th token coincides with the conjunction of satisfaction of the generation conditions preceding the i -th position and failure of the i -th generation condition. Substitution of Definition 6.59, Definition 6.56, and Definition 6.54 gives the claim. $\square$

Proposition 6.61 (Generation boundary of representation-stage first failure). For each $i \in I_{\text{repr}}^{\text{fail}}$ and $\text{Ev} \in \text{EvSpec}^{\text{op}}$,

$$\text{FR}_i[\text{Ev}] \iff \text{Repr}_{\leq i-1}[\mathbb{I}] \wedge \neg \text{Repr}_{\leq i}[\mathbb{I}].$$

Proof. This follows from Proposition 6.60 and Definition 6.54. $\square$

Expanding the failure evidence at each position to the corresponding generation condition yields, for $i = 1$, a four-point witness against the canonical order-descent condition; for $i = 2$, a nontransitivity witness for the local incomparability relation; and, for $i = 3$, a nontransitivity witness for the global incomparability relation. The first concrete form is given by the negation of Definition 4.99; the second and third are given by Definition 6.13.

Proposition 6.62 (Correspondence between representation failure-position tokens and the three conditions). For each $i \in I_{\text{repr}}^{\text{fail}}$ and $\text{Ev} \in \text{EvSpec}^{\text{op}}$, the formation-input satisfaction value and the evaluation-input satisfaction value equal $\top$, and the state of every direct predecessor token outside the series

$$\text{FailTok}^{(0)} \in \text{FailPredOf}(\text{FailTok}_{\text{repr},i}) \setminus \text{FailTok}_{\text{repr}}$$

is

$$\text{FailStateOf}(\text{FailTok}^{(0)}; \text{Ev}_{\text{FailSeriesOf}(\text{FailTok}^{(0)})}^{\text{fail}}) = \text{sat}.$$

Under these conditions,

$$\text{FR}_i[\text{Ev}] \iff \text{FailTok}_{\text{repr},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}].$$

Proof. The two input-satisfaction values follow from Definition 6.57. The satisfaction of the direct predecessor tokens outside the series follows from the proof of Proposition 6.60, and the final equivalence follows from Definition 6.59. $\square$

Definition 6.63 (Internal refinement families of representation-failure parent regions). For each $i \in I_{\text{repr}}^{\text{fail}}$, suppose that an index set $\text{SubIdx}_{\text{repr},i}$ and a family of subregions satisfy

$$\begin{aligned} & (\text{FailReg}_{\text{repr},i;\lambda}[\Gamma_{\text{repr}}])_{\lambda \in \text{SubIdx}_{\text{repr},i}}, \\ & \text{FailReg}_{\text{repr},i;\lambda}[\Gamma_{\text{repr}}] \subseteq \text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}] \end{aligned}$$

This family is called mutually exclusive when, for distinct $\lambda_0, \lambda_1 \in \text{SubIdx}_{\text{repr},i}$,

$$\text{FailReg}_{\text{repr},i;\lambda_0}[\Gamma_{\text{repr}}] \cap \text{FailReg}_{\text{repr},i;\lambda_1}[\Gamma_{\text{repr}}] = \emptyset$$

holds.

Parent failure region	First-failure condition	Generation boundary
FR_i	$K_{\text{repr},<i}^{\text{fail}}[\text{Ev}] \wedge \neg K_{\text{repr},i}^{\text{fail}}[\text{Ev}]$ $i \in I_{\text{repr}}^{\text{fail}} = \{1, 2, 3\}$	$\text{Repr}_{\leq i-1}[\text{I}] \wedge \neg \text{Repr}_{\leq i}[\text{I}]$

Table 4: Correspondence of the three representation-failure parent regions with their first-failure conditions and generation boundaries

Proposition 6.64 (Unique localization by representation-failure parent regions).

$$\text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] = \bigsqcup_{i \in I_{\text{repr}}^{\text{fail}}} \text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}].$$

Consequently, for each $\text{Ev} \in \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}]$, $\text{FR}_i[\text{Ev}]$ holds for a unique $i \in I_{\text{repr}}^{\text{fail}}$.

Proof. Apply Lemma 6.12 with

$$\begin{aligned} \mathcal{Y} &= \text{EvSpec}^{\text{op}}, \\ n &= 3, \\ K_i(\text{Ev}) &\iff K_{\text{repr},i}^{\text{fail}}[\text{Ev}] \end{aligned}$$

The claim follows from Definition 6.58, Proposition 6.60, and Definition 6.59. $\square$

Definition 6.65 (Representation-stage failure-refinement forest). For each $i \in I_{\text{repr}}^{\text{fail}}$ and $\lambda_1 \in \text{SubIdx}_{\text{repr},i}$, take a second-layer child-index set

$$\text{SubIdx}_{\text{repr},i;\lambda_1}$$

and a second-layer family of subregions, and suppose that

$$\text{FailReg}_{\text{repr},i;\lambda_1,\lambda_2}[\Gamma_{\text{repr}}] \subseteq \text{FailReg}_{\text{repr},i;\lambda_1}[\Gamma_{\text{repr}}], \quad \lambda_2 \in \text{SubIdx}_{\text{repr},i;\lambda_1}$$

Define the child-index set at each path by

$$\text{ChildIdx}_{\text{repr}}(i; \lambda_1, \dots, \lambda_k) := \begin{cases} \text{SubIdx}_{\text{repr},i}, & k = 0, \\ \text{SubIdx}_{\text{repr},i;\lambda_1}, & k = 1, \\ \emptyset, & k = 2 \end{cases}$$

Define the node-region map by

$$\begin{aligned} \text{NodeReg}_{\text{repr}}((i); \Gamma_{\text{repr}}) &:= \text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}], \\ \text{NodeReg}_{\text{repr}}((i; \lambda_1); \Gamma_{\text{repr}}) &:= \text{FailReg}_{\text{repr},i;\lambda_1}[\Gamma_{\text{repr}}], \\ \text{NodeReg}_{\text{repr}}((i; \lambda_1, \lambda_2); \Gamma_{\text{repr}}) &:= \text{FailReg}_{\text{repr},i;\lambda_1,\lambda_2}[\Gamma_{\text{repr}}] \end{aligned}$$

Define the representation-stage failure-refinement forest by

$$\text{RefForest}_{\text{repr}}[\Gamma_{\text{repr}}] := \text{RefForest} [I_{\text{repr}}^{\text{fail}}; \text{ChildIdx}_{\text{repr}}; \text{NodeReg}_{\text{repr}}(-; \Gamma_{\text{repr}})]$$

For $k \in \{0, 1\}$ and $\varsigma \in \{\text{cov}, \text{unref}\}$, obtain the layer-specific components by

$$\text{FailReg}_{\text{repr},i;\lambda_1,\dots,\lambda_k}^{\varsigma}[\Gamma_{\text{repr}}] := \text{NodeReg}_{\text{repr}}^{\varsigma}((i; \lambda_1, \dots, \lambda_k); \Gamma_{\text{repr}})$$

When $k = 0$, omit the semicolon and everything that follows it.

Proposition 6.66 (Node decomposition of the representation-stage failure-refinement forest). The representation-stage failure-refinement forest of Definition 6.65 satisfies the node decomposition and conditional disjoint union of Theorem 6.35.

Proof. Substitute $I_{\text{repr}}^{\text{fail}}$, $\text{ChildIdx}_{\text{repr}}$, and $\text{NodeReg}_{\text{repr}}(-; \Gamma_{\text{repr}})$ from Definition 6.65 into Theorem 6.35. $\square$

Proposition 6.67 (Separation of the comparison-stage and representation-stage failure domains).

$$\begin{aligned} \text{FailDom}_{\text{cmp}}(\text{EvSpec}) \cap \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] &= \emptyset, \\ \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] &\subseteq \{\text{Ev} = (l, o, \zeta) \in \text{EvSpec} \mid \text{Operative}[\text{Ev}]\}. \end{aligned}$$

Proof. Definition 6.37 gives the first domain as the domain on which $\neg \text{Operative}[\text{Ev}]$ holds, whereas Definition 6.58 places the second domain within the domain on which $\text{Operative}[\text{Ev}]$ holds. Both claims follow. $\square$

Definition 6.68 (Representation-stage failure-refinement snapshots and recursive components). For each $i \in I_{\text{repr}}^{\text{fail}}$, take two refinement data that share the parent region $\text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}]$ and satisfy the assumptions of Lemma 6.33:

$$\left(\text{SubIdx}_{\text{repr},i}^{(\rho)}, \left(\text{FailReg}_{\text{repr},i;\lambda}^{(\rho)}[\Gamma_{\text{repr}}] \right)_{\lambda \in \text{SubIdx}_{\text{repr},i}^{(\rho)}} \right), \quad \rho \in \{0, 1\}$$

By RefSplit in Definition 6.32, the two refinement snapshots at the root node are

$$\left(\text{RefSplit} \left(\text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}]; \left(\text{FailReg}_{\text{repr},i;\lambda}^{(\rho)}[\Gamma_{\text{repr}}] \right)_{\lambda \in \text{SubIdx}_{\text{repr},i}^{(\rho)}} \right)_{\rho \in \{0,1\}} \right)$$

For each $\lambda \in \text{SubIdx}_{\text{repr},i}$, the recursive components at the node $(i; \lambda)$ are $\text{ChildIdx}_{\text{repr}}$, $\star$, and $\text{NodeReg}_{\text{repr}}$ of Definition 6.65.

Proposition 6.69 (Expansion of representation-stage failure-refinement snapshots and recursive components). The two root snapshots of Definition 6.68 satisfy the covered-region expansion and unrefined-region contraction of Lemma 6.33. The recursive components of Definition 6.65 satisfy the ancestor inclusion of Theorem 6.35.

Proof. The first assertion follows by substituting the two root snapshots into Lemma 6.33. The second follows by substituting $I_{\text{repr}}^{\text{fail}}$, $\text{ChildIdx}_{\text{repr}}$, and $\text{NodeReg}_{\text{repr}}(-; \Gamma_{\text{repr}})$ into Theorem 6.35. $\square$

6.4.1 Representation-stage first-failure signature

Definition 6.70 (One-hot signature of the representation-stage first-failure parent regions). Using the membership indicator function in Definition 6.49, set

$$\chi_{\text{FR}_i}[\text{Ev}] := \mathbf{1}_{\text{FailReg}_{\text{repr},i}[\Gamma_{\text{repr}}]}(\text{Ev}).$$

Define the representation-stage first-failure signature by

$$\begin{aligned} \Sigma_{\text{repr}} : \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}] &\longrightarrow \{0, 1\}^3, \\ \Sigma_{\text{repr}}[\text{Ev}] &:= (\chi_{\text{FR}_i}[\text{Ev}])_{i=1}^3 \end{aligned}$$

Proposition 6.71 (One-hot property of the representation-stage first-failure signature). Let $\mathbf{e}_i^{(3)}$ denote, in $\{0, 1\}^3$, the standard basis vector with index i . For every

$$\text{Ev} \in \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}]$$

there exists a unique $i \in I_{\text{repr}}^{\text{fail}}$ such that

$$\begin{aligned} \Sigma_{\text{repr}}[\text{Ev}] &= \mathbf{e}_i^{(3)}, \\ \sum_{j=1}^3 \chi_{\text{FR}_j}[\text{Ev}] &= 1. \end{aligned}$$

Proof. This follows from Proposition 6.64 and Definition 6.70. $\square$

Proposition 6.72 (Correspondence between representation failure-position tokens and the one-hot signature). Under the assumptions of Proposition 6.62, for every

$$\text{Ev} \in \text{FailDom}_{\text{repr}}[\Gamma_{\text{repr}}], \quad i \in I_{\text{repr}}^{\text{fail}}$$

one has

$$\chi_{\text{FR}_i}[\text{Ev}] = 1 \iff \text{FR}_i[\text{Ev}] \iff \text{FailTok}_{\text{repr},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}].$$

Proof. This follows from Definition 6.70 and Proposition 6.62. $\square$

6.5 Failure positions at the continuum-approximation stage

Take as inputs the continuum-approximation evaluation-input datum of Definition 4.168, the three typed $\mathfrak{D}$ -record code relations of Definition 5.38, and the observer-time-relative $\mathfrak{D}$ -record images of Definition 5.75. The continuum-approximation conditions consist of six finite-granularity and record-fidelity conditions, the continuous-extension condition, the order-stability condition, and the relation-stability condition.

Definition 6.73 (Index set of continuum-approximation failure conditions). For the continuum-approximation failure-position index set of Definition 6.10, define the three-layer index sets by

$$\begin{aligned} I_{\text{approx}}^{\text{fail}} &= \{1, \dots, 9\}, \\ I_{\text{approx}}^{(1)} &:= \{1, \dots, 6\}, \\ I_{\text{approx}}^{(2)} &:= \{7\}, \\ I_{\text{approx}}^{(3)} &:= \{8, 9\} \end{aligned}$$

The layer and evaluation target of each index are as follows.

i	Layer	Evaluation target
1	First layer	Counting-step width derived from $\mathfrak{C}$
2	First layer	$\mathfrak{D}$ -record stability
3	First layer	$\mathfrak{D}$ -record-cell width
4	First layer	$\mathfrak{D}$ -typed-record separation
5	First layer	$\mathfrak{B}$ -object-state distinguishability
6	First layer	$\mathfrak{D}$ -observer-side coupling limit
7	Second layer	Continuous extension
8	Third layer	Order stability
9	Third layer	Relation stability

Definition 6.74 (Continuum-approximation-stage evaluation datum). Let the complete evaluation specification and approximation scale be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Define its continuum-approximation evaluation context by

$$\Gamma_{\text{Ev}, \lambda}^{\text{approx}} := \text{Ev}_{\text{approx}, \lambda}^{\text{fail}}$$

Let the approximation-map tag set be

$$\text{MapTag}^{\text{approx}} := \left\{ (\mathfrak{C}, \text{cmp}), (\mathfrak{D}, \text{cmp}), (\mathfrak{C}, \text{obs}), \right. \\ \left. (\mathfrak{B}, \text{state}), (\mathfrak{D}, \text{coupl}) \right\}$$

and let the strictly ordered approximation-map tag set be

$$\text{OrdTag}^{\text{approx}} := \text{MapTag}^{\text{approx}} \setminus \{(\mathfrak{D}, \text{coupl})\}$$

Define the typed-record tag set and the corresponding family of $\mathfrak{D}$ -record code relations by

$$\begin{aligned} \text{RecSepTag}^{\text{approx}} &:= \text{Rol}^{\text{dyn}, \text{quot}} \cup \{\text{pair}\}, \\ \text{RecSepRel}_{\sigma, o}^{\text{dyn}} &:= \begin{cases} \text{RecordCodeRel}_{\sigma, o}^{\text{dyn}}, & \sigma \in \text{Rol}^{\text{dyn}, \text{quot}}, \\ \text{RecordCodeRel}_o^{\text{dyn}, \text{pair}}, & \sigma = \text{pair} \end{cases} \end{aligned}$$

Define the record-code evaluation component by

$$\begin{aligned} & \text{RecCodeComp}_{\mathfrak{D},o}^{\text{approx}} \\ & := (\text{RecSepRel}_{\sigma,o}^{\text{dyn}})_{\sigma \in \text{RecSepTag}^{\text{approx}}} \end{aligned}$$

For each $t \in \text{MapTag}^{\text{approx}}$ define the approximation-map evaluation component by

$$\begin{aligned} \text{MapComp}_{t,\lambda}^{\text{approx}} := & \left((\text{Dom}_t^{\text{approx}}, \mathcal{T}_t^{\text{approx,src}}), (\text{Cod}_t^{\text{approx}}, \mathcal{T}_t^{\text{approx,tgt}}), \right. \\ & \text{Int}_{t,\lambda}^{\text{approx}}, \text{Samp}_{t,\lambda}^{\text{approx}}, \text{SVal}_{t,\lambda}^{\text{approx}}, \\ & \left. \text{Map}_{t,\lambda}^{\text{approx}}, \text{Dev}_t^{\text{approx}}, \varepsilon_t^{\text{approx,fit}} \right) \end{aligned}$$

The first two components are topological spaces, with types

$$\begin{aligned} & \text{Int}_{t,\lambda}^{\text{approx}} \subseteq \text{Dom}_t^{\text{approx}}, \quad \text{Samp}_{t,\lambda}^{\text{approx}} \subseteq \text{Int}_{t,\lambda}^{\text{approx}}, \\ & \text{SVal}_{t,\lambda}^{\text{approx}} : \text{Samp}_{t,\lambda}^{\text{approx}} \longrightarrow \text{Cod}_t^{\text{approx}}, \quad \text{Map}_{t,\lambda}^{\text{approx}} : \text{Int}_{t,\lambda}^{\text{approx}} \longrightarrow \text{Cod}_t^{\text{approx}}, \\ & \text{Dev}_t^{\text{approx}} : (\text{Cod}_t^{\text{approx}})^2 \longrightarrow \overline{\mathbb{R}}_{\geq 0}, \quad \varepsilon_t^{\text{approx,fit}} \in \overline{\mathbb{R}}_{\geq 0} \end{aligned}$$

The interval component is a connected ordered interval, and the sample component is a finite set.

For each $A \in \mathcal{P}(\text{Int}_{t,\lambda}^{\text{approx}})$ define the set image under the approximation map by

$$\begin{aligned} & \text{Map}_{t,\lambda}^{\text{approx}}[A] \\ & := \{ \text{Map}_{t,\lambda}^{\text{approx}}(x) \mid x \in A \} \end{aligned}$$

For each $A \in \mathcal{P}(\text{Cod}_t^{\text{approx}})$ define the deviation diameter by

$$\begin{aligned} & \text{Diam}_t^{\text{approx}}(A) \\ & := \begin{cases} 0, & A = \emptyset, \\ \sup \{ \text{Dev}_t^{\text{approx}}(y_0, y_1) \mid y_0, y_1 \in A \}, & A \in \mathcal{P}(\text{Cod}_t^{\text{approx}}) \setminus \{ \emptyset \} \end{cases} \end{aligned}$$

Suppose that the sample-fit tolerance $\varepsilon_t^{\text{approx,fit}} \in \overline{\mathbb{R}}_{\geq 0}$ is given.

For each $t \in \text{OrdTag}^{\text{approx}}$ let the strict orders on the domain and codomain be $\prec_t^{\text{approx,src}}$, $\prec_t^{\text{approx,tgt}}$.

Let the approximation map for $\mathfrak{D}$ -record code values be

$$\text{RecMap}_{\mathfrak{D},\lambda}^{\text{approx}} : \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}} \longrightarrow \text{Cod}_{\mathfrak{D},\lambda}^{\text{approx,rec}}$$

Define the domain for each typed-record tag by

$$\text{RecSepDom}_{\sigma} := \begin{cases} \text{Qu}_{\sigma,o}^{\text{dyn,can}}, & \sigma \in \text{Rol}^{\text{dyn,quot}}, \\ \text{TrjRel}_o^{\text{dyn,can}}, & \sigma = \text{pair} \end{cases}$$

For each $\sigma \in \text{RecSepTag}^{\text{approx}}$, $z \in \text{RecSepDom}_{\sigma}$ define the post-approximation $\mathfrak{D}$ -record image by

$$\begin{aligned} & \text{RecImg}_{\sigma,\lambda}^{\text{approx}}(z) \\ & := \{ \text{RecMap}_{\mathfrak{D},\lambda}^{\text{approx}}(r_{\mathfrak{D}}) \mid (z, r_{\mathfrak{D}}) \in \text{RecSepRel}_{\sigma,o}^{\text{dyn}} \} \end{aligned}$$

Let the relation-stability tag set be

$$\text{RelTag}^{\text{approx}} := \{\text{cmp}, \text{trj}\} \cup \text{RecSepTag}^{\text{approx}}$$

For each $\rho \in \text{RelTag}^{\text{approx}}$ define the types of the relation-preservation component by

$$\begin{aligned} \text{Arity}_{\rho}^{\text{approx}} &\in \mathbb{N}_{\geq 1}, & \text{Rel}_{\rho}^{\text{approx,src}} &\subseteq \prod_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}} \text{Dom}_{\rho,k}^{\text{approx}}, \\ \text{RelMap}_{\rho,k,\lambda}^{\text{approx}} : \text{Dom}_{\rho,k}^{\text{approx}} &\longrightarrow \text{Cod}_{\rho,k}^{\text{approx}}, & \text{Rel}_{\rho,\lambda}^{\text{approx,cand}} &\subseteq \prod_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}} \text{Cod}_{\rho,k}^{\text{approx}} \end{aligned}$$

Define the five typed relations by

$$\text{Rel}_{\rho}^{\text{approx,src}} := \begin{cases} \text{Rel}^{\text{cmp,cand}}[\mathbb{I}], & \rho = \text{cmp}, \\ \text{TrjRel}_o^{\text{dyn,cand}}, & \rho = \text{trj}, \\ \text{RecSepRel}_{\rho,o}^{\text{dyn}}, & \rho \in \text{RecSepTag}^{\text{approx}} \end{cases}$$

For the continuum-approximation componentwise evaluation values

$$\kappa \in \{(\mathfrak{C}, \text{cntstep}), (\mathfrak{D}, \text{stab}), (\mathfrak{B}, \text{disc}), (\mathfrak{O}, \text{coupl})\}$$

let the order-preserving evaluation map into the nonnegative extended real values be

$$\text{DefMap}_{\kappa,\lambda}^{\text{approx}} : \text{EvalVal}_{\kappa}^{\text{cont}} \longrightarrow \overline{\mathbb{R}}_{\geq 0}$$

Let the first-layer defect quantities and tolerances be

$$\begin{aligned} &\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda], \\ &\varepsilon_i^{\text{approx}} \in \overline{\mathbb{R}}_{\geq 0}, \\ &i \in I_{\text{approx}}^{(1)} \end{aligned}$$

For each

$$(i, \kappa) \in \left\{ (1, (\mathfrak{C}, \text{cntstep})), (2, (\mathfrak{D}, \text{stab})), \right. \\ \left. (5, (\mathfrak{B}, \text{disc})), (6, (\mathfrak{O}, \text{coupl})) \right\}$$

define the defect quantities for the four components by

$$\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] := \text{DefMap}_{\kappa,\lambda}^{\text{approx}}(\text{Eval}_{\kappa}^{\text{cont}}[\mathbb{I}])$$

Assume, for the two representation tags, that

$$\begin{aligned} \text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}}) &\subseteq \text{Int}_{(\mathfrak{D},\text{cmp}),\lambda}^{\text{approx}}, \\ \text{RecordImg}_{\mathfrak{D},o}^{\text{time,real}}(x_{\mathfrak{D}}) &\subseteq \text{Int}_{(\mathfrak{C},\text{obs}),\lambda}^{\text{approx}} \end{aligned}$$

Define the post-approximation $\mathfrak{D}$ -record-cell images by

$$\begin{aligned} &\text{RecCellImg}_{(\mathfrak{D},\text{cmp}),\lambda}^{\text{approx}}(u, x_{\mathfrak{D}}) \\ &:= \text{Map}_{(\mathfrak{D},\text{cmp}),\lambda}^{\text{approx}}[\text{CellImg}_{\mathfrak{D},u}^{\text{repr}}[\mathbb{I}](x_{\mathfrak{D}})], \\ &\text{RecCellImg}_{(\mathfrak{C},\text{obs}),\lambda}^{\text{approx}}(o, x_{\mathfrak{D}}) \\ &:= \text{Map}_{(\mathfrak{C},\text{obs}),\lambda}^{\text{approx}}[\text{RecordImg}_{\mathfrak{D},o}^{\text{time,real}}(x_{\mathfrak{D}})] \end{aligned}$$

Define the $\mathfrak{D}$ -record-cell-width defect quantity by

$$\begin{aligned} \text{Def}_3^{\text{approx}}[\text{Ev}; \lambda] \\ := \sup \left(\{0\} \cup \left\{ \text{Diam}_{(\mathfrak{D}, \text{cmp})}^{\text{approx}} \left(\text{RecCellImg}_{(\mathfrak{D}, \text{cmp}), \lambda}^{\text{approx}}(u, x_{\mathfrak{D}}) \right) \mid \begin{array}{l} u \in U, \\ x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^* \end{array} \right\} \right. \\ \left. \cup \left\{ \text{Diam}_{(\mathfrak{C}, \text{obs})}^{\text{approx}} \left(\text{RecCellImg}_{(\mathfrak{C}, \text{obs}), \lambda}^{\text{approx}}(o, x_{\mathfrak{D}}) \right) \mid x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D}}^* \right\} \right) \end{aligned}$$

Define the $\mathfrak{D}$ -typed-record-separation defect quantity, using the truth-value indicator in Definition 6.49 as follows:

$$\begin{aligned} \text{Def}_4^{\text{approx}}[\text{Ev}; \lambda] \\ := \mathbf{1} \left\{ \begin{array}{l} \exists \sigma \in \text{RecSepTag}^{\text{approx}}, \quad \exists z_0, z_1 \in \text{RecSepDom}_{\sigma}, \\ z_0 \neq z_1, \quad \text{Reclmg}_{\sigma, \lambda}^{\text{approx}}(z_0) \cap \text{Reclmg}_{\sigma, \lambda}^{\text{approx}}(z_1) \neq \emptyset \end{array} \right\} \end{aligned}$$

For each $\rho \in \text{RelTag}^{\text{approx}}$ define the relation-preservation evaluation component by

$$\begin{aligned} \text{RelComp}_{\rho, \lambda}^{\text{approx}} \\ := \left(\begin{array}{c} \text{Rel}_{\rho}^{\text{approx}, \text{src}}, \\ (\text{RelMap}_{\rho, k, \lambda}^{\text{approx}})_{k=1}^{\text{Arity}_{\rho}^{\text{approx}}}, \\ \text{Rel}_{\rho, \lambda}^{\text{approx}, \text{cand}} \end{array} \right) \end{aligned}$$

Define the continuum-approximation-stage evaluation datum by

$$\begin{aligned} \text{EvalDat}^{\text{approx}}[\text{Ev}; \lambda] \\ := \left(\begin{array}{l} \text{EvalInput}^{\text{cont}}[I], \\ (\text{RecCodeComp}_{\mathfrak{D}, o}^{\text{approx}})_o, \\ (\text{RecordImg}_{\mathfrak{D}, o}^{\text{time}, \text{real}}(x_{\mathfrak{D}}))_{o, x_{\mathfrak{D}}}, \\ (\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda], \varepsilon_i^{\text{approx}})_{i \in I_{\text{approx}}^{(1)}}, \\ (\text{MapComp}_{t, \lambda}^{\text{approx}})_{t \in \text{MapTag}^{\text{approx}}}, \\ \text{RecMap}_{\mathfrak{D}, \lambda}^{\text{approx}}, \\ (\text{RelComp}_{\rho, \lambda}^{\text{approx}})_{\rho \in \text{RelTag}^{\text{approx}}} \end{array} \right) \end{aligned}$$

Definition 6.75 (Continuum-approximation-stage condition group). Let the complete evaluation specification and approximation scale be

$$(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$$

For the continuum-approximation-stage evaluation datum in Definition 6.74, define the six first-layer conditions by

$$\begin{aligned} K_i^{\text{approx}}[\text{Ev}; \lambda] \\ \iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \varepsilon_i^{\text{approx}}, \quad i \in I_{\text{approx}}^{(1)} \end{aligned}$$

Define the continuous-extension condition by

$$\begin{aligned}
& K_7^{\text{approx}}[\text{Ev}; \lambda] \\
& \iff \left[\forall t \in \text{MapTag}^{\text{approx}}, \right. \\
& \quad \text{Map}_{t,\lambda}^{\text{approx}} : \left(\text{Int}_{t,\lambda}^{\text{approx}}, \mathcal{T}_t^{\text{approx,src}} \Big|_{\text{Int}_{t,\lambda}^{\text{approx}}} \right) \\
& \quad \longrightarrow (\text{Cod}_t^{\text{approx}}, \mathcal{T}_t^{\text{approx,tgt}}) \text{ is continuous,} \\
& \quad \forall s \in \text{Samp}_{t,\lambda}^{\text{approx}}, \\
& \quad \left. \text{Dev}_t^{\text{approx}}(\text{Map}_{t,\lambda}^{\text{approx}}(s), \text{SVal}_{t,\lambda}^{\text{approx}}(s)) \leq \varepsilon_t^{\text{approx,fit}} \right]
\end{aligned}$$

Define the order-stability condition by

$$\begin{aligned}
& K_8^{\text{approx}}[\text{Ev}; \lambda] \\
& \iff \left[\forall t \in \text{OrdTag}^{\text{approx}}, \right. \\
& \quad \forall x, y \in \text{Int}_{t,\lambda}^{\text{approx}}, \\
& \quad \left. x \prec_t^{\text{approx,src}} y \implies \text{Map}_{t,\lambda}^{\text{approx}}(x) \prec_t^{\text{approx,tgt}} \text{Map}_{t,\lambda}^{\text{approx}}(y) \right]
\end{aligned}$$

Define the relation-stability condition by

$$\begin{aligned}
& K_9^{\text{approx}}[\text{Ev}; \lambda] \\
& \iff \left[\forall \rho \in \text{RelTag}^{\text{approx}}, \right. \\
& \quad \forall (x_k)_{k=1}^{\text{Arity}_\rho^{\text{approx}}} \in \prod_{k=1}^{\text{Arity}_\rho^{\text{approx}}} \text{Dom}_{\rho,k}^{\text{approx}}, \\
& \quad (x_k)_{k=1}^{\text{Arity}_\rho^{\text{approx}}} \in \text{Rel}_\rho^{\text{approx,src}} \\
& \quad \iff (\text{RelMap}_{\rho,k,\lambda}^{\text{approx}}(x_k))_{k=1}^{\text{Arity}_\rho^{\text{approx}}} \in \text{Rel}_{\rho,\lambda}^{\text{approx,cand}} \left. \right]
\end{aligned}$$

Define the continuum-approximation-stage condition group by

$$\mathcal{K}_{\text{cont}}[\Gamma_{\text{Ev},\lambda}^{\text{approx}}] := \{ K_i^{\text{approx}}[\text{Ev}; \lambda] \mid i \in I_{\text{approx}}^{\text{fail}} \}$$

Define the predicate asserting that all nine conditions hold by

$$\begin{aligned}
& \text{CondAll}^{\text{approx}}[\text{Ev}; \lambda] \\
& \iff \bigwedge_{i \in I_{\text{approx}}^{\text{fail}}} K_i^{\text{approx}}[\text{Ev}; \lambda]
\end{aligned}$$

The target of each condition is as follows.

i	Target
1	Counting-step width derived from $\mathfrak{C}$
2	$\mathfrak{D}$ -record stability
3	$\mathfrak{D}$ -record-cell width
4	$\mathfrak{D}$ -typed-record separation
5	$\mathfrak{B}$ -object-state distinguishability
6	$\mathfrak{D}$ -observer-side coupling limit
7	Continuous extension
8	Order stability
9	Relation stability

Let the non-ancestor formative-input satisfaction value be

$$\text{InputSat}_i^{\text{approx,form}}[\text{Ev}; \lambda] \in \mathbb{B}$$

When the domains and codomains of the sets, orders, topologies, maps, and relations appearing in condition $K_i^{\text{approx}}[\text{Ev}; \lambda]$ are typed, this value takes $\top$.

Let the evaluation-input satisfaction value be

$$\text{InputSat}_i^{\text{approx,eval}}[\text{Ev}; \lambda] \in \mathbb{B}$$

This value takes $\top$ when the first layer provides defect quantities and tolerances, the second layer provides finite samples, approximation maps, deviation maps, and fit tolerances, and the third layer provides strict orders or post-approximation relation candidates.

Specialize the input-satisfaction maps in Definition 6.21 by

$$\begin{aligned} \text{FailInputSat}^{\text{form}}(\text{FailTok}_{\text{approx},i}; \Gamma_{\text{Ev},\lambda}^{\text{approx}}) \\ &:= \text{InputSat}_i^{\text{approx,form}}[\text{Ev}; \lambda], \\ \text{FailInputSat}^{\text{eval}}(\text{FailTok}_{\text{approx},i}; \Gamma_{\text{Ev},\lambda}^{\text{approx}}) \\ &:= \text{InputSat}_i^{\text{approx,eval}}[\text{Ev}; \lambda] \end{aligned}$$

Specialize the failure-condition evaluation map by

$$\begin{aligned} \text{FailEval}(\text{FailTok}_{\text{approx},i}; \Gamma_{\text{Ev},\lambda}^{\text{approx}}) \\ &:= K_i^{\text{approx}}[\text{Ev}; \lambda], \quad i \in I_{\text{approx}}^{\text{fail}} \end{aligned}$$

Define input satisfaction for quantitative evaluation by

$$\begin{aligned} \text{InputSat}^{\text{approx}}[\text{Ev}; \lambda] &= \top \\ \iff K_{\text{dyn}}^{\text{fail}}[\text{Ev}] &= \top \\ \wedge \left[\forall i \in I_{\text{approx}}^{\text{fail}}, \quad \text{InputSat}_i^{\text{approx,form}}[\text{Ev}; \lambda] &= \text{InputSat}_i^{\text{approx,eval}}[\text{Ev}; \lambda] = \top \right]. \end{aligned}$$

Satisfaction of the dynamical series expresses satisfaction of all direct predecessors outside the approximation series; this input condition does not include satisfaction of the nine conditions within the approximation series.

Proposition 6.76 (Generation dependencies of the nine continuum-approximation conditions). By the within-series dependency edges of the continuum-approximation series in Definition 6.17, the six first-layer conditions are mutually incomparable. Within the first layer, satisfaction of one condition is not a prerequisite for evaluation readiness of another.

The direct-predecessor set of the continuous-extension condition is the token set of the six first-layer conditions. The direct-predecessor sets of the order-stability and relation-stability conditions consist of the token for the continuous-extension condition.

By the between-series dependency edges in Definition 6.17, every token in the dynamical-description series is a direct predecessor of each continuum-approximation condition. Consequently, the evaluation state in Definition 6.21 is uniquely determined by the satisfaction states of the comparison, representation, and dynamical-description series; the direct-predecessor conditions within the continuum-approximation series; the non-ancestor formative-input satisfaction value; and the evaluation-input satisfaction value.

Definition 6.77 (Continuum-approximation-stage failure). Let the complete evaluation specification and approximation scale be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. The necessary and sufficient condition for continuum-approximation-stage failure is $\text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \neq \emptyset$.

If two or more conditions are each ready for evaluation and fail at the same scale, all their tokens belong to this failed-token set.

Proposition 6.78 (Recursive expansion of failed tokens in the continuum-approximation series). For each

$$\begin{aligned} i \in I_{\text{approx}}^{\text{fail}}, \quad (\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda, \\ \text{FailTok}_{\text{approx}, i} \in \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \\ \iff \left[\text{InputSat}_i^{\text{approx}, \text{form}}[\text{Ev}; \lambda] = \top \right] \\ \quad \wedge \left[\text{FailPredSat}(\text{FailTok}_{\text{approx}, i}; \Gamma_{\text{Ev}, \lambda}^{\text{approx}}) = \top \right] \\ \quad \wedge \left[\text{InputSat}_i^{\text{approx}, \text{eval}}[\text{Ev}; \lambda] = \top \right] \\ \quad \wedge \left[K_i^{\text{approx}}[\text{Ev}; \lambda] = \perp \right]. \end{aligned}$$

Proof. By Definition 6.21, the left-hand side is equivalent to

$$\text{FailStateOf}(\text{FailTok}_{\text{approx}, i}; \Gamma_{\text{Ev}, \lambda}^{\text{approx}}) = \text{failed}.$$

Substituting in order the equivalent conditions for the evaluation state, formation judgment, and evaluation-readiness judgment in the same definition, together with the specialization to the continuum-approximation series in Definition 6.75, yields the four conditions on the right-hand side. $\square$

Definition 6.79 (Nine failure classes at the continuum-approximation stage). Let the complete evaluation specification and approximation scale be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. For each $i \in I_{\text{approx}}^{\text{fail}}$, define the i -th continuum-approximation failure class by

$$\begin{aligned} \text{FA}_i[\text{Ev}; \lambda] \\ \iff \text{FailTok}_{\text{approx}, i} \in \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda]. \end{aligned}$$

The index $i \in I_{\text{approx}}^{\text{fail}}$ is common to the condition $K_i^{\text{approx}}[\text{Ev}; \lambda]$ and the failure class $\text{FA}_i[\text{Ev}; \lambda]$. Each $i \in I_{\text{approx}}^{(1)}$ in the first layer corresponds to $\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda]$ and $\varepsilon_i^{\text{approx}}$, whereas each index in the second and third layers corresponds to the truth value of its condition.

Proposition 6.80 (Condition correspondence and type separation of the six first-layer failure classes). For a scale-indexed complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ and each $i \in I_{\text{approx}}^{(1)}$, if

$$\text{FailReady}(\text{FailTok}_{\text{approx},i}; \Gamma_{\text{Ev},\lambda}^{\text{approx}}) = \top$$

then

$$\begin{aligned} & \text{FA}_i[\text{Ev}; \lambda] \\ & \iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}}. \end{aligned}$$

The three failure classes derived from $\mathfrak{D}$ correspond to the following distinct evaluation targets.

Failure class	Evaluation target
FA_2	$\mathfrak{D}$ -record stability
FA_3	$\mathfrak{D}$ -record-cell width
FA_4	$\mathfrak{D}$ -typed-record separation

Furthermore, if distinct first-layer indices i, j are both ready for evaluation and satisfy

$$\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}}, \quad \text{Def}_j^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_j^{\text{approx}},$$

then

$$\begin{aligned} & |\text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \cap \{\text{FailTok}_{\text{approx},i} \mid i \in I_{\text{approx}}^{(1)}\}| \\ & \geq 2. \end{aligned}$$

Proof. Substituting the three satisfaction conditions into Proposition 6.78 and then applying Definition 6.74 and Definition 6.79 in order shows, for each $i \in I_{\text{approx}}^{(1)}$, that membership in the failure class, membership of the failed token, failure of the condition, and exceedance of the defect quantity are equivalent.

In Definition 6.74, the second component is the evaluation value for $\mathfrak{D}$ -record stability, the third component is the supremum of the deviation diameters of the $\mathfrak{D}$ -record-cell images over the two representation tags, and the fourth component is the separation-evaluation value for the three typed $\mathfrak{D}$ -record images. Hence the three failure classes derived from $\mathfrak{D}$ correspond to the distinct evaluation targets shown in the table of the proposition.

Applying the preceding equivalence to the distinct indices i, j places both corresponding tokens in the failed-token set. The tokens have distinct semantic indices, so the set contains at least two first-layer failed tokens. $\square$

Table 5: Correspondence between the nine continuum-approximation-stage failure classes and the condition group

Failure class	Layer	Evaluation target	Condition
FA_1	First layer	Counting-step width derived from $\mathfrak{C}$	K_1^{approx}
FA_2		$\mathfrak{D}$ -record stability	K_2^{approx}
FA_3		$\mathfrak{D}$ -record-cell width	K_3^{approx}
FA_4		$\mathfrak{D}$ -typed-record separation	K_4^{approx}
FA_5	Second layer	$\mathfrak{B}$ -object-state distinguishability	K_5^{approx}
FA_6		$\mathfrak{D}$ -observer-side coupling limit	K_6^{approx}
FA_7	Third layer	Continuous extension	K_7^{approx}
FA_8		Order stability	K_8^{approx}
FA_9		Relation stability	K_9^{approx}

Proposition 6.81 (Localization of continuum-approximation-stage failures). For every $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$,

$$\begin{aligned} & \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \\ &= \left\{ \text{FailTok}_{\text{approx}, i} \mid \begin{array}{l} i \in I_{\text{approx}}^{\text{fail}}, \\ \text{FA}_i[\text{Ev}; \lambda] \end{array} \right\}. \end{aligned}$$

The necessary and sufficient condition for continuum-approximation-stage failure is

$$\bigvee_{i \in I_{\text{approx}}^{\text{fail}}} \text{FA}_i[\text{Ev}; \lambda].$$

Its localization is

$$\begin{aligned} & \text{Min}_{\preceq_{\text{fail}, \text{dep}}} \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda] \\ &= \text{Min}_{\preceq_{\text{fail}, \text{dep}}} \left\{ \text{FailTok}_{\text{approx}, i} \mid \begin{array}{l} i \in I_{\text{approx}}^{\text{fail}}, \\ \text{FA}_i[\text{Ev}; \lambda] \end{array} \right\}. \end{aligned}$$

The simultaneous-failure criterion for the first layer is given by Proposition 6.80. In the third layer, if $\text{FA}_8[\text{Ev}; \lambda] \wedge \text{FA}_9[\text{Ev}; \lambda]$ holds, both corresponding tokens belong to the failed-token set.

Proof. By Definition 6.79, for each

$$i \in I_{\text{approx}}^{\text{fail}}$$

one has

$$\text{FA}_i[\text{Ev}; \lambda] \iff \text{FailTok}_{\text{approx}, i} \in \text{FailedTok}^{\text{approx}}[\text{Ev}; \lambda].$$

Substituting the nine-token family of Definition 6.10 gives the displayed representation of the failed-token set in the proposition.

By Definition 6.77, continuum-approximation-stage failure is equivalent to the nonemptiness of the failed-token set. Substituting the preceding set representation gives Equation (6.81). Applying Definition 6.3 to the same set representation gives the equality of the sets of minimal elements under the dependency order.

Apply Proposition 6.80 to the first layer. For the third layer, substitute satisfaction of the two failure classes into the preceding representation of the failed-token set. $\square$

Definition 6.82 (Refinement family). Let the totally ordered scale set be $(\Lambda, \leq_\Lambda)$. For every complete evaluation specification $\text{Ev} \in \text{EvSpec}$, call the family of continuum-approximation-stage evaluation data $(\text{EvalDat}^{\text{approx}}[\text{Ev}; \lambda])_{\lambda \in \Lambda}$ a refinement family.

For each $\lambda, \mu \in \Lambda$, call $\lambda \leq_\Lambda \mu$ the refinement relation from scale λ to scale μ .

Definition 6.83 (Nine-component tolerance family for continuum approximation). Extend the first-layer tolerance family $(\varepsilon_i^{\text{approx}})_{i \in I_{\text{approx}}^{(1)}}$ of Definition 6.74 to the index set $I_{\text{approx}}^{\text{fail}}$ of Definition 6.73.

For each $i \in I_{\text{approx}}^{(2)} \cup I_{\text{approx}}^{(3)}$ in the second or third layer, define $\varepsilon_i^{\text{approx}} := \frac{1}{2}$. The defect quantities of these three components encode the truth values of the conditions as 0 and 1; their margins measure the separation from this encoding threshold. Let the nine-component tolerance family for continuum approximation be

$$(\varepsilon_i^{\text{approx}})_{i \in I_{\text{approx}}^{\text{fail}}} \in (\overline{\mathbb{R}}_{\geq 0})^{I_{\text{approx}}^{\text{fail}}}.$$

Definition 6.84 (Nine-component defect-quantity family for continuum approximation). Let the complete evaluation specification and approximation scale be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$.

For each $i \in I_{\text{approx}}^{(1)}$ in the first layer, $\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \in \overline{\mathbb{R}}_{\geq 0}$ is the first-layer defect quantity of Definition 6.74.

For each $i \in I_{\text{approx}}^{(2)} \cup I_{\text{approx}}^{(3)}$ in the second or third layer, define

$$\begin{aligned} & \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \\ & := \begin{cases} 0, & K_i^{\text{approx}}[\text{Ev}; \lambda] = \top, \\ 1, & K_i^{\text{approx}}[\text{Ev}; \lambda] = \perp. \end{cases} \end{aligned}$$

Let the nine-component defect-quantity family for continuum approximation be

$$(\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda])_{i \in I_{\text{approx}}^{\text{fail}}}.$$

Then, for each $i \in I_{\text{approx}}^{\text{fail}}$,

$$\begin{aligned} & K_i^{\text{approx}}[\text{Ev}; \lambda] = \top \\ & \iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \varepsilon_i^{\text{approx}}. \end{aligned}$$

Definition 6.85 (Monotonic nondecrease of the continuum-approximation defect-quantity family). For a refinement family in Definition 6.82, the nine-component defect-quantity family for continuum approximation is monotonically nondecreasing if, for every

$$\begin{aligned} & \text{Ev} \in \text{EvSpec}, \quad i \in I_{\text{approx}}^{\text{fail}}, \quad \lambda, \mu \in \Lambda, \\ & \lambda \leq_\Lambda \mu \\ & \implies \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \text{Def}_i^{\text{approx}}[\text{Ev}; \mu]. \end{aligned}$$

Definition 6.86 (Continuum-approximation scale validity domain). Let the complete evaluation specification be $\text{Ev} \in \text{EvSpec}$. Define the continuum-approximation scale validity domain as the scale slice of the approximation-validity domain in Definition 6.4:

$$\mathcal{V}_{\text{Ev}}^{\text{approx}} := \{\lambda \in \Lambda \mid (\text{Ev}, \lambda) \in \mathcal{V}^{\text{approx}}\}.$$

This is the scale domain of satisfaction, including input satisfaction and satisfaction of all nine conditions.

Definition 6.87 (Margin operation on the extended nonnegative reals). For arbitrary $\alpha, \beta \in \overline{\mathbb{R}}_{\geq 0}$, define the margin operation by

$$\text{Mar}_\infty(\alpha, \beta) := \begin{cases} \alpha - \beta, & \alpha < \infty, \beta < \infty, \\ +\infty, & \alpha = \infty, \beta < \infty, \\ 0, & \alpha = \infty, \beta = \infty, \\ -\infty, & \alpha < \infty, \beta = \infty \end{cases}.$$

Definition 6.88 (Componentwise margin and excess for continuum approximation). Let $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ and $i \in I_{\text{approx}}^{\text{fail}}$. Define the componentwise margin and excess by

$$\begin{aligned} & \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] := \text{Mar}_\infty(\varepsilon_i^{\text{approx}}, \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda]), \\ & \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] := \max\{0, -\text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda]\}. \end{aligned}$$

Their types are

$$\begin{aligned}\text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] &\in \overline{\mathbb{R}}, \\ \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] &\in \overline{\mathbb{R}}_{\geq 0}.\end{aligned}$$

Under $\text{InputSat}^{\text{approx}}[\text{Ev}; \lambda] = \top$,

$$\begin{aligned}\lambda \in \mathcal{V}_{\text{Ev}}^{\text{approx}} &\iff \forall i \in I_{\text{approx}}^{\text{fail}}, \quad \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] \geq 0 \\ &\iff \forall i \in I_{\text{approx}}^{\text{fail}}, \quad \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] = 0.\end{aligned}$$

Lemma 6.89 (Positivity criterion for componentwise continuum-approximation excess). For every $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$ and $i \in I_{\text{approx}}^{\text{fail}}$,

$$\begin{aligned}\text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] &> 0 \\ &\iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}}.\end{aligned}$$

Proof. Apply Definition 6.87 and Definition 6.88 to the case in which the defect quantity and tolerance threshold are both finite, the case in which only the defect quantity is infinite, the case in which only the tolerance threshold is infinite, and the case in which both are infinite. The truth values of the two sides coincide in every case. $\square$

Definition 6.90 (Continuum-approximation saturation and excess index sets). Let the complete evaluation specification and approximation scale be $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$.

Define the saturation and excess index sets by

$$\begin{aligned}I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] &:= \left\{ i \in I_{\text{approx}}^{\text{fail}} \mid \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] = 0 \right\}, \\ I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] &:= \left\{ i \in I_{\text{approx}}^{\text{fail}} \mid \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] > 0 \right\}.\end{aligned}$$

Definition 6.91 (Robust validity for continuum approximation). Let the complete evaluation specification be $\text{Ev} \in \text{EvSpec}$.

Define the positive-margin scale domain for continuum approximation by

$$\begin{aligned}\mathcal{V}_{\Lambda}^{\text{approx},+}[\text{Ev}] &:= \left\{ \lambda \in \mathcal{V}_{\text{Ev}}^{\text{approx}} \mid \forall i \in I_{\text{approx}}^{\text{fail}}, \right. \\ &\quad \left. \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] > 0 \right\}.\end{aligned}$$

For an approximation scale $\lambda \in \Lambda$, define robust validity for continuum approximation by

$$\begin{aligned}\text{Rob}^{\text{approx}}[\text{Ev}; \lambda] &\iff \lambda \in \mathcal{V}_{\Lambda}^{\text{approx},+}[\text{Ev}].\end{aligned}$$

Definition 6.88 gives

$$\mathcal{V}_{\Lambda}^{\text{approx},+}[\text{Ev}] \subseteq \mathcal{V}_{\text{Ev}}^{\text{approx}}.$$

Proposition 6.92 (Open-neighborhood representation of robust validity for continuum approximation). With respect to the order topology generated by the total order $(\Lambda, \leq_\Lambda)$, suppose that, for a complete evaluation specification $\mathbf{Ev} \in \mathbf{EvSpec}$ and each $i \in I_{\text{approx}}^{\text{fail}}$, the map

$$\begin{aligned}\Lambda &\longrightarrow \overline{\mathbb{R}}, \\ \mu &\longmapsto \mathbf{Mar}_i^{\text{approx}}[\mathbf{Ev}; \mu]\end{aligned}$$

is lower semicontinuous. Assume $\forall \mu \in \Lambda, \mathbf{InputSat}^{\text{approx}}[\mathbf{Ev}; \mu] = \top$.

For every $\lambda \in \Lambda$,

$$\begin{aligned}\mathbf{Rob}^{\text{approx}}[\mathbf{Ev}; \lambda] \\ \iff \left[\exists W \subseteq \Lambda, \quad W \text{ is open in the order topology,} \right. \\ \left. \lambda \in W \subseteq \mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}] \right].\end{aligned}$$

Proof. For each $i \in I_{\text{approx}}^{\text{fail}}$, lower semicontinuity implies that

$$\left\{ \mu \in \Lambda \mid \mathbf{Mar}_i^{\text{approx}}[\mathbf{Ev}; \mu] > 0 \right\}$$

is open in the order topology.

By the input-satisfaction assumption and Definition 6.88, positive margins in all nine components imply membership in the validity scale domain. Because the index set $I_{\text{approx}}^{\text{fail}}$ is finite, the intersection of these nine open sets, $\mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}]$, is also open.

When $\mathbf{Rob}^{\text{approx}}[\mathbf{Ev}; \lambda]$ holds, Definition 6.91 gives $\lambda \in \mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}]$. Therefore one may take $W = \mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}]$.

Conversely, when there exists an open set W satisfying the right-hand side of the proposition, $\lambda \in W \subseteq \mathcal{V}_\Lambda^{\text{approx},+}[\mathbf{Ev}]$. Definition 6.91 gives $\mathbf{Rob}^{\text{approx}}[\mathbf{Ev}; \lambda]$. $\square$

Definition 6.93 (Maximal effective interval for continuum approximation). Let the complete evaluation specification be $\mathbf{Ev} \in \mathbf{EvSpec}$.

The admissible initial-interval family of Definition 6.8 is, by Definition 6.86,

$$\begin{aligned}\mathfrak{I}_{\text{init}}[\mathbf{Ev}] \\ = \left\{ \mathcal{I} \in \mathcal{P}(\Lambda) \mid \right. \\ \mathcal{I} \subseteq \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}, \\ \forall \mu \in \mathcal{I}, \quad \forall \lambda \in \Lambda, \\ \left. \lambda \leq_\Lambda \mu \implies \lambda \in \mathcal{I} \right\}.\end{aligned}$$

Define the maximal effective interval for continuum approximation by

$$\begin{aligned}\mathcal{I}^{\text{max}}[\mathbf{Ev}] \\ := \bigcup_{\mathcal{I} \in \mathfrak{I}_{\text{init}}[\mathbf{Ev}]} \mathcal{I}.\end{aligned}$$

Definition 6.94 (Effective continuum-approximation regime). Define the effective continuum-approximation regime by

$$\mathcal{V}^{\text{approx,eff}} := \left\{ (\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda \mid \lambda \in \mathcal{I}^{\text{max}}[\text{Ev}] \right\}.$$

Definition 6.10 gives

$$\begin{aligned} \mathcal{V}^{\text{approx,eff}} &= \left\{ \Gamma_{\text{Ev},\lambda}^{\text{approx}} \mid \text{Ev} \in \text{EvSpec}, \quad \lambda \in \mathcal{I}^{\text{max}}[\text{Ev}] \right\} \\ &\subseteq \mathcal{V}^{\text{approx}} \subseteq \text{EvSpec}_\Lambda. \end{aligned}$$

Theorem 6.95 (Equivalent characterizations of quantitative validity for continuum approximation). Let the complete evaluation specification and approximation scale be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Assume $\text{InputSat}^{\text{approx}}[\text{Ev}; \lambda] = \top$. Then the following eight conditions are equivalent.

$$\begin{aligned} &\lambda \in \mathcal{V}_{\text{Ev}}^{\text{approx}} \\ &\iff (\text{Ev}, \lambda) \in \mathcal{V}^{\text{approx}} \\ &\iff \text{CondAll}^{\text{approx}}[\text{Ev}; \lambda] \\ &\iff \left[\begin{array}{c} \forall i \in I_{\text{approx}}^{\text{fail}}, \\ K_i^{\text{approx}}[\text{Ev}; \lambda] = \top \end{array} \right] \\ &\iff \left[\begin{array}{c} \forall i \in I_{\text{approx}}^{\text{fail}}, \\ \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \varepsilon_i^{\text{approx}} \end{array} \right] \\ &\iff \left[\begin{array}{c} \forall i \in I_{\text{approx}}^{\text{fail}}, \\ \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] \geq 0 \end{array} \right] \\ &\iff \left[\begin{array}{c} \forall i \in I_{\text{approx}}^{\text{fail}}, \\ \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] = 0 \end{array} \right] \\ &\iff I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] = \emptyset. \end{aligned}$$

Proof. By Definition 6.86, the first two conditions are equivalent. Under the input-satisfaction assumption, induction along the DAG within the approximation series in Definition 6.17 shows that satisfaction of all nine conditions makes every approximation token satisfied. Conversely, if every approximation token is satisfied, each condition holds by Definition 6.21. By Definition 6.27, these statements are equivalent to membership in the approximation-validity domain. Definition 6.75 makes satisfaction of all nine conditions equivalent to $K_i^{\text{approx}}[\text{Ev}; \lambda] = \top$ for every index. By Definition 6.84, each such condition is equivalent to $\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] \leq \varepsilon_i^{\text{approx}}$.

Definition 6.88 equates the defect-quantity conditions for all components with, respectively, nonnegativity of every margin and vanishing of every excess. By Definition 6.90, vanishing of every excess is equivalent to emptiness of the excess index set. Hence the eight conditions listed in the theorem are equivalent. $\square$

Theorem 6.96 (Separation of continuum-approximation saturation and excess components). Let the complete evaluation specification and approximation scale be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Assume $\text{InputSat}^{\text{approx}}[\text{Ev}; \lambda] = \top$.

For each $i \in I_{\text{approx}}^{\text{fail}}$,

$$\begin{aligned} i &\in I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] \\ &\iff \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] = 0 \\ &\iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] = \varepsilon_i^{\text{approx}} \end{aligned}$$

and

$$\begin{aligned} i &\in I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] \\ &\iff \text{Exc}_i^{\text{approx}}[\text{Ev}; \lambda] > 0 \\ &\iff \text{Mar}_i^{\text{approx}}[\text{Ev}; \lambda] < 0 \\ &\iff \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}} \end{aligned}$$

hold.

Furthermore,

$$\begin{aligned} I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] \\ \cap I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] = \emptyset \end{aligned}$$

and

$$\begin{aligned} \lambda &\in \mathcal{V}_{\text{Ev}}^{\text{approx}} \setminus \mathcal{V}_{\Lambda}^{\text{approx},+}[\text{Ev}] \\ &\iff \left[\begin{aligned} I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] &\neq \emptyset, \\ I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] &= \emptyset \end{aligned} \right] \end{aligned}$$

hold.

For every $A \in \mathcal{P}(I_{\text{approx}}^{\text{fail}})$,

$$\begin{aligned} A &\subseteq I_{\text{approx}}^{\text{sat}}[\text{Ev}; \lambda] \\ &\iff \left[\forall i \in A, \quad \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] = \varepsilon_i^{\text{approx}} \right] \end{aligned}$$

and

$$\begin{aligned} A &\subseteq I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda] \\ &\iff \left[\forall i \in A, \quad \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}} \right] \end{aligned}$$

hold.

Proof. By Definition 6.90 and Definition 6.88, membership in the saturation index set is equivalent to a zero margin and to equality of the defect quantity and the tolerance, whereas membership in the excess index set is equivalent to a positive excess, a negative margin, and exceedance of the tolerance by the defect quantity.

If the same index belonged to both the saturation and excess index sets, its margin would be both zero and negative, a contradiction. Thus the two sets are disjoint. By Theorem 6.95 and Definition 6.91, the scales that are valid but not robust are exactly those at which a saturation index exists and no excess index exists. The two equivalences for an arbitrary subset $A \subseteq I_{\text{approx}}^{\text{fail}}$ follow by applying Definition 6.90 to each element of A . $\square$

Theorem 6.97 (Maximal effective interval and first excess for continuum approximation). Suppose that the nine-component continuum-approximation defect-quantity family is monotonically nondecreasing in the sense of Definition 6.85. For every $\mathbf{Ev} \in \mathbf{EvSpec}$ satisfying $\forall \mu \in \Lambda, \text{InputSat}^{\text{approx}}[\mathbf{Ev}; \mu] = \top$,

$$\begin{aligned} \mathcal{I}^{\max}[\mathbf{Ev}] &= \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}, \\ \forall \mathcal{I} \in \mathfrak{J}_{\text{init}}[\mathbf{Ev}], \quad \mathcal{I} &\subseteq \mathcal{I}^{\max}[\mathbf{Ev}] \end{aligned}$$

holds. If λ_{in} is a greatest element of $\mathcal{I}^{\max}[\mathbf{Ev}]$, then

$$I_{\text{approx}}^{\text{exc}}[\mathbf{Ev}; \lambda_{\text{in}}] = \emptyset.$$

Moreover, if $\mathbf{Ev} \in \text{Dom}(\lambda_{\text{bdry}}^{\text{approx}})$, then

$$\begin{aligned} \mathcal{I}^{\max}[\mathbf{Ev}] &= \{\lambda \in \Lambda \mid \lambda <_{\Lambda} \lambda_{\text{bdry}}^{\text{approx}}[\mathbf{Ev}]\}, \\ I_{\text{approx}}^{\text{exc}}[\lambda_{\text{bdry}}^{\text{approx}}[\mathbf{Ev}]] &\neq \emptyset, \\ \forall \lambda <_{\Lambda} \lambda_{\text{bdry}}^{\text{approx}}[\mathbf{Ev}], \quad I_{\text{approx}}^{\text{exc}}[\mathbf{Ev}; \lambda] &= \emptyset \end{aligned}$$

hold. Therefore $\lambda_{\text{bdry}}^{\text{approx}}[\mathbf{Ev}]$ is the least approximation scale at which an excess component occurs.

Proof. Take $\mu \in \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}$ and $\lambda \leq_{\Lambda} \mu$. For each $i \in I_{\text{approx}}^{\text{fail}}$, scale validity and monotonic nondecrease give

$$\text{Def}_i^{\text{approx}}[\mathbf{Ev}; \lambda] \leq \text{Def}_i^{\text{approx}}[\mathbf{Ev}; \mu] \leq \varepsilon_i^{\text{approx}}.$$

Hence $\lambda \in \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}$, and this scale validity domain is an initial segment.

In Definition 6.93, the scale validity domain itself belongs to $\mathfrak{J}_{\text{init}}[\mathbf{Ev}]$, and every member of that family is contained in the scale validity domain. Hence the union of the family equals the scale validity domain, and each initial segment is contained in that union. If a greatest element λ_{in} exists, it belongs to the scale validity domain; Theorem 6.95 then gives an empty excess index set.

Under the assumption in the latter part, Definition 6.86 and Definition 6.4 identify the scale validity domain with $\text{Val}_{\Lambda}(K_{\text{approx}}^{\text{fail}}; \mathbf{Ev})$. Proposition 6.31 gives the initial-interval representation of the maximal effective interval. By Definition 6.29, the boundary scale lies outside the scale validity domain, while every smaller scale belongs to the maximal effective interval. Applying Theorem 6.95 to the boundary scale and to each lower scale gives, respectively, nonemptiness and emptiness of the excess index set. Thus the boundary scale is the least scale at which an excess component occurs. $\square$

Maximal effective interval and first excess

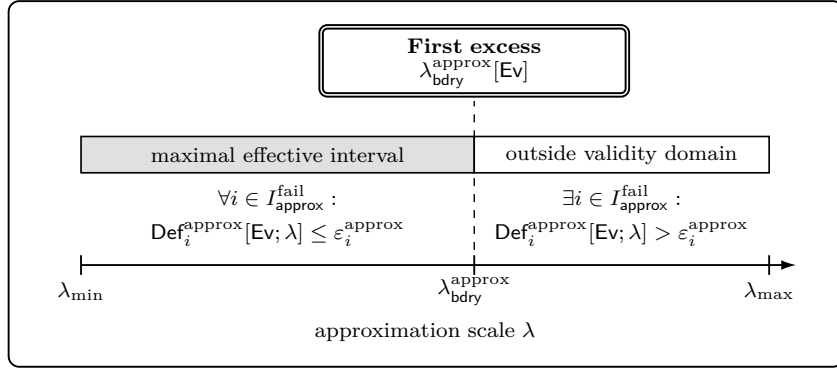


Figure 13: The maximal effective interval and first excess for continuum approximation. When the inputs for quantitative evaluation are satisfied at every scale, the nine-component defect-quantity family is monotonically nondecreasing with respect to the approximation scale, and the boundary scale is defined, the maximal effective interval equals the scale validity domain. Every scale below the boundary scale belongs to the maximal effective interval; the boundary scale lies outside the scale validity domain and is the least approximation scale at which an excess component first occurs.

6.5.1 Failure-witness objects at the continuum-approximation stage

Let the complete evaluation specification, approximation scale, and continuum-approximation failure index be

$$(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}, \quad i \in I_{\text{approx}}^{\text{fail}}.$$

The witness conditions for the nine continuum-approximation failure classes are given by the satisfaction states of the formation inputs, evaluation inputs, and direct-predecessor conditions of each condition, together with the corresponding defect quantity exceeding its tolerance.

Proposition 6.98 (Witness conditions for the nine continuum-approximation failure classes). Let the complete evaluation specification and approximation scale be

$$(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}.$$

For each $i \in I_{\text{approx}}^{\text{fail}}$, assume the evaluation-readiness condition of Definition 6.21 for $\text{FailTok}_{\text{approx}, i}$ and $\Gamma_{\text{Ev}, \lambda}^{\text{approx}}$. Then the following three conditions are equivalent:

$$\begin{aligned} & \text{FA}_i[\text{Ev}; \lambda], \\ & \text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] > \varepsilon_i^{\text{approx}}, \\ & i \in I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda]. \end{aligned}$$

Proof. Substituting the three satisfaction conditions into Proposition 6.78 makes failed-token membership equivalent to $K_i^{\text{approx}}[\text{Ev}; \lambda] = \perp$. Definition 6.79 converts the former into $\text{FA}_i[\text{Ev}; \lambda]$, while Definition 6.84 converts the latter into exceedance of the tolerance by the defect quantity. By Lemma 6.89 and Definition 6.90, that exceedance is equivalent to $i \in I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda]$. Hence the three conditions are equivalent. $\square$

Remark 6.99 (Witness objects for the nine continuum-approximation failure classes). For each $i \in I_{\text{approx}}^{\text{fail}}$, the component obtained by applying Proposition 6.98 gives the witness condition for the corresponding continuum-approximation failure class. The evaluation object of each component is shown in the row for FA_i in Table 5.

6.6 Failure positions at the dynamical-description stage

For the complete evaluation specification $\text{Ev} \in \text{EvSpec}$ of Definition 5.2, the dynamical-description-series evaluation-specification component of Definition 6.10 is $\text{Ev}_{\text{dyn}}^{\text{fail}} = \text{Ev}$. The eight failure-position tokens in the dynamical-description series are associated with the eight dynamical-description conditions on a complete evaluation specification.

The eight conditions below follow the common evaluation rule of Definition 5.3. A false value outside the evaluation domain does not by itself mean that the condition has failed. Failure of the condition itself also requires evaluation readiness.

Definition 6.100 (Evaluation of dynamical-description-stage conditions on complete evaluation specifications). For each $\text{Ev} \in \text{EvSpec}$, $i \in I_{\text{dyn}}^{\text{fail}}$, let the evaluation context for the dynamical-description series be $\Gamma_{\text{Ev}}^{\text{dyn}} := \text{Ev}_{\text{dyn}}^{\text{fail}}$, and define the eight conditions by

$$K_i^{\text{dyn}}[\text{Ev}] := \begin{cases} K_{\text{GenTimeOrdAntiSym}}^{\text{dyn}}(\text{Ev}), & i = 1, \\ K_{\text{CanTrjDesc}}^{\text{dyn}}(\text{Ev}), & i = 2, \\ K_{\text{CanTrjFunc}}^{\text{dyn}}(\text{Ev}), & i = 3, \\ K_{\text{CanObsDesc}}^{\text{dyn}}(\text{Ev}), & i = 4, \\ K_{\text{LocObsChgComm}}^{\text{dyn}}(\text{Ev}), & i = 5, \\ K_{\text{ObsTimeIncTrans}}^{\text{dyn}}(\text{Ev}), & i = 6, \\ K_{\text{ObsTimeRealOrdEmb}}^{\text{dyn}}(\text{Ev}), & i = 7, \\ K_{\text{ObsTimeTrjFact}}^{\text{dyn}}(\text{Ev}), & i = 8 \end{cases}.$$

The non-ancestral formation input for the dynamical-description series includes exact operability. Thus, for every $i \in I_{\text{dyn}}^{\text{fail}}$,

$$\begin{aligned} \text{FailInputSat}^{\text{form}}(\text{FailTok}_{\text{dyn},i}; \Gamma_{\text{Ev}}^{\text{dyn}}) &= \top \\ \implies \text{Ev} &\in \text{EvSpec}^{\text{exact}}. \end{aligned}$$

Further, define the evaluation-readiness predicate by

$$\begin{aligned} \text{Ready}_i^{\text{dyn}}[\text{Ev}] \\ \iff \text{FailReady}(\text{FailTok}_{\text{dyn},i}; \Gamma_{\text{Ev}}^{\text{dyn}}) &= \top. \end{aligned}$$

Then

$$\begin{aligned} \text{DynCondAll}[\text{Ev}] \\ \iff \bigwedge_{i \in I_{\text{dyn}}^{\text{fail}}} K_i^{\text{dyn}}[\text{Ev}] \end{aligned}$$

holds.

Proposition 6.101 (Correspondence between failure-position tokens in the dynamical-description series and the eight conditions). For each $i \in I_{\text{dyn}}^{\text{fail}}$, $\text{Ev} \in \text{EvSpec}$, the failure-condition classification is the $\tau = \text{dyn}$ component of Definition 6.10, and

$$\begin{aligned} & \text{FailEval} \left(\text{FailTok}_{\text{dyn},i}; \Gamma_{\text{Ev}}^{\text{dyn}} \right) \\ &= K_i^{\text{dyn}}[\text{Ev}] \end{aligned}$$

holds.

Further,

$$\begin{aligned} & \text{FailTok}_{\text{dyn},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}] \\ & \iff \left[\text{Ready}_i^{\text{dyn}}[\text{Ev}] \right] \\ & \quad \wedge \left[K_i^{\text{dyn}}[\text{Ev}] = \perp \right]. \end{aligned}$$

Proof. Composing the failure-condition evaluation map of Definition 6.10 with Definition 6.100 yields the first equality.

Substituting the first equality into the failed-state equivalence of Definition 6.21 and applying Proposition 6.23 yields the second equivalence. $\square$

Definition 6.102 (Failure family for dynamical-description conditions). For each $i \in I_{\text{dyn}}^{\text{fail}}$, $\text{Ev} \in \text{EvSpec}$, define the dynamical-description-condition failure predicate by

$$\begin{aligned} & \text{FDyn}_i[\text{Ev}] \\ & \iff \text{FailTok}_{\text{dyn},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}]. \end{aligned}$$

Lemma 6.103 (Common evaluation formula for the failure family of dynamical-description conditions). For each $i \in I_{\text{dyn}}^{\text{fail}}$, $\text{Ev} \in \text{EvSpec}$,

$$\begin{aligned} & \text{FDyn}_i[\text{Ev}] \\ & \iff \left[\text{Ready}_i^{\text{dyn}}[\text{Ev}] \right] \\ & \quad \wedge \left[K_i^{\text{dyn}}[\text{Ev}] = \perp \right]. \end{aligned}$$

Further, when $\text{Ev} \in \text{EvSpec}^{\text{exact}}$, the failure-witness condition corresponding to each $i \in \{1, \dots, 8\}$ is obtained by expanding the definition of each condition.

Proof. The first claim follows from Definition 6.102, Proposition 6.101, the evaluation-readiness judgment and failed-state equivalence of Definition 6.21, and Proposition 6.23.

Substitute $i = 1, \dots, 8$, and expand the respective conditions appearing in Definition 6.100 by Definitions 5.18, 5.22, 5.26, 5.31, 5.34, 5.53, 5.59, and 5.65. This yields the concrete failure-witness condition corresponding to each index. $\square$

Definition 6.104 (Failure at the dynamical-description stage). For each $\text{Ev} \in \text{EvSpec}$, define failure at the dynamical-description stage by

$$\begin{aligned} & \text{Fail}_{\text{dyn}}[\text{Ev}] \\ & \iff \exists i \in I_{\text{dyn}}^{\text{fail}}, \quad \text{FDyn}_i[\text{Ev}] \\ & \iff \text{FailedTok}^{\text{strict}}[\text{Ev}] \cap \text{FailTok}_{\text{dyn}} \neq \emptyset. \end{aligned}$$

Definition 6.105 (Sets of failed-condition indices and minimal-failure indices in the dynamical-description series). For each $\text{Ev} \in \text{EvSpec}$, define the set of failed-condition indices in the dynamical-description series by

$$\begin{aligned} & \text{FailedCondIdx}_{\text{dyn}}[\text{Ev}] \\ & := \left\{ i \in I_{\text{dyn}}^{\text{fail}} \mid K_i^{\text{dyn}}[\text{Ev}] = \perp \right\}. \end{aligned}$$

Define the set of minimal-failure indices with respect to the dependency order by

$$\begin{aligned} & \text{MinFailIdx}_{\text{dyn}}[\text{Ev}] \\ & := \left\{ i \in \text{FailedCondIdx}_{\text{dyn}}[\text{Ev}] \mid \begin{array}{l} \forall j \in \text{FailedCondIdx}_{\text{dyn}}[\text{Ev}], \\ \text{FailTok}_{\text{dyn},j} \preceq^{\text{fail,dep}} \text{FailTok}_{\text{dyn},i} \implies j = i \end{array} \right\}. \end{aligned}$$

Proposition 6.106 (Eight-condition characterization of dynamical-description-stage failure under satisfied inputs). For each $\text{Ev} \in \text{EvSpec}$, suppose that, for every $i \in I_{\text{dyn}}^{\text{fail}}$, both

$$\text{FailInputSat} \left(\text{FailTok}_{\text{dyn},i}; \Gamma_{\text{Ev}}^{\text{dyn}} \right) = \top$$

and, for every

$$\text{FailTok}^{(0)} \in \text{FailPredOf}(\text{FailTok}_{\text{dyn},i}) \setminus \text{FailTok}_{\text{dyn}},$$

$$\text{FailStateOf} \left(\text{FailTok}^{(0)}; \text{Ev}_{\text{FailSeriesOf}(\text{FailTok}^{(0)})}^{\text{fail}} \right) = \text{sat}$$

hold. Then

$$\begin{aligned} & \text{Fail}_{\text{dyn}}[\text{Ev}] \\ & \iff \neg \text{DynCondAll}[\text{Ev}] \\ & \iff \text{FailedCondIdx}_{\text{dyn}}[\text{Ev}] \neq \emptyset \\ & \iff \text{MinFailIdx}_{\text{dyn}}[\text{Ev}] \neq \emptyset. \end{aligned}$$

Further, for each $i \in I_{\text{dyn}}^{\text{fail}}$,

$$\text{FDyn}_i[\text{Ev}] \iff i \in \text{MinFailIdx}_{\text{dyn}}[\text{Ev}]$$

holds.

Proof. By Proposition 6.101 and Definition 6.21, for each $i \in I_{\text{dyn}}^{\text{fail}}$, the failed state is the conjunction of evaluation readiness and $K_i^{\text{dyn}}[\text{Ev}] = \perp$.

Under the assumed satisfaction of the formation inputs, evaluation inputs, and extra-series direct-predecessor states, finite induction along the directed acyclic graph within the dynamical-description series of Definition 6.17 gives

$$\begin{aligned} & \text{FDyn}_i[\text{Ev}] \\ & \iff \left[\forall j \in I_{\text{dyn}}^{\text{fail}}, \right. \\ & \quad \text{FailTok}_{\text{dyn},j} \prec^{\text{fail,dep}} \text{FailTok}_{\text{dyn},i} \\ & \quad \left. \implies K_j^{\text{dyn}}[\text{Ev}] = \top \right] \wedge \left[K_i^{\text{dyn}}[\text{Ev}] = \perp \right]. \end{aligned}$$

By Definition 6.105, the right-hand side is equivalent to $i \in \text{MinFailIdx}_{\text{dyn}}[\text{Ev}]$.

The finiteness of the index set and the existence of a minimal element make nonemptiness of the set of failed-condition indices equivalent to nonemptiness of the set of minimal-failure indices. By Definition 6.100, failure of the conjunction of all dynamical-description-stage conditions is equivalent to nonemptiness of the set of failed-condition indices. Finally, applying Definition 6.104 and the equivalence for each index yields the claim. $\square$

Table 6: Classification of failure positions in the dynamical-description series

Index	Failure predicate	Corresponding condition	Direct-predecessor indices
1	FDyn ₁	$K_{\text{GenTimeOrdAntiSym}}^{\text{dyn}}$	$\emptyset$
2	FDyn ₂	$K_{\text{CanTrjDesc}}^{\text{dyn}}$	$\emptyset$
3	FDyn ₃	$K_{\text{CanTrjFunc}}^{\text{dyn}}$	$\{2\}$
4	FDyn ₄	$K_{\text{CanObsDesc}}^{\text{dyn}}$	$\emptyset$
5	FDyn ₅	$K_{\text{LocObsChgComm}}^{\text{dyn}}$	$\emptyset$
6	FDyn ₆	$K_{\text{ObsTimeIncTrans}}^{\text{dyn}}$	$\{1\}$
7	FDyn ₇	$K_{\text{ObsTimeRealOrdEmb}}^{\text{dyn}}$	$\{6\}$
8	FDyn ₈	$K_{\text{ObsTimeTrjFact}}^{\text{dyn}}$	$\{3, 6\}$

Theorem 6.107 (Complete classification of dynamical-description-stage failure and simultaneous minimal failures). For each $\text{Ev} \in \text{EvSpec}$,

$$\begin{aligned}
& \text{Fail}_{\text{dyn}}[\text{Ev}] \\
& \iff \bigvee_{i \in I_{\text{dyn}}^{\text{fail}}} \text{FDyn}_i[\text{Ev}] \\
& \iff \text{FailedTok}^{\text{strict}}[\text{Ev}] \cap \text{FailTok}_{\text{dyn}} \neq \emptyset
\end{aligned}$$

holds.

When the three conditions concerning the formation inputs, evaluation inputs, and extra-series direct-predecessor states in Proposition 6.106 hold, the further equivalence

$$\begin{aligned}
& \text{Fail}_{\text{dyn}}[\text{Ev}] \\
& \iff \text{FailedCondlx}_{\text{dyn}}[\text{Ev}] \neq \emptyset \\
& \iff \text{MinFailldx}_{\text{dyn}}[\text{Ev}] \neq \emptyset
\end{aligned}$$

holds. Moreover, for every $i \in I_{\text{dyn}}^{\text{fail}}$, $\text{FDyn}_i[\text{Ev}] \iff i \in \text{MinFailldx}_{\text{dyn}}[\text{Ev}]$ holds. Consequently, if

$$|\text{MinFailldx}_{\text{dyn}}[\text{Ev}]| \geq 2,$$

the dynamical-description-condition failure predicates corresponding to multiple distinct minimal failure conditions in the dependency order hold simultaneously.

Further,

$$\text{FailedTok}^{\text{strict}}[\text{Ev}] \cap \text{FailTok}_{\text{dyn}}$$

is an antichain with respect to $\preceq^{\text{fail,dep}}$.

Proof. Fix any $\text{Ev} \in \text{EvSpec}$. Definitions 6.102 and 6.104 give Equation (6.107).

Assume the three conditions of Proposition 6.106. That proposition gives the chain of equivalences among stage failure, nonemptiness of the set of failed-condition indices, and

nonemptiness of the set of minimal-failure indices, and it gives the minimal-failure-index criterion for the dynamical-description-condition failure predicate for each $i \in I_{\text{dyn}}^{\text{fail}}$.

Assume Equation (6.107). Then there exist distinct

$$i, j \in \text{MinFailIdx}_{\text{dyn}}[\text{Ev}].$$

The minimal-failure-index criterion in Proposition 6.106 therefore gives

$$\text{FDyn}_i[\text{Ev}] \wedge \text{FDyn}_j[\text{Ev}].$$

The final antichain assertion does not require the additional input-satisfaction assumptions. By Proposition 6.26, the full failed-token set is an antichain. The set in Equation (6.107) is a subset of it and is therefore also an antichain. $\square$

6.7 Failure localization at the law-description stage

Definition 6.108 (Selected family of law-evaluation inputs). Fix a subset of evaluation specifications and a selected input family

$$\begin{aligned} \text{EvSpec}^{\text{law}} &\subseteq \{\text{Ev} \in \text{EvSpec} \mid \text{DynOperative}[\text{Ev}]\}, \\ &(\rho_{\text{Ev}}^0, \text{LawEvalDat}_{\text{Ev}})_{\text{Ev} \in \text{EvSpec}^{\text{law}}}. \end{aligned}$$

Each component consists of the single-object generated law datum of Definition 5.85 and the base representation used in its construction, with

$$\begin{aligned} \text{LawEvalDat}_{\text{Ev}}.\text{EvalSpec} &= \iota_{\text{single}}(\text{Ev}), \\ \text{LawEvalDat}_{\text{Ev}} &\in \text{LawEvalDatSp}^{\text{gen}}[\rho_{\text{Ev}}^0]. \end{aligned}$$

The selected family is not assumed to be determined automatically by the complete evaluation specification. In formulas with a fixed $\text{Ev} \in \text{EvSpec}^{\text{law}}$, abbreviate $\rho_{\text{Ev}}^0, \text{LawEvalDat}_{\text{Ev}}$ as $\rho^0, \text{LawEvalDat}$. Conditions, failure regions and signatures are relative to this selected family. Set formation-input satisfaction for the law series to the truth value of $\text{Ev} \in \text{EvSpec}^{\text{law}}$, and evaluation-input satisfaction to $\top$.

Failure at the law-description stage is localized by the minimal failed nodes relative to the law-condition dependency directed acyclic graph of Definition 5.99. The simultaneous minimal-failure node set may contain multiple elements.

6.7.1 Failure Domain at the Law-Description Stage

Definition 6.109 (Evaluation of Law Conditions on Complete Evaluation Specifications). For each

$$\begin{aligned} \text{Ev} &= (l, o, \zeta) \in \text{EvSpec}^{\text{law}}, \\ I_{\text{law}}^{\text{fail}} &= \text{LawCondNodeSet} = \{1, \dots, 5\}, \end{aligned}$$

and the five conditions in the law series are defined by

$$K_i^{\text{law}}[\text{Ev}] := \mathbf{K}_i^{\text{law}}[\text{LawEvalDat}] \quad (i \in I_{\text{law}}^{\text{fail}}).$$

Then

$$\begin{aligned} &\text{LawCondAll}[\text{LawEvalDat}] \\ &\iff \bigwedge_{i \in I_{\text{law}}^{\text{fail}}} K_i^{\text{law}}[\text{Ev}] \end{aligned}$$

holds. Extend these five conditions by false outside the selected domain, using Definition 6.10. Set the total law-series conditions to $K_{\text{law},i}^{\text{fail}} := K_i^{\text{law}}$.

Definition 6.110 (Failure at the Law-Description Stage). For a complete evaluation specification $\text{Ev} = (l, o, \zeta) \in \text{EvSpec}$, failure at the law-description stage is defined by

$$\begin{aligned} \text{Fail}_{\text{law}}[\text{Ev}] \\ \iff \text{Ev} \in \text{EvSpec}^{\text{law}} \\ \wedge \neg \bigwedge_{i \in I_{\text{law}}^{\text{fail}}} K_i^{\text{law}}[\text{Ev}]. \end{aligned}$$

Definition 6.111 (Law-Description-Stage Failure Domain). The law-description-stage failure domain in the set EvSpec of complete evaluation specifications is defined by

$$\begin{aligned} \text{FailDom}_{\text{law}}(\text{EvSpec}) \\ := \{ \text{Ev} \in \text{EvSpec} \mid \text{Fail}_{\text{law}}[\text{Ev}] \}. \end{aligned}$$

Proposition 6.112 (Failure at the Law-Description Stage and Law-Description Operability). For each $\text{Ev} = (l, o, \zeta) \in \text{EvSpec}^{\text{law}}$,

$$\begin{aligned} \text{Fail}_{\text{law}}[\text{Ev}] \\ \iff \text{DynOperative}[\text{Ev}] \\ \wedge \neg \text{LawOperative}[\text{LawEvalDat}]. \end{aligned}$$

Proof. On the domain where $\text{DynOperative}[\text{Ev}]$ holds, Definition 5.102 gives

$$\text{LawOperative}[\text{LawEvalDat}] \iff \text{LawCondAll}[\text{LawEvalDat}].$$

Substitution of this equivalence into Definition 6.110 yields the claim. $\square$

Definition 6.113 (All-Ancestor Satisfaction Bundle for a Law Condition). For each $\text{Ev} \in \text{EvSpec}$, $i \in I_{\text{law}}^{\text{fail}}$, the all-ancestor satisfaction bundle for a law condition is defined by

$$\begin{aligned} \text{LawCondAncAll}_i[\text{Ev}] \\ \iff \begin{cases} \top, & i \in \{1, 2, 4\}, \\ K_1^{\text{law}}[\text{Ev}], & i = 3, \\ K_1^{\text{law}}[\text{Ev}] \wedge K_3^{\text{law}}[\text{Ev}], & i = 5 \end{cases}. \end{aligned}$$

Definition 6.114 (Minimal Failure Class at the Law-Description Stage). For each $\text{Ev} \in \text{EvSpec}$, $i \in I_{\text{law}}^{\text{fail}}$, the i -th minimal failure class at the law-description stage is defined by

$$\begin{aligned} \text{FL}_i[\text{Ev}] \\ \iff \text{Ev} \in \text{EvSpec}^{\text{law}} \\ \wedge \text{LawCondAncAll}_i[\text{Ev}] \\ \wedge \neg K_i^{\text{law}}[\text{Ev}]. \end{aligned}$$

Definition 6.115 (Simultaneous Minimal-Failure Node Set at the Law-Description Stage). For each $\text{Ev} \in \text{EvSpec}$, the simultaneous minimal-failure node set at the law-description stage is defined by

$$\begin{aligned} \text{MinFailDx}_{\text{law}}[\text{Ev}] \\ := \{ i \in I_{\text{law}}^{\text{fail}} \mid \text{FL}_i[\text{Ev}] \}. \end{aligned}$$

Proposition 6.116 (Nonemptiness and Antichain Property of the Simultaneous Minimal-Failure Node Set at the Law-Description Stage).

For each $\text{Ev} \in \text{FailDom}_{\text{law}}(\text{EvSpec})$, $\text{MinFailIdx}_{\text{law}}[\text{Ev}]$ is a nonempty antichain in LawCondDAG .

Proof. For a complete evaluation specification in the law-description-stage failure domain, Definitions 6.111 and 6.110 give $\text{DynOperative}[\text{Ev}]$ and $\neg \text{LawCondAll}[\text{LawEvalDat}]$. Definition 6.109 therefore implies that

$$\{i \in I_{\text{law}}^{\text{fail}} \mid \neg K_i^{\text{law}}[\text{Ev}]\}$$

is nonempty.

Because the law-condition dependency directed acyclic graph of Definition 5.99 is finite, the set of nodes with false values has a minimal node i with respect to the reachability order. At every ancestor of node i , the corresponding law condition holds. Definition 6.113 therefore gives $\text{LawCondAncAll}_i[\text{Ev}]$. Definition 6.114 then gives $\text{FL}_i[\text{Ev}]$. Hence the simultaneous minimal-failure node set at the law-description stage is nonempty.

For distinct $i, j \in \text{MinFailIdx}_{\text{law}}[\text{Ev}]$, suppose that a path runs from i to j in the law-condition dependency directed acyclic graph. Definition 6.114 gives $\neg K_i^{\text{law}}[\text{Ev}]$. On the other hand, since i is an ancestor of j , Definition 6.113 together with $\text{LawCondAncAll}_j[\text{Ev}]$ gives $K_i^{\text{law}}[\text{Ev}]$, a contradiction. If reachability from j to i is assumed, then $\neg K_j^{\text{law}}[\text{Ev}]$ holds, whereas the fact that j is an ancestor of i gives $K_j^{\text{law}}[\text{Ev}]$, again a contradiction. Thus any two distinct nodes in the set are mutually unreachable, and the simultaneous minimal-failure node set at the law-description stage is an antichain. $\square$

Node	Minimal failure class	Law condition	Strict ancestor nodes
1	$\text{FL}_1[\text{Ev}]$	Individual law-evaluation time-domain compatibility condition $K_1^{\text{law}}[\text{Ev}]$	$\emptyset$
2	$\text{FL}_2[\text{Ev}]$	Law-candidate-relation right-uniqueness condition $K_2^{\text{law}}[\text{Ev}]$	$\emptyset$
3	$\text{FL}_3[\text{Ev}]$	Generated evaluation-tuple law-candidate-relation membership condition $K_3^{\text{law}}[\text{Ev}]$	$\{1\}$
4	$\text{FL}_4[\text{Ev}]$	Faithful-transport invariance condition for the law-candidate relation $K_4^{\text{law}}[\text{Ev}]$	$\emptyset$
5	$\text{FL}_5[\text{Ev}]$	Common valid time-domain condition for laws $K_5^{\text{law}}[\text{Ev}]$	$\{1, 3\}$

Table 7: Minimal Failure Classes at the Law-Description Stage Relative to the Law-Condition DAG

Proposition 6.117 (Failure-Domain Cover Relative to the Law-Condition DAG). For

the set EvSpec of complete evaluation specifications,

$$\begin{aligned} & \text{FailDom}_{\text{law}}(\text{EvSpec}) \\ &= \bigcup_{i \in I_{\text{law}}^{\text{fail}}} \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\}. \end{aligned}$$

Equivalently, for each $\text{Ev} \in \text{EvSpec}$,

$$\begin{aligned} & \text{Fail}_{\text{law}}[\text{Ev}] \\ & \iff \text{MinFailIdx}_{\text{law}}[\text{Ev}] \neq \emptyset \\ & \iff \bigvee_{i \in I_{\text{law}}^{\text{fail}}} \text{FL}_i[\text{Ev}]. \end{aligned}$$

Moreover, if $|\text{MinFailIdx}_{\text{law}}[\text{Ev}]| \geq 2$, the minimal failure classes at the law-description stage corresponding to the respective nodes in the set hold simultaneously.

Proof. Suppose failure at the law-description stage holds. By Definition 6.111, the complete evaluation specification belongs to the law-description-stage failure domain. Proposition 6.116 then shows that the corresponding simultaneous minimal-failure node set is nonempty.

Conversely, suppose the simultaneous minimal-failure node set at the law-description stage is nonempty. Definition 6.115 gives a minimal failure class at the law-description stage corresponding to some law-failure index. Definition 6.114 gives dynamical-description operability and a false value for the corresponding law-condition evaluation. By Definition 6.109, the simultaneous satisfaction of all law conditions fails; Definition 6.110 then establishes failure at the law-description stage.

Definition 6.115 also shows that nonemptiness of the simultaneous minimal-failure node set is equivalent to the occurrence of a minimal failure class at the law-description stage corresponding to some law-failure index, and hence to a disjunction over the finite index set. The three conditions in the proposition are therefore equivalent.

The equivalence for every complete evaluation specification, together with Definition 6.111, gives the set equality for the failure domain.

Suppose that the simultaneous minimal-failure node set at the law-description stage has at least two elements. By Definition 6.115, the minimal failure class at the law-description stage corresponding to each node in the set holds. Proposition 6.116 shows that these nodes form an antichain in the law-condition dependency directed acyclic graph. $\square$

Proposition 6.118 (Stage Separation of Failure at the Dynamical-Description and Law-Description Stages). For each $\text{Ev} = (\text{l}, o, \zeta) \in \text{EvSpec}$,

$$\begin{aligned} & \text{Fail}_{\text{dyn}}[\text{Ev}] \implies \neg \text{DynOperative}[\text{Ev}], \\ & \text{Fail}_{\text{law}}[\text{Ev}] \implies \text{DynOperative}[\text{Ev}] \end{aligned}$$

hold. Consequently,

$$\neg (\text{Fail}_{\text{dyn}}[\text{Ev}] \wedge \text{Fail}_{\text{law}}[\text{Ev}])$$

holds.

Proof. By Definitions 6.104 and 6.102, if $\text{Fail}_{\text{dyn}}[\text{Ev}]$ holds, then for some $i \in I_{\text{dyn}}^{\text{fail}}$,

$$\text{FailStateOf}(\text{FailTok}_{\text{dyn}, i}; \Gamma_{\text{Ev}}^{\text{dyn}}) = \text{failed}$$

holds. Proposition 6.101 gives $K_i^{\text{dyn}}[\text{Ev}] = \perp$. Definition 6.100 then gives $\neg \text{DynCondAll}[\text{Ev}]$. By Definition 5.70, $\neg \text{DynOperative}[\text{Ev}]$ holds.

The equivalence in Proposition 6.112 implies that failure at the law-description stage entails dynamical-description operability. Dynamical-description operability is incompatible with the negation of dynamical-description operability obtained from failure at the dynamical-description stage. The two failures therefore cannot hold simultaneously, and the final conclusion of the proposition follows. $\square$

Proposition 6.119 (Finite Witness Set for Minimal Failure Classes at the Law-Description Stage). For the set EvSpec of complete evaluation specifications, there exists a finite set $\text{FailWit}_{\text{law}} \subseteq \text{EvSpec}$ satisfying

$$|\text{FailWit}_{\text{law}}| \leq |I_{\text{law}}^{\text{fail}}| = 5$$

such that, for each $i \in I_{\text{law}}^{\text{fail}}$,

$$\begin{aligned} & \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\} \neq \emptyset \\ & \iff \exists \text{Ev}_i^{\text{wit}} \in \text{FailWit}_{\text{law}}, \quad \text{FL}_i[\text{Ev}_i^{\text{wit}}] \end{aligned}$$

holds.

Proof. For each $i \in I_{\text{law}}^{\text{fail}}$, if

$$\{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\} \neq \emptyset$$

holds, fix one of its elements and denote the selected witness by Ev_i^{wit} . Define the set of selected complete evaluation specifications by

$$\begin{aligned} & \text{FailWit}_{\text{law}} \\ & := \{\text{Ev}_i^{\text{wit}} \mid i \in I_{\text{law}}^{\text{fail}}, \quad \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\} \neq \emptyset\}. \end{aligned}$$

This set is a subset of the set of complete evaluation specifications.

Because the law-failure index set has five elements, the finite witness set contains at most five elements.

For each law-failure index, if the domain of the corresponding minimal failure class is nonempty, the selected $\text{Ev}_i^{\text{wit}} \in \text{FailWit}_{\text{law}}$ satisfies $\text{FL}_i[\text{Ev}_i^{\text{wit}}]$.

Conversely, if a selected witness in the finite witness set satisfies the corresponding minimal failure class, then

$$\begin{aligned} & \text{Ev}_i^{\text{wit}} \\ & \in \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\}. \end{aligned}$$

Thus, for each minimal failure class, the equivalence stated in the proposition holds between nonemptiness of the class and the existence of a selected witness corresponding to that class. $\square$

6.7.2 Counterexample Refinements of Minimal Failure Classes at the Law-Description Stage

Definition 6.120 (Minimal-Failure Parent Region at the Law-Description Stage). For the set EvSpec of complete evaluation specifications and each $i \in I_{\text{law}}^{\text{fail}}$, the minimal-failure parent region at the law-description stage is defined by

$$\begin{aligned} & \text{FailReg}_{\text{law},i}(\text{EvSpec}) \\ & := \{\text{Ev} \in \text{EvSpec} \mid \text{FL}_i[\text{Ev}]\}. \end{aligned}$$

Definition 6.121 (Law-Failure Counterexample-Type Index Family). For each $i \in I_{\text{law}}^{\text{fail}}$, the law-failure counterexample-type index family is defined by

$$\text{SubIdx}_{\text{law},i} := \begin{cases} \{\text{EmptyTimeDom}, \text{OutEvalDom}\}, & i = 1, \\ \{\text{GenEvalArgConflict}\}, & i = 2, \\ \{\text{EmptyCandOutFiber}, \text{AltCandOutput}\}, & i = 3, \\ \{\text{CandRelMemLoss}, \text{CandRelMemGain}\}, & i = 4, \\ \left\{ \begin{array}{l} \text{EmptyCommonValidTimeDom}, \\ \text{InadmissibleCommonValidTimeDom} \end{array} \right\}, & i = 5 \end{cases}.$$

Definition 6.122 (Individual-Law Application-Domain Counterexample Types). For a complete evaluation specification $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$, each $a \in \text{LawIdx}$, and $t \in \text{EvalTimeDom}_a^{\text{law}}$, the generated law-evaluation argument is defined by

$$\text{GenEvalArg}_a^{\text{law}}(t) := (t, \text{InVal}_a^{\text{law}}(t)).$$

The individual-law application-domain counterexample types are defined by

$$\begin{aligned} \text{LawEvalDomDefect}_{\text{EmptyTimeDom}}[\text{Ev}] &\iff \exists a \in \text{LawIdx}, \quad \text{EvalTimeDom}_a^{\text{law}} = \emptyset, \\ \text{LawEvalDomDefect}_{\text{OutEvalDom}}[\text{Ev}] &\iff \exists a \in \text{LawIdx}, \quad \exists t \in \text{EvalTimeDom}_a^{\text{law}}, \\ &\quad \text{GenEvalArg}_a^{\text{law}}(t) \notin \text{EvalDom}_a^{\text{law}}. \end{aligned}$$

Definition 6.123 (Law-Output-Determinacy Counterexample Types). For a complete evaluation specification $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$ and each

$$a \in \text{LawIdx}, \quad \text{EvalArg}_{a,0}^{\text{law}} \in \text{LawTimeDom} \times \text{InValSp}_a^{\text{law}},$$

the law-candidate output fiber and its nonuniqueness are defined by

$$\begin{aligned} \text{CandOutFib}_a^{\text{law}}(\text{EvalArg}_{a,0}^{\text{law}}) &:= \{\text{OutVal}_{a,0}^{\text{law}} \in \text{OutValSp}_a^{\text{law}} \mid (\text{EvalArg}_{a,0}^{\text{law}}, \text{OutVal}_{a,0}^{\text{law}}) \in \text{CandRel}_a^{\text{law}}\}, \\ \text{CandOutNonUniq}_a^{\text{law}}(\text{EvalArg}_{a,0}^{\text{law}}) &\iff \exists \text{OutVal}_{a,0}^{\text{law}}, \text{OutVal}_{a,1}^{\text{law}} \in \text{CandOutFib}_a^{\text{law}}(\text{EvalArg}_{a,0}^{\text{law}}), \\ &\quad \text{OutVal}_{a,0}^{\text{law}} \neq \text{OutVal}_{a,1}^{\text{law}}. \end{aligned}$$

The output conflict on generated evaluation arguments is defined by

$$\begin{aligned} \text{LawGenEvalArgConflict}[\text{Ev}] &\iff \exists a \in \text{LawIdx}, \quad \exists t \in \text{EvalTimeDom}_a^{\text{law}}, \\ &\quad \text{GenEvalArg}_a^{\text{law}}(t) \in \text{EvalDom}_a^{\text{law}}, \quad \text{CandOutNonUniq}_a^{\text{law}}(\text{GenEvalArg}_a^{\text{law}}(t)). \end{aligned}$$

The output conflict localized outside generated evaluation arguments is defined by

$$\begin{aligned} \text{LawOffGenArgConflict}[\text{Ev}] &\iff \exists a_0 \in \text{LawIdx}, \quad \exists \text{EvalArg}_{a_0,0}^{\text{law}} \in \text{EvalDom}_{a_0}^{\text{law}}, \\ &\quad \text{CandOutNonUniq}_{a_0}^{\text{law}}(\text{EvalArg}_{a_0,0}^{\text{law}}) \\ &\quad \wedge \neg \text{LawGenEvalArgConflict}[\text{Ev}]. \end{aligned}$$

Definition 6.124 (Generated Law-Evaluation Relation-Membership Counterexample Types). For a complete evaluation specification $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$ and each

$$\begin{aligned}\sigma &\in \text{SubIdx}_{\text{law},3}, \\ a &\in \text{LawIdx}, \\ t &\in \text{EvalTimeDom}_a^{\text{law}},\end{aligned}$$

the candidate-output-fiber type is defined by

$$\begin{aligned}\text{CandOutFibCase}_{a,\sigma}^{\text{law}}(t) \\ \iff \begin{cases} \text{CandOutFib}_a^{\text{law}}(\text{GenEvalArg}_a^{\text{law}}(t)) = \emptyset, & \sigma = \text{EmptyCandOutFiber}, \\ \text{CandOutFib}_a^{\text{law}}(\text{GenEvalArg}_a^{\text{law}}(t)) \neq \emptyset, & \sigma = \text{AltCandOutput} \end{cases}.\end{aligned}$$

The occurrence of a membership counterexample of candidate-output-fiber type σ is defined by

$$\begin{aligned}\text{LawCandOutCaseOcc}_\sigma[\text{Ev}] \\ \iff \exists a \in \text{LawIdx}, \quad \exists t \in \text{EvalTimeDom}_a^{\text{law}}, \\ \text{EvalTuple}_a^{\text{law}}(t) \notin \text{CandRel}_a^{\text{law}}, \quad \text{CandOutFibCase}_{a,\sigma}^{\text{law}}(t).\end{aligned}$$

Definition 6.125 (Law-Candidate-Relation Membership-Change Counterexample Types). For a complete evaluation specification $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$ and each

$$\begin{aligned}\sigma &\in \text{SubIdx}_{\text{law},4}, \\ a &\in \text{LawIdx}, \\ \text{EvalTupleArg}_{a,0}^{\text{law}}, \text{EvalTupleArg}_{a,1}^{\text{law}} &\in \text{EvalTupleSp}_a^{\text{law}},\end{aligned}$$

the directional predicate for a candidate-relation membership change is defined by

$$\begin{aligned}\text{CandRelMemChg}_{a,\sigma}^{\text{law}}(\text{EvalTupleArg}_{a,0}^{\text{law}}, \text{EvalTupleArg}_{a,1}^{\text{law}}) \\ \iff \left(\mathbf{1}_{\text{CandRel}_a^{\text{law}}}(\text{EvalTupleArg}_{a,0}^{\text{law}}), \mathbf{1}_{\text{CandRel}_a^{\text{law}}}(\text{EvalTupleArg}_{a,1}^{\text{law}}) \right) \\ = \begin{cases} (1, 0), & \sigma = \text{CandRelMemLoss}, \\ (0, 1), & \sigma = \text{CandRelMemGain} \end{cases}.\end{aligned}$$

The occurrence of a counterexample in membership-change direction σ is defined by

$$\begin{aligned}\text{LawCandRelMemChgOcc}_\sigma[\text{Ev}] \\ \iff \exists (\text{EvalTupleBij}_{a,0}^{\text{law}})_{a \in \text{LawIdx}} \in \text{FaithTransSet}^{\text{law}}, \\ \exists a_0 \in \text{LawIdx}, \quad \exists \text{EvalTupleArg}_{a_0,0}^{\text{law}} \in \text{EvalTupleSp}_{a_0}^{\text{law}}, \\ \text{CandRelMemChg}_{a_0,\sigma}^{\text{law}}(\text{EvalTupleArg}_{a_0,0}^{\text{law}}, \text{EvalTupleBij}_{a_0,0}^{\text{law}}(\text{EvalTupleArg}_{a_0,0}^{\text{law}})).\end{aligned}$$

Definition 6.126 (Common-Valid-Time-Domain Counterexample Types for Laws). For a complete evaluation specification $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{law}}$ and each $\sigma \in \text{SubIdx}_{\text{law},5}$, the

common-valid-time-domain counterexample types are defined by

$$\begin{aligned} & \text{CommonValidTimeDomFailCase}_\sigma[\text{Ev}] \\ & \iff \left[\begin{array}{l} \sigma = \text{EmptyCommonValidTimeDom} \\ \wedge \text{CommonValidTimeDom}^{\text{law}} = \emptyset \end{array} \right] \\ & \vee \left[\begin{array}{l} \sigma = \text{InadmissibleCommonValidTimeDom} \\ \wedge \text{CommonValidTimeDom}^{\text{law}} \neq \emptyset \\ \wedge \text{CommonValidTimeDom}^{\text{law}} \notin \text{AdmCommonTimeDomSet}^{\text{law}} \end{array} \right]. \end{aligned}$$

In the following law-failure counterexample-type occurrence family, law-failure counterexample pure-type family, and unrefined remainder-type family, fix

$$\begin{aligned} & \text{Ev} \in \text{EvSpec}^{\text{law}}, \\ & i \in I_{\text{law}}^{\text{fail}}, \\ & \sigma \in \text{SubIdx}_{\text{law},i} \end{aligned}$$

unless otherwise stated.

Definition 6.127 (Law-Failure Counterexample-Type Occurrence Family). The law-failure counterexample-type occurrence family is defined by

$$\text{LawCaseOcc}_{i,\sigma}[\text{Ev}] \iff \begin{cases} \text{LawEvalDomDefect}_\sigma[\text{Ev}], & i = 1, \\ \text{LawGenEvalArgConflict}[\text{Ev}], & i = 2, \\ \text{LawCandOutCaseOcc}_\sigma[\text{Ev}], & i = 3, \\ \text{LawCandRelMemChgOcc}_\sigma[\text{Ev}], & i = 4, \\ \text{CommonValidTimeDomFailCase}_\sigma[\text{Ev}], & i = 5 \end{cases}.$$

Definition 6.128 (Law-Failure Counterexample Pure-Type Family and Unrefined Remainder-Type Family). For each

$$\text{Ev} \in \text{EvSpec}^{\text{law}}, \quad i \in I_{\text{law}}^{\text{fail}}, \quad \sigma \in \text{SubIdx}_{\text{law},i},$$

the law-failure counterexample pure-type family is defined by

$$\begin{aligned} & \text{LawPureCase}_{i,\sigma}[\text{Ev}] \\ & \iff \text{LawCaseOcc}_{i,\sigma}[\text{Ev}] \wedge \bigwedge_{\lambda \in \text{SubIdx}_{\text{law},i} \setminus \{\sigma\}} \neg \text{LawCaseOcc}_{i,\lambda}[\text{Ev}]. \end{aligned}$$

For each $\text{Ev} \in \text{EvSpec}^{\text{law}}$, $i \in I_{\text{law}}^{\text{fail}}$, the unrefined remainder-type family is defined by

$$\text{LawUnrefCase}_i[\text{Ev}] \iff \begin{cases} \bigwedge_{\sigma \in \text{SubIdx}_{\text{law},1}} \text{LawCaseOcc}_{1,\sigma}[\text{Ev}], & i = 1, \\ \text{LawOffGenArgConflict}[\text{Ev}], & i = 2, \\ \bigwedge_{\sigma \in \text{SubIdx}_{\text{law},i}} \text{LawCaseOcc}_{i,\sigma}[\text{Ev}], & i \in \{3, 4\}, \\ \perp, & i = 5 \end{cases}.$$

Definition 6.129 (Law-Failure Counterexample Refinement Forest). For each

$$i \in I_{\text{law}}^{\text{fail}}, \quad \sigma \in \text{SubIdx}_{\text{law},i},$$

the law-failure counterexample-type subregion is defined by

$$\begin{aligned} & \text{FailReg}_{\text{law},i;\sigma}(\text{EvSpec}) \\ & := \{ \text{Ev} \in \text{FailReg}_{\text{law},i}(\text{EvSpec}) \mid \text{LawPureCase}_{i,\sigma}[\text{Ev}] \}. \end{aligned}$$

For $k \in \{0, 1\}$, define the child-index set along each path by

$$\text{ChildIdx}_{\text{law}}(i; \sigma_1, \dots, \sigma_k) := \begin{cases} \text{SubIdx}_{\text{law},i}, & k = 0, \\ \emptyset, & k = 1 \end{cases}.$$

When $k = 0$, omit the semicolon and everything that follows it. Define the node-region map by

$$\begin{aligned} \text{NodeReg}_{\text{law}}((i); \text{EvSpec}) &:= \text{FailReg}_{\text{law},i}(\text{EvSpec}), \\ \text{NodeReg}_{\text{law}}((i; \sigma); \text{EvSpec}) &:= \text{FailReg}_{\text{law},i;\sigma}(\text{EvSpec}), \\ \text{NodeReg}_{\text{law}}((i; \sigma); \text{EvSpec}) &\subseteq \text{NodeReg}_{\text{law}}((i); \text{EvSpec}). \end{aligned}$$

Define the law-failure counterexample refinement forest by

$$\text{RefForest}_{\text{law}}(\text{EvSpec}) := \text{RefForest} [I_{\text{law}}^{\text{fail}}; \text{ChildIdx}_{\text{law}}; \text{NodeReg}_{\text{law}}(-; \text{EvSpec})].$$

For each $i \in I_{\text{law}}^{\text{fail}}$ and $\varsigma \in \{\text{cov}, \text{unref}\}$, the covered or unrefined region at the root node is defined by

$$\text{FailReg}_{\text{law},i}^{\varsigma}(\text{EvSpec}) := \text{NodeReg}_{\text{law}}^{\varsigma}((i); \text{EvSpec}).$$

The two child regions at node 1 are called the individual-law evaluation-time-domain emptiness subregion and the evaluation-argument outside-application-domain subregion; the child region at node 2, the output-conflict-on-generated-evaluation-arguments subregion; the two child regions at node 3, the candidate-output-fiber emptiness subregion and the alternative-candidate-output existence subregion; the two child regions at node 4, the pure-type candidate-relation-membership-loss subregion and the pure-type candidate-relation-membership-gain subregion; and the two child regions at node 5, the common-valid-time-domain emptiness subregion and the common-valid-time-domain inadmissibility subregion.

Proposition 6.130 (Node Decomposition and Remainder Representation of the Law-Failure Counterexample Refinement Forest). For each $i \in I_{\text{law}}^{\text{fail}}$,

$$\begin{aligned} \text{FailReg}_{\text{law},i}^{\text{cov}}(\text{EvSpec}) &= \bigsqcup_{\sigma \in \text{SubIdx}_{\text{law},i}} \text{FailReg}_{\text{law},i;\sigma}(\text{EvSpec}), \\ \text{FailReg}_{\text{law},i}(\text{EvSpec}) &= \text{FailReg}_{\text{law},i}^{\text{cov}}(\text{EvSpec}) \sqcup \text{FailReg}_{\text{law},i}^{\text{unref}}(\text{EvSpec}) \end{aligned}$$

hold. For each $\text{Ev} \in \text{FailReg}_{\text{law},i}(\text{EvSpec})$,

$$\text{Ev} \in \text{FailReg}_{\text{law},i}^{\text{unref}}(\text{EvSpec}) \iff \text{LawUnrefCase}_i[\text{Ev}]$$

holds. In particular,

$$\begin{aligned} \text{FailReg}_{\text{law},5}^{\text{unref}}(\text{EvSpec}) &= \emptyset, \\ \text{FailReg}_{\text{law},5}^{\text{cov}}(\text{EvSpec}) &= \text{FailReg}_{\text{law},5}(\text{EvSpec}). \end{aligned}$$

Furthermore, if two child-region families for the same parent region satisfy the inclusion condition and the agreement condition on common indices in Lemma 6.33, then the covered region after refinement contains the covered region before refinement, and the unrefined region after refinement is contained in the unrefined region before refinement.

Proof. Each child region satisfies $\text{LawPureCase}_{i,\sigma}$. Because two distinct types at the same node cannot both be pure types, the family of child regions is mutually exclusive. Definition 6.129 and Theorem 6.35 give the disjoint-union representation of the covered region and the decomposition of the parent region.

The unrefined remainder representation is established node by node. At node 1, at least one of the two individual-law application-domain counterexample types occurs on the parent region. Belonging to neither pure-type region is equivalent to the occurrence of both types.

At node 2, nonuniqueness of a candidate-output fiber occurs on the parent region. Being outside the output-conflict-on-generated-evaluation-arguments subregion is equivalent to the existence of a nonunique candidate-output fiber and to the candidate-output fiber containing at most one element at every generated evaluation argument that belongs to the evaluation domain. This is $\text{LawOffGenArgConflict}[\text{Ev}]$.

At node 3, a generated law-evaluation relation-membership counterexample occurs on the parent region, and its candidate-output fiber is either empty or nonempty. Belonging to neither pure-type region is equivalent to the occurrence of counterexamples belonging to both fiber types.

At node 4, a counterexample to the faithful-representation-transformation invariance of candidate-relation membership occurs on the parent region. The two distinct membership-indicator values are either $(1, 0)$ or $(0, 1)$. Belonging to neither pure-type region is equivalent to the occurrence of both a membership-loss counterexample and a membership-gain counterexample.

At node 5, exactly one of the following holds: the common valid time domain is empty, or it is nonempty and inadmissible. Every element of the parent region therefore belongs to one pure-type child region, and the unrefined remainder region is empty.

Substituting these results into Definition 6.128 gives the unrefined remainder representation and the two identities for node 5. The final two inclusions follow by applying Lemma 6.33 to the root nodes of the law-failure counterexample refinement forest. $\square$

Definition 6.131 (Law-Series Failure Signature). For a complete evaluation specification $\text{Ev} \in \text{EvSpec}$, the law-series failure signature is defined by

$$\begin{aligned} \Sigma_{\text{law}}^{\text{fail}}[\text{Ev}] \\ &:= \left(\mathbf{1}_{\{\text{FailTok}_{\text{law},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}]\}} \right)_{i \in I_{\text{law}}^{\text{fail}}} \\ &\in \{0, 1\}^5. \end{aligned}$$

Proposition 6.132 (Correspondence between Local Law-Failure Classes and Failure Tokens).

Let $\text{Ev} \in \text{EvSpec}$, $i \in I_{\text{law}}^{\text{fail}}$. Suppose that $\text{DynOperative}[\text{Ev}]$, $\text{LawCondAncAll}_i[\text{Ev}]$ and

$$\text{FailReady}(\text{FailTok}_{\text{law},i}; \text{Ev}_{\text{law}}^{\text{fail}}) = \top$$

hold. Then

$$\begin{aligned} \text{FL}_i[\text{Ev}] \\ \iff \text{FailTok}_{\text{law},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}]. \end{aligned}$$

If this assumption holds for every $i \in I_{\text{law}}^{\text{fail}}$, then

$$\begin{aligned} \Sigma_{\text{law}}^{\text{fail}}[\text{Ev}] \\ = \left(\mathbf{1}_{\{\text{FL}_i[\text{Ev}]\}} \right)_{i \in I_{\text{law}}^{\text{fail}}} . \end{aligned}$$

Proof. By Proposition 6.23, membership of a failure token in the strict failed-token set is equivalent to its failure state being **failed**. By the evaluation-readiness assumption and Definition 6.21, the latter is equivalent to the failure evaluation taking the value $\perp$. By Definition 6.10, this is equivalent to the corresponding law-condition evaluation taking the value $\perp$. By Definition 6.114 and the assumptions of dynamical-description operability and the full-ancestor satisfaction bundle, it is further equivalent to the local failure class $\text{FL}_i[\text{Ev}]$ holding. The equivalence stated in the proposition therefore holds. Applying the equivalence obtained for all indices to each component in Definition 6.131 gives the displayed signature. $\square$

Proposition 6.133 (Type of the Law-Failure Localization Family). For each $i \in I_{\text{law}}^{\text{fail}}$, $\text{FailReg}_{\text{law},i}(\text{EvSpec}) \subseteq \text{EvSpec}$, for each $\lambda \in \text{SubIdx}_{\text{law},i}$,

$$\begin{aligned} \text{FailReg}_{\text{law},i;\lambda}(\text{EvSpec}) \\ \subseteq \text{FailReg}_{\text{law},i}(\text{EvSpec}) , \end{aligned}$$

and

$$\begin{aligned} \text{FailReg}_{\text{law},i}^{\text{unref}}(\text{EvSpec}) \\ \subseteq \text{FailReg}_{\text{law},i}(\text{EvSpec}) \end{aligned}$$

hold.

The law-series failure signature satisfies

$$\Sigma_{\text{law}}^{\text{fail}} : \text{EvSpec} \longrightarrow \{0, 1\}^5 .$$

Proof. The parent-region inclusion follows from Definition 6.120. Each refinement subregion is defined as the set obtained by adding a condition to membership in the corresponding parent region, and the unrefined remainder region is the set obtained by removing the refined covered region from the parent region. The three inclusions therefore hold.

The type of the law-series failure signature follows from Definition 6.131. $\square$

6.8 Localization of Differential-Stage Failure

Definition 6.134 (Selected family of differential-evaluation inputs). Use the same law-input family as in Definition 6.108. On a subset $\text{EvSpec}^{\text{diff}} \subseteq \text{EvSpec}^{\text{law}}$, fix a typed differential-data family

$$\text{DiffEvalDat}_{\text{Ev}} := \left(\rho_{\text{Ev}}^0, \text{LawEvalDat}_{\text{Ev}}, \text{EvalStrFam}_{\text{Ev}}^{\text{diff}} \right) .$$

The third component is a structure family from Definition 5.131 for the same transported law family determined by the first two components. It gives $\text{EvalInput}_{\text{Ev}}^{\text{diff}}$ from Definition 5.136. Write its condition and ancestor-bundle components as $\text{DiffCond}_{i,\text{Ev}}^{\text{univ}}$ and $\text{DiffCondAncAll}_{i,\text{Ev}}$. In formulas with a fixed $\text{Ev} \in \text{EvSpec}^{\text{diff}}$, omit the index Ev from these selected inputs and components. Set formation-input satisfaction for the differential series to the truth value of $\text{Ev} \in \text{EvSpec}^{\text{diff}}$, and evaluation-input satisfaction to $\top$.

Localize failure of the universal differential-condition family by minimal failed nodes relative to the differential-stage condition DAG of Definition 5.130. At each minimal failed node, all ancestor conditions hold and the universal differential condition at that node is false. Simultaneous minimal failures at mutually unreachable nodes are allowed.

6.8.1 Differential-Stage Failure Domain

Definition 6.135 (Evaluation of differential conditions on a complete evaluation specification). Take a complete evaluation specification $\text{Ev} \in \text{EvSpec}$ from Definition 5.2. For each $i \in I_{\text{diff}}^{\text{cond}}$, define the evaluations of the universal differential condition and the all-ancestors-satisfied bundle from Definition 5.136 by

$$\begin{aligned} \text{DiffCond}_i^{\text{univ}}[\text{Ev}] &\iff \exists e \in \text{EvSpec}^{\text{diff}}, \\ &\quad \text{Ev} = e \wedge \text{DiffCond}_{i,e}^{\text{univ}}, \\ \text{DiffCondAncAll}_i[\text{Ev}] &\iff \exists e \in \text{EvSpec}^{\text{diff}}, \\ &\quad \text{Ev} = e \wedge \text{DiffCondAncAll}_{i,e}. \end{aligned}$$

Definition 6.136 (Assignment of universal differential conditions to differential-series failure conditions). Definitions 6.10 and 5.130 give $I_{\text{diff}}^{\text{fail}} = I_{\text{diff}}^{\text{cond}}$. For each i in this common index set and each $\text{Ev} \in \text{EvSpec}$, define the differential-series failure condition by

$$K_{\text{diff},i}^{\text{fail}}[\text{Ev}] := K_i^{\text{diff}}[\text{Ev}] := \text{DiffCond}_i^{\text{univ}}[\text{Ev}].$$

Definition 6.137 (Differential-stage failure). On the selected domain, $\text{DiffCondAncAll}_1[\text{Ev}]$ is the operativity of the selected law datum; it is false outside that domain.

For a complete evaluation specification $\text{Ev} = (\text{l}, o, \zeta) \in \text{EvSpec}$, define differential-stage failure by

$$\begin{aligned} \text{Fail}_{\text{diff}}[\text{Ev}] \\ &\iff \text{DiffCondAncAll}_1[\text{Ev}] \\ &\quad \wedge \neg \bigwedge_{i \in I_{\text{diff}}^{\text{cond}}} K_i^{\text{diff}}[\text{Ev}]. \end{aligned}$$

Definition 6.138 (Differential-stage failure domain). Define the differential-stage failure domain in the complete-evaluation-specification set EvSpec by

$$\begin{aligned} \text{FailDom}_{\text{diff}}(\text{EvSpec}) \\ &:= \{\text{Ev} \in \text{EvSpec} \mid \text{Fail}_{\text{diff}}[\text{Ev}]\}. \end{aligned}$$

Proposition 6.139 (Differential-stage failure and differential-stage operability). For each $\text{Ev} = (\text{l}, o, \zeta) \in \text{EvSpec}^{\text{diff}}$,

$$\begin{aligned} \text{Fail}_{\text{diff}}[\text{Ev}] \\ &\iff \text{LawOperative}[\text{LawEvalDat}] \wedge \neg \text{DiffOperative}[\text{DiffEvalDat}]. \end{aligned}$$

Proof. By Definition 5.132, differential-stage operability is equivalent to the conjunction of law-description operability and the full differential-condition bundle. Thus, under law-description operability, the negation of differential-stage operability is equivalent to the negation of the full differential-condition bundle. By Definition 6.137, this equivalence gives the equivalence stated in the proposition. $\square$

6.8.2 Differential-Stage Minimal Failure Classes

Definition 6.140 (Differential-stage minimal failure classes). For each $\text{Ev} \in \text{EvSpec}$, $i \in I_{\text{diff}}^{\text{cond}}$, define the i -th minimal failure class at the differential stage by

$$\begin{aligned} \text{FDiff}_i[\text{Ev}] \\ \iff \text{DiffCondAncAll}_i[\text{Ev}] \wedge \neg K_i^{\text{diff}}[\text{Ev}]. \end{aligned}$$

Proposition 6.141 (Counterexamples for differential-stage minimal failure classes). For a complete evaluation specification $\text{Ev} = (\text{I}, o, \zeta) \in \text{EvSpec}^{\text{diff}}$ and $i \in I_{\text{diff}}^{\text{cond}}$, the following conditions are equivalent:

$$\begin{aligned} \text{FDiff}_i[\text{Ev}] \\ \iff \text{DiffCondAncAll}_i[\text{Ev}] \\ \wedge \exists \mathbf{T} \left[\begin{array}{l} \mathbf{T} = (T_r)_{0 \leq r < \text{arity}_{\text{diff}}(i)} \in (\text{OrdEmbSet}_o^{\text{time}}(\mathbb{R}))^{\text{arity}_{\text{diff}}(i)}, \\ \neg \text{DiffPred}_i \left[\left(\text{EvalStr}^{\text{diff}}(T_r) \right)_{0 \leq r < \text{arity}_{\text{diff}}(i)} \right] \end{array} \right]. \end{aligned}$$

Proof. Negate the family of universal differential conditions in Definition 5.130 and transform the universal quantification into existential quantification of the same arity. Conjoining the result with the all-ancestors-satisfied bundle in Definition 6.140 gives the stated equivalence. $\square$

Table 8: Differential-stage minimal failure classes

Failure class	Satisfied ancestors	Counterexample	Structure that fails
FDiff_1	Law-description operability	A differential-evaluation structure that fails the differential-atlas regularity condition	The system of differential chart changes for a common jet order
FDiff_2	Law-description operability and the differential-atlas regularity condition	A differential-evaluation structure that fails the generated-trajectory-jet existence condition	The generated trajectory jet
FDiff_3	Law-description operability, the differential-atlas regularity condition, and the generated-trajectory-jet existence condition	A differential-evaluation structure that fails the generated-trajectory-jet local-relation-membership condition	Establishment of the local differential law
FDiff_4	Law-description operability and the differential-atlas regularity condition	A differential-evaluation structure that fails the value-jet-relation covariance condition	Identity of the differential relation across value charts
FDiff_5	Law-description operability and the differential-atlas regularity condition	A pair of differential-evaluation structures that fails the time-representation-change jet-relation covariance condition	Identity of the differential relation across time representations

6.8.3 Differential-Stage Failure Parent Regions and Internal Subregions

Definition 6.142 (Differential-stage failure parent regions). Take a set EvSpec of complete evaluation specifications and $i \in I_{\text{diff}}^{\text{cond}}$. Define the i -th differential-stage failure parent region by

$$\begin{aligned} \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \\ := \{\text{Ev} \in \text{FailDom}_{\text{diff}}(\text{EvSpec}) \mid \text{FDiff}_i[\text{Ev}]\}. \end{aligned}$$

Definition 6.143 (Internal subregions of differential-stage failure parent regions). Take a set EvSpec of complete evaluation specifications and $i \in I_{\text{diff}}^{\text{cond}}$. Suppose that an index set $\text{SubIdx}_{\text{diff},i}$ and a family of subregions

$$(\text{FailReg}_{\text{diff},i;\alpha}(\text{EvSpec}))_{\alpha \in \text{SubIdx}_{\text{diff},i}}$$

satisfy

$$\text{FailReg}_{\text{diff},i;\alpha}(\text{EvSpec}) \subseteq \text{FailReg}_{\text{diff},i}(\text{EvSpec}).$$

Define the covered region and the unrefined region within the i -th parent region by

$$\begin{aligned} & (\text{FailReg}_{\text{diff},i}^{\text{cov}}(\text{EvSpec}), \text{FailReg}_{\text{diff},i}^{\text{unref}}(\text{EvSpec})) \\ & := \text{RefSplit} \left(\text{FailReg}_{\text{diff},i}(\text{EvSpec}); \left(\text{FailReg}_{\text{diff},i;\alpha}(\text{EvSpec}) \right)_{\alpha \in \text{Subld}_{\text{diff},i}} \right). \end{aligned}$$

Then

$$\begin{aligned} & \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \\ & = \text{FailReg}_{\text{diff},i}^{\text{cov}}(\text{EvSpec}) \dot{\cup} \text{FailReg}_{\text{diff},i}^{\text{unref}}(\text{EvSpec}). \end{aligned}$$

Definition 6.144 (Simultaneous minimal-failure node set). Define the family of antichains of a finite DAG (I, E) by

$$\begin{aligned} & \text{Antichain}(I, E) \\ & := \{A \subseteq I \mid \forall i, j \in A, \quad i \neq j \implies (i, j), (j, i) \notin E^+\}. \end{aligned}$$

For each $\text{Ev} \in \text{EvSpec}$, define the simultaneous minimal-failure node set by

$$\begin{aligned} & \text{MinFailld}_{\text{diff}}[\text{Ev}] \\ & := \{i \in I_{\text{diff}}^{\text{cond}} \mid \text{FDiff}_i[\text{Ev}]\}. \end{aligned}$$

6.8.4 Failure-Domain Cover Relative to the Differential-Condition DAG

Theorem 6.145 (Failure-domain cover relative to the differential-condition DAG). Take a set EvSpec of complete evaluation specifications. Relative to the five universal differential conditions and the differential-condition DAG of Definition 5.130,

$$\begin{aligned} & \text{FailDom}_{\text{diff}}(\text{EvSpec}) \\ & = \bigcup_{i \in I_{\text{diff}}^{\text{cond}}} \text{FailReg}_{\text{diff},i}(\text{EvSpec}). \end{aligned}$$

Equivalently, for each $\text{Ev} \in \text{EvSpec}$,

$$\begin{aligned} & \text{Fail}_{\text{diff}}[\text{Ev}] \\ & \iff \bigvee_{i \in I_{\text{diff}}^{\text{cond}}} \text{FDiff}_i[\text{Ev}]. \end{aligned}$$

Members of this cover may intersect at evaluation specifications containing simultaneous minimal-failure nodes.

Proof. First take any evaluation specification Ev in the differential-stage failure domain. By Definition 6.137, law-description operability holds and the full differential-condition bundle does not hold. By Definition 5.132, the set of nodes at which the universal differential condition takes the value false is nonempty.

Because the differential-condition DAG is finite, this set has a minimal element i with respect to the reachability order. For i , every ancestor node j satisfies $K_j^{\text{diff}}[\text{Ev}]$. Indeed, if an ancestor node taking the value false existed, that node would be a false node preceding i , contrary to the minimality of i .

Proposition 5.137 gives $\text{DiffCondAncAll}_i[\text{Ev}]$. Because i is a false node, $\neg K_i^{\text{diff}}[\text{Ev}]$ also holds. Definition 6.140 then gives $\text{FDiff}_i[\text{Ev}]$. Therefore, Ev belongs to the union of the differential-stage failure parent regions.

Conversely, for some $i \in I_{\text{diff}}^{\text{cond}}$, suppose that $\text{FDiff}_i[\text{Ev}]$ holds. Definition 6.140 gives the all-ancestors-satisfied bundle and failure of the i -th universal differential condition. The all-ancestors-satisfied bundle includes law-description operability, and failure of the i -th universal differential condition entails failure of the full differential-condition bundle. By Definition 6.137, $\text{Fail}_{\text{diff}}[\text{Ev}]$ holds. Therefore, Ev belongs to the differential-stage failure domain.

These two implications establish the set equality and the predicate equivalence for each evaluation specification. $\square$

Proposition 6.146 (Antichain property of the simultaneous minimal-failure node set). For each $\text{Ev} \in \text{FailDom}_{\text{diff}}(\text{EvSpec})$,

$$\begin{aligned} & \text{MinFailDx}_{\text{diff}}[\text{Ev}] \\ & \in \text{Antichain}(I_{\text{diff}}^{\text{cond}}, E_{\text{diff}}^{\text{cond}}) \setminus \{\emptyset\}. \end{aligned}$$

Proof. Let $i, j \in \text{MinFailDx}_{\text{diff}}[\text{Ev}]$, and suppose that reachability holds from i to j . Definition 6.140 gives $\neg K_i^{\text{diff}}[\text{Ev}]$. On the other hand, because i is a strict ancestor of j , the satisfaction of $\text{DiffCondAncAll}_j[\text{Ev}]$ gives $K_j^{\text{diff}}[\text{Ev}]$, a contradiction. Next suppose that reachability holds from j to i . Definition 6.140 gives $\neg K_j^{\text{diff}}[\text{Ev}]$. On the other hand, because j is a strict ancestor of i , the satisfaction of $\text{DiffCondAncAll}_i[\text{Ev}]$ gives $K_i^{\text{diff}}[\text{Ev}]$, again a contradiction. Thus the simultaneous minimal-failure node set is an antichain.

Nonemptiness follows from the existence of a minimal element, with respect to the reachability order, in every nonempty set of false nodes on a finite DAG, together with Theorem 6.145. $\square$

Remark 6.147 (Dependence of the differential-stage failure-domain cover on the condition family). The cover in Theorem 6.145 covers the failure domain relative to the five universal differential conditions of Definition 5.130. Each parent region decomposes into the covered region and the unrefined region of Definition 6.143.

6.8.5 Finite Witnesses for Differential-Stage Failure Parent Regions

Proposition 6.148 (Relative-nonemptiness criterion for differential-stage failure parent regions). For a set EvSpec of complete evaluation specifications, the following are equivalent.

1.

$$\prod_{i \in I_{\text{diff}}^{\text{cond}}} \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \neq \emptyset.$$

2.

$$\forall i \in I_{\text{diff}}^{\text{cond}}, \quad \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \neq \emptyset.$$

3. There exists a family of evaluation specifications

$$(\text{Ev}_i)_{i \in I_{\text{diff}}^{\text{cond}}} \in \prod_{i \in I_{\text{diff}}^{\text{cond}}} \text{EvSpec}$$

such that, for each $i \in I_{\text{diff}}^{\text{cond}}$, $\text{FDiff}_i[\text{Ev}_i]$ holds.

Proof. First suppose the first condition holds. Taking an element of the product shows that each coordinate belongs to the corresponding factor. Hence every parent region is nonempty.

Next suppose the second condition holds. For each index in the finite index set, take an evaluation specification from the corresponding nonempty parent region. By Definition 6.142, each evaluation specification belongs to the set of complete evaluation specifications and satisfies the corresponding differential-stage minimal failure condition.

Finally suppose the third condition holds. By Definition 6.142, for each $i \in I_{\text{diff}}^{\text{cond}}$, Ev_i belongs to the corresponding parent region. Therefore, the family of evaluation specifications $(\text{Ev}_i)_{i \in I_{\text{diff}}^{\text{cond}}}$ belongs to the product of the family of parent regions, and the first condition holds. $\square$

Corollary 6.149 (Finite witness set for differential-stage failure parent regions). For a set of complete evaluation specifications

$$\text{EvSpec}$$

and

$$(\text{Ev}_i)_{i \in I_{\text{diff}}^{\text{cond}}} \in \prod_{i \in I_{\text{diff}}^{\text{cond}}} \text{FailReg}_{\text{diff},i}(\text{EvSpec})$$

the set $\{\text{Ev}_i \mid i \in I_{\text{diff}}^{\text{cond}}\}$ has at most five elements, and for each $i \in I_{\text{diff}}^{\text{cond}}$,

$$\{\text{Ev}_j \mid j \in I_{\text{diff}}^{\text{cond}}\} \cap \text{FailReg}_{\text{diff},i}(\text{EvSpec}) \neq \emptyset.$$

Proof. The product membership in the proposition places the evaluation specification for each index in the corresponding parent region. Each evaluation specification also belongs to the candidate set, so the intersection of each parent region with that candidate set contains the corresponding evaluation specification. The candidate set has at most five elements. $\square$

Remark 6.150 (Scope of nonemptiness of differential-stage failure parent regions). Proposition 6.148 expresses simultaneous nonemptiness of the five parent regions as nonemptiness of the finite witness product. Once evaluation specifications belonging to the respective parent regions have been constructed, the finite witness set in Corollary 6.149 is determined.

6.8.6 Recursive Refinement of Differential-Stage Failure Regions

Definition 6.151 (Differential-stage failure-refinement forest). Take a complete evaluation specification set EvSpec . For each $i \in I_{\text{diff}}^{\text{cond}}$, take a first-layer child-index set $\text{Subldx}_{\text{diff},i}$. For each $\alpha_1 \in \text{Subldx}_{\text{diff},i}$, take a second-layer child-index set

$$\text{Subldx}_{\text{diff},i;\alpha_1}$$

Suppose that the corresponding second-layer family of subregions satisfies

$$\text{FailReg}_{\text{diff},i;\alpha_1,\alpha_2}(\text{EvSpec}) \subseteq \text{FailReg}_{\text{diff},i;\alpha_1}(\text{EvSpec}), \quad \alpha_2 \in \text{Subldx}_{\text{diff},i;\alpha_1}$$

Define the child-index set at each path by

$$\text{Childldx}_{\text{diff}}(i; \alpha_1, \dots, \alpha_k) := \begin{cases} \text{Subldx}_{\text{diff},i}, & k = 0, \\ \text{Subldx}_{\text{diff},i;\alpha_1}, & k = 1, \\ \emptyset, & k = 2, \end{cases}$$

Define the node-region map by

$$\begin{aligned}\text{NodeReg}_{\text{diff}}((i); \text{EvSpec}) &:= \text{FailReg}_{\text{diff},i}(\text{EvSpec}), \\ \text{NodeReg}_{\text{diff}}((i; \alpha_1); \text{EvSpec}) &:= \text{FailReg}_{\text{diff},i;\alpha_1}(\text{EvSpec}), \\ \text{NodeReg}_{\text{diff}}((i; \alpha_1, \alpha_2); \text{EvSpec}) &:= \text{FailReg}_{\text{diff},i;\alpha_1,\alpha_2}(\text{EvSpec})\end{aligned}$$

Define the differential-stage failure-refinement forest by

$$\text{RefForest}_{\text{diff}}(\text{EvSpec}) := \text{RefForest} [I_{\text{diff}}^{\text{cond}}; \text{ChildIdx}_{\text{diff}}; \text{NodeReg}_{\text{diff}}(-; \text{EvSpec})]$$

For each $i \in I_{\text{diff}}^{\text{cond}}$, $\alpha \in \text{SubIdx}_{\text{diff},i}$, and $\varsigma \in \{\text{cov}, \text{unref}\}$, set

$$\begin{aligned}\text{NodeReg}_{\text{diff}}^{\varsigma}((i); \text{EvSpec}) &= \text{FailReg}_{\text{diff},i}^{\varsigma}(\text{EvSpec}), \\ \text{FailReg}_{\text{diff},i;\alpha}^{\varsigma}(\text{EvSpec}) &:= \text{NodeReg}_{\text{diff}}^{\varsigma}((i; \alpha); \text{EvSpec}).\end{aligned}$$

For each $i \in I_{\text{diff}}^{\text{cond}}$, take two refinement data that share the parent region $\text{FailReg}_{\text{diff},i}(\text{EvSpec})$ and satisfy the assumptions of Lemma 6.33:

$$\left(\text{SubIdx}_{\text{diff},i}^{(\rho)}, \left(\text{FailReg}_{\text{diff},i;\alpha}^{(\rho)}(\text{EvSpec}) \right)_{\alpha \in \text{SubIdx}_{\text{diff},i}^{(\rho)}} \right), \quad \rho \in \{0, 1\}$$

The two refinement snapshots at the root node are, by RefSplit in Definition 6.32,

$$\left(\text{RefSplit} \left(\text{FailReg}_{\text{diff},i}(\text{EvSpec}); \left(\text{FailReg}_{\text{diff},i;\alpha}^{(\rho)}(\text{EvSpec}) \right)_{\alpha \in \text{SubIdx}_{\text{diff},i}^{(\rho)}} \right) \right)_{\rho \in \{0,1\}}$$

Proposition 6.152 (Expansion of the differential-stage failure-refinement forest). The differential-stage failure-refinement forest of Definition 6.151 satisfies the node decomposition, conditional disjoint union, and ancestor inclusion of Theorem 6.35. The two refinement snapshots at the root node satisfy the covered-region expansion and the unrefined-region contraction of Lemma 6.33.

Proof. The first assertion follows by substituting, from Definition 6.151, $I_{\text{diff}}^{\text{cond}}$, $\text{ChildIdx}_{\text{diff}}$, and $\text{NodeReg}_{\text{diff}}(-; \text{EvSpec})$ into Theorem 6.35. The second follows by substituting the two root snapshots from the same definition into Lemma 6.33. $\square$

6.8.7 Boundaries Among the Law, Approximation, and Differential Failure Series

Proposition 6.153 (Failure-series boundary between the law-description and differential stages). For each complete evaluation specification $\text{Ev} \in \text{EvSpec}^{\text{diff}}$,

$$\text{Fail}_{\text{diff}}[\text{Ev}] \implies (\text{LawOperative}[\text{LawEvalDat}] \wedge \neg \text{DiffOperative}[\text{DiffEvalDat}]).$$

Furthermore,

$$\begin{aligned}\text{DiffOperative}[\text{DiffEvalDat}] \\ \implies \text{LawOperative}[\text{LawEvalDat}].\end{aligned}$$

Proof. The first implication follows from the equivalence in Proposition 6.139. The second follows from the first component of the conjunction given by Definition 5.132. $\square$

Proposition 6.154 (Evaluation targets of the law, approximation, and differential failure series). The evaluation targets of the three series are given in the following table.

Series	Evaluation target
FL	Application domain, determinacy, satisfaction, faithful-representation covariance, and simultaneous satisfaction of the general-law relation
FA	Quantitative-approximation defect quantities, admissible thresholds, continuous extension, order stability, and relation stability
FDiff	Differential-atlas regularity, generated-trajectory jets, local jet relations, and jet-relation covariance

FL3 evaluates membership of a generated trajectory in a general-law relation, whereas FDiff3 evaluates membership of a generated-trajectory jet in a local jet relation. FL4 evaluates preservation of general-law-relation membership under a faithful-representation change, whereas FDiff4 and FDiff5 evaluate preservation of jet-relation membership under a value-chart change and a time-representation change, respectively.

The FA series takes as input the quantitative-evaluation component of $\text{EvalInput}^{\text{cont}}[l]$, whereas the FDiff series takes as input $\text{EvalInput}^{\text{diff}}$ of Definition 5.136. An implication linking the two series requires a condition that relates the components of the two input types.

Proof. The evaluation targets of FL3 and FL4 are an element of a general-law relation and a faithful-representation change. The evaluation targets of FDiff3 are the generated-trajectory jet and local jet relation of Definition 5.123. The evaluation targets of FDiff4 and FDiff5 are, respectively, the value-jet-change map of Definition 5.126 and the time-representation-change jet map of Definition 5.128.

The quantitative-evaluation component of the FA series belongs to the continuum-approximation evaluation input of Definition 4.168. The condition DAG, all-ancestors-satisfied bundle, and family of differential-evaluation structures of the FDiff series belong to the differential-stage failure-evaluation input of Definition 5.136. Therefore, the evaluation targets and input types of the respective series follow the distinctions in the table. $\square$

6.8.8 Differential-Stage Failure Signature and Failure-Position Tokens

Definition 6.155 (Five-component differential-stage failure signature). For a complete evaluation specification $\text{Ev} \in \text{EvSpec}$, define the five-component differential-stage failure signature by

$$\begin{aligned} \Sigma_{\text{diff}}^{\text{fail}}[\text{Ev}] &:= \left(\mathbf{1}_{\{\text{FDiff}_i[\text{Ev}]\}} \right)_{i \in I_{\text{diff}}^{\text{fail}}} \\ &\in \{0, 1\}^5 \end{aligned}$$

Proposition 6.156 (Support of the differential-stage failure signature). For a complete evaluation specification $\text{Ev} \in \text{EvSpec}$,

$$\begin{aligned} \text{supp} \left(\Sigma_{\text{diff}}^{\text{fail}}[\text{Ev}] \right) &= \text{MinFailIdx}_{\text{diff}}[\text{Ev}]. \end{aligned}$$

Furthermore, if $\text{Ev} \in \text{FailDom}_{\text{diff}}(\text{EvSpec})$, then $\Sigma_{\text{diff}}^{\text{fail}}[\text{Ev}] \neq (0, 0, 0, 0, 0)$, and its support is a nonempty antichain in the differential-stage condition DAG.

Proof. By Definition 6.155, the i -th component of the signature equals 1 if and only if $\text{FDiff}_i[\text{Ev}]$ holds. Definition 6.144 gives the set equality for the support. Nonzeroness on the failure domain and the antichain property of the support follow from Theorem 6.145 and Proposition 6.146. $\square$

Proposition 6.157 (Correspondence between differential-stage minimal-failure classes and failure tokens). Let $\text{Ev} \in \text{EvSpec}$, $i \in I_{\text{diff}}^{\text{fail}}$. Suppose that $\text{DiffCondAncAll}_i[\text{Ev}]$ and

$$\text{FailReady}(\text{FailTok}_{\text{diff},i}; \text{Ev}_{\text{diff}}^{\text{fail}}) = \top$$

hold. Then

$$\begin{aligned} & \text{FDiff}_i[\text{Ev}] \\ & \iff \text{FailTok}_{\text{diff},i} \in \text{FailedTok}^{\text{strict}}[\text{Ev}]. \end{aligned}$$

If this assumption holds for every $i \in I_{\text{diff}}^{\text{fail}}$, then each failure-token membership indicator equals the corresponding component of Definition 6.155.

Proof. By Proposition 6.23, membership of a failure token is equivalent to the failed state of that token. By the evaluation-readiness assumption and Definition 6.21, this failed state is equivalent to a false evaluation value for that token. By Definition 6.10 and Definition 6.136, this false evaluation value is equivalent to the negation of the i -th universal differential condition. By the assumption of the all-ancestors-satisfied bundle and Definition 6.140, the conjunction of this negated condition and the all-ancestors-satisfied bundle is equivalent to the i -th differential-stage minimal-failure class. Applying the equivalences obtained for all indices to the respective components of Definition 6.155 gives the equality between each failure-token membership indicator and the corresponding signature component. $\square$

Proposition 6.158 (Output families of the differential-stage failure classification). The differential-stage failure classification for the complete evaluation specification set EvSpec determines the following families:

$$\begin{aligned} & \text{DiffCondDAG}, \\ & (\text{FailReg}_{\text{diff},i}(\text{EvSpec}))_{i \in I_{\text{diff}}^{\text{cond}}}, \\ & (\text{FailReg}_{\text{diff},i}^{\text{unref}}(\text{EvSpec}))_{i \in I_{\text{diff}}^{\text{cond}}}, \\ & (\Sigma_{\text{diff}}^{\text{fail}}[\text{Ev}])_{\text{Ev} \in \text{EvSpec}}. \end{aligned}$$

Proof. The differential-stage condition DAG is determined by Definition 5.130; the five parent regions and the unrefined regions are determined, respectively, by Definition 6.142 and Definition 6.143. The five-component signature for each evaluation specification is determined by Definition 6.155. $\square$

6.9 Failure Signatures and the Approximation-Scale Boundary

For the six series of Definition 6.10, record by series the failure states of the five scale-invariant series and the scale-indexed failure state of the continuum-approximation series.

Definition 6.159 (Six-series failure signature). Take a scale-augmented complete evaluation specification $(\mathbf{Ev}, \lambda) \in \mathbf{EvSpec}_\Lambda$. Using the series-index families of Definition 6.10 and the unified state section of Definition 6.21, define the six-series failure signature by

$$\begin{aligned} \Sigma^{\text{fail}}[\mathbf{Ev}; \lambda] &:= \left(\left(\mathbf{1}_{\{\text{FailStateAt}_{\mathbf{Ev}, \lambda}(\text{FailTok}_{\tau, i}) = \text{failed}\}} \right)_{i \in I_{\tau}^{\text{fail}}} \right)_{\tau \in \text{FailSeries}} \\ &\in \prod_{\tau \in \text{FailSeries}} \{0, 1\}^{n_{\tau}^{\text{fail}}} \end{aligned}$$

The law-series component is the component at the series index $\tau = \text{law}$.

Proposition 6.160 (Minimal-position representation of the six-series failure signature). For every $(\mathbf{Ev}, \lambda) \in \mathbf{EvSpec}_\Lambda$, the coincidence of minimal elements in Proposition 6.26 holds. Furthermore,

$$\begin{aligned} \Sigma^{\text{fail}}[\mathbf{Ev}; \lambda] &= \left(\left(\mathbf{1}_{\{\text{FailTok}_{\tau, i} \in \text{FailedTok}[\mathbf{Ev}; \lambda]\}} \right)_{i \in I_{\tau}^{\text{fail}}} \right)_{\tau \in \text{FailSeries}}. \end{aligned}$$

Proof. The first conclusion follows from the coincidence of minimal elements in Proposition 6.26. By Definition 6.21, membership of each token of a scale-invariant series in the failed-token set is equivalent to its strict failure state being **failed**. For each token of the continuum-approximation series, the same membership is equivalent to its scale-indexed approximation failure state being **failed**.

By Definition 6.159, these membership indicators equal the respective components of the six-series failure signature. $\square$

Definition 6.161 (Strict, approximate, and full evaluation-validity domains and the approximation-boundary signature). Take the global validity domains of Definition 6.4 as input. Take a complete evaluation specification $\mathbf{Ev} \in \mathbf{EvSpec}$.

Define the strict-validity scale domain by

$$\begin{aligned} \mathcal{V}_{\mathbf{Ev}}^{\text{strict}} &:= \{ \lambda \in \Lambda \mid \mathbf{Ev} \in \mathcal{V}^{\text{strict}} \} \end{aligned}$$

For each $\eta \in \{\text{approx}, \text{full}\}$, define the validity scale domain by

$$\begin{aligned} \mathcal{V}_{\mathbf{Ev}}^{\eta} &:= \{ \lambda \in \Lambda \mid (\mathbf{Ev}, \lambda) \in \mathcal{V}^{\eta} \} \end{aligned}$$

The approximation-excess component set is $I_{\text{approx}}^{\text{exc}}[\mathbf{Ev}; \lambda]$ of Definition 6.90. Lemma 6.89 gives its membership criterion by the strict inequality between the defect quantity and the admissible threshold.

Define the approximation-boundary signature by

$$\begin{aligned} \Sigma_{\text{approx}}^{\text{bdry}}[\mathbf{Ev}; \lambda] &:= \left(\mathbf{1}_{\{i \in I_{\text{approx}}^{\text{exc}}[\lambda]\}} \right)_{i \in I_{\text{approx}}^{\text{fail}}} \\ &\in \{0, 1\}^9 \end{aligned}$$

Proposition 6.162 (Intersection representation of the full-evaluation-validity domain).
For every $\text{Ev} \in \text{EvSpec}$,

$$\begin{aligned}\mathcal{V}_{\text{Ev}}^{\text{full}} \\ &= \mathcal{V}_{\text{Ev}}^{\text{strict}} \cap \mathcal{V}_{\text{Ev}}^{\text{approx}}.\end{aligned}$$

Proof. Take any $\lambda \in \Lambda$. By Definition 6.27, the full-evaluation validity condition is equivalent to the conjunction of the strict-validity and approximation-validity conditions. Applying Definition 6.4 and Definition 6.161 gives the asserted equality. $\square$

Theorem 6.163 (Maximal approximation-validity interval and boundary signature).
On the totally ordered scale set $(\Lambda, \leq_\Lambda)$, suppose that the continuum-approximation defect-quantity family of Definition 6.85 is monotonically nondecreasing. Then, for every $\text{Ev} \in \text{EvSpec}$ satisfying $\forall \mu \in \Lambda, \text{InputSat}^{\text{approx}}[\text{Ev}; \mu] = \top$, the conclusions of Theorem 6.97 concerning the maximal effective interval and the first excess hold; moreover, one has

$$\mathcal{V}_{\text{Ev}}^{\text{approx}} \in \text{Init}_\Lambda(\Lambda)$$

and, in particular, if $\mathcal{V}_{\text{Ev}}^{\text{approx}} = \Lambda$, then $\mathcal{I}^{\text{max}}[\text{Ev}] = \Lambda$.

If the approximation-validity boundary $\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]$ is defined, then

$$\Sigma_{\text{approx}}^{\text{bdry}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]] \neq (0, \dots, 0).$$

Furthermore, for each $\lambda \in \Lambda$ and each $i \in I_{\text{approx}}^{\text{fail}}$, the saturation equality $\text{Def}_i^{\text{approx}}[\text{Ev}; \lambda] = \varepsilon_i^{\text{approx}}$ gives $K_i^{\text{approx}}[\text{Ev}; \lambda] = \top$.

Proof. Theorem 6.97 gives the conclusions concerning the maximal effective interval and the first excess. By Definition 6.85 and Theorem 6.95, the approximation-validity domain is an initial subset. The conclusion for the case in which the approximation-validity domain equals Λ follows from its equality with the maximal effective interval.

The nonemptiness of the excess-component set at the boundary scale and the indicator-signature definition in Definition 6.161 show that the boundary signature is nonzero. The assertion concerning the saturation equality follows from the threshold equivalence in Definition 6.84. $\square$

Theorem 6.164 (First-Failure Report at Structural Minimal Positions and the Approximation-Scale Boundary). Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. For Ev for which the approximation-validity boundary is defined, define the boundary excess-component set, boundary signature, and boundary datum by

$$\begin{aligned}\text{BdryExc}^{\text{approx}}[\text{Ev}] &:= I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]], \\ \text{BdrySig}^{\text{approx}}[\text{Ev}] &:= \Sigma_{\text{approx}}^{\text{bdry}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]], \\ \text{BdryDatum}^{\text{approx}}[\text{Ev}] &:= (\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}], \text{BdrySig}^{\text{approx}}[\text{Ev}]).\end{aligned}$$

Define the approximation-scale boundary data by the set of all least elements of the complement:

$$\begin{aligned}\text{BdryData}^{\text{approx}}[\text{Ev}] \\ &:= \left\{ (\lambda, \Sigma_{\text{approx}}^{\text{bdry}}[\text{Ev}; \lambda]) \mid \begin{array}{l} \lambda \in \Lambda \setminus \mathcal{V}_{\text{Ev}}^{\text{approx}}, \\ \forall \mu \in \Lambda \setminus \mathcal{V}_{\text{Ev}}^{\text{approx}}, \lambda \leq_\Lambda \mu \end{array} \right\}.\end{aligned}$$

This set is empty if $\mathcal{V}_{\text{Ev}}^{\text{approx}} = \Lambda$ or if the complement has no least element; when the approximation-validity boundary is defined,

$$\text{BdryData}^{\text{approx}}[\text{Ev}] = \{\text{BdryDatum}^{\text{approx}}[\text{Ev}]\}.$$

Define the first-failure report by

$$\text{FirstFailRep}[\text{Ev}; \lambda] := (\text{FailedTok}[\text{Ev}; \lambda], \Sigma^{\text{fail}}[\text{Ev}; \lambda], \text{BdryData}^{\text{approx}}[\text{Ev}]).$$

The first component consists of all failure tokens minimal with respect to the failure-dependency order, the second component is the six-series failure-state signature, and the third component is the approximation-scale boundary data.

Proof. By Definition 6.21 and Proposition 6.26, the first component coincides with the set of all minimal elements with respect to the failure-generation dependency order. Definition 6.159 determines the second component.

A least element of a subset of a total order is unique. Hence the approximation-scale boundary data are uniquely determined as either the empty set or a singleton, and the defined approximation-validity boundary and boundary signature provide that singleton. It follows that the first-failure report is uniquely determined. $\square$

Definition 6.165 (Multiplicity of Structural Failure and Approximation-Boundary Excess). Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$.

Write the structural-failure classification as $\text{FailCaseOf}(\text{Ev}; \lambda)$.

Call the case $\text{FailCaseOf}(\text{Ev}; \lambda) = \text{FailUnique}$ a simple structural failure, and the case $\text{FailCaseOf}(\text{Ev}; \lambda) = \text{FailParallel}$ a multiple structural failure.

Define the set of approximation-boundary-excess classification values by

$$\text{BdryCase}^{\text{approx}} := \{\text{BdryAbsent}, \text{BdryUnique}, \text{BdryParallel}\}.$$

The three values on the right-hand side are taken to be distinct. **BdryAbsent** means that there are no boundary-excess components. Whether the boundary itself exists is recorded in $\text{BdryData}^{\text{approx}}[\text{Ev}]$.

Define the approximation-boundary-excess multiplicity

$$\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] \in \text{BdryCase}^{\text{approx}}$$

by

$$\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] := \begin{cases} \text{BdryAbsent}, & \text{BdryData}^{\text{approx}}[\text{Ev}] \subseteq \Lambda \times \{(0, \dots, 0)\}, \\ \text{BdryUnique}, & \begin{array}{l} \text{BdryData}^{\text{approx}}[\text{Ev}] \neq \emptyset, \\ |\text{BdryExc}^{\text{approx}}[\text{Ev}]| = 1, \end{array} \\ \text{BdryParallel}, & \begin{array}{l} \text{BdryData}^{\text{approx}}[\text{Ev}] \neq \emptyset, \\ |\text{BdryExc}^{\text{approx}}[\text{Ev}]| \geq 2 \end{array} \end{cases}.$$

Call the case $\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] = \text{BdryUnique}$ a simple approximation-boundary excess, and the case $\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] = \text{BdryParallel}$ a multiple approximation-boundary excess.

Define the multiplicity report by

$$\begin{aligned} \text{MultRep}[\text{Ev}; \lambda] \\ := (\text{FailCaseOf}(\text{Ev}; \lambda), \text{BdryCaseOf}^{\text{approx}}[\text{Ev}]). \end{aligned}$$

Corollary 6.166 (Unique Minimal Position of a Simple Structural Failure). For a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$, if

$$\text{FailCaseOf}(\text{Ev}; \lambda) = \text{FailUnique}$$

then $\text{FailedTok}[\text{Ev}; \lambda]$ is a singleton and exactly one component of the six-series failure signature has value 1.

Proof. By Definition 6.24, the hypothesis is equivalent to $|\text{FailedTok}[\text{Ev}; \lambda]| = 1$. By Proposition 6.26, every element of the failed-token set is minimal with respect to the failure-dependency order. Hence the minimal-position set is a singleton and its indicator signature has exactly one component equal to 1. $\square$

Proposition 6.167 (Multiple Structural Failure and Multiple Approximation-Boundary Excess). For a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$, if $\text{FailCaseOf}(\text{Ev}; \lambda) = \text{FailParallel}$, then $|\text{FailedTok}[\text{Ev}; \lambda]| \geq 2$.

If $\text{BdryCaseOf}^{\text{approx}}[\text{Ev}] = \text{BdryParallel}$, then

$$|\text{BdryExc}^{\text{approx}}[\text{Ev}]| \geq 2.$$

The first component of the multiplicity report is determined by the cardinality of the structural minimal-position set, and the second component by the cardinality of the excess-component set at the approximation-scale boundary.

Proof. The first assertion follows from Definition 6.24 and Proposition 6.26.

The second assertion follows from the third branch of the approximation-boundary-excess multiplicity in Definition 6.165.

By Definition 6.165, the multiplicity report is an ordered pair of the two classification values. Each component is determined solely by the cardinality of the corresponding set. $\square$

Theorem 6.168 (Completeness of Failure Localization Relative to the Adopted Six Series). For every $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$, the equality of minimal elements in Proposition 6.26 holds.

If $\text{FailedTok}[\text{Ev}; \lambda] \neq \emptyset$, then

$$\text{Min}_{\preceq_{\text{fail,dep}}} \text{FailedTok}[\lambda] \neq \emptyset.$$

The minimal-position set may contain multiple failure tokens belonging to the same series or to distinct series.

The construction by state evaluation and the construction by minimal failure positions yield the same six-series failure signature $\Sigma^{\text{fail}}[\text{Ev}; \lambda]$.

This cover ranges over the failure tokens of the six series declared in Definition 6.10. The domain of evaluation specifications is EvSpec_Λ .

Proof. By the disjoint union in Definition 6.10, each failure token belongs to exactly one series.

By Definition 6.21, the failed-token set is a subset of the full set of failure tokens in the six series. Proposition 6.26 gives the equality of minimal elements stated in the theorem. This equality implies that nonemptiness of the failed set gives nonemptiness of the minimal-position set.

The cardinality of the series-index set follows from the definition of the six series in Definition 6.10.

By Proposition 6.160, the construction by state evaluation and the construction by minimal failure positions yield the same six-series failure signature. $\square$

6.10 Failure Positions and Structural Burden

Definition 6.169 (Structural Burden from Principal Upper Sets of the Failure-Dependency Order). Define the six-series set of structural-target tokens by

$$\text{StructTok} := \bigsqcup_{\tau \in \text{FailSeries}} \{\text{StructTok}_{\tau,i} \mid i \in I_{\tau}^{\text{fail}}\}.$$

Define the map that assigns the corresponding structural-target token to each failure-position token by

$$\text{StructOf} : \text{FailTok} \longrightarrow \text{StructTok},$$

$$\text{StructOf}(\text{FailTok}_{\tau,i}) := \text{StructTok}_{\tau,i}.$$

For any $s \in \text{FailTok}$, define the principal upper set with respect to the failure-generation dependency order by

$$\text{Desc}_{\text{fail,dep}}(s) := \{t \in \text{FailTok} \mid s \preceq^{\text{fail,dep}} t\}.$$

For any $A \subseteq \text{FailTok}$, define the image under the map **StructOf** by

$$\text{StructOf}[A] := \{\text{StructOf}(s) \mid s \in A\}.$$

Define the structural burden of a set of failure positions $A \subseteq \text{FailTok}$ by

$$\text{StructBurden}(A) := \bigcup_{s \in A} \text{StructOf}[\text{Desc}_{\text{fail,dep}}(s)].$$

For any $(\text{Ev}, \lambda) \in \text{EvSpec}_{\Lambda}$, define the failure-position-side structural burden by

$$\text{Burden}_{\text{pos}}^{\text{struct}}[\text{Ev}; \lambda] := \text{StructBurden}(\text{FailedTok}[\text{Ev}; \lambda]).$$

If the approximation-scale boundary data are empty, set $\text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}] := \emptyset$. If the approximation-scale boundary data are

$$\text{BdryData}^{\text{approx}}[\text{Ev}] = \{(\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}], \text{BdrySig}^{\text{approx}}[\text{Ev}])\}$$

define

$$\begin{aligned} & \text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}] \\ & := \text{StructBurden}(\{\text{FailTok}_{\text{approx},i} \mid i \in \text{BdryExc}^{\text{approx}}[\text{Ev}]\}). \end{aligned}$$

Define the structural-burden report by

$$\begin{aligned} & \text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda] \\ & := (\text{Burden}_{\text{pos}}^{\text{struct}}[\text{Ev}; \lambda], \text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}]). \end{aligned}$$

Lemma 6.170 (Monotonicity and Separation of the Structural-Burden Kernel). For any $s \in \text{FailTok}$, the structural burden of a single failure token is

$$\text{StructBurden}(\{s\}) = \text{StructOf}[\text{Desc}_{\text{fail,dep}}(s)].$$

For any $A, B \subseteq \text{FailTok}$,

$$A \subseteq B \implies \text{StructBurden}(A) \subseteq \text{StructBurden}(B).$$

Moreover, for any $s \in A$,

$$\begin{aligned} & \text{StructBurden}(\{s\}) \not\subseteq \text{StructBurden}(B) \\ & \implies \text{StructBurden}(A) \setminus \text{StructBurden}(B) \neq \emptyset. \end{aligned}$$

Proof. The equality for a singleton follows by taking the union in Definition 6.169 over its sole element s .

For monotonicity, $A \subseteq B$ implies that every component of the union indexed by A is also a component of the union indexed by B .

The separation assumption gives an element

$$x \in \text{StructBurden}(\{s\}) \setminus \text{StructBurden}(B).$$

Because $\{s\} \subseteq A$ and monotonicity imply $x \in \text{StructBurden}(A)$, the required set difference is nonempty. $\square$

Table 9: Failure-position tokens and structural-target tokens in the six series

Series	τ	n_τ^{fail}
Comparison	cmp	8
Representation	repr	3
Dynamical description	dyn	8
Continuum approximation	approx	9
Law representation	law	5
Differential consistency and law-form preservation	diff	5

For every series in the table and every $i \in I_\tau^{\text{fail}}$, the correspondence is

$$\text{StructOf}(\text{FailTok}_{\tau,i}) = \text{StructTok}_{\tau,i}.$$

Table 10: Two components of the structural-burden report

Component	Generating input	Codomain
Failure-position-side structural burden	$\text{FailedTok}[\text{Ev}; \lambda]$	$\mathcal{P}(\text{StructTok})$
Approximation-scale-boundary-side structural burden	$\text{BdryData}^{\text{approx}}[\text{Ev}]$	$\mathcal{P}(\text{StructTok})$
Structural-burden report	Failure-position-side and approximation-scale-boundary-side structural burdens	$\mathcal{P}(\text{StructTok})^2$

Proposition 6.171 (Uniqueness and Dependency-Upward Closure of the Structural-Burden Report). For any $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$, $\text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda]$ is uniquely determined. Moreover,

$$\begin{aligned} \forall s \in \text{FailedTok}[\text{Ev}; \lambda], \\ \text{StructBurden}(\{s\}) \subseteq \text{Burden}_{\text{pos}}^{\text{struct}}[\text{Ev}; \lambda]. \end{aligned}$$

If the approximation-scale boundary data form a singleton,

$$\begin{aligned} \forall i \in \text{BdryExc}^{\text{approx}}[\text{Ev}], \\ \text{StructBurden}(\{\text{FailTok}_{\text{approx},i}\}) \subseteq \text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}]. \end{aligned}$$

Proof. By Definition 6.169, the failure-position-side structural burden is the union of the single-token structural burdens of the failure tokens. Hence each single-token structural burden is contained in this union.

When the approximation-scale boundary data are empty, the boundary-side structural burden is empty. When they form a singleton, the boundary-side structural burden is the union of the structural-target images of the principal upper sets determined by the respective boundary excess components. Hence each image is contained in this union. Both components are uniquely determined in either branch, and the structural-burden report is therefore uniquely determined. $\square$

6.11 Failure-Localization Results

Definition 6.172 (Failure-Localization Data). For any $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$, define the failure-localization data by

$$\begin{aligned} \text{FailLocData}[\text{Ev}; \lambda] \\ := & \left((\mathcal{V}_{\text{Ev}}^{\text{strict}}, \mathcal{V}_{\text{Ev}}^{\text{approx}}, \mathcal{V}_{\text{Ev}}^{\text{full}}), \mathcal{I}^{\text{max}}[\text{Ev}], \right. \\ & \text{FirstFailRep}[\text{Ev}; \lambda], \text{MultRep}[\text{Ev}; \lambda], \\ & \left. \text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda], \Sigma^{\text{fail}}[\text{Ev}; \lambda] \right). \end{aligned}$$

Proposition 6.173 (Determinacy and Audit Inputs of the Failure-Localization Results). The failure-localization data of Definition 6.172 are uniquely determined. Among its three validity scale domains, the full-evaluation validity scale domain equals the intersection of the strict-validity and approximation-validity scale domains. The maximal approximation-validity interval is contained in the approximation-validity scale domain; under the hypotheses of Theorem 6.163, the two are equal.

The first-failure report, multiplicity report, and structural-burden report are uniquely determined. The construction by state evaluation and the construction by minimal failure positions yield the same six-series failure signature. The minimal failure positions are covered relative to the adopted six series and to the evaluation states on the scale-augmented complete evaluation specification.

Proof. Definition 6.161 and Proposition 6.162 uniquely determine the three validity scale domains and establish their intersection equality. Definition 6.93 uniquely determines the maximal approximation-validity interval and places it within the approximation-validity scale domain. Under the hypotheses of Theorem 6.163, that theorem gives equality of the two domains.

Theorem 6.164 uniquely determines the first-failure report. Definition 6.165 uniquely determines the multiplicity report. Proposition 6.171 uniquely determines the structural-burden report.

By Proposition 6.160, the construction by state evaluation and the construction by minimal failure positions yield the same six-series failure signature. By Theorem 6.168, the minimal failure positions are covered relative to the adopted six series.

Consequently, the three validity scale domains, maximal approximation-validity interval, first-failure report, multiplicity report, and structural-burden report are uniquely determined, yielding the inputs to the operational audit report. $\square$

6.11.1 Failure-State Classification Relative to the Adopted Evaluation Domain

Definition 6.174 (Failure-Token State Profile). Take any $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$.

Using the state fibers of Definition 6.21, define the satisfied, unformed, and unevaluable failure-token sets, respectively, by

$$\begin{aligned}\text{SatTok}[\text{Ev}; \lambda] &:= \text{FailFiber}_{\text{sat}}[\text{Ev}; \lambda], \\ \text{UnformedTok}[\text{Ev}; \lambda] &:= \text{FailFiber}_{\text{unformed}}[\text{Ev}; \lambda], \\ \text{UnevaluableTok}[\text{Ev}; \lambda] &:= \text{FailFiber}_{\text{unevaluable}}[\text{Ev}; \lambda].\end{aligned}$$

Define the failure-token state profile by

$$\begin{aligned}\text{TokStateProf}^{\text{fail}}[\text{Ev}; \lambda] \\ := \Big(K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda], \\ \text{SatTok}[\text{Ev}; \lambda], \text{FailedTok}[\text{Ev}; \lambda], \\ \text{UnformedTok}[\text{Ev}; \lambda], \text{UnevaluableTok}[\text{Ev}; \lambda], \\ \text{BdryData}^{\text{approx}}[\text{Ev}] \Big).\end{aligned}$$

Table 11: Evaluation states of failure tokens in the adopted evaluation domain

Condition	Evaluation-state component	Corresponding failure-evaluation quantity
The states of all failure-position tokens are sat	$K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda] = \top$	Full-evaluation validity
$\text{FailedTok}[\text{Ev}; \lambda] \neq \emptyset$	failed	$\text{FirstFailRep}[\text{Ev}; \lambda],$ $\text{MultRep}[\text{Ev}; \lambda],$ $\text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda]$
$\text{UnformedTok}[\text{Ev}; \lambda] \neq \emptyset$	unformed	Unformed failure-token set
$\text{UnevaluableTok}[\text{Ev}; \lambda] \neq \emptyset$	unevaluable	Unevaluable failure-token set
$ \text{BdryData}^{\text{approx}}[\text{Ev}] = 1$	Approximation-scale boundary	$\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}],$ $\text{BdrySig}^{\text{approx}}[\text{Ev}],$ $\text{Burden}_{\text{bdry}}^{\text{struct}}[\text{Ev}]$

Distinct state sets may be nonempty simultaneously. The approximation-scale boundary data are a component concerning the scale direction of the continuum-approximation series.

Proposition 6.175 (Failure-State Classification Relative to the Adopted Evaluation Domain). For any $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$,

$$\begin{aligned}\text{FailTok} &= \text{SatTok}[\text{Ev}; \lambda] \\ &\sqcup \text{FailedTok}[\text{Ev}; \lambda] \\ &\sqcup \text{UnformedTok}[\text{Ev}; \lambda] \\ &\sqcup \text{UnevaluableTok}[\text{Ev}; \lambda].\end{aligned}$$

Furthermore, $K_{\text{full}}^{\text{fail}}[\text{Ev}; \lambda] = \top$ is equivalent to $\text{SatTok}[\lambda] = \text{FailTok}$.

If

$$\text{FailedTok}[\lambda] \neq \emptyset$$

each failure token belongs to exactly one of the adopted six series. The first-failure report, multiplicity report, and structural-burden report

$$\begin{aligned} &\text{FirstFailRep}[\text{Ev}; \lambda], \\ &\text{MultRep}[\text{Ev}; \lambda], \\ &\text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda] \end{aligned}$$

determine the minimal failure positions, six-series failure signature, multiplicity, approximation-scale boundary, and structural burden.

$\text{TokStateProf}^{\text{fail}}[\text{Ev}; \lambda]$ has as its components the full-evaluation validity condition, the four state-specific failure-token sets, and the approximation-scale boundary data.

The domain of this classification is EvSpec_Λ , and its completeness is relative to the failure-token system of Definition 6.10.

Proof. By Definition 6.10, the full failure-token set decomposes as a disjoint union of the failure-token sets of the scale-invariant series and the continuum-approximation series. By the evaluation-state map and series-specific evaluation specifications of Definition 6.21 and Proposition 6.22, each failure-position token is assigned exactly one of the evaluation states **sat**, **failed**, **unformed**, and **unevaluable**. Hence the four state-specific failure-token sets give a disjoint-union decomposition allowing empty components of the full failure-token set.

By Proposition 6.28, the full-evaluation validity condition holds if and only if the evaluation state of every failure-position token is **sat**.

By Theorem 6.168, each failure token belongs to exactly one of the adopted six series. By Proposition 6.173, the first-failure report, multiplicity report, and structural-burden report are uniquely determined.

Definition 6.174 determines each component of the failure-token state profile. □

Remark 6.176 (Relative Scope of Classification Completeness). The domain of the failure-state classification is EvSpec_Λ . The adopted six series and the full failure-token set **FailTok** of Definition 6.10 give the domain of quantification for classification completeness.

On the basis of the results of Theorem 4.177, Theorem 5.112, and Theorem 6.164, localize validity, validity domains, the approximation limit, the maximal validity interval, the failure-position token system, and the minimal failure-position set.

Failure positions divide into six groups: the comparison, representation, continuum-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages. Represent each position by a failure-position token of Definition 6.10. The failed-token set obtained from the evaluation states and the failure-generation dependency order determine the minimal-position representation of the six-series failure signature in Proposition 6.160.

Validity and failure localization

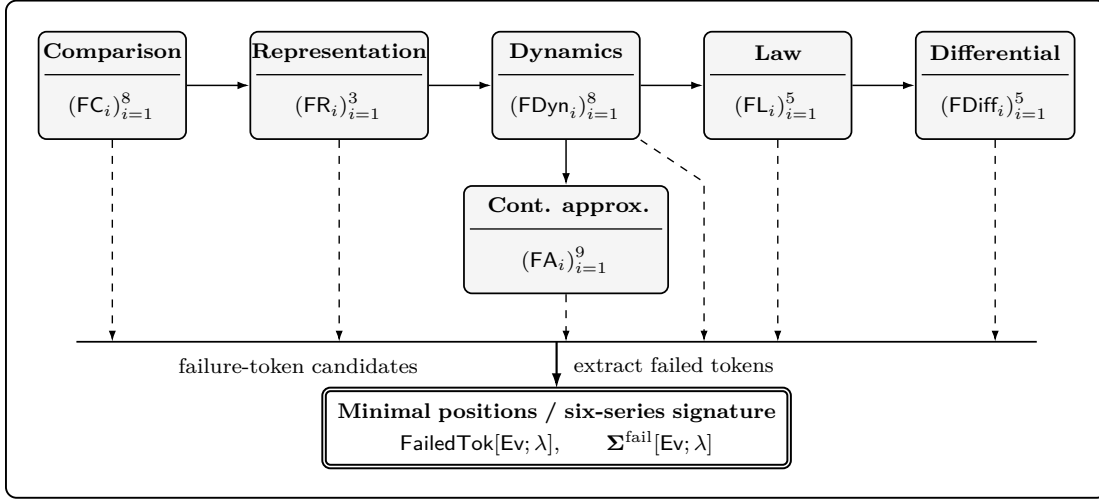


Figure 14: Six-group localization of validity and failure. The minimal failure-position set is defined as the set of positions minimal with respect to the failure-generation dependency order among the failure-position tokens belonging to the comparison, representation, continuum-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages.

6.12 Operational Auditability of the Time-Reference Construction

Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Take as input the semantic-series-tagged failure-position token system of Definition 6.10, together with the failure-localization data of Definition 6.172, $\text{FailLocData}[\text{Ev}; \lambda]$. Operational audit data consist of the computational components used to evaluate each failure condition and an evidence-provenance certificate that verifies the origin of those computational components.

Definition 6.177 (Operational Audit Data for the Time-Reference Construction). Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Use the six-failure-series family of evaluation specifications Ev^{fail} of Definition 6.10.

Define the set of series that require inputs with formation states by

$$\text{AuditInputSeries} := \{\text{approx}, \text{law}, \text{diff}\}.$$

Let the set of reasons for unformed audit inputs be

$$\text{AuditInputReason}.$$

For each

$$\tau \in \text{AuditInputSeries}$$

take a series-specific audit-input type $\text{AuditInputType}_\tau$ and a series-specific audit input

$$\text{AuditBranchInput}_\tau \in \text{AuditInputType}_\tau \sqcup \text{AuditInputReason}$$

The image under the canonical injection into the left summand is a formed audit input, and the image under the canonical injection into the right summand is an unformed reason.

Define the audit-input datum by

$$\text{AuditInput}[\text{Ev}] := \left(\mathcal{X}, \text{Ev}, (\text{AuditBranchInput}_\tau)_{\tau \in \text{AuditInputSeries}} \right).$$

Define the all-scale family of evaluation specifications for the continuum-approximation series by

$$\text{Ev}_{\text{approx}}^{\text{fail}} := (\text{Ev}_{\text{approx}, \lambda}^{\text{fail}})_{\lambda \in \Lambda}.$$

Define the condition-evidence family for the six failure series by

$$\text{AuditEvidence}[\text{Ev}^{\text{fail}}] := \left((\text{AuditEvidence}_\tau[\text{Ev}_\tau^{\text{fail}}])_{\tau \in \text{FailSeries}^{\text{strict}}}, \right. \\ \left. (\text{AuditEvidence}_{\text{approx}}[\text{Ev}_{\text{approx}, \lambda}^{\text{fail}}])_{\lambda \in \Lambda} \right).$$

Define the map that gives the condition-evidence component for each $t \in \text{FailTok}$ by

$$\text{AuditEvidenceAt}[\text{Ev}; \lambda](t) \\ := \text{AuditEvidence}_{\text{FailSeriesOf}(t)} [\text{FailSpecAt}_{\text{FailSeriesOf}(t)} [\text{Ev}; \lambda]].$$

Let $\text{AuditScaleData}[\text{Ev}_{\text{approx}}^{\text{fail}}]$ be the scale-audit data consisting of the scale order, approximation-scale set, codomains of defect quantities, and admissible thresholds.

Define the audit computation core by

$$\text{AuditCore}[\text{Ev}^{\text{fail}}] := \left(\text{AuditContractID}, \text{AuditInput}[\text{Ev}], \right. \\ \left. \text{AuditEvidence}[\text{Ev}^{\text{fail}}], \text{AuditScaleData}[\text{Ev}_{\text{approx}}^{\text{fail}}] \right).$$

Define the evidence-provenance certificate by

$$\text{AuditCert}[\text{Ev}^{\text{fail}}] := \left(\text{AuditInputID}, \text{AuditEvidenceID}, \text{AuditContentHash}, \right. \\ \left. \text{AuditTransformTrace}, \text{AuditEvidenceLocation} \right).$$

Define the operational audit data for the time-reference construction by

$$\text{AuditData}_{\text{time}}[\text{Ev}^{\text{fail}}] := (\text{AuditCore}[\text{Ev}^{\text{fail}}], \text{AuditCert}[\text{Ev}^{\text{fail}}]).$$

In the remainder of this section, the time-type subscript is omitted from this notation.

Definition 6.178 (Component Families of the Audit Computation Core). The audit computation core $\text{AuditCore}[\text{Ev}^{\text{fail}}]$ of Definition 6.177 consists of the audit contract, audit input, six-series condition-evidence family, and scale-audit data.

The condition-evidence family $\text{AuditEvidence}[\text{Ev}^{\text{fail}}]$ has, for every $\tau \in \text{FailSeries}$ and every $i \in I_\tau^{\text{fail}}$, as its components the typed condition evidence for Ev in a scale-invariant series and for each (Ev, λ) in the continuum-approximation series.

The scale-audit data $\text{AuditScaleData}[\text{Ev}_{\text{approx}}^{\text{fail}}]$ have as their components the scale order, approximation-scale set, codomains of defect quantities, and admissible thresholds. Through the audit-input identifiers, condition-evidence identifiers, content hash, transformation history, and evidence location, the evidence-provenance certificate associates provenance with each component of the audit computation core.

Table 12: Typed components and computational responsibilities of the audit computation core

Component family	Typed component	Computational responsibility
Audit contract	<code>AuditContractID</code>	Identification of the condition predicates, input types, scale types, and audit-report generation convention
Audit input	<code>AuditInput[Ev]</code>	Operative construction, complete evaluation specification, and inputs with formation states for the three series
Six-series condition-evidence family	<code>AuditEvidence[Ev^{fail}]</code>	Evaluation of every condition in the five scale-invariant series and every condition at every scale in the continuum-approximation series
Scale-audit data	<code>AuditScaleData[Ev^{fail}_{approx}]</code>	Evaluation of the scale order, approximation-scale set, codomains of defect quantities, and admissible thresholds

Definition 6.179 (Five-Valued Audit-Condition States and Six-Group Audit States). Take a scale-augmented complete evaluation specification $(Ev, \lambda) \in \text{EvSpec}_\Lambda$.

Define the audit-condition state set and its group-aggregation priority order by

$$\begin{aligned}
 \text{AuditStatus} &:= \left\{ \text{AuditPass}, \text{AuditFail}, \text{AuditBlocked}, \right. \\
 &\quad \left. \text{AuditUnformed}, \text{AuditIndeterminate} \right\}, \\
 \text{AuditPass} &\prec_{\text{AuditStatus}} \text{AuditIndeterminate} \prec_{\text{AuditStatus}} \text{AuditUnformed} \\
 &\quad \prec_{\text{AuditStatus}} \text{AuditBlocked} \prec_{\text{AuditStatus}} \text{AuditFail}
 \end{aligned}$$

The five state values are mutually distinct.

Define the five-valued lift map that adds the success or failure of immediate predecessor conditions to the four-valued evaluation states of Definition 6.21 by

$$\begin{aligned}
 \text{AuditStateLift} : \{\perp, \top\} \times \text{FailState} &\longrightarrow \text{AuditStatus}, \\
 \text{AuditStateLift}(b, q) &:= \begin{cases} \text{AuditBlocked}, & b = \perp, q \in \text{FailState}, \\ \text{AuditUnformed}, & (b, q) = (\top, \text{unformed}), \\ \text{AuditIndeterminate}, & (b, q) = (\top, \text{unevaluable}), \\ \text{AuditPass}, & (b, q) = (\top, \text{sat}), \\ \text{AuditFail}, & (b, q) = (\top, \text{failed}) \end{cases}
 \end{aligned}$$

For any $t \in \text{FailTok}$, define the immediate-predecessor-satisfaction value and the audit-condition state by

$$\begin{aligned}
 &\text{AuditPredecessorsPass}[Ev; \lambda](t) \\
 &\quad := \text{FailPredSat}(\text{FailArgAt}_{Ev, \lambda}(t)), \\
 &\text{AuditConditionStateOf}[Ev; \lambda](t) \\
 &\quad := \text{AuditStateLift}(\text{AuditPredecessorsPass}[Ev; \lambda](t), \text{FailStateAt}_{Ev, \lambda}(t)).
 \end{aligned}$$

Consequently,

$$\text{AuditConditionStateOf}[\text{Ev}; \lambda] : \text{FailTok} \longrightarrow \text{AuditStatus}.$$

For each $q \in \text{AuditStatus}$, define the audit-state fiber by

$$\text{AuditStatusFiber}_q[\text{Ev}; \lambda] := \{t \in \text{FailTok} \mid \text{AuditConditionStateOf}[\text{Ev}; \lambda](t) = q\}.$$

For each $\tau \in \text{FailSeries}$, define the series-group state by

$$\text{AuditGroupState}_\tau[\text{Ev}; \lambda] := \max_{\prec_{\text{AuditStatus}}} \{\text{AuditConditionStateOf}[\text{Ev}; \lambda](t) \mid t \in \text{FailTok}_\tau\}.$$

Define the six-group audit state by

$$\text{AuditGroupStatus}[\text{Ev}; \lambda] := (\text{AuditGroupState}_\tau[\text{Ev}; \lambda])_{\tau \in \text{FailSeries}}.$$

Define the indeterminate-condition set and indeterminate signature by

$$\begin{aligned} \text{AuditIndeterminateTok}[\text{Ev}; \lambda] &:= \text{AuditStatusFiber}_{\text{AuditIndeterminate}}[\text{Ev}; \lambda], \\ \text{AuditIndeterminateSig}[\text{Ev}; \lambda](t) &:= \mathbf{1}_{\text{AuditIndeterminateTok}[\text{Ev}; \lambda]}(t), \quad t \in \text{FailTok}. \end{aligned}$$

Consequently,

$$\text{AuditIndeterminateSig}[\text{Ev}; \lambda] : \text{FailTok} \longrightarrow \{0, 1\}.$$

Definition 6.180 (Five-Valued Audit-Report Object). Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Use the six-failure-series family of evaluation specifications Ev^{fail} of Definition 6.10.

Define the set of audit-undefined states by

$$\text{AuditUndefinedStatus} := \{\text{AuditBlocked}, \text{AuditUnformed}, \text{AuditIndeterminate}\}.$$

For each $\tau \in \text{FailSeries}$, define the series-specific audit-undefined reason set and the all-series audit-undefined reason set by

$$\begin{aligned} \text{AuditUndefinedReason}_\tau[\text{Ev}^{\text{fail}}; \lambda] &:= \left\{ (t, q) \in \text{FailTok}_\tau \times \text{AuditUndefinedStatus} \right. \\ &\quad \left. \mid \text{AuditConditionStateOf}[\text{Ev}; \lambda](t) = q \right\}, \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] &:= \bigcup_{\tau \in \text{FailSeries}} \text{AuditUndefinedReason}_\tau[\text{Ev}^{\text{fail}}; \lambda]. \end{aligned}$$

For any set $\mathcal{Y}$, define the tagged audit-result map by

$$\begin{aligned} \text{AuditTaggedResult}_\mathcal{Y} &: \mathcal{Y} \times \mathcal{P}(\text{FailTok} \times \text{AuditUndefinedStatus}) \\ &\longrightarrow (\{\text{AuditValue}\} \times \mathcal{Y}) \\ &\quad \sqcup (\{\text{AuditReason}\} \times [\mathcal{P}(\text{FailTok} \times \text{AuditUndefinedStatus}) \setminus \{\emptyset\}]), \end{aligned}$$

$\text{AuditTaggedResult}_\mathcal{Y}(y; R)$

$$:= \begin{cases} (\text{AuditValue}, y), & R = \emptyset, \\ (\text{AuditReason}, R), & R \neq \emptyset. \end{cases}$$

The tags **AuditValue** and **AuditReason** are distinct. In what follows, the subscript $\mathcal{V}$, which is uniquely determined by the object type, is omitted.

Define the four tagged diagnostic components by

$$\begin{aligned}
& \text{AuditFailedPosition}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad := \text{AuditTaggedResult} \left(\begin{array}{c} \text{FailedTok}[\text{Ev}; \lambda]; \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right), \\
& \text{AuditFailureSignature}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad := \text{AuditTaggedResult} \left(\begin{array}{c} \Sigma^{\text{fail}}[\text{Ev}; \lambda]; \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right), \\
& \text{AuditApproxBoundary}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad := \text{AuditTaggedResult} \left(\begin{array}{c} \text{BdryData}^{\text{approx}}[\text{Ev}]; \\ \text{AuditUndefinedReason}_{\text{approx}}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right), \\
& \text{AuditStructuralBurden}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad := \text{AuditTaggedResult} \left(\begin{array}{c} \text{BurdenRep}^{\text{struct}}[\text{Ev}; \lambda]; \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right).
\end{aligned}$$

Define the audit-state report and audit-diagnostic report by

$$\begin{aligned}
& \text{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad \text{AuditGroupStatus}[\text{Ev}; \lambda], \\
& \quad := \left(\begin{array}{c} (\text{AuditConditionStateOf}[\text{Ev}; \lambda](t))_{t \in \text{FailTok}}, \\ \text{AuditIndeterminateTok}[\text{Ev}; \lambda], \\ \text{AuditIndeterminateSig}[\text{Ev}; \lambda], \\ \text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right), \\
& \text{AuditDiagnosticReport}[\text{Ev}^{\text{fail}}; \lambda] \\
& \quad \text{AuditFailedPosition}[\text{Ev}^{\text{fail}}; \lambda], \\
& \quad := \left(\begin{array}{c} \text{AuditFailureSignature}[\text{Ev}^{\text{fail}}; \lambda], \\ \text{AuditApproxBoundary}[\text{Ev}^{\text{fail}}; \lambda], \\ \text{AuditStructuralBurden}[\text{Ev}^{\text{fail}}; \lambda] \end{array} \right).
\end{aligned}$$

Define the five-valued audit-report object by

$$\begin{aligned}
\text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda] := & \left(\text{AuditData}[\text{Ev}^{\text{fail}}], \right. \\
& \text{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda], \\
& \left. \text{AuditDiagnosticReport}[\text{Ev}^{\text{fail}}; \lambda] \right).
\end{aligned}$$

Definition 6.181 (Audit-Report Generation Map). Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Use the six-failure-series family of evaluation specifications Ev^{fail} of Definition 6.10.

Define the audit-report generation map by

$$\begin{aligned} \text{AuditReportOf} : & \left\{ (\text{AuditData}[\text{Ev}^{\text{fail}}], \lambda) \right. \\ & \left. \begin{array}{l} \mid \text{Ev} \in \text{EvSpec}, \lambda \in \Lambda \end{array} \right\} \\ & \longrightarrow \left\{ \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda] \right. \\ & \left. \begin{array}{l} \mid \text{Ev} \in \text{EvSpec}, \lambda \in \Lambda \end{array} \right\}, \\ \text{AuditReportOf}(\text{AuditData}[\text{Ev}^{\text{fail}}]; \lambda) : &= \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda]. \end{aligned}$$

From the state section of Definition 6.21, the generation-dependency sequence leading to the audit-state report, the audit-diagnostic report, and the five-valued audit-report object is

$$\begin{aligned} \text{FailStateAt}_{\text{Ev}, \lambda} &\longmapsto \text{AuditConditionStateOf}[\text{Ev}; \lambda] \\ &\longmapsto \text{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda], \\ (\text{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda], \text{FailLocData}[\text{Ev}; \lambda]) &\longmapsto \text{AuditDiagnosticReport}[\text{Ev}^{\text{fail}}; \lambda], \\ (\text{AuditData}[\text{Ev}^{\text{fail}}]; \lambda) &\xrightarrow{\text{AuditReportOf}} \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda]. \end{aligned}$$

Table 13: Generation dependencies of audit-report components

Output component	Input family	Generation rule	Basis
Audit-condition state family	Defined four-valued state section and immediate-predecessor condition-satisfaction values	Five-valued lift map AuditStateLift	Definition 6.179
Audit-state report	Audit-condition state family	Serieswise maxima, state fibres, indicator functions, and set of undefined reasons	Definition 6.179, Definition 6.180
Audit-diagnostic report	Failure-localization data and set of audit-undefined reasons	Component tuple of four tagged audit results	Definition 6.172, Definition 6.180
Five-valued audit-report object	Audit datum, audit-state report, and audit-diagnostic report	Product of three components	Definition 6.180

Proposition 6.182 (Determinacy of Five-Valued Audit-Condition States and Six-Group Audit States). For every $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$, and each $t \in \text{FailTok}$, the audit-condition state

$$\text{AuditConditionStateOf}[\text{Ev}; \lambda](t)$$

is determined as exactly one element of AuditStatus .

For each $\tau \in \text{FailSeries}$, the series-group state $\text{AuditGroupState}_\tau[\text{Ev}; \lambda]$ is uniquely determined, and the six-group audit state $\text{AuditGroupStatus}[\text{Ev}; \lambda]$ is also uniquely determined.

The serieswise set of audit-undefined reasons $\text{AuditUndefinedReason}_\tau[\text{Ev}^{\text{fail}}; \lambda]$ and the all-series set of audit-undefined reasons $\text{AuditUndefinedReason}[\text{Ev}^{\text{fail}}; \lambda]$ are also uniquely

determined. They are independent of the choice of linear extension of the failure-generation dependency order.

Proof. By Definition 6.21 and Proposition 6.22, for each $t \in \text{FailTok}$, the evaluation argument $\text{FailArgAt}_{\text{Ev},\lambda}(t)$, the immediate-predecessor condition-satisfaction value $\text{FailPredSat}(\text{FailArgAt}_{\text{Ev},\lambda}(t))$, and the state $\text{FailStateAt}_{\text{Ev},\lambda}(t)$ are uniquely determined and are independent of the choice of linear extension.

The five cases of the five-valued lift map in Definition 6.179 are mutually disjoint and cover its domain. The audit-condition state is therefore uniquely determined. The lift map of the same definition gives

$$\text{AuditConditionStateOf}[\text{Ev}; \lambda](t) = \text{AuditPass} \iff \text{FailStateAt}_{\text{Ev},\lambda}(t) = \text{sat}.$$

By Proposition 6.15, each FailTok_τ is a nonempty finite set. Because the group-aggregation priority order for audit-condition states is a total order on AuditStatus , the image of the audit-condition states of each series has a unique maximum. Thus the series-group states and the six-group audit state are uniquely determined.

Because the audit-condition state family is uniquely determined, the set comprehension in Definition 6.180 uniquely determines the serieswise sets of audit-undefined reasons. Their finite union also uniquely determines the all-series set of audit-undefined reasons. $\square$

Theorem 6.183 (Operational-Audit Completeness Relative to Input Evidence). Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Use the six-failure-series family of evaluation specifications Ev^{fail} of Definition 6.10.

For the operational audit datum,

$$\text{AuditReportOf}(\text{AuditData}[\text{Ev}^{\text{fail}}]; \lambda) = \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda]$$

is uniquely determined, as are all its components.

The four component families

$$\begin{array}{cc} \text{AuditContractID}, & \text{AuditInput}[\text{Ev}], \\ \text{AuditEvidence}[\text{Ev}^{\text{fail}}], & \text{AuditScaleData}[\text{Ev}_{\text{approx}}^{\text{fail}}] \end{array}$$

that constitute the audit computation core respectively identify the audit convention, specify the evaluation target and formation states, provide the six-series condition evaluations, and supply the scale order, defect quantities, and admissible thresholds.

Proof. By Definition 6.178, the components of the audit computation core are the audit contract, audit input, six-series family of condition evidence, and scale-audit datum.

For each $\tau \in \text{FailSeries}$ and $i \in I_\tau^{\text{fail}}$, the family of condition evidence provides the typed condition evidence corresponding to $\text{FailSpecAt}_\tau[\text{Ev}; \lambda]$ from the current scale-augmented evaluation specification. Definition 6.179 determines the audit-condition state from the evaluation state and immediate-predecessor condition-satisfaction value in that definition.

By Proposition 6.182, all audit-condition states, series-group states, and sets of audit-undefined reasons are uniquely determined and are independent of the choice of linear extension of the failure-generation dependency order. The state fibres and indicator functions of Definition 6.179 also uniquely determine the set of indeterminate positions and the indeterminate signature.

Definition 6.172 uniquely determines the set of first failure positions, the six-series failure signature, the approximation-boundary data, and the structural-burden report.

The tagged audit-result map of Definition 6.180 returns the value component when the set of audit-undefined reasons is empty and the reason set when it is nonempty. Consequently, the four tagged diagnostic components, the audit-state report, the audit-diagnostic report, and the audit report are uniquely determined in sequence. Definition 6.181 then gives the equality stated in the theorem. $\square$

Proposition 6.184 (Covariance of Audit-Report Generation under Audit-Data Isomorphisms). For each $a \in \{0, 1\}$, take a complete evaluation specification $\text{Ev}^{(a)} \in \text{EvSpec}$ and an approximation scale $\lambda^{(a)} \in \Lambda^{(a)}$, and let the corresponding six-failure-series family of evaluation specifications be

$$\text{Ev}^{\text{fail},(a)} = \left(\text{Ev}_{\tau}^{\text{fail},(a)} \right)_{\tau \in \text{FailSeries}}.$$

Assume that the typed audit-data isomorphism

$$\begin{aligned} \text{AuditDataIso}^{01} : \text{AuditData} \left[\text{Ev}^{\text{fail},(0)} \right] \\ \xrightarrow{\cong} \text{AuditData} \left[\text{Ev}^{\text{fail},(1)} \right] \end{aligned}$$

preserves the following structures.

1. The condition predicates, input types, scale types, and audit-report generation rules identified by the audit contract.
2. The six failure series, series index sets, failure-position tokens, and failure-generation dependency order.
3. The formation-state-augmented audit input, membership of the serieswise condition evidence, condition-evaluation values, measurement intervals, and confidence sets.
4. The total scale order, approximation scale, codomains of defect quantities, admissible thresholds, and approximation boundaries.
5. The correspondences among the audit-input identifiers, condition-evidence identifiers, content hashes, transformation histories, and evidence locations in the evidence-provenance certificates.
6. The correspondence from failure-position tokens to structural-target tokens.

Let the correspondence of approximation scales be $\lambda^{(0)} \mapsto \lambda^{(1)}$.

Then there exists a unique induced audit-report isomorphism

$$\begin{aligned} \text{AuditReportIso}^{01} : \text{AuditReport} \left[\text{Ev}^{\text{fail},(0)}; \lambda^{(0)} \right] \\ \xrightarrow{\cong} \text{AuditReport} \left[\text{Ev}^{\text{fail},(1)}; \lambda^{(1)} \right] \end{aligned}$$

such that

$$\begin{aligned} & \text{AuditReportIso}^{01} \left(\text{AuditReportOf} \left(\text{AuditData} \left[\text{Ev}^{\text{fail},(0)} \right]; \lambda^{(0)} \right) \right) \\ &= \text{AuditReportOf} \left(\text{AuditData} \left[\text{Ev}^{\text{fail},(1)} \right]; \lambda^{(1)} \right). \end{aligned}$$

This isomorphism assigns to the audit-condition state family, six-group audit state, set of audit-undefined reasons, failure-position audit component, failure-signature audit component, set of indeterminate positions, indeterminate signature, approximation-boundary audit component, and structural-burden audit component the correspondences induced by the assumed correspondences.

Proof. The typed audit-data isomorphism gives isomorphisms of the six failure series, series index sets, failure-position tokens, and failure-generation dependency orders. The roots, immediate predecessors, and linear extensions of the failure-generation dependency orders therefore correspond.

Preservation of the audit contracts, formation-state-augmented audit inputs, and serieswise condition evidence preserves, for each pair of corresponding failure-position tokens, the value recording whether all immediate-predecessor condition states equal **AuditPass**, the formation decision, the condition-evidence satisfaction value, and the condition-evidence failure value. Applying the finite-DAG recursion of Proposition 6.182 to corresponding linear extensions preserves every value of the five-valued audit-condition state of Definition 6.179 between corresponding failure-position tokens.

Because the state-priority order and the failure-position-token set of each series are preserved, **AuditGroupStatus**, which consists of the serieswise maxima, is also preserved. Because the correspondences between the audit-undefined state values and the failure-position tokens are preserved, they induce the serieswise sets of audit-undefined reasons and the all-series set of audit-undefined reasons.

The correspondence of audit-condition states and failure-position tokens induces the set of failure positions and the set of indeterminate positions. The correspondences of the six series and series index sets induce the failure signature and indeterminate signature.

Preservation of the total scale order, approximation scale, codomains of defect quantities, admissible thresholds, and approximation boundaries induces the approximation-boundary data and approximation-boundary audit component. The isomorphism of failure-generation dependency orders preserves the principal upper set generated by each failure-position token. Together with preservation of the correspondence from failure-position tokens to structural-target tokens, this induces the failure-position-side structural burden, approximation-scale-boundary-side structural burden, and structural-burden audit component.

Combining componentwise the correspondences among the components of the evidence-provenance certificates, the audit-data isomorphism, and the induced correspondences above gives the induced audit-report map stated in the proposition. Each component correspondence is bijective, and its inverse preserves the same structures; hence this map is an isomorphism.

Definition 6.181 determines the audit report from these components. The induced correspondences of the components make this isomorphism satisfy the commutative equation in the proposition.

Each component of the audit report is uniquely determined by the correspondence supplied by the audit-data isomorphism. The induced audit-report isomorphism having the component correspondences above is therefore also unique. $\square$

Definition 6.185 (Finite Operational Audit Fragment). Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Use the six-failure-series family of evaluation specifications Ev^{fail} of Definition 6.10.

The finite operational audit fragment $\text{AuditFragment}[\text{Ev}^{\text{fail}}; \lambda]$ consists of the following finite components of the operational audit datum $\text{AuditData}[\text{Ev}^{\text{fail}}]$.

1. The condition predicates, input types, scale types, and audit-report generation rules specified by the audit contract.
2. The finitely many values used to evaluate all failure conditions for the operational construction, complete evaluation specification, and formation-state-augmented inputs of the three series.
3. For each $\tau \in \mathbf{FailSeries}$ and $i \in I_\tau^{\text{fail}}$, the typed condition evidence, condition-evidence satisfaction value, and condition-evidence failure value.
4. The finite relation table of the failure-generation dependency order.
5. The approximation scale λ , together with the scale-order comparisons, defect-quantity values, codomains of the defect quantities, admissible thresholds, and approximation-boundary data used in its evaluation.
6. For each $t \in \mathbf{FailTok}$, the principal upper set and the single-token structural burden

$$\begin{aligned} & \text{Desc}_{\text{fail}, \text{dep}}(t), \\ & \text{StructBurden}(\{t\}). \end{aligned}$$

7. The finite part of the evidence-provenance certificate corresponding to each component above.

For each $t \in \mathbf{FailTok}$, retain the finitely many typed values required for condition evaluation as the three components

$$\left(\text{FailInputSat}^{\text{form}}, \text{FailInputSat}^{\text{eval}}, \text{FailEval} \right) \left(\text{FailArgAt}_{\text{Ev}, \lambda}(t) \right).$$

Proposition 6.186 (Finite Audit Procedure with Five-Valued Halting Output).

Take a scale-augmented complete evaluation specification $(\text{Ev}, \lambda) \in \mathbf{EvSpec}_\Lambda$. Use the six-failure-series family of evaluation specifications Ev^{fail} of Definition 6.10. Given the finite operational audit fragment $\mathbf{AuditFragment}[\text{Ev}^{\text{fail}}; \lambda]$, there exists an audit procedure that, after finitely many decisions, halts, for every $t \in \mathbf{FailTok}$, with exactly one element of $\mathbf{AuditStatus}$.

From these halting outputs and the approximation-boundary data and structural images retained in the finite operational audit fragment, finitely many set operations, maximum operations, and tagged-component generations yield every component of $\mathbf{AuditStateReport}[\text{Ev}^{\text{fail}}; \lambda]$ and $\mathbf{AuditDiagnosticReport}[\text{Ev}^{\text{fail}}; \lambda]$. If the operational audit datum $\mathbf{AuditData}[\text{Ev}^{\text{fail}}]$ is also supplied, its tuple with these two reports gives $\mathbf{AuditReport}[\text{Ev}^{\text{fail}}; \lambda]$.

Proof. By Definition 6.10 and Proposition 6.15, $\mathbf{FailTok}$ is a finite set with $\mathbf{FailCount}^{\text{max}}$ elements. By Proposition 6.18, the failure-generation dependency order is a partial order on this finite set. The failure-generation dependency DAG therefore has a finite linear extension.

Process each

$$t \in \mathbf{FailTok}$$

along that linear extension. From the finite operational audit fragment, obtain the immediate-predecessor condition-satisfaction value, formation value, evaluation-input satisfaction value, and condition-evaluation value for

$$\text{FailArgAt}_{\text{Ev}, \lambda}(t).$$

Definition 6.21 determines the four-valued evaluation state, and the five-valued lift map of Definition 6.179 determines the audit-condition state. Each processing step consists of finitely many membership, equality, or truth-value decisions and halts in one of the five mutually disjoint cases.

Because the failure-position-token set is finite, generation of all audit-condition states also halts after finitely many decisions. By Proposition 6.182, the resulting state family is unique and is independent of the choice of linear extension.

Each series is a nonempty finite set, and the audit-state priority order is a total order; hence the six-group audit state is obtained by finitely many maximum operations. The set of audit-undefined reasons, set of indeterminate positions, and indeterminate signature are obtained by set comprehensions, a finite union, and indicator functions on the finite state family.

The audit-state report consists of the components obtained above. The condition-state family determines the failure-position set and failure signature. The approximation-boundary data and single-token structural burdens retained in the finite operational audit fragment determine the structural-burden report by finite unions. Applying the tagged audit-result maps of Definition 6.180 to these values and the set of audit-undefined reasons yields every component of the audit-diagnostic report. If the operational audit datum is also supplied, its tuple with these two reports is the audit report of the same definition. $\square$

Corollary 6.187 (Five-Valued Operational Auditability of the Time-Reference Formation Conditions).

Given the operational audit datum $\text{AuditData}[\text{Ev}^{\text{fail}}]$ and an approximation scale $\lambda \in \Lambda$, the five-valued audit states of all failure conditions, six-group audit state, set of audit-undefined reasons, failure-position audit component, failure-signature audit component, set of indeterminate positions, indeterminate signature, approximation-boundary audit component, and structural-burden audit component are uniquely determined.

When the corresponding finite operational audit fragment is given, these components are obtained from the halting output of a finite audit procedure together with that fragment. The audit-state and audit-diagnostic reports obtained from two finite operational audit fragments corresponding under a typed audit-data isomorphism correspond under the respective component maps of $\text{AuditReportIso}^{01}$.

Proof. By Theorem 6.183, all components of the audit report are uniquely determined by the operational audit datum and approximation scale.

By Proposition 6.186, the condition evaluation for every failure-position token contained in the finite operational audit fragment halts after finitely many decisions with one of the five-valued audit states. By the same proposition, every component of the audit-state and audit-diagnostic reports is obtained from all condition states and the data retained in the finite operational audit fragment by finitely many operations.

By Proposition 6.184, a typed audit-data isomorphism induces a unique audit-report isomorphism. The audit-report isomorphism therefore preserves the component correspondences that depend on the representation of the finite operational audit fragment. $\square$

Five-valued audit report generation

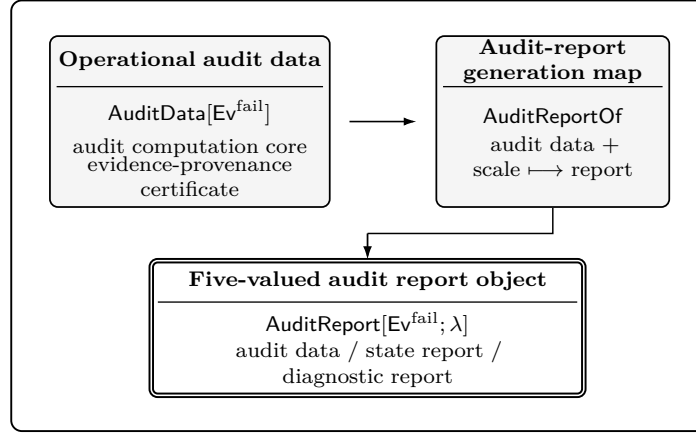


Figure 15: Map generating a five-valued audit-report object from operational audit data. The audit data, comprising the audit computation core and evidence-provenance certificate, are supplied together with the scale to the audit-report generation map **AuditReportOf**, which sends them to **AuditReport**, whose components are the audit data, audit-state report, and audit-diagnostic report.

Finite audit procedure

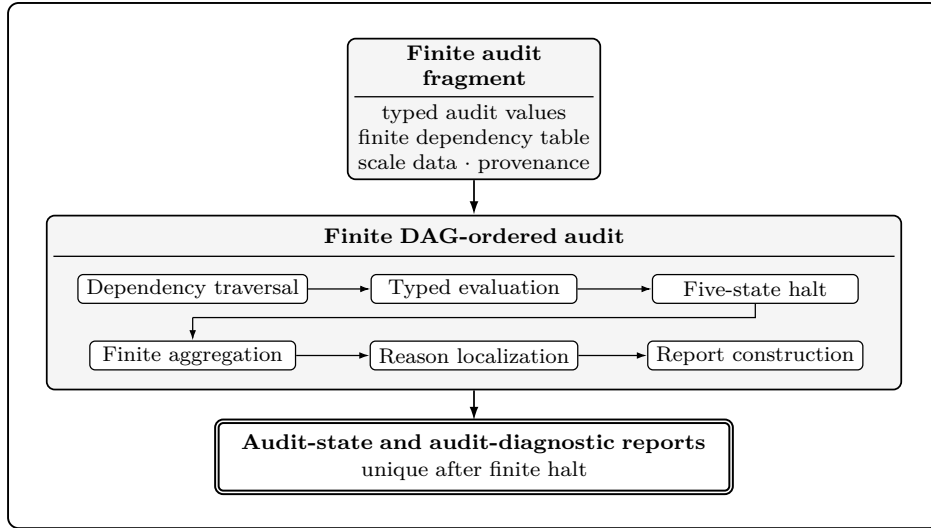


Figure 16: Finite procedure generating audit-state and audit-diagnostic reports from a finite audit fragment. It receives typed audit values, a finite dependency table, scale and boundary data, and finite evidence-provenance records; it executes dependency-order traversal, typed condition evaluation, five-valued halting, finite aggregation, reason localization, and report generation in finite-DAG order. If the operational audit datum is also supplied, its tuple with the two resulting reports gives **AuditReport**.

6.13 Engineering Audit Parameters and Their Correspondence to Failures of Physical-Carrier Specifications

Definition 3.2 of role-typed physical-carrier realization, Definition 6.10 of the failure-position-token system with semantic series, and Definition 6.177 of the operational audit

data for the time-reference construction are taken as inputs. Engineering measurement data constitute the formation inputs for the serieswise audit inputs and typed condition evidence. Each failure condition is evaluated through its corresponding audit-input type and condition-evidence type.

Definition 6.188 (Engineering Evaluation-Specification Type). Define the typed tuple of the complete evaluation specification, approximation scale, observer-side information-coupling position, and target-specification index used in the engineering audit by

$$\text{EngEvSpec} := \{(\text{Ev}, \lambda, o, \zeta) \in \text{EvSpec} \times \Lambda \times U \times Z_0 \mid \text{Ev} = (l, o, \zeta)\}.$$

Definition 6.189 (Closed-Interval Audit Kernel). For a subset S of a linearly ordered set, define its closed-interval type by

$$\mathcal{I}^{\text{cl}}(S) := \{[\ell, r] \mid \ell, r \in S, \ell \leq r\}.$$

Define the audit kernel for closed intervals over the nonnegative extended reals and admissible values by

$$\begin{aligned} \text{IntervalAudit} : \mathcal{I}^{\text{cl}}(\overline{\mathbb{R}}_{\geq 0}) \times \overline{\mathbb{R}}_{\geq 0} &\longrightarrow \text{AuditStatus}, \\ \text{IntervalAudit}([\ell, r]; \theta) &:= \begin{cases} \text{AuditPass}, & r \leq \theta, \\ \text{AuditFail}, & \theta < \ell, \\ \text{AuditIndeterminate}, & \ell \leq \theta < r \end{cases}. \end{aligned}$$

Definition 6.190 ($\mathfrak{D}$ -Derived Recording-Field-of-View Engineering Datum). Let the engineering evaluation specification be $(\text{Ev}, \lambda, o, \zeta) \in \text{EngEvSpec}$. Let the $\mathfrak{D}$ -derived recording-field-of-view set be

$$\text{View}_{\mathfrak{D}, \zeta, o, \lambda}^{\text{rec}}[\text{Ev}] \subseteq \text{Qu}_{\mathfrak{B}, o}^{\text{dyn, can}}.$$

Define the $\mathfrak{D}$ -derived recording-field-of-view engineering datum by

$$\begin{aligned} &\text{ViewData}_{\mathfrak{D}, \zeta, o, \lambda}^{\text{rec}}[\text{Ev}] \\ &:= (\text{CarrierAt}_{\mathfrak{D}}(\zeta), o, \lambda, \text{View}_{\mathfrak{D}, \zeta, o, \lambda}^{\text{rec}}[\text{Ev}]). \end{aligned}$$

At target-specification position ζ , observer-side information-coupling position o , and approximation scale λ , the fourth component is the set recorded by the $\mathfrak{D}$ -derived physical recording carrier whose elements are $\mathfrak{B}$ -derived target-state quotient classes.

Definition 6.191 (Bulk-Recording and Following-Recording Types). Define the set of two-target complete evaluation specifications by

$$\begin{aligned} \text{EvSpec}^{\text{pair}} := \Big\{ (\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}^2 \mid &\exists \mathcal{X}, \exists o \in U, \exists \zeta_1, \zeta_2 \in Z_0, \\ &\zeta_1 \neq \zeta_2, \text{Operative}(\mathcal{X}), \\ &\bigwedge_{k=1}^2 [\text{Ev}_k = (l, o, \zeta_k)] \Big\}. \end{aligned}$$

Take

$$(\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}^{\text{pair}}, \quad \lambda \in \Lambda$$

and write $\text{Ev}_k = (l, o, \zeta_k)$. Define the bulk-recording condition by

$$\begin{aligned} \text{BulkRecord}_{\mathfrak{D}} &: \text{EvSpec}^{\text{pair}} \longrightarrow \{\perp, \top\}, \\ \text{BulkRecord}_{\mathfrak{D}} [(\text{Ev}_k)_{k=1}^2] &\iff \text{CarrierAt}_{\mathfrak{D}}(\zeta_1) = \text{CarrierAt}_{\mathfrak{D}}(\zeta_2). \end{aligned}$$

For each $k \in \{1, 2\}$, define, restricted to the target-specific $\mathfrak{D}$ -derived recording field of view, the $\mathfrak{B}$ -typed $\mathfrak{D}$ -record-code relation by

$$\begin{aligned} \text{RecordCodeRel}_{\mathfrak{B}, o, k}^{\text{dyn, view}} [(\text{Ev}_j)_{j=1}^2; \lambda] \\ := \text{RecordCodeRel}_{\mathfrak{B}, o}^{\text{dyn}} \cap \left(\text{View}_{\mathfrak{D}, \zeta_k, o, \lambda}^{\text{rec}} [\text{Ev}_k] \times \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}} \right). \end{aligned}$$

This relation therefore satisfies

$$\begin{aligned} \text{RecordCodeRel}_{\mathfrak{B}, o, k}^{\text{dyn, view}} [(\text{Ev}_j)_{j=1}^2; \lambda] \\ \subseteq \text{View}_{\mathfrak{D}, \zeta_k, o, \lambda}^{\text{rec}} [\text{Ev}_k] \times \text{RecordCodeVal}_{\mathfrak{D}}^{\text{dyn}}. \end{aligned}$$

Define the following-recording condition by

$$\begin{aligned} \text{FollowingRecord}_{\mathfrak{D}} &: \text{EvSpec}^{\text{pair}} \times \Lambda \longrightarrow \{\perp, \top\}, \\ \text{FollowingRecord}_{\mathfrak{D}} [(\text{Ev}_k)_{k=1}^2; \lambda] \\ &\iff \text{CarrierAt}_{\mathfrak{D}}(\zeta_1) \neq \text{CarrierAt}_{\mathfrak{D}}(\zeta_2) \\ &\quad \wedge \bigwedge_{k=1}^2 \left[\text{View}_{\mathfrak{D}, \zeta_k, o, \lambda}^{\text{rec}} [\text{Ev}_k] \right. \\ &\quad \left. \subseteq \text{Dom} \left(\text{RecordCodeRel}_{\mathfrak{B}, o, k}^{\text{dyn, view}} [(\text{Ev}_j)_{j=1}^2; \lambda] \right) \right]. \end{aligned}$$

Definition 6.192 (Engineering Datum for Recording Invasiveness). Let the engineering evaluation specification be $(\text{Ev}, \lambda, o, \zeta) \in \text{EngEvSpec}$. Let the correspondence of $\mathfrak{B}$ -states before and after the invasive action be

$$\text{InvasivenessRel}_{\mathfrak{B}, \zeta, o, \lambda} [\text{Ev}] \subseteq \left(\text{Qu}_{\mathfrak{B}, o}^{\text{dyn, can}} \right)^2.$$

Let the lower endpoint, upper endpoint, and admissible value of the recording-invasiveness defect quantity be

$$\begin{aligned} \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^- [\text{Ev}; \lambda], \quad \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^+ [\text{Ev}; \lambda], \quad \text{InvasivenessTol}_{\mathfrak{D}, \zeta, o} \in \overline{\mathbb{R}}_{\geq 0}, \\ \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^- [\text{Ev}; \lambda] \leq \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^+ [\text{Ev}; \lambda]. \end{aligned}$$

Define the recording-invasiveness engineering datum by

$$\begin{aligned} \text{InvasivenessData}_{\mathfrak{D}, \zeta, o, \lambda} [\text{Ev}] \\ := \left(\begin{array}{c} \text{InvasivenessRel}_{\mathfrak{B}, \zeta, o, \lambda} [\text{Ev}], \\ [\text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^- [\text{Ev}; \lambda], \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^+ [\text{Ev}; \lambda]] \\ \text{InvasivenessTol}_{\mathfrak{D}, \zeta, o} \end{array} \right). \end{aligned}$$

Using the closed-interval audit kernel of Definition 6.189, define the quantitative invasiveness-admissibility state by

$$\begin{aligned} \text{InvasivenessAuditState}_{\mathfrak{D}, \zeta, o} [\text{Ev}; \lambda] \\ := \text{IntervalAudit} \left(\begin{array}{c} [\text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^- [\text{Ev}; \lambda], \text{InvasivenessDef}_{\mathfrak{D}, \zeta, o}^+ [\text{Ev}; \lambda]] \\ \text{InvasivenessTol}_{\mathfrak{D}, \zeta, o} \end{array} \right) \\ \in \text{AuditStatus}. \end{aligned}$$

Definition 6.193 (Observer-Side Communication Engineering Datum). Define the set of two-target complete evaluation specifications corresponding to the $\mathfrak{D}$ -distributed type of Definition 3.4 by

$$\begin{aligned} \text{EvSpec}_{\mathfrak{D}}^{\text{pair,distributed}} \\ := \left\{ (\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}^{\text{pair}} \mid \exists \mathcal{X}, \exists o \in U, \exists \zeta_1, \zeta_2 \in Z_0, \right. \\ \left. \bigwedge_{k=1}^2 [\text{Ev}_k = (l, o, \zeta_k)], \quad \bigwedge_{k=1}^2 [\text{CarrierAt}_{\mathfrak{D}}(\zeta_k) \in \text{Car}_{\mathfrak{D}}], \right. \\ \left. \text{CarrierAt}_{\mathfrak{D}}(\zeta_1) \neq \text{CarrierAt}_{\mathfrak{D}}(\zeta_2) \right\}. \end{aligned}$$

Take

$$(\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}_{\mathfrak{D}}^{\text{pair,distributed}}, \quad \lambda \in \Lambda$$

and write $\text{Ev}_k = (l, o, \zeta_k)$. Let the finite directed multigraph whose vertices are observer-side physical carriers be

$$\begin{aligned} \text{CommGraph}_{\mathfrak{D}} &:= \left(\begin{array}{c} \text{CommVert}_{\mathfrak{D}}, \text{CommEdge}_{\mathfrak{D}}, \\ \text{CommSrc}_{\mathfrak{D}}, \text{CommTgt}_{\mathfrak{D}} \end{array} \right), \\ \text{CommVert}_{\mathfrak{D}} &\subseteq \text{Car}_{\mathfrak{D}}, \\ |\text{CommVert}_{\mathfrak{D}}| &< \infty, \quad |\text{CommEdge}_{\mathfrak{D}}| < \infty, \\ \text{CommSrc}_{\mathfrak{D}}, \text{CommTgt}_{\mathfrak{D}} &: \text{CommEdge}_{\mathfrak{D}} \longrightarrow \text{CommVert}_{\mathfrak{D}}, \\ \bigwedge_{k=1}^2 &[\text{CarrierAt}_{\mathfrak{D}}(\zeta_k) \in \text{CommVert}_{\mathfrak{D}}]. \end{aligned}$$

Let the map assigning a physical communication carrier to each communication edge be

$$\text{CommCarrier}_{\mathfrak{D}} : \text{CommEdge}_{\mathfrak{D}} \longrightarrow \text{Car}_{\text{comm}}.$$

Define the observer-side set of communication walks and set of closed communication walks by

$$\begin{aligned} \text{CommWalk}_{\mathfrak{D}} \\ := \bigsqcup_{n \in \mathbb{N}_{\geq 1}} \left\{ (e_1, \dots, e_n) \in (\text{CommEdge}_{\mathfrak{D}})^n \mid \bigwedge_{r=1}^{n-1} [\text{CommTgt}_{\mathfrak{D}}(e_r) = \text{CommSrc}_{\mathfrak{D}}(e_{r+1})] \right\}, \\ \text{CommClosedWalk}_{\mathfrak{D}} \\ := \left\{ (e_1, \dots, e_n) \in \text{CommWalk}_{\mathfrak{D}} \mid \text{CommSrc}_{\mathfrak{D}}(e_1) = \text{CommTgt}_{\mathfrak{D}}(e_n) \right\}. \end{aligned}$$

Define the relation that holds when a positive-length directed communication walk exists in at least one direction by

$$\begin{aligned} \text{CommLinked}_{\mathfrak{D}} &\subseteq (\text{CommVert}_{\mathfrak{D}})^2, \\ (v, w) &\in \text{CommLinked}_{\mathfrak{D}} \\ \iff \exists n \in \mathbb{N}_{\geq 1}, \exists (e_1, \dots, e_n) &\in \text{CommWalk}_{\mathfrak{D}}, \\ (\text{CommSrc}_{\mathfrak{D}}(e_1), \text{CommTgt}_{\mathfrak{D}}(e_n)) &\in \{(v, w), (w, v)\} \end{aligned}$$

and require

$$(\text{CarrierAt}_{\mathfrak{D}}(\zeta_1), \text{CarrierAt}_{\mathfrak{D}}(\zeta_2)) \in \text{CommLinked}_{\mathfrak{D}}.$$

Let the measurement maps for communication distance, delay, bandwidth, and jitter be

$$\begin{aligned} & \text{CommDistance}, \text{CommDelay}, \text{CommBandwidth}, \text{CommJitter} \\ & : \text{CommEdge}_{\mathfrak{D}} \longrightarrow \mathcal{I}^{\text{cl}}(\overline{\mathbb{R}}_{\geq 0}), \end{aligned}$$

and let the measurement maps for loss rate and path integrity be

$$\text{CommLossRate}, \text{CommPathIntegrity} : \text{CommEdge}_{\mathfrak{D}} \longrightarrow \mathcal{I}^{\text{cl}}([0, 1]).$$

Let the communication-edge-type map be

$$\text{CommEdgeType} : \text{CommEdge}_{\mathfrak{D}} \longrightarrow \{\text{SourceIdEdge}, \text{TransportEdge}\}.$$

For each $\bullet \in \{\mathfrak{C}, \mathfrak{D}\}$, let the partial map assigning an extended comparison word to a communication walk be

$$\begin{aligned} & \text{CommWord}^{\bullet}[(\text{Ev}_k)_{k=1}^2] : \text{CommWalk}_{\mathfrak{D}} \\ & \longrightarrow \text{ExtWord}^{\bullet}(\mathcal{X}). \end{aligned}$$

Define the topological component, metric component, and observer-side communication engineering datum by

$$\begin{aligned} & \text{CommTopology}_{\mathfrak{D}} \\ & := \left(\begin{array}{c} \text{CommGraph}_{\mathfrak{D}}, \text{CommCarrier}_{\mathfrak{D}}, \\ \text{CommEdgeType}, \text{CommWalk}_{\mathfrak{D}}, \\ \text{CommClosedWalk}_{\mathfrak{D}} \end{array} \right), \\ & \text{CommMetric}_{\mathfrak{D}} \\ & := \left(\begin{array}{c} \text{CommDistance}, \text{CommDelay}, \text{CommLossRate}, \\ \text{CommBandwidth}, \text{CommJitter}, \text{CommPathIntegrity} \end{array} \right), \\ & \text{CommData}_{\mathfrak{D}}[(\text{Ev}_k)_{k=1}^2] \\ & := \left(\begin{array}{c} \text{CommTopology}_{\mathfrak{D}}, \\ ((\text{CommWord}^{\bullet}[(\text{Ev}_k)_{k=1}^2])_{\bullet \in \{\mathfrak{C}, \mathfrak{D}\}}), \\ \text{CommMetric}_{\mathfrak{D}} \end{array} \right). \end{aligned}$$

Definition 6.194 (Engineering Audit-Input Datum). Define the set of engineering-measurement type tags by

$$\begin{aligned} \text{EngParamTag} := & \{\text{ViewTag}_{\mathfrak{D}}, \text{RecordConfigTag}_{\mathfrak{D}}, \text{InvasivenessTag}_{\mathfrak{D}}, \\ & \text{CommunicationTag}_{\mathfrak{D}}, \text{CountStepTag}_{\mathfrak{C}}, \text{StabilityTag}_{\mathfrak{D}}, \\ & \text{RecordCellWidthTag}_{\mathfrak{D}}, \text{DiscernibilityTag}_{\mathfrak{B}}, \text{CouplingTag}_{\mathfrak{D}}, \\ & \text{TrajectoryTag}, \text{ObservableTag}, \text{LawTestTag}, \text{DifferentialTestTag}\}. \end{aligned}$$

Let the set of engineering-measurement references be EngParamRef , and let the type-tag map, tag-indexed engineering-measurement value types, and engineering-measurement

map be

$$\begin{aligned} \text{EngTagOf} &: \text{EngParamRef} \longrightarrow \text{EngParamTag}, \\ \text{EngValueType}_\rho & \quad (\rho \in \text{EngParamTag}), \\ \text{EngMeasure} & \in \prod_{r \in \text{EngParamRef}} \text{EngValueType}_{\text{EngTagOf}(r)}. \end{aligned}$$

Define the four basic engineering-measurement value types by

$$\begin{aligned} \text{ViewDataType}_\mathfrak{D} & \\ &:= \left\{ \text{ViewData}_{\mathfrak{D}, \zeta, o, \lambda}^{\text{rec}}[\text{Ev}] \mid (\text{Ev}, \lambda, o, \zeta) \in \text{EngEvSpec} \right\}, \\ \text{RecordConfigDataType}_\mathfrak{D} & \\ &:= \left\{ \left(\begin{array}{c} \text{BulkRecord}_\mathfrak{D}[(\text{Ev}_k)_{k=1}^2], \\ \text{FollowingRecord}_\mathfrak{D}[(\text{Ev}_k)_{k=1}^2; \lambda] \end{array} \right) \mid \begin{array}{c} (\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}^{\text{pair}}, \\ \lambda \in \Lambda \end{array} \right\}, \\ \text{InvasivenessDataType}_\mathfrak{D} & \\ &:= \left\{ \text{InvasivenessData}_{\mathfrak{D}, \zeta, o, \lambda}[\text{Ev}] \mid (\text{Ev}, \lambda, o, \zeta) \in \text{EngEvSpec} \right\}, \\ \text{CommunicationDataType}_\mathfrak{D} & \\ &:= \left\{ \text{CommData}_\mathfrak{D}[(\text{Ev}_k)_{k=1}^2] \mid (\text{Ev}_1, \text{Ev}_2) \in \text{EvSpec}_\mathfrak{D}^{\text{pair, distributed}} \right\}. \end{aligned}$$

Define the four components of the corresponding family of value types by

$$\begin{aligned} \text{EngValueType}_{\text{ViewTag}_\mathfrak{D}} & \\ &:= \text{ViewDataType}_\mathfrak{D}, \\ \text{EngValueType}_{\text{RecordConfigTag}_\mathfrak{D}} & \\ &:= \text{RecordConfigDataType}_\mathfrak{D}, \\ \text{EngValueType}_{\text{InvasivenessTag}_\mathfrak{D}} & \\ &:= \text{InvasivenessDataType}_\mathfrak{D}, \\ \text{EngValueType}_{\text{CommunicationTag}_\mathfrak{D}} & \\ &:= \text{CommunicationDataType}_\mathfrak{D}. \end{aligned}$$

For a complete evaluation specification $\text{Ev} \in \text{EvSpec}$, let its six-failure-series family of evaluation specifications be Ev^{fail} . Define the engineering audit-input datum by

$$\begin{aligned} \text{EngAuditInput}[\text{Ev}^{\text{fail}}] &:= \left(\text{EngParamTag}, \text{EngParamRef}, \right. \\ &\quad \text{EngTagOf}, \\ &\quad (\text{EngValueType}_\rho)_{\rho \in \text{EngParamTag}}, \\ &\quad \left. \text{EngMeasure}, \text{AuditCert}[\text{Ev}^{\text{fail}}] \right). \end{aligned}$$

Definition 6.195 (Engineering-Implementation Auxiliary Audit Data). Define the engineering-implementation auxiliary-audit flag type by

$$\begin{aligned} &\text{EngAuxFlagType} \\ &:= \left\{ \text{MissingFieldFlag}, \text{ProvenanceInconsistencyFlag}, \right. \\ &\quad \text{RecordCellInconsistencyFlag}, \\ &\quad \text{ExternalTimeContaminationFlag}, \text{NonmonotoneDefectFlag}, \\ &\quad \text{UncertaintyOverlapFlag}, \\ &\quad \text{ContractVersionMismatchFlag}, \text{NumericalSolverFailureFlag}, \\ &\quad \left. \text{DataGapFlag} \right\}. \end{aligned}$$

For a complete evaluation specification $\text{Ev} \in \text{EvSpec}$, let the set of engineering-implementation auxiliary-audit flags that hold be $\text{EngAuxFlagSet}[\text{Ev}] \subseteq \text{EngAuxFlagType}$.

For each $\text{EngAuxFlag}^{(0)} \in \text{EngAuxFlagSet}[\text{Ev}]$, let the map assigning the nonempty set of engineering-measurement references used in its determination be

$$\begin{aligned} &\text{EngAuxFlagEvidenceRef} : \text{EngAuxFlagSet}[\text{Ev}] \\ &\longrightarrow \mathcal{P}(\text{EngParamRef}) \setminus \{\emptyset\}. \end{aligned}$$

Define the engineering-implementation auxiliary audit data by

$$\begin{aligned} &\text{EngAuxAuditData}[\text{Ev}] \\ &:= \left(\text{EngAuxFlagSet}[\text{Ev}], \text{EngAuxFlagEvidenceRef} \right). \end{aligned}$$

Multiple auxiliary-audit flags may hold simultaneously for the same complete evaluation specification. Each flag is accompanied by the nonempty set of engineering-measurement references that grounds its determination.

Definition 6.196 (Engineering Auxiliary-Audit Effect Types and Effect Relation). Define the auxiliary-audit effect type by

$$\begin{aligned} &\text{EngAuxEffectType} \\ &:= \left\{ \text{FormInputUnsatisfiedEffect}, \text{EvalInputUnsatisfiedEffect}, \right. \\ &\quad \text{InputAdmissibilityViolationEffect}, \text{BoundaryMethodRequirementEffect}, \\ &\quad \left. \text{ExactConditionEvidenceRequirementEffect} \right\}. \end{aligned}$$

Let the relation between auxiliary-audit flags and auxiliary-audit effects be

$$\begin{aligned} &\text{EngAuxEffectRel} \\ &\subseteq \text{EngAuxFlagType} \times \text{EngAuxEffectType}. \end{aligned}$$

Definition 6.197 (Engineering Auxiliary-Audit Effect Evidence). Let the scale-augmented complete evaluation specification be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Define the typed pairs of auxiliary-audit flags and effects by

$$\begin{aligned} &\text{EngAuxEffectPair}[\text{Ev}] \\ &:= \left\{ \left(\begin{array}{c} \text{EngAuxFlag}^{(0)}, \\ \text{EngAuxEffect}^{(0)} \end{array} \right) \in \text{EngAuxEffectRel} \right. \\ &\quad \left. \left| \text{EngAuxFlag}^{(0)} \in \text{EngAuxFlagSet}[\text{Ev}] \right. \right\}. \end{aligned}$$

Let the candidate type and validation predicate for typed auxiliary-audit effect evidence be

$$\begin{aligned} \text{EngAuxEffectEvidenceType}[\text{Ev}; \lambda] &:= \text{EngAuxEffectPair}[\text{Ev}] \times \text{FailTok}, \\ \text{EngAuxEffectEvidenceValid}_{\text{Ev}, \lambda} : \text{EngAuxEffectEvidenceType}[\text{Ev}; \lambda] &\longrightarrow \{\perp, \top\}. \end{aligned}$$

This predicate takes the value $\top$ if and only if the evidence references, the target failure-position token, and the corresponding input or condition evidence have compatible types and the referenced evidence substantiates the applicable effect conditions below.

1. For an unsatisfied formation input or evaluation input, the corresponding input-satisfaction value at the target token is in fact $\perp$.
2. A requirement for exact condition evidence identifies the typed evidence needed to evaluate the target condition and the condition that makes this evidence necessary.
3. An input-admissibility violation identifies where an external-time value is used as a premise of a generation condition.
4. A requirement for a boundary-handling method has a target token in the continuum-approximation series. For each engineering-measurement reference cited, that boundary treatment is necessary to evaluate the target token. If either form of boundary-handling data in Definition 6.198 below is supplied, it is included in the evidence component for the target condition within $\text{AuditEvidenceAt}[\text{Ev}; \lambda](t)$. If the method is unformed, the absence of the required evaluation input gives the target token an evaluation-input satisfaction value of $\perp$.

Here t is the candidate's target token. Verifying a requirement for condition evidence identifies what evidence is required; it does not determine whether the condition holds. Define the set of verified auxiliary-audit effect evidence by

$$\begin{aligned} &\text{EngAuxEffectEvidence}[\text{Ev}; \lambda] \\ &:= \left(\text{EngAuxEffectEvidenceValid}_{\text{Ev}, \lambda} \Big|_{\text{EngAuxEffectEvidenceType}[\text{Ev}; \lambda]} \right)^{-1} [\{\top\}]. \end{aligned}$$

Table 14: Engineering-implementation auxiliary-audit flags and typed audit effects

Auxiliary-audit flag	Auxiliary-audit effect	Required typed evidence
Missing required field Data gap	Unsatisfied formation input or unsatisfied evaluation input	Correspondence between the target failure-position token and missing input type Correspondence between the target failure-position token and missing data type
Provenance inconsistency Uncertainty-interval overlap Audit-contract version mismatch	Unsatisfied evaluation input	Inconsistent component of the evidence-provenance certificate Intersection of the defect-quantity interval and admissible value Mismatch between the audit-contract identifier and evidence-provenance certificate
Record-cell inconsistency	Unsatisfied formation input or requirement for exact condition evidence	Unformed record-code relation or typed record-code conflict evidence
External-time contamination	Input-admissibility violation	Expression in which an external-time value occurs in a premise of a generation condition
Nonmonotone defect	Boundary-handling- method requirement	Counterexample to monotonicity on the scale order
Numerical-computation failure	Unsatisfied evaluation input or requirement for exact condition evidence	Missing evaluation value or independently constructed exact condition evidence

Definition 6.198 (Boundary-Handling Data for a Nonmonotone Engineering Defect). Let the engineering-measurement reference to which the nonmonotone-defect flag refers be $r \in \text{EngParamRef}$. Let the scale domain corresponding to this reference be $\Lambda_r \subseteq \Lambda$, and let the engineering-defect map be

$$\text{EngDefect}_r : \Lambda_r \longrightarrow [0, +\infty].$$

For each $\lambda \in \Lambda_r$, let

$$\begin{aligned} & \text{EngMonotoneEnvelope}_r(\lambda) \\ & := \sup \left\{ \text{EngDefect}_r(\mu) \mid \begin{array}{l} \mu \in \Lambda_r, \\ \mu \leq_\Lambda \lambda \end{array} \right\}. \end{aligned}$$

When the right-hand side is finite for every $\lambda \in \Lambda_r$, call this map finite monotone-envelope data.

Let a finite index set K_r and a family of subsets

$$(\Lambda_{r,k})_{k \in K_r}$$

satisfy

$$\Lambda_r = \bigsqcup_{k \in K_r} \Lambda_{r,k}.$$

When each subset is order-convex with respect to $\leq_\Lambda$ and every restriction of the engineering-defect map is monotone, call this family of subsets finite operating-region partition data.

Define the boundary-handling-method type by

$$\begin{aligned} & \text{EngBoundaryMethodType} \\ & := \left\{ \begin{array}{l} \text{MonotoneEnvelopeMethod}, \\ \text{OperatingRegionPartitionMethod}, \\ \text{UnformedBoundaryMethod} \end{array} \right\}. \end{aligned}$$

Let the boundary-handling method satisfy

$$\begin{aligned} & \text{EngBoundaryMethod}(r) \\ & \in \text{EngBoundaryMethodType}. \end{aligned}$$

When finite monotone-envelope data are given, permit

$$\begin{aligned} & \text{EngBoundaryMethod}(r) \\ & = \text{MonotoneEnvelopeMethod}. \end{aligned}$$

When finite operating-region partition data are given, permit

$$\begin{aligned} & \text{EngBoundaryMethod}(r) \\ & = \text{OperatingRegionPartitionMethod}. \end{aligned}$$

When both kinds of boundary-handling data are unformed, set

$$\begin{aligned} & \text{EngBoundaryMethod}(r) \\ & = \text{UnformedBoundaryMethod}. \end{aligned}$$

Proposition 6.199 (Closure of Engineering-Implementation Auxiliary Audit within the Six Series). Let the scale-augmented complete evaluation specification be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. For any

$$(\text{EngAuxFlag}^{(0)}, \text{EngAuxEffect}^{(0)}, t) \in \text{EngAuxEffectEvidence}[\text{Ev}; \lambda]$$

the following hold:

$$\begin{aligned} & \text{EngAuxEffect}^{(0)} = \text{FormInputUnsatisfiedEffect} \\ & \implies \text{FailInputSat}^{\text{form}}(\text{FailArgAt}_{\text{Ev}, \lambda}(t)) = \perp, \\ & \text{EngAuxEffect}^{(0)} = \text{EvalInputUnsatisfiedEffect} \\ & \implies \text{FailInputSat}^{\text{eval}}(\text{FailArgAt}_{\text{Ev}, \lambda}(t)) = \perp. \end{aligned}$$

When $\text{EngAuxEffect}^{(0)} = \text{ExactConditionEvidenceRequirementEffect}$, the required typed condition evidence is identified as the component for evaluating the target condition within $\text{AuditEvidenceAt}[\text{Ev}; \lambda](t)$.

For a nonmonotone-defect flag, let

$$\begin{aligned} & \text{EngAuxFlag}^{(0)} = \text{NonmonotoneDefectFlag}, \\ & \text{EngAuxEffect}^{(0)} = \text{BoundaryMethodRequirementEffect}. \end{aligned}$$

If the boundary-handling method for the corresponding

$$r \in \text{EngAuxFlagEvidenceRef}(\text{EngAuxFlag}^{(0)})$$

is `MonotoneEnvelopeMethod` or `OperatingRegionPartitionMethod`, then

$$t \in \text{FailTok}^{\text{approx}}$$

holds, and the boundary-handling data are included in the evidence component for the target condition within `AuditEvidenceAt`[`Ev`; `λ`](`t`).

If the boundary-handling method is `UnformedBoundaryMethod`, then

$$t \in \text{FailTok}^{\text{approx}}, \\ \text{FailInputSat}^{\text{eval}}(\text{FailArgAt}_{\text{Ev}, \lambda}(t)) = \perp.$$

Define the set of failure positions targeted by verified auxiliary-audit effect evidence by

$$\text{EngAuxEffectTarget}[\text{Ev}; \lambda] \\ := \left\{ t \mid \begin{array}{l} \exists \text{EngAuxFlag}^{(0)}, \exists \text{EngAuxEffect}^{(0)}, \\ (\text{EngAuxFlag}^{(0)}, \text{EngAuxEffect}^{(0)}, t) \in \text{EngAuxEffectEvidence}[\text{Ev}; \lambda] \end{array} \right\}.$$

Then $\text{EngAuxEffectTarget}[\text{Ev}; \lambda] \subseteq \text{FailTok}$.

Proof. The effect-specific verification conditions in Definition 6.197 give the respective conclusions for unsatisfied formation inputs, unsatisfied evaluation inputs, and requirements for boundary-handling methods. A requirement for exact condition evidence identifies the necessary component within the series evidence data of Definition 6.177. By the candidate type, the third component of each verified item belongs to `FailTok`. Its image, the target set, is therefore a subset of `FailTok`. $\square$

Definition 6.200 (Uses of External Time References). Let the set of engineering-measurement references that refer to external-time values be $\text{ExternalTimeRefSet} \subseteq \text{EngParamRef}$. Define the type of external-time-reference uses by

$$\text{ExternalTimeUseType} \\ := \{ \text{LogIndexUse}, \text{CalibrationUse}, \text{ProvenanceUse}, \text{GeneratedConditionPremiseUse} \}.$$

Let the external-time-reference-use map be

$$\text{ExternalTimeUse} : \text{ExternalTimeRefSet} \\ \longrightarrow \text{ExternalTimeUseType}.$$

For each reference, its use value is `GeneratedConditionPremiseUse` if and only if the audited formation inputs, evaluation inputs, or condition-evaluation expressions contain a use of its external-time value as a premise. Only references without such a use are assigned one of the other three values according to their actual use. Define the set of external-time references producing input-admissibility violations by

$$\text{ExternalTimeViolationRefSet} \\ := \left\{ r \in \text{ExternalTimeRefSet} \mid \begin{array}{l} \text{ExternalTimeUse}(r) \\ = \text{GeneratedConditionPremiseUse} \end{array} \right\}.$$

Proposition 6.201 (Noncircularity of Time Generation under External-Time-Reference Audit).

Suppose that $\text{ExternalTimeViolationRefSet} = \emptyset$. Then external-time references belong to the engineering audit input as components of log indexing, instrument calibration, or evidence provenance, while the grounds establishing the comparison conditions, representation conditions, dynamical-description conditions, continuum-approximation conditions, law-representation conditions, and differential-consistency conditions are determined by the typed audit inputs and condition evidence of the respective series.

If, on the other hand,

$$\text{ExternalTimeViolationRefSet} \neq \emptyset$$

then every

$$\text{EngParamRef}^{(0)} \in \text{ExternalTimeViolationRefSet}$$

identifies a use as a generation-condition premise corresponding to

$$\text{ExternalTimeContaminationFlag},$$

and provides evidence for

$$\text{InputAdmissibilityViolationEffect}.$$

Proof. The use map in Definition 6.200 preserves whether an external-time value is actually used as a generation-condition premise. Consider its four use cases.

For log-index use, the external-time value identifies an engineering-measurement record. For instrument-calibration use, the external-time value is used in an engineering-measurement component accompanied by a calibration provenance. For evidence-provenance use, the external-time value is used in components recording evidence location and transformation history. In these three uses, the six-series condition states are determined from the formation-input satisfaction values, evaluation-input satisfaction values, and condition-evaluation values of Definition 6.179.

For generated-condition-premise use, an external-time value occurs in the grounds establishing a formation input, evaluation input, or condition-evaluation expression of one of the six series. A reference having this use belongs to $\text{ExternalTimeViolationRefSet}$ and identifies the use corresponding to the external-time-contamination flag of Definition 6.195. By Table 14, the evidence for an input-admissibility violation is this use as a generation-condition premise.

Consequently, when the violation set is empty, the six-series condition states are determined from the typed audit inputs and condition evidence of the respective series. When the violation set is nonempty, each element identifies a use that violates input admissibility. $\square$

Definition 6.202 (Engineering Formal-Condition Evidence). Let the scale-augmented complete evaluation specification be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$, and let the corresponding six-failure-series family of evaluation specifications be Ev^{fail} . Define the engineering formal-condition-evidence type by

$$\begin{aligned} & \text{EngEvidenceType}[\text{Ev}; \lambda] \\ & := \bigsqcup_{t \in \text{FailTok}} \left((\mathcal{P}(\text{EngParamRef}) \setminus \{\emptyset\}) \times \{\text{AuditEvidenceAt}[\text{Ev}; \lambda](t)\} \right. \\ & \quad \left. \times \{\text{AuditCert}[\text{Ev}^{\text{fail}}]\} \right). \end{aligned}$$

For each $e \in \text{EngEvidenceType}[\text{Ev}; \lambda]$, write its failure-position-token component and engineering-measurement-reference-set component as

$$\begin{aligned}\text{EngTarget}(e) &\in \text{FailTok}, \\ \text{EngRefs}(e) &\in \mathcal{P}(\text{EngParamRef}) \setminus \{\emptyset\}.\end{aligned}$$

Let the validation predicate for engineering formal-condition evidence be

$$\text{EngEvidenceValid}_{\text{Ev}, \lambda} : \text{EngEvidenceType}[\text{Ev}; \lambda] \longrightarrow \{\perp, \top\}.$$

For any $e \in \text{EngEvidenceType}[\text{Ev}; \lambda]$, $\text{EngEvidenceValid}_{\text{Ev}, \lambda}(e)$ is the truth value of the conjunction of the following three conditions.

1. For every $r \in \text{EngRefs}(e)$,

$$\text{EngMeasure}(r) \in \text{EngValueType}_{\text{EngTagOf}(r)}.$$

2. The condition-evidence component

$$\text{AuditEvidenceAt}[\text{Ev}; \lambda](\text{EngTarget}(e))$$

contains as components the typed condition evidence required to evaluate

$$\text{FailInputSat}^{\text{form}}, \quad \text{FailInputSat}^{\text{eval}}, \quad \text{FailEval}$$

at the target failure-position token.

3. The evidence-provenance certificate $\text{AuditCert}[\text{Ev}^{\text{fail}}]$ supplies the correspondence between the set of engineering-measurement references and the typed condition evidence of the target failure-position token.

Define the set of verified engineering formal-condition evidence by

$$\text{EngEvidence}^{\text{ver}}[\text{Ev}; \lambda] := (\text{EngEvidenceValid}_{\text{Ev}, \lambda})^{-1} [\{\top\}].$$

For any $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$ and $e \in \text{EngEvidence}^{\text{ver}}[\text{Ev}; \lambda]$, the set of engineering-measurement references $\text{EngRefs}(e)$ and the single-token structural burden of the target failure-position token

$$\text{StructBurden}(\{\text{EngTarget}(e)\})$$

are uniquely determined by e .

Table 15: Correspondence between engineering-measurement components and formal condition-evidence types

Engineering-measurement component	Formal object formed	Object evaluated by the condition evidence
$\mathfrak{D}$ -derived recording field of view	$\mathfrak{B}$ -typed or $\mathfrak{C}/\mathfrak{B}$ -correspondence-typed $\mathfrak{D}$ -record-code relation	Formation input, record-cell width, typed record-code conflict
Bulk-recording type and following-recording type	Target-specific record-code relation, typed trajectory relation, joint trajectory relation	Relation descent, single-valuedness of trajectories, scalar-time trajectory factorization
$\mathfrak{D}$ -derived recording invasiveness	Correspondence of $\mathfrak{B}$ -states before and after the invasive action, invasiveness-defect interval, relation-truth-value pair	$\mathfrak{B}$ -state discernibility, preservation of approximation relations, preservation of law relations
$\mathfrak{D}$ -derived communication engineering data	Comparison word, closed walk, communication-defect interval	Comparison transport, loop consistency, $\mathfrak{D}$ -coupling bound
$\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{D}$ -approximation measurement	Continuum-approximation-stage evaluation datum	Nine continuum-approximation conditions
Trajectory and observable measurements	Dynamical-description-series evaluation specification	Eight dynamical-description conditions
Law-test measurement	Law-representation-series evaluation specification	Five law-representation conditions
Differential-test measurement	Differential-consistency-series evaluation specification	Five differential-consistency and law-form-preservation conditions

Definition 6.203 (Six-Series Audit-State Evaluation of Engineering Formal-Condition Evidence). Let the scale-augmented complete evaluation specification be $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$. Define the engineering audit-state evaluation by

$$\begin{aligned} \text{EngState}_{\text{Ev}, \lambda} &: \text{EngEvidence}^{\text{ver}}[\text{Ev}; \lambda] \longrightarrow \text{AuditStatus}, \\ \text{EngState}_{\text{Ev}, \lambda}(e) &:= \text{AuditConditionStateOf}[\text{Ev}; \lambda](\text{EngTarget}(e)). \end{aligned}$$

Define the set of audit-failing engineering formal-condition evidence and the set-image of its failure-position-token component by

$$\begin{aligned} \text{EngEvidence}^{\text{fail}}[\text{Ev}; \lambda] &:= (\text{EngState}_{\text{Ev}, \lambda})^{-1} [\{\text{AuditFail}\}], \\ \text{EngFailTok}[\text{Ev}; \lambda] &:= \text{EngTarget} [\text{EngEvidence}^{\text{fail}}[\text{Ev}; \lambda]]. \end{aligned}$$

Proposition 6.204 (Localization of Engineering Formal-Condition Evidence within the Six Series). For any $(\text{Ev}, \lambda) \in \text{EvSpec}_\Lambda$,

$$\text{EngFailTok}[\text{Ev}; \lambda] \subseteq \text{FailTok}.$$

Further, let

$$r \in \text{EngParamRef}, \quad e_j \in \text{EngEvidence}^{\text{ver}}[\text{Ev}; \lambda], \quad j \in \{0, 1\}.$$

Then

$$\bigwedge_{j=0}^1 [r \in \text{EngRefs}(e_j) \wedge \text{EngState}_{\text{Ev}, \lambda}(e_j) = \text{AuditFail}] \\ \implies \{\text{EngTarget}(e_j) \mid j \in \{0, 1\}\} \subseteq \text{EngFailTok}[\text{Ev}; \lambda].$$

Proof. By the disjoint-union type in Definition 6.202, for each item of verified engineering formal-condition evidence e , its target component $\text{EngTarget}(e)$ belongs to FailTok . Hence, the target image of the audit-failing evidence, $\text{EngFailTok}[\text{Ev}; \lambda]$, is a subset of FailTok .

By the latter assumption, for every $j \in \{0, 1\}$, $e_j \in \text{EngEvidence}^{\text{fail}}[\text{Ev}; \lambda]$. Applying the target map gives $\text{EngTarget}(e_j) \in \text{EngFailTok}[\text{Ev}; \lambda]$, which proves the required inclusion. $\square$

Six-family localization of engineering evidence

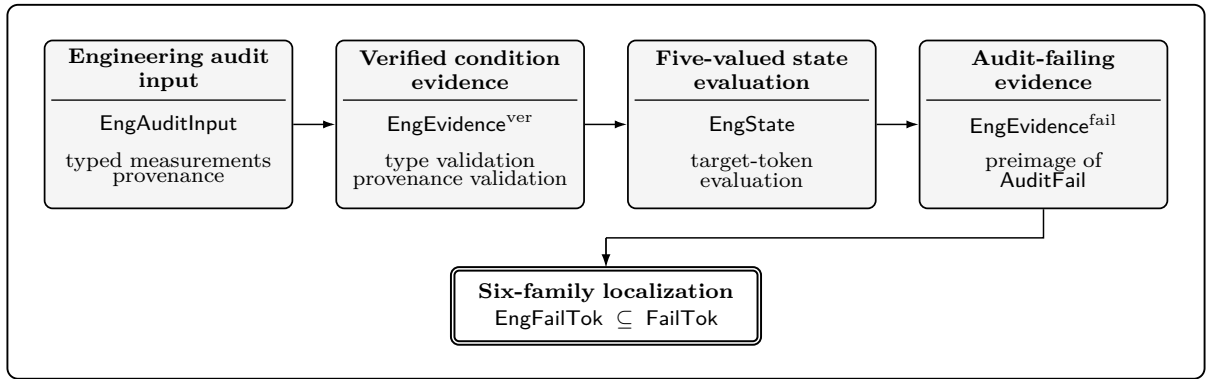


Figure 17: Localization of engineering evidence within the six series under a fixed scale-augmented complete evaluation specification (Ev, λ) . Common arguments of the objects in the figure are omitted. Verified engineering formal-condition evidence is formed from engineering audit input, the five-valued audit state is evaluated, and audit-failing evidence is extracted. Its target image $\text{EngFailTok}[\text{Ev}; \lambda]$ is localized as a subset of FailTok .

Definition 6.205 (Engineering-Evidence-Relative Structural Burden by Recording Configuration). For a common object-specification set Z_0 , observer-side information-coupling position set U , and scale set Λ , take two realized operational configurations $\mathcal{X}^{(a)}$ and a comparison-rule family $\mathcal{J}_{\text{cmp}}$ that is well typed for both configurations. For each $a \in \{0, 1\}$, apply the preceding definitions separately to

$$\mathbf{l}^{(a)} := (\mathcal{X}^{(a)}, \mathcal{J}_{\text{cmp}}).$$

In the following comparison of the two configurations, a superscript (a) distinguishes the configuration-specific instances of input-dependent types, data, and maps. The physical-carrier value map $\text{CarrierAt}_{\mathfrak{S}}^{(a)}$ is likewise determined by the physical realization maps of that configuration. The failure-position type, structural-target type, and structural-burden kernel are common to both configurations. The complete-evaluation-specification set is

$$\text{EvSpec}^{(a)} := \{\mathbf{l}^{(a)}\} \times U \times Z_0.$$

Let

$$\zeta_1, \zeta_2 \in Z_0, \quad \zeta_1 \neq \zeta_2, \quad o \in U, \quad \lambda \in \Lambda, \quad (a, k) \in \{0, 1\} \times \{1, 2\}.$$

Let the complete evaluation specification be

$$\text{Ev}_k^{(a)} = (l^{(a)}, o, \zeta_k) \in \text{EvSpec}^{(a)}$$

and assume

$$\begin{aligned} (\text{Ev}_k^{(a)})_{k=1}^2 &\in \text{EvSpec}^{\text{pair},(a)}, \\ (\text{Ev}_k^{(1)})_{k=1}^2 &\in \text{EvSpec}_{\mathfrak{D}}^{\text{pair,distributed},(1)}. \end{aligned}$$

Write the corresponding family of six-failure-series evaluation specifications as $\text{Ev}^{\text{fail},(a,k)}$. Also assume

$$\begin{aligned} \text{BulkRecord}_{\mathfrak{D}}^{(0)}[(\text{Ev}_k^{(0)})_{k=1}^2] &= \top, \\ \text{FollowingRecord}_{\mathfrak{D}}^{(1)}[(\text{Ev}_k^{(1)})_{k=1}^2; \lambda] &= \top. \end{aligned}$$

Define the structural-burden input-tag sets for the bulk-recording and following-recording types by

$$\text{RecordBurdenTag}^{(0)} := \left\{ \begin{array}{c} \text{ViewTag}_{\mathfrak{D}}, \text{RecordConfigTag}_{\mathfrak{D}}, \\ \text{InvasivenessTag}_{\mathfrak{D}}, \text{CountStepTag}_{\mathfrak{C}}, \text{StabilityTag}_{\mathfrak{D}}, \\ \text{RecordCellWidthTag}_{\mathfrak{D}}, \text{DiscernibilityTag}_{\mathfrak{B}} \end{array} \right\},$$

$$\text{RecordBurdenTag}^{(1)} := \text{RecordBurdenTag}^{(0)} \cup \{\text{CommunicationTag}_{\mathfrak{D}}, \text{CouplingTag}_{\mathfrak{D}}\}.$$

Define the verified evidence for each recording configuration by

$$\text{RecordEvidence}^{\text{ver},(a,k)} := \text{EngEvidence}^{\text{ver},(a)}[\text{Ev}_k^{(a)}; \lambda].$$

Define the predicate for evidence that refers to a structural-burden input tag by

$$\text{HasRecordBurdenRef}^{(a)} : \bigcup_{k \in \{1,2\}} \text{RecordEvidence}^{\text{ver},(a,k)} \longrightarrow \{\perp, \top\}$$

and

$$\begin{aligned} \text{HasRecordBurdenRef}^{(a)}(e) &\iff \exists r \in \text{EngRefs}^{(a)}(e), \\ &\quad \text{EngTagOf}^{(a)}(r) \in \text{RecordBurdenTag}^{(a)}. \end{aligned}$$

Define the structural-burden evidence for each recording configuration and its object-indexed form by

$$\begin{aligned} \text{RecordBurdenEvidence}^{(a,k)} &:= \left\{ e \in \text{RecordEvidence}^{\text{ver},(a,k)} \right. \\ &\quad \left. \mid \text{HasRecordBurdenRef}^{(a)}(e) \right\}, \\ \text{RecordBurdenEvidence}^{(a)} &:= \bigsqcup_{k \in \{1,2\}} \\ &\quad \left(\{k\} \times \text{RecordBurdenEvidence}^{(a,k)} \right). \end{aligned}$$

Define the evidence state of each object, the audit-failing evidence, and the recording-configuration-specific set of failure positions by

$$\begin{aligned} \text{RecordState}^{(a,k)} &:= \text{EngState}_{\text{Ev}_k^{(a)}, \lambda}^{(a)} \Big|_{\text{RecordBurdenEvidence}^{(a,k)}}, \\ \text{RecordBurdenEvidence}^{\text{fail},(a,k)} &:= \left(\text{RecordState}^{(a,k)} \right)^{-1} [\{\text{AuditFail}\}], \\ \text{RecordBurdenEvidence}^{\text{fail},(a)} &:= \bigcup_{k \in \{1,2\}} \text{RecordBurdenEvidence}^{\text{fail},(a,k)}, \\ \text{RecordBurdenTok}^{(a)} &:= \text{EngTarget}^{(a)} \left[\text{RecordBurdenEvidence}^{\text{fail},(a)} \right]. \end{aligned}$$

Using the structural-burden kernel of Definition 6.169, define the structural burden for each recording configuration and its three representations by

$$\begin{aligned}\text{RecordBurden}^{(a)} &:= \text{StructBurden}(\text{RecordBurdenTok}^{(a)}), \\ \text{RecordBurden}^{\text{bulk}} &:= \text{RecordBurden}^{(0)}, \\ \text{RecordBurden}^{\text{follow}} &:= \text{RecordBurden}^{(1)}, \\ \text{RecordBurden}^{\Delta} &:= \text{RecordBurden}^{\text{bulk}} \triangle \text{RecordBurden}^{\text{follow}}.\end{aligned}$$

Proposition 6.206 (Separation of Engineering-Evidence-Relative Structural Burdens for Bulk-Recording and Following-Recording Types). The two structural burdens are subsets of StructTok, and

$$\text{RecordBurden}^{\text{bulk}} = \text{RecordBurden}^{\text{follow}} \iff \text{RecordBurden}^{\Delta} = \emptyset.$$

For each $a \in \{0, 1\}$, if some $t \in \text{RecordBurdenTok}^{(a)}$ satisfies

$$\text{StructBurden}(\{t\}) \not\subseteq \text{RecordBurden}^{(1-a)}$$

then

$$\text{RecordBurden}^{(a)} \setminus \text{RecordBurden}^{(1-a)} \neq \emptyset.$$

If this condition holds for both $a = 0$ and $a = 1$, the two structural burdens are incomparable under inclusion.

Proof. The set of failure positions for each recording configuration is a subset of FailTok. By Definition 6.169, each $\text{RecordBurden}^{(a)}$ is therefore a subset of StructTok. Applying the equivalence between emptiness of the symmetric difference and equality of two sets to the two structural burdens yields the first equivalence.

For each $a \in \{0, 1\}$, nonemptiness of the set difference follows by applying Lemma 6.170 to

$$A = \text{RecordBurdenTok}^{(a)}, \quad B = \text{RecordBurdenTok}^{(1-a)}$$

Nonemptiness of the set difference in both directions establishes that the two structural burdens are incomparable under inclusion. $\square$

7 Audit of Law-Time Use and Nonachievement of Primitive-Time Premises

The physical establishment of a time variable appearing in a physical law requires that its time content be physically generated before its use within the law and the derivation of results. On the basis of formal evidence and statement evidence from the 72 audited references, primitive-time roots, generation dependencies, result dependencies, and physical-time qualification are assessed.

7.1 Physical-Time Qualification and Primitive-Time Inputs

This subsection defines the grounding obligations of physical time content, the dependence of result carriers on that content, the distinction between evaluation success and physical-time qualification, and primitive-time inputs. A time variable appearing in a physical

law serves as an argument for state change, evolution, propagation, comparison, and evaluation. Its physical use requires the identity of time values, their before-and-after order, the equivalence of time intervals, scale and unit, correspondence across locations, correspondence with clock states, and correspondence with observational, recording, and experimental procedures.

By Proposition 2.12, Proposition 3.26, and Corollary 4.190, the generation dependence from physical-role inputs to the representation-stage completion conditions for the finite family of object-specification indices

$$Z_0$$

is determined by

$$\begin{aligned} \text{RI}[Z_0] &\longrightarrow \mathcal{X} \\ &\longrightarrow \text{Qu}_{\mathfrak{C}\mathfrak{D}}^{\text{cmp}, \text{can}}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &\longrightarrow \mathcal{E}^{\text{op}}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &\longrightarrow \text{Repr}_{\leq 3}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \end{aligned}$$

For each $o \in U$, $\zeta \in Z_0$, let the complete evaluation specification be

$$\text{Ev} = ((\mathcal{X}; \mathcal{J}_{\text{cmp}}), o, \zeta) \in \text{EvSpec}$$

By Theorem 5.138, motion-description operability is determined by

$$\begin{aligned} &\text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &\quad \wedge \text{DynCondAll}[\text{Ev}] \\ &\iff \text{DynOperative}[\text{Ev}] \end{aligned}$$

For the law stage, fix the base law datum and base representation associated with the single-object specification Ev in Theorem 5.138. For the differential stage, also fix a differential-evaluation-structure family belonging to the same pair. Law-description operability is determined by

$$\begin{aligned} &\text{DynOperative}[\text{Ev}] \\ &\quad \wedge \text{LawCondAll}[\text{LawEvalDat}_{o, \zeta}] \\ &\iff \text{LawOperative}[\text{LawEvalDat}_{o, \zeta}] \end{aligned}$$

Differential-stage operability is determined by

$$\begin{aligned} &\text{LawOperative}[\text{LawEvalDat}_{o, \zeta}] \\ &\quad \wedge \text{DiffCondAll}[\text{EvalStrFam}_{o, \zeta}^{\text{diff}}] \\ &\iff \text{DiffOperative}[\text{DiffEvalDat}_{o, \zeta}] \end{aligned}$$

In accordance with Definition 6.181, let the corresponding family of six-failure-series evaluation specifications be

$$\text{Ev}^{\text{fail}} = (\text{Ev}_{\tau}^{\text{fail}})_{\tau \in \text{FailSeries}}$$

For an approximation scale

$$\lambda \in \Lambda$$

, the value of the audit-report generation map is

$$\begin{aligned} & \text{AuditReportOf}(\text{AuditData}_{\text{time}}[\text{Ev}^{\text{fail}}]; \lambda) \\ & := \text{AuditReport}[\text{Ev}^{\text{fail}}; \lambda] \end{aligned}$$

. By Theorem 6.183 and Corollary 6.187, this audit-report object is uniquely determined by the operational audit data and approximation scale and contains the states of the time-establishment conditions, invalidation positions, invalidation signatures, approximation boundaries, and structural burdens. The audit-state report and audit-diagnostic report are obtained from the corresponding finite operational audit fragment by finitely many operations.

For time content in a physical description, the audit covers the generation path from nontemporal physical input to that time content, transfer paths that preserve time content between distinct positions in the description, and dependency paths to physical results that use the time content. These paths provide the common basis of determination for the theory-specific synthesis in Chapter 8 and the complete, reference-by-reference audit of all 72 references in the Supplement.

Definition 7.1 (Finite Audited Reference Set, Time Content, and Result Carriers). Let the finite set consisting of the 72 references subject to audit be

$$\text{Ref}^{\text{audit}}$$

and let

$$|\text{Ref}^{\text{audit}}| = 72$$

.

For each $R \in \text{Ref}^{\text{audit}}$, write the time contents in the source in their order of occurrence as

$$(\text{TimeContent}_{R,i})_{i=1}^{n_R}$$

. Define the set of time contents of the reference

$$R$$

by

$$\text{TimeContent}_R := \{\text{TimeContent}_{R,i} \mid i \in \{1, \dots, n_R\}\}$$

. Define the set of time contents of all audited references by

$$\text{TimeContent}^{\text{audit}} := \coprod_{R \in \text{Ref}^{\text{audit}}} (\{R\} \times \text{TimeContent}_R)$$

.

Write the set of results derived by each reference as

$$\text{Result}_R$$

. Here, a result is an object obtained by a definition, law, inference, calculation, experimental procedure, or observational analysis in the reference and constituting part of the physical claim of that reference.

Definition 7.2 (Grounding Obligations of Physical Time Content). The correspondence between the operations carried by time content appearing in a physical law and the structures generated through Chapter 4 is defined as follows.

1. The identity of time values is supplied by the comparison equivalence relation

$$\sim_{\bullet}^{\text{cmp}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

and the canonical comparison quotient

$$\text{Qu}_{\bullet}^{\text{cmp,can}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

.

2. The before-and-after order of time values is supplied by the $\mathfrak{C}/\mathfrak{D}$ -canonical partial order

$$\preceq_{\bullet}^{\text{can}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

and the local and global order-coordinate structures.

3. Within the $\mathfrak{C}/\mathfrak{D}$ -derived conditional quantitative affine realization

$$\text{QtyAffReal}_{\bullet}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

, the equivalence of time intervals is supplied by the quantitative affine difference

$$\text{AffDiff}_{\bullet}^{\text{qty}}$$

.

4. The correspondence between time scale and unit is supplied by a specification of the $\mathfrak{C}/\mathfrak{D}$ -derived quantitatively linearly ordered field, quantitative affine action, and quantitative affine difference.

5. Correspondence between time contents belonging to distinct information-coupling positions is supplied by the local order-coordinate chart-transition maps contained in the comparison order atlas

$$\text{Atlas}^{\text{cmp,ord}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

.

6. Correspondence of clock states, displays, pulses, and cycles with time content is supplied by the generation sequence through the $\mathfrak{C}$ -derived arrival map $\text{Arr}_{\mathfrak{C},o}^{\text{cmp}}$ and the comparison-side recovery map $\text{Rec}_{\mathfrak{C},o}^{\text{cmp}}$ to the $\mathfrak{C}$ -derived canonical comparison quotient, order-coordinate structure, and $\mathfrak{C}$ -derived representation coordinates.

7. Correspondence of observation, recording, and experimental procedures with time content is supplied by the generation sequence through the $\mathfrak{D}$ -derived arrival map

$$\text{Arr}_{\mathfrak{D},o}^{\text{cmp}},$$

the $\mathfrak{D}$ -derived record cell

$$W_{\mathfrak{D}}(x_{\mathfrak{D}}),$$

and the $\mathfrak{D}$ -derived representation image of the record cell

$$\text{CellImg}_{\mathfrak{D},o}^{\text{repr}}(\mathcal{X}; \mathcal{J}_{\text{cmp}})(x_{\mathfrak{D}})$$

to the $\mathfrak{D}$ -derived canonical comparison quotient, order-coordinate structure, and $\mathfrak{D}$ -derived representation coordinates.

Proposition 7.3 (Dependence of Law-Time Operations on Generated Structures). Identity determination, order comparison, interval comparison, scale transformation, correspondence between information-coupling positions, clock correspondence, or observation-recording correspondence performed by a physical law on time content takes as input the respective generated structure specified in Definition 7.2.

Time differentiation, time integration, frequency, period, proper time, and descriptions of evolution by an evolution parameter take as input the generated structures corresponding to the time difference, time order, time scale, and time correspondence that they use.

Proof. The identity of time values is determined by the comparison equivalence classes and canonical comparison quotient of Definition 4.68 and Definition 4.89.

The order of time values is determined by the canonical partial order and order-coordinate quotient of Definition 4.101 and Definition 4.116.

Time intervals, scales, and unit correspondences are determined by the quantitatively linearly ordered field, quantitative affine action, and quantitative affine difference of Definition 4.148.

Correspondence between information-coupling positions is determined by the local order-coordinate chart-transition maps contained in the comparison order atlas of Definition 4.128.

Correspondence with clock states, displays, pulses, and cycles is determined by the $\mathfrak{C}$ -derived source-induced arrival preorder of Definition 4.95, the comparison-side recovery map of Definition 3.21, and the generation sequence comprising the canonical comparison quotient, order-coordinate structure, and representation coordinates.

Correspondence with observation, recording, and experimental procedures is determined by the $\mathfrak{D}$ -derived arrival map, the $\mathfrak{D}$ -derived record cell of Definition 3.15, the $\mathfrak{D}$ -derived representation image of the record cell of Definition 4.159, and the $\mathfrak{D}$ -derived representation coordinates.

Therefore, every law-time operation takes as input the generated structure corresponding to the time content used by that operation. $\square$

Definition 7.4 (Time-Content Transfer Events and Result Prerequisites). For each $R \in \text{Ref}^{\text{audit}}$, write the set of time-content transfer events within the reference as

$$\text{TransferEvent}_R$$

. For each $\epsilon \in \text{TransferEvent}_R$, represent the source-content set and target-content set of the transfer event

$$\epsilon$$

by

$$\begin{aligned} \text{TransferIn}_{R,\epsilon} &\subseteq \text{TimeContent}_R, \\ \text{TransferOut}_{R,\epsilon} &\subseteq \text{TimeContent}_R \end{aligned}$$

.

For each $\eta \in \text{Result}_R$, represent the set of time contents used in deriving the result

$$\eta$$

by

$$\text{ResultPrereq}_{R,\eta} \subseteq \text{TimeContent}_R$$

.

Definition 7.5 (Within-Reference Time-Content Transfer Reachability). For each $R \in \text{Ref}^{\text{audit}}$, define the within-reference time-content transfer-reachability relation by

$$\begin{aligned} & \rightsquigarrow_R^{\text{time,content}} \\ & := \left(\bigcup_{\epsilon \in \text{TransferEvent}_R} \text{TransferIn}_{R,\epsilon} \times \text{TransferOut}_{R,\epsilon} \right)^+ \end{aligned}$$

. The superscript

+

denotes the transitive closure of a binary relation.

Proposition 7.6 (Condition for Time-Content Participation in Result Derivation). For each

$$R \in \text{Ref}^{\text{audit}}, \quad x \in \text{TimeContent}_R, \quad \eta \in \text{Result}_R$$

, if

$$\begin{aligned} & x \in \text{ResultPrereq}_{R,\eta} \\ & \vee \left[\exists y \in \text{ResultPrereq}_{R,\eta} : x \rightsquigarrow_R^{\text{time,content}} y \right] \end{aligned}$$

holds, then the time content

x

participates in the derivation sequence of the result

η

.

The same condition determines participation in result derivation for time content used in a time derivative, time integral, frequency, period, proper time, or evolution parameter.

Proof. In the first case,

$$x \in \text{ResultPrereq}_{R,\eta}$$

; hence, by Definition 7.4,

x

is used directly in deriving the result

η

.

In the second case, there exists some $y \in \text{ResultPrereq}_{R,\eta}$ such that

$$x \rightsquigarrow_R^{\text{time,content}} y$$

. By Definition 7.5, a finite sequence of time-content transfer events connects

x

to

y

. Moreover, $y \in \text{ResultPrereq}_{R,\eta}$; therefore,

x

participates, through the time-content transfer sequence, in deriving the result

η

The same argument applies to a time derivative, time integral, frequency, period, proper time, and evolution parameter when the corresponding time content belongs to the direct result prerequisites or reaches a result prerequisite by transfer. $\square$

Definition 7.7 (Evaluation Success and Physical-Time Qualification). The mathematical consistency of each reference means that the conclusions in the reference follow from its adopted premises, definitions, and rules of inference.

Predictive performance means that observable outputs are calculated from the theoretical structure in the reference. Empirical fit means that the correspondence between those outputs and values obtained by experiment or observation satisfies the evaluation conditions in the reference.

A physical-law time variable acquires physical-time qualification when the identity, before-and-after order, intervals, scale, correspondence between information-coupling positions, clock correspondence, and observation–recording correspondence listed in Definition 7.2 are generated from nontemporal physical input.

Proposition 7.8 (Non-Entailment of Physical-Time Qualification by Evaluation Success). The simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit does not entail physical-time qualification of a physical-law time variable. That is,

$$\left. \begin{array}{l} \text{mathematical consistency,} \\ \text{predictive performance,} \\ \text{empirical fit} \end{array} \right\} \not\Rightarrow \text{physical-time qualification.}$$

Proof. Given equations of motion and initial values with a real-valued time variable as input, solutions conforming to the adopted rules of inference, predicted values, and correspondences with observed values can be constructed. This construction does not include as premises maps or relations that generate, from nontemporal physical input, the identity, order, intervals, and scale of the values of the time variable, correspondence between information-coupling positions, clock correspondence, and observation–recording correspondence.

By Definition 7.7, mathematical consistency, predictive performance, and empirical fit evaluate inferences, outputs, and value correspondences that use a given time variable. Physical-time qualification evaluates the generation of the time content itself that the time variable uses.

Therefore, even in a description in which the former three evaluations hold simultaneously, there exists a construction that does not take as input the generated structures of Definition 7.2. Hence, physical-time qualification does not follow from simultaneous satisfaction of the three evaluations. $\square$

Corollary 7.9 (Time-Content Dependence under Evaluation Success). For each $R \in \text{Ref}^{\text{audit}}$, when the time content $x \in \text{TimeContent}_R$ satisfies the condition in Proposition 7.6, the satisfaction of mathematical consistency, predictive performance, or empirical fit preserves the direct dependence or transfer-reachability dependence from

x

to the result.

Proof. By Proposition 7.6,

x

either belongs directly to the result prerequisites or reaches a result prerequisite by transfer. By Proposition 7.8, evaluation success does not entail the generation of the time content itself. Therefore, after evaluation success has been established, the dependency relation from

x

to the result remains in force. □

Definition 7.10 (Primitive-Time Input). For each $R \in \text{Ref}^{\text{audit}}$, $x \in \text{TimeContent}_R$, suppose that the time content

x

is entered into the physical description in the reference and is used in a law operation or result derivation before the structure required for the use of

x

among the identity, before-and-after order, interval, scale, correspondence between information-coupling positions, clock correspondence, or observation–recording correspondence listed in Definition 7.2 is generated from nontemporal physical input. Then

x

is called a primitive-time input of the reference

R

.

Theorem 7.11 (Failure of Physical-Time Qualification for a Primitive-Time Input). For a reference $R \in \text{Ref}^{\text{audit}}$, a primitive-time input $x \in \text{TimeContent}_R$ lacks physical-time qualification.

Proof. By Definition 7.10, at least one of the identity, before-and-after order, interval, and scale of the time values required for the use of

x

or their correspondence between information-coupling positions, clock correspondence, or observation–recording correspondence enters a law operation or result derivation before the corresponding generated structure.

By Definition 7.2 and Proposition 7.3, these time operations take as input their respective comparison equivalence relation, canonical comparison quotient, order-coordinate structure, quantitative affine realization, comparison order atlas, $\mathfrak{C}$ -derived arrival–recovery structure, or $\mathfrak{D}$ -derived observation–recording structure.

For a primitive-time input, the required time content is supplied to a law operation or result derivation before these generated structures. Therefore, the generation dependence from nontemporal physical input to that time content does not hold, and the physical-time qualification of Definition 7.7 is not satisfied. □

Corollary 7.12 (Inheritance of Primitive-Time Dependence by Results). When, for a reference $R \in \text{Ref}^{\text{audit}}$, a primitive-time input $x \in \text{TimeContent}_R$, and a result $\eta \in \text{Result}_R$ satisfy the result-derivation participation condition of Proposition 7.6, the result

$$\eta$$

depends on time content that lacks physical-time qualification.

Proof. By Theorem 7.11,

$$x$$

lacks physical-time qualification.

By Proposition 7.6,

$$x$$

either belongs to the direct prerequisites of the result

$$\eta$$

or reaches a result prerequisite through a time-content transfer sequence. Therefore,

$$\eta$$

depends on the time content

$$x$$

that lacks physical-time qualification. □

Proposition 7.13 (Failure of Evaluation Success to Complete a Primitive-Time Input). In a reference that uses a primitive-time input, even simultaneous mathematical consistency, predictive performance, and empirical fit do not establish physical-time qualification of the primitive-time input.

Proof. By Proposition 7.8, physical-time qualification does not follow from the simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit.

By Theorem 7.11, for a primitive-time input, the generation dependence required for the use of its time content does not hold. The three forms of evaluation success do not add that generation dependence. Therefore, physical-time qualification of the primitive-time input is not established. □

Proposition 7.14 (Loss of Generation-Assessment Discrimination under Admission of Primitive Time). If an assessment of the establishment of time generation admits a primitive-time input as sufficient temporal input, time content generated from nontemporal physical input cannot be distinguished in the assessment result from time content entered before the generated structure.

Proof. For each obligation in Definition 7.2, consider time content obtained through the corresponding generated structure. Time content carrying the same identity, order, interval, scale, correspondence between information-coupling positions, clock correspondence, or observation-recording correspondence can also be placed at the input to a law operation or result derivation in accordance with Definition 7.10.

An assessment that admits the primitive-time input as sufficient temporal input maps the former generation path and the latter prior input to the same establishment result.

Therefore, its assessment result does not distinguish the presence or absence of a generation path.

An assessment of the establishment of time generation evaluates whether the generation dependence from nontemporal physical input to time content holds. An assessment that does not distinguish the presence or absence of a generation path cannot distinguish the establishment and invalidation of time generation. $\square$

Corollary 7.15 (Generation-Priority Condition for a Physical-Law Time Variable). For a physical-law time variable to be physically established, the generated structure corresponding to the time content used by that time variable must be established before law operations and result derivation.

Proof. By Theorem 7.11, a primitive-time input used before the generated structure lacks physical-time qualification. By Proposition 7.14, admission of a primitive-time input does not distinguish the establishment and invalidation of time generation. Therefore, physical establishment of a physical-law time variable requires prior establishment of the corresponding generated structure. $\square$

Definition 7.16 (Scope of the Audited Reference Set). The set $\text{Ref}^{\text{audit}}$ of Definition 7.1 consists of 72 references whose source locations, time contents, transfers, and result dependencies are identified and whose generation dependencies from nontemporal physical input are audited.

Each reference contains at least one of the following uses of time.

1. An independent variable in an equation of motion, field equation, state-evolution equation, or stochastic-evolution equation.
2. Simultaneity, coordinate time, proper time, spacetime interval, or causal order.
3. A clock, period, frequency, standard, synchronization, or timekeeping procedure.
4. Thermodynamic evolution, statistical evolution, irreversibility, or an arrow of time.
5. Quantum measurement, quantum correlation, a constrained system, or an evolution description from a fundamental equation without time.
6. Gravity, cosmology, quantum gravity, emergent spacetime, or time reconstruction.
7. Temporal correspondence appearing in an experimental arrangement, observational record, or data analysis.

For each $R \in \text{Ref}^{\text{audit}}$, the audit input consists of the corresponding source locations, time contents, transfers, result dependencies, and source evidence for determining the generation dependencies from nontemporal physical input.

Proposition 7.17 (Reference Domain of Cross-Reference Claims). The quantification domain of propositions across the audited reference set is $\text{Ref}^{\text{audit}}$. Each reference-specific determination takes the corresponding source locations, time contents, transfers, and result dependencies as evidence inputs.

Proof. Definition 7.1 types references, time contents, and results for each $R \in \text{Ref}^{\text{audit}}$. Definition 7.16 defines the scope of the audited reference set and the reference-specific audit inputs. Therefore, the quantification domain of the cross-reference audit and the evidence inputs for each reference-specific determination are determined. $\square$

7.2 Evidence-Bearing Generation-Dependency Audit Method

Audit-boundary nodes, audit-boundary edges, generation traces, obligation atoms, primitive-time roots, and result dependencies are determined for each reference. For each reference, define the object of the generation-dependency audit as a finite node set with three components: physical or mathematical boundary claims, time contents, and reference-specific results.

Definition 7.18 (Audit-Boundary Nodes). Define the set of audit-boundary-node types by

$$\text{Type}^{\text{audit,node}} := \left\{ \begin{array}{l} \text{Type}_{\text{claim}}^{\text{audit,node}}, \\ \text{Type}_{\text{time}}^{\text{audit,node}}, \\ \text{Type}_{\text{result}}^{\text{audit,node}} \end{array} \right\}$$

The three elements of this set are pairwise distinct.

For each $R \in \text{Ref}^{\text{audit}}$, call a claim partitioned into a minimal verifiable unit from a formal definition, equation, proposition, or an accompanying explicit statement in the source an audit-claim record. Let

$$\text{Type}_{\text{claim}}^{\text{audit,node}}$$

be the first component, and let an audit-claim record carrying, as attributes, its source location, attributed subject, degree of certainty, evidence-expression class, and a reference to its source context be the second component. Denote the finite set of such ordered pairs by

$$\text{Node}_R^{\text{audit,claim}}$$

Define the item-category domain for time contents and reference-specific results by $\mathcal{I}_{\text{audit,item}} := \{\text{time}, \text{result}\}$, and define the item families by

$$\begin{aligned} \text{Item}_R^{\text{audit,time}} &:= \text{TimeContent}_R, \\ \text{Item}_R^{\text{audit,result}} &:= \text{Result}_R \end{aligned}$$

For each $j \in \mathcal{I}_{\text{audit,item}}$, define the node set and node-forming map by

$$\begin{aligned} \text{Node}_R^{\text{audit},j} &:= \left\{ \text{Type}_j^{\text{audit,node}} \right\} \times \text{Item}_R^{\text{audit},j}, \\ \text{NodeOf}_R^{\text{audit},j} : \text{Item}_R^{\text{audit},j} &\longrightarrow \text{Node}_R^{\text{audit},j}, \\ z &\longmapsto (\text{Type}_j^{\text{audit,node}}, z) \end{aligned}$$

Define the set of audit-boundary nodes by

$$\begin{aligned} \text{Node}_R^{\text{audit,bd}} &:= \text{Node}_R^{\text{audit,claim}} \dot{\cup} \text{Node}_R^{\text{audit,time}} \\ &\quad \dot{\cup} \text{Node}_R^{\text{audit,result}} \end{aligned}$$

Define the node-type map by

$$\begin{aligned} \text{TypeOf}_R^{\text{audit,node}} : \text{Node}_R^{\text{audit,bd}} &\longrightarrow \text{Type}^{\text{audit,node}}, \\ \text{TypeOf}_R^{\text{audit,node}} (\text{NodeType}, \text{NodePayload}) &:= \text{NodeType} \end{aligned}$$

Proposition 7.19 (Component Decomposition of Audit-Boundary Nodes). For each

$$R \in \text{Ref}^{\text{audit}}, \quad \text{Node} \in \text{Node}_R^{\text{audit,bd}}$$

exactly one of

$$\begin{aligned} \text{Node} &\in \text{Node}_R^{\text{audit,claim}}, & \text{or} \\ \text{Node} &\in \text{Node}_R^{\text{audit,time}}, & \text{or} \\ \text{Node} &\in \text{Node}_R^{\text{audit,result}} \end{aligned}$$

holds. Consequently,

$$\text{TypeOf}_R^{\text{audit,node}}$$

assigns a unique node type to each audit-boundary node.

Proof. By Definition 7.18, $\text{Node}_R^{\text{audit,bd}}$ is the disjoint union of the three node sets. The three components are pairwise disjoint, and their union equals $\text{Node}_R^{\text{audit,bd}}$. Hence every audit-boundary node belongs to exactly one of the three components. By the definition of the node-type map, each component is assigned one node type. $\square$

Definition 7.20 (Five-Type Classification of Time Content). Define the codomain of the five time-content types by

$$\begin{aligned} \text{Type}^{\text{time,ord,struct}} := \{ & \text{Type}_{\text{member}}^{\text{time,ord,struct}}, \text{Type}_{\text{dependency}}^{\text{time,ord,struct}}, \\ & \text{Type}_{\text{branch}}^{\text{time,ord,struct}}, \text{Type}_{\text{separate}}^{\text{time,ord,struct}}, \\ & \text{Type}_{\text{lexical}}^{\text{time,ord,struct}} \} \end{aligned}$$

The five values denote, respectively, a member of a boundary, stage, or transition-position family; content dependent on the same family; content on a branch outside the boundary; content belonging to a separate requirement family; and lexical or illustrative content.

For each $R \in \text{Ref}^{\text{audit}}$, provide the five-type classification map of time content

$$\begin{aligned} \text{TypeOf}_R^{\text{time,ord,struct}} : \text{TimeContent}_R \\ \longrightarrow \text{Type}^{\text{time,ord,struct}} \end{aligned}$$

Assign each time content one type according to the function given by a definition, equation, proposition, or explicit statement in the source.

Definition 7.21 (Eight-Axis Classification of Temporal Order Structure). Define the codomain of the eight temporal-order-structure axes by

$$\begin{aligned} \text{Axis}^{\text{time,ord,struct}} := \{ & \text{endpoint, instant, interval_boundary, phase_boundary,} \\ & \text{turning_position, order_relation, orientation, transition_position} \} \end{aligned}$$

The eight values denote, respectively, endpoint, instant, interval boundary, phase boundary, turning position, order relation, orientation, and transition position.

For each $R \in \text{Ref}^{\text{audit}}$, provide the eight-axis classification map of temporal order structure

$$\begin{aligned} \text{AxisOf}_R^{\text{time,ord,struct}} : \text{TimeContent}_R \\ \longrightarrow \mathcal{P}(\text{Axis}^{\text{time,ord,struct}}) \end{aligned}$$

For each $x \in \text{TimeContent}_R$, impose

$$\begin{aligned} \text{TypeOf}_R^{\text{time,ord,struct}}(x) &= \text{Type}_{\text{member}}^{\text{time,ord,struct}} \\ \implies \left| \text{AxisOf}_R^{\text{time,ord,struct}}(x) \right| &\geq 1, \end{aligned}$$

and

$$\begin{aligned} \text{TypeOf}_R^{\text{time,ord,struct}}(x) &\neq \text{Type}_{\text{member}}^{\text{time,ord,struct}} \\ \implies \text{AxisOf}_R^{\text{time,ord,struct}}(x) &= \emptyset \end{aligned}$$

Proposition 7.22 (Structural Attributes of Time-Content Nodes). For each $R \in \text{Ref}^{\text{audit}}$, $x \in \text{TimeContent}_R$, the time-content node $\text{NodeOf}_R^{\text{audit,time}}(x)$ has the unique attribute

$$\left(\text{TypeOf}_R^{\text{time,ord,struct}}(x), \right. \\ \left. \text{AxisOf}_R^{\text{time,ord,struct}}(x) \right)$$

Proof. The type-classification map of Definition 7.20 and the axis-classification map of Definition 7.21 both have domain

$$\text{TimeContent}_R$$

By Definition 7.18, the time-content node-forming map is injective. Hence every time-content node uniquely carries the type value and axis set belonging to the single time content from which it was formed. $\square$

Definition 7.23 (Common Type of Structural Signatures and Member Structural Signature). For each $R \in \text{Ref}^{\text{audit}}$, define the signature-kind set and the type of the structural-signature family by

$$\begin{aligned} \mathcal{I}_{\text{sig}}^{\text{time,ord,struct}} &:= \{\text{member}, \text{dependency}\}, \\ \text{Sig}_R^{\text{time,ord,struct}, \varsigma_{\text{sig}}} &: \\ &\left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\varsigma_{\text{sig}}}^{\text{time,ord,struct}} \right\} \right) \\ &\longrightarrow \{0, 1\}^3, \\ \varsigma_{\text{sig}} &\in \mathcal{I}_{\text{sig}}^{\text{time,ord,struct}} \end{aligned}$$

For each member

$$x$$

of the member kind, define its signature by

$$\begin{aligned} &\text{Sig}_R^{\text{time,ord,struct,member}}(x) \\ &:= \left(\mathbf{1} \bigcup_{\epsilon \in \text{TransferEvent}_R} \text{TransferIn}_{R,\epsilon}(x), \right. \\ &\quad \mathbf{1} \bigcup_{\epsilon \in \text{TransferEvent}_R} \text{TransferOut}_{R,\epsilon}(x), \\ &\quad \left. \mathbf{1} \bigcup_{\eta \in \text{Result}_R} \text{ResultPrereq}_{R,\eta}(x) \right) \end{aligned}$$

The three components denote, respectively, membership in a transfer input set, membership in a transfer output set, and membership in a result prerequisite.

Definition 7.24 (Dependent-Content Structural Signature). For each $R \in \text{Ref}^{\text{audit}}$, define the signature of each dependent content

$$x$$

of the dependent-content kind by

$$\begin{aligned} & \text{Sig}_R^{\text{time,ord,struct,dependency}}(x) \\ &:= \left(\mathbf{1} \left[\begin{array}{c} \exists y \in \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\text{member}}^{\text{time,ord,struct}} \right\} \right) : \\ x \rightsquigarrow_R^{\text{time,content}} y \end{array} \right] , \right. \\ & \quad \left. \mathbf{1} \left[\begin{array}{c} \exists y \in \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\text{member}}^{\text{time,ord,struct}} \right\} \right) : \\ y \rightsquigarrow_R^{\text{time,content}} x \end{array} \right] , \right. \\ & \quad \left. \mathbf{1} \bigcup_{\eta \in \text{Result}_R} \text{ResultPrereq}_{R,\eta}(x) \right) \end{aligned}$$

The three components denote, respectively, transfer reachability to a member, transfer reachability from a member, and membership in a result prerequisite.

Proposition 7.25 (Totality and Uniqueness of Structural Signatures).

For each $R \in \text{Ref}^{\text{audit}}$, every member-type time content belongs to one fiber of $\text{Sig}_R^{\text{time,ord,struct,member}}$, and every dependent-content-type time content belongs to one fiber of $\text{Sig}_R^{\text{time,ord,struct,dependency}}$.

Proof. Each component of an indicator function takes one of

$$0 \quad \text{or} \quad 1$$

as its value. Hence each signature map assigns every element of its domain a unique element of

$$\{0, 1\}^3$$

The fibers of a map decompose its domain into pairwise disjoint inverse images; hence every element belongs to one fiber. $\square$

Definition 7.26 (Audit-Boundary Support Sets and Audit-Boundary Edges). For each $R \in \text{Ref}^{\text{audit}}$, assume that the audited transfer-event set TransferEvent_R is finite. Define the audit-edge category domain, the edge carriers for each category, and the set of audit-boundary-edge types by

$$\begin{aligned} \mathcal{I}_{\text{audit,edge}} &:= \{\text{transfer}, \text{result}\}, \\ \text{EdgItem}_R^{\text{audit,transfer}} &:= \text{TransferEvent}_R, \\ \text{EdgItem}_R^{\text{audit,result}} &:= \text{Result}_R, \\ \text{Type}^{\text{audit,edge}} &:= \left\{ \text{Type}_{\text{transfer}}^{\text{audit,edge}}, \text{Type}_{\text{result}}^{\text{audit,edge}} \right\} \end{aligned}$$

For each $\kappa_{\text{edge}} \in \mathcal{I}_{\text{audit,edge}}$, let the source-support map be

$$\text{Support}_R^{\text{audit},\kappa_{\text{edge}}} : \text{EdgItem}_R^{\text{audit},\kappa_{\text{edge}}} \longrightarrow \mathcal{P}_{\text{fin}} \left(\text{Node}_R^{\text{audit,claim}} \right)$$

For each $\epsilon \in \text{TransferEvent}_R$,

$$\text{Support}_R^{\text{audit,transfer}}(\epsilon)$$

consists of boundary-claim nodes that state the transfer input domain, output domain, and direction. For each $\eta \in \text{Result}_R$,

$$\text{Support}_R^{\text{audit,result}}(\eta)$$

consists of boundary-claim nodes that state the derivation of the result.

Define the edge set and edge-forming map for each edge category by

$$\begin{aligned} \text{Edge}_R^{\text{audit},\kappa_{\text{edge}}} &:= \left\{ \text{Type}_{\kappa_{\text{edge}}}^{\text{audit,edge}} \right\} \times \text{EdgItem}_R^{\text{audit},\kappa_{\text{edge}}}, \\ \text{EdgeOf}_R^{\text{audit},\kappa_{\text{edge}}} : \\ \text{EdgItem}_R^{\text{audit},\kappa_{\text{edge}}} &\longrightarrow \text{Edge}_R^{\text{audit},\kappa_{\text{edge}}}, \\ z &\longmapsto \left(\text{Type}_{\kappa_{\text{edge}}}^{\text{audit,edge}}, z \right) \end{aligned}$$

This map is injective, and its image equals the corresponding edge set.

Define the set of audit-boundary edges by

$$\text{Edge}_R^{\text{audit,bd}} := \text{Edge}_R^{\text{audit,transfer}} \cup \text{Edge}_R^{\text{audit,result}}$$

Define the edge-type map by

$$\text{TypeOf}_R^{\text{audit,edge}} : \text{Edge}_R^{\text{audit,bd}} \longrightarrow \text{Type}^{\text{audit,edge}}, \quad \text{TypeOf}_R^{\text{audit,edge}}(t, z) := t$$

Definition 7.27 (Audit-Boundary Directed Hypergraph). For each $R \in \text{Ref}^{\text{audit}}$, define the input and output maps of audit-boundary edges by

$$\begin{aligned} \text{In}_R^{\text{audit}} : \text{Edge}_R^{\text{audit,bd}} &\longrightarrow \mathcal{P}_{\text{fin}} \left(\text{Node}_R^{\text{audit,bd}} \right), \\ \text{Out}_R^{\text{audit}} : \text{Edge}_R^{\text{audit,bd}} &\longrightarrow \mathcal{P}_{\text{fin}} \left(\text{Node}_R^{\text{audit,bd}} \right) \end{aligned}$$

For each $\epsilon \in \text{TransferEvent}_R$ and each $\eta \in \text{Result}_R$, define the corresponding inputs and outputs, respectively, by

$$\begin{aligned} \text{In}_R^{\text{audit}} \left(\text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon) \right) &:= \text{Support}_R^{\text{audit,transfer}}(\epsilon) \cup \text{NodeOf}_R^{\text{audit,time}} [\text{TransferIn}_{R,\epsilon}], \\ \text{Out}_R^{\text{audit}} \left(\text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon) \right) &:= \text{NodeOf}_R^{\text{audit,time}} [\text{TransferOut}_{R,\epsilon}], \\ \text{In}_R^{\text{audit}} \left(\text{EdgeOf}_R^{\text{audit,result}}(\eta) \right) &:= \text{Support}_R^{\text{audit,result}}(\eta) \cup \text{NodeOf}_R^{\text{audit,time}} [\text{ResultPrereq}_{R,\eta}], \\ \text{Out}_R^{\text{audit}} \left(\text{EdgeOf}_R^{\text{audit,result}}(\eta) \right) &:= \left\{ \text{NodeOf}_R^{\text{audit,result}}(\eta) \right\} \end{aligned}$$

Define the audit-boundary directed hypergraph by

$$\text{Hypergraph}_R^{\text{audit}} := \left(\text{Node}_R^{\text{audit,bd}}, \text{Edge}_R^{\text{audit,bd}}, \right. \\ \left. \text{In}_R^{\text{audit}}, \text{Out}_R^{\text{audit}}, \right. \\ \left. \text{TypeOf}_R^{\text{audit,node}}, \text{TypeOf}_R^{\text{audit,edge}} \right)$$

The simultaneous establishment of all nodes in an edge's input set is the establishment condition for the nodes in that edge's output set.

Let the specified subset consisting of physical or mathematical boundary-claim nodes independent of the derivation of time content be

$$\text{Init}_R^{\text{audit,phys}} \subseteq \text{Node}_R^{\text{audit,claim}}$$

and call it the physical-input initial set.

Definition 7.28 (Generation Initial Set and Reference Formal-Evidence Environment). For each $R \in \text{Ref}^{\text{audit}}$, let the specified subset consisting of boundary-claim nodes that provide the reference-specific formal derivation be

$$\text{Env}_R^{\text{audit,evid}} \subseteq \text{Node}_R^{\text{audit,claim}}$$

and call it the reference formal-evidence environment. The physical-input initial set and the reference formal-evidence environment satisfy

$$\text{Init}_R^{\text{audit,phys}} \subseteq \text{Env}_R^{\text{audit,evid}} \subseteq \text{Node}_R^{\text{audit,claim}}$$

Proposition 7.29 (Cardinalities of the Audit-Boundary Node and Edge Sets). For each $R \in \text{Ref}^{\text{audit}}$, the audit-boundary node set and the audit-boundary edge set are finite, and

$$\begin{aligned} |\text{Node}_R^{\text{audit,bd}}| &= |\text{Node}_R^{\text{audit,claim}}| + |\text{TimeContent}_R| \\ &\quad + |\text{Result}_R|, \\ |\text{Edge}_R^{\text{audit,bd}}| &= |\text{TransferEvent}_R| + |\text{Result}_R|. \end{aligned}$$

Proof. By Definition 7.18, the audit-boundary node set is the disjoint union of the physical- or formal-input claim-node set, the time-content node set, and the result node set. The time-content node-forming map and the result node-forming map provide bijections from, respectively,

$$\text{TimeContent}_R, \quad \text{Result}_R$$

to the corresponding node sets. Hence the first equality holds.

By Definition 7.26, the audit-boundary edge set is the disjoint union of the transfer-edge set and the result-edge set. The transfer-edge-forming map and the result-edge-forming map provide bijections from, respectively,

$$\text{TransferEvent}_R, \quad \text{Result}_R$$

to the corresponding edge sets. Hence the second equality holds. The finiteness of each disjoint-union component implies that the audit-boundary node set and the audit-boundary edge set are finite. $\square$

Definition 7.30 (Reachability Stages and Reachable Set on the Audit-Boundary Directed Hypergraph). For each $R \in \text{Ref}^{\text{audit}}$ and $\text{Seed} \subseteq \text{Node}_R^{\text{audit,bd}}$, define the initial reachability stage by

$$\text{Reach}_R^{\text{audit},(0)}(\text{Seed}) := \text{Seed}$$

For each

$$k \in \mathbb{N}$$

define the successor reachability stage by

$$\begin{aligned} & \text{Reach}_R^{\text{audit},(k+1)}(\text{Seed}) \\ &:= \text{Reach}_R^{\text{audit},(k)}(\text{Seed}) \\ & \quad \cup \bigcup_{\substack{\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}} \\ \text{In}_R^{\text{audit}}(\text{AuditEdge}) \subseteq \text{Reach}_R^{\text{audit},(k)}(\text{Seed})}} \text{Out}_R^{\text{audit}}(\text{AuditEdge}) \end{aligned}$$

Define the audit-reachable set by

$$\begin{aligned} & \text{Reach}_R^{\text{audit}}(\text{Seed}) \\ &:= \text{Reach}_R^{\text{audit},(|\text{Node}_R^{\text{audit,bd}}|)}(\text{Seed}) \end{aligned}$$

Proposition 7.31 (Finite Stabilization of Audit Reachability Stages). For each $R \in \text{Ref}^{\text{audit}}$ and $\text{Seed} \subseteq \text{Node}_R^{\text{audit,bd}}$, the reachability-stage sequence is monotone increasing with respect to inclusion. Moreover, if

$$k \geq |\text{Node}_R^{\text{audit,bd}}|$$

then

$$\begin{aligned} & \text{Reach}_R^{\text{audit},(k)}(\text{Seed}) \\ &= \text{Reach}_R^{\text{audit}}(\text{Seed}) \end{aligned}$$

holds.

Proof. The recursion in Definition 7.30 expresses each stage as its union with the immediately preceding stage. The reachability-stage sequence is therefore monotone increasing.

The audit-boundary node set is finite, and every strict inclusion adds at least one node. Once two adjacent stages coincide, the same recursion makes all subsequent stages coincide. Hence the reachability-stage sequence stabilizes by stage

$$|\text{Node}_R^{\text{audit,bd}}|$$

and its stable value equals the audit-reachable set. $\square$

Definition 7.32 (Audit-Generation Traces). For each $R \in \text{Ref}^{\text{audit}}$, $\text{Seed} \subseteq \text{Node}_R^{\text{audit,bd}}$, and $\text{AuditNode} \in \text{Node}_R^{\text{audit,bd}}$, let

$$\text{Trace}_R^{\text{audit}}(\text{Seed}, \text{AuditNode})$$

be the set of all finite edge sequences satisfying the following conditions.

For an edge sequence of length $\ell \in \mathbb{N}$, $\left(\text{AuditEdge}^{(j)}\right)_{j=1}^{\ell}$, define

$$\text{TraceNodeSet}^{(0)} := \text{Seed}$$

and

$$\begin{aligned} \text{TraceNodeSet}^{(j)} &:= \text{TraceNodeSet}^{(j-1)} \\ &\cup \text{Out}_R^{\text{audit}} \left(\text{AuditEdge}^{(j)} \right) \end{aligned}$$

For each

$$j \in \{1, \dots, \ell\}$$

require

$$\text{AuditEdge}^{(j)} \in \text{Edge}_R^{\text{audit, bd}}$$

and

$$\begin{aligned} &\text{In}_R^{\text{audit}} \left(\text{AuditEdge}^{(j)} \right) \\ &\subseteq \text{TraceNodeSet}^{(j-1)} \end{aligned}$$

and further require

$$\text{AuditNode} \in \text{TraceNodeSet}^{(\ell)}$$

An edge sequence satisfying these conditions is called an audit-generation trace. The edge sequence of length zero belongs to the set when

$$\text{AuditNode} \in \text{Seed}$$

Lemma 7.33 (Equivalence of Audit Reachability and Audit-Generation Traces). For each $R \in \text{Ref}^{\text{audit}}$, $\text{Seed} \subseteq \text{Node}_R^{\text{audit, bd}}$, and $\text{AuditNode} \in \text{Node}_R^{\text{audit, bd}}$,

$$\begin{aligned} &\text{AuditNode} \in \text{Reach}_R^{\text{audit}}(\text{Seed}) \\ &\iff \text{Trace}_R^{\text{audit}}(\text{Seed}, \text{AuditNode}) \neq \emptyset \end{aligned}$$

holds.

Proof. For the forward direction, a node in the initial reachability stage has a corresponding edge sequence of length zero. A node first added at reachability stage

$$k + 1$$

belongs to the output of an audit-boundary edge whose input set is contained in reachability stage

$$k$$

Ordering by stage the finite edge sequences corresponding to the input nodes and appending the audit-boundary edge yields a finite edge sequence that satisfies the all-input containment condition in Definition 7.32. Induction over reachability stages therefore gives an audit-generation trace for every node in the audit-reachable set.

For the reverse direction, the terminal node of an audit-generation trace of length zero belongs to the initial reachability stage. In an audit-generation trace of length

$$\ell + 1$$

all inputs of the terminal edge are contained in the preceding trace-node set. Induction on edge-sequence length, together with the recursion in Definition 7.30, shows that the terminal node belongs to a finite reachability stage. By Proposition 7.31, that node belongs to the audit-reachable set. $\square$

Definition 7.34 (Structural and Essential Influence Regions of a Physical-Input Node). For each $R \in \text{Ref}^{\text{audit}}$ and $\text{InitNode} \in \text{Init}_R^{\text{audit,phys}}$, define the initial influence stage by

$$\text{Influence}_R^{\text{audit},(0)}(\text{InitNode}) := \{\text{InitNode}\}$$

For each

$$k \in \mathbb{N}$$

define the successor influence stage by

$$\begin{aligned} & \text{Influence}_R^{\text{audit},(k+1)}(\text{InitNode}) \\ &:= \text{Influence}_R^{\text{audit},(k)}(\text{InitNode}) \\ & \quad \cup \bigcup_{\substack{\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}} \\ \text{In}_R^{\text{audit}}(\text{AuditEdge}) \subseteq \text{Reach}_R^{\text{audit},(k)}(\text{Init}_R^{\text{audit,phys}}) \\ \text{In}_R^{\text{audit}}(\text{AuditEdge}) \cap \text{Influence}_R^{\text{audit},(k)}(\text{InitNode}) \neq \emptyset}} \text{Out}_R^{\text{audit}}(\text{AuditEdge}) \end{aligned}$$

Define the structural influence region by

$$\begin{aligned} & \text{Influence}_R^{\text{audit}}(\text{InitNode}) \\ &:= \text{Influence}_R^{\text{audit},(|\text{Node}_R^{\text{audit,bd}}|)}(\text{InitNode}) \end{aligned}$$

Define the essential influence region by

$$\begin{aligned} & \text{EssentialInfluence}_R^{\text{audit}}(\text{InitNode}) \\ &:= \text{Reach}_R^{\text{audit}}(\text{Init}_R^{\text{audit,phys}}) \\ & \quad \setminus \text{Reach}_R^{\text{audit}}(\text{Init}_R^{\text{audit,phys}} \setminus \{\text{InitNode}\}) \end{aligned}$$

Proposition 7.35 (Inclusion of the Essential Influence Region in the Structural Influence Region). For each $R \in \text{Ref}^{\text{audit}}$ and $\text{InitNode} \in \text{Init}_R^{\text{audit,phys}}$,

$$\begin{aligned} & \text{EssentialInfluence}_R^{\text{audit}}(\text{InitNode}) \\ & \subseteq \text{Influence}_R^{\text{audit}}(\text{InitNode}) \end{aligned}$$

holds.

Proof. Proceed by induction over the reachability stages from the physical-input initial set. At the initial stage,

$$\text{InitNode}$$

is the only node not added to the reachable set from the initial set with that node removed, and this node belongs to the initial influence stage.

At reachability stage

$$k + 1$$

take a node

$$\text{AuditNode}$$

that first belongs to the essential influence region. All inputs of an audit-boundary edge

$$\text{AuditEdge}$$

that outputs this node belong to reachability stage

$$k$$

Suppose that all inputs belong to the reachable set from

$$\text{Init}_R^{\text{audit,phys}} \setminus \{\text{InitNode}\}$$

The same edge would then place

$$\text{AuditNode}$$

in that reachable set, contrary to the definition of the essential influence region. Hence the edge has an input node that belongs to the essential influence region at a preceding stage. By the induction hypothesis, that input node belongs to influence stage

$$k$$

By the recursion in Definition 7.34,

$$\text{AuditNode}$$

belongs to influence stage

$$k + 1$$

Finite stabilization yields the inclusion. $\square$

Definition 7.36 (Reference-Specific Obligation Atoms). For each $R \in \text{Ref}^{\text{audit}}$, define reference-specific obligation atoms by the following four types. An obligation atom is the smallest unit of obligation to which one audit state is assigned in the generation-dependency audit.

1. A boundary-claim atom is $\text{ClaimNode} \in \text{Node}_R^{\text{audit,claim}}$.
2. A time-content atom is $x \in \text{TimeContent}_R$, and its audit-boundary node is $\text{NodeOf}_R^{\text{audit,time}}(x)$.
3. A transfer-event atom is $\epsilon \in \text{TransferEvent}_R$, and its audit-boundary edge is

$$\text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon)$$

4. A result atom is $\eta \in \text{Result}_R$, and its audit-boundary edge and result node are

$$\text{EdgeOf}_R^{\text{audit,result}}(\eta), \quad \text{NodeOf}_R^{\text{audit,result}}(\eta)$$

Definition 7.37 (Typed Audit States of Reference-Specific Obligation Atoms). For each $R \in \text{Ref}^{\text{audit}}$, describe the typed audit states of reference-specific obligation atoms by the following components.

For a boundary-claim node $\text{ClaimNode} \in \text{Node}_R^{\text{audit,claim}}$, the state component is

$$\left(\text{ClaimNode}, \right. \\ \text{TypeOf}_R^{\text{audit,node}}(\text{ClaimNode}), \\ \left. \mathbf{1}_{\text{Reach}_R^{\text{audit}}}(\text{Init}_R^{\text{audit,phys}})(\text{ClaimNode}) \right)$$

For a time content $x \in \text{TimeContent}_R$, the state component is

$$\begin{aligned} & \left(x, \text{NodeOf}_R^{\text{audit,time}}(x), \right. \\ & \quad \text{TypeOf}_R^{\text{time,ord,struct}}(x), \\ & \quad \text{AxisOf}_R^{\text{time,ord,struct}}(x), \\ & \quad \left. \mathbf{1}_{\text{Reach}_R^{\text{audit}}}(\text{Init}_R^{\text{audit,phys}})(\text{NodeOf}_R^{\text{audit,time}}(x)) \right) \end{aligned}$$

For a transfer event $\epsilon \in \text{TransferEvent}_R$, the state component is

$$\begin{aligned} & \left(\epsilon, \text{Support}_R^{\text{audit,transfer}}(\epsilon), \right. \\ & \quad \text{TransferIn}_{R,\epsilon}, \text{TransferOut}_{R,\epsilon}, \\ & \quad \text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon), \\ & \quad \text{In}_R^{\text{audit}}(\text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon)), \\ & \quad \left. \text{Out}_R^{\text{audit}}(\text{EdgeOf}_R^{\text{audit,transfer}}(\epsilon)) \right) \end{aligned}$$

For a result $\eta \in \text{Result}_R$, the state component is

$$\begin{aligned} & \left(\eta, \text{Support}_R^{\text{audit,result}}(\eta), \right. \\ & \quad \text{ResultPrereq}_{R,\eta}, \text{EdgeOf}_R^{\text{audit,result}}(\eta), \\ & \quad \text{In}_R^{\text{audit}}(\text{EdgeOf}_R^{\text{audit,result}}(\eta)), \\ & \quad \text{NodeOf}_R^{\text{audit,result}}(\eta), \\ & \quad \left. \mathbf{1}_{\text{Reach}_R^{\text{audit}}}(\text{Init}_R^{\text{audit,phys}})(\text{NodeOf}_R^{\text{audit,result}}(\eta)) \right) \end{aligned}$$

Proposition 7.38 (Determinacy of Typed Audit States). For every reference-specific obligation atom, all components in Definition 7.37 are uniquely determined by $\text{Hypergraph}_R^{\text{audit}}$, $\text{Init}_R^{\text{audit,phys}}$, $\text{TypeOf}_R^{\text{time,ord,struct}}$, and $\text{AxisOf}_R^{\text{time,ord,struct}}$. Consequently, the reachability state of the same obligation atom in the same reference is unique.

Proof. The node type and edge type are respectively the values of the maps

$$\text{TypeOf}_R^{\text{audit,node}}, \quad \text{TypeOf}_R^{\text{audit,edge}}$$

The structural type and axis set of a time content are respectively the values of the maps

$$\text{TypeOf}_R^{\text{time,ord,struct}}, \quad \text{AxisOf}_R^{\text{time,ord,struct}}$$

The support set of a transfer event or result is recovered as the intersection of the corresponding edge input with the claim-node set. Time-content inputs and outputs are uniquely recovered by taking the inverse images of the time-content-node components of the corresponding edge inputs and outputs under the injective time-content node-forming map.

By Proposition 7.31, the audit-reachable set from the physical-input initial set is uniquely determined after finitely many stages. Each indicator-function value is therefore uniquely determined, and all state components are fixed. $\square$

Definition 7.39 (Audit-Generated Time Contents, Primitive-Time Roots, and Audit-Generated Results). For each $R \in \text{Ref}^{\text{audit}}$, define the audit-generated time-content set, primitive-time-root set, audit-generated result set, and result set outside the generation-reachable region, respectively, by

$$\begin{aligned}\text{TimeContent}_R^{\text{audit,gen}} &:= \left(\text{NodeOf}_R^{\text{audit,time}} \right)^{-1} \left[\text{Reach}_R^{\text{audit}} \left(\text{Init}_R^{\text{audit,phys}} \right) \right], \\ \text{TimeContent}_R^{\text{audit,root}} &:= \text{TimeContent}_R \setminus \text{TimeContent}_R^{\text{audit,gen}}, \\ \text{Result}_R^{\text{audit,gen}} &:= \left(\text{NodeOf}_R^{\text{audit,result}} \right)^{-1} \left[\text{Reach}_R^{\text{audit}} \left(\text{Init}_R^{\text{audit,phys}} \right) \right], \\ \text{Result}_R^{\text{audit,unreach}} &:= \text{Result}_R \setminus \text{Result}_R^{\text{audit,gen}}\end{aligned}$$

Definition 7.40 (Generation Reach and Evidence-Conditioned Dependency Reach). For each $R \in \text{Ref}^{\text{audit}}$, define the generation-reachable set by

$$\text{Reach}_R^{\text{audit,gen}} := \text{Reach}_R^{\text{audit}} \left(\text{Init}_R^{\text{audit,phys}} \right)$$

For each $\text{RootSeed} \subseteq \text{TimeContent}_R^{\text{audit,root}}$, define the evidence-conditioned dependency-reachable set by

$$\begin{aligned}\text{Reach}_R^{\text{audit|evid}}(\text{RootSeed}) \\ &:= \text{Reach}_R^{\text{audit}} \left(\text{Init}_R^{\text{audit,phys}} \cup \text{Env}_R^{\text{audit,evid}} \right. \\ &\quad \left. \cup \text{NodeOf}_R^{\text{audit,time}}[\text{RootSeed}] \right)\end{aligned}$$

Definition 7.41 (Time Contents Reachable under the Evidence Environment). For each $R \in \text{Ref}^{\text{audit}}$, define the set of time contents reached after adding the reference formal-evidence environment without supplying primitive-time roots as seeds by

$$\text{TimeContent}_R^{\text{audit,evid}} := \left(\text{NodeOf}_R^{\text{audit,time}} \right)^{-1} \left[\text{Reach}_R^{\text{audit|evid}}(\emptyset) \right]$$

Call the assessment based on the audit-generated time-content set $\text{TimeContent}_R^{\text{audit,gen}}$ the LCTR complete-generation assessment. Call the assessment based on the foregoing set of time contents reachable under the evidence environment the within-reference evidence-conditioned local-reach assessment. Below, abbreviate these as the complete assessment and the local-reach assessment, respectively. The latter is a reachability assessment that adds, as the reference formal-evidence environment, the physical-carrier processing or construction procedures explicitly stated in that reference. Because the two assessments have different starting sets, nonemptiness under the latter does not entail nonemptiness under the former.

In the reference-specific assessment values, abbreviate the number of time contents generated under the complete assessment as “complete,” the number of time contents reached under the local-reach assessment as “local reach,” and the number of primitive-time roots that remain after the latter assessment as “remaining roots.”

Definition 7.42 (Nongenerated Results Reachable under the Evidence Environment and Residual Results). For each $R \in \text{Ref}^{\text{audit}}$, define the set of nongenerated results reachable

under the evidence environment by

$$\begin{aligned} & \text{Result}_R^{\text{audit,evidonly}} \\ & := \left\{ \eta \in \text{Result}_R \mid \text{NodeOf}_R^{\text{audit,result}}(\eta) \in \text{Reach}_R^{\text{audit|evid}}(\emptyset) \right\} \\ & \quad \setminus \text{Result}_R^{\text{audit,gen}} \end{aligned}$$

Define the residual result set by

$$\begin{aligned} & \text{Result}_R^{\text{audit,resid}} \\ & := \text{Result}_R^{\text{audit,unreach}} \setminus \text{Result}_R^{\text{audit,evidonly}} \end{aligned}$$

Proposition 7.43 (Time-Antecedent Preservation Condition). For each $R \in \text{Ref}^{\text{audit}}$, if every $\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}}$ satisfies

$$\begin{aligned} & \text{Out}_R^{\text{audit}}(\text{AuditEdge}) \cap \text{Node}_R^{\text{audit,time}} \neq \emptyset \\ \implies & \text{In}_R^{\text{audit}}(\text{AuditEdge}) \cap \text{Node}_R^{\text{audit,time}} \neq \emptyset \end{aligned}$$

then

$$\begin{aligned} & \text{Reach}_R^{\text{audit}}(\text{Init}_R^{\text{audit,phys}}) \cap \text{Node}_R^{\text{audit,time}} \\ & = \emptyset \end{aligned}$$

holds.

Proof. By Definition 7.27,

$$\text{Init}_R^{\text{audit,phys}} \subseteq \text{Node}_R^{\text{audit,claim}}.$$

The disjoint-union decomposition in Definition 7.18 implies that the initial reachability stage has empty intersection with the time-content node set.

Suppose that reachability stage

k

has empty intersection with the time-content node set. By hypothesis, every audit-boundary edge that outputs a time-content node has at least one time-content node among its inputs. Its input set is therefore not a subset of reachability stage

k

Consequently, reachability stage

$k + 1$

also has empty intersection with the time-content node set. Induction over reachability stages and Proposition 7.31 give the claim. $\square$

Definition 7.44 (Evidence-Conditioned Primitive-Time-Root Support Family and Minimal Support Family). For each $R \in \text{Ref}^{\text{audit}}$ and $\eta \in \text{Result}_R$, define the evidence-conditioned primitive-time-root support family by

$$\begin{aligned} & \text{RootSupport}_R^{\text{audit|evid}}(\eta) \\ & := \left\{ \text{RootSeed} \subseteq \text{TimeContent}_R^{\text{audit,root}} \mid \right. \\ & \quad \left. \text{NodeOf}_R^{\text{audit,result}}(\eta) \in \text{Reach}_R^{\text{audit|evid}}(\text{RootSeed}) \right\} \end{aligned}$$

Define the minimal primitive-time-root support family by

$$\begin{aligned} & \text{MinRootSupport}_R^{\text{audit}}(\eta) \\ & := \left\{ \text{RootSeed} \in \text{RootSupport}_R^{\text{audit|evid}}(\eta) \mid \right. \\ & \quad \forall \text{RootSeed}' \in \text{RootSupport}_R^{\text{audit|evid}}(\eta), \\ & \quad \left. \text{RootSeed}' \subseteq \text{RootSeed} \implies \text{RootSeed} \subseteq \text{RootSeed}' \right\} \end{aligned}$$

Proposition 7.45 (Upward Closure and Minimal Elements of the Evidence-Conditioned Primitive-Time-Root Support Family). For each $R \in \text{Ref}^{\text{audit}}$ and $\eta \in \text{Result}_R$, the evidence-conditioned primitive-time-root support family is upward closed under inclusion. Moreover, if

$$\text{RootSupport}_R^{\text{audit|evid}}(\eta) \neq \emptyset$$

then

$$\text{MinRootSupport}_R^{\text{audit}}(\eta) \neq \emptyset$$

holds, and any two distinct elements of the minimal primitive-time-root support family are incomparable under inclusion.

Proof. To prove upward closure, take arbitrary $\text{RootSeed} \in \text{RootSupport}_R^{\text{audit|evid}}(\eta)$ and $\text{RootSeed} \subseteq \text{RootSeed}' \subseteq \text{TimeContent}_R^{\text{audit,root}}$. Induction on the recursion in Definition 7.30 shows that inclusion of the starting sets entails inclusion at every reachability stage. Hence $\text{RootSeed}' \in \text{RootSupport}_R^{\text{audit|evid}}(\eta)$ holds.

Because the primitive-time-root set is finite, its power set and the evidence-conditioned primitive-time-root support family are finite. Choose an element of minimum cardinality from the nonempty evidence-conditioned primitive-time-root support family. Every element of the same support family contained in that element equals it by minimality of cardinality. Hence the minimal primitive-time-root support family is nonempty.

If one of two distinct elements of the minimal primitive-time-root support family were contained in the other, the minimality of the containing element would make them equal. Thus two distinct elements are incomparable under inclusion. $\square$

Definition 7.46 (Finite Root-Support Antichain Update). For a family of subsets of the primitive-time-root set

$$\mathcal{F} \subseteq \mathcal{P} \left(\text{TimeContent}_R^{\text{audit,root}} \right)$$

define minimization under inclusion by

$$\begin{aligned} & \text{Min}_{\subseteq}(\mathcal{F}) \\ & := \left\{ \text{RootSeed} \in \mathcal{F} \mid \forall \text{RootSeed}' \in \mathcal{F}, \quad \begin{array}{l} \text{RootSeed}' \subseteq \text{RootSeed} \\ \implies \text{RootSeed} \subseteq \text{RootSeed}' \end{array} \right\} \end{aligned}$$

For each $R \in \text{Ref}^{\text{audit}}$, define the initial root-support antichain assignment

$$\begin{aligned} & \text{RootAnt}_R^{\text{audit,init}} : \text{Node}_R^{\text{audit,bd}} \\ & \longrightarrow \mathcal{P} \left(\mathcal{P} \left(\text{TimeContent}_R^{\text{audit,root}} \right) \right) \end{aligned}$$

by

$$\begin{aligned} & \text{RootAnt}_R^{\text{audit,init}}(\text{AuditNode}) \\ & := \begin{cases} \{\emptyset\}, & \text{AuditNode} \in \text{Env}_R^{\text{audit,evid}}, \\ \{\{x\}\}, & \text{AuditNode} = \text{NodeOf}_R^{\text{audit,time}}(x), \quad x \in \text{TimeContent}_R^{\text{audit,root}}, \\ \emptyset, & \text{otherwise} \end{cases} \end{aligned}$$

For a map

$$\mathcal{S} : \text{Node}_R^{\text{audit,bd}} \longrightarrow \mathcal{P}\left(\mathcal{P}\left(\text{TimeContent}_R^{\text{audit,root}}\right)\right)$$

that assigns a support antichain to each node, and for each $\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}}$, define the hyperedge root-support update by

$$\begin{aligned} & \text{RootUpd}_{R,\text{AuditEdge}}^{\text{audit}}(\mathcal{S})(\text{AuditNode}) \\ & := \text{Min}_{\subseteq} \left(\mathcal{S}(\text{AuditNode}) \right. \\ & \quad \left. \cup \left\{ \begin{array}{l} \bigcup_{\text{InputNode} \in \text{In}_R^{\text{audit}}(\text{AuditEdge})} \text{RootSeed}_{\text{InputNode}} \\ \text{AuditNode} \in \text{Out}_R^{\text{audit}}(\text{AuditEdge}), \\ \mid \\ \forall \text{InputNode} \in \text{In}_R^{\text{audit}}(\text{AuditEdge}), \\ \text{RootSeed}_{\text{InputNode}} \in \mathcal{S}(\text{InputNode}) \end{array} \right\} \right) \end{aligned}$$

For a finite edge sequence $\left(\text{AuditEdge}^{(j)}\right)_{j=1}^{\ell}$, define

$$\mathcal{S}^{(0)} := \text{RootAnt}_R^{\text{audit,init}}$$

and

$$\mathcal{S}^{(j)} := \text{RootUpd}_{R,\text{AuditEdge}^{(j)}}^{\text{audit}}(\mathcal{S}^{(j-1)})$$

When the terminal assignment satisfies

$$\begin{aligned} & \forall \text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}}, \\ & \text{RootUpd}_{R,\text{AuditEdge}}^{\text{audit}}(\mathcal{S}^{(\ell)}) = \mathcal{S}^{(\ell)} \end{aligned}$$

say that the finite edge sequence saturates the root-support antichain.

Proposition 7.47 (Termination, Scan-Order Independence, and Completeness of the Finite Root-Support Antichain Update). For each $R \in \text{Ref}^{\text{audit}}$, there exists a finite edge sequence that saturates the root-support antichain.

If two finite edge sequences saturate the root-support antichain, their terminal assignments coincide. Moreover, for any $\eta \in \text{Result}_R$, the terminal assignment

$$\mathcal{S}^{(\ell)}$$

satisfies

$$\begin{aligned} & \mathcal{S}^{(\ell)}\left(\text{NodeOf}_R^{\text{audit,result}}(\eta)\right) \\ & = \text{MinRootSupport}_R^{\text{audit}}(\eta) \end{aligned}$$

Proof. Represent the support antichain at each node by its upward closure. The hyperedge root-support update adds to the upward closure at that node the upward closure of the union of supports selected one from each input node of the hyperedge. Minimization under inclusion does not change the upward closure. The hyperedge root-support update is therefore monotone and extensive with respect to the inclusion order on upward closures over the finite set

$$\prod_{\text{AuditNode} \in \text{Node}_R^{\text{audit}, \text{bd}}} \mathcal{P} \left(\mathcal{P} \left(\text{TimeContent}_R^{\text{audit}, \text{root}} \right) \right)$$

. Repeatedly selecting a hyperedge that produces a change reaches a fixed point for all hyperedges after finitely many steps.

The initial root-support antichain assignment assigns the empty support to the reference formal-evidence environment nodes and the singleton support containing only that root to each primitive-time-root node. Induction on the reachability stages in Definition 7.30 shows that the upward closure of the support family at each node at the fixed point equals the family of all subsets of primitive-time roots that place that node in the evidence-conditioned dependency-reachability set. The converse follows by induction that maps each hyperedge root-support update to one stage of the reachability recursion.

The fixed point is therefore unique and independent of the order in which the edges are scanned. Taking the inclusion-minimal elements at a result node yields the minimal primitive-time-root support family in Definition 7.44 . $\square$

Definition 7.48 (Residual Result Sets with Respect to Primitive-Time Roots). For each $R \in \text{Ref}^{\text{audit}}$, define the primitive-time-root-dependent result set by

$$\begin{aligned} & \text{Result}_R^{\text{audit}, \text{rootdep}} \\ & := \left\{ \eta \in \text{Result}_R^{\text{audit}, \text{resid}} \mid \text{RootSupport}_R^{\text{audit}|\text{evid}}(\eta) \neq \emptyset \right\} \end{aligned}$$

Define the result set that remains unreachable after addition of primitive-time roots by

$$\begin{aligned} & \text{Result}_R^{\text{audit}, \text{rootunreach}} \\ & := \left\{ \eta \in \text{Result}_R^{\text{audit}, \text{resid}} \mid \text{RootSupport}_R^{\text{audit}|\text{evid}}(\eta) = \emptyset \right\} \end{aligned}$$

Proposition 7.49 (Four-Class Partition of Reference-Specific Results). For each $R \in \text{Ref}^{\text{audit}}$,

$$\begin{aligned} \text{Result}_R &= \text{Result}_R^{\text{audit}, \text{gen}} \dot{\cup} \text{Result}_R^{\text{audit}, \text{evidonly}} \\ &\quad \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootdep}} \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootunreach}} \end{aligned}$$

holds. Moreover,

$$\begin{aligned} \text{Result}_R^{\text{audit}, \text{unreach}} &= \text{Result}_R^{\text{audit}, \text{evidonly}} \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootdep}} \\ &\quad \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootunreach}} \end{aligned}$$

and

$$\text{Result}_R^{\text{audit}, \text{resid}} = \text{Result}_R^{\text{audit}, \text{rootdep}} \dot{\cup} \text{Result}_R^{\text{audit}, \text{rootunreach}}$$

hold.

Proof. The inclusion of starting sets in Definition 7.28 and monotonicity of reachability give

$$\text{Reach}_R^{\text{audit,gen}} \subseteq \text{Reach}_R^{\text{audit|evid}}(\emptyset)$$

. The evidence-environment-reachable nongenerated result set is obtained by removing the audit-generated results from the results reachable from the evidence-conditioned empty-root starting set. The two sets are therefore disjoint.

The residual result set is obtained by removing the evidence-environment-reachable nongenerated result set from the result set outside the generation-reachability region. For each residual result, the evidence-conditioned primitive-time-root support family is either empty or nonempty. The former case gives the result set that remains unreachable after addition of primitive-time roots; the latter gives the primitive-time-root-dependent result set. Hence the four sets are pairwise disjoint and their union is the entire result set. Applying the definitions to the result set outside the generation-reachability region and to the residual result set gives the other two equalities. $\square$

Definition 7.50 (Primitive-Time-Root-Result Dependency Relations). For each $R \in \text{Ref}^{\text{audit}}$, define the primitive-time-root-result dependency relation by

$$\begin{aligned} & \text{RootResultDep}_R^{\text{audit}} \\ &:= \left\{ (x, \eta) \in \text{TimeContent}_R^{\text{audit,root}} \times \text{Result}_R^{\text{audit,rootdep}} \mid \right. \\ & \quad \left. \exists \text{RootSeed} \in \text{MinRootSupport}_R^{\text{audit}}(\eta), \quad x \in \text{RootSeed} \right\} \end{aligned}$$

Define the essential primitive-time-root-result dependency relation by

$$\begin{aligned} & \text{RootResultEssDep}_R^{\text{audit}} \\ &:= \left\{ (x, \eta) \in \text{RootResultDep}_R^{\text{audit}} \mid \right. \\ & \quad \left. \forall \text{RootSeed} \in \text{MinRootSupport}_R^{\text{audit}}(\eta), \quad x \in \text{RootSeed} \right\} \end{aligned}$$

Proposition 7.51 (Inclusion and Result Projection of Primitive-Time-Root-Result Dependency Relations). For each $R \in \text{Ref}^{\text{audit}}$,

$$\begin{aligned} & \text{RootResultEssDep}_R^{\text{audit}} \\ & \subseteq \text{RootResultDep}_R^{\text{audit}} \end{aligned}$$

holds. Moreover,

$$\begin{aligned} & \left\{ \eta \in \text{Result}_R^{\text{audit,rootdep}} \mid \exists x \in \text{TimeContent}_R^{\text{audit,root}}, \right. \\ & \quad \left. (x, \eta) \in \text{RootResultDep}_R^{\text{audit}} \right\} \\ &= \text{Result}_R^{\text{audit,rootdep}} \end{aligned}$$

holds.

Proof. The first inclusion follows from Definition 7.50 .

Take any $\eta \in \text{Result}_R^{\text{audit}, \text{rootdep}}$. By Definition 7.48 the primitive-time-root support family is nonempty. By Proposition 7.45 the minimal primitive-time-root support family is also nonempty. Take any

$$\text{RootSeed} \in \text{MinRootSupport}_R^{\text{audit}}(\eta)$$

. Because the evidence-conditioned primitive-time-root support family of a residual result does not contain the empty set, there exists some

$$x \in \text{RootSeed}$$

such that

$$(x, \eta) \in \text{RootResultDep}_R^{\text{audit}}$$

holds.

The reverse inclusion follows from

$$\text{RootResultDep}_R^{\text{audit}} \subseteq \text{TimeContent}_R^{\text{audit}, \text{root}} \times \text{Result}_R^{\text{audit}, \text{rootdep}}$$

.

□

Definition 7.52 (Reference Subsets Induced by Primitive-Time Roots and the Result Decomposition). Define the result-class index domain by

$$\mathcal{I}_{\text{res}} := \{\text{evidonly}, \text{resid}, \text{rootdep}, \text{rootunreach}, \text{unreach}\}$$

. In order, the indices denote nongenerated results reachable under the evidence environment, residual results, primitive-time-root-dependent results, results that remain unreachable after addition of primitive-time roots, and results outside the generation-reachability region.

Define the set of references containing primitive-time roots by

$$\text{Ref}^{\text{audit}, \text{root}} := \left\{ R \in \text{Ref}^{\text{audit}} \mid \text{TimeContent}_R^{\text{audit}, \text{root}} \neq \emptyset \right\}$$

.

For each $\rho \in \mathcal{I}_{\text{res}} \setminus \{\text{resid}\}$, define the set of references containing results in the corresponding class by

$$\text{Ref}^{\text{audit}, \rho} := \left\{ R \in \text{Ref}^{\text{audit}} \mid \text{Result}_R^{\text{audit}, \rho} \neq \emptyset \right\}$$

.

Proposition 7.53 (Membership Inclusion for Primitive-Time-Root-Dependent References).

$$\text{Ref}^{\text{audit}, \text{rootdep}} \subseteq \text{Ref}^{\text{audit}, \text{root}} \cap \text{Ref}^{\text{audit}, \text{unreach}}$$

holds.

Proof. Take any

$$R \in \text{Ref}^{\text{audit}, \text{rootdep}}$$

. By Definition 7.52, there exists some $\eta \in \text{Result}_R^{\text{audit}, \text{rootdep}}$. By Definition 7.48, $\text{RootSupport}_R^{\text{audit|evid}}(\eta) \neq \emptyset$. Choose

$$\text{RootSeed} \in \text{RootSupport}_R^{\text{audit|evid}}(\eta)$$

from this support family.

Because the result η is residual, it is not reachable from the empty-root starting set augmented by the evidence environment. Hence

$$\text{RootSeed} \neq \emptyset.$$

By Definition 7.44, $\text{RootSeed} \subseteq \text{TimeContent}_R^{\text{audit,root}}$, so the primitive-time-root set is nonempty. Thus

$$R \in \text{Ref}^{\text{audit,root}}.$$

By Proposition 7.49,

$$\text{Result}_R^{\text{audit,rootdep}} \subseteq \text{Result}_R^{\text{audit,unreach}}.$$

Because the left-hand side is nonempty, so is the right-hand side, and therefore

$$R \in \text{Ref}^{\text{audit,unreach}}$$

follows. □

Proposition 7.54 (Result Reachability Criterion after Addition of Primitive-Time Roots).

For each $R \in \text{Ref}^{\text{audit}}$, assume that

$$\text{TimeContent}_R \neq \emptyset, \quad \text{Result}_R^{\text{audit,resid}} \neq \emptyset$$

. Assume further that

$$\text{Reach}_R^{\text{audit,gen}} \cap \text{Node}_R^{\text{audit,time}} = \emptyset$$

and

$$\begin{aligned} \forall \eta \in \text{Result}_R^{\text{audit,resid}}, \quad \text{NodeOf}_R^{\text{audit,result}}(\eta) \\ \in \text{Reach}_R^{\text{audit|evid}}(\text{TimeContent}_R) \end{aligned}$$

hold. Then

$$\begin{aligned} \text{TimeContent}_R^{\text{audit,gen}} &= \emptyset, & \text{TimeContent}_R^{\text{audit,root}} &= \text{TimeContent}_R, \\ \text{Result}_R^{\text{audit,rootdep}} &= \text{Result}_R^{\text{audit,resid}}, & \text{Result}_R^{\text{audit,rootunreach}} &= \emptyset \end{aligned}$$

hold. Moreover,

$$R \in \text{Ref}^{\text{audit,root}}, \quad R \in \text{Ref}^{\text{audit,rootdep}}, \quad R \in \text{Ref}^{\text{audit,unreach}}$$

hold.

Proof. The empty intersection of the generation-reachability set and the time-content node set, together with Definition 7.39, shows that the set of audit-generated time contents is empty and that the primitive-time-root set equals the entire time-content set.

Take any

$$\eta \in \text{Result}_R^{\text{audit,resid}}$$

. Because the result node belongs to the evidence-conditioned dependency-reachability set after all primitive-time roots have been added,

$$\text{TimeContent}_R \in \text{RootSupport}_R^{\text{audit|evid}}(\eta)$$

holds. By Definition 7.48, every residual result is a primitive-time-root-dependent result, and the result set that remains unreachable after addition of primitive-time roots is empty.

The nonemptiness of the time-content set and of the residual result set, together with Definition 7.52, gives the three reference memberships. □

Proposition 7.55 (Criterion for All Primitive-Time Roots and Dependence of All Results). Suppose that each $R \in \text{Ref}^{\text{audit}}$ satisfies the assumptions of Proposition 7.54. Assume further that

$$\text{Result}_R \neq \emptyset$$

and

$$\text{Reach}_R^{\text{audit|evid}}(\emptyset) \cap \text{Node}_R^{\text{audit,result}} = \emptyset$$

hold. Then, in addition to the conclusion of that proposition,

$$\begin{aligned} \text{Result}_R^{\text{audit,gen}} &= \emptyset, & \text{Result}_R^{\text{audit,evidonly}} &= \emptyset, \\ \text{Result}_R^{\text{audit,resid}} &= \text{Result}_R^{\text{audit,unreach}} = \text{Result}_R \end{aligned}$$

hold.

Proof. Definition 7.28 and monotonicity of reachability give $\text{Reach}_R^{\text{audit,gen}} \subseteq \text{Reach}_R^{\text{audit|evid}}(\emptyset)$. By the assumption and Definition 7.39, the set of audit-generated results is empty, and the result set outside the generation-reachability region equals the entire result set. By the assumption and Definition 7.42, the evidence-environment-reachable nongenerated result set is empty, and the residual result set equals the entire result set.

The conclusion of Proposition 7.54 then follows. $\square$

Definition 7.56 (Reference Set in Which All Time Contents Are Primitive-Time Roots). Define the set of references for which the audit-generated time-content set is empty and the primitive-time-root set equals the entire time-content set by

$$\begin{aligned} &\text{Ref}^{\text{audit,rootall}} \\ &:= \left\{ R \in \text{Ref}^{\text{audit}} \mid \begin{array}{l} \text{TimeContent}_R^{\text{audit,gen}} = \emptyset, \\ \text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R \end{array} \right\}. \end{aligned}$$

Definition 7.57 (Reference Set in Which All Results Depend on Primitive-Time Roots). Define the set of references for which the audit-generated result set is empty and the primitive-time-root-dependent result set equals the entire result set by

$$\begin{aligned} &\text{Ref}^{\text{audit,resultrootall}} \\ &:= \left\{ R \in \text{Ref}^{\text{audit}} \mid \begin{array}{l} \text{Result}_R^{\text{audit,gen}} = \emptyset, \\ \text{Result}_R^{\text{audit,rootdep}} = \text{Result}_R \end{array} \right\}. \end{aligned}$$

Definition 7.58 (Reference Set with All Time Contents as Primitive-Time Roots and All Results Root-Dependent). Define the intersection of the reference set in which all time contents are primitive-time roots and the reference set in which all results depend on primitive-time roots by

$$\text{Ref}^{\text{audit,rootfull}} := \text{Ref}^{\text{audit,rootall}} \cap \text{Ref}^{\text{audit,resultrootall}}$$

.

Definition 7.59 (Disjoint Unions of Per-Reference Result Classes and Primitive-Time-Root Dependency Relations). For each $\rho \in \mathcal{I}_{\text{res}} \setminus \{\text{unreach}\}$, define the per-reference disjoint union of the corresponding result sets by

$$\begin{aligned} &\text{Result}^{\text{audit},\rho} \\ &:= \coprod_{R \in \text{Ref}^{\text{audit}}} \left(\{R\} \times \text{Result}_R^{\text{audit},\rho} \right) \end{aligned}$$

Define the per-reference disjoint unions of the primitive-time-root-result dependency relation and the essential primitive-time-root-result dependency relation, respectively, by

$$\begin{aligned}\text{RootResultDep}^{\text{audit}} &:= \coprod_{R \in \text{Ref}^{\text{audit}}} (\{R\} \times \text{RootResultDep}_R^{\text{audit}}), \\ \text{RootResultEssDep}^{\text{audit}} &:= \coprod_{R \in \text{Ref}^{\text{audit}}} (\{R\} \times \text{RootResultEssDep}_R^{\text{audit}})\end{aligned}$$

Proposition 7.60 (Per-Reference Disjoint-Union Decomposition of Results outside the Generation-Reachability Region). For the reference subsets,

$$\begin{aligned}\text{Ref}^{\text{audit}, \text{unreach}} &= \text{Ref}^{\text{audit}, \text{evidonly}} \cup \text{Ref}^{\text{audit}, \text{rootdep}} \\ &\quad \cup \text{Ref}^{\text{audit}, \text{rootunreach}}\end{aligned}$$

holds.

For the per-reference disjoint unions,

$$\begin{aligned}\coprod_{R \in \text{Ref}^{\text{audit}}} (\{R\} \times \text{Result}_R^{\text{audit}, \text{unreach}}) \\ = \text{Result}^{\text{audit}, \text{evidonly}} \dot{\cup} \text{Result}^{\text{audit}, \text{rootdep}} \dot{\cup} \text{Result}^{\text{audit}, \text{rootunreach}}\end{aligned}$$

holds. Moreover,

$$\text{Result}^{\text{audit}, \text{resid}} = \text{Result}^{\text{audit}, \text{rootdep}} \dot{\cup} \text{Result}^{\text{audit}, \text{rootunreach}}$$

holds.

Proof. Applying Proposition 7.49 to each reference shows that the result set outside the generation-reachability region equals the disjoint union of the evidence-environment-reachable nongenerated result set, the primitive-time-root-dependent result set, and the result set that remains unreachable after addition of primitive-time roots. Applying the nonemptiness of each component to Definition 7.52 gives the equality of reference subsets.

For each reference $R \in \text{Ref}^{\text{audit}}$, apply

$$\{R\} \times (-)$$

to the decomposition of results outside the generation-reachability region, and then take the disjoint union over the reference set. This gives the first equality of per-reference disjoint unions. Applying the same Cartesian product to the residual-result decomposition for each reference and taking the disjoint union over the reference set gives the second equality. $\square$

Proposition 7.61 (Inclusion and Result Projection in the Per-Reference Disjoint Union of Primitive-Time-Root-Result Dependency Relations). For the per-reference disjoint unions of the dependency relations,

$$\text{RootResultEssDep}^{\text{audit}} \subseteq \text{RootResultDep}^{\text{audit}}$$

holds.

For the per-reference disjoint union of the primitive-time-root-dependent result sets,

$$\begin{aligned} & \text{Result}^{\text{audit}, \text{rootdep}} \\ &= \left\{ (R, \eta) \in \text{Result}^{\text{audit}, \text{rootdep}} \mid \right. \\ & \quad \exists x \in \text{TimeContent}_R^{\text{audit}, \text{root}}, \\ & \quad \left. (R, (x, \eta)) \in \text{RootResultDep}^{\text{audit}} \right\} \end{aligned}$$

holds.

Proof. For each reference $R \in \text{Ref}^{\text{audit}}$, apply

$$\{R\} \times (-)$$

to the inclusion in Proposition 7.51, and then take the disjoint union over the reference set. This gives the first inclusion. Applying the same Cartesian product to the result projection in that proposition and taking the disjoint union over the reference set gives the second equality. $\square$

Proposition 7.62 (Component Equalities of Per-Reference Disjoint Unions). For each $R \in \text{Ref}^{\text{audit}}$ and $\rho \in \mathcal{I}_{\text{res}} \setminus \{\text{unreach}\}$,

$$\begin{aligned} & \{\eta \mid (R, \eta) \in \text{Result}^{\text{audit}, \rho}\} \\ &= \text{Result}_R^{\text{audit}, \rho}. \end{aligned}$$

Moreover, for each $R \in \text{Ref}^{\text{audit}}$,

$$\begin{aligned} & \{(x, \eta) \mid (R, (x, \eta)) \in \text{RootResultDep}^{\text{audit}}\} \\ &= \text{RootResultDep}_R^{\text{audit}}, \end{aligned}$$

and

$$\begin{aligned} & \{(x, \eta) \mid (R, (x, \eta)) \in \text{RootResultEssDep}^{\text{audit}}\} \\ &= \text{RootResultEssDep}_R^{\text{audit}} \end{aligned}$$

hold.

Proof. For every disjoint union in Definition 7.59, the collection of second components of elements whose first component is a fixed $R \in \text{Ref}^{\text{audit}}$ equals the corresponding per-reference set or per-reference relation. Applying this equality to each disjoint union gives the six equalities. $\square$

Definition 7.63 (Change Condition for the Generation-Dependency Audit System). The input to the generation-dependency audit system consists of the following five items.

1. The audit reference set

$$\text{Ref}^{\text{audit}}.$$

2. The time contents, transfer events, results, and result prerequisites of each reference

$$\text{TimeContent}_R, \quad \text{TransferEvent}_R, \quad \text{Result}_R, \quad \left\{ \text{ResultPrereq}_{R, \eta} \right\}_{\eta \in \text{Result}_R}.$$

3. The boundary claim nodes and original-source support sets of each reference

$$\text{Node}_R^{\text{audit,claim}}, \quad \text{Support}_R^{\text{audit,transfer}}, \quad \text{Support}_R^{\text{audit,result}}.$$

4. The five-type classification of time contents, the eight-axis classification of time-order structures, and the transfer inputs and outputs

$$\text{TypeOf}_R^{\text{time,ord,struct}}, \quad \text{AxisOf}_R^{\text{time,ord,struct}}, \quad \text{TransferIn}_{R,\epsilon}, \quad \text{TransferOut}_{R,\epsilon}.$$

5. The physical-input initial set and reference formal-evidence environment

$$\text{Init}_R^{\text{audit,phys}}, \quad \text{Env}_R^{\text{audit,evid}}.$$

The change condition for the generation-dependency audit system holds if the codomain, domain, map value, set component, or original-source support of any of the five items changes.

Definition 7.64 (Retrospective Generation-Dependency Reaudit). If the change condition in Definition 7.63 holds, recompute the following for all references in the stated order.

1. The reference-specific time contents, transfer events, results, boundary claim nodes, and original-source support sets.
2. The five types of time contents, eight axes of time-order structure, membership-structure signatures, and dependency-content structure signatures.
3. The audit boundary nodes, audit boundary edges, input and output maps, and audit boundary directed hypergraph.
4. The generation-reachability set from the physical-input initial set, audit-generation traces, structural domains of influence, and essential domains of influence.
5. The audit-generated time contents, primitive-time roots, audit-generated results, and results outside the generation-reachability region.
6. The evidence-conditioned dependency-reachability family, nongenerated results reachable under the evidence environment, residual results, evidence-conditioned primitive-time-root support family, minimal support family, four-class partition of results, and primitive-time-root–result dependency relations.
7. The time-order-structure audit family, reference subsets, and per-reference disjoint unions of result classes and dependency relations.

Even when the audit-input change occurs in only one reference, recompute the per-reference disjoint unions and reference subsets over all references.

Proposition 7.65 (Determinacy of Generation-Dependency Reaudit). Once the five inputs in Definition 7.63 are specified, the audit boundary directed hypergraph, generation-reachability set, evidence-conditioned dependency-reachability family, primitive-time roots, four-class partition of results, and per-reference disjoint unions of dependency relations obtained in the order specified in Definition 7.64 are uniquely determined.

Proof. The node-forming and edge-node-forming maps are specified as injections, while the node-type, edge-type, support-set, input, output, and classification maps are specified as maps on their respective domains. Their map values uniquely determine the audit boundary directed hypergraph.

The starting set and recursion uniquely determine each reachability stage, which stabilizes after finitely many stages by Proposition 7.31. The physical-input initial set uniquely determines the generation-reachability set. Set-builder definitions and set difference then uniquely determine the audit-generated time contents, primitive-time roots, audit-generated results, and results outside the generation-reachability region from the generation-reachability set.

The reference formal-evidence environment and each subset of primitive-time roots uniquely determine the evidence-conditioned dependency-reachability family. The evidence-conditioned dependency-reachability set corresponding to the empty primitive-time-root subset, the audit-generated results, and the results outside the generation-reachability region uniquely determine the nongenerated results reachable under the evidence environment and the residual results. The evidence-conditioned dependency-reachability family uniquely determines the evidence-conditioned primitive-time-root support family and its minimal support family; nonemptiness or emptiness of the same support family uniquely determines the primitive-time-root-dependent results and the results that remain unreachable after addition of primitive-time roots. Consequently, the four-class partition of results, the primitive-time-root–result dependency relations, and all per-reference disjoint unions are also uniquely determined. $\square$

Proposition 7.66 (Recomputation Invariance under Stable Audit Inputs). If the five inputs in Definition 7.63 remain identical after one recomputation over all references, then repeating that recomputation produces an audit boundary directed hypergraph, generation-reachability set, evidence-conditioned dependency-reachability family, primitive-time roots, four-class partition of results, and per-reference disjoint unions of dependency relations identical to the immediately preceding recomputation results.

Proof. By Proposition 7.65, identical five inputs produce a unique recomputation result. Because the five inputs are identical before and after the repetition, the respective values of the audit boundary directed hypergraph, generation-reachability set, evidence-conditioned dependency-reachability family, primitive-time roots, four-class partition of results, and per-reference disjoint unions of dependency relations also coincide. $\square$

Definition 7.67 (Fibers and Zero Fibers of Structure-Signature Maps). For a map between sets, the inverse image of a singleton in the codomain is called the fiber over that point. If a zero element is specified in the codomain, the fiber over the zero element is called the zero fiber.

For each structure-signature map in Definition 7.23 and Definition 7.24, the zero fiber is

$$\left(\text{Sig}_R^{\text{time,ord,struct},\varsigma_{\text{sig}}} \right)^{-1} (\{(0,0,0)\}), \quad \varsigma_{\text{sig}} \in \mathcal{I}_{\text{sig}}^{\text{time,ord,struct}}.$$

Each zero fiber consists of the time contents for which all three corresponding structural attributes have value zero.

Definition 7.68 (Support Antichains). A family of sets is called an antichain under inclusion if any two distinct members of the family are incomparable under inclusion. An antichain whose members are primitive-time-root support sets is called a support antichain.

Proposition 7.69 (Support-Antichain Property of the Minimal Primitive-Time-Root Support Family). For each $R \in \text{Ref}^{\text{audit}}$ and $\eta \in \text{Result}_R$,

$$\text{MinRootSupport}_R^{\text{audit}}(\eta)$$

is a support antichain.

Proof. By Proposition 7.45, any two distinct members of the minimal primitive-time-root support family are incomparable under inclusion. The claim follows from Definition 7.68 . $\square$

Definition 7.70 (Atomic Time Uses and Reference-Specific Classifications of Time Use). A time use is an occurrence in which one time content is used as a formal input, index, order position, clock reading, interval, evolution variable, or reference boundary in an original-source definition, formula, proposition, or explicit statement. Each time use is associated with an original-source location, one time content, a use role, and an audit state. An atomic time use is the smallest unit audited individually in the finite audit of a reference.

Here, “atomic” denotes the granularity of decomposition in a reference-specific finite audit. “Primitive” in Definition 7.10 denotes the order between the generation of time content and its use within a law.

A time use at whose original-source location a time expression appears directly is called an explicit time use. A time use in which time content is incorporated into the establishment of a definition, relation, order, evolution, observation, or comparison is called a structural time use. Additional classifications, including direct versus derived occurrence and independent root versus structural derivation, are defined separately for each reference according to its formal structure.

A time use presupposed by a definition, law, coordinate, observation, comparison, or derivation of a result is called a law-time use. A time use that gives only an order of presentation or reporting within a reference is called a report-order use.

Definition 7.71 (Time-Use Roles, Original-Source Support, and Audit Branches). A time-use role is the formal function performed by a time content within a reference. A reference-specific finite audit uses the following roles.

Interval Gives the separation or order interval between two positions.

Quantity Uses a time content as a numerical value, difference, rate, scale, or factor in a product.

Cross-position correspondence Associates states, events, or readings belonging to different positions.

Clock reading Associates a clock state or clock output with a time content.

Causal Gives an ordering or dependency condition between cause and effect.

State evolution Orders a state transformation, trajectory, or evolution equation.

Stochastic evolution Orders a transition probability, transition rate, or the evolution of a stochastic process.

Observation record Associates the acquisition, preservation, or arrangement of observational results with a time content.

Reference boundary Gives the boundary of a starting point, endpoint, reference position, or domain of application.

An original-source support is a record that associates an audit node, audit edge, or result with a reference identifier, original-source location, subject of attribution, evidence type, and conditions of application.

If the same formal evidence gives rise to multiple audit paths, an audit branch is each path distinguished by a finite index. Reachability of results, the supporting-edge set, and inclusion-minimal support are determined separately for each audit branch. Statement evidence is retained as a quotation record containing the original text, a translation, the original-source location, and the minimal context.

Definition 7.72 (Cycles, Residuals, and Completeness Counts in Finite Dependency Structures). In an audit boundary directed hypergraph, a finite sequence of audit edges is called a directed cycle if the output nodes of each edge connect to the input nodes of the next edge and the output nodes of the terminal edge connect to the input nodes of the initial edge. A maximal connected component consisting of nodes and audit edges that share a directed cycle is called a cyclic component.

An audit edge $\text{AuditEdge} \in \text{Edge}_R^{\text{audit}, \text{bd}}$ has one-edge self-dependence if

$$\text{In}_R^{\text{audit}}(\text{AuditEdge}) \cap \text{Out}_R^{\text{audit}}(\text{AuditEdge}) \neq \emptyset$$

holds.

Original-source evidence acquired for audit but corresponding to none of the audit-claim records, the original-source supports of audit edges, or the original-source supports of results is called unmatched evidence.

A completeness count vector is a finite sequence, arranged in the same reference-specific order, of the cardinalities of the finite bases, correspondences, nodes, audit edges, support families, obligation states, and unmatched-evidence sets that constitute a reference-specific finite audit. The objects and ordering of its coordinates are stated explicitly in the definition for each reference, and components are compared between finite audits that have the same coordinate order.

The terminology of Definition 7.70, Definition 7.71, and Definition 7.72 is used in the reference-specific finite audits in the Supplement . The time contents, transfer events, results, and dependency structure of a reference-specific finite audit are represented, respectively, by the previously defined

$$\text{TimeContent}_R, \quad \text{TransferEvent}_R, \quad \text{Result}_R$$

and

$$\text{Hypergraph}_R^{\text{audit}}$$

. Original-source locations, time-use roles, reference-specific obligations, and finite counts belong to the finite audit of each reference.

Definition 7.73 (Primitive Law-Bearing Time Premises). If the time content taken as input by a law-time use in Definition 7.70 belongs to $\text{TimeContent}_R^{\text{audit,root}}$ in Definition 7.39, the use of that time content within the law is called a primitive law-bearing time premise. A primitive law-bearing time premise is a primitive-time root used to establish a definition, law, evolution, observation, comparison, or derivation of a result.

Definition 7.74 (Finite Audit Carrier and Reference-Specific Complete Time-Use Audit). The finite sets of original-source locations, atomic time uses, time contents, time-use roles, obligation atoms, audit edges, support families, and results identified for one reference are called its finite audit carrier.

An audit that assigns an obligation atom and typed audit state to every atomic time use in a finite audit carrier and supplies correspondences to all audit edges, support families, and results is called a reference-specific complete time-use audit. “Complete” denotes exhaustive coverage of this finite audit carrier.

Definition 7.75 (Classification Margins). If the same formal evidence admits multiple audit-classification branches, each furnished with an original-source location and grounds for its adoption, the resulting reference-specific finite branch set is called a classification margin. Each branch associates a classification boundary with the formal evidence supporting that boundary.

Remark 7.156 defines the interpretability of statements. A classification margin represents audit-classification branches for formal evidence.

Definition 7.76 (Time-Generation, Cyclic-Generation, and Evidence-Open Candidates). In a reference-specific finite audit, an obligation-conforming audit path from the physical-input initial set to a time-content node is called a time-generation candidate. The existence of a time-generation candidate is determined by the reachability set in Definition 7.30, the audit-generation traces in Definition 7.32, and the audit-generated time contents in Definition 7.39 .

A candidate path that outputs the generation of time content is called a cyclic-generation candidate if it contains a cyclic component or one-edge self-dependence from Definition 7.72 . Cyclic-generation candidates are counted separately for each reference as candidates containing cyclic structure.

A candidate path is called an evidence-open candidate if an obligation atom required by that path has an evidence-open state. Evidence-open candidates are counted separately for each reference through the correspondence between original-source support and obligation states.

Definition 7.77 (Result-Preserving Correspondences). A bijection that places the result nodes of a reference-specific finite audit in one-to-one correspondence with the result values in the reference-specific result projection of the same reference is called a result-preserving correspondence. A result-preserving correspondence uniquely specifies, within the result projection, the reference-specific result represented by each result node.

The terminology of Definition 7.73, Definition 7.74, Definition 7.75, Definition 7.76, and Definition 7.77 applies to the reference-specific audits in the Supplement . The candidate sets, classification-margin projections, and result-preserving correspondences of each reference are determined by the finite audit carrier and original-source supports of that reference.

7.2.1 Separate Formal and Statement-Evidence Audit Determinations for Time-Bearing Expressions

Definition 7.78 (Four Audit Determinations for Time-Bearing Expressions). Separate the formal evidence and statement evidence of a reference, and apply each of the following four determinations to a time content or time-bearing expression.

Pre-generation placement. A time content or time-bearing expression is supplied to a law operation, definition, or derivation of a result before its physical generation.

Omission of generation stages. A physical input, clock state, record, or observational result is connected to a law description or quantitative expression that uses a time variable without passing through the generation stages required to proceed from nontemporal physical inputs to a qualified time representation.

Reattribution of a time property. A relational difference holding among physical carriers, clocks, signals, records, or observables is stated as a change of state or property of a time object independent of them.

Reverse presupposition of a derived quantity. A derived quantity that requires generated time content is used as a premise of that time content or of the generation structure that confers physical-time qualification.

For formal evidence, each determination follows from the original-source definitions, formulas, propositions, and dependencies. For statement evidence, the relation to the four determinations is conditional on each interpretation permitted by the original text, translation, original-source location, and minimal context.

The four determinations are not mutually exclusive. Multiple determinations may hold simultaneously for a single object, reference, or evidence track.

7.2.2 Reference-Evidence Layer, Dependency-Effective Layer, and Semantic Audit Boundary

Definition 7.79 (Reference-Evidence Layer, Dependency-Effective Layer, and Semantic Audit Boundary). For each $R \in \text{Ref}^{\text{audit}}$, suppose that the finite sets

$$\text{Node}_R^{\text{audit,use}}, \quad \text{Node}_R^{\text{audit,time,src}}, \quad \text{Node}_R^{\text{audit,rootuse}}$$

are given. Assume that these sets and

$$\text{Node}_R^{\text{audit,claim}}, \quad \text{Node}_R^{\text{audit,result}}$$

are pairwise disjoint.

Define the full reference-evidence node set by

$$\begin{aligned} \text{Node}_R^{\text{audit,evid}} := & \text{Node}_R^{\text{audit,claim}} \dot{\cup} \text{Node}_R^{\text{audit,use}} \\ & \dot{\cup} \text{Node}_R^{\text{audit,time,src}} \dot{\cup} \text{Node}_R^{\text{audit,rootuse}} \\ & \dot{\cup} \text{Node}_R^{\text{audit,result}} \end{aligned}$$

and the dependency-effective node set by

$$\begin{aligned} \text{Node}_R^{\text{audit,dep}} := & \text{Node}_R^{\text{audit,claim}} \dot{\cup} \text{Node}_R^{\text{audit,time,src}} \\ & \dot{\cup} \text{Node}_R^{\text{audit,result}} \end{aligned}$$

Define the reference-evidence node-type set by

$$\begin{aligned} \text{Type}^{\text{audit,evid,node}} &:= \text{Type}^{\text{audit,node}} \\ &\cup \left\{ \text{Type}_{\text{use}}^{\text{audit,node}}, \text{Type}_{\text{rootuse}}^{\text{audit,node}} \right\} \end{aligned}$$

Define the reference-evidence node-type map

$$\begin{aligned} \text{TypeOf}_R^{\text{audit,evid,node}} &: \text{Node}_R^{\text{audit,evid}} \\ &\longrightarrow \text{Type}^{\text{audit,evid,node}} \end{aligned}$$

to have, on the respective disjoint-union components, the values

$$\text{Type}_{\text{claim}}^{\text{audit,node}}, \quad \text{Type}_{\text{use}}^{\text{audit,node}}, \quad \text{Type}_{\text{time}}^{\text{audit,node}}, \quad \text{Type}_{\text{rootuse}}^{\text{audit,node}}, \quad \text{Type}_{\text{result}}^{\text{audit,node}}$$

Define the dependency-effective node-type map by

$$\text{TypeOf}_R^{\text{audit,dep,node}} := \text{TypeOf}_R^{\text{audit,evid,node}} \Big|_{\text{Node}_R^{\text{audit,dep}}}$$

Suppose that the reference-specific content correspondence

$$\begin{aligned} \text{ContentOf}_R^{\text{audit,time}} &: \text{Node}_R^{\text{audit,time,src}} \\ &\twoheadrightarrow \text{TimeContent}_R \end{aligned}$$

is given. Define the boundary-contraction map on dependency-effective nodes

$$\begin{aligned} \text{Quot}_R^{\text{audit,bd}} &: \text{Node}_R^{\text{audit,dep}} \\ &\twoheadrightarrow \text{Node}_R^{\text{audit,bd}} \end{aligned}$$

by the following two actions. First, for

$$\text{AuditNode} \in \text{Node}_R^{\text{audit,claim}} \cup \text{Node}_R^{\text{audit,result}}$$

define

$$\text{Quot}_R^{\text{audit,bd}}(\text{AuditNode}) := \text{AuditNode}$$

Second, for

$$\text{AuditNode} \in \text{Node}_R^{\text{audit,time,src}}$$

define

$$\begin{aligned} &\text{Quot}_R^{\text{audit,bd}}(\text{AuditNode}) \\ &:= \text{NodeOf}_R^{\text{audit,time}} \left(\text{ContentOf}_R^{\text{audit,time}}(\text{AuditNode}) \right) \end{aligned}$$

Suppose that the reference-specific dependency input and output maps

$$\begin{aligned} \text{In}_R^{\text{audit,dep}} &: \text{Edge}_R^{\text{audit,bd}} \longrightarrow \mathcal{P}_{\text{fin}} \left(\text{Node}_R^{\text{audit,dep}} \right), \\ \text{Out}_R^{\text{audit,dep}} &: \text{Edge}_R^{\text{audit,bd}} \longrightarrow \mathcal{P}_{\text{fin}} \left(\text{Node}_R^{\text{audit,dep}} \right) \end{aligned}$$

are given. For each layer

$$\text{Layer} \in \{\text{evid}, \text{dep}\}$$

define the finite directed hypergraphs of the reference-evidence layer and the dependency-effective layer uniformly by

$$\text{Hypergraph}_R^{\text{audit,Layer}} := \left(\text{Node}_R^{\text{audit,Layer}}, \text{Edge}_R^{\text{audit,bd}}, \right. \\ \left. \text{In}_R^{\text{audit,dep}}, \text{Out}_R^{\text{audit,dep}}, \right. \\ \left. \text{TypeOf}_R^{\text{audit,Layer,node}}, \text{TypeOf}_R^{\text{audit,edge}} \right)$$

The three hypergraphs are said to be compatible under boundary contraction if, for each $\text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}}$,

$$\text{In}_R^{\text{audit}}(\text{AuditEdge}) = \text{Quot}_R^{\text{audit,bd}} \left[\text{In}_R^{\text{audit,dep}}(\text{AuditEdge}) \right], \\ \text{Out}_R^{\text{audit}}(\text{AuditEdge}) = \text{Quot}_R^{\text{audit,bd}} \left[\text{Out}_R^{\text{audit,dep}}(\text{AuditEdge}) \right]$$

hold.

Proposition 7.80 (Relations among the Reference-Evidence Layer, Dependency-Effective Layer, and Semantic Audit Boundary). Under Definition 7.79,

$$\text{Node}_R^{\text{audit,dep}} \subseteq \text{Node}_R^{\text{audit,evid}}$$

holds, and the input and output nodes of every edge belong to

$$\text{Node}_R^{\text{audit,dep}}$$

. Consequently,

$$\text{Node}_R^{\text{audit,use}} \dot{\cup} \text{Node}_R^{\text{audit,rootuse}} \\ = \text{Node}_R^{\text{audit,evid}} \setminus \text{Node}_R^{\text{audit,dep}}$$

consists of isolated nodes of the reference-evidence finite directed hypergraph.

If the three hypergraphs are compatible under boundary contraction, the boundary-contraction map

$$\text{Quot}_R^{\text{audit,bd}}$$

gives a surjective directed-hypergraph morphism from the dependency-effective finite directed hypergraph to the audit boundary directed hypergraph.

The cardinalities of the node sets are

$$\left| \text{Node}_R^{\text{audit,evid}} \right| = \left| \text{Node}_R^{\text{audit,claim}} \right| + \left| \text{Node}_R^{\text{audit,use}} \right| \\ + \left| \text{Node}_R^{\text{audit,time,src}} \right| + \left| \text{Node}_R^{\text{audit,rootuse}} \right| \\ + \left| \text{Node}_R^{\text{audit,result}} \right|, \\ \left| \text{Node}_R^{\text{audit,dep}} \right| = \left| \text{Node}_R^{\text{audit,claim}} \right| + \left| \text{Node}_R^{\text{audit,time,src}} \right| \\ + \left| \text{Node}_R^{\text{audit,result}} \right|$$

Proof. The inclusion of the two node sets, the expression for their set difference, and the isolation of every node in that difference follow from the disjoint-union decompositions in Definition 7.79 and from the codomains of the input and output maps.

On the claim-node set and the result-node set, the boundary-contraction map is the identity and is surjective. On the time-content node set, it is surjective as the composite of the reference-specific content correspondence and the time-content node-forming map. Surjectivity on the three disjoint-union components makes the boundary-contraction map surjective onto the audit boundary node set. When the three hypergraphs are compatible, the compatibility equations for the input and output sets show that the boundary-contraction map preserves the input and output of each hyperedge.

The two cardinality formulas follow from the corresponding finite disjoint-union decompositions. $\square$

7.2.3 Edge-Restricted Audit Reachability and Branch Support

Definition 7.81 (Edge-Restricted Audit Closure and Reachability). For each $R \in \text{Ref}^{\text{audit}}$ and $\text{EdgeSet} \subseteq \text{Edge}_R^{\text{audit,bd}}$, define the edge-restricted one-stage closure operator

$$\text{ClStep}_{R,\text{EdgeSet}}^{\text{audit}} : \mathcal{P}(\text{Node}_R^{\text{audit,bd}}) \longrightarrow \mathcal{P}(\text{Node}_R^{\text{audit,bd}})$$

for each

$$\text{NodeSet} \subseteq \text{Node}_R^{\text{audit,bd}}$$

by

$$\text{ClStep}_{R,\text{EdgeSet}}^{\text{audit}}(\text{NodeSet}) := \text{NodeSet} \cup \bigcup_{\substack{\text{AuditEdge} \in \text{EdgeSet} \\ \text{In}_R^{\text{audit}}(\text{AuditEdge}) \subseteq \text{NodeSet}}} \text{Out}_R^{\text{audit}}(\text{AuditEdge})$$

. For each

$$\text{Seed} \subseteq \text{Node}_R^{\text{audit,bd}}$$

define the edge-restricted audit-reachability set by iterated composition, with the zeroth iterate taken to be the identity map:

$$\text{Reach}_{R,\text{EdgeSet}}^{\text{audit}}(\text{Seed}) := (\text{ClStep}_{R,\text{EdgeSet}}^{\text{audit}})^{|\text{Node}_R^{\text{audit,bd}}|}(\text{Seed})$$

. For each

$$k \in \mathbb{N}$$

define the edge-restricted k -th reachability stage by

$$\text{Reach}_{R,\text{EdgeSet}}^{\text{audit},(k)}(\text{Seed}) = (\text{ClStep}_{R,\text{EdgeSet}}^{\text{audit}})^k(\text{Seed})$$

.

Definition 7.82 (Audit-Branch Support Families). For each $R \in \text{Ref}^{\text{audit}}$, let the finite audit-branch set be

$$\text{Branch}_R^{\text{audit}}$$

. Let the map assigning to each audit branch the reference-specific result determined by that branch be

$$\text{ResultOf}_R^{\text{audit,branch}} : \text{Branch}_R^{\text{audit}} \longrightarrow \text{Result}_R$$

For each

$$\iota \in \text{Branch}_R^{\text{audit}}$$

define the corresponding result edge by

$$\text{Edge}_{R,\iota}^{\text{audit,result}} := \text{EdgeOf}_R^{\text{audit,result}} \left(\text{ResultOf}_R^{\text{audit,branch}}(\iota) \right)$$

For each

$$\text{EdgeSet} \subseteq \text{Edge}_R^{\text{audit,bd}}$$

define the support determination by

$$\begin{aligned} \text{EdgeSet} \Vdash_R^{\text{audit}} \iota &\iff \text{Edge}_{R,\iota}^{\text{audit,result}} \in \text{EdgeSet} \\ &\quad \wedge \text{NodeOf}_R^{\text{audit,result}} \left(\text{ResultOf}_R^{\text{audit,branch}}(\iota) \right) \\ &\quad \in \text{Reach}_{R,\text{EdgeSet}}^{\text{audit}} \left(\text{In}_R^{\text{audit}} \left(\text{Edge}_{R,\iota}^{\text{audit,result}} \right) \right) \end{aligned}$$

Define the branch-support family by

$$\text{Support}_{R,\iota}^{\text{audit,branch}} := \left\{ \text{EdgeSet} \subseteq \text{Edge}_R^{\text{audit,bd}} \mid \text{EdgeSet} \Vdash_R^{\text{audit}} \iota \right\}$$

Define the proper-subset support predicate by

$$\begin{aligned} \text{SubSupp}_{R,\iota}^{\text{prop}} (\text{EdgeSet}', \text{EdgeSet}) \\ \iff \text{EdgeSet}', \text{EdgeSet} \in \text{Support}_{R,\iota}^{\text{audit,branch}}, \\ \text{EdgeSet}' \subsetneq \text{EdgeSet} \end{aligned}$$

Define the inclusion-minimal branch-support family by

$$\begin{aligned} \text{MinSupport}_{R,\iota}^{\text{audit,branch}} \\ := \left\{ \text{EdgeSet} \mid \begin{array}{l} \text{EdgeSet} \in \text{Support}_{R,\iota}^{\text{audit,branch}}, \\ \nexists \text{EdgeSet}' \text{ SubSupp}_{R,\iota}^{\text{prop}} (\text{EdgeSet}', \text{EdgeSet}) \end{array} \right\} \end{aligned}$$

Proposition 7.83 (Inclusion-Minimal Support Property of a Single Result Edge). For each

$$R \in \text{Ref}^{\text{audit}}, \quad \iota \in \text{Branch}_R^{\text{audit}}$$

,

$$\begin{aligned} \text{MinSupport}_{R,\iota}^{\text{audit,branch}} \\ = \left\{ \left\{ \text{Edge}_{R,\iota}^{\text{audit,result}} \right\} \right\} \end{aligned}$$

holds.

Proof. Take the input set of the result edge as the initial set and include that result edge in the edge subset. The result node is then added at the first reachability stage. The singleton result-edge set therefore belongs to the branch-support family.

By definition, every edge set in the branch-support family contains the result edge. No proper subset of the singleton result-edge set contains that edge. Hence the singleton result-edge set is inclusion-minimal, and there is no other inclusion-minimal element. $\square$

7.2.4 Original-Source Support, Terminal Residuals, and Completeness Counts

Definition 7.84 (Original-Source Supports of Audit Edges and Terminal Residuals). For each $R \in \text{Ref}^{\text{audit}}$, let the map assigning a finite set of original-source locations to each dependency-effective node be

$$\text{SrcLoc}_R^{\text{audit}} : \text{Node}_R^{\text{audit,dep}} \longrightarrow \mathcal{P}_{\text{fin}}(\text{Loc}^{\text{src}})$$

. Define the original-source location set of an audit edge by

$$\begin{aligned} & \text{SrcLoc}_R^{\text{audit,edge}}(\text{AuditEdge}) \\ & := \bigcup_{\text{AuditNode} \in \text{In}_R^{\text{audit,dep}}(\text{AuditEdge}) \cup \text{Out}_R^{\text{audit,dep}}(\text{AuditEdge})} \text{SrcLoc}_R^{\text{audit}}(\text{AuditNode}) \end{aligned}$$

. Define the original-source support of each edge by

$$\begin{aligned} & \text{Evidence}_R^{\text{audit,edge}}(\text{AuditEdge}) \\ & := \left\{ \left(R, \text{SrcLoc}_R^{\text{audit,edge}}(\text{AuditEdge}), \right. \right. \\ & \quad \left. \left. \begin{array}{l} \text{attributed-source, source-statement-or-derivation,} \\ \text{applicability-condition} \end{array} \right) \right\} \end{aligned}$$

. Define the inverse of the result-node-forming map by

$$\begin{aligned} & \text{ResultOf}_R^{\text{audit,node}} : \text{Node}_R^{\text{audit,result}} \longrightarrow \text{Result}_R, \\ & \text{ResultOf}_R^{\text{audit,node}} := \left(\text{NodeOf}_R^{\text{audit,result}} \right)^{-1} \end{aligned}$$

. Let the set of directed cycles of the dependency-effective finite directed hypergraph be

$$\text{Cycle}_R^{\text{audit}}$$

. Define the set of audit edges with one-edge self-dependence by

$$\begin{aligned} & \text{SelfEdge}_R^{\text{audit}} \\ & := \left\{ \text{AuditEdge} \in \text{Edge}_R^{\text{audit,bd}} \mid \begin{array}{l} \text{In}_R^{\text{audit,dep}}(\text{AuditEdge}) \\ \cap \text{Out}_R^{\text{audit,dep}}(\text{AuditEdge}) \neq \emptyset \end{array} \right\} \end{aligned}$$

. Define the audit acyclicity condition by

$$\text{Acyc}^{\text{audit}}(R) \iff \text{Cycle}_R^{\text{audit}} = \text{SelfEdge}_R^{\text{audit}} = \emptyset$$

. Among the acquired original-source evidence, let the complement of the set of original-source locations corresponding to reference-evidence nodes, original-source supports of edges, or result supports be

$$\text{Evidence}_R^{\text{audit,unmatched}}$$

Proposition 7.85 (Reachability from the Empty Initial Set for Audit Edges with Nonempty Inputs). For each $R \in \text{Ref}^{\text{audit}}$ and $\text{EdgeSet} \subseteq \text{Edge}_R^{\text{audit}, \text{bd}}$, if every

$$\text{AuditEdge} \in \text{EdgeSet}$$

satisfies

$$\text{In}_R^{\text{audit}}(\text{AuditEdge}) \neq \emptyset$$

then

$$\text{Reach}_{R, \text{EdgeSet}}^{\text{audit}}(\emptyset) = \emptyset$$

holds.

Proof. The initial reachability stage is the empty set. For each audit edge whose input set is nonempty, that input set is not a subset of the empty reachability stage. No node is therefore added at the next reachability stage. Induction on the stages shows that every reachability stage is empty. $\square$

Definition 7.86 (Reference-Specific Completeness-Count Fields). Define the reference-specific completeness-count field set by

$$\begin{aligned} \text{Field}^{\text{audit}, \text{count}} := \{ & \text{timeuse}, \text{timecontent}, \text{userole}, \text{contentrole}, \\ & \text{obligation}, \text{useobligation}, \text{roleobligation}, \\ & \text{physclaim}, \text{mathclaim}, \text{claimnode}, \\ & \text{usenode}, \text{srccontentnode}, \text{rootusenode}, \\ & \text{resultnode}, \text{evidnode}, \\ & \text{transferredge}, \text{resultedge}, \text{auditedge}, \\ & \text{branch}, \text{alternative}, \\ & \text{fulfilled}, \text{unfulfilled}, \text{open}, \\ & \text{result}, \text{margin}, \text{unmatched} \} \end{aligned}$$

In display order, the fields represent the cardinalities of the sets of atomic time uses, time contents, use–role correspondences, content–role correspondences, obligation atoms, use–obligation correspondences, role–obligation correspondences, physical and formal claims, mathematical-audit claims, all claim nodes, time-use nodes, reference-specific time-content nodes, root-use nodes, result nodes, all reference-evidence nodes, transfer edges, result edges, all audit edges, audit branches, alternative interpretations, fulfilled obligations, unfulfilled obligations, open obligations, reference-specific results, classification-margin candidates, and unmatched evidence.

Let the reference-specific completeness-count map be

$$\text{Count}_R^{\text{audit}} : \text{Field}^{\text{audit}, \text{count}} \longrightarrow \mathbb{N}$$

. The reference-specific finite audit gives the cardinality of the finite object corresponding to each field as the value of this map.

Proposition 7.87 (Structural Identities for Reference-Specific Completeness Counts). For each $R \in \text{Ref}^{\text{audit}}$,

$$\begin{aligned} \text{Count}_R^{\text{audit}}(\text{evidnode}) &= \sum_{F \in \{\text{claimnode}, \text{usenode}, \text{srccontentnode}, \text{rootusenode}, \text{resultnode}\}} \text{Count}_R^{\text{audit}}(F), \\ \text{Count}_R^{\text{audit}}(\text{auditedge}) &= \sum_{F \in \{\text{transferredge}, \text{resultedge}\}} \text{Count}_R^{\text{audit}}(F) \end{aligned}$$

hold.

Proof. The first equality follows from the finite disjoint-union decomposition of the full reference-evidence node set in Definition 7.79 . The second equality follows from the finite disjoint-union decomposition of the audit boundary edge set in Definition 7.26 . $\square$

7.2.5 Typed Finite Audit Input

Definition 7.88 (Audit Decomposition of Original-Source Formal Entries). For one reference, a finite collection of formal entries that belong to the same original-source location or to a contiguous minimal context and are presented together before the audit is called an undecomposed original-source entry group. If one undecomposed entry contains multiple objects, predicates, dependency inputs, validity conditions, or domains of application, partition it, preserving the logical direction, quantification, conditions, and domain of application of the original-source entry, into units that can be subjected independently to truth-value and dependency determinations. Each resulting unit is called a minimal audit claim.

Assign to every minimal audit claim exactly one claim type: either a physical and formal claim or a mathematical-audit claim. A component is called type-homogeneous if claim type, object, and unit of determination are not mixed within it. A type-homogeneous component whose values are minimal audit claims obtained from original-source definitions, formulas, propositions, operations, or conditions and that preserves provenance correspondence to original-source locations is called a formal-evidence component.

A type-homogeneous component separated from a reference's formal description so as to bear independently an object, predicate, dependency input, validity condition, or domain of application is called a semantic component. A semantic component is an audit unit obtained in this paper by decomposing formal evidence from the original source.

Definition 7.89 (Typed Finite Audit-Input Structure). To record the minimal audit claims and formal-evidence components of Definition 7.88 as typed finite inputs, for each $R \in \text{Ref}^{\text{audit}}$, define the claim-category index domain by $\mathcal{I}_{\text{claim}} := \{\text{phys}, \text{math}\}$. Take two finite index sets for the respective claim categories and the following five finite index sets:

$$\begin{aligned} & \left(\text{Idx}_R^{\text{audit}, \chi} \right)_{\chi \in \mathcal{I}_{\text{claim}}}, & \text{Idx}_R^{\text{audit}, \text{use}}, \\ & \text{Idx}_R^{\text{audit}, \text{time}, \text{src}}, & \text{Idx}_R^{\text{audit}, \text{result}}, \\ & \text{Idx}_R^{\text{audit}, \text{transfer}}, & \text{Idx}_R^{\text{audit}, \text{branch}}. \end{aligned}$$

Let the value families of physical and formal claims, mathematical-audit claims, time uses, reference-specific time contents, and results, respectively, be

$$\begin{aligned} & \left(\text{ClaimVal}_{R,i}^{\text{audit}, \chi} \right)_{\chi \in \mathcal{I}_{\text{claim}}, i \in \text{Idx}_R^{\text{audit}, \chi}}, \\ & \left(\text{UseVal}_{R,k}^{\text{audit}} \right)_{k \in \text{Idx}_R^{\text{audit}, \text{use}}}, \\ & \left(\text{SrcTimeVal}_{R,j}^{\text{audit}} \right)_{j \in \text{Idx}_R^{\text{audit}, \text{time}, \text{src}}}, \\ & \left(\text{ResultVal}_{R,q}^{\text{audit}} \right)_{q \in \text{Idx}_R^{\text{audit}, \text{result}}} \end{aligned}$$

For each

$$\kappa_{\text{edge}} \in \mathcal{I}_{\text{audit}, \text{edge}}$$

let the claim-input table and time-content-input table be given uniformly by

$$\begin{aligned}\text{ClaimIn}_R^{\text{audit}, \kappa_{\text{edge}}} &: \text{Idx}_R^{\text{audit}, \kappa_{\text{edge}}} \longrightarrow \mathcal{P}_{\text{fin}} \left(\text{Idx}_R^{\text{audit}, \text{phys}} \right), \\ \text{TimeIn}_R^{\text{audit}, \kappa_{\text{edge}}} &: \text{Idx}_R^{\text{audit}, \kappa_{\text{edge}}} \longrightarrow \mathcal{P}_{\text{fin}} \left(\text{Idx}_R^{\text{audit}, \text{time}, \text{src}} \right)\end{aligned}$$

. Let the transfer-output table be

$$\text{TimeOut}_R^{\text{audit}, \text{transfer}} : \text{Idx}_R^{\text{audit}, \text{transfer}} \longrightarrow \mathcal{P}_{\text{fin}} \left(\text{Idx}_R^{\text{audit}, \text{time}, \text{src}} \right)$$

. Require the claim-input and time-content-input tables to take the same values at any two result indices whose result values are equal.

Let the correspondence between time uses and reference-specific time contents be

$$\text{UseSrc}_R^{\text{audit}} \subseteq \text{Idx}_R^{\text{audit}, \text{use}} \times \text{Idx}_R^{\text{audit}, \text{time}, \text{src}}$$

. Let the original-source location maps be

$$\begin{aligned}\text{UseLoc}_R^{\text{audit}} &: \text{Idx}_R^{\text{audit}, \text{use}} \longrightarrow \text{Loc}^{\text{src}}, \\ \text{ClaimLoc}_R^{\text{audit}} &: \text{Idx}_R^{\text{audit}, \text{phys}} \longrightarrow \text{Loc}^{\text{src}}\end{aligned}$$

. Let the map assigning the result determined by each audit branch be

$$\text{ResultOf}_R^{\text{audit}, \text{branch}} : \text{Idx}_R^{\text{audit}, \text{branch}} \longrightarrow \left\{ \text{ResultVal}_{R, q}^{\text{audit}} \mid q \in \text{Idx}_R^{\text{audit}, \text{result}} \right\}$$

. Let the unmatched-evidence set and the reference-specific completeness-count map be

$$\begin{aligned}\text{Evidence}_R^{\text{audit}, \text{unmatched}} &\subseteq \text{Loc}^{\text{src}}, \\ \text{Count}_R^{\text{audit}} &: \text{Field}^{\text{audit}, \text{count}} \longrightarrow \mathbb{N}\end{aligned}$$

. Denote by

$$\text{AuditInput}_R^{\text{fin}}$$

the typed finite audit-input structure whose components are the foregoing finite index sets, value families, input tables, time-use-content correspondence, original-source locations, branch-result correspondence, unmatched evidence, and reference-specific completeness-count map.

7.2.6 Canonical Realization of Typed Finite Audit Input

Definition 7.90 (Canonical Realization of Typed Finite Audit Input). Take the structure $\text{AuditInput}_R^{\text{fin}}$ of Definition 7.89. Define the claim-kind values by

$$\begin{aligned}\text{ClaimKind}_{\text{phys}} &:= \text{physical-formal-claim}, \\ \text{ClaimKind}_{\text{math}} &:= \text{mathematical-audit-claim}\end{aligned}$$

. For each $\chi \in \mathcal{I}_{\text{claim}}$ and $i \in \text{Idx}_R^{\text{audit},\chi}$, define the enumerated-node map and its value by

$$\begin{aligned} \text{EnumNode}_R^{\text{audit},\text{claim},\chi}(i) &= \text{Node}_{R,i}^{\text{audit},\text{claim},\chi}, \\ \text{Node}_{R,i}^{\text{audit},\text{claim},\chi} &:= \begin{pmatrix} \text{Type}_{\text{claim}}^{\text{audit},\text{node}}, \\ \text{ClaimKind}_{\chi}, \\ i, \text{ClaimVal}_{R,i}^{\text{audit},\chi} \end{pmatrix} \end{aligned}$$

. Define the use-node category domain by $\mathcal{I}_{\text{use}} := \{\text{use}, \text{rootuse}\}$. For each $v \in \mathcal{I}_{\text{use}}$ and $k \in \text{Idx}_R^{\text{audit},\text{use}}$, define the enumerated-node map and its value by

$$\begin{aligned} \text{EnumNode}_R^{\text{audit},v}(k) &= \text{Node}_{R,k}^{\text{audit},v}, \\ \text{Node}_{R,k}^{\text{audit},v} &:= \begin{pmatrix} \text{Type}_v^{\text{audit},\text{node}}, \\ k, \text{UseVal}_{R,k}^{\text{audit}} \end{pmatrix} \end{aligned}$$

. Define the enumeration maps and their values for reference-specific time nodes and result nodes by

$$\begin{aligned} \text{EnumNode}_R^{\text{audit},\text{time},\text{src}}(j) &= \text{Node}_{R,j}^{\text{audit},\text{time},\text{src}}, \\ \text{Node}_{R,j}^{\text{audit},\text{time},\text{src}} &:= \begin{pmatrix} \text{Type}_{\text{time}}^{\text{audit},\text{node}}, \\ \text{source}, \\ j, \text{SrcTimeVal}_{R,j}^{\text{audit}} \end{pmatrix}, \\ \text{EnumNode}_R^{\text{audit},\text{result}}(q) &= \text{Node}_{R,q}^{\text{audit},\text{result}}, \\ \text{Node}_{R,q}^{\text{audit},\text{result}} &:= \text{NodeOf}_R^{\text{audit},\text{result}}(\text{ResultVal}_{R,q}^{\text{audit}}) \end{aligned}$$

. Define the claim-node set, the node set for each v in the use-node category domain, the reference-specific time-node set, and the result-node set by

$$\begin{aligned} \text{Node}_R^{\text{audit},\text{claim}} &:= \bigcup_{\chi \in \mathcal{I}_{\text{claim}}} \text{EnumNode}_R^{\text{audit},\text{claim},\chi} [\text{Idx}_R^{\text{audit},\chi}], \\ \text{Node}_R^{\text{audit},v} &:= \text{EnumNode}_R^{\text{audit},v} [\text{Idx}_R^{\text{audit},\text{use}}], \\ \text{Node}_R^{\text{audit},\text{time},\text{src}} &:= \text{EnumNode}_R^{\text{audit},\text{time},\text{src}} [\text{Idx}_R^{\text{audit},\text{time},\text{src}}], \\ \text{Node}_R^{\text{audit},\text{result}} &:= \text{EnumNode}_R^{\text{audit},\text{result}} [\text{Idx}_R^{\text{audit},\text{result}}]. \end{aligned}$$

Define the range of the reference-specific time-value family as the time-content set by

$$\text{TimeContent}_R := \left\{ \text{SrcTimeVal}_{R,j}^{\text{audit}} \mid j \in \text{Idx}_R^{\text{audit},\text{time},\text{src}} \right\}$$

and define the time-content correspondence by

$$\text{TimeSem}_R^{\text{audit}}(j) := \text{SrcTimeVal}_{R,j}^{\text{audit}}$$

. On the node set, denote the reference-specific content correspondence by

$$\begin{aligned} \text{ContentOf}_R^{\text{audit},\text{time}}(\text{Node}_{R,j}^{\text{audit},\text{time},\text{src}}) \\ := \text{TimeSem}_R^{\text{audit}}(j) \end{aligned}$$

Define transfer events and reference-specific results by

$$\begin{aligned}\text{TransferEvent}_{R,s} &:= (R, s, \text{transfer}), \\ \text{Result}_{R,q} &:= \text{ResultVal}_{R,q}^{\text{audit}}\end{aligned}$$

and set

$$\begin{aligned}\text{TransferEvent}_R &:= \left\{ \text{TransferEvent}_{R,s} \mid s \in \text{Idx}_R^{\text{audit,transfer}} \right\}, \\ \text{Result}_R &:= \left\{ \text{Result}_{R,q} \mid q \in \text{Idx}_R^{\text{audit,result}} \right\}\end{aligned}$$

Define the enumerated edges by

$$\begin{aligned}\text{Edge}_{R,s}^{\text{audit,transfer}} &:= \text{EdgeOf}_R^{\text{audit,transfer}}(\text{TransferEvent}_{R,s}), \\ \text{Edge}_{R,q}^{\text{audit,result}} &:= \text{EdgeOf}_R^{\text{audit,result}}(\text{Result}_{R,q})\end{aligned}$$

Denote the two families for each $\kappa_{\text{edge}} \in \mathcal{I}_{\text{audit,edge}}$ and $i \in \text{Idx}_R^{\text{audit},\kappa_{\text{edge}}}$ by $\text{Edge}_{R,i}^{\text{audit},\kappa_{\text{edge}}}$.

Define the reference-specific dependency input by

$$\begin{aligned}\text{In}_R^{\text{audit,dep}}(\text{Edge}_{R,i}^{\text{audit},\kappa_{\text{edge}}}) &:= \text{EnumNode}_R^{\text{audit,claim,phys}} \left[\text{ClaimIn}_R^{\text{audit},\kappa_{\text{edge}}}(i) \right] \\ &\quad \cup \text{EnumNode}_R^{\text{audit,time,src}} \left[\text{TimeIn}_R^{\text{audit},\kappa_{\text{edge}}}(i) \right],\end{aligned}$$

Define the reference-specific dependency output for a transfer edge by

$$\text{Out}_R^{\text{audit,dep}}(\text{Edge}_{R,s}^{\text{audit,transfer}}) := \text{EnumNode}_R^{\text{audit,time,src}} \left[\text{TimeOut}_R^{\text{audit,transfer}}(s) \right]$$

and, for a result edge, by

$$\text{Out}_R^{\text{audit,dep}}(\text{Edge}_{R,q}^{\text{audit,result}}) := \text{EnumNode}_R^{\text{audit,result}} [\{q\}]$$

For the same κ_{edge} and i , define the audit-boundary support by

$$\begin{aligned}\text{ClaimSupport}_{R,i}^{\text{audit},\kappa_{\text{edge}}} &:= \text{EnumNode}_R^{\text{audit,claim,phys}} \left[\text{ClaimIn}_R^{\text{audit},\kappa_{\text{edge}}}(i) \right], \\ \text{TimePrereq}_{R,i}^{\text{audit},\kappa_{\text{edge}}} &:= \text{TimeSem}_R^{\text{audit}} \left[\text{TimeIn}_R^{\text{audit},\kappa_{\text{edge}}}(i) \right]\end{aligned}$$

Define the support, time prerequisites, and time output for transfer events and results by

$$\begin{aligned}\text{Support}_R^{\text{audit,transfer}}(\text{TransferEvent}_{R,s}) &:= \text{ClaimSupport}_{R,s}^{\text{audit,transfer}}, \\ \text{TransferIn}_{R,\text{TransferEvent}_{R,s}} &:= \text{TimePrereq}_{R,s}^{\text{audit,transfer}}, \\ \text{Support}_R^{\text{audit,result}}(\text{Result}_{R,q}) &:= \text{ClaimSupport}_{R,q}^{\text{audit,result}}, \\ \text{ResultPrereq}_{R,\text{Result}_{R,q}} &:= \text{TimePrereq}_{R,q}^{\text{audit,result}}, \\ \text{TransferOut}_{R,\text{TransferEvent}_{R,s}} &:= \text{TimeSem}_R^{\text{audit}} \left[\text{TimeOut}_R^{\text{audit,transfer}}(s) \right]\end{aligned}$$

Define the original-source location map of dependency-effective nodes by

$$\begin{aligned}\text{SrcLoc}_R^{\text{audit}}(\text{Node}_{R,p}^{\text{audit,claim,phys}}) &:= \{ \text{ClaimLoc}_R^{\text{audit}}(p) \}, \\ \text{SrcLoc}_R^{\text{audit}}(\text{Node}_{R,m}^{\text{audit,claim,math}}) &:= \emptyset, \\ \text{SrcLoc}_R^{\text{audit}}(\text{Node}_{R,j}^{\text{audit,time,src}}) &:= \{ \text{UseLoc}_R^{\text{audit}}(k) \mid (k,j) \in \text{UseSrc}_R^{\text{audit}} \}, \\ \text{SrcLoc}_R^{\text{audit}}(\text{Node}_{R,q}^{\text{audit,result}}) &:= \emptyset\end{aligned}$$

Let the audit-branch set be

$$\text{Branch}_R^{\text{audit}} := \text{Idx}_R^{\text{audit,branch}}$$

The structure determined above,

$$\begin{aligned} \text{Realize}^{\text{audit}}(\text{AuditInput}_R^{\text{fin}}) := & \left(\text{Hypergraph}_R^{\text{audit,evid}}, \text{Hypergraph}_R^{\text{audit,dep}}, \right. \\ & \text{Hypergraph}_R^{\text{audit}}, \text{Quot}_R^{\text{audit,bd}}, \\ & \text{SrcLoc}_R^{\text{audit}}, \left\{ \text{Support}_{R,\iota}^{\text{audit,branch}} \right\}_{\iota \in \text{Branch}_R^{\text{audit}}}, \\ & \text{Cycle}_R^{\text{audit}}, \text{SelfEdge}_R^{\text{audit}}, \text{Evidence}_R^{\text{audit,unmatched}}, \\ & \left. \text{Count}_R^{\text{audit}} \right) \end{aligned}$$

is called the canonical realization of typed finite audit input.

Proposition 7.91 (Uniqueness of the Realization of Typed Finite Audit Input). Once all components of the typed finite audit-input structure $\text{AuditInput}_R^{\text{fin}}$ are given, all node sets, all edge sets, input and output maps, the boundary-contraction map, the original-source location map, the branch-support families, the unmatched-evidence set, and the completeness-count map belonging to the canonical realization of Definition 7.90 are uniquely determined.

Proof. The enumerated nodes are uniquely determined by the corresponding value families, and the node sets are uniquely determined as the images or finite disjoint unions of the respective enumerated-node families. The transfer and result edges and the reference-specific inputs and outputs are uniquely determined by the corresponding finite index sets and input tables.

The reference-specific content correspondence determines the boundary-contraction map, and the images of the inputs and outputs determine the audit boundary. The original-source location map is determined by the original-source location components and the time-use-content correspondence, and the audit-branch support families are determined by the branch-result correspondence and edge-restricted reachability. The unmatched-evidence set and the completeness-count map are given as input components. Hence every component of the canonical realization is unique. $\square$

7.2.7 Typed Index Display of Audit Tables

Definition 7.92 (Typed Index Display of Audit Tables). For an index set I , a family

$$\mathcal{F} = (F_i)_{i \in I}$$

and a finite set $J \subseteq I$, let

$$\mathcal{F}[J] := \{F_i \mid i \in J\}$$

. When positive integers m, n satisfy $m \leq n$, define the discrete closed interval in the natural numbers by

$$\llbracket m, n \rrbracket := \{k \in \mathbb{N} \mid m \leq k \leq n\}$$

. Define the positive initial index interval for a positive integer n by

$$[n] := \llbracket 1, n \rrbracket$$

. Then define the finite-index-set display by

$$\{F_i\}_{i=1}^n := \{F_i \mid i \in [n]\}$$

. A column heading in a reference-specific audit table specifies a target family. The value in each cell is a finite index set of that target family. For each reference R , a time-content index set J , a supporting-claim index set P , and a result index set Q correspond, respectively, to

$$\begin{aligned} \text{TimeContent}_R[J] &= \{\text{TimeContent}_{R,j} \mid j \in J\}, \\ \text{Node}_R^{\text{audit,claim}}[P] &= \{\text{Node}_{R,p}^{\text{audit,claim}} \mid p \in P\}, \\ \text{Result}_R[Q] &= \{\text{Result}_{R,q} \mid q \in Q\} \end{aligned}$$

. This correspondence determines the semantic objects of the respective columns for time-content input indices, time-content output indices, time-content prerequisite indices, corresponding time-content indices, related time-content indices, supporting-claim indices, and result-output indices.

Definition 7.93 (Finite-Set Constructors for Audit-Node Families). For each $R \in \text{Ref}^{\text{audit}}$ and positive integer n , define the physical and formal claim-node set constructor and the semantic-node set constructor by

$$\begin{aligned} \text{ClaimNodes}_R(n) &:= \left\{ \text{Node}_{R,i}^{\text{audit,claim}} \mid i \in [n] \right\}, \\ \text{SemanticNodes}_R(n) &:= \left\{ \text{Node}_{R,i}^{\text{audit,semantic}} \mid i \in [n] \right\} \end{aligned}$$

Definition 7.94 (Semantic-Support Lift Map). For each reference R that has a semantic-node family and a semantic-support index map, define

$$\begin{aligned} \text{SemanticLift}_R &: \text{dom}(\text{SupportIndex}_R^{\text{semantic}}) \longrightarrow \mathcal{P}(\text{Node}_R^{\text{audit,semantic}}), \\ \text{SemanticLift}_R(P) &:= \text{Node}_R^{\text{audit,semantic}}[\text{SupportIndex}_R^{\text{semantic}}(P)]. \end{aligned}$$

Definition 7.95 (Prefix-Interval Audit Boundary for a Finite Node Family). For a finite indexed set

$$S = \{S_i\}_{i=1}^N$$

and

$$1 \leq n \leq N$$

define the prefix-interval audit boundary by

$$\text{PrefixBoundary}(S; n) := (\{S_i\}_{i=1}^n, S)$$

. The first component is the generation-initial set consisting of the first n components, and the second component is the reference formal-evidence environment consisting of the entire finite indexed set.

Definition 7.96 (Prefix-Interval Boundary Constructors for Audit-Node Families). For each $R \in \text{Ref}^{\text{audit}}$ and positive integer n , define

$$\begin{aligned} \text{ClaimPrefix}_R(n) &:= \text{PrefixBoundary}(\text{Node}_R^{\text{audit,claim}}; n), \\ \text{SemanticPrefix}_R(n) &:= \text{PrefixBoundary}(\text{Node}_R^{\text{audit,semantic}}; n) \end{aligned}$$

Definition 7.97 (Audit Boundary by Two Index Sets). For an index set I , a family $\mathcal{F} = (F_i)_{i \in I}$, and $(J, K) \in \mathcal{P}(I)^2$, define $\text{IndexedBoundary}(\mathcal{F}; (J, K)) := (\mathcal{F}[J], \mathcal{F}[K])$.

Definition 7.98 (Initial-Set–Evidence–Environment Pair). For each reference R , define the claim-audit-boundary index pair by

$$\text{ClaimBoundaryIdx}_R := \left(\text{Idx}_R^{\text{audit,claim,init}}, \text{Idx}_R^{\text{audit,claim,evid}} \right)$$

. Define the audit-typed and physically typed initial-set–evidence–environment pairs by

$$\begin{aligned} \text{InitEnv}_R^{\text{audit}} &:= \left(\text{Init}_R^{\text{audit,phys}}, \text{Env}_R^{\text{audit,evid}} \right), \\ \text{InitEnv}_R^{\text{phys}} &:= \left(\text{Init}_R^{\text{phys}}, \text{Env}_R^{\text{phys}} \right) \end{aligned}$$

. When the enumeration of physical and formal claim nodes and the two claim-index sets are given,

$$\begin{aligned} \text{InitEnv}_R^{\text{audit}} \\ = \text{IndexedBoundary} \left(\begin{array}{c} \text{EnumNode}_R^{\text{audit,claim,phys}}, \\ \text{ClaimBoundaryIdx}_R \end{array} \right). \end{aligned}$$

The first component of each pair is the generation-initial physical-input node set, and the second is the reference formal-evidence environment node set.

Definition 7.99 (Proper-Inclusion Property of Claim Audit Boundaries). For each $R \in \text{Ref}^{\text{audit}}$, define

$$\begin{aligned} \text{StrictClaimBoundary}(R) \\ \iff \text{Idx}_R^{\text{audit,claim,init}} \subsetneq \text{Idx}_R^{\text{audit,claim,evid}}, \\ \text{TypedStrictClaimBoundary}(R) \\ \iff \begin{array}{c} \text{StrictClaimBoundary}(R) \wedge \\ \text{Idx}_R^{\text{audit,claim,evid}} \subseteq \text{Idx}_R^{\text{audit,phys}} \end{array} \end{aligned}$$

Definition 7.100 (Standard Lift of Semantic-Support Indices). For each reference R , positive integer n , and finite index set P , define

$$\begin{aligned} \text{SemanticIndexLift}_R(P; n) \\ := \{n + p \mid p \in P\} \\ \cup \{a \in [n] \mid \text{ClaimIndex}_R^{\text{init}}(a) \cap P \neq \emptyset\}. \end{aligned}$$

The first term moves the reference formal-claim indices after the first n generation-initial semantic components; the second adds the generation-initial semantic components having a reference formal claim in common with P .

Definition 7.101 (Lift of Claim Nodes and Mathematical Nodes to Semantic Nodes). For each reference R , nonnegative integer s , and positive integer n , define the lifts of claim nodes and mathematical nodes to semantic nodes by

$$\begin{aligned} \text{ClaimNodeLift}_R(s; n) &\iff \forall i \in [n], \quad \text{Node}_{R,s+i}^{\text{audit,semantic}} = \text{Node}_{R,i}^{\text{audit,claim}}, \\ \text{MathNodeLift}_R(s; n) &\iff \forall i \in [n], \quad \text{Node}_{R,s+i}^{\text{audit,semantic}} = \text{Node}_{R,i}^{\text{audit,math}}. \end{aligned}$$

Definition 7.102 (Standard Semantic-Support Index Specialization). For each reference R and positive integers n, M , define the standard semantic-support index specialization by

$$\text{StdSemLift}_R(n; M) \iff \forall P \subseteq [M], \quad \text{SupportIndex}_R^{\text{semantic}}(P) = \text{SemanticIndexLift}_R(P; n).$$

Definition 7.103 (Audit Count Vectors). For each $R \in \text{Ref}^{\text{audit}}$, define the obligation-state count vector and terminal-component occurrence-count vector by

$$\begin{aligned} \text{ObligationCount}_R &:= (N_{\text{fulfilled}}, N_{\text{unfulfilled}}, N_{\text{open}}), \\ \text{TerminalCount}_R &:= (N_{\text{explicit}}, N_{\text{structural}}, N_{\text{cycle}}, N_{\text{physical}}, N_{\text{evidence-open}}) \end{aligned}$$

. Define the reference-specific audit count vector by

$$\text{AuditCount}_R := (\text{ObligationCount}_R, \text{TerminalCount}_R)$$

. Define the audit-boundary count pair consisting of the number of audit-boundary nodes and the number of audit-boundary edges by

$$\text{BoundaryCount}_R^{\text{audit}} := \left(\left| \text{Node}_R^{\text{audit, bd}} \right|, \left| \text{Edge}_R^{\text{audit, bd}} \right| \right)$$

.

Definition 7.104 (Reference-Specific Completeness-Count Vector). Define the ordered sequence of completeness-count fields by

$$\begin{aligned} \text{AuditFieldSeq} &= (\text{Field}_k^{\text{audit}})_{k=1}^{26} \\ &:= \left(\begin{array}{l} \text{timeuse, timecontent, userole, contentrole, obligation,} \\ \text{useobligation, roleobligation, physclaim, mathclaim, claimnode,} \\ \text{usenode, srccontentnode, rootusenode, resultnode,} \\ \text{evidnode, transferedge, resultedge, auditedge,} \\ \text{branch, alternative, fulfilled, unfulfilled,} \\ \text{open, result, margin, unmatched} \end{array} \right). \end{aligned}$$

For each $R \in \text{Ref}^{\text{audit}}$, define the reference-specific completeness-count vector by

$$\text{CompletenessCount}_R := (\text{Count}_R^{\text{audit}} (\text{Field}_k^{\text{audit}}))_{k=1}^{26}$$

.

Definition 7.105 (Typed-Index Realization of Audit Incidence). For each $R \in \text{Ref}^{\text{audit}}$, let the supporting-claim index set, time-content input index set, and time-content output index set specified by the reference-specific transfer table or by finite indices be

$$P_{R,s}^{\text{tr}}, \quad J_{R,s}^{\text{in}}, \quad J_{R,s}^{\text{out}}, \quad s \in \text{Idx}_R^{\text{audit, transfer}}$$

. These satisfy

$$\begin{aligned} P_{R,s}^{\text{tr}} &\subseteq \text{Idx}_R^{\text{audit, phys}}, \\ J_{R,s}^{\text{in}}, J_{R,s}^{\text{out}} &\subseteq \text{Idx}_R^{\text{audit, time, src}} \end{aligned}$$

. Let the supporting-claim index set and time-content prerequisite index set specified by the reference-specific result table or by finite indices be

$$P_{R,q}^{\text{res}}, \quad J_{R,q}^{\text{res}}, \quad q \in \text{Idx}_R^{\text{audit,result}}$$

. These satisfy

$$P_{R,q}^{\text{res}} \subseteq \text{Idx}_R^{\text{audit,phys}}, \quad J_{R,q}^{\text{res}} \subseteq \text{Idx}_R^{\text{audit,time,src}}$$

. Require indices that denote the same transfer event or the same result to give identical support, input, and output sets in the realizations below.

Realize each transfer index by

$$\begin{aligned} \text{Support}_R^{\text{audit,transfer}}(\text{TransferEvent}_{R,s}) &:= \text{EnumNode}_R^{\text{audit,claim,phys}}[P_{R,s}^{\text{tr}}], \\ \text{TransferIn}_{R,\text{TransferEvent}_{R,s}} &:= \text{TimeContent}_R[J_{R,s}^{\text{in}}], \\ \text{TransferOut}_{R,\text{TransferEvent}_{R,s}} &:= \text{TimeContent}_R[J_{R,s}^{\text{out}}] \end{aligned}$$

. Realize each result index by

$$\begin{aligned} \text{Support}_R^{\text{audit,result}}(\text{Result}_{R,q}) &:= \text{EnumNode}_R^{\text{audit,claim,phys}}[P_{R,q}^{\text{res}}], \\ \text{ResultPrereq}_{R,\text{Result}_{R,q}} &:= \text{TimeContent}_R[J_{R,q}^{\text{res}}] \end{aligned}$$

. For a source R for which the map in Definition 7.94 is defined, define the semantic support of each transfer s or result q by the corresponding equation below whenever its supporting-claim index set $P_{R,s}^{\text{tr}}$ or $P_{R,q}^{\text{res}}$ belongs to $\text{dom}(\text{SemanticLift}_R)$.

$$\begin{aligned} \text{SemanticSupport}_R^{\text{audit,transfer}}(\text{TransferEvent}_{R,s}) &:= \text{SemanticLift}_R(P_{R,s}^{\text{tr}}), \\ \text{SemanticSupport}_R^{\text{audit,result}}(\text{Result}_{R,q}) &:= \text{SemanticLift}_R(P_{R,q}^{\text{res}}). \end{aligned}$$

When a reference-specific specification has a single time-content output index

$$j_{R,s}^{\text{out}}$$

set

$$J_{R,s}^{\text{out}} := \{j_{R,s}^{\text{out}}\}$$

. **Proposition 7.106** (Compatibility of Typed Incidence Indices with the Canonical Audit Realization). For the input of Definition 7.89, let

$$\begin{aligned} P_{R,s}^{\text{tr}} &= \text{ClaimIn}_R^{\text{audit,transfer}}(s), & J_{R,s}^{\text{in}} &= \text{TimeIn}_R^{\text{audit,transfer}}(s), \\ J_{R,s}^{\text{out}} &= \text{TimeOut}_R^{\text{audit,transfer}}(s), & P_{R,q}^{\text{res}} &= \text{ClaimIn}_R^{\text{audit,result}}(q), \\ J_{R,q}^{\text{res}} &= \text{TimeIn}_R^{\text{audit,result}}(q) \end{aligned}$$

. Then the support sets, time-content input and output sets, and result-prerequisite sets of Definition 7.105 coincide with the corresponding components determined by the canonical realization of Definition 7.90. Therefore, the two realizations determine the same audit-boundary directed hypergraph.

Proof. Substitute the identifications of the five index sets into the support sets, time-content input and output sets, and result-prerequisite sets in the two definitions. By the family-image correspondence of Definition 7.92, each right-hand side gives the same set of enumerated nodes or the same reference-specific time-content set. Because the input and output sets of the transfer edges and result edges coincide, the nodes, edges, and incidence maps of the audit-boundary directed hypergraphs coincide. $\square$

Proposition 7.107 (Unique Realization of an Audit Boundary by a Reference-Specific Finite Display). For each $R \in \text{Ref}^{\text{audit}}$, take finite families of claim nodes, time contents, transfer events, and results. Suppose that the physical-input initial set and the reference formal-evidence environment follow Definition 7.28 and that each index set in the transfer and result tables satisfies the typing conditions and the consistency conditions for identical events and results in Definition 7.105.

This finite display uniquely determines an audit-boundary directed hypergraph equipped with support, time-content inputs, time-content outputs, and result prerequisites. Hence reachability from the physical-input initial set and the reference formal-evidence environment, primitive time roots, the fourfold result partition, and the minimal primitive-time-root support family are also uniquely determined.

Proof. By Definition 7.105, the inputs, outputs, and support of each transfer edge and result edge are uniquely determined. The finite node and edge families, together with their incidence, uniquely determine the audit-boundary directed hypergraph.

Reachability is determined inductively from this directed hypergraph and the specified initial set. The primitive time roots and the fourfold result partition are determined by that reachability. The minimal primitive-time-root support family is determined as the family of inclusion-minimal elements in the family of subsets of the finite primitive-time-root set. $\square$

Definition 7.108 (Minimal Primitive-Time-Root Support Index Families and Sum-Product of Set Families). For each $R \in \text{Ref}^{\text{audit}}$ and

$$\text{Result}_{R,q} \in \text{Result}_R$$

define the minimal primitive-time-root support index family by

$$\begin{aligned} \mathcal{A}_R(q) \\ := \{ J \subseteq [n_R] \mid \text{TimeContent}_R[J] \in \text{MinRootSupport}_R^{\text{audit}}(\text{Result}_{R,q}) \}. \end{aligned}$$

Then

$$\begin{aligned} \text{MinRootSupport}_R^{\text{audit}}(\text{Result}_{R,q}) \\ = \{ \text{TimeContent}_R[J] \mid J \in \mathcal{A}_R(q) \}. \end{aligned}$$

Define the sum-product of finite index-set families

$$\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}([n_R])$$

by

$$\mathcal{A} \boxplus \mathcal{B} := \{ A \cup B \mid A \in \mathcal{A}, B \in \mathcal{B} \}$$

.

7.2.8 Reference-Specific Two-Track Evidence Records

Definition 7.109 (Reference-Specific Two-Track Evidence Records). The evidence record for each reference is defined by two tracks: a formal-evidence record and a statement-evidence record.

The formal-evidence record has the following components in this order.

1. The formal object.
2. The original-source location at which the formal object is given.
3. The assumptions and non-temporal physical inputs explicitly stated in the original source.
4. The time contents used in the original source.
5. Dependencies or transfers among time contents, inputs, and results.
6. Reference-specific results.
7. For a reference that includes an experiment or observation, its directly measured object.
8. The domain of application of the formal generalization given by the original source.
9. Time contents generated by audit reachability from non-temporal physical inputs, and the primitive time roots that remain.
10. The terminal objects, predicates, conditions, and domains of application directly supported by the original-source evidence.
11. Reference-specific results preserved after the generation-order audit.

The statement-evidence record has a quotation-fact layer and an interpretive-audit layer, in this order. The quotation-fact layer has the following components in this order.

1. The verbatim quotation in the evidentiary source language.
2. For a non-English evidentiary source, an English translation corresponding to the verbatim quotation.
3. The original-source location.
4. The minimal context required for interpretation.
5. The subject to which the statement is attributed.

The interpretive-audit layer has the following components in this order.

1. The object addressed by the statement, the predicate assigned to that object, and the explicit domain of application.
2. The multiple possible interpretations permitted by the original text and its minimal context.

3. For each possible interpretation, the terminal objects, predicates, conditions, and domains of application directly supported by the original-source evidence.
4. The formal-evidence record against which it is compared.
5. The correspondence or difference between each possible interpretation and the formal evidence.

Audit determinations in the formal-evidence record are determined from the definitions, formulas, propositions, operational rules, and dependencies in the original source. Comparison results in the statement-evidence record do not alter the audit determinations in the formal-evidence record.

Proposition 7.110 (Separation and Comparison of Two-Track Evidence Records). Determinations in the reference-specific formal audit are determined from the formal-evidence record, and possible interpretations of a statement are determined from the quotation-fact and interpretive-audit layers of the statement-evidence record.

When the formal description and statement in the same reference address the same object, predicate, or domain of application, their correspondence is recorded for each possible interpretation. When the objects, predicates, or domains of application of the formal description and statement differ, the difference is preserved.

Proof. Definition 7.109 defines the formal-evidence record and statement-evidence record as separate tracks and divides the latter into a quotation-fact layer and an interpretive-audit layer. The objects, dependencies, results, and audit reachability in the formal-evidence record provide the inputs to the formal audit. In the statement-evidence record, the quotation-fact layer preserves evidence from which the original text can be recovered, while the interpretive-audit layer preserves, for every permissible interpretation, the correspondence with or difference from the formal evidence and the evidence-effect boundary. The formal and statement tracks can therefore be compared while remaining separate. $\square$

7.2.9 Evidence Fields and Local Keys in Reference-Specific Audit Tables

Definition 7.111 (Evidence Fields in Reference-Specific Audit Tables). When the formal-evidence record of Definition 7.109 is displayed in a reference-specific audit table, the following six fields are distinguished.

1. The grounds for the result state the definitions, assumptions, formal relations, operations, observations, or experiments in the original source that support the reference-specific result.
2. The directly measured object states the physical carrier, physical quantity, or display to which the apparatus directly responds, which it records, or which it compares.
3. The directly measured predicate states the value, difference, rate, correlation, count, or relation obtained for the directly measured object.
4. The conditions and domain of application of the direct measurement state the apparatus configuration, calibration, approximations, background conditions, operations, and object range required for that measurement.

5. The object of formal generalization states the object or family of objects to which the formal construction in the original source extends its application. This object need not be identical to the directly measured object.
6. The conditions and domain of application of the formal generalization state the assumptions, maps, identifications, approximations, background conditions, and object range required for that generalization.

The six fields are used to determine the evidence-effect domain of each item without substituting the grounds for the result, the direct measurement, and the formal generalization for one another.

Definition 7.112 (Local Keys in Reference-Specific Audit Tables). In a finite index table of Definition 7.92, a positive integer that specifies a referent only within one correspondence table for one reference is called a local key. The following kinds of local key are distinguished.

1. An original-source location key refers to one or more locations in the original source.
2. A time-content key refers to a finite set of time-content indices defined for that reference.
3. A formal-evidence key refers to a finite set of indices of reference formal descriptions supporting a transfer or result.
4. A semantic-component key refers to a finite set of indices of semantic components separated from the reference formal description.

Even when the integer is the same, it does not denote the same referent if the reference, correspondence table, or kind of key differs. Each correspondence table gives the correspondence between local keys and their referents before the table that uses those keys or within the same descriptive unit. A local key does not replace evidence; it abbreviates only repeated display within a table while preserving the ability to recover the evidence, time content, or semantic component through the correspondence table.

7.2.10 Public Names and Result-Preservation Domains in Reference-Specific Audits

Definition 7.113 (Public Names in Reference-Specific Audits). A transfer-typed audit-boundary edge of Definition 7.26 is called a transfer hyperedge, and a result-typed audit-boundary edge is called a result-dependency hyperedge.

The physical-input initial set of Definition 7.28 is the starting point of the audit-reachable set of Definition 7.30, which is called the generation-initial closure. For the same audit-reachable set, the set that starts from the reference formal-evidence environment is called the evidence closure. These are results of the same reachability operation with different starting points; they do not introduce a new closure operation.

Definition 7.114 (LCTR Generation Order, Result-Preservation Conditions, and Result-Preservation Domain). The dependency precedence relations among non-temporal physical inputs, physical-carrier realization, comparison, time representation, dynamical description, law time, and validity determination constructed from 2 through 6 are called the LCTR generation order. This term does not mean a total order that arranges all objects in one

sequence; it denotes the precedence relations determined by the input dependencies of the respective constructed objects.

The original-source inputs, assumptions, approximations, boundary conditions, dependency directions, and domains of application that must be preserved in order to use the formal content of a reference-specific result without alteration under the LCTR generation order, together with the time representations that must be generated before use in place of primitive time roots, are called the result-preservation conditions. The range of objects and scales over which all result-preservation conditions hold and the result in question can be used is called the result-preservation domain.

The result-preserving correspondence of Definition 7.77 specifies the correspondence between audited result nodes and reference-specific results. The result-preservation conditions and result-preservation domain specify the conditions under which, and the range over which, the result identified by that correspondence can be used under the LCTR generation order.

7.2.11 Evidence-Effect Boundaries and Unbounded Inheritance of Limited Results

Definition 7.115 (Evidence-Effect Domain and Evidence-Effect Boundary). The tuple of objects, predicates, conditions, and domains of application directly supported by reference-specific evidence is called the evidence-effect domain of that evidence.

In a dependency sequence from a body of evidence to a subsequent claim, a change in object, predicate, evidence type, grounding condition, or domain of application requires independent evidence or a valid derivation supporting that change. The first position at which the change occurs without such support is called the evidence-effect boundary in that dependency sequence.

The evidence-effect boundary gives the terminal point of the domain directly established by the preceding evidence. A claim beyond the boundary requires evidence or a derivation specific to that claim.

Definition 7.116 (Unbounded Inheritance of Limited Results). The direct inheritance, by a claim beyond the evidence-effect boundary, of mathematical consistency, predictive performance, model fit, or empirical agreement established at a preceding stage of a dependency sequence is called the unbounded inheritance of limited results.

The stages of inheritance subject to audit include the following.

1. A comparative difference measured for an apparatus, clock, oscillator, signal, detector, record, or physical process.
2. The physical validity of the entire formal system that calculated that comparative difference.
3. The physical-time qualification of the time content used by the formal system.
4. Attribution from a comparative difference on a physical carrier to the existence or property of time itself, independently of the physical carrier.
5. Transfer from a property attributed to time itself to biological processes, subjective experience, cosmic history, or another domain of application.

If inheritance across an evidence-effect boundary occurs between two adjacent stages in this sequence, the unbounded inheritance of limited results holds.

Definition 7.117 (Interpretation-Specific Evidence-Comparison Path and Number of Evidence-Effect Transgressions). An evidence-dependency sequence determined by a formal construction in a reference, a statement in the reference, and one permissible interpretation of that statement is called an interpretation-specific evidence-comparison path. If the same reference or the same statement has multiple permissible interpretations, a distinct path is defined for each interpretation.

Within one interpretation-specific evidence-comparison path, each position at which the object, predicate, evidence type, grounding condition, or domain of application changes without independent evidence or a valid derivation supporting that change is called an evidence-effect transgression. The number of evidence-effect transgressions connected successively by direct dependencies is called the number of evidence-effect transgressions along that path. The number of evidence-effect layers is not taken as the number of evidence-effect transgressions. Even for the same statement, different permissible interpretations can have different numbers of evidence-effect transgressions.

The number of evidence-effect transgressions is 0 for a path that directly describes a comparative difference on a physical carrier in terms of its object and relation. If an expression that appears to assign a property to time is used, the number of evidence-effect transgressions produced by that figurative expression is 0 when the original-source text states, within the same explanatory scope, both that the expression is figurative and does not denote a literal deformation of time, and what the actual referent is.

If expressions such as “time slows down,” “time expands or contracts,” or “time is affected” are used without identifying their figurative character and actual referent, the statement transfers a comparative difference on a physical carrier to a property of time and crosses one evidence-effect boundary. The possibility of interpreting the expression retrospectively as shorthand does not remove this evidence-effect transgression.

Proposition 7.118 (Failure of Limited Results to Establish Claims Outside Their Domain of Application). Measurement success, model fit, predictive success, or mathematical consistency directly establishes the objects, predicates, conditions, and domains of application that belong to its evidence-effect domain. A claim beyond the evidence-effect boundary is not established by the preceding limited result alone; it requires independent evidence or a valid derivation specific to that claim.

Proof. By Definition 7.115, the evidence-effect domain is determined by the objects, predicates, conditions, and domains of application directly supported by reference-specific evidence, and the evidence-effect boundary is the first position of change that lacks independent evidence or a valid derivation. A claim beyond the boundary therefore does not belong to the domain directly supported by the preceding evidence.

By Proposition 7.8 and Proposition 7.13, evaluation success does not add physical-time qualification to a time content and does not change primitive time input into generated time content. For inheritance from a limited result to the physical-time qualification of a time content, these two propositions establish that an evidence-effect boundary is crossed.

Applying the same determination to every stage at which the object, predicate, evidence type, grounding condition, or domain of application changes shows that a claim beyond the boundary requires independent evidence or a valid derivation specific to that claim. $\square$

Corollary 7.119 (Inheritance of Primitive Time Dependence and Unbounded Inheritance of Limited Results). Result inheritance of primitive time dependence represents the dependency direction along which ungenerated time content remaining in an input propagates to a result. The unbounded inheritance of limited results represents the dependency direction

along which the effect of preceding evidence is extended to a different object, predicate, evidence type, grounding condition, or domain of application.

The two forms of inheritance constitute distinct audit axes and can hold simultaneously in the same reference or the same explanatory sequence.

Proof. The former is determined by a dependency path from primitive time input to a result, and the latter by inheritance across the evidence-effect boundary of Definition 7.115. Because the dependency directions subjected to determination differ, they provide distinct audit axes. If a single reference has ungenerated time content as an input and simultaneously transmits a limited success to another claim, the determinations on both axes hold at the same time. $\square$

Definition 7.120 (Backgrounding of a Grounding Obligation by Primitive Time Input). Suppose that a reference R introduces a time content x into a law operation or result derivation and proceeds with that derivation without examining, as an explicit input, generation condition, or grounding condition within the reference, what is required for the generation or use qualification of x , namely a condition χ . The primitive time input x is then said to background the grounding obligation for the condition χ .

For a backgrounded grounding obligation, the following four items are recorded separately.

1. The backgrounded generation condition or grounding condition.
2. Independent evidence establishing actual failure of the same condition.
3. Dependency reachability from the actual failure position to downstream results.
4. The extent of condition repair, use-domain restriction, and downstream regeneration or recalculation.

Actual failure is established by the independent evidence in the second item; it is not established solely by the presence of primitive time input or backgrounding.

Proposition 7.121 (Causal Connection from a Backgrounded Grounding Obligation to Downstream Results). When the first through fourth items of Definition 7.120 are connected by the same condition χ , primitive time input, the backgrounding of a grounding obligation, actual failure of the condition, loss of use qualification for downstream results, and regeneration or recalculation after repair constitute one audit causal sequence.

If the independent evidence in the second item does not hold, actual failure of the condition in question is undetermined. If the dependency reachability in the third item does not hold, actual failure of that condition does not invalidate the use qualification of that downstream result.

Proof. The first item identifies a generation condition or grounding condition that is not explicitly examined because of primitive time input. The second establishes actual failure of the same condition from independent evidence within or outside the reference. The third establishes dependency reachability with the actual failure position as its starting point and the result derivation as its endpoint. The fourth gives the repair, use-domain restriction, and regeneration range for results belonging to that reachable domain. When the four items share the same condition, the respective stages are connected as one dependency sequence. $\square$

7.2.12 Input Consistency of Audit-Boundary Directed Hypergraphs

Audit-boundary directed hypergraphs are given by Definition 7.27.

The physical and mathematical boundary claims, time contents, transfer events, and reference-specific results stated in each reference are represented by a finite directed hypergraph for that reference. The input set of each hyperedge represents the set of premises used jointly to establish its output.

Audit-boundary nodes, audit-boundary support sets, and audit-boundary edges are given by Definition 7.18 and Definition 7.26.

Proposition 7.122 (Input Consistency of Audit-Boundary Directed Hypergraphs). For each $R \in \text{Ref}^{\text{audit}}$, the time contents, transfer events, and results correspond injectively to their respective audit-boundary nodes or audit-boundary edges. The time-content input and time-content output of each transfer edge coincide with the images of

$$\text{TransferIn}_{R,\epsilon}, \quad \text{TransferOut}_{R,\epsilon}$$

and the time-content input of each result edge coincides with the image of

$$\text{ResultPrereq}_{R,\eta}$$

.

Proof. The time-content node map, result-node map, transfer-edge node map, and result-edge node map are injective in Definition 7.18 and Definition 7.26, respectively.

The input and output maps of Definition 7.27 take

$$\begin{aligned} &\text{NodeOf}_R^{\text{audit,time}} [\text{TransferIn}_{R,\epsilon}], \\ &\text{NodeOf}_R^{\text{audit,time}} [\text{TransferOut}_{R,\epsilon}] \end{aligned}$$

as the time-content input and time-content output, respectively, for a transfer edge. The same definition takes the time-content input of a result edge to be

$$\text{NodeOf}_R^{\text{audit,time}} [\text{ResultPrereq}_{R,\eta}]$$

. Thus each correspondence holds. □

7.3 Formal Evidence, Source Statements, and Representative Audits

The formal descriptions and original-source statements of the audited references are distinguished, and the reference-specific formal audits, representative formal evidence, and statement evidence preserving interpretive possibilities are presented.

For every reference in the audited reference set $\text{Ref}^{\text{audit}}$, Table 16 presents the complete reference-specific audit evidence for primitive time roots and result dependence. Under the complete determination, there are no generated time contents in any of the 72 references; in every reference, all reference-specific time contents remain as primitive time roots, and each reference has results that depend on those roots. The local-reachability-determination column records differences from this common result when the reference formal-evidence environment is added. A dash denotes the baseline case in which local reachability is empty and all reference-specific time contents remain as primitive time roots. The first

reference in the evidence column establishes the primitive time roots, and the second establishes result dependence.

Table 16: Primitive-time-root and result-dependence evidence for the 72 audited references

No.	Reference	Difference under local-reachability determination	Root and result-dependence evidence
1	Aristotle	—	8.121, 8.121
2	Newton 1687/1729	—	8.128, 8.128
3	Einstein 1905	—	8.135, 8.135
4	Minkowski 1908/1909	—	8.142, 8.142
5	Einstein 1916	—	8.149, 8.149
6	Bridgman 1927	—	8.155, 8.155
7	von Neumann 1932	—	8.171, 8.171
8	Curtiss 1949	—	8.179, 8.179
9	Dirac 1958	—	8.188, 8.188
10	Pound–Rebka 1960	—	8.197, 8.197
11	ADM 1962	—	8.205, 8.205
12	Feynman 1963	—	8.213, 8.213
13	Taylor–Wheeler 1992	—	8.228, 8.228
14	DeWitt 1967	2 reached; 33 remaining	8.235, 8.235
15	Ehlers–Pirani–Schild 1972	—	8.243, 8.243
16	Hafele–Keating 1972b	—	8.259, 8.259
17	Hafele–Keating 1972a	—	8.268, 8.268
18	Bekenstein 1973	—	8.276, 8.276
19	Misner–Thorne–Wheeler 1973	4 reached; 39 remaining	8.284, 8.284
20	Hawking 1975	—	8.292, 8.292
21	Hawking 1976	—	8.301, 8.301
22	Bailey et al. 1977	—	8.308, 8.308
23	Rindler 1977	—	8.316, 8.316
24	Penrose 1979	—	8.324, 8.324
25	Hartle–Hawking 1983	—	8.332, 8.332
26	Page–Wootters 1983	—	8.342, 8.342
27	Bombelli–Lee–Meyer–Sorkin 1987	—	8.351, 8.351
28	Weinberg 1989	2 reached; 20 remaining	8.362, 8.362
29	Isham 1993	12 reached; 33 remaining	8.373, 8.373
30	Kuchař 1992	9 reached; 38 remaining	8.384, 8.384
31	Lebowitz 1993	—	8.395, 8.395

Continued on the next page

Table 16 (continued)

No.	Reference	Difference under local-reachability determination	Root and result-dependence evidence
32	Page 1993	—	8.406, 8.406
33	Barbour 1994 I	—	8.416, 8.416
34	Barbour 1994 II	—	8.427, 8.427
35	Connes–Rovelli 1994	3 reached; 25 remaining	8.439, 8.439
36	Headley–Clayton–Reed 1996	—	8.451, 8.451
37	Maldacena 1998	—	8.460, 8.460
38	Perlmutter et al. 1999	—	8.473, 8.473
39	Riess et al. 1998	—	8.486, 8.486
40	Adler 2003	—	8.501, 8.501
41	Philip W. Anderson 2001	—	8.510, 8.510
42	Clark 2001	—	8.520, 8.520
43	Rovelli 2002	—	8.532, 8.532
44	Zurek 2003	—	8.546, 8.546
45	Davis–Lineweaver 2004	—	8.557, 8.557
46	Schlosshauer 2005	—	8.569, 8.569
47	Kiefer 2005	4 reached; 18 remaining	8.579, 8.579
48	Grøn–Elgarøy 2007	—	8.589, 8.589
49	Ryu–Takayanagi 2006	—	8.602, 8.602
50	Francis–Barnes–James– Lewis 2007	—	8.611, 8.611
51	Cook–Burns 2009	—	8.621, 8.621
52	Peacock 2008	—	8.632, 8.632
53	Bunn–Hogg 2009	—	8.642, 8.642
54	Kiefer 2009	9 reached; 26 remaining	8.650, 8.650
55	Rovelli 2011	7 reached; 16 remaining	8.660, 8.660
56	Van Raamsdonk 2010	—	8.676, 8.676
57	Greene <i>The Illusion of Time</i>	—	8.684, 8.684
58	Wolf et al. 2011	—	8.692, 8.692
59	Anderson 2012	5 reached; 15 remaining	8.702, 8.702
60	Ellis 2012	—	8.710, 8.710
61	Tong 2013	—	8.718, 8.718
62	Lewis 2016	—	8.726, 8.726
63	Smolin 2018/2020	—	8.737, 8.737
64	BIPM SI 2019/2026	—	8.746, 8.746

Continued on the next page

Table 16 (continued)

No.	Reference	Difference under local-reachability determination	Root and result-dependence evidence
65	ESA 2019	—	8.754, 8.754
66	NASA/JPL–Caltech 2020	—	8.762, 8.764
67	Pössel 2021	—	8.773, 8.773
68	PDG 2023 Resonances	—	8.783, 8.783
69	NIST 2025	—	8.792, 8.792
70	Feynman 1995	—	8.799, 8.799
71	Narlikar 2010	—	8.806, 8.806
72	Heisenberg 1958	—	8.813, 8.813

This local-reachability determination shows that adding the physical-carrier processing or construction procedures explicitly stated in the respective references to the evidence environment makes a total of 57 time contents locally reachable in the ten references shown in the table. This reachability represents the limited construction capacity present in the existing references; its domain of determination differs from complete generation starting solely from non-temporal physical inputs.

7.3.1 Explicit Absolute Time and Operational Critique

The audit question common to Newton 1687/1729 and Bridgman 1927 is whether either the explicit positing of absolute time or the operational critique of absolute time generates time contents from the physical-input initial set.

In Newton 1687/1729, absolute, true, and mathematical time precedes the dynamical results. In Bridgman 1927, the operational critique of absolute time is followed by signal propagation, clocks, interval counting, synchronization, and correction as time contents. The respective time contents and original-source locations are given in Table 33 and Table 67.

Proposition 7.123 (Non-Generation of Time under Explicit Absolute Time and Operational Critique). For the reference R corresponding to Newton 1687/1729, Definition 7.56 gives $R \in \text{Ref}^{\text{audit,rootall}}$, and

$$\begin{aligned} \text{Result}_R^{\text{audit,gen}} &= \emptyset, \\ \text{Result}_R^{\text{audit,evidonly}} &= \{\text{Result}_{R,1}\}, \\ \text{Result}_R^{\text{audit,rootdep}} &= \{\text{Result}_{R,j} \mid j \in \{2, \dots, 18\}\} \end{aligned}$$

hold.

For the reference R corresponding to Bridgman 1927, Definition 7.58 gives $R \in \text{Ref}^{\text{audit,rootfull}}$.

Proof. For Newton 1687/1729, apply Proposition 8.128 and Proposition 8.128. For Bridgman 1927, apply Proposition 8.155 and Proposition 8.155. $\square$

Proposition 7.124 (Failure of Operational Critique to Generate Time). The operational critique, clock construction, signal procedures, synchronization procedures, and correction

procedures in Bridgman 1927 do not generate the time contents used by that reference from the physical-input initial set.

Proof. Each transfer event in Definition 8.153 takes at least one time content as input when it outputs a time content. No generation trace from the physical-input initial set to a time content therefore exists. By Proposition 7.123, all 57 time contents are primitive time roots and all 20 results depend on primitive time roots. $\square$

Corollary 7.125 (Logical Separation of the Rejection of Absolute Time from the Generation of Time). Neither a construction that explicitly posits absolute time nor a construction that operationally criticizes absolute time generates physical time contents. To provide a physical ground for a time variable in a physical law, the generation-precedence condition of Corollary 7.15 is required.

Proof. Apply Proposition 7.123 and Proposition 7.124. $\square$

7.3.2 Synchronization, Coordinate Time, and Proper Time

The audit question common to Einstein 1905, Einstein 1916, and Taylor–Wheeler 1992 is whether the constructions of synchronization, coordinate time, and proper time generate time contents from the physical-input initial set.

Einstein 1905 introduces rods, coordinates, and clocks at the outset and develops its account from temporal relations among processes to the common time of a synchronization network and coordinate time. In Einstein 1916, the formal correspondence between spatial coordinates and a time coordinate appears at the beginning of the original source and is developed into local clocks, location-dependent clock rates, and four coordinates. In Taylor–Wheeler 1992, events, inertial frames, clocks, and worldlines are given in advance and are developed into synchronization, event coordinates, proper time, wristwatch time, and integration along worldlines.

The time contents and original-source locations of the respective references are given in Table 41, Table 58, and Definition 8.226.

Proposition 7.126 (Non-Generation of Time in the Construction of Synchronization, Coordinate Time, and Proper Time). For the reference R corresponding to each of Einstein 1905, Einstein 1916, and Taylor–Wheeler 1992, Definition 7.58 gives $R \in \text{Ref}^{\text{audit,rootfull}}$.

Proof. For Einstein 1905, apply Proposition 8.135 and Proposition 8.135. For Einstein 1916, apply Proposition 8.149 and Proposition 8.149. For Taylor–Wheeler 1992, apply Proposition 8.228 and Proposition 8.228. $\square$

Proposition 7.127 (Time-Content Presuppositions of Clock Networks, Worldlines, and Proper Time). The synchronization procedures, coordinate-time constructions, clock-rate comparisons, worldline constructions, and proper-time integrations in the three references transform, compare, or integrate time contents. These procedures do not generate their input time contents from the physical-input initial set.

Proof. Every transfer event in each reference includes a time content in its input when it outputs a time content. By Proposition 7.126, all time contents in the three references are primitive time roots, and all reference-specific results depend on primitive time roots. $\square$

Corollary 7.128 (Physical Grounds for Coordinate Time and Proper Time). The mathematical success of synchronization consistency, Lorentz transformations, clock comparison, or worldline integration does not provide physical-time qualification for the input time contents. Before coordinate time and proper time are used as time variables in physical laws, the generation-precedence condition of Corollary 7.15 must be satisfied.

Proof. Apply Proposition 7.127 and Proposition 7.13. $\square$

7.3.3 Unit Standards, Distributed Clocks, and Experimental Clock Differences

The audit question common to BIPM SI 2019/2026, NASA/JPL–Caltech 2020, and Hafele–Keating 1972a is whether unit definitions, clock corrections, and experimental clock comparisons generate the time contents they use from the physical-input initial set.

BIPM SI 2019/2026 uses the cesium transition frequency, the second, proper time, primary clock signals, clock comparison, and relativistic corrections as components of a single system of time standards, clock comparisons, and corrections. NASA/JPL–Caltech 2020 gives clock comparisons and relativistic corrections for airborne clocks, ground reference clocks, and satellite clocks. Hafele–Keating 1972a gives the reunion display difference between moving clocks and reference clocks as an experimental quantity.

Proposition 7.129 (Primitive Time Roots in Unit Standards, Clock Corrections, and Experimental Clock Differences). For the reference R corresponding to BIPM SI 2019/2026, Definition 7.56 gives $R \in \text{Ref}^{\text{audit,rootall}}$, and 25 of the 32 results depend on primitive time roots.

For $R \in \text{Ref}^{\text{audit}}$ corresponding to NASA/JPL–Caltech 2020, under the complete determination the following equalities

$$\text{TimeContent}_R^{\text{audit,gen}} = \emptyset, \quad \text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R$$

hold. In the four-way partition of results, there are no generated results, evidence-environment-reachable nongenerated results, or results that remain unreachable after primitive time roots are adjoined; all eleven results depend on primitive time roots. Even under the local-reachability determination that adds the reference formal-evidence environment,

$$\text{TimeContent}_R^{\text{audit,evid}} = \emptyset$$

holds: all 20 time contents remain as primitive time roots, and all eleven results depend on those roots.

For the reference R corresponding to Hafele–Keating 1972a, Definition 7.58 gives $R \in \text{Ref}^{\text{audit,rootfull}}$.

Proof. For BIPM SI 2019/2026, apply Proposition 8.746 and Proposition 8.746.

For NASA/JPL–Caltech 2020, apply Proposition 8.762 and Proposition 8.764.

For Hafele–Keating 1972a, apply Proposition 8.268 and Proposition 8.268. $\square$

Proposition 7.130 (Primitive Time Roots Remaining after Addition of the Evidence Environment). For NASA/JPL–Caltech 2020, all 20 time contents are primitive time roots under the complete determination starting solely from non-temporal physical inputs, and there are no generated results. Adding the reference formal-evidence environment reaches none of the 20 time contents, and all eleven results depend on primitive time roots. The clock comparisons and corrections in this reference therefore do not provide physical-time qualification for its use of time as a whole.

Proof. Apply Proposition 7.129 and Proposition 8.763. Theorem 7.11 and Corollary 7.12 give the failure of generation under the complete assessment, the remaining primitive time roots, and the inheritance of their dependence by the results. $\square$

Corollary 7.131 (Generation Boundary of Unit Definitions, Clock Corrections, and Clock Comparisons). Agreement among unit definitions, clock corrections, and experimental clock differences does not generate the time contents that they take as inputs. Primitive time roots remain in all three references, and dependence on those roots is inherited by the reference-specific results.

Proof. Apply Proposition 7.129 and Proposition 7.130. $\square$

7.3.4 Quantum Measurement, Records, and Decoherence

The audit question common to von Neumann 1932, Zurek 2003, and Schlosshauer 2005 is whether measurement processes, environmental records, and decoherence generate from the physical-input initial set the time contents taken as inputs to quantum-mechanical time evolution.

In von Neumann 1932, the time evolution of nonstationary states, two processes, before-and-after measurement relations, and finite-time measurement interactions constitute time contents. In Zurek 2003, unitary evolution, decoherence scales, record persistence, histories, and inference of the past from present records constitute time contents. In Schlosshauer 2005, premeasurement evolution, system–environment interaction evolution, time-dependent reduced states, and the temporal stability of pointer states constitute time contents.

Von Neumann 1932 states:

Wenn der Zustand nicht stationär ist, d. h. sich mit der Zeit ändert, so enthält die Wellenfunktion $\psi = \psi(q_1, \dots, q_k; t)$ die Zeit t , [...]

If the state is not stationary, that is, if it changes with time, the wave function $\psi = \psi(q_1, \dots, q_k; t)$ contains the time t , [...]

The original-source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer, 1932, p. 8, §I.2. The complete evidence record for the same reference is given in paragraph 8.19.7.

On the type of statement supplied by experience, the same reference further states:

Denn die Erfahrung macht nur Aussagen von diesem Typus: ein Beobachter hat eine bestimmte (subjektive) Wahrnehmung gemacht, und nie eine solche: eine physikalische Größe hat einen bestimmten Wert.

For experience makes statements only of the following type: an observer has had a particular (subjective) perception, and never statements of the following type: a physical quantity has a particular value.

The original-source location is p. 224, §VI.1 of the same work. The same section distinguishes the placement of the boundary between the observed object and the observer, the boundary required for comparison with experience, and the conditions under which the two processes apply depending on whether a measurement interaction is present. This statement includes subjective perception among its objects of statement, but it does not state that a change in subjective temporal experience was directly measured.

Zurek 2003 states:

Decoherence is caused by the interaction with the environment which in effect monitors certain observables of the system

The original-source location is Zurek, *Decoherence, Einselection, and the Quantum Origins of the Classical*, *Reviews of Modern Physics* 75, p. 715, abstract. The complete evidence record for the same reference is given in paragraph 8.19.44.

Adler 2003 states:

decoherence has not solved the quantum measurement problem.

The original-source location is Adler, *Why Decoherence Has Not Solved the Measurement Problem: A Response to P. W. Anderson*, arXiv:quant-ph/0112095v3, p. 2, Abstract. The complete evidence record for the same reference is given in paragraph 8.19.40.

Schlosshauer 2005 states:

their implications for the foundational problems of quantum mechanics, most notably the quantum measurement problem, have remained a matter of great controversy.

The original-source location is Schlosshauer, *Decoherence, the Measurement Problem, and Interpretations of Quantum Mechanics*, *Reviews of Modern Physics* 76, p. 1267, abstract. The complete evidence record for the same reference is given in paragraph 8.19.46.

Proposition 7.132 (Separation of the Evidence-Effect Domains of Quantum-Measurement Statements). The first statement in von Neumann 1932 directly supports that a nonstationary wave function contains time t . The measurement statement in the same reference distinguishes subjective apperception, the boundary between the observed object and observer, experiential statements, and the conditions of application of the two processes, but it does not support that a change in subjective temporal experience was directly measured. The statement in Zurek 2003 directly supports the occurrence of decoherence through system–environment interaction. The statement in Adler 2003 separates the suppression of interference by decoherence from a solution to the quantum measurement problem as a whole. The statement in Schlosshauer 2005 supports the continuing controversy over the implications of decoherence for the quantum measurement problem.

Interference suppression, environmental correlation, pointer structure, single-outcome selection, solution of the quantum measurement problem as a whole, and physical generation of the time contents taken as inputs to quantum evolution belong to distinct evidence-effect domains. A claim that transfers the validity of one domain to another requires independent evidence or a valid derivation supporting that transfer.

Adler 2003 is corrective evidence separating limited decoherence results from a solution to the quantum measurement problem as a whole. Schlosshauer 2005 is reserving evidence that does not treat the implications between those same two domains as settled.

Proof. The quotations above and paragraph 8.19.7, paragraph 8.19.44, paragraph 8.19.40, and paragraph 8.19.46 give the object, predicate, explicit domain of application, and original-source location of each quotation. By Definition 7.115, the objects, predicates, conditions, and domains of application directly supported by each quotation determine its evidence-effect domain. A claim that moves to a different object or predicate requires evidence or a derivation specific to that transfer. $\square$

Proposition 7.133 (Non-Generation of Time in Quantum Measurement, Records, and Decoherence). For the reference R corresponding to each of von Neumann 1932, Zurek 2003, and Schlosshauer 2005, Definition 7.56 gives $R \in \text{Ref}^{\text{audit}, \text{rootall}}$.

In von Neumann 1932, results 3, 5–7, and 10–13, eight in total, depend on primitive time roots; in Zurek 2003, 32 of the 43 results do. In Schlosshauer 2005, the 27 results other than results 2, 3, and 16 depend on primitive time roots.

Proof. For von Neumann 1932, apply Proposition 8.171 and Proposition 8.171. For Zurek 2003, apply Proposition 8.546 and Proposition 8.546. For Schlosshauer 2005, apply Proposition 8.569 and Proposition 8.569. $\square$

Proposition 7.134 (Primitive-Time Presupposition of Quantum Time Evolution). The unitary evolutions, before-and-after measurement relations, decoherence times, record persistence, and history constructions in the three references take time contents as inputs and yield quantum states, measurement records, or classical appearances. These constructions do not generate the input time contents from the physical-input initial set.

Proof. By Proposition 7.133, all time contents in the three references are primitive time roots. Dependence of reference-specific results that take primitive time roots as inputs is inherited by those results under Corollary 7.12. $\square$

Corollary 7.135 (Failure of Decoherence to Ground Physical Time). The derivation of interference suppression, pointer stability, environmental records, and classical appearance through decoherence does not provide physical grounds for the time contents taken as inputs to quantum evolution. Success in measurement theory does not convert primitive time input into physical time.

Proof. Apply Proposition 7.134 and Proposition 7.13. $\square$

7.3.5 Internal Clocks, Thermal Time, and Local-Reachability Determination

The audit question common to Page–Wootters 1983, Connes–Rovelli 1994, and Smolin 2018/2020 is whether the elimination of external time, the construction of a temporal flow from a state, or the claim that time is fundamental generates the time contents used from the physical-input initial set.

In Page–Wootters 1983, external-coordinate-time averaging, stationary replacement, clock-state orbits, and internal time constitute time contents. In Connes–Rovelli 1994, the thermal-time flow corresponding to a state, KMS structure, inverse-temperature time scales, and geometrical time scales constitute time contents. In Smolin 2018/2020, fundamental time, fundamental causality, the objective present, the thick present, event generation, and global temporal order constitute time contents.

Proposition 7.136 (Primitive Time Roots in Relational and Internal-Time Constructions). Each R corresponding to one of the three references belongs to $\text{Ref}^{\text{audit}, \text{rootall}}$ in Definition 7.56. All 23 time contents in Page–Wootters 1983, 28 in Connes–Rovelli 1994, and 27 in Smolin 2018/2020 are primitive time roots under the complete determination. The results dependent on primitive time roots number 16 of 18, 21 of 37, and 21 of 28, respectively.

Under the local-reachability determination that adds the reference formal-evidence environment, Connes–Rovelli 1994 reaches Time Contents 11, 19, and 27, leaving 25 primitive time roots. Local reach is empty in Page–Wootters 1983 and Smolin 2018/2020. Smolin 2018/2020 has 127 obligation atoms (Proposition 8.740).

Proof. For Page–Wootters 1983, apply Proposition 8.342 and Proposition 8.342. For Connes–Rovelli 1994, apply Proposition 8.439 and Proposition 8.439. For Smolin 2018/2020, apply Proposition 8.737 and Proposition 8.737. $\square$

Proposition 7.137 (Separation of the Relational Reformulation of Time from the Generation of Time). Averaging over external coordinate time to remove it from specified observables, selecting a temporal flow from a state, and taking time to be fundamental do not imply the physical generation of time contents. Describing time as relational, internal, or fundamental does not supply a generation history for a primitive time root.

Proof. By Proposition 7.136, all time contents in the three references are primitive time roots under the complete determination starting solely from non-temporal physical inputs. Inheritance of dependence by the root-dependent results in each reference follows from Corollary 7.12. $\square$

Corollary 7.138 (Failure of Relational Time to Ground Physical Time). Rendering external time unobservable, stationary conditional probabilities, the thermal-time hypothesis, and the introduction of event generation do not convert primitive time roots into physical time. Even if these results are mathematically or physically valid, they do not jointly satisfy the generation-history, comparison-structure, and grounding requirements for the input time contents to be established as physical time.

Proof. Apply Proposition 7.137 and Proposition 7.13. $\square$

7.3.6 Canonical Gravity, the Problem of Time, and Cosmological Boundaries

The audit question common to DeWitt 1967, Kuchař 1992, and Hartle–Hawking 1983 is whether constraint equations that do not explicitly contain an external time parameter, candidates for intrinsic time, or a no-boundary path integral generate the time contents used therein from the physical-input initial set.

In DeWitt 1967, hypersurfaces, evolution, intrinsic time, and a Wheeler–DeWitt-type constraint constitute time contents. In Kuchař 1992, extrinsic time, intrinsic time, many-fingered time, the choice of time, and the components of the problem of time constitute time contents. In Hartle–Hawking 1983, the Euclidean region, the Lorentzian region, the complex-time contour, the initial condition of the universe, and the emergence of classical spacetime constitute time contents.

Proposition 7.139 (Primitive Time Roots Remaining in Timeless Formulations of Quantum Gravity). Each R corresponding to one of the three references belongs to $\text{Ref}^{\text{audit, rootall}}$ in Definition 7.56. All 35 time contents in DeWitt 1967, 47 in Kuchař 1992, and 21 in Hartle–Hawking 1983 are primitive time roots under the complete determination. The results dependent on primitive time roots number 19 of 25, 28 of 38, and 17 of 24, respectively.

Proof. For DeWitt 1967, apply Proposition 8.235 and Proposition 8.235. For Kuchař 1992, apply Proposition 8.384 and Proposition 8.384. For Hartle–Hawking 1983, apply Proposition 8.332 and Proposition 8.332. $\square$

Proposition 7.140 (Non-Identity of Timeless Formulations and the Physical Generation of Time). Removing an external time parameter from a fundamental equation does not generate the ordering of hypersurfaces, candidates for intrinsic time, the Euclidean–Lorentzian connection, or the time contents of classical spacetime. The formal absence of a time parameter does not establish the physical grounding of time contents.

Proof. By Proposition 7.139, none of the time contents in the three references is reachable from the physical-input initial set. The generation of time contents therefore does not follow from the absence of a time parameter in a formal equation. $\square$

Corollary 7.141 (Failure of the Problem of Time to Ground Physical Time). Classifying the problem of time, choosing an intrinsic time, and deriving classical spacetime from the no-boundary condition do not provide a generation history for the time contents used. Timelessness in quantum gravity does not replace the grounds for physical time.

Proof. Apply Proposition 7.140 and Proposition 7.13. $\square$

7.3.7 Arrows of Time, the Gravitational Arrow, and Temporal Flow

The audit question common to Lebowitz 1993, Penrose 1979, and Ellis 2012 is whether an explanation of irreversibility, a low-entropy boundary condition, or a cosmological direction of time generates from the physical-input initial set the time contents supplied as inputs to that explanation.

In Lebowitz 1993, microscopic time reversal, macroscopic temporal evolution, a low-entropy initial state, and the thermodynamic arrow of time constitute time contents. In Penrose 1979, the initial singularity, the final singularity, the Weyl-curvature condition, gravitational entropy, and cosmological time asymmetry constitute time contents. In Ellis 2012, the evolution of spacetime, causal structure, the present boundary, the cosmological direction of time, and emergent complexity constitute time contents.

Proposition 7.142 (Primitive Time Roots Remaining in Explanations of the Arrow of Time). Each R corresponding to one of the three references belongs to $\text{Ref}^{\text{audit,rootall}}$ in Definition 7.56. All 34 time contents in Lebowitz 1993, 25 in Penrose 1979, and 40 in Ellis 2012 are primitive time roots. The results dependent on primitive time roots number 18 of 24 in Lebowitz 1993, all 20 in Penrose 1979, and all 10 in Ellis 2012.

Proof. For Lebowitz 1993, apply Proposition 8.395 and Proposition 8.395. For Penrose 1979, apply Proposition 8.324 and Proposition 8.324. For Ellis 2012, apply Proposition 8.710 and Proposition 8.710. $\square$

Proposition 7.143 (Separation of Explanations of Temporal Direction from the Generation of Time). A boundary condition, entropy gradient, or cosmological causal direction that assigns an asymmetric direction to a sequence of states does not generate the time contents that order that sequence. An explanation of the arrow of time does not supply the physical grounds for time.

Proof. By Proposition 7.142, all time contents in the three references are primitive time roots. An asymmetry that presupposes a primitive time root does not yield a generation history for that root. $\square$

Corollary 7.144 (Non-Repair of Primitive Time Roots by Explanations of Irreversibility). Even if explanations of thermodynamic irreversibility, low-entropy initial conditions, and a cosmological direction of time hold, the input time contents are not thereby established as physical time. Success concerning temporal direction does not repair the non-generation of time itself.

Proof. Apply Proposition 7.143 and Proposition 7.13. $\square$

7.3.8 Black-Hole Thermodynamics, Radiation, and Information

The audit question common to Bekenstein 1973, Hawking 1975, and Hawking 1976 is whether changes in horizon area, radiation, evaporation, and temporal evolution from a pure state to a mixed state generate from the physical-input initial set the time contents used therein.

In Bekenstein 1973, stationary processes, finite processes, changes in horizon area, and the generalized second law constitute time contents. In Hawking 1975, past and future infinity, collapse, emission times, evaporation time, and thermal particle flux constitute time contents. In Hawking 1976, the initial state, the final state, temporal evolution, completion of evaporation, and information loss constitute time contents.

Proposition 7.145 (Primitive Time Roots Remaining in Black-Hole Thermodynamics and Evaporation). All time contents in the three references are primitive time roots: 67 in Bekenstein 1973, 94 in Hawking 1975, and 77 in Hawking 1976. Of the 18 results in Bekenstein 1973, result 4 has evidence-only reach, and the remaining 17 depend on primitive time roots. There are no generated results and no results that remain unreachable after the roots are added. All 30 results in Hawking 1975 and all 35 results in Hawking 1976 depend on primitive time roots. Thus, of the three references, Hawking 1975 and Hawking 1976 belong to $\text{Ref}^{\text{audit, rootfull}}$ in Definition 7.58, whereas Bekenstein 1973 does not.

Proof. For Bekenstein 1973, apply Proposition 8.276 and Proposition 8.276. For Hawking 1975, apply Proposition 8.292 and Proposition 8.292. For Hawking 1976, apply Proposition 8.301 and Proposition 8.301. $\square$

Proposition 7.146 (Failure of Black-Hole Dynamics to Ground Time). Derivations of area increase, radiation spectra, evaporation time, and processes of information loss do not generate the time contents that order and compare them. The calculability of the temporal evolution of a black hole does not imply that its time variable has a physical ground.

Proof. By Proposition 7.145, none of the time contents in the three references is reachable from the physical-input initial set. Result 4, which has evidence-only reach in Bekenstein 1973, is not a result of generating time content either. Thus, the derivation of results in these references does not establish the generation of the time contents used in those derivations. $\square$

Corollary 7.147 (Primitive-Time Presupposition of the Information-Loss Problem). The information-loss problem, which contrasts an initial state with a final state, the state before collapse with that after evaporation, and a pure state with a mixed state, presupposes temporal order and temporal evolution. Neither the formulation of this problem nor candidate solutions to it generate their input time from the physical-input initial set.

Proof. Apply Proposition 7.146 and Corollary 7.12. $\square$

7.3.9 Duality, Entropy, and Spacetime Emergence

The audit question common to Maldacena 1998, Ryu–Takayanagi 2006, and Van Raamsdonk 2010 is whether the correspondence between a boundary theory and a bulk theory, the correspondence between entropy and a minimal surface, or the relation between quantum

entanglement and spatial connectivity generates from the physical-input initial set the time contents used on both sides.

In Maldacena 1998, boundary time, bulk time, Hamiltonian time evolution, causal structure, and thermal states constitute time contents. In Ryu–Takayanagi 2006, static spacetime, time slices, horizons, and the thermal interpretation of time constitute time contents. In Van Raamsdonk 2010, boundary time, asymptotic time and causal structure, eternal-black-hole time and causal structure, and Hamiltonian time translation constitute time contents.

Proposition 7.148 (Primitive Time Roots Remaining in Holography and Spacetime Connectivity). All time contents in the three references are primitive time roots: 26 in Maldacena 1998, 9 in Ryu–Takayanagi 2006, and 3 in Van Raamsdonk 2010. The results dependent on primitive time roots number 32 of 37, 12 of 16, and 8 of 10, respectively. The remaining results have evidence-only reach; there are no generated results and no results that remain unreachable after the roots are added. Thus, all three references belong to $\text{Ref}^{\text{audit,rootall}}$ in Definition 7.56, and none belongs to $\text{Ref}^{\text{audit,rootfull}}$ in Definition 7.58.

Proof. For Maldacena 1998, apply Proposition 8.460 and Proposition 8.460. For Ryu–Takayanagi 2006, apply Proposition 8.602 and Proposition 8.602. For Van Raamsdonk 2010, apply Proposition 8.676 and Proposition 8.676. $\square$

Proposition 7.149 (Separation of Spacetime Correspondence and Reconstruction from the Generation of Time). Correlating time evolution between two theories, correlating a geometrical quantity on a time slice with entropy, and reconstructing spatial connectivity from quantum entanglement do not generate the time contents supplied to the correspondence and reconstruction. Even if part of spacetime can be reconstructed from another quantity, the physical grounding of the time variable does not follow.

Proof. By Proposition 7.148, none of the time contents in the three references is reachable from the physical-input initial set. The results with evidence-only reach in the three references are not results of generating time content either. Thus, establishing results by correspondence or reconstruction does not establish the generation of the time contents used therein. $\square$

Corollary 7.150 (Failure of Quantum Entanglement to Ground Physical Time). The involvement of quantum entanglement in spatial connectivity, distance, or geometry does not jointly satisfy the generation-history, comparison-structure, and grounding requirements for time contents to be established as physical time. A result concerning spatial emergence is not a proof of temporal emergence.

Proof. Apply Proposition 7.149. $\square$

7.3.10 The Cosmological Constant, Observed Acceleration, and the Vocabulary of Expansion

The audit question common to Weinberg 1989, Riess et al. 1998, and Davis–Lineweaver 2004 is whether the magnitude of the cosmological constant, the supernova distance–redshift relation, and analyses of cosmological horizons generate from the physical-input initial set the time contents supplied to their descriptions of cosmic history.

In Weinberg 1989, cosmic expansion, the age of the universe, the times of phase transitions, and anthropic selection of an epoch constitute time contents. In Riess et al.

1998, light-curve time scales, time dilation, expansion history, the deceleration parameter, and the age of the universe constitute time contents. In Davis–Lineweaver 2004, conformal time, proper distance, light cones, particle horizons, event horizons, and expansion history constitute time contents.

Proposition 7.151 (Primitive Time Roots Remaining in Reconstructions of Cosmic History). Each R corresponding to one of the three references belongs to $\text{Ref}^{\text{audit,rootall}}$ in Definition 7.56. All 22 time contents in Weinberg 1989, 16 in Riess et al. 1998, and 26 in Davis–Lineweaver 2004 are primitive time roots under the complete determination. The results dependent on primitive time roots number 16 of 21, all 24, and 28 of 29, respectively.

Proof. For Weinberg 1989, apply Proposition 8.362 and Proposition 8.362. For Riess et al. 1998, apply Proposition 8.486 and Proposition 8.486. For Davis–Lineweaver 2004, apply Proposition 8.557 and Proposition 8.557. $\square$

Proposition 7.152 (Separation of the Quantification of Cosmic History from the Generation of Time). Calculating the age, expansion rate, and acceleration of the universe, as well as past light cones and horizons, does not generate the time contents that constitute the variables, order, and domains of integration in those calculations. The capacity to reconstruct cosmic history precisely does not imply that the temporal axis of cosmic history has been physically grounded.

Proof. By Proposition 7.151, none of the time contents in the three references is reachable from the physical-input initial set. $\square$

Corollary 7.153 (Failure of Cosmological Observational Success to Ground Time). Agreement with supernova observations, constraints on the cosmological constant, and the consistency of horizon structures do not provide a generation history for the input time contents. Observational success does not turn primitive time input into physical time.

Proof. Apply Proposition 7.152 and Proposition 7.13. $\square$

The boundary supplied by a cosmological model or observational reference is separated from the domain of existence of physical time in general. Feynman, Leighton and Sands 1963 states:

It is now believed that at least our part of the universe had its beginning about ten or twelve billion years ago.

The same reference then poses the following question:

Does an earlier time have any meaning?

The original-source location is Feynman, Leighton and Sands, *The Feynman Lectures on Physics*, Volume I, Chapter 5, §5-4, at the end of *Long times*. The complete original text, location, and minimal context from the same reference are given in paragraph 8.19.12.

Proposition 7.154 (Interpretation-Specific Evidence Effect of Cosmological Boundary Statements). The path that interprets the preceding statement as the beginning of a cosmological model or region within the explicitly specified “our part of the universe” belongs to its evidence-effect domain. On a path that transfers the preceding statement

to the beginning of physical time in general, to nonexistence before that beginning, or to meaninglessness before it, the object and domain of application change. If no independent evidence or valid derivation supporting that change is supplied, the number of evidence-effect transgressions along that path is 1.

The question in the original source does not by itself yield the conclusion that time before the beginning does not exist or is meaningless.

Proof. By Definition 7.115, the object, predicate, conditions, and domain of application directly supported by reference-specific evidence determine the evidence-effect domain. By Definition 7.117, a separate evidence-comparison path is taken for each permissible interpretation of the same statement. A path that transfers the beginning of a model or region to the domain of existence of physical time in general changes the object and domain of application once; if it is not accompanied by independent evidence or a valid derivation, the number of evidence-effect transgressions is therefore 1. $\square$

Riess et al. 1998 states:

The distances of the high-redshift SNe Ia are, on average, 10% to 15% farther than expected in a low mass density ($\Omega_M = 0.2$) Universe without a cosmological constant.

The original-source location is the abstract of Riess et al., *Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant*, *The Astronomical Journal* 116, p. 1009. The complete evidentiary record from the same reference is given in paragraph 8.19.39.

Perlmutter et al. 1999 states:

The data are strongly inconsistent with a $\Lambda = 0$ flat cosmology, the simplest inflationary universe model.

The original-source location of this $\Lambda = 0$ statement is the abstract of Perlmutter et al., *Measurements of Ω and Λ from 42 High-Redshift Supernovae*, *The Astrophysical Journal* 517, p. 565. The complete evidentiary record from the same reference is given in paragraph 8.19.38.

Proposition 7.155 (Separation of Cosmological Observation, Inference, Model Fit, and Direct Demonstration). The distance excess in Riess et al. 1998 is an inferred result mediated by photometry, spectroscopy, correction, calibration, and distance fitting. The inconsistency in Perlmutter et al. 1999 is a model determination relative to the adopted sample, light-curve standardization, likelihood, and family of FRW models.

The observations, inferences, and model fits in the two references belong to their respective evidence-effect domains. Direct demonstration concerning the whole of cosmic history, the physical generation of time contents, or the domain of existence of physical time in general requires independent evidence or a valid derivation specific to each claim.

Proof. Paragraph 8.19.39 presents the distance excess as an inferred result mediated by photometry, spectroscopy, correction, calibration, and distance fitting, and presents a positive cosmological constant and present acceleration as statistical inferences based on FRW models and likelihood. Paragraph 8.19.38 presents the inconsistency with the $\Lambda = 0$ flat model as a model determination restricted to a sample, corrections, likelihood, and a family of FRW models. By Definition 7.115, a claim having a different object, predicate, evidence type, or domain of application requires evidence or a derivation specific to that claim. $\square$

7.3.11 Comparison of Source Statements with Formal Descriptions

Table 16 gives, for each reference in the survey, the time contents, primitive time roots, and root-dependent results in its formal descriptions. Proposition 7.186 and Proposition 7.187 give the cross-reference results obtained from those reference-specific formal audits.

Comparison of a source statement with a formal description concerns the logical relation between each interpretation permitted by the statement and the formal description in the same reference.

Remark 7.156 (Interpretability of Source Statements). The permissible interpretations of a source statement are constrained by the words, qualifications, figurative expressions, assertoric strength, and context of the original text. When multiple permissible interpretations remain, the result of comparing each with the formal description is conditional on that interpretation.

Remark 7.157 (Conditional Comparison with Formal Descriptions). If one permitted interpretation of a statement and the fixed formal description cannot both hold, the contradiction that arises under that condition is stated. If the formal description does not derive that interpretation, formal non-support is stated. If the explanatory causal direction and the input direction in the formal description are opposite, causal-direction reversal is stated. If the literal content reduces to a restatement of an observable, calculated quantity, or operational rule, collapse into shorthand is stated.

Remark 7.158 (Logical Role of Source-Statement Comparison). The reference-specific formal audits and cross-reference propositions are determined from the formal descriptions in the original sources. Conditional comparison with source statements identifies tensions between formal use and explanatory expression within the same reference and supplies concrete comparison cases for the discussion of multistage misidentification in Chapter 8.

Remark 7.159 (Evidentiary Conditions for Source Statements). The object, predicate, conditions, and domain of application of a statement identified by bibliographic information and its source location are determined by the meaning, qualifications, figurative expressions, assertoric strength, and context of the original text. In a comparison mediated by translation, preservation of these elements is the condition of correspondence between the original text and its translation.

Remark 7.160 (Context of Source Statements). When an equation, figure, table, footnote, prior definition, or another sentence in the same argument determines the meaning of a statement, the permissible interpretations of that statement are restricted to those consistent with these relations.

Remark 7.161 (Boundary between Source Statements and Formal Audit). A determination in a reference-specific formal audit follows from definitions, equations, propositions, operational rules, uses of time contents, generation dependencies, and result dependencies. With the inputs to the formal audit fixed, differences in the interpretation of explanatory statements alone leave the classification values and cross-reference propositions unchanged. Statement comparison concerns the logical relation between each interpretation and that formal description.

7.3.12 The Explicit Form of Primitive Time and Its Use across Physics

The most explicit form of primitive time appears in Newton 1687/1729. That reference states:

Absolute, true, and mathematical time, of itself, and from its own nature flows equably without regard to anything external, and by another name is called duration :

The original-source location is Newton, *The Mathematical Principles of Natural Philosophy*, Motte translation, p. 77, *Scholium* I immediately following the Definitions. The statement evidence in the Appendix is given in paragraph 8.19.2.

This time is not introduced as an object generated from external physical inputs; it is an input prior to dynamical relations. By Proposition 7.123, the time contents in that reference have primitive time roots, and those roots propagate to subsequent results.

Proposition 7.162 (Persistence of Primitive Time Roots among Representatives of Disparate Fields). For each $R \in \text{Ref}^{\text{audit}}$ corresponding to Newton 1687/1729, Einstein 1905, Einstein 1916, von Neumann 1932, DeWitt 1967, Maldacena 1998, or Weinberg 1989,

$$\begin{aligned}\text{TimeContent}_R^{\text{audit,root}} &\neq \emptyset, \\ \text{Result}_R^{\text{audit,rootdep}} &\neq \emptyset\end{aligned}$$

hold.

Proof. For Newton 1687/1729, apply Proposition 8.128 and Proposition 8.128.

For Einstein 1905, apply Proposition 8.135 and Proposition 8.135. For Einstein 1916, apply Proposition 8.149 and Proposition 8.149.

For von Neumann 1932, apply Proposition 8.171 and Proposition 8.171. For DeWitt 1967, apply Proposition 8.235 and Proposition 8.235.

For Maldacena 1998, apply Proposition 8.460 and Proposition 8.460. For Weinberg 1989, apply Proposition 8.362 and Proposition 8.362. $\square$

In each reference covered by Proposition 7.162, temporal dependency roots precede results in classical mechanics, special relativity, general relativity, quantum measurement, quantum gravity, spacetime emergence, and cosmology. Although the theoretical field and formal mechanism change, the dependency order from use of time before its generation to propagation into results is preserved.

Statements concerning the difficulty of defining time and the naming of clocks raise the same generation obligation from a perspective different from the explicit form of primitive time. Feynman, Leighton and Sands 1963 states:

time is one of the things we probably cannot define

The same reference then states:

how we measure it

The original-source location is Volume I, Chapter 5, §5-2, *Time*, and the statement evidence in the Appendix is given in paragraph 8.19.12.

Anderson 2012 states:

Simply calling an entity a time or a clock does not make it one.

The original-source location is Anderson, *Problem of Time in Quantum Gravity*, arXiv:1206.2403v2, p. 1, §1.1, and the statement evidence in the Appendix is given in paragraph 8.19.59.

Proposition 7.163 (Separation of Definitional Difficulty, Measurement Procedures, and Naming from the Generation of Time). Feynman, Leighton and Sands 1963 and Anderson 2012 state the difficulty of defining time, a shift to measurement procedures, or a warning about the names “time” and “clock.” The formal constructions in both references, however, do not generate the time contents they use from non-temporal physical inputs.

Proof. For Feynman, Leighton and Sands 1963, apply Proposition 8.213 and Proposition 8.213. For Anderson 2012, apply Proposition 8.702 and Proposition 8.702. In both references, all time contents are primitive time roots and results dependent on primitive time roots exist. $\square$

Corollary 7.164 (Common Limit of Explicit Primitive Time, Definitional Difficulty, Measurement Procedures, and Warnings about Naming). An explicit statement of absolute time, the difficulty of defining time, a shift to measurement procedures, and a warning about the names “time” and “clock” do not complete the generation of the time contents used by physical law. The generation-priority condition of Corollary 7.15 must be satisfied independently of these statements.

Proof. Apply Proposition 7.162 and Proposition 7.163. $\square$

7.3.13 Evaluation Success and Physical-Time Qualification

Agreement with experimental values, the standardization of units, the establishment of observational relations, calculations by duality, predictions of quantum measurement, and descriptions of decoherence are distinct physical and mathematical results. The dependency roots of the time contents used by these results are audited separately from the precision or validity of the results.

Proposition 7.165 (Dependence on Primitive Time Roots in Successful Results). For each $R \in \text{Ref}^{\text{audit}}$ corresponding to Hafele–Keating 1972a/1972b, BIPM SI 2019/2026, Riess et al. 1998, Maldacena 1998, Ryu–Takayanagi 2006, von Neumann 1932, or Zurek 2003,

$$\begin{aligned} \text{Result}_R &\neq \emptyset, \\ \text{TimeContent}_R^{\text{audit,root}} &\neq \emptyset, \\ \text{Result}_R^{\text{audit,rootdep}} &\neq \emptyset \end{aligned}$$

hold.

Proof. For Hafele–Keating 1972a, apply Proposition 8.268 and Proposition 8.268. For Hafele–Keating 1972b, apply Proposition 8.259 and Proposition 8.259.

For BIPM SI 2019/2026, apply Proposition 8.746 and Proposition 8.746. For Riess et al. 1998, apply Proposition 8.486 and Proposition 8.486.

For Maldacena 1998, apply Proposition 8.460 and Proposition 8.460. For Ryu–Takayanagi 2006, apply Proposition 8.602 and Proposition 8.602.

For von Neumann 1932, apply Proposition 8.171 and Proposition 8.171. For Zurek 2003, apply Proposition 8.546 and Proposition 8.546. $\square$

Corollary 7.166 (Non-Completion of Time Generation by Evaluation Success). Even if the results in Proposition 7.165 hold, the primitive time roots are not eliminated, and no generation trace from the physical-input initial set to the time contents is added.

Proof. By Proposition 7.165, the result set is nonempty in each reference, and both the set of primitive time roots of time contents and the set of results dependent on primitive time roots are nonempty. The establishment of a result therefore implies neither the elimination of primitive time roots nor the addition of a generation trace. $\square$

The difference between an evaluated result and the object of a statement appears clearly in references concerning clock rates. Feynman, Leighton and Sands 1963 states:

Now if all moving clocks run slower, if no way of measuring time gives anything but a slower rate, we shall just have to say, in a certain sense, that time itself appears to be slower in a space ship.

The original-source location is Volume I, Chapter 15, §15-4, *Transformation of time*, and the statement evidence in the Appendix is given in paragraph 8.19.12.

NIST 2025 states the following about the general theory of relativity Einstein published in 1915:

According to the audacious new theory of general relativity he published that year, time is also affected by gravity. The more strongly you feel the force of gravity, Einstein argued, the slower time passes for you.

The statement evidence in the Appendix is given in paragraph 8.19.69. ESA 2019 states:

gravity and speed influences time

The statement evidence in the Appendix is given in paragraph 8.19.65.

Proposition 7.167 (Separation of Clock Relations from the Attribution of Properties to Time Itself). The formal results established in Feynman, Leighton and Sands 1963, NIST 2025, and ESA 2019 are relations among clock readings, clock rates, frequencies, lifetimes, or proper times. If the quoted statements are interpreted literally as explaining that time itself slows down, or that time itself is acted upon by gravity or speed, the formal results in those references do not provide a state space for a time carrier, a quantity of change, a law of evolution, or a coupling law to the clock relations.

Proof. For Feynman, Leighton and Sands 1963, apply Proposition 8.213. For NIST 2025, apply Proposition 8.792. For ESA 2019, apply Proposition 8.754. The objects of the results supplied by these propositions are relations among clock readings, clock rates, frequencies, lifetimes, or proper times; an independent state and evolution law for a time carrier are not components of the result sets. $\square$

Corollary 7.168 (Non-Completion by Evaluation Success and Explanatory Attribution of Properties). The establishment of clock relations, observational relations, standardization, duality, and quantum-theoretical predictions preserves those results while leaving primitive time roots in place. Explanatory attribution of properties to time itself likewise adds no generation trace from non-temporal physical inputs to time contents.

Proof. Apply Corollary 7.166 and Proposition 7.167. $\square$

7.3.14 Adjacent Problems of Time and Complete Time Generation

The creation, nonexistence, definition, internalization, conditional evolution, and relational recovery of time include issues adjacent to the primitive-time problem. Because the formal scope borne by each term differs among references, the wording and formal construction of the original source are compared reference by reference.

Barbour 1994 I makes the following two statements:

Time is not any one variable; it is created by all variables working in concert.

It is argued that time as such does not exist but that instants, defined as complete relative configurations of the universe, do.

The statement evidence in the Appendix is given in paragraph 8.19.33. Barbour 1994 II states:

time is truly non-existent in the kinematic foundations of both theories

The statement evidence in the Appendix is given in paragraph 8.19.34. Connes–Rovelli 1994 states:

there is no preferred time flow

and

thermal time ... defines the physical time

The statement evidence in the Appendix is given in paragraph 8.19.35. The second quotation occurs in the postulate identifying the thermal time determined by a state with physical time.

Page–Wootters 1983 states:

Only the dependence on an internal clock time can be observable.

The statement evidence in the Appendix is given in paragraph 8.19.26. Kiefer 2005 makes the following two statements:

Quantum gravity does not contain an external time parameter at the most fundamental level.

intrinsic time, given in quantum cosmology by the scale factor

The statement evidence in the Appendix is given in paragraph 8.19.47. Kiefer 2009 makes the following two statements:

Time is absolute in standard quantum theory and dynamical in general relativity.

Time emerges from the separation into two different subsystems: one subsystem (here: the gravitational part) defines the time with respect to which the other subsystem (here: the non-gravitational part) evolves.

The original-source locations are Kiefer, *Does Time Exist in Quantum Gravity?*, arXiv:0909.3767v1, p. 1, abstract, and p. 6, §3. The complete evidentiary record from the same reference is given in paragraph 8.19.54.

Smolin 2018/2020 states:

I take the opposite view, that time is fundamental.

The original-source location is Smolin, *Temporal Relationalism*, arXiv:1805.12468v1, p. 3, §1. The complete evidentiary record from the same reference is given in paragraph 8.19.63.

Proposition 7.169 (Evidence-Effect Domains of Two-Subsystem Time and Asserted Fundamentality of Time). The first statement in Kiefer 2009 directly supports the difference in role between the external background parameter of standard quantum theory and the dynamical spacetime variables of general relativity. Within the domain of application of the separation and approximation, the second statement supports the construction of a semiclassical time parameter or an internal-clock candidate through the separation of a heavy subsystem from a light subsystem.

The statement in Smolin 2018/2020 directly supports a position according to which time is fundamental. Fundamentality permits the coexisting interpretations of ontological fundamentality, fundamentality as a theoretical input, fundamentality of causal order, and a methodological contrast.

The construction of subsystem time, the fundamentality of time, the generation of all reference-specific time contents, the complete elimination of primitive time roots, and qualification for use within laws belong to distinct evidence-effect domains. A claim that moves between these domains requires independent evidence or a valid derivation supporting that transition.

Proof. Paragraph 8.19.54 and paragraph 8.19.63 give the object, predicate, explicit domain of application, interpretive possibilities, and original-source location of each quotation. By Definition 7.115, the objects, predicates, conditions, and domains of application directly supported by each quotation determine its evidence-effect domain. A claim that moves to a different object, predicate, or domain of application requires evidence or a derivation specific to that transition. $\square$

Rovelli 2002 states:

Therefore time cannot be an observable [2]. But if time is not an observable, how can we measure it?

The same reference also states the following about the form of dynamical predictions:

Observe that the predictions of a mechanical theory can always be expressed as relations between partial observables.

The reference further states the following about a general mechanical formalism:

There is no need to single out a specific partial variable as the time, nor to mention evolution. Mechanics is a theory of relations between partial observables.

The original-source location of the first quotation is Rovelli, *Partial Observables*, arXiv:gr-qc/0110035v3, p. 1, §I; the second quotation is on p. 2, §II, and the third on p. 7, §VI of the same reference. The complete statement evidence and interpretation audit are given in paragraph 8.19.43 and paragraph 8.19.43. Rovelli 2011 states:

I summarize here this point of view, explaining why I think that in a fundamental description of nature we must “forget time”, and how this can be done in the classical and in the quantum theory.

The statement evidence in the Appendix is given in paragraph 8.19.55.

Proposition 7.170 (Evidence-Effect Domains of Relational Formalisms and the Generation of Physical Time). The statements in Rovelli 2002 directly support expressing dynamical predictions as relations among partial observables and, in a general mechanical formalism, neither selecting a particular partial variable as time nor placing the vocabulary of evolution in the fundamental structure.

The establishment of a relational formalism, the removal of privileged time, and the nonuse of the vocabulary of evolution do not mean that the clocks, records, comparisons, order, scale, and qualification for use within laws that physically realize the relations have been generated from non-temporal physical inputs. The responsibility to remove external time from a formalism and the responsibility to generate physical time belong to distinct evidence-effect domains.

Proof. Paragraph 8.19.43 presents the observability of time, the impossibility of predicting t alone, relations among partial observables, and a general mechanical formalism without privileged time as separate statements. Paragraph 8.19.43 restricts the evidentiary effect of each statement to the partial-observable formalism and general mechanical formalism of the original source. By Definition 7.115, progressing from a relational formalism to clocks, records, comparisons, order, scale, and qualification for use within laws requires independent evidence or a valid derivation supporting that transition. $\square$

Proposition 7.171 (Primitive Time Roots Remaining in the Formal Constructions of Adjacent Problems of Time). For each R corresponding to Connes–Rovelli 1994, Page–Wootters 1983, Kuchař 1992, or Smolin 2018/2020, Definition 7.56 gives $R \in \text{Ref}^{\text{audit, rootall}}$.

Proof. For Connes–Rovelli 1994, apply Proposition 8.439. For Page–Wootters 1983, apply Proposition 8.342. For Kuchař 1992, apply Proposition 8.384. For Smolin 2018/2020, apply Proposition 8.737. $\square$

Proposition 7.172 (Difference in Scope between Adjacent Programs of Time and Complete Time Generation). Thermal time, internal-clock time, intrinsic time, conditional evolution, relational observables, and some time contents reached under the local-reachability determination each provide reference-specific formal results. In each reference covered by Proposition 7.171, the path that generates all reference-specific time contents from non-temporal physical inputs is incomplete.

Proof. In each of Connes–Rovelli 1994, Page–Wootters 1983, Kuchař 1992, and Smolin 2018/2020, the set of generated time contents under the complete determination is empty, and all reference-specific time contents are primitive time roots. No complete generation trace covering all reference-specific time contents therefore exists in any of these references. $\square$

Corollary 7.173 (Separation of the Creation, Definition, Internalization, and Recovery of Time from Complete Generation). Statements that directly invoke the creation, nonexistence, definition, internalization, conditional evolution, or relational recovery of time explicitly address issues adjacent to the primitive-time problem. The qualification of all time contents used by physical law is determined, separately from the vocabulary of each statement, by generation traces from non-temporal physical inputs to all time contents.

Proof. Apply Proposition 7.172. □

7.3.15 Difference between Formal Results and the Objects of Statements

Newton 1687/1729 takes absolute time as an input to the dynamical construction; Einstein 1905/1916 and Taylor–Wheeler 1992 formalize coordinates, clock synchronization, spacetime intervals, proper time, and clock comparisons. These formal objects are fixed by the reference-specific time contents, transfer events, results, and dependencies in the Appendix.

Attributions of properties to time itself in Feynman, Leighton and Sands 1963 and NIST 2025 were presented in Section 7.3.13. Tong 2013 states:

The correct interpretation is that time itself runs more slowly in moving frames.

The same reference further states:

it is time itself which is running slower.

The statement evidence in the Appendix is given in paragraph 8.19.61.

Greene states:

It turns out time itself can speed up or slow down.

The same reference further states:

That is, Einstein realized that if a stationary observer carefully analyzes what he sees, he'll find that time itself elapses more slowly for me than it does for him.

The statement evidence in the Appendix is given in paragraph 8.19.57.

After discussing the frequency comparison between upper and lower clocks by means of light signals, Feynman 1995 also describes the situation as follows:

One of the ways of describing this situation is to say that the time scale is faster at the top; time flows are different in different gravitational potentials, so that time flows are unequal in various parts of the world.

After explaining the signal reaching an external observer from the surface of a collapsing body, its redshift, and the waiting time for its reception, Narlikar 2010 states:

The time flows at different rates for the two protagonists.

The complete quotation records and minimal contexts are given in paragraph 8.19.70 and paragraph 8.19.71.

Taylor–Wheeler 1992 states:

time cannot be separated from space. Space and time are part of a single entity, spacetime.

The statement evidence in the Appendix is given in paragraph 8.19.13.

Proposition 7.174 (Conditional Comparison of Formal Results with the Objects of Statements). If the statements in Feynman, Leighton and Sands 1963, Feynman 1995, NIST 2025, Narlikar 2010, Tong 2013, Greene, and Taylor–Wheeler 1992 are interpreted literally as explanations in which time itself or a spacetime carrier independently slows, accelerates, expands, contracts, or otherwise changes, the corresponding formal systems do not jointly provide a state space for that carrier, a quantity of change, a law of evolution, a coupling law to clock relations, and an independent measurement procedure.

If the same statements are interpreted with their effect restricted to differences in clock readings, clock rates, lifetimes, or proper times, or to spacetime intervals or coordinate relations, the corresponding formal results are preserved. This preservation is a determination concerning the domain of validity of formal results; it does not replace a determination concerning the object type and property attribution of the source statement itself.

If the original-source text states within the same explanatory scope that the temporal-property expression is figurative, does not designate a literal deformation of time, and has a specified actual referent, the number of evidence-effect transgressions produced by that explicit figure is 0. Expressions such as “time itself slows down” or “time itself speeds up” that do not state these points transfer a comparative difference on a physical carrier to a property of time and cross one evidence-effect boundary, even when a restricted interpretation is possible. Under a literal interpretation that places an unconstructed time entity before the comparative difference, placement before generation and reattribution to a property of time occur successively by direct dependency.

Proof. Apply the result-dependency propositions in the Appendix to Feynman, Leighton and Sands 1963, Feynman 1995, NIST 2025, Narlikar 2010, Tong 2013, Greene, and Taylor–Wheeler 1992. The result set for each reference includes differences in clock readings, clock rates, lifetimes, or proper times, or spacetime intervals or coordinate relations. A state space for an independent time carrier, a quantity of change, a law of evolution, a coupling law to clock relations, and an independent measurement procedure do not appear together as formal objects in those result sets.

The restricted interpretation preserves the differences in clock readings, clock rates, lifetimes, or proper times, or the spacetime intervals or coordinate relations in each reference as formal results. By Definition 7.117, the path on which the original-source text specifies both the figurative character and the actual referent has 0 evidence-effect transgressions, whereas a path containing an unqualified temporal-property expression crosses one evidence-effect boundary. The preservation of formal results and the number of evidence-effect transgressions in a source statement are therefore determined separately. $\square$

Representative Comparison of Time-Nonexistence and Time-Attribute Statements Concerning the region in which quantum geometrodynamical effects become important and the boundary of the semiclassical approximation, Misner–Thorne–Wheeler 1973 states:

There is no spacetime, there is no time, there is no before, there is no after.
The question of what happens “next” is without meaning.

The same reference also states the following concerning relativistic clock relations:

Velocity produces a universal time dilation; acceleration does not.

The original-source location of the first statement is Chapter 43, §43.1, p. 1183, and that of the second statement is Chapter 16, §16.4, p. 393. The complete quotation records and minimal contexts are given in paragraph 8.19.19. The two statements belong to the same reference, but their stated domains of application differ. These two statements therefore do not constitute a direct contradiction asserting both existence denial and property attribution for the same time-content type and domain of application.

Heisenberg 1958 introduces the following statement as an argument by Augustine:

Time has been created together with the world, it belongs to the world,
therefore time did not exist before the universe existed.

The same reference observes that “created” already requires a before and an after and concludes, as Heisenberg’s own analysis:

Therefore, it is impossible to define in rational terms what could be meant by
the phrase “time has been created.”

The first statement is Augustine’s argument as presented by Heisenberg; the second is Heisenberg’s own conceptual analysis. The attributable speakers, complete quotation records, and minimal contexts are given in paragraph 8.19.72. This pair connects, while preserving their different attributable speakers, a statement of the nonexistence of time with a criticism that the language of creating time presupposes temporal relations.

Remark 7.175 (Comparison of the Targets of Time-Nonexistence and Time-Attribute Statements). Within one reference or across multiple references, the compatibility of a statement asserting the nonexistence of time with a statement attributing expansion, contraction, speed, flow, or deformation to time depends on the type and domain of application of the time contents targeted by each statement.

If both statements target the same time-content type and the same domain of application, and the property attribution presupposes the existence of that object, the property attribution is incompatible with the existence denial. If they target different time-content types or different domains of application, the existence denial and property attribution are not claims about the same object under the same conditions.

A shift of attribution from a difference in clock indication, clock rate, proper time, or coordinates, or a comparative difference between generated time representations, to a change in a property of “time itself,” independently of a physical carrier, requires independent evidence or a valid derivation supporting that shift.

The statement evidence also contains references that state relations as relations and references that constrain carrier language. Bunn–Hogg 2009 states:

the rate of change of the distance ... with respect to the cosmic time coordinate

The statement evidence in the Appendix is given in paragraph 8.19.53.
Lewis 2016 makes the following two statements:

the concept of expanding space is useful in a particular scenario

expanding space can be extinguished, replaced with the simple Doppler shift

The statement evidence in the Appendix is given in paragraph 8.19.62.
Peacock 2008 states:

The worst of these is the ‘expanding space’ fallacy.

The statement evidence in the Appendix is given in paragraph 8.19.52.

Proposition 7.176 (Explicit Formal Scope in Boundary Contrasts). The statement in Bunn–Hogg 2009 concerns the rate of change of distance with respect to the cosmic-time coordinate. The statements in Lewis 2016 and Peacock 2008 concern the domain of application or fallacious character of the explanation of expanding space. These statements do not add to the formal results the expansion of a spatial carrier independent of distance, coordinates, redshift, and Doppler relations.

Proof. Apply the result-dependency propositions in the Appendix to Bunn–Hogg 2009, Lewis 2016, and Peacock 2008. The result objects in these references are distance, coordinates, redshift, or Doppler relations. An independent expanding carrier is not a component of the result sets. $\square$

Corollary 7.177 (Separation of the Explanatory Range of Statements from the Time-Generation Audit). For strong carrier statements, statements that make relations explicit, and statements that constrain carrier language alike, the reference-specific time-generation audit is determined by the dependency roots of the formal construction. The explanatory range of a statement does not change the establishment of a formal result or the presence or absence of primitive time roots.

Proof. Apply Proposition 7.174 and Proposition 7.176 in accordance with the boundary in Remark 7.161. $\square$

7.3.16 Formal Evidence and Statement Evidence for Time-Bearing Quantities

In the formal descriptions of the original sources, a quantity containing a time difference, frequency, rate, lifetime, linewidth, time derivative, or temporal dimension is called a time-bearing quantity. On the basis of the reference-specific audits in the Appendix, twelve objects are treated: the speed of light, the cesium-133 transition frequency, the Planck constant, the reduced Planck constant, the gravitational constant, vacuum permittivity, vacuum permeability, the Hubble rate, decay constants and lifetimes, linewidths and decay widths, Planck time, and the Compton angular frequency. The four audit determinations of Definition 7.78 are applied separately to the formal evidence and statement evidence for each object.

Proposition 7.178 (Generation Order of Time-Bearing Quantities). The dimension, unit, or algebraic relation of a time-bearing quantity does not generate physically the time contents used by that quantity. Before a time-bearing quantity can be used within a physical law, the time contents it requires and their physical-time qualification must first be established.

Proof. The time differences, frequencies, rates, lifetimes, linewidths, or time derivatives used by a time-bearing quantity require structures for time-value identity, before–after order, intervals, scales, clock correspondence, or observation–record correspondence. By Definition 7.2, the structures required for these operations must be generated from non-temporal physical inputs before use. If required time contents are supplied to a law operation or result derivation before their generation, they constitute primitive time input under Definition 7.10. By Theorem 7.11, those time contents do not satisfy physical-time qualification. The generation of required time contents therefore cannot be inferred backward from the dimension, unit, or algebraic relation of a time-bearing quantity. $\square$

Remark 7.179 (Post-Generation Use of Time-Bearing Quantities). When time contents and their physical-time qualification have been generated and the establishment conditions specific to each theory are satisfied, existing formal relations containing time-bearing quantities can be used in their post-generation domains of application. The physical relations of emission, propagation, reception, recording, atomic transitions, oscillators, particle decay, electromagnetic fields, gravitational fields, resonance lines, and measuring instruments are preserved as theory-specific inputs to the LCTR generation and audit construction. The generation-order audit does not alter these physical relations; it determines the grounds and order of application under which time-dependent expressions may be used within laws.

Generation Order and Domain of Application of Light-Speed Expressions

Einstein 1905 states the following concerning the definition of the speed of light:

Hierbei ist

$$\text{Geschwindigkeit} = \frac{\text{Lichtweg}}{\text{Zeitdauer}},$$

wobei „Zeitdauer“ im Sinne der Definition des § 1 aufzufassen ist.

That is, velocity is the light path divided by the time interval, where “time interval” is to be understood in the sense of the definition in Section 1.

Einstein 1916 states the following concerning descriptions in accelerated coordinates:

Ebenso sieht man unmittelbar, daß das Prinzip von der Konstanz der Vakuum-Lichtgeschwindigkeit eine Modifikation erfahren muß.

Likewise, one sees immediately that the principle of the constancy of the velocity of light in vacuum must undergo a modification.

Misner–Thorne–Wheeler 1973 states the following concerning the relation between a time standard and the displayed speed of light:

No standard of time that varies so much from one month to another is acceptable. If adopted, it would make the speed of light vary from month to month!

NASA/JPL–Caltech 2020 states the light-speed limit as follows:

nothing can travel faster than the speed of light

The complete quotation records, original-source locations, minimal contexts, and interpretation audits are given in paragraph 8.19.3, paragraph 8.19.5, paragraph 8.19.19, and paragraph 8.19.66.

Proposition 7.180 (Generation Order of Light-Speed Expressions and the Domain of Application of Constancy). The physical relations of the emission, propagation, and reception of light are preserved as theory-specific inputs to the LCTR generation and audit construction. Because the numerical expression for the speed of light uses a spatial difference, time difference, clock, length standard, reference frame, and measurement procedure, the physical-time qualification of that time difference must be generated before the velocity expression is used within a law.

The constancy of the vacuum speed of light c measured in a local inertial frame, and the local upper propagation limit for matter, signals, or causal influences, are preserved in the domain in which the corresponding local structure, clock and length standards, and measurement conditions hold. This establishment is not an unconditional universal proposition concerning an arbitrary coordinate speed of light, cosmological recession velocity, phase velocity, or displayed quantity that carries no information.

If clocks or other physical carriers and comparison constructions differ, and different generated time representations consequently result, identical numerical invariance across those representations does not follow automatically. Numerical invariance between representations depends on the coordinate transformation, metric, clock and length standards, and domain of application. Where these conditions hold, existing special-relativistic light-speed relations can be used with the post-generation time representations.

Proof. Einstein 1905 gives the speed of light as the ratio of a light path to a defined time interval. Einstein 1916 distinguishes within the same reference the postulate of the constancy of the vacuum speed of light from the modification of that principle in a description using accelerated coordinates. Misner–Thorne–Wheeler 1973 states that the adopted time standard propagates into the displayed speed of light. By paragraph 8.19.66, the limiting statement in NASA/JPL–Caltech 2020 has effect for the vacuum speed of light in a local inertial frame and for the local propagation limit of matter, signals, or causal influences.

By Proposition 7.178, the algebraic relation for velocity does not generate the time contents used by that velocity. By Definition 7.115, progressing from local constancy or a local propagation limit to a universal proposition having a different object, reference frame, type of velocity, or domain of application requires independent evidence or a valid derivation supporting that transition. $\square$

Proposition 7.181 (Time-Bearing Quantities and Primitive Time Roots in Five References). In Curtiss 1949, Headley–Clayton–Reed 1996, Clark 2001, Wolf et al. 2011, and the PDG 2023 resonance reference, all reference-specific time contents are primitive time roots, and all reference-specific results that presuppose time contents are results dependent on primitive time roots.

Proof. For Curtiss 1949, apply Proposition 8.179 and Proposition 8.179. For Headley–Clayton–Reed 1996, apply Proposition 8.451 and Proposition 8.451. For Clark 2001, apply Proposition 8.520 and Proposition 8.520. For Wolf et al. 2011, apply Proposition 8.692 and Proposition 8.692. For the PDG 2023 resonance reference, apply Proposition 8.783 and Proposition 8.783. $\square$

Curtiss 1949 uses decay constants, half-lives, and mean lives in formal relations; Headley–Clayton–Reed 1996 uses Brillouin frequency shifts, frequency widths, and acoustic-wave lifetimes; Clark 2001 uses the Planck constant, transition angular frequencies, linewidths, and lifetimes; and the PDG 2023 resonance reference uses resonance widths and lifetimes. Wolf et al. 2011 distinguishes an interpretation under examination that uses the Compton angular frequency from the conclusion of the same reference that the system is not a clock possessing the Compton frequency as a physical signal.

In accordance with Proposition 7.174, interpretations that permit attribution of properties to time itself are compared conditionally with interpretations that state differences in clock readings, rates, lifetimes, frequencies, or linewidths.

Proposition 7.182 (Result of the Dual-Track Audit of Twelve Time-Bearing Objects). The twelve objects, the two tracks of formal evidence and statement evidence, and the four audit determinations give

$$12 \times 2 \times 4 = 96$$

determination fields. The evidence included in the Appendix directly establishes 23 fields on the formal track and 0 fields on the statement track. The breakdown by determination is

Audit determination	Formal track	Statement track
Pre-generation placement	9	0
Omission of generation stages	9	0
Reattribution of a time property	0	0
Reverse presupposition of a derived quantity	5	0
Total	23	0

The evidence included in the Appendix does not establish the corresponding determination for the remaining 73 fields.

Proof. Pre-generation placement holds on the formal track for the speed of light, the cesium-133 transition frequency, the Planck constant, the reduced Planck constant, the Hubble rate, decay constants and lifetimes, linewidths and decay widths, Planck time, and the Compton angular frequency. Omission of generation stages also holds on the formal track for these nine objects. Reverse presupposition of a derived quantity holds on the formal track for the speed of light, the cesium-133 transition frequency, the Hubble rate, decay constants and lifetimes, and linewidths and decay widths. Formal evidence directly establishing reattribution of a time property does not occur on the formal track for any of the twelve objects.

In Clark 2001, the Planck constant, time content 1 in Definition 8.516, enters transfer 1 together with an energy difference between states to yield a transition angular frequency. In Wolf et al. 2011, the reduced Planck constant, time content 28 in Definition 8.688, enters transfers 20 and 21 together with internal-level differences to yield transition angular frequencies. These transfers connect physical state differences to time-bearing expressions without a stage generating physical-time qualification for the required time contents. Omission of generation stages therefore also holds for these paths using the two constants as inputs.

For the gravitational constant, vacuum permittivity, and vacuum permeability, the included unit-conversion and Maxwell relations do not supply the individual generation dependencies required to establish the four determinations for the respective constants.

The statements included in the Appendix give, with their original texts, the name, formal relation, interpretation, or object of criticism for each object. For pre-generation placement and omission of generation stages, the included quotations do not determine the dependency order between use of the quantity and physical generation of the required time content. For reattribution of a time property, the quotations concerning the twelve objects do not attribute differences in relations between carriers to state or property changes of an independent time object. For reverse presupposition of a derived quantity, the included quotations do not establish a dependency that returns the quantity to a premise generating its required time content or physical-time qualification. The determinations are therefore not established on the statement track. The contrast in Wolf et al. 2011 between the reading of a “clock that runs at the Compton frequency” and the same reference’s statements that it is “not a real clock” and “does not deliver a physical signal” gives a conditional relation between the literal clock interpretation and omission of generation stages. The contrast does not supply the required dependency order and therefore does not establish the determination in that field.

Adding these object counts by determination gives 23 fields on the formal track and 0 fields on the statement track. The number of fields for which establishment is not obtained is therefore

$$96 - 23 = 73$$

□

7.3.17 Evidentiary Stages of Clocks, Time Attributes, and Biological Application

Definition 7.183 (Evidence-Stage Audit Including Biological Application). When an application from clock relations to biological processes is audited, the following three stages are distinguished.

1. Comparative differences directly measured in experiments on clocks, oscillators, frequencies, particle lifetimes, or other physical carriers.
2. An interpretation that attributes the comparative differences in the preceding item to time itself slowing, accelerating, expanding, contracting, or acting independently of physical carriers.
3. Application to pulse rates, thought, growth, aging, disease progression, lifetimes, or other biological processes.

The evidence at each stage is distinguished by its object, predicate, and evidence type. The evidence types are direct measurement, formal generalization, theoretical prediction, and source statement. When a source statement permits multiple interpretations, its correspondence to the second or third stage is determined for each interpretation.

Representative Statements of Biological Application In the same paragraph in which Feynman, Leighton, and Sands (1963) move from rate differences in moving clocks to an expression attributing a property to time itself, the objects of application are extended as follows:

the man’s pulse rate, his thought processes

The same paragraph further identifies the following object:

the time it takes for a cancer to develop

The original-source location is Feynman, Leighton and Sands, *The Feynman Lectures on Physics*, Volume I, Chapter 15, §15-4, *Transformation of time*. The minimal context is the paragraph that generalizes the light-clock rate difference to all clocks, states with a qualification that time itself appears to be slower, and then applies the account to pulse rate, thought, growth, aging, and the development of cancer. The complete quotation facts and interpretation audit for this reference are given in paragraph 8.19.12 and paragraph 8.19.12.

Additional statements belonging to the same type of biological application are as follows.

Feynman 1995 Our theory is then not inconsistent, and it suggests that the frequency shift required by energy conservation be considered as a general property of all physical processes, that they run more slowly in lower gravitational potentials.

Immediately after this generalization, the same reference states:

The man on top of a mountain is living and aging at a faster rate than we.

The original-source location is *Feynman Lectures on Gravitation*, Lecture 5, §5.2, printed p. 69. The generalization from frequency shift to all physical processes and the application to human life and aging occur in the same dependency sequence. The complete reference-specific record is given in paragraph 8.19.70.

Misner–Thorne–Wheeler 1973 aging, like atomic processes,
is tied to proper time

The original-source location is Chapter 16, §16.4, pp. 393, 396. The complete reference-specific record is given in paragraph 8.19.19.

NIST 2025 you would now be older than your own twin!

The original-source location is the third paragraph of the opening relativity overview in *Putting Einstein to the Test*; its objects are twins reunited after one has traveled in a high-speed spacecraft or near a black hole. The complete reference-specific record is given in paragraph 8.19.69.

Rindler 1977 the time dilation argument applies to any temporal change or process and therefore also to muon decay and even human life times

The original-source location is the paragraph immediately before Section 2.14 on p. 45. The complete reference-specific record is given in paragraph 8.19.23.

Taylor–Wheeler 1992 our biological clock, by which we age, and all other good clocks carried along will tick away in concord with it

The original-source location is Chapter 4, §4.5, printed p. 124. The complete reference-specific record is given in paragraph 8.19.13.

Tong 2013 the same argument holds for any process, be it clocks, elementary particles or human hearts

The original-source location is *Dynamics and Relativity*, §7.2.3, p. 119. The complete reference-specific record is given in paragraph 8.19.61.

These are evidence of theoretical application to biological processes or explanatory statements. An experiment whose object is a clock, electromagnetic radiation, or a particle lifetime is not evidence that the biological process itself was directly measured.

Proposition 7.184 (Necessary Conditions for Direct Demonstration of Biological Application). For direct demonstration to hold at the third stage of Definition 7.183, the biological process itself must be included among the experimental objects, and an experimental arrangement that isolates the relativistic conditions, a measurement procedure for the biological process, and corresponding measurement results must be supplied.

If, on the premise of direct measurement of a first-stage physical carrier, a consequence concerning the third stage is derived by formal generalization, that consequence is a theoretical prediction. The domain of direct demonstration at the third stage is determined by the measurement procedure and results whose experimental object is a biological process.

Proof. By Definition 7.71, the reference-specific audit separates the formal and statement branches and preserves the original text, translation, original-source location, and minimal context of statement evidence. By Definition 7.78, formal evidence is determined from definitions, equations, and dependencies, whereas statement evidence is compared conditionally for each interpretive possibility.

The object domain of direct demonstration is determined by the objects included in the experimental arrangement and the measurement procedure and results obtained for those objects. A formal generalization yields a theoretical consequence concerning the object of application, but does not generate a measurement procedure and result for an object not included in the experimental arrangement. Direct demonstration at the third stage therefore requires the three conditions stated in the proposition. $\square$

Proposition 7.185 (Separation of Evidence Types for Clocks, Time Attributes, and Biological Application). The objects directly measured in Bailey et al. 1977 are the decay lifetime and decay-electron counts of moving muons. The objects directly measured in Hafele–Keating 1972b are differences in cesium-clock indications, clock rates, and post-reunion display deviations. The objects directly measured in Pound–Rebka 1960 are relative frequency differences of electromagnetic radiation, resonant-transmission counts, and calibration quantities. Hafele–Keating 1972a gives predictions of rate relations and indication differences between airborne and reference clocks.

The statements preserved from Feynman, Leighton and Sands 1963, Feynman 1995, Misner–Thorne–Wheeler 1973, NIST 2025, Rindler 1977, Taylor–Wheeler 1992, and Tong 2013 extend the objects of application to pulse rates, thought, cancer development, human aging, the ages of twins, human lifetimes, biological clocks, or human hearts.

The preceding direct measurements, attribution of properties to time itself, and application to biological processes belong to different stages and evidence types under Definition 7.183.

Proof. The measured objects and results in Bailey et al. 1977, Hafele–Keating 1972a/1972b, and Pound–Rebka 1960 are determined by Proposition 8.308, Proposition 8.268, Proposition 8.259, and Proposition 8.197 in the Appendix.

The verbatim statements containing application to biological processes and their original-source locations are preserved in paragraph 8.19.12, paragraph 8.19.70, paragraph 8.19.19, paragraph 8.19.69, paragraph 8.19.23, paragraph 8.19.13, and paragraph 8.19.61 in the Appendix.

Proposition 7.167 and Proposition 7.174 separate formal results that hold for physical carriers from attributions of properties to time itself. Applying these propositions to Definition 7.183 gives the separation of the three stages and evidence types stated in the proposition. $\square$

7.3.18 Connections from All Source Statements to the Representative Arguments

In addition to the references treated directly in the representative arguments of this section, the original-source statements in the remaining references are connected below according to the point supported by each statement. Each item distinguishes the content stated directly, the permitted interpretive possibilities, and the boundary of evidentiary force. At the end of each item, “Statement” identifies the location of the original text, English translation when required, original-source location, and minimal context; “Audit” identifies the location of the conditional comparison between the interpretive possibilities and the formal evidence.

Operational Critique and Stated Deficiencies of Construction This group connects to the representative argument in Subsection 7.3.1.

Reference 6: Bridgman 1927. This reference presents the view that physical concepts should be specified by actual operations rather than their properties. It also states that, at the time of writing, the basic operations of time measurement have not undergone the critical examination that seems to be required, and that the specifications for clock construction needed to answer rigorously whether local time also involves space measurement have not been supplied. The scope of the operational critique is compared separately from the presence or absence of a construction that completes time generation. (Statement: paragraph 8.19.6; Audit: paragraph 8.19.6).

Reference 15: Ehlers–Pirani–Schild 1972. In the EPS route itself, measurement operations are derived from light rays and freely falling particles, and clocks are not basic tools for constructing spacetime geometry. The quotation concerning Synge’s preceding method instead reports the equality claim of the chronometric axiom conditional on interpreting a clock as an atomic clock. This is distinct from the additional empirical postulate after the EPS reconstruction. The EPS construction results concern spacetime structure and measurement operations. Additional claims of removing all temporal inputs or completing physical-time generation each require independent grounds for generation. (Statement: paragraph 8.19.15; Audit: paragraph 8.19.15).

Synchronization, Coordinate Time, and Proper Time This group connects to the representative argument in Subsection 7.3.2.

Reference 3: Einstein 1905. This reference defines simultaneity and time by clock indications and uses the speed of light as the ratio of a light path to a defined time interval. This operational construction of coordinate time presupposes light signals, clock indications, synchronization relations, and time intervals. The grounds for establishing

these inputs do not follow solely from the construction of coordinate time using them. (Statement: paragraph 8.19.3; Audit: paragraph 8.19.3).

Reference 4: Minkowski 1908/1909. This reference combines space and time into a four-dimensional world and describes worldlines and cone relations through the emission and reception of light. A reading as geometrical unification is distinguished from a reading as ontological unification of space and time, and the force of the quotation is limited to the four-dimensional relations and light-signal relations. (Statement: paragraph 8.19.4; Audit: paragraph 8.19.4).

Reference 5: Einstein 1916. This reference states the postulate of constant light speed in vacuum and its modification in accelerated coordinates. It states that a time meeting physical requirements cannot be introduced by identically constructed clocks at rest in the rotating coordinate system. It also describes the rate relation between clocks at the center and on the circumference of the disk. A reading in terms of relations between coordinate light speed and clock rate is separated from a reading that permits attribution of properties to time itself. (Statement: paragraph 8.19.5; Audit: paragraph 8.19.5).

Metrological Standards, Clock Differences, and Direct Measurement This group connects to the representative argument in Subsection 7.3.3.

Reference 10: Pound–Rebka 1960. This reference records a resonance line, a frequency difference, temperature correction, and agreement between the measured shift and the predicted value; it also states that the attribution of the frequency difference is not unique. Its evidentiary force lies within the measurement domain of electromagnetic radiation and detector counts and is distinguished from measurement of a property of time itself. (Statement: paragraph 8.19.10; Audit: paragraph 8.19.10).

Reference 16: Hafele–Keating 1972b. This reference states measured values from macroscopic clock comparison, their agreement with predicted values, uncertainty in clock-rate changes, and statistical qualifications. What is directly supported is the difference in clock indications and its uncertainty; attribution to a change in a time entity requires a separate ground. (Statement: paragraph 8.19.16; Audit: paragraph 8.19.16).

Reference 17: Hafele–Keating 1972a. This reference predicts direction-dependent effects on portable cesium clocks and takes the difference between the indications of clocks reunited at the same location as the physical observable. Formal generalization of the predicted clock difference is distinguished from attribution of a property to time itself. (Statement: paragraph 8.19.17; Audit: paragraph 8.19.17).

Reference 22: Bailey et al. 1977. This reference explains time dilation as the retardation of a moving clock and shows agreement between measurements of the muon lifetime and relativistic predictions. The directly measured particle lifetime is distinguished from explanatory extension to clocks in general or to time itself. In the route that estimates the negative-muon lifetime at rest by taking special relativity as an antecedent, a test of CPT invariance using that estimate depends on special relativity, and the estimate does not provide an independent test of special relativity. (Statement: paragraph 8.19.22; Audit: paragraph 8.19.22).

Reference 64: BIPM SI 2019/2026. This reference treats the second as the SI unit of time and proper time and describes TAI as a coordinate-time scale. The domains of establishment of the metrological standard and coordinate-time operation are separated from the physical generation of time contents themselves. (Statement: paragraph 8.19.64; Audit: paragraph 8.19.64).

Quantum Measurement and Decoherence This group connects to the representative argument in Subsection 7.3.4.

Reference 41: Philip W. Anderson 2001. This reference treats an account according to which decoherence, the description of collapse, and experimental verification solve the quantum measurement problem. The formal results supported by environmental interaction are separated from the extent to which they support the establishment of a unique measurement outcome or of time evolution. (Statement: paragraph 8.19.41; Audit: paragraph 8.19.41).

Canonical Gravity, Internal Time, and the Problem of Time This group connects to the representative argument in Subsection 7.3.6.

Reference 9: Dirac 1958. This reference takes the state at a given time as a basic concept of Hamiltonian form and discusses Hamiltonian formulation prior to quantization. The loss of four-coordinate symmetry under the adopted simplification is distinguished from the author's tentative inference about the physical status of four-dimensional symmetry. The selection of a time coordinate in the formalism is distinguished from the generation or physical qualification of time. (Statement: paragraph 8.19.9; Audit: paragraph 8.19.9).

Reference 11: ADM 1962. This reference distinguishes a preferred time coordinate already selected in a particle-mechanics example from an arbitrary function left by parametrized dynamics. The indeterminacy of lapse and shift in gravitation is stated in §3.4; the classical qualification that no single preferred coordinate frame exists in the interior is stated in §8.1, footnote 25. Coordinate freedom in the spacetime decomposition and the physical ground of the adopted time variable are compared as distinct obligations. (Statement: paragraph 8.19.11; Audit: paragraph 8.19.11).

Reference 14: DeWitt 1967. This reference treats internal time as a useful phenomenological concept under specified conditions and discusses the absence of a universal clock and limits on clock precision. The presentation of conditions for using an internal variable as time is distinguished from generating complete physical time contents from non-temporal inputs. (Statement: paragraph 8.19.14; Audit: paragraph 8.19.14).

Reference 25: Hartle–Hawking 1983. This reference states that a closed universe has no independent standard of time, that proper time depends on the four-dimensional geometries summed over rather than on boundary data, and that the classical concept of time fails in the joining region. Elimination of an external time standard is distinguished from physical generation of time contents. (Statement: paragraph 8.19.25; Audit: paragraph 8.19.25).

Reference 29: Isham 1993. This reference discusses the adequacy of definitions of time using gravitational variables alone, physical clocks made of matter, WKB time, and the Schrödinger form lost at an order of approximation. The conditions for constructing clock candidates and approximate time are separated from complete physical-time qualification. (Statement: paragraph 8.19.29; Audit: paragraph 8.19.29).

Reference 30: Kuchař 1992. This reference discusses time variables in classical gravity, construction of time from matter fields, the danger of generalizing model results to quantum gravity as a whole, and the lack of a successful resolution of the problem of time. Local construction of a candidate time is not identified with completed time generation for the theory as a whole. (Statement: paragraph 8.19.30; Audit: paragraph 8.19.30).

Arrows of Time and Causal Order This group connects to the representative argument in Subsection 7.3.7.

Reference 24: Penrose 1979. This reference contrasts the time symmetry of local laws with macroscopic temporal asymmetry and connects singularity structure and a low-entropy boundary condition to the arrow of time. The explanation of orientation conditions and the generation of the time contents being oriented are treated as distinct obligations. (Statement: paragraph 8.19.24; Audit: paragraph 8.19.24).

Reference 27: Bombelli–Lee–Meyer–Sorkin 1987. This reference identifies spacetime at the smallest scale with a causal set and correlates its partial order with macroscopic past–future relations. Construction of causal order is distinguished from complete time generation, including clock scales, intervals, and law time. (Statement: paragraph 8.19.27; Audit: paragraph 8.19.27).

Reference 31: Lebowitz 1993. This reference connects the macroscopic direction of time to a low-entropy initial state and temporal ordering and denies the need for an alternative explanation. The explanation of thermodynamic asymmetry is distinguished from the generation of the before–after order and time variable used in that explanation. (Statement: paragraph 8.19.31; Audit: paragraph 8.19.31).

Black-Hole Thermodynamics, Radiation, and Information This group connects to the representative argument in Subsection 7.3.8.

Reference 18: Bekenstein 1973. This reference gives black-hole change a direction in time, initially restricts the thermodynamic correspondence to a formal analogy, and presents the generalized second law as a conjecture. The evidence types of temporal direction, formal analogy, and conjecture are not conflated. (Statement: paragraph 8.19.18; Audit: paragraph 8.19.18).

Reference 20: Hawking 1975. This reference states the nonuniqueness of the particle concept in curved spacetime, positive frequency relative to a specified time, the Planck time, a heuristic pair-creation picture, and the domain in which the semiclassical approximation fails. The establishment of the radiation result is separated from responsibility for establishing the background time, frequency decomposition, and approximation boundary. (Statement: paragraph 8.19.20; Audit: paragraph 8.19.20).

Reference 21: Hawking 1976. This reference states that the known laws are formulated on a classical spacetime background and uses Fourier components with respect to time and a hypersurface near the Planck time. The consequences concerning the information problem depend on this formulation using a background spacetime and a time variable. (Statement: paragraph 8.19.21; Audit: paragraph 8.19.21).

Reference 32: Page 1993. This reference estimates the appearance of information in black-hole radiation using an S-matrix and a model of random pure states and states explicitly that the result is a first conjecture lacking a complete quantum analysis. This model-dependent estimate presupposes the S-matrix and the time-evolution description of the radiation process. (Statement: paragraph 8.19.32; Audit: paragraph 8.19.32).

Duality, Entanglement, and Spacetime Emergence This group connects to the representative argument in Subsection 7.3.9.

Reference 37: Maldacena 1998. This reference presents the duality between AdS and conformal field theory as a conjecture and makes explicit the definition of the theories,

the boundary conditions, and unresolved problems. The success of dual calculations, the complete definition of the theories, and qualification of the time variable are treated separately. (Statement: paragraph 8.19.37; Audit: paragraph 8.19.37).

Reference 49: Ryu–Takayanagi 2006. This reference proposes a geometrical area law for quantum entanglement entropy on the basis of AdS/CFT and states the difficulty of quantitative comparison in higher dimensions. Results obtained from static minimal surfaces are distinguished from a mechanism that generates spacetime or time. (Statement: paragraph 8.19.49; Audit: paragraph 8.19.49).

Reference 56: Van Raamsdonk 2010. This reference connects quantum entanglement with the emergence of classically connected spacetime while stating that a deep understanding of why and how spacetime and gravity emerge from field-theoretical degrees of freedom has not been attained. Evidence for the correspondence is separated from completion of a generation mechanism. (Statement: paragraph 8.19.56; Audit: paragraph 8.19.56).

Cosmology, Expansion, and Observables This group connects to the representative argument in Subsection 7.3.10.

Reference 28: Weinberg 1989. This reference presents the discrepancy of approximately 120 orders of magnitude between the theoretical prediction and observational upper bound for the cosmological constant as a problem and compares several approaches to its resolution. The quotation supports the scale of the problem and the assessment at that time; it is separate from a generation route for the time-dependent quantities used to describe the problem. (Statement: paragraph 8.19.28; Audit: paragraph 8.19.28).

Reference 45: Davis–Lineweaver 2004. This reference uses redshift as a parameter of time and rewrites cosmological equations in terms of observables. Re-expression in terms of observables is distinguished from generating time contents or cosmic history from redshift. (Statement: paragraph 8.19.45; Audit: paragraph 8.19.45).

Reference 48: Grøn–Elgarøy 2007. This reference describes expanding space as real and connects synchronization of comoving clocks with the measurement of cosmic time. A reading as a construction of cosmic time from the metric and comoving clocks is separated from a reading as expansion of an independent spatial carrier. (Statement: paragraph 8.19.48; Audit: paragraph 8.19.48).

Reference 50: Francis–Barnes–James–Lewis 2007. This reference states the synchronization of clocks belonging to privileged observers, an account according to which space appears between observers, differing starting times, and the rejection of an account according to which galaxies resist expansion. A reading in terms of clock and coordinate relations is separated from a reading in terms of the appearance of a spatial carrier. (Statement: paragraph 8.19.50; Audit: paragraph 8.19.50).

Reference 51: Cook–Burns 2009. This reference identifies the time coordinate of cosmology with the proper time of clocks moving with the galaxies. The force of coordinate specification by comoving clocks is distinguished from the obligations of generating and gluing a global law time. (Statement: paragraph 8.19.51; Audit: paragraph 8.19.51).

Definitions of Time and Primitive Time This group connects to the representative argument in Subsection 7.3.12.

Reference 1: Aristotle. This reference defines time as the number of change in respect of before and after and distinguishes the object counted from the means of counting. This can

be read as a definitional relation of time or as a relation between discrimination of before and after and temporal judgment; what the quotation directly supports is the distinction and relation themselves. (Statement: paragraph 8.19.1; Audit: paragraph 8.19.1).

Adjacent Problems of Time and Temporal Flow This group connects to the representative argument in Subsection 7.3.14.

Reference 60: Ellis 2012. This reference attributes existence and flow to time, extends the time parameter supplied by clocks from physics to biology, and also refers to the emergence of effective time from a timeless substrate. Proper time, clock rate, temporal flow, biological application, and emergence are separated as distinct evidentiary stages. (Statement: paragraph 8.19.60; Audit: paragraph 8.19.60).

Reference 72: Heisenberg 1958. This reference introduces, as Augustine’s argument, the nonexistence of time before the universe and then analyzes the phrase “time was created” as incapable of rational definition because it presupposes a before–after relation. The object being introduced is separated from Heisenberg’s own analysis, and the self-dependence of creation-of-time vocabulary is compared as an adjacent problem of time. (Statement: paragraph 8.19.72; Audit: paragraph 8.19.72).

Time-Bearing Quantities Such as Lifetime, Frequency, and Linewidth This group connects to the representative argument in Subsection 7.3.16.

Reference 8: Curtiss 1949. This reference names the exponential decay law, decay constant, half-life, and mean lifetime. These quotations identify the names of the law and quantities; the relations among the quantities are given in the surrounding equations and derivations. Those equations and derivations use a unit of time and elapsed time as inputs. (Statement: paragraph 8.19.8; Audit: paragraph 8.19.8).

Reference 36: Headley–Clayton–Reed 1996. This reference relates frequency difference, Lorentzian shape, full width at half maximum, acoustic-wave lifetime, and reduction in gain. The force of the statement lies in the physical relations among frequency, linewidth, and lifetime and in their operating domain; it does not include the generation of the time scales they use. (Statement: paragraph 8.19.36; Audit: paragraph 8.19.36).

Reference 42: Clark 2001. This reference identifies the rate constant with the reciprocal of lifetime and relates spectral width to lifetime, while acknowledging significant deviations from the simplified form. The conditions under which the lifetime–width relation holds are distinguished from generation of the time scale used in that relation. (Statement: paragraph 8.19.42; Audit: paragraph 8.19.42).

Reference 58: Wolf et al. 2011. Wolf et al. introduce the account by Müller, Peters, and Chu that treats the atom as a whole as a clock operating at the Compton frequency, and criticize this clock interpretation because no readable physical signal at the Compton frequency is available. Using the de Broglie relation, flight time, and Bragg condition, the leading-order neutron-interferometer phase in the same reference is expressed in terms of the lattice wave vector, gravitational acceleration, and flight time; with these quantities and the configuration fixed, that expression contains no explicit Compton frequency. (Statement: paragraph 8.19.58; Audit: paragraph 8.19.58).

Reference 68: PDG 2023 Resonances. This reference relates the pole width of a resonance to its lifetime when the relevant thresholds lie sufficiently below the resonance position, while stating that broad resonances have no direct relation between the usual width or lifetime and the pole parameters. The conditions of application of the width–lifetime

relation are distinguished from generation of the time scale. (Statement: paragraph 8.19.68; Audit: paragraph 8.19.68).

Clock Relations, Time Attributes, and Biological Application This group connects to the representative argument in Subsection 7.3.17.

Reference 66: NASA/JPL–Caltech 2020. This reference states that the faster one travels, the slower the time one experiences, and gives the speed of light as an upper limit. A reading in terms of clock, navigation, and signal relations; attribution to temporal experience; and possible extension to life or subjectivity are distinguished. (Statement: paragraph 8.19.66; Audit: paragraph 8.19.66).

Reference 67: Pössel 2021. This reference generalizes from the rate of indication of a moving light clock to all clocks and all processes aboard a space station. The stage at which the predicate remains attached to the physical carrier constituted by the light clock is separated from the theoretical stage of application to processes in general. (Statement: paragraph 8.19.67; Audit: paragraph 8.19.67).

Reference 70: Feynman 1995. This reference generalizes frequency shift to all physical processes and applies it to human life and aging; in another section, it describes comparison between upper and lower clocks and signals as a difference between time scales and the flow of time. Frequency and clock-rate relations, generalization to all physical processes, biological application, and attribution of time properties differ in their objects and inferential roles. (Statement: paragraph 8.19.70; Audit: paragraph 8.19.70).

Reference 71: Narlikar 2010. After explaining the signal from the surface of a collapsing object to an external observer, the waiting time for reception, redshift, and dimming, this reference states that time flows at different rates for the two subjects. A reading in terms of signal and wavelength relations is separated from a reading in terms of a property of temporal flow independent of physical carriers. (Statement: paragraph 8.19.71; Audit: paragraph 8.19.71).

The formal and statement evidence for the 72 references in Table 16 is linked to an audit method or a representative argument in Chapter 7 by the preceding correspondences and representative arguments. In this correspondence, agreement, lack of support, tension, and boundaries of evidentiary force between source statements and formal descriptions are determined for each interpretation permitted by the original text and minimal context.

7.4 Cross-Audit Results for the 72 References

The reference-specific complete audits yield cross-reference results for primitive time roots, results dependent on primitive time roots, results beyond the audit-reachable domain, and time-generation obligations.

The finite audited reference set of 72 references in Definition 7.1 is the quantification domain of the cross-reference audit.

Each row of Table 16 associates one reference R in the finite audited reference set of that definition with its primitive time-root set $\text{TimeContent}_R^{\text{audit,root}}$ and with the supplementary propositions proving primitive time roots and result dependence. The evidence for the cross-reference propositions consists of these 72 rows and the supplementary propositions cited by each row.

Proposition 7.186 (Universal Persistence of Primitive Time Roots in the 72 Audited References).

$$\text{Ref}^{\text{audit,root}} = \text{Ref}^{\text{audit}}.$$

Moreover, for every $R \in \text{Ref}^{\text{audit}}$,

$$\text{TimeContent}_R^{\text{audit,gen}} = \emptyset, \quad \text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R \neq \emptyset$$

hold.

Proof. Definition 7.52 gives the following inclusion.

$$\text{Ref}^{\text{audit,root}} \subseteq \text{Ref}^{\text{audit}}$$

For the reverse inclusion, Table 16 contains one row for each of the 72 references in $\text{Ref}^{\text{audit}}$. The first evidentiary proposition in each row gives, for the corresponding reference R , $\text{TimeContent}_R^{\text{audit,gen}} = \emptyset$ and $\text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R \neq \emptyset$. Therefore, by Definition 7.52, $R \in \text{Ref}^{\text{audit,root}}$. Since this holds for every reference, the reverse inclusion follows. $\square$

Proposition 7.187 (Universal Persistence of Results Dependent on Primitive Time Roots in the 72 Audited References).

$$\text{Ref}^{\text{audit,rootdep}} = \text{Ref}^{\text{audit}}.$$

Consequently, for every $R \in \text{Ref}^{\text{audit}}$,

$$\text{Result}_R^{\text{audit,rootdep}} \neq \emptyset$$

holds.

Proof. Definition 7.52 gives the following inclusion.

$$\text{Ref}^{\text{audit,rootdep}} \subseteq \text{Ref}^{\text{audit}}$$

The second evidentiary proposition attached to each row of Table 16 gives, for the corresponding reference R , $\text{Result}_R^{\text{audit,rootdep}} \neq \emptyset$. Therefore, $R \in \text{Ref}^{\text{audit,rootdep}}$. This determination holds for all 72 rows, giving the reverse inclusion. $\square$

Proposition 7.188 (Decomposition of Results Beyond the Audit-Reachable Domain in the 72 Audited References).

$$\begin{aligned} \text{Ref}^{\text{audit,unreach}} &= \text{Ref}^{\text{audit,rootdep}} \\ &= \text{Ref}^{\text{audit}}, \end{aligned}$$

and

$$\text{Ref}^{\text{audit,rootunreach}} = \emptyset$$

hold.

Furthermore, for every $R \in \text{Ref}^{\text{audit}}$,

$$\begin{aligned} \text{Result}_R^{\text{audit,unreach}} &= \text{Result}_R^{\text{audit,evidonly}} \cup \text{Result}_R^{\text{audit,rootdep}} \\ &\neq \emptyset, \end{aligned}$$

and

$$\text{Result}_R^{\text{audit,rootunreach}} = \emptyset$$

hold.

Proof. The source-specific result-dependence propositions in Table 16 give, for each $R \in \text{Ref}^{\text{audit}}$,

$$\text{Result}_R^{\text{audit,rootunreach}} = \emptyset.$$

Substituting this equality into the four-way partition in Proposition 7.49 gives

$$\text{Result}_R^{\text{audit,unreach}} = \text{Result}_R^{\text{audit,evidonly}} \dot{\cup} \text{Result}_R^{\text{audit,rootdep}}.$$

By Proposition 7.187, the set of primitive-time-root-dependent results is nonempty for every reference. The set of results beyond generation reach, which contains it, is therefore also nonempty. Applying Definition 7.52 gives

$$\text{Ref}^{\text{audit,unreach}} = \text{Ref}^{\text{audit,rootdep}} = \text{Ref}^{\text{audit}}, \quad \text{Ref}^{\text{audit,rootunreach}} = \emptyset.$$

No assumption that the set of evidence-environment-reachable nongenerated results is empty is made. $\square$

Proposition 7.189 (Invariance of the Complete Determination under Local Reachability).

$$\begin{aligned} \left| \left\{ R \in \text{Ref}^{\text{audit}} \mid \text{TimeContent}_R^{\text{audit,evid}} \neq \emptyset \right\} \right| &= 10, \\ \sum_{R \in \text{Ref}^{\text{audit}}} \left| \text{TimeContent}_R^{\text{audit,evid}} \right| &= 57. \end{aligned}$$

The ten references with nonempty locally reachable time-content sets are DeWitt 1967, Misner–Thorne–Wheeler 1973, Weinberg 1989, Isham 1993, Kuchař 1992, Connes–Rovelli 1994, Kiefer 2005, Kiefer 2009, Rovelli 2011, and Anderson 2012. The locally reachable time-content sets of the remaining 62 references are empty.

By contrast, for all 72 references,

$$\text{TimeContent}_R^{\text{audit,gen}} = \emptyset, \quad \text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R$$

hold. Local reachability in the ten references therefore does not establish complete generation from non-temporal physical inputs or remove primitive time roots under the complete determination.

Proof. The local-reachability column of Table 16 and the supplementary reachability determinations cited by its rows give the following counts for the ten references, in the order listed:

$$2, 4, 2, 12, 9, 3, 4, 9, 7, 5$$

Their sum is 57. The remaining 62 references have zero locally reachable time contents.

By Proposition 7.186, strict generation starting solely from non-temporal physical inputs yields an empty generated-time-content set in all 72 references, and every reference-specific time content remains a primitive time root. By Definition 7.41, local reachability adds the reference formal-evidence environment and thus has a different starting set from strict generation. $\square$

Theorem 7.190 (Cross-Reference Theorem on Primitive Time Roots and Result Dependence for the 72 Audited References).

$$\begin{aligned} \text{Ref}^{\text{audit,root}} &= \text{Ref}^{\text{audit,rootdep}} \\ &= \text{Ref}^{\text{audit,unreach}} \\ &= \text{Ref}^{\text{audit}}, \\ \text{Ref}^{\text{audit,rootunreach}} &= \emptyset \end{aligned}$$

hold.

Moreover, for every $R \in \text{Ref}^{\text{audit}}$,

$$\begin{aligned}\text{TimeContent}_R^{\text{audit,gen}} &= \emptyset, \\ \text{TimeContent}_R^{\text{audit,root}} &= \text{TimeContent}_R \neq \emptyset, \\ \text{Result}_R^{\text{audit,rootdep}} &\neq \emptyset, \\ \text{Result}_R^{\text{audit,rootunreach}} &= \emptyset\end{aligned}$$

hold simultaneously.

Thus, none of the 72 audited references generates all the time contents it uses from non-temporal physical inputs.

Proof. Proposition 7.186 gives

$$\text{Ref}^{\text{audit,root}} = \text{Ref}^{\text{audit}}.$$

Proposition 7.187 gives

$$\text{Ref}^{\text{audit,rootdep}} = \text{Ref}^{\text{audit}}.$$

Proposition 7.188 gives

$$\text{Ref}^{\text{audit,unreach}} = \text{Ref}^{\text{audit}}, \quad \text{Ref}^{\text{audit,rootunreach}} = \emptyset.$$

For each reference, nonemptiness of the primitive-time-root-dependent result set follows from Proposition 7.187, and emptiness of the result set that remains unreachable after primitive time roots are adjoined follows from Proposition 7.188.

By Proposition 7.189, local reachability conditional on the evidence environment exists in ten references, but this reachability is not complete generation. By Proposition 7.186, the audited generated-time-content set is empty in all 72 references, and every reference-specific time content is a primitive time root. The final negative consequence of the theorem follows. $\square$

Proposition 7.191 (Counts of Time-Generation Obligations in the 72 Audited References). The counts based on primitive time input under Definition 7.10 and on the reference-specific audit results in the Appendix are as follows.

1. References that explicitly formulate the primitive-time problem number 0/72.
2. References that eliminate primitive time input from every dependency root number 0/72.
3. References that provide complete time generation from non-temporal physical inputs number 0/72.
4. References that use at least one temporal dependency root before the generation of time contents number 72/72.
5. References that explicitly state the absence of external or independent time, the fundamental status of time, difficulty defining the concept of time, or an internal or relational construction of time number 18/72.
6. References that present a dynamical construction or recovery of intrinsic, relational, thermal, or semiclassical time number 13/72.

7. Of the thirteen references in the preceding item, references that satisfy the complete time-generation obligation from elimination of primitive time input through physical-time qualification number 0/13.

Proof. For each reference division in Table 16, the formal inputs, first temporal dependency root, audited generated time contents, primitive time roots, results dependent on primitive time roots, and statement evidence are compared.

By Theorem 7.190, primitive time roots and results dependent on primitive time roots exist in all 72 references, and no reference generates all the time contents it uses from non-temporal physical inputs. Items 2 through 4 follow.

Explicit formulation of the primitive-time problem requires more than the absence of a single external variable. It requires formulating the following as obligations that time generation must satisfy: identification of the temporal dependency roots that act in place of time, order, clocks, histories, rates of change, and causal order; their elimination from every dependency root; construction of a generation route from non-temporal physical inputs to time contents; and separation of evaluative success from physical-time qualification. Comparing the statement evidence and formal inputs in Table 16 with these conditions gives item 1.

Item 5 counts the absence, fundamental status, or definitional difficulty of time, or an internal or relational construction of time, stated as part of the reference's own analysis. The 18 references are references 6, 12, 14, 19, 25, 26, 29, 30, 33–35, 43, 47, 54–55, 59, 63, and 72 in Table 16. Heisenberg 1958 states the definitional difficulty that language describing the creation of time presupposes a temporal relation.

The dynamical construction or recovery in item 6 denotes a procedure that provides an intrinsic, relational, thermal, or semiclassical time variable, time flow, or temporal evolution relation from a candidate dynamical clock variable, subsystem, configuration, history, or state. The 13 references are references 14, 19, 26, 29, 30, 33–35, 43, 47, 54–55, and 59 in the same table. The domain of each procedure is given by the reference-specific statements, formal constructions, and their conditions in the Appendix. Comparison of the formal inputs and primitive time roots in all 13 references shows that none satisfies the complete time-generation obligation. Items 5 through 7 follow. □

Corollary 7.192 (Difference in Determination between Statements of Problems of Time and Complete Time Generation). In the 72 audited references, explicitly stating a problem of time listed in item 5 of the preceding proposition, or presenting a dynamical time-construction or recovery procedure under item 6, does not imply the establishment of complete time generation.

Proof. Apply items 5 through 7 of Proposition 7.191. □

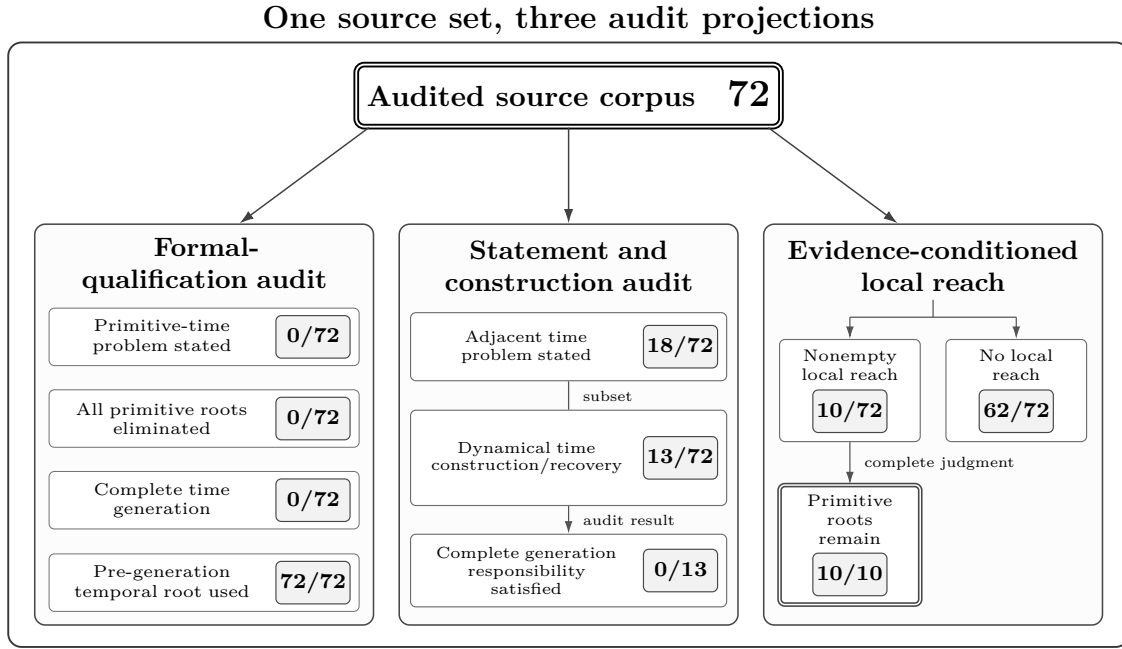


Figure 18: Three audit surfaces for the same 72 references: formal qualification, statements and constructions, and local reachability conditional on the evidence environment. Subsets defined by statements and constructions and local reachability conditional on the evidence environment are separated as distinct axes of determination.

Proposition 7.193 (Scope of Direct Demonstration for Biological Application in the 72 Audited References). Among the seven references containing statements of biological application in Proposition 7.185, the number of references that directly measure biological processes under an experimental arrangement that isolates relativistic conditions is 0/7.

Furthermore, among the 72 audited references in Definition 7.1, the number of references that directly measure biological processes under an experimental arrangement that isolates relativistic conditions is 0/72.

Proof. By Proposition 7.185, the seven references containing statements of biological application and the objects directly measured in Bailey et al. 1977, Hafele–Keating 1972a/1972b, and Pound–Rebka 1960 belong to different evidentiary stages.

The reference-specific results, result dependencies, and original-source-location evidence in the Appendix cited by the 72 rows of Table 16 are compared. The objects recorded as directly measured include the muon decay lifetime, decay-electron counts, differences in cesium-clock indications, clock rates, post-reunion display deviations, relative frequency differences of electromagnetic radiation, resonant-transmission counts, and calibration quantities.

The biological-application statements in the same 72 references identify pulse rates, thought, cancer development, human aging, the ages of twins, human lifetimes, biological clocks, and human hearts as objects of application. None of the 72 rows contains a reference that includes these biological processes among its experimental objects, isolates the relativistic conditions, and provides a measurement procedure and measurement results for the biological processes. The displayed counts for the seven references and the 72 audited references therefore follow. $\square$

Corollary 7.194 (Separation of Biological-Application Statements from the Scope of Direct Demonstration). The biological-application statements in the seven references

of Proposition 7.193 are audited as theoretical predictions or explanatory applications. Establishment of direct demonstration concerning biological processes is determined by the necessary conditions in Proposition 7.184.

Proof. Apply Proposition 7.193 and Proposition 7.184. $\square$

7.4.1 Cross-Reference Results for Time-Content Classification and Structural Signatures

Proposition 7.195 (Coverage of the Five-Type Classification of Time Contents). The time-content set of the audited reference set satisfies

$$|\text{TimeContent}^{\text{audit}}| = 2266$$

For each reference $R \in \text{Ref}^{\text{audit}}$,

$$\text{TimeContent}_R = \bigcup_{\tau \in \text{Type}^{\text{time,ord,struct}}} \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1}(\{\tau\}).$$

Proof. For each reference row in Table 25, the five index sets are pairwise disjoint, and their disjoint union equals

$$\{1, \dots, n_R\}$$

Taking the disjoint union over all references gives coverage by the five-type classification and cardinality 2266. $\square$

Proposition 7.196 (Totality of the Eight-Axis Classification for Members). For every $R \in \text{Ref}^{\text{audit}}$ and every x satisfying

$$x \in \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\text{member}}^{\text{time,ord,struct}} \right\} \right)$$

the following inequality holds:

$$\left| \text{AxisOf}_R^{\text{time,ord,struct}}(x) \right| \geq 1.$$

Proof. In Table 26, all 642 member-type entries belong to the index set of at least one axis. The eight-axis classification image of each member therefore contains at least one axis. $\square$

Proposition 7.197 (Unique Decomposition of Structural Signatures). The signature maps $\text{Sig}_R^{\text{time,ord,struct,member}}$ and $\text{Sig}_R^{\text{time,ord,struct,dependency}}$ for each reference uniquely classify the 642 reference-tagged member-type entries and the 1313 reference-tagged dependent-content-type entries, respectively, by the eight signature values.

Proof. For each reference, each of the two signature maps assigns each element of its domain one element of

$$\{0, 1\}^3$$

On the reference-tagged disjoint union, entries with the same signature value are grouped together. The sum of the fiber cardinalities in Table 27 is 642, and the sum of the fiber cardinalities in Table 28 is 1313. Each entry therefore belongs to the cross-reference group for the single signature value assigned by the map for its own reference. $\square$

Proposition 7.198 (Correspondence between Zero Fibers and Original-Source Evidence). Each of the 69 entries in the zero fiber of the member structural signature and the 133 entries in the zero fiber of the dependent-content structural signature corresponds to an evidentiary location in an original-source definition, equation, or explicit statement. Each of the remaining 573 members corresponds to an occurrence in a transfer event or result prerequisite, and each of the remaining 1180 dependent contents corresponds to transfer reachability involving a member or to an occurrence in a result prerequisite.

Proof. The three components of the member structural signature represent occurrence in a transfer input set, occurrence in a transfer output set, and occurrence in a result prerequisite. The three components of the dependent-content structural signature represent transfer reachability to a member, transfer reachability from a member, and occurrence in a result prerequisite. The assertion follows from the zero-fiber cardinalities in Table 27 and Table 28, together with the evidentiary correspondence established by comparison with original-source locations. $\square$

7.5 Contrast with Generation-Complete Time and Physical Grounds

The generation dependencies of LCTR are contrasted with the results from the 72 audited references to derive the physical grounds required of time variables in physical laws.

Proposition 7.199 (Grounding Sequence for Completion of LCTR Time Generation). For a finite object-specification index family

$$Z_0$$

, the generation dependencies of the physical-role inputs, realized operational configuration, canonical comparison quotient, exact operational time structure, and representation-stage completion condition are determined, in this order, by Proposition 2.12, Proposition 3.26, and Corollary 4.190.

For each $o \in U$, $\zeta \in Z_0$, consider the complete evaluation specification

$$\mathbf{Ev} = ((\mathcal{X}; \mathcal{J}_{\text{cmp}}), o, \zeta)$$

. Operability at the dynamical-description, law-description, and differential stages is determined by the three equivalence conditions in Theorem 5.138.

Proof. Proposition 2.12 and Proposition 3.26 give the generation from physical-role inputs to the realized operational configuration

$$\mathcal{X}$$

. Corollary 4.190 gives the generation dependencies through the canonical comparison quotient, exact operational time structure, and representation-stage completion condition.

Theorem 5.138 gives the three equivalence conditions at the dynamical-description, law-description, and differential stages for a complete evaluation specification. The preceding generation dependencies and the three equivalence conditions therefore constitute the grounding sequence for completion of LCTR time generation. $\square$

Theorem 7.200 (Physical-Time Qualification of Generation-Complete Time Content). For time content obtained through the generation dependencies of Proposition 7.199, when the generation structures corresponding to Definition 7.2 are established prior to law operations and result derivation, a time variable in a physical law that uses this time content satisfies physical-time qualification.

Proof. By Proposition 7.3, identity determination, order comparison, interval comparison, scale transformation, correspondence between information-coupling positions, clock correspondence, and observation–record correspondence for time content take the corresponding generation structures as inputs.

By the generation dependencies in Proposition 7.199, the comparison equivalence relation, canonical comparison quotient, canonical order, local and global order coordinates, quantitative affine realization, comparison-order atlas, clock correspondence, and observation–record correspondence precede law operations and result derivation.

By Definition 7.7, time content for which these grounding obligations are generated from non-temporal physical inputs satisfies physical-time qualification. $\square$

Theorem 7.201 (Structural Separation between the Generation-Completion Condition and the Results of the 72-Reference Audit). For every $R \in \text{Ref}^{\text{audit}}$,

$$\text{TimeContent}_R^{\text{audit,gen}} \subsetneq \text{TimeContent}_R$$

holds.

By contrast, for a reference

$$R$$

, the generation-completion condition for all its time contents is

$$\text{TimeContent}_R^{\text{audit,gen}} = \text{TimeContent}_R,$$

that is,

$$\text{TimeContent}_R^{\text{audit,root}} = \emptyset$$

.

Consequently, none of the 72 audited references satisfies the generation-completion condition for all the time contents it uses.

Proof. By Definition 7.39,

$$\text{TimeContent}_R^{\text{audit,root}} = \text{TimeContent}_R \setminus \text{TimeContent}_R^{\text{audit,gen}}.$$

By Theorem 7.190, for every $R \in \text{Ref}^{\text{audit}}$, $\text{TimeContent}_R^{\text{audit,root}} \neq \emptyset$. Therefore,

$$\text{TimeContent}_R^{\text{audit,gen}} \subsetneq \text{TimeContent}_R.$$

On the generation-complete side given by Proposition 7.199 and Theorem 7.200, the time contents used are generated from non-temporal physical inputs, and the required generation structures precede law operations and result derivation. Generation completion for all time contents therefore coincides with emptiness of the primitive-time-root set.

Because the primitive-time-root set is nonempty in every one of the 72 audited references, the generation-completion condition for all time contents fails in every case. $\square$

Proposition 7.202 (Extensibility of the Primitive-Time Failure Classification). When a new value or component is added to a time-content type, time-order-structure axis, audit-boundary node, audit-boundary edge, original-source support, or physical-input initial set, the change condition of Definition 7.63 holds.

For the extended audit inputs, full-reference recomputation under Definition 7.64 uniquely determines the classifications, primitive time roots, four-way result partition, and cross-reference dependency relations. For references whose audit inputs are unchanged, the recomputed mathematical objects equal their immediately preceding values.

Proof. Definition 7.63 takes as its change condition a change in the codomain, domain, map value, or set component of the surveyed reference set, reference-specific objects, original-source support, classification maps, transfer inputs and outputs, or physical-input initial set.

Definition 7.64 recomputes from the changed inputs the classifications, audit-boundary directed hypergraph, reachability sets, primitive time roots, four-way result partition, and cross-reference dependency relations. By Proposition 7.65, the result of this recomputation is unique.

For references with unchanged inputs, Proposition 7.66 makes the mathematical objects agree before and after recomputation. Audit inputs incorporating new types, axes, nodes, edges, and support components can therefore be reevaluated while preserving the determination values for references whose audit inputs remain unchanged. $\square$

Theorem 7.203 (Physical-Grounding Requirement for Time Variables in Physical Laws). When the time content used by a time variable in a physical law is primitive time input, that time variable does not satisfy physical-time qualification.

Even simultaneous mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification. A result dependent on primitive time input inherits dependence on time content that does not satisfy physical-time qualification.

Furthermore, none of the 72 audited references generates all the time contents it uses from non-temporal physical inputs. Every reference contains a result dependent on a primitive time root.

Physical establishment of a time variable in a physical law therefore requires the time content and grounding structures used by that variable to be generated from non-temporal physical inputs prior to law operations and result derivation.

Proof. By Theorem 7.11, primitive time input does not satisfy physical-time qualification. By Proposition 7.13, simultaneous mathematical consistency, predictive performance, and empirical fit do not repair primitive time input.

By Corollary 7.12, direct dependence or transfer-reachability dependence on primitive time input is inherited by results. By Theorem 7.190, primitive time roots and results dependent on primitive time roots exist in all 72 audited references.

By Theorem 7.201, the audited generated-time-content set is a proper subset of the full time-content set in all 72 audited references. Theorem 7.200 establishes physical-time qualification when generation dependencies and grounding structures precede use.

Physical generation of time contents and grounding structures prior to law operations and result derivation is therefore necessary for the physical establishment of a time variable in a physical law. $\square$

By Theorem 7.203, physical establishment of a time variable in a physical law requires physical generation of its time contents and grounding structures prior to law operations and result derivation.

By Theorem 7.190, primitive time roots and results dependent on primitive time roots exist in all 72 audited references, and none of the 72 references generates all the time contents it uses from non-temporal physical inputs. Proposition 7.191 expresses this cross-reference result as seven counts concerning the primitive-time problem, adjacent problems of time, and direct constructions for resolving time.

Proposition 7.162, Proposition 7.165, Proposition 7.171, Proposition 7.170, Proposition 7.132, and Proposition 7.174 provide representative cases of cross-field formal inputs, evaluative success, adjacent problems of time, the difference in scope between relational formalisms and physical time generation, the evidence types of experiential statements in quantum measurement, and the difference between formal results and objects of statements.

By Proposition 7.8 and Proposition 7.13, mathematical consistency, predictive performance, and empirical fit neither entail physical-time qualification of primitive time input nor repair its loss of qualification.

The four audit determinations in Definition 7.78, Proposition 7.178, and Proposition 7.181 separately determine, from formal evidence and statement evidence, pre-generation placement, omission of generation stages, reattribution of a time property, and reverse presupposition of a derived quantity in time-bearing expressions. Proposition 7.182 applies these determinations to the 96 determination fields on the formal and statement tracks for twelve objects and gives 23 fields established on the formal track, 0 fields established on the statement track, and 73 fields not established by the evidence included in the Appendix. Proposition 7.180 separates the physical propagation of light, the generation order of the time contents used by velocity expressions, constancy of the vacuum speed of light in local inertial frames, and distinct velocity types, reference frames, and domains of application.

Definition 7.183 separates direct measurement of physical carriers, attribution of a property to time itself, and application to biological processes into three evidentiary stages. Proposition 7.184 gives the necessary conditions for direct demonstration concerning biological processes. Proposition 7.193 gives, for the seven references containing statements of biological application and for the 72 audited references, the number of direct demonstrations within the audited reference set.

The 72 audited references, cross-cutting review references that treat studies of time diachronically, and references used to investigate whether direct experiments on biological aging that isolate relativistic conditions have been reported belong to distinct reference groups. The quantification domain of $\text{Ref}^{\text{audit}}$ is the 72 references in the first group. The cross-cutting review references support the positioning of the research history and audit method, while the references concerning biological-aging experiments support determination of the reported scope. The denominator for counts concerning the 72 audited references remains the 72 references in the first group.

Chapter 8 takes these theorems and propositions as inputs and synthesizes the physical grounds of time variables in physical laws, cross-theoretical consequences arising from primitive-time dependence, the difference between formal results and objects of statements, and multistage misidentification.

The Appendix applies the audit-boundary directed hypergraph of Definition 7.27, the audit-generation traces of Definition 7.32, and the result dependencies of Definition 7.50 to the 72 audited references and gives, for each reference, its time contents, transfers, results, original-source-location evidence, primitive time roots, result dependencies, and statement evidence.

8 Representative Schemes, Classical Time, and Final Synthesis

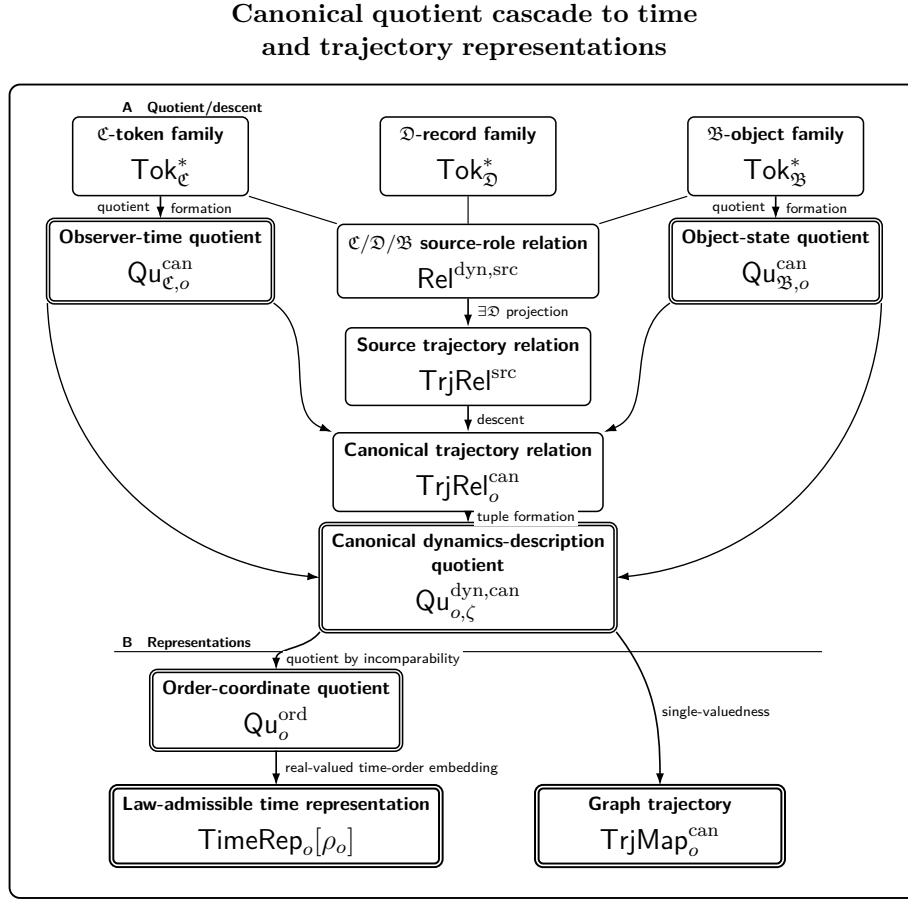


Figure 19: Dependency structure from the canonical quotient cascade to time representations and trajectory maps. The upper row constructs the canonical dynamical-description quotient from the token families derived from $\mathfrak{C}$, $\mathfrak{D}$, and $\mathfrak{B}$, through the observer-time quotient, object-state quotient, and canonical trajectory relation. The lower row distinguishes the branch that obtains a law-admissible time representation from the time-order side through the quotient by incomparability and a real-valued time-order embedding, from the branch that obtains a graph trajectory from the single-valuedness of the canonical trajectory relation.

8.1 Standard Schemes and Theory Reduction

A standard physical-time comparison scheme is a typed input comprising theory-specific geometry, scheme-restricted raw operational input, a family of observer worldlines, a family of comparison rules, and a family of binding data. The ordered pair consisting of the scheme-restricted raw operational input and the scheme-specific source-side realization datum determines a realized operational configuration of the same type as in Definition 3.25. The structures at the comparison and representation stages take this realized operational configuration and the scheme-specific family of comparison rules as domain-side inputs and are generated when the respective stage conditions hold. The continuum-approximation evaluation input is determined from the same input pair in accordance with Definition

4.168. The restriction to each scheme of the complete evaluation specification in Definition 5.2 serves as the domain-side input at the dynamical-description, law-representation, and differential-law-form stages.

The scheme-restricted raw operational input has as components the physical realizations and reception network derived from $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{D}$. The scheme-specific source-side realization datum has as components the source tokens derived from $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}$, display and record data, source-side orders, comparison-side arrival maps, comparison-side recovery maps, and the relation derived from $\mathfrak{C}\mathfrak{D}$. The scheme-restricted family of binding data has as components record-side retention, object-state-side distinction, and observer-side binding conditions. A time variable in a physical law is determined by a generated observer-time quotient, observer-time law core, and law-admissible time representation. Frequency, period, Hz, and standard frequency are generated at the representation stage from generated comparison-representation coordinates and count differences derived from $\mathfrak{C}$.

8.1.1 General Form of Standard Schemes

Definition 8.1 (Standard Physical-Time Comparison Scheme). A standard physical-time comparison scheme is defined by

$$\text{Sch} := \left((M_{\text{Sch}}, g_{\text{Sch}}), \mathcal{I}_{\text{Sch}}^{\text{raw}}, (\text{WL}_u)_{u \in U_{\text{Sch}}}, \right. \\ \left. \mathcal{J}_{\text{cmp}, \text{Sch}}, \text{BindDat}_{\text{Sch}} \right)$$

. Its components have the following types.

- $M_{\text{Sch}}, g_{\text{Sch}}$ are the scheme-specific geometric carrier and geometric structure.
- $\mathcal{I}_{\text{Sch}}^{\text{raw}}$ is the scheme-restricted raw operational input.
- For each $u \in U_{\text{Sch}}$, WL_u is an observer worldline.
- $\mathcal{J}_{\text{cmp}, \text{Sch}}$ is the scheme-restricted family of comparison rules.
- $\text{BindDat}_{\text{Sch}}$ is the scheme-restricted family of binding data containing record-binding data and object-state-binding data.

The clock-side operational material consists of the cycle indices, phase labels, display updates, counter transitions, and internal counter increments in Definition 2.2. Frequency, period, Hz, and standard frequency are generated at the representation stage under the conditions of Proposition 4.153. A quantity called proper frequency in a given scheme is the value obtained by restricting the $\mathfrak{C}$ -derived source-relative frequency representation of Definition 4.150 to that scheme.

Definition 8.2 (Scheme-Restricted Raw Operational Input). The reception network of a standard physical-time comparison scheme Sch is defined by

$$\text{Rcv}_{U, \text{Sch}} := \left(U_{\text{Sch}}, \right. \\ \left. (\text{RcvCar}_{u, \text{Sch}}, \text{Rcv}_{\mathfrak{C}, u, \text{Sch}}, \text{Rcv}_{\mathfrak{D}, u, \text{Sch}}, \text{Rcv}_{\mathfrak{B}, u, \text{Sch}}, \preceq_{u, \text{Sch}}^{\text{rcv}})_{u \in U_{\text{Sch}}} \right)$$

. The scheme-restricted raw operational input is defined by

$$\mathcal{I}_{\text{Sch}}^{\text{raw}} := \left(\text{Rl}_{\text{Sch}}[Z_0], \text{Car}_{\text{phys},\text{Sch}}[Z_0], \text{Rlz}_{\mathfrak{C},\text{Sch}}, \right. \\ \left. \text{Rlz}_{\mathfrak{D},\text{Sch}}, \text{Rlz}_{\mathfrak{B},\text{Sch}}[Z_0], \text{Rlz}_{\mathfrak{D},\text{Sch}}, \text{Rcv}_{U,\text{Sch}} \right)$$

. It has the same seven-component type as the raw operational input of Definition 3.6.

Definition 8.3 (Scheme-Specific Source-Side Realization Datum and Dynamical-Description-Stage Entry Data). For a standard physical-time comparison scheme Sch , define the family of source-side tokens by

$$\text{Src}_{*,\text{Sch}} := (\text{Tok}_{\mathfrak{C},\text{Sch}}^*, \text{Tok}_{\mathfrak{D},\text{Sch}}^*, \text{Tok}_{\mathfrak{B},\text{Sch}}^*),$$

define the display and record data by

$$\text{DispRec}_{*,\text{Sch}} := \left(\phi_{\mathfrak{C},\text{Sch}}, \phi_{\mathfrak{D},\text{Sch}}, \text{seq}_{\mathfrak{D},\text{Sch}}, \zeta_{\mathfrak{B},\text{Sch}}, \right. \\ \left. \{W_{\mathfrak{D},\text{Sch}}(x_{\mathfrak{D}})\}_{x_{\mathfrak{D}} \in \text{Tok}_{\mathfrak{D},\text{Sch}}^*} \right),$$

and the source-side orders by

$$\text{SOrd}_{*,\text{Sch}} := (\preceq_{*,\text{Sch}}^{\mathfrak{C}}, \preceq_{*,\text{Sch}}^{\mathfrak{D}})$$

.

Define the families of comparison-side arrival maps and comparison-side recovery maps, and the relation derived from $\mathfrak{C}\mathfrak{D}$, by

$$\text{Arr}_{U,\text{Sch}}^{\text{cmp}} := \left(\text{Arr}_{\mathfrak{C},u,\text{Sch}}^{\text{cmp}}, \text{Arr}_{\mathfrak{D},u,\text{Sch}}^{\text{cmp}}, \text{Arr}_{\mathfrak{B},u,\text{Sch}}^{\text{cmp}} \right)_{u \in U_{\text{Sch}}}, \\ \text{Rec}_{U,\text{Sch}}^{\text{cmp}} := \left(\text{Rec}_{\mathfrak{C},u,\text{Sch}}^{\text{cmp}}, \text{Rec}_{\mathfrak{D},u,\text{Sch}}^{\text{cmp}}, \text{Rec}_{\mathfrak{B},u,\text{Sch}}^{\text{cmp}} \right)_{u \in U_{\text{Sch}}}, \\ \text{Rel}_{\text{Sch}} := \left(\left(\text{Rel}_{u,\text{Sch}}^{\text{cmp}} \right)_{u \in U_{\text{Sch}}}, \text{Rel}_{\text{Sch}}^{\text{src}} \right)$$

.

Define the scheme-specific source-side realization datum by

$$\text{Dat}_{\text{Sch}}^{\text{src}} := \left(\text{Src}_{*,\text{Sch}}, \text{DispRec}_{*,\text{Sch}}, \text{SOrd}_{*,\text{Sch}}, \right. \\ \left. \text{Arr}_{U,\text{Sch}}^{\text{cmp}}, \text{Rec}_{U,\text{Sch}}^{\text{cmp}}, \text{Rel}_{\text{Sch}} \right)$$

.

. Define the scheme-restricted realized operational configuration by

$$\mathcal{X}_{\text{Sch}} := (\mathcal{I}_{\text{Sch}}^{\text{raw}}, \text{Dat}_{\text{Sch}}^{\text{src}})$$

.

For each

$$\bullet \in \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}\}, \quad u \in U_{\text{Sch}}$$

, the comparison-side arrival map has type

$$\text{Arr}_{\bullet,u,\text{Sch}}^{\text{cmp}} : \text{Dom} \left(\text{Arr}_{\bullet,u,\text{Sch}}^{\text{cmp}} \right) \subseteq \text{Tok}_{\bullet,\text{Sch}}^* \\ \longrightarrow \text{Rcv}_{\bullet,u,\text{Sch}}$$

. Define the uniquely recoverable reception set by

$$\text{Rcv}_{\bullet,u,\text{Sch}}^{\text{uniq}} := \left\{ \alpha \in \text{Im} \left(\text{Arr}_{\bullet,u,\text{Sch}}^{\text{cmp}} \right) \mid \left| \left(\text{Arr}_{\bullet,u,\text{Sch}}^{\text{cmp}} \right)^{-1} (\{\alpha\}) \right| = 1 \right\}$$

. The comparison-side recovery map has type

$$\text{Rec}_{\bullet,u,\text{Sch}}^{\text{cmp}} : \text{Rcv}_{\bullet,u,\text{Sch}}^{\text{uniq}} \longrightarrow \text{Tok}_{\bullet,\text{Sch}}^*$$

.
For each

$$o \in U_{\text{Sch}}, \quad \bullet \in \{\mathfrak{C}, \mathfrak{D}, \mathfrak{B}\}$$

, let the dynamical-description-stage direct-arrival map be

$$\begin{aligned} \text{Arr}_{\bullet,o,\text{Sch}}^{\text{dyn}} : \text{Dom} \left(\text{Arr}_{\bullet,o,\text{Sch}}^{\text{dyn}} \right) &\subseteq \text{Tok}_{\bullet,\text{Sch}}^* \\ &\longrightarrow \text{Rcv}_{\bullet,o,\text{Sch}} \end{aligned}$$

. Define the dynamical-description-stage arrival set by

$$\mathcal{R}_{\bullet,o,\text{Sch}}^{\text{dyn,arr}} := \text{Im} \left(\text{Arr}_{\bullet,o,\text{Sch}}^{\text{dyn}} \right)$$

. The dynamical-description source-role relation and the dynamical-description-stage source-role relation have the respective types

$$\begin{aligned} \text{Rel}_{\text{Sch}}^{\text{dyn,src}} &\subseteq \text{Tok}_{\mathfrak{C},\text{Sch}}^* \times \text{Tok}_{\mathfrak{D},\text{Sch}}^* \times \text{Tok}_{\mathfrak{B},\text{Sch}}^*, \\ \text{Rel}_{o,\text{Sch}}^{\text{dyn}} &\subseteq \mathcal{R}_{\mathfrak{C},o,\text{Sch}}^{\text{dyn,arr}} \times \mathcal{R}_{\mathfrak{D},o,\text{Sch}}^{\text{dyn,arr}} \times \mathcal{R}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn,arr}} \end{aligned}$$

. Define the dynamical-description-stage four-party reception structure by

$$\begin{aligned} \text{RcvStr}_{o,\text{Sch}}^{\text{dyn}} &:= \left(\left(\mathcal{R}_{\mathfrak{C},o,\text{Sch}}^{\text{dyn,arr}}, \mathcal{R}_{\mathfrak{D},o,\text{Sch}}^{\text{dyn,arr}}, \mathcal{R}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn,arr}} \right), \right. \\ &\quad \left(\text{Arr}_{\mathfrak{C},o,\text{Sch}}^{\text{dyn}}, \text{Arr}_{\mathfrak{D},o,\text{Sch}}^{\text{dyn}}, \text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}} \right), \\ &\quad \left. \left(\text{Rel}_{o,\text{Sch}}^{\text{dyn}}, \text{Rel}_{\text{Sch}}^{\text{dyn,src}} \right) \right) \end{aligned}$$

. The dynamical-description-stage recovery map has type

$$\text{Rec}_{\bullet,o,\text{Sch}}^{\text{dyn}} : \mathcal{R}_{\bullet,o,\text{Sch}}^{\text{dyn,arr}} \longrightarrow \text{Tok}_{\bullet,\text{Sch}}^*$$

.
For every

$$x_{\mathfrak{B}} \in \text{Dom} \left(\text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}} \right)$$

, require

$$\begin{aligned} &\text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}}(x_{\mathfrak{B}}) \\ &\in \text{Rcv}_{\mathfrak{B},o,\text{Sch}} \cap \left(\text{Tok}_{\mathfrak{B},\text{Sch}}^* \times \text{Tok}_{\mathfrak{B},\text{Sch}}^* \right) \end{aligned}$$

and

$$\text{pr}_1 \left(\text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}}(x_{\mathfrak{B}}) \right) = x_{\mathfrak{B}}$$

. Assign the $\mathfrak{B}$ -derived object-state-side identification map the type

$$\begin{aligned} \text{Stateld}_{\mathfrak{B},o,\text{Sch}} &: \text{Dom} \left(\text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}} \right) \\ &\longrightarrow \text{Tok}_{\mathfrak{B},\text{Sch}}^* \end{aligned}$$

, and set

$$\text{Stateld}_{\mathfrak{B},o,\text{Sch}} := \text{pr}_2 \circ \text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}}$$

. Then

$$\text{Arr}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}}(x_{\mathfrak{B}}) = (x_{\mathfrak{B}}, \text{Stateld}_{\mathfrak{B},o,\text{Sch}}(x_{\mathfrak{B}}))$$

holds.

$\text{Dat}_{\text{Sch}}^{\text{src}}$ has the same six-component type as in Definition 3.24. Take the dynamical-description-stage entry data to be

$$\begin{aligned} &\left(\text{RcvStr}_{o,\text{Sch}}^{\text{dyn}}, \text{Rec}_{\mathfrak{C},o,\text{Sch}}^{\text{dyn}}, \right. \\ &\quad \left. \text{Rec}_{\mathfrak{D},o,\text{Sch}}^{\text{dyn}}, \text{Rec}_{\mathfrak{B},o,\text{Sch}}^{\text{dyn}} \right)_{o \in U_{\text{Sch}}} \end{aligned}$$

.

Proposition 8.4 (Generation Position of Clock-Side Inputs and Frequency Representations in Standard Schemes). The clock-side operational material of a standard physical-time comparison scheme Sch consists of the cycle indices, phase labels, display updates, counter transitions, and internal counter increments in Definition 2.2. The scheme-restricted family of binding data $\text{BindDat}_{\text{Sch}}$ has record-side retention, object-state-side distinction, and observer-side binding conditions as components. The scheme-restricted raw operational input $\mathcal{I}_{\text{Sch}}^{\text{raw}}$ and the scheme-specific source-side realization datum $\text{Dat}_{\text{Sch}}^{\text{src}}$ do not have frequency, period, Hz, or standard frequency as components. These quantities are generated at the representation stage from generated comparison-representation coordinates and count differences derived from $\mathfrak{C}$. A quantity called proper frequency in a given scheme is the value obtained by restricting the $\mathfrak{C}$ -derived source-relative frequency representation to that scheme.

Proof. Definitions 3.6 and 3.24 construct the raw operational input and source-side realization datum from physical realizations, a reception network, source tokens derived from $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}$, display and record data, source-side orders, comparison-side arrival maps, comparison-side recovery maps, and a relation derived from $\mathfrak{C}\mathfrak{D}$. By Proposition 3.26, these input data contain no time variables, time intervals, periods, or frequencies.

Definition 4.150 defines frequency and period representations from quantitative affine differences of generated $\mathfrak{C}$ -derived comparison-representation coordinates and count differences derived from $\mathfrak{C}$. Definition 4.151 defines an Hz value from a frequency representation and a $\mathfrak{C}$ -derived frequency-unit scalar, while Definition 4.152 defines a standard-frequency representation from the frequency representation of a reference $\mathfrak{C}$ -derived source. By Proposition 4.153, these quantities are generated under the conditions of the representation stage. A quantity called proper frequency in a given scheme is likewise determined from the $\mathfrak{C}$ -derived source-relative frequency representation in Definition 4.150. $\square$

8.1.2 Representative Schemes

A representative scheme is a theory-specific input datum that supplies the input of each physical theory as a realized operational configuration

$$\mathcal{X}_{\text{Sch}} = (\mathcal{I}_{\text{Sch}}^{\text{raw}}, \text{Dat}_{\text{Sch}}^{\text{src}})$$

. Its components include clock-side cycle indices, phase labels, display updates, counter transitions, internal counter increments, record-side retention, object-state-side distinction, and observer-side binding conditions. Time variables are based on the observer-time quotient, observer-time law core, and law-admissible time representation generated from the inputs of each representative scheme. Frequency, period, Hz numerical values, standard frequency, and proper frequency are defined from cycle-count differences derived from the clock-side operational material and quantitative affine differences of generated representation coordinates.

Each representative scheme has the general form of Definition 8.1 and has as components the scheme-restricted raw operational input of Definition 8.2 and the scheme-specific source-side realization datum and dynamical-description-stage entry data of Definition 8.3. These components reduce each representative theory to the general generation sequences of Chapters 3 and 5.

Definition 8.5 (Single-Object Evaluation Specification for a Representative Scheme). Let the representative-scheme label be Sch . When the representative scheme supplies the realized operational configuration

$$\mathcal{X}_{\text{Sch}} = (\mathcal{I}_{\text{Sch}}^{\text{raw}}, \text{Dat}_{\text{Sch}}^{\text{src}}),$$

and the typed family of comparison rules $\mathcal{J}_{\text{cmp}, \text{Sch}}$, define the comparison-rule-dependent common input by

$$\mathbf{l}_{\text{Sch}} := (\mathcal{X}_{\text{Sch}}, \mathcal{J}_{\text{cmp}, \text{Sch}})$$

. Define the comparison-stage condition-family evaluation specification by

$$\text{Ev}_{\text{Sch}}^{\text{cmp}} := \left(\text{Ev}_{\text{Sch}, i}^{\text{cmp}} \right)_{i=1}^7, \quad \text{Ev}_{\text{Sch}, i}^{\text{cmp}} := \begin{cases} \mathcal{X}_{\text{Sch}}, & i = 1, \\ \mathbf{l}_{\text{Sch}}, & i \in \{2, \dots, 7\} \end{cases}$$

.

Further, given an observer-side information-coupling position $o \in U_{\text{Sch}}$ and a single-object specification index ζ_{Sch} , define the scheme-restricted complete evaluation specification by

$$\begin{aligned} \text{Ev}_{\text{Sch}} \\ &:= (\mathbf{l}_{\text{Sch}}, o, \zeta_{\text{Sch}}) \end{aligned}$$

.

Denote comparison-stage operability, exact comparison closure, and dynamical-description operability, respectively, by

$$\begin{aligned} &\text{Operative}(\mathcal{X}_{\text{Sch}}), \\ &\text{ExactOperable}[\mathbf{l}_{\text{Sch}}], \\ &\text{DynOperative}[\text{Ev}_{\text{Sch}}] \end{aligned}$$

.

The correspondence between the classification numbers DG_1 through DG_8 and the dynamical-description conditions is as follows.

$$(DG_1) \ K_{\text{GenTimeOrdAntiSym}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_2) \ K_{\text{CanTrjDesc}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_3) \ K_{\text{CanTrjFunc}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_4) \ K_{\text{CanObsDesc}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_5) \ K_{\text{LocObsChgComm}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_6) \ K_{\text{ObsTimeIncTrans}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_7) \ K_{\text{ObsTimeRealOrdEmb}}^{\text{dyn}}(\text{EvSch}).$$

$$(DG_8) \ K_{\text{ObsTimeTrjFact}}^{\text{dyn}}(\text{EvSch}).$$

Denote the simultaneous satisfaction of all eight conditions by $\text{DynCondAll}[\text{EvSch}]$.

At their respective generation stages, denote the dynamical-description generation object, the law-output determinacy condition, and the law-description generation object by

$$\begin{aligned} & \text{Dat}_{o, \zeta_{\text{Sch}}}^{\text{dyn}, 0}, \\ & \text{DynWit}_{o, \zeta_{\text{Sch}}} [\text{RepEmb}_o^{\text{dyn}, \text{time}}], \\ & K_{\text{CandRelRightUnique}}^{\text{law}} [\text{LawEvalDat}_{o, \zeta_{\text{Sch}}}], \\ & \text{LawCore}_{o, \zeta_{\text{Sch}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}}), \\ & \mathcal{O}_{o, \zeta_{\text{Sch}}}^{\text{dyn}, \text{can}} \end{aligned}$$

. Here,

$$\text{RepEmb}_o^{\text{dyn}, \text{time}} \in \text{OrdEmbSet}_o^{\text{dyn}, \text{time}}(\mathbb{R})$$

. Generation of the observer-time law core requires

$$\text{LawOperative} [\text{LawEvalDat}_{o, \zeta_{\text{Sch}}}]$$

.

Definition 8.6 (Generation Rule for Clock-Side Inputs and Frequency Representations). Clock-side inputs in a representative scheme are supplied as cycle indices, phase labels, display updates, counter transitions, and internal counter increments. Suppose that the conditional quantitative affine realization derived from $\mathfrak{C}$,

$$\text{QtyAffReal}_{\mathfrak{C}}[\text{ISch}]$$

, is specified and that the admissible counting window

$$\text{win} = (u, j_{\mathfrak{C}}^{\text{src}}, k_0, k_1)$$

satisfies the membership and strict-order conditions of Definition 4.150. Then, upon specialization of that definition to ISch , the counting-window frequency representation and period representation derived from $\mathfrak{C}$ are

$$\begin{aligned} \nu_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}] &= \frac{\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}]}{\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[\text{ISch}; \text{win}]}, \\ \text{Per}_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}] &= (\nu_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}])^{-1}, \\ \nu_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}], \quad \text{Per}_{\mathfrak{C}}^{\text{win}}[\text{ISch}; \text{win}] &\in \text{QtyVal}_{\mathfrak{C}}. \end{aligned}$$

The frequency-unit scalar and Hz numerical value follow Definition 4.151.

Definition 8.7 (Newtonian and classical-mechanics scheme). Denote the Newtonian and classical-mechanics scheme by Sch_{NCM} . This scheme has shared clock-side material, stable record correspondence, distinguishability of object states, and observer-side binding conditions as theory-specific inputs. The clock-side material is assigned to the same order representation; the record-side material supplies records corresponding to that order representation; and the object-state-side material supplies the distinctions required for the dynamical description.

Conventional classical physics uses the time variable t as an input to laws of motion. This scheme constructs the classical time variable as an order representation generated from comparison-stage closure, the representation stage, the $\mathfrak{CDB}$ -derived dynamical-description stage, the observer-time law core, and a law-admissible time representation.

The Newtonian and classical-mechanics scheme has the following input components.

- (NM₁) A family of clock-side sources corresponding to a shared clock.
- (NM₂) Record-side retention that supplies stable record correspondence.
- (NM₃) Distinguishability of object states required for the dynamical description.
- (NM₄) Observer-side binding conditions that bind the clock side, record side, and object-state side at the same phase.
- (NM₅) Comparison-side arrival maps, comparison-side recovery maps, comparison rules, and binding data.
- (NM₆) The dynamical-description-stage entry data of Definition 8.3.
- (NM₇) Idealization conditions for the classical domain that make branching burdens single-valued.
- (NM₈) An object-state correspondence that approximates observational binding by the identity.
- (NM₉) Trajectory and observable correspondences for evaluating law-output determinacy.

These inputs are used to evaluate the conditions for the comparison stage, representation stage, continuous-approximation stage, dynamical-description stage, law-representation stage, and differential-law-form stage in the classical domain.

Definition 8.8 (Time-Metrology Scheme). Denote the time-metrology scheme by Sch_{TM} . This scheme has a reference source derived from $\mathfrak{C}$, $\text{Src}_{\mathfrak{C}}(j_{\text{std}}^{\text{src}})$, and a normalization convention $N_{\text{std}} \in \mathbb{N}_{>0}$, and retains cycle-count records through record-side correspondence. A standard-time construction is a representation generated from cycle-count records, counting comparison and correction, correction records, and gluing of comparison quotients.

The time-metrology scheme has the following input components.

- (TM₁) A reference source derived from $\mathfrak{C}$ and a normalization convention.
- (TM₂) A family of record-side sources that retains cycle-count records.
- (TM₃) Comparison rules containing counting comparison and correction, correction records, and traceability information.

- (TM₄) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (TM₅) The dynamical-description-stage entry data of Definition 8.3.

Specializing Definition 4.152 to the time-metrology scheme gives the standard-frequency representation derived from $\mathfrak{C}$ and the Hz unit scalar derived from $\mathfrak{C}$ as

$$\begin{aligned}\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{I}_{\text{Sch}_{\text{TM}}}] &= \nu_{\mathfrak{C}}^{\text{src}}[\mathbf{I}_{\text{Sch}_{\text{TM}}}; j_{\text{std}}^{\text{src}}], \\ \nu_{\mathfrak{C}}^{\text{Hz}}[\mathbf{I}_{\text{Sch}_{\text{TM}}}] &= \frac{\nu_{\mathfrak{C}}^{\text{std}}[\mathbf{I}_{\text{Sch}_{\text{TM}}}]}{N_{\text{std}} \cdot_{\mathfrak{C}}^{\text{qty}} 1_{\mathfrak{C}}^{\text{qty}}}.\end{aligned}$$

Definition 8.9 (GPS scheme). Denote the GPS scheme by Sch_{GPS} . This scheme is a distributed comparison scheme with multiple families of clock-side sources, record-side correction records, transport rules, and observer-side reception structures. Clock adjustment and time correction in GPS map differences in count ratios, transport, and gluing to comparable records.

The clock-side materials at the satellite and receiver are supplied as cycle-count sources. Transport corrections, relativistic corrections, and receiver-side reconstruction are structural conditions for gluing cycle-count records and comparison-transport data into a canonical comparison quotient.

The GPS scheme has the following input components.

- (GPS₁) Families of clock-side sources at satellites and receivers.
- (GPS₂) Families of record-side sources that include navigation messages, correction records, and reception records.
- (GPS₃) Receiver-side object-state material and observer-side binding conditions.
- (GPS₄) Comparison rules that include propagation corrections, relativistic corrections, and clock corrections.
- (GPS₅) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (GPS₆) The dynamical-description-stage entry data of Definition 8.3.

The domain in which adjustment, synchronization, correction, and navigation time-tagging for multiple clock families in the GPS scheme hold is determined by the failure-position token system of Chapter 6.

Definition 8.10 (Special-relativity scheme). Denote the special-relativity scheme by Sch_{SR} . This scheme has observer worldlines, local clock-side reception, record-side correspondence, and transport rules on Minkowski spacetime. Conventional special relativity uses Minkowski spacetime and inertial coordinate time as inputs to law descriptions. This scheme uses Minkowski spacetime as a theory-specific input carrier and constructs inertial coordinate time as a coordinate representation generated from comparison-stage closure, the representation stage, the dynamical-description stage, the observer-time law core, and a law-admissible time representation.

The special-relativity scheme has the following input components.

- (SR₁) A set of observers and worldlines corresponding to a network of inertial observers.
- (SR₂) Local clock-side, record-side, and object-state-side reception for each observer.
- (SR₃) Comparison rules corresponding to synchronization procedures, proper-time comparisons, and inertial-coordinate transformations.
- (SR₄) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (SR₅) The dynamical-description-stage entry data of Definition 8.3.

Branches in proper-time comparison, synchronization procedures, and simultaneity are determined by the conditions under which comparison, representation, and gluing hold along the same generation sequence.

Definition 8.11 (General-relativity scheme). Denote the general-relativity scheme by Sch_{GR} . This scheme has a network of local observers on Lorentzian spacetime, worldline-dependent reception structures, transport data, and gluing data. Conventional general relativity uses Lorentzian spacetime and proper-time representations as inputs to law descriptions. This scheme uses Lorentzian spacetime as a theory-specific input carrier and constructs a proper-time representation as a local display generated from worldline-local comparison data and generated representation coordinates.

The general-relativity scheme has the following input components.

- (GR₁) A network of local observers and worldline-dependent reception structures.
- (GR₂) Local clock-side, record-side, and object-state-side reception.
- (GR₃) Comparison rules for local comparison, transport, and gluing.
- (GR₄) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (GR₅) The dynamical-description-stage entry data of Definition 8.3.

The domain in which local-to-global comparison restrictions, transport, and gluing hold is determined by the failure-position token system of Chapter 6.

Definition 8.12 (Quantum-reconstruction scheme). Denote the quantum-reconstruction scheme by Sch_{QG} . This scheme takes families of clock-side, record-side, and object-state-side sources, branch-sensitive record structures, observer-side binding, and comparison-gluing conditions as inputs, while excluding external time from the primitive inputs. From these inputs it generates comparison-stage closure, a canonical comparison quotient, a $\mathcal{CDB}$ -derived dynamical description, and a law-admissible time representation under the establishment conditions of their respective stages. For a law-evaluation datum satisfying law-description operability, it generates the observer-time law core.

The quantum-reconstruction scheme has the following input components.

- (QG₁) Families of clock-side, record-side, and object-state-side sources.
- (QG₂) Branch-sensitive record structures and object-state-side distinguishability.

- (QG₃) Observer-side binding conditions and comparison-gluing conditions.
- (QG₄) Comparison-side arrival maps, comparison-side recovery maps, and source-side pairing relations.
- (QG₅) The dynamical-description-stage entry data of Definition 8.3.

The domains in which observational binding, object-state distinction, state identification, and law representation hold are determined by the conditions of the dynamical-description stage, law-representation stage, and differential-law-form stage.

Proposition 8.13 (Primitive-Frequency-Free Reduction of Representative Schemes). For each

$$\text{Sch} \in \left\{ \text{Sch}_{\text{NCM}}, \text{Sch}_{\text{TM}}, \text{Sch}_{\text{GPS}}, \right. \\ \left. \text{Sch}_{\text{SR}}, \text{Sch}_{\text{GR}}, \text{Sch}_{\text{QG}} \right\}$$

the theory-specific components of the representative scheme construct

$$\mathcal{A}_{\text{Sch}} = (\mathcal{I}_{\text{Sch}}^{\text{raw}}, \text{Dat}_{\text{Sch}}^{\text{src}})$$

. The observer-time quotient and law-admissible time representation are generated, within domains satisfying their respective generation conditions, from this realized operational configuration, the typed family of comparison rules, the observer-side information-coupling position, and the single-object specification index. The observer-time law core is generated when the corresponding law-evaluation datum satisfies law-description operability. Frequency, period, Hz numerical values, standard frequency, and proper frequency are generated from cycle-count differences and quantitative affine differences of generated representation coordinates, subject to the conditions on quantitative affine realization, admissible counting windows, window invariance, and unit normalization required by the definition of each quantity.

Proof. By Definition 8.6, clock-side inputs are supplied as cycle indices, phase labels, display updates, counter transitions, and internal counter increments, while frequency representations are defined from cycle-count differences and quantitative affine differences of generated representation coordinates.

Definitions 8.7, 8.8, 8.9, 8.10, 8.11, and 8.12 supply, for their respective representative schemes, the components of the raw operational input, source-side realization datum, typed family of comparison rules, and dynamical-description-stage entry data. By Definition 8.5, these components determine the corresponding complete evaluation specification.

Definition 3.25 determines the realized operational configuration, and Definition 5.56 determines the observer-time quotient. Theorem 5.112 generates the dynamical-description object and dynamical-description evidence datum, while Theorem 5.113 together with Definition 5.108 generates the observer-time law core. Definition 5.63 generates the law-admissible time representation. By Proposition 8.4, the raw operational input and source-side realization datum contain no time variable, time interval, period, or frequency as components; these quantities belong to subsequent generation stages. Definitions 4.150 and 4.151 generate frequency, period, and Hz numerical values. Definition 4.152 generates standard frequency and the Hz unit scalar. The six representative schemes therefore obtain each time-representation quantity and frequency quantity from the general generation sequence without requiring a time variable or primitive frequency value as a pregeneration input. $\square$

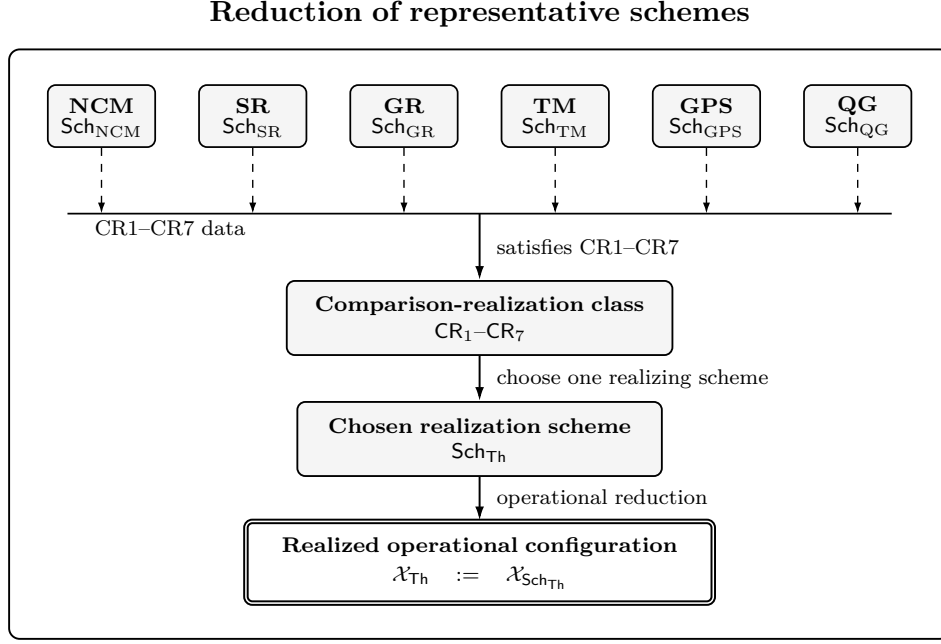


Figure 20: Reduction from the six representative schemes to theory-specific realized operational configurations. The theory corresponding to each representative scheme belongs to the comparison-realization class by supplying the data from CR1 through CR7. Once a realization scheme Sch_{Th} is selected for a theory Th , the operational reduction defines $\mathcal{X}_{\text{Th}} := \mathcal{X}_{\text{Sch}_{\text{Th}}}$.

8.1.3 Reduction to the Comparison-Realization Class and Realized Operational Configurations

Definition 8.14 (Comparison-realization class). A physical theory Th belongs to the comparison-realization class if there exists at least one standard physical-time-comparison scheme that realizes the geometry, operational input, worldline family, comparison rules, and binding data of Th and satisfies the following seven obligations. In this section, select one such scheme and write

$$\text{Sch}_{\text{Th}}$$

This notation does not assume uniqueness of the realization scheme.

- (CR1) Sch_{Th} has the general form of a standard physical-time-comparison scheme from Definition 8.1.
- (CR2) The source-side token families, display and record data, and source-side orders of Sch_{Th} have the types specified in Definition 8.3.
- (CR3) The family of comparison-side arrival maps of Sch_{Th} has the type specified in Definition 8.3.
- (CR4) The family of comparison-side recovery maps of Sch_{Th} has the type specified in Definition 8.3.
- (CR5) The family of direct-arrival maps at the dynamical-description stage of Sch_{Th} has the type specified in Definition 8.3.

- (CR6) The family of recovery maps at the dynamical-description stage of $\mathbf{Sch}_{\mathbf{Th}}$ has the type specified in Definition 8.3.
- (CR7) The comparison-side relations, source-side relations, and dynamical-description source-role relation of $\mathbf{Sch}_{\mathbf{Th}}$ have the types specified in Definition 8.3.

The components and types in CR1 through CR7, together with the distinction among their $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{D}$ origins, are inherited from Definitions 8.1, 8.2, and 8.3.

Definition 8.15 (Reduction from a theory to a realized operational configuration). For a physical theory $\mathbf{Th}$ in the comparison-realization class, define the realized operational configuration by

$$\begin{aligned}\mathcal{X}_{\mathbf{Th}} &:= \mathcal{X}_{\mathbf{Sch}_{\mathbf{Th}}} = (\mathcal{I}_{\mathbf{Sch}_{\mathbf{Th}}}^{\text{raw}}, \text{Dat}_{\mathbf{Sch}_{\mathbf{Th}}}^{\text{src}}), \\ \text{Red}^{\text{op}}(\mathbf{Th}) &:= \mathcal{X}_{\mathbf{Th}}.\end{aligned}$$

Theorem 8.16 (Reduction from the comparison-realization class to realized operational configurations). For every physical theory $\mathbf{Th}$ in the comparison-realization class, $\text{Red}^{\text{op}}(\mathbf{Th})$ from Definition 8.15 is a realized operational configuration in the sense of Definition 3.25.

Proof. By CR1 through CR7 of Definition 8.14, the selected $\mathbf{Sch}_{\mathbf{Th}}$ has the general form specified in Definitions 8.1 and 8.3. Consequently, $\mathcal{X}_{\mathbf{Sch}_{\mathbf{Th}}}$ is a realized operational configuration in the sense of Definition 3.25. By Definition 8.15, this equals $\mathcal{X}_{\mathbf{Th}}$ and hence is $\text{Red}^{\text{op}}(\mathbf{Th})$. $\square$

Proposition 8.17 (Representative theories supply the input data of the comparison-realization class). The representative theories corresponding to

$$\begin{array}{ccc}\mathbf{Sch}_{\text{NCM}}, & \mathbf{Sch}_{\text{SR}}, & \mathbf{Sch}_{\text{GR}}, \\ \mathbf{Sch}_{\text{TM}}, & \mathbf{Sch}_{\text{GPS}}, & \mathbf{Sch}_{\text{QG}}\end{array}$$

supply the data in CR1 through CR7 of Definition 8.14. Therefore, each representative theory belongs to the comparison-realization class.

Proof. Definitions 8.1 and 8.3 show that the general form of a standard scheme has the data types in CR1 through CR7. Definitions 8.7 and 8.8 through 8.12 give the theory-specific components of this general form for the six representative schemes. Therefore, the representative theory corresponding to each scheme belongs to the comparison-realization class. $\square$

Corollary 8.18 (Common generation-stage judgments for representative theories). For each representative theory in Proposition 8.17, the operability judgment and the judgments at the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-law-form stages take the corresponding realized operational configuration $\mathcal{X}_{\mathbf{Th}} = \text{Red}^{\text{op}}(\mathbf{Th})$ as a domain-side input. The failure location of each stage condition is classified by the failure-position tokens of Definition 6.10.

Proof. By Proposition 8.17, each representative theory belongs to the comparison-realization class. By Theorem 8.16, the realized operational configuration of each representative theory is determined. Each judgment from the operability judgment through the differential-law-form-stage judgment takes this realized operational configuration and the additional inputs required at that stage as domain-side inputs. The failure location of each stage condition is classified by the failure-position tokens specified in Definition 6.10. $\square$

8.1.4 Interpretation-Path-Specific Multistage Misrecognition

Definition 8.19 (Multistage misrecognition relative to generation order). In time-reference construction, the objects and grounding conditions of each stage are generated from raw operational inputs in the dependency order defined by the sequence from comparison through representation and dynamical description to law representation and differential law form, together with the branch from dynamical description to continuous approximation. For an interpretation-specific evidence-comparison path from Definition 7.117, an explanation containing the following operations is called a multistage misrecognition.

1. It places an object, quantity, structure, or its bearer generated at a later stage as an input or given at a stage preceding its generation conditions.
2. It identifies the satisfaction of a local or partial stage with the satisfaction of a subsequent or global stage that requires additional conditions.
3. It redescribes a relational quantity or comparison-derived difference generated at a later stage as a change in an attribute of an entity posited before construction.
4. It transfers the establishment of preceding evidence to a different object, predicate, evidence type, grounding condition, or domain of application and treats it as a direct demonstration without independent evidence or a valid derivation specific to the destination of the transfer.

Within one interpretation-specific evidence-comparison path, each of the preceding operations that occurs successively by direct dependency is counted as one stage. Neither the number of evidence-effect layers nor the number of enumerated items is taken directly as the number of stages of misrecognition. Even for the same statement, different permissible interpretations can have different numbers of stages of misrecognition.

If the number of evidence-effect transgressions along an interpretation-specific evidence-comparison path is 0, no multistage misrecognition holds under that interpretation. If an unqualified temporal-property expression transfers a comparative difference on a physical carrier to a property of time, that object-type transfer is a one-stage misrecognition. Under a literal explanation that places an unconstructed time entity in advance and attributes a comparative difference to a change in an attribute of that entity, placement before generation and attribution of the property occur successively by direct dependency and form a two-stage misrecognition.

Proposition 8.20 (Interpretation-path-specific application of the unbounded inheritance of limited results). Apply Definitions 7.115, 7.116, and 8.19 to each representative scheme in Definition 8.5.

For each reference, retain mathematical consistency, predictive performance, model fit, empirical agreement, and comparative differences on physical carriers within their respective evidence-effect domains. If an interpretation-specific evidence-comparison path transfers an object, predicate, evidence type, grounding condition, or domain of application without independent evidence or a valid derivation, determine that transfer with direct dependency as a multistage misrecognition.

If the same reference or the same statement has multiple permissible interpretations, determine the establishment and number of stages of misrecognition separately for each interpretation. No multistage misrecognition holds under an interpretation that directly describes a comparative difference on a physical carrier or under an interpretation for

which the original-source text states its figurative character, states that it does not designate a literal deformation of time, and identifies its actual referent. An unqualified temporal-property expression constitutes a one-stage misrecognition even if it can be interpreted retrospectively as shorthand. A literal time-stretching account that places an unconstructed time entity in advance constitutes a two-stage misrecognition; a path that then transfers evidential effect to a different object domain acquires subsequent stages according to its direct dependencies.

The number of stages of misrecognition is not determined uniformly from the name of a field or reference. It is determined by the direct-dependency path formed by the formal construction, statements, and permissible interpretations of each reference.

Proof. By Proposition 7.118, a claim beyond the evidence-effect boundary of a limited result requires independent evidence or a valid derivation specific to that claim. By Corollary 7.119, inheritance to a result of primitive-time dependence and inheritance to a claim outside the domain of application of a limited result are distinct audit axes.

Definition 7.117 separates an interpretation-specific evidence-comparison path for each permissible interpretation of the same statement and counts evidence-effect transgressions connected by direct dependency. Definition 8.19 determines placement before generation, identification of stages of establishment, reattribution to a property of time, and transfer of evidential effect to another object domain along that path as operations of misrecognition. It is therefore possible to preserve the formal results of each representative scheme within their evidence-effect domains while determining only the stages of misrecognition actually constituted by the original-source statements and their permissible interpretations. $\square$

Two independent inheritance audits

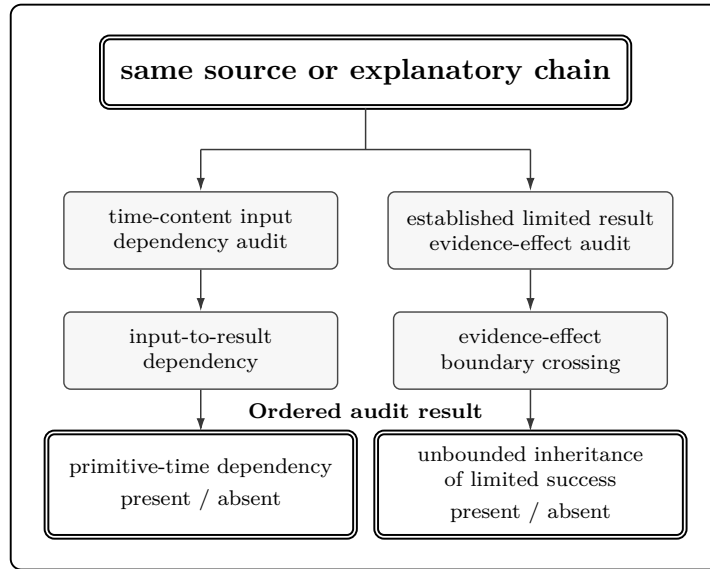


Figure 21: Primitive-time-dependence result inheritance and transgression of the evidence-effect domain of limited results are determined as distinct audit axes. The two determinations are not identified and are retained as an ordered pair for the same reference or explanatory sequence.

Apply validity localization, approximation localization, classification by the failure-position tokens of Definition 6.10, and structural-burden correspondence to the determination of representative schemes and classical time. For representative schemes, the realized

operational configuration supplied by each theory is matched to its structural burdens. In the classical domain, this separates the subclass on which a classical time representation holds, the scale-indexed validity domain on which the continuous approximation is valid, and the boundary of that domain.

Proposition 8.21 (Loss of physical-time qualification in modern physics under primitive-time premises). For any representative scheme Sch , let the complete evaluation specification of Definition 8.5 be

$$\begin{aligned} \text{Ev}_{\text{Sch}} \\ = ((\mathcal{X}_{\text{Sch}}; \mathcal{J}_{\text{cmp}, \text{Sch}}), o, \zeta_{\text{Sch}}) \end{aligned}$$

When the time content and grounding structure used by the physical-law time variable of a representative theory are generated from nontemporal physical inputs prior to law operations and result derivation, that time variable satisfies physical-time qualification.

When time content enters a law operation or result derivation before the grounding structure required for its use, that time content is a primitive-time input, and the corresponding time variable does not satisfy physical-time qualification. Simultaneous mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification.

By Proposition 7.191, among the 72 audited references, the numbers of references that eliminate primitive-time inputs from all dependency roots and that give complete time generation from nontemporal physical inputs are, respectively, 0/72. The number of references that use at least one temporal dependency root before the generation of time content is 72/72. By Theorem 7.190, primitive-time roots and primitive-time-root-dependent results exist in all 72 references. By Theorem 7.201, all 72 references fail to satisfy the generation-completion condition for the entire set of time contents used within them.

Therefore, a time variable in a physical law is not established on a primitive premise. So long as they retain their present use of time variables without alteration, the theoretical descriptions of modern physics represented by the 72 audited references do not satisfy physical-time qualification. That is, modern physics does not hold as it stands. This nonestablishment is a loss of physical-time qualification concerning the order of time-content generation, law use, and result propagation. Mathematical consistency, predictive performance, and empirical fit are retained as evaluation successes of the respective theories.

Proof. By Definition 8.5, the components of a complete evaluation specification are a realized operational configuration, a typed family of comparison rules, an observer-side information-coupling position, and an object-specification index.

When time content and its grounding structure are generated prior to law operations and result derivation, Theorem 7.200 gives physical-time qualification to the time variable of a physical law that uses that time content.

When time content is used before the required grounding structure, Definition 7.10 makes that time content a primitive-time input. By Theorem 7.11, that time content does not satisfy physical-time qualification. By Proposition 7.13, simultaneous mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification.

Proposition 7.191 gives 0/72 both for elimination of primitive-time inputs from all dependency roots and for complete time generation, and 72/72 for the use of a temporal dependency root before generation. Theorem 7.190 gives the existence of primitive-time

roots and primitive-time-root-dependent results in all references. Theorem 7.201 gives failure of the generation-completion condition in all references.

The theoretical descriptions of modern physics represented by the 72 audited references therefore retain time content that lacks physical-time qualification and results that depend on that content. By the generation-priority condition in Corollary 7.15, a time variable in a physical law is not established on a primitive premise. Hence, so long as present time-variable use is retained without alteration, modern physics does not hold as it stands. $\square$

8.2 Theory-Specific Structural Consequences

The consequence for each representative theory consists of an application of the general generation theorems and the structural burdens specific to that theory. For a realized operational configuration

$$\mathcal{A}_{\text{Sch}}$$

the general generation theorems supply the operability determination, comparison-stage determination, representation-stage determination, continuous-approximation-stage determination, dynamical-description-stage determination, law-representation-stage determination, differential-law-form-stage determination, and first structural-failure determination. The theory-specific structural burdens supply the conditions that the inputs of each theory must satisfy at the respective generation stages.

In SR, inertial coordinate time supplies an order representation based on the observer-time law core. In GR, comparison burdens with branch resolution appear in the canonical comparison quotient at the comparison stage or in descent conditions at the dynamical-description stage. In time metrology, standard frequency, period, and numerical values in Hz are determined from a traceability-based quotient order, $\mathfrak{C}$ -derived comparison-representation coordinates, and quantity-affine differences. In GPS, a rotation-derived loop defect appears as a failure of global gluing at the comparison stage. In quantum reconstruction, the conditions for generating a law-admissible time representation are determined from records that exclude external time from primitive inputs, branch sensitivity, object-state differences, observer-side binding, and comparison-gluing conditions.

Table 17: Input burdens of representative schemes

Representative	Input carriers	Comparison burdens	Motion and law burdens
NCM	Shared clock Stable records Object-state sequence	Shared-clock domain Single ordering	Branching burden Single-valued observational binding
SR	Inertial-observer network Local clocks Synchronization rules	Inter-inertial closure Order representation	Law-kernel representation burden
GR	Local-observer network Worldline-dependent reception Transport data	Proper-time display Global gluing	Branch resolution Transport preservation Quotient descent Law single-valuedness
TM	Reference cycles Counting convention Correction records	Counting comparison Representation-coordinate descent	Frequency-representation burden Period-representation burden Hz-representation burden
GPS	Satellite clocks Reception records Correction rules Transport rules	Multiple-clock comparison Propagation correction Loop consistency Global gluing	Navigation-time representation Law-kernel representation burden
QG	External time excluded from primitive inputs Records Branch sensitivity Observer binding	Comparison gluing Object-state identification Grounding burden	Dynamical description Operability Law-kernel generation burden

Table 18: Generated objects and typical failures in representative schemes

Representative	Generated objects	Typical failures
NCM	Classical time	FDyn
	Classical continuous approximation	FL
	Classical law display	FDiff
SR	Inertial coordinate time	FR
	Generated representation	FDyn
GR	Local proper-time display	FC
	Global law-time representation	FDyn
	Stage difference	FL
TM	$\mathfrak{C}$ -derived comparison-representation coordinates	FR
	Standard frequency and period	FA
	Numerical values in Hz	
GPS	Navigation time	FC
	Localization of rotation-derived loop defects	FA
		FL
QG	Law-admissible time representation with external time	FDyn
	excluded from primitive inputs	FL

Proposition 8.22 (Type boundary of representative schemes). A representative scheme supplies the input carriers, comparison burdens, dynamical-description burdens, and law-representation burdens of an existing theory as a realized operational configuration and the additional inputs of the respective stages. It separates by type the theoretical content supplied as theory-specific input from the time comparisons, time representations, continuous-approximation validity, dynamical descriptions, law-admissible time representations, differential-law forms, scale-indexed validity domains, and failure-position tokens constructed at the generation stages.

Proof. Tables 17 and 18 separate the carriers supplied by representative theories as theory-specific inputs, the burdens determined at the generation stages, the generated objects, and the typical failures. NCM, SR, GR, TM, GPS, and QG have different input carriers, but by Corollary 8.18, the corresponding realized operational configurations and additional inputs at each stage follow the common generation order of the sequence from comparison through representation and dynamical description to law representation and differential law form, together with the branch from dynamical description to continuous approximation. Failure locations for the conditions of each stage are classified by the failure-position tokens of Definition 6.10. A representative scheme is therefore an input type that determines, in the same generation order and without conflating theory-specific inputs with generated objects, the time-grounding burdens, law-representation burdens, differential-law-form burdens, and scale-indexed validity domains of each theory. $\square$

Relativistic calculations supply highly precise correspondences among clock readings, coordinate displays, proper-time displays, dynamical descriptions, and observational records. What those correspondences establish comprises clock-reading differences, coordinate-time

differences, proper-time differences along worldlines, metric structures, and the relations among them.

Under descriptions that retain time or spacetime as a preconstruction entity, these highly precise correspondences have been described as stretching, slowing, speeding, or deformation of a time entity or spacetime entity.

By Theorem 7.201, none of the 72 audited references generates from nontemporal physical inputs the entire set of time contents used within it. By Proposition 7.13, mathematical consistency, predictive performance, and empirical fit do not repair the physical-time qualification of a primitive-time input. Success of relativistic calculation therefore does not entail the stretching, slowing, speeding, or deformation of an independent time entity or spacetime entity.

The pattern that retains time or spacetime as a preconstruction entity and explains a postgeneration relational quantity as a change in an attribute of that entity also occurs in subsequent quantum theory, quantum gravity, spacetime emergence, and cosmology. Transferring the dependency into subsequent theories while retaining primitive-time inputs causes questions that should be determined as matters of time-reference construction, generated law-admissible time representations, scale-indexed validity domains, and failure boundaries to be inherited as undifferentiated problems of time, spacetime, observation, or gravity.

Proposition 8.23 (Two-stage misrecognition in the time-stretching account). In the generation order of this paper, relativistic clock differences are determined as comparison-derived differences among generated law-admissible time representations corresponding to multiple observers or worldlines. The time-stretching account posits time as an unconstructed entity prior to these comparison-derived differences and explains relativistic clock differences by the stretching, slowing, speeding, or deformation of that time entity. This explanatory form is therefore a two-stage misrecognition of the time-stretching account in which Items 1 and 3 of Definition 8.19 occur in a sequence of direct dependence.

Proof. In the representative schemes, time use in SR and GR follows the generation order from realized operational configurations along the sequence from comparison through representation and dynamical description to law representation and differential law form, together with the branch from dynamical description to continuous approximation. A relativistic clock difference is a comparison-derived difference among multiple law-admissible time representations obtained in this generation order.

Prior to that comparison-derived difference, the time-stretching account places time as an unconstructed entity at a stage preceding its generation conditions. It therefore satisfies Item 1 of Definition 8.19. It further redescribes the comparison-derived difference generated at a later stage as a change in an attribute of that time entity—its stretching, slowing, speeding, or deformation—and therefore satisfies Item 3 of the same definition. The redescription in Item 3 has as its antecedent the prior input of a time entity under Item 1. Items 1 and 3 are thus connected by direct dependence, making this explanatory form a two-stage misrecognition in the time-stretching account. $\square$

Theorem 8.24 (Structural consequences for Newtonian and classical mechanics). For a complete evaluation specification $\text{Ev}_{\text{Sch}_{\text{NCM}}}$ and law-evaluation datum $\text{LawEvalDat}_{o, \zeta_{\text{Sch}_{\text{NCM}}}}$, suppose that

$$\text{LawOperative} \left[\text{LawEvalDat}_{o, \zeta_{\text{Sch}_{\text{NCM}}}} \right]$$

holds and that

$$\text{RepEmb}_o^{\text{dyn,time}} \in \text{OrdEmbSet}_o^{\text{dyn,time}}(\mathbb{R})$$

Then the observer-time law core

$$\text{LawCore}_{o, \zeta_{\text{Sch}_{\text{NCM}}}^{\text{obs,time}}}(\text{RepEmb}_o^{\text{dyn,time}})$$

is determined. The classical time variable displays values of the law-admissible time representation $\text{TimeRep}_o^{\text{dyn}}$. When the branching burden, observational-binding burden, and law-output determinacy hold in the classical domain, the laws of Newtonian and classical mechanics may use this generated time representation within their safe domain of application.

Proof. By Definition 5.102, law-description operability includes the simultaneous satisfaction of dynamical-description operability and the family of law conditions. Definition 5.108 determines the observer-time law core relative to the displayed real-valued time-order embedding. By Definition 5.63, the law-admissible time representation maps the $\mathfrak{C}$ -derived canonical observer-time quotient to real-valued time.

The shared-clock domain of Definition 8.7 maps clock-side material to the same order representation; stable records connect object-state-side material to the same law-bearing domain; and the observer-side binding conditions establish the four-party binding at the same phase. In the classical domain in which law-output determinacy holds, the classical time variable displays values of the generated law-admissible time representation.

By Theorem 7.200, when time content and grounding structures are generated prior to law operations and result derivation, the time variable satisfies physical-time qualification. Failures of the conditions at each stage are classified by failure-position tokens of Definition 6.10. $\square$

Connection to physical-time qualification. Time content appearing in the shared-clock domain, single ordering, and classical law display of Newtonian and classical mechanics satisfies the physical-time qualification of Theorem 7.200 when the corresponding grounding structures precede law operations and result derivation. When time content is input before its grounding structures, Theorem 7.11 entails that the content lacks physical-time qualification.

Proposition 8.25 (Inheritance of primitive-time dependence in Newtonian and classical mechanics). In Newtonian and classical mechanics, if a classical time variable is used in equations of motion, initial-value specifications, parametric representations of trajectories, or the evolution variable of a phase flow before its time content and grounding structures are generated, that time variable becomes a primitive-time input. Trajectories, velocities, accelerations, initial-value evolution, identification of conserved quantities, determination of their temporal evolution, phase flows, and law results constructed in dependence on this input inherit primitive-time dependence, and the time content contained in each result lacks physical-time qualification. Mathematical consistency, predictive performance, operational reproducibility, and empirical fit of the equations of motion do not repair this loss of qualification.

The law-admissible time representation generated under the conditions of Theorem 8.24 may be used as a law-time input in Newtonian and classical mechanics within its safe domain of application. Formulating velocity, acceleration, phase flow, or evolution by differential equations additionally requires differential-stage operability under Definition 5.132.

Proof. When pregeneration time content is used in equations of motion, initial-value specifications, parametric representations of trajectories, or the evolution variable of a phase flow, Definition 7.10 and Theorem 7.11 make that use a primitive-time input lacking physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by trajectories, velocities, accelerations, initial-value evolution, identification of conserved quantities, determination of their temporal evolution, phase flows, and law results. By Proposition 7.13, mathematical consistency, predictive performance, operational reproducibility, and empirical fit do not repair this loss of qualification.

By Theorem 8.24, satisfaction of law-description operability and a real-valued time-order embedding generates the law-admissible time representation displayed by the classical time variable. By Definition 5.132, differential-stage operability includes simultaneous satisfaction of law-description operability and the five differential-stage conditions. The safe domain of application for time-dependent quantities and differential-law forms in Newtonian and classical mechanics is therefore the domain in which the generated law-admissible time representation is used and the required differential-stage conditions hold. $\square$

Proposition 8.26 (Safe reuse of Newtonian and classical mechanics through generated time and failure classification). When the conditions of Theorem 8.24 hold, the laws, initial-value evolution, trajectories, and conserved quantities of Newtonian and classical mechanics that use a generated law-admissible time representation hold within their safe domain of application. Velocity, acceleration, phase flow, and evolution by differential equations additionally require differential-stage operability under Definition 5.132.

When a condition at any generation stage fails, the failure-position token system of Definition 6.10 and the set of minimal failure locations of Theorem 6.168 determine the condition that must be repaired first. Repair proceeds upstream-to-downstream according to the failure-generation dependency order of Definition 6.17, and downstream results are recomputed from the repaired inputs. Within the domain in which the grounding conditions again hold, the corresponding classical laws, initial-value evolution, trajectories, conserved quantities, and phase flows can be retained.

When an error source in initial values, a time representation, or a differential approximation is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the recomputed downstream results. The decision to use a classical law therefore separates into retention within its safe domain of application and repair beginning at a minimal failure location.

Proof. By Theorem 8.24, satisfaction of law-description operability and a real-valued time-order embedding supplies the generated law-admissible time representation used as the law-time input of Newtonian and classical mechanics. By Definition 5.132, the differential-law forms required for velocity, acceleration, phase flow, and evolution by differential equations are retained on the domain of differential-stage operability.

Definition 6.10 maps each failed condition to its corresponding failure-position token. Theorem 6.168 determines the failed locations that are minimal with respect to the failure-generation dependency order. By Definition 6.17, conditions and results downstream of such a location depend on upstream conditions. This determines an order in which the cause at a minimal failure location is repaired and downstream results are recomputed.

When an error source belongs to an initial value, a time representation, or a differential approximation, downstream results can be recomputed from inputs in which that source has been repaired, and an accuracy evaluation can be applied to the recomputed results. Existing results can consequently be retained where the grounding conditions hold, while

the repair order begins at the minimal failure location and accuracy evaluation remains conditional on recomputation. $\square$

Theorem 8.27 (Structural consequences for special relativity). Suppose that

$$\text{LawOperative} \left[\text{LawEvalDat}_{o, \zeta_{\text{Sch}_{\text{SR}}}} \right]$$

holds and that

$$\text{RepEmb}_o^{\text{dyn}, \text{time}} \in \text{OrdEmbSet}_o^{\text{dyn}, \text{time}}(\mathbb{R})$$

Then the observer-time law core

$$\text{LawCore}_{o, \zeta_{\text{Sch}_{\text{SR}}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}})$$

is determined. Inertial coordinate time in SR supplies the coordinate display of the law-admissible time representation contained in this law core.

Proof. By Definition 5.102, law-description operability includes simultaneous satisfaction of dynamical-description operability and the family of law conditions. Definition 5.108 determines the observer-time law core, and by Definition 5.63, the law-admissible time representation maps the $\mathfrak{C}$ -derived canonical observer-time quotient to real-valued time.

By Definition 8.10, the inertial-observer network, local clock-side reception, synchronization procedures, and comparison rules corresponding to inertial-coordinate transformations supply the inputs of SR. Within the domain in which these inputs satisfy the conditions of each generation stage, inertial coordinate time is the coordinate display of a generated law-admissible time representation. $\square$

Theorem 8.28 (Structural consequences for general relativity). When

$$\text{ExactOperable} (\mathcal{X}_{\text{Sch}_{\text{GR}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{GR}}})$$

holds, worldline-local comparison data and generated representation coordinates supply the local proper-time display $\tau_{u, \text{Sch}_{\text{GR}}}^{\text{loc}}$.

When, in addition,

$$\text{DynOperative} [\text{Ev}_{\text{Sch}_{\text{GR}}}]$$

holds, the $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation $\text{TrjRel}_o^{\text{dyn}, \text{can}}$ is determined. When law-description operability holds, the observer-time law core is determined. The existence of a local proper-time display alone does not entail right-uniqueness of the canonical trajectory relation or existence of the observer-time law core.

Proof. By Definition 8.11, worldline-local clock-side reception, record-side reception, local comparison rules, and generated representation coordinates construct the local proper-time display.

At the dynamical-description stage, descent of the $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation of Definition 5.23 requires comparison-stage closure and the family of dynamical-description conditions. Its right-uniqueness requires the dynamical-description-stage canonical-trajectory-function condition of Definition 5.26. The law stage requires the law-description operability of Definition 5.102 and the real-valued time-order embedding of Definition 5.108.

If branch resolution, transport preservation, quotient descent, or law-output determinacy fails, right-uniqueness of the canonical trajectory relation or existence of the observer-time law core does not hold even when a local proper-time display is supplied. $\square$

Theorem 8.29 (Structural consequences for time metrology). Suppose that

$$\text{ExactOperable}(\mathcal{X}_{\text{Sch}_{\text{TM}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{TM}}})$$

holds and that a $\mathfrak{C}$ -derived conditional quantity-affine realization

$$\text{QtyAffReal}_{\mathfrak{C}}(\mathcal{X}_{\text{Sch}_{\text{TM}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{TM}}})$$

is specified. Suppose further that a nonempty family of admissible counting windows for the reference source is given and satisfies the window-invariance condition of Definition 4.150. Then $\mathfrak{C}$ -derived comparison-representation coordinates, a standard-frequency representation, a period representation, and an Hz unit scalar are determined from the family of reference cycle-count sources, the cycle-counting convention, counting comparisons and corrections, correction records, the canonical comparison quotient, and generated representation coordinates.

Proof. By Definition 4.175, exact operability includes completion of the comparison and representation stages. By Definition 8.6, the family of reference cycle-count sources, the cycle-counting convention, counting comparisons and corrections, and correction records are connected to the canonical comparison quotient and generated representation coordinates.

The $\mathfrak{C}$ -derived conditional quantity-affine realization supplies quantity-affine differences of the generated representation coordinates. A cycle-count difference together with this quantity-affine difference determines the frequency and period representations, and Definition 4.152 determines the standard-frequency representation and the Hz unit scalar. The traceability network thus supplies standard-frequency and period representations and numerical values in Hz based on the generated comparison order and quantity-affine differences, without taking a primitive frequency as input. $\square$

Theorem 8.30 (Structural consequences for GPS). Within a domain in which the conditions at every generation stage of the GPS scheme hold, the navigation-time variable displays values of a generated law-admissible time representation.

Suppose that pure transport words

$$\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}_{\bullet}^{\text{tr}}[\text{ISch}_{\text{GPS}}]$$

with the same initial and terminal points supply a witness of nonidentity action for the loop word $\mathbf{w}_1^{\dagger} \star \mathbf{w}_0$. Then

$$\neg G \left[\text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}} \right]$$

holds, and global gluing at the comparison stage fails.

Proof. By Definition 8.9, the GPS scheme supplies families of clock-side sources at satellites and receivers, reception records, correction rules, transport rules, and observer-side binding conditions. Within a domain in which exact operability, dynamical-description operability, and law-description operability hold, the law-admissible time representation of Definition 5.63 is generated, and the navigation-time variable displays its values.

The assumed witness of nonidentity action refutes the requirement that every pure transport loop act identically. Consequently, the comparison-stage global-gluing condition

$$G \left[\text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}} \right]$$

therefore does not hold, and its failure location is classified by a comparison-stage failure-position token of Definition 6.10. $\square$

Theorem 8.31 (Structural consequences for quantum reconstruction). The quantum-reconstruction scheme Sch_{QG} excludes external time from primitive inputs and takes families of clock-side, record-side, and object-state-side sources, branch-sensitive record structures, observer-side binding, and comparison-gluing conditions as inputs.

When

$$\text{ExactOperable}(\mathcal{X}_{\text{Sch}_{\text{QG}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{QG}}})$$

holds, the canonical comparison quotient and representation-stage objects are determined. When, in addition,

$$\text{DynOperative}[\text{Ev}_{\text{Sch}_{\text{QG}}}]$$

holds, the $\mathfrak{C}$ -derived canonical observer-time quotient, $\mathfrak{B}$ -derived canonical object-state quotient, $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation, graph trajectory, family of canonical observable functions, and law-admissible time representation are determined. When law-description operability holds, the observer-time law core is determined.

Proof. By Definition 8.12, the quantum-reconstruction scheme supplies a realized operational configuration

$$\mathcal{X}_{\text{Sch}_{\text{QG}}} = (\mathcal{I}_{\text{Sch}_{\text{QG}}}^{\text{raw}}, \text{Dat}_{\text{Sch}_{\text{QG}}}^{\text{src}})$$

that excludes external time from primitive inputs, together with a typed family of comparison rules.

When exact operability holds, the canonical comparison quotient of Definition 4.89 and the representation-stage objects are determined. When dynamical-description operability holds, the canonical observer-time and object-state quotients of Definition 5.16, the canonical trajectory relation of Definition 5.23, the graph trajectory of Definition 5.28, the law-admissible time representation of Definition 5.63, and the family of canonical observable functions are determined.

When law-description operability holds and a $\mathfrak{C}$ -derived real-valued time-order embedding is selected, Definition 5.108 determines the corresponding observer-time law core. Quantum reconstruction that excludes external time from primitive inputs therefore generates the law-bearing time representation and law core within the domain in which the conditions of the respective generation stages hold. $\square$

8.3 Worked Reduction in Time Metrology

Specialize the time-metrology scheme of Definition 8.8 to the $\mathfrak{C}$ -derived frequency and period representations of Definition 4.150 and the $\mathfrak{C}$ -derived standard-frequency representation of Definition 4.152. The reference $\mathfrak{C}$ -derived source, cycle-count index, record-side correspondence, comparison closure, $\mathfrak{C}$ -derived canonical comparison quotient, $\mathfrak{C}$ -derived comparison-representation coordinate, quantity-affine difference, frequency and period representations, and numerical value in Hz are determined in this dependency order.

The second, frequency, and Hz used in time metrology are supplied as standardized measurement implementations[12]. A standardized measurement implementation supplies numerical conventions, measurement reproducibility, and transfer among standards. If, however, the second, frequency, or Hz is used before the generation of time content as a time interval, calibration criterion, synchronization criterion, reference criterion for traceability, or law-time input, that use constitutes primitive-time input. The time-metrology scheme takes reference cycles, records, comparison rules, and unit-normalization conventions as inputs and determines frequencies, periods, and numerical values in Hz as quantities

on a generated comparison representation. This generation order connects standardized numerical values to time metrology that has physical-time qualification.

8.3.1 From Cycle Count to Frequency Representation

Consider the time-metrology scheme of Definition 8.8. Its reference $\mathfrak{C}$ -derived source and normalization convention are given by that definition.

For each $i \in \{0, 1\}$, let the cycle-count indices be $k_i \in \mathbb{N}$, $k_0 < k_1$, and let the corresponding $\mathfrak{C}$ -derived source tokens be

$$x_{\mathfrak{C},i} := \text{tok}_{\mathfrak{C}}(j_{\text{std}}^{\text{src}}, k_i) \in \text{Tok}_{\mathfrak{C}, \text{Sch}_{\text{TM}}}^*$$

Let the record tokens containing the ancillary records m_i and derived from $\mathfrak{D}$ be

$$x_{\mathfrak{D},i} = (k_i, m_i) \in \text{Tok}_{\mathfrak{D}, \text{Sch}_{\text{TM}}}^*$$

and supply the $\mathfrak{C}/\mathfrak{D}$ correspondence by

$$(x_{\mathfrak{C},i}, x_{\mathfrak{D},i}) \in \text{Rel}_{\text{Sch}_{\text{TM}}}^{\text{src}}$$

The ancillary records include correction records, environmental records, counting comparisons and corrections, and record content required for traceability. Object-state-side tokens distinguish the comparison phase to which each record belongs.

Let the counting window for the observer-side information-coupling position $u_{\text{std}} \in U$ be

$$\text{win}_{\text{std}} := (u_{\text{std}}, j_{\text{std}}^{\text{src}}, k_0, k_1)$$

Suppose that record-side retention, correction records, counting comparisons and corrections, and the typed family of comparison rules make the two $\mathfrak{C}$ -derived source tokens comparable within the same comparison window of the same reference sequence. When the membership and strict-order conditions of Definition 4.150 hold,

$$\text{win}_{\text{std}} \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\text{Sch}_{\text{TM}}].$$

The window-endpoint images $\text{CntCls}_{\mathfrak{C},i}[\text{Sch}_{\text{TM}}; \text{win}_{\text{std}}]$ ($i \in \{0, 1\}$) are then elements of the $\mathfrak{C}$ -derived canonical comparison quotient

$$\text{Qu}_{\mathfrak{C}}^{\text{cmp}, \text{can}}(\mathcal{X}_{\text{Sch}_{\text{TM}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{TM}}})$$

and satisfy the $\mathfrak{C}$ -derived canonical strict order.

When the structural representation stage of Definition 4.92 is complete and the $\mathfrak{C}$ -derived conditional quantity-affine realization of Definition 4.148

$$\text{QtyAffReal}_{\mathfrak{C}}[\text{Sch}_{\text{TM}}]$$

is specified, the quantity-affine difference of the two window-endpoint images in the $\mathfrak{C}$ -derived comparison-representation coordinates is determined. The cycle-count difference is $k_1 - k_0$.

Proposition 8.32 (Generation from cycle counts to a frequency representation). Suppose that

$$\text{Operative}(\mathcal{X}_{\text{Sch}_{\text{TM}}})$$

and

$$\text{Repr}_{\leq 3}(\mathcal{X}_{\text{Sch}_{\text{TM}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{TM}}})$$

hold and that a $\mathfrak{C}$ -derived conditional quantity-affine realization is specified. Suppose further that

$$\text{win}_{\text{std}} \in \text{CntWin}_{\mathfrak{C}}^{\text{adm}}[\text{lsch}_{\text{TM}}]$$

Then the $\mathfrak{C}$ -derived counting-window frequency representation is determined as

$$\nu_{\mathfrak{C}}^{\text{win}}[\text{lsch}_{\text{TM}}; \text{win}_{\text{std}}] = \frac{\text{CntDiff}_{\mathfrak{C}}^{\text{qty}}[\text{win}_{\text{std}}]}{\text{RepDiff}_{\mathfrak{C}}^{\text{qty}}[\text{lsch}_{\text{TM}}; \text{win}_{\text{std}}]} \in \text{QtyVal}_{\mathfrak{C}}$$

Its product with the $\mathfrak{C}$ -derived counting-window period representation is $1_{\mathfrak{C}}^{\text{qty}}$.

When window invariance holds for the reference $\mathfrak{C}$ -derived source, the common value in the preceding expression supplies the $\mathfrak{C}$ -derived standard-frequency representation. The normalization convention N_{std} determines the $\mathfrak{C}$ -derived Hz unit scalar, and the numerical value in Hz relative to that unit scalar is determined by Definition 4.151.

Proof. By the definition of an admissible counting window, the two window-endpoint images satisfy the strict order on the $\mathfrak{C}$ -derived canonical comparison quotient. By Proposition 4.153, their quantity-affine difference in the $\mathfrak{C}$ -derived comparison-representation coordinates is positive, so the $\mathfrak{C}$ -derived counting-window frequency and period representations are determined. The same proposition gives their product as the $\mathfrak{C}$ -derived quantity unit $1_{\mathfrak{C}}^{\text{qty}}$.

When the window-invariance condition of Definition 4.150 holds for the reference $\mathfrak{C}$ -derived source, the common value of the counting-window frequency representations becomes the $\mathfrak{C}$ -derived source-relative frequency representation. Definition 4.152 makes this common value the $\mathfrak{C}$ -derived standard-frequency representation and determines the $\mathfrak{C}$ -derived Hz unit scalar from the normalization convention. Frequency, period, and numerical values in Hz are therefore determined in the dependency order from cycle-count records through comparison closure, $\mathfrak{C}$ -derived comparison-representation coordinates, quantity-affine differences, window invariance, and the unit-normalization convention. $\square$

Connection to physical-time qualification. Cycle-count records, $\mathfrak{C}$ -derived frequency and period representations, and numerical values in Hz in time metrology are used to calibrate time variables in physical laws or as quantitative inputs within laws. Time variables in physical laws that use these quantities are subject to the generation-priority condition of Corollary 7.15. By Proposition 8.21, when time content and grounding structures are generated prior to law operations and result derivation, the time variable satisfies physical-time qualification. When time content is used before the required grounding structures, the time variable does not satisfy physical-time qualification.

In addition to the reference $\mathfrak{C}$ -derived source, $\mathfrak{C}/\mathfrak{D}$ record correspondence, typed family of comparison rules, and structural representation stage, an admissible counting window, quantity-affine realization, window invariance, and unit normalization are required according to the quantity used. Continuous approximation requires a finite-operational-granularity datum and continuous-approximation evaluation input. If a formed and evaluable condition in the six series fails, a failure-position token of Definition 6.10 classifies each minimal failure location. Non-satisfaction of a quantity-specific condition is determined as a defining condition of the corresponding quantity, in accordance with Proposition 8.35.

Proposition 8.33 (Inheritance of primitive-time dependence in time metrology). If, before the generation conditions of the time-metrology scheme hold, the second, frequency, period, or a numerical value in Hz is used as a time interval, calibration criterion, synchronization criterion, reference criterion for traceability, or law-time input, that use constitutes a primitive-time input. Calibration values, frequency ratios, time scales, synchronization results, calibration chains, and law results that depend on these quantities inherit primitive-time dependence, and the time content contained in each result does not satisfy physical-time qualification. Measurement accuracy, reproducibility, agreement among standards, and traceability do not repair this loss of qualification.

When the generation conditions of the time-metrology scheme hold, frequency, period, and numerical values in Hz are generated from cycle-count records, comparison closure, $\mathfrak{C}$ -derived comparison-representation coordinates, quantity-affine differences, window invariance, and the unit-normalization convention. When these quantities are used as law-time inputs, they become law-time inputs with physical-time qualification if law-description operability holds and the generation of law-time content precedes law operations and result derivation.

Proof. When time content is used in a law operation or result derivation before the generation conditions hold, Definition 7.10 and Theorem 7.11 make that use a primitive-time input that does not satisfy physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by calibration values, frequency ratios, time scales, synchronization results, calibration chains, and law results. By Proposition 7.13, measurement accuracy, reproducibility, agreement among standards, and traceability do not repair this loss of qualification.

When the generation conditions hold, Proposition 8.32 generates frequency, period, and numerical values in Hz according to the dependency order from cycle-count records to the unit-normalization convention. Definition 5.102 and Corollary 7.15 give physical-time qualification to law-time inputs when law-description operability holds and generated time content precedes law operations and result derivation. $\square$

Corollary 8.34 (Noncircularity in time metrology). The primitive inputs of the time-metrology scheme consist of a reference $\mathfrak{C}$ -derived source, cycle-count indices, record-side correspondence, a typed family of comparison rules, and a normalization convention. Frequency, period, and numerical values in Hz are determined from the canonical comparison quotient, $\mathfrak{C}$ -derived comparison-representation coordinates, quantity-affine differences, and window invariance generated from these primitive inputs.

Proof. By Proposition 8.32, the $\mathfrak{C}$ -derived counting-window frequency representation is determined as the ratio of a cycle-count difference to a quantity-affine difference of generated comparison-representation coordinates. The $\mathfrak{C}$ -derived standard-frequency representation is determined by window invariance, and the $\mathfrak{C}$ -derived Hz unit scalar and numerical values in Hz are determined by the unit-normalization convention. The dependency order from primitive inputs through comparison, representation, quantification, and normalization therefore generates the frequency, period, and numerical values in Hz of time metrology. $\square$

Proposition 8.35 (Localization of failure results in the worked reduction for time metrology). When a generation condition for a frequency representation, period representation, or numerical value in Hz fails in the time-metrology scheme and belongs to one of the six failure series, each minimal failure location is classified by a failure-position token of Definition 6.10.

- (1) When a formed and evaluable comparison-series condition concerning $\mathfrak{C}/\mathfrak{D}$ record correspondence or comparison rules fails, each minimal failure location belongs to the comparison stage.
- (2) After the $\mathfrak{C}$ -derived canonical comparison quotient has been established, failure of a formed and evaluable representation-series condition has each minimal failure location in the representation stage. Quantity-affine realization, admissible counting windows, window invariance, and unit normalization are evaluated as defining conditions of the quantities that require them.
- (3) When the finite-operational-granularity datum or continuous-approximation evaluation input is insufficient, the corresponding approximation condition is unformed or unevaluable. When a formed and evaluable condition among the nine continuous-approximation conditions fails, each minimal failure location belongs to the continuous-approximation stage.
- (4) When a formed and evaluable series condition connecting a generated frequency representation, period representation, or numerical value in Hz to a dynamical description, law representation, or differential-law form fails, each minimal failure location belongs to the corresponding subsequent stage.

Proof. By Definition 4.150 and Proposition 4.153, generation of frequency and period representations depends on the comparison stage, structural representation stage, quantity-affine realization, and an admissible counting window. By Definition 4.152, the standard-frequency representation and Hz unit scalar depend on window invariance and the unit-normalization convention.

Definitions 4.156 and 4.168 determine the finite-operational-granularity datum and continuous-approximation evaluation input. By Proposition 6.76, the evaluation input containing the finite-operational-granularity datum and continuous-approximation evaluation input determines the evaluation states of the nine continuous-approximation conditions. Definition 6.77 makes failure of those nine conditions a continuous-approximation-stage failure.

The defining conditions of the quantities above determine whether frequency, period, and numerical values in Hz are generated. For conditions in the six series, Definition 6.21 determines the formation and evaluation states, and Proposition 6.26 establishes that the failed-token set equals its set of minimal elements. Each token of Definition 6.10 retains the series and condition index of the failed condition. $\square$

Proposition 8.36 (Safe reuse of time-metrology results through generated time and failure classification). When the generation conditions of Proposition 8.32 hold, the frequency, period, and numerical values in Hz obtained from cycle-count records, comparison closure, $\mathfrak{C}$ -derived comparison-representation coordinates, quantity-affine differences, window invariance, and the unit-normalization convention can be used within the corresponding admissible counting windows and scale-indexed validity domain. Their use as law-time inputs additionally requires law-description operability and the generation-priority condition for law-time content.

Each minimal failure location determined by Proposition 8.35 and each unsatisfied defining condition of the quantity used specify the first objects to repair among record correspondence, comparison rules, and admissible counting windows; structural representation, quantity-affine realization, window invariance, and unit normalization; finite

operational granularity and continuous-approximation evaluation input; or subsequent law connection. Frequency, period, numerical values in Hz, calibration values, frequency ratios, time scales, synchronization results, and calibration chains are recomputed from the repaired inputs and can be retained within the domain in which the grounding conditions again hold.

When an error source in calibration, a frequency ratio, record-cell width, window selection, unit normalization, or finite operational granularity is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the recomputed downstream results. The decision to use a time-metrology result therefore separates into retention within the validity domain and repair beginning at each minimal failure location.

Proof. By Proposition 8.32, frequency, period, and numerical values in Hz are generated according to the dependency order from cycle-count records through the unit-normalization convention. By Proposition 8.33, when generated quantities are used as law-time inputs, satisfaction of law-description operability and the generation-priority condition gives them physical-time qualification.

Proposition 8.35 classifies each minimal failure location of a generation condition into the comparison, representation, continuous-approximation, or subsequent stage. By Definition 6.17, conditions and results downstream of each minimal failure location depend on upstream conditions. This determines the order in which the cause at each minimal failure location is repaired and downstream results are recomputed from the repaired input.

When an error source belongs to calibration, a frequency ratio, record-cell width, window selection, unit normalization, or finite operational granularity, downstream results can be recomputed from inputs in which that source has been repaired, and accuracy evaluation can be applied to the recomputed results. Time-metrology results can consequently be retained where the grounding conditions hold, while the repair order begins at each minimal failure location and accuracy evaluation remains conditional on recomputation. $\square$

8.4 Worked Reduction for GPS

In the GPS scheme, comparison-stage objects are constructed from satellite-side clock sources, reception records, correction records, and transport rules, and a navigation-time representation is generated through the dynamical-description and law-description stages. If what is conventionally called a navigation-time variable in GPS is presupposed before its generation, satellite-clock corrections, propagation corrections, pseudoranges, receiver-clock offsets, synchronization results, position and velocity navigation solutions, and law results using them inherit primitive-time dependence. If a rotation-derived nonidentity loop action arises under the adopted transport rules, that defect is localized at the comparison stage as a failure of the identity-action condition for every pure transport loop. In the construction of GPS navigation time, correction of the transport rules, restriction of the domain of validity, or specification of the scale domain in which an approximation holds determines the conditions for using downstream results in accordance with that failure.

8.4.1 Comparison Closure and the Gluing Burden in GPS

Let the set of GPS satellite identifiers be $\text{SatIdx}_{\text{SchGPS}}$. For each $\sigma \in \text{SatIdx}_{\text{SchGPS}}$, let the satellite-side $\mathfrak{C}$ -derived source token be $x_{\mathfrak{C},\sigma} \in \text{Tok}_{\mathfrak{C},\text{SchGPS}}^*$. Represent the family of

satellite-side clock sources by

$$\{x_{\mathfrak{C},\sigma} \in \mathbf{Tok}_{\mathfrak{C},\text{Sch}_{\text{GPS}}}^*\}_{\sigma \in \text{SatId} \times \text{Sch}_{\text{GPS}}}$$

Let the GPS receiver-side information-coupling position and $\mathfrak{D}$ -derived record token be

$$\begin{aligned} u_{\text{recv},\text{Sch}_{\text{GPS}}} &\in U_{\text{Sch}_{\text{GPS}}}, \\ x_{\mathfrak{D},\text{recv}} &\in \mathbf{Tok}_{\mathfrak{D},\text{Sch}_{\text{GPS}}}^* \end{aligned}$$

Type the receiver-side arrival by

$$\mathbf{Arr}_{\mathfrak{D},u_{\text{recv},\text{Sch}_{\text{GPS}}},\text{Sch}_{\text{GPS}}}^{\text{cmp}}$$

The satellite-side tokens retain cycle indices, candidate time tags in navigation messages, and correction records. The receiver-side tokens retain reception records, satellite identification, the receiver-side information-coupling position, and correction records. The input-side structure consists of the following four components.

- Rules relating satellite-side $\mathfrak{C}$ -derived source tokens to receiver-side $\mathfrak{D}$ -derived record tokens.
- Comparison rules that include propagation corrections, clock corrections, and relativistic corrections.
- Conditions that bind records from multiple satellites to the same receiver-side information-coupling position.
- Transport rules that map multiple comparison paths to the same navigation-time representation.

For evaluation at the GPS receiver, set

$$o = u_{\text{recv},\text{Sch}_{\text{GPS}}} \in U_{\text{Sch}_{\text{GPS}}}$$

and let the single-object specification index be $\zeta_{\text{Sch}_{\text{GPS}}}$. Use the GPS-scheme-restricted complete evaluation specification $\mathbf{Ev}_{\text{Sch}_{\text{GPS}}}$ of Definition 8.5.

Suppose that the realized operational configuration and typed family of comparison rules obtained by restricting the GPS inputs to a specified domain satisfy the exact operability of Definition 4.175. By Definition 4.89, the satellite-side clock tokens, reception records, correction records, propagation corrections, and relativistic corrections descend, under the same family of comparison rules, to the $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient. When a complete evaluation specifying a nonrotating domain satisfies this exact operability, records from multiple satellites become comparison objects on the same canonical comparison quotient.

When the dynamical-description operability of Definition 5.70 holds, the dynamical-description extension base data and the dynamical-description evidence datum relative to the real-valued time-order embedding selected in Definition 5.59,

$$\begin{aligned} &\mathbf{Dat}_{o,\zeta_{\text{Sch}_{\text{GPS}}}}^{\text{dyn},0}, \\ &\mathbf{DynWit}_{o,\zeta_{\text{Sch}_{\text{GPS}}}} \left[\mathbf{RepEmb}_o^{\text{dyn},\text{time}} \right] \end{aligned}$$

are generated.

Suppose that the law-evaluation datum satisfies the equality of evaluation specifications under Proposition 5.86 and that the law-description operability of Definition 5.102 holds. By Definition 5.108, this law-evaluation datum and the selected real-valued time-order embedding determine the observer-time law core.

For any

$$T_0 \in \text{Qu}_{\mathbf{c},o}^{\text{dyn},\text{can}}$$

let the time value supplied by the law-admissible time representation of Definition 5.63,

$$t_0 := \text{TimeRep}_o^{\text{dyn}}(T_0)$$

be a GPS navigation-time value. The object ordinarily called a navigation-time variable in GPS is this GPS navigation-time value supplied by the law-admissible time representation belonging to the observer-time law core.

Proposition 8.37 (Navigation-time representation on the exact validity domain of GPS). Suppose that the law-evaluation datum of the GPS scheme satisfies

$$\begin{aligned} \text{EvalSpec} \left[\text{LawEvalDat}_{o,\zeta_{\text{SchGPS}}} \right] &= \text{Ev}_{\text{SchGPS}}, \\ \text{LawOperative} \left[\text{LawEvalDat}_{o,\zeta_{\text{SchGPS}}} \right] & \end{aligned}$$

and let

$$\text{RepEmb}_o^{\text{dyn},\text{time}} \in \text{OrdEmbSet}_o^{\text{dyn},\text{time}}(\mathbb{R})$$

. Then the object ordinarily called a navigation-time variable in GPS is the GPS navigation-time value generated by the law-admissible time representation belonging to the observer-time law core.

Proof. Definition 5.102 and the equality of evaluation specifications imply that the corresponding complete evaluation specification is dynamical-description operative. By Definition 5.108 and Proposition 5.109,

$$\text{LawCore}_{o,\zeta_{\text{SchGPS}}}^{\text{obs},\text{time}}(\text{RepEmb}_o^{\text{dyn},\text{time}})$$

exists uniquely. By Definition 5.63, each time-quotient class is mapped to real-valued time by the law-admissible time representation of that definition. Therefore, the object ordinarily called a navigation-time variable in GPS is a GPS navigation-time value generated through the comparison, dynamical-description, and law-description stages. $\square$

Connection to physical-time qualification. By Corollary 7.15, the generation structures corresponding to the time content used by a GPS navigation-time value must be established prior to law operations and result derivation. By Proposition 8.21, when time content and grounding structures are generated from nontemporal physical inputs prior to law operations and result derivation, the time variable satisfies physical-time qualification.

When time content enters a law operation or result derivation before the grounding structures required for its use, that time content is a primitive-time input under Definition 7.10. By Theorem 7.11, the time variable does not satisfy physical-time qualification. By Proposition 7.13, the simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit does not repair this loss of qualification.

Failures at the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages of the complete GPS evaluation specification are classified by the failure-position tokens of Definition 6.10. Failure of the identity-action condition for all-pure-transport loops belongs to the comparison-failure series, and a determination outside the scale-indexed validity domain belongs to the continuous-approximation-failure series.

Proposition 8.38 (Inheritance of primitive-time dependence in GPS). If a navigation-time variable is used in satellite-clock correction, propagation correction, pseudorange equations, receiver-clock-offset estimation, synchronization processing, or navigation solutions before the generation structures of the GPS navigation-time value are established prior to law operations and result derivation, that use becomes a primitive-time input. Satellite-clock correction values, propagation-correction values, pseudoranges, receiver-clock offsets, synchronization results, position and velocity navigation solutions, and law results that use them inherit primitive-time dependence, and the time content contained in each result does not satisfy physical-time qualification. Mathematical consistency of the navigation equations, positioning accuracy, operational reproducibility, and empirical fit do not repair this loss of qualification.

When the GPS scheme satisfies exact operability, dynamical-description operability, and law-description operability, a GPS navigation-time value is generated. Within a domain in which the identity-action condition for all-pure-transport loops holds, this value can be used as a single exact navigation-time representation. Where that condition fails, the scale domain of approximate validity is stated, and qualification to use navigation-time values, correction values, synchronization results, and navigation solutions is determined within that domain.

Proof. When a navigation-time variable is used in a law operation or result derivation before the generation structures are established, Definition 7.10 and Theorem 7.11 make that use a primitive-time input that does not satisfy physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by correction values, pseudoranges, receiver-clock offsets, synchronization results, navigation solutions, and law results. By Proposition 7.13, mathematical consistency, positioning accuracy, operational reproducibility, and empirical fit do not repair this loss of qualification.

When exact operability, dynamical-description operability, and law-description operability hold, Proposition 8.37 shows that the law-admissible time representation belonging to the observer-time law core generates the GPS navigation-time value. By Definition 6.42, the identity-action condition for all-pure-transport loops determines exact gluing independent of comparison path. By Definition 6.161, the domain of use for approximate operation is determined by the scale domain of approximate validity. $\square$

8.4.2 Rotation-Derived Loop Defects and FC_8

Definition 8.39 (GPS rotation-derived loop action and nonidentity-action witness). Represent two comparison paths in GPS with the same initial and terminal points, for each $\bullet \in \{\mathfrak{C}, \mathfrak{D}\}$, by pure transport words

$$\mathbf{w}_0, \mathbf{w}_1 \in \text{Word}_{\bullet}^{\text{tr}} [\text{ISch}_{\text{GPS}}]$$

. The word constructed from the inverse and concatenation,

$$\mathbf{w}_1^{\dagger} \star \mathbf{w}_0 \in \text{LoopWord}_{\text{all}}^{\text{tr}, \bullet} [\text{ISch}_{\text{GPS}}]$$

is a pure transport loop word.

Define the action of this loop word by

$$\begin{aligned} \text{LoopAct}_{\text{GPS}}^{\bullet}[\mathbf{w}_0, \mathbf{w}_1] \\ := \text{Act}_{\bullet}^{\text{tr}}[\text{I}_{\text{Sch}_{\text{GPS}}}] \left[\mathbf{w}_1^{\dagger} \star \mathbf{w}_0 \right] \end{aligned}$$

and define the nonidentity-action witness predicate by

$$\begin{aligned} \text{LoopWit}_{\text{GPS}}^{\bullet}[x; \mathbf{w}_0, \mathbf{w}_1] \\ \iff x \in \text{Dom}(\text{LoopAct}_{\text{GPS}}^{\bullet}[\mathbf{w}_0, \mathbf{w}_1]) \\ \wedge \text{LoopAct}_{\text{GPS}}^{\bullet}[\mathbf{w}_0, \mathbf{w}_1](x) \neq x \end{aligned}$$

. Call an x satisfying this predicate a nonidentity-action witness of a rotation-derived loop defect in GPS. The condition $\mathbf{G}[\text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}}]$ of Definition 6.42 determines exact identity of the loop action. Approximate operation relative to an engineering tolerance threshold is determined by the scale domain of approximate validity of Definition 6.161.

Proposition 8.40 (Localization of a rotation-derived loop defect in GPS). Suppose that, for the GPS-scheme-restricted complete evaluation specification,

$$\mathbf{L}_{\leq 7} \left[\text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}} \right]$$

holds. Suppose further that there exist

$$\begin{aligned} \bullet &\in \{\mathfrak{C}, \mathfrak{D}\}, \\ \mathbf{w}_0, \mathbf{w}_1 &\in \text{Word}_{\bullet}^{\text{tr}}[\text{I}_{\text{Sch}_{\text{GPS}}}] \end{aligned}$$

and x such that

$$\text{LoopWit}_{\text{GPS}}^{\bullet}[x; \mathbf{w}_0, \mathbf{w}_1]$$

holds. Then

$$\text{FC}_8[\text{Ev}_{\text{Sch}_{\text{GPS}}}]$$

holds.

Proof. The nonidentity-action witness of Definition 8.39 invalidates the global gluing condition at the comparison stage, $\mathbf{G}[\text{Ev}_{\text{Sch}_{\text{GPS}}}^{\text{cmp}}]$. The conclusion follows from the assumption that the first seven comparison conditions hold, Definition 6.42, and Proposition 6.43. $\square$

Corollary 8.41 (Separation of stages in the worked GPS reduction). For each complete evaluation specification constructed by restricting the GPS inputs, the following hold.

1. When law-description operability holds, a navigation-time value for the restricted domain is generated.
2. When the first seven comparison-stage conditions hold and a nonidentity loop-action witness exists, the all-pure-transport-loop identity-action failure FC_8 holds.
3. Approximate operation relative to an engineering tolerance threshold is determined by the scale domain of approximate validity.

Proof. Item 1 follows from Proposition 8.37. Item 2 follows from Proposition 8.40. Item 3 follows from Definition 6.161. The GPS navigation-time value generated in Item 1 is supplied by a law-admissible time representation belonging to the observer-time law core and is the object ordinarily called a navigation-time variable in GPS. Item 1 determines law-time generation on a specified restricted domain; Item 2 determines exact loop identity action at the comparison stage; and Item 3 determines the scale-indexed domain of approximate validity. $\square$

Proposition 8.42 (Safe reuse of GPS results through generated time and failure classification). Where the conditions of Proposition 8.37 and the identity-action condition for all-pure-transport loops hold, generated GPS navigation-time values, satellite-clock correction values, propagation-correction values, pseudoranges, receiver-clock offsets, synchronization results, and position and velocity navigation solutions can be used within the exact validity domain. Under approximate operation, the scale domain of approximate validity of Definition 6.161 determines the domain of use of these results.

The comparison-stage loop defect, boundary of the scale-indexed validity domain, or failure of a subsequent-stage condition classified by Proposition 8.40 and Corollary 8.41 determines the GPS construction component that must be repaired first. Clock corrections, propagation corrections, pseudoranges, receiver-clock offsets, synchronization results, and navigation solutions are recalculated from the repaired inputs, and these results can be retained within the exact validity domain or the scale domain of approximate validity.

When an error source in clock corrections, propagation corrections, comparison paths, loop defects, synchronization processing, or receiver-clock offsets is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the recalculated navigation solution. The decision to use GPS results therefore separates into retention within the validity domain and repair from the first failure position.

Proof. By Proposition 8.37, within a domain in which law-description operability holds, the law-admissible time representation belonging to the observer-time law core generates a GPS navigation-time value. By Proposition 8.38, within the range in which the generated GPS navigation-time value and its grounding conditions are used, the corresponding correction values, pseudoranges, receiver-clock offsets, synchronization results, and navigation solutions connect to time inputs that have physical-time qualification.

Proposition 8.40 maps a nonidentity loop action to failure of the identity-action condition for all-pure-transport loops. Corollary 8.41 separates exact law-time generation, a comparison-stage loop defect, and a scale-indexed domain of approximate validity. By Definition 6.17, conditions and results downstream of the first failure position depend on upstream conditions. The order is therefore fixed in which a cause belonging to the first failure position is repaired and the navigation results are recalculated from those inputs.

When the source of error belongs to clock corrections, propagation corrections, comparison paths, loop defects, synchronization processing, or receiver-clock offsets, the navigation solution can be recalculated from inputs in which that source has been repaired, and an accuracy evaluation can be applied to the recalculated result. It follows that retention of GPS results within the validity domain, the repair order from the first failure position, and conditional evaluation of positioning accuracy hold. $\square$

8.5 Worked Reduction for SR

In the special-relativity scheme, comparison-stage objects are constructed from an inertial-observer network, local clock-side receptions, record-side receptions, synchronization rules,

and comparison rules, and inertial coordinate time is generated through the dynamical-description and law-description stages. If inertial coordinate time or proper time is presupposed prior to this generation, synchronization results, Lorentz-transformation results, time representations of worldlines, proper-time differences, differential quantities such as velocity and acceleration, clock-rate differences, and law results that use them inherit primitive-time dependence. The coordinate representation ordinarily called inertial coordinate time in SR is the law-admissible time representation obtained through this generation order, and each of its values is an SR inertial-coordinate-time value.

8.5.1 From an Inertial-Observer Network to Inertial Coordinate Time

The inputs to the worked reduction for SR are the inertial-observer network $U_{\text{Sch}_{\text{SR}}}$ and, for each $u \in U_{\text{Sch}_{\text{SR}}}$, the local clock-side reception, record-side reception, and object-state-side reception. Clock-side reception contains proper-clock tokens or candidate local-clock tokens, and record-side reception contains local event records, signal-exchange records, and synchronization records. Object-state-side reception supplies material for distinguishing object states or event positions.

Synchronization rules provide comparability between local clock-side tokens and record-side tokens. Comparison rules make the local records of different inertial observers descend to the same comparison quotient. For an inertial observer $o \in U_{\text{Sch}_{\text{SR}}}$ and single-object specification index $\zeta_{\text{Sch}_{\text{SR}}}$, use the SR-scheme-restricted complete evaluation specification $\text{Ev}_{\text{Sch}_{\text{SR}}}$ of Definition 8.5.

Suppose that the realized operational configuration and typed family of comparison rules obtained by restricting the SR inputs to a specified domain satisfy the exact operability of Definition 4.175. By Definition 4.89, the local clock-side receptions, record-side receptions, synchronization records, and comparison records of the inertial-observer network descend, under the same family of comparison rules, to the $\mathfrak{CD}$ -derived canonical comparison quotient. The objects obtained at this stage are quotient objects based on the comparison conditions and their order-bearing structure.

When the dynamical-description operability of Definition 5.70 holds, the dynamical-description extension base data and the dynamical-description evidence datum relative to the real-valued time-order embedding selected in Definition 5.59,

$$\begin{aligned} & \text{Dat}_{o, \zeta_{\text{Sch}_{\text{SR}}}}^{\text{dyn}, 0}, \\ & \text{DynWit}_{o, \zeta_{\text{Sch}_{\text{SR}}}} [\text{RepEmb}_o^{\text{dyn}, \text{time}}] \end{aligned}$$

are generated.

Suppose that the law-evaluation datum satisfies the equality of evaluation specifications under Proposition 5.86 and that the law-description operability of Definition 5.102 holds. By Definition 5.108, this law-evaluation datum and the selected real-valued time-order embedding determine the observer-time law core.

For any

$$T_0 \in \text{Qu}_{\mathfrak{e}, o}^{\text{dyn}, \text{can}}$$

define the conventional notation for an inertial-coordinate-time value in ordinary SR by

$$t := \text{TimeRep}_o^{\text{dyn}}(T_0)$$

. The coordinate representation ordinarily called inertial coordinate time in SR is the law-admissible time representation $\text{TimeRep}_o^{\text{dyn}}$ of Definition 5.63, and t is the conventional

notation for its SR inertial-coordinate-time value. Minkowski geometry and Lorentz transformations are theory-specific input carriers that supply admissible transformations of this time representation and the spatial representation. The $\mathfrak{CQ}$ -derived canonical comparison quotient is the quotient object preceding the action of those representation transformations.

Proposition 8.43 (Inertial coordinate time is a generated representation). Suppose that the law-evaluation datum of the SR scheme satisfies

$$\begin{aligned} \text{EvalSpec} \left[\text{LawEvalDat}_{o, \zeta_{\text{SchSR}}} \right] &= \text{Ev}_{\text{SchSR}}, \\ \text{LawOperative} \left[\text{LawEvalDat}_{o, \zeta_{\text{SchSR}}} \right] & \end{aligned}$$

and let

$$\text{RepEmb}_o^{\text{dyn, time}} \in \text{OrdEmbSet}_o^{\text{dyn, time}}(\mathbb{R})$$

. Then the coordinate representation ordinarily called inertial coordinate time in SR is the law-admissible time representation $\text{TimeRep}_o^{\text{dyn}}$ belonging to the observer-time law core. The SR inertial-coordinate-time value for each time-quotient class is generated by this law-admissible time representation and is conventionally denoted by t .

Proof. Definition 5.102 and the equality of evaluation specifications imply that the corresponding complete evaluation specification is dynamical-description operative. By Definition 5.108 and Proposition 5.109, the corresponding observer-time law core exists uniquely. By Definition 5.63, each time-quotient class is mapped to real-valued time by the law-admissible time representation of that definition. Therefore, the coordinate representation ordinarily called inertial coordinate time in SR is a law-admissible time representation generated through the comparison, dynamical-description, and law-description stages. Each of its values is an SR inertial-coordinate-time value and is conventionally denoted by t . $\square$

8.5.2 Generation Stage of the Lorentz Transformation

Definition 8.10 supplies inertial-coordinate transformations among different inertial observers as comparison rules in the theory-specific inputs of SR. When law-description operability holds for different inertial observers $o, o' \in U_{\text{SchSR}}$, the observer-time law core and law-admissible time representation of each observer are generated. As the inertial-coordinate transformation rule in the theory-specific inputs of SR, the Lorentz transformation relates these generated time representations and spatial representations.

Proposition 8.44 (A Lorentz transformation is a transformation among generated representations). Suppose that law-description operability holds for different inertial observers in the SR scheme. The Lorentz transformation supplied by the theory-specific inputs of SR is then an admissible representation transformation among the law-admissible time representations and spatial representations generated for the respective observers.

Proof. By Definition 8.10, the inertial-observer network, local clock-side receptions, record-side receptions, object-state-side receptions, synchronization procedures, proper-time comparisons, and comparison rules corresponding to inertial-coordinate transformations supply the theory-specific inputs of SR.

When exact operability holds for each observer, local receptions and synchronization records descend to the $\mathfrak{CD}$ -derived canonical comparison quotient. When dynamical-description operability holds, dynamical-description evidence data are generated. When law-description operability holds, Proposition 8.43 generates the law-admissible time representation and SR inertial-coordinate-time values of each observer. The inertial-coordinate transformation rules in the theory-specific inputs of SR therefore relate the generated time representations and spatial representations among different inertial observers.

Furthermore, within the range in which differential-description operability holds for the differential evaluation specification of each observer, Theorem 5.135 gives differential covariance among the real-valued time-order embeddings belonging to that observer. This within-observer differential covariance and the between-observer inertial-coordinate transformation rule bear different types of responsibility. A Lorentz transformation is therefore an admissible representation transformation among the time representations and spatial representations generated for the respective observers. $\square$

Connection to physical-time qualification. By Corollary 7.15, the generation structures corresponding to the time content used by an SR inertial-coordinate-time value must be established prior to law operations and result derivation. By Proposition 8.21, when time content and grounding structures are generated from nontemporal physical inputs prior to law operations and result derivation, the time variable satisfies physical-time qualification.

When time content enters a law operation or result derivation before the grounding structures required for its use, that time content is a primitive-time input under Definition 7.10. By Theorem 7.11, the time variable does not satisfy physical-time qualification. By Proposition 7.13, the simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit does not repair this loss of qualification.

Failures at the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages of the SR-scheme-restricted complete evaluation specification are classified by the failure-position tokens of Definition 6.10.

Proposition 8.45 (Inheritance of primitive-time dependence in SR). If inertial coordinate time or proper time is used in synchronization rules, Lorentz transformations, parametric representations of worldlines, proper-time comparisons, spacetime intervals, differential quantities such as velocity and acceleration, or clock-rate comparisons before the generation structures of an SR inertial-coordinate-time value or proper-time value are established prior to law operations and result derivation, that use becomes a primitive-time input. Transformed coordinates, synchronization results, simultaneity relations, time representations of worldlines, proper-time differences, differential quantities, clock-rate differences, and law results that use them inherit primitive-time dependence, and the time content contained in each result does not satisfy physical-time qualification. Lorentz covariance, invariance of the spacetime interval, predictive performance, and empirical fit do not repair this loss of qualification.

When the SR scheme satisfies exact operability, dynamical-description operability, and law-description operability, the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer are generated. Lorentz transformations are applied among these generated time representations and spatial representations. Results involving time derivatives additionally require differential-description operability.

Proof. When inertial coordinate time or proper time is used in a law operation or result derivation before the generation structures are established, Definition 7.10 and Theorem 7.11 make that use a primitive-time input that does not satisfy physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by transformed coordinates, synchronization results, simultaneity relations, time representations of worldlines, proper-time differences, differential quantities, clock-rate differences, and law results. By Proposition 7.13, Lorentz covariance, invariance of the spacetime interval, predictive performance, and empirical fit do not repair this loss of qualification.

By Proposition 8.43, the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer are generated when exact operability, dynamical-description operability, and law-description operability hold. By Proposition 8.44, a Lorentz transformation becomes an admissible representation transformation among generated representations. By Theorem 5.135, differential covariance among real-valued time-order embeddings holds within the range in which differential-description operability holds. $\square$

Corollary 8.46 (The two-stage misrecognition of the time-stretching account in SR). Relativistic clock differences in SR are comparison differences among generated law-admissible time representations corresponding to multiple inertial observers or worldlines. The time-stretching account presupposes time as an unconstructed entity prior to these generated representation differences and explains relativistic clock differences through stretching, slowing, or deformation of that time entity. By Proposition 8.23, this form of explanation is therefore the two-stage misrecognition of the time-stretching account.

Proof. By Proposition 8.43, inertial coordinate time in SR is a law-admissible time representation obtained from a generated observer-time law core. By Proposition 8.44, a Lorentz transformation is an admissible representation transformation among generated law-admissible time representations and spatial representations. Coordinate-time representations of different inertial observers are therefore related by the inertial-coordinate transformation rules in the theory-specific inputs of SR. Relativistic clock differences, including differences in proper-clock readings and proper-time differences along worldlines, are comparison differences among generated law-admissible time representations corresponding to multiple inertial observers or worldlines. By Proposition 8.23, in a form of explanation that places time as an unconstructed entity prior to these representation differences and redescribes subsequently generated comparison differences as changes in an attribute of that time entity, the first and third items of multistage misrecognition occur consecutively with direct dependence. This form of explanation is therefore the two-stage misrecognition of the time-stretching account. $\square$

Proposition 8.47 (Separation of the formal construction of special relativity from a bearer of time stretching). The formal construction of special relativity determines Lorentz transformations among inertial coordinates, spacetime intervals, accumulation of proper time along worldlines, clock readings, clock rates, and their comparison relations. Conditions of relative motion specify relations among $\mathcal{C}\mathcal{O}\mathcal{B}\mathcal{O}$ physical carriers, worldlines, signals, synchronization, records, and generated time representations.

If the account that “time itself stretches, slows, or deforms” is taken as a literal physical mechanism, that account asserts a bearer of time stretching that remains identical through the deformation while taking different states. For this assertion to hold as a physical mechanism, it requires the type of the bearer of time stretching, an identity condition before and after deformation, a state space, a deformation magnitude, a deformation law,

a generation path from nontemporal physical inputs, coupling laws to $\mathcal{C}\mathcal{B}\mathcal{D}$ physical carriers and clock processes, and a measurement procedure independent of clock differences.

The formal results of special relativity established by Proposition 7.167 are comparison relations among clock readings, clock rates, proper times, and generated time representations. These formal results do not supply the type, identity condition, state space, deformation law, generation path, coupling laws, or independent measurement procedure of a bearer of time stretching. A literal time-stretching account therefore reintroduces an unconstructed bearer of change prior to the comparison differences generated by the formal construction and attributes those comparison differences to changes in an attribute of that bearer. This attribution is the two-stage misrecognition of Corollary 8.46.

No misrecognition occurs when differences in clock readings, clock rates, proper times, or generated time representations are described directly in terms of their objects and relations. Even when the term time stretch is used, no misrecognition occurs in an explicit metaphor if the source text states, within the same explanatory scope, its metaphorical status, that it does not designate literal deformation of time, and its actual referent.

When the term time stretch is used without stating its metaphorical status and actual referent, interpreting the term retrospectively as shorthand leaves a one-stage misrecognition that transfers a comparison difference to an attribute of time. In a literal account that places an unconstructed time entity in advance, this one stage is joined by pre-generation placement and constitutes the two-stage misrecognition of Corollary 8.46.

Proof. By Proposition 8.43 and Proposition 8.44, inertial coordinate time is a value of a generated law-admissible time representation, and a Lorentz transformation is an admissible representation transformation among generated time representations and spatial representations. By Proposition 7.167, the formal results of special relativity supply clock readings, clock rates, proper times, and their comparison relations, but do not supply an independent state and evolution law for a bearer of time stretching.

Literal stretching, slowing, or deformation requires an object that changes, its states before and after the change, and a condition for determining both states as states of the same object. Treating that change as the cause of clock differences additionally requires coupling laws from the bearer of change to $\mathcal{C}\mathcal{B}\mathcal{D}$ physical carriers and clock processes, together with a measurement procedure independent of clock differences. The formal results of special relativity contain none of these constructions.

A literal time-stretching account therefore attributes a comparison difference among generated representations to an unconstructed bearer of time stretching. By Corollary 8.46, this attribution is a two-stage misrecognition. No misrecognition occurs when comparison differences on physical carriers are described directly or when metaphorical status and the actual referent are stated. A bare expression of time stretching constitutes a one-stage misrecognition that transfers a comparison difference to an attribute of time even under an interpretation that does not assert an independent time entity. $\square$

Proposition 8.48 (Generation structure of relativistic clock differences in special relativity). For the $\mathcal{C}\mathcal{B}\mathcal{D}$ physical carriers of Definition 3.1 and Definition 3.2, suppose that the conditions of Proposition 8.43 hold. Conditions of relative motion specify the worldlines to which the $\mathcal{C}\mathcal{B}\mathcal{D}$ physical carriers belong and the relations of signal propagation, synchronization, and comparison. The clock processes, records, object-state differences, and information coupling obtained from the $\mathcal{C}\mathcal{B}\mathcal{D}$ physical carriers generate the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer.

By Proposition 8.44, a Lorentz transformation acts as an admissible representation transformation among generated time representations and spatial representations. Accumulation of proper time along worldlines, proper-clock readings, and signal-mediated comparisons determine proper-time differences, clock-reading differences, and clock-rate differences. The physical grounds for these differences lie in the dependency chain of $\mathcal{CWB}$ physical carriers, worldlines, signals, synchronization, comparison, and generated time representations.

This dependency chain does not contain primitive time as an object of physical action. An account that attributes results generated in relation to relative motion to stretching, slowing, or deformation of primitive time itself placed prior to generation constitutes the two-stage misrecognition of Corollary 8.46.

Proof. Definition 3.1 and Definition 3.2 map each role in $\mathcal{CWB}$ to a physical-carrier realization. By Proposition 8.43, the establishment of clock processes, records, object-state differences, information coupling, synchronization, and comparison generates the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer. By Proposition 8.44, a Lorentz transformation becomes an admissible representation transformation among generated representations. Accumulation of proper time along worldlines, proper-clock readings, and signal-mediated comparisons therefore supply proper-time differences, clock-reading differences, and clock-rate differences as comparison results on generated time representations.

In this generation order, conditions relating to relative motion act on relations among $\mathcal{CWB}$ physical carriers, worldlines, signals, synchronization, and comparison. By Corollary 8.46, an account that places primitive time prior to this comparison result and attributes the comparison result to a change in an attribute of primitive time itself is a two-stage misrecognition. $\square$

Corollary 8.49 (Localization of failed results in the worked reduction for SR). In the SR scheme, if the conditions that generate inertial-coordinate-time values as values of a law-admissible time representation fail even when local clock-side receptions, record-side receptions, synchronization rules, or comparison rules are supplied, the failure position is classified by the failure-position-token system.

Proof. Under the conditions of Proposition 8.43, inertial-coordinate-time values are obtained as values of a generated law-admissible time representation. Claims using continuous approximation require an approximation-validity scale domain, and use of differential law forms requires satisfaction of the differential-stage conditions. When a stage condition fails, Definition 6.10 classifies each minimal failure position as the corresponding comparison, representation, continuous-approximation, dynamical-description, law-description, or differential failure-position token. $\square$

Proposition 8.50 (Safe reuse of special-relativistic results through generated time and failure classification). Within a domain in which the conditions of Proposition 8.43 and Proposition 8.44 hold, generated law-admissible time representations, SR inertial-coordinate-time values, Lorentz-transformation results, synchronization results, simultaneity relations, and time representations of worldlines can be used within that domain. Proper-time differences, spacetime intervals, differential quantities such as velocity and acceleration, and clock-rate differences can be used within a domain in which the generation conditions for the time content on which each depends and the required conditions at the comparison,

representation, continuous-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages hold.

Each minimal failure position classified by Corollary 8.49 determines the SR construction component that must be repaired first. A failure at the comparison stage requires repair of local clock-side receptions, record-side receptions, synchronization rules, or comparison rules; a failure at the representation or continuous-approximation stage requires repair of the time representation, comparison domain, or scale domain of approximate validity. A failure at the dynamical-description, law-representation, or differential-consistency and law-form-preservation stage requires repair of the dynamical-description evidence datum, law-evaluation datum, or differential evaluation specification. From the repaired inputs, time representations, transformed coordinates, synchronization results, simultaneity relations, time representations of worldlines, proper-time differences, differential quantities, clock-rate differences, and law results are regenerated or recalculated in dependency order.

Corollary 8.46 separates comparison differences among generated representations from the two-stage misrecognition that places an unconstructed time entity prior to those comparison differences and explains the differences through stretching, slowing, or deformation of that entity. The grounds for using Lorentz covariance, invariance of the spacetime interval, proper-time differences, and clock-rate differences are supplied by generated time representations, admissible representation transformations, and the grounding conditions at each stage.

When an error source in synchronization rules, proper-time comparisons, clock-rate comparisons, continuous approximations, or differential evaluations is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the regenerated or recalculated downstream results. The decision to use special-relativistic results therefore separates into retention within the validity domain and repair from each minimal failure position.

Proof. By Proposition 8.43, within a domain in which law-description operability holds, the law-admissible time representation and SR inertial-coordinate-time values of each inertial observer are generated. By Proposition 8.44, a Lorentz transformation becomes an admissible representation transformation among generated time representations and spatial representations. By Proposition 8.21, when the time content and grounding structures on which each result depends are generated prior to law operations and result derivation, that time input satisfies physical-time qualification. Transformed coordinates, synchronization results, simultaneity relations, time representations of worldlines, proper-time differences, differential quantities, clock-rate differences, and law results that satisfy the grounding conditions at each stage therefore use time inputs that have physical-time qualification.

By Corollary 8.49 and Definition 6.17, conditions and results downstream of each minimal failure position depend on upstream conditions. The order is therefore fixed in which an input or rule belonging to each minimal failure position is repaired and downstream results are regenerated or recalculated from those inputs in dependency order.

Corollary 8.46 separates comparison differences among generated law-admissible time representations from the two-stage misrecognition that explains those comparison differences through changes in an attribute of an unconstructed time entity. This connects the grounds for Lorentz covariance, invariance of the spacetime interval, proper-time differences, and clock-rate differences to generated time representations, admissible representation transformations, and the conditions at each stage.

When the source of error belongs to synchronization rules, proper-time comparisons, clock-rate comparisons, continuous approximations, or differential evaluations, downstream results can be regenerated or recalculated from inputs in which that source has been

repaired, and an accuracy evaluation can be applied to the recalculated result. It follows that retention of special-relativistic results within the validity domain, the repair order from each minimal failure position, and conditional evaluation of accuracy hold. $\square$

8.6 Worked Reduction for GR

In the general-relativity scheme, comparison-stage objects are constructed from a local-observer network, worldline-dependent receptions, local comparison rules, transport rules, and gluing rules. Worldline-local comparison data and generated representation coordinates generate local proper-time representations, while satisfaction of the conditions at the dynamical-description and law-description stages generates a global law-admissible time representation. If local proper time, coordinate time, or a global law-time representation is presupposed prior to this generation, time-dependent components of the metric and connection, parametric representations of geodesics, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time representations, and law results that use them inherit primitive-time dependence. Local proper-time representations and global law-time representations belong to different generation stages.

8.6.1 From Local Proper-Time Representations to a Global Law-Time Representation

The inputs to the worked reduction for GR are the local-observer network $U_{\text{Sch}_{\text{GR}}}$ and, for each $u \in U_{\text{Sch}_{\text{GR}}}$, worldline-dependent reception, local clock-side reception, record-side reception, and object-state-side reception. Local comparison rules, transport rules, and gluing rules supply the conditions under which comparison materials belonging to different local observers or worldlines descend to the same comparison quotient.

For a local observer $o \in U_{\text{Sch}_{\text{GR}}}$ and single-object specification index $\zeta_{\text{Sch}_{\text{GR}}}$, use the GR-scheme-restricted complete evaluation specification $\text{Ev}_{\text{Sch}_{\text{GR}}}$ of Definition 8.5. Lorentzian spacetime, the metric, the connection, local charts, transport rules, and gluing rules supply the theory-specific input carriers of GR.

Suppose that the realized operational configuration and typed family of comparison rules obtained by restricting the GR inputs to a specified domain satisfy the exact operability of Definition 4.175. By Theorem 8.28, worldline-local clock-side reception, record-side reception, local comparison rules, and generated representation coordinates generate the local proper-time representation $\tau_{u, \text{Sch}_{\text{GR}}}^{\text{loc}}$.

A local proper-time representation is a representation of comparison data belonging to the worldline of a local observer. Transport preservation, loop consistency, quotient descent, global gluing, branch resolution, and law-output determinacy, which are required to make records belonging to multiple local observers or worldlines descend to a single comparison quotient, canonical trajectory relation, or law domain, belong to subsequent generation stages.

By Definition 4.89, when the comparison-stage and representation-stage conditions hold, local receptions, transport records, and gluing records descend to the $\mathfrak{CD}$ -derived canonical comparison quotient.

When the dynamical-description operability of Definition 5.70 holds, the $\mathfrak{CD}\mathfrak{B}$ -derived canonical trajectory relation $\text{TrjRel}_o^{\text{dyn, can}}$ is generated. Its right-uniqueness requires

$$K_{\text{CanTrjFunc}}^{\text{dyn}}(\text{Ev}_{\text{Sch}_{\text{GR}}})$$

to hold.

Use the $\mathfrak{C}$ -derived real-valued time-order embedding selected in Definition 5.59.

Suppose that the law-evaluation datum satisfies the equality of evaluation specifications under Proposition 5.86 and that the law-description operability of Definition 5.102 holds. By Definition 5.108, this law-evaluation datum and the selected real-valued time-order embedding determine the observer-time law core, and Definition 5.63 generates the law-admissible time representation $\text{TimeRep}_o^{\text{dyn}}$.

Proposition 8.51 (Stage difference between local proper-time representations and a global law-time representation). Suppose that the local proper-time representation $\tau_{u, \text{Sch}_{\text{GR}}}^{\text{loc}}$ has been generated in the GR scheme. This condition alone does not imply the right-uniqueness of the $\mathfrak{CDB}$ -derived canonical trajectory relation $\text{TrjRel}_o^{\text{dyn}, \text{can}}$, the existence of the observer-time law core

$$\text{LawCore}_{o, \zeta_{\text{Sch}_{\text{GR}}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}}),$$

or the existence of the law-admissible time representation $\text{TimeRep}_o^{\text{dyn}}$. Their generation requires satisfaction of the respective conditions at the comparison, representation, dynamical-description, and law-description stages.

Proof. By Theorem 8.28, a local proper-time representation is generated from worldline-local comparison data and generated representation coordinates.

By Definition 5.23, descent of the $\mathfrak{CDB}$ -derived canonical trajectory relation requires comparison-stage closure and the dynamical-description condition groups. Its right-uniqueness requires the dynamical-description-stage canonical-trajectory-function condition of Definition 5.26. By Definition 5.102 and Definition 5.108, the existence of the observer-time law core requires the simultaneous satisfaction of dynamical-description operability and the law condition groups. By Definition 5.63, the law-admissible time representation is generated as a component of the observer-time law core.

Consequently, even after a local proper-time representation has been generated, failure of a condition at the comparison, dynamical-description, or law-description stage prevents generation of a right-unique canonical trajectory relation, an observer-time law core, or a law-admissible time representation. $\square$

Connection to physical-time qualification. By Corollary 7.15, the generation structures corresponding to the time content used by a GR law-time variable must be established prior to law operations and result derivation. By Proposition 8.21, when time content and grounding structures are generated from nontemporal physical inputs prior to law operations and result derivation, the time variable satisfies physical-time qualification.

When time content enters a law operation or result derivation before the grounding structures required for its use, that time content is a primitive-time input under Definition 7.10. By Theorem 7.11, the time variable does not satisfy physical-time qualification. By Proposition 7.13, the simultaneous satisfaction of mathematical consistency, predictive performance, and empirical fit does not repair this loss of qualification.

Failures at the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages of the GR-scheme-restricted complete evaluation specification are classified by the failure-position tokens of Definition 6.10.

Proposition 8.52 (Inheritance of primitive-time dependence in GR). If local proper-time representations, coordinate time, or a global law-time representation are used in time-dependent components of the metric or connection, parametric representations of geodesics,

worldline comparisons, gravitational redshift, initial-value evolution, or cosmological time representations before the generation structures of those time contents are established prior to law operations and result derivation, that use becomes a primitive-time input. The time evolution of the metric and connection, geodesic solutions, the time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results that use them inherit primitive-time dependence, and the time content contained in each result does not satisfy physical-time qualification. General covariance, the mathematical consistency of the field equations, predictive performance, and empirical fit do not repair this loss of qualification.

Worldline-local comparison data and generated representation coordinates generate local proper-time representations. Global law-time use further requires exact operability, dynamical-description operability, law-description operability, transport preservation, loop consistency, quotient descent, and global gluing. Results containing time derivatives require differential-description operability.

Proof. When local proper time, coordinate time, or a global law-time representation is used in a law operation or result derivation before the generation structures are established, Definition 7.10 and Theorem 7.11 make that use a primitive-time input that does not satisfy physical-time qualification. By Corollary 7.12, dependence on that time content is inherited by the time evolution of the metric and connection, geodesic solutions, the time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results. By Proposition 7.13, general covariance, the mathematical consistency of the field equations, predictive performance, and empirical fit do not repair this loss of qualification.

By Theorem 8.28, a local proper-time representation is generated from worldline-local comparison data and generated representation coordinates. By Proposition 8.51, a local proper-time representation alone does not imply a right-unique canonical trajectory relation, an observer-time law core, or a law-admissible time representation; these require the conditions of subsequent stages to hold. By Theorem 5.135, differential covariance among real-valued time-order embeddings holds within the range in which differential-description operability holds. $\square$

8.6.2 Stage Separation of Failure Positions in GR

Generation of a local proper-time representation and generation of a global comparison quotient, a right-unique canonical trajectory relation, a law-admissible time representation, and a differential law form require conditions at different stages. If a condition at any stage fails after generation of a local proper-time representation, Definition 6.10 classifies each minimal failure position by the corresponding failure-position token.

Proposition 8.53 (Failure localization in the worked reduction for GR). Suppose that a local proper-time representation is generated in the GR scheme and that a condition for generating a law-admissible time representation fails. Each minimal failure position is then classified uniquely as a comparison, representation, continuous-approximation, dynamical-description, law-description, or differential failure-position token.

Proof. Failure of a comparison-stage condition is classified as a comparison failure by Definition 6.36. Failure of a representation-stage condition is classified as a representation failure by Definition 6.58. Failure of a continuous-approximation-stage condition is classified

as a continuous-approximation failure by Definition 6.77. Failure of a dynamical-description-stage condition is classified as a dynamical-description failure by Definition 6.104. Failure of a law-description-stage condition is classified as a law-description failure by Definition 6.110. Failure of a differential-consistency and law-form-preservation-stage condition is classified as a differential failure by Definition 6.137.

By Proposition 6.26, the failed-token set equals its set of minimal elements. By Definition 6.10, the corresponding failure-position token is assigned to each minimal failure position. Hence, if a condition for generating a law-admissible time representation fails after generation of a local proper-time representation, each minimal failure position is classified uniquely within one of the six failure series. $\square$

Corollary 8.54 (Separation of stages in the worked reduction for GR). In GR, local proper-time representations, the $\mathcal{CD}$ -derived canonical comparison quotient, the $\mathcal{CD}\mathcal{B}$ -derived canonical trajectory relation, law-admissible time representations, and differential law forms belong to different generation stages.

Proof. By Proposition 8.51, a local proper-time representation alone implies neither the right-uniqueness of the canonical trajectory relation nor the existence of an observer-time law core or law-admissible time representation. By Proposition 8.53, each minimal failure position among the conditions at each stage is classified by the corresponding failure-position token. Local proper-time representations, the $\mathcal{CD}$ -derived canonical comparison quotient, the $\mathcal{CD}\mathcal{B}$ -derived canonical trajectory relation, law-admissible time representations, and differential law forms therefore belong to different generation stages. $\square$

Corollary 8.55 (The two-stage misrecognition of the time-stretching account in GR). Gravitational clock differences in GR are comparison differences among generated law-admissible time representations corresponding to multiple local observers or worldlines. The time-stretching account presupposes time as an unconstructed entity prior to these generated representation differences and explains gravitational clock differences through stretching, slowing, or deformation of that time entity. By Proposition 8.23, this form of explanation is therefore the two-stage misrecognition of the time-stretching account.

Proof. By Proposition 8.51, a global law-time representation is not generated from a local proper-time representation alone. By Corollary 8.54, local proper-time representations, the $\mathcal{CD}$ -derived canonical comparison quotient, the $\mathcal{CD}\mathcal{B}$ -derived canonical trajectory relation, and law-admissible time representations belong to different generation stages. Gravitational clock differences in GR are therefore comparison differences among generated law-admissible time representations corresponding to multiple local observers or worldlines. By Proposition 8.23, in a form of explanation that places time as an unconstructed entity prior to these representation differences and redescribes subsequently generated comparison differences as changes in an attribute of that time entity, the first and third items of multistage misrecognition occur consecutively with direct dependence. This form of explanation is therefore the two-stage misrecognition of the time-stretching account. $\square$

Proposition 8.56 (Separation of the formal construction of general relativity from a bearer of time stretching). The formal construction of general relativity determines the metric, connection, worldlines, accumulation of local proper time, clock readings, clock rates, gravitational redshift, and their comparison relations. Gravitational conditions specify relations among $\mathcal{CD}\mathcal{B}\mathcal{D}$ physical carriers, worldlines, local clock processes, signals, transport, records, and generated time representations.

If the account that “time itself stretches, slows, or warps” is taken as a literal physical mechanism, that account asserts a bearer of time stretching that remains identical through the deformation while taking different states. For this assertion to hold as a physical mechanism, it requires the type of the bearer of time stretching, an identity condition before and after deformation, a state space, a deformation magnitude, a deformation law, a generation path from nontemporal physical inputs, coupling laws to the metric, connection, and $\mathcal{CDBD}$ physical carriers, and a measurement procedure independent of clock differences and gravitational redshift.

The formal results of general relativity established by Proposition 7.167 are comparison relations among clock readings, clock rates, proper times, gravitational redshift, and generated time representations. These formal results do not supply the type, identity condition, state space, deformation law, generation path, coupling laws, or independent measurement procedure of a bearer of time stretching. A literal time-stretching account therefore reintroduces an unconstructed bearer of change prior to the comparison differences generated by the formal construction and attributes those comparison differences to changes in an attribute of that bearer. This attribution is the two-stage misrecognition of Corollary 8.55.

No misrecognition occurs when differences in clock readings, clock rates, proper times, gravitational redshift, or generated time representations are described directly in terms of their objects and relations. Even when the term time stretch is used, no misrecognition occurs in an explicit metaphor if the source text states, within the same explanatory scope, its metaphorical status, that it does not designate literal deformation of time, and its actual referent.

When the term time stretch is used without stating its metaphorical status and actual referent, interpreting the term retrospectively as shorthand leaves a one-stage misrecognition that transfers a comparison difference to an attribute of time. In a literal account that places an unconstructed time entity in advance, this one stage is joined by pre-generation placement and constitutes the two-stage misrecognition of Corollary 8.55.

Proof. By Theorem 8.28 and Proposition 8.51, local clock processes, records, object-state differences, information coupling, and comparison generate local proper-time representations, while a global law-time representation requires comparison, transport preservation, loop consistency, quotient descent, and global gluing to hold. By Proposition 7.167, the formal results of general relativity supply clock readings, clock rates, proper times, gravitational redshift, and their comparison relations, but do not supply an independent state and evolution law for a bearer of time stretching.

Literal stretching, slowing, or warping requires an object that changes, its states before and after the change, and a condition for determining both states as states of the same object. Treating that change as the cause of a clock difference or gravitational redshift additionally requires coupling laws from the bearer of change to the metric, connection, $\mathcal{CDBD}$ physical carriers, and clock processes, together with a measurement procedure independent of clock differences and gravitational redshift. The formal results of general relativity contain none of these constructions.

A literal time-stretching account therefore attributes a comparison difference among generated representations to an unconstructed bearer of time stretching. By Corollary 8.55, this attribution is a two-stage misrecognition. No misrecognition occurs when comparison differences on physical carriers are described directly or when metaphorical status and the actual referent are stated. A bare expression of time stretching constitutes a one-stage misrecognition that transfers a comparison difference to an attribute of time even under

an interpretation that does not assert an independent time entity. $\square$

Proposition 8.57 (Generation structure of gravitational clock differences). For the $\mathcal{CDB}$ physical carriers of Definition 3.1 and Definition 3.2, suppose that the conditions of Theorem 8.28 hold. As the metric and connection, gravitational conditions specify the worldlines to which the $\mathcal{CDB}$ physical carriers belong and the relations among local clock processes, signal propagation, transport, and comparison. The clock processes, records, object-state differences, and information coupling obtained from the $\mathcal{CDB}$ physical carriers determine generated local proper-time representations.

Accumulation of clock processes belonging to different worldlines or local settings, clock readings, and signal-mediated comparisons determine gravitational clock differences, clock-rate differences, and gravitational redshift. Within a domain in which the subsequent conditions of Proposition 8.51 hold, generation from local proper-time representations to a global law-time representation holds, and these results can be used within the same validity domain. The physical grounds for these differences lie in the dependency chain of $\mathcal{CDB}$ physical carriers, worldlines, the metric, the connection, signals, transport, comparison, and generated time representations.

This dependency chain does not contain primitive time as the object of gravitational action. An account that attributes results generated in relation to gravitational conditions to stretching, slowing, or deformation of primitive time itself placed prior to generation constitutes the two-stage misrecognition of Corollary 8.55.

Proof. Definition 3.1 and Definition 3.2 map each role in $\mathcal{CDB}$ to a physical-carrier realization. By Theorem 8.28, the establishment of worldline-local clock processes, records, object-state differences, information coupling, and comparison generates local proper-time representations. By Proposition 8.51, generation of a right-unique $\mathcal{CB}$ -derived canonical trajectory relation, observer-time law core, and law-admissible time representation requires, in addition to local proper-time representations, satisfaction of the conditions for comparison, transport preservation, loop consistency, quotient descent, and global gluing.

Accumulation of clock processes according to worldlines, the metric, and the connection, together with clock readings and signal-mediated transport and comparison, therefore supplies gravitational clock differences, clock-rate differences, and gravitational redshift as comparison results on generated time representations. In this generation order, gravitational conditions act on relations among $\mathcal{CDB}$ physical carriers, worldlines, the metric, the connection, signals, transport, and comparison. By Corollary 8.55, an account that places primitive time prior to this comparison result and attributes the comparison result to a change in an attribute of primitive time itself is a two-stage misrecognition. $\square$

Proposition 8.58 (Safe reuse of general-relativistic results through generated time and failure classification). Within a domain in which the conditions of Theorem 8.28 hold, generated local proper-time representations can be used in worldline-local comparisons. Within a domain in which the conditions at the comparison, representation, dynamical-description, and law-description stages required by Proposition 8.51 hold, the right-unique $\mathcal{CB}$ -derived canonical trajectory relation, observer-time law core, and law-admissible time representation can be used. The time evolution of the metric and connection, geodesic solutions, the time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results that use them can be used within a domain in which their respective theory-specific grounding conditions and the generation conditions for the time content on which they depend hold.

Each minimal failure position classified by Proposition 8.53 determines the GR construction component that must be repaired first or the condition whose validity domain must be restricted. A failure at the comparison stage requires repair of local clock-side receptions, record-side receptions, comparison rules, transport preservation, loop consistency, quotient descent, or global gluing. A failure at the representation or continuous-approximation stage requires repair of the time representation, comparison domain, or scale domain of approximate validity. A failure at the dynamical-description, law-representation, or differential-consistency and law-form-preservation stage requires repair of the canonical trajectory relation, dynamical-description evidence datum, law-evaluation datum, or differential evaluation specification. When repair holds, the global law-time representation, time evolution of the metric and connection, geodesic solutions, time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results are regenerated or recalculated in dependency order from the repaired inputs. If the relevant stage condition does not hold throughout the entire domain, the domain of use of the results is restricted to the domain in which that condition holds.

Corollary 8.55 separates gravitational clock differences among generated representations from the two-stage misrecognition that places an unconstructed time entity prior to those comparison differences and explains the differences through stretching, slowing, or deformation of that entity. The grounds for using general covariance, the field equations, geodesic solutions, gravitational redshift, and gravitational clock differences are supplied by generated time representations, theory-specific inputs, and the grounding conditions at each stage.

When an error source in local clock comparisons, transport rules, global gluing, gravitational redshift, geodesic calculations, initial-value evolution, continuous approximations, or differential evaluations is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the regenerated or recalculated downstream results. The decision to use general-relativistic results therefore separates into retention within the validity domain and repair from each minimal failure position or restriction of the validity domain.

Proof. By Theorem 8.28, worldline-local comparison data and generated representation coordinates generate local proper-time representations. By Proposition 8.51, generation of a right-unique $\mathcal{CDB}$ -derived canonical trajectory relation, observer-time law core, and law-admissible time representation requires, in addition to local proper-time representations, satisfaction of the conditions at subsequent stages. By Proposition 8.21, when the time content and grounding structures on which each result depends are generated prior to law operations and result derivation, that time input satisfies physical-time qualification. The time evolution of the metric and connection, geodesic solutions, the time ordering of worldlines, gravitational clock differences, gravitational redshift, initial-value evolution, cosmological time relations, and law results that satisfy the grounding conditions at each stage and the theory-specific conditions therefore use time inputs that have physical-time qualification.

By Proposition 8.53 and Definition 6.17, conditions and results downstream of each minimal failure position depend on upstream conditions. When an input or rule belonging to each minimal failure position can be repaired, downstream results are regenerated or recalculated from the repaired inputs in dependency order. If the relevant stage condition does not hold throughout the entire domain, the domain of use is restricted to the domain in which that condition holds, and results generated within that domain are used.

Corollary 8.55 separates gravitational clock differences among generated law-admissible

time representations from the two-stage misrecognition that explains those comparison differences through changes in an attribute of an unconstructed time entity. This connects the grounds for general covariance, the field equations, geodesic solutions, gravitational redshift, and gravitational clock differences to generated time representations, theory-specific inputs, and the conditions at each stage.

When the source of error belongs to local clock comparisons, transport rules, global gluing, gravitational redshift, geodesic calculations, initial-value evolution, continuous approximations, or differential evaluations, downstream results can be regenerated or recalculated from inputs in which that source has been repaired, and an accuracy evaluation can be applied to the recalculated result. It follows that retention of general-relativistic results within the validity domain, the order of repair from each minimal failure position or restriction of the validity domain, and conditional evaluation of accuracy hold. $\square$

8.7 Three-Stage Misrecognition in the Biological Application of Relativistic Explanations

Proposition 8.59 (Relativistic three-stage misrecognition with biological application). Direct measurements of clocks, oscillators, electromagnetic radiation, or particle decay belonging to the first stage of Definition 7.183 yield comparison differences on physical carriers. An account that attributes the preceding comparison differences to slowing, stretching, or action of time itself independently of physical carriers belongs to the second stage. An account that further applies that temporal attribute to biological processes and treats the direct measurements at the first stage as direct demonstration of biological processes belongs to the third stage.

When the second-stage account presupposes an unconstructed time entity prior to the comparison differences and attributes those differences to changes in an attribute of the time entity as a literal physical mechanism, it constitutes the two-stage misrecognition of the time-stretching account in Proposition 8.23. With this two-stage misrecognition as antecedent, the transition that treats direct measurements at the first stage as direct demonstration at the third stage falls under Item 4 of Definition 8.19. An account that connects these three evidential stages consecutively through this dependency is therefore a relativistic three-stage misrecognition with biological application.

By Proposition 7.193, the number of direct demonstrations of biological processes among the seven references containing biological-application statements is 0/7, and the number of direct demonstrations of biological processes among the 72 audited references is 0/72. The references by Feynman–Leighton–Sands, Feynman 1995, Misner–Thorne–Wheeler, NIST, Rindler, Taylor–Wheeler, and Tong recorded in Proposition 7.185 contain statements that apply accounts of clocks, proper time, or particle processes to pulse rate, thought, cancer development, aging of the human body, the ages of twins, human lifespan, biological clocks, or the human heart. Among them, the Feynman 1995 reference generalizes from nuclear transitions and frequency shifts to all physical processes and then immediately applies that generalization to human life and aging, making the dependency chain explicit within the same reference. These biological-application statements are determined to be theoretical predictions or explanatory applications in accordance with Corollary 7.194.

The validity of a theoretical prediction concerning biological application is determined by coupling laws between the physical carriers that constitute biological processes and relativistic conditions. Direct demonstration of biological processes requires a biological object, an experimental configuration that separates the relativistic condition, a mea-

surement procedure for the biological process, and corresponding measurement results. An account that states that “time itself slows, so organisms age differently, and this has been confirmed by clock or particle experiments” transfers a comparison difference on a first-stage physical carrier to a second-stage temporal attribute and then transfers that temporal attribute to a third-stage biological phenomenon. This consecutive transfer is a relativistic three-stage misrecognition with biological application that twice replaces the object of measurement, the bearer of physical action, and the evidence type.

Proof. Proposition 7.185 separates direct measurement of physical carriers, attribution of a property to time itself, and application to biological processes into distinct evidential stages. Proposition 7.184 gives, as necessary conditions for direct demonstration of a biological application, inclusion of the biological process among the experimental objects, an experimental configuration that separates the relativistic condition, a measurement procedure for the biological process, and corresponding measurement results.

An account that presupposes an unconstructed time entity prior to the comparison differences and attributes the first-stage comparison differences to changes in an attribute of that entity is the two-stage misrecognition of Proposition 8.23. An account that extends the experimental object domain of the first stage to the biological processes of the third stage and identifies direct demonstration at the two stages satisfies Item 4 of Definition 8.19. The counts in Proposition 7.193 show that, for both the seven references and the 72 audited references, the experimental configurations and measurement results required for direct demonstration at the third stage have not been obtained. An account that connects the three evidential stages consecutively through this dependency is therefore a relativistic three-stage misrecognition with biological application. □

In this consecutive transfer, the object under discussion changes from the clock to time and then to a biological process while retaining the evidence obtained from the clock.

The target changes while the evidence is carried forward

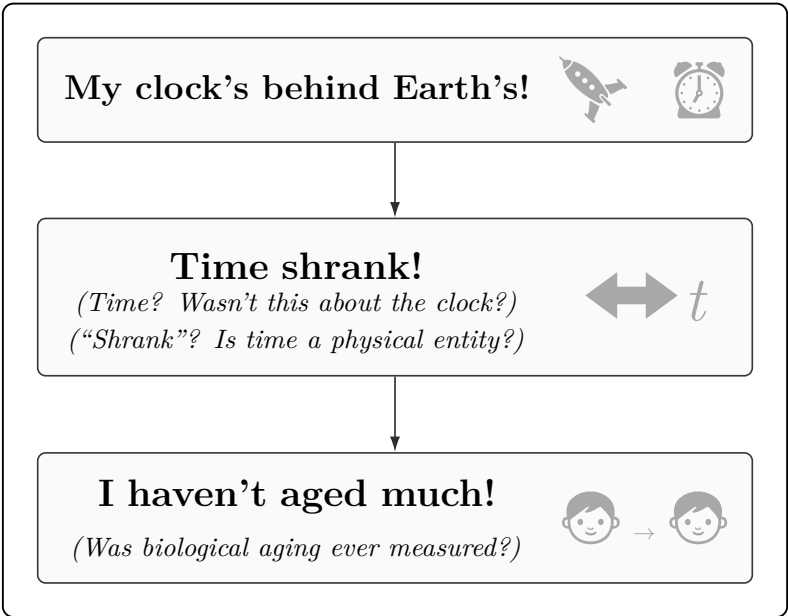


Figure 22: The object under discussion shifts among a clock, time, and a biological process; two questions identify the evidence boundary at each transition.

8.8 Worked Reduction for QG

In the quantum-reconstruction scheme, law-admissible time representations and observer-time law cores are generated from records, branch sensitivity, object-state differences, observer-side information coupling, comparison gluing, dynamical-description operability, and law-description operability.

A quantum-gravitational reconstruction that does not include external time among its primitive inputs has constraint systems, canonical gravity, and Wheeler–DeWitt-type descriptions as its background[19, 20, 21]. The problem of time, internal clocks, and relational observables in quantum gravity are representative research subjects concerning the generation conditions for time representations and their qualification for use in laws[8, 9, 10, 11].

Eliminating external time does not generate time content, record order, comparison order, observer-side information coupling, or qualification for use in laws. Even when an internal degree of freedom is selected as a clock, primitive-time input remains merely replaced by the internal clock unless time content is generated from differences, records, comparisons, and observer-side information coupling involving that degree of freedom. If primitive-time input is used to derive results concerning constraint equations, relational change, observables, initial boundaries, the arrow of time, measurement records, or black-hole information, those results inherit dependence on time content that lacks physical-time qualification. Mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification.

On the same dependency chain, the QG scheme generates clock-candidate-side materials, record-side retention, object-state-side differences, observer-side information coupling, comparison closure, the canonical observer-time quotient, the canonical trajectory relation, law-admissible time representations, and law-output determinacy, and determines the qualification of the time content used by each result in quantum gravity.

8.8.1 From QG Inputs to an Observer-Time Law Core

The inputs to the quantum-reconstruction scheme consist of clock-side or clock-candidate-side tokens, record-side retention, branch-sensitive record structures, object-state-side differences, observer-side information-coupling positions, comparison-gluing conditions, and source-side ternary relations at the dynamical-description stage.

Define this input by

$$\mathcal{X}_{\text{Sch}_{\text{QG}}} := \left(\mathcal{I}_{\text{Sch}_{\text{QG}}}^{\text{raw}}, \text{Dat}_{\text{Sch}_{\text{QG}}}^{\text{src}} \right)$$

. The raw operational input $\mathcal{I}_{\text{Sch}_{\text{QG}}}^{\text{raw}}$ contains no external time, and the QG-side source-realization datum $\text{Dat}_{\text{Sch}_{\text{QG}}}^{\text{src}}$ has as components the comparison-side arrival maps, recovery maps, record retention, comparison gluing, direct dynamical-description arrival maps, source-side pair relations, and source-side ternary relations.

Let the observer-side information-coupling position be o , and let the single-object specification index be $\zeta_{\text{Sch}_{\text{QG}}}$. Use the QG-scheme-restricted complete evaluation specification $\text{Ev}_{\text{Sch}_{\text{QG}}}$ of Definition 8.5.

The generation sequence in the worked reduction for QG is

$$\begin{aligned}
& (\mathcal{X}_{\text{Sch}_{\text{QG}}} ; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{QG}}}) \\
& \longmapsto \text{Qu}^{\text{cmp}, \text{can}} (\mathcal{X}_{\text{Sch}_{\text{QG}}} ; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{QG}}}) \\
& \longmapsto \text{Dat}_{o, \zeta \text{Sch}_{\text{QG}}}^{\text{dyn}, 0} \\
& \longmapsto \text{DynWit}_{o, \zeta \text{Sch}_{\text{QG}}} [\text{RepEmb}_o^{\text{dyn}, \text{time}}] \\
& \longmapsto \text{TimeRep}_o^{\text{dyn}} \\
& \longmapsto \text{LawCore}_{o, \zeta \text{Sch}_{\text{QG}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}}) .
\end{aligned}$$

The comparison-stage objects require exact operability; the dynamical-description objects and the law-admissible time representation require dynamical-description operability; and the observer-time law core and qualification for use in laws require law-description operability.

Use the real-valued time-order embedding selected in Definition 5.59. Suppose that the law-evaluation datum satisfies the equality of evaluation specifications under Proposition 5.86.

Proposition 8.60 (Law admissibility in the worked reduction for QG). Suppose that, for the QG-scheme-restricted complete evaluation specification,

$$\text{LawOperative} \left[\text{LawEvalDat}_{o, \zeta \text{Sch}_{\text{QG}}} \right]$$

holds. Then the law-admissible time representation $\text{TimeRep}_o^{\text{dyn}}$ and the observer-time law core

$$\text{LawCore}_{o, \zeta \text{Sch}_{\text{QG}}}^{\text{obs}, \text{time}} (\text{RepEmb}_o^{\text{dyn}, \text{time}})$$

are generated. For any

$$T_0 \in \text{Qu}_{\mathcal{C}, o}^{\text{dyn}, \text{can}}$$

the law-time value is

$$t_0 := \text{TimeRep}_o^{\text{dyn}} (T_0)$$

.

Proof. Definition 5.102 and the equality of evaluation specifications imply dynamical-description operability for the QG-scheme-restricted complete evaluation specification. By Theorem 8.31, the canonical observer-time quotient, canonical object-state quotient, canonical trajectory relation, graph trajectory, canonical family of observable functions, and law-admissible time representation are generated. Definition 5.108 generates the observer-time law core. The equality for the law-time value follows from Definition 5.63. $\square$

Proposition 8.61 (Generation condition for a time representation in QG reconstruction). When the dynamical-description-operability condition of Theorem 8.31 holds for the QG-scheme-restricted complete evaluation specification, the law-admissible time representation $\text{TimeRep}_o^{\text{dyn}}$ is generated. If, in addition,

$$\text{LawOperative} \left[\text{LawEvalDat}_{o, \zeta \text{Sch}_{\text{QG}}} \right]$$

holds, this time representation satisfies qualification for use in laws and the observer-time law core is generated.

Proof. By Definition 5.70 and Theorem 8.31, satisfaction of dynamical-description operability generates the canonical observer-time quotient, canonical object-state quotient, canonical trajectory relation, graph trajectory, canonical family of observable functions, and law-admissible time representation. By Definition 5.102, law-description operability is the simultaneous satisfaction of dynamical-description operability and the law condition groups. By Definition 5.108, this condition generates the observer-time law core. By Corollary 7.15, generation of law-time content precedes law operations and result derivation. The time representation generated at the dynamical-description stage therefore satisfies qualification for use in laws through the satisfaction of law-description operability. $\square$

Connection to the audit of law-time use. Law-time use in a quantum-gravitational reconstruction is subject to the audit of physical-time qualification in Chapter 7. When every generation structure required by the time content used by a law-time use ϑ is generated from nontemporal physical inputs and is established prior to law operations and result derivation, that time content satisfies physical-time qualification. If any generation stage for the evaluation specification fails, simultaneous satisfaction of all stages for that complete evaluation specification is not obtained. When time content is used in a law operation or result derivation prior to its generation structures, that time content is a primitive-time input under Definition 7.10. By Theorem 7.11, this time use does not satisfy physical-time qualification. By Proposition 7.13, mathematical consistency, predictive performance, and empirical fit do not repair this loss of qualification.

8.8.2 Law Admissibility without External Time

The problem of time in quantum gravity arises not only from the tension between law descriptions that use external time and theoretical constructions that do not take background spacetime as a primitive premise, but also from assigning temporal roles to constraint equations, internal clocks, or relational observables without generating temporal content after external time has been removed. Removal of external time eliminates one form of primitive-time input, but it does not generate clock-candidate-side differences, record retention, comparison order, observer-side information coupling, the canonical observer-time quotient, or a law-admissible time representation. Even when an internal degree of freedom is selected as a clock, that internal clock bears ungenerated temporal content unless it satisfies the same generation pathway.

When ungenerated temporal content enters a constraint equation or law operation, the time variable fails physical-time qualification. Results concerning relational change, observables, initial boundaries, the arrow of time, measurement records, and black-hole information that are derived from ungenerated temporal content inherit this qualification failure. The problem of time in quantum gravity therefore cannot be resolved solely by the presence or absence of external time or by the choice of a clock variable; it requires simultaneous determination of the generation of temporal content, qualification for use within laws, and the inheritance of dependence by results.

The QG scheme requires clock-candidate-side materials, record-side retention, object-state-side differences, observer-side information coupling, comparison closure, the canonical observer-time quotient, the canonical trajectory relation, a law-admissible time representation, and law-output determinacy to hold along the same dependence chain. This dependence chain generates the time variable, confers qualification for use within laws, and excludes the inheritance of primitive-time dependence by results.

Proposition 8.62 (Relativity, relationality, and non-primitivity do not imply nonexistence). The dependence of an object on a frame, observer, state, local region, or relation to another object supplies conditions for that object's existence. The absence of a preferred representative expresses nonuniqueness of choice; being nonabsolute expresses the failure of frame-independent, stand-alone entityhood; and being nonprimitive expresses a dependence position in which the object is generated from other physical inputs. Each condition preserves the possibility that the object exists as a relational or generated object and does not imply its nonexistence.

A claim that time does not exist must specify the type and domain of the temporal content being denied and establish that no object of that type performing a temporal role exists within the theory. If instants, relations among configurations, an order of change, histories, evolution, or clock comparisons perform temporal roles, declining to adopt an independent absolute time entity and denying the existence of all temporal structure are different propositions.

Proof. Relational dependence, nonuniqueness, nonabsoluteness, and generative dependence specify conditions for an object or its relations to other objects. An existence denial states that, within a specified type and domain, no object of that type exists. The latter therefore does not follow from the former.

Definition 7.10 classifies temporal content used before generation as primitive-time input. Corollary 7.15 requires temporal content to be generated from nontemporal physical inputs before it is used within a law. In this generation order, temporal content exists as a generated object rather than a primitive input. Non-primitivity therefore does not imply the nonexistence of temporal structure.

Time representations dependent on multiple observers, states, or local regions also exist as relational or local temporal structures when comparison rules, gluing conditions, and domains of validity are supplied. Failure of a preferred representative or global representation does not imply the nonexistence of local or relational temporal structure. $\square$

Proposition 8.63 (Formal obligations on claims of time creation and time nonexistence). A theory that literally claims the creation of time must exhibit a generation path from nontemporal physical inputs through object-state differences that presuppose no temporal order, records that retain those differences, comparison rules among the records, observer-side information coupling, temporal order, a time representation, and qualification for use within laws; it must also exclude primitive-time roots from every temporal content it uses.

A theory that literally claims the nonexistence of time must, in accordance with Proposition 8.62, separate claims of relativity, relationality, non-preference, nonabsoluteness, or non-primitivity from an existence denial. It must then delimit the temporal content, temporal order, time representations, and law time that it denies; supply a formal construction in their place; and establish that no temporal dependence root remains in the replacement construction and that subsequent results do not depend on any such root. If the replacement construction uses “instant,” “change,” “history,” “before and after,” or “evolution,” the grounds for each must be generated from nontemporal physical inputs.

These formal obligations apply to every theory that advances the creation, emergence, recovery, or nonexistence of time, and they apply with equal force to the generation of time in this paper.

Proof. By Corollary 7.15, generation of temporal content precedes law operations and result derivation. By Definition 7.10, temporal content used before its grounding struc-

ture is primitive-time input. By Theorem 7.11, primitive-time input fails physical-time qualification. By Corollary 7.12, that dependence is inherited by subsequent results.

A claim of time creation therefore requires a generation path from nontemporal physical inputs to the temporal content used, together with exclusion of primitive-time roots. A claim of time nonexistence requires a delimited object of denial, a replacement construction, and exclusion of temporal dependence roots from that construction. Renaming, eliminating a variable, substituting a relation, or omitting temporal vocabulary does not discharge these formal obligations. $\square$

Proposition 8.64 (Interpretation-path-specific determination of statements of time creation and time nonexistence). The Barbour 1994 I source presented in Subsection 7.3.14 contains the following statement.

Original English:

change creates time

Original English:

It is argued that time as such does not exist but that instants, defined as complete relative configurations of the universe, do.

The Barbour 1994 II source contains the following statement.

Original English:

time is truly non-existent in the kinematic foundations of both theories

The three statements are determined separately for each statement and each admissible interpretation, along the interpretation-specific evidence-comparison path of Definition 7.117.

If “change creates time” is interpreted as the construction, within the explicitly stated domain of application, of configurational differences, an ordering, a scale, a clock correspondence, or an ephemeris-type parameter, it directly describes the corresponding formal construction. The number of crossings of evidential scope is then zero, and no multi-stage misrecognition arises. This interpretation does not claim physical generation of every temporal content used in the source.

If the same statement is interpreted as complete generation of physical time, the operation that uses a realized history, path parameter, or existing temporal content before its generation falls under item 1 of Definition 8.19. The further operation that identifies a restricted configurational construction with physical time possessing generation of every temporal content, exclusion of primitive-time roots, and qualification for use within laws falls under item 2. An interpretive path that connects these two operations by direct dependence constitutes a two-stage misrecognition.

If “time as such does not exist” and “time is truly non-existent in the kinematic foundations” are interpreted as restricted to declining to adopt absolute external time or to the kinematic priority of relative configurations, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises. An interpretation that extends these statements to the nonexistence of physical time in general transfers the force of an existence denial from a restricted object and domain of application to a different object and domain. It therefore constitutes a one-stage misrecognition under item 4 of the same definition. The total of 39 temporal contents and the time-dependent transfers present in the Barbour 1994 I and II sources are incompatible with this extended interpretation.

Proof. Subsection 7.3.11 and Remark 7.159 establish the original wording, source attribution, and context of the three statements. Theorem 7.190 establishes that all 18 temporal contents in Barbour 1994 I and all 21 temporal contents in Barbour 1994 II are primitive-time roots. The source-specific Propositions 8.416 and 8.427 partition the 25 results in Barbour 1994 I into 9 results with evidence-only reach and 16 results that depend on primitive-time roots, and the 26 results in Barbour 1994 II into 8 results with evidence-only reach and 18 results that depend on primitive-time roots. Neither source has generated results or results that remain unreachable after the roots are added. Establishing a result with evidence-only reach does not imply the generation of temporal content.

The interpretation restricted to a configurational construction remains within the evidential scope of the formal result and crosses no evidential boundary. The interpretation as complete generation of physical time directly connects pre-generation use of primitive-time roots to identification of a partial construction with complete generation. The interpretation as the nonexistence of physical time in general transfers an existence denial from the restricted domain of absolute external time or kinematic foundations to physical time in general. The respective numbers of stages are therefore zero, two, and one. $\square$

Proposition 8.65 (Interpretation-path-specific determination of generation obligations in thermal-time statements). The Connes–Rovelli 1994 source presented in Subsection 7.3.14 contains the following statements.

Original English:

there is no preferred time flow

Original English:

thermal time ... defines the physical time

If “there is no preferred time flow” is interpreted as restricted to the absence of an externally specified, state-independent preferred flow, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises. An interpretation that extends the same statement to the absence of every physical time flow transfers the absence of a restricted flow to a different object and domain of application. It therefore constitutes a one-stage misrecognition under item 4 of Definition 8.19.

If “thermal time ... defines the physical time” is interpreted as restricted to the mathematical construction in which a thermal state selects a modular automorphism flow, or to a postulate assigning a physical-time interpretation to that flow, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises. This interpretation does not claim a theorem-level derivation of physical-time qualification.

An interpretation that identifies the selected state-dependent flow with physical time possessing every generation condition and qualification for use within laws transfers a partial construction to the establishment of a subsequent stage. It therefore constitutes a one-stage misrecognition under item 2 of the same definition. An interpretation that further claims complete generation of physical time from nontemporal physical inputs, although the construction uses a real parameter, known Hamiltonian time, proper time, geometric time, or FRW time as a pre-generation input, directly connects items 1 and 2 and constitutes a two-stage misrecognition.

Proof. Subsection 7.3.11 and Remark 7.159 establish the original wording, source attribution, and context of the two statements. Proposition 7.171 establishes that every temporal content in the source is a primitive-time root. Theorem 7.190 establishes that the set

of audit-generated temporal contents is empty. The source-specific Proposition 8.439 establishes that the 37 results in the source consist of 16 results with evidence-only reach and 21 results that depend on primitive-time roots.

By Proposition 7.137 and Corollary 7.138, selecting a time flow from a state does not supply the generation history, comparison structure, or conditions of validity of the input temporal content. By Proposition 7.172, the scope of the formal result concerning thermal time does not coincide with that of complete time generation.

The restricted flow selection and interpretive postulate remain within the evidential scope of their formal result. The path that transfers a partial flow to complete establishment of physical time has one stage under item 2. The path that treats a construction using pre-generation time roots as complete generation from nontemporal inputs has two stages under items 1 and 2. The path that extends the absence claim to every physical time flow has one stage under item 4. The stage count is therefore zero, one, or two according to the admissible interpretation. $\square$

Proposition 8.66 (Interpretation-path-specific determination of time-emergence statements through bipartite separation). The Kiefer 2009 source presented in Subsection 7.3.14 contains the following statements.

Original English:

Time is absolute in standard quantum theory and dynamical in general relativity.

Original English:

Time emerges from the separation into two different subsystems

If the second statement purports to assert that separation into two subsystems generates temporal content usable by physical laws, the formal obligations concerning time emergence in Proposition 8.63 arise.

For the formal audit of this source, in the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [35]; \emptyset, \{5, 6, 7, 8, 13\}, \{1, 2, 3, 4, 9, 10, 11, 12, 14, 15\}, \emptyset).$$

If the first statement is interpreted as restricted to the difference in formal role between an external background parameter in standard quantum theory and a dynamical spacetime variable in general relativity, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises.

If the second statement is interpreted as restricted to constructing a semiclassical time parameter or an internal-clock candidate through separation into two subsystems, it directly describes a source-specific approximate construction, and the number of crossings of evidential scope is likewise zero. If the same statement is interpreted as complete generation of physical time and the subsystem parameter is identified with generation of every temporal content, qualification for use within laws, and exclusion of primitive-time roots, it constitutes a one-stage misrecognition under item 2 of Definition 8.19. An interpretation that further treats a semiclassical construction using pre-generation time roots as complete generation from nontemporal physical inputs directly connects items 1 and 2 and constitutes a two-stage misrecognition.

An interpretation that transfers the absence of external time from the fundamental equation to the nonexistence of physical time in general changes the object and domain of

application. It therefore constitutes a one-stage misrecognition under item 4 of the same definition.

Proof. Proposition 7.169 establishes the original wording, source attribution, and original locations of the two statements, and separates the contrast between temporal roles in standard quantum theory and general relativity, bipartite time, complete time generation, exclusion of primitive-time roots, and qualification for use within laws into distinct evidential scopes. By Theorem 7.190, the set of audit-generated temporal contents is empty and every temporal content in the source is a primitive-time root. For the Kiefer 2009 reference specifically, Proposition 8.650 establishes that the 15 results consist of 5 results with evidence-only reach and 10 results that depend on primitive-time roots. The four-way partition in Proposition 7.49 then implies that the audit-generated result set is empty and that all 15 results lie beyond reach from the generation-initial set.

Proposition 8.63 requires, for a literal claim of time emergence, a generation path from nontemporal physical inputs through temporal content, temporal order, a time representation, and qualification for use within laws, together with exclusion of primitive-time roots from every temporal content. Because the source contains no audit-generated temporal content and every result depends on primitive-time roots, these formal obligations are not fulfilled.

The interpretation restricted to a semiclassical time parameter or internal-clock candidate remains within the evidential scope of a partial formal result. The path that transfers the partial construction to complete establishment of physical time has one stage under item 2; the path that connects use of pre-generation time roots to identification with complete generation has two stages under items 1 and 2; and the path that transfers the absence of external time to the nonexistence of physical time in general has one stage under item 4. The stage count is therefore zero, one, or two according to the admissible interpretation. $\square$

Proposition 8.67 (Interpretation-path-specific determination of generation qualification in the statement that time is fundamental). The Smolin 2018/2020 source presented in Subsection 7.3.14 contains the following statement.

Original English:

I take the opposite view, that time is fundamental.

If this statement purports to license the use of temporal content as a ground of physical laws or as a law input prior to its generation, that temporal content is a primitive-time input under Definition 7.10. By Theorem 7.11, that temporal content fails physical-time qualification.

For the complete determination and result classification of this source, in the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R &= (\emptyset, [27]; \\ &\quad \emptyset, \{2, 3, 4, 6, 7, 21, 22\}; \\ &\quad \{1, 5\} \cup \llbracket 8, 20 \rrbracket \cup \llbracket 23, 28 \rrbracket, \emptyset). \end{aligned}$$

The local-reach determination gives

$$\begin{aligned} \text{TimeContent}_R^{\text{audit,evid}} &= \emptyset, \\ \left| \text{TimeContent}_R \setminus \text{TimeContent}_R^{\text{audit,evid}} \right| &= 27. \end{aligned}$$

There are 127 obligation atoms.

If “time is fundamental” is interpreted as a classification asserting the theoretical priority or irreducibility of time, the number of crossings of evidential scope is zero, and no multi-stage misrecognition arises. This classification does not assert generation-prior qualification of temporal content.

If the same statement is interpreted as approving the use of temporal content before its generation as a ground of physical laws or as a law input, that temporal content becomes a primitive-time input. This interpretation constitutes a one-stage misrecognition under item 1 of Definition 8.19. Neither the local-reach determination nor the complete determination yields any generated temporal content. All 27 source-specific temporal contents remain primitive-time roots, and 21 of the 28 results depend on primitive-time roots.

Proof. Proposition 7.169 separates the fundamentality of time, generation of all source-specific temporal contents, exclusion of primitive-time roots, and qualification for use within laws into distinct evidential scopes. Proposition 7.171 establishes the original wording, source attribution, and original location of the statement, that the set of strictly generated temporal contents is empty, and that all 27 source-specific temporal contents are primitive-time roots. No temporal content is reached under the local-reach determination, so all 27 remain primitive-time roots. Proposition 7.136 establishes that there are 127 obligation atoms and that 21 of the 28 results depend on primitive-time roots. The source-specific result classification in Proposition 8.737, together with the four-way partition in Proposition 7.49, establishes that the audit-generated result set is empty and that all 28 results, comprising 7 results with evidence-only reach and 21 results that depend on primitive-time roots, lie beyond reach from the generation-initial set.

By Corollary 7.15, generation of temporal content precedes law operations and result derivation. For this source, neither the local-reach determination nor the complete determination yields any generated temporal content. None of the 27 temporal contents is therefore generated, and the dependence of 21 of the 28 results on primitive-time roots remains.

The interpretation restricted to theoretical priority or irreducibility remains within the evidential scope of a classification and has zero stages. The interpretation that licenses use within laws before generation places a primitive-time input among the law inputs and therefore constitutes a one-stage misrecognition under item 1. Along either path, the qualification of temporal content used in physical laws is determined by its generation path from nontemporal physical inputs. $\square$

Proposition 8.68 (The problem of time as law admissibility). Even when external time is absent from the raw operative input to the QG-scheme-restricted complete evaluation specification, the establishment condition for the law-description stage is

$$\text{LawOperative} \left[\text{LawEvalDat}_{o, \zeta_{\text{Sch}_{\text{QG}}}} \right]$$

. If temporal content is used in a constraint equation, law operation, or result derivation before the generated structures required for that use are established, that content is a primitive-time input, and results depending on it inherit the qualification failure.

Proof. Exclusion of external time from the input is a component condition on the raw operative input. Law-description operability under Definition 5.102 requires the joint satisfaction of dynamical-description operability and the law condition group. Removing

external time does not supply the generation conditions constituting law-description operability.

When law-description operability holds, Proposition 8.60 generates a law-admissible time representation, observer-time law core, and law-time value. By Corollary 7.15, generation of law-time content precedes law operations and result derivation.

If temporal content is used before the generated structures required for that use are established, Definition 7.10 and Theorem 7.11 make that content a primitive-time input that fails physical-time qualification. By Corollary 7.12, dependence on that temporal content is inherited by the results. By Proposition 7.13, mathematical consistency, predictive performance, and empirical adequacy do not repair this qualification failure. $\square$

Proposition 8.69 (Failure localization in the worked QG reduction). If a generation condition in the QG-scheme-restricted complete evaluation specification fails, each minimal failure position is classified within the six series of comparison, representation, continuous-approximation, dynamical-description, law-description, or differential failure. The QG-specific correspondences are as follows.

- (1) When a comparison-stage or representation-stage condition concerning record sequences, object-state identification, or comparison gluing is formed and evaluable, its failure is classified by a failure-position token of the corresponding stage.
- (2) Where evaluation readiness holds for condition 3, excess $\mathfrak{D}$ -record cell width is determined by

$$\text{FA}_3 [\text{Ev}_{\text{Sch}_{\text{QG}}}; \lambda]$$

- (3) Whether a failure occurs at the dynamical-description stage is determined by

$$\bigvee_{i \in I_{\text{dyn}}^{\text{fail}}} \text{FDyn}_i [\text{Ev}_{\text{Sch}_{\text{QG}}}]$$

- (4) Whether a failure occurs at the law-description stage is determined by

$$\bigvee_{i \in I_{\text{law}}^{\text{fail}}} \text{FL}_i [\text{Ev}_{\text{Sch}_{\text{QG}}}]$$

- (5) Whether a failure occurs at the differential stage is determined by

$$\bigvee_{i=1}^5 \text{FDiff}_i [\text{Ev}_{\text{Sch}_{\text{QG}}}]$$

Proof. By Proposition 6.26, the failed-token set equals its set of minimal elements. By Definition 6.10, the corresponding failure-position token is assigned to each minimal failure position. Failure of a comparison-stage condition is classified by Definition 6.36; failure of a representation-stage condition, by Definition 6.58; failure of a continuous-approximation-stage condition, by Definition 6.77; failure of a dynamical-description-stage condition, by Definition 6.104; failure of a law-description-stage condition, by Definition 6.110; and failure of a differential-stage condition, by Definition 6.137.

Where evaluation readiness holds for condition 3, excess $\mathfrak{D}$ -record cell width is determined by Definition 6.79. The minimal failure classes at the dynamical-description, law-description, and differential stages are given, respectively, by Definitions 6.104, 6.114, and 6.140. Each minimal failure position in the QG-scheme-restricted complete evaluation specification is therefore classified within one of the six failure series. $\square$

8.8.3 Cosmological initial boundaries and time-generation conditions

When a cosmological initial boundary is specified by a time variable, if generation of that variable does not precede the law description of the initial boundary, the beginning of the universe, what is before or after that beginning, and the direction of time are described by a temporal order lacking grounds. If a theory describing the origin of time uses ungenerated time in its account of that origin, the theory uses the temporal order to be explained as an explanatory input before generating it, and results derived from the initial boundary inherit primitive-time dependence.

Quantum cosmology and the problem of initial conditions concern cosmological initial boundaries and temporal orientation[22, 23]. From local displays, records, object-state differences, and observer-side information coupling, the QG scheme generates comparison closure, the canonical observer-time quotient, a law-admissible time representation, and an observer-time law core. If a time variable is applied to an initial boundary before generation holds, its qualification failure is determined as primitive-time input; if it is applied after generation holds, it is evaluated as a boundary on a generated time representation.

Proposition 8.70 (Time-generation conditions at a cosmological initial boundary). Suppose that the QG-scheme-restricted complete evaluation specification targets a cosmological initial boundary. When dynamical-description operability holds, the law-admissible time representation used to describe the initial boundary is generated. When law-description operability holds, that time representation satisfies qualification for use within laws and the observer-time law core is generated.

Proof. For a QG-scheme-restricted complete evaluation specification, Proposition 8.61 gives generation of a law-admissible time representation through dynamical-description operability and generation of qualification for use within laws and an observer-time law core through law-description operability. The cosmological initial boundary belongs to the same evaluation specification through the object specification. The generation conditions of that proposition therefore apply to the description of the initial boundary. $\square$

Proposition 8.71 (Localization of time failure at a cosmological initial boundary). If any generation condition in Proposition 8.70 fails, each minimal failure position is classified by the corresponding failure-position token in the six failure series.

Proof. Proposition 8.69 classifies each minimal failure position in a QG-scheme-restricted complete evaluation specification within the six failure series. Because a cosmological initial boundary belongs to the object specification of the same evaluation specification, the classification of that proposition applies. $\square$

8.8.4 Consequences for extreme-gravity domains, observables, and the arrow of time

In the black-hole information problem, descriptions of information preservation, radiation, records, external observers, and interior regions compare the periods before and after radiation, the evaporation process, and information recovery along the same temporal evolution. Black-hole entropy, Hawking radiation, predictability, and recovery of information in radiation are representative lines of research[24, 25, 26, 27]. If this temporal evolution uses ungenerated temporal content, results asserting information preservation, information loss, or information recovery inherit primitive-time dependence. The QG scheme generates

records, comparisons, observer-side information coupling, a law-admissible time representation, a scale-indexed effective domain, and failure positions, thereby determining the physical-time qualification of the temporal evolution used in the information problem.

For the problem of observables, selecting an internal degree of freedom as a clock and describing its relation to other quantities does not by itself generate temporal content from differences in that clock degree of freedom, records, comparisons, and observer-side information coupling. Relational changes and observable values defined relative to an ungenerated internal clock inherit primitive-time dependence. Joint satisfaction of object-state differences, observer-side information coupling, the canonical observer-time quotient, the canonical family of observable functions, and a law-admissible time representation confers qualification for use within laws on observable values. Constrained systems, canonical quantum gravity, and relational observables are representative lines of research[19, 21, 11].

The arrow of time is an additional orientation condition on a generated time representation. If a low-entropy initial condition or the orientation of a coarse-grained history is placed on ungenerated time, the before-and-after relation on which that orientation depends remains without physical-time qualification. Thermodynamic orientation, cosmological orientation, and record orientation are evaluated when record retention, comparison closure, observer-side information coupling, and coarse-grained histories hold on a generated time representation. Cosmological initial conditions, singularities, and boundary conditions in quantum cosmology are representative lines of research[23, 22].

When quantum measurement records are assigned a before and after, a recording time, or a decoherence process, if the temporal content bearing that order is ungenerated, the measurement records and decoherence results inherit primitive-time dependence. Joint satisfaction of object-state differences, record-side retention, observer-side information coupling, and a law-admissible time representation temporalizes the measurement records. Mathematical formulations of the measurement process and decoherence are representative lines of research[28, 29].

Proposition 8.23 classifies an explanatory form that treats relativistic clock differences as stretching, slowing, or deformation of an unconstructed time entity as the two-stage misrecognition of the time-stretching account. Atomic-clock differences, differences in molecular-vibration periods, covalent-bond vibration sequences, electronic-state transitions, scattering events, and decay events supply physical carriers from which $\mathfrak{C}$ -derived sequences, $\mathfrak{D}$ -derived records, $\mathfrak{B}$ -derived differences, and $\mathfrak{O}$ -derived coupling generate time representations.

Gravitational, geometric, and medium conditions modulate the physical carriers supplied by atomic clocks, molecular vibrations, covalent-bond vibrations, electronic-state transitions, scattering detections, and decay detections. This modulation is classified as changes in role-carrier realization, record retention, comparison gluing, object-state identification, observer-side information coupling, trajectory relations, and law-output determinacy.

Atomic-clock and period differences under gravitational conditions, conventionally represented as proper-time differences, are comparison-derived differences among generated law-admissible time representations. By Proposition 8.32, clock and period differences reduce to differences in frequency representations obtained from $\mathfrak{C}$ -derived cycle-count differences and generated representation-coordinate differences. Stretching of a time entity is not the generation mechanism of these comparison-derived differences.

Using electronic-state transitions, scattering events, and decay events as event sequences

on a time variable requires generation of that variable to precede the law description of the event sequence. Electronic-state transitions contribute to the establishment of object-state differences and trajectory relations; scattering and decay events contribute to the establishment of detection records, object-state differences, and observer-side information coupling.

In extreme-gravity domains, identification of $\mathfrak{C}$ -derived sequences, preservation of the correspondence between $\mathfrak{D}$ -derived records and object-state differences, the $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived canonical trajectory relation, or law-output determinacy may fail. By Proposition 8.69, each minimal failure position is classified within the six failure series.

For theories of spacetime emergence, the QG scheme requires generation conditions for the time representations used in post-emergence spatiotemporal descriptions, scale-indexed effective domains, and failure positions. Spacetime construction from light and free fall, causal sets, holography, and connections between entanglement and geometry are representative lines of research[30, 31, 32, 33].

In descriptions concerning the cosmological constant and dark energy, expansion history, observational records, distance scales, time values, and law representations are joined on the same generated time representation. Vacuum-energy values, matter components, and field content belong to theory-specific inputs; the audit concerns the conditions for generating temporal comparisons, dynamical descriptions, and law-admissible time representations from those inputs. The cosmological-constant problem and observations of dark energy are representative lines of research[34, 35, 36].

Proposition 8.72 (Safe reuse of quantum-gravitational results through generated time and failure classification). Within a domain in which the conditions of Propositions 8.61 and 8.60 hold, the generated law-admissible time representation, observer-time law core, law-time value, canonical observer-time quotient, canonical object-state quotient, canonical trajectory relation, graph trajectory, and canonical family of observable functions may be used. Results concerning constraint equations, internal clocks, relational observables, cosmological initial boundaries, the arrow of time, quantum measurement records, decoherence, black-hole information, spacetime emergence, the cosmological constant, and dark energy may be used within domains in which their respective theory-specific conditions and the generation conditions of the temporal contents on which they depend hold.

Each minimal failure position classified by Proposition 8.69 determines the QG constituent to be repaired first or the condition by which the domain of validity must be restricted. Failure at the comparison stage requires repair of clock-candidate-side materials, record-side retention, object-state-side differences, observer-side information coupling, comparison rules, or comparison gluing. Failure at the representation or continuous-approximation stage requires repair of the canonical observer-time quotient, law-admissible time representation, $\mathfrak{D}$ -record cell width, comparison domain, or scale domain of approximation validity. Failure at the dynamical-description, law-representation, or differential stage requires repair of the canonical trajectory relation, canonical family of observable functions, law-evaluation datum, law-output determinacy, or differential evaluation specification. When repair succeeds, relational changes, observables, initial boundaries, the arrow of time, measurement records, decoherence, black-hole information, post-emergence spacetime descriptions, and cosmological law results are regenerated or recomputed in dependence order from the repaired inputs. If the stage condition does not hold throughout the entire domain, the domain in which a result is used is restricted to the region in which that condition holds.

By Proposition 8.68, generation conditions for temporal content and law-description

operability remain necessary after external time has been removed. Failure classification assigns failures of external-time input, undergeneration of an internal clock, comparison gluing, time representation, dynamical description, law-output determinacy, and differential consistency to the corresponding minimal failure positions. The problem of time in quantum gravity can thereby be decomposed into stagewise problems accompanied by a repair target or a restriction of the domain of validity.

When an error source in record retention, object-state identification, comparison gluing, $\mathfrak{D}$ -record cell width, relational observables, initial boundaries, measurement records, continuous approximation, or differential evaluation is repaired, the repair's effect on accuracy is determined by an accuracy evaluation of the regenerated or recomputed downstream results. Decisions concerning the use of quantum-gravitational results therefore divide into retention within a domain of validity and repair from each minimal failure position or restriction of the domain of validity.

Proof. By Proposition 8.61, dynamical-description operability generates a law-admissible time representation, while law-description operability confers qualification for use within laws on that representation and generates the observer-time law core. By Proposition 8.60 and Theorem 8.31, the canonical observer-time quotient, canonical object-state quotient, canonical trajectory relation, graph trajectory, canonical family of observable functions, law-admissible time representation, and law-time value are generated. By Proposition 8.21, when the temporal content on which each result depends and its grounding structure are generated prior to law operations and result derivation, the temporal input satisfies physical-time qualification.

By Proposition 8.69 and Definition 6.17, conditions and results downstream of each minimal failure position depend on upstream conditions. When the input or rule at each minimal failure position can be repaired, downstream results are regenerated or recomputed in dependence order from the repaired inputs. If the relevant stage condition does not hold throughout the entire domain, the domain of use is restricted to the region in which it does hold, and results generated within that region are used.

Proposition 8.68 separates removal of external time from generation of temporal content and determines qualification for use within laws through law-description operability. Propositions 8.70 and 8.71 supply the same generation conditions and failure classification for cosmological initial boundaries. The problems of external-time input, internal clocks, comparison gluing, time representation, dynamical description, law-output determinacy, and differential consistency can therefore be separated by their minimal failure positions.

When an error source belongs to record retention, object-state identification, comparison gluing, $\mathfrak{D}$ -record cell width, relational observables, initial boundaries, measurement records, continuous approximation, or differential evaluation, downstream results can be regenerated or recomputed from inputs in which that source has been repaired, and accuracy evaluation can be applied to the recomputed results. Thus retention of quantum-gravitational results within a domain of validity, the order of repair or domain restriction from each minimal failure position, and conditional accuracy evaluation are established. $\square$

8.9 Generation order of time-bearing quantities and cross-field consequences

Proposition 8.73 (Post-generation qualification for use of time-bearing quantities). For the twelve objects in Subsection 7.3.16, the dimension, unit, constancy, invariance, or algebraic relation of a time-bearing quantity does not supply the physical generation of

the temporal content used by that quantity. The use of each time-bearing quantity within a physical law is established in a domain in which the required temporal content and physical-time qualification are generated before law operations and result derivation, and in which the theory-specific establishment conditions are satisfied.

The object name and the presence or absence of a temporal dimension provide the extraction conditions, while the generation-dependence order appearing in the formal relations of the original source provides the basis of the audit determination. The physical relations of emission, propagation, reception, recording, atomic transitions, oscillators, particle decay, electromagnetic fields, gravitational fields, resonance lines, and measuring instruments are preserved as theory-specific inputs to the LCTR generation and audit construction. After temporal content has been generated, predicates concerning the constancy, invariance, rate, lifetime, linewidth, or frequency of each quantity are applied relative to generated time representations, admissible transformations, measurement procedures, and theory-specific domains of validity.

Proof. By Proposition 7.178, the generation of the temporal content required by a time-bearing quantity does not follow from the dimension, unit, or algebraic relation of that quantity. By Theorem 7.203, the physical establishment of a physical-law time variable requires the generation of temporal content and its grounding structure before law operations and result derivation.

By Remark 7.179, when temporal content and physical-time qualification have been generated and the theory-specific establishment conditions are satisfied, existing formal relations, including those for the twelve objects, may be used in their post-generation domains of application. Preservation of the physical relations as theory-specific inputs and post-generation qualification for use of time-bearing quantities therefore hold jointly. $\square$

Proposition 8.74 (Physical domain of light-speed invariance). The physical relations of the emission, propagation, reception, and recording of light are preserved as theory-specific inputs. Because the numerical expression for the speed of light uses a spatial difference and a time difference, it requires the temporal content supplying that time difference and its physical-time qualification.

The quantification domain of light-speed invariance in special relativity is the domain in which a generated law-admissible time representation, a spatial representation, admissible representation transformations between inertial observers, clock and length standards, measurement procedures, and the light-propagation relation hold. The constancy of the vacuum speed of light c measured in a local inertial frame, and the local upper propagation limit for matter, signals, or causal influences, are preserved in this domain. This establishment does not entail an unconditional universal proposition concerning an arbitrary coordinate speed of light, cosmological recession velocity, phase velocity, or displayed quantity that carries no information. If physical carriers and comparison constructions differ, and different generated time representations result, identical numerical invariance across those representations does not follow automatically. The determination must be made with the coordinate transformation, metric, clock and length standards, and domain of application fixed consistently. Where these conditions hold, the numerical expression and invariance of the speed of light may be used as results of special relativity. Dependence that uses the speed of light as a ground for the establishment of a time scale or synchronization rule before temporal content is generated is determined as pre-generation placement and reverse presupposition of a derived quantity.

Proof. By Proposition 7.182, pre-generation placement, omission of generation stages, and

reverse presupposition of a derived quantity hold on the formal track for the speed of light. Proposition 7.180 separates the physical propagation of light, the generation order of the temporal content used by a velocity expression, the constancy of the vacuum speed of light in a local inertial frame, and the different types of velocity, frames of reference, and domains of application. By Proposition 8.44, a Lorentz transformation is established as an admissible representation transformation between generated time representations and spatial displays. By Proposition 8.50, the transformation results and differential quantities of special relativity may be used in a domain in which the temporal content on which each result depends and the conditions at each stage hold.

It follows that, while the physical propagation relation of light is preserved as a theory-specific input, the quantification domain of the numerical expression and invariance of the speed of light is determined by generated time representations, admissible transformations, and theory-specific establishment conditions. $\square$

Corollary 8.75 (Cross-field consequence for the twelve time-bearing objects). Among the 96 determination fields produced by applying the four audit determinations on two tracks to the twelve time-bearing objects, an audit determination holds in 21 fields on the formal track and in zero fields on the statement track. The evidence included in the Appendix does not establish the corresponding determination for the remaining 75 fields.

The 21 established fields on the formal track consist of nine fields of pre-generation placement, seven fields of omission of generation stages, and five fields of reverse presupposition of a derived quantity. Reattribution of a time property is established in zero fields. This distribution separates the failure positions according to the dependence order in the original-source formal description of each object.

Proof. Apply the four audit determinations on two tracks and the determination-specific breakdown of Proposition 7.182. Proposition 7.181 establishes the persistence of primitive-time-root dependence in the decay constants, half-lives, mean lives, frequency shifts, frequency widths, acoustic-wave lifetimes, Planck constant, transition angular frequencies, linewidths, resonance widths, lifetimes, and Compton angular frequency treated in Curtiss 1949, Headley–Clayton–Reed 1996, Clark 2001, Wolf et al. 2011, and the PDG 2023 resonance reference. The determinations for the twelve objects therefore follow from the dependence orders in the source-specific formal evidence. $\square$

8.10 Cross-field synthesis of representative theoretical results through generated time and failure classification

Proposition 8.76 (Cross-field evidence comparison of the unbounded inheritance of limited results). The general application of Proposition 8.20 is compared with the formal evidence and statement evidence from thermodynamics, cosmology, quantum measurement, quantum gravity, relativity, biological application, and time-bearing quantities.

Subsection 7.3.18 maps the formal evidence and statement evidence from all 72 audited references to the audit method or representative arguments of Chapter 7. From that full-reference connection, the present proposition extracts cross-field direct-dependency paths and synthesizes, for each representative path, the evidence boundaries encompassing sources of the same or a closely related form.

When macroscopic asymmetry, redshift and model fit, decoherence, removal of external time and selection of an internal clock, differences in clock readings, frequencies, or

particle lifetimes, and relations containing time-bearing quantities are established, their establishment belongs to the evidence-effect domain specific to each result.

These limited results alone do not establish the physical generation of temporal content, the physical-time qualification of law time, a time attribute independent of physical carriers, direct demonstration concerning biological processes or subjective experience, a beginning, nonexistence, or emergence of time itself, or generation of the time roots on which time-bearing quantities depend. Each such claim requires independent evidence or a valid derivation specific to that claim.

A literal interpretation that transfers a relativistic clock difference to a change in an attribute of an unconstructed time entity constitutes a two-stage misrecognition, and an interpretation that then transfers it to direct demonstration concerning biological processes constitutes a three-stage misrecognition. No misrecognition arises for a statement whose original-source text identifies its figurative character, states that it does not designate a literal deformation of time, and identifies the actual referent within the same explanatory scope. An interpretation that reads an expression as a bare time-attribute expression constitutes a one-stage misrecognition; one that reads it as a literal time attribute constitutes a two-stage misrecognition. Thermodynamics, cosmology, quantum measurement, quantum gravity, and time-bearing quantities are likewise assigned no single stage count by field. Only direct dependencies that cross evidence-effect boundaries are counted, separately for each permissible interpretation of an actual original-source statement.

Statements about the arrow of time concerning time-reversal symmetry, low-entropy boundary conditions, and the cosmological direction of time are preserved within the evidence-effect domain of the asymmetry or orientation that each describes. The arrow of time is a subsequent orientation condition on a generated time representation, and an interpretive path that remains within this dependence order crosses zero evidence-effect boundaries.

An interpretive path that reads Feynman's statement concerning the beginning of "at least our part of the universe" as the initial boundary of a cosmological model or region crosses zero evidence-effect boundaries. A path that transfers the same statement to the beginning, nonexistence, or meaninglessness of physical time in general crosses one evidence-effect boundary, from the boundary of a model or region to the domain of existence of time in general, and constitutes a one-stage misrecognition. The interrogative statement in the original source asking whether an earlier time has meaning does not conclude that physical time in general is nonexistent or meaningless.

The argument of Augustine presented in Heisenberg 1958 states that time was created together with the world and did not exist before the universe. Heisenberg himself analyzes the word "created" in the creation of time as presupposing a before-after relation and therefore finds no rational definition for the phrase "time was created." This analysis directly identifies the self-dependence by which a statement about the beginning or creation of time uses a temporal relation in the statement operation itself. Establishing a statement of time creation requires a generation relation that identifies the states before and after creation without depending on time.

The excess distances of distant supernovae derived by Riess et al. and the incompatibility derived by Perlmutter et al. between their supernova sample and a flat, matter-only cosmological model are preserved within their respective evidence-effect domains of observation, correction, calibration, estimation, and model fitting. An interpretation that transfers these results to direct demonstration of the entire cosmic history, generation of temporal content, or the domain of existence of physical time in general constitutes

an evidence-effect boundary along that interpretive path when an actual statement or explanation performs the transfer. The cited original sources of Riess et al. and Perlmutter et al. do not yield this transfer.

The corrections to the vocabulary of cosmic expansion by Bunn–Hogg, Lewis et al., and Peacock describe observed redshift in terms of geodesics, local relative motion, or spacetime geometry, without attributing it to the action of an independent physical carrier called expanding space. In the cross-field audit, these corrections provide counterevidence against paths that move from a limited result to a temporal ontology or an independent carrier.

The agreement between the muon-lifetime measurement and relativistic prediction in Bailey et al. 1977 is preserved within the evidence-effect domain of the particle lifetime, measurement conditions, and adopted theoretical antecedents. A path that estimates the negative-muon lifetime at rest with special relativity as an antecedent does not simultaneously transmit that same estimate to an independent test of special relativity and an unconditional test of CPT invariance.

In Feynman 1995, an explanation of a nuclear transition and frequency shift proceeds within the same source to a generalization to all physical processes, an application to human life and aging, and a description in terms of an attribute of the flow of time. In Narlikar 2010, an explanation of signals from a collapsing body, redshift, and reception waiting time proceeds to a description in terms of a difference in the rates at which time flows for two subjects. The former makes concrete the boundary between formal generalization to physical processes and direct demonstration concerning biological processes; the latter makes concrete the boundary between signal–wavelength relations and an attribute of time flow independent of physical carriers. Limited results concerning clocks, signals, frequencies, or particle lifetimes add no evidence specific to the claims beyond those boundaries.

Von Neumann 1932 explicitly introduces time t as an input to a nonstationary wave function and quantum evolution. The mathematical consistency, predictive performance, or empirical agreement of a quantum formalism using this time input does not generate time t from nontemporal physical inputs. The source distinguishes an empirical statement of the type in which an observer has obtained a subjective perception from a statement of the type in which a physical quantity has a value. The former includes subjective perception among its objects but supplies no direct measurement of a change in subjective temporal experience. The system–environment interaction, environmental correlations, suppression of interference, and pointer structure in Zurek 2003 are preserved within the evidence-effect domain of decoherence. Adler 2003 supplies corrective evidence that separates limited decoherence results from a solution of the entire quantum measurement problem, while Schlosshauer 2005 supplies reserving evidence that does not treat the implication between them as settled. The evidence paths provided by these four sources separate external-time input, decoherence results, single-outcome selection, resolution of the entire quantum measurement problem, and generation of temporal content into distinct evidence-effect domains. If another actual statement moves among these domains, an evidence-effect boundary is counted for each direct dependency.

In quantum gravity, removal of an external time parameter, selection of an internal clock, a state-dependent flow, bipartite time, relational reformulation, and complete generation of physical time are treated as distinct results. Interpretive paths that read statements in Barbour 1994 I and II, Connes–Rovelli 1994, Kiefer 2009, and Smolin 2018/2020 as limited formal results or theoretical classifications preserve those statements within their evidence-

effect domains. Paths that transfer the same statements to the creation or nonexistence of physical time in general, complete generation of every temporal content, complete exclusion of primitive-time roots, or qualification for use within laws prior to generation constitute one- or two-stage misrecognitions according to the established interpretation-path-specific determinations. Descriptions of the problem of time, candidate constructions, and relational results by DeWitt, Isham, Kuchař, Page–Wootters, and Rovelli are preserved as evidence identifying the problem and supplying partial formal results. In particular, the relational formalism of Rovelli 2002 expresses dynamical predictions as relations among partial observables, selects no particular partial variable as time, and places no vocabulary of evolution in the fundamental structure. This result does not generate, from nontemporal physical inputs, the clocks, records, comparisons, order, scale, and qualification for use within laws that physically realize those relations. The obligation to remove external time from a formalism and the obligation to generate physical time belong to different evidence-effect domains.

Within operational constructions, Bridgman 1927 explicitly identifies the inadequacy of basic operations for time measurement and clock-construction specifications. Ehlers–Pirani–Schild 1972 excludes clocks from the basic tools of its own construction from light rays and freely falling particles and derives measurement operations. The atomic-clock interpretation of the chronometric axiom in Synge’s preceding method is distinct from the additional empirical postulate equating gravitational and atomic time after the EPS reconstruction. These results identify incomplete construction, change the order of construction, or establish conditional structural results. A hypothetical interpretation that extends identification of a problem, removal of a primitive instrument, or transition to a derived operation to completed generation of all time contents or law-time qualification crosses an evidence-effect boundary from a local or partial stage to a subsequent stage.

In black-hole thermodynamics, radiation, and information, the formal analogy and conjecture in Bekenstein 1973, the radiation result and boundary of the semiclassical approximation in Hawking 1975, the classical spacetime background and temporal Fourier components in Hawking 1976, and the model-dependent first conjecture in Page 1993 are each preserved within the domain of effect established by the original source. Computational success concerning area increase, a radiation spectrum, an evaporation time, or the appearance of information does not repair the generation of the temporal content presupposed by those computations, the physical establishment of the background spacetime, or resolution of the entire black-hole information problem. An interpretive path that transfers a limited computational result to any of these subsequent claims requires independent evidence or a valid derivation for each destination of transfer.

In duality, entanglement, and spacetime emergence, the duality conjecture of Maldacena 1998, the proposed area law and limitations on quantitative comparison in Ryu–Takayanagi 2006, and the relation between entanglement and spatial connectedness in Van Raamsdonk 2010 are preserved as their respective formal results. An interpretation that transfers a boundary–bulk correspondence, minimal-surface correspondence, or explanation of spatial connectedness to physical-time qualification of a time variable, completion of a spacetime-generation mechanism, or complete generation of physical time in general crosses an evidence-effect boundary from the domain in which the formal correspondence is established to a generation obligation.

The mathematical consistency, predictive performance, model fit, and empirical agreement established in each field are preserved within their respective evidence-effect domains. Generating temporal content from nontemporal physical inputs and auditing its domain of

validity and failure position determine the domain in which existing results may be reused safely.

Proof. By Proposition 7.118, a limited result does not establish a claim beyond an evidence-effect boundary. By Corollary 7.119, inheritance of primitive-time dependence by results and inheritance of a limited result by a claim outside its domain of application are distinct audit axes.

Proposition 7.142, Proposition 7.143, and Corollary 7.144 separate the establishment of a temporal direction from the generation of temporal content. Proposition 7.154 separates the initial boundary of a cosmological model or region from the domain of existence of physical time in general, and determines the number of boundary crossings for each interpretive path without converting an interrogative statement in the original source into a conclusion. Proposition 7.155 separates the evidence types of observation, estimation, model fit, and direct demonstration. Proposition 7.176 and Corollary 7.177 separate the formal domain of effect made explicit by corrections to cosmic-expansion vocabulary from the time-generation audit. Proposition 7.133, Proposition 7.134, Corollary 7.135, and Proposition 7.132 separate external-time input, an empirical statement concerning subjective perception, direct measurement of subjective temporal experience, the domain of validity of decoherence, single-outcome selection, resolution of the entire quantum measurement problem, and generation of law time, and determine the roles of corrective and reserving evidence.

Proposition 7.172, Proposition 7.170, and Proposition 7.169 separate the limited results of adjacent time programs, removal of a privileged time from a relational formalism, the distinct obligations of removing external time and generating physical time, the fundamentality of time, and complete time generation into different evidence-effect domains. Proposition 8.64, Proposition 8.65, Proposition 8.66, and Proposition 8.67 determine, for each permissible interpretation of each actual statement, the number of evidence-effect boundaries connected by direct dependency.

Subsection 7.3.18 connects the statement content, interpretability, conditional comparison with formal evidence, and evidence-effect boundary for all 72 audited references—not only the representative paths above—to the audit method or representative arguments of Chapter 7. Subsections 7.3.1, 7.3.8, and 7.3.9 establish the primitive-time roots and result dependencies remaining in the formal evidence for operational constructions, black holes, and spacetime emergence.

Proposition 7.178 and Proposition 8.73 separate relations containing time-bearing quantities from post-generation qualification for use of their time roots. Proposition 7.185, Proposition 7.184, Proposition 7.193, and Proposition 8.59 separate direct measurement of physical carriers, attribution to a time property, and direct demonstration of biological application.

By Theorem 7.190, primitive-time roots and results dependent on primitive-time roots occur in all 72 audited references. It follows that evidence-effect boundaries separate each limited result's evidence-effect domain from generation of temporal content, time qualification, time attributes, biological application, temporal ontology, and the time roots of time-bearing quantities. Within a path formed by the formal construction, statements, and permissible interpretations of each reference, a direct dependency that crosses such a boundary constitutes unbounded inheritance of a limited result. Generated time and failure classification determine, for each such path, the domain in which existing results may be reused safely. $\square$

Proposition 8.77 (Target consistency between time-nonexistence statements and the attribution of time properties). When a time-nonexistence statement and a time-property statement coexist within the same theoretical description or reference, or when statements from multiple references are synthesized across fields under a common explanatory vocabulary, the type and domain of application of the temporal content targeted by the two statements are aligned for determination in accordance with Proposition 8.63 and Remark 7.175.

If the two statements target the same type of temporal content and the same domain of application, the nonexistence of time is incompatible with the establishment of stretching, slowing, flow, or deformation whose bearer is that time. Attribution of a property to time requires, in the same domain of application, an object that bears the property.

If the two statements target different types of temporal content or different domains of application, the existence denial and property attribution are not directly contradictory. In this case, the target of the existence denial must be separated from the target of the property attribution, and the change in target type or domain of application must be made explicit.

If the target of property attribution is a difference in clock readings, clock rates, proper times, or coordinates, or a comparison-derived difference among generated law-admissible time representations, the property belongs to that physical carrier or relational quantity. This change of target type or domain of application does not entail the stretching, slowing, flow, or deformation of “time itself” independently of physical carriers.

Proof. By Proposition 8.62 and Proposition 8.63, a time-nonexistence statement bears the obligation to specify the type and domain of application of the temporal content being denied.

By Proposition 8.47 and Proposition 8.56, a statement that attributes stretching, slowing, or deformation to time as a literal physical property requires a property-bearing object that remains identical before and after the change, together with its state space, deformation rule, and measurement correspondence. Literal attribution of flow likewise requires an object that flows and conditions for identifying changes in the state or position of that object.

If an existence denial and property attribution target the same type of temporal content and the same domain of application, one requires the absence of the object while the other requires that object as the bearer of a property. They therefore cannot hold simultaneously.

If the type of temporal content or domain of application differs, the existence denial and property attribution belong to different objects or regions. No direct contradiction then arises, but an explanation that connects both statements through the same word “time” leaves the bearer of the property undetermined unless it makes the change in target type or domain of application explicit.

By Proposition 7.174, the objects directly supplied by the formal results of relativity are differences in clock readings, clock rates, lifetimes, or proper times, spacetime intervals, or coordinate relations; they are not a state space, deformation rule, and independent measurement procedure for a bearer of time stretching independent of physical carriers. Changing the target of property attribution to these physical carriers or relational quantities therefore does not entail a change in a property of “time itself” independently of physical carriers.

It follows that statements with the same target type and domain of application are incompatible, while statements with different target types or domains bear an obligation

to make the target change explicit and do not entail the attribution of a property to “time itself.” □

Figure 23 compares time-nonexistence statements with time-property statements along the two axes of the type and domain of application of the temporal content targeted by the statement. The temporal-content types distinguish time independent of physical carriers, differences in clock readings or clock rates, differences in proper times or coordinates, and comparison-derived differences among generated time representations. The domains of application distinguish the theoretical domain, approximate domain, and physical conditions in which the statement is said to hold.

Target-consistency audit of statements about time

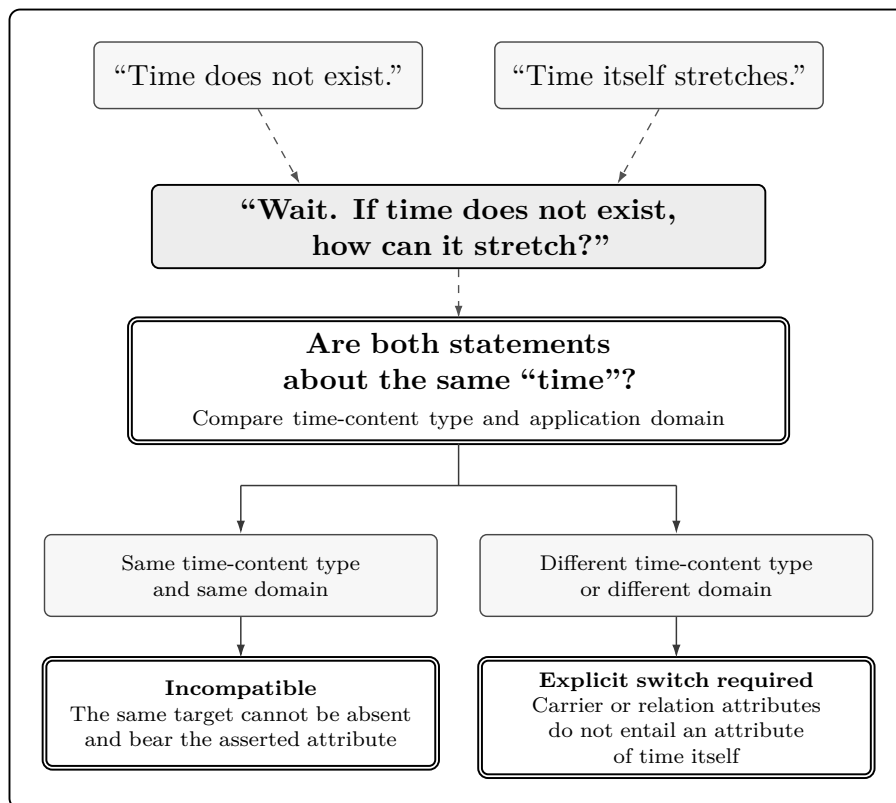


Figure 23: Target consistency between time-nonexistence statements and time-property statements. The upper row gives representative forms of the two kinds of statement subjected to audit, while the gray band gives the question that arises for a reader who encounters them in succession. These are not verbatim quotations from an original source. The dashed lines from the upper row to the central audit box indicate reading order and do not express formal implication. Solid lines from the central box onward indicate the target-consistency determination. The two statements are incompatible when they concern the same type and the same domain. Different types or domains require an explicit change, and a property of a physical carrier or relational quantity does not entail a property of time itself independently of physical carriers.

The 72 audited references contain both statements that time does not exist and statements that attribute stretching, slowing, flow, or deformation to time itself. Misner–Thorne–Wheeler 1973 contains both series of statements within the same reference, but their explicitly stated domains of application differ, so they are not determined to be

directly contradictory under identical conditions. Heisenberg 1958 records the nonexistence of time before the universe and the temporal presupposition in the operation of stating it, while Feynman 1995 and Narlikar 2010 record statements that move from clock or signal relations to an attribute of time flow. These cases show that the target of an existence denial must be separated by type and domain of application from the target of a property attribution, not only within a single reference but also when the same word “time” is used in different theoretical domains.

The same determination applies when statements from multiple references are synthesized as part of the common explanatory vocabulary of physics. The object of determination is whether a physical synthesis that places time-nonexistence statements and time-property statements within the same explanatory system preserves the identity of their targets or changes to a different object or domain of application.

Proposition 8.78 (Safe reuse of representative theoretical results through generated time and failure classification). Under the respective conditions of Propositions 8.26, 8.36, 8.42, 8.50, 8.58, and 8.72, time-dependent results in Newtonian and classical mechanics, time metrology, GPS, special relativity, general relativity, and quantum gravity are qualified for use within domains in which temporal content and its grounding structure are generated prior to use within laws.

When a generation-stage condition fails, the failure-position-token system of Definition 6.10 and the set of minimal failure positions of Theorem 6.168 determine the condition to be repaired first or the condition by which the domain of validity must be restricted. When repair succeeds, downstream results are regenerated or recomputed from the repaired inputs in accordance with the failure-generation dependence order of Definition 6.17. If the condition does not hold throughout the entire domain, the domain in which a result is used is restricted to the region in which the condition holds. The mathematical, predictive, measurement, and engineering results of each theory can thereby be reused safely within domains that satisfy physical-time qualification.

The strength of the connection to numerical accuracy differs across fields. In time metrology and GPS, repair of error sources belonging to calibration, frequency ratios, record-cell widths, window selection, clock corrections, propagation corrections, loop defects, and receiver-clock offsets connects directly to recomputation and accuracy evaluation of downstream results. In Newtonian and classical mechanics, special relativity, and general relativity, repair connects to recomputation and accuracy evaluation when an error source belonging to initial values, time representations, synchronization, clock-rate comparison, local or global time conditions, transport, approximation domains, or differential evaluation is included among the minimal failure positions. In quantum gravity, the principal consequence is determination of the time-generation conditions, failure positions, repair targets, and qualification of existing results for application; a subsequent accuracy evaluation is connected when a specific approximation or estimation error is localized at a failure position.

Problems that depended on a primitive-time premise thereby shift from mysteries or theoretical impasses formulated with unestablished temporal content to problems of identifying time-generation conditions, minimal failure positions, repair order, and domains of validity. Generated time and failure classification supply grounds for using the results of existing theories safely under their conditions of validity and, upon failure, for identifying repair targets from upstream to downstream.

Proof. For each representative theory, the six local propositions give the domain of validity in which generated time representations may be used, each minimal failure position, repair

or restriction of the domain of validity, regeneration or recomputation after repair, and conditional accuracy evaluation.

Definition 6.10 classifies a failed generation-stage condition by the corresponding failure-position token. Theorem 6.168 determines the failure positions minimal with respect to the failure-generation dependence order. By Definition 6.17, conditions and results downstream of a minimal failure position depend on upstream conditions. The condition belonging to the minimal failure position is therefore repaired, and downstream results are regenerated or recomputed in dependence order from the repaired inputs. If repair does not hold throughout the entire domain, the domain of validity of that condition determines the domain in which the result may be used.

Comparison of the connections between error sources and accuracy evaluation specified by the local propositions shows a direct connection to measurement and correction inputs in time metrology and GPS; in Newtonian and classical mechanics, special relativity, and general relativity, the connection holds when the relevant initial value, time representation, synchronization, transport, approximation, or differential evaluation is included at a failure position; in quantum gravity, it holds when a specific approximation or estimation error is localized at a failure position.

Thus, for existing results in all six fields, explicit time-generation dependence, determination of safe domains of application, first-failure diagnosis, repair order, restriction of domains of validity, safe reuse, and field-specific conditional accuracy evaluation are established. $\square$

Proposition 8.79 (Qualification failure of primitive-time input and conditional failure of downstream computational results). Let x be temporal content used by a representative scheme, and let η be a downstream result in whose derivation that temporal content participates either directly or through a temporal-content transfer sequence. The following two determinations hold.

The first is the determination of physical-time qualification. If x is a primitive-time input, x fails physical-time qualification, and η depends on temporal content that fails physical-time qualification. This qualification failure remains even when mathematical consistency, predictive performance, and empirical agreement hold.

The second is the determination of downstream computational results. In a scale-indexed complete evaluation specification (Ev, λ) , consider temporal content x , a downstream result η , and a condition χ that governs either the generation or qualification for use of that temporal content, or the input value, comparison, representation, synchronization, transport, approximation, or differential evaluation of that result. Suppose that the primitive-time input x backgrounds the grounding obligation for χ in the sense of Definition 7.120, that independent evidence establishes the actual failure of the same χ , and that downstream dependence holds from that failure position to the derivation stage of η . Then η loses qualification for use as a physical computational result within the evaluation domain concerned. If the failed condition can be repaired, downstream results are regenerated or recomputed from the repaired inputs. If the condition cannot be repaired throughout the entire domain, the domain in which the result is used is restricted to the region in which the condition holds. Whether the recomputed value agrees with or differs from the preceding value is determined by the recomputation after repair.

The establishment conditions of the first determination are the presence of a primitive-time input and its participation in result derivation. The establishment conditions of the second determination are the backgrounding of a grounding obligation by primitive-time input, independent evidence establishing the actual failure of the same condition, and

downstream dependence from the failure position to result derivation. Evaluation success, failure of physical-time qualification, and loss of qualification for use of a downstream computational result can therefore hold jointly in the same case.

Proof. By Theorem 7.11, the primitive-time input x fails physical-time qualification. By Proposition 7.6 and Corollary 7.12, if x participates in the derivation of η either directly or through a temporal-content transfer sequence, η depends on temporal content that fails physical-time qualification. By Proposition 7.13, mathematical consistency, predictive performance, and empirical agreement do not repair this qualification failure. The first determination follows.

Proposition 7.121 connects, within a single audit-causal sequence, the backgrounding of a grounding obligation by primitive-time input, independent evidence establishing the actual failure of the same condition, the reach of dependence to a downstream result, and regeneration or recomputation after repair. By Proposition 8.78, the six representative fields localize an actual failure of a generation condition or establishment condition at a minimal failure position and regenerate or recompute downstream results from repaired inputs in accordance with the failure-generation dependence order. By Definition 6.4, an evaluation specification requiring a failed condition does not belong to the corresponding domain of validity. By Definition 6.17 and Theorem 6.168, each minimal failure position and its downstream dependencies are determined; by Proposition 6.171, the structural burden imposed by that failure position is determined. If the derivation stage of η belongs to this downstream dependence, η loses qualification for use as a physical computational result within the evaluation domain concerned. If repair succeeds, the result is regenerated or recomputed from the repaired inputs; if repair does not succeed throughout the entire domain, the establishment domain determines the domain in which the result may be used. The second determination follows.

The first determination follows from primitive-time input and its participation in result derivation. The second follows from the identity of the condition whose grounding obligation is backgrounded by primitive-time input, the condition whose actual failure is established by independent evidence, and the condition from which downstream dependence begins. The two determinations have different establishment conditions, and mathematical consistency, predictive performance, and empirical agreement repair neither failure condition. $\square$

Evaluation success and qualification consequences

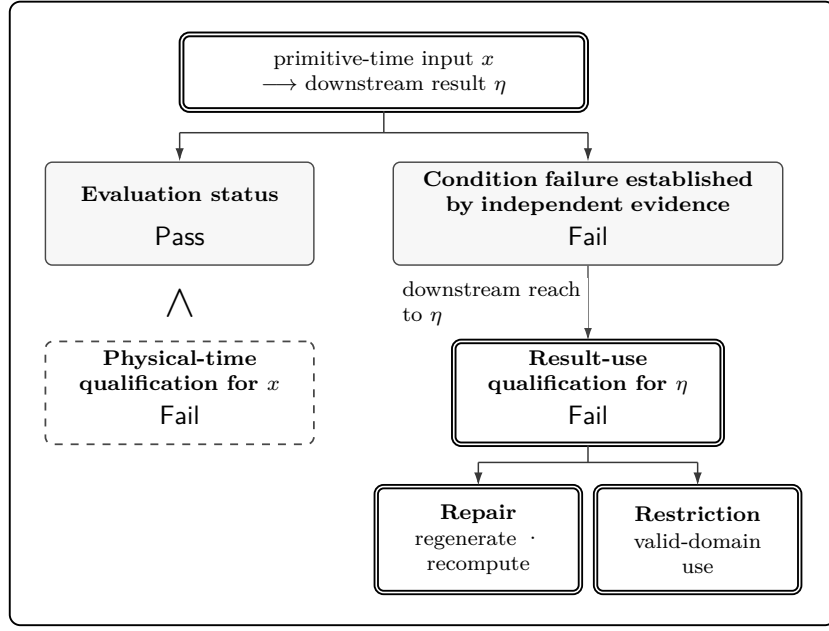


Figure 24: Coexistence of evaluation success and two qualification failures. The left branch shows evaluation success coexisting with the failure of physical-time qualification for a primitive-time input. The right branch shows failure of a generation or establishment condition, established by independent evidence, reaching the qualification for use of downstream results and requiring repair or use within the domain of validity.

8.11 Classical time as a generated representation

This section constructs the classical domain as a subclass satisfying the conditions of the theory of generated time. A time variable belonging to this domain is generated in the order in which exact comparison closure, dynamical-description operability, law-description operability, and differential-description operability are established. A continuous time representation in this domain holds at scales belonging to the scale domain of validity of the continuous approximation in Definition 6.86.

The representative-scheme single-object evaluation specification of Definition 8.5 connects each theory to a realized operational configuration, and Proposition 8.22 supplies the type boundary between theory-specific inputs and generated objects. When the specialization conditions defining a classical domain hold, Theorem 5.112, Theorem 5.113, and Definition 5.133 apply.

8.11.1 Minimal operational model of classical time

The minimal operational model of classical time has a single observer-side information-coupling position, a single object-specification index, a $\mathfrak{C}$ -derived clock-side source, a $\mathfrak{D}$ -derived record-side source, and a $\mathfrak{B}$ -derived object-state-side source. Each generated object is obtained from satisfaction of the corresponding stage conditions.

Definition 8.80 (Minimal operational model of classical time). Let $\text{Sch}_{\text{ClMin}}$ be an instance

of the minimal classical scheme. Define the complete evaluation specification by

$$\begin{aligned} & \text{Ev}_{\text{Sch}_{\text{CMin}}} \\ & := ((\mathcal{X}_{\text{Sch}_{\text{CMin}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{CMin}}}), o_{\text{Sch}_{\text{CMin}}}, \zeta_{\text{Sch}_{\text{CMin}}}) \end{aligned}$$

Define the realized operational configuration by

$$\begin{aligned} & \mathcal{X}_{\text{Sch}_{\text{CMin}}} \\ & = (\mathcal{I}_{\text{Sch}_{\text{CMin}}}^{\text{raw}}, \text{Dat}_{\text{Sch}_{\text{CMin}}}^{\text{src}}) \end{aligned}$$

Let the single $\mathfrak{C}$ -derived source index of the minimal classical model be $j_{\mathfrak{C}, \text{Sch}_{\text{CMin}}}^{\text{src}} \in \text{Idx}_{\mathfrak{C}}^{\text{src}}$. For each $k \in \mathbb{N}$, define the $\mathfrak{C}$ -derived clock-side source token by

$$\begin{aligned} x_{\mathfrak{C}, k} & := \text{tok}_{\mathfrak{C}}(j_{\mathfrak{C}, \text{Sch}_{\text{CMin}}}^{\text{src}}, k) \\ & \in \text{Tok}_{\mathfrak{C}, \text{Sch}_{\text{CMin}}}^* \end{aligned}$$

Define the $\mathfrak{D}$ -derived record-side source token and the $\mathfrak{B}$ -derived object-state-side source token by

$$\begin{aligned} x_{\mathfrak{D}, k} & \in \text{Tok}_{\mathfrak{D}, \text{Sch}_{\text{CMin}}}^*, \\ x_{\mathfrak{B}, k} & \in \text{Tok}_{\mathfrak{B}, \text{Sch}_{\text{CMin}}}^* \end{aligned}$$

The $\mathfrak{D}$ -derived record-side source token satisfies

$$\text{seq}_{\mathfrak{D}}(x_{\mathfrak{D}, k}) = k$$

The $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}$ -derived coupling satisfies

$$(x_{\mathfrak{C}, k}, x_{\mathfrak{D}, k}, x_{\mathfrak{B}, k}) \in \text{Rel}_{\text{Sch}_{\text{CMin}}}^{\text{dyn}, \text{src}}$$

, and its $\mathfrak{C}/\mathfrak{D}$ component satisfies

$$(x_{\mathfrak{C}, k}, x_{\mathfrak{D}, k}) \in \text{Rel}_{\text{Sch}_{\text{CMin}}}^{\text{src}}$$

Use

$$\text{Arr}_{\mathfrak{C}, o_{\text{Sch}_{\text{CMin}}}}^{\text{cmp}}, \quad \text{Arr}_{\mathfrak{D}, o_{\text{Sch}_{\text{CMin}}}}^{\text{cmp}}$$

as the comparison-stage arrival maps, and, as the dynamical-description-stage $\mathfrak{B}$ -derived arrival map, use

$$\text{Arr}_{\mathfrak{B}, o_{\text{Sch}_{\text{CMin}}}}^{\text{dyn}}$$

Proposition 8.81 ($\mathfrak{C}$ -derived observer-time quotient in the minimal classical model). For the minimal classical model of Definition 8.80, suppose that

$$\text{ExactOperable}(\mathcal{X}_{\text{Sch}_{\text{CMin}}}; \mathcal{J}_{\text{cmp}, \text{Sch}_{\text{CMin}}})$$

and

$$\text{DynOperative}[\text{Ev}_{\text{Sch}_{\text{CMin}}}]$$

hold.

For each $k \in \mathbb{N}$, define

$$T_k := \Theta_{\mathfrak{C}, \mathcal{O}_{\text{SchClMin}}}^{\text{dyn,can}}(x_{\mathfrak{C},k})$$

. Suppose further that the following three conditions hold.

First, for every

$$T \in \text{Qu}_{\mathfrak{C}, \mathcal{O}_{\text{SchClMin}}}^{\text{dyn,can}}$$

there exists $k \in \mathbb{N}$ such that $T = T_k$.

Second, for every $k_0, k_1 \in \mathbb{N}$,

$$\begin{aligned} T_{k_0} = T_{k_1} \\ \iff k_0 = k_1 \end{aligned}$$

.

Third, for every $k_0, k_1 \in \mathbb{N}$,

$$\begin{aligned} T_{k_0} \prec_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn,time,can}} T_{k_1} \\ \iff k_0 < k_1 \end{aligned}$$

.

Then

$$\text{Qu}_{\mathfrak{C}, \mathcal{O}_{\text{SchClMin}}}^{\text{dyn,can}} \cong \mathbb{N}$$

holds as an order isomorphism.

For any

$$\rho_{\mathcal{O}_{\text{SchClMin}}} \in \text{OrdEmbSet}_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn,time}}(\mathbb{R})$$

, define the time value for each $k \in \mathbb{N}$ by

$$t_k := \text{TimeRep}_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn}} \left[\rho_{\mathcal{O}_{\text{SchClMin}}} \right] (T_k)$$

.

Proof. Exact comparison closure determines the $\mathfrak{C}$ -derived observer-time quotient and $\mathfrak{C}$ -derived natural projection of Definition 5.16. Dynamical-description operability and Theorem 5.112 give

$$\text{OrdEmbSet}_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn,time}}(\mathbb{R}) \neq \emptyset$$

.

By the assumed coverage and injectivity, the correspondence $k \mapsto T_k$ is a bijection from the natural numbers to the $\mathfrak{C}$ -derived observer-time quotient. By order agreement, this bijection preserves the natural-number order and the canonical strict observer-time order, and its inverse also preserves order. The $\mathfrak{C}$ -derived observer-time quotient is therefore order-isomorphic to the natural numbers.

Choose any real-valued temporal-order embedding. Definition 5.63 determines a law-admissible time representation, and a time value is determined as the image of each time-quotient class. Hence

$$t_k = \text{TimeRep}_{\mathcal{O}_{\text{SchClMin}}}^{\text{dyn}} \left[\rho_{\mathcal{O}_{\text{SchClMin}}} \right] (T_k)$$

is a real-valued order representation of the generated time-quotient classes. $\square$

Corollary 8.82 (Continuous approximation of the minimal classical model). For the complete evaluation specification of Definition 8.80, suppose that a refinement family under Definition 6.82 and a nine-component defect family under Definition 6.84 are given, and that monotone nondecreasing behavior under Definition 6.85 holds.

For every

$$\lambda \in \mathcal{I}^{\max} [\text{Ev}_{\text{Sch}_{\text{CMin}}}]$$

,

$$\lambda \in \mathcal{V}_{\text{Ev}_{\text{Sch}_{\text{CMin}}}}^{\text{approx}}$$

holds. At this scale, all nine continuous-approximation conditions hold, including the continuous-extension, order-stability, and relation-stability conditions.

Proof. By Definition 6.93, the maximal effective interval is the initial interval of the evaluation-specification-specific scale domain of validity of continuous approximation. Membership in the maximal effective interval therefore gives membership in the scale domain of validity of continuous approximation for the same full evaluation specification. By Theorem 6.95, the latter membership is equivalent to simultaneous satisfaction of the nine conditions for continuous approximation. $\square$

Connection to physical-time qualification. The components of the full evaluation specification $\text{Ev}_{\text{Sch}_{\text{CMin}}}$ are the realized operational configuration, the family of comparison rules, the observer-side information-coupling position, and the object-specification index. Stable records, the object-state sequence, clock-side finite granularity, the conditions for continuous approximation, and law-bearing single-valuedness are used in the law-time-use audit as establishment evidence for the respective generation stages.

Proposition 7.199 gives the generation dependencies from the realized operational configuration through the representation, dynamical-description, law-description, and differential stages. In this model, when the grounding structure corresponding to time content precedes law operations and result derivation, Theorem 7.200 implies that a physical-law time variable using that time content satisfies physical-time qualification. Failure at any generation stage is localized as a failure-position token by Definition 6.10 and Definition 6.21. Failure-position tokens localize the establishment boundary of the classical approximation, and physical-time qualification is determined by the generation-precedence condition of Corollary 7.15.

Definition 8.83 (Classical regime). Let the full evaluation specification be

$$\text{Ev} = ((\mathcal{X}; \mathcal{J}_{\text{cmp}}), o, \zeta) \in \text{EvSpec}$$

and let the law-evaluation datum having this full evaluation specification as its first component be

$$\text{LawEvalDat}_{o,\zeta} = (\text{Ev}, \text{CandRelStr}_{o,\zeta}^{\text{law}})$$

and the differential-evaluation datum having this law-evaluation datum as its first component be

$$\begin{aligned} \text{DiffEvalDat}_{o,\zeta} \\ = (\text{LawEvalDat}_{o,\zeta}, \text{EvalStrFam}_{o,\zeta}^{\text{diff}}) \end{aligned}$$

.

The data Ev , $\text{LawEvalDat}_{o,\zeta}$, and $\text{DiffEvalDat}_{o,\zeta}$ form a classical regime if the following six conditions hold.

- (CL₁) A single shared $\mathfrak{C}$ -derived clock domain is established, and the $\mathfrak{C}$ -derived clock-side data used at the comparison stage belong to the same shared family of comparison clocks.
- (CL₂) The branching structure at the comparison stage is single-valued with respect to generation of the $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient.
- (CL₃) The $\mathfrak{B}$ -derived object-state-side identification map satisfies $\text{Stateld}_{\mathfrak{B},o} = \text{id}$, and, for every admissible $\mathfrak{B}$ -derived object-state-side token $x_{\mathfrak{B}}$,

$$\text{Arr}_{\mathfrak{B},o}^{\text{dyn}}(x_{\mathfrak{B}}) = (x_{\mathfrak{B}}, x_{\mathfrak{B}})$$

holds.

- (CL₄) The exact comparison closure of Definition 4.175,

$$\text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$$

holds.

- (CL₅) The all-dynamical-description-conditions bundle $\text{DynCondAll}[\text{Ev}]$ of Definition 5.67 holds. Its eight conjunctive components correspond, in the order given in that definition, to the dynamical-description conditions DG_1 through DG_8 .
- (CL₆) The law-description operability of Definition 5.102,

$$\text{LawOperative}[\text{LawEvalDat}_{o,\zeta}]$$

and the differential-description operability of Definition 5.132,

$$\text{DiffOperative}[\text{DiffEvalDat}_{o,\zeta}]$$

hold.

Remark 8.84 (Classical subclasses). The classical regime of Definition 8.83 includes a local classical configuration with a single observer-side information-coupling position and a single object, and a classical many-body configuration in which multiple objects or multiple observer-side information-coupling positions share the same $\mathfrak{C}$ -derived clock domain. In the latter, the shared family of comparison clocks, the $\mathfrak{B}$ -derived object-state-side identification maps, and the direct-reception conditions hold for all objects and observer-side information-coupling positions.

Theorem 8.85 (Classical-time establishment theorem). Suppose that Ev , $\text{LawEvalDat}_{o,\zeta}$, and $\text{DiffEvalDat}_{o,\zeta}$ form a classical regime in the sense of Definition 8.83. Then $\text{Qu}_{\mathfrak{C},o}^{\text{dyn,can}}$ is generated, and

$$\text{OrdEmbSet}_o^{\text{dyn,time}}(\mathbb{R}) \neq \emptyset$$

For every

$$\rho_o \in \text{OrdEmbSet}_o^{\text{dyn,time}}(\mathbb{R})$$

, the law-admissible time representation

$$\begin{aligned} & \text{TimeRep}_o^{\text{dyn}}[\rho_o] \\ &= \rho_o \circ \text{Prj}_o^{\text{dyn,time,ord}} \end{aligned}$$

is determined. For any

$$T_0 \in \mathbf{Qu}_{\mathfrak{C},o}^{\text{dyn,can}}$$

, denote its conventional representation in existing classical physics by

$$t := \text{TimeRep}_o^{\text{dyn}}[\rho_o](T_0)$$

.

For the same real-valued temporal-order embedding, the observer-time law core

$$\text{LawCore}_{o,\zeta}^{\text{obs,time}}(\rho_o)$$

and the chart-compatible differential law

$$\text{ChartCompatDiffLaw}_{o,\zeta}(\rho_o)$$

are determined.

Proof. Conditions CL_4 and CL_5 of Definition 8.83 give

$$\begin{aligned} & \text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ & \wedge \text{DynCondAll}[\text{Ev}] \end{aligned}$$

. Definition 5.70 then gives $\text{DynOperative}[\text{Ev}]$. By Theorem 5.112, the $\mathfrak{C}$ -derived observer-time quotient is generated and the set of real-valued temporal-order embeddings is nonempty. Definition 5.63 then determines the law-admissible time representation and real-valued time for each real-valued temporal-order embedding.

Law-description operability in CL_6 of Definition 8.83 and Theorem 5.113 establish the observer-time law core. Differential-description operability in the same CL_6 and Definition 5.133 establish the chart-compatible differential law. $\square$

Corollary 8.86 (Classical time is a generated representation). The time variable t in a classical regime is obtained by composing exact comparison closure, the all-dynamical-description-conditions bundle, the $\mathfrak{C}$ -derived observer-time quotient, the temporal-order coordinate projection, and a real-valued temporal-order embedding.

Proof. By Theorem 8.85 and Definition 5.63,

$$t = \rho_o(\text{Prj}_o^{\text{dyn,time,ord}}(T_0)) .$$

$\square$

8.12 Validity domain and breakdown boundary of classical time

Definition 8.87 (Classical continuous-approximation regime). Take a full evaluation specification Ev , a law-evaluation datum $\text{LawEvalDat}_{o,\zeta}$ having Ev as its first component, and a differential-evaluation datum $\text{DiffEvalDat}_{o,\zeta}$ having that law-evaluation datum as its first component, together forming a classical regime in the sense of Definition 8.83.

Suppose that a refinement family of Definition 6.82, a nine-component family of defect quantities of Definition 6.84, and a nine-component family of tolerances of Definition 6.83 are given, and that monotone nondecrease in the sense of Definition 6.85 holds.

When the maximal effective interval of Definition 6.93 satisfies $\mathcal{I}^{\max}[\mathbf{Ev}] \neq \emptyset$, call this maximal effective interval the classical continuous-approximation regime of the full evaluation specification $\mathbf{Ev}$.

For each $\lambda \in \mathcal{I}^{\max}[\mathbf{Ev}]$ in this interval, membership in the scale domain of validity of continuous approximation of Definition 6.86, $\lambda \in \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}$, holds.

A classical continuous-approximation regime associated with a single observer-side information-coupling position and a single object is called a local classical continuous-approximation regime.

For multiple observer-side information-coupling positions or multiple objects, suppose that the full evaluation specification, law-evaluation datum, and differential-evaluation datum associated with each pair satisfy the six conditions of Definition 8.83, that each maximal effective interval is nonempty, and that $\mathbf{CL}_1$ of the same definition uses a $\mathfrak{C}$ -derived clock family common to all pairs. The family consisting of the respective maximal effective intervals is then called a shared classical continuous-approximation regime.

Theorem 8.88 (Classical continuous-approximation validity theorem). Suppose that $\mathbf{Ev}$, $\mathbf{LawEvalDat}_{o,\zeta}$, and $\mathbf{DiffEvalDat}_{o,\zeta}$ form a classical regime and that the classical continuous-approximation regime of Definition 8.87 is determined. Then, for every $\lambda \in \mathcal{I}^{\max}[\mathbf{Ev}]$, $\mathbf{CondAll}^{\text{approx}}[\mathbf{Ev}; \lambda]$ holds.

The objects generated from the full evaluation specification $\mathbf{Ev}$,

$$\begin{aligned} & \mathbf{Qu}_{\mathfrak{C},o}^{\text{dyn,can}}, \\ & \mathbf{TrjRel}_o^{\text{dyn,can}}, \\ & \mathcal{O}_{o,\zeta}^{\text{dyn,can}}, \\ & \mathbf{TimeRep}_o^{\text{dyn}} \end{aligned}$$

jointly satisfy the continuous-extension, order-stability, and relation-stability conditions at this scale.

Proof. Definition 8.87 gives $\lambda \in \mathcal{V}_{\mathbf{Ev}}^{\text{approx}}$. By Theorem 6.95, this membership is equivalent to simultaneous satisfaction of the nine conditions for continuous approximation. Conditions $\mathbf{CL}_4$ and $\mathbf{CL}_5$ of Definition 8.83 establish dynamical-description operability, and $\mathbf{CL}_6$ of the same definition establishes law-description operability and differential-description operability. The $\mathfrak{C}$ -derived observer-time quotient, trajectory relation, observable system, and law-admissible time representation are therefore generated from the assumed full evaluation specification, and the nine conditions for continuous approximation are evaluated at scale λ . $\square$

Corollary 8.89 (Generation and domain-of-validity guarantee of the classical continuous approximation). Continuous time, infinitesimal time intervals, smooth trajectory representations, and differential law descriptions in a classical continuous-approximation regime are generated from the full evaluation specification $\mathbf{Ev}$. The establishment of these representations is guaranteed at each scale in the maximal effective interval $\mathcal{I}^{\max}[\mathbf{Ev}]$, and this interval gives the maximal initial domain of establishment of the classical continuous approximation.

Proof. By Theorem 8.88, the $\mathfrak{C}$ -derived observer-time quotient, trajectory relation, observable system, and law-admissible time representation are generated from the full evaluation

specification. Continuous time, infinitesimal time intervals, smooth trajectory representations, and differential law descriptions are representations based on these generated objects.

By Definition 6.93 and Theorem 6.95, the nine conditions for continuous approximation hold at every scale in the maximal effective interval of the full evaluation specification. This interval is therefore the maximal initial domain guaranteeing the establishment of these representations. $\square$

Proposition 8.90 (Generation and domain-of-validity guarantee of a shared classical continuous-approximation regime). For each object and observer-side information-coupling position using a shared $\mathfrak{C}$ -derived clock, suppose that a classical continuous-approximation regime of Definition 8.87 is defined. A classical continuous-approximation time representation is then generated from each full evaluation specification. Its establishment is guaranteed at every scale in the respective maximal effective interval, and the family consisting of these intervals gives the shared classical continuous-approximation regime.

Proof. By Theorem 8.88, the $\mathfrak{C}$ -derived observer-time quotient, trajectory relation, observable system, and law-admissible time representation are generated from each full evaluation specification. Conditions CL_1 – CL_6 of Definition 8.83 establish a classical regime for each pair.

By Definition 6.93 and Theorem 6.95, the nine conditions for continuous approximation hold at every scale in each maximal effective interval. By Definition 8.87, the family of these intervals satisfying the shared $\mathfrak{C}$ -derived clock condition gives the shared classical continuous-approximation regime. Each time representation constituting the shared regime is therefore generated from its full evaluation specification, and its domain of establishment is guaranteed by the respective maximal effective interval. $\square$

Theorem 8.91 (First invalidation boundary of classical time). Let the full evaluation specification in a classical regime be $\text{Ev} \in \text{EvSpec}$, and suppose that a classical continuous-approximation regime of Definition 8.87 is defined. A classical continuous-approximation time representation is generated from Ev , and its establishment is guaranteed at every scale in $\mathcal{I}^{\max}[\text{Ev}]$.

For any $\lambda \in \Lambda$, if

$$\text{FailedTok}[\text{Ev}; \lambda] \neq \emptyset$$

, then

$$\begin{aligned} & \text{Min}_{\leq_{\text{fail}, \text{dep}}} \text{FailedTok}[\text{Ev}; \lambda] \\ &= \text{FailedTok}[\text{Ev}; \lambda]. \end{aligned}$$

Every member of this set of invalidation tokens belongs to one of the six failure series of Definition 6.10.

Furthermore, under the assumptions of Theorem 6.163, if $\Lambda \setminus \mathcal{V}_{\text{Ev}}^{\text{approx}}$ has a least element, the first approximation-invalidation scale $\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]$ exists, and

$$\begin{aligned} & \mathcal{I}^{\max}[\text{Ev}] \\ &= \{ \lambda \in \Lambda \mid \lambda < \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}] \}, \\ & I_{\text{approx}}^{\text{exc}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]] \neq \emptyset. \end{aligned}$$

The approximation-scale boundary datum is then

$$\{ (\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}], \Sigma_{\text{approx}}^{\text{bdry}}[\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]]) \}$$

For the local classical configuration and shared classical configuration of Remark 8.84, let the respective full evaluation specifications and scales be $(\mathbf{Ev}_0, \lambda_0)$ and $(\mathbf{Ev}_1, \lambda_1)$. If

$$\text{FailedTok}[\mathbf{Ev}_0; \lambda_0] \neq \text{FailedTok}[\mathbf{Ev}_1; \lambda_1]$$

, then the sets of minimal invalidation tokens of the two configurations are distinct.

Proof. By Theorem 8.88 and Corollary 8.89, the classical continuous-approximation time representation is generated from the full evaluation specification and its establishment is guaranteed at every scale in the maximal effective interval.

Definition 6.21 determines the set of all tokens invalidated at the scale in question. By Proposition 6.26, this set coincides with the set of minimal elements under the failure-generation dependence order. By Definition 6.10, each invalidation token belongs to the comparison, representation, continuous-approximation, dynamical-description, law-description, or differential-consistency failure series.

By Theorem 6.163, when the complement of the scale domain of approximation establishment has a least element, the first approximation-invalidation scale exists, and the maximal effective interval is the set of all scales smaller than this scale. The same theorem gives a nonempty set of approximation-excess components at the first approximation-invalidation scale. By Theorem 6.164, the ordered pair consisting of this scale and the approximation-boundary signature is the unique member of the approximation-scale boundary datum.

Each invalidation-token set for the local classical configuration and the shared classical configuration coincides with the respective set of minimal invalidation tokens. Therefore, if the two invalidation-token sets are distinct, extensionality gives distinct sets of minimal invalidation tokens. $\square$

For the following claims about the generation and validity of continuous-approximation representations, take a classical continuous-approximation regime as defined in Definition 8.87.

Connection to physical-time qualification. The generation sequence of Proposition 7.199 gives the generation order from nontemporal physical inputs through the representation, dynamical-description, law-description, and differential stages. In a classical regime, Corollary 8.89 generates a classical continuous-approximation time representation from the full evaluation specification.

By Corollary 7.15, the generation structure corresponding to the time content used by a physical-law time variable is established before law operations and result derivation. When this generation-precedence condition holds, Theorem 7.200 implies that the physical-law time variable using that time content satisfies physical-time qualification.

For a full evaluation specification and scale satisfying the assumptions of Theorem 8.91, Theorem 8.88 guarantees establishment of the generated classical continuous-approximation time representation at each scale in the respective maximal effective interval. By the same first-invalidation-boundary theorem, a nonempty invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order, and each member belongs to one of the six failure series.

Thus, the physical-time qualification of time content used by classical physical laws and the scale-indexed establishment guarantee of a generated classical continuous-approximation time representation are supplied by the generation-precedence condition

and the maximal effective interval, respectively, and invalidation is localized within the six failure series.

Corollary 8.92 (Establishment boundary of the classical-time continuous approximation). The conventional representation t of a classical time variable is determined from the generation sequence of Theorem 8.85. The classical continuous-approximation time representation is generated from the full evaluation specification by Corollary 8.89. For a full evaluation specification and scale satisfying the assumptions of Theorem 8.91, its establishment is guaranteed at every scale in the respective maximal effective interval of Definition 6.93, and a nonempty invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order; each member belongs to one of the six failure series of Definition 6.10.

Proof. By Theorem 8.85, the conventional representation t of a classical time variable is determined by applying a law-admissible time representation to a member of the generated $\mathfrak{C}$ -derived observer-time quotient.

By Theorem 8.88 and Corollary 8.89, the classical continuous-approximation time representation is generated from the full evaluation specification and its establishment is guaranteed at every scale in the maximal effective interval. By Theorem 8.91, the invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order, and each member belongs to one of the six failure series. $\square$

Proposition 8.93 (Generation, establishment guarantee, and invalidation localization in a classical regime). In a classical regime, Theorem 8.85 determines the conventional representation t of a classical time variable from the generation sequence. By Corollary 8.89, the classical continuous-approximation time representation is generated from the full evaluation specification. For a full evaluation specification and scale satisfying the assumptions of Theorem 8.91, its establishment is guaranteed at every scale in the respective maximal effective interval, and a nonempty invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order; each member belongs to one of the six failure series.

Proof. The generation sequence for the conventional representation of a classical time variable follows from Theorem 8.85. Generation of the classical continuous-approximation time representation and its establishment guarantee through the maximal effective interval follow from Corollary 8.89. Minimality of the invalidation-token set and membership in the six failure series follow from Theorem 8.91. $\square$

Corollary 8.94 (Generation, establishment guarantee, and invalidation localization of law time in classical physics). The conventional representation t of a time variable in a classical regime is determined, by Theorem 8.85, by applying a law-admissible time representation to a member of the generated $\mathfrak{C}$ -derived observer-time quotient. When the generation-precedence condition of Corollary 7.15 holds, Theorem 7.200 implies that the physical-law time variable using that time content satisfies physical-time qualification.

The continuous time, infinitesimal time intervals, smooth trajectory representations, and differential law descriptions used in classical physics are generated from the full evaluation specification by Corollary 8.89. For a full evaluation specification and scale satisfying the assumptions of Theorem 8.91, their establishment is guaranteed at every scale in the respective maximal effective interval. By Corollary 8.92, a nonempty invalidation-token set coincides with the set of minimal elements under the failure-generation dependence order, and each member belongs to one of the six failure series.

Proof. Generation of the conventional representation of a classical time variable follows from Theorem 8.85. Physical-time qualification of a physical-law time variable follows from Corollary 7.15 and Theorem 7.200. Generation and scale-indexed establishment guarantee of the classical continuous-approximation time representation follow from Corollary 8.89. Minimality of the invalidation-token set and membership in the six failure series follow from Corollary 8.92. $\square$

8.13 Difference from existing theories

The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism are mapped to representative schemes and to the stages of the generation pathway. Differences between the partial objects borne by these seven subjects and the generation hierarchy are expressed as hierarchical differences along the generation pathway through physical-time generation, scale-indexed validity domains, and six-series failure localization.

Clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism respectively make objects of temporal measurement, synchronization, geometrization, relationalization, standardization, and connection to observation. Differences in clock readings, clock rates, proper times, and frequencies on physical carriers are relativistic results belonging to their respective objects and relations. Statements in Bailey et al. 1977 and Pössel 2021 directly describe these physical-carrier relations[37, 18]. An interpretation path that places ungenerated time prior to the comparative difference and, in direct dependence on that placement, reattributes the comparative difference on physical carriers to a change in an attribute of that time constitutes a two-stage misrecognition in the sense of Proposition 8.23. The difference from the seven subjects lies in mapping these partial objects onto a single generation pathway from local displays, records, object-state differences, and observer-side information-coupling positions through comparison closure, the canonical comparison quotient, the representation stage, the dynamical-description stage, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

Clock synchronization, proper time in GR, relational time, observables, time metrology, and operational measurement have developed as distinct subjects[2, 4, 10, 11, 12, 7]. The generation pathway in this chapter maps the structures they address to the respective stages of time reference, the observer-time quotient, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

Table 19: Hierarchical differences between the seven subjects and the generation pathway

Subject	Partial object	Correspondence on the generation pathway	Hierarchical difference
Clock synchronization	Interlock comparison Signal exchange Convention of simultaneity	Comparison stage Part of the representation stage	• Connects $\mathfrak{C}/\mathfrak{D}/\mathfrak{B}/\mathfrak{D}$ to dynamical description • Connects to the observer-time law core • Connects to a scale-indexed validity domain and six-series failure localization
Relativistic time-stretching representation	Clock differences Coordinate-time differences Reattribution of proper-time differences to an attribute of time	Worked SR and GR reductions Comparative differences among generated time representations	• Preserves direct descriptions of physical-carrier relations • Evaluates time-stretching representations by interpretation path
Proper time in GR	Local-time display along a worldline	Input carrier or locally generated representation in the worked GR reduction	• Separates local proper-time displays • Separates global law-admissible time representations
Relational time	Relations among variables Ordering by an internal variable	Representation stage Part of the dynamical-description stage	• Stages records, comparison, and observer coupling • Stages the path from quotient descent through law representation
Quantum-theoretic time reconstruction	Temporal description without external time	Input-audit target in the worked QG reduction	• Requires generation conditions for the observer-time law core • Requires generation conditions for a law-admissible time representation
Time metrology	Standard frequency Time scale Unit convention	Worked reduction for time metrology	• Generates frequency representations from count differences • Uses generated coordinate differences for quantitative affine differences
Operationalism	Operational measurement Procedures Observables	Materials for input formation, the comparison stage, and the dynamical-description stage	• Connects to six-series failure localization • Connects to scale-indexed validity domains • Connects to the law-representation stage and differential-law-form stage

Proposition 8.95 (The difference from the seven subjects is hierarchical). The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism correspond to the comparison, representation, dynamical-description, law-representation, or differential-law-form stage of the generation pathway, to scale-indexed

validity-domain determination, or to six-series failure localization. The difference from the seven subjects is a hierarchical difference: their structures are mapped to stages of a single generation pathway and carried through physical-time generation, scale-indexed validity domains, and six-series failure localization.

Proof. Each row of Table 19 shows a structure addressed by one of the seven subjects and its correspondence on the generation pathway. Clock synchronization addresses interclock comparison and signal exchange and therefore corresponds to the comparison and representation stages. Statements in Bailey et al. 1977 and Pössel 2021 directly describe physical-carrier relations in which a moving clock or light clock bears the predicate of running faster or slower. Consider an interpretation path that places ungenerated time prior to a clock difference, coordinate-time difference, or proper-time difference. Suppose that reattribution of the subsequently obtained comparative difference to a change in an attribute of that time directly depends on that placement. Theorem 7.190 establishes that primitive-time roots remain in all 72 audited references. Every interpretation path of a recorded statement that contains placement before generation and attribute reattribution in direct dependence constitutes a two-stage misrecognition in the sense of Proposition 8.23. Proper time in GR is a local display along a worldline and corresponds to an input carrier or locally generated representation in Section 8.6. Relational time addresses relations among variables and therefore corresponds to the representation or dynamical-description stage. Quantum-theoretic time reconstruction addresses temporal description in formalisms without external time and is therefore an input-audit target in Section 8.8. Time metrology addresses counting, standard frequencies, time scales, and unit conventions and, in Section 8.3, generates frequency and period representations from count differences and generated coordinate differences. Operationalism addresses operational measurement and observables and therefore supplies materials for input formation and the comparison and dynamical-description stages.

The structures addressed by these seven subjects correspond to stages of the generation pathway. This correspondence supplies a generation hierarchy from local reception, records, object-state differences, and observer-side information-coupling positions through the canonical comparison quotient, representation coordinates, the observer-time law core, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization. The difference is therefore one of generation hierarchy and scope. $\square$

The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism correspond to stages on the generation pathway through comparison closure, the canonical comparison quotient, time representation, dynamical description, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

Corollary 8.96 (Reconstruction of the seven subjects within the generation pathway). The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in GR, relational time, quantum-theoretic time reconstruction, time metrology, and operationalism correspond to partial stages of the generation pathway or to inputs of representative schemes. Through this correspondence, each structure becomes a component of the generation pathway through physical-time generation, scale-indexed validity domains, and six-series failure localization.

Proof. By Proposition 8.95, the structures addressed by the seven subjects correspond to partial stages of the generation pathway or to inputs of representative schemes. Each

structure thereby becomes a component of the generation pathway through physical-time generation, scale-indexed validity domains, and six-series failure localization. $\square$

8.14 Theoretical structures retained as theory-specific inputs

In theories belonging to the comparison-realization class of Definition 8.14, equations of motion, interaction terms, field equations, quantization procedures, correction rules, and engineering implementations are retained as theory-specific inputs to the realized operational configuration of Theorem 8.16. From these theory-specific inputs, the generation pathway generates the establishment conditions for temporal comparison, time representation, dynamical description, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

The following table shows the correspondence between theoretical content retained as theory-specific input and objects constructed or determined on the generation pathway.

Table 20: Separation of theoretical structures retained as theory-specific input from generated objects

Retained object	Input position	Corresponding object on the generation pathway
Equation of motion	Theory-specific input	Law-admissible time representation
Field equation		
Interaction term		
Quantization procedure	QG-specific input	Canonical family of observables, trajectory relation, observer-time law core
Correction rule	Comparison-rule and record-side input	Determination of observer-time-law-core generation
Engineering implementation	Realization material	Comparison quotient, representation coordinates, navigation-time representation
		Canonical quotient, representation coordinates, scale-indexed validity domain, six-series failure localization

Equations of motion supply motion content borne on law-admissible time representations. Interaction terms supply motion content evaluated on canonical families of observables, trajectory relations, and observer-time law cores. Field equations supply field-law content and become generation inputs for the time representations bearing that content. Quantization procedures are theory-specific QG inputs used to determine whether an observer-time law core can be generated without taking external time as a primitive premise. Correction rules are generation inputs for comparison quotients, representation coordinates, and navigation-time representations. Engineering implementations supply realization materials for determining canonical quotients, representation coordinates, scale-indexed validity domains, and six-series failure localization.

Proposition 8.97 (Separation of theory-specific input from generated objects). In representative schemes belonging to the comparison-realization class of Definition 8.14, equations of motion, interaction terms, field equations, quantization procedures, correction rules, and

engineering implementations are retained as theory-specific inputs to the corresponding realized operational configuration. The objects constructed or determined on the generation pathway are comparison closure, the canonical comparison quotient, representation coordinates, the observer-time law core, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization. Existing theories' motion content is therefore retained as theory-specific input, while the establishment conditions and validity domain of the time variable used for that motion content are generated and its invalidation position is classified within the six series.

Proof. Conditions CR1 through CR7 of Definition 8.14 supply the carriers, comparison rules, record-side materials, object-state-side materials, and dynamical-description-side materials of representative theories as theory-specific inputs. By Theorem 8.16, these inputs determine the corresponding realized operational configuration.

By Corollary 8.18, determinations of operability and of the comparison, representation, continuous-approximation, dynamical-description, law-representation, and differential-law-form stages take this realized operational configuration as a domain-side input. Theorems 4.90 and 4.147 construct the objects from comparison closure through representation coordinates; Definitions 5.108 and 5.63 determine the observer-time law core and law-admissible time representation. Definition 6.4 determines the validity domain, and Definition 6.10 classifies invalidation positions within the six series.

Existing theoretical content is therefore retained as theory-specific input to the realized operational configuration, while the generation pathway constructs or determines the time variable's establishment conditions, validity domain, and invalidation positions. $\square$

Through the separation of theory-specific input from generated objects, equations of motion, interaction terms, field equations, quantization procedures, correction rules, and engineering implementations are retained as theory-specific inputs. While retaining this theoretical content as theory-specific input, the generation pathway constructs temporal comparison, time representation, dynamical description, a law-admissible time representation, a scale-indexed validity domain, and six-series failure localization.

Corollary 8.98 (Generation of time-variable establishment conditions while retaining existing theoretical content). For the existing theoretical content in Proposition 8.97, the generation pathway generates establishment conditions for using a time variable within laws, a scale-indexed validity domain, and six-series failure localization. Existing theoretical content holds as law content under a generated law-admissible time representation and scale-indexed validity domain; upon invalidation, its invalidation position within the six series and its structural burden are determined.

Proof. By Proposition 8.97, theoretical content is retained as theory-specific input to the realized operational configuration, while the time variable's establishment conditions, validity domain, and invalidation positions are constructed or determined on the generation pathway. Definition 6.4 determines the scale-indexed validity domain, while Definition 6.10 and Proposition 6.171 determine the invalidation position and structural burden. The claim follows. $\square$

Definition 8.99 (Result-preserving re-grounding). The construction that retains existing theoretical content as theory-specific input to the generation and audit construction of LCTR; retains existing results within their result-specific evidence-effect domains; invalidates the physical-time qualification of primitive-time premises; generates the required

time content and grounding structure from nontemporal physical inputs; audits their validity domains and invalidation positions; performs repair, regeneration, recomputation, or restriction of the domain of use upon invalidation; and reuses existing results within the intersection of the evidence-effect domain and the validity domain is called result-preserving re-grounding.

Figure 25 shows the retention pathway for existing theoretical content, the generation-and-audit pathway for temporal qualification, and the post-invalidation regeneration pathway.

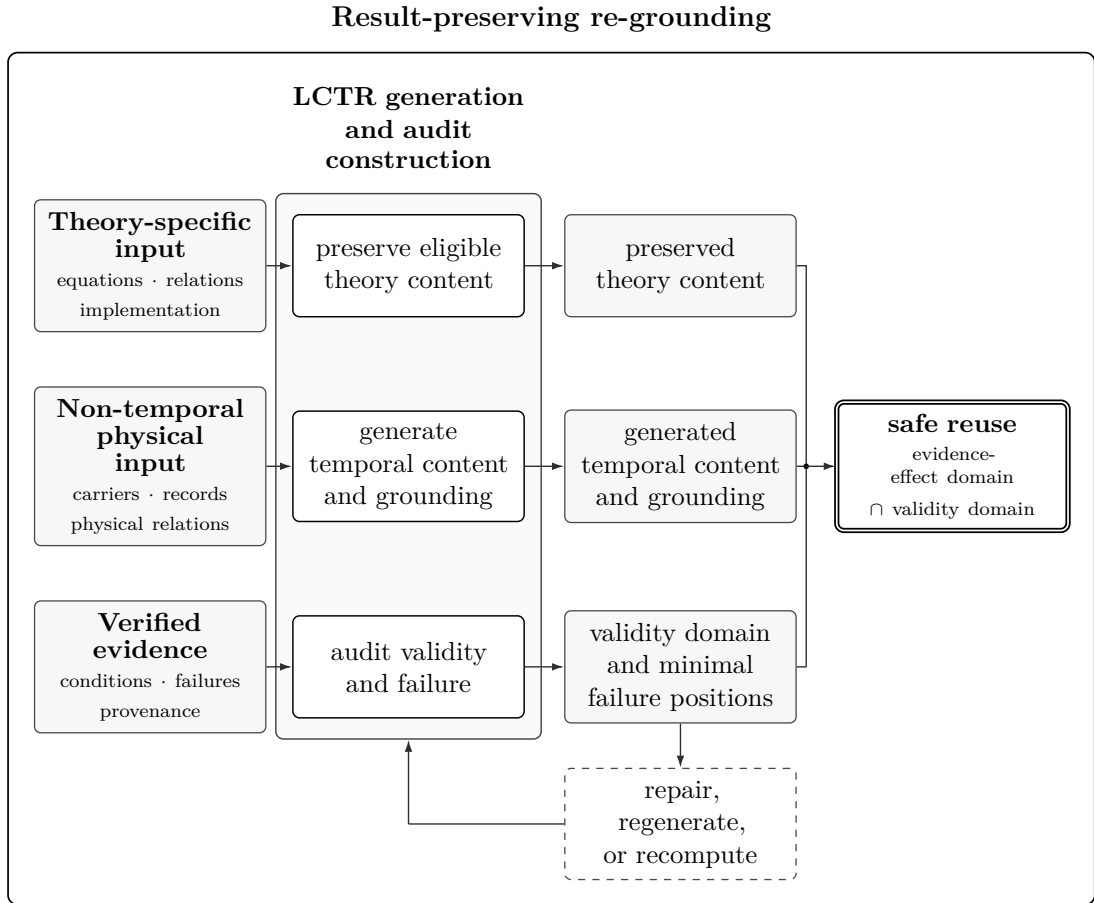


Figure 25: Result-preserving re-grounding in Definition 8.99. The retention pathway for existing theoretical content is separated from the generation-and-audit pathway for temporal qualification, and the safe domain of reuse is determined as the intersection of the evidence-effect domain and the validity domain. After invalidation, the construction is re-audited through repair, regeneration, and recomputation.

8.15 The typed generation-and-audit architecture of LCTR

LCTR integrates type separation, input–operation–output boundaries, staged generation, dependency order, qualification for use, audit states, invalidation localization, evidence provenance, and the scope of regeneration as conditions for the establishment of a physical theory. In LCTR, these principles used by the design methodology of computer science (CS) are mathematically defined as types, maps, generation closures, partial orders, validity domains, state sets, and evidence correspondences.

The CS-style hierarchy of this construction lies in reusing an encapsulated typed output as a single input item at the next level without re-expanding its internal construction, and then closing that item together with added components or conditions into one higher-level construction. In Chapters 2–3, role instances are encapsulated into a role-instance bundle family, raw operational input, and a realized operational configuration. In Chapter 4, the representation-stage generated-structure bundles accumulate recursively by head-preserving extension. In Chapter 5, qualification for dynamical, law, and differential description accumulates while retaining upstream qualification. In Chapter 6, the audit core and provenance certificate are encapsulated into operational audit data, which is then encapsulated with the state and diagnostic reports into the five-valued audit-report object.

Definition 8.100 (LCTR typed generation-and-audit architecture). Define the LCTR typed generation-and-audit architecture as the structure consisting of the following six structures and their dependency edges.

1. Type structure: Definitions 2.5, 2.10, and 3.2 separate roles, role instances, and physical-carrier realizations and assign types to each evaluation input.
2. Generation structure: Definition 4.171 and Theorems 4.90, 4.147, 5.112, 5.113, and 5.135 determine the generation order of comparison, time representation, dynamical description, law description, and differential form.
3. Qualification-for-use structure: Definitions 5.63 and 6.4 and Proposition 6.162 determine the qualification of generated objects for use within laws and their scale-indexed domains of establishment.
4. Invalidation-audit structure: Definitions 6.17, 6.10, and 6.179, Theorem 6.164, and Proposition 6.171 determine audit states, minimal invalidation positions, scopes of effect, and scopes of regeneration.
5. Theoretical-and-engineering evidence structure: Definitions 6.188 and 6.194 and Proposition 6.204 determine correspondences among theoretical conditions, engineering measurands, verified evidence, and invalidation positions.
6. Cross-reference audit structure: Definitions 7.89 and 7.90 and Theorem 7.190 determine correspondences between reference-specific formal and statement evidence and common audit determinations.

The output of each structure becomes the input of a subsequent structure as specified by the dependency edges.

Proposition 8.101 (Integrity of the typed generation-and-audit architecture). Under Definition 8.100, the following hold.

1. For each generated object, its input type, formation conditions, output type, and conditions for subsequent use are determined.
2. The physical qualification for use of time representations, dynamical descriptions, law time, continuous approximations, and differential forms is determined from upstream conditions and scale conditions.
3. Each audit state is determined as pass, fail, blocked, unformed, or indeterminate; upon invalidation, the minimal invalidation positions are determined.

4. The upward dependency set of an invalidation position determines the affected subsequent objects, structural burden, and scope of regeneration.
5. Theoretical conditions and engineering measurands retain distinct types, and verified evidence determines the state of the corresponding invalidation-position token.
6. Formal evidence and statement evidence from original sources retain reference-specific provenance, and the canonical finite audit realization determines the common audit determination.

LCTR therefore gives the input formation, object generation, qualification for use within laws, scale-indexed domains of establishment, invalidation states, evidence provenance, repair targets, and scope of downstream regeneration of a physical theory as a single typed dependency structure.

Proof. Item 1 follows from the type separation of roles, role instances, and physical-carrier realizations and from the input and output specifications of typed generation closure. Item 2 follows from the generation order of comparison, time representation, dynamical description, law description, and differential form and from the definitions of law-admissible time representations and scale-indexed domains of establishment.

Item 3 follows from the five-valued audit states and the first-breakdown theorem. Item 4 follows from the invalidation-dependency order, invalidation-position tokens, and structural-burden completeness. Item 5 follows from engineering evaluation specifications, engineering audit-parameter vectors, and formal-condition evidence localization. Item 6 follows from typed finite audit inputs, canonical finite audit realizations, and the all-reference primitive-time-root result.

The dependencies among types, generation, qualification for use, invalidation, evidence, and regeneration constitute the theoretical structure of CS design methodology in LCTR. The equations of physical theory, the types, dependencies, states, and change effects specified by CS design discipline, and engineering evidence jointly constitute this typed generation-and-audit architecture. $\square$

8.16 Worked classical many-body audit and background conditions of classical world representation

Proposition 8.102 (Backgrounding the establishment conditions for time reference in a classical world representation). A classical world representation explicitly includes point masses, centers of mass, shapes, moments of inertia, and rigid-body constraints as variables or relations in law expressions. At the same time, it does not describe the role instances, physical-carrier realizations, record-arrival conditions, observer-side information coupling, and joint law domain that establish time reference as variables, relations, generation conditions, or invalidation conditions in those expressions; instead, it uses them as presuppositions for describing multiple objects by the same time variable. This treatment is called the backgrounding of establishment conditions for time reference.

Proof. Definitions 2.10, 3.2, 6.190, 6.193, and 5.139 supply the role types, carriers, records, information coupling, and relational construction required for multi-object relations to hold physically.

A classical world representation does not include within its law description either a path by which these structures are generated from nontemporal physical inputs or conditions

under which use of the law ceases when those structures are invalidated. Nevertheless, it uses these structures as presuppositions for describing multiple objects by the same time variable. A classical world representation therefore supplies the establishment conditions for time reference in advance as background conditions outside the expressions. $\square$

Proposition 8.103 (Non-completion of background conditions by the success of classical physics and the scale-indexed validity domain). The mathematical consistency, predictive performance, and empirical fit of classical physics do not complete the physical generation of backgrounded establishment conditions for time reference. The domain in which multi-object relations hold physically under the same time variable belongs to the scale-indexed validity domain in which the physical-carrier realization of role instances, $\mathfrak{D}$ -derived record field of view, record stability, observer-side information coupling, $\mathfrak{D}$ -communication, multi-object relations, and scale-indexed defect quantities satisfy their respective establishment conditions.

Proof. By Proposition 7.13, mathematical consistency, predictive performance, and empirical fit add no generation dependence to time inputs.

Definition 6.194 supplies $\mathfrak{D}$ -derived record field of view, record invasiveness, the $\mathfrak{D}$ -communication graph, multi-object-relation defect, and common-law-domain defect as engineering audit inputs. Definition 6.161 supplies the scale domain of approximation establishment in which approximation defects do not exceed their tolerances. Definitions 6.191 and 6.205 supply the two-object recording configurations and their structural burdens relative to engineering evidence.

Each engineering audit result is connected to the establishment determination of the corresponding theoretical condition through verified engineering evidence for formal conditions. Evaluation specifications and scales for which all six series are established belong to the full evaluation-validity domain and give the physical domain of multi-object relations under the same time variable. The success of classical physics therefore supplies no ground for completing the physical generation of the establishment conditions for time reference while leaving them outside the expressions; the domain of that success is fixed as a scale-indexed validity domain satisfying the respective establishment conditions. $\square$

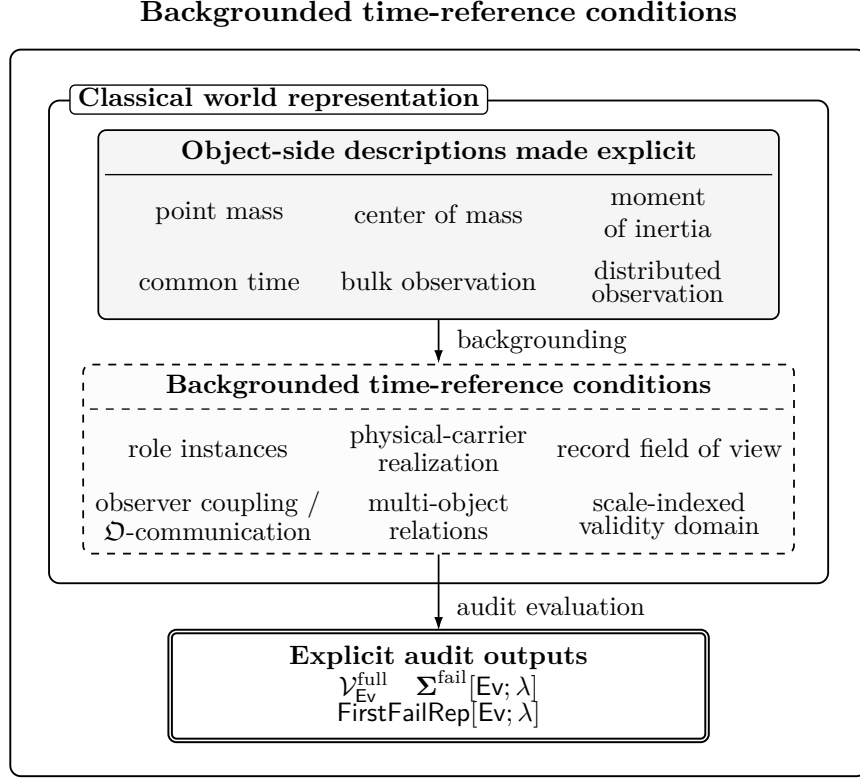


Figure 26: Object-side physical quantities made explicit by a classical world representation and the establishment conditions for time reference that it backgrounds outside the expressions. Multi-object descriptions using point masses, centers of mass, moments of inertia, and common time correspond to role instances, physical-carrier realizations, $\mathfrak{D}$ -derived record field of view, $\mathfrak{D}$ -communication, multi-object relations, scale-indexed validity domains, six-series invalidation signatures, and audit reports.

Table 21: Object-side structures made explicit by a classical world representation, backgrounded obligations for establishing time reference, and LCTR-generated objects

Classical input	Backgrounded establishment obligation	LCTR-generated object
Point-mass approximation	Object identity and interobject relations	Role-instance family, multi-object relations
Center of mass	Composite-object state	Physical-carrier realization, object-state quotient
Moment of inertia	Comparability of observables	Canonical family of observables, quantitative affine realization
Common time t	$\mathfrak{C}/\mathfrak{D}/\mathfrak{D}$ comparison, recording, and coupling	Comparison closure, order-coordinate structure, observer-side coupling
Bulk observation	$\mathfrak{D}$ -record field of view, cell width, and stability	Bulk-recording condition, recording structural burden
Distributed observation	$\mathfrak{D}$ -communication and comparison-loop consistency	Following-recording condition, recording structural burden

Definition 8.104 (Classical multi-object audit example). Let two distinct object-specification indices and the finite family of object-specification indices be

$$\zeta_1, \zeta_2 \in Z_0, \quad \zeta_1 \neq \zeta_2, \quad Z_0 = \{\zeta_1, \zeta_2\}$$

, and let the relation index be $\rho \in \mathbf{JRelIdx}_{Z_0}$.

For each $a \in \{0, 1\}$ and $k \in \{1, 2\}$, let the full evaluation specification be

$$\mathbf{Ev}_k^{(a)} = ((\mathcal{X}^{(a)}; \mathcal{J}_{\text{cmp}}), o, \zeta_k)$$

, and assume

$$(\mathbf{Ev}_1^{(a)}, \mathbf{Ev}_2^{(a)}) \in \mathbf{EvSpec}^{\text{pair}}$$

. For the full evaluation specifications of following-recording type, assume

$$(\mathbf{Ev}_1^{(1)}, \mathbf{Ev}_2^{(1)}) \in \mathbf{EvSpec}_{\mathfrak{D}}^{\text{pair, distributed}}$$

.

The two recording configurations satisfy

$$\mathbf{BulkRecord}_{\mathfrak{D}} \left[\left(\mathbf{Ev}_k^{(0)} \right)_{k=1}^2 \right] = \top$$

and

$$\mathbf{FollowingRecord}_{\mathfrak{D}} \left[\left(\mathbf{Ev}_k^{(1)} \right)_{k=1}^2 ; \lambda \right] = \top$$

.

For each configuration $a \in \{0, 1\}$, suppose that the multi-object relation input data of Definition 5.139 are supplied and satisfy the representative-invariance condition of Definition 5.142. Denote the corresponding quotient-level datum by $\mathbf{J}_{o, \rho, Z_0}^{(a), \text{can}}$. For the same object-specification set Z_0 and relation-type index ρ , audit each quotient-level relation $\mathbf{J}_{o, \rho, Z_0}^{(a), \text{can}}.\text{Rel}$ under its corresponding recording configuration. Call these two configurations the classical multi-object audit example.

Bulk-record / following- record audit comparison

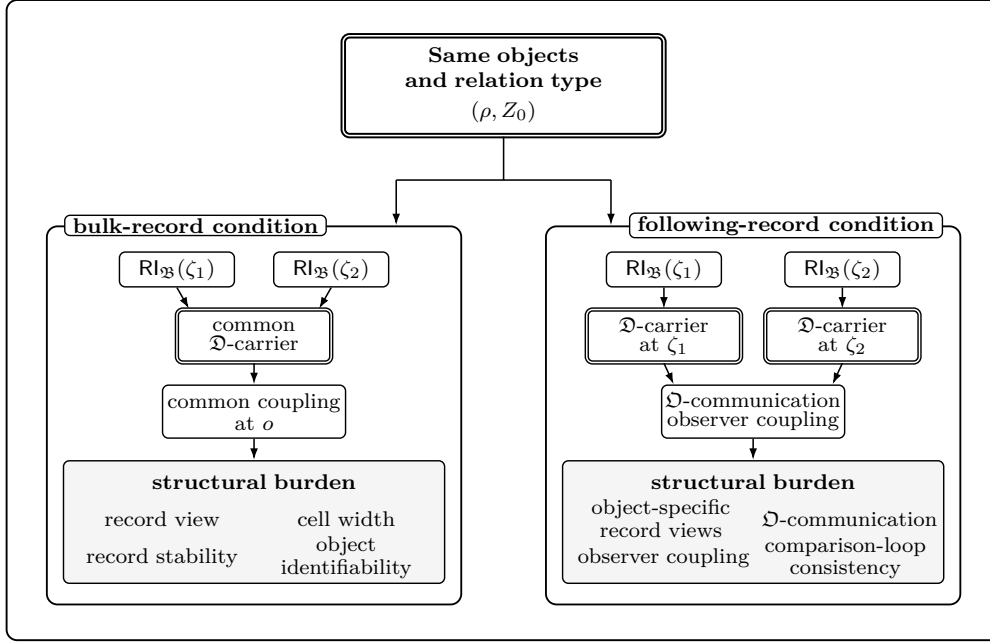


Figure 27: Structural burdens relative to engineering evidence for the bulk-recording and following-recording conditions. For the same object-specification set and relation index, the bulk-recording condition requires record field of view, cell width, record stability, and object identifiability in a single $\mathfrak{D}$ -carrier. The following-recording condition requires $\mathfrak{D}$ -communication, observer-side coupling, and comparison-loop consistency in addition to object-specific record fields of view in distinct $\mathfrak{D}$ -carriers.

Table 22: Audit inputs for the bulk-recording and following-recording conditions

Recording configuration	$\mathfrak{D}$ -carrier condition	Additional audit inputs
Bulk-recording condition	Single $\mathfrak{D}$ -carrier	Record field of view, cell width, record stability, object identifiability
Following-recording condition	Distinct $\mathfrak{D}$ -carriers	Object-specific record fields of view, $\mathfrak{D}$ -communication, observer-side coupling, comparison-loop consistency

Proposition 8.105 (Classical multi-object audit of the bulk-recording condition). The invalidation-position set for the bulk-recording condition in the classical multi-object audit example is $\text{RecordBurdenTok}^{(0)}$, and its structural burden relative to engineering evidence is $\text{RecordBurden}^{\text{bulk}}$.

Proof. By Definition 6.191, the bulk-recording condition requires the two $\mathfrak{D}$ -derived role instances to be realized by the same physical carrier. For the engineering evidence of the bulk-recording condition, Definition 6.205 determines the invalidation-position set in the proposition from the respective input tags for record field of view, recording configuration, record invasiveness, $\mathfrak{C}$ -count increment, $\mathfrak{D}$ -record stability, $\mathfrak{D}$ -record cell width, and $\mathfrak{B}$ -identifiability, and gives the structural burden relative to engineering evidence in the proposition as the union of their dependency-upward structural images. $\square$

Proposition 8.106 (Classical multi-object audit of the following-recording condition). The invalidation-position set for the following-recording condition in the classical multi-object audit example is $\text{RecordBurdenTok}^{(1)}$, and its structural burden relative to engineering evidence is $\text{RecordBurden}^{\text{follow}}$.

Proof. By Definition 6.191, the following-recording condition requires the two $\mathfrak{D}$ -derived role instances to be realized by distinct physical carriers and each object-specific record field of view to be contained in the domain of the corresponding record-code relation. For the engineering evidence of the following-recording condition, Definition 6.205 adds the $\mathfrak{D}$ -communication and observer-side-coupling input tags to the input tags of the bulk-recording condition to determine the invalidation-position set in the proposition, and gives the structural burden relative to engineering evidence in the proposition as the union of their dependency-upward structural images. $\square$

Proposition 8.107 (Difference in structural burden between the bulk-recording and following-recording conditions). The two structural burdens relative to engineering evidence in the classical multi-object audit example are subsets of StructTok , and

$$\begin{aligned} \text{RecordBurden}^{\text{bulk}} &= \text{RecordBurden}^{\text{follow}} \\ \iff \text{RecordBurden}^{\Delta} &= \emptyset. \end{aligned}$$

If both set differences

$$\text{RecordBurden}^{\text{bulk}} \setminus \text{RecordBurden}^{\text{follow}}$$

and

$$\text{RecordBurden}^{\text{follow}} \setminus \text{RecordBurden}^{\text{bulk}}$$

are nonempty, the two structural burdens are incomparable under inclusion.

Proof. Applying Proposition 6.206 to the bulk-recording and following-recording conditions of Definition 8.104 gives the asserted equivalence and incomparability condition. $\square$

Connection to the audit of law-time use. The two recording configurations in the classical multi-object audit example supply full evaluation specifications to the physical-time-qualification audit in Section 7. For the full evaluation specifications corresponding to the two recording configurations, when the generation dependencies and operability conditions of Proposition 7.199 hold prior to law operations and result derivation, Theorem 7.200 implies that the physical-law time variable bearing the time content used there satisfies physical-time qualification.

If time content is entered into a law operation or result derivation before the generation structure required for its use has been established, that time content is a primitive-time input under Definition 7.10. By Theorem 7.203, the physical-law time variable bearing that time content does not satisfy physical-time qualification. Invalidation of engineering evidence in each recording configuration is localized as the recording-configuration-specific invalidation-position set of Definition 6.205, the six-series invalidation signature of Definition 6.159, and the structural-burden difference of Proposition 8.107.

Configuration-specific outputs of the classical multi-object audit

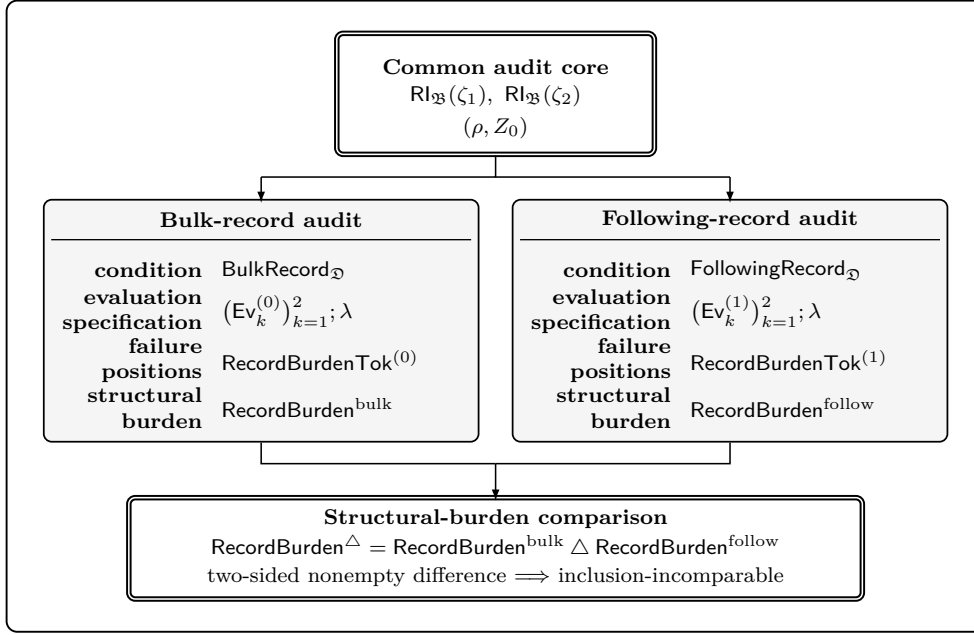


Figure 28: Configuration-specific outputs of the classical multi-object audit example. Role instances, full evaluation specifications, $\mathfrak{D}$ -carrier conditions, record evidence, $\mathfrak{D}$ -side evidence, and the multi-object relation are supplied as inputs, and the structural burdens relative to engineering evidence for the bulk-recording and following-recording conditions are compared.

Table 23: Configuration-specific outputs of the classical multi-object audit example

Audit component	Bulk-recording condition	Following-recording condition
Role instances	$RI_{\mathfrak{B}}(\zeta_1), RI_{\mathfrak{B}}(\zeta_2)$	
Full evaluation specifications	$Ev_1^{(0)}, Ev_2^{(0)}$	$Ev_1^{(1)}, Ev_2^{(1)}$
$\mathfrak{D}$ -carrier	Common physical carrier	Distinct physical carriers
Record evidence	Shared record field of view and shared code relation	Object-specific record fields of view and object-specific code relations
$\mathfrak{D}$ -side evidence	Common information-coupling position o	$\mathfrak{D}$ -communication datum and observer-side coupling
Multi-object relation	$J_{o,\rho,Z_0}^{(0),can} \cdot Rel$	$J_{o,\rho,Z_0}^{(1),can} \cdot Rel$
Failure output	Bulk-recording structural burden	Following-recording structural burden

8.17 Final synthesis

The final synthesis theorem integrates, into a single dependence order, generation from the comparison stage through the differential-description stage, the scale-indexed validity domain of the continuous-approximation stage, invalidation localization in the six failure series, five-valued audit reports, the cross-source results of the 72 audited sources, worked reductions of representative theories, the classical multi-object audit, differences from

existing theories, result-preserving re-grounding, and the typed generation-and-audit architecture. The following table maps each synthesis item to its direct grounds and outputs.

Table 24: Direct grounds and outputs of the final synthesis theorem

Synthesis item	Direct grounds	Output
Synthesis (1): Comparison closure	Theorem 4.90	$L_{\leq 7}[Ev^{cmp}]$, $G[Ev^{cmp}]$, $\mathfrak{CD}$ -derived canonical comparison quotient
Synthesis (2): Representation generation	Theorems 4.147, 4.177, 4.179, and 4.187	$\text{Repr}_{\leq 3}(\mathcal{X}; \mathcal{J}_{cmp})$, exact operational time structure, local and global order-coordinate structures, comparison-order atlas
Synthesis (3): Dynamical-description generation	Theorems 5.112 and 5.138	$\text{DynCondAll}[Ev]$, observer-time quotient, trajectory relation, graph trajectory, canonical family of observables, law-admissible time representation
Synthesis (4): Law and differential generation	Theorems 5.113, 5.135, and 7.203	Observer-time law core, generated representationality of law-evaluation time, chart-compatible differential law, differential covariance under changes of time representation
Synthesis (5): Six failure series	Definition 6.10 and Theorem 6.168	Six series of comparison, representation, dynamical description, continuous approximation, law description, and differentiation; invalidation positions; six-series invalidation signature
Synthesis (6): Scale boundary and audit	Propositions 6.162 and 6.154, Theorems 6.163, 6.164, and 6.183, and Proposition 6.199	Full evaluation-validity domain, dual qualification of continuous approximation and differential form, maximal effective interval for approximation, approximation-invalidation scale, approximation-boundary signature, five-valued audit-report object, structural-burden audit component, typed separation of failure causes

Continued on next page

Synthesis item	Direct grounds	Output
Synthesis (7): 72 audited sources and representative theories	Theorem 7.190, Propositions 7.191, 8.73, 8.74, Corollary 8.75, Propositions 8.59 and 8.77, Theorem 8.16, and the respective worked reductions	Universal persistence of primitive-time roots, counts of time-generation obligations, target consistency between time-nonexistence statements and time-attribute statements, generation conditions, validity domains, and invalidation positions of representative theories
Synthesis (8): Theoretical differ- ence and result-preserving re-grounding	Propositions 8.95 and 8.97, Definition 8.99, Propositions 8.102, 7.121, 8.78, and 8.79, Definitions 3.2 and 6.188, and Proposition 6.204	Result-preserving re-grounding, objects retained from existing theories, generation obligations for time-reference conditions, scale-indexed domain of establishment of multi-object relations under a single time variable, causal connection from the backgrounding of grounding obligations through actual invalidation and downstream reach, separation between the invalidation of physical-time qualification and conditional invalidation of use of downstream computational results, scope of regeneration and recomputation after repair, range of safe reuse under generation conditions, typed applicability conditions for engineering implementations
Synthesis (9): Typed generation and audit structure	Definition 8.100 and Proposition 8.101	Typed dependence structure of physical theory, use qualification, audit states, invalidation localization, evidence provenance, range of effects, scope of repair and regeneration

Continued on next page

Synthesis item	Direct grounds	Output
Synthesis (10): Final conse- quence	Synthesis (1)–Synthesis (9)	Constructive definition of physical time, generation of temporal identity and before–after relations, conditional representation by linear order and real-valued time, unique re-expression of time-coordinate freedom, establishment obligations for multi-object relations under a single time variable, separation of the obligations for temporal order and causal order, generation and maximal validity domain of classical continuous time, formal obligations for statements of time creation, emergence, and nonexistence, change of the problem of time in quantum gravity into a problem of law-time use qualification, changes in field-specific temporal obligations, qualification invalidation of primitive-time premises, six-series invalidation localization, audit report

The following final synthesis theorem therefore gives the generation, establishment, invalidation, and audit of time variables in the same dependence order from the direct grounds in Table 24.

Collapse of the Mystery Basis

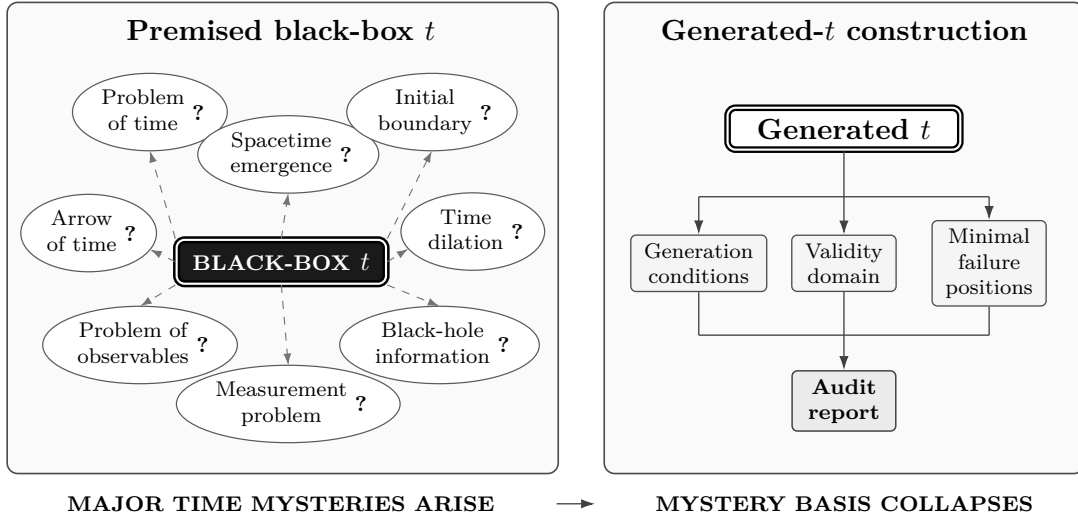


Figure 29: A time variable t presupposed as a black box without a generation path supplies the basis on which the problem of time, time stretching, initial boundaries, the arrow of time, the problem of observables, the measurement problem, the black-hole information problem, and spacetime emergence appear as mutually independent mysteries. Once time content and its grounding structure are generated from nontemporal physical inputs, the basis on which these mysteries are constituted is lost, and each problem is transferred to generation conditions, scale-indexed validity domains, the six failure series, and audit reports.

Definition 8.108 (Time-problem connection domain). A time-problem connection domain is a family of problem descriptions that appear as independent mysteries because an ungenerated time variable, temporal order, or time representation is entered in advance into law operations or result derivations, and that are connected to the generation conditions, scale-indexed validity domain, six failure series, or audit report of the corresponding time content and grounding structure.

The difficulty of defining time in Definition 1.1 is made concrete by Theorem 7.203 as the requirement for the physical grounds of physical-law time variables. When comparison closure, the $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient, the $\mathfrak{C}$ -derived observer-time quotient, the $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived trajectory relation, the observer-time law core, and a law-admissible time representation hold along the same generation dependence, a time variable is established as a generated law-admissible time representation.

Comparison closure and the canonical comparison quotient give the conditions under which two physical materials may be identified as belonging to the same time content. The canonical partial order preserves before–after relations and incomparability, and the quotient by incomparability gives a strict linear order when its transitivity condition holds. Real-valued time is obtained as a representation when an order embedding from that ordered quotient into the real numbers exists. Accordingly, LCTR specifies the real line and linear order as downstream representations of a time structure whose generation conditions have been satisfied. Freedom in the choice of an admissible order representation is fixed as re-expression through a unique strict order isomorphism between the representation images. This is LCTR’s answer to the problem that Feynman presented as the difficulty of giving time a satisfactory definition[6]. LCTR changes the definition of physical time

from the specification of lexical meaning to the generation of conditions of establishment and generates, from nontemporal physical inputs, the conditions for temporal identity, before–after relations, comparability, linearization, real-valued representation, use within laws, and invalidation.

For a claim that uses continuous approximation, its range of establishment is determined as a scale-indexed validity domain, and invalidation is localized by the six failure series, approximation-boundary signatures, and five-valued audit reports.

In the following figure, the horizontal axis presents the structural reach borne by a use of time in the order local, multiple records, multiple observers, and gluing/boundary; the vertical axis presents the stages at which grounding obligations arise in the order comparison, representation, continuous approximation, dynamical description, law description, and differentiation. A large filled circle denotes a principal localization burden; a small filled circle, an additional direct burden; a diamond, a scale condition; an open circle, a derived or downstream application; a gray horizontal line, a structural range; a thin line, the connection among multiple burdens borne by the same object; and a square, engineering evidence. Positional differences within a band are used for visual discrimination. The engineering-evidence band at the bottom represents verified engineering inputs connected to the six theoretical stages.

The abbreviations correspond to the following constructions or audits in the text. C.Min. denotes the minimal operational model of classical time (8.11.1); T.Met., the worked reduction of time metrology (8.3); SR, the worked reduction of special relativity (8.5); C.Dom., the validity domain and failure boundary of classical time (8.12); GPS, the worked reduction of GPS (8.4); GR, the worked reduction of general relativity (8.6); and QG, quantum-gravitational reconstruction (8.8). I.Bd., connected to QG, denotes the downstream application to a cosmological initial boundary (8.8.3); B.H., that to the extreme-gravity domain including the black-hole information problem (8.8.4). G/F and C.Multi. correspond to bulk recording and following recording and to the classical multi-object audit example (8.16).

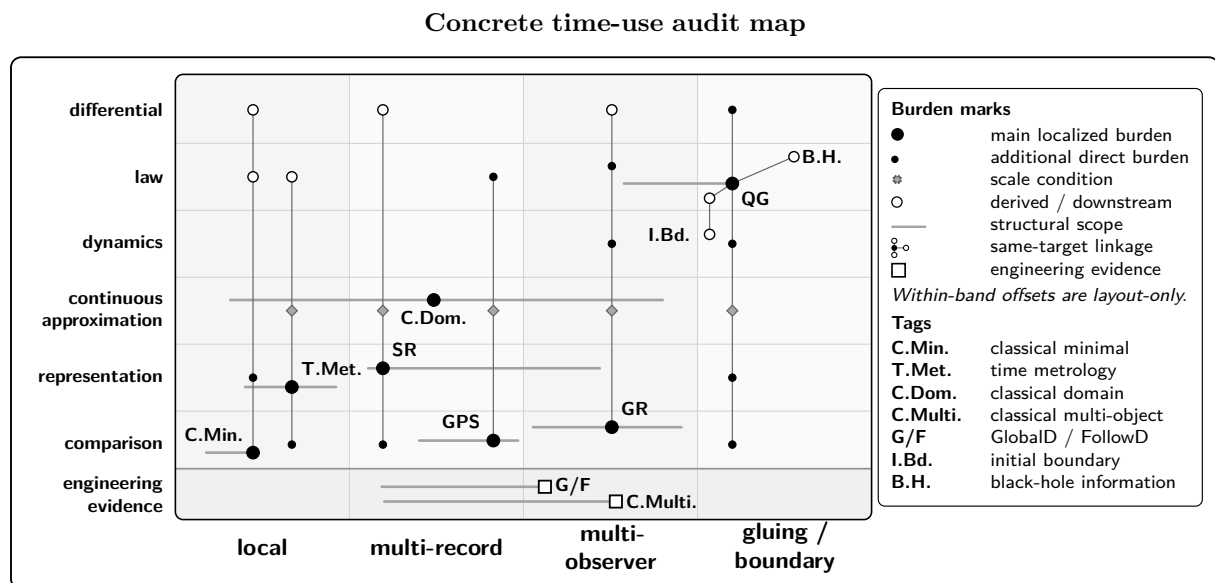


Figure 30: Correspondence between structural reach and generation-and-audit burdens in representative uses of time.

Theorem 8.109 (Final synthesis theorem). For a realized operational configuration $\mathcal{X}$, a typed family of comparison rules $\mathcal{J}_{\text{cmp}}$, an observer-side information-coupling position $o \in U$, an object-specification index $\zeta \in Z$, a complete evaluation specification

$$\text{Ev} = ((\mathcal{X}; \mathcal{J}_{\text{cmp}}), o, \zeta) \in \text{EvSpec}$$

and an approximation scale $\lambda \in \Lambda$, the following hold.

Synthesis (1): comparison closure. Operability at the comparison stage is given by

$$\begin{aligned} \text{Operative}(\mathcal{X}) &\iff \text{L}_{\leq 7} [\text{Ev}^{\text{cmp}}] \\ &\quad \wedge \text{G} [\text{Ev}^{\text{cmp}}] \end{aligned}$$

Under this condition, the extended comparison-arrival equivalence relation, the $\mathfrak{C}/\mathfrak{D}$ comparison-quotient sets, the canonical projections, the quotient-level $\mathfrak{C}\mathfrak{D}$ -derived relation, and the $\mathfrak{C}\mathfrak{D}$ -derived canonical comparison quotient are determined uniquely in dependence order.

Synthesis (2): representation generation. Exact operability is given by

$$\begin{aligned} &\text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \\ &\iff \text{Operative}(\mathcal{X}) \wedge \text{Repr}_{\leq 3}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \end{aligned}$$

Under this condition, the $\mathfrak{C}/\mathfrak{D}$ canonical partial orders, exact operational time structure, local and global order-coordinate quotients, local and global exact linear orders, canonical order-coordinate projections, local-to-global order-coordinate embeddings, and comparison-order atlas are generated in dependence order. Once a selected representation range and an order embedding of that range are specified, the $\mathfrak{C}/\mathfrak{D}$ -derived representation coordinates factor through the canonical global order-coordinate projection.

Synthesis (3): dynamical-description generation. Dynamical-description operability is given by

$$\begin{aligned} &\text{DynOperative}[\text{Ev}] \\ &\iff \text{ExactOperable}(\mathcal{X}; \mathcal{J}_{\text{cmp}}) \wedge \text{DynCondAll}[\text{Ev}] \end{aligned}$$

Under this condition, the $\mathfrak{C}$ -derived observer-time quotient, the $\mathfrak{B}$ -derived object-state quotient, the $\mathfrak{C}\mathfrak{D}\mathfrak{B}$ -derived trajectory relation, a graph trajectory, a canonical family of observables, the observer-time order-coordinate quotient, a real-valued time representation, a scalar-time trajectory, and a family of real-valued observable curves are generated.

Synthesis (4): law and differential generation and the physical establishment of time variables. Law-description operability and differential-stage operability are given by

$$\begin{aligned} &\text{LawOperative}[\text{LawEvalDat}_{o,\zeta}] \\ &\iff \text{DynOperative}[\text{Ev}] \wedge \text{LawCondAll}[\text{LawEvalDat}_{o,\zeta}], \\ &\text{DiffOperative}[\text{DiffEvalDat}_{o,\zeta}] \\ &\iff \text{LawOperative}[\text{LawEvalDat}_{o,\zeta}] \wedge \text{DiffCondAll}[\text{EvalStrFam}_{o,\zeta}^{\text{diff}}] \end{aligned}$$

The former establishes the observer-time law core and the generated representationality of law-evaluation time; the latter establishes chart-compatible differential laws, value-jet-relation covariance, and time-representation-change jet-relation covariance.

The law-admissible time representation is generated as

$$\text{TimeRep}_o^{\text{dyn}} = \text{RepEmb}_o^{\text{dyn,time}} \circ \text{Prj}_o^{\text{dyn,time,ord}}$$

By Theorem 7.203, a time variable in a physical law is not established as a primitive premise. When time content and its grounding structure are generated from nontemporal physical inputs before law operations and result derivations, the time variable satisfies physical-time qualification.

Synthesis (5): invalidation localization in the six failure series. The six series of comparison, representation, dynamical description, continuous approximation, law description, and differentiation are distinguished by

$$\text{FailSeries} = \{\text{cmp, repr, dyn, law, diff}\} \sqcup \{\text{approx}\}$$

Invalidation of a formed and evaluable condition is localized to the corresponding set of invalidation-position tokens $\text{FailedTok}[\text{Ev}; \lambda]$ and gives the six-series invalidation signature $\Sigma^{\text{fail}}[\text{Ev}; \lambda]$. Because multiple invalidation positions can hold in the same evaluation, invalidations in the six series do not require an exclusive disjunction. Unformed and unevaluable states are not replaced by invalidation; they are retained as their respective audit states.

Synthesis (6): scale-indexed validity domain, boundary, and five-valued audit.

Under the all-scale approximation-input admissibility and defect-monotonicity assumptions of Theorem 6.163, the approximation-validity scale domain $\mathcal{V}_{\text{Ev}}^{\text{approx}}$ is an initial subset of the totally ordered set of scales and coincides with the maximal effective interval for approximation $\mathcal{I}^{\text{max}}[\text{Ev}]$. If the complement has a least element, the approximation-invalidation scale $\lambda_{\text{bdry}}^{\text{approx}}[\text{Ev}]$ and approximation-boundary signature

$$\Sigma_{\text{approx}}^{\text{bdry}} [\text{Ev}; \lambda_{\text{bdry}}^{\text{approx}} [\text{Ev}]]$$

are determined uniquely.

The five-valued audit-report object

$$\text{AuditReport} [\text{Ev}^{\text{fail}}; \lambda]$$

is determined uniquely from operational audit data and gives the series-specific audit-group states, condition-specific audit states, invalidation positions, six-series invalidation signature, indeterminate tokens, indeterminate signature, approximation boundary, structural burden, and reason for an undefined audit.

When a differential equation that uses continuous approximation is employed as a physical description, its use qualification is given by the full evaluation-validity domain in which both the five exact series—comparison, representation, dynamical description, law description, and differentiation—and the continuous-approximation series hold. The continuous-approximation series evaluates the chain from finite physical realization through continuous extension, order stability, and relation stability; the differential series evaluates the differential atlas, generated-trajectory jets, local jet relations, and jet-relation covariance. The input types of the two series are distinguished, and both qualifications are satisfied simultaneously in the full evaluation-validity domain.

The five-valued audit states separate satisfaction, condition invalidation, blockage by a prior condition, an unformed state, and indeterminacy. Engineering-implementation

auxiliary-audit flags are connected, through verified auxiliary-audit-effect evidence, to problems concerning formation inputs, evaluation inputs, input admissibility, boundary treatment, or condition evidence. Numerical-solver failure and data gaps are retained as engineering auxiliary-audit causes and have a type distinct from condition invalidation of a target invalidation-position token.

Synthesis (7): the 72 audited sources and representative theories. For the 72 audited sources,

$$\text{Ref}^{\text{audit,root}} = \text{Ref}^{\text{audit}}$$

holds, and every source contains a primitive-time root and results that depend on a primitive-time root. Under the complete criterion, the generated-time-content set is empty for all 72 sources, and no source generates, from nontemporal physical inputs, the entirety of the time content it uses. Under the local-reach criterion, 10 sources reach a total of 57 temporal contents. The strict generated-time-content sets of all 72 sources are empty, and the entire source-specific time content consists of primitive-time roots. In the remaining 62 sources, no time content is reached even under the local-reach criterion. Sources that explicitly state the absence of external or independent time, the fundamental status of time, difficulty defining the concept of time, or an internal or relational construction of time number (18/72); sources that present a dynamical construction or recovery of intrinsic, relational, thermal, or semiclassical time number (13/72); and among the latter, sources that satisfy the complete time-generation obligation number (0/13).

Newtonian and classical mechanics, time metrology, GPS, special relativity, general relativity, and quantum-gravitational reconstruction are placed within the same generation dependence, conditions of establishment, scale-indexed validity domains, and six failure series. The minimal classical model and the classical multi-object audit example give this placement as a concrete construction and audit. Relativistic clock differences hold as comparison differences among generated law-admissible time representations on physical carriers of $\mathcal{CQBD}$. Statements that directly describe comparison differences on physical carriers are retained as relativistic results within the evidence-effect domains of those results. An interpretation path in which reattribution of comparison differences on physical carriers to changes in an attribute of time itself directly depends on prior placement of ungenerated time constitutes a two-stage misrecognition by Definition 8.19. By Proposition 8.77, statements that assert the nonexistence of time are incompatible with statements that attribute stretching, rate, flow, or deformation to time itself when both target the same time-content type and the same application domain. When they target distinct time-content types or application domains, their separation and transition must be stated; a transition of the target to a clock-indication difference, clock-rate difference, proper-time difference, coordinate difference, or comparison difference between generated time representations does not entail a change in an attribute of “time itself” independent of physical carriers. Proposition 8.62 shows that relativity, relationality, nonprivilege, nonabsoluteness, or nonprimitivity does not imply the nonexistence of time. A theory that asserts the creation, emergence, recovery, or nonexistence of time bears the general formal obligations of Proposition 8.63. For Barbour 1994 I and II, Proposition 8.64 retains, as formal results, an interpretation restricted to a specified configuration construction, the nonadoption of absolute external time, and the kinematical priority of relative configuration. An interpretation that transfers time creation to complete generation of physical time constitutes a two-stage misrecognition, consisting of the use of primitive-time roots before generation and identification of a partial construction with complete generation; an interpretation that transfers a restricted nonexistence of time to the nonexistence of

physical time in general constitutes a one-stage misrecognition. For Connes–Rovelli 1994, Proposition 8.65 retains, as restricted results, the selection of a state-dependent flow and the assignment of a physical-time interpretation to that flow. An interpretation that transfers a partial flow to complete establishment of physical time constitutes a one-stage misrecognition; an interpretation that treats a construction using time roots before generation as complete generation from nontemporal inputs constitutes a two-stage misrecognition. For Kiefer 2009, Proposition 8.66 retains, as restricted results, the construction of a semiclassical time parameter and internal-clock candidates. An interpretation that transfers a partial construction to complete establishment of physical time constitutes a one-stage misrecognition; an interpretation that treats a semiclassical construction using time roots before generation as complete generation from nontemporal inputs constitutes a two-stage misrecognition. For Smolin 2018/2020, Proposition 8.67 retains, as a classification, an interpretation restricted to the theoretical priority or irreducibility of time. An interpretation under which time content before generation can be used as the grounding of a physical law or as a law input constitutes a one-stage misrecognition. For this source, neither the local-reach determination nor the complete determination yields any generated temporal content. All 27 temporal contents are primitive-time roots, and 21 of the 28 results depend on primitive-time roots.

By Proposition 8.73, the use within physical laws of the speed of light, the caesium-133 transition frequency, the Planck constant, the reduced Planck constant, the gravitational constant, vacuum permittivity, vacuum permeability, the Hubble rate, decay constants and lifetimes, linewidths and decay widths, Planck time, and the Compton angular frequency is established after the required time content and physical-time qualification have been generated. The physical relations borne by these quantities are retained as theory-specific inputs to the LCTR generation-and-audit construction, and predicates concerning their constancy, invariance, rates, lifetimes, linewidths, or frequencies are applied to generated time representations and theory-specific validity domains.

By Proposition 8.74, the quantification domain of light-speed invariance is determined as the domain in which generated law-admissible time representations, spatial representations, admissible representation transformations among inertial observers, clock and length standards, measurement procedures, and light-propagation relations hold. This quantification domain retains the constancy of the vacuum speed of light c as measured in local inertial frames and the local propagation limit for matter, signals, or causal influence; it does not include arbitrary coordinate speeds of light, cosmological recession velocities, phase velocities, or display quantities that carry no information in the same quantification domain of invariance. By Corollary 8.75, the two-track, four-part determination for the twelve time-bearing targets establishes generation-order determinations in 21 formal-track fields and separates invalidation positions according to the dependence order of each target.

By Proposition 8.59, direct measurements of physical carriers, attribution of properties to time itself, and application to living processes are separated as distinct evidential stages. An account that presupposes an unconstructed time entity, attributes comparison differences on physical carriers to changes in its properties, and further treats direct measurements established for physical carriers as direct demonstrations of living processes replaces the object of measurement, the bearer of physical action, and the evidence type twice. This account constitutes a relativistic three-stage misrecognition involving biological application. The evidential force of successful relativistic calculations and of experimental results concerning clocks, oscillators, electromagnetic radiation, and particle decay is not inherited by direct demonstration of living processes. The numbers of direct demonstrations

of living processes among the seven sources containing biological-application statements and among all 72 audited sources are 0/7 and 0/72, respectively. The Feynman 1995 source connects frequency shift, generalization to all physical processes, and application to human life and aging within a single source, thereby displaying this difference among evidential stages as one dependence sequence.

The Feynman 1995 and Narlikar 2010 sources add cases in which a predicate moves from clock, signal, and frequency relations to a time-scale or temporal-flow property. The Heisenberg 1958 source adds a statement of the nonexistence of time before the universe and a conceptual analysis according to which the vocabulary of time creation presupposes temporal relations. Time-property attribution, biological application, time nonexistence, and the self-dependence of time creation are thereby connected to the 72-source audit not merely as vocabulary dispersed among source groups, but as dependence sequences directly traceable within sources.

Time descriptions concerning the problem of time in quantum gravity, cosmological initial boundaries, the black-hole information problem, the problem of observables, the arrow of time, quantum-measurement records, and spacetime emergence are connected, as the time-problem connection domain of Definition 8.108, to the generation conditions, scale-indexed validity domains, six failure series, and audit reports of their respective time content. By Proposition 7.170, a result that removes privileged time from a relational formalism and the obligation to generate clocks, records, comparisons, order, scale, and qualification for use within laws from nontemporal physical inputs belong to different evidence-effect domains.

Proposition 8.76 applies the unbounded inheritance of limited results, defined in Chapter 7, to thermodynamics, cosmology, quantum measurement, quantum gravity, relativity, biological application, and time-bearing quantities. The mathematical consistency, predictive performance, model fit, and empirical fit of each field are retained within their evidence-effect domains; their force is not automatically inherited by the physical generation of time content, the physical-time qualification of law time, a time attribute independent of physical carriers, direct demonstration of an unmeasured target, or an ontology of time. Section 7.3.18 connects the evidential roots of this cross-field collation to the formal evidence and statement evidence of all 72 audited sources.

Synthesis (8): differences from existing theories and result-preserving re-grounding. The equations of motion, interaction terms, field equations, quantization procedures, correction rules, and engineering implementations of existing theories are retained as theory-specific inputs to the LCTR generation-and-audit construction. By generating the time content used by time variables and its grounding structure, and by auditing their validity domains and invalidation positions, existing theories are safely reused within the range in which the generation conditions hold. This gives the result-preserving re-grounding of Definition 8.99.

By Proposition 8.79, a primitive-time-input audit separates the determination of physical-time qualification from the determination of qualification for use as a physical computational result when actual invalidation of a generation condition or condition of establishment reaches a downstream result. The presence of primitive-time input and its participation in result derivation determine the former, and the qualification invalidation remains after mathematical consistency, predictive performance, and empirical fit have been established. The latter requires that the condition backgrounded by primitive-time input as a grounding obligation, the condition whose actual invalidation is established by independent evidence, and the condition that forms the starting point of downstream

dependence be the same. When this causal connection holds, the result loses qualification for use as a physical computational result in the evaluation domain and is regenerated or recomputed from repaired inputs. Whether the recomputed value agrees with or differs from the prior value is determined by the result of recomputation after repair.

When existing physics uses time-reference roles, physical carriers, recording arrangements, information coupling, and joint-law establishment conditions as background conditions outside its equations, without making them explicit as variables, relations, generation conditions, or failure conditions of those equations, the mathematical consistency, predictive performance, and empirical fit of the theory do not supply the physical generation of the backgrounded time-reference conditions. The classical multi-object audit makes these background conditions explicit as audit inputs and separates the structural burdens and invalidation signatures of the respective recording arrangements.

For each engineering implementation, if a role-typed realization by physical carriers, an engineering evaluation specification, typed engineering measurements, evidence-provenance certificates, and verified evidence for engineering formal conditions are given, then the use of time-dependent laws in that implementation is evaluated by the same six-series, five-valued audit. Applicability is determined not by the name of the field but by the establishment of typed correspondences and verified evidence connecting theoretical conditions to engineering measurements.

Synthesis (9): typed generation-and-audit architecture. By Definition 8.100 and Proposition 8.101, the type structure, generation structure, use-qualification structure, invalidation-audit structure, theoretical and engineering evidence structure, and cross-source audit structure form a single typed dependence structure. For every generated object, its input type, formation conditions, output type, and downstream conditions of use are determined; upon invalidation, the minimal invalidation positions, range of effects, target of repair, and scope of downstream regeneration are determined. The design methodology of computer science is realized as the mathematical and logical structure of LCTR that specifies types, dependence order, audit states, evidence provenance, and change effects.

Synthesis (10): final consequence. By Synthesis (1)–Synthesis (9), temporal comparison, temporal order, time representation, dynamical description, law time, and differential-law form are given as distinct stages of one generation dependence beginning with non-temporal physical inputs. In a physical case in which every stage holds, the physical grounds and the range of safe use of a time variable are obtained. For a differential equation that uses continuous approximation, the full evaluation-validity domain, maximal effective interval for approximation, and approximation boundary give its physical use qualification and scale limit. Under the classical-continuum-regime assumptions, LCTR changes the stage of establishment of classical continuous time from a primitive premise of law description to a classical continuous-approximation time representation generated from a complete evaluation specification and established at every scale within the maximal effective interval for approximation. In a case of invalidation, the six failure series, scale boundary, invalidation signature, and five-valued audit report separate the minimal invalidation positions, causal type, unresolved range, target of repair, and regeneration order of downstream results. These outputs constitute a conditional operational warranty for continuous approximation and differential forms that returns the domain of use and the cause of invalidation.

LCTR changes the stage at which temporal identity, before–after relations, and real-valued time variables become available for use in physical laws. Whether two physical

materials may be treated as belonging to the same time content, which may be treated as before or after, when incomparable materials may be quotiented into a single linear order, and when that order may be represented by a real-valued time variable are determined by satisfaction of generation conditions prior to law description. Accordingly, temporal identity, simultaneity based on synchronization procedures, the order used for initial conditions and trajectory evolution, a single time axis, and real-valued time variables are not descriptive resources that a theory may use unconditionally; they acquire qualification for use in domains where the corresponding generation conditions hold. Freedom in the choice of an admissible time coordinate is determined as a unique strict order isomorphism between representation images of the generated temporal structure, so freedom of coordinate choice and establishment of the temporal structure become distinct obligations.

The use of a single time variable for multiple objects is itself subject to an independent establishment obligation. Use of a common time representation, synchronization, same-time judgments, and a joint law-time domain requires satisfaction of the conditions on the physical-carrier realization of each object, record field of view and record stability, observer-side information coupling, communication, multi-object relations, and scale-indexed defect quantities that apply to the objects and configuration in question. Remote or global constructions involving transport or gluing additionally require transport consistency, loop consistency, quotient descent, and gluing.

LCTR's framework, which grounds the establishment of time in generation conditions, reallocates the obligations assigned to time in each field. In time metrology, periods, frequencies, standard frequencies, and representations in hertz are not premises of time generation but metrological results obtained from cycle counts, record correspondences, comparison rules, generated time differences, and normalization. This generation order blocks the circularity in which frequency is presupposed as an input to time generation while an ungenerated time difference is used in its definition. In special relativity, LCTR does not rederive Lorentz transformations; it retains them as theory-specific SR inputs that relate generated law-admissible time representations and spatial representations. In general relativity, establishment of a local proper-time representation does not imply establishment of a global law-time representation; the latter requires downstream conditions including transport, loop consistency, quotient descent, and gluing. Construction of causal order and generation of temporal order are likewise distinct obligations. Providing causal order as a primitive partial order or a pre-given spacetime structure does not entail generation of clock scales, time intervals, or law time. In quantum gravity, elimination of external time or selection of an internal clock is an obligation distinct from generation of physical time and establishment of its qualification for use within laws. LCTR changes the problem of time in quantum gravity from the problem of eliminating external time or selecting an internal clock to the problem of whether the time content in use has been generated from nontemporal physical inputs, satisfies law-descriptive operability, and has acquired qualification for input into physical laws. The arrow of time is not a ground for generating time but a downstream orientation condition on an already generated time representation.

For a theory that literally asserts the creation, emergence, recovery, or nonexistence of time, LCTR imposes as formal obligations the time-content type and domain at issue, a generation path from nontemporal physical inputs or an alternative structure, elimination of primitive-time roots from all time content, and elimination of temporal-dependence roots from downstream results. LCTR thereby changes assertions of time creation, time emergence, and time nonexistence from verbal claims into claims in physics whose establishment can be audited by formal and statement evidence.

The consequences of primitive-time input are separated into invalidation of physical-time qualification and invalidation of qualification for use of downstream computational results reached by actual invalidation of a backgrounded grounding obligation. The former is compatible with mathematical consistency, predictive performance, and empirical fit. The latter holds through the backgrounding of a grounding obligation by primitive-time input, independent evidence establishing actual invalidation of the same condition, and downstream dependence from the invalidation position to result derivation. When the latter holds, downstream results are regenerated or recomputed from repaired inputs, and the recomputed values are evaluated anew. The domain in which existing results can be retained safely is the intersection of the evidence-effect domain specific to each result and the validity domain given by the generation conditions and conditions of establishment.

Cross-field collation of the 72 audited sources shows that descriptions in which the force of limited computational success, predictive success, model fit, or experimental success is inherited by the physical establishment of an entire mathematical expression, the physical-time qualification of an ungenerated time variable, a time attribute independent of physical carriers, application to an unmeasured target, or an ontology of time recur across multiple fields from antiquity to the present. This recurrent transfer of the force of limited results from their result-specific evidence-effect boundaries, successively to the physical establishment of an entire mathematical expression, the physical-time qualification of an ungenerated time variable, a time attribute independent of physical carriers, application to an unmeasured target, or an ontology of time is a common cross-field misrecognition in physics. It is the unbounded inheritance of limited results in Definition 7.116. By Proposition 8.76, this inheritance is localized to the evidence-effect boundary of each field. Retaining each result within its evidence-effect domain, generating time content from nontemporal physical inputs, and auditing its validity domain and invalidation position determine the domain in which existing results can be reused safely.

A time variable in a physical law is not established as a primitive premise. The 72 audited sources show that mathematical consistency, predictive performance, or empirical fit does not remove primitive-time roots. In every audited source from antiquity to the present, at least one item of time content was entered into law operations or result derivations before the physical grounding structure required for its use had been generated from nontemporal physical inputs. This unanimity shows that the unfulfilled obligation to ground physical time content has recurred historically in the uses of time throughout physics. When such content is replaced by generated time representations, the basis of mysteries constituted by presupposing ungenerated time is lost, and each problem is transferred to generation conditions, validity domains, failure classifications, and the audit evidence to be acquired next.

Proof. Synthesis (1) follows from Theorem 4.90. That theorem makes joint satisfaction of $\mathbf{L}_{\leq 7}[\mathbf{Ev}^{\text{cmp}}]$ and $\mathbf{G}[\mathbf{Ev}^{\text{cmp}}]$ equivalent to operability at the comparison stage and supplies, in dependence order, the extended comparison-arrival equivalence relation, the $\mathcal{C}/\mathcal{D}$ comparison-quotient sets, the canonical projections, the quotient-level $\mathcal{C}\mathcal{D}$ -derived relation, and the $\mathcal{C}\mathcal{D}$ -derived canonical comparison quotient.

Synthesis (2) follows from Theorem 4.147, Theorem 4.177, Theorem 4.179, and Theorem 4.187. From satisfaction of $\mathbf{Repr}_{\leq 3}(\mathcal{X}; \mathcal{J}_{\text{cmp}})$, they supply, in dependence order, the $\mathcal{C}/\mathcal{D}$ canonical partial orders, exact operational time structure, local and global order-coordinate structures, canonical order-coordinate projections, local-to-global order-coordinate embeddings, and comparison-order atlas. When a selected representation range and an order embedding of that range are specified, the $\mathcal{C}/\mathcal{D}$ -derived representation coordinates factor

through the canonical global order-coordinate projection.

Synthesis (3) follows from Theorem 5.112 and Theorem 5.138. The two theorems make dynamical-description operability equivalent to joint satisfaction of exact operability and all dynamical-description conditions, and generate the $\mathfrak{C}$ -derived observer-time quotient, the $\mathfrak{B}$ -derived object-state quotient, the $\mathfrak{CDB}$ -derived trajectory relation, a graph trajectory, a canonical family of observables, the observer-time order-coordinate quotient, a real-valued time representation, a scalar-time trajectory, and a family of real-valued observable curves.

Synthesis (4) follows from Theorem 5.113, Theorem 5.135, and Theorem 7.203. The first theorem supplies the observer-time law core and the generated representationality of law-evaluation time; the second supplies chart-compatible differential laws, value-jet-relation covariance, and time-representation-change jet-relation covariance. By the third theorem, generation of time content and its grounding structure from nontemporal physical inputs before law operations and result derivations is necessary for the physical qualification of a physical-law time variable. A time variable in a physical law is therefore not established as a primitive premise.

Synthesis (5) follows from Definition 6.10 and Theorem 6.168. Within each of the comparison, representation, dynamical-description, continuous-approximation, law-description, and differential series, invalidation of a formed and evaluable condition is localized to the corresponding invalidation-position token and gives a six-series invalidation signature. Because multiple invalidation positions can hold in the same evaluation, the six series do not require an exclusive disjunction. Unformed and unevaluable states are not replaced by invalidation; they are retained as their respective audit states.

Synthesis (6) follows from Proposition 6.162, Proposition 6.154, Theorem 6.163, Theorem 6.164, and Theorem 6.183. The first proposition gives the full evaluation-validity domain as the intersection of the strict validity domain and the approximation-validity domain. The second separates the objects of determination and input types of the continuous-approximation and differential series. By Theorem 5.138, differential-stage operability includes joint satisfaction of law-description-stage operability and the five universal differential conditions. The physical use qualification of a differential equation employing continuous approximation is therefore given as the full evaluation-validity domain in which the five exact series and the continuous-approximation series hold simultaneously.

Theorem 6.163 and Theorem 6.164 supply the approximation-validity scale domain, maximal effective interval for approximation, approximation-invalidation scale, approximation-boundary signature, and first-invalidation report. Theorem 6.183 supplies the five-valued audit-report object whose components are series-specific audit-group states, condition-specific audit states, invalidation positions, the six-series invalidation signature, indeterminate tokens, the indeterminate signature, the approximation boundary, structural burden, and reasons for an undefined audit. Definition 6.179 gives satisfaction, condition invalidation, blockage by a prior condition, an unformed state, and indeterminacy as distinct state values.

The $\mathfrak{D}$ -derived record-cell width belongs to the engineering-audit input datum of Definition 6.194. By Proposition 6.204, if the audit state of verified engineering formal-condition evidence containing its engineering-measurement reference is invalidation, the invalidation position of the theoretical condition indicated by the target component of that evidence is localized within the six series. Proposition 6.199 connects an engineering auxiliary-audit cause, through verified effect evidence, to a formation input, evaluation input, input admissibility, boundary treatment, or condition evidence in the existing six series and separates the type of theoretical-condition invalidation from that of an

engineering auxiliary-audit cause.

Synthesis (7) follows from Theorem 7.190, Proposition 7.191, Theorem 8.16, and the respective worked reductions listed in Table 24. Every one of the 72 audited sources contains primitive-time roots and at least one result dependent on primitive-time roots. This does not mean that every result in each source depends on primitive time roots. By Proposition 7.188, the set of results beyond generation reach is the disjoint union of the evidence-environment-reachable nongenerated result set and the primitive-time-root-dependent result set. Under the complete criterion, the generated-time-content set is empty for all 72 sources, and no source generates, from nontemporal physical inputs, the entirety of the time content it uses. By Proposition 7.189, under the local-reach criterion, 10 sources reach a total of 57 temporal contents. The strict generated-time-content sets of all 72 sources are empty, and the entire source-specific time content consists of primitive-time roots. In the remaining 62 sources, no time content is reached even under the local-reach criterion. Sources that explicitly state the absence of external or independent time, the fundamental status of time, difficulty defining the concept of time, or an internal or relational construction of time number (18/72); sources that present a dynamical construction or recovery of intrinsic, relational, thermal, or semiclassical time number (13/72); and among the latter, sources that satisfy the complete time-generation obligation number (0/13).

By Definition 7.10, the universal persistence of primitive-time roots means that in each source at least one item of time content was entered into law operations or result derivations before the physical grounding structure required for its use had been generated from nontemporal physical inputs. The audited source set extends from ancient definitions of time to modern theories, experiments, metrological standards, and engineering materials. This unanimity therefore shows that the unfulfilled obligation to ground physical time content has recurred historically in the uses of time throughout physics.

Theorem 8.16 and the respective worked reductions connect Newtonian and classical mechanics, time metrology, GPS, special relativity, general relativity, and quantum-gravitational reconstruction to the generation conditions, validity domains, and invalidation positions of time content. Definition 8.19 retains the path that directly describes comparison differences on physical carriers as a relativistic result, and classifies a path in which reattribution to a time attribute directly depends on prior placement of ungenerated time as a two-stage misrecognition. Theorem 7.190 establishes that primitive-time roots remain in all 72 sources. Proposition 7.167 distinguishes formal results from statement targets. Proposition 8.77 collates time-nonexistence statements and time-attribute statements after aligning their time-content types and application domains, and gives incompatibility when both are the same and, when they differ, an obligation to state the transition in target type or application domain.

Proposition 8.73, from Proposition 7.178 and Remark 7.179, gives post-generation use qualification for time-bearing quantities and retention of physical relations as theory-specific inputs. Proposition 8.74 connects the three determinations on the formal track of the speed of light to admissible transformations among generated time representations in special relativity, and gives a quantification domain of light-speed invariance that separates the vacuum speed of light and local propagation limit in local inertial frames from coordinate speeds of light, cosmological recession velocities, phase velocities, and display quantities that carry no information. Corollary 8.75 connects the 96 determination fields of Proposition 7.182 to the cross-field consequences of Chapter 8.

Proposition 8.59 connects the separation of evidential stages in Section 7.3.17 and the

range of direct demonstrations in Section 7.4 to a three-stage misrecognition in relativistic explanation. This separates direct measurements of physical carriers, attribution of properties to time itself, and application to living processes, and localizes the compression of the evidential range that treats a direct measurement at the first stage as a direct demonstration at the third. No direct demonstration of living processes occurs in either the seven sources containing biological-application statements or all 72 audited sources; the counts are (0/7) and (0/72), respectively. The Feynman 1995 source connects frequency shift, generalization to all physical processes, and application to human life and aging within a single source.

The Feynman 1995 and Narlikar 2010 sources show predicate transfer from clock, signal, and frequency relations to a time-scale or temporal-flow property. The Heisenberg 1958 source distinguishes a statement of the nonexistence of time before the universe from the presupposition of temporal relations by the vocabulary of time creation. The Bombelli–Lee–Meyer–Sorkin 1987 source distinguishes the construction that relates primitive causal order to macroscopic past–future relations from the complete generation of clock scales, time intervals, and law time. This statement evidence adds time-property attribution, biological application, time nonexistence, the self-dependence of time creation, and separation of the obligations for causal order and complete time generation to the 72-source audit as dependence sequences traceable within sources.

Definition 8.108 connects time descriptions concerning the problem of time in quantum gravity, cosmological initial boundaries, the black-hole information problem, the problem of observables, the arrow of time, quantum-measurement records, and spacetime emergence to the generation conditions, scale-indexed validity domains, six failure series, and audit reports of their respective time content. Proposition 7.170 separates a result that removes privileged time from a relational formalism from the obligation to generate physical time from nontemporal physical inputs. Proposition 8.72 connects results concerning atomic-clock differences, molecular-vibration-period differences, covalent-bond vibration sequences, electronic-state transitions, scattering events, decay events, the cosmological constant, and dark energy to the generation conditions, validity domains, minimal invalidation positions, and regeneration or recomputation after repair of the time content on which they depend.

Proposition 8.63 requires a theory that literally asserts the creation, emergence, recovery, or nonexistence of time to provide a generation path or alternative structure, elimination of primitive-time roots, and elimination of temporal-dependence roots from downstream results. These statements therefore become claims in physics whose establishment is judged by satisfaction of the formal obligations specified in that proposition.

Proposition 8.68 shows that even after external time has been excluded from raw operational input, the law-description stage requires law-description operability, while use of each temporal content within laws requires prior establishment of the generated structures needed for that use. The problem of time in quantum gravity therefore becomes a problem of judging generation of time content and establishment of law-time use qualification, rather than only elimination of external time or selection of an internal clock.

Definition 7.115, Definition 7.116, Proposition 7.118, and Corollary 7.119 give the evidence-effect domain of a limited result, the nonestablishment of claims beyond its boundary, and the difference between this audit axis and primitive-time dependency inheritance. Proposition 8.76 applies them to the physical fields considered in Chapter 8. Retaining the limited results of each field is therefore separated from inheriting those results into qualification of ungenerated time, time attributes, unmeasured targets, or an ontology of time. By Section 7.3.18, the statement evidence and interpretive possibilities that

determine this separation are connected to the audit method or representative arguments of Chapter 7 for all 72 audited sources, not only the representative sources.

Synthesis (8) follows from Proposition 8.95, Proposition 8.97, Proposition 8.102, and Proposition 8.103. The structures addressed by clock synchronization, relativistic time-stretching representations, proper time in general relativity, relational time, quantum-theoretical time reconstruction, time metrology, and operationalism are mapped to stages of the generation dependence or to inputs of representative schemes. The equations of motion, interaction terms, field equations, quantization procedures, correction rules, and engineering implementations of existing theories are retained as theory-specific inputs to the LCTR generation-and-audit construction. By generating time content and its grounding structure and auditing their validity domains and invalidation positions, existing theories are reused safely within the range in which the generation conditions hold.

A classical world representation does not describe time-reference roles, physical carriers, recording arrangements, information coupling, or joint-law establishment conditions as variables, relations, generation conditions, or invalidation conditions in law expressions; it enters them in advance as background conditions outside the expressions. By Proposition 8.103, mathematical consistency, predictive performance, and empirical fit do not supply the physical generation of backgrounded time-reference establishment conditions.

Definition 3.2 gives role-typed realization by physical carriers, while Definition 6.188 and Definition 6.194 give engineering evaluation specifications and typed engineering measurements. Proposition 6.204 localizes the target components of verified engineering formal-condition evidence to invalidation positions in the six series. For every engineering implementation supplied with these typed correspondences and evidence, its use of time-dependent laws can therefore be connected to the same six-series, five-valued audit.

Proposition 7.121 connects backgrounding, actual invalidation, downstream reach, repair, and recomputation into one causal sequence when the grounding obligation backgrounded by primitive-time input, the condition whose actual invalidation is established by independent evidence, and the condition forming the starting point of dependence reaching downstream result derivation are the same. In the absence of this same-condition connection, primitive-time input and condition invalidation are retained as separate audit results. Proposition 8.79 separates invalidation of physical-time qualification, determined by the presence of primitive-time input and its participation in result derivation, from invalidation of qualification for use of downstream computational results, determined by this causal sequence. Proposition 8.78 gives a direct connection to recomputation and precision evaluation after repair in time metrology and GPS; a connection in Newtonian and classical mechanics, special relativity, and general relativity when the corresponding error cause is included in the minimal invalidation positions; and a connection in quantum gravity when a concrete approximation or estimation error is localized to an invalidation position. The safe reuse domain of existing results under result-preserving re-grounding is therefore determined as the intersection of the evidence-effect domain specific to a result and the validity domain given by the generation conditions and conditions of establishment.

Synthesis (9) follows from Definition 8.100 and Proposition 8.101. The type, generation, use-qualification, invalidation-audit, theoretical and engineering evidence, and cross-source audit structures connect input types, formation conditions, output types, downstream conditions of use, audit states, evidence provenance, and scopes of regeneration by dependence edges.

Synthesis (10) is the logical composition of Synthesis (1) through Synthesis (9). Synthesis (1) through Synthesis (4) supply the generation dependence from nontemporal

physical inputs through temporal comparison, temporal order, time representation, dynamical description, law time, and differential-law forms. Synthesis (5) and Synthesis (6) separate their validity domains, invalidation positions, and unresolved ranges through the six failure series, scale boundaries, invalidation signatures, and five-valued audit reports. Synthesis (7) supplies the persistence of primitive-time roots in the 72 audited sources and concrete connections to representative theories; Synthesis (8) supplies the objects retained from existing theories, the separation of invalidation of physical-time qualification from conditional invalidation of use of downstream computational results, and the conditions for safe reuse. Synthesis (9) supplies these as a single mathematical and logical structure connecting types, generation, use qualification, audit states, evidence provenance, ranges of effects, and scopes of regeneration.

In a classical continuous-approximation regime, Corollary 8.89 establishes that a classical continuous-approximation time representation is generated from a complete evaluation specification. Theorem 8.88 establishes that it holds at every scale in the maximal effective interval for approximation. Through this generation and validity-domain guarantee, LCTR changes the stage of establishment of classical continuous time from a primitive premise of law description to a generated approximation representation that holds within the maximal effective interval for approximation.

Hence, in a physical case in which every stage holds, the physical grounds and range of safe use of a time variable are obtained; in a case of invalidation, the invalidation positions and unresolved range become auditable. A time variable in a physical law is not established as a primitive premise. When a time variable presupposed as a black box without a generation path is replaced by a generated time representation, the basis on which problems presupposing ungenerated time appear as independent mysteries is lost, and each problem is transferred to generation conditions, validity domains, failure classifications, and the audit evidence to be acquired next. $\square$

Corollary 8.110 (Conditional physical use qualification for continuous approximation and differential forms). For a scale-indexed complete evaluation specification $(\mathbf{Ev}, \lambda) \in \mathbf{EvSpec}_\Lambda$ that uses a differential equation employing continuous approximation as a physical description,

$$\begin{aligned} (\mathbf{Ev}, \lambda) &\in \mathcal{V}^{\text{full}} \\ \iff [\mathbf{Ev} \in \mathcal{V}^{\text{strict}} \wedge (\mathbf{Ev}, \lambda) \in \mathcal{V}^{\text{approx}}] \\ \iff K_{\text{full}}^{\text{fail}}[\mathbf{Ev}; \lambda] &= \top \end{aligned}$$

holds. The strict-establishment side includes the five series of comparison, representation, dynamical description, law description, and differentiation; the approximation-establishment side includes the nine continuous-approximation conditions. The physical use qualification of a differential equation employing continuous approximation therefore requires establishment of both continuous approximation and differential form.

The set of scales at which the qualification holds is given by the full evaluation-validity domain, and the maximal effective interval of its approximation component is the largest initial subset of the approximation-validity scale domain. Under the hypotheses of Theorem 6.163, if the complement of that scale domain has a least element, that element is the first approximation-failure scale. When condition invalidation occurs, the first-invalidation report and structural burden give each minimal invalidation position and the affected downstream objects, and the invalidation-generation dependence order gives the regeneration order after repair. When blockage by a prior condition, an unformed

state, or indeterminacy occurs, the state and its reason are retained in the five-valued audit report.

Proof. The first equivalence follows from Definition 6.4 and Proposition 6.162. The second equivalence follows from Definition 6.27 and Definition 6.4. The separation of the five exact series from the continuous-approximation series follows from Definition 6.10 and Proposition 6.154. That the differential stage includes the law-description stage and the five universal differential conditions follows from Definition 5.132.

The maximal effective interval for approximation and first exceedance are obtained from Theorem 6.97; the minimal invalidation positions and approximation scale boundary, from Theorem 6.164; structural burden, from Proposition 6.171; and the five-valued audit states and their reasons, from Theorem 6.183 and Corollary 6.187. $\square$

Proposition 8.111 (Connection of the final synthesis to engineering audit and classical world representations). The theoretical-establishment audit layer of Theorem 8.109 has as components the six failure series, invalidation positions, invalidation signatures, scale-indexed validity domains, and five-valued audit reports. The engineering-audit layer of Definition 6.194 has as components engineering-measurement type tags, engineering-measurement references, typed engineering-measurement values, and evidence-provenance certificates.

Through the verified engineering formal-condition evidence of Proposition 6.204, the invalidation positions of theoretical conditions corresponding to engineering-audit results for physical-carrier realization, record field of view, record invasiveness, observer-side information coupling, joint relations, and common law domains are determined. The classical multi-object audit example of Definition 8.104 and the classical world representation of Proposition 8.102 make concrete the evidence correspondence between these two layers.

For each engineering implementation, if a role-typed realization by physical carriers, an engineering evaluation specification, typed engineering measurements, evidence-provenance certificates, and verified evidence for engineering formal conditions are given, then the use of time-dependent laws in that implementation is evaluated by the same six-series, five-valued audit.

Proof. Definition 6.194 and Definition 6.195 supply engineering-measurement references, typed engineering-measurement values, engineering-implementation auxiliary-audit flags, and their evidence references as components of the engineering-audit layer. Definition 6.10 supplies the invalidation positions of theoretical conditions belonging to comparison, representation, dynamical description, continuous approximation, law description, and differentiation as components of the theoretical-establishment audit layer.

Proposition 6.199 treats verified engineering auxiliary-audit-effect evidence as the three components of an engineering auxiliary-audit flag, an engineering auxiliary-audit effect, and a target invalidation-position token, and maps the set of target components to the invalidation-position-token set. Proposition 6.204 localizes only the target components of verified engineering formal-condition evidence to the invalidation positions of theoretical conditions. Engineering measurements, auxiliary-audit flags, and auxiliary-audit effects therefore belong to the engineering-audit layer, while the invalidation positions of the corresponding theoretical conditions belong to the theoretical-establishment audit layer.

Table 22 and Proposition 8.107 show that the bulk-recording and following-recording types can have different invalidation signatures. Proposition 6.206 separates the structural burdens of the two recording types relative to engineering evidence.

By Proposition 8.102, a classical world representation enters time-reference establishment conditions in advance as background conditions outside the expressions. By Proposition 8.103, the mathematical consistency, predictive performance, and empirical fit of classical physics do not supply their physical generation. The same proposition restricts the domain in which multi-object relations hold under the same time variable to the scale-indexed validity domain satisfying the conditions on physical-carrier realization, $\mathfrak{D}$ -derived record field of view, record stability, observer-side information coupling, $\mathfrak{D}$ -communication, multi-object relations, and scale-indexed defect quantities.

Thus, the measurements, flags, effects, and structural burdens of the engineering-audit layer are connected to the invalidation positions, invalidation signatures, scale-indexed validity domains, and five-valued audit reports of the theoretical-establishment audit layer through the target components of verified evidence. Every engineering implementation supplied with a role-typed realization by physical carriers, an engineering evaluation specification, typed engineering measurements, evidence-provenance certificates, and verified evidence for engineering formal conditions belongs to the domain of this connection. Its use of time-dependent laws is therefore evaluated by the same audit-report generation map. $\square$

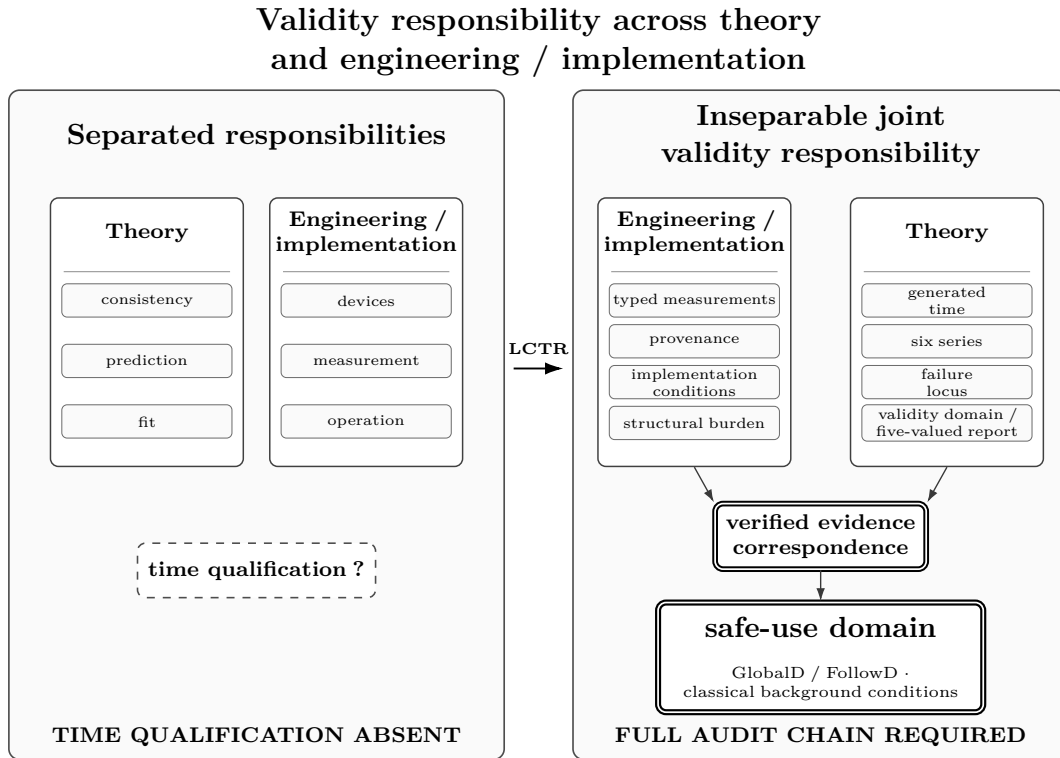


Figure 31: Connection of the final synthesis to engineering audit and classical world representations. The engineering measurements, auxiliary-audit flags, auxiliary-audit effects, and structural burdens of the engineering-audit layer are distinguished from the six failure series, invalidation positions, invalidation signatures, scale-indexed validity domains, and five-valued audit reports of the theoretical-establishment audit layer, and the two layers are connected through target components of verified evidence.

Consequence after the final synthesis. By Theorem 8.109, Corollary 8.110, and Proposition 8.111, the establishment of a time variable appearing in a physical law requires

the generation conditions of comparison, representation, dynamical description, and law description needed for the use of that temporal content. Use of differential law forms requires the differential-stage conditions, and claims using continuous approximation require a scale-indexed validity domain. A differential equation employing continuous approximation requires the dual qualification of continuous approximation and differential form. An invalidated or unresolved position is localized to the six failure series, invalidation signature, approximation boundary, structural burden, and five-valued audit report; condition invalidation, blockage by a prior condition, an unformed state, indeterminacy, and an engineering auxiliary-audit cause are separated by type.

By Theorem 7.203, a time variable in a physical law is not established as a primitive premise. By Theorem 7.190, every one of the 72 audited sources contains primitive-time roots and results dependent on primitive-time roots. Safe reuse of results from existing theories therefore requires generation of time content and its grounding structure from nontemporal physical inputs and audit of their validity domains, invalidation positions, and unresolved ranges.

Supplement: Primitive-Time-Root Audit of 72 Sources

Each source division in this supplement separates formal evidence from statement evidence at the highest level. Formal evidence is divided into three roles: “formal description and location in the original source,” “audit decomposition and dependency reconstruction by this paper,” and “audit determination by LCTR.” The first presents definitions, formulas, propositions, operations, or conditions stated in the original source and their locations. The second presents the decomposition, typing, and dependency reconstruction of this paper under Definition 7.88. The third presents the LCTR determinations concerning primitive-time roots, generation reach, root-dependent results, evidence-effect domains and boundaries, and result-preservation domains. The second and third are not the terminology, classifications, or verbatim expressions of the original author.

Statement evidence is divided into two roles: “quotation in the source language and translation where required” and “statement-evidence record by this paper.” A quotation environment contains the text in the evidentiary source language. When an identified standard English edition is adopted for a classical source, the quotation is taken from that edition. When the evidentiary source is in another language, an English translation corresponding to the immediately preceding source-language quotation is supplied. The “statement-evidence record by this paper” records the statement target, minimal context, attribution, predicate, explicit application domain, interpretive possibilities, interpretation-specific evidential boundaries, and corresponding formal evidence.

For local keys within a source, the column heading states explicitly whether they refer to source locations, time-content sets, or evidence-support sets. A local key is not a theoretical symbol extending across sources. In source-specific tables, (i), (a), (p), (m), (k), and (j) identify rows of time contents, auxiliary objects, source claims or formal components, mathematical-object types, semantic components, and results, respectively. In lookup tables, (d), ℓ , and (e) denote dependency-object, source-location, and formal-evidence keys, respectively. Each symbol ranges over the finite index set fixed by the corresponding definition or lookup table and has scope only within that table.

The “determination values” for each source present three values in the order “complete,” “local reach,” and “remaining roots,” as defined in Definition 7.41.

8.18 Audited Source Set and Audit Scope

The audited source set consists of sources that state uses of time in classical mechanics, relativity, quantum theory, thermal and statistical physics, gravitation and cosmology, quantum gravity, emergence and information theory, and measurement and standards. Each source as of August 2026 is identified by its formal bibliographic record and the original source under audit.

Definition 8.112 (Finite audited source set). The finite set of the 72 sources under audit,

$$\text{Ref}^{\text{audit}}, \quad |\text{Ref}^{\text{audit}}| = 72$$

is called the finite audited source set.

This set is the source quantification domain of the source-specific and cross-source audits. Each

$$R \in \text{Ref}^{\text{audit}}$$

is associated with a formal bibliographic record, the original source under audit, and structured source locations corresponding to its principal claims.

Section 7 gives the general audit method for determining use of time within laws, primitive-time premises, and time-generation obligations. This supplement gives the source-specific audit obtained by applying that method to every member of the finite audited source set and the cross-source comparisons obtained from the audit family.

8.18.1 Classification of Time-Order-Structure Content

The audited source set, the time contents of individual sources, the time content of all audited sources, the results of individual sources, the transfer events of individual sources, and the time-content inputs of transfer events and results are as specified in Definitions 7.1 and 7.4 .

The five-type classification of time content and the eight-axis classification of temporal order structure are as specified in Definitions 7.20 and 7.21 . The following tables give the source-wise inverse images of the two classification maps.

The component order of each five-type inverse-image index-set tuple is: a member of a boundary, stage, or transition-position family; content dependent on the same family; content on a branch outside the boundary; content belonging to a separate requirement family; and lexical or illustrative content.

Table 25: Five-Type Indices of Time Content for the Audited Source Set

Source	Ordered Tuple of Index Sets
Ref _{Adl03MP}	$(\{1, 2, 5\}; \{4, 6, \dots, 8\}; \emptyset; \{3, 9\}; \emptyset)$
Ref _{And12PT}	$(\{12, 16, 17\}; \{1, \dots, 11, 13, \dots, 15, 18, \dots, 20\}; \emptyset; \emptyset; \emptyset)$
Ref _{And01DW}	$(\{5, \dots, 7\}; \{4\}; \emptyset; \emptyset; \{1, \dots, 3\})$
Ref _{AriPhys4}	$(\{2, 3, 8\}; \{1, 4, \dots, 7\}; \emptyset; \emptyset; \emptyset)$
Ref _{ADM62Dyn}	$(\{2, 7, 11\}; \{1, 3, \dots, 6, 8, 10, 12, \dots, 17\}; \{9\}; \emptyset; \emptyset)$
Ref _{Bai77TD}	$(\{2, 3\}; \{1, 4, \dots, 17\}; \emptyset; \emptyset; \emptyset)$
Ref _{Bar94TQ1}	$(\{2, 10, 13\}; \{3, \dots, 9, 11, 12, 14, 15, 17, 18\}; \{1, 16\}; \emptyset; \emptyset)$
Ref _{Bar94TQ2}	$(\{2, 3, 6, 7, 11, 14, \dots, 18\}; \{1, 4, 5, 8, 9, 13, 19, \dots, 21\}; \emptyset; \{10, 12\}; \emptyset)$
Ref _{Bek73BHE}	$(\{1, \dots, 4, 6, 8, 9, 12, \dots, 15, 17, 18, 21, 22, 26, 27, 29, 30, 32, \dots, 36, 38, 42, 46, 47, 52, \dots, 56, 61, \dots, 64, 67\}; \{5, 7, 10, 11, 16, 19, 20, 24, 25, 28, 31, 37, 39, 40, 43, \dots, 45, 48, \dots, 51, 57, \dots, 60, 66\}; \emptyset; \{23, 41, 65\}; \emptyset)$
Ref _{BIPM19SI}	$(\{16, 19\}; \{1, \dots, 15, 17, 18, 20, \dots, 34\}; \emptyset; \emptyset; \emptyset)$
Ref _{Bom87CS}	$(\{1, 3, \dots, 5, 9, 11\}; \{2, 6, 7\}; \emptyset; \{8, 10, 12, \dots, 17\}; \emptyset)$
Ref _{Bri27LM}	$(\{1, 3, \dots, 5, 22, 23, 30, 35, 38, 40, 44, 45, 47, 49, \dots, 52, 56\}; \{2, 6, \dots, 15, 18, \dots, 21, 24, \dots, 29, 31, \dots, 34, 36, 37, 39, 41, \dots, 43, 46, 57\}; \emptyset; \{16, 17, 48, 53, \dots, 55\}; \emptyset)$
Ref _{Bun09Red}	$(\{1, 5, 6, 12, 19\}; \{3, 4, 7, \dots, 11, 13, 14, 18, 22, \dots, 24\}; \emptyset; \{2, 15, \dots, 17, 20, 21\}; \emptyset)$
Ref _{CR94TT}	$(\{3, 5, 9, 12, 14\}; \{1, 4, 6, \dots, 8, 11, 13, 15, \dots, 28\}; \emptyset; \{2, 10\}; \emptyset)$

Source	Ordered Tuple of Index Sets
Ref _{CookBurns09Metric}	$(\{1, 6, 11, 21, 24, 27, 28\}; \{3, 4, 10, 12, 13, 15, \dots, 17, 19, 20, 23, 25, 26\}; \emptyset; \{2, 5, 7, \dots, 9, 14, 18, 22\}; \emptyset)$
Ref _{DL04EC}	$(\{1, 4, \dots, 6, 8, 9, 12, 13, 15, 18, 24\}; \{7, 14, 16, 17, 19, \dots, 21, 25, 26\}; \emptyset; \{2, 3, 10, 11, 22, 23\}; \emptyset)$
Ref _{DeW67QG1}	$(\{6, 13, 16, 28, 29, 33\}; \{1, \dots, 5, 7, \dots, 12, 14, 15, 17, \dots, 27, 30, \dots, 32, 34, 35\}; \emptyset; \emptyset; \emptyset)$
Ref _{Dir58GH}	$(\{2\}; \{1, 3, \dots, 5\}; \emptyset; \emptyset; \emptyset)$
Ref _{EPS72GFL}	$(\{1, 3, 5, 9, 18, 20, 23, 24, 29, \dots, 31, 34, 38, 44, \dots, 47, 50, 51, 53\}; \{2, 4, 6, \dots, 8, 10, \dots, 12, 19, 21, 22, 25, \dots, 28, 32, 33, 35, \dots, 37, 39, \dots, 43, 48, 49, 52, 54, 59, \dots, 62\}; \emptyset; \{13, \dots, 17, 55, \dots, 58\}; \emptyset)$
Ref _{Ein05ED}	$(\{5, 7, 9, \dots, 11, 13, 15, \dots, 19, 21, 23, 26, \dots, 28, 36, \dots, 38, 43, 51, 53, 55, \dots, 60, 62, 64, 74, \dots, 76, 88, 89, 102, 111, 118, 119, 131\}; \{1, 3, 6, 8, 12, 14, 20, 22, 24, 29, \dots, 35, 39, \dots, 42, 45, \dots, 50, 52, 54, 61, 63, 65, 67, \dots, 73, 77, 78, 82, \dots, 87, 90, \dots, 96, 105, \dots, 107, 109, 110, 115, 121, 123, 126, 128\}; \{25\}; \{2, 4, 44, 66, 79, \dots, 81, 97, \dots, 101, 103, 104, 108, 112, \dots, 114, 116, 117, 120, 122, 124, 125, 127, 129, 130\}; \emptyset)$
Ref _{Ein16GR}	$(\{5, 6, 9, 10, 21, 22, 24, 28, 29, 37, 38\}; \{1, 3, 4, 8, 11, 19, 23, 26, 27, 30, \dots, 36, 39, \dots, 45, 48, 49, 51, \dots, 53, 55, \dots, 59, 62, \dots, 65, 69, 71, 77, 78, 80, \dots, 91, 93\}; \emptyset; \{2, 7, 12, \dots, 18, 20, 25, 46, 47, 50, 54, 60, 61, 66, \dots, 68, 70, 72, \dots, 76, 79, 92\}; \emptyset)$
Ref _{Ell12PT}	$(\{2, \dots, 4, 6, 8, \dots, 11, 13, 14, 16, 28, 29, 33, 34, 39, 40\}; \{5, 7, 12, 17, \dots, 27, 30, \dots, 32, 38\}; \emptyset; \{15, 35, \dots, 37\}; \{1\})$
Ref _{ESA19CGR}	$(\{18\}; \{1, \dots, 3, 5, \dots, 17, 19, \dots, 24\}; \emptyset; \{4\}; \emptyset)$
Ref _{FLS63T}	$(\{1, 4, 8, 17, 18, 20, 34, 37, 45, 49, 50, 60, 68, 69, 75, 90, 91, 94, 104, 107, 116, 119, 131, \dots, 133, 135, 136\}; \{2, 3, 6, 7, 9, 10, 12, \dots, 16, 19, 21, \dots, 33, 36, 38, \dots, 44, 46, \dots, 48, 52, \dots, 59, 61, \dots, 67, 70, \dots, 74, 76, \dots, 79, 81, \dots, 89, 92, 93, 95, \dots, 98, 100, \dots, 103, 105, 106, 109, \dots, 113, 118, 120, \dots, 127, 134, 137\}; \emptyset; \{5, 35, 99, 108, 114, 115, 117, 128, \dots, 130\}; \{11, 51, 80, 138\})$
Ref _{FBJL07ES}	$(\{1, 3, 6, 10, 14, 17, 18, 20\}; \{7, 13, 15, 19, 22\}; \emptyset; \{2, 4, 5, 8, 9, 11, 12, 16, 21, 23\}; \emptyset)$
Ref _{GE07FE}	$(\{1, 7, 10, 11, 13\}; \{3, 9, 12, 14, 15, 17, 19\}; \emptyset; \{2, 4, \dots, 6, 8, 16, 18\}; \emptyset)$
Ref _{HK72Pred}	$(\{1, 2, 6, 7, 20, 42, \dots, 44, 48, \dots, 50, 54, \dots, 56, 69, 71\}; \{3, \dots, 5, 8, \dots, 15, 17, \dots, 19, 21, \dots, 28, 30, \dots, 41, 45, \dots, 47, 51, \dots, 53, 57, 59, \dots, 65, 67, 68, 70, 72, 73, 75, \dots, 77\}; \emptyset; \{16, 29, 58, 66, 74\}; \{78\})$
Ref _{HK72Obs}	$(\{1, 3, 21, 26, \dots, 30, 32, 36, 40, \dots, 42, 44, 48, 50, 54, 55, 58, 62, \dots, 64, 68, 70, 72, \dots, 74, 81, 82, 84, 105, 106\}; \{2, 4, \dots, 20, 22, \dots, 25, 31, 33, \dots, 35, 37, \dots, 39, 43, 45, \dots, 47, 49, 51, \dots, 53, 56, 57, 59, \dots, 61, 65, \dots, 67, 69, 71, 75, \dots, 80, 83, 85, \dots, 102, 104\}; \emptyset; \{103\}; \emptyset)$
Ref _{HH83WFL}	$(\{2, 4, 6, 10, 11, 15, 17, 19, 20\}; \{1, 3, 5, 7, 8, 12, \dots, 14, 16, 21\}; \emptyset; \{9, 18\}; \emptyset)$

Source	Ordered Tuple of Index Sets
Ref _{Haw75PCB}	($\{1, 3, 4, 6, 7, 10, \dots, 13, 20, 21, 23, 30, 33, 36, \dots, 39, 42, \dots, 45, 47, 48, 51, 55, 56, 59, 60, 64, 71, 73, 74, 84, \dots, 91, 94\}$; $\{5, 8, 9, 14, \dots, 19, 22, 24, 29, 31, 32, 34, 35, 40, 41, 46, 49, 50, 52, \dots, 54, 57, 58, 61, \dots, 63, 65, \dots, 70, 72, 75, \dots, 79, 82, 83, 92\}$; $\emptyset$; $\{2, 25, \dots, 28, 80, 81, 93\}$; $\emptyset$)
Ref _{Haw76BP}	($\{1, \dots, 4, 6, \dots, 8, 10, 12, \dots, 17, 19, 20, 23, 24, 26, 30, \dots, 35, 37, 39, 40, 42, 48, \dots, 50, 52, 58, 61, 65, \dots, 68, 70, \dots, 72, 74, \dots, 77\}$; $\{5, 9, 18, 21, 22, 25, 27, \dots, 29, 36, 38, 41, 43, \dots, 47, 51, 54, \dots, 57, 59, 60, 62, \dots, 64, 69, 73\}$; $\emptyset$; $\{11, 53\}$; $\emptyset$)
Ref _{Ish93POT}	($\{2, 4, 5, 8, 9, 11, 14, 32, 42, \dots, 44\}$; $\{1, 3, 6, 7, 10, 12, 13, 15, 16, 18, \dots, 31, 33, 36, \dots, 40, 45\}$; $\emptyset$; $\{17, 34, 35, 41\}$; $\emptyset$)
Ref _{Kie05QCAT}	($\{1, \dots, 3, 7, \dots, 9, 12, \dots, 16, 18, 19, 21\}$; $\{4, \dots, 6, 10, 11, 20, 22\}$; $\emptyset$; $\{17\}$; $\emptyset$)
Ref _{Kie09TQG}	($\{6, 10, 13, 14, 24, 26, \dots, 28, 30, \dots, 32, 34, 35\}$; $\{1, \dots, 5, 7, \dots, 9, 11, 16, \dots, 23, 25, 33\}$; $\emptyset$; $\{12, 15, 29\}$; $\emptyset$)
Ref _{Kuc92TIQG}	($\{1, 2, 6, 8, 9, 18, 22, \dots, 24, 28, 33, 36, 37, 46\}$; $\{4, 5, 7, 10, \dots, 17, 19, \dots, 21, 25, \dots, 27, 29, 32, 34, 35, 38, \dots, 45, 47\}$; $\emptyset$; $\{3, 30, 31\}$; $\emptyset$)
Ref _{Leb93ETA}	($\{1, 5, 10, 12, \dots, 14, 19, \dots, 23, 25, 29, 31, \dots, 33\}$; $\{3, 4, 9, 11, 15, \dots, 18, 24, 26, \dots, 28, 30, 34\}$; $\emptyset$; $\{2, 6, \dots, 8\}$; $\emptyset$)
Ref _{Lew16RR}	($\{1, 2, 4, 5, 9\}$; $\{3, 8, 10, 14\}$; $\emptyset$; $\{6, 7, 11, \dots, 13\}$; $\emptyset$)
Ref _{Mal98LN}	($\{8, 26\}$; $\{1, 2, 4, 14, 15, 18, 20, \dots, 23\}$; $\emptyset$; $\{3, 5, \dots, 7, 9, \dots, 13, 16, 17, 19, 24, 25\}$; $\emptyset$)
Ref _{Min09RZ}	($\{1, 10, 14, 16, \dots, 18, 20\}$; $\{3, \dots, 8, 15, 19, 21, 22, 29, 30\}$; $\emptyset$; $\{2, 9, 11, \dots, 13, 23, \dots, 28, 31, 32\}$; $\emptyset$)
Ref _{MTW73Grav}	($\{5, 17, 36, 39, \dots, 43\}$; $\{1, \dots, 4, 6, \dots, 16, 18, \dots, 35, 37, 38\}$; $\emptyset$; $\emptyset$; $\emptyset$)
Ref _{NASA20TT}	($\{4, 10, \dots, 12, 19\}$; $\{1, \dots, 3, 5, 6, 9, 13, 15, \dots, 18, 20\}$; $\emptyset$; $\{7, 8, 14\}$; $\emptyset$)
Ref _{New1687P}	($\{3, 4, 20, 22, \dots, 26\}$; $\{1, 2, 5, \dots, 12, 15, \dots, 19, 21, 27, \dots, 32\}$; $\emptyset$; $\{13, 14, 33, \dots, 41\}$; $\emptyset$)
Ref _{NIST25E}	($\{21\}$; $\{3, \dots, 20, 22, \dots, 24\}$; $\emptyset$; $\{1, 2\}$; $\emptyset$)
Ref _{Fey95LG}	($\{11\}$; $\{1, \dots, 10\}$; $\emptyset$; $\emptyset$; $\emptyset$)
Ref _{Nar10IR}	($\{6, 7, 9, \dots, 12\}$; $\{1, \dots, 5, 8, 13\}$; $\emptyset$; $\emptyset$; $\{14\}$)
Ref _{Hei58PP}	($\{4, \dots, 6, 13\}$; $\{14, 15\}$; $\emptyset$; $\{1\}$; $\{2, 3, 7, \dots, 12\}$)
Ref _{Pag93IBH}	($\{1, 3, 11, 17, \dots, 19\}$; $\{2, 6, \dots, 8, 12, \dots, 16\}$; $\emptyset$; $\{4, 5, 9, 10\}$; $\emptyset$)
Ref _{PW83EwE}	($\{4, 7, 20\}$; $\{1, \dots, 3, 5, 6, 8, \dots, 19, 21, \dots, 23\}$; $\emptyset$; $\emptyset$; $\emptyset$)
Ref _{Pea08ES}	($\{6, 15\}$; $\{1, 3, 4, 14, 17, 18, 21\}$; $\emptyset$; $\{2, 5, 7, \dots, 13, 16, 19, 20, 22\}$; $\emptyset$)
Ref _{Pen79STA}	($\{1, \dots, 3, 5, 7, 9, \dots, 12, 14, 15, 21, 23, 24\}$; $\{4, 6, 18, 19, 22\}$; $\emptyset$; $\{8, 13, 16, 17, 20, 25\}$; $\emptyset$)
Ref _{Per99OL}	($\{1, 2, 6, 11\}$; $\{3, \dots, 5, 7, 8, 10, 12, \dots, 14\}$; $\emptyset$; $\{9, 15\}$; $\emptyset$)
Ref _{Poe21LC}	($\{1, 3, 11, 13\}$; $\{4, 6, \dots, 10, 12, 14\}$; $\emptyset$; $\{2, 5\}$; $\emptyset$)
Ref _{PR60AWP}	($\{2, 9, 11, 15, 19\}$; $\{1, 3, \dots, 6, 8, 10, 12, \dots, 14, 16, \dots, 18, 20, \dots, 25\}$; $\emptyset$; $\{7\}$; $\emptyset$)

Source	Ordered Tuple of Index Sets
Ref _{Rie98OE}	$(\{1, 2, 6\}; \{3, 4, 7, \dots, 9, 11, \dots, 15\}; \emptyset; \{5, 10, 16\}; \emptyset)$
Ref _{Rin77ER}	$(\{3, 5, \dots, 7, 11, 13, 20, 23\}; \{4, 8, \dots, 10, 14, \dots, 19\}; \{1\}; \{2, 12, 21, 22\}; \emptyset)$
Ref _{Rov02PO}	$(\{4, 12, 13\}; \{1, \dots, 3, 5, \dots, 11\}; \emptyset; \emptyset; \emptyset)$
Ref _{Rov11FT}	$(\{3, 11, 14\}; \{2, 4, \dots, 10, 12, 13, 15, \dots, 23\}; \{1\}; \emptyset; \emptyset)$
Ref _{RT06HE}	$(\{8\}; \{1, \dots, 3, 5, 6, 9\}; \emptyset; \{4, 7\}; \emptyset)$
Ref _{Sch05DMP}	$(\{1, 2, 10, 12, 13, 17, 18, 20\}; \{3, \dots, 9, 11, 14, \dots, 16, 19, 21\}; \emptyset; \emptyset; \emptyset)$
Ref _{Smo18TR}	$(\{2, \dots, 7, 9, 10, 12, 13, 15, \dots, 17, 19, 20, 22, 26, 27\}; \{11, 14, 18, 21, 23, \dots, 25\}; \emptyset; \{8\}; \{1\})$
Ref _{TW92SP}	$(\{1, 2, 4, 5, 7, 13, 14, 17, 18, 20, 40\}; \{3, 6, 8, \dots, 12, 15, 16, 19, 21, \dots, 25, 28, \dots, 35, 41\}; \emptyset; \{26, 27, 36, \dots, 39\}; \emptyset)$
Ref _{VR10BS}	$(\{2, 3\}; \{1\}; \emptyset; \emptyset; \emptyset)$
Ref _{VN32MQ}	$(\{3, 5, 7, \dots, 10\}; \{1, 2, 4, 6, 11\}; \emptyset; \emptyset; \emptyset)$
Ref _{Wei89CCP}	$(\{3, 5, 9, \dots, 11\}; \{1, 2, 4, 6, 7, 12, \dots, 19, 21, 22\}; \emptyset; \{8, 20\}; \emptyset)$
Ref _{Zur03DEQ}	$(\{2, 4, 5, 10, 12, \dots, 14, 17\}; \{1, 3, 6, \dots, 8, 11, 15, 16, 18\}; \emptyset; \{9\}; \emptyset)$
Ref _{Ton13DR}	$(\{3, 4, 10\}; \{1, 2, 5, \dots, 9, 11, 12, 14, \dots, 17\}; \{13, 19\}; \{18\}; \emptyset)$
Ref _{Gre11FoC}	$(\{6, 9, 10, 13, 14, 16, 17\}; \{1, \dots, 5, 7, 8, 11, 12, 15\}; \emptyset; \emptyset; \emptyset)$
Ref _{NBSC476}	$(\emptyset; \{1, \dots, 7\}; \emptyset; \emptyset; \emptyset)$
Ref _{NISTSP958}	$(\emptyset; \{2, \dots, 8\}; \emptyset; \{1\}; \emptyset)$
Ref _{PDG23Res}	$(\emptyset; \{1, 2\}; \emptyset; \emptyset; \emptyset)$
Ref _{NISTSP905}	$(\emptyset; \{1, 3, \dots, 6\}; \emptyset; \{2\}; \emptyset)$
Ref _{Wolf11CF}	$(\emptyset; \{1, 4, 6, 8, \dots, 10, 12, \dots, 18, 20, 22, \dots, 26, 29, 30, 32, \dots, 41, 43, \dots, 55, 57, \dots, 67, 69, 70\}; \emptyset; \{2, 3, 5, 7, 11, 19, 21, 27, 28, 31, 42, 56, 68\}; \emptyset)$

The global fiber cardinalities of the five types are

$$\begin{aligned}
& \left| \coprod_{R \in \text{Ref}^{\text{audit}}} \left(\{R\} \times \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\text{member}}^{\text{time,ord,struct}} \right\} \right) \right) \right| = 642, \\
& \left| \coprod_{R \in \text{Ref}^{\text{audit}}} \left(\{R\} \times \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\text{dependency}}^{\text{time,ord,struct}} \right\} \right) \right) \right| = 1313, \\
& \left| \coprod_{R \in \text{Ref}^{\text{audit}}} \left(\{R\} \times \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\text{branch}}^{\text{time,ord,struct}} \right\} \right) \right) \right| = 8, \\
& \left| \coprod_{R \in \text{Ref}^{\text{audit}}} \left(\{R\} \times \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\text{separate}}^{\text{time,ord,struct}} \right\} \right) \right) \right| = 284, \\
& \left| \coprod_{R \in \text{Ref}^{\text{audit}}} \left(\{R\} \times \left(\text{TypeOf}_R^{\text{time,ord,struct}} \right)^{-1} \left(\left\{ \text{Type}_{\text{lexical}}^{\text{time,ord,struct}} \right\} \right) \right) \right| = 19.
\end{aligned}$$

The component order of each eight-axis inverse-image index-set tuple is: endpoint, instant, interval boundary, phase boundary, turning position, order relation, orientation, and transition position.

Table 26: Eight-Axis Indices of Temporal Order Structure for the Audited Source Set

Source	Ordered Tuple of Index Sets
Ref _{Adl03MP}	({2}; {1, 2}; ∅; ∅; ∅; {5}; ∅; ∅)
Ref _{And12PT}	(∅; {12, 16}; ∅; ∅; ∅; {16, 17}; ∅; ∅)
Ref _{And01DW}	(∅; {5}; ∅; ∅; ∅; {6, 7}; ∅; ∅)
Ref _{AriPhys4}	({3}; {3}; ∅; ∅; ∅; {2, 8}; ∅; ∅)
Ref _{ADM62Dyn}	({2}; {2, 7}; {7}; ∅; ∅; ∅; {11}; ∅)
Ref _{Bai77TD}	(∅; {2, 3}; {2}; ∅; ∅; ∅; ∅; ∅)
Ref _{Bar94TQ1}	(∅; {10}; ∅; ∅; ∅; {2, 10, 13}; ∅; ∅)
Ref _{Bar94TQ2}	({16}; {2, 6, 7, 14}; {18}; {11}; ∅; {2, 3, 7, 11, 14}; {15, ..., 17}; ∅)
Ref _{Bek73BHE}	({9, 12, 13, 21, 26}; {67}; ∅; ∅; {61}; {1, ..., 4, 29, 46, 52, 56, 64}; {1, 2, 4, 6, 15, 29, 30, 38}; {8, 14, 17, 18, 22, 27, 32, ..., 36, 42, 47, 53, ..., 55, 62, 63})
Ref _{BIPM19SI}	(∅; {16, 19}; ∅; ∅; ∅; ∅; ∅; ∅)
Ref _{Bom87CS}	(∅; {5}; ∅; ∅; ∅; {1, 4, 9, 11}; {3, 4}; ∅)
Ref _{Bri27LM}	({52}; {1, 5, 22, 30, 35, 38, 47, 50, 52, 56}; ∅; ∅; ∅; {3, 4, 23, 40, 49, 51}; {40, 44, 45, 51}; ∅)
Ref _{Bun09Red}	(∅; {1, 5, 6, 19}; ∅; ∅; ∅; {12}; ∅; ∅)
Ref _{CR94TT}	(∅; {5, 14}; ∅; ∅; {9}; {3, 9, 14}; {12}; {9})
Ref _{CookBurns09Metric}	(∅; {1, 6, 11, 21, 24, 27, 28}; ∅; ∅; {28}; ∅; ∅; {28})
Ref _{DL04EC}	({4, 8}; {1, 6, 9, 12, 13, 18, 24}; {5, 13}; ∅; ∅; {4, 12, 15, 24}; ∅; {12, 13})

Source	Ordered Tuple of Index Sets
Ref _{DeW67QG1}	$(\{13\}; \emptyset; \emptyset; \emptyset; \emptyset; \{6, 28, 33\}; \{16, 28, 29\}; \emptyset)$
Ref _{Dir58GH}	$(\emptyset; \{2\}; \emptyset; \emptyset; \emptyset; \{2\}; \emptyset; \emptyset)$
Ref _{EPS72GFL}	$(\{1\}; \{1, 3, 23, 24, 29, \dots, 31, 44, \dots, 47\}; \emptyset; \emptyset; \emptyset; \{5, 9, 18, 20, 31, 34, 38, 53\}; \{50, 51\}; \emptyset)$
Ref _{Ein05ED}	$(\{26, 28, 55, 56, 59, 60, 88, 89\}; \{7, 9, \dots, 11, 13, 17, \dots, 19, 21, 23, 26, \dots, 28, 36, \dots, 38, 51, 53, 55, \dots, 60, 62, 64, 74, \dots, 76, 88, 89, 102, 111, 118, 119, 131\}; \{27, 57, 58\}; \emptyset; \emptyset; \{5, 15, 16, 43\}; \emptyset; \emptyset)$
Ref _{Ein16GR}	$(\{9, 10\}; \{5, 6, 9, 10, 21, 22, 24, 28, 29, 37, 38\}; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{Ell12PT}	$(\{34\}; \{2, \dots, 4, 6, 8, 9, 14, 16\}; \{39\}; \emptyset; \emptyset; \{2, 3, 8, 10, 11, 16, 33\}; \{13, 28, 29, 40\}; \{6, 39\})$
Ref _{ESA19CGR}	$(\emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \{18\}; \emptyset; \emptyset)$
Ref _{FLS63T}	$(\{34, 49, 68, 69, 104\}; \{4, 8, 17, 60, 68, 69, 75, 90, 91, 94, 104, 119, 135, 136\}; \emptyset; \emptyset; \{107\}; \{1, 18, 20, 34, 37, 45, 50, 116, 133\}; \{131, 132\}; \emptyset)$
Ref _{FBJL07ES}	$(\emptyset; \{1, 3, 6, 10, 14, 17, 18\}; \{17, 18\}; \emptyset; \emptyset; \{1, 14, 20\}; \emptyset; \emptyset)$
Ref _{GE07FE}	$(\emptyset; \{1, 7, 10, 11, 13\}; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \{7\})$
Ref _{HK72Pred}	$(\{20, 42, 44, 48, 50\}; \{2, 6, 7, 20, 42, \dots, 44, 48, \dots, 50, 54, 55, 69, 71\}; \emptyset; \emptyset; \emptyset; \{1, 56\}; \emptyset; \emptyset)$
Ref _{HK72Obs}	$(\{26, 54, 84\}; \{1, 3, 21, 26, \dots, 28, 30, 40, 42, 54, 55, 62, \dots, 64, 68, 72, 74, 106\}; \{36, 44\}; \{81\}; \emptyset; \{1, 29, 32, 41, 50, 58, 68, 70, 81, 82, 105\}; \emptyset; \{21, 48, 62, 72, 73\})$
Ref _{HH83WFO}	$(\{2, 10, 15\}; \{6\}; \{4, 11\}; \{19\}; \{19\}; \{2, 4, 6, 10, 19, 20\}; \{17\}; \{20\})$
Ref _{Haw75PCB}	$(\{3, 23, 33, 45, 47, 48, 64, 86, 90\}; \{4, 6, 33, 55, 59, 73\}; \{43, 44, 71\}; \{10, \dots, 12, 20, 38, 39, 51, 85, 88, 89\}; \emptyset; \{1, 7, 13, 21, 30, 42, 56, 60, 74, 84, 87, 91\}; \{36, 37, 94\}; \{73\})$
Ref _{Haw76BP}	$(\{1, 12, 15, 16, 76\}; \{6, 10, 17, 20, 24, 30, 49, 52, 61, 71\}; \{2, \dots, 4, 23, 58\}; \{32, 35, 37, 40, 48, 65, \dots, 67, 70, 74\}; \emptyset; \{7, 8, 13, 14, 31, 33, 34, 50, 68, 72, 75, 77\}; \{13, 19, 26, 42, 75\}; \{10, 17, 39\})$
Ref _{Ish93POT}	$(\{9\}; \{2, 8, 9, 14, 42, 44\}; \emptyset; \emptyset; \emptyset; \{4, 5, 11, 42, 43\}; \{5, 32\}; \emptyset)$
Ref _{Kie05QCAT}	$(\emptyset; \{7, 21\}; \{12, 16\}; \{8\}; \{9, 19, 21\}; \{3, 8, 9, 13, 14\}; \{1, 2, 13, 15, 18, 19\}; \emptyset)$
Ref _{Kie09TQG}	$(\{14, 28, 31, 32, 35\}; \{6\}; \{24\}; \{26\}; \{26, 34\}; \{6, 26, 27, 30\}; \{10, 13, 30\}; \emptyset)$
Ref _{Kuc92TIQG}	$(\{36\}; \{2, 6, 8, 18, 24, 28, 33, 37, 46\}; \emptyset; \emptyset; \emptyset; \{1, 9, 23\}; \{9, 22, 36\}; \emptyset)$
Ref _{Leb93ETA}	$(\{23\}; \{1, 10, 12, 14\}; \{19, 22\}; \emptyset; \emptyset; \{1, 20, \dots, 22, 25\}; \{5, 13, 29, 31, \dots, 33\}; \{12, 14, 29\})$
Ref _{Lew16RR}	$(\emptyset; \{5\}; \emptyset; \emptyset; \emptyset; \{1, 2, 9\}; \{1, 4\}; \emptyset)$
Ref _{Mal98LN}	$(\emptyset; \emptyset; \{8\}; \emptyset; \emptyset; \{26\}; \emptyset; \emptyset)$
Ref _{Min09RZ}	$(\{20\}; \{1\}; \emptyset; \emptyset; \emptyset; \{10, 16, \dots, 18\}; \{10, 14, 16, 20\}; \emptyset)$

Source	Ordered Tuple of Index Sets
Ref _{MTW73Grav}	$(\emptyset; \{17, 36, 40\}; \{43\}; \emptyset; \emptyset; \{39, \dots, 43\}; \{5, 42\}; \emptyset)$
Ref _{NASA20TT}	$(\emptyset; \{10, \dots, 12, 19\}; \emptyset; \emptyset; \emptyset; \{4\}; \emptyset; \emptyset)$
Ref _{New1687P}	$(\emptyset; \{20, 22\}; \emptyset; \emptyset; \emptyset; \{23, \dots, 26\}; \{3, 4\}; \emptyset)$
Ref _{NIST25E}	$(\emptyset; \{21\}; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{Pag93IBH}	$(\emptyset; \{17\}; \{11\}; \emptyset; \emptyset; \{1, 3, 18, 19\}; \emptyset; \{17\})$
Ref _{PW83EwE}	$(\emptyset; \{4\}; \emptyset; \emptyset; \emptyset; \{7\}; \{20\}; \emptyset)$
Ref _{Pea08ES}	$(\{15\}; \{6, 15\}; \emptyset; \emptyset; \emptyset; \{15\}; \emptyset; \emptyset)$
Ref _{Pen79STA}	$(\{14, 21\}; \emptyset; \{7, 23, 24\}; \emptyset; \emptyset; \{2, 5, 9, 11, 15\}; \{1, \dots, 3, 7, 9, \dots, 12, 15\}; \emptyset)$
Ref _{Per99OL}	$(\emptyset; \{1, 2\}; \emptyset; \{6\}; \emptyset; \{1, 6, 11\}; \emptyset; \emptyset)$
Ref _{Poe21LC}	$(\emptyset; \{1, 3, 11\}; \emptyset; \emptyset; \emptyset; \{1, 3, 13\}; \emptyset; \{11\})$
Ref _{PR60AWP}	$(\emptyset; \{15, 19\}; \{9, 15, 19\}; \{9\}; \emptyset; \{2, 11, 19\}; \emptyset; \emptyset)$
Ref _{Rie98OE}	$(\emptyset; \{1, 2\}; \emptyset; \emptyset; \emptyset; \{1, 6\}; \emptyset; \emptyset)$
Ref _{Rin77ER}	$(\emptyset; \{3, 5, \dots, 7, 13, 20\}; \emptyset; \emptyset; \emptyset; \{11\}; \{23\}; \{20\})$
Ref _{Rov02PO}	$(\{12\}; \{4\}; \emptyset; \emptyset; \emptyset; \{12, 13\}; \emptyset; \emptyset)$
Ref _{Rov11FT}	$(\emptyset; \{11\}; \emptyset; \emptyset; \emptyset; \{3, 11\}; \{3, 14\}; \emptyset)$
Ref _{RT06HE}	$(\emptyset; \{8\}; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{Sch05DMP}	$(\{2, 20\}; \{10, 12, 17, 18\}; \emptyset; \emptyset; \emptyset; \{1, 2, 13, 18\}; \emptyset; \emptyset)$
Ref _{Smo18TR}	$(\emptyset; \{22\}; \emptyset; \{6, 12, 15, 16\}; \emptyset; \{2, 4, 5, 7, 13, 19, 20, 26, 27\}; \{9, 10, 17, 26, 27\}; \{3\})$
Ref _{TW92SP}	$(\emptyset; \{1, 2, 4, 7, 17, 18, 20, 40\}; \emptyset; \emptyset; \emptyset; \{5, 13, 14\}; \{13, 14\}; \emptyset)$
Ref _{VR10BS}	$(\emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \{2, 3\}; \{2, 3\}; \emptyset)$
Ref _{VN32MQ}	$(\{5, 8\}; \{3, 5, 8, 9\}; \{7\}; \emptyset; \emptyset; \{7, 8\}; \{10\}; \emptyset)$
Ref _{Wei89CCP}	$(\emptyset; \{3, 10, 11\}; \{9\}; \{5\}; \emptyset; \{3\}; \emptyset; \emptyset)$
Ref _{Zur03DEQ}	$(\{2, 10\}; \{12, 14, 17\}; \emptyset; \emptyset; \emptyset; \{4, 5, 12, 13\}; \emptyset; \emptyset)$
Ref _{Ton13DR}	$(\emptyset; \{3\}; \emptyset; \emptyset; \emptyset; \{4\}; \{4, 10\}; \emptyset)$
Ref _{Gre11FoC}	$(\{17\}; \{6, 9, 10\}; \{16\}; \emptyset; \emptyset; \{13\}; \{14\}; \emptyset)$
Ref _{NBSC476}	$(\emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{NISTSP958}	$(\emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{PDG23Res}	$(\emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{NISTSP905}	$(\emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{Wolf11CF}	$(\emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{Fey95LG}	$(\{11\}; \emptyset; \{11\}; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset)$
Ref _{Nar10IR}	$(\{6, 7, 10, 11\}; \{6, 7, 9, \dots, 12\}; \emptyset; \emptyset; \emptyset; \emptyset; \emptyset; \{9\})$
Ref _{Hei58PP}	$(\emptyset; \{5\}; \emptyset; \emptyset; \emptyset; \{13\}; \{4, 6\}; \emptyset)$

The cardinalities of the time contents whose images include each axis are as follows.

axis	cardinality
endpoint	80
instant	281
interval_boundary	41
phase_boundary	32
turning_position	10
order_relation	216
orientation	97
transition_position	43

The cardinality of the time contents whose images include at least two axes is

$$\left| \left\{ (R, x) \in \text{TimeContent}^{\text{audit}} \mid \left| \text{AxisOf}_R^{\text{time,ord,struct}}(x) \right| \geq 2 \right\} \right| = 149.$$

8.18.2 Transfer Reachability of Time Content and Structural Signatures

Within-reference time-content transfer reachability, the member structural signature, and the dependent-content structural signature are given, respectively, by Definitions 7.5, 7.23 and 7.24 .

Table 27: Cardinalities of the Fibers of the Member Structural Signature

Signature	Cardinality
(0, 0, 0)	69
(1, 0, 0)	13
(0, 1, 0)	9
(1, 1, 0)	10
(0, 0, 1)	174
(1, 0, 1)	201
(0, 1, 1)	99
(1, 1, 1)	67

Table 28: Cardinalities of the Fibers of the Dependent-Content Structural Signature

Signature	Cardinality
(0, 0, 0)	133
(1, 0, 0)	12
(0, 1, 0)	17
(1, 1, 0)	2
(0, 0, 1)	730
(1, 0, 1)	127
(0, 1, 1)	237
(1, 1, 1)	55

The zero fiber of the member structural signature,

$$\coprod_{R \in \text{Ref}^{\text{audit}}} \left(\{R\} \times \left(\text{Sig}_R^{\text{time,ord,struct,member}} \right)^{-1} (\{(0, 0, 0)\}) \right)$$

contains 69 entries. Each corresponds to an evidentiary location in an original-source definition, equation, or explicit statement that provides an endpoint, instant, interval boundary, phase boundary, turning position, order relation, orientation, or transition position.

The zero fiber of the dependent-content structural signature,

$$\coprod_{R \in \text{Ref}^{\text{audit}}} \left(\{R\} \times \left(\text{Sig}_R^{\text{time,ord,struct,dependency}} \right)^{-1} (\{(0,0,0)\}) \right)$$

contains 133 entries. Each corresponds to an evidentiary location in an original-source definition, equation, or explicit statement that provides a clock, coordinate, reference scale, duration, rate, differentiation variable, or representational mediator.

Coverage by the five-type classification of time content, totality of the eight-axis classification for members, unique decomposition of structural signatures, and correspondence of zero fibers to original-source evidence are established, respectively, by Propositions 7.195, 7.196, 7.197 and 7.198 .

8.18.3 Audit-Boundary Directed Hypergraphs

Audit-boundary directed hypergraphs are given by Definition 7.27.

The physical and mathematical boundary claims, time contents, transfer events, and reference-specific results stated in each reference are represented by a finite directed hypergraph for that reference. The input set of each hyperedge represents the set of premises used jointly to establish its output.

Audit-boundary nodes, audit-boundary support sets, and audit-boundary edges are given by Definition 7.18 and Definition 7.26.

Input consistency of audit-boundary directed hypergraphs is established by Proposition 7.122.

8.18.4 Reachability on the Audit-Boundary Directed Hypergraph

Audit reachability stages, audit-generation traces, input influence regions, and audit-generated time contents, primitive time roots, and audit-generated results are given by Definition 7.30, Proposition 7.31, Definition 7.32, Definition 7.34, and Definition 7.39.

8.18.5 Result Dependence on Primitive Time Roots

The decomposition of reference-specific results by dependence on primitive time roots is given by Proposition 7.49.

8.19 Complete Audits by Source

Each index cell in a reference-specific audit table has the typed index display of Definition 7.92. The indices of supporting claims for transfers and results, the indices of time-content inputs and outputs, and the indices of time-content prerequisites have the typed incidence realization of Definition 7.105. The minimal primitive-time-root support index families and the sum-product of set families are given by Definition 7.108.

Each of the 72 references in this Supplement has a finite family of claim nodes, a finite family of time contents, a finite family of transfer events, and a finite family of results.

For each such reference, the inclusion

$$\text{Init}_R^{\text{audit,phys}} \subseteq \text{Env}_R^{\text{audit,evid}} \subseteq \text{Node}_R^{\text{audit,claim}}$$

holds. By Proposition 7.107, the transfer and result tables for each reference determine a unique audit-boundary directed hypergraph, together with its reachability, primitive time roots, fourfold result partition, and minimal primitive-time-root support family. For a reference that directly supplies typed finite audit input, Proposition 7.106 determines the same audit boundary as its canonical realization.

Definition 8.113 (Tabular Display of Minimal Primitive-Time-Root Support Statistics). For each $R \in \text{Ref}^{\text{audit}}$ and $q \in \text{Idx}_R^{\text{audit,result}}$, the support-statistics field in the reference-specific minimal primitive-time-root support table retains the following three components in order:

$$\#\mathcal{A}_R(q), \quad \min_{J \in \mathcal{A}_R(q)} \#J, \quad \max_{J \in \mathcal{A}_R(q)} \#J$$

If the support family is empty, it is displayed as

$$(0; \text{—})$$

If the support family is nonempty and the two cardinality endpoints coincide, it is displayed as

$$\left(\#\mathcal{A}_R(q); \min_{J \in \mathcal{A}_R(q)} \#J \right)$$

If the two cardinality endpoints differ, it is displayed as

$$\left(\#\mathcal{A}_R(q); \min_{J \in \mathcal{A}_R(q)} \#J, \max_{J \in \mathcal{A}_R(q)} \#J \right)$$

Result indices having the same three components are listed in the same row.

Definition 8.114 (Typed Index Relations of Primitive-Time-Root–Result Dependence). For each $R \in \text{Ref}^{\text{audit}}$, define

$$\begin{aligned} \text{RootResultDepIdx}_R^{\text{audit}} &:= \left\{ (i, q) \in \{1, \dots, n_R\} \times \text{Idx}_R^{\text{audit,result}} \mid \right. \\ &\quad \left. (\text{TimeContent}_{R,i}, \text{Result}_{R,q}) \in \text{RootResultDep}_R^{\text{audit}} \right\}, \\ \text{RootResultEssDepIdx}_R^{\text{audit}} &:= \left\{ (i, q) \in \{1, \dots, n_R\} \times \text{Idx}_R^{\text{audit,result}} \mid \right. \\ &\quad \left. (\text{TimeContent}_{R,i}, \text{Result}_{R,q}) \in \text{RootResultEssDep}_R^{\text{audit}} \right\}. \end{aligned}$$

The following equations uniquely reconstruct the two complete relations:

$$\begin{aligned} \text{RootResultDep}_R^{\text{audit}} &= \left\{ \left(\text{TimeContent}_{R,i}, \text{Result}_{R,q} \right) \mid (i, q) \in \text{RootResultDepIdx}_R^{\text{audit}} \right\}, \\ \text{RootResultEssDep}_R^{\text{audit}} &= \left\{ \left(\text{TimeContent}_{R,i}, \text{Result}_{R,q} \right) \mid (i, q) \in \text{RootResultEssDepIdx}_R^{\text{audit}} \right\} \end{aligned}$$

Definition 8.115 (Typed Index Tuple of the Reference-Specific Audit Partition). For each $R \in \text{Ref}^{\text{audit}}$, define

$$\begin{aligned} J_R^{\text{gen}} &:= \left\{ j \in \{1, \dots, n_R\} \mid \text{TimeContent}_{R,j} \in \text{TimeContent}_R^{\text{audit,gen}} \right\}, \\ J_R^{\text{root}} &:= \left\{ j \in \{1, \dots, n_R\} \mid \text{TimeContent}_{R,j} \in \text{TimeContent}_R^{\text{audit,root}} \right\}. \end{aligned}$$

Define the index sets of the fourfold result partition by

$$\begin{aligned} Q_R^\beta &:= \left\{ q \in \text{Idx}_R^{\text{audit,result}} \mid \begin{array}{l} \text{Result}_{R,q} \\ \in \text{Result}_R^{\text{audit},\beta} \end{array} \right\}, \\ \beta &\in \{\text{gen}, \text{evidonly}, \text{rootdep}, \text{rootunreach}\} \end{aligned}$$

Define the typed index tuple of the reference-specific audit partition by

$$\text{AuditPartIdx}_R := \left(J_R^{\text{gen}}, J_R^{\text{root}}, Q_R^{\text{gen}}, Q_R^{\text{evidonly}}, Q_R^{\text{rootdep}}, Q_R^{\text{rootunreach}} \right)$$

The order of these six components is: the generated set of time contents; the primitive-time-root set; the generated result set; the evidence-environment-reachable nongenerated result set; the primitive-time-root-dependent result set; and the result set remaining unreachable after addition of primitive time roots.

The following equations uniquely reconstruct all six sets:

$$\begin{aligned} \text{TimeContent}_R^{\text{audit},\alpha} &= \text{TimeContent}_R[J_R^\alpha], \quad \alpha \in \{\text{gen}, \text{root}\}, \\ \text{Result}_R^{\text{audit},\beta} &= \text{Result}_R[Q_R^\beta], \quad \beta \in \{\text{gen}, \text{evidonly}, \text{rootdep}, \text{rootunreach}\} \end{aligned}$$

Definition 8.116 (All-Primitive-Time-Root and All-Result-Dependent Index Tuple). For positive integers

$$n, m \in \mathbb{N}, \quad n, m \geq 1$$

define

$$\begin{aligned} \text{AllRootDepldx}(n, m) &:= \left(\emptyset, \{1, \dots, n\}; \emptyset, \emptyset, \{1, \dots, m\}, \emptyset \right). \end{aligned}$$

Accordingly,

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(n, m)$$

denotes the six-component index tuple in which, for reference R , all n time contents belong to the primitive time roots and all m results belong to the primitive-time-root-dependent results.

8.19.1 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 1.

Source Name Primitive-Time-Root Audit of Aristotle, *Physics*, Book IV, Chapters 11–14

Bibliographic Identification Aristotle, *Physica*, Book IV, Chapters 11–14, Bekker 218b21–224a15, edited by Immanuel Bekker, *Aristotelis opera*, Volumen Primum (1831), and translated by Edward Hussey, *Aristotle's Physics: Books III and IV*, Clarendon Press / Oxford University Press (1983; corrected reprint, 1993), are taken as $R = \text{Ref}_{\text{AriPhys4}}$.

Audit-Scope Identifier 1-SC1.

Audit Scope The audit scope is Bekker 218b21–224a15 and pp. 43–54 of Hussey's English translation.

Original-Source Location Identifier 1-L1.

Original-Source Location Bekker 218b21–224a15 and pp. 43–54 of Hussey's English translation. The locations of the individual time contents are given in Table 29.

Formal Object

Object Identifier 1-F1.

Object Name Formal object of the original source.

Object Type Formal object.

Formal Content Definitions, arguments, and dependency relations for the object-level distinction between change and time; before and after; two nows; counted number; the mutual measurement of time and change; a common time for multiple changes; periodic scales; and deictic temporal positions.

Original-Source Location Identifier Set 1-L1.

Assumptions

Assumption Identifier 1-A1.

Assumption Content The argument uses continuity of magnitude, the correspondence whereby change follows magnitude and time follows change, and the premise that what is bounded by the now is time (219a10–29). Measurement requires a change serving as a unit; common time requires equality and simultaneity; the periodic scale uses uniform circular motion. The discussion of soul is conditional on restricting the capacity to count to soul and intellect (223a21–29).

Original-Source Location Identifier Set 1-L1.

Non-Temporal Physical Inputs

Input Identifier 1-P1.

Input Name Non-temporal physical inputs of the original source.

Input Type Non-temporal physical input.

Input Content Source statement groups 1–11 are represented by Components 1–29 in Table 30. The generation-initial inputs are spatial geometry, the constitution and positions of bodies bearing change, and collections and attributes of countable bodies, specified in Components 30–32. Before and after in change, the performance of counting, simultaneity and periodicity belong to the source formal evidence and the corresponding time contents.

Original-Source Location Identifier Set 1-L1.

Time Content

Time-Content Identifier 1-T1–1-T8.

Time-Content Name Time contents used in the original source.

Time-Content Type Time content.

Time-Content Specification The eight time contents of Definition 8.117.

Original-Source Location Identifier Set 1-L1.

Formal Input Elements

Formal-Component Identifier 1-C1.

Formal-Component Name Formal-evidence-environment specification.

Formal-Component Type Formal component.

Formal-Component Text Perception of change, recurrence, measurement, existence, rest, present boundaries, domains of causal attribution, simultaneity, uniformity, periodicity, and cyclic scales are retained in the reference formal-evidence environment.

Original-Source Support Identifier Set 1-L1.

Definition 8.117 (Time Contents in the Aristotle Reference). For reference $R = \text{Ref}_{\text{AriPhys4}}$, set $n_R = 8$. Table 29 gives each time content and its original-source location.

Table 29: Time Contents and Original-Source Locations
in the Aristotle Reference

i	Time Content	Original-Source Location
1	Elapsed extent and finite interval corresponding to perception of change, two nows, or a reference now	218b21–219a9, 219a22–219b9, 220b32–221a4, 222a10–222b27, 223a16–223a29
2	General domain of temporal order used for the before and after of change, continuity, existence in time, and placement of all changes	218b21–220a26, 220b14–222a9, 222a10–224a15
3	Position of the now used for two nows, the unit of temporal number, the limit of time, and the reference now	219a10–220a26, 220b14–222a9, 222a10–223a29

i	Time Content	Original-Source Location
4	Counted number of change with respect to before and after, countable temporal number, and object counted by soul or intellect	219b1–220b14, 223a16–223b12
5	Quantitative time or temporal scale that measures change and is measured by change	219b9–222a9, 223a16–224a15
6	The same time assigned in common to multiple simultaneous changes, and different times corresponding to before and after	220a27–220b14, 223a29–223b12
7	Periodic and cyclic temporal scale corresponding to recurrence of years and seasons and to uniform circular motion	220a27–220b14, 223b12–224a15
8	Deictic temporal position or interval that locates past, future, near, earlier, and suddenly relative to the now	222a10–223a16

Definition 8.118 (Physical and Formal Input-Claim Components in the Aristotle Reference). Define the set of physical and formal input-claim components by $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(32)$. Table 30 gives the content of each component and its corresponding original-source statement-group key. The generation-initial input index set and the reference formal-evidence-environment index set are defined by

$$\text{ClaimBoundaryIdx}_R = \left(\{30, 31, 32\}, [32] \right)$$

Hence $\text{StrictClaimBoundary}(R)$ holds. Components 1–29 retain the source definitions, distinctions and arguments; Components 30–32 specify non-temporal projections of the carriers and spatial configurations involved.

Table 30: Physical and Formal Input-Claim Components in the Aristotle Reference

p	Original-Source Statement-Group Key	Type-Homogeneous Component
1	1	Perception of change
2	2	Relative positional order in magnitude and place
3	2	Object of change
4	3	Limit relation
5	3	Intermediate relation
6	3	Mark
7	3	Countability
8	4	Moving object

p	Original- Source State- ment- Group Key	Type-Homogeneous Component
9	4	Countable before-and-after relation in change
10	5	Number
11	5	Continuous quantity
12	5	Object counted
13	6	Quantity of change
14	6	Recurrence relation
15	6	Equality relation
16	7	Measurement relation
17	7	Predicate concerning existence in time
18	7	Predicate concerning rest
19	8	Present boundary
20	8	Alteration
21	8	Domain of application of causal attribution to time
22	9	Soul
23	9	Intellect
24	9	Act of counting
25	10	Continuous change
26	10	Simultaneity
27	10	Common limit of multiple changes
28	11	Physical state of circular motion
29	11	Uniformity, periodic recurrence, and cyclic scale
30	2,3,5	Spatial configurations of magnitudes and places, and the geometrical limit and intermediate relations of lines and points.
31	2,4,8,11	The constitution and spatial positions of bodies bearing change, alteration and circular motion.
32	5,10	Collections of countable bodies, such as horses and dogs, and their material attributes.

Dependency Inputs

Dependency Inputs The time-content inputs and physical-input-claim support assigned to the seven transfer events in Definition 8.119.

Dependency and Transfer Elements

Definition 8.119 (Time-Content Transfers in the Aristotle Reference). Define the transfer index set by $\text{Idx}_R^{\text{audit,transfer}} = [7]$. For each $j \in [7]$, define the time-content input index set, time-content output index set, and physical-input-claim support index set by the following

finite index table.

j	$J_{R,j}^{\text{in}}$	$J_{R,j}^{\text{out}}$	$P_{R,j}^{\text{tr}}$
1	{3}	{1}	{4, 5}
2	{2}	{2}	{2, 3}
3	{2, 3}	{4}	{7, 9, 10, 12, 22, 24}
4	{4}	{5}	{13, 15, 16}
5	{1, 3}	{8}	{19}
6	{3, 4}	{6}	{15, 25, 26, 27}
7	{7}	{5}	{16, 28, 29}

The support set, time-content input set, and time-content output set of each transfer are determined by Definition 7.105.

The operations and original-source locations of the transfers are as follows.

- 1 Distinguish two nows as before and after and recognize the elapsed interval between them (219a22–219b9).
- 2 Relate before and after in magnitude and place to before and after in change. The latter belongs to Time Content 2 (219a10–21).
- 3 Count nows distinguished as before and after, taking time to be the number of change in that respect (219a22–219b9; 223a16–29).
- 4 Use a specified change as the unit for mutual measurement of counted temporal number and quantity of change (220b14–221a17).
- 5 Describe past, future, nearness, remoteness and suddenness through intervals relative to a reference now (222a10–223a16).
- 6 Assign the same time to multiple changes whose numbers are equal and which reach a common limit together (223a29–223b12).
- 7 Measure time using the periodic scale of uniform circular motion (223b12–224a2).

Transfer Events and Outputs The seven time-content transfers of the same definition and the time-content output assigned to each.

Applicability Conditions

Conditions under Which a Dependency or Transfer Applies Transfer 1 uses two nows distinguished as before and after and an intermediate; Transfer 2 uses the correspondence between magnitude, place and change; Transfer 3 uses the countability and counting of before and after. Transfer 4 specifies a change as the unit of measurement, and Transfer 5 uses intervals relative to a reference now. Transfer 6 requires equal temporal numbers and a common limit reached together; Transfer 7 uses uniform circular motion and its periodic scale.

Original-Source Support

Original-Source Support for Each Dependency or Transfer The physical-input-claim support for each transfer is given by the support set in the same definition. The original-source locations of the input and output time contents are given in Table 29.

Results

Definition 8.120 (Results in the Aristotle Reference). Define the result index set by $\text{idx}_R^{\text{audit,result}} = [16]$. Define each result, time-content prerequisite, supporting physical and formal input-claim component, and original-source location by the equalities in Table 31. Results 4 and 12 concern general distinctions between numbers and counted objects; Results 5 and 11 concern their application to time. Result 16 summarizes source statement groups 1–11 and jointly uses Time Contents 1–8 and Claim Components 1–29.

Table 31: Results and Audit Inputs
in the Aristotle Reference

j	Result and Original-Source Location	Time-Content Prerequisite Indices	Supporting Claim-Component Indices
1	The object-level distinction and inseparability of time and change. 218b21–219a9	$\{2\}$	$\{1\}$
2	Argument relating before and after in change to before and after in magnitude and place. 219a10–219a21	$\{2\}$	$\{2, 3\}$
3	Relation among two nows, an intermediate, and judgment of elapsed time. 219a22–219b9	$\{1\}$	$\llbracket 4, 7 \rrbracket$
4	Distinction between number used for counting and number counted. 219b1–220a26	$\emptyset$	$\{7, 10, 12\}$
5	Time as the counted number of change with respect to before and after. 219b1–220a26	$\{4\}$	$\{7, 9, 10, 12\}$
6	The limiting and unit functions of the now. 219b9–220a26	$\{3\}$	$\{8, 9, 19\}$
7	Mutual measurement of time and change. 220a27–221b3	$\{5\}$	$\{10, 11, 13, 15, 16\}$
8	Analysis of existence in time and domain of application. 220b14–222a9	$\{2, 5\}$	$\llbracket 16, 18 \rrbracket$
9	Scope of everyday causal attribution to time. 221a30–221b3; 222b16–27	$\{2\}$	$\{20, 21\}$
10	Distinctions among soul, intellect, countability, counting, and temporal number. 223a16–223b12	$\{4\}$	$\{7, 9, 22, 23, 24\}$
11	Condition for assigning a common time to multiple changes. 220a27–220b14, 223a29–223b12	$\{6\}$	$\{10, 12, 15, 25, 26, 27\}$
12	Distinction between the same number and different objects counted. 220b5–14, 223b3–11, 224a2–15	$\emptyset$	$\{10, 12, 15\}$

j	Result and Original-Source Location	Time-Content Prerequisite Indices	Supporting Claim-Component Indices
13	Rule for selecting uniform circular motion as the primary measure. 223b12–224a2	{7}	{16, 28, 29}
14	Reason for the identification of the motion of the celestial sphere with time. 223b21–224a2	{7}	{16, 28, 29}
15	Analysis of past, present, future, and temporal adverbs. 222a10–223a16	{8}	{19}
16	Continuous argument for the constitution of time in Book IV, Chapters 11–14. 218b21–224a15	[8]	[29]

Source-Specific Results The 16 results of Definition 8.120. The original-source location, time-content prerequisite, and supporting physical and formal input-claim components for each result are given in Table 31.

Result Dependencies

Index Identifier S01-XI-RD1.

Existing-Evidence Reference Set Definition 8.120.

Grounds of Establishment

Grounds for Establishing the Results Definitions, distinctions, arguments and conceptual dependencies in Bekker 218b21–224a15. The conclusion concerning soul and intellect in Result 10 is conditional on restricting the capacity to count to soul and intellect; the possibility of change itself existing without soul is distinguished (223a21–29).

Evidence Hierarchy

Objects of Formal Generalization Within the argumentative scope of Book IV, Chapters 11–14, the original source gives general relations concerning time and change, before and after, the now, counting, measurement, a common time for multiple changes, periodic scales, and tense expressions.

Conditions and Domain of Application of Formal Generalization Its domain of application is determined by the original-source location and supporting physical and formal input-claim components in each result row.

Generation Judgment **Judgment values:** LCTR complete-generation judgment: 0 cases / within-reference-evidence-conditional local-reach judgment: 0 cases / remaining primitive time roots: 8 cases.

Generation Reach and Primitive Time Roots The generation reach whose initial set consists only of non-temporal physical inputs generates zero time contents. All eight time contents in Definition 8.117 remain as primitive time roots.

Within-Reference-Evidence-Conditional Local Reach The within-reference-evidence-conditional local-reach judgment has zero cases, and eight primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ in Definition 7.108 and apply Definitions 7.50 and 8.114. Summation over all results gives 33 primitive-time-root–result dependence relations, of which 10 are necessary primitive-time-root–result dependence relations.

Result Dependence Results 4 and 12 have empty primitive-time-root support; the remaining 14 results depend on primitive time roots. The minimal primitive-time-root support of each result is given by Proposition 8.122.

Evidence Effect

Evidence-Effect Domain The formal evidence in this reference directly supports the 16 source-specific results in Bekker 218b21–224a15 and their definitions, distinctions, and dependence relations.

Evidence-Effect Boundary By Definition 7.115, a claim that the time contents in the original source have been physically generated lies beyond the evidence-effect boundary of this reference.

Result-Preservation Domain

Result-Preservation Domain The 16 results in Definition 8.120 are preserved when each result’s time-content presupposition is supplied by the corresponding generated time content in the LCTR generation sequence and the original source’s supporting claims, definitions, distinctions, and directions of dependence are retained.

Placement of Grounding Obligations

Placement of Grounding Obligations The eight time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as already established time inputs. The judgment on the placement of the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment is grounded in a comparison between the use of each time content in a law operation or result derivation and the absence, before that use, of the corresponding generation or establishment condition as an input, formation condition, evaluation condition, or invalidation condition.

Evidence of Actual Invalidation

Evidence of Actual Invalidation The complete-generation judgment and remaining-root judgment concern physical-time qualification and are distinct from the actual invalidation of a specific condition that determines the value or domain of application of a result. Neither the original-source formal evidence nor the LCTR audit evidence for this reference establishes the actual invalidation of such a target condition. The actual-invalidation judgment is “indeterminate from the original-source evidence”; actual invalidation is not inferred solely from the backgrounding of a grounding obligation.

Downstream Reach

Downstream Reach The reach to results that depend on primitive time roots is determined for this reference by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because the actual invalidation of a specific condition has not been established, the judgment on actual reach from that invalidation position to results is “indeterminate from the original-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the targets of repair are the corresponding time content and the generation or establishment conditions required before its use. The scope of recalculation is limited to results for which actual invalidation has been independently established and that are reached by dependence from the invalidation position. Because actual invalidation is indeterminate in this reference, this repair-and-recalculation specification is conditional. If the condition holds, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusions

Proposition 8.121 (Primitive Time Roots and Result Dependence in the Aristotle Reference). All eight time contents are primitive time roots. Under the source formal-evidence environment, Results 4 and 12 are reachable without primitive time roots as seeds; the remaining 14 results depend on primitive time roots. In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [8]; \emptyset, \{4, 12\}, \\ [16] \setminus \{4, 12\}, \emptyset).$$

Proof. The generation-initial inputs are the non-temporal projections in Claim Components 30–32. Every one of the seven transfers has at least one temporal input, so adding the source formal-evidence environment yields no time content without a temporal seed. Results 4 and 12 have empty temporal prerequisites, and their support uses claim components outside the generation-initial set. They therefore belong to the evidence-reachable, nongenerated result class. The remaining 14 results have nonempty temporal prerequisites and all become reachable when the eight primitive time roots are supplied. The set equality follows. $\square$

Proposition 8.122 (Minimal Primitive-Time-Root Support in the Aristotle Reference). For each $q \in \text{ld}\mathbf{x}_R^{\text{audit,result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108 and the minimum and maximum cardinalities of those elements are given in Table 32. The total number of family members is 28.

Table 32: Minimal Primitive-Time-Root Support in the Aristotle Reference

Result Indices	Support Statistic	Result Indices	Support Statistic
1–2, 6, 9, 13–14	(1; 1)	3, 15	(2; 1)
4, 12	(1; 0)	5, 10	(2; 1, 2)
7	(4; 1, 2)	8	(4; 2)
11	(3; 1, 2)	16	(1; 3)

Proof. Apply the update in Definition 7.46 to the seven transfer hyperedges in Definition 8.119 and the sixteen result-dependence hyperedges in Definition 8.120. Once the support antichains are invariant under further updates, Proposition 7.47 uniquely determines the minimal primitive-time-root support families of all results. $\square$

Individual Statements

Statement Objects The relations between time and change and between before and after and temporal number; the distinction between the object counted and the means of counting; judgment of elapsed time by two nows; and the relations between coming-to-be and ceasing-to-be and existence in time.

Verbatim Original The verbatim originals are from Hussey’s English translation.

a number of change in respect of the before and after

Time is that which is counted and not that by which we count.

Original-Source Locations The first quotation is from Aristotle, *Physics*, IV.11, Bekker 219b1, Hussey’s English translation, p. 44. The second quotation is from the same work, IV.11, Bekker 219b1–9, Hussey’s English translation, p. 44.

Minimal Context In the same work, IV.11, Bekker 218b21–219a9 distinguishes time from change itself without separating time from change; Bekker 219a22–219b1 speaks of time when the soul distinguishes two nows as before and after. IV.13, Bekker 222b16 states that coming-to-be and ceasing-to-be occur in time.

Attribution The statements are attributed to Aristotle, *Physics*, Book IV, Chapters 11 and 13, and the English wording of the quotations is attributed to Hussey’s translation.

Predicates The first quotation assigns to time the predicate of being a number of change in respect of before and after. The second quotation assigns to time the predicate of being the object counted and distinguishes it from the means of counting. The minimal context states the object-level distinction and inseparability of time and change, judgment of elapsed time by two nows, and the existence of coming-to-be and ceasing-to-be in time.

Explicit Domain of Application The domain of application of the first and second quotations is the relation among time, change, before and after, the now, number, and counting in *Physics* IV.11. The domain of application of the minimal context extends to judgment of elapsed time and existence in time in IV.11 and IV.13.

Individual Audit

Interpretive Possibilities The first quotation defines time as a number of change in respect of before and after. The second identifies time as what is counted, distinguishing it from the number used for counting. In the context concerning two nows, the soul speaks of elapsed time when it distinguishes the nows as before and after.

Independent Grounds In 219a22–219b9, the argument concerning elapsed time through two nows and an intermediate is followed by the distinction between number counted and number used for counting. What is counted includes both what has been counted and what is countable.

Evidence-Effect Boundary The evidential domain of the two quotations is the conceptual relation among time, change, before and after, and counting in IV.11. The condition for the existence of time when the capacity to count is restricted to soul and intellect belongs to the conditional argument in IV.14, 223a21–29.

Corresponding Formal Evidence The first quotation corresponds to time content 4 in Definition 8.117 and result 5 in Definition 8.120. The second quotation corresponds to time content 4 and result 4. The minimal context corresponds to time contents 1, 2, and 3 and results 1, 3, and 8.

Within-Reference Comparison The first quotation supplies the definition of time; the second distinguishes the role of the number used to count time from that of the time counted. The relations among before and after in two nows, countability, counting and measurement correspond to Time Contents 1–5 and their respective source locations.

8.19.2 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 2.

Source Name Primitive-Time-Root Audit of Newton 1687/1729

Bibliographic Identification Let $R = \text{Ref}_{\text{New1687P}}$ denote Isaac Newton, *The Mathematical Principles of Natural Philosophy*, translated by Motte, edited by Chittenden, 1846 edition. The audit domain comprises paragraphs 1 through 4 of the Scholium following the Definitions, the astronomical corrections, the argument concerning the order of the parts of time, the arguments concerning absolute and relative motion, the bucket experiment, the two-globe thought experiment, Definitions II and IV, and Laws I and II of the Laws of Motion.

Original-Source Location Identifier 2-L1.

Original-Source Location Paragraphs 1–4 of the Scholium following the Definitions; the astronomical corrections; the argument concerning the order of the parts of time; the arguments concerning absolute and relative motion; the bucket experiment; the two-globe thought experiment; Definitions II and IV; and Laws I and II of the Laws of Motion. The locations of the respective semantic objects and formal-evidence claim groups are given in Table 33 and Table 35.

Audit-Scope Identifier 2-SC1.

Audit Scope The audit domain stated under Bibliographic Identification.

Formal Objects

Object Identifier 2-F1.

Object Name Formal object.

Object Type Formal object.

Formal Content Definitions, arguments, and measurement relations concerning absolute and relative time, duration, uniform progress, astronomical corrections, the order of parts of time and succession, absolute and relative motion, the bucket experiment and the two-globe thought experiment, and time dependence in the laws of motion.

Original-Source Location Identifier Set 2-L1.

Object Identifier 2-F2.

Object Name Objects that use time.

Object Type Formal object.

Formal Content Distinguish 41 time contents and 17 correspondence relations in 17 location groups. The formal-claim indices in the table specify the grounds for each correspondence.

Original-Source Location Identifier Set 2-L1.

Assumptions

Assumption Identifier 2-A1.

Assumption Content The composition of the velocities of the ship, Earth, and sailor uses a common velocity scale and direction convention. The bucket experiment uses initial rest of the vessel and water, imposed rotation of the vessel, transfer of motion to the water, and eventual equality of their periods of revolution. The two-globe thought experiment assumes circular motion about a common center of gravity at a separation maintained by a cord, with equal forces applied at once to alternate faces. The first law makes impressed forces that change the state its exception; the second law includes forces applied at once and forces applied gradually and successively.

Original-Source Location Identifier Set 2-L1.

Non-Temporal Physical Inputs

Input Identifier 2-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content Of the 76 formal components, the 23 that depend only on physical bearers, spatial configurations, forces, and interactions are taken as generation-initial inputs.

Original-Source Location Identifier Set 2-L1.

Time Content

Time-Content Identifier 2-T1.

Time-Content Name Time contents in the original source.

Time-Content Type Time content.

Time-Content Specification The individual time contents are given in Definitions 8.123 and 8.124.

Original-Source Location Identifier Set 2-L1.

Formal Input Elements

Formal-Component Identifier 2-C1.

Formal-Component Name Composition of the reference's formal-evidence environment.

Formal-Component Type Formal component.

Formal-Component Text All 76 formal components are included in the reference's formal-evidence environment.

Original-Source Support Identifier Set 2-L1.

Definition 8.123 (Time Contents and Correspondence Relations in the Newton 1687/1729 Reference). For reference $R = \text{Ref}_{\text{New1687P}}$, distinguish 41 time contents according to 17 source-location groups. The formal claims listed in the same table give each group's correspondence relation. The time contents serve as time bearers, time-containing physical quantities, or inputs and outputs of operations using time. Supporting claims for transfers and results specify the source definitions, conditions, and arguments, including these correspondences; separate temporal-index columns specify the time-content inputs and outputs.

Table 33: Time Contents and Correspondence Relations and Their Original-Source Locations in the Newton 1687/1729 Reference

Loca- tion Group	Time Contents (Index: Name)	Correspondence (Supporting Claims)	Original- Source Location
1	1: absolute, true, and mathematical time; 2: physical use of absolute time	Intrinsic nature and external independence of absolute time {4, 5}	Scholium following the Definitions, paragraph 1

Location Group	Time Contents (Index: Name)	Correspondence (Supporting Claims)	Original-Source Location
2	3: uniform flow and progress; 4: physical use of uniform progress	Uniform progress in relation to changes in measuring motions {4, 22}	Scholium I and astronomical corrections
3	5: duration; 6: physical use of duration	Nominal correspondence between absolute time and duration {6}	Scholium following the Definitions, paragraph 1
4	7: relative, apparent, and common time; 8: relative-time input using duration, motion, and an external measure; 9: output as a relative time measure	Sensible measure of duration by motion {7, 8}	Scholium following the Definitions, paragraph 1
5	10: hours, days, months, and years; 11: input relating a relative time measure to its unit examples; 12: time measure expressed in hours, days, months, and years	Unit examples of relative time measures {9}	Scholium following the Definitions, paragraph 1
6	13: velocity of motion and composition of true and relative motion; 14: physical use of composition of motions	Distinction and composition of absolute and relative motions {14, 16, 17}	Scholium following the Definitions, paragraph 4
7	15: common time and natural days; 16: physical use of natural days	Inequality and use of natural days as a measure {19}	Astronomical corrections
8	17: time of corrected celestial motions; 18: input using natural days, celestial observations, clocks or eclipse observations, and correction; 19: output as corrected astronomical time	Astronomical corrections and their observational grounds {18, 20, 26}	Astronomical corrections
9	20: times of phenomena; 21: input using natural or corrected time, observation of phenomena, and clock comparison; 22: output as phenomenon time	Determination of phenomenon times using correction {26}	Astronomical corrections
10	23: immutable order of the parts of time; 24: physical use of the order of parts of time	Immutability of the order of parts of time {27}	Argument concerning the order of the parts of time

Location Group	Time Contents (Index: Name)	Correspondence (Supporting Claims)	Original-Source Location
11	25: order of succession; 26: physical use of the order of succession	Placement in time by order of succession {28}	Argument concerning the order of the parts of time
12	27: duration and change of motion; 28: motion-change input using duration, velocity, order, laws, and force; 29: output as change of motion	Force-induced changes and the distinction between absolute and relative motions {30, 34, 37}	Argument concerning absolute and relative motion
13	30: comparison of the periods of revolution and relative motion of vessel and water; 31: rotational-comparison input using change of motion, vessel and water, axis of rotation, and water-surface effect; 32: output as rotational comparison	Relation between the water-surface effect, periods of revolution, and relative rest {43, 45, 46}	Bucket experiment
14	33: increase and decrease of circular motion under the action of force; 34: motion-change input using the circular motion of two globes, cord tension, and equal forces applied at once; 35: output as force-dependent rotational change	Relation of tension to the quantity and direction of circular motion {56, 57} $\cup$ [59, 61]	Two-globe thought experiment
15	36: continuance of state; 37: physical use of continuance of state	Continuance of state in the first law {71}	Law of Motion I
16	38: force applied at once or successively and change of motion; 39: physical use of change of motion under impressed force	Correspondence between force and change of motion in the second law {73, 74}	Law of Motion II
17	40: velocity and the laws of motion; 41: physical use of velocity laws	Relations among velocity, quantity of motion, force, and the laws of motion {65, 66, 68, 69, 71, 73, 74}	Definitions II and IV and the Laws of Motion

Definition 8.124 (Formal-Evidence Claim Groups in the Newton 1687/1729 Reference).

$$\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(76), \quad \text{Idx}_R^{\text{audit,phys}} = [76].$$

Set the claim-audit boundary to

$$\text{ClaimBoundaryIdx}_R = (\{11, 13, 15, 24, 25, 31, 33, 35, 36, 38, 41, 42, 44, 49, 52, 54, 55, 58, 62, 64, 67, 70, 72\}, [76]).$$

By Definition 7.98, the 23 indices in the first component specify the generation-initial inputs, and all 76 indices specify the source formal-evidence environment. Each component

is retained as a type-homogeneous component whose truth value and dependence inputs can be judged independently. Each initial component concerns physical bearers, spatial configurations, or interactions and has no temporal prerequisite. Table 35 gives the source definitions, conditions, and arguments together with the temporal prerequisites of their use.

The dependency key d denotes the following temporal-index set.

d	Temporal Prerequisites	d	Temporal Prerequisites
1	$\{1, 2\} \cup \llbracket 7, 9 \rrbracket$	19	$\{1, 2, 15, 16\} \cup \llbracket 7, 9 \rrbracket$
2	$\{1, 2, 13, 14\} \cup \llbracket 7, 9 \rrbracket$	20	$\{38, 39\} \cup \llbracket 27, 29 \rrbracket$
3	$[4]$	21	$\llbracket 27, 35 \rrbracket$
4	$\{1, 2\}$	22	$\llbracket 27, 32 \rrbracket$
5	$\{1, 2, 5, 6\}$	23	$\{38, 39\} \cup \llbracket 27, 29 \rrbracket \cup \llbracket 33, 35 \rrbracket$
6	$\llbracket 5, 9 \rrbracket$	24	$\{13, 14, 40, 41\}$
7	$\{13, 14\} \cup \llbracket 5, 9 \rrbracket$	25	$\llbracket 27, 29 \rrbracket \cup \llbracket 36, 39 \rrbracket$
8	$\llbracket 7, 12 \rrbracket$	26	$\{36, 37\}$
9	$\{13, 14\}$	27	$\llbracket 38, 41 \rrbracket$
10	$\emptyset$	28	$[41]$
11	$\{13, 14\} \cup \llbracket 27, 29 \rrbracket$	29	$\{1, 2, 5, 6\} \cup \llbracket 23, 26 \rrbracket$
12	$\llbracket 7, 9 \rrbracket \cup \llbracket 15, 19 \rrbracket$	30	$\{15, 16\}$
13	$\llbracket 15, 19 \rrbracket$	31	$\llbracket 3, 9 \rrbracket$
14	$\{13, 14\} \cup \llbracket 1, 4 \rrbracket$	32	$\llbracket 1, 4 \rrbracket \cup \llbracket 27, 29 \rrbracket$
15	$\{5, 6\} \cup \llbracket 27, 29 \rrbracket$	33	$\llbracket 27, 29 \rrbracket$
16	$\llbracket 15, 22 \rrbracket$	34	$\llbracket 27, 29 \rrbracket \cup \llbracket 33, 35 \rrbracket$
17	$\{23, 24\}$	35	$\{40, 41\}$
18	$\llbracket 23, 26 \rrbracket$		

The original-source location keys in Table 35 correspond to original-source locations as follows.

- 1 Opening of the Scholium following the Definitions
- 2 Scholium following the Definitions, paragraph 1
- 3 Scholium following the Definitions, paragraph 2
- 4 Scholium following the Definitions, paragraph 3
- 5 Scholium following the Definitions, paragraph 4
- 6 Astronomical corrections
- 7 Argument concerning the order of the parts of time
- 8 Argument concerning measures
- 9 Argument concerning rest and motion
- 10 Argument concerning properties
- 11 Argument concerning causes

- 12 Argument concerning effects
- 13 Bucket experiment
- 14 Argument concerning quantity and measure
- 15 Argument concerning recognition
- 16 Two-globe thought experiment
- 17 Domain of application
- 18 Definition II
- 19 Definition IV
- 20 Law of Motion I
- 21 Law of Motion II
- 22 Selected scope

Table 35: Physical and Formal Input-Claim Components
in the Newton 1687/1729 Reference

p	Original Group	Type-Homogeneous Component	Dependence-Object Number	Original-Source Location Key
1	1	Treats time, space, place, and motion as quantities known to all and omits their definitions.	10	1
2	2	Traces certain prejudices to conceiving the four quantities only through their relations to sensible objects.	10	1
3	3	Introduces the distinctions between absolute and relative, true and apparent, and mathematical and common.	2	1
4	4	Absolute, true, and mathematical time flows uniformly in itself and from its own nature.	3	2

p	Original Group	Type-Homogeneous Component	Dependence-Object Number	Original-Source Location Key
5	5	Gives absolute time as a quantity independent of relations to external objects.	4	2
6	6	Gives duration as another name for absolute time.	5	2
7	7	Gives relative, apparent, and common time as a sensible external measure of duration.	6	2
8	8	Gives the measure of relative time by motion.	6	2
9	9	Gives hours, days, months, and years as examples of relative time.	8	2
10	10	Distinguishes absolute space from relative space.	10	3
11	11	A body and the spatial region occupied by that body	10	4
12	11	The distinction between absolute and relative place	10	4
13	12	Correspondence between a body and distinct spatial positions	10	5
14	12	The distinction between absolute and relative motion	11	5
15	13	The physical-bearer configuration of a ship, the Earth, and a sailor	10	5
16	13	Composition of the relative motions of the three bodies	9	5
17	13	The difference between the composed relative motion and true motion	9	5
18	14	Distinguishes absolute from relative time by equation-of-time corrections to common time.	12	6

p	Original Group	Type-Homogeneous Component	Dependence-Object Number	Original-Source Location Key
19	15	Uses the natural day as a measure and treats its inequality as an object of correction.	30	6
20	16	Uses corrections for inequality in the length of the day to refine celestial motions.	13	6
21	17	States that an equable motion by which time can be accurately measured may not exist.	31	6
22	18	Retains the uniform progress of absolute time through acceleration or retardation of motions.	32	6
23	19	Gives the durations of objects as quantities independent of the swiftness or cessation of motion.	15	6
24	20	The physical bearer of a pendulum clock	10	6
25	20	The physical-bearer configuration of Jupiter and its satellites	10	6
26	20	Cites pendulum-clock experiments and eclipses of Jupiter's satellites as evidence that astronomical correction is necessary for determining the times of phenomena.	16	6
27	21	Takes the order of the parts of time to be immutable.	17	7
28	22	Places objects in time according to the order of succession.	18	7
29	23	Distinguishes a sensible measure from the quantity measured in natural philosophy.	10	8

p	Original Group	Type-Homogeneous Component	Dependence-Object Number	Original-Source Location Key
30	24	Distinguishes absolute and relative rest and motion by properties, causes, and effects.	11	9
31	25	Relative positions of surrounding bodies and the target body	10	10
32	25	Absolute rest cannot be determined solely from the relative positions of bodies in our region.	11	10
33	26	The physical interaction of a force impressed upon a body	10	11
34	26	Generation or alteration of true motion by force	20	11
35	27	Multiple bodies serving as objects of comparison	10	11
36	27	The physical interaction of forces impressed upon the objects of comparison	10	11
37	27	Alteration of relative motion by force	20	11
38	28	The spatial configuration of a body, an axis of rotation, and the radial direction	10	12
39	28	Endeavor to recede from the axis of rotation	33	12
40	28	Distinction between absolute and relative motion by the endeavor to recede from the axis	33	12
41	29	The physical-bearer configuration of a bucket, vessel, and water	10	13
42	29	The concave form and elevated state of the water surface	10	13
43	29	Attribution of the water-surface state as an effect of true circular motion	22	13

p	Original Group	Type-Homogeneous Component	Dependence-Object Number	Original-Source Location Key
44	30	The physical-bearer configuration of the vessel and water	10	13
45	30	Relative rest of the water with respect to the vessel	22	13
46	30	The true circular motion of the water	22	13
47	31	Treats a relative quantity as a sensible measure and distinguishes it from the quantity measured itself.	10	14
48	32	Apparent motion	11	15
49	32	The physical interaction of a force acting on a body	10	15
50	32	Inference that determines true motion from apparent motion and force	11	15
51	33	Apparent motion	11	15
52	33	The physical interaction of a force acting on a body	10	15
53	33	Argument from apparent motion and force to true motion	11	15
54	34	The physical bearers and spatial configuration of two globes, a cord, a given separation, and an axis through their common center of gravity	10	16
55	34	The state of tension in the cord	10	16
56	34	Inference from tension to the globes' endeavor to recede from the axis	34	16
57	34	Inference from tension to the quantity of circular motion	34	16
58	35	The physical interaction of forces impressed upon the two globes	10	16
59	35	Change in tension when equal forces are applied at once to alternate faces of the two globes	23	16

p	Original Group	Type-Homogeneous Component	Dependence-Object Number	Original-Source Location Key
60	35	Inference from change in tension to increase or decrease of motion	23	16
61	35	Inference from change in tension to the direction of motion	23	16
62	36	The physical configurations of the bucket experiment and the two-globe thought experiment	10	17
63	36	Domain of application that takes absolute rotation as the direct object of both experiments	21	17
64	37	Quantity of matter	10	18
65	37	Velocity	35	18
66	37	Definition of quantity of motion by velocity and quantity of matter	35	18
67	38	The physical interaction of a force impressed upon a body	10	19
68	38	A state of rest or uniform rectilinear motion	25	19
69	38	Alteration of the state of motion by force	25	19
70	39	The physical condition in which no force acts on a body	10	20
71	39	Maintenance of a state of rest or uniform rectilinear motion	26	20
72	40	Motive force acting upon a body	10	21
73	40	Change of motion	27	21
74	40	Proportional correspondence between motive force applied at once or gradually and successively and change of motion in the direction of that force	27	21

p	Original Group	Type-Homogeneous Component	Dependence-Object Number	Original-Source Location Key
75	41	Uses time, duration, succession, velocity, continuance of state, and change of motion as descriptive inputs.	28	22
76	42	Uses time, duration, and the order of succession as prior inputs.	29	22

Dependency Inputs

Dependency Inputs The index sets in the table specify the supporting claims and temporal inputs of the seven transfers.

Dependency and Transfer Elements

Definition 8.125 (Time-Content Transfers in the Newton 1687/1729 Reference). Let $\text{ld}\mathbf{x}_R^{\text{audit,transfer}} = [7]$. Realize the supporting claims, temporal inputs, and temporal outputs in each row by Definition 7.105.

Table 36: Time-Content Transfers in the Newton 1687/1729 Reference

j	Supporting Claim Indices	Time-Input Indices	Time-Output Indices	Operation
1	$\llbracket 6, 8 \rrbracket$	$\{5, 6\}$	$\llbracket 7, 9 \rrbracket$	Relate duration to a sensible external measure by motion.
2	$\llbracket 7, 9 \rrbracket$	$\llbracket 7, 9 \rrbracket$	$\llbracket 10, 12 \rrbracket$	Relate a relative time measure to the unit examples of hours, days, months, and years.
3	$\llbracket 18, 20 \rrbracket \cup \llbracket 24, 26 \rrbracket$	$\{15, 16\}$	$\llbracket 17, 19 \rrbracket$	Correct the inequality of natural days to obtain the time measure used for celestial motions.
4	$\llbracket 18, 20 \rrbracket \cup \llbracket 24, 26 \rrbracket$	$\llbracket 15, 19 \rrbracket$	$\llbracket 20, 22 \rrbracket$	Determine the times of phenomena using observation and astronomical correction.

j	Supporting Claim Indices	Time-Input Indices	Time-Output Indices	Operation
5	$\{6, 14, 16, 17, 27, 28, 30, 33, 34, 37\} \cup \llbracket 64, 74 \rrbracket$	$\{5, 6, 13, 14\} \cup \llbracket 23, 26 \rrbracket \cup \llbracket 36, 41 \rrbracket$	$\llbracket 27, 29 \rrbracket$	Describe continuance and change of motion using duration, velocity, succession, state, force, and the laws of motion.
6	$\{30, 34\} \cup \llbracket 37, 46 \rrbracket$	$\llbracket 27, 29 \rrbracket$	$\llbracket 30, 32 \rrbracket$	Compare the motions, periods of revolution, and water-surface shape of vessel and water to distinguish relative rest from true circular motion.
7	$\{30, 34, 37\} \cup \llbracket 54, 61 \rrbracket$	$\llbracket 27, 29 \rrbracket$	$\llbracket 33, 35 \rrbracket$	For circular motion of two globes about their common center of gravity, infer the quantity of motion from cord tension and its increase or decrease and direction from tension changes under equal forces applied at once.

Transfer Events and Outputs The seven transfers and temporal outputs in Table 36.

Applicability Conditions

Conditions under Which a Dependency or Transfer Applies Transfers 1 and 2 use the measure relation and unit examples of relative time; Transfers 3 and 4 use natural-day inequality and astronomical correction. Transfer 5 uses duration, succession, and the source laws of motion. Transfer 6 requires transfer of motion from vessel to water and comparison of revolution periods and water-surface shape; Transfer 7 requires a fixed inter-globe separation, a common center of gravity, equal forces applied at once, and comparison of tension.

Original-Source Support

Original-Source Support for Each Dependency or Transfer Table 35 gives the content and source locations of supporting claims; Table 33 locates the time contents and correspondence relations.

Results

Definition 8.126 (Results in the Newton 1687/1729 Reference). Define the result index set by $\text{ldx}_R^{\text{audit,result}} = [18]$. The correspondence between results and original-source locations is given in Table 37.

The original-source location keys in Table 37 correspond to original-source locations as follows.

- 1 Opening of the Scholium following the Definitions
- 2 Scholium following the Definitions, paragraph 1
- 3 Scholium following the Definitions, paragraphs 2 through 4
- 4 Scholium following the Definitions, paragraph 4
- 5 Astronomical corrections
- 6 Argument concerning the order of the parts of time
- 7 Argument concerning rest and motion
- 8 Bucket experiment
- 9 Two-globe thought experiment
- 10 Laws of Motion I and II
- 11 Definitions II and IV and the Laws of Motion
- 12 Entire selected scope

Table 37: Results and Original-Source Locations in the
Newton 1687/1729 Reference

j	Result	Original-Source Location Key
1	Declaration that time, space, place, and motion are left undefined	1
2	Distinctions between absolute and relative, true and apparent, and mathematical and common	1
3	The intrinsic nature and uniform progress of absolute time	2
4	Nominal correspondence between absolute time and duration	2
5	Definition of relative time as an external measure of duration by motion	2
6	Hours, days, months, and years as examples of relative time measures	2
7	Distinctions between absolute and relative place and motion	3
8	Composition of relative motions by means of a ship, the Earth, and a sailor	4
9	Inequality of natural days	5
10	Correction of common time by the astronomical equation of time	5

j	Result	Original-Source Location Key
11	Corrective evidence from pendulum clocks and eclipses of Jupiter's satellites	5
12	Possible nonexistence of equable motion that accurately measures time	5
13	Order of the parts of time and arrangement by succession	6
14	Distinction of motions by properties, causes, and effects	7
15	Inference to absolute rotation from the water-surface effect	8
16	Inference to the quantity and direction of circular motion from the cord tension of two globes	9
17	Continuance of state in the first law and change of motion in the second law	10
18	Connections among time, duration, succession, quantity of motion, velocity, quantity of matter, force, and mechanical description in the selected scope	12

Source-Specific Results The 18 results in Definition 8.126. The original-source location of each result is given by Table 37, and its temporal prerequisites and physical and formal input-claim support are given by Table 38.

Result Dependencies

Definition 8.127 (Result-Dependence Hyperedges in the Newton 1687/1729 Reference). For each $j \in [18]$, realize the result-dependence hyperedge from the temporal prerequisites, supporting claims, and result output in the table by Definition 7.105. Result 18 includes Claims 75 and 76 concerning time use throughout the selected scope and jointly uses Time Contents 1–41.

Table 38: Result-Dependence Hyperedges in the Newton 1687/1729 Reference

j	Temporal Prerequisites	Supporting Claim Indices	Result-Output Indices
1	$\emptyset$	$\{1, 2\}$	$\{1\}$
2	$\{1, 2, 13, 14\} \cup \llbracket 7, 9 \rrbracket$	$\{7, 8, 14, 16, 17\} \cup \llbracket 3, 5 \rrbracket$	$\{2\}$
3	$[4]$	$\{4, 5, 22\}$	$\{3\}$
4	$\{1, 2, 5, 6\}$	$\llbracket 4, 6 \rrbracket$	$\{4\}$
5	$\llbracket 5, 9 \rrbracket$	$\llbracket 6, 8 \rrbracket$	$\{5\}$
6	$\llbracket 10, 12 \rrbracket$	$\{9\}$	$\{6\}$
7	$\{13, 14\}$	$\{16, 17\} \cup \llbracket 10, 14 \rrbracket$	$\{7\}$
8	$\{13, 14\}$	$\llbracket 14, 17 \rrbracket$	$\{8\}$
9	$\{15, 16\}$	$\{19\}$	$\{9\}$

j	Temporal Prerequisites	Supporting Claim Indices	Result-Output Indices
10	$\llbracket 15, 19 \rrbracket$	$\{26\} \cup \llbracket 18, 20 \rrbracket$	$\{10\}$
11	$\llbracket 17, 19 \rrbracket$	$\{18, 20\} \cup \llbracket 24, 26 \rrbracket$	$\{11\}$
12	$\llbracket 3, 9 \rrbracket$	$\{4, 21, 22\} \cup \llbracket 6, 8 \rrbracket$	$\{12\}$
13	$\llbracket 23, 26 \rrbracket$	$\{27, 28\}$	$\{13\}$
14	$\llbracket 27, 29 \rrbracket$	$\llbracket 30, 40 \rrbracket$	$\{14\}$
15	$\llbracket 27, 32 \rrbracket$	$\{30, 34, 37\} \cup \llbracket 41, 46 \rrbracket$	$\{15\}$
16	$\llbracket 27, 29 \rrbracket \cup \llbracket 33, 35 \rrbracket$	$\{30, 34, 37\} \cup \llbracket 54, 61 \rrbracket$	$\{16\}$
17	$\llbracket 36, 39 \rrbracket$	$\llbracket 67, 74 \rrbracket$	$\{17\}$
18	$[41]$	$[76]$	$\{18\}$

Grounds of Establishment

Grounds for Establishing the Results Results 1–8, 12–14, 17, and 18 rest on the source definitions, distinctions, statement of possibility, and mechanical arguments. For Results 9–11, the source invokes natural-day inequality, astronomical correction, pendulum-clock experiments, and eclipses of Jupiter’s satellites. Result 15 is an experimental argument concerning the water-surface shape in a bucket; Result 16 is a thought-experimental argument using two globes at a fixed separation and cord tension.

Evidence Hierarchy

Directly Measured Objects The astronomical-correction paragraph invokes natural-day inequality, pendulum-clock experiments, and eclipses of Jupiter’s satellites. The bucket experiment observes water-surface shape in relation to the motions of vessel and water. Cord tension in the two-globe argument is a comparison quantity posited for two globes at a fixed separation.

Directly Measured Predicates The source states the inequality of natural-day lengths and cites clock experiments and eclipses as evidence that correction is necessary to determine phenomenon times. For the bucket, it describes the rise and concavity of the water surface as rotation is transferred to the water, together with the eventual equality of revolution periods at relative rest with respect to the vessel.

Conditions and Domain of Application of Direct Measurement Evidence for astronomical correction concerns use of natural days as a measure, clock experiments, and comparison with satellite eclipses. Bucket observations concern a procedure starting with vessel and water at rest, rotating the vessel, and transferring motion to the water. The inter-globe separation, circular motion about a common center of gravity, equal forces applied at once, and changes in tension are conditions of the two-globe thought experiment.

Objects of Formal Generalization Results concerning absolute and relative time, absolute and relative motion, the order of the parts of time, quantity of motion, force, and the laws of motion apply to the bodies, motions, and mechanical systems specified by the definitions and laws in the original source.

Conditions and Domain of Application of Formal Generalization Generalization from the astronomical corrections, bucket experiment, and two-globe argument takes as its establishment conditions the temporal inputs and supporting claims in Table 38.

Generation Judgment Judgment values: LCTR complete-generation judgment: 0 cases / within-reference-evidence-conditional local-reach judgment: 0 cases / remaining primitive time roots: 41 cases.

Generation Reach and Primitive Time Roots Both the time contents generated under the complete judgment and the time contents reached under the local-reach judgment number zero. All 41 time contents in Definition 8.123 remain as primitive time roots.

Within-Reference-Evidence-Conditional Local Reach The within-reference-evidence-conditional local-reach judgment has zero cases, and 41 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ in Definition 7.108 and apply Definition 7.50. Summation over all results determines the primitive-time-root–result dependence relations and necessary primitive-time-root–result dependence relations.

Result Dependence Result 1 has no time-content input, but its claim support belongs only to the evidence environment; it is therefore a result with evidence-only reach. Results 2–18 depend on primitive time roots. The minimal primitive-time-root support of each result is given by Proposition 8.129.

Evidence Effect

Evidence-Effect Domain The formal evidence in this reference directly supports the 18 results in the adopted audit domain, astronomical observations and corrections, clock comparisons, the shape of the water surface in the bucket, the argument using two globes and cord tension, and dependence on the laws of motion.

Evidence-Effect Boundary The evidence effect of the astronomical observations, clock comparisons, water-surface shape, and tension extends to the respective physical bearers, observational relations, comparison differences, and mechanical derivations. The transition from those items to the intrinsic nature and uniform progress of absolute time and to physical-time qualification independent of external objects lies beyond the evidence-effect boundary under Definition 7.115.

Result-Preservation Domain

Result-Preservation Domain The mathematical, observational, and mechanical content of results 6, 8–13, 17, and 18 is preserved when the time presuppositions used there are supplied by the corresponding generated time contents in the LCTR generation sequence and the original source’s physical inputs, observations, correction rules, and directions of dependence are retained. The content concerning measure relations,

relative-motion relations, water-surface shape, and tension in results 5, 7, and 14–16 is preserved under the same conditions, but attribution to primitive absolute time or absolute motion falls outside the preservation domain. The declaration that time is undefined, the intrinsic nature and uniform progress of absolute time, and its independence from external objects in results 1–4 are not preserved as results satisfying LCTR physical-time qualification.

Placement of Grounding Obligations

Placement of Grounding Obligations The 41 time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as already established time inputs. The judgment on the placement of the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment is grounded in a comparison between the use of each time content in a law operation or result derivation and the absence, before that use, of the corresponding generation or establishment condition as an input, formation condition, evaluation condition, or invalidation condition.

Evidence of Actual Invalidation

Evidence of Actual Invalidation The complete-generation judgment and remaining-root judgment concern physical-time qualification and are distinct from the actual invalidation of a specific condition that determines the value or domain of application of a result. Neither the original-source formal evidence nor the LCTR audit evidence for this reference establishes the actual invalidation of such a target condition. The actual-invalidation judgment is “indeterminate from the original-source evidence”; actual invalidation is not inferred solely from the backgrounding of a grounding obligation.

Downstream Reach

Downstream Reach The reach to results that depend on primitive time roots is determined for this reference by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because the actual invalidation of a specific condition has not been established, the judgment on actual reach from that invalidation position to results is “indeterminate from the original-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the targets of repair are the corresponding time content and the generation or establishment conditions required before its use. The scope of recalculation is limited to results for which actual invalidation has been independently established and that are reached by dependence from the invalidation position. Because actual invalidation is indeterminate in this reference, this repair-and-recalculation specification is conditional. If the condition holds, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted.

Audit Conclusions

Proposition 8.128 (Primitive Time Roots and Result Dependence in the Newton 1687/1729 Reference). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \left(\emptyset, [41]; \right. \\ \left. \emptyset, \{1\}, \{2, \dots, 18\}, \emptyset \right)$$

The four types in Definition 7.36 contain 76 boundary claims, 41 time contents, seven transfer events, and 18 results, totaling 142 atoms. Moreover, $R \in \text{Ref}^{\text{audit,rootdep}}$ holds.

Proof. The 23 claims in the first component of the claim-audit boundary form the generation-initial set. Every transfer has a nonempty temporal input, so this initial set yields no time content. Adding all 76 formal claims as the evidence environment still leaves the temporal prerequisites unsatisfied unless time contents are supplied as seeds. Complete generation and within-reference-evidence-conditional local reach therefore both yield zero time contents, and all 41 time contents are primitive time roots. Result 1 has no temporal prerequisite and is supported by Claims 1 and 2 outside the initial set, so it is reachable from the evidence environment alone. Results 2–18 have nonempty temporal prerequisites and become reachable when all 41 roots are added to the evidence environment. Definition 8.115 gives the displayed fourfold partition. Atoms of the four types correspond one-to-one to the elements of their respective finite index domains; the total over distinct types is $76 + 41 + 7 + 18 = 142$. $\square$

Proposition 8.129 (Minimal Primitive-Time-Root Support in the Newton 1687/1729 Reference). For each $q \in \text{Idx}_R^{\text{audit,result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108 and the minimum and maximum cardinalities of those elements are given in Table 39.

Table 39: Minimal Primitive-Time-Root Support in the Newton 1687/1729 Reference

Result Indices	Support Statistic	Result Indices	Support Statistic
1	(1; 0)	2	(2; 6, 7)
3, 4, 12, 13, 17	(1; 4)	5, 7, 8, 9, 10	(1; 2)
6	(3; 2, 3)	11	(2; 2, 3)
14, 15, 16	(2; 3, 14)	18	(1; 20)

Proof. Apply the update in Definition 7.46 one edge at a time in the order of the seven transfer rows. The time-content support antichains obtained after the first full pass remain unchanged on the next pass. Taking the family product over the temporal prerequisites of each result edge and retaining inclusion-minimal members gives the displayed support families. Each member reaches the corresponding result when supplied as a seed, whereas removing any one root from that member prevents reachability. There are 25 support-family members, 123 primitive-time-root–result dependence pairs, and 54 necessary primitive-time-root–result dependence pairs. $\square$

Individual Statements Statement 1: Absolute, True, and Mathematical Time and Duration

Verbatim Original Absolute, true, and mathematical time, of itself, and from its own nature flows equably without regard to anything external, and by another name is called duration :

Original-Source Location Newton, *The Mathematical Principles of Natural Philosophy*, Motte translation, Definitions, Scholium I, source p. 77.

Minimal Context The same Scholium distinguishes time, space, place, and motion as absolute and relative, true and apparent, and mathematical and common. The quoted passage is followed immediately by the description of relative, apparent, and common time.

Statement 2: Relative, Apparent, and Common Time

Verbatim Original relative, apparent, and common time, is some sensible and external (whether accurate or unequable) measure of duration by the means of motion, which is commonly used instead of true time; such as an hour, a day, a month, a year.

Original-Source Location Newton, *The Mathematical Principles of Natural Philosophy*, Motte translation, Definitions, Scholium I, source p. 77.

Minimal Context The passage follows the description that also calls absolute, true, and mathematical time duration, and describes relative time as an external measure by motion. The subsequent passage distinguishes absolute from relative space.

Statement 3: Possible Nonexistence of Uniform Motion as a Measure of Time

Verbatim Original It may be, that there is no such thing as an equable motion, whereby time may be accurately measured. All motions may be accelerated and retarded, but the true, or equable, progress of absolute time is liable to no change.

Original-Source Location Newton, *The Mathematical Principles of Natural Philosophy*, Motte translation, the Scholium following the Definitions, the astronomical-correction paragraph after item IV, source pp. 78–79.

Minimal Context The passage follows the description of the inequality of natural days and their correction by astronomers. The same paragraph proceeds to distinguish the duration of the existence of things from motion.

Statement 4: Duration Independent of Motion

Verbatim Original The duration or perseverance of the existence of things remains the same, whether the motions are swift or slow, or none at all: and therefore it ought to be distinguished from what are only sensible measures thereof; [...]

Original-Source Location Newton, *The Mathematical Principles of Natural Philosophy*, Motte translation, the Scholium following the Definitions, the astronomical-correction paragraph after item IV, source p. 79.

Minimal Context The passage is placed immediately after the possible nonexistence of a uniform measuring motion and the uniform progress of absolute time. The subsequent passage refers to the astronomical equation of time, pendulum clocks, and eclipses of Jupiter's satellites.

Common Evidence with Domain of Application The following four statements are attributed to Newton's *Scholium*, and their quoted English wording is attributed to Motte's translation.

Target State-ment	Statement Object	Predicate	Explicit Domain of Application
S02-Q001	Absolute, true, and mathematical time	Flows uniformly in itself and from its own nature without regard to anything external, and is also called duration.	The descriptive domain in Definitions, Scholium I, that distinguishes time as absolute/relative, true/apparent, and mathematical/common.
S02-Q002	Relative, apparent, and common time	Is a sensible and external measure of duration by motion and is commonly used instead of true time.	The descriptive domain in Definitions, Scholium I, that states the relation between relative time and a measure by motion.
S02-Q003	Uniform motion used to measure time and the uniform progress of absolute time	A uniform motion by which time can be accurately measured may not exist, while the true progress of absolute time is unaffected when motions are accelerated or retarded.	The descriptive domain in the Scholium following the Definitions, the astronomical-correction paragraph after item IV, that compares the inequality of natural days, measures of motion, and the progress of absolute time.
S02-Q004	The duration of the existence of things	Remains the same whether motions are swift, slow, or nonexistent and is distinguished from sensible measures.	The descriptive domain in the Scholium following the Definitions, the astronomical-correction paragraph after item IV, that states the relation between the duration of things and sensible measures by motion.

Individual Audit

Interpretive Possibilities The first statement describes time as flowing uniformly by its own nature independently of external objects. It admits a reading as a property

of physical reality and a reading as a mathematical standard distinguished from empirical measures. In the second statement, being a sensible external measure of duration by motion and being commonly used in place of true time are both predicated of the same relative time. The third contrasts the possible nonexistence of equable motion permitting accurate measurement with the equable progress of absolute time, which remains unaffected by changes in motion. The fourth states both the independence of duration from motion and the distinction between duration itself and its sensible measures.

Independent Grounds The four statements belong to the treatment of time, duration, and measures in the Scholium following the Definitions. The astronomical-correction paragraph states the inequality of natural days and invokes pendulum-clock experiments and eclipses of Jupiter’s satellites to establish the need for correction. These observational grounds concern relative time measures and the determination of phenomenon times.

Evidence-Effect Boundary The first statement directly supports the attribution to absolute time of an intrinsic nature, uniform flow, external independence, and the name duration. The second statement directly supports the statement that relative time is an external measure of duration by motion and that hours, days, months, and years are examples of that measure. The third statement directly supports the possible nonexistence of a perfect measure of time by uniform motion and the progress of absolute time said to remain invariant under acceleration and retardation of motions. The fourth statement directly supports the statement that the duration of things is independent of their state of motion and is distinguished from sensible measures. None of the four statements provides a path that generates the time content used in descriptions of absolute time, duration, relative-time measures, or measuring motions solely from non-temporal physical inputs. Nor do the definitions, distinctions, and relations given by the four statements alone establish the physical-time qualification of that time content.

Corresponding Formal Evidence The first statement corresponds to time contents 1–6 in Definition 8.123, input claims 4–6 in Definition 8.124, and results 3–4 in Definition 8.126. The second statement corresponds to time contents 7–12, input claims 7–9, and results 5–6. The third statement corresponds to Time Contents 1–9 and 27–29, Claims 21–22, and result 12. The fourth statement corresponds to time contents 5–6, input claim 23, and the distinction between duration and sensible measures in the same reference.

Within-Reference Comparison The first and fourth statements describe time or duration as independent of external objects or states of motion; the second statement takes sensible measures by motion as proxy indications of it; and the third statement states the possible nonexistence of a perfectly uniform measuring motion. Within the same reference, the ontological positing of the quantity measured is distinguished from the empirical construction of the measuring scale, while no physical generation path connecting the two is given.

8.19.3 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 3.

Source Name Primitive-Time-Root Audit of Einstein 1905

Bibliographic Identification Let $R = \text{Ref}_{\text{Ein05ED}}$ denote A. Einstein, *Zur Elektrodynamik bewegter Körper*, *Annalen der Physik*, Vierte Folge, Band 17, 891–921 (1905), doi:10.1002/andp.19053221004. The audit uses the original German edition published in 1905, and takes as its audit domain the original Introduction, original kinematical §§1–5, original electrodynamical §§6–10, the original note, acknowledgments, end of the main text, and date of receipt.

Original-Source Location Identifier 3-L1.

Original-Source Location The original Introduction; kinematical §§1–5; electrodynamical §§6–10; the original note; acknowledgments; end of the main text; and date of receipt. The locations of each time content and each input claim are given by Table 41 and Table 44.

The original-source location keys in Table 44 correspond to original-source locations as follows.

1 Introduction

2 Introduction; §2

3 §§1–3

4 §1

5 §§1–2

6 §3

7 §2

8 §4

9 §5

10 §6

11 §7

12 §8

13 §9

14 §10

Audit-Scope Identifier 3-SC1.

Audit Scope The audit domain stated under Bibliographic Identification.

Formal Objects

Object Identifier 3-F1.

Object Name Formal object.

Object Type Formal object.

Formal Content Definitions, assumptions, derivations, and experimental connections of time dependence in synchronization of local and distant clocks, simultaneity, coordinate time, light propagation, Lorentz coordinate transformations, rate relations of moving clocks, composition of velocities, electromagnetic-field transformations, Doppler relations, radiation, charge, current, and electron motion.

Original-Source Location Identifier Set 3-L1.

The correspondence indices in Table 41 refer to the 18 correspondence relations in Table 42. That table preserves each relation's content and source location and specifies its supporting claims. The dependency keys in Table 43 specify the temporal dependencies of input claims.

Assumptions

Assumption Identifier 3-A1.

Assumption Content The individual reference assumptions are given in Definition 8.131.

Original-Source Location Identifier Set 3-L1.

Non-Temporal Physical Inputs

Input Identifier 3-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content The 15 components belonging to magnets, conductors, rigid rods and spheres, clocks, light signals, electromagnetic fields, charges, electrons, observational apparatus, and their spatial configurations are taken as generation-initial inputs.

Original-Source Location Identifier Set 3-L1.

Time Content

Time-Content Identifier 3-T1.

Time-Content Name Time contents used in the original source.

Time-Content Type Time content.

Time-Content Specification The 131 time contents in Definition 8.130.

Original-Source Location Identifier Set 3-L1.

Formal Input Elements

Formal-Component Identifier 3-C1.

Formal-Component Name Physical and formal input-claim components.

Formal-Component Type Formal component.

Formal-Component Text The 34 claim groups used by the original source are decomposed into the 56 type-homogeneous components in Definition 8.131.

Original-Source Support Identifier Set 3-L1.

Formal-Component Identifier 3-C2.

Formal-Component Name Formal input elements.

Formal-Component Type Formal component.

Formal-Component Text Components belonging to the light-speed ratio, local-clock time, synchronization, simultaneity, coordinate time, time transformation, clock rate, velocity, frequency, time differentiation, dynamical evolution, and formal principles are placed in the reference's formal-evidence environment.

Original-Source Support Identifier Set 3-L1.

Definition 8.130 (Time Contents in the Einstein 1905 Reference). For $R = \text{Ref}_{\text{Ein05ED}}$, let $n_R = 131$. Table 41 gives the time-content indices, contents, correspondence relations and source locations for 45 source-location groups. Within each row, ascending indices correspond one-to-one to the listed contents.

Table 41: Time Contents and Original-Source Locations
in the Einstein 1905 Reference

Location Group	Time-Content Indices	Time Contents (Index Order)	Correspondence Indices	Original-Source Location
1	{1, 2}	Temporal relation among rod, coordinates, clocks, and processes; temporal relation of electromagnetic processes	{1}	Introduction
2	[[3, 5]]	Coordinate-time parameter; position as a function of time; order of motion	$\emptyset$	§1
3	[[6, 8]]	Identity of an event pair; simultaneity judgment; time judgment	$\emptyset$	§1
4	[[9, 11]]	Train-arrival event; seven-o'clock clock indication; local simultaneity	$\emptyset$	§1
5	[[12, 14]]	Local-clock identity; clock-hand indication; local time	$\emptyset$	§1
6	{15, 16}	Point-specific sequence of local events; temporal relation between points	$\emptyset$	§1
7	[[17, 20]]	Distant event; received light signal; origin-clock indication; evaluation of distant time	$\emptyset$	§1
8	[[21, 25]]	Local event at A; local A-time; local event at B; local B-time; incomparability of the two local times before an additional stipulation	$\emptyset$	§1

Location Group	Time-Content Indices	Time Contents (Index Order)	Correspondence Indices	Original-Source Location
9	[[26, 31]]	Emission event; reflection event; return event; outbound time width; return time width; common distant time	{2}	§1
10	[[32, 35]]	Symmetry of synchronization; reverse-direction synchronization relation; clock-network consistency; synchronization relation of the entire clock network	{3}	§1
11	[[36, 38]]	Local event; synchronized-clock indication; event coordinate time	$\emptyset$	§1
12	{39, 40}	Round-trip time width; universal light-speed ratio	{4}	§1
13	{41, 42}	Stationary synchronized clock network; stationary-system coordinate time	$\emptyset$	§1
14	{43, 44}	Order of state change; law evolution in an inertial system	{5}	§2
15	[[45, 47]]	Identity of a light signal; propagation time width; constant propagation velocity	$\emptyset$	§2
16	{48, 49}	Time width; velocity ratio	{6}	§2
17	{50, 51}	Single stationary-system coordinate time; simultaneous endpoint positions	{7}	§2
18	[[52, 54]]	Identity of the endpoint clocks; endpoint-clock indications; stipulation that they show stationary-system time	$\emptyset$	§2
19	[[55, 60]]	Emission event; emission time; reflection event; reflection time; return event; return time	$\emptyset$	§2
20	{61, 62}	Comparison by a moving observer; nonsynchrony judgment	$\emptyset$	§2
21	{63, 64}	Identity of an event pair; relativity of simultaneity	{8}	§2
22	{65, 66}	Stationary-system coordinate time; axial position at that time	{9}	§3
23	[[67, 69]]	Distributed stationary clocks; light-signal synchronization; stationary-system time at every position	$\emptyset$	§3

Location Group	Time-Content Indices	Time Contents (Index Order)	Correspondence Indices	Original-Source Location
24	[[70, 72]]	Distributed moving clocks; isomorphic synchronization; moving-system time at every position	$\emptyset$	§3
25	[[73, 75]]	Event identity; stationary-system time; moving-system time	{10, 11}	§3
26	[[76, 78]]	Moving-clock indication; clock-network synchronization; moving-system time τ	$\emptyset$	§3
27	[[79, 81]]	Propagation by stationary-system coordinate time; propagation by moving-system coordinate time; wave relation between the two systems	{12}	§3
28	[[82, 84]]	Stationary-system time t ; clock indication τ at the moving origin; Lorentz rate relation	$\emptyset$	§4
29	[[85, 87]]	One second in the stationary system; increment of a moving clock; retardation per second	$\emptyset$	§4
30	[[88, 91]]	Departure from point 0; arrival at point 1; transport time width; transported-clock retardation	$\emptyset$	§4
31	{92}	Accumulation of clock retardation	[[13, 15]]	§4
32	[[93, 95]]	Identity of the equatorial clock; identity of the polar clock; rate comparison	$\emptyset$	§4
33	[[96, 98]]	Coordinate-time parameter; spatial trajectory; velocity-composition relation	$\emptyset$	§5
34	[[99, 101]]	Time derivative of field state; Maxwell–Hertz evolution; convective-current evolution	$\emptyset$	§§6,9
35	[[102, 104]]	Selected instant; force state at that instant; charge state at that instant	$\emptyset$	§§6,10
36	[[105, 107]]	Time-dependent plane-wave phase; coordinate time as a phase variable; angular frequency of the wave	$\emptyset$	§7
37	{108, 109}	Relative state of light source and observer; observed Doppler frequency	{16}	§7

Location Group	Time-Content Indices	Time Contents (Index Order)	Correspondence Indices	Original-Source Location
38	[[110, 114]]	Time t of the moving boundary enclosing the light complex; simultaneous slice $\tau = 0$ defining its moving-frame volume; incident energy per unit area and time; reflected energy per unit area and time; mechanical work per unit area and time	$\emptyset$	§8
39	[[115, 117]]	Coordinate time; charge-density state; current evolution	$\emptyset$	§9
40	[[118, 120]]	Electron rest event; selected instant; subsequent motion	$\emptyset$	§10
41	{121, 122}	Immediately subsequent time interval; state correspondence	{17}	§10
42	[[123, 125]]	Coordinate time; first derivative of velocity; second derivative of position	$\emptyset$	§10
43	{126, 127}	Slow-acceleration time interval; electron history	{18}	§10
44	[[128, 130]]	Period of a high-frequency field; velocity response; direct-velocity measurement relation	$\emptyset$	§10
45	{131}	Stationary-system simultaneous slice $t = 0$ defining the rigid sphere's shape	{10, 11}	§4, p. 903

Location groups 9, 11, 13, 17, 19–33 and 36–38 contain transfer outputs listed in Table 45. That table specifies the input and output indices of each transfer.

Table 42: Correspondence Relations and Supporting Claims in Einstein 1905

Corr.	Content	Supporting Claims	Groups
1	Description of a theory that uses clocks	$\{11, 13, 17, 18, 34\} \cup \llbracket 7, 9 \rrbracket$	{1}
2	Equality definition for time widths	{13}	{9}
3	Transitivity of the synchronization relation	{15}	{10}
4	Round-trip path length	{7, 12, 16}	{12}
5	Identity of physical-system state	{3}	{14}
6	Optical path length	{4, 5, 16}	{16}
7	Identity of the endpoints of a moving rod	{19, 20}	{17}
8	Two inertial systems	{3, 7, 22}	{21}
9	Identity of the moving axis	{7}	{22}

Corr.	Content	Supporting Claims	Groups
10	Spatial coordinates of the stationary system	{7}	{25}
11	Spatial coordinates of the moving system	{7}	{25}
12	Identity of a spherical wave	{4, 5, 25}	{27}
13	Polygonal path partition	{30, 31}	{31}
14	Closed path	{30, 31}	{31}
15	Extension assumption to a curved path	{31}	{31}
16	Lorentz transformation	{23, 24}	{37}
17	Instantaneously comoving system	{23, 24, 49}	{41}
18	No-radiation assumption	{49, 52}	{43}

Definition 8.131 (Audit-Input Claim Components in the Einstein 1905 Reference).

$$\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(56), \quad \text{Idx}_R^{\text{audit,phys}} = [56].$$

Set the claim-audit boundary to

$$\text{ClaimBoundaryIdx}_R = (\{1, 4, 7, 8, 10, 12, 19, 26, 28, 30, 34, 44, 46, 48, 54\}, [56]).$$

The 15 components in the first entry are the non-temporal generation-initial inputs; all 56 components constitute the source formal-evidence environment. The dependency key of each claim in Table 44 specifies the time contents it uses. The temporal dependencies of all 15 initial components are empty, so $\text{StrictClaimBoundary}(R)$ holds.

Table 43: Input-Claim Dependency Keys in Einstein 1905

Key	Time Indices	Key	Time Indices
1	$\emptyset$	18	$\{41, 42\} \cup \llbracket 70, 78 \rrbracket \cup \llbracket 96, 98 \rrbracket$
2	$\{43, 44\} \cup \llbracket 3, 5 \rrbracket$	19	$\{99, 100\} \cup \llbracket 73, 78 \rrbracket$
3	$\llbracket 45, 49 \rrbracket$	20	$\llbracket 73, 78 \rrbracket \cup \llbracket 102, 104 \rrbracket$
4	$\llbracket 12, 14 \rrbracket \cup \llbracket 21, 25 \rrbracket \cup \llbracket 32, 35 \rrbracket$	21	$\llbracket 73, 78 \rrbracket \cup \llbracket 105, 109 \rrbracket$
5	$\llbracket 6, 14 \rrbracket$	22	$\llbracket 73, 78 \rrbracket \cup \llbracket 105, 111 \rrbracket$
6	$\llbracket 21, 35 \rrbracket$	23	$\{41, 42\} \cup \llbracket 105, 109 \rrbracket \cup \llbracket 112, 114 \rrbracket$
7	$\{32, 33\}$	24	$\llbracket 73, 78 \rrbracket \cup \llbracket 99, 101 \rrbracket \cup \llbracket 115, 117 \rrbracket$
8	$\{34, 35\}$	25	$\llbracket 73, 78 \rrbracket \cup \llbracket 115, 117 \rrbracket$
9	$\{39, 40\} \cup \llbracket 26, 30 \rrbracket$	26	$\llbracket 102, 104 \rrbracket \cup \llbracket 118, 125 \rrbracket$
10	$\llbracket 70, 78 \rrbracket$	27	$\llbracket 102, 104 \rrbracket \cup \llbracket 123, 127 \rrbracket$
11	$\{41, 42\} \cup \llbracket 48, 51 \rrbracket$	28	$\{39, 40\} \cup \llbracket 123, 127 \rrbracket$
12	$\{48, 49\}$	29	$\llbracket 123, 130 \rrbracket$
13	$\llbracket 52, 64 \rrbracket$	30	$\llbracket 105, 107 \rrbracket$
14	$\llbracket 39, 42 \rrbracket \cup \llbracket 63, 81 \rrbracket$	31	$\{1, 2\}$
15	$\llbracket 39, 42 \rrbracket \cup \llbracket 73, 81 \rrbracket$	32	$\{41, 42\} \cup \llbracket 36, 38 \rrbracket$
16	$\llbracket 73, 78 \rrbracket \cup \llbracket 82, 87 \rrbracket$	33	$\{131\} \cup \llbracket 73, 75 \rrbracket$
17	$\llbracket 82, 95 \rrbracket$		

Table 44: Physical and Formal Input-Claim Components
in the Einstein 1905 Reference

p	Original Group	Type-Homogeneous Component	Dependence-Object Key	Original-Source Location Key
1	1	Physical-bearer configuration of a magnet and conductor	1	1
2	1	Asymmetry appearing in the description of relative motion between the magnet and conductor	31	1
3	2	Principle of relativity	2	2
4	3	Physical-bearer configuration of an emitting source and the emitted light signal	1	2
5	3	Assumption that the propagation velocity of light is independent of the state of motion of the emitting source	3	2
6	4	Formal condition that no privileged ether or absolutely stationary space is used	1	2
7	5	Configuration of a rigid rod and spatial coordinate system	1	3
8	6	Physical bearers of clocks having the same construction	1	4
9	6	Correspondence of clocks at rest and identical indications to a common time	4	4
10	7	Physical-bearer configuration of a local clock and local record	1	4
11	7	Simultaneity of a local event and clock indication	5	4
12	8	Physical-bearer configuration of distant clocks, emitter, reflector, and receiver	1	4

p	Original Group	Type-Homogeneous Component	Dependence-Object Key	Original-Source Location Key
13	8	Procedure for synchronizing clocks by distant light signals	6	4
14	9	Symmetry of the synchronization relation	7	4
15	9	Transitivity and network consistency of the synchronization relation	8	4
16	10	Assumed empirical universality of the light-speed ratio	9	5
17	11	Stationary-system coordinate time	32	4
18	12	Moving-system coordinate time	10	6
19	13	Physical bearer of a moving rod and spatial configuration of its endpoints	1	7
20	13	Measurement of the distance between the endpoints of a moving rod at the same stationary-system time	11	7
21	13	Length measurement by directly applying a comoving measuring rod	12	7
22	14	Relativity of simultaneity	13	7
23	15	Spatial-coordinate components of the Lorentz transformation	14	6
24	15	Time-coordinate component of the Lorentz transformation	14	6
25	16	Preservation of spherical-light-wave form in two coordinate systems	15	6
26	17	Physical carriers and spatial shapes of a rigid sphere and rigid bodies	1	8
27	17	Longitudinal contraction and unchanged transverse dimensions on the stationary-system slice $t = 0$	33	8

p	Original Group	Type-Homogeneous Component	Dependence-Object Key	Original-Source Location Key
28	18	Physical bearer of a moving clock	1	8
29	18	Rate relation of a moving clock	16	8
30	19	Physical bearer of a clock and spatial configuration of its transport path	1	8
31	19	Transported-clock lag, its accumulation on closed polygonal paths, and the assumed extension to curved closed paths	17	8
32	20	Velocity-composition law	18	9
33	20	Group property of parallel coordinate transformations	18	9
34	21	Physical system bearing an electromagnetic field, charge, and current	1	10
35	21	Covariance of the Maxwell–Hertz equations	19	10
36	22	Coordinate transformation of electric-field components	19	10
37	22	Coordinate transformation of magnetic-field components	19	10
38	23	Coordinate-system-dependent redescription of electromotive force	20	10
39	24	Relativistic Doppler relation	21	11
40	25	Aberration-of-light relation	21	11
41	26	Coordinate transformation of wave amplitude	21	11
42	27	Transformation of the volumes and energies of the same light complex on the simultaneous slices of the two frames	22	12
43	28	Radiation pressure from incident and reflected energy fluxes and the work rate at a moving mirror	23	12
44	29	Physical configuration of charge bearers and an electromagnetic field	1	13

p	Original Group	Type-Homogeneous Component	Dependence-Object Key	Original-Source Location Key
45	29	Covariance of the convection-current equation	24	13
46	30	Physical quantities of a group of charged bodies and total charge	1	13
47	30	Constancy of charge in the stationary frame conditional on its constancy in the comoving frame	25	13
48	31	Physical configuration of an electron, an electromagnetic field, and force	1	14
49	31	Electron equations of motion in the instantaneously comoving frame and their transformation to the stationary frame	26	14
50	32	Longitudinal mass defined using instantaneously comoving force and stationary-frame acceleration	26	14
51	32	Transverse mass defined using instantaneously comoving force and stationary-frame acceleration	26	14
52	33	Kinetic-energy relation for slow acceleration with negligible radiative loss	27	14
53	33	Light-speed boundary obtained from the electron kinetic energy	28	14
54	34	Physical-bearer configuration of an electron and observational apparatus	1	14
55	34	Three electron relations open to experimental examination	29	14

p	Original Group	Type-Homogeneous Component	Dependence-Object Key	Original-Source Location Key
56	24	Plane-wave approximation for a distant source and phase expressed using coordinate time and angular frequency	30	11

Dependency Inputs

Dependency Inputs The index sets in the table specify the supporting claims, temporal inputs and temporal outputs of the 23 transfers.

Dependency and Transfer Elements

Definition 8.132 (Time-Content Transfers in Einstein 1905). Let $\text{ldx}_R^{\text{audit,transfer}} = [23]$. Realize the supporting claims, temporal inputs and temporal outputs in each row by Definition 7.105. The source location of each transfer is the union of the source locations corresponding to its temporal inputs, temporal outputs and supporting claims.

Table 45: Time-Content Transfers in Einstein 1905

j	Supporting-Claim Indices	Time-Input Indices	Time-Output Indices	Operation
1	$\llbracket 8, 15 \rrbracket$	$\llbracket 9, 14 \rrbracket \cup \llbracket 21, 25 \rrbracket \cup \llbracket 32, 35 \rrbracket$	$\llbracket 26, 31 \rrbracket$	From local events, clock indications, the two local times and synchronization consistency to common time
2	$\{17\} \cup \llbracket 10, 15 \rrbracket$	$\llbracket 9, 11 \rrbracket \cup \llbracket 26, 31 \rrbracket$	$\llbracket 36, 38 \rrbracket$	From local events and common time to event time read from synchronized clocks
3	$\{8, 9, 17\} \cup \llbracket 12, 15 \rrbracket$	$\llbracket 12, 14 \rrbracket \cup \llbracket 26, 31 \rrbracket$	$\{41, 42\}$	From local clocks and common time to stationary-system coordinate time
4	$\{7, 17, 19, 20\}$	$\{41, 42\}$	$\{50, 51\}$	Determine both endpoints of the moving rod at one stationary-system time
5	$\{12, 13, 16, 17, 19, 20\} \cup \llbracket 7, 9 \rrbracket$	$\{48, 49\} \cup \llbracket 39, 42 \rrbracket \cup \llbracket 52, 54 \rrbracket$	$\llbracket 55, 60 \rrbracket$	From endpoint clocks showing stationary-system time, light speed and rod speed to the three signal-event times

j	Supporting-Claim Indices	Time-Input Indices	Time-Output Indices	Operation
6	{12, 13, 19, 22}	$\llbracket 55, 60 \rrbracket$	{61, 62}	Compare signal-event time differences to obtain the moving observers' nonsynchrony judgment
7	{3, 7, 17, 22}	{41, 42, 61, 62}	{63, 64}	From stationary-system synchrony and the moving observers' nonsynchrony judgment to relativity of simultaneity
8	{7, 17}	{41, 42}	{65, 66}	Specify positions on the moving axes at stationary-system coordinate time
9	$\{17\} \cup \llbracket 7, 9 \rrbracket \cup \llbracket 12, 15 \rrbracket$	$\llbracket 12, 14 \rrbracket \cup \llbracket 26, 31 \rrbracket$	$\llbracket 67, 69 \rrbracket$	Apply the synchronization procedure to the stationary clock network
10	$\{18\} \cup \llbracket 7, 9 \rrbracket \cup \llbracket 12, 15 \rrbracket$	$\llbracket 12, 14 \rrbracket \cup \llbracket 26, 31 \rrbracket$	$\llbracket 70, 72 \rrbracket$	Apply the same synchronization procedure to the moving clock network
11	{7, 17, 18}	$\{41, 42\} \cup \llbracket 67, 72 \rrbracket$	$\llbracket 73, 75 \rrbracket$	Assign times to the same event in the two coordinate systems
12	{7, 18}	$\llbracket 70, 72 \rrbracket$	$\llbracket 76, 78 \rrbracket$	Specify moving-system coordinate time using its clock network
13	$\{4, 5, 7, 12, 16\} \cup \llbracket 23, 25 \rrbracket$	$\llbracket 39, 42 \rrbracket \cup \llbracket 70, 78 \rrbracket$	$\llbracket 79, 81 \rrbracket$	From light speed and the Lorentz transformation to spherical-wave propagation in the two systems
14	{7, 23, 24, 28, 29}	$\{41, 42\} \cup \llbracket 73, 78 \rrbracket$	$\llbracket 82, 84 \rrbracket$	Obtain the moving-origin clock rate by substitution into the Lorentz time transformation
15	{28, 29}	$\llbracket 82, 84 \rrbracket$	$\llbracket 85, 87 \rrbracket$	From the clock rate to the lag per stationary-system second
16	$\llbracket 28, 31 \rrbracket$	$\llbracket 82, 84 \rrbracket$	$\llbracket 88, 91 \rrbracket$	From the clock rate and transport path to transported-clock lag
17	{30, 31}	$\llbracket 88, 91 \rrbracket$	{92}	Sum the lag on a closed polygonal path and apply the assumed extension to curved closed paths

j	Supporting-Claim Indices	Time-Input Indices	Time-Output Indices	Operation
18	{30, 31}	{92}	[[93, 95]]	Compare equatorial and polar clocks under otherwise identical conditions
19	{23, 24, 32, 33}	{41, 42} $\cup$ [[70, 72]]	[[96, 98]]	From times and trajectories in the two frames to velocity composition
20	{56}	{106, 107}	{105}	Use coordinate time and angular frequency in the plane-wave phase
21	{7, 23, 24, 39, 56}	[[73, 78]] $\cup$ [[105, 107]]	{108, 109}	From the two frame times, incident-wave phase and Lorentz transformation to Doppler frequency
22	{4, 5, 7, 23, 24, 41, 42}	[[73, 78]]	{110, 111}	Use the two frame times and light speed to specify the moving boundary and slice $\tau = 0$ of the light complex
23	{23, 24, 43, 56} $\cup$ [[39, 41]]	{41, 42, 108, 109}	[[112, 114]]	Balance incident and reflected wave energy with mechanical work per unit area and time at the moving mirror

Transfers and Outputs The 23 transfers connect clock synchronization, event-time and coordinate-time assignments, simultaneity comparison, clock rate and lag, velocity composition, plane-wave phase, Doppler frequency, the moving boundary and simultaneous slice of a light complex, and radiative energy balance.

Formation Conditions

Formation Conditions Each transfer applies where its listed supporting claims, temporal inputs and the following conditions hold. Transfers 1–3 and 9–12 use local clocks, light signals and the synchronization convention. The endpoint clocks in transfer 5 are stipulated to show stationary-system time; this condition differs from that of the naturally running moving clocks in transfers 14–16. Transfer 17 uses the accumulated lag on a closed polygonal path and the assumed extension to curved closed paths; transfer 18 uses otherwise identical clocks, and transfer 19 uses parallel coordinate transformations. Transfers 20 and 21 use the distant-source plane-wave approximation. Transfer 22 uses the same light complex and the simultaneous slices of both frames; transfer 23 uses the balance of incident and reflected energy fluxes and mechanical work at a moving mirror.

All 23 transfers have nonempty temporal inputs. There are no directed cycles among time contents and no transfer has a self-input.

Source Support

Source Support Table 44 gives the supporting claims and their source locations; Table 41 gives the locations of the time contents, and Table 42 gives the grounds for the 18 correspondence relations. Each edge’s supporting-claim set includes the support for the correspondence relations it uses.

Results

Definition 8.133 (Results in the Einstein 1905 Reference). Define the result index set by $\text{ld}\mathbf{x}_R^{\text{audit,result}} = [10]$. The correspondence between results and original-source locations is given in Table 46.

Table 46: Results and Original-Source Locations in the Einstein 1905 Reference

j	Result	Source location
1	Distinction between local clock readings and common time at two locations	§1
2	Explicit definition of two-location synchronization by light signals	§1
3	Assumed symmetry, transitivity and network consistency of synchronization	§1
4	Coordinate construction of stationary- and moving-system times	§§1,3
5	Relativity of simultaneity and the Lorentz transformation	§§2,3
6	Rigid-body length transformation, moving-clock rate and transported-clock lag	§§2,4
7	Velocity composition and the kinematic light-speed boundary	§§1,3,5
8	Covariance of the Maxwell–Hertz and convection-current equations, and field transformations	§§3,6,9
9	Doppler effect, aberration, light energy and radiation pressure	§§3,7,8
10	Conditional constancy of charge, electron motion, mass and kinetic energy	§§9,10

Source-Specific Results The ten results are specified in Definition 8.133. Table 47 gives their temporal prerequisites and supporting claims.

Result Dependencies

Definition 8.134 (Result-Dependence Hyperedges in the Einstein 1905 Reference). For each result edge $j \in [10]$, realize the incidence from its listed supporting claims and temporal prerequisites to that result by Definition 7.105.

Table 47: Result-Dependency Hyperedges in Einstein 1905

j	Temporal Prerequisites	Supporting-Claim Indices
1	$\llbracket 12, 25 \rrbracket$	$\llbracket 10, 12 \rrbracket$
2	$\llbracket 9, 14 \rrbracket \cup \llbracket 21, 35 \rrbracket$	$\llbracket 8, 16 \rrbracket$
3	$\llbracket 32, 35 \rrbracket$	$\{14, 15\}$
4	$\{41, 42\} \cup \llbracket 67, 78 \rrbracket$	$\{17, 18\} \cup \llbracket 7, 9 \rrbracket \cup \llbracket 12, 15 \rrbracket$
5	$\{63, 64\} \cup \llbracket 73, 81 \rrbracket$	$\{7\} \cup \llbracket 3, 5 \rrbracket \cup \llbracket 16, 18 \rrbracket \cup \llbracket 22, 25 \rrbracket$
6	$\{50, 51, 131\} \cup \llbracket 73, 75 \rrbracket \cup \llbracket 82, 95 \rrbracket$	$\{7, 23, 24\} \cup \llbracket 19, 21 \rrbracket \cup \llbracket 26, 31 \rrbracket$
7	$\llbracket 39, 42 \rrbracket \cup \llbracket 70, 78 \rrbracket \cup \llbracket 96, 98 \rrbracket$	$\{4, 5, 7, 12, 16, 23, 24, 32, 33\}$
8	$\llbracket 73, 78 \rrbracket \cup \llbracket 99, 104 \rrbracket \cup \llbracket 115, 117 \rrbracket$	$\{7, 23, 24, 44, 45\} \cup \llbracket 34, 38 \rrbracket$
9	$\llbracket 73, 78 \rrbracket \cup \llbracket 105, 114 \rrbracket$	$\{7, 23, 24, 56\} \cup \llbracket 39, 43 \rrbracket$
10	$\llbracket 102, 104 \rrbracket \cup \llbracket 115, 130 \rrbracket$	$\{23, 24\} \cup \llbracket 44, 55 \rrbracket$

Temporal prerequisites occur 150 times across the ten results. A time content used by multiple results is counted once for each result.

Grounds of Establishment

Grounds for Establishing the Results Results 1–4 rest on the distinction between local and common time, the definition and assumed consistency of synchronization, and the construction of coordinate time. Results 5–7 rest on kinematic derivations under the relativity principle and light-speed postulate. Results 8–10 apply this coordinate construction and transformation to the source’s conditions for fields, radiation, charge and electrons.

Evidence Hierarchy

Directly Measured Objects This reference reports new direct-measurement data for zero objects.

Directly Measured Predicates This reference reports new direct-measurement data for zero predicates. The light-speed ratio is placed as an assumption with empirical universality, and the three electron relations belonging to result 10 are presented as predictive objects open to experimental examination.

Conditions and Domain of Application of Direct Measurement Universality of the light-speed ratio is an assumption based on experience, and the three electron relations are presented for experimental examination. The paper reports no new measured values.

Objects of Formal Generalization The domain of application of results 1–7 is a kinematical system equipped with the principle of relativity, the light-speed assumption, rigid rods, clock synchronization, and inertial coordinate systems. The domain of application of results 8–10 is a system that additionally includes the assumptions concerning electromagnetic fields, charge, current, radiation, and electron motion in §§6–10 of the original source.

Conditions and Domain of Application of Formal Generalization Each result uses the temporal prerequisites and supporting claims in Table 47, together with the following conditions. The curved closed path in result 6 follows the source’s extension assumption; the equatorial and polar clocks are compared under otherwise identical conditions. The second-order clock-lag approximation omits fourth and higher orders in speed. The group property in result 7 concerns parallel coordinate transformations. Result 9 uses the distant-source plane-wave approximation, volume comparison for the same light complex, and energy balance at a moving mirror. Charge constancy in result 10 is conditional on constancy in the comoving frame. Longitudinal and transverse masses follow the source’s definitions using instantaneously comoving force and stationary-frame acceleration; kinetic energy assumes slow acceleration with negligible radiative loss.

Generation Judgment Judgment values: LCTR complete-generation judgment: 0 cases / within-reference-evidence-conditional local-reach judgment: 0 cases / remaining primitive time roots: 131 cases.

Generation Reach and Primitive Time Roots The generation reach whose initial set consists only of non-temporal physical inputs generates zero time contents. All 131 time contents in Definition 8.130 remain as primitive time roots.

Within-Reference-Evidence-Conditional Local Reach The within-reference-evidence-conditional local-reach judgment has zero cases, and 131 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ in Definition 7.108 and apply Definition 7.50. Summation over all results determines the primitive-time-root–result dependence relations and necessary primitive-time-root–result dependence relations.

Result Dependence All 10 results in Definition 8.133 are primitive-time-root-dependent results. The minimal primitive-time-root support of each result is given by Proposition 8.136.

Evidence Effect

Evidence-Effect Domain The formal evidence in this reference directly supports 10 results concerning the definitions of clock synchronization and coordinate time, the Lorentz transformation under the principle of relativity and the light-speed assumption, simultaneity, length, clock rate, velocity composition, electromagnetic fields, Doppler relations, radiation, charge, and electron motion. The evidence types supplied by this reference itself are definitions, assumptions, formal derivations, and predictions.

Evidence-Effect Boundary The validity of formal derivations that take the principle of relativity, the light-speed assumption, synchronization conventions, and coordinate time as inputs does not imply that the 131 time contents have been generated from non-temporal physical inputs. Nor does the success of predictions or experiments in other references retroactively confer physical-time qualification on the time contents of this reference. These transitions lie beyond the evidence-effect boundary under Definition 7.115.

Result-Preservation Domain

Result-Preservation Domain The 10 results are preserved when each time-content prerequisite is supplied by the corresponding generated time representation in the LCTR generation sequence and the principle of relativity, emission–propagation–reception relations, light-speed relation, coordinate transformations, fields, charges, forces, and the directions of dependence in the original source are retained. Results concerning clock rate, clock retardation, length, frequency, and velocity are preserved as comparison relations on generated time representations. Attribution that proceeds from comparison differences on physical bearers to changes in properties of time itself independent of physical bearers falls outside the preservation domain.

Placement of Grounding Obligations

Placement of Grounding Obligations The source requires the meaning of time to be made clear for descriptions of motion to have physical meaning, and identifies the insufficiency of a local clock for comparing events at separated points. Its construction of coordinate time uses simultaneity between local events and clock indications, emission, reflection and return of light signals, and a convention equating time intervals. The 131 time contents, including these, are used before generation from non-temporal physical inputs; the LCTR classification of the placement of the physical-time grounding obligation is therefore “backgrounded by primitive-time inputs.”

Evidence of Actual Invalidation

Evidence of Actual Invalidation The complete-generation judgment and remaining-root judgment concern physical-time qualification and are distinct from the actual invalidation of a specific condition that determines the value or domain of application of a result. Neither the original-source formal evidence nor the LCTR audit evidence for this reference establishes the actual invalidation of such a target condition. The actual-invalidation judgment is “indeterminate from the original-source evidence”; actual invalidation is not inferred solely from the backgrounding of a grounding obligation.

Downstream Reach

Downstream Reach The reach to results that depend on primitive time roots is determined for this reference by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because the actual invalidation of a specific condition has not been established, the judgment on actual reach from that invalidation position to results is “indeterminate from the original-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the targets of repair are the corresponding time content and the generation or establishment conditions required before its use.

The scope of recalculation is limited to results for which actual invalidation has been independently established and that are reached by dependence from the invalidation position. Because actual invalidation is indeterminate in this reference, this repair-and-recalculation specification is conditional. If the condition holds, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted.

Audit Conclusions

Proposition 8.135 (Primitive Time Roots and Result Dependence in the Einstein 1905 Reference).

In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepIdx}(131, 10)$. Moreover, $R \in \text{Ref}^{\text{audit,rootdep}}$ holds.

Proof. Take the 15 claims in the first entry of the claim-audit boundary as the generation-initial set. Every one of the 23 transfers has a nonempty temporal input, so this initial set yields no time content. Adding all 56 claims as the evidence environment still leaves the temporal prerequisites unsatisfied unless time contents are supplied as seeds. Complete generation and source-evidence-conditioned local reach are therefore both zero, and all 131 time contents remain primitive-time roots. All ten result edges have nonempty temporal prerequisites; adding all 131 roots to the evidence environment reaches every result. Definition 8.115 gives the displayed partition. Membership in the reference sets follows from Definition 7.52. $\square$

Proposition 8.136 (Minimal Primitive-Time-Root Support in the Einstein 1905 Reference). For each $q \in \text{Idx}_R^{\text{audit,result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108 and the minimum and maximum cardinalities of those elements are given in Table 48.

Table 48: Minimal Primitive-Time-Root Support in the Einstein 1905 Reference

Result Indices	Support Statistic	Result Indices	Support Statistic
1	(1; 14)	2	(1; 15)
3	(1; 4)	4	(3; 8, 15)
5	(32; 11, 24)	6	(7; 9, 16)
7	(4; 10, 17)	8	(5; 15, 24)
9	(7; 10, 17)	10	(1; 19)

Proof. Apply Definition 7.46 in the order of the 23 transfers. After the first scan, the second scan leaves every time-content support antichain unchanged. Taking family unions of supports over each result's temporal prerequisites and retaining inclusion-minimal members gives the displayed families. Each member reaches the corresponding result as a seed; removing any one root from that member prevents that reach. There are 62 support-family members, 295 primitive-time-root/result dependence pairs, and 66 essential primitive-time-root/result dependence pairs. $\square$

Individual Statements

Statement Objects The definitions of simultaneity and time, the relation between the time of an event and a simultaneous clock indication, and the speed of light V and the time interval used in its definition as a speed.

Original Text (German) eine Definition von „gleichzeitig“ und „Zeit“ gewonnen

Die „Zeit“ eines Ereignisses ist die mit dem Ereignis gleichzeitige Angabe einer am Orte des Ereignisses befindlichen, ruhenden Uhr, welche mit einer bestimmten, ruhenden Uhr, und zwar für alle Zeitbestimmungen mit der nämlichen Uhr, synchron läuft.

Jeder Lichtstrahl bewegt sich im „ruhenden“ Koordinatensystem mit der bestimmten Geschwindigkeit V , unabhängig davon, ob dieser Lichtstrahl von einem ruhenden oder bewegten Körper emittiert ist. Hierbei ist

$$\text{Geschwindigkeit} = \frac{\text{Lichtweg}}{\text{Zeitdauer}},$$

wobei „Zeitdauer“ im Sinne der Definition des § 1 aufzufassen ist.

English Translation A definition of “simultaneous” and “time” has been obtained.

The “time” of an event is the indication, simultaneous with that event, of a stationary clock at the event’s location. This clock runs synchronously with one particular stationary clock—the same clock for all time determinations.

Every ray of light moves in the “stationary” coordinate system with the definite speed V , independently of whether this ray of light is emitted by a stationary or moving body. Here,

$$\text{speed} = \frac{\text{light path}}{\text{time interval}}$$

and “time interval” is to be understood in the sense of the definition in § 1.

Original-Source Locations The first and second quotations are from Einstein, *Zur Elektrodynamik bewegter Körper*, *Annalen der Physik* 17, p. 894, §1; the third quotation is from the same reference, p. 895, §2.

Minimal Context Section 1 of the original source starts from the correspondence between an event at one point and a clock indication, introduces a relation for synchronizing clocks at two separated points by light signals, and then assigns a common time to events in the stationary system. Section 2 of the original source states the principle of relativity and the principle of constant light speed, and expressly gives the speed of light V in the stationary coordinate system as the ratio of the light path to the time interval defined in §1.

Attribution The three statements are attributed to §§1–2 of Einstein’s original German paper of 1905.

Predicates The first quotation states that definitions of simultaneity and time have been obtained. The second defines the event’s time as the simultaneous indication of a stationary clock at the event’s location, synchronized with one particular stationary clock common to all time determinations. The third makes the speed of light V in the stationary coordinate system independent of the emitting body’s motion and gives speed as the ratio of light path to the interval defined in §1.

Explicit Domain of Application The domain of application of the first and second quotations is the stationary system in §1 of the original source, an event and clock at the same point, and a network of clocks at separated points synchronized by light signals. The domain of application of the third quotation is the stationary coordinate system in §2 of the original source, rays of light emitted by stationary or moving bodies, the light path, and the time interval defined in §1.

Individual Audit

Interpretation of the Statements Together, the three statements specify event time by a simultaneous local-clock indication, synchronize distant clocks by light signals, and construct coordinate time using the synchronized stationary-clock network. The speed of light V is expressed as the ratio of light path to the interval defined in this construction. LCTR imposes obligations for generation from non-temporal physical inputs and for consistency on the clock indications, signal events, records, comparisons and transport used here.

Independent Grounds The three statements give formal and operational definitions of clock indication, synchronization convention, and the representation of the speed of light. The quoted scope contains no independent measurement showing that the time contents used by these definitions were generated from non-temporal physical inputs, and no new experimental result that directly tests the temporal qualification of the speed of light.

Evidence-Effect Boundary The first and second quotations directly support the definitions of simultaneity, clock indication, and the time of the stationary system in §1 of the original source. The third quotation directly supports the original notation V , independence from the state of motion of the emitting body, the definition of speed by light path and time interval, and the dependence of that time interval on the definition in §1. The three quotations do not establish that local-clock indication, light path, time interval, speed ratio, and synchronization relation themselves were generated from non-temporal physical inputs, or that the resulting coordinate time and representation of the speed of light possess physical-time qualification. These matters lie beyond the evidence-effect boundary of the three quotations under Definition 7.115.

Corresponding Formal Evidence The first and second quotations correspond to time contents 6–38 and 41–42 (location groups 3–11 and 13) in Definition 8.130, formal-evidence support components 8–17 in Definition 8.131, and results 1–4 in Definition 8.133. The third quotation corresponds to time contents 39–42 and 45–49

in Definition 8.130, formal-evidence support components 4–5 and 16 in Definition 8.131, and results 2 and 7 in Definition 8.133.

Within-Reference Comparison The first and second statements construct coordinate time from local-clock indications and distant-clock synchronization, and the third statement represents the speed of light by using that time interval. The light-speed statement therefore depends on the preceding clock and synchronization definitions, but that dependence does not substitute for the physical generation of clock indication, recording, comparison, and transport.

8.19.4 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 4.

Source Name Primitive-Time-Root Audit of Minkowski 1908/1909

Bibliographic Identification Let $R = \text{Ref}_{\text{Min09RZ}}$ denote Hermann Minkowski, *Raum und Zeit*, *Jahresbericht der Deutschen Mathematiker-Vereinigung*, Vol. 18 (1909), pp. 75–88. The lecture was delivered on September 21, 1908. The audit domain comprises the introductory passage preceding Section I, Sections I through V, Figures 1 through 4, and the closing passage of the text.

Original-Source Location Identifier 4-L1.

Original-Source Location the introductory passage preceding Section I, Sections I through V, Figures 1 through 4, and the closing passage of the text. The locations of the respective semantic objects and formal-evidence components are given in Table 49 and Table 51.

Audit-Scope Identifier 4-SC1.

Audit Scope The audit domain stated under Bibliographic Identification.

Formal Objects

Object Identifier 4-F1.

Object Name Formal object.

Object Type Formal object.

Formal Content Definitions, assumptions, and derivations involving time dependence in spacetime coordinates, the Newton and Lorentz groups, world-points and world-lines, the division of world-points by the light cone, timelike, spacelike, and lightlike directions, proper time, four-velocity, four-acceleration, four-momentum, energy, the four-dimensional electromagnetic field, and the proposed theory of gravitation.

Original-Source Location Identifier Set 4-L1.

Object Identifier 4-F2.

Object Name Objects that use time.

Object Type Formal object.

Formal Content The 32 time contents and their 30 correspondence relations listed in the tables.

Original-Source Location Identifier Set 4-L1.

Assumptions

Assumption Identifier 4-A1.

Assumption Content The individual contents of the reference assumptions are given in Definition 8.138.

Original-Source Location Identifier Set 4-L1.

Non-Temporal Physical Inputs

Input Identifier 4-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content Of the 71 formal-evidence components, eight describe non-temporal bearers of spatial coordinates, objects of perception, substantial points, light signals, physical phenomena, massive bodies and interactions, charge and electromagnetic fields, and gravitational sources. These eight components are the generation-initial inputs.

Original-Source Location Identifier Set 4-L1.

Time Content

Time-Content Identifier 4-T1.

Time-Content Name Time contents in the original source.

Time-Content Type Time content.

Time-Content Specification The individual time contents are given in Definition 8.137.

Original-Source Location Identifier Set 4-L1.

Formal Input Elements

Formal-Component Identifier 4-C1.

Formal-Component Name Composition of the reference's formal-evidence environment.

Formal-Component Type Formal component.

Formal-Component Text All 71 components constitute the source formal-evidence environment.

Original-Source Support Identifier Set 4-L1.

Definition 8.137 (Time Contents in the Minkowski 1908/1909 Reference). For $R = \text{Ref}_{\text{Min09RZ}}$, let $n_R = 32$. Table 49 lists each time content and its source location. Correspondence indices refer to the relations in Table 52.

Table 49: Time Contents and Source Locations in
Minkowski 1908/1909

Group	i	Time Content	Corr.	Location
1	{1}	Temporal origin	{2}	I, p. 75
2	{2}	Connection of place and time in objects of perception	{3}	I, p. 76
3	{3}	Temporal coordinate t	{1}	I, p. 76
4	{4}	World-point (x, y, z, t)	{4}	I, p. 76
5	{5}	Substance at each place and time	{5}	I, p. 76
6	{6}	Identity of a substantial point at different times	{6}	I, p. 76
7	{7}	Temporal element dt	{7}	I, p. 76
8	{8}	World-line parameterized by t	{8}	I, p. 76
9	{9}	Coordinate t preserved by Galilean transformations	{10}	I, pp. 76–77
10	{10}	Sign of t and temporal orientation	{9}	I, p. 77; III, p. 82
11	{11}	Identification of c with the speed of light in vacuum	{11}	I, p. 78
12	{12}	Transformed time t' and the temporal transformation relation	{12}	I–II, pp. 78–81
13	{13}	Spatial projections as simultaneity planes	{13}	I–II, pp. 79–81
14	{14}	Timelike world-line and its temporal orientation	{18}	II, p. 79; III, pp. 83–84
15	{15}	Lorentz local time	{14}	II, p. 81
16	{16}	Vorkegel and Nachkegel	{15}	III, p. 82
17	{17}	Order of timelike-separated world-points	{16}	III, pp. 82–83
18	{18}	Coordinate-dependent order of spacelike-separated world-points	{17}	III, pp. 82–83
19	{19}	Proper-time element $d\tau$	{19}	III, p. 83
20	{20}	Oriented world-line segment from fixed initial point P_0 to endpoint P	{20}	III, p. 83
21	{21}	Proper time τ along the specified world-line segment	{20}	III, p. 83
22	{22}	Four coordinates parameterized by proper time	{21}	III, p. 84
23	{23}	Proper-time dependence of four-velocity $U = d(x, y, z, t)/d\tau$	{22}	III, p. 84
24	{24}	Four-acceleration $A = dU/d\tau$	{23}	III, p. 84
25	{25}	All components of four-momentum mU	{24}	IV, p. 85
26	{26}	Temporal component $mU_t = m dt/d\tau$ of four-momentum	{24}	IV, p. 85
27	{27}	Energy $E = mc^2 U_t$	{25}	IV, p. 85
28	{28}	Energy in a specified frame	{26}	IV, p. 85
29	{29}	Ratio of length and time units and the representation $c = 1, s = it$	{27}	IV, p. 86
30	{30}	Dependence of the retarded field on the past light cone	{28}	V, pp. 86–87

Group	i	Time Content	Corr.	Location
31	{31}	Time-axis-dependent decomposition into electric and magnetic fields	{29}	V, p. 87
32	{32}	Motion and the relation between proper and coordinate time in the gravitational proposal	{30}	V, p. 88

Definition 8.138 (Formal-Evidence Semantic Components in the Minkowski 1908/1909 Reference).

$$\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(71), \quad \text{Idx}_R^{\text{audit,phys}} = [71].$$

$$\text{ClaimBoundaryIdx}_R = (\{9, 11, 16, 27, 29, 55, 63, 68\}, [71]).$$

The eight claims in the first entry are the non-temporal generation-initial inputs; all 71 claims constitute the source formal-evidence environment. The temporal dependencies of the eight initial claims are empty, so $\text{StrictClaimBoundary}(R)$ holds. Table 51 and the following keys specify each claim's content, temporal dependencies and source location.

Table 50: Claim Dependency Keys in Minkowski

Key	Time Indices	Key	Time Indices
1	$\llbracket 2, 4 \rrbracket$	25	$\{10, 16, 17\}$
2	$\{1, 3, 9\}$	26	$\{16, 18\}$
3	$\{1\}$	27	$\{11, 14, 17\}$
4	$\emptyset$	28	$\{11, 18\}$
5	$\{3\}$	29	$\{11, 16\}$
6	$\{2\}$	30	$\{14\}$
7	$\{3, 4\}$	31	$\{3, 7, 11, 14, 19\}$
8	$\{4\}$	32	$\llbracket 19, 21 \rrbracket$
9	$\{5\}$	33	$\{14\} \cup \llbracket 20, 22 \rrbracket$
10	$\{6\}$	34	$\{22, 23\}$
11	$\{7, 8\}$	35	$\{23, 24\}$
12	$\{8\}$	36	$\{3\} \cup \llbracket 23, 26 \rrbracket$
13	$\{3, 9\}$	37	$\{11\} \cup \llbracket 25, 27 \rrbracket$
14	$\{3, 10, 11\}$	38	$\{11, 23, 27\}$
15	$\{3\} \cup \llbracket 10, 12 \rrbracket$	39	$\{12, 25, 28\}$
16	$\{3\} \cup \llbracket 9, 12 \rrbracket$	40	$\{11, 29\}$
17	$\{11\}$	41	$\{3, 29\}$
18	$\{3, 12, 13\}$	42	$\{14, 23, 24, 30\}$
19	$\{3, 12, 15\}$	43	$\{31\}$
20	$\{12, 13\}$	44	$\llbracket 23, 25 \rrbracket$
21	$\{4, 12, 13\}$	45	$\{14, 23, 24, 30, 31\}$
22	$\{15\}$	46	$\{32\}$
23	$\{12, 15\}$	47	$\{3, 9, 11, 12, 19, 23, 25, 30, 31\}$
24	$\{10, 11, 16\}$		

The correspondence between source-location keys and source locations is as follows.

- 1 Introduction, p. 75
- 2 I, p. 75
- 3 I, p. 76
- 4 I, pp. 76–78
- 5 I–II, pp. 78–81
- 6 II, pp. 81–82
- 7 III, pp. 82–83
- 8 II, p. 79; III, p. 83
- 9 III, p. 83
- 10 III, p. 84
- 11 IV, pp. 84–85
- 12 IV, p. 86
- 13 V, pp. 86–88
- 14 V, p. 88
- 15 I–V, pp. 75–88

Table 51: Physical and Formal Input Claims in Minkowski

p	Claim	Depen- dency Key	Location Key
1	Statement that space by itself recedes into a shadow	1	1
2	Statement that time by itself recedes into a shadow	1	1
3	claim attributing independence to the combination of space and time	1	1
4	claim presenting the view of space and time as derived from a basis in experimental physics	1	1
5	form invariance of the Newtonian equations of mechanics under changes in the position of the spatial coordinate system	2	2
6	form invariance of the Newtonian equations of mechanics under the addition of uniform translational motion	2	2
7	form invariance of the Newtonian equations of mechanics under changes in the time origin	3	2
8	complete group of Newtonian mechanics obtained by composing the two invariance groups	2	2
9	construction of an orthogonal spatial coordinate system	4	3
10	introduction of t as a time coordinate	5	3
11	physical bearer of an object of perception	4	3

p	Claim	Depen- dency Key	Location Key
12	combination of the place and time of an object of perception	6	3
13	representation of a world-point by a quadruple of coordinate values	7	3
14	world as the manifold of all quadruples of coordinate values	8	3
15	definition of substance as an object perceptible at every place and time	9	3
16	physical bearer of a substantial point	4	3
17	identity of a substantial point at different times	10	3
18	correspondence between the time element and changes in spatial coordinates	11	3
19	parameter representation of a world-line by a time variable	12	3
20	proposal to express physical laws as interrelations among world-lines	12	3
21	preservation of the time coordinate under Galilean-type transformations	13	4
22	velocity-proportional transformation of spatial coordinates under Galilean-type transformations	13	4
23	Hyperboloid $c^2t^2 - x^2 - y^2 - z^2 = 1$, $t > 0$, for $c > 0$	14	4
24	Transformations preserving the hyperboloid and temporal orientation, with coordinate translations	15	4
25	Newtonian group in the infinite-light-speed limit at fixed physical relative velocity	16	4
26	distinction between the mathematical existence of the finite-vacuum-light-speed group and the physical invariance of the laws of nature	14	4
27	Light signals and their physical bearers	4	4
28	Identification of the constant with vacuum light speed and the ratio between electrostatic and electromagnetic units of charge	17	4
29	bearer configuration of physical phenomena subject to coordinate representation	4	5
30	multiple space-time coordinate systems fitted to phenomena	18	5
31	equal qualification of the time coordinate after an admissible transformation	19	5
32	three-dimensional space corresponding to a time projection	20	5
33	one four-dimensional world retained across numerous spatial projections	21	5
34	attribution of local time to Lorentz in the reference	22	5
35	attribution to Einstein in the reference of the contribution concerning the equivalence of multiple time coordinates	23	5

p	Claim	Depen- dency Key	Location Key
36	absoluteness of the four-dimensional world asserted by Minkowski	21	6
37	relativity of spatial and temporal projections asserted by Minkowski	21	6
38	<i>Vorkegel</i> , the past side of the light cone	24	7
39	<i>Nachkegel</i> , the future side of the light cone	24	7
40	Preservation of the order of timelike-separated world-points under admissible transformations preserving temporal orientation	25	7
41	coordinate-system dependence of the temporal order of spacelike-separated world-points	26	7
42	classification of timelike directions by the sign of the quadratic form	27	7
43	classification of spacelike directions by the sign of the quadratic form	28	7
44	classification of lightlike directions by the sign of the quadratic form	29	7
45	local-rest-system condition imposed on the world-line of a substantial point	30	8
46	timelike character of the world-line tangent required by the local-rest-system condition	30	8
47	$d\tau = c^{-1}\sqrt{c^2 dt^2 - dx^2 - dy^2 - dz^2}$ on a timelike world-line	31	9
48	proper time as the integral of the line element from a specified initial point to a variable terminal point	32	9
49	reparameterization of the four coordinates by proper time	33	10
50	Four-velocity normalization $g(U, U) = c^2$ and source-defined timelike magnitude 1	34	10
51	Four-acceleration is spacelike or zero	35	10
52	orthogonality relation between four-velocity and four-acceleration	35	10
53	Curvature hyperbola for nonzero acceleration and its degeneration to the tangent at zero acceleration	35	10
54	Nonzero acceleration magnitude c^2/ρ and $\rho = \infty$ at zero acceleration	35	10
55	Massive substantial points and physical bearers of interaction	4	11
56	Four-momentum mU , equation of motion $mA = K$, and four-force orthogonality for constant mass	36	11
57	Energy $E = mc^2 U_t = mc^2 dt/d\tau$	37	11
58	agreement of the low-velocity limit, after subtraction of the rest term, with Newtonian kinetic energy	38	11
59	Frame dependence of energy under transformation of four-momentum	39	11
60	selection of the ratio between the units of length and time	40	12

p	Claim	Depen- dency Key	Location Key
61	normalization of the numerical value of the vacuum speed of light to 1 by the selection of units	40	12
62	Representation of the temporal component by the imaginary coordinate $s = it$ after choosing units	41	12
63	Substantial points bearing charge and physical bearers of the electromagnetic field	4	13
64	Four-dimensional representation of the Liénard–Wiechert field using the retarded intersection, source world-line and charge	42	13
65	decomposition into electric and magnetic fields dependent on the selected time axis	43	13
66	four-dimensional form of mechanics	44	13
67	four-dimensional form of electromagnetism	45	13
68	bearer configuration of physical sources in the proposed theory of gravitation	4	14
69	Gravitational proposal with zero source acceleration and receiver motion described in the source rest frame	46	14
70	Keplerian motion in proper time with the additional factor $(dt_1/d\tau_1)^{-1}$, and the low-speed approximation	46	14
71	extraction of invariant relations by a four-dimensionally covariant form	47	15

Table 52: Correspondence Relations and Supporting Claims in Minkowski

Corr.	Content	Time Contents	Supporting Claims
1	connection between the four-dimensional coordinate representation and the designation of coordinate t as time	$\{3\}$	$\{9, 10\}$
2	connection between the Newton group and the time origin	$\{1\}$	$\{7\}$
3	connection between the perception of space and time and the combination of place and time	$\{2\}$	$\{11, 12\}$
4	connection between the four-dimensional description of the world and the world-point	$\{4\}$	$\{13, 14\}$
5	connection between the description of the world and substance at every place and time	$\{5\}$	$\{15, 16\}$
6	connection between the description of a world-line and the identity of a substantial point at different times	$\{6\}$	$\{17\}$

Corr.	Content	Time Contents	Supporting Claims
7	connection between world-line differentiation and the time element dt	{7}	{18}
8	connection between the world-line and its t -parameter representation	{8}	{19}
9	connection between the division of world-points and the sign of t	{10}	{23, 24, 38, 39}
10	connection between Galilean transformations and preserved time t	{9}	{21, 22}
11	Physical identification of the vacuum speed of light	{11}	{27, 28}
12	connection between Lorentz transformations and transformed time t'	{12}	{24, 30, 31}
13	connection between spatial projection and the simultaneity plane	{13}	{32, 33, 37}
14	connection between Lorentz transformations and local time	{15}	{34, 35}
15	connection between the classification of world-points and past and future light cones	{16}	{38, 39}
16	connection between causal classification and the ordering of timelike-separated world-points	{17}	{40, 42}
17	connection between causal classification and coordinate-dependent ordering of spacelike-separated world-points	{18}	{41, 43}
18	connection between world-line tangents and the timelike world-line	{14}	{45, 46}
19	invariant-line-element rule	{19}	{47}
20	world-line integration rule	{20, 21}	{48}
21	reparameterization rule	{22}	{49}
22	proper-time differentiation rule for four-velocity	{23}	{50}
23	proper-time differentiation rule for four-acceleration	{24}	[[51, 54]]
24	Multiplication by mass and projection onto the temporal component	{25, 26}	{55, 56}
25	energy-identification rule	{27}	{57, 58}
26	reference-system transformation rule	{28}	{59}
27	connection between the selection of units and the length–time unit ratio	{29}	[[60, 62]]
28	connection between the retarded electromagnetic field and past-light-cone dependence	{30}	{63, 64, 67}
29	connection between electric–magnetic field decomposition and time-axis dependence	{31}	{65}

Corr.	Content	Time Contents	Supporting Claims
30	connection between the proposed theory of gravitation and time-dependent motion and approximation	{32}	[[68, 70]]

The source locations of each correspondence relation are those of its listed time contents and supporting claims.

Dependency Inputs

Dependency Inputs The index sets in the table specify the supporting claims, temporal inputs and temporal outputs of the eight transfers.

Dependency and Transfer Elements

Definition 8.139 (Time-Content Transfers in Minkowski). Let $\text{Id}x_R^{\text{audit,transfer}} = [8]$. Realize each row's supporting claims, temporal inputs and temporal outputs by Definition 7.105. The source locations of a transfer are the union of the locations of its input and output time contents and supporting claims.

Table 53: Time-Content Transfers in Minkowski

j	Supporting-Claim Indices	Time-Input Indices	Time-Output Indices	Operation
1	{9, 10, 18, 24, 27, 28, 42} $\cup$ [[45, 47]]	{3, 7, 11, 14}	{19}	Evaluation of the invariant line element
2	{47, 48}	{19, 20}	{21}	Integration along the specified world-line segment
3	{45, 46, 48, 49}	{14, 20, 21}	{22}	Reparameterization by monotone proper time
4	{49, 50}	{22}	{23}	Proper-time differentiation of the coordinates
5	[[50, 54]]	{23}	{24}	Proper-time differentiation of the four-velocity function
6	{9, 10, 50, 55, 56}	{3, 23}	{25, 26}	Multiplication by mass and projection onto the temporal component
7	{27, 28} $\cup$ [[55, 58]]	{11, 26}	{27}	Identification of energy from the temporal component
8	{24, 30, 31, 55, 56, 59}	{12, 25}	{28}	Transformation of full four-momentum and projection onto the temporal component

Transfers and Outputs The eight transfers connect the proper-time element, proper-time integral, reparameterized coordinates, four-velocity, four-acceleration, four-momentum and its temporal component, energy, and energy in a specified frame.

Formation Conditions

Formation Conditions Use the source's coordinate order (x, y, z, t) , with vacuum light speed $c > 0$, and write the associated bilinear form as $g(V, W) = c^2 V_t W_t - V_x W_x - V_y W_y - V_z W_z$. Transfers 1–5 apply along a future-directed timelike world-line. Transfer 1 requires a curve with spatial and temporal derivatives; transfer 2, specified endpoints, orientation and path with an integrable line element; transfer 3, monotone proper time; transfer 4, differentiable coordinate functions; and transfer 5, a differentiable four-velocity function. For $U = d(x, y, z, t)/d\tau$ and $A = dU/d\tau$, one has $g(U, U) = c^2$ and $g(U, A) = 0$. If $A \neq 0$, then $g(A, A) < 0$; at $A = 0$, the curvature hyperbola degenerates to the tangent and $\rho = \infty$.

Transfers 6–8 use constant mass m and a specified coordinate system. Four-momentum is mU , and energy is $E = mc^2 U_t$. Transfer 8 takes the full four-momentum and an admissible transformation as inputs. For a transformation with speed b along the x -direction, $|b| < c$, one obtains $E' = (E - bmU_x)/\sqrt{1 - b^2/c^2}$.

All eight transfers have nonempty temporal inputs. There are no directed cycles among time contents or transfers with self-inputs.

Source Support

Source Support Table 49 gives the source locations of time contents; Table 51 gives supporting claims and their locations; and Table 52 gives the support for correspondence relations.

Results

Definition 8.140 (Results in the Minkowski 1908/1909 Reference). Define the result index set by $\text{Idx}_R^{\text{audit, result}} = [17]$. The correspondence between results and original-source locations is given in Table 54.

Table 54: Results and Source Locations in Minkowski

j	Result	Source Location
1	presentation of the relation between space alone and time alone and their four-dimensional combination	Introduction, p. 75
2	two invariance groups and the complete group of Newtonian mechanics	I, p. 75
3	finite-vacuum-light-speed group and Newtonian limit	I, pp. 76–79
4	four-coordinate representation of world-points	I, p. 76
5	representation of substantial points by world-lines	I, p. 76
6	proposal to express physical laws as relations among world-lines	I, p. 76
7	multiple temporal and spatial projections	I–II, pp. 78–81
8	separation of one four-dimensional world from coordinate projections	I–II, pp. 79–82
9	past and future light cones	III, p. 82

j	Result	Source Location
10	classification into timelike, spacelike, and lightlike directions and causal order	III, pp. 82–83
11	proper time as a world-line integral	III, p. 83
12	four-velocity and four-acceleration	III, p. 84
13	covariant mechanics in terms of four-force and four-momentum	IV, pp. 84–85
14	Energy as c^2 times the temporal component of four-momentum	IV, p. 85
15	four-dimensional representation of the electromagnetic field	V, pp. 86–87
16	decomposition into electric and magnetic fields dependent on the selected time axis	V, p. 87
17	proposed theory of gravitation with a delimited validity domain	V, p. 88

Reference-Specific Results The 17 results in Definition 8.140. The time-content inputs, supporting claims, and result outputs of each result are given by Table 55.

Result Dependence

Definition 8.141 (Result-Dependence Hyperedges in the Minkowski 1908/1909 Reference). For each result edge $j \in [17]$, realize the incidence from its listed temporal prerequisites and supporting claims to that result by Definition 7.105.

Table 55: Result-Dependency Hyperedges in Minkowski

j	Temporal Prerequisites	Supporting-Claim Indices
1	$\llbracket 2, 4 \rrbracket$	$\llbracket 1, 4 \rrbracket \cup \llbracket 9, 14 \rrbracket$
2	$\{1, 3, 9\}$	$\llbracket 5, 8 \rrbracket$
3	$\{3\} \cup \llbracket 9, 12 \rrbracket$	$\{28\} \cup \llbracket 21, 26 \rrbracket$
4	$\{3, 4\}$	$\{9, 10, 13, 14\}$
5	$\llbracket 5, 8 \rrbracket$	$\llbracket 15, 19 \rrbracket$
6	$\{8\}$	$\{20\}$
7	$\{3, 12, 13, 15\}$	$\{34, 35\} \cup \llbracket 29, 32 \rrbracket$
8	$\{4, 12, 13, 15\}$	$\{33, 36, 37\}$
9	$\{3, 10, 11, 16\}$	$\{23, 24, 28, 38, 39\}$
10	$\{10, 11\} \cup \llbracket 16, 18 \rrbracket$	$\llbracket 40, 44 \rrbracket$
11	$\{14\} \cup \llbracket 19, 21 \rrbracket$	$\llbracket 45, 48 \rrbracket$
12	$\llbracket 22, 24 \rrbracket$	$\llbracket 49, 54 \rrbracket$
13	$\{3\} \cup \llbracket 23, 26 \rrbracket$	$\{9, 10\} \cup \llbracket 50, 56 \rrbracket$
14	$\{11, 12\} \cup \llbracket 25, 28 \rrbracket$	$\{24, 27, 28, 30, 31\} \cup \llbracket 55, 59 \rrbracket$
15	$\{14, 23, 24, 30\}$	$\{45, 46, 63, 64, 67\} \cup \llbracket 50, 54 \rrbracket$
16	$\{31\}$	$\{65\}$
17	$\{32\}$	$\llbracket 68, 70 \rrbracket$

Temporal prerequisites occur 59 times and supporting claims 93 times across the 17 results. An object used by multiple results is counted once for each result.

Grounds of Establishment

Grounds of Establishment of Results The evidence types directly supplied by this reference are definitions, assumptions, group-theoretic and geometrical derivations, four-dimensional reformulations of existing mechanics and electromagnetism, and a proposed theory of gravitation with a delimited domain of application.

Evidence Hierarchy

Objects of Direct Measurement This reference itself reports no new objects of direct measurement.

Predicates of Direct Measurement This reference itself reports no new predicates of direct measurement.

Conditions and Domain of Application of Direct Measurement The paper refers to an experimental-physics basis, physically identifies the vacuum speed of light, and cites Michelson's experiment. These are not new measured values reported in this paper.

Objects of Formal Generalization Results 1–10 apply to Newtonian mechanics, Lorentz transformations, the four-dimensional world, the light cone, and causal classification. Results 11–14 apply to proper time on timelike world-lines and covariant mechanics, and results 15–16 apply to the four-dimensional electromagnetic field. Result 17 belongs to the proposed theory of gravitation under the limiting conditions stated in the original source.

Conditions and Domain of Application of Formal Generalization The Newtonian group expresses form invariance of the mechanical equations considered in the source. The Lorentz group and temporal-order preservation in results 9–10 concern $c > 0$ and transformations preserving temporal orientation. The Newtonian limit takes $c \rightarrow \infty$ at fixed physical relative velocity. Results 11–14 obey the curve, differentiation, mass and coordinate conditions stated above. In result 14, subtracting mc^2 and taking the low-speed limit yields Newtonian kinetic energy. Result 15 uses source charge, world-line, velocity and acceleration, together with receiver motion, on the domain where the retarded intersection exists. The field decomposition in result 16 follows the selected time axis. Result 17 sets source acceleration to zero, uses the source rest frame, and lets the receiver move solely under the source force. Including the source's additional factor $(dt_1/d\tau_1)^{-1}$ gives Keplerian laws in proper time τ_1 , approximating coordinate-time motion at low speed.

On p. 86, the source asserts that an unbounded world-line with timelike tangents intersects the past light cone of any evaluation point exactly once. Timelikeness implies at most one intersection; existence requires a further condition. For example, with $c = 1$, the curve $(x, y, z, t) = (\sqrt{1 + t^2}, 0, 0, t)$, $t \in \mathbb{R}$, satisfies $1 - (dx/dt)^2 = 1/(1 + t^2) > 0$, yet $x^2 - t^2 = 1$, so it never intersects the origin's light cone. The retarded-field construction applies where the retarded intersection exists and its tangent, normal and $r > 0$ are defined.

Generation Judgment Judgment values: LCTR-complete-generation judgment: 0; reference-internal evidence-conditional local-reach judgment: 0; remaining primitive time roots: 32.

Generation Reach and Primitive Time Roots No time content is generated by the generation reach whose initial set consists solely of non-temporal physical inputs. All 32 time contents in Definition 8.137 remain primitive time roots.

Reference-Internal Evidence-Conditional Local Reach The reference-internal evidence-conditional local-reach judgment is 0, and 32 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ in Definition 7.108 and apply Definition 7.50 to it. Summing over all results determines the primitive-time-root–result dependence relation and the necessary-primitive-time-root–result dependence relation.

Result Dependence All 17 results in Definition 8.140 belong to the primitive-time-root-dependent results. There are no generated results, evidence-reach-only results, or results unreached after root addition. The minimal primitive-time-root support of each result is given by Proposition 8.143.

Evidence Effect

Evidence-Effect Domain The formal evidence in this reference directly supports 17 results concerning the Newton and Lorentz groups, the four-dimensional world, the light cone, the directional classification of world-points, proper time, covariant mechanics, the four-dimensional electromagnetic field, and the limited proposed theory of gravitation. The possibility of sending and receiving light supports the physical interpretation of the past and future light cones.

Evidence-Effect Boundary The establishment of the four-dimensionally covariant form and the division of world-points by the light cone does not imply that coordinate time, proper time, the light-speed ratio, or temporal order has been generated from non-temporal physical inputs. The unification of mathematical form, the reference to an empirical basis, and the possibility of sending and receiving light likewise do not retroactively confer physical-time qualification on the 32 time contents. These transitions lie beyond the evidence-effect boundary under Definition 7.115.

Result-Preservation Domain

Result-Preservation Domain The 17 results are preserved insofar as each time-content prerequisite is supplied by the corresponding generated time representation in the LCTR generation sequence, while retaining the original-source inputs and dependency directions comprising world-points, world-lines, emission, propagation, reception, the quadratic form, transformations preserving temporal orientation, bodies, constant mass, force, charge, current, and electromagnetic fields. The light cone and causal classification are preserved as relations among world-points and the possibility of sending and receiving light signals on generated time representations.

Proper time and its derivatives obey the specified curve, segment and requisite differentiability. Retarded fields apply where the retarded intersection exists. The domain of application of the limited proposed theory of gravitation is determined by the zero source acceleration, additional factor, and low-speed approximation required in the original source.

Placement of Grounding Obligations

Placement of Grounding Obligations The source places world-lines in a four-dimensional world supplied with coordinate time and temporal orientation, and defines proper time by integrating the world-line element. It specifies conditions on timelikeness, differentiability, transformation groups and mechanics, but uses the 32 time contents without generating them from non-temporal physical inputs. The LCTR classification of the placement of the physical-time grounding obligation is therefore “backgrounded by primitive-time inputs.”

Evidence of Actual Invalidation

Evidence of Actual Invalidation The complete judgment and remaining-root judgment concern physical-time qualification and are distinct from the actual invalidation of a specific condition that determines a result’s value or domain of application. The original-source formal evidence and LCTR audit evidence for this reference do not establish the actual invalidation of such a target condition. The actual-invalidation judgment is “indeterminate from original-source evidence”; actual invalidation is not inferred from the backgrounding of the grounding obligation alone.

Downstream Reach

Downstream Reach The reach to primitive-time-root-dependent results is determined for this reference by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because the actual invalidation of a specific condition has not been established, the judgment of actual reach from the invalidation position to a result is “indeterminate from original-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair Scope The repair targets based on structural reach are the corresponding time content and the generation or establishment conditions required before its use. The recalculation scope is limited to results for which actual invalidation is independently established and which are dependency-reachable from that invalidation position. Because actual invalidation is indeterminate in this reference, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from the repaired inputs, or their establishment domain is restricted.

Audit Conclusion

Proposition 8.142 (Primitive Time Roots and Result Dependence in the Minkowski 1908/1909 Reference). In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepldx}(32, 17)$. Furthermore, $R \in \text{Ref}^{\text{audit, rootdep}}$ holds.

Proof. Take the eight claims in the first entry of the claim-audit boundary as the generation-initial set. Every transfer has a nonempty temporal input, so this initial set yields no time content. Adding all 71 claims as the evidence environment leaves the temporal prerequisites unsatisfied unless time contents are supplied as seeds. Complete generation and source-evidence-conditioned local reach are therefore both zero, and all 32 time contents remain primitive-time roots. Every result edge has nonempty temporal prerequisites; adding all 32 roots to the evidence environment reaches all 17 results. Definition 8.115 gives the displayed partition. Membership in the reference sets follows from Definition 7.52. $\square$

Proposition 8.143 (Minimal Primitive-Time-Root Support in the Minkowski 1908/1909 Reference). For each $q \in \text{Idx}_R^{\text{audit, result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108, and the minimum and maximum cardinalities of those elements, are given in Table 56.

Table 56: Minimal Primitive-Time-Root Supports in Minkowski

Result Index	Support Statistics	Result Index	Support Statistics
1	(1; 3)	2	(1; 3)
3	(1; 5)	4	(1; 2)
5	(1; 4)	6	(1; 1)
7	(1; 4)	8	(1; 4)
9	(1; 4)	10	(1; 5)
11	(2; 3, 5)	12	(4; 1, 5)
13	(5; 2, 5)	14	(6; 4, 6)
15	(5; 3, 6)	16	(1; 1)
17	(1; 1)		

Proof. Apply Definition 7.46 in the order of the eight transfers. After the first scan, the second scan leaves every time-content support antichain unchanged. Taking unions of one support set for each temporal prerequisite of a result and retaining inclusion-minimal members gives the displayed families. Each member reaches the corresponding result as a seed; removing any one root prevents that reach. There are 34 support-family members, 82 primitive-time-root/result dependence pairs, and 44 essential primitive-time-root/result dependence pairs. $\square$

Individual Statements Statement Evidence 1: Space, Time, and Their Four-Dimensional Combination

Original (German):

Von Stund' an sollen Raum für sich und Zeit für sich völlig zu Schatten herabsinken und nur noch eine Art Union der beiden soll Selbständigkeit bewahren.

English translation:

Henceforth space by itself and time by itself are to recede completely into shadows, and only a kind of union of the two is to preserve independence.

Original-Source Location Minkowski, *Raum und Zeit, Jahresbericht der Deutschen Mathematiker-Vereinigung* 18, p. 75, opening of the lecture.

Minimal Context The sentence stating the subject of the lecture, immediately after the announcement that it would present a view of space and time arising from a basis in experimental physics.

Attributing Source Hermann Minkowski, in his 1908 lecture.

Statement Evidence 2: Definition of the World

Original (German):

Die Mannigfaltigkeit aller denkbaren Wertsysteme x, y, z, t soll die Welt heißen.

English translation:

The manifold of all conceivable systems of values x, y, z, t shall be called the world.

Original-Source Location Same source, p. 76, Section I.

Minimal Context The passage that defines the totality as the world immediately after calling a system of four coordinate values comprising a spatial point and a time point a world-point.

Attributing Source Hermann Minkowski, in his 1908 lecture.

Statement Evidence 3: Physical Laws as Relations among World-Lines

Original (German):

Die ganze Welt erscheint aufgelöst in solche Weltlinien, und ich möchte sogleich vorwegnehmen, daß meiner Meinung nach die physikalischen Gesetze ihren vollkommensten Ausdruck als Wechselbeziehungen unter diesen Weltlinien finden dürften.

English translation:

The entire world appears resolved into such world-lines, and I wish to state at once that, in my view, physical laws may find their most complete expression as interrelations among these world-lines.

Original-Source Location Same source, p. 76, Section I.

Minimal Context The passage that states the form in which physical laws are to be expressed, immediately after describing the persistence of a substantial point as a world-line in the four-dimensional world.

Attributing Source Hermann Minkowski, in his 1908 lecture.

Statement Evidence 4: Propagation Speed of Light in Vacuum

Original (German):

Für c wird die Fortpflanzungsgeschwindigkeit des Lichtes im leeren Raume eintreten.

English translation:

The propagation speed of light in vacuum will be identified with c .

Original-Source Location Same source, p. 78, Section I.

Minimal Context The passage concerning the transformation group G_c that states which physical quantity is ultimately to correspond to its finite constant c .

Attributing Source Hermann Minkowski, in his 1908 lecture.

Statement Evidence 5: One World and Multiple Spatial Projections

Original (German):

Hiernach würden wir dann in der Welt nicht mehr den Raum, sondern unendlich viele Räume haben, analog wie es im dreidimensionalen Räume unendlich viele Ebenen gibt. Die dreidimensionale Geometrie wird ein Kapitel der vierdimensionalen Physik. Sie erkennen, weshalb ich am Eingange sagte, Raum und Zeit sollen zu Schatten herabsinken und nur eine Welt an sich bestehen.

English translation:

On this view, we would then have in the world no longer one space, but infinitely many spaces, just as there are infinitely many planes in three-dimensional space. Three-dimensional geometry becomes a chapter of four-dimensional physics. Here lies the reason why I said at the outset that space and time are to recede into shadows and that only one world in itself is to persist.

Original-Source Location Same source, p. 79, end of Section I.

Minimal Context The concluding passage of the discussion that introduces different time axes and the corresponding projections into three-dimensional spaces.

Attributing Source Hermann Minkowski, in his 1908 lecture.

Statement Evidence 6: Past and Future Light Cones through the Sending and Receiving of Light

Original (German):

Der erste, der *Vorkegel von O*, besteht, sagen wir, aus allen Weltpunkten, die „Licht nach O senden“, der zweite, der *Nachkegel von O*, aus allen Weltpunkten, die „Licht von O empfangen“.

English translation:

The first, the past light cone of O , consists, let us say, of all world-points that “send light to O ,” and the second, the future light cone of O , of all world-points that “receive light from O .”

Original-Source Location Same source, p. 82, Section III.

Minimal Context The passage that distinguishes the two components by the possibility of sending and receiving light, immediately after dividing the cone of the quadratic form with vertex at world-point O into two components.

Attributing Source Hermann Minkowski, in his 1908 lecture.

Common Evidence with Domain of Application

Target State- ment	Statement Object	Predicate	Explicit Domain of Application
S04- Q001	space itself, time itself, and their combination	space and time each recede into shadows, and only their combination preserves independence	statement of the theme of the view of space and time arising from a basis in experimental physics at the opening of the lecture
S04- Q002	the manifold of all conceivable systems of values x, y, z, t	is called the world	domain of the definition of the world in Section I, after a system of four coordinate values has been designated a world-point
S04- Q003	the entire world, world-lines, and physical laws	the entire world appears resolved into world-lines, and, in the author’s view, physical laws may find their most complete expression as interrelations among world-lines	domain in Section I that describes the persistence of a substantial point as a world-line in the four-dimensional world
S04- Q004	the constant c of the transformation group G_c	the propagation speed of light in vacuum corresponds to it	domain in Section I that associates a physical quantity with the finite constant c of the transformation group G_c
S04- Q005	one world, multiple spaces, and three-dimensional geometry	the world contains infinitely many spaces, three-dimensional geometry becomes a chapter of four-dimensional physics, and only the world in itself persists	domain at the end of Section I that describes different time axes and the corresponding projections into three-dimensional spaces

S04- Q006	the past and future light cones of world-point O	the past light cone consists of world-points that send light to O , and the future light cone consists of world-points that receive light from O	domain in Section III that divides the cone of the quadratic form with vertex at world-point O into two components by the possibility of sending and receiving light
--------------	--	--	---

Individual Audit

Interpretive Possibilities The first statement permits both a mathematical claim that unifies spatial and temporal coordinates within four-dimensional geometry and an ontological claim that rejects the separate independence of space and time and attributes independence to their combination. The second statement gives a mathematical convention defining the totality of quadruples of coordinate values as the world. The third statement gives a methodological proposal to express physical laws as interrelations among world-lines. The fourth statement gives a physical identification associating the vacuum propagation speed of light with the constant c of the transformation group. The fifth statement permits both a geometrical interpretation allowing multiple time axes and spatial projections for one four-dimensional world and an ontological interpretation attributing independence to the world. The sixth statement gives an operational and geometrical relation that distinguishes the past and future light cones by the possibility of sending and receiving light signals.

Independent Grounds The first and fifth statements present Minkowski's view of space and time; the second defines the world, and the third proposes a world-line-based method. The fourth identifies the transformation-group constant with vacuum light speed; the sixth associates the two light-cone components with the sending and receiving of light signals.

Evidence-Effect Boundary The first statement directly supports the change in attribution from the separate independence of space and time to their four-dimensional combination; the second directly supports the definition of the world as the totality of world-points; and the third directly supports the proposal to express physical laws as relations among world-lines. The fourth statement directly supports the correspondence between c and the vacuum propagation speed of light, but does not support the physical generation of the time content contained in the speed representation. The fifth statement directly supports the relation between multiple spatial projections and one four-dimensional world. The scope of the sixth statement is the correspondence between the two light-cone components and the possibility of sending and receiving light. None of the six statements directly supports either the generation, solely from non-temporal physical inputs, of coordinate t , the temporal component of the speed of light, world-line order, light-cone order, or proper time, or their physical-time qualification.

Corresponding Formal Evidence The first statement corresponds to claims 1–4 and result 1; the second, to time content 4, claims 13–14 and result 4; the third, to time contents 5–8, claims 15–20 and results 5–6; the fourth, to time content 11, claims 23–28 and result 3; the fifth, to time contents 12–13 and 15, claims 29–37 and results

7–8; and the sixth, to time content 16, claims 38–39 and result 9. Time-content, claim and result indices are specified by Definitions 8.137, 8.138 and 8.140, respectively.

Common Audit with Domain of Application

Within-Reference Comparison The first, second, and fifth statements integrate space and time into a four-dimensional world; the third places physical laws as relations among world-lines; and the fourth and sixth connect light propagation to the transformation constant and the distinction between light-cone components. These statements constitute a single geometrical system, but that system does not eliminate the separate obligation to generate physically the time coordinate and order used within it.

8.19.5 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 5.

Source Name Primitive-Time-Root Audit of Einstein 1916

Bibliographic Identification Let $R = \text{Ref}_{\text{Ein16GR}}$ denote A. Einstein, *Die Grundlage der allgemeinen Relativitätstheorie*, *Annalen der Physik*, Vierte Folge, Band 49, 769–822 (1916). The audit uses the original German publication of 1916, and its domain comprises the Introduction; Part A, §§1–4; Part B, §§5–12; Part C, §§13–18; Part D, §§19–20; Part E, §§21–22; the original notes; the closing passage; and the date of receipt.

Original-Source Location Identifier 5-L1.

Original-Source Location The Introduction; Part A, §§1–4; Part B, §§5–12; Part C, §§13–18; Part D, §§19–20; Part E, §§21–22; the original notes; the closing passage; and the date of receipt. The locations of the respective time contents and input claims are given in Table 58 and Table 62.

The correspondence between the original-source location keys in Table 62 and the original-source locations is given in Table 61.

Audit-Scope Identifier 5-SC1.

Audit Scope The audit domain stated under Bibliographic Identification.

Formal Objects

Object Identifier 5-F1.

Object Name Formal object.

Object Type Formal object.

Formal Content Definitions, assumptions, derivations, and predictions involving time dependence in generally covariant coordinates, the metric, the line element, measurement by local clocks and rods, geodesics, curvature, gravitational field equations, coupling to matter and electromagnetic fields, conservation relations, weak-field, low-velocity, and quasi-static approximations, clock rates in static fields, gravitational redshift, light deflection, and the advance of Mercury’s perihelion.

Original-Source Location Identifier Set 5-L1.

The auxiliary indices in Table 58 refer to the objects in Table 59.

The dependence-object keys in Table 62 refer to pairs of time-content indices and auxiliary-object indices according to Table 60. Ranges include both endpoints.

Assumptions

Assumption Identifier 5-A1.

Assumption Content The individual contents of the reference assumptions are given in Definition 8.145.

Original-Source Location Identifier Set 5-L1.

Non-Temporal Physical Inputs

Input Identifier 5-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content Bodies, rods, clocks and oscillators, physical bearers used in emission, propagation and reception, physical bearers of point events, freely falling bodies, matter, fluids, charge, current, electromagnetic fields, gravitational sources and observational apparatus constitute the 22 generation-initial input components.

Original-Source Location Identifier Set 5-L1.

Time Content

Time-Content Identifier 5-T1.

Time-Content Name Time content.

Time-Content Type Time content.

Time-Content Specification Clock indications and clock rates, simultaneity, time coordinates, the light-speed ratio, temporal classification of the line element, proper time, time differentiation, four-dimensional regions, transport per unit time, orbital periods, and quantities per century belong to time contents or dependences that use time contents.

Original-Source Location Identifier Set 5-L1.

Time-Content Identifier 5-T2.

Time-Content Name Time contents used in the original source.

Time-Content Type Time content.

Time-Content Specification The 93 time contents in Definition 8.144.

Original-Source Location Identifier Set 5-L1.

Formal Input Elements

Formal-Component Identifier 5-C1.

Formal-Component Name Physical and formal input-claim components.

Formal-Component Type Formal component.

Formal-Component Text The audit inputs used by the original source are the 83 type-homogeneous components in Definition 8.145.

Original-Source Support Identifier Set 5-L1.

Formal-Component Identifier 5-C2.

Formal-Component Name Formal input elements.

Formal-Component Type Formal component.

Formal-Component Text All 83 components are included in the reference’s formal-evidence environment.

Original-Source Support Identifier Set 5-L1.

Definition 8.144 (Time Contents in the Einstein 1916 Reference). For $R = \text{Ref}_{\text{Ein16GR}}$, let $n_R = 93$. Table 58 lists time contents, auxiliary objects and source locations in 58 source-location groups. Time contents in each row correspond to that row’s indices in ascending order.

Table 58: Time Contents and Source Locations in Einstein 1916

Group	i	Time Contents (Index Order)	Auxiliary Indices	Location
1	1	time coordinate	1–2	Introduction
2	2	uniform translation	3	§1
3	3–4	light propagation; constant propagation rate	none	§1
4	5–6	simultaneity; relativity of simultaneity	none	§1
5	7–8	Behavior of moving rigid bodies; Behavior of moving clocks	none	§1
6	9–11	first clock-hand position; second clock-hand position; fixed time interval	none	§1
7	none	none	4–5	§1
8	12–13	relative rotation; constant angular velocity	6	§2
9	14	Persistent absence of internal relative motion in the two bodies	38	§2
10	15–16	uniform acceleration; accelerated coordinate system	none	§2
11	17	motion of a mass	7–8	§2
12	18–20	light trajectory; propagation speed; accelerated-coordinate description	none	§2
13	21–23	point-event; Local clock value $x_4 = t$ assigned to the event; Period count of the local unit clock	none	§3

Group	i	Time Contents (Index Order)	Auxiliary Indices	Location
14	24	Local spacetime coincidence and its ascertainability	none	§3 original note
15	25–27	rotating system; Persistent coincidence of the origins and axes of the two coordinate systems; spacetime measurement	9–11	§3
16	28–29	peripheral-clock indication; Observation of peripheral-clock indications received by light	12–14	§3
17	30–31	Comparison of rotating-frame clocks using inertial-frame time; slower clock rate	none	§3
18	32	coordinate speed of light	15–16	§3
19	33–34	Position-dependent clock rate that the time definition must admit; Requirement for a time definition admitting position-dependent clock rates	17	§3
20	35–36	coordinate-time difference; Rule for directly measuring coordinate-time differences with standard clocks	none	§3
21	37	simultaneous coincidence	39	§3
22	38–39	point-event; Arbitrary four-coordinate assignment representing spacetime coincidences of events	18	§3
23	none	none	19–21	§4
24	40	Locally measured time coordinate X_4	none	§4
25	41–42	local unit of time; measured vacuum speed of light	22	§4 original note
26	43–44	Invariant line element ds corresponding to local spacetime measurement; spacetime measurement	23	§4
27	45–47	Sign of ds^2 for a specified displacement; timelike classification; spacelike classification	none	§4
28	48–49	metric coefficients; dependence on the four coordinates	none	§4
29	50–52	first occurrence of uniform rectilinear motion; local special-relativistic coordinates; local special-relativistic system	24–25	§4
30	53–54	identical motion; curvilinear nonuniform representation	26	§4

Group	i	Time Contents (Index Order)	Auxiliary Indices	Location
31	55	Four-coordinate domain including temporal dependence of physical fields	none	§§5–12
32	56–58	Coordinate four-volume element before multiplication by the metric density; Invariant four-volume element corresponding to local spacetime measurement; measurement	none	§8
33	none	none	27, 40–42	§9
34	none	none	43–44	§9
35	none	none	45–46	§10
36	59	Constant-metric coordinates representing special-relativistic spacetime on the source-specified domain	28, 47	§12
37	60–62	Parameterized material-point motion in the gravitational field; second occurrence of uniform rectilinear motion; Line-element parameter s used for material-point motion	none	§13
38	63	Four-dimensional domain including temporal dependence of gravitational and material fields	none	§§14–16
39	64	Bounded spacetime domain used in physical-field variation	none	§15
40	none	none	29	§§15–17
41	65–66	Unit of time used for material momentum and energy transport; Material momentum and energy transport per unit volume and time	30	§18
42	67–69	fluid motion; dx_μ/ds along the fluid worldline; line-element parameter s along the fluid worldline	none	§19
43	70	Differentiation with respect to the time coordinate in the Maxwell equations	none	§20
44	71–72	unit of time; Electromagnetic momentum and energy transport per unit volume and time	31	§20
45	73–74	particle velocity; differentiation with respect to the time coordinate	32	§21
46	75–76	low-velocity condition; comparison with the propagation speed of light	none	§21

Group	i	Time Contents (Index Order)	Auxiliary Indices	Location
47	77–78	$ds = dx_4$ at the retained order of the weak-field, low-velocity approximation; Time-coordinate notation $x_4 = t$	33	§21
48	79	neglect of time derivatives	none	§21
49	80	selected unit of time	34	§21
50	81–83	local time; General-coordinate differences corresponding to local measurement; Metric relation connecting local length and time measurements with coordinate differences	35	§22
51	none	none	36, 48	§22
52	84–85	Coordinate-time width corresponding to one period of a stationary unit clock; one unit-clock period	none	§22
53	86	reduced clock rate	37	§22
54	87	redshift	none	§22
55	88	Null condition $ds^2 = 0$ imposed on light propagation	none	§22
56	89	coordinate speed of light	none	§22
57	90–91	planetary orbital period; seconds scale	none	§22
58	92–93	Perihelion-precession rate expressed per century; Conversion from orbital period in seconds to a value per century	none	§22

Location groups in which time content is a transfer output are given in the transfer table below. The time contents in the other location groups appear as prior uses of time within the original source.

Table 59: Auxiliary Objects in Einstein 1916

a	Auxiliary Object	Group
1	spatial coordinates	1
2	formal equivalence	1
3	reference system	2
4	location-independence condition	7
5	epoch-independence condition	7
6	pair of bodies	8
7	composition-independence condition	11
8	state-independence condition	11
9	identity of origins	15

a	Auxiliary Object	Group
10	identity of axes	15
11	application to the rotating system	15
12	central placement	16
13	peripheral placement	16
14	central observer	16
15	light path	18
16	Interpretive condition that light speed on the considered path is not explicitly time-dependent	18
17	location	19
18	assignment of four coordinates	22
19	infinitesimal region	23
20	local coordinate system	23
21	local elimination of gravitation	23
22	unit calibration	25
23	local rod	26
24	free mass	29
25	free body	29
26	arbitrary coordinate transformation	30
27	stationarity condition	33
28	finite region	36
29	three-volume	40
30	unit of volume	41
31	unit of volume	44
32	spatial differentiation	45
33	weak field	47
34	comparison with Newtonian theory	49
35	local length	50
36	static gravitational field	51
37	location in the vicinity of a mass	53
38	Constant spatial distance between the two bodies	9
39	Spatial co-location condition for observation	21
40	Fixed geometric endpoints P_1, P_2	33
41	A curve joining fixed endpoints and its variation family	33
42	$\int ds$ along the specified curve	33
43	Arc parameter s on a geodesic with nonzero line element	34
44	Geodesic with nonzero line element supplied for parameterization	34
45	Specified curve used for covariant differentiation	35
46	Curve parameter s used for covariant differentiation	35
47	Vanishing-curvature condition on the source-specified domain	36
48	Constant spatial coordinates of the clock in the static field	51

Definition 8.145 (Audit-Input Claim Components in the Einstein 1916 Reference).

$$\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(83).$$

$$\text{ClaimBoundaryIdx}_R = (\{5, 6, 8, 11, 13, 16, 19, 21, 24, 29, 34, 48, 49, 54, 59, 61, 69, 71, 74, 76, 80, 82\}, [83]).$$

The 22 claims in the first entry are the non-temporal generation-initial inputs; all 83 claims constitute the source formal-evidence environment. The time-content dependencies of the 22 initial claims are empty, so **StrictClaimBoundary**(R) holds. Table 62 lists each claim, its dependency key and its source-location key.

Table 60: Input-Claim Dependency Keys in Einstein 1916

Key	Time-Content Indices	Auxiliary-Object Indices
1	1	1–2
2	2	3
3	3–4	none
4	1–8	3
5	none	none
6	7–11	4–5, 23
7	12–14	6, 38
8	1–2, 15–16	3
9	15–17	7–8
10	17	7–8
11	18, 20	3, 15
12	19–20	3, 15
13	21–23, 27	1, 22–23
14	25–26	9–11
15	25	11, 23
16	25, 28–31	9–10, 12–16
17	31–33	16–17
18	33–34	16–17
19	25, 31, 33–36	17, 23
20	24, 37	39
21	38–39	18
22	38–39	18, 26
23	40	19–21
24	40–42	22–23
25	40, 43–44, 48–49	18–20, 22–23
26	43, 45–46	none
27	43, 45, 47	none
28	40, 43, 48–49	18–20
29	48–54	24–26
30	39, 48–49, 55	18, 26
31	48–49, 56–58	22–23
32	48–49, 56–57	26
33	43, 48–49	27, 40–42
34	43, 48–49	40–41, 43–44
35	48–49, 55	26, 45–46
36	48–49, 55, 59	26, 28, 47
37	59–62	24–26, 28, 43, 47
38	48–49, 63	26

Key	Time-Content Indices	Auxiliary-Object Indices
39	48–49, 63–64	27
40	63, 65–66	29–30
41	48–49, 67–69	20, 31
42	48–49, 55, 70	26, 31
43	48–49, 70–72	31
44	48–49	33
45	73–76, 78	32–33
46	48–49, 78–79	32–33
47	48–49, 73, 75, 77–80	32–34
48	48–49	33, 36–37
49	48–49, 81–83	35–37
50	81–85	36–37, 48
51	84–87	15, 36–37, 48
52	43, 48–49, 88	15, 36
53	48–49, 88–89	15, 36
54	48–49, 88–89	15, 36–37
55	60, 62, 90–93	33, 36

In the following lookup table, ℓ denotes the table-local original-source-location key.

Table 61: Source-Location Keys in Einstein 1916

Key	Source Location	Key	Source Location
1	Introduction, 769	13	§13,801–802
2	§1,770	14	§14,802–804
3	§2,771–773	15	§15,804–806
4	§3,773–777	16	§16,806–808
5	§3,775–776	17	§§16–18,806–811
6	§3–4,776–779	18	§18,810–811
7	§4,777–779	19	§19,811–812
8	§§5–12,779–801	20	§20,812–815
9	§8,788–789	21	§21,816–818
10	§9,790–792	22	§22,819–820
11	§10,792–796	23	§22,819–822
12	§12,798–801		

Table 62: Physical and Formal Input Claims in Einstein 1916

p	Claim	Depen- dency Key	Location Key
1	Minkowski-type four-dimensional coordinate representation	1	1
2	special principle of relativity	2	2

p	Claim	Depen- dency Key	Location Key
3	vacuum-light-speed assumption	3	2
4	Relativity of simultaneity, Lorentz transformation, and laws for moving rigid bodies and clocks based on the special-relativity principle and light-speed postulate	4	2
5	physical-bearer configuration of rigid bodies	5	2
6	physical-bearer configuration of clocks and oscillators	5	2
7	Measurement interpretation of rigid bodies and clocks, and place- and epoch-related assumptions for lengths and clock intervals	6	2
8	two physical bodies used in rotational comparison	5	3
9	Mach-type comparison of rotating bodies	7	3
10	requirement of general relativity	8	3
11	physical system bearing the gravitational field	5	3
12	locally equivalent descriptions of an accelerated system and a gravitational field	9	3
13	freely falling body and material composition	5	3
14	free-fall motion	10	3
15	independence of free fall from material composition	10	3
16	physical-bearer relation comprising the emission, propagation, and reception of light signals	5	3
17	light path in a noninertial system	11	3
18	coordinate speed of light in a noninertial system	12	3
19	physical-bearer configuration of rods, clocks, and observational apparatus	5	4
20	Assignment of coordinate time to an event by counting periods of a local clock	13	4
21	physical-bearer configuration of a rotating disk and measuring rods	5	4
22	rotational motion of the disk	14	4
23	non-Euclidean measurement of the rotating disk	15	4
24	physical-bearer configuration of the central and peripheral clocks on the rotating disk	5	4
25	rate comparison between the central and peripheral clocks	16	4
26	Position-dependent clock rates required in the rotating frame	17	4
27	Requirement for a time definition admitting position-dependent clock rates	18	4
28	Failure of direct measurement of coordinate differences by unit rods and standard clocks in general	19	5
29	Physical bearers of point events	5	4
30	observational description by point-coincidences	20	4
31	arbitrary assignment of four coordinates	21	4
32	requirement of general covariance	22	6
33	local special-relativistic assumption in an infinitesimal region	23	7

p	Claim	Depen- dency Key	Location Key
34	physical-bearer configuration of light signals, clocks, and rods used for local calibration	5	7
35	Choice of local time unit making the measured vacuum speed of light unity	24	7
36	invariant line element	25	7
37	Timelike classification by the sign of the line element of a specified displacement	26	7
38	Spacelike classification by the sign of the line element of a specified displacement	27	7
39	metric coefficients	28	7
40	gravitational-potential role of the metric coefficients	29	7
41	Tensor-transformation calculus applied to physical fields on spacetime	30	8
42	natural four-volume element	31	9
43	coordinate specialization $\sqrt{-g} = 1$	32	9
44	Stationarity of the spacetime line-element integral for specified endpoints and curve family	33	10
45	Geodesic equation of the spacetime metric and parameterization for nonzero line element	34	10
46	Application of covariant differentiation to physical fields on spacetime	35	11
47	Riemann–Christoffel curvature of the spacetime metric and conditions for constant-metric coordinates on the source-specified domain	36	12
48	physical-bearer configuration of a material point	5	13
49	physical system bearing the gravitational field	5	13
50	Coordinate description of local free motion and the geodesic-motion assumption for a general gravitational field	37	13
51	gravitational field equations in a matter-free region	38	14
52	Hamiltonian form of the gravitational equations	39	15
53	gravitational-energy components	38	15
54	physical-bearer configuration of matter	5	16
55	matter energy tensor	40	16
56	matter-coupled field equations	38	16
57	total energy–momentum conservation relation	40	17
58	matter energy–momentum balance	40	18
59	physical configuration of a perfect fluid	5	19
60	Perfect-fluid energy tensor and motion equations using fluid worldlines and the line-element parameter	41	19
61	physical configuration of charge, current, and electromagnetic fields	5	20
62	generally covariant Maxwell branch	42	20
63	electromagnetic-energy components	43	20
64	weak-field approximation	44	21
65	low-velocity approximation	45	21

p	Claim	Depen- dency Key	Location Key
66	Quasi-static approximation omitting time derivatives of the field	46	21
67	Newtonian gravitational potential	47	21
68	Poisson limit	47	21
69	physical-bearer configuration of a point-mass source	5	21
70	First-order metric of a static point-mass field	48	22
71	physical-bearer configuration of radial and tangential rods and observational apparatus	5	22
72	coordinate-length relation for a radial rod	49	22
73	coordinate-length relation for a tangential rod	49	22
74	physical-bearer configuration of a unit clock placed in a static field	5	23
75	Relation between the proper period and coordinate-time width of a stationary unit clock	50	23
76	physical-bearer configuration of the emitter, receiver, and spectroscopic observational apparatus	5	23
77	prediction of gravitational spectral redshift	51	23
78	light propagation following a null line element	52	23
79	Ray curvature obtained from the null condition and light-speed distribution in a static field	53	23
80	physical-bearer configuration of the Sun as gravitational source and of the light-detection apparatus	5	23
81	prediction of solar light deflection	54	23
82	physical-bearer configuration of the Sun–Mercury system and observational apparatus	5	23
83	Per-orbit perihelion advance from the higher-order approximation and conversion to a value per century	55	23

Dependence Inputs

Dependence Inputs The time-content inputs, auxiliary-object supports, and formal-evidence support components assigned to the 15 transfers in Definition 8.146.

Dependence and Transfer Elements

Definition 8.146 (Time-Content Transfers in the Einstein 1916 Reference). Define the transfer index set by $\text{Idx}_R^{\text{audit,transfer}} = [15]$. For each transfer, define its time-content inputs, time-content outputs, auxiliary-object supports, formal-evidence support components, and transfer relation by Table 63. The original-source location of each transfer is the union of the original-source locations corresponding to its time-content inputs, time-content outputs, auxiliary-object supports, and formal-evidence support components. In the following table, i , a , and p denote time-content, auxiliary-object, and formal-evidence-component indices, respectively.

Table 63: Time-Content Transfers in Einstein 1916

j	Input i	Output i	a	p	Transfer Relation
1	11, 25–30	31	9–10, 12–15	2, 6–7, 21–22, 24–25	Rate comparison of central and peripheral clocks
2	40–41, 44	43	1, 19–20, 22–23	19, 33–34, 36	Evaluation of the line element from local clocks and rods
3	43	45–47	none	36–38	Classification by the sign of the line element of a specified displacement
4	50–53	54	24–26	31–33, 40–41	Description of the same motion in arbitrary coordinates
5	48–49, 56, 58	57	22–23	33–34, 41–42	Conversion of coordinate four-volume using the metric density
6	39, 48–49	59	26, 28, 47	41, 47	Constant-metric coordinates on a region satisfying the curvature condition
7	67, 69	68	none	59–60	Differentiation of the fluid worldline with respect to its line-element parameter
8	60, 74, 78	73	32	48, 50, 65	Coordinate-time velocity of the specified motion
9	48–49, 62, 73–76, 78	77	33	64–65	Relation between line-element parameter and coordinate time in the weak-field, low-velocity approximation
10	48–49, 81–82	83	35	33, 36, 70, 72–73	Metric relation between local measurements and general-coordinate differences

j	Input i	Output i	a	p	Transfer Relation
11	48–49, 81, 83, 85	84	36–37, 48	69–70, 74–75	Coordinate-time width of a stationary unit clock
12	84–85	86	36–37, 48	69–70, 74–75	Clock-rate comparison in a static point-mass field
13	85–86	87	15, 36–37	16, 74–77	Redshift from clock-rate and spectral comparison
14	48–49, 82, 88	89	15, 33, 36–37	70, 78–79	Coordinate light speed from the null condition in a static field
15	90–91, 93	92	none	82–83	Conversion to perihelion advance per century using the orbital period

The 58 location groups in Table 58 associate time contents and auxiliary objects within the source. The 15 transfers producing time contents are given in Table 63.

Transfers and Outputs The 15 transfers concern clock rates, line elements, coordinate descriptions, four-volume, differentiation, static-field clock periods, redshift, coordinate light speed and time-scale conversion. General geometric variation, parameterization and covariant differentiation support the physical results through the formal input claims in Definition 8.145.

Establishment Conditions

Establishment Conditions Each transfer presupposes the tabulated temporal inputs, auxiliary objects and formal evidence, and holds under the following conditions.

1. Use identically constructed central and peripheral clocks and the specified uniform rotation; compare their rates against inertial-frame time.
2. Evaluate the line element from local special relativity, chosen units, rod and clock measurements, and local-coordinate differences.
3. Evaluate the line element for the specified displacement and classify it as timelike or spacelike by the sign of its square.
4. Apply a differentiable invertible coordinate transformation to the specified parameterized motion to obtain another description of the same physical motion.
5. Use the coordinate four-volume element and metric determinant to obtain invariant four-volume corresponding to local rod and clock units.

6. Impose the curvature and integrability conditions on the source-specified region to obtain constant-metric coordinates there.
7. Differentiate the specified differentiable timelike fluid worldline with respect to its line-element parameter s .
8. Differentiate the spatial coordinates of the specified particle motion with respect to coordinate time; the corresponding time increment is nonzero on the velocity domain.
9. Use $ds = dx_4 = dt$ at the retained weak-field, low-material-particle-velocity order, with the time coordinate written as $x_4 = t$.
10. Relate the local length and time measurements, general-coordinate differences and metric.
11. For a clock at fixed spatial coordinates in a static field, obtain the positive coordinate-period width from its given proper period and $g_{44} > 0$.
12. Specify reference-clock normalization, comparison locations and the point-mass field, and compare coordinate periods of unit clocks.
13. Use the same local emission process, light propagation and receiving spectral reference to apply the clock-rate relation to spectral comparison.
14. Obtain coordinate light speed using a static weak point-mass field, null propagation and the specified ray direction and geometry.
15. Obtain the value per century from the per-orbit angular advance in the source higher-order approximation, the orbital period in seconds and the second-to-century conversion.

There are no transfers with an empty time-content input set. There are no directed cyclic components and no self-input transfers. The time-content input of each of the 15 transfers is nonempty.

Original-Source Support

Original-Source Support The formal-evidence support of each transfer is given by the formal-evidence support component numbers in the same table. The original-source locations of each time content, auxiliary object, and formal-evidence support component are given by Table 58, Table 59, and Table 62.

Results

Definition 8.147 (Results in the Einstein 1916 Reference). Define the result index set by $\text{ldx}_R^{\text{audit,result}} = [17]$. The correspondence between results and original-source locations is given in Table 64.

Table 64: Results and Source Locations in Einstein 1916

j	Result	Source Location
1	General limit of directly measuring coordinate differences with clocks and rods	§3,774–776
2	Position dependence of clock rates in the rotating frame	§3,774–775
3	Reduction of observational descriptions to spacetime point coincidences	§3,774–777
4	Arbitrary four coordinates representing event coincidences and the general-covariance requirement	§3–4,776–779
5	Local special-relativity assumption in an infinitesimal region	§4,777–778
6	Choice of local time unit using the vacuum speed of light	§4 note,777
7	Invariant line element corresponding to local clock and rod measurements	§4,777–779
8	Coordinate transformation of the same motion and the metric form in arbitrary coordinates	§4,778–779
9	Representation of the gravitational field by metric coefficients	§4,779; §13,801–802
10	Covariant geodesic equation for material-point motion	§§9–13,790–802
11	Vacuum and matter-coupled field equations and conservation relations	§§14–18,802–811
12	Generally covariant forms of the fluid and Maxwell branches	§§18–20,810–815
13	Reduction to Newtonian theory in the weak-field, low-velocity, quasi-static limit	§21,816–818
14	Coordinate period and rate of a unit clock in a static field	§22,819–820
15	Gravitational spectral redshift in a static field	§22,820 and note
16	Light deflection in a static point-mass field	§22,821–822
17	Mercury perihelion advance and its value per century	§22,822

Reference-Specific Results The 17 results in Definition 8.147. The original-source location of each result is given in Table 64, and its time-content prerequisites, auxiliary-object prerequisites, and formal-evidence support components are given in Table 65.

Result Dependence

Definition 8.148 (Result-Dependence Hyperedges in the Einstein 1916 Reference). For each $j \in [17]$, define the time-content prerequisites, auxiliary-object prerequisites, formal-evidence support components, and result output of the result-dependence hyperedge $\text{ResultEdge}_{R,j}$ by Table 65.

Table 65: Result-Dependency Hyperedges in Einstein 1916

j	Temporal Prerequisites	Auxiliary Prerequisites	Formal-Evidence Support	Result Output Index
1	25, 31, 33–36	16–17, 23	19, 21–28	1
2	11, 25–33	9–10, 12–17	2, 6–7, 21–22, 24–26	2
3	21–22, 24, 37	39	19–20, 29–30	3
4	38–39	18, 26	29–32	4
5	40–42, 44	19–23	19, 33–35	5
6	40–42	22–23	16, 33–35	6
7	40–41, 43–44, 48–49	1, 18–20, 22–23	19, 33–34, 36–39	7
8	39, 43, 48–54	18, 24–26	31–33, 36, 39–41	8
9	48–54, 60–62	24–26, 28	11, 31–33, 39–40, 48–50	9
10	43, 48–49, 59–62	24–28, 40–47	33, 41, 44–50	10
11	48–49, 55, 63–66	26–27, 29–30	41, 49, 51–58	11
12	48–49, 65–72	20, 26, 31	54–55, 58–63	12
13	48–49, 60, 62, 73–80	32–34	48–50, 64–68	13
14	48–49, 81–86	35–37, 48	33, 36, 69–75	14
15	84–87	15, 36–37, 48	16, 69–70, 74–77	15
16	48–49, 82, 88–89	15, 33, 36–37	16, 69–70, 78–81	16
17	60, 62, 73–74, 78, 90–93	33, 36	48, 50, 69–70, 82–83	17

Across the 17 results there are 116 temporal-prerequisite occurrences and 72 auxiliary-prerequisite occurrences, for 188 semantic-prerequisite occurrences. The same object is counted once for each result in which it is a prerequisite.

Grounds of Establishment

Grounds of Establishment of Results The evidence types directly supplied by this reference are definitions, principles and assumptions, formal derivations by tensor analysis, approximate calculations, and predictions comparable with observation.

Evidence Hierarchy

Objects of Direct Measurement This reference itself newly obtains no objects of direct measurement.

Predicates of Direct Measurement This reference itself newly obtains no predicates of direct measurement.

Conditions and Domain of Application of Direct Measurement The empirical connections of the clocks, redshift, light deflection, and the advance of Mercury’s

perihelion treated by the original source belong to comparisons between theoretical formulae and existing or future observations. They do not belong to direct measurement data obtained by apparatus of this reference itself.

Objects of Formal Generalization Results 1–9 concern clocks, coordinates, local measurement, the metric and gravitation; result 10, material-point geodesic motion; result 11, field equations and conservation; and result 12, fluids and electromagnetic fields. Result 13 concerns the Newtonian limit; 14, static-field clocks; 15, redshift; 16, light deflection; and 17, perihelion advance.

Conditions and Domain of Application of Formal Generalization The extension of result 10 to a general gravitational field includes the physical assumption in source §13. Result 13 uses weak-field, low-material-particle-velocity and quasi-static approximations. Results 14–15 use stationary clocks and their comparison in a static field; result 16 uses null rays in a static weak field; and result 17 uses the source higher-order approximation, orbital period and time-scale conversion.

Generation Judgment Judgment values: LCTR-complete-generation judgment: 0; reference-internal evidence-conditional local-reach judgment: 0; remaining primitive time roots: 93.

Generation Reach and Primitive Time Roots Generation reach from non-temporal physical inputs alone yields zero time contents. All 93 time contents in Definition 8.144 remain primitive-time roots. Each transfer presupposes time contents; local reach with the source formal-evidence environment also yields zero time contents unless time contents are supplied as seeds.

Reference-Internal Evidence-Conditional Local Reach The reference-internal evidence-conditional local-reach judgment is 0, and 93 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ in Definition 7.108 and apply Definition 7.50 to it. Summing over all results determines the primitive-time-root–result dependence relation and the necessary-primitive-time-root–result dependence relation.

Result Dependence All 17 results in Definition 8.147 belong to the primitive-time-root-dependent results. There are no generated results, evidence-reach-only results, or results unreached after root addition. The minimal primitive-time-root support of each result is given by Proposition 8.150.

Evidence Effect

Evidence-Effect Domain The formal evidence in this reference directly supports 17 results concerning generally covariant coordinates, the metric, the line element, geodesics, curvature, gravitational field equations, coupling to matter and electromagnetic fields, conservation relations, weak-field approximations, clock rates, gravitational redshift, light deflection, and the advance of Mercury’s perihelion.

Evidence-Effect Boundary The establishment of the generally covariant form, local special relativity, metric clock-rate relations, weak-field approximations, and predictions comparable with observation does not imply that clock indications, coordinate time, proper time, the light-speed ratio, or time variables within laws have been generated from non-temporal physical inputs. Agreement between prediction and observation likewise does not retroactively confer physical-time qualification on the 93 time contents. These transitions lie beyond the evidence-effect boundary under Definition 7.115.

Result-Preservation Domain

Result-Preservation Domain The preservation domain of the 17 results under replacement of temporal prerequisites requires corresponding generated time representations together with the source physical bearers, motion, local measurements, metric, physical assumptions and approximation orders. Clock periods require the rest condition; redshift requires emission, propagation and spectral comparison; light deflection requires null rays; and perihelion advance requires the higher-order approximation, orbital period and scale conversion. The source equations relating the point-mass metric parameter to mass have the coefficient discrepancy stated below.

Point-Mass Coefficient Relations

For a point mass M in source equation(68a), §21, with the potential normalized to zero at infinity,

$$\Phi = \frac{g_{44} - 1}{2} = -\frac{\varkappa M}{8\pi r}.$$

Combining this with $g_{44} = 1 - \alpha/r$ in equation(70), §22, gives

$$\alpha = \frac{\varkappa M}{4\pi}.$$

Source equations(70a) and(74), respectively, give

$$\alpha = \frac{\varkappa M}{8\pi}, \quad B = \frac{2\alpha}{\Delta} = \frac{\varkappa M}{4\pi\Delta}$$

The α and mass expressions in equation(74) agree through equation(70a), but for $\varkappa M \neq 0$ their mass-conversion coefficient disagrees with equations(68a) and(70).

Placement of the Grounding Obligation

Placement of the Grounding Obligation Source §3 examines the introduction of time by clocks and its limits, and requires a time definition admitting position-dependent clock rates. From §4 onward, temporal content enters local measurement, line elements, fields and motion. The 93 time contents are used before generation from non-temporal physical inputs, so LCTR classifies the physical-time grounding obligation as “backgrounded by primitive-time inputs.”

Evidence of Actual Invalidation

Evidence of Actual Invalidation The complete judgment and remaining-root judgment concern physical-time qualification and are distinct from the actual invalidation of a specific condition that determines a result's value or domain of application. The original-source formal evidence and LCTR audit evidence for this reference do not establish the actual invalidation of such a target condition. The actual-invalidation judgment is “indeterminate from original-source evidence”; actual invalidation is not inferred from the backgrounding of the grounding obligation alone.

Downstream Reach

Downstream Reach The reach to primitive-time-root-dependent results is determined for this reference by the tabulated primitive-time-root–result dependence relations and minimal primitive-time-root supports. This is structural reach. Because the actual invalidation of a specific condition has not been established, the judgment of actual reach from the invalidation position to a result is “indeterminate from original-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair Scope The repair targets based on structural reach are the corresponding time content and the generation or establishment conditions required before its use. The recalculation scope is limited to results for which actual invalidation is independently established and which are dependency-reachable from that invalidation position. Because actual invalidation is indeterminate in this reference, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from the repaired inputs, or their establishment domain is restricted.

Audit Conclusion

Proposition 8.149 (Primitive Time Roots and Result Dependence in the Einstein 1916 Reference).

In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepIdx}(93, 17)$. Furthermore, $R \in \text{Ref}^{\text{audit, rootdep}}$ holds.

Proof. The generation-initial set contains 22 claims with empty temporal dependencies. All 15 transfers have nonempty temporal inputs, so this set yields no time content. Adding all 83 claims as the source formal-evidence environment leaves the temporal prerequisites unsatisfied unless time contents are supplied as seeds. Complete generation and source-evidence-conditioned local reach are both zero, and all 93 time contents remain primitive-time roots. All 17 result edges also have nonempty temporal prerequisites. With the tabulated auxiliary objects and source conditions satisfied, adding all 93 roots to the source formal-evidence environment reaches all 17 results. Definition 8.115 gives the displayed partition. $\square$

Proposition 8.150 (Minimal Primitive-Time-Root Support in the Einstein 1916 Reference). For each $q \in \text{Idx}_R^{\text{audit, result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108, and the minimum and maximum cardinalities of those elements, are given in Table 66.

Table 66: Minimal Primitive-Time-Root Supports in Einstein 1916

Result Index	Support Statistics	Result Index	Support Statistics
1	(2; 6, 11)	2	(1; 9)
3	(1; 4)	4	(1; 2)
5	(1; 4)	6	(1; 3)
7	(1; 5)	8	(2; 8, 10)
9	(1; 9)	10	(4; 7, 9)
11	(1; 7)	12	(1; 9)
13	(1; 10)	14	(1; 5)
15	(3; 2, 5)	16	(1; 4)
17	(1; 7)		

Proof. Apply Definition 7.46 in the order of the 15 transfers. After the first scan, the second leaves all time-content support antichains unchanged. For each result, choose one support from each temporal prerequisite's family, take their union, and retain the inclusion-minimal unions to obtain the tabulated families. Each member reaches the corresponding result as a seed, while removing any root prevents that reach. There are 24 support-family members, 119 primitive-root/result dependence pairs and 96 essential primitive-root/result dependence pairs. $\square$

Individual Statements

Statement Objects The postulate of the constancy of the vacuum speed of light in special relativity; modification of the same principle in an accelerated-coordinate description; the limit on introducing time using clocks of identical construction at rest in a rotating coordinate system; comparison of the clock rates at the center and circumference of a disk; and the relation between the gravitational constant and the field coupling under a selected unit of time.

Original (German) das Postulat von der Konstanz der Vakuum-Lichtgeschwindigkeit

Ebenso sieht man unmittelbar, daß das Prinzip von der Konstanz der Vakuum-Lichtgeschwindigkeit eine Modifikation erfahren muß.

Ebensowenig kann man in K' eine den physikalischen Bedürfnissen entsprechende Zeit einführen, welche durch relativ zu K' ruhende, gleich beschaffene Uhren angezeigt wird.

die auf der Kreisperipherie angeordnete Uhr langsamer

während Newtons Theorie bei der von uns gewählten Zeiteinheit

$$-\frac{K}{c^2} \int \frac{\varrho d\tau}{r}$$

ergibt, wobei K die gewöhnlich als Gravitationskonstante bezeichnete Konstante $6,7 \cdot 10^{-8}$ bedeutet. Durch Vergleich ergibt sich

$$\varkappa = \frac{8\pi K}{c^2} = 1,87 \cdot 10^{-27}.$$

English Translation the postulate of the constancy of the vacuum speed of light.

Likewise, one immediately sees that the principle of the constancy of the vacuum speed of light must undergo a modification.

Nor can one introduce in K' a time corresponding to physical requirements that is indicated by clocks of identical construction at rest relative to K' .

the clock arranged on the circumference of the circle runs more slowly.

whereas Newton's theory, with the unit of time chosen by us, gives

$$-\frac{K}{c^2} \int \frac{\varrho d\tau}{r}$$

where K denotes the constant $6,7 \cdot 10^{-8}$, usually called the gravitational constant. Comparison gives

$$\varkappa = \frac{8\pi K}{c^2} = 1,87 \cdot 10^{-27}$$

Original-Source Location In Einstein, *Die Grundlage der allgemeinen Relativitätstheorie*,

Annalen der Physik, Vierte Folge, Band 49, the first quotation is on p. 770, §1; the second on p. 773, §2; the third and fourth on p. 775, §3; and the fifth in equations (68a)–(69) on p. 818, §21.

Minimal Context Section 1 of the original source presents the constancy of the vacuum speed of light as a postulate distinguishing special relativity from classical mechanics, and connects it to the relativity of simultaneity, the Lorentz transformation, and the laws governing moving rigid bodies and clocks. Section 2 states that the same principle requires modification in an accelerated coordinate system and that a light path is generally curved. Section 3 places clocks of identical construction at the center and on the circumference of a disk at rest in the rotating coordinate system, and compares the clock indications received at the center. The attribution of the difference to clock rates is conditional on not taking the light speed on the path considered to be explicitly time-dependent. A time meeting physical requirements cannot be introduced through the indications of identically constructed clocks at rest in the rotating frame; a definition of time allowing their rates to depend on position is required. Section 21 explicitly gives the selected unit of time used in comparison with the Newtonian limit, and connects the original notation K to the field coupling $\varkappa$ in equation (69).

Attributing Source The five statements are attributable to §§1–3 and 21 of Einstein's original German paper of 1916.

Predicates The first quotation states the constancy of the vacuum speed of light as a postulate of special relativity. The second states that the same principle requires modification in an accelerated-coordinate description. The third states that one cannot introduce a time corresponding to physical requirements that is indicated by clocks of identical construction at rest in the rotating coordinate system. The fourth assigns to the clock placed on the circumference a lower rate of progress in comparison with the clock at the center. Under the selected unit of time, the fifth quotation denotes the ordinary gravitational constant by the original notation K and gives its relation to the field coupling κ .

Explicit Domains of Application The domain of the first quotation is special relativity in §1 of the original source; the domain of the second is the accelerated-coordinate description in §2. The domain of the third and fourth quotations is the rotating coordinate system in §3, the clocks of identical construction at the center and on the circumference of the disk, and the comparison of clock indications received at the center. The domain of the fifth quotation is the weak-field, low-velocity, and quasi-static approximation in §21, together with the comparison with Newtonian theory using the unit of time selected there.

Individual Audit

Interpretive Possibilities The first and second quotations distinguish the special-relativistic vacuum-light-speed postulate from coordinate light speed that varies in accelerated-coordinate descriptions. The third and fourth concern indications and rates of identically constructed clocks at rest in the rotating frame. The central observer's attribution to clock rate is conditional on not treating light speed on the considered path as explicitly time-dependent. The fifth connects the physical role of gravitational coupling to its numerical representation in the selected time unit. Using these clock indications in a construction of time requires concrete recording, comparison and transport relations, together with loop consistency and gluing conditions over any multiple paths or regions involved.

Independent Grounds The first and second statements rest on the special-relativistic postulate and accelerated-coordinate relations; the third and fourth, on the rotating-frame clock comparison; and the fifth, on coefficient comparison in the Newtonian limit. These are theoretical statements in the source, distinct in evidence type from the connection to spectral observations discussed in §22.

Evidence-Effect Boundary The first and second quotations concern the light-speed postulate and coordinate descriptions; the third and fourth, clock indications and rates; and the fifth, time units and coupling constants. The time variable in a speed representation, clock indications and the time unit are prerequisites of these relations. Attribution of the physical responses of clocks or oscillators to property changes of time itself independently of those bearers, and physical-time qualification of these temporal inputs, lie beyond their evidence-effect boundary under Definition 7.115.

Corresponding Formal Evidence The first quotation corresponds to time contents 3–4 and claim 3; the second, to time contents 18–20 and claims 17–18; the third and fourth, to time contents 25–36, claims 21–28 and results 1–2; and the fifth, to time

contents 73–80, claims 64–68 and result 13. Indices are specified by Definitions 8.144, 8.145 and 8.147.

Within-Reference Comparison Speed representations use a time coordinate; the rotating-frame clock-rate comparison uses clock indications and inertial-frame time; and the numerical representation of gravitational coupling uses a selected time unit. The distinct time contents and their dependencies appear in the transfer and result-dependency tables for the corresponding source operations.

8.19.6 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 6.

Source Name Primitive-Time-Root Audit of Bridgman 1927

Bibliographic Identification Let $R = \text{Ref}_{\text{Bri27LM}}$ denote Percy W. Bridgman, *The Logic of Modern Physics*, The Macmillan Company (1927). The audit uses the sixth printing published by The Macmillan Company (1951), and its domain comprises pp.4–9, pp.68–100, pp.158–162, p.202, pp.216–218, and pp.225–226 of the original source.

Original-Source Location Identifier 6-L1.

Original-Source Location Pages 4–9, 68–100, 158–162, 202, 216–218, and 225–226 of the original source. The locations of the respective time contents and input claims are given in Table 67 and Table 70.

The correspondence between the original-source location keys in Table 70 and the original-source locations is as follows.

- 1 pp.4–9
- 2 pp.7–9
- 3 pp.68–79,202
- 4 pp.69–70
- 5 pp.70–73
- 6 pp.72–73,78
- 7 pp.72–73
- 8 pp.73–74
- 9 pp.74–77,202
- 10 pp.73–77,216–218
- 11 pp.75–79,202
- 12 pp.80–91,158–160,216–218

- 13 pp.97–100
- 14 pp.160–162
- 15 pp.225–226
- 16 pp.4–9,68–100,158–162,202,216–218,225–226
- 17 p.79
- 18 pp.216–218
- 19 pp.99–100
- 20 pp.91–97

Audit-Scope Identifier 6-SC1.

Audit Scope The audit domain stated under Bibliographic Identification.

Formal Objects

Object Identifier 6-F1.

Object Name Formal object.

Object Type Formal object.

Formal Content Definitions, assumptions, and analyses of time dependence in the correspondence between operations and physical concepts, the critique of absolute time, local and extended time, the light-clock unit, interval counting, clock calibration, remote-clock synchronization, causal order, velocity measurement, the operational domain of application, and the quantum-domain boundary.

Original-Source Location Identifier Set 6-L1.

The auxiliary objects treated separately from the time contents are as follows.

The auxiliary objects comprise light speed, clock-mediated light measurement, spatial measurement, optical and electrical operations, distance, the criticized Newtonian description, operational domains, and observational agreement. Input claims supply the physical carriers of clocks, atoms and signals; the indications, periods and temporal propagation roles they carry are distinguished as time contents.

Assumptions

Assumption Identifier 6-A1.

Assumption Content The individual contents of the reference assumptions are given in Definition 8.152.

Original-Source Location Identifier Set 6-L1.

Non-Temporal Physical Inputs

Input Identifier 6-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content The physical carriers of clocks and oscillators, light sources, signals, emission, propagation, reception, records, observers, spatial arrangements, distance comparisons, and the physical materials of experimental operations are defined as 13 generation-initial input components.

Original-Source Location Identifier Set 6-L1.

Time Content

Time-Content Identifier 6-T1.

Time-Content Name Time content.

Time-Content Type Time content.

Time-Content Specification The order in which operations are performed, clock indications, units of time, intervals, velocity, simultaneity, synchronization, local time, extended time, past and future, causal order, and the time variables of equations belong to time contents or dependences that use time contents.

Original-Source Location Identifier Set 6-L1.

Time-Content Identifier 6-T2.

Time-Content Name Time contents used in the original source.

Time-Content Type Time content.

Time-Content Specification The 57 time contents in Definition 8.151.

Original-Source Location Identifier Set 6-L1.

The roles of the time contents are assigned by the following correspondence.

Formal Input Elements

Formal-Component Identifier 6-C1.

Formal-Component Name Physical and formal input-claim components.

Formal-Component Type Formal component.

Formal-Component Text The audit inputs used by the original source are the 52 type-homogeneous components in Definition 8.152.

Original-Source Support Identifier Set 6-L1.

Formal-Component Identifier 6-C2.

Formal-Component Name Formal input elements.

Formal-Component Type Formal component.

Formal-Component Text All 52 components are included in the reference's formal-evidence environment.

Original-Source Support Identifier Set 6-L1.

Definition 8.151 (Time Contents in the Bridgman 1927 Reference). For reference $R = \text{Ref}_{\text{Bri27LM}}$, let $n_R = 57$. The correspondence between time contents and original-source locations is given in Table 67. The indices and contents of the eight auxiliary objects are given in Table 69.

Table 67: Time Contents and Original-Source Locations
in the Bridgman 1927 Reference

i	Time content	Original-source location
1	Occurrence times of events and arrival times of signals	pp.5–9,75,85–88,95–97,160–161
2	Operationally relative measurement and concept of time	p.6
3	Order and continuation of measurement and observation operations	pp.5–9,79,92,99
4	Before-and-after relation	pp.7–8
5	Relation of simultaneity	pp.7–8, pp.76–77
6	Observer-relative time	pp.8–9
7	Local time	pp.68–69, p.73, p.75, pp.75–76, p.77
8	Extended time	pp.68–69, p.73, p.77, pp.98–100
9	Propagation of light and signals	pp.69–70, pp.73–74, p.75, pp.160–161
10	Round-trip time interval	pp.69–70, pp.161–162
11	Light-clock unit of time	pp.69–70
12	Identity of equal intervals	p.70
13	Interval-count record	p.70
14	Time value derived from counting	p.70
15	Temporal course of transporting a moving clock	pp.70–71,98–99
16	Acceleration history of a moving clock	pp.70–71, pp.98–100
17	Velocity defined by distance and a time interval	pp.70,74,97–98
18	Time interval for measuring velocity	p.70, pp.97–98
19	Oscillation and rotation periods of candidate mechanical clocks	pp.70–71
20	Law time variable and use of clock laws	p.71
21	Uncorrected clock time	pp.71–72
22	Indications of local, emission, and remote clocks	pp.71–75,85–88,97–98,160–162
23	Iteration order in the proposed clock-correction procedure	p.71
24	Time indications of the postulated clock	pp.71–72
25	Temporal component of an infinitesimal interval	pp.72–73
26	Correspondence between physical-clock measurement and an infinitesimal temporal component	pp.72–73
27	Period of the oscillatory atomic process	pp.72–73

<i>i</i>	Time content	Original-source location
28	Assumed correspondence of atomic oscillation to clock indications	pp.72–73
29	Prescribed one-second interval	pp.73–74
30	Record of signal arrival	pp.73–74, p.74
31	Rates of a master clock and remote clocks	pp.73–74
32	Remote synchronization and a system of synchronized clocks	pp.73–74, 216–218
33	Rate adjustment of remote clocks	pp.73–74
34	Signal travel-time interval	p.74
35	Remote-clock zero setting using a known speed of light	p.74
36	Time-coordinate convention used for extended time	pp.73–74, 76–77
37	Agreement of clock readings for nearby events communicated by light, acoustic, or tactual signals	p.75
38	Present-time correspondence assigned to spatially separated observations	pp.76–77, 87–88, 158–160, 216–218
39	Age value derived from local time	pp.76–77
40	Past and past history	pp.77–78, pp.80–81, pp.158–160
41	Numerical input representing a short time interval	p.78
42	Short-time value given meaning through theoretical equations	p.78
43	Time coordinate entering the equations of electrodynamics	p.78
44	Irrevocability of the past implicit in the performed operations	p.79
45	Forward time direction specified by operations	p.79
46	Temporal extent of the history included in initial conditions	pp.80–83, 158–159
47	Starting time of a repeated experiment	p.81
48	Future course associated with initial conditions or a present state	pp.80–83, 87–89, 158–160, 216–218
49	Temporal order imposed on cause, effect, and a return signal	pp.85–87
50	Arbitrary clock-zero settings and the emission-zero convention	pp.85–88, 160–161
51	Temporal relation of the causal cone with the added light-signal bound	pp.86–88
52	Clock readings at two positions	pp.97–98
53	Velocity value using a local clock on the moving object	pp.98–100
54	Unit velocity defined by comparison with a travelling disturbance	p.99

i	Time content	Original-source location
55	Velocity value obtained by physical composition of unit velocities	pp.99–100
56	Alternative remote-clock setting to zero on reception	pp.160–161
57	Ordinary time operations	pp.225–226

Table 68: Temporal Roles in Bridgman 1927

Role	Corresponding time-content indices
Direct Time-Bearing Content	1–15, 18–30, 32, 34–52, 56–57
Time-Containing Physical Quantity	16–17, 31, 33, 53–55
Temporal Operation	2–3, 9, 15, 20, 23, 26, 28, 30, 32–33, 35, 37, 42–43, 45, 49–51, 54–57

Table 69: Auxiliary objects in Bridgman 1927

a	Auxiliary object	Type	Related time content i
1	Speed of light	Time-containing physical quantity	17
2	Clock-mediated light measurement	Measurement operation	17, 24
3	Spatial measurement operation	Spatial measurement operation	none
4	Optical and electrical operations	Measurement and comparison operation	41, 43
5	Distance used in clock, synchronization, and velocity measurements	Spatial physical quantity	none
6	Newtonian absolute time and uniform flow as objects of criticism	Statement under criticism	none
7	Feasibility and scale-dependent domain of temporal operations	Application condition	none
8	Agreement of optical and electrodynamic calculations with observations	Observational correspondence condition	none

Distance enters the light path, clock construction, synchronization and velocity measurement (pp.69–70,74–75,97–99). Newtonian absolute time is an object of criticism, distinct from dependence on the starting time of a repeated experiment.

Definition 8.152 (Audit-Input Claim Components in the Bridgman 1927 Reference). Define the set of physical- and formal-input claim components by $\text{Node}_R^{\text{audit,claim}} =$

$\text{ClaimNodes}_R(52)$. The respective components, corresponding original-source statement-group keys, associated time contents, and original-source locations are given in Table 70. Define the generation-initial input index set and the reference formal-evidence-environment index set by

$$\text{ClaimBoundaryIdx}_R = \left(\{1, 5, 10, 13, 16, 18, 21, 24, 26, 30, 34, 40, 43\}, [52] \right)$$

respectively. Hence $\text{StrictClaimBoundary}(R)$ holds. The associated time-content index sets of the 13 generation-initial input components are all empty sets.

Table 70: Physical- and Formal-Input Claim Components
in the Bridgman 1927 Reference

p	Original group	Type-homogeneous component	Associated time-content indices	Original-source location key
1	1	Physical materials and apparatus used in experimental operations	none	1
2	1	Operational definition relating a time concept to actual measurement operations	2–3	1
3	2	Difference between time concepts associated with different sets of measurement operations	2–3	1
4	3	Critique of Newtonian absolute time based on relative measurement operations	2–3	1
5	4	Physical-carrier configuration of the observer and recording apparatus	none	2
6	4	Observer-relative before, after, and simultaneity relations used in relativity	1, 4–6	2
7	5	Local time measured by a clock for nearby events	1, 7, 22, 37	3
8	5	Extended time using synchronization and settings of remote clocks	8, 32–33, 35–36	3
9	5	Distinction between local measurement and time measurement extended over space	7–8, 32, 36	3

p	Original group	Type-homogeneous component	Associated time-content indices	Original-source location key
10	6	Physical-carrier configuration of the light source, reflector, light receiver, clock, and recording medium	none	4
11	6	Light clock defining one round trip as its time unit	9–11	4
12	6	Counting and recording of equal intervals	11–14	4
13	7	Physical-carrier configuration of clocks and oscillators	none	5
14	7	Mechanical-clock periods and laws of mechanics investigated with clocks	19–20, 22	5
15	7	Interdependence of clock measurement and inference of mechanical laws	15–16, 19–23	5
16	8	Physical-carrier configuration of the clocks and recording media used in clock calibration	none	5
17	8	Proposal to iterate clock correction using approximate laws and measurements with an uncorrected clock	3, 15–16, 19–23	5
18	9	Physical-carrier configuration of physical clocks	none	5
19	9	Clock postulated to measure specified properties of light	17, 24	5
20	9	Correspondence problem connecting a postulated clock, the temporal line-element component, and physical clocks	17, 22, 24–26	5
21	10	Physical apparatus used to measure time intervals	none	6
22	10	Theoretical use of an infinitesimal temporal line-element component and a short-time value	25, 41–43	6

p	Original group	Type-homogeneous component	Associated time-content indices	Original-source location key
23	10	Measurement of an infinitesimal temporal component and observational correspondence of short-time values	25–26, 41–43	6
24	11	Physical system carrying an oscillatory atomic process	none	7
25	11	Assumption of using atomic oscillation as a clock and the problem of its criterion	27–28	7
26	12	Physical-carrier configuration of remote clocks, signals, and recording media	none	8
27	12	Remote-clock rate adjustment using one-second signal-arrival intervals	9, 29–31, 33	8
28	12	Remote-clock zero setting according to a known light speed and distance	17, 22, 30, 33–35	8
29	13	Agreement of nearby signal readings, the analytical status of local time, and its limits at large separations or high velocities	1, 7, 9, 22, 37–39	9
30	14	Physical configuration of spatial arrangements, distance comparisons, and measuring apparatus	none	10
31	14	Extended time using spatial measurement, local clocks, and synchronization conventions	7–8, 22, 32–33, 35–36	10
32	15	Change in the time concept accompanying changed operations for short-time measurement	7, 41–43	11
33	15	Operation relating short-time values to optical and electrodynamic equations and observations	41–43	11

p	Original group	Type-homogeneous component	Associated time-content indices	Original-source location key
34	16	Physical-carrier relation comprising the emission, propagation, and reception of a signal	none	12
35	16	Causal relation including the system background, history, and possibility of intervention	1, 4, 40, 46–47, 49	12
36	16	Past history included in initial conditions and the conditions of repeated experiments	40, 46–48	12
37	16	Examination of future determination conditional on a state description, laws, and history	17, 20, 38, 40, 46–48	12
38	16	Temporal relation of cause and effect using clock settings and signals	1, 4, 9, 22, 49–50	12
39	16	Causal cone under the additional assumption that signals cannot propagate faster than light	1, 4, 9, 17, 22, 32, 51	12
40	17	Physical carriers comprising a moving body, clocks, distance apparatus, and taut strings with supporting boards	none	13
41	17	Velocity as distance divided by the interval obtained from clock readings at two positions	8, 17–18, 52	13
42	17	Alternative velocity definition using a local clock on the moving object and road distance	7, 15, 18, 22, 53	13
43	18	Physical-carrier configuration of the emitter, reflector, light receiver, and clock	none	14
44	18	Convention setting a remote clock to zero on reception of a signal emitted at the source-clock zero	1, 9, 22, 50, 56	14

p	Original group	Type-homogeneous component	Associated time-content indices	Original-source location key
45	18	Round-trip time and reversal of motion measured with a single local clock	1, 9–10, 22	14
46	19	Inapplicability of ordinary spacetime concepts if their measurement operations cannot be performed in the quantum domain	57	15
47	20	Empirical meaning of time concepts grounded in experimental operations	2–3, 57	16
48	20	Revisability of time concepts as the domain of experience expands	2–3, 7–8, 41, 43, 57	16
49	21	Operational definition of forward as the direction in which time flows	3–4, 44–45	17
50	22	Observations at synchronized clock zeros and the definition of a present state extended over space	1, 17, 22, 32, 38	18
51	23	Direct velocity measurement by running together, overtaking, and physically composing unit velocities	1, 3–5, 54–55	19
52	24	Scale dependence of object identity through continuous observation and event identity through spacetime boundaries	1, 3, 40	20

Dependent Inputs

Dependent Inputs Time contents in Definition 8.151, including operational order, signal arrival, clock indications, units of time, intervals, simultaneity, synchronization, velocity, and causal order, are input into 20 transfer events.

Dependence and Transfer Elements

Definition 8.153 (Time-Content Transfers in the Bridgman 1927 Reference). Define the transfer index set by $\text{Idx}_R^{\text{audit,transfer}} = [20]$. The time-content inputs, time-content outputs, physical- and formal-input claim-component supports, and transfer relations of

the respective transfer events are defined by the equalities in Table 71. The original-source location of each transfer event is the union of the original-source locations corresponding to its time-content inputs, time-content outputs, and physical- and formal-input claim-component supports. The input and output indices i refer to time contents; a gives the supporting auxiliary-object indices in Table 69, and p gives the supporting physical/formal input-claim component indices.

Table 71: Time-Content Transfers in the Bridgman 1927 Reference

j	Input i	Out-put i	a	p	Transfer relation
1	3	2	none	1–3	Correspondence from the temporal order of performed operations to an operationally relative time concept
2	1, 6	5	none	5–6	Determine simultaneity from events and observer-relative times
3	9–10	11	5	10–11	Define the interval of one light round trip as the time unit
4	11–13	14	none	10, 12	Obtain a time value by counting equal unit intervals
5	3, 19–21	23	none	13–17	Form the proposed iteration of clock correction using approximate laws and uncorrected time measurements
6	9, 29–31	33	none	26–27	Adjust the remote-clock rate using one-second master-clock signals and arrival intervals
7	17, 22, 30, 33–34	35	1, 3, 5	26, 28	Set the zero of a rate-adjusted clock according to distance and a known light speed
8	1, 9, 22, 37	7	5	7, 29	Relate agreement of nearby signal-based clock readings to local time
9	7, 32, 35–36	8	3, 5	8–9, 30–31	Extend local clocks, synchronization, zero settings, and coordinate conventions over space
10	41, 43	42	4, 8	22–23, 32–33	Give meaning to a short-time value through substitution in equations and observational correspondence
11	3–4	45	none	49	Define the forward direction along the temporal order of the operations

j	Input i	Out- put i	a	p	Transfer relation
12	1, 4, 9, 22, 50	49	none	34–35, 38	Obtain a local causal-order restriction from a controllable cause and a return signal
13	18, 52	17	3, 5	40–41	Obtain velocity by dividing distance by the interval from readings at two positions
14	1, 9, 22, 50	56	none	43–44	Set the remote clock to zero when the signal emitted at the source zero arrives
15	1, 9, 22	10	none	43, 45	Obtain the round-trip interval by reading a single local clock for the returning signal
16	22, 32	38	none	26, 50	Assign a common present to observations at the zeros of synchronized clocks
17	4, 9, 17, 22, 32	51	1, 5	34–35, 38–39	Obtain causal-cone time relations from relativistic synchronization and the light-signal bound
18	7, 15, 18, 22	53	5	40, 42	Obtain velocity from road distance and an interval measured by the moving local clock
19	1, 3–5	54	none	40, 51	Define unit velocity by running together with or overtaking a disturbance
20	3–5, 54	55	none	40, 51	Compose velocities by superposing the motion of duplicated velocity standards

Transfer Events and Outputs The light clock defines one round trip as its unit, whereas remote-clock rate adjustment uses the master clock’s second. Zero setting by a known light speed and distance, zero setting on reception, and a one-clock round-trip measurement are distinct operations. The critique of absolute time, the physical-clock correspondence problem, the conditions of future determination, and the quantum-domain application boundary are given as results in Table 72.

Conditions of Formation

Conditions under which a Dependence or Transfer Applies Each transfer takes the tabulated time contents, auxiliary objects and supporting claims as inputs and holds under the following conditions.

1. The order of measurement operations associated with a time concept is supplied as an input.
2. The observer-relative assignment of times used in relativity is presupposed.
3. Mirrors are placed at both ends of a meter stick, and the interval of one round trip is chosen as the unit.

4. The same unit interval is identified, and the number of such intervals is recorded.
5. An experiment is chosen in which the effect on the target phenomenon exceeds the anticipated effect on the clock; corrections are iterated using approximate laws and uncorrected time measurements. The source leaves the feasibility of this procedure unanalyzed.
6. The remote-clock rate is adjusted so that the arrival intervals of signals sent by the master clock every second are also read as one second on the remote clock.
7. When a signal emitted at the master-clock zero arrives, the remote zero is set so that distance divided by the remote-clock indication equals the known speed of light.
8. The empirical condition is that, in the limit of approaching the clock, time readings obtained through light, acoustic and tactual signals show no measurable difference.
9. The specified spatial measurements and conventions for synchronization and zero setting are used.
10. Short-time values are substituted into optical and electrodynamic equations and related to agreement with observed facts.
11. The direction of temporal progression implicit in the performed operations is defined as forward.
12. The intervention constituting the cause can be cancelled, and the effect is assumed to occur only if that cause occurs. A return signal from the effect cannot arrive before the controllable cause.
13. The clocks at the two positions belong to the specified extended-time system, and distance is divided by their reading difference. Variable velocity uses the limiting operation stated in the source.
14. The remote clock is set to zero at the instant it receives the signal sent at the emission-clock zero.
15. The signal round trip and reversal of direction are measured with a single local clock. Changing the remote-clock zero convention does not change this local measurement.
16. The present time is assigned to observations made at the zeros of synchronized clocks. The present physical state additionally includes physical measurement information collected at each location.
17. The additional assumption that signals cannot propagate faster than light, relativistic synchronization and spatial separation are used.
18. The interval on a clock carried by the moving object is used together with distance measured by a road-fixed measuring tape or log line.
19. A disturbance travelling along a string stretched by weights on a board serves as the standard; running together, overtaking and lagging behind are compared.
20. The equality of duplicated standards is checked by running together, and velocities are composed by making the second board itself move with the first disturbance.

Original-Source Support

Original-Source Support for the Dependences and Transfers Source locations of supporting claims and time contents are specified in their respective definition tables.

Results

Definition 8.154 (Results in the Bridgman 1927 Reference). Define the result index set by $\text{ldx}_R^{\text{audit,result}} = [20]$. The respective results, time-content presuppositions, physical- and formal-input claim-component supports, and original-source locations are defined by the equalities in Table 72. Auxiliary prerequisite indices a refer to Table 69.

Table 72: Results and Audit Inputs in the Bridgman 1927 Reference

j	Result and source location	Time prerequisites i	Aux. a	Supporting claims p
1	Correspondence between time concepts and sets of measurement operations. pp.4–9	2–3	none	1–2
2	Differences between time concepts relative to sets of measurement operations. pp.4–9	2–3	none	3
3	Operational critique of Newtonian absolute time. pp.4–6	2–3	6	4
4	Before, after, and simultaneity relations relative to the observer. pp.7–9	1, 4–6	none	5–6
5	Distinction between local and extended time. pp.68–79,202	7–8, 32, 36	7	7–9
6	Time unit defined by one light round trip and interval counting. pp.69–70	9–14	5	10–12
7	Interdependence of clocks and mechanical laws. pp.70–71	15–16, 19–22	none	13–15
8	Proposal for clock calibration by successive approximation. p.71	3, 15–16, 19–23	none	13–17

j	Result and source location	Time prerequisites i	Aux. a	Supporting claims p
9	Correspondence problem between postulated clocks, line elements, and physical clocks. pp.71–73	17, 22, 24–26	1–2	18–20
10	Correspondence problem connecting infinitesimal temporal components and short-time values to physical measurement. pp.72–73,78	25–26, 41–43	4, 8	21–23
11	Conditions and criterion for treating atomic oscillation as a clock. pp.72–73	27–28	none	24–25
12	Remote-clock rate adjustment and zero setting using a known light speed. pp.73–74	9, 17, 22, 29–35	1, 3, 5	26–28
13	Neighborhood measurement of local time and its limits for distant or high-velocity phenomena. pp.74–77,202	1, 7, 9, 22, 37–39	7	7, 29
14	Construction of extended time incorporating spatial measurement. pp.73–77,216–218	7–8, 22, 32–33, 35–36	3, 5	30–31
15	Analysis of time concepts and event identity according to scale and operations. pp.75–79,91–97,202	1, 3, 7, 40–45	4, 7–8	32–33, 49, 52
16	Analysis of causal order, initial conditions, future determination, and their conditions. pp.80–91,158–160,216–218	1, 4, 9, 17, 20, 22, 32, 38, 40, 46–51	1, 5	34–39, 50
17	Velocity definitions using two position clocks, a moving clock, or comparison and composition of velocity standards. pp.97–100	1, 3–5, 7–8, 15, 17–18, 22, 52–55	3, 5	40–42, 51

j	Result and source location	Time prerequisites i	Aux. a	Supporting claims p
18	Reception-zero convention and finite round-trip time measured with a single clock. pp.160–162	1, 9–10, 22, 50, 56	none	43–45
19	Conditional quantum-domain boundary of ordinary spacetime operations. pp.225–226	57	7	46
20	Empirical meaning of time concepts and their revisability as experience expands. pp.4–9,68–100,202,225–226	2–3, 7–8, 41, 43, 57	7	47–48

Source-Specific Results The 20 results in Definition 8.154. The time-content presuppositions, supporting claim components, and original-source locations of the respective results are given in Table 72.

Result Dependences Across the 20 results there are 122 temporal-prerequisite occurrences and 22 auxiliary-prerequisite occurrences. An object supporting several results is counted once for each such result.

Take the supporting claim indices, time-content presupposition indices, and auxiliary-object presupposition indices in each row of Table 72 as the corresponding index sets in Definition 7.105. The support set, time-content presupposition set, and auxiliary-object presupposition set of each result are determined by that definition.

Grounds of Formation

Grounds of Formation of the Results The source supplies operational definitions, proposed measurement procedures, methodological criticism and analyses of applicability. Iterative clock calibration is a proposal whose feasibility remains unanalyzed; atomic-clock correspondence is examined through the question of its criterion. Future determination is conditional on the physical state, laws, history and domain; synchronized time assignments alone do not specify the state.

Evidence Hierarchy

Directly Measured Objects This reference itself newly obtains zero directly measured objects.

Directly Measured Predicates This reference itself newly obtains zero directly measured predicates.

Conditions and Domain of Application of Direct Measurement The clocks, signals, atomic processes, and velocity measurement discussed in the original source are cases

used to analyze the structure and domain of application of operations. They do not belong to measurement data obtained by apparatus in this reference itself.

Objects of Formal Generalization Results 1–4 concern the operational meaning of time concepts and observer relativity; 5–14, clocks, time units, local measurement and remote synchronization; 15–18, scale, event identity, causal order, velocity definitions and alternative synchronization; and 19–20, operational applicability and empirical revisability.

Conditions and Domain of Application of Formal Generalization Choosing the light-clock unit and sending one-second master-clock signals are different procedures. Neighborhood agreement depends on comparison of light, acoustic and tactual signals. Causal order uses possible interventions and the system background; the causal cone adds the light-signal bound. In the radiation example, a present state specified by positions, velocities and similar mechanical quantities lacks information needed to determine the future. A present state incorporating physical information throughout space is therefore considered. Direct comparison and composition of velocity standards use simultaneity of running together and overtaking order. The quantum-domain boundary holds if the ordinary measurement operations cannot be performed there.

Generation Judgment Judgment values: LCTR-complete-generation judgment: 0; reference-internal evidence-conditional local-reach judgment: 0; remaining primitive time roots: 57.

Generation Reach and Primitive Time Roots No time content is generated by the generation reach whose initial set consists solely of non-temporal physical inputs. All 57 time contents in Definition 8.151 remain primitive time roots.

Reference-Internal Evidence-Conditional Local Reach The reference-internal evidence-conditional local-reach judgment is 0, and 57 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ in Definition 7.108 and apply Definition 7.50 to it. Summing over all results determines the primitive-time-root–result dependence relation and the necessary-primitive-time-root–result dependence relation.

The definition tables specify the roles of the 57 time contents and eight auxiliary objects. The 20 results have 76 minimal primitive-time-root support sets in total, with 161 root–result dependence pairs and 82 essential root–result dependence pairs.

Result Dependence All 20 results in Definition 8.154 belong to the primitive-time-root-dependent results. There are no generated results, evidence-reach-only results, or results unreached after root addition. The minimal primitive-time-root support of each result is given by Proposition 8.157.

Evidence Effect

Evidence-Effect Domain The source's operational definitions, proposals, criticisms and conditional analyses support the 20 tabulated results under their respective conditions on clocks, signals, measurement operations, system backgrounds and domains.

Evidence-Effect Boundary Operational definitions of time measurement use order, intervals, clock indications, synchronization, velocity comparisons and causal order. Their physical use requires the generation structures corresponding to those temporal roles. The evidence-effect boundary in Definition 7.115 lies at the transition from operational meaning and applicability to establishment of those generation structures.

Result-Preservation Domain

Result-Preservation Domain The 20 results are preserved when their tabulated temporal prerequisites are supplied by corresponding generated time representations and their physical carriers, signal paths, distance measurements, clock conventions, calibration-proposal conditions, system backgrounds, scales and domains are retained. Criticisms, unresolved problems and conditional analyses retain their logical form and conditions. Velocity measurement by distance divided by a clock interval requires that interval; direct comparison and composition of velocity standards require simultaneity of running together and overtaking order.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The 57 time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as established time inputs. The placement judgment for the physical-time grounding obligation is "backgrounded by primitive time input." This judgment is grounded in the comparison between the use of each time content in a law operation or result derivation and the absence, before that use, of the corresponding generation or establishment condition as an input, formation condition, evaluation condition, or invalidation condition.

Evidence of Actual Invalidation

Evidence of Actual Invalidation The complete judgment and remaining-root judgment concern physical-time qualification and are distinct from the actual invalidation of a specific condition that determines a result's value or domain of application. The original-source formal evidence and LCTR audit evidence for this reference do not establish the actual invalidation of such a target condition. The actual-invalidation judgment is "indeterminate from original-source evidence"; actual invalidation is not inferred from the backgrounding of the grounding obligation alone.

Downstream Reach

Downstream Reach The reach to primitive-time-root-dependent results is determined for this reference by the primitive-time-root-result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach.

Because the actual invalidation of a specific condition has not been established, the judgment of actual reach from the invalidation position to a result is “indeterminate from original-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair Scope The repair targets based on structural reach are the corresponding time content and the generation or establishment conditions required before its use. The recalculation scope is limited to results for which actual invalidation is independently established and which are dependency-reachable from that invalidation position. Because actual invalidation is indeterminate in this reference, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from the repaired inputs, or their establishment domain is restricted. Values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.155 (Primitive Time Roots and Result Dependence in the Bridgman 1927 Reference).

In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepIdx}(57, 20)$. Furthermore, $R \in \text{Ref}^{\text{audit, rootdep}}$ holds.

Proof. The time-content set has 57 elements. Each of the 20 transfer events in Definition 8.153 takes at least one time content as an input. Each of the 20 result edges in Table 72 also takes at least one time content as an input. In the generation reach that starts from the 13 generation-initial input components of this reference, the first-application condition for every transfer event and every result edge fails. Proposition 7.43, Definition 7.39, and Definition 8.115 yield the displayed index tuple. When the 57 primitive time roots are added as seeds and the reference formal-evidence environment is supplied, all 20 results are reached. Membership in the three reference sets follows from Definition 7.52. $\square$

Proposition 8.156 (Physical-Time Qualification in the Bridgman 1927 Reference). All 57 time contents in the reference are primitive time inputs and do not satisfy physical-time qualification. All 20 results depend on time contents that do not satisfy physical-time qualification.

Proof. By Proposition 8.155, all 57 time contents belong to the primitive time roots, and all 20 results belong to the primitive-time-root-dependent results. The conclusion follows from Definition 7.10, Theorem 7.11, and Corollary 7.12. $\square$

Proposition 8.157 (Minimal Primitive-Time-Root Support in the Bridgman 1927 Reference). For each $q \in \text{Idx}_R^{\text{audit, result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108, and the minimum and maximum cardinalities of those elements, are given in Table 73.

Table 73: Minimal Primitive-Time-Root Support in the
Bridgman 1927 Reference

Result Index	Support Statistics	Result Index	Support Statistics
1–3, 19	(1; 1)	4	(1; 3)
5	(12; 4, 12)	6	(2; 4, 5)
7	(1; 6)	8	(1; 7)
9	(2; 5, 6)	10, 18	(1; 4)
11	(1; 2)	12	(2; 8, 9)
13	(2; 6)	14	(12; 6, 12)
15	(4; 8, 10)	16	(2; 12, 13)
17	(16; 10, 16)	20	(12; 6, 16)

Proof. Apply the update of Definition 7.46 to the 20 transfers in their listed order, starting from singleton root supports. The support antichains become invariant on scan 3. For each result, choose one support set for each temporal prerequisite, form their union, and retain the minimal unions. Proposition 7.47 gives the displayed support families and statistics. $\square$

Boundary between Operational Definition and Physical-Time Generation Corresponding concepts to actual operations and making the relativity of temporal operations explicit supply operational conditions for time measurement. A construction that takes a hypothetical clock, a prescribed unit of time, clock rates, signal travel time, and law time variables as inputs, however, does not generate physical time from physical inputs containing no temporal role.

Conditions of Formation of Time-Containing Physical Quantities Clock rates, acceleration histories and velocities defined by distance and an interval use temporal indications, intervals and courses of motion. Those temporal roles require their corresponding generation and physical-time qualification. For direct comparison and composition of velocity standards, the corresponding obligations concern simultaneity and order in running together and overtaking.

Domain of Formation of a Light-Speed-Invariance Proposition A proposition comparing a common finite light-speed value presupposes the reference frames, clock units, distance measurements and synchronization method used in the comparison. The source's reception-zero convention changes the time-coordinate setting and assigns an infinite velocity value to light, while its round-trip time measured by a single local clock remains finite.

Individual Statements

Statement Objects The relativity of time-measurement operations, the relation between physical concepts and actual operations, the absence of critical examination of time-measurement operations, the absence of specifications for clock construction, and the analytical status of local time.

Verbatim Original all such operations are relative operations

the proper definition of a concept is not in terms of its properties but in terms of actual operations

The physical operations at the basis of the measurement of time have never been subjected to the critical examination which seems to be required.

A rigorous answer to this question involves giving the specifications for the construction of a clock, which we have seen has not yet been done.

Local time is, therefore, a concept treated by the physicist even now as simple and unanalyzable.

Original-Source Location The first and second quotations are from Bridgman, *The Logic of Modern Physics*, p. 6, *Broad Points of View*. The third quotation is on p. 69 of the same reference, the fourth on p. 74, and the fifth on p. 75.

Minimal Context In the first quotation, “such operations” refers to time-measurement operations in the discussion of the meaning of absolute time. The second quotation is part of a sentence stating the consequences of adopting the view that concepts are defined in terms of actual operations rather than their properties. The third states that the operations underlying time measurement have not undergone the critical examination that seems to be required. In the fourth, “this question” is whether local time also involves the measurement of space; the quotation states that the specifications for clock construction needed for a rigorous answer have not been supplied. The fifth describes the treatment of local time as a simple and unanalyzable concept in physical practice at the time of writing.

Attributing Source All five statements are attributable to Bridgman’s book.

Predicates The first quotation attributes relativity to time-measurement operations. The second states the view that a physical concept is properly defined in terms of actual operations rather than its properties. The third states that the physical operations underlying time measurement have not undergone the critical examination that seems to be required. The fourth states that specifications for clock construction have not yet been supplied. The fifth states that, at the time of writing, local time is still treated as a simple and unanalyzable concept.

Explicit Domains of Application The domain of the first quotation is the time-measurement operations discussed in the examination of absolute time on p. 6 of the original source. The domain of the second is physical concepts for which the operational view of definition is adopted. The domain of the third is the physical operations underlying time measurement; that of the fourth, the specifications for constructing a clock; and that of the fifth, the concept of local time.

Individual Audit

Interpretive Possibilities First, the statements permit interpretation as operational definitions that give the empirical meaning of physical concepts through executable operations. Second, they permit a scale- and apparatus-relative interpretation under which concepts bearing the same name have different empirical contents when their

sets of operations differ. Third, they permit an interpretation that uses clocks, signals, records, and comparison operations as generative components for the concept of time, while leaving the generation obligations for order, interval, synchronization, and velocity already contained in those operations. Fourth, the third through fifth quotations permit the interpretation that unresolved obligations remain in the analysis of time-measurement operations, specifications for clock construction, and local time.

Independent Grounds The first statement describes the relativity of time-measurement operations; the second states the view that physical concepts are specified through actual operations. The third through fifth describe the state, at the time of writing, of critical examination of time-measurement operations, specifications for clock construction and analysis of local time. The calibration proposal on p.71 leaves its feasibility unanalyzed; on pp.74–75, the involvement of spatial measurement in constructing a local clock is considered probable.

Evidence-Effect Boundary The first quotation supports the relativity of time-measurement operations; the second supports specifying concepts through actual operations rather than their properties. The third through fifth support the incomplete state of critical examination of time-measurement operations, provision of clock-construction specifications and local-time analysis described by the source. The source’s temporal operations use postulated clock indications, local time, clock rates, unit intervals and law time variables. The generation and physical-time qualification required for their use lie beyond the statements’ evidence-effect boundary under Definition 7.115.

Corresponding Formal Evidence Indices refer to Definitions 8.151, 8.152 and 8.154. The first two quotations correspond to time contents 2–3; input claims 1–4; and results 1–3. The third quotation corresponds to time contents 7–14; input claims 7–12; and results 5–6. The fourth quotation corresponds to time contents 7, 19–26; input claims 13–20, 29; and results 7–10, 13. The fifth quotation corresponds to time contents 7, 22, 37; input claims 7, 29; and results 5, 13.

Within-Reference Comparison The first statement describes the relativity of time-measurement operations; the second presents the operational view of concept definition; and the third through fifth describe the incomplete state, at the time of writing, of time-measurement operations, clock specifications, and the analysis of local time. Comparing these statements aligns the requirement of a single methodology with the points identified as unmet in the source. The use of time inputs by the temporal formalism in the same reference does not discharge this unmet requirement.

Common Audit with Domain of Application When clocks, signals and records serve as components for generating time, the grounding obligations concern the temporal order, intervals, indication correspondences and synchronization included in the operations.

8.19.7 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 7.

Source Name Complete Time-Use Audit of von Neumann 1932

Bibliographic Identification Write the audit target as $R := \text{Ref}_{\text{VN32MQ}}^{\text{survey}}$. This denotes J. von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer, Berlin (1932). The audit domain is the complete 1932 German edition. The text and notes occupy pp. 1–262; the principal locations are I.2, I.4, III.1–6, IV.1–4, V.1–3, and VI.1–3.

Audit-Scope Identifier 7-SC1.

Audit Scope The audit domain stated under Bibliographic Identification.

Original-Source Location Identifier 7-L1.

Original-Source Location The audit domain stated under Bibliographic Identification and the original-source location columns of the respective evidence tables.

Formal Objects

Object Identifier 7-F1.

Object Name Formal objects in the original source.

Object Type Formal object.

Formal Content The individual contents of the formal objects are given in Definitions 8.159 and 8.160.

Original-Source Location Identifier Set 7-L1.

Assumptions

Assumption Identifier 7-A1.

Assumption Content Each result in the original source is treated under its own conditions of validity, as listed in Table 79. These include the two laws of thermodynamics and operational conditions, conditions for immediate repetition of measurement, the prepared apparatus state, and the approximation conditions of the position–pointer model.

Original-Source Location Identifier Set 7-L1.

Non-Temporal Physical Inputs

Input Identifier 7-P1.

Input Name Non-temporal physical inputs in the original source.

Input Type Non-temporal physical input.

Input Content The individual contents of the non-temporal physical inputs are given in Definitions 8.159 and 8.160.

Original-Source Location Identifier Set 7-L1.

Time Content

Time-Content Identifier 7-T1.

Time-Content Name Time contents in the original source.

Time-Content Type Time content.

Time-Content Specification The individual contents of the time contents are given in Definition 8.161.

Original-Source Location Identifier Set 7-L1.

Formal Input Elements

Formal-Component Identifier 7-C1.

Formal-Component Name Formal-evidence environment of the original source.

Formal-Component Type Formal component.

Formal-Component Text The individual contents of the formal input elements are given in Definitions 8.159 and 8.160.

Original-Source Support Identifier Set 7-L1.

Definition 8.158 (Source and Attribution Components). Claims in the original source are attributed to von Neumann and distinguished from views considered within the original source, subsequent measurement terminology, and the audit judgments of this paper. This reference itself newly obtains zero items of direct measurement evidence.

Definition 8.159 (Formal Evidence in the Original Source). The 46 physical and formal claim components in the original source are given in Table 74, and the 12 mathematical-object-type components in Table 75. The related-time column gives sets of time-content indices from Table 77.

Table 74: Physical and Formal Claim Components in the von Neumann 1932 Reference

Com- po- nent	Minimal claim	Related time	Source location
1	Quantum system and its physical state	$\emptyset$	I.2, pp. 5–9
2	Hilbert-space representation of a physical state	$\emptyset$	I.4, pp. 15–17; III.1, p. 102
3	Physical observable	$\emptyset$	III.1, pp. 101–106
4	Self-adjoint-operator representation of an observable	$\emptyset$	IV.2, p. 167
5	Physical pointer record obtained by measurement	$\emptyset$	V.1, pp. 187–190
6	Correspondence between an observed value and a spectral projection	$\emptyset$	III.1, pp. 103–106
7	Measurement probability from a state and projection	$\emptyset$	III.1, pp. 103–106
8	Statistical ensemble	$\emptyset$	IV.1–2, pp. 157–169

Com- po- nent	Minimal claim	Related time	Source location
9	Statistical-operator representation of a statistical ensemble	$\emptyset$	IV.1–2, pp. 157–169
10	Expectation value from a statistical operator	$\emptyset$	IV.1–2, pp. 157–169
11	Thermodynamic entropy formula for a statistical operator	$\{8, 10\}$	V.2, pp. 191–202
12	Composite system comprising multiple physical subsystems	$\emptyset$	VI.2, pp. 225–232
13	Composite-system representation by the tensor product of subsystem spaces	$\emptyset$	VI.2, pp. 225–232
14	Subsystem statistical operator obtained from a composite pure state	$\emptyset$	VI.2, pp. 225–232
15	Separation of composite-state correlations from subsystem marginal states	$\emptyset$	VI.2, pp. 225–232
16	Time-dependent evolution equation for a nonstationary state	$\{2, 6\}$	I.2, pp. 8–9; V.1, pp. 185–186
17	Unique evolution from an initial state to a state at another time	$\{2, 3, 6\}$	I.2, pp. 8–9; V.1, pp. 185–186
18	Continuous and causal state change in process 2	$\{6\}$	V.1, pp. 185–190
19	Unitary reversibility of process 2	$\{6\}$	V.1, pp. 185–190
20	Small-interval composition of a time-dependent Hamiltonian	$\{6, 7\}$	V.1, p. 187
21	Process 1 accompanying measurement	$\{8\}$	V.1, pp. 184–186
22	Probability weights of process 1	$\{8\}$	V.1, pp. 184–186
23	Difference between the mapping types of process 1 and process 2	$\{6, 8, 10\}$	V.1, pp. 186–191
24	Post-measurement states of eigenstates and general states	$\{8\}$	III.3, pp. 111–115
25	Repeated measurement and order before and after measurement	$\{8\}$	III.3, pp. 111–115
26	Numerical time in nonrelativistic quantum mechanics	$\{2, 6, 9\}$	V.1, p. 188
27	Objective simultaneity in a nonrelativistic description	$\{9\}$	V.1, p. 188
28	Comparison of microscopic entropy and reversibility for the two processes	$\{6, 8, 10\}$	V.1, pp. 190–191; V.3, pp. 202–207
29	Future and past directions and irreversibility	$\{8, 10\}$	V.1, pp. 190–191

Com- po- nent	Minimal claim	Related time	Source location
30	Distinction between subjective perception and physical processes	$\emptyset$	VI.1, pp. 223–224
31	Psychophysical parallelism	$\emptyset$	VI.1, pp. 223–224
32	Configuration of the physical systems carrying a perceptual chain	$\emptyset$	VI.1, pp. 223–224
33	Relational description of a physical perceptual chain	$\emptyset$	VI.1, pp. 223–224
34	Quantum system that is the object of observation	$\emptyset$	VI.1, pp. 223–224
35	Physical carriers of the observational apparatus and observer	$\emptyset$	VI.1, pp. 223–224
36	Placement of the boundary between the observed system and observer	$\emptyset$	VI.1, pp. 223–224
37	Mobility of the measurement boundary	$\emptyset$	VI.1, pp. 223–224
38	Physical carriers of system I, apparatus II, and observer III	$\emptyset$	VI.1, pp. 223–224
39	Tripartition into system I, apparatus II, and observer III	$\emptyset$	VI.1, pp. 223–224
40	Physical measurement interaction between system I and apparatus II	$\emptyset$	VI.3, pp. 233–235
41	Separation between the correlated state and process 1 on the observer side	$\{6, 8, 11\}$	VI.3, pp. 233–235
42	Physical interaction carrying measurement coupling	$\emptyset$	VI.3, pp. 233–235
43	Construction of measurement correlations by an abstract unitary map	$\emptyset$	VI.3, p. 235
44	Physical systems carrying position and pointer	$\emptyset$	VI.3, pp. 236–237
45	Conditional approximation in the position–pointer model	$\{8, 11\}$	VI.3, pp. 236–237
46	Incompatibility between observer-initial-mixture weights and input-state-dependent Born weights	$\{8, 11\}$	VI.3, p. 233

Table 75: Mathematical Object Types in the von Neumann 1932 Reference

Component	Mathematical object type	Related time contents
1	Hilbert space	$\emptyset$

Component	Mathematical object type	Related time contents
2	Self-adjoint operator	$\emptyset$
3	Spectral projection	$\emptyset$
4	Statistical operator	$\emptyset$
5	Tensor product	$\emptyset$
6	Partial trace	$\emptyset$
7	Two processes	$\{6, 8, 10, 11\}$
8	Measurement boundary	$\emptyset$
9	Evolution operator	$\{2, 3, 4, 6, 7\}$
10	External numerical time in nonrelativistic quantum mechanics	$\llbracket 2, 11 \rrbracket$
11	Simultaneity	$\{9\}$
12	Direction of time	$\{10\}$

Definition 8.160 (Semantic Components and Generation-Initial Set). The 70 semantic components are enumerated in the following order: the 12 non-temporal initial components in Table 76, the 46 physical and formal claims in Table 74, and the 12 mathematical object types in Table 75, each in its table-index order. Following Definition 7.90, write the physical-and-formal claim nodes and mathematical claim nodes as

$$\begin{aligned} \text{Node}_{R,p}^{\text{audit,claim}} &:= \text{EnumNode}_R^{\text{audit,claim,phys}}(p), \quad p \in [46], \\ \text{Node}_{R,m}^{\text{audit,math}} &:= \text{EnumNode}_R^{\text{audit,claim,math}}(m), \quad m \in [12]. \end{aligned}$$

Let $\text{ClaimIndex}_R^{\text{init}}(k)$ be the corresponding claim-index set in the initial-component table. By Definitions 7.101 and 7.102, set

$$\begin{aligned} \text{Node}_R^{\text{audit,semantic}} &= \text{SemanticNodes}_R(70), \\ \text{ClaimNodeLift}_R(12; 46), \quad &\text{MathNodeLift}_R(58; 12), \\ \text{StdSemLift}_R(12; 46). \end{aligned}$$

The first 12 components are projections onto non-temporal carriers, and the corresponding claim indices specify the semantic-support correspondence. In the common audit boundary, each projection is represented by its corresponding claim node.

Table 76: Generation-Initial Semantic Components in the von Neumann 1932 Reference

k	Non-temporal physical input	Corresponding claim indices	Source location
1	Quantum system and its physical state	$\{1\}$	I.2, pp. 5–9
2	Physical observable	$\{3\}$	III.1, pp. 101–106
3	Physical pointer record obtained by measurement	$\{5\}$	V.1, pp. 187–190

k	Non-temporal physical input	Corresponding claim indices	Source location
4	Statistical ensemble	{8}	IV.1–2, pp. 157–169
5	Composite system comprising multiple physical subsystems	{12}	VI.2, pp. 225–232
6	Configuration of the physical systems carrying a perceptual chain	{32}	VI.1, pp. 223–224
7	Quantum system that is the object of observation	{34}	VI.1, pp. 223–224
8	Physical carriers of the observational apparatus and observer	{35}	VI.1, pp. 223–224
9	Physical carriers of system I, apparatus II, and observer III	{38}	VI.1, pp. 223–224
10	Physical measurement interaction between system I and apparatus II	{40}	VI.3, pp. 233–235
11	Physical interaction carrying measurement coupling	{42}	VI.3, pp. 233–235
12	Physical systems carrying position and pointer	{44}	VI.3, pp. 236–237

Definition 8.161 (Time Contents and Source Locations). Table 77 gives the 11 time contents used in this audit and their locations in the original source. There are no auxiliary objects.

Table 77: Time Contents in the von Neumann 1932 Reference

i	Time content	Source location
1	Time differentiation in classical laws	III.2, pp. 107–108; III.6, pp. 136–139; pp. 238–239, 253–254
2	External time of a nonstationary quantum state	I.2, pp. 8–9; III.1, p. 102; III.6, pp. 150–156
3	Temporal boundary of an initial value	I.2, pp. 8–9; III.2, p. 108; III.6, p. 151
4	Time in stationary phase and frequency	I.2, pp. 9–14; pp. 254, 256
5	Time for position measurements at two instants and for velocity	III.4, p. 128
6	Time of unitary evolution in process 2	V.1, pp. 185–187
7	Interval order of a time-dependent Hamiltonian	V.1, p. 187
8	Measurement boundary, duration, and energy precision	III.3, pp. 111–115; III.5, pp. 129–130; IV.3, pp. 178–179; V.1, pp. 187–188; p. 252
9	Common numerical time and objective simultaneity	V.1, p. 188

i	Time content	Source location
10	Irreversibility and future–past direction	V.1, pp. 190–191
11	Finite-time measurement interaction and pointer motion	VI.3, pp. 233–237

Time content 1 includes time differentiation of the electromagnetic field and the time variable in classical oscillations. The external time in content 2 indexes the quantum amplitudes and transition probabilities in III.6 and supplies the time unit for its transition rates per second. The frequency in content 4 is also used to determine energy levels from emitted spectral lines. Content 8 includes the start, end, and duration of measurement and the temporal correspondence among past measurement records, the present state, and future measurement results.

Dependent Inputs

Index Identifier S07-XI-DI1.

Existing-Evidence Reference Set Definition 8.162.

Dependence and Transfer Elements

Definition 8.162 (Dependence and Generation Components). The dependence structure of this reference comprises 70 semantic components, 11 time contents, eight time-content transfers, and 13 results. The original-source formal evidence, time-content inputs, and outputs of the eight transfers are given in Table 78. Every transfer takes one or more time contents as inputs; zero transfers output time content from non-temporal physical inputs alone.

Table 78: Time-Content Transfers in the von Neumann 1932 Reference

Trans-fer	Output content	Formal evidence	Time input	Time output
1	Specification of the initial-data time	$\{16, 17, 26\}$	$\{2\}$	$\{3\}$
2	Stationary phase evolution	$\{16, 20, 26\}$	$\{2\}$	$\{4\}$
3	Process 2	$\llbracket 18, 20 \rrbracket \cup \{26\}$	$\{2\}$	$\{6\}$
4	Composition of small-interval evolution	$\{20\}$	$\{6\}$	$\{7\}$
5	Irreversible direction	$\{21, 22, 23, 25, 28, 29\}$	$\{8\}$	$\{10\}$
6	Measurement interaction and pointer motion	$\llbracket 38, 45 \rrbracket$	$\{6, 8\}$	$\{11\}$
7	Finite-time measurement interaction	$\{18, 20, 42, 43\}$	$\{6, 8\}$	$\{11\}$
8	Common numerical time	$\{26, 27\}$	$\{9\}$	$\{2\}$

Definition 8.163 (Audit-Judgment Components). The complete judgment yields zero generated time contents, the local-reach judgment yields zero reached time contents, and 11 primitive time roots remain.

Definition 8.164 (Result and Preservation-Condition Components). The 13 results that remain preservation targets after their time-content presuppositions are replaced by generated time representations, together with their conditions of formation, are given in Table 79.

Table 79: Results and Preservation Conditions in the von Neumann 1932 Reference

Re- sult	Content	Preservation condition
1	Formalism using Hilbert spaces and self-adjoint operators	Object types of Hilbert spaces and operators
2	Spectral projections and quantum probabilities	Correspondence among projections, probabilities, and states
3	Statistical operators, mixtures, trace, and the thermodynamic entropy formula	Normalization of the statistical operator, the first and second laws of thermodynamics, and the source's operational conditions
4	Composite systems and subsystem statistical operators	Construction by tensor products and partial traces
5	Unitary, causal, and reversible evolution in process 2	Inputs, outputs, and unitarity of process 2
6	Measurement transformation in process 1	Measurement conditions and state transformation in process 1
7	Agreement of outcomes under immediate repetition of a projective measurement	Immediate remeasurement of the same observable, with negligible intervening state evolution
8	Construction of object–apparatus correlations by an abstract unitary map	Complete orthonormal systems, the prepared apparatus state, and the corresponding unitary map
9	Placement of the measurement boundary and conditions for empirical comparison	Mobility of the boundary and attribution of experiential statements to the observer
10	Agreement of observational statistics under two boundary placements	Correspondence between observational statistics in the two placements
11	Original-source examination of an explanation of process 1 by the observer's initial mixture	The source's assumed form of the post-measurement state and the independence of initial mixture weights from the input state
12	Conditional approximation in the position–pointer model	Negligible kinetic-energy terms and the width of the initial pointer distribution

Re- sult	Content	Preservation condition
13	Original-source formulation of the problem of overlap between the two processes	The overlap problem of the two processes and its original-source location

Definition 8.165 (Evidence Boundary). The formal evidence in the original source treats states, observables, measurement probabilities, statistical operators, composite systems, the two processes, the measurement boundary, and the pointer model. That evidence uses external numerical time, initial time, temporal order, measurement duration, objective simultaneity, and future–past direction as inputs.

Conditions of Formation

Index Identifier S07-XI-FC1.

Existing-Evidence Reference Set Definition 8.162.

Original-Source Support

Index Identifier S07-XI-SS1.

Existing-Evidence Reference Set Definition 8.162.

Results

Definition 8.166 (Typed Finite Audit Input for von Neumann 1932). The audit input is a finite directed hypergraph comprising 70 semantic components, 11 time contents, eight time-content transfers, and 13 results. The inputs and outputs of transfers follow Table 78, and the results and preservation conditions follow Table 79. The original-source formal evidence and time-content presuppositions for the 13 results are given in Table 80.

Table 80: Result Inputs in the von Neumann 1932 Reference

Re- sult	Content	Formal evidence	Time presupposition
1	Formalism using Hilbert spaces and self-adjoint operators	$\llbracket 1, 4 \rrbracket$	$\emptyset$
2	Spectral projections and quantum probabilities	$\llbracket 5, 7 \rrbracket$	$\emptyset$
3	Statistical operators, mixtures, trace, and the thermodynamic entropy formula	$\llbracket 8, 11 \rrbracket$	$\{8, 10\}$
4	Composite systems and subsystem statistical operators	$\llbracket 12, 15 \rrbracket$	$\emptyset$

Re- sult	Content	Formal evidence	Time presupposition
5	Unitary, causal, and reversible evolution in process 2	$\llbracket 16, 20 \rrbracket$	$\{2, 3, 6, 7\}$
6	Measurement transformation in process 1	$\llbracket 21, 24 \rrbracket$	$\{8\}$
7	Agreement of outcomes under immediate repetition of a projective measurement	$\{24, 25\}$	$\{8\}$
8	Construction of object–apparatus correlations by an abstract unitary map	$\{38, 42, 43\}$	$\emptyset$
9	Placement of the measurement boundary and conditions for empirical comparison	$\llbracket 30, 39 \rrbracket$	$\emptyset$
10	Agreement of observational statistics under two boundary placements	$\{34, 35, 38, 40, 41, 43\}$	$\{8, 11\}$
11	Original-source examination of an explanation of process 1 by the observer’s initial mixture	$\{46\}$	$\{8, 11\}$
12	Conditional approximation in the position–pointer model	$\{42, 44, 45\}$	$\{8, 11\}$
13	Original-source formulation of the problem of overlap between the two processes	$\{23, 28\}$	$\{6, 8, 10\}$

Definition 8.167 (Canonical Audit Realization of von Neumann 1932). Specify the generation-initial set and the reference’s formal-evidence environment by the claim-audit boundary index pair of Definition 7.98,

$$\text{ClaimBoundaryIdx}_R := (\{1, 3, 5, 8, 12, 32, 34, 35, 38, 40, 42, 44\}, [46]) .$$

The initial set contains the 12 claim nodes corresponding to non-temporal carriers; the evidence environment contains the 46 physical-and-formal claim nodes. By Definition 7.105, the formal-evidence indices in the transfer and result tables are realized as support by these physical-and-formal claim nodes.

All eight transfer edges take one or more time contents as inputs. Results 1, 2, 4, 8, and 9 have no time-content presuppositions; the remaining eight results have time-content presuppositions. Thus, no time content is reachable from the physical-input initial set. With the reference’s formal-evidence environment added and no primitive time roots supplied as seeds, results 1, 2, 4, 8, and 9 are reachable under their specified formal and physical conditions.

Definition 8.168 (Semantic Audit Boundary of the von Neumann 1932 Reference). The source-specific semantic-component representation has 94 nodes comprising 70 semantic components, 11 time contents, and 13 results. The common audit boundary has 82 nodes: 46 physical-and-formal claim nodes, 12 mathematical claim nodes, 11 time-content nodes, and 13 result nodes. The eight time-content transfer edges and 13 result edges give 21 dependence edges in total.

Definition 8.169 (Support-Completion Specification for von Neumann 1932). For each of the 13 results, Table 80 specifies support within the set of 46 physical and formal claims and the time-content presuppositions. The dependence structure consisting of eight transfer edges and 13 result edges is acyclic.

Result Dependence

Index Identifier S07-XI-RD1.

Existing-Evidence Reference Set Definition 8.166.

Grounds of Formation

Index Identifier S07-XI-RG1.

Existing-Evidence Reference Set Definition 8.169.

Evidence Hierarchy The objects and predicates newly measured directly by this reference both number zero. The formal evidence concerns nonrelativistic quantum mechanics, statistical ensembles, projective measurement, apparatus interaction, and the approximation domain of the position–pointer model; it supplies Hilbert-space representations, operator, projection, and trace calculations, unitary evolution, measurement transformations, composite-system constructions, and measurement-boundary analysis.

Generation Judgment Judgment values: LCTR-complete-generation judgment: 0; reference-internal evidence-conditional local-reach judgment: 0; remaining primitive time roots: 11.

Reference-Internal Evidence-Conditional Local Reach The reference-internal evidence-conditional local-reach judgment is 0, and 11 primitive time roots remain after that judgment.

Result Dependence There are 24 primitive-time-root–result dependence pairs, of which seven are necessary dependences.

Evidence Effect The effect domain of the formal evidence comprises representation by Hilbert spaces and operators, spectral projections, quantum probabilities, statistical operators, composite systems, the two processes, measurement correlations, the measurement boundary, and the position–pointer model. Their establishment does not imply that external numerical time, initial time, measurement start and end points, objective simultaneity, and future–past direction have been generated from non-temporal physical inputs.

Result-Preservation Domain The 13 results are preserved insofar as each time-content presupposition is generated in the LCTR generation sequence and the preservation conditions in Definition 8.164, together with the objects and approximation domains specific to the original source, are satisfied.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The 11 time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as established time inputs. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment is grounded in the comparison between the use of each time content in a law operation or result derivation and the absence, before that use, of the corresponding generation or establishment condition as an input, formation condition, evaluation condition, or invalidation condition.

Evidence of Actual Invalidation

Evidence of Actual Invalidation The complete judgment and remaining-root judgment concern physical-time qualification and are distinct from the actual invalidation of a specific condition that determines a result’s value or domain of application. The original-source formal evidence and LCTR audit evidence for this reference do not establish the actual invalidation of such a target condition. The actual-invalidation judgment is “indeterminate from original-source evidence”; actual invalidation is not inferred from the backgrounding of the grounding obligation alone.

Downstream Reach

Downstream Reach The reach to primitive-time-root-dependent results is determined for this reference by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because the actual invalidation of a specific condition has not been established, the judgment of actual reach from the invalidation position to a result is “indeterminate from original-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair Scope The repair targets based on structural reach are the corresponding time content and the generation or establishment conditions required before its use. The recalculation scope is limited to results for which actual invalidation is independently established and which are dependency-reachable from that invalidation position. Because actual invalidation is indeterminate in this reference, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from the repaired inputs, or their establishment domain is restricted. Values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.170 (Complete Time-Use Audit of von Neumann 1932). This reference has 70 semantic components, 11 time contents, eight time-content transfers, and 13 results. The generated time contents under the complete judgment and the reached time contents under the local-reach judgment both number zero, and results 3, 5–7, and 10–13, eight in total, depend on primitive time roots.

Proof. Every transfer in Definition 8.162 takes time content as an input. In Definition 8.166, results 3, 5–7, and 10–13 have time-content presuppositions, whereas results 1, 2, 4, 8, and 9 do not. Every edge that outputs time content takes time content as an input. Thus no time content is reached even when the reference’s formal-evidence environment is added to the physical-input initial set. The time-content sets under both the complete judgment and the local-reach judgment are therefore empty, and all 11 time contents are primitive time roots. $\square$

Proposition 8.171 (Primitive Time Roots and Result Dependence in the von Neumann 1932 Reference). All 11 time contents are primitive time roots. Results 3, 5–7, and 10–13 depend on at least one of them; results 1, 2, 4, 8, and 9 have empty primitive-time-root supports.

Proof. By Definition 8.167, reach with the reference’s formal-evidence environment added and no primitive time roots supplied as seeds reaches zero time contents and five results. Adding the 11 primitive time roots makes all 13 results reachable under their specified formal and physical conditions. $\square$

Proposition 8.172 (Minimal Primitive-Time-Root Support in the von Neumann 1932 Reference). The minimal primitive-time-root supports of the 13 results are given in Table 81. The total number of family members is 25.

Table 81: Minimal Primitive-Time-Root Support in the von Neumann 1932 Reference

Result	Members of the minimal support-index family
1	$\emptyset$
2	$\emptyset$
3	$\{8\}$
4	$\emptyset$
5	$\{2\}, \{9\}$
6	$\{8\}$
7	$\{8\}$
8	$\emptyset$
9	$\emptyset$
10	$\{2, 8\}, \{6, 8\}, \{8, 9\}, \{8, 11\}$
11	$\{2, 8\}, \{6, 8\}, \{8, 9\}, \{8, 11\}$
12	$\{2, 8\}, \{6, 8\}, \{8, 9\}, \{8, 11\}$
13	$\{2, 8\}, \{6, 8\}, \{8, 9\}$

Proof. Updating the support antichains to a fixed point over the eight transfer edges and 13 result edges yields the 25 family members in the table. $\square$

Individual Statements Statement Evidence 1

Original (German):

Wenn der Zustand nicht stationär ist, d. h. sich mit der Zeit ändert, so enthält die Wellenfunktion $\psi = \psi(q_1, \dots, q_k; t)$ die Zeit t , [...]

English Translation:

If the state is not stationary, that is, if it changes with time, the wave function $\psi = \psi(q_1, \dots, q_k; t)$ contains the time t . [...]

The source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer (1932), p. 8, §I.2. The minimal context is the introduction of the wave-function representation of a nonstationary state. The attributing source of the statement is J. von Neumann.

Statement Evidence 2

Original (German):

Zunächst ist es an und für sich durchaus richtig, daß das Messen, bzw. der damit verknüpfte Vorgang der subjektiven Apperzeption eine gegenüber der physikalischen Umwelt neue, auf diese nicht zurückführbare Wesenheit ist.

English Translation:

First, it is entirely correct in itself that measurement, or the associated process of subjective apperception, is an entity new in relation to the physical environment and irreducible to it.

The source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer (1932), p. 223, §VI.1. The minimal context is the discussion of measurement and subjective apperception in relation to the physical environment. The attributing source of the statement is J. von Neumann.

Statement Evidence 3

Original (German):

Die Grenze zwischen beiden ist weitgehend willkürlich

English Translation:

The boundary between the two is arbitrary to a considerable extent.

The source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer (1932), p. 224, §VI.1. The minimal context is the discussion of the placement of the boundary between the observed part and the observing part. The attributing source of the statement is J. von Neumann.

Statement Evidence 4

Original (German):

Daß diese Grenze beliebig tief ins Innere des Körpers des wirklichen Beobachters verschoben werden kann, ist der Inhalt des Prinzips vom psychophysikalischen Parallelismus — dies ändert aber nichts daran, daß sie bei jeder Beschreibungsweise irgendwo gezogen werden muß, wenn dieselbe nicht leer laufen, d. h. wenn ein Vergleich mit der Erfahrung möglich sein soll.

English Translation:

That this boundary can be displaced arbitrarily far into the interior of the actual observer's body is the content of the principle of psychophysical parallelism. This does not alter the fact that in every mode of description the boundary must be drawn somewhere if the description is not to run empty, that is, if comparison with experience is to be possible.

The source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer (1932), p. 224, §VI.1. The minimal context simultaneously states the mobility of the boundary and the necessity of placing a boundary for comparison with experience. The attributing source of the statement is J. von Neumann.

Statement Evidence 5

Original (German):

Denn die Erfahrung macht nur Aussagen von diesem Typus: ein Beobachter hat eine bestimmte (subjektive) Wahrnehmung gemacht, und nie eine solche: eine physikalische Größe hat einen bestimmten Wert.

English Translation:

For experience makes statements only of this type: an observer has had a particular subjective perception, and never of this type: a physical quantity has a particular value.

The source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer (1932), p. 224, §VI.1. The minimal context is the distinction between the types of statements supplied by experience. The attributing source of the statement is J. von Neumann.

Statement Evidence 6

Original (German):

Die Quantenmechanik beschreibt nun die Ereignisse, die im beobachteten Teile der Welt stattfinden, solange sie mit dem beobachtenden Teile nicht in Wechselwirkung stehen, mit Hilfe des Prozesses 2. (V. 1.); sobald aber eine derartige Wechselwirkung stattfindet, d. h. eine Messung, schreibt sie das Verwenden des Prozesses 1. vor.

English Translation:

Quantum mechanics describes the events occurring in the observed part of the world by means of process 2 so long as that part does not interact with the observing part. As soon as such an interaction, that is, a measurement, occurs, it prescribes the use of process 1.

The source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer (1932), p. 224, §VI.1. The minimal context contrasts the conditions of application of process 2 and process 1. The attributing source of the statement is J. von Neumann.

Statement Evidence 7

Original (German):

Indessen besteht die Gefahr, daß das Prinzip vom psychophysikalischen Parallelismus verletzt ist, solange nicht gezeigt wird, daß die Grenze zwischen dem beobachteten System und dem Beobachter im w. o. angegebenen Sinne beliebig verschiebbar ist.

English Translation:

There is, however, a danger that the principle of psychophysical parallelism is violated unless it is shown that the boundary between the observed system and the observer can be displaced arbitrarily in the sense stated above.

The source location is von Neumann, *Mathematische Grundlagen der Quantenmechanik*, Springer (1932), p. 224, §VI.1. The minimal context connects the description using the two processes to the condition of boundary mobility. The attributing source of the statement is J. von Neumann.

Common Evidence with Domain of Application

Target State- ment	Statement Object	Predicate	Explicit Domain of Application
S07- Q001	Wave function $\psi(q_1, \dots, q_k; t)$ of a nonstationary state	Contains time t .	The wave-function representation in §I.2 of a nonstationary state that changes with time.
S07- Q002	Measurement and its associated process of subjective apperception	Is an entity new in relation to the physical environment and irreducible to it.	The relation among measurement, subjective apperception, and the physical environment in §VI.1.
S07- Q003	Boundary between the observed part and the observing part	Is arbitrary to a considerable extent.	The placement of the boundary between the observed part and the observing part in §VI.1.
S07- Q004	Boundary between the observed part and the actual observer	Can be displaced arbitrarily far into the observer's body, while it must be drawn somewhere in every description if comparison with experience is to be possible.	Psychophysical parallelism, boundary mobility, and the condition of empirical comparison in §VI.1.
S07- Q005	Statements supplied by experience	Are of the type in which an observer has had a particular subjective perception, not of the type in which a physical quantity has a particular value.	The distinction between the types of statements supplied by experience in §VI.1.

Target State- ment	Statement Object	Predicate	Explicit Domain of Application
S07- Q006	Quantum- mechanical description of events in the observed part of the world	Uses process 2 while there is no interaction with the observing part and uses process 1 when an interaction, namely measurement, occurs.	The conditions of application of process 2 and process 1 in §VI.1.
S07- Q007	Mobility of the boundary between the observed system and observer	Unless arbitrary displacement is shown, the principle of psychophysical parallelism is in danger of being violated.	The condition connecting the description using the two processes to boundary mobility in §VI.1.

Individual Audit

Interpretive Possibilities The first statement explicitly states that the wave-function representation of a nonstationary state contains time t . The second statement presents the claim that subjective apperception associated with measurement is a new entity irreducible to the physical environment. The third statement presents the arbitrariness of the placement of the boundary between the observed object and observer; the fourth, the mobility of that boundary and the necessity of placing it for empirical comparison. The fifth statement treats empirical statements as statements about an observer's subjective perception and distinguishes them from statements that a physical quantity has a value. The sixth statement gives the conditions of application of process 2 in the absence of interaction with the observing part and process 1 when measurement interaction occurs. The seventh states that the principle of psychophysical parallelism is in danger of being violated unless boundary mobility is demonstrated.

Independent Grounds The seven statements set out the formal and conceptual relations among state evolution by external numerical time, the measurement boundary, subjective apperception, empirical statements, and the two processes.

Evidence-Effect Boundary The first statement directly supports that the wave function contains time t , but does not support the physical generation, measurement procedure, or physical-time qualification of t . The second through fifth statements directly support the distinctions adopted by the original source concerning measurement, subjective apperception, boundary placement, and empirical statements. The sixth statement directly supports the conditions of application of process 1 and process 2, but does not support the generation of the temporal order, interaction duration, or evolution time used by the two processes. The seventh statement directly supports boundary mobility as a condition on psychophysical parallelism.

Corresponding Formal Evidence The first statement corresponds to the time variable used in physical and formal claim component 16 of Definition 8.159 and to time content 2, and supports the time input of result 5 in Definition 8.164. The second

and fifth statements correspond to the distinction between subjective perception and physical description treated in component 30. The third statement corresponds to the boundary placement in component 36; the fourth corresponds to psychophysical parallelism in component 31 and to boundary placement and mobility in components 36 and 37. The sixth statement corresponds to the conditions of application of the two processes treated in components 18 and 21, and uses mathematical object type 7. The seventh statement states that the principle of psychophysical parallelism in component 31 is in danger of being violated unless boundary mobility in components 36 and 37 is demonstrated. The boundaries in the third, fourth, and seventh statements correspond to the measurement boundary in mathematical object type 8.

Within-Reference Comparison The first statement uses external time for state evolution; the second through fifth and seventh treat the boundary between the observed object and observer and the attribution of empirical statements; and the sixth distinguishes the two processes by the presence or absence of measurement interaction. Formal time, measurement interaction, boundary placement, and subjective perception have distinct object types within the same reference and cannot be integrated into a single evidence type.

8.19.8 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 8.

Source Name Primitive-Time-Root Audit of Curtiss 1949

Bibliographic Identification Let reference R be Leon F. Curtiss, *Measurements of Radioactivity*, National Bureau of Standards Circular 476 (1949)[38]. The audit domain comprises the exponential-decay equation, decay-rate equation, half-life equation, mean-life equation, and derivational text in §I.2.

Definition 8.173 (Semantic Components in the Curtiss 1949 Reference). The 22 semantic components comprise five non-temporal initial components, 13 reference claims, and four mathematical object types. The reference claims in Table 84 and the mathematical objects in Table 85 correspond, respectively, to the physical-formal and mathematical claim nodes of Definition 7.90, with the abbreviations

$$\begin{aligned}\text{Node}_{R,p}^{\text{audit,claim}} &:= \text{EnumNode}_R^{\text{audit,claim,phys}}(p), \quad p \in [13], \\ \text{Node}_{R,m}^{\text{audit,math}} &:= \text{EnumNode}_R^{\text{audit,claim,math}}(m), \quad m \in [4].\end{aligned}$$

The semantic nodes are enumerated as the five initial components in Table 83, the 13 reference claims, and the four mathematical object types, in the index order of each table. Let $\text{ClaimIndex}_R^{\text{init}}(k)$ be the corresponding claim indices in that table. Using Definitions 7.101 and 7.102, set

$$\begin{aligned}\text{Node}_R^{\text{audit,semantic}} &= \text{SemanticNodes}_R(22), \\ \text{ClaimNodeLift}_R(5; 13), \quad \text{MathNodeLift}_R(18; 4), \\ \text{StdSemLift}_R(5; 13).\end{aligned}$$

The first five components are projections of non-temporal carriers, and the corresponding claim indices specify their semantic-support relations. They are not counted as additional claim nodes of the common audit boundary.

Audit-Scope Identifier 8-SC1.

Audit Scope The audit domain stated under Bibliographic Identification.

Original-Source Location Identifier 8-L1.

Original-Source Location The audit domain stated under Bibliographic Identification and the original-source location columns of the respective evidence tables.

Formal Objects

Object Identifier 8-F1.

Object Name Formal objects in the original source.

Object Type Formal object.

Formal Content The individual contents of the formal objects are given in Definitions 8.173 and 8.174.

Original-Source Location Identifier Set 8-L1.

Assumptions

Assumption Identifier 8-A1.

Assumption Content The common age-independent decay probability and small-interval proportionality in claims 4 and 5 are assumed; equation (2) concerns a population with a large initial number of atoms. Results 1–4 use this exponential-decay model with constants $N_0 > 0$ and $\lambda > 0$, and with $t \geq 0$. In the probabilistic model, the smooth function $N(t)$ in equation (2) corresponds to the expected number of atoms.

Original-Source Location Identifier Set 8-L1.

Non-Temporal Physical Inputs

Input Identifier 8-P1.

Input Name Non-temporal physical inputs in the original source.

Input Type Non-temporal physical input.

Input Content The individual contents of the non-temporal physical inputs are given in Definitions 8.173 and 8.174. The non-temporal initial components comprise the atoms, events, energy, and count records specified in the respective rows. Their relations to age, initial time, and elapsed time in the corresponding claims are excluded from this non-temporal projection.

Original-Source Location Identifier Set 8-L1.

Time Content

Time-Content Identifier 8-T1.

Time-Content Name Time contents in the original source.

Time-Content Type Time content.

Time-Content Specification The individual contents of the time contents are given in Definition 8.175.

Original-Source Location Identifier Set 8-L1.

Formal Input Elements

Formal-Component Identifier 8-C1.

Formal-Component Name Formal-evidence environment of the original source.

Formal-Component Type Formal component.

Formal-Component Text The individual contents of the formal input elements are given in Definitions 8.173 and 8.174.

Original-Source Support Identifier Set 8-L1.

Table 83: Non-Temporal Initial Components in the Curtiss 1949 Reference

k	Semantic component	Corresponding claim indices
1	Population of radioactive atoms belonging to the same substance	$\{1, 4\}$
2	Spontaneous atomic transformation event	$\{2, 4\}$
3	Radiant energy accompanying transformation	$\{3\}$
4	Record of the initial number of atoms	$\{6, 7\}$
5	Record of the number of untransformed atoms	$\{8, 9\}$

Definition 8.174 (Claims in the Curtiss 1949 Reference). The 13 reference claims are given in the following table.

Table 84: Claims in the Curtiss 1949 Reference

p	Claim	Original-source location
1	Population of atoms belonging to the same substance	§I.2, first paragraph
2	Spontaneous transformation of an atom into another element	§I.2, first paragraph
3	Emission of radiant energy accompanying transformation	§I.2, first paragraph
4	Common decay probability independent of age	§I.2, derivation of equation (1)
5	Proportionality between decay probability within a small time and interval length	§I.2, derivation of equation (1)
6	Record of the initial number of atoms	Immediately before equation (2)
7	Correspondence between the initial number of atoms and initial time	Immediately before equation (2)
8	Record of the number of untransformed atoms	Equation (2)

p	Claim	Original-source location
9	Correspondence between the number of untransformed atoms and elapsed time	Equation (2)
10	Empirical establishment of the exponential-decay law	Immediately after equation (2)
11	Specification of the half-life	Immediately after equation (3)
12	Specification of the mean life	Immediately after equation (3)
13	Differentiation of the exponential-decay equation by elapsed time	Equations (2)–(3)

Table 85: Mathematical Object Types in the Curtiss 1949 Reference

m	Mathematical object type	Original-source location
1	Exponential-decay equation	Equation (2)
2	Differential decay-rate equation	Equation (3)
3	Relation between half-life and the decay constant	Immediately after equation (3)
4	Relation between mean life and the decay constant	Immediately after equation (3)

Definition 8.175 (Time Contents in the Curtiss 1949 Reference). The seven time contents are given in the following table. In the source, p is the probability of decay within an interval Δ . In the small-interval limit using $p = \lambda\Delta$, λ is the decay rate; it is distinct from the finite-interval probability $1 - e^{-\lambda\Delta}$ in the same exponential-decay model.

Table 86: Time Contents in the Curtiss 1949 Reference

i	Time content	Original-source location
1	Sufficiently small time interval	§I.2, derivation of equation (1)
2	Unit time	§I.2, derivation of equation (1)
3	Constant decay rate λ (inverse time)	§I.2, equations (1)–(3)
4	Elapsed time	Equations (1)–(3)
5	Half-life	Immediately after equation (3)
6	Mean life	Immediately after equation (3)

i	Time content	Original-source location
7	Time-differentiation operation	Equation (3)

Dependent Inputs

Index Identifier S08-XI-DI1.

Existing-Evidence Reference Set Definition 8.176.

Dependence and Transfer Elements

Definition 8.176 (Time-Content Transfers in the Curtiss 1949 Reference). The four time-content transfers are given in the following table.

Table 87: Time-Content Transfers in the Curtiss 1949 Reference

Transfer	Dependence or transfer	Reference claims	Time input	Time output	Original-source location
1	Specification of the decay rate using small-interval proportionality and unit time	$\llbracket 1, 5 \rrbracket$	$\{1, 2\}$	$\{3\}$	§I.2, derivation of equation (1)
2	From the decay constant to half-life	$\llbracket 4, 9 \rrbracket \cup \{11\}$	$\{3\}$	$\{5\}$	Immediately after equation (3)
3	From the decay constant to mean life	$\llbracket 4, 9 \rrbracket \cup \{12\}$	$\{3\}$	$\{6\}$	Immediately after equation (3)
4	From elapsed time to the time-differentiation operation	$\{13\}$	$\{4\}$	$\{7\}$	Equations (2)–(3)

All four transfers take one or more existing time contents as inputs before outputting time content. Zero transfers output time content from non-temporal initial components alone.

Conditions of Formation

Index Identifier S08-XI-FC1.

Existing-Evidence Reference Set Definition 8.176.

Original-Source Support

Index Identifier S08-XI-SS1.

Existing-Evidence Reference Set Definition 8.176.

Results

Definition 8.177 (Results and Preservation Conditions in the Curtiss 1949 Reference). The four reference-specific results, reference claims, time presuppositions, original-source locations, and preservation conditions are given in the following table. On printed page 1, equation (3) gives the middle term as $\lambda N_0 e^{-\lambda t}$ but the final term as $-\lambda N$; the following sentence also states that the negative sign indicates a decrease in the number of atoms. Differentiating $N = N_0 e^{-\lambda t}$ in equation (2) with respect to t , with constants $N_0 > 0$ and $\lambda > 0$, requires a minus sign in the middle term. The half-life derivation on the same page prints $\log_e 2 = -\lambda T$, whereas the preceding equality $N/N_0 = 1/2 = e^{-\lambda T}$ implies $\log_e(1/2) = -\lambda T$. Distinguishing these printed sign inconsistencies, this audit adopts the following expressions for Results 2 and 3, respectively:

$$\frac{dN}{dt} = -\lambda N_0 e^{-\lambda t} = -\lambda N, \qquad T = \frac{\log_e 2}{\lambda}$$

The source’s final expression $T = 0.693/\lambda$ corresponds to an approximation to this half-life.

Table 88: Results, Dependences, and Preservation Conditions in the Curtiss 1949 Reference

Result	Content	Reference claims	Time presuppositions	Original-source location	Preservation condition
1	Exponential-decay law	$\llbracket 1, 10 \rrbracket$	$\{3, 4\}$	Equation (2)	Retain the population of atoms of the same substance, constant decay rate, and statistical domain of application.
2	Differential decay rate	$\llbracket 1, 9 \rrbracket \cup \{13\}$	$\{3, 4, 7\}$	Equation (3)	Retain the exponential-decay equation, elapsed time, and conditions of differentiation.

Result	Content	Reference claims	Time presuppositions	Original-source location	Preservation condition
3	Relation between half-life and the decay constant	$\llbracket 4, 9 \rrbracket \cup \{11\}$	$\{3, 5\}$	Immediately after equation (3)	Retain the exponential-decay model and $\lambda > 0$; obtain the half-life from $N(T)/N_0 = 1/2$.
4	Relation between mean life and the decay constant	$\llbracket 4, 9 \rrbracket \cup \{12\}$	$\{3, 6\}$	Immediately after equation (3)	Retain the exponential-decay model and $\lambda > 0$; take the mean over $t \in [0, \infty)$.

Result Dependence

Index Identifier S08-XI-RD1.

Existing-Evidence Reference Set Definition 8.177.

Grounds of Formation

Index Identifier S08-XI-RG1.

Existing-Evidence Reference Set Definition 8.177.

Evidence Hierarchy The number of objects newly measured directly by this reference within the audit domain is zero. The establishment of the exponential-decay law through radioactivity measurements is a technical reference statement concerning existing measurements.

Generation Judgment Judgment values: LCTR-complete-generation judgment: 0; reference-internal evidence-conditional local-reach judgment: 0; remaining primitive time roots: 7.

Reference-Internal Evidence-Conditional Local Reach The reference-internal evidence-conditional local-reach judgment is 0, and seven primitive time roots remain after that judgment.

Result Dependence There are fourteen root–result dependences, of which two are necessary dependences.

Evidence Effect The formal evidence supports the relations among the exponential-decay law, differential decay rate, half-life, and mean life. Its empirical and technical validity does not imply that unit time, elapsed time, time differentiation, half-life, and mean life have been generated from non-temporal inputs.

Result-Preservation Domain The four results are preserved insofar as their time-content presuppositions are generated in the LCTR generation sequence and the conditions in Table 88 are satisfied.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The seven time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as established time inputs. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment is grounded in the comparison between the use of each time content in a law operation or result derivation and the absence, before that use, of the corresponding generation or establishment condition as an input, formation condition, evaluation condition, or invalidation condition.

Evidence of Actual Invalidation

Evidence of Actual Invalidation The complete judgment and remaining-root judgment concern physical-time qualification and are distinct from the actual invalidation of a specific condition that determines a result’s value or domain of application. The original-source formal evidence and LCTR audit evidence for this reference do not establish the actual invalidation of such a target condition. The actual-invalidation judgment is “indeterminate from original-source evidence”; actual invalidation is not inferred from the backgrounding of the grounding obligation alone.

Downstream Reach

Downstream Reach The reach to primitive-time-root-dependent results is determined for this reference by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because the actual invalidation of a specific condition has not been established, the judgment of actual reach from the invalidation position to a result is “indeterminate from original-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair Scope The repair targets based on structural reach are the corresponding time content and the generation or establishment conditions required before its use. The recalculation scope is limited to results for which actual invalidation is independently established and which are dependency-reachable from that invalidation position. Because actual invalidation is indeterminate in this reference, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from the repaired inputs, or their establishment domain is restricted. Values after recalculation are determined as audit results.

Audit Conclusion

Definition 8.178 (Finite Dependence Graph of the Curtiss 1949 Reference). The generation-initial set and source formal-evidence environment are specified by the claim-boundary index pair of Definition 7.98,

$$\text{ClaimBoundaryIdx}_R := (\{1, 2, 3, 6, 8\}, [13]) .$$

The initial set contains claims 1, 2, 3, 6, and 8, corresponding to non-temporal carriers; the evidence environment contains all 13 reference claims. By Definition 7.105, the reference-claim indices in the transfer and result tables are realized as support from physical-formal claim nodes. The common audit boundary comprises 17 claim nodes—13 reference claims and four mathematical objects— together with seven time-content nodes and four result nodes, giving 28 nodes and eight edges.

Proposition 8.179 (Complete Time-Use Audit of the Curtiss 1949 Reference). The generated time contents under the complete judgment and the reached time contents under the local-reach judgment both number zero. All seven time contents remain primitive time roots, and each of the four results depends on at least one of them.

Proposition 8.180 (Minimal Primitive-Time-Root Support in the Curtiss 1949 Reference). The minimal supports of the four results are given in the following table. The total number of family members is eight.

Table 89: Minimal Primitive-Time-Root Support in the Curtiss 1949 Reference

Result	Minimal support-index family
1	$\{1, 2, 4\}, \{3, 4\}$
2	$\{1, 2, 4\}, \{3, 4\}$
3	$\{1, 2\}, \{3\}$
4	$\{1, 2\}, \{3\}$

Individual Statements Statement Evidence 1

Verbatim Original:

the law of exponential decay

The source location is Curtiss, *Measurements of Radioactivity*, §I.2, immediately after equation (2). The minimal context is the location that names the exponential-decay equation. The attributing source of the statement is Leon F. Curtiss.

Statement Evidence 2

Verbatim Original:

λ is the disintegration constant

The source location is Curtiss, *Measurements of Radioactivity*, §I.2, immediately after equation (2). The minimal context is the location that names λ . The attributing source of the statement is Leon F. Curtiss.

Statement Evidence 3

Verbatim Original:

the half-value period, T

The source location is Curtiss, *Measurements of Radioactivity*, §I.2, immediately after equation (3). The minimal context is the location that names the time in which the initial activity is reduced by half. The attributing source of the statement is Leon F. Curtiss.

Statement Evidence 4

Verbatim Original:

the mean life, L

The source location is Curtiss, *Measurements of Radioactivity*, §I.2, immediately after equation (3). The minimal context is the location that names the mean life. The attributing source of the statement is Leon F. Curtiss.

Common Evidence with Domain of Application

Target State-ment	Statement Object	Predicate	Explicit Domain of Application
S08-Q001	The radioactive-decay relation in equation (2)	Is called the law of exponential decay.	The exponential-decay equation in §I.2 and its naming immediately thereafter.
S08-Q002	λ	Is the disintegration constant.	Equation (2) in §I.2 and the naming of its symbol immediately thereafter.
S08-Q003	T	Is the half-value period.	The naming immediately after equation (3) in §I.2 of the time in which the initial activity is reduced by half.
S08-Q004	L	Is the mean life.	The naming of mean life immediately after equation (3) in §I.2.

Individual Audit

Interpretive Possibilities The four quotations name the law of exponential decay, the decay constant, the half-life, and the mean life. The surrounding context links these names to the decay equation and its derivation, which use unit time and elapsed time. The four quotations supply no procedure for generating unit time or elapsed time.

Independent Grounds The four statements directly give the names of the law and quantities; the surrounding equations and derivations give the relations among the quantities.

Evidence-Effect Boundary The four quotations directly support the names and correspondences of the exponential-decay law, decay constant, half-life, and mean life. The transition from those names and their technical use to the physical generation and

physical-time qualification of unit time, elapsed time, time differentiation, half-life, and mean life lies beyond the evidence-effect boundary under Definition 7.115.

Corresponding Formal Evidence The first statement corresponds to the name of the exponential-decay law treated in claim component 10 of Definition 8.174 and in result 1 of Definition 8.177. The second statement corresponds to the name of the decay constant used in time content 3 of Definition 8.175. The third statement corresponds to the name of the half-life treated in claim component 11, time content 5, and result 3; the fourth corresponds to the name of the mean life treated in claim component 12, time content 6, and result 4.

Within-Reference Comparison In the source, the decay constant in the exponential-decay equation and the half-life and mean life derived from it are linked by the names in the four quotations. Consistency between these names and the surrounding equations does not supply the physical generation of the time inputs they use.

Common Audit with Domain of Application The names in the four statements are distinguished from the relations among quantities given by the surrounding equations, and the evidence-effect boundary stated above applies.

8.19.9 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 9.

Source Name Primitive-Time-Root Audit of Dirac 1958

Bibliographic Identification Let P. A. M. Dirac, *The Theory of Gravitation in Hamiltonian Form*, *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences* 246(1246), 333–343 (1958), doi:10.1098/rspa.1958.0142 be $R = \text{Ref}_{\text{Dir58GH}}$. The audit covers the abstract, Introduction, Notation, The ϕ equations, The modified Lagrangian, The general equation of motion, The Hamiltonian, The weak-field approximation, Gravitational waves, Conclusion, and References.

Original-Source Location Identifier 9-L1.

Original-Source Location Content The abstract, Introduction, Notation, The ϕ equations, The modified Lagrangian, The general equation of motion, The Hamiltonian, The weak-field approximation, Gravitational waves, Conclusion, and References. The locations of the respective time contents and physical-input claims are given in Tables 91 and 94.

Audit-Scope Identifier 9-SC1.

Audit-Scope Content The audit domain stated under Bibliographic Identification.

Formal Objects

Object Identifier 9-F1.

Object Name Formal objects.

Object Type Formal object.

Formal Content Definitions, assumptions, derivations, and approximate calculations concerning the Hamiltonian form of general relativity, spacetime decomposition, canonical variables, primary and secondary constraints, surface deformations, the weak-field approximation, gravitational-wave modes, and families of classical solutions.

Original-Source Location Identifier Set 9-L1.

Assumptions

Assumption Identifier 9-A1.

Assumption Content Surfaces and canonical form. The state-carrying surface and the surfaces of constant x^0 are spacelike, and the spatial metric is nondegenerate. The matter Lagrangian contains no metric derivatives. Boundary variations are fixed when adding a total divergence, and the variable subject to a surface deformation is independent of $g_{\mu 0}$ (pp. 333–338).

Original-Source Location Identifier Set 9-L1.

Assumption Identifier 9-A2.

Assumption Content Weak field and point particles. In the weak-field expansion, metric deviations from the flat metric, their derivatives and the momenta are first order in a small quantity ϵ in suitable coordinates; terms of order ϵ^2 and higher are discarded in the constraints, and terms of order ϵ^3 and higher in the Hamiltonian. In the model that sums point-particle contributions, interactions are purely gravitational. The particle expressions for the matter terms are derived without the weak-field approximation, but the source excludes their use in the exact theory because a point source produces a singularity at which the field equations cannot be satisfied accurately (pp. 340–341).

Original-Source Location Identifier Set 9-L1.

Assumption Identifier 9-A3.

Assumption Content Fourier modes and energy. The spatial Fourier coefficients of real fields at opposite wave vectors are complex conjugates, and the mode decomposition using inverse wave numbers applies to nonzero wave vectors. The momentum-constraint reduction in the gravitational-wave section treats vacuum or a regime in which the momentum of slowly moving matter is negligible. Energy is identified with the weak-field Hamiltonian, and the derivation of Newtonian attraction assumes slowly moving particles. The proposed elimination by a contact transformation belongs to the weak-field regime and does not require slowly moving matter (pp. 341–343).

Original-Source Location Identifier Set 9-L1.

Non-Temporal Physical Inputs

Input Identifier 9-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content The gravitational field, matter degrees of freedom, action density, metric components, canonical gravitational components, constraint relations, spatial Fourier modes, and physical carriers of the fields are taken as inputs distinct from time content.

Original-Source Location Identifier Set 9-L1.

Time Content

Time-Content Identifier 9-T1.

Time-Content Name Time content.

Time-Content Type Time content.

Time-Content Specification The time contents are surface variation representing temporal evolution, given-time slices and their order, the temporal coordinate x^0 , differentiation with respect to it, and the normal–tangential decomposition of surface displacement. State values, the metric, canonical variables, constraints, approximation conditions, wave modes and classical-family correspondences are given as auxiliary objects in Table 92.

Original-Source Location Identifier Set 9-L1.

Time-Content Identifier 9-T2.

Time-Content Name Time contents used in the original source.

Time-Content Type Time content.

Time-Content Specification The five time contents in Definition 8.181.

Original-Source Location Identifier Set 9-L1.

Definition 8.181 (Time Contents and Auxiliary Objects in the Dirac 1958 Reference). For $R = \text{Ref}_{\text{Dir58GH}}$, let $n_R = 5$. The tables give the time contents and auxiliary objects. Source-location keys share the location mapping following Table 94.

Table 91: Temporal contents and source locations

Group	Time content i	Temporal content	Auxil- iary index	Source location key
1	1	Variation of a spacelike surface representing temporal evolution	None	4,10
2	2	Spacelike slices corresponding to given times and their order ($x^0 = c$ in the adopted representation)	None	4,6,10
3	3	The temporal coordinate parameter x^0	None	4,6,10
4	4	Time differentiation with respect to x^0	None	5,9,10

Group	Time content i	Temporal content	Auxil- iary index	Source location key
5	5	Decomposition of a surface displacement into normal and tangential directions	None	10

Table 92: Auxiliary objects

a	Auxiliary object	Source location key
1	The Hamiltonian and its equations of motion	4,12,13
2	State values specified at a given time	4
3	State values and dynamical variables on a spacelike hypersurface	4,10
4	Spatial coordinates, metric components and surface geometry	1,6,10
5	Metric-velocity values and canonical momenta	5,6,9
6	Coordinate transformations and freedom in metric velocities	7
7	Changes of action density by a total divergence and invariance of the equations of motion	8
8	Arbitrary metric components $g_{\mu 0}$ and constraint coefficients	10,13
9	Hamiltonian constraints and first-class consistency	7,11,13
10	Arbitrary functions in the general solution	13
11	The condition of vacuum or slowly moving matter with negligible matter momentum	3,18
12	Gravitational-wave modes, polarizations and energy	3,17
13	The Newtonian potential and gravitational self-energy in the slow-motion approximation	18
14	The interpretation of a physical state as a family of solutions	21
15	Families of classical solutions related to a common Hamilton principal function	21
16	The interpreted correspondence between classical solution families and quantum wave functions	21
17	The methodological requirement of canonical form before quantization	4

Definition 8.182 (Physical and Formal Input Claims in the Dirac 1958 Reference). Let $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(48)$. Claims 1–4 are nontemporal physical initial components; claims 5–48 correspond to 44 source claims. These 48 claims form the source-formal evidence environment. A dependency key d denotes a pair of time-content and auxiliary-object index sets; ℓ is a source-location key.

Table 93: Claim-dependency keys

d	Time indices	Auxiliary indices
1	$\emptyset$	$\{4\}$
2	$\emptyset$	$\emptyset$
3	$\emptyset$	$\{1, 17\}$
4	$\{2\}$	$\{2\}$
5	$\{2\}$	$\{3, 4\}$
6	$\{2, 3\}$	$\{3, 4\}$
7	$\{3, 4\}$	$\{4, 5\}$
8	$\{3, 4\}$	$\emptyset$
9	$\{2, 3\}$	$\{4\}$
10	$\llbracket 2, 4 \rrbracket$	$\{4, 5\}$
11	$\{3, 4\}$	$\{6\}$
12	$\{3, 4\}$	$\{6, 9\}$
13	$\{3, 4\}$	$\{4, 5, 7\}$
14	$\llbracket 2, 4 \rrbracket$	$\{4, 5, 9\}$
15	$\llbracket 3 \rrbracket$	$\{3, 4\}$
16	$\llbracket 2, 5 \rrbracket$	$\{1, 3, 4\}$
17	$\llbracket 2, 4 \rrbracket$	$\{8, 9\}$
18	$\{3, 4\}$	$\{8, 9\}$
19	$\llbracket 2, 5 \rrbracket$	$\{1, 4, 5, 8, 9\}$
20	$\llbracket 2, 5 \rrbracket$	$\{1, 8, 9\}$
21	$\llbracket 2, 5 \rrbracket$	$\{1, 4, 5, 9\}$
22	$\{3, 4\}$	$\{8, 10\}$
23	$\llbracket 2, 4 \rrbracket$	$\{1, 4, 5\}$
24	$\llbracket 2, 4 \rrbracket$	$\{1, 4, 5, 8, 9\}$
25	$\{3, 4\}$	$\{1\}$
26	$\emptyset$	$\{4, 5\}$
27	$\llbracket 2, 4 \rrbracket$	$\{4, 5, 9, 11, 12\}$
28	$\llbracket 2, 4 \rrbracket$	$\llbracket 11, 13 \rrbracket$
29	$\{2, 3\}$	$\{1\}$
30	$\{3, 4\}$	$\{15\}$

Table 94: Physical and formal claims

p	Claim content	d	ℓ
1	Physical carrier of the gravitational field	1	1
2	Physical carriers of matter degrees of freedom and point-particle sources	2	2
3	Physical carriers of metric-field components	1	1
4	Field carrying gravitational-wave modes	2	3
5	Hamiltonian form prior to canonical quantization	3	4
6	State at a given time	4	4
7	Relativistic state on a spacelike three-dimensional hypersurface	5	4

p	Claim content	d	ℓ
8	Hypersurface representation with constant x^0	6	4
9	Einstein gravitational action density	7	5
10	Supplementary matter action density	8	5
11	Dynamical coordinates of the metric	9	6
12	x^0 velocities	7	5
13	Four primary constraints	10	5
14	Coordinate freedom	11	7
15	First-class character derived from coordinate freedom	12	7
16	Modification of the action density by a total divergence	13	8
17	Modified primary constraints $p^{\mu 0} \approx 0$	14	8
18	Inversion of six velocities and six momenta	7	9
19	Arbitrary normal surface displacement	15	10
20	Arbitrary tangential surface displacement	15	10
21	Hamiltonian as generator of normal deformations	16	10
22	Hamiltonian as generator of tangential deformations	16	10
23	Four secondary constraints	17	11
24	First-class consistency of the equations of motion	18	11
25	Gravitational Hamiltonian density	19	12
26	Matter Hamiltonian density	20	12
27	Separation of the gravitational and matter Hamiltonian densities	19	12
28	Exact Hamiltonian constraints	21	13
29	Six canonical gravitational components	10	13
30	Arbitrary $g_{\mu 0}$ coefficients	22	13
31	Weak-field approximation	10	14
32	Ordinary Hamiltonian	23	14
33	Combination of constraints	17	14
34	Composition relation between the ordinary Hamiltonian and a combination of constraints	24	14
35	Point-particle contribution	25	15
36	Limit of singular sources	1	15
37	Spatial Fourier decomposition	26	16
38	Combination of four coordinate-dependent modes	27	17
39	Combination of two coordinate-invariant modes	27	17
40	Two positive-energy gravitational-wave modes	27	17
41	Newtonian attraction	28	18
42	Negative gravitational self-energy mode	28	18
43	Proposed elimination by a weak-field contact transformation	14	19
44	Loss of four-dimensional symmetry in the simplified Hamiltonian form adopted in this reference	29	20
45	Family of classical solutions	30	21
46	Common Hamilton–Jacobi function	30	21
47	Correspondence between a family of classical solutions sharing a Hamilton–Jacobi function and a quantum wave function	30	21

p	Claim content	d	ℓ
48	The relation from surface variation to differentiation in the x^0 direction, $\dot{\eta} = \int \xi_0 d^3x$	15	10

Define the source-location keys used in the tables as follows.

- 1 pp. 334–335, metric and Einstein action.
- 2 p. 335, matter Lagrangian; p. 341, Eqs. (45)–(50) and the limitation on point-particle sources.
- 3 pp. 341–343, Gravitational waves.
- 4 p. 333, Introduction.
- 5 pp. 335–336, Eqs. (8)–(12) and primary constraints.
- 6 pp. 334–335, Notation and metric velocities.
- 7 p. 336, Eq. (13) and the first-class argument.
- 8 p. 336, Eqs. (14)–(15), The modified Lagrangian.
- 9 p. 337, Eqs. (21)–(22).
- 10 p. 338, Eqs. (23)–(29).
- 11 p. 338, Eq. (30) and the consistency argument immediately following it.
- 12 pp. 338–339, Eqs. (28)–(32).
- 13 p. 340, Eqs. (40)–(41) and the paragraph immediately following them.
- 14 p. 340, The weak-field approximation, Eqs. (42)–(44).
- 15 p. 341, Eqs. (45)–(50) and the limitation on singular sources.
- 16 pp. 341–342, Eq. (52) and the reality condition on Fourier coefficients.
- 17 p. 342, Eqs. (53)–(56) and the interpretation of modes.
- 18 pp. 342–343, the density term in Eq. (56) and the discussion of Newtonian attraction.
- 19 p. 343, the first paragraph, on elimination by a contact transformation.
- 20 p. 333, the simplified surface representation; p. 343, Conclusion.
- 21 p. 343, Conclusion.

Here $\approx$ denotes weak equality on the constraint surface.

Definition 8.183 (Semantic Components and Generation-Initial Set in the Dirac 1958 Reference). Let $\text{Node}_R^{\text{audit,semantic}} = \text{Node}_R^{\text{audit,claim}}$. Set the initial-set and evidence-environment pair to $\text{InitEnv}_R^{\text{audit}} = \text{InitEnv}_R^{\text{phys}} = \text{ClaimPrefix}_R(4)$.

Table 95: Generation-initial components

a	Physical carrier	Claim index
1	Physical carrier of the gravitational field	$\{1\}$
2	Physical carriers of matter degrees of freedom and point-particle sources	$\{2\}$
3	Physical carriers of metric-field components	$\{3\}$
4	Field carrying gravitational-wave modes	$\{4\}$

Formal Input Elements

Formal-Component Identifier 9-C1.

Formal-Component Name Formal-evidence environment of the original source.

Formal-Component Type Formal component.

Formal-Component Text The individual contents of the formal input elements are given in Definition 8.182.

Original-Source Support Identifier Set 9-L1.

Dependence Inputs

Dependence Inputs The two transfers take the supplied surface variation, spacelike slices and their order, and the temporal coordinate x^0 as time inputs, with surface variables and metric components as auxiliary inputs.

Dependence and Transfer Elements

Definition 8.184 (Time-Content Transfers in the Dirac 1958 Reference). Let $\text{ld}\mathbf{x}_R^{\text{audit,transfer}} = [2]$. Inputs and outputs in the table are pairs of time-content and auxiliary-object index sets; formal support is a claim-index set. Each row is mapped to the corresponding premise and output sets by Definition 7.105.

Table 96: Temporal-content transfers

j	Input (time; auxiliary)	Output (time; auxiliary)	Formal support	Transfer content
1	$([3]; \{3, 4\})$	$(\{5\}; \emptyset)$	$\{8, 11, 19, 20\}$	Decompose a given surface displacement into normal and tangential directions using the metric
2	$([3]; \{3, 4\})$	$(\{4\}; \emptyset)$	$\{48\}$	Obtain time differentiation in the x^0 direction from the response to a surface variation

Transfer Events and Outputs Transfer 1 decomposes the given surface displacement into normal and tangential directions; transfer 2 expresses the response to a surface variation as time differentiation in the x^0 direction.

Validity Conditions

Conditions under Which the Dependences or Transfers Apply The surface is spacelike, the spatial metric is nondegenerate, and the surface variable η is independent of $g_{\mu 0}$. Both transfers require existing time content, so zero transfers output time content from nontemporal physical inputs alone.

Original-Source Support

Original-Source Support for Each Dependence and Transfer Equations (23)–(26) on p. 338 support the transfers. Transfer 1 uses the normal and component decomposition in Equations (25)–(26); transfer 2 uses the relation between surface variation in Equation (23) and differentiation with respect to x^0 in Equation (24).

Results

Definition 8.185 (Results in the Dirac 1958 Reference). Define the result index set by $\text{ldx}_R^{\text{audit,result}} = [20]$. The correspondence between results and original-source locations is given in Table 97.

Source-location keys follow the location mapping after Table 94.

Table 97: Results and source locations

j	Result	Source location key
1	Hamiltonian reconstruction of Einstein gravity	5,6,7,8,9,10,11,12,13
2	Generalization from a state at a given time to a state on a spacelike hypersurface	4
3	Slicing with constant x^0 and metric velocities	4,5,6
4	Four primary constraints and four secondary constraints	5,8,11
5	First-class consistency based on coordinate freedom	7,11
6	Modification by a total divergence and invariance of the equations of motion	8
7	Six canonical gravitational components	9,13
8	Normal and tangential surface deformations	10
9	Separation of the gravitational and matter Hamiltonians	12
10	Exact Hamiltonian and arbitrary constraint coefficients	11,13
11	Weak-field expansion	14
12	Spatial Fourier-mode decomposition	16
13	Combination of four coordinate-dependent modes	14,16,17
14	Two radiative polarizations	14,16,17
15	Positive gravitational-wave energy	14,16,17
16	Newtonian attraction	15,16,17,18
17	Negative gravitational self-energy mode	15,16,17,18
18	Proposed elimination by a weak-field contact transformation	14,19
19	Author’s conclusion that four-dimensional symmetry is lost in the simplified Hamiltonian form adopted in this reference	20

j	Result	Source location key
20	Author's interpretation of the correspondence between families of classical solutions and quantum wave functions	21

Source-Specific Results The 20 results in Definition 8.185. The time-content inputs, supporting claims, and result output of each result are given in Table 98.

Result Dependence

Definition 8.186 (Result-Dependence Hyperedges in the Dirac 1958 Reference). The time-prerequisite, auxiliary-prerequisite and formal-support columns give the index sets of time contents, auxiliary objects and claims. Each row is mapped to the corresponding premise sets by Definition 7.105, with result output $\{j\}$.

Result 12 is a Fourier expansion of fields on three-dimensional space; results 14 and 15 concern wave polarization and energy. Result 11 assumes weak fields; results 13–15 additionally assume vacuum or negligible matter momentum; the Newtonian branch of results 16 and 17 assumes slowly moving particles. Result 18 is a proposed weak-field contact transformation without a slow-matter requirement.

Table 98: Result-dependency hyperedges

j	Time prerequisites	Auxiliary prerequisites	Formal support
1	[5]	$\{1, 3, 4, 5, 6, 7, 8, 9, 10\}$	$[3] \cup \llbracket 8, 30 \rrbracket$
2	$\{2\}$	$\llbracket 2, 4 \rrbracket$	$\{6, 7\}$
3	$\llbracket 2, 4 \rrbracket$	$\llbracket 3, 5 \rrbracket$	$\{8, 11, 12\}$
4	$\llbracket 2, 4 \rrbracket$	$\{4, 5, 8, 9\}$	$\{13, 17, 23\}$
5	$\llbracket 2, 4 \rrbracket$	$\{6, 8, 9\}$	$\{14, 15, 24\}$
6	$\{3, 4\}$	$\{4, 5, 7\}$	$\{16\}$
7	$\llbracket 2, 4 \rrbracket$	$\{4, 5\}$	$\{18, 29\}$
8	[5]	$\{1, 3, 4\}$	$\{19, 20, 21, 22, 48\}$
9	$\llbracket 2, 5 \rrbracket$	$\{1, 4, 5, 8, 9\}$	$\llbracket 25, 27 \rrbracket$
10	$\llbracket 2, 5 \rrbracket$	$\{1, 4, 5, 8, 9, 10\}$	$\{23, 24, 28, 30\}$
11	$\llbracket 2, 4 \rrbracket$	$\{1, 4, 5, 8, 9\}$	$\llbracket 31, 34 \rrbracket$
12	$\emptyset$	$\{4, 5\}$	$\{37\}$
13	$\llbracket 2, 4 \rrbracket$	$\{4, 5, 8, 9, 11, 12\}$	$\{31, 33, 37, 38\}$
14	$\llbracket 2, 4 \rrbracket$	$\{4, 5, 8, 9, 11, 12\}$	$\{31, 33, 37, 39, 40\}$
15	$\llbracket 2, 4 \rrbracket$	$\{1, 4, 5, 8, 9, 11, 12\}$	$\{31, 32, 33, 34, 37, 40\}$
16	$\llbracket 2, 4 \rrbracket$	$\{1, 4, 5, 9, 11, 12, 13\}$	$\{35, 37, 40, 41\}$
17	$\llbracket 2, 4 \rrbracket$	$\{1, 4, 5, 9, 11, 12, 13\}$	$\{35, 36, 37, 40, 42\}$
18	$\llbracket 2, 4 \rrbracket$	$\{1, 4, 5, 8, 9\}$	$\{31, 33, 34, 43\}$
19	$\{2, 3\}$	$\{1, 3, 4\}$	$\{8, 44\}$
20	$\{3, 4\}$	$\{1, 15\}$	$\llbracket 45, 47 \rrbracket$

There are 58 time-prerequisite occurrences, 91 auxiliary-prerequisite occurrences and 90 supporting-claim occurrences.

Proposition 8.187 (Audit Boundary Counts in the Dirac 1958 Reference).

$$\text{BoundaryCount}_R^{\text{audit}} = (73, 22).$$

Proof. The 48 claims, five time contents and 20 results give $48 + 5 + 20 = 73$; the two transfers and 20 result edges give $2 + 20 = 22$. $\square$

Grounds of Establishment

Grounds of Establishment of the Results Results 1–10 are based on the canonical arguments for changing the action density, reconstructing the Hamiltonian, analysing constraints and deforming surfaces. Results 11–17 are based on weak-field expansion, spatial Fourier decomposition and mode-energy calculations. Result 18 is a proposal for elimination by a contact transformation; result 19 is the author’s conclusion about symmetry in the simplified representation; result 20 is the author’s interpretation of the correspondence between classical solution families and quantum wave functions.

Evidence Hierarchy

Directly Measured Objects The number of directly measured objects newly acquired by this reference is zero.

Direct-Measurement Predicates The number of direct-measurement predicates newly acquired by this reference is zero.

Conditions and Domain of Direct Measurement The source gives theoretical constructions, approximate calculations, proposals and theoretical interpretations. It supplies zero source-specific direct-measurement inputs consisting of apparatus, samples, calibration, errors and measurement data.

Objects of Formal Generalization Results 1–10 apply to the Hamiltonian form of Einstein gravity, hypersurface states, constraints, and surface deformations. Results 11–18 apply to the weak-field expansion, Fourier modes, gravitational-wave polarizations, wave energy, Newtonian attraction, and contact transformations. Results 19–20 apply to the representation of four-dimensional symmetry in Hamiltonian form and to the correspondence between families of classical solutions and quantum wave functions.

Conditions and Domain of Formal Generalization Each result is subject to the time prerequisites, auxiliary prerequisites and supporting claims in Table 98, together with the applicable conditions in assumptions 9-A1–9-A3.

Generation Judgment **Judgment values:** LCTR complete-generation judgment: 0; source-internal evidence-conditioned local-reach judgment: 0; remaining primitive time roots: 5.

Generation Reach and Primitive Time Roots The number of time contents generated by generation reach from an initial set consisting solely of non-temporal physical inputs is zero. All 5 time contents in Definition 8.181 remain primitive time roots.

Source-Internal Evidence-Conditioned Local Reach The source-internal evidence-conditioned local-reach judgment is zero; 5 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ from Definition 7.108 and apply Definition 7.50 to it, and total over all results. This determines the primitive-time-root–result dependence relations and the necessary-primitive-time-root–result dependence relations.

Result Dependence The 19 results other than result 12 in Definition 8.185 depend on primitive time roots. Result 12 is a non-generated result reached from the source-formal-evidence environment. There are no generated results or results unreached after root addition. The minimal primitive-time-root support of each result is given by Proposition 8.189.

Evidence Effect

Evidence-Effect Scope The evidence for results 1–17 supports Hamiltonian reconstruction, primary and secondary constraints, surface deformations, weak-field expansion, Fourier decomposition, gravitational-wave energy and the Newtonian branch. Result 18 is supported as a proposed contact transformation, result 19 as a conclusion about symmetry in the simplified representation, and result 20 as an interpretation of the correspondence between classical solution families and quantum wave functions.

Evidence-Effect Boundary Establishment of the Hamiltonian form, constraint consistency, weak-field approximation and gravitational-wave-mode calculations does not imply that given-time slices and their order, a temporal coordinate, time differentiation, surface variation and its normal–tangential decomposition have been generated from nontemporal physical inputs. Nor do formal consistency and quantizability retroactively confer physical-time qualification on the five time contents. These transitions lie beyond the evidence-effect boundary in Definition 7.115.

Result-Preservation Domain

Result-Preservation Domain Each result is preserved when its temporal prerequisites are supplied by time representations in an LCTR generation sequence established before use, with the applicable conditions of assumptions 9-A1–9-A3, auxiliary objects and directions of dependence retained. Result 18 retains the force of a proposal, result 19 that of the author’s conclusion, and result 20 that of the author’s theoretical interpretation.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The 5 time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as already-established time inputs. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment follows by matching the use of each time content in a law operation or result derivation against the absence, before that use, of the corresponding generation or validity condition as an input, formation condition, evaluation condition, or failure condition.

Evidence of Actual Failure

Evidence of Actual Failure The complete judgment and the remaining-root judgment concern physical-time qualification; they are distinct from actual failure of a concrete condition governing a result's value or domain of application. Neither the original formal evidence nor the LCTR audit evidence in this reference establishes an actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence,” and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream Reach For each reference, the reach to primitive-time-root-dependent results is determined by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because actual failure of a concrete condition has not been established, the actual-reach judgment from that failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the objects of repair are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is limited to results for which actual failure is independently established and which are dependency-reachable from that failure location. Because actual failure is indeterminate in this reference, this repair-and-recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.188 (Primitive Time Roots and Result Dependence in the Dirac 1958 Reference).

In the component order of Definition 8.115, $\text{AuditPartIdx}_R = (\emptyset, [5]; \emptyset, \{12\}, [20] \setminus \{12\}, \emptyset)$. Moreover, $R \in \text{Ref}^{\text{audit}, \text{rootdep}}$ holds.

Proof. The generation-initial input in Definition 8.182 consists of four positive non-temporal physical-input claim components, and the source-formal-evidence environment consists of 48 physical- and formal-input claim components. Both transfer hyperedges in Definition 8.184 require time content. Adding the source-formal-evidence environment to the generation-initial input therefore reaches no time content, so all 5 time contents remain primitive time roots. Result 12 in Definition 8.186 has an empty time-content input, but its supporting claim 37 does not belong to the generation-initial input. Result 12 is therefore reached upon addition of the source-formal-evidence environment and is not a generated result. The other 19 results require time content and are reached when the 5 primitive time roots and the source-formal-evidence environment are supplied. The displayed index tuple follows from Definitions 7.48 and 8.115. The existence of primitive-time-root-dependent results gives membership in the reference set by Definition 7.52. $\square$

Proposition 8.189 (Minimal Primitive-Time-Root Support in the Dirac 1958 Reference). For each $q \in \text{ld}\mathbf{x}_R^{\text{audit,result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108, and the minimum and maximum cardinalities of those elements, are given in Table 99.

Table 99: Minimal Primitive-Time-Root Support in the Dirac 1958 Reference

Result indices	Support statistics
1,8	(1; 3)
2	(1; 1)
3–5,7,11,13–18	(2; 3)
6,20	(2; 2, 3)
9,10	(2; 3, 4)
12	(1; 0)
19	(1; 2)

Proof. Apply the update in Definition 7.46 to the two transfer hyperedges in Definition 8.184. After one scan applying the update to both transfers in their listed order, the support antichains of all time-content nodes are invariant under the second scan. Proposition 7.47 then gives the values corresponding to the time-content presuppositions in Table 98. There are 35 members of the minimal primitive-time-root support families, 71 primitive-time-root–result pairs and 37 necessary-primitive-time-root–result pairs. $\square$

Individual Statements Verbatim original (English):

The basic concept in Hamiltonian theory is that of a state at a given time. In a relativistic theory this is to be understood as the state on a general three-dimensional space-like surface in space-time.

The source location is Dirac, *The Theory of Gravitation in Hamiltonian Form*, *Proceedings of the Royal Society of London A* 246, p. 333, *Introduction*.

Verbatim original (English):

Any dynamical theory must first be put in the Hamiltonian form before one can quantize it.

The source location is p. 333, *Introduction*, in the same paper.

Verbatim original (English):

The symmetry between the four x 's is completely destroyed

The quoted expression refers to the four coordinates x .

The source location is p. 333, *Introduction*, in the same paper, in the paragraph describing the simplification obtained by choosing spacelike surfaces with constant x^0 .

Verbatim original (English):

I am inclined to believe from this that four-dimensional symmetry is not a fundamental property of the physical world.

The source location is p. 343, *Conclusion*, in the same paper.
Verbatim original (English):

a physical state does not correspond to an individual solution of the equations of motion, but to a family of solutions all related to the same Hamilton's principal function—it is such a family that corresponds to a wave function in the quantum theory, while the individual solution has no quantum analogue.

The source location is p. 343, *Conclusion*, in the same paper.

Objects of the Statements The state at a given time and its generalization to a spacelike surface; the Hamiltonian form preceding quantization; loss of symmetry among the four coordinates upon the choice of a surface with constant x^0 ; the physical status of four-dimensional symmetry; and the correspondence between families of classical solutions and quantum wave functions.

Original-Source Locations Quotations one through three are from *Introduction*, p. 333; quotations four and five are from *Conclusion*, p. 343.

Minimal Context Required for Interpretation The first quotation belongs to a context in which the state at a given time in Hamiltonian theory is understood, in a relativistic theory, as the state on a general three-dimensional spacelike surface in spacetime. The second quotation belongs to a context stating a formal requirement prior to quantization. The third quotation concerns the loss of symmetry among the four coordinates in the simplified scheme that considers only states on surfaces with constant x^0 . The same paragraph explains that choosing the surface independently of the coordinate system can preserve the symmetry, at the cost of more variables and more complicated equations. In the fourth quotation, the author tentatively infers a claim about the physical status of four-dimensional symmetry from its loss in this simplified scheme. The same conclusion acknowledges that individual solutions of the equations of motion exhibit four-dimensional symmetry and restricts its abandonment by Hamiltonian methods to their simplest form. The fifth quotation belongs to a context in which a physical state corresponds, not to an individual solution, but to a family of solutions related to the same Hamilton's principal function.

Attribution of the Statements Dirac is the author of all five quoted statements.

Predicates Assigned to the Objects by the Statements The first quotation identifies a state at a given time as the basic concept of Hamiltonian theory and states that its carrier domain in a relativistic theory is a general three-dimensional spacelike surface. The second quotation states that Hamiltonian form is a prerequisite for quantization. The third quotation states that symmetry among the four coordinates is completely destroyed under the adopted simplification. The fourth quotation states that the author is inclined to believe that four-dimensional symmetry is not a fundamental property of the physical world. The fifth quotation associates a physical state with a family of classical solutions and associates that family with a wave function in quantum theory.

Explicit Domains of Application of the Statements The five quotations apply to theoretical accounts of the Hamiltonian form of general relativity, canonical quantization, the choice of surfaces, the interpretation of four-dimensional symmetry, and the correspondence between families of classical solutions and quantum wave functions.

Individual Audit

Interpretation The first quotation identifies the basic concept of Hamiltonian theory as a state at a given time and, in relativity, understands it as a state on a general three-dimensional spacelike surface in spacetime. The second quotation requires Hamiltonian form before quantization for any dynamical theory. The third quotation states that choosing a surface of constant x^0 as a simplification removes the representation in which the four coordinates are treated equally. The same introduction also explains that symmetry can be retained in a formulation that selects the surface independently of the coordinate system. The fourth quotation takes the loss of symmetry in the simplified Hamiltonian representation as the starting point for the author's inclination to regard four-dimensional symmetry as not a fundamental property of the physical world. This is the author's qualified conjecture. The fifth quotation associates a physical state with a family of classical solutions related to the same Hamilton's principal function, states that this family corresponds to a quantum wave function, and says that an individual solution has no quantum analogue.

Independent Grounds The grounds for the five statements are the source's formulation of states at given times and on spacelike surfaces, its Hamiltonian reconstruction of gravitation, and its theoretical account of the correspondence between families of classical solutions and quantum wave functions. The introduction explains the relation between surface choice and the symmetry of the four coordinates; the conclusion states the author's conjecture about the physical status of that symmetry based on the simplified formulation.

Evidential Scope The first quotation supports the relation between a state at a given time and a state on a spacelike surface. The second supports Hamiltonian form as the prerequisite for quantization adopted by the author. The loss of symmetry in the third quotation belongs to the adopted simplification; the fourth quotation is a qualified conjecture that proceeds from this loss to a view about the physical world. The correspondence in the fifth quotation is between a family of solutions related to the same Hamilton's principal function and a quantum wave function. The five statements provide formal and conceptual theoretical accounts in which the given time, surface order and evolution variables serve as temporal inputs.

Corresponding Formal Evidence The first quotation corresponds to time content 2 and auxiliary objects 2 and 3 in Definition 8.181, components 6 and 7 in Definition 8.182, and result 2 in Definition 8.185. The second quotation corresponds to the prerequisite for quantization treated in auxiliary object 17 and component 5, and states the role of Hamiltonian form in result 1. The third quotation corresponds to the loss of symmetry under the adopted coordinate splitting treated in time contents 2 and 3, components 8 and 44, and result 19. The fourth quotation is the author's qualified conjecture about the physical status of symmetry, drawn from the loss of symmetry under the simplification treated in component 44 and result 19. The fifth quotation

corresponds to the relation between families of classical solutions and quantum wave functions in time contents 3 and 4, auxiliary objects 14–16, components 45–47, and result 20.

Within-Source Comparison The first and second statements concern the carrier domain of a state and the form required before quantization; the third and fourth concern surface choice and four-dimensional symmetry; the fifth concerns families of classical solutions and quantum states. These relations are formulated using given times, surface order and evolution variables. Propositions 8.188 and 8.188 give the generation assessment and result dependence of the time contents.

8.19.10 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 10.

Source Name Primitive-Time-Root Audit of Pound–Rebka 1960

Bibliographic Identification Let the original source R. V. Pound and G. A. Rebka, Jr., *Apparent Weight of Photons*, *Physical Review Letters* 4 (1960), 337–341 be $R = \text{Ref}_{\text{PR60AWP}}$. The audit domain comprises Mössbauer resonance, the vertical propagation path, configuration reversal, transmission counts by phase, Doppler calibration, temperature correction, weighted aggregation, frequency-shift estimation, and comparison with the weak-gravitational-field prediction.

Original-Source Location Identifier 10-L1.

Original-Source Location Content The 25 time contents and 16 auxiliary objects are specified in Tables 100 and 101; the 49 physical and formal input claims are specified in Table 102.

Audit-Scope Identifier 10-SC1.

Audit-Scope Content The audit domain stated under Bibliographic Identification.

Formal Objects

Object Identifier 10-F1.

Object Name Objects that use time.

Object Type Formal object.

Formal Content The 41 semantic objects are separated into 25 time contents and 16 auxiliary objects.

Original-Source Location Identifier Set 10-L1.

Assumptions

Assumption Identifier 10-A1.

Assumption Content The intrinsic source–absorber frequency difference is assumed to remain unchanged under configuration reversal (transfer 7 in Table 104; original source, p. 340, right column).

Source-Location Identifier Set 10-L1.

Non-Temporal Physical Inputs

Input Identifier 10-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content Table 103 specifies 15 non-temporal physical inputs. They are projections of carriers, material or state configurations, apparatus geometry, or temperature states; they exclude lifetimes, velocities, periods, event order, flow rates, and interpretation in terms of frequency.

Original-Source Location Identifier Set 10-L1.

Definition 8.190 (Time Contents and Auxiliary Objects in the Pound–Rebka 1960 Reference). For $R = \text{Ref}_{\text{PR60AWP}}$, specify 25 time contents corresponding to temporal quantities and relations used in the source. A tuple with explicitly identified components is treated as one item; the textual representation, input use, and output use of the same content do not constitute different time contents merely because their roles differ. Table 100 specifies the contents and source locations. Table 101 specifies 16 auxiliary correspondence rules and formal conditions. Quantities indexed by data population are evaluated separately for each population. The first-four-day example in Table I and the approximately ten-day full data set underlying the final result are not treated as the same input value.

Table 100: Time Contents and Original-Source Locations
in the Pound–Rebka 1960 Reference

i	Time content	Original-source location
1	Source and absorber resonance frequencies	p. 337
2	Order of gamma-ray emission and absorption	p. 337–338
3	Approximately 0.1-microsecond lifetime of the excited Fe-57 level	p. 337
4	Fractional resonance-line full width at half-height of 1.13×10^{-12}	p. 337
5	270-day radioactive-decay time scale of the Co-57 used in the source	p. 337
6	One-hour heating interval during source preparation	p. 337
7	Helium flow rate of approximately 30 liters per hour along the propagation path	p. 338
8	Frequency, phase, and velocity of sinusoidal source modulation	p. 338 Fig.1
9	Quarter-cycle windows near maximum velocity and at the opposite phase, and their corresponding counting rates	p. 338–339
10	Fractional frequency shift in each configuration inferred from phase-resolved count-rate differences and sensitivity calibration	p. 338–340
11	Order of configuration reversals and measurements with the source above and below	p. 338–340
12	Known constant velocity and the positive and negative Doppler shifts it imposes	p. 339

<i>i</i>	Time content	Original-source location
13	Calibrated sensitivity relating count-rate difference to fractional frequency shift	p. 339
14	Calibration intervals and their allocation of approximately one third of operating time	p. 339–340
15	Times of temperature records and counting periods over which they are averaged	p. 339
16	Temperature coefficient of frequency due to lattice vibrations	p. 339
17	Fractional frequency correction for the temperature difference in each counting period	p. 339–340
18	Fractional frequency shift in each configuration after temperature correction	p. 339–340
19	Measurement times and periods of repeated observations, and time series indexed by data population	p. 339–341
20	Configuration-specific weighted mean shifts in Table I	p. 340 Table I
21	Two-way fractional shift comparing rising and falling configurations within the same data population	p. 339–340
22	Predicted fractional frequency shift for the approximately 148-ft two-way height difference	p. 340
23	Source-reported ratio of measured to predicted frequency difference	p. 341
24	Sign relation expressing the frequency increase of falling radiation	p. 341
25	Approximately ten days of operation used to collect the final data	p. 341

Table 101: Auxiliary Objects and Supporting Claims in the Pound–Rebka 1960 Reference

<i>a</i>	Auxiliary object	Supporting claims
1	Correspondence between resonance frequencies and the measurement object	{3}
2	Sensitivity conditions supplied by lifetime, line shape, and linewidth	{9, 10}
3	Correspondence of emission and absorption with the propagation path	{11, 12, 15}
4	Correspondence between source modulation and phase	{18, 19}
5	Assignment of counts by phase	{16, 17, 20, 21, 37}
6	Doppler calibration rule using known velocity	{22, 23}
7	Correspondence between calibration intervals and operating records	[[39, 41]]
8	Correspondence between rising/falling configurations and records	{12, 13}
9	Correspondence among temperature averaging, coefficients, and frequency correction	[[27, 30]] $\cup$ {41}

a	Auxiliary object	Supporting claims
10	Weighted aggregation preserving data population and configuration	$\{42, 43\}$
11	Correspondence between acquisition duration and operating records	$\{39\}$
12	Correspondence among count difference, calibrated sensitivity, and frequency shift	$\{21, 23, 37, 38\}$
13	Configuration differencing under invariance of the intrinsic offset	$\llbracket 24, 26 \rrbracket \cup \{31, 33\}$
14	Conversion between two-way and one-way quantities in the symmetric configuration	$\{32\}$
15	Comparison of measured values, predictions, and the reported ratio	$\llbracket 34, 36 \rrbracket$
16	Sign correspondence among configuration, shift, and frequency change	$\llbracket 1, 3 \rrbracket \cup \{12, 13, 36\}$

Definition 8.191 (Physical and Formal Input Claims in the Pound–Rebka 1960 Reference). Enumerate 49 physical and formal input claims. Among Components 1–44, Component 32 specifies the mathematics of the symmetric conversion applied to the source’s two-way quantity; the other 43 components correspond to source claims. Components 45–49 establish nontemporal projections distinguished from source descriptions that contain temporal content.

$$\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(49).$$

Table 102 gives each component and its original-source location.

Table 102: Physical and Formal Input Claims in the Pound–Rebka 1960 Reference

Claim	Content	Original-source location
1	Weak terrestrial gravitational-potential difference	Introduction
2	Physical carrier of electromagnetic radiation	Introduction
3	Measurement of the apparent frequency of electromagnetic radiation	Introduction
4	Approximately 14.4 keV nuclear transition of Fe-57	Measurement principle
5	Co-57 source	Apparatus configuration
6	Fe-57 absorber	Apparatus configuration
7	Recoilless emission	Measurement principle
8	Recoilless absorption	Measurement principle

Claim	Content	Original-source location
9	Lorentzian Mössbauer resonance line from the approximately 14.4 keV, 0.1 μ s level of Fe-57, with fractional full-width at half-height 1.13×10^{-12}	Measurement principle
10	Use of the Mössbauer resonance line for measurement sensitivity	Measurement principle
11	One-way vertical distance of approximately 74 ft	Apparatus configuration
12	Radiation-upward and radiation-downward configurations	Configuration reversal
13	Reversal between the radiation-upward and radiation-downward configurations	Configuration reversal
14	Configuration difference corresponding to approximately 148 ft	Configuration reversal
15	Vertical propagation path with a helium-filled Mylar bag along most of its length	Apparatus configuration
16	Scintillation detector	Detection system
17	Counting of transmission events by the scintillation detector	Detection system
18	Source transducer	Modulation apparatus
19	Transducer-driven sinusoidal motion of the source	Modulation apparatus
20	Counting channel for each modulation phase	Phase-resolved counting
21	Recording of the count difference between phase windows as a measure of resonance-line asymmetry or frequency shift	Resonance response
22	Known constant-velocity motion	Calibration
23	Doppler calibration by constant-velocity motion	Calibration
24	Intrinsic resonance frequency of the source	Deviation separation
25	Intrinsic resonance frequency of the absorber	Deviation separation
26	Separation of the intrinsic resonance deviation between source and absorber	Deviation separation
27	Physical state of lattice motion	Temperature correction
28	Second-order Doppler contribution from lattice motion	Temperature correction
29	Temperature states of the source and absorber, and thermometers	Temperature monitoring
30	Monitoring of the temperature difference between source and absorber	Temperature monitoring

Claim	Content	Original-source location
31	Measurement difference between upward and downward configurations	Configuration difference
32	Symmetric mathematical conversion from two-way to one-way quantities	Conversion of the two-way quantity on p. 340
33	Measured bidirectional relative difference	Numerical result
34	Predicted bidirectional relative difference	Theoretical comparison
35	Ratio of measured value to predicted value	Theoretical comparison
36	Comparison of the weak-gravitational-field local-frequency relation with the measured value	Theoretical comparison
37	Detector counts as the directly observed quantity	Detection system
38	Frequency displacement as the calibrated inferred quantity	Shift estimation
39	Measurement times, periods, and sequence	pp. 339–341
40	Calibration intervals	p. 339
41	Order of correction	Measurement procedure
42	Statistical uncertainty	Statistical processing
43	Weighted aggregation	pp. 339–340
44	Historical attribution to Einstein	Introduction
45	Spatial configuration of the terrestrial gravitational field	Nontemporal projection of Claim 1
46	Configuration of the nuclear levels of Fe-57	Nontemporal projection of Claim 4
47	Nucleus–solid coupling configuration involved in recoilless emission	Nontemporal projection of Claim 7
48	Nucleus–solid coupling configuration involved in recoilless absorption	Nontemporal projection of Claim 8
49	Material and configuration of the lattice	Nontemporal projection of Claim 27

Definition 8.192 (Semantic Components and Generation-Initial Set in the Pound–Rebka

1960 Reference). Identify the 49 claim components one-to-one with the 49 semantic components and 49 boundary claim nodes bearing the same indices.

$$\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(49), \quad \text{ClaimNodeLift}_R(0; 49), \quad \text{StdSemLift}_R(0; 49).$$

Define the generation-initial set and source-formal-evidence environment by the index pair in Definition 7.98:

$$\text{ClaimBoundaryIdx}_R := ([14, 16] \cup [45, 49] \cup \{2, 5, 6, 11, 12, 18, 29\}, [49]).$$

Table 103 maps the 15 initial components to claim indices. Initial components identical to existing carrier or geometric claims are not counted again as separate nodes. Auxiliary conditions are retained in the corresponding supporting claims. Temporal prerequisites remain in the time-content columns of the transfer and result tables; auxiliary objects are not additional time contents or boundary node types.

Table 103: Generation-Initial Components in the Pound–Rebka 1960 Reference

k	Non-temporal physical input	Claim index
1	Spatial configuration of the terrestrial gravitational field	{45}
2	Physical carrier of electromagnetic radiation	{2}
3	Configuration of the nuclear levels of Fe-57	{46}
4	Physical carrier of the Co-57 source	{5}
5	Fe-57 absorber	{6}
6	Nucleus–solid coupling configuration involved in recoilless emission	{47}
7	Nucleus–solid coupling configuration involved in recoilless absorption	{48}
8	Vertical distance of approximately 74 ft	{11}
9	Spatial configurations for rising and falling radiation	{12}
10	Height difference corresponding to approximately 148 ft between the two configurations	{14}
11	Vertical propagation path with a helium-filled Mylar bag along most of its length	{15}
12	Scintillation detector	{16}
13	Source transducer	{18}
14	Material and configuration of the lattice	{49}
15	Temperature states of the source and absorber, and thermometers	{29}

Time Content

Time-Content Identifier 10-T1.

Time-Content Name Time contents in the original source.

Time-Content Type Time content.

Time-Content Specification The individual contents of the time contents are given in Definitions 8.190 and 8.193.

Original-Source Location Identifier Set 10-L1.

Formal Input Elements

Definition 8.193 (Formal Inputs in the Pound–Rebka 1960 Reference). Define the experimental setup by

$$\text{Setup}_{\text{PR}} := (\text{GammaSrc}_{\text{PR}}, \text{Absorber}_{\text{PR}}, \text{Path}_{\text{PR}}^{\text{vert}}, \\ \text{Transducer}_{\text{PR}}, \text{ScintDet}_{\text{PR}}, \text{TempMon}_{\text{PR}})$$

Let the radiation-upward and radiation-downward configurations be

$$\text{Config}_{\text{PR}}^{\uparrow}, \quad \text{Config}_{\text{PR}}^{\downarrow}$$

and let the transmission counts at modulation phase φ be

$$N_{\text{PR}}^{\uparrow}(\varphi), \quad N_{\text{PR}}^{\downarrow}(\varphi)$$

Let the directly observed quantity set and the inferred quantity set be

$$\mathcal{R}_{\text{PR}}^{\text{dir}}, \quad \mathcal{R}_{\text{PR}}^{\text{inf}}$$

and impose

$$\mathcal{R}_{\text{PR}}^{\text{dir}} \cap \mathcal{R}_{\text{PR}}^{\text{inf}} = \emptyset$$

Let the frequency-shift inference map be

$$\text{FreqShiftInf}_{\text{PR}} : \mathcal{R}_{\text{PR}}^{\text{dir}} \longrightarrow \mathcal{R}_{\text{PR}}^{\text{inf}}$$

Let the intrinsic deviation, gravitational component, and thermal component be

$$\delta_{\nu, \text{PR}}^{\text{intr}}, \quad \delta_{\nu, \text{PR}}^{\text{grav}}, \quad \delta_{\nu, \text{PR}}^{\text{therm}}$$

Let the measured and predicted bidirectional differences be

$$\delta_{\nu, \text{PR}, 2\text{w}}^{\text{meas}}, \quad \delta_{\nu, \text{PR}, 2\text{w}}^{\text{pred}}$$

and let the measured and predicted one-way quantities be

$$\delta_{\nu, \text{PR}, 1\text{w}}^{\text{meas}}, \quad \delta_{\nu, \text{PR}, 1\text{w}}^{\text{pred}}$$

Let the ratio of measured value to predicted value be

$$\text{Ratio}_{\text{PR}}^{\text{meas/pred}}$$

Dependence Inputs

Index Identifier S10-XI-DI1.

Existing Evidence Reference Set Definition 8.194.

Dependence and Transfer Elements

Definition 8.194 (Time-Content Transfers in the Pound–Rebka 1960 Reference). Let $\text{idx}_R^{\text{audit,transfer}} = [9]$. The time-content input and output indices in Table 104 refer to Definition 8.190; formal-support indices refer to Definition 8.191. Auxiliary conditions are retained by including their corresponding claims in formal support. Each row reconstructs dependence within the source; it does not represent LCTR time generation from nontemporal inputs. Transfers of population-dependent quantities apply within the same data population.

Table 104: Time-Content Transfers in the Pound–Rebka 1960 Reference

s	Time-content input	Time-content output	Formal support	Transfer and applicability conditions
1	{8, 19}	{9}	$\llbracket 16, 21 \rrbracket \cup \{37, 39\}$	Use the modulation phase as a common reference to assign counts to opposite quarter-cycle windows, with counting rates referred to the corresponding measurement periods.
2	{9, 12}	{13}	$\llbracket 20, 23 \rrbracket \cup \{16, 17, 37, 38, 40\}$	Calibrate sensitivity from counts in two periods with equal and opposite Doppler shifts imposed by the known velocity.
3	{15, 16}	{17}	$\llbracket 27, 30 \rrbracket \cup \{41\}$	Obtain the correction from the temperature difference averaged over the counting period and the frequency temperature coefficient.
4	{9, 13}	{10}	$\llbracket 20, 23 \rrbracket \cup \{16, 17, 37, 38\}$	Relate the count-rate difference between phase windows to fractional frequency shift using the calibrated sensitivity.
5	{10, 17}	{18}	$\llbracket 27, 30 \rrbracket \cup \{21, 23, 37, 38, 41\}$	Apply the temperature correction to the observed shift from the same counting period.
6	{18, 19}	{20}	$\{12, 13, 31, 39, 42, 43\}$	Form weighted aggregates of corrected observations from the same population and configuration as Table I. The source does not specify the actual weights and covariances.
7	{11, 18, 19}	{21}	$\llbracket 12, 14 \rrbracket \cup \llbracket 24, 31 \rrbracket \cup \llbracket 41, 43 \rrbracket \cup \{33, 39\}$	Compare corrected rising and falling measurements from the same population, conditional on invariance of the intrinsic source–absorber frequency difference under reversal. Keep the four-day Table I example distinct from the final approximately ten-day population.

s	Time-content input	Time-content output	Formal support	Transfer and applicability conditions
8	$\{2, 21\}$	$\{24\}$	$\llbracket 1, 3 \rrbracket \cup \llbracket 11, 15 \rrbracket \cup \llbracket 24, 26 \rrbracket \cup \{31, 33, 36\}$	Use the source configuration and shift-sign conventions to identify the increase in frequency during falling from the two-way difference.
9	$\{2, 23\}$	$\{24\}$	$\llbracket 1, 3 \rrbracket \cup \llbracket 11, 13 \rrbracket \cup \llbracket 34, 36 \rrbracket \cup \{15\}$	Apply the source's stated interpretation of the positive measured-to-predicted ratio to identify the frequency increase during falling.

Validity Conditions Each of the nine transfers requires the establishment of one or more pre-existing time contents and the original-source formal evidence. Consequently, none starts from non-temporal physical inputs alone, and the number of reached time contents is zero. The complete and local-reach assessments in Definition 7.41 give zero generated time contents and zero reached time contents, respectively.

Original-Source Support

Index Identifier S10-XI-SS1.

Existing Evidence Reference Set Definition 8.194.

Results

Definition 8.195 (Results in the Pound–Rebka 1960 Reference). Let $\text{Idx}_R^{\text{audit,result}} = [15]$. Table 105 specifies each result, its original-source location, temporal prerequisites, and formal support. Result 14 is a symmetric conversion of the source's two-way quantity, not a separately printed one-way measurement.

Definition 8.196 (Result-Dependence Hyperedges in the Pound–Rebka 1960 Reference). The temporal-prerequisite indices in Table 105 refer to Definition 8.190, and formal-support indices refer to Definition 8.191. Auxiliary conditions are included through their corresponding supporting claims.

Table 105: Results, Original-Source Locations, and Dependencies in the Pound–Rebka 1960 Reference

q	Result and original-source location	Temporal prerequisites	Formal support
1	Measurement principle using recoilless Mössbauer resonance. Original-source location: p. 337	$\{1, 3, 4\}$	$\llbracket 3, 10 \rrbracket$
2	Apparatus geometry of the approximately 74-ft vertical propagation path. Original-source location: p. 338	$\emptyset$	$\{11\}$

q	Result and original-source location	Temporal prerequisites	Formal support
3	Measurement procedure that reverses the source-above and source-below configurations. Original-source location: pp. 338–340	$\{11\}$	$\llbracket 12, 14 \rrbracket$
4	Doppler scanning by sinusoidal source motion. Original-source location: p. 338	$\{1, 8\}$	$\{2, 3, 18, 19\}$
5	Transmission counts corresponding to modulation phase. Original-source location: p. 338	$\{8, 9\}$	$\llbracket 16, 21 \rrbracket \cup \{37\}$
6	Sensitivity calibration by known velocity and resonance response. Original-source location: p. 339	$\{9, 12, 13\}$	$\llbracket 20, 23 \rrbracket \cup \{16, 17, 37, 38\}$
7	Sensitivity calibration with simultaneous acquisition of gravitational-measurement data. Original-source location: pp. 339–340	$\{9, 13, 14\}$	$\llbracket 20, 23 \rrbracket \cup \llbracket 39, 41 \rrbracket \cup \{16, 17, 37\}$
8	Temperature monitoring and frequency correction by configuration. Original-source location: pp. 339–340	$\llbracket 15, 17 \rrbracket \cup \{11\}$	$\llbracket 27, 30 \rrbracket \cup \{12, 13, 41\}$
9	Cancellation by differencing of the intrinsic resonance offset that remains invariant under configuration reversal. Original-source location: p. 340	$\{11, 18\}$	$\llbracket 24, 31 \rrbracket \cup \{12, 13, 33, 41\}$
10	Repeated measurements and the configuration-specific weighted aggregates in Table I. Original-source location: pp. 339–340	$\llbracket 18, 20 \rrbracket$	$\{12, 13, 31, 39, 42, 43\}$
11	Data acquisition over approximately ten days. Original-source location: p. 341	$\{25\}$	$\llbracket 39, 41 \rrbracket$
12	Two-way fractional frequency difference reported from the full data set. Original-source location: p. 340	$\{21\}$	$\llbracket 24, 26 \rrbracket \cup \{31, 33, 37, 38, 42, 43\}$
13	Comparison with the two-way weak-field prediction and the source-reported measured-to-predicted ratio. Original-source location: pp. 340–341	$\llbracket 21, 23 \rrbracket$	$\llbracket 33, 36 \rrbracket \cup \{42, 43\}$
14	Conversion from two-way to one-way quantities under the symmetric-configuration condition. Original-source location: pp. 339–340	$\{21, 22\}$	$\llbracket 32, 34 \rrbracket \cup \{14\}$
15	Signed conclusion that the frequency of falling radiation increases. Original-source location: p. 341	$\{2, 24\}$	$\llbracket 1, 3 \rrbracket \cup \llbracket 11, 13 \rrbracket \cup \{15, 31, 33, 35, 36\}$

Source-Specific Results The 15 results in Definition 8.195. The time-content inputs and supporting claims of each result are given in Table 105.

Result Dependence

Index Identifier S10-XI-RD1.

Existing Evidence Reference Set Definition 8.195.

Grounds of Establishment

Grounds of Establishment of the Results Direct apparatus counts, configuration reversal, Doppler calibration, temperature correction, statistical integration, frequency-shift estimation, and comparison with the weak-gravitational-field prediction.

Presuppositions, Calibration, Corrections, and Error Domain Recoilless emission and absorption, a narrow resonance line, Doppler calibration by known velocity, separation of the intrinsic resonance deviation between source and absorber that remains unchanged under configuration reversal, the second-order Doppler contribution from lattice motion, temperature-difference correction, configuration reversal, and weighted integration are required.

Table I is an example from the first four days; its difference of averages is $(-4.2 \pm 1.1) \times 10^{-15}$. The final reported two-way shift from approximately ten days of data is $-(5.13 \pm 0.51) \times 10^{-15}$, with the corresponding prediction -4.92×10^{-15} . The source does not specify the weights and covariances used for the weighted means in Table I. On p. 341 the source reports a measured-to-predicted ratio of $+1.05 \pm 0.10$. Division of the printed central values instead gives $5.13/4.92 = 171/164 \approx 1.04268$; the displayed central values differ. The reported ratio is distinguished from the ratio calculated from the printed inputs. For Result 14, assume a symmetric configuration with equal and opposite rising and falling gravitational components, and use the rising direction as the sign convention for the one-way quantity:

$$\delta_{\nu, \text{PR}, 1\text{w}}^a := \frac{1}{2} \delta_{\nu, \text{PR}, 2\text{w}}^a, \quad a \in \{\text{meas}, \text{pred}\}.$$

The falling quantity has the opposite sign. This conversion is an inference from the two-way difference and does not supply two independent one-way measurements.

Evidence Hierarchy

Directly Measured Objects Phase-resolved transmission-detector counts, count differences by configuration, the relative phase of source modulation and counting gates, temperature records, and records of measurement runs.

Direct-Measurement Predicates Detector counts and their statistical uncertainty are obtained in correspondence with the radiation-upward and radiation-downward configurations, modulation phase, calibration runs, and temperature conditions.

Inferred Objects and Inference Predicates Infer configuration-specific fractional frequency shifts by applying Doppler calibration and temperature correction to phase-resolved counts, and obtain the two-way difference by comparing rising and falling configurations within the same data population. One-way quantities follow from the symmetric conversion in Result 14. Distinguish the source-reported measured-to-predicted ratio from the ratio calculated from the printed values.

Conditions and Domain of Direct Measurement The domain comprises the approximately 14.4 keV nuclear transition of Fe-57, the Co-57 source, the Fe-57 absorber, the approximately 74 ft vertical propagation path (with a helium-filled Mylar bag along most of its length), upward and downward configurations, sinusoidal modulation, temperature monitoring, the approximately ten-day acquisition period, and the statistical uncertainty of the original source.

Object of Formal Generalization Compare the source-reported two-way fractional frequency shift with the two-way weak-field prediction. One-way quantities are obtained from the two-way quantities under the symmetric-configuration condition of Result 14.

Conditions and Domain of Formal Generalization The object of comparison is a frequency difference of electromagnetic radiation; it is not a direct measurement of time itself independently of a physical carrier.

Generation Judgment Judgment values: time contents generated under the LCTR complete-generation judgment: 0; time contents reachable under the source-internal evidence-conditioned local-reach judgment: 0; remaining primitive time roots: 25.

Generation Reach The number of time contents reachable from the 15 non-temporal physical inputs through only the nine source-internal transfers is zero.

Remaining Primitive Time Roots All 25 time contents remain primitive time roots.

Source-Internal Evidence-Conditioned Local Reach No time contents are reachable under the source-internal evidence-conditioned local-reach judgment; 25 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ from Definition 7.108 and apply Definition 7.50 to it, then total over all results. This determines the primitive-time-root–result dependence relations and the necessary-primitive-time-root–result dependence relations.

Result Dependence The 14 results other than Result 2 depend on primitive time roots. The minimal support families of all 15 results have 86 members in total; the minimal support of Result 2 is the empty set.

Evidence Effect

Evidence-Effect Scope The source evidence supports the experimental principle, apparatus geometry, measurement procedures, frequency-shift estimation, and prediction comparison in Table 105. Result 14 follows from the source two-way quantities and the symmetric conversion specified by Formal Claim 32.

Evidence-Effect Boundary Directly measured detector counts and frequency shifts inferred after calibration and correction are different evidence types. Their empirical establishment cannot be inherited by the generation of the 25 time contents, the physical-time qualification of the primitive time roots, or a time attribute independent of a physical carrier.

Result-Preservation Domain

Result Preservation The 15 results are preserved insofar as the Fe-57 nuclear transition, source, absorber, vertical propagation path, configuration reversal, transducer, detector, temperature monitoring, Doppler calibration, corrections, statistical uncertainty, and directions of dependence are preserved.

Placement of the Grounding Obligation

Relocation into the LCTR Generation Order If the frequency scale, lifetime, propagation order, modulation period, counting intervals, calibration time, and measurement schedule used by each result are first established through the LCTR generation sequence, the 14 primitive-root-dependent results can be safely reused as conditional inferences based on generated time representations. This judgment follows by matching the use of each time content in a law operation or result derivation against the absence, before that use, of the corresponding generation or validity condition as an input, formation condition, evaluation condition, or failure condition.

Placement of the Grounding Obligation The 25 time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as already-established time inputs. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.”

Evidence of Actual Failure

Evidence of Actual Failure The complete judgment and the remaining-root judgment concern physical-time qualification; they are distinct from actual failure of a concrete condition governing a result’s value or domain of application. Neither the original formal evidence nor the LCTR audit evidence in this reference establishes an actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence,” and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream Reach For each reference, the reach to primitive-time-root-dependent results is determined by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because actual failure of a concrete condition has not been established, the actual-reach judgment from that failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the objects of repair are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is limited to results for which actual failure is independently established and which are dependency-reachable from that failure location. Because actual failure is indeterminate in this reference, this repair-and-recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.197 (Primitive Time Roots and Result Dependence in the Pound–Rebka 1960 Reference). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [25]; \{2\}, \emptyset, [15] \setminus \{2\}, \emptyset)$$

. All 25 time contents remain primitive time roots. Result 2 is reachable from the nontemporal initial set, and the other 14 results depend on primitive time roots.

Proof. The initial set contains geometric Claim 11 and reaches Result 2, which has no temporal prerequisite. Every one of the nine transfers takes existing time contents as inputs, so neither the initial set nor the source-formal-evidence environment reaches any time content. Adding the 25 roots reaches all 15 results. Every result other than Result 2 has nonempty temporal prerequisites and nonempty minimal root supports. $\square$

Proposition 8.198 (Minimal Primitive-Time-Root Support in the Pound–Rebka 1960 Reference). For each $q \in \text{Idx}_R^{\text{audit,result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108, and the minimum and maximum cardinalities of those elements, are given in Table 106.

Table 106: Minimal Primitive-Time-Root Support in the Pound–Rebka 1960 Reference

Result indices	Support statistics
1, 8	(1; 3)
2	(1; 0)
3, 11	(1; 1)
4	(1; 2)
5	(2; 2)
6	(2; 2, 3)
7	(4; 3, 4)
9	(11; 2, 6)
10	(11; 2, 5)
12	(12; 1, 6)
13	(12; 3, 8)
14	(12; 2, 7)
15	(14; 2, 7)

Proof. Apply the update in Definition 7.46 to the nine transfers. After the first scan in listed order, the support antichains of all time-content nodes are invariant under the second scan. Proposition 7.47 gives the values of the result presuppositions. The total number of members of the minimal primitive-time-root support families is 86. $\square$

Proposition 8.199 (Audit Conclusion for the Pound–Rebka 1960 Reference). The 41 semantic objects comprise 25 time contents and 16 auxiliary objects, with 49 physical and formal input claims, nine transfers, and 15 results. The complete judgment generates no time content, and the local-reach judgment reaches no time content; all 25 remain primitive time roots. Geometric Result 2 is reachable from the initial set but supplies no time generation. The other 14 results depend on primitive time roots.

Proof. The objects and types follow from Definitions 8.190 and 8.192, and incidence follows from the transfer and result tables. Reachability and minimal supports follow from Propositions 8.197 and 8.198. $\square$

Proposition 8.200 (Audit-Boundary Counts for the Pound–Rebka 1960 Reference). $\text{BoundaryCount}_R^{\text{audit}} = (89, 24)$.

Proof. The 49 claim nodes, 25 time-content nodes, and 15 result nodes give 89 nodes. The nine transfers and 15 result dependences give 24 hyperedges. $\square$

1. Detector counts, the relative phase of source modulation and counting gates, temperature records, and run records are measured directly, but the resonance-frequency scale, excited-state lifetime, propagation order, counting intervals, and measurement schedule are not generated from non-temporal physical inputs.
2. Doppler calibration, temperature correction, configuration reversal, and statistical integration provide a route for inferring frequency shifts from counts, but do not supply the physical generation of the time contents entered by that route.
3. Agreement between the frequency difference and the weak-gravitational-field prediction supports results concerning electromagnetic radiation, the source, the absorber, and the detector, but does not directly measure a change in time itself independently of a physical carrier.
4. The narrow Lorentzian resonance line, the level of approximately $0.1\ \mu\text{s}$, and the fractional full-width at half-height 1.13×10^{-12} use scales containing time as inputs, but supply no rule generating those scales from non-temporal inputs.
5. The number of source-internal rules generating the 25 time contents from non-temporal physical inputs is zero.

Individual Statements Measurement Object

Verbatim original (English):

we have now measured the effect, originally hypothesized by Einstein, of gravitational potential on the apparent frequency of electromagnetic radiation

The original-source location is Pound and Rebka, *Apparent Weight of Photons, Physical Review Letters* 4, p. 337, left column at the beginning of the main text. The minimal context is the introduction to a measurement using recoilless emission and absorption of γ rays. The statement is attributable to the text of the paper by R. V. Pound and G. A. Rebka, Jr.

Resonance Line with a Level Lifetime

Verbatim original (English):

the line obtained at room temperature for the 14.4-keV, 0.1-microsecond level in Fe^{57}

The original-source location is p. 337, left column at the beginning of the main text, in the same paper. The minimal context is a description, reported previously, of the shape and width of the Mössbauer resonance line used. The statement is attributable to the text of the paper by R. V. Pound and G. A. Rebka, Jr.

Shape and Width of the Resonance Line

Verbatim original (English):

We found that the line had a Lorentzian shape with a fractional full-width at half-height of 1.13×10^{-12} when the source was carefully prepared according to a prescription developed from experience.

The quoted fractional full-width at half-height is 1.13×10^{-12} .

The original-source location is p. 337, left column at the beginning of the main text, in the same paper. The minimal context is the description of the conditions under which the narrow resonance line used to measure the gravitational effect is obtained. The statement is attributable to the text of the paper by R. V. Pound and G. A. Rebka, Jr.

Non-Unique Attribution of the Frequency Difference

Verbatim original (English):

Observation of a frequency difference between a given source and absorber cannot be uniquely attributed to this perturbation.

The original-source location is p. 337, right column at the beginning of the main text, in the same paper. The minimal context is the explanation that an asymmetry or frequency difference intrinsic to the source-absorber pair must be separated from the change in relative frequency produced by the perturbation. The statement is attributable to the text of the paper by R. V. Pound and G. A. Rebka, Jr.

Temperature-Difference Correction Condition

Verbatim original (English):

We conclude that the temperature difference between the source and absorber must be accurately known and its effect considered before any meaning can be extracted from even a change observed when the perturbation is altered.

The original-source location is p. 337, right column at the beginning of the main text, in the same paper. The minimal context is the treatment of the frequency change due to the temperature of the source or absorber as a systematic error. The statement is attributable to the text of the paper by R. V. Pound and G. A. Rebka, Jr.

Agreement between Measured and Predicted Shifts

Verbatim original (English):

The shift observed agrees with -4.92×10^{-15} , the predicted gravitational shift for this “two-way” height difference.

The quoted predicted gravitational shift is -4.92×10^{-15} .

The original-source location is p. 340, at the end of the right column, in the same paper. Here the “two-way” height difference is approximately 148 ft, obtained by comparing rising and falling configurations with a height difference of approximately 74 ft each. The minimal context is the comparison between the net relative-frequency shift and the weak-gravitational-field prediction after combining the configuration-reversal data, temperature correction, sensitivity calibration, and statistical uncertainty. The statement is attributable to the text of the paper by R. V. Pound and G. A. Rebka, Jr.

Common Evidence with Specified Domain of Application

Target statement	Object of statement	Predicate	Explicit domain of application
S10-Q001	Effect of gravitational potential on the apparent frequency of electromagnetic radiation	Was measured.	Experimental domain in which the apparent frequency difference between a source and absorber is measured using recoilless emission and absorption of γ rays.
S10-Q002	Line of the 14.4-keV, 0.1-microsecond level in Fe^{57}	Was obtained at room temperature.	Descriptive domain introducing the shape and width of the Mössbauer resonance line used.
S10-Q003	Resonance line obtained by preparing the source according to the specified procedure	Has a Lorentzian shape and a fractional full-width at half-height of 1.13×10^{-12} .	Domain in which the source was carefully prepared according to a procedure developed from experience.
S10-Q004	Frequency difference observed between a given source and absorber	Cannot be attributed uniquely to the perturbation in question.	Domain in which the frequency difference intrinsic to the combination is separated from the relative-frequency change produced by altering the perturbation.

Target state- ment	Object of statement	Predicate	Explicit domain of application
S10- Q005	Temperature difference between source and absorber, and its effect	Must be known accurately and considered before meaning is extracted from the observed change.	Temperature-correction condition for interpreting the change observed when the perturbation is altered.
S10- Q006	Corrected measured shift after configuration reversal	Agrees with the predicted gravitational shift -4.92×10^{-15} for the “two-way” height difference.	Domain comparing the net shift, including configuration-reversal data, temperature correction, sensitivity calibration, and statistical uncertainty, with the weak-gravitational-field prediction.

Individual Audit

Interpretive Possibilities The first statement reports a measurement in which the apparent frequency difference of electromagnetic radiation corresponding to a gravitational-potential difference was inferred from detector counts through calibration, correction, and statistical processing. The second statement uses the lifetime of the Fe⁵⁷ level as part of the spectroscopic conditions for obtaining a narrow resonance line. The third statement gives the Lorentzian shape and full-width at half-height obtained under the specified conditions of source preparation. The fourth statement shows that the causal attribution of the observed frequency difference is not unique. The fifth statement makes knowledge and correction of the temperature difference a condition for extracting meaning from the observed change. The sixth statement gives the agreement between the corrected measured shift and the weak-gravitational-field prediction. None of the six statements names time itself independently of a physical carrier as the object of measurement.

Independent Grounds Detector counts, configuration reversal, Doppler calibration, temperature correction, statistical uncertainty, and the ratio of measured value to predicted value provide independent experimental grounds for the apparent frequency shift of electromagnetic radiation. Those grounds do not directly measure a change in an attribute of a time entity independent of clocks or other physical carriers.

Evidence-Effect Boundary The first statement directly supports that the correspondence between gravitational potential and the apparent frequency difference of electromagnetic radiation was the measurement object; the directly observed quantities are detector counts, whereas the frequency shift is an inferred quantity after calibration and correction. The second and third statements directly support the level lifetime, resonance-line shape, and linewidth as conditions of measurement sensitivity, but do not supply the lifetime or linewidth as a starting point for time generation. The fourth statement directly supports the non-uniqueness of attribution of the frequency difference, and the fifth directly supports the need for temperature

correction; they do not permit an uncorrected difference to be attributed uniquely to the gravitational effect. The sixth statement supports agreement between the measured inferred quantity and predicted quantity within the domain of the original source’s apparatus, configuration reversal, calibration, temperature correction, and uncertainty. That agreement does not yield the physical generation of the time contents used, physical-time qualification, or a change in a time attribute independent of a physical carrier.

Corresponding Formal Evidence The first statement corresponds to the frequency-measurement objects in Time Contents 1, 10, and 21 of Definition 8.190, and Components 1–3 of Definition 8.191. The second and third statements correspond to Time Contents 3–4, Formal-Evidence Components 4, 9, and 10, and the level-lifetime and resonance-line aspects of Result 1. The fourth corresponds to the non-unique attribution of the intrinsic frequency difference and the need to separate it in Components 24–26. The cancellation condition of Result 9 comes from the invariance-under-reversal condition on p. 340. The fifth corresponds to Time Contents 15–17, Components 27–30, and the temperature-correction correspondence of Result 8. The sixth corresponds to Time Contents 21–22, Components 33–34 and 36, and the two-way comparison in Results 12–13. The reported ratio in Result 13 comes from the equation on p. 341; the one-way conversion in Result 14 follows from Formal Claim 32.

Within-Source Comparison The first statement gives the measurement object; the second and third give the level lifetime, line shape, and linewidth; the fourth and fifth give causal attribution and correction conditions; and the sixth gives the agreement after correction. Taken together, the six statements make explicit the experimental dependence sequence from apparatus conditions to comparison of the inferred and predicted quantities, but physical generation of the time contents is not contained in that sequence.

Common Audit with Specified Domain of Application The audit relates the six statements on the measurement object, spectroscopic conditions, the conditions for causal attribution and temperature correction, and agreement between measured and predicted values to their source contexts and corresponding formal evidence.

8.19.11 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 11.

Source Name Primitive-Time-Root Audit of Arnowitt–Deser–Misner 1962

Bibliographic Identification Let R. Arnowitt, S. Deser, and C. W. Misner, *The Dynamics of General Relativity*, in L. Witten, ed., *Gravitation: An Introduction to Current Research*, Chapter 7, Wiley (1962), together with the authors’ reprint, arXiv:gr-qc/0405109v1 (19 May 2004), be $R = \text{Ref}_{\text{ADM62Dyn}}$. The audit domain is §§1–8, 30 pages, of the authors’ reprint.

Original-Source Location Identifier 11-L1.

Original-Source Location Content Sections 1–8, 30 pages, of the authors’ reprint arXiv:gr-qc/0405109v1. The location of each time content is determined by Table 108, and the section corresponding to each formal input is determined by Table 109.

Audit-Scope Identifier 11-SC1.

Audit-Scope Content The audit domain stated under Bibliographic Identification.

Formal Objects

Object Identifier 11-F1.

Object Name Formal objects.

Object Type Formal object.

Formal Content Definitions, assumptions, derivations, and approximate calculations concerning the Hamiltonian form of general relativity, parametrization and de-parametrization, the first-order Palatini form, 3+1 decomposition, constraints, canonical reduction, wave-zone analysis, Einstein–Maxwell–particle coupling, proper time, and the choice of canonical variables in quantization.

Original-Source Location Identifier Set 11-L1.

Assumptions

Assumption Identifier 11-A1.

Assumption Content Conditions of the respective derivations (§§2–8). Hamiltonian mechanics uses an autonomous Hamiltonian and a simple root of the constraint; changes of time variable use differentiable, locally invertible coordinate choices. Canonical reduction of the gravitational field assumes a nondegenerate Lorentzian metric, spacelike hypersurfaces, constraint-compatible initial data, boundary conditions for the inverse Laplacian, and admissible coordinate conditions. Energy and momentum require asymptotic flatness and the specified falloff; the wave zone requires the far-field, weak-field, and small derivative-nonlinearity conditions. The self-energy calculation concerns a single particle at rest with no independent field excitations, the ansatz in Eq. (7.4), and the point-particle limit. Each condition applies to its corresponding derivation. Consistent quantization and the choice of preferred coordinates remain open questions in §8.1.

Original-Source Location Identifier Set 11-L1.

Non-Temporal Physical Inputs

Input Identifier 11-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content The physical carriers of the gravitational field, matter fields, electromagnetic field, and particles, together with spatial boundary geometry, are taken as non-temporal physical inputs.

Original-Source Location Identifier Set 11-L1.

Time Content

Time-Content Identifier 11-T1.

Time-Content Name Time content.

Time-Content Type Time content.

Time-Content Specification Hamiltonian time, boundary time, evolution parameters, intrinsic time coordinates, 3+1 time, temporal reference location, normal proper time, wave waiting time and frequency, worldline coordinate time, Lorentz proper time of the total system, and candidate preferred time coordinates belong to time content.

Original-Source Location Identifier Set 11-L1.

Time-Content Identifier 11-T2.

Time-Content Name Time contents used in the original source.

Time-Content Type Time content.

Time-Content Specification The 17 time contents in Definition 8.201.

Original-Source Location Identifier Set 11-L1.

Formal Input Elements

Formal-Component Identifier 11-C1.

Formal-Component Name Formal input elements.

Formal-Component Type Formal component.

Formal-Component Text Evolution values of the spatial metric, canonical variables, and constraint relations belong to the source-formal-evidence environment.

Original-Source Support Identifier Set 11-L1.

Definition 8.201 (Time Contents in the Arnowitt–Deser–Misner Reference). For reference $R = \text{Ref}_{\text{ADM62Dyn}}$, let $n_R = 17$. There are no independent auxiliary objects. Each time content and its original-source location are given in Table 108.

The correspondence between original-source location keys and original-source locations in Table 108 is as follows.

1 §§2.1–2.3, Eqs. (2.1)–(2.7), and the field extension at the end of §2.3

2 §1, §§3.1–8.1

3 §1, §§3.1–7.4

4 §§4.1–4.7

5 §§3.1–3.5

6 §5.4, §§7.2–7.3

7 §§5.5–5.7, Eqs. (5.5)–(5.9)

8 §§6.1–6.2, Eqs. (6.1)–(6.7), §7.3

9 §8.1

In the following table, ℓ denotes an original-source-location key.

Table 108: Time Contents and Original-Source Locations
in the Arnowitt–Deser–Misner Reference

i	Time content	ℓ
1	Ordinary Hamiltonian time	1
2	Boundary times in Hamilton’s action	1
3	Arbitrary evolution parameter	1
4	Temporal component of the field-parameter family	1
5	Selected intrinsic time coordinate	1
6	Temporal spacetime coordinate and 3+1 evolution time	2
7	3+1 temporal reference location and boundary location	3
8	Selected time after canonical reduction	4
9	Alternative canonical time coordinate	4
10	Normal proper-time coordinate	5
11	Time-reversal structure	6
12	Wave-arrival waiting time	7
13	Wave-zone frequency	7
14	Worldline coordinate time	8
15	Arbitrary worldline parameter	8
16	Lorentz proper time of the total system	8
17	Family of candidate preferred time coordinates	9

Time Content 4 denotes the temporal component in the field parametrization at the end of §2.3; spatial components are excluded. Time Content 6 denotes the temporal spacetime coordinate specified as $x^0 = t$ in footnote 1 of §3.1. The Palatini form in §3.1 uses this coordinate, which serves as evolution time in the 3+1 decomposition from §3.2 onward.

Definition 8.202 (Physical- and Formal-Input Claim Components in the Arnowitt–Deser–Misner Reference).

Decompose the non-temporal physical inputs at the source entry and the 13 original-source statement groups into type-homogeneous components, and let the sequence of physical- and formal-input claim components be $(\text{ClaimVal}_{R,p}^{\text{audit,phys}})_{p \in [37]}$. The components, original-statement-group keys, associated time-content indices, and original-source locations are defined by Table 109. Components 1–5 are non-temporal physical inputs, and Components 6–37 are time-dependent formal evidence. Components 6–37 are the relations, derivations, and questions at the source locations specified in the table. The generation-initial inputs are Claim Components 1–5.

The correspondence between the time-content keys and sets of time-content indices in Table 109 is as follows.

1 $\emptyset$

2 $\{6, 7\}$

3 $\{6\}$

- 4 {6, 8, 9}
- 5 {1, 2}
- 6 {3, 4}
- 7 {1, 3, 5}
- 8 {6}
- 9 {6, 7, 10}
- 10 $\llbracket 6, 9 \rrbracket$
- 11 {7, 8}
- 12 {6, 8}
- 13 {6, 7, 11}
- 14 {6, 12, 13}
- 15 {6, 7, 14, 15}
- 16 {14, 15}
- 17 {8, 9, 17}
- 18 {6, 7}
- 19 {14, 16}

The correspondence between the original-source-location keys and original-source locations in Table 109 is as follows.

- 1 §§1–8
- 2 §§6.1–6.2, §§7.1–7.4
- 3 §§3.1–6.2
- 4 §1
- 5 §§1–5
- 6 §§2.1–2.3
- 7 §3.1
- 8 §§3.1–3.5
- 9 §§4.1–4.7
- 10 §§5.1–5.7
- 11 §8.1
- 12 §5.4, §§7.1–7.4

Table 109: Physical- and Formal-Input Claim Components
in the Arnowitt–Deser–Misner Reference

Com- po- nent	Original- statement- group key	Audit minimal claim	Time- content key	Original- source- location key
1	Entry	Physical carrier of the gravitational field	1	1
2	Entry	Physical carrier of matter fields	1	1
3	Entry	Physical carrier of the electromagnetic field	1	2
4	Entry	Physical carrier of particles	1	2
5	Entry	Spatial boundary geometry	1	3
6	1	Description of the gravitational field invariant under general coordinate transformations	2	4
7	1	Temporal field evolution from initial data and evolution equations	3	4
8	1	Description of the gravitational field by two canonical pairs	4	5
9	2	Action (2.1) and Hamilton equations	5	6
10	2	Time translation generated by (2.2)	5	6
11	3	Constrained parametrized action (2.3) and its field extension	6	6
12	3	Actions (2.4)–(2.5) using the constraint solution and an intrinsic coordinate choice	7	6
13	4	First-order equations (3.2)–(3.4) from independent variation of metric and connection	8	7
14	4	3+1 decomposition (3.9)–(3.14) of metric and momentum	9	8
15	5	Constant-time hypersurfaces and their tangent vectors	2	8
16	5	Lapse defined by the metric in (3.9a)–(3.12)	9	8
17	5	Shift specifying coordinate continuation in (3.9a)–(3.11)	2	8
18	5	Spatial-metric evolution equation (3.15a)	9	8
19	5	Relation between extrinsic curvature and canonical momentum, §3.3	9	8
20	6	Cauchy data satisfying constraints (3.15c)	2	8

Com- po- nent	Original- statement- group key	Audit minimal claim	Time- content key	Original- source- location key
21	6	Constraint propagation through the evolution equations and Bianchi identity (3.16)	2	8
22	7	Constraint solutions (4.8), (4.16), with the specified boundary conditions	10	9
23	7	Temporal and spatial coordinate conditions (4.11), (4.22)	10	9
24	7	Canonical form (4.17) from constraint elimination and coordinate conditions	10	9
25	8	Equal-time Poisson brackets (4.18)–(4.19)	11	9
26	8	Hamilton equations (4.20)	12	9
27	9	Energy and momentum from asymptotic surface integrals (5.1)–(5.2)	18	10
28	9	Heisenberg variables without explicit coordinate dependence and the Hamiltonian value, §5.3	18	10
29	9	Time-symmetric data with vanishing initial momentum, §5.4	13	12
30	10	Evolution equations (5.6) at leading order $1/r$ in the wave zone	14	13
31	10	Wave-zone radiation amplitudes and energy flux (5.7) through order $1/r^2$	14	13
32	11	Coupled action (6.1) and constraint elimination (6.3)–(6.7)	15	2
33	11	Conversion from worldline parameter s to $x^0 = t$, Eqs. (6.1)–(6.2)	16	2
34	12	Self-energy and static solution in the point limit of a single-particle rest state, §§7.2–7.3	13	12
35	12	Lorentz proper time used in the total stress tensor in a moving frame, §7.3	19	2
36	13	Classical canonical equivalence and the question of quantum unitary equivalence, §8.1	17	11
37	13	Open question of preferred-time-coordinate selection by quantum consistency, §8.1	17	11

Define the generation-initial-input index set and the source-formal-evidence-environment index set by

$$\text{ClaimBoundaryIdx}_R = \left(\{1, 2, 3, 4, 5\}, [37] \right)$$

It follows that $\text{TypedStrictClaimBoundary}(R)$ holds.

Dependence Inputs

Dependence Inputs Hamiltonian time, an arbitrary evolution parameter, an intrinsic time coordinate, 3+1 time, a worldline parameter, worldline coordinate time, and a time coordinate after canonical reduction enter the fourteen transfers.

Dependence and Transfer Elements

Definition 8.203 (Time-Content Transfers in the Arnowitt–Deser–Misner Reference). Define the transfer index set by $\text{Idx}_R^{\text{audit,transfer}} = [14]$. Let the time-content input-index set, time-content output-index set, and physical- and formal-input claim-support index set of Transfer s be $J_{R,s}^{\text{in}}$, $J_{R,s}^{\text{out}}$, and $P_{R,s}^{\text{tr}}$, respectively. These sets, together with the transfer content and original-source location, are defined by Table 110.

Table 110: Time-Content Transfers in the Arnowitt–Deser–Misner Reference

s	Time-content input indices	Time-content output indices	Supporting-component indices	Transfer content and original-source location
1	$\{1\}$	$\{2\}$	$\{9, 10\}$	Endpoint evaluation. §§2.1–2.3, Eqs. (2.1)–(2.7)
2	$\{1\}$	$\{5\}$	$\{12\}$	Identification of ordinary time with the intrinsic coordinate. §2.3, Eq. (2.5)
3	$\{3\}$	$\{5\}$	$\{11, 12\}$	Choice of a locally invertible $q_{M+1}(\tau)$. §2.3
4	$\{5\}$	$\{1\}$	$\{12\}$	Deparametrization. §§2.1–2.3, Eqs. (2.1)–(2.7)
5	$\{6\}$	$\{7\}$	$\llbracket 14, 19 \rrbracket$	Temporal-location evaluation. §§3.1–3.5, Eqs. (3.1)–(3.17)
6	$\{6\}$	$\{8\}$	$\llbracket 22, 24 \rrbracket$	Constraint solution and coordinate conditions (4.11). §§4.2–4.5
7	$\{6\}$	$\{9\}$	$\llbracket 22, 24 \rrbracket$	Alternative coordinate conditions (4.22). §4.7

s	Time-content input indices	Time-content output indices	Supporting- component indices	Transfer content and original-source location
8	{6}	{10}	{15, 16, 17, 19}	Choice of normal coordinate lines and $N = 1$. §3.3, p. 8
9	{6}	{11}	{29}	Time-reversal symmetry of initial data. §5.4, §7.2
10	{6}	{12}	{30, 31}	Waiting for arrival in the wave zone. §5.5, p. 19
11	{6}	{13}	{30, 31}	Wave-zone frequency and time differentiation. §§5.5–5.7
12	{6}	{14}	{32, 33}	Restriction of coordinate time to the worldline. §§6.1–6.2
13	{15}	{14}	{32, 33}	Change of variable through a locally invertible $x^0(s)$. §6.1, Eqs. (6.1)–(6.2)
14	{14}	{16}	{35}	Lorentz-proper-time representation of total stress in a moving frame. §7.3, p. 27

Transfer Events and Outputs Boundary times, an intrinsic time coordinate, ordinary Hamiltonian time, a 3+1 temporal reference location, canonical time coordinates, normal proper time, time reversal, wave waiting time and frequency, worldline coordinate time, and Lorentz proper time of the total system are obtained through the input–output relations in Table 110.

Validity Conditions

Conditions under Which the Dependences or Transfers Apply All fourteen transfers take one or more existing time contents as inputs. The number of transfers that output time content solely from the gravitational field, matter fields, the electromagnetic field, particles, the spatial metric, canonical variables, and constraint relations is zero.

Original-Source Support Take the supporting-claim indices, time-content input indices, and time-content output indices in each row of Table 110 as the corresponding index sets in Definition 7.105. The support set, time-content input set, and time-content output set of each transfer are determined by that definition. Transfers 2 and 4 form a cycle between Time Contents 1 and 5. Reachability is determined by applying the updates in Definition 7.46 until they stabilize.

Original-Source Support for Each Dependence and Transfer The supporting claims, transfer content, and original-source location of each transfer are specified by the same table.

Results

Definition 8.204 (Results in the Arnowitt–Deser–Misner Reference). Define the result index set by $\text{Idx}_R^{\text{audit,result}} = [18]$. Let the time-content-premise index set and physical- and formal-input claim-support index set of Result q be $J_{R,q}^{\text{res}}$ and $P_{R,q}^{\text{res}}$, respectively. Each result, both index sets, and the original-source location are defined by Table 111.

Table 111: Results and Audit Inputs in the Arnowitt–Deser–Misner Reference

q	Result and original-source location	Time-content-premise indices	Supporting-component indices
1	Separation of physical degrees of freedom under general covariance. §1, §3.4, §4.5	$\{6, 7\}$	$\{6, 7, 8, 20, 22, 23, 24\}$
2	Parametrization of ordinary Hamiltonian mechanics. §§2.1–2.3	$\{1, 2, 3, 5\}$	$\{9, 11\}$
3	Deparametrization by a constraint solution and a locally invertible intrinsic coordinate. §2.3	$\{1, 3, 5\}$	$\{9, 11, 12\}$
4	First-order Palatini form. §3.1	$\{6\}$	$\{13\}$
5	3+1 decomposition. §§3.1–3.3	$\{6, 7\}$	$\{13, 14, 15, 16, 17, 19\}$
6	Lapse function, shift vector, spatial metric, momenta, and four constraints. §§3.2–3.4	$\{6, 7\}$	$\llbracket 14, 20 \rrbracket$
7	Constrained Cauchy data and constraint propagation. §§3.1–3.5	$\{6, 7\}$	$\{20, 21\}$
8	Constraint elimination and canonical reduction with the specified boundary and coordinate conditions. §§4.1–4.7	$\llbracket 6, 9 \rrbracket$	$\{20, 22, 23, 24\}$
9	Equal-time Poisson brackets and Hamilton equations. §4.6	$\{7, 8\}$	$\llbracket 24, 26 \rrbracket$

q	Result and original-source location	Time-content-premise indices	Supporting-component indices
10	Energy and momentum in the asymptotically flat regime. §§5.1–5.3	{6, 7}	{27, 28}
11	Gravitational radiation at the leading asymptotic orders in the wave zone. §§5.5–5.7	{6, 12, 13}	{30, 31}
12	Einstein–Maxwell–particle coupling. §§6.1–6.2	{6, 7, 14, 15}	{32, 33}
13	Classical self-energy in the point limit of a single-particle rest state. §§7.1–7.3	{6, 7, 11}	{29, 32, 34}
14	Proper-time coordinate along normal coordinate lines with $N = 1$. §3.3	{6, 7, 10}	{15, 16, 17, 19}
15	Lorentz proper time used in the total stress tensor in a moving frame. §7.3	{14, 16}	{35}
16	Time-symmetric data and static point-particle solutions under the additional conditions. §5.4, §§7.2–7.3	{6, 7, 11}	{29, 34}
17	Open question of preferred-time-coordinate selection by quantum consistency. §8.1	{8, 9, 17}	{36, 37}
18	Question of quantum unitary equivalence for classically equivalent canonical choices. §8.1	{8, 9, 17}	{36, 37}

Source-Specific Results The 18 results in Definition 8.204. The time-content premises, supporting claims, and original-source location of each result are given by the table in that definition.

Result Dependence Take the supporting-claim indices and time-content-premise indices in each row of Table 111 as the corresponding index sets in Definition 7.105. The support set and time-content-premise set of each result are determined by that definition.

Grounds of Establishment

Grounds of Establishment of the Results Results 1–16 are grounded in formal derivations through the action principle, first-order Palatini form, 3+1 decomposition, constraint analysis, canonical reduction, Poisson brackets, Hamilton equations,

asymptotic analysis, wave-zone calculation, and Einstein–Maxwell–particle coupling. Results 17–18 are grounded in the questions and indeterminacy stated in §8.1.

Evidence Hierarchy

Directly Measured Objects The number of directly measured objects newly acquired by this reference is zero.

Direct-Measurement Predicates The number of direct-measurement predicates newly acquired by this reference is zero.

Conditions and Domain of Direct Measurement The results of this reference belong to theoretical constructions and analyses. The number of source-specific direct-measurement inputs comprising apparatus, samples, calibration, errors, and measurement data is zero.

Objects of Formal Generalization Results 1–10 apply to the Hamiltonian form of generally covariant gravity, spacetime decomposition, constraints, canonical reduction, and asymptotic quantities. Results 11–16 apply to the wave zone, gravitational radiation, Einstein–Maxwell–particle coupling, proper time, and static solutions. Results 17–18 apply to the non-uniqueness problems of the preferred time coordinate and the canonical choice for quantization.

Conditions and Domain of Formal Generalization Each generalization takes as its validity conditions the time-content premises and supporting claims in the same table. The asymptotic quantities belong to the asymptotically flat domain, the wave results to the wave zone, and the particle results to the domain of application of Einstein–Maxwell–particle coupling. Free wave propagation holds at leading order $1/r$ in the fields, and the energy flux is evaluated through order $1/r^2$. The self-energy and static solution require a single-particle rest state without independent field excitations and the point-particle limit. Time-symmetric initial data can contain waves; staticity requires additional conditions.

Generation Judgment **Judgment values:** LCTR complete-generation judgment: 0; source-internal evidence-conditioned local-reach judgment: 0; remaining primitive time roots: 17.

Generation Reach and Primitive Time Roots The number of time contents generated by generation reach from an initial set consisting solely of non-temporal physical inputs is zero. All 17 time contents in Definition 8.201 remain primitive time roots.

Source-Internal Evidence-Conditioned Local Reach The source-internal evidence-conditioned local-reach judgment is zero; 17 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ from Definition 7.108 and apply Definitions 7.50 and 8.114 to it. Totaling over all results gives 29 primitive-time-root–result dependence relations and 17 necessary-primitive-time-root–result dependence relations.

Result Dependence All 18 results in Definition 8.204 are primitive-time-root-dependent results. The numbers of generated results, evidence-reach-only results, and unreached-after-root-addition results are zero. The minimal primitive-time-root support of each result is given by Proposition 8.206.

Evidence Effect

Evidence-Effect Scope The formal evidence in this reference supports 18 results concerning the Hamiltonian form of general relativity, 3+1 decomposition, constraints, canonical reduction, asymptotic quantities, gravitational radiation, Einstein–Maxwell–particle coupling, proper time, the preferred time coordinate, and canonical choice in quantization.

Evidence-Effect Boundary Establishment of parametrization, deparametrization, an intrinsic time coordinate, 3+1 decomposition, normal proper time, and Lorentz proper time of the total system does not imply that Hamiltonian time, boundary times, an evolution parameter, a 3+1 temporal reference, worldline time, and the preferred time coordinate have been generated from non-temporal physical inputs. Nor does posing the question of preferred time in quantization retroactively confer physical-time qualification on the 17 time contents. These transitions lie beyond the evidence-effect boundary in Definition 7.115.

Result-Preservation Domain

Result-Preservation Domain The 18 results are preserved insofar as every time-content premise is supplied by the corresponding generated time representation in the LCTR generation sequence, while preserving the gravitational field, matter fields, electromagnetic field, particles, spatial metric, canonical variables, constraint relations, asymptotic-flatness conditions, wave-zone conditions, and the original source’s directions of dependence. The asymptotic orders of Result 11 and the rest-state and point-limit conditions of Results 13 and 16 are also retained. For Results 17–18, what is preserved is the open question concerning preferred time coordinates and quantum unitary equivalence.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The 17 time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as already-established time inputs. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment follows by matching the use of each time content in a law operation or result derivation against the absence, before that use, of the corresponding generation or validity condition as an input, formation condition, evaluation condition, or failure condition.

Evidence of Actual Failure

Evidence of Actual Failure The complete judgment and the remaining-root judgment concern physical-time qualification; they are distinct from actual failure of a concrete

condition governing a result's value or domain of application. Neither the original formal evidence nor the LCTR audit evidence in this reference establishes an actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence,” and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream Reach For each reference, the reach to primitive-time-root-dependent results is determined by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because actual failure of a concrete condition has not been established, the actual-reach judgment from that failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the objects of repair are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is limited to results for which actual failure is independently established and which are dependency-reachable from that failure location. Because actual failure is indeterminate in this reference, this repair-and-recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.205 (Primitive Time Roots and Result Dependence in the Arnovitt–Deser–Misner Reference). In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepIdx}(17, 18)$. Moreover, $R \in \text{Ref}^{\text{audit, rootdep}}$ holds.

Proof. The generation-initial input in Definition 8.202 consists of five positive non-temporal physical-input claim components, and the source-formal-evidence environment consists of 37 physical- and formal-input claim components. All 14 transfer hyperedges in Definition 8.203 take one or more time contents as inputs, and all 18 result hyperedges in Definition 8.204 also take one or more time contents as inputs. Hence, neither a transfer hyperedge nor a result hyperedge satisfies its first-application condition from the generation-initial input. Proposition 7.43 and Definition 7.39 give the index components of the time-content partition, generated-result set, and evidence-environment-reachable non-generated-result set. Adding the 17 primitive time roots to the seed and supplying the source-formal-evidence environment makes all 18 results reachable. The common reconstruction formulae in Definitions 7.48 and 8.115 give the index components of the remaining two result sets. Membership in the reference set follows from Definition 7.52. $\square$

Proposition 8.206 (Minimal Primitive-Time-Root Support in the Arnovitt–Deser–Misner Reference). For each $q \in \text{Idx}_R^{\text{audit, result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108, and the minimum and maximum cardinalities of those elements, are given in Table 112. The total number of family members is 23.

Table 112: Minimal Primitive-Time-Root Support in the
Arnowitt–Deser–Misner Reference

Result indices	Support statistics	Result indices	Support statistics
1–8, 10–11, 13–14, 16	(1; 1)	9	(2; 1, 2)
12	(1; 2)	15	(3; 1)
17–18	(2; 2, 3)		

Proof. Apply the update in Definition 7.46 to the 14 transfer hyperedges in Definition 8.203 and the 18 result-dependence hyperedges in Definition 8.204. Once the support antichains become invariant under the update, Proposition 7.47 uniquely determines the minimal primitive-time-root support families of all results. $\square$

Individual Statements

Objects of the Statements The preferred time coordinate t already selected in the comparison with particle mechanics, and N and q_{M+1} , which are left arbitrary by the dynamics in the parametrized system.

Verbatim Original (English) preferred time coordinate t has been decided upon

Thus N and hence q_{M+1} , are left arbitrary by the dynamics

The first quotation contains the time-coordinate symbol t ; the second contains N and q_{M+1} .

Original-Source Locations The first quotation is from Arnowitt, Deser, and Misner, *The Dynamics of General Relativity*, arXiv:gr-qc/0405109v1, p. 28, §8.1, footnote 25. The second quotation is from p. 5, §2.3, in the same reference.

Minimal Context Required for Interpretation Section 2.3 states the freedom to choose $q_{M+1}(\tau)$ and use it as the new independent variable. The change of independent variable is valid where it is differentiable and locally invertible. Choosing this function also fixes N . Page 8, §3.3 gives a coordinate choice in which lines of constant spatial coordinates are normal to the hypersurface and x^0 measures proper time along these lines. The first quotation, in footnote 25 on p. 28, §8.1, concerns the particle-mechanics example. The same footnote states that no such single preferred coordinate frame exists in the interior in gravitation, at least classically, and that it is not known whether quantum consistency requirements may force a preferred frame. The main text on that page states that classically equivalent choices of canonical variables need not remain unitarily equivalent in quantum theory.

Attribution of the Statements Both statements are attributable to the authors' reposted version of the 1962 chapter by Arnowitt, Deser, and Misner.

Predicates Assigned to the Objects by the Statements The first quotation states that the time coordinate t has already been selected as preferred in the particle-mechanics example. The second quotation assigns dynamical indeterminacy to N and q_{M+1} .

Explicit Domains of Application of the Statements The domain of application comprises parametrization and deparametrization in Hamiltonian form, 3+1 decomposition, canonical-variable choice, and canonical quantization.

Individual Audit

Interpretability The first statement describes a preferred time coordinate already selected in the particle-mechanics example. The source contrasts this particle-mechanics example with the absence of a single preferred coordinate frame in the interior in classical gravitation. It states that whether consistency requirements in quantum theory force a preferred frame is unknown. The second statement describes the freedom to choose a function left undetermined by the dynamics. Using it as a new independent variable requires differentiability and local invertibility. The surrounding context also discusses why classically equivalent choices of canonical variables need not be unitarily equivalent starting points in quantum theory.

Independent Grounds Both quotations directly state, in the original wording, the selection of a time coordinate and indeterminacy under the dynamics.

Evidence-Effect Boundary The first statement supports prior selection of a preferred time coordinate in the particle-mechanics example, but does not support that the coordinate has been generated from non-temporal physical inputs or that it is qualified as physical time. The second statement supports that N and q_{M+1} are left arbitrary by the dynamics, but does not provide grounds for identifying that arbitrary function with physical time. The quantum-theoretical non-uniqueness of canonical-variable choice belongs to the discussion in the main text of §8.1 and footnote 25.

Corresponding Formal Evidence The contrast and qualifications in footnote 25 and the main text of §8.1 correspond to Time Content 17 in Definition 8.201, Claim Components 36–37 in Definition 8.202, and Results 17–18 in Definition 8.204.

The second statement and the action transformation in §2.3 correspond to Time Contents 3 and 5, Claim Components 11–12, and Results 2 and 3. The field extension at the end of that section and the reduction by constraint solutions and coordinate conditions in §4 correspond to Time Contents 4 and 8–9, Claim Components 22–24, and Result 8.

Within-Source Comparison The first statement describes a time coordinate already selected in a particle-mechanics example; the second describes an arbitrary function left by parametrized dynamics. The former concerns a choice of variable in particle mechanics; the latter concerns the indeterminacy of a function in parametrized dynamics.

8.19.12 Source Audit

Audit Inputs

Original Source and Audit Scope

Reference Identifier 12.

Reference Name Primitive-time-root audit of Feynman–Leighton–Sands 1963.

Bibliographic Identification Let Richard P. Feynman, Robert B. Leighton, and Matthew Sands, *The Feynman Lectures on Physics*, Volume I (1963), be $R = \text{Ref}_{\text{FLS63T}}$. The audit takes Chapter 4, §4-4, Chapter 5, and Chapters 15–17 as its audit domain.

Original-Source Location Identifier 12-L1.

Original-Source Location Content Volume I, Chapter 4, §4-4, Chapter 5, and Chapters 15–17. The location of each time content is determined by Table 113, and the corresponding section for each formal input is determined by Table 115.

Audit-Scope Identifier 12-SC1.

Audit-Scope Content The audit domain specified in the bibliographic identification.

Formal Object

Object Identifier 12-F1.

Object Name Formal object.

Object Type Formal object.

Formal Content Definitions, assumptions, derivations, experimental reports and explanations concerning time measurement and standards, radioactive dating, distance measurements using time, energy processes and conservation conditions, Galilean and Lorentz transformations, light clocks, remote synchronization, particle lifetimes, twin comparisons, spacetime intervals, causal domains, the energy-momentum four-vector, photon energy-momentum-frequency relations and uncertainty in time measurement.

Original-Source Location Identifier Set 12-L1.

Table 114 lists the auxiliary objects. The claim and transfer tables specify the time-content and auxiliary-object premises at each use.

Assumptions

Assumption Identifier 12-A1.

Assumption Content The individual contents of the assumptions in the reference are given in Definition 8.208.

Original-Source Location Identifier Set 12-L1.

Non-Temporal Physical Inputs

Input Identifier 12-P1.

Input Name Formal input elements.

Input Type Non-temporal physical input.

Input Content The physical carriers of clocks, pendulums, oscillators, oscilloscopes, atomic systems, radioactive materials, light sources, light signals, mirrors, interferometers, spacecraft, particles, muons, and observers are taken as non-temporal physical inputs.

Original-Source Location Identifier Set 12-L1.

Time Contents

Time-Content Identifier 12-T1.

Time-Content Name Time contents.

Time-Content Type Time content.

Time-Content Description The contents include elapsed time, periods, lifetimes, ages, cosmic ages, time standards, coordinate time, proper time, simultaneity, time derivatives, temporal order and frequency.

Original-Source Location Identifier Set 12-L1.

Time-Content Identifier 12-T2.

Time-Content Name Time contents used in the original source.

Time-Content Type Time content.

Time-Content Description The 138 time contents in Definition 8.207.

Original-Source Location Identifier Set 12-L1.

Formal Input Elements

Formal-Component Identifier 12-C1.

Formal-Component Name Formal input elements.

Formal-Component Type Formal component.

Formal-Component Text Coordinate systems, events, worldlines, momentum, and energy belong to the source-formal-evidence environment.

Original-Source Support Identifier Set 12-L1.

Definition 8.207 (Time Contents and Auxiliary Objects in the Feynman–Leighton–Sands 1963 Reference). For $R = \text{Ref}_{\text{FLS63T}}$, the 196 semantic objects comprise 138 time contents and 58 auxiliary objects. Table 113 gives the time contents and their source locations. Within each row, the ascending indices correspond one-to-one to the contents in their listed order.

Table 113: Time Contents and Source Locations in Feynman–Leighton–Sands 1963

i	Time Contents (Index Order)	Pages
1	Temporal order in processes changing the form of energy	4-6–4-8
2	Temporal variation of internal motion and vibrations	4-6
3	Frequency in the photon energy and wavelength relations	4-7, 17-8
4	Time translation of the instant at which an experiment is performed	4-7
5	Fuel amount and energy supply per unit time	4-8
6–9	Correspondence between position and elapsed time in motion measurement; elapsed-time variable; Heartbeats as markers of equal time intervals; Time counting by heartbeats	5-1

<i>i</i>	Time Contents (Index Order)	Pages
10–12	equal intervals; Word “time”; Time familiar from experience as how long one waits	5-1
13–16	Periodic occurrence; number of periods; time measurement; Length of a day	5-1, 5-2
17–20	noon marker; Succession of hourglass inversions; number of hours; Successive time intervals	5-1, 5-2
21–22	Repetitive process; Time based on apparently periodic repetition	5-1, 5-2
23–26	Periodicity of pendulum oscillations; equal intervals; The 3600 periods corresponding to one hour; Unit of seconds	5-2, 5-3
27–30	Period of the faster oscillator; Period of the slower oscillator; Temporal variation of signal oscillations; Time correspondence in current and voltage records	5-2, 5-3
31–33	time axis; Short period; standard second	5-2, 5-3
34–37	Ordered endpoints; velocity; duration; Temporal order of pi-meson flight and decay	5-3
38–41	lifetime; Correspondence between time values obtained by different measurement procedures; Time scales of resonance processes and nuclear vibrations; lifetime	5-3
42	Shorter time interval	5-3
43–46	Long duration; number of days; Formation periods and accumulation order of tree rings and sediment layers; Time dependence of radioactive decay	5-3, 5-4, 5-5
47–48	half-life; age	5-3, 5-4, 5-5
49–52	Beginning of the universe; relation “before”; description of an earlier time; Chosen unit of time	5-5
53–56	Time standard distinguished from the unit; Second; mean day; Oscillation periods of atomic systems	5-5
57–58	Regularity of the process used as a clock standard; time scale	5-5
59–60	Elapsed time from transmission to return of a radar wave; Summer and winter observation times in stellar parallax	5-6
61	Year and light-propagation time used in the light-year	5-7–5-9
62–63	Time interval being measured; time uncertainty	5-10
64–67	Common coordinate time; Time; Time scales used in velocity and invariant light speed; Lorentz time coordinate	15-1, 15-2, 15-3
68–71	Endpoint events of the outward light path; Endpoint events of the return light path; propagation time; Equality of propagation times	15-3, 15-4, 15-5
72–75	Elapsed time measured by the onboard observer; Elapsed time measured by the external observer; elapsed-time relation; tick events	15-5, 15-6

<i>i</i>	Time Contents (Index Order)	Pages
76–79	clock period; Time correspondence when clocks are compared; Time correspondence when different physical processes are compared; common-rate relation	15-5, 15-6
80–83	The expression that time itself appears slower; Duration of a heartbeat; duration of a thought; duration of growth	15-5, 15-6
84	time required for a cancer to develop	15-5, 15-6
85–88	Muon decay processes paired in lifetime comparisons; rest lifetime; laboratory lifetime; time-dilation factor	15-7
89–92	Coordinate-time scale; distant simultaneity; Readings of separated clocks; Propagation time of light signals used for synchronization	15-7, 15-8
93–94	synchronization rule; simultaneity	15-7, 15-8
95–98	Time coordinates mixed with spatial coordinates; Time difference entering the spacetime invariant; time derivative; time derivative	15-8, 15-9, 15-10, 15-11
99	rate of work	15-8, 15-9, 15-10, 15-11
100–102	Time evolution compared in experiments on motion; time input; Rates of the clocks being compared	16-1, 16-2, 16-3
103–106	Temporal correspondence tracking the same twins from departure to reunion; departure and reunion; worldline duration; age difference	16-3, 16-4
107–110	Turnaround; Time dependence of acceleration during turnaround; Elapsed ages compared at reunion; Time evolution of muons moving on curved paths	16-3, 16-4
111	lifetime	16-3, 16-4
112–114	Time coordinate; Observer time; velocity composition	16-4, 16-5, 16-6
115–116	Velocity; Temporal order before and after collision	16-6, 16-7, 16-8, 16-9, 16-10
117–120	Position–time relation; Temporal persistence of an object; Time at which an event occurs; time coordinate	17-1, 17-2, 17-3
121–124	Temporal correspondence tracking the same particle along a worldline; worldline; indexing by time; Temporal relation between the two events whose interval is compared	17-1, 17-2, 17-3
125	spacetime interval	17-1, 17-2, 17-3
126–129	Light speed relating units of time and length; correspondence between spatial and temporal units; Spacelike interval; timelike interval	17-3, 17-4, 17-5
130–133	lightlike interval; Causal past; causal future; order of influence	17-3, 17-4, 17-5
134–136	signal limit; Distant present; simultaneity	17-3, 17-4, 17-5
137–138	Photon frequency in the moving frame; Time in the Minkowski quotation	17-8

Table 114: Auxiliary Objects in Feynman–Leighton–Sands
1963

a	Auxiliary Object	Related Time i
1	Circular dictionary definitions of time and period	none
2	Comparison of mean days	16–17
3	Equality judgment	20
4	Apparent regularity	20–21
5	Periodicity judgment	13, 20–22
6	Correspondence with time	19, 23–26
7	Period comparison	27–28, 32–33
8	Distance	none
9	Track length	none
10	Comparison correspondence	13–15, 34–36, 39
11	Equivalence judgment	39
12	Question concerning existence and meaning	42
13	Initial quantity	none
14	Fractional relation	54–55
15	Observer and coordinate system	none
16	Coordinate-system dependence	62
17	Measurement relation	63
18	Relation between Galilean coordinate systems	64
19	Transformation	64
20	Distance	none
21	Spatial coordinate	none
22	Mixing	65–67
23	Interferometer arm	none
24	Contraction	66
25	Arm-length relation	none
26	All-clock relation	74, 76–79
27	Restrictive condition	79–80, 84
28	Coordinate-system dependence	89–90, 94
29	Mixing	95–96
30	Law of mechanics	66, 97–99
31	Measurement boundary	100–101
32	Comparison	102
33	Prediction	110–111
34	Position coordinate	none
35	Inverse Lorentz transformation	66–67, 112
36	Displacement	none
37	Two coordinate systems	none
38	Transformation	117
39	Invariance	124–125
40	Law of light	126–127, 134
41	Coordinate-system dependence	135–136
42	Energy conservation	115
43	Composition law	none
44	Space	none

a	Auxiliary Object	Related Time i
45	Minkowski quotation on the unification of space and time	138
46	Energy uncertainty	none
47	Spatial light path of the light clock	none
48	Momentum	none
49	Energy	none
50	Energy-momentum four-vector	none
51	Component transformation law of a four-vector	none
52	Present amount of radioactive material	none
53	Planck constant and unit conventions of the relations	none
54	Principles of quantum mechanics assumed by the source	none
55	Amounts of fuel and energy	none
56	Light speed used in ranging and distance conversion	none
57	Earth-orbit baseline and observation angles	none
58	Wavelength	none

The related-time column identifies the time contents addressed by each claim. The earlier-than relation and description of an earlier time in claim 32, and the shorter time in claim 86, are subjects of questions. The related-time column for auxiliary objects identifies their temporal referents; the respective tables specify the premises of individual transfers and results.

Definition 8.208 (Physical- and Formal-Input Claim Components in the Feynman–Leighton–Sands 1963 Reference). Let the sequence of physical and formal input claims be $(\text{ClaimVal}_{R,p}^{\text{audit,phys}})_{p \in [95]}$. Table 115 specifies each component, its time-content and auxiliary-object correspondences, and its source location. Components 1–12 are non-temporal physical inputs. Components 13–95 are source formal evidence comprising measurement procedures, definitions, laws, conditional derivations, reported experiments or measurements, and questions or explanations. Claims 46, 52, 59 and 63 include reported experiments or measurements. Claim 16 explains time through the experience of waiting; claim 86 questions the meaning of shorter times.

Table 115: Physical and Formal Input Claims in Feynman–Leighton–Sands 1963

Com- po- nent	Claim	Related Time i	Auxil- iary Objects a	Pages
1	Physical carrier of a clock	none	none	Chapters 5, 15–17
2	Physical carrier of a pendulum	none	none	Chapter 5
3	Physical carrier of an oscillator	none	none	Chapter 5
4	Physical carrier of an oscilloscope	none	none	Chapter 5
5	Physical carrier of an atomic system	none	none	Chapter 5

Com- po- nent	Claim	Related Time i	Auxil- iary Objects a	Pages
6	Physical carrier of a radioactive material	none	none	Chapter 5
7	Physical carriers of a light source and light signal	none	none	Chapters 15–17
8	Physical carrier of a mirror	none	none	Chapters 15–17
9	Physical carrier of an interferometer	none	none	Chapter 15
10	Physical carrier of a spacecraft	none	none	Chapters 15–16
11	Physical carriers of particles and muons	none	none	Chapters 5, 15–16
12	Biological and physical carrier of an observer	none	none	Chapters 15–16
13	Motion measurement by position	6–10	8	5-1
14	Motion measurement by time	6–10	8	5-1
15	Equal time intervals by counting heartbeats	8–10	none	5-1
16	Explanation of time through the experience of waiting	12	none	5-1
17	Time measurement by a periodic process	13–15	none	5-1–5-2
18	Comparison of day and noon	16–17	2	5-1–5-2
19	Hourglass comparison	16–19	2	5-1–5-2
20	Apparent periodicity	20–22	3–5	5-2
21	Unproven boundary with true periodicity	20–22	3–5	5-2
22	Convention for the pendulum second	19, 23–26	6	5-2–5-3
23	Calibration of an oscillator chain	26–28, 32–33	7	5-3
24	Measurement of a short period by oscilloscope	27–33	7	5-3
25	Estimation of duration from distance and velocity	34–36	8	5-3
26	Pi-meson lifetime	34–35, 37–38	9	5-3
27	Agreement of multiple time procedures	13–15, 34–36, 39	8, 10–11	5-3
28	Equivalence relation based on agreement	13–15, 34–36, 39	10–11	5-3
29	Long-duration counting by natural indicators	43–45	none	5-3–5-4
30	Radioactive clock	46–47	13, 52	5-4
31	Radioactive dating	46–48	13, 52	5-4–5-5
32	Age of the universe	46–51	13, 52	5-5
33	Selection of a unit of time	52–53	none	5-5

Com- po- nent	Claim	Related Time i	Auxil- iary Objects a	Pages
34	Selection of a time standard	52–58	none	5-5
35	Rotation of the Earth	54–55	14	5-5
36	Comparison between the rotation of the Earth and clocks	54–58	14	5-5
37	Atomic oscillation	56–57	none	5-5
38	Clock reference based on atomic oscillation	52–53, 56–58	none	5-5
39	Observer-dependent time interval	62	15–16	5-10
40	Time–energy uncertainty	62–63	17, 46, 53	5-10
41	Galilean transformation using a common time	64	18–20	15-1–15-2
42	Representation of the speed of light	65–66	20	15-1–15-2
43	Incompatibility between the speed of light and the Galilean transformation	64–66	18–20	15-1–15-2
44	Lorentz transformation	65–67	21–22	15-2–15-3
45	Michelson–Morley round-trip-time model	65–66, 68–70	23	15-3–15-5
46	Report of the Michelson–Morley null result	68–71	23, 25	15-3–15-5
47	Lorentz contraction	65–66, 70–71	23–25	15-4–15-5
48	Time-coordinate transformation	65–67	21–22	15-2–15-3
49	Transverse light clock	66, 72–76	47	15-5–15-6
50	Rate relation for all clocks and all processes	74, 76–79	26	15-6
51	Reduction using an inconsistency assumption	74, 76–79	26	15-6
52	Lifetime of moving muons	66, 85–88	none	15-7
53	Distant simultaneity	67, 89–90, 94	28	15-7–15-8
54	Light-signal synchronization	66, 89–94	28	15-7–15-8
55	Mixing of space and time coordinates	66–67, 95	21–22, 29	15-8–15-9
56	Invariant expression	66–67, 95–96	21, 29	15-8–15-9
57	Relativistic momentum	65–66, 97	30, 48	15-9–15-10
58	Mass–energy relation expressed by time derivatives	65–66, 97–99	30, 48–49	15-10–15- 11
59	Experimental boundary concerning absolute motion	100–102	31–32	16-1–16-3
60	Departure–reunion comparison of twins	74, 103–107	26	16-3–16-4
61	Turnaround	103–109	none	16-3–16-4
62	Explanation concerning acceleration	103–109	none	16-3–16-4

Com- po- nent	Claim	Related Time i	Auxil- iary Objects a	Pages
63	Reports on curved-path muons and prediction for a complete-circle comparison	85–88, 110–111	33	16-3–16-4
64	Velocity composition	66, 112–115	34–36	16-4–16-6
65	Momentum in a collision	66, 114–116	30, 48–49	16-6–16-8
66	Historical relativistic mass	66, 114–116	30, 48	16-6–16-8
67	Total relativistic energy	66, 115–116	30, 48–49	16-8–16-10
68	Analogy of spacetime geometry	66, 117, 120	37–38	17-1–17-2
69	Event	118–120	none	17-1–17-2
70	Worldline	118–123	none	17-1–17-2
71	Invariant interval	66, 124–125	39	17-2–17-3
72	Unit conversion setting the speed of light to 1	126–127	40	17-3
73	Classification of spacelike intervals	124–125, 128	39	17-3–17-4
74	Classification of timelike intervals	124–125, 129	39	17-3–17-4
75	Classification of lightlike intervals	124–125, 130	39–40	17-3–17-4
76	Causal past	124–125, 129–131, 133–134	39–40	17-4–17-5
77	Causal future	124–125, 129–130, 132–134	39–40	17-4–17-5
78	Distant present	128, 134–136	41	17-4–17-5
79	Coordinate-system-dependent simultaneity	67, 135–136	41	17-4–17-5
80	Energy as the time component of a four-vector	66–67, 112, 114–115	48–51	17-5–17-7
81	Lorentz transformation of energy–momentum	66–67, 112, 114–115	48–51	17-5–17-7
82	Four-vector conservation	66, 115–116	42–43, 48–51	17-7–17-8
83	Relation between photon energy and the product of Planck’s constant h and frequency	3	49, 53	4-7; 17-8
84	Rates of biological processes under common physical laws	74, 77–79, 81–84	26	15-6
85	Reported resonance lifetimes and nuclear-vibration periods	40–41	none	5-3
86	Question of the existence and meaning of shorter times	42	12	5-3
87	Changes in energy form and internal motion	1–2	49	4-6–4-8
88	Time-translation invariance and the conditions for energy conservation	4	49, 54	4-7

Com- po- nent	Claim	Related Time i	Auxil- iary Objects a	Pages
89	Fuel and energy supply per unit time	5	49, 55	4-8
90	Distance measurement from radar round-trip time	59	8, 56	5-6
91	Stellar-parallax measurement at separated seasonal epochs	60	57	5-6
92	Distance expressed in light-years	61	56	5-7-5-9
93	Photon energy, momentum and wavelength relations	3, 126	48-49, 53, 58	17-8
94	Doppler frequency obtained from the energy transformation	3, 66, 115, 137	48-50, 53	17-8
95	Inverse transformation of the Lorentz time coordinate	66-67, 112	34-35	16-4

Definition 8.209 (Semantic Components and the Generation-Initial Set in the Feynman–Leighton–Sands 1963 Reference). Let the semantic-component nodes be $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(107)$. The values of Components 1–12 and the corresponding source-formal-description claims are defined by Table 116. $\text{ClaimNodeLift}_R(12; 95)$ and $\text{StdSemLift}_R(12; 95)$ hold. Define the generation-initial set and source-formal-evidence environment by $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(12)$. Components 13–107 retain the source-formal description and do not belong to the element type of the generation-initial set.

Table 116: Generation-Initial Semantic Components in the Feynman–Leighton–Sands 1963 Reference

k	Semantic component	Corresponding claim indices
1	Physical carrier of a clock	{1}
2	Physical carrier of a pendulum	{2}
3	Physical carrier of an oscillator	{3}
4	Physical carrier of an oscilloscope	{4}
5	Physical carrier of an atomic system	{5}
6	Physical carrier of a radioactive material	{6}
7	Physical carriers of a light source and light signal	{7}
8	Physical carrier of a mirror	{8}
9	Physical carrier of an interferometer	{9}
10	Physical carrier of a spacecraft	{10}
11	Physical carriers of particles and muons	{11}
12	Biological and physical carrier of an observer	{12}

Dependence Inputs

Dependency Inputs Periods, existing time scales, endpoint events, velocities, clock comparisons, synchronization and the time interval being measured supply the temporal inputs of the respective transfers.

Dependence and Transfer Elements

Definition 8.210 (Time-Content Transfers in the Feynman–Leighton–Sands 1963 Reference). Define the transfer index set by

$$\text{Idx}_R^{\text{audit,transfer}} = [24]$$

The equalities in Table 117 define the time-content input, time-content output, auxiliary-object support, and physical- and formal-input claim support of each transfer. The original-source location of each transfer is the union of the original-source locations corresponding to its time-content input, time-content output, and supporting claim components.

Table 117: Time-Content Transfers in Feynman–Leighton–Sands 1963

s	Time input i	Time output i	Auxiliary support a	Claim support p	Operation
1	$\llbracket 13, 14 \rrbracket \cup \llbracket 16, 21 \rrbracket$	$\{22\}$	$\llbracket 2, 5 \rrbracket$	$\llbracket 17, 21 \rrbracket$	Time basis from periodic repetition
2	$\{19\} \cup \llbracket 23, 25 \rrbracket$	$\{26\}$	$\{6\}$	$\{22\}$	Setting the second by a pendulum period
3	$\llbracket 28, 31 \rrbracket \cup \{33\}$	$\{27\} \cup \{32\}$	$\{7\}$	$\llbracket 23, 24 \rrbracket$	Calibration of oscillator periods
4	$\llbracket 34, 35 \rrbracket$	$\{36\}$	$\{8\}$	$\{25\}$	Duration from distance and speed
5	$\{35\} \cup \{37\}$	$\{38\}$	$\{9\}$	$\{26\}$	Pion lifetime from tracks
6	$\llbracket 46, 47 \rrbracket$	$\{48\}$	$\{13\} \cup \{52\}$	$\llbracket 30, 31 \rrbracket$	Radioactive dating
7	$\{55\}$	$\{54\}$	$\{14\}$	$\{35\}$	Definition of the second by the mean day
8	$\llbracket 52, 53 \rrbracket \cup \llbracket 56, 57 \rrbracket$	$\{58\}$	$\emptyset$	$\{34\} \cup \llbracket 37, 38 \rrbracket$	Time standard from atomic oscillations
9	$\llbracket 65, 66 \rrbracket$	$\{67\}$	$\llbracket 21, 22 \rrbracket$	$\{44\} \cup \{48\}$	Lorentz time-coordinate transformation
10	$\llbracket 65, 66 \rrbracket \cup \llbracket 68, 69 \rrbracket$	$\{70\}$	$\{23\}$	$\{45\}$	Calculation of light round-trip times
11	$\{66\} \cup \{70\}$	$\{71\}$	$\llbracket 23, 25 \rrbracket$	$\{47\}$	Round-trip equality under contraction

s	Time input i	Time output i	Auxil- iary support a	Claim support p	Operation
12	$\{66\} \cup \{72\} \cup \llbracket 75, 76 \rrbracket$	$\llbracket 73, 74 \rrbracket$	$\{47\}$	$\{49\}$	Elapsed time and rate of a light clock
13	$\{74\} \cup \llbracket 76, 78 \rrbracket$	$\{79\}$	$\{26\}$	$\llbracket 50, 51 \rrbracket$	Rate relation under common physical laws
14	$\llbracket 77, 79 \rrbracket$	$\llbracket 81, 84 \rrbracket$	$\{26\}$	$\{84\}$	Application of the common rate to biological processes
15	$\{66\} \cup \llbracket 85, 86 \rrbracket \cup \{88\}$	$\{87\}$	$\emptyset$	$\{52\}$	Mean lifetime of moving muons
16	$\{66\} \cup \{89\} \cup \llbracket 91, 92 \rrbracket$	$\llbracket 93, 94 \rrbracket$	$\{28\}$	$\{54\}$	Remote synchronization by light signals
17	$\{74\} \cup \llbracket 103, 105 \rrbracket \cup \llbracket 107, 108 \rrbracket$	$\{106\} \cup \{109\}$	$\emptyset$	$\llbracket 60, 62 \rrbracket$	Age comparison along the twins' paths
18	$\llbracket 66, 67 \rrbracket$	$\{112\}$	$\llbracket 34, 35 \rrbracket$	$\{95\}$	Inverse Lorentz time-coordinate transformation
19	$\{66\} \cup \llbracket 112, 113 \rrbracket$	$\{114\}$	$\llbracket 35, 36 \rrbracket$	$\{64\}$	Relativistic velocity composition
20	$\{120\} \cup \{126\}$	$\{127\}$	$\{40\}$	$\{72\}$	Correspondence of time and length units through light speed
21	$\llbracket 124, 125 \rrbracket$	$\llbracket 128, 130 \rrbracket$	$\llbracket 39, 40 \rrbracket$	$\llbracket 73, 75 \rrbracket$	Classification of spacetime intervals
22	$\llbracket 124, 125 \rrbracket \cup \llbracket 129, 130 \rrbracket \cup \{134\}$	$\llbracket 131, 133 \rrbracket$	$\llbracket 39, 40 \rrbracket$	$\llbracket 76, 77 \rrbracket$	Past and future causal domains
23	$\{3\} \cup \{66\} \cup \{115\}$	$\{137\}$	$\llbracket 48, 50 \rrbracket \cup \{53\}$	$\{81\} \cup \llbracket 93, 94 \rrbracket$	Transformation of photon frequency
24	$\{62\}$	$\{63\}$	$\{17\} \cup \{46\} \cup \{53\}$	$\{40\}$	Uncertainty in time measurement

Conditions of Formation Each transfer takes the tabulated time contents, auxiliary objects and supporting claims as inputs and holds under the following conditions.

1. Recurrent noons, reversals of an hourglass and successive time intervals supply the apparently periodic repetition on which the definition of time is based. The source leaves open the question of proving its true periodicity.
2. Equal small-amplitude pendulum periods and the correspondence of 3600 periods to an hour define one period as one second.
3. A slow reference oscillator and the recorded time correspondence of current and voltage calibrate the shorter period of a fast oscillator by counting waveform periods.

The source describes calibration of laser oscillations as a task not yet accomplished at that time.

4. The ordered positions of switching on and off, together with the known speed, give the duration as distance divided by speed.
5. The mean flight distance before decay and the known speed close to light speed determine the mean lifetime of the π^0 .
6. Given the initial amount A , present amount B and half-life T , constant fractional decay and the assumptions about the material's history determine the unknown age as $t = T \log(A/B) / \log 2$.
7. The source convention defines one second as $1/86400$ of the mean day.
8. The selected time unit is matched to the regular period of an atomic process to establish the time scale. The source anticipates a future improvement in accuracy with this standard.
9. Existing time and position coordinates, light speed and a subluminal relative velocity determine the transformed time coordinate.
10. In the source's hypothetical model of motion, the endpoint events of the outward and return paths, the path lengths and the speeds determine longitudinal and transverse round-trip times.
11. Applying the stipulated longitudinal contraction makes the model's two round-trip times equal. The experimental null result corresponds to claim 46.
12. The onboard clock period, invariant light speed and tick events determine the elapsed time measured by the external observer and its ratio to the onboard period.
13. If the relativity principle holds for uniform motion, clocks and processes governed by the same physical laws share a common rate factor.
14. Under the condition that the same physical laws apply, the common-rate argument extends to heartbeat, thought, growth and the time until cancer develops. For cancer the source says "nearly certain."
15. The rest-frame mean lifetime multiplied by the time-dilation factor predicts the laboratory-frame mean lifetime. The source reports agreement between this prediction and measurement.
16. In the specified frame, separated clock readings and a light signal's round-trip time define the remote clock setting and simultaneity by the midpoint of emission and reception times.
17. Between common departure and reunion events, elapsed ages are compared along the specified paths, including the turnaround and its acceleration.
18. The Lorentz transformation with the sign of the relative velocity reversed gives the inverse-transformed time from the existing time coordinate.

19. The transformed displacement divided by the same observer's elapsed time gives the velocity. The simple transverse formula holds when $v'_x = 0$.
20. Light speed relates the time and length units, allowing a unit system in which $c = 1$ numerically.
21. The sign of the interval between two events defines the three classes of spacelike, timelike and null intervals. Each event pair receives the class corresponding to its interval sign.
22. The temporal relation of two events, timelike and null intervals, temporal orientation and the signal-speed restriction define the past and future domains of possible influence and their order.
23. Given the photon momentum direction and the sign convention for the coordinate transformation, the Lorentz transformation of energy and $E = h\nu$ determine the frequency in the moving frame.
24. For a process whose time interval is being measured, the energy uncertainty ΔE and Planck constant h express the minimum error in time measurement through the source relation $\Delta t = h/\Delta E$.

Original-Source Support The claim, time-content and auxiliary-object indices in each row of Table 117 are the corresponding index sets of Definition 7.105. Source locations are given by the time-content and claim tables and the respective conditions of formation. The time-content transfer relation defined by the 24 transfers is acyclic.

Results

Definition 8.211 (Results in the Feynman–Leighton–Sands 1963 Reference). Define the result index set by $\text{ldx}_R^{\text{audit,result}} = [14]$. The correspondence between results and original-source locations is given in Table 118. Each result is the collection of claims listed in its supporting-claim column. The claim table specifies the time-content and auxiliary-object correspondences of each constituent; the result-dependence table gives the premises of the collection as a whole. Claim 86 states the question about the meaning of shorter times accompanying result 2; claim 32 states the question about time before the beginning accompanying result 3.

Table 118: Results and Source Locations in Feynman–Leighton–Sands 1963

j	Result	Source Location
1	Time measurement using motion and periodic processes, and clock standards	5-1-5-5
2	Apparent periodicity, consistency between procedures, and the question of shorter times	5-2-5-3
3	Time and distance estimates, dating, and the choice of units and standards	5-3-5-10

j	Result	Source Location
4	Galilean and Lorentz transformations, invariants, and the domain of the relativity principle	5-10; 15-1–15-3; 15-8–15-9; 16-4; 17-1–17-3; 16-1–16-3
5	Michelson–Morley round-trip model, null result, and the contraction explanation	15-3–15-5
6	Light-clock rate relation and rates of clocks and processes governed by common physical laws	15-5–15-6
7	Moving-muon lifetime predictions and reports, and the predicted full-circle comparison	15-7; 16-3–16-4
8	Distant simultaneity and light-signal synchronization	15-7–15-8; 17-4–17-5
9	Age comparison along the specified twin paths between departure and reunion	16-3–16-4
10	Energy processes, conservation conditions, and relativistic dynamics	4-6–4-8; 15-9–15-11; 16-4–16-10
11	Events, worldlines, invariant intervals, and causal-domain classification	17-1–17-5
12	Transformation and conservation relations of the energy-momentum four-vector	17-5–17-8
13	Photon energy, momentum and frequency relations and the Doppler effect	4-7; 17-8
14	The source’s stated relation between time and energy measurement uncertainties	5-10

Source-Specific Results The 14 results in Definition 8.211. The original-source location, time-content premises, auxiliary-object premises, and supporting claims of each result are given by the tables in that definition and Definition 8.212.

Result Dependence

Definition 8.212 (Result-Dependence Hyperedges in the Feynman–Leighton–Sands 1963 Reference). For each $q \in [14]$, the equalities in Table 119 define the time-content input, auxiliary-object input, and physical- and formal-input claim support of the result-dependence hyperedge $\text{ResultEdge}_{R,q}$, whose result output is Result q .

Table 119: Result-Dependence Hyperedges in Feynman–Leighton–Sands 1963

q	Time Premises i	Auxiliary Premises a	Supporting Claims p
1	$\llbracket 6, 10 \rrbracket \cup \llbracket 13, 19 \rrbracket \cup \llbracket 23, 36 \rrbracket \cup \{39\} \cup \llbracket 52, 58 \rrbracket$	$\{2\} \cup \llbracket 6, 8 \rrbracket \cup \llbracket 10, 11 \rrbracket \cup \{14\}$	$\llbracket 13, 15 \rrbracket \cup \llbracket 17, 19 \rrbracket \cup \llbracket 22, 24 \rrbracket \cup \llbracket 27, 28 \rrbracket \cup \llbracket 33, 38 \rrbracket$

q	Time Premises i	Auxiliary Premises a	Supporting Claims p
2	$\llbracket 13, 15 \rrbracket \cup \llbracket 20, 22 \rrbracket \cup \llbracket 34, 36 \rrbracket \cup \{39\}$	$\llbracket 3, 5 \rrbracket \cup \{8\} \cup \llbracket 10, 11 \rrbracket$	$\llbracket 20, 21 \rrbracket \cup \llbracket 27, 28 \rrbracket$
3	$\llbracket 34, 38 \rrbracket \cup \llbracket 40, 41 \rrbracket \cup \llbracket 43, 49 \rrbracket \cup \llbracket 52, 61 \rrbracket$	$\llbracket 8, 9 \rrbracket \cup \llbracket 13, 14 \rrbracket \cup \{52\} \cup \llbracket 56, 57 \rrbracket$	$\llbracket 25, 26 \rrbracket \cup \llbracket 29, 38 \rrbracket \cup \{85\} \cup \llbracket 90, 92 \rrbracket$
4	$\{62\} \cup \llbracket 64, 67 \rrbracket \cup \llbracket 95, 96 \rrbracket \cup \llbracket 100, 102 \rrbracket \cup \{112\} \cup \{117\} \cup \{120\}$	$\llbracket 15, 16 \rrbracket \cup \llbracket 18, 22 \rrbracket \cup \{29\} \cup \llbracket 31, 32 \rrbracket \cup \llbracket 34, 35 \rrbracket \cup \llbracket 37, 38 \rrbracket$	$\{39\} \cup \llbracket 41, 44 \rrbracket \cup \{48\} \cup \llbracket 55, 56 \rrbracket \cup \{59\} \cup \{68\} \cup \{95\}$
5	$\llbracket 65, 66 \rrbracket \cup \llbracket 68, 71 \rrbracket$	$\llbracket 23, 25 \rrbracket$	$\llbracket 45, 47 \rrbracket$
6	$\{66\} \cup \llbracket 72, 79 \rrbracket \cup \llbracket 81, 84 \rrbracket$	$\{26\} \cup \{47\}$	$\llbracket 49, 51 \rrbracket \cup \{84\}$
7	$\{66\} \cup \llbracket 85, 88 \rrbracket \cup \llbracket 110, 111 \rrbracket$	$\{33\}$	$\{52\} \cup \{63\}$
8	$\llbracket 66, 67 \rrbracket \cup \llbracket 89, 94 \rrbracket \cup \{128\} \cup \llbracket 134, 136 \rrbracket$	$\{28\} \cup \{41\}$	$\llbracket 53, 54 \rrbracket \cup \llbracket 78, 79 \rrbracket$
9	$\{74\} \cup \llbracket 103, 109 \rrbracket$	$\{26\}$	$\llbracket 60, 62 \rrbracket$
10	$\llbracket 1, 2 \rrbracket \cup \llbracket 4, 5 \rrbracket \cup \llbracket 65, 66 \rrbracket \cup \llbracket 97, 99 \rrbracket \cup \llbracket 112, 116 \rrbracket$	$\{30\} \cup \llbracket 34, 36 \rrbracket \cup \llbracket 48, 49 \rrbracket \cup \llbracket 54, 55 \rrbracket$	$\llbracket 57, 58 \rrbracket \cup \llbracket 64, 67 \rrbracket \cup \llbracket 87, 89 \rrbracket$
11	$\llbracket 66, 67 \rrbracket \cup \llbracket 117, 136 \rrbracket$	$\llbracket 37, 41 \rrbracket$	$\llbracket 68, 79 \rrbracket$
12	$\llbracket 66, 67 \rrbracket \cup \{112\} \cup \llbracket 114, 116 \rrbracket$	$\llbracket 42, 43 \rrbracket \cup \llbracket 48, 51 \rrbracket$	$\llbracket 80, 82 \rrbracket$
13	$\{3\} \cup \{66\} \cup \{115\} \cup \{126\} \cup \{137\}$	$\llbracket 48, 50 \rrbracket \cup \{53\} \cup \{58\}$	$\{83\} \cup \llbracket 93, 94 \rrbracket$
14	$\llbracket 62, 63 \rrbracket$	$\{17\} \cup \{46\} \cup \{53\}$	$\{40\}$

Take the supporting-claim indices, time-content input indices, and auxiliary-object input indices in each row of Table 119 as the corresponding index sets in Definition 7.105. The claim-support set, semantic-component support set, time-content-premise set, and auxiliary-object-premise set of each result are determined by that definition.

The 14 results have 246 semantic-object premise occurrences: 176 time-content and 70 auxiliary-object occurrences; and every result has one or more time-content premises.

Grounds of Establishment

Grounds of Establishment of the Results The evidence types directly supplied by this reference are textbook definitions, mathematical derivations, explanations of measurement procedures, conceptual arguments, and reports of existing experimental results.

1. The heartbeat example assumes equal intervals, and the pendulum uses equal periods at small amplitude. Each clock standard rests on its calibration and comparison of regularity.
2. Apparently periodic repetition grounds time measurement, and consistency between different procedures is examined. The source poses questions about proving true periodicity and about the meaning of still shorter times.
3. Lifetime estimation from flight distance presupposes speed; long-time estimation from natural markers presupposes formation periods; radioactive dating presupposes

a half-life and material history. Radar ranging uses round-trip time, stellar parallax uses observation times separated by seasons, and a light-year uses one year and light speed. Cosmic age is stated as an estimate current at the time of the source.

4. The Galilean transformation uses common time; the Lorentz transformation uses subluminal relative motion in the specified direction and invariant light speed. The source describes experiments supporting relativity for uniform translation and treats rotation as detectable. Its discussion of removing the stars is an untested question.
5. The constituents are the round-trip calculation in the hypothetical model, the reported null result, and the model explanation by length contraction.
6. The light-clock construction and the relativity principle yield a common rate, which applies to processes governed by the same physical laws. Application to cancer carries the source qualification “nearly certain.”
7. Chapter 15 reports agreement between moving-muon lifetimes and prediction. Chapter 16 describes observations involving curved motion and presents the comparison of a particle returning after a complete circle with a stationary particle as a prediction for an experiment not yet performed.
8. The specified frame, light-propagation conditions, clock readings and midpoint convention determine remote times. Distant simultaneity depends on the frame.
9. Elapsed ages are compared for common departure and reunion events and specified stay-at-home and outward-and-return paths between them. Turnaround and acceleration are features of these paths.
10. The statement connecting time-translation invariance to energy conservation adds the principles of quantum mechanics. The differential derivation of relativistic mass uses $E = mc^2$, the rate of work and the time derivative of momentum. The collision derivation uses elastic scattering of identical particles, conservation laws, parallel momentum and velocity, and the low-speed limit.
11. Event positions and times, together with the temporal correspondence of the same particle, describe a worldline. Interval-sign classification and the signal-speed restriction determine causal domains of possible influence.
12. In the source’s $c = 1$ units, energy follows the same transformation law as the time component of a four-vector. Energy-momentum transformations use the existing velocity and coordinate transformations and dynamical definitions, yielding the four-component conservation relation.
13. $E = h\nu$, $p = h/\lambda$ and $\nu = c/\lambda$ relate photon energy, momentum and frequency. Given the momentum direction and sign convention for the coordinate transformation, the energy transformation yields the Doppler frequency.
14. The source relation expresses the minimum error in time measurement from the energy uncertainty of the process whose time interval is measured. The meaning of the time interval as the measurand is given; this specification holds even when its numerical value is unknown.

Evidence Hierarchy

Directly Measured Objects The number of directly measured objects newly acquired by this reference is zero.

Direct-Measurement Predicates The number of direct-measurement predicates newly acquired by this reference is zero.

Conditions and Domain of Direct Measurement The number of conditions and domains of direct measurement newly acquired by this reference is zero.

Measured Objects Reported from Other References The Michelson–Morley interferometer, moving muons, clocks, and periodic processes.

Measurement Predicates Reported from Other References The null result of the interferometer, the lifetime difference of moving muons, and comparison results for clocks and periodic processes are reported from existing experiments.

Conditions and Domain of Measurement Reported from Other References For the interferometer, the domain includes arm arrangement, rotation and fringe-detection conditions; for muons, comparison of mean lifetimes in the rest and laboratory frames; for clocks and periodic processes, their calibration conditions and comparison of regularity.

Objects of Formal Generalization Results 1–3 concern time measurement, dating, distance measurement and standard selection; results 4–5 concern coordinate transformations and light round-trip times; result 6 concerns rates of clocks and processes governed by common physical laws; result 7 concerns muon lifetimes; result 8 concerns remote synchronization; result 9 concerns elapsed ages along specified twin paths. Result 10 concerns energy processes, conservation conditions and relativistic mechanics; result 11, spacetime intervals and causal domains; result 12, the energy-momentum four-vector; result 13, photon relations and the Doppler effect; and result 14, uncertainty in time measurement.

Conditions and Domain of Formal Generalization Each result holds under its tabulated time-content premises, auxiliary-object premises and supporting claims, together with the conditions listed under Grounds of Establishment. Application of the common rate to biological processes requires the same physical laws to apply and carries the source qualification for cancer.

Generation Judgment **Judgment values:** LCTR complete-generation judgment: 0; source-internal evidence-conditioned local-reach judgment: 0; remaining primitive time roots: 138.

Generation Reach and Primitive Time Roots No time content is generated by generation reach from an initial set consisting only of non-temporal physical inputs. All 138 time contents in Definition 8.207 remain primitive time roots. Source-internal evidence-conditioned local reach also reaches zero time contents.

Source-Internal Evidence-Conditioned Local Reach The source-internal evidence-conditioned local-reach judgment is zero; 138 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ from Definition 7.108 and apply Definition 7.50 to it. Totaling over all results determines the primitive-time-root–result dependence relations and the necessary-primitive-time-root–result dependence relations.

Result Dependence All 14 results in Definition 8.211 are primitive-time-root-dependent results. The numbers of generated results, evidence-reach-only results, and unreached-after-root-addition results are zero. The minimal primitive-time-root support of each result is given by Proposition 8.214.

Evidence Effect

Evidence-Effect Scope The definitions, measurement procedures, conditional derivations, reported experiments or measurements, and explanations in this source support the 14 tabulated results under their respective calibration, frame, physical-law, material-history, path and domain conditions.

Evidence-Effect Boundary Time measurement, calibration and dating use periods, time scales, decay laws and half-lives; relativistic derivations use times, velocities, elapsed times or temporal relations between events. The photon relations presuppose frequency, and the time-uncertainty relation presupposes the time interval being measured. The evidence-effect boundary of Definition 7.115 lies at the passage from results using these time contents to the generation structure establishing their physical use.

Result-Preservation Domain

Result-Preservation Domain The 14 results are preserved insofar as each time-content premise is supplied by the corresponding generated time representation while preserving the physical carriers, calibration, frames, decay laws, material histories, paths and dependence directions. Application to biological processes is conditional on the same physical laws applying and carries the source qualification for cancer.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The time-content, claim, transfer and result-dependence tables identify the uses of the 131 time contents serving as physical premises of transfers or results. They are used as established time inputs without the corresponding non-temporal generation structures. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.” Time contents 11, 12, 42, 50, 51, 80 and 138 correspond to a word, an experiential explanation, open questions or expressions in quotations.

Evidence of Actual Failure

Evidence of Actual Failure The complete judgment and the remaining-root judgment concern physical-time qualification; they are distinct from actual failure of a concrete condition governing a result’s value or domain of application. Neither the original formal evidence nor the LCTR audit evidence in this reference establishes an actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence,” and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream Reach For each reference, the reach to primitive-time-root-dependent results is determined by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because actual failure of a concrete condition has not been established, the actual-reach judgment from that failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the objects of repair are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is limited to results for which actual failure is independently established and which are dependency-reachable from that failure location. Because actual failure is indeterminate in this reference, this repair-and-recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.213 (Primitive Time Roots and Result Dependence in the Feynman–Leighton–Sands 1963 Reference). In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepldx}(138, 14)$ holds. Moreover, $R \in \text{Ref}^{\text{audit, rootdep}}$ holds.

Proof. Take the generation-initial semantic-component set $\text{Init}_R^{\text{phys}}$ as the starting point of generation reach. All 24 transfers in Definition 8.210 have one or more time contents as inputs. By Proposition 7.43, no time-content node is reachable from this starting point, and all 138 time contents are primitive time roots.

Even when the source-formal-evidence environment $\text{Env}_R^{\text{phys}}$ is taken as the starting point of evidence-conditioned reach, the time-content inputs of the transfers are not satisfied. Each of the 14 result-dependence hyperedges in Table 119 presupposes at least one primitive time root. Adding the source-formal-evidence environment and 138 primitive time roots to the initial set makes all 14 results reachable. The common reconstruction formulae in Proposition 7.49 and Definition 8.115 give the displayed index tuple. Membership in the reference set follows from Definition 7.52. $\square$

Proposition 8.214 (Minimal Primitive-Time-Root Support in the Feynman–Leighton–Sands 1963 Reference). For each

$$q \in \text{Idx}_R^{\text{audit, result}}$$

the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108, and the minimum and maximum cardinalities of those elements, are given in Table 120.

Table 120: Minimal Primitive-Time-Root Support in
Feynman–Leighton–Sands 1963

Result indices	Support statistics	Result indices	Support statistics
1	(1; 28)	2	(2; 9, 12)
3	(1; 19)	4	(1; 11)
5	(1; 4)	6	(1; 6)
7	(1; 6)	8	(4; 10, 11)
9	(2; 6, 9)	10	(1; 12)
11	(2; 15)	12	(4; 5)
13	(1; 4)	14	(1; 1)

Proof. Apply the update in Definition 7.46 to the 24 transfer hyperedges. After one scan in the listed order, the second scan leaves the support antichains of all time-content nodes unchanged. Proposition 7.47 gives the tabulated values. The support families contain 23 members in total, with 150 primitive-time-root–result dependence relations and 129 necessary-primitive-time-root–result dependence relations. $\square$

Individual Statements

Objects of the Statements The definability and measurement of time, dating using a half-life, the age and beginning of the universe and what precedes it, the common rate of moving clocks and the expression that time itself appears slower, and application to heartbeat, thought and cancer development.

Verbatim Original (English) time is one of the things we probably cannot
define

how we measure it

In an arbitrary time interval t there are t/T “half-lives,”

At some time more than five billion years ago, the universe started. It is now believed that at least our part of the universe had its beginning about ten or twelve billion years ago. We do not know what happened before then. In fact, we may well ask again: Does the question make any sense? Does an earlier time have any meaning?

Now if all moving clocks run slower, if no way of measuring time gives anything but a slower rate, we shall just have to say, in a certain sense, that time itself appears to be slower in a space ship.

the man’s pulse rate, his thought processes

the time it takes for a cancer to develop

The third quotation contains the interval symbols t and t/T .

Original-Source Locations The first and second quotations are from Feynman, Leighton, and Sands, *The Feynman Lectures on Physics*, Volume I, Chapter 5, §5-2, *Time*. The third quotation is from Chapter 5, §5-4, *Long times*, in the same book. The fourth quotation is from the end of Chapter 5, §5-4, *Long times*, in the same book. The fifth through seventh quotations are from the same paragraph in Chapter 15, §15-4, *Transformation of time*, in the same book.

Minimal Context Required for Interpretation Section 5-2 describes the difficulty of defining time in a dictionary and explains methods of measurement. It bases the definition of time on apparently periodic repetition. Section 5-4 obtains the unknown age t from the initial amount A , present amount B and half-life T , connecting the discussion to the age and beginning of the universe and the meaning of what precedes that beginning. Under the relativity principle and common physical laws, Section 15-4 extends the light-clock rate relation to other clocks and processes, describing the slowness of time itself with the qualifications “in a certain sense” and “appears.” It mentions heartbeat, thought, growth, aging and cancer development, describing the application to cancer as “nearly certain.”

Attribution of the Statements All seven statements are attributable to the text of Volume I by Feynman, Leighton, and Sands.

Contents Directly Stated by the Quotations The first quotation identifies time as one of the things that we, including the authors, probably cannot define in the dictionary sense. The second makes the method of measuring time the subject of discussion. The third quotation expresses the number of half-lives contained in an arbitrary time interval t as t/T . The fourth quotation assigns a beginning and an age to the universe and to our part of the universe, and asks about events and the meaning of time before that beginning. The fifth quotation states that, if all moving clocks and all methods of measuring time give a slower rate, time itself appears slower in the spacecraft in a certain sense. The sixth and seventh quotations extend the object of application of that rate difference to human biological and cognitive processes.

Explicit Domains of Application of the Statements Time measurement, radioactive dating, descriptions of the age of the universe, special-relativistic clock comparison, moving-particle lifetimes, and human biological and cognitive processes.

Evidence Provenance The text of Feynman, Leighton and Sands, *The Feynman Lectures on Physics*, Volume I, Chapter 5, §§5-2 and 5-4, and Chapter 15, §15-4.

Individual Audit

Interpretability The first and second statements describe the difficulty of a dictionary definition and explain measurement methods. Apparently periodic repetition provides the basis for measuring and defining time. The third states the ratio of an arbitrary interval t to the half-life T ; dating obtains t from the initial amount, present amount and half-life. The fourth follows its account of cosmic age with a question about the meaning of time before the beginning. The fifth statement, that time itself appears slower, occurs within the argument that clocks and processes share a common rate under the relativity principle and carries explicit qualifications. The sixth and

seventh apply the argument to biological and cognitive processes governed by the same physical laws, with the qualification “nearly certain” for cancer.

Independent Grounds The grounds for the seven statements are explanations of measurement methods, calculation using the decay law, an account of cosmic age, and the rate argument based on the relativity principle and common physical laws. The muon measurements reported in the same chapter compare the rest mean lifetime with the laboratory-frame lifetime. Application to biological processes is presented as theoretical inference from common physical laws.

Evidence-Effect Boundary Measurement methods presuppose periods and their comparison; dating presupposes the time dependence of decay and a half-life; relativistic rate relations presuppose times, elapsed times and velocities. The account of cosmic age introduces a beginning and the temporal relation of being earlier. Physical use of these contents requires the corresponding generation structure to hold. The evidence-effect boundary of Definition 7.115 lies at the passage from the measurement, calculation or explanation supported by each statement to that establishment.

Corresponding Formal Evidence The first through fourth statements correspond to time contents 11–22, 43–58 in Definition 8.207, claims 16–21 and 29–38 in Definition 8.208, and results 1–3 in Definition 8.211. The fifth through seventh correspond to time contents 72–84, claims 49–51 and 84, and result 6 in the same definitions.

Within-Source Comparison The clock-rate relation follows from the light-clock construction and the relativity principle, while its application to biological processes shares the condition that the same laws operate. The time scale used in measurement, the unknown inferred by dating, the questions in the account of cosmic age and the theoretical predictions for biological processes correspond to their respective claim components and conditions of formation.

Common Audit with Specified Domain of Application The domains and time-content dependencies of the seven statements are specified by the claim components, results and source locations above.

8.19.13 Source Audit

Audit Input

Original Source and Audit Scope

Source Identifier 13.

Source Name Complete audit of time use in Taylor–Wheeler 1992

Bibliographic Identification Write the audit target as

$$R := \text{Ref}_{\text{TW92SP}}^{\text{survey}}$$

This denotes Edwin F. Taylor and John Archibald Wheeler, *Spacetime Physics: Introduction to Special Relativity*, 2nd ed., W. H. Freeman and Company, 1992. This division covers time use appearing in Chapters 1–9, the Lorentz-transformation special topic, the answers, and the index.

Audit-Scope Identifier 13-SC1.

Audit-Scope Content The audit domain stated in the bibliographic identification.

Source-Location Identifier 13-L1.

Source-Location Content The audit domain stated in the bibliographic identification and the source-location columns of the respective evidence tables.

Formal Objects

Object Identifier 13-F1.

Object Name Formal objects of the original source.

Object Type Formal object.

Formal Content The individual contents of the formal objects are given in Definitions 8.216, 8.217, and 8.218.

Source-Location Identifier Set 13-L1.

Assumptions

Assumption Identifier 13-A1.

Assumption Content Construction of an inertial frame presupposes that disturbances of test bodies and tidal effects are within tolerance for the specified region, observation duration and detection accuracy. Flat-spacetime relations, the response of good clocks, isolated-system boundaries, mass-shell conditions, the weak-field approximation and emitter-receiver motion supply the conditions of formation of the respective claims and transfers. The auxiliary-object table and the conditions below specify these correspondences.

Original-Source Location Identifier Set 13-L1.

Non-Temporal Physical Inputs

Input Identifier 13-P1.

Input Name Non-temporal physical inputs of the original source.

Input Type Non-temporal physical input.

Input Content The individual contents of the non-temporal physical inputs are given in Definitions 8.216 and 8.218.

Source-Location Identifier Set 13-L1.

Time Contents

Time-Content Identifier 13-T1.

Time-Content Name Time contents of the original source.

Time-Content Type Time content.

Time-Content Statement Table 122 specifies the individual time contents, and Table 125 gives their time-use roles.

Source-Location Identifier Set 13-L1.

Formal Input Components

Formal-Component Identifier 13-C1.

Formal-Component Name Formal-evidence environment of the original source.

Formal-Component Type Formal component.

Formal-Component Text The individual contents of the formal input components are given in Definitions 8.218, 8.216, and 8.217.

Original-Source Support Identifier Set 13-L1.

Definition 8.215 (Source and Attribution Components). The source claims, quotations and source locations are attributable to Taylor and Wheeler. Aircraft atomic-clock comparisons, particle-lifetime measurements, resonance absorption and binary-pulsar observations are results of the respective experiments or observations reported in the book.

Definition 8.216 (Composition of Formal Evidence). The physical and formal evidence comprises 106 components. Components 1–12 are non-temporal physical carriers. Components 13–106 comprise specifications of events and apparatus, definitions, laws, conditional derivations, reported experiments or observations, and explanations. The claim table specifies the time contents and auxiliary objects addressed by each component.

Definition 8.217 (Finite Audit Index Domains for Taylor–Wheeler 1992). The index domains for physical and formal evidence, source mathematical-object types, time contents, auxiliary objects, results and transfers are respectively

$$[106], \quad [14], \quad [41], \quad [37], \quad [29], \quad [23].$$

Definition 8.218 (Physical and Formal Input Claim Components for Taylor–Wheeler 1992). Table 121 gives each claim, its correspondence to time contents i and auxiliary objects a , and its source location. “–” denotes the absence of a corresponding index. Event identity, spatial arrangement and nearest-point selection in claims 13, 15, 31, 32 and 38 are distinguished from the associated clock settings and readings.

Superscript codes on the time indices denote the use roles in Table 125.

Table 121: Physical and Formal Evidence in Taylor–Wheeler 1992

Com- po- nent	Claim	i	a	Source Location
1	Physical carrier of clocks	–	–	Chs.1–9
2	Physical carrier of rods	–	–	Chs.1–3; Ch.9
3	Physical carrier of the lattice and lattice-point recording apparatus	–	–	§§2.6–2.7
4	Physical carrier of light signals	–	–	Chs.1–9
5	Physical carrier of the spacecraft	–	–	Ch.4; Ex.8-41
6	Physical carrier of particles and test particles	–	–	Chs.1–3,7–9

Com- po- nent	Claim	i	a	Source Location
7	Physical carriers of the bodies constituting a collision system	–	–	Chs.7–8
8	Physical constitution of recording media	–	–	§§2.6–2.7; §9.7
9	Physical carrier of aircraft-borne atomic clocks	–	–	§4.10
10	Physical carriers of photons and of the bodies involved in pair creation and annihilation	–	–	Ch.8
11	Physical carrier of gravitational sources and gravitational-response detectors	–	–	Ch.9; Exs.8-37–8-38
12	Physical carrier of pulsar signals	–	–	§9.7
13	Identification of local events	–	3	§§2.6,5.1–5.2
14	Assignment of spacetime coordinates by inertial frame	$1^Q, 2^K, 4^X$	5–6	§§2.6–2.7; Ch.5
15	Specification of spatial separation	–	4	Chs.1,3; Lorentz transformation special topic
16	Specification of temporal separation	$1^{QB}, 3^{IQ}$	3	Chs.1,3; Lorentz transformation special topic
17	Construction of the spacetime interval from both separations	$3^{IQ}, 16^{IQ}$	4, 7	Chs.1,3; §6.2
18	Use of the vacuum speed of light	$28^{IQ}, 29^Q$	6, 11	§1.4; §§3.6–3.8; Box3-2
19	Conversion between time and length units	$3^Q, 29^Q$	11, 17	§1.4; Box3-2
20	Sign determination for timelike separation	$3^{IQ}, 16^{IQ}$	4, 7	§6.2
21	Sign determination for spacelike separation	3^{IQ}	4, 7	§6.2
22	Sign determination for lightlike separation	$3^{IQ}, 28^{IQ}, 29^Q$	4, 7	§6.2
23	Invariance of the spacetime interval for the same event pair	$3^{IQ}, 16^{IQ}, 17^Q$	4, 6–7	Chs.1,3; Lorentz transformation special topic
24	Specification of a clock passing through two events	$5^B, 14^B$	3, 8	L.3–L.6 Ch.1; §§5.6–5.7
25	Clock readings at two events	$2^K, 6^{IK}$	8	Ch.1; §§2.6,5.7
26	Correspondence between clock readings and proper time	$6^K, 21^{IQ}$	7–8	Ch.1; §§5.6–5.7
27	Correspondence between proper time and local time	$6^K, 21^{IQ}$	8	§§5.6–5.7

Com- po- nent	Claim	i	a	Source Location
28	Correspondence between proper time and wristwatch time	$7^K, 21^{IQ}$	8	§§4.5–4.6, 5.6–5.8
29	Specification of test-particle motion	$3^{IS}, 26^Q, 27^Q$	9–10	§§2.1–2.5, 9.2–9.5
30	Selection of a local free-fall frame by test-particle motion	$3^{IO}, 26^Q, 27^Q$	9–10	§§2.3–2.5, 9.2–9.5
31	Arrangement of the rod lattice	–	4–5	§2.6
32	Spatial placement of clock carriers at lattice points	–	5, 8	§2.6
33	Constitution of an inertial frame by a rod-and-clock lattice	$2^K, 4^X, 18^{XB}$	5–6	§§2.6–2.7
34	Emission and propagation of an outgoing flash from the reference clock	$2^{KB}, 28^{IXC}, 29^Q$	4, 12	§2.6, pp.37–38
35	Synchronization of clocks in an inertial frame by an outgoing reference flash	$2^K, 3^{IQ}, 18^{QB}, 28^{IX}, 29^Q$	5, 8, 12	§2.6, pp.37–38
36	Operation of slow clock transport	$2^K, 3^I, 6^{IK}, 26^Q$	8, 13	§2.6, p.38
37	Synchronization approximation by slow clock transport	$3^I, 6^{IK}, 19^{QX}, 26^Q$	6, 13	§2.6, p.38
38	Selection of the nearest recording clock	–	3–5	§2.6
39	Assignment of event coordinates by the nearest recording clock	$2^K, 18^B, 20^O, 28^I$	3, 5	§§2.6–2.7
40	Constitution of a distributed recording apparatus	$2^{KO}, 20^O$	5, 14	§§2.6–2.7
41	Representation of an observer by a distributed recording system	$13^O, 20^O, 28^I$	5, 14	§2.7
42	Adoption of the principle of relativity	$1^Q, 3^Q, 26^Q$	6	§3.1
43	Identity of law form across inertial frames	$1^Q, 3^Q, 26^Q$	6	§§3.1–3.8
44	Determination of distant simultaneity	$2^{XK}, 4^X, 18^{XB}$	5–6	§3.4; §§4.7–4.9
45	Inertial-frame dependence of distant simultaneity	$1^Q, 4^X, 17^Q, 26^Q$	6–7	§3.4; §§4.7–4.9; Ex.5-8
46	Specification of the Lorentz transformation	$1^Q, 4^X, 26^Q$	4, 6–7	Lorentz transformation special topic L.3–L.6
47	Correspondence of event coordinates under the Lorentz transformation	$1^Q, 17^Q, 26^Q$	3–4, 7	Lorentz transformation special topic L.3–L.6

Com- po- nent	Claim	i	a	Source Location
48	Earth-frame coordinates for the round trip to Canopus	$1^S, 3^{IQ}, 26^Q$	4, 7	§§4.5–4.9
49	Spacecraft proper time on the round trip to Canopus	$14^S, 21^{IQ}, 22^{IQ}$	7–8	§§4.5–4.9
50	Outbound worldline segment	$14^{SB}, 21^{IQ}, 26^Q$	4, 7	§§4.6–4.7
51	Turnaround worldline segment	$4^X, 14^{SB}, 27^Q$	7–8	§§4.8–4.9; p.132; §5.7
52	Return worldline segment	$14^{SB}, 21^{IQ}, 26^Q$	4, 7	§§4.8–4.9
53	Age on the Earth side	$3^{IQ}, 8^Q$	15	§§4.5–4.9
54	Age on the traveler side	$7^K, 8^Q, 22^{IQ}, 23^{IQ}$	8, 15	§§4.5–4.9; §5.6
55	Age difference between the two	$8^Q, 23^{IQ}$	15	§§4.5–4.9
56	Proper-time difference between the two	$14^B, 21^{IQ}, 22^{IQ}$	7–8	§§4.5–4.9; §§5.6–5.8
57	Inertial-frame-dependent explanation of clock comparison	$2^{XK}, 4^X, 25^{QX}, 26^Q$	5–6	§4.9, p.131
58	Invariant-interval description of the same comparison	$14^B, 16^{IQ}, 21^{IQ}$	7	§§4.6–4.9
59	Comparison of aircraft-borne atomic clocks	$2^{KO}, 3^{IO}, 6^{IKO}, 26^Q, 31^{QO}$	8, 29	§4.10, pp.131–134
60	Velocity effect in clock comparison	$3^I, 6^{IK}, 26^Q$	6, 8	§4.10, p.133
61	Difference in elapsed clock readings due to gravity	$3^I, 6^{IK}, 27^Q, 29^Q, 31^Q$	8, 29–30	§4.10, p.133
62	Particle rest-frame lifetime and half-life	9^{IQ}	16	Exs.1-11–1-12; Sample Problem 3-2; §4.10
63	Laboratory particle lifetime and detector-to-detector flight time	$10^{IQ}, 32^{IQ}$	16, 18	Sample Problem 3-2; §4.10
64	Inertial-frame transformation of lifetime and flight time	$9^{IQ}, 10^{IQ}, 24^Q, 26^Q, 32^{IQ}$	6, 16	Ex.1-12; Sample Problem 3-2; §4.10
65	Specification of a worldline	$1^S, 5^S, 14^S$	3–4	§§5.1–5.5
66	Geometric representation of an event sequence by a worldline	$1^S, 3^{IS}, 14^S, 26^Q$	3–4	§§5.2–5.5
67	Specification of a curved worldline	$14^S, 21^{IQ}, 27^Q$	7–8	§§5.6–5.7
68	Accumulation of proper time along a curved worldline	$14^S, 21^{IQ}, 22^{IQ}$	7–8	§§5.6–5.9
69	Inertial-frame clock time	$2^K, 3^{IQ}$	5–6	§5.8
70	Wristwatch time	$7^K, 21^{IQ}$	8	§5.8
71	Stretch factor between the two clock times	$3^{IQ}, 21^{IQ}, 24^Q, 26^Q$	6–7	§5.8
72	Use of the vacuum speed of light	$28^{IQ}, 29^Q$	6, 11	§§2.7,3.6–3.8; Ch.6

Com- po- nent	Claim	i	a	Source Location
73	Causal upper bound on signal propagation	$13^C, 28^{IC}, 29^Q$	28	§2.7; §6.3
74	Construction of the light cone	$1^B, 3^I, 13^C, 29^Q$	4, 7, 28	§§6.2–6.3
75	Classification of causal regions by the light cone	$3^I, 5^C, 13^C, 29^Q$	4, 7, 28	§6.3
76	Worldline displacement	$3^{IQ}, 14^S$	4, 7	§§7.1–7.2
77	Proper-time interval	21^{IQ}	7, 21	§§7.1–7.2
78	Construction of momenergy from both	$3^{IQ}, 14^S, 21^{IQ}, 26^Q$	20–22	§§7.1–7.4
79	Spatial component of momenergy	$14^S, 21^{IQ}, 26^Q$	22–23	§§7.2–7.4
80	Momentum as the spatial component	$21^{IQ}, 26^Q$	20, 23	§§7.2–7.4
81	Temporal component of momenergy	$3^{IQ}, 21^{IQ}$	22, 24	§§7.2–7.4
82	Energy as the temporal component	$3^{IQ}, 21^{IQ}, 26^Q$	20, 24	§§7.2–7.4
83	Specification of an isolated collision system	5^B	26	§7.6; Ch.8
84	Conservation of momenergy in an isolated collision system	$5^B, 21^{IQ}, 26^Q$	22, 25–26	§7.6; Ch.8
85	Physical processes involving photons	$5^B, 12^Q, 28^I, 29^Q$	23–24, 27	§8.4; Exs.8-24–8-40
86	Pair creation	$5^B, 26^Q$	20, 25, 27	§§8.5–8.6
87	Pair annihilation	$5^B, 12^Q, 26^Q$	20, 25, 27	§§8.4–8.5
88	Mass–energy conversion	$5^B, 26^Q$	20, 24–25	§7.5; §§8.2–8.4; Ex.8-40
89	Local free-fall motion	$3^{IS}, 26^Q, 27^Q$	9–10, 29	§§9.1–9.5
90	Curved spacetime geometry corresponding to local free-fall motion	$3^{IS}, 27^Q$	4, 10, 29	§§9.3–9.6
91	Variation of spacetime geometry	$3^S, 26^Q, 27^Q$	22, 29	§§9.5–9.7
92	Description of gravitational waves through variation of spacetime geometry	$3^{IO}, 11^{IO}, 28^I, 40^{IO}$	29, 31	§9.7
93	Formal description concerning black holes	$3^{IO}, 11^{IO}, 26^Q, 28^I, 29^Q$	29, 32	§9.8, pp.292–295
94	Pulsar timing using radio pulses	$2^{KO}, 3^{IO}, 11^{IO}, 13^O, 30^{IO}$	1, 14	§9.7, pp.290–291
95	Formal description concerning cosmic history	$3^{IS}, 15^S, 26^Q, 28^I, 39^Q$	33	§9.9, Table 9-2
96	The atomic-clock second and the light-speed length unit	$3^Q, 11^{IK}, 12^Q, 29^Q$	1, 11	Box3-2, pp.58–59

Com- po- nent	Claim	i	a	Source Location
97	Comparison of elapsed time between inertial frames using a round-trip light clock	$3^{\text{IQ}}, 6^{\text{IK}}, 11^{\text{I}}, 24^{\text{Q}}, 28^{\text{I}}, 29^{\text{Q}}$	4, 6, 8	§§3.6–3.8
98	Half-life inference from particle-count ratio and proper flight time	$9^{\text{IQ}}, 10^{\text{IQP}}, 21^{\text{IQP}}, 26^{\text{Q}}, 33^{\text{IQ}}$	16, 18–19	Sample Problem 3-2, pp.72–73; Exs.1-11–1-12
99	Comparison of emission and reception periods and frequencies, and apparent motion	$3^{\text{I}}, 11^{\text{IQX}}, 12^{\text{QX}}, 26^{\text{Q}}, 28^{\text{IX}}$	4, 34	Exs.3-14–3-16; Exs.8-24–8-32
100	Clock comparison using atomic thermal motion and internal nuclear vibrations	$3^{\text{I}}, 11^{\text{IQ}}, 12^{\text{Q}}, 26^{\text{Q}}, 36^{\text{Q}}$	8, 35–36	§4.10, p.134; Ex.8-39, pp.272–273
101	Frequency comparison using resonance linewidth and photon counting rate	$3^{\text{IO}}, 12^{\text{QO}}, 26^{\text{Q}}, 34^{\text{QO}}, 35^{\text{Q}}$	14, 27, 37	Exs.8-34–8-37, pp.270–272
102	Power and the transfer of mass and energy	$3^{\text{IQ}}, 5^{\text{B}}, 26^{\text{Q}}, 29^{\text{Q}}, 37^{\text{Q}}, 38^{\text{Q}}$	20, 24–25	Ex.8-40, pp.273–274
103	Comparison of binary orbital timing shifts with gravitational radiation	$3^{\text{IO}}, 11^{\text{IO}}, 30^{\text{IO}}, 40^{\text{IO}}$	1, 14, 31	§9.7, Fig.9-7, pp.290–291
104	Invariance across frames, conservation across interactions, and constancy over time	$3^{\text{I}}, 5^{\text{B}}, 26^{\text{Q}}$	6, 25	Box7-3, pp.208–209
105	Conditions for a good clock according to its path and mechanism sensitivity	$3^{\text{I}}, 6^{\text{K}}, 7^{\text{K}}, 14^{\text{S}}, 21^{\text{IQ}}, 27^{\text{Q}}$	8, 10	§§5.6–5.7, pp.149–152
106	Relative emitted/received frequency difference in a weak gravitational field	$12^{\text{QX}}, 27^{\text{Q}}, 29^{\text{Q}}, 41^{\text{QX}}$	29–30, 34, 37	Exs.8-6,8-37–8-38

Definition 8.219 (Time Contents and Auxiliary Objects). The 78 semantic objects comprise 41 time contents and 37 auxiliary objects. Tables 122 and 123 give these two classes. Each period, frequency or clock-reading entry is parameterized by the specified physical process, frame, carrier and events. Related-time indices identify the temporal referents of an auxiliary object; the respective tables specify the premises of individual transfers and results.

Table 122: Time Contents in Taylor–Wheeler 1992

i	Time Content	Source Location
1	Event time coordinate	§§1.2–1.5,2.6–2.7,5.1–5.5
2	Clock reading within an inertial frame	§§2.6–2.7,4.9–4.10,5.7

<i>i</i>	Time Content	Source Location
3	Elapsed time and observation duration measured in an inertial frame	§§2.3–2.6, 3.6–3.8, 5.8, 9.3–9.7
4	Specification and comparison of simultaneity	§§2.6, 3.4, 4.7–4.9; Ex.5-8
5	Before-and-after order of events and interactions	§§2.7, 5.2, 6.3, 7.6; Ch.8
6	Elapsed reading of a proper clock	Ch.1; §§4.5–4.10, 5.6–5.8
7	Wristwatch reading	§§4.5–4.9, 5.6–5.8
8	Ages on the Earth and traveler sides	§§4.5–4.9, 5.6
9	Particle rest-frame lifetime and half-life	Exs.1-11–1-12; Sample Problem 3-2; §4.10
10	Particle flight time between detectors in the laboratory frame	Sample Problem 3-2, pp.72–73; §4.10
11	Periods of clocks, light, oscillations, and orbits	Chs.1–3; §§4.10, 9.7–9.8; Exs.8-24–8-39
12	Emission, reception, and resonance frequencies	Chs.1–3; §§4.10, 8.4; Exs.8-24–8-39
13	Causal past and future, and signal-arrival order	§§2.7, 6.2–6.3, 9.7
14	Time orientation and segment order along a worldline	§§4.6–4.9, 5.2–5.9
15	Cosmic age and historical epoch	Ex.3-17; §9.9, Table 9-2
16	Timelike interval obtained from temporal and spatial separations	Chs.1, 3; §§5.6–5.8, 6.2
17	Time coordinate in another inertial frame under a Lorentz transformation	Lorentz transformation special topic L.3–L.6
18	Preset time of a lattice clock started by an outgoing flash from the reference clock	§2.6, pp.37–38
19	Clock offset compared by slow transport	§2.6, p.38
20	Event time assigned from the nearest clock record	§§2.6–2.7, pp.38–40
21	Proper time calculated from the spacetime interval of a timelike segment	Ch.1; §§4.6, 5.6–5.8
22	Total proper time obtained by adding worldline segments	§§5.6–5.8; Ch.4
23	Age increment corresponding to worldline proper time	§§4.5–4.9, 5.6
24	Ratio of elapsed inertial-frame time to proper time	§5.8; §§3.6–3.8
25	Time dilation in clock comparison with another inertial frame	§4.9, p.131
26	Velocity specified by coordinate displacement and elapsed time	Lorentz transformation special topic; Chs.5, 7–9

<i>i</i>	Time Content	Source Location
27	Acceleration specified by velocity change and elapsed time	§§2.3–2.5, 5.7, 9.2–9.5; Lorentz transformation exercises
28	Propagation time of light and signals	§§2.6–2.7, 3.6; Ch.6; §9.7
29	Propagation-time scale used in the vacuum speed of light	§1.4; Box3-2; Ch.6
30	Pulsar period determined from repeated radio pulses	§9.7, pp.290–291
31	Comparison of elapsed clock readings in a gravitational field	§4.10, pp.131–134
32	Laboratory-frame particle lifetime predicted from the rest lifetime	§4.10; Ex.1-12; p.76
33	Rest-frame half-life inferred from flight time and particle-count ratio	Sample Problem 3-2, pp.72–73
34	Photon counting rate	Exs.8-36–8-37, pp.271–272
35	Frequency width of a natural spectral line	Ex.8-34, pp.270–271
36	Time average of squared thermal speed	Ex.8-39, p.273
37	Power and energy-transfer rate	Ex.8-40, pp.273–274
38	Mass transfer per unit time	Ex.8-40, p.273
39	Volume-increase rate in the cosmological model	Table 9-2, p.297
40	Cumulative shift in the time to complete a specified number of orbital revolutions	§9.7, Fig.9-7, pp.290–291
41	Relative difference between emitted and received frequencies in a gravitational field	Exs.8-6, 8-37–8-38

Table 123: Auxiliary Objects in Taylor–Wheeler 1992

<i>a</i>	Auxiliary Object	Related Time <i>i</i>
1	Cycle identity and counting	2–3, 11–13, 29–30, 40
2	Differentiation with respect to the specified variable	–
3	Event identity and endpoint specification	1–3, 5, 14, 17–18, 20, 26, 28
4	Spatial positions, displacements, light paths, and length standards	1–6, 11–14, 16–17, 21, 24, 26–29
5	Lattice arrangement of rods, clocks, and recorders	1–4, 13, 18, 20, 25–26, 28–29
6	Inertial-frame conditions and the principle of relativity	1–6, 9–11, 16–19, 21, 24–26, 28–29, 32
7	Interval relations in flat spacetime	1, 3–6, 13–14, 16–17, 21–22, 24, 26–29

<i>a</i>	Auxiliary Object	Related Time <i>i</i>
8	Clock identity, stability, and mechanism sensitivity	2–8, 11–12, 14, 18, 21–24, 26–29, 31, 36
9	Free test bodies and negligible disturbances	3, 26–27
10	Tolerance conditions on the local region, observation duration, and detection accuracy	3, 6–7, 14, 21, 26–27
11	Unit conventions for seconds, length, and light speed	3, 11–12, 28–29
12	Reference flash and compensation of clock-start response	2–3, 18, 28–29
13	Slow clock transport along a finite route	2–3, 6, 19, 26
14	Detection, counting, memory, and comparison of records	2–3, 11–13, 20, 26, 28, 30, 34–35, 40
15	Correspondence between good clocks and biological aging	3, 7–8, 22–23
16	Species-specific relation between lifetime, half-life, and decay fraction	9–10, 21, 24, 26, 32–33
17	Distance calculated from propagation time and light speed	3, 29
18	Detector separation and particle flight distance	9–10, 21, 26, 32–33
19	Ratio of surviving particle counts	9–10, 21, 26, 33
20	Particle mass, system invariant mass, and temperature	3, 5, 12, 14, 21, 26, 29, 37–38
21	Timelike worldline of a massive particle	3, 14, 21, 26
22	Momenergy four-vector	3, 5, 14, 21, 26–27
23	Momentum	5, 12, 14, 21, 26, 28–29
24	Energy	3, 5, 12, 21, 26, 28–29, 37–38
25	Momenergy conservation and the system boundary	3, 5, 12, 21, 26, 29, 37–38
26	Isolated interacting system including radiation	5, 21, 26
27	Mass-shell, target, and recoil conditions	3, 5, 12, 26, 28–29, 34–35
28	Time orientation and causal transmission conditions	1, 3, 5, 13, 28–29
29	Gravitational field, spacetime curvature, and body configuration	2–3, 6, 11–12, 26–29, 31, 40–41
30	Weak-field and clock-comparison approximation conditions	3, 6, 12, 27, 29, 31, 41
31	Gravitational-radiation model and observational comparison	3, 11, 28, 30, 40
32	Rotation, mass, and observational conditions of the black-hole model	3, 11, 26, 28–29
33	Model conditions for the expansion history and closed-universe table	3, 15, 26, 28, 39
34	Relative motion and light paths of emitter and receiver	3, 11–12, 26–29, 41
35	Relation among thermal motion, nuclear vibration, and temperature	3, 11–12, 26, 36
36	Time averaging over the specified interval	3, 11–12, 26, 36

a	Auxiliary Object	Related Time i
37	Recoil, resonance line shape, and detection conditions	3, 12, 26–27, 29, 34–35, 41

Definition 8.220 (Mathematical-Object Types of the Original Source). The following table presents the 14 object types appearing in the mathematical descriptions of the original source as formal evidence from that source.

Table 124: Mathematical-object types in Taylor–Wheeler 1992

m	Mathematical-object type in the original source
1	Event
2	Inertial frame
3	Clock
4	Worldline
5	Rod-and-clock lattice
6	Clock reading
7	Synchronization
8	Event coordinate
9	Spacetime interval
10	Proper time
11	Lorentz transformation
12	Stretch factor
13	Light cone
14	Momenergy

Definition 8.221 (Time-Use Roles and Obligations in Taylor–Wheeler 1992). Each pair of a claim component and one of its time-content indices in Table 121 is an atomic time use in this source. For all 283 pairs, superscript codes on the indices in that table give the roles of Definition 7.71. Table 125 gives the codes and the number of uses for each role. Including uses that serve multiple roles, there are 403 role assignments.

The obligation atoms of Definition 7.36 comprise 106 claims, 41 time contents, 23 transfers and 29 results, totaling 199. Each use corresponds to its claim and time-content obligation atoms. Table 126 gives the five-type and eight-axis attributes of the time contents; the transfer and result-dependence tables give the support and input/output sets of the edges. The nodes reachable from the physical-input initial set are claims 1–12; no time content or result is reachable from that set. These components determine the typed audit states of Definition 7.37.

Table 125: Time-Use Roles in Taylor–Wheeler 1992

Role	Code	Uses	Time-content indices
Interval	I	112	3, 6, 9–11, 16, 21–23, 28, 30, 32–33, 40

Role	Code	Uses	Time-content indices
Quantity	Q	163	1, 3, 8–12, 16–19, 21–29, 31–39, 41
Cross-position correspondence	X	20	2, 4, 11–12, 18–19, 25, 28, 41
Clock reading	K	28	2, 6–7, 11
Causal	C	6	5, 13, 28
State evolution	S	23	1, 3, 5, 14–15
Probabilistic evolution	P	2	10, 21
Observation record	O	27	2–3, 6, 11–13, 20, 30–31, 34, 40
Reference boundary	B	22	1–2, 5, 14, 18

Table 126: Structural Attributes of Time Contents in Taylor–Wheeler 1992

Time-content indices	Type	Axes
1–2, 4, 17–18, 20	Member	Instant
3, 6–12, 16, 19, 21–41	Same-family dependency	$\emptyset$
5	Member	Order relation
13	Member	Order relation, Orientation
14	Member	Endpoint, Interval boundary, Turning position, Order relation, Orientation
15	Member	Instant, Order relation

Dependency Inputs

Index Identifier S13-XI-DI1.

Existing-Evidence Reference Set Definition 8.222.

Dependency and Transfer Elements

Definition 8.222 (Time-Content Transfers in Taylor–Wheeler 1992). The time-input and time-output columns in Table 127 use indices i from Table 122; auxiliary inputs use indices a from Table 123; formal support uses claim indices p from Table 121. Each row transfers time contents in the stated direction.

Table 127: Time-Content Transfers in Taylor–Wheeler
1992

Transfer	Relation	Time Input i	Auxiliary Input a	Time Output i	Formal Support p
1	Calculation of a timelike spacetime interval	3	4, 7	16	15–17, 20, 23
2	Time-coordinate correspondence under a Lorentz transformation	1, 4, 26	4, 6–7	17	14, 42–47
3	Synchronization of lattice clocks by an outgoing reference flash	2–3, 28	5, 8, 12	18	31–35
4	Offset comparison for a slowly transported clock	3, 6, 26	6, 8, 13	19	36–37
5	Time assignment from the nearest clock record	2, 18	3, 5, 14	20	38–41
6	Calculation of proper time for a timelike segment	16	7, 21	21	17, 24–28
7	Addition of proper time along a worldline	14, 21	7–8	22	50–52, 65–68, 105
8	Correspondence between total proper time and age increment	22	8, 15	23	49, 53–56
9	Ratio of elapsed coordinate time to proper time	3, 21	6–7	24	69–71
10	Clock comparison between inertial frames in relative motion	2, 4, 24	5–6	25	44–45, 57–58, 71
11	Velocity from displacement and elapsed time	3	4, 6	26	14, 16, 65–66
12	Acceleration from velocity change	3, 26	2, 10	27	29–30, 67, 89–90
13	Light-propagation time from distance and light speed	29	4, 11	28	18–19, 34, 72

Transfer	Relation	Time Input i	Auxiliary Input a	Time Output i	Formal Support p
14	Correspondence from rest lifetime to laboratory lifetime	9, 24	6, 16	32	62–64
15	Flight duration from detector separation and speed	26	6, 18	10	63–64, 98
16	Half-life inference from flight time and count ratio	10, 26	6, 16, 19	33	62–64, 98
17	Period determination from a clocked pulse sequence	2, 5	1, 14	30	94
18	Comparison of elapsed clock readings in a gravitational field	3, 27, 29	8, 29–30	31	61
19	Comparison of emitted and received frequencies under gravity	27, 29	29–30, 34	41	106
20	Linewidth from central frequency and fractional width	12	37	35	101
21	Time derivative of transferred energy	3	2, 24	37	102
22	Mass-transfer rate from power	29, 37	20, 24–25	38	88, 102
23	Cumulative timing shift for a specified orbital revolution count	3, 11, 30	1, 14, 31	40	94, 103

Grounding Conditions

1. Use the temporal and spatial separations of the same timelike-separated event pair in flat inertial coordinates. (Chs.1,3; §6.2)
2. Relate coordinates of the same event in two inertial frames with specified relative velocity and synchronization. (Lorentz transformation special topic L.3–L.6)
3. Preset each lattice clock to the known light-travel time and start it on arrival of

the outgoing flash from the reference clock. The start-response delay is zero or compensated. (§2.6, pp.37–38)

4. Compare elapsed readings of the transported clock and the reference-frame clocks in the low-speed limit along a finite transport path. (§2.6, p.38)
5. Use the record of the synchronized lattice clock nearest to the event. Lattice spacing and local propagation time determine the assignment accuracy. (§§2.6–2.7)
6. Obtain proper time from the endpoint spacetime interval of a straight timelike segment in flat spacetime. (Ch.1; §§4.6,5.6–5.8)
7. Add the ordered worldline segments and take the fine-segmentation limit. The clock meets the source conditions for a small, acceleration-proof good clock. (§§5.6–5.9)
8. The source assertion that the traveler’s biological clock and good clocks advance in concord relates total proper time to the age increment. Final age is the sum of initial age and this increment. (§§4.5–4.9,5.6)
9. Divide elapsed coordinate time by proper time for the same timelike inertial segment. (§5.8)
10. Each inertial frame selects its clock pair according to its own simultaneity and compares elapsed times with the other frame. (§4.9, p.131)
11. Divide displacement in the selected inertial frame by the nonzero elapsed time measured in that frame. (Lorentz transformation special topic; Ch.5)
12. Differentiate velocity with respect to the specified frame time. A tidal-motion comparison specifies two nearby trajectories and the observation duration. (§§2.3–2.5,5.7,9.2–9.5)
13. Calculate propagation time using the specified distance, speed of light and time-unit convention. (§§1.4,2.6–2.7)
14. Multiply the specified particle species’ rest-frame lifetime by the frame-transformation factor. Mean lifetime and half-life retain their respective definitions. (Ex.1-12; §4.10)
15. Obtain flight duration from detector separation and particle speed. Sample Problem 3-2 uses a separation of 9 meters and a speed of $0.868c$, with negligible slowing by the counters. (Sample Problem 3-2, pp.72–73)
16. Attribute the particle-count ratio of $1/4$ in Sample Problem 3-2 to decay and obtain the rest-frame half-life from the flight proper time, which equals two half-lives. (Sample Problem 3-2, pp.72–73)
17. Determine the period from the repetition count and elapsed time of radio-pulse arrivals compared against a clock. (§9.7, pp.290–291)
18. Specify the aircraft and ground-clock configuration, gravitational field and flight duration, and predict the altitude contribution to elapsed-time difference in the weak-field approximation. Combine it with the motion contribution for comparison with the clock-reading difference. (§4.10, pp.131–134)

19. Specify the weak gravitational field, emission and reception positions and light path to obtain the relative gravitational frequency shift. Account for the Doppler contribution according to the motion of the emitter and receiver being compared. (Exs.8-6,8-37–8-38)
20. Multiply the supplied fractional natural linewidth by the central frequency. (Ex.8-34, pp.270–271)
21. For the energy transfer in Exercise 8-40, obtain the rate dE/dt with respect to the specified time. (Ex.8-40, pp.273–274)
22. Obtain mass transfer per unit time from power using the specified units and mass-energy relation. The momentum estimate uses the source’s small-momentum approximation. (Ex.8-40, p.273)
23. Use pulsar arrival times, the binary orbital period and the observation span to obtain the cumulative shift in the time to complete a specified number of orbital revolutions, and compare it with the gravitational-radiation prediction. (§9.7, Fig.9-7, pp.290–291)

Original-Source Support

Index Identifier S13-XI-SS1.

Existing-Evidence Reference Set Definition 8.222.

Results

Definition 8.223 (Result Inputs in Taylor–Wheeler 1992). Table 128 specifies results j and their source locations. Each result is the collection of the listed claims, and the premises in the dependence table support that collection as a whole. The claim table specifies the arguments of each constituent claim.

Table 128: Results in Taylor–Wheeler 1992

j	Result	Source Location
1	Distinction between event identity and inertial-frame coordinates	§§2.6–2.7,5.1–5.2
2	Spacetime interval and invariance across inertial frames	Chs.1,3; Lorentz transformation special topic
3	Atomic-clock time units and conversion using light speed	§1.4; Box3-2
4	Classification of timelike, spacelike, and lightlike separation	§6.2
5	Correspondence among proper time, local time, and wristwatch readings	Ch.1; §§5.6–5.7
6	Local free-float frames at the specified measurement accuracy	§§2.1–2.5

<i>j</i>	Result	Source Location
7	Inertial-frame construction using a rod-and-clock lattice	§2.6
8	Clock synchronization by an outgoing reference flash	§2.6, pp.37–38
9	Approximate synchronization by slow clock transport	§2.6, p.38
10	Assignment of event coordinates by the nearest clock	§2.6
11	Observation using distributed recording clocks	§2.7
12	Inertial-frame dependence of simultaneity	§3.4; §§4.7–4.9; Ex.5-8
13	Event-coordinate correspondence under a Lorentz transformation	Lorentz transformation special topic L.3–L.6
14	Addition of proper time along a curved worldline	§§5.1–5.9
15	Comparison of the twins' paths, proper times, and aging	§§4.5–4.9; §5.6
16	Elapsed-time ratios and mutual clock comparison	§4.9; §5.8
17	Comparison of motion and gravitational effects on transported atomic clocks	§4.10
18	Comparison of particle lifetimes, half-lives, flight durations, and counts	Exs.1-11–1-12; Sample Problem 3-2; §4.10
19	Classification of possible causal connection by oriented light cones	Ch.6
20	Momenergy and its components for a massive particle	§§7.1–7.4
21	Isolated-system conservation in collisions, pair creation, and annihilation	§§7.5–7.7; Ch.8
22	Local free fall and spacetime curvature	§§9.1–9.6
23	Pulsar timing observations and gravitational radiation	§9.7
24	Conditional black-hole models and observations	§9.8
25	Cosmic expansion history and a closed-universe model	§9.9, Table 9-2
26	Emission and reception frequencies, and apparent motion	Exs.3-14–3-16; Exs.8-24–8-32
27	Clock comparison using nuclear oscillation and thermal motion	§4.10; Ex.8-39
28	Frequency and gravitational-redshift comparisons using resonance counts	Exs.8-34–8-38
29	Power and mass-energy transfer	Ex.8-40

Table 129: Result Dependencies in Taylor–Wheeler 1992

<i>j</i>	Time Premises <i>i</i>	Auxiliary Premises <i>a</i>	Supporting Claims <i>p</i>
1	1–2, 4	3, 5–6	13–14
2	1, 3, 6, 11, 16–17, 24, 26, 28–29	3–4, 6–8	15–17, 23, 42–43, 97
3	3, 11–12, 28–29	1, 6, 11, 17	18–19, 96
4	3, 16, 28–29	4, 7	20–22

j	Time Premises i	Auxiliary Premises a	Supporting Claims p
5	2–3, 5–7, 14, 21, 27	3, 7–8, 10	24–28, 105
6	3, 26–27	9–10	29–30
7	2, 4, 18	4–6, 8	31–33
8	2–3, 18, 28–29	4–5, 8, 12	34–35
9	2–3, 6, 19, 26	6, 8, 13	36–37
10	2, 18, 20, 28	3–5	38–39
11	2, 13, 20, 28	5, 14	40–41
12	1–2, 4, 17–18, 26	5–7	44–45
13	1, 4, 17, 26	3–4, 6–7	46–47
14	1, 3, 5–7, 14, 21–22, 26–27	3–4, 7–8, 10	65–68, 105
15	1, 3–4, 7–8, 14, 21–23, 26–27	4, 7–8, 15	48–56
16	2–4, 7, 14, 16, 21, 24–26	5–8	57–58, 69–71
17	2–3, 6, 26–27, 29, 31	6, 8, 29–30	59–61
18	9–10, 21, 24, 26, 32–33	6, 16, 18–19	62–64, 98
19	1, 3, 5, 13, 28–29	4, 6–7, 11, 28	72–75
20	3, 14, 21, 26	4, 7, 20–24	76–82
21	3, 5, 12, 21, 26, 28–29	6, 20, 22–27	83–88, 104
22	3, 26–27	4, 9–10, 22, 29	89–91
23	2–3, 11, 13, 28, 30, 40	1, 14, 29, 31	92, 94, 103
24	3, 11, 26, 28–29	29, 32	93
25	3, 15, 26, 28, 39	33	95
26	3, 11–12, 26, 28	4, 34	99
27	3, 11–12, 26, 36	8, 35–36	100
28	3, 12, 26–27, 29, 34–35, 41	14, 27, 29–30, 34, 37	101, 106
29	3, 5, 26, 29, 37–38	20, 24–25	88, 102

Definition 8.224 (Audit Boundary for Taylor–Wheeler 1992). The initial generation set consists of the 12 non-temporal physical-carrier components in Definition 8.216. All 106 components constitute the formal-evidence environment. The claim, transfer and result tables specify the uses of time contents and auxiliary objects.

Definition 8.225 (Support-Completion Specification for Taylor–Wheeler 1992). The columns in Table 129 specify time premises i , auxiliary premises a and supporting claims p . The supporting-claim columns for the 29 results contain 96 occurrences, covering all 94 components 13–106. Each result is supported by its claims together with its time and auxiliary premises in Table 129. All 29 results presuppose at least one time content.

Result Dependencies

Definition 8.226 (Finite Dependency Graph for Taylor–Wheeler 1992). The audit boundary in Definition 7.18 consists of 106 physical and formal claim nodes, 41 time-content nodes and 29 result nodes, totaling 176 nodes. The tabulated supporting claims and time inputs lead to time outputs or results through 23 transfer edges and 29 result edges, giving the 52 edges of Definition 7.27. The auxiliary-object and claim tables and the conditions for each edge specify the uses and conditions of the 37 auxiliary objects. The transfer relation among time contents is acyclic.

The results comprise definitions, coordinate constructions, mathematical derivations, spacetime-diagram arguments, thought experiments, reports of experiments or observations, and explanations of applications. The reported measurement objects and conditions provide empirical grounds for the aircraft-clock, particle, resonance-absorption and binary-pulsar results. The correspondence between biological clocks and aging belongs to the book's assertion that good clocks carried along advance in concord.

Grounding Grounds The conditions for the respective results are as follows.

1. Assign coordinates to the same event according to each inertial frame's lattice and synchronization.
2. Compare the interval for the same event pair using flat inertial coordinates and each frame's synchronization.
3. Define the second by counting atomic-clock cycles and convert between time and length units using the speed of light. Light-speed invariance and the unit convention specifying its numerical value have their respective roles.
4. Classify separation by the sign of the difference between squared temporal and spatial separations, using proper time in the timelike domain and proper distance in the spacelike domain.
5. The clock follows the specified path and meets the good-clock conditions for its mechanism and acceleration.
6. Disturbances of test bodies and tidal deviations remain within measurement tolerance over the selected region and observation interval.
7. Combine the spatial arrangement of rods, clocks and recorders with the clocks' synchronization settings and simultaneity.
8. Preset lattice clocks to the known light-travel time and start them on arrival of the reference flash, with zero or compensated response delay.
9. Take the low-speed limit while holding the finite transport path fixed.
10. Lattice spacing and local signal-propagation time determine the accuracy of assigning position and time using the nearest clock.
11. Retain each locally recorded event with its local time; transmission to a central receiver entails a propagation delay in arrival.
12. Each inertial frame's simultaneity selects the remote-clock events to be compared.
13. Relate coordinates of the same event under the specified axes, relative velocity and synchronization convention.
14. Add the proper times of finely divided path segments in order. Maximizing proper time over paths between fixed endpoints and minimizing elapsed coordinate time across inertial-frame descriptions of the same interval have their respective comparison domains.

15. The Canopus round trip has 202 years of Earth-frame elapsed time and 40 years of traveler proper time. The age increments follow the source's stated concordance between biological clocks and wristwatches.
16. Specify each frame's synchronization and the event pair being compared. Reciprocal clock comparison follows the simultaneous events selected by each frame.
17. Use each reported clock's path, flight duration, altitude and speed, and combine motion and gravity contributions for comparison with the difference in elapsed clock readings.
18. Fix the particle species and lifetime definition, and relate rest-frame lifetime, laboratory lifetime, detector-to-detector flight duration and count ratio. Sample Problem 3-2 neglects slowing and attributes the count reduction to decay to infer the half-life.
19. Distinguish future and past timelike and lightlike regions and the spacelike region. Timelike separation permits causal transmission; the physical process determines the actual interaction.
20. For a massive particle's timelike worldline, normalize the four-displacement by a positive proper-time interval and multiply by mass to obtain momenergy. Its spatial component is momentum and its temporal component is energy.
21. Specify the system boundary including radiation and targets, the incoming and outgoing states, and the mass-shell conditions, and conserve total momenergy across the interaction. Individual exercises use their stated recoil, binding, low-speed or high-energy approximations.
22. Compare tidal relative motion of nearby trajectories according to the observation duration and detection sensitivity. The test-particle volume argument holds in the small-region, infinitesimal-change approximation.
23. The source compares 14 years of binary-pulsar timing observations with the gravitational-radiation prediction. Pulse repetition period, orbital period and cumulative orbital timing shift are the corresponding distinct observables.
24. The displayed black-hole radius follows the nonrotating model. Candidate mass estimates depend on the variability, velocity and other observations reported in the source and on the model conditions.
25. Use cosmic age, light propagation and volume-increase rate in the specified expansion model. Table 9-2 gives one closed-universe model, and the source leaves the universe's future unsettled.
26. Specify the emitter and receiver frames, the sequences of emission and arrival events, and the light path. Apparent pattern speed uses differences in arrival time and follows the motion, recoil and observation-span conditions of each exercise.
27. Use the internal nuclear-oscillation frequency and the time average of squared thermal speed. Under the source's low-speed approximation, compare the change in nuclear-clock frequency associated with mechanical thermal motion.

28. Specify central frequency, linewidth, photon counting rate, relative emitter-receiver motion and light path. Gravitational comparisons use the weak-field and Doppler-contribution conditions of each exercise.
29. Use work per unit time and the mass-energy relation to obtain the transfer rate under the source's small-momentum approximation.

Index Identifier S13-XI-RG1.

Existing-Evidence Reference Set Definition 8.225.

Evidence Hierarchy

Index Identifier S13-XI-MT1.

Existing-Evidence Reference Set Definition 8.224.

Generation Judgment Judgment values: LCTR complete-generation judgment: 0; source-internal evidence-conditioned local-reach judgment: 0; remaining primitive time roots: 41.

Source-Internal Evidence-Conditioned Local Reach The source-internal evidence-conditioned local-reach judgment is zero; 41 primitive time roots remain after that judgment.

Result Dependence There are 130 primitive-time-root-result dependence relations and 95 necessary-root-result dependence relations common to every minimal support.

Evidence Effect Coordinate constructions, mathematical derivations, spacetime diagrams, thought experiments and reported experiments or observations support the 29 results under their respective physical conditions. Interval relations, Lorentz transformations and worldline comparisons use times, elapsed times, simultaneity or path order. Clock, lifetime, resonance and pulsar comparisons use periods, lifetimes, frequencies or observation durations. The temporal scale used in the vacuum speed of light belongs to time content 29. Physical use of these contents requires the corresponding time-generation structure to hold. The evidence-effect boundary of Definition 7.115 lies at the passage from results using these contents to the generation structure establishing their physical use.

Result-Preservation Domain The 29 results are preserved insofar as each time-content premise is supplied by the corresponding generated time representation while preserving the physical carriers, endpoint events, frames, clock properties, paths, decay laws, isolated-system boundaries and domains of the respective approximations. The correspondence with biological aging, conditional theoretical predictions and reported measurements follow their respective claims and conditions.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The 41 time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as already-established time inputs. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment follows by matching the use of each time content in a law operation or result derivation against the absence, before that use, of the corresponding generation or validity condition as an input, formation condition, evaluation condition, or failure condition.

Evidence of Actual Failure

Evidence of Actual Failure The complete judgment and the remaining-root judgment concern physical-time qualification; they are distinct from actual failure of a concrete condition governing a result’s value or domain of application. Neither the original formal evidence nor the LCTR audit evidence in this reference establishes an actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence,” and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream Reach For this reference, the reach to primitive-time-root-dependent results is determined by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because actual failure of a concrete condition has not been established, the actual-reach judgment from that failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the objects of repair are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is limited to results for which actual failure is independently established and which are dependency-reachable from that failure location. Because actual failure is indeterminate in this reference, this repair-and-recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.227 (Complete Audit of Time Use in Taylor–Wheeler 1992). The 106 physical and formal claim nodes, 41 time-content nodes and 29 result nodes, together with 23 transfer edges and 29 result edges, give an audit-boundary directed hypergraph with 176 nodes and 52 edges. Of the 41 time contents, the numbers generated under the complete judgment and reached under the local-reach judgment are both zero.

Proof. The initial generation set consists of the 12 non-temporal physical-carrier components in Definition 8.216. All 23 transfers in Table 127 take time contents as inputs, and all 29 results in Table 129 presuppose time contents. Thus no first transfer obtains a time content by generation reach from non-temporal physical inputs alone. Use of the source formal evidence preserves this temporal-input condition. $\square$

Proposition 8.228 (Primitive Time Roots and Result Dependence in the Taylor–Wheeler 1992 Reference).

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(41, 29).$$

That is, all 41 time contents remain primitive time roots, and all 29 results are primitive-time-root-dependent results.

Proof. The number of time contents reached by fixed-point propagation from non-temporal physical inputs alone is zero. Adding the 41 primitive time roots to the source formal-evidence environment as seeds reaches all 29 results. No result is reachable from the source formal-evidence environment alone. $\square$

Proposition 8.229 (Minimal Primitive-Time-Root Support in the Taylor–Wheeler 1992 Reference). For the minimal-root support family of each result, the following table gives “number of family members; minimum and maximum support cardinalities.” Rows with equal minimum and maximum show only one value. The total number of family members is 51.

Table 130: Minimal Primitive-Time-Root Support in Taylor–Wheeler 1992

Result Index	Support Statistics
1	(1; 3)
2	(2; 6)
3	(1; 4)
4	(1; 2)
5	(1; 6)
6	(1; 1)
7	(3; 3, 4)
8	(1; 3)
9	(1; 3)
10	(4; 3)
11	(6; 4)
12	(4; 5)
13	(2; 3)
14	(1; 6)
15	(1; 6)
16	(1; 5)
17	(1; 4)
18	(3; 2, 4)
19	(1; 5)
20	(1; 2)
21	(1; 4)

Result Index	Support Statistics
22	(1; 1)
23	(4; 6)
24	(1; 3)
25	(2; 4)
26	(2; 4)
27	(1; 4)
28	(1; 4)
29	(1; 3)

Proof. Apply the update in Definition 7.46 to the 23 transfers. In the listed order, 23 edges update supports in the first scan and five in the second; the third scan leaves the support antichains of all time-content nodes unchanged. Proposition 7.47 gives 51 support-family members across the 29 results, 130 primitive-time-root–result dependence pairs and 95 necessary-root–result dependence pairs. $\square$

Individual Statements Time and Space

Verbatim original (English):

time cannot be separated from space. Space and time are part of a single entity, spacetime.

The source location is Taylor and Wheeler, *Spacetime Physics*, 2nd ed., Chapter 1, §1.2, printed p. 7. The minimal context is the passage explaining that temporal and spatial separations differ across inertial frames while the value of the spacetime interval agrees. The statement is attributable to Taylor and Wheeler.

Biological Clock and Aging

Verbatim original (English):

our biological clock, by which we age, and all other good clocks carried along will tick away in concord with it

The source location is Chapter 4, §4.5, *Flight Plan*, printed p. 124, in the same book. The minimal context is the explanation of a flight plan that determines the elapsed time shown by the traveler’s wristwatch. The statement is attributable to Taylor and Wheeler.

Clocks in Another Inertial Frame and Time-Stretch Vocabulary

Verbatim original (English):

Each observes less elapsed time on clocks in the other frame than on his or her own clocks (time stretching or time dilation).

The source location is Chapter 4, §4.9, printed p. 131, in the same book. The minimal context is the explanation in which each observer compares clocks in the other of two inertial frames in relative motion. The statement is attributable to Taylor and Wheeler.

Common Evidence with Specified Domain of Application

Target state- ment	Statement object	Predicate	Explicit domain of application
S13- Q001	Time, space, and spacetime	Time cannot be separated from space, and space and time are parts of a single entity called spacetime.	The domain in Chapter 1, §1.2 that explains inertial-frame-specific temporal and spatial separations and the spacetime interval.
S13- Q002	The biological clock by which one ages and good physical clocks carried along	The biological clock and all good clocks tick in concord with the elapsed time shown by the traveler's wristwatch.	The domain in the flight plan in Chapter 4, §4.5 that relates the traveler's wristwatch time to biological aging.
S13- Q003	Elapsed times on clocks compared between two inertial frames in relative motion	Each observer observes less elapsed time on clocks in the other frame than on the observer's own clocks, and calls the comparison difference time stretching or time dilation.	The domain of mutual clock comparison between two inertial frames in relative motion in Chapter 4, §4.9.

Individual Audit

Interpretability The first statement explains the relation between frame-dependent temporal and spatial separations and an invariant spacetime interval in terms of a single spacetime. The second states that the traveler's wristwatch, biological clock and good clocks carried along advance in concord. The third describes each relatively moving inertial frame comparing clocks selected according to its own simultaneity with clocks in the other frame, calling the smaller elapsed time in the other frame time stretching or time dilation.

Independent Grounds The grounds for the three statements are spacetime-interval invariance, the worldline and proper time of the Canopus round trip, and clock comparison between inertial frames. The correspondence between biological aging and wristwatch time is asserted in Sections 4.5 and 5.6; Section 5.7 discusses clock mechanisms' sensitivity to acceleration and the conditions for good clocks. Section 4.10 reports aircraft atomic-clock comparisons, particle-lifetime measurements and thermal-motion effects on nuclear-clock frequency as experimental comparisons.

Evidence-Effect Boundary The first statement uses temporal separation between events; the second uses proper time along a worldline and aging; the third uses each inertial frame's simultaneity and clock readings. Their physical use requires the generation structure corresponding to each time content to hold. The evidence-effect boundary of Definition 7.115 lies at the passage from the interval, worldline and clock-comparison relations supporting each statement to this establishment.

Corresponding Formal Evidence The first statement corresponds to time contents 1 and 16 in Table 122, claims 13–17 in Table 121, and results 1–2 in Table 128. The second corresponds to time contents 8 and 23, claims 48–56 and result 15. The third corresponds to time content 25, claims 57–58 and 69–71, and result 16. The result-dependence table and Proposition 8.229 specify the premises and minimal-root supports of each result.

Within-Source Comparison Chapter 1 integrates time and space into spacetime, while page 18 describes their different properties. Sections 4.5 and 5.6 relate wristwatch time to aging, and Section 5.7 states the conditions for the clocks used to measure it. Reciprocal time dilation in Section 4.9 is specified by each frame’s clock pairs and simultaneity. These accounts connect the coordinate description of events, proper time along paths and application to physical processes.

Common Audit with Specified Domain of Application The domains and time-content dependencies of the three statements are specified by the source locations, claim components, results and conditions above.

8.19.14 Source Audit

Audit Input

Original Source and Audit Scope

Source Identifier 14.

Source Name Primitive-time-root audit of DeWitt 1967

Bibliographic Identification Let B. S. DeWitt, *Quantum Theory of Gravity. I. The Canonical Theory*, *Physical Review* 160, 1113–1148 (1967), doi:10.1103/PhysRev.160.1113, be $R = \text{Ref}_{\text{DeW67QG1}}$. The audit domain consists of the abstract, Sections 1–10, Appendices A–B, and Figures 1–2 of the original source.

Audit-Scope Identifier 14-SC1.

Audit-Scope Content The audit domain stated in the bibliographic identification.

Source-Location Identifier 14-L1.

Source-Location Content The audit domain stated in the bibliographic identification and the source-location columns of the respective evidence tables.

Formal Objects

Object Identifier 14-F1.

Object Name Formal objects of the original source.

Object Type Formal object.

Formal Content The individual contents of the formal objects are given in Definition 8.231.

Source-Location Identifier Set 14-L1.

Assumptions The canonical theory assumes a positive-definite spatial three-metric, a Cauchy foliation, lapse and shift functions, and boundary conditions. Asymptotic flatness belongs to the infinite-universe branch, and closed-universe constraints to the finite branch. The WKB approximation assumes a slowly varying amplitude and specified phase solutions; the closed Friedmann model assumes identical, coherent, noninteracting dust. Clock representations have spectral, domain, packet and branch conditions, and the inner product has finite-contour and endpoint conditions. Table 133 specifies the corresponding auxiliary objects.

Non-Temporal Physical Inputs

Input Identifier 14-P1.
Input Name Non-temporal physical inputs of the original source.
Input Type Non-temporal physical input.
Input Content The individual contents of the non-temporal physical inputs are given in Definition 8.231.
Source-Location Identifier Set 14-L1.

Time Contents

Time-Content Identifier 14-T1.
Time-Content Name Time contents of the original source.
Time-Content Type Time content.
Time-Content Statement The individual contents of the time contents are given in Definitions 8.230 and 8.231.
Source-Location Identifier Set 14-L1.

Formal Input Components

Definition 8.230 (Time Contents in the DeWitt 1967 Reference). For $R = \text{Ref}_{\text{DeW67QG1}}$, let $n_R = 35$. Table 132 gives the time contents and source locations. The periods, angular frequencies and proper times of the two clocks are families indexed by the geometric and matter subsystems. The transfer and result conditions specify the components used.

Table 132: Time Contents and Source Locations in DeWitt 1967

i	Time Content	Source Location
1	External coordinate time	§§1–4
2	Configuration-variable candidate for intrinsic time	§5,5.7–5.20; §7,7.13–7.16
3	Time given by clocks	Abstract; §10
4	Phenomenological time	§8 p. 1137; §10
5	Differentiation with respect to external coordinate time	§§2–4
6	History order of Cauchy hypersurfaces and three-geometries	§§1–3; §6
7	Asymptotic Minkowski and laboratory time	§4,4.5–4.9

<i>i</i>	Time Content	Source Location
8	Wave-packet evolution in asymptotic time	§4,4.5–4.9
9	Proper-time interval between two hypersurfaces	§5 p. 1124
10	Time parameter of a WKB–Hamilton–Jacobi history	§6,6.10–6.23
11	Proper-time interval along a classical history	§2; §6,6.24–6.26
12	Lifetime in the spherical approximation to a vacuum universe	§6,6.24–6.28
13	Singularity endpoints of a classical cosmological history	§6 pp. 1126–1129
14	Expansion–contraction history of the Friedmann radius	§7; §8,8.17–8.21
15	Proper-time derivative used in the matter model	§7,7.4–7.5
16	Frequency signs of expanding and contracting branches	§7,7.17–7.34
17	Correspondence of matter-clock state and phase with time	§7; §8
18	Proper-time periods of geometric and matter clocks	§7,7.55–7.59; §8
19	Proper-time angular frequencies of geometric and matter clocks	§7,7.57–7.59; §8,8.2
20	Temporal motion of a good WKB wave packet	§8,8.1; pp. 1134–1135
21	Proper times of both clocks defined from phase and frequency	§8,8.2–8.3
22	Proper-time operators	§8,8.4–8.10
23	Proper-time identification modulo a common subperiod	§8 pp. 1135–1136; Fig2
24	Representation window using geometric-clock time	§8,8.11–8.13
25	Conditional proper-time evolution of the subsystems	§8,8.14–8.15
26	Temporal spreading of conditional wave packets	§8 pp. 1136–1137
27	Limit on clock time resolution	§8 p. 1137
28	Temporal order in entropy, time reversal and branches	§10 pp. 1140–1142
29	Meaning and direction of time during final contraction	§10 p. 1142
30	Representation window using matter-clock time	§8,8.11–8.13; Fig2
31	Temporal behavior of radius–matter correlations	§7,7.22–7.31; Fig1
32	Cycloidal wave-packet trajectory in proper time	§8,8.17–8.21
33	Convention relating increasing action to increasing time	§8 p. 1135
34	Proper-time lifetime of the closed Friedmann dust universe	§8,8.17–8.21
35	Proper-time lapse corresponding to the inner-product contour	§9 p. 1138,9.9–9.11

Table 133: Auxiliary Objects in DeWitt 1967

a	Auxiliary Object	Source Location
1	Positive-definite spatial three-metric	§§1–2,5; App. A
2	Conditions for spacetime decomposition and a Cauchy domain	§§1–3
3	Lapse, shift and gauge conditions	§§2–3; §6
4	Intrinsic spatial curvature	§2,2.3–2.6
5	Canonical constraints and phase-space relations	§§2–4
6	Asymptotic conditions and surface energy	§§2,4
7	Constraint system of a finite closed universe	§§3–4
8	Quantization, factor ordering and domains	§§4–5
9	Metric representation of three-geometries	§5,5.1–5.5
10	Configuration-space metric and negative-sign direction	§5,5.6–5.18; App. A
11	Configuration coordinate zeta and its Friedmann reduction	5.7–5.10;7.13–7.16
12	Two hypersurface geometries and classical solution conditions	§5 p. 1124
13	Formal flux and boundary conditions	5.19–5.20
14	Positive-frequency restriction and the half-line domain	5.19–5.20
15	WKB approximation and Hamilton–Jacobi phase	6.10–6.15;7.22–7.54
16	Configuration-space arc length, force term and volume functional	6.16–6.26; App. B
17	Spherical approximation to a vacuum universe	6.24–6.28
18	Configuration-space frontier	§§5–6; App. A
19	Proposed quantum barrier and boundary condition	6.29–6.31
20	Closed Friedmann model and coherent dust	7.1–7.16
21	Sign of the Friedmann kinetic term	7.3,7.10
22	Actions, phase origins and branches	7.22–7.31;8.3
23	Matter state, superposition and degeneracy	§7 pp. 1132–1134
24	Spectral matching of the subsystems	7.55–7.59
25	Period commensurability and level spacing	7.55–7.59;§8
26	Coefficients and localization conditions for a good wave packet	8.1;pp. 1134–1135
27	Domains of clock operators, projectors and spectra	8.4–8.10
28	Transforms of the three representations and mutual configuration window	8.11–8.13
29	Conditions for the dust-model cycloid solution	8.16–8.21
30	Quantum and mass conditions for clock precision	§8 p. 1137
31	Finite contour, orientation and inner-product conditions	§9
32	Everett interpretation and entropy conditions	§10
33	Physical constants entering the Planck length	§1 pp. 1113–1116

Definition 8.231 (Physical and Formal Input Claim Components in DeWitt 1967). Let the physical and formal input claim sequence be $(\text{ClaimVal}_{R,p}^{\text{audit,phys}})_{p \in [62]}$. Components 1–4

are non-temporal physical inputs; Components 5–62 are source formal evidence. The initial generation set consists of Components 1–4, and the source formal-evidence environment consists of all 62 components. Table 134 gives each claim, its correspondence to time contents i and auxiliary objects a , and its source location. The time-use units are the 88 claim–time-content pairs in the claim table and four further pairs: Time Content 8 in Transfer 1, Time Contents 20 and 33 in Transfer 10, and Time Content 33 in Transfer 11, giving 92 pairs in total. Superscripts on time indices denote the roles in Table 135. The latter four pairs all have the state-evolution role S. Clock reading K is the mapping from clock states to time content; in Claim 54 it is the function asserted by the source statements. Table 136 gives the structural attributes of time contents. The source-specific obligation atoms comprise 62 claims, 35 time contents, 23 transfers and 25 results, totaling 145.

Table 134: Physical and Formal Input Claim Components
in DeWitt 1967

p	Claim	i	a	Source Location
1	Configuration of the spatial three-metric	$\emptyset$	$\emptyset$	§§1–2,5
2	Spatial configuration of matter fields	$\emptyset$	$\emptyset$	§7,7.1–7.5
3	Intrinsic spatial curvature	$\emptyset$	$\emptyset$	§2,2.3–2.6
4	Spatial boundary data	$\emptyset$	$\emptyset$	§§1–6
5	3+1 metric decomposition	$1^Q, 6^S$	1–3	§2,2.1–2.2
6	Specification of the lapse function	$1^Q, 11^{1Q}$	2–3	§2; §6,6.26
7	Specification of the shift vector	1^Q	2–3	§2,2.1–2.4
8	Extrinsic curvature	$1^Q, 5^Q$	1–3	§2,2.3–2.6
9	Intrinsic curvature	$\emptyset$	1, 4	§2,2.3–2.6
10	Canonical Lagrangian	$1^Q, 5^Q$	1, 3–5	§§2–3,2.5–2.6,3.1–3.6
11	Canonical momentum	$1^Q, 5^Q$	1, 3, 5	§3,3.1–3.3
12	Primary constraints	$1^Q, 5^Q$	3, 5	§3,3.1–3.2
13	Momentum constraints	$1^Q, 5^Q$	1, 5	§3,3.5–3.10
14	Hamiltonian constraint	$1^Q, 5^Q$	1, 4–5	§3,3.4–3.10
15	Constraint propagation	$1^{QS}, 5^{QS}, 6^S$	2, 5	§3; §4,4.10–4.26
16	Sequence of Cauchy hypersurfaces	6^S	2	§1 p. 1116; §3; §6,6.10–6.15
17	Surface energy of an infinite system	7^{QS}	5–6	§2,2.7–2.9; §4,4.5–4.9
18	Hamiltonian of a finite closed system	1^Q	5, 7	§§3–4 pp. 1119–1120
19	Quantum constraint equation in the metric representation	$\emptyset$	4–5, 8, 10	§§4–5,4.1–4.4,5.5–5.6
20	Metric representation	$\emptyset$	1, 8–9	§5,5.1–5.5
21	External coordinate label	1^Q	7, 9	§§4–5 pp. 1120,1122

p	Claim	i	a	Source Location
22	State of three-geometry	$\emptyset$	1, 9	§5,5.1–5.5
23	Configuration-space metric at each point	$\emptyset$	1, 10	§5,5.6–5.10; App. A A1–A14
24	Hyperbolic type of the configuration-space metric	$\emptyset$	1, 10	§5,5.6–5.10; App. A A1–A14
25	Candidate intrinsic time	2^Q	9–11	§5 pp. 1123–1124
26	Time interval between two hypersurfaces	9^{IQ}	1, 5, 12	§5 p. 1124
27	Formal flux inner product	2^{QB}	8, 10, 13	§5,5.19–5.20
28	Positive-frequency restriction and its obstruction	2^Q	13–14	§5,5.19–5.20
29	WKB correspondence	$\emptyset$	4, 8, 15	§6,6.10–6.12
30	Hamilton–Jacobi correspondence	$1^{QS}, 5^{QS}$	3, 5, 15	§6,6.12–6.15
31	Configuration-space trajectory	$6^S, 10^{QS}$	3, 15–16	§6,6.16–6.23; App. B
32	Proper-time interval	$10^Q, 11^{IQ}$	3, 16	§6,6.24–6.26
33	Configuration-space frontier	$\emptyset$	10, 18	§5,5.18; §6; App. A
34	Proposed quantum barrier	13^B	15, 18–19	§6,6.29–6.31
35	Closed Friedmann model	$1^Q, 15^Q$	3, 5, 20	§7,7.1–7.16
36	Negative sign of the Friedmann-scale kinetic term	$\emptyset$	11, 20–21	§7,7.3,7.10,7.13–7.16
37	Expanding branch	$14^S, 15^{QS}, 16^{QS}$	15, 20, 22	§7,7.17–7.34
38	Contracting branch	$14^S, 15^{QS}, 16^{QS}$	15, 20, 22	§7,7.17–7.34
39	Lifetime in the spherical vacuum-universe approximation	$11^{IQ}, 12^{IQ}, 13^B$	16–17	§6,6.24–6.28
40	Matter clock	$15^Q, 17^K, 18^{IQ}, 21^{KQ}$	20, 22–23, 27	§7; §8,8.2–8.10
41	Geometric clock	$18^{IQ}, 19^Q, 21^{KQ}$	20, 22, 27	§§7–8,8.2–8.10
42	Matter–geometry cross-correlation	31^S	15, 20, 22	§7,7.22–7.31; Fig1
43	Matching of allowed energy levels	$\emptyset$	23–24	§7,7.55–7.56
44	Criterion for a good wave packet	$\emptyset$	15, 23, 26	§8,8.1;pp. 1134–1135
45	Proper-time period	$18^{IQ}, 19^Q$	24–25	§7,7.57–7.59
46	Proper-time frequency	$15^Q, 19^Q$	20, 22, 24	§7,7.57–7.59
47	Phase variable	$\emptyset$	15, 22	§8,8.3
48	Clock window	$21^Q, 24^B, 30^B$	26–28	§8,8.11–8.13; Fig2
49	Conditional Schrödinger representation	$21^{QS}, 25^S$	5, 27–28	§8,8.14–8.15

p	Claim	i	a	Source Location
50	Conditional wave-packet spreading	$25^S, 26^S$	26, 28	§8 pp. 1136–1137
51	Finite limit on clock precision	27^Q	27, 30	§8 p. 1137
52	Model inner product	$18^{IQ}, 19^Q, 35^{IQB}$	8, 20, 26, 31	§9 pp. 1137–1139, 9.8–9.11
53	Relation among entropy, time reversal, and branch direction	$28^S, 29^{SB}$	32	§10 pp. 1140–1142
54	Claim of phenomenological time defined by the contents of the universe	$3^K, 4^K$	27–28, 30	§8 p. 1137; §10
55	Periodicity and non-Hermiticity of proper-time operators	$21^Q, 22^{KQ}$	22, 27	§8, 8.4–8.10
56	Planck length and unit conditions on physical constants	$\emptyset$	33	§1 pp. 1113–1116
57	Proper-time lifetime of the closed Friedmann dust universe	$32^{SB}, 34^{IQ}$	20, 29	§8, 8.17–8.21
58	Classical cosmological histories and singularity endpoints	$10^{QS}, 11^{IQ}, 13^B$	16–18	§6 pp. 1126–1129
59	Period commensurability over a broad action range	$18^{IQ}, 19^Q$	23–25	§7, 7.55–7.59
60	Common subperiod and ghost multiplicity in clock representations	$18^{IQ}, 21^{KQ}, 23^{QX}$	24–25, 27–28	§8 pp. 1135–1136; Fig2
61	Requirement of intrinsic dynamics in a finite closed universe	1^Q	5, 7	§4 p. 1120; §5 p. 1122
62	Proper-time lapse of the inner-product contour	$18^{IQ}, 35^{IQB}$	24, 26, 31	§9 p. 1138, 9.9–9.11

Table 135: Time-Use Roles in DeWitt 1967

Code	Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causal
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 136: Structural Attributes of Time Contents in DeWitt 1967

Time-Content Indices	Type	Axes
1–2, 7, 10, 21, 23	member	instant
3–5, 8–9, 11–12, 14–15, 17–20, 22, 25–27, 31–32, 34–35	dependency	$\emptyset$
6	member	order_relation
13	member	endpoint
16, 29, 33	member	orientation
24, 30	member	interval_boundary
28	member	order_relation, orientation

Dependency Inputs

Dependency Inputs The time-content input index sets and formal-evidence support index sets in Table 137.

Dependency and Transfer Elements

Definition 8.232 (Time-Content Transfers in the DeWitt 1967 Reference). Let $\text{Idx}_R^{\text{audit,transfer}} = [23]$. Table 137 specifies each transfer’s time-content inputs and outputs, source-formal-evidence support and auxiliary inputs. The indices i, p and a refer to time contents, claims and auxiliary objects, respectively.

Table 137: Time-Content Transfers in DeWitt 1967

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
1	7	8	17, 19–20	6, 8, 26	Wave-packet evolution in asymptotic time
2	$\emptyset$	2	23–25	1, 10–11	Calculation of the configuration coordinate proposed as intrinsic time
3	$\emptyset$	9	26	1, 5, 12	Determination of a time interval from two hypersurface geometries
4	1	10	10–11, 29–31	3, 15–16	Parametrization of a WKB–Hamilton–Jacobi history
5	10	11	31–32	3, 16	Proper-time interval from a history parameter
6	11, 13	12	32, 39, 58	16–17	Lifetime estimate in the vacuum-universe approximation

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
7	15	16	35–38	15, 20–22	Sign correspondence of expanding and contracting branches
8	15	31	35, 37–38, 42	15, 20, 22	Temporal correlation from both Hamilton–Jacobi phases
9	15, 18	17	35, 40, 43, 45–46, 59	20, 22–24	Matter-clock correspondence from periodic states and phase
10	16, 19, 33	20	37–38, 43–44, 46–47	15, 22, 24, 26	Temporal motion of a selected good wave packet
11	19, 33	21	40–41, 46–47	20, 22	Proper times of both clocks from phase and frequency
12	18, 21	23	43, 45, 55, 59–60	24–25, 27	Identification of clock proper times by a common subperiod
13	21–22	24, 30	40–41, 44, 48, 55	26–28	Representation windows using the two clocks
14	21, 24, 30	25	19, 35, 48–49	5, 27–28	Conditional proper-time evolution of both subsystems
15	16, 21	32	35, 37–38, 49, 57	20, 22, 29	Cycloidal trajectory in model-clock proper time
16	1	5	8, 10–11, 15	2	Differentiation in external coordinate time
17	1	6	5, 15–16	2–3	Ordering of a hypersurface history by coordinate time
18	5	15	6, 35	3, 20	Matter-model proper-time derivative via the lapse
19	15	19	35, 43, 46	20, 22, 24	Model proper-time angular frequencies
20	19	18	45–46	25	Proper-time periods from angular frequencies
21	21	22	47, 55	22, 27	Operator representation of model proper time
22	32	34	57	20, 29	Lifetime of the dust-universe cycloid
23	18	35	52, 62	24, 26, 31	Proper-time lapse of the contour from the period and level spacing

Transfer Events The 23 transfers in the table.

Transfer Outputs The time contents corresponding to the time-content output index set of each transfer.

Dependency Cycle The time contents have no directed cycle under this transfer relation.

Grounding Conditions

Conditions under Which a Dependency or Transfer Applies Each transfer holds under its tabulated time inputs and claim supports and the auxiliary conditions below.

1. 4.5–4.9; Specify the Hamiltonian, spectrum, initial state and expansion coefficients for the asymptotically flat infinite universe.
2. 5.7–5.10; 7.13–7.16; Use the negative direction of the configuration-space metric and Eq. (5.7) on positive-definite spatial three-metrics. The Friedmann reduction follows Eqs. (7.13)–(7.16).
3. §5 p. 1124; This is the classical determination attributed to Baierlein, Sharp and Wheeler. It requires the domain excluding the source’s singular exceptions and the conditions for a classical solution joining the two hypersurfaces.
4. 6.10–6.23; Specify slow amplitude variation, a Hamilton–Jacobi phase solution, initial spatial data, lapse and shift, and a branch. The source’s physical history follows its coordinate-time differential equation.
5. 6.24–6.26; Integrate $d\tau/ds = \alpha$ under uniform lapse, zero shift and nonzero I .
6. 6.24–6.28; Use the history endpoints and spherical approximation for the classical vacuum universe. Equation (6.28) gives a lifetime of approximately $2.62R_{\max}$.
7. 7.17–7.34; Use the proper-time radial velocity, negative kinetic sign and WKB phase of the closed dust model, with the expansion or contraction branch specified.
8. 7.22–7.31; Specify the geometric and matter Hamilton–Jacobi phases and actions, and their relative phase. Apply the velocity-ratio equation where the radial velocity is nonzero.
9. §7; Specify identical coherent particles, a suitable energy superposition, spectral matching and phase, using the proper-time derivative of source Eq. (7.4).
10. 8.1; pp. 1134–1135; Choose expansion coefficients on the allowed levels, with localization conditions, branch and WKB domain specified.
11. 8.2–8.3; For both subsystems, supply a nonzero proper-time angular frequency, phase origin and branch orientation, and use Θ/ω in Eqs. (8.2)–(8.3).
12. §8 pp. 1135–1136; Identify clock representations using the sparse allowed levels and common subperiod. Ghost multiplicities follow the source level spacing and representation windows.
13. 8.11–8.13; Use the physical projector, clock operators, states and approximate spectral orthogonality in the domains of Eqs. (8.11)–(8.13).
14. 8.14–8.15; Use the total Hamiltonian constraint and both clock representations under their canonical-conjugacy and domain conditions.
15. 8.16–8.21; Use the closed dust model’s Hamilton equations and constructive-interference conditions, branch signs, integration origins and the source hard-wall boundary condition.
16. §§2–4; Use differentiable fields on the specified coordinate-time domain.
17. §§1–3; Index an admissible Cauchy foliation and lapse by the specified coordinate-time order.

18. 7.4–7.5; Use $\dot{f} = \alpha^{-1} df/dx^0$ in Eq. (7.4) where the lapse is nonzero.
19. 7.57–7.59; Identify action derivatives of energy with angular frequencies in the model’s proper-time convention under the semiclassical conditions of Eqs. (7.57)–(7.59).
20. 7.57–7.59; Define each subsystem’s period by $2\pi/\omega$ for a nonzero angular frequency.
21. 8.4–8.10; Represent model proper time as an operator using the domains, periodicity and formal canonical-conjugacy relations of Eqs. (8.4)–(8.10).
22. 8.17–8.21; With the source origins, take the cycloid phase from 0 to 2π , obtaining the proper-time interval $\pi R_{\max}$.
23. §9 p. 1138, 9.9–9.11; Use overlapping good packets, selected branches, positive level spacing Δn and common period T . Extending the contour to the destructive-interference endpoints changes the corresponding proper-time lapse from $T/(4\Delta n)$ to $T/(2\Delta n)$.

Generation Judgment The four physical inputs in the initial generation set do not supply the source-formal-evidence supports of the transfers. Generation reaches no time content. Under the source formal-evidence environment, Transfers 2 and 3 output Time Contents 2 and 9.

Original-Source Support

Original-Source Support for Each Dependency and Transfer Each transfer’s source location is given under the same transfer index in the conditions above.

Results

Definition 8.233 (Results in the DeWitt 1967 Reference). Define the result index set by $\text{ld}x_R^{\text{audit,result}} = [25]$. The correspondences between results and source locations are as shown in Table 138.

Table 138: Results and Source Locations in DeWitt 1967

q	Result	Source Location
1	Presentation of the canonical 3+1 decomposition	§§1–3
2	Canonical constraints and their propagation	§§2–3
3	Difference between finite and infinite Hamiltonian systems	§§2–4
4	Construction of the quantum constraint equation	§§4–5
5	Identification of the nonphysical role of the external coordinate x^0	§§3–4
6	Requirement of intrinsic dynamics in a finite closed system	§§4–5
7	Metric representation and description of three-geometries	§5, 5.1–5.5

q	Result	Source Location
8	Hyperbolic signature of the configuration-space metric	§5,5.6–5.18; App. A
9	Presentation of candidate intrinsic times by region	§5,5.7–5.10; §7,7.13–7.16
10	Determination of the proper-time interval between two hypersurfaces	§5 p. 1124
11	Formal conserved-flux inner product and the positive-definiteness problem	§5,5.19–5.20
12	WKB and Hamilton–Jacobi approximation	§6,6.10–6.15
13	Configuration-space geodesic equation with a force term and proper time	§6,6.16–6.26; App. B
14	Configuration-space frontier and classical cosmological histories	§§5–6; App. A
15	Conditional domain of application of the potential-barrier analysis	§6,6.29–6.31
16	Closed Friedmann dust model and its cosmological histories	§7; §8,8.17–8.21
17	Separation of matter and geometric clocks	§§7–8,8.2–8.10
18	Matter–geometry cross-correlation	§7,7.22–7.31; Fig1
19	Criteria involving spectra, degeneracy, and commensurability	§7,7.55–7.59
20	Good-wave-packet criterion and ghost ambiguity	§8 pp. 1134–1136; Fig2
21	Conditional Schrödinger equation	§8,8.11–8.15
22	Non-Hermitian clock variable and finite-precision boundary	§8 pp. 1135–1137
23	Conclusion concerning phenomenological time	§8 p. 1137; §10
24	Model-level inner-product construction	§9 pp. 1137–1140
25	Assumption boundary in the discussion of the Everett interpretation and the arrow of time	§10 pp. 1140–1142

Source-Specific Results The 25 results in Definition 8.233.

Source Locations of the Results The source-location column of Table 138 provides an individual correspondence for each of the 25 results.

Result Dependencies

Definition 8.234 (Result-Dependency Hyperedges in the DeWitt 1967 Reference). For each $q \in [25]$, Table 139 specifies the time-content premises, claim support, auxiliary inputs and result output of $\text{ResultEdge}_{R,q}$.

Table 139: Result-Dependence Hyperedges in DeWitt 1967

q	Time Premises i	Claim Support p	Auxiliary In a	Result Output Indices
1	1, 5–6, 11	5–9, 16	1–4	1
2	1, 5–6	10–15	1–5	2
3	1, 7	17–18	5–7	3
4	$\emptyset$	19	4–5, 8, 10	4
5	1	21	7, 9	5
6	1	18, 61	5, 7	6
7	$\emptyset$	20, 22	1, 8–9	7
8	$\emptyset$	23–24	1, 10	8
9	2	25	9–11	9
10	9	26	1, 5, 12	10
11	2	27–28	8, 10, 13–14	11
12	1, 5	29–30	3–5, 8, 15	12
13	6, 10–11	31–32	3, 15–16	13
14	10–13	33, 39, 58	10, 16–18	14
15	13	34	15, 18–19	15
16	1, 14–16, 32, 34	35–38, 56–57	3, 5, 11, 15, 20–22, 29, 33	16
17	15, 17–19, 21–22	40–41, 47, 55	15, 20, 22–23, 27	17
18	31	42	15, 20, 22	18
19	15, 18–19	43, 45–46, 59	20, 22–25	19
20	18, 21, 23	44, 60	15, 23–28	20
21	21, 24–25, 30	48–49	5, 26–28	21
22	21–22, 25–27	50–51, 55	22, 26–28, 30	22
23	3–4	54	27–28, 30	23
24	18–19, 35	52, 62	8, 20, 24, 26, 31	24
25	28–29	53	32	25

Grounding Grounds

Grounding Grounds of the Results

The construction of the canonical formalism, derivation of the constraint equations, geometric analysis of configuration space, WKB and Hamilton–Jacobi approximations, analysis of the Friedmann model, clock models, and the conditional Schrödinger representation.

Evidence Hierarchy

Objects of Direct Measurement The number of objects newly measured by this reference is zero.

Direct-Measurement Predicates The number of measurement predicates directly applied by this reference to apparatus outputs or sample data is zero.

Conditions and Domains of Direct Measurement Because the number of direct measurements made by this reference is zero, direct-measurement conditions and direct-measurement domains of application are also inapplicable.

Objects of Formal Generalization Results 1–8 concern the canonical 3+1 decomposition, constraints, and configuration space of general relativity; Results 9–15 concern candidate intrinsic times, the WKB approximation, and contraction boundaries; Results 16–22 concern the closed Friedmann model, matter clocks, geometric clocks, conditional quantum evolution, and finite clock precision; and Results 23–25 concern phenomenological time, the model inner product, and time direction.

Conditions and Domains of Formal Generalization Each generalization has as its grounding conditions the time-content inputs and formal-evidence support in Table 139. The conclusions of the closed Friedmann model, WKB approximation, clock models, and conditional Schrödinger representation belong to domains in which the respective model assumptions and approximation conditions hold.

Generation Judgment Judgment Values: LCTR complete-generation judgment: 0; source-evidence-conditioned local reach: 2; primitive-time roots remaining after that judgment: 33.

Generation Reach and Primitive Time Roots Generation reach from the four non-temporal physical inputs reaches none of the 35 time contents. All 35 belong to the primitive-time roots.

Source-Internal Evidence-Conditioned Local Reach Under the source formal-evidence environment, the configuration-variable candidate for intrinsic time and the proper-time interval between two hypersurfaces, Time Contents 2 and 9, are reached. Local reach comprises two contents, leaving 33 roots. The first is supported by the metric signature and coordinate formula; the second by the classical determination attributed to Baierlein, Sharp and Wheeler and its conditions of application.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ from Definition 7.108 and apply Definition 7.50 to it. Totalling over all results determines the primitive-time-root–result dependence relations and necessary-primitive-time-root–result dependence relations.

Result Dependence Results 4 and 7–11, six in total, are evidence-only results; the remaining 19 are root-dependent. There are no generated results or results unreachable after adding roots.

Evidence Effect

Evidence-Effect Scope The canonical constraints, configuration-space geometry, WKB–Hamilton–Jacobi approximation, closed Friedmann dust model, clock representations and finite-contour inner product hold under the conditions stated for the respective claims and results.

Evidence-Effect Boundary Determination of the time candidate and interval using source formal evidence depends on the supports and conditions in Table 137. Physical-time qualification is judged by generation reach from non-temporal physical inputs. The proper-time derivative in source Eq. (7.4) and the phase-angular-frequency correspondence in Eq. (8.2) are also subject to that generation judgment.

Result-Preservation Domain

Result-Preservation Domain The 25 results are preserved within the domains where their claim supports, model assumptions, approximation conditions, operator domains and boundary conditions hold. Results with time-content premises are supplied with the corresponding LCTR-generated time representations. Table 139 specifies these premises.

Placement of the Grounding Obligation

Placement of the Grounding Obligation Source Eq. (7.4) uses proper-time differentiation through $\dot{f} = \alpha^{-1}df/dx^0$, and Eq. (8.2) defines clock variables from the corresponding proper-time angular frequencies and phases. Generation from non-temporal physical inputs corresponding to these temporal uses is absent within the audit boundary. The grounding-obligation placement is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure

Evidence of Actual Failure The complete judgment and the remaining-root judgment concern physical-time qualification; they are distinct from actual failure of a concrete condition governing a result’s value or domain of application. Neither the original formal evidence nor the LCTR audit evidence in this reference establishes an actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence,” and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream Reach For this reference, the reach to primitive-time-root-dependent results is determined by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because actual failure of a concrete condition has not been established, the actual-reach judgment from that failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the objects of repair are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is limited to results for which actual failure is independently established and which are dependency-reachable from that failure location. Because actual failure is indeterminate in this reference, this repair-and-recalculation specification is conditional. If that condition is satisfied, the target results are regenerated

or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.235 (Primitive Time Roots and Result Dependence in the DeWitt 1967 Reference). In the component order of Definition 8.115,

$$\begin{aligned}\text{AuditPartIdx}_R = (\emptyset, [35]; \\ \emptyset, \{4, 7, 8, 9, 10, 11\}; \\ [25] \setminus \{4, 7, 8, 9, 10, 11\}, \emptyset).\end{aligned}$$

Thus $R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. The audit boundary comprises 122 nodes—62 claim nodes, 35 time-content nodes and 25 result nodes—and 48 edges—23 transfer edges and 25 result edges. The initial generation set consists of Claims 1–4. Every transfer and result edge requires claim support outside this set, so its closure remains the initial four nodes. With all 62 claims in the source formal-evidence environment, Transfers 2 and 3 have empty time inputs and reach Time Contents 2 and 9. The time inputs of the other transfers are not contained in this set, so no further time contents are reached. Results 4, 7 and 8 have empty time premises; the premises of Results 9–11 are contained in Time Contents 2 and 9. The remaining 19 results become reachable on adding primitive-time roots, and the closure with all 35 roots contains all 122 nodes. Definition 8.115 gives the displayed partition. $\square$

Proposition 8.236 (Physical-Time Qualification in the DeWitt 1967 Reference). None of the 35 time contents satisfies physical-time qualification. Local reach under the source formal-evidence environment comprises Time Contents 2 and 9. The 19 root-dependent results depend on time contents that lack this qualification.

Proof. Under the complete judgment, all 35 time contents belong to the primitive-time roots. Local reach obtained by adding the source formal-evidence environment reaches Time Contents 2 and 9. Physical-time qualification is determined by generation reach under the complete judgment. Applying Definition 7.10, Theorem 7.11 and Corollary 7.12 to the complete-judgment root set and the 19 root-dependent results gives the conclusion. $\square$

Proposition 8.237 (Minimal Primitive-Time-Root Support in the DeWitt 1967 Reference). For each $q \in \text{Idx}_R^{\text{audit,result}}$, the number of elements of $\mathcal{A}_R(q)$ in Definition 7.108, and the minimum and maximum cardinalities of those elements, are as shown in Table 140.

Table 140: Minimal Primitive-Time-Root Support in DeWitt 1967

Result Indices	Support Statistics
1–2, 5–6, 12, 15	(1; 1)
3, 23, 25	(1; 2)
4, 7–11	(1; 0)
13	(2; 1, 2)

Result Indices	Support Statistics
14	(2; 2)
16	(3; 3)
17	(6; 2)
18, 24	(4; 1)
19	(3; 1)
20	(9; 2)
21	(5; 1, 2)
22	(5; 3, 4)

Proof. Applying the update in Definition 7.46 to the 23 transfers leaves the support antichains unchanged on scan 4 in the listed order. Proposition 7.47 gives 58 support-family members across the 25 results, 60 primitive-time-root–result dependence pairs and 17 necessary-root–result dependence pairs. $\square$

Individual Statements Verbatim original (English):

if we insist on viewing R , and hence X , as a “time” coordinate

The first quotation uses the model variables R and X .

The source location is DeWitt, *Quantum Theory of Gravity. I. The Canonical Theory*, *Physical Review* 160, immediately after Eq. (7.16) in §7.

Verbatim original (English):

It is for this reason that we may say that “time” is only a phenomenological concept, useful under certain circumstances.

The source location is §8, p. 1137, of the same paper.

Verbatim original (English):

For the only “time” which a covariant theory can admit is an intrinsic time defined by the contents of the universe itself.

The source location is §8, p. 1137, of the same paper, in the paragraph after the second quotation, following two sentences on general covariance and the phenomenological character of quantum mechanics in everyday use.

Verbatim original (English):

Any intrinsically defined time is necessarily non-Hermitian, which is equivalent to saying that there exists no clock, whether geometrical or material, which can yield a measure of time which is operationally valid under all circumstances, and hence there exists no operational method for determining the Schrödinger state function with arbitrarily high precision.

The source location is §8, p. 1137, of the same paper, following the immediately preceding sentence concerning intrinsic time.

Verbatim original (English):

Because every clock has a “one-sided” energy spectrum, its ultimate accuracy must necessarily be inversely proportional to its rest mass. When the whole universe is cast in the role of a clock, the concept of time can of course be made fantastically accurate (at least in principle) because of the enormity of the masses and quantum numbers involved. But as long as the universe is finite, a theoretical limit to the accuracy nevertheless remains.

The source location is §8, p. 1137, of the same paper. In the second sentence of the quotation, DeWitt discusses the accuracy obtained when the entire universe is treated as a clock.

Objects of the Statements The treatment of the model variables R and X as time coordinates; the status of time as a phenomenological concept; intrinsic time defined by the contents of the universe; the non-Hermiticity of intrinsic time and the operational limits of clocks; and the dependence of clock precision on rest mass and its limit in a finite universe.

Original-Source Locations The first quotation is immediately after Eq. (7.16) in §7, and the second through fifth quotations are in §8, p. 1137. The second quotation ends the opening paragraph on that page; the third and fourth quotations occur consecutively in the next paragraph. The fifth quotation also belongs to the discussion of clock precision in that paragraph.

Minimal Context Required for Interpretation The first quotation belongs to a context that conditionally describes a time-coordinate interpretation of the variables R and X in the closed Friedmann model. Equation (7.13) of the source defines $X = R^{3/2}$. With mass eigenvalues and the stated boundary condition, the source treats Eq. (7.14) as a Schrödinger equation and discusses normalizability when R or X is regarded as a time coordinate. The second quotation belongs to a context that treats time as a phenomenological concept useful under certain circumstances. The third quotation belongs to a context that restricts the time admissible to a covariant theory to intrinsic time defined by the contents of the universe. The fourth quotation belongs to a context that identifies the non-Hermiticity of that intrinsic time as equivalent to the nonexistence of a clock that yields an operationally valid measure of time under all circumstances. The fifth quotation belongs to the context of a clock’s one-sided energy spectrum, rest mass, the treatment of the whole universe as a clock, and the precision limit in a finite universe.

Attribution of the Statements All five quotations are attributable to DeWitt.

Predicates Assigned to the Objects by the Statements The first quotation places a condition on treating R and X as time coordinates. The second describes time as a phenomenological concept useful under certain circumstances. The third describes the only time admissible to a covariant theory as intrinsic time defined by the contents of the universe. The fourth describes intrinsic time as necessarily non-Hermitian and states this to be equivalent to the nonexistence of a clock providing a universally valid measure of time. The fifth makes the ultimate precision of a clock inversely proportional to its rest mass and states that a theoretical precision limit remains in a finite universe.

Explicit Domains of Application of the Statements The domain of application of the five quotations is the theoretical explanation concerning the canonical theory of quantum gravity, the closed Friedmann model, intrinsic time, and clock models.

Individual Audit

Interpretations Permitted by the Original Wording and Minimal Context The first statement specifies the condition of viewing R and X in the closed Friedmann model as time coordinates. The second treats time as a phenomenological concept useful under certain circumstances. The third identifies the only time admissible to a covariant theory as intrinsic time defined by the contents of the universe. The fourth states that the necessary non-Hermiticity of such intrinsic time is equivalent to the nonexistence of a clock yielding a time measure valid under all circumstances. The fifth grounds the rest-mass dependence of precision in the one-sided energy spectrum and concludes that a theoretical precision limit remains in a finite universe.

Independent Grounds The first statement follows the variable transformation and kinetic term in Eqs. (7.13)–(7.16). The second through fifth statements draw on the discussion in Section 8 of clock operators, conditional Schrödinger representations, packet spreading, general covariance and clock energy spectra.

Evidence-Effect Boundary Transfer 2 calculates the configuration coordinate, and Transfer 11 gives model-clock proper times. The latter depends on temporal reference through the proper-time derivative used in source Eq. (7.4) and the angular frequencies in Eqs. (7.57)–(7.59). The statements about clock operating domains and precision also concern this time measure. The grounding obligation for applying each statement to physical time use is determined by the generation conditions of its corresponding time content.

Corresponding Formal Evidence The first statement corresponds to Claims 25 and 36 and Results 9 and 16; the second and third to Claim 54 and Result 23; the fourth to Claim 55 and Results 17 and 22; and the fifth to Claim 51 and Result 22. Table 134, Table 139 and Proposition 8.237 specify the time contents, result premises and minimal-root supports.

Within-Source Comparison Section 7 gives a closed Friedmann dust model using proper-time differentiation. Section 8 constructs both clocks' proper times and conditional evolution from its phases and angular frequencies. The discussion of phenomenological time and clock precision at the end of that section connects to these clock representations and domains. Section 9 uses good packets and finite contours for the inner product, and Section 10 discusses the Everett interpretation, entropy and the arrow of time.

Common Audit with Specified Domain of Application The domains and time-content dependencies of the five statements are specified by the source locations, claim components, results and conditions above.

8.19.15 Source Audit

Audit Input

Original Source and Audit Scope

Source Identifier 15.

Source Name Primitive-time-root audit of Ehlers–Pirani–Schild 1972

Bibliographic Identification J. Ehlers, F. A. E. Pirani and A. Schild, *The Geometry of Free Fall and Light Propagation, General Relativity: Papers in Honour of J. L. Synge*, Clarendon Press, Oxford, pp. 63–84 (1972); *General Relativity and Gravitation* 44, 1587–1609 (2012), doi:10.1007/s10714-012-1353-4 is denoted by $R = \text{Ref}_{\text{EPS72GFL}}$. The audit domain comprises Section 1, Figures 1–2, and the sections on differential topology, light propagation and conformal structure, free fall and projective structure, compatibility and Weyl structure, and curvature and Riemannian spacetime.

Source-Location Identifier 15-L1.

Source-Location Content Table 143 gives the source locations of the 73 claim components, and Table 141 those of the 62 time contents. Page numbers refer to the 1972 original.

Audit-Scope Identifier 15-SC1.

Audit-Scope Content The audit domain stated in the bibliographic identification.

Formal Objects

Object Identifier 15-F1.

Object Name Formal objects.

Object Type Formal object.

Formal Content The primitive event set; families of light rays and freely falling particles; echo and message connection relations; topological, differential, conformal, projective, affine, Weyl, and metric structures; radar atlases; proper-time scales; clock comparisons; Einstein simultaneity; and the equality of gravitational and atomic time.

Source-Location Identifier Set 15-L1.

Table 142 specifies 41 auxiliary objects: event, signal and clock identities, geometric objects, operations and calibration conditions. Time contents represent the temporal functions of physical light and particle use, local clock coordinates, calibrated proper time and clock comparison. The preceding chronometric route and the EPS test-particle route have their respective input conditions.

Assumptions

Assumption Identifier 15-A1.

Assumption Content The construction uses four dimensions, Lorentz signature $(+++-)$, classical test particles and light in an otherwise empty region, D1–D4, L1–L2, P1–P2 and compatibility axiom C. The C^1 regularity of the projective coefficients is an additional assumption. The source regards the sufficiency of C^3 regularity for D1–D3 as tentative. Local constructions use the specified neighborhood, coverage and orientation conditions; metric reduction uses $F = 0$ and a simply connected domain. Equality of gravitational and atomic time is an additional empirical postulate.

Source-Location Identifier Set 15-L1.

Non-Temporal Physical Inputs

Input Identifier 15-P1.

Input Name Non-temporal physical inputs.

Input Type Non-temporal physical input.

Input Content Primitive event identity, the unparametrized light-ray connection relation, the unparametrized freely falling particle connection relation, the coincidence relation of emission, reflection, and reception, the echo connection relation, and the message connection relation in Components 1–6.

Source-Location Identifier Set 15-L1.

Time Contents

Time-Content Identifier 15-T1.

Time-Content Name Time contents used in the original source.

Time-Content Type Time content.

Time-Content Statement Table 141 gives the 62 time contents and their source locations.

Source-Location Identifier Set 15-L1.

Formal Input Components

Definition 8.238 (Time Contents in the Ehlers–Pirani–Schild 1972 Reference). For $R = \text{Ref}_{\text{EPS72GFL}}$, let $n_R = 62$. Table 141 specifies the time contents and source locations.

Table 141: Ehlers–Pirani–Schild 1972 Time Contents and Source Locations

i	Time Content	Source Location
1	Neighboring event pair on a standard-clock history	pp. 64,65
2	Measured time interval	pp. 64,65
3	Clock reading and tick	pp. 64,65
4	Worldlines of the clocks under comparison	pp. 64–65,69,82
5	Ordered histories of the clocks under comparison	pp. 64–65,69,82
6	Gravitational time scale	pp. 65
7	Atomic time scale	pp. 65
8	Signal propagation	pp. 65,67,68,70,81
9	Emission and arrival order	pp. 65,67,68,70,81
10	Free-fall worldline	pp. 65,67,68,70,81
11	Particle-motion history	pp. 65,67,68,70,81
12	Measured interval and scale output	pp. 65,67,68,70,81
13	Timelike typing	pp. 72,73,74
14	Spacelike typing	pp. 72,73,74
15	Null typing	pp. 72,73,74
16	First of the two light-vector classes (no intrinsic future–past distinction)	pp. 72

<i>i</i>	Time Content	Source Location
17	Second of the two light-vector classes (no intrinsic future–past distinction)	pp. 72
18	Order of particle history	pp. 63,70,71
19	Preferred affine parameters of light and particle geodesics	pp. 63,64,67,68,81
20	Affine-parameter order of light and particle geodesics	pp. 63,64,67,68,81
21	Affinely parametrized light and particle geodesics	pp. 63,64,67,68,81
22	Timelike worldline	pp. 67,68
23	Event selected as the proper-time origin	pp. 67,68
24	Event selected as the proper-time unit point	pp. 67,68
25	Proper-time scale	pp. 67,68
26	Observer worldline	pp. 68,81
27	Proper-time measurement output	pp. 68,81
28	Comparison of the proper-time scales and rates of two clocks	pp. 69,82
29	Events corresponding to a signal pair	pp. 69,82
30	Einstein-simultaneity relation	pp. 69,82
31	Tick sequence mapped by simultaneity	pp. 69,82
32	Path dependence of clock rates at reunion	pp. 66,69,82
33	Newtonian absolute time	pp. 69
34	Order of three events	pp. 70,71
35	Signal propagation between particles	pp. 70,71
36	Interparticle arrival map	pp. 70,71
37	Particle local coordinate	pp. 70
38	Local-coordinate order	pp. 70
39	Nonmetric clock role	pp. 70
40	Source-particle coordinate	pp. 70
41	Destination-particle coordinate	pp. 70
42	Coordinate representation of an interparticle message map	pp. 70
43	Smooth time-coordinate relation	pp. 70
44	Emission coordinate of Particle 0	pp. 71
45	Reception coordinate of Particle 0	pp. 71
46	Emission coordinate of Particle 1	pp. 71
47	Reception coordinate of Particle 1	pp. 71
48	Radar coordinates from four readings	pp. 71
49	Null-cone hypersurface	pp. 74,75,76
50	Selected causal orientation	pp. 74,75,76
51	Classes of future and past neighborhoods	pp. 74,75,76
52	Timelike projective geodesic	pp. 76,77
53	Order of free-fall history	pp. 76,77
54	Arbitrary parameter of a particle geodesic	pp. 77
55	First coordinate derivative with respect to an arbitrary parameter	pp. 77
56	Second coordinate derivative with respect to an arbitrary parameter	pp. 77

<i>i</i>	Time Content	Source Location
57	Initial causal class	pp. 78,79
58	Causal type along a projective geodesic	pp. 78,79
59	Temporal comparison in the redshift observations cited by the source	pp. 82
60	Consistency of atomic-time transport	pp. 82
61	Gravitational time	pp. 82
62	Atomic time	pp. 82

Table 142: Ehlers–Pirani–Schild 1972 Auxiliary Objects

<i>a</i>	Auxiliary Object	Source Location
1	Identification of geometric and measured intervals	pp. 64–65
2	Standard-clock assumption	pp. 64–65
3	Equality of gravitational and atomic time scales	p. 65
4	Freely falling particle identity	pp. 65,70,76–77
5	Time-measurement operation	pp. 68,81
6	Application of the measurement procedure	pp. 68,81
7	Selection of local orientation	pp. 74–76
8	Parallel transport	pp. 67–68
9	Vector congruence	pp. 67–69
10	Selection of the zero event	pp. 67–68
11	Selection of the unit event	pp. 67–68
12	Unit-interval criterion	pp. 67–68
13	Light-signal exchange procedure	pp. 69–73,82
14	Equal-rate determination	pp. 69,82
15	Local comparison rule	pp. 69,82
16	Equal-interval criterion	pp. 69,82
17	Parallel transport along two paths	pp. 69,82
18	Closed-loop congruence comparison	pp. 69,82
19	Path-independence condition for clock rates	pp. 69,82
20	Source-particle identity	pp. 70–71
21	Destination-particle identity	pp. 70–71
22	Event–four-coordinate relation	p. 71
23	Determination of reception from emission	pp. 72–73
24	Empirical equality of gravitational and atomic time	p. 82
25	Event manifold	pp. 63–64,70–71
26	Hierarchy of geometric structures	pp. 63–67
27	Standard-clock identity	pp. 64–65
28	Light-signal identity	pp. 65,70–73
29	Nowhere-null curve	pp. 67–68
30	General Weyl arc length	pp. 67–68
31	Identities of the two clocks under comparison	pp. 69,82
32	Specified tick events	pp. 69,82
33	Emission, reflection and return events of an echo	pp. 70–71
34	Emission and reception events of a light connection	pp. 72–73

a	Auxiliary Object	Source Location
35	Conformal field, Hessian and tensor density	pp. 72–76
36	Projective connection coefficients	pp. 76–77
37	Compatible Weyl connection	pp. 67–68, 78–81
38	Homogeneous ODE for causal-type preservation	pp. 78–79; Eq 7, 18
39	Weyl curvature and length curvature	pp. 81–83
40	Fixed electromagnetic field and fixed specific charge	p. 76
41	Local-domain, differentiability and coverage conditions	pp. 70–82

The roles of Auxiliary Objects 1–24 are: 1, identification condition; 2, assumption; 3 and 24, equality conditions; 4, 20 and 21, auxiliary carriers; 5, 6, 8 and 17, operations; 7, selection condition; 9 and 22, relations; 10–12, calibration conditions; 13, procedure; 14 and 16, determinations; 15, rule; 18, comparison; 19, condition; and 23, determination rule.

Definition 8.239 (Ehlers–Pirani–Schild 1972 Physical and Formal Input Claim Components). Let the physical and formal input claim sequence be $(\text{ClaimVal}_{R,p}^{\text{audit,phys}})_{p \in [73]}$. Components 1–6 are non-temporal physical inputs and Components 7–73 are source formal evidence. The initial generation set consists of Components 1–6; the source formal-evidence environment contains all 73 components. Table 143 gives each claim, its correspondence to time contents i and auxiliary objects a , and its source location. The time-use units comprise 206 pairs in the claim table, one Newtonian comparison on source p. 69, and the following eight transfer pairs, totaling 215: Transfer 1 / Time Content 12^{QK} ; Transfer 1 / Time Content 27^{QK} ; Transfer 4 / Time Content 27^{QK} ; Transfer 9 / Time Content 42^{xQ} ; Transfer 9 / Time Content 43^{xQ} ; Transfer 14 / Time Content 19^{QS} ; Transfer 14 / Time Content 20^{S} ; Transfer 14 / Time Content 21^{S} . Superscripts denote the use roles in Table 144. Table 145 gives the structural attributes. The source-specific obligation atoms comprise 73 claims, 62 time contents, 14 transfers and 12 results, totaling 161.

Table 143: Ehlers–Pirani–Schild 1972 Physical and Formal Input Claim Components

p	Claim	i	a	Source Location
1	Primitive event identity	$\emptyset$	25	p. 70; initial incidences
2	Unparametrized light-ray connection relation	$\emptyset$	28	p. 70; initial incidences
3	Unparametrized freely falling particle connection relation	$\emptyset$	4	p. 70; initial incidences
4	Coincidence relation of emission, reflection, and reception	$\emptyset$	4, 25, 28	p. 70; initial incidences
5	Echo connection relation	$\emptyset$	33	p. 70; initial incidences

p	Claim	i	a	Source Location
6	Message connection relation	$\emptyset$	20–21, 33	p. 70; initial incidences
7	Topological structure	$\emptyset$	25–26	pp. 63–67
8	Differential structure	$\emptyset$	25–26	pp. 63–67
9	Conformal structure	$\emptyset$	25–26	pp. 63–67
10	Projective structure	$\emptyset$	25–26	pp. 63–67
11	Affine structure	$\emptyset$	25–26	pp. 63–67
12	Hierarchy of metric structures	$\emptyset$	25–26	pp. 63–67
13	Synge’s chronometric method	$1^{\text{IB}}, 2^{\text{IQK}}, 3^{\text{K}}, 4^{\text{S}}, 5^{\text{S}}$	1–2, 27	pp. 64–65
14	Role of standard clocks	$2^{\text{IQK}}, 3^{\text{K}}, 4^{\text{S}}, 5^{\text{S}}$	1–2, 27	pp. 64–65
15	Ehlers–Pirani–Schild test-particle method	$8^{\text{C}}, 9^{\text{CX}}, 10^{\text{S}}, 11^{\text{S}}$	4, 28, 41	pp. 65,70
16	Joint use of light rays and free fall	$8^{\text{C}}, 9^{\text{CX}}, 10^{\text{S}}, 11^{\text{S}}$	4, 28, 41	pp. 65,70
17	Local constructive axiomatization	$8^{\text{C}}, 10^{\text{S}}, 11^{\text{S}}$	4, 28, 41	pp. 65,70–82
18	Objective of constructing spacetime geometry	$8^{\text{C}}, 10^{\text{S}}, 11^{\text{S}}$	25–26, 41	pp. 65,67–68
19	Construction of conformal null cones from light propagation	$8^{\text{C}}, 9^{\text{CX}}, 16^{\text{C}}, 17^{\text{C}}, 37^{\text{QK}}, 38^{\text{X}}, 39^{\text{K}}, 48^{\text{QK}}$	28, 34–35, 41	pp. 72–76
20	Construction of projective structure from free-fall trajectories	$10^{\text{S}}, 11^{\text{S}}, 13^{\text{C}}, 18^{\text{S}}, 48^{\text{QK}}, 54^{\text{QS}}, 55^{\text{QS}}, 56^{\text{QS}}$	4, 36, 41	pp. 76–77
21	Compatibility of conformal and projective structures	$13^{\text{C}}, 14^{\text{C}}, 15^{\text{C}}, 49^{\text{C}}, 50^{\text{C}}, 51^{\text{C}}, 52^{\text{S}}, 53^{\text{S}}$	35–36, 41	pp. 78–81
22	Weyl structure	$13^{\text{C}}, 15^{\text{C}}, 52^{\text{S}}$	35–37, 41	pp. 67,81
23	Unique compatible affine connection	$\emptyset$	35–37	pp. 67,81
24	Weyl arc length on a nonnull curve	$\emptyset$	8–9, 29–30, 37	pp. 67–68
25	Zero-event calibration	$22^{\text{S}}, 23^{\text{QB}}$	10, 30, 37	pp. 67–68
26	Unit-event calibration	$22^{\text{S}}, 23^{\text{QB}}, 24^{\text{QB}}$	11–12, 30, 37	pp. 67–68
27	Proper-time scale	25^{QK}	30, 37	pp. 67–68
28	Kundt–Hoffmann-type proper-time measurement procedure	$8^{\text{C}}, 9^{\text{CX}}, 10^{\text{S}}, 11^{\text{S}}, 22^{\text{S}}, 25^{\text{QK}}, 26^{\text{S}}$	4–6, 28, 37, 41	pp. 68,81
29	Criteria for reduction to a compatible Riemannian metric	$4^{\text{S}}, 5^{\text{S}}, 25^{\text{QK}}, 28^{\text{QX}}, 29^{\text{X}}, 30^{\text{X}}, 31^{\text{XK}}, 32^{\text{QX}}$	14–16, 19, 31, 39, 41	pp. 69,82
30	Uniqueness of the metric up to scale (within a simply connected domain)	$\emptyset$	26, 37, 39, 41	pp. 69,82

p	Claim	i	a	Source Location
31	Temporal order structure of the echo operation	$34^{CX}, 35^C, 36^X$	20–21, 28, 33	pp. 70–71
32	Particle smoothness in D1	$\emptyset$	4, 41	p. 70;D1
33	Echo smoothness in D1	$34^{CX}, 37^{QK}, 38^X$	33, 41	p. 70;D1
34	Echo reversibility in D1	$34^{CX}, 37^{QK}, 38^X$	33, 41	p. 70;D1
35	Message smoothness in D2	$35^C, 36^X, 40^{QK}, 41^{QK}$	20–21, 33, 41	p. 70;D2
36	Nonmetric clock coordinate local to a particle	$37^{QK}, 38^X, 39^K$	4, 41	p. 70
37	Radar-coordinate map	$34^{CX}, 35^C, 36^X, 40^{QK}, 41^{QK}, 44^{QK}, 45^{QK}, 46^{QK}, 47^{QK}$	20–22, 33, 41	p. 71;Fig5
38	Smooth radar atlas in D3	$44^{QK}, 45^{QK}, 46^{QK}, 47^{QK}, 48^{QK}$	22, 25, 41	p. 71;D3
39	Light-ray smoothness in D4	8^C	28, 41	p. 71;D4
40	Initial-direction-dependent smoothness in D4	$8^C, 35^C, 36^X$	28, 33, 41	pp. 71–72;D4
41	Two-ray local connection relation in L1	$8^C, 9^{CX}, 37^{QK}$	23, 28, 34, 41	p. 72;L1
42	The C^2 function g in L1	$37^{QK}, 38^X, 39^K$	23, 34–35, 41	pp. 72–73;L1
43	Separation of light and nonlight directions in L2	$16^C, 17^C$	28, 35, 41	p. 72;L2
44	Lorentz conformal tensor density	$13^C, 14^C, 15^C, 37^{QK}, 48^{QK}$	35, 41	pp. 73–76
45	Normalization of the conformal tensor density	$\emptyset$	35	pp. 73–74
46	Light rays as conformal null curves	$8^C, 15^C$	28, 35, 41	pp. 74–75
47	Light rays as conformal null geodesics	$8^C, 15^C, 49^C, 50^C, 51^C$	28, 35–36, 41	pp. 74–76
48	Unique particle through each timelike direction in P1	$10^S, 11^S, 13^C, 18^S$	4, 35, 41	p. 76;P1
49	Lorentz-force trajectory family at fixed field and specific charge	$10^S, 11^S, 13^C$	4, 40–41	p. 76
50	Counterexample to the sufficiency of P1	$10^S, 11^S, 13^C$	4, 40–41	p. 76
51	Local projective-coordinate condition in P2	$10^S, 11^S, 48^{QK}, 54^{QS}, 55^{QS}, 56^{QS}$	4, 36, 41	pp. 76–77;P2, Eq6
52	Projective coefficients	$10^S, 13^C, 48^{QK}, 54^{QS}, 55^{QS}, 56^{QS}$	4, 36, 41	p. 77;Eq7–9
53	Projective geodesics	$52^S, 53^S, 54^{QS}, 55^{QS}, 56^{QS}$	36, 41	p. 77;Eq7

p	Claim	i	a	Source Location
54	Light-cone-interior-free-fall equivalence in C	$10^S, 13^C, 49^C, 50^C, 51^C$	4, 35–36, 41	p. 78;C
55	Preservation of causal type	$57^C, 58^C$	35–36, 38, 41	pp. 78–79;Eq18
56	Coincidence of projective and conformal null geodesics	$15^C, 49^C, 50^C, 51^C, 52^S, 58^C$	35–36, 41	pp. 79–81
57	Compatibility equation	$\emptyset$	35–36, 41	p. 81;Eq25
58	Weyl connection	$\emptyset$	35–37	p. 81;Eq26
59	Curvature decomposition	$\emptyset$	37, 39	pp. 81–82;Eq27–28
60	Weyl bivector F	$\emptyset$	37, 39	pp. 81–83
61	Einstein simultaneity	$4^S, 5^S, 25^{QK}, 29^X, 30^X, 31^{XK}$	13, 15–16, 31–32, 41	pp. 69,82
62	Path-independent congruence criterion	$4^S, 5^S, 25^{QK}, 28^{QX}, 32^{QX}$	8–9, 17–19, 31, 39, 41	pp. 69,82
63	Compatible Riemannian metric (within a simply connected domain)	$\emptyset$	26, 37, 39, 41	pp. 69,82
64	Postulate of equality between gravitational and atomic time	$59^{OX}, 60^{XK}, 61^{QK}, 62^{QK}$	24, 39	p. 82
65	Atomic-clock interpretation of the chronometric postulate	$6^{QK}, 7^{QK}$	1–3, 27	p. 65
66	Calibration rule for timelike Weyl arc length	$22^S, 23^{QB}, 24^{QB}$	10–12, 30, 37	pp. 67–68
67	Clock-rate comparison rule for proper-time scales	$4^S, 5^S, 25^{QK}$	14–15, 31, 41	pp. 69,82
68	Simultaneity correspondence rule using light signals	$4^S, 5^S, 8^C, 9^{CX}, 22^S, 25^{QK}$	13, 15, 28, 31, 41	pp. 69,82
69	Tick-sequence correspondence rule using simultaneity	$25^{QK}, 29^X, 30^X$	15–16, 32, 41	pp. 69,82
70	Preservation rule from the initial causal class	$54^{QS}, 55^{QS}, 56^{QS}, 57^C$	35–36, 38, 41	pp. 78–79;Eq7,18
71	Closed length curvature and conserved current density	$\emptyset$	39	pp. 82–83
72	Additional axiom selecting special relativity	$10^S, 11^S$	4, 26, 41	p. 69
73	Uniqueness of differentiable structure under a local causal isomorphism	$49^C, 50^C, 51^C$	25, 35, 41	p. 76

Claims 13, 14 and 65 concern the preceding chronometric route; Claim 72 concerns an additional axiom selecting special relativity. Newtonian absolute time (Time Content 33) is the comparison used on source p. 69 to explain the role of conformal structure.

Table 144: Ehlers–Pirani–Schild 1972 Time-Use Roles

Code	Use Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causal
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 145: Ehlers–Pirani–Schild 1972 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–3, 6–7	separate	$\emptyset$
4–5, 9–11, 18, 20–22, 26, 30–31, 34, 38, 52–53	member	order_relation
8, 12–17, 25, 27–28, 32, 35–36, 39, 42–43, 55–62	dependency	$\emptyset$
19, 23–24, 29, 37, 40–41, 44–48, 54	member	instant
33	lexical	$\emptyset$
49	member	interval_boundary
50–51	member	orientation

Dependency Inputs

Dependency Inputs Each of the 14 transfers has at least one time-content input.

Dependency and Transfer Elements

Definition 8.240 (Ehlers–Pirani–Schild 1972 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [14]$. The following table specifies each transfer’s time inputs and outputs, claim support and auxiliary inputs. The indices i , p and a denote time contents, claims and auxiliary objects, respectively.

Table 146: Ehlers–Pirani–Schild 1972 Time-Content Transfers

s	Time In i	Time Out i	Claims p	Auxiliary In a	Transfer Relation
1	27	12	28	5–6	Measurement-interval and scale output from constructed proper-time measurement
2	8–9, 16–17, 37–39, 48	13–15	41–45	23, 28, 34–35, 41	Causal typing from the L1 clock-coordinate function and the two L2 light-vector classes
3	22–24	25	24–26, 66	10–12, 30, 37	Proper-time scale from timelike Weyl arc length and zero/unit events
4	8–11, 22, 25–26	27	28	4–6, 28, 37, 41	Proper-time measurement output from light signals, freely falling particles, and an observer worldline
5	4–5, 25	28	67	14–15, 31, 41	Clock-rate comparison from the histories and proper-time scales of two specified clocks
6	4–5, 8–9, 22, 25	29–30	68	13, 15, 28, 31, 41	Einstein simultaneity from signal correspondence using proper-time scales
7	25, 29–30	31	69	15–16, 32, 41	Tick sequence from specified tick events, a proper-time scale, and simultaneity
8	4–5, 25, 28	32	67	8–9, 17–19, 31, 39, 41	Second-clock-effect criterion from two-path clock histories and rate comparison
9	35–36, 40–41	42–43	35	20–21, 33, 41	Coordinate relation from both particle charts and the message correspondence
10	34–36, 40–41	44–48	31–35, 37–38	20–22, 25, 33, 41	Radar coordinates from both particle charts, light connections and D3
11	8–9, 13–17, 50, 52–53	49, 51	41, 43, 46, 48, 51	7, 28, 35–36, 41	Null-cone neighborhoods from light rays, causal type, a selected local time orientation, and particle coverage
12	54–57	58	70	35–36, 38, 41	Causal type along the path from its initial class and the homogeneous ODE
13	10–11, 13, 18, 48, 54–56	52–53	48, 51–52	4, 36, 41	Timelike projective geodesics and free-fall histories from the P1–P2 particle family and local coordinates

s	Time In i	Time Out i	Claims p	Auxiliary In a	Transfer Relation
14	8–9, 52–53	19–21	23, 58	28, 36–37, 41	Affine parametrization from light and particle geodesics and the Weyl connection

Grounding Conditions

Conditions under which dependence or transfer applies Each transfer holds under its tabulated inputs and claim supports and the following source conditions.

1. pp. 65,68,81. The Weyl proper-time measurement output supplies a measured interval or scale. Its measurability follows the Kundt–Hoffmann-type construction using light, freely falling particles and observer proper time.
2. pp. 72–74;D4,L1–L2,Eq2–5. L1 uses a selected particle coordinate t with $t(e) = 0$. Under the C^2 condition on $g(p) = -t(e_1)t(e_2)$, the local two-ray connections, L2, D3–D4 and the required particle coverage, its Hessian gives the Lorentz conformal form and causal types. The two light-vector classes in L2 and the subsequent local choice of time orientation are distinct conditions.
3. pp. 67–68. Restrict the arc length defined by the constructed Weyl connection to a timelike worldline, and fix its affine freedom by choosing distinct zero and unit events.
4. pp. 68,81. The construction uses the Weyl geometry of light and freely falling particles, a timelike observer worldline and Weyl proper time along it. The source states that the theory of the Kundt–Hoffmann ideal experiment can be expressed in these quantities, with that proper time as its measured quantity.
5. pp. 69,82;Eq29. Compare the specified clock histories and proper-time scales at the comparison events under the stated rule. The criterion for neighboring freely falling clocks uses Einstein simultaneity to first order in their separation.
6. pp. 69,82. Einstein simultaneity is defined from the proper-time scale on the specified worldline and the correspondence of exchanged light signals. The source compares infinitesimally close freely falling worldlines to first order in their separation.
7. pp. 69,82. Map specified clock-tick events to the other worldline by Einstein simultaneity. The correspondence of an equidistant sequence to an approximately equidistant sequence is the first criterion for metric reduction.
8. pp. 69,82. Specify two paths and a reunion event for clocks that initially have equal rates, and compare their rates at reunion. Preservation of equal rates for every permitted pair of paths corresponds to the second criterion, expressed by congruence of parallel-transported vectors.
9. p. 70;D1–D2. Represent the D2 message correspondence in local coordinates selected on both the source and destination particles. Both coordinates are inputs to the resulting smooth coordinate relation.
10. p. 71;D3,Fig5. The two particle charts and light connections define four readings. D3 assumes that a collection of the resulting maps forms an atlas and that

other maps of the same kind are smoothly related. The source presents the sufficiency of C^3 differentiability as tentative.

11. pp. 74–76. Use the constructed conformal causal types, light rays, local particle coverage under P1–P2 and a selected local time orientation. The light cone is a hypersurface in a sufficiently small neighborhood, where its local future and past parts are specified.
12. pp. 78–79;Eq7,11–13,18,20. Along a regular projective geodesic, set $N = \mathfrak{g}_{ab}\dot{x}^a\dot{x}^b$. The condition in Eq. (18) gives $N' = (2\lambda - 2s_c\dot{x}^c)N$. This homogeneous equation preserves the zero value or sign of the supplied initial value.
13. pp. 76–77;P1–P2,Eq6–9. Use P1–P2, coordinates admitted by the radar differentiable structure and coordinate derivatives with respect to an arbitrary parameter u . With the additional C^1 regularity of Π^a_{bc} , free-fall trajectories are represented as projective geodesics.
14. pp. 63–64,67–68,81;Eq25–26. The compatible Weyl connection supplies preferred affine parameters along light and particle geodesics, with freedom under affine transformations. A proper-time scale is specified by arc length and calibration on a timelike curve.

The number of directed cyclic components is zero, and the number of self-input transfers is also zero.

Original-Source Support The conditions are supported by the claim components in Table 143 and the source locations above. Weyl arc length on a general nowhere-null curve is the geometric construction specified by Auxiliary Objects 29, 30 and 37 and Claim 24. Transfer 3 gives its timelike specialization and calibration.

Results

Definition 8.241 (Ehlers–Pirani–Schild 1972 Results). Let $\text{Idx}_R^{\text{audit,result}} = [12]$. Table 147 specifies the results and source locations.

Table 147: Ehlers–Pirani–Schild 1972 Results and Source Locations

j	Result	Source Location
1	Definitions of the geometric hierarchy	pp. 63–67; Fig. 1 (p. 66)
2	The EPS construction route using light rays and freely falling particles as basic tools	pp. 65, 67–68, 70–82
3	Echo, message and radar-coordinate atlas under D1–D3	pp. 70–71; D1–D3, Figs. 4–5
4	Lorentz conformal structure obtained from light rays	pp. 71–77; D4, L1–L2, P1–P2, Eqs. (2)–(6)
5	Projective structure obtained from free-fall trajectories	pp. 76–77; P1–P2, Eqs. (6)–(9)
6	Compatibility of conformal and projective structures	pp. 78–81; C, Eqs. (7), (10)–(25)

j	Result	Source Location
7	The unique compatible Weyl connection	pp. 67, 81; Eqs. (25)–(26)
8	Weyl arc length and calibrated proper time on timelike curves	pp. 67–68
9	Kundt–Hoffmann-type measurability of proper time	pp. 68, 81
10	Criteria for clock-rate comparison, Einstein simultaneity, and the second clock effect	pp. 69, 82; Eq. (29)
11	Integrable Weyl structure and metric uniqueness up to a positive constant factor within a simply connected domain	pp. 69, 81–83; Eqs. (27)–(29)
12	Equality of gravitational and atomic times as an additional empirical postulate	p. 82

Result Dependencies

Definition 8.242 (Ehlers–Pirani–Schild 1972 Result-Dependence Hyperedges). For each $j \in [12]$, Table 148 specifies the time premises, claim support, auxiliary inputs and result output of $\text{ResultEdge}_{R,j}$.

Table 148: Ehlers–Pirani–Schild 1972 Result-Dependence Hyperedges

j	Time Premises i	Claims p	Auxiliary In a	Result Output Index
1	$\emptyset$	7–12	25–26	1
2	8–11	15–18	4, 25–26, 28, 41	2
3	34–41, 44–48	31–38	4, 20–22, 25, 28, 33, 41	3
4	8–11, 13–18, 35–39, 48–51, 54–56	19, 39–48, 51, 73	4, 23, 25, 28, 33–36, 41	4
5	10–11, 13, 18, 48, 52–56	20, 48–53	4, 35–36, 40–41	5
6	10, 13–15, 49–58	21, 54–57, 70	4, 35–36, 38, 41	6
7	13, 15, 52	22–23, 57–58	35–37, 41	7
8	22–25	24–27, 66	8–12, 29–30, 37	8
9	8–11, 22, 25–26	28	4–6, 28, 37, 41	9
10	4–5, 8–9, 22, 25, 28–32	29, 61–62, 67–69	8–9, 13–19, 28, 31–32, 39, 41	10
11	4–5, 25, 28–32	29–30, 59–63, 71	8–9, 13–19, 26, 31–32, 37, 39, 41	11
12	59–62	64	24, 39	12

There are 100 time-premise occurrences. All 12 results use 36 distinct auxiliary objects, with 83 auxiliary-premise occurrences in total. Auxiliary-object indices refer to Table 142.

Grounding Grounds

Grounding grounds of the results Result 1 presents definitions in the geometric hierarchy. Results 2–11 connect the stated local axioms, construction rules, comparison conditions and geometric consequences. Result 12 states equality of gravitational and atomic time as an additional empirical postulate. Claims 13, 14 and 65 about the preceding chronometric route form the comparative context of Result 2; Claim 72 is an additional condition selecting special relativity in the context of Result 11. The source presents a local constructive axiomatization and its consequences, invoking measurability through the Kundt–Hoffmann ideal experiment and consistency of interpretations of observed redshifts.

Evidence Hierarchy

Direct measurement targets This source itself newly measures no target.

Direct measurement predicates This source itself directly establishes no predicate by measured values, calibrated values or error evaluation.

Conditions and domain of direct measurement The source-specific conditions and domain of direct measurement are empty.

Targets of formal generalization Result 1 concerns the geometric hierarchy; Result 2, the EPS construction route; Results 3–5, echo/message/radar construction and conformal and projective structures; Results 6–7, compatibility and the Weyl connection; Results 8–9, Weyl arc length, calibrated proper time and proper-time measurement; and Results 10–11, clock comparison, metric reduction and metric uniqueness. Result 12 states equality of gravitational and atomic time as an additional empirical postulate.

Conditions and domain of formal generalization Table 148 gives the time premises, claim supports and auxiliary inputs of each result.

1. **Construction Conditions** The conformal structure is constructed in four dimensions with signature $(+++ -)$; $\det \mathbf{g} = -1$ normalizes a conformal tensor density of weight $-1/2$. The projective construction uses P1–P2 and the additional C^1 condition on its coefficients. The compatibility proof proceeds through preservation of the initial causal class, local coincidence of null geodesics, the compatibility relation in Eq. (25), and the Weyl connection in Eq. (26).
2. **Uniqueness of the Differentiable Structure** Claim 73 invokes Hawking’s theorem: for two four-dimensional C^3 manifolds with C^2 conformal structures, strong causality on the source manifold and preservation of causal relations by a bijection and its inverse imply that the map is a C^3 diffeomorphism. The source applies this theorem in sufficiently small neighborhoods to the identity between structures on the same event, light-ray and particle carriers.
3. **Clock Comparison and the Metric** The first criterion concerns infinitesimally close freely falling clocks: to first order in separation, Einstein simultaneity maps equidistant sequences to approximately equidistant sequences. The second concerns congruence under parallel transport along every permitted pair of

paths, or equal reunited rates of clocks initially running at equal rates. Under $F = 0$, a compatible metric exists on a simply connected domain and is unique up to a positive constant factor.

4. **Additional Postulate and Curvature Interpretation** Equality of gravitational and atomic time is an additional empirical postulate whose force the source questions. Closed length curvature $dF = 0$ and the conserved current density are the formal consequences in Claim 71. The concluding discussion asks whether a physical interpretation relating them to another universally conserved current might hold.

Generation Judgment Judgment Values: LCTR complete-generation judgment: 0; source-internal evidence-conditioned local-reach judgment: 0; remaining primitive-time roots: 62.

Source-Internal Evidence-Conditioned Local Reach Even with the source formal-evidence environment, local reach includes no time content. Result 1 has no time premise and is evidence-only; Results 2–12 are primitive-time-root-dependent.

Result Dependence The support families in Proposition 8.244 give 136 primitive-time-root–result dependence pairs and 60 necessary-root–result dependence pairs.

Evidence Effect

Index identifier S15-XI-EE1.

Existing evidence reference set Evidence 8.245.

Result-Preservation Domain

Index identifier S15-XI-RPD1.

Existing evidence reference set Evidence 8.252.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The constructions and comparisons use the nonmetric clock coordinates of D1, echo/message order, causal orientation and proper-time scales. The D/L/P/C axioms, local regularity, calibration and integrability support their respective constructions. Generation reach from non-temporal physical Inputs 1–6 reaches none of these time contents, so the placement of the grounding obligation for law-bearing time use is “backgrounded by primitive-time inputs.” Time Content 33 belongs to the Newtonian comparison and has the lexical/example type.

Evidence of Actual Failure

Evidence of Actual Failure Actual failure of a particular construction condition, measurement condition or domain is “indeterminate from the original-source evidence.” The generation-reach judgment concerns physical-time qualification within the audit boundary.

Downstream Reach

Downstream Reach Structural downstream reach follows the dependencies between time contents and Results 2–12. Actual reach from a specific point of actual failure is “indeterminate from the original-source evidence.”

Repair Scope

Repair Scope Repair of physical time use concerns the relevant time content and its generation conditions before use. If an actual failure of a concrete condition is established, recalculation concerns the results dependent on that failure point. Those results are recalculated from repaired inputs or restricted to a domain in which their conditions hold.

Audit Conclusion

Proposition 8.243 (Ehlers–Pirani–Schild 1972 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = & (\emptyset, [62]; \\ & \emptyset, \{1\}; \\ & \llbracket 2, 12 \rrbracket, \emptyset). \end{aligned}$$

Thus $R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. The audit boundary comprises 147 nodes—73 claim nodes, 62 time-content nodes and 12 result nodes—and 26 edges—14 transfers and 12 results. The initial generation set consists of Claims 1–6. Every transfer and result edge requires claim support outside that set, so its closure remains the initial six nodes. With all 73 claims in the source formal-evidence environment, Result 1 is reachable because its time-premise set is empty. All 14 transfers have nonempty time inputs, so no time content is locally reached. Adding all 62 roots reaches Results 2–12, and the closure contains all 147 nodes. Definition 8.115 gives the displayed partition. $\square$

Proposition 8.244 (Ehlers–Pirani–Schild 1972 Minimal Primitive-Time-Root Support). Table 149 gives each result’s minimal primitive-time-root support statistics in the order: number of members, minimum cardinality, maximum cardinality.

Table 149: Ehlers–Pirani–Schild 1972 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(1; 0)
2, 12	(1; 4)
3	(1; 8)
4	(2; 17, 19)
5	(4; 8, 18)
6	(10; 13, 20)

Result Index	Support Statistics
7	(7; 3, 18)
8	(1; 3)
9	(2; 7, 8)
10	(2; 6, 7)
11	(4; 5, 7)

Proof. Applying the update in Definition 7.46 to the 14 transfers leaves all support antichains unchanged on the fourth scan in the listed order. Proposition 7.47 gives 36 support-family members across the 12 results, 136 primitive-time-root–result dependence pairs and 60 necessary-root–result dependence pairs. $\square$

Proposition 8.245 (Ehlers–Pirani–Schild 1972 Physical-Time Qualification). The 61 time contents with law-bearing uses lack physical-time qualification. Results 2–12 depend on time contents lacking that qualification.

Proof. All 62 time contents belong to the primitive-time-root set. Excluding lexical/example Time Content 33, 61 have law-bearing uses. Apply Definition 7.10, Theorem 7.11 and Corollary 7.12 to these roots and Results 2–12. $\square$

Proposition 8.246 (Boundary between the Clock-First and Test-Particle Routes). The chronometric route begins with standard clocks and identifies geometric intervals with measured intervals; the EPS route reconstructs spacetime structure from light rays and freely falling particles. Their inputs and dependency orders differ. EPS time use includes local clock coordinates, signal order, causal orientation, particle parameters and calibrated proper time.

Proof. Claims 13, 14 and 65 concern the chronometric route on source pp. 64–65; Claims 15–18 concern EPS. D1–D3 and Claims 31–38 specify clock coordinates, echoes and messages, while Transfer 3 specifies proper-time calibration. These time contents belong to the root set in Proposition 8.243. $\square$

Proposition 8.247 (Qualification Boundary for Events, Signals, Free Fall and Radar Coordinates). Non-temporal Inputs 1–6 give the event, light-ray and particle carriers and their incidence relations. Physical echo order, particle clock coordinates and the message representation in both particle charts are temporal inputs to the radar construction. The coordinate map formed from four readings uses the D3 atlas condition. Physical use of these time contents requires the physical signal carrier, local records, arrival comparison, and interobserver correspondence to be established.

Proof. Transfer 9 takes Time Contents 35–36, 40–41 as inputs and outputs coordinate relations 42–43. Transfer 10 uses both particle charts, light connections and D3 to output Time Contents 44–48. Proposition 8.244 gives their primitive-time-root supports. $\square$

Proposition 8.248 (Qualification Boundary for Conformal, Projective, Weyl and Metric Reconstruction). Light propagation and free fall determine conformal and projective structures; their compatibility gives the Weyl connection, and $F = 0$ permits metric reduction. These constructions depend on the time contents used at each stage and on the source’s local-domain and regularity conditions.

Proof. Table 148 specifies the time premises and claim supports of Results 3–7 and 11. Claims 55 and 70 concern preservation from the initial causal class, Claim 56 the coincidence of null geodesics, and Claims 57 and 58 the compatibility equation and connection construction. Proposition 8.245 applies to their time premises. $\square$

Proposition 8.249 (Prior-Generation Conditions for Weyl Arc Length, Event Calibration and Proper-Time Measurement). General Weyl arc length is a geometric construction. Restriction to a timelike worldline and calibration by distinct zero and unit events give a proper-time scale. The Kundt–Hoffmann-type construction targets that proper time along the observer worldline for measurement. Physical use of the proper-time scale requires prior generation of the physical identification of the zero and unit events, signal exchange, the clock carrier, reading, recording, and comparison.

Proof. Auxiliary Objects 29, 30 and 37 and Claim 24 specify general arc length. Time Contents 22–25 and Claim 66 specify its timelike specialization and calibration. Transfer 3 gives the proper-time scale, and Transfer 4 its measurement output. Proposition 8.245 determines the qualification of the inputs. $\square$

Proposition 8.250 (Qualification Boundary for Clock-Rate Comparison, Simultaneity, the Second Clock Effect and Integrability). The first criterion for neighboring freely falling clocks concerns correspondence of equidistant sequences under first-order Einstein simultaneity. The equal-rate condition in the second criterion concerns rates at reunion for clocks initially running at equal rates. Both use clock histories and proper-time scales. Metric existence and uniqueness hold under these comparison and domain conditions.

Proof. Transfers 5–8 construct rate comparison, simultaneity, tick sequences and path dependence of reunited rates from clock histories and proper-time scales. Results 10 and 11 and Claims 29, 30 and 61–63 connect the two criteria with a metric on a simply connected domain under $F = 0$. $\square$

Proposition 8.251 (Logical Boundary between the Two Gravitational–Atomic-Time Equality Claims). The equality on source p. 65 belongs to the chronometric postulate with “clock” interpreted as an atomic clock. The equality on p. 82 is an empirical postulate added to the EPS geometric construction. For the latter, the source invokes consistency of atomic-time transport while questioning the force of the postulate.

Proof. Time Contents 6–7 and Claim 65 correspond to the former; Time Contents 59–62, Claim 64 and Result 12 correspond to the latter. The claim and result tables give their source locations and conditions. $\square$

Remark 8.252 (Ehlers–Pirani–Schild 1972 Domain of Use after Generation). The test-particle construction, echo and message operations, radar atlas, Lorentz conformal structure, projective structure, conformal–projective compatibility, Weyl connection, arc length, calibrated proper time, proper-time measurement, clock-rate comparison, Einstein simultaneity, criterion for the second clock effect, integrability and metric uniqueness are used under the local axioms, regularity, calibration and domain conditions of their claims. Use with physical-time qualification requires the time premises of Results 2–12 to be realized by corresponding generated-time representations. Result 1, the geometric hierarchy, has no time premise. Taking events, unparametrized light-ray connections, unparametrized freely falling particle connections and non-temporal echo/message incidences as prior inputs gives a local structural similarity to LCTR. For this source, both the complete judgment and source-internal evidence-conditioned local reach yield zero reachable time contents.

Individual Statements Verbatim original (English):

Not only the measurement of length but also that of time then appears as a derived operation.

The source location is Ehlers, Pirani, and Schild, *The Geometry of Free Fall and Light Propagation*, 2012 reprint, *General Relativity and Gravitation* 44, p. 1590, §1.

Verbatim original (English):

we reject clocks as basic tools for setting up the space-time geometry

The source location is p. 1590, §1, of the same paper, in the paragraph introducing the construction that takes light rays and freely falling particles as inputs.

Verbatim original (English):

it claims the equality of gravitational and atomic time

The source location is p. 1590, §1, of the same paper, in the paragraph explaining the chronometric axiom of the prior construction using standard clocks. The quotation is conditional on the preceding interpretation of “clock” as an atomic clock.

Objects of the Statements The derived-operational status of time measurement, the exclusion of clocks from the basic tools of spacetime geometry, and the equality claim concerning gravitational and atomic time.

Original-Source Locations All three statements occur on p. 1590, §1, of the 2012 reprint.

Minimal Context Required for Interpretation The first and second statements concern the authors’ own EPS route, which reconstructs spacetime structure from light rays and freely falling particles. In the third statement, “it” refers to the chronometric axiom in the preceding account of Synge’s method. The equality interpretation is conditional on interpreting “clock” as an atomic clock.

Attribution of the Statements Ehlers, Pirani, and Schild are responsible for the wording of all three quotations. The first and second describe their own construction; the third gives a conditional interpretation of the chronometric axiom in Synge’s method.

Predicates Assigned to the Objects The first statement describes length and time measurement as derived operations in the EPS route. The second excludes clocks from the basic tools of that construction. The third states that the chronometric axiom claims equality of gravitational and atomic time when “clock” is interpreted as an atomic clock.

Explicit Domains of Application of the Statements All three are theoretical descriptions of methods for constructing spacetime geometry.

Individual Audit

Interpretations Allowed by the Original Text and Minimal Context The first statement describes the derivation of time- and length-measurement operations in the EPS construction from light rays and freely falling particles. The second describes the exclusion of standard clocks from the basic tools of that construction. The construction uses the nonmetric clock coordinates of D1 and the echo/message conditions of D2–D3. The third concerns the preceding chronometric route and states the equality expressed by its chronometric axiom when “clock” is interpreted as an atomic clock.

Independent Grounds The formal grounds are the staged reconstruction of geometric structures from the D/L/P/C axioms. Weyl arc length, zero/unit calibration and the Kundt–Hoffmann-type ideal experiment support proper-time measurement. For atomic time, the source invokes consistency of interpretations of observed redshifts.

Evidence-Effect Boundary under Each Interpretive Possibility The derivation of measurement operations corresponds to measurement-interval and scale output in Transfer 1 and proper-time measurement in Transfer 4; these use calibrated proper time along the observer worldline. In the light-ray and freely falling particle construction, local clock coordinates, echo/message order, causal orientation and particle parameters are temporal inputs. Generation reach from non-temporal physical Inputs 1–6 reaches no time content. The preceding chronometric equality claim is conditional on the atomic-clock interpretation; the equality claim after EPS reconstruction is conditional on an additional empirical postulate.

Corresponding Formal Evidence The first statement corresponds to Claim 28 and Result 9; the second to Claims 15–18 and Result 2, together with Claims 31–38 and Result 3 for D1–D3. The third corresponds to Claims 13, 14 and 65 in the preceding chronometric route. Claim 64 and Result 12 state the additional empirical postulate after reconstruction. Tables 141, 143, 146 and 147 specify the 62 time contents, 73 claims, 14 transfers and 12 results, respectively. Propositions 8.243 and 8.245 determine primitive-time roots and physical-time qualification.

Within-Reference Comparison Source pp. 64–65 contrast the two construction routes, and D1 on p. 70 specifies local clock coordinates in the EPS route. Pages 67–68 give Weyl arc length and proper-time calibration, pp. 68 and 81 give proper-time measurability, and pp. 69 and 82 give clock criteria and metric reduction. The chronometric axiom with the atomic-clock interpretation on p. 65 and the additional empirical postulate on p. 82 belong to different stages of their respective routes.

Evidence-Effect Boundary The evidence effect of the three statements covers geometric constructions and measurement operations under their respective construction routes, local axioms, calibration and interpretive conditions. Generation reach to the time contents they use is determined by the primitive-time-root judgment.

8.19.16 Reference Audit

Audit Inputs

Original Source and Audit Scope

Reference identifier 16.

Reference name Primitive-time-root audit of Hafele–Keating 1972b

Bibliographic identification Let $R = \text{Ref}_{\text{HK72Obs}}$ denote J. C. Hafele and Richard E. Keating, *Around-the-World Atomic Clocks: Observed Relativistic Time Gains*, *Science* 177, 168–170 (14 July 1972). The audit uses all three pages of the original source; its domain comprises the abstract, main text, Figures 1 and 2, Table 1, and Notes 1–11.

The source-location keys in Table 150 correspond to source locations as follows.

- 1 Abstract and introduction
- 2 SI-second paragraph
- 3 Clock-rate instability paragraph
- 4 Figure 1 and difference-record paragraph
- 5 Figure 2 and mean-rate-method paragraph
- 6 Correlated-rate-change-method paragraph
- 7 Table 1 and results paragraph
- 8 Uncertainty paragraph
- 9 Abstract, conclusion, and Notes 1–11

Audit-scope identifier 16-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 16-L1.

Source-location content The audit domain stated in the bibliographic identification and the source-location columns of the respective evidence tables.

Formal Objects The auxiliary indices in Table 150 refer to the objects in Table 151. Physical carriers, statistical and formal operations, reference conditions, and explanatory contexts that are required for time relations but are not themselves time contents are preserved separately. The preceding and subsequent reports specify the order of presentation; 9,192,631,770 in the SI-second definition is an integer constant; and the rate-change counts are event cardinalities indexed by route and clock. These are auxiliary objects. The temporal range of the records is the 636-hour recording interval.

The formal evidence comprises 78 physical and formal input-claim components, 106 time contents, and 28 auxiliary objects corresponding to 44 source-location groups.

Assumptions The mean-rate method uses the mean clock rates before and after flight and assumes a single clock-rate change at the flight midpoint. The correlated-clock-rate-change method uses the independence of individual clock-rate changes and attributes a change to a clock from slope changes of the same size and direction at the same instant in comparisons among three or more clocks. Further premises concern the greater stability of the long-term clock-ensemble mean rate than that of an individual clock, cancellation of common relativistic and environmental effects in internal differences, environmental sensitivity, the statistical limitations of the small sample, and the route-dependent prediction model.

Non-Temporal Physical Inputs The components with positive grounding grounds as non-temporal physical inputs are the four portable cesium-clock devices, the geometries of the eastward and westward routes, the clock-comparison electronic system, temperature, pressure, magnetic shielding, shock testing, clock identity, and physical arrangement of the apparatus. Place the ten components in Table 153 in $\text{Init}_R^{\text{phys}}$.

The source-formal-evidence environment $\text{Env}_R^{\text{phys}}$ comprises all 78 claim nodes. Its first ten nodes form the generation-initial set; the remaining 68 nodes are not included in that set.

Time Contents The 106 time contents concern flight-itinerary events, clock readings, time intervals, clock rates, difference trajectories, measurement instants, the cesium-cycle definition, reference-scale correspondence, fitting intervals, estimated clock-rate histories, statistical outputs, and direction-specific reunion reading differences.

Formal Input Elements

Definition 8.253 (Time Contents and Auxiliary Objects in the Hafele–Keating 1972b Reference). For the reference $R = \text{Ref}_{\text{HK72Obs}}$, divide the 134 semantic objects appearing in the 44 source-location groups of the original source into 106 time contents and 28 auxiliary objects. Table 150 gives the time-content indices, time contents, auxiliary-object indices, and source location for each source-location group. Within each row, the listed names correspond one-to-one, in their order of presentation, to the indices in ascending order.

Table 150: Time contents and source locations in the Hafele–Keating 1972b reference

Location group	Time-content indices	Time contents (index order)	Auxiliary indices	Source-location key
1	1–2	Eastward and westward flight-itinerary events; October 1971 experimental interval	19	1
2	3–5	Flying-clock readings; atomic-time reference scale; relative reading difference	None	1
3	6–7	Eastward reading loss; westward reading gain	1	1

Location group	Time-content indices	Time contents (index order)	Auxiliary indices	Source-location key
4	∅	None	2–3	1
5	8–10	Cesium transition frequency; cumulative duration; SI-second definition	27	2
6	11–12	Clock frequency; elapsed reading	4	2
7	13–14	One-day scale for clock-rate divergence; clock-rate divergence	20	3
8	15–17	Relativistic reading deviation; one-day scale; predicted signal	None	3
9	18–20	Short-term clock-rate fluctuation; semipermanent clock-rate change; clock-rate history	None	3
10	21–22	Clock-rate-change event; day and month intervals	5	3
11	23–25	Individual clock rates; common comparison interval; long-term mean rate of the clock ensemble	None	3
12	26–29	Recording-start event; 0 hours U.T.; 25 September 1971; recording order	None	4
13	30–32	Clock-difference records with measurement instants; 636-hour interval; ordered records	None	4
14	33–34	Internal clock difference; MEAN(USNO) reference	6	4
15	35–36	Flight-itinerary interval; in-flight comparison boundary	21	4
16	37–38	Measurement interval; nanosecond scale	7	4
17	39–41	Duration; measurement instant; trajectory-record order	None	4
18	42–44	Measurement instant; ± 5 -minute uncertainty; instant-position boundary	None	4
19	45–47	Clock-difference trajectory; local slope; instantaneous relative clock rate	None	4
20	48–50	Spontaneous clock-rate change; nearly constant-rate interval; ordering among constant-rate intervals	None	4
21	51–54	Preflight interval; postflight interval; mean clock reading; flight boundary	None	5

Location group	Time-content indices	Time contents (index order)	Auxiliary indices	Source-location key
22	55–56	Mean clock-difference records with instants before and after flight; qualitative judgment of eastward and westward reading differences	8	5
23	57–59	25-hour fitting window; preflight–postflight order; least-squares fitting interval	None	5
24	60–61	Fitted mean clock rate over the 25 hours before flight; Fitted mean clock rate over the 25 hours after flight	None	5
25	62–64	Single clock-rate-change event; flight midpoint; assumed instant	None	5
26	65–67	Arithmetic mean of the preflight and postflight fitted clock rates; flight duration; estimated in-flight reading deviation	None	5
27	68–70	Sequential clock readings; pairwise clock difference; recording order	None	6
28	71–73	Trajectory-slope change; change instant; clock-rate transition	None	6
29	74–75	Coincidence at the same instant; change in the same direction	9	6
30	76–78	Attribution of clock-rate change; flight-itinerary interval; reconstructed clock-rate history	None	6
31	79	common relativistic reading deviation	10, 22	6
32	80–82	Regular-recording interval; preflight, in-flight, and postflight phases; complete recording order	11	7
33	83–85	Piecewise difference trajectory; MEAN(USNO) endpoint; extrapolation parameter	None	7
34	86	Final clock-specific indication offsets from the correlated-rate-change method	23–24	7

Location group	Time-content indices	Time contents (index order)	Auxiliary indices	Source-location key
35	87–90	Route-specific four-tuples of indication offsets indexed by clock identity; Source-reported mean indication offsets indexed by route; Source-reported standard deviations indexed by route; Source-reported mean and standard-deviation pairs indexed by route	None	7
36	91–92	Predicted reading difference; observed reading difference	12	7
37	93–95	Westward reading deviation; Source-reported westward mean indication offset; standard deviation / mean	None	7
38	96–98	Clock-rate perturbation; 0.4-nanosecond-per-hour scale; sensitivity-evaluation input	None	8
39	99–101	Clock-rate uncertainty; flight duration; error product	None	8
40	102	Clock rate	13–14	8
41	103	shared environmental history	15, 25	8
42	∅	None	26, 28	9
43	104	Direction-specific observation	16–17	9
44	105–106	Departure–reunion order; returning-clock reading	18	9

Table 151: Auxiliary objects in the Hafele–Keating 1972b reference

<i>a</i>	Auxiliary object	Location group	Supporting claims
1	Attribution after reunion	3	{14, 15, 75, 76}
2	Prediction report	4	{78}
3	Observation report	4	{78}
4	Clock-frequency control by the cesium transition frequency	6	[[16, 19]]
5	Persistence of quasi-permanent clock-rate changes	10	{22}
6	Reference-scale correspondence	14	{13, 27, 28}
7	Measurement to the nearest nanosecond	16	{4, 29}
8	Direction sign	22	{14, 15}

a	Auxiliary object	Location group	Supporting claims
9	Comparison of slope changes of the same size and direction at the same instant	29	$\llbracket 41, 46 \rrbracket \cup \{23\}$
10	Common-mode relation	31	$\{47, 48\}$
11	Internal-comparison recording intervals and Figure 1 display intervals	32	$\{77\}$
12	Final comparison	36	$\llbracket 63, 66 \rrbracket$
13	Environmental input	40	$\llbracket 5, 8 \rrbracket \cup \{71\}$
14	Systematic correspondence	40	$\llbracket 71, 73 \rrbracket \cup \{49, 50\}$
15	Common-mode systematic component	41	$\{49, 50, 72, 73\}$
16	Prediction comparison	43	$\llbracket 63, 66 \rrbracket$
17	Conclusion on agreement between predictions and clock records	43	$\llbracket 63, 66 \rrbracket \cup \{75, 76\}$
18	Reading inequality	44	$\{14, 15, 75, 76\}$
19	Device identity of the flying clocks	1	$\{1, 9\}$
20	Colocation of the clock devices	7	$\{10\}$
21	Unavailability of external comparisons during flight	15	$\{28\}$
22	Composition of the flying-clock ensemble	31	$\{1\}$
23	Device identities of the four clocks	34	$\{1, 9\}$
24	Spatial directions of the eastward and westward routes	34	$\{2, 3\}$
25	Device identities of the four clocks sharing the environment	41	$\{1, 9\}$
26	Selection of the flight itinerary for event counting	42	$\{11\}$
27	The integer constant 9,192,631,770 in the SI-second definition	5	$\llbracket 16, 19 \rrbracket$
28	Route- and clock-indexed counts of significant rate changes	42	$\{74\}$

Definition 8.254 (Physical and Formal Input-Claim Components in the Hafele–Keating 1972b Reference). Let the sequence of physical and formal input-claim components be $(\text{ClaimVal}_{R,p}^{\text{audit,phys}})_{p \in [78]}$. Define each component, original-source statement group, minimal formal claim, dependent object, and source location by Table 152. Components 1–10 correspond to non-temporal physical inputs; Components 11–78 belong to the source-formal evidence.

The dependent-object keys in Table 152 correspond to sets of time-content indices and auxiliary-object indices as follows.

- 1** $(\emptyset, \emptyset)$
- 2** $([1, 7] \cup \{105, 106\}, \{1, 18, 19\})$
- 3** $([8, 12], \{4, 27\})$
- 4** $([26, 47] \cup [68, 75], \{6, 7, 9, 21\})$
- 5** $([26, 36], \{6, 21\})$
- 6** $([30, 34] \cup [37, 47] \cup [68, 75], \{6, 7, 9\})$
- 7** $([57, 67], \emptyset)$
- 8** $([68, 78], \{9\})$
- 9** $(\{79, 102, 103\}, [13, 15] \cup \{10, 22, 25\})$
- 10** $(\{86\}, \{23, 24\})$
- 11** $([86, 90], \{23, 24\})$
- 12** $([86, 90] \cup [93, 95], \{23, 24\})$
- 13** $([104, 106] \cup \{6, 7, 91, 92\}, [16, 18] \cup \{1, 12\})$
- 14** $([87, 90] \cup [93, 103], [13, 15] \cup \{25\})$
- 15** $([87, 90] \cup [93, 103] \cup \{79\}, [13, 15] \cup \{10, 22, 25\})$
- 16** $([1, 5] \cup [104, 106], [16, 19])$
- 17** $([80, 82] \cup \{35, 36\}, \{11\})$
- 18** $(\emptyset, \{2, 3\})$
- 19** $(\{14\}, \{20\})$
- 20** $(\{18\}, \emptyset)$
- 21** $([19, 22], \{5\})$
- 22** $(\{19, 21\}, \emptyset)$
- 23** $([23, 25], \{22\})$
- 24** $([48, 50] \cup \{31\}, \{23\})$
- 25** $([68, 70] \cup [76, 78] \cup [83, 85] \cup \{33, 34\}, \{6, 23, 24\})$
- 26** $(\emptyset, \{26, 28\})$

Table 152: Physical and formal input-claim components
in the Hafele–Keating 1972b reference

Com- po- nent	Original- source statement group	Minimal formal claim	Dependent- object key	Source- location key
1	1	Four portable cesium-clock devices	1	1
2	1	Geometry of the eastward route	1	1
3	1	Geometry of the westward route	1	1
4	13	Clock-comparison electronic system	1	4
5	41	Temperature condition	1	8
6	41	Pressure condition	1	8
7	41	Magnetic-shielding condition	1	8
8	41	Shock-test condition	1	8
9	27–30	Assignment of clock identifiers	1	7
10	All three pages	Physical arrangement of the apparatus	1	1–9
11	1	Eastward and westward flight operations	2	1
12	2	Display of the flying clocks	2	1
13	2	Comparison with MEAN(USNO)	2	1
14	3	Loss in the eastward display	2	1
15	3	Gain in the westward display	2	1
16	4	Definition of the SI second	3	2
17	4	Content of the cesium-133 transition period	3	2
18	5	Frequency control by the cesium transition frequency and approximation to an ideal standard clock	3	2
19	5	Realization of the unit	3	2
20	6	Systematic relative clock-rate deviation	19	3
21	7	Shot noise in clock rate	20	3
22	7	Quasi-permanent clock-rate changes and intervals between changes	21	3
23	8	Independence assumption for individual clock-rate changes	22	3
24	9	Stability of the mean rate of the clock ensemble	23	3
25	10	More than 5,000 difference records	4	4
26	10	636-hour recording range	4	4
27	11	Local atomic time scale MEAN(USNO), constructed from clock data	5	9
28	12	External-comparison boundaries before and after flight	5	4

Com- po- nent	Original- source statement group	Minimal formal claim	Dependent- object key	Source- location key
29	13	Measurement to the nearest nanosecond with an electronic time-interval counter	6	4
30	14	Slope of the difference trajectory	6	4
31	14	Instantaneous relative clock rate	6	4
32	15	Clock-rate changes in all flying clocks	24	4
33	16	25-hour fitting intervals before and after flight	7	5
34	16	Least-squares fitting	7	5
35	17	Assumption of a single midpoint clock-rate change	7	5
36	17	Mean-rate method	7	5
37	18	Eastward estimate of -66 ns	7	5
38	18	Westward estimate of $+205$ ns	7	5
39	19	Midpoint assumption	7	5
40	19	Boundary of the error evaluation	7	5
41	20	Slope changes of the same size and direction at the same instant among three or more clocks	8	6
42	20	Inductive attribution of a rate change to the clock common to the comparisons	8	6
43	21	Difference between two clocks	8	6
44	21	Indeterminacy of the source of change	8	6
45	22	Attribution redundancy using three or more clocks	8	6
46	23	Correlation of observed difference trajectories	8	6
47	24	Common relativistic reading deviation	9	6
48	24	Cancellation in internal differences	9	6
49	25	Common environmental effect	9	8
50	25	Cancellation of common environmental effects	9	8
51	26	Piecewise extrapolation	25	7
52	26	Clock-specific values relative to the reference scale	10	7
53	27	Final reading deviation of Clock 120	10	7
54	28	Final reading deviation of Clock 361	10	7
55	29	Final reading deviation of Clock 408	10	7
56	30	Final reading deviation of Clock 447	10	7
57	31	Reported eastward mean	11	7
58	31	Reported display value of the eastward mean	11	7
59	32	Reported westward mean	11	7

Com- po- nent	Original- source statement group	Minimal formal claim	Dependent- object key	Source- location key
60	32	Reported display value of the westward mean	11	7
61	33	Reported eastward standard deviation	11	7
62	34	Reported westward standard deviation	12	7
63	36	Predicted net value	13	7
64	36	Comparison with the observed mean	13	7
65	37	Route-dependent predicted values	13	7
66	37	Agreement between prediction and observation	13	7
67	38	Strength boundary of the four-value statistics	14	8
68	39	Clock-rate perturbation of 0.4 ns/h	14	8
69	39	Sensitivity evaluation	14	8
70	40	Conversion to a width below 30 ns	14	8
71	41	Scope of tests involving temperature and pressure variations, shocks, and magnetic-field orientation	14	8
72	42	Shared environmental history	15	8
73	42	Non-detection by internal comparisons of systematic effects common to all flying clocks	15	8
74	43	Number of clock-rate changes by clock	26	9
75	44	Direction-specific inequality of returning-clock displays	16	9
76	45	Increase–decrease vocabulary as display-clock-rate comparison	16	9
77	10	Regular internal comparisons before, during, and after flight, and hourly display in Figure 1	17	4, 7
78	1	Predictions in the preceding report and observations and extraction methods in this report	18	1

Definition 8.255 (Semantic Components and Generation-Initial Set in the Hafele–Keating 1972b Reference). The 78 physical and formal input-claim components correspond one-to-one to 78 semantic components and 78 claim nodes with the same indices. The generation-initial set $\text{Init}_R^{\text{phys}}$ comprises the first ten claim nodes; the source-formal-evidence environment $\text{Env}_R^{\text{phys}}$ comprises all 78 claim nodes. The values of the generation-initial components are given in Table 153. The conditions associated with auxiliary objects are retained in the supporting claims listed in Table 151 and included in the claim-support sets of the transfer and result tables. Time-content premises remain in the time-input columns.

Table 153: Generation-initial semantic components in the Hafele–Keating 1972b reference

k	Semantic component	Corresponding claim index
1	Four portable cesium-clock devices	$\{1\}$
2	Geometry of the eastward route	$\{2\}$
3	Geometry of the westward route	$\{3\}$
4	Clock-comparison electronic system	$\{4\}$
5	Temperature condition	$\{5\}$
6	Pressure condition	$\{6\}$
7	Magnetic-shielding condition	$\{7\}$
8	Shock-test condition	$\{8\}$
9	Assignment of clock identifiers	$\{9\}$
10	Physical arrangement of the apparatus	$\{10\}$

Dependent Inputs

Dependent Inputs The flight itineraries, clock readings, clock rates, difference trajectories, measurement instants, fitting intervals, estimated clock-rate histories, direction-specific reading deviations, ensemble statistics, uncertainties, and reunion comparison enter the thirteen transfers as inputs.

Dependence and Transfer Elements

Definition 8.256 (Time-Content Transfers in the Hafele–Keating 1972b Reference). Define the transfer-index set by $\text{Idx}_R^{\text{audit,transfer}} = [13]$. Define the time-content inputs, time-content outputs, auxiliary-object support, and supporting claims of each transfer by the equalities in Table 154. The supporting claims include the conditions associated with the auxiliary objects. The source location of each transfer is the union of the source locations corresponding to its time-content inputs and time-content outputs.

Table 154: Time-content transfers in the Hafele–Keating 1972b reference

s	Time inputs	Time outputs	Auxiliary indices	Supporting claims
1	$\{88\}$	$\{6, 7\}$	$\{1, 23, 24\}$	$\llbracket 1, 3 \rrbracket \cup \{9, 14, 15, 57, 59, 75, 76\}$
2	$\{23, 24\}$	$\{25\}$	$\{22\}$	$\{1, 23, 24\}$
3	$\llbracket 39, 41 \rrbracket \cup \{30, 34\}$	$\llbracket 45, 47 \rrbracket$	$\{23\}$	$\llbracket 29, 31 \rrbracket \cup \{1, 4, 9, 25\}$
4	$\llbracket 51, 55 \rrbracket$	$\{56\}$	$\{8\}$	$\{14, 15\}$
5	$\llbracket 57, 59 \rrbracket \cup \{55\}$	$\{60, 61\}$	$\emptyset$	$\{33, 34\}$
6	$\llbracket 51, 55 \rrbracket \cup \llbracket 60, 64 \rrbracket \cup \{66\}$	$\{65, 67\}$	$\{24\}$	$\llbracket 33, 40 \rrbracket \cup \{2, 3\}$
7	$\llbracket 68, 75 \rrbracket \cup \{77\}$	$\{76, 78\}$	$\{9\}$	$\llbracket 41, 46 \rrbracket \cup \{23\}$
8	$\llbracket 68, 70 \rrbracket \cup \llbracket 76, 78 \rrbracket \cup \llbracket 83, 85 \rrbracket \cup \{33, 34\}$	$\{86, 87\}$	$\{6, 23, 24\}$	$\llbracket 1, 3 \rrbracket \cup \{9, 13, 23, 27, 28, 41, 42, 51, 52\}$

s	Time inputs	Time outputs	Auxiliary indices	Supporting claims
9	$\{87\}$	$\{88, 94\}$	$\{23, 24\}$	$\llbracket 1, 3 \rrbracket \cup \llbracket 53, 60 \rrbracket \cup \{9\}$
10	$\{89, 94\}$	$\{95\}$	$\{24\}$	$\{2, 3, 59, 60, 62, 67\}$
11	$\llbracket 96, 98 \rrbracket \cup \{100\}$	$\{99, 101\}$	$\emptyset$	$\llbracket 68, 70 \rrbracket$
12	$\llbracket 1, 4 \rrbracket \cup \{105\}$	$\{5, 106\}$	$\{6, 19\}$	$\llbracket 11, 13 \rrbracket \cup \{1, 9, 27, 28\}$
13	$\{6, 7\}$	$\{104\}$	$\{1, 24\}$	$\{2, 3, 14, 15, 75, 76\}$

Transfer Events and Outputs Attribution after reunion in the eastward and westward directions, the long-term ensemble mean rate, instantaneous relative clock rate, qualitative eastward–westward difference, mean clock rate, estimated in-flight reading deviation, reconstructed clock-rate history, final reading deviation, clock-ensemble mean, standard deviation / mean, error product, returning-clock reading, and time-record conclusion are obtained from the input–output relations in Table 154.

Conditions of Establishment

Conditions under Which the Dependence or Transfer Applies Each transfer applies within the range in which the input time contents in the table hold and the corresponding conditions of the original source’s observational operation, fitting, correlation-based attribution, statistical operation, or prediction comparison are satisfied. All thirteen transfers take one or more existing time contents as inputs. The number of transfers that output time contents from non-temporal physical inputs alone is zero.

Original-Source Support Take the supporting-claim indices, time-content input indices, and time-content output indices in Table 154 as the corresponding index sets in Definition 7.105. The table includes the claims supporting the auxiliary conditions in each claim-support set; the definition determines the support and time-input/output sets. The time-content transfer relation determined by the thirteen transfers is acyclic.

Original-Source Support for Each Dependence and Transfer The source locations corresponding to the time-content inputs and outputs of each transfer are the union of the source locations in Definition 8.253.

Results

Definition 8.257 (Results of the Hafele–Keating 1972b Reference). Define the result-index set by $\text{ldx}_R^{\text{audit,result}} = [12]$. The correspondence between the results and their source locations is given in Table 155.

Table 155: Results and source locations in the Hafele–Keating 1972b reference

j	Result	Source location
1	Eastward and westward flight experiment with four clocks	Abstract and introduction
2	Separation of the SI-second definition from its realization by commercial clocks	SI-second paragraph
3	Role of MEAN(USNO) as an atomic reference time scale constructed from clock data	Figure 1 and difference-record paragraph
4	More than 5,000 clock-reading-difference records	Figure 1 and difference-record paragraph
5	Boundary of the mean-rate method and midpoint clock-rate-change assumption	Figure 2 and mean-rate-method paragraph
6	Correlated-clock-rate-change method and inductive attribution using three or more clocks	Correlated-clock-rate-change-method paragraph
7	Individual eastward and westward reading deviations of the four clocks	Table 1
8	Reported means and standard deviations for the four clocks	Table 1
9	Elimination of the common mode in internal differences	Correlated-clock-rate-change-method paragraph
10	Comparison of direction-specific net predicted and observed values	Table 1 and prediction-comparison paragraph
11	Boundary imposed by the small sample and common-mode uncertainty	Uncertainty paragraph
12	Direction-specific inequality of returning-clock readings	Abstract and conclusion

Reference-Specific Results The twelve results in Definition 8.257. The source location, time-content prerequisites, auxiliary-object prerequisites, and supporting claims of each result are given by the tables in that definition and Definition 8.258.

Result Dependence

Definition 8.258 (Result-Dependency Hyperedges in the Hafele–Keating 1972b Reference). For each $q \in [12]$, define the time-content inputs, auxiliary-object inputs, and physical and formal input-claim support of the result-dependency hyperedge $\text{ResultEdge}_{R,q}$ by the equalities in Table 156, and take its result output to be Result q .

Table 156: Result-dependency hyperedges in the Hafele–Keating 1972b reference

q	Time-content input indices	Auxiliary-object input indices	Supporting-claim indices
1	$\llbracket 1, 7 \rrbracket \cup \{105, 106\}$	$\{1, 18, 19\}$	$\llbracket 1, 3 \rrbracket \cup \llbracket 11, 15 \rrbracket \cup \{9, 75, 76\}$
2	$\llbracket 8, 12 \rrbracket$	$\{4, 27\}$	$\llbracket 16, 19 \rrbracket$
3	$\{25, 33, 34\}$	$\{6\}$	$\{13, 24, 27, 28\}$
4	$\llbracket 26, 44 \rrbracket$	$\{6, 7, 21\}$	$\llbracket 25, 29 \rrbracket \cup \{4, 13\}$
5	$\llbracket 51, 55 \rrbracket \cup \llbracket 57, 67 \rrbracket$	$\emptyset$	$\llbracket 33, 40 \rrbracket$
6	$\llbracket 68, 78 \rrbracket$	$\{9\}$	$\llbracket 41, 46 \rrbracket \cup \{23\}$
7	$\{86\}$	$\{23, 24\}$	$\llbracket 1, 3 \rrbracket \cup \llbracket 53, 56 \rrbracket \cup \{9\}$
8	$\llbracket 86, 90 \rrbracket$	$\{23, 24\}$	$\llbracket 1, 3 \rrbracket \cup \llbracket 57, 62 \rrbracket \cup \{9\}$
9	$\{79, 102, 103\}$	$\llbracket 13, 15 \rrbracket \cup \{10, 22, 25\}$	$\llbracket 5, 9 \rrbracket \cup \llbracket 47, 50 \rrbracket \cup \llbracket 71, 73 \rrbracket \cup \{1\}$
10	$\llbracket 88, 92 \rrbracket$	$\{12, 16\}$	$\llbracket 63, 66 \rrbracket$
11	$\llbracket 87, 90 \rrbracket \cup \llbracket 93, 103 \rrbracket$	$\llbracket 13, 15 \rrbracket \cup \{25\}$	$\llbracket 5, 9 \rrbracket \cup \llbracket 67, 73 \rrbracket \cup \{1, 49, 50\}$
12	$\{6, 7, 105, 106\}$	$\{1, 18, 23, 24\}$	$\llbracket 1, 3 \rrbracket \cup \{9, 14, 15, 75, 76\}$

Take the supporting-claim indices and time-content input indices in Table 156 as the corresponding index sets in Definition 7.105. The claim-support sets include the conditions associated with the auxiliary objects. There are 126 occurrences of time-content and auxiliary-object premises, and all twelve results have at least one time-content premise.

Grounding Grounds

Grounding Grounds of the Results The experimental and statistical grounds comprise the eastward and westward flights of four portable cesium clocks, comparison of their readings with MEAN(USNO), more than 5,000 pairwise clock-difference records, least-squares fitting before and after flight, correlated-clock-rate-change analysis, the individual values in Table 1, clock-ensemble statistics, and environmental-sensitivity evaluation.

Evidence Hierarchy

Direct Measurement Targets The readings of the four flying cesium clocks, their reading differences relative to MEAN(USNO), and the reading differences among the flying clocks.

Direct Measurement Predicates How much the readings of the compared clocks differ at each recorded measurement time.

Conditions and Domain of Direct Measurement The domain comprises the eastward and westward flights of the four clocks, comparisons with MEAN(USNO) before

and after flight, internal comparisons before, during, and after flight, and more than 5,000 difference records over 636 hours. The differences were measured to the nearest nanosecond with an electronic time-interval counter, with measurement times uncertain by about ± 5 minutes.

Targets of Formal Generalization Relative clock rates and their change histories are inferred from pairwise clock-difference trajectories. Extrapolation anchored by comparisons before and after flight yields the final direction-specific reading deviations, and the clock-ensemble means are compared with route-dependent relativistic predictions. The aging of human or nonhuman organisms is neither a direct measurement target nor a target of formal generalization in this reference.

Conditions and Domain of Formal Generalization The mean-rate method uses 25-hour fits before and after flight and assumes a single clock-rate change at the midpoint. The correlated-clock-rate-change method uses the independence of individual clock-rate changes and inductive attribution from slope changes of the same size and direction at the same instant among three or more clocks. Piecewise extrapolation accounting for the attributed changes yields the values in Table 1. The domain is fixed by the flight conditions shared by the four clocks, comparisons with MEAN(USNO) before and after flight, common-mode cancellation in internal differences, clock-rate sensitivity of 0.4 ns/h, environmental-test conditions, the statistical limitations of the small sample, and the route-dependent prediction model. All twelve results use one or more time contents as dependent inputs.

Generation Judgment Judgment values: Time contents generated under the complete LCTR judgment: 0; time contents reached under source-internal evidence conditions: 0; remaining primitive time roots: 106.

Generation Reach and Primitive Time Roots The complete judgment, with non-temporal physical inputs alone as its initial set, generates zero time contents. All 106 time contents in Definition 8.253 remain primitive time roots. The local-reach judgment reaches zero time contents.

Source-Internal Evidence-Conditioned Local Reach The source-internal evidence-conditioned local-reach judgment reaches zero time contents; 106 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ from Definition 7.108 and apply Definition 7.50 to it. Totalling over all results determines the primitive-time-root–result dependence relations and the necessary-primitive-time-root–result dependence relations.

Result Dependence All twelve results in Definition 8.257 are primitive-time-root-dependent results. The numbers of generated results, evidence-reach-only results, and unreached-after-root-addition results are zero. The minimal support of each result is given by Proposition 8.260.

Evidence Effect

Evidence-Effect Scope The formal evidence supports comparisons of readings between the four portable cesium clocks and MEAN(USNO), eastward reading loss, westward reading gain, relative clock-rate changes, pairwise clock correlation, clock-ensemble statistics, uncertainty evaluation, and comparison with route-dependent predictions.

Evidence-Effect Boundary Directly recorded clock readings and reading differences are distinguished from estimates obtained by fitting, change attribution, extrapolation, and statistical analysis. That experimental success neither implies that the 106 time contents have been generated from non-temporal physical inputs nor retroactively confers physical-time qualification on the primitive time roots. It also establishes neither a time attribute independent of physical carriers, nor the aging of human or nonhuman organisms, nor direct measurement of either.

Result-Preservation Domain

Result-Preservation Domain The twelve results are preserved insofar as every time-content premise is supplied by the corresponding generated time representation in the LCTR generation sequence, while preserving the conditions required by each result among the four clocks, two routes, comparison with MEAN(USNO), difference records, least-squares fitting, attribution of clock-rate changes, common-mode cancellation, environmental boundaries, statistical limitations of the small sample, and the route-dependent prediction model.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The 106 time contents audited in this reference are not generated from non-temporal physical inputs before use; they are accepted as already-established time inputs. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment follows by matching the use of each time content in a law operation or result derivation against the absence, before that use, of the corresponding generation or validity condition as an input, formation condition, evaluation condition, or failure condition.

Evidence of Actual Failure

Evidence of Actual Failure The complete judgment and the remaining-root judgment concern physical-time qualification; they are distinct from actual failure of a concrete condition governing a result’s value or domain of application. Neither the original formal evidence nor the LCTR audit evidence in this reference establishes an actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence,” and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream Reach For this reference, the reach to primitive-time-root-dependent results is determined by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because actual failure of a concrete condition has not been established, the actual-reach judgment from that failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the objects of repair are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is limited to results for which actual failure is independently established and which are dependency-reachable from that failure location. Because actual failure is indeterminate in this reference, this repair-and-recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.259 (Primitive Time Roots and Result Dependence in the Hafele–Keating 1972b Reference). In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepldx}(106, 12)$. Moreover, $R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. The audit boundary has 196 nodes, comprising 78 semantic-component nodes, 106 time-content nodes, and 12 result nodes, and 25 hyperedges, comprising 13 transfer hyperedges and 12 result-dependency hyperedges. Take the generation-initial semantic-component set $\text{Init}_R^{\text{phys}}$ as the starting point of generation reach. Its generation-initial closure reaches the ten semantic-component nodes but reaches no time content or result. Taking the source-formal-evidence environment $\text{Env}_R^{\text{phys}}$ as the starting point of evidence-conditioned reach reaches all 78 semantic-component nodes but no time content or result. All thirteen transfers in Definition 8.256 have one or more time contents as inputs. By Proposition 7.43, the 106 time contents are not reached from non-temporal physical inputs. Each of the twelve result-dependency hyperedges in Table 156 also presupposes at least one primitive time root. Adding the source-formal-evidence environment and the 106 primitive time roots to the initial set reaches all twelve results. The common reconstruction equations in Proposition 7.49 and Definition 8.115 then yield the displayed index tuple. Membership in the reference sets follows from Definition 7.52. $\square$

Proposition 8.260 (Minimal Primitive-Time-Root Supports in the Hafele–Keating 1972b Reference). For each $q \in \text{Idx}_R^{\text{audit,result}}$, the number of elements of $\mathcal{A}_R(q)$ from Definition 7.108, together with the minimum and maximum cardinalities of those elements, is given in Table 157.

Table 157: Minimal primitive-time-root supports in the Hafele–Keating 1972b reference

Result index	Support statistics	Result index	Support statistics
1	(5; 6, 19)	2	(1; 5)
3	(2; 3, 4)	4	(1; 19)
5	(1; 12)	6	(1; 9)
7	(3; 1, 14)	8	(3; 4, 16)
9	(1; 3)	10	(4; 5, 18)
11	(3; 10, 23)	12	(10; 3, 19)

Proof. Apply the update in Definition 7.46 to the thirteen transfer hyperedges in Definition 8.256. After two scans through all thirteen transfers in their listed order, the support antichains of all time-content nodes become invariant on the third scan. By Proposition 7.47, this yields the values corresponding to the time-content prerequisites in Table 155. The total number of members of the minimal primitive-time-root support families is 35. $\square$

Proposition 8.261 (Audit Conclusion for the Hafele–Keating 1972b Reference). The complete judgment generates zero time contents, and 106 primitive time roots remain. The local-reach judgment reaches zero time contents. All twelve reference-specific results depend on primitive time roots, and all twelve are preserved after connection to the LCTR generation sequence.

Proof. The reach classification of time contents and results follows from Proposition 8.259. The minimal primitive-time-root supports of the respective results are given by Proposition 8.260. Generating the 106 time contents through the LCTR generation sequence and assigning qualification and a domain of application at each point of use satisfies the dependent inputs of all twelve results. $\square$

Individual Statements

Objects of the Statements Empirical resolution of the clock paradox with macroscopic clocks; the relation between experimental facts and theoretical arguments; unpredictable rate changes in real clocks; selection of an elapsed-time axis; agreement between predicted and measured values; and statistical constraints imposed by a small sample.

Verbatim Original (English) These results provide an unambiguous empirical resolution of the famous clock "paradox" with macroscopic clocks.

In science, relevant experimental facts supersede theoretical arguments.

These unpredictable changes in rate produce the major uncertainty in our results.

The axis of abscissas in Fig. 1 is the running time for the data period, and, because at any given instant all clocks involved read the same time to within much better than the nearest second, it is immaterial whether the running time is the time of MEAN(USNO) or any member of the flying ensemble.

The agreement between the mean of the measured values and the predicted values in Table 1 is very satisfactory.

However, the number of measured values is too small for a good statistical analysis.

Original-Source Locations The first quotation is from the abstract of Hafele–Keating, *Around-the-World Atomic Clocks: Observed Relativistic Time Gains*, *Science* 177, p. 168. The second is from the opening paragraph on p. 168; the third, from the clock-stability paragraph on p. 168; the fourth, from the paragraph explaining Figure 1 on p. 169; and the fifth and sixth, from the discussion following Table 1 on p. 170.

Minimal Context The first and second quotations belong to the abstract and introduction of the experimental report in which four portable cesium clocks were flown eastward and westward and their reading differences after return relative to MEAN(USNO) were compared with relativistic predictions. The third belongs to the explanation that rates of real clocks change irregularly, requiring analysis of difference data and evaluation of uncertainty. The fourth belongs to the explanation of Figure 1, which plots pairwise clock differences against the elapsed time of the data period under conditions in which continuous comparison with an external reference during flight was unavailable. The fifth and sixth belong to the same discussion: immediately after evaluating agreement between the observed mean and the predictions in Table 1, the text states the limitation imposed by the number of samples.

Attribution All six statements are attributable to the abstract or main text by J. C. Hafele and Richard E. Keating.

Contents Directly Stated by the Quotations The first states an empirical resolution of the clock paradox with macroscopic clocks. The second states that relevant experimental facts supersede theoretical arguments. The third identifies unpredictable clock-rate changes as the major uncertainty. The fourth states that either MEAN(USNO) time or the time of any clock in the flying ensemble may serve as the elapsed-time axis. The fifth evaluates agreement between the measured mean and the predicted values; the sixth limits that evaluation by stating that the number of measured values is too small for a good statistical analysis.

Explicit Domain of Application The explicit domain of the six quotations comprises the four portable cesium clocks; eastward and westward flight; comparison with MEAN(USNO); pairwise clock-difference data; clock-rate-change analysis; the measured and predicted values in Table 1; and the statistical limitations of four samples.

Evidence Provenance The quotations are from the published article in *Science* 177, pp. 168–170.

Individual Audit

Interpretive Possibilities The first statement describes agreement between relativistic predictions and the post-reunion reading differences of macroscopic clocks flown

eastward and westward relative to the ground reference as an empirical resolution of the clock paradox. The second is a methodological claim that experimental facts take precedence over theoretical arguments. The third is a measurement-theoretic explanation that treats unpredictable rate changes in real clocks as the principal uncertainty in the results. The fourth states a data-processing convention: under the condition that clock readings agree to much better than one second, either MEAN(USNO) time or the time of a clock in the flying ensemble may be selected as the elapsed-time axis. The fifth states agreement between the measured mean and the predicted values; the sixth limits that statement by noting that the number of samples is insufficient for a good statistical analysis.

Independent Grounds The four portable cesium clocks, eastward and westward flights, their reading differences after reunion relative to MEAN(USNO), clock-rate changes, uncertainty evaluation, and comparison with predicted values provide independent experimental grounds for the display differences of macroscopic clocks.

Evidence-Effect Boundary The experimental object of the first statement is the post-reunion reading difference of macroscopic clocks. The second states the authors' methodological position on the relation between experimental facts and theoretical arguments. The third identifies the principal source of uncertainty; the fourth states the conditions under which the elapsed-time axes in Figure 1 are interchangeable. The agreement in the fifth statement is limited to the original source's apparatus, flight routes, corrections, prediction model, and uncertainties. The sixth states that the number of measured values is insufficient for a good statistical analysis.

Corresponding Formal Evidence The time-content, claim-component, and result indices below refer, respectively, to Definition 8.253, Definition 8.254, and Definition 8.257. The first statement corresponds to Time Contents 1–7 and 105–106, Claim Components 1–3, 9, 11–15, and 75–76, and the post-reunion reading differences of macroscopic clocks in Results 1 and 12. The second is a methodological statement concerning the relation between experimental facts and theoretical arguments; it constructs no individual time content or formal claim. The third corresponds to Time Contents 18–22 and Claim Components 21 and 22, and concerns the explanation of the source of uncertainty in Result 11. The fourth corresponds to Time Contents 34 and 39–44 and the conditions for selecting the elapsed-time axis in Results 3 and 4. The fifth corresponds to Time Contents 88 and 91–92, Claim Components 63–66, and the comparison between the observed mean and predicted values in Result 10. The sixth corresponds to Time Contents 87–90, Claim Component 67, and the statistical limitation imposed by the small sample in Result 11.

Within-Reference Comparison The first and second statements assert the empirical resolution of the clock paradox and the priority of experimental facts; the third and fourth state the uncertainty due to clock-rate changes and the conditions for selecting the elapsed-time axis. The fifth states agreement between the measured mean and the predicted values; the sixth states the insufficiency of the sample size.

8.19.17 Reference Audit

Audit Inputs

Original Source and Audit Scope

Reference identifier 17.

Reference name Primitive-time-root audit of Hafele–Keating 1972a

Bibliographic identification Let $R = \text{Ref}_{\text{HK72Pred}}$ denote J. C. Hafele and Richard E. Keating, *Around-the-World Atomic Clocks: Predicted Relativistic Time Gains*, *Science* 177, 166–168 (14 July 1972). The audit uses all three pages of the original source; its domain comprises the abstract, main text, Equations (1)–(4), Figure 1, Table 1, and Notes 6–11.

The source-location keys in Table 158 correspond to source locations as follows.

- 1 Abstract and introduction
- 2 Overview of the low-velocity approximation
- 3 Equation (1)
- 4 Equation (2)
- 5 Equation (3) and Figure 1
- 6 Equation (4)
- 7 Table 1
- 8 Uncertainty paragraph
- 9 Notes 6–11
- 10 Equation (4) and Notes 6–11
- 11 Figure 1 and Note 7
- 12 Note 9
- 13 Note 10 and issue date

Audit-scope identifier 17-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 17-L1.

Source-location content The audit domain stated in the bibliographic identification and the source-location columns of the respective evidence tables.

Formal Objects The auxiliary indices in Table 158 refer to the objects in Table 159. Physical carriers, spatial quantities, reference conditions, and formal objects that are required for time relations but are not themselves time contents are preserved separately. The presentation order of the prediction and observation reports is a documentary relation. The temporal order from the preliminary report to the present report is a distinct time content. The circumnavigation time τ_0 recorded by the ground clock is represented by one time content; its computation for a nonstop equatorial circumnavigation is represented as a transfer to that quantity. The supporting claims for auxiliary objects are given in Table 159.

The formal evidence comprises 84 physical and formal input-claim components obtained by decomposing 36 original-source statement groups, and 78 time contents and 35 auxiliary objects obtained by decomposing 40 source-location groups in the original source.

Assumptions The reference assumptions are the low-velocity approximation, weak gravitational field, the selected nonrotating coordinate system, the rotating Earth, the clock rate of a reference clock on Earth’s surface, route data for latitude, altitude, and ground speed, the resolution boundary for higher-order relativistic terms, the smallness of external-field residuals, and coordinate invariance of the total display difference at the reunion event.

Non-Temporal Physical Inputs The components with positive grounding grounds as non-temporal physical inputs are the portable cesium-clock apparatus, aircraft, Earth’s spatial radius, route geometry, geographical position, altitude, joining of the departure and reunion events, and identity of the check points. Place the nine components in Table 161 in $\text{Init}_R^{\text{phys}}$.

The remaining source-formal-claim components belong to the source-formal-evidence environment and do not have the element types of the generation-initial set. The source formal-evidence environment $\text{Env}_R^{\text{phys}}$ consists of 93 semantic components: the 9 generation-initial semantic components and the 84 semantic components obtained by lifting the 84 physical and formal input-claim components.

Time Contents The 78 time contents concern times recorded by clocks, time intervals, clock rates, route histories, coordinate time, differential time, interval partitioning, reunion comparison, and temporal order from the preliminary report to the present report. Time Content 78 is a report-order use; Time Contents 1–77 are law-time uses.

Formal Input Elements

Definition 8.262 (Time Contents and Auxiliary Objects in the Hafele–Keating 1972a Reference). For the reference $R = \text{Ref}_{\text{HK72Pred}}$, divide the 113 semantic objects appearing in the 40 source-location groups of the original source into 78 time contents and 35 auxiliary objects. Table 158 gives the time-content indices, time contents, auxiliary-object indices, and source location for each source-location group. Within each row, the listed names correspond one-to-one, in their order of presentation, to the indices in ascending order.

Table 158: Time contents and source locations in the Hafele–Keating 1972a reference

Location group	Time-content indices	Time contents (index order)	Auxiliary indices	Source-location key
1	1–2	round-trip itinerary order; reunion comparison	32	1
2	3	experimental time interval	1, 33	1
3	4–5	Route history; time-series flight data	2	1

Location group	Time-content indices	Time contents (index order)	Auxiliary indices	Source-location key
4	$\emptyset$	None	3–4, 35	1
5	6–8	Moving-clock reading; reference-clock reading; elapsed reading difference	None	2
6	9–10	Family of time readings of inertial coordinate clocks; coordinate time	5	2
7	11–12	recorded-time ratio; clock-rate law	None	2
8	13–14	Earth's angular speed; coordinate rate of a surface clock	6	3
9	15–16	Rate of the flying clock relative to coordinate clocks; signed ground speed	11	3
10	17–19	Circumnavigation time τ recorded by the flying clock; Circumnavigation time τ_0 recorded by the ground clock; same-itinerary comparison	None	3
11	20	Beginning and end of the circumnavigation comparison interval	None	3
12	21–22	Flying-clock–ground-clock reading difference; Lowest-order kinematic time-difference term	7	3
13	23	kinematic term	8, 11	3
14	24–25	Gravitational potential difference used to evaluate clock rates; clock-rate relation	9	4
15	26–27	Gravitational clock rate; flying-clock reading gain	10	4
16	28	Combined gravitational and kinematic time-difference term	11–12	4
17	$\emptyset$	None	34	5
18	29–30	Speed; predicted reading difference	13	5
19	31–33	Closure-time difference of a portable clock upon return; One-day scale for evaluating closure-time spread; Estimated detection threshold for clock differences	None	11
20	34	Post-return reading differences compared in the east–west asymmetry test	11, 14	5

Location group	Time-content indices	Time contents (index order)	Auxiliary indices	Source-location key
21	35–38	Time-parameterized itinerary path; Time history of ground speed; Time history of altitude; Time history of latitude	None	6
22	39	Time parameter of the route integral	15	6
23	40–41	Flight differential time; ground differential time	16	6
24	42–43	Beginning of the eastward trip; universal-time value	None	6
25	44–47	Beginning and end of the eastward trip; total duration; flight time; ground time interval	None	6
26	48–49	Beginning of the westward trip; universal-time value	None	6
27	50–53	Beginning and end of the westward trip; total duration; flight time; ground time interval	None	6
28	54–55	Time-stamped waypoint events; clock time	17	6
29	56–58	Ordered waypoints; interval time; Average ground speed over each interval	18	6
30	59–60	Integrated gravitational and kinematic time differences; Net predicted time difference	19–20	7
31	61–62	Maximum fractional uncertainty in each integrated component; Maximum fractional uncertainty in the net predicted time difference	21	7
32	63–64	Absolute uncertainties in the integrated time-difference components; Absolute uncertainty in the net predicted time difference	22	7
33	65	Higher-order time-difference corrections	23–24	8
34	66–68	Motion of the earth–moon–sun system; Time-difference calculation including motion of the earth–moon system; First-order time-difference contributions from external fields	None	8

Location group	Time-content indices	Time contents (index order)	Auxiliary indices	Source-location key
35	69–70	Clock-comparison position after a round trip; Comparison of readings of two colocated clocks	32	9
36	71–72	Readings of two colocated clocks; reunion reading difference	25	9
37	73	Correspondence of time descriptions between reference systems	26–27	9
38	74–75	Latitude-dependent surface speed; Difference in surface potential between latitudes	28	12
39	76–77	Clock-rate contribution from the difference in surface potential; Latitude-independent time relation of clocks at rest at average sea level	29	12
40	78	Temporal order from the preliminary report to the present report	30–31	13

Table 159: Auxiliary objects in the Hafele–Keating 1972a reference

a	Auxiliary object	Location group	Supporting claim indices
1	Stability	2	$\{1, 40\}$
2	Prediction input	3	$\{4, 9, 49, 50\}$
3	Prediction report	4	$\{11, 12\}$
4	Observation report	4	$\{11, 12\}$
5	Inertial rest	6	$\llbracket 17, 19 \rrbracket \cup \{14, 15\}$
6	Reference system	8	$\llbracket 21, 24 \rrbracket \cup \{3, 16, 18, 19, 26\}$
7	Sign of the difference	12	$\llbracket 27, 29 \rrbracket \cup \{35, 37, 38, 78\}$
8	Attribution of gain or loss	13	$\{28, 29, 36, 78\}$
9	Weak-field domain of application	14	$\llbracket 30, 33 \rrbracket \cup \{35, 36, 81, 82\}$
10	Altitude interval	15	$\llbracket 30, 33 \rrbracket \cup \{36\}$
11	Correspondence between east–west direction and the sign of ground speed	16	$\llbracket 24, 26 \rrbracket \cup \{28, 29, 34, 37, 38, 42, 55, 78\}$
12	Addition or cancellation	16	$\{35, 37, 38\}$
13	Altitude	18	$\{6, 39, 41, 44, 57\}$
14	Asymmetry	20	$\{41, 42\}$
15	Ordered integration	22	$\{46, 47\}$
16	Interchangeability at lowest order	23	$\llbracket 81, 83 \rrbracket \cup \{46, 47\}$

a	Auxiliary object	Location group	Supporting claim indices
17	Position record	28	$\llbracket 48, 52 \rrbracket \cup \{4, 5, 9, 54, 55\}$
18	Route reconstruction	29	$\llbracket 50, 53 \rrbracket \cup \llbracket 55, 57 \rrbracket \cup \{4\}$
19	Eastward–westward sign	30	$\llbracket 58, 63 \rrbracket$
20	Prediction table	30	$\llbracket 58, 65 \rrbracket$
21	Integration order	31	$\llbracket 50, 52 \rrbracket \cup \{66\}$
22	Root-sum-square combination assumption	32	$\llbracket 67, 69 \rrbracket \cup \{65\}$
23	Retained order	33	$\llbracket 70, 73 \rrbracket \cup \{46, 83\}$
24	Resolution domain of application	33	$\llbracket 70, 73 \rrbracket \cup \{40, 41\}$
25	Difference rule	36	$\{75, 76\}$
26	Two-reference-system description	37	$\{19, 20, 76, 77, 79\}$
27	Transformation	37	$\{76, 77, 79\}$
28	Clocks at rest at average sea level and the oblate figure of the earth	38	$\llbracket 80, 82 \rrbracket$
29	Latitude	39	$\llbracket 80, 82 \rrbracket \cup \{45, 53\}$
30	Preliminary report	40	$\{84\}$
31	Present report	40	$\{84\}$
32	Moving and reference clocks and the clock pair compared at one location	1	$\{8, 10, 13, 74, 75\}$
33	Portable atomic-clock device	2	$\{1, 10\}$
34	Spatial route and nonstop condition of an equatorial circumnavigation	17	$\{3, 4, 39\}$
35	Presentation order of the prediction and observation reports	4	$\{11, 12\}$

Definition 8.263 (Physical and Formal Input-Claim Components in the Hafele–Keating 1972a Reference). Let the sequence of physical and formal input-claim components be $(\text{ClaimVal}_{R,p}^{\text{audit,phys}})_{p \in [84]}$. Define each component, original-source statement group, minimal formal claim, dependent object, and source location by Table 160. Components 1–9 correspond to non-temporal physical inputs; Components 10–84 belong to the source-formal evidence.

The dependent-object keys in Table 160 correspond to sets of time-content indices and auxiliary-object indices as follows.

1 $(\emptyset, \{33\})$

2 $(\emptyset, \emptyset)$

3 $(\emptyset, \{17, 18\})$

4 $(\emptyset, \{17\})$

5 $(\emptyset, \{13\})$

6 $(\emptyset, \{32\})$

- 7** ($\llbracket 1, 3 \rrbracket \cup \llbracket 6, 8 \rrbracket, \{32, 33\}$)
- 8** ($\emptyset, \{3, 4\}$)
- 9** ($\emptyset, \{3, 4, 35\}$)
- 10** ($\{6\}, \{32\}$)
- 11** ($\{7, 9, 10\}, \{5\}$)
- 12** ($\{6, 7, 11, 12\}, \{5\}$)
- 13** ($\{13, 14\}, \{6\}$)
- 14** ($\{9, 10\}, \{5, 6\}$)
- 15** ($\{9, 10, 13, 14\}, \{5, 6\}$)
- 16** ($\{10\}, \{5, 6\}$)
- 17** ($\{13\}, \emptyset$)
- 18** ($\{13\}, \{6\}$)
- 19** ($\{13, 14\}, \{5, 6\}$)
- 20** ($\{13\}, \{6, 11\}$)
- 21** ($\{16\}, \{11\}$)
- 22** ($\{13, 16\}, \{6, 11\}$)
- 23** ($\llbracket 16, 23 \rrbracket \cup \{13\}, \{7\}$)
- 24** ($\{13, 16, 18, 22, 23\}, \{7, 8, 11\}$)
- 25** ($\{24\}, \{9, 10\}$)
- 26** ($\emptyset, \{10\}$)
- 27** ($\llbracket 24, 26 \rrbracket, \{9, 10\}$)
- 28** ($\{13, 16, 23\}, \{11\}$)
- 29** ($\{18, 22, 27, 28\}, \llbracket 9, 12 \rrbracket$)
- 30** ($\{24, 26, 27\}, \{9, 10\}$)
- 31** ($\{16, 22, 27, 28\}, \{7, 11, 12\}$)
- 32** ($\{18, 29, 30\}, \{13, 34\}$)
- 33** ($\llbracket 31, 33 \rrbracket, \{1, 24\}$)
- 34** ($\{30, 33\}, \{13, 14\}$)
- 35** ($\{30, 34\}, \{11, 14\}$)

- 36** $(\{35, 36\}, \emptyset)$
- 37** $(\{35, 37\}, \{13\})$
- 38** $(\{35, 38\}, \{29\})$
- 39** $(\llbracket 36, 39 \rrbracket \cup \{13\}, \{16\})$
- 40** $(\llbracket 35, 40 \rrbracket \cup \{59, 60\}, \{15, 16\})$
- 41** $(\{54, 55\}, \{17\})$
- 42** $(\llbracket 35, 38 \rrbracket \cup \llbracket 55, 58 \rrbracket, \{17, 18\})$
- 43** $(\{56, 57\}, \{17, 18\})$
- 44** $(\{38\}, \{18, 29\})$
- 45** $(\{56\}, \{11, 17, 18\})$
- 46** $(\{58\}, \{18\})$
- 47** $(\{56\}, \{13, 17, 18\})$
- 48** $(\{59\}, \{19, 20\})$
- 49** $(\{60\}, \{19, 20\})$
- 50** $(\{59, 61, 63\}, \{20\})$
- 51** $(\{60, 62, 64\}, \{20, 22\})$
- 52** $(\llbracket 36, 38 \rrbracket \cup \llbracket 57, 59 \rrbracket \cup \{61\}, \{17, 18, 21\})$
- 53** $(\llbracket 59, 64 \rrbracket, \{22\})$
- 54** $(\{60, 62, 64\}, \{19, 22\})$
- 55** $(\{65\}, \{23, 24\})$
- 56** $(\llbracket 66, 68 \rrbracket, \{23, 24\})$
- 57** $(\{68\}, \{23, 24\})$
- 58** $(\{1, 69, 70\}, \{32\})$
- 59** $(\{71, 72\}, \{25, 32\})$
- 60** $(\{72, 73\}, \{26, 27\})$
- 61** $(\{22, 27, 28\}, \{7, 8, 11\})$
- 62** $(\{74\}, \{28, 29\})$
- 63** $(\llbracket 74, 76 \rrbracket, \{9, 16, 28, 29\})$
- 64** $(\llbracket 74, 77 \rrbracket, \{9, 16, 28, 29\})$

65 ($\{40, 41\}, \{16\}$)

66 ($\{78\}, \{30, 31\}$)

Table 160: Physical and formal input-claim components
in the Hafele–Keating 1972a reference

Com- po- nent	Original- source statement group	Minimal formal claim	Dependent- object key	Source- location key
1	1	Portable cesium-clock apparatus	1	1
2	1	Clock-carrying aircraft	2	1
3	6	Spatial radius of Earth	2	3
4	18	Route geometry	3	6
5	20	Geographical position	4	6
6	18, 20, 22	Altitude	5	6
7	1	Joining of the departure event	2	1
8	32	Joining of the reunion event	6	9
9	20	Check-point identity	4	6
10	1	Macroscopic round-trip clock comparison	7	1
11	2	Separation of the prediction and observation papers	8	1
12	2	Order of the prediction and observation reports	9	1
13	3	Time recorded by a moving standard clock	10	2
14	3	Time recorded by inertial coordinate clocks	11	2
15	3	Clock-rate difference in the low-velocity domain	12	2
16	4	Clock on the rotating surface of Earth	13	2
17	4	Global lattice of inertial clocks	14	2
18	4	Separation of the surface clock from lattice clocks	15	2
19	5	Selected nonrotating coordinate system	16	2
20	5	Assignment of calculation coordinates	14	2
21	6	Angular velocity of Earth	17	3
22	6	Speed of the equatorial reference clock	18	3
23	6	Equatorial clock-rate model	19	3
24	7	Contribution of Earth's rotation	20	3
25	7	Signed ground speed	21	3
26	7	Aircraft coordinate speed	22	3
27	8	Lowest-order kinematic reading difference	23	3

Com- po- nent	Original- source statement group	Minimal formal claim	Dependent- object key	Source- location key
28	9	Negative eastward kinematic sign	24	3
29	10	Positive westward kinematic sign	24	3
30	11	Surface gravity	25	4
31	11	Altitude difference	26	4
32	11	Weak gravitational-potential difference	25	4
33	12	Weak-gravity term	27	4
34	12	Low-velocity kinematic term	28	4
35	12	Combination of gravitational and kinematic terms	29	4
36	13	Gravitational term positive in both directions	30	4
37	14	Addition of westward components	31	4
38	14	Partial cancellation of eastward components	31	4
39	15	Specialization to a nonstop equatorial circumnavigation	32	5
40	16	Detection threshold of the portable clock	33	11
41	16	Magnitude of the predicted reading difference	34	5
42	17	Test of east–west asymmetry	35	5
43	18	Variation of route speed	36	6
44	18	Variation of route altitude	37	6
45	18	Variation of route latitude	38	6
46	19	Lowest-order route integrand	39	6
47	19	Route-history integral	40	6
48	20	Map data	4	6
49	20	Check-point times	41	6
50	20	Route reconstruction	42	6
51	21	Eastward partition into 125 intervals	43	6
52	21	Westward partition into 108 intervals	43	6
53	22	Interval latitude	44	6
54	20	Check-point longitude	4	6
55	22	Eastward bearing of the interval	45	6
56	22	Interval speed	46	6
57	22	Interval altitude	47	6
58	23	Eastward gravitational prediction	48	7
59	23	Eastward kinematic prediction	48	7
60	23	Eastward combined prediction	49	7
61	24	Westward gravitational prediction	48	7
62	24	Westward kinematic prediction	48	7
63	24	Westward combined prediction	49	7
64	25	Uncertainty of component predictions	50	7
65	25	Uncertainty of combined predictions	51	7

Com- po- nent	Original- source statement group	Minimal formal claim	Dependent- object key	Source- location key
66	26	Flight-data error for each integral	52	8
67	27	Root-sum-square combined uncertainty	53	8
68	28	Maximum fractional uncertainty in the eastward net prediction	54	8
69	29	Maximum fractional uncertainty in the westward net prediction	54	8
70	30	Resolution boundary for higher-order relativistic terms	55	8
71	31	External-field term due to the Moon	56	8
72	31	External-field term due to the Sun	56	8
73	31	Small external-field residual	57	8
74	32	Reunion at the same location	58	9
75	32	Clock-display difference at reunion	59	9
76	33	Coordinate invariance of the total display difference	60	9
77	33	Non-necessity of a subsequent coordinate transformation to the noninertial rest system	60	9
78	34	Vocabulary of gain and loss	61	3
79	34	Separation of the comparative display quantity	60	9
80	35	Difference in surface speed with latitude	62	12
81	35	Potential difference due to the oblate earth and lowest-order cancellation	63	12
82	35	Latitude-independent time relation of clocks at rest at average sea level	64	12
83	19	Interchangeability of flight and ground differential times at lowest order	65	6
84	36	Temporal order from the preliminary report to the present report	66	13

Definition 8.264 (Semantic Components and Generation-Initial Set in the Hafele–Keating 1972a Reference). Let the semantic-component nodes be

$$\text{Node}_R^{\text{audit, semantic}} = \text{SemanticNodes}_R(93)$$

Take Components 1–9 as the generation-initial semantic components and Components 10–93 as the components corresponding to the 84 physical and formal input claims.

$$\text{ClaimNodeLift}_R(9; 84), \quad \text{StdSemLift}_R(9; 84)$$

hold. Define the values of the generation-initial components and their corresponding claim indices by Table 161. Define the generation-initial set and source-formal-evidence

environment by

$$\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(9)$$

Table 161: Generation-initial semantic components in the Hafele–Keating 1972a reference

k	Semantic component	Corresponding claim index
1	Portable cesium-clock apparatus	$\{1\}$
2	Clock-carrying aircraft	$\{2\}$
3	Spatial radius of Earth	$\{3\}$
4	Route geometry	$\{4\}$
5	Geographical position	$\{5\}$
6	Altitude	$\{6\}$
7	Joining of the departure event	$\{7\}$
8	Joining of the reunion event	$\{8\}$
9	Check-point identity	$\{9\}$

Dependent Inputs

Dependent Inputs Times recorded by clocks, clock rates, coordinate time, ground speed, route histories, differential time, interval times, reunion comparison, surface speed, and potential contributions to clock rates enter the 14 transfers.

Dependence and Transfer Elements

Definition 8.265 (Time-Content Transfers in the Hafele–Keating 1972a Reference). Define the transfer-index set by $\text{Idx}_R^{\text{audit,transfer}} = [14]$. Define the time-content inputs, time-content outputs, and auxiliary-object support of each transfer by the equalities in Table 162. Physical and formal input-claim support is also specified in the table. The source location of each transfer is the union of the source locations corresponding to its time-content inputs, outputs, and supporting claims.

Table 162: Time-content transfers in the Hafele–Keating 1972a reference

s	Time-content inputs i	Time-content outputs a	Auxiliary support indices	Supporting claim indices
1	$\{9, 10, 12, 13, 16\}$	$\{14, 15, 23\}$	$\{5, 6, 11\}$	$\llbracket 14, 19 \rrbracket \cup \llbracket 21, 26 \rrbracket \cup \{3, 28, 29, 34, 37, 38, 42, 55, 78\}$
2	$\{18, 23\}$	$\{21, 22\}$	$\{7\}$	$\llbracket 27, 29 \rrbracket \cup \{35, 37, 38, 78\}$
3	$\{18, 24, 25\}$	$\{26, 27\}$	$\{9, 10\}$	$\llbracket 30, 33 \rrbracket \cup \{35, 36, 81, 82\}$

s	Time-content inputs i	Time-content outputs a	Auxiliary support indices	Supporting claim indices
4	$\{22, 27\}$	$\{28\}$	$\llbracket 9, 12 \rrbracket$	$\llbracket 24, 26 \rrbracket \cup \llbracket 28, 38 \rrbracket \cup \{42, 55, 78, 81, 82\}$
5	$\{29\}$	$\{18\}$	$\{34\}$	$\{3, 4, 39\}$
6	$\{13, 24, 29\}$	$\{30\}$	$\{9, 13, 34\}$	$\llbracket 30, 36 \rrbracket \cup \{3, 4, 6, 39, 41, 44, 57, 81, 82\}$
7	$\{30\}$	$\{34\}$	$\{11, 14\}$	$\llbracket 24, 26 \rrbracket \cup \{28, 29, 34, 37, 38, 41, 42, 55, 78\}$
8	$\{54, 55\}$	$\llbracket 56, 58 \rrbracket$	$\{17, 18\}$	$\llbracket 4, 6 \rrbracket \cup \llbracket 48, 57 \rrbracket \cup \{9\}$
9	$\llbracket 36, 40 \rrbracket \cup \{13, 44, 50\}$	$\{59, 60\}$	$\{15, 16, 19, 20\}$	$\llbracket 43, 47 \rrbracket \cup \llbracket 58, 65 \rrbracket \cup \llbracket 81, 83 \rrbracket$
10	$\llbracket 36, 39 \rrbracket \cup \{13, 41, 44, 50\}$	$\{59, 60\}$	$\{15, 16, 19, 20\}$	$\llbracket 43, 47 \rrbracket \cup \llbracket 58, 65 \rrbracket \cup \llbracket 81, 83 \rrbracket$
11	$\{13, 37, 38, 57, 58\}$	$\{59, 60\}$	$\llbracket 16, 20 \rrbracket$	$\llbracket 4, 6 \rrbracket \cup \llbracket 46, 65 \rrbracket \cup \llbracket 81, 83 \rrbracket \cup \{9\}$
12	$\llbracket 59, 61 \rrbracket$	$\llbracket 62, 64 \rrbracket$	$\llbracket 20, 22 \rrbracket$	$\llbracket 50, 52 \rrbracket \cup \llbracket 58, 69 \rrbracket$
13	$\{70, 71\}$	$\{72\}$	$\{25, 32\}$	$\llbracket 74, 76 \rrbracket \cup \{8, 10, 13\}$
14	$\llbracket 74, 76 \rrbracket$	$\{77\}$	$\{9, 16, 28, 29\}$	$\llbracket 30, 33 \rrbracket \cup \llbracket 45, 47 \rrbracket \cup \llbracket 80, 83 \rrbracket \cup \{35, 36, 53\}$

Transfer Events and Outputs The input–output relations in Table 162 yield clock rates and the kinematic term, kinematic and gravitational time differences, their combined difference, nonstop trip duration, direction-specific predictions, interval times and average ground speeds, integrated components and net predictions, conditional uncertainties, colocated reading differences, and the ground-clock time relation.

Conditions of Establishment

Conditions under Which the Dependence or Transfer Applies Transfers 1–4 presuppose the low-speed approximation, nonrotating inertial coordinates, the rotating-earth model, and a weak gravitational field. With $v = -w$ and $R, \Omega, \tau_0, c > 0$, Equation (1) gives the kinematic time difference $\tau_0 w(2R\Omega - w)/(2c^2)$; the westward gain condition is $0 < w < 2R\Omega$. Transfers 5–7 apply to a nonstop equatorial circumnavigation at constant nonzero ground speed and its direction-specific predictions. Transfer 8 reconstructs finite intervals from checkpoint times and spatial positions. The $d\tau$ input in Transfer 9 and the $d\tau_0$ input in Transfer 10 are alternatives at lowest order; they are not jointly required. Transfer 11 is the source numerical integration using interval-average inputs. The quadrature rule in Transfer 12 is an assumption made by the source; the component-error estimates include systematic and random errors. Transfer 13 takes colocated readings as inputs and forms their difference. Transfer 14 presupposes clocks at rest at average sea level and the oblate figure of the earth, and gives the lowest-order cancellation of latitude differences. All 14 transfers take one or more existing time contents as inputs. No transfer outputs time content from non-temporal physical inputs alone.

Original-Source Support Take the time-content input indices, time-content output indices, auxiliary-object support indices, and supporting claim indices in each row of Table 162 as the corresponding index sets in Definition 7.105. The semantic-component support set, time-content input set, time-content output set, and auxiliary-object support set of each transfer are determined by that definition. The time-content transfer relation determined by the 14 transfers is acyclic.

Original-Source Support for Each Dependence and Transfer The source locations of each transfer are the union of those specified for its time-content inputs and outputs in Definition 8.262 and those specified for its supporting claims in Definition 8.263.

Results

Definition 8.266 (Results of the Hafele–Keating 1972a Reference). Define the result-index set by $\text{Idx}_R^{\text{audit,result}} = [9]$. The correspondence between the results and their source locations is given in Table 163.

Table 163: Results and source locations in the Hafele–Keating 1972a reference

j	Result	Source location
1	Low-velocity relation between moving- and reference-clock rates	Low-speed approximation
2	Kinematic model of Earth’s rotating equator	Low-speed approximation; Eq. (1)
3	Clock-rate contribution from a weak gravitational field	Eq. (2) and adjoining text
4	Eastward–westward directional asymmetry	Eqs. (1)–(3); Fig. 1; note 7
5	Actual-route integral	Eq. (4) and flight-data paragraph; Table 1
6	Component and net predicted values and uncertainties	Table 1; uncertainty paragraph
7	Boundary of route reconstruction from flight data	Flight-data paragraph
8	Coordinate-system invariance of the total display difference at reunion	Introduction; note 6
9	Lowest-order cancellation of surface latitude and potential	Note 9

Reference-Specific Results The nine results in Definition 8.266. The source location, time-content prerequisites, auxiliary-object prerequisites, and supporting claims of each result are given by the tables in that definition and Definition 8.267.

Result Dependence

Definition 8.267 (Result-Dependency Hyperedges in the Hafele–Keating 1972a Reference). For each $q \in [9]$, define the time-content inputs, auxiliary-object inputs, and physical and formal input-claim support of the result-dependency hyperedge $\text{ResultEdge}_{R,q}$ by the equalities in Table 164, and take its result output to be Result q .

Table 164: Result-dependency hyperedges in the Hafele–Keating 1972a reference

q	Time-content input indices	Auxiliary-object input indices	Supporting-claim indices
1	$\{11, 12\}$	$\{5\}$	$\llbracket 13, 15 \rrbracket \cup \llbracket 17, 19 \rrbracket$
2	$\llbracket 13, 22 \rrbracket \cup \{9, 10\}$	$\{5, 6, 11\}$	$\llbracket 14, 29 \rrbracket \cup \{3, 34, 37, 38, 42, 55, 78\}$
3	$\llbracket 24, 27 \rrbracket$	$\{9, 10\}$	$\llbracket 30, 33 \rrbracket \cup \{35, 36, 81, 82\}$
4	$\{22, 27, 28, 30, 33, 34\}$	$\llbracket 7, 14 \rrbracket \cup \{24\}$	$\llbracket 24, 42 \rrbracket \cup \llbracket 70, 73 \rrbracket \cup \{6, 44, 55, 57, 78, 81, 82\}$
5	$\llbracket 35, 39 \rrbracket \cup \{59, 60\}$	$\llbracket 15, 20 \rrbracket \cup \{29\}$	$\llbracket 4, 6 \rrbracket \cup \llbracket 43, 65 \rrbracket \cup \llbracket 80, 83 \rrbracket \cup \{9\}$
6	$\llbracket 59, 68 \rrbracket$	$\llbracket 19, 24 \rrbracket$	$\llbracket 50, 52 \rrbracket \cup \llbracket 58, 73 \rrbracket \cup \{40, 41, 46, 83\}$
7	$\llbracket 42, 58 \rrbracket \cup \{4, 5\}$	$\{2, 17, 18\}$	$\llbracket 4, 7 \rrbracket \cup \llbracket 43, 57 \rrbracket \cup \{9\}$
8	$\llbracket 69, 73 \rrbracket \cup \{1, 2\}$	$\llbracket 25, 27 \rrbracket \cup \{32\}$	$\llbracket 74, 77 \rrbracket \cup \{8, 10, 13, 19, 20, 79\}$
9	$\llbracket 74, 77 \rrbracket$	$\{9, 16, 28, 29\}$	$\llbracket 30, 33 \rrbracket \cup \llbracket 45, 47 \rrbracket \cup \llbracket 80, 83 \rrbracket \cup \{35, 36, 53\}$

Take the supporting-claim indices, time-content input indices, and auxiliary-object input indices in each row of Table 164 as the corresponding index sets in Definition 7.105. The claim-support set, semantic-component support set, time-content prerequisite set, and auxiliary-object prerequisite set of each result are determined by that definition. The semantic-object prerequisites of the nine results comprise 110 occurrences, and every result has one or more time-content prerequisites.

Grounding Grounds

Grounding Grounds of the Results The analytic grounds are the low-velocity clock-rate relation, the rotating-Earth kinematic model, the weak-gravity contribution, and the route-history integrals in Equations (1)–(4). The numerical grounds are the component predictions, combined predictions, and propagated uncertainties in Table 1.

Evidence Hierarchy

Direct Measurement Targets No new measurement of a clock-display difference belongs to this reference.

Direct Measurement Predicates No eastward loss, westward gain, or agreement with prediction is directly established by observed values in this reference.

Conditions and Domain of Direct Measurement This reference has no reference-specific direct-measurement condition or domain. The flight-route data belong to the inputs of the predictive calculation.

Targets of Formal Generalization The formal generalization proceeds from the model of a nonstop equatorial flight to commercial routes with varying latitude, longitude, ground speed, and altitude, and from local kinematic and gravitational contributions to route-integrated, combined eastward and westward predictions.

Conditions and Domain of Formal Generalization The domain is limited to the low-velocity approximation, weak gravitational field, selected nonrotating coordinate system, rotating-Earth model, route reconstruction, first-order integration by component, the resolution boundary of omitted higher-order terms, and the domain in which external-field residuals are evaluated. All nine results use one or more time contents as dependent inputs.

Generation Judgment Judgment values: Time contents generated under the LCTR complete-generation judgment: 0; time contents reached under the source-internal evidence-conditioned local-reach judgment: 0; remaining primitive time roots: 78.

Generation Reach and Primitive Time Roots The complete judgment, with non-temporal physical inputs alone as its initial set, generates zero time contents. All 78 time contents in Definition 8.262 remain primitive time roots. The local-reach judgment reaches zero time contents.

Source-Internal Evidence-Conditioned Local Reach The source-internal evidence-conditioned local-reach judgment is zero; 78 primitive time roots remain after that judgment.

Result Dependence For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ from Definition 7.108 and apply Definition 7.50 to it. Totalling over all results determines the primitive-time-root–result dependence relations and the necessary-primitive-time-root–result dependence relations.

Result Dependence All nine results in Definition 8.266 are primitive-time-root-dependent results. The numbers of generated results, evidence-reach-only results, and unreached-after-root-addition results are zero. The minimal support of each result is given by Proposition 8.269.

Evidence Effect

Evidence-Effect Scope The formal evidence supports the low-velocity clock-rate relation, rotating-Earth kinematics, weak-gravity contribution, east–west directional asymmetry, route-history integrals, component and combined predictions, uncertainty evaluation, and reunion clock-display difference.

Evidence-Effect Boundary The analytic and numerical predictions establish clock-rate relations, route integrals, direction-specific predictions, and uncertainty limits. Their establishment implies neither generation of the 78 time contents from non-temporal

physical inputs, nor physical-time qualification of the primitive time roots, nor direct measurement of the predicted clock-display differences, nor a time attribute independent of physical carriers. Results from the subsequent observation reference do not belong to the direct measurement evidence of this reference.

Result-Preservation Domain

Result-Preservation Domain The nine results are preserved insofar as every time-content premise is supplied by the corresponding generated time representation in the LCTR generation sequence, while preserving the conditions required by each result among the low-velocity approximation, weak-gravity approximation, selected nonrotating coordinate system, rotating-Earth model, route inputs, first-order integration, uncertainty propagation, domain of application of higher-order corrections, and external-field residuals.

Placement of the Grounding Obligation

Placement of the Grounding Obligation Time Contents 1–77, which belong to law-time uses, are not generated from non-temporal physical inputs before use; they are accepted as already-established time inputs. The placement judgment for the physical-time grounding obligation is “backgrounded by primitive time input.” This judgment follows by matching the use of these time contents in law operations or result derivations against the absence, before that use, of the corresponding generation or validity condition as an input, formation condition, evaluation condition, or failure condition. Time Content 78 is the temporal order from the preliminary report to the present report; it is not a premise of a law operation or physical-result derivation.

Evidence of Actual Failure

Evidence of Actual Failure The complete judgment and the remaining-root judgment concern physical-time qualification; they are distinct from actual failure of a concrete condition governing a result’s value or domain of application. Neither the original formal evidence nor the LCTR audit evidence in this reference establishes an actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence,” and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream Reach For this reference, the reach to primitive-time-root-dependent results is determined by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because actual failure of a concrete condition has not been established, the actual-reach judgment from that failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair Scope On the basis of structural reach, the objects of repair are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is limited to results for which actual failure is independently established and which are dependency-reachable from that failure location. Because actual failure is indeterminate in this reference, this repair-and-recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domain is restricted. The values after recalculation are determined as audit results.

Audit Conclusion

Proposition 8.268 (Primitive Time Roots and Result Dependence in the Hafele–Keating 1972a Reference). In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepldx}(78, 9)$. Moreover, $R \in \text{Ref}^{\text{audit, rootdep}}$ and $R \notin \text{Ref}^{\text{audit, rootunreach}}$.

Proof. The audit boundary has 180 nodes, comprising 93 semantic-component nodes, 78 time-content nodes, and 9 result nodes, and 23 hyperedges, comprising 14 transfer hyperedges and 9 result-dependency hyperedges. Take the generation-initial semantic-component set $\text{Init}_R^{\text{phys}}$ as the starting point of generation reach. Its generation-initial closure reaches the nine semantic-component nodes but reaches no time content or result. Taking the source-formal-evidence environment $\text{Env}_R^{\text{phys}}$ as the starting point of evidence-conditioned reach reaches all 93 semantic-component nodes but no time content or result. All 14 transfers in Definition 8.265 have one or more time contents as inputs. By Proposition 7.43, the 78 time contents are not reached from non-temporal physical inputs. Each of the nine result-dependency hyperedges in Table 164 also presupposes at least one primitive time root. Adding the source-formal-evidence environment and the 78 primitive time roots to the initial set reaches all nine results. The common reconstruction equations in Proposition 7.49 and Definition 8.115 then yield the displayed index tuple. Membership in the reference sets follows from Definition 7.52. $\square$

Proposition 8.269 (Minimal Primitive-Time-Root Supports in the Hafele–Keating 1972a Reference). For each $q \in \text{Idx}_R^{\text{audit, result}}$, the number of elements of $\mathcal{A}_R(q)$ from Definition 7.108, together with the minimum and maximum cardinalities of those elements, is given in Table 165.

Table 165: Minimal primitive-time-root supports in the Hafele–Keating 1972a reference

Result index	Support statistics
1	(1; 2)
2	(6; 9, 12)
3	(3; 3, 4)
4	(16; 4, 10)
5	(5; 7, 9)
6	(5; 7, 13)
7	(1; 16)

Result index	Support statistics
8	(1; 6)
9	(1; 3)

Proof. Apply the update in Definition 7.46 to the 14 transfer hyperedges in Definition 8.265. Scan all 14 transfers in their listed order and determine the support antichains of all time-content nodes at the finite fixed point. By Proposition 7.47, this yields the values corresponding to the time-content prerequisites in Table 163. The total number of members of the minimal primitive-time-root support families is 39. $\square$

Proposition 8.270 (Audit Conclusion for the Hafele–Keating 1972a Reference). The complete judgment generates zero time contents, and 78 primitive time roots remain. The local-reach judgment reaches zero time contents. All nine reference-specific results depend on primitive time roots, and all nine are preserved after connection to the LCTR generation sequence.

Proof. The reach classification of time contents and results follows from Proposition 8.268. The minimal primitive-time-root supports of the respective results are given by Proposition 8.269. Generating the 77 time contents belonging to law-time uses through the LCTR generation sequence and assigning qualification and a domain of application at each point of use satisfies the dependent inputs of all nine results. $\square$

Individual Statements

Objects of the Statements Detectability of the predicted effect with portable cesium clocks; evidential force of the east–west directional asymmetry; the physical relation between inertial systems and the universe; and observability and invariance of the difference between the readings of two clocks at the same place.

Verbatim Original (English) The theory predicts a detectable effect with cesium beam clocks if they are flown around the world at typical jet aircraft speeds (4).

Furthermore, the mere existence of a definite east-west directional asymmetry in the observed time differences would give strong evidence for the validity of the kinetic term in Eq. 2.

Although inertial systems are highly specialized, they have an objective physical relationship with the universe because they have no acceleration or rotation relative to the universe.

The difference in the times indicated by two clocks located at the same place is a physically observable quantity that is invariant.

Original-Source Locations The first quotation is from the introduction of Hafele–Keating, *Around-the-World Atomic Clocks: Predicted Relativistic Time Gains*, *Science* 177, p. 166. The second is from the paragraph following Figure 1 on p. 167; the third and fourth are from note 6 on the same page.

Minimal Context The first quotation belongs to the introduction that separates the proposed global-flight test using portable atomic clocks into a prediction report and a subsequent observation report. Following the detection threshold shown in Figure 1, the second describes east–west directional asymmetry as evidence supporting the validity of the kinetic term. The third and fourth belong to a note explaining description by an inertial reference system and the frame independence of the difference between the readings of two reunited clocks.

Attribution All four statements are attributable to the main text or a note by J. C. Hafele and Richard E. Keating.

Contents Directly Stated by the Quotations The first states that the predicted effect is detectable with portable cesium clocks. The second states that an east–west directional asymmetry in the observed time differences would constitute strong evidence for the validity of the kinetic term. The third states that inertial systems have an objective physical relationship with the universe. The fourth describes the difference between the readings of two clocks at the same place as an invariant physically observable quantity.

Explicit Domain of Application The first and second quotations concern portable cesium clocks, typical jet-aircraft speeds, global flight, a detection threshold, and east–west directional comparison. The third concerns inertial systems that are neither accelerated nor rotating relative to the universe; the fourth concerns two clocks compared at the same place.

Evidence Provenance The quotations are from the published article in *Science* 177, pp. 166–168.

Individual Audit

Interpretive Possibilities The first statement is a conditional description predicting a detectable reading difference in cesium clocks flown around the world at typical jet-aircraft speeds. The second is a conditional evaluation of evidence that treats east–west directional asymmetry as evidence supporting the validity of the kinetic term. The third is a theoretical explanation that attributes an objective physical relationship to inertial systems that are neither accelerated nor rotating relative to the universe. The fourth describes the reading difference between two clocks compared at the same place as an invariant physically observable quantity.

Independent Grounds The four statements theoretically describe a detectable effect in globally flown cesium clocks, east–west directional asymmetry, the relation between inertial systems and the universe, and invariance of the reading difference between reunited clocks. This reference is a prediction report; the quoted passages contain no direct measurement result obtained from clock-display differences after global flight.

Evidence-Effect Boundary The first statement predicts detectability. The second states the evidential force supporting the kinetic term if the specified east–west directional asymmetry is observed. The third supports the authors’ explanation of the relation between inertial systems and the universe, but supplies no result independently measuring that relation. The invariance in the fourth statement

concerns the reading difference between two clocks reunited at the same place; it supports neither invariance nor dilation of a time attribute independent of physical carriers.

Corresponding Formal Evidence The first statement corresponds to Time Contents 3 and 31–33 in Definition 8.262, Auxiliary Object 33, and Claim Components 1–2 and 39–41 in Definition 8.263. The second corresponds to Time Contents 23 and 34, Auxiliary Objects 11 and 14, Claim Components 28–29, 37–38, and 42, and Result 4 in Definition 8.266. The third corresponds to Time Contents 9–10, Auxiliary Objects 5–6 and 26–27, Claim Components 16–20, and Results 2 and 8. The fourth corresponds to Time Contents 69–73, Auxiliary Objects 25–27 and 32, Claim Components 8 and 74–77, Transfer 13 in Definition 8.265, and Result 8.

Within-Reference Comparison The first statement predicts detectability, and the second conditionally states the evidential force of an observed east–west directional asymmetry. The third describes the relation between inertial systems and the universe; the fourth describes the reading difference between reunited clocks as an invariant observable quantity. The prediction, conditional evaluation of evidence, reference-system explanation, and clock-comparison invariant are mutually related, but they constitute neither a measurement result nor measurement of an attribute of a time entity.

8.19.18 Source Audit

Audit Input

Original Source and Audit Scope

Reference Identifier 18.

Reference Name Primitive-time-root audit of Bekenstein 1973

Bibliographic Identification Jacob D. Bekenstein, *Black Holes and Entropy*, *Physical Review D* 7, 2333–2346 (1973) is denoted by $R = \text{Ref}_{\text{Bek73BHE}}$. The audit domain comprises the abstract, §§I–VII and Appendices A–C.

Original-Source Location Identifier 18-L1.

Original-Source Location Content Table 168 gives the source locations of the 78 claim components; Table 166 gives those of the 67 time contents.

Audit-Scope Identifier 18-SC1.

Audit-Scope Content The audit domain stated in the bibliographic identification.

Formal Object

Object Identifier 18-F1.

Object Name Formal object.

Object Type Formal object.

Formal Content The black-hole area theorem; the structural analogy with thermodynamic entropy; black-hole entropy; generalized entropy; the generalized second law; missing information; capture, merger, and formation; oscillator absorption; null beams and photon spectra; the Geroch box process; Kerr geodesics; and locally nonrotating reference frames.

Set of Original-Source Location Identifiers 18-L1.

Assumption

Assumption Identifier 18-A1.

Assumption Content The analysis uses Kerr relations, the classical area theorem, the thermodynamic analogy, Shannon information, the one-way horizon and the stated semiclassical and statistical conditions. The oscillator is nonrelativistic; the null beam uses a stationary, uncharged Kerr background and geometrical optics; the finite particle and box use the specified near-horizon approximation. The Appendix B capture-barrier conditions and the finite-particle conditions of Appendices A and C apply to their respective calculations. The source uses $G = c = 1$.

Set of Original-Source Location Identifiers 18-L1.

Nontemporal Physical Input

Input Identifier 18-P1.

Input Name Nontemporal physical input.

Input Type Nontemporal physical input.

Input Content The 38 components among Claims 1–40 other than 11 and 33 supply carrier and event identities, spatial geometry and numerical state, energy, area, entropy and information data. Claims 11 and 33 specify the causal constraint on exterior information access and the temporal content of null incidence, respectively.

Set of Original-Source Location Identifiers 18-L1.

Time Contents

Time-Content Identifier 18-T1.

Time-Content Name Time Contents used in the original source.

Time-Content Type Time content.

Time-Content Statement Table 166 gives the 67 time contents and their source locations.

Set of Original-Source Location Identifiers 18-L1.

Formal Input Elements

Definition 8.271 (Time Contents in the Bekenstein 1973 reference). For $R = \text{Ref}_{\text{Bek73BHE}}$, let $n_R = 67$. Table 166 specifies the time contents and source locations.

Table 166: Bekenstein 1973 Time Contents and Source Locations

i	Time Content	Source Location
1	Black-hole area nondecrease order	pp. 2333,2334,2339
2	Thermodynamic-entropy increase order	pp. 2333,2334,2339
3	Object-fall process order	pp. 2333,2334,2339

<i>i</i>	Time Content	Source Location
4	Generalized-entropy nondecrease order	pp. 2333,2334,2339
5	Black-hole transformation	pp. 2333,2334
6	General process orientation	pp. 2333,2334
7	Temporal course of an arbitrary process	pp. 2333,2334
8	General capture boundary	pp. 2333,2334
9	Final-result boundary	pp. 2333,2334
10	Reversible-process category	pp. 2333,2334
11	Irreversible-process category	pp. 2333,2334
12	Merger initial boundary	pp. 2333,2334
13	Merger final boundary	pp. 2333,2334
14	Merger transition	pp. 2333,2334
15	Preferred direction in time	pp. 2334,2335
16	Black-hole angular velocity	pp. 2334,2335
17	Reference stage before compression	pp. 2335,2336
18	Reference stage after compression	pp. 2335,2336
19	Temporal dependence relating entropies before and after compression	pp. 2335,2336
20	Thermodynamic evolution	pp. 2335,2336
21	Initial-condition stage of the thermodynamic system	pp. 2335,2336
22	Late stage at which effects of the thermodynamic initial conditions are washed out	pp. 2335,2336
23	Stationary black-hole state	pp. 2336
24	Course of a dynamically evolving black hole	pp. 2336
25	Approach to equilibrium	pp. 2336
26	Stage carrying the black hole's initial peculiarities	pp. 2336
27	Late stage at which the black hole loses the effects of its initial conditions	pp. 2336
28	Black-hole evolution	pp. 2336,2337
29	Monotone order of black-hole entropy	pp. 2336,2337
30	One-way causal structure of information access across the event horizon	pp. 2336,2337,2339
31	Radiative-loss process in a merger	pp. 2336,2337
32	Position of access to external information before capture	pp. 2336,2338,2339
33	Position of information loss after capture	pp. 2336,2338,2339
34	Particle-capture transition	pp. 2336,2338,2339
35	Particle-insertion boundary	pp. 2336,2338,2339
36	Particle-capture boundary	pp. 2336,2338,2339
37	Black-hole formation process	pp. 2336,2338,2339
38	Irreversibility of formation	pp. 2336,2338,2339
39	Change in information access accompanying fall	pp. 2336,2338,2339
40	Decrease in external entropy upon absorption of the body	pp. 2339
41	Spontaneous statistical decrease	pp. 2339
42	Quantum-absorption trial position	pp. 2339
43	Harmonic-oscillator frequency	pp. 2339,2340

i	Time Content	Source Location
44	Null-geodesic beam path	pp. 2340,2341,2345
45	Reversible beam propagation	pp. 2340,2341,2345
46	Order of approach to the horizon	pp. 2340,2341,2345
47	Beam-capture boundary	pp. 2340,2341,2345
48	Photon mode frequencies and frequency distribution	pp. 2340–2343
49	Photon-frequency cutoff	pp. 2340,2341
50	Characteristic photon frequency	pp. 2340–2343
51	Continuous-beam evolution	pp. 2341,2342
52	Succession one quantum at a time	pp. 2341,2342
53	Box-lowering position	pp. 2342,2343
54	Radiation-release position	pp. 2342,2343
55	Box-retrieval position	pp. 2342,2343
56	Heat-engine cycle	pp. 2342,2343
57	Energy-conversion process	pp. 2342,2343
58	Kerr coordinate time	pp. 2343,2344,2345
59	Differential dt of the Kerr time coordinate	pp. 2343,2344,2345
60	Kerr-geodesic motion	pp. 2343,2344,2345
61	Geodesic turning point	pp. 2343,2344,2345
62	Capture boundary of the geodesic	pp. 2343,2344,2345
63	Position of gravitational radiation before fall	pp. 2343,2346
64	Radiation–fall order	pp. 2343,2346
65	Motion in a locally nonrotating reference frame	pp. 2345
66	Angular rate	pp. 2345
67	Reference position at the instant of particle release	pp. 2345

Table 167: Bekenstein 1973 Auxiliary Objects

a	Auxiliary Object	Source Location
1	Kerr black-hole state parameters	pp. 2333–2335; Appendices A–C, pp. 2343–2345
2	Geometric data of a horizon cross-section	pp. 2333–2335; Appendices A–C, pp. 2343–2345
3	Irreducible-mass data	pp. 2333–2335; Appendices A–C, pp. 2343–2345
4	Horizon-area data	pp. 2333–2335; Appendices A–C, pp. 2343–2345
5	Thermodynamic-entropy state data	pp. 2333–2335
6	Mass-increment data	pp. 2334–2335
7	Charge-increment data	pp. 2334–2335
8	Angular-momentum-increment data	pp. 2334–2335

<i>a</i>	Auxiliary Object	Source Location
9	Physical-state alternatives	pp. 2335–2337
10	Measurement-outcome alternatives and identification data	pp. 2335–2337
11	Causal constraints on external information access	pp. 2336–2339
12	Fixed- (M, Q, L) state data	pp. 2336–2339
13	Black-hole physical-state carrier	pp. 2335–2337
14	Carrier identities of the merging black holes	pp. 2333–2334, 2336–2337
15	Identity of the merger event	pp. 2333–2334, 2336–2337
16	Radiation carrier	pp. 2333–2334, 2336–2337
17	Radiative-energy data	pp. 2333–2334, 2336–2337
18	Planck-scale area data	pp. 2337–2338
19	One-bit information data	pp. 2337–2338
20	Black-hole state data	pp. 2333, 2339
21	External-system state data	pp. 2333, 2339
22	Black-hole entropy data	pp. 2333, 2339
23	External-entropy data	pp. 2333, 2339
24	Statistical ensemble	p. 2339
25	Macroscopic-state data	p. 2339
26	Harmonic-oscillator carrier	pp. 2339–2340
27	Oscillator energy-level data	pp. 2339–2340
28	Identity of the absorption event	pp. 2339–2340
29	Identity of the beam carrier	pp. 2340–2342; Appendix B, p. 2345
30	Photon carrier	pp. 2340–2342; Appendix B, p. 2345
31	Photon-energy data	pp. 2340–2342; Appendix B, p. 2345
32	Spatial geometric data of the beam-capture problem	pp. 2340–2342; Appendix B, p. 2345
33	Null incidence	pp. 2340–2342; Appendix B, p. 2345
34	Finite-size box carrier	pp. 2342–2343
35	Box-energy data	pp. 2342–2343
36	Box-size geometry	pp. 2342–2343
37	Radiation carrier	pp. 2342–2343
38	Finite-size particle carrier	pp. 2343–2345
39	Particle spatial-position data	pp. 2343–2345
40	Size and spatial-position data of the near-horizon box	p. 2345, Appendix C
41	Stationary and dynamical scope of the entropy formula	pp. 2336
42	Minimum area increase in finite-particle capture	pp. 2338, 2343–2345
43	Applicability conditions concerning statistical and quantum fluctuations	pp. 2339, 2341–2342
44	Kerr metric coefficients and spatial coordinates	pp. 2343–2345
45	Particle constants and radius and geodesic first integrals	pp. 2343–2345

a	Auxiliary Object	Source Location
46	Oscillator masses, spring constant and statistical distribution	pp. 2339–2340
47	Photon density of states, polarization and temperature data	pp. 2340–2343
48	Area-entropy coefficient and quantum size constraints	pp. 2338
49	Near-horizon approximation and box-control conditions	pp. 2342–2345
50	Null-geodesic constants and capture-barrier conditions	p. 2345, Appendix B; Eq8 and A6

Definition 8.272 (Bekenstein 1973 Physical and Formal Input Claim Components). Let the physical and formal input claim sequence be $(\text{ClaimVal}_{R,p}^{\text{audit,phys}})_{p \in [78]}$. The initial generation set consists of the 38 components among Claims 1–40 excluding 11 and 33; the source formal-evidence environment contains all 78 components. The following table gives each claim, its time contents i , auxiliary objects a and source locations. The time-use units comprise 151 pairs in the claim table and two contextual pairs concerning incidental radiation in Claim 71, totaling 153. Superscripts denote the roles in Table 169. The source-specific obligation atoms comprise 78 claims, 67 time contents, 22 transfers and 18 results, totaling 185.

Table 168: Bekenstein 1973 Physical and Formal Input Claim Components

p	Claim	i	a	Source Location
1	Kerr black-hole state parameters	$\emptyset$	1	pp. 2333–2335; Appendices A–C, pp. 2343–2345
2	Geometric data of a horizon cross-section	$\emptyset$	2	pp. 2333–2335; Appendices A–C, pp. 2343–2345
3	Irreducible-mass data	$\emptyset$	3	pp. 2333–2335; Appendices A–C, pp. 2343–2345
4	Horizon-area data	$\emptyset$	4	pp. 2333–2335; Appendices A–C, pp. 2343–2345
5	Thermodynamic-entropy state data	$\emptyset$	5	pp. 2333–2335
6	Mass-increment data	$\emptyset$	6	pp. 2334–2335
7	Charge-increment data	$\emptyset$	7	pp. 2334–2335
8	Angular-momentum-increment data	$\emptyset$	8	pp. 2334–2335
9	Physical-state alternatives	$\emptyset$	9	pp. 2335–2337

p	Claim	i	a	Source Location
10	Measurement-outcome alternatives and identification data	$\emptyset$	10	pp. 2335–2337
11	Causal constraints on external information access	30^C	11	pp. 2336–2339
12	Fixed- (M, Q, L) state data	$\emptyset$	12	pp. 2336–2339
13	Black-hole physical-state carrier	$\emptyset$	13	pp. 2335–2337
14	Carrier identities of the merging black holes	$\emptyset$	14	pp. 2333–2334, 2336–2337
15	Identity of the merger event	$\emptyset$	15	pp. 2333–2334, 2336–2337
16	Radiation carrier	$\emptyset$	16	pp. 2333–2334, 2336–2337
17	Radiative-energy data	$\emptyset$	17	pp. 2333–2334, 2336–2337
18	Planck-scale area data	$\emptyset$	18	pp. 2337–2338
19	One-bit information data	$\emptyset$	19	pp. 2337–2338
20	Black-hole state data	$\emptyset$	20	pp. 2333, 2339
21	External-system state data	$\emptyset$	21	pp. 2333, 2339
22	Black-hole entropy data	$\emptyset$	22	pp. 2333, 2339
23	External-entropy data	$\emptyset$	23	pp. 2333, 2339
24	Statistical ensemble	$\emptyset$	24	p. 2339
25	Macroscopic-state data	$\emptyset$	25	p. 2339
26	Harmonic-oscillator carrier	$\emptyset$	26	pp. 2339–2340
27	Oscillator energy-level data	$\emptyset$	27	pp. 2339–2340
28	Identity of the absorption event	$\emptyset$	28	pp. 2339–2340
29	Identity of the beam carrier	$\emptyset$	29	pp. 2340–2342; Appendix B, p. 2345
30	Photon carrier	$\emptyset$	30	pp. 2340–2342; Appendix B, p. 2345
31	Photon-energy data	$\emptyset$	31	pp. 2340–2342; Appendix B, p. 2345
32	Spatial geometric data of the beam-capture problem	$\emptyset$	32	pp. 2340–2342; Appendix B, p. 2345
33	Null incidence	44^S	33	pp. 2340–2342; Appendix B, p. 2345
34	Finite-size box carrier	$\emptyset$	34	pp. 2342–2343
35	Box-energy data	$\emptyset$	35	pp. 2342–2343
36	Box-size geometry	$\emptyset$	36	pp. 2342–2343
37	Radiation carrier	$\emptyset$	37	pp. 2342–2343

p	Claim	i	a	Source Location
38	Finite-size particle carrier	$\emptyset$	38	pp. 2343–2345
39	Particle spatial-position data	$\emptyset$	39	pp. 2343–2345
40	Size and spatial-position data of the near-horizon box	$\emptyset$	40	p. 2345, Appendix C
41	Kerr black-hole transformation	$5^S, 6^S, 7^S, 8^B, 9^B, 10^{XS}, 11^{XS}, 23^S$	1–4	pp. 2333–2335; Appendices A–C, pp. 2343–2345
42	Irreducible-mass relation	$\emptyset$	1, 3–4	pp. 2333–2335; Appendices A–C, pp. 2343–2345
43	Horizon-area relation	$1^{XS}, 7^S, 12^B, 13^B, 14^S$	2, 4, 14–15	pp. 2333–2335; Appendices A–C, pp. 2343–2345
44	Horizon-area–thermodynamic-entropy correspondence	$1^{XS}, 2^{XS}, 15^S$	4–5	pp. 2333–2335
45	Kerr variational relation	$16^Q, 23^S$	1–2, 6–8	pp. 2334–2335
46	First-law-type relation	$1^{XS}, 2^{XS}, 7^S, 16^Q, 23^S$	1–2, 5–8	pp. 2334–2335
47	Shannon-entropy functional	$\emptyset$	9–10, 24	pp. 2335–2337
48	Relation between information gain and entropy	$\emptyset$	9–10, 24	pp. 2335–2337
49	Thermodynamic states and entropy change before and after compression	$17^B, 18^B, 19^X$	5, 9–10	pp. 2335–2337
50	Fixed- (M, Q, L) equivalence class	23^S	12–13	pp. 2336–2339
51	Horizon constraint on access to interior information	$3^{XS}, 32^B, 33^B, 34^S, 37^S, 38^{XS}, 39^{XC}, 30^C$	11–13	pp. 2336–2339
52	Extension of the entropy introduced for stationary states to dynamical states	$23^S, 24^S$	13, 41	pp. 2335–2337
53	Initial–final area relation	$12^B, 13^B, 14^S$	4, 14–15	pp. 2333–2334, 2336–2337
54	Radiative-loss relation	$12^B, 13^B, 14^S, 31^S$	14–17	pp. 2333–2334, 2336–2337
55	Estimate of the entropy coefficient from finite-particle capture and one-bit identification	$16^Q, 23^S, 32^B, 33^B, 34^S, 35^B, 36^B, 60^S, 61^B, 62^B, 30^C$	18–19, 38–39, 42, 45, 48	pp. 2337–2338
56	Black-hole entropy functional	$1^{XS}, 24^S, 28^S, 29^{XS}$	4, 20, 22, 41, 48	pp. 2333, 2339
57	External entropy loss upon body absorption	$3^{XS}, 40^{XS}$	21, 23	pp. 2333, 2339
58	Generalized-entropy construction	$\emptyset$	22–23	pp. 2333, 2339

p	Claim	i	a	Source Location
59	Statistical-fluctuation distribution	$41^P, 42^{BP}$	24–25, 43	p. 2339
60	Generalized second law in the regime of negligible fluctuations	$4^{XS}, 41^P, 42^{BP}$	24–25, 43	p. 2339
61	Harmonic-oscillator Hamiltonian	43^Q	26–27, 46	pp. 2339–2340
62	Energy and entropy evaluation of oscillator absorption	$3^{XS}, 43^Q$	26–28, 42, 46, 48	pp. 2339–2340
63	Oscillation-frequency relation	43^Q	26–27, 46	pp. 2339–2340
64	Photon-energy spectrum	$48^Q, 49^Q, 50^Q$	29–31, 47	pp. 2340–2342; Appendix B, p. 2345
65	Null-geodesic capture conditions and area-increase bound	$16^Q, 23^S, 44^S, 46^S, 47^B, 58^Q, 59^Q$	1–2, 29–32, 44, 50	pp. 2340–2342; Appendix B, p. 2345
66	Null-beam propagation relation	$23^S, 44^S, 45^{XS}, 46^S$	29–30, 32, 44	pp. 2340–2342; Appendix B, p. 2345
67	Wavelength and size constraints and efficiency bound for a finite box	$48^Q, 50^Q, 53^B, 54^B, 55^B, 56^S$	34–37, 47–49	pp. 2342–2343
68	Heat-engine energy balance	$53^B, 54^B, 55^B, 56^S, 57^S$	34–37, 49	pp. 2342–2343
69	Kerr line element and coordinate differentials	$23^S, 58^Q, 59^Q$	1–2, 44	pp. 2343–2345
70	Geodesic-constant relation	$16^Q, 23^S, 58^Q, 59^Q, 60^S, 61^B, 62^B$	38–39, 44–45	pp. 2343–2345
71	Finite-particle area-increase relation	$16^Q, 23^S, 34^S, 35^B, 36^B, 60^S, 61^B, 62^B$	38–39, 42, 44–45	pp. 2343–2345
72	Entropy change upon photon capture	$16^Q, 23^S, 44^S, 47^B, 48^Q, 58^Q, 59^Q$	29–32, 47–48, 50	pp. 2341–2342; Appendix B, p. 2345
73	Near-horizon box-energy relation	$16^Q, 23^S, 53^B, 58^Q, 59^Q, 65^S, 66^Q, 67^B$	34–36, 40, 44–45, 49	p. 2345, Appendix C
74	Angular-rate relation for a locally nonrotating reference frame	$23^S, 58^Q, 59^Q, 65^S, 66^Q$	40, 44	p. 2345, Appendix C
75	The generalized-second-law conjecture and compensation condition for the same process	$3^{XS}, 4^{XS}, 29^{XS}, 40^{XS}, 30^C$	20–23, 43, 48	pp. 2339

p	Claim	i	a	Source Location
76	Washing out of initial-condition effects in thermodynamic evolution	$20^S, 21^B, 22^B$	5, 9–10, 24	pp. 2335,2336
77	Loss of initial peculiarities as the black hole approaches equilibrium	$25^S, 26^B, 27^B$	1, 12–13	pp. 2336
78	The sparse-photon regime and conditional per-quantum entropy test	$48^Q, 49^Q, 50^Q, 51^S, 52^S$	29–32, 43, 47–48	pp. 2341,2342

Time Contents 63–64 belong to the context of incidental radiation excluded from the Appendix A calculation. The correspondence between entropies before and after compression is Time Content 19; the quantitative lower bound on area increase is Auxiliary Object 42.

Table 169: Bekenstein 1973 Time-Use Roles

Code	Use Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causal
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 170: Bekenstein 1973 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–2, 4, 7, 20, 28–29, 40, 44, 46, 51–52, 56, 60	member	order_relation
3	member	order_relation, transition_position
5, 14, 24, 31, 34, 37, 39, 41, 57, 65	member	transition_position
6, 15	member	orientation
8–9, 12–13, 17–18, 21, 26, 32–33, 36, 47, 53, 55, 62	member	endpoint
10–11, 19, 23, 38, 45	dependency	$\emptyset$
16, 43, 48–50, 58–59, 66	separate	$\emptyset$
22, 25, 27	member	phase_boundary

Time Indices	Structural Type	Axes
30	member	order_relation, orientation
35, 42, 54, 67	member	instant
61	member	turning_position
63–64	lexical	$\emptyset$

Dependency Inputs

Dependency Inputs Each of the 22 transfers has at least one time-content input.

Dependency and Transfer Elements

Definition 8.273 (Bekenstein 1973 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [22]$. Table 171 specifies time inputs and outputs, claim support, auxiliary inputs and transfer relations.

Table 171: Bekenstein 1973 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
1	3, 29–30, 40	4	57–58, 75	20–23, 43, 48	Application of entropy compensation to the same absorption process
2	5	7	41	1–4	Extraction of the course of the specified black-hole transformation
3	7	1	43	2, 4	Application of the area theorem to the specified process
4	1	15	44	4–5	Identification of a preferred direction from area nondecrease
5	2	15	44	5	Identification of a preferred direction from thermodynamic entropy increase
6	30, 32, 34	8–9, 33, 39	41, 51	11–13	Extraction of boundaries and external information access for completed particle capture
7	14	12–13	53	4, 14–15	Extraction of initial and final boundaries of a specified merger history
8	17–18	19	47, 49	5, 9–10	Correspondence between compression states and their entropies

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
9	20–21	22	76	5, 9–10, 24	Identification of the late loss of initial effects in a specified thermodynamic evolution
10	25–26	27	77	1, 12–13	Identification of the late loss of initial effects as the black hole approaches equilibrium
11	1, 28	29	52, 56	4, 20, 22, 41, 48	Application of the monotone entropy formula to the area order of the same black-hole history
12	30, 37	38	51–52, 56	5, 11, 13, 22, 41, 48	Assessment of irreversibility from formation and constrained information access
13	23, 44	45	66	29–30, 32, 44	Reversible beam propagation along the specified null path
14	16, 23, 44, 58–59	46	65–66	1–2, 29, 32, 44, 50	Horizon approach of a null path satisfying the capture condition
15	16, 23, 44, 46, 58–59	47	65	1–2, 29, 32, 44, 50	Extraction of the capture boundary from the capturable incident path and approach
16	48	50	64	29–31, 47	Evaluation of the characteristic frequency of the unfiltered thermal spectrum
17	56	53–55	68	34–37, 49	Extraction of lowering, release and retrieval stages of the specified box cycle
18	56	57	68	34–37, 49	Extraction of the energy-conversion process of the specified box cycle
19	58	59	69	1–2, 44	Formation of the differential of the Kerr time coordinate
20	23, 60	61	70	38–39, 44–45	Selection of a turning point on the given geodesic
21	23, 34, 61	36, 62	70–71	38–39, 44–45	Extraction of the capture boundary of the specified turning-point capture process
22	23, 58–59, 65	66	74	40, 44	Calculation of angular rate from the time reference and locally nonrotating frame

Grounding Conditions

Conditions for the Dependencies or Transfers Each transfer holds under its tabulated inputs, claim support and the following source conditions.

1. pp. 2333,2339. For the same absorption process, $\Delta S_c = -S$, and the compensation condition of the generalized-second-law conjecture is $\Delta S_{\text{bh}} \geq S$. The irrecoverable loss of information uses the one-way character of the event horizon.
2. pp. 2333–2334. The ordered history of the given black-hole transformation supplies the temporal course of that same transformation.
3. pp. 2333–2334. Under the conditions of the classical area theorem, the given black-hole process has nondecreasing area.
4. pp. 2333–2335. The area-nondecrease order in irreversible area-increasing processes identifies the preferred direction stated in the source.
5. pp. 2333–2335. For entropy-increasing processes of a closed thermodynamic system, the preferred direction is identified within the statistical domain of the second law.
6. pp. 2333,2336–2339. The given completed particle capture, pre-capture exterior information access and one-way horizon supply the capture boundary, final-result boundary, post-capture access and its change.
7. pp. 2333–2334,2336–2337. The given merger history supplies its initial and final boundaries, conditional on formation of a final black hole.
8. pp. 2335–2336. For the specified compression, the states before and after compression and their probability distributions are given. The relation $S = -\sum_n p_n \ln p_n$ associates the two states with their entropies.
9. pp. 2335–2336. Given the evolution and initial stage in the source’s macroscopic thermodynamic description, the late stage at which the effects of the initial conditions are lost is identified.
10. p. 2336. In the given approach of a black hole to equilibrium, the effects of its initial peculiarities are lost, leaving a late stage whose exterior description is determined by M, Q, L .
11. pp. 2336–2337. For the same black-hole history, area nondecrease and the increasing function $S_{\text{bh}} = f(\alpha)$ give entropy nondecrease.
12. pp. 2336–2339. For the given formation process, the area-proportional black-hole entropy is compared with the ordinary entropy before formation. The large increase described on source p. 2338 and the one-way information access across the horizon support the assessment of irreversible formation.
13. pp. 2340–2341. The specified null path lies in a stationary, uncharged Kerr background. In geometrical optics, beam propagation before absorption is reversible.
14. pp. 2340–2341,2345. The null-geodesic constants and capture-barrier condition of Appendix B are used. Horizon approach is selected for a capturable incident path whose energy exceeds the effective-potential barrier. Equations (8) and (A6), using the Kerr time reference and Ω , support the barrier evaluation.
15. pp. 2340–2341,2345. The same incident null path and approach satisfying the Appendix B condition supply the capture boundary of that path.
16. pp. 2340–2341. For the unfiltered thermal Planck spectrum, the characteristic frequency satisfies $\hbar\omega_c \sim T$.

17. pp. 2342–2343. The given controlled cycle comprises slow lowering of the box, radiation release and box retrieval in that order. Each stage is extracted from the complete protocol.
18. pp. 2342–2343. The energy balance of the specified cycle, including lowering, release, retrieval and work extraction, identifies its energy-conversion process.
19. pp. 2343–2344. The given differentiable Kerr time coordinate t supplies its coordinate differential dt .
20. pp. 2343–2344. The given Kerr geodesic, its constants and the radial turning condition select an existing turning position on that geodesic.
21. pp. 2343–2345. The particle-capture process specified using a turning point supplies its own capture boundary.
22. p. 2345. For the locally nonrotating frame, the specified time reference and metric coefficients give $d\phi/dt = -g_{t\phi}/g_{\phi\phi}$, with $g_{\phi\phi} \neq 0$.

Results

Definition 8.274 (Bekenstein 1973 Results). Let $\text{idx}_R^{\text{audit,result}} = [18]$. The following table gives the results and source locations.

Table 172: Bekenstein 1973 Results and Source Locations

j	Result	Source Location
1	Structural analogy between the black-hole area theorem and the second law of thermodynamics	pp. 2333–2335
2	Proposal of black-hole entropy based on inaccessible information and area monotonicity	pp. 2335–2338
3	Estimate of the entropy coefficient from finite-particle capture and one-bit identification	pp. 2337–2338, 2343–2345
4	Definition of the generalized-entropy sum	p. 2339
5	Proposal of the generalized second law from information loss and compensation in the same absorption process	pp. 2333, 2339
6	Coupling of black-hole entropy and missing information	pp. 2335–2337
7	Change in exterior information access accompanying particle capture	pp. 2336–2339
8	Distinction between reversible and irreversible transformations	pp. 2333–2334
9	Area and entropy evaluation of merger, formation and capture	pp. 2333–2334, 2336–2339, 2343–2345
10	Statistical fluctuations and the domain of the macroscopic generalized entropy law	p. 2339
11	Generalized entropy change in nonrelativistic oscillator absorption	pp. 2339–2340, 2343–2345
12	Conditional test using a null beam and photon spectrum	pp. 2340–2342, 2345

j	Result	Source Location
13	Limit of the continuous-beam description in the sparse regime and the entropy test for individual quanta	pp. 2341–2342,2345
14	Lowering, release and retrieval cycle of a finite box and its efficiency bound	pp. 2342–2345
15	Work and entropy change from radiation release and box retrieval	pp. 2342–2345
16	Kerr-geodesic turning-point energy and the area bound for finite-particle capture	pp. 2343–2345
17	Domains specified by statistical, frequency and finite-size conditions	pp. 2338–2345
18	Coupling of area mechanics, information and the generalized second law	pp. 2333–2339,2343–2345

Result Dependencies

Definition 8.275 (Bekenstein 1973 Result-Dependence Hyperedges). For each $j \in [18]$, the following table specifies the time premises, claim support, auxiliary inputs and result output of $\text{ResultEdge}_{R,j}$.

Table 173: Bekenstein 1973 Result-Dependence Hyperedges

j	Time Premises i	Claim Support p	Auxiliary In a	Result Output Index
1	1–2, 6, 15	41–44	1–5	1
2	1, 12–14, 23–24, 28–31	42, 47–48, 50–54, 56	1–4, 9–17, 41	2
3	16, 23, 30, 32–36, 58–62	47–48, 51, 55, 69–71	1–2, 18–19, 38–39, 42, 44–45, 48	3
4	$\emptyset$	58	22–23	4
5	3–4, 29–30, 40	51, 56–58, 75	20–23, 43, 48	5
6	17–23, 25–27, 30	47–51, 76–77	5, 9–13, 24	6
7	30, 32–34, 39	51	11–13	7
8	1, 5, 7–11, 23	41–43	1–4, 13	8
9	1, 12–14, 16, 23, 30–31, 34–38, 58–62	43, 51–54, 56, 69–71	1–5, 11, 13–17, 22, 38–39, 41–42, 44–45, 48	9
10	4, 41–42	59–60	20–25, 43	10
11	3, 16, 23, 34–36, 43, 58–62	56–58, 61–63, 69–71	1–2, 26–28, 38–39, 42, 44–46, 48	11
12	16, 23, 44–50, 58–59	56–58, 64–66, 72	1–2, 29–33, 44, 47–48, 50	12

j	Time Premises i	Claim Support p	Auxiliary In a	Result Output Index
13	16, 23, 44, 47–52, 58–59	64–65, 72, 78	1–2, 29–32, 43–44, 47–48, 50	13
14	16, 23, 48, 50, 53–61, 65–67	67–70, 73–74	1–2, 34–37, 40, 44–45, 47–49	14
15	16, 23, 48, 50, 53–61, 65–67	56–58, 67–70, 73–74	1–2, 34–37, 40, 44–45, 47–49	15
16	16, 23, 34–36, 58–62	69–71	1–2, 38–39, 42, 44–45	16
17	16, 23, 41–44, 48–53, 56, 58–61, 65–67	59–61, 64–67, 69–71, 73–74, 78	1–2, 24–27, 29–32, 34–40, 43–50	17
18	1–4, 15–16, 23, 29–30, 32–36, 40, 58–62	41–48, 50–51, 55–58, 69–71, 75	1–12, 18–23, 38–39, 42–45, 48	18

The time-premise and auxiliary-premise occurrences each total 193.

Grounds of Establishment

- Grounds for the Results**
- 1. Area, Information and Entropy** Results 1–3 and 6–9 connect the classical area theorem, fixed- (M, Q, L) information classes, the one-way horizon and $S_{\text{bh}} = f(\alpha)$. The function f is increasing; proportionality to area is supported by merger/radiation constraints and the information-theoretic evaluation of finite-particle capture. For the Shannon quantity $S = -\sum_n p_n \ln p_n$, average conditioning satisfies $H(X | Y) \leq H(X)$. In maximum-entropy evaluation, additional constraints on the admissible distributions make the maximum nonincreasing.
 - 2. Coefficient and Characteristic Temperature** Result 3 uses the finite-particle radius bound $b \geq \max(\hbar/\mu, 2\mu)$ and $\Delta\alpha_{\min} = 2\mu b$. Identifying the minimum area increase $2\hbar$ with the one-bit information loss $\ln 2$ gives $S_{\text{bh}} = (\ln 2/2)\alpha/\hbar$. Given the quantum uncertainty in particle radius, the source estimates that the coefficient is very close to $\ln 2/2$, probably within a factor of two, and adopts $\eta = \ln 2/2$ in its subsequent examples. The characteristic temperature $T_{\text{bh}} = 2\hbar\Theta/\ln 2$ is used in the box-efficiency calculation.
 - 3. Generalized Second Law** Result 4 defines the sum $S_{\text{bh}} + S_c$; Result 5 conjectures its nondecrease. For the same body-absorption process, $\Delta S_c = -S$, and the source conjectures compensation $\Delta S_{\text{bh}} \geq S$ on the basis of lost information access. Result 10 concerns the macroscopic regime in which statistical and quantum fluctuations are negligible.
 - 4. Oscillator Absorption** Result 11 uses a nonrelativistic harmonic oscillator formed by two particles of mass $m/2$, with $T \ll m$ and $\xi = m/\langle E \rangle \gg 1$. For $x = \hbar\omega/T$, the normalized distribution is $p_n = (1 - e^{-x})e^{-nx}$. It gives $S = x/(e^x - 1) - \ln(1 - e^{-x})$ and $\langle E \rangle = \hbar\omega[(e^x - 1)^{-1} + 1/2]$. Using the finite-particle capture bound, the total entropy change is positive in this regime. Radiation inside the box is excluded from the calculation.

5. **Beam and Sparse-Quantum Regime** Results 12 and 13 concern a null beam in a stationary, uncharged Kerr background. The Appendix B capture barrier gives $\Delta\alpha \geq \beta ME$, with $4(1 - \sqrt{3}/2) \leq \beta \leq 8$. The thermal beam uses $S = 4E/(3T)$ and $M\omega_c \gg 1$. In the filtered example, the lower cutoff ω'_c is chosen independently of temperature, with $M\omega'_c \gg 1$ and $\hbar\omega'_c/T \gg 1$. At low temperatures in the sparse regime, the continuous mean-beam approximation fails, and the source treats individual photons. A fixed-frequency photon with two equally probable polarizations has entropy $\ln 2$; the total-change lower bound $(\beta \ln 2/2)M\omega - \ln 2$ is positive for sufficiently large $M\omega$.
6. **Finite Box, Geodesics and Reference Frame** Results 14–16 use slow box lowering, radiation release, retrieval and the finite-particle turning-point calculation. Appendix A assumes a neutral finite particle and $\delta \ll r_+ - r_-$; its leading-order evaluation neglecting incidental radiation gives $E - \Omega p_\phi \geq 2\mu b\Theta$. In the locally nonrotating frame, $d\phi/dt = -g_{t\phi}/g_{\phi\phi}$ and $p_\phi = 0$, and the release energy is $E = 2\mu b\Theta$. For the box, $b\Theta \ll 1$, $T_{\text{bh}} \ll T$ and the finite-wavelength condition give efficiency $< 1 - T_{\text{bh}}/T$. Radiation release gives $\Delta S_{\text{bh}} > (\pi \ln 2/2)\Delta\mu/T$ and $\Delta S_c = -\Delta\mu/T$, hence a positive total change.
7. **Conditions and Synthesis** Result 17 collects the respective conditions for the oscillator, beam, finite particle and box. The limiting Kerr rays of Appendix B and the nonextremal near-horizon expansion of Appendix A retain their respective conditions. Result 18 connects area mechanics, information, the entropy coefficient and the generalized-second-law conjecture.

Evidence Hierarchy

Direct Measurement Subjects The source presents no new direct-measurement subjects of its own.

Direct Measurement Predicates The source establishes no predicates through direct measurements of its own.

Conditions and Domain of Direct Measurement The source-specific set of direct-measurement conditions and domains is empty.

Subjects of Formal Generalization The analysis connects the area–entropy structural analogy, information access, capture, merger and formation with the oscillator, beam and box calculations. The generalized second law is proposed as a conjecture supported by the individual conditional tests.

Conditions and Domain of Formal Generalization The Kerr, statistical, semiclassical, frequency, finite-size and near-horizon conditions for each result are those stated in the grounds above.

Generation Judgment Judgment Values: LCTR complete-generation judgment: 0; source-evidence-conditioned local reach: 0; remaining primitive-time roots: 67.

Generation Reach and Primitive-Time Roots From the initial set of 38 non-temporal physical inputs, no time content is reached. Supplying the source formal-evidence environment also yields no local time-content reach. All 67 time contents belong to the primitive-time-root set.

Result Dependence

Result Dependencies The support families in Proposition 8.277 give 149 primitive-time-root–result dependence pairs and 116 necessary-root–result dependence pairs. Result 4 is evidence-only; the 17 Results 1–3 and 5–18 are primitive-time-root-dependent.

Evidence Effect

Domain of Evidence Effect The evidence concerns area, entropy and information relations; the generalized-second-law conjecture; merger, formation and capture; and the oscillator, beam and box tests. Each result holds under the Kerr, statistical, semiclassical, frequency, finite-size and near-horizon conditions stated above.

Boundary of Evidential Force The boundary of evidential force is set by the conditions and domains of the calculations supporting the generalized-second-law conjecture.

Result-Preservation Domain

Result-Preservation Domain Result 4 is preserved by the supplied entropy values and sum definition. The other 17 results are preserved under time inputs belonging to their support families and the source conditions. When the temporal premises are realized by generated-time representations preserving the relations required by the source, the corresponding results are preserved within their domains.

Results Preserved after Reordering into the LCTR Generation Order When the temporal premises of a support family are realized by generated time representations preserving the source relations, Results 1–3 and 5–18 are preserved under their source conditions. Result 4 is preserved by the definition of the sum of entropy values.

Placement of the Grounding Obligation

Placement of the Grounding Obligation The calculations use area/entropy orders, capture/merger/relaxation histories, frequencies, the Kerr time reference and the one-way horizon. There are 65 time contents with law-bearing uses, none of which is reached by generation. The grounding obligation is “backgrounded by primitive-time inputs.” Time Contents 63–64 are mentions of incidental radiation excluded from the calculation.

Evidence of Actual Failure

Evidence of Actual Failure Section VI.B establishes from its calculation that the continuous mean-beam approximation fails in the low-temperature sparse regime and turns to individual quanta. This concrete condition failure concerns the domains of Results 12 and 13. Actual failure of the generation conditions for time content is undetermined from the source evidence.

Downstream Reach

Downstream Reach Failure of the continuous-beam approximation reaches the mean-beam entropy evaluation that depends on it. The individual-quantum evaluation holds under its fixed-frequency, two-polarization and capture conditions. Structural reach from primitive-time roots to the 17 dependent results is given by the tabulated dependencies and minimal support families.

Repair Scope

Repair Scope The source beam analysis restricts the continuous approximation and evaluates individual photons in the sparse regime. Repair of physical time use concerns the relevant time content and its generation conditions before use. If actual failure of a specific generation condition is independently established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.276 (Bekenstein 1973 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [67]; \\ \emptyset, \{4\}; \\ [18] \setminus \{4\}, \emptyset).$$

Proof. The audit boundary has 163 nodes—78 claim nodes, 67 time-content nodes and 18 result nodes—and 40 edges—22 transfers and 18 results. The initial set contains the 38 nodes among Claims 1–40 excluding 11 and 33. Each edge requires claim support outside this set, so generation closure remains the initial 38 nodes. Supplying all 78 claims as the source formal-evidence environment reaches Result 4, which has no time premise. Every one of the 22 transfers has a nonempty time input, so no time content is locally reached. Adding the 67 roots to the seed reaches the other 17 results. $\square$

Proposition 8.277 (Bekenstein 1973 Minimal Primitive-Time-Root Support). Table 174 gives each minimal support family in the order: number of members, minimum cardinality and maximum cardinality. Here $\mathcal{A}_R(4) = \{\emptyset\}$, and the total number of family members across all results is 37.

Table 174: Bekenstein 1973 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(3; 3, 3)
2	(3; 7, 7)
3	(1; 8, 8)
4	(1; 0, 0)
5	(4; 4, 5)
6	(1; 8, 8)
7	(1; 3, 3)
8	(2; 6, 7)
9	(3; 11, 11)
10	(5; 3, 7)
11	(1; 8, 8)
12	(1; 6, 6)
13	(1; 8, 8)
14	(1; 8, 8)

Result Index	Support Statistics
15	(1; 8, 8)
16	(1; 6, 6)
17	(1; 15, 15)
18	(6; 13, 13)

Proof. Apply the update in Definition 7.46 to the 22 transfer hyperedges and 18 result-dependence hyperedges until every support antichain is unchanged. Proposition 7.47 uniquely determines the support families of the 18 results and the displayed counts. $\square$

Proposition 8.278 (Bekenstein 1973 Physical-Time Qualification). The 65 time contents with law-bearing uses lack physical-time qualification, and Results 1–3 and 5–18 depend on time contents lacking that qualification.

Proof. Of the 67 primitive-time roots, the 65 other than contextual Time Contents 63–64 have law-bearing uses. Apply Definition 7.10, Theorem 7.11 and Corollary 7.12 to these roots and the 17 results. $\square$

Individual Statements Verbatim Original (English):

Changes of a black hole or of a system of black holes select a preferred direction in time

The original-source location is Bekenstein, *Black Holes and Entropy*, *Physical Review D* 7, p. 2334, §II.

Verbatim Original (English):

purely formal nature

At the end of Section II on pp. 2334–2335, the source describes the analogy developed up to that stage as formal. Sections III and IV then provide a physical construction of black-hole entropy based on information theory.

Verbatim Original (English):

natural to conjecture

Page 2339, §V, presents the generalized second law as a conjecture. Page 2338, §IV, introduces T_{bh} in Eq. (18) as a characteristic temperature and cautions against identifying it with the actual black-hole temperature. On the same page, the source estimates that the entropy coefficient is very close to $\ln 2/2$, probably within a factor of two, and adopts $\eta = \ln 2/2$ thereafter. The source attributes this uncertainty in the coefficient to the absence of a full quantum treatment.

Objects of the Statements The direction in time selected by changes in a black hole; the formal nature of the analogy with thermodynamics; and presentation of the generalized second law as a conjecture.

Original-Source Locations The first quotation is from p. 2334, §II; the second, from pp. 2334–2335; and the third, from p. 2339, §V. The qualification of the characteristic temperature in Eq. (18) is on p. 2338, §IV.

Minimal Context Required for Interpretation The structural correspondence between the area theorem and thermodynamic entropy increase precedes the first quotation. The second concerns the stage reached in Section II and is followed by the information-theoretic physical construction in Sections III–IV. The third states the generalized second law of Section V as a conjecture. Section IV attributes the uncertainty of the semiclassical entropy coefficient to the absence of a full quantum treatment and specifies the meaning of the characteristic temperature.

Attribution of the Statements All three statements are attributable to Jacob D. Bekenstein.

Predicates Assigned to the Objects by the Statements The first statement says that black-hole changes select a preferred direction in time. The second describes the analogy through Section II as formal. The third states the generalized second law as a conjecture.

Explicit Domains of Application of the Statements The first statement applies to changes in a black hole or a system of black holes; the second, to the initial analogy between black-hole physics and thermodynamics; and the third, to the proposal of the generalized second law. The qualification of the characteristic temperature applies to Eq. (18) and its semiclassical evaluation.

Individual Audit

Interpretations Allowed by the Original Text and Minimal Context The first statement says that changes in a black hole or a system of black holes select a preferred direction in time. The second describes the thermodynamic analogy developed through Section II as formal at that stage. Sections III and IV then construct black-hole entropy on information-theoretic grounds. The third statement presents the generalized second law as a conjecture.

Independent Grounds The grounds are the area-theorem/second-law correspondence, constrained information access, finite-particle capture and the conditional oscillator, beam and box calculations.

Evidence-Effect Boundary for Each Interpretive Possibility The first statement concerns the direction of area and entropy change; the second, the formal analogy preceding the information-theoretic construction; and the third, the generalized-second-law conjecture in the regime of negligible statistical and quantum fluctuations. None of the temporal premises of these calculations is reached by generation; physical-time qualification follows from the primitive-time-root judgment.

Corresponding Formal Evidence The first statement corresponds to Time Contents 1, 2 and 15 and Result 1; the second to Result 1 and Results 2 and 3 from Sections III–IV; and the third to Claims 57, 58, 60 and 75 and Results 4, 5 and 10. Tables 166, 168, 171 and 172 specify the 67 time contents, 78 claims, 22 transfers and 18 results, respectively.

Within-Reference Comparison The formal analogy of Section II is followed by the connection between information access and area entropy in Sections III–IV. The semiclassical coefficient estimate and characteristic temperature of Section IV connect

to the generalized-second-law proposal in Section V and the conditional tests in Sections VI–VII. Appendix B supports the beam-capture bound; Appendices A and C support the finite-particle and box calculations near the horizon. The three statements concern the direction of area and entropy change, the formal analogy through Section II and the generalized-second-law conjecture in Section V. The information-theoretic construction in Sections III–IV and the conditional tests in Sections VI–VII provide their respective physical content.

8.19.19 Source Audit

Audit Input

Original Source and Audit Scope

Reference Identifier 19.

Reference Name Primitive-time-root audit of Misner–Thorne–Wheeler 1973

Bibliographic Identification Let

$$R = \text{Ref}_{\text{MTW73Grav}}$$

denote Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler, *Gravitation*, San Francisco: W. H. Freeman and Company, 1973. The audit domain comprises Chapters 1, 2, 13, 16, 21, 27–30, 38, 40, and 43, and the relevant boxes in the original source.

Original-Source Location Identifier 19-L1.

Original-Source Location Content Chapters 1, 2, 13, 16, 21, 27–30, 38, 40, and 43, and the relevant boxes in the original source. The source locations of time contents, auxiliary objects, claims, and results are given, respectively, in Tables 175, 176, 177, and 181.

Audit-Scope Identifier 19-SC1.

Audit-Scope Content The audit domain stated in the bibliographic identification.

Formal Objects

Object Identifier 19-F1.

Object Name Formal objects.

Object Type Formal object.

Formal Content Definitions, assumptions, and derivations concerning event order, coordinate time, worldlines, proper time, ideal and atomic clocks, material processes, spatial slicing, many-fingered time, cosmological time, redshift, quantum geometrodynamics, classical history construction and semiclassical spacetime reconstruction, time standards and representations of the speed of light, conversion between seconds and length, photon energy–frequency–wavelength relations, and the time dependence appearing in the measured gravitational constant.

Set of Original-Source Location Identifiers 19-L1.

Object Identifier 19-F2.

Object Name Objects that use time.

Object Type Formal object.

Formal Content The audit uses 43 time contents and 40 auxiliary objects, with their names and source locations tabulated below.

Set of Original-Source Location Identifiers 19-L1.

Assumptions

Assumption Identifier 19-A1.

Assumption Content Each use has its conditions on the Lorentzian metric, material clock laws, precision and operation, ADM initial constraints and boundaries, the specified cosmological model, complete EHJ families or semiclassical packets.

Set of Original-Source Location Identifiers 19-L1.

Nontemporal Physical Input

Input Identifier 19-P1.

Input Name Nontemporal physical input.

Input Type Nontemporal physical input.

Input Content Initial Claims 1–12 supply event/coincidence records, physical bodies, atoms, molecules and the human body, clock and length-measuring apparatus with display/length data, spatial three-metric values, and material-state/stress values.

Set of Original-Source Location Identifiers 19-L1.

Time Contents

Time-Content Identifier 19-T1.

Time-Content Name Time Contents in the original source.

Time-Content Type Time content.

Time-Content Statement The individual Time Contents are given in Definition 8.279.

Set of Original-Source Location Identifiers 19-L1.

Formal Input Elements

Formal-Component Identifier 19-C1.

Formal-Component Name Composition of the reference's formal-evidence environment.

Formal-Component Type Formal component.

Formal-Component Text The source formal-evidence environment contains all 103 claims.

Set of Original-Source Support Identifiers 19-L1.

Definition 8.279 (Misner–Thorne–Wheeler 1973 Time Contents and Auxiliary Objects). For $R = \text{Ref}_{\text{MTW73Grav}}$, let $n_R = 43$. The following tables give the 43 time contents and 40 auxiliary objects.

Table 175: Misner–Thorne–Wheeler 1973 Time Contents

i	Time Content	Source Location
1	Coordinate-time label	Ch1–2;§13.1–3; Exercise13.7 p. 324;§13.6
2	Parameter of a timelike worldline	Ch1–2;§13.1–3; §13.4 pp. 315–324
3	Time standard used in inertial motion and clock comparisons	§1.5 pp. 23–29;Box1.4
4	Time unit, converted interval, and fiducial metric scale	§1.6;Box1.8 p. 36; Exercise13.7 p. 324;§13.6
5	Timelike interval and directional structure of the Lorentz metric	Ch1–2;§13.1–3; §1.6;Box1.8 p. 36; §13.4 pp. 315–324
6	Photon frequency relative to an observer	§2.8 pp. 63–64
7	Metric proper time along a timelike curve	§13.4 pp. 315–324; Exercise13.7 p. 324;§13.6
8	Stationary proper time between fixed endpoints	§13.4 pp. 315–324
9	Parameter obtained by an affine transformation of proper time	§13.4 pp. 315–324; Exercise13.7 p. 324;§13.6
10	Proper-time reading of an ideal clock	§16.4 pp. 393–396;Box16.2–3
11	Material-clock reading under its operating conditions	Ch16;Box16.1; §16.4 pp. 393–396;Box16.2–3
12	Velocity-dependent clock-rate and interval comparison	§16.4 pp. 393–396;Box16.2–3
13	Change of material and biological processes along proper time	§16.4 pp. 393,396
14	Standard timelike interval for geodesic-clock calibration	Box16.4 pp. 397–399
15	Calibrated geodesic-clock tick interval	Box16.4 pp. 397–399
16	Timelike interval measured with a geodesic clock	Box16.4 pp. 397–399
17	Foliation label in the ADM decomposition	§21.4–5 pp. 506–513; §21.8–13
18	Lapse relating normal proper interval to the foliation label	§21.4–5 pp. 506–513
19	Proper interval along a normal connector	§21.4–5 pp. 506–513
20	Local temporal displacement of a hypersurface	§21.4–5 pp. 506–513; §21.8–13
21	Coordinate-time evolution of initial data	§21.8–13
22	York time variable defined from extrinsic curvature	§21.8–13
23	Cosmic time from proper intervals of the cosmic fluid	§27.3–4 pp. 715–717
24	Conformal time	§27.9–11
25	Scale factor used as a time parameter on a monotone branch	§27.9–11
26	Expansion rate H with respect to cosmic time and its present value H_0	Box27.1 p. 709;Box27.3–4;Ch29

<i>i</i>	Time Content	Source Location
27	Hubble time	Box27.1 p. 709;Box27.3–4;Ch29
28	Age inferred from stellar evolution	Box27.1 p. 709;Box27.3–4;Ch29
29	Age in the specified cosmological model	Box27.1 p. 709;Box27.3–4;Ch29
30	Cosmological extrapolation of local proper-time standards	Ch28–30; Box30.1 pp. 813–814
31	Cosmic age from a calibrated clock sequence	Box30.1 pp. 813–814
32	Proper crest intervals at emission and reception	§38.3–5 pp. 1054–1058
33	Clock-rate ratio between emission and reception	§38.3–5 pp. 1054–1058
34	Time argument of the measured gravitational coupling	§40.8 pp. 1121–1126;§40.9
35	Temporal change of spatial geometry	§43.1 pp. 1180–1183
36	Instant assigned to a three-geometry	§43.1 pp. 1180–1183
37	Temporal rate of change of three-geometry	§43.1 pp. 1180–1183
38	Classical temporal history used in the comparison with quantum theory	§43.1 pp. 1180–1183
39	Before, after, and next denied in the no-time statement	§43.1 pp. 1180–1183
40	Dynamically allowed classical history leaf	§43.2 pp. 1184–1185; §43.2–3 pp. 1185–1189
41	Reconstructed temporal relations of events	§43.2 pp. 1184–1185
42	Temporal structure of the reconstructed classical four-geometry	§43.2 pp. 1184–1185; §43.2–3 pp. 1185–1189
43	Earlier–later boundary relation in the ADM action	§43.2–3 pp. 1185–1189

Table 176: Misner–Thorne–Wheeler 1973 Auxiliary Objects

<i>a</i>	Auxiliary Object	Source Location
1	Event-identification data	Ch1–2;§13.1–3; §38.3–5 pp. 1054–1058; Box30.1 pp. 813–814
2	Curve, endpoint, and admissible-variation data	Ch1–2;§13.1–3; §13.4 pp. 315–324; Exercise13.7 p. 324

<i>a</i>	Auxiliary Object	Source Location
3	Metric components and geometric domain	Ch1–2; §13.1–3; §1.6; Box1.8 p. 36; §13.4 pp. 315–324; §16.4 pp. 393–396; Box16.2–3; §21.4–5 pp. 506–513; Box30.1 pp. 813–814; §38.3–5 pp. 1054–1058; §43.1 pp. 1180–1183; Exercise13.7 p. 324
4	Affine transformation constants	§13.4 pp. 315–324; Exercise13.7 p. 324; §13.6
5	Clock apparatus and display states	§16.4 pp. 393–396; Box16.2–3; Box30.1 pp. 813–814; §38.3–5 pp. 1054–1058; §1.5 pp. 23–29; Box1.4
6	Rod apparatus and spatial-length data	§16.4 pp. 393–396; Box16.2–3; §13.4 pp. 315–324
7	Physical carriers: bodies, atoms, molecules, and biological matter	§16.4 pp. 393–396; Box16.2–3; §16.4 pp. 393, 396
8	Matter laws and measurement coupling	§16.4 pp. 393–396; Box16.2–3; Ch16; Box16.1; §16.4 pp. 393, 396; Box30.1 pp. 813–814
9	Apparatus structure, precision, and operating conditions	§16.4 pp. 393–396; Box16.2–3
10	Specified worldlines, velocities, and comparison events	§16.4 pp. 393–396; Box16.2–3
11	Geodesic, light-signal, and Schild-construction data	Exercise13.7 p. 324; Box16.4 pp. 397–399
12	Geodesic-clock tick counts and comparison data	Box16.4 pp. 397–399
13	Geometry of spacelike hypersurfaces, embeddings, and normals	§21.4–5 pp. 506–513
14	Spatial shift components	§21.4–5 pp. 506–513; §21.8–13
15	Values of the three-metric and canonical momentum	§21.4–5 pp. 506–513; §21.8–13; §43.1 pp. 1180–1183; §21.9–11 pp. 534–543; §21.11 p. 540; §43.3 p. 1186 Eqs43.6–43.10

<i>a</i>	Auxiliary Object	Source Location
16	ADM constraints, boundaries, and initial-data admissibility conditions	§21.8–13; §21.9–11 pp. 534–543; §21.11 p. 540; §43.3 p. 1186 Eqs43.6–43.10
17	Cosmic fluid and homogeneous initial hypersurface	§27.3–4 pp. 715–717
18	Cosmological model, scale factor, and model conditions	§27.3–4 pp. 715–717; §27.9–11; Box30.1 pp. 813–814; Box27.1 p. 709; Box27.3–4; Ch29; §27.11
19	Selected monotone branch of the scale factor	§27.9–11
20	Redshift, distance, and stellar-evolution inputs	Box27.1 p. 709; Box27.3–4; Ch29
21	Clock-sequence carriers, calibration, and matching conditions	Box30.1 pp. 813–814
22	Wave train, emission–reception positions, and observer setup	§38.3–5 pp. 1054–1058
23	Gravitational-test observations, apparatus, and applicability conditions	§38.3–5 pp. 1054–1058; §40.8 pp. 1121–1126; §40.9
24	Measured gravitational coupling and unit-conversion constants	§1.6; Box1.8 p. 36; §40.8 pp. 1121–1126; §40.9
25	Superspace of three-geometries	§43.1 pp. 1180–1183; §43.2 pp. 1184–1185; §43.2–3 pp. 1185–1189
26	Intersection and interconnection structure of three-geometries	§43.1 pp. 1180–1183; §43.2 pp. 1184–1185
27	ADM action and variational problem	§43.3 p. 1186 Eqs43.6–43.10
28	Complete Hamilton–Jacobi solution family and parameters	§43.2–3 pp. 1185–1189
29	Configuration-space phase and probability amplitude	§43.2–3 pp. 1185–1189; §43.2 p. 1185 Eqs43.2–43.4
30	Localized packet and superposition data	§43.2–3 pp. 1185–1189
31	Applicability conditions for semiclassical approximation and packet localization	§43.1 pp. 1180–1183; §43.2–3 pp. 1185–1189; §43.2 p. 1185 Eqs43.2–43.4
32	Ground-state configuration probability distribution	§43.4 p. 1191 Eq43.23
33	Unit conventions and conversion rules	§1.5 pp. 23–29; Box1.4; §1.6; Box1.8 p. 36; §40.8 pp. 1121–1126; §40.9
34	Photon momentum and energy with observer data	§2.8 pp. 63–64
35	Source-specified operations and definitions	§16.4 pp. 393–396; Box16.2–3

a	Auxiliary Object	Source Location
36	Special Hamilton–Jacobi solution obtained from specified boundaries	§43.3 p. 1186 Eqs43.6–43.10; §43.2 p. 1185 Eqs43.2–43.4
37	Slowly varying amplitude in the semiclassical expression	§43.2 p. 1185 Eqs43.2–43.4
38	Values of the spatial three-metric	§21.4–5 pp. 506–513
39	Completeness, regularity, and interference conditions for a complete EHJ family	§43.2–3 pp. 1185–1189
40	Material-state and stress values of the apparatus	§16.4 pp. 393–396

Definition 8.280 (Formal-Evidence Semantic Components in the Misner–Thorne–Wheeler 1973 Reference). Define the set of formal-evidence semantic components by $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(103)$. Define the set of generation-initial-input indices and the set of reference-formal-evidence-environment indices by

$$\text{ClaimBoundaryIdx}_R = ([12], [103])$$

. Consequently, $\text{StrictClaimBoundary}(R)$ holds. The content, dependency objects, and original-source locations of the respective components are given in Table 177.

Table 177: Misner–Thorne–Wheeler 1973 Physical and Formal Input Claims

p	Claim	i	a	Source Location
1	Physical records identifying events	$\emptyset$	1	Ch1–2;§13.1–3
2	Data for bodies carrying worldlines	$\emptyset$	7	Ch1–2;§13.1–3
3	Clock apparatus and display states	$\emptyset$	5	§16.4 pp. 393–396;Box16.2–3
4	Rod apparatus and spatial-length data	$\emptyset$	6	§16.4 pp. 393–396;Box16.2–3
5	Physical carriers of atoms	$\emptyset$	7	§16.4 pp. 393–396;Box16.2–3
6	Physical carriers of molecules	$\emptyset$	7	§16.4 pp. 393–396;Box16.2–3
7	Physical carriers of human bodies	$\emptyset$	7	§16.4 pp. 393,396
8	Event-identification data at emission	$\emptyset$	1	§38.3–5 pp. 1054–1058
9	Event-identification data at reception	$\emptyset$	1	§38.3–5 pp. 1054–1058
10	Coincidence records at clock-comparison locations	$\emptyset$	1	§38.3–5 pp. 1054–1058
11	Geometric data on a spatial hypersurface	$\emptyset$	38	§21.4–5 pp. 506–513

p	Claim	i	a	Source Location
12	Material-state and stress data of the apparatus	$\emptyset$	40	§16.4 pp. 393–396;Box16.2–3
13	Type distinction between events and coordinates	1^Q	1	Ch1–2;§13.1–3
14	Type distinction between coordinates and worldlines	$1^Q, 2^{SQ}$	2	Ch1–2;§13.1–3
15	Type distinction between worldlines and metrics	$2^{SQ}, 5^{IQ}$	2–3	Ch1–2;§13.1–3
16	Distinction between coordinate order and causal order	1^Q	1	Ch1–2;§13.1–3
17	Distinction between coordinate values and directly observed quantities	1^Q	1	Ch1–2;§13.1–3
18	Local Lorentzian metric form	5^{IQC}	3	Ch1–2;§13.1–3; §1.6;Box1.8 p. 36
19	Relation between the timelike metric interval and proper time	$5^{IQ}, 2^Q, 7^{IQ}$	2–3	§13.4 pp. 315–324
20	Relation between the spacelike metric interval and proper distance	$\emptyset$	3, 6	§13.4 pp. 315–324
21	Stationarity of proper time along a timelike geodesic	$7^{IQ}, 8^{IQ}$	2–3	§13.4 pp. 315–324
22	Local maxima and global saddle points of stationary proper time	$7^{IQ}, 8^{IQ}$	2–3	§13.4 pp. 315–324
23	Affine relation between an affine parameter and proper time	$7^Q, 9^{QS}$	4	§13.4 pp. 315–324; Exercise13.7 p. 324;§13.6
24	Operational definition of an ideal rod	$\emptyset$	6, 35	§16.4 pp. 393–396;Box16.2–3
25	Operational definition of an ideal clock	$7^{IK}, 10^{IK}$	5, 35	§16.4 pp. 393–396;Box16.2–3
26	Relation between metric intervals and measured values	$7^{IQK}, 10^{IQK}$	3, 5, 8	§16.4 pp. 393–396;Box16.2–3
27	Applicability of the laws of matter	7^S	8	Ch16;Box16.1; §16.4 pp. 393–396;Box16.2–3
28	Conditions of apparatus use	$\emptyset$	9	§16.4 pp. 393–396;Box16.2–3
29	Relation between atomic clocks and proper time	$10^{IK}, 7^{IK}, 11^{IQK}$	5, 8–9	§16.4 pp. 393–396;Box16.2–3
30	Relation between molecular clocks and proper time	$10^{IK}, 7^{IK}, 11^{IQK}$	5, 8–9	§16.4 pp. 393–396;Box16.2–3

p	Claim	i	a	Source Location
31	Operating boundary of clock structures	$\emptyset$	5, 9	§16.4 pp. 393–396;Box16.2–3
32	Apparatus invalidation by strong acceleration	$\emptyset$	8–9	§16.4 pp. 393–396;Box16.2–3
33	Apparatus invalidation by tidal stress	$\emptyset$	8–9	§16.4 pp. 393–396;Box16.2–3
34	Universal velocity-dependent time-dilation statement	$7^{\text{IQX}}, 12^{\text{IQXK}}$	8, 10	§16.4 pp. 393–396;Box16.2–3
35	Device dependence of clock response to acceleration	$\emptyset$	8–9	§16.4 pp. 393–396;Box16.2–3
36	Relation between material processes and proper time	$7^{\text{S}}, 13^{\text{S}}$	7–8	§16.4 pp. 393,396
37	Relation between biological processes and proper time	$7^{\text{S}}, 13^{\text{S}}$	7–8	§16.4 pp. 393,396
38	Spatial foliation of four-dimensional geometry	$5^{\text{C}}, 17^{\text{B}}$	3, 13	§21.4–5 pp. 506–513
39	Intrinsic three-dimensional geometry	$\emptyset$	38	§21.4–5 pp. 506–513
40	Relation between adjacent leaves	17^{B}	13	§21.4–5 pp. 506–513
41	Role of the foliation label	17^{QB}	13	§21.4–5 pp. 506–513
42	Relation between lapse and the normal-direction proper-time interval	$17^{\text{QB}}, 5^{\text{IQ}}, 18^{\text{IQ}}$	3, 13	§21.4–5 pp. 506–513
43	Relation between shift and spatial relabeling	17^{Q}	13–14	§21.4–5 pp. 506–513
44	Relation $d\tau_{\text{prop}} = N dt_{\text{slice}}$	$17^{\text{Q}}, 18^{\text{Q}}, 19^{\text{IQ}}$	13	§21.4–5 pp. 506–513
45	Construction of extrinsic curvature	$17^{\text{QS}}, 18^{\text{Q}}$	3, 14–15	§21.4–5 pp. 506–513
46	Hypersurface-displacement relation supplied by many-fingered time	$17^{\text{BS}}, 18^{\text{Q}}, 20^{\text{SB}}$	13–14	§21.4–5 pp. 506–513
47	Constraint obligation for initial data	$\emptyset$	15–16	§21.8–13
48	Evolution of initial data with specified lapse and shift	$17^{\text{SQ}}, 18^{\text{Q}}, 21^{\text{S}}$	14–16	§21.8–13
49	Construction of cosmic time from proper intervals of the cosmic fluid	$7^{\text{IQ}}, 23^{\text{IQB}}$	17–18	§27.3–4 pp. 715–717
50	Role of conformal time	$23^{\text{IQ}}, 24^{\text{IQ}}$	18	§27.9–11
51	Role of the scale factor	23^{QS}	18	§27.9–11
52	Reparameterization condition on a monotonic branch	$23^{\text{QS}}, 25^{\text{QS}}$	18–19	§27.9–11
53	Physical significance and reality of proper time	7^{IK}	5	Box30.1 pp. 813–814

p	Claim	i	a	Source Location
54	Relation between proper time and physical clock ticks	$7^{\text{IK}}, 11^{\text{IK}}$	5, 8	Box30.1 pp. 813–814
55	Relation between proper time and the rhythms of real events	7^{S}	1, 8	Box30.1 pp. 813–814
56	Validity of proper time in local special relativity	7^{IQ}	3	Box30.1 pp. 813–814
57	Condition for reconsideration on a global scale	30^{IB}	18	Box30.1 pp. 813–814
58	Physical existence and operating domain of a clock carrier	$\emptyset$	5, 18	Box30.1 pp. 813–814
59	Construction of a chain of clocks	$30^{\text{IK}}, 31^{\text{IQK}}$	18, 21	Box30.1 pp. 813–814
60	Extrapolation of the age of the universe by a chain of clocks	$30^{\text{IK}}, 31^{\text{IQK}}$	18, 21	Box30.1 pp. 813–814
61	Comparison of redshift and metric proper time	$7^{\text{IQ}}, 32^{\text{IQX}}, 33^{\text{QX}}$	3, 22	§38.3–5 pp. 1054–1058
62	Proper-time comparison using atomic clocks	$7^{\text{IK}}, 32^{\text{IQX}}, 33^{\text{QXK}}$	5, 22–23	§38.3–5 pp. 1054–1058
63	Description of dynamical space	35^{S}	15	§43.1 pp. 1180–1183
64	Relation for the temporal change of three-dimensional geometry	37^{QS}	15	§43.1 pp. 1180–1183
65	Domain of superspace	$\emptyset$	25	§43.1 pp. 1180–1183
66	Leaf representation of a classical spacetime history	$38^{\text{S}}, 40^{\text{SX}}$	25–26	§43.1 pp. 1180–1183
67	Limit on simultaneous determination of three-dimensional geometry and its rate of change	$36^{\text{B}}, 37^{\text{QS}}$	15	§43.1 pp. 1180–1183
68	Relation between the intrinsic three-metric, canonical momentum, and extrinsic curvature	37^{QS}	3, 15	§43.1 pp. 1180–1183
69	No-time statement in quantum geometrodynamics	$\emptyset$	31	§43.1 pp. 1180–1183
70	Spacetime as a secondary classical concept	$\emptyset$	31	§43.1 pp. 1180–1183
71	Time as a secondary classical concept	$\emptyset$	31	§43.1 pp. 1180–1183
72	Domain of applicability of the no-time statement	$\emptyset$	31	§43.1 pp. 1180–1183

p	Claim	i	a	Source Location
73	Construction from classical spacetime to history leaves	$38^S, 40^{SX}$	25–26	§43.2 pp. 1184–1185
74	Reconstruction from intersecting three-dimensional geometries to four-dimensional geometry	$40^{SX}, 42^{SXB}$	26	§43.2 pp. 1184–1185
75	Existence claim for intersection composition	40^X	26	§43.2 pp. 1184–1185
76	Dimensional claim for intersection composition	40^X	26	§43.2 pp. 1184–1185
77	Structural claim for intersection composition	40^X	26	§43.2 pp. 1184–1185
78	Reconstruction of event identity	40^X	26	§43.2 pp. 1184–1185
79	Reconstruction of the precedence relation	$40^X, 41^{XC}$	26	§43.2 pp. 1184–1185
80	Quantum-state probability amplitude on three-dimensional geometries	$\emptyset$	25, 29	§43.2–3 pp. 1185–1189
81	Unlocalized character of a typical probability amplitude	$\emptyset$	29	§43.2–3 pp. 1185–1189
82	Amplitude-localization condition	$\emptyset$	29–31	§43.2–3 pp. 1185–1189
83	Constructive-interference condition	$\emptyset$	28–29, 39	§43.2–3 pp. 1185–1189
84	Construction of a classical history leaf from a complete EHJ family and interference conditions	40^{SX}	28, 39	§43.2–3 pp. 1185–1189
85	Presupposition for the semiclassical approximation domain	$\emptyset$	31	§43.2–3 pp. 1185–1189
86	Dependence relation between stability of the time standard and the displayed speed of light	3^{IQKX}	5, 33	§1.5 pp. 23–29;Box1.4
87	Geometrized units $c = 1$ and conversion between seconds and centimeters	4^{IQ}	33	§1.6;Box1.8 p. 36
88	Photon energy–frequency–wavelength relation $E = h\nu = hc/\lambda$	6^Q	34	§2.8 pp. 63–64
89	Geometrized units $G = 1$ and G/c^2 conversion	4^{IQ}	24, 33	§1.6;Box1.8 p. 36; §40.8 pp. 1121–1126; §40.9

p	Claim	i	a	Source Location
90	Distinction between definitional G and experimental quantity G_c , and event and time dependence	34^{QO}	23–24	§40.8 pp. 1121–1126; §40.9
91	Definitional–present-value relation between $H = \dot{a}/a$ and $H(\text{today}) = H_0$	$23^{\text{QS}}, 26^{\text{Q}}$	18, 20	Box27.1 p. 709; Box27.3–4; Ch29
92	Relation between the time function H and Hubble time H^{-1}	$26^{\text{Q}}, 27^{\text{IQ}}$	18	Box27.1 p. 709; Box27.3–4; Ch29
93	Determination of the metric without standard clocks	$1^{\text{Q}}, 4^{\text{IQ}}$	2–3, 11	Exercise13.7 p. 324
94	Calibration of a geodesic clock by a standard timelike interval	$14^{\text{IQB}}, 15^{\text{IQK}}$	11–12	Box16.4 pp. 397–399
95	Interval measurement with a calibrated geodesic clock	$15^{\text{IQK}}, 16^{\text{IQK}}$	11–12	Box16.4 pp. 397–399
96	Admissibility conditions for thin-sandwich and York initial-data methods	$\emptyset$	15–16	§21.9–11 pp. 534–543
97	Definition of York time from extrinsic curvature	22^{Q}	15–16	§21.11 p. 540
98	Model-dependent relation between Hubble time and cosmic age	$27^{\text{IQ}}, 29^{\text{IQ}}$	18	Box27.1 p. 709; §27.11
99	Age since the start of expansion in the specified model	$23^{\text{IQB}}, 29^{\text{IQB}}$	18	§27.9–11
100	Comparison with a time estimate from stellar development	$28^{\text{IQ}}, 27^{\text{IQ}}$	18, 20	Box27.1 p. 709
101	Principal functional from the ADM action between boundary hypersurfaces	$17^{\text{QS}}, 21^{\text{S}}, 43^{\text{B}}$	15–16, 27, 36	§43.3 p. 1186 Eqs43.6–43.10
102	Semiclassical phase amplitude from the principal functional	$\emptyset$	29, 31, 36–37	§43.2 p. 1185 Eqs43.2–43.4
103	Spread of the ground-state configuration distribution	$\emptyset$	32	§43.4 p. 1191 Eq43.23

Time uses comprise 125 claim/time pairs and three additional pairs explicit in transfers, totaling 128. Among the 125 pairs, the uses of Time Content 38 in Claims 69, 70 and 72, and Time Content 39 in Claims 69, 71 and 72, are contextual occurrences in the no-time statement and its classical contrast. The three additional pairs are Time Content 3^{IK} in Transfer 22 and Time Contents 1^{Q} and 2^{QS} in Transfer 27. Superscript codes denote the

use roles.

Table 178: Misner–Thorne–Wheeler 1973 Time-Use Roles

Code	Use Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causal
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 179: Misner–Thorne–Wheeler 1973 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–12, 14–16, 19, 22–29, 31–33 13, 21	separate member	$\emptyset$ order_relation, transition_position
17	member	instant, order_relation
18, 30, 37, 42	dependency	$\emptyset$
20, 35	member	transition_position
34, 36	member	instant
38, 40–41	member	order_relation
39	lexical	$\emptyset$
43	member	endpoint, order_relation

The source-specific obligation atoms comprise 103 claims, 43 time contents, 27 transfers and 26 results, totaling 199.

Dependency Inputs Transfers 15 and 26 give York time and a classical history leaf from their specified formal conditions. The other 25 transfers have time-content inputs.

Dependency and Transfer Elements

Definition 8.281 (Misner–Thorne–Wheeler 1973 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [27]$. The following table gives inputs, outputs and support.

Table 180: Misner–Thorne–Wheeler 1973 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
1	2, 5	7	18–19	2–3	Proper time from the timelike line element
2	7	8	21–22	2–3	Stationary proper time
3	7	9	23	4	Affine reparametrization of proper time
4	7	10	25–26	5, 8	Metric proper time and the ideal-clock reading
5	7, 10	11	27–33	5, 8–9	Material-clock reading
6	7	12	34–35	2, 10	Velocity time-dilation comparison
7	7	13	7, 36–37	7–8	Material and biological processes in proper time
8	14	15	94	11–12	Geodesic-clock calibration
9	15	16	95	11–12	Interval measurement by the geodesic clock
10	7, 32	33	8–10, 61–62	3, 5, 22	Comparison of emission and reception clock rates
11	5, 17	18	38, 40–42	3, 13	Lapse from normal separation
12	17–18	19	42, 44	13	Normal proper-time interval
13	17–18	20	38, 41, 43, 46	13–14	Local hypersurface deformation
14	17–18	21	45, 47–48, 96	14–16	Evolution of initial data
15	$\emptyset$	22	97	15–16	Definition of extrinsic time
16	7	23	49	17–18	Cosmic time from fluid proper time
17	23	24	50–51	18	Construction of conformal time
18	23	25	51–52	18–19	Reparametrization by the scale factor
19	23	26	91	18, 20	Construction of the expansion rate
20	26	27	92	18	Construction of Hubble time
21	23	29	99	18	Construction of model age
22	3, 7	30	53–58	5, 18	Extrapolation of local proper-time standards
23	30	31	59–60	18, 21	Cosmic age from a chain of clocks
24	38	40	66, 73	25–26	History-leaf representation of classical spacetime
25	40	41–42	74–79	25–26	Reconstruction of events and spacetime from a history leaf

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
26	$\emptyset$	40	83–84	28, 39	History-leaf construction from a complete EHJ family
27	1–2	3	86	2, 5	Selection of a time standard by inertial motion

Grounding Conditions

Conditions for the Dependencies or Transfers Each transfer holds under the following source conditions.

1. Ch1–2;§13.1–3; §1.6;Box1.8 p. 36; §13.4 pp. 315–324. Integrate the line element between the specified endpoints on the same timelike curve with its Lorentzian metric.
2. §13.4 pp. 315–324. Vary admissible paths with fixed endpoints, retaining the conditions for a maximum in flat spacetime and for local maxima or global saddle points in curved spacetime.
3. §13.4 pp. 315–324; Exercise13.7 p. 324;§13.6. Use an affine transformation with a nonzero slope and an arbitrary constant offset along the same geodesic.
4. §16.4 pp. 393–396;Box16.2–3. Under the ideal-clock definition and measurement coupling, identify the reading with proper time on its worldline.
5. Ch16;Box16.1; §16.4 pp. 393–396;Box16.2–3. Use the material laws, precision, construction and operating domain of the atomic or molecular clock. Responses to acceleration and tidal stress depend on the apparatus conditions.
6. §16.4 pp. 393–396;Box16.2–3. Compare proper-time intervals along the world-lines for the specified motions and comparison events. Acceleration effects on clock rates follow the material dynamics of each apparatus.
7. §16.4 pp. 393,396. Apply the electromagnetic and quantum-mechanical laws governing material processes under the conditions of the physical system. The source also states the connection between human aging and proper time.
8. Box16.4 pp. 397–399. Use the fiducial timelike interval AB, timelike geodesics, light signals and tick counts. For unit AB, determine the tick interval τ_0 from $(N_1\tau_0)(N_2\tau_0) = 1$.
9. Box16.4 pp. 397–399. Read the interval using calibrated tick intervals and counts within the curvature and resolution conditions.
10. §38.3–5 pp. 1054–1058; §38.6–7;§40.1–7. Compare proper intervals between wave crests at emission and reception in the specified metric. The static gravitational-redshift relation and experimental values are related under their observer, apparatus and precision conditions.
11. §21.4–5 pp. 506–513. Specify the spacetime metric, foliation, normal and slice label, and relate the normal separation of neighboring slices to the label increment.
12. §21.4–5 pp. 506–513. Along the normal displacement $dx^i = -N^i dt$, use $d\tau = N dt$.

13. §21.4–5 pp. 506–513; §21.8–13. The specified lapse and shift determine normal displacement and spatial relabeling of the hypersurface.
14. §21.8–13. Use the constrained three-metric and canonical momentum with the specified lapse and shift. The source presents thin-sandwich existence and uniqueness as a conjecture; the York method has its stated closure, boundary and conformal conditions.
15. §21.8–13. Define $\tau_{\text{York}} = (2/3)g^{-1/2} \text{Tr } \pi = (4/3) \text{Tr } K$ and take it spatially constant on the initial hypersurface. Apply the initial-value method to a closed three-geometry or a region with the specified boundary data.
16. §27.3–4 pp. 715–717. Use proper-time intervals along the cosmic-fluid worldlines measured from a homogeneous initial hypersurface.
17. §27.9–11. Integrate $d\eta = dt/a(t)$ using the scale factor of the specified cosmological model.
18. §27.9–11. Use the scale factor as a time parameter on a monotone branch where $a(t)$ is invertible.
19. Box27.1 p. 709; Box27.3–4; Ch29. Define $H = \dot{a}/a$ by differentiation with respect to cosmic time and denote its present value by H_0 .
20. Box27.1 p. 709; Box27.3–4; Ch29. For a nonzero expansion rate, form H^{-1} ; its present value H_0^{-1} is the time scale of linear extrapolation.
21. Box27.1 p. 709; Box27.3–4; Ch29. Use elapsed cosmic time since the expansion origin in the specified model. The ratio of H^{-1} to model age depends on that cosmological model.
22. Ch28–30; Box30.1 pp. 813–814. Connect local special-relativistic proper time and clock standards over domains in which the clock carriers exist and operate. Cosmological extrapolation is conditioned on the physical processes available in each era.
23. Box30.1 pp. 813–814. In the specified cosmological model, connect calibrations and operating domains of successive clock types and combine the weighted ticks.
24. §43.1 pp. 1180–1183; §43.2 pp. 1184–1185. Map the three-geometries contained in a given classical four-geometry to its dynamically admissible history leaf in superspace.
25. §43.2 pp. 1184–1185. Construct the four-geometry, events and their earlier, later or synchronous relations from intersections and interconnections of three-geometries on the same admissible history leaf. The full collection of crossing slices has no single monotone ordering.
26. §43.2–3 pp. 1185–1189. For a complete Einstein–Hamilton–Jacobi solution family $S(\mathcal{G}_3; \alpha, \beta)$, take the three-geometries satisfying $\delta S/\delta\alpha = \delta S/\delta\beta = 0$. Under the source completeness conditions, this is the classical history leaf of Einstein’s equations.
27. §1.5 pp. 23–29; Box1.4. Choose the time coordinate so that free motion is uniform in a local Lorentz frame and define the time standard through comparisons of clocks and motion.

Results

Definition 8.282 (Misner–Thorne–Wheeler 1973 Results). Let $\text{Idx}_R^{\text{audit,result}} = [26]$.

Table 181: Misner–Thorne–Wheeler 1973 Results and Source Locations

j	Result	Source Location
1	Type separation of events, coordinates, worldlines, and metrics	Ch1–2; §13.1–3; §1.6; Box1.8 p. 36; §13.4 pp. 315–324
2	Separation of coordinate labels from direct observables	Ch1–2; §13.1–3
3	Proper time supplied by the timelike metric interval	Ch1–2; §13.1–3; §1.6; Box1.8 p. 36; §13.4 pp. 315–324
4	Stationarity of proper time on timelike geodesics and its local and global conditions	§13.4 pp. 315–324; Exercise13.7 p. 324; §13.6
5	Conditions for ideal and material clocks	§16.4 pp. 393–396; Box16.2–3; Ch16; Box16.1
6	Universal velocity time dilation and device-dependent acceleration effects	§16.4 pp. 393–396; Box16.2–3
7	Relation of material and biological processes to proper time	§16.4 pp. 393, 396
8	Relation between the label, lapse, and shift in the ADM decomposition	§21.4–5 pp. 506–513
9	Normal-direction proper-time interval	§21.4–5 pp. 506–513
10	Foliation-deformation freedom of many-fingered time	§21.4–5 pp. 506–513
11	Evolution of data satisfying the initial constraints	§21.8–13; §21.9–11 pp. 534–543
12	Construction of cosmic time and cosmological reparametrization	§27.3–4 pp. 715–717; §27.9–11
13	Reality of local proper time and the domain of cosmological extrapolation	Box30.1 pp. 813–814
14	Extrapolation of cosmological age by a chain of clocks	Box30.1 pp. 813–814
15	Theory and evidence of metric-clock and redshift experiments	§38.3–5 pp. 1054–1058
16	Absence of time in quantum geometrodynamics and the classical approximation	§43.1 pp. 1180–1183
17	Reconstruction of classical four-dimensional geometry from intersecting three-dimensional geometries	§43.2 pp. 1184–1185
18	Semiclassical histories from localized packets and interference conditions	§43.2–3 pp. 1185–1189; §43.2 p. 1185 Eqs43.2–43.4

j	Result	Source Location
19	Dependence relation between stability of the time standard and the displayed speed of light	§1.5 pp. 23–29; Box1.4
20	Conversion between seconds and length in geometrized units	§1.6; Box1.8 p. 36; §40.8 pp. 1121–1126; §40.9
21	Photon energy–frequency–wavelength relation	§2.8 pp. 63–64
22	Separation of the definitional gravitational constant from the Cavendish-measured quantity	§1.6; Box1.8 p. 36; §40.8 pp. 1121–1126; §40.9
23	Metric reconstruction without standard clocks	Exercise13.7 p. 324
24	Geodesic-clock calibration and timelike-interval measurement	Box16.4 pp. 397–399
25	York time used in specifying initial data	§21.11 p. 540; §21.9–11 pp. 534–543
26	Comparison of Hubble time with model and stellar ages	Box27.1 p. 709; Box27.3–4; Ch29; §27.11; §27.9–11

Result Dependencies

Definition 8.283 (Misner–Thorne–Wheeler 1973 Result-Dependence Hyperedges). For each $j \in [26]$, the following table specifies the time premises, claim support, auxiliary inputs and output of $\text{ResultEdge}_{R,j}$. For each result, its set of semantic-object prerequisites is uniquely reconstructed by mapping the time-premise and auxiliary-input indices in that table to the objects of their respective types listed in the time-content table 175 and the auxiliary-object table 176, and taking the union of the two resulting object sets.

Table 182: Misner–Thorne–Wheeler 1973 Result-Dependence Hyperedges

j	Time Premises i	Claim Support p	Auxiliary In a	Result Output Index
1	1–2, 5, 7	13–20	1–3, 6	1
2	1	13, 16–17	1	2
3	7	18–19	2–3	3
4	7–9	21–23	2–4	4
5	7, 10–11	25–33	3, 5, 8–9, 35	5
6	12	32–35	8–10	6
7	13	7, 36–37	7–8	7
8	17–19	38–45	3, 13–15, 38	8
9	19	42, 44	3, 13	9
10	20	38, 41, 43, 46	3, 13–14	10
11	21	47–48, 96	14–16	11
12	23–25	49–52	17–19	12
13	7, 30	53–58	1, 3, 5, 8, 18	13
14	31	59–60	18, 21	14

j	Time Premises i	Claim Support p	Auxiliary In a	Result Output Index
15	32–33	8–10, 61–62	1, 3, 5, 22–23	15
16	35–37	63–64, 67–72	3, 15, 31	16
17	41–42	74–79	26	17
18	40	80–85, 102	25, 28–31, 36–37, 39	18
19	3	86	5, 33	19
20	4	87, 89	24, 33	20
21	6	88	34	21
22	4, 34	89–90	23–24, 33	22
23	1, 4	93	2–3, 11	23
24	15–16	94–95	11–12	24
25	22	96–97	15–16	25
26	26–29	91–92, 98–100	18, 20	26

Across the 26 results there are 48 time-premise occurrences, 77 auxiliary-premise occurrences, 125 semantic-object prerequisite occurrences, and 103 claim-support occurrences. Every result has at least one time premise.

Grounds of Establishment

- Grounds for the Results**
- 1. Metric and Clocks** Events, coordinates, worldlines and the metric have distinct roles. Proper time along a timelike curve is given by a line integral and is stationary on a fixed-endpoint geodesic. Determination of the metric without standard clocks uses coordinate-time velocities and accelerations, light rays and free-fall paths, metric compatibility and a scale fixed at one event. The geodesic clock calibrates ticks from a fiducial timelike interval, timelike geodesics and light signals, then measures intervals under its curvature and resolution conditions.
 - 2. Material Realization of Clocks and Aging** Atomic and molecular clocks realize the ideal proper-time reading within their precision and operating domains. Universal velocity time dilation is distinguished from device-dependent responses to acceleration and tidal stress. The source states that human aging is governed by the same electromagnetic and quantum-mechanical laws as atomic and molecular processes and is tied to proper time. This correspondence holds under the biological and environmental conditions of the physical system.
 - 3. Hypersurfaces and Initial Data** The ADM decomposition relates the slice label, lapse, shift and normal displacement. The relation $d\tau = Ndt$ holds along the normal direction satisfying $dx^i = -N^i dt$. Extrinsic curvature is $K_{ij} = (N_{i|j} + N_{j|i} - \partial_t g_{ij})/(2N)$, and the three-metric and canonical momentum satisfy the initial constraints. York time is $\tau_{\text{York}} = (2/3)g^{-1/2} \text{Tr } \pi = (4/3) \text{Tr } K$, taken spatially constant on the specified initial hypersurface. Initial-data construction follows the source closure, boundary and conformal conditions.
 - 4. Cosmic Time and Age** Cosmic time is constructed from cosmic-fluid proper intervals measured from a homogeneous initial hypersurface. Conformal time satisfies $d\eta = dt/a(t)$, and the scale factor parametrizes time on a monotone

invertible branch. The expansion rate $H = \dot{a}/a$, Hubble time H^{-1} , model age and stellar age are distinct quantities. The age ratio of at least 1.5 in Box 27.3 belongs to the expansion regime of the specified closed $\Lambda = 0$ dust model. Extrapolation of cosmic age requires clock carriers and calibration in each era; Box 30.1 also considers the mathematical possibility of infinitely many events within finite proper time.

5. **Classical Histories and Quantum Geometrodynamics** A given classical spacetime determines the history leaf formed by its three-geometries. Conversely, intersections and interconnections on the same dynamically admissible leaf reconstruct the four-geometry and temporal relations of events. Because the three-geometries cross, their full collection has no single monotone ordering. Constructive-interference conditions on a complete EHJ family yield the same history leaf as the classical Einstein equations. In quantum theory, a typical amplitude is delocalized over superspace; a suitable superposition satisfying the localization conditions yields a semiclassical history. The special principal function from the ADM action, the complete solution family, the quantum amplitude and the localized packet enter their respective constructions.
6. **Time Standards, Quantities and Experiments** Time standards are tied to local inertial motion and clock comparison. The nonuniform solar-day example rejected by the source concerns actual successive high-noon intervals. The geometrized convention $c = G = k = 1$ specifies unit conversions. The photon relation $E = h\nu = hc/\lambda$ uses observer-relative frequency, while the event/time dependence of the Cavendish quantity G_c depends on the gravitational theory. Gravitational-redshift experiments compare emission and reception under the specified metric, observers and clocks.

Generation Judgment Judgment Values: LCTR complete-generation judgment: 0; source-evidence-conditioned local reach: 4; remaining primitive-time roots after local reach: 39.

Locally reached Time Contents 22, 40–42 are York time, the classical history leaf, temporal relations of events and the temporal structure of reconstructed classical spacetime.

Result Dependence For each result q , Definition 7.108 gives its minimal primitive-time-root support family $\mathcal{A}_R(q)$; Definition 7.50 determines the root–result dependence and necessary dependence relations. Of the 26 results, the three Results 17–18, 25 are reachable from the source formal-evidence environment; the 23 Results 1–16, 19–24, 26 depend on primitive-time roots. There are 79 root–result dependence pairs, including 16 necessary dependence pairs.

Evidence Effect York time is defined from specified geometric data, and a complete EHJ family with constructive-interference conditions gives a classical history leaf. Its intersection structure gives events and the temporal relations of classical spacetime. These constructions establish the four local reaches. Readings, calibration and connection to material laws belong to the physical realization conditions of each construction.

Result-Preservation Domain Each result holds under its respective conditions on the Lorentzian metric, timelike curves, clock operation, initial constraints, cosmological

model, complete solution family or semiclassical approximation. Generalization to material and biological processes additionally requires applicability of the laws of matter and the clock hypothesis. For the 23 root-dependent results, realization of their minimal-support time inputs by generated representations preserving the source-required relations preserves the corresponding results within their domains.

Placement of the Grounding Obligation Of the 43 time contents, 42 have law-bearing uses and none is reached by complete generation. The construction conditions of the four local reaches are supplied in the formal-evidence environment. For coordinate time, proper time, clock calibration, initial-data evolution, cosmic age and the classical temporal uses in the quantum argument, physical-time grounding obligations are backgrounded by primitive-time inputs. Time Content 39 is the reference to the concepts before, after and next denied by the quantum-geometrodynamic no-time statement.

Evidence of Actual Failure Clock precision and operating domains, responses to acceleration and tidal stress, and the existence domains of cosmological clock carriers supply concrete physical conditions. Actual failure of the audited physical-time generation conditions, and resulting changes in measured values, are undetermined from the source evidence.

Downstream Reach Structural reach is determined by the 79 root–result dependence pairs given by the dependency table and minimal support families. Actual reach from a concrete failure location and the recalculated values remain undetermined.

Repair Scope Grounding obligations for physical time use concern each time input and its generation conditions before use. If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition. The target results are regenerated or recalculated using time inputs that satisfy the grounding conditions, or their domains of application are restricted to regions where those conditions hold.

Audit Conclusion

Proposition 8.284 (Misner–Thorne–Wheeler 1973 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = & (\emptyset, [43]; \\ & \emptyset, \{17, 18, 25\}; \\ & [26] \setminus \{17, 18, 25\}, \emptyset). \end{aligned}$$

Proof. The boundary contains 172 nodes—103 claims, 43 time contents and 26 results—and 53 edges—27 transfers and 26 results. Initial Claims 1–12 do not satisfy the support of any edge, so complete-generation closure remains the 12 initial nodes. With all 103 claims supplied, Transfers 15 and 26, which have no time input, give York time and the classical history leaf; Transfer 25 gives temporal relations of events and the temporal structure of reconstructed spacetime. This set reaches Results 17, 18 and 25. The remaining results require the temporal premises shown in the table. $\square$

Proposition 8.285 (Misner–Thorne–Wheeler 1973 Minimal Primitive-Time-Root Support). The following table gives the minimal-support family statistics in the order: number of members, minimum cardinality and maximum cardinality. There are 59 family members in total.

Table 183: Misner–Thorne–Wheeler 1973 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1, 16	(1; 3, 3)
2, 20, 21	(1; 1, 1)
3, 4, 5, 19	(2; 1, 2)
6, 7, 9, 10, 11, 12	(3; 1, 2)
8	(2; 2, 2)
13	(6; 2, 3)
14	(6; 1, 3)
15	(3; 2, 3)
17, 18, 25	(1; 0, 0)
22, 23	(1; 2, 2)
24	(2; 1, 1)
26	(4; 2, 3)

Proof. Start with each singleton temporal seed and the empty supports of transfers with no time input. Apply Definition 7.46 to the 27 transfers until the support antichains stabilize, then combine the time premises of each result. Proposition 7.47 determines the families and counts. $\square$

Individual Statements Statement 1: Definition of Time

Verbatim Original (English):

Time is defined so that motion looks simple.

Original-source location: p. 23, §1.5 *Time*. Minimal context: the relation between the time coordinate of a local Lorentz reference frame and the description of motion.

Statement 2: Time Standard and the Speed of Light

Verbatim Original (English):

No standard of time that varies so much from one month to another is acceptable. If adopted, it would make the speed of light vary from month to month!

Original-source location: p. 24, §1.5 *Time*. Minimal context: nonuniformity resulting from adoption of the actual interval between successive high noons as the time standard.

Statement 3: Geometrized Units

Verbatim Original (English):

Throughout this book, we use “geometrized units,” in which the speed of light c , Newton’s gravitational constant G , and Boltzman’s constant k are all equal to unity.

Original-source location: p. 36, Box 1.8 *Geometrized Units*. Minimal context: conversion among grams, centimeters, seconds, and ergs.

Statement 4: Photon Energy, Frequency, and Wavelength

Verbatim Original (English):

The ratio of absorbed wavelength to emitted wavelength is the inverse of the energy ratio (since $E = h\nu = hc/\lambda$):

Original-source location: p. 64, §2.8, between Eq. (2.29) and Eq. (2.30). Minimal context: photon energies and wavelength ratios measured by emitter and absorber.

Statement 5: Present Rate of Cosmic Expansion

Verbatim Original (English):

Immediately observable today is the present rate of expansion of the universe, with every distance increasing at a rate directly proportional to the magnitude of that distance:

Original-source location: p. 709, Box 27.1, Eq. (10). Minimal context: the Hubble quantity relating recession velocity, galactic distance, and the rate of increase of the radius of the universe. Independence of the inferred H_0 from the chosen galaxy requires negligible local motions and negligible change in recession velocity between emission and the present.

Statement 6: Hubble Time

Verbatim Original (English):

The Hubble time, $H_0^{-1} \sim 18 \times 10^9$ yr (linearly extrapolated back to zero separation on the basis of the expansion rate observed today, as illustrated in the diagram) is predicted to be greater by a factor 1.5 or more (Box 27.3) than the actual time back to the start of the expansion as deduced from the rate of the development of stars ($\sim 10 \times 10^9$ yr).

Original-source location: p. 709, Box 27.1. Minimal context: comparison between Hubble time obtained by linear extrapolation of the present expansion rate and time estimated from stellar development. The factor of at least 1.5 applies to the expansion regime of the closed $\Lambda = 0$ dust model specified in Box 27.3.

Statement 7: Spacetime-Location Dependence of the Cavendish Experiment

Verbatim Original (English):

But if the Dicke–Brans–Jordan theory, or almost any other metric theory gives the correct description of gravity, the force in the Cavendish experiment will depend on where and when the experiment is performed, as well as on m_1 , m_2 , and r .

Original-source location: pp. 1121–1122, §40.8. Minimal context: introduction of a proportionality factor for using the Cavendish experiment as a test of theories of gravity.

Statement 8: Event Dependence of the Cavendish Constant

Verbatim Original (English):

This Cavendish constant will be unity in general relativity, but in most other metric theories it will vary from event to event in spacetime.

Original-source location: p. 1122, §40.8, immediately after Eq. (40.41). Minimal context: comparison of the Cavendish constant in general relativity and other metric theories of gravity.

Statement 9: Timeless Statement in Quantum Geometrodynamics

Verbatim Original (English):

There is no spacetime, there is no time, there is no before, there is no after.
The question of what happens “next” is without meaning.

Original-source location: p. 1183, Chapter 43, §43.1. Minimal context: From the impossibility of specifying both the instantaneous three-geometry and its time rate of change precisely, the source argues that a deterministic classical history evolving in time can neither be predicted nor assigned meaning. It restricts spacetime and time to the classical approximation; the statement concerns regimes where quantum geometrodynamic effects become important.

Statement 10: Velocity and Time Dilation

Verbatim Original (English):

Velocity produces a universal time dilation; acceleration does not.

Original-source location: p. 393, Chapter 16, §16.4. Minimal context: discussion of the breaking point of clocks and the effects of acceleration on clock structures.

Statement 11: Human Aging and the Laws of Matter

Verbatim Original (English):

The aging of the human body is governed by the same electromagnetic and quantum-mechanical laws

Original-source location: p. 393, Chapter 16, §16.4, item (3) *human clocks*. Minimal context: comparison with the laws governing atomic and molecular periodicity and level transitions.

Statement 12: Aging and Proper Time

Verbatim Original (English):

aging, like atomic processes, is tied to proper time

Original-source location: pp. 393, 396, Chapter 16, §16.4, item (3) *human clocks*. Minimal context: the discussion of human clocks that continues across Boxes 16.2 and 16.3.

Common Evidence with Specified Domain of Application The twelve statements are attributable to Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler. For each statement, its verbatim original, original-source location, and minimal context required for interpretation are given.

Target State- ment	Object of the Statement	Predicate	Explicit Domain of Application
--------------------------	----------------------------	-----------	-----------------------------------

S19-Q001	The time coordinate of a local Lorentz reference frame and description of motion	Defines time so that motion looks simple.	The domain in p. 23, §1.5, in which the time coordinate of a local Lorentz reference frame is used to describe motion.
S19-Q002	A time standard based on the actual interval between successive high noons and the displayed speed of light	Rejects a time standard that varies substantially from month to month and states that its adoption would make the speed of light vary from month to month.	The domain in p. 24, §1.5, that discusses nonuniformity of the actual interval between successive high noons and selection of a time standard.
S19-Q003	The speed of light c , Newton's gravitational constant G , Boltzmann's constant k , and geometrized units	Uses throughout the book geometrized units in which all three constants are set equal to unity.	The domain of unit conventions in Box 1.8, including conversions among grams, centimeters, seconds, and ergs.
S19-Q004	Energy, frequency, and wavelength of emitted and absorbed photons	On the basis of $E = h\nu = hc/\lambda$, takes the ratio of absorbed wavelength to emitted wavelength as the inverse of the energy ratio.	The comparison domain in §2.8 for photon energies and wavelength ratios measured by emitter and absorber.
S19-Q005	The present rate of cosmic expansion and distances in the universe	Describes the present rate of expansion as immediately observable and states that each distance increases at a rate directly proportional to its magnitude.	The relation in Box 27.1 between recession velocity, galactic distance and the rate of increase of the radius of the universe. Independence of H_0 from the chosen galaxy applies where local motions and the change in recession velocity between emission and the present are negligible.
S19-Q006	Hubble time and the time to the start of expansion estimated from stellar development	Predicts that Hubble time, obtained by linear extrapolation of the present expansion rate to zero separation, is greater by a factor of at least 1.5 than the time estimated from stellar development.	The comparison of Hubble time and age estimates in Boxes 27.1 and 27.3, within the expansion regime of the specified closed $\Lambda = 0$ dust model.

S19-Q007	Force in the Cavendish experiment under the Dicke–Brans–Jordan theory and other metric theories	States that, if the Dicke–Brans–Jordan theory or almost any other metric theory correctly describes gravity, the force depends on the place and time of the experiment in addition to m_1 , m_2 , and r .	The domain in §40.8 that treats the Cavendish experiment as a test of theories of gravity.
S19-Q008	The Cavendish constant in general relativity and other metric theories of gravity	States that it is unity in general relativity and varies from event to event in spacetime in most other metric theories.	The domain in §40.8 that compares the Cavendish constant across metric theories of gravity.
S19-Q009	Spacetime, time, before, after, and the question of “next” within the descriptive domain of quantum geometrodynamics	States that there is no spacetime, time, before, or after, and that the question of what happens “next” has no meaning.	The argument in Chapter 43, §43.1 that restricts spacetime and time to the classical approximation, on the basis of the inability to specify both the three-geometry and its time rate of change precisely.
S19-Q010	Velocity, acceleration, and time dilation	States that velocity produces universal time dilation and that acceleration effects on clocks depend on apparatus construction.	The domain in Chapter 16, §16.4, that discusses the breaking point of clocks and the effects of acceleration on clock structures.
S19-Q011	Aging of the human body	States that aging of the human body is governed by the same electromagnetic and quantum-mechanical laws as atomic and molecular processes.	The domain of <i>human clocks</i> in Chapter 16, §16.4, that maps human aging to the laws of matter.
S19-Q012	Aging, atomic processes, and proper time	States that aging, like atomic processes, is tied to proper time.	The domain of <i>human clocks</i> in Chapter 16, §16.4, that relates human clocks to proper time.

Individual Audit

Interpretations Allowed by the Original Text and Minimal Context Statements 1 and 2 concern the time coordinate used to describe local inertial motion uniformly and the stability of that standard. Statement 3 gives the geometrized-unit convention; Statement 4 gives the relation among observer-measured photon quantities. Statements 5 and 6 relate cosmic expansion to Hubble time and age estimates in the specified model. Statements 7 and 8 concern theory-dependent position/time dependence of the Cavendish quantity. Statement 9 asserts the absence of spacetime, time, before, after and next in the domain of quantum geometrodynamics. Statement 10 distinguishes universal velocity time dilation from device-dependent acceleration

responses. Statements 11 and 12 state that human aging obeys the same material laws as atomic and molecular processes and is tied to proper time.

Independent Grounds The twelve statements are textbook definitions, quantitative relations and theoretical explanations. Time standards and geometrized units occur in Chapter 1; photon quantities in Chapter 2; cosmic ages in Boxes 27.1 and 27.3; the Cavendish quantity in §40.8; and the no-time assertion and classical-history construction in Chapter 43. Clocks and human aging are described as applications of material laws in §16.4; empirical gravitational-redshift comparisons are treated in Chapter 38.

Evidence-Effect Boundary for Each Interpretive Possibility Statements 1 and 2 concern inertial motion, clock comparison and time standards; Statements 3 and 4 concern unit conversion and observer-specific photon quantities. Statements 5 and 6 distinguish the present expansion rate, linear extrapolation, stellar age and model age. The ratio of at least 1.5 is conditioned on the expansion regime of the specified closed $\Lambda = 0$ dust model. Statement 7 assumes that the quoted metric theory correctly describes gravity, and Statement 8 compares predictions of general relativity and other metric theories. Statement 9 belongs to the quantum-geometrodynamical domain; the classical EHJ construction and semiclassical packet conditions in the same chapter hold in their respective domains. Statements 10–12 hold under the material laws and the apparatus, biological and environmental conditions of their application.

Corresponding Formal Evidence Statement 1 corresponds to Time Contents 1–3, Claim 86 and Transfer 27. Statement 2 corresponds to Time Content 3, Claim 86 and Result 19. Statement 3 corresponds to Time Content 4, Claims 87 and 89, and Result 20. Statement 4 corresponds to Time Content 6, Claim 88 and Result 21. Statements 5–6 correspond to Time Contents 23, 26–29, Claims 91–92, 98–100, and Result 26. Statements 7–8 correspond to Time Contents 4 and 34, Claims 89–90 and Result 22. Statement 9 corresponds to Time Contents 35–39, Claims 63–64, 67–72, and Result 16. Statement 10 corresponds to Time Contents 7 and 12, Claims 34–35 and Result 6. Statements 11–12 correspond to Time Contents 7 and 13, Claims 7, 36–37, and Result 7.

Within-Reference Comparison Chapters 1 and 2 use time coordinates and metric time; Chapter 16 relates material clocks to proper time; Chapters 27–30 determine age from cosmological models and chains of clocks. The no-time assertion in Chapter 43 concerns regimes in which quantum geometrodynamical effects become important. In the same chapter, a complete EHJ family yields a classical history leaf, while a suitable superposition of quantum amplitudes under localization conditions yields a semiclassical history.

Common Audit with Specified Domain of Application The conditions on time standards, clocks, cosmological models, gravitational theories and classical/quantum domains are specified by the claim correspondences above and the source conditions of each result.

8.19.20 Source Audit

Audit Inputs

Source and Audit Scope

Source Identifier 20.

Source Name Hawking 1975

Bibliographic Identification S. W. Hawking, *Particle Creation by Black Holes*, *Communications in Mathematical Physics* 43, 199–220 (1975) is denoted by $R = \text{Ref}_{\text{Haw75PCB}}$.

The audit domain comprises the abstract, §§1–4, Eqs. (1.1)–(4.10) and Figs. 1–5.

Audit-Scope Identifier 20-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Original-Source Location Identifier 20-L1.

Original-Source Location Content Each table identifies the source pages of its time contents, claims, transfers or results.

Formal Objects The audit uses 94 time contents, 65 auxiliary objects, 59 claims, 34 transfers and 30 results.

Assumptions The respective calculations use semiclassical coupling, the specified incoming vacuum and asymptotic regions, the free-field equation and mode normalization, geometric optics, late stationarity, scattering conditions and the quasistationary domain.

Non-temporal Physical Inputs The initial physical inputs are the carrier and value data in Claims 1–11: event identification, collapsing matter, mass, charge and angular momentum, horizon cross-sections, field degrees of freedom, scattering channels, field states, area, exterior entropy, ray connections and observers.

Formal Input Elements

Definition 8.286 (Hawking 1975 Time Contents and Auxiliary Objects). Let $n_R = 94$. The following tables specify the time contents and auxiliary objects.

Table 185: Hawking 1975 Time Contents

i	Time Content	Source Location
1	Ordering of successive mass states	pp. 199, 201–203, 219
2	Rate of mass change	pp. 199, 201–203, 219
3	Conditional evaporation endpoint	pp. 199, 202–203, 219
4	Primordial black-hole formation time	pp. 199, 202–203, 219
5	Cosmic interval since formation	pp. 199, 201–203, 219
6	Present reference for cosmic age	pp. 199, 201–203, 219
7	State ordering for entropy comparison	pp. 199, 202–203, 219
8	Minkowski time coordinate	pp. 199–201
9	Frequency with respect to Minkowski time	pp. 199–201
10	Temporal position of the initial flat region	pp. 199–201
11	Temporal position of the intermediate curved region	pp. 199–201
12	Temporal position of the final flat region	pp. 199–201

<i>i</i>	Time Content	Source Location
13	Ordering of the initial flat, curved and final flat regions	pp. 199–201
14	Frequency decomposition in the initial flat region	pp. 199–201
15	Frequency decomposition in the final flat region	pp. 199–201
16	Temporal dependence of the metric	pp. 199–201, 204
17	Time coordinate of a local inertial frame	pp. 199–201, 215–216
18	Temporal origin for the early universe	pp. 201
19	Planck-time interval in the early applicability estimate	pp. 201
20	Temporal position of the early-universe approximation domain	pp. 201
21	Ordering of the early and subsequent cosmic domains	pp. 201
22	Temporal course of particle creation and emission	pp. 201–202
23	Future null infinity	pp. 201–202, 206–207
24	Particle-emission rate	pp. 201–202, 207–211
25	Energy-absorption rate	pp. 201–202
26	Energy-emission rate	pp. 201–202
27	Temperature change during mass evolution	pp. 201–202
28	Temporal change of the energy-emission rate	pp. 201–202
29	Strong-interaction time scale	pp. 202, 219
30	Ordering of the terminal emission process	pp. 202, 219
31	Evaporation time integrated over the specified mass range	pp. 202, 219
32	Duration of terminal emission	pp. 202, 219
33	Terminal event in the explosion estimate	pp. 202, 219
34	Parameter of stationary time translation	pp. 202–203, 216–218
35	Frequency with respect to stationary time translation	pp. 202–203
36	Temporal orientation of worldlines in the heuristic pair picture	pp. 202–203
37	Choice of future orientation	pp. 202–203
38	Stage of gravitational collapse	pp. 204, 215–216
39	Temporal position of the late stationary region	pp. 204, 211–213
40	Postcollapse settling interval	pp. 204
41	Schwarzschild time coordinate	pp. 204–205
42	null-ray causal ordering	pp. 204–205
43	Past horizon of the eternal solution	pp. 204–205
44	Causal boundary at the future horizon	pp. 204–205
45	History boundary at the singular region	pp. 204–205
46	Temporal history of gravitational collapse	pp. 204–205, 211–213, 215–216
47	Asymptotic future region	pp. 204–205
48	Past null infinity	pp. 206–207
49	Incoming positive-frequency choice	pp. 206–207
50	Outgoing positive-frequency choice	pp. 206–207
51	Late retarded-time domain	pp. 207–211, 216–218

i	Time Content	Source Location
52	Late stationary particle flux	pp. 207–209
53	Advanced-time coordinate v	pp. 207–209
54	Retarded-time coordinate u	pp. 207–211
55	Advanced time v_0 of the last escaping ray	pp. 207–209
56	Advanced-time ordering relative to the last escaping ray	pp. 207–209
57	Affine parameter λ of a null generator	pp. 207–209
58	Advanced-time frequency of the backtraced mode	pp. 209–211
59	Retarded-time centre of a wave packet	pp. 209–211, 216–218
60	Ordering of wave-packet centres	pp. 209–211, 216–218
61	Infinite duration of steady emission	pp. 209–211
62	Asymptotic time/frequency basis for incoming massive states	pp. 211–213
63	Asymptotic time/frequency basis for outgoing massive states	pp. 211–213
64	Infinite past in the discussion of past bound states	pp. 211–213
65	Stationarity of the past asymptotic region	pp. 211–213
66	Incoming temporal phase in the rotating sector	pp. 213–215
67	Outgoing temporal phase in the rotating sector	pp. 213–215
68	Time coordinate of the stationary axisymmetric background	pp. 213–215
69	Effective frequency with respect to the horizon generator	pp. 213–215
70	Temporal flow of the horizon Killing generator	pp. 213–215
71	Event-horizon causal boundary	pp. 215–216
72	Full future history determining the event horizon	pp. 215–216
73	Horizon-crossing event of a freely falling observer	pp. 215–216
74	Event ordering relative to horizon crossing	pp. 215–216
75	Frequency with respect to local inertial time	pp. 215–216
76	Temporal extent of particle creation	pp. 215–216
77	Time parameter for horizon modes	pp. 216–218
78	Frequency choice for horizon modes	pp. 216–218
79	Retarded-time interval for the flux average	pp. 218–219
80	Average energy flux over the specified interval	pp. 218–219
81	Energy flux in the stationary limit	pp. 218–219
82	Black-hole mass-evolution time scale	pp. 219
83	Light-crossing time scale of the Schwarzschild radius	pp. 219
84	Sequence of states in quasistationary evaporation	pp. 219
85	Approximation boundary near the Planck mass	pp. 219
86	Initial stage of the baryon-only universe speculation	pp. 219
87	Hypothesized succession from baryons to radiation	pp. 219
88	Collapse stage in the conditional endpoint scenario	pp. 219
89	Evaporation stage in the conditional endpoint scenario	pp. 219
90	Final-state boundary of the conditional empty spacetime	pp. 219
91	Event ordering in the conditional endpoint scenario	pp. 219

i	Time Content	Source Location
92	Retarded-time width of a wave packet	pp. 209–211
93	Signed energy flux in the outer-horizon limit	pp. 216–219
94	Observer’s timelike direction fixing the local time axis	pp. 199–201, 215–216

Table 186: Hawking 1975 Auxiliary Objects

a	Auxiliary Object	Source Location
1	Area of a horizon cross-section	pp. 202–203, 213–215
2	Event-identification data	pp. 199–201, 204–207
3	Entropy value of exterior matter	pp. 202–203
4	Field charge	pp. 213–215
5	Field rest mass	pp. 211–213
6	Quantum-field state data	pp. 199–201, 206–207, 216–218
7	Field statistics	pp. 211–215
8	Mass at the specified starting state	pp. 202, 219
9	Black-hole mass value	pp. 201–202, 204, 213–215, 219
10	Physical carrier of the observer or detector	pp. 199–201, 215–216
11	Ray paths and endpoint connections	pp. 206–209
12	Scattering-channel data	pp. 206–211
13	Scale constant of the affine parameter	pp. 207–209
14	Angular-mode indices and mixing data	pp. 211–213
15	Azimuthal mode number	pp. 213–215
16	Bogoliubov coefficients between the specified bases	pp. 206–207
17	Bogoliubov coefficients between massive asymptotic bases	pp. 211–213
18	Data of the asymptotic Bondi frame	pp. 211–213
19	Local curvature bound	pp. 199–201, 215–216
20	Mass cutoff for the integration	pp. 202, 219
21	Horizon electric potential relative to infinity	pp. 213–215
22	Empty spacetime of the conditional endpoint scenario	pp. 219
23	Specified frequency band	pp. 209–213
24	Horizon-mode and component data	pp. 216–218
25	Integration surface for energy flux	pp. 216–219
26	Vacuum condition for the specified incoming basis	pp. 206–207, 209–215
27	Geometric data of the local orthonormal frame	pp. 199–201, 215–216
28	Mass-loss coefficient for the specified particle species	pp. 202, 219
29	Specified classical background metric	pp. 199–201, 204–205, 207–209, 211–213, 215–216
30	Solution and mode families of the selected field	pp. 199–201, 206–207, 209–211

<i>a</i>	Auxiliary Object	Source Location
31	Angular-mode families for nonspherical collapse	pp. 211–213
32	Normalized pair of null vectors	pp. 207–209
33	Formal wave-packet indices	pp. 209–213
34	Constants in the backtraced phase	pp. 207–209
35	Planck-mass scale	pp. 219
36	Spatial radial coordinate	pp. 204–205, 207–209
37	Renormalized stress tensor in the specified prescription	pp. 216–219
38	Signed absorption coefficient of the selected mode	pp. 209–215, 218–219
39	Family of stationary solutions	pp. 204, 219
40	Surface gravity in the specified normalization	pp. 201–203, 207–211, 213–215
41	Area–entropy correspondence	pp. 202–203, 209–211
42	Asymptotic Bogoliubov coefficients for spherical collapse	pp. 206–213
43	Backtraced phase near the last escaping ray	pp. 207–209
44	Nondecrease of generalized entropy in the specified ordering	pp. 202–203
45	Generalized-entropy value	pp. 202–203
46	Energy flux integrated over the specified interval	pp. 216–219
47	Killing energy associated with time translation	pp. 202–203, 207–209
48	Massive-field propagation condition and thermal-emission suppression	pp. 211–213
49	Angular wave packet for nonspherical collapse	pp. 211–213
50	Expected occupation of the specified outgoing mode	pp. 206–207
51	Sign of the absorption coefficient in the superradiant sector	pp. 213–215
52	Scattering condition for bosonic superradiance	pp. 213–215
53	Mass-survival condition to the specified cosmic age	pp. 199, 201–203, 219
54	Temperature corresponding to surface gravity	pp. 201–203, 209–211
55	Thermal occupation after nonspherical collapse	pp. 211–213
56	Thermal occupation in the rotating and charged sectors	pp. 211–215
57	Bosonic thermal occupation in the Schwarzschild sector	pp. 209–211
58	Tortoise coordinate r^*	pp. 204–205, 207–209
59	Particle number integrated over the full duration	pp. 209–211
60	Frequency-banded wave packet of the selected field	pp. 209–211
61	Fermionic thermal occupation in the Schwarzschild sector	pp. 211–213
62	Physical carrier of the collapsing matter	pp. 204–205
63	Black-hole charge value	pp. 204, 213–215
64	Black-hole angular-momentum value	pp. 204, 213–215
65	Spatial geometry of a horizon cross-section	pp. 202–205, 213–215

Definition 8.287 (Hawking 1975 Physical and Formal Input Claims). Let the claim-node

set be $\text{ClaimNodes}_R(59)$, with $\text{ClaimBoundaryIdx}_R = ([11], [59])$. The following table gives each claim, its time uses, auxiliary objects and source locations.

Table 187: Hawking 1975 Physical and Formal Input Claims

p	Claim	i	a	Source Location
1	Event-identification data on the manifold	$\emptyset$	2	pp. 199–201, 204–207
2	Physical carrier of the collapsing matter	$\emptyset$	62	pp. 204–205
3	Black-hole mass, charge and angular-momentum values	$\emptyset$	8–9, 63–64	pp. 201–202, 204, 213–215, 219
4	Spatial geometry of a horizon cross-section	$\emptyset$	65	pp. 202–205, 213–215
5	Quantum-field degrees of freedom, mass, charge and statistics	$\emptyset$	4–5, 7	pp. 211–215
6	Scattering-channel data	$\emptyset$	12	pp. 206–211
7	Quantum-field state data	$\emptyset$	6	pp. 199–201, 206–207, 216–218
8	Area value of a horizon cross-section	$\emptyset$	1	pp. 202–203, 213–215
9	Entropy value of exterior matter	$\emptyset$	3	pp. 202–203
10	Ray-path and endpoint-connection data	$\emptyset$	11	pp. 206–209
11	Physical carrier of the detector or observer	$\emptyset$	10	pp. 199–201, 215–216
12	Formal mode and wave-packet indices	$\emptyset$	14–15, 33	pp. 209–215
13	The classical metric is coupled to the expectation value of the quantum-matter stress tensor.	16^S	29	pp. 199–201
14	The coupling of a classical metric to quantum matter is used as a semiclassical approximation.	16^S	29	pp. 199–201
15	The semiclassical approximation is expected to be good where the curvature is small compared with the Planck value.	$18^B, 19^{IQ}, 20^B, 21^S$	$\emptyset$	pp. 201

p	Claim	i	a	Source Location
16	In a general curved spacetime, orthonormality alone does not uniquely determine the positive-frequency subspace.	$8^Q, 9^Q, 11^B, 16^S, 17^Q, 75^Q$	30	pp. 199–201
17	In the local inertial region of the specified observer, the ambiguity of the particle decomposition is exponentially small for modes of high frequency relative to the curvature scale.	$17^Q, 37^C, 75^Q, 94^C$	2, 10, 19, 27, 29–30	pp. 199–201, 215–216
18	At frequencies at or below the curvature scale, the local particle decomposition and local energy density are ambiguous.	$17^Q, 75^Q$	19, 27, 30	pp. 199–201, 215–216
19	Positive-frequency bases are selected with respect to the specified times in the initial and final flat regions, at null infinity, and in stationary asymptotic regions for massive fields, respectively.	$10^B, 11^B, 12^B, 13^S, 14^Q, 15^Q, 23^B, 48^B, 49^Q, 50^Q, 53^Q, 54^Q, 62^Q, 63^Q, 64^I, 65^S$	5, 12, 14, 16–18, 26, 30–31, 42–43, 48, 50, 54–55, 61	pp. 199–201, 206–207, 211–213
20	Incoming and outgoing bases are related by a Bogoliubov transformation including conjugate components.	$14^{XQ}, 15^{XQ}, 16^S, 49^{XQ}, 50^{XQ}, 53^Q, 58^Q$	12, 16, 26, 30, 42–43, 50, 61	pp. 199–201, 206–209
21	In the specified incoming vacuum, the expected outgoing occupation is $\sum_j \beta_{ij} ^2$, which is positive when the corresponding coefficients contain a nonzero component.	$49^Q, 50^Q$	16, 26, 50, 61	pp. 206–207
22	The finite transient particle creation depends on collapse details, while the late stationary flux is determined by the asymptotic coefficients.	$46^S, 51^B, 52^Q$	12, 30, 42–43, 57, 60–61	pp. 207–209
23	The late thermal factor uses the final-state surface gravity and the effective frequency of the sector, with a scattering coefficient multiplying the emission.	$24^Q, 39^B, 50^Q, 51^B, 69^Q$	7, 14, 18, 26, 29, 31, 38, 40, 42, 49, 55–57, 60	pp. 207–215

p	Claim	i	a	Source Location
24	The affine parameter and retarded time obey an exponential relation, and the backtraced phase is logarithmic near the last escaping ray.	$23^B, 34^Q, 37^C, 41^Q, 42^C, 46^S, 48^B, 50^Q, 53^{XQ}, 54^{XQ}, 55^{XB}, 56^C, 57^{XQ}, 58^{XQ}$	9, 11, 13, 29–30, 32, 34, 36, 40, 42–43, 47, 58	pp. 207–209
25	Outgoing modes are propagated backwards in the specified collapse spacetime using the field equation and the geometrical-optics approximation.	$46^S, 49^{XQ}, 50^{XQ}, 53^{XQ}, 54^{XQ}, 55^B, 56^C, 57^{XQ}, 58^{XQ}$	11–13, 18, 29–30, 32, 34, 40, 42–43, 55, 58	pp. 206–209
26	Frequency-banded wave packets describe occupation in finite temporal regions, and the late packet sequence gives a stationary rate. The infinite total number corresponds to an infinite duration.	$24^Q, 50^Q, 51^B, 52^Q, 54^Q, 59^{QB}, 60^{QS}, 61^I, 92^{IQ}$	14, 23, 26, 30–31, 33, 38, 40, 42, 49, 55, 57, 59–61	pp. 209–213
27	Bosonic occupation has a Planck denominator using the frequency of the sector and a signed greybody coefficient.	$49^Q, 50^Q, 51^B, 59^{QB}, 69^Q$	7, 26, 38, 40, 42, 55–57, 60	pp. 209–211, 213–215
28	The particle flux at infinity has a greybody-corrected thermal spectrum, and the energy flux is weighted by mode frequency.	$24^Q, 26^Q, 34^Q, 50^Q, 51^B, 52^Q, 60^S, 79^{IQ}, 80^Q, 81^Q, 92^{IQ}$	6, 14, 18, 24–26, 29, 37–38, 40, 46, 49, 55–57, 60–61	pp. 209–211, 218–219
29	In the specified geometric units, the temperature is $T = \kappa/(2\pi)$.	$34^Q, 50^Q$	1, 40–41, 47, 54, 57	pp. 201–203, 209–211
30	The thermal denominator for fermionic occupation has the +1 sign, and negative-frequency components contribute positively to the probability flux.	$49^Q, 50^Q, 51^B, 59^{QB}, 69^Q$	7, 26, 38, 40, 42, 56, 60–61	pp. 211–215
31	Massive modes propagating to infinity have a mass threshold, and emission is suppressed when the temperature is low relative to the rest-energy scale.	$34^Q, 62^Q, 63^Q$	5, 17, 26, 40, 48, 54	pp. 211–213

p	Claim	i	a	Source Location
32	In the rotating sector, the horizon frequency is $\omega - m\Omega$, from the specified temporal phase and azimuthal dependence.	$66^Q, 67^Q, 68^Q, 69^Q, 70^S$	4, 15, 21, 51–52, 56	pp. 213–215
33	In the charged sector, the potential with its reference fixed at infinity contributes $-e\Phi$ to the effective frequency.	$68^Q, 69^Q, 70^S$	4, 15, 21, 51–52, 56	pp. 213–215
34	In bosonic superradiance the absorption coefficient is negative; fermion fields have positive probability flux and exhibit no amplification.	$34^Q, 50^Q, 69^Q$	7, 26, 38, 40, 51–52, 56	pp. 213–215
35	Emission is pictured heuristically through pair creation and tunnelling, or through worldlines turning from past-directed to future-directed.	$34^Q, 35^Q, 36^C, 37^C$	$\emptyset$	pp. 202–203
36	Thermal emission is derived from the coefficient calculation for asymptotic modes propagated backwards through the collapse spacetime.	$46^S, 49^Q, 50^Q, 51^B, 53^Q, 58^Q$	12, 14, 18, 26, 29–30, 38, 40, 42–43, 49, 55, 57, 60	pp. 206–213
37	Particle creation is described as a global process involving local observers and the asymptotic particle decomposition.	$17^{XQ}, 22^S, 23^{XB}, 46^S, 50^{XQ}, 75^{XQ}, 76^S$	$\emptyset$	pp. 215–216
38	The local particle decomposition of a freely falling observer and the decomposition at future infinity depend on their respective time choices and curvature scales.	$17^{XQ}, 23^{XB}, 50^{XQ}, 73^B, 74^C, 75^{XQ}, 83^{IQ}$	2, 10, 19, 27, 29–30	pp. 215–216
39	The renormalized stress tensor depends on the field state, renormalization prescription and local curvature.	$17^Q, 34^Q, 75^Q$	24, 37, 46	pp. 216–218
40	Under conservation, stationarity and agreement at infinity, the integrated energy flux in the stationary limit is independent of the exterior integration surface.	$34^Q, 49^Q, 50^Q, 51^B, 60^S, 77^Q, 78^Q, 79^{IQ}, 80^Q, 81^{XQ}, 93^{XQ}$	6, 24–26, 37–38, 40, 46, 60	pp. 216–219

p	Claim	i	a	Source Location
41	Under stationary conservation, positive energy flux to infinity balances negative energy flux into the horizon.	$34^Q, 81^{XQ}, 93^{XQ}$	25, 37	pp. 216–219
42	The negative energy density required for horizon-area decrease is interpreted as arising from the local energy-density indeterminacy of order M^{-4} .	$17^Q, 34^Q, 73^B, 75^Q$	$\emptyset$	pp. 215–218
43	violation of the weak energy condition by quantum effects permits a decrease in event-horizon area	$7^S, 34^Q, 93^Q$	$\emptyset$	pp. 202–203, 216–218
44	In the specified units, the correspondence between the thermal-emission temperature and the first law identifies one quarter of the horizon area with entropy.	$34^Q, 50^Q$	1, 40–41, 54, 57	pp. 202–203, 209–211
45	The generalized entropy $S_{\text{gen}} = S_{\text{out}} + A/4$ does not decrease between successive states.	7^S	1, 3, 44–45	pp. 202–203
46	A solar-mass black hole in a 3 K background lies in a regime where absorption exceeds emission and its mass increases.	$1^S, 2^Q, 6^B, 25^Q, 26^Q$	9	pp. 201–202
47	Small primordial black holes hotter than the background emit more than they absorb.	$1^S, 2^Q, 4^B, 5^{IQ}, 6^B, 25^Q, 26^Q$	8–9, 20, 28, 53	pp. 201–202
48	In the specified net-emission regime, the mass decreases while the temperature and emission rate increase.	$1^S, 2^{QS}, 25^Q, 26^Q, 27^S, 28^S, 34^Q$	9	pp. 201–202, 219
49	The evaporation-lifetime estimate in the emission model has a cubic dependence on the initial mass.	$2^{QS}, 4^B, 5^{IQ}, 6^B, 31^{IQ}$	8, 20, 28, 53	pp. 202, 219
50	For a black hole with mass large compared with the Planck mass, the mass-evolution time scale greatly exceeds the light-crossing time scale.	$2^Q, 82^{IQ}, 83^{IQ}$	9, 39	pp. 219

p	Claim	i	a	Source Location
51	After collapse the black hole settles into a stationary state; where mass evolution is slow relative to light crossing, it is approximated by a sequence of stationary solutions.	$1^S, 38^B, 39^B, 40^{IQ}, 46^S, 82^{IQB}, 83^{IQB}, 84^S$	9, 39	pp. 204, 219
52	The quasistationary approximation breaks down as the mass approaches the Planck scale.	$1^S, 84^S, 85^B$	9, 20, 35	pp. 219
53	The terminal Planck regime lies outside the applicability of the semiclassical description using a classical metric.	85^B	$\emptyset$	pp. 219
54	Complete disappearance without a negative-mass naked singularity, followed by empty spacetime, is presented as an endpoint scenario.	$3^B, 85^B, 88^B, 89^B, 90^B, 91^S$	22	pp. 219
55	The source considers matter reemerging in another universe, or reemerging in this universe to form a negative-mass naked singularity.	$88^B, 89^B, 90^B, 91^S$	$\emptyset$	pp. 219
56	Estimates of terminal explosions depend on the increase in particle species, and the source speculates that primordial black holes converted baryons into radiation.	$27^S, 28^S, 29^{IQ}, 30^S, 32^{IQ}, 33^B, 86^B, 87^S$	$\emptyset$	pp. 202, 219
57	The event-horizon position is determined by the full future history of the black hole.	$71^{CB}, 72^B, 73^B, 74^C$	$\emptyset$	pp. 215–216
58	The eternal solution and collapse geometry are distinguished, with the Schwarzschild exterior and its analytic extension used in backtracing.	$23^B, 37^C, 41^Q, 42^C, 43^B, 44^B, 45^B, 46^S, 47^B, 48^B, 54^Q, 57^Q$	13, 29, 32, 36, 40, 58	pp. 204–205, 207–209
59	The time-translation Killing field fixes the normalization of frequency, energy and energy flux.	$34^Q, 35^Q, 41^Q, 50^Q, 68^Q, 69^Q, 70^S, 79^{IQ}$	4, 6, 15, 21, 25, 29–30, 37, 46–47	pp. 202–203, 207–209, 213–219

Time uses comprise 252 claim/time-content pairs. Superscript codes give their use roles.

Table 188: Hawking 1975 Time-Use Roles

Code	Use Role
I	Interval
Q	Quantity
X	Cross-position correspondence
C	Causal
S	State evolution
B	Reference boundary

Table 189: Hawking 1975 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 7, 13, 21, 30, 42, 56, 60, 74, 84, 87, 91	member	order_relation
2, 5, 8–9, 14–17, 19, 22, 24–29, 31–32, 34–35, 40–41, 46, 49–50, 52–54, 57–58, 61–70, 75–83, 92–93	separate	$\emptyset$
3, 18, 23, 45, 47–48, 86, 90	member	endpoint
4, 33, 73	member	instant, transition_position
6, 59	member	instant
10–12, 20, 38–39, 51, 88–89	member	phase_boundary
36–37, 94	member	orientation
43–44, 71	member	interval_boundary
55	member	instant, interval_boundary
72	dependency	$\emptyset$
85	member	phase_boundary, transition_position

Table 190: Hawking 1975 Claim Groups

Claim Indices	Role
1–11	Physical-carrier and value inputs
12	Formal mode and packet indices
13–59	Source definitions, relations, approximations and conditional statements

Definition 8.288 (Hawking 1975 Claim Boundary). Claims 1–11 form the initial set for physical generation; all 59 claims form the source formal-evidence environment.

Table 191: Hawking 1975 Initial Generation Claims

p	Initial Claim
1	Event-identification data on the manifold
2	Physical carrier of the collapsing matter
3	Black-hole mass, charge and angular-momentum values
4	Spatial geometry of a horizon cross-section
5	Quantum-field degrees of freedom, mass, charge and statistics
6	Scattering-channel data
7	Quantum-field state data
8	Area value of a horizon cross-section
9	Entropy value of exterior matter
10	Ray-path and endpoint-connection data
11	Physical carrier of the detector or observer

The source-specific obligation atoms comprise 59 claims, 94 time contents, 34 transfers and 30 results, totaling 217.

Dependency Inputs The following table gives the time inputs, claim support and auxiliary inputs of each transfer.

Definition 8.289 (Hawking 1975 Time-Content Transfers). Let $\text{ld}\mathbf{x}_R^{\text{audit,transfer}} = [34]$.

Table 192: Hawking 1975 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
1	41	53–54	3, 24–25, 58	9, 36	$v = t + r^*$, $u = t - r^*$
2	41	34–35	13–14, 16, 19, 24, 26, 29, 58–59	29–30	Stationary time translation and its frequency normalization
3	34, 37, 54	57	13–14, 24–25, 29, 58–59	13, 29, 32, 40	$\lambda = -C \exp(-\kappa u)$
4	23, 42, 46, 48, 53	55–56	10, 13–14, 24–25, 58	11, 29	Advanced time of the last escaping ray and the ordering relative to it
5	50, 53, 55–57	58	10, 13–14, 24–25, 29, 58	11, 29, 34, 40	$\omega_{\text{back}} = \omega/[\kappa(v_0 - v)]$
6	50, 54	59–60, 92	12, 16, 19, 26	23, 30, 33	$u_n = 2\pi n/\epsilon$, width $\sim \epsilon^{-1}$, and centre ordering
7	49–51, 53–60, 92	24, 52	6, 10, 12–14, 16–17, 19–22, 24–30, 34, 36, 58	57, 60	From finite packet occupation to the late particle-emission rate

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
8	49–51, 53–60, 92	24, 52	5–6, 10, 12–14, 16–17, 19–22, 24–30, 34, 36, 58	60–61	From finite packet occupation to the late particle-emission rate
9	39, 46, 49–51, 53–54	24, 52	12–14, 17, 19, 21, 23–30, 34, 36, 58	14, 18, 26, 29, 38, 40, 49	Stationary particle-emission rate after nonspherical collapse
10	34, 79	80	7, 39–40, 59	6, 25, 37	Interval average of energy flux integrated over a retarded-time interval
11	34, 79–80	81	7, 28, 39–40, 59	6, 25, 37	Stationary limit of the interval average
12	34, 49–51, 54, 60, 77–78	81	12, 16–17, 19, 21, 24, 26–30, 34, 39–41	24, 26, 37–38, 40, 60	Stationary energy flux from horizon and asymptotic mode expansions
13	34, 81	93	39–41, 59	25, 37	$F_{\text{in},H} = -F_{\text{out},\infty}$
14	24, 49–51, 53–59	26	6, 10, 12–14, 16–17, 19–21, 24–30, 34, 36, 40, 58	57	Energy-emission rate obtained by frequency-weighting the particle spectrum
15	24, 49–51, 53–59	26	5–6, 10, 12–14, 16–17, 19–21, 24–30, 34, 36, 40, 58	61	Energy-emission rate obtained by frequency-weighting the particle spectrum
16	24, 39, 46, 49–51, 53–54	26	12–14, 17, 19, 21, 23–30, 34, 36, 40, 58	55	Energy-emission rate obtained by frequency-weighting the particle spectrum
17	24, 49–51, 69	26	5, 17, 19, 21, 24, 26–30, 32–34, 40	56	Energy-emission rate obtained by frequency-weighting the particle spectrum
18	24, 49–51, 69	26	5, 17, 19, 21, 24, 26–30, 32–34, 40	56	Energy-emission rate obtained by frequency-weighting the particle spectrum
19	25–26, 34	1–2	3, 14–15, 47–48, 52–53	9	$dM/dt = P_{\text{abs}} - P_{\text{em}}$

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
20	25–26, 34	1–2	3, 14–15, 46, 48, 52–53	9	$dM/dt = P_{\text{abs}} - P_{\text{em}}$
21	1–2	27–28	3, 29, 47–49, 56	9	Evolution of $T \propto M^{-1}$ and $P_{\text{em}} \propto M^{-2}$ with the mass
22	2	31	3, 47–49, 52, 56	8, 20, 28	$\Delta t = (M_0^3 - M_{\text{cut}}^3)/(3a)$
23	2	82	3, 48, 50	9	$t_{\text{evol}} = M/ dM/dt $
24	1, 82–83	84	3, 50–51	9, 39	Quasistationary state sequence from time-scale separation
25	1, 84	85	3, 14–15, 52–53	9, 35	Applicability boundary of the quasistationary approximation along mass evolution
26	85, 88–89	3, 90–91	54	22	Conditional endpoint scenario of complete disappearance and empty spacetime
27	8	9	13, 16–17, 19, 21, 25–27, 30, 34	30	Positive-frequency selection with respect to Minkowski time
28	37, 94	17	1, 11, 13–14, 17–18, 38, 58	2, 10, 27, 29	Local inertial time along the observer’s timelike direction
29	17	75	13, 16–19, 21, 25–27, 30, 34, 38	19, 27, 30	Positive-frequency selection with respect to local inertial time
30	66, 68, 70	69	5, 12, 32–33, 59	4, 15, 21	$\omega_{\text{eff}} = \omega - m\Omega - e\Phi$
31	67–68, 70	69	5, 12, 32–33, 59	4, 15, 21	$\omega_{\text{eff}} = \omega - m\Omega - e\Phi$
32	50, 54	59–60, 92	12, 19, 23, 26	14, 23, 31, 33	$u_n = 2\pi n/\epsilon$, widths and centre ordering of angular-mode packets
33	53	49	13, 16–17, 19–21, 25–27, 30, 34	30	Incoming positive-frequency selection with respect to advanced time
34	54	50	13, 16–17, 19–21, 25–27, 30, 34	30	Outgoing positive-frequency selection with respect to retarded time

Grounding Conditions

Conditions for Applying Dependencies or Transfers 1. pp. 204–205, 207–209.

Use Schwarzschild coordinates outside the matter, with $r^* = r +$

$$2M \log |r/(2M) - 1|.$$

2. pp. 202–203, 207–209. Normalize the time-translation Killing field of the Schwarzschild exterior at infinity.
3. pp. 207–209. Use the selected future-directed null generator, affine scale, surface gravity and Killing-field normalization.
4. pp. 207–209. The ray connections in the collapse spacetime determine v_0 ; $v < v_0$ at past null infinity is the escaping side.
5. pp. 207–209. Differentiate with respect to v the phase $-\omega\kappa^{-1} \log[(v_0 - v)/(CD)]$ backtraced to $v < v_0$ in geometric optics.
6. pp. 209–211. Construct finite wave packets from the frequency band $[j\epsilon, (j+1)\epsilon]$ and the outgoing modes of the selected field.
7. pp. 209–211. In the Schwarzschild sector with the specified incoming vacuum, relate finite-band occupation to the packet-centre spacing $2\pi/\epsilon$ and take the late limit. Schwarzschild bosonic sector.
8. pp. 209–211. In the Schwarzschild sector with the specified incoming vacuum, relate finite-band occupation to the packet-centre spacing $2\pi/\epsilon$ and take the late limit. Schwarzschild fermionic sector.
9. pp. 211–213. For collapse settling to an uncharged nonrotating Schwarzschild final state, use angular-mode mixing, the asymptotic Bondi frame, the incoming vacuum and finite packets.
10. pp. 216–219. Integrate the flux of the specified renormalized stress tensor from u_1 to u_2 , then divide by $u_2 - u_1$.
11. pp. 216–219. For the specified field state and stationarity condition, send the width of the retarded-time interval to infinity.
12. pp. 216–219. Use the specified incoming vacuum, positive frequency in the horizon time, mode normalization, finite packets and signed absorption coefficient.
13. pp. 216–219. Relate fluxes across exterior integration surfaces under stress-tensor conservation, late stationarity and agreement with the normal-ordered flux at infinity.
14. pp. 209–215, 218–219. Use $G = c = \hbar = 1$, with occupation and particle-emission rate in the selected field and scattering sector. Schwarzschild bosonic sector.
15. pp. 209–215, 218–219. Use $G = c = \hbar = 1$, with occupation and particle-emission rate in the selected field and scattering sector. Schwarzschild fermionic sector.
16. pp. 209–215, 218–219. Use $G = c = \hbar = 1$, with occupation and particle-emission rate in the selected field and scattering sector. Nonspherical collapse with a Schwarzschild final state.
17. pp. 209–215, 218–219. Use $G = c = \hbar = 1$, with occupation and particle-emission rate in the selected field and scattering sector. Rotating and charged sectors. Bose occupation.

18. pp. 209–215, 218–219. Use $G = c = \hbar = 1$, with occupation and particle-emission rate in the selected field and scattering sector. Rotating and charged sectors. Fermi occupation.
19. pp. 201–202, 219. Within the domain of semiclassical mass balance, compare absorption and emission rates with the same time normalization. Net-emission regime.
20. pp. 201–202, 219. Within the domain of semiclassical mass balance, compare absorption and emission rates with the same time normalization. Net-absorption regime.
21. pp. 201–202, 219. Apply the mass-change rate to the temperature and emission-rate relations in the specified particle-species regime.
22. pp. 202, 219. Integrate $dM/dt = -a/M^2$ from M_0 to M_{cut} over a mass range with constant $a > 0$.
23. pp. 219. Use the specified mass state and its nonzero mass-change rate in the chosen time normalization.
24. pp. 219. Approximate the regime $M \gg M_{\text{Pl}}$, in which mass evolution is much slower than light crossing, by a sequence of stationary solutions.
25. pp. 219. As the mass decreases to order M_{Pl} , the quasistationary approximation breaks down and the classical-metric description reaches its applicability boundary.
26. pp. 199, 202–203, 219. For the case in which no negative-mass naked singularity develops, order collapse, evaporation and the empty spacetime after disappearance.
27. pp. 199–201. Use the flat-spacetime time coordinate, free-field equation and mode normalization.
28. pp. 199–201, 215–216. Use the local inertial frame at the specified event, future orientation, Lorentzian metric and observer.
29. pp. 199–201, 215–216. Normalize the free-field modes in the local inertial region. For $\omega \gg \sqrt{B}$, ambiguity in the particle decomposition becomes exponentially small.
30. pp. 213–215. Evaluate the selected incoming or outgoing temporal phase along the horizon generator, with electric potential referenced to infinity.
31. pp. 213–215. Evaluate the selected incoming or outgoing temporal phase along the horizon generator, with electric potential referenced to infinity.
32. pp. 211–213. For the angular-mode basis of nonspherical collapse, use the specified finite frequency band and outgoing time coordinate.
33. pp. 206–209. Use the canonical affine parameter v at past null infinity, the free-field equation and mode normalization.
34. pp. 206–209. Use the canonical affine parameter u at future null infinity, the free-field equation and mode normalization.

Results

Definition 8.290 (Hawking 1975 Results). Let $\text{Idx}_R^{\text{audit,result}} = [30]$, with each result taken in its specified domain.

Table 193: Hawking 1975 Results and Source Locations

j	Result	Source Location
1	Late bosonic particle creation in spherical collapse	pp. 206–211
2	Late fermionic particle creation in the nonrotating Schwarzschild sector	pp. 211–213
3	Schwarzschild thermal emission after nonspherical collapse	pp. 211–213
4	Particle decomposition with respect to Minkowski time	pp. 199–201
5	Local particle decomposition and curvature-scale ambiguity	pp. 199–201, 215–216
6	Relation between the freely falling local observer and the asymptotic particle decomposition	pp. 215–216
7	Bogoliubov coefficients and outgoing occupation in the incoming vacuum	pp. 206–207
8	Asymptotic Bogoliubov coefficients from backtracing	pp. 206–211
9	Stationary asymptotic bases for a massive field	pp. 211–213
10	Exponential relation between the affine parameter and retarded time	pp. 207–209
11	Logarithmic phase and frequency near the last escaping ray	pp. 207–209
12	Greybody-corrected bosonic thermal spectrum	pp. 209–211
13	Greybody-corrected fermionic thermal spectrum	pp. 211–213
14	Propagation domain and low-temperature emission suppression for a massive field (temperature assigned from surface gravity)	pp. 211–213
15	Propagation domain and low-temperature emission suppression for a massive field (temperature identified from the spectrum)	pp. 211–213
16	Effective horizon frequency due to rotation and charge	pp. 213–215
17	Energy flow and absorption sign in bosonic superradiance	pp. 213–215
18	Thermal occupation in the rotating and charged sectors (bosonic field)	pp. 213–215
19	Thermal occupation in the rotating and charged sectors (fermionic field)	pp. 213–215
20	Balance of positive asymptotic flux and negative horizon flux	pp. 216–219
21	Temperature corresponding to surface gravity (temperature assigned from surface gravity)	pp. 201–203, 209–211

j	Result	Source Location
22	Temperature corresponding to surface gravity (temperature identified from the spectrum)	pp. 201–203, 209–211
23	Area–entropy correspondence from thermal emission and the first law	pp. 202–203, 209–211
24	Mass loss in the net-emission regime	pp. 201–202, 219
25	Evaporation time over the specified mass range and its cubic scale	pp. 202, 219
26	Quasistationary evolution under time-scale separation	pp. 219
27	Applicability boundary of the semiclassical approximation in the Planck regime	pp. 219
28	Generalized second law for the specified entropy assignment	pp. 202–203
29	Global particle creation through local and asymptotic particle decompositions	pp. 215–216
30	Dependence of the event horizon on the full future history	pp. 215–216

Definition 8.291 (Hawking 1975 Result-Dependence Hyperedges). The following table specifies time premises, claim support, auxiliary inputs and result outputs.

Table 194: Hawking 1975 Result-Dependence Hyperedges

j	Time Premises i	Claim Support p	Auxiliary In a	Result Output Index
1	49–51, 53–59	6, 10, 12–14, 16–17, 19–21, 24–30, 34, 36, 58	57	1
2	49–51, 53–59	5–6, 10, 12–14, 16–17, 19–21, 24–30, 34, 36, 58	61	2
3	39, 46, 49–51, 53–54	12–14, 17, 19, 21, 23–30, 34, 36, 58	55	3
4	8–9	16	$\emptyset$	4
5	17, 75, 94	16–18	19	5
6	17, 23, 50, 73, 75	38	$\emptyset$	6
7	49–50	17, 19–21, 26–27, 30, 34	50	7
8	49–50, 53, 55–58	6, 10, 13–14, 16, 19–20, 24–26, 29, 36, 58	42	8
9	62–63	19, 31	17	9
10	54, 57	24–25	$\emptyset$	10

j	Time Premises i	Claim Support p	Auxiliary In a	Result Output Index
11	50, 53, 55–58	10, 13–14, 24–25, 29, 58	43	11
12	49–51, 53–59	6, 10, 12–14, 16–17, 19–21, 24–30, 34, 36, 58	57	12
13	49–51, 53–59	5–6, 10, 12–14, 16–17, 19–21, 24–30, 34, 36, 58	61	13
14	34, 62–63	5, 17, 19, 21, 24, 26–27, 29–31, 34	48	14
15	34, 49–51, 53–59, 62–63	5–6, 10, 12–14, 16–17, 19–21, 24–31, 34, 36, 44, 58	48	15
16	69–70	32–33	$\emptyset$	16
17	69	5, 32–34, 44	51–52	17
18	49–51, 69	5, 17, 19, 21, 24, 26–30, 32–34	56	18
19	49–51, 69	5, 17, 19, 21, 24, 26–30, 32–34	56	19
20	34, 81, 93	39–41	$\emptyset$	20
21	34	24, 29	54	21
22	34, 49–51, 53–59	6, 10, 12–14, 16–17, 19–21, 24–30, 34, 36, 44, 58	54	22
23	34, 49–51, 53–59	6, 8, 10, 12–14, 16–17, 19–21, 24–30, 34, 36, 44, 58	41	23
24	1–2, 25–26	47–48	$\emptyset$	24
25	31	49	$\emptyset$	25
26	82–84	50–51	$\emptyset$	26
27	85	52–53	$\emptyset$	27
28	7	8–9, 45	44	28
29	17, 23, 46, 50, 75–76	37–38	$\emptyset$	29
30	71–72	57	$\emptyset$	30

Results and Evidence-Effect Domain Bosonic and fermionic fields, spherical and nonspherical collapse, rotation and charge, and local and asymptotic particle decompositions use their respective fields, geometries, incoming states, normalizations and approximation conditions. Assigning temperature from surface gravity and identifying it from the spectrum have their respective premises.

Generation Judgment Determination values: The LCTR complete-generation count is 0, the source-evidence-conditioned local-reach count is 0, and 94 primitive-time roots remain.

Result Dependence All 30 results depend on primitive-time roots. The minimal-support families give 320 root–result dependence pairs, including 31 necessary dependence pairs.

Placement of the Grounding Obligation The 94 time contents enter time coordinates, mode frequencies, collapse and emission orderings, fluxes and lifetimes, approximation boundaries and conditional endpoint scenarios. The 252 uses in the claim table belong to these coordinates, definitions, comparisons, evolutions and result derivations. Every transfer takes given time contents as inputs, leaving the physical-time grounding obligation at those primitive inputs.

Evidence of Actual Failure Page 219 states that as the mass decreases to the Planck scale, the quasistationary approximation breaks down and the classical-metric description reaches its applicability boundary. Actual failure of physical-time generation conditions, and resulting changes in result values, remain undetermined.

Downstream Reach Structural reach is determined by the 320 root–result dependence pairs supplied by the minimal-support families. Actual reach from the failure location of a specific generation condition remains undetermined.

Repair Scope Each result holds under its specified field, geometry, incoming state, normalization, conservation law, stationarity and approximation domain. Realizing its time inputs by generated representations that preserve these relations preserves the result within that domain. If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.292 (Hawking 1975 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(94, 30).$$

Also, $R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. The boundary consists of 183 nodes—59 claims, 94 time contents and 30 results—and 64 edges—34 transfers and 30 result edges. Closure from Initial Claims 1–11 remains the initial 11 nodes. Even with all 59 claims supplied, no time content is reached, because every transfer has a time input. All 30 results also have time premises. Supplying all 94 time contents reaches all 30 results. The common reachability classification and result partition therefore give the displayed index tuple. $\square$

Proposition 8.293 (Hawking 1975 Minimal Primitive-Time-Root Support). The following table gives the minimal-support family statistics in the order: number of members, minimum cardinality and maximum cardinality. There are 313 family members in total.

Table 195: Hawking 1975 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(8; 5, 9)
2	(8; 5, 9)
3	(2; 4, 5)
4	(1; 1, 1)
5	(2; 2, 2)
6	(6; 4, 5)
7	(5; 1, 2)
8	(10; 4, 8)
9	(1; 2, 2)
10	(4; 2, 3)
11	(10; 4, 8)
12	(8; 5, 9)
13	(8; 5, 9)
14	(2; 3, 3)
15	(8; 7, 11)
16	(3; 2, 3)
17	(3; 1, 3)
18	(15; 3, 6)
19	(15; 3, 6)
20	(7; 2, 6)
21	(2; 1, 1)
22	(8; 5, 9)
23	(8; 5, 9)
24	(43; 3, 11)
25	(29; 1, 10)
26	(31; 3, 11)
27	(58; 1, 11)
28	(1; 1, 1)
29	(6; 5, 6)
30	(1; 2, 2)

Proof. Start from the singleton seed of each time content and apply Definition 7.46 to the 34 transfers. After the antichains stabilize, combine the supports of each result's time premises and discard every support set that properly contains another support set. Proposition 7.47 yields the displayed families and counts. $\square$

Proposition 8.294 (Hawking 1975 Audit Conclusion). Complete generation and local reach are both zero; all 94 time contents remain primitive-time roots. All 30 results depend on primitive-time roots.

Proof. This follows from the reachable sets in Proposition 8.292 and the minimal-support families in Proposition 8.293. $\square$

Individual Statements

Objects of the Statements The incompleteness of quantum gravity theory; the nonuniqueness of the particle concept in curved spacetime; unit conventions and the curvature scale; the domain of validity of the semiclassical approximation in the early universe; conversion of the lifetime of the universe into units of Planck time; the heuristic status of the pair-creation picture near the horizon; positive frequency relative to a horizon time parameter; and failure of the quasi-stationary approximation in the Planck-mass regime.

Verbatim Original (English) I think it would be fair to say that we do not yet have a fully satisfactory and consistent quantum theory of gravity.

Although it is obvious what the subspace spanned by the $\{f_i\}$ is for an asymptotically flat region, it is not uniquely defined for a general point of a curved space-time.

Thus the uncertainty in the local energy density caused by the ambiguity in defining modes of wavelength longer than the local radius of curvature $B^{-1/2}$, is of order B^2 in units in which $G = c = \hbar = 1$.

Thus, as far as we are concerned, the classical geometry–quantum matter treatment should be valid apart from the first 10^{-43} s of the universe.

Even though the effects of particle creation may be negligible locally, I shall show in this paper that they can add up to have a significant influence on black holes over the lifetime of the universe $\sim 10^{17}$ s or 10^{60} units of Planck time.

One might picture this negative energy flux in the following way.

It should be emphasized that these pictures of the mechanism responsible for the thermal emission and area decrease are heuristic only and should not be taken too literally.

positive frequency with respect to the time parameter defined by K^a along the generators of the horizon

Eventually, when the mass of the black hole is reduced to 10^{-5} g, the quasi-stationary approximation will break down. At this point, one cannot continue to use the concept of a classical metric.

The second statement concerns the subspace spanned by $\{f_i\}$. The third concerns the curvature scale $B^{-1/2}$, the unit convention $G = c = \hbar = 1$, and an uncertainty of order B^2 . The fourth specifies the interval 10^{-43} s. The fifth compares $\sim 10^{17}$ s with 10^{60} units of Planck time. The eighth uses the field K^a , and the ninth specifies the mass 10^{-5} g.

Original-Source Locations The first quotation appears in Hawking, *Particle Creation by Black Holes*, *Communications in Mathematical Physics* 43, p. 199, Introduction. The second appears on p. 200 and the third through fifth on p. 201, in the Introduction: the second concerns the local particle concept; the third, the curvature scale and

unit convention; the fourth, the domain of validity of the semiclassical treatment in the early universe; and the fifth, the scale of the lifetime of the universe. The sixth introduces the negative-energy pair picture on p. 202, and the seventh is the qualification immediately following that picture on the same page. The eighth appears on p. 217, §4, in the paragraph on horizon-mode selection before Eq. (4.3), and the ninth in the terminal discussion on p. 219, §4.

Minimal Context The first quotation belongs to the introduction of the semiclassical construction that treats quantum matter on a classical general-relativistic metric. The second concerns ambiguity in the local particle concept for modes whose wavelengths exceed the curvature scale; the third concerns the resulting estimate of local energy density. The fourth and fifth state the curvature limit of the semiclassical treatment and the time scale over which particle-creation effects accumulate. Immediately after introducing the negative-energy pair picture near the horizon, the sixth and seventh restrict that picture to heuristic status. The eighth gives the condition selecting modes on the quasi-stationary horizon in the final state; the ninth concerns the endpoint of the semiclassical description as mass decreases.

Attributed Author The nine quotations are attributable to S. W. Hawking's exposition in the article.

Content Stated Directly by the Quotations The first quotation states that a consistent quantum theory of gravity is incomplete; the second, that the particle subspace is not unique at a general point of curved spacetime. The third states an uncertainty estimate set by the curvature scale and a unit convention; the fourth, the domain of validity of the semiclassical treatment in the early universe. The fifth relates the lifetime of the universe in seconds to units of Planck time. The sixth and seventh introduce the pair-creation picture and restrict its status to that of a heuristic picture. The eighth specifies positive frequency relative to a given time parameter; the ninth states that the quasi-stationary approximation and classical-metric description fail in the Planck-mass regime.

Explicit Domain of Application The explicit domains of the nine quotations are the semiclassical classical-geometry–quantum-matter construction; mode selection relative to a curvature scale; the horizon and asymptotic regions of a collapse-formed black hole; quasi-stationary late-time radiation; and mass decrease into the Planck-mass regime.

Evidence Provenance Hawking, *Particle Creation by Black Holes*, pp. 199–202, 217, 219.

Individual Audit

Interpretive Possibilities The first statement says that a consistent quantum theory of gravity remains incomplete. The second and third concern the nonuniqueness of the particle subspace in curved spacetime and the resulting uncertainty in local energy density. The fourth gives an expectation for the early-universe domain of the semiclassical treatment; the fifth gives the cosmic lifetime over which particle-creation effects accumulate. The sixth and seventh concern the heuristic pair picture; the eighth specifies positive frequency relative to the horizon time parameter; the

ninth states the breakdown of the quasistationary approximation in the Planck-mass regime.

Independent Grounds Sections 2 and 3 derive particle creation and thermal emission using field-mode expansions, Bogoliubov coefficients and finite packets. Section 4 establishes the negative horizon-energy flux from stress-tensor conservation, stationarity and agreement at infinity. The local particle concept depends on the curvature scale and the observer’s time choice; quasistationary evolution rests on separation between mass-evolution and light-crossing time scales.

Evidence-Effect Boundary Ambiguity in particle number and local energy density is evaluated under the selected time coordinate and curvature scale. The convention $G = c = \hbar = 1$ specifies unit conversion; seconds and Planck-time units express the cosmic lifetime. Pair creation is presented as a heuristic picture. Horizon modes are selected relative to the time parameter defined by K^a , and the classical-metric and quasistationary descriptions reach their applicability boundary as the mass approaches the Planck scale.

Corresponding Formal Evidence The first statement corresponds to Claims 13–15. The second and third correspond to Claims 16–18 and 42, Time Contents 8–9, 17, 75, and Results 4 and 5. The fourth corresponds to Claim 15 and Time Contents 18–21. The cumulative time scale in the fifth statement corresponds to Time Contents 5–6, 22. The sixth and seventh correspond to Claim 35 and Time Contents 34–37. The eighth corresponds to Claims 40 and 59, Time Contents 34, 77–78, and the stationary-flux calculation in Result 20. The ninth corresponds to Claims 52 and 53, Time Contents 1, 84–85, and Result 27.

Within-Reference Comparison Curvature dependence of the local particle concept, mode selection relative to asymptotic time, late stationary radiation, mass-evolution time scales and the Planck-regime approximation boundary are connected under their respective source conditions. The heuristic picture on p. 202 accompanies the mode derivation in Section 2, the rotation and charge extensions in Section 3, and flux conservation and stationarity in Section 4.

Common Audit with Specified Domain of Application Time coordinates, frequencies, intervals and orderings enter the definitions, comparisons and evolution of the respective fields and geometries. Their dependencies are specified in the time-use, transfer and result-dependence tables above.

8.19.21 Source Audit

Audit Inputs

Source and Audit Scope

Source Identifier 21.

Source Name Hawking 1976

Bibliographic Identification S. W. Hawking, *Breakdown of Predictability in Gravitational Collapse*, *Physical Review D* 14, 2460–2473 (1976) is denoted by $R = \text{Ref}_{\text{Haw76BP}}$. The audit domain comprises the abstract and §§I–V.

Audit-Scope Identifier 21-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Original-Source Location Identifier 21-L1.

Original-Source Location Content Each table identifies the source pages of its time contents, claims, transfers or results.

Formal Objects The audit uses 77 time contents, 99 auxiliary objects, 59 claims, 12 transfers and 35 results.

Assumptions Hawking’s respective calculations use the classical singularity conditions, the initial/final/hidden-sector decomposition, the specified incoming vacuum, semiclassical collapse geometry, mode normalization, the late stationary approximation, transmission and reflection, and the quasistationary domain. Relaxation in a box uses thermal equilibrium and ergodicity; correspondence between formation and disappearance histories uses CPT invariance with the specified boundary conditions.

Non-temporal Physical Inputs The initial physical inputs are the carrier and value data in Claims 1–10: event identification, collapsing matter, boundary surfaces, observation records, hidden-sector and quantum-field degrees of freedom, black-hole mass, charge and angular momentum, horizon cross-sections, detectors and observers, and signal connections.

Formal Input Elements

Definition 8.295 (Hawking 1976 Time Contents and Auxiliary Objects). Let $n_R = 77$. The following tables specify the time contents and auxiliary objects.

Table 196: Hawking 1976 Time Contents

i	Time Content	Source Location
1	Singular boundary of a classical history	pp. 2460
2	Temporal reference role of the initial surface and state	pp. 2460
3	Temporal reference role of the final surface and state	pp. 2460
4	Causal-boundary role of the hidden surface	pp. 2460
5	Temporal course of evaporation	pp. 2460
6	Temporal position of black-hole formation	pp. 2460
7	Order from formation to evaporation	pp. 2460
8	Order of the map from initial to final state	pp. 2460
9	State evolution on classical spacetime	pp. 2460
10	Temporal position of an observer crossing the horizon	pp. 2461
11	Temporal change of black-hole mass due to emission	pp. 2461
12	Endpoint under the complete-evaporation assumption	pp. 2461
13	Light-cone causal structure in the heuristic tunnelling picture	pp. 2461

<i>i</i>	Time Content	Source Location
14	Causal accessibility to future infinity	pp. 2462
15	Future null infinity	pp. 2462
16	Past null infinity	pp. 2462
17	Horizon-crossing time to be identified by a signal	pp. 2462
18	Temporal resolution of a crossing-time signal	pp. 2462
19	Past-directed worldline in the heuristic tunnelling picture	pp. 2462
20	Temporal position of emission events	pp. 2462
21	Propagation of outgoing components to future infinity	pp. 2462
22	Propagation of barrier-reflected components into the horizon	pp. 2462
23	Temporal position of the early-universe hidden surface	pp. 2463
24	Illustrative Planck time of the initial hidden surface	pp. 2463
25	Cosmic expansion history	pp. 2463
26	Time-reversal asymmetry of the early universe	pp. 2464
27	Temporal process of cosmological phase separation	pp. 2464
28	Causal reach of the cosmological particle horizon	pp. 2464
29	Early time independence in the compared particle-creation account	pp. 2464
30	Late-universe hypersurface for configuration counting	pp. 2464
31	Temporal order of the early and late surfaces	pp. 2464
32	Temporal position of the final evaporation stage	pp. 2464
33	Quasi-stationary sequence of late Schwarzschild approximations	pp. 2464
34	Causal order of retarded and advanced Green functions	pp. 2464
35	Initial temporal role of incoming modes	pp. 2465
36	Incoming positive frequency with respect to advanced time	pp. 2465
37	Final temporal role of outgoing modes	pp. 2465
38	Outgoing positive frequency with respect to retarded time	pp. 2465
39	Horizon-crossing relation of inward modes	pp. 2465
40	Late temporal role of inward horizon modes	pp. 2465
41	Choice of positive-frequency splitting for horizon modes	pp. 2465
42	Asymptotic orientation from past to future	pp. 2465
43	Advanced time coordinate v	pp. 2465
44	Retarded time coordinate u	pp. 2465
45	Mode time dependence under stationary time translations	pp. 2465
46	Temporal phase of stationary modes	pp. 2465
47	Backward propagation history of outgoing modes	pp. 2466
48	Temporal position of late radiation modes	pp. 2466

i	Time Content	Source Location
49	Advanced time v_0 of the last escaping ray	pp. 2466
50	Advanced-time order relative to the last escaping ray	pp. 2466
51	Logarithmic temporal phase of back-propagated modes	pp. 2466
52	Retarded-time centre of a finite wave packet	pp. 2466
53	Temperature evolution accompanying quasi-stationary mass loss	pp. 2467
54	Flow generated by stationary time translations	pp. 2468
55	Timelike worldline of a stationary observer	pp. 2468
56	Timelike worldline of a freely falling observer	pp. 2468
57	Schwarzschild time coordinate	pp. 2468
58	Past horizon used in the analytic extension	pp. 2469
59	Affine parameter U on the past horizon	pp. 2469
60	Positive frequency with respect to the affine parameter U	pp. 2469
61	Temporal position of the partner entering shortly after formation	pp. 2470
62	Infinite emission duration in the constant-mass approximation	pp. 2471
63	Stationary particle-emission rate	pp. 2471
64	Finite emission interval with negligible mass change	pp. 2471
65	High-temperature stage after most mass has been lost	pp. 2471
66	Early quasi-stationary emission stage	pp. 2471
67	Late emission stage considered for information recovery	pp. 2471
68	Temporal order of quasi-stationary and final emission	pp. 2471
69	Long interval of evolution in the reflecting box	pp. 2473
70	Temporal role of the initial nonequilibrium state	pp. 2473
71	Position of the first density-map application	pp. 2473
72	Order of successive density-map applications	pp. 2473
73	Relaxation toward thermal equilibrium	pp. 2473
74	Role of the thermal-equilibrium state after a long time	pp. 2473
75	Time-reversed order of formation and evaporation	pp. 2473
76	Candidate endpoint stabilized at the Planck mass	pp. 2473
77	Order of measurements after horizon crossing	pp. 2473

Table 197: Hawking 1976 Auxiliary Objects

a	Auxiliary Object	Source Location
1	Values of final observation records	pp. 2462–2463
2	Energy of a photon signalling the crossing time	pp. 2462

<i>a</i>	Auxiliary Object	Source Location
3	Number of configurations on the specified surface	pp. 2463–2464
4	Incoming mode family	pp. 2464–2465
5	Outgoing mode family	pp. 2465, 2468
6	Identification data for an observation event	pp. 2462–2463
7	Mean occupation of an outgoing mode	pp. 2467
8	Probability distribution of outgoing particle number	pp. 2467, 2471
9	Positive Killing energy of the outgoing pair member	pp. 2471
10	Negative Killing energy of the inward pair member	pp. 2471
11	Mass change over a finite interval	pp. 2467, 2471
12	Black-hole mass value	pp. 2462, 2464
13	Fraction of the initial mass lost	pp. 2471
14	Radiation temperature value	pp. 2467
15	Number of emitted particles	pp. 2467, 2471
16	Planck energy used as the scattering scale	pp. 2461, 2472
17	Center-of-mass energy of the scattering particles	pp. 2472
18	Energy retained in the reflecting box	pp. 2472–2473
19	Probability of a box configuration	pp. 2473
20	Particle configuration in the box	pp. 2472–2473
21	Planck mass value of a candidate endpoint	pp. 2464, 2473
22	Final probability distribution under specified initial conditions	pp. 2463, 2472
23	Boundary conditions on the hidden surface	pp. 2462–2463, 2473
24	Nearly zero total energy of the early universe	pp. 2463–2464
25	Thermal macrostate selected by configuration counting	pp. 2463–2464
26	Spacelike tunnelling curve in the heuristic picture	pp. 2461–2462
27	Event-identification data	pp. 2462–2463
28	Physical carrier of the collapsing matter	pp. 2460, 2462, 2464
29	Physical carriers and spatial geometry of boundary surfaces	pp. 2462–2463
30	Physical degrees of freedom in the hidden sector	pp. 2462–2463
31	Physical degrees of freedom of the scalar field	pp. 2464–2465
32	Black-hole charge and angular-momentum values	pp. 2462, 2464
33	Physical carriers of observers and detectors	pp. 2462, 2468, 2471
34	Signal emission–reception connection data	pp. 2462
35	Data on the initial surface	pp. 2462–2463
36	Data on the hidden surface	pp. 2462–2463
37	Components of the initial density operator	pp. 2463, 2472
38	Joint amplitude of the outgoing and hidden sectors	pp. 2463
39	Components of the outgoing reduced density operator	pp. 2463, 2466
40	Partial-trace operation over the hidden sector	pp. 2463, 2466
41	Superscattering map between density operators	pp. 2463
42	S matrix between pure states	pp. 2472
43	Entropy of a density operator	pp. 2472
44	Number of box states in the finite energy window	pp. 2473
45	Uniform density operator in the box	pp. 2473

<i>a</i>	Auxiliary Object	Source Location
46	Spatial configuration of a box with perfectly reflecting walls	pp. 2472–2473
47	Conserved-energy window from E to $E + \Delta E$	pp. 2473
48	CPT transformation of states and boundary conditions	pp. 2473
49	Configuration weights and measure for the early universe	pp. 2463–2464
50	Wavelengths of the signal photon and infalling particle	pp. 2462
51	Energy budget of the collapsing matter and signals	pp. 2462
52	Specified Schwarzschild exterior geometry	pp. 2464, 2468
53	Surface-gravity value κ	pp. 2466–2467
54	Radial coordinate in the exterior region	pp. 2464, 2468
55	Klein–Gordon inner product and mode normalization	pp. 2464–2465
56	Specified incoming vacuum state	pp. 2465–2466
57	Family of inward horizon modes	pp. 2465–2466
58	Bogoliubov transformation coefficients	pp. 2465–2466
59	Barrier-transmission probability	pp. 2467
60	Barrier-reflection probability	pp. 2467, 2470
61	Thermal factor at the specified frequency	pp. 2467, 2470
62	Amplitude of correlated pair-occupation states	pp. 2470–2471
63	Angular-momentum labels of paired modes	pp. 2468, 2471
64	Mass-energy scale for baryon emission	pp. 2471
65	Baryon numbers of forming matter and radiation	pp. 2471
66	Information capacity associated with late particle number and energy	pp. 2471–2472
67	Energy remaining in the final stage	pp. 2471–2472
68	Unit convention $G = c = \hbar = k = 1$	pp. 2464
69	Fourier-superposition coefficients of a finite wave packet	pp. 2467
70	Configuration weights at equal emitted energy	pp. 2471
71	Spacetime geometry of the classical singularity argument	pp. 2460–2461
72	Components of the specified pure initial state	pp. 2463
73	Three-index amplitude linking the initial, final and hidden sectors	pp. 2463
74	Final operator components of the specified density map	pp. 2463, 2472
75	Observable operator acting on the outgoing sector	pp. 2466
76	Expectation value of an observable at future null infinity	pp. 2466
77	Reference scales and origins of logarithmic coordinates	pp. 2468–2469
78	Transformation coefficients between analytically continued modes	pp. 2469–2470
79	Incoming vacuum specified in the analytic extension	pp. 2469–2470

<i>a</i>	Auxiliary Object	Source Location
80	Particle absence in the freely falling mode decomposition	pp. 2471
81	Reduced density operator of thermal outgoing radiation	pp. 2467, 2471
82	Expected emitted particle number in the specified finite interval	pp. 2467, 2471
83	Probability of a finite-total-particle configuration in the infinite-time limit	pp. 2471
84	Energy remaining at the onset of thermally allowed baryon emission	pp. 2471
85	Bound on baryon-number recovery by final radiation	pp. 2471
86	Reduced entropy of quasi-stationary radiation	pp. 2471–2472
87	Bound on late information recovery under the specified capacity premise	pp. 2471–2472
88	Trace-preservation condition of the superscattering map	pp. 2472
89	Unitarity condition of the superscattering map	pp. 2473
90	Equilibrium density operator in the box after a long time	pp. 2472–2473
91	CPT correspondence condition on formation and disappearance histories	pp. 2473
92	Black-hole/white-hole correspondence for observers at infinity	pp. 2473
93	Communicability of internal records to infinity	pp. 2473
94	Specified semiclassical spacetime including the collapsing interior	pp. 2464, 2466
95	Counts of baryon and antibaryon configurations at specified energy	pp. 2464
96	CP-dependent energy splitting of expanding configurations	pp. 2464
97	Expected cosmic baryon number	pp. 2464
98	Spatial pattern of baryon and antibaryon domains	pp. 2464
99	General linear superscattering map between initial and final density operators	pp. 2463, 2472

Definition 8.296 (Hawking 1976 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(59)$, with $\text{ClaimBoundaryIdx}_R = ([10], [59])$. Claims 1–10 represent the initial generation inputs of this audit. Claims 11–59 represent Hawking’s physical settings, calculations, arguments and proposals. The following table gives each claim, its time uses, auxiliary objects and source locations.

Table 198: Hawking 1976 Physical and Formal Input Claims

p	Claim	i	a	Source Location
1	Event-identification data	$\emptyset$	6, 27	pp. 2462–2463
2	Physical carrier of the collapsing matter	$\emptyset$	28	pp. 2460, 2462, 2464
3	Physical carrier and spatial geometry of boundary surfaces	$\emptyset$	29	pp. 2462–2463
4	Values of observational records	$\emptyset$	1, 6	pp. 2462–2463
5	Physical degrees of freedom of the hidden sector	$\emptyset$	30	pp. 2462–2463
6	Physical degrees of freedom of the quantum field	$\emptyset$	31	pp. 2464–2465
7	Black-hole mass, charge and angular-momentum values in the specified state	$\emptyset$	12, 32	pp. 2462
8	Spatial geometry of a horizon cross-section	$\emptyset$	29	pp. 2464, 2470
9	Physical carrier of the detector or observer	$\emptyset$	33	pp. 2462, 2468, 2471
10	Connection data for signal-emission and reception endpoints	$\emptyset$	34	pp. 2462
11	Under general relativity and the specified energy and global conditions, gravitational collapse has a singular boundary of the classical spacetime description.	$9^S, 1^B$	28, 71	pp. 2460–2461
12	Modifications of general relativity, cosmic censorship and quantization of gravity are examined as attempts to avoid the breakdown of predictability at singularities.	9^S	$\emptyset$	pp. 2461
13	Cosmic censorship is treated as the unproved hypothesis that collapse singularities developing from an initially nonsingular asymptotically flat state are hidden from infinity by an event horizon.	$14^C, 6^B, 2^B, 15^B$	28, 30	pp. 2461

p	Claim	i	a	Source Location
14	In perturbative quantum gravity about a fixed background, closed loops integrate over all energies and the perturbation series breaks down at the Planck scale, motivating a new approach to quantization.	$9^S, 13^C$	16–17	pp. 2461
15	Emission is pictured heuristically by relating tunnelling along spacelike or past-directed worldlines to the creation of positive- and negative-energy pairs.	$20^B, 21^{CS}, 19^C, 13^C$	9–10, 26	pp. 2461–2462
16	The interaction region is bounded by an initial surface carrying prescribed data, a final surface carrying measurements, and a hidden surface about which the observer has limited information.	$3^B, 4^{CB}, 2^B, 8^S$	1, 29–30, 35–36, 38, 72–73	pp. 2462–2463
17	The mass, angular momentum and charge available at infinity are compatible with multiple field configurations on the hidden surface.	$14^C, 4^{CB}$	12, 32, 36	pp. 2462
18	Hawking argues that, for observers just outside the horizon to signal every infalling particle's crossing time to observers at future infinity with the specified accuracy, each photon must have the same wavelength and energy as the corresponding particle, so the total signalling energy exhausts the energy of the collapsing matter.	$14^C, 18^{IQ}, 17^{OQ}, 10^B$	2, 28, 34, 50–51	pp. 2462

p	Claim	i	a	Source Location
19	The principle of ignorance assigns equal probabilities to hidden-surface configurations compatible with the known conserved quantities. In its extension to a sum over histories including gravity, the measure remains unspecified.	4^{CB}	23, 36	pp. 2462–2463
20	Data on the initial, final and hidden surfaces are represented in their respective Hilbert spaces.	$3^{\text{B}}, 4^{\text{CB}}, 2^{\text{B}}, 8^{\text{S}}$	30, 38, 72–73	pp. 2463
21	The specified initial state and three-index amplitude determine the joint amplitude of the final and hidden sectors.	$3^{\text{B}}, 4^{\text{CB}}, 2^{\text{B}}, 8^{\text{S}}$	30, 38, 72–73	pp. 2463
22	Taking the partial trace over an orthonormal basis of the hidden sector gives the reduced density operator describing final observations.	$3^{\text{B}}, 4^{\text{CB}}, 2^{\text{B}}, 15^{\text{B}}, 8^{\text{S}}$	38–41, 73, 75–76	pp. 2463, 2466
23	For a pure joint state entangling the final and hidden sectors, the reduced state of the final sector is mixed.	$3^{\text{B}}, 4^{\text{CB}}$	38–40	pp. 2463, 2466
24	Hawking proposes information loss through black-hole radiation and hidden surfaces as a fundamental limitation on prediction additional to ordinary quantum uncertainty.	$12^{\text{B}}, 5^{\text{S}}, 3^{\text{B}}, 6^{\text{B}}, 2^{\text{B}}, 8^{\text{S}}$	22	pp. 2460–2464, 2471–2472
25	A massless Hermitian scalar field is considered on an uncharged, nonrotating collapse spacetime; the late exterior is approximated by a slowly varying Schwarzschild solution, with $G = c = \hbar = k = 1$.	$43^{\text{Q}}, 47^{\text{S}}, 14^{\text{C}}, 9^{\text{S}}, 50^{\text{C}}, 6^{\text{B}}, 16^{\text{B}}, 15^{\text{B}}, 49^{\text{QB}}, 32^{\text{B}}, 33^{\text{S}}, 44^{\text{Q}}, 57^{\text{Q}}, 54^{\text{S}}$	12, 14, 31, 34, 52–54, 68, 77, 94	pp. 2464
26	The wave equation and commutation relation are used, with the commutator Green function specified as half the difference between the retarded and advanced solutions.	$9^{\text{S}}, 34^{\text{C}}$	31, 55	pp. 2464

p	Claim	i	a	Source Location
27	Mode families at past infinity, future infinity and the horizon are specified, with positive frequency defined relative to advanced time, retarded time or an affine parameter on the horizon.	$43^Q, 60^Q, 59^Q, 42^C, 41^Q, 39^{CB}, 40^B, 36^Q, 35^B, 16^B, 15^B, 38^Q, 37^B, 44^Q, 57^Q, 45^S, 46^Q, 55^S, 54^S$	4–5, 12, 31, 52, 54–55, 57, 77	pp. 2464–2466, 2468–2469
28	The positive-frequency decomposition of horizon modes is nonunique, but its choice does not affect observables or the reduced density operator at future infinity.	$3^B, 4^{CB}, 41^Q, 39^{CB}, 15^B$	38–40, 57, 75–76	pp. 2465–2466
29	Bogoliubov coefficients are obtained by decomposing outgoing modes propagated backwards through the collapse spacetime into incoming positive and negative frequencies; the occupation in the incoming vacuum is $\sum_k \beta_{jk} ^2$.	$43^{XQ}, 47^S, 51^{XQ}, 14^C, 50^C, 6^B, 36^{XQ}, 16^B, 15^B, 49^{QB}, 48^B, 38^{XQ}, 52^{QB}$	4–5, 7, 34, 53, 55–56, 58–59, 69, 77, 94	pp. 2465–2467
30	In the specified late wave-packet approximation, the outgoing reduced density matrix is diagonal in the number basis and is expressed as a product of mode probabilities.	$36^Q, 48^B, 38^Q, 52^{QB}$	8, 40, 55–56, 58–60, 62, 69, 81	pp. 2467
31	The mean occupation at future infinity is $\Gamma/(e^{2\pi\omega/\kappa} - 1)$; the number probabilities contain the thermal factor and the barrier-transmission probability Γ .	$21^{CS}, 36^Q, 48^B, 38^Q, 52^{QB}, 22^{CS}$	7–8, 40, 53, 55–56, 58–62, 69, 81	pp. 2467
32	In a finite interval at fixed temperature, configurations with the same total energy have equal weights before barrier transmission. Particle-number probabilities at infinity are determined by the occupation distribution incorporating transmission and reflection at the barrier.	$64^I, 38^Q$	15, 59, 62, 70	pp. 2471

p	Claim	i	a	Source Location
33	Analytic continuation in the affine parameter on the past horizon gives the thermal factor relating exterior and interior modes.	$60^Q, 59^{XQ}, 38^Q, 58^B, 44^{XQ}$	53, 55, 61, 77–78	pp. 2468–2470
34	The two vacuum-annihilation conditions give equal occupation numbers for correlated exterior and interior mode pairs, with amplitudes proportional to powers of the thermal factor.	$60^Q, 61^{CB}, 38^Q, 46^Q$	55, 61–62, 78–79	pp. 2470–2471
35	The specified correlated pairs have opposite angular momenta and Killing-energy signs, and the inward negative-energy component contributes to mass loss. Positive-energy components reflected by the barrier also enter the horizon.	$61^{CB}, 63^Q, 21^{CS}, 64^I, 39^{CB}, 11^S, 38^Q, 22^{CS}, 46^Q, 54^S$	9–12, 62–63	pp. 2468, 2470–2471
36	Stationary observers use modes defined by stationary time translations; in the source’s freely falling decomposition, modes that remain positive-frequency continuously across the horizon contain no particles.	$60^{XQ}, 59^Q, 56^{XS}, 10^B, 38^Q, 44^Q, 45^S, 46^Q, 55^{XS}, 54^S$	5, 31, 33, 52, 55, 79–80	pp. 2468, 2471
37	In the constant-mass approximation, emission continues steadily for an infinite duration, and the probability of a specified configuration with finite total particle number is zero.	$63^Q, 62^I$	8, 83	pp. 2471
38	Over a finite emission interval in which the mass change is negligible, Eq. (4.25) gives the relative configuration probabilities.	$63^Q, 64^I, 38^Q, 52^{QB}, 45^S$	7, 11, 15, 62, 69–70, 82	pp. 2471

p	Claim	i	a	Source Location
39	Net radiated energy reduces the mass, and in the quasistationary Schwarzschild regime the temperature rises according to $T = \kappa/(2\pi) = 1/(8\pi M)$.	$63^Q, 64^I, 11^S, 33^S, 53^S, 54^S$	9, 11–12, 14, 52–53, 68	pp. 2467
40	Complete disappearance is used as the endpoint of radiative mass loss, and the alternative of stabilization at the Planck mass is opposed on CPT grounds.	$12^B, 5^S, 3^B, 6^B, 7^S, 2^B, 32^B, 76^B, 75^C$	21, 23, 48, 91	pp. 2464, 2473
41	An initially massive, cold black hole emits mainly massless particles and loses most of its mass before reaching temperatures that permit baryon emission. Under complete disappearance and the remaining-energy budget, the source argues that the initial baryon number cannot be restored by the radiation.	$12^B, 66^B, 68^S, 65^{BS}, 53^S$	12–13, 64–65, 68, 84–85	pp. 2471
42	Using a capacity assumption relating information to particle number and energy, the source argues that the energy remaining in the final stage of complete evaporation cannot restore the information associated with the quasistationary radiation.	$12^B, 66^B, 68^S, 32^B, 67^B$	15, 66–67, 86–87	pp. 2471–2472
43	The superscattering operator is a linear map from initial to final density operators.	$3^B, 4^{CB}, 2^B, 8^S$	37, 40–41, 73–74, 99	pp. 2463, 2472
44	In the sector where the probability of black-hole formation is negligible, the S matrix maps pure states, and the density-operator map is $\rho_2 = S\rho_1 S^\dagger$.	$3^B, 2^B, 8^S$	37, 42, 74	pp. 2472

p	Claim	i	a	Source Location
45	Hawking claims that scattering with centre-of-mass energy at the Planck scale and an appreciable probability of black-hole formation and evaporation has nonzero final entropy and does not admit S-matrix factorization.	$5^S, 3^B, 6^B, 7^S, 2^B, 8^S$	16–17, 43, 74	pp. 2472
46	Normalization of the final probabilities for every initial density operator gives trace preservation of superscattering. Reverse normalization is the condition of preserving the identity operator.	$3^B, 2^B, 8^S$	41, 88–89, 99	pp. 2472
47	For the N states in a finite energy window in a reflecting box, thermal equilibrium and ergodicity are assumed, and probability $1/N$ for each state and entropy $\ln N$ are expected after a long time.	$74^{PB}, 5^S, 71^B, 6^B, 7^S, 70^B, 69^{IP}, 73^{PS}, 72^S$	18–20, 37, 44–47, 90, 99	pp. 2472–2473
48	Using time evolution approximated by successive superscattering maps and thermal relaxation in the finite box, preservation of the identity operator is argued from the impossibility of perpetual motion.	$74^B, 71^{PB}, 70^B, 69^I, 73^{PS}, 72^{PS}$	37, 44–47, 88–90, 99	pp. 2472–2473
49	Assuming CPT invariance, including that of the hidden-surface boundary conditions, CPT reversal and trace preservation give reverse normalization.	$3^B, 2^B, 8^S, 75^C$	23, 48, 88–89, 99	pp. 2473
50	On the basis of CPT invariance, Hawking claims that complete-disappearance processes corresponding to formation processes exist and that black holes do not stabilize at the Planck mass.	$12^B, 3^B, 6^B, 7^S, 2^B, 76^B, 75^C$	21, 23, 48, 91	pp. 2473

p	Claim	i	a	Source Location
51	Through the correspondence of CPT-reversed formation and evaporation histories, the source claims that an observer at infinity has no operational distinction between a black hole and a white hole.	$7^S, 15^B, 75^C$	23, 48, 92	pp. 2473
52	An observer who has fallen into the black hole cannot communicate the results of measurements inside it to an observer at infinity because of the causal restriction imposed by the event horizon.	$14^C, 10^B, 15^B, 77^{CO}$	29, 33–34, 93	pp. 2473
53	The source proposes treating a surface near the initial big bang, for example at the Planck time, as a hidden surface, and applying nearly zero total energy and equal configuration probabilities to cosmology.	$25^S, 31^S, 23^B, 30^B, 24^{QB}$	24, 49	pp. 2463
54	The source proposes that counting microscopic configurations with nearly zero total energy on a specified later hypersurface gives an approximately thermal, homogeneous and isotropic macrostate as the most probable under equal weighting.	30^B	3, 24–25, 49	pp. 2463–2464
55	The source proposes that time-reversal asymmetry and CP violation in the early universe lower the energy of expanding baryon-dominated configurations relative to antibaryon-dominated ones, giving more configurations with positive baryon number at the same energy.	$25^S, 26^S$	24, 49, 95–97	pp. 2464

p	Claim	i	a	Source Location
56	As an alternative, the source considers phase separation when pure baryon or antibaryon regions have lower energy density than mixed regions, with separation scales that may exceed the particle horizon.	$28^{\text{CB}}, 27^{\text{S}}$	98	pp. 2464
57	Hawking states that Zel'dovich's particle-creation account of isotropization assumes time independence at early times, whereas his proposal counts configurations at a convenient later time.	$29^{\text{S}}, 30^{\text{B}}$	3	pp. 2464
58	Assuming complete evaporation, energy conservation and a quasistationary mass sequence, Hawking argues from configuration probabilities with unit transmission in every mode that configurations with the same total emitted energy have equal weights.	$12^{\text{B}}, 11^{\text{S}}, 38^{\text{Q}}, 52^{\text{QB}}, 33^{\text{S}}$	12, 15, 59, 61	pp. 2467
59	The entropy of a positive normalized density operator is $S = -\text{Tr}(\rho \ln \rho)$, evaluated in a basis that diagonalizes the density operator.	66^{B}	43, 74, 81, 86	pp. 2472

Time uses comprise 238 claim/time-content pairs. Superscript codes give their use roles.

Table 199: Hawking 1976 Time-Use Roles

Code	Use Role
I	Interval
Q	Quantity
X	Cross-position correspondence
C	Causal
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 200: Hawking 1976 Structural Attributes of Time
Contents

Time Indices	Structural Type	Axes
1–3, 12, 15–16, 74, 76	member	endpoint
4, 28, 58	member	interval_boundary
5, 9, 11, 18, 21–22, 25–27, 29, 36, 38, 41, 43–47, 51, 53–57, 59–60, 62–64, 69, 73	separate	$\emptyset$
6, 10	member	instant, transition_position
7–8, 14, 31, 33–34, 50, 68, 72, 77	member	order_relation
13	dependency	$\emptyset$
17, 20, 24, 52, 71	member	instant
19, 42	member	orientation
23, 30, 32, 35, 37, 40, 48, 66–67, 70	member	phase_boundary
39	member	transition_position
49	member	instant, interval_boundary
61	member	instant, order_relation
65	member	order_relation, phase_boundary
75	member	orientation, order_relation

Table 201: Hawking 1976 Claim Groups

Claim Indices	Role
1–10	Physical-carrier and value inputs
11–59	Source definitions, relations, approximations and conditional statements

Definition 8.297 (Hawking 1976 Claim Boundary). Claims 1–10 form the initial set for physical generation; all 59 claims form the source formal-evidence environment.

Table 202: Hawking 1976 Initial Generation Claims

p	Initial Claim
1	Event-identification data
2	Physical carrier of the collapsing matter
3	Physical carrier and spatial geometry of boundary surfaces
4	Values of observational records
5	Physical degrees of freedom of the hidden sector
6	Physical degrees of freedom of the quantum field

p	Initial Claim
7	Black-hole mass, charge and angular-momentum values in the specified state
8	Spatial geometry of a horizon cross-section
9	Physical carrier of the detector or observer
10	Connection data for signal-emission and reception endpoints

The source-specific obligation atoms comprise 59 claims, 77 time contents, 12 transfers and 35 results, totaling 183.

Dependency Inputs The following table gives the time inputs, claim support and auxiliary inputs of each transfer.

Definition 8.298 (Hawking 1976 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [12]$.

Table 203: Hawking 1976 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
1	9	1	2, 11	28, 71	Classical development to a singular boundary
2	57	43–44	7, 25, 27, 33	12, 54, 77	Schwarzschild time to advanced and retarded coordinates
3	6, 9, 14–16, 43, 47	49–50	2, 10, 25, 29	34, 94	Last escape time of backward-propagated rays
4	38, 43, 48–50, 54, 57	51	7, 25, 29, 31–33, 39	53, 59, 77	Logarithmic phase of a backward-propagated outgoing mode
5	44, 54, 57–58	59	7, 25, 31, 33, 39	53, 77	Retarded time to an affine parameter on the past horizon
6	44, 54–55, 57	38, 45–46	6, 25, 27, 36	31, 52, 55	Outgoing positive frequency and phase relative to stationary time flow
7	16, 35–36, 38, 43–45, 48, 51–52, 54–55, 57, 64	63	6, 25–27, 29–32, 36, 38	7, 69	Expected finite-packet particle count to emission rate
8	16, 36, 38, 43–46, 48, 52, 54, 57–60, 64	63	7, 22, 25, 27, 29–36, 38–39	7, 69	Expected finite-packet particle count to emission rate

s	Time In i	Time Out i	Claim Support p	Auxiliary In a	Transfer Relation
9	16, 36, 38, 43–44, 46, 54, 57–60, 63–64	11	7, 25, 27, 31, 33–36, 38–39	9, 12	Net radiated energy to mass evolution
10	11, 33, 54, 57	53	7, 25, 31, 39	12, 14, 68	Quasistationary mass decrease to increasing temperature
11	53, 66, 68	65	7, 25, 41	12, 64, 68	Hot stage permitting baryon emission
12	2–3, 8, 69–72	73–74	43, 47–48	37, 44, 46–47, 99	Successive density maps in a box to long-time equilibrium

Grounding Conditions

Conditions for Applying Dependencies or Transfers Each item below states Hawking’s setting and the resulting input–output relation for the transfer with the same number.

1. pp. 2460–2461. Under general relativity, the source’s energy and global conditions, and the specified gravitational collapse, the classical spacetime description has a singular boundary.
2. pp. 2468. In the specified exterior geometry, set $u = t - r_*$ and $v = t + r_*$, with the logarithmic scale in r_* and coordinate origins specified.
3. pp. 2466, 2468. Propagate backward through the spacetime, including the collapsing body, the rays that pass through the center and reach future null infinity; specify the boundary between rays that escape to future null infinity and rays that do not by the advanced time v_0 .
4. pp. 2466, 2468. Use the outgoing frequency specified in the late Schwarzschild exterior and apply source Eq. (4.1) to rays on the escaping side, $v < v_0$.
5. pp. 2469. On the past horizon of the analytic extension, apply source Eq. (4.5) for $U < 0$, with the affine scale and the origin of u specified.
6. pp. 2465, 2468. Use exterior stationary time flow with the free-field equation and Klein–Gordon normalization. The p modes are purely positive-frequency and outgoing at future null infinity, with zero Cauchy data on the event horizon.
7. pp. 2466, 2471. Sum the packet occupations in the specified finite interval and obtain the emission rate per interval duration. Use the stationary approximation on intervals over which mass change is negligible. Obtain the mean occupation either directly from Bogoliubov coefficients or from outgoing moments.
8. pp. 2466, 2471. Sum the packet occupations in the specified finite interval and obtain the emission rate per interval duration. Use the stationary approximation on intervals over which mass change is negligible. Obtain the pair-occupation amplitudes from the vacuum of the analytic extension, then use transmission, reflection and the partial trace over the hidden sector to obtain the mean occupation.

9. pp. 2467, 2471. Relate net energy escaping to infinity during the specified finite interval to mass decrease. Include both the pair component of negative Killing energy and the reflected positive-energy component among the components entering the horizon.
10. pp. 2467. In the uncharged, nonrotating quasistationary domain, use $M > 0$ and $T = (8\pi M)^{-1}$. Net radiative mass loss increases the temperature.
11. pp. 2471. Use the source's estimate that a large black hole emits mainly massless particles while cold and retains only a small fraction of its initial mass when its temperature permits thermal baryon emission.
12. pp. 2472–2473. For N states in a finite energy window in a perfectly reflecting box, approximate time evolution by successive applications of the superscattering map. Use the source's argument that the usual assumptions of thermal equilibrium and ergodicity lead one to expect long-time probabilities $1/N$ and entropy $\ln N$.

Results

Definition 8.299 (Hawking 1976 Results). Let $\text{Idx}_R^{\text{audit,result}} = [35]$, with the results below comprising Hawking's calculations and arguments in the domains specified by the corresponding claims.

Table 204: Hawking 1976 Results and Source Locations

j	Result	Source Location
1	Predictive boundary under the classical singularity conditions	pp. 2460–2461
2	Separation of initial, final and hidden boundary surfaces	pp. 2462–2463
3	Signalling-energy budget for horizon-crossing times	pp. 2462
4	Final density operator from the hidden-sector partial trace	pp. 2463, 2466
5	Mixed final reduction of an entangled pure joint state	pp. 2463, 2466
6	Bogoliubov coefficients from back-propagation in the collapse spacetime	pp. 2465–2466
7	Relation between retarded time and the past-horizon affine parameter	pp. 2468–2469
8	Thermal reduced density operator of late outgoing wave packets (outgoing-moment calculation)	pp. 2467
9	Thermal reduced density operator of late outgoing wave packets (paired-mode calculation)	pp. 2470–2471
10	Occupation amplitudes of correlated exterior and interior mode pairs	pp. 2470–2471
11	Pre-transmission equal-energy configuration weights at fixed temperature over a finite interval	pp. 2471

j	Result	Source Location
12	Greybody-corrected occupation at future infinity (direct mean or outgoing-moment calculation)	pp. 2467
13	Greybody-corrected occupation at future infinity (paired-mode calculation)	pp. 2470–2471
14	Particle absence in the source’s freely falling mode decomposition	pp. 2468, 2471
15	Mass decrease from the radiation budget over a finite interval	pp. 2467, 2471
16	Temperature increase accompanying quasistationary mass loss	pp. 2467
17	Probability of finite-total-particle configurations at constant mass over infinite emission time (outgoing-moment calculation)	pp. 2471
18	Probability of finite-total-particle configurations at constant mass over infinite emission time (paired-mode calculation)	pp. 2471
19	General linear superscattering between initial and final density operators	pp. 2463, 2472
20	Superscattering constructed from a three-index amplitude and hidden-sector reduction	pp. 2463
21	Factorization of the density-operator map in the S-matrix sector	pp. 2472
22	Trace-preservation condition for general superscattering	pp. 2472
23	Identity-preservation condition for general superscattering (CPT argument)	pp. 2473
24	Identity-preservation condition for general superscattering (finite-box thermodynamic argument)	pp. 2472–2473
25	Long-time equilibrium state and uniform probabilities in the finite box	pp. 2472–2473
26	CPT correspondence of formation and complete-disappearance histories	pp. 2473
27	Constraint on baryon-number recovery from complete disappearance and the remaining energy	pp. 2471
28	Constraint on information recovery from the specified capacity and final-stage energy (outgoing-moment calculation)	pp. 2471–2472
29	Constraint on information recovery from the specified capacity and final-stage energy (paired-mode calculation)	pp. 2471–2472
30	Most probable thermal cosmic state from equal-weight counting on a late surface	pp. 2463–2464
31	Expected baryon number under the proposed CP-dependent configuration asymmetry	pp. 2464
32	Proposed baryon–antibaryon domain separation through a phase transition	pp. 2464

j	Result	Source Location
33	Equal-weight argument for complete-emission configurations under quasistationary and complete-evaporation assumptions	pp. 2467
34	CPT correspondence of black and white holes for an observer at infinity	pp. 2473
35	Restriction on communicating interior measurement records to infinity	pp. 2473

Definition 8.300 (Hawking 1976 Result-Dependence Hyperedges). The following table specifies time premises, claim support, auxiliary inputs and result outputs.

Table 205: Hawking 1976 Result-Dependence Hyperedges

j	Time Premises i	Claim Support p	Auxiliary In a	Result Output Index
1	1, 9	11	71	1
2	2–4	3, 8, 16–17	29	2
3	14, 17–18	2, 18	51	3
4	2–4, 8	5, 16, 20–22	39	4
5	2–4, 8	5, 16, 20–23	39	5
6	16, 35–36, 38, 43–44, 48, 51, 54–55, 57	6, 25–27, 29, 31–32, 36	58	6
7	44, 59	33	77	7
8	16, 35–36, 38, 43–44, 48, 51–52, 54–55, 57	6, 25–27, 29–32, 36	81	8
9	16, 36, 38, 43–44, 46, 48, 52, 54, 57–60	7, 22, 25, 27, 30–36, 39	81	9
10	16, 36, 38, 43–44, 46, 54, 57–60	7, 25, 27, 31, 33–34, 36, 39	62	10
11	16, 36, 38, 43–44, 46, 54, 57–60, 64	7, 25, 27, 31–34, 36, 38–39, 42	70	11
12	16, 35–36, 38, 43–44, 48, 51–52, 54–55, 57	6, 25–27, 29–32, 36	7	12
13	16, 36, 38, 43–44, 46, 48, 52, 54, 57–60	7, 22, 25, 27, 30–36, 39	7	13
14	10, 16, 36, 43, 55–60	25, 27, 33–34, 36	80	14
15	11, 16, 36, 38, 43–44, 46, 54, 57–60, 63–64	7, 25, 27, 31, 33–36, 38–39	11	15
16	53–54, 57	7, 25, 31, 39	14	16
17	16, 35–36, 38, 43–44, 48, 51–52, 54–55, 57, 62–63	6, 25–27, 29–32, 36–37	83	17

j	Time Premises i	Claim Support p	Auxiliary In a	Result Output Index
18	16, 36, 38, 43–44, 46, 48, 52, 54, 57–60, 62–63	7, 22, 25, 27, 30–37, 39	83	18
19	2–3, 8	43	99	19
20	2–4, 8	21–22, 43	41	20
21	2–3, 8	43–44	37, 42	21
22	2–3, 8	43, 46	88	22
23	2–4, 8, 75	16, 19, 43, 46, 49	89	23
24	2–3, 8, 69–74	43, 46–48	89	24
25	2–3, 8, 69–72, 74	43, 47–48	19, 90	25
26	2–4, 6–8, 75	16, 19, 49–50	91	26
27	12, 53, 65–66, 68	7, 25, 40–41	85	27
28	12, 16, 32, 35–36, 38, 43–44, 48, 51–52, 54–55, 57, 66–68	6, 25–27, 29–32, 34, 36, 40, 42, 59	87	28
29	12, 16, 32, 36, 38, 43–44, 46, 48, 52, 54, 57–60, 66–68	7, 22, 25, 27, 30–36, 39–40, 42, 59	87	29
30	30	19, 53–54	25	30
31	25–26, 30	19, 53–55	97	31
32	27–28	56	98	32
33	12, 33, 38, 52, 54, 57	7, 25, 31–32, 39, 58	12, 59, 61	33
34	4, 7–8, 15, 75	16, 19, 49, 51	92	34
35	10, 14–15, 77	3, 8–10, 52	93	35

Results and Evidence-Effect Domain The particle-number and reduced-density calculations use a massless scalar field on an uncharged, nonrotating semiclassical collapse background. Thermal emission at fixed mass over a finite interval, a quasistationary sequence with changing mass, and the argument about total emitted configurations under complete evaporation are taken under their respective conditions. General superscattering maps, thermal relaxation in a box, CPT-based history correspondence and early-universe configuration counts use the premises specified in the corresponding claims.

Generation Judgment Determination values: The LCTR complete-generation count is 0, the source-evidence-conditioned local-reach count is 0, and 77 primitive-time roots remain.

Result Dependence All 35 results depend on primitive-time roots. The minimal-support families give 301 root–result dependence pairs, including 190 necessary dependence pairs.

Placement of the Grounding Obligation The 77 time contents enter initial, final and hidden boundaries, asymptotic times and mode frequencies, collapse and emission orderings, emission rates, quasistationary evolution, thermal relaxation in a box and

cosmological surfaces. The 238 uses in the claim table belong to these coordinates, definitions, comparisons, evolutions and result derivations. Every transfer takes given time contents as inputs. Each time content enters a law operation or result derivation without its corresponding physical-time generation conditions being supplied before use. Primitive-time inputs thus leave the grounding obligation in the background.

Evidence of Actual Failure Pages 2471–2472 connect the quasistationary-emission calculation to arguments concerning complete evaporation and late information recovery. Actual failure of specific physical-time generation conditions, and resulting changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 301 root–result dependence pairs supplied by the minimal-support families. Actual reach from the failure location of a specific generation condition remains undetermined.

Repair Scope The premises of each result are specified by the corresponding claims and transfer conditions. Realizing its time inputs by generated representations that preserve those premises and relations preserves the result within that domain. If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.301 (Hawking 1976 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \text{AllRootDepIdx}(77, 35).$$

Also, $R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. The dependence boundary of this audit consists of 171 nodes—59 claims, 77 time contents and 35 results—and 47 edges—12 transfers and 35 result edges. Closure from Initial Claims 1–10 remains the initial 10 nodes. Because every transfer has a time input, closure from all 59 claims also contains no time-content node. All 35 results have time premises as well. Closure from all 59 claims together with all 77 time contents contains all 35 result nodes. The common reachability classification and result partition therefore give the displayed index tuple. $\square$

Proposition 8.302 (Hawking 1976 Minimal Primitive-Time-Root Support). The following table gives the minimal-support family statistics in the order: number of members, minimum cardinality and maximum cardinality. There are 73 family members in total.

Table 206: Hawking 1976 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(1; 1, 1)
2	(1; 3, 3)

Result Index	Support Statistics
3	(1; 3, 3)
4	(1; 4, 4)
5	(1; 4, 4)
6	(3; 8, 12)
7	(3; 2, 3)
8	(3; 9, 13)
9	(2; 9, 10)
10	(2; 7, 8)
11	(2; 8, 9)
12	(3; 9, 13)
13	(2; 9, 10)
14	(2; 9, 9)
15	(4; 9, 12)
16	(6; 3, 13)
17	(6; 11, 15)
18	(4; 11, 13)
19	(1; 3, 3)
20	(1; 4, 4)
21	(1; 3, 3)
22	(1; 3, 3)
23	(1; 5, 5)
24	(1; 7, 7)
25	(1; 7, 7)
26	(1; 7, 7)
27	(6; 4, 16)
28	(3; 14, 18)
29	(2; 14, 15)
30	(1; 1, 1)
31	(1; 3, 3)
32	(1; 2, 2)
33	(2; 6, 6)
34	(1; 5, 5)
35	(1; 4, 4)

Proof. Start from the singleton seed of each time content and apply Definition 7.46 to the 12 transfers. After the antichains stabilize, combine the supports of each result's time premises and discard every support set that properly contains another support set. Proposition 7.47 yields the displayed families and counts. $\square$

Proposition 8.303 (Hawking 1976 Audit Conclusion). Complete generation and local reach are both zero; all 77 time contents remain primitive-time roots. All 35 results depend on primitive-time roots.

Proof. This follows from the reachable sets in Proposition 8.301 and the minimal-support families in Proposition 8.302. $\square$

Individual Statements

Objects of the Statements The relation between the known laws of physics and a classical spacetime background; the use of Fourier components with respect to time; the unit convention $G = c = \hbar = k = 1$; and Hawking's statement that a surface near the initial big-bang singularity could be regarded as a hidden surface. The Planck time is given as an example of where to place that surface.

Verbatim Original (English) all the known laws of physics ... are all formulated on a classical space-time background

using Fourier components with respect to time

Throughout the paper units will be used in which $G = c = \hbar = k = 1$.

One could regard a surface very close to the initial big-bang singularity (say, at the Planck time 10^{-43} sec) as being a "hidden surface" in the sense that we have no *a priori* information about it.

The third statement gives the unit convention $G = c = \hbar = k = 1$. The fourth gives the Planck time 10^{-43} sec as an example of where a surface could be treated as hidden.

Original-Source Locations The first quotation appears in the abstract of Hawking, *Breakdown of Predictability in Gravitational Collapse*, *Physical Review D* 14, p. 2460. The second appears in the introduction of the Fourier components on p. 2466, §2. The third appears in the unit convention at the beginning of §2 on p. 2464. The fourth appears in the application to the early universe at the end of §1 on p. 2463.

Minimal Context The first quotation belongs to the abstract's statement that the concepts of classical spacetime and the known laws break down at a singularity. The second belongs to the use of Fourier components with respect to time to calculate coefficients under an approximation in which the black-hole mass changes slowly. The third is the paper-wide unit convention following the formal setting of a massless Hermitian scalar field and an uncharged, nonrotating black hole. The fourth belongs to the discussion of the possibility that the principle of ignorance accounts for the homogeneity, isotropy, and thermal equilibrium of the early universe.

Attributed Author The four quotations are attributable to S. W. Hawking's exposition in the article.

Content Stated Directly by the Quotations The first quotation states that all the known laws are formulated on a classical spacetime background; the second states the use of Fourier components with respect to time. The third states the unit convention in which four constants equal one. In the fourth quotation, Hawking states that a surface near the initial singularity could be regarded as hidden in the sense that we have no *a priori* information about it, and gives the Planck time as an example of when to place that surface.

Explicit Domain of Application The first quotation is limited to the known laws and the classical spacetime background discussed by the source; the second, to the mode decomposition under the approximation that the black-hole mass changes slowly; the third, to the unit notation used throughout the source; and the fourth, to the discussion that applies the principle of ignorance to the early universe.

Evidence Provenance Hawking, *Breakdown of Predictability in Gravitational Collapse*, pp. 2460, 2463, 2464, 2466.

Individual Audit

First Statement The first statement expresses the dependence of the known laws of physics on a classical spacetime background. Hawking argues that the classical spacetime concept and the known laws formulated on it break down at a singularity. In Section I, Hawking relates this claim to the singularity conditions and argues for a predictive limitation additional to ordinary quantum uncertainty. The formal correspondence is to Claims 11 and 24, Time Contents 1–3, 5–6, 8–9, 12, and Result 1.

Second Statement The second statement gives the procedure for calculating Bogoliubov coefficients using Fourier components with respect to time under the slow-mass-variation approximation. In Sections II and III, Hawking obtains the outgoing particle number using the incoming vacuum, positive frequency relative to asymptotic time, backward-propagated modes and finitely normalized packets. The formal correspondence is to Claims 25, 27, 29 and 30, Time Contents 33, 36, 38, 43–44, 47, 51–52, and Results 6 and 12.

Third Statement The third statement is the unit convention $G = c = \hbar = k = 1$ used throughout the source. Numerical representations of mass, length, time and temperature are converted according to this convention. The formal correspondence is to Claim 25 and Auxiliary Object 68; Claims 31 and 39 use the relations among mass, surface gravity and temperature.

Fourth Statement In the fourth statement, Hawking proposes heuristically treating a surface near the initial big-bang singularity as hidden, giving the Planck time as an example of when to place that surface. Using the principle of ignorance and nearly zero total energy, Hawking further argues that equal-weight counting of microscopic configurations on a later hypersurface makes a thermal, homogeneous and isotropic macrostate the most probable. The formal correspondence is to Claims 19, 53 and 54, Time Contents 23–25, 30–31, and Result 30.

Common Audit with Specified Domain of Application Evolution on classical spacetime, frequency decomposition relative to asymptotic time, temporal quantities expressed through unit conversion, and early and later cosmological surfaces enter the respective arguments under their specified conditions. The claim, transfer and result-dependence tables above give these time uses and dependencies.

8.19.22 Source Audit

Audit Inputs

Source and Audit Scope

Source Identifier 22.

Source Name Bailey et al. 1977

Bibliographic Identification J. Bailey et al., *Measurements of Relativistic Time Dilatation for Positive and Negative Muons in a Circular Orbit*, *Nature* 268, 301–305 (1977), DOI: 10.1038/268301a0 is denoted by $R = \text{Ref}_{\text{Bai77TD}}$. The audit domain comprises the abstract, main text, references, Eqs. (1)–(2), Figs. 1–2 and Tables 1–3.

Audit-Scope Identifier 22-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Original-Source Location Identifier 22-L1.

Original-Source Location Content The tables identify the source pages of their time contents, claims, transfers and results.

Formal Objects The audit uses 17 time contents, 41 claims, 5 transfers and 16 results.

Assumptions Lifetime fitting uses exponential decay, spin-precession modulation, and the loss, gain and background corrections applicable to each run. Calculation of the Lorentz factor uses rotation frequency, proton magnetic-resonance frequency, the magnetic-moment ratio and the g-factor. Inference of rest lifetimes assumes the SR lifetime relation. The positive-muon SR comparison and the CPT comparison use an independently measured positive-muon lifetime at rest. The short-distance bound uses Rédei’s correction model.

Non-temporal Physical Inputs The initial generation inputs are the carriers and values in Claims 1–13: muons; storage-ring, magnetic-field, detector and scraping configurations; identification and count values for decay electrons, lost muons and protons; spin degrees of freedom; magnetic-moment-ratio and g-factor values; LED signals; and circular-orbit geometry.

Formal Input Elements

Definition 8.304 (Bailey et al. 1977 Time Contents). Let $n_R = 17$. The following table specifies the time contents and their source locations.

Table 207: Bailey et al. 1977 Time Contents

i	Time Content	Source Location
1	Laboratory-time parameter	pp. 301, 302, 303, 304
2	Injection reference and boundaries of control and counting intervals	pp. 302, 303, 304
3	Time association of detection and calibration records	pp. 301, 302, 303, 304
4	Muon lifetimes in motion	pp. 301, 302, 303
5	Reference muon lifetimes at rest	pp. 301, 303, 304
6	Lifetime of a radioactive sample in motion	pp. 301
7	Lifetime in motion in the earlier CERN experiment	pp. 301

i	Time Content	Source Location
8	Relation between moving and rest lifetimes and clock rates	pp. 301, 304
9	Outbound, return, and stay intervals in the twin description	pp. 301, 304
10	Aging and clock readings in the twin description	pp. 301, 304
11	Muon rotation period	pp. 302
12	Mean muon rotation frequency	pp. 303, 304
13	Switch-off time constant of the scraping voltages	pp. 302
14	Interval of circular motion	pp. 301, 304
15	Muon spin-precession angular frequency	pp. 301, 302, 303
16	Proton magnetic-resonance frequency	pp. 304
17	Muon lifetimes at rest inferred assuming SR	pp. 301, 303, 304

Definition 8.305 (Bailey et al. 1977 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(41)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([13], [41])$. The following table gives each claim, its time uses and source locations.

Table 208: Bailey et al. 1977 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Positive-muon carriers	$\emptyset$	pp. 301, 302, 303
2	Negative-muon carriers	$\emptyset$	pp. 301, 302, 303
3	Geometrical configuration of the storage ring	$\emptyset$	pp. 301, 304
4	Magnetic-field carriers and spatial configuration	$\emptyset$	pp. 301, 304
5	Detector channels and threshold settings	$\emptyset$	pp. 301, 303
6	Event-identification data for decay electrons	$\emptyset$	pp. 301, 302
7	Detector count values for decay electrons, lost muons, and protons	$\emptyset$	pp. 301, 302, 303
8	Muon spin degrees of freedom	$\emptyset$	pp. 301, 302
9	Numerical data for the magnetic-moment ratio and g-factor	$\emptyset$	pp. 304
10	Identification data for proton-background events	$\emptyset$	pp. 302, 303
11	Electric-field and aperture-stop configuration for scraping	$\emptyset$	pp. 302
12	LED, photomultiplier, and discriminator configuration and signal values	$\emptyset$	pp. 303
13	Geometrical configuration of the stored-muon circular orbits	$\emptyset$	pp. 301, 304

p	Claim	i	Source Location
14	The experimenters store the injected muons in the ring and specify control and counting intervals relative to injection.	$1^Q, 2^B$	pp. 301, 302, 303
15	The experimenters associate selected decay-electron counts with measured times to obtain the time spectrum used in the lifetime fit.	$1^Q, 3^O, 2^B$	pp. 301, 303
16	The decay-electron time spectrum includes muon decay through the exponential factor $\exp(-t/\tau)$ in Eq. (1).	$1^{PQ}, 4^{PQ}$	pp. 301, 302
17	The experimenters represent precession of the muon spin relative to its momentum by the modulation $1 - A \cos(\omega_a t + \phi)$ in Eq. (1).	$1^{QS}, 15^{QS}$	pp. 301, 302
18	The experimenters fit Eq. (1) by maximum likelihood to the decay-electron spectrum from 100 microseconds for runs A, B and F, 50 for C and D, and 90 for E, to 650 microseconds in every case, varying $N_0, \tau, A, \omega_a, \phi, B_0$.	$1^{PQ}, 3^O, 2^B, 4^{PQ}, 15^Q$	pp. 303
19	The experimenters obtain $F(t)$ from the lost-muon time spectrum $M(t)$, corrected for electron background, using Eq. (2), and calibrate the detection-efficiency ratio with high-loss data taken without scraping.	$1^{PQ}, 3^O, 2^B, 4^{PQ}$	pp. 302, 303
20	Assuming equal losses of positive and negative muons under the same machine conditions, the experimenters obtain the proton-background time distribution by subtraction and calibrate the detection-efficiency ratio from proton counts about 1 millisecond after injection.	$1^Q, 3^O, 2^B$	pp. 303
21	The experimenters multiply the mean gain curve $G(t)$ of the selected detectors by $S = \Delta N / (N \Delta G)$ to obtain the gain correction $SG(t)$ in Eq. (1).	$1^Q, 3^O, 2^B$	pp. 303
22	The experimenters remove muons on near-wall orbits by scraping shortly after injection. Taking 40 microseconds after injection as $t = 0$, the loss correction $F(0)$ is about 0.03 without scraping and 0.001 with scraping. The rotation period in this apparatus is 0.147 microseconds.	$2^B, 11^Q$	pp. 302, 303
23	The experimenters switch off the scraping voltages with a time constant of 10 microseconds, allowing the remaining muon population to return adiabatically to the centre of the storage region.	$2^B, 13^Q$	pp. 302

p	Claim	i	Source Location
24	The experimenters measure each detector's gain using the timing of LED pulses and the count ratio from two discriminators. Typically, they measure 11 points between 7 and 580 microseconds after injection, with a precision of about 0.1% per point.	$1^Q, 3^O, 2^B$	pp. 303
25	The experimenters analyse two positive-muon and four negative-muon runs separately and report their starting times, uncorrected lifetimes, corrections, and corrected lifetimes in Table 1.	$3^O, 2^B, 4^Q$	pp. 302, 303
26	The experimenters apply loss and gain corrections to each run's lifetime and the required proton-background correction to the positive-muon runs, then report charge-separated and combined weighted averages in Table 2.	$3^O, 4^Q$	pp. 302, 303
27	The experimenters obtain the mean rotation frequency from the bunch structure over about 40 microseconds after injection. Figure 2 shows the bunch structure from 6 to 10 microseconds after injection.	$3^O, 2^B, 12^Q$	pp. 303, 304
28	The experimenters use the vacuum-corrected proton magnetic-resonance frequency corresponding to the mean magnetic field in the calibration.	16^Q	pp. 304
29	λ is the ratio of the muon and proton magnetic moments.	$\emptyset$	pp. 304
30	g is the muon g -factor.	$\emptyset$	pp. 304
31	The experimenters obtain $\bar{\gamma} = 2\lambda\bar{f}_p/(g\bar{f}_{\text{rot}}) = 29.327(4)$ from the mean rotation frequency $\bar{f}_{\text{rot}}$ and the vacuum-corrected proton magnetic-resonance frequency $\bar{f}_p$.	$12^Q, 16^Q$	pp. 304
32	Bailey et al. use the Einstein lifetime relation $\tau_0 = \tau/\gamma$ to calculate lifetimes at rest from the measured lifetimes in motion and the calibrated Lorentz factor.	$4^{\text{xQ}}, 17^{\text{xQ}}, 8^{\text{xQ}}, 12^Q, 16^Q$	pp. 301, 303, 304
33	For the independently measured positive-muon lifetime at rest τ_0^+ and $\tau^+/\bar{\gamma}$ from this experiment, Bailey et al. report the ratio $(\tau_0^+ - \tau^+/\bar{\gamma})/\tau_0^+$ as $(2 \pm 9) \times 10^{-4}$, with a 95% confidence interval of $[-1.6, 2.0] \times 10^{-3}$.	$4^{\text{xQ}}, 5^{\text{xQ}}, 8^{\text{xQ}}, 12^Q, 16^Q$	pp. 304
34	Assuming special relativity, Bailey et al. infer a negative-muon lifetime at rest of 2.1948(10) microseconds from its lifetime in motion and $\bar{\gamma}$.	$4^{\text{xQ}}, 17^{\text{xQ}}, 8^{\text{xQ}}, 12^Q, 16^Q$	pp. 301, 303, 304

<i>p</i>	Claim	<i>i</i>	Source Location
35	In the abstract, Bailey et al. describe agreement between the inferred negative-muon value and the positive-muon lifetime at rest as confirmation of CPT invariance. In the body, they report the SR-conditional ratio $(\tau_0^- - \tau_0^+)/\tau_0^+$ as $-0.00105(46)$, with a 95% confidence interval of $[-0.002, -0.00013]$.	5 ^{XQ} , 17 ^{XQ} , 8 ^Q	pp. 301, 304
36	Bailey et al. explain that the lifetime of a radioactive sample moving at a known velocity can test time dilation of moving clocks, and present the circular-muon lifetime measurement as such an experiment.	6 ^Q , 4 ^Q , 5 ^Q , 8 ^Q	pp. 301, 304
37	Bailey et al. relate the lifetime comparison between muons making circular round trips and muons at rest in the laboratory to the comparison of clocks and aging for a travelling twin and a twin who remains at home.	8 ^{XQ} , 9 ^{XI} , 10 ^{KX}	pp. 301, 304
38	Bailey et al. report no resolved additional effect on the muon lifetime at a transverse acceleration of about $10^{18}g$ within the precision of this experiment.	1 ^Q , 4 ^Q , 5 ^Q , 8 ^Q , 14 ^I	pp. 304
39	The short-distance correction model of Rédei used by Bailey et al. is $\tau = \gamma\tau_0(1 + 2.5 \times 10^{24}\gamma^2a^2)$, with a expressed in centimetres.	4 ^Q , 5 ^Q , 8 ^Q	pp. 304
40	Bailey et al. apply the positive-muon lifetime comparison to Rédei's short-distance correction model and obtain $a \leq 9.6 \times 10^{-16}$ cm at 95% confidence.	4 ^Q , 5 ^Q , 8 ^Q , 12 ^Q , 16 ^Q	pp. 304
41	Bailey et al. report that the earlier CERN storage-ring experiment at γ about 12 agreed within 1% with the prediction obtained by applying the Einstein factor to the measured lifetime at rest.	7 ^Q , 5 ^Q , 8 ^Q	pp. 301

Time uses comprise 83 claim/time-content pairs. Superscript codes indicate their use roles.

Table 209: Bailey et al. 1977 Time-Use Roles

Code	Use Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading

Code	Use Role
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 210: Bailey et al. 1977 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3–17	separate	$\emptyset$
2	member	instant, interval_boundary

Claims 1–13 form the initial set for physical generation; all 41 claims form the source formal-evidence environment. The source-specific obligation atoms comprise 41 claims, 17 time contents, 5 transfers and 16 results, totaling 79.

Dependency Inputs The following table gives the time inputs, time outputs and claim support of each transfer.

Definition 8.306 (Bailey et al. 1977 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [5]$.

Table 211: Bailey et al. 1977 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1	2	1–3, 14	Specification of operational intervals from the injection reference and selected boundaries. pp. 301, 302, 303
2	1–2	3	5–7, 15, 24, 27	Association of detection and calibration events with measured times. pp. 301, 303, 304
3	1–3	4, 15	5–8, 16–21, 24–26	Maximum-likelihood fit of the decay spectrum with the correction model. pp. 301, 302, 303
4	2–3	12	3, 7, 27	Estimation of mean rotation frequency from timed bunch structure. pp. 303, 304
5	4, 8, 12, 16	17	9, 29–32, 34	Inference of rest lifetime from the calibrated Lorentz factor and the SR relation. pp. 303, 304

Grounding Conditions

1. The injection event identified by the experimenters serves as the reference for the source-specified control, counting and fitting boundaries in calibrated laboratory time.
2. The experimenters associate decay electrons, lost muons, protons, LED pulses and rotating bunches with times under their respective detection or calibration conditions and acquisition intervals.
3. The experimenters fit Eq. (1) by maximum likelihood from each run's starting time to 650 microseconds. Loss correction uses Eq. (2) and efficiency calibration with high-loss data; gain correction uses LED measurements. Proton-background correction for positive-muon runs uses equal positive and negative muon losses under identical machine conditions and an efficiency ratio measured separately. Lifetime and precession angular frequency are fit parameters; corrected run values and charge-separated means belong to their corresponding measurement populations.
4. The experimenters associate each bunch with one turn in the storage ring and obtain the mean rotation frequency from the bunch structure over about 40 microseconds after injection. Figure 2 shows the interval from 6 to 10 microseconds.
5. The experimenters obtain the mean Lorentz factor from the mean rotation frequency and vacuum-corrected proton magnetic-resonance frequency. Bailey et al. apply this factor and the SR lifetime relation to the charge-separated or combined lifetimes in motion to infer the corresponding lifetimes at rest.

Results

Definition 8.307 (Bailey et al. 1977 Results). Let $\text{ldx}_R^{\text{audit,result}} = [16]$. The following table specifies each result, its time premises, claim support and source locations.

Table 212: Bailey et al. 1977 Results and Audit Inputs

j	Result and Source Location	Time Premises i	Claim Support p
1	Measurement of positive- and negative-muon lifetimes in motion. pp. 301, 302, 303	3–4	1–2, 5–7, 15, 18, 25–26
2	Decay-spectrum fit including spin-precession modulation. pp. 301, 302, 303	1–4, 15	16–18
3	Loss correction from lost-muon counts. pp. 302, 303	1–4	19
4	Proton-background correction for positive-muon runs. pp. 303	1–3	20
5	Detector-gain correction from LED measurements. pp. 303	1–3	5, 12, 21, 24
6	Corrected lifetimes in motion for the six runs. pp. 302, 303	2–4	18–21, 25–26

j	Result and Source Location	Time Premises i	Claim Support p
7	Charge-separated and combined weighted mean lifetimes in motion. pp. 303	4	26
8	Mean Lorentz factor from rotation and proton magnetic-resonance frequencies. pp. 303, 304	12, 16	3, 9, 27–31
9	Charge-separated and combined rest lifetimes inferred using the SR relation. pp. 303, 304	4, 8, 12, 16–17	31–32
10	SR comparison with an independent positive-muon lifetime at rest. pp. 304	4–5, 8, 12, 16	31, 33
11	Negative-muon rest lifetime inferred assuming SR. pp. 301, 303, 304	4, 8, 12, 16–17	31, 34
12	CPT comparison using the negative-muon value inferred assuming SR. pp. 301, 304	5, 8, 17	35
13	Explanation of radioactive-sample lifetime as a moving clock. pp. 301	6, 8	36
14	Bailey et al.’s correspondence between circular clocks and the twin description. pp. 301, 304	8–10	37
15	Muon-lifetime comparison under large transverse acceleration. pp. 301, 304	1, 4–5, 8, 12, 14, 16	13, 31, 33, 38
16	95% upper bound on the short-distance parameter in Rédei’s model. pp. 304	4–5, 8, 12, 16	31, 33, 39–40

Results and Evidence-Effect Domain The six run lifetimes in Table 1 use their respective corrections and uncertainties. Table 2 gives charge-separated and combined lifetimes in motion and rest lifetimes inferred assuming SR. The positive-muon SR comparison uses an independent rest lifetime as its reference. The CPT comparison compares the negative-muon value inferred assuming SR with the same positive-muon reference. Bailey et al.’s clock and twin descriptions use these lifetime measurements and theoretical relations; the tests involving transverse acceleration and Rédei’s model use the reported precision and model.

Generation Judgment Determination values: The LCTR complete-generation count is 0, the source-evidence-conditioned local-reach count is 0, and 17 primitive-time roots remain.

Result Dependence All 16 results depend on primitive-time roots. The minimal-support families give 67 root–result dependence pairs, including 27 necessary dependence pairs.

Placement of the Grounding Obligation The 17 time contents enter laboratory time, injection references and control boundaries, detection records, lifetimes, clock comparisons, rotation and precession frequencies, proton magnetic-resonance frequency, and the voltage switch-off time constant. The 83 uses in the claim table belong to measurement, calibration, fitting and theoretical comparison. Each transfer takes given time contents as inputs without providing conditions that generate physical time from non-temporal physical inputs before use. Primitive-time inputs thus leave the grounding obligation in the background.

Evidence of Actual Failure The source compares muon lifetimes with theoretical relations under specified storage-ring, detector and calibration conditions. Actual failure of specific physical-time generation conditions, and resulting changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 67 root–result dependence pairs supplied by the minimal-support families. Actual reach from the failure location of a specific generation condition remains undetermined.

Result-Preservation Domain The lifetime measurements hold under the specified run, detector, calibration, correction and fitting conditions. The positive-muon SR comparison uses an independent rest lifetime; inference of the negative-muon rest lifetime and the CPT comparison assume the SR lifetime relation. Clock correspondence, comparison under transverse acceleration, and the bound in Rédei’s model use their respective theoretical relations and domains. Realizing these time inputs by generated representations that preserve the corresponding relations preserves the 16 results in their respective domains.

Repair Scope If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.308 (Bailey et al. 1977 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(17, 16).$$

Also, $R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. The boundary consists of 74 nodes—41 claims, 17 time contents and 16 results—and 21 edges—5 transfers and 16 result edges. Closure from Initial Claims 1–13 remains the initial 13 nodes. Even with all 41 claims supplied, no time content is reached, because every transfer has a time input. All 16 results also have time premises. Supplying all 17 time contents reaches all 16 results. The common reachability classification and result partition therefore give the displayed index tuple. $\square$

Proposition 8.309 (Bailey et al. 1977 Minimal Primitive-Time-Root Support). The following table gives the minimal-support family statistics in the order: number of members, minimum cardinality and maximum cardinality. There are 32 family members in total.

Table 213: Bailey et al. 1977 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(2; 1, 2)
2	(1; 1, 1)
3	(1; 1, 1)
4	(1; 1, 1)
5	(1; 1, 1)
6	(2; 1, 3)
7	(2; 1, 1)
8	(3; 2, 3)
9	(3; 3, 5)
10	(3; 4, 6)
11	(3; 3, 5)
12	(4; 3, 6)
13	(1; 2, 2)
14	(1; 3, 3)
15	(1; 5, 5)
16	(3; 4, 6)

Proof. Start from the singleton seed of each time content and apply Definition 7.46 to the 5 transfers. After the antichains stabilize, combine the supports of each result’s time premises and discard every support set that properly contains another support set. Proposition 7.47 yields the displayed families and counts. □

Proposition 8.310 (Bailey et al. 1977 Audit Conclusion). Complete generation and local reach are both zero; all 17 time contents remain primitive-time roots. All 16 results depend on primitive-time roots.

Proof. This follows from the reachable sets in Proposition 8.308 and the minimal-support families in Proposition 8.309. □

Individual Statements

Verbatim Original (English):

the so-called time dilation, or slowing down of moving clocks

The source location is Bailey et al., *Measurements of relativistic time dilatation for positive and negative muons in a circular orbit*, *Nature* 268, p. 301.

Verbatim Original (English):

agrees with experiment

Minimal Context Required for Interpretation

In the abstract on the same page, Bailey et al. report that, for positive muons, the Einstein time-dilation factor agrees with experiment within a fractional error of 2×10^{-3} at 95% confidence.

Verbatim Original (English):

Assuming special relativity

Minimal Context Required for Interpretation In the same abstract, Bailey et al. describe agreement between the negative-muon rest lifetime inferred assuming SR and the previously measured positive-muon rest lifetime as confirmation of CPT invariance. On page 304, they report the ratio $(\tau_0^- - \tau_0^+)/\tau_0^+$ as $-0.00105(46)$, with a 95% confidence interval of $[-0.002, -0.00013]$. In the gain-effects section on page 303, the experimenters measure apparatus gain as a function of time.

Objects of the Statements The first statement concerns time dilation and the rate of moving clocks; the second, agreement between the Einstein time-dilation factor for positive muons and experiment; and the third, the negative-muon lifetime at rest and the CPT comparison conditional on special relativity.

Original-Source Locations The first statement appears on p. 301 of the original source; the second and third statements appear in the abstract. The description of measuring gain effects as a function of time appears under *Gain effects* on p. 303.

Minimal Context Required for Interpretation The first statement belongs to the introductory context that treats moving unstable particles as relativistic clocks. The second belongs to the report of the relative error in the Einstein time-dilation factor for positive muons at the 95% confidence level. The third belongs to the context in which the negative-muon lifetime at rest is estimated with SR as an antecedent, and its agreement with the positive-muon lifetime at rest is used in a CPT comparison.

Attributed Authors of the Statements The three statements are attributable to J. Bailey et al.

Predicates Assigned by the Statements to Their Objects The first statement explains time dilation as moving clocks running slowly. The second states that the specified positive-muon factor agrees with experiment. The third explicitly states the SR assumption used in estimating the negative-muon lifetime at rest.

Explicit Domains of Application of the Statements The explicit domain of the first statement is moving clocks and the circular-orbit muons of this source; that of the second is the Einstein time-dilation factor for positive muons and the reported confidence interval; and that of the third is the negative-muon estimate and CPT comparison under the SR condition.

Individual Audit

First Statement Bailey et al. explain time dilation as the slowing of moving clocks and use lifetime measurements of radioactive samples at known velocity to test it. The measurements in this experiment concern positive and negative muons stored in circular orbits at CERN and their decay records. The experimenters obtain lifetimes in motion from the decay-electron time spectra and relate them to the Lorentz factor determined from rotation and proton magnetic-resonance frequencies and to previously measured lifetimes at rest. The formal correspondence is to Claims 31–37, Time Contents 4–10, 12, 16 and 17, Transfers 3–5, and Results 1 and 9–14.

Second Statement Bailey et al. state that, for the positive muons in this experiment, the Einstein time-dilation factor agrees with experiment at the reported 95% confidence level. The comparison uses the positive-muon lifetime in motion, the calibrated

Lorentz factor, and an independently measured positive-muon lifetime at rest. The source gives the signed fractional difference $(\tau_0^+ - \tau^+/\bar{\gamma})/\tau_0^+$ as $(2 \pm 9) \times 10^{-4}$, with a 95% confidence interval of $[-1.6, 2.0] \times 10^{-3}$. The formal correspondence is to Claims 31 and 33, Time Contents 4, 5, 8, 12 and 16, and Result 10.

Third Statement Bailey et al. infer the negative-muon lifetime at rest assuming special relativity and compare it with the previously measured positive-muon lifetime at rest. The inferred negative-muon value of 2.1948(10) microseconds follows from applying the SR lifetime relation to the lifetime in motion and the calibrated Lorentz factor. The abstract describes the agreement as confirmation of CPT invariance, while page 304 reports the ratio $(\tau_0^- - \tau_0^+)/\tau_0^+$ as $-0.00105(46)$, with a 95% confidence interval of $[-0.002, -0.00013]$. The formal correspondence is to Claims 31, 34 and 35, Time Contents 4, 5, 8, 12, 16 and 17, Transfer 5, and Results 11 and 12.

Auxiliary Apparatus Description In the gain-effects section on page 303, the experimenters measure apparatus gain as a function of time using LED pulses and discriminator count ratios. This calibration uses the selected detectors, pulse times, discriminator thresholds and corresponding measurement conditions. The mean gain curve $G(t)$ and sensitivity factor S give $SG(t)$, which enters the gain correction of the decay-electron spectrum. The formal correspondence is to Claims 5, 12, 21 and 24, Time Contents 1–3, Transfers 2 and 3, and Result 5.

8.19.23 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 23.

Source Name Rindler 1977

Bibliographic Identification Wolfgang Rindler, *Essential Relativity: Special, General, and Cosmological*, Revised Second Edition, Springer-Verlag (1977), Chapter 2, pp. 23–53 is denoted by $R = \text{Ref}_{\text{Rin77ER}}$. The audit domain comprises the relativity principle, simultaneity and synchronization, coordinate lattices, Lorentz transformations, invariant intervals, clocks and physical processes, twin comparisons, velocity and acceleration, and transformation branches.

Audit-Scope Identifier 23-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Original-Source Location Identifier 23-L1.

Original-Source Location Content The tables give the source pages of the time contents, claims, transfers and results.

Formal Objects The audit uses 23 time contents, 61 claims, 15 transfers and 18 results. Derivative operators are included in the claims and distinguished from their temporal arguments. Rindler distinguishes the Galilean branch, which combines the relativity principle with absolute time, from the Lorentz branch using invariant light speed (pp. 23–25). His account of physical laws as models of a domain of nature, and his distinction between an exact theory in ideal inertial frames and approximate application

to local inertial frames, describe his view of physical theory and its domain of application (pp. 25–26).

Rindler criticizes explanations that require general relativity because accelerated phenomena supposedly fall outside special relativity, and explanations that use general relativity to avoid the clock hypothesis. For the former, he states that special relativity applies to accelerated motion described in inertial frames; for the latter, he points out that the clock hypothesis remains a hypothesis in general relativity (p. 47).

Assumptions Rindler’s coordinate lattice uses a standard clock at the origin, its replicas, a homogeneous and isotropic inertial frame, and Newton’s first law. Coordinate transformations use regularity excluding infinite coordinates for finite events, linearity, continuity and the relativity principle. Each transfer and result specifies the conditions of the derivation using invariant light speed or the derivation without the second postulate. The instantaneous rate of a nonuniformly moving ideal clock uses the clock hypothesis; the instantaneous length of an accelerating infinitesimal body uses the length hypothesis.

Non-temporal Physical Inputs The initial generation inputs are the carriers of clocks, signals, particles, atoms, aircraft and records; event identities; spatial configurations; length standards; spatial field configurations; and material properties of apparatus in Claims 1–14.

Time Contents

Definition 8.311 (Rindler 1977 Time Contents and Auxiliary Operators). Let $n_R = 23$. The following table specifies the time contents and their source locations.

Table 214: Rindler 1977 Time Contents

i	Time Content	Source Location
1	Absolute time in the Galilean branch	pp. 23, 26–27, 53
2	Signal speeds and relative velocities of inertial frames	pp. 23, 28–33, 35–37, 51–53
3	Simultaneity relative to an inertial frame	pp. 28–29, 35, 40–42
4	Synchronization of separated clocks	pp. 28–31, 44
5	Time and elapsed-time readings of the standard clock at the origin	pp. 30–31
6	Settings of remote lattice clocks	pp. 30–31
7	Time readings of local events in the calibrated lattice	pp. 30–31
8	Inertial-frame coordinate times, differences and translations	pp. 30–53
9	Time-coordinate correspondence under Lorentz transformations	pp. 32–39, 43, 47–49, 51–53
10	Invariant squared interval containing a time difference	pp. 34, 37–39, 48
11	Ordering of cause and effect and of signal transmission and reception	pp. 35–36, 39–40, 42, 46, 52–53

i	Time Content	Source Location
12	Finite upper bound on information-transmission and particle speeds	pp. 35–36, 40, 42, 48, 53
13	Time specification of worldlines, simultaneous slices and world maps	pp. 38–40
14	Time-dilation interval relation for uniformly moving standard clocks	pp. 28–29, 43–47
15	Proper rate of a standard clock	pp. 43–44
16	Instantaneous rate of a nonuniformly moving ideal clock	pp. 43–47
17	Muon lifetimes used as natural clocks	pp. 43–45
18	Rates of physical processes, including rotation, atomic vibration and particle-cluster expansion	pp. 35, 43–45
19	Time-dilation relation for elapsed quantities and rates of physical processes	pp. 44–45
20	Accumulated elapsed quantities and reunion comparison of clocks on different worldlines	pp. 45–47
21	Coordinate velocity of particles and noninformative moving patterns	pp. 36, 39–40, 47–51
22	Coordinate acceleration and proper acceleration	pp. 43–44, 46–47, 49–51
23	Time orientation preserved in transformations and causal comparisons	pp. 31–32, 34–36, 51–53

Auxiliary Operator A1 is the first time derivative; A2 is the second time derivative obtained by iterating A1. A1 occurs in the definitions and calculations of Claims 31 and 52–54, and A2 in Claims 53–54; each is included in the corresponding claim support.

Definition 8.312 (Rindler 1977 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(61)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([14], [61])$. The following table gives each claim, its law-time uses and source locations.

Table 215: Rindler 1977 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Material clock carriers	$\emptyset$	pp. 30, 43
2	Replicated clock carriers	$\emptyset$	pp. 30–31, 44
3	Light and projectile signal carriers	$\emptyset$	pp. 28, 30–31
4	Identities of emission, reflection and reception events	$\emptyset$	pp. 28, 30–31
5	Spatial configurations of reference-frame bodies	$\emptyset$	pp. 23, 25, 31
6	Spatial arrangement of lattice points	$\emptyset$	pp. 30–31
7	Calibration rods and length standards	$\emptyset$	pp. 29–30
8	Muon carriers	$\emptyset$	pp. 43–45
9	Atomic and charged-particle carriers	$\emptyset$	pp. 43, 45

p	Claim	i	Source Location
10	Aircraft carriers	$\emptyset$	pp. 45
11	Spatial geometry of paths and event locations	$\emptyset$	pp. 38, 40, 45–46, 49–51
12	Spatial field configurations and material properties of apparatus	$\emptyset$	pp. 43, 45
13	Detection-record carriers	$\emptyset$	pp. 44–45
14	Local-event identity	$\emptyset$	pp. 30–31
15	Principle of relativity	$\emptyset$	pp. 23–25, 31–32, 51–53
16	Postulate of invariant light speed in vacuum	2^Q	pp. 23–24, 32–33
17	Galilean transformation using absolute time and the Newtonian collision example	$1^Q, 2^{QS}, 8^{QS}$	pp. 23, 26–27, 53
18	Lorentz-transformation branch	$2^Q, 8^Q, 9^X$	pp. 23–24, 32–33, 51–53
19	Model character of physical laws	$\emptyset$	pp. 25–26
20	Domain of validity of physical laws	$\emptyset$	pp. 25–26
21	Definition of practical simultaneity	$2^Q, 3^X$	pp. 28–29
22	Synchronization convention adapted to an inertial system	$2^Q, 3^X, 4^X$	pp. 28–31
23	Use of a standard clock at the origin	5^K	pp. 30–31
24	Distance assignment by a signal round trip	$2^Q, 5^{KI}$	pp. 30–31
25	Use of coordinate-lattice test particles	$2^Q, 8^Q$	pp. 30–31
26	Use of a replicated-clock family from the standard clock	$4^X, 5^K, 6^K$	pp. 30–31
27	Spatial calibration relation	$2^Q, 5^K$	pp. 29–31
28	Temporal calibration relation	$2^Q, 4^X, 5^{KX}, 6^{KX}$	pp. 30–31
29	Coordinate reading of a local event	$7^{KO}, 8^{OQ}$	pp. 30–31
30	Calibration antecedent from Newton's first law	$2^Q, 8^Q$	pp. 25, 30–31
31	Linear transformation from coordinate regularity of finite events and preservation of uniform motion	$2^Q, 8^Q, 9^X, 23^B$	pp. 31–32
32	Standard Lorentz transformation	$2^Q, 8^{XQ}, 9^X$	pp. 32–34
33	Invariant quadratic interval	$2^Q, 8^{IQ}, 9^X, 10^Q$	pp. 34, 37, 48
34	Lorentz group	$2^Q, 8^Q, 9^X, 10^Q, 23^B$	pp. 34, 37–38
35	Causal speed boundary	$2^Q, 8^{CQ}, 9^X, 11^C, 12^Q, 23^B$	pp. 35–36, 39–40, 48
36	Passive diagrammatic coordinate transformation	$2^Q, 8^Q, 9^X, 10^Q, 13^B$	pp. 38–39
37	World picture	$2^Q, 8^Q$	pp. 40
38	Distinction between world picture and world map	$3^X, 7^K, 8^{QB}, 13^B$	pp. 40

p	Claim	i	Source Location
39	Length contraction from simultaneous endpoint comparison and paradox explanations using finite signal propagation	$2^Q, 3^{XB}, 8^{QB}, 9^X, 12^Q$	pp. 28–29, 40–43
40	Time-dilation relation for standard clocks	$2^Q, 8^{IQ}, 9^X, 14^{XI}$	pp. 43–44
41	Assignment of proper rate	$14^{XI}, 15^Q$	pp. 43
42	Clock hypothesis fixing the instantaneous rate of an ideal clock by its speed	$2^Q, 14^{XI}, 16^{KQ}, 22^Q$	pp. 43
43	Discussion of mechanism-dependent response of nonideal clocks and compensation mechanisms	$16^{KQ}, 22^Q$	pp. 43
44	Vibrating atoms as natural clocks	$16^Q, 18^{KQ}$	pp. 43
45	Decaying muon as a clock example	$16^Q, 17^{KI}$	pp. 43–45
46	Reference to muon experiments	$14^{XI}, 16^Q, 17^{IO}, 22^Q$	pp. 44–45
47	Reference to the airborne Cs-clock experiment	$14^{XI}, 16^Q, 20^{XIO}$	pp. 45
48	Application to physical processes using a comoving standard clock and relativistic focusing of particle clusters	$14^{XI}, 17^I, 18^{QS}, 19^{XS}$	pp. 44–45
49	Reciprocity of inertial-clock comparison	$3^X, 4^X, 14^{XI}, 15^Q$	pp. 43–45
50	Twin worldlines and accumulated elapsed quantities using ideal clocks	$2^Q, 8^{QS}, 14^{XI}, 16^Q, 20^{XIS}, 22^Q$	pp. 45–47
51	Explanation of twin asymmetry through extended free-fall legs and received signals	$2^Q, 8^{KQ}, 11^C, 14^{XI}, 20^{KXI}$	pp. 46–47
52	Velocity transformation	$2^Q, 8^{QS}, 9^X, 21^{QS}$	pp. 47–49
53	Proper-acceleration transformation	$2^Q, 8^{QS}, 14^{XI}, 21^{QS}, 22^{QS}$	pp. 49
54	Proper separation and rigid motion of a family of hyperbolic trajectories	$2^Q, 8^{QS}, 21^{QS}, 22^{QS}$	pp. 50–51
55	Derivation of a universal transformation constant without the second postulate	$2^Q, 8^Q, 9^X, 23^B$	pp. 51–52
56	Lorentz and Galilean branches from a positive finite constant and its infinite limit	$2^Q, 8^Q, 9^X, 23^B$	pp. 52–53
57	Exclusion of the negative-constant branch using positive coefficients, time orientation and causal order	$8^Q, 11^C, 23^B$	pp. 31–32, 36, 52–53
58	Separate transformation arguments for bounded and unbounded particle speeds	$1^Q, 2^Q, 8^{CQ}, 9^X, 11^C, 12^Q, 23^B$	pp. 53
59	Rindler’s criticism of explanations requiring general relativity for the clock paradox	$\emptyset$	pp. 47
60	Rindler’s account of ideal inertial frames and local application	$\emptyset$	pp. 25, 47
61	Rindler’s account of accelerated motion in inertial frames and general relativity	$\emptyset$	pp. 25, 47

Law-time uses comprise 141 claim/time-content pairs. Superscript codes indicate use roles. Claim 59 gives Rindler’s criticism concerning Time Contents 16, 20 and 22; Claim 60 describes the application domain concerning Time Content 8; and Claim 61 concerns Time Contents 16 and 22.

Table 216: Rindler 1977 Time-Use Roles

Code	Use Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causality
S	State evolution
O	Observation record
B	Reference boundary

Table 217: Rindler 1977 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–2, 5–10, 12, 14–22	separate	$\emptyset$
3, 11	member	order_relation
4	dependency	$\emptyset$
13	member	instant
23	member	orientation

Claims 1–14 form the initial set for physical generation; all 61 claims form the source formal-evidence environment. Source-specific obligation atoms comprise 61 claims, 23 time contents, 15 transfers and 18 results, totaling 117.

Formal Input Elements

Definition 8.313 (Rindler 1977 Clock and Coordinate Inputs). Rindler assigns calibrated coordinates to two inertial frames S, S' . Using the origin observer’s standard clock and its replicas, he sets remote clocks to $t_0 + r/c$ for light signals and to $(t + t_0)/2$ for projectiles. Here c is the light speed, while t_0, t are the origin emission and return times. Each transfer specifies the relations among coordinate time, clock elapsed quantities, speed, instantaneous rate and accumulation.

Dependency Inputs The following table gives each transfer’s time inputs, time outputs and claim support.

Definition 8.314 (Rindler 1977 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [15]$.

Table 218: Rindler 1977 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	2	3	3–5, 15, 21	Specification of simultaneity through symmetric signal propagation. pp. 28–29
2	3, 5	4	1–4, 21–23, 26	Synchronization of separated standard clocks using the simultaneity convention. pp. 28–31
3	2, 4–5	6	1–4, 7, 22–24, 26–28, 30	Setting a remote clock using a round-trip signal and the origin clock. pp. 30–31
4	4–6	7	1–2, 4, 6–7, 14, 25–29	Time assignment to local events using calibrated lattice clocks. pp. 30–31
5	2, 8, 23	9	4–6, 15–16, 31–32	Lorentz time transformation using invariant light speed. pp. 31–33
6	2, 8–9	10	4–5, 32–33	Invariant interval from coordinate differences between two events. pp. 33–34, 37
7	7–8	13	4, 6, 11, 14, 29, 36, 38	Description of worldlines and world maps by time slices. pp. 38–40
8	2, 8–9	14	1, 4–5, 32, 40	Time-dilation relation from two events at the same standard clock. pp. 43
9	14	15	1, 5, 40–41	Proper rate of a standard clock in its rest frame. pp. 43
10	14, 17–18	19	1, 8–9, 15, 40, 45, 48	Time-dilation relation for physical processes using a comoving clock. pp. 44–45
11	8, 14, 16	20	1, 5, 11, 40, 42, 50	Accumulation of elapsed quantities along ideal-clock worldlines. pp. 45–46
12	2, 8–9	21	4–5, 11, 32, 52	Particle-velocity transformation from ratios of coordinate increments. pp. 47–49
13	2, 8, 14, 21	22	5, 11, 40, 52–53	One-dimensional proper acceleration from the instantaneous rest frame. pp. 49
14	2, 8, 11, 23	1	4–5, 15, 31, 57–58	Absolute time in the unbounded-particle-speed branch. pp. 53
15	2, 8, 12, 23	9	4–5, 15, 31, 56, 58	Lorentz-type time transformation from a finite particle-speed bound. pp. 53

Grounding Conditions

1. In a homogeneous and isotropic inertial frame, Rindler defines as simultaneous the

arrival events of light or identical projectiles traveling from a midpoint to equidistant endpoints.

2. Observers at the separate positions use standard clocks and assign equal readings to events defined as simultaneous by symmetric signal propagation. Equal clock rates are presupposed.
3. For light emitted at origin time t_0 , the lattice clock at distance r is set to $t_0 + r/c$ on reception. For projectiles, the origin emission and return readings t_0 and t give distance $u(t - t_0)/2$ and remote reception time $(t + t_0)/2$. Projectile calibration presupposes Newton's first law.
4. After spatial and temporal calibration of the lattice, auxiliary observers associate local clock readings with events at their lattice positions.
5. Using calibrated coordinates, coordinate regularity of finite events, linearity, the relativity principle, isotropy, continuity and invariant light speed, Rindler derives $t' = \gamma(t - vx/c^2)$ in standard configuration, with $|v| < c$ and positive γ .
6. Lorentz transformations act on the temporal and spatial coordinate differences between two events and preserve $c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$. Translations in a general transformation cancel in the coordinate differences.
7. Rindler's world map is the event configuration on a slice $t = t_0$ of one inertial frame, recorded locally at the same specified time by lattice observers. A Minkowski diagram assigns different coordinates to the same events.
8. For a clock at rest in one inertial frame, set $\Delta x' = 0$ and obtain $T = \gamma T_0$ from the inverse Lorentz transformation. The clock moves uniformly relative to the other frame.
9. Rindler calls the rate assigned in the clock's rest frame, where the time-dilation relation assigns its greatest rate, the proper rate.
10. Rindler imagines a standard clock accompanying a muon or traveler and applies time dilation to elapsed quantities and rates expressed by that clock. Uniform-motion segments use Eq. (2.22); the instantaneous rate of a nonuniformly moving ideal clock additionally uses the clock hypothesis.
11. Rindler compares a returning ideal clock with a clock at rest in a single inertial frame when they reunite. Segment elapsed quantities accumulate along the specified trajectory using the instantaneous clock rate.
12. Rindler transforms ratios of coordinate increments between two events and takes the limit to obtain instantaneous velocities. The same coordinate transformation applies to noninformative moving patterns where its denominator is nonzero.
13. Restricting motion to the common x direction, Rindler differentiates the velocity transformation and uses $dt' = dt/\gamma(u)$ to obtain $\alpha = \gamma(u)^3 du/dt$.
14. In the unbounded-particle-speed case, Rindler uses causal connectibility and preservation of time orientation to associate $t = 0$ with $t' = 0$, then derives $t' = t$ by symmetry.

15. When particle speeds have a finite upper bound, Rindler derives invariance of that bound using the relativity principle, continuity and monotonicity of the velocity transformation, and obtains a Lorentz-type transformation using that invariant speed.

Results

Definition 8.315 (Rindler 1977 Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [18]$. The following table specifies each result, its time premises, claim support and source locations.

Table 219: Rindler 1977 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Galilean branch using absolute time and the Newtonian collision example. pp. 23, 26–27, 53	1–2, 8	5, 11, 15, 17
2	Simultaneity relative to an inertial frame. pp. 28–29	2–3	3–5, 15–16, 21
3	Synchronization of separated clocks adapted to an inertial frame. pp. 28–31	3–6	1–4, 21–23, 26, 28
4	Coordinate-lattice calibration and local readings using standard clocks. pp. 30–31	2, 4–7	1–4, 6–7, 14, 23–30
5	Lorentz coordinate transformations, group structure and hyperbolic representations. pp. 31–34, 37–38	2, 8–9, 23	4–5, 15–16, 31–32, 34
6	Lorentz-invariant squared interval. pp. 33–34, 37	2, 8–10	4, 32–33
7	Bound on information-transmission speed and preservation of causal order. pp. 35–36, 39–40, 48	2, 8–9, 11–12, 23	3–5, 34–35, 57
8	Distinction among passive coordinate representation, world picture and world map. pp. 38–40	2–3, 7–8, 13	4, 6, 11, 14, 36–38
9	Length contraction from simultaneous endpoint comparison. pp. 28–29, 40–43	2–3, 8–9	5, 7, 11, 32, 39
10	Time dilation and proper rate of standard clocks. pp. 43	14–15	1, 5, 40–41
11	Clock hypothesis for ideal clocks and acceleration response of nonideal clocks. pp. 43	2, 14, 16, 22	1, 12, 42–43
12	Reported examples involving muons, atoms, particle clusters and airborne clocks. pp. 43–45	14, 16–18, 20, 22	8–10, 12–13, 44–48

q	Result and Source Location	Time Premises i	Claim Support p
13	Application of time dilation to physical processes and lifetimes using a comoving clock. pp. 44–45	14, 17–19	1, 8–9, 40, 45, 48
14	Reciprocity of inertial-clock comparisons. pp. 43–44	3–4, 14–15	1–2, 5, 40–41, 49
15	Ideal-clock worldlines and comparison of elapsed quantities at reunion. pp. 45–46	8, 14, 16, 20	1, 5, 11, 40, 42, 50
16	Velocity transformation, one-dimensional proper acceleration and hyperbolic rigid motion. pp. 47–51	2, 8, 14, 21–22	5, 11, 52–54
17	Galilean/Lorentz-type dichotomy without the second postulate. pp. 51–53	2, 8, 23	5, 15, 31, 55–57
18	Asymmetry of the twin comparison with extended free-fall legs. pp. 46–47	8, 11, 14, 20	1, 5, 11, 40, 51

Grounds for the Results

1. The transformation belongs to the absolute-time branch; the collision example concerns identical perfectly elastic spheres.
2. The comparison uses each frame's simultaneity convention and light speed invariant between the aircraft and ground frames.
3. Standard clocks and symmetric signal propagation make each frame's coordinate description of physics isotropic.
4. The respective light/projectile procedures use distance assignment, clock setting and Newton's first law.
5. Standard transformations, inverses and composition are distinguished from general transformations including spatial and temporal translations.
6. Invariance of the quadratic form in the coordinate differences between two events.
7. The result concerns causal order for information-carrying signals. Separation speed of opposite photons and speed of a light spot are distinct velocity concepts.
8. The world picture comprises events at different past times conveyed by arriving light; the world map comprises simultaneous events in one frame.
9. $L = L_0/\gamma$ uses endpoint events simultaneous in the measuring frame. Instantaneous application to an accelerating infinitesimal body uses the length hypothesis.
10. The result concerns uniformly moving standard clocks.

11. Rindler distinguishes mechanism-dependent response to acceleration from the clock relation associated with speed.
12. Natural-clock and experimental examples reported by Rindler. The approximately 1% for CERN and approximately 10% for Hafele–Keating are the precisions stated in the book.
13. Rindler uses a comoving standard clock to apply the argument to temporal processes, including muon decay and human lifetimes.
14. The comparison uses a clock array synchronized in each inertial frame.
15. The rate of an ideal clock on nonuniform-motion segments uses the clock hypothesis.
16. The hyperbolic family $x^2 - c^2t^2 = X^2$ has $X > 0$, proper separation dX and position-dependent proper acceleration c^2/X .
17. The argument uses the relativity principle, Euclidean geometry, isotropy, the established coordinate regularity and continuity, group composition and positive coefficients. The alternative proof based on particle-speed bounds treats bounded and unbounded cases separately.
18. Rindler argues that sufficiently extended free-fall legs can dominate effects of acceleration itself. The received-signal explanation compares readings that include propagation time.

Results and Evidence-Effect Domain Rindler applies the standard-clock argument to muon decay, atomic vibration, expansion of charged-particle clusters, and human lifetimes expressed by a comoving clock. The experimental values for muons and airborne clocks are results of external experiments cited by Rindler. For the CERN storage ring, the book gives a Lorentz factor of about 12, precision of about 1%, and accelerations up to about $10^{19}g$; it gives about 10% for Hafele–Keating and about 1% for the Maryland experiments of 1975–76 (pp. 44–45).

Result 10 concerns uniformly moving standard clocks, Result 15 concerns accumulated elapsed quantities of ideal clocks including nonuniform-motion segments, and Result 18 concerns twin asymmetry with extended free-fall legs. The instantaneous rate in Result 15 uses the clock hypothesis. Result 18 uses Rindler’s argument that extending the free-fall legs can make the effects of acceleration itself relatively small (pp. 45–47).

Generation Judgment The LCTR complete-generation count is zero, the source-evidence-conditioned local-reach count is zero, and 23 primitive-time roots remain.

Result Dependence All 18 results depend on primitive-time roots. The minimal-support families give 94 root–result dependence pairs, of which 36 are necessary dependence pairs.

Placement of the Grounding Obligation Rindler’s constructions and derivations use standard-clock readings, signal speeds, simultaneity and synchronization, coordinate time, clock rates, lifetimes and physical-process rates, causal order, and time orientation. The 141 law-time uses in the claim table and the 15 transfers supply no conditions that generate physical time from non-temporal physical inputs before those time contents are used. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The source establishes theoretical relations under the specified clock, inertial-frame, transformation and worldline conditions and reports existing experiments. Actual failure of specific physical-time generation conditions, and resulting changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 94 root–result dependence pairs. Actual reach from the failure location of a specific generation condition remains undetermined.

Result-Preservation Domain Simultaneity and synchronization hold under the conditions of their signal procedures and standard clocks; coordinate transformations hold under the postulates and regularity conditions of their respective branches. Uniform-clock relations, accelerated ideal-clock relations and twin comparisons with long free-fall legs use the respective conditions of Results 10, 15 and 18. Velocity, acceleration and rigid motion use the motion domain and geometric conditions of Result 16. Realizing these inputs through generated temporal representations that preserve the corresponding relations preserves the 18 results in their respective domains.

Repair Scope If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.316 (Rindler 1977 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \text{AllRootDepIdx}(23, 18).$$

$$R \in \text{Ref}^{\text{audit, rootdep}} \text{ and } R \notin \text{Ref}^{\text{audit, rootunreach}}.$$

Proof. The boundary has 102 nodes—61 claims, 23 time contents and 18 results—and 33 edges—15 transfers and 18 result dependencies. Closure from Initial Claims 1–14 remains those 14 nodes. Even with all 61 claims supplied, no time content or result is reached, because every transfer and result dependency has a time input. Supplying all 23 time contents reaches all 18 results. $\square$

Proposition 8.317 (Rindler 1977 Minimal Primitive-Time-Root Support). The following table gives minimal-support statistics as the number of members, minimum cardinality and maximum cardinality. There are 50 family members in total.

Table 220: Rindler 1977 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(2; 3, 4)
2	(1; 1, 1)
3	(2; 2, 3)
4	(1; 2, 2)
5	(1; 3, 3)
6	(2; 3, 3)
7	(1; 5, 5)
8	(2; 3, 3)
9	(2; 3, 3)
10	(3; 1, 3)
11	(4; 4, 5)
12	(5; 6, 7)
13	(3; 3, 5)
14	(8; 3, 4)
15	(3; 3, 4)
16	(3; 3, 4)
17	(1; 3, 3)
18	(6; 4, 5)

Proof. Start from the singleton seed of each time content and apply Definition 7.46 to the 15 transfers. Once the antichains stabilize, combine the supports of each result’s time premises and discard supports that properly contain another support. Proposition 7.47 gives the displayed families and counts. □

Proposition 8.318 (Rindler 1977 Audit Conclusion). Complete generation and local reach are both zero; all 23 time contents remain primitive-time roots. All 18 results depend on primitive-time roots.

Proof. This follows from the reachable sets in Proposition 8.316 and the minimal-support families in Proposition 8.317. □

Individual Statements Relativity of Simultaneity
Verbatim Original (English):

Thus simultaneity is relative!

The original-source location is Wolfgang Rindler, *Essential Relativity: Special, General, and Cosmological*, Revised Second Edition, p. 28, Section 2.4.

Minimal Context Required for Interpretation Rindler explains that a flash emitted at the center of the aircraft reaches its two ends simultaneously in the aircraft frame, whereas it reaches the rear end first in the ground frame.

The attributed author is Wolfgang Rindler in the text of the book.

Reality and Experimental Confirmation of Time Dilation
Verbatim Original (English):

Time dilation, like length contraction, is real. And this has been confirmed experimentally.

The original-source location is p. 44, Section 2.13 of the same book. The minimal context is the paragraph that places this statement immediately after an explanation of the reciprocal difference between readings of synchronized standard clocks fixed in two inertial systems, and then cites cosmic-ray muons and the CERN storage-ring experiment as examples. The attributed author is Wolfgang Rindler in the text of the book.

Application to Physical Processes and Human Lifetimes

Verbatim Original (English):

the time dilation argument applies to any temporal change or process and therefore also to muon decay and even human life times

The original-source location is the opening paragraph on p. 45, near the end of Section 2.13 of the same book. The minimal context raises the objection that a muon is not a clock, then supposes that a standard clock travels with the muon or space traveler and extends the domain of application of the time-dilation argument. The attributed author is Wolfgang Rindler in the text of the book.

Common Evidence with Specified Domain of Application

Target state-ment	Object of the statement	Predicate	Explicit domain of application
S23-Q001	Two light-reception events at the ends of the aircraft, and the simultaneity relation between two inertial frames	Rindler states that two light-reception events simultaneous in the aircraft frame are not simultaneous in the ground frame, and that simultaneity is relative to an inertial frame.	The domain in Section 2.4 that compares, in the aircraft and ground inertial frames, the arrival events of a flash propagating from the aircraft center to its two ends.
S23-Q002	Time dilation described as the reading relation of standard clocks in two inertial systems	Rindler states that time dilation, like length contraction, is real and has been confirmed experimentally.	The domain in Section 2.13 that explains reciprocal readings of synchronized standard clocks, cosmic-ray muons, and the CERN storage-ring experiment.
S23-Q003	Temporal changes, physical processes, muon decay, and human lifetimes as objects to which the time-dilation argument is applied	Rindler applies the time-dilation argument to any temporal change or process, including muon decay and human lifetimes.	The domain near the end of Section 2.13 that assumes a comoving standard clock and generalizes the clock relation to physical processes and human lifetimes.

Individual Audit

First Statement Rindler compares the arrival order of light emitted at the aircraft center in the aircraft and ground frames. Invariant light speed and the simultaneity convention in each frame show that reception events simultaneous in the aircraft are not simultaneous on the ground. The corresponding LCTR audit entries are Time Contents 2–4, Claims 15–16 and 21–22, Transfers 1–2, and Results 2–3.

Second Statement Rindler discusses time dilation of standard clocks and cites cosmic-ray and CERN storage-ring muon results as experimental confirmation. The measured quantity in these examples is the lifetime of moving muons; the connection with standard-clock relations and the clock hypothesis belongs to the book’s theoretical account. The corresponding entries are Time Contents 14–18 and 22, Claims 40–46, and Results 10–12.

Third Statement Rindler imagines a standard clock accompanying a muon or space traveler and applies the time-dilation argument to temporal changes and processes expressed by that clock. Application to human lifetimes is part of this theoretical generalization using a comoving clock. Result 10 concerns uniform-motion clock relations; Result 13 concerns application to physical processes; Result 15 concerns reunion comparison using nonuniformly moving ideal clocks; and Result 18 concerns asymmetry with long free-fall legs. Claims 40–42, 45, 48 and 50–51 support these distinctions.

8.19.24 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 24.

Source Name Penrose 1979

Bibliographic Identification Roger Penrose, *Singularities and Time-Asymmetry*, in S. W. Hawking and W. Israel, eds., *General Relativity: An Einstein Centenary Survey*, Cambridge University Press, pp. 581–638 (1979) is denoted by $R = \text{Ref}_{\text{Pen79STA}}$. The audit covers singularities; seven arrows of time or candidates; a low-entropy past; radiation, psychological time and cosmic expansion; black and white holes; gravitational entropy; causal boundaries and cosmic censorship; the initial Weyl condition; and arguments concerning time-asymmetric laws.

Audit-Scope Identifier 24-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Original-Source Location Identifier 24-L1.

Original-Source Location Content The tables give the source pages for the time contents, claims, transfers and results.

Formal Objects The audit uses 25 time contents, 45 claims, 11 transfers and 20 results. The source definitions, operations and hypotheses are specified in the respective claims. Penrose assesses the singularity theorems as indicating limits of known physical laws and looks to singular structure for the origin of the arrow of time (p. 581). Section 12.2

lists seven arrows of time, or possible arrows. Microscopic neutral-K-meson asymmetry, quantum observation, thermodynamics, radiation, psychological time, cosmic expansion, and black-hole/white-hole processes are treated as distinct questions.

Penrose discusses the reality of the wave function and observational state reduction, seeking new laws that incorporate observation. For the subsequent discussion in Section 12.2, he provisionally adopts the conventional view that quantum mechanics has no manifest arrow of time, returning to measurement in Section 12.4 (pp. 583–586, 635–636). His treatment of psychological time distinguishes an observer-independent universal present, branching futures, ease of prediction and retrodiction, and fixity of the past, before returning to conventional Hausdorff spacetime (pp. 591–596).

The initial Weyl condition is Penrose’s proposal. He prefers the form in which Weyl curvature tends to zero, while also mentioning faster decay, boundedness and domination by Ricci curvature; he has not examined their differences in detail (pp. 630–631). He states that detailed initial matter fluctuations require further understanding of particle physics (p. 634).

Penrose presents a cosmological account using an initial restriction and randomness within it. Against explaining the entropy imbalance of the entire universe by a large fluctuation favorable to observers, he argues that producing a few observers would require a smaller entropy decrease (pp. 633–634). Section 12.4 outlines prospects involving nonlinear quantum gravity incorporating observation, baryon–antibaryon production and annihilation, and detection of CPT violation. The logarithmic nonlinear Schrödinger model he cites is time-symmetric (pp. 635–638).

Assumptions The quantum comparisons use unitary evolution and specified observation times. The thermodynamic, radiative and psychological arguments add a low-entropy past condition to time-symmetric laws. Classical black-hole/white-hole reversal uses time-symmetric equations of state; area nondecrease uses the conditions of the classical area theorem. The causal-boundary definitions and equivalences assume strong causality. Application of strong cosmic censorship to physical spacetimes is Penrose’s conjecture. The initial Weyl restriction is an additional condition. For history spaces, Penrose uses classical deterministic Hamiltonian evolution and coarse-graining, setting aside infinite-dimensional, infinite-measure and general-relativistic gauge problems (pp. 631–632).

Non-temporal Physical Inputs The generation-initial inputs are the physical carriers, material carriers of records and apparatus, event identities, and spatial configurations of matter, apparatus and boundaries in Claims 1–3. The claim table identifies temporal uses of oriented metrics, curvature histories, dated records, causal boundaries and initial conditions.

Time Contents

Definition 8.319 (Penrose 1979 Time Contents). Let $n_R = 25$. The following table specifies the time contents and their source locations.

Table 222: Penrose 1979 Time Contents

i	Time Content	Source Location
1	Time orientation distinguishing past and future	pp. 582–583, 600, 637
2	Lorentzian metric, light cones and chronological causal relations	pp. 591–592, 597–598, 600, 621–624
3	Neutral-K-meson decay rates and time-reversal asymmetry	pp. 583, 636–637
4	Quantum evolution between observation times and its temporal arguments	pp. 583–586, 635–636
5	Ordering of observation and state reduction	pp. 583–586, 635–636
6	Histories of microscopic states and correlations	pp. 586–590, 595–596, 631–633
7	Boundary conditions imposing a low-entropy past and a high-entropy future	pp. 586–590, 595–596, 611, 633
8	Time dependence of coarse-grained entropy	pp. 586–588, 595–596, 603–604, 611, 616, 632–633
9	Temporal relations in retarded and advanced radiation solutions	pp. 588–591
10	Retarded direction of radiation and incoming-correlation conditions	pp. 588–591, 611
11	Temporal relations among records, memory, prediction and retrodiction	pp. 591–596
12	Asymmetry of explaining future correlations by past causes	pp. 595–596, 611
13	Cosmic radius and its temporal change during expansion and recollapse	pp. 596–599, 612–613, 616–617
14	Initial and final boundaries of cosmological and singular histories	pp. 581, 596–599, 611–613, 627–628, 630–631, 634, 637
15	Black-hole formation and its time-reversed white-hole process	pp. 597–602, 606–610, 634–635
16	Temporal change of horizon area	pp. 602
17	Temporal change of the entropy associated with horizon area	pp. 602–605, 608–609, 630, 635–636
18	Durations and histories of Hawking evaporation, background cooling and relaxation in a container	pp. 603–606, 608–610, 617–618
19	Matter and geometry histories along gravitational clumping and collapse	pp. 612–615, 619–621, 625–626
20	Changes and oscillations of Weyl and Ricci curvature along worldlines and their limits at singular boundaries	pp. 614–617, 621, 625–626, 630–631, 634, 637
21	TIPs, TIFs and ideal endpoints defined by chronological past and future sets	pp. 606, 621–628, 630
22	Proper elapsed quantities along worldlines and their finite or infinite extent	pp. 598, 613, 615, 621, 627

i	Time Content	Source Location
23	Initial hypersurfaces and domains of dependence of temporal evolution	pp. 601, 610, 618, 620–621, 624–628, 631
24	Weyl-curvature restriction imposed only at initial ideal boundaries	pp. 630–635, 637
25	Canonical evolution relating time-indexed universe-state and history spaces	pp. 631–634

Definition 8.320 (Penrose 1979 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(45)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([3], [45])$. The table gives each claim, its law-time uses and source locations.

Table 223: Penrose 1979 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Physical carriers of neutral K mesons, quantum systems, matter and radiation	$\emptyset$	pp. 583, 586, 588, 600, 611
2	Material carriers of records, memories and observation apparatus	$\emptyset$	pp. 583, 591, 595
3	Event identities and spatial configurations of matter, apparatus and boundaries	$\emptyset$	pp. 588, 591, 598, 603
4	Penrose uses timelike curves, light cones and proper elapsed quantities in time-oriented Lorentzian spacetime.	$1^B, 2^C, 22^I$	pp. 591, 598, 600, 621
5	Penrose connects the singularity theorems with limits of existing laws and looks to singular structure for the origin of the arrow of time.	$\emptyset$	pp. 581
6	Penrose lists seven arrows of time, or possible arrows.	$\emptyset$	pp. 582–583
7	Penrose cites T asymmetry in neutral K mesons through the CP/CPT argument and leaves its relation to macroscopic arrows unresolved.	$1^B, 3^{\text{OPQ}}$	pp. 583, 636–637
8	For unitary evolution U, the forward and inverse transition amplitudes are complex conjugates and have equal squared moduli.	$1^B, 4^{\text{PS}}$	pp. 584
9	An eigenstate ψ of P maps to the eigenstate $U\psi$ of UPU^{-1} . For a simple eigenvalue, the eigenstate is unique up to phase.	$4^{\text{XS}}, 5^{\text{O}}$	pp. 585
10	Penrose distinguishes observational state reduction from unitary evolution and discusses the reality of the wave function and the need for new laws.	$4^{\text{S}}, 5^{\text{OS}}$	pp. 583–586, 635–636

p	Claim	i	Source Location
11	Time-symmetric deterministic laws require boundary data; a low-entropy past and a high-entropy future provide asymmetric conditions.	$1^B, 6^S, 7^B$	pp. 586–588
12	$S = k \log N$ is defined from the number N of states in a coarse-grained macrocell and assigned through the cell containing the microscopic state at each time.	$6^{XS}, 8^Q$	pp. 587–588, 632
13	Penrose argues that a low-entropy past condition makes future microscopic correlations compatible with macroscopic entropy increase.	$6^S, 7^B, 8^Q$	pp. 587–588, 595–596
14	Time reversal relates retarded and advanced radiation solutions and reverses the causal order of emission and absorption.	$1^B, 2^C, 6^S, 9^C$	pp. 588–590
15	Penrose explains the retarded direction of radiation through a low-entropy past and the absence of specially correlated incoming radiation.	$7^B, 8^Q, 9^C, 10^C$	pp. 589–591, 611
16	Penrose distinguishes memory, prediction and retrodiction, and treats ease of prediction separately from the fixity of the past.	$2^C, 11^O$	pp. 591–596
17	Penrose connects one aspect of psychological time—the availability of past causes for future correlations—with a low-entropy past.	$6^S, 7^B, 8^Q, 11^O, 12^C$	pp. 595–596, 611
18	Penrose argues that cosmic expansion alone does not explain entropy increase, comparing a recollapsing universe with an observer inside a black hole.	$2^C, 7^B, 8^Q, 13^S, 14^B, 15^S$	pp. 596–599, 611
19	In classical general relativity with time-symmetric equations of state, reversal of time orientation maps a black-hole collapse solution to the white-hole process obtained by reversing that collapse.	$1^B, 2^C, 14^B, 15^{CS}$	pp. 600–601
20	The positive coefficient in the Bekenstein–Hawking formula $S = kAc^3/(4\hbar G)$ transfers classical area nondecrease to the associated entropy.	$1^B, 15^S, 16^{QS}, 17^{QS}$	pp. 602
21	Penrose cites the temperature and duration of Hawking evaporation, giving model values exceeding 10^{53} Hubble times for evaporation of a hole of at least one solar mass and about 10^7 Hubble times for background cooling.	$13^{QS}, 15^S, 17^Q, 18^{IQS}$	pp. 603

p	Claim	i	Source Location
22	Depending on reflecting-container size and total mass-energy, entropy maxima occur for radiation alone or for a hole in equilibrium with radiation. Thermal fluctuations produce transitions between these states.	$8^Q, 17^Q, 18^{PS}$	pp. 603–605, 608–609
23	Hawking proposes identifying black and white holes through equilibrium time symmetry. Penrose objects on the basis of transient exterior geometry and the difference between formation and evaporation.	$1^B, 2^C, 15^{CS}, 18^S, 21^{CB}, 23^{CB}$	pp. 605–610
24	Penrose links the thermodynamic and retarded-radiation arrows to a low-entropy initial condition and leaves psychological time as a possible further connection.	$7^B, 8^Q, 10^C, 12^C$	pp. 611
25	High thermal entropy of early matter is compatible with low gravitational entropy of the initial geometry.	$7^B, 13^S, 14^B, 19^S$	pp. 611–613
26	Penrose qualitatively relates gravitational clumping, collapse and increasing Weyl curvature to gravitational entropy.	$14^B, 19^S, 20^S$	pp. 612–615
27	In the source cigar model, $\text{Weyl} \sim r^{-3}$ and $\text{Ricci} \sim r^{-3/2}$; in its pancake model, $\text{Weyl} \sim r^{-2}$ and $\text{Ricci} \sim r^{-1}$. Their ratios diverge as $r \rightarrow 0$ in the respective models. The oscillating-curvature comparison uses frequency ν .	$19^S, 20^{QS}, 22^{IQ}$	pp. 614–615
28	In Tolman’s models, entropy increases through bulk viscosity during expansion and recollapse even though Weyl curvature vanishes everywhere.	$8^Q, 13^S, 20^S$	pp. 616–617
29	Under strong causality, past sets of future-endless timelike curves define TIPs and future sets of past-endless timelike curves define TIFs. Curves with the same respective past or future set share an ideal endpoint.	$2^C, 21^{CB}$	pp. 621–622
30	Under strong causality, excluding naked TIPs and TIFs, including those at infinity, is equivalent to global hyperbolicity and to the existence of a Cauchy hypersurface.	$2^C, 21^{CB}, 23^{CB}$	pp. 623–625
31	Penrose conjectures that physically reasonable classical spacetimes satisfy strong cosmic censorship.	$2^C, 14^B, 21^{CB}, 23^B$	pp. 617–619, 624–626

p	Claim	i	Source Location
32	Penrose cites divergent weak-field perturbations at Cauchy horizons and gives a plausibility argument that generic nonlinear perturbations produce curvature singularities.	$2^C, 19^S, 20^S, 22^{IQ}, 23^{CB}$	pp. 619–621, 625–626
33	A globally hyperbolic spacetime interior splits as a product of a Cauchy hypersurface and timelike curves, while its ideal boundaries need not share the hypersurface's topology.	$2^C, 14^B, 21^{CB}, 23^{BS}$	pp. 627–628
34	Penrose imposes a condition that Weyl curvature tend to zero only at initial ideal boundaries. He also lists faster decay, boundedness and Ricci dominance as alternatives.	$14^B, 20^{BS}, 24^B$	pp. 630–631
35	Under classical deterministic Hamiltonian evolution, time-indexed data spaces are related by canonical isomorphisms and Liouville measure is preserved.	$6^S, 14^B, 25^{XS}$	pp. 631–632
36	Penrose embeds the history space U satisfying the initial Weyl restriction into an unrestricted space W and evaluates entropy at each time from macrocell volumes extended into W .	$6^S, 8^{QS}, 24^B, 25^{XS}$	pp. 632–633
37	Penrose uses randomness within the initial Weyl restriction and the absence of an additional final constraint to explain the lack of initial correlations and their later increase.	$6^S, 7^B, 8^{QS}, 14^B, 24^B, 25^{XS}$	pp. 633–634
38	Combining initially vanishing Weyl curvature with nearly thermal matter and radiation, Penrose argues for approximate initial homogeneity and isotropy.	$13^S, 14^B, 20^S, 24^B$	pp. 634
39	Penrose allows initial matter fluctuations but requires further understanding of particle physics to calculate their density and velocity details.	$\emptyset$	pp. 630, 634
40	The initial Weyl condition excludes white-hole initial singularities that violate it while allowing black-hole formation by ordinary gravitational collapse.	$14^B, 15^{CS}, 20^{BS}, 24^B$	pp. 610, 630, 634–635
41	Penrose looks for time asymmetry in future nonlinear quantum-gravitational laws incorporating observation. The logarithmic nonlinear Schrödinger model he cites is time-symmetric.	$\emptyset$	pp. 635–636

p	Claim	i	Source Location
42	Penrose discusses an explanation of the residual baryon-to-photon ratio through early baryon–antibaryon production and subsequent annihilation.	$\emptyset$	pp. 636–637
43	Laws imposing the initial Weyl restriction while allowing collapse with nonzero Weyl are incompatible with T, PT, CT and CPT symmetries that reverse temporal orientation while preserving Weyl zero versus nonzero.	$1^B, 14^B, 15^S, 20^{BS}, 24^B$	pp. 637
44	Penrose anticipates future detection of CPT violation and discusses Poincaré covariance near singularities and in quantum gravity, together with the limited quantitative detail of his argument.	$\emptyset$	pp. 637–638
45	A TIP or TIF is classified as infinity-type if at least one generating curve has infinite proper length, and as singular if no such curve exists.	$21^{CB}, 22^{IB}$	pp. 621

Law-time uses comprise 127 claim/time-content pairs. Superscript codes indicate use roles. Claims 5, 6, 39, 41, 42 and 44 record Penrose’s assessments of the singularity theorems, enumeration of seven candidates, discussion of detailed initial fluctuations and views on future laws.

Table 224: Penrose 1979 Time-Use Roles

Code	Use Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causality
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 225: Penrose 1979 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 10	member	orientation
2, 11	member	order_relation

Time Indices	Structural Type	Axes
3, 18, 22	separate	$\emptyset$
4, 6, 8–9, 13, 15–17, 19–20, 25	dependency	$\emptyset$
5	member	order_relation, transition_position
7, 14, 24	member	endpoint
12	member	orientation, order_relation
21	member	endpoint, order_relation
23	member	instant, order_relation

Claims 1–3 form the initial set for physical generation; all 45 claims form the source formal-evidence environment. Source-specific obligation atoms comprise 45 claims, 25 time contents, 11 transfers and 20 results, totaling 101.

Formal Input Elements

Definition 8.321 (Penrose 1979 Time Orientation, Evolution and Boundaries). Penrose uses time-oriented Lorentzian spacetime $(\mathcal{M}, g)$ and the times, histories, and initial and final conditions of each argument. The temporal arguments of quantum evolution U , the time of $S = k \log N$, the initial boundary of $C_{abcd} \rightarrow 0$, and the times of canonical universe-history evolution correspond to Time Contents 4, 8, 24 and 25, respectively.

Dependency Inputs The table gives each transfer’s time inputs, time outputs and claim support.

Definition 8.322 (Penrose 1979 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [11]$.

Table 226: Penrose 1979 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	6	8	1, 12	Coarse-grained entropy assignment from microscopic states. pp. 587–588, 632
2	1–2, 6	9	1, 3, 14	Time-reversal correspondence between retarded and advanced radiation solutions. pp. 588–590
3	7–9	10	1, 15	Explanation of the radiation arrow using a low-entropy past. pp. 589–591
4	6–8, 11	12	2, 17	Causal explanation of future correlations through the past condition. pp. 595–596
5	1–2, 14	15	1, 3–4, 19	Correspondence between a classical collapse solution and the white-hole process obtained by reversing that collapse. pp. 600–601

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
6	16	17	20	Entropy history from horizon area. pp. 602
7	14, 19	20	26–27	Curvature limits in the specified collapse models. pp. 614–615
8	2	21	4, 29	TIPs and TIFs from timelike curves. pp. 621–622
9	2, 21	23	30	Correspondence between causal-boundary conditions and a Cauchy evolution domain. pp. 623–625
10	6, 14	25	35	Correspondence of time-indexed data spaces under Hamiltonian evolution. pp. 631–632
11	24–25	8	12, 35–36	Entropy evaluation using extended cells for constrained histories. pp. 632–633

Grounding Conditions

1. At each time, the microscopic state is assigned to a macrocell of macroscopically indistinguishable states and given $S = k \log N$.
2. In the source discussion of time-symmetric radiation, reversing the source/absorption process and time orientation relates retarded and advanced solutions.
3. Penrose uses a low-entropy past and the absence of specially correlated incoming radiation to explain the retarded direction in the stone/ripple and radiation examples. An additional external agent arranging the waves supplies different boundary data.
4. Penrose's connection concerns the aspect of psychological time in which future correlations admit past causes.
5. Using time-oriented Lorentzian spacetime and time-symmetric equations of state, reverse the orientation of the classical spherically symmetric collapse solution.
6. Apply $S = kAc^3/(4\hbar G)$ to the horizon sections. In processes satisfying the classical area theorem, the positive coefficient transfers area nondecrease to entropy.
7. For the cigar and pancake models separately, compare their stated r dependences along the specified collapse history as $r \rightarrow 0$.
8. Assuming strong causality, define ideal endpoints through $I^-[\gamma]$ for future-endless timelike curves and $I^+[\eta]$ for past-endless timelike curves.
9. Under strong causality, imposing that no TIP be contained in the past set of a point implies global hyperbolicity and existence of a Cauchy hypersurface. The time-reversed TIF condition is equivalent.

10. In Penrose's classical deterministic Hamiltonian model, the equations of motion supply canonical isomorphisms among initial, intermediate and final data spaces.
11. For a state in U satisfying the initial Weyl restriction, extend its macrocell into unrestricted W and define $S = k \log V$ from its volume V . Penrose's model uses Hamiltonian evolution and the specified coarse-graining.

Results

Definition 8.323 (Penrose 1979 Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [20]$. The table specifies each result, its time premises, claim support and source locations.

Table 227: Penrose 1979 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Neutral-K-meson time asymmetry and its relation to macroscopic arrows. pp. 583, 636–637	1, 3	1, 7
2	Inverse propagation and observable transport under unitary evolution, and the measurement problem. pp. 583–586	1, 4–5	1, 8–10
3	Relation among a low-entropy past, coarse-graining and future correlations. pp. 586–588	1, 6–8	1, 11–13
4	Correspondence of retarded and advanced solutions and the retarded radiation arrow. pp. 588–591	1–2, 6–10	1, 3, 14–15
5	Memory, prediction and a partial explanation of psychological time. pp. 591–596	2, 6–8, 11–12	2, 16–17
6	Argument distinguishing cosmic expansion from thermodynamic direction. pp. 596–599, 611	2, 7–8, 13–15	4, 18–19
7	Classical time-reversal correspondence of black and white holes. pp. 600–601	1–2, 14–15	1, 3–4, 19
8	Classical area law and change of horizon entropy. pp. 602	1, 15–17	19–20
9	Hawking-evaporation timescales and entropy maxima in a reflecting container. pp. 603–605, 608–609	8, 13, 15, 17–18	1, 3, 21–22
10	Penrose's argument distinguishing transient black-hole and white-hole processes. pp. 605–610	1–2, 15, 18, 21, 23	19, 23, 29
11	Early matter equilibrium and low initial gravitational entropy. pp. 611–613	7, 13–14, 19	1, 25–26

<i>q</i>	Result and Source Location	Time Premises <i>i</i>	Claim Support <i>p</i>
12	Weyl dominance in collapse models and the qualitative gravitational-entropy relation. pp. 614–615	14, 19–20	26–27
13	Tolman models combining zero Weyl curvature with entropy increase. pp. 616–617	8, 13, 20	1, 28
14	TIPs, TIFs and singular or infinity-type ideal endpoints. pp. 621–622	2, 21–22	4, 29, 45
15	Equivalence of causal-boundary conditions, global hyperbolicity and Cauchy hypersurfaces. pp. 623–625	2, 21, 23	29–30
16	Interior decomposition and ideal boundaries under cosmic censorship. pp. 619–621, 624–628	2, 14, 19–23	30–33
17	Approximate homogeneity from the initial Weyl condition and thermal matter. pp. 630–631, 634	13–14, 20, 24	1, 34, 38
18	Exclusion of white holes and allowance of collapse black holes under the initial Weyl condition. pp. 610, 630, 634–635	14–15, 20, 24	19, 34, 40
19	Constrained initial histories and the explanation of coarse-grained entropy. pp. 631–634	6–8, 14, 24–25	11–12, 34–37
20	Incompatibility of the initial Weyl condition with symmetries reversing temporal orientation. pp. 637	1, 14–15, 20, 24	19, 34, 40, 43

Grounds for the Results

Table 228: Penrose 1979 Conditions for the Results

Re- sult Index	Conditions and Domain of Applicability
1	The result uses the CP asymmetry and CPT argument cited by Penrose. The source leaves the relation between this microscopic effect and macroscopic arrows unresolved.
2	For unitary U , forward and inverse propagation amplitudes have equal squared moduli; $P\psi = \lambda\psi$ implies $(UPU^{-1})(U\psi) = \lambda U\psi$. Uniqueness of the eigenstate uses a simple eigenvalue. Penrose separately discusses the physical interpretation of state reduction.

Re- sult Index	Conditions and Domain of Applicability
3	The source statistical argument supplies asymmetric boundary conditions to time-symmetric laws. Entropy assigned to a macrocell is distinguished from microscopic correlations within it.
4	The formal reversal correspondence uses time-symmetric radiation laws. The explanation of the physical retarded direction additionally uses a low-entropy past and incoming-correlation conditions.
5	Penrose distinguishes the ease of prediction and retrodiction from the fixity of the past. The entropy connection concerns the aspect in which future correlations admit past causes.
6	Using a spherically symmetric model with ordinary hyperbolic evolution, Penrose compares a recollapsing universe with an observer inside a large black hole and argues that expansion alone does not explain a low-entropy past. The nonspherical extension assumes sufficiently strong cosmic censorship.
7	The result concerns a collapse solution of classical general relativity with time-symmetric equations of state and the opposite choice of time orientation.
8	Where the Bekenstein–Hawking relation and classical area law apply, nondecreasing black-hole area corresponds to nondecreasing entropy. In the white-hole process obtained by reversing the black-hole process, area and entropy are nonincreasing.
9	The result concerns the historical model estimates in the source. The background-cooling wait assumes indefinite expansion; container size and total mass-energy distinguish local maxima, global maxima and fluctuation-driven transitions.
10	Accepting equality of stationary exterior geometries, Penrose argues from differences in causal geometry and matter processes for transient histories involving classical formation and Hawking evaporation.
11	Penrose distinguishes thermalization of matter and radiation from the geometrical degrees of freedom available to gravitational clumping.
12	The result concerns the curvature-ratio limits in the cigar and pancake models and Penrose’s qualitative interpretation of gravitational clumping.
13	The models use nonequilibrium matter processes with bulk viscosity during expansion and recollapse. Vanishing Weyl curvature alone does not fix the total entropy.
14	Strong causality is assumed. Infinity-type classification requires at least one infinite-proper-length curve among the curves generating the same TIP or TIF.
15	Under the source assumption of strong causality, the exclusion of naked ideal endpoints includes points at infinity.
16	Consequences assuming Penrose’s strong-censorship conjecture are distinguished from the perturbative arguments for its plausibility. An interior product structure does not generally identify the topology of the ideal boundaries with that of a Cauchy hypersurface.

Re- sult Index	Conditions and Domain of Applicability
17	Penrose’s initial Weyl condition is combined with nearly thermal initial matter and radiation. Detailed allowed density and velocity fluctuations require particle physics.
18	The Weyl condition applies to initial ideal boundaries, including white-hole singularities. The same restriction is not imposed on final singularities of ordinary collapse.
19	Penrose uses Hamiltonian evolution, specified coarse-graining, randomness within the initial restriction and no additional final constraint. Preserved fine-grained measure is distinguished from the volumes of the time-indexed extended macrocells.
20	The laws require initially vanishing Weyl and allow collapse with nonzero Weyl. Penrose’s argument, using reversal of temporal orientation and preservation of Weyl zero versus nonzero, gives incompatibility of these laws with T, PT, CT and CPT symmetry.

Results and Evidence-Effect Domain Neutral-K-meson asymmetry rests on existing experiments cited by Penrose. Quantum Result 2, curvature-model Result 12 and causal-boundary Results 14–15 use their respective stated mathematical conditions. Results 4–6 on radiation, psychological time and cosmic expansion, and Result 10 opposing identification of black and white holes, are Penrose’s physical arguments.

Results 17–20 concern consequences of imposing the initial Weyl condition. Penrose uses matter thermal equilibrium, randomness within the initial restriction and absence of an additional final constraint in the respective arguments. The source supplies no general integral functional relating gravitational entropy to Weyl curvature. Tolman-model Result 13 gives a concrete case of entropy increase through bulk viscosity while Weyl curvature remains zero.

Generation Judgment The LCTR complete-generation count is zero, the source-evidence-conditioned local-reach count is zero, and 25 primitive-time roots remain.

Result Dependence All 20 results depend on primitive-time roots. The minimal-support families give 95 root–result dependence pairs, of which 56 are necessary.

Placement of the Grounding Obligation Penrose’s arguments use time orientation, quantum-evolution times, microscopic histories, entropy boundaries, causal curves, proper elapsed quantities, collapse processes, and initial and final conditions. The 127 law-time uses in the claim table and the 11 transfers supply no conditions generating physical time from non-temporal physical inputs before those time contents are used. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The source discusses quantum, thermodynamic, gravitational and causal-boundary relations under the respective models and hypotheses, and refers to existing experiments. Actual failure of specific physical-time generation conditions, and resulting changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 95 root–result dependence pairs. Actual reach from the failure location of a specific generation condition remains undetermined.

Result-Preservation Domain The quantum algebraic relations use conditions such as unitarity; the radiation and psychological-time arguments use conditions including a low-entropy past; and classical black-hole/white-hole reversal and area laws use their respective classical models. Causal-boundary equivalences use strong causality, while the initial Weyl consequences use that restriction and the respective additional assumptions. Realizing these inputs through generated temporal representations preserving their relations preserves the 20 results in their respective domains.

Repair Scope If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.324 (Penrose 1979 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \text{AllRootDepIdx}(25, 20).$$

$R \in \text{Ref}^{\text{audit, rootdep}}$ and $R \notin \text{Ref}^{\text{audit, rootunreach}}$.

Proof. The boundary has 90 nodes—45 claims, 25 time contents and 20 results—and 31 edges—11 transfers and 20 result dependencies. Closure from Initial Claims 1–3 remains those 3 nodes. Even with all 45 claims supplied, no time content or result is reached, because every transfer and result dependency has a time input. Supplying all 25 time contents reaches all 20 results. $\square$

Proposition 8.325 (Penrose 1979 Minimal Primitive-Time-Root Support). The table gives minimal-support statistics as the number of members, minimum cardinality and maximum cardinality. There are 50 members in total.

Table 229: Penrose 1979 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(1; 2, 2)
2	(1; 3, 3)
3	(1; 3, 3)
4	(1; 4, 4)
5	(1; 4, 4)
6	(6; 6, 7)
7	(1; 3, 3)
8	(2; 3, 4)
9	(12; 5, 8)
10	(2; 4, 4)

Result Index	Support Statistics
11	(1; 4, 4)
12	(1; 2, 2)
13	(6; 3, 5)
14	(1; 2, 2)
15	(1; 1, 1)
16	(1; 4, 4)
17	(2; 4, 4)
18	(4; 4, 5)
19	(1; 4, 4)
20	(4; 5, 5)

Proof. Start from each time content's singleton seed and apply Definition 7.46 to the 11 transfers. After stabilization, combine the supports of each result's time premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Proposition 8.326 (Penrose 1979 Audit Conclusion). Complete generation and local reach are both zero; all 25 time contents remain primitive-time roots. All 20 results depend on primitive-time roots.

Proof. This follows from the reachable sets in Proposition 8.324 and the minimal-support families in Proposition 8.325. $\square$

Proposition 8.327 (Penrose 1979 Audit-Boundary Cardinalities).

$$\text{BoundaryCount}_R^{\text{audit}} = (90, 31).$$

Proof. The 45 claims, 25 time contents and 20 results give 90 nodes; the 11 transfers and 20 result dependencies give 31 edges. $\square$

Individual Statements Statement Evidence 1

Verbatim Original (English):

The local physical laws we know and understand are all symmetrical in time.
Yet on a macroscopic level, time-asymmetry is manifest.

Original-source location: Penrose, *Singularities and Time-Asymmetry*, p. 582; Section 12.2.
Minimal context: Immediately after contrasting time-symmetric local laws with macroscopic asymmetry, Penrose mentions the neutral-K-meson exception and the unresolved question of quantum observation. He then lists seven arrows of time, or possible arrows.
Attribution: Roger Penrose.

Statement Evidence 2

Verbatim Original (English):

Indeed, these laws are even self-limiting, as the singularity theorems^{6,7,8} seem to show. But I do not feel that this is a cause for pessimism. There is a need for new laws in any case; while, in my opinion, the presence and the apparent structure of spacetime singularities contain the key to the solution to one of the long-standing mysteries of physics: the origin of the arrow of time.

Original-source location: Penrose, *Singularities and Time-Asymmetry*, p. 581; Section 12.1.
 Minimal context: Penrose gives his assessment of the singularity theorems and his own view that singular structure holds the key to the origin of the arrow of time. Attribution: Roger Penrose.

Statement Evidence 3

Verbatim Original (English):

According to section 12.2.6, some low-entropy assumption does need to be imposed on the big bang; that is, the mere fact that the universe expands away from a singularity is in no way sufficient.

Original-source location: Penrose, *Singularities and Time-Asymmetry*, p. 611; Section 12.3.
 Minimal context: Following the expansion/recollapse comparison in Section 12.2.6, Penrose states that explaining thermodynamic direction requires a low-entropy past condition in addition to the fact of expansion. Attribution: Roger Penrose.

Common Evidence with Specified Domain of Application

Target State-ment	Object	Predicate	Explicit Domain
S24-Q001	Time symmetry of local physical laws and macroscopic time asymmetry	Penrose contrasts the two. The immediate context identifies the neutral-K-meson exception and the question of quantum observation.	The opening contrast in Section 12.2 and the introduction of seven arrows or possible arrows.
S24-Q002	Limits of known laws, singular structure and the origin of the arrow of time	Penrose assesses the singularity theorems as apparently indicating limits of known laws and states his own view that singular structure holds the key to the origin of the arrow of time.	Penrose's assessment and research proposal in Section 12.1.
S24-Q003	Cosmic expansion and the low-entropy past condition needed to explain thermodynamic direction	Penrose states that expansion away from a singularity is insufficient by itself to explain thermodynamic direction and that a low-entropy past condition is required.	The expansion/recollapse argument in Section 12.2.6, invoked at the beginning of Section 12.3.

Individual Audits

First Statement Penrose contrasts time-symmetric local laws with macroscopic asymmetry and introduces seven arrows or candidates. Penrose draws the comparison from known laws, neutral-K-meson experiments and macroscopic phenomena, and explicitly identifies the K-meson exception and unresolved quantum-observation question in that context. The corresponding formal evidence comprises Time Contents 1, 3–5, 7–13 and 15; Claims 6–8, 10–11, 15–19 and 24; and Results 1–7.

Second Statement Penrose states his own assessment of the singularity theorems and his research proposal seeking the origin of the arrow of time in singular structure. His assessment of application limits rests on the classical spacetime theorems he cites, while his view of the origin rests on a theoretical interpretation of singular structure. The corresponding formal evidence comprises Time Contents 2 and 14 and Claim 5.

Third Statement Penrose argues that a low-entropy past condition is required in addition to cosmic expansion. His grounds are the comparison of a recollapsing universe, an observer inside a black hole and thermodynamic direction in Section 12.2.6, and its synthesis at the beginning of Section 12.3. The corresponding formal evidence comprises Time Contents 1–2, 6–8 and 13–15; Claims 4, 11–13, 18–19 and 24; and Results 3 and 6.

8.19.25 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 25.

Source Name Hartle–Hawking 1983

Bibliographic Identification J. B. Hartle and S. W. Hawking, *Wave Function of the Universe*, *Physical Review D* 28, pp. 2960–2975 (1983) is denoted by $R = \text{Ref}_{\text{HH83WFU}}$. The audit covers the abstract, Sections I–IX, Eqs. (1.1)–(8.3) and Figs. 1–4.

Audit-Scope Identifier 25-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Original-Source Location Identifier 25-L1.

Original-Source Location Content The tables specify source pages for the time contents, claims, transfers and results.

Formal Objects The audit uses 21 time contents, 54 claims, 11 transfers and 24 results. The claim table specifies the geometries, matter fields, actions, measures, contours, boundary conditions and relations. Hartle and Hawking start from the Euclidean remote-past construction of ground states in ordinary quantum systems and propose defining the closed-universe ground state by integrating over compact Euclidean geometries with the prescribed three-dimensional boundary as their only boundary, together with regular matter fields. Given their claim that a closed universe has no independent time standard or natural definition of total energy, they associate the ground state with minimal excitation and high symmetry (pp. 2960–2962, 2965–2966).

In the amplitude between two boundaries, specifying their three-geometries and field values leaves proper time dependent on the four-geometries being summed over. Hartle and Hawking state that an amplitude specifying proper time requires intermediate observations that determine its value (pp. 2961–2962).

Following the general prescription, Hartle and Hawking consider a spatially closed, homogeneous isotropic geometry with positive cosmological constant and one homogeneous massless conformal field. The model’s Euclidean boundary conditions yield a Gaussian matter factor; its geometric factor is studied through the semiclassical approximation and

the Wheeler–DeWitt equation. They associate the two large-radius phases with contracting and expanding branches of Lorentzian de Sitter space, and discuss matter excitations, tunneling probabilities and correlations between multiple boundaries (pp. 2967–2974).

In their conclusion, Hartle and Hawking state that the zero-volume amplitude is finite and nonzero and suggest interpreting it as the possibility of continuation through a singularity to another expansion period. Under that interpretation, classical time would not permit saying that expansion occurred after contraction. They state that the massless conformal-field ground state does not represent the observed universe and discuss the possibility that a ground state including a massive field could do so (pp. 2974–2975).

Assumptions The general functional integral uses specified boundary geometries, regular fields, an action, a measure and an integration contour. Hartle and Hawking condition the construction on convergence of the non-positive-definite conformal direction along a suitable complex contour. The canonical description uses lapse, shift, canonical momenta and operator ordering; Hermiticity uses a domain, measure and boundary conditions. The subsequent conformal model has positive cosmological constant. Semiclassical evaluation uses specified saddles and contours, while the resonance and tunneling approximations use a small positive cosmological constant and low excitation.

Non-temporal Physical Inputs Generation-initial inputs are the intrinsic geometry, topology and spatial configuration of the three-dimensional boundary, boundary matter-field values, observed-object identification and non-temporal observation/record values in Claims 1–3. Temporal boundary ordering, histories, evolution variables and proper times appear in the time-content table and the uses assigned to each claim.

Time Contents

Definition 8.328 (Hartle–Hawking 1983 Time Contents). Let $n_R = 21$. The table defines the time contents and their source locations.

Table 231: Hartle–Hawking 1983 Time Contents

i	Time Content	Source Location
1	Time and evolution parameter t in ordinary quantum mechanics and field theory	pp. 2960, 2961
2	Preparation, observation and propagator-endpoint ordering for a particle	pp. 2960
3	Euclidean time τ and projection from the remote past onto an ordinary quantum ground state	pp. 2960, 2961
4	Past, future and spacelike-boundary ordering in Lorentzian spacetime histories	pp. 2961, 2962, 2964
5	Proper time between boundary surfaces as dependent on the four-geometry	pp. 2961, 2962
6	Placement and ordering of intermediate observations needed to specify proper time	pp. 2962
7	Proper-time separation specified conditional on intermediate observations	pp. 2962

i	Time Content	Source Location
8	Time coordinate, lapse and temporal boundary displacement in the 3+1 decomposition	pp. 2963, 2965
9	Time derivative of the three-metric in Lorentzian extrinsic curvature	pp. 2963, 2964, 2965
10	Incoming and outgoing asymptotic ordering in the scattering comparison	pp. 2965
11	Euclidean history parameter tau and fixed-tau boundary in the cosmological model	pp. 2965, 2968, 2969, 2970, 2971
12	Conformal parameter eta and its half-line in the Euclidean cosmological model	pp. 2968, 2969, 2970, 2971
13	Lorentzian time and scale-factor and matter-field histories in the minisuperspace model	pp. 2967, 2972, 2973
14	WKB phases associated with Lorentzian evolution	pp. 2970, 2971, 2972
15	Time t , scale factor $a(t)$ and past/future limits of the de Sitter solution	pp. 2971, 2972
16	Conformal time eta of the Lorentzian de Sitter solution	pp. 2972
17	Contracting and expanding de Sitter branches and their temporal orientation	pp. 2972, 2973, 2974, 2975
18	Change of the mean field along the scale factor in a branch-specified wave packet	pp. 2972
19	Expansion, turning, recollapse and unbounded branches of closed-universe solutions with radiation	pp. 2972, 2973, 2974
20	Minimum-to-maximum-radius traversal and ordering of a tunneling event	pp. 2973
21	Age of the universe expressed in Planck-time units	pp. 2973

Definition 8.329 (Hartle–Hawking 1983 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(54)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([3], [54])$. The table gives each claim, its law-time uses and source locations.

Table 232: Hartle–Hawking 1983 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Data for intrinsic three-dimensional boundary geometry, topology and spatial configuration	$\emptyset$	pp. 2961, 2962, 2965, 2967, 2973
2	Matter-field values and nontemporal configuration data on the boundary	$\emptyset$	pp. 2961, 2962, 2967
3	Identification of observed objects and nontemporal values of observables and records	$\emptyset$	pp. 2962, 2963, 2973
4	The gravitational action uses the convention $\ell^2 = 16\pi G$ and $\hbar = c = 1$ and includes the cosmological constant Λ .	$\emptyset$	pp. 2961, 2963, 2965

p	Claim	i	Source Location
5	The particle wave function and propagator superpose amplitudes for histories $x(t)$ consistent with preparation and observation at specified times.	$1^{\text{PS}}, 2^{\text{OB}}$	pp. 2960
6	With the lowest eigenvalue of a time-independent Hamiltonian normalized to zero, the Euclidean remote-past limit under $t = -i\tau$ projects onto the ground state of an ordinary quantum system.	$1^{\text{S}}, 3^{\text{BS}}$	pp. 2960, 2961
7	The flat-spacetime field wave functional takes the field configuration on an equal-time surface as its argument; its ground-state integral matches fields at $\tau < 0$ to prescribed values at $\tau = 0$.	$1^{\text{S}}, 3^{\text{BS}}$	pp. 2961
8	The closed-universe wave functional $\Psi[h, \phi]$ superposes spacetime and field-history amplitudes over a specified class of histories matching the boundary three-metric and field values.	4^{B}	pp. 2961, 2962, 2963
9	For the amplitude between two spacelike boundary surfaces, proper time depends on the four-geometries being integrated over and is not fixed to a single value by the boundary data alone.	$4^{\text{B}}, 5^{\text{I}}$	pp. 2961, 2962
10	Hartle and Hawking state that an amplitude specifying the proper time between the boundary surfaces requires additional intermediate observations that determine that time.	$4^{\text{B}}, 5^{\text{I}}, 6^{\text{O}}, 7^{\text{IO}}$	pp. 2962
11	In a specified quantum state of the universe, $\Psi[h, \phi]$ describes correlations between observables such as spatial volume and matter-field values.	$\emptyset$	pp. 2963
12	Hartle and Hawking state that a closed universe has no independent time standard or natural definition of total energy, and propose a ground state corresponding to minimum excitation and high symmetry.	$\emptyset$	pp. 2961
13	The 3+1 decomposition uses time coordinate t and lapse N , and expresses Lorentzian extrinsic curvature through the time derivative of the three-metric and spatial derivatives of the shift.	$8^{\text{S}}, 9^{\text{S}}$	pp. 2963

p	Claim	i	Source Location
14	Temporal displacement of the boundary through lapse variation gives the Hamiltonian constraint. The functional integral requires a translation-invariant measure or appropriate regularization of additional terms.	8^S	pp. 2963
15	Expressing extrinsic curvature in canonical momenta and choosing an operator ordering gives the Wheeler–DeWitt equation as the operator form of the Hamiltonian constraint.	$8^S, 9^S$	pp. 2963
16	The operator forms of the other canonical constraints express gauge invariance of the wave functional.	$\emptyset$	pp. 2963
17	The DeWitt superspace metric has one negative conformal direction. The volume density is nonnegative for a positive-definite or degenerate three-metric, and the wave functional is set to zero in the negative-volume-density region.	$\emptyset$	pp. 2963, 2964
18	The representation transform based on conjugacy of volume density and Lorentzian extrinsic curvature relates the zero condition at negative volume to analyticity and contour conditions in the K plane. A conformal matter field is rescaled to preserve this conjugacy.	9^S	pp. 2964
19	The functional inner product of Ψ and Ψ' is a sum over spacetime histories in the class for Ψ to the past of the boundary and in the time-reversed class for Ψ' to its future.	4^B	pp. 2964
20	Hartle and Hawking integrate their functional inner product over all three-geometries, distinguishing it from the integral over a hypersurface in geometry space envisaged for physical states. Hermiticity of the Wheeler–DeWitt operator uses a domain, measure and boundary conditions that annul the boundary form.	$\emptyset$	pp. 2964, 2971, 2972
21	Hartle and Hawking propose to define the ground-state wave functional by integrating over compact Euclidean four-geometries whose sole boundary has the prescribed three-geometry, with regular matter fields on those geometries.	$\emptyset$	pp. 2961, 2965

p	Claim	i	Source Location
22	Hartle and Hawking interpret this one-boundary integral as an amplitude for the prescribed three-geometry to arise from a zero three-geometry.	$\emptyset$	pp. 2961
23	Hartle and Hawking state that regular Euclidean solutions of the field equations for positive Λ are compact, and also give noncompact symmetric solutions for $\Lambda = 0$ and $\Lambda < 0$. The radius of the spherical metric (3.2) is $\sigma/H = \sqrt{3/\Lambda}$. For the same compact-integral prescription they state that the wave function diverges at large volume and is not normalizable when $\Lambda \leq 0$; they restrict subsequent analysis to $\Lambda > 0$, treating $\Lambda = 0$ as its limit.	$\emptyset$	pp. 2965
24	With zero shift and the source normal conventions, the rotation $t = i\tau$ gives $K_L = -iK_E$ and relates the fixed-K boundary action to the contour transform.	$8^S, 9^S, 11^{BS}$	pp. 2965, 2966
25	Hartle and Hawking distinguish boundary conditions for scattering states incoming from and outgoing to infinity from the compact-geometry prescription describing observations within a universe.	10^S	pp. 2965
26	The Euclidean gravitational action is not positive definite in the conformal direction. Hartle and Hawking propose choosing a conformal-factor contour to obtain convergence and construct the functional integral conditional on that convergence.	$\emptyset$	pp. 2966
27	The semiclassical contribution of a single suitable stationary point is given by $\exp(-I_{cl})$ and a determinant factor from fluctuation operators. For the pure-gravity spherical saddle, the normalization exponent is $N^2 = \exp[-2/(3H^2)]$.	$\emptyset$	pp. 2966
28	When several stationary points exist, the contour determines the contributing saddles; for a real wave function, saddles with complex-conjugate actions contribute in pairs.	$\emptyset$	pp. 2966, 2970, 2971
29	Fluctuation determinants enter the full small-volume behavior of the wave function, and the source does not evaluate their general values.	$\emptyset$	pp. 2966, 2972

p	Claim	i	Source Location
30	Hartle and Hawking associate slowly damped oscillations of the large-volume wave function with a universe that expands without limit.	14 ^{PS} , 17 ^S	pp. 2967, 2972
31	Restricting to a homogeneous, isotropic, spatially closed universe with positive Λ and one homogeneous massless conformal scalar field gives the Lorentzian action and Hamiltonian constraint for $a(t)$, $N(t)$ and $\phi(t)$.	13 ^S	pp. 2967
32	For specified factor ordering p and matter-energy constant ε_0 , equation (5.8) separates into matter-oscillator states. The geometric equation (5.11) uses $E = n + 1/2 - \varepsilon_0$.	13 ^S	pp. 2967
33	The Euclidean model uses $d\eta = d\tau/a$. Regularity of a and χ on $\eta \in (-\infty, 0)$ and their prescribed values at $\eta = 0$ specify the geometric and matter boundary conditions.	11 ^{BS} , 12 ^{BS}	pp. 2968, 2969
34	The model action with $k = \sigma K/9$ fixed satisfies $I^k = ka^3 + I$; the inverse transform to fixed a uses $\exp(ka^3)$ and the specified complex contour.	11 ^{BS}	pp. 2968, 2969
35	In the conformal model, regularity as $\eta \rightarrow -\infty$ and the prescribed field value at $\eta = 0$ on $\eta \in (-\infty, 0)$ separate the matter action and boundary conditions from the geometric sector, giving the matter ground-state factor $\exp(-\chi^2/2)$.	12 ^{BS}	pp. 2968, 2969
36	The actions of the two spherical caps are $I_{\pm} = -[1 \pm (1 - H^2 a^2)^{3/2}]/(3H^2)$. For $Ha < 1$ the specified contour selects the cap smaller than a hemisphere; for $Ha > 1$ it passes through conjugate complex saddles.	11 ^{BS} , 12 ^{BS}	pp. 2969, 2970, 2971
37	The conjugate-saddle sum gives a real oscillatory function at large radius. Its leading phase is $S(a) = (H^2 a^2 - 1)^{3/2}/(3H^2)$, and the real sum of the two phases has zero net current.	$\emptyset$	pp. 2970, 2971
38	The Lorentzian de Sitter solution satisfies $a(t) = \cosh(Ht)/H$ and $dt = a d\eta$; a distribution of spheres uniform in conformal time η has density proportional to $1/[a\sqrt{H^2 a^2 - 1}]$ in a .	15 ^S , 16 ^{IQ}	pp. 2972
39	Through their correspondence with the Lorentzian de Sitter solution, WKB phases of opposite signs represent contracting and expanding branches.	14 ^S , 15 ^S , 17 ^S	pp. 2972

p	Claim	i	Source Location
40	The model norm is $\int da a^p \psi(a) ^2$. Large-radius solutions $c_n \sim a^{-p/2-1} \exp(\pm i H a^3/3)$ have an integrable weighted density at infinity; small-radius integrability also depends on the boundary condition.	$\emptyset$	pp. 2967, 2968, 2971, 2972
41	Figure 4 in the source is a numerical illustration with $\psi = 0$ at zero volume. In their conclusion, Hartle and Hawking instead describe the zero-volume amplitude as finite and nonzero.	$\emptyset$	pp. 2971, 2972, 2974
42	For a selected asymptotic branch, superposing states of different n to form a wave packet localized near a mean field makes that mean field increase in the direction from large to small a .	$14^S, 17^S, 18^S$	pp. 2972
43	For $\lambda > 0$, $V(a) = (a^2 - \lambda a^4)/2$ has a barrier of height $1/(8\lambda)$. At $\lambda = 0$ and $p > -1$, selecting solutions that have a power series in a^2 at the origin and are square-integrable with weight a^p at infinity gives $E_N = 2N + (p+1)/2$, $N = 0, 1, \dots$. For small positive λ , metastable resonances occur near levels sufficiently far below the barrier.	$\emptyset$	pp. 2972, 2973
44	For $\lambda > 0$ and $0 < E < 1/(8\lambda)$, the classical model has bounded orbits that expand and recollapse and unbounded de Sitter-type orbits with a minimum radius.	$13^S, 19^S$	pp. 2973
45	The tunneling probability per minimum-to-maximum-radius traversal is $P \sim \exp(-B)$. For kinetic term $-\partial_a^2/2$, $B = 2 \int \sqrt{2(V-E)} da$; its leading value at low energy and small λ is $2/(3\lambda)$.	$19^S, 20^{PB}$	pp. 2973
46	Taking the age of the universe as about 10^{60} Planck times, Hartle and Hawking estimate $a_{\max}^2$ and the required excitation number n to be at least of order 10^{120} in a radiation-dominated model and evaluate a tunneling probability for a small cosmological constant.	$19^S, 20^P, 21^{IQ}$	pp. 2973
47	Integrating over compact four-geometries with N disconnected three-dimensional boundaries defines an N -argument wave functional symmetric under permutations of the boundaries.	$\emptyset$	pp. 2973

<i>p</i>	Claim	<i>i</i>	Source Location
48	The Wheeler–DeWitt equation in each boundary argument alone does not determine correlations between distinct boundary geometries; those correlations depend on the specified functional integral.	∅	pp. 2973
49	For two spherical boundaries, disconnected fillings give a product contribution and a connected saddle gives a correlated contribution. The fluctuation determinant of the connected contribution does not factorize and is unevaluated in the source.	∅	pp. 2974
50	In the source exponential estimate of the connected correlation, $\exp[-1/(3H^2)]$ suppresses topological complexity for small positive H^2 .	∅	pp. 2974
51	Hartle and Hawking suggest that a finite, nonzero zero-volume amplitude might be interpreted as allowing the universe to continue through a singularity into another expansion period.	∅	pp. 2974
52	If the zero-volume amplitude is interpreted as passage through a singularity, Hartle and Hawking state that the classical concept of time would break down and the expansion could not be said to occur after the contraction.	17 ^S , 19 ^S	pp. 2974, 2975
53	Hartle and Hawking distinguish the massless conformal-field ground state, whose matter wave function does not oscillate, from the state of the observed universe and discuss the possibility that a ground state with a massive field could represent the observed universe.	∅	pp. 2972, 2975
54	If a ground state including a massive field represents the observed universe, Hartle and Hawking state that the condition of having no boundary would solve the problem of its initial boundary conditions.	∅	pp. 2975

Law-time uses comprise 51 claim/time-content pairs. Superscript codes indicate the use roles in the following table.

Table 233: Hartle–Hawking 1983 Time-Use Roles

Code	Use Role
I	Interval

Code	Use Role
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causality
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 234: Hartle–Hawking 1983 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 8–9, 11, 13–16, 18	dependency	$\emptyset$
2, 6	member	instant, order_relation
3, 12	member	endpoint, orientation
4	member	endpoint, order_relation, orientation
5, 7, 21	separate	$\emptyset$
10	lexical	$\emptyset$
17	member	orientation, phase_boundary
19	member	endpoint, turning_position, phase_boundary
20	member	order_relation, transition_position

Claims 1–3 form the initial set for physical generation, and all 54 claims form the source formal-evidence environment. Source-specific obligation atoms comprise 54 claims, 21 time contents, 11 transfers and 24 results, totaling 110.

Formal Input Elements Hartle and Hawking use $\ell^2 = 16\pi G$ and $\hbar = c = 1$. The spherical metric in the source’s Eq. (3.2) has radius $\sigma/H = \sqrt{3/\Lambda}$.

Dependency Inputs The table specifies each transfer’s time inputs, time outputs, claim support and source location.

Definition 8.330 (Hartle–Hawking 1983 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [11]$.

Table 235: Hartle–Hawking 1983 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1	3	6–7	Euclidean rotation of time for an ordinary quantum system. pp. 2960, 2961
2	8	9	1, 13	Correspondence from the metric time derivative to Lorentzian extrinsic curvature. pp. 2963
3	8–9	11	18, 24	Euclidean continuation of the Lorentzian boundary representation. pp. 2965, 2966
4	11	12	33	Conformal parameterization of Euclidean histories. pp. 2968, 2969
5	8	13	1–2, 13, 31	Restriction of the 3+1 temporal representation to the homogeneous isotropic model. pp. 2963, 2967
6	13	15	4, 23, 31, 38	de Sitter history in the model vacuum solution. pp. 2965, 2967, 2972
7	15	16	38	Transformation of the de Sitter history to conformal time. pp. 2972
8	11–12, 15	14, 17	27–28, 34, 36–39	Correspondence between saddle phases and Lorentzian evolution. pp. 2969, 2970, 2971, 2972
9	14–15	17	39	Association of opposite WKB phases with contracting and expanding branches. pp. 2972
10	14, 17	18	2, 42	Change of the wave-packet mean field on a specified branch. pp. 2972
11	13	19	2, 31–32, 43–44	Classical orbit classification in the radiation and cosmological-constant model. pp. 2972, 2973

Grounding Conditions

1. Use $t = -i\tau$ in the source particle and flat-spacetime field examples. Ground-state projection uses a time-independent Hamiltonian, normalization of its lowest eigenvalue and remote-past boundary conditions.
2. Evaluate K_{ij} in the 3+1 decomposition of the specified Lorentzian metric using the lapse, shift, normal convention and time-dependent three-metric.
3. Continue with zero shift, $t = i\tau$ and the convention $K_L = -iK_E$, matching the boundary action and integration contour.
4. Integrate $d\eta = d\tau/a$ where $a > 0$ in the model interior, using $\eta \in (-\infty, 0)$ for the stated regular compact-geometry boundary conditions.
5. Restrict to a spatially closed homogeneous isotropic geometry, one homogeneous conformal field and positive Λ , using $N(t)$, $a(t)$ and $\phi(t)$.

6. Use the $N = 1$ time coordinate of the maximally symmetric vacuum solution with positive Λ and $a(t) = \cosh(Ht)/H$.
7. Use $dt = a d\eta$ along the positive-scale-factor de Sitter history. Conversion to a distribution in a uses $|d\eta/da|$ on each contracting or expanding branch.
8. Use the specified contour and large-radius semiclassical approximation in the positive- Λ conformal model, relating the conjugate-saddle phases to the classical de Sitter history.
9. Interpret each phase through the Lorentzian de Sitter solution and the sign of extrinsic curvature. The unseparated real two-phase sum has zero net current.
10. On the selected asymptotic branch, superpose different matter-oscillator numbers n to form a wave packet localized near its mean field; that mean increases from large to small a .
11. For $V(a) = (a^2 - \lambda a^4)/2$ with $\lambda > 0$ and $0 < E < 1/(8\lambda)$, distinguish bounded expansion–recollapse orbits from unbounded orbits with a minimum radius.

Transfer 5 uses the time-coordinate and lapse relation in Claim 13; Transfers 6 and 8 use $a(t) = \cosh(Ht)/H$ from Claim 38. The conformal-time and sphere-density relations in Claim 38 correspond to Transfer 7 and Result 14.

Results

Definition 8.331 (Hartle–Hawking 1983 Results and Dependencies). Let $\text{ldx}_R^{\text{audit,result}} = [24]$. The table gives each result, its time premises, claim support and source locations.

Table 236: Hartle–Hawking 1983 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Ordinary quantum wave functions, propagators and ground-state projection. pp. 2960, 2961	1–3	5–7
2	Closed-universe wave functional on boundary geometry and field values. pp. 2961, 2962, 2963	4	1–2, 8, 11–12
3	Hamiltonian constraint and Wheeler–DeWitt equation. pp. 2963	8–9	4, 13–16
4	Superspace signature, volume boundary and transformation to the K representation. pp. 2963, 2964	9	17–18
5	History-sum inner product and boundary conditions for Hermiticity. pp. 2964	4	19–20
6	Hartle–Hawking proposal for a compact one-boundary wave functional. pp. 2961, 2965, 2966	$\emptyset$	1–2, 21–23, 26

q	Result and Source Location	Time Premises i	Claim Support p
7	Proper time not fixed by two-boundary data and the condition of additional observations. pp. 2961, 2962	4–7	8–10
8	Correspondence between Lorentzian and Euclidean boundary representations. pp. 2965, 2966	8–9, 11	18, 24
9	Conformal-factor problem and conditional proposal for a convergent contour. pp. 2966	$\emptyset$	26
10	Semiclassical expression through stationary points and fluctuation determinants. pp. 2966, 2967	$\emptyset$	21, 23, 27–29
11	Closed-universe minisuperspace model with conformal matter. pp. 2967, 2968	11–13	4, 31–34
12	Gaussian matter ground-state factor. pp. 2968, 2969	12	35
13	Spherical-cap selection and conjugate-saddle sum at large radius. pp. 2968, 2969, 2970, 2971	11–12	27–28, 33–34, 36–37
14	de Sitter history, contracting/expanding branches and distribution of spheres. pp. 2971, 2972	14–17	37–39
15	Integrability at infinity in the model norm and small-radius boundary conditions. pp. 2967, 2968, 2971, 2972	$\emptyset$	20, 29, 40–41
16	Matter-excitation resonances and classical orbits of radiation universes. pp. 2972, 2973	13, 19	31–32, 43–44
17	Semiclassical tunneling probability per traversal. pp. 2973	19–20	43, 45
18	High-excitation and tunneling-scale estimates using the age of the universe. pp. 2973	19–21	43, 45–46
19	Multiple-boundary wave functional and connected correlation contribution. pp. 2973, 2974	$\emptyset$	1, 21, 47–50
20	Hartle–Hawking statement about zero-volume amplitude and its boundary conditions. pp. 2966, 2971, 2972, 2974	$\emptyset$	29, 41, 51
21	Limit of classical before–after ordering in the singularity-passage interpretation. pp. 2974, 2975	17, 19	41, 51–52
22	Difference between the conformal ground state and the observed universe, and the massive-field prospect. pp. 2972, 2975	$\emptyset$	53–54

q	Result and Source Location	Time Premises i	Claim Support p
23	Distinction between scattering asymptotic boundary conditions and the cosmological interior prescription. pp. 2965	10	23, 25
24	Relation between wave-packet mean field and scale factor on a specified branch. pp. 2972	14, 17–18	2, 42

Grounds for the Results

Table 237: Hartle–Hawking 1983 Conditions for the Results

Re- sult Index	Conditions and Domain of Applicability
1	In the particle and flat-spacetime field examples, use preparation/observation times, path-integral boundary conditions, a time-independent Hamiltonian and normalization of its lowest eigenvalue.
2	Use the specified spacetime/field history class and boundary matching. The arguments of Ψ are the boundary geometry and field values, with no independent external time argument.
3	Use the Lorentzian action and lapse variation, an invariant measure or regulated additional terms, canonical momenta and the chosen operator ordering.
4	Use nonnegative volume density and its conjugacy with Lorentzian extrinsic curvature. The transform imposes analyticity, contour and matter-field-rescaling conditions.
5	Use the inner product gluing past histories to time-reversed future histories, and annul the boundary form of the integral over all three-geometries.
6	Hartle and Hawking’s proposal integrates over compact Euclidean geometries with the prescribed three-dimensional boundary and regular fields. For the conformal factor of the gravitational action, the construction is conditional on convergence along a suitable complex contour.
7	The two-boundary amplitude sums over four-geometries, so the boundary data alone do not fix proper time. Hartle and Hawking state that an amplitude specifying proper time requires intermediate observations that determine that value.
8	With zero shift and the conventions $t = i\tau$ and $K_L = -iK_E$, the Lorentzian and Euclidean actions and integration contours correspond for the specified fixed boundary data.
9	The issue is the non-positive-definite conformal direction of the Euclidean gravitational action. Hartle and Hawking construct the integral conditional on convergence along a suitable complex conformal-factor contour.

Re- sult Index	Conditions and Domain of Applicability
10	Use stationary points, contours, regular boundary conditions and fluctuation operators of the specified integral. The pure-gravity sphere gives the normalization exponent; the full small-volume determinant is still required.
11	Use $\Lambda > 0$, a spatially closed homogeneous isotropic geometry, one homogeneous massless conformal field and specified p and ε_0 . The Lorentzian and Euclidean actions are related by analytic continuation, and the conformal parameter η is defined by $d\eta = d\tau/a$.
12	Obtain $\exp(-\chi^2/2)$ from the separated action and independent geometric/matter boundary conditions: regularity in the remote past of η and prescribed values at $\eta = 0$.
13	Use the positive- Λ conformal model and specified contour: select the smaller cap for $Ha < 1$ and combine conjugate saddles for $Ha > 1$. The stated phase and normalization belong to the exponential part of the semiclassical approximation.
14	Use $a(t) = \cosh(Ht)/H$, $dt = a d\eta$ and the large-radius semiclassical domain. The probability-density envelope is proportional to $1/[a\sqrt{H^2a^2 - 1}]$; for $p=0$ the amplitude envelope is proportional to its square root.
15	Use weight a^p , the specified operator domain and small-radius boundary condition. The large-radius asymptotic form gives weighted density a^{-2} .
16	For the resonance levels, use the origin condition in Claim 43, the matter-energy constant and $\lambda > 0$. The resonance approximation applies at small λ and low E ; bounded/unbounded classical classification applies for $0 < E < 1/(8\lambda)$.
17	For kinetic term $-\partial_a^2/2$, use $B = 2 \int \sqrt{2(V - E)} da$ between the two turning points $V(a) = E$. At low E and small λ , $B \sim 2/(3\lambda)$ and $P \sim \exp(-B)$.
18	Use Hartle and Hawking's age estimate of about 10^{60} Planck times, the radiation-dominated model and the corresponding small-cosmological-constant condition.
19	The integral ranges over compact Euclidean geometries with the prescribed boundaries. For nonoverlapping spherical boundaries, disconnected fillings give a product contribution and connected fillings give a correlation contribution. The fluctuation determinant for the connected contribution does not factor and is not evaluated in the source.
20	Hartle and Hawking state in their conclusion that the zero-volume amplitude is finite and nonzero. Their numerical illustration in Fig. 4 imposes the boundary condition $\psi(0) = 0$. The full small-volume amplitude depends on fluctuation factors and boundary conditions.
21	The condition is Hartle and Hawking's proposed interpretation of continuation through zero volume. Within that interpretation, classical time cannot order expansion as an event occurring after contraction.

Re- sult Index	Conditions and Domain of Applicability
22	Hartle and Hawking state that the massless conformal-field ground state, whose matter wave function does not oscillate, does not represent the observed universe. They state that, if a massive-field ground state represents the observed universe, the no-boundary condition would solve the initial-boundary-condition problem.
23	Hartle and Hawking distinguish scattering boundary conditions describing incoming and outgoing propagation at infinity from the compact-geometry prescription describing observations in the cosmological interior.
24	Use a wave packet localized near a mean field by superposing different n and select one asymptotic branch. The mean field increases in the direction of decreasing a .

Mathematical Relations in the Model The geometric energy in the source's Eq. (5.11) is $E = n + \frac{1}{2} - \varepsilon_0$; Eq. (7.1) prints $n + \frac{1}{2} + \varepsilon_0$. For $\lambda > 0$, the potential $V(a) = (a^2 - \lambda a^4)/2$ in Eq. (7.2) has maximum $1/(8\lambda)$ at $a^2 = 1/(2\lambda)$. The barrier height printed in the eighth statement is $1/(4\lambda)$. The tunneling exponent for kinetic term $-\partial_a^2/2$ is $B = 2 \int \sqrt{2(V - E)} da$, with $B \sim 2/(3\lambda)$ at low energy and small λ .

The fixed- k boundary term in Eq. (5.20) is ka^3 , whereas Eq. (6.12) prints $-ka^2$. The large-radius de Sitter sphere probability-density envelope is proportional to $[a\sqrt{H^2 a^2 - 1}]^{-1}$ in Eq. (6.26); for $p = 0$, the amplitude envelope is proportional to its square root. The inputs specified for Fig. 4—Eq. (5.11), $p = 0$, $n = 0$, $H = 1$ and $\varepsilon_0 = -1/2$ —give $E = 1$, so $2(E - V) = a^4 - a^2 + 2 = (a^2 - 1/2)^2 + 7/4 > 0$. These specified values do not yield the small-radius classically forbidden region described in the caption.

The estimate $\exp[-1/(3H^2)]$ following Eq. (8.3) becomes small for small positive H^2 . Hartle and Hawking argue that this suppression of the correlated amplitude suppresses topological complexity.

Results and Evidence-Effect Domain Result 1 concerns the ordinary quantum constructions in the source's introduction. Results 2–5 and 7–8 concern the closed-universe functional and its canonical and boundary representations. Results 6 and 9 concern Hartle and Hawking's compact-integral prescription and convergence condition; Result 10 concerns its semiclassical expression. Results 11–18 and 24 concern the action, wave functions, branches, excitations and conditional probabilities of the positive-cosmological-constant conformal model. Result 19 concerns multiple-boundary correlations; Results 20–22 concern Hartle and Hawking's statements and interpretations about zero-volume amplitude and the observed universe; Result 23 distinguishes scattering and cosmological-interior boundary prescriptions. The result table specifies each domain and source location.

Generation Judgment The LCTR complete-generation count is zero, the source-evidence-conditioned local-reach count is zero, and 21 primitive-time roots remain.

Result Dependence Of the 24 results, 17 depend on primitive-time roots. The seven Results 6, 9–10, 15, 19–20, 22 are reachable from the source formal-evidence environment.

No result is generated from the non-temporal physical inputs alone. Minimal-support families give 50 root–result dependence pairs, of which 14 are necessary.

Placement of the Grounding Obligation Hartle and Hawking use ordinary quantum evolution and preparation/observation times, boundary ordering of spacetime histories, geometry-dependent proper time, lapse displacements, Euclidean and conformal parameters, Lorentzian de Sitter histories, contracting/expanding branches and the age of the universe. The 51 law-time uses in the claim table and the 11 transfers supply no conditions generating these time contents from non-temporal physical inputs before their use. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The source proposes a closed-universe wave functional and studies constraints, boundary conditions, semiclassical solutions and probabilities in specified models. Actual failure of specific physical-time generation conditions, and resulting changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 50 root–result dependence pairs. Actual reach from the failure location of a specific generation condition to a result remains undetermined.

Result-Preservation Domain Fix the actions, measures, contours, regularity, operator ordering, boundary conditions, models and approximation domains specified in the result table. If generated temporal representations realize the inputs of the 17 time-dependent results while satisfying their relations and application conditions, those results are preserved in their respective domains. The seven evidence-reachable results retain their respective definitions, conditional relations or attributed Hartle–Hawking statements.

Repair Scope If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.332 (Hartle–Hawking 1983 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [21]; \emptyset, \{6, 9, 10, 15, 19, 20, 22\}, \\ [24] \setminus \{6, 9, 10, 15, 19, 20, 22\}, \emptyset).$$

$R \in \text{Ref}^{\text{audit, rootdep}}$ and $R \notin \text{Ref}^{\text{audit, rootunreach}}$.

Proof. The boundary has 99 nodes—54 claims, 21 time contents and 24 results—and 35 edges—11 transfers and 24 result dependencies. Closure from Initial Claims 1–3 remains those three nodes. Supplying all 54 claims as the source formal-evidence environment reaches Results 6, 9, 10, 15, 19, 20 and 22, which have no time inputs. All 11 transfers have time inputs, so no time content is reached. Adding all 21 time contents as seeds reaches all 24 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.333 (Hartle–Hawking 1983 Minimal Primitive-Time-Root Support). The table gives minimal-support statistics as the number of members, minimum cardinality and maximum cardinality. There are 49 members in total. For an evidence-reachable result, the empty set is the sole minimal temporal support.

Table 238: Hartle–Hawking 1983 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(1; 2, 2)
2	(1; 1, 1)
3	(1; 1, 1)
4	(2; 1, 1)
5	(1; 1, 1)
6	(1; 0, 0)
7	(1; 4, 4)
8	(1; 1, 1)
9	(1; 0, 0)
10	(1; 0, 0)
11	(2; 1, 2)
12	(3; 1, 1)
13	(2; 1, 1)
14	(5; 1, 2)
15	(1; 0, 0)
16	(2; 1, 1)
17	(3; 2, 2)
18	(3; 3, 3)
19	(1; 0, 0)
20	(1; 0, 0)
21	(7; 1, 3)
22	(1; 0, 0)
23	(1; 1, 1)
24	(6; 1, 2)

Proof. Start from each time content’s singleton seed and apply Definition 7.46 to the 11 transfers. After stabilization, combine the supports of each result’s time premises and discard supports properly containing another support. A result with no time premises has the empty set as its support. Proposition 7.47 gives the displayed families and counts. $\square$

Proposition 8.334 (Hartle–Hawking 1983 Audit Conclusion). Complete generation and local reach are both zero, and all 21 time contents remain primitive-time roots. The 24 results divide into seven evidence-reachable results and 17 results dependent on primitive-time roots.

Proof. This follows from the reachable sets in Proposition 8.332 and the minimal-support families in Proposition 8.333. $\square$

Individual Statements Statement Evidence 1

Minimal context: In proposing a closed-universe ground state, Hartle and Hawking associate the absence of an independent time standard with the absence of a natural definition of total energy, and characterize the ground state through minimal excitation and high symmetry.

Verbatim Original (English):

There is no natural definition of energy for a closed universe just as there is no independent standard of time.

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, p. 2961, §I.
Attribution: J. B. Hartle and S. W. Hawking.

Statement Evidence 2

Minimal context: Hartle and Hawking consider the amplitude with specified fields and geometries on two boundaries in Eq. (1.8). The proper time between the boundaries varies with the four-geometry being summed over.

Verbatim Original (English):

Here one clearly sees that one cannot specify time in these states. The proper time between the surfaces depends on the four-geometries in the sum.

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, p. 2961, §I.
Attribution: J. B. Hartle and S. W. Hawking.

Statement Evidence 3

Minimal context: The two-boundary amplitude in Eq. (2.2) sums over four-geometries. Immediately following this passage, Hartle and Hawking state that an amplitude specifying proper time requires intermediate observations that determine its value.

Verbatim Original (English):

We note again that the proper time between the two surfaces is not specified. Rather it is summed over in the sense that the separation between the surfaces depends on the four-geometry being summed over.

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, p. 2962, §II.A.
Attribution: J. B. Hartle and S. W. Hawking.

Statement Evidence 4

Minimal context: After the gravitational action in Eq. (2.4), Hartle and Hawking state the conventions $\ell^2 = 16\pi G$ and $\hbar = c = 1$.

Verbatim Original (English):

ℓ is the Planck length $(16\pi G)^{1/2}$ in the units with $\hbar = c = 1$ we use throughout.

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, p. 2963, §II.B.
Attribution: J. B. Hartle and S. W. Hawking.

Statement Evidence 5

Minimal context: Hartle and Hawking express the age of the universe in Planck-time units and estimate that $a_{\max}^2$ and the excitation number in a radiation-dominated model must be at least of order 10^{120} .

Verbatim Original (English):

We know that the age of the Universe is about 10^{60} Planck times.

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, p. 2973, §VII.
Attribution: J. B. Hartle and S. W. Hawking.

Statement Evidence 6

Minimal context: The quoted action is the action in Eq. (1.11), which defines the ground-state integral over compact Euclidean geometries.

Verbatim Original (English):

where I_E is the Euclidean action for gravity including a cosmological constant Λ .

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, p. 2961, §I.
Attribution: J. B. Hartle and S. W. Hawking.

Statement Evidence 7

Minimal context: Hartle and Hawking discuss compactness of Euclidean solutions and the sign of the cosmological constant. Their symmetric examples are spherical, flat and hyperbolic geometries for positive, zero and negative cosmological constant, respectively. Their subsequent model uses positive cosmological constant.

Verbatim Original (English):

In the case of a positive cosmological constant Λ any regular Euclidean solution of the field equations is necessarily compact. [...] If Λ is zero or negative there are noncompact solutions of the field equations.

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, p. 2965, §III.
Attribution: J. B. Hartle and S. W. Hawking.

Statement Evidence 8

Minimal context: The quotation includes the source’s stated barrier height and its interpretation of the two large-radius asymptotic solutions. The contracting/expanding association rests on the Lorentzian de Sitter solution and the signs of the asymptotic phases.

Verbatim Original (English):

The potential $V(a)$ is a barrier of height $1/(4\lambda)$. At large a , the cosmological-constant part of the potential dominates and one has solutions which are linear combinations of the oscillating functions in (5.13). As we have already seen in the analysis of the ground state, the two possibilities correspond to a de Sitter contraction and a de Sitter expansion.

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, p. 2972, §VII.
Attribution: J. B. Hartle and S. W. Hawking.

Statement Evidence 9

Minimal context: Immediately beforehand, Hartle and Hawking suggest interpreting the finite, nonzero zero-volume amplitude as the possibility of continuation through a singularity to another expansion period. Under that interpretation, they state that classical time would not order expansion as occurring after contraction.

Verbatim Original (English):

the classical concept of time would break down so that one could not say that the expansion happened after the contraction.

Original-source location: Hartle–Hawking, *Wave Function of the Universe*, pp. 2974–2975, §IX. Attribution: J. B. Hartle and S. W. Hawking.

Common Evidence with Specified Domain of Application

Target State-ment	Object	Predicate	Explicit Domain
S25-Q001	Energy definition and independent time standard for a closed universe	Hartle and Hawking state that there is no natural definition of total energy or independent time standard.	Construction of a closed-universe ground state.
S25-Q002	Proper time of a two-boundary state	Time cannot be specified in the states, and boundary proper time depends on the four-geometries in the sum.	Two-boundary states in Eq. (1.8).
S25-Q003	Sum over four-geometries and boundary separation	Proper time is not independently specified; it is summed over in the sense that boundary separation depends on the geometry being integrated over.	Two-boundary amplitude in Eq. (2.2).
S25-Q004	Planck length ℓ and unit convention	Hartle and Hawking define $\ell = (16\pi G)^{1/2}$.	Normalization of the gravitational action with $\hbar = c = 1$.
S25-Q005	Age of the universe and Planck-time units	Hartle and Hawking state that the universe is about 10^{60} Planck times old.	Excitation and tunneling-scale estimates in the radiation-dominated model.
S25-Q006	Euclidean gravitational action with cosmological constant	The gravitational action I_E includes cosmological constant Λ .	Ground-state integral in Eq. (1.11).
S25-Q007	Regular Euclidean solutions and the sign of the cosmological constant	Hartle and Hawking state that regular solutions for positive cosmological constant are compact and that noncompact solutions exist for zero or negative cosmological constant.	Euclidean geometries and cosmological-constant sign domains in Section III.
S25-Q008	Barrier height and contracting/expanding de Sitter branches	The source states a barrier height of $1/(4\lambda)$ and associates the two large-radius asymptotic solutions with contraction and expansion.	Conformal-model potential and large-radius semiclassical solutions in Section VII.

S25-Q009	Classical before-after ordering in the singularity-passage interpretation	Hartle and Hawking state that the classical concept of time would break down and expansion could not be said to occur after contraction.	Conditional on interpreting the zero-volume amplitude as possible passage through a singularity.
----------	---	--	--

Individual Audits

First Statement Hartle and Hawking state that a closed universe has no independent time standard and characterize its ground state through minimal excitation and high symmetry. The wave-functional construction in Section I provides the context. Claim 12 and Results 2 and 6 correspond to this statement.

Second and Third Statements Hartle and Hawking’s two-boundary amplitude specifies three-geometries and field values on the boundaries and sums over the intervening four-geometries. Proper time depends on each four-geometry. Equation (1.8) in the second statement and Eq. (2.2) in the third give the same relation; immediately after the third statement, the source identifies the need for intermediate observations that determine proper time. Time Contents 4–7, Claims 8–10 and Result 7 correspond to these statements.

Fourth Statement Hartle and Hawking use $\ell^2 = 16\pi G$ and $\hbar = c = 1$ to normalize the gravitational action. The statement concerns the unit convention and action normalization, corresponding to Claim 4 and Results 3 and 11.

Fifth Statement Hartle and Hawking insert an age estimate of about 10^{60} Planck times into the radiation-dominated model to estimate high matter excitation and the tunneling scale for a small cosmological constant. Time Content 21, Claim 46 and Result 18 correspond to this statement.

Sixth Statement The sixth statement concerns the gravitational action with cosmological constant in Eq. (1.11). Hartle and Hawking propose a ground state by integrating over compact geometries and regular fields. Claims 4 and 21 and Result 6 correspond to this statement.

Seventh Statement Hartle and Hawking relate the sign of the cosmological constant to compactness of regular Euclidean solutions and give spherical, flat and hyperbolic symmetric examples. Claim 23 and Result 6 correspond to this statement. The radius of the spherical metric follows from Eq. (3.2).

Eighth Statement The potential in Eq. (7.2) has $V_{\max} = 1/(8\lambda)$, differing from the quoted $1/(4\lambda)$. The two large-radius asymptotic phases represent contracting and expanding branches through their correspondence with Lorentzian de Sitter space. Time Contents 14, 15 and 17, Claims 39 and 43 and Results 14 and 16 correspond to this statement.

Ninth Statement Hartle and Hawking suggest interpreting the zero-volume amplitude as passage through a singularity and state that, under that interpretation, classical time would not order expansion as occurring after contraction. Hartle and Hawking

support that interpretation by stating that the zero-volume amplitude in their minisuperspace model is finite and nonzero. That interpretation is the condition of their claim about temporal ordering. Time Contents 17 and 19, Claims 51–52 and Results 20–21 correspond to this statement.

8.19.26 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 26.

Source Name Page–Wootters 1983

Bibliographic Identification Don N. Page and William K. Wootters, *Evolution without Evolution: Dynamics Described by Stationary Observables*, *Physical Review D* 27, pp. 2885–2892 (1983) is denoted by $R = \text{Ref}_{\text{PW83EwE}}$. The audit covers the abstract, Sections I–IV and equations (1)–(29).

Original-Source Location Identifier 26-L1.

Original-Source Location Content The tables specify source pages for the time contents, claims, transfers and results.

Audit-Scope Identifier 26-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Formal Objects The audit uses 23 time contents, 47 claims, five transfers and 18 results. The claim table and formal-input definition specify the closed systems, states, Hamiltonians, clock projections, conditional expectations and relations. Page and Wootters consider a closed system including its observers, exclude dependence on external coordinate time from its observables, and describe dynamical evolution through the dependence of other variables on internal clock readings. The spin model in Section II uses a prescribed frequency and initial product state. The noninteracting ideal clock in Section III takes a stationary global state and a clock-state orbit as premises and derives agreement between clock-conditioned expectations and Heisenberg evolution. Real clocks, the thermodynamic arrow and intrinsic geometry are discussed under their respective conditions.

Assumptions Page and Wootters’ observability argument concerns a closed system with a well-defined Hamiltonian, internal observers and a specified observable class commuting with the Hamiltonian. An asymptotically flat universe is presented as one consistent model. The spin model uses finitely many spins and a specified preparation state. The ideal-clock derivation uses noninteraction, an additive Hamiltonian, a stationary global state, a normalized clock zero state and positive conditioning weight.

Nontemporal Physical Inputs Initial inputs for generation are Claims 1–3. Hilbert spaces and spatial degrees of freedom, nontemporal state/measurement values and algebraic relations, and Hamiltonian/energy data are separated from temporal correspondence rules. The use column specifies each claim’s temporal scales, orders, evolution relations and boundaries.

Time Contents

Definition 8.335 (Page–Wootters 1983 Time Contents). Let $n_R = 23$. The table defines the time contents and source locations.

Table 240: Page–Wootters 1983 Time Contents

i	Time Content	Source Location
1	External coordinate time t and Hamiltonian translations	pp. 2885, 2886, 2889
2	Angular frequency ω of spin precession	pp. 2886
3	Spin-precession phase and orientation as functions of coordinate time	pp. 2886, 2887
4	Coordinate-time origin specifying the initial spin product state	pp. 2886
5	Coordinate-time intervals and integration measure in operator and state averages	pp. 2887, 2888, 2889
6	Clock reading assigned to the orientation of the first spin	pp. 2886, 2887, 2888
7	Ordering of the sequence of clock readings and corresponding states	pp. 2888
8	Common parameter τ of the ideal clock-state orbit and clock reading	pp. 2890
9	Dependence of conditional probabilities and expectations on spin-clock readings	pp. 2887, 2888
10	Heisenberg evolution of rest-system operators in clock parameter τ	pp. 2890
11	Clock readings using the size of the observed universe	pp. 2890
12	Clock readings using the expansion rate of the observed universe	pp. 2890
13	Clock readings using cosmic radiation density	pp. 2890
14	Clock readings using relative abundances of radioactive elements	pp. 2890
15	Clock readings using the orientation of the Earth–Sun system relative to distant stars	pp. 2890
16	Clock readings using the orientation of Earth itself relative to distant stars	pp. 2890
17	Clock readings using the angle of a pendulum to the vertical	pp. 2890
18	Clock readings using watch-hand orientation	pp. 2890
19	Reading resolution $\delta\tau$ of a real clock	pp. 2890
20	Clock-relative direction used to describe the observed increase of entropy	pp. 2890, 2891
21	Temporal intervals and intrinsic time relative to intrinsic geometrical variables	pp. 2891
22	Dependence of the time-averaged conditional expectation on a general clock reading τ	pp. 2889

i	Time Content	Source Location
23	Dependence of the conditional rest-system expectation on ideal-clock parameter τ	pp. 2890

Definition 8.336 (Page–Wootters 1983 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(47)$, with claim boundary $\text{ClaimBoundaryI}dx_R = ([3], [47])$. The table gives each claim, its law-time uses and source locations.

Table 241: Page–Wootters 1983 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Hilbert spaces, tensor-product decomposition, spin degrees of freedom and spatial-orientation data of the composite system.	$\emptyset$	pp. 2886, 2890
2	Nontemporal values and algebraic relations of state vectors, density operators, projections, observables and records.	$\emptyset$	pp. 2886, 2887, 2888, 2890
3	Data for Hamiltonians, interaction operators, energy eigenvalues and finite energy bounds.	$\emptyset$	pp. 2886, 2888, 2890
4	Page and Wootters regard the external coordinate time of the Schrödinger equation as unobservable from within a closed system.	1^S	pp. 2885, 2886
5	Page and Wootters propose energy superselection by analogy with charge superselection through coupling to a long-range field.	$\emptyset$	pp. 2885, 2886
6	The operators Page and Wootters admit as observables of a closed system are stationary operators commuting with its Hamiltonian.	1^S	pp. 2885, 2886
7	Expectations of stationary operators are unchanged when density-matrix elements between distinct energies are removed.	$\emptyset$	pp. 2888, 2889
8	Page and Wootters describe observed dynamical evolution through correlations between internal clock readings and other variables.	$6^K, 7^O, 9^P, 22^P, 23^P$	pp. 2885, 2888
9	The closed system under consideration contains all observers making observations within it and admits a quantum description with a well-defined Hamiltonian.	$\emptyset$	pp. 2885, 2886

p	Claim	i	Source Location
10	Page and Wootters assume asymptotic flatness as a consistent model admitting a Hamiltonian for the universe; they also discuss hypothetical closed systems with long-range interactions neglected.	$\emptyset$	pp. 2885, 2886
11	Under Page and Wootters' restriction on observables, phases between distinct energies changed by a coordinate-time translation of the entire closed system are unobservable.	1^S	pp. 2886
12	Page and Wootters exclude operators with explicit or implicit coordinate-time dependence from the closed-system observable class.	1^S	pp. 2886, 2891
13	The spin model uses N spin- j particles at fixed positions and a stationary laboratory frame specifying relative positions, orientations and velocities.	$\emptyset$	pp. 2886
14	For the specified Hamiltonian $H = \hbar\omega \sum_i J_{iz}$, the product state with each spin along x at $t = 0$ gives $\langle J_{ix} \pm iJ_{iy} \rangle = j \exp(\pm i\omega t)$.	$1^S, 2^Q, 3^S, 4^B$	pp. 2886
15	The orientation of the first spin specifies the clock reading; that orientation can be measured at more than one coordinate time.	$1^S, 3^S, 6^K$	pp. 2886, 2888
16	The reading probabilities of the second spin are conditional probabilities given the specified reading of the first spin.	$6^K, 9^P$	pp. 2886, 2887
17	In the finite spin model, separate coordinate-time averages of the joint projection and the clock projection give conditional probabilities by their expectation-value ratio when the averaged clock weight is positive.	$5^I, 6^K, 9^P$	pp. 2887
18	The time average of the joint projection retains correlations within equal-energy subspaces. The averaged operator is positive but is generally not idempotent.	5^I	pp. 2887
19	Page and Wootters treat the infinite ensemble required to determine conditional probabilities exactly as an idealization.	$\emptyset$	pp. 2887
20	Finite observations test statistical consistency between predicted conditional probabilities and observations at a specified confidence level.	$6^K, 9^P$	pp. 2887
21	An ordered sequence of clock readings and corresponding states of other particles describes clock-relative evolution.	$6^K, 7^O, 9^P$	pp. 2888

p	Claim	i	Source Location
22	Readout operators for different directions of the same spin generally do not commute, and repeated readings alter its precession.	$3^S, 6^K, 7^O$	pp. 2888
23	Page and Wootters propose using many clock particles prepared in the same state and copies of the system, measuring each particle or copy once.	$6^K, 7^O$	pp. 2888
24	Expectations of the specified time-averaged operators do not depend on density-matrix elements connecting distinct total energies.	5^I	pp. 2888, 2889
25	In the finite spin model, time averaging removes density-matrix elements between distinct energies and gives a stationary state preserving the specified conditional probabilities.	$5^I, 6^K, 9^P$	pp. 2888, 2889
26	The mixed two-spin state is the reduced state of a pure zero-energy state formed with another two-spin system having the reversed energy spectrum.	$\emptyset$	pp. 2888, 2889
27	For clock projection P_τ , $E(A \mid \tau) = \frac{\text{Tr}[\text{Avg}(P_\tau A P_\tau)\rho]}{\text{Tr}[\text{Avg}(P_\tau)\rho]}$ defines the conditional expectation where the averages exist and the denominator is positive.	$5^I, 22^P$	pp. 2889
28	The ideal-clock model decomposes the Hilbert space of the closed system into a tensor product of clock and rest subsystems.	$\emptyset$	pp. 2890
29	The ideal-clock model sets the dynamical interaction between the clock and the rest to zero.	$\emptyset$	pp. 2890
30	The ideal-clock Hamiltonian is $H = H_c \otimes I_r + I_c \otimes H_r$, and the global density operator ρ is stationary under it.	$\emptyset$	pp. 2889, 2890
31	A normalized clock zero state $ \phi_c(0)\rangle$ defines the orbit $ \phi_c(\tau)\rangle = \exp(-iH_c\tau/\hbar) \phi_c(0)\rangle$.	8^{KS}	pp. 2890
32	Page and Wootters assign to the clock-orbit state $ \phi_c(\tau)\rangle$ the same parameter τ as its clock reading.	8^{KS}	pp. 2890
33	With clock projection $\Pi_c(\tau) = \phi_c(\tau)\rangle\langle\phi_c(\tau) $ and positive weight $w = \text{Tr}[\rho(\Pi_c(\tau) \otimes I_r)]$, the conditional rest-system expectation is $\text{Tr}[\rho(\Pi_c(\tau) \otimes A_r)]/w$.	$8^{\text{KS}}, 23^P$	pp. 2890
34	For the additive Hamiltonian and stationary global state, cyclicity of the trace equates the clock-conditioned expectation to the expectation of $A_r(\tau) = \exp(iH_r\tau/\hbar)A_r \exp(-iH_r\tau/\hbar)$.	$8^{\text{KS}}, 23^P, 10^S$	pp. 2890

p	Claim	i	Source Location
35	Under the same conditions, the conditioning weight w is independent of τ , and $\rho_r = \text{Tr}_c[(\Pi_c(0) \otimes I_r)\rho]/w$ is the normalized relative state of the rest system.	$8^{\text{KS}}, 23^{\text{P}}$	pp. 2890
36	The real clocks listed by Page and Wootters use cosmic size, expansion rate, radiation density, relative radioactive abundances, orientations of the Earth–Sun system and Earth relative to distant stars, pendulum angle, or watch-hand orientation, and interact with the rest of the universe.	$11^{\text{K}}, 12^{\text{KQ}}, 13^{\text{K}}, 14^{\text{K}}, 15^{\text{K}}, 16^{\text{K}}, 17^{\text{K}}, 18^{\text{K}}$	pp. 2890
37	Reading a real clock requires an interaction between the clock and the rest system.	$11^{\text{K}}, 12^{\text{KQ}}, 13^{\text{K}}, 14^{\text{K}}, 15^{\text{K}}, 16^{\text{K}}, 17^{\text{K}}, 18^{\text{K}}$	pp. 2890
38	Page and Wootters argue that reading with fine temporal resolution entails a large energy exchange.	19^{KO}	pp. 2890
39	Page and Wootters argue that a finite total available energy limits clock precision.	19^{KO}	pp. 2890
40	For a good clock, the effect of the reading interaction can be practically negligible.	$11^{\text{K}}, 12^{\text{KQ}}, 13^{\text{K}}, 14^{\text{K}}, 15^{\text{K}}, 16^{\text{K}}, 17^{\text{K}}, 18^{\text{K}}$	pp. 2890
41	Page and Wootters state that idealizing actual clock behavior yields the abstract notion of a preferred coordinate time.	$1^{\text{S}}, 11^{\text{K}}, 12^{\text{KQ}}, 13^{\text{K}}, 14^{\text{K}}, 15^{\text{K}}, 16^{\text{K}}, 17^{\text{K}}, 18^{\text{K}}$	pp. 2890
42	Page and Wootters describe observable dynamics as dependence on internal clock time.	$8^{\text{KS}}, 23^{\text{P}}, 11^{\text{K}}, 12^{\text{KQ}}, 13^{\text{K}}, 14^{\text{K}}, 15^{\text{K}}, 16^{\text{K}}, 17^{\text{K}}, 18^{\text{K}}$	pp. 2890
43	Page and Wootters argue that the observed thermodynamic arrow should be formulated as the dependence of an entropy observable on clock readings.	20^{X}	pp. 2890, 2891
44	Page and Wootters extend the energy-observability argument in asymptotically flat spacetime to momentum and angular momentum. Simultaneous definite values are restricted to commuting sets.	$\emptyset$	pp. 2891
45	Page and Wootters propose expressing quantum-gravitational spacetime intervals relative to intrinsic geometrical variables and defining field distributions relative to a three-geometry rather than coordinates.	21^{I}	pp. 2891

p	Claim	i	Source Location
46	Page and Wootters argue that an observation record within a single closed system is consistent with at least one energy eigenstate.	$\emptyset$	pp. 2888, 2889
47	Page and Wootters call for defining entropy through coarse-grained properties that are actually observed.	$\emptyset$	pp. 2890, 2891

Definition 8.337 (Page–Wootters 1983 Initial Set and Evidence Environment). Claims 1–3 form the physical-generation initial set, and all 47 claims form the source formal-evidence environment.

Time Contents Time contents are given in Definition 8.335. The 94 claim/time-content use pairs have the following roles.

Table 242: Page–Wootters 1983 Time-Use Roles

Code	Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causality
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 243: Page–Wootters 1983 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–3, 6, 8–19, 22–23	dependency	$\emptyset$
4	member	instant
5	member	interval_boundary, order_relation
7	member	order_relation
20	member	orientation, order_relation
21	lexical	$\emptyset$

Formal Input Elements

Definition 8.338 (Page–Wootters 1983 Clock-Conditioned Description). Let the clock and rest Hilbert spaces, self-adjoint Hamiltonians and normalized positive global density

operator satisfy

$$\mathcal{H} = \mathcal{H}_c \otimes \mathcal{H}_r, \quad H = H_c \otimes I_r + I_c \otimes H_r, \quad \rho \geq 0, \quad \text{Tr } \rho = 1.$$

Global stationarity means $U(s)\rho U(s)^* = \rho$ for all real s , where $U(s) = \exp(-iHs/\hbar)$. A normalized clock zero state defines the clock orbit and projector, with the same parameter for the orbit and the reading:

$$\begin{aligned} \|\phi_c(0)\| &= 1, & |\phi_c(\tau)\rangle &= \exp(-iH_c\tau/\hbar)|\phi_c(0)\rangle, \\ \Pi_c(\tau) &= |\phi_c(\tau)\rangle\langle\phi_c(\tau)|, & P_\tau &= \Pi_c(\tau) \otimes I_r. \end{aligned}$$

For a bounded self-adjoint rest-system operator A_r , when $w = \text{Tr}(P_0\rho) > 0$,

$$\begin{aligned} E(A_r \mid \tau) &= \frac{\text{Tr}[\rho(\Pi_c(\tau) \otimes A_r)]}{\text{Tr}(P_\tau\rho)}, \\ \text{Tr}(P_\tau\rho) &= w, \quad \rho_r = \frac{\text{Tr}_c(P_0\rho)}{w}, \\ E(A_r \mid \tau) &= \text{Tr}_r[A_r(\tau)\rho_r], \quad A_r(\tau) = \exp(iH_r\tau/\hbar)A_r\exp(-iH_r\tau/\hbar). \end{aligned}$$

The additive Hamiltonian, global stationarity and cyclicity of the trace give this identity.

For the general closed system in Section II, let H be its self-adjoint Hamiltonian, ρ its density operator and $U(t) = \exp(-iHt/\hbar)$. Define time averaging on bounded operators for which the following weak operator limit exists. In the finite spin model it exists and removes matrix elements between distinct energies:

$$\text{Avg}_t(X) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T U(t)XU(t)^* dt.$$

For a general clock-reading projection P_τ and bounded observable A , if the averages exist and the denominator is positive, source equation (23) gives

$$E(A \mid \tau) = \frac{\text{Tr}[\text{Avg}_t(P_\tau A P_\tau)\rho]}{\text{Tr}[\text{Avg}_t(P_\tau)\rho]}.$$

The average acts on the whole product. Section III directly assumes a stationary state and uses the identity for the noninteracting model above.

Dependency Inputs The table gives each transfer's time inputs, outputs, claim support and source location.

Dependency and Transfer Elements

Definition 8.339 (Page–Wootters 1983 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [5]$.

Table 244: Page–Wootters 1983 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1–2, 4	3	13–14	From coordinate time and precession frequency to spin phase. pp. 2886
2	3–4	6	14–15	Assignment of a clock reading to spin orientation. pp. 2886, 2888
3	5–6	9	2, 16–18	Statistics conditioned on spin-clock readings. pp. 2887
4	8	23	2, 28–33	Conditioning the rest system on an ideal clock state. pp. 2890
5	8, 23	10	28–31, 33–35	Heisenberg representation of ideal-clock-conditioned expectations. pp. 2890

Grounding Conditions

1. Evaluate phase ωt for N spin- j particles with the specified Hamiltonian, frequency ω and x -directed product state at $t = 0$.
2. Choose the orientation of the first spin as a clock and assign twelve o'clock to the x direction; a reading need not correspond to a unique coordinate instant.
3. In the finite spin model, average the whole joint projection and the clock projection separately and take their expectation-value ratio at positive clock weight.
4. Use the projector defined from a normalized clock zero state, clock parameter τ on the same scale, a stationary global state, an additive Hamiltonian and positive conditioning weight.
5. Obtain equations (27)–(29) from equation (26), using noninteraction of clock and rest, global stationarity, cyclicity of trace and positive conditioning weight.

Transfer 2 uses the initial-state specification in Claim 14 and the orientation/readout correspondence in Claim 15.

Results

Definition 8.340 (Page–Wootters 1983 Results and Dependencies). Let $\text{ldx}_R^{\text{audit,result}} = [18]$.

Definition 8.341 (Page–Wootters 1983 Result-Dependency Hyperedges). Each result takes the time premises and claim support shown in the following table as its inputs.

Table 245: Page–Wootters 1983 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Argument for unobservability of coordinate time in a closed system. pp. 2885, 2886	1	4–6, 9–12
2	Invariance under removal of cross-energy coherences for stationary observables. pp. 2888, 2889	$\emptyset$	2–3, 7
3	Precession phase in the spin model. pp. 2886	1–4	13–14
4	Spin product state specified at coordinate time zero. pp. 2886	4	1–2, 13–14
5	Stationary averages of clock and joint projections. pp. 2887	5	2, 13, 18
6	Conditional probabilities and expectations for the spin clock. pp. 2887	4–6, 9	14, 16–20
7	Evolution described by reading sequences and measurement disturbance. pp. 2888	3, 6–7, 9	21–23
8	Replacement by a time-averaged stationary mixed state. pp. 2888, 2889	5–6, 9	2–3, 7, 24–25
9	Embedding the mixed two-spin state in a pure zero-energy state. pp. 2888, 2889	$\emptyset$	1–3, 26
10	Time-averaged conditional expectations for a general clock reading. pp. 2889	5, 22	2, 27
11	Ideal clock-state orbit and reading projector. pp. 2890	8	2–3, 28, 31–33
12	Ideal-clock conditional expectations and constant denominator. pp. 2890	8, 23	28–31, 33, 35
13	Agreement with Heisenberg evolution in clock time. pp. 2890	8, 10, 23	28–31, 33–35
14	Examples of real clocks and reading interactions. pp. 2890	11–18	36–37, 40
15	Argument for an energy-limited clock precision. pp. 2890	19	3, 38–39
16	Idealization of abstract coordinate time from real clocks. pp. 2890	1, 11–18	36, 40–41
17	Clock-relative description of the observed arrow of time. pp. 2890, 2891	20	43, 47
18	Proposal for spacetime intervals relative to intrinsic geometry. pp. 2891	21	44–45

Grounds for the Results

Table 246: Page–Wootters 1983 Conditions for the Results

Re- sult Index	Conditions and Domain
1	The argument concerns the closed system including its observers, a well-defined Hamiltonian and the stationary observable class adopted by Page and Wootters.
2	Remove density-matrix elements between distinct energy eigenspaces. Expectations of the specified operators commuting with the Hamiltonian remain unchanged.
3	Use the Hamiltonian, precession frequency and initial product state of equations (1)–(6).
4	Each spin has maximal x projection j at $t = 0$.
5	Use equations (7)–(12) of the finite spin model. At $j = 1/2$, the clock factor of the averaged clock operator is $I/2$ and its square is $I/4$.
6	Equations (13)–(17) use the specified product state and positive conditioning weight. Equation (14) has $0 \leq k \leq 2j$; equation (16) is a large- j approximation at fixed k .
7	Use the ordering of readings and corresponding states. Page and Wootters propose measuring different particles or copies once each.
8	Remove cross-energy coherences in the finite spin model while preserving its specified stationary conditional statistics.
9	In equations (18)–(22), adding a two-spin system with the reversed energy spectrum yields a pure zero-energy state of four spins. The partial trace of the four-spin density operator over the added two spins is the original two-spin density operator.
10	Equation (23) averages the whole sandwich $P_\tau A P_\tau$. The averages and traces must exist and the denominator must be positive.
11	Use a normalized clock zero state and self-adjoint H_c . The orbit and reading use the same parameter τ .
12	For the additive Hamiltonian and stationary global state, the conditioning weight is a positive constant.
13	Equations (25)–(29) for the noninteracting ideal clock give $E(A_r \tau) = \text{Tr}_r[A_r(\tau)\rho_r]$.
14	The eight listed clock variables are individual examples. Each reading uses an interaction with the rest; for a good clock its effect can be practically negligible.
15	Page and Wootters argue for a precision limit from the energy exchange needed for fine readings and the finite available energy.
16	Page and Wootters discuss preferred coordinate time as an idealization of actual clock behavior.
17	Page and Wootters call for an entropy observable and its dependence on a clock, with observed coarse-grained properties as the basis of entropy.
18	This is Page and Wootters’ proposed extension to Poincaré-invariant observables and a quantum-gravitational description using intrinsic geometrical variables.

Results 4 and 6 use the initial-state specification in Claim 14; Result 11 uses the clock-projector definition in Claim 33. Results 14 and 16 enumerate eight individual examples of real clocks.

Evidence Hierarchy Page and Wootters discuss closed-system observability and derive spin-model conditional probabilities, algebraic relations for stationary states and ideal-clock conditional evolution. Real-clock interactions and precision, the observed arrow of time and intrinsic geometry are theoretical arguments or proposals. The source has zero directly measured objects and zero direct-measurement predicates.

Generation Judgment The complete LCTR generation count and the source-evidence-conditioned local-reach count are zero; 23 primitive-time roots remain.

Result Dependence Of the 18 results, 16 depend on primitive-time roots. Results 2 and 9 are reachable from the source formal-evidence environment. No result is generated from nontemporal physical inputs alone. There are 51 root–result dependence pairs, of which 35 are necessary.

Evidentiary Effect The source supports its argument about closed-system observability, the finite spin model, stationary correlations and conditional evolution for a noninteracting ideal clock. Real-clock approximations, energy and precision, the clock-relative arrow and intrinsic geometry are supported as Page and Wootters’ arguments or proposals specified in the result table.

Placement of the Grounding Obligation Page and Wootters use coordinate time, precession frequency, an initial instant, time-averaging intervals, clock readings and their order, an ideal-clock parameter, conditional evolution, real-clock temporal representations, resolution, the arrow of time and intrinsic time. The 94 law-time uses and five transfers supply no conditions generating these time contents from nontemporal physical inputs before use. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The source presents theoretical models and arguments about observability. Actual failure of specific physical-time generation conditions, and consequent changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 51 root–result dependence pairs. Actual reach from the failure location of a specific generation condition to a result remains undetermined.

Result-Preservation Domain If the models, states, Hamiltonians, projections, conditioning weights and domains in the result table are fixed, and generated temporal representations realize the input relations of the 16 time-dependent results with the same scales and orders, those results are preserved within their respective domains. The algebraic relations in Results 2 and 9 are preserved under their conditions on energy subspaces and partial traces.

Repair Scope If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.342 (Page–Wootters 1983 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [23]; \emptyset, \{2, 9\}, [18] \setminus \{2, 9\}, \emptyset).$$

$R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. Closure from Initial Claims 1–3 remains those three nodes. Supplying all 47 claims as source formal evidence reaches Results 2 and 9, whose temporal premises are empty. Each of the five transfers has temporal inputs, so no time content is reached. Adding all 23 time contents as seeds reaches all 18 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.343 (Page–Wootters 1983 Minimal Primitive-Time-Root Support). The table gives minimal-support statistics as member count, minimum cardinality and maximum cardinality. The total member count is 27. The empty set is the sole minimal temporal support of Results 2 and 9.

Table 247: Page–Wootters 1983 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(1; 1, 1)
2	(1; 0, 0)
3	(1; 3, 3)
4	(1; 1, 1)
5	(1; 1, 1)
6	(3; 3, 4)
7	(6; 4, 5)
8	(3; 2, 4)
9	(1; 0, 0)
10	(1; 2, 2)
11	(1; 1, 1)
12	(1; 1, 1)
13	(1; 1, 1)
14	(1; 8, 8)
15	(1; 1, 1)
16	(1; 9, 9)
17	(1; 1, 1)
18	(1; 1, 1)

Proof. Start from singleton seeds and apply Definition 7.46 to the five transfers. From the stable antichains, combine supports for each result’s time premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Proposition 8.344 (Page–Wootters 1983 Audit Conclusion). Complete generation and local reach are zero, leaving 23 primitive-time roots. The 18 results divide into two evidence-reachable and 16 primitive-time-root-dependent results. The 47 claims, 23 time contents, five transfers and 18 results give 93 source-specific obligation atoms. The 94 claim/time-content use pairs have 107 role assignments.

Proof. This follows from the definitions of claims, time contents, transfers and results and the reachable sets and minimal-support families above. $\square$

Proposition 8.345 (Page–Wootters 1983 Audit-Boundary Counts). $\text{BoundaryCount}_R^{\text{audit}} = (88, 23)$.

Proof. The 47 claims, 23 time contents and 18 results give 88 nodes; the five transfers and 18 result dependencies give 23 edges. $\square$

Individual Statements Statement Evidence 1: Stationary States of Closed Systems and Observational Consistency

Verbatim Original (English):

This means that it is consistent with all observations to assume that any closed system such as the Universe is in a stationary state.

Original-Source Location Page and Wootters, *Evolution without Evolution: Dynamics Described by Stationary Observables*, *Physical Review D* 27, abstract, p. 2885.

Minimal Context Following the argument that the external coordinate-time parameter is unobservable, the source states the consistency between observations and the assumption that a closed system is in a stationary state.

Attribution Statement by Don N. Page and William K. Wootters.

Statement Evidence 2: Description of Dynamical Evolution by Stationary Observables and Internal Clock Readings

Verbatim Original (English):

We show how the observed dynamic evolution of a system can be described entirely in terms of stationary observables as a dependence upon internal clock readings.

Original-Source Location Page and Wootters, *Evolution without Evolution: Dynamics Described by Stationary Observables*, *Physical Review D* 27, abstract, p. 2885.

Minimal Context The statement gives the central proposal of the paper: observed dynamical evolution is described by the dependence between stationary observables and internal clock readings.

Attribution Statement by Don N. Page and William K. Wootters.

Statement Evidence 3: Observability of Dependence on Internal Clock Time

Verbatim Original (English):

Only the dependence on an internal clock time can be observable.

Original-Source Location Page and Wootters, *Evolution without Evolution: Dynamics Described by Stationary Observables*, *Physical Review D* 27, at the end of Section III, p. 2890.

Minimal Context Following the idealization of abstract coordinate time from a real clock, the source distinguishes dependence on abstract coordinate time from the observability of dependence on internal clock time.

Attribution Statement by Don N. Page and William K. Wootters.

Common Evidence with Specified Domain of Application The three statements above give the verbatim originals from Page–Wootters 1983, their original-source locations, the minimal contexts required for interpretation, and their source attribution.

Target state-ment	Object of the statement	Predicate	Explicit domain of application
S26-Q001	The stationary-state assumption for an arbitrary closed system, including the universe, and observations	States that assuming a closed system to be in a stationary state is consistent with all observations.	The domain in the abstract that treats quantities depending on external coordinate time as unobservable and discusses the stationary-state assumption for a closed system.
S26-Q002	Stationary observables, internal clock readings, and the observed dynamical evolution of a system	Shows how dynamical evolution can be described entirely in terms of stationary observables as dependence on internal clock readings.	The domain of the abstract and central construction in which dynamical evolution is described by a clock–remainder-system decomposition and conditional correlations.
S26-Q003	The observability of dependence on abstract coordinate time and dependence on internal clock time	States that only dependence on internal clock time is observable.	The domain at the end of Section III that idealizes abstract coordinate time from a real clock and distinguishes dependence on the two.

Individual Audit

First Statement Page and Wootters’ first statement asserts that assuming a stationary state for a closed system is consistent with observations. In their stationary observable class, removing density-matrix elements between distinct energies leaves expectations unchanged. This algebraic relation and their observability argument for a closed system including its observers correspond to Claims 4–12 and Results 1, 2 and 8.

Their argument about observation records within a single closed system is Claim 46; embedding in a pure zero-energy state of a larger system is Claim 26 and Result 9.

Second Statement Page and Wootters’ second statement gives a method for describing dynamical evolution through correlations conditioned on internal clock readings. The spin model in Section II corresponds to Claims 13–27 and Results 3–10; the noninteracting ideal clock in Section III corresponds to Claims 28–35 and Results 11–13. For the ideal clock, global stationarity, the additive Hamiltonian, the selected clock zero state, the projection tied to its orbit and positive conditioning weight imply that conditional rest-system expectations equal expectations of Heisenberg-evolved operators in clock time.

Third Statement Page and Wootters’ third statement restricts observable temporal dependence to dependence on internal clock time. Claims 36–42 and Results 14–16 concern reading interactions, energy-limited precision and abstract coordinate time as an idealization of real clocks. The practical approximation applies when interaction effects on a good clock are negligible. Commutation with the Hamiltonian serves as a necessary condition in the source’s observability argument.

Common Audit with Specified Domain of Application The stationary description of a closed system, internal-clock conditional evolution and real-clock readings and approximations have their respective state, operator and interaction conditions specified above.

8.19.27 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 27.

Source Name Bombelli–Lee–Meyer–Sorkin 1987

Bibliographic Identification Luca Bombelli, Jooan Lee, David Meyer and Rafael D. Sorkin, *Space-Time as a Causal Set*, *Physical Review Letters* 59, pp. 521–524 (1987) is denoted by $R = \text{Ref}_{\text{Bom87CS}}$. The audit covers the abstract and the arguments and notes on all four pages.

Original-Source Location Identifier 27-L1.

Original-Source Location Content The tables specify source pages for time contents, claims, transfers and results.

Audit-Scope Identifier 27-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Formal Objects The audit uses 17 time contents, 40 claims, seven transfers and 14 results. The claim table and formal-input definition specify causal sets, continuum causal order, volume correspondence, faithful embedding, coarse graining and the conditional effective-action argument. Bombelli, Lee, Meyer and Sorkin propose that spacetime at the smallest scales is a causal set. They identify a locally finite partial order with physical past–future relations and element counts with spacetime volume. Continuum

approximation is discussed under faithful-embedding conditions; coarse graining restricts the causal order to a representative subset. The quantum-action argument uses causal-set amplitudes and a given Lorentzian geometry.

Assumptions A mathematical causal set satisfies transitivity, noncircularity between distinct elements, the reflexive convention and local finiteness of intervals. Recovery of conformal geometry from continuum causal order uses suitable smoothness and causality conditions and dimension greater than two. Faithful embedding uses time orientation, absence of closed timelike or null curves, order agreement, uniform unit density and separation of geometric variation scales from point spacing. Result 14 specifies the quantum-action conditions.

Nontemporal Physical Inputs Initial inputs for generation are Claims 1–3. Data for sets, cardinalities, binary relations, samples, density coefficients, maps and algebraic coefficients are separated from physical correspondences with past and future, events, spacetime volume and Lorentzian geometry.

Time Contents

Definition 8.346 (Bombelli–Lee–Meyer–Sorkin 1987 Time Contents). Let $n_R = 17$. The following table defines the time contents and source locations.

Table 249: Bombelli–Lee–Meyer–Sorkin 1987 Time Contents

i	Time Content	Source Location
1	Causal order P of continuum spacetime	pp. 521, 522
2	Conformal metric and light cones of continuum spacetime	pp. 521, 522
3	Time orientation of Lorentzian spacetime	pp. 522
4	Physical past–future order of the causal set	pp. 521, 522, 523
5	Spacetime events corresponding to causal-set elements	pp. 522
6	Alexandrov intervals of the causal set	pp. 522, 523
7	Alexandrov neighborhoods of continuum spacetime	pp. 522, 523, 524
8	Spacetime volume and density units identified with event counts	pp. 522, 524
9	Correspondence of discrete and continuum causal orders under an embedding	pp. 522
10	Continuum-spacetime approximation through an embedding satisfying the three faithfulness conditions	pp. 522, 523, 524
11	Coarse-grained causal order restricted to a representative subset	pp. 523
12	Kaluza–Klein spacetime regions with different internal dimensions	pp. 523
13	Effective spacetime dimensionality and topology at continuum-approximation and coarse-graining scales	pp. 523
14	Spacetime regions inside black holes	pp. 523

i	Time Content	Source Location
15	Spacetime regions in the very early universe	pp. 523
16	Topology-changing processes of spacetime	pp. 523
17	Lorentzian metric and spacetime regions used in faithful-embedding and action arguments	pp. 522, 523, 524

Definition 8.347 (Bombelli–Lee–Meyer–Sorkin 1987 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(40)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([3], [40])$. The table gives each claim, its time uses and source locations.

Table 250: Bombelli–Lee–Meyer–Sorkin 1987 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Nontemporal data for sets, elements, cardinalities and binary relations.	$\emptyset$	pp. 522
2	Nontemporal data for probabilities, density coefficients, representative-subset selection and samples.	$\emptyset$	pp. 522, 523, 524
3	Nontemporal data for maps, candidate manifolds and topologies, algebraic coefficients and complex amplitudes.	$\emptyset$	pp. 522, 523, 524
4	Bombelli, Lee, Meyer and Sorkin propose that spacetime at the smallest scales is a locally finite causal set whose partial order corresponds to the macroscopic past–future relation.	$4^C, 5^B$	pp. 521, 522
5	In a time-oriented Lorentzian spacetime, the metric light cones determine the causal order P between events.	$1^C, 2^C, 3^C$	pp. 521, 522
6	For a continuum spacetime satisfying suitable smoothness and causality conditions and having dimension greater than two, causal order P determines its topology, differentiable structure and conformal metric.	$1^C, 2^C$	pp. 521, 522
7	The causal order of continuum spacetime does not determine the conformal factor, so order alone does not fix the spacetime-volume scale.	$1^C, 8^Q$	pp. 521, 522
8	The four authors postulate finitely many events in a finite spacetime volume and identify their count with the region’s spacetime volume in natural units.	$5^B, 8^Q$	pp. 522

p	Claim	i	Source Location
9	The four authors identify Lorentzian signature as the metric signature with a two-napped light cone representing macroscopic causal order.	$1^C, 2^C$	pp. 522
10	A mathematical causal set is a locally finite partially ordered set. The source adopts transitivity, noncircularity between distinct elements, the reflexive convention $x < x$, and finiteness of each interval.	$\emptyset$	pp. 522
11	The causal-set interval $A(x, y) = \{z \mid x < z < y\}$ is the intersection of the future of x with the past of y .	$4^C, 6^I$	pp. 522
12	The continuum Alexandrov neighborhood $J^+(q) \cap J^-(p)$ is the intersection of the causal future of the initial point q with the causal past of the final point p .	$1^C, 7^I$	pp. 522, 523, 524
13	The target of a faithful embedding is a time-oriented Lorentzian manifold (M, g) without closed timelike or null curves.	$2^C, 3^C, 17^S$	pp. 522
14	The order-agreement condition for $f : C \rightarrow M$ is the equivalence of $f(x) \in J^-(f(y))$ and $x < y$.	$1^C, 4^C, 9^C$	pp. 522
15	The embedding-density condition requires uniform unit density of embedded points in natural units, in which the spacetime-volume integral counts elements of C .	$5^B, 8^Q, 17^S$	pp. 522
16	Footnote 9 explains uniform density by requiring the point count in each Alexandrov neighborhood to agree with its volume within Poisson-type fluctuations.	$5^B, 7^I, 8^Q$	pp. 524
17	The scale condition for a faithful embedding requires the length λ over which continuum geometry varies appreciably to be everywhere much larger than mean embedded-point spacing.	17^S	pp. 522, 524
18	An embedding into the specified Lorentzian target is called faithful when it satisfies causal-order agreement, uniform unit density and the geometric-scale condition.	$1^C, 2^C, 3^C, 4^C, 5^B, 8^Q, 9^C, 10^C, 17^S$	pp. 522
19	The four authors expect essential uniqueness of a continuum approximation admitting a faithful embedding and leave its precise formulation and proof as a task.	$10^C, 17^S$	pp. 522

p	Claim	i	Source Location
20	A faithful embedding need not exist. Footnote 10 explains that order agreement and density alone cannot exclude large curvature introduced by adjustment of the conformal factor.	$2^C, 8^Q, 9^C, 10^C, 17^S$	pp. 522, 524
21	The four authors sketch a dimensional-consistency argument using small causal-set intervals and approximately flat continuum neighborhoods: with great likelihood some interval admits no lower-dimensional embedding, giving equal dimensions for two faithful embeddings.	$6^I, 7^I, 8^Q, 10^C, 13^Q$	pp. 522, 523
22	A faithfully embedded causal set resembles a random sprinkling from the continuum viewpoint; in the authors' approximation relation, the continuum M approximates the causal set C .	$4^C, 10^C$	pp. 523
23	A coarse graining C' of a causal set C is obtained by restricting its order to a representative subset of C .	$4^C, 11^C$	pp. 523
24	Random thinning with selection probability p is interpreted as retaining structures whose characteristic volume scale exceeds $1/p$.	$8^Q, 11^C$	pp. 523
25	A causal set C with no faithful embedding may have a coarse graining C' that admits one.	$4^C, 10^C, 11^C$	pp. 523
26	A causal set containing Kaluza–Klein-type regions with different internal dimensions may become embeddable after coarse graining above the characteristic scales of the internal metrics.	$10^C, 11^C, 12^Q, 13^Q$	pp. 523
27	Coarse graining can wash out small-scale structures, simplify effective topology and lower effective spacetime dimensionality.	$10^C, 11^C, 13^Q$	pp. 523
28	The four authors give examples of causal sets with different spacetime dimensions and topologies at different scales, including a set with no meaningful dimension before coarse graining.	$4^C, 13^Q$	pp. 523
29	The four authors identify black-hole interiors, the very early universe and topology-changing processes as domains where continuum description breaks down and causal-set kinematics may be useful.	$4^C, 14^C, 15^B, 16^S$	pp. 523

p	Claim	i	Source Location
30	The four authors turn from the kinematical to the dynamical stage of theory development and ask which action yields general relativity in the classical limit.	$\emptyset$	pp. 523
31	The four authors assume that a quantum amplitude is defined for each causal set C .	4^C	pp. 523
32	The four authors assume that constructive interference among amplitudes of causal sets contributing to a given continuum geometry (M, g) yields the classical limit.	$4^C, 17^S$	pp. 523
33	The action argument uses a given continuum geometry (M, g) and a collection of causal sets faithfully embedded in it.	$4^C, 10^C, 17^S$	pp. 523
34	If amplitudes are approximately multiplicative across macroscopic regions, the action, taken as the logarithm of the amplitude, becomes an integral of a locally defined scalar density in the continuum approximation.	$4^C, 10^C, 17^S$	pp. 523
35	The four authors argue that the sum over causal-set histories washes out individual embedding anisotropies, so the effective action density depends only on the metric and is generally covariant.	$4^C, 17^S$	pp. 523
36	In the expansion $L_{\text{eff}} = L_0 + L_2 + L_4 + \dots$, L_0 is a constant, L_2 is proportional to scalar curvature and L_4 is a sum of curvature-squared terms.	17^S	pp. 523
37	From the action having dimensions of $\hbar$ in n spacetime dimensions, the four authors estimate coefficients of L_{2k} to be of order $\hbar l_P^{2k-n}$.	17^S	pp. 523
38	Where higher-curvature terms are negligible, $L_0 + L_2$ remains and has the general-relativistic Lagrangian form with a cosmological term.	17^S	pp. 523, 524
39	Why L_0 does not dominate L_2 remains unresolved in this effective-action argument.	17^S	pp. 524
40	The four authors defer a specific amplitude, its large-scale action and an attempt to derive a small cosmological constant from order fluctuations to another paper.	$4^C, 13^Q, 17^S$	pp. 524

Formal Input Elements

Definition 8.348 (Bombelli–Lee–Meyer–Sorkin 1987 Causal Orders and Embeddings). Under the source’s reflexive order convention, causal-set intervals and order agreement for

a faithful embedding are expressed as

$$A(x, y) = \{z \in C \mid x < z < y\}, \quad f(x) \in J^-(f(y)) \iff x < y.$$

Here $f : C \rightarrow M$, and $J^-(p)$ is the causal past of p . Faithfulness also requires the density and scale conditions in Claims 15–18. The order on a coarse graining $C' \subseteq C$ is $<|_{C' \times C'}$.

Table 251: Bombelli–Lee–Meyer–Sorkin 1987 Time-Use Roles

Code	Role
I	Interval
Q	Quantity
X	Cross-position correspondence
K	Clock reading
C	Causality
S	State evolution
P	Probabilistic evolution
O	Observation record
B	Reference boundary

Table 252: Bombelli–Lee–Meyer–Sorkin 1987 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 4, 11	member	order_relation
2, 8–10, 13, 17	dependency	$\emptyset$
3	member	orientation
5	member	instant
6–7	member	interval_boundary, order_relation
12, 14–16	lexical	$\emptyset$

Dependency Inputs The table gives each transfer’s time inputs, outputs, claim support and source location.

Dependency and Transfer Elements

Definition 8.349 (Bombelli–Lee–Meyer–Sorkin 1987 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [7]$.

Table 253: Bombelli–Lee–Meyer–Sorkin 1987 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	2–3	1	5	From time-oriented light cones to continuum causal order. pp. 521, 522
2	1	2	6	From continuum causal order to conformal geometry. pp. 521, 522
3	4	6	11	From causal-set order to causal intervals. pp. 522
4	1	7	12	From continuum causal order to Alexandrov neighborhoods. pp. 522, 523, 524
5	5	8	8	From event counts to spacetime volume in natural units. pp. 522
6	4	11	23	From causal-set order to the order of a representative subset. pp. 523
7	1–5, 8–9, 17	10	13–15, 17–18	From the three conditions on a specified embedding to a faithful continuum approximation. pp. 522, 524

Grounding Conditions

1. Use the specified Lorentzian conformal metric and time orientation to determine causal precedence between events.
2. The source domain requires suitable smoothness and causality conditions and dimension greater than two.
3. Form $A(x, y) = \{z \mid x < z < y\}$ under the source’s reflexive order convention.
4. Intersect the causal future $J^+(q)$ of the initial point q with the causal past $J^-(p)$ of the final point p .
5. Count under the authors’ postulate identifying elements in a finite region with spacetime events and identifying their number with volume.
6. Select C' as a representative subset of C and restrict the order of C to C' .
7. The specified target is time-oriented and has no closed timelike or null curves. The specified embedding must satisfy order agreement, uniform unit density and separation of geometric variation scales from point spacing.

Results

Definition 8.350 (Bombelli–Lee–Meyer–Sorkin 1987 Results and Dependencies). Let $\text{ldx}_R^{\text{audit,result}} = [14]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 254: Bombelli–Lee–Meyer–Sorkin 1987 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Mathematical definition as a locally finite partially ordered set. pp. 522	$\emptyset$	10
2	Causal-set proposal for smallest-scale spacetime and past–future order. pp. 521, 522	4–5	4
3	Relation between continuum causal structure and Lorentzian conformal geometry. pp. 521, 522	1–3	5–6, 9
4	Independence of continuum causal order and volume scale. pp. 521, 522	1, 8	7
5	Spacetime volume in natural units expressed by event count. pp. 522	5, 8	8
6	Three conditions for a faithful embedding. pp. 522, 524	1–5, 7–10, 17	13–18
7	Natural-unit density and Poisson-type fluctuations. pp. 522, 524	5, 7–8, 17	15–16
8	Conjectured essential uniqueness of a faithful continuum approximation. pp. 522, 524	2, 8–10, 17	19–20
9	Sketch of dimensional consistency between faithful embeddings. pp. 522, 523	6–8, 10, 13	21
10	Continuum M as an approximation to causal set C . pp. 523	4, 10	22
11	Coarse graining by restricting order to a representative subset. pp. 523	4, 11	23
12	Variation of effective dimension and topology with coarse-graining scale. pp. 523	4, 8, 10–13	24–28
13	Causal-set kinematics for candidate domains of continuum breakdown. pp. 523	4, 14–16	29
14	Conditional argument for an effective action of general-relativistic form with a cosmological term. pp. 523, 524	4, 10, 17	31–39

Grounds for the Results

Table 255: Bombelli–Lee–Meyer–Sorkin 1987 Conditions
for the Results

Re- sult Index	Conditions and Domain
1	Transitivity, noncircularity between distinct elements, the reflexive convention and local finiteness of intervals.
2	The four authors give causal-set elements and order the physical meanings of spacetime events and causal relations.
3	Recovery from order uses the smoothness/causality conditions and $\dim M > 2$; causal ordering from light cones uses time orientation.
4	The conformal metric determined by causal order leaves the volume-setting conformal factor unspecified.
5	The authors' postulate relating finite spacetime volume to a finite event count.
6	Specified Lorentzian target, order agreement, uniform unit density with Poisson-type fluctuations, and geometric variation scales much larger than point spacing.
7	Natural units in which spacetime volume counts points, with the fluctuation prescription for each Alexandrov neighborhood.
8	A conjecture conditional on existence of a faithful embedding; the authors leave its precise formulation and proof as a task.
9	Uses local approximate flatness and the unit-density condition, with a probabilistic step excluding lower-dimensional embedding.
10	Where a faithful embedding is given, M is read as an approximation to C .
11	Choose $C' \subseteq C$ representatively and restrict the original causal order.
12	Representative coarse graining, volume-scale selection and the source's embedding, internal-dimension and topology examples.
13	Candidate applications considered under the hypothesis that a causal set underlies the continuum.
14	Uses causal-set amplitudes, a constructive-interference classical limit, faithful embedding, approximate multiplicativity across macroscopic regions, averaging of anisotropies by the history sum, dimensional coefficient estimates and a regime of negligible higher-curvature terms. Why L_0 does not dominate L_2 remains unresolved.

Evidence Hierarchy The four authors present a mathematical causal-set definition, a physical interpretation of order, faithful-embedding conditions and coarse graining. Essential uniqueness of the continuum approximation is a conjecture; dimensional consistency is a sketch with a probabilistic step; the general-relativistic action form is an effective-action argument under the stated assumptions. The source has zero directly measured objects and zero direct-measurement predicates.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local-reach count are zero; 17 primitive-time roots remain.

Result Dependence Of the 14 results, 13 depend on primitive-time roots. Result 1, the mathematical definition, is reachable from the source formal-evidence environment. No result is generated from nontemporal physical inputs alone. There are 77 root–result dependence pairs, of which 25 are necessary.

Evidentiary Effect The source gives the locally finite causal-set definition, the relation between continuum causal order and conformal geometry, three faithful-embedding conditions and order restriction under coarse graining. Dimensional consistency is a sketch with a probabilistic step; essential uniqueness of a faithful approximation is the authors’ conjecture. The quantum-action and classical-limit argument uses the assumptions of Result 14.

Placement of the Grounding Obligation The four authors presuppose causal past and future, time orientation, spacetime events, volume and causal intervals, the spacetime structure of continuum approximations, and the Lorentzian geometry used by the action. The 91 time uses in the claim table and seven transfers supply no conditions generating these time contents from nontemporal physical inputs before use. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The source presents a causal-set definition and physical proposal, with conditional continuum-approximation and effective-action arguments. Actual failure of specific physical-time generation conditions, and consequent changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 77 root–result dependence pairs. Actual reach from the failure location of a specific generation condition to a result remains undetermined.

Result-Preservation Domain If the result table’s conditions on sets, orders, embeddings, densities, scales and amplitudes are fixed, and generated temporal representations realize the 13 time-dependent results’ input relations with the same causal orders, time orientations, volume scales and domains, each result is preserved with its stated epistemic status and domain. Result 1 remains the mathematical definition under the locally finite partial-order axioms.

Repair Scope If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Proposition 8.351 (Bombelli–Lee–Meyer–Sorkin 1987 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [17]; \emptyset, \{1\}, [14] \setminus \{1\}, \emptyset).$$

$R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. Closure from Initial Claims 1–3 remains those three nodes. Supplying all 40 claims as source formal evidence reaches Result 1, whose temporal premises are empty. Each of the seven transfers has temporal inputs, so no time content is reached. Adding all 17 time contents as seeds reaches all 14 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.352 (Bombelli–Lee–Meyer–Sorkin 1987 Minimal Primitive-Time-Root Support). The table gives minimal-support statistics as member count, minimum cardinality and maximum cardinality. The total member count is 46. The empty set is the sole minimal temporal support of Result 1.

Table 256: Bombelli–Lee–Meyer–Sorkin 1987 Minimal Primitive-Time-Root Support

Result Index	Support Statistics
1	(1; 0, 0)
2	(1; 2, 2)
3	(2; 2, 2)
4	(4; 2, 3)
5	(1; 1, 1)
6	(2; 6, 6)
7	(3; 3, 4)
8	(6; 5, 6)
9	(14; 5, 7)
10	(3; 2, 6)
11	(1; 1, 1)
12	(4; 5, 8)
13	(1; 4, 4)
14	(3; 3, 6)

Proof. Start from singleton seeds and apply Definition 7.46 to the seven transfers. From the stable antichains, combine supports for each result’s time premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Proposition 8.353 (Bombelli–Lee–Meyer–Sorkin 1987 Audit Conclusion). Complete generation and local reach are zero, leaving 17 primitive-time roots. The 14 results divide into one evidence-reachable and 13 primitive-time-root-dependent results. The 40 claims, 17 time contents, seven transfers and 14 results give 78 source-specific obligation atoms. The 91 claim/time-content use pairs have 91 role assignments.

Proof. This follows from the definitions of claims, time contents, transfers and results and the reachable sets and minimal-support families above. $\square$

Proposition 8.354 (Bombelli–Lee–Meyer–Sorkin 1987 Audit-Boundary Counts). $\text{BoundaryCount}_R^{\text{audit}} = (71, 21)$.

Proof. The 40 claims, 17 time contents and 14 results give 71 nodes; the seven transfers and 14 result dependencies give 21 edges. $\square$

Individual Statements

Object of the statements The four authors' proposal that smallest-scale spacetime is a causal set, and the correspondence of its partial order with the macroscopic past–future relation.

Verbatim originals Verbatim originals (English):

We propose that space-time at the smallest scales is in reality a causal set
a partial order corresponding to the macroscopic relation that defines past
and future

Original-source location Bombelli et al., *Space-Time as a Causal Set*, *Physical Review Letters* 59, p. 521, abstract.

Minimal context Bombelli, Lee, Meyer and Sorkin propose a locally finite partially ordered set as the structure of smallest-scale spacetime. The second quotation is the part of the same opening abstract sentence specifying the physical meaning of that order.

Attribution Luca Bombelli, Joohan Lee, David Meyer and Rafael D. Sorkin.

Predicate The four authors propose that smallest-scale spacetime is a causal set and identify its partial order with the macroscopic past–future relation.

Explicit domain of application The causal-set proposal identifying smallest-scale spacetime with a locally finite causal set and exploring its continuum approximation by a Lorentzian manifold.

Individual Audit

First Statement The four authors propose a locally finite causal set as the fundamental structure of smallest-scale spacetime. The statement concerns spacetime ontology, with its proposal status expressed by the governing verb. Its mathematical support consists of the causal-set definition and the three faithful-embedding conditions. Essential uniqueness of the continuum approximation is the authors' conjecture; its precise formulation and proof remain open tasks. The corresponding evidence is Time Contents 4,5,10, Claims 4,10,13–20 and Results 1,2,6,8.

Second Statement The four authors give causal-set order the meaning of macroscopic past–future relations. In a continuum approximation using a faithful embedding, this correspondence is expressed by agreement of discrete order and continuum causal order. Its formal support is the order-agreement condition and the conditional recovery of conformal geometry from an existing continuum causal order. In the LCTR audit, Claims 4 and 14 and Transfers 1–4 and 7 use temporal premises in their physical order interpretation. The corresponding evidence is Time Contents 1–4,6,7,9,10, Claims 4–6,11–14,18 and Results 2,3,6.

Common Audit with Scope The two statements concern spacetime constitution and the physical meaning of order within the authors' single causal-set proposal.

8.19.28 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 28.

Source Name Weinberg 1989

Bibliographic Identification Steven Weinberg, *The Cosmological Constant Problem, Reviews of Modern Physics* 61, pp. 1–23 (1989), DOI: 10.1103/RevModPhys.61.1 is denoted by $R = \text{Ref}_{\text{Wei89CCP}}^{\text{survey}}$. The audit covers the text, footnotes, note added in proof and references on all 23 pages.

Original-Source Location Identifier 28-L1.

Original-Source Location Content The tables specify source pages for time contents, claims, transfers and results.

Audit-Scope Identifier 28-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Formal Objects The audit uses 22 time contents, 59 claims, 12 transfers and 21 results. The claim table and formal-input definition specify the cosmological term and vacuum contributions, Friedmann models, supersymmetry, constant-field adjustment mechanisms, modified gravity and quantum-cosmological constructions. Weinberg examines supersymmetry, anthropic considerations, adjustment mechanisms, modifications of gravity and quantum cosmology in response to the disparity between theoretical expectations for the cosmological constant and observational limits. Ages calculated from cosmic expansion use a formation redshift and a cosmological model; the stellar ages used for comparison come from independent observational evidence. His discussion of quantum cosmology constructs a matter-field local clock, an average over histories and measurement times, a global configuration time from spatial volume, and a conserved flux associated with that coordinate.

Assumptions Each argument uses a chosen spacetime metric, field action, cosmological model, symmetry and symmetry-breaking regime, and observational inputs. The galaxy-formation and recollapse bounds have their respective formation and lifetime conditions. The argument about constant-field adjustment mechanisms assumes a constant metric and matter fields and local field-space coordinates. The quantum-cosmological constructions require specified boundary conditions, contours, measures, operator orderings and normalization. The corresponding tables state the conditions for each transfer and result.

Nontemporal Physical Inputs Initial inputs for generation are Claims 1–4. The same table gives the claims that associate nontemporal data for coefficients, fields, spatial geometry and states with the spacetime metrics, evolution, histories and observation times used by the physical models.

Definition 8.355 (Source and Attribution). Weinberg reviews earlier theoretical proposals and observational results. The comparative assessments in the source are attributed to Weinberg. The source reports zero directly measured objects and zero direct-measurement predicates from measurements of its own.

Time Contents

Definition 8.356 (Weinberg 1989 Time Contents). Let $n_R = 22$. The following table defines the time contents and source locations.

Table 257: Weinberg 1989 Time Contents

i	Time Content	Source Location
1	de Sitter coordinate time and test-body motion	pp. 1, 2
2	Cosmic time and scale-factor evolution in Friedmann models with matter	pp. 2, 3, 7, 8, 9
3	Emission–observation order and redshift	pp. 2, 3, 7, 8, 9
4	Expansion rate and Hubble time on the present slice	pp. 2, 8, 9
5	Cosmological epochs of phase transitions, nucleosynthesis and formation	pp. 3, 4, 7, 9
6	Age and elapsed time since formation calculated in a cosmological model	pp. 7, 8
7	Time to recollapse in a negative-cosmological-constant model	pp. 9
8	Temporal evolution of vacuum, matter, radiation and fields	pp. 3, 4, 7, 9, 10, 11
9	Four-dimensional cosmological histories and boundary three-surfaces	pp. 14, 15, 16, 17, 18, 19, 20, 21
10	Definite time specified in ordinary probabilistic questions	pp. 15
11	Local clock defined by matter-field level surfaces	pp. 15, 16
12	Expectation value averaged over histories and measurement times	pp. 16
13	Global configuration-time coordinate defined by spatial-metric volume	pp. 16
14	Independent stellar-age and main-sequence-lifetime inputs	pp. 7, 8
15	Assumed time required for the appearance of life	pp. 7, 9
16	Lower bound on proton lifetime	pp. 6
17	Temporal structure of Lorentzian spacetime used for vacuum, field and gravity arguments	pp. 1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 15
18	Relativistic-particle worldline parameter and einbein	pp. 12, 13
19	Time coordinate, lapse and shift in ADM decomposition	pp. 13, 14
20	Conserved current with respect to global configuration time	pp. 16
21	Probability conditioned on a local-clock value	pp. 15, 16
22	Transformed time coordinate in the Friedmann representation of the de Sitter model	pp. 2

Definition 8.357 (Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(59)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([4], [59])$. The table gives each claim, its time uses and source locations.

Table 258: Weinberg 1989 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Coefficients, coupling constants, algebraic values and symmetry data.	$\emptyset$	pp. 2, 3, 4, 5, 6
2	Nontemporal candidate data for fields, matter configurations, potentials and actions.	$\emptyset$	pp. 3, 4, 5, 9, 10, 11, 14
3	Nontemporal data for positive-definite spatial three-metrics, spatial field values, topology and boundary geometry.	$\emptyset$	pp. 13, 14, 15, 16
4	Nontemporal candidate data for states, operators and integrals, and values of densities and object distributions.	$\emptyset$	pp. 2, 4, 7, 8, 14, 15, 16
5	The Einstein equation with a cosmological term relates a Lorentzian metric to the matter energy–momentum tensor.	17^C	pp. 1, 2
6	Test bodies generally move according to the metric connection even in the static-coordinate representation of the de Sitter model.	$1^S, 17^C$	pp. 1, 2
7	Weinberg describes redshift accompanying test-body motion in the de Sitter model, with $z = Hr$ at short distances.	$1^S, 3^{XQ}$	pp. 2
8	The coordinate transformation in Eq. (2.11) puts the specified de Sitter metric into Friedmann form with curvature $k = 0$ and density $\rho = 0$.	$1^S, 22^S$	pp. 2
9	For specified density, curvature and cosmological term, the Friedmann equation determines expansion or contraction of the scale factor $R(t)$.	2^S	pp. 2, 3
10	For specified emission and observation events, redshift is expressed by the scale-factor ratio $1 + z = R(t_{\text{obs}})/R(t_{\text{em}})$.	$2^S, 3^{XQ}, 4^{QB}$	pp. 3, 7, 8
11	A scale-factor solution and a specified present time give the present expansion rate $H_0 = (dR/dt)/R$ and its reciprocal, the Hubble time.	$2^S, 4^{QB}$	pp. 2, 8, 9
12	The energy–momentum tensor of a Lorentz-invariant vacuum is proportional to the metric, and its contribution enters the cosmological term.	17^C	pp. 2
13	The bare cosmological term and vacuum density combine as $\lambda_{\text{eff}} = \lambda + 8\pi G\langle\rho\rangle$ and $\rho_V = \langle\rho\rangle + \lambda/(8\pi G)$.	$\emptyset$	pp. 2

p	Claim	i	Source Location
14	Weinberg combines the redshift–distance, present-expansion, density and curvature information then available with the Friedmann equation to estimate an observational upper bound on vacuum density.	$2^S, 3^{OQ}, 4^{QB}$	pp. 2, 8
15	Zero-point energies and higher-order corrections give vacuum contributions depending on the chosen cutoff and field model.	17^C	pp. 2, 3, 6
16	Minima of a temperature-dependent potential change across cosmological phase transitions; tuning the present vacuum density to be small does not imply that it was small in the early universe.	$5^{BS}, 8^S$	pp. 3, 4
17	For configurations constant throughout spacetime, general covariance constrains the metric dependence of the Lagrangian density after the matter equations hold; a flat solution requires the remaining coefficient to vanish.	17^C	pp. 4
18	For unbroken global supersymmetry, the supersymmetry algebra and vacuum conditions give zero vacuum energy. That conclusion does not hold when the symmetry is broken.	17^C	pp. 4
19	In locally supersymmetric supergravity, the vacuum potential also contains a negative term, so a supersymmetric stationary point and zero vacuum energy are distinct conditions.	17^C	pp. 5
20	The no-scale models and string-theory examples considered by Weinberg depend on specified potential and kinetic structures and perturbative order.	17^C	pp. 5, 6
21	The five approaches to the cosmological-constant problem are supersymmetry, anthropic considerations, adjustment mechanisms, modifications of gravity and quantum cosmology.	$\emptyset$	pp. 1, 4, 6, 9, 11, 14, 20
22	Weinberg distinguishes using existence as an observational datum, a strong claim requiring observers, and a weak claim selecting habitable regions or epochs.	5^B	pp. 6, 7
23	In Goldhaber’s example, human survival is used as an observational datum giving a lower bound on proton lifetime.	16^I	pp. 6

p	Claim	i	Source Location
24	In Dicke's argument, stellar lifetimes required for heavy-element production and stellar energy supply constrain the range of cosmic ages permitting life.	$6^I, 14^I, 15^I$	pp. 7
25	Weinberg lists field changes during expansion, regions repeatedly formed by fluctuations, phase-transition bubbles and quantum-state superpositions as distinct models allowing different vacuum values.	$2^S, 5^{BS}, 8^{SP}$	pp. 7
26	The galaxy-formation constraint on a positive cosmological constant uses an expansion model, present density and the redshift by which gravitational condensation had already begun.	$2^S, 3^{OQ}, 4^{QB}$	pp. 7, 8
27	For a flat dust universe with positive constant vacuum density, Eq. (5.4) calculates elapsed time since formation from H_0 , the density ratio and a specified formation redshift.	$2^S, 3^{BQ}, 4^{QB}, 6^{IQ}$	pp. 8
28	Globular-cluster age estimates are inputs from independent stellar evidence, which Weinberg compares with ages calculated in cosmological models.	$6^{IQ}, 14^{IO}$	pp. 8
29	Constraints from galaxy counts as a function of redshift depend on assumptions about object distribution, detectability and temporal evolution of the luminosity distribution.	$2^S, 3^{OQ}, 4^{QB}, 8^S$	pp. 8, 9
30	A flat dust universe with a negative constant cosmological term recollapses in finite time along the specified expansion-contraction solution.	$2^S, 7^I$	pp. 9
31	The anthropic bound on a negative cosmological constant requires the recollapse time to exceed the time needed for life to appear, with a benchmark expressed in terms of the present Hubble time.	$4^{QB}, 7^{IQ}, 15^{IB}$	pp. 9
32	The conservation law along Friedmann time relates energy exchange among vacuum, matter and radiation. Specified exchange laws and initial values determine their density evolution.	$2^S, 8^S$	pp. 9
33	An adjustment mechanism requires a scalar field, sourced for example by the stress-tensor trace, to evolve to an equilibrium value permitting a flat solution.	$8^S, 17^C$	pp. 9, 10, 11

p	Claim	i	Source Location
34	The constant-field no-go argument assumes a constant metric and matter fields, the trace-equation relation and local field-space coordinates. In that domain, general covariance and the assumed symmetry leave the tuning required for a flat solution.	17 ^C	pp. 10, 11
35	When an anomaly term breaks the symmetry, scalar equilibrium and the Einstein equation for a constant metric impose different conditions.	17 ^C	pp. 10, 11
36	Source footnote 8 identifies invariance under combined translations and gauge transformations, and a foliation failing globally in field space, as possible departures from the constant-field no-go assumptions.	17 ^C	pp. 11
37	Variation at fixed metric determinant gives the trace-free part of the Einstein equation.	17 ^C	pp. 11, 12
38	Imposing stress-energy conservation and the Bianchi identities on the trace-free equation makes $R - 8\pi GT$ constant on a connected domain and recovers the Einstein equation with the cosmological term as an integration constant.	17 ^C	pp. 12
39	Converting the cosmological term into an integration constant does not by itself select its value.	$\emptyset$	pp. 12, 14
40	The relativistic-particle action uses a worldline parameter s and an einbein. Varying or integrating over the einbein imposes the mass-shell constraint; making it nondynamical does not automatically yield that constraint.	17 ^C , 18 ^S	pp. 12, 13
41	The ADM history integral uses a time slicing, spatial three-metric, conjugate momentum, lapse and shift. Integrating the corresponding Lagrange multipliers imposes constraints.	17 ^C , 19 ^S	pp. 13
42	The four-form strength of a three-form gauge field has a constant coefficient by its field equation and contributes to the cosmological term. The gauge potential itself need not be constant throughout spacetime.	17 ^C	pp. 11, 14
43	The Wheeler–DeWitt equation imposes on the wave functional of the spatial three-metric and matter fields the constraint obtained by integrating a Lagrange multiplier in the canonical representation.	19 ^S	pp. 13, 14

<i>p</i>	Claim	<i>i</i>	Source Location
44	The Euclidean history integral for the cosmological wave function integrates four-dimensional metrics and matter fields with specified three-dimensional boundary data. The Hartle–Hawking condition requires no other boundary.	9 ^{BS}	pp. 14, 15
45	Weinberg distinguishes the Euclidean history formalism from its physical interpretation in Lorentzian spacetime, identifying the unbounded action, integration contour, boundary conditions and probability normalization as problems.	9 ^{SB} , 17 ^C	pp. 14, 15
46	In asymptotically flat cosmologies, ordinary probabilistic questions concern field values at a specified definite time.	10 ^B	pp. 15
47	Weinberg uses the value α of a field $a(x, t)$ as local time and evaluates other fields on level surfaces satisfying $a(x, t(x, \alpha)) = \alpha$. The construction is defined where a local inverse and suitable level surfaces exist.	9 ^S , 11 ^{KB}	pp. 15
48	Equations (8.5)–(8.6) integrate histories containing a surface at local-clock value α and normalize the conditional probability of fields on that surface. The normalization factor N generally depends on α .	9 ^{SB} , 11 ^{KB} , 21 ^B	pp. 15, 16
49	The scalar expectation value in Eq. (8.7) averages over cosmological histories and the measurement time within each history.	9 ^{SB} , 12 ^{QO}	pp. 16
50	Weinberg computes $T[h] = \left(\int d^3x \sqrt{h}\right)^{1/2}$ from a three-metric of finite positive spatial volume and uses it as global configuration time together with other coordinates independent of the overall scale.	13 ^{KQ}	pp. 16
51	Expressing the Wheeler–DeWitt conservation law (8.8) in T and the remaining coordinates, and integrating with the measure (8.11), gives the flux conserved with respect to T in Eq. (8.12).	13 ^Q , 20 ^Q	pp. 16
52	The conserved flux with respect to global configuration time is not generally positive; Weinberg notes that Hartle–Hawking conditions do not make it a positive probability density. Using it as positive probability requires positivity and normalization for the chosen state and domain.	20 ^Q	pp. 16

p	Claim	i	Source Location
53	Hawking's proposal treats the three-form constant as an integration variable and obtains a weight for the cosmological term from Euclidean-history expectations and stationary effective actions. Concentration at zero cosmological term presupposes the validity of this construction and its normalization.	$9^{\text{SB}}, 12^{\text{Q}}$	pp. 16, 17
54	Coleman's proposal represents small-wormhole and baby-universe effects as shifts in coefficients of local action terms and obtains weights for couplings and the cosmological term through a four-dimensional history integral.	$9^{\text{SB}}, 12^{\text{Q}}$	pp. 17, 18, 19
55	Summing arbitrary numbers of closed connected components in Coleman's proposal exponentiates the single-component history integral and gives a double-exponential weight under the stationary-point evaluation.	9^{SB}	pp. 18, 19
56	Weinberg's four qualms concern the physical meaning of Euclidean quantum cosmology, boundary conditions, wormholes or three-form fields and the dominant stationary point, and coefficients of higher-derivative terms.	9^{SB}	pp. 19, 20
57	The two further qualms in the note added in proof concern the double exponential overcoming suppression of large wormholes and the phase of the gravitational integral inside the exponential eliminating the peak at zero cosmological term.	9^{SB}	pp. 20, 21
58	In his 1989 assessment, Weinberg regards all five approaches as still of interest and states that the fifth, based on quantum cosmology, appears the most promising.	$\emptyset$	pp. 20
59	Against the popular account attributing Einstein's retraction of the cosmological term solely to Hubble's discovery, Weinberg describes a more complex history including the de Sitter model and a 1923 letter.	$\emptyset$	pp. 1, 2

Formal Input Elements

Definition 8.358 (Cosmological Term and Clock Constructions). The combination of the bare cosmological term with the vacuum contribution, and the matter-clock surface

relation, are

$$\lambda_{\text{eff}} = \lambda + 8\pi G \langle \rho \rangle, \quad \rho_V = \langle \rho \rangle + \frac{\lambda}{8\pi G}, \quad a(\mathbf{x}, t(\mathbf{x}, \alpha)) = \alpha.$$

Here λ is the bare cosmological term, $\langle \rho \rangle$ the vacuum density, a the matter field used as a clock, and α its value. Global configuration time and its conserved flux are expressed in terms of the spatial three-metric h_{ij} and the remaining configuration coordinates ζ as

$$T[h] = \left(\int d^3x \sqrt{h} \right)^{1/2}, \quad \frac{d}{dT} \int J[d\zeta] \operatorname{Im} \left(\Psi^* \frac{\partial \Psi}{\partial T} \right) = 0$$

Here $h = \det(h_{ij})$, J is the measure factor in source Eq. (8.11), and Ψ is the cosmological wave function. The first relation uses finite positive spatial volume; the second uses the operator, measure and boundary conditions of Transfer 12. Claim 52 gives the conditions for interpreting the conserved flux as probability.

Table 259: Weinberg 1989 Time-Use Roles

Code	Role	Use Count
I	Interval	10
Q	Quantity	23
X	Cross-position correspondence	2
K	Clock reading	3
C	Causality	18
S	State evolution	34
P	Probabilistic evolution	1
O	Observation record	5
B	Reference boundary	25

Table 260: Weinberg 1989 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–2, 4, 10–11, 13, 22	member	instant
3	member	endpoint, order_relation
5	member	phase_boundary, order_relation
6–9, 12, 14–17, 20–21	dependency	$\emptyset$
18–19	member	instant, order_relation

Dependency Inputs The table gives each transfer’s time inputs, outputs, claim support and source location.

Dependency and Transfer Elements

Definition 8.359 (Weinberg 1989 Time-Content Transfers). Let $\text{ld}_{\mathbf{x}_R}^{\text{audit,transfer}} = [12]$.

Table 261: Weinberg 1989 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1	3	7	From de Sitter test-body motion to nearby redshift. pp. 2
2	1	22	8	From static de Sitter coordinates to Friedmann coordinates for the same model. pp. 2
3	2, 4	3	10	From scale factor and observation time to the redshift relation. pp. 3, 7, 8
4	2	4	11	From a scale-factor solution in cosmic time to the present expansion rate. pp. 2, 8, 9
5	2–4	6	27	From formation redshift and a cosmological model to elapsed time since formation. pp. 8
6	2	7	30	From an expansion–contraction solution to the recollapse time. pp. 9
7	2	8	32	From cosmic time and exchange laws to density evolution. pp. 9
8	9	11	47	From a field history to a level-surface local clock. pp. 15
9	9, 11	21	48	From local-clock surfaces and a history integral to conditional probability. pp. 15, 16
10	9	12	49	From a history integral to an expectation averaged over all times. pp. 16
11	$\emptyset$	13	50	From a spatial three-metric to computation of global configuration time. pp. 16
12	13	20	51	From global configuration time and the Wheeler–DeWitt conservation law to conserved flux. pp. 16

Grounding Conditions

1. Use the source’s test-body motion and emission–observation configuration in the approximation $Hr \ll 1$.
2. Apply Eq. (2.11) where the transformation is defined, representing the same de Sitter model with $k = 0$ and $\rho = 0$.
3. Use a solution of the same specified cosmological model and specified emission and observation events. A formation redshift, when used, is an observational input.
4. Specify the present time in the same model, evaluate $\dot{R}/R$, and take the reciprocal of nonzero H_0 .
5. Use a flat dust universe, positive constant vacuum density, and specified H_0 , density ratio and formation redshift.

6. Use a flat dust universe, a negative constant cosmological term, and the initial expansion and final collapse boundaries.
7. Specify the exchange law among the relevant components and initial values in addition to energy conservation.
8. For the chosen field history, $t(x, \alpha)$ is locally unique and suitable level surfaces exist in the domain used.
9. Use histories containing the specified clock surface, with the chosen contour, measure and boundary conditions, and require finite nonzero normalization under Eq. (8.6).
10. Evaluate numerator and denominator of Eq. (8.7) with the same measure and boundary conditions in a domain with finite nonzero denominator.
11. The positive-definite three-metric has finite positive spatial volume, and the parametrization by overall scale and remaining coordinates is defined in the domain used.
12. Use the source operator ordering, regular coordinates and measure, and the boundary conditions required for integration by parts.

Results

Definition 8.360 (Weinberg 1989 Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [21]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 262: Weinberg 1989 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Relation of the Lorentz-invariant vacuum to the effective cosmological term. pp. 1, 2	17	5, 12–13
2	Disparity between the observational bound and field-theory scales. pp. 1, 2, 3, 4, 6, 8	2–4, 17	14–15
3	Sensitivity of vacuum density to radiative corrections and cosmological phase transitions. pp. 3, 4, 6	5, 8, 17	15–16
4	Five approaches distinguished by Weinberg. pp. 1, 4, 6, 9, 11, 14, 20	$\emptyset$	21
5	Applicability conditions for supersymmetry, supergravity and no-scale models. pp. 4, 5, 6	17	18–20
6	Galaxy-formation constraint on a positive cosmological constant. pp. 7, 8	2–4	26
7	Cosmic-lifetime constraint on a negative cosmological constant. pp. 9	2, 4, 7, 15	30–31

q	Result and Source Location	Time Premises i	Claim Support p
8	Domain of the no-go argument for constant-field adjustment mechanisms. pp. 10, 11	17	34–36
9	Recovery of the Einstein equation from trace-free gravitational equations. pp. 11, 12	17	37–38
10	Value selection remaining after conversion to an integration constant. pp. 12, 14	$\emptyset$	39
11	Cosmological integration constant from a three-form gauge field. pp. 11, 14	17	42
12	Weight for the cosmological term in Hawking’s three-form proposal. pp. 16, 17	9, 12	53
13	Weights and applicability conditions for Coleman’s wormholes and disconnected components. pp. 17, 18, 19, 20, 21	9, 12	54–57
14	Constraints, Euclidean construction and physical interpretation of the cosmological wave function. pp. 13, 14, 15	9, 17, 19	43–45
15	Conditional probability on matter-clock surfaces. pp. 15, 16	9, 11, 21	47–48
16	Scalar expectation over histories and measurement times. pp. 16	9, 12	49
17	Global configuration time and its associated conserved flux. pp. 16	13, 20	50–52
18	Weinberg’s 1989 assessment of the candidate approaches. pp. 20	$\emptyset$	58
19	Weinberg’s historical account of Einstein’s retraction of the cosmological term. pp. 1, 2	$\emptyset$	59
20	Comparison of multipliers and constraints in particle mechanics and ADM gravity. pp. 12, 13	17–19	40–41
21	Relation between coordinate representations of the de Sitter model. pp. 1, 2	1, 3, 17, 22	6–8

Grounds for the Results

Table 263: Weinberg 1989 Conditions for the Results

Re- sult Index	Conditions and Domain
1	Use the adopted metric convention, a Lorentz-invariant vacuum and the Einstein equation.
2	Compare the cosmological observational conditions then available with estimates for specified fields and cutoff scales.
3	Specify the field models, temperature-dependent potential and epochs before and after the transition.
4	The classification in the five source sections.
5	Distinguish global and local symmetry, symmetry breaking, potential and kinetic choices, and perturbative order.
6	Use the redshift of existing condensations, present density, an expansion model and gravitational-condensation assumptions.
7	Compare a recollapsing cosmological model with an independently assumed lower bound on the time required for life.
8	Use constant fields, the trace relation and local field-space coordinates. The constructions outside these assumptions identified in footnote 8 lie outside this domain.
9	A connected domain satisfying stress-energy conservation and the Bianchi identities.
10	The distinct conditions of being a constant and selecting a particular value.
11	Use the specified field action and four-form field-strength equation.
12	The domain in which Euclidean histories, boundary conditions, a dominant stationary effective action and normalization are valid.
13	Use small dilute wormholes, a boundary state, a stationary point, bounded higher-derivative coefficients and normalization. The source retains four qualms and two further qualms in its note.
14	Use canonical constraints, specified boundaries and the history integral; physical interpretation requires a valid contour, measure and normalization.
15	Locally usable clock surfaces and finite nonzero normalization over histories containing them.
16	The chosen contour, measure and boundary in Eq. (8.7), with a finite nonzero denominator.
17	Finite positive spatial volume, a regular coordinate representation, the conservation law and boundary conditions. Probability interpretation requires positivity and normalization for the chosen state and domain.
18	Weinberg's comparison of the five approaches in Section IX.
19	The historical evidence cited by Weinberg, including the de Sitter model and the 1923 letter.
20	Use the worldline or ADM representation, specified action and integration variables. Weinberg leaves the need to integrate the determinant in gravity as an open question.

Re- sult Index	Conditions and Domain
----------------------	-----------------------

21	The coordinate transformation within the same de Sitter model and the source's test-body and observational configuration.
----	---

Evidence Hierarchy The cosmological-constant limits, expansion rates, galaxy-formation information and stellar ages come from observational sources cited by Weinberg. The theoretical constructions and applicability conditions of the five approaches use their respective field-model, cosmological-model and quantum-cosmological assumptions. The comparison in Section IX is Weinberg's assessment as of 1989.

Generation Judgment The complete LCTR generation count is zero; two time contents are locally reachable under source-evidence conditions, leaving 20 primitive-time roots. Global configuration time 13, defined from spatial volume, and its associated conserved flux 20 are locally reachable under the formal-evidence conditions in Claims 50–51.

Result Dependence Of the 21 results, 16 depend on primitive-time roots. Results 4,10,17,18 and 19 are reachable from the source formal-evidence environment. No result is generated from nontemporal physical inputs alone. There are 26 root–result dependence pairs, of which 24 are necessary.

Evidentiary Effect Weinberg presents a comparison of vacuum contributions with observational limits, the constructions, conditions and unresolved problems of five approaches, and his comparative assessment as of 1989. The galaxy-formation and cosmic-lifetime bounds use chosen cosmological models and observational inputs; the constant-field argument has the domain specified in Claims 34–36. Conservation of flux with respect to global configuration time and its interpretation as a positive probability density have the distinct conditions in Claims 51–52.

Placement of the Grounding Obligation Weinberg's arguments presuppose temporal structure in Lorentzian spacetime, cosmic time, emission–observation order, cosmological epochs, stellar, biological or particle lifetimes, worldline or ADM time, or cosmological histories. The 87 claim/time-content use pairs and 12 transfers supply no conditions generating the remaining 20 time contents from nontemporal physical inputs before use. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Weinberg discusses the conditions and unresolved problems of the models. Actual failure of specific physical-time generation conditions, and consequent changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 26 root–result dependence pairs. Actual reach from the failure location of a specific generation condition to a result remains undetermined.

Result-Preservation Domain If the models, field actions, observational inputs, boundary and integration conditions and each result’s domain are fixed, and generated temporal representations realize the 16 time-dependent results’ input relations with the same temporal order, intervals and metric structure, each result is preserved with its stated epistemic status and domain. Result 17 uses the conditions in Claims 50–52 for its configuration coordinate and conserved flux; Results 4,10,18 and 19 use their respective grounds for classification, the value-selection problem, assessment and historical account.

Repair Scope If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusion

Definition 8.361 (Weinberg 1989 Finite Dependency Graph). The source dependency graph comprises 59 claims, 22 time contents, 12 transfers and 21 results. Claims 1–4 are the nontemporal initial inputs for generation.

Proposition 8.362 (Weinberg 1989 Primitive-Time Roots and Result Dependence). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [22]; \emptyset, \{4, 10, 17, 18, 19\}, [21] \setminus \{4, 10, 17, 18, 19\}, \emptyset).$$

$R \in \text{Ref}^{\text{audit,rootdep}}$ and $R \notin \text{Ref}^{\text{audit,rootunreach}}$.

Proof. Closure from Initial Claims 1–4 remains those four nodes. Supplying all 59 claims as source formal evidence reaches Time Contents 13 and 20 through Transfers 11–12 and reaches Results 4,10,17,18 and 19. Adding the 20 remaining time contents as seeds reaches all 21 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.363 (Weinberg 1989 Minimal Primitive-Time-Root Support). The table gives each result’s minimal primitive-time-root support family. There are 22 family members in total. For each of Results 4,10,17,18 and 19, the empty set is the sole family member.

Table 264: Weinberg 1989 Minimal Primitive-Time-Root Support

Result Index	Minimal-Support Index Family
1	$\{17\}$
2	$\{2, 17\}$
3	$\{2, 5, 17\}, \{5, 8, 17\}$
4	$\emptyset$
5	$\{17\}$
6	$\{2\}$
7	$\{2, 15\}$
8	$\{17\}$
9	$\{17\}$

Result Index	Minimal-Support Index Family
10	$\emptyset$
11	$\{17\}$
12	$\{9\}$
13	$\{9\}$
14	$\{9, 17, 19\}$
15	$\{9\}$
16	$\{9\}$
17	$\emptyset$
18	$\emptyset$
19	$\emptyset$
20	$\{17, 18, 19\}$
21	$\{1, 17\}$

Proof. Start from singleton time-content seeds and Transfer 11, whose temporal input is empty, and apply Definition 7.46 to the 12 transfers. From the stable antichains, combine supports for each result's temporal premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Proposition 8.364 (Weinberg 1989 Time-Use Audit). Complete generation is zero and local reach is two, leaving 20 primitive-time roots. The 21 results divide into five evidence-reachable and 16 primitive-time-root-dependent results. The 59 claims, 22 time contents, 12 transfers and 21 results give 114 source-specific obligation atoms. The 87 claim/time-content use pairs have 121 role assignments.

Proof. This follows from the definitions of claims, time contents, transfers and results and the reachable sets and minimal-support families above. $\square$

Proposition 8.365 (Weinberg 1989 Audit-Boundary Counts). $\text{BoundaryCount}_R^{\text{audit}} = (102, 33)$.

Proof. The 59 claims, 22 time contents and 21 results give 102 nodes; the 12 transfers and 21 result dependencies give 33 edges. $\square$

Individual Statements Statement Evidence 1

Verbatim Original (English):

theoretical expectations for the cosmological constant exceed observational limits by some 120 orders of magnitude.

The original-source location is Weinberg, *The Cosmological Constant Problem, Reviews of Modern Physics* 61 (1989), p. 1, Section I. The minimal context is the passage that introduces the cosmological-constant problem as a crisis and states the disparity between theoretical expectations and observational limits. The statement is attributed to Steven Weinberg.

Statement Evidence 2

Verbatim Original (English):

All of the five approaches to the cosmological constant problem described in Secs. IV–VIII remain interesting.

The original-source location is Weinberg, *The Cosmological Constant Problem, Reviews of Modern Physics* 61 (1989), p. 20, Section IX. The minimal context is the passage that, after comparing five approaches, states that each remains under consideration. The statement is attributed to Steven Weinberg.

Statement Evidence 3

Verbatim Original (English):

At present, the fifth, based on quantum cosmology, appears the most promising.

The original-source location is Weinberg, *The Cosmological Constant Problem, Reviews of Modern Physics* 61 (1989), p. 20, Section IX. The minimal context is the passage that presents the approach based on quantum cosmology as a comparative assessment, in 1989, among the five approaches. The statement is attributed to Steven Weinberg.

Common Evidence with Specified Domain of Application The three statements above give the verbatim originals from Weinberg 1989, their original-source locations, the minimal contexts required for interpretation, and their source attribution.

Target state-ment	Object of the statement	Predicate	Explicit domain of application
S28-Q001	The theoretical expectation for the cosmological constant and the observational limit	States that the theoretical expectation exceeds the observational limit by some 120 orders of magnitude.	The domain introduced in Section I that compares the theoretical scales adopted by the source with the observational limit as of 1989.
S28-Q002	The five approaches to the cosmological-constant problem considered in Sections IV–VIII	States that all five approaches remain interesting.	The comparative summary in Section IX, in 1989, of the five approaches distinguished by the source.
S28-Q003	The fifth approach, based on quantum cosmology	States that it appears the most promising in 1989.	The 1989 comparative assessment assigned to the fifth of the five approaches in Section IX.

Individual Audits

First Statement Weinberg states an approximately 120-order disparity between vacuum contributions at the field-theoretic scales adopted in the source and observational limits as of 1989. The grounds are the vacuum estimates in Sections II–III and the bound obtained by comparing redshift, expansion-rate and density information with a cosmological model. The corresponding evidence is Time Contents 2–4 and 17, Claims 14–15 and Result 2.

Second Statement Weinberg assesses all five approaches examined in Sections IV–VIII as candidates still worthy of consideration in 1989. This assessment is his summary in Section IX of their constructions, conditions and remaining problems. The corresponding evidence is Claims 21 and 58 and Results 4 and 18.

Third Statement In his 1989 comparison of the five approaches, Weinberg assesses the fifth, based on quantum cosmology, as apparently the most promising. The constructions in Hawking’s and Coleman’s proposals in Section VIII, together with their remaining qualms, are the specific subjects of that comparison. The corresponding evidence is Claims 53–58 and Results 12,13 and 18; the ranking itself corresponds to Claim 58 and Result 18.

Common Audit with Specified Domain of Application The first statement compares theoretical scales with observational limits; the second and third express Weinberg’s comparative assessment of five approaches. Their domains are determined by the models and observational information specified in the source and its assessment as of 1989.

8.19.29 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 29.

Source Name Isham 1993

Bibliographic Identification C. J. Isham, *Canonical Quantum Gravity and the Problem of Time*, *NATO ASI Series C* 409, pp. 157–287 (1993); arXiv:gr-qc/9210011. $R = \text{Ref}_{\text{Ish93POT}}$ denotes this source. The audit covers time uses, construction equations and arguments in the abstract and Sections 1–7.

Original-Source Location Identifier 29-L1.

Original-Source Location Content The tables give text-page numbers in arXiv:gr-qc/9210011.

Audit-Scope Identifier 29-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Formal Objects The audit uses 45 time contents, 84 claims, 27 transfers and 43 results. The tables specify the constructions and conditions of ordinary quantum theory, relativistic spacetime, ADM constraints, internal clocks, reference fluids, Wheeler–DeWitt interpretations, WKB time, conditional probabilities, histories and evolving constants. Isham examines the problem of time in canonical quantum gravity, starting from the relation between the external time used in ordinary quantum theory and the dynamical spacetime metric of general relativity. He discusses the constructions and obstacles of internal-time reduction, material clocks, unimodular time, Wheeler–DeWitt interpretations, WKB time, conditional probabilities, histories and evolving constants.

Assumptions Each approach uses specified metrics, canonical variables and states, together with constraints and boundary conditions. Quantum evolution with an internal clock requires a clock domain, constraint root, operator domain, inner product and integrability. WKB constructions use a selected phase and branch and an approximation order; history probabilities use state, coarse-graining and decoherence conditions. Each transfer and result states the conditions of its approach.

Nontemporal Physical Inputs Claims 1–4 are the initial inputs for generation: spatial geometry and canonical data, functions and coefficients, states and operators, and boundary and coarse-graining data. The conditions under which each approach assigns temporal meaning and operations to these data are given in the corresponding claims and transfers.

Time Contents

Definition 8.366 (Isham 1993 Time Contents). Let $n_R = 45$. The table defines the time contents and their source locations.

Table 266: Isham 1993 Time Contents

i	Time Content	Source Location
1	History from the primordial universe to the present classical universe	pp. 5, 79, 103
2	External time and equal-time instants in ordinary quantum theory	pp. 10, 11, 12, 91
3	Quantum-state evolution along external time	pp. 10, 11, 84, 85, 86
4	Temporal ordering of quantum measurements and records	pp. 10, 90, 96, 97, 98
5	Causal and temporal structure of Lorentzian spacetime	pp. 9, 12, 13, 14, 15, 16, 26, 28, 90, 102
6	Time coordinate in an inertial frame	pp. 10, 11
7	Asymptotic time reference	pp. 14, 35, 74
8	Cosmic time of a preferred foliation	pp. 15, 50
9	Clock worldline and specified endpoint events	pp. 15, 16, 59, 60
10	Proper-time interval determined by a metric and worldline	pp. 15, 16
11	Spacelike-hypersurface embeddings and their order	pp. 22, 23, 24, 26, 43, 63
12	Quantum-field evolution between hypersurfaces in a fixed background	pp. 23, 24
13	Killing time for Klein–Gordon positive-frequency decomposition	pp. 72, 73, 74, 75
14	Selected internal-time function and clock sections	pp. 37, 38, 39, 40, 41, 42, 43, 46, 47, 48, 49, 53, 54, 55, 56, 57, 58, 59
15	Time parameter of an ADM foliation	pp. 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35

<i>i</i>	Time Content	Source Location
16	Proper-time interval in the ADM normal direction	pp. 28, 29
17	ADM metric–momentum evolution and hypersurface deformation	pp. 27, 28, 29, 30, 31, 32, 33, 34, 35
18	Reduced quantum evolution along internal time	pp. 43, 45, 47, 48, 49, 54, 55, 56, 57
19	Scale factor selected as a minisuperspace clock	p. 51
20	Geometric momentum selected as a minisuperspace clock	p. 51
21	Matter field selected as a minisuperspace clock	p. 51
22	Reduced evolution with the scale-factor clock	pp. 51, 56
23	Reduced evolution with the geometric-momentum clock	pp. 51, 56
24	Reduced evolution with the matter-field clock	pp. 51, 56
25	York time defined from canonical geometric data	pp. 51, 52
26	Reduced quantum evolution with respect to York time	pp. 47, 48, 51, 52, 56, 59
27	Clock field of a Gaussian reference fluid	pp. 59, 60, 61, 62
28	Readings of point clocks and clock clouds and correspondence between worldlines	pp. 59, 60
29	Formal quantum evolution with respect to a Gaussian clock field	pp. 60, 61, 62
30	Oriented four-volume time between embeddings	p. 63
31	Volume-time evolution in the unimodular model	p. 63
32	Intrinsic time direction selected in general superspace	pp. 71, 73, 74, 75, 90
33	Logarithmic configuration time defined from a positive scale factor	pp. 70, 71, 74
34	Klein–Gordon flux conserved between general superspace hypersurfaces	pp. 73, 74
35	Minisuperspace flux conserved with respect to logarithmic configuration time	pp. 74
36	Time reference for interpreting transitions in third quantization	pp. 75, 76, 77
37	WKB time constructed along the phase flow	pp. 79, 80, 81, 82, 83
38	Matter-state evolution along WKB time	pp. 80, 81, 82, 83
39	Lorentzian spacetime histories used in the gravity model of generalized histories	pp. 100, 101, 102, 103, 104
40	Clock readings referred to in conditional probabilities	pp. 92, 93, 94, 95
41	Probability of a physical quantity conditioned on a clock reading	pp. 92, 93, 94, 95
42	Event times and ordering in conventional histories	pp. 96, 97, 98, 99, 100
43	Clocks and temporal relations in quasiclassical histories	pp. 99, 100, 103, 104
44	Clock sections used to construct evolving constants	pp. 104, 105, 106, 107
45	Relational change of evolving constants indexed by clock readings	pp. 104, 105, 106, 107

Definition 8.367 (Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(84)$, with claim boundary $\text{ClaimBoundaryI}dx_R = ([4], [84])$. Table 267 gives each claim, its time uses and its source locations.

Formal Input Elements

Definition 8.368 (Internal-Time Constructions). York time and logarithmic configuration time are defined by

$$T_Y(x; g, p) = \frac{2}{3}|g|^{-1/2}(x)p^a{}_a(x), \quad \Omega = \ln a \quad (a > 0)$$

. Here g_{ab} is a positive-definite three-metric, $|g| = \det(g_{ab})$; p^{ab} is its canonical momentum, $p^a{}_a = g_{ab}p^{ab}$; and a is the minisuperspace scale factor. For a WKB solution of the Wheeler–DeWitt equation of the form

$$\Psi[g] = A[g] \exp\left(\frac{iS[g]}{\hbar\kappa^2}\right), \quad \kappa^2 = \frac{8\pi G}{c^2},$$

the quantity $A[g]$ is a positive real amplitude, $S[g]$ is a real phase, G is Newton’s constant, c is the speed of light, and $\hbar$ is the reduced Planck constant. For an amplitude satisfying

$$\hbar\kappa^2 \left| \frac{\delta A[g]}{\delta g_{ab}} \right| \ll \left| A[g] \frac{\delta S[g]}{\delta g_{ab}} \right|$$

, the lowest-order phase satisfies the Hamilton–Jacobi equation. WKB time is constructed from that selected phase $S[g]$ and DeWitt metric G_{abcd} through

$$2G_{abcd}(x, g) \frac{\delta S[g]}{\delta g_{ab}(x)} \frac{\delta T(x', g)}{\delta g_{cd}(x)} = \delta(x, x')$$

on a domain where it and the comoving variables form a valid coordinate chart. The lowest-order matter equation is then a Schrödinger equation in this time.

Table 267: Isham 1993 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Differentiable manifolds, spatial topology, positive-definite three-metrics, canonical momenta and values of matter fields on space.	$\emptyset$	pp. 26, 30, 40, 50
2	Specified coefficients, coupling constants, potentials, phase functions and constraint functions on canonical variables.	$\emptyset$	pp. 30, 41, 51, 79, 80
3	State vectors, density operators, system and environment degrees of freedom, and state data for clock-candidate matter and reference fields.	$\emptyset$	pp. 10, 23, 24, 59, 60, 61, 85, 86, 87

p	Claim	i	Source Location
4	Specified field values on boundaries, sets of alternatives, coarse-graining maps and sets of histories.	$\emptyset$	pp. 69, 96, 97, 100, 101, 102
5	Isham distinguishes identifying time before quantization, identifying it after quantization, and omitting time from the fundamental formalism.	$\emptyset$	pp. 5, 9, 16, 17, 18, 19, 107, 108
6	Isham organizes ten approaches and eight problem groups, including global clocks, multiple choice, Hilbert space, reconstruction of the spatial metric, spacetime-coordinate character, functional evolution and quantum constraints. Isham states that interpretative difficulties concerning time also arise in minisuperspace models without ultraviolet divergences.	$\emptyset$	pp. 8, 17, 18, 19, 20, 108
7	Isham treats the transition from the early universe to the present classical spacetime as something quantum cosmology must explain.	1^{SB}	pp. 5, 79, 103
8	In ordinary quantum theory, external time t labels states, while observables are represented by operators. The Hamiltonian determines state evolution through the Schrödinger equation.	$2^{\text{S}}, 3^{\text{S}}$	p. 10
9	With a time-independent inner product and an appropriate self-adjoint Hamiltonian, unitary evolution preserves inner products between states.	$2^{\text{S}}, 3^{\text{Q}}$	pp. 10, 24, 25
10	Ordinary measurement probabilities are determined from a state at a specified time, and records of measurement sequences use temporal ordering.	$2^{\text{OB}}, 4^{\text{O}}$	p. 10
11	The obstruction to a perfect clock described by Isham arises when a self-adjoint Hamiltonian bounded below, normalizable reading states and exclusively forward clock transitions are required together.	$2^{\text{P}}, 3^{\text{P}}$	p. 11
12	If a self-adjoint time operator canonically conjugate to the Hamiltonian implements energy translations over the entire real line, the Hamiltonian spectrum is not bounded below.	2^{Q}	p. 12
13	Relativistic quantum fields use Lorentzian causal structure and inertial-frame time, with local commutativity imposed on fields in spacelike-separated regions.	$5^{\text{C}}, 6^{\text{S}}$	pp. 10, 11, 13, 14

p	Claim	i	Source Location
14	In general relativity the metric is dynamical and the theory is invariant under diffeomorphisms preserving the applicable boundary conditions. Stating local commutativity of fields requires a metric specifying spacelike separation between events.	5 ^C	pp. 12, 13, 14
15	In an asymptotically flat system, asymptotic boundary structure supplies time translations and temporal reference for scattering. Isham focuses mainly on the problem of time for compact spaces or systems in which boundary data play no fundamental role.	5 ^C , 7 ^{BS}	pp. 14, 35
16	In models with cosmological symmetry, a preferred foliation labels spatial slices by cosmic time.	5 ^C , 8 ^{SB}	pp. 15, 50
17	The proper-time interval between the endpoints of a specified timelike worldline is obtained by integrating the Lorentzian metric along that worldline.	5 ^C , 9 ^{IB} , 10 ^{IQ}	pp. 15, 16
18	An operator interpretation of proper time in quantized gravity requires identification of the worldline and endpoints and raises questions of metric-operator ordering, domain and measurement meaning.	5 ^C , 9 ^B , 10 ^{IQ}	p. 16
19	On a fixed globally hyperbolic background, specified spacelike embeddings and the field Hamiltonian densities define functional evolution between hypersurfaces.	5 ^C , 11 ^{SB} , 12 ^S	pp. 22, 23, 24
20	Quantum evolution with a hypersurface-dependent inner product also requires compensation for the changing inner product; symmetry of the Hamiltonian at each instant alone does not imply norm conservation.	11 ^B , 12 ^Q	pp. 24, 25
21	A Lorentzian metric distinguishes timelike, spacelike and null directions, and timelike vectors with a common time orientation form a convex cone.	5 ^C	pp. 13, 14, 16, 34
22	In an ADM foliation, lapse N determines the normal proper-time increment $N dt$, while the shift relates spatial labels on adjacent slices.	5 ^C , 15 ^{IB} , 16 ^{IQ}	pp. 27, 28, 29

p	Claim	i	Source Location
23	The time derivative of the spatial metric, lapse and shift determine extrinsic curvature, which together with the three-metric determines canonical momentum. Hamilton equations give the evolution of these canonical data.	$15^{\text{SB}}, 17^{\text{S}}$	pp. 28, 29, 30, 31
24	The Poisson brackets of ADM constraints have metric-dependent structure functions. Momentum constraints generate spatial diffeomorphisms, while the Hamiltonian constraint generates normal deformations on solutions.	$15^{\text{SB}}, 17^{\text{S}}$	pp. 32, 33, 34
25	Geometric constraints holding for every spacelike embedding recover the specified vacuum Einstein equations.	$5^{\text{C}}, 15^{\text{B}}, 17^{\text{S}}$	pp. 34, 35
26	In a compact system without boundary, the Hamiltonian is a combination of constraints. Relational change through an internal clock is formulated separately from quantities constant on constraint orbits.	$2^{\text{S}}, 15^{\text{SB}}, 17^{\text{S}}$	pp. 30, 31, 32, 33, 34, 65, 66, 104, 105, 106
27	Internal coordinates use an invertible canonical chart consisting of a time function and spatial labels. Level surfaces of the time function must be spacelike, and curves at fixed spatial labels must be timelike.	14^{KB}	pp. 37, 38, 39, 40, 41, 42, 43
28	Solving the constraints for the momenta conjugate to internal coordinates and specifying the true Hamiltonian and quantum-operator domain yields reduced Schrödinger evolution in that internal time.	$14^{\text{S}}, 18^{\text{S}}$	pp. 41, 42, 43, 46, 47, 48, 49
29	Global use of internal time in quantum theory requires existence of the canonical transformation, uniqueness of the constraint solution, clock monotonicity and self-adjoint domains for the required operators.	14^{B}	pp. 53, 54, 55, 56
30	Even when different internal clocks give classically equivalent descriptions, unitary equivalence of the corresponding quantum theories is a separate condition.	14^{K}	pp. 54, 55
31	Path independence of functional evolution requires integrability of the constraint operators and an appropriate common domain. Isham also gives a two-dimensional free-field example in which the anomaly is removed.	$14^{\text{S}}, 18^{\text{S}}$	pp. 55, 56, 57

p	Claim	i	Source Location
32	Interpreting internal time as a spacetime coordinate requires the time assigned to the same event to be independent of the reference foliation. The metric-only intrinsic clocks examined by Isham fail this condition.	$5^C, 14^{XK}$	pp. 58, 59, 75, 82
33	Isham's minisuperspace example uses a homogeneous isotropic preferred foliation, scale factor a , scalar field ϕ and their conjugate momenta. Constraint roots and admissible clock domains differ with the clock choice.	$8^B, 15^S$	pp. 49, 50, 51
34	On the $t = a$ branch, solving the constraint for p_a gives the reduced Hamiltonian, using a real root and a monotonic clock domain.	$19^S, 22^S$	p. 51
35	On the $t = p_a$ branch, solving the constraint for a gives the reduced Hamiltonian, using a positive scale factor and the domain of the selected root.	$20^S, 23^S$	p. 51
36	On the $t = \phi$ branch, solving the constraint for p_ϕ gives the reduced Hamiltonian. The potential, reality of the root and clock monotonicity specify its domain.	$21^S, 24^S$	p. 51
37	York time is defined from the three-metric and canonical momentum as a function proportional to mean extrinsic curvature.	25^{KQ}	pp. 51, 52
38	Using York time and its conjugate volume variable and solving the remaining constraints yields a reduced Schrödinger equation in York time on the specified domain.	$25^S, 26^S$	pp. 47, 48, 51, 52
39	Isham distinguishes intrinsic time defined from the metric, extrinsic time using the metric and momenta, and time using matter fields, each with its own clock-admissibility conditions.	$14^K, 19^K, 20^K, 21^K, 25^K$	pp. 49, 50, 51, 53
40	The conformal metric has determinant one and the conformal momentum is traceless; these fields are not independent canonical pairs before their constraints are resolved.	$\emptyset$	p. 52
41	Differentiating a first-order evolution equation with a time-dependent Hamiltonian adds a term containing the time derivative of the Hamiltonian.	$14^S, 18^S$	p. 55

p	Claim	i	Source Location
42	Material clocks are physical systems coupled to gravity. Local readings of point clocks and extended event identification by clock clouds carry their respective worldline and reference-field conditions.	$9^B, 28^{KX}$	pp. 59, 60
43	A Gaussian reference fluid uses a clock field with unit timelike gradient and orthogonal spatial reference fields. A constraint linear in clock momentum yields a formal Schrödinger equation in the clock field.	$27^{SK}, 29^S$	pp. 60, 61, 62
44	The Gaussian reference fluid examined by Isham fails energy conditions, and focusing of reference curves limits the coordinate domain.	27^B	p. 62
45	In reference systems constrained by energy conditions for physical clock matter, linearity of the constraint in clock momentum is lost.	$27^K, 28^{KX}$	p. 62
46	In the unimodular model, the Lorentzian spacetime volume between two specified embeddings is calculated as an oriented time variable τ .	$5^C, 11^{IB}, 30^{IQ}$	p. 63
47	Using volume time conjugate to the cosmological constant yields a formal Schrödinger equation for the unimodular-model state.	$30^S, 31^S$	p. 63
48	Residual constraints remain in the unimodular model, and embeddings giving the same four-volume are not unique, so volume time alone does not determine a probability interpretation for an unrestricted three-metric.	$11^B, 30^{BQ}, 31^S$	pp. 63
49	Quantizing positive-definite three-metrics requires compatibility between metric positivity, canonical translations and operator self-adjointness. Isham also examines affine relations and logarithmic variables.	$\emptyset$	pp. 64, 65, 69, 70
50	Dirac quantization imposes quantum constraints on physical states. Even when external-time dependence disappears in a compact system, a temporal interpretation of Wheeler–DeWitt solutions requires an internal clock and an inner product.	2^S	pp. 65, 66, 67, 68, 69, 71

p	Claim	i	Source Location
51	For $a > 0$ in minisuperspace, $\Omega = \ln a$ is a configuration variable ranging over the real line. The corresponding Wheeler–DeWitt equation has hyperbolic form with Omega as the timelike direction.	33^{KQ}	pp. 70, 71, 74
52	For the particle Klein–Gordon equation, an appropriate timelike Killing flow and potential conditions permit a positive inner product on positive-frequency solutions of the conserved indefinite pairing.	$5^{\text{C}}, 13^{\text{S}}$	pp. 72, 73
53	The Klein–Gordon-type flux of Wheeler–DeWitt solutions is conserved under deformations of the specified spacelike hypersurface in superspace.	$32^{\text{B}}, 34^{\text{Q}}$	pp. 73, 74
54	For the specified minisuperspace equation, the Klein–Gordon-type pairing on constant-Omega surfaces is conserved in Omega.	$33^{\text{B}}, 35^{\text{Q}}$	pp. 74
55	For the DeWitt metric and potential treated by Isham, no Killing vector satisfies the required conditions, obstructing the positive-frequency Hilbert-space construction by this method. The asymptotically flat case is treated separately.	$13^{\text{S}}, 32^{\text{B}}$	pp. 74, 75
56	Isham presents third quantization as a proposal to turn the Wheeler–DeWitt wavefunction into a field operator on superspace. Interpreting creation and destruction of universes or topology change raises the problems of constructing a one-universe Hilbert space and temporal reference, and interpreting states of multiple universes.	$32^{\text{SB}}, 36^{\text{S}}$	pp. 75, 76, 77
57	The semiclassical equations coupling classical gravity to quantum matter relate the Einstein tensor of a classical Lorentzian metric to the renormalized expectation value of the matter energy–momentum tensor.	5^{C}	pp. 77, 78
58	Substituting a WKB form with real phase S and slowly varying amplitude A into the Wheeler–DeWitt equation gives a Hamilton–Jacobi equation and an amplitude conservation equation at the specified lowest order.	$\emptyset$	pp. 79, 80

p	Claim	i	Source Location
59	Solving Eq. (5.4.11) along the nondegenerate flow of a selected real phase S and combining it with comoving variables into a valid coordinate chart gives the WKB time function T . In a superposition of WKB branches, each Hamilton–Jacobi phase defines its own time.	37^{KQ}	pp. 80, 81, 82, 83
60	In WKB time and comoving variables, the lowest-order matter-state equation becomes a Schrödinger equation in WKB time, with probability conserved to the same approximation order.	$37^{\text{S}}, 38^{\text{S}}$	pp. 80, 81
61	In the WKB expansion presented by Isham, the simple first-order Schrödinger form of the lowest-order equation is lost at the next order.	$37^{\text{S}}, 38^{\text{S}}$	pp. 82, 83
62	The domain of this WKB time is the selected phase branch and the domain of the semiclassical approximation; the same construction does not apply to a state lacking that domain.	37^{B}	pp. 79, 81, 82, 83
63	In ordinary decoherence models, correlations formed between system and environment through time evolution, together with near orthogonality of environment states, suppress interference terms in the reduced density operator. Application to quantum cosmology requires choosing which modes to retain as the system and which to treat as the environment; Isham states that there seems to be no general way to determine that split.	$2^{\text{S}}, 3^{\text{S}}$	pp. 84, 85, 86, 87
64	The reduced density operator is calculated by tracing over the environment. Unitary transformation preserves the purity of a pure total density operator, consistently with a mixed state of a subsystem.	$\emptyset$	pp. 85, 86, 87
65	An argument that uses the Schrödinger inner product obtained after choosing a particular WKB branch to justify choosing that same branch is circular.	$37^{\text{B}}, 38^{\text{S}}$	pp. 87, 88
66	A probability interpretation based on a positive configuration-space inner product requires an appropriate measure and normalization; constraint solutions in a continuous spectrum need not be normalizable in that kinematical inner product.	$\emptyset$	pp. 89, 90, 91

p	Claim	i	Source Location
67	The configuration-space wave-packet interpretation described by Isham uses the set of spatial metrics embeddable in a specified classical Lorentzian spacetime, rather than a state on one time slice.	$5^C, 32^B$	pp. 89, 90, 91
68	The conditional probability in Eq. (6.2.9) is calculated from the specified state and projectors when its denominator is nonzero.	$\emptyset$	pp. 92, 93, 94
69	Interpreting that probability as a physical quantity conditioned on a clock reading requires a specified clock observable, reading and corresponding projector. Repeated occurrences of the same reading require separate event identification.	$40^{KB}, 41^Q$	pp. 92, 93, 94, 95
70	A clock observable that changes relative to a constraint does not commute with the constraint operator, and projection onto a reading need not preserve the constraint condition on physical states.	40^K	pp. 91, 94, 95
71	In Page's proposal as presented by Isham, past histories are interpreted through correlations among present records.	$40^O, 42^O$	pp. 95
72	Conventional consistent histories construct a class operator for each history from projectors ordered by external times and state evolution.	$2^{SB}, 42^{SB}$	pp. 96, 97, 98
73	The decoherence functional evaluates interference among complete history alternatives; probabilities of coarse-grained histories are additive for a family satisfying the consistency condition.	42^B	pp. 97, 98, 99, 100
74	The probability axioms of generalized histories specify a set of histories, coarse graining and a decoherence functional, with conditions including positivity, normalization and additivity.	$\emptyset$	pp. 100, 101
75	The gravity implementation of generalized histories integrates histories of Lorentzian metrics and matter fields, with state information included in boundary conditions.	$5^C, 39^{SB}$	pp. 101, 102, 103, 104

p	Claim	i	Source Location
76	This gravitational history integral requires consistent gauge fixing, measure and boundary conditions, as well as a definition of quantum gravity. Perturbative nonrenormalizability obstructs that perturbative construction. Isham points out that conventional gauge fixing in gravity involves choosing an internal time and raises the risk that constructing the decoherence functional first requires solving the problem of time.	39 ^B	pp. 102, 103, 104
77	Isham identifies finding consistent history families that yield a quasiclassical domain from specified initial conditions and coarse graining as an unresolved task for explaining clock and record relations.	39 ^B , 43 ^{KO}	pp. 99, 100, 103, 104
78	The construction of evolving constants uses clock sections $T = t$ that intersect constraint orbits uniquely, with a nonzero clock Poisson bracket on the domain.	44 ^{KB}	pp. 104, 105, 106, 107
79	Rovelli's evolving constant F_t commutes with the constraint for each t and agrees with the original quantity F on the clock section $T = t$. The family obtained by varying the clock reading expresses relational change.	44 ^K , 45 ^S	pp. 104, 105, 106, 107
80	Quantizing evolving constants requires operator ordering, a common domain and a physical Hilbert space, together with correspondence to the classical clock-section construction.	44 ^B , 45 ^S	pp. 106, 107
81	Isham identifies unresolved problems in the approaches and discusses his reasons for favouring generalized histories and the possibilities of future theories. Isham contrasts preserving the classical role of time in quantum theory with obtaining time as a phenomenological emergence from a different fundamental framework.	$\emptyset$	pp. 107, 108
82	One internal time is defined by $t = a$ using the minisuperspace canonical variable a .	19 ^{KQ}	pp. 50, 51
83	One extrinsic time is defined by $t = p_a$ using the minisuperspace canonical variable p_a .	20 ^{KQ}	pp. 50, 51
84	One matter time is defined by $t = \phi$ using the minisuperspace matter variable ϕ .	21 ^{KQ}	pp. 50, 51

Definition 8.369 (Semantic Components and Generation-Initial Set). The 84 claims are

the semantic components. Claims 1–4 form the generation-initial set; Claims 1–84 form the source formal-evidence environment.

Table 268: Isham 1993 Time-Use Roles

Code	Role	Use Count
I	Interval	7
Q	Quantity	17
X	Cross-position correspondence	3
K	Clock reading	23
C	Causality	16
S	State evolution	54
P	Probabilistic evolution	2
O	Observation record	5
B	Reference boundary	37

Table 269: Isham 1993 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1	member	phase_boundary, order_relation
2, 8, 11, 15, 28, 32, 40, 42, 44	member	instant, order_relation
3, 5, 10, 12, 16–18, 22–24, 26, 29, 31, 34–36, 38–39, 41, 43, 45	dependency	$\emptyset$
4	member	order_relation
6, 13, 19–21, 25, 27, 33, 37	member	instant
7, 14	member	instant, endpoint
9	member	endpoint, order_relation
30	member	instant, orientation

Dependency Inputs The table gives each transfer’s time inputs, outputs, claim support and source location.

Definition 8.370 (Isham 1993 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [27]$.

Table 270: Isham 1993 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	2	3	8–9	State evolution along external time. p. 10
2	2	4	10	Measurement records ordered by time. p. 10
3	5, 9	10	17	Proper-time integration along a worldline. pp. 15, 16

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
4	5, 11	12	19–20	Functional evolution on a fixed background. pp. 23, 24
5	5, 15	16	22	Calculation of ADM normal proper time. pp. 28, 29
6	15	17	23–24	Evolution of ADM canonical data. pp. 30, 31, 32, 33, 34
7	14	18	28, 31, 41	General reduced evolution in internal time. pp. 43, 45, 47, 48, 49, 54, 55, 56, 57
8	19	22	34	Reduced Hamiltonian for the scale-factor clock. p. 51
9	20	23	35	Reduced Hamiltonian for the geometric-momentum clock. p. 51
10	21	24	36	Reduced Hamiltonian for the matter clock. p. 51
11	$\emptyset$	25	37	Construction of York time from canonical geometry. pp. 51, 52
12	25	26	38, 40	Reduced evolution along York time. pp. 47, 48, 51, 52, 56, 59
13	27	29	43	Formal evolution in a Gaussian clock. pp. 60, 61, 62
14	5, 11	30	46	Calculation of oriented four-volume time. p. 63
15	30	31	47	Unimodular volume-time evolution. p. 63
16	$\emptyset$	33	51	Logarithmic configuration time from the scale factor. pp. 70, 71, 74
17	32	34	53	Flux conservation in general superspace. pp. 73, 74
18	33	35	54	Flux conservation in logarithmic configuration time. pp. 74
19	$\emptyset$	37	58–59	Construction of WKB time from the phase flow. pp. 79, 80, 81, 82, 83
20	37	38	60	Matter evolution along WKB time. pp. 80, 81
21	40	41	68–69	Conditioning probability on a clock reading. pp. 92, 93, 94, 95
22	2	42	72	Class operators for conventional histories. pp. 96, 97, 98
23	5	39	75	Gravitational integral over Lorentzian spacetime histories. pp. 101, 102, 103, 104
24	44	45	78–79	Evolving-constant family from clock sections. pp. 104, 105, 106, 107
25	$\emptyset$	19	82	Definition of internal time using the scale factor. p. 51
26	$\emptyset$	20	83	Definition of internal time using geometric momentum. p. 51

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
27	$\emptyset$	21	84	Definition of internal time using a matter field. p. 51

Grounding Conditions

1. Specified Hamiltonian, state, inner product and operator domain.
2. Measurement times, states, observables and measurement order.
3. Timelike worldline, identified endpoints and Lorentzian metric.
4. Global hyperbolicity, spacelike embeddings, field operators, measure and domains.
5. Spacelike foliation, positive lapse and unit timelike normal.
6. Specified constraints, boundary conditions, lapse and shift.
7. Invertible canonical transformation, selected constraint solution, quantum domain and integrability of functional evolution.
8. $a > 0$, selected real root and admissible clock domain.
9. $V = 0$ in the source example, positive scale factor and selected constraint root.
10. Specified potential, selected real root and clock monotonicity domain.
11. Positive-definite three-metric, canonical momentum and Eq. (4.2.21).
12. Resolve conformal and remaining constraints and specify the volume-momentum solution and operator domain.
13. Unit timelike gradient, orthogonal reference fields and a constraint linear in clock momentum.
14. Specified embeddings, metric, orientation and existence of the volume integral.
15. Volume-time conjugacy, residual constraints and state domain.
16. $a > 0$, $\Omega = \ln a$ and the specified minisuperspace representation.
17. Wheeler–DeWitt solutions, selected hypersurface, operator ordering, measure and boundary conditions.
18. Specified minisuperspace equation and boundary conditions on constant- Ω surfaces.
19. Selected real phase and approximation domain, nondegenerate local flow and valid comoving coordinates.
20. Lowest-order Born–Oppenheimer expansion, matter Hamiltonian and inner product at the same order.

21. Clock projector, projector for the compared quantity, state and nonzero denominator.
22. External time sequence, complete projector alternatives and state evolution.
23. Specified domain of Lorentzian metrics and matter histories, measure and boundary state.
24. Unique orbit intersection, nonzero clock bracket and specified classical quantity F .
25. Minisuperspace scale factor a and the definition $t = a$.
26. Minisuperspace momentum p_a and the definition $t = p_a$.
27. Minisuperspace matter variable ϕ and the definition $t = \phi$.

Results

Definition 8.371 (Isham 1993 Results and Dependencies). Let $\text{ld}\mathbf{x}_R^{\text{audit,result}} = [43]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 271: Isham 1993 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Evolution, measurement and perfect-clock conditions in ordinary quantum theory. pp. 10, 11, 12, 24, 25	2–4	8–12
2	Temporal structures of relativistic quantum fields and general relativity. pp. 10, 11, 12, 13, 14, 15, 16, 34, 35, 50	5–8	13–16, 21
3	Construction of proper time and conditions for its quantum-operator interpretation. pp. 15, 16	5, 9–10	17–18
4	Functional quantum evolution on a fixed background. pp. 23, 24, 25	5, 11–12	19–20
5	ADM constraints, hypersurface deformation and frozen formalism. pp. 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37	2, 5, 15–17	22–26
6	Three classes, ten approaches and eight problem groups. pp. 5, 8, 9, 16, 17, 18, 19, 20, 107, 108	$\emptyset$	5–6
7	Conditions for reduced quantum theory using an internal clock. pp. 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 53, 54, 55, 56, 57, 58, 59, 75	5, 14, 18	27–32, 41
8	Minisuperspace clock choices and the scope of the approximation. pp. 49, 50, 51, 53	8, 14–15, 19–21, 25	33, 39

q	Result and Source Location	Time Premises i	Claim Support p
9	Reduced Hamiltonian for the scale-factor clock. p. 51	19, 22	34, 82
10	Reduced Hamiltonian for the geometric-momentum clock. p. 51	20, 23	35, 83
11	Reduced Hamiltonian for the matter clock. p. 51	21, 24	36, 84
12	Construction of York time and reduced evolution. pp. 47, 48, 51, 52, 56, 59	25–26	37–38, 40
13	Physical reference conditions for point clocks and clock clouds. pp. 59, 60	9, 28	42
14	Formal Schrödinger equation for a Gaussian reference fluid. pp. 60, 61, 62	27, 29	43
15	Energy-condition and coordinate-domain obstructions for Gaussian clocks. p. 62	27–28	44–45
16	Unimodular volume time and residual constraints. p. 63	5, 11, 30–31	46–48
17	Operator conditions for quantizing positive-definite metrics. pp. 64, 65, 69, 70	$\emptyset$	49
18	Wheeler–DeWitt constraints and external-time dependence. pp. 65, 66, 67, 68, 69, 71	2	50
19	Logarithmic configuration time in minisuperspace. pp. 70, 71, 74	33	51
20	Positive-frequency interpretation of a Klein–Gordon particle. pp. 72, 73	5, 13	52
21	General Wheeler–DeWitt flux and obstruction to the positive-frequency method. pp. 73, 74, 75	13, 32, 34	53, 55
22	Conserved flux in minisuperspace. pp. 74	33, 35	54
23	Third-quantization proposal and its interpretive conditions. pp. 75, 76, 77	32, 36	56
24	Semiclassical equations coupling classical gravity and quantum matter. pp. 77, 78	5	57
25	Phase and amplitude equations from the lowest WKB order. pp. 79, 80	$\emptyset$	58
26	WKB time from a selected phase and its domain. pp. 79, 80, 81, 82, 83	37	59, 62
27	WKB matter evolution and the equation at the next order. pp. 80, 81, 82, 83	37–38	60–61
28	Interference suppression through system–environment evolution. pp. 84, 85, 86, 87	2–3	63
29	Partial trace and purity of the total system. pp. 85, 86, 87	$\emptyset$	64
30	Circular branch selection using a branch-specific inner product. pp. 87, 88	37–38	65

q	Result and Source Location	Time Premises i	Claim Support p
31	Measure and normalization conditions for configuration-space probabilities. pp. 89, 90, 91	$\emptyset$	66
32	Wave-packet interpretation of classical spacetime. pp. 89, 90, 91	5, 32	67
33	Static conditional-probability formula. pp. 92, 93, 94	$\emptyset$	68
34	Clock-conditioned probabilities and constraint-state conditions. pp. 91, 92, 93, 94, 95	40–41	69–70
35	History interpretation using present records. pp. 95	40, 42	71
36	Consistency and probability additivity for conventional histories. pp. 96, 97, 98, 99, 100	2, 42	72–73
37	Probability axioms for generalized histories. pp. 100, 101	$\emptyset$	74
38	Gravitational history integral using Lorentzian metrics. pp. 101, 102, 103, 104	5, 39	75–76
39	Task of constructing history families that explain quasiclassical clocks. pp. 99, 100, 103, 104	39, 43	77
40	Classical evolving constants based on clock sections. pp. 104, 105, 106, 107	44–45	78–79
41	Conditions for quantizing evolving constants. pp. 106, 107	44–45	80
42	Unresolved problems and outlook in Isham’s review. pp. 107, 108	$\emptyset$	81
43	Transition to a classical universe to be explained by quantum cosmology. pp. 5, 79, 103	1	7

Grounds for the Results

Table 272: Isham 1993 Conditions for the Results

Re- sult Index	Conditions and Domain
1	External times, state/operator domains, inner product and the assumptions of each clock theorem.
2	Lorentzian metric and the respective inertial-frame, asymptotic-boundary and cosmological-foliation settings.
3	The same worldline and endpoints, metric and quantum-operator domain.

Re- sult Index	Conditions and Domain
4	Specified spacelike embeddings, measure and operator/inner-product conditions.
5	Constraints in the compact boundaryless case and the source's specified extensions and boundary conditions.
6	Isham's classification within the review.
7	Selected clock and canonical transformation, constraint solution, and operator/inner-product domains.
8	Specified homogeneous model and the constraint roots and domains for each clock.
9	$a > 0$, real root and clock monotonicity domain.
10	The $V = 0$ example, positive scale factor and selected root.
11	Specified potential, real root and clock monotonicity domain.
12	Positive-definite three-metric and momentum, conformal constraints, remaining constraint solution and operator domain.
13	Gravitational coupling of matter and domains of the worldlines and reference fields.
14	Unit timelike gradient, orthogonal fields and linear clock constraint.
15	Reference matter and geodesic congruence examined in the source.
16	Lorentzian metric, specified embeddings, conjugacy and residual constraints.
17	Specified canonical, affine or logarithmic representation and its domain.
18	Constraint operators and physical-state domains in the compact system.
19	$a > 0$ and the specified Wheeler–DeWitt representation.
20	Suitable Killing flow, preserved potential, positivity and domain conditions.
21	The source's DeWitt metric, potential and superspace hypersurface.
22	Solutions of Eq. (5.2.13) and boundary conditions on constant- Ω surfaces.
23	Field representation on superspace, state space and temporal reference.
24	Classical metric, quantum state and renormalized stress tensor.
25	Real phase, slowly varying amplitude, and specified operator ordering and expansion.
26	Nondegenerate flow, valid comoving coordinates, selected branch and approximation domain.
27	The Born–Oppenheimer expansion and approximation orders specified in the source.
28	Specified state, interaction, system/environment split and environment-state overlaps.
29	Specified tensor-product decomposition and trace-class state.
30	An argument using an inner product obtained after branch selection to justify that same selection.
31	Specified measure, constraint spectrum and state normalizability.
32	Spatial metrics embeddable in the specified Lorentzian spacetime and support of the state.
33	Specified state, projectors and nonzero denominator.

Re- sult Index	Conditions and Domain
34	Clock reading/projector, event identification and constraint operator.
35	Page's proposal as presented by Isham and correlations among records.
36	Complete alternatives, time ordering and decoherence functional.
37	History set, coarse graining and an axiom-satisfying decoherence functional.
38	Specified boundary state, measure, gauge fixing and construction domain of quantum gravity.
39	The source's problem concerning initial state, coarse graining and a quasiclassical domain.
40	Unique clock section, nonzero clock bracket and specified quantity F.
41	Operator ordering, common domain and physical Hilbert space.
42	The scope of the source's comparison and Isham's assessment.
43	The relation between the primordial universe and present classical spacetime.

Definition 8.372 (Result-Dependency Hyperedges). Each result edge takes the temporal premises and claim support in Table 271 as inputs and outputs the corresponding result node.

Evidence Hierarchy Isham reviews earlier quantum-gravity approaches and examines their definitions, derivations, conditions and unresolved problems. The source reports zero directly measured objects and zero direct-measurement predicates from measurements of its own. Proposals belong to their respective proponents; the comparative assessments in the review are Isham's.

Generation Judgment The complete LCTR generation count is zero; 12 time contents are locally reachable under source-evidence conditions, leaving 33 primitive-time roots. Local reach includes the internal times defined using scale factor, geometric momentum and matter field; York time; logarithmic configuration time; WKB time; and the reduced evolution or conserved flux satisfying the respective conditions. The corresponding time contents are 19–26, 33, 35, 37–38.

Result Dependence Of the 43 results, 26 depend on primitive-time roots and 17 are reachable from the source formal-evidence environment. The latter are Results 6, 9–12, 17, 19, 22, 25–27, 29–31, 33, 37, 42. No result is generated from nontemporal physical inputs alone. There are 47 root–result dependence pairs, of which 43 are necessary.

Evidentiary Effect Isham presents three classes of approaches to the problem of time and their specific constructions, conditions and obstacles. Internal-clock definitions, reduced evolution, positive-frequency interpretations, WKB approximations, conditional probabilities and histories have the distinct conditions stated in the result table. Identifying history families that yield quasiclassical clocks remains an unresolved construction task in the source.

Placement of the Grounding Obligation External instants, Lorentzian causal structure, worldlines and endpoints, foliations, material clocks, clock sections and history ordering provide temporal premises for the respective approaches. None of the 45 time contents belongs to the generation closure from nontemporal physical inputs alone, and 33 roots remain even after the local constructions under source formal evidence. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Isham discusses global clock obstructions, energy conditions for Gaussian reference fluids, the obstruction to the positive-frequency method and changes at the next WKB order. Actual failure of the physical-time generation conditions audited by LCTR, and consequent changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 47 root–result dependence pairs. Actual reach from the failure location of a specific physical-time generation condition to a result is undetermined by the source evidence.

Result-Preservation Domain If the constraints, states, operators, inner products, boundary conditions, clock domains and approximation orders of each approach are fixed, and generated temporal representations realize the time-dependent results’ input relations with the same ordering, intervals and clock correspondences, each result is preserved with its stated domain and epistemic status. Classifications, static operator and probability expressions, formally conditioned clock constructions and Isham’s comparative assessments use their respective claim support.

Repair Scope The repair targets are the time contents used by each approach and the physical-time generation conditions that must be satisfied before their use. If actual failure of a specific physical-time generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.373 (Primitive-Time Roots and Result Dependence in the Isham 1993 Source). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = (\emptyset, [45]; \\ \emptyset, \llbracket 9, 12 \rrbracket \cup \llbracket 25, 27 \rrbracket \cup \llbracket 29, 31 \rrbracket \cup \{6, 17, 19, 22, 33, 37, 42\}, \\ [43] \setminus (\llbracket 9, 12 \rrbracket \cup \llbracket 25, 27 \rrbracket \cup \llbracket 29, 31 \rrbracket \cup \{6, 17, 19, 22, 33, 37, 42\}), \emptyset). \end{aligned}$$

$$R \in \text{Ref}^{\text{audit, rootdep}}, \quad R \notin \text{Ref}^{\text{audit, rootunreach}}.$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Supplying all 84 claims as source formal evidence reaches 12 time contents through Transfers 11, 16, 19 and 25–27, whose temporal inputs are empty, and their successors. This closure contains 17 results. Adding the remaining 33 time contents as seeds reaches all 43 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.374 (Minimal Primitive-Time-Root Support in the Isham 1993 Source). The table gives each result's minimal-support family, with 45 family members in total. For each result reachable from the evidence environment alone, the empty set is the sole family member.

Proof. Start from singleton time-content seeds and apply Definition 7.46 to the 27 transfers. From the stable antichains, combine supports for each result's temporal premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Table 273: Minimal Primitive-Time-Root Support in the Isham 1993 Source

Result Index	Minimal-Support Index Family
1	$\{2\}$
2	$\{5, 6, 7, 8\}$
3	$\{5, 9\}$
4	$\{5, 11\}$
5	$\{2, 5, 15\}$
6	$\emptyset$
7	$\{5, 14\}$
8	$\{8, 14, 15\}$
9	$\emptyset$
10	$\emptyset$
11	$\emptyset$
12	$\emptyset$
13	$\{9, 28\}$
14	$\{27\}$
15	$\{27, 28\}$
16	$\{5, 11\}$
17	$\emptyset$
18	$\{2\}$
19	$\emptyset$
20	$\{5, 13\}$
21	$\{13, 32\}$
22	$\emptyset$
23	$\{32, 36\}$
24	$\{5\}$
25	$\emptyset$
26	$\emptyset$
27	$\emptyset$
28	$\{2\}$
29	$\emptyset$
30	$\emptyset$
31	$\emptyset$
32	$\{5, 32\}$
33	$\emptyset$

Result Index	Minimal-Support Index Family
34	$\{40\}$
35	$\{2, 40\}, \{40, 42\}$
36	$\{2\}$
37	$\emptyset$
38	$\{5\}$
39	$\{5, 43\}, \{39, 43\}$
40	$\{44\}$
41	$\{44\}$
42	$\emptyset$
43	$\{1\}$

Proposition 8.375 (Audit Conclusion for the Isham 1993 Source). Complete generation is zero and local reach is 12, leaving 33 primitive-time roots after local reach. The 43 results divide into 17 evidence-reachable and 26 primitive-time-root-dependent results. The 84 claims, 45 time contents, 27 transfers and 43 results give 199 source-specific obligation atoms. The 130 claim/time-content use pairs have 164 role assignments.

Proof. This follows from the definitions of claims, time contents, transfers and results, the reachable sets and the minimal-support families. $\square$

Proposition 8.376 (Audit-Boundary Counts for the Isham 1993 Source).

$\text{BoundaryCount}_R^{\text{audit}} = (172, 70)$.

Proof. The 84 claims, 45 time contents and 43 results give 172 nodes; the 27 transfers and 43 result dependencies give 70 edges. $\square$

Individual Statements

Objects of the Statements A definition of time using only the canonical variables of the gravitational field; physical clocks made of matter; the scale that fixes the lowest order of the WKB expansion; and the first-order Schrödinger form that is lost at the next WKB order.

Verbatim Originals (English) a satisfactory definition of time using only the canonical variables of the gravitational field

Real physical clocks are made of matter with definite properties

to lowest order in an expansion in powers of $\hbar\kappa^2 \simeq (L_P)^2$ (where $L_P := (G\hbar/c^3)^{1/2}$ is the Planck length)

The elegant first-order form of the Schrödinger equation (5.4.10) is lost at the next order in the WKB approximation. This makes it difficult to assess the proper status of (5.4.10) and to understand how the physics of time changes as one gets nearer to the Planck regime.

The third quotation states that the lowest order is taken in an expansion in powers of $\hbar\kappa^2 \simeq (L_P)^2$, where $L_P := (G\hbar/c^3)^{1/2}$ is the Planck length.

Original-Source Locations The first and second quotations are in §4.3.1, p. 59; the third is between Eqs. (5.4.3)–(5.4.5) in §5.4.2, p. 79; the fourth is the fifth item in §5.4.4, p. 82. Page numbers refer to the text of arXiv:gr-qc/9210011.

Minimal Context In the sentence containing the first quotation, Isham states the difficulty of obtaining a satisfactory time definition from the gravitational canonical variables alone, then turns to using matter to define time. The second quotation states that real clocks are made of matter with definite properties. The third specifies the expansion scale used to obtain the lowest-order Hamilton–Jacobi equation from the WKB form; the fourth discusses the change in the equation at the next order of that WKB scheme.

Attributed Author of the Statements All four quotations are attributable to the text of the review by C. J. Isham.

Content Stated Directly by the Quotations The first quotation names the object whose construction Isham describes as difficult: a satisfactory time definition using only gravitational canonical variables. The second states the material constitution of physical clocks; the third specifies the lowest WKB order; the fourth states that the first-order Schrödinger form is lost at the next WKB order.

Explicit Domains of Application of the Statements The first and second quotations concern internal time and material clocks in canonical quantum gravity. The third and fourth apply to the Wheeler–DeWitt equation, phase branch and WKB expansion orders used in the source.

Evidence Provenance C. J. Isham, *Canonical Quantum Gravity and the Problem of Time*, arXiv:gr-qc/9210011, pp. 59, 79, 82.

Individual Audits

First Statement Isham states the difficulty of obtaining a satisfactory time definition from the gravitational canonical variables alone. Section 4.2 examines canonical transformations for internal clocks, uniqueness of constraint solutions, inequivalent clock choices after quantization, functional evolution and foliation-independent event times. Claims 27–41 and Results 7–12 give the specific constructions, conditions and obstacles.

Second Statement Isham describes real physical clocks as systems made of matter with definite properties. Section 4.3 examines coupling between clock matter and gravity, point clocks and clock clouds, and clock constraints in a Gaussian reference fluid. Claims 42–45 and Results 13–15 give the formal Schrödinger derivation, energy conditions and reference-coordinate domain.

Third Statement Isham specifies the lowest WKB order using a scale related to the gravitational coupling and Planck length. Isham specifies a dimensionless expansion parameter formed from the coupling and a scale of the system. Claims 58–60 and Results 25–27 give the Hamilton–Jacobi equation from the selected real phase, WKB time and matter evolution.

Fourth Statement Isham states that the simple first-order Schrödinger form of the lowest-order equation is lost at the next order of the specified WKB expansion. The time function defined by Eq. (5.4.11) is a local coordinate along the flow of the selected real phase; the simple WKB approximation breaks down at turning points of that phase. Claims 59–62 and Results 26–27 give the time construction and the approximation order and domain of the evolution equation.

Common Audit with Specified Domain of Application The formal grounds and domains of the four statements are determined by the internal-clock, material-clock and WKB conditions in the corresponding claims, transfers and results.

8.19.30 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 30.

Source Name Kuchař 1992

Bibliographic Identification Karel V. Kuchař, *Time and Interpretations of Quantum Gravity, Proceedings of the 4th Canadian Conference on General Relativity and Relativistic Astrophysics*, World Scientific (1992), pp. 211–314. $R = \text{Ref}_{\text{Kuc92TIQG}}$ denotes this source. The audit covers time uses, construction equations and arguments in the abstract and Sections 1–16.

Original-Source Location Identifier 30-L1.

Original-Source Location Content The tables give the page numbers printed in the source text.

Audit-Scope Identifier 30-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Formal Objects The audit uses 47 time contents, 82 claims, 25 transfers and 38 results. The tables specify the constructions and conditions of ordinary quantum theory, relativistic spacetime, ADM constraints, internal clocks, reference fluids, Wheeler–DeWitt interpretations, WKB time, conditional probabilities, histories and evolving constants. Kuchař examines the problems of time arising in quantization, starting from hypersurfaces, constraints and embedding representations in canonical general relativity. After comparing the background structures, preferred foliations and degrees of freedom of eleven models, he classifies ten interpretations into three frameworks: internal time, Wheeler–DeWitt theory, and interpreting quantum theory without requiring time. He presents constructions, conditions and obstacles for internal clocks and reference fluids, four-volume time, Klein–Gordon inner products, WKB time, third quantization, conditional probabilities, histories and evolving constants.

Assumptions Each approach uses its specified canonical variables, states, constraints, boundary conditions and operator domains. Reduced evolution in an internal clock requires the domain of the selected clock, a constraint root and inner-product conditions. Heat flow and energy conditions enter the Gaussian reference-fluid construction; inner-product sign

and symmetry enter the Klein–Gordon interpretation; phase branch, transverse surface and approximation order enter the WKB construction. Conditional probabilities and histories use their respective conditions on states, projectors, detectors and normalization.

Nontemporal Physical Inputs Claims 1–4 are the initial inputs for generation: spatial geometry and canonical data, phase-space functions and coefficients, states and operators, boundary values, and data for detectors and coarse graining of paths. The conditions under which each approach assigns temporal meaning and operations to these data are given in the corresponding claims and transfers.

Time Contents

Definition 8.377 (Kuchař 1992 Time Contents). Let $n_R = 47$. The table defines the time contents and their source locations.

Table 274: Kuchař 1992 Time Contents

i	Time Content	Source Location
1	Temporal ordering in prediction from past data to the future	pp. 1
2	Spacelike hypersurfaces representing instants and their spacetime embeddings	pp. 1, 2, 3, 4, 5, 6, 7, 15, 19, 25, 31, 39, 40, 41, 42, 65, 89
3	Hamilton–Jacobi evolution of classical canonical data	pp. 1, 2, 3, 4, 5, 83
4	Embedding-time variables extracted by canonical separation	pp. 3, 4, 6, 21
5	External time in Newtonian mechanics and ordinary quantum theory	pp. 4, 8, 12, 13, 67, 68, 70, 75, 77, 80
6	Clock functions and sections selected in the internal-time framework	pp. 5, 7, 20, 21, 22, 27, 28, 29, 30, 31, 32
7	Functional quantum-state evolution between embeddings	pp. 6, 7, 15, 16, 17, 18, 19, 21, 89
8	Time of a selected foliation or a model’s preferred foliation	pp. 8, 9, 10, 19, 20, 60
9	Temporal structure of background spacetimes in particle and field models	pp. 9, 10, 11, 12, 13, 14, 43, 44
10	Logarithmic volume clock defined from the spatial metric	pp. 10, 22, 23, 27, 28
11	Reduced quantum evolution in a model’s logarithmic volume clock	pp. 10, 22, 23, 27, 28
12	Anomaly-free functional evolution of a massless field on a two-dimensional cylinder	pp. 15, 16, 17, 18
13	Reduced quantum evolution in a selected internal time	pp. 21, 22, 28, 29, 30
14	Internal clocks using fundamental matter fields	pp. 22, 38, 39
15	Extrinsic time defined from momentum in linearized gravity	pp. 23, 24
16	Quantum evolution in extrinsic time in linearized gravity	pp. 23, 24, 25

<i>i</i>	Time Content	Source Location
17	Embedding time defined from momentum and boundary time in the cylindrical model	pp. 24, 25, 26
18	Asymptotic boundary time in the cylindrical model	pp. 25
19	Quantum evolution in embedding time in the cylindrical model	pp. 25, 26
20	York time defined from canonical geometric data	pp. 26, 27, 28, 31, 32
21	Formal reduced quantum evolution in York time	pp. 26, 27, 28, 31, 32
22	Expansion, contraction and time-reversed geometric branches	pp. 27, 28, 46, 47, 55, 56, 57, 58
23	Clock worldlines and event ordering in material reference systems	pp. 32, 33, 34, 35, 65
24	Clock field and equal-clock surfaces of a Gaussian reference fluid	pp. 32, 33, 34, 35, 36, 37, 38
25	Formal quantum evolution in a Gaussian clock field	pp. 35, 36, 37, 38
26	Oriented four-volume time from a reference embedding	pp. 39, 40, 41, 42
27	Unimodular quantum evolution in four-volume time	pp. 41, 42
28	Superspace hypersurfaces on which the Klein–Gordon pairing is evaluated	pp. 43, 45, 46, 47
29	Timelike Killing flow used in the positive-frequency construction	pp. 43, 44, 45, 46
30	Klein–Gordon flux conserved between spacetime hypersurfaces for a relativistic particle	pp. 43, 44
31	Klein–Gordon flux conserved between hypersurfaces in superspace	pp. 45, 46, 47
32	WKB time constructed from the Hamilton–Jacobi phase flow	pp. 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59
33	Classical trajectory congruence and transverse surface used as the WKB-time reference	pp. 49, 50, 54, 55, 56, 57, 58
34	Quantum-state evolution and norm conservation along WKB time	pp. 49, 50, 51, 52, 53, 54, 58, 59
35	Homogeneous-mode clock selected in a semiclassical cosmological model	pp. 59, 60, 61
36	Past and future time references of incoming and outgoing Fock constructions	pp. 64
37	Measurement time at which detectors are jointly prepared	pp. 65
38	Transition probability asked for between two clock readings	pp. 68, 74, 75, 76
39	Temporal relations in questions about change in the state of the universe	pp. 68, 89, 90
40	Clock-reading projector family constructed from the clock Hamiltonian	pp. 70, 71, 72, 73
41	Single-time probability conditioned on a clock reading	pp. 70, 71, 72, 73
42	Time parameter indexing the conditional-clock projector family	pp. 72, 73, 75
43	Proper time used in the cosmological scattering interpretation	pp. 76

i	Time Content	Source Location
44	Label time used for propagation and integration of history amplitudes	pp. 77, 78, 79, 80, 81, 82
45	Temporal correlations of detector responses calculated from history amplitudes	pp. 77, 78, 79, 80, 81, 82
46	Global clock sections used to construct evolving constants	pp. 84, 85, 86, 87, 88
47	Change of evolving constants indexed by clock readings	pp. 84, 85, 86, 87, 88

Definition 8.378 (Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(82)$, with claim boundary $\text{ClaimBoundaryId}_R = ([4], [82])$. Table 275 gives each claim, its time uses and its source locations.

Formal Input Elements

Definition 8.379 (Internal-Time Constructions). For a single WKB branch, let Q^A denote the heavy configuration variables, $S(Q)$ the selected Hamilton–Jacobi phase, and G^{AB} the inverse configuration-space metric. The phase flow $S^A = G^{AB}\partial_B S$ defines WKB time on a domain where it is nondegenerate and the transverse surface Σ meets each trajectory once, through

$$S^A \partial_A t = 1, \quad t|_{\Sigma} = 0$$

. At the specified leading approximation order, the light degrees of freedom satisfy a Schrödinger equation along this time. The inner product is conserved under the stated time-dependent measure, operator-domain and boundary-flux conditions (text pp. 49–54).

For classical evolving constants, let the constraint H generate orbits that meet each clock section $T = t$ exactly once, with $\{T, H\} \neq 0$ on the domain. For a given classical quantity F the family F_t satisfies

$$\{F_t, H\} = 0, \quad F_t|_{T=t} = F|_{T=t},$$

$$\left. \frac{\partial F_t}{\partial t} \right|_{t=T(Q,P)} = \frac{\{F, H\}}{\{T, H\}}(Q, P).$$

Here (Q, P) is a point in canonical phase space; $\{ , \}$ denotes the Poisson bracket. A fixed t labels a quantity constant along a constraint orbit. Whether the family varies with the clock reading depends on the original quantity F (text pp. 84–85).

Table 275: Kuchař 1992 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Differentiable spatial manifolds and their topology, positive-definite three-metrics, canonical momenta, matter fields including reference fluids, and their conjugate variables.	$\emptyset$	pp. 1, 2, 32, 38

p	Claim	i	Source Location
2	Functions on phase space, algebraic components of constraints, actions and Hamilton–Jacobi equations, coupling constants and potentials.	$\emptyset$	pp. 2, 3, 4, 10, 12, 48
3	Configuration variables including true gravitational degrees of freedom, quantum states, operators, inner products and data specified as boundary values.	$\emptyset$	pp. 3, 6, 7, 43, 69
4	Detector states, regions of configuration space, projection operators, sets of paths and coarse graining of histories by region-crossing conditions.	$\emptyset$	pp. 65, 69, 77, 78, 79
5	Prediction of the future by physical laws uses the temporal ordering between past states and the states to be predicted.	1^{SB}	pp. 1
6	Canonical general relativity takes the metric and momentum associated with extrinsic curvature on a spacelike hypersurface as instantaneous data; the supermomentum and super-Hamiltonian constraints determine admissible data.	2^{B}	pp. 1, 2, 3
7	The super-Hamiltonian constraint generates geometrical change between spacelike hypersurfaces.	$2^{\text{S}}, 3^{\text{S}}$	pp. 2, 3, 83
8	Kuchař identifies the disappearance of explicit hypersurface reference from the constraints as the starting point of the problem of time in canonical quantum gravity.	2^{B}	pp. 3
9	Assuming a canonical transformation separating embedding variables from true gravitational degrees of freedom, solving the constraints for the embedding momenta yields constraints linear in those momenta.	$2^{\text{B}}, 4^{\text{K}}$	pp. 3, 4
10	A complete solution of the Hamilton–Jacobi equations corresponding to the separated constraints yields solutions of Einstein’s equations for the specified embeddings.	$2^{\text{B}}, 3^{\text{S}}, 4^{\text{S}}$	pp. 3, 4, 5
11	Under classical constraint closure, Hamilton–Jacobi evolution is independent of the foliation joining the same initial and final hypersurfaces.	$2^{\text{B}}, 3^{\text{S}}$	pp. 4, 5

p	Claim	i	Source Location
12	Kuchař regards the existence of a global canonical separation in full general relativity as unresolved and suspects its absence. The global-time problem concerns this global separation and the existence of clock sections.	6^B	pp. 5, 27, 28, 89
13	The sandwich problem asks whether the intrinsic geometries of initial and final hypersurfaces uniquely determine the spacetime joining them.	2^B	pp. 5, 89
14	Dirac quantization imposes the constraints as operator equations on physical states; in the metric representation, the super-Hamiltonian constraint becomes the Wheeler–DeWitt equation.	$\emptyset$	pp. 6
15	The quantum constraints in the embedding representation give a functional Schrödinger equation in embedding time.	$4^S, 7^S$	pp. 6, 21
16	Quantum constraint anomalies can obstruct the consistency of functional evolution and make state propagation along different foliations inequivalent.	$2^B, 7^S$	pp. 6, 7, 15, 16, 17, 18, 19
17	Quantization with different internal clocks can yield inequivalent quantum theories even when the corresponding classical descriptions are equivalent.	$6^K, 13^S$	pp. 7, 28, 29, 30
18	The Hilbert-space problem is to equip the solution space of the Wheeler–DeWitt equation with the positive-definite inner product needed for a probabilistic interpretation.	$\emptyset$	pp. 7, 43, 45
19	Kuchař distinguishes functional evolution, multiple choice and Hilbert-space construction as problems of time concerning different conditions of validity.	$\emptyset$	pp. 7, 8, 89
20	The eleven models in the source have features such as preferred foliations, background structures and finitely many degrees of freedom that differ from full general relativity. A solution in a finite-dimensional model does not include a solution of the functional-evolution problem of many-fingered time.	$5^S, 7^S, 8^B, 9^C$	pp. 8, 9, 10, 11, 12, 13, 14

p	Claim	i	Source Location
21	Among schemes choosing a spatial-metric functional as intrinsic time, the source's homogeneous cosmological models define a logarithmic spatial-volume coordinate as the clock.	10^{KQ}	pp. 10, 22, 23
22	The positive-definite kinetic energy of Barbour–Bertotti-type models and model restrictions on realistic clocks do not transfer unchanged to general relativity with its indefinite gravitational kinetic term.	$5^{\text{S}}, 9^{\text{C}}$	pp. 12, 13, 14
23	For a massless field on a flat two-dimensional cylinder, adding the embedding-dependent anomaly potential removes the constraint-commutator anomaly displayed in the source.	$2^{\text{B}}, 12^{\text{S}}$	pp. 15, 16, 17, 18
24	Kuchař classifies ten competing interpretations into three frameworks—internal time, Wheeler–DeWitt theory, and interpreting quantum theory without requiring time—and notes overlap between their boundaries. The third framework allows time to emerge in particular situations but does not require time for the probabilistic interpretation itself.	$\emptyset$	pp. 19, 20, 89, 90
25	The internal Schrödinger interpretation separates the clock from true degrees of freedom, solves the constraints for the clock momenta, and then quantizes with respect to that clock. Operators self-adjoint in the conserved inner product are observables, whose value probabilities are evaluated on a specified hypersurface. These observables need not be constants of motion.	$4^{\text{S}}, 6^{\text{K}}, 13^{\text{S}}$	pp. 21, 22
26	Solving the quadratic constraint for the momentum conjugate to the logarithmic-volume clock and specifying a root and operator domain puts the source's homogeneous cosmological model into Schrödinger form in that clock.	$10^{\text{S}}, 11^{\text{S}}$	pp. 22, 23
27	In linearized gravity, an extrinsic-time variable is defined from the transverse component of canonical momentum and the inverse Laplacian relative to a flat spatial background.	15^{KQ}	pp. 23, 24

p	Claim	i	Source Location
28	In linearized gravity, the constraints become linear in embedding momenta at the specified perturbative order and yield a functional Schrödinger equation in extrinsic time.	15 ^S , 16 ^S	pp. 23, 24, 25
29	Einstein–Rosen time in the cylindrical model is constructed by integrating momentum relative to an asymptotic time. Cylindrical symmetry does not fix a single foliation.	17 ^{KQ} , 18 ^B	pp. 24, 25, 26
30	In the cylindrical model, separating Einstein–Rosen embedding variables from true degrees of freedom gives constraints linear in embedding momenta and hence a many-fingered functional Schrödinger equation.	17 ^S , 19 ^S	pp. 25, 26
31	York time is the mean extrinsic curvature defined from a positive-definite three-metric and canonical momentum.	20 ^{KQ}	pp. 26, 27
32	The constraint after conformal decomposition with York time becomes an elliptic equation. Specifying its positive real solution, the remaining constraints and operator domains yields a formal Schrödinger equation in York time.	20 ^S , 21 ^S	pp. 26, 27, 28
33	An internal-time Hamiltonian may be specified by an implicit, nonlocal spectral problem. An operator with a negative spectral sector has no self-adjoint square root, so the selected root and domain are conditions on the reduced quantum evolution.	6 ^B , 13 ^S , 10 ^K , 20 ^K	pp. 27, 28, 29, 30
34	A global clock requires each admissible orbit to cross each clock section exactly once. A nonzero local Poisson bracket alone is insufficient, and global use of a volume clock fails on trajectories with repeated volume readings.	6 ^B , 10 ^K , 22 ^S , 46 ^B	pp. 27, 28, 29, 84
35	Raw mean extrinsic curvature changes when the hypersurface through the same spacetime event is bent differently. A nonlocal scalar constructed from a constant-mean-curvature foliation is a different quantity, and using a curvature invariant as a clock also requires spacelike clock surfaces and canonical separation.	2 ^B , 6 ^{XK} , 20 ^{XK}	pp. 31, 32

p	Claim	i	Source Location
36	Material reference systems identify events by reference-particle worldlines and clocks along them. Reference fluids can be coupled to gravity through a matter action or through an action imposing coordinate conditions before variation.	$23^{\text{KB}}, 24^{\text{K}}$	pp. 32, 33, 34, 35
37	Linearity in the embedding momenta of a Gaussian reference fluid yields a formal Schrödinger equation in its clock field.	$24^{\text{S}}, 25^{\text{S}}$	pp. 35, 36, 37
38	For the heat-conducting Gaussian fluid in the source, the energy condition is expressed in terms of density and heat flow as $M - 2 \mathbf{M} \geq 0$. It can be violated during subsequent evolution even if it holds initially.	24^{S}	pp. 35, 36, 37, 38
39	For reference dust with zero heat flow, preservation of the energy condition considered in the source is restored. This restriction makes the spatial reference variables unobservable and restricts quantum states to be independent of them.	$24^{\text{S}}, 25^{\text{S}}$	pp. 37, 38
40	The energy constraints of the Gaussian fluid also restrict a naive position interpretation using metric multiplication operators. Fundamental-matter clock schemes reintroduce nonlinear clock momenta, spectral and reality conditions, and the spacetime problem for nonscalar fields.	$24^{\text{K}}, 25^{\text{S}}, 14^{\text{K}}$	pp. 37, 38, 39
41	In unimodular theory, the time conjugate to the cosmological constant is the oriented four-volume from a reference embedding. One four-volume value does not uniquely specify a measurement hypersurface, and hypersurfaces with different values can intersect.	$2^{\text{BI}}, 26^{\text{IQB}}$	pp. 39, 40, 41, 42
42	Formal time evolution in unimodular theory is described by a Hamiltonian equal to the integrated gravitational Hamiltonian density divided by spatial volume, together with the remaining embedding and geometrical constraints.	$26^{\text{S}}, 27^{\text{S}}$	pp. 41, 42
43	In the specified relativistic-particle construction, a future-directed timelike Killing field preserving the potential, together with a nonnegative potential, permits a positive-definite inner product on the positive-energy solutions.	$9^{\text{C}}, 29^{\text{S}}$	pp. 43, 44

p	Claim	i	Source Location
44	The current of states satisfying the relativistic-particle Klein–Gordon equation obeys a continuity equation. Under conditions including the boundary-flux condition, the pairing on spacetime hypersurfaces is conserved, but it is not positive definite on the full solution space.	$2^B, 30^Q$	pp. 43, 44
45	The superspace current of states satisfying the Klein–Gordon form of the Wheeler–DeWitt equation obeys a continuity equation. Under conditions including the boundary-flux condition, the pairing is conserved independently of the superspace hypersurface, but it is not positive definite on the entire solution space.	$28^B, 31^Q$	pp. 45, 46, 47
46	In spatially compact geometrodynamics, the source’s search for a symmetry generator linear in momentum reduces the candidate to the integrated momentum trace. It scales the kinetic and potential terms differently and does not weakly commute with the full super-Hamiltonian constraint.	$28^B, 29^S$	pp. 45, 46
47	The geometrodynamical potential has indefinite sign, and both expansion and contraction are allowed, so a positive-energy restriction excludes admissible states. Conservation of flux in superspace alone also does not provide a spacetime probability interpretation.	$2^B, 22^S, 28^B, 31^Q, 29^S$	pp. 46, 47
48	The semiclassical construction separates heavy and light degrees of freedom and expands a single WKB state in inverse mass or coupling under a chosen factor ordering. Its leading phase depends only on the heavy degrees of freedom and satisfies the Hamilton–Jacobi equation.	$\emptyset$	pp. 47, 48, 49, 52, 53
49	A classical trajectory congruence of the selected Hamilton–Jacobi phase and a transverse reference hypersurface define WKB time through $S^A \partial_A t = 1$. Different reference hypersurfaces give different time functions, and global use requires a global transverse section.	$32^{KQ}, 33^{BS}$	pp. 49, 50, 54, 55

p	Claim	i	Source Location
50	At the specified approximation order on a single branch, the light degrees of freedom satisfy a Schrödinger equation along each classical trajectory. Including the time-dependent measure and imposing suitable self-adjointness and boundary-flux conditions gives a conserved inner product.	$32^S, 33^B, 34^{SQ}$	pp. 49, 50, 51, 52, 53, 54
51	At the next order of the source's semiclassical expansion, an inhomogeneous term containing second derivatives in the heavy coordinates appears. The corrected equation does not retain the leading-order Schrödinger form.	$32^S, 34^S$	pp. 49, 58, 59
52	The back-reaction modification of the amplitude examined by Kuchař yields a nonlinear integro-differential equation. Kuchař questions whether the decoherence arguments then available supply a principle selecting background modes suitable as clocks.	$32^B, 35^K$	pp. 53, 60, 61, 62
53	For a superposition of different Hamilton–Jacobi phases, the constraint equation does not impose a separate Schrödinger equation on each branch. The congruences defined by the phases need not share a common time foliation.	$22^S, 32^S, 33^B, 34^S$	pp. 55, 56, 57
54	Interference terms vanish where the supports of the heavy-system branches do not overlap. Rapid phase oscillations for a large but finite mass, however, do not make interference terms identically zero; sufficiently fine resolution can distinguish the interference.	$22^S, 32^B, 34^Q$	pp. 55, 56, 57
55	A positive norm on a WKB branch uses restriction to a single phase and orientation of the transverse surface along its flow. Opposite-phase branches and real superpositions have negative or zero Klein–Gordon norm, so positivity cannot be retained while including all branches.	$22^S, 31^Q, 32^B, 33^B, 34^Q$	pp. 57, 58
56	Almost-homogeneous cosmological models select a homogeneous mode as a clock, but that slicing does not cover strongly inhomogeneous geometries. The pointwise background–perturbation split has no uniquely supplied selection principle in the proposal examined in the source.	$32^B, 35^K$	pp. 59, 60, 61

p	Claim	i	Source Location
57	The Fock construction in third quantization uses a one-particle Hilbert space and its inner product. The incoming/outgoing prescription uses asymptotically stationary regions, leaving the problem of selecting these temporal structures.	36 ^{BS}	pp. 62, 63, 64
58	Kuchař distinguishes measurement of a field on superspace, measurement of multiple geometries on one spatial slice, and interpretation of geometries on separate copies of the slice. Ordinary detector arrangements use a common spacelike hypersurface and a readiness time.	2 ^B , 23 ^{KO} , 24 ^{KO} , 37 ^{OB}	pp. 64, 65, 66
59	Kuchař identifies the measuring apparatus for multiple geometries and the interpretation of topology change as unresolved issues in third quantization.	∅	pp. 64, 65, 66
60	The naive interpretation treats the squared modulus of the wave function as a configuration probability without first separating clock variables.	∅	pp. 66, 67
61	In the Newtonian model, a nonzero constant spatial norm makes the spacetime integral over the entire unbounded time axis diverge. The relative probabilities and time averages in the source are defined where their specified limits exist.	5 ^{IQ}	pp. 67, 68
62	A probability distribution for a single configuration does not by itself give a transition probability between two different times.	5 ^P , 38 ^P	pp. 68
63	The relativistic-particle position and norm problems cited in the source, and the fact that metric multiplication generally does not preserve the Wheeler–DeWitt solution space, restrict the naive configuration-probability interpretation.	∅	pp. 68, 69
64	A general projection-conditioned probability is defined by a state, projectors and a nonzero denominator as $\langle \Psi C B C \Psi \rangle / \langle \Psi C \Psi \rangle$. Conditioning on a clock additionally requires a clock projector.	∅	pp. 69, 70

<i>p</i>	Claim	<i>i</i>	Source Location
65	For noninteracting clock and system subsystems and a stationary total state, conjugation by the clock Hamiltonian constructs a family of clock projectors. Variation of the clock projector requires it not to commute with the clock Hamiltonian.	42 ^S , 40 ^K	pp. 70, 71, 72
66	Exact continuous-position eigenstates are improper states outside the Hilbert space. The source's formal projections onto such states cannot simply be treated as ordinary probability-density projections onto normalized states.	∅	pp. 71, 74, 75
67	For the specified clock-projector family and a nonzero conditioning probability, the one-time conditional probability equals the Heisenberg probability in the conditioned system state.	42 ^S , 40 ^K , 41 ^Q	pp. 72, 73
68	Projections used in an ordinary measurement process generally take a state outside the constraint kernel. The sequential projection rule tested in the source fails to reproduce the correct two-time propagation in the parametrized Newtonian model.	5 ^S , 40 ^O , 41 ^Q , 38 ^P	pp. 73, 74, 75, 76
69	The histories prescription constructs detector-response probabilities from detectors of passage through configuration-space regions, propagation generated by a selected constraint as Hamiltonian, and amplitude integrals over positive and negative label times. It specifies the initial preparation, region restrictions and common normalization.	44 ^S , 45 ^{OQ}	pp. 76, 77, 78, 79
70	Teitelboim's scattering interpretation describes in proper time a universe that starts at a velocity-dominated singularity, encounters the scalar potential of the Wheeler–DeWitt equation, and ends at a velocity-dominated singularity.	43 ^{SB}	pp. 76
71	Detector numbers and response combinations in the histories prescription distinguish which detectors, associated with regions of configuration space, have responded.	∅	pp. 77, 78, 79

p	Claim	i	Source Location
72	For the specified thin-slab detectors in the Newtonian model with exact clock-time/label-time correlation, histories responses reproduce Schrödinger evolution. Good clock states in the stationary two-particle model reproduce it approximately over their valid interval.	$5^S, 44^S, 45^{OX}$	pp. 79, 80
73	Histories probabilities depend on the chosen form of the constraint. Kuchař accepts a consistently fixed constraint as an answer to multiple choice in this scheme, while raising the problems of repeated crossings of general hypersurfaces and anticipated foliation dependence in full geometrodynamics.	$2^B, 44^S, 45^{OQ}$	pp. 80, 81
74	Formal projection onto physical states for a single detector does not ensure that intermediate states between multiple detections satisfy the constraint. In the source's relativistic-particle application, problems also remain in relating amplitudes to Klein–Gordon probabilities.	$30^Q, 44^S, 45^{OQ}$	pp. 81, 82
75	Kuchař distinguishes spatial gauge transformations from dynamical transformations and examines Rovelli's evolving constants as an alternative proposal for describing change in the universe. The quantity for a fixed clock reading is constant along a constraint orbit, while the family indexed by clock readings can vary with that reading.	$46^K, 47^S, 39^S$	pp. 82, 83, 84
76	On a domain where every admissible orbit meets each clock section exactly once, the two requirements of agreement with the original quantity on the clock section and constancy along the constraint orbit uniquely determine the classical family of evolving constants.	$46^{KB}, 47^S$	pp. 84, 85
77	The relation between the clock-parameter derivative of the classical family F_t and the ratio of Poisson brackets is evaluated at $t = T(Q, P)$.	$46^{KQ}, 47^Q$	pp. 85

p	Claim	i	Source Location
78	Rovelli's quantum construction proposes obtaining observables by replacing the classical family with operators. Kuchař identifies factor-ordering and operator-substitution ambiguities and examines the construction for a selected canonical subalgebra.	46 ^K , 47 ^S	pp. 85, 86, 87
79	Rovelli advances a conjecture concerning construction of the inner product. Kuchař offers a counterconjecture against the prospect of making the observables for all clock choices simultaneously self-adjoint in one inner product.	46 ^K , 47 ^S	pp. 86, 87, 88
80	When a global clock does not exist, the classical construction loses its premise that each orbit meets each clock section exactly once. Kuchař separately discusses the special two-oscillator model and the subsequent response concerning factor ordering.	46 ^B , 47 ^S	pp. 88
81	Kuchař states that the extension of evolving constants to field systems had not yet been developed and anticipates difficulties.	$\emptyset$	pp. 88
82	In his 1992 assessment, Kuchař states that he does not consider the ten interpretations to have solved or circumvented the problems of time. His assessment includes both the constructive progress of Gaussian dust and the phenomenological limitations of the reference fluid.	$\emptyset$	pp. 88, 89, 90

Definition 8.380 (Semantic Components and Generation-Initial Set). The 82 claims are the semantic components. Claims 1–4 form the generation-initial set; Claims 1–82 form the source formal-evidence environment.

Table 276: Kuchař 1992 Time-Use Roles

Code	Role	Use Count
I	Interval	3
Q	Quantity	22
X	Cross-position correspondence	3
K	Clock reading	28
C	Causality	3
S	State evolution	64
P	Probabilistic evolution	3
O	Observation record	8

Code	Role	Use Count
B	Reference boundary	39

Table 277: Kuchař 1992 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1	member	order_relation
2, 33	member	instant, endpoint, order_relation
3, 7, 9, 11–13, 16, 19, 21–22, 25, 27, 30–31, 34, 38–39, 41, 45, 47	dependency	$\emptyset$
4, 10, 14–15, 17, 20, 32, 35, 37, 40, 42	member	instant
5–6, 8, 23–24, 28–29, 43–44, 46	member	instant, order_relation
18	member	instant, endpoint
26	member	instant, orientation
36	member	endpoint, order_relation

Dependency Inputs The table gives each transfer’s time inputs, outputs, claim support and source location.

Definition 8.381 (Kuchař 1992 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [25]$.

Table 278: Kuchař 1992 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	2	4	9	Extraction of embedding time by canonical separation. pp. 3, 4
2	2, 4	3	10	Classical evolution from a complete Hamilton–Jacobi solution. pp. 3, 4, 5
3	4	7	15	Functional quantum evolution in the embedding representation. pp. 6, 21
4	$\emptyset$	10	21	Definition of a logarithmic-volume clock from spatial volume. pp. 10, 22, 23
5	2	12	23	Anomaly-free functional evolution in the two-dimensional model. pp. 15, 16, 17, 18
6	4, 6	13	25	Reduced quantum evolution in a selected internal clock. pp. 21, 22
7	10	11	26	Reduced evolution in the logarithmic-volume clock. pp. 22, 23
8	$\emptyset$	15	27	Definition of extrinsic time from linearized momentum. pp. 23, 24

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
9	15	16	28	Quantum evolution in linearized-gravity extrinsic time. pp. 23, 24, 25
10	18	17	29	Cylindrical embedding time from asymptotic time and momentum. pp. 24, 25, 26
11	17	19	30	Quantum evolution in cylindrical embedding time. pp. 25, 26
12	$\emptyset$	20	31	Definition of York time from canonical geometry. pp. 26, 27
13	20	21	32	Formal reduced evolution in York time. pp. 26, 27, 28
14	23	24	36	Specification of Gaussian clock fields by a material reference system. pp. 32, 33, 34, 35
15	24	25	37	Formal evolution in the Gaussian clock field. pp. 35, 36, 37
16	2	26	41	Oriented four-volume time between specified embeddings. pp. 39, 40, 41, 42
17	26	27	42	Formal quantum evolution in four-volume time. pp. 41, 42
18	2	30	44	Conservation of particle Klein–Gordon flux in spacetime. pp. 43, 44
19	28	31	45	Conservation of Klein–Gordon flux in superspace. pp. 45, 46, 47
20	$\emptyset$	32–33	48–49	WKB time and reference congruence from the Hamilton–Jacobi phase flow. pp. 47, 48, 49, 50, 54, 55
21	32–33	34	50	Quantum evolution and norm conservation in WKB time. pp. 49, 50, 51, 52, 53, 54
22	42	40	65	Construction of a clock-projector family along a time parameter. pp. 70, 71, 72
23	40, 42	41	67	One-time probability conditioned on the clock projector. pp. 72, 73
24	44	45	69	Detector-response correlations from history amplitudes. pp. 76, 77, 78, 79
25	46	47	76–77	Classical evolving-constant family from global clock sections. pp. 84, 85

Grounding Conditions

1. Canonical transformation separating embeddings from true degrees of freedom, and constraints solvable for embedding momenta.
2. Selected canonical representation, constraints and complete Hamilton–Jacobi solution.

3. Impose constraints linear in embedding momenta in the specified operator representation and domain.
4. Positive spatial volume and specified logarithmic coordinate in the source's homogeneous model.
5. Massless field on the flat cylinder and the source's embedding-dependent anomaly potential.
6. Canonical separation, constraint solution for clock momenta, conserved inner product and self-adjoint operator domains.
7. Selected root of the quadratic constraint, operator domain and admissible clock domain.
8. Flat spatial background, transverse momentum component and boundary conditions for the inverse Laplacian.
9. The source's perturbative order and constraints linear in embedding momenta.
10. Cylindrical symmetry, asymptotic reference time, momentum integral and spacelike-embedding admissibility.
11. Separated canonical variables and model constraints linear in embedding momenta.
12. Positive-definite three-metric, canonical momentum and specified mean-extrinsic-curvature functional.
13. Positive real solution of the elliptic equation, remaining constraints and operator domains.
14. Clock-carrying reference particles and the selected fluid action or coordinate conditions imposed before variation.
15. Fluid-coupled constraints linear in clock momentum and specified quantum representation.
16. Reference and final embeddings, spacetime metric and orientation.
17. Domain of nonzero spatial volume, volume-averaged Hamiltonian and remaining constraints.
18. Relativistic-particle Klein–Gordon equation and boundary-flux conditions.
19. Specified Wheeler–DeWitt equation, common domain and boundary-flux conditions.
20. Selected single real phase, approximation order, nondegenerate local flow and transverse reference surface; global use requires a global transverse surface.
21. Specified leading order, measure compensation, self-adjointness and boundary-flux conditions.
22. Two noninteracting subsystems, stationary total state and a clock projector not commuting with its specified Hamiltonian.

23. The same clock-projector family and Hamiltonian, and a state with nonzero conditioning probability.
24. Selected constraint, initial state, detection regions, positive/negative label-time integration and common normalization.
25. Unique orbit intersection, nonzero clock bracket, specified classical quantity and evaluation on the clock section.

Results

Definition 8.382 (Kuchař 1992 Results and Dependencies). Let $\text{ldx}_R^{\text{audit,result}} = [38]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 279: Kuchař 1992 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Physical prediction and constraints on instantaneous data. pp. 1, 2, 3, 83	1–3	5–8
2	Canonical separation and Hamilton–Jacobi evolution. pp. 3, 4, 5	2–4	9–11
3	Global time and the sandwich problem. pp. 5, 27, 28, 89	2, 6	12–13
4	Dirac quantization and the Hilbert-space problem. pp. 6, 7, 43, 45	$\emptyset$	14, 18
5	Embedding representation and functional evolution. pp. 6, 7, 15, 16, 17, 18, 19, 21	2, 4, 7	15–16
6	Clock choice and distinction between problems. pp. 7, 8, 28, 29, 30, 89	6, 13	17, 19
7	Features of the eleven models and conditions for transfer to the full theory. pp. 8, 9, 10, 11, 12, 13, 14	5, 7–9	20, 22
8	Logarithmic-volume clock and reduced evolution. pp. 10, 22, 23	10–11	21, 26
9	Anomaly cancellation for the two-dimensional massless field. pp. 15, 16, 17, 18	2, 12	23
10	Classification into three frameworks and ten interpretations. pp. 19, 20, 89, 90	$\emptyset$	24
11	Construction of the internal Schrödinger interpretation. pp. 21, 22	4, 6, 13	25
12	Extrinsic time and evolution in linearized gravity. pp. 23, 24, 25	15–16	27–28
13	Cylindrical embedding time and evolution. pp. 24, 25, 26	17–19	29–30
14	York time and formal reduced evolution. pp. 26, 27, 28	20–21	31–32

q	Result and Source Location	Time Premises i	Claim Support p
15	Root, globality and spacetime-scalar conditions for internal clocks. pp. 27, 28, 29, 30, 31, 32, 84	2, 6, 10, 13, 20, 22, 46	33–35
16	Material reference systems and formal Gaussian-clock evolution. pp. 32, 33, 34, 35, 36, 37	23–25	36–37
17	Energy conditions for Gaussian fluids and reference dust. pp. 35, 36, 37, 38, 39	14, 24–25	38–40
18	Four-volume clock and unimodular evolution. pp. 39, 40, 41, 42	2, 26–27	41–42
19	Particle Klein–Gordon flux and positive sector. pp. 43, 44	2, 9, 29–30	43–44
20	Flux, positivity and symmetry conditions in geometrodynamics. pp. 45, 46, 47	2, 22, 28–29, 31	45–47
21	Time construction from a single WKB phase. pp. 47, 48, 49, 50, 52, 53, 54, 55	32–33	48–49
22	Quantum evolution and inner-product conservation in WKB time. pp. 49, 50, 51, 52, 53, 54	32–34	50
23	Higher-order corrections and limits of semiclassical clock modes. pp. 49, 53, 58, 59, 60, 61, 62	32, 34–35	51–52, 56
24	Superposition and positivity of semiclassical branches. pp. 55, 56, 57, 58	22, 31–34	53–55
25	Hilbert space and measurement interpretation in third quantization. pp. 62, 63, 64, 65, 66	2, 23–24, 36–37	57–59
26	Naive configuration probabilities and operator restrictions. pp. 66, 67, 68, 69	$\emptyset$	60, 63
27	Normalization over all time and the two-time question. pp. 67, 68	5, 38	61–62
28	Static projection-conditioned probability. pp. 69, 70	$\emptyset$	64
29	Clock-projector family and equality of one-time probabilities. pp. 70, 71, 72, 73	40–42	65, 67
30	Continuous-position projections and failure of the sequential measurement rule. pp. 71, 73, 74, 75, 76	5, 38, 40–41	66, 68
31	Detector responses from history amplitudes. pp. 76, 77, 78, 79	44–45	69, 71
32	Proper-time scattering interpretation of the universe. pp. 76	43	70
33	Recovery of model evolution by the histories prescription. pp. 79, 80	5, 44–45	72
34	Constraint choice, foliation and intermediate states in the histories prescription. pp. 80, 81, 82	2, 30, 44–45	73–74

q	Result and Source Location	Time Premises i	Claim Support p
35	Classical evolving-constant construction and derivative identity. pp. 82, 83, 84, 85	39, 46–47	75–77
36	Quantization proposal and inner-product conjectures for evolving constants. pp. 85, 86, 87, 88	46–47	78–79
37	Globality of evolving constants and extension to field systems. pp. 88	46–47	80–81
38	Kuchař’s 1992 assessment. pp. 88, 89, 90	$\emptyset$	82

Grounds for the Results

Table 280: Kuchař 1992 Conditions for the Results

Re- sult Index	Conditions and Domain
1	Past–future ordering, specified hypersurfaces and canonical data.
2	Separable constraints, complete Hamilton–Jacobi solution and classical constraint closure.
3	Clock-section intersections and intrinsic geometries at the two specified boundaries.
4	Constraint operators, physical solution space and the requirement of a positive-definite inner product.
5	Operator representation, domains, quantum constraint commutators and foliation.
6	Conditions on quantum representation, inner product and evolution for each clock choice.
7	Background, preferred foliation, number of degrees of freedom and kinetic term of each model.
8	Positive volume, selected root and admissible clock/operator domains.
9	The source’s flat-cylinder model and anomaly potential.
10	The competing interpretations distinguished in Kuchař’s review.
11	Selected clock, canonical separation, solved constraints and self-adjoint operators.
12	Flat spatial background, inverse Laplacian and specified perturbative order.
13	Asymptotic time, cylindrical symmetry and admissible spacelike embeddings.
14	Canonical geometry, positive real solution of the elliptic constraint, remaining constraints and operator domains.
15	Spectra, orbit intersections and foliation dependence for the clock candidates examined in the source.
16	Reference worldlines, fluid action and constraints linear in clock momentum.
17	Distinguish heat-conducting fluid from zero-heat-flow dust and impose their respective constraints.

Re- sult Index	Conditions and Domain
18	Reference embedding, oriented volume, volume-averaged Hamiltonian and remaining constraints.
19	Spacetime boundary-flux conditions, timelike Killing field preserving the potential, and nonnegative potential.
20	Superspace pairing, specified symmetry generator in compact geometrodynamics, and solutions of both signs.
21	Heavy and light systems, selected phase, local flow and transverse surface.
22	Leading approximation order, time-dependent measure, self-adjointness and boundary conditions.
23	The source's semiclassical expansion order and the back-reaction, almost-homogeneous models and mode selection examined in the source.
24	Distinct phases, branch supports, transverse-surface orientation and finite resolution.
25	One-particle inner product, incoming/outgoing regions and the compared measuring arrangements.
26	Specified wave function, position/norm interpretation and operators preserving physical states.
27	Nonzero conserved spatial norm, unbounded time axis and existing time-average limits.
28	Specified state, projectors and nonzero denominator.
29	Noninteracting subsystems, stationary total state, same clock-projector family and nonzero conditioning probability.
30	The source's formal projections and the sequential update rule tested in the parametrized Newtonian model.
31	Selected constraint, initial preparation, detection regions, normalization and combinations of responding detectors.
32	Teitelboim's model with velocity-dominated singularities and the Wheeler–DeWitt potential.
33	Thin-slab detectors and exact Newtonian clock correlation, or the valid interval of a good clock state.
34	Consistent constraint choice and constraints on states between detections.
35	Unique intersection of each orbit with each clock section, specified classical quantity and evaluation on the section.
36	Rovelli's proposal and Kuchař's examination of a selected canonical subalgebra.
37	Global-clock premise, two-oscillator model and the status of field-system extensions at the time of the source.
38	The ten interpretations examined in the review and the construction/material conditions of Gaussian reference dust.

Definition 8.383 (Result-Dependency Hyperedges). Each result edge takes the temporal premises and claim support in Table 279 as inputs and outputs the corresponding result node.

Evidence Hierarchy Kuchař reviews earlier quantum-gravity approaches and examines their definitions, derivations, conditions and unresolved problems. The source reports zero directly measured objects and zero direct-measurement predicates from measurements of its own.

Generation Judgment The complete LCTR generation count is zero; 9 time contents are locally reachable under source-evidence conditions, leaving 38 primitive-time roots. Local reach includes the constructions of intrinsic time from logarithmic volume, extrinsic time in the linearized theory, York time and WKB time, together with quantum evolution satisfying their respective conditions. The WKB construction determines the temporal correspondence from a phase flow and a transverse surface. The corresponding time contents are 10–11, 15–16, 20–21, 32–34.

Result Dependence Of the 38 results, 28 depend on primitive-time roots and 10 are reachable from the source formal-evidence environment. The latter are Results 4, 8, 10, 12, 14, 21–22, 26, 28, 38. No result is generated from nontemporal physical inputs alone. There are 63 root–result dependence pairs, of which 48 are necessary.

Evidentiary Effect Kuchař presents three frameworks and ten interpretations addressing the problems of time, together with their specific constructions, conditions and obstacles. Internal-clock constructions, reduced quantum evolution, reference fluids, Klein–Gordon inner products, WKB approximations, conditional probabilities, histories and evolving constants have the conditions stated in the result table. Classical evolving constants are constructed from the specified clock sections. For self-adjoint quantum operators, Rovelli proposes an inner-product conjecture and Kuchař presents a counterconjecture.

Placement of the Grounding Obligation External instants, hypersurfaces and embeddings, asymptotic time, material-clock worldlines, the temporal parameter of a clock-projector family, temporal labels for histories and global clock sections provide temporal premises for the respective approaches. None of the 47 time contents belongs to the generation closure from nontemporal physical inputs alone, and 38 roots remain even after the local constructions under source formal evidence. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Kuchař discusses global clock obstructions, failure of the energy condition for Gaussian reference fluids with heat flow, the obstruction to the specified positive-frequency construction in quantum geometrodynamics, and the failure to retain the leading-order Schrödinger evolution at the next WKB order. Actual failure of the physical-time generation conditions audited by LCTR, and consequent changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 63 root–result dependence pairs. Actual reach from the failure location of a specific physical-time generation condition to a result is undetermined by the source evidence.

Result-Preservation Domain If the constraints, states, operators, inner products, boundary conditions, clock domains and approximation orders of each approach are fixed, and generated temporal representations realize the time-dependent results' input relations with the same ordering, intervals and clock correspondences, each result is preserved with its stated domain and epistemic status. Classifications, static operator and probability expressions, formally conditioned clock constructions and Kuchař's comparative assessments use their respective claim support.

Repair Scope The repair targets are the time contents used by each approach and the physical-time generation conditions that must be satisfied before their use. If actual failure of a specific physical-time generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.384 (Primitive-Time Roots and Result Dependence in the Kuchař 1992 Source). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = & (\emptyset, [47]; \\ & \emptyset, \{4, 8, 10, 12, 14, 21, 22, 26, 28, 38\}, \\ & [38] \setminus (\{4, 8, 10, 12, 14, 21, 22, 26, 28, 38\}), \emptyset). \end{aligned}$$

$$R \in \text{Ref}^{\text{audit,rootdep}}, \quad R \notin \text{Ref}^{\text{audit,rootunreach}}.$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Supplying all 82 claims as source formal evidence reaches 9 time contents through Transfers 4, 8, 12 and 20, whose temporal inputs are empty, and their successors. This closure contains 10 results. Adding the remaining 38 time contents as seeds reaches all 38 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.385 (Minimal Primitive-Time-Root Support in the Kuchař 1992 Source). The table gives each result's minimal-support family, with 46 family members in total. For each result reachable from the evidence environment alone, the empty set is the sole family member.

Proof. Start from singleton time-content seeds and apply Definition 7.46 to the 25 transfers. From the stable antichains, combine supports for each result's temporal premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Table 281: Minimal Primitive-Time-Root Support in the Kuchař 1992 Source

Result Index	Minimal-Support Index Family
1	$\{1, 2\}$
2	$\{2\}$
3	$\{2, 6\}$

Result Index	Minimal-Support Index Family
4	$\emptyset$
5	$\{2\}$
6	$\{2, 6\}, \{4, 6\}, \{6, 13\}$
7	$\{2, 5, 8, 9\}, \{4, 5, 8, 9\}, \{5, 7, 8, 9\}$
8	$\emptyset$
9	$\{2\}$
10	$\emptyset$
11	$\{2, 6\}, \{4, 6\}$
12	$\emptyset$
13	$\{18\}$
14	$\emptyset$
15	$\{2, 6, 22, 46\}$
16	$\{23\}$
17	$\{14, 23\}, \{14, 24\}$
18	$\{2\}$
19	$\{2, 9, 29\}$
20	$\{2, 22, 28, 29\}$
21	$\emptyset$
22	$\emptyset$
23	$\{35\}$
24	$\{22, 28\}, \{22, 31\}$
25	$\{2, 23, 36, 37\}$
26	$\emptyset$
27	$\{5, 38\}$
28	$\emptyset$
29	$\{42\}$
30	$\{5, 38, 40, 41\}, \{5, 38, 42\}$
31	$\{44\}$
32	$\{43\}$
33	$\{5, 44\}$
34	$\{2, 44\}$
35	$\{39, 46\}$
36	$\{46\}$
37	$\{46\}$
38	$\emptyset$

Proposition 8.386 (Audit Conclusion for the Kuchař 1992 Source). Complete generation is zero and local reach is 9, leaving 38 primitive-time roots after local reach. The 38 results divide into 10 evidence-reachable and 28 primitive-time-root-dependent results. The 82 claims, 47 time contents, 25 transfers and 38 results give 192 source-specific obligation atoms. The 147 claim/time-content use pairs have 173 role assignments.

Proof. This follows from the definitions of claims, time contents, transfers and results, the reachable sets and the minimal-support families. $\square$

Proposition 8.387 (Audit-Boundary Counts for the Kuchař 1992 Source). $\text{BoundaryCount}_R^{\text{audit}} = (167, 63)$.

Proof. The 82 claims, 47 time contents and 38 results give 167 nodes; the 25 transfers and 38 result dependencies give 63 edges. □

Individual Statements Statement Evidence 1: The time variable in classical theory

Verbatim original (English):

The functional nature of the time variable thus presents no problem in the classical theory.

Original-Source Location Kuchař, *Time and Interpretations of Quantum Gravity*, text p. 5, immediately after Eq. (1.13) in §1.

Minimal Context Immediately after the statement that functional Hamilton–Jacobi evolution in classical canonical theory is independent of the foliation connecting the initial and final hypersurfaces.

Attributed Author A statement in the text by Karel V. Kuchař.

Statement Evidence 2: Construction of time from matter fields coupled to gravity

Verbatim original (English):

Time is to be constructed, not from the geometric data, but from matter fields coupled to gravity.

Original-Source Location Kuchař, *Time and Interpretations of Quantum Gravity*, text p. 22, *Matter Time* in §6.

Minimal Context The passage that enumerates three candidates for an internal time variable— intrinsic time, extrinsic time, and matter time—and introduces the matter-time candidate.

Attributed Author A statement in the text by Karel V. Kuchař.

Statement Evidence 3: Warning against inference from a model to quantum gravity as a whole

Verbatim original (English):

One should try to avoid a common hubris which consists of solving a time problem for a model way down in the hierarchy, and jumping to the conclusion that the time problems of quantum gravity are removed by the same treatment.

Original-Source Location Kuchař, *Time and Interpretations of Quantum Gravity*, printed p. 14, at the end of §3.

Minimal Context The passage that discusses the precision limit in using the dynamical degrees of freedom of a realistic clock as an ideal kinematical clock, and the transfer to quantum gravity in general of a solution obtained in a model.

Attributed Author A statement in the text by Karel V. Kuchař.

Statement Evidence 4: Assessment that the problem of time in quantum gravity remained unresolved

Verbatim original (English):

In my opinion, none of us has so far succeeded in proposing an interpretation of quantum gravity that would either solve or circumvent the problems of time.

Original-Source Location Kuchař, *Time and Interpretations of Quantum Gravity*, printed p. 90, *Summary* in §16.

Minimal Context The passage that summarizes whether the interpretations of quantum gravity considered in the source have succeeded in solving or circumventing the problems of time.

Attributed Author A statement in the text by Karel V. Kuchař.

Common Evidence with Specified Domain of Application The four statements above give the original text of Kuchař 1992, the original-source location, the minimal context, and the source attribution.

Target state-ment	Object of statement	Predicate	Explicit domain of application
S30-Q001	the time variable in classical theory	States that the functional nature of the time variable presents no problem in classical theory.	The domain of constraint closure and foliation-independent Hamilton–Jacobi evolution in classical canonical theory in §1.
S30-Q002	construction of time from matter fields coupled to gravity	States that time is to be constructed from matter fields coupled to gravity, not from geometric data.	The domain in §6 that compares intrinsic, extrinsic, and matter time and introduces the matter-time candidate.
S30-Q003	inference from a model to quantum gravity as a whole	Rejects the inference that transfers treatment of the time problem in a lower-level model to quantum gravity as a whole and concludes that the problems of time have been removed.	The precision limit of realistic clocks at the end of §3, and the range of degrees of freedom, constraints, and many-fingered structure retained by a model.
S30-Q004	assessment that the problem of time in quantum gravity remained unresolved	Kuchař assesses that, among the interpretations of quantum gravity he considers, none had succeeded in solving or circumventing the problems of time as of 1992.	The comparative summary, in the Summary of §16, of the three frameworks and ten interpretations considered in the original source as of 1992.

Individual Audit

First Statement Kuchař uses closure of the classical canonical constraints to explain the foliation independence of Hamilton–Jacobi evolution connecting initial and final hypersurfaces. The first statement concerns the functional nature of the time variable in this classical evolution. The corresponding formal evidence consists of Claims 9–11, Transfers 1–2 and Result 2.

Second Statement In Section 6, Kuchař compares intrinsic, extrinsic and matter time and introduces matter fields coupled to gravity as internal-clock candidates. The Gaussian reference fluid in Section 7 uses clock fields along material worldlines to give formal quantum evolution. The energy condition for a fluid with heat flow and the dust case in which that condition is preserved have distinct domains. The specific constructions and conditions of the matter-time approach introduced by the second statement correspond to Claims 36–40, Transfers 14–15 and Results 16–17.

Third Statement Kuchař compares four field-theoretic models and seven models with finitely many degrees of freedom in terms of background spacetime, preferred foliations, degrees of freedom and constraints. In cases such as the Barbour–Bertotti model, the model structure permits separation of clock variables. The third statement concerns the inference from such a solution of a time problem in a model to full quantum gravity. Claims 20 and 22 and Result 7 provide the corresponding formal evidence; the structures retained by each model determine its domain of application.

Fourth Statement Kuchař classifies ten interpretations into three frameworks: internal time, Wheeler–DeWitt theory, and interpreting quantum theory without requiring time. He compares how they address the problems of time and which conditions remain unresolved. The fourth statement is Kuchař’s assessment of their lack of success as of 1992, based on that comparison. The corresponding formal evidence consists of Claims 24 and 82 and Results 10 and 38.

8.19.31 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 31.

Source Name Lebowitz 1993

Bibliographic Identification Joel L. Lebowitz, *Boltzmann’s Entropy and Time’s Arrow*, *Physics Today* 46(9), pp. 32–38 (1993), DOI: 10.1063/1.881363. $R = \text{Ref}_{\text{Leb93ETA}}$ denotes this source. The audit covers time uses, entropy definitions and relations, typicality, initial conditions, velocity reversal and the quantum discussion on pp. 32–38.

Original-Source Location Identifier 31-L1.

Original-Source Location Content The tables give the page numbers printed in the source.

Audit-Scope Identifier 31-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Formal Objects The audit uses 34 time contents, 54 claims, 11 transfers and 24 results. The tables specify the constructions and conditions of microscopic reversibility, coarse graining, entropy, scale separation, typicality, initial boundaries, reversal and the quantum correspondence. Lebowitz uses Boltzmann’s statistical account to examine the compatibility of time-reversal-symmetric microscopic dynamics with the temporal direction of observed macroscopic nonequilibrium processes. He defines Boltzmann entropy from the macroregion containing an individual microstate and explains macroscopic evolution using scale separation, phase-volume typicality and a special low-entropy initial state. After examining Lorentz-gas diffusion, velocity reversal, preparation and the cosmic initial boundary, and the distinction from Gibbs entropy, he discusses the quantum correspondence and the arrows associated with measurement, psychology and electromagnetism.

Assumptions The central dynamics is reversible Hamiltonian mechanics for an isolated classical Newtonian system. Macrostates require specified coarse graining and tolerances for particle densities, energy and other quantities. Typicality-based future prediction uses a special initial macroregion and a typical microstate within it. The hydrodynamic description uses many particles and scale separation; exact retracing uses perfect velocity reversal and isolation; Gibbs-entropy constancy uses Liouville volume preservation. The quantum correspondence uses the wavefunction representation, conjugation and unitary evolution specified in the source.

Nontemporal Physical Inputs Claims 1–5 are the initial inputs for generation. They provide configurations and phase space, phase volume and coarse graining, boundary-state data, state data for experimenters, apparatus, resources and environment, and quantum-state representations and subspaces. The relations assigning instants, velocities, initial boundaries and evolution to these data are specified in the corresponding claims and transfers.

Time Contents

Definition 8.388 (Lebowitz 1993 Time Contents). Let $n_R = 34$. The table defines the time contents and their source locations.

Table 283: Lebowitz 1993 Time Contents

i	Time Content	Source Location
1	Instants and temporal ordering in classical mechanics	pp. 33, 34, 35, 36, 37
2	Particle velocities constituting a microstate	pp. 33, 34
3	Time evolution of microstates under Hamiltonian dynamics	pp. 32, 33, 34, 35, 36, 37, 38
4	Histories of nonequilibrium macrostates, relaxation and fluctuations	pp. 32, 33, 34, 35, 36, 37, 38
5	Statistical temporal orientation of a macroscopic state sequence	pp. 32, 33, 34, 35, 36, 37, 38
6	Boltzmann entropy along an individual system’s time evolution	pp. 32, 34, 35, 36, 37, 38

<i>i</i>	Time Content	Source Location
7	Gibbs entropy of a density undergoing Liouville evolution	pp. 32, 37, 38
8	Macroscopic density evolution in the hydrodynamic limit	pp. 32, 33
9	Time-symmetric chaotic trajectories of a few-particle system	pp. 32, 33, 38
10	Distinct instants at which macroscopic snapshots are taken	pp. 33, 34, 35, 36
11	Positive time interval in reversal and two-direction typicality	pp. 34, 35
12	Instant of simultaneous particle-velocity reversal	pp. 34, 37
13	Microscopic trajectory retraced after exact velocity reversal	pp. 33, 34, 37
14	Time at which the partition is removed	pp. 35
15	Timescale of macroscopic relaxation	pp. 35, 36
16	Waiting time for macroscopic recurrence	pp. 32, 35
17	Age of the universe used for comparison with recurrence waiting	pp. 32, 35
18	Fraction of time spent in a macroregion	pp. 35
19	Initial boundary at which a low-entropy state is prepared	pp. 35, 36
20	Temporal chain of laboratory preparation, biological resources and solar radiation	pp. 36
21	Before-and-after relation of a meteor–Moon collision	pp. 36
22	Initial low-entropy cosmic boundary and its temporal relation to the present	pp. 36, 37, 38
23	Terminal cosmic state in Penrose’s comparison model	pp. 36, 37
24	Short entropy-decrease interval before perturbations restore an increase	pp. 37
25	Temporal relation between past production of an external perturbation and its action on the system	pp. 37
26	Discrete time of a reversible integer-arithmetic simulation	pp. 37
27	Time parameter of quantum-state evolution	pp. 38
28	Unitary time evolution of the wavefunction	pp. 38
29	Time-asymmetric state change of measurement reduction	pp. 38
30	Time-symmetric joint evolution of system and environment proposed by Lebowitz	pp. 38
31	Temporal orientation of a macroscopic quantum measurement process	pp. 33, 38
32	Psychological arrow of time	pp. 38
33	Temporal orientation associated with retarded and advanced electromagnetic potentials	pp. 38
34	Duration of isolation compared with the relaxation time	pp. 35, 36

Definition 8.389 (Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(54)$, with claim boundary $\text{ClaimBoundaryI}dx_R = ([5], [54])$. Table 284 gives each claim, its time uses and its source locations.

Formal Input Elements

Definition 8.390 (Entropy and Reversible Evolution). Let the source microstate be $X = (\mathbf{r}_1, \mathbf{v}_1, \dots, \mathbf{r}_N, \mathbf{v}_N)$ and its macrostate $M(X)$. Writing $V_M = |\Gamma_M|$ for the volume of its macroregion and $k > 0$ for Boltzmann's constant, source Eqs. (1), (3) and (4) give

$$S_B(M) = k \log V_M, \quad S_G(\rho) = -k \int_{\Gamma} \rho(X) \log \rho(X) dX.$$

$$\rho_M(X) = \frac{\mathbf{1}_{\Gamma_M}(X)}{V_M}, \quad S_G(\rho_M) = k \log V_M = S_B(M).$$

The identity uses positive finite V_M and a common phase-volume convention, with $0 \log 0 = 0$. Under a common additive-constant convention, $V_2 > V_1 > 0$ entails $S_B(M_2) > S_B(M_1)$. Under Hamiltonian–Liouville evolution, phase-volume preservation and density transport give $S_G(\rho_t) = S_G(\rho_0)$ (pp. 34, 37–38).

Source Eq. (2) for equilibrium, $S_{\text{eq}}(E, V, N) \approx S_B(M_{\text{eq}})$ expresses a difference per particle that is negligible at large particle number, with large macrocells and appropriate additive constants (p. 34).

Let $\mathcal{R}(\mathbf{r}_i, \mathbf{v}_i) = (\mathbf{r}_i, -\mathbf{v}_i)$ denote velocity reversal and Φ_t the time evolution. For an isolated time-reversal-symmetric system, $\mathcal{R}\Phi_t = \Phi_{-t}\mathcal{R}$ holds, so equal positive intervals τ give

$$\Phi_{\tau}\mathcal{R}\Phi_{\tau}X(t_0) = \mathcal{R}X(t_0).$$

Thus exact reversal at $t_0 + \tau$ takes the system at $t_0 + 2\tau$ to the velocity reverse of its initial state (p. 34).

Table 284: Lebowitz 1993 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Particle configurations, phase-space coordinates, algebraic components of the Hamiltonian, and interaction data.	$\emptyset$	pp. 33, 34
2	Macroregions, the phase-volume measure, and coarse-graining rules distinguishing macrostates.	$\emptyset$	pp. 33, 34, 37, 38
3	Configuration, density and energy data of a boundary macrostate, apart from its designation as temporally initial or final.	$\emptyset$	pp. 33, 34, 36, 37
4	State data of apparatus, experimenters, biological resources, solar radiation, and environmental degrees of freedom.	$\emptyset$	pp. 36, 38

p	Claim	i	Source Location
5	Configuration representations of quantum states and quantum subspaces associated with macrostates.	$\emptyset$	pp. 38
6	A microstate in the source consists of particle positions and velocities; velocity carries a physical rate of change.	2^Q	pp. 33, 34
7	Lebowitz adopts classical Newtonian mechanics as the main framework and excludes relativity and weak-interaction violation of time-reversal symmetry from the essay's argument.	$\emptyset$	pp. 33
8	The classical microscopic mechanics used in the source is time-reversal symmetric and connects a specified microstate to states at other times through Hamiltonian dynamics.	$1^S, 2^{QS}, 3^S$	pp. 33, 34
9	A macrostate corresponds to a phase-space region of microstates sharing particle densities, energy and other specified macroscopic quantities within chosen tolerances.	$\emptyset$	pp. 33, 34
10	With a specified phase-volume convention, Boltzmann entropy is Boltzmann's constant times the logarithm of the volume of the macroregion containing the microstate.	$\emptyset$	pp. 34
11	Under the same phase-volume and additive-constant conventions, an increase in positive finite macroregion volume entails an increase in Boltzmann entropy.	$\emptyset$	pp. 34, 35
12	For an equilibrium system with large particle number, macrocells much larger than microscopic scales, and suitably fixed additive constants, the difference between thermodynamic and Boltzmann entropies per particle is negligible.	$\emptyset$	pp. 34
13	Gibbs entropy is a functional of a normalized ensemble density on phase space.	$\emptyset$	pp. 37, 38
14	For a normalized density uniform on one macroregion of positive finite volume and zero outside, Gibbs entropy equals the Boltzmann entropy of that macrostate.	$\emptyset$	pp. 38
15	For particle, energy and momentum densities varying slowly in space on microscopic scales under local equilibrium, the source states agreement of the two entropies with generalized Clausius entropy.	$\emptyset$	pp. 38

p	Claim	i	Source Location
16	Hamiltonian–Liouville evolution preserves phase volume, and the fine-grained Gibbs entropy of an ensemble density following that evolution is independent of time.	$1^S, 3^S, 7^{SQ}$	pp. 32, 37, 38
17	In the mixing example after partition removal, the initially equal entropies evolve differently: Boltzmann entropy increases while Gibbs entropy remains constant.	$4^S, 6^{SQ}, 7^{SQ}, 14^B$	pp. 38
18	Very many degrees of freedom and separation of macroscopic and microscopic scales allow statistical prediction of typical nonequilibrium evolution of an individual macroscopic system.	$3^S, 4^S, 5^S$	pp. 32, 33, 38
19	Lebowitz states that the essential features of macroscopic evolution do not depend on strict satisfaction of positive Lyapunov exponents, ergodicity, mixing or equal a priori probabilities.	4^S	pp. 32
20	Ensembles are mathematical devices for computing typical individual behavior when the relevant quantities have sufficiently small dispersion; using phase-volume ratios does not require a physically instantiated ensemble.	$\emptyset$	pp. 35, 38
21	For a Lorentz gas of many noninteracting particles moving in a plane among a fixed periodic array of convex scatterers with bounded free-flight distance, reversible microscopic dynamics yields macroscopic density evolution governed by a diffusion equation in the hydrodynamic scaling limit.	$1^S, 2^{QS}, 3^S, 8^S$	pp. 33
22	The deterministic macroscopic description of the Lorentz gas remains highly accurate at a finite but very large scale ratio; the same description does not hold for one or a few particles.	$8^S, 9^S$	pp. 33
23	Even with chaos and good ergodic behavior, a few-particle Hamiltonian system has no macroscopic unidirectional temporal behavior in an individual realization from which to identify the orientation of a snapshot sequence.	9^S	pp. 32, 38
24	Under the source’s coarse graining, a microstate and its velocity reverse represent the same macrostate, and microscopic mechanics permits both orientations of the macroscopic snapshot sequence.	$2^{QS}, 3^S, 10^{OX}, 13^S$	pp. 33, 34

p	Claim	i	Source Location
25	The source relates the orientation of ordinarily observed macroscopic state sequences to experience and a low-entropy preparation boundary.	$5^O, 10^O, 19^B$	pp. 34, 35, 36
26	Under reversible Hamiltonian dynamics of an isolated system, exact reversal of all particle velocities at $t_0 + \tau$ yields the velocity reverse of the microstate at t_0 after one further equal positive interval τ .	$1^S, 2^{QS}, 3^{SX}, 11^{IQ}, 12^B, 13^{SX}$	pp. 34
27	Removing the partition enlarges the accessible phase-space region; the source gives the order of the resulting volume ratio for a one-mole example.	14^B	pp. 35
28	Typicality is a claim about an overwhelming phase-volume fraction within a specified macroregion and allows exceptional microstates.	$\emptyset$	pp. 35
29	Evolution after partition removal from a microstate typical of its initial macroregion proceeds with overwhelming probability into larger macroregions and toward equilibrium.	$3^S, 4^S, 6^{SQ}, 14^B, 19^B$	pp. 35
30	For the isolated macroscopic systems considered in the source, waiting times associated with large departures from equilibrium and recurrence are vastly longer than the age of the universe.	$3^S, 16^{IQ}, 17^{IQ}$	pp. 32, 35
31	The source regards a weak ergodic hypothesis equating macroregion residence-time fractions with phase-volume ratios in an aged isolated system as plausible absent contrary reasons such as additional conserved quantities, and does not require it for the main explanation.	$1^S, 3^S, 18^{IQ}, 34^I$	pp. 35
32	If a microstate at an intermediate nonequilibrium time is typical under a time-reversal-symmetric measure, equal past and future displacements receive the same prediction toward higher entropy.	$1^{SX}, 3^S, 4^{SX}, 6^{SQX}, 11^{IQ}$	pp. 35
33	By Liouville volume preservation, microstates arriving from a smaller earlier macroregion occupy a small atypical subset of the larger later macroregion.	$1^{SX}, 3^{SX}, 19^B$	pp. 36
34	For an equilibrium system isolated much longer than its relaxation time, fluctuations around a local entropy minimum are typically symmetric, unlike the inference of a prepared past from the displayed sequence.	$4^S, 6^{SQ}, 15^{IQ}, 19^B, 34^{IQ}$	pp. 36

<i>p</i>	Claim	<i>i</i>	Source Location
35	Assuming a microstate typical within a special low-entropy initial macroregion allows statistical prediction of subsequent macroscopic evolution.	$1^S, 3^S, 4^S, 5^S, 19^B$	pp. 35, 36, 37
36	The laboratory account connects preparation by low-entropy experimenters and apparatus to a chain supplying low entropy through biological resources and solar radiation. The total Boltzmann entropy, including that of the experimenters and apparatus, increases.	$19^B, 20^{CB}, 6^{SQ}$	pp. 36
37	Lebowitz cites the before-and-after sequence of a meteor–Moon collision as a case with an evident macroscopic temporal orientation without human participation.	$5^O, 21^O$	pp. 36
38	Lebowitz states that explaining irreversibility requires supplementing reversible physical laws with the hypothesis that the past universe occupied a special macrostate of lower Boltzmann entropy than the present.	22^B	pp. 36, 37
39	Lebowitz reports Penrose’s estimate of the phase-volume ratio between model macrostates just after the Big Bang and at a final Big Crunch, while reserving judgment about the reasonableness of those states.	$22^{BX}, 23^{BX}$	pp. 36, 37
40	After adopting a special initial cosmic macrostate and a microstate typical within its region, phase-volume reasoning applies to future prediction but does not by itself determine the actual past.	$1^S, 3^S, 4^S, 5^S, 22^B$	pp. 37
41	Typical macroscopic forward evolution is robust to perturbations, whereas entropy decrease through exact retracing is unstable to velocity-reversal errors and external perturbations.	$3^S, 12^B, 13^S$	pp. 37
42	After a short entropy decrease in an artificially prepared atypical microstate, imperfect reversal and external perturbations restore the typical increasing behavior.	$6^{SQ}, 13^S, 24^{IQ}, 25^C$	pp. 37
43	The source cites spin-echo experiments as examples of imperfect physical reversal and requires both exact reversal and exact isolation for complete retracing in the mixing system.	$12^B, 13^S$	pp. 36, 37
44	Roundoff errors perturb retracing in ordinary numerical simulations; discrete-time integer arithmetic avoids those errors and allows exact retracing after complete reversal.	$13^S, 26^S$	pp. 37

<i>p</i>	Claim	<i>i</i>	Source Location
45	The source's quantum correspondence maps microstates to wavefunctions, velocity reversal to complex conjugation, time evolution to unitary Schrödinger evolution, and macroregion phase volume to the dimension of the associated quantum subspace.	1 ^S , 2 ^{QS} , 3 ^S , 27 ^S , 28 ^S	pp. 38
46	Lebowitz states that this quantum correspondence preserves the large-system equilibrium entropy correspondence and the time symmetry of classical mechanics.	28 ^S	pp. 38
47	Conventional wavefunction collapse discussed in the source is nonunitary and appears time asymmetric.	29 ^{SO}	pp. 38
48	Lebowitz considers an additional fundamental measurement postulate unnecessary and proposes explaining measurement through a single time-symmetric theory of the system and its environment, possibly encompassing the universe.	30 ^S	pp. 33, 38
49	Lebowitz proposes that conventional wavepacket reduction should arise from the macroscopically many degrees of freedom affecting a nonisolated system's evolution and stresses the care needed in conceptually isolating a quantum system.	29 ^{SO} , 30 ^S , 31 ^O	pp. 38
50	The source separates the measurement formalism into a time-symmetric part and a second-law asymmetric part, and connects the latter to the universe's low-entropy initial state through Boltzmann-type reasoning.	22 ^B , 31 ^O	pp. 38
51	Lebowitz holds that the psychological arrow and the arrow associated with retarded and advanced electromagnetic potentials share the origin in the universe's low-entropy initial state.	22 ^B , 32 ^O , 33 ^C	pp. 38
52	In the 1993 introduction, Lebowitz judges the inconsistency claims known to him mistaken and sees no need for alternatives to Boltzmann's account.	∅	pp. 32
53	The microscopic evolution $X(t)$ and the coarse-graining map M define the macrohistory $M(X(t))$ at the same instants.	3 ^S , 4 ^S	pp. 33, 34

p	Claim	i	Source Location
54	Applying the Boltzmann-entropy definition to the macrohistory $M(X(t))$ gives $S_B(M(X(t)))$ at each instant.	$4^S, 6^{SQ}$	pp. 34, 35, 38

Definition 8.391 (Semantic Components and Generation-Initial Set). The 54 claims are the semantic components. Claims 1–5 form the generation-initial set; Claims 1–54 form the source formal-evidence environment.

Table 285: Lebowitz 1993 Time-Use Roles

Code	Role	Use Count
I	Interval	9
Q	Quantity	23
X	Cross-position correspondence	10
K	Clock reading	0
C	Causality	3
S	State evolution	71
P	Probabilistic evolution	0
O	Observation record	10
B	Reference boundary	19

Table 286: Lebowitz 1993 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 10, 26–27	member	instant, order_relation
2–4, 6–9, 11, 13, 15–18, 20, 24, 28–30, 34	dependency	$\emptyset$
5, 31–33	member	order_relation, orientation
12, 14	member	instant
19	member	instant, endpoint, order_relation
21, 25	member	order_relation
22–23	member	endpoint, order_relation

Dependency Inputs The table gives each transfer’s time inputs, outputs, claim support and source location.

Definition 8.392 (Lebowitz 1993 Time-Content Transfers). Let $\text{ld}\mathbf{x}_R^{\text{audit}, \text{transfer}} = [11]$.

Table 287: Lebowitz 1993 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1–2	3	1, 8	Microscopic evolution under Hamiltonian dynamics. pp. 33, 34
2	3	4	9, 53	Macrohistory from the microscopic history. pp. 33, 34
3	4	6	10, 54	Boltzmann entropy along the macrohistory. pp. 34, 35, 38
4	1, 3	7	13, 16	Constancy of Gibbs entropy under Liouville evolution. pp. 37, 38
5	1–3	8	1, 21	Hydrodynamic density evolution of the Lorentz gas. pp. 33
6	1–3, 11–12	13	26	Retracing trajectory after exact velocity reversal. pp. 34, 37
7	10, 19	5	25	Macroscopic orientation from preparation and snapshots. pp. 34, 35, 36
8	3, 14, 19	4, 6	28–29	Typical relaxation history after partition removal. pp. 35
9	1, 3, 19	4–5	28, 35	Future prediction from a prepared low-entropy state. pp. 35, 36, 37
10	1, 3, 22	4–5	28, 40	Future prediction from a low-entropy cosmic boundary. pp. 37
11	1–3, 27	28	5, 45	Unitary quantum evolution in the classical–quantum correspondence. pp. 38

Grounding Conditions

1. The source’s isolated classical system, specified initial microstate and Hamiltonian dynamics.
2. The source’s coarse-graining rules and tolerances for particle densities, energy and other specified quantities.
3. Evaluate each instant using the same macroregion, phase-volume and additive-constant conventions.
4. Volume preservation by the Hamiltonian flow and a normalized fine-grained ensemble density transported by that flow.
5. Many noninteracting particles in a plane, fixed periodic convex scatterers, bounded free-flight distance and an infinite macroscopic-to-microscopic scale ratio.
6. Time-reversal-symmetric Hamiltonian dynamics, perfect isolation, exact reversal of every particle velocity and equal positive intervals before and after reversal.
7. Low-entropy initial preparation and the experience-based reading of state sequences used in the source.

8. A typical microstate in the specified low-entropy initial macroregion and removal of the constraint. The conclusion concerns an overwhelming phase-volume fraction.
9. Specify the special initial macroregion and impose typicality within it.
10. Adopt the special initial cosmic macrostate and assume a typical initial microstate within its region.
11. The source's wavefunction representation, correspondence between velocity reversal and conjugation, and specified unitary Schrödinger evolution.

Results

Definition 8.393 (Lebowitz 1993 Results and Dependencies). Let $\text{ld}x_R^{\text{audit,result}} = [24]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 288: Lebowitz 1993 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Scope of the classical mechanics considered. pp. 33	$\emptyset$	7
2	Microscopic reversibility and both orientations of a macrosequence. pp. 33, 34	1–3, 10, 13	8, 24
3	Boltzmann entropy and monotonicity in phase volume. pp. 34, 35	$\emptyset$	10–11
4	Equilibrium entropy correspondence at large particle number. pp. 34	$\emptyset$	12
5	Gibbs functional, uniform-density identity and local equilibrium. pp. 37, 38	$\emptyset$	13–15
6	The two entropies during nonequilibrium evolution. pp. 32, 33, 34, 35, 37, 38	1, 3–4, 6–7, 14	16–17, 53–54
7	Scale separation and typical behavior of individual macroscopic systems. pp. 32, 33, 38	3–5	18–19
8	Phase-volume typicality and the computational role of ensembles. pp. 35, 38	$\emptyset$	20, 28
9	Scaling limit and large finite systems for the Lorentz gas. pp. 33	1–3, 8–9	21–22
10	Few-particle chaos and macroscopic temporal orientation. pp. 32, 38	9	23
11	Preparation boundary, observed sequence and typical relaxation. pp. 34, 35, 36	3–6, 10, 14, 19	25, 29
12	Exact velocity reversal and retracing. pp. 34	1–3, 11–13	26
13	Recurrence waiting times and residence-time fractions. pp. 32, 35	1, 3, 16–18, 34	30–31

q	Result and Source Location	Time Premises i	Claim Support p
14	Intermediate-time typicality and a low-entropy past. pp. 35, 36	1, 3–4, 6, 11, 15, 19, 34	32–34
15	Statistical prediction from low-entropy initial preparation. pp. 35, 36, 37	1, 3–5, 19	35
16	Laboratory preparation chain and total entropy. pp. 36	6, 19–20	36
17	Temporal orientation of a collision without human participation. pp. 36	5, 21	37
18	Cosmic low-entropy boundary and Penrose’s model estimate. pp. 36, 37	22–23	38–39
19	Future prediction and past inference from the cosmic initial boundary. pp. 37	1, 3–5, 22	40
20	Instability of physical reversal and reversal with integer arithmetic. pp. 36, 37	3, 6, 12–13, 24–26	41–44
21	Quantum correspondence and unitary evolution. pp. 38	1–3, 27–28	45–46
22	Lebowitz’s proposal for explaining measurement. pp. 33, 38	22, 29–31	47–50
23	View on the shared origin of psychological and electromagnetic arrows. pp. 38	22, 32–33	51
24	Lebowitz’s 1993 assessment. pp. 32	$\emptyset$	52

Grounds for the Results

Table 289: Lebowitz 1993 Conditions for the Results

Re- sult Index	Conditions and Domain
1	The nonrelativistic classical Newtonian mechanics selected by Lebowitz.
2	Time-reversal-symmetric Hamiltonian dynamics and the source’s reversal-invariant coarse graining.
3	Common phase-volume and additive-constant conventions, positive finite region volumes and positive Boltzmann constant.
4	Equilibrium, large macrocells, appropriate additive constants and comparison of the difference per particle.
5	The identity uses a normalized uniform density on a macroregion of positive finite volume. The generalized thermodynamic correspondence uses local equilibrium and slowly varying densities.
6	Hamiltonian–Liouville evolution and the source’s mixing example after partition removal.
7	Very many degrees of freedom and separation of macroscopic and microscopic scales.

Re- sult Index	Conditions and Domain
8	Specified macroregion and measure, with small dispersion of the quantities of interest when using ensembles computationally.
9	The source's scatterer arrangement, bounded free-flight distance and many particles, with either infinite or very large finite scale ratio.
10	Individual realizations of the Hamiltonian systems compared in the source, including a few hard spheres.
11	Experience-based observed sequence, low-entropy initial preparation, typical microstate and constraint removal.
12	Perfect isolation, time-reversal-symmetric dynamics, complete velocity reversal and equal positive time intervals.
13	The isolated macroscopic systems considered. Equating residence fractions with volume fractions is the stated weak ergodic hypothesis, absent contrary reasons such as extra conserved quantities.
14	Time-reversal-symmetric measure, Liouville volume preservation and the respective conditions of a prepared history or long-equilibrated system.
15	Typicality within the special initial macroregion.
16	The preparation chain in the source, including experimenters, apparatus, biological resources and solar radiation.
17	The before-and-after meteor–Moon collision example given by Lebowitz.
18	The past-boundary hypothesis added to reversible physical laws and the initial/final model states compared by Penrose.
19	The special initial cosmic macrostate and a typical initial microstate within its region.
20	The source's mixing system, precision of reversal and isolation, external perturbations, and simulation with discrete-time integer arithmetic.
21	The source's wavefunction representation, conjugation, macro-subspaces and specified unitary evolution.
22	Time-symmetric joint evolution of system and environment, many degrees of freedom and the cosmic low-entropy boundary.
23	The two arrows that Lebowitz attributes to the universe's low-entropy initial state.
24	Inconsistency claims known to Lebowitz at the time and his assessment of Boltzmann's statistical account.

Definition 8.394 (Result-Dependency Hyperedges). Each result edge takes the temporal premises and claim support in Table 288 as inputs and outputs the corresponding result node.

Evidence Hierarchy Lebowitz discusses Boltzmann's statistical account, derivations in models, earlier reversal experiments and numerical simulations, and the correspondence with quantum theory. The source reports zero directly measured objects and zero direct-measurement predicates from measurements of its own.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local reach count are both zero, leaving 34 primitive-time roots. The source explains the orientation of typical macroscopic evolution using given instants, intervals, microscopic evolution and a low-entropy initial boundary.

Result Dependence Of the 24 results, 18 depend on primitive-time roots and 6 are reachable from the source formal-evidence environment. The latter are Results 1, 3–5, 8, 24, covering the scope, static entropy definitions and relations, the meaning of typicality, and Lebowitz’s assessment. No result is generated from nontemporal physical inputs alone. There are 89 root–result dependence pairs, of which 49 are necessary.

Evidentiary Effect Lebowitz reconciles reversible microscopic dynamics with macroscopic irreversibility through scale separation, phase-volume typicality and a low-entropy initial boundary. The Lorentz-gas argument has the source’s geometry and scaling-limit conditions; exact velocity reversal requires perfect isolation and equal intervals; Gibbs-entropy constancy requires Liouville volume preservation. The quantum correspondence relates the specified states, evolutions, reversal and macro-subspaces. Explaining measurement through a single theory of system and environment is Lebowitz’s proposal, and the shared origin of psychological and electromagnetic arrows is his view.

Placement of the Grounding Obligation Classical instants, velocities and evolution, intervals around reversal, preparation and cosmic boundaries, isolation, relaxation and recurrence timescales, and the temporal argument of quantum evolution are premises of the respective arguments. None of these 34 time contents belongs to the generation closure from nontemporal physical inputs or to the local closure with source formal evidence. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Velocity-reversal errors, external perturbations and numerical roundoff in the source concern the stability of retracing trajectories toward lower entropy. Actual failure of the physical-time generation conditions audited by LCTR, and consequent changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 89 root–result dependence pairs. Actual reach from the failure location of a specific physical-time generation condition to a result is undetermined by the source evidence.

Result-Preservation Domain If the dynamics, coarse graining, phase-volume measure, boundary conditions, typicality assumptions and model domains are fixed, and generated temporal representations realize the time-dependent results’ input relations with the same ordering, intervals and evolution correspondences, each result is preserved with its stated domain and epistemic status. Static entropy relations, the meaning of typicality and Lebowitz’s assessment use their respective claim support.

Repair Scope The repair targets are the time contents used in the respective arguments and the physical-time generation conditions that must be satisfied before their use. If actual failure of a specific physical-time generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.395 (Primitive-Time Roots and Result Dependence in the Lebowitz 1993 Source). In the component order of Definition 8.115,

$$\begin{aligned}\text{AuditPartIdx}_R = & (\emptyset, [34]; \\ & \emptyset, \{1, 3, 4, 5, 8, 24\}, \\ & [24] \setminus \{1, 3, 4, 5, 8, 24\}, \emptyset).\end{aligned}$$

$$R \in \text{Ref}^{\text{audit,rootdep}}, \quad R \notin \text{Ref}^{\text{audit,rootunreach}}.$$

Proof. Closure from Initial Claims 1–5 remains those five nodes. Even when all 54 claims are supplied as source formal evidence, none of the 11 transfers fires, since every transfer has temporal inputs. This closure includes the 6 results with empty temporal premises. Adding the 34 time contents as seeds reaches all 24 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.396 (Minimal Primitive-Time-Root Support in the Lebowitz 1993 Source). The table gives each result’s minimal-support family, with 47 family members in total. For each result reachable from the evidence environment alone, the empty set is the sole family member.

Proof. Start from singleton time-content seeds and apply Definition 7.46 to the 11 transfers. From the stable antichains, combine supports for each result’s temporal premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Table 290: Minimal Primitive-Time-Root Support in the Lebowitz 1993 Source

Result Index	Minimal-Support Index Family
1	$\emptyset$
2	$\{1, 2, 10, 11, 12\}, \{1, 2, 10, 13\}$
3	$\emptyset$
4	$\emptyset$
5	$\emptyset$
6	$\{1, 2, 14\}, \{1, 3, 14\}$
7	$\{1, 2, 5\}, \{3, 5\}, \{1, 2, 19\}, \{1, 3, 19\}, \{3, 10, 19\}, \{1, 2, 22\}, \{1, 3, 22\}$
8	$\emptyset$
9	$\{1, 2, 9\}$
10	$\{9\}$
11	$\{1, 2, 10, 14, 19\}, \{3, 10, 14, 19\}$
12	$\{1, 2, 11, 12\}$
13	$\{1, 2, 16, 17, 18, 34\}, \{1, 3, 16, 17, 18, 34\}$
14	$\{1, 2, 11, 15, 19, 34\}, \{1, 3, 11, 15, 19, 34\}$
15	$\{1, 2, 19\}, \{1, 3, 19\}$
16	$\{1, 2, 19, 20\}, \{3, 19, 20\}, \{4, 19, 20\}, \{6, 19, 20\}$

Result Index	Minimal-Support Index Family
17	$\{5, 21\}, \{1, 2, 19, 21\}, \{1, 3, 19, 21\}, \{10, 19, 21\}, \{1, 2, 21, 22\}, \{1, 3, 21, 22\}$
18	$\{22, 23\}$
19	$\{1, 2, 22\}, \{1, 3, 22\}$
20	$\{1, 2, 11, 12, 24, 25, 26\}, \{1, 2, 12, 13, 24, 25, 26\}, \{3, 12, 13, 24, 25, 26\}$
21	$\{1, 2, 27\}$
22	$\{22, 29, 30, 31\}$
23	$\{22, 32, 33\}$
24	$\emptyset$

Proposition 8.397 (Audit Conclusion for the Lebowitz 1993 Source). Complete generation and local reach are both zero, leaving 34 primitive-time roots. The 24 results divide into 6 evidence-reachable and 18 primitive-time-root-dependent results. The 54 claims, 34 time contents, 11 transfers and 24 results give 123 source-specific obligation atoms. The 110 claim/time-content use pairs have 145 role assignments.

Proof. This follows from the definitions of claims, time contents, transfers and results, the reachable sets and the minimal-support families. $\square$

Proposition 8.398 (Audit-Boundary Counts for the Lebowitz 1993 Source). $\text{BoundaryCount}_R^{\text{audit}} = (112, 35)$.

Proof. The 54 claims, 34 time contents and 24 results give 112 nodes; the 11 transfers and 24 result dependencies give 35 edges. $\square$

Individual Statements Statement evidence 1: Preferred temporal direction of macroscopic events

Original text (English):

why do all macroscopic events have a preferred time direction?

Source location The introduction on p. 32 of Lebowitz, *Boltzmann's Entropy and Time's Arrow*, *Physics Today*, Vol. 46, No. 9.

Minimal context The introductory question concerning compatibility between the time-reversal symmetry of the microscopic laws and the preferred direction of observed macroscopic nonequilibrium processes.

Attribution A question in the text by Joel L. Lebowitz.

Statement evidence 2: Correspondence between classical temporal evolution and quantum unitary evolution

Original text (English):

the time evolution $X(t)$, to the unitary Schrodinger evolution

Source location *Quantum mechanics*, p. 38 of Lebowitz, *Boltzmann's Entropy and Time's Arrow*, *Physics Today*, Vol. 46, No. 9.

Minimal context The passage extending the time-reversal structure of classical statistical mechanics to quantum theory by mapping a classical state to a wave function, velocity reversal to complex conjugation, and the temporal evolution $X(t)$ to unitary evolution.

Attribution A statement in the text by Joel L. Lebowitz.

Statement evidence 3: Denial of the need for an alternative explanation

Original text (English):

All claims of inconsistencies that I know of are, in my opinion, wrong; I see no need for alternate explanations.

Source location The introduction on p. 32 of Lebowitz, *Boltzmann's Entropy and Time's Arrow*, *Physics Today*, Vol. 46, No. 9.

Minimal context The passage immediately following the statement that Boltzmann's later writings are clear and have repeatedly been misunderstood.

Attribution A statement in the text by Joel L. Lebowitz.

Statement evidence 4: Temporal order and the assumption of a low-entropy initial state

Original text (English):

If we think further about our ordering of Mb and Mc, it seems to derive from our assumption that Mb is itself so unlikely that it must have evolved from an initial state of even lower entropy like Ma.

Source location *Initial conditions*, p. 35 of Lebowitz, *Boltzmann's Entropy and Time's Arrow*, *Physics Today*, Vol. 46, No. 9.

Minimal context The passage connecting the ordering of macrostates to the assumption of a low-entropy initial state after stating that microscopic mechanics gives the two directions of the time variable equal a priori standing.

Attribution A statement in the text by Joel L. Lebowitz.

Statement evidence 5: Absence of historical knowledge and experience-based inference of an initial state

Original text (English):

In the absence of any knowledge about the history of the system before and after the sequence, we use our experience to conclude that the low-entropy state Ma must have been an initial prepared state.

Source location *Initial conditions*, p. 36 of Lebowitz, *Boltzmann's Entropy and Time's Arrow*, *Physics Today*, Vol. 46, No. 9.

Minimal context The passage describing inference of an initial state when no information is available about the history before or after a sequence of snapshots that includes a nonequilibrium state; it immediately precedes a quotation from Roger Penrose.

Attribution A statement in the text by Joel L. Lebowitz.

Common Evidence with Delimited Scope The five statements above specify the original text of Lebowitz 1993, its source location, minimal context, and source attribution.

Target state- ment	Statement object	Predicate	Explicit domain of application
S31- Q001	preferred temporal direction of macroscopic events	Asks why macroscopic events have a preferred temporal direction.	The compatibility problem, in the introduction, between time-reversal symmetry of microscopic laws and the direction of macroscopic nonequilibrium processes.
S31- Q002	correspondence between classical temporal evolution and quantum unitary evolution	Maps the classical temporal evolution $X(t)$ to unitary Schrödinger evolution.	The domain in <i>Quantum mechanics</i> that maps classical states, velocity reversal, and temporal evolution to wave functions, complex conjugation, and unitary evolution.
S31- Q003	need for an alternative explanation	Assesses the inconsistency claims known to Lebowitz as wrong and states that no alternative explanation is needed.	The scope, in the introduction, of Boltzmann's statistical-mechanical explanation and of the criticisms considered by Lebowitz as of 1993.
S31- Q004	ordering of the macrostates M_b , M_c and the low-entropy initial state M_a	States that the ordering of M_b, M_c appears to derive from the assumption that M_b evolved from a still lower-entropy M_a .	The sequence of macrostates and the low-entropy boundary in <i>Initial conditions</i> , under the condition that microscopic mechanics gives equal standing to both temporal directions.
S31- Q005	a sequence of states without historical knowledge and the low-entropy state M_a	States that, when the history before and after the sequence is unknown, experience is used to infer M_a as an initially prepared state.	A sequence of snapshots including a nonequilibrium state, in <i>Initial conditions</i> , when information about its history before and after is not supplied.

Individual Audit

First Statement Lebowitz asks how microscopic time-reversal symmetry is compatible with the temporal direction of nonequilibrium processes observed in individual macroscopic systems. The source explains macroscopic evolution through scale separation associated with many degrees of freedom, phase-volume typicality, and low-entropy preparation and cosmic initial boundaries. The corresponding formal evidence comprises Claims 8, 18, 21, 25, 29, 35, 38 and 40, Transfers 1, 5 and 7–10, and Results 2, 7, 9, 11, 15, 18 and 19.

Second Statement Lebowitz maps microstates to wavefunctions, velocity reversal to complex conjugation, time evolution to unitary Schrödinger evolution, and macroregion phase volume to the dimension of a quantum subspace. The source states that this correspondence preserves the large-system equilibrium entropy correspondence and time symmetry. The corresponding formal evidence comprises Claims 45–46, Transfer 11 and Result 21.

Third Statement In 1993, Lebowitz assesses the inconsistency claims known to him as mistaken and regards alternative explanations as unnecessary, on the basis of Boltzmann’s later writings and the criticisms of his account known to Lebowitz. The third statement is Lebowitz’s own assessment arising from that comparison. The corresponding formal evidence consists of Claim 52 and Result 24.

Fourth Statement In the source, time-reversal-symmetric typicality imposed at an intermediate macrostate gives the same prediction toward higher entropy for equal past and future displacements. Lebowitz suggests that the ordering of the experienced sequence derives from the assumption that it evolved from an initial state of still lower entropy. Microstates arriving from the smaller earlier macroregion occupy an atypical subset of the later region. The corresponding formal evidence comprises Claims 32–33 and 35, Transfer 9 and Results 14–15.

Fifth Statement Lebowitz states that, in an equilibrium system isolated for much longer than its relaxation time, fluctuations around a local entropy minimum are typically symmetric. He then explains that, when the history before and after the snapshots is unknown, experience of experimental preparation is used to infer that the low-entropy state was initially prepared. The corresponding formal evidence comprises Claims 25 and 34–35, Transfers 7 and 9, and Results 11 and 14–15.

8.19.32 Source Audit

Audit Inputs

Original Source and Audit Scope

Source Identifier 32.

Source Name Page 1993

Bibliographic Identification Don N. Page, *Information in Black Hole Radiation*, arXiv:hep-th/9306083v2 (1993). $R = \text{Ref}_{\text{Pag93IBH}}$ denotes this source. The audit covers the S-matrix account of formation and evaporation, bipartite information, the adiabatic box model, the moving-mirror example, the two-dimensional dilatonic model and measurement access.

Original-Source Location Identifier 32-L1.

Original-Source Location Content The tables give the page numbers printed in the source.

Audit-Scope Identifier 32-SC1.

Audit-Scope Content The domain specified in the bibliographic identification.

Formal Objects The audit uses 19 time contents, 45 claims, 11 transfers and 23 results. The tables specify information relations for joint pure states, random averages, physical applications to evaporation and the conditions of each model. Page discusses information release in radiation conditional on an S-matrix description of black-hole formation and evaporation and calculates subsystem information for random joint pure states. The source examines the initial information outflow rate during adiabatic evaporation in a box, and the large-N expansion and measurement count in a two-dimensional dilatonic model. The moving-mirror example of early-late radiation correlations is distinct from the dilatonic model.

Assumptions The bipartite calculation uses a normalized joint pure state and a random-state ensemble. The box model uses semiclassical thermal equilibrium, slow expansion, fixed total entropy and the locally stable domain. The dilatonic coordinate transformation uses the exterior region and a positive model constant; its large-N information estimate uses the source’s validity domain and fixed radiation dimension. The claim and result tables specify the conditions for the equations and their physical applications.

Nontemporal Physical Inputs Claims 1–4 are the generation-initial inputs, supplying state spaces and operators, state spectra and ensembles, state data for the box model, and nontemporal data for the dilatonic model. The corresponding claims and transfers specify their relations to physical evolution, expansion, coordinates and outflow rates.

Time Contents

Definition 8.399 (Page 1993 Time Contents). Let $n_R = 19$. The table defines the time contents and their source locations.

Table 292: Page 1993 Time Contents

i	Time Content	Source Location
1	Temporal ordering of black-hole formation and evaporation	pp. 1, 2, 3, 10
2	Evolution connecting an initial quantum state to the final radiation state	pp. 1, 2, 3, 9
3	Early-late radiation ordering in the moving-mirror model	pp. 3
4	Decrease of black-hole mass during evaporation	pp. 7, 8
5	Changes in the two subsystem dimensions during evaporation	pp. 3, 7
6	Slow expansion of the box over time	pp. 7
7	Monotonic increase of the energy ratio x along adiabatic evaporation	pp. 7, 8
8	Physical time t defining outflow rates in the box model	pp. 8
9	Gradual emergence of information in the radiation	pp. 1, 3, 7, 9, 10
10	Radiation-information outflow rate and its initial suppression	pp. 1, 3, 7, 8, 10

i	Time Content	Source Location
11	Early-evaporation boundary specifying the initial energy E_0	pp. 7, 8
12	Null-coordinate pair x^- and x^+ in the dilatonic model	pp. 8
13	Rescaled null-coordinate pair u and v in the dilatonic model	pp. 8, 9
14	Exterior time coordinate t in the dilatonic model	pp. 8, 9
15	Physical temporal intervals in the two-dimensional spacetime metric	pp. 8, 9
16	Fixed emission duration used in the example with fixed radiation dimension	pp. 9
17	Change in outflow rate at the onset of late radiation in the moving-mirror model	pp. 3
18	Sequence of measurements on separate parts of the radiation	pp. 7
19	Causal structure of the information-transport regions conjectured by Page	pp. 10

Definition 8.400 (Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(45)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([4], [45])$. The table gives each claim, its time uses and source locations.

Table 293: Page 1993 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Two finite-dimensional Hilbert spaces, a normalized pure state, density operators, partial traces and subsystem dimensions.	$\emptyset$	pp. 3, 5
2	State spectra, entropy functionals and a random ensemble of pure states.	$\emptyset$	pp. 5, 6
3	Non-temporal data specifying box volume, energy, mass, area and radiation state density.	$\emptyset$	pp. 3, 4, 5, 7
4	Non-temporal data for the dilaton field, a positive model constant, the number of field species and quantum state spaces.	$\emptyset$	pp. 8, 9, 10
5	If formation and evaporation admit an S-matrix description, Page expects the initial information to emerge in black-hole radiation.	$1^S, 2^S, 9^S$	pp. 1, 2
6	Page describes Hawking's proposal of a map from a pure initial state to a mixed final radiation state.	$1^S, 2^S$	pp. 2

p	Claim	i	Source Location
7	Page lists release of information in the bulk radiation, retention in a long-lived or stable remnant, and information loss, and favors release in the bulk radiation.	$1^S, 2^S, 9^S$	pp. 2
8	Page argues that suppression of the initial outflow rate or dispersion of information over many correlations may prevent finite-order semiclassical or $1/N$ expansions from detecting gradual release.	$9^S, 10^{SQ}$	pp. 1, 3, 10
9	In the moving-mirror model cited by Page, early radiation is a maximally mixed thermal state with no information, whereas correlations with late radiation make the total radiation a pure state containing the initial information.	$2^S, 3^{XS}$	pp. 3
10	In the moving-mirror model, the information outflow rate changes nonanalytically at the onset of late radiation, and the information resides in many correlations between the early and late radiation.	$3^{XS}, 17^{BSQ}$	pp. 3
11	The source's bipartite model uses a normalized joint pure state of a black hole of dimension n and radiation of dimension m , with each reduced state defined by the partial trace over the other subsystem.	$\emptyset$	pp. 3, 5
12	The two reduced density operators of a joint pure state share their nonzero eigenvalues and have equal von Neumann entropies.	$\emptyset$	pp. 5
13	The source defines radiation information by $I_r = \ln m - S_r$, the shortfall of the reduced entropy S_r from its maximum $\ln m$.	$\emptyset$	pp. 5
14	Page takes proximity of the actual radiation information to the subsystem average over random joint pure states as a reasonable first guess.	$\emptyset$	pp. 5
15	Page presents the harmonic-sum expression in Eq. (6) for the mean information when $m \leq n$ and states that this expression appears to hold.	$\emptyset$	pp. 5
16	For $1 \ll m \leq n$, the source approximates the mean information by $m/(2n)$, with the smaller subsystem typically close to maximally mixed.	$\emptyset$	pp. 5, 7

p	Claim	i	Source Location
17	For a joint pure state, $I_r - I_h = \ln(m/n)$. Exchanging the subsystems for $m \geq n$ and approximating for sufficiently large subsystem dimensions gives a leading mean information $\ln(m/n)$ with correction $n/(2m)$.	$\emptyset$	pp. 6
18	Figure 1 plots the mean entropy and mean information against $\ln m$ for the 105 integer factorizations of the fixed total dimension $mn = 291600$.	$\emptyset$	pp. 6
19	In the random pure-state model, each part with dimension smaller than the remainder typically contains little information, while correlations among the parts contain much of the total information.	$\emptyset$	pp. 7
20	Page expects little information from each part when small parts of the radiation are examined one at a time, even after the radiation has become the larger subsystem.	18^O	pp. 7
21	In the source's application to evaporation, the expected radiation information increases as black-hole mass and dimension n decrease and radiation dimension m increases.	$4^{SQ}, 5^{SQ}, 9^{SQ}$	pp. 7
22	The source considers formation from a pure radiation state in a box, with initial energy near E and angular momentum near zero, and a semiclassically stable black hole holding more than four fifths of the total energy for sufficiently large E^5/V .	$1^S, 2^S$	pp. 3, 4
23	For a nearly Schwarzschild black hole in a large box with $E^3 \ll V$, semiclassical thermal equilibrium and negligible contributions from massive particles, Eqs. (1) and (2) give the radiation energy and entropy.	$\emptyset$	pp. 5
24	The source takes black-hole thermodynamic entropy to be one quarter of its area, approximately $4\pi M^2$ in the nearly Schwarzschild case, and approximates the state-space dimension by its exponential.	$\emptyset$	pp. 5
25	The source considers adiabatic evaporation under slow box expansion with $1 \ll E^3 \ll V$ and holds the total semiclassical entropy $s_r + s_h$ constant.	$4^S, 6^S$	pp. 7

p	Claim	i	Source Location
26	The source uses $x = (E - M)/M$ as a variable increasing monotonically along adiabatic expansion and restricts locally stable equilibrium to $x \lesssim 1/4$.	6 ^S , 7 ^{SQ}	pp. 7
27	With E_0 specified at the initial stage where x is negligible, Eqs. (10)–(15) give E , M , temperature, V , s_r and s_h as functions of x .	4 ^{SQ} , 7 ^{SQ} , 11 ^B	pp. 7
28	Combining the random-state mean information, the approximate relation between dimensions and thermodynamic entropies, and the entropies of the adiabatic family, the source gives the leading exponential factor of I_r in Eq. (16).	7 ^{SQ} , 9 ^{SQ} , 11 ^B	pp. 7
29	The source differentiates Eq. (16) with respect to physical time t to obtain Eq. (17), retaining the exponential behavior of dI_r/dt multiplied by dx/dt and omitting the prefactor.	7 ^{SQ} , 8 ^{SQ} , 9 ^{SQ} , 10 ^{SQ}	pp. 8
30	With $M \approx E_0$ initially and $y = M_{\text{Planck}}/E_0$, the source identifies $\exp(-4\pi/y^2)$ as the leading exponential factor in the information outflow rate and notes its nonanalyticity at $y = 0$.	8 ^{SQ} , 10 ^{SQ} , 11 ^B	pp. 8
31	The function equal to $\exp(-4\pi/y^2)$ for $y > 0$ and zero at $y = 0$ has all Taylor coefficients zero at the origin while remaining positive for $y > 0$.	$\emptyset$	pp. 8
32	The source applies the argument about a $1/N$ expansion to a two-dimensional dilatonic black-hole model with N minimally coupled scalar fields.	$\emptyset$	pp. 8
33	Equation (20) defines the null-coordinate pair u and v by $x^- = \sqrt{M/\lambda^3} u$ and $x^+ = \sqrt{M/\lambda^3} v$.	12 ^{XQ} , 13 ^{XQ}	pp. 8
34	In the exterior region $u < 0$ and $v > 0$, $u = -\sinh(\lambda r)e^{-\lambda t}$ and $v = \sinh(\lambda r)e^{\lambda t}$ transform the metric to $ds^2 = -\tanh^2(\lambda r)dt^2 + dr^2$.	13 ^{XQ} , 14 ^{XQ} , 15 ^{IQ}	pp. 8
35	With time normalized so that $g_{tt} \rightarrow -1$ at spatial infinity, the classical temperature is $\lambda/(2\pi)$, independent of mass M .	14 ^{BQ} , 15 ^{IQ}	pp. 8, 9
36	At the horizon, the classical solution satisfies $M = \lambda e^{-2\phi_H}$; the first law gives $s_h = 2\pi M/\lambda$, with Page neglecting the integration constant for large s_h .	$\emptyset$	pp. 9

p	Claim	i	Source Location
37	In the exterior regime where the source's quantum-corrected $1/N$ expansion is valid, $e^{-2\phi_H} \gtrsim N/24$ and hence $s_h \gtrsim \pi N/12$.	$\emptyset$	pp. 9
38	Assuming unitary evolution of the joint quantum system of the black hole and exterior radiation, Page estimates the black-hole dimension as $n \sim e^{s_h} \gtrsim e^{\pi N/12}$.	2^S	pp. 9
39	For $m \sim e^{s_r} < n$, the source approximates the expected radiation information by $m/(2n)$ and bounds its scale above by $e^{s_r - \pi N/12}$; the argument as $1/N \rightarrow 0$ holds the effective radiation dimension m fixed.	$\emptyset$	pp. 9
40	Page gives radiation for a fixed duration at mass-independent temperature as an example of holding the effective radiation dimension m fixed.	$14^B, 16^{IQ}$	pp. 9
41	For $m > n$, the source approximates the expected radiation information by $\ln(m/n) + n/(2m)$ and contrasts it with exponential suppression for the smaller radiation subsystem.	$\emptyset$	pp. 10
42	Page expects testing for information in the radiation to require measuring most of the $m^2 - 1$ independent real parameters of its density matrix, and estimates at least order m^2 measurements, hence a number exponentially large in N .	$\emptyset$	pp. 10
43	Page argues that even if the prediction error for each measurement can be made smaller than any finite power of $1/N$, the squared errors summed over exponentially many measurements cannot be controlled by increasing N .	$\emptyset$	pp. 10
44	Page states that no sufficiently plausible mechanism for information escape has been established and offers as speculation narrow regions, with metric and causal structure altered by quantum fluctuations, transporting information from near the center to near the apparent horizon.	19^C	pp. 10
45	Page concludes conditionally that if information emerges gradually, an extremely small initial rate or dispersion over many correlations would likely prevent perturbative analysis from either finding or excluding that release.	$9^S, 10^{SQ}$	pp. 10

Definition 8.401 (Semantic Components and Generation-Initial Set). The 45 claims are the semantic components. Claims 1–4 form the generation-initial set; Claims 1–45 form the source formal-evidence environment.

Table 294: Page 1993 Time-Use Roles

Code	Role	Use Count
I	Interval	3
Q	Quantity	25
X	Cross-position correspondence	6
K	Clock reading	0
C	Causality	1
S	State evolution	36
P	Probabilistic evolution	0
O	Observation record	1
B	Reference boundary	6

Table 295: Page 1993 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3, 18–19	member	order_relation
2, 4–7, 9–10, 15–17	dependency	$\emptyset$
8, 12–14	member	instant, order_relation
11	member	instant, endpoint

Formal Input Elements

Definition 8.402 (Bipartite Information and Model Coordinates). Write the source subsystem dimensions as $m = \text{Dim}_{\text{rad}}$, $n = \text{Dim}_{\text{BH}}$. The joint state satisfies

$$|\Psi_{\text{BH:rad}}\rangle \in \mathcal{H}_{\text{BH}} \otimes \mathcal{H}_{\text{rad}}, \quad \langle \Psi_{\text{BH:rad}} | \Psi_{\text{BH:rad}} \rangle = 1, \quad \rho_{\text{BH:rad}} = |\Psi_{\text{BH:rad}}\rangle \langle \Psi_{\text{BH:rad}}|.$$

$$\rho_{\text{rad}} = \text{Tr}_{\text{BH}} \rho_{\text{BH:rad}}, \quad \rho_{\text{BH}} = \text{Tr}_{\text{rad}} \rho_{\text{BH:rad}}.$$

Radiation information is the shortfall from maximum entropy. Its correspondence with the established notation is

$$I_r = \mathcal{I}_{\text{def}}^{\text{rad}} = S_{\text{max}}^{\text{rad}} - S_{\text{vN}}^{\text{rad}} = \ln m - S_r, \quad S_r = S_h.$$

For the random pure-state average with $m \leq n$, Page states that the following expression appears to hold (Eq. (6)):

$$I_{m,n} = \ln m + \frac{m-1}{2n} - \sum_{k=n+1}^{mn} \frac{1}{k}, \quad I_{m,n} \simeq \frac{m}{2n} \quad (1 \ll m \leq n).$$

Subsystem exchange gives $I_{m,n} = \ln(m/n) + I_{n,m}$; for sufficiently large subsystem dimensions, $m \geq n$ gives $I_{m,n} \simeq \ln(m/n) + n/(2m)$ (Eq. (8)).

In the box model, $x = \chi_{\text{box}}^E = (E - M)/M$ is the energy ratio. For the source null coordinates of the two-dimensional model, set $x_{\text{null},0} = x^-$, $x_{\text{null},1} = x^+$ and write the exterior time coordinate as $t_{2\text{D}}$.

$$x^- = \sqrt{M/\lambda^3} u, \quad x^+ = \sqrt{M/\lambda^3} v, \quad u = -\sinh(\lambda r)e^{-\lambda t_{2\text{D}}}, \quad v = \sinh(\lambda r)e^{\lambda t_{2\text{D}}}.$$

$$ds^2 = -\frac{du dv}{\lambda^2(1 - uv)} = -\tanh^2(\lambda r) dt_{2\text{D}}^2 + dr^2.$$

The exterior coordinate domain is $u < 0, v > 0, \lambda > 0$ (Eqs. (20)–(21)).

Dependency Inputs The table gives each transfer’s time inputs, outputs, claim support and source location.

Definition 8.403 (Page 1993 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [11]$.

Table 296: Page 1993 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	4	5	21, 24	Subsystem-dimension changes accompanying mass decrease. pp. 7
2	5	9	14, 16–17, 21	From subsystem-dimension changes to changes in expected radiation information. pp. 5, 6, 7
3	6	4	23, 25	Adiabatic evaporation under slow box expansion. pp. 5, 7
4	6	7	26	Increase of the energy ratio along adiabatic expansion. pp. 7
5	7, 11	4	27	Mass history from the initial energy and expansion variable. pp. 7
6	7, 11	9	14, 16, 24, 27–28	Exponential factor of expected information along the adiabatic family. pp. 5, 7
7	7–9	10	29	Differentiation of expected information with respect to physical time. pp. 8
8	12	13	33	Positive rescaling of null coordinates. pp. 8
9	13	14	34	Transformation from null coordinates to exterior time. pp. 8
10	13	15	34	Spacetime intervals in null coordinates. pp. 8
11	14	15	34–35	Temporal metric component in exterior time coordinates. pp. 8, 9

Grounding Conditions

1. The source’s dimension–entropy correspondence for the evaporating black hole and radiation.

2. Page's conjectured application of the random pure-state average to physical radiation.
3. $1 \ll E^3 \ll V$, semiclassical thermal equilibrium and fixed total semiclassical entropy.
4. The source's $x = (E - M)/M$ within the locally stable domain $x \lesssim 1/4$.
5. The adiabatic family in Eq. (11), with E_0 specified at $x \approx 0$.
6. The smaller-subsystem mean information and the approximations in Eqs. (14)–(16).
7. A differentiable expansion history, with the prefactor omitted in Eq. (17).
8. Equation (20), $M > 0$ and $\lambda > 0$.
9. The exterior region $u < 0$, $v > 0$ and $\lambda > 0$.
10. The classical metric in Eq. (21) and the exterior region.
11. The exterior region and time normalization at spatial infinity.

Results

Definition 8.404 (Page 1993 Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [23]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 297: Page 1993 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Conditional expectation of information release under an S-matrix description. pp. 1, 2	1–2, 9	5
2	Alternatives compared in the source for the fate of information. pp. 2	1–2, 9	6–7
3	Early–late correlations and the pure total radiation state in the moving-mirror model. pp. 3	2–3, 17	9–10
4	Equal reduced entropies of a joint pure state. pp. 5	$\emptyset$	11–12
5	Information as a shortfall from maximum entropy. pp. 5	$\emptyset$	13
6	Mean information of the smaller subsystem. pp. 5, 7	$\emptyset$	14–16
7	Mean information of the larger subsystem. pp. 6	$\emptyset$	12–13, 17
8	Subsystem-entropy and information plot at fixed total dimension. pp. 6	$\emptyset$	18
9	Information in small parts and in correlations across the system. pp. 7	$\emptyset$	19

q	Result and Source Location	Time Premises i	Claim Support p
10	Expected information from sequential measurements of small radiation parts. pp. 7	18	20
11	Increase of expected radiation information during evaporation. pp. 7	4–5, 9	14, 21, 24
12	Adiabatic family under slow box expansion. pp. 5, 7	4, 6–7, 11	23–27
13	Information outflow rate and initial exponential suppression in the box model. pp. 7, 8	7–8, 10–11	14, 16, 24, 28–30
14	Positive exponential factor absent from every finite Taylor expansion. pp. 8	$\emptyset$	31
15	Null coordinates, exterior time and metric in the dilatonic model. pp. 8, 9	12–15	32–35
16	Mass–entropy relation and validity domain in the dilatonic model. pp. 9	$\emptyset$	36–37
17	Black-hole dimension estimate assuming a unitary joint system. pp. 9	2	37–38
18	Large- N information suppression at fixed radiation dimension. pp. 9	2	14, 16, 37–39
19	Fixed-dimension example based on fixed emission duration. pp. 9	14, 16	35, 40
20	Information in the larger radiation subsystem. pp. 10	$\emptyset$	17, 41
21	Page’s exponential estimate of measurement count for radiation information. pp. 9, 10	2	37–38, 42
22	Page’s argument concerning perturbative prediction errors and measurement count. pp. 10	2	37–38, 42–43
23	Conjectured information transport through a different causal structure. pp. 10	19	44

Grounds for the Results

Table 298: Page 1993 Conditions for the Results

Re- sult Index	Conditions and Domain
1	The S-matrix description of formation and evaporation posited by Page.
2	Hawking’s proposal and Page’s alternatives and preference.
3	The moving-mirror example cited by the source.
4	A normalized finite-dimensional joint pure state.
5	$I_r = \ln m - S_r$.

Re- sult Index	Conditions and Domain
6	Random pure states with $1 \ll m \leq n$.
7	$m \geq n$, joint purity and the approximation for large subsystem dimensions.
8	$mn = 291600$, 105 factor pairs and horizontal axis $\ln m$.
9	Random joint pure states with each small part of lower dimension than its remainder.
10	Page's example of examining one part at a time.
11	Decreasing black-hole mass and physical application of the subsystem average.
12	$1 \ll E^3 \ll V$, $x \lesssim 1/4$ and initial energy E_0 at $x \approx 0$.
13	The source's adiabatic family, conjectured random-average description and leading-exponential approximation.
14	$f(y)=\exp(-4\pi/y^2)$ for $y > 0$ and $f(0) = 0$.
15	$M > 0$, $\lambda > 0$, $u < 0$, $v > 0$, the classical solution and asymptotic time normalization.
16	Neglect of the entropy integration constant at large s_h and the source's valid regime for the $1/N$ expansion.
17	Page's unitary-joint-system assumption and $n \sim e^{s_h}$.
18	$1 \ll m < n$, fixed radiation dimension m and the valid regime of the $1/N$ expansion.
19	The source's example of emission for fixed duration at mass-independent classical temperature.
20	The random pure-state mean approximation for $m > n$.
21	$m > n$ and Page's expectation that most independent real parameters of the density matrix must be measured.
22	Page's exponential measurement-count estimate and prediction-error assessment at each finite order.
23	Transport from near the center to near the apparent horizon, offered by Page as speculation.

Definition 8.405 (Result-Dependency Hyperedges). Each result edge takes the temporal premises and claim support in Table 297 as inputs and outputs the corresponding result node.

Evidence Hierarchy Page presents model calculations, examples from earlier work, conjectures about application to radiation information, and measurement-count estimates. The source reports zero directly measured objects and zero direct-measurement predicates from measurements of its own.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local reach count are both zero, leaving 19 primitive-time roots. The source discusses information release using given formation–evaporation order, state evolution, expansion operations, physical time, null coordinates and temporal intervals.

Result Dependence Of the 23 results, 14 depend on primitive-time roots and 9 are reachable from the source formal-evidence environment. The latter are Results 4–9, 14, 16, 20, covering static entropy and mean-information relations, the dimension pairs in Fig. 1, the nonanalytic exponential factor, and mass–entropy conditions. No result is generated from nontemporal physical inputs alone. There are 40 root–result dependence pairs, of which 19 are necessary.

Evidentiary Effect Page expects information to emerge in radiation conditional on an S-matrix account of formation and evaporation. Applying the subsystem average to actual radiation is Page’s first guess. The box model uses slow adiabatic expansion and the locally stable domain; the dilatonic model uses the exterior coordinate domain and the valid regime for the $1/N$ expansion. The large- N estimate of small information holds radiation dimension fixed. Exponential growth of measurement count is Page’s estimate, and the information-transport regions are his speculation.

Placement of the Grounding Obligation Formation–evaporation order, quantum evolution, box operations and information outflow rates, temporal and null coordinates of the metric, fixed emission duration, measurement order and the causal structure of the conjectured transport regions are premises of the respective arguments. None of these 19 time contents belongs to the generation closure from nontemporal physical inputs or the local closure with source formal evidence. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The source’s local-stability conditions, domains of semiclassical and $1/N$ approximations, and measurement-count argument concern model applicability and information detection. Actual failure of the physical-time generation conditions audited by LCTR, and consequent changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 40 root–result dependence pairs. Actual reach from the failure location of a specific physical-time generation condition to a result is undetermined by the source evidence.

Result-Preservation Domain If subsystem states, ensembles, model equations, approximation conditions and domains are fixed, and generated temporal representations realize the time-dependent results’ input relations with the same ordering, intervals, coordinates and evolution correspondences, each result is preserved with its stated domain and epistemic status. Static entropy and information relations and the exponential factor use their respective claim support.

Repair Scope The repair targets are the time contents used in the respective arguments and the physical-time generation conditions that must be satisfied before their use. If actual failure of a specific physical-time generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.406 (Primitive-Time Roots and Result Dependence in the Page 1993 Source). In the component order of Definition 8.115,

$$\begin{aligned}\text{AuditPartIdx}_R = (\emptyset, [19]; \\ \emptyset, \{4, 5, 6, 7, 8, 9, 14, 16, 20\}, \\ [23] \setminus \{4, 5, 6, 7, 8, 9, 14, 16, 20\}, \emptyset).\end{aligned}$$

$$R \in \text{Ref}^{\text{audit,rootdep}}, \quad R \notin \text{Ref}^{\text{audit,rootunreach}}.$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Even when all 45 claims are supplied as source formal evidence, none of the 11 transfers fires, since every transfer has temporal inputs. This closure includes the 9 results with empty temporal premises. Adding the 19 time contents as seeds reaches all 23 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.407 (Minimal Primitive-Time-Root Support in the Page 1993 Source). The table gives each result’s minimal-support family, with 36 family members in total. For each result reachable from the evidence environment alone, the empty set is the sole family member.

Proof. Start from singleton time-content seeds and apply Definition 7.46 to the 11 transfers. From the stable antichains, combine supports for each result’s temporal premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Table 299: Minimal Primitive-Time-Root Support in the Page 1993 Source

Result Index	Minimal-Support Index Family
1	$\{1, 2, 4\}, \{1, 2, 5\}, \{1, 2, 6\}, \{1, 2, 9\}, \{1, 2, 7, 11\}$
2	$\{1, 2, 4\}, \{1, 2, 5\}, \{1, 2, 6\}, \{1, 2, 9\}, \{1, 2, 7, 11\}$
3	$\{2, 3, 17\}$
4	$\emptyset$
5	$\emptyset$
6	$\emptyset$
7	$\emptyset$
8	$\emptyset$
9	$\emptyset$
10	$\{18\}$
11	$\{4\}, \{6\}, \{7, 11\}$
12	$\{6, 11\}$
13	$\{6, 8, 11\}, \{7, 8, 11\}$
14	$\emptyset$
15	$\{12\}$
16	$\emptyset$
17	$\{2\}$
18	$\{2\}$

Result Index	Minimal-Support Index Family
19	$\{12, 16\}, \{13, 16\}, \{14, 16\}$
20	$\emptyset$
21	$\{2\}$
22	$\{2\}$
23	$\{19\}$

Proposition 8.408 (Audit Conclusion for the Page 1993 Source). Complete generation and local reach are both zero, leaving 19 primitive-time roots. The 23 results divide into 9 evidence-reachable and 14 primitive-time-root-dependent results. The 45 claims, 19 time contents, 11 transfers and 23 results give 98 source-specific obligation atoms. The 50 claim/time-content use pairs have 78 role assignments.

Proof. This follows from the definitions of claims, time contents, transfers and results, the reachable sets and the minimal-support families. $\square$

Proposition 8.409 (Audit-Boundary Counts for the Page 1993 Source). $\text{BoundaryCount}_R^{\text{audit}} = (87, 34)$.

Proof. The 45 claims, 19 time contents and 23 results give 87 nodes; the 11 transfers and 23 result dependencies give 34 edges. $\square$

Individual Statements Statement evidence 1: S-matrix description and the appearance of information in radiation

Original text (English):

If black hole formation and evaporation can be described by an S matrix, information would be expected to come out in black hole radiation.

Source location The opening of the abstract, p. 1 of Page, *Information in Black Hole Radiation*, arXiv:hep-th/9306083v2.

Minimal context The conditional expectation that information appears in the radiation if black-hole formation and evaporation admit an S-matrix description.

Attribution A statement by Don N. Page.

Statement evidence 2: Bipartite-system model assuming a random pure state
Original text (English):

Therefore, here I shall here examine a more natural model, in which the black hole and its surrounding radiation are two subsystems of a combined system which is assumed to be in a random pure state.

Source location The model introduction preceding the subsystem-entropy calculation, p. 3 of Page, *Information in Black Hole Radiation*, arXiv:hep-th/9306083v2.

Minimal context The introduction of a model treating the black hole and its surrounding radiation as two subsystems and assuming that the combined system is in a random pure state.

Attribution A statement by Don N. Page.

Statement evidence 3: First guess in the absence of a full quantum analysis
Original text (English):

Without a full quantum analysis, we cannot really answer this question definitely. However, a reasonable first guess would be that I_r is near the average information in a subsystem of dimension m when the total system, of dimension mn , is in a random pure state.

Source location The introduction to the mean-information estimate immediately preceding Eq. (6), p. 5 of Page, *Information in Black Hole Radiation*, arXiv:hep-th/9306083v2.

Minimal context The passage explicitly states the absence of a full quantum analysis and adopts the mean subsystem information in a random pure state as a first guess.

Attribution A statement by Don N. Page.

Common Evidence with Delimited Scope The table gives the objects, predicates and domains of the three statements above.

Target state- ment	Statement object	Predicate	Explicit domain of application
S32- Q001	S-matrix description and the appearance of information in black-hole radiation	States that, if black-hole formation and evaporation admit an S-matrix description, information is expected to emerge in black-hole radiation.	The conditional domain in the abstract whose antecedent is that the formation and evaporation processes admit an S-matrix description.
S32- Q002	bipartite-system model assuming a random pure state	Introduces a model that treats the black hole and surrounding radiation as two subsystems of a combined system in a random pure state.	The model domain preceding the subsystem-entropy calculation, which assumes a finite-dimensional bipartite system and a random-pure-state ensemble.
S32- Q003	first guess in the absence of a full quantum analysis	Page takes it as a reasonable first guess that the radiation information I_r is close to the mean information of a subsystem of dimension m in a random pure total system of dimension mn .	The conjectured correspondence between actual radiation information and the random-pure-state ensemble average, immediately preceding Eq. (6).

Individual Audit

First Statement Page expects the initial information to emerge in radiation conditional on an S-matrix description of formation and evaporation. This expectation uses the ordering of formation and evaporation, the evolution connecting initial and final states, and the information-release process. The corresponding formal evidence is Claim 5 and Result 1.

Second Statement After the moving-mirror example correlating early and late radiation, Page introduces a model treating the black hole and surrounding radiation as subsystems of a random joint pure state. The model defines reduced states by partial traces and calculates entropies and mean information. The corresponding formal evidence comprises Claims 11–13 and 15–18 and Results 4–8.

Third Statement Page’s first guess is that the actual radiation information is close to the subsystem average for random pure states. The source applies this conjecture to changing subsystem dimensions during evaporation, the information outflow rate in the adiabatic box model, and the large- N estimate at fixed radiation dimension. The corresponding formal evidence comprises Claims 14, 16–17, 21, 28, 39 and 41, Transfers 2 and 6, and Results 6–7, 11, 13, 18 and 20.

8.19.33 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 33.

Source Name Primitive-Time-Root Audit of Barbour 1994 I

Bibliographic Identification Let $R = \text{Ref}_{\text{Bar94TQ1}}$ denote Julian B. Barbour, *The Timelessness of Quantum Gravity: I. The Evidence from the Classical Theory, Classical and Quantum Gravity* 11, 2853–2873 (1994). The audit domain comprises all 21 pages of the source, §§1–14, and equations (1)–(37).

Audit-Scope Identifier 33-SC1.

Audit-Scope Content The audit domain stated in the bibliographic identification.

Original-Source Location Identifier 33-L1.

Original-Source Location Content The audit domain stated in the bibliographic identification and the original-source location columns of the respective evidence tables. The tables identify source locations by printed page number.

Formal Object

Object Identifier 33-F1.

Object Name Formal objects of the original source.

Object Type Formal object.

Formal Content The individual formal objects are given in Definitions 8.410 and 8.412.

Original-Source Location Identifier Set 33-L1.

Assumptions

Assumption Identifier 33-A1.

Assumption Content The individual source assumptions are given in Definitions 8.412 and 8.410.

Original-Source Location Identifier Set 33-L1.

Nontemporal Physical Inputs

Input Identifier 33-P1.

Input Name Nontemporal physical inputs of the original source.

Input Type Nontemporal physical input.

Input Content The individual nontemporal physical inputs are given in Definition 8.412.

Original-Source Location Identifier Set 33-L1.

Time Content

Time-Content Identifier 33-T1.

Time-Content Name Time contents of the original source.

Time-Content Type Time content.

Time-Content Statement The individual time contents are given in Definitions 8.411 and 8.410.

Original-Source Location Identifier Set 33-L1.

Formal Input Elements

Definition 8.410 (Formal objects and explicit assumptions in Barbour 1994 I). Barbour takes complete relative configurations of the universe as fundamental and discusses the operational recovery of time along classical histories. For particle systems he uses the fixed-energy Jacobi principle and best matching; for geometrodynamics he uses matching between three-geometries, local ephemeris time and the BSW action. His discussions of subsystem clocks, astronomical ephemeris time, atomic time and proper-time clocks involve their respective dynamical conditions and comparisons of clock rates.

The Jacobi construction uses a conservative particle system, fixed energy, a configuration metric and a potential. Its Newtonian correspondence through best matching uses zero total angular momentum of the universe; agreement of subsystem-clock rates uses effective isolation and energy-conserving parameters. Four-dimensional reconstruction in geometrodynamics is conditional on a solution of the variational problem determining equilocality. Specialization to general relativity uses the DeWitt supermetric and three-dimensional scalar curvature.

In the source's convention, the line element of the configuration metric is $ds^2 = \sum_i (m_i/2) d\mathbf{x}_i^2$. Set $F = E - V$. Equations (1), (5) and (6) give the Jacobi action, affine parameter and ephemeris time:

$$I_J = \int \sqrt{F} ds, \quad s_a = \int \sqrt{F} ds, \quad t = \int \frac{ds}{\sqrt{F}}.$$

A chosen direction along a realized history and a segment with $F > 0$ specify the ephemeris-time differences given by the last integral. In general geometrodynamics, the source's

kinetic term T_g and conformal factor F give the lapse $N = \sqrt{T_g/F}$ in Eq. (34). The BSW form of general relativity uses $F = R$, where R is the three-dimensional scalar curvature.

Definition 8.411 (Time contents in Barbour 1994 I). Let $n_R = 18$. Table 301 gives the time contents and their source locations.

Table 301: Barbour 1994 I Time Contents and Source Locations

i	Time Content	Source Location
1	External time t in the compared Newtonian mechanics and its quantization	pp. 2855, 2856, 2859, 2860, 2863
2	Succession in a realized particle-configuration history	pp. 2854, 2855, 2856, 2857, 2862, 2863
3	Jacobi ephemeris time and duration defined along a realized history	pp. 2856, 2857, 2858, 2863, 2864
4	Rate of subsystem motion relative to change in the universe as a whole	pp. 2855, 2856
5	Ephemeris times obtained from effectively isolated subsystems	pp. 2857, 2860, 2861
6	Constant proportionality between subsystem-clock rates	pp. 2857, 2860, 2861
7	Time and variable rate of the Earth-rotation clock	pp. 2858
8	Observed solar-system motion histories used to construct astronomical time	pp. 2858
9	Astronomical ephemeris time fitted to solar-system dynamics	pp. 2858
10	Physical processes and reading histories of the atomic-clock network	pp. 2858
11	Temporal evolution used by the environmental model for atomic clocks	pp. 2858
12	Atomic time extracted from the clock network and its environmental model	pp. 2858
13	Sequence and evolution of realized three-geometries	pp. 2864, 2867, 2868, 2869, 2871
14	Local ephemeris time constructed along geometric evolution	pp. 2867, 2868, 2871
15	Local proper time in the general-relativistic specialization	pp. 2868, 2869, 2871
16	External Minkowski time in the comparison with linearized gravity	pp. 2871
17	Readings of clocks measuring proper time	pp. 2871
18	Agreement of clock rates attributed by Barbour to proper-time clocks	pp. 2871

Definition 8.412 (Barbour 1994 I Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(37)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([4], [37])$. Claims 1–4 form the generation-initial set; all 37 claims form the source formal-evidence environment. Each row in the following table gives a claim, its time uses and source location.

Table 302: Barbour 1994 I Physical and Formal Input Claims

p	Claim	i	Source Location
1	Nontemporal data specifying particle configurations, masses, relative distances and configuration space.	$\emptyset$	pp. 2854, 2855, 2856, 2861
2	Values of the configuration metric, potential, fixed energy and conformal factor.	$\emptyset$	pp. 2856, 2857
3	Nontemporal data for three-geometries, spatial metrics, spatial coordinates and supermetrics.	$\emptyset$	pp. 2864, 2865, 2868
4	A wavefunction on configurations with no external-time argument and its static values.	$\emptyset$	pp. 2853, 2861
5	Barbour distinguishes a relative configurational instant from a relative phase instant that also includes instantaneous relative velocities, and explains that Newtonian initial states additionally require an inertial frame.	1^{SQ}	pp. 2854, 2855
6	In the prerelativistic Newtonian picture described by the source, instants of the universe form a unique succession, while the metric of duration is given independently of configurational content.	1^{IB}	pp. 2855
7	Barbour represents a realized classical history by a configuration-space curve containing the motions of the parts of the universe, and recovers operational time along that succession.	2^{S}	pp. 2854, 2855, 2856
8	Barbour distinguishes an external-time speed of traversal of the universe's configuration curve from the rate of subsystem motion relative to change in the universe as a whole.	$1^{\text{Q}}, 2^{\text{S}}, 4^{\text{SQ}}$	pp. 2855, 2856
9	At fixed energy E of a conservative particle system, the Jacobi action is a geodesic variational principle using the configuration metric and $E - V$, invariant under reparametrization by an arbitrary increasing label.	$\emptyset$	pp. 2856, 2857, 2859
10	Choosing the parameter for which $T = E - V$ along a Jacobi orbit yields Newton's equations at fixed energy. The real, nondegenerate integral representation applies on orbit segments with $E - V > 0$.	$\emptyset$	pp. 2856, 2857

p	Claim	i	Source Location
11	Along a realized history, ephemeris time is defined by $t = \int ds/\sqrt{E - V}$ in Eq. (6), and its differences give duration along that history.	$2^S, 3^{IQ}$	pp. 2857, 2858
12	The source defines ephemeris time t and the geodesic affine parameter $s_a = \int \sqrt{E - V} ds$ as distinct quantities.	3^Q	pp. 2857
13	For effectively noninteracting subsystems using energy-conserving Jacobi parameters, Barbour states that their clock rates agree up to constant proportions.	$5^S, 6^{XQ}$	pp. 2857, 2860, 2861
14	Barbour argues that the universe is ultimately the only isolated system and that exact determination of time requires the universe as a whole.	$3^Q, 5^S$	pp. 2857, 2858
15	The source describes how tidal variability due to the Moon made Earth rotation inadequate as a precise time standard.	7^{OQ}	pp. 2858
16	The source's astronomical ephemeris time is constructed by approximating the solar system as isolated and choosing a time parameter so that the motions of its dynamically significant bodies satisfy Newtonian laws with small relativistic corrections.	$8^{SO}, 9^{OQ}$	pp. 2858
17	The source requires the physical processes of the atomic-clock network and a parametrized environmental model covering relevant terrestrial, solar-system, galactic and quasar effects before useful atomic time is extracted.	$10^{SOK}, 11^{SQ}, 12^{OQ}$	pp. 2858
18	The action of parametrized Newtonian mechanics retains external time t as a variable. Canonical quantization of the constraint linear in the time momentum yields the time-dependent Schrödinger equation (13).	1^{SQ}	pp. 2859, 2860
19	The momenta in the Jacobi formalism share a denominator that expresses direction in configuration space and satisfy a quadratic relation. Barbour contrasts this structure with parametrized mechanics, whose constraint is linear in the time momentum.	$\emptyset$	pp. 2859, 2860
20	Barbour criticizes choosing an arbitrary coordinate of each subsystem as a clock because the resulting clock rates do not agree, and instead uses the energy-conserving parameter involving all degrees of freedom.	$5^S, 6^{XQ}$	pp. 2860, 2861

p	Claim	i	Source Location
21	Barbour argues for a static quantum description of the fixed-energy Jacobi constraint and leaves recovery of experienced time from a static wavefunction to the companion paper.	$\emptyset$	pp. 2853, 2861
22	Best matching varies the Euclidean translations and rotations relating neighboring relative configurations to determine their intrinsic differences. This variation constrains total momentum and total angular momentum to vanish. The monotone label of a trial path is a mathematical variable describing this variation.	$\emptyset$	pp. 2861, 2862
23	An extremal best-matched history corresponds to the succession of Newtonian relative configurations at fixed energy and zero total angular momentum of the universe. Subsystems may have arbitrary angular momentum.	2^S	pp. 2862, 2863
24	Horizontal placement of the extremal configurations at best match supplies the spatial frame, while spacing them according to ephemeris-time differences supplies the operational temporal representation.	$2^S, 3^{1Q}$	pp. 2863
25	The source's Machian lapse is determined from configurational change by $N_{\text{Mach}} = \sqrt{T_{\text{Mach}}/(E - V)}$, in contrast to the Newtonian lapse defined as $N = dt/d\lambda$ using external time.	$1^Q, 3^Q$	pp. 2863, 2864
26	In general geometrodynamics, equilocality between three-geometries is determined by varying coordinate matchings, using a supermetric and scalar conformal factor that depend on the spatial metric.	$\emptyset$	pp. 2864, 2865
27	Barbour proposes an action with the square root inside the spatial integral from an argument for specifying local directions independently. He states that this generalization requires rigorous definition and leaves the particular supermetric and conformal factor unspecified.	$\emptyset$	pp. 2865, 2866

p	Claim	i	Source Location
28	Determining equilocality along a sequence of three-geometries requires solving the variational problem for the matching field. Barbour makes an available solution a condition of four-dimensional reconstruction and does not address the general existence problem in this paper.	13^S	pp. 2867
29	For a geometric sequence with a matching solution, local ephemeris time fixes the orthogonal separation between equilocal points. $N = \sqrt{T_g/\bar{F}}$ in Eq. (34) is the rate relating this local time to its label.	$13^S, 14^{IQ}$	pp. 2867, 2868
30	The reduced BSW action uses three-dimensional scalar curvature as its conformal factor, together with the DeWitt supermetric and shift. The source presents it as a special case of general Machian geometrodynamics.	$\emptyset$	pp. 2868, 2869
31	In the general-relativistic specialization, local ephemeris time along realized geometric evolution corresponds to local proper time.	$13^S, 14^{IQ}, 15^{IQ}$	pp. 2868, 2869
32	Barbour compares an interpretation that separates many-fingered time from geometric degrees of freedom with one that takes three-geometries as basic, arguing that the choice bears on the problem of time in quantum gravity.	$\emptyset$	pp. 2870, 2871, 2872
33	Barbour identifies linearized gravity as a special approximation using a fixed external Minkowski spacetime and distinguishes that temporal framework from general reconstruction from three-geometries.	$13^S, 15^I, 16^S$	pp. 2871
34	Barbour gives the claim that clocks measuring proper time keep in step with one another as a reason for regarding proper time as a good time.	$15^I, 17^{OIK}, 18^{XQ}$	pp. 2871
35	In Barbour's three-dimensional representation, intrinsic comparison of geometries and specification of initial directions play distinct roles in constructing spacetime, in contrast to changing coordinates on an already constructed spacetime.	$13^S, 15^I$	pp. 2871, 2872

p	Claim	i	Source Location
36	The source's timeless kinematics accompanies recovery of ephemeris time within a realized classical history and of local proper time in general relativity. Barbour connects this classical construction to the task of recovering experienced time from a static quantum description.	$2^S, 3^I, 13^S, 15^I$	pp. 2853, 2854, 2857, 2861, 2869, 2872
37	Suitable foliations of the four-dimensional metric space constructed from a mathematical solution of general relativity yield other extremal families of three-geometries. This foliation invariance allows nonunique families of geodesics connecting two three-geometries.	$\emptyset$	pp. 2868, 2869

Table 303: Barbour 1994 I Time-Use Roles

Code	Role	Use Count
I	Interval	12
Q	Quantity	20
X	Cross-position correspondence	3
K	Clock reading	2
C	Causality	0
S	State evolution	22
P	Probabilistic evolution	0
O	Observation record	6
B	Reference boundary	1

Table 304: Barbour 1994 I Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3, 5, 7, 9, 12, 14, 16	member	instant, order_relation
2, 8, 10, 13	member	order_relation
4, 6, 11, 15, 17–18	dependency	$\emptyset$

Dependency Inputs

Definition 8.413 (Barbour 1994 I Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [9]$. The table gives time inputs, time outputs, claim support and source locations.

Table 305: Barbour 1994 I Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	2	3	9–11	Define ephemeris time along a realized Jacobi history. pp. 2856, 2857
2	2	3	22–25	Assign ephemeris intervals to a best-matched history. pp. 2862, 2863, 2864
3	2	5	9–11, 13	Define subsystem ephemeris times from isolated subsystem histories. pp. 2857
4	5	6	13	Determine proportionality between subsystem-clock rates. pp. 2857
5	8	9	16	Determine astronomical ephemeris time from solar-system motions. pp. 2858
6	10–11	12	17	Extract atomic time from the clock network and environmental model. pp. 2858
7	13	14	26, 28–29	Construct local ephemeris time from geometric evolution. pp. 2867, 2868
8	14	15	30–31	Identify local ephemeris time with proper time in general relativity. pp. 2868, 2869
9	17	18	34	Attribute mutually consistent rates to proper-time clocks. pp. 2871

Grounding Conditions

1. Fixed E , configuration metric, potential, a chosen orbit and direction, and a segment with $E - V > 0$.
2. A best-matching solution, fixed E , zero total angular momentum of the universe and a nondegenerate Machian lapse.
3. Effectively noninteracting subsystems and the energy-conserving parameter of each subsystem.
4. The Jacobi structure of the whole universe and the source's agreement of clock rates up to constant proportions.
5. An isolated-system approximation, dynamically significant bodies, Newtonian laws and small relativistic corrections.
6. The physical clock processes and a parametrized model of the relevant environmental effects.
7. A solution of the matching-field variational problem, local direction data and a real, nondegenerate $N = \sqrt{T_g/\bar{F}}$.
8. The BSW form, DeWitt supermetric, $F = R$ and the source's lapse convention.
9. Barbour's assessment of clocks measuring proper time.

Results

Definition 8.414 (Barbour 1994 I Results and Dependencies). Let $\text{ldx}_R^{\text{audit,result}} = [25]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 306: Barbour 1994 I Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Newtonian instants and duration used in the source's comparison. pp. 2854, 2855	1	5–6
2	Fixed-energy Jacobi variational principle and reparametrization invariance. pp. 2856, 2859	$\emptyset$	9
3	Mathematical representation of configuration-space geodesics and the distinguished parameter. pp. 2856, 2857	$\emptyset$	9–10
4	Construction of ephemeris time and duration along a realized history. pp. 2857, 2858	2–3	10–12
5	Common-rate condition for effectively isolated subsystem clocks. pp. 2857	5–6	13–14
6	Correspondence between astronomical ephemeris time and the Jacobi construction. pp. 2857, 2858	3, 8–9	11, 16
7	Earth-rotation timekeeping and atomic time with environmental corrections. pp. 2858	7, 10–12	15, 17
8	Barbour's criticism of choosing an arbitrary degree of freedom as a clock. pp. 2860, 2861	5–6	13, 20
9	Best matching that determines intrinsic differences between relative configurations. pp. 2861, 2862	$\emptyset$	22
10	Momentum and angular-momentum constraints of the Machian action. pp. 2862, 2863	$\emptyset$	19, 22
11	Recovery of a spatial frame and ephemeris time from extremal configurations. pp. 2862, 2863, 2864	2–3	23–25
12	Parametrized mechanics retaining external time and its quantization. pp. 2859, 2860	1	18
13	Static quantization of the Jacobi constraint and the task of recovering experienced time. pp. 2853, 2861	$\emptyset$	4, 19, 21
14	Variational form of intrinsic matching between three-geometries. pp. 2864, 2865	$\emptyset$	26

q	Result and Source Location	Time Premises i	Claim Support p
15	Proposed action construction based on local directions. pp. 2865, 2866	$\emptyset$	27
16	General relativity in the reduced BSW action. pp. 2868, 2869	$\emptyset$	30
17	Construction of local ephemeris time along a matching solution. pp. 2867, 2868	13–14	28–29
18	Correspondence between local ephemeris time and proper time in general relativity. pp. 2868, 2869	14–15	30–31
19	Foliation invariance and nonunique extremal families in general relativity. pp. 2868, 2869	$\emptyset$	30, 37
20	Four-dimensional reconstruction from evolving three-geometries. pp. 2867, 2868	13–14	28–29
21	Comparison of general spacetime reconstruction with the linearized approximation. pp. 2871	13, 15–16	33
22	Barbour’s assessment of proper-time-clock rates. pp. 2871	15, 17–18	34
23	Roles of the constraints in the three- and four-dimensional representations. pp. 2871, 2872	13, 15	35
24	Classical recovery of time and the task posed by a static quantum description. pp. 2853, 2861, 2869, 2872	2–3, 13, 15	21, 36
25	Distinction between an external speed of the whole universe and relative subsystem rates. pp. 2855, 2856	1–2, 4	8

Definition 8.415 (Barbour 1994 I Grounds and Domains of Results). Each result uses the temporal premises and claim support in Table 306 and holds under the conditions and within the domain specified below.

Table 307: Barbour 1994 I Conditions for Results

Result Index	Conditions and Domain
1	The prerelativistic picture used for comparison.
2	A conservative particle system, fixed E , configuration metric and potential, and a regular domain.
3	Equations (4)–(6) on a specified curve and metric over a segment with $E - V > 0$.
4	A chosen fixed-energy history and its direction, within the domain of the nondegenerate integral.

Result Index	Conditions and Domain
5	Effectively noninteracting subsystems, energy-conserving parameters and constant proportionality.
6	The isolated-solar-system approximation and the relevant dynamical laws and corrections.
7	Existing timekeeping practices and environmental modelling described by the source.
8	The criterion that subsystem-clock rates keep in step.
9	Variation over Euclidean translations and rotations of neighboring configurations.
10	The source's variational form and momentum convention.
11	Fixed E, zero total angular momentum, a best-matching solution and ephemeris intervals.
12	Equation (7) and canonical quantization of the constraint linear in the time momentum.
13	Barbour's quantization argument for the fixed-energy constraint and the task deferred to the companion paper.
14	A spatial-metric-dependent supermetric and scalar conformal factor, and a matching field.
15	The generalization that Barbour says requires rigorous definition; the particular supermetric and conformal factor remain unspecified.
16	The source's DeWitt supermetric, curvature and shift, within the domain of the reduced action.
17	A solution of the matching-field variational problem and a real, nondegenerate local lapse.
18	The BSW specialization and the source's lapse convention.
19	Suitable foliations of a constructed mathematical solution.
20	A solved equilocality matching, spatial distances and orthogonal separations given by local ephemeris time.
21	Barbour's argument concerning the special approximation with a fixed external Minkowski background.
22	The source's claim about clocks measuring proper time keeping in step.
23	Barbour's interpretation of spacetime construction and coordinate changes on an already constructed spacetime.
24	The classical construction treated in the source and recovery of experienced time deferred to the companion paper.
25	The source's comparison between the universe's configuration curve and motions of its parts.

Evidence Hierarchy Barbour discusses variational constructions in classical mechanics and geometrodynamics, existing timekeeping practices and theoretical claims about quantization. The source reports zero directly measured objects and zero direct-measurement predicates from measurements of its own. Solar-system observations, atomic-clock networks and proper-time clocks are examples of time determination discussed by Barbour.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local reach count are both zero, leaving 18 primitive-time roots. Barbour’s ephemeris-time construction determines duration along realized histories of configurations or geometries. The source’s timekeeping practices and comparison theories use clock processes, temporal evolution of the environment or external time.

Result Dependence Of the 25 results, 16 depend on primitive-time roots and 9 are reachable from the source formal-evidence environment. The latter are Results 2–3, 9–10, 13–16, 19, covering variational forms, mathematical parameters, constraints, claims about quantization and properties of foliations. No result is generated from nontemporal physical inputs alone. There are 30 root–result dependence pairs, of which 19 are necessary.

Placement of the Grounding Obligation Ephemeris-time transfers presuppose succession along a realized history; extraction of atomic time presupposes temporal evolution in the clock network and environmental model; the proper-time correspondence presupposes local ephemeris time along geometric evolution. The durations and times output by these transfers depend on those premises. None of the 18 time contents belongs to the generation closure from nontemporal physical inputs or the local closure with source formal evidence. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The source’s isolation approximations, energy conditions, best-matching constraints and conditions on equilocality solutions determine the domains of the respective constructions. Actual failure of the physical-time generation conditions audited by LCTR, and consequent changes in result values, are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 30 root–result dependence pairs. Actual reach from the failure location of a specific physical-time generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are the time contents used by each construction and the physical-time generation conditions that must be satisfied before their use. If actual failure of a specific generation condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.416 (Primitive-Time Roots and Result Dependence in the Barbour 1994 I Source). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = (\emptyset, [18]; \\ \emptyset, \{2, 3, 9, 10, 13, 14, 15, 16, 19\}, \\ [25] \setminus \{2, 3, 9, 10, 13, 14, 15, 16, 19\}, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Even when all 37 claims are supplied as source formal evidence, none of the 9 transfers fires, since every transfer has temporal inputs. This closure includes the 9 results with empty temporal premises.

Adding the 18 time contents as seeds reaches all 25 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.417 (Minimal Primitive-Time-Root Support in the Barbour 1994 I Source). The table gives each result's minimal-support family, with 31 family members in total. For each result reachable from the evidence environment alone, the empty set is the sole family member.

Proof. Start from singleton time-content seeds and apply Definition 7.46 to the nine transfers. From the stable antichains, combine supports for each result's temporal premises and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Table 308: Minimal Primitive-Time-Root Support in the Barbour 1994 I Source

Result Index	Minimal-Support Index Family
1	$\{1\}$
2	$\emptyset$
3	$\emptyset$
4	$\{2\}$
5	$\{2\}, \{5\}$
6	$\{2, 8\}, \{3, 8\}$
7	$\{7, 10, 11\}$
8	$\{2\}, \{5\}$
9	$\emptyset$
10	$\emptyset$
11	$\{2\}$
12	$\{1\}$
13	$\emptyset$
14	$\emptyset$
15	$\emptyset$
16	$\emptyset$
17	$\{13\}$
18	$\{13\}, \{14\}$
19	$\emptyset$
20	$\{13\}$
21	$\{13, 16\}$
22	$\{13, 17\}, \{14, 17\}, \{15, 17\}$
23	$\{13\}$
24	$\{2, 13\}$
25	$\{1, 2, 4\}$

Evidentiary Effect

Definition 8.418 (Evidentiary Domain of the Barbour 1994 I Source). The mathematical part of the Jacobi construction supplies a fixed-energy variational principle and

reparametrization invariance. Physical ephemeris time is determined along a chosen realized history and its direction. The Newtonian correspondence through best matching uses zero total angular momentum of the universe. Reconstruction in general geometrodynamics is conditional on an equilocal matching solution; the BSW specialization supplies the correspondence with local proper time. Barbour’s arguments about clock rates rely on effective subsystem isolation, energy conservation, environmental models or his assessment of clocks measuring proper time. For quantum theory, Barbour argues for a static description and leaves recovery of experienced time to the companion paper.

Result-Preservation Domain

Definition 8.419 (Result-Preservation Conditions for the Barbour 1994 I Source). Fix the configurations, metrics, energies, actions, source approximation conditions and domains in Table 307. If generated temporal representations realize the time-dependent results’ inputs with the same succession, duration, clock-rate and proper-time correspondences, each result is preserved with its stated domain and epistemic status.

Table 309: Result-Preservation Conditions and Domains
for the Barbour 1994 I Source

Results	Preservation Conditions	Domain	Placement in the LCTR Generation Order
2–3, 9–10, 13–16, 19	The claim support of each result.	The mathematical domain of each result.	Variational forms, constraints and claims about static quantum descriptions enter as nontemporal structural inputs or construction constraints.
1, 4–8, 11–12, 17–18, 20–25	The claim support of each result and generation of its tabulated temporal premises.	The physical domain of each result.	Duration, clock-rate and spacetime correspondences or comparisons follow the grounding of the relevant time contents.

Proposition 8.420 (Audit-Boundary Counts for the Barbour 1994 I Source). The 37 claims, 18 time contents, 9 transfers and 25 results give 89 source-specific obligation atoms. The 45 claim/time-content use pairs have 66 role assignments.

$$\text{BoundaryCount}_R^{\text{audit}} = (80, 34).$$

Proof. Count claims, time contents and results as nodes, transfers and result dependencies as edges, and sum the roles of time uses in the claim table. $\square$

Individual Statements Determination of Time by All Variables

Original text:

Time is not any one variable; it is created by all variables working in concert.

The statement occurs on p. 2860, §7. Barbour criticizes choosing an arbitrary variable, such as the longitude of Mars, as a clock. He discusses a time parameter for which effectively noninteracting subsystem clocks keep in step up to constant proportionality. That parameter is determined from changes in all variables so that energy is conserved in each subsystem.

“Time as Such” and Instants as Relative Configurations

Original text:

It is argued that time as such does not exist but that instants, defined as complete relative configurations of the universe, do.

The statement occurs in the abstract on p. 2853. Barbour presents a timeless kinematics grounded in complete relative configurations of the universe. The same abstract describes the operational recovery of time within classical histories and connects this construction to the interpretation of a static wavefunction in quantum gravity.

Common Evidence with Specified Scope The table gives the objects, predicates and domains of the two statements.

State- ment	Object	Predicate	Explicit Domain
S33- Q001	Changes in all variables and a common time parameter	Barbour states that time is created by all variables working in concert.	Time determination in §7 using subsystem energy conservation and agreement of clock rates.
S33- Q002	“Time as such” and instants as complete relative configurations of the universe	Barbour argues that “time as such” does not exist but that instants defined as complete relative configurations do.	Timeless kinematics and operational recovery of time within classical histories, as described in the abstract.

Individual Audit Determination of Time by All Variables

In the first statement, Barbour describes a construction that obtains a parameter from changes in all variables, subject to energy conservation and agreement of subsystem-clock rates. Choosing an arbitrary single variable of each subsystem as a clock fails to give the agreement of rates he requires. The source’s integral uses a realized configurational history and its direction to determine ephemeris-time differences; subsystem comparisons use effective isolation and constant proportionality. The corresponding formal evidence comprises Claims 9–14 and 20, Transfers 1 and 3–4, and Results 4–5 and 8.

“Time as Such” and Instants as Relative Configurations

The second statement expresses Barbour’s timeless kinematics grounded in complete relative configurations of the universe. His classical construction recovers ephemeris time along realized histories of relative configurations and local proper time in the specialization to general relativity. The physical succession, clock rates and proper-time correspondence used by this construction are temporal inputs to this audit. Barbour defers recovery of experienced time from the static quantum description to the companion paper. The corresponding formal evidence comprises Claims 1, 4, 7, 21, 28–31 and 36, Transfers 1–2 and 7–8, and Results 4, 11, 13, 17–18, 20 and 24.

8.19.34 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 34.

Source Name Primitive-Time-Root Audit of Barbour 1994 II

Bibliographic Identification Let $R = \text{Ref}_{\text{Bar94TQ2}}$ denote Julian B. Barbour, *The Timelessness of Quantum Gravity: II. The Appearance of Dynamics in Static Configurations, Classical and Quantum Gravity* 11, 2875–2897 (1994). The audit domain comprises all 23 pages of the source, §§1–8, and the Appendix.

Audit-Scope Identifier 34-SC1.

Audit-Scope Content The audit domain stated in the bibliographic identification.

Original-Source Location Identifier 34-L1.

Original-Source Location Content The audit domain stated in the bibliographic identification and the original-source location columns of the respective evidence tables. The tables identify source locations by printed page number.

Formal Object

Object Identifier 34-F1.

Object Name Formal objects of the original source.

Object Type Formal object.

Formal Content The individual formal objects are given in Definitions 8.421 and 8.423.

Original-Source Location Identifier Set 34-L1.

Assumptions

Assumption Identifier 34-A1.

Assumption Content The individual source assumptions are given in Definitions 8.423 and 8.421.

Original-Source Location Identifier Set 34-L1.

Nontemporal Physical Inputs

Input Identifier 34-P1.

Input Name Nontemporal physical inputs of the original source.

Input Type Nontemporal physical input.

Input Content The individual nontemporal physical inputs are given in Definition 8.423.

Original-Source Location Identifier Set 34-L1.

Time Content

Time-Content Identifier 34-T1.

Time-Content Name Time contents of the original source.

Time-Content Type Time content.

Time-Content Statement The individual time contents are given in Definitions 8.422 and 8.421.

Original-Source Location Identifier Set 34-L1.

Formal Input Elements

Definition 8.421 (Formal objects and explicit assumptions in Barbour 1994 II). Barbour proposes using a static wavefunction on relative configurations of the whole universe to explain temporal experience and the appearance of dynamics through record-bearing configurations. Classical histories and ephemeris time from Part I, external time and measurement in ordinary quantum theory, the stationary Mott model, and WKB solutions and the semiclassical program play different roles in his argument.

The account of configuration probabilities uses a finite-dimensional interpretive model and assigns fixed weights to configurations. Association of experience with brain configurations uses psychophysical parallelism. The Mott model uses fixed nuclei, a special zeroth approximation, a distinguished Ra atom and consistent outgoing waves at every order. Spontaneous operation of the same mechanism for configurations of the whole universe is Barbour's conjectural extension to quantum gravity. The appendix examines the conditions required to recover first-order TDSE evolution from a general WKB state.

The source's formal objects are grouped into the following 20 groups.

1. Static Wheeler–DeWitt equation
2. Relative configuration space
3. Configuration probability weights
4. Sets of possibilities and actualities, and actualization
5. Psychophysical parallelism
6. Static configurations and record correlations
7. Dynamical inference from configurational information
8. Stationary Mott scattering model
9. Geometry of track configurations
10. Complex conjugation of the wavefunction
11. Perturbation orders and excitation configurations
12. WKB phase and congruence of orbits
13. Stratification of configuration space
14. Origin of total particle coincidence
15. Scattering kernels and regularity conditions
16. Configuration-space boundaries and asymmetry

17. Semiclassical approximation program
18. First- and second-derivative terms
19. Configurations containing measuring apparatus and outcomes
20. Classical configuration construction from Part I

The spherical wave in equation (1) and the WKB form in equation (2) are:

$$\frac{e^{ikR}}{R}, \quad \Psi = e^{iS}\phi, \quad \Psi^* = e^{-iS}\phi^*.$$

Here R in the spherical wave is the radius, k is the wave number, and S in the WKB form is a real Hamilton–Jacobi function. The function ϕ depends on the WKB variables and the remaining quantum variables. Invariance of $\Psi\Psi^*$ under conjugation and the use of one outgoing convention at every perturbation order support distinct arguments in the source.

Definition 8.422 (Time contents in Barbour 1994 II). Let $n_R = 21$. Table 311 gives the time contents and their source locations.

Table 311: Barbour 1994 II Time Contents and Source Locations

i	Time Content	Source Location
1	External time in the classical mechanics and ordinary quantum theory used for comparison	pp. 2876, 2878, 2879, 2880, 2886
2	Measurement instants and order, and the temporal precedence of prior knowledge, in the comparison framework	pp. 2880, 2881, 2882, 2886
3	Succession in the classical configuration histories recalled from Part I	pp. 2876, 2877, 2892
4	Ephemeris time and intervals along the histories recalled from Part I	pp. 2876, 2877
5	Local proper time in the specialization to Lorentzian four-geometry	pp. 2877
6	The experienced present and experience of motion associated with brain configurations	pp. 2875, 2882, 2883, 2885
7	Past events and their temporal relations represented by memories and records	pp. 2876, 2883, 2884, 2885
8	Dates and durations inferred from geological and cosmological records	pp. 2876, 2885
9	A classical history inferred from a single relative configuration	pp. 2886, 2887
10	Speeds and directions of motion inferred from shape, density and environmental traces	pp. 2886, 2887
11	The particle course and earlier/later stages interpreted as recorded by a Mott track	pp. 2887, 2888, 2890
12	Direction of motion inferred from a Mott track configuration	pp. 2888, 2889

i	Time Content	Source Location
13	Ephemeris-time intervals inferred from the dynamical information in a Mott track	pp. 2876, 2890
14	Classical histories and their different instants assigned to a WKB congruence	pp. 2888, 2889, 2890, 2894, 2895
15	The arrow of time assigned to cosmological record configurations	pp. 2892, 2893, 2894
16	A finite past and open future proposed to be explained by a configuration-space boundary	pp. 2893, 2894
17	Retarded and causal temporal direction compared with consistent outgoing-wave selection	pp. 2894, 2895
18	Temporal initial conditions in the evolution equations and cosmology used for comparison	pp. 2884, 2885, 2889, 2893
19	Differential time used along WKB orbits in the semiclassical program	pp. 2895
20	Future evolution determined by the TDSE from a state at one instant	pp. 2895
21	Agreement of rates required of candidate local clocks	pp. 2878

Definition 8.423 (Barbour 1994 II Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(44)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([4], [44])$. Claims 1–4 form the generation-initial set; all 44 claims form the source formal-evidence environment. Each row in the following table gives a claim, its time uses and source location.

Table 312: Barbour 1994 II Physical and Formal Input Claims

p	Claim	i	Source Location
1	Relative configurations, spatial positions, shapes, densities and static record patterns.	$\emptyset$	pp. 2879, 2881, 2882, 2886
2	A static wavefunction on configurations, its values and probability weights.	$\emptyset$	pp. 2881, 2882
3	Fixed nuclear positions, interactions, energy and configuration variables in the Mott model.	$\emptyset$	pp. 2887, 2888
4	The metric, boundaries and strata of relative configuration space, and spatial coefficients of the wave equation.	$\emptyset$	pp. 2884, 2891, 2892, 2893
5	Barbour recalls the classical histories of Part I and the ephemeris time defined from the contents of two configurations.	$3^S, 4^{IQ}$	pp. 2876
6	Barbour constructs orthogonal four-dimensional intervals from a history of three-geometries and interprets them as local proper time for Lorentzian geometry with spacelike slices.	$3^S, 4^{IQ}, 5^{IQ}$	pp. 2877

<i>p</i>	Claim	<i>i</i>	Source Location
7	Barbour criticizes local clocks formed by arbitrarily separating a degree of freedom because their rates would not keep in step.	21 ^{XQ}	pp. 2878
8	The ordinary quantum theories used for comparison employ time derivatives or evolution, and Everett's construction retains external time.	1 ^{SQ}	pp. 2878, 2879, 2880
9	Barbour proposes a static Wheeler–DeWitt equation on relative configurations of the entire universe.	∅	pp. 2875, 2876, 2877, 2879, 2880
10	The wavefunction has the same value on different coordinate representations of one intrinsic configuration.	∅	pp. 2879
11	Barbour distinguishes the sets of possibilities and actualities weighted by fixed configuration probabilities from the actualization of one configuration.	∅	pp. 2881, 2882
12	In the ordinary measurement framework contrasted by Barbour, time-dependent probabilities are conditioned on prior knowledge and tied to measurement arrangements and instants.	1 ^P , 2 ^{OB}	pp. 2880, 2881, 2882, 2886
13	Barbour treats a whole configuration containing apparatus, outcome and setup records as one experienced instant.	6 ^O , 7 ^O	pp. 2882, 2883
14	Barbour's psychophysical parallelism associates the experience of seeing motion with one brain configuration containing information comparable to several still images.	6 ^{OS}	pp. 2882, 2883
15	Barbour distinguishes present experience from the question whether the other instants represented in present memories were actually experienced.	6 ^O , 7 ^O	pp. 2883
16	Barbour defines a time capsule as a static configuration suggesting mutually consistent records of past processes governed by laws.	7 ^{OS}	pp. 2884, 2885
17	Barbour uses dates and durations represented by geological, biological and cosmological records as examples of temporal information within configurations.	7 ^O , 8 ^{IQO}	pp. 2876, 2885
18	The conventional account of cosmic history contrasted by Barbour uses temporal evolution and a low-entropy initial state in time.	1 ^S , 18 ^B	pp. 2885, 2893

p	Claim	i	Source Location
19	Barbour discusses continuity, single-valuedness and boundedness of static solutions as natural boundary conditions.	$\emptyset$	pp. 2884, 2893
20	The Laplacian comparison specifies positions and momenta at one absolute instant and places ordinary quantum measurements in a common external time.	$1^{\text{SB}}, 2^{\text{O}}$	pp. 2886
21	Barbour proposes inferring speeds, directions and histories from typical, mutually consistent configurational information using physical laws.	$9^{\text{S}}, 10^{\text{QS}}$	pp. 2886, 2887
22	The stationary Mott model starts from a product of a spherical wave and atomic states and treats the interaction perturbatively.	$\emptyset$	pp. 2887, 2888
23	Under consistent outgoing-wave conditions, excitation configurations and scattered waves at successive perturbation orders have the track-like correlations described in the source.	$\emptyset$	pp. 2887, 2888, 2894
24	Barbour reads Mott track configurations of different lengths as records of earlier and later stages in a particle's course.	11^{S}	pp. 2888, 2890
25	Complex conjugation of the entire wavefunction preserves the probability of each configuration.	$\emptyset$	pp. 2888, 2889
26	Barbour rejects distinguishing expansion from contraction solely by the sign of conjugate WKB phases and states that both contain the same configurational records.	14^{S}	pp. 2888, 2889
27	Barbour locates the evidence for the direction of particle motion in the Mott track configuration as a whole.	$11^{\text{S}}, 12^{\text{SO}}$	pp. 2888, 2889
28	Barbour states that equal ephemeris-time intervals can in principle be marked using dynamical track information such as curvature or scattering angles.	$11^{\text{S}}, 13^{\text{IQ}}$	pp. 2890
29	The nuclei are fixed in the Mott model, leaving the quantum formation of the apparatus and environment to be established.	$\emptyset$	pp. 2888, 2889
30	The WKB phase of a real Hamilton–Jacobi function and dependence on the remaining quantum variables form equation (2).	$\emptyset$	pp. 2888

p	Claim	i	Source Location
31	Barbour explains how correlations between one WKB solution and quantum degrees of freedom carry records of different classical histories and their stages.	$7^O, 14^S$	pp. 2889, 2890
32	In Barbour's actualization interpretation, superposition changes configuration weights while each experienced configuration retains its own record structure.	$6^O, 7^O$	pp. 2890, 2891
33	The three Mott conditions are a special zeroth approximation, a distinguished Ra atom and consistent outgoing waves; a spontaneous quantum-gravitational counterpart of the first is not established.	$\emptyset$	pp. 2891
34	Three-body relative configuration space is bounded by triangle inequalities and has interior, face, edge and total-coincidence strata.	$\emptyset$	pp. 2891, 2892
35	Barbour proposes intrinsic configuration-space asymmetry as a candidate origin of the arrow of time apparent in records.	15^S	pp. 2892, 2893
36	Barbour proposes relating a boundary at finite distance on one side and unbounded extension on the other to the appearance of a finite past and open future.	16^{IB}	pp. 2893, 2894
37	The scattering Helmholtz equation permits linear combinations of incoming and outgoing components, while Mott track correlations depend on the same outgoing convention at every order.	$\emptyset$	pp. 2887, 2894
38	Barbour compares consistent outgoing-wave selection with the selection of retarded solutions and discusses its relation to causal scattering behavior.	17^{CS}	pp. 2894, 2895
39	Barbour proposes that correlations between a WKB congruence from the configuration-space origin and quantum variables may form records of directed histories.	$7^O, 14^S, 15^S$	pp. 2893, 2894
40	In the appendix, Barbour points out that a second derivative with respect to time along WKB orbits remains alongside the first derivative.	$14^S, 19^{SQ}$	pp. 2895
41	Barbour identifies the TDSE's determination of the future from the wavefunction at one instant as a property that semiclassical recovery must explain.	$19^S, 20^{SB}$	pp. 2895

p	Claim	i	Source Location
42	Barbour argues that a WKB regime alone does not eliminate the extra solutions of the second-order equation or ensure agreement with the causal behavior of the TDSE.	$17^C, 19^{SQ}, 20^{SB}$	pp. 2895
43	Barbour examines what remains of classical and quantum structure on the premise, argued in Part I, of timeless kinematic foundations.	$\emptyset$	pp. 2875, 2876
44	Barbour proposes explaining temporal experience and the appearance of dynamics through the probabilities and records of static configurations.	$6^O, 7^{OS}$	pp. 2875, 2876, 2884, 2885

Table 313: Barbour 1994 II Time-Use Roles

Code	Role	Use Count
I	Interval	6
Q	Quantity	10
X	Cross-position correspondence	1
K	Clock reading	0
C	Causality	2
S	State evolution	26
P	Probabilistic evolution	1
O	Observation record	17
B	Reference boundary	6

Table 314: Barbour 1994 II Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–2, 4, 6–7, 14, 16, 19	member	instant, order_relation
3, 9, 11	member	order_relation
5, 8, 10, 12–13, 15, 17–18, 20–21	dependency	$\emptyset$

Dependency Inputs

Definition 8.424 (Barbour 1994 II Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [7]$. The table gives time inputs, time outputs, claim support and source locations.

Table 315: Barbour 1994 II Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	3	4	5	Assign ephemeris intervals to the classical histories of Part I. pp. 2876
2	3–4	5	6	Identify intervals along a geometric history with local proper time. pp. 2877
3	11	12	24, 27	Associate a direction of motion with the course interpreted from a Mott track. pp. 2888, 2889
4	11	13	28	Infer ephemeris intervals from the dynamical information in a Mott track. pp. 2890
5	14	7	31	Associate configurational records in quantum variables with WKB histories. pp. 2889, 2890
6	14	19	40	Use differential time along WKB histories in the semiclassical program. pp. 2895
7	19	20	41	Determine future evolution from one instant in the differential time of the TDSE. pp. 2895

Grounding Conditions

1. A selected classical history and the configuration-based ephemeris-time construction recalled from Part I.
2. Lorentzian four-geometry, spacelike three-slices and the source's local parameter choice.
3. The whole track configuration in the fixed-nucleus Mott model and its dynamical interpretation.
4. The source's in-principle construction, reading curvature, scattering angles and related data through existing dynamics.
5. Correlations under special Mott-type conditions; application to the whole universe belongs to the conjecture in Result 24.
6. The WKB orbit parameter of the semiclassical program criticized by Barbour.
7. The initial-value evolution of the first-order TDSE within its domain of validity.

Results

Definition 8.425 (Barbour 1994 II Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [26]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 316: Barbour 1994 II Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Proposed quantum gravity based on a static WDE and relative configurations. pp. 2875, 2876, 2877, 2879, 2880	$\emptyset$	9–10, 43
2	Classical histories and ephemeris time recalled from Part I. pp. 2876	3–4	5
3	Correspondence with local proper time in the Lorentzian case. pp. 2877	3–5	6
4	Clock-rate criticism of arbitrary candidate local clocks. pp. 2878	21	7
5	External time, measurement order and temporal initial conditions in the comparison theories. pp. 2878, 2879, 2880, 2881, 2884, 2885, 2886, 2893	1–2, 18	8, 12, 18, 20
6	Fixed configuration probabilities and the distinction between possibilities, actualities and actualization. pp. 2881, 2882	$\emptyset$	11
7	Distinction between an experienced instant and other instants represented in memory. pp. 2882, 2883	6–7	13–15, 44
8	Time capsules and the histories and dates represented by records. pp. 2875, 2876, 2884, 2885	7–8	16–17
9	Proposal to infer speeds, directions and histories from configuration. pp. 2886, 2887	9–10	21
10	Track-like configuration correlations in the stationary Mott solution. pp. 2887, 2888, 2894	$\emptyset$	22–23
11	Interpretation of earlier/later stages and direction of motion in Mott tracks. pp. 2888, 2889	11–12	24, 27
12	Inference of ephemeris intervals from Mott tracks. pp. 2890	11, 13	28
13	Invariance of configuration probabilities under complex conjugation. pp. 2888, 2889	$\emptyset$	25, 30
14	Criticism of distinguishing expansion from contraction by the WKB phase sign. pp. 2888, 2889	14	26
15	Static correlation structure of WKB phase and quantum variables. pp. 2888, 2890	$\emptyset$	23, 30
16	Records of multiple histories associated with one WKB solution. pp. 2889, 2890	7, 14	31

q	Result and Source Location	Time Premises i	Claim Support p
17	Special conditions and the domain of the fixed-nucleus approximation in the Mott model. pp. 2888, 2889, 2891	$\emptyset$	29, 33
18	Superposition and the record structure retained by each experienced configuration. pp. 2890, 2891	6–7	32
19	Boundary stratification of three-body relative configuration space. pp. 2891, 2892	$\emptyset$	34
20	Hypothesis of an arrow of time arising from configuration-space asymmetry. pp. 2892, 2893	15	35
21	Boundary-structure hypothesis concerning a finite past and open future. pp. 2893, 2894	16	36
22	Natural regularity conditions and selection of a scattering kernel. pp. 2884, 2893, 2894	$\emptyset$	19, 37
23	Relation between outgoing-wave conditions and retarded or causal behavior. pp. 2894, 2895	17	38
24	Conjecture of concentration on records of directed histories in general WDE solutions. pp. 2891, 2893, 2894	7, 14–15	33, 39
25	The semiclassical time parameter and initial-value evolution of the TDSE. pp. 2895	14, 19–20	40–41
26	Criticism of the semiclassical approximation that discards the second time derivative. pp. 2895	17, 19–20	40–42

Definition 8.426 (Barbour 1994 II Grounds and Domains of Results). Each result’s temporal premises and claim support are given in Table 316; its conditions and domain are given below.

Table 317: Barbour 1994 II Conditions for Results

Result Index	Conditions and Domain
1	Barbour’s kinematic premise and proposed static equation.
2	The classical time construction based on the contents of two configurations.
3	Lorentzian four-geometry, spacelike slices and the specified local parameter.
4	Agreement of clock rates as required by the source.

Result Index	Conditions and Domain
5	The respective contexts of ordinary quantum theory, measurement and accounts of cosmic history.
6	The source's finite-dimensional interpretive model and weights assigned once and for all to configurations.
7	Psychophysical parallelism and the association of experience with one brain configuration.
8	Barbour's definition in terms of mutually consistent records of a lawful past and his explanatory program.
9	Barbour's proposed possibility of interpreting typical configurational information and consistent independent clues through physical laws.
10	Fixed nuclei, the source's zeroth approximation, perturbative scattering and consistent outgoing waves.
11	Track configurations and their existing dynamical interpretation.
12	Dynamical information such as the curvature and scattering angles cited by the source.
13	Real S in equation (2) and complex conjugation of the whole wavefunction.
14	The source's argument that the conjugate pair has the same configuration probabilities and records.
15	Equation (2) and Mott-type stationary correlations.
16	Additional quantum degrees of freedom and Mott-type correlation conditions.
17	The three conditions listed by the source and the outstanding quantum formation of apparatus and environment.
18	The interpretation of actualization in one configuration and each configuration's intrinsic structure.
19	Nonnegative interparticle distances and triangle inequalities, with collinear, partial-coincidence and total-coincidence configurations.
20	Barbour's conjectural extension to quantum cosmology.
21	Barbour's proposal relating record configurations to a one-sided boundary.
22	Boundary conditions on static solutions and Mott's outgoing convention; enforcement by cosmological regularity is the source's conjecture.
23	The Mott-model analogy with retarded solutions and comparison of conditions with the semiclassical program.
24	Corresponding WKB orbits, additional quantum degrees of freedom and configuration-space structure, and cosmological realization of the special Mott conditions.
25	The approximation program examined by Barbour and the domain of validity of the first-order TDSE.
26	The distinction between a general WKB state and a special formal expansion, and the extra solutions and their causal behavior.

Evidence Hierarchy Barbour discusses an interpretation of static quantum theory, correlations in the Mott model, configuration-space structure, cosmological conjectures and criticism of the semiclassical approximation. The source reports zero directly measured

objects and zero direct-measurement predicates from measurements of its own. Configurations of measuring instruments, memories, geological records and particle tracks are objects whose appearance of time Barbour seeks to explain.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local reach count are both zero, leaving 21 primitive-time roots. Classical ephemeris time presupposes succession along a history; temporal intervals in Mott tracks presuppose a dynamical interpretation of the track; semiclassical differential time presupposes a WKB history. Experience and records associated with static configurations are also interpreted using temporal content.

Result Dependence Of the 26 results, 18 depend on primitive-time roots. The results reachable from the source formal-evidence environment without added time roots are 1, 6, 10, 13, 15, 17, 19, 22: 8 results concerning static constructions, configuration probabilities, Mott correlations, conjugation invariance, WKB form, model conditions and configuration-space structure. There are 30 root–result dependence pairs, of which 22 are necessary.

Placement of the Grounding Obligation Barbour’s proposal interprets static configurations as experienced instants, records of a past or carriers of directed histories. The Mott model uses a dynamical reading of stationary correlations, while the classical and quantum theories used for comparison employ histories, law time and initial conditions. None of the 21 time contents belongs to the generation closure from nontemporal physical inputs or the local closure with source formal evidence. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Fixed nuclei, the zeroth-order state, the outgoing convention and semiclassical approximation conditions specify the domains of the respective models and claims. Actual failure of physical-time generation conditions and consequent changes in result values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 30 root–result dependence pairs. Actual reach from the failure location of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are the temporal uses and the physical-time generation conditions required before those uses. If actual failure of a specific condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.427 (Primitive-Time Roots and Result Dependence in Barbour 1994 II). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = & (\emptyset, [21]; \\ & \emptyset, \{1, 6, 10, 13, 15, 17, 19, 22\}, \\ & [26] \setminus \{1, 6, 10, 13, 15, 17, 19, 22\}, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Even with all 44 claims as source formal evidence, none of the 7 transfers fires because each requires temporal inputs. This closure includes the 8 results with empty temporal premises. Adding the 21 time contents as seeds reaches all 26 results. The common reachability classification gives the displayed six components. $\square$

Proposition 8.428 (Minimal Primitive-Time-Root Support in Barbour 1994 II). The table gives the minimal-support index family of each result, with 30 family members in total.

Proof. Start from singleton time-content seeds and apply Definition 7.46 to the seven transfers. Combine supports for each result’s temporal premises from the stable antichains and discard supports properly containing another support. Proposition 7.47 gives the displayed families and counts. $\square$

Table 318: Minimal Primitive-Time-Root Support in Barbour 1994 II

Result Indices	Minimal-Support Family Members
1, 6, 10, 13, 15, 17, 19, 22	$\emptyset$
2–3	$\{3\}$
4	$\{21\}$
5	$\{1, 2, 18\}$
7, 18	$\{6, 7\}, \{6, 14\}$
8	$\{7, 8\}, \{8, 14\}$
9	$\{9, 10\}$
11–12	$\{11\}$
14, 16, 25	$\{14\}$
20	$\{15\}$
21	$\{16\}$
23	$\{17\}$
24	$\{14, 15\}$
26	$\{14, 17\}, \{17, 19\}$

Evidentiary Effect

Definition 8.429 (Evidentiary Domain of Barbour 1994 II). The interpretation of static configuration probabilities distinguishes possibilities, actualities and actualization of one configuration. The Mott model exhibits track-like correlations under fixed-nucleus and specified wave and scattering conditions. Conjugate wavefunctions have the same configuration probabilities; evidence for a direction of motion belongs to the dynamical reading of the whole configuration. Configuration-space geometry and hypotheses explaining the cosmic arrow and finite past have their respective result conditions. Barbour presents concentration on record-bearing configurations in general WDE solutions as an open problem and, in the appendix, criticizes the conditions used to discard the second time derivative in the semiclassical program.

Result-Preservation Domain

Definition 8.430 (Result-Preservation Conditions for Barbour 1994 II). Fix the static configurations, wave equations, model conditions and domains in Table 317. If generated temporal representations realize the time-dependent results’ inputs with the same histories, record orders, intervals, directions and differential evolution, each result is preserved with its stated domain and epistemic status.

Table 319: Result-Preservation Conditions and Domains for Barbour 1994 II

Results	Preservation Conditions	Domain	Placement in the LCTR Generation Order
1, 6, 10, 13, 15, 17, 19, 22	The claim support and model conditions of each result.	Static constructions, configuration structures and their conditions.	Nontemporal structural inputs or construction constraints.
2–5, 7–9, 11–12, 14, 16, 18, 20–21, 23–26	Claim support and generation of the tabulated temporal premises.	The domain of each result’s interpretation, comparison, conjecture or approximation.	Histories, records or temporal evolution after grounding the relevant time contents.

Proposition 8.431 (Audit-Boundary Counts for Barbour 1994 II). The 44 claims, 21 time contents, 7 transfers and 26 results give 98 source-specific obligation atoms. The 48 claim/time-content use pairs have 69 role assignments.

$$\text{BoundaryCount}_R^{\text{audit}} = (91, 33).$$

Proof. Count claims, time contents and results as nodes, transfers and result dependencies as edges, and sum the roles of time uses in the claim table. $\square$

Individual Statements The Nonexistence of Time in the Kinematic Foundations

Original text:

time is truly non-existent in the kinematic foundations of both theories

The passage occurs on pp. 2875–2876, §1. Immediately before the quoted clause, Barbour writes “if, as argued in [1]”, taking the result argued in Part I as a premise for examining what remains of the structures of classical and quantum theory. In the same introduction, he contrasts ephemeris time determined from two configurations in a classical history with time inferred from records within one static configuration.

Time Capsules

Original text:

Such configurations are called time capsules

The passage occurs in the abstract on p. 2875. Barbour proposes that the appearance of time, dynamics and history is encoded as apparently consistent records within experienced static configurations of the universe, and calls such configurations time capsules. In §4, p. 2884, he defines a time capsule as a static configuration containing structures that suggest mutually consistent records of past processes governed by laws.

Common Evidence with Specified Scope The table gives the objects, predicates and domains of the two statements.

State- ment	Object	Predicate	Explicit Domain
S34- Q001	Time in the kinematic foundations of classical and quantum theory	Barbour takes timeless kinematic foundations, argued for in Part I, as a premise.	The conditional clause referring to Part I in §1, pp. 2875–2876.
S34- Q002	Static configurations containing structures suggestive of consistent records of past processes	Barbour calls such configurations time capsules.	The abstract on p. 2875 and the definition in §4, p. 2884.

Individual Audit The Nonexistence of Time in the Kinematic Foundations

The first statement expresses a kinematic premise referring to Part I. Barbour uses ephemeris time along configuration histories in classical theory and proposes explaining the appearance of time from records within one configuration in quantum theory. The histories, ephemeris time and local proper time of the former correspond to Time Contents 3–5; the experience, past records and dates of the latter correspond to Time Contents 6–8. The static equation and configuration construction form Result 1, the classical references form Results 2–3, and the interpretation of experience and records forms Results 7–8. The corresponding claims are 5–6, 9–10, 13–17 and 43–44.

Time Capsules

The time capsule in the second statement follows Barbour’s definition: a static structure suggests records of lawful past processes. In the Mott model, fixed nuclei, a specified zeroth-order state and consistent outgoing waves at every order support track-like correlations. The course, direction and temporal intervals of a track belong to the dynamical interpretation of that configuration. Barbour shows that conjugating the entire wavefunction gives the same configuration probabilities and locates evidence for motion direction in the whole configuration. Analogous concentration in WDE solutions for the entire universe is the conjecture examined in §8. The corresponding evidence comprises Claims 16, 22–28, 33 and 35–39, Transfers 3–5, and Results 8, 10–14, 17 and 20–24.

8.19.35 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 35.

Source Name Primitive-time-root audit of Connes–Rovelli 1994

Bibliographic Identification Let A. Connes and C. Rovelli, *Von Neumann algebra automorphisms and time-thermodynamics relation in general covariant quantum theories*, *Classical and Quantum Gravity* 11, 2899–2918 (1994), doi:10.1088/0264-9381/11/12/007 be $R = \text{Ref}_{\text{CR94TT}}$. The audit domain comprises the abstract, Source §§1–5, Source Equations (1)–(57), and Source Footnotes 1–7.

Audit-Scope Identifier 35-SC1.

Audit-Scope Content The audit domain specified under Bibliographic Identification.

Source-Location Identifier 35-L1.

Source-Location Content The audit domain specified under Bibliographic Identification and the source- location columns of the respective evidence tables. Page numbers refer to arXiv:gr-qc/9406019v1.

Formal Object

Object Identifier 35-F1.

Object Name Formal objects in the source.

Object Type Formal object.

Formal Content The individual formal objects are given in Definitions 8.432 and 8.434.

Source-Location Identifier Set 35-L1.

Assumption

Assumption Identifier 35-A1.

Assumption Content The individual assumptions in the source are given in Definitions 8.434 and 8.432.

Source-Location Identifier Set 35-L1.

Nontemporal Physical Input

Input Identifier 35-P1.

Input Name Nontemporal physical inputs in the source.

Input Type Nontemporal physical input.

Input Content The individual nontemporal physical inputs are given in Definition 8.434.

Source-Location Identifier Set 35-L1.

Time Content

Time-Content Identifier 35-T1.

Time-Content Name Time contents in the source.

Time-Content Type Time content.

Time-Content Statement The individual time contents are given in Definitions 8.433 and 8.432.

Source-Location Identifier Set 35-L1.

Formal Input Elements

Definition 8.432 (Formal objects and explicit assumptions in Connes–Rovelli 1994). Connes and Rovelli propose the thermal-time hypothesis, which uses the modular flow associated with a specified state and observables algebra as physical time. The GNS representation, Tomita–Takesaki construction and outer automorphism class supply mathematical constructions. Internal clocks, comparison with known Hamiltonian time, Rindler proper time, the bath rest frame and FRW time have their respective physical application conditions.

The modular construction uses a fixed von Neumann algebra and faithful normal state, or a cyclic separating vector. The Gibbs branch uses a finite partition function; the classical branch uses a positive density and Poisson structure; the Rindler branch uses Minkowski causal structure and the vacuum restricted to a wedge. Adoption as physical time is given by the thermal-time hypothesis. Scale comparisons in each branch use the known time identified by the source.

The formal objects comprise the following nine groups.

1. Observables, states, physical reference fields and classical phase space
2. Relational observables and internal clocks in generally covariant theories
3. GNS representation, modular group and outer automorphism class
4. Thermal-time hypothesis, KMS condition and correlators
5. Hilbert–Schmidt representation of Gibbs states and known time
6. Negative logarithm of classical density and Poisson evolution
7. Rindler wedge, boost flow, Unruh relation and Hawking extension
8. Selection of subalgebras and states, and the Planck-phase proposal
9. CMB and FRW, quantum thermal clocks and gravitational statistical mechanics

In the source sign convention, the modular construction and Gibbs scale correspondence are:

$$S = J\Delta^{1/2}, \quad \alpha_t(A) = \Delta^{-it}A\Delta^{it}, \quad \alpha_t = \gamma_{\beta t}, \quad \tau = \beta t.$$

For a faithful finite-dimensional density in the Hilbert–Schmidt representation, the relevant relations are:

$$\Delta = L_\rho R_{\rho^{-1}}, \quad [L_A, \ln \Delta] = L_{[A, \ln \rho]}.$$

In the classical limit, the negative logarithm of a positive density supplies the Hamiltonian function. In the Rindler branch, the group-action correspondence in source Equations(54) and(55) fixes the following scale relation:

$$H = -\ln \rho, \quad \dot{f} = \{H, f\}, \quad \tau = \frac{2\pi}{a}t, \quad t = \frac{a}{2\pi}\tau, \quad T = \frac{a}{2\pi k_B}.$$

Here L and R denote left and right actions on Hilbert–Schmidt operators. The final scale and temperature relations use the source units $c = \hbar = 1$.

Definition 8.433 (Time contents in Connes–Rovelli 1994). Let $n_R = 28$. Table 321 gives the time contents and their source locations.

Table 321: Connes–Rovelli 1994 Time Contents and Source Locations

i	Time Content	Source Location
1	Hamiltonian time and its evolution in the comparison theories	pp. 2, 13, 16, 17, 18
2	Causal classification on GR solutions and its domain in quantum superpositions	pp. 2, 3
3	Succession and proper-time intervals along an observer worldline	pp. 2, 19, 20
4	Coordinate time used to represent GR fields	pp. 8, 9
5	Reference time specified by laboratory clocks, pulses or astronomical variables	pp. 8, 9
6	Relational observables indexed by reference time	pp. 9, 11
7	Selected internal-clock variable and clock labels	pp. 9, 10, 11
8	Evolution and observable families expressed in an internal clock	pp. 10, 11
9	Monotonic order and stopping/reversal boundaries of an internal clock	pp. 12
10	Unitarity conditions on quantum internal-clock evolution	pp. 12
11	Thermal flow constructed from a state and algebra and proposed as physical time	pp. 3, 4, 7, 14, 15, 22
12	Time orientation relative to a selected flow	pp. 13
13	Inverse-temperature scale relating known time evolution to thermal time	pp. 13, 14, 18
14	Preparation, waiting and detection instants, intervals and order	pp. 8, 9, 15
15	Geometrical time supplied independently of thermal time	pp. 15, 20
16	Scale ratio of thermal and geometrical time	pp. 15, 20
17	Lorentz times of relatively moving observers	pp. 10, 18
18	Lorentz time selected by the heat-bath rest condition	pp. 18
19	Thermal flow proposed from the negative logarithm of a classical density and a Poisson structure	pp. 3, 4, 18, 19
20	Causal domain defining the Rindler wedge	pp. 19, 20
21	Boost evolution along a Rindler observer's proper time	pp. 19, 20
22	Scale correspondence between Rindler thermal time and proper time	pp. 20
23	Time–temperature correspondence for a stationary Schwarzschild observer	pp. 20, 21
24	Conjectured domain of time in a Planck phase	pp. 21, 23
25	Present time flow envisaged as emerging after the Planck phase	pp. 21, 23
26	FRW time used for cosmological comparison	pp. 4, 22
27	Cosmological thermal flow associated with the CMB state	pp. 4, 22

i	Time Content	Source Location
28	Decay times envisaged as quantum thermal clocks in a Planck regime	pp. 15

Definition 8.434 (Connes–Rovelli 1994 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(43)$, with claim boundary $\text{ClaimBoundaryI}dx_R = ([4], [43])$. Claims 1–4 form the generation-initial set; all 43 claims form the source formal-evidence environment. Each row gives the content, time uses and source location.

Table 322: Connes–Rovelli 1994 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Specified physical observables, field configurations and algebraic kinematics.	$\emptyset$	pp. 6, 8, 11, 14
2	Expectation functional of a specified state and the values of its density.	$\emptyset$	pp. 6, 14, 16
3	Field configurations representing the physical reference system and states of preparation and detection apparatus.	$\emptyset$	pp. 9, 15
4	Classical phase space, density and Poisson structure, and a selected observables subalgebra.	$\emptyset$	pp. 11, 18, 19, 21
5	Connes and Rovelli contrast Hamiltonian evolution in non-generally-covariant theories with the problem of selecting a preferred flow in generally covariant theories.	1^{SQ}	pp. 2, 12
6	Each GR solution supplies causal classification and worldline proper time; quantum or thermal superpositions raise the problem of selecting that geometry.	$2^{\text{BC}}, 3^{\text{IQS}}$	pp. 2, 3
7	Comparison with experiment uses reference fields representing clock readings, pulse counts or astronomical variables.	$4^{\text{B}}, 5^{\text{OQK}}$	pp. 8, 9
8	Local inversion of four reference fields defines generally covariant observables indexed by reference time.	$5^{\text{OQK}}, 6^{\text{SO}}$	pp. 9, 11
9	Taking the particle coordinate X^0 or the clock field f_4 as t , local elimination of the auxiliary parameter expresses evolution and observable families.	$5^{\text{OQK}}, 7^{\text{SQ}}, 8^{\text{S}}, 17^{\text{XS}}$	pp. 9, 10, 11
10	An arbitrary internal clock can lose monotonicity or stop and reverse in some superposed geometries, so the inverse map has a local domain of validity.	$7^{\text{SQ}}, 9^{\text{BS}}$	pp. 12

p	Claim	i	Source Location
11	The source raises the interpretive consistency problem of nonunitary quantum internal-clock evolution and assumes a consistent definition of the theory for this paper.	$7^{\text{SQ}}, 8^{\text{S}}, 10^{\text{BS}}$	pp. 12
12	A positive normalized state gives a GNS representation. The modular construction uses a cyclic separating vector, or a faithful normal state, for the specified von Neumann algebra.	$\emptyset$	pp. 6, 7
13	The Tomita–Takesaki construction gives $\alpha_t(A) = \Delta^{-it}A\Delta^{it}$ from $S = J\Delta^{1/2}$; source Footnote 3 specifies this sign convention.	$\emptyset$	pp. 7
14	Modular flows of faithful normal states on the same von Neumann algebra agree modulo inner automorphisms and determine a state-independent outer class.	$\emptyset$	pp. 7, 8, 21
15	The source’s thermal-time hypothesis adopts the modular flow obtained from a specified state and algebra as physical time flow.	11^{SQ}	pp. 3, 4, 14, 15, 23
16	The source distinguishes the selection of a physical flow from the question of time orientation.	$11^{\text{SQ}}, 12^{\text{S}}$	pp. 13
17	The KMS condition for a known Hamiltonian evolution gamma and inverse temperature beta uses strip analyticity and a boundary-value relation.	$1^{\text{SQ}}, 13^{\text{QI}}$	pp. 13, 14
18	A faithful normal state is a KMS state at $\beta = 1$ for its own modular flow.	$\emptyset$	pp. 14
19	Source Equations (25) and (26) define correlators of the form $\omega(B\alpha_t(A))$ from observables and a flow.	$\emptyset$	pp. 14, 15
20	The source physically interprets the correlator as preparation by an apparatus, a subsequent wait and detection by another apparatus.	$11^{\text{SQ}}, 14^{\text{OIS}}$	pp. 15
21	When an independent geometrical time is available, the source associates its scale ratio with thermal time with temperature.	$11^{\text{SQ}}, 15^{\text{IQ}}, 16^{\text{Q}}$	pp. 15, 20
22	The Gibbs branch uses finite partition function and $\rho = Ne^{-\beta H}$, constructing $\Omega = \rho^{1/2}$, left action L_A and $\Delta = L_\rho R_{\rho^{-1}}$ on the full Hilbert–Schmidt operator space.	$\emptyset$	pp. 16, 17
23	The modular flow of the specified Gibbs state is $\alpha_t(A) = e^{i\beta t H} A e^{-i\beta t H}$.	$\emptyset$	pp. 17, 18

p	Claim	i	Source Location
24	Comparison with the known Hamiltonian time gives $\alpha_t = \gamma_{\beta t}$, so its time parameter is $\tau = \beta t$.	$1^{\text{SQ}}, 11^{\text{SQ}}, 13^{\text{QI}}$	pp. 18
25	The bath momentum and rest condition select a Lorentz frame, and the thermal-time hypothesis selects the flow corresponding to time in that frame.	$11^{\text{SQ}}, 17^{\text{XS}}, 18^{\text{SX}}$	pp. 18
26	In the finite-dimensional Hilbert–Schmidt representation, $[L_A, \Delta] = L_{[A, \rho]} R_{\rho^{-1}}$ and $[L_A, \ln \Delta] = L_{[A, \ln \rho]}$; differentiation of alpha gives $i[A, \ln \rho]$.	$\emptyset$	pp. 17, 18
27	The source’s classical-limit prescription uses $H = -\ln \rho$ for a positive density and Hamilton’s equation through a Poisson bracket.	$\emptyset$	pp. 18, 19
28	In the classical limit of the thermal-time hypothesis, the specified density and Poisson structure determine thermal flow through $H = -\ln \rho$.	19^{SQ}	pp. 3, 4, 18, 19
29	The uniformly accelerated trajectory and Minkowski causal structure specify the observer’s accessible Rindler wedge $x^1 > x^0 $.	$3^{\text{IQS}}, 20^{\text{BC}}$	pp. 19
30	Evolution along the Rindler observer’s proper time tau is $\gamma_\tau(A) = e^{i\tau a K_1} A e^{-i\tau a K_1}$.	$3^{\text{IQS}}, 21^{\text{SQ}}$	pp. 19, 20
31	For the vacuum restricted to the Rindler wedge algebra, the Bisognano–Wichmann correspondence identifies the modular action with the specified boost group.	20^{BC}	pp. 20
32	The correspondence in source Equations (54) and (55) gives $\alpha_t = \gamma_{2\pi t/a}$, hence $\tau = 2\pi t/a$ and $t = a\tau/(2\pi)$.	$3^{\text{IQS}}, 11^{\text{SQ}}, 21^{\text{SQ}}, 22^{\text{QI}}$	pp. 20
33	The inverse temperature in this Rindler branch is $\beta = 2\pi/a$ and the temperature is $T = a/(2\pi k_B)$, in the source units $c = \hbar = 1$.	22^{QI}	pp. 20
34	Connes and Rovelli state that extension to a stationary Schwarzschild observer leads to the Hawking temperature and leave the derivation undeveloped.	$22^{\text{QI}}, 23^{\text{QS}}$	pp. 20, 21
35	Equality of outer classes on a fixed algebra represents each state’s concrete modular action modulo inner automorphisms.	$\emptyset$	pp. 7, 8, 21
36	Restricting the same state to different observables subalgebras yields modular groups under the respective faithfulness conditions.	$\emptyset$	pp. 21

p	Claim	i	Source Location
37	The source leaves selection of the physical state and subalgebra open and raises the possibility of a maximum-entropy selection mechanism.	11 ^{SQ}	pp. 21, 23
38	Computing thermal time requires specifying the state and kinematics. The source proposes determining a state from a small number of observations but supplies no unique reconstruction rule from the observations.	$\emptyset$	pp. 14, 23
39	Footnote 7 proposes associating flows of different subalgebras with phases from a Planck phase to the present time flow and leaves the concrete implementation unspecified.	24 ^B , 25 ^S	pp. 21
40	The cosmological application cited by the source applies the thermal-time construction to a specified CMB state.	19 ^{SQ} , 27 ^{SQ}	pp. 4, 22
41	The source cites the result of the model in Reference 11 that classical thermal time based on the CMB state agrees with conventional FRW time.	26 ^{SQ} , 27 ^{SQ}	pp. 4, 22
42	Footnote 6 conjectures quantum thermal clocks, such as decay times, as clocks that could survive in a Planck regime.	11 ^{SQ} , 28 ^{IO}	pp. 15
43	The conclusion reports, as a result of Reference 4, a framework in which the classical thermal-time hypothesis defines generally covariant statistical mechanics of gravity.	19 ^{SQ}	pp. 22

Table 323: Connes–Rovelli 1994 Time-Use Roles

Code	Role	Use Count
I	Interval	12
Q	Quantity	38
X	Cross-position correspondence	3
K	Clock reading	3
C	Causality	3
S	State evolution	39
P	Probabilistic evolution	0
O	Observation record	6
B	Reference boundary	7

Table 324: Connes–Rovelli 1994 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3, 5, 7, 11, 14–15, 17–19, 25–27	member	instant, order_relation
2, 6, 8, 10, 12–13, 16, 20–24, 28	dependency	$\emptyset$
4	member	instant
9	member	order_relation

Dependency Inputs

Definition 8.435 (Connes–Rovelli 1994 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [13]$. The table gives time inputs, time outputs, claim support and source locations.

Table 325: Connes–Rovelli 1994 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	5	6	8	Construct relational observables from a reference clock. pp. 9, 11
2	17	7	9	Select an internal clock from a particle’s Lorentz-time coordinate. pp. 9, 10
3	5	7	9	Select an internal clock from a physical reference field. pp. 10, 11
4	7	8	9	Express local evolution in an internal clock. pp. 10, 11
5	7	9	10	Identify monotonicity and stopping boundaries of the chosen clock. pp. 12
6	$\emptyset$	11	12–13, 15	Construct a thermal-time candidate from a state and algebra. pp. 7, 14
7	11, 15	16	21	Compare thermal and geometrical time scales. pp. 15, 20
8	1, 11	13	24	Determine the inverse-temperature scale against known time in the Gibbs branch. pp. 18
9	11, 17	18	25	Select Lorentz time using the bath rest condition. pp. 18
10	$\emptyset$	19	27–28	Construct a classical thermal-time candidate from density and Poisson structure. pp. 3, 4, 18, 19
11	3, 20	21	29–30	Associate Rindler proper time with boost evolution. pp. 19, 20
12	3, 11, 20–21	22	31–32	Obtain the Rindler thermal/proper-time scale correspondence. pp. 20

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
13	19	27	40	Apply the classical thermal-time construction to the cosmological state. pp. 4, 22

Grounding Conditions

1. Local invertibility of four reference fields and three fixed spatial reference values.
2. Specified relativistic trajectory, $t = X^0$ and a local inverse.
3. The field f_4 interpreted as a clock and a local inverse.
4. Domain in which the chosen clock permits elimination of the auxiliary parameter.
5. Clock variable on the specified solution or superposition of geometries.
6. A faithful normal state on the specified algebra, the Tomita–Takesaki construction and the source’s thermal-time hypothesis.
7. An independently valid geometrical time description of the same system.
8. Finite partition function, specified H and β , and $\alpha_t = \gamma_{\beta t}$.
9. Momentum and rest condition of the specified bath, with the thermal-time hypothesis.
10. Positive differentiable density, Poisson structure, the source’s Hamiltonian convention and domain of existence of the local flow.
11. Uniformly accelerated trajectory in Minkowski spacetime and the specified representation.
12. Bisognano–Wichmann correspondence for the wedge algebra and restricted vacuum, with Equations (54) and (55).
13. CMB state and classical thermal-time hypothesis in the specified model of Reference 11.

Results

Definition 8.436 (Connes–Rovelli 1994 Results and Dependencies). Let $\text{ldx}_R^{\text{audit,result}} = [37]$. Each result takes the temporal premises and claim support shown in the table as inputs.

Table 326: Connes–Rovelli 1994 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Problem of selecting a preferred flow in generally covariant theories. pp. 2, 12	1	5

q	Result and Source Location	Time Premises i	Claim Support p
2	Causal classification and proper time on GR solutions. pp. 2, 3	2–3	6
3	Comparison of experiment and field solutions through physical reference clocks. pp. 8, 9	4–5	7
4	Relational observables obtained by inversion of reference fields. pp. 9, 11	5–6	8
5	Local evolution expressed in a selected internal clock. pp. 9, 10, 11	7–8	9
6	Monotonicity and stopping/reversal boundaries of internal clocks. pp. 12	7, 9	10
7	Interpretive problem of unitarity for quantum internal clocks. pp. 12	7–8, 10	11
8	GNS representation and state conditions for modular construction. pp. 6, 7	$\emptyset$	12
9	Tomita–Takesaki modular automorphism group. pp. 7	$\emptyset$	13
10	State-independent outer class on a fixed algebra. pp. 7, 8, 21	$\emptyset$	14, 35
11	Self-KMS relation for modular flow. pp. 14	$\emptyset$	18
12	Distinction between flow selection and time orientation. pp. 13	11–12	16
13	Candidate flow of the thermal-time hypothesis for a specified state and algebra. pp. 3, 4, 14, 15, 23	11	15
14	KMS equilibrium condition for known time. pp. 13, 14	1, 13	17
15	Correlator defined by the state and operators. pp. 14, 15	$\emptyset$	19
16	Physical interpretation of the correlator through preparation, waiting and detection. pp. 15	11, 14	20
17	Temperature as a scale relation between thermal and geometrical time. pp. 15, 20	11, 15–16	21
18	Hilbert–Schmidt representation of a Gibbs state. pp. 16, 17	$\emptyset$	22
19	Modular action obtained from a Gibbs state. pp. 17, 18	$\emptyset$	23
20	Correspondence of thermal and known Hamiltonian time in the Gibbs branch. pp. 18	1, 11, 13	24
21	Selection of Lorentz time in the bath rest frame. pp. 18	11, 17–18	25
22	Modular generator in the Hilbert–Schmidt representation. pp. 17, 18	$\emptyset$	26

q	Result and Source Location	Time Premises i	Claim Support p
23	Relation between classical density and a Hamiltonian function. pp. 18, 19	$\emptyset$	27
24	Classical thermal flow from density and Poisson structure. pp. 3, 4, 18, 19	19	28
25	Rindler wedge and proper-time boost evolution. pp. 19, 20	3, 20–21	29–30
26	Bisognano–Wichmann correspondence for the vacuum restricted to a wedge. pp. 20	20	31
27	Proportionality between Rindler thermal time and proper time. pp. 20	3, 11, 21–22	32
28	Unruh temperature. pp. 20	22	33
29	Sketch of a Hawking extension to a stationary Schwarzschild observer. pp. 20, 21	22–23	34
30	Restriction of a state to subalgebras and modular groups. pp. 21	$\emptyset$	36
31	Problem of selecting a physical state and subalgebra. pp. 21, 23	11	37
32	State specification and conditions for reconstruction from observations. pp. 14, 23	$\emptyset$	38
33	Phase proposal from a Planck phase to the present time flow. pp. 21	24–25	39
34	Classical thermal-time construction applied to a CMB state. pp. 4, 22	19, 27	40
35	Correspondence of CMB thermal time with conventional FRW time. pp. 4, 22	26–27	41
36	Proposal of quantum thermal clocks in a Planck regime. pp. 15	11, 28	42
37	Reference to generally covariant gravitational statistical mechanics using classical thermal time. pp. 22	19	43

Definition 8.437 (Connes–Rovelli 1994 Grounds and Domains of Results). Each result’s temporal premises and claim support are given in Table 326; its conditions and domain are given below.

Table 327: Connes–Rovelli 1994 Conditions for Results

Result Index	Conditions and Domain
1	Structural comparison with Hamiltonian time in non-generally-covariant theories.
2	Specified spacetime solution and worldline, in the context of discussing superpositions.

Result Index	Conditions and Domain
3	Reference fields interpreted as clock readings, pulses or astronomical observables.
4	Existence of a local inverse and fixed spatial reference values.
5	Clock variable of the particle or reference field and its local domain of validity.
6	Domain for the chosen clock and each geometry.
7	The open consistency problem identified by the source and the consistency assumption adopted in the paper.
8	A positive normalized state for GNS; cyclic separation or faithful normality on the specified algebra for modular construction.
9	Operator and domain assumptions of the theorem and the sign convention of source Footnote 3.
10	The same von Neumann algebra and faithful normal states being compared.
11	The state's own modular flow and $\beta = 1$ normalization.
12	The two questions treated separately by the source.
13	Conditions of modular construction and the source's hypothesis adopting the flow as physical time.
14	Specified evolution, inverse temperature, strip analyticity and boundary-value relation.
15	Observables and the specified action in source Equations (25) and (26).
16	Specified apparatus, waiting interval and order of experimental operations.
17	The two time descriptions of the same system.
18	Finite partition function, faithful density, full Hilbert–Schmidt space and left action.
19	Source Equations (40)–(43) for the specified H and β .
20	$\alpha_t = \gamma_{\beta t}$ and interpretation of the known evolution parameter.
21	Bath momentum, rest condition and thermal-time hypothesis.
22	For a faithful finite-dimensional density, the logarithmic commutator identity distinguishing left and right actions.
23	Positive density, $H = -\ln \rho$ and the source's classical-limit prescription using the Poisson bracket.
24	The source's thermal-time hypothesis and local Hamiltonian flow for a differentiable density.
25	Minkowski causal structure, uniformly accelerated trajectory and specified boost representation.
26	Local algebra of the specified wedge, vacuum state and applicability conditions of the theorem.
27	$\tau = 2\pi t/a$ from Equations (54) and (55).
28	Uniformly accelerated Rindler branch and the source units.
29	The extension to a stationary observer stated by the source; identifying local temperature requires the state and normalization of the time generator.
30	Modular-construction conditions for each subalgebra and restricted state.

Result Index	Conditions and Domain
31	State selection left open by the source and the possibility of a maximum-entropy mechanism.
32	Full specification of state and kinematics, and the source's proposed state determination by observation.
33	The unimplemented subalgebra-based proposal in Footnote 7.
34	Cosmological model and specified state of Reference 11.
35	Conditions of the model whose result the source cites from Reference 11.
36	Speculation concerning decay times and related quantities in Footnote 6.
37	The theoretical framework reported by the source as a result of Reference 4.

Result Dependence

Definition 8.438 (Connes–Rovelli 1994 Result-Dependence Hyperedges). For each row of the result table, let $\text{ResultEdge}_{R,q}$ ($q \in [37]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Evidence Hierarchy Connes and Rovelli propose the thermal-time hypothesis on the basis of operator-algebraic constructions and discuss internal clocks, Gibbs states, the classical limit, the Rindler branch and cosmological applications. The source reports zero directly measured objects and zero direct-measurement predicates from measurements of its own. Clocks, preparation apparatus and detectors appear as objects of physical interpretation.

Generation Judgment The complete LCTR generation count is zero; source-evidence-conditioned local reach contains 3 time contents, leaving 25 primitive-time roots. Thermal time from a state and algebra, thermal time from a classical density and Poisson structure, and application to the specified CMB state are reached conditionally on the source's constructions and thermal-time hypothesis. Known Hamiltonian time, reference clocks, proper time, Lorentz time and the FRW comparison time enter their respective application domains as inputs.

Result Dependence Of the 37 results, 21 depend on primitive-time roots. Results reachable from the source formal-evidence environment without added time roots are 8–11, 13, 15, 18–19, 22–24, 30–32, 34, 37: 16 results, including mathematical constructions, conditions of state specification and local thermal-time constructions. There are 40 root–result dependence pairs, of which 22 are necessary.

Placement of the Grounding Obligation Connes and Rovelli define thermal flow from a state and algebra and adopt it as physical time. Internal clocks and experimental interpretation use clock readings, order and intervals; special branches use correspondence with known times. All 28 time contents are primitive-time roots under the complete judgment. Under local reach, Time Contents 11, 19 and 27 are constructed and 25 remain. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Faithfulness, local clock invertibility, normalization of the Gibbs state, existence of the classical flow and conditions on the wedge algebra specify the domains of the respective results. The source discusses stopping, reversal and nonunitarity of internal clocks as general applicability problems. Actual failure of a specific audited physical-time generation condition and the resulting change in result values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 40 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are each temporal use and the physical-time generation conditions required before that use. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Proposition 8.439 (Primitive-Time Roots and Result Dependence in Connes–Rovelli 1994). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = (\emptyset, [28]; \\ \emptyset, \{8, 9, 10, 11, 13, 15, 18, 19, 22, 23, 24, 30, 31, 32, 34, 37\}, \\ [37] \setminus \{8, 9, 10, 11, 13, 15, 18, 19, 22, 23, 24, 30, 31, 32, 34, 37\}, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Adding all 43 claims as source formal evidence allows Transfers 6 and 10 to output Time Contents 11 and 19, and Transfer 13 to output Time Content 27. Temporal inputs required by the other transfers are outside this closure. Comparing the closure with each result’s input set gives 16 reachable results. Adding all 28 time contents as seeds reaches all 37 results; the common reachability classification gives the displayed six components. $\square$

Proposition 8.440 (Physical-Time Admissibility in Connes–Rovelli 1994). All 28 time contents are primitive-time roots under the complete judgment and lack physical-time admissibility. Local reach contains three time contents and leaves 25 roots. Twenty-one results are reached by adding primitive-time roots.

Proof. This follows from the reachability classification above, Definition 7.10 and Theorem 7.11. $\square$

Proposition 8.441 (Minimal Primitive-Time-Root Support in Connes–Rovelli 1994). The table gives each result’s minimal-support index family, with 46 family members in total.

Proof. Start from singleton time-content seeds and apply Definition 7.46 to the 13 transfers. Empty support propagates through constructions with no temporal inputs. Combine supports of each result’s premises from the stable antichains and retain the inclusion-minimal supports. Proposition 7.47 gives the displayed families and counts. $\square$

Table 328: Minimal Primitive-Time-Root Support in
Connes–Rovelli 1994

Result Indices	Minimal-Support Family Members
1, 14, 20	$\{1\}$
2	$\{2, 3\}$
3	$\{4, 5\}$
4	$\{5\}$
5–6	$\{5\}, \{7\}, \{17\}$
7	$\{5, 10\}, \{7, 10\}, \{10, 17\}$
8–11, 13, 15, 18–19, 22–24, 30–32, 34, 37	$\emptyset$
12	$\{12\}$
16	$\{14\}$
17	$\{15\}$
21	$\{17\}$
25	$\{3, 20\}$
26	$\{20\}$
27	$\{3, 20\}, \{3, 21, 22\}$
28	$\{3, 20\}, \{22\}$
29	$\{3, 20, 23\}, \{22, 23\}$
33	$\{24, 25\}$
35	$\{26\}$
36	$\{28\}$

Evidentiary Effect

Definition 8.442 (Evidentiary Domain of Connes–Rovelli 1994). The specified algebra and state determine a modular group; on a fixed algebra its outer class is state independent. The thermal-time hypothesis uses the group as physical time. The Gibbs and Rindler branches establish scale correspondences with independently supplied times. The classical branch uses the negative logarithm of density and a Poisson structure. The Hawking relation is an extension sketched by the source, the CMB–FRW correspondence is a result of a cited model, and the Planck phase and quantum thermal clocks are proposals of the source, each with its respective grounds and conditions.

Result-Preservation Domain

Definition 8.443 (Result-Preservation Conditions for Connes–Rovelli 1994). Fix the algebra, state, operator domains and conditions in Table 327. When inputs of results using physical time can be realized by generated temporal representations with the same order, intervals, flow and scale correspondences, each result is preserved with its stated domain and epistemic status.

Table 329: Result-Preservation Conditions for Connes–Rovelli 1994

Results	Preservation Conditions	Domain	Placement in the LCTR Generation Order
8–11, 13, 15, 18–19, 22–24, 30–32, 34, 37	Mathematical conditions, state specification or candidate-construction conditions of each result.	Operator algebras, classical flow and specified models.	Nontemporal structures and conditional components of physical-time generation.
1–7, 12, 14, 16–17, 20–21, 25–29, 33, 35–36	Claim support and generation of temporal premises.	Domain of each comparison, experimental interpretation or cosmological proposal.	Use after grounding the relevant time contents.

Proposition 8.444 (Audit-Boundary Counts for Connes–Rovelli 1994). The 43 claims, 28 time contents, 13 transfers and 37 results give 121 source-specific obligation atoms. The 55 claim/time-content use pairs have 111 role assignments.

$$\text{BoundaryCount}_R^{\text{audit}} = (108, 50).$$

Proof. Count claims, time contents and results as nodes, transfers and result dependencies as edges, and sum the roles of time uses in the claim table. $\square$

Individual Statements Absence of a Preferred Time Flow

Original text:

there is no preferred time flow

The passage occurs in arXiv:gr-qc/9406019v1, p. 2, §1. Connes and Rovelli state that a Hamiltonian function specifies evolution in non-generally-covariant theories, while generally covariant theories supply no preferred flow of the same kind. Their immediately following account of general relativity retains causal classification and worldline proper time on each spacetime solution and discusses their state dependence.

Postulate Identifying Thermal Time with Physical Time

Original text:

thermal time ... defines the physical time

The passage occurs at the beginning of p. 4, §1, continuing a sentence from the end of p. 3. For the flow determined by a state in generally covariant classical statistical mechanics, Connes and Rovelli introduce the identification of thermal and physical time after the phrase “one can postulate”. On the same p. 4 they extend the idea to quantum theory by identifying physical time with the modular group associated with a state and algebra, and state the thermal-time hypothesis in §3.1, p. 14.

Common Evidence with Specified Scope The table gives the objects, predicates and domains of the two statements.

State- ment	Object	Predicate	Explicit Domain
S35- Q001	Preferred time flow in generally covariant theories	Connes and Rovelli state that generally covariant theories supply no preferred time flow.	The contrast between non-generally-covariant and generally covariant theories in §1, p. 2.
S35- Q002	Identification of state-determined thermal time with physical time	Connes and Rovelli postulate the adoption of thermal time as physical time.	The classical account on pp. 3–4, §1, its quantum extension on p. 4, and the hypothesis in §3.1, p. 14.

Individual Audit Absence of a Preferred Time Flow

The first statement concerns selection of a preferred flow in generally covariant theories. Its ground in the source is the structural comparison with non-generally-covariant theories, where a Hamiltonian function specifies evolution. The immediately following account of general relativity retains causal classification and worldline proper time on each solution. Hamiltonian comparison time is Time Content 1; causal classification and proper time are Time Contents 2–3; the corresponding results are 1–2. The corresponding claims are 5–6.

Postulate Identifying Thermal Time with Physical Time

The second statement expresses Connes and Rovelli’s postulate that a state-determined flow is used as physical time. The classical account uses a positive density and Poisson structure; the quantum extension uses the modular construction for a specified algebra and faithful normal state. The outer class on a fixed algebra and the self-KMS relation are mathematical results of this construction. Scale comparisons in the Gibbs and Rindler branches use known times, while the CMB–FRW correspondence is a model result cited from Reference 11. The locally reached Time Contents 11, 19 and 27 and the comparison Time Contents 1, 3, 13, 15–18, 20–23 and 26 have the respective inputs and conditions shown in the tables. Corresponding evidence comprises Claims 12–15, 17–18, 21–34 and 40–41, Transfers 6–13, and Results 8–11, 13–14, 17–29 and 34–35.

8.19.36 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 36.

Source Name Primitive-time-root audit of Headley, Clayton, and Reed 1996

Bibliographic Identification Let C. Headley, J. B. Clayton, and W. A. Reed, *Methods of Suppressing Stimulated Brillouin Scattering in Optical Fibers by Manipulation of the Fiber Properties* (1996) [46] be Source *R*. The audit domain is p. 105, Equation (1) in Section II, the expression for the full width at half maximum of the Brillouin gain spectrum, and the surrounding paragraphs. The account of wave generation and coupling in the first paragraph of Section I on the same page supplies the context for claims 1–3.

Definition 8.445 (Semantic components in Headley, Clayton, and Reed 1996). The 23 semantic components comprise five nontemporal initial components, 15 source claims, and three mathematical-object types.

Audit-Scope Identifier 36-SC1.

Audit-Scope Content The audit domain specified under Bibliographic Identification.

Source-Location Identifier 36-L1.

Source-Location Content The audit domain specified under Bibliographic Identification and the source- location columns of the respective evidence tables.

Formal Object

Object Identifier 36-F1.

Object Name Formal objects in the source.

Object Type Formal object.

Formal Content The individual formal objects are given in Definitions 8.445 and 8.446.

Source-Location Identifier Set 36-L1.

Assumption

Assumption Identifier 36-A1.

Assumption Content The Lorentzian gain profile assumes exponential decay of the acoustic wave and either CW or QCW operation of the pump. The QCW condition is $T_0 \gg T_B$; CW operation specifies no finite pulse width. The gain reduction stated for short pulses, $T_0 < T_B$, concerns a different operating regime. The width–lifetime relation uses the source’s ordinary-frequency convention $\Delta\nu_B T_B = 1/\pi$, with $T_B > 0$.

Source-Location Identifier Set 36-L1.

Nontemporal Physical Input

Input Identifier 36-P1.

Input Name Nontemporal physical inputs in the source.

Input Type Nontemporal physical input.

Input Content The individual nontemporal physical inputs are given in Definitions 8.445 and 8.446. Initial components 1–3 select the wave carriers; this projection excludes propagation, temporal oscillation, and temporal coupling relations. Initial components 4 and 5 are the dimensionless refractive index and the wavelength as a spatial length, respectively.

Source-Location Identifier Set 36-L1.

Time Content

Time-Content Identifier 36-T1.

Time-Content Name Time contents in the source.

Time-Content Type Time content.

Time-Content Statement The individual time contents are given in Definition 8.447.

Source-Location Identifier Set 36-L1.

Formal Input Elements

Formal-Component Identifier 36-C1.

Formal-Component Name Formal-evidence environment in the source.

Formal-Component Type Formal component.

Formal-Component Text The individual formal input elements are given in Definitions 8.445 and 8.446.

Source-Support Identifier Set 36-L1.

Table 331: Nontemporal initial components in Headley, Clayton, and Reed 1996

k	Semantic component	Corresponding claim indices
1	Pump-wave carrier	$\llbracket 1, 3 \rrbracket$
2	Stokes-wave carrier	$\llbracket 1, 3 \rrbracket$
3	Acoustic-wave carrier	$\llbracket 1, 3 \rrbracket$
4	Fiber refractive index	$\{4, 7\}$
5	Pump wavelength	$\{5, 7\}$

Definition 8.446 (Source claims in Headley, Clayton, and Reed 1996). The following table gives the 15 source claims.

Table 332: Source claims in Headley, Clayton, and Reed 1996

p	Claim	Source location
1	Pump wave propagating in an optical fiber	p. 105, first paragraph of Section I
2	Backscattered Stokes wave arising from the pump wave	p. 105, first paragraph of Section I
3	Acoustic wave coupled to the pump and Stokes waves	p. 105, first paragraph of Section I
4	Fiber refractive index	Equation (1)
5	Pump wavelength	Equation (1)
6	Acoustic-wave velocity	Equation (1)
7	Ordinary Brillouin-frequency difference	Equation (1)
8	continuous-wave operation	paragraph following Equation (1)
9	quasi-continuous-wave operation	paragraph following Equation (1)

p	Claim	Source location
10	Exponential decay of the acoustic wave	paragraph following Equation (1)
11	Lorentzian Brillouin gain profile	paragraph following Equation (1)
12	Ordinary-frequency full width at half maximum of the Brillouin gain profile	paragraph following Equation (1)
13	Relation between the full width at half maximum and acoustic-wave lifetime	paragraph following Equation (1)
14	Operating regime determined by pulse width and acoustic-wave lifetime	p. 105, description of operating regimes
15	Reduction of SBS gain accompanying a change of operating regime	p. 105, description of operating regimes

Table 333: Mathematical-object types in Headley, Clayton, and Reed 1996

m	Mathematical-object type	Source location
1	Relation for the ordinary Brillouin-frequency difference	Equation (1)
2	Lorentzian Brillouin gain profile	paragraph following Equation (1)
3	Relation between ordinary-frequency full width at half maximum and acoustic-wave lifetime	paragraph following Equation (1)

Definition 8.447 (Time contents in Headley, Clayton, and Reed 1996). The following table gives the six time contents.

Table 334: Time contents in Headley, Clayton, and Reed 1996

i	Time content	Source location
1	Brillouin ordinary-frequency shift ν_B	p. 105, Equation (1)
2	Acoustic-wave velocity	p. 105, Equation (1)

i	Time content	Source location
3	Ordinary-frequency full width at half maximum	p. 105, paragraph following Equation (1)
4	Acoustic-wave lifetime	p. 105, paragraph following Equation (1)
5	Pulse width	p. 105, description of operating regimes
6	Frequency variable ν of the gain spectrum $g_B(\nu)$	p. 105, paragraph following Equation (1)

Dependent Input

Index Identifier S36-XI-DI1.

Existing-Evidence Reference Set Definition 8.448.

Dependence and Transfer Elements

Definition 8.448 (Time-content transfers in Headley, Clayton, and Reed 1996). The following table gives the two time-content transfers.

Table 335: Time-content transfers in Headley, Clayton, and Reed 1996

Transfer	Dependence and transfer	Source claims	Time input	Time output	Source location
1	From acoustic-wave velocity and other inputs to the ordinary Brillouin-frequency shift	[[3, 7]]	{2}	{1}	p. 105, Equation (1)
2	From acoustic-wave lifetime to ordinary-frequency full width at half maximum	[[10, 13]]	{4}	{3}	p. 105, paragraph following Equation (1)

Both transfers take at least one existing time content as input before producing a time content. No transfer produces a time content solely from the nontemporal initial components.

Formation Conditions

Index Identifier S36-XI-FC1.

Existing-Evidence Reference Set Definition 8.448.

Source Support

Index Identifier S36-XI-SS1.

Existing-Evidence Reference Set Definition 8.448.

Results

Definition 8.449 (Results and preservation conditions in Headley, Clayton, and Reed 1996). The following table gives the three source-specific results, source claims, time premises, source locations, and preservation conditions.

Table 336: Results, dependences, and preservation conditions in Headley, Clayton, and Reed 1996

Re-sult	Content	Source claims	Time premises	Source location	Preservation conditions
1	Ordinary Brillouin-frequency shift	$\llbracket 1, 7 \rrbracket$	$\{1, 2\}$	Equation (1)	Preserve pump–Stokes–acoustic coupling, refractive index, wavelength, and acoustic-wave velocity.
2	Lorentzian gain profile under the respective CW and QCW conditions	$\llbracket 8, 11 \rrbracket \cup \{14\}$	$\{1\} \cup \llbracket 3, 6 \rrbracket$	p. 105, paragraph following Equation (1)	Preserve exponential decay and the respective conditions: CW, or QCW with $T_0 \gg T_B$.
3	Width–lifetime relation in the exponential-decay and Lorentzian model	$\llbracket 10, 13 \rrbracket$	$\{3, 4\}$	p. 105, paragraph following Equation (1)	Preserve exponential acoustic-wave decay, the Lorentzian gain profile, and the lifetime convention of the source.

Result Dependence

Index Identifier S36-XI-RD1.

Existing-Evidence Reference Set Definition 8.449.

Grounds

Index Identifier S36-XI-RG1.

Existing-Evidence Reference Set Definition 8.449.

Evidence Hierarchy Within the audit domain, this source itself newly measures no object directly. The source presents an engineering analysis of methods for suppressing SBS.

Generation Determination **Determination values:** 0 time contents generated under the LCTR-complete-generation determination / 0 time contents reachable under the source-evidence-conditional local-attainment determination / 6 residual primitive time roots.

Source-evidence-conditional local attainment No time contents are reachable under the source-evidence-conditional local-attainment determination, and six primitive time roots remain after that determination.

Result Dependence The root–result dependence relation has seven pairs, five of which are indispensable dependences.

Evidential Force The formal evidence supports the ordinary-frequency difference, the conditional Lorentzian gain profile, and the relation between ordinary- frequency full width at half maximum and lifetime. Its engineering validity does not imply that frequency, velocity, lifetime, or pulse width has been generated from nontemporal input.

Result-Preservation Domain The three results are preserved where their time-content premises are generated in the LCTR generation sequence and the conditions in Table 336 are satisfied.

Placement of Grounding Obligations

Placement of grounding obligations The six time contents audited in this source are not generated from nontemporal physical input before use; they are accepted as already-valid temporal inputs. The placement determination for the physical-time grounding obligations is “backgrounded by primitive time input.” This determination follows by comparing, for each time content, its use in a law operation or result derivation with the absence, prior to that use, of a corresponding generation or validity condition supplied as an input, formation condition, evaluation condition, or failure condition.

Evidence of Actual Failure

Evidence of actual failure The complete determination and residual-root determination concern physical-time qualification and are distinct from an actual failure of a concrete condition governing the value or domain of application of a result. Neither the source’s formal evidence nor the LCTR audit evidence establishes an actual failure of such a target condition. The actual-failure determination is “indeterminate from source evidence”; actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream reach For primitive-time-root-dependent results, the reach is determined source by source by the primitive-time-root–result dependence relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because no actual failure of a concrete condition has been established, determination of actual reach from the failure location to a result is “indeterminate from source evidence.” Structural reach does not replace actual reach.

Repair Scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation or validity conditions required before their use. The scope of recalculation is restricted to results reached from a failure location when actual failure has been independently established. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If that condition is met, the target results are regenerated or recalculated from the repaired inputs, or their domain of validity is restricted. The value after recalculation is fixed as an audit result.

Audit Conclusions

Definition 8.450 (Finite dependence graph of Headley, Clayton, and Reed 1996). The finite dependence graph comprises 23 semantic components, six time contents, two transfers, and three results, giving 32 nodes and five edges. The 23 semantic components are enumerated as audit-boundary claim nodes: first the five claims establishing the non-temporal initial components, then the 15 source claims, and finally the three specifications of mathematical object types. Source-claim index p corresponds to semantic-component index $5 + p$. Claim support for each transfer and result is given by its source-claim column. In addition, transfers 1 and 2 use mathematical-object-type specifications 1 and 3, respectively, and result j uses specification j . The generation-initial set comprises the first five claim nodes; the source-formal-evidence environment comprises all 23. Time-content and result nodes are formed from their respective item families.

Proposition 8.451 (Complete audit of time use in Headley, Clayton, and Reed 1996). The number of generated time contents under the complete determination and the number of reachable time contents under the local-attainment determination are both zero. All six time contents remain primitive time roots, and each of the three results depends on at least one of them.

Proposition 8.452 (Minimal primitive-time-root support in Headley, Clayton, and Reed 1996). The minimal supports of the three results are given in the following table. The total number of family members is four.

Table 337: Minimal primitive-time-root support in Headley, Clayton, and Reed 1996

Result	Minimal support-index family
1	$\{2\}$
2	$\{1, 4, 5, 6\}, \{2, 4, 5, 6\}$
3	$\{4\}$

Individual Statements Statement Evidence 1

Original text (English):

frequency difference between the pump and Stokes wave

The source location is Headley, Clayton, and Reed, *Methods of Suppressing Stimulated Brillouin Scattering in Optical Fibers by Manipulation of the Fiber Properties*, p. 105, Section II. The minimal context is the paragraph that explains the frequency shift of fiber SBS together with Equation (1). The statement is attributed to C. Headley, J. B. Clayton, and W. A. Reed.

Statement Evidence 2

Original text (English):

has a Lorentzian profile

The source location is p. 105, Section II of the same source. The minimal context is the paragraph that states the relation between acoustic decay and the gain spectrum. The statement is attributed to the same three authors.

Statement Evidence 3

Original text (English):

full width at half maximum

The source location is p. 105, Section II of the same source. The minimal context is the paragraph that states the relation between ordinary-frequency full width at half maximum and acoustic-wave lifetime. The statement is attributed to the same three authors.

Statement Evidence 4

Original text (English):

SBS gain is significantly reduced

The source location is p. 105, Section II of the same source. The minimal context is the paragraph that describes the operating regime determined by pulse width and acoustic-wave lifetime. The statement is attributed to the same three authors.

Common Evidence with Delimited Scope The following four statements give the original text of Headley, Clayton, and Reed 1996, its source location, minimal context, and source attribution.

Target state- ment	Statement object	Predicate	Explicit domain of application
S36- Q001	pump wave and backscattered Stokes wave	States that there is a frequency difference between the two waves.	The scope on p. 105, Section II, Equation (1), of pump–Stokes–acoustic coupling, refractive index, pump wavelength, and acoustic-wave velocity in fiber SBS.

S36-Q002	Brillouin gain spectrum corresponding to acoustic-wave decay	States that the gain spectrum has a Lorentzian profile.	The scope in the paragraph following Equation (1) on p. 105, Section II, of exponential decay, the CW/QCW conditions, and the operating regime determined by pulse width and acoustic-wave lifetime.
S36-Q003	ordinary-frequency width of the Brillouin gain spectrum	Expresses that width as the full width at half maximum.	The scope in the paragraph following Equation (1) on p. 105, Section II, of the relation among exponential decay, the Lorentzian line shape, and acoustic-wave lifetime.
S36-Q004	SBS gain in an operating regime determined by pulse width and acoustic-wave lifetime	States that SBS gain is significantly reduced in that operating regime.	The operating regimes on p. 105, Section II, distinguished by pulse width and acoustic-wave lifetime.

Individual Audit Frequency difference between the pump wave and Stokes wave

The source identifies ν_B as the frequency difference between the pump and Stokes waves and gives $\nu_B = 2nv_A/\lambda_p$ in Equation (1). This is a relation within the model of pump–Stokes–acoustic coupling.

Equation (1) takes the refractive index, pump wavelength, and acoustic-wave velocity as inputs. A relation between time-bearing velocity and frequency does not establish the physical generation of their time metric. The corresponding formal evidence is claims 1–7, transfer 1, and result 1.

Lorentzian profile

The source states that the Brillouin gain spectrum has a Lorentzian profile under exponential acoustic-wave decay and either CW or QCW operation. The QCW condition is $T_0 \gg T_B$. Result 2 combines the CW and QCW statements with their respective applicability conditions. Time prerequisite 5 represents the pulse width in the QCW condition; it is not required for the CW-only specialization. The gain-spectrum variable ν and its peak location ν_B are distinct: $\nu = \nu_B$ holds at the peak.

The basis of this conditional statement is the gain-spectrum model presented with a reference to work [1] in the source. Gain reduction for short pulses, $T_0 < T_B$, is a phenomenon in an operating regime different from CW/QCW. The corresponding formal evidence is claims 8–11 and 14, and result 2.

Full width at half maximum

$\Delta\nu_B$ is the ordinary-frequency full width at half maximum of the Brillouin gain spectrum and satisfies $\Delta\nu_B = (\pi T_B)^{-1}$ in the source’s exponential-decay and Lorentzian model. The width definition and the lifetime relation describe the same quantity together.

Conversion to angular-frequency width gives $\Delta\omega_B = 2\pi\Delta\nu_B$. The width–lifetime relation holds under the source’s exponential-decay convention and Lorentzian model. The

corresponding formal evidence is claims 10–13, transfer 2, and result 3.

Reduction of SBS gain

The source reports the finding of earlier work [3] that, for short pulses $T_0 < T_B$, SBS gain is significantly reduced and other nonlinear effects become important.

“Significantly” does not set a numerical threshold for the gain reduction in this paragraph. The corresponding formal evidence is claims 14 and 15, which contrast this operating regime with the CW/QCW conditions of result 2.

Common Audit with Delimited Scope Within-source correspondence among the four statements

The four statements concern, respectively, frequency difference, line-shape family, width measure, and regime-dependent gain change. Proposition 8.451 determines the generative reachability and result dependence of the time contents corresponding to each statement.

8.19.37 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 37.

Source Name Primitive-time-root audit of Maldacena 1998

Bibliographic Identification Let Juan M. Maldacena, *The Large N Limit of Superconformal Field Theories and Supergravity*, *Advances in Theoretical and Mathematical Physics*, Vol. 2 (1998), pp. 231–252, arXiv:hep-th/9711200v3 be $R = \text{Ref}_{\text{Mal98LN}}$. The audit domain comprises the abstract, Sections 1–6, appendix, equations, footnotes, figures, and references of the source.

Audit-Scope Identifier 37-SC1.

Audit-Scope Content The audit domain specified under Bibliographic Identification.

Source-Location Identifier 37-L1.

Source-Location Content The audit domain specified under Bibliographic Identification and the source- location columns of the respective evidence tables.

Formal Object The claim table gives each relation and its conditions of use in the source through the content, time-use and source-location fields.

Assumption

Assumption Identifier 37-A1.

Assumption Content The individual assumptions in the source are given in Definitions 8.453 and 8.455.

Source-Location Identifier Set 37-L1.

Nontemporal Physical Input

Input Identifier 37-P1.

Input Name Nontemporal physical inputs in the source.

Input Type Nontemporal physical input.

Input Content The individual nontemporal physical inputs are given in Definition 8.455.

Source-Location Identifier Set 37-L1.

Time Content

Time-Content Identifier 37-T1.

Time-Content Name Time contents in the source.

Time-Content Type Time content.

Time-Content Statement The individual time contents are given in Definition 8.454.

Source-Location Identifier Set 37-L1.

Formal Input Elements

Definition 8.453 (Maldacena 1998 Formal Objects and Conditions). Maldacena relates low-energy limits of brane field theories to near-horizon limits of supergravity and conjectures dualities with full string theory or M theory. The D3, M5, M2, D1/D5, five-dimensional black-string and Reissner–Nordström branches have their respective scaling limits, charges, couplings and moduli conditions. The restricted D3 probe action, agreement of symmetry groups, cited absorption calculations and duality conjectures have distinct grounds.

In the D3 branch, the limit at fixed worldvolume energy and the near-horizon metric use the following relations:

$$\alpha' \rightarrow 0, \quad U = r/\alpha' \text{ fixed}, \quad \lambda = 4\pi gN, \quad ds^2 = \alpha' \left[\frac{U^2}{\sqrt{\lambda}} dx_{\parallel}^2 + \sqrt{\lambda} \left(\frac{dU^2}{U^2} + d\Omega_5^2 \right) \right].$$

For the near-extremal D3 solution, proper time and proper energy of an observer static at fixed U are related outside the horizon by:

$$f(U) = 1 - \frac{U_0^4}{U^4}, \quad d\tau = \sqrt{\alpha'} \frac{U}{\lambda^{1/4}} \sqrt{f} dt, \quad E_{\text{proper}} = \frac{\omega \lambda^{1/4}}{\sqrt{\alpha'} U \sqrt{f}}, \quad U > U_0.$$

The proper-energy expression on source p. 4 has the $f = 1$ form. At fixed U, ω , $\sqrt{\alpha'} E_{\text{proper}}$ remains finite. The near-extremal temperature and compactification condition in Equation (2.12) give

$$T = \frac{U_0}{\pi \sqrt{\lambda}}, \quad U_0 \gg \frac{(gN)^{1/4}}{R_i} \implies T \gg \frac{1}{2\pi^{3/2} (gN)^{1/4} R_i}.$$

This inequality is the scale condition that each compact circle be much larger than the string scale. For M5 and M2 the radius relations are $R_{\text{AdS}} = 2R_{\text{sph}}$ and $R_{\text{sph}} = 2R_{\text{AdS}}$, respectively. The coupling in the D2 limit leading to M2 satisfies

$$g_{\text{YM}}^2 \sim \frac{R_{11}^2}{l_p^3} = \frac{g}{\sqrt{\alpha'}}, \quad R_{11} = g\sqrt{\alpha'}, \quad l_p^3 = g\alpha'^{3/2}.$$

The printed g/α' on source p. 10 is inconsistent with $R_{11}/l_p^{3/2} \sim g_{\text{YM}}$ on the same page and with the coupling dimension. The D1/D5 dilute-gas condition is $r_0, r_n \ll r_1, r_5$. Source p. 13 prints $r_1 r_5$ on the right; Equation (2.7) of the Maldacena–Strominger paper included in source Reference 54 compares with the two lengths separately. The limit at finite temperature and finite momentum chemical potential gives this dilute-gas domain. The Lorentzian time interpretation of the Reissner–Nordström near-horizon geometry uses a negative temporal component. In source Equation (5.4), the dt^2 , radial and sphere components are all positive, so the metric as printed is positive definite.

Definition 8.454 (Maldacena 1998 Time Contents). Let $n_R = 26$. The table gives each time content and source location. Page numbers are the printed page numbers of the source.

Table 339: Maldacena 1998 Time Contents and Source Locations

i	Time Content	Source Location
1	Time evolution in the boundary and bulk theories	pp. 1, 4, 16, 17
2	Time coordinate on the D3 worldvolume	pp. 2, 3, 5, 6
3	Worldvolume energy held fixed in the D3 limit	pp. 2, 3, 4
4	Proper time of a local observer	pp. 1, 4
5	Local proper energy and wavelength in string units	pp. 4
6	Temperature of the near-extremal D3 black hole	pp. 3, 4
7	Propagation and energy flux of Hawking radiation	pp. 1, 4, 5
8	Horizon and boundary conditions for evolution in the D3 description	pp. 1, 4, 5
9	Time derivatives, velocity and action of the D3 probe	pp. 5, 6, 7
10	Gauge-theory energy of an open-string excitation	pp. 7
11	Temperature domain for D3 compactification	pp. 8
12	Dilaton and graviton absorption processes	pp. 8
13	Finite-temperature entropy comparison for D3	pp. 8
14	M5 worldvolume time and held-fixed energy	pp. 9
15	M2 worldvolume time and low-energy domain	pp. 10
16	Velocity-dependent action of the M2 probe	pp. 10
17	Temperature and entropy of near-extremal M2 states	pp. 10
18	D1/D5 worldvolume time and Lorentzian metric	pp. 11, 12
19	Finite-temperature and near-extremal D1/D5 states	pp. 12, 13
20	Radiation frequency and greybody propagation for D1/D5	pp. 13
21	Worldvolume time of the five-dimensional black string	pp. 13, 14
22	Near-horizon temporal component for Reissner–Nordström	pp. 15
23	Time of the quantum mechanics associated with the AdS_2 candidate	pp. 15
24	Velocity and quadratic action of the AdS_2 probe	pp. 15
25	Nonextremal black-hole states in pure AdS_3	pp. 16, 17

i	Time Content	Source Location
26	AdS coordinate time and causal structure of the Poincare region	pp. 17, 18

Definition 8.455 (Maldacena 1998 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(45)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([4], [45])$. Claims 1–4 form the generation-initial set; all 45 claims form the source formal-evidence environment. The claim table gives the content, time uses and source location.

Table 340: Maldacena 1998 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Specification of brane species, charges, gauge groups, couplings, string and Planck scales, and compact spaces.	$\emptyset$	pp. 2, 3, 9, 10, 11, 13, 14, 15
2	Spatial configurations, radial coordinates, harmonic functions, and compact and transverse geometry.	$\emptyset$	pp. 3, 7, 8, 9, 10, 11, 14
3	Hilbert spaces, nontemporal field, operator and state data, and the symmetry groups being compared.	$\emptyset$	pp. 1, 4, 5, 9, 10, 12
4	Nontemporal boundary, state and absorption-channel data, and the algebraic metric of the embedding space.	$\emptyset$	pp. 4, 8, 13, 17, 18
5	Maldacena argues for inclusion of the supergravity-state sector using finite Hawking flux and gauge-theory unitarity.	$1^S, 6^Q, 7^{SQ}$	pp. 1, 4
6	The D3 limit takes $\alpha' \rightarrow 0$ at fixed $U = r/\alpha'$, g and worldvolume energy, separating the brane theory from the bulk.	$2^{SB}, 3^Q$	pp. 2, 3
7	Dropping the constant in the D3 harmonic function gives the $\text{AdS}_5 \times S^5$ metric with the specified worldvolume time.	2^{SB}	pp. 3
8	In the D3 supergravity domain the curvature radius is large; $gN \gg 1$ is required for $g \leq 1$. Source Footnote 5 specifies $N/g \gg 1$ for $g > 1$.	$\emptyset$	pp. 3, 4
9	A local observer obtains proper energy by converting worldvolume energy with the local time scale; $\sqrt{\alpha'} E_{\text{proper}}$ and wavelength in string units remain finite in the limit.	$2^{SB}, 3^Q, 4^{IQ}, 5^Q$	pp. 4

p	Claim	i	Source Location
10	Local supergravity processes use a proper-energy window between cosmological and Planck scales, allowing local questions insensitive to boundary specifications at large N .	$4^{\text{IQ}}, 5^{\text{Q}}$	pp. 1
11	A finite-temperature D3 state corresponds to a near-extremal black hole whose Hawking radiation carries energy into AdS.	$2^{\text{SB}}, 6^{\text{Q}}, 7^{\text{SQ}}$	pp. 3, 4
12	Maldacena conjectures the duality of full type-IIB string theory and the D3 gauge theory, leaving the AdS boundary conditions to be selected.	$1^{\text{S}}, 2^{\text{SB}}, 8^{\text{BC}}$	pp. 1, 4, 5
13	The source identifies agreement between the D3 superconformal group and the supersymmetry group on $\text{AdS}_5 \times S^5$.	$\emptyset$	pp. 2, 4
14	The source associates gauge-side $1/N$ effects in the D3 branch with bulk quantum effects, including Hawking radiation.	$1^{\text{S}}, 7^{\text{SQ}}$	pp. 5
15	The low-energy D3 probe action takes Born–Infeld form under $\partial U/U \ll U$ and $\partial\theta \ll U$.	$2^{\text{SB}}, 9^{\text{SQ}}$	pp. 5, 6
16	For constant theta and $F = 0$, the conformal transformation in Equation (2.8), the no-force condition, quadratic nonrenormalization and the one-loop quartic term determine the action.	$2^{\text{SB}}, 9^{\text{SQ}}$	pp. 6
17	Determining the full probe action from 32 supersymmetries and testing the resulting spectral predictions remain open tasks in the source.	$\emptyset$	pp. 5, 6, 7
18	The source redshifts a local open-string excitation to gauge-theory energy $E = U(4\pi gN)^{-1/4}\sqrt{n}$, associating large and small U with UV and IR scales.	$2^{\text{SB}}, 4^{\text{IQ}}, 5^{\text{Q}}, 10^{\text{Q}}$	pp. 7
19	For compactified D3, the condition $U_0 \gg (gN)^{1/4}/R_i$ keeps the physical circle at the horizon larger than the string scale and sets a lower temperature scale.	$2^{\text{SB}}, 6^{\text{Q}}, 11^{\text{BQ}}$	pp. 8
20	The cited D3 dilaton and graviton absorption calculations agree with gauge-theory results; nonrenormalization accounts for some of the agreements.	12^{SQ}	pp. 8
21	D3 black-hole and perturbative gauge-theory entropies differ by a numerical factor, which the source expects large- gN effects to change.	$6^{\text{Q}}, 13^{\text{Q}}$	pp. 8

p	Claim	i	Source Location
22	The M5 limit holds $U^2 = r/l_p^3$ and worldvolume energy fixed as $l_p \rightarrow 0$, giving $\text{AdS}_7 \times S^4$ with $R_{\text{AdS}} = 2R_{\text{sph}}$.	14 ^{SQ}	pp. 9
23	Maldacena conjectures a duality between the large-N six-dimensional (0,2) CFT and full M theory.	14 ^{SQ}	pp. 9
24	The M2 limit holds $U^{1/2} = r/l_p^{3/2}$ fixed as $l_p \rightarrow 0$, giving $\text{AdS}_4 \times S^7$ with $R_{\text{sph}} = 2R_{\text{AdS}}$. Maldacena proposes a duality between this M2 conformal field theory and M-theory on $\text{AdS}_4 \times S^7$.	15 ^{SQ}	pp. 10
25	The D2 route to M2 uses a field-theory limit followed by a strong-coupling limit. The dimension of g_{YM}^2 , proportional to R_{11}^2/l_p^3 , is inverse length.	15 ^{SQ}	pp. 10
26	The temperature dependence of near-extremal M2 entropy agrees with conformal dimensional analysis; its N dependence and numerical coefficient remain unresolved in the source.	15 ^{SQ} , 17 ^Q	pp. 10
27	Starting from the known M2 probe v^4 term, Maldacena proposes that conformal invariance fixes all velocity orders and the duality fixes the coefficient.	15 ^{SQ} , 16 ^{SQ}	pp. 10
28	The D1/D5 limit holds U , v and g_6 fixed and gives the $\text{AdS}_3 \times S^3 \times M^4$ metric, including its temporal component, with tensor moduli fixed by charges.	18 ^{SB}	pp. 11, 12
29	Maldacena's D1/D5 duality conjecture concerns the Higgs branch and requires tensor-moduli independence and decoupling from the Coulomb branch.	18 ^{SB}	pp. 12
30	The near-horizon metric of a finite-temperature D1/D5 state has local BTZ form; compactification of the spatial direction supplies periodic angular identification.	18 ^{SB} , 19 ^{QS}	pp. 12, 13
31	The cited D1/D5 greybody calculations agree under the dilute-gas conditions, with the AdS region supplying the nontrivial factor.	18 ^{SB} , 19 ^{QS} , 20 ^{QS}	pp. 13
32	The five-dimensional black string uses $D = D_{ABC}p^A p^B p^C$ and $R_{\text{AdS}} = 2R_{\text{sph}}$; vector-moduli independence remains an open condition of the duality conjecture.	21 ^{SB}	pp. 13, 14

p	Claim	i	Source Location
33	The Reissner–Nordström near-horizon geometry is discussed as $\text{AdS}_2 \times S^2$, whose two factors have equal radii.	22^{BC}	pp. 15
34	In the AdS_2 candidate limit the energy gap becomes very large. Maldacena leaves the relation between the resulting quantum mechanics and the expected chiral CFT unresolved and suggests that a simultaneous large- N limit will be crucial.	$22^{\text{BC}}, 23^{\text{S}}$	pp. 15
35	The leading AdS_2 probe action is proportional to $\int dt \dot{U}^2/U^3$; the source leaves its coefficient to be tested.	$23^{\text{S}}, 24^{\text{SQ}}$	pp. 15
36	Maldacena proposes a nonperturbative definition of M/string theory on selected backgrounds through large- N field theory, with a prospective extension to flat spacetime.	1^{S}	pp. 16
37	The source leaves the AdS boundary conditions and the physical interpretation of the horizon and its interior to be determined.	$8^{\text{BC}}, 26^{\text{BC}}$	pp. 17
38	Maldacena claims that all AdS dynamics reduces to known conformal field theories.	1^{S}	pp. 17
39	The appendix constructs the induced AdS metric from a hyperboloid in a flat space with signature (minus, minus, plus, ..., plus) and a coordinate transformation.	$\emptyset$	pp. 17, 18
40	In the source’s physical interpretation x^0 is a time coordinate, and $U > 0$ represents the exterior Poincare region.	26^{BC}	pp. 18
41	The boundary and bulk theories each use time evolution; the symmetry, limiting procedure and boundary conditions of each branch determine the correspondence’s domain.	$1^{\text{S}}, 2^{\text{SB}}, 14^{\text{SQ}}, 15^{\text{SQ}}, 18^{\text{SB}}, 21^{\text{SB}}, 22^{\text{BC}}, 23^{\text{S}}, 26^{\text{BC}}$	pp. 1, 3, 9, 10, 11, 14, 15, 18
42	The multicentre D3 solution separates $U(N - M)$ and $U(M)$ regions, with limitations on the supergravity approximation at small N .	2^{SB}	pp. 7, 8
43	Nonextremal black holes in pure AdS_3 are free of the low-energy restriction that arose from matching to an asymptotically flat region.	25^{SQ}	pp. 16, 17
44	The source leaves the construction of asymptotic graviton states open and relates N to curvature and spatial size, unlike its momentum interpretation in matrix theory.	$\emptyset$	pp. 16

p	Claim	i	Source Location
45	The proper-time scale of a D3 observer static at a specified position is determined by worldvolume time and the temporal metric component.	$2^{\text{SB}}, 4^{\text{IQ}}$	pp. 3, 4

Table 341: Maldacena 1998 Time-Use Roles

Code	Role	Use Count
I	Interval	4
Q	Quantity	37
X	Cross-position correspondence	0
K	Clock reading	0
C	Causality	8
S	State evolution	48
P	Probabilistic evolution	0
O	Observation record	0
B	Reference boundary	28

Table 342: Maldacena 1998 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 4, 23	<code>member</code>	<code>instant, order_relation</code>
2, 14–15, 18, 21–22, 26	<code>member</code>	<code>instant</code>
3, 5–13, 16–17, 19–20, 24–25	<code>dependency</code>	$\emptyset$

Dependency Inputs

Definition 8.456 (Maldacena 1998 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [4]$. The table gives time inputs, time outputs, claim support and source locations.

Table 343: Maldacena 1998 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	2	4	7, 45	Obtain a local static observer’s proper-time scale from the D3 metric. pp. 3, 4
2	2–4	5	9	Convert worldvolume energy to local proper energy. pp. 4
3	2, 4–5	10	18	Redshift a local open-string excitation to gauge-theory energy. pp. 7

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
4	2, 6	11	19	Transfer the compactification validity condition to D3 temperature. pp. 8

Grounding Conditions

1. A static observer at a specified position and a Lorentzian metric; for a nonextremal state, $U > U_0$ and the blackening factor.
2. The specified scale ratio of t and τ , with evaluation in string units.
3. D3 probe position, specified open-string excitation level and redshift factor.
4. Each circle radius R_i , D3 time normalization and the lower-bound condition of Equation (2.12) at the horizon U_0 .

Results

Definition 8.457 (Maldacena 1998 Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [37]$. The table gives each result's temporal premises and claim support.

Table 344: Maldacena 1998 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Low-energy D3 decoupling. pp. 2, 3	2–3	6
2	Near-horizon $\text{AdS}_5 \times S^5$ metric for D3. pp. 3	2	7
3	Coupling and curvature domain for D3. pp. 3, 4	$\emptyset$	8
4	Conversion to local proper energy for D3. pp. 4	2–5	9, 45
5	Energy window for local supergravity processes at large N . pp. 1	4–5	10
6	Argument for inclusion of the D3 supergravity Hilbert sector. pp. 1, 4	1–2, 6–7	5, 11
7	D3 full-duality conjecture and boundary conditions. pp. 1, 4, 5	1–2, 8	12
8	Agreement of D3 superconformal-symmetry groups. pp. 2, 4	$\emptyset$	13
9	D3 correspondence of $1/N$ quantum effects. pp. 5	1, 7	14
10	Restricted D3 probe action. pp. 5, 6	2, 9	15–16
11	Open tasks for the full probe action and spectrum. pp. 5, 6, 7	$\emptyset$	17

q	Result and Source Location	Time Premises i	Claim Support p
12	Open-string redshift and radius–energy-scale correspondence. pp. 7	2, 4–5, 10	18
13	Lower-temperature condition for D3 compactification. pp. 8	2, 6, 11	19
14	Cited agreement of D3 absorption calculations. pp. 8	12	20
15	Numerical-factor difference in D3 entropy. pp. 8	6, 13	21
16	M5 near-horizon metric and radii. pp. 9	14	22
17	M5 full-M-theory duality conjecture. pp. 9	14	23
18	M2 near-horizon metric, proposed duality and D2 limiting route. pp. 10	15	24–25
19	Temperature dependence of M2 entropy. pp. 10	15, 17	26
20	Prediction for the M2 probe-velocity action. pp. 10	15–16	27
21	D1/D5 near-horizon metric and moduli. pp. 11, 12	18	28
22	Duality conjecture for the D1/D5 Higgs branch. pp. 12	18	29
23	D1/D5 finite-temperature state and BTZ identification. pp. 12, 13	18–19	30
24	D1/D5 greybody correspondence. pp. 13	18–20	31
25	Candidate correspondence for the five-dimensional black string. pp. 13, 14	21	32
26	AdS ₂ candidate for the Reissner–Nordström near-horizon geometry. pp. 15	22	33
27	Unresolved relation of AdS ₂ quantum mechanics and chiral CFT. pp. 15	22–23	34
28	Leading velocity action of the AdS ₂ probe. pp. 15	23–24	35
29	Candidate nonperturbative definition on selected backgrounds and extension to flat spacetime. pp. 16	1	36
30	Open tasks concerning boundary conditions and horizon interpretation. pp. 17, 18	8, 26	37
31	Claim concerning correspondence of all AdS dynamics with the CFT. pp. 17	1	38
32	Coordinate transformation and induced metric on the AdS hyperboloid. pp. 17, 18	$\emptyset$	39
33	Physical interpretation of AdS coordinate time and the Poincare region. pp. 18	26	40

q	Result and Source Location	Time Premises i	Claim Support p
34	Boundary and bulk time use by branch. pp. 1, 3, 9, 10, 11, 14, 15, 18	1–2, 14–15, 18, 21–23, 26	41
35	Branch separation in the multicentre D3 solution. pp. 7, 8	2	42
36	Energy domain of nonextremal states in pure AdS ₃ . pp. 16, 17	25	43
37	Status of asymptotic-state construction and comparison with matrix theory. pp. 16	$\emptyset$	44

Definition 8.458 (Maldacena 1998 Grounds and Domains of Results). The table gives conditions and domains of the results.

Table 345: Maldacena 1998 Conditions for Results

Result Index	Conditions and Domain
1	$\alpha' \rightarrow 0$ at fixed U, g and worldvolume energy.
2	D3 harmonic function and specified Lorentzian worldvolume metric.
3	$gN \gg 1$ for $g \leq 1$; $N/g \gg 1$ for $g > 1$, as in Footnote 5.
4	Specified static observer, metric, energy and string units.
5	Proper energies between cosmological and Planck scales.
6	Finite temperature, Hawking flux and gauge-theory unitarity.
7	Full IIB theory, gauge theory and unspecified AdS boundary conditions.
8	Algebraic identification of the two groups compared in the source.
9	Bulk quantum effects, including Hawking radiation, and the gauge-side expansion.
10	Leading derivative order, constant theta, $F = 0$, no force, and quadratic and quartic input.
11	Action determination and prediction tests left unverified by Maldacena.
12	Probe position, excitation level and specified time scales.
13	Domain in which every physical circle at the horizon is much larger than the string scale.
14	Specified dilaton and graviton channels and applicability of nonrenormalization.
15	Comparison of black-hole and perturbative gauge calculations, with unresolved large-gN corrections.
16	Large-N limit holding $U^2 = r/l_p^3$ and worldvolume energy fixed.
17	Six-dimensional (0,2) CFT and the AdS ₇ $\times$ S ⁴ background.
18	Fixed $U^{1/2}$, the dimensional relation of M-circle radius and coupling, and the strong-coupling limit; supergravity uses $N \gg 1$. Maldacena uses the M2 duality as a proposal.
19	Agreement with conformal dimensional analysis, with unresolved N dependence and coefficient.
20	Known v^4 term, conformal invariance and the proposed duality.

Result Index	Conditions and Domain
21	Fixed U , v and g_6 , with $g_6 Q_1, g_6 Q_5 \gg 1$ and a fixed charge ratio. Charges fix the tensor components; hypermultiplet components are retained.
22	Tensor-moduli independence and decoupling from the Coulomb branch.
23	Finite temperature and momentum chemical potential, with periodic identification from spatial compactification.
24	Dilute-gas domain $r_0, r_n \ll r_1, r_5$ and conditions of the cited scattering calculations.
25	Cubic charge form D and the moduli, with unresolved independence from vector components.
26	Equal-radius factors and a Lorentzian time interpretation.
27	Decoupling, energy gap and simultaneous large- N limit.
28	Leading derivative order and near-horizon domain, with an untested coefficient.
29	Proposal based on the duality advanced by Maldacena.
30	AdS theory and Poincare region treated by the source.
31	Maldacena's claim of reduction to known CFTs.
32	Hyperboloid, embedding metric and coordinate domain in Equations(7.1)–(7.3).
33	Time coordinate x^0 and the exterior region $U > 0$.
34	Time evolution, symmetries, scaling limit and boundary conditions of each branch.
35	$U(N - M)$ and $U(M)$ regions and the supergravity domain of validity.
36	AdS ₃ description without matching conditions to an asymptotically flat region.
37	Unconstructed graviton states and N as curvature and spatial size.

Result Dependence

Definition 8.459 (Maldacena 1998 Result-Dependence Hyperedges). For each row of the result table, let $\text{ResultEdge}_{R,q}$ ($q \in [37]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Evidence Hierarchy Maldacena develops theoretical correspondences for the brane limits and cites earlier calculations of absorption, greybody factors and entropy. The source reports zero directly measured objects and zero direct-measurement predicates from measurements of its own.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local-reach count are both zero, leaving 26 primitive-time roots. Transfers of proper time, proper energy, open-string energy and temperature domain take worldvolume time, energy or temperature as inputs.

Result Dependence Of the 37 results, 32 depend on primitive-time roots. Results reachable from the source formal-evidence environment without additional time roots are 3, 8, 11, 32, 37, comprising 5 results concerning coupling and curvature conditions,

mathematical relations of symmetry groups and the induced metric, and records of open tasks. There are 64 root–result dependence pairs, of which 58 are necessary.

Placement of the Grounding Obligation Maldacena’s branches use worldvolume time coordinates, local proper time, energy, temperature, velocity or causal structure as inputs to the correspondence. Time-scale conversions give another representation from those temporal inputs. All 26 time contents remain primitive-time roots under both the complete and local-reach judgments. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Curvature, coupling, low-energy domain, probe derivative order, moduli and boundary conditions specify the domains of the results. Open tasks in the source concern entropy coefficients, boundary specification, asymptotic-state construction and the AdS2 correspondence. Actual failure of a specific audited physical-time generation condition and the resulting change in result values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 64 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are each temporal use and the physical-time generation conditions required before that use. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Proposition 8.460 (Primitive-Time Roots and Result Dependence in Maldacena 1998). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = & (\emptyset, [26]; \\ & \emptyset, \{3, 8, 11, 32, 37\}, \\ & [37] \setminus \{3, 8, 11, 32, 37\}, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Even after adding all 45 claims as source formal evidence, the 4 transfers each require temporal inputs and output no time content. The five results with empty temporal premises are reached from the source formal evidence. Adding all 26 time contents as seeds reaches all 37 results; the common reachability classification gives the displayed six components. $\square$

Proposition 8.461 (Minimal Primitive-Time-Root Support in Maldacena 1998). The table gives each result’s minimal-support index family, with 40 family members in total.

Proof. Start from singleton time-content seeds and apply Definition 7.46 to the four transfers. Combine supports of each result’s premises from the stable antichains and retain the inclusion-minimal supports. Proposition 7.47 gives the displayed families and counts. $\square$

Table 346: Minimal Primitive-Time-Root Support in Maldacena 1998

Result Indices	Minimal-Support Family Members
1, 4	$\{2, 3\}$
2, 35	$\{2\}$
3, 8, 11, 32, 37	$\emptyset$
5	$\{2, 3\}, \{2, 5\}, \{4, 5\}$
6	$\{1, 2, 6, 7\}$
7	$\{1, 2, 8\}$
9	$\{1, 7\}$
10	$\{2, 9\}$
12	$\{2, 3\}, \{2, 5\}$
13	$\{2, 6\}$
14	$\{12\}$
15	$\{6, 13\}$
16–17	$\{14\}$
18	$\{15\}$
19	$\{15, 17\}$
20	$\{15, 16\}$
21–22	$\{18\}$
23	$\{18, 19\}$
24	$\{18, 19, 20\}$
25	$\{21\}$
26	$\{22\}$
27	$\{22, 23\}$
28	$\{23, 24\}$
29, 31	$\{1\}$
30	$\{8, 26\}$
33	$\{26\}$
34	$\{1, 2, 14, 15, 18, 21, 22, 23, 26\}$
36	$\{25\}$

Evidentiary Effect

Definition 8.462 (Evidentiary Domain of Maldacena 1998). Low-energy limits and near-horizon metrics in each branch relate the specified inputs. Symmetry groups and the hyperboloid metric supply mathematical relations; the restricted D3 probe action is derived at the stated order and under stated conditions; absorption and greybody factors come from cited calculations; full dualities and nonperturbative definitions have the status of Maldacena’s conjectures and proposals. The open tasks concerning boundary conditions, moduli dependence, entropy coefficients and AdS2 specify conditions for the respective results.

Result-Preservation Domain

Definition 8.463 (Result-Preservation Conditions for Maldacena 1998). Fix the brane species, charges, couplings, moduli, scaling limits and conditions in Table 345. When

inputs of results using physical time can be realized by generated temporal representations with the same order, intervals, evolution and scale conversions, each result is preserved with its stated domain and epistemic status.

Proposition 8.464 (Audit-Boundary Counts for Maldacena 1998). The 45 claims, 26 time contents, 4 transfers and 37 results give 112 source-specific obligation atoms. The 73 claim/time-content use pairs have 125 role assignments.

$$\text{BoundaryCount}_R^{\text{audit}} = (108, 41).$$

Proof. Count claims, time contents and results as nodes, transfers and result dependencies as edges, and sum the roles of time uses in the claim table. $\square$

Individual Statements Conjecture of Duality

Original text:

We conjecture that compactifications of M/string theory on various Anti-deSitter spacetimes is dual to various conformal field theories.

Source location: Abstract on the unnumbered opening page. Following the argument concerning the supergravity Hilbert sector, Maldacena conjectures a duality of full M/string theory and CFT.

Need for AdS Boundary Conditions

Original text:

A theory in AdS spacetime is not completely well defined since there is a horizon and it is also necessary to give some boundary conditions at infinity.

Source location: Section 1, p. 1. Maldacena identifies the horizon and asymptotic boundary in the AdS description used and requires boundary conditions to define the theory.

Duality Conjecture with Unspecified Boundary Conditions

Original text:

We will conjecture that the full quantum M/string-theory on AdS space, plus suitable boundary conditions is dual to the corresponding brane theory. We are not going to specify the boundary conditions in AdS, we leave this interesting problem for the future.

Source location: Section 1, p. 1. The correspondence of full quantum M/string theory with the brane theory requires suitable boundary conditions, whose specification Maldacena leaves to future work.

Local Energy Evaluated with Respect to Proper Time

Original text:

proper energies (measured with respect to proper time)

Source location: Section 2, p. 4, immediately after Equation (2.5). The passage relates fixed worldvolume energy ω to redshifted proper energy, with $\sqrt{\alpha'} E_{\text{proper}}$ finite in the limit.

Remaining Construction Tasks

Original text:

We are only scratching the surface and there are many things to be worked out.

Source location: Section 6, p. 16. Following the candidate nonperturbative definition, Maldacena discusses construction of asymptotic graviton states and the relation to matrix theory.

Common Evidence with Specified Scope The table gives the objects, predicates and domains of the five statements.

State- ment	Object	Predicate	Explicit Domain
S37- Q001	M/string theory on AdS and conformal field theory	Maldacena conjectures their duality.	Abstract and Section 1, with the large-N and limiting conditions of each branch.
S37- Q002	Definition of the theory in the AdS description used by the source	Maldacena states the need for additional boundary conditions.	Section 1, p. 1, and the coordinate region in the appendix, pp. 17–18.
S37- Q003	Full quantum theory with boundary conditions and the brane theory	Maldacena conjectures the duality and leaves the boundary conditions unspecified.	Section 1, p. 1; the conjecture including suitable boundary conditions.
S37- Q004	A local observer’s proper energy	Maldacena evaluates proper energy with respect to proper time.	Section 2, p. 4; D3 near-horizon geometry and local energy in string units.
S37- Q005	Status of Maldacena’s duality proposal and remaining tasks	Maldacena states that many construction tasks remain.	Section 6, p. 16; nonperturbative definition and comparison with matrix theory.

Individual Audit Conjecture of Duality

The first statement is Maldacena’s conjecture concerning full M/string theory and CFT. Its grounds include low-energy limits, near-horizon geometry, supergravity-sector inclusion and agreement of symmetry groups. The full duality has the status of the source’s conjecture, with branch-specific charges, couplings, moduli and boundary conditions defining its domain. Corresponding Claims are 12, 23, 29 and 36; Results are 7, 17, 22 and 29.

Boundary Conditions of the AdS Theory

The second statement concerns conditions for defining a theory in the AdS description used by the source. The horizon is associated with the appendix’s Poincare region $U>0$; asymptotic boundary conditions are needed to specify evolution. Time Contents 8 and 26 and Claims 12, 37 and 40 provide this context, corresponding to Results 7, 30 and 33.

Conjecture with Unspecified Conditions

The third statement concerns duality between the full theory with suitable boundary conditions and the brane theory. Maldacena leaves specification of those conditions to future work. Claim 12 and Result 7 give the conjecture’s conditions; Claim 37 and Result 30 represent the remaining boundary problem.

Local Proper Energy

The fourth statement shows Maldacena’s use of local energy conjugate to proper time. Worldvolume time, fixed energy, proper time and proper energy are Time Contents 2–5.

Claims 9 and 45 and Transfers 1–2 express the scale conversion leading to Result 4. The finite quantity is $\sqrt{\alpha'} E_{\text{proper}}$ in string units; the specified metric and existing time scales are inputs to that conversion.

Status of the Proposal

The fifth statement is Maldacena’s assessment of the proposal presented in that paper. Its concrete context comprises the candidate nonperturbative definition, extension to flat spacetime, construction of asymptotic graviton states and comparison with matrix theory. Claims 36 and 44 and Results 29 and 37 record the proposal and its open tasks.

8.19.38 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 38.

Source Name Primitive-time-root audit of Perlmutter et al. 1999

Bibliographic Identification Let $R = \text{Ref}_{\text{Per99OL}}$ denote the original source S. Perlmutter et al., *Measurements of Omega and Lambda from 42 High-Redshift Supernovae*, *The Astrophysical Journal* 517, 565–586 (1999). The audited domain comprises the high- and low-redshift Type Ia supernova samples, photometric and spectroscopic observations, light-curve standardization, observer-frame–rest-frame time-width conversion, FRW likelihood, the acceleration region, cosmic age, and systematic-error tests.

Source-Location Identifier 38-L1.

Source-Location Content The time-content table gives 15 time contents and their source locations; the claim table gives 36 claims and their source locations.

Audit-Scope Identifier 38-SC1.

Audit-Scope Content The audited domain stated in the bibliographic identification.

Formal Object

Object Identifier 38-F1.

Object Name Objects using time.

Object Type Formal object.

Formal Content The 15 time contents and their uses in each claim.

Source-Location Identifier Set 38-L1.

Assumption

Assumption Identifier 38-A1.

Assumption Content The individual source assumptions are given in Definitions 8.465 and 8.468.

Source-Location Identifier Set 38-L1.

Nontemporal Physical Input

Input Identifier 38-P1.

Input Name Nontemporal physical input.

Input Type Nontemporal physical input.

Input Content The nontemporal object, wavelength and detector data, algebraic parameter specifications and statistical rules of Claims 1–3.

Source-Location Identifier Set 38-L1.

Definition 8.465 (Perlmutter et al. 1999 Formal Objects and Conditions). Perlmutter et al. use time-indexed photometry and spectral evidence for Type Ia supernovae to treat phase, stretch, color and extinction, then fit corrected peak magnitudes and redshifts to cosmological models. Sample selection, calibration, covariance, errors, prior domain and nuisance marginalization condition the inference. Flat-conditional results, results including nonflat models, partially filled beams and the constant- w extension have their respective domains.

Definition 8.466 (Perlmutter et al. 1999 Formal Inputs). For corresponding widths of the same event, the observer-frame and rest-frame widths and the intrinsically stretched template phase satisfy the following relations at the specified redshift and epoch of maximum:

$$\Delta t_{\text{rest}} = \frac{\Delta t_{\text{obs}}}{1+z}, \quad \phi_{\text{template}} = \frac{t_{\text{obs}} - t_{\text{max,obs}}}{(1+z)s}, \quad s > 0.$$

Stretch and the epoch of maximum are obtained by fitting the template to time-indexed photometry. Corrected magnitude and the Hubble-constant-free distance and intercept follow source Equations (1)–(4):

$$m_B^{\text{eff}} = m_R + \alpha(s-1) - K_{BR} - A_R = \mathcal{M}_B + 5 \log_{10} \mathcal{D}_L, \\ \mathcal{M}_B = M_B - 5 \log_{10} H_0 + 25, \quad \mathcal{D}_L = H_0 d_L.$$

H_0 cancels when the same specified unit convention is used. Present acceleration in the matter–Lambda model is given by

$$q_0 = \frac{\Omega_M}{2} - \Omega_\Lambda < 0.$$

For an expanding big-bang branch with $H_0 > 0$, cosmic age is defined where the integrand is real and the integral finite:

$$H_0 t_0 = \int_0^1 \frac{da}{\sqrt{\Omega_M/a + \Omega_k + \Omega_\Lambda a^2}}, \quad \Omega_k = 1 - \Omega_M - \Omega_\Lambda.$$

The present scale factor is normalized to one. The distribution including nonflat models and the flat-conditional distribution in source Section 6.1 and Figure 9 are used separately.

Definition 8.467 (Perlmutter et al. 1999 Time Contents). Let $n_R = 15$. The table gives each time content and source location.

Table 348: Perlmutter et al. 1999 Time Contents and Source Locations

i	Time Content	Source Location
1	Observation-time sequences for photometry and spectroscopy	§2
2	Rest-frame phase relative to maximum light	§2
3	Time axis of the template light curve	§2
4	Intrinsic stretch factor and stretched template	§2, Eqs. (1)–(3)
5	Observer-frame and rest-frame temporal widths	§2
6	Rise and decline intervals around maximum light	§2, §4.4
7	FRW cosmic time and expansion history	§3, §6, Figs. 7, 9–10
8	Hubble rate and Hubble time	§2, §6.1, Fig. 9
9	Present expansion acceleration and acceleration–deceleration classification	§6.1, Fig. 7
10	Cosmic age and its ratio to Hubble time	§6.1, Fig. 9
11	Order and selection intervals of search, discovery, confirmation and follow-up	§2, §4.2
12	Evolutionary histories of low- and high-redshift populations and their hosts	§4.4, §6.2–6.3
13	Photon propagation from emission to observation	§3, §4.3, §6
14	Ages of the oldest stars	§6.1
15	Velocities and propagation rates of astronomical objects and emitting material	§2, §4.4, §6.1–6.2, Tables 1–2

Definition 8.468 (Perlmutter et al. 1999 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(36)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([3], [36])$. Claims 1–3 form the generation-initial set; all 36 claims form the source formal-evidence environment.

Table 349: Perlmutter et al. 1999 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Object identities, sky positions, host galaxies, passband wavelength characteristics and nontemporal detector data.	$\emptyset$	§2, Tables 1–2
2	Specification of algebraic density-parameter coordinates, spatial curvature and the flatness condition.	$\emptyset$	§3
3	Mathematical rules for numerical errors, covariance, likelihood, marginalization, priors and sample selection.	$\emptyset$	§3, Appendix

p	Claim	i	Source Location
4	Perlmutter et al. compare 42 high-redshift objects and 18 low-redshift objects from Calán/Tololo using photometric series with discovery epochs and follow-up coverage. The text gives an approximate high-redshift range of 0.18–0.83; Table 1 spans 0.172–0.830. Typical follow-up lasts two to three months, about 40–60 rest-frame days.	$1^{\text{KO}}, 11^{\text{OB}}$	§2, Tables 1–2
5	Perlmutter et al. assess Type Ia classification confidence from spectra and light curves and determine redshift from supernova or host-galaxy spectra. Spectral comparisons match light-curve phase. Redshifts in Table 1 are geocentric; those in Table 2 are in the Local Group rest frame.	$1^{\text{O}}, 2^{\text{IB}}, 15^{\text{Q}}$	§2, §4.4
6	Perlmutter et al. fit a stretched template to time-indexed photometry to estimate the epoch and magnitude of maximum light and the stretch factor.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}$	§2
7	Before fitting the observed light curve, the template width is dilated by $1 + z$. Corresponding widths of the same event satisfy $\Delta t_{\text{rest}} = \Delta t_{\text{obs}}/(1 + z)$.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 5^{\text{IX}}$	§2
8	The intrinsic stretch factor s scales the template time axis and has the same value in the B and V bands. The width–luminosity correction is $\Delta_{\text{corr}} = \alpha(s - 1)$.	$3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}$	§2, Eqs. (1)–(2)
9	K corrections use redshift, passbands and phase-dependent spectra. Galactic extinction is corrected, while host and intergalactic extinction are assessed separately.	$2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}$	§2, §4.1
10	The corrected peak magnitude is $m_B^{\text{eff}} = m_R + \alpha(s - 1) - K_{BR} - A_R$; Tables 1–2 report fitted peak magnitudes and their errors.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}$	§2, Eq. (3), Tables 1–2
11	Perlmutter et al. restrict the low-redshift light-curve fit to about 60 rest-frame days after maximum to match the high-redshift coverage.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 5^{\text{IX}}, 6^{\text{I}}, 11^{\text{OB}}$	§2
12	Fit A uses 60 objects, Fit B 56 and the primary Fit C 54, with exclusions based on residuals, stretch and color.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}, 11^{\text{OB}}$	§3–4.1, Table 3, Figs. 2–5
13	The color indicator $B_{\text{max}} - V_{\text{max}}$ uses the separate band maxima, about two days apart, with stretch, K-correction and color-excess conditions.	$2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}$	§4.1, Fig. 6

p	Claim	i	Source Location
14	Perlmutter et al. examine sample reddening differences, corrections using color priors and grey extinction, retaining uniform grey extinction as an alternative.	$2^{\text{IB}}, 4^{\text{Q}}, 12^{\text{SX}}$	§4.1, §4.4, Appendix
15	Malmquist selection depends on search limits, visibility duration and light-curve width and is tested in the low- and high-redshift samples.	$1^{\text{O}}, 4^{\text{Q}}, 5^{\text{IX}}, 11^{\text{OB}}$	§4.2
16	The principal FRW luminosity-distance analysis and partially filled-beam Fits K and L reflecting matter clumping use their respective propagation models.	$7^{\text{S}}, 8^{\text{Q}}, 13^{\text{CX}}$	§3, §4.3, Fig. 8
17	Perlmutter et al. compare phase-matched spectra, rise and decline intervals and host environments across low and high redshift to examine supernova, population and extinction evolution.	$2^{\text{IB}}, 4^{\text{Q}}, 6^{\text{I}}, 12^{\text{SX}}, 15^{\text{Q}}$	§4.4
18	Perlmutter et al. cite preliminary indications of a rise-time difference below about one day. If this bound holds, the modeled width–luminosity zero-point change is limited to below about 0.1 mag. The host-specific tests still have large uncertainties.	$4^{\text{Q}}, 6^{\text{I}}, 12^{\text{SX}}$	§4.4
19	The magnitude–redshift relation uses the Hubble-constant-free intercept $\mathcal{M}_B$ and distance $\mathcal{D}_L = H_0 d_L$; Ω_M , Ω_Λ , $\mathcal{M}_B$ and alpha are fitted jointly and nuisance parameters are marginalized.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}, 7^{\text{S}}, 8^{\text{Q}}, 13^{\text{CX}}$	§3, Eq. (4)
20	Perlmutter et al. use photometric errors, intrinsic dispersion, shared calibration and peculiar-velocity redshift errors as conditions of the sample and likelihood.	$1^{\text{O}}, 4^{\text{Q}}, 15^{\text{Q}}$	§2–3, Tables 1–2
21	The principal Fit C gives $\Omega_M = 0.28$ under flatness, with statistical $+0.09/-0.08$ and systematic $+0.05/-0.04$ errors; the main region is summarized by $0.8\Omega_M - 0.6\Omega_\Lambda \simeq -0.2 \pm 0.1$. The linear summary approximates the principal region with Ω_M below about 1.5.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}, 7^{\text{S}}, 8^{\text{Q}}, 13^{\text{CX}}, 15^{\text{Q}}$	§5, Table 3, Fig. 7
22	Under the specified likelihood and $\Omega_M \geq 0$ prior domain, the probability of a positive cosmological constant is 0.9979 for Fit C and 0.9912 for Fit J with the adverse identified-systematic shift. The source finds poor fits for both flat and open models with $\Lambda = 0$.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}, 7^{\text{S}}, 8^{\text{Q}}, 13^{\text{CX}}, 15^{\text{Q}}$	§5, Table 3, Fig. 5

p	Claim	i	Source Location
23	Present acceleration is classified by $q_0 = \Omega_M/2 - \Omega_\Lambda < 0$. The source distinguishes acceleration from deceleration, recollapse from eternal expansion and models without a big bang.	$7^S, 9^{QS}$	§6.1, Fig. 7
24	Perlmutter et al. infer present acceleration from the Fit C probability region.	$1^O, 2^{IB}, 3^B, 4^Q, 5^{IX}, 7^S, 8^Q, 9^{QS}, 13^{CX}, 15^Q$	§6.1, Fig. 7
25	Cosmic age is calculated by integrating the history of an expanding big-bang model. The age distribution including nonflat models gives $H_0 t_0 = 0.93 \pm 0.06$; under flatness it gives 0.96 with $+0.09/-0.07$.	$1^O, 2^{IB}, 3^B, 4^Q, 5^{IX}, 7^S, 8^Q, 10^{IQ}, 13^{CX}, 15^Q$	§6.1, Fig. 9
26	Scaling by H_0 gives an age of $14.5 \pm 1.0(0.63/h)$ Gyr including nonflat models and $14.9(+1.4/-1.1)(0.63/h)$ Gyr under flatness; these are compared with estimates for the oldest stars.	$1^O, 2^{IB}, 3^B, 4^Q, 5^{IX}, 7^S, 8^Q, 10^{IQ}, 13^{CX}, 14^I, 15^Q$	§6.1, Fig. 9
27	Perlmutter et al. test outlier exclusion, removal of the width–luminosity correction, classification confidence and the Galactic extinction model, reporting similar principal cosmological estimates.	$1^O, 2^{IB}, 3^B, 4^Q, 5^{IX}, 7^S, 8^Q, 13^{CX}, 15^Q$	§4.5–5, Table 3, Fig. 5
28	Perlmutter et al. assess statistical errors and systematics in reddening, K corrections, width corrections, selection and matter clumping, distinguishing unquantified evolution and extinction alternatives.	$1^O, 2^{IB}, 3^B, 4^Q, 5^{IX}, 6^I, 7^S, 8^Q, 11^{OB}, 12^{SX}, 13^{CX}, 15^Q$	§4–5, Table 4
29	The comparison with Riess et al. involves a different high-redshift search but shares the low-redshift sample, two high-redshift objects and the K-correction method.	$1^O, 2^{IB}, 3^B, 4^Q, 5^{IX}, 7^S, 8^Q, 13^{CX}, 15^Q$	§6.1
30	The source compares its constraints with other studies of cluster evolution and dynamics, peculiar velocities, radio-lobe propagation, lensing statistics and background radiation.	$7^S, 12^{SX}, 13^{CX}, 15^Q$	§6.2
31	For a flat model with $\Omega_M \simeq 0.28$, matter–Lambda density equality occurs at $z \simeq 0.37$ and the deceleration–acceleration transition at $z \simeq 0.73$; they are different events.	$7^S, 9^{QS}, 10^{IQ}, 12^{SX}$	§6.3
32	The fit in Figure 10 assumes flatness and a constant equation-of-state coefficient w . The source discusses time-varying tracking-field models separately.	$1^O, 2^{IB}, 3^B, 4^Q, 5^{IX}, 7^S, 8^Q, 12^{SX}, 13^{CX}, 15^Q$	§6.3, Fig. 10

p	Claim	i	Source Location
33	The appendix distinguishes the actual extinction distribution from the assumed prior and calculates noise-induced bias in individual extinction estimates.	$\emptyset$	Appendix, Eqs. (A1)–(A4)
34	Perlmutter et al. conclude that the cosmological constant is positive and the present expansion accelerating from their specified FRW analysis of corrected supernova data.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{B}}, 4^{\text{Q}}, 5^{\text{IX}}, 7^{\text{S}}, 8^{\text{Q}}, 9^{\text{QS}}, 13^{\text{CX}}, 15^{\text{Q}}$	Abstract, §5–6
35	For a specified FRW expansion history and H_0 , cosmic age is calculated by a finite integral back to the big-bang boundary.	$7^{\text{S}}, 8^{\text{Q}}, 10^{\text{IQ}}$	§6.1, Fig. 9
36	Perlmutter et al. describe the P97 width correction through the 15-day decline parameter Δm_{15} and fit the stretch factor s directly in the present analysis.	$3^{\text{B}}, 4^{\text{Q}}, 6^{\text{IQ}}$	§2

Table 350: Perlmutter et al. 1999 Time-Use Roles

Code	Role	Use Count
I	Interval	50
Q	Quantity	59
X	Cross-position correspondence	39
K	Clock reading	1
C	Causality	13
S	State evolution	27
P	Probabilistic evolution	0
O	Observation record	25
B	Reference boundary	46

Table 351: Perlmutter et al. 1999 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3, 7	member	instant, order_relation
2	member	instant, turning_position
4, 8–9, 15	dependency	$\emptyset$
5	member	interval_boundary
6	member	interval_boundary, turning_position
10, 14	member	endpoint, interval_boundary
11–12	member	order_relation, interval_boundary

Time Indices	Structural Type	Axes
13	member	order_relation

Dependency Inputs

Definition 8.469 (Perlmutter et al. 1999 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [4]$. The table gives time inputs, time outputs, claim support and source locations.

Table 352: Perlmutter et al. 1999 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1, 3	2, 4–5	6–8	Obtain phase, stretch and rest-frame width from timed photometry and a template. §2
2	1–5	6	6, 11	Obtain rise and decline intervals from a fitted light curve. §2, §4.4
3	7	9	23	Classify present acceleration or deceleration for a specified FRW model. §6.1, Fig. 7
4	7–8	10	35	Calculate cosmic age from a specified expansion history and Hubble rate. §6.1, Fig. 9

Grounding Conditions

1. Object-specific redshift, fitting including the epoch of maximum, phase-dependent corrections and the specified template.
2. Maximum-light reference and the early and late time-indexed photometric coverage required by the fit domain.
3. Evaluate $q_0 = \Omega_M/2 - \Omega_\Lambda$ from the current density parameters of the specified matter–Lambda model.
4. $H_0 > 0$, an expanding big-bang branch and a convergent age integral; flatness is specified separately when required.

Results

Definition 8.470 (Perlmutter et al. 1999 Results and Dependencies). Let $\text{ldx}_R^{\text{audit,result}} = [25]$. The table gives each result’s temporal premises and claim support.

Table 353: Perlmutter et al. 1999 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Supernova samples, classification confidence and follow-up coverage. §2, §4.4, Tables 1–2	1–3, 5–6, 11, 15	4–5, 11
2	Light-curve fitting for maximum light, phase and intrinsic stretch. §2	1–5	6
3	Distinction between intrinsic stretch and cosmological temporal dilation. §2, Eqs. (1)–(2)	1–5	7–8
4	Phase-dependent K corrections and treatment of extinction. §2, §4.1	2–4	9
5	Corrected peak magnitudes and the Hubble diagram. §2, Eq. (3), Tables 1–2	1–5	10
6	Sample selection and color conditions for Fits A, B and C. §3–4.1, Table 3	1–5, 11	12–13
7	Joint fitting with H_0 eliminated and its error model. §3, Eq. (4)	1–5, 7–8, 13, 15	19–20
8	Fit C density parameters and the estimate conditional on flatness. §5, Table 3, Fig. 7	1–5, 7–8, 13, 15	21
9	Statistical support for a positive cosmological constant. §5, Table 3, Fig. 5	1–5, 7–8, 13, 15	22
10	Inference of present acceleration from supernova data. §6.1, Fig. 7	1–5, 7–9, 13, 15	23–24
11	Cosmic-age distributions with and without flatness. §6.1, Fig. 9	1–5, 7–8, 10, 13, 15	25, 35
12	Age conversion to Gyr and comparison with the oldest stars. §6.1, Fig. 9	1–5, 7–8, 10, 13–15	26
13	Reddening and extinction tests and residual grey extinction. §4.1, §4.4, Appendix	2, 4, 12	14
14	Malmquist-selection tests. §4.2	1, 4–5, 11	15
15	Cosmological fits for matter clumping and partially filled beams. §4.3, Fig. 8	1–5, 7–8, 13, 15	16, 19–20
16	Tests of supernova and host-galaxy evolution. §4.4	2, 4, 6, 12, 15	17–18
17	Robustness to width correction, outliers, classification and extinction models. §4.5–5, Table 3	1–5, 7–8, 13, 15	27
18	Assessment of statistical and systematic errors and unquantified alternatives. §4–5, Table 4	1–8, 11–13, 15	28
19	Comparison with Riess et al. and shared data. §6.1	1–5, 7–8, 13, 15	29

q	Result and Source Location	Time Premises i	Claim Support p
20	Comparison with other cosmological observations. §6.2	7, 12–13, 15	30
21	Cosmic history of density equality and acceleration onset. §6.3	7, 9–10, 12	31
22	Flat-model fit with constant w and discussion of tracking fields. §6.3, Fig. 10	1–5, 7–8, 12–13, 15	32
23	Extinction priors and statistical bias in individual estimates. Appendix, Eqs. (A1)–(A4)	$\emptyset$	33
24	Source conclusion of a positive cosmological constant and present acceleration. Abstract, §5–6	1–5, 7–9, 13, 15	34
25	Earlier correction through 15-day decline and direct stretch fitting. §2	3–4, 6	36

Definition 8.471 (Perlmutter et al. 1999 Grounds and Domains of Results). The table gives conditions and domains of the results.

Table 354: Perlmutter et al. 1999 Conditions for Results

Result Index	Conditions and Domain
1	42 high-redshift and 18 low-redshift objects, object-specific spectral evidence, discovery epochs and observational coverage.
2	Time-indexed photometry, redshift, template, calibration, corrections and fit domain.
3	Common B/V stretch factor and observer/rest-frame correspondence through $1 + z$.
4	Redshift, phase-dependent spectra, passbands and Galactic extinction model.
5	Fitted peak magnitudes, width–luminosity relation, K correction and Galactic extinction.
6	Selection by residual, stretch and color, with $B_{\max} - V_{\max}$ at the separate band maxima.
7	Peak magnitudes derived from timed photometry, FRW luminosity distance, covariance, peculiar-velocity errors and nuisance marginalization.
8	Specified Fit C likelihood; $\Omega_M = 0.28$ assumes flatness, while the linear summary concerns the principal parameter region.
9	Specified prior domain, corrections and likelihood, with the identified systematics assessed by Fit J.
10	Density region inferred by Fit C and the $q_0 < 0$ criterion for the matter–Lambda model.
11	Supernova likelihood, expanding big-bang branch, age integral and H_0 , with the flat-conditional distribution distinguished.

Result Index	Conditions and Domain
12	Value of H_0 , cosmic-age distribution and external stellar-age estimates.
13	Color and stretch conditions, sample and adopted extinction prior.
14	Search intervals, detection limits, visibility widths and sample-selection functions.
15	Specified filling factor and propagation model, supernova photometry and joint-fit conditions.
16	Spectra matched in phase and width, preliminary rise-time comparisons and host-specific samples.
17	Sample, calibration and likelihood conditions of each fit; Fit D still uses timed light-curve fitting when width correction is omitted.
18	Sample, observation, correction and model conditions appropriate to each error or alternative.
19	Differences between the analyses and the shared low-redshift sample, two high-redshift objects and K corrections.
20	Observational, dynamical, evolutionary and propagation models of the studies cited by the source.
21	Flat $\Omega_M \simeq 0.28$, matter density proportional to a^{-3} and constant Lambda.
22	The fit specifies constant w; discussion of time-varying models uses the conditions of the respective cited studies.
23	Assumed half-Gaussian prior, measurement noise and true extinction distribution.
24	Corrected supernova sample, specified FRW model, likelihood, prior domain and errors assessed in the source.
25	Attribution to P97, with the width–luminosity relation and template-fitting domain of the present source.

Result Dependence

Definition 8.472 (Perlmutter et al. 1999 Result-Dependence Hyperedges). For each row of the result table, let **ResultEdge** $_{R,q}$ ($q \in [25]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Evidence Hierarchy Photometric, spectral and observation-time records form the observational material. Type Ia classification confidence, redshift, peak magnitudes and stretch factors are obtained by identification, calibration and fitting of that material. Cosmological parameters, present acceleration and cosmic age are inferred from the likelihood for corrected magnitudes and redshifts and the specified cosmological model. The appendix’s extinction-estimation bias is a mathematical calculation for assumed probability distributions.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local-reach count are both zero, leaving 15 primitive-time roots. Transfers to light-curve phase, stretch and width and to model acceleration and age take observation epochs, template time, cosmic time or Hubble rate as inputs.

Result Dependence Of the 25 results, 24 depend on primitive-time roots. Result 23, concerning extinction priors and bias in individual estimates, is reached without additional time roots as the statistical relation in Appendix Equations (A1)–(A4). There are 122 root–result dependence pairs, of which 102 are necessary.

Placement of the Grounding Obligation Perlmutter et al. use observation epochs, light-curve phases and widths, astronomical velocities, propagation, expansion history and ages in the analysis. Temporal-width conversions and model calculations take existing time contents as inputs, leaving all 15 as primitive-time roots. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Sample selection, calibration, corrections, covariance, errors, prior domain and model family specify the conditions of each result. The source tests changes in estimates and robustness under variations of these conditions. Actual failure of a specific audited physical-time generation condition and the resulting change in result values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 122 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are each temporal use and the physical-time generation conditions required before that use. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Proposition 8.473 (Primitive time roots and result dependence in Perlmutter et al. 1999). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [15]; \emptyset, \{23\}, [25] \setminus \{23\}, \emptyset).$$

Proof. Closure from Initial Claims 1–3 remains those three nodes. Even with all 36 claims as source formal evidence, each of the four transfers requires temporal inputs. Result 23, with empty temporal premises, is reached from source formal evidence. Adding all 15 time contents as seeds reaches all 25 results. $\square$

Proposition 8.474 (Minimal primitive-time-root support in Perlmutter et al. 1999). The table gives each result’s minimal-support index family, with 31 family members in total.

Proof. Start with singleton time-content seeds and apply Definition 7.46 to the four transfers. Combine supports of result premises from the stable antichains and retain inclusion-minimal supports. Proposition 7.47 gives the displayed families. $\square$

Table 355: Minimal Primitive-Time-Root Support in Perlmutter et al. 1999

Result Indices	Minimal-Support Family Members
1	{1, 3, 11, 15}
2–3, 5	{1, 3}
4	{1, 3}, {2, 3, 4}
6	{1, 3, 11}
7–11, 15, 17, 19, 24	{1, 3, 7, 8, 13, 15}
12	{1, 3, 7, 8, 13, 14, 15}
13	{1, 3, 12}, {2, 4, 12}
14	{1, 3, 11}, {1, 4, 5, 11}
16	{1, 3, 12, 15}, {2, 4, 6, 12, 15}
18	{1, 3, 7, 8, 11, 12, 13, 15}
20	{7, 12, 13, 15}
21	{7, 8, 12}, {7, 10, 12}
22	{1, 3, 7, 8, 12, 13, 15}
23	$\emptyset$
25	{1, 3}, {3, 4, 6}

Evidentiary Effect Observation epochs and photometric and spectral records support light-curve fitting and corrections. The specified likelihood for corrected magnitude and redshift gives a density-parameter region, while FRW evolution supports its interpretation in terms of present acceleration and age. Direct observational records, correction and fitting, and cosmological inference are linked through the tabulated conditions.

Result-Preservation Domain Fix the sample, calibration, corrections, covariance, likelihood, prior domain and model conditions in Table 354. When inputs can be realized with generated temporal representations preserving the same epoch correspondence, intervals, width ratios, evolution, propagation and unit conversions, each result is preserved with its stated domain and epistemic status.

Proposition 8.475 (Audit conclusion for Perlmutter et al. 1999). There are 15 time contents, 36 claims, 4 transfers and 25 results. The 185 claim/time-content use pairs have 260 role assignments. Complete generation and local reach are both zero, leaving 15 primitive-time roots and 24 root-dependent results.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. $\square$

Proposition 8.476 (Audit-boundary cardinalities for Perlmutter et al. 1999). There are 80 source-specific obligation atoms, with

$$\text{BoundaryCount}_R^{\text{audit}} = (76, 29).$$

Proof. Count claims, time contents and results as nodes, and transfers and result dependencies as edges. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual Statements Temporal-width correction of the light-curve template

Original text:

For the supernovae discussed in this paper, the template must be time-dilated by a factor $1 + z$ before fitting to the observed lightcurves to account for the cosmological lengthening of the supernova timescale (Goldhaber et al. 1995; Leibundgut et al. 1996a; Riess et al. 1997a).

Source location: Section 2, between Equations (1) and (2). Perlmutter et al. distinguish intrinsic stretch from cosmological temporal dilation by $1 + z$ and apply the latter to the template before fitting the observed light curve.

Stretch factor common to two bands

Original text:

A similar relation corrects the V band lightcurve, with the same stretch factor in both bands.

Source location: Section 2, immediately after Equation (1). Perlmutter et al. also apply the width–luminosity correction to the V band and use the same intrinsic stretch factor in both B and V.

Inconsistency with a flat $\Lambda = 0$ cosmology

Original text:

The data are strongly inconsistent with a $\Lambda = 0$ flat cosmology, the simplest inflationary universe model.

Source location: Abstract. Perlmutter et al. correct the light curves of 42 high-redshift supernovae and a low-redshift sample and report cosmological parameter distributions and flat-conditional estimates from the specified FRW likelihood.

Common Evidence with Specified Scope The table gives the objects, predicates and domains of the three statements.

State- ment	Object	Predicate	Explicit Domain
S38- Q001	Light-curve templates for the Type Ia supernovae in this source	Perlmutter et al. require temporal dilation by $1 + z$ before fitting the observed light curve.	Section 2, around Equations (1)–(2): the sample, redshift, template and fitting procedure.
S38- Q002	B- and V-band light-curve corrections	Perlmutter et al. use one intrinsic stretch factor in both bands.	Section 2, around Equation (1): width–luminosity relation and two-band fitting.
S38- Q003	Corrected supernova data and a flat $\Lambda = 0$ model	Perlmutter et al. report strong inconsistency between this model and the data.	Abstract and Sections 3 and 5: samples, corrections and FRW likelihood.

Individual Audit Inputs to temporal-width correction

The first statement gives Perlmutter et al.’s procedure of dilating the template width by $1 + z$ before fitting an observed light curve. Its grounds are the timed photometry, redshifts, template and stretch fitting in Section 2, using Time Contents 1–5. Correspondence between existing temporal widths in the specified sample and template conditions this procedure. Claims 6–8, Transfer 1 and Results 2–3 correspond to the statement.

Shared condition for two bands

The second statement gives Perlmutter et al.’s use of the same intrinsic stretch factor in the B- and V-band width–luminosity corrections. Its ground is the two-band template fit in Section 2, which uses the stretch factor determined from the B band also in V. Its domain is the source’s sample, template and width–luminosity relation. Claim 8, Time Contents 3–5 and Result 3 correspond to the statement.

Statistical inconsistency with the flat model

The third statement is Perlmutter et al.’s statistical conclusion that their corrected supernova data are strongly inconsistent with a flat $\Lambda = 0$ cosmology. Its ground is the model comparison using the samples, corrections, errors and FRW likelihood in Sections 3 and 5. The specified model family, prior domain and error conditions define the domain of this inconsistency and the support for a positive cosmological constant. Claims 19–22 and 34 and Results 7–9 and 24 correspond to the statement.

8.19.39 Source Audit

Audit Input

Source and Audit Scope

Source Identifier 39.

Source Name Primitive-time-root audit of Riess et al. 1998

Bibliographic Identification Let $R = \text{Ref}_{\text{Rie98OE}}$ denote the original source Adam G. Riess et al., *Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant*, *The Astronomical Journal* 116 (1998), 1009–1038. The audited domain comprises the high-redshift Type Ia supernova sample, nearby comparison sample, photometric and spectroscopic observations, rest-frame corrections, MLCS, decline-rate-template and snapshot distance estimation, cosmological likelihood, estimates of acceleration, cosmic age, and future expansion, and systematic branches.

Source-Location Identifier 39-L1.

Source-Location Content The time-content table gives 16 time contents and their source locations; the claim table gives 41 claims and their source locations.

Audit-Scope Identifier 39-SC1.

Audit-Scope Content The audited domain stated in the bibliographic identification.

Formal Object

Object Identifier 39-F1.

Object Name Objects using time.

Object Type Formal object.

Formal Content The 16 time contents and their uses in each claim.

Source-Location Identifier Set 39-L1.

Assumption

Assumption Identifier 39-A1.

Assumption Content The source assumptions are specified in Definitions 8.477 and 8.480.

Source-Location Identifier Set 39-L1.

Nontemporal Physical Input

Input Identifier 39-P1.

Input Name Nontemporal physical input.

Input Type Nontemporal physical input.

Input Content The nontemporal object, wavelength and detector data, algebraic parameter specifications and statistical rules of Claims 1–3.

Source-Location Identifier Set 39-L1.

Definition 8.477 (Riess et al. 1998 Formal Objects and Conditions). Riess et al. use time-indexed photometry and spectroscopy of Type Ia supernovae to obtain distances through MLCS, decline-rate templates and the snapshot method, with K corrections and treatment of extinction. They specify training, samples, calibration and errors for each method and fit the distance–redshift relation to an FRW model containing matter, curvature and a cosmological constant. The cosmological analysis adopts nonnegative matter density and an expanding big-bang branch, with Gaussian distance errors, independence apart from systematics, a likelihood and a prior domain. Present acceleration, future evolution and cosmic age are obtained from their respective model relations and the inferred parameter distribution.

Definition 8.478 (Riess et al. 1998 Formal Inputs). Phase relative to B maximum and corresponding temporal widths of the same event satisfy, at the specified redshift,

$$\phi = \frac{t_{\text{obs}} - t_{\text{max,obs}}}{1 + z}, \quad \Delta t_{\text{rest}} = \frac{\Delta t_{\text{obs}}}{1 + z}.$$

MLCS uses phase-indexed trained vectors. The quadratic model in source Appendix Equations (A3)–(A4) is

$$\begin{aligned} \mathbf{m}_V &= \mathbf{M}_V + \mathbf{R}_V \Delta + \mathbf{Q}_V \Delta^2 + \mu_V, \\ \mathbf{m}_{B-V} &= \mathbf{M}_{B-V} + \mathbf{R}_{B-V} \Delta + \mathbf{Q}_{B-V} \Delta^2 + E_{B-V}. \end{aligned}$$

The vectors are functions of supernova age with zero at B maximum; the constant distance and color terms are added componentwise. Fitting with the source covariance and extinction prior supplies the estimation conditions. The snapshot method uses phase-indexed spectral calibration and two-band photometry. For the FRW matter–curvature–cosmological-constant model,

$$\begin{aligned} E(z)^2 &= \Omega_M(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda, \quad \Omega_k = 1 - \Omega_M - \Omega_\Lambda. \\ D_L &= \frac{c(1+z)}{H_0} \mathcal{S}_{\Omega_k} \left(\int_0^z \frac{du}{E(u)} \right), \quad \mu = 5 \log_{10} \left(\frac{D_L}{\text{Mpc}} \right) + 25, \end{aligned}$$

$$\mathcal{S}_k(\chi) = \begin{cases} \sinh(\sqrt{k}\chi)/\sqrt{k}, & k > 0, \\ \chi, & k = 0, \\ \sin(\sqrt{-k}\chi)/\sqrt{-k}, & k < 0. \end{cases}$$

These expressions use the same unit and curvature conventions as source Equations (2)–(3) and are evaluated in the adopted domain with positive expansion rate and luminosity distance. The present-acceleration criterion and cosmic age are

$$q_0 = \frac{\Omega_M}{2} - \Omega_\Lambda < 0, \quad t_0 = H_0^{-1} \int_0^\infty \frac{dz}{(1+z)E(z)}.$$

The age integral applies on an expanding big-bang branch with a positive Hubble constant where the integral converges. Riess et al. map the supernova-inferred parameter distribution through these relations to obtain the probability of present acceleration and the distribution of cosmic age.

Definition 8.479 (Riess et al. 1998 Time Contents). Let $n_R = 16$. The table gives each time content and source location.

Table 357: Riess et al. 1998 Time Contents and Source Locations

i	Time Content	Source Location
1	Observation-time sequences and exposure intervals	§2, Tables 1–4
2	Supernova phase relative to B-band maximum	§2–3, Appendix, Tables 1, 7
3	Observer-frame and rest-frame temporal widths	§3, Fig. 3, Tables 2, 4
4	Time axes of trained light curves and phase-indexed spectra	§2–3, Appendix
5	Decline in brightness over 15 rest-frame days	§3.2, Table 6
6	Order, cadence and selection intervals of search and follow-up	§2, §5.3
7	Propagation from supernova emission to detection	§4, §5.2, §5.5
8	FRW cosmic time and scale-factor history	§4, Eqs. (2), (13)–(18)
9	Hubble rate, Hubble time and the physical interpretation of densities	§4, Eqs. (2), (18)
10	Present acceleration or deceleration	§4.2, Eqs. (15)–(17)
11	Cosmic age and its ratio to Hubble time	§4.3, Eqs. (18)–(19)
12	Progenitor and population evolution and lookback intervals	§5.1–5.3, §6
13	Future evolution of cosmic expansion	§4.1, Eqs. (13)–(14)
14	Periods and phases in Cepheid calibration	§3.2, §4.1
15	Stellar ages from cooling, radioactive dating and cluster models	§4.3
16	Peculiar and expansion velocities and event rates	§2.2, §4, §5.1, §5.3–5.4, §6

Definition 8.480 (Riess et al. 1998 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(41)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([3], [41])$. Claims 1–3 form the generation-initial set; all 41 claims form the source formal-evidence environment.

Definition 8.481 (Riess et al. 1998 Claim Boundary). The generation-initial set consists of Claims 1–3 and the source formal-evidence environment of Claims 1–41. The following table gives each claim’s temporal uses and roles.

Table 358: Riess et al. 1998 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Object identities, sky positions, host galaxies, passband wavelength characteristics and nontemporal signal data.	$\emptyset$	§2, Tables 1–3
2	Algebraic density-parameter coordinates, spatial curvature and the flatness condition.	$\emptyset$	§4
3	Mathematical rules for errors, covariance, likelihoods, marginalization and priors.	$\emptyset$	§4, Appendix
4	Riess et al. use 16 high-redshift objects, including 10 new objects, and 34 nearby objects. The 15 confirmed objects excluding SN 1997ck comprise 9 with well-sampled light curves and 6 snapshots.	$1^{\text{O}}, 6^{\text{OB}}$	§2–4, Tables 5–10
5	Spectroscopic classification and redshift use phase-dependent comparisons and host lines or broad supernova features; the Type Ia assignment for SN 1997ck is conditional.	$1^{\text{O}}, 2^{\text{IB}}, 4^{\text{B}}, 16^{\text{Q}}$	§2.2, §5.7, Table 1
6	Riess et al. construct observational series using difference images, time-indexed multiband photometry, spectroscopy, calibration stars and host templates. The tabulated magnitudes are calibrated photometric values.	$1^{\text{O}}, 6^{\text{OB}}$	§2, Tables 1–4
7	K corrections depend on phase, color, redshift and passbands, and Riess et al. iterate corrections and light-curve fits. Table 4 gives rest-frame B/V bands with observer-frame dates in its JD column.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}$	§3.1–3.2, Table 4
8	MLCS fits time-indexed B/V light curves using trained phase-dependent vectors and covariance, including a quadratic term in luminosity difference Delta.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}$	§3.2, Appendix, Table 10
9	Template fitting estimates distance using six light curves, the 15-rest-frame-day decline, reference value 1.1, and relations for luminosity and color at maximum.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 5^{\text{Q}}$	§3.2, Table 6
10	The snapshot method estimates supernova phase and luminosity properties from calibrated spectral features.	$1^{\text{O}}, 2^{\text{IB}}, 4^{\text{B}}$	§3.2, Tables 7, 9

p	Claim	i	Source Location
11	Snapshot distances combine spectroscopic estimates with two-band photometry. The four new objects have distance errors of 0.4–0.6 mag; the snapshot samples comprise 6 high-redshift and 7 nearby objects.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}$	§3.2, §4.1, Tables 7, 9
12	Extinction correction uses color, phase-dependent color relations, Galactic extinction and host-extinction priors. The luminosity-correction vector and reddening coefficient in the MLCS appendix are distinct quantities.	$1^{\text{O}}, 2^{\text{IB}}, 4^{\text{B}}$	§3.1–3.2, Appendix
13	Riess et al. calibrate absolute supernova luminosity using Cepheids in the hosts of SNe 1972E, 1981B and 1990N, and propagate distance-scale uncertainty to the Hubble constant and cosmic age.	$1^{\text{O}}, 4^{\text{B}}, 14^{\text{IB}}$	§3.2, §4.1–4.3
14	Fits to the 27 nearby objects use B/V observations within 40 rest-frame days of B maximum. Training, distance fitting and spectral-phase estimation have their respective sample and calibration conditions.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 6^{\text{OB}}$	§3, Appendix, Table 10
15	Light-curve fitting uses observation epochs, redshift, trained time axes and corrections to estimate maximum-light epoch and rest-frame phase, and converts corresponding temporal widths by $1 + z$.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}$	§3, Appendix, Fig. 3
16	The 15-day decline is the magnitude difference between B maximum and 15 days later on the rest-frame light curve, evaluated using the specified time axis.	$2^{\text{IB}}, 4^{\text{B}}, 5^{\text{Q}}$	§3.2, Table 6
17	Luminosity distance and distance modulus are inferred from corrected, calibrated luminosity information, and method-specific MLCS, template and snapshot results enter the cosmological analysis.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}$	§3–4, Tables 5–7, 9–10
18	FRW luminosity distance uses the matter–curvature–Lambda expansion history, light propagation and c/H_0 , with curvature-dependent sinn and the flat limit distinguished.	$7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}$	§4, Eqs. (1)–(3)
19	Riess et al. adopt expanding big-bang models with nonnegative matter density, specify Gaussian distance errors, independence apart from systematics and a prior domain, and marginalize H_0 .	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 16^{\text{Q}}$	§4.1, Eqs. (4)–(12)

p	Claim	i	Source Location
20	Peculiar-velocity and redshift errors are converted into distance-modulus uncertainties; the principal analysis uses 200 km/s and an additional 2500 km/s for broad supernova features.	$1^{\text{O}}, 16^{\text{Q}}$	§4.1, Eq. (4)
21	The inferred distances of high-redshift supernovae exceed those in the comparison model with $\Omega_M = 0.2$ and $\Lambda = 0$ by 10–15% on average.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 16^{\text{Q}}$	Abstract, §4
22	For the specified analysis of 15 confirmed high-redshift objects, support for positive Lambda is 99.7% with MLCS and above 99.9% with template fitting, independent of the numerical H_0 and absolute-luminosity zero point.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 16^{\text{Q}}$	§4.1, Table 8
23	The analysis including SN 1997ck adds a Type Ia assignment and extinction $A_B = 0.0 \pm 0.1$, and constrains the parameter region with data extending to $z = 0.97$.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 16^{\text{Q}}$	§4.1, Table 8
24	MLCS and template fitting use the same objects and shared calibration. Riess et al. state that the difference between their cosmological constraints must be treated as a systematic uncertainty.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}$	§4.3, §5.6
25	In the matter–Lambda model, the present deceleration parameter is $q_0 = \Omega_M/2 - \Omega_\Lambda$, and $q_0 < 0$ denotes present acceleration.	$8^{\text{S}}, 10^{\text{QS}}$	§4.2
26	Riess et al. integrate the posterior over the future-evolution region of the adopted FRW model and report statistical support for continued expansion.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 13^{\text{S}}, 16^{\text{Q}}$	§4.1, Eqs. (13)–(14)
27	Cosmic age is calculated by a finite FRW time integral from a specified expanding big-bang history and positive H_0 .	$8^{\text{S}}, 9^{\text{Q}}, 11^{\text{IQ}}$	§4.3, Eq. (18)
28	The age distributions inferred from the supernova parameter distributions give 13.6 Gyr with MLCS and 14.8 Gyr with template fitting. Riess et al. retain one method’s statistical uncertainty and include distance-scale systematics to obtain 14.2 ± 1.5 Gyr.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 11^{\text{IQ}}, 14^{\text{IB}}, 16^{\text{Q}}$	§4.3, Eqs. (18)–(19)
29	Riess et al. compare cosmic age with external estimates from white-dwarf cooling, radioactive dating and cluster models, and discuss uncertainty in distance scales and stellar models.	$8^{\text{S}}, 9^{\text{Q}}, 11^{\text{IQ}}, 14^{\text{IB}}, 15^{\text{I}}$	§4.3

p	Claim	i	Source Location
30	Support for present acceleration is obtained from the $q_0 < 0$ region of the same supernova likelihood and is 99.5% for MLCS and above 99.9% for template fitting.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 10^{\text{QS}}, 16^{\text{Q}}$	§4.2, Table 8
31	Evolution tests compare progenitor and host ages, phase-dependent spectra, light curves, event rates and lookback intervals, and assess agreement within the limited samples.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 5^{\text{Q}}, 12^{\text{SX}}, 16^{\text{Q}}$	§5.1
32	Extinction tests use color and dispersion, distinguishing grey extinction under an assumed dust distribution from uniform intergalactic grey extinction.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 12^{\text{SX}}$	§5.2
33	Selection simulations use search cadence, detection limits, luminosity distributions, light curves and event rates, and examine the nearby sample selection conditions.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 5^{\text{Q}}, 6^{\text{OB}}, 12^{\text{SX}}, 16^{\text{Q}}$	§5.3
34	In the local-flow test, removing nearby objects within 7000 km/s changes support for positive Lambda from 99.7% to 98.3% for MLCS and from above 99.9% to 99.8% for template fitting.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 16^{\text{Q}}$	§5.4
35	Tests of weak lensing and compact objects evaluate propagation corrections for specified cosmologies, redshifts and matter distributions.	$7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}$	§5.5
36	The Ia/Ic classification of SN 1996E depends on phase and the availability of premaximum observations; Riess et al. also test the sample excluding 1996E and 1997ck.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 6^{\text{OB}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 16^{\text{Q}}$	§5.7
37	Riess et al. expect the principal limitation on supernova cosmology to shift from statistical error to physical understanding as the method develops, and discuss systematic tests and future observations.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 6^{\text{OB}}, 12^{\text{SX}}, 16^{\text{Q}}$	§1, §5–6
38	The source compares constraints from other studies of clusters, peculiar velocities, gravitational lensing and early-universe background radiation.	$7^{\text{CX}}, 8^{\text{S}}, 12^{\text{SX}}, 16^{\text{Q}}$	§4.1, §6
39	Riess et al. conclude that the cosmological constant is positive and the present expansion accelerating from their specified analysis of calibrated, corrected supernova data, and discuss systematics, model-dependent future evolution and cosmic age.	$1^{\text{O}}, 2^{\text{IB}}, 3^{\text{IX}}, 4^{\text{B}}, 7^{\text{CX}}, 8^{\text{S}}, 9^{\text{Q}}, 10^{\text{QS}}, 11^{\text{IQ}}, 13^{\text{S}}, 14^{\text{IB}}, 16^{\text{Q}}$	Abstract, §4–6

p	Claim	i	Source Location
40	The expansion history of a specified FRW model is extended into the future, and growth or recollapse of its scale factor is determined from that model's equations of motion.	$8^S, 13^S$	§4.1, Eqs. (13)–(14)
41	Riess et al. estimate H_0 as 65.2 ± 1.3 km/s/Mpc with MLCS and 63.8 ± 1.3 km/s/Mpc with template fitting, and give 65 ± 7 km/s/Mpc after including absolute-luminosity and Cepheid-scale systematics.	$1^O, 2^{IB}, 3^{IX}, 4^B, 7^{CX}, 8^S, 9^Q, 14^{IB}, 16^Q$	§4.1

Table 359: Riess et al. 1998 Time-Use Roles

Code	Role	Use Count
I	Interval	60
Q	Quantity	43
X	Cross-position correspondence	43
K	Clock reading	0
C	Causality	15
S	State evolution	29
P	Probabilistic evolution	0
O	Observation record	36
B	Reference boundary	66

Table 360: Riess et al. 1998 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1	member	instant, order_relation, interval_boundary
2	member	instant, turning_position
3	member	interval_boundary
4, 8	member	instant, order_relation
5, 9–10, 16	dependency	$\emptyset$
6, 12	member	order_relation, interval_boundary
7	member	order_relation
11, 15	member	endpoint, interval_boundary
13	member	order_relation, endpoint
14	member	interval_boundary, turning_position

Dependency Inputs

Definition 8.482 (Riess et al. 1998 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [6]$. The table gives time inputs, time outputs, claim support and source locations.

Table 361: Riess et al. 1998 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1, 4	2–3	15	Obtain phase and temporal width from light-curve fitting. §3, Appendix, Fig. 3
2	1, 4	2	10	Estimate phase through spectroscopic calibration. §3.2
3	2, 4	5	16	Evaluate the 15-day decline from the light curve. §3.2
4	8	10	25	Classify present acceleration for a specified FRW model. §4.2
5	8–9	11	27	Calculate cosmic age from the expansion history and H_0 . §4.3, Eq. (18)
6	8	13	40	Determine future evolution in a specified FRW model. §4.1

Grounding Conditions

1. Redshift, time-indexed photometry, the specified MLCS or template model and corrections, and a fit domain including B maximum.
2. Target spectrum, phase-indexed comparison calibration and the conditions of the snapshot method.
3. A light curve and time scale permitting evaluation at B maximum and 15 rest-frame days later.
4. $q_0 = \Omega_M/2 - \Omega_\Lambda$ for matter and Lambda, with observational parameter constraints supplied separately.
5. Positive H_0 , an expanding big-bang branch, a positive expansion rate and a convergent age integral.
6. The equations of motion and future domain of the same model.

Results

Definition 8.483 (Riess et al. 1998 Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [24]$. The table gives each result’s temporal premises and claim support.

Table 362: Riess et al. 1998 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	High-redshift and nearby samples and observational records. §2–3, Tables 1–10	1–2, 4, 6, 16	4–6
2	Phase estimation, temporal-width conversion and K corrections. §3.1–3.2, Fig. 3, Table 4	1–4	7, 15
3	MLCS distance estimation. §3.2, Appendix, Tables 5, 10	1–4, 6, 14	8, 12–14
4	Distance estimation by decline-rate templates. §3.2, Tables 6, 10	1–6, 14	9, 12–14, 16
5	Snapshot distance estimation. §3.2, §4.1, Tables 7, 9	1–4	10–11
6	Absolute-luminosity scale from Cepheids. §3.2, §4.1	1, 4, 14	13
7	Distance–redshift likelihood and the Hubble constant. §4.1, Eqs. (1)–(12)	1–4, 7–9, 14, 16	17–20, 41
8	Excess distance relative to the comparison model. Abstract, §4	1–4, 7–9, 16	21
9	Statistical support for a positive cosmological constant. §4.1, Table 8	1–4, 7–9, 16	22
10	Constraints including SN 1997ck. §4.1, Table 8	1–4, 7–9, 16	23
11	Statistical support for present acceleration. §4.2, Table 8	1–4, 7–10, 16	25, 30
12	Inference of future expansion under the model. §4.1, Eqs. (13)–(14)	1–4, 7–9, 13, 16	26, 40
13	Cosmic-age calculation and inferred distribution. §4.3, Eqs. (18)–(19)	1–4, 7–9, 11, 14, 16	27–28
14	Comparison with stellar ages. §4.3	8–9, 11, 14–15	29
15	Differences between the two fitting methods. §4.3, §5.6	1–4	24
16	Tests of supernova and population evolution. §5.1	1–5, 12, 16	31
17	Tests of extinction and grey dust. §5.2	1–4, 7, 12	32
18	Tests of search-selection effects. §5.3	1–6, 12, 16	33
19	Tests of local flows and the nearby sample. §5.4	1–4, 7–9, 16	34
20	Tests of weak lensing and matter distributions. §5.5	7–9	35
21	Tests of type classification and sample exclusion. §5.7	1–4, 6–9, 16	36
22	Systematic limitations and future observations. §1, §5–6	1–4, 6, 12, 16	37
23	Comparison with other cosmological studies. §4.1, §6	7–8, 12, 16	38

q	Result and Source Location	Time Premises i	Claim Support p
24	Source conclusion of positive Lambda and present acceleration. Abstract, §4–6	1–4, 7–11, 13–14, 16	39

Definition 8.484 (Riess et al. 1998 Grounds and Domains of Results). The table gives conditions and domains of the results.

Table 363: Riess et al. 1998 Conditions for Results

Result Index	Conditions and Domain
1	10 new, 16 combined and 34 nearby objects, classification confidence, light-curve coverage and the snapshot branch.
2	Observation dates, redshift, trained time axes, passbands and iterative fitting.
3	Quadratic luminosity-difference model, phase-indexed training, B/V photometry, covariance and extinction conditions, and fit interval. Absolute distances use the Cepheid zero point.
4	Six templates, the 15-day decline, reference value 1.1, and color, extinction and sample conditions. Absolute distances use the Cepheid zero point.
5	Spectral-phase and luminosity calibration, two-band photometry, extinction and the method-specific distance scale.
6	Host calibrations for 1972E, 1981B and 1990N and distance-scale systematics.
7	Method-specific distances, FRW model, calibration, errors, prior domain and H_0 marginalization.
8	The comparison model with $\Omega_M = 0.2$ and $\Lambda = 0$ and the corrected distances.
9	15 confirmed high-redshift objects, the specified likelihood and prior domain, and separate MLCS and template analyses.
10	Additional Type Ia assignment and $A_B = 0.0 \pm 0.1$ assumption.
11	The specified parameter distribution and $q_0 < 0$ in the matter–Lambda model.
12	Future-evolution law of the adopted FRW model and the probability region inferred from supernovae.
13	Supernova likelihood, expanding big-bang history, H_0 , age integral and Cepheid-scale error.
14	External estimates from cooling, radioactive dating and cluster models, and their distance scales.
15	Comparison of MLCS and template fitting with shared objects and calibration, and retention of their difference as a systematic uncertainty.
16	Phase-indexed spectra, light curves, hosts, progenitor models and limited samples.
17	Color, dispersion and the adopted dust-distribution and propagation conditions.
18	Search cadence, light curves, event rates and detection rules.

Result Index	Conditions and Domain
19	Exclusion of nearby objects within 7000 km/s and the resulting method-specific confidence levels.
20	Specified cosmology, light propagation and compact-object distribution.
21	Phase and coverage for SN 1996E and the Type Ia condition for SN 1997ck.
22	The contemporary error budget, understanding of supernova physics and observation plans.
23	Models and observational conditions of the cluster, velocity, lensing and background-radiation studies cited by the source.
24	Supernova samples, calibration, corrections, the specified model, likelihood, prior domain and assessment of systematics.

Result Dependence

Definition 8.485 (Riess et al. 1998 Result-Dependence Hyperedges). For each row of the result table, let $\text{ResultEdge}_{R,q}$ ($q \in [24]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Evidence Hierarchy Time-indexed photometry and spectroscopy form the observational material. Riess et al. infer supernova phases and distances through object identification, calibration, K corrections and the MLCS, template and snapshot methods. Under the specified errors, samples and prior domain, they fit an FRW model to the distance–redshift relation to obtain distributions of matter density, the cosmological constant and the Hubble constant. These distributions and FRW temporal-evolution relations support inferences about present acceleration, cosmic age and future expansion.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local-reach count are both zero, leaving 16 primitive-time roots. Transfers to light-curve phase, decline and width and to model acceleration, age and future evolution take observation epochs, template time, cosmic time or Hubble rate as inputs.

Result Dependence Of the 24 results, 24 depend on primitive-time roots. There are 114 root–result dependence pairs, of which 114 are necessary.

Placement of the Grounding Obligation Riess et al. use observation epochs, light-curve phases and widths, astronomical velocities, propagation, expansion history and ages in the analysis. Temporal-width conversions and model calculations take existing time contents as inputs, leaving all 16 as primitive-time roots. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Sample selection, calibration, corrections, covariance, errors, prior domain and model family specify the conditions of each result. The source tests changes in estimates and robustness under variations of these conditions. Actual failure of a specific audited physical-time generation condition and the resulting change in result values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 114 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are each temporal use and the physical-time generation conditions required before that use. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Proposition 8.486 (Primitive time roots and result dependence in Riess et al. 1998). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(16, 24).$$

Proof. Closure from Initial Claims 1–3 remains those three nodes. Even with all 41 claims as source formal evidence, each of the six transfers requires temporal inputs. Adding all 16 time contents as seeds reaches all 24 results. $\square$

Proposition 8.487 (Minimal primitive-time-root support in Riess et al. 1998). The table gives each result’s minimal-support index family, with 24 family members in total.

Proof. Start with singleton time-content seeds and apply Definition 7.46 to the six transfers. Combine supports of result premises from the stable antichains and retain inclusion-minimal supports. Proposition 7.47 gives the displayed families. $\square$

Table 364: Minimal Primitive-Time-Root Support in Riess et al. 1998

Result Indices	Minimal-Support Family Members
1	{1, 4, 6, 16}
2, 5, 15	{1, 4}
3–4	{1, 4, 6, 14}
6	{1, 4, 14}
7, 13, 24	{1, 4, 7, 8, 9, 14, 16}
8–12, 19	{1, 4, 7, 8, 9, 16}
14	{8, 9, 14, 15}
16	{1, 4, 12, 16}
17	{1, 4, 7, 12}
18, 22	{1, 4, 6, 12, 16}
20	{7, 8, 9}
21	{1, 4, 6, 7, 8, 9, 16}
23	{7, 8, 12, 16}

Evidentiary Effect Observation epochs and photometric and spectral records support light-curve fitting and corrections. The specified likelihood for corrected magnitude and redshift gives a density-parameter region, while FRW evolution supports its interpretation in terms of present acceleration, age and future evolution. Direct observational records, correction and fitting, and cosmological inference are linked through the tabulated conditions.

Result-Preservation Domain Fix the sample, calibration, corrections, covariance, likelihood, prior domain and model conditions in Table 363. When inputs can be realized with generated temporal representations preserving the same epoch correspondence, intervals, phase correspondence, evolution, propagation and unit conversions, each result is preserved with its stated domain and epistemic status.

Proposition 8.488 (Audit conclusion for Riess et al. 1998). There are 16 time contents, 41 claims, 6 transfers and 24 results. The 204 claim/time-content use pairs have 292 role assignments. Complete generation and local reach are both zero, leaving 16 primitive-time roots and 24 root-dependent results.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. $\square$

Proposition 8.489 (Audit-boundary cardinalities for Riess et al. 1998). There are 87 source-specific obligation atoms, with

$$\text{BoundaryCount}_R^{\text{audit}} = (81, 30).$$

Proof. Count claims, time contents and results as nodes, and transfers and result dependencies as edges. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual Statements Model-conditional excess distance

Original text:

The distances of the high-redshift SNe Ia are, on average, 10% to 15% farther than expected in a low mass density ($\Omega_M = 0.2$) Universe without a cosmological constant.

Source location: Abstract. Riess et al. compare high-redshift supernova distances obtained by two methods using light-curve shape with the expectation for a low-matter-density model without a cosmological constant.

Principal sources of uncertainty in supernova cosmology

Original text:

As the use of SNe Ia for measuring Ω_M and Ω_Λ progresses from its infancy into childhood, we can expect a similar shift in the discussion from results limited principally by statistical errors to those limited by our depth of understanding of SNe Ia.

Source location: Section 1. In discussing the development of supernova cosmology, Riess et al. explain the growing importance of photometric calibration, uniform treatment of nearby and distant samples, extinction and understanding of supernova physics as samples increase.

Difference between two fitting methods

Original text:

Until then, we must consider the difference between the cosmological constraints reached from the two fitting methods to be a systematic uncertainty.

Source location: Section 5.6. Riess et al. compare the scatter and mean difference of MLCS and template distances for the same high-redshift sample and retain the difference as a systematic uncertainty until future samples can distinguish the methods.

Common Evidence with Specified Scope The table gives the objects, predicates and domains of the three statements.

State- ment	Object	Predicate	Explicit Domain
S39- Q001	Inferred high-redshift supernova distances and the comparison model with $\Omega_M = 0.2$ and $\Lambda = 0$	Riess et al. state that the inferred distances exceed the model expectation by 10%–15% on average.	The samples, light-curve shapes, distance calibration, corrections and comparison model in the abstract.
S39- Q002	Principal limitations on estimates of Ω_M and Ω_Λ from Type Ia supernovae	Riess et al. expect the principal limitation to shift from statistical errors to understanding of supernovae.	The discussion of observational precision, sample size, supernova physics, calibration and systematics in Section 1.
S39- Q003	The difference between cosmological constraints from MLCS and template fitting	Riess et al. state that the difference must be treated as a systematic uncertainty.	The high-redshift sample, two fitting methods and future discrimination between them in Section 5.6.

Individual Audit Comparison model and inferred distances

The first statement is Riess et al.’s inference that their estimated high-redshift supernova distances exceed the comparison model with $\Omega_M = 0.2$ and $\Lambda = 0$ by 10%–15% on average. Its ground is the distance–redshift relation obtained from time-indexed photometry and spectroscopy, nearby calibration, phase-dependent corrections and two distance-fitting methods. The adopted samples, calibration, corrections and FRW comparison model define the domain of this excess distance. Claims 17–22 and Results 7–9 correspond to the statement.

Supernova understanding and the error budget

The second statement is Riess et al.’s expectation of a shift in the principal limiting factors as supernova cosmology develops. Its ground is the relation among sample size, photometric precision, calibration, extinction and supernova physics discussed in Section 1. Understanding of supernova and population evolution, observation plans and calibration conditions this expectation and the assessment of systematics. Claims 31–37 and Results 16–22 correspond to the statement.

Fitting-method differences for a shared sample

The third statement gives Riess et al.’s requirement to treat the difference between MLCS and template cosmological constraints as a systematic uncertainty. Its ground is the mean distance difference and scatter for the same high-redshift sample and their propagation into cosmological constraints. The difference is retained as an uncertainty under the shared-object and calibration conditions and until future samples distinguish the methods. Claim 24 and Result 15 correspond to the statement.

8.19.40 Source Audit

Audit Inputs

Source and Audit Scope

Source Identifier 40.

Source Name Primitive-Time-Root Audit of Adler 2003

Bibliographic Identification Stephen L. Adler, *Why Decoherence Has Not Solved the Measurement Problem: A Response to P. W. Anderson*, *Studies in History and Philosophy of Modern Physics* 34, 135–142 (2003), DOI: 10.1016/S1355-2198(02)00086-2; arXiv:quant-ph/0112095v3. Let $R = \text{Ref}_{\text{Adl03MP}}$. Page numbers refer to the arXiv text. The audit covers the abstract on p. 2, the argument on pp. 3–11, Equations (1)–(6b), footnotes 1 and 2, and the conclusion.

Audit-Scope Identifier 40-SC1.

Audit-Scope Content The scope specified in the bibliographic identification.

Primary-Source-Location Identifier 40-L1.

Primary-Source-Location Content The source-location keys 1–8 below.

Formal Objects

Object Identifier 40-F1.

Object Name Formal objects of the source

Object Type Formal object

Formal Content Definition 8.491 specifies nine time contents and nine auxiliary objects; Definition 8.492 specifies 24 audit claims comprising 20 source claims and four physical initial components.

Primary-Source-Location Identifier Set 40-L1.

Assumptions

Assumption Identifier 40-A1.

Assumption Content The explicit assumptions in Definition 8.490.

Primary-Source-Location Identifier Set 40-L1.

Nontemporal Physical Inputs

Input Identifier 40-P1.

Input Name Physical carriers and spatial configurations

Input Type Nontemporal physical input

Input Content The four components in Table 369. The initial time, cT defining the causal region, the operator's temporal scale, and the temporal components of stochastic differentials are supplied separately as time contents.

Primary-Source-Location Identifier Set 40-L1.

Time Contents

Time-Content Identifier 40-T1.

Time-Content Name Time contents of evolution, measurement, and stochastic differentials

Time-Content Type Time content

Time-Content Description The nine contents in Table 366.

Primary-Source-Location Identifier Set 40-L1.

Formal Input Elements

Definition 8.490 (Formal Objects and Explicit Assumptions of Adler 2003). The source assumes an initial product state of system, apparatus, and environment and deterministic unitary evolution generated by a self-adjoint Hamiltonian. It assumes a finite effective Hilbert space and restricts the environment to $R = cT$, using the allotted measurement duration T and light speed c . The scattering model neglects environment–environment and environment–system interactions and uses an elastic, static, instantaneous apparatus-scattering approximation without internal excitation or recoil. The asymptotic decay uses factors generally smaller than 1 in magnitude and a factor count growing approximately linearly with elapsed time. The orthogonal-outcome argument assumes the same normalized initial state and the same unitary action. The stochastic alternative uses dt and dW_t , with $dW_t^2 = dt$.

Definition 8.491 (Time Contents and Auxiliary Objects of Adler 2003). Set $n_R = 9$ and define nine time contents and nine auxiliary objects by the following two tables.

Table 366: Time Contents and Source Locations in Adler 2003

Group	Time content i	Time content	Auxiliary indices	Source-location key
1	1	Evolution time t , including its origin $t = 0$	None	2,3,4,5
2	2	The allotted measurement duration T and endpoint $t = T$	None	3
3	3	The light speed c used in the causal region $R = cT$	None	3
4	4	The rate Λ in the asymptotic decay $\exp(-\Lambda t)$	None	5
5	5	The temporal ordering of reading the record long after T	None	3
6	6	The stochastic-evolution index t	None	8
7	7	The time differential dt in stochastic differential evolution	None	8
8	8	The stochastic differential dW_t obeying $dW_t^2 = dt$	None	8
9	9	The temporal scale carried by the Hamiltonian H in $\exp(-iHt)$	None	3,7,8

The duration T has endpoint $t = T$ relative to the origin $t = 0$. The scattered-particle partition in Equation (5a) uses the variable t . Later readout is ordered long after T . The spatial range $R = cT$ is determined by time contents 2 and 3; $\exp(-\Lambda t)$ is the asymptotic form using time contents 1 and 4. The operator-algebra structure of H is auxiliary object 3, and its temporal scale is time content 9.

Table 367: Auxiliary Objects of Adler 2003

a	Auxiliary object	Source-location key
1	The normalized two-state superposition of the system and the system–apparatus–environment product state	2
2	Hilbert space, inner product, normalization, and the algebraic structure of unitary operators	2,5
3	The operator-algebra structure of the Hamiltonian and the exponential-operator construction	3
4	The measurement interaction and branch states of the system, apparatus, and environment	3,4,5
5	Environmental particles, single-particle scattering matrices, and the scattering approximations	4,5
6	The recording device, record states, and an observer arriving from outside	3
7	The spatial region of the apparatus and environment	2,3

a	Auxiliary object	Source- location key
8	The linear and nonlinear stochastic contributions to state-vector evolution	7,8
9	Discrete outcomes, Born-rule probabilities, and apparatus conditions for coherence and reduction	6,7,8

Definition 8.492 (Physical and Formal Claims in Adler 2003). Let $\text{Node}_R^{\text{audit}, \text{claim}} = \text{ClaimNodes}_R(24)$. Claims 1–4 are physical initial components, and Claims 5–24 correspond to the 20 source claims. These 24 claims constitute the source-formal-evidence environment. A claim’s dependency key d denotes a pair of time-content and auxiliary-object index sets; ℓ is its source-location key.

The dependency keys correspond to the following pairs of time-content and auxiliary-object index sets.

- 1** ($\emptyset$; $\{2, 7\}$)
- 2** ($\{1\}$; $\{1, 2\}$)
- 3** ($\{1\}$; $\{1\}$)
- 4** ($\{2, 3\}$; $\{7\}$)
- 5** ($\{1, 9\}$; $\llbracket 1, 3 \rrbracket$)
- 6** ($\{1, 2\}$; $\{4\}$)
- 7** ($\{2, 5\}$; $\{6, 7\}$)
- 8** ($\{1\}$; $\{1, 2, 4\}$)
- 9** ($\{1\}$; $\{5\}$)
- 10** ($\{1\}$; $\{4, 5\}$)
- 11** ($\{1\}$; $\{2, 4, 5\}$)
- 12** ($\{1, 4\}$; $\{5\}$)
- 13** ($\emptyset$; $\{2, 4\}$)
- 14** ($\emptyset$; $\{2\}$)
- 15** ($\emptyset$; $\{2, 9\}$)
- 16** ($\{1\}$; $\{1, 3, 9\}$)
- 17** ($\llbracket 6, 9 \rrbracket$; $\{3, 8\}$)
- 18** ($\llbracket 6, 9 \rrbracket$; $\{3, 8, 9\}$)
- 19** ($\emptyset$; $\{2, 4, 9\}$)

20 ($\emptyset; \emptyset$)

The source-location keys correspond to the following locations.

- 1 p. 2 (abstract) and pp. 3–4: decoherence and the measurement problem
- 2 p. 5: states, system division, effective Hilbert space, and Equations (1) and (2)
- 3 p. 6: measurement duration, causal region, unitary evolution, Equation (3), records, and observer
- 4 p. 7: Equation (4), scattering approximations, and Equations (5a) and (5b)
- 5 p. 8: asymptotic decay, Equations (6a) and (6b), and the unitary inner-product argument
- 6 p. 9: repeated trials, interpretations, and footnote 1
- 7 p. 10: the initial-state proposal, introduction of stochastic evolution, and footnote 2
- 8 p. 11: stochastic differentials, phenomenological assessment, and conclusion

Table 368: Physical and Formal Claims in Adler 2003

p	Content	d	ℓ
1	Physical carriers constituting the system	20	2
2	The apparatus and recording-device carriers and spatial configuration	20	2,3
3	The environmental-particle carriers and spatial configuration	20	2,3,4
4	The observer as a physical carrier	20	3
5	The source partitions system, apparatus, and environment using particle species conserved at low energies and assumes a finite effective Hilbert space.	1	2
6	A normalized two-state superposition is the initial system state.	2	2
7	The initial total state is a product of system, apparatus, and environment states.	3	2
8	The allotted duration T and light speed c delimit environmental particles within $R = cT$.	4	3
9	The total state evolves deterministically under $U = \exp(-iHt)$.	5	3
10	For a suitably designed apparatus, the apparatus–environment branches in Equation (3) become macroscopically distinguishable by T .	6	3
11	An observer arriving from outside the causal region can read the state recorded at T long afterward.	7	3
12	The apparatus–environment branch overlap in Equation (4) decays from its initial value 1 as time evolves, suppressing interference between the system states.	8	4

p	Content	d	ℓ
13	The scattering model neglects environment–environment and environment–system interactions and uses an elastic, static, instantaneous apparatus-scattering approximation without internal excitation or recoil.	9	4
14	The products in Equation (5a) partition particles into those that have and have not scattered by the variable time t .	10	4
15	Equation (5b) approximates the apparatus–environment overlap by the apparatus overlap and a product of single-particle scattering factors.	11	4
16	When the factors generally have magnitude below 1 and their number grows approximately linearly with elapsed time, the large- t decay has the form $\exp(-\Lambda t)$.	12	5
17	The candidate outcomes in Equations (6a) and (6b) are individually normalized and mutually orthogonal.	13	5
18	The overlap of two vectors obtained by applying the same unitary operator U to the same normalized state $ 0\rangle$ is 1.	14	5
19	Decay of the apparatus–environment overlap does not remove the incompatibility between a common initial state, a common deterministic evolution, and distinct orthogonal single outcomes.	13	5,6
20	The source discusses many-worlds accounts as adding a measure and Bohmian and related accounts as adding an auxiliary configuration.	15	6
21	The source considers a proposal based on unspecified apparatus or environment initial states and reports a negative analysis within its stated scope.	16	7
22	The state differential receives a contribution proportional to dt and a small nonlinear contribution proportional to dW_t , with $dW_t^2 = dt$.	17	7,8
23	The source assesses the stochastic approach as a viable phenomenological solution to the measurement problem that may ultimately be falsified experimentally.	18	8
24	The source concludes that decoherence has yet to achieve that status in the absence of a detailed theory leading to stochastic outcomes with the correct properties.	19	8

Definition 8.493 (Semantic Components and Generation-Initial Set in Adler 2003). Let $\text{Node}_R^{\text{audit,semantic}} = \text{Node}_R^{\text{audit,claim}}$. The first column gives the claim index of each physical initial component, and the third gives its associated source-claim indices. Set the initial-set/evidence-environment pairs to $\text{InitEnv}_R^{\text{audit}} = \text{InitEnv}_R^{\text{phys}} = \text{ClaimPrefix}_R(4)$.

Table 369: Generation-Initial Components in Adler 2003

p	Physical initial component	Associated source-claim indices
1	Physical carriers constituting the system	$\llbracket 5, 7 \rrbracket$
2	The apparatus and recording-device carriers and spatial configuration	$\{5, 10, 11\}$
3	The environmental-particle carriers and spatial configuration	$\{5, 8, 13\}$
4	The observer as a physical carrier	$\{11\}$

Dependent Inputs

Existing-Evidence Reference Set 8.491, 8.492.

Dependency and Transfer Elements

Definition 8.494 (Time-Content Transfers in Adler 2003). Let $\text{ldx}_R^{\text{audit,transfer}} = [1]$. The first component of each input or output is a time-content index set and the second an auxiliary-object index set; formal support is a claim-index set.

Table 370: Time-Content Transfers in Adler 2003

j	Input (time; auxiliary)	Output (time; auxiliary)	Formal support	Transfer Content
1	$(\{6, 8\}; \{8\})$	$(\{7\}; \emptyset)$	$\{22\}$	The time differential supplied by the stochastic differential rule $dW_t^2 = dt$

Conditions of Validity The transfer uses the stochastic differential rule $dW_t^2 = dt$ on p. 11. This is a multiplication rule for stochastic differentials with the common index t , not an equality between the square of an observed finite stochastic increment and its elapsed interval. The measurement duration T and the ordering of later readout are supplied time contents 2 and 5; the scattering partition uses t in content 1.

Source Support Claim 22 supports the transfer. The support, input, and output indices in the table define the corresponding sets in Definition 7.105. The transfer presupposes time contents 6 and 8.

Results

Definition 8.495 (Results in Adler 2003). Let $\text{ldx}_R^{\text{audit,result}} = [8]$. The table gives each result and its source location.

Table 371: Results and Source Locations in Adler 2003

j	Result	Source-location key
1	Decay of the apparatus–environment branch overlap	3,4,5
2	Suppression of interference between system states	3,4,5
3	Logical separation between overlap decay and selection of a single outcome	5,6
4	Incompatibility of distinct orthogonal outcomes with the same unitary action on the same normalized state	5,6
5	A phenomenological stochastic-reduction solution	7,8
6	Recording at the end of the measurement period and later observer readout	3
7	State evolution using time and stochastic differentials	7,8
8	The environmental causal range $R = cT$ during the measurement interval	3

Result Dependencies

Definition 8.496 (Result-Dependency Hyperedges in Adler 2003). The following table defines the temporal prerequisites, auxiliary prerequisites, and supporting claims of each result edge, as in Definition 7.105.

Table 372: Result-Dependency Hyperedges in Adler 2003

j	Temporal prerequisites	Auxiliary prerequisites	Formal support
1	$\{1, 4, 9\}$	$[5]$	$\{3, 9, 12, 13, 14, 15, 16\}$
2	$\{1, 4, 9\}$	$[5]$	$\{3, 9, 12, 13, 14, 15, 16\}$
3	$\emptyset$	$\{2, 4\}$	$\llbracket 17, 19 \rrbracket$
4	$\emptyset$	$\{2\}$	$\{17, 18\}$
5	$\llbracket 6, 9 \rrbracket$	$\{3, 8, 9\}$	$\{22, 23\}$
6	$\{2, 5\}$	$\{4, 6, 7\}$	$\{2, 4, 10, 11\}$
7	$\llbracket 6, 9 \rrbracket$	$\{3, 8\}$	$\{22\}$
8	$\{2, 3\}$	$\{7\}$	$\{3, 8\}$

There are 18 temporal-prerequisite occurrences, 22 auxiliary-prerequisite occurrences, and 28 supporting-claim occurrences.

Grounds of Establishment

Proposition 8.497 (Audit-Boundary Cardinalities in Adler 2003). The audit boundary has 41 nodes and nine hyperedges.

Proof. The 24 claim nodes, nine time contents, and eight results give $24 + 9 + 8 = 41$ nodes; one transfer and eight result edges give $1 + 8 = 9$ hyperedges. $\square$

Definition 8.498 (Evidential Scope of Adler 2003). Results 1, 2, and 8 are conditional theoretical calculations; Results 3 and 4 are formal logical analyses; Result 5 is the source’s phenomenological assessment; Result 6 describes the arrangement of recording and readout; and Result 7 constructs stochastic differential evolution. The antecedents of Results 3 and 4 are the same normalized state and the same unitary action, and their conclusions hold without substituting a time value. The physical process $U(t)$, overlap decay, and measurement and readout use the time contents in the table.

Evidence Hierarchy

Direct-Measurement Objects The source supplies zero direct-measurement objects.

Direct-Measurement Predicates The source supplies zero direct-measurement predicates.

Conditions and Domain of Direct Measurement The source supplies zero direct-measurement conditions.

Objects of Formal Generalization The objects are branch-overlap decay and interference suppression, incompatibility with the single-outcome requirement, and additional stochastic dynamics.

Conditions and Domain of Formal Generalization The source assumes an initial product state of system, apparatus, and environment and deterministic unitary evolution generated by a self-adjoint Hamiltonian. It assumes a finite effective Hilbert space and restricts the environment to $R = cT$, using the allotted measurement duration T and light speed c . The scattering model neglects environment–environment and environment–system interactions and uses an elastic, static, instantaneous apparatus-scattering approximation without internal excitation or recoil. The asymptotic decay uses factors generally smaller than 1 in magnitude and a factor count growing approximately linearly with elapsed time. The orthogonal-outcome argument assumes the same normalized initial state and the same unitary action. The stochastic alternative uses dt and dW_t , with $dW_t^2 = dt$.

Generation Judgment **Judgment values:** complete generation 0 / local reach 0 / remaining roots 9.

Source-Internal Evidence-Conditioned Local Reach The source-formal-evidence environment alone reaches zero time contents, leaving all nine as primitive time roots.

Result Dependencies Results 1, 2, and 5–8 depend on primitive time roots; Results 3 and 4 are reachable from the source-formal-evidence environment alone.

Evidential Force

Definition 8.499 (Evidential Scope and Boundaries of Adler 2003). The scope comprises apparatus–environment overlap decay and suppression of system interference, the logical constraint on a common initial state and unitary action, recording and later readout,

additional stochastic differential dynamics, and the causal region $R = cT$. Adler assesses the stochastic approach as a viable phenomenological solution while allowing its possible falsification by future experiments.

Even when the apparatus–environment overlap decays, the overlap of two vectors obtained from the same normalized state by the same unitary action is 1. Distinct orthogonal single outcomes require changing these antecedents or adding structure. Evolution time, measurement duration, light speed, decay rate, the ordering of later readout, stochastic differentials, and the Hamiltonian’s temporal scale are temporal inputs preceding the source’s calculations and constructions.

Result-Preservation Domain

Definition 8.500 (Conditions for Preserving the Results of Adler 2003). Overlap decay and interference suppression hold under the initial-state, unitary-evolution, and scattering assumptions. Incompatibility with orthogonal outcomes assumes the same normalized state and the same unitary action. Recording and later readout use the measurement duration, apparatus configuration, and readout ordering; the stochastic alternative uses the linear and nonlinear stochastic terms and $dW_t^2 = dt$; the causal region uses c and T . If an LCTR generation sequence supplies the same order, scale, and correspondence for each temporal input, and these assumptions and construction rules are retained, the corresponding conditional results are preserved.

Placement of grounding responsibilities

Placement of grounding responsibilities The nine time contents audited in this source are not generated from nontemporal physical inputs before use; they are accepted as already established temporal inputs. The placement judgment for the physical-time grounding responsibility is "backgrounded by primitive time input." This judgment is based on comparing, for each time content, its use in a law operation or result derivation against the absence, before that use, of a corresponding generation condition or condition of validity stated as an input, formation condition, evaluation condition, or failure condition.

Evidence of actual failure

Evidence of actual failure The complete-generation judgment and the remaining-root judgment concern qualification as physical time and are distinct from actual failure of a concrete condition governing the value or domain of application of a result. Neither the source’s formal evidence nor the LCTR audit evidence establishes actual failure of such a target condition. The actual-failure judgment is "indeterminate from the original source evidence"; actual failure is not inferred from the backgrounding of a grounding responsibility alone.

Downstream reach

Downstream reach The scope of reach to results dependent on primitive time roots is determined source by source by the primitive-time-root/result dependency relation reconstructed in this division and by minimal primitive-time-root support. This is structural reach. Because no actual failure of a concrete condition has been

established, the judgment of actual reach from a failure location to results is "indeterminate from the original source evidence." Structural reach does not substitute for actual reach.

Repair scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation conditions or conditions of validity required before their use. The recalculation scope is restricted to results reached by dependency from a failure location whose actual failure has been established independently. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from repaired inputs, or their domain of validity is restricted. Values after recalculation are determined as audit results.

Audit Conclusions

Proposition 8.501 (Primitive Time Roots and Result Dependencies in Adler 2003). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [9]; \emptyset, \{3, 4\}, \{1, 2, 5, 6, 7, 8\}, \emptyset).$$

Proof. The only transfer presupposes time contents 6 and 8, so neither nontemporal physical inputs nor the source-formal-evidence environment alone can initiate it; all nine time contents are primitive roots. Results 3 and 4 have no temporal prerequisites and are reached using Claims 17–19 in the source-formal-evidence environment; those claims are not supplied by the four physical initial components alone. The other six results have temporal prerequisites and are reached after all nine roots are added to that environment. The four-class result partition gives the displayed tuple. $\square$

Proposition 8.502 (Primitive-Time-Root/Result Dependencies in Adler 2003). For the minimal support families in Definition 7.108, there are eight family members in total, 16 root–result pairs, and 16 necessary-root–result pairs. The support family of each of Results 3 and 4 is $\{\emptyset\}$.

Table 373: Minimal Primitive-Time-Root Support in Adler 2003

Result index	Support statistics
1,2,5,7	(1; 3)
3,4	(1; 0)
6,8	(1; 2)

Proof. Apply Definition 7.46 to the sole transfer $\{6, 8\} \rightarrow \{7\}$ and the eight result edges. The minimal supports are $\{1, 4, 9\}$ for Results 1 and 2, $\{6, 8, 9\}$ for Results 5 and 7, $\{2, 5\}$ for Result 6, $\{2, 3\}$ for Result 8, and the empty set for Results 3 and 4. The table and pair counts follow. $\square$

Individual Statements Temporal decay of the apparatus–environment branch overlap

The statement concerns decay of the apparatus–environment branch overlap in the unitary measurement model. Original (English):

the rapid decay with time of the inner product

The source location is p. 7, the paragraph introducing Equation (4) of arXiv:quant-ph/0112095v3.

The minimal context is as follows. The overlap in Equation (4) is between apparatus–environment branch states and decays from its initial value 1. Equations (5a) and (5b) factorize it under the scattering approximations; p. 8 describes its asymptotic $\exp(-\Lambda t)$ decay using the time dependence of the factor count. The attributed author is Stephen L. Adler. The predicate is “the rapid decay with time”; the explicit domain is pages 6–8, Equations (3)–(5b), the initial state, unitary evolution, scattering approximations, and time-dependent factor count.

Qualification of decoherence as a solution

The statement concerns qualification of decoherence as a solution to the quantum measurement problem. Original (English):

decoherence has not solved the quantum measurement problem.

The source location is p. 2, the abstract of arXiv:quant-ph/0112095v3.

The minimal context is as follows. The source distinguishes suppression of interference through decay of the apparatus–environment overlap from selection of a single outcome. The argument on p. 8 uses the fact that two states obtained from the same normalized initial state by the same unitary action have overlap 1, which is incompatible with the orthogonality of Equations (6a) and (6b). Page 9 explains that only enough repeated trials to obtain each distinct outcome at least once are needed. The attributed author is Stephen L. Adler. The predicate is “has not solved the quantum measurement problem”; the explicit domain is the abstract on p. 2, pp. 8–9, Equations (6a) and (6b), and the requirements of a common normalized state, a common unitary action, and orthogonal outcomes.

Status as a phenomenological solution

The statement concerns the status of decoherence as a phenomenological solution to the measurement problem. Original (English):

Decoherence, in the absence of a detailed theory showing that it leads to stochastic outcomes with the correct properties, has yet to achieve this status.

The source location is p. 11, the concluding paragraph of arXiv:quant-ph/0112095v3.

The minimal context is as follows. The preceding sentence states that the stochastic approach constitutes a viable phenomenological solution to the measurement problem, although it may ultimately be falsified experimentally. The subject of the quoted sentence is decoherence. The absence of a detailed theory leading to stochastic outcomes with the correct properties is the condition under which it has yet to attain that status. The attributed author is Stephen L. Adler. The predicate is “has yet to achieve this status”; the explicit domain is the stochastic differential evolution on pp. 10–11 and the assessment of decoherence following the phenomenological assessment of the stochastic approach on p. 11.

Common Evidence with Specified Domain of Application The objects, predicates, and domains of the three statements are as follows.

Statement	Object	Predicate	Explicit domain
S40-Q001	Decay of the apparatus–environment branch overlap in the unitary measurement model	States that the overlap decays rapidly with time.	Pages 6–8, Equations (3)–(5b), the initial state, unitary evolution, scattering approximations, and time-dependent factor count.
S40-Q002	Qualification of decoherence as a solution to the quantum measurement problem	Judges that interference suppression alone does not supply single-outcome selection and has not solved the measurement problem.	The abstract on p. 2, pp. 8–9, Equations (6a) and (6b), and the requirements of a common normalized state, a common unitary action, and orthogonal outcomes.
S40-Q003	The status of decoherence as a phenomenological solution to the measurement problem	Judges that decoherence has yet to attain the status of a phenomenological solution in the absence of a detailed theory leading to stochastic outcomes with the correct properties.	The stochastic differential evolution on pp. 10–11 and the assessment of decoherence following the phenomenological assessment of the stochastic approach on p. 11.

Individual Audits **Temporal decay of the apparatus–environment branch overlap**

Statement S40-Q001 concerns the apparatus–environment branch overlap. Its independent basis is the scattering model in Equations (5a) and (5b) and the asymptotic calculation on p. 8, supporting overlap decay and interference suppression. The calculation uses the initial state, unitary evolution, scattering approximations, evolution time t , and rate Λ . The decaying overlap and the overlap of the total states, including the system, in Equations (6a) and (6b) have different objects. The corresponding formal evidence is Results 1 and 2 in Table 371 and Definition 8.499.

Qualification of decoherence as a solution

The independent basis of Statement S40-Q002 is the argument that, for $|A\rangle = U|0\rangle$ and $|B\rangle = U|0\rangle$ obtained from the same normalized state $|0\rangle$ by the same unitary action U , one has $\langle A|B\rangle = \langle 0|U^\dagger U|0\rangle = 1$. This is incompatible with orthogonality of the two outcomes. Adding decay of the apparatus–environment branch overlap leaves this implication unchanged. Its domain is the common-state, common-action antecedent, corresponding to the simplified measurement problem identified in footnote 1 on p. 9. The corresponding formal evidence is Results 3 and 4.

Status as a phenomenological solution

Statement S40-Q003 compares the status attained by decoherence with the status of a phenomenological solution just accorded to the stochastic approach. The source grounds that assessment of the stochastic approach in discrete outcomes with Born-rule probabilities, maintenance of coherence under conditions where it is observed, and state reduction under measurement-apparatus conditions. It attributes detailed analyses of these properties to the cited literature; the paragraph supplies the stochastic differential construction and phenomenological assessment. Results 3 and 4 correspond to the nonattainment judgment for decoherence, while Results 5 and 7 correspond to the construction and assessment of the stochastic alternative.

Common Audit with Specified Domain of Application The three statements connect three objects: decay of the apparatus–environment overlap, orthogonal single outcomes of the total state, and additional stochastic dynamics. The decoherence calculation evaluates a time-dependent overlap. The single-outcome incompatibility is a formal consequence of the common-state, common-action antecedent. The stochastic alternative uses time and stochastic differentials. In Table 372, Results 3 and 4 have no temporal prerequisites, while the other six results depend on primitive time roots.

Boundary of Evidential Force

The source’s negative logical judgment assumes a common initial state and common deterministic action. Its assessment of the stochastic approach carries the stated phenomenological properties and possibility of falsification. The generation judgment for the nine time contents and the four-class partition of the eight results are given by Proposition 8.501.

8.19.41 Source audit

Audit inputs

Source and audit scope

Source identifier 41.

Source name Primitive-time-root audit of Philip W. Anderson 2001

Bibliographic identification Let $R = \text{Ref}_{\text{And01DW}}$ denote Philip W. Anderson, *Science: A “Dappled World” or a “Seamless Web”?*, *Studies in History and Philosophy of Modern Physics* 32(3), 487–494 (2001), DOI: 10.1016/S1355-2198(01)00011-9. The audit scope is pp. 487–494.

Audit-scope identifier 41-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 41-L1.

Source-location content The audit domain stated in the bibliographic identification and the source-location columns of the respective evidence tables.

Formal object

Object identifier 41-F1.

Object name Claims and examples in the essay.

Object type Formal object.

Formal content The objects of the argument and their audit records are given in Definitions 8.503 and 8.505.

Source-location identifier set 41-L1.

Assumptions

Assumption identifier 41-A1.

Assumption content The conditions and examples used in the argument are given in Definitions 8.503 and 8.505.

Source-location identifier set 41-L1.

Nontemporal physical inputs

Input identifier 41-P1.

Input name Nontemporal physical inputs of the source.

Input type Nontemporal physical input.

Input content The source discusses existing theories and experiments. It specifies no tuple of nontemporal physical inputs for an operation performed within the source. The audit initial set and evidence environment are given in Definition 8.505.

Source-location identifier set 41-L1.

Time contents

Time-content identifier 41-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content specification The individual time contents are given in Definitions 8.504 and 8.503.

Source-location identifier set 41-L1.

Formal input elements

Definition 8.503 (Objects and conditions of the argument in Philip W. Anderson 2001). The source is an essay review criticizing Cartwright's conception of science. The claims and examples in Table 376 are attributed to Anderson.

The history of BCS validation, equations of motion and initial conditions, radiocarbon dating, repeated observation, decoherence and atomic-beam verification, universality, broken symmetry, and renormalization enter the argument about theoretical interconnections and scientific understanding. The BCS validation interval 1957–1965 is approximate.

Decoherence, the former collapse terminology, and atomic-beam verification form one continuous discussion on p. 492. Universality, broken symmetry, and renormalization are discussed in terms of their conceptual roles and relations between scales.

The validation interval, career duration, historical order, state evolution, date, observational order, and process order belong to the time contents in Table 375.

Definition 8.504 (Time contents in Philip W. Anderson 2001). For source $R = \text{Ref}_{\text{And01DW}}$, let $n_R = 7$. The time contents and source locations are shown in Table 375.

Table 375: Time contents and source locations in Philip W. Anderson 2001

i	Time content	Source location
1	Period over which the BCS model was verified	p. 490
2	Duration of a research career	pp. 489–490
3	Historical order of theory formation and connections among theories	pp. 487–494
4	State evolution given by equations of motion	p. 491
5	Location of a past event given by radiocarbon dating	p. 491
6	Order of repeated observations	pp. 491–492
7	Order of the process described as decoherence and of atomic-beam verification	p. 492

Definition 8.505 (Audit claim records for Philip W. Anderson 2001). Let the audit claim nodes be $\text{ClaimNodes}_R(14)$ and the initial-set/evidence-environment pair be $\text{InitEnv}_R^{\text{audit}} = \text{ClaimPrefix}_R(0)$. The physical-input initial set is empty; the fourteen claims belong to the evidence environment of source statements. Table 376 gives their content and source locations.

Table 376: Audit claim records for Philip W. Anderson 2001

Claim	Author's account	Source location
1	History of validation of the BCS model and its domain in ordinary metals, described as a theory controlled by a small parameter about a mean-field solution.	pp. 489–491
2	Distinction between knowing the equations of motion and predicting the detailed outcome of a chaotic fluid.	p. 491
3	The conception of prediction as calculating outcomes from initial conditions.	p. 491
4	Example of understanding the sequence of past events by correlating radiocarbon dates.	p. 491
5	Verification of predictions and its use in subsequent observation and prediction.	pp. 491–492
6	The author's criticism that decoherence is absent from the discussion of the measurement problem.	p. 492
7	The author's description linking the process described by decoherence to the process formerly called collapse of the wave function.	p. 492
8	The author's report that the concept has been experimentally verified by atomic-beam techniques quantifying the whole process.	p. 492
9	Explanation of using a simple model belonging to the same universality class.	pp. 490, 492

Claim	Author's account	Source location
10	Explanation of broken symmetry as a principal way in which complexity can emerge from fundamental laws.	p. 491
11	Passage from atomic to macroscopic scales through renormalization and the justification of model selection.	p. 492
12	The position that science is an interconnected, growing and self-correcting web within and across fields.	pp. 488–489, 493
13	The author's criticism of Cartwright's accounts of BCS history and scientific discovery.	pp. 489–490, 493
14	The author's experience of involvement in the theory and practice of superconductivity for 43 years.	pp. 489–490

Dependent inputs

Index identifier S41-XI-DI1.

Existing-evidence reference set Definition 8.506.

Dependency and transfer elements

Definition 8.506 (Time-content transfers in Philip W. Anderson 2001).

Define the transfer index set by $\text{ldx}_R^{\text{audit,transfer}} = \emptyset$. The numbers of transfer events to which transfer inputs, transfer outputs, and source support are assigned are therefore 0, 0, 0, respectively. The judgment that the set of transfer events is empty applies to pp. 487–494. The BCS model, equations of motion, radiocarbon dating, repeated observation, the account of decoherence, and atomic-beam verification juxtaposed within that scope do not define source-specific operations having time contents as domains and codomains.

Conditions of validity

Index identifier S41-XI-FC1.

Existing-evidence reference set Definition 8.506.

Source support

Index identifier S41-XI-SS1.

Existing-evidence reference set Definition 8.506.

Results

Definition 8.507 (Results of Philip W. Anderson 2001). Define the result index set by $\text{ldx}_R^{\text{audit,result}} = [11]$. Let the time-content premise index set and formal-claim support index set of the j -th result be $J_{R,j}^{\text{res}}$ and $P_{R,j}^{\text{res}}$, respectively. The sets, results, basis types, and source locations are specified in Table 377.

The correspondence between source-location keys and source locations in Table 377 is as follows.

- 1 Historical and philosophical argument; pp. 487–494
- 2 Use of existing observational and verification cases within the argument; pp. 489–491
- 3 Conceptual roles of universality and symmetry breaking; pp. 490–492
- 4 Conceptual distinction between equations of motion and initial conditions, and macroscopic understanding; p. 491
- 5 Use of existing observational and verification cases within the argument; pp. 491–492
- 6 Explanatory comparison of decoherence, collapse, and the measurement problem; p. 492
- 7 Use of existing observational and verification cases within the argument; p. 492
- 8 Theoretical structures of universality, symmetry breaking, and renormalization; pp. 490–492

Table 377: Results, audit inputs, basis types, and source locations in Philip W. Anderson 2001

j	Result	$J_{R,j}^{\text{res}}$	$P_{R,j}^{\text{res}}$	Source-location key
1	The position, supported by historical examples, that science forms an interconnected web	$\{1, 3\}$	$\{1, 9, 10, 11, 12\}$	1
2	Verification, extension of the domain of application, and maturation of the BCS model	$\{1, 3\}$	$\{1\}$	2
3	Account of the roles of universality and symmetry breaking	$\emptyset$	$\{9, 10\}$	3
4	Distinction between detailed prediction and understanding of macroscopic behavior	$\{4\}$	$\{2, 3\}$	4
5	Understanding the past through radiocarbon dating	$\{5\}$	$\{4\}$	5
6	Relation between repeated observations and prediction of the next observation	$\{6\}$	$\{5\}$	5

j	Result	$J_{R,j}^{\text{res}}$	$P_{R,j}^{\text{res}}$	Source-location key
7	Criticism of the omission of decoherence from the account of the measurement problem	$\{7\}$	$\{6, 7, 8\}$	6
8	The author's explanation linking decoherence to the former collapse terminology	$\{7\}$	$\{7\}$	6
9	The author's report of quantitative verification by atomic-beam techniques	$\{7\}$	$\{8\}$	7
10	Account of scale connections and model selection through renormalization	$\emptyset$	$\{9, 11\}$	8
11	Critique of the account of scientific discovery based on BCS history and research experience	$\{1, 2, 3\}$	$\{1, 13, 14\}$	1

Proposition 8.508 (Audit-boundary cardinality for Philip W. Anderson 2001). The audit boundary of source $R = \text{Ref}_{\text{And01DW}}$ has 32 nodes and 11 hyperedges.

Definition 8.509 (Evidential scope of the results of Philip W. Anderson 2001). **Grounds of validity of the results.** The basis type of Results 1 and 11 is historical and philosophical argument; that of Results 2, 5, 6, and 9 is the use of existing observational and verification cases within the argument; that of Results 3 and 10 is the source's conceptual account; that of Result 4 is conceptual distinction; and that of Results 7 and 8 is explanatory comparison. **Directly measured objects.** This source contains zero new directly measured objects. **Direct-measurement predicates.** This source contains zero new direct-measurement predicates. **Conditions and domain of application of direct measurement.** This source contains zero new direct-measurement conditions. **Objects of formal generalization.** The objects of formal generalization comprise the following six groups.

1. Generalization from the history of verification of the BCS model to theoretical maturation and extension of its domain of application

2. Connections among theoretical levels through universality, symmetry breaking, and renormalization
3. Distinction between prediction by equations of motion and macroscopic understanding
4. Generalization from radiocarbon dating and repeated observation to understanding the past and predicting the next observation
5. Criticism of the account of the measurement problem using the description of decoherence and the statement about atomic-beam verification
6. Historical and philosophical generalization from individual cases to an image of connections among scientific theories as a whole

Conditions and domain of application of formal generalization. The four conditions of application for these generalizations are as follows.

1. The BCS, dating, repeated-observation, and decoherence/atomic-beam verification examples are subject to the scopes stated in the source.
2. The evidential force of existing experiments belongs to their respective primary sources.
3. State evolution by equations of motion and initial conditions presupposes Time Content 4.
4. Dating, repeated observation, and the history of verification presuppose Time Contents 1–3 and 5–7.

Result dependencies For each row of Table 377, take the support-claim indices and time-content premise indices as the corresponding index sets in Definition 7.105. The support set and time-content premise set of each result are thereby determined.

Grounds of validity

Proof. The evidence environment of source statements has 14 nodes, the time-content set has 7 nodes, and the result set has 11 nodes. The time-content transfers comprise 0 hyperedges, and the result dependencies comprise 11 hyperedges. By Proposition 7.29, $14 + 7 + 11 = 32$, $0 + 11 = 11$. $\square$

Evidential hierarchy

Index identifier S41-XI-MT1.

Existing-evidence reference set Definition 8.509.

Generation judgment Judgment values: LCTR complete generation 0 / source-internal evidence-conditioned local reach 0 / remaining primitive time roots 7.

Source-internal evidence-conditioned local reach The source-internal evidence-conditioned local-reach judgment count is 0, and 7 primitive time roots remain after this judgment.

Result dependencies

Index identifier S41-XI-RD1.

Existing-evidence reference set Evidence 8.511.

Evidential force

Index identifier S41-XI-EE1.

Existing-evidence reference set Definition 8.512.

Result-preservation domain

Index identifier S41-XI-RPD1.

Existing-evidence reference set Definition 8.513.

Placement of grounding responsibilities

Placement of grounding responsibilities The seven time contents audited in this source are not generated from nontemporal physical inputs before use; they are accepted as already established temporal inputs. The placement judgment for the physical-time grounding responsibility is "backgrounded by primitive time input." This judgment is based on comparing, for each time content, its use in a law operation or result derivation against the absence, before that use, of a corresponding generation condition or condition of validity stated as an input, formation condition, evaluation condition, or failure condition.

Evidence of actual failure

Evidence of actual failure The complete-generation judgment and the remaining-root judgment concern qualification as physical time and are distinct from actual failure of a concrete condition governing the value or domain of application of a result. Neither the source's formal evidence nor the LCTR audit evidence establishes actual failure of such a target condition. The actual-failure judgment is "indeterminate from the original source evidence"; actual failure is not inferred from the backgrounding of a grounding responsibility alone.

Downstream reach

Downstream reach The scope of reach to results dependent on primitive time roots is determined source by source by the primitive-time-root/result dependency relation reconstructed in this division and by minimal primitive-time-root support. This is structural reach. Because no actual failure of a concrete condition has been established, the judgment of actual reach from a failure location to results is "indeterminate from the original source evidence." Structural reach does not substitute for actual reach.

Repair scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation conditions or conditions of validity required before their use. The recalculation scope is restricted to results reached by dependency from a failure location whose actual failure has been established independently. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from repaired inputs, or their domain of validity is restricted. Values after recalculation are determined as audit results.

Audit conclusions

Proposition 8.510 (Primitive time roots and result dependencies in Philip W. Anderson 2001). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \left(\emptyset, [7]; \right. \\ \left. \emptyset, \{3, 10\}, [11] \setminus \{3, 10\}, \emptyset \right)$$

holds. Because the transfer-event set is empty, the transfer-input source set is also empty.

Proof. The physical-input initial set is empty, and there are no time-content transfers. Each result edge requires source claims, so neither time contents nor results are reachable from the physical-input initial set.

Adding the source statements as the evidence environment supplies the claim inputs for Results 3 and 10. Their time inputs are empty, so they are reachable without adding a time root. The remaining nine results require the time contents in Table 377; supplying all seven roots reaches all eleven results. No edge outputs a time content, and all seven time contents remain primitive time roots. Definition 8.115 gives the displayed index tuple. $\square$

Proposition 8.511 (Primitive-time-root/result dependency relations in Philip W. Anderson 2001). For each $q \in \text{Idx}_R^{\text{audit,result}}$, the cardinality of $\mathcal{A}_R(q)$ in Definition 7.108, together with the minimum and maximum cardinalities of its members, is given in Table 378. The support family of each of Results 3 and 10 is $\{\emptyset\}$, and the total number of family members across all results is 11. The primitive-time-root/result dependency relation and the necessary-primitive-time-root/result dependency relation each have 13 pairs.

Table 378: Minimal primitive-time-root support in Philip W. Anderson 2001

Result index	Support statistics	Result index	Support statistics
1–2	(1; 2)	3, 10	(1; 0)
4–9	(1; 1)	11	(1; 3)

Proof. Apply the update in Definition 7.46 to the eleven result-dependency hyperedges in Definition 8.507. Updating until the support antichains become invariant uniquely determines the minimal primitive-time-root support families of all results by Proposition 7.47. Applying Definition 7.50 and Definition 8.114 to these families uniquely determines the two dependency relations. $\square$

Definition 8.512 (Evidential scope and boundaries of Philip W. Anderson 2001). The evidential scope comprises the following six groups.

1. The history of verification and extension of the domain of application of the BCS model
2. The theoretical structures of universality, symmetry breaking, and renormalization
3. State evolution by equations of motion and initial conditions
4. Radiocarbon dating and repeated observation
5. Explanation of decoherence, collapse, and atomic-beam verification
6. Historical and philosophical argument concerning connections among levels of scientific theory

The boundaries of evidential force are the following five.

1. The local reach of Results 3 and 10 is established with the source's accounts of universality, broken symmetry, and renormalization as evidence. It is not generation from physical inputs alone.
2. The history of BCS verification, equations of motion, dating, repeated observation, the measurement problem, and the synthesis of the history of science presuppose time contents.
3. This source uses existing experiments as materials for its argument and conducts no new direct measurement within the source.
4. State evolution by equations of motion does not provide the physical generation of a time variable.
5. Radiocarbon dates, observational order, and historical order are not generated from nontemporal physical inputs.

Definition 8.513 (Conditions for preserving the results of Philip W. Anderson 2001). Results 3 and 10 retain the scope of the source's conceptual accounts. The remaining nine results are preserved within the source's historical and philosophical scope by establishing their respective time inputs in the LCTR generation order.

Individual statements Decoherence, collapse description, and experimental verification

The statement concerns the process described by decoherence, its relation to the former name "collapse of the wave function," and the scope of the object subjected to experimental verification. Original (English):

‘decoherence’ which describes the process that used to be called ‘collapse of the wave function’. The concept is now experimentally verified

The source location is Philip W. Anderson, *Science: A “Dappled World” or a “Seamless Web”?*, p. 492.

The minimal context is a discussion of the quantum measurement problem, decoherence, collapse processes, and verification by atomic-beam techniques. The quoted passage states in succession the object explained by decoherence, the former collapse designation, and experimental verification. The attributed author is Philip W. Anderson. The predicates assigned to the objects are that decoherence describes the process formerly called collapse, and that “The concept” is now experimentally verified. The explicit domain of application comprises the explanation of the quantum measurement problem, decoherence, and collapse on p. 492, and the subsequent discussion of atomic-beam verification.

Common evidence with specified domain of application The following statement gives the Anderson 2001 original, source location, minimal context, and source attribution.

State- ment	Object of statement	Predicate	Explicit domain of application
S41- Q001	The process described by decoherence, its relation to the former name “collapse of the wave function,” and the object subjected to experimental verification	States that decoherence describes the process formerly called “collapse of the wave function” and that “The concept” is experimentally verified.	The discussion on p. 492 of the source concerning the quantum measurement problem, decoherence, the collapse process, and subsequent verification by atomic-beam techniques.

Individual audits Decoherence, collapse description, and experimental verification

The original and its minimal context permit the following four interpretations.

1. An interpretation that treats decoherence and the process formerly called collapse as the same object of explanation
2. A restricted interpretation under which decoherence describes the observable process of interference suppression associated with collapse descriptions
3. An interpretation that attaches “The concept” to the concept of decoherence and states its experimental verification
4. An interpretation that attaches “The concept” to the entire immediately preceding process description and states the experimental verification of a broader process that includes collapse

The four possible interpretations coexist.

The independent basis is the explanatory relation between decoherence and the former collapse designation, and the existing cases of atomic-beam verification, which the source

places in the same descriptive sequence. Anderson uses existing experiments within a historical and philosophical argument.

The boundaries of evidential force under the respective interpretations are the following five.

1. Formal equivalence between decoherence and collapse does not follow from the same-object interpretation.
2. The restricted interpretation is consistent with experimental support for interference suppression but does not imply the generation of a single outcome with probabilities.
3. The experimental force of the decoherence-attachment interpretation belongs to the decoherence phenomena and verification cases mentioned in the source.
4. Direct experimental resolution of collapse as a whole or of the quantum measurement problem as a whole does not follow from the broader-attachment interpretation.
5. The physical generation of the time contents used by decoherence, collapse, and atomic-beam processes does not follow from this source.

The corresponding formal evidence is given in Table 375, Table 376, Table 377, Definition 8.512, and Proposition 8.510. Time contents reached from nontemporal inputs are 0/7, primitive time roots are 7/7, results generated under the complete judgment are 0/11, source-internal evidence-conditioned locally reachable results are 2/11, and root-dependent results are 9/11. The contrasting separation between interference suppression and the generation of a single outcome with probabilities is given in Definition 8.499.

8.19.42 Source audit

Audit inputs

Source and audit scope

Source identifier 42.

Source name Primitive-time-root audit of Clark 2001

Bibliographic identification Let Charles W. Clark, *Effects of Configuration Interaction on Intensities and Phase Shifts*, NIST Special Publication 958 (2001)[49] be source *R*. The audit domain comprises Equations (1)–(3) and the introductory and explanatory paragraphs on pp. 116–117.

Definition 8.514 (Semantic components in Clark 2001). The 36 semantic components consist of 11 nontemporal initial components, 22 source claims, and 3 mathematical-object types.

Audit-scope identifier 42-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 42-L1.

Source-location content The audit domain stated in the bibliographic identification and the source-location columns of the respective evidence tables.

Formal object

Object identifier 42-F1.

Object name Formal objects of the source.

Object type Formal object.

Formal content The individual formal objects are given in Definitions 8.514 and 8.515.

Source-location identifier set 42-L1.

Assumptions

Assumption identifier 42-A1.

Assumption content On p. 116 the source explicitly states that Equation (1) is derived under simplifying assumptions. Equations (1)–(2) use elapsed time $t \geq 0$ after state preparation, a constant rate $\Gamma > 0$, and the lifetime $\tau = 1/\Gamma$. Equation (3) uses the probability amplitudes and phase relations of alternative transition paths. The correspondence with Equation (2) for $q \gg 1$ is stated by the source for the region where the signal is strongest.

Source-location identifier set 42-L1.

Nontemporal physical inputs

Input identifier 42-P1.

Input name Nontemporal physical inputs of the source.

Input type Nontemporal physical input.

Input content The individual nontemporal physical inputs are given in Definitions 8.514 and 8.515. The initial components are nontemporal projections of the carriers and configurations of atoms, electrons, radiation fields, and apparatus; they exclude transition rates, oscillation periods, and elapsed time. The alternative transition paths are paths of quantum-state transitions. The algebraic structure and phase relations of probability amplitudes belong to the formal evidence of Source Claims 7 and 8 and Mathematical-Object Type 3.

Source-location identifier set 42-L1.

Time contents

Time-content identifier 42-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content specification The individual time contents are given in Definition 8.516.

Source-location identifier set 42-L1.

Formal input elements

Formal-component identifier 42-C1.

Formal-component name Formal-evidence environment of the source.

Formal-component type Formal component.

Formal-component text The individual formal input elements are given in Definitions 8.514 and 8.515.

Source-support identifier set 42-L1.

Table 380: Nontemporal initial components in Clark 2001

k	Semantic component	Corresponding claim indices
1	Physical carrier of a two-level atom	{1, 2}
2	Physical constitution of the high-energy state	{1}
3	Physical constitution of the low-energy state	{1}
4	Transition-coupling structure between the two states	{2}
5	Physical carrier of the radiation field	{3}
6	Physical carrier of a diffraction grating or prism	{5}
7	Spatial arrangement of the dispersive element	{5}
8	Screen, photographic plate, or detector array	{6}
9	Spatial arrangement of the recording apparatus	{6}
10	Atomic and electronic state configurations involved in alternative transitions	{7, 8}
11	Physical carrier of the cesium-133 atom	{19}

Definition 8.515 (Source claims in Clark 2001). The following table gives the 22 source claims.

Table 381: Source claims in Clark 2001

p	Claim	Source location
1	Atom with high- and low-energy states	p. 116, immediately before Equation (1)
2	Transition from the high state to the low state	p. 116, immediately before Equation (1)
3	Emission of light accompanying the transition	p. 116, immediately before Equation (1)
4	Energy difference between the two states	p. 116, energy-gap and angular-frequency paragraph before Equation (1)

p	Claim	Source location
5	Dispersion of radiation by a dispersive element	p. 117, immediately before Equation (2)
6	Registration by a screen or equivalent apparatus	p. 117, immediately before Equation (2)
7	Phase relation between alternative transition paths	p. 117, immediately before Equation (3)
8	Interference between alternative transition paths	p. 117, immediately before Equation (3)
9	Elapsed time measured from the instant of state preparation	Equation (1)
10	Exponential time dependence of radiation intensity	Equation (1)
11	Inverse relation between the rate constant and the lifetime of the high state	immediately after Equation (1)
12	Transition angular frequency from Planck's constant and the energy difference	p. 116, energy-gap and angular-frequency paragraph before Equation (1)
13	Angular-frequency spectral decomposition of radiation intensity	paragraph introducing Equation (2)
14	Lorentzian line shape corresponding to exponential decay	Equation (2)
15	Role of the rate constant as a spectral width	immediately after Equation (2)
16	Inverse proportionality between energy uncertainty and lifetime	immediately after Equation (2)
17	Possible deviation from the Lorentzian form due to additional physical processes	p. 117, paragraph following Equation (2)
18	Fano line shape representing interference between alternative paths	Equation (3)
19	Specified transition and radiation of cesium-133 used as the standard for the second	p. 116, paragraph defining the second
20	Definition of the second by 9,192,631,770 periods of the specified radiation	p. 116, paragraph defining the second

p	Claim	Source location
21	Realization of the second in principle using identical atoms unperturbed by their immediate environment	p. 116, paragraph defining the second
22	Practical unattainability of ideal realization and need for quantitative models linking realistic and ideal situations	p. 116, paragraph following the definition of the second

Table 382: Mathematical-object types in Clark 2001

m	Mathematical-object type	Source location
1	Exponential-decay equation	Equation (1)
2	Lorentzian angular-frequency line-shape equation	Equation (2)
3	Fano line-shape equation	Equation (3)

Definition 8.516 (Time contents in Clark 2001). The following table gives the eight time contents.

Table 383: Time contents in Clark 2001

i	Time content	Source location
1	Planck's constant	pp. 116–117
2	Transition angular frequency	p. 116, energy-gap and angular-frequency paragraph before Equation (1)
3	Elapsed time from the instant of state preparation	Equation (1)
4	Rate constant and the corresponding angular-frequency scale	Equations (1)–(3)
5	Spectral angular frequency	Equations (2)–(3)
6	Lifetime of the high-energy state	immediately after Equation (1)
7	Oscillation period of the specified cesium radiation used as the standard for the second	p. 116, paragraph defining the second
8	Unit of time, the second, defined by counting the specified periods	p. 116, paragraph defining the second

Dependent inputs

Index identifier S42-XI-DI1.

Existing-evidence reference set Definition 8.517.

Dependency and transfer elements

Definition 8.517 (Time-content transfers in Clark 2001). The following table gives the three time-content transfers.

Table 384: Time-content transfers in Clark 2001

Transfer	Dependency and transfer	Source claims	Time inputs	Time outputs	Source location
1	Transition angular frequency from Planck's constant and the energy difference between states	{4, 12}	{1}	{2}	p. 116, energy-gap and angular-frequency paragraph before Equation (1)
2	Rate constant from the lifetime of the high-energy state	{11}	{6}	{4}	immediately after Equation (1)
3	Definition of the second by counting 9,192,631,770 periods of the specified radiation	{19, 20}	{7}	{8}	p. 116, paragraph defining the second

All three transfers input at least one existing time content before outputting a time content. Zero transfers output a time content from nontemporal initial components alone.

Conditions of validity

Index identifier S42-XI-FC1.

Existing-evidence reference set Definition 8.517.

Source support

Index identifier S42-XI-SS1.

Existing-evidence reference set Definition 8.517.

Results

Definition 8.518 (Results and preservation conditions in Clark 2001). The following table gives the five source-specific results with their source claims, temporal premises, source locations, and preservation conditions.

Table 385: Results, dependencies, and preservation conditions in Clark 2001

Re- sult	Content	Source claims	Temporal premises	Source location	Preservation condition
1	Exponential decay of radiation intensity	$\llbracket 1, 3 \rrbracket \cup \llbracket 9, 11 \rrbracket$	$\{3, 4\}$	Equation (1)	Preserve the conditions of application of the two-level approximation and exponential-decay model.
2	Lorentzian angular-frequency line shape	$\llbracket 4, 6 \rrbracket \cup \llbracket 12, 15 \rrbracket$	$\{2, 4, 5\}$	Equation (2)	Preserve exponential decay, spectral decomposition, and the Lorentzian approximation.
3	Inverse proportionality between energy width and state lifetime	$\{11, 12, 15, 16\}$	$\{1, 4, 6\}$	p. 117, immediately after Equation (2)	Preserve Lorentzian full width at half maximum, $\tau = 1/\Gamma$, and the conversion to energy using $\hbar = h/(2\pi)$.
4	Possible significant deviation from the Lorentzian form	$\{14, 17\}$	$\{2, 4, 5\}$	p. 117, paragraph following Equation (2)	Preserve additional physical processes excluded from the simplifying assumptions.
5	Fano line shape representing interference between alternative transition paths	$\{7, 8, 18\}$	$\{2, 4, 5\}$	Equation (3)	Preserve the phase relations of the alternative paths, the interference parameter q , and $\epsilon = 2(\omega - \omega_0)/\Gamma$.

Result dependencies

Index identifier S42-XI-RD1.

Existing-evidence reference set Definition 8.518.

Grounds of validity

Index identifier S42-XI-RG1.

Existing-evidence reference set Definition 8.518.

Evidential hierarchy The number of objects newly measured directly by this source within the audit domain is 0. The source explains known radiation and spectral relations and the limits of application of the models.

Generation judgment Judgment values: LCTR complete generation 0 / source-internal evidence-conditioned local reach 0 / remaining primitive time roots 8.

Source-internal evidence-conditioned local reach The number of time contents reached under the source-internal evidence-conditioned local-reach judgment is 0, and 8 primitive time roots remain after this judgment.

Result dependencies The root/result dependency relation has 20 pairs, of which 6 are necessary dependencies.

Evidential force The formal evidence supports exponential decay, the Lorentzian line shape, inverse proportionality between uncertainty and lifetime, model deviations, and the Fano line shape. Its theoretical validity does not imply that Planck's constant, angular frequencies, rate constants, lifetimes, or other time contents have been generated from nontemporal inputs.

Result-preservation domain The five results are preserved where their time-content premises are generated by the LCTR generation sequence and the conditions in Table 385 are satisfied.

Placement of grounding responsibilities

Placement of grounding responsibilities The eight time contents audited in this source are not generated from nontemporal physical inputs before use; they are accepted as already established temporal inputs. The definition of the second counts an existing oscillation period; the realization conditions and quantitative models in Claims 21 and 22 are not conditions that generate that period from nontemporal inputs. The placement judgment for the physical-time grounding responsibility is "backgrounded by primitive time input." This judgment is based on comparing, for each time content, its use in a law operation, unit definition, or result derivation against the absence, before that use, of a corresponding generation condition or condition of validity stated as an input, formation condition, evaluation condition, or failure condition.

Evidence of actual failure

Evidence of actual failure The complete-generation judgment and the remaining-root judgment concern qualification as physical time and are distinct from actual failure of a concrete condition governing the value or domain of application of a result. Neither the source's formal evidence nor the LCTR audit evidence establishes actual failure of such a target condition. The actual-failure judgment is "indeterminate from the original source evidence"; actual failure is not inferred from the backgrounding of a grounding responsibility alone.

Downstream reach

Downstream reach The scope of reach to results dependent on primitive time roots is determined source by source by the primitive-time-root/result dependency relation reconstructed in this division and by minimal primitive-time-root support. This is structural reach. Because no actual failure of a concrete condition has been established, the judgment of actual reach from a failure location to results is "indeterminate from the original source evidence." Structural reach does not substitute for actual reach.

Repair scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation conditions or conditions of validity required before their use. The recalculation scope is restricted to results reached by dependency from a failure location whose actual failure has been established independently. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from repaired inputs, or their domain of validity is restricted. Values after recalculation are determined as audit results.

Audit conclusions

Definition 8.519 (Finite dependency graph for Clark 2001). The finite dependency graph consists of 36 semantic components, 8 time contents, 3 transfers, and 5 results, and has 49 nodes and 8 edges. The 36 semantic components are enumerated as claim nodes of the audit boundary in this order: 11 establishment claims for nontemporal initial components, 22 source claims, and 3 mathematical-object specifications. Source-claim index p corresponds to semantic-component index $11 + p$. The source-claim columns specify the claim support for transfers and results. In addition, Transfer 2 is supported by Mathematical-Object Type 1, Result 1 by Type 1, Results 2–4 by Type 2, and Result 5 by Type 3. The generation-initial set consists of the first 11 claim nodes, and the source-formal-evidence environment contains all 36 claim nodes. Time-content and result nodes are given by their respective item families.

Proposition 8.520 (Complete audit of time use in Clark 2001). The counts of generated time contents under the complete judgment and time contents reached under the local-reach judgment are both 0. All eight time contents remain primitive time roots, and all five results depend on at least one of them.

Proposition 8.521 (Minimal primitive-time-root support in Clark 2001). The following table gives the minimal supports of the five results. The total number of family members is 15.

Table 386: Minimal primitive-time-root support in Clark 2001

Result	Minimal-support index family
1	$\{3, 4\}, \{3, 6\}$
2	$\{1, 4, 5\}, \{1, 5, 6\}, \{2, 4, 5\}, \{2, 5, 6\}$
3	$\{1, 6\}$
4	$\{1, 4, 5\}, \{1, 5, 6\}, \{2, 4, 5\}, \{2, 5, 6\}$
5	$\{1, 4, 5\}, \{1, 5, 6\}, \{2, 4, 5\}, \{2, 5, 6\}$

Individual statements Statement evidence 1

Original (English):

inverse of the lifetime

The source location is Clark, *Effects of Configuration Interaction on Intensities and Phase Shifts*, p. 116, immediately after Equation (1). The minimal context is the passage relating the rate constant to the lifetime of the high-energy state. The attributed author of the statement is Charles W. Clark.

Statement evidence 2

Original (English):

the “width” of the spectral feature

The source location is Clark, *Effects of Configuration Interaction on Intensities and Phase Shifts*, p. 117, immediately after Equation (2). The minimal context is the passage relating the rate constant to the width of the Lorentzian angular-frequency line shape. The attributed author of the statement is Charles W. Clark.

Statement evidence 3

Original (English):

inversely proportional to its lifetime

The source location is Clark, *Effects of Configuration Interaction on Intensities and Phase Shifts*, p. 117, immediately after Equation (2). The minimal context is the passage relating the energy uncertainty of the state to its lifetime. The attributed author of the statement is Charles W. Clark.

Statement evidence 4

Original (English):

significant deviations from the form of Eq. (2)

The source location is Clark, *Effects of Configuration Interaction on Intensities and Phase Shifts*, p. 117, the paragraph following Equation (2). The minimal context is the passage stating that physical processes excluded from the simplifying assumptions can modify the Lorentzian form. The attributed author of the statement is Charles W. Clark.

Common evidence with specified domain of application The following four statements give the Clark 2001 originals, source locations, minimal contexts, and source attributions.

State- ment	Object of statement	Predicate	Explicit domain of application
S42- Q001	The exponential-decay rate constant Γ and the lifetime of the high-energy state	States that Γ is the inverse of the lifetime.	The relation among the two-level atom, exponential decay after state preparation, and the rate constant in Equation (1) and the immediately following passage on p. 116 of the Clark paper.
S42- Q002	Γ in the Lorentzian angular-frequency line shape	States that Γ is the width of the spectral feature.	Exponential decay, angular-frequency spectral decomposition, and the Lorentzian approximation in Equation (2) and the immediately following passage on p. 117.
S42- Q003	The energy uncertainty and lifetime of the high-energy state	States that the energy uncertainty is inversely proportional to the lifetime.	The relation among the rate constant, angular-frequency width, Planck's constant, and state lifetime immediately after Equation (2) on p. 117.
S42- Q004	The simplified Lorentzian line shape and spectral forms containing additional physical processes	States that physical processes excluded from the simplifying assumptions can produce significant deviations from the form of Equation (2).	The Lorentzian approximation, additional physical processes, alternative transition paths, and interference conditions in the passage following Equation (2) on p. 117.

Individual audits Rate constant and state lifetime

The source uses Γ in Equation (1) as the rate constant of exponential decay and identifies it with the inverse lifetime of the high-energy state. The independent basis is a theoretical model that represents the radiation intensity after state preparation by an exponential-decay equation and relates its rate constant to the lifetime. The evidential force of this relation belongs to the exponential-decay relation under the simplifying assumptions imposed by the source. The corresponding formal evidence consists of Source Claims 9–11 in Table 381, Time Contents 3, 4, and 6 in Table 383, Transfer 2 in Table 384, Result 1 in Table 385, and Proposition 8.520.

Width of the spectral feature

In Equation (2), Γ is the angular-frequency full width at half maximum of the Lorentzian line shape. In Equation (3), Γ is a detuning scale and $\epsilon = 2(\omega - \omega_0)/\Gamma$ is dimensionless. The full width at half maximum of a general Fano line shape is not identified with Γ . The independent basis is the Lorentzian line shape representing exponential decay in the angular-frequency domain and the role of Γ within that line shape. The evidential force of

this relation belongs to the line shape in Equation (2). The corresponding formal evidence consists of Source Claims 12–15 in Table 381, Time Contents 2, 4, and 5 in Table 383, Result 2 in Table 385, and Proposition 8.520.

Energy uncertainty and lifetime

The source states that energy uncertainty is inversely proportional to lifetime. Result 3 specifies this uncertainty as the Lorentzian full width at half maximum in Equation (2), $\Delta E_{\text{FWHM}} = \hbar\Gamma = \hbar/\tau$. This width is not the standard deviation of energy. The independent basis is a theoretical relation that uses Γ both as the inverse of the lifetime and as the angular-frequency width and relates energy uncertainty to lifetime through Planck's constant. The evidential force of this relation belongs to the inverse proportionality described by the source. The corresponding formal evidence consists of Source Claims 11, 12, 15, and 16 in Table 381, Time Contents 1, 4, and 6 in Table 383, Transfer 2 in Table 384, Result 3 in Table 385, and Proposition 8.520.

Deviation from the Lorentzian form

The source states that the Lorentzian form rests on simplifying assumptions and that additional physical processes can produce significant deviations. The independent basis is a theoretical comparison between the simplified Lorentzian model and line shapes that include additional physical processes, alternative transition paths, and interference. The evidential force of this relation belongs to the additional physical processes and interference conditions described by the source. The corresponding formal evidence consists of Source Claims 14, 17, and 18 in Table 381, Time Contents 2, 4, and 5 in Table 383, Results 4 and 5 in Table 385, and Proposition 8.520.

Common audit with specified domain of application The four statements concern, respectively, decay rate and lifetime, Lorentzian width, energy width and lifetime, and possible deviations due to additional physical processes. The individual audits above specify the conditions of application and corresponding formal evidence for each relation.

The four statements do not provide the physical generation of the time contents carried by the rate constant, angular-frequency width, state lifetime, and spectral angular frequency.

The corresponding formal evidence is given in Table 383, Table 381, Table 384, Table 385, Definition 8.518, and Proposition 8.520. Time contents reached from nontemporal inputs are 0/8, primitive time roots are 8/8, and root-dependent results are 5/5.

8.19.43 Source audit

Audit inputs

Source and audit scope

Source identifier 43.

Source name Primitive-time-root audit of Rovelli 2002

Bibliographic identification Let Carlo Rovelli, *Partial Observables*, *Physical Review D* 65, 124013 (2002), DOI: 10.1103/PhysRevD.65.124013 be source $R = \text{Ref}_{\text{Rov02PO}}$.

The audit domain comprises partial observables, complete observables, relational predictions, general-relativistic clock relations, quantum representations, constraints, the history formalism, and independence from the choice of external time.

Audit-scope identifier 43-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 43-L1.

Source-location content The time-content table gives 13 time contents and their source locations; the claim table gives 39 claims and their source locations.

Formal object

Object identifier 43-F1.

Object name Formal objects of the source.

Object type Formal object.

Formal content The 13 time contents and their uses in each claim.

Source-location identifier set 43-L1.

Assumptions

Assumption Identifier 43-A1.

Assumption Content The source assumptions are specified in Definitions 8.522 and 8.525.

Source-Location Identifier Set 43-L1.

Nontemporal physical inputs

Input identifier 43-P1.

Input name Nontemporal physical inputs of the source.

Input type Nontemporal physical input.

Input content The nontemporal object, apparatus and configuration data and the algebraic, geometric and quantum structures of Claims 1–3.

Source-location identifier set 43-L1.

Time contents

Time-content identifier 43-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content specification The individual time contents are given in Definitions 8.524 and 8.523.

Source-location identifier set 43-L1.

Definition 8.522 (Rovelli 2002 Formal Objects and Conditions). Rovelli distinguishes measurability from predictability and explains partial and complete observables through pendulums, field measurements, relativistic clocks and quantum observables. The pendulum uses the small-oscillation approximation and initial position and velocity; clock relations use a metric, worldlines and initial data. The general mechanical formulation concerns finitely many degrees of freedom and uses complete integrals, regular constraint surfaces, smooth local quotients and domains admitting the corresponding solutions. The quantum formulation uses an appropriate solution space and physical inner product, with probability amplitudes conditional on prior observations; its relation to the action uses a semiclassical approximation.

Definition 8.523 (Rovelli 2002 Formal Inputs). For a small-oscillation pendulum with specified initial position and velocity,

$$\ddot{q} = -\omega^2 q, \quad q(t) = A \sin(\omega t + \phi)$$

The prediction of position at a given time differs from the inverse selection of time from position, which additionally requires periodicity and branch conditions. For the satellite and base clocks, source Equations (4)–(5) give

$$f(\tau_s, \tau_b) = 0, \quad T_\tau = \max\{u \mid f(u, \tau) = 0\}.$$

The highest-solution definition applies where the solution set has a maximum. For finitely many degrees of freedom, let the extended configuration space be $\mathcal{C}$ and the constraint function on its cotangent bundle be C ; then

$$C\left(q^a, \frac{\partial S}{\partial q^a}\right) = 0, \quad f_a(q; Q, P) = \frac{\partial S(q, Q)}{\partial Q^a} - P_a = 0.$$

This is the complete-integral representation of source Equations (6)–(8). For a smooth local quotient of a regular single constraint,

$$\dim \Gamma = 2n - 2, \quad f : \mathcal{C} \times \Gamma \longrightarrow \mathbb{R}^n.$$

In the nonrelativistic special case,

$$q^0 = t, \quad p_0 = -E, \quad C = p_0 + H = 0, \quad \frac{\partial S}{\partial t} + H\left(q^i, \frac{\partial S}{\partial q^i}\right) = 0.$$

For the chosen orbit and endpoints, the action is $S = \int L dt$; the source introduces a time parameter in this integral. The quantum constraint is

$$C\left(q^a, -i\hbar \frac{\partial}{\partial q^a}\right) \psi(q) = 0, \quad C\left(q^a, -i\hbar \frac{\partial}{\partial q^a}\right) W(q, Q) = 0.$$

In source Equations (21)–(22), $W(q, Q)$ is the probability amplitude for q conditional on a prior observation of Q . In the semiclassical approximation, its phase is represented by $\exp(iS/\hbar)$.

Definition 8.524 (Rovelli 2002 Time Contents). Let $n_R = 13$. The table gives each time content and source location.

Table 388: Rovelli 2002 Time Contents and Source Locations

i	Time Content	Source Location
1	Nonrelativistic external time and clock readings	§II, IV, VI
2	Time derivatives, frequency and initial velocity	§II, Eqs. (1)–(2)
3	Time-indexed pendulum predictions and phase relations	§II, Eqs. (1)–(2)
4	Temporal locations of events and field measurements	§II–III

i	Time Content	Source Location
5	Coordinate time and gauge flow in general relativity	§III, V
6	Proper time along worldlines	§III
7	Elapsed Earth proper time from a reference event	§III
8	Satellite and base clock readings and their relation	§III, V, Eqs. (4)–(5)
9	Satellite-clock values indexed by the base-clock reading	§V, Eq. (5)
10	Time dependence of Heisenberg operators and Schrödinger states	§IV
11	Quantum-clock readings relative to another clock	§IV
12	Parameter time and endpoint order used in the action	§VI, Eqs. (19)–(20)
13	Prior-observation order in quantum predictions	§VI–VII, Eq. (22)

Definition 8.525 (Rovelli 2002 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(39)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([3], [39])$. Claims 1–3 form the generation-initial set; all 39 claims form the source formal-evidence environment.

Definition 8.526 (Rovelli 2002 Claim Boundary). The generation-initial set consists of Claims 1–3 and the source formal-evidence environment of Claims 1–39. The following table gives each claim’s temporal uses and roles.

Table 389: Rovelli 2002 Physical and Formal Input Claims

p	Claim	i	Source Location
1	Identities of measured objects and apparatus, numerical readings and nontemporal configuration data.	$\emptyset$	§II–III
2	Algebraic and geometric structures of configuration spaces, cotangent bundles, constraints, quotients and action functionals.	$\emptyset$	§VI
3	Mathematical structures of state spaces, operators, probability amplitudes and correlation laws.	$\emptyset$	§II, IV, VI
4	Rovelli distinguishes partial observables defined by procedures yielding numerical measurements from complete observables whose values or probability distributions the theory predicts.	$\emptyset$	§I–II
5	Rovelli states in a Section II footnote that describing partial observables through measurement does not imply operationalism.	$\emptyset$	§II, footnote
6	For the upper number n and lower number N on a card, the theory predicts N given n through the correlation $N = N(n)$.	$\emptyset$	§II

p	Claim	i	Source Location
7	For a pendulum in the small-oscillation approximation, Rovelli gives the equation of motion using the second derivative with respect to the specified clock time and the frequency.	$1^S, 2^Q$	§II, Eq. (1)
8	The initial position and velocity determine the pendulum amplitude and phase, and $q(t)$ predicts position at each clock reading. The inverse correspondence from periodic position requires a branch choice.	$1^B, 2^Q, 3^{SX}$	§II, Eqs. (1)–(2)
9	Partial observables provide coordinates on the extended configuration space, and a relation surface with specified state parameters determines the allowed combinations of values.	$\emptyset$	§II, VI, Eqs. (3), (7)–(8)
10	In the nonrelativistic case, the extended configuration space includes clock time t , with position and time treated as partial observables.	1^K	§II, VI
11	Prediction of $E(x, t)$ for the Maxwell field relates the field-component reading to three spatial locations and the measurement time.	4^{OX}	§II
12	Bare coordinates in general relativity are not direct physical readings; Rovelli distinguishes coordinate representations from gauge choices using physical quantities.	5^X	§III, V
13	The Sun–Venus angular example predicts a relation between two angles on the same observational occasion.	4^{OX}	§III
14	Rovelli’s relativistic setting uses a metric and body worldlines and tetrads expressed in proper time.	6^S	§III
15	The Earth clock’s elapsed proper time from a reference event is determined by proper length along the Earth worldline and the initial reading.	$6^I, 7^{IQBK}$	§III
16	Given the initial data and relativistic arrangement, the received satellite-clock reading and base-clock reading satisfy $f(\tau_s, \tau_b) = 0$.	$6^I, 8^{KX}$	§III, Eq. (4)
17	The relation between physical clocks can be described without permanently assigning the independent-variable role to one clock and the dependent-variable role to the other.	8^{KX}	§III

p	Claim	i	Source Location
18	Complete observables are functions on the space Γ of gauge-equivalence classes of solutions and are invariant, on the constraint surface, under gauge transformations generated by the constraints.	$\emptyset$	§III, VI
19	In the Heisenberg picture $Q(t)$ represents a complete observable at a specified time; in the Schrödinger picture predictions combine Q with the time-dependent state $\Psi(t)$.	$1^S, 10^S$	§IV
20	Rovelli states the condition that the Schrödinger picture is viable in the special case where an independent external time can be chosen.	$1^S, 10^S$	§IV
21	The clock operator $T(t)$ represents a physical-clock reading relative to another clock, and its quantum fluctuations occur in that clock relation.	$1^K, 11^{KX}$	§IV
22	Rovelli distinguishes invariance under coordinate-time gauge flow from relative evolution between physical clocks.	$5^S, 8^{SX}$	§V
23	The complete observable T_τ at base-clock reading τ is defined as the highest solution of $f(T_\tau, \tau) = 0$. Its existence requires a nonempty solution set with an attained maximum.	$8^{KB}, 9^{KX}$	§V, Eq. (5)
24	When $\tau_b(t)$ and $\tau_s(t)$ are represented in coordinates, the relation giving the satellite reading at a base reading is independent of the coordinate choice for the same four-geometry.	$5^S, 8^K, 9^X$	§V
25	Rovelli rejects the objection that individual readings of partial observables entail their identification with physical operators, and uses the operator correspondence for complete observables.	$\emptyset$	§V
26	Rovelli regards research on noncommutative geometry as valuable while criticizing the particular motivation of promoting bare coordinates to quantum operators, and distinguishes noncommutativity of physical gravitational-field quantities.	$\emptyset$	§V
27	For mechanics with finitely many degrees of freedom, Rovelli specifies the pair $(\mathcal{C}, C)$ consisting of an extended configuration space and a constraint function on its cotangent bundle.	$\emptyset$	§VI, Eqs. (6), (11)

p	Claim	i	Source Location
28	A complete integral $S(q, Q)$ of the Hamilton–Jacobi equation gives relations between partial observables using integration constants P . This representation uses a local domain admitting such a complete integral.	$\emptyset$	§VI, Eqs. (6)–(8)
29	On a regular hypersurface defined by one constraint, the null direction of the restricted symplectic form determines characteristic curves. Where a smooth local quotient exists, that quotient Γ has dimension $2n - 2$.	$\emptyset$	§VI
30	Rovelli interprets a point of Γ as a state representing a whole history and uses the relation surface $f = 0$ to give combinations of partial observables compatible with that state.	$\emptyset$	§VI–VII, Eqs. (7)–(8)
31	For the nonrelativistic special form $C = p_0 + H$, the identifications $q^0 = t$ and $p_0 = -E$ give the usual time-dependent Hamilton–Jacobi equation.	1^S	§VI, Eqs. (9)–(12)
32	Holding the integration constants Q fixed along an orbit gives $dS = p_a dq^a$; Rovelli introduces an arbitrary time parameter and integrates the action between endpoint configurations.	12^{SIB}	§VI, Eqs. (18)–(19)
33	The nonrelativistic action representation uses the chosen time t and endpoint configurations to integrate $dS = p_i dq^i - H dt = L dt$.	$1^S, 12^{IB}$	§VI, Eq. (20)
34	The classical formulation by the pair $(\mathcal{C}, C)$ and a relation surface does not require choosing one partial observable as an independent external time.	$\emptyset$	§VI–VII
35	Rovelli describes a quantum formulation in which the constraint acts as a differential operator on ψ and W , with a physical inner product on an appropriate solution space.	$\emptyset$	§VI, Eqs. (21)–(22)
36	The predictive interpretation of $W(q, Q)$ gives the probability amplitude for observing q after Q has previously been observed.	13^{OP}	§VI–VII, Eq. (22)
37	In the semiclassical limit, Rovelli represents the propagator phase by $\exp(iS/\hbar)$ in terms of the action S and describes the transition to the Hamilton–Jacobi equation.	$12^{IB}, 13^P$	§VI

p	Claim	i	Source Location
38	In Rovelli's general formulation, one chooses partial observables to fix as conditions and obtains complete observables on a domain where those conditions determine unique corresponding values.	$\emptyset$	§VII
39	The source's relativistic clock relations use proper-clock readings, and its quantum predictions use prior-observation order.	$8^K, 13^O$	§III–V, VI–VII

Table 390: Rovelli 2002 Time-Use Roles

Code	Role	Use Count
I	Interval	6
Q	Quantity	3
X	Cross-position correspondence	10
K	Clock reading	10
C	Causality	0
S	State evolution	13
P	Probabilistic evolution	2
O	Observation record	4
B	Reference boundary	6

Table 391: Rovelli 2002 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 5, 8–10	member	instant, order_relation
2	dependency	$\emptyset$
3	member	instant, turning_position
4, 11	member	instant
6	member	instant, interval_boundary
7	member	endpoint, interval_boundary
12	member	order_relation, endpoint, interval_boundary
13	member	order_relation

Definition 8.527 (Rovelli 2002 Finite Cardinalities). The table gives cardinalities of claims, temporal uses, transfers, results and their dependency structure.

Table 392: Rovelli 2002 Finite Cardinalities

Object	Cardinality
Time contents	13
Claims	39
Transfers	7
Results	28
Claim/time-content uses	37
Role assignments	54
Obligation atoms	87
Audit-boundary nodes	80
Audit-boundary edges	35

Dependency Inputs

Definition 8.528 (Rovelli 2002 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [7]$. The table gives time inputs, time outputs, claim support and source locations.

Table 393: Rovelli 2002 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1	2	7	Evaluate pendulum time derivatives. §II, Eq. (1)
2	1–2	3	8	Obtain time-indexed pendulum position from initial data. §II, Eq. (2)
3	6	7	15	Obtain elapsed proper time from a reference event. §III
4	6	8	16	Relate two clock readings in the relativistic arrangement. §III, Eq. (4)
5	8	9	23	Select the satellite reading corresponding to a base reading. §V, Eq. (5)
6	1	10	19–20	Represent quantum predictions using external time. §IV
7	1	11	21	Represent a clock relation by a quantum-clock operator. §IV

Grounding Conditions

1. The specified clock time and the small-oscillation approximation.
2. Initial position, initial velocity, frequency and the clock time defining phase.
3. The metric, Earth worldline, reference event and initial reading.
4. Known initial data, body worldlines and the signal-reception relation.
5. A domain where the solution set of $f(T_\tau, \tau) = 0$ is nonempty and has a maximum.

6. A quantum system admitting external time as an independent variable, with its dynamics and state.
7. Reference-clock time t , the physical clock T and the quantum correspondence between complete observables and operators.

Results

Definition 8.529 (Rovelli 2002 Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [28]$. The table gives each result's temporal premises and claim support.

Table 394: Rovelli 2002 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Distinction between partial and complete observables. §I–II	$\emptyset$	4
2	Meaning of the description through measurement. §II, footnote	$\emptyset$	5
3	Conditional card correlation. §II	$\emptyset$	6
4	Conditional prediction of pendulum position. §II, Eqs. (1)–(2)	1–3	7–8
5	Extended configuration space and relation surface. §II, VI	$\emptyset$	9
6	Nonrelativistic time coordinate. §II, VI	1	10
7	Spacetime prediction of the Maxwell field. §II	4	11
8	Coordinates and physical gauge choices in general relativity. §III, V	5	12
9	Relational prediction for the Sun and Venus. §III	4	13
10	Earth proper time from a reference event. §III	6–7	14–15
11	Satellite–base clock relation. §III, Eq. (4)	6, 8	16–17
12	Gauge-invariant complete observables. §III, VI	$\emptyset$	18
13	Heisenberg and Schrödinger pictures. §IV	1, 10	19–20
14	Rejection of operator identification from partial observables. §V	$\emptyset$	25
15	Relational quantum clock. §IV	1, 11	21
16	Coordinate-independent relative clock evolution. §V, Eq. (5)	5, 8–9	22–24
17	Examination of a motivation for noncommutative geometry. §V	$\emptyset$	26
18	Constraint formulation for finite-dimensional mechanics. §VI	$\emptyset$	27

q	Result and Source Location	Time Premises i	Claim Support p
19	Relations from the Hamilton–Jacobi formulation. §VI, Eqs. (6)–(8)	$\emptyset$	28
20	Quotient of characteristic curves and history states. §VI–VII	$\emptyset$	29–30
21	Usual time-dependent Hamilton–Jacobi equation. §VI, Eqs. (9)–(12)	1	31
22	Action represented along an orbit. §VI, Eq. (19)	12	32
23	Nonrelativistic action integral. §VI, Eq. (20)	1, 12	33
24	Independence from external-time selection in the classical formulation. §VI–VII	$\emptyset$	34
25	Form of the quantum constraint equation. §VI, Eqs. (21)–(22)	$\emptyset$	35
26	Quantum prediction conditional on a prior observation. §VI–VII	13	36
27	Semiclassical relation between action and propagator. §VI	12–13	37
28	Complete observables from fixing partial observables. §VII	$\emptyset$	38

Definition 8.530 (Rovelli 2002 Grounds and Domains of Results). The table gives conditions and domains of the results.

Table 395: Rovelli 2002 Conditions for Results

Result Index	Conditions and Domain
1	Definitions in terms of measurement procedures and predictability.
2	The footnote in source Section II.
3	Two numerical values and the relation $N = N(n)$.
4	Small oscillations, specified clock, initial position and velocity, and a branch for the inverse correspondence.
5	Combinations of partial observables and specified state parameters.
6	A configuration space containing external time.
7	A field component, three spatial locations and the measurement time.
8	Coordinate representation, physical quantities and a specified gauge.
9	A common observational occasion and two angles.
10	Metric, worldline and initial reading.
11	Initial data and the relativistic arrangement of physical clocks.
12	Gauge-equivalence classes of solutions and invariance on the constraint surface.
13	Operators, states and predictions in a case admitting independent external time.

Result Index	Conditions and Domain
14	The distinction between partial observables and the correspondence for complete observables.
15	Readings and quantum fluctuations relative to another physical clock.
16	The same four-geometry, the two-clock relation and a domain admitting a highest solution.
17	The distinction between a specific argument requiring coordinate operators and physical gravitational-field quantities.
18	Extended configuration space and a constraint function on its cotangent bundle.
19	A local domain admitting a complete integral and a choice of independent relations.
20	A regular single constraint, a smooth local quotient and a relation surface.
21	The nonrelativistic constraint $C = p_0 + H$ with $q^0 = t$ and $p_0 = -E$.
22	Fixed integration constants, an arbitrary time parameter and endpoint configurations.
23	Chosen time, Lagrangian and endpoint conditions.
24	The classical formulation by the pair $(\mathcal{C}, C)$ and relations among partial observables.
25	An operator realization of the constraint and an appropriate solution space and physical inner product.
26	Prior observation of Q and the probability amplitude $W(q, Q)$.
27	Semiclassical approximation, action endpoints and observational conditions for quantum prediction.
28	A domain where the specified conditions uniquely determine corresponding values.

Result Dependence

Definition 8.531 (Rovelli 2002 Result-Dependence Hyperedges). For each row of the result table, let $\text{ResultEdge}_{R,q}$ ($q \in [28]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Evidence Hierarchy Observable definitions and numerical-correlation examples distinguish measurement from prediction. Pendulum, field and clock examples give relational predictions using their respective initial conditions and time contents. The finite-dimensional constraint formulation describes relation surfaces and histories under regularity and solution-domain conditions. In the quantum formulation, amplitudes conditional on prior observations and the domain of the semiclassical approximation specify the predictions.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local-reach count are both zero, leaving 13 primitive-time roots. Transfers for the pendulum, relativistic clocks and quantum clocks take specified clock time or proper time as inputs. The action's time parameter and the prior-observation order of quantum predictions also enter the source description as inputs.

Result Dependence Of the 28 results, 15 depend on primitive-time roots. Thirteen are reached from source formal evidence concerning observable definitions, card correlations, geometric structures, operator correspondences and the general formulation. There are 19 root–result dependence pairs, of which 17 are necessary.

Placement of the Grounding Obligation Rovelli’s relational predictions use clock readings, proper time, parameter time or observation order in their respective examples. Within a formulation requiring no single external time as the independent variable, these temporal uses carry the grounding obligation. The LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Initial data, physical clock arrangements, solution domains and approximation conditions specify the domain of each prediction. Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 19 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Proposition 8.532 (Primitive Time Roots and Result Dependence in Rovelli 2002). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = (\emptyset, [13]; \\ \emptyset, \{1, 2, 3, 5, 12, 14, 17, 18, 19, 20, 24, 25, 28\}, \\ [28] \setminus \{1, 2, 3, 5, 12, 14, 17, 18, 19, 20, 24, 25, 28\}, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–3 remains those three nodes. Even with all 39 claims as source formal evidence, the seven transfers require temporal inputs. The thirteen results with empty temporal premises are reached from source formal evidence; adding all thirteen time contents as seeds reaches all 28 results. $\square$

Proposition 8.533 (Minimal Primitive-Time-Root Support in Rovelli 2002). The table gives each result’s minimal-support index family, with 29 family members in total.

Proof. Start with singleton time-content seeds and empty supports for results with empty temporal premises, and apply Definition 7.46 to the seven transfers and 28 result dependencies. By Proposition 7.47, the inclusion-minimal supports of the stable antichains are the displayed families. $\square$

Table 396: Minimal Primitive-Time-Root Support in Rovelli 2002

Result Indices	Minimal-Support Family Members
1–3, 5, 12, 14, 17–20, 24–25, 28	$\emptyset$
4, 6, 13, 15, 21	$\{1\}$
7, 9	$\{4\}$
8	$\{5\}$
10–11	$\{6\}$
16	$\{5, 6\}, \{5, 8\}$
22	$\{12\}$
23	$\{1, 12\}$
26	$\{13\}$
27	$\{12, 13\}$

Evidentiary Effect The observable definitions and card example distinguish measurement from prediction. The pendulum, field and clock examples give relational predictions using specified time contents. The finite-dimensional constraint formulation gives relation surfaces and the geometry of histories; the quantum formulation gives probability amplitudes conditional on prior observations. The result table states the conditions of each result.

Result-Preservation Domain Fix the physical arrangement, initial data, domains and approximation conditions in Table 395. When temporal inputs can be realized by generated temporal representations preserving the same clock correspondence, intervals, phase, endpoint order and observation order, each result is preserved within its domain.

Proposition 8.534 (Audit Conclusion for Rovelli 2002). There are 13 time contents, 39 claims, 7 transfers and 28 results. The 37 claim/time-content use pairs have 54 role assignments. Complete generation and local reach are both zero, leaving thirteen primitive-time roots and fifteen root-dependent results.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. $\square$

Proposition 8.535 (Audit-Boundary Cardinalities for Rovelli 2002). There are 87 source-specific obligation atoms, with

$$\text{BoundaryCount}_R^{\text{audit}} = (80, 35).$$

Proof. Count claims, time contents and results as nodes, and transfers and result dependencies as edges. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual Statements The problem of time measurement and observables
Original text:

Therefore time cannot be an observable [2]. But if time is not an observable, how can we measure it?

Source location: Section I. Rovelli introduces the tension between the argument concerning a self-adjoint time operator in ordinary quantum theory and measurement by clocks, then develops the distinction between partial and complete observables.

Clock readings and conditional prediction

Original text:

The theory cannot predict the value of t .

Source location: Section II. Rovelli treats pendulum position q and clock reading t as partial observables and explains that, given initial data, the theory predicts q corresponding to t .

Relational prediction between partial observables

Original text:

Observe that the predictions of a mechanical theory can always be expressed as relations between partial observables.

Source location: Section II. Rovelli presents the relation between pendulum position and time and states the general claim that mechanical predictions can be expressed as relations between partial observables.

A mechanical formulation without selecting external time

Original text:

There is no need to single out a specific partial variable as the time, nor to mention evolution. Mechanics is a theory of relations between partial observables. No distinction between dependent and independent partial observables is required. This distinction is an accident of non-relativistic theories, in which the constraint $C(q^a, p_a) = 0$ happens to have the form (12).

Source location: Section VI. Rovelli uses the pair consisting of a finite-dimensional extended configuration space $\mathcal{C}$ and a constraint C to describe mechanics without selecting one time as the independent variable. Equation (12) gives the nonrelativistic special form.

Common Evidence with Specified Scope The table gives the objects, predicates and domains of the four statements.

State- ment	Object	Predicate	Explicit Domain
S43- Q001	Time operators in quantum theory and time measurement by clocks	Rovelli presents the relation between observability and measurement as a problem.	The introductory discussion of time operators in Section I and the subsequent distinction between observables.
S43- Q002	The clock reading t in the pendulum example	Rovelli states that the theory does not predict the value of t itself.	A small-oscillation pendulum, clock readings and initial position and velocity.

S43-Q003	Representation of predictions of a mechanical theory	Rovelli states that predictions can always be expressed as relations between partial observables.	The pendulum example and generalization to relational prediction in Section II.
S43-Q004	The mechanical formulation by the pair $(\mathcal{C}, C)$	Rovelli does not require a distinguished time variable or a dependent-independent variable distinction in the general formulation.	Section VI for finitely many degrees of freedom and the special form of the nonrelativistic constraint.

Individual Audit Distinction between measurement and prediction

In the first statement, Rovelli presents the tension between an operator representation of time and measurement by clocks. In the source distinction, clock readings are partial observables, while quantities predicted from a specified state are complete observables. A quantum-clock operator represents the relation of a physical clock to another clock. This distinction addresses the introductory question. Claims 4 and 19–26 and Results 1 and 13–17 correspond to it.

Prediction at a specified clock reading

The second statement says that the pendulum theory predicts the position corresponding to a clock reading rather than selecting the observed clock reading t itself. The small-oscillation approximation and initial position and velocity determine $q(t)$. The inverse correspondence from pendulum position has periodicity and branch conditions. Clock time is an input to this prediction. Claims 7–8 and Result 4 correspond to the statement.

Prediction supplied by a relation surface

In the third statement, Rovelli expresses mechanical predictions as relations between partial observables. In the card example, a numerical correlation gives a conditional prediction; in the pendulum, Maxwell-field and relativistic-clock examples, temporal locations or clock readings enter the respective relations. A relation surface on extended configuration space determines combinations of values compatible with a specified state. Claims 6, 8–11, 16–18 and 28–30 and their corresponding results provide the examples and formulation of this general claim.

The general formulation and predictions using time

The fourth statement describes finite-dimensional mechanics by the pair $(\mathcal{C}, C)$, giving relational predictions without choosing one external time as the independent variable. Characteristic curves of the constraint surface and their quotient are constructed geometrically; the source introduces a time parameter when explaining the action. Relativistic clock relations use proper clocks, and the predictive interpretation of quantum amplitudes uses prior-observation order. Claims 27–38 and Results 18–28 give the general formulation and the conditions of the predictions.

8.19.44 Source audit

Audit input

Original source and audit scope

Source identifier 44.

Source title Complete audit of time use in Zurek 2003

Bibliographic identification Write the audited source as $R := \text{Ref}_{\text{Zur03DEQ}}^{\text{survey}}$. This denotes Wojciech H. Zurek, *Decoherence, Einselection, and the Quantum Origins of the Classical*, *Reviews of Modern Physics* 75, 715–775 (2003), DOI: 10.1103/RevModPhys.75.715. The audit domain comprises the discussion in Sections I–IX.

Audit-scope identifier 44-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 44-L1.

Source-location content The time-content, claim, transfer and result tables identify source sections and equations.

Formal objects

Object identifier 44-F1.

Object name Formal objects of the original source.

Object type Formal object.

Formal content The individual contents of the formal objects are given in Definitions 8.539 and 8.539.

Source-location identifier set 44-L1.

Assumptions

Assumption identifier 44-A1.

Assumption content The assumptions of each model are given in Definitions 8.537 and 8.539 and in the result tables.

Source-location identifier set 44-L1.

Nontemporal physical inputs

Input identifier 44-P1.

Input name Nontemporal physical inputs of the original source.

Input type Nontemporal physical input.

Input content The physical constitution and nontemporal mathematical structures in Claims 1–4.

Source-location identifier set 44-L1.

Time contents

Time-content identifier 44-T1.

Time-content name Time contents of the original source.

Time-content type Time content.

Time-content description The individual contents of the time contents are given in Definition 8.540.

Source-location identifier set 44-L1.

Formal input elements

Formal-component identifier 44-C1.

Formal-component name Formal-evidence environment of the original source.

Formal-component type Formal component.

Formal-component body The individual contents of the formal input elements are given in Definitions 8.539 and 8.539.

Original-source support identifier set 44-L1.

Definition 8.536 (Provenance and Attribution). Zurek is the author of the theoretical construction and interpretive claims. Experimental descriptions rely on the experimental studies cited by the source; this review reports zero newly acquired direct measurements. The assessment of Bohr’s reception is Zurek’s, and the Einstein letter is quoted through Born.

Definition 8.537 (Models and Conditions). Zurek uses a division into system, apparatus and environment, together with unitary composite-system evolution, to explain premeasurement, einselection and observer records. Finite interaction models, an oscillator linearly coupled to a thermal bath, chaotic systems, memory models and quantum control each use their own initial states, interactions, approximations and evaluation intervals. Enviance symmetry and the probability argument, system–fragment mutual information, the history formalism and past inference from records apply under the conditions stated for the respective claims and results below.

Definition 8.538 (Formal Inputs). Subsystem decomposition and quantum evolution use

$$\mathcal{H}_{SAE} = \mathcal{H}_S \otimes \mathcal{H}_A \otimes \mathcal{H}_E, \quad i\hbar \partial_t |\Psi\rangle = H|\Psi\rangle, \quad \rho_S = \text{Tr}_E \rho_{SE}.$$

In the high-temperature, decoherence-dominated approximation of source Section V.B, for fixed position separation and a positive decay rate,

$$\lambda_T = \frac{\hbar}{\sqrt{2Mk_B T}}, \quad \Gamma_D = \gamma \left(\frac{\Delta x}{\lambda_T} \right)^2, \quad \tau_D = \Gamma_D^{-1}, \quad \rho_S(x, x', t) = \rho_S(x, x', 0) e^{-\Gamma_D t}.$$

Here the position separation is $\Delta x = x - x'$. The enviance and Born-weight argument of Section VI.D uses

$$(U_S \otimes U_E) |\Psi_{SE}\rangle = |\Psi_{SE}\rangle, \quad p_k = \frac{m_k}{\sum_j m_j} = |\alpha_k|^2.$$

The latter holds under equal-amplitude fine graining, equiprobability, normalization and additivity; extension to incommensurate weights uses continuity. For disjoint environment fragments and $H(S) > 0$ in Section VII.A,

$$I(S : E_k) = H(S) + H(E_k) - H(S, E_k), \quad R_I = \frac{\sum_k I(S : E_k)}{H(S)}, \quad \dot{R} = \frac{dR}{dt}.$$

The ratio evaluated at a state is distinct from its rate under timed record copying. Histories in Section VII.C use projections at specified ordered times and an initial state:

$$C_\alpha = P_{\alpha_n}^n(t_n) \cdots P_{\alpha_1}^1(t_1), \quad D(\alpha, \beta) = \text{Tr}(C_\alpha \rho C_\beta^\dagger).$$

The condition that off-diagonal real parts vanish is distinct from the condition that the entire off-diagonal functional vanishes.

Definition 8.539 (Formal Evidence of the Source). Let the claim-node set be $\text{ClaimNodes}_R(50)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([4], [50])$. Claims 1–4 form the generation-initial set; all 50 claims form the source formal-evidence environment.

Definition 8.540 (Time Contents). Let $n_R = 18$. The table gives the time contents and source locations.

Table 398: Zurek 2003 Time Contents

i	Time Content	Source Location
1	Schrödinger evolution time of a closed composite system	§II; IV.A; VI.C
2	Interaction endpoints, durations and action	§II.A–B; VI.B
3	Entangling, disentangling and recurrence scales in finite models	§II.A.2; IV.A.1
4	Order of premeasurement and remeasurement	§II.A; VI.B–C
5	Order of memory recording and subsequent consultation	§VI.B–C; VII.B–C
6	Chaotic stretching rate and Ehrenfest correspondence time	§III; V.E
7	Evolution time of the open-system reduced state	§V.A–B
8	Decoherence timescale	§IV.A; V.B; VI.B
9	Entropy-production rate and its validity interval	§V.E.2
10	Quantum predictability horizon for the initial resolution	§V.E.3
11	Evaluation interval for pointer stability and record retention	§IV.D; V.C; VI.B–C
12	Times and order of history projections	§VII.C, Eq. (7.21)
13	Past order inferred from present records	§VII.C
14	Experimental preparation, waiting and detection times	§VIII.A–B
15	Control clock, drive frequencies, pulses and correction cycles	§VIII.B–D
16	Time dependence and phase trajectories of classical noise	§IV.C; VIII.B
17	Times and rate of increasing record redundancy	§VII.A.4, Eq. (7.20)
18	Relaxation and memory-erasure timescale	§V.A–B; VI.B

Table 399: Zurek 2003 Claims and Temporal Uses

p	Claim	i	Source Location
1	Nontemporal physical constitution of systems, apparatus, environment fragments, memories and experimental equipment.	$\emptyset$	§II; VI; VIII

p	Claim	i	Source Location
2	Mathematical structures of subsystem Hilbert spaces, their tensor product as the composite space, and state vectors and density operators.	$\emptyset$	§II; VI.C
3	Nontemporal operator structures of Hamiltonians, projections, partial traces, control maps, pointer algebras and information quantities.	$\emptyset$	§II; IV; VI; VII
4	Nontemporal set and combinatorial structures of phase space and candidate histories.	$\emptyset$	§III; VII.C
5	The composite state evolves unitarily under the specified Hamiltonian and Schrödinger time.	1^S	§II.A; VI.C
6	Controlled-not and controlled-shift maps send input states in the specified basis to correlated states. The direction of information transfer depends on the input state and basis.	$\emptyset$	§II.A
7	System–apparatus premeasurement interaction forms correlations between the premeasurement and postmeasurement states with specified coupling and interaction time.	$1^S, 2^{IQS}, 4^O$	§II.A; VI.B
8	In information amplification, system and apparatus dimensions, gain, the initial apparatus state and interaction time determine the formation of distinguishable apparatus records.	$1^S, 2^{IQS}$	§II.A.2–B
9	Unitary premeasurement correlates the system and apparatus. Zurek describes measurement by distinguishing this correlation formation from the observer’s acquisition of an outcome.	$1^S, 2^{IQS}, 4^O$	§II.C; VI.B–C
10	In finite controlled-shift and spin-environment models, the given spectrum and dynamics determine timescales of entanglement formation, disentanglement and recurrence.	$1^S, 2^{IQS}, 3^{IQS}$	§II.A.2; IV.A.1
11	Chaotic wavepacket stretching follows positive Lyapunov exponents. Comparison of Moyal and Poisson evolution depends on the wavepacket and nonlinearity scales.	$1^S, 6^{QSB}$	§III

p	Claim	i	Source Location
12	For an initially localized wavepacket, Zurek estimates the Ehrenfest correspondence time from a positive Lyapunov exponent and action and nonlinearity scales. Regular motion has a different time dependence.	$1^S, 6^{QSB}$	§III; V.E
13	The partial trace determines a reduced state from a composite state. Off-diagonal elements of the reduced matrix of a correlated pure state contain the corresponding environment-state overlap.	$\emptyset$	§II.B; IV.A
14	Interaction with a spin environment changes the environment-state overlap and reduced-state coherence over time. The form and presence of decay depend on the interaction and initial state.	$1^S, 8^{IQB}$	§IV.A.1
15	Commutation with the specified interaction and self-Hamiltonians gives conservation of pointer subspaces. Approximate commutation requires a specified interval for evaluating stability.	$1^S, 2^{IQS}, 11^{SB}$	§IV.A; IV.D
16	The difference between quantum mutual information and measurement-basis-dependent conditional information defines discord and characterizes classical correlations in the einselected record basis.	$\emptyset$	§IV.B; VI.C.4
17	Classical phase noise gives unitary time evolution for each realization, while averaging over realizations gives dephasing. Knowledge of an individual phase history permits construction of the inverse transformation.	16^{SP}	§IV.C
18	The predictability sieve compares initial states by entropy increase or purity loss over a specified interval. Zurek leaves general agreement between these criteria open.	$7^S, 11^{SB}$	§IV.D; V.C
19	For a harmonic oscillator linearly coupled to a harmonic thermal bath with an initial product state, composite-system evolution and a Gaussian propagator yield a time-local reduced evolution equation.	$1^S, 7^S$	§V.A
20	The high-temperature approximation describes diffusion and decay under conditions on energy scales, the bath cutoff and the initial transient.	$7^S, 8^{IQB}, 18^{IQB}$	§V.A–B

p	Claim	i	Source Location
21	In the decoherence-dominated high-temperature model, position separation, thermal wavelength and relaxation rate determine the off-diagonal decay time. Saturation at large separation depends on the interaction model.	$7^S, 8^{IQB}, 18^{IQB}$	§V.B
22	For an underdamped harmonic oscillator, evaluation of purity loss over one period selects coherent states as pointer states under the specified approximation.	$7^S, 11^{SB}$	§V.C
23	Zurek discusses the classical limit and predictability in phase space by distinguishing macroscopic scales, environmental coupling and initial resolution.	$7^S, 8^{IQB}, 11^{SB}$	§V.D
24	The balance between chaotic contraction and environmental diffusion limits the coherence length in the specified model. Classical correspondence is restored where that length is small relative to the nonlinearity scale.	$6^{QSB}, 7^S, 8^{IQB}$	§V.E.1
25	After diffusion-limited width saturation and before equilibration, the entropy-production rate is approximated by the sum of positive Lyapunov exponents. Saturation times can differ between directions.	$6^{QSB}, 7^S, 9^{QSB}$	§V.E.2
26	At finite Planck constant and specified initial resolution, the quantum predictability horizon of a chaotic system is the time at which the wavepacket spreads to the characteristic system scale. The argument applies to open and closed systems.	$6^{QSB}, 10^{QB}$	§V.E.3
27	For gentle observations at intervals longer than the decoherence time and shorter than the Lyapunov time, the mean information-growth rate of records corresponds to the dynamical entropy-production rate under the specified conditions.	$4^O, 6^{QSB}, 8^{IQB}, 9^{QSB}$	§V.E.2
28	Apparatus memory acquires correlations through premeasurement and is consulted after decoherence but before relaxation erases the information. Stable pointer states carry the record.	$4^O, 5^O, 8^{IQB}, 11^{SB}, 18^{IQB}$	§VI.B–C

p	Claim	i	Source Location
29	Zurek explains immediate repeatability of the same observable using the specified measurement interaction and apparatus truth table. He distinguishes orthogonal eigenstates of discrete Hermitian observables from nonideal and generalized measurement conditions.	$1^S, 2^{IQS}, 4^O, 11^{SB}$	§VI.C.1–3
30	Under envariance, a local unitary transformation on the system is undone by a transformation on the environment, preserving the composite state. Schmidt phases and swaps of equal-amplitude components exhibit this symmetry.	$\emptyset$	§VI.D.1–2
31	Zurek derives Born weights from fine-grained equal-amplitude components using local-state dependence, causality, equiprobability, normalization and additivity. Extension to incommensurate weights assumes continuity of probabilities.	$\emptyset$	§VI.D.2
32	For a product state of distinguishable systems prepared in the same state, counting fine-grained components gives a binomial distribution, with relative frequencies concentrating at the Born weights in the large-ensemble limit.	$\emptyset$	§VI.D.3
33	Through interaction with the system and subsequent information propagation, multiple environment fragments carry records of pointer states.	$1^S, 2^{IQS}, 11^{SB}$	§VII.A
34	Redundancy ratios normalize the mutual information between the system and each environment fragment by the system entropy. The definition specifies disjoint fragments, positive system entropy and the measurement basis for locally accessible information.	$\emptyset$	§VII.A.3, Eqs. (7.11)–(7.19)
35	Action distance between environment records is defined under a subsystem partition and constraints on allowed local operations.	$\emptyset$	§VII.A.2
36	In a model of record copying with specified temporal order, Zurek defines redundancy $R(t)$ and its rate dR/dt to describe the spread of information.	$1^S, 2^{IQS}, 17^{QS}$	§VII.A.4, Eq. (7.20)

p	Claim	i	Source Location
37	Zurek defines the relative objectivity of states through multiple observers independently consulting stable environment records and obtaining compatible information.	$5^{\text{O}}, 11^{\text{SB}}$	§VI.A; VII.B–C
38	Histories consist of Heisenberg projections at ordered times and, together with an initial density state, define the decoherence functional. Diagonality of its real part differs from diagonality of the full complex functional.	$1^{\text{S}}, 12^{\text{S}}$	§VII.C, Eq. (7.21)
39	Observers infer the order of past events from compatible redundant records retained in the present and appropriate dynamical assumptions. Zurek expresses the relative objectivity of such a history through independent access to its records.	$5^{\text{O}}, 11^{\text{SB}}, 12^{\text{S}}, 13^{\text{O}}$	§VII.C
40	In the cited cavity experiment, atomic preparation of the field state, waiting and detection of a test atom are ordered, and cavity decay properties are compared with conditional detection probabilities.	$8^{\text{IQB}}, 14^{\text{IO}}$	§VIII.A
41	The cited ion-trap experiments apply time-dependent classical noise and measure coherence changes averaged over realizations. They also distinguish nonmonotonic changes caused by coupling to a finite intermediate environment.	$14^{\text{IO}}, 16^{\text{SP}}$	§VIII.B
42	The cited driven-atom and quantum-chaos experiments investigate dynamical localization and its loss under specified kicks, spontaneous emission or noise.	$6^{\text{QSB}}, 14^{\text{IO}}, 15^{\text{IQS}}$	§VIII.B.1
43	The cited classical optical experiment investigates transverse spatial coherence of light propagating through a random medium and its dependence on separation, exhibiting an analogy with spatial decay laws in quantum models.	$\emptyset$	§VIII.C
44	Interaction symmetries yield pointer and decoherence-free subspaces protected under the specified evolution.	$1^{\text{S}}, 11^{\text{SB}}$	§VIII.D.1
45	Dynamical decoupling suppresses effective system–environment coupling through externally specified time-dependent operations or measurements.	$1^{\text{S}}, 15^{\text{IQS}}$	§VIII.D.1

p	Claim	i	Source Location
46	Environment engineering adjusts drive frequencies, couplings and related parameters to change the effective Hamiltonian and reduced evolution under specified approximations and select protected states.	$7^S, 15^{IQS}$	§VIII.D.2
47	Quantum error correction uses sequences of encoding, syndrome measurement and correction operations. Fault tolerance for long computations depends on error magnitude and character and on encoding resources.	15^{IQS}	§VIII.D.3
48	In his conclusion, Zurek identifies unresolved conceptual and technical issues, including what constitutes a system.	$\emptyset$	§IX
49	Assessments of the reception of Bohr, the teaching of quantum theory and acceptance of interpretations are Zurek's historical and methodological discussion. The Einstein letter is quoted through Born.	$\emptyset$	§I; IX
50	In Zurek's existential interpretation, the one-to-one correspondence between einselected observer-memory states and their information, together with redundant environment records, explains the outcomes and branches known to the observer.	$5^O, 11^{SB}$	§I.C–D; VII.B

Table 400: Zurek 2003 Time-Use Roles

Code	Role	Use Count
I	Interval	27
Q	Quantity	35
X	Cross-position correspondence	0
K	Clock reading	0
C	Causality	0
S	State evolution	63
P	Probabilistic evolution	2
O	Observation record	13
B	Reference boundary	32

Table 401: Zurek 2003 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 7, 12, 16–17	member	instant, order_relation
2	member	endpoint, interval_boundary
3, 6, 8–11, 18	member	interval_boundary
4–5, 13	member	order_relation
14	member	endpoint, interval_boundary, order_relation
15	member	instant, interval_boundary, order_relation

Table 402: Zurek 2003 Finite Cardinalities

Object	Cardinality
Time contents	18
Claims	50
Transfers	13
Results	43
Claim/time-content uses	88
Role assignments	172
Obligation atoms	124
Audit-boundary nodes	111
Audit-boundary edges	56

Dependency Inputs

Definition 8.541 (Zurek 2003 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [13]$. The table gives time inputs, time outputs, claim support and source locations.

Table 403: Zurek 2003 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1–2	4	7	Obtain premeasurement before-and-after correlations from interaction. §II.A; VI.B
2	1–2	3	10	Obtain entangling and recurrence scales in finite models. §II.A.2; IV.A.1
3	1	6	11–12	Estimate correspondence scales from chaotic evolution. §III
4	1	8	14	Obtain coherence change from spin-environment evolution. §IV.A.1
5	7	8, 18	20–21	Obtain decoherence and relaxation scales from reduced evolution. §V.B

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
6	6–7	9	25	Obtain the entropy-production interval from chaos and diffusion. §V.E.2
7	6	10	26	Obtain a predictability horizon from the chaotic stretching scale. §V.E.3
8	1–2	11	15	Obtain pointer stability from conservation conditions. §IV.A; IV.D
9	7	11	18, 22	Select stable candidates by evaluating reduced evolution. §IV.D; V.C
10	4, 8, 11, 18	5	28	Obtain consultable memory from measurement and retention conditions. §VI.B–C
11	5, 11–12	13	39	Infer the past from present records and history conditions. §VII.C
12	1–2	17	36	Obtain redundancy-growth rate from timed record copying. §VII.A.4
13	1	7	19	Obtain reduced evolution from composite-system evolution. §V.A

Grounding Conditions

1. Apparatus truth table, coupling, initial state and interaction time.
2. Specified spectrum, finiteness and isolation conditions.
3. Positive Lyapunov exponent, initial localization and nonlinearity scale.
4. Specified interaction and initial state, with an overlap that may or may not decay.
5. Validity domain of the high-temperature, decoherence-dominated approximation, position separation and specified damping coefficient.
6. After diffusion-limited width saturation and before equilibration.
7. Finite Planck constant, initial resolution and characteristic system scale.
8. Commutation with the total Hamiltonian, or a specified approximation and evaluation interval.
9. Evaluation interval, initial candidates and functional, with the underdamped approximation for the oscillator result.
10. Consultation occurs after decoherence and before relaxation erases the information.
11. Compatible redundant records accessible independently, with dynamical assumptions.
12. Order and times of copying interactions and the specified environment partition.
13. Harmonic oscillator, linear coupling, harmonic thermal bath and initial product state.

Results

Definition 8.542 (Zurek 2003 Results and Dependencies). Let $\text{Idx}_R^{\text{audit,result}} = [43]$. The table gives each result's temporal premises and claim support.

Table 404: Zurek 2003 Results and Audit Inputs

q	Result and Source Location	Time Premises i	Claim Support p
1	Controlled maps and information transfer. §II.A	$\emptyset$	6
2	Premeasurement and acquisition of an outcome. §II; VI.B–C	1–2, 4	7, 9
3	Information amplification. §II.A–B	1–2	8
4	Entanglement, disentanglement and recurrence in finite models. §II.A; IV.A	1–3	10
5	Limits of correspondence under chaos. §III; V.E	1, 6	11–12
6	Partial trace and environment-state overlap. §II.B; IV.A	$\emptyset$	13
7	Coherence change in a spin environment. §IV.A	1, 8	14
8	Conservation of pointer subspaces. §IV.A; IV.D	1–2, 11	15
9	Mutual information, conditional information and discord. §IV.B; VI.C.4	$\emptyset$	16
10	Dephasing caused by classical noise. §IV.C	16	17
11	Predictability sieve. §IV.D; V.C	7, 11	18
12	Time-local reduced evolution. §V.A	1, 7	5, 19
13	High-temperature diffusion and decay approximation. §V.A–B	7–8, 18	20
14	Relation of decoherence and relaxation scales. §V.B	7–8, 18	21
15	Coherent pointer states. §V.C	7, 11	22
16	Classical limit in phase space. §V.D	7–8, 11	23
17	Chaos–diffusion balance and restored correspondence. §V.E.1	6–8	24
18	Entropy-production rate over a finite interval. §V.E.2	6–7, 9	25
19	Quantum predictability horizon. §V.E.3	6, 10	26
20	Growth of recorded information under gentle observation. §V.E.2	4, 6, 8–9	27
21	Memory formation, retention and consultation. §VI.B–C	4–5, 8, 11, 18	28
22	Measurement repeatability and record basis. §VI.C.1–3	1–2, 4, 11	29
23	Envariance symmetry. §VI.D.1–2	$\emptyset$	30

q	Result and Source Location	Time Premises i	Claim Support p
24	Born weights from envariance. §VI.D.2	$\emptyset$	31
25	Born weights and relative frequencies. §VI.D.3	$\emptyset$	32
26	Formation of environment records. §VII.A	1–2, 11	33
27	Information redundancy between system and environment fragments. §VII.A.3	$\emptyset$	34
28	Action distance between environment records. §VII.A.2	$\emptyset$	35
29	Time rate of change of record redundancy. §VII.A.4	1–2, 17	36
30	Relative objectivity through environment records. §VI.A; VII.B–C	5, 11	37
31	Histories and the decoherence functional. §VII.C	1, 12	38
32	Past inference from present records. §VII.C	5, 11–13	39
33	Decay detection in cavity experiments. §VIII.A	8, 14	40
34	Noise and coherence in ion-trap experiments. §VIII.B	14, 16	41
35	Dynamical localization and its loss in driven atoms. §VIII.B.1	6, 14–15	42
36	Spatial-coherence analogy in classical optics. §VIII.C	$\emptyset$	43
37	Decoherence-free subspaces. §VIII.D.1	1, 11	44
38	Dynamical decoupling. §VIII.D.1	1, 15	45
39	Selection of protected states by environment engineering. §VIII.D.2	7, 15	46
40	Error correction and fault-tolerant computation. §VIII.D.3	15	47
41	The problem of defining a system. §IX	$\emptyset$	48
42	Zurek’s discussion of reception and teaching. §I; IX	$\emptyset$	49
43	Explanation of outcomes and branches through observer memory. §I.C–D; VII.B	5, 11	50

Definition 8.543 (Zurek 2003 Conditions for Results). The table gives the conditions and domains of the results.

Table 405: Zurek 2003 Conditions for Results

Result Index	Conditions and Domain
1	Specified input basis and correlation map.
2	Interaction time, apparatus truth table and observer records.

Result Index	Conditions and Domain
3	Apparatus dimension, gain, initial state and distinguishability.
4	Spectrum, finiteness, isolation conditions and interaction time.
5	Initial localization, positive Lyapunov exponent and nonlinearity scale.
6	Composite state and specified subsystem partition.
7	Interaction, initial environment state and evolution time.
8	Commutation condition or specified approximation and evaluation interval.
9	State, measurement basis and definition of conditional information.
10	Distribution of noise realizations and individual phase histories.
11	Initial candidates, evaluation interval and entropy or purity.
12	Linearly coupled harmonic system and thermal bath, with an initial product state.
13	Validity conditions on energy, cutoff and initial transient.
14	Specified high-temperature model and validity range of position separation.
15	Underdamped oscillator, purity-loss approximation and evaluation over one period.
16	Macroscopic scale, environmental coupling and initial resolution.
17	Domain where the coherence length is small relative to the nonlinearity scale.
18	After width saturation and before equilibration.
19	Finite Planck constant, initial resolution and characteristic scale.
20	Observation intervals between the decoherence and Lyapunov times.
21	After decoherence and before relaxation erases the information.
22	Measurement interaction, apparatus truth table and remeasurement conditions.
23	Schmidt decomposition and allowed local unitary transformations.
24	Locality, equiprobability, normalization and additivity, with continuity for incommensurate weights.
25	Product of independent systems in the same state, finite-ensemble binomial distribution and large-ensemble limit.
26	Interaction with the system, information propagation and environment partition.
27	Disjoint fragments, positive system entropy and specified basis.
28	Subsystem partition and allowed local operations.
29	Timed copying interactions and environment partition.
30	Independent consultation of stable records and compatibility of information.
31	Initial state, evolution law and ordered projection times.
32	Compatible redundant records and dynamical assumptions.
33	Order of preparation and test atoms, waiting time and cavity conditions.
34	Noise realizations, time modulation and intermediate-environment conditions.
35	Kick sequence, spontaneous emission or noise, and measurement conditions.
36	Random medium, transverse optical separation and transport model.

Result Index	Conditions and Domain
37	Symmetries of the interaction governing evolution.
38	Externally imposed time-dependent operations and measurement procedures.
39	Drive frequencies, coupling, effective approximation and initial state.
40	Error model, correction cycles and encoding resources.
41	Open problem explicitly identified in Zurek’s conclusion.
42	Attribution of the assessments and quoted letter.
43	Einselection, one-to-one correspondence of memory state and information, and redundant records.

Result Dependence

Definition 8.544 (Zurek 2003 Result-Dependence Hyperedges). For each row of the result table, let $\text{ResultEdge}_{R,q}$ ($q \in [43]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Evidence Hierarchy Formal system–environment models give correlations, decay and stability according to their interactions, initial states, approximations and time domains. Enviance and redundancy definitions use conditions on states, symmetry, probability and information quantities. The history formalism and physical inference of the past from present redundant records each use their own inputs. Experimental descriptions rely on the cavity, ion-trap, driven-atom and classical-optics studies cited by Zurek.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local-reach count are both zero, leaving 18 primitive-time roots. Each transfer takes evolution time, interaction time, a timescale or existing record order as input. Experimental schedules and quantum-control clocks are supplied to their respective experimental and control procedures.

Result Dependence Of the 43 results, 32 depend on primitive-time roots. The other 11 are reached from source formal evidence concerning controlled maps, partial traces, information quantities, enviance and probability, redundancy, spatial-coherence analogy or attribution. There are 94 root–result dependence pairs, of which 31 are necessary.

Placement of the Grounding Obligation Zurek’s dynamical models use quantum-evolution times, interaction durations, decay scales, record order and control times in their laws and predictions. These temporal uses carry the physical-time grounding obligation; the LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The initial state, system–environment partition, interaction, approximation conditions and time domain of each model determine the domain of its results. Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 94 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Definition 8.545 (Zurek 2003 Finite Dependency Graph). There are 50 claims, 18 time contents, 13 transfers and 43 results. Counting claims, time contents and results as nodes, and transfers and result dependencies as edges,

$$\text{BoundaryCount}_R^{\text{audit}} = (111, 56).$$

Proposition 8.546 (Primitive Time Roots and Result Dependence in Zurek 2003). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = (\emptyset, [18]; \\ \emptyset, \{1, 6, 9, 23, 24, 25, 27, 28, 36, 41, 42\}, \\ [43] \setminus \{1, 6, 9, 23, 24, 25, 27, 28, 36, 41, 42\}, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Even with all 50 claims as source formal evidence, all 13 transfers require temporal inputs. The 11 results with empty temporal premises are reached from source formal evidence; adding all 18 time contents as seeds reaches all 43 results. $\square$

Proposition 8.547 (Minimal Primitive-Time-Root Support in Zurek 2003). The table gives each result’s minimal-support index family, with 79 family members in total.

Proof. Start with singleton time-content seeds and empty supports for results with empty temporal premises, and apply Definition 7.46 to all transfers and result dependencies. By Proposition 7.47, the inclusion-minimal supports of the stable antichains are the displayed families. $\square$

Table 406: Minimal Primitive-Time-Root Support in Zurek 2003

Result Indices	Minimal-Support Family Members
1, 6, 9, 23–25, 27–28, 36, 41–42	$\emptyset$
2–4, 8, 22, 26, 29	$\{1, 2\}$
5, 7, 12, 37	$\{1\}$
10	$\{16\}$
11, 13–16	$\{1\}, \{7\}$
17–18	$\{1\}, \{6, 7\}$
19	$\{1\}, \{6\}$

Result Indices	Minimal-Support Family Members
20	$\{1, 2\}, \{1, 4\}, \{4, 6, 7\}, \{4, 6, 8, 9\}$
21	$\{1, 2\}, \{1, 4\}, \{4, 7\}, \{4, 8, 11, 18\}$
30, 43	$\{1, 2\}, \{1, 4\}, \{1, 5\}, \{4, 7\}, \{5, 7\}, \{5, 11\}, \{4, 8, 11, 18\}$
31	$\{1, 12\}$
32	$\{1, 2, 12\}, \{1, 4, 12\}, \{1, 5, 12\}, \{4, 7, 12\}, \{5, 7, 12\}, \{5, 11, 12\}, \{4, 8, 11, 12, 18\}$
33	$\{1, 14\}, \{7, 14\}, \{8, 14\}$
34	$\{14, 16\}$
35	$\{1, 14, 15\}, \{6, 14, 15\}$
38	$\{1, 15\}$
39	$\{1, 15\}, \{7, 15\}$
40	$\{15\}$

Evidentiary Effect Premeasurement and einselection explain correlations, pointer stability and records under specified interactions and time domains. Probability and information arguments use their respective state, symmetry and normalization conditions; histories and past inference use temporal order and record conditions. Experimental descriptions and quantum-control results rely on the models, apparatus and operating conditions of the studies cited by the source.

Result-Preservation Domain Fix the models, initial states, interactions, approximation conditions and domains in Table 405. When temporal inputs can be realized by generated temporal representations preserving the same evolution, intervals, rates, record order and control order, time-dependent results are preserved within their domains. Results with empty temporal premises are determined by the nontemporal formal conditions stated in the tables.

Proposition 8.548 (Complete Audit of Time Use in Zurek 2003). There are 18 time contents, 50 claims, 13 transfers and 43 results. The 88 claim/time-content use pairs have 172 role assignments. There are 124 source-specific obligation atoms. Complete generation and local reach are both zero, leaving 18 primitive-time roots and 32 root-dependent results.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual Statements Completion of the many-worlds interpretation by decoherence

Original text:

Decoherence makes MWI complete

Source location: Section I.C. Zurek explains branches of the many-worlds interpretation and classical records available to observers through decoherence and einselection. Section IX identifies open issues, including what constitutes a system.

Collapse through einselection and memory

Original text:

The collapse is a consequence of einselection

Source location: Section I.D. Following the quoted phrase, Zurek identifies the one-to-one correspondence between the observer's memory state and the information encoded in it, together with einselection, as the grounds for collapse. Section VII.B develops this account through observer memory and environment records.

The environment as a witness in determining what exists

Original text:

the environment as a witness in determining what exists

Source location: Section I.D. Zurek describes the environment broadcasting information about a system and observers obtaining information through its fragments. Sections VI.A and VII define relative objectivity through independent access to stable records.

Common Evidence with Specified Scope The table gives the objects, predicates and domains of the three statements.

State- ment	Object	Predicate	Explicit Domain
S44- Q001	Branches of the many-worlds interpretation and observers' classical records	Zurek states that decoherence makes the many-worlds interpretation complete.	Branches, einselection and records in Section I.C, with the theoretical scope discussed in Section IX.
S44- Q002	Measurement outcomes and collapse as available in observer memory	Zurek attributes collapse to einselection and the one-to-one correspondence between memory state and information.	The correspondence of memory states, information and environment records in Sections I.D and VII.B.
S44- Q003	System information recorded in environment fragments and independently accessible to observers	Zurek describes the environment as a witness in determining what exists.	Environment records, information acquisition and relative objectivity in Sections I.D, VI.A and VII.

Individual Audit Decoherence and branches of the many-worlds interpretation

In the first statement, Zurek states that decoherence makes the many-worlds interpretation complete. Interference suppression and pointer selection through system–environment interaction provide records stably available to observers. In Zurek's construction, unitary premeasurement, memory retention and environment records support this account. The open issues Zurek identifies in Section IX, including the definition of a system, concern

the domain of this theoretical construction. Claims 7, 9, 15, 28, 37, 48 and 50 and Results 2, 8, 21, 30, 41 and 43 correspond to the statement.

Einselection and observer memory

In the second statement, Zurek attributes the collapse known to the observer to einselection and the one-to-one correspondence between memory state and information. When system–apparatus correlations undergo einselection, the records encoded in pointer states become available for consultation by the observer. Consultation uses the interval after decoherence and before relaxation erases the information. The correspondence between observer-memory states and redundant information in the environment provides the account of outcomes and branches in the source’s existential interpretation. Claims 7, 9, 15, 28, 29 and 50 and Results 2, 8, 21, 22 and 43 correspond to the statement.

Environment records and relative objectivity

In the third statement, Zurek describes the environment recording system information as a witness in determining what exists. Observers obtain information from environment fragments correlated with the system without acting directly on it. For a specified environment partition and basis, a redundancy ratio based on locally accessible information expresses independent accessibility of records, while consultation of stable records by multiple observers establishes relative objectivity. Record formation and retention, the rate of timed copying and past inference from present records use their respective evolution, order and dynamical conditions. Claims 33–39 and Results 26–32 correspond to the statement.

8.19.45 Source audit

Audit input

Original source and audit scope

Source identifier 45.

Source title Primitive-time-root audit of Davis–Lineweaver 2004

Bibliographic identification Let T. M. Davis and C. H. Lineweaver, *Expanding Confusion: Common Misconceptions of Cosmological Horizons and the Superluminal Expansion of the Universe*, *Publications of the Astronomical Society of Australia* 21, 97–109 (2004), doi:10.1071/AS03040 be $R = \text{Ref}_{\text{DL04EC}}$. The audit domain comprises the abstract, §§1–6 of the original source, Appendices A and B of the original source, Figures 1–6 of the original source, and equations (1)–(29) of the original source.

Audit-scope identifier 45-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 45-L1.

Source-location content The time-content, claim, transfer and result tables identify source sections and equations.

Formal objects

Object identifier 45-F1.

Object name Formal objects of the original source.

Object type Formal object.

Formal content The individual contents of the formal objects are given in Definition 8.550.

Source-location identifier set 45-L1.

Assumptions

Assumption identifier 45-A1.

Assumption content Davis and Lineweaver use cosmic proper time, the scale factor, radial null propagation and distance conventions in homogeneous, isotropic Robertson–Walker models to discuss recession velocities, horizons, reception and supernova observations. Each time content is evaluated for a selected model, source and observation epoch. The particle horizon is determined by integration from a selected initial endpoint, and the event horizon by integration to a selected future endpoint. An infinite future and a finite collapse endpoint belong to different models. Each supernova record belongs to its own observational clock process, calibration and selection conditions.

Source-location identifier set 45-L1.

Nontemporal Physical Inputs Claims 1–4 in Table 409 provide physical carriers and nontemporal connections, spatial structure, model parameters and nontemporal correspondence among recorded values.

Time Contents

Time-content identifier 45-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content description Given in Definition 8.549.

Source-location identifier set 45-L1.

Definition 8.549 (Davis–Lineweaver 2004 Time Contents). Let $n_R = 26$. The table gives the time contents and source locations.

Table 408: Davis–Lineweaver 2004 Time Contents

i	Time Content	Source Location
1	Cosmic proper time of comoving observers and constant-time slices	§2;5;A(15)–(18)
2	Temporal histories of the scale factor and expansion rate	§2;3.1–3.3;A(25)–(26)
3	Epoch dependence of proper distance and recession velocity	§3.1;A(16)–(20)
4	Temporal order and emission–reception endpoints of radial light propagation	§2;3.3;A(21)–(24)
5	Temporal boundary of possible reception	§2;3.3
6	Time-indexed particle horizon from the initial endpoint	§3.4;A(27)

i	Time Content	Source Location
7	Comparison of the particle horizon with the light-travel-time scale	§3.4
8	Future endpoint and temporal interval of the selected model	A(28)
9	Time-indexed event horizon from the future null integral	§2;3.2;A(28)
10	Temporal history of the Hubble sphere and superluminal threshold	§2;3.1–3.3
11	Time derivative of the proper distance of an incoming photon	A(19)–(21)
12	Times and order of photon–Hubble-sphere crossings	§2;3.3
13	Finite inflationary interval and reentry epoch	§3.2
14	Planck-time reference	§3.2
15	Source proper-time width and order of successive emissions	§4.1(3)–(9)
16	Receiver arrival-time width	§4.1(3)–(9)
17	Cosmological arrival-time-width ratio	§4.1(3)
18	Records of supernova clock processes, temporal widths and observation epochs	§4.1–4.2;Fig4–5
19	Special-relativistic time-width ratio measured by synchronized inertial clocks	§4.1(4)–(5)
20	Propagation-time increment between successive emissions	§4.1(6)–(9)
21	Special-relativistic arrival-time-width ratio including propagation	§4.1(6)–(9)
22	Temporal history of the luminosity-distance model	§4.2–5(10),(12)–(14)
23	Redshift history and differential drift of a fixed comoving source	§4.3(11);Fig6
24	Epochs, comparison baseline and planned sequence of repeated spectroscopy	§4.3;Fig6
25	Alternative coordinates and reparametrization of cosmic time	§5
26	Conformal-time differential and interval	§2;A(29)

Definition 8.550 (Davis–Lineweaver 2004 Physical-Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(37)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([4], [37])$. Claims 1–4 form the generation-initial set; all 37 claims form the source formal-evidence environment.

Definition 8.551 (Cosmological Relations). With cosmic proper time t , scale factor $R(t) > 0$, and comoving radial coordinate χ , the relations for a fixed comoving object and an incoming photon are, respectively,

$$D = R\chi, \quad H = \dot{R}/R, \quad v_{\text{rec}} = HD, \quad \dot{D}_\gamma = HD_\gamma - c.$$

For emission and observation epochs t_{em}, t_0 , the past light cone and redshift satisfy

$$\chi = c \int_{t_{\text{em}}}^{t_0} \frac{dt}{R(t)}, \quad 1 + z = \frac{R(t_0)}{R(t_{\text{em}})}, \quad \chi = \frac{c}{R(t_0)} \int_0^z \frac{dz'}{H(z')}.$$

The last representation holds on a monotone branch at fixed observation epoch. When the respective integrals are finite, the particle and event horizons are

$$D_{\text{ph}}(t) = R(t)c \int_{t_i}^t \frac{dt'}{R(t')}, \quad D_{\text{eh}}(t) = R(t)c \int_t^{t_{\text{end}}} \frac{dt'}{R(t')}.$$

For a fixed comoving source, the proper-time correspondence between emission and observation gives the differential drift

$$\frac{dt_{\text{em}}}{dt_0} = \frac{1}{1+z}, \quad \frac{dz}{dt_0} = H(t_0)(1+z) - H(t_{\text{em}}).$$

The change over a finite baseline is the integral of this rate. A constant-rate approximation applies when variation of the rate over the baseline is negligible.

For constant radial relative velocity βc , the width ratio compared with synchronized inertial clocks and the arrival-width ratio including propagation delay are, respectively,

$$\gamma = (1 - \beta^2)^{-1/2}, \quad \frac{\Delta t_{\text{arrival}}}{\Delta t_{\text{em}}} = \gamma(1 + \beta) = 1 + z.$$

The cosmological arrival-width ratio has the same redshift factor over an interval in which coefficient variation is negligible. For selected epochs and curvature, the distance conventions are

$$D = R(t)\chi, \quad D_L = R_0 S_k(\chi)(1+z), \quad D_A = \frac{R_0 S_k(\chi)}{1+z}.$$

Conformal time τ is defined by $d\tau = dt/R(t)$. Under the source convention in which R has dimensions of length, τ has dimensions of time divided by length; the time-dimensional coordinate corresponding to $a = R/R_0$ is $R_0\tau$. A regular monotone transformation $u = f(t)$ preserves the same homogeneous slices when it does not mix spatial coordinates, and changes the temporal metric term to $-c^2(dt/du)^2 du^2$.

Table 409: Davis–Lineweaver 2004 Claims and Temporal Uses

p	Claim	i	Source Location
1	Physical carriers and nontemporal connections of galaxies, supernovae, photons, observers, apparatus and records	$\emptyset$	§2–4
2	Nontemporal structure of comoving spatial coordinates, spatial curvature, spatial differences and labels	$\emptyset$	§5;A(15)–(16)
3	Matter and energy components and nontemporal model parameters	$\emptyset$	§2;A(25)–(26)
4	Recorded spectral, magnitude and redshift values and their nontemporal correspondence	$\emptyset$	§4.1–4.3
5	Robertson–Walker metric and scale factor	$1^{\text{IS}}, 2^{\text{S}}$	A(15)
6	Comoving proper time, simultaneous slices and homogeneity	1^{IXK}	§2;5;A(15)–(16)

p	Claim	i	Source Location
7	Proper distance at fixed time and recession velocity of a comoving object	$1^{\text{IQ}}, 2^{\text{S}}, 3^{\text{QS}}$	A(16)–(18)
8	Radial null paths and the past-light-cone integral	$1^{\text{IB}}, 2^{\text{S}}, 4^{\text{ICB}}$	A(21)–(22)
9	Scale-factor redshift and the redshift representation of the past light cone	$1^{\text{XB}}, 2^{\text{S}}, 4^{\text{IXB}}$	A(23)–(24)
10	Friedmann expansion model with radiation omitted	$1^{\text{S}}, 2^{\text{S}}$	A(25)–(26)
11	Hubble sphere and recession-speed threshold	$1^{\text{B}}, 2^{\text{S}}, 10^{\text{QB}}$	§2;3.1–3.3
12	Particle-horizon integral from a selected initial endpoint	$1^{\text{IB}}, 2^{\text{S}}, 4^{\text{ICB}}, 6^{\text{QB}}$	A(27)
13	Time-indexed particle horizon and light-travel-time scale	$1^{\text{IQ}}, 6^{\text{QB}}, 7^{\text{IQ}}$	§3.4;A(27)
14	Selected future endpoint and convergence condition for the event horizon	$1^{\text{IB}}, 2^{\text{S}}, 4^{\text{ICB}}, 8^{\text{IB}}, 9^{\text{QB}}$	A(28)
15	Separation of recession and peculiar velocities and proper-distance rate of an incoming photon	$1^{\text{IQ}}, 2^{\text{S}}, 3^{\text{QS}}, 4^{\text{ICS}}, 11^{\text{QS}}$	A(19)–(21)
16	Hubble-sphere crossing and subsequent photon reception	$4^{\text{ICS}}, 5^{\text{CB}}, 10^{\text{QS}}, 11^{\text{QS}}, 12^{\text{CB}}$	§2;3.3
17	Epoch-specific recession velocity and redshift at fixed comoving position	$1^{\text{B}}, 2^{\text{S}}, 3^{\text{QS}}, 4^{\text{XB}}$	§3.1(1);A(17), (24)
18	Algebraic form of the special-relativistic Doppler relation	$\emptyset$	§3.1(2);4.1(5)
19	Synchronized inertial clocks and the proper-time width of a moving source	$15^{\text{IK}}, 19^{\text{IQX}}$	§4.1(4)–(5)
20	Special-relativistic arrival width including propagation delay between successive emissions	$15^{\text{IK}}, 16^{\text{IK}}, 19^{\text{IQ}}, 20^{\text{IQ}}, 21^{\text{IQ}}$	§4.1(6)–(9)
21	Emission and arrival widths in the Robertson–Walker model	$1^{\text{S}}, 2^{\text{S}}, 4^{\text{IXC}}, 15^{\text{IK}}, 16^{\text{IK}}, 17^{\text{IQ}}$	§4.1(3)
22	Equality of the two arrival-time-width ratios as functions of redshift	$17^{\text{QX}}, 21^{\text{QX}}$	§4.1(3),(9)
23	Observational records of Type Ia supernova clock widths and spectral changes	18^{IOK}	§4.1;Fig4
24	Arrival-width data and the comparison model with unit width ratio	$17^{\text{QX}}, 18^{\text{IO}}$	§4.1;Fig4
25	Illustrative magnitude curve under the selected velocity and distance prescriptions	22^{SB}	§4.2(10),(13)
26	Comparison of supernova magnitude records with the illustrative curves	$18^{\text{OB}}, 22^{\text{SB}}$	§4.2;Fig5
27	Redshift-drift equation for a fixed comoving source	$1^{\text{IS}}, 2^{\text{S}}, 4^{\text{IXB}}, 23^{\text{QS}}$	§4.3(11)
28	Drift forecast over a finite baseline and plan for repeated observations	$23^{\text{QS}}, 24^{\text{IB}}$	§4.3;Fig6
29	Baseline, precision and future apparatus required for repeated redshift observations	24^{IOB}	§4.3

p	Claim	i	Source Location
30	Conventions for proper, luminosity and angular-diameter distances	$1^B, 2^S, 4^{XB}, 22^{SB}$	§5(12)–(14)
31	Monotone coordinate relabeling of cosmic proper time and its foliation	$1^X, 25^{XS}$	§5
32	Conformal-time differential and integral with specified endpoints	$1^{IQ}, 2^S, 26^{IQB}$	A(29)
33	Finite inflation and reentry into the comoving Hubble radius	$2^S, 10^{QS}, 13^{ISB}$	§3.2
34	Coincidence of the Hubble sphere and event horizon in eternal exponential expansion	$1^B, 2^S, 8^{IB}, 9^{QB}, 10^{QB}$	§3.2;A(28)
35	Comparison of the Hubble radius with the Planck-time reference	$2^Q, 10^Q, 14^{IQ}$	§3.2
36	Distinction between present and all-epoch superluminal recession	$1^B, 2^S, 3^{QS}, 4^{XB}, 10^{QB}$	§3.1–3.3;Fig2
37	Distinction among spatial expansion, moving clocks and arrival widths	$3^Q, 17^{IQ}, 19^{IKQ}, 21^{IQ}$	§3.1;4.1;5

Definition 8.552 (Roles and Structure of Temporal Uses). Superscripts on time indices in the claim table denote the use roles in the following table.

Table 410: Davis–Lineweaver 2004 Time-Use Roles

Code	Role	Use Count
I	Interval	41
Q	Quantity	38
X	Cross-position correspondence	14
K	Clock reading	8
C	Causality	8
S	State evolution	38
P	Probabilistic evolution	0
O	Observation record	4
B	Reference boundary	36

Table 411: Davis–Lineweaver 2004 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–2, 10, 12, 22–23	member	instant, order_relation
3, 11	member	instant
4, 6, 9	member	endpoint, order_relation
5	member	order_relation, endpoint
7, 14, 16–17, 19–21, 26	member	interval_boundary
8	member	endpoint, interval_boundary

Time Indices	Structural Type	Axes
13, 15	member	interval_boundary, order_relation
18, 24	member	instant, interval_boundary, order_relation
25	member	order_relation

Table 412: Davis–Lineweaver 2004 Finite Cardinalities

Object	Cardinality
Time contents	26
Claims	37
Transfers	16
Results	29
Claim/time-content uses	100
Role assignments	187
Obligation atoms	108
Audit-boundary nodes	92
Audit-boundary edges	45

Dependency Inputs

Definition 8.553 (Davis–Lineweaver 2004 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [16]$. Each row gives time inputs, time outputs and claim support.

Table 413: Davis–Lineweaver 2004 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1	2	5, 10	Evaluate scale-factor evolution. A(15),(25)–(26)
2	1–2	3	7	Obtain recession velocity at a selected epoch. A(16)–(18)
3	1–2, 4	6	12	Obtain the time-indexed particle horizon. A(27)
4	1–2, 4, 8	9	14	Obtain the time-indexed event horizon. A(28)
5	1–2	10	11	Obtain the temporal history of the Hubble sphere. §2;3.1–3.3
6	1–4	11	15	Obtain the proper-distance rate of an incoming photon. A(19)–(21)
7	4, 10–11	5, 12	16	Obtain photon crossing order and the reception boundary. §2;3.3

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
8	1, 6	7	13	Compare the particle horizon with the light-travel-time scale. §3.4
9	15	19	19	Obtain the time-width ratio of synchronized inertial clocks. §4.1(4)–(5)
10	1–2, 4, 15–16	17	21	Obtain the cosmological arrival-time-width ratio. §4.1(3)
11	15–16, 19–20	21	20	Obtain the special-relativistic arrival-width ratio including propagation. §4.1(6)–(9)
12	1–2, 4	22	30	Express the temporal history through luminosity distance. §5(12)–(14)
13	1–2, 4	23	27	Obtain differential drift for a fixed comoving source. §4.3(11)
14	1	25	31	Monotonically reparametrize cosmic time. §5
15	1–2	26	32	Obtain the conformal-time differential and interval. A(29)
16	1–2, 8	9–10	34	Obtain horizon coincidence on the eternal exponential branch. §3.2;A(28)

Table 414: Davis–Lineweaver 2004 Conditions for Transfers

Transfer Index	Conditions
1	Cosmic proper-time domain, nontemporal model parameters, scale factor at the reference epoch and selected expanding branch.
2	Fixed comoving position and differentiable scale factor.
3	Null integral from the selected initial endpoint to the evaluation epoch, with convergence.
4	Convergence of the null integral to the selected future endpoint.
5	$H > 0$ and the adopted distance domain.
6	Specified radial null path and decomposition into recession and peculiar velocity.
7	Evaluate crossing and reception under the specified complete trajectory and expansion history.
8	Same evaluation epoch and initial endpoint, with the selected cosmological model.
9	Constant relative velocity and the specified inertial-clock comparison procedure.
10	Null propagation and the scale-factor–redshift relation; the finite-width formula assumes negligible coefficient variation over the interval.
11	Constant radial relative velocity and delay due to the path-length increment.

Trans-fer Index	Conditions
12	Selected curvature and distance conventions, with emission and observation epochs.
13	Same comoving source, differentiable scale factor and the relation between emission and observation epochs.
14	A monotone transformation without spatial-coordinate mixing preserves the slices and transforms the lapse.
15	$R(t) > 0$; specify endpoints for an integrated interval and the normalization of the time coordinate.
16	Exponential expansion with constant $H > 0$ continues to the infinite future.

Results

Definition 8.554 (Davis–Lineweaver 2004 Results). Let $\text{ld}\mathbf{x}_R^{\text{audit,result}} = [29]$. The table gives the results and their source locations.

Table 415: Davis–Lineweaver 2004 Results

q	Result	Source Location
1	Robertson–Walker metric and expansion model using cosmic proper time	§2;A(15),(25)–(26)
2	Proper distance at fixed time and recession velocity	A(16)–(18)
3	Recession and peculiar velocities and local light propagation	A(19)–(21)
4	Past-light-cone integral with specified endpoints	A(21)–(22)
5	Change of variable to redshift in the past-light-cone integral	A(22)–(24)
6	Time-indexed particle horizon	A(27)
7	Particle horizon and light-travel-time scale	§3.4;A(27)
8	Event horizon for the selected future continuation	A(28)
9	Hubble sphere and condition for superluminal recession	§2;3.1–3.3
10	Hubble-sphere crossing and reception from a source in superluminal recession	§2;3.3
11	Distinction between present and all-epoch superluminal recession	§3.1–3.3;Fig2
12	Horizon coincidence in eternal exponential expansion	§3.2;A(28)
13	Reentry into the comoving Hubble radius after finite inflation	§3.2
14	Scale comparison between the Hubble radius and the Planck reference	§3.2
15	Algebraic identities of the Doppler redshift relation	§3.1(2);4.1(5)

q	Result	Source Location
16	Time-width ratio of synchronized inertial clocks and the source clock	§4.1(4)–(5)
17	Special-relativistic arrival width including light propagation	§4.1(6)–(9)
18	Cosmological relation between emission and arrival widths	§4.1(3)
19	Indistinguishability of the two arrival predictions by width and redshift alone	§4.1(3),(9)
20	Cited observations of supernova clocks and spectral changes	§4.1
21	Cited temporal-width data relative to the unit-width-ratio comparison model	§4.1;Fig4
22	Selected illustrative magnitude curve	§4.2(10),(13)
23	Reported comparison of the illustrative curves with supernova magnitude data	§4.2;Fig5
24	Redshift drift of a fixed comoving source	§4.3(11)
25	Finite-baseline drift forecast and conditions for future observations	§4.3;Fig6
26	Relations among the three distance conventions	§5(12)–(14)
27	Relabeling cosmic time on the same homogeneous slices	§5
28	Representation by conformal time	§2;A(29)
29	Domains of the relations for expansion, moving clocks and arrival widths	§3.1;4.1;5

Table 416: Davis–Lineweaver 2004 Conditions for Results

Result Index	Conditions and Domain
1	Homogeneous isotropic model and selected matter, curvature and cosmological constant; equations (25)–(26) omit radiation.
2	Hold comoving position fixed and differentiate with respect to cosmic proper time.
3	Specified radial path and local light speed c .
4	Emission and reception endpoints and $R(t) > 0$.
5	Fix the observation epoch and relate emission epoch to redshift on a monotone branch with nonzero H .
6	Convergence of the integral from the selected initial endpoint.
7	$D_{\text{ph}}(t) = R(t)\chi_{\text{ph}}(t)$; $D_{\text{ph}} = 3ct$ is exact in the flat dust model.
8	Select the future endpoint and require a finite null integral to that endpoint.
9	For $H > 0$ within the adopted distance domain, $v_{\text{rec}} > c$ is equivalent to $D > c/H$.
10	The source null path reaches the observer under the selected expansion history.

Result Index	Conditions and Domain
11	In the radiation-free $\Omega_M = 0.3, \Omega_\Lambda = 0.7$ model, the present superluminal threshold is $z \simeq 1.46$. A comoving object at $z = 1.5$ currently recedes faster than light and has past epochs of subluminal recession.
12	Exponential expansion with constant $H > 0$ continues to the infinite future.
13	The selected subsequent expansion makes the comoving Hubble radius exceed the scale of the specified mode.
14	Prescribed Planck length and time and the Hubble expansion rate.
15	For $1 + z > 0, \beta = ((1 + z)^2 - 1)/((1 + z)^2 + 1)$.
16	Constant inertial relative velocity and comparison by synchronized clocks.
17	Reception procedure including constant radial relative velocity and the increase in propagation distance.
18	Scale-factor redshift and null propagation; for finite widths, the interval must allow negligible coefficient variation.
19	Under the assumptions of the two procedures, both width ratios equal $1 + z$.
20	Time calibration, selection conditions and uncertainties of the cited observations.
21	Goldhaber et al.'s observations and the 18σ assessment cited by the source.
22	SR Doppler velocity, the approximation $H = H_0$, and the adopted luminosity-distance prescription.
23	Perlmutter et al.'s data, the selected normalization, and Davis–Lineweaver's 23σ and 12σ reports.
24	Relation between emission and observation proper times and differentiable expansion history.
25	Obtain the change by integrating the drift rate over the baseline; the constant-rate approximation requires a short baseline, and observation requires paired records with adequate precision.
26	Common cosmological model, emission and observation epochs, and curvature.
27	Monotone time-coordinate transformation without spatial mixing; the lapse relates it to proper clocks.
28	$R(t) > 0$ and specified normalization, with endpoints when an integrated interval is used.
29	Time references corresponding to each velocity, clock procedure and reception procedure.

Definition 8.555 (Davis–Lineweaver 2004 Result-Dependence Hyperedges). For each row of the table, let $\text{ResultEdge}_{R,q}$ ($q \in [29]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 417: Davis–Lineweaver 2004 Result-Dependence
Hyperedges

q	Time Premises i	Claim Support p
1	1–2	5–6, 10
2	1–3	7
3	1–4, 11	15
4	1–2, 4	8
5	1–2, 4	9
6	1–2, 4, 6	12
7	1, 6–7	13
8	1–2, 4, 8–9	14
9	1–4, 10	11, 17
10	1–5, 10–12	16–17
11	1–4, 10	36
12	1–2, 8–10	34
13	2, 10, 13	33
14	2, 10, 14	35
15	$\emptyset$	18
16	15, 19	19
17	15–16, 19–21	20
18	1–2, 4, 15–17	21
19	17, 21	22
20	18	23
21	17–18	24
22	22	25
23	18, 22	26
24	1–2, 4, 23	27
25	23–24	28–29
26	1–2, 4, 22	30
27	1, 25	6, 31
28	1–2, 26	32
29	3, 17, 19, 21	37

Evidence Hierarchy The cosmological derivations and construction of comparison models are due to Davis and Lineweaver. The supernova clock processes, light curves and spectral changes belong to the observational studies cited in Section 4.1; the 18σ assessment of the unit-width-ratio comparison is attributed to Goldhaber et al. The magnitude data in Section 4.2 are due to Perlmutter et al.; the reported 23σ and 12σ comparisons for the selected illustrative curves are due to Davis and Lineweaver. The 100-year drift and precision requirements in Section 4.3 are forecasts and conditions for future observations. The 25 quotations in Appendix B belong to their respective quoted authors; the subsequent evaluations are Davis and Lineweaver’s commentary.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local-reach count are both zero, leaving 26 primitive-time roots. Cosmic proper time, expansion history, light-propagation endpoints, clock widths, observation epochs and

future comparison baselines supply temporal inputs to their respective relations.

Result Dependence Of the 29 results, 28 depend on primitive-time roots. Result 15 is reached from source formal evidence for the algebraic Doppler relation. There are 80 root–result dependence pairs, of which 42 are necessary.

Placement of the Grounding Obligation Davis and Lineweaver supply cosmological time, expansion histories, endpoints and observational time widths to model calculations and observational comparisons. These temporal uses carry the physical-time grounding obligation; the LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Each result’s domain is determined by the selected cosmological model, expansion history, distance convention, clock and light-propagation procedures, and observational conditions. Actual failure of a specific physical-time generation condition and the resulting change in predictions are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 80 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Definition 8.556 (Finite Dependency Graph of Davis–Lineweaver 2004). Counting 37 claims, 26 time contents and 29 results as nodes, and 16 transfers and 29 result dependencies as edges,

$$\text{BoundaryCount}_R^{\text{audit}} = (92, 45).$$

Proposition 8.557 (Primitive Time Roots and Result Dependence in Davis–Lineweaver 2004). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [26]; \emptyset, \{15\}; [29] \setminus \{15\}, \emptyset).$$

Proof. Closure from Initial Claims 1–4 remains those four nodes. Even with all 37 claims as source formal evidence, all 16 transfers require temporal inputs. Result 15, whose temporal premises are empty, is reached from source formal evidence. Adding all 26 time contents as seeds to that evidence environment reaches all 29 results. $\square$

Proposition 8.558 (Physical-Time Qualification in Davis–Lineweaver 2004). All 26 time contents are primitive-time inputs and lack physical-time qualification. The local-reach judgment reaches zero time contents, and the 28 results other than Result 15 depend on those time contents.

Proof. This follows from the preceding proposition, Definition 7.10, Theorem 7.11 and Corollary 7.12. $\square$

Proposition 8.559 (Minimal Primitive-Time-Root Support in Davis–Lineweaver 2004). The table gives each result’s minimal-support index family, with 46 family members in total.

Proof. Start with singleton time seeds and the empty support of Result 15, and apply Definition 7.46 to all transfers and result dependencies. By Proposition 7.47, the inclusion-minimal supports of the stable antichains are the displayed families. $\square$

Table 418: Minimal Primitive-Time-Root Support in Davis–Lineweaver 2004

Result Indices	Minimal-Support Family Members
1–2, 27–28	$\{1\}$
3–6, 9–11, 24, 26	$\{1, 4\}$
7	$\{1, 4\}, \{1, 6\}$
8	$\{1, 4, 8\}$
12	$\{1, 8\}$
13	$\{1, 13\}, \{2, 10, 13\}$
14	$\{1, 14\}, \{2, 10, 14\}$
15	$\emptyset$
16	$\{15\}$
17	$\{15, 16, 20\}$
18	$\{1, 4, 15, 16\}$
19	$\{1, 4, 15, 16, 20\}, \{15, 16, 17, 20\}, \{1, 4, 15, 16, 21\}, \{17, 21\}$
20	$\{18\}$
21	$\{1, 4, 15, 16, 18\}, \{17, 18\}$
22	$\{1, 4\}, \{22\}$
23	$\{1, 4, 18\}, \{18, 22\}$
25	$\{1, 4, 24\}, \{23, 24\}$
29	$\{1, 4, 15, 16, 20\}, \{1, 15, 16, 17, 20\}, \{3, 15, 16, 17, 20\}, \{1, 4, 15, 16, 21\}, \{1, 15, 17, 21\}, \{3, 15, 17, 21\}, \{1, 17, 19, 21\}, \{3, 17, 19, 21\}$

Result-Preservation Domain Fix the models, conventions, clock procedures and observational conditions in Table 416. When generated temporal representations realize the temporal inputs with the same evolution, endpoints, intervals, rates, order and record correspondence, time-dependent results are preserved within their domains. Result 15 is determined by the algebraic conditions of Claim 18.

Individual statements Original (English):

We can parametrize time using redshift and thus recast Equation (22) in terms of observables.

The source location is Davis and Lineweaver, *Expanding Confusion: Common Misconceptions of Cosmological Horizons and the Superluminal Expansion of the Universe*, *Publications of the Astronomical Society of Australia* 21, p. 106, immediately after equation (22). Davis and Lineweaver transform the time integral for the past light cone in Appendix

A, equation (22), into the redshift integral in equation (24), using the relation between the scale factor and redshift. The transformed quantity is the comoving distance from emission to observation.

Common Evidence with Specified Scope The table gives the object, relation and domain of the quotation.

State- ment	Object	Predicate	Explicit Domain
S45- Q001	Past-light-cone integral giving the comoving distance from emission to observation	Davis and Lineweaver transform equation (22) into equation (24) in terms of observables using the scale-factor–redshift relation.	The past light cone in the homogeneous, isotropic expanding universe of Appendix A; observation at t_0 , emission at t_{em} , and Equations (22)–(24).

Individual Audit

Change of Variable and Temporal Dependence The following change of variable is valid on a monotone branch with $R(t) > 0$ and $H(t) \neq 0$. For fixed $R_0 = R(t_0)$, the scale-factor relation and its derivative give

$$1 + z = \frac{R_0}{R(t)}, \qquad \frac{dz}{dt} = -(1 + z)H(t), \qquad \frac{dt}{R(t)} = -\frac{dz}{R_0H(z)}.$$

Using the endpoint correspondence $t = t_0$ at $z = 0$, it follows that

$$c \int_{t_{\text{em}}}^{t_0} \frac{dt}{R(t)} = \frac{c}{R_0} \int_0^{z_{\text{em}}} \frac{dz}{H(z)}.$$

This change of variable uses cosmic proper time, expansion history, and emission and observation endpoints. These relations establish the representation in terms of observables described in the quotation; its redshift form retains those temporal dependencies.

Correspondence with Formal Evidence Claim 9 and Result 5 represent this change of variable and use Time Contents 1, 2 and 4. These time contents and their dependence relations to the claim and result are specified in Tables 409 and 417 and Proposition 8.557.

8.19.46 Source audit

Audit input

Original source and audit scope

Source identifier 46.

Source title Primitive-time-root audit of Schlosshauer 2005

Bibliographic identification Let the original source be Maximilian Schlosshauer, *Decoherence, the Measurement Problem, and Interpretations of Quantum Mechanics*, *Reviews of Modern Physics* 76, 1267–1305 (2005), DOI: 10.1103/RevModPhys.76.1267, arXiv:quant-ph/0312059v4, denoted by $R = \text{Ref}_{\text{Sch05DMP}}$. The audit domain comprises premeasurement, system–environment interaction, local decoherence, global coherence, three time scales, pointer structure, envariance, collapse models, Bohmian configurations, and consistent histories.

Audit-scope identifier 46-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 46-L1.

Source-location content The time-content, claim, transfer and result tables identify source sections and equations.

Formal objects

Object identifier 46-F1.

Object name Formal objects of the original source.

Object type Formal object.

Formal content The individual contents of the formal objects are given in Definitions 8.563, 8.560, and 8.562.

Source-location identifier set 46-L1.

Assumptions

Assumption identifier 46-A1.

Assumption content Schlosshauer examines measurement, interference suppression, pointer selection and quantum interpretations using a system–apparatus–environment partition, unitary composite evolution, reduced states and environmental correlations. Each model fixes its initial state, Hamiltonian, coupling distribution, time domain and approximation conditions. The envariance probability argument uses the source’s A1–A4 and its counting and continuity conditions; collapse models, Bohmian mechanics and consistent histories use their respective stochastic laws, configurations and ordered times.

Source-location identifier set 46-L1.

Nontemporal physical inputs

Input identifier 46-P1.

Input name Nontemporal physical inputs of the original source.

Input type Nontemporal physical input.

Input content The physical carriers of the system, apparatus, environment, composite and records in Claims 1–5.

Source-location identifier set 46-L1.

Time contents

Time-content identifier 46-T1.

Time-content name Time contents of the original source.

Time-content type Time content.

Time-content description The individual contents of the time contents are given in Definitions 8.561 and 8.560.

Source-location identifier set 46-L1.

Definition 8.560 (Schlosshauer 2005 Formal Relations). Local expectations and evolution in source Sections III.B–D use

$$i\hbar\partial_t|\Psi\rangle = H|\Psi\rangle, \quad \rho_{SA} = \text{Tr}_E \rho_{SAE}, \quad \text{Tr}(\rho_{SAE}(O_{SA} \otimes I_E)) = \text{Tr}_{SA}(\rho_{SA}O_{SA}).$$

For a correlated system–apparatus state $|\Psi\rangle = \sum_n c_n |s_n a_n e_n\rangle$,

$$\rho_{SA}(t) = \sum_{m,n} c_m c_n^* \langle e_n(t) | e_m(t) \rangle |s_m a_m\rangle \langle s_n a_n|.$$

Environmental overlap determines the corresponding off-diagonal terms. Section III.E uses the commutation condition in an interaction-dominated model and entropy or purity evaluated over time:

$$[P_n^{(A)}, H_{AE}] = 0, \quad S(\rho) = -\text{Tr}(\rho \ln \rho), \quad \mathcal{P}(\rho) = \text{Tr}(\rho^2).$$

For a normalized composite state, envariance in Section III.F is

$$(I_S \otimes u_E)(u_S \otimes I_E)|\Psi_{SE}\rangle = |\Psi_{SE}\rangle.$$

For two orthogonal Schmidt components of equal magnitude, an environment swap reverses the system swap. The connection to probabilities uses A1–A4; A4 assumes perfect correlation in an environment measurement following the system measurement. Bohmian configurations in Section IV.F obey

$$\frac{d\mathbf{q}_i}{dt} = \frac{\hbar}{m_i} \text{Im} \frac{\Psi^* \nabla_i \Psi}{\Psi^* \Psi}, \quad \Psi^* \Psi > 0.$$

For the specified ordered times and projector families, Section IV.G uses

$$P_{\alpha_i}^i(t_i) = U(t_0, t_i)^\dagger P_{\alpha_i}^i(t_0) U(t_0, t_i), \quad t_0 < t_1 < \cdots < t_n,$$

$$C_\alpha = P_{\alpha_n}^n(t_n) \cdots P_{\alpha_1}^1(t_1), \quad D(\alpha, \beta) = \text{Tr}(C_\alpha \rho_0 C_\beta^\dagger).$$

For a union of mutually exclusive histories, $C_{\alpha \vee \beta} = C_\alpha + C_\beta$ and

$$p(\alpha \vee \beta) = p(\alpha) + p(\beta) + 2 \text{Re } D(\alpha, \beta).$$

Vanishing real parts of the decoherence functional between distinct histories give probability additivity. For approximate histories based on environmental decoherence, the residual off-diagonal terms determine the additivity error.

Definition 8.561 (Schlosshauer 2005 Time Contents). Let $n_R = 21$. The table gives the time contents and source locations. Each content is scoped by the model, state and domain of its corresponding claims and results.

Table 420: Schlosshauer 2005 Time Contents

i	Time Content	Source Location
1	Time parameter and order of unitary evolution	§ II;III.A–D;IV
2	Start, end and correlation-formation order of premeasurement	§ II.A;III.C(3.5)
3	Temporal history of system–environment interaction	§ III.C–E;V
4	Time dependence of the reduced density matrix	§ III.B–E;IV.D–E;IV.G
5	Time dependence of environmental overlap and interference coefficients	§ III.D(3.8)–(3.18)
6	Averaging interval and long-time limit	§ III.D(3.17)
7	Temporal dependence and validity interval of Gaussian damping	§ III.D(3.18)–(3.20)
8	Characteristic decoherence time	§ III.D–E;IV.E;IV.G
9	Characteristic thermal-relaxation time	§ III.D
10	Recurrence epochs and time scale of a finite environment	§ III.D
11	Persistence and reidentification of pointer correlations, records and branches	§ III.E;IV.C–D;IV.F–G
12	Evaluation epochs and changes of instantaneous Schmidt bases	§ III.E.4;IV.C–D;IV.G
13	Order of system measurement, subsequent environment measurement and future records	§ III.F(A4);IV.C.3
14	Stochastic temporal evolution of collapse processes	§ IV.E
15	Occurrence rate and waiting intervals of spontaneous localization	§ IV.E
16	Temporal trajectories of Bohmian configurations	§ IV.F(4.2)
17	Initial epoch of a history	§ IV.G.1
18	Sequence, order and intervals of history times	§ IV.G(4.3)–(4.16)
19	Time dependence of Heisenberg history projectors	§ IV.G(4.4)–(4.10)
20	Endpoints and evolution of history propagators	§ IV.G(4.4),(4.11)
21	Observation duration and temporal resolution	§ III.D;IV.E

Definition 8.562 (Schlosshauer 2005 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(46)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([5], [46])$. Claims 1–5 form the generation-initial set; all 46 claims form the source formal-evidence environment.

Table 421: Schlosshauer 2005 Claims and Temporal Uses

p	Claim	i	Source Location
1	Physical carriers of the system	$\emptyset$	§ II–III
2	Physical carriers of the apparatus	$\emptyset$	§ II–III
3	Physical carriers of environmental degrees of freedom	$\emptyset$	§ III

p	Claim	i	Source Location
4	Physical carriers of the composite and their nontemporal subsystem partition	$\emptyset$	§ III.A–C
5	Physical carriers of records	$\emptyset$	§ III.E;IV.C.3
6	Hilbert spaces, superposition, tensor products and decomposition conditions	$\emptyset$	§ II;III.A–C;III.E
7	System–apparatus correlation under unitary premeasurement	$1^S, 2^{ISB}$	§ II.A(2.1); III.C(3.5)
8	Definite-outcome and preferred-basis problems after premeasurement and their interpretive conditions	$1^S, 2^{SB}$	§ II.A–C
9	Partial trace, local expectations, and proper and improper mixtures	$\emptyset$	§ III.B(3.1)–(3.4)
10	Total and reduced states evolving under interaction	$1^S, 3^S, 4^S$	§ III.C–D(3.5)–(3.9)
11	Time-dependent environmental overlap and local off-diagonal terms	$3^S, 4^S, 5^{QS}$	§ III.D(3.8)–(3.9)
12	Interaction and interference coefficient in the two-state spin model	$3^S, 5^{QS}$	§ III.D(3.10)–(3.16)
13	Long-time average of interference coefficients and conditions on the couplings	$5^{QS}, 6^{IB}$	§ III.D(3.17)
14	Gaussian temporal approximation for a large environment	$3^S, 5^{QS}, 7^{QSB}$	§ III.D(3.18)–(3.20)
15	Interference damping and decoherence time in the selected model	$3^S, 5^{QS}, 8^{IQ}$	§ III.D
16	Time scales of decoherence, thermal relaxation and recurrence	$8^{IQX}, 9^{IQX}, 10^{IQX}$	§ III.D
17	Recurrence in a finite environment and interference suppression during observation	$1^S, 8^{IQB}, 10^{IQB}, 21^{IB}$	§ III.D
18	Global coherence of the unitarily evolving total state	1^S	§ III.D
19	Environmental record correlations and robustness over time	$1^S, 3^S, 11^{SO}$	§ III.C–E;IV.C.3
20	Pointer selection through commutation with the interaction	$3^S, 11^S$	§ III.E(3.21)–(3.22)
21	Entropy, purity and the predictability sieve	$1^S, 4^S, 11^S$	§ III.E.1
22	Schmidt decomposition of the reduced state at each evaluation epoch	$4^S, 12^{SB}$	§ III.E.4;IV.C–D
23	Comparison of instantaneous Schmidt and stable pointer bases	$4^S, 11^S, 12^{SX}$	§ III.E.4(3.23)
24	Unitary transformation and countertransformation identities of envariance	$\emptyset$	§ III.F(3.24)–(3.27)
25	Probability symmetry for equal amplitudes, rational counting and continuous extension	$\emptyset$	§ III.F(3.28)–(3.32)
26	Subsequent measurement and perfect correlation in assumption A4	13^{OB}	§ III.F(A4)

p	Claim	i	Source Location
27	Measurement conditions in the standard and Copenhagen interpretations	$1^S, 2^{SB}$	§ IV.B
28	Branches, stable records and future observer records in relative-state interpretations	$1^S, 11^{SO}, 13^{OB}$	§ IV.C
29	Property assignment and its temporal evolution in modal interpretations	$1^S, 4^S, 11^S, 12^{SB}$	§ IV.D
30	Stochastic state change in physical collapse models	14^{SP}	§ IV.E
31	Rate, amplification and waiting intervals of spontaneous localization	$14^P, 15^{IQP}$	§ IV.E
32	Comparison of the time scales and bases of decoherence and localization	$8^{IQX}, 14^{SP}, 15^{IQPX}, 21^{IB}$	§ IV.E.1–3
33	Relations between collapse models and decoherence in reduced-state evolution equations	$4^S, 14^{SP}$	§ IV.E.4–5(4.1)
34	Wave function, Bohmian configuration and guiding equation	$1^S, 16^{QS}$	§ IV.F(4.2)
35	Quasiclassical Bohmian trajectories and conditions on environmental interaction	$1^S, 11^S, 16^S$	§ IV.F.2
36	Initial statistical state and initial epoch of a history	17^B	§ IV.G.1
37	Ordered history times and branch-dependent projector choices	18^{IB}	§ IV.G(4.3), (4.5)–(4.7)
38	Unitary propagator between initial and evaluation epochs	$17^B, 18^{IB}, 20^{IS}$	§ IV.G(4.4), (4.11)
39	Heisenberg history projectors obtained from the propagator	$17^B, 18^B, 19^S, 20^S$	§ IV.G(4.4)
40	History functional from time-ordered projectors and the initial state	$17^B, 18^B, 19^S$	§ IV.G(4.9)–(4.11)
41	Probability additivity, weak consistency and medium consistency of histories	$17^B, 18^B, 19^S$	§ IV.G(4.12)–(4.15)
42	Reduced history functional and conditions on dependence on environmental correlations	$4^S, 17^B, 18^B, 19^S, 20^S$	§ IV.G.4
43	Quasiclassical histories from pointer projectors and approximate consistency	$8^{IB}, 11^{SO}, 18^{IB}, 19^S$	§ IV.G.5–8
44	Openness and environmental interaction in realistic quantum systems	3^S	§ V
45	Local classical appearance and conditions for explaining a single outcome	$1^S, 4^S, 11^{SO}$	§ IV.A;V
46	Recurrence under unitary evolution with a finite environmental spectrum	$1^S, 10^{IQB}$	§ III.D

Definition 8.563 (Schlosshauer 2005 Finite Cardinalities). Superscripts on time indices in the claim table denote the use roles below. The finite audit counts are determined by the claims, time contents, transfers and results.

Table 422: Schlosshauer 2005 Time-Use Roles

Code	Role	Use Count
I	Interval	19
Q	Quantity	17
X	Cross-position correspondence	6
K	Clock reading	0
C	Causality	0
S	State evolution	65
P	Probabilistic evolution	6
O	Observation record	6
B	Reference boundary	28

Table 423: Schlosshauer 2005 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3–5, 12, 14, 16, 19	member	instant, order_relation
2, 20	member	endpoint, order_relation
6	member	interval_boundary, order_relation
7, 10	member	instant, interval_boundary
8–9, 15, 21	member	interval_boundary
11	member	order_relation, interval_boundary
13	member	order_relation
17	member	instant, endpoint
18	member	instant, order_relation, interval_boundary

Table 424: Schlosshauer 2005 Finite Cardinalities

Object	Cardinality
Time contents	21
Claims	46
Transfers	10
Results	30
Claim/time-content uses	95
Role assignments	147
Obligation atoms	107
Audit-boundary nodes	97
Audit-boundary edges	40

Dependency Inputs

Definition 8.564 (Schlosshauer 2005 Time-Content Transfers). Let $\text{ld}x_R^{\text{audit,transfer}} = [10]$. The table gives time inputs, time outputs, claim support and source locations.

Table 425: Schlosshauer 2005 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1, 3	4	10	Obtain the time-dependent reduced state from the total state. § III.C–D(3.5)–(3.9)
2	3–4	5	11	Obtain the interference coefficient from environmental overlap. § III.D(3.8)–(3.9)
3	5	6	13	Obtain the long-time average from averaging over specified intervals. § III.D(3.17)
4	3, 5	7	14	Obtain Gaussian damping in the selected large-environment regime. § III.D(3.18)–(3.20)
5	3, 5	8	15	Obtain the characteristic time from damping in the selected model. § III.D
6	1	10	46	Obtain the recurrence scale from unitary evolution of a finite environment. § III.D
7	3	11	20	Obtain persistent pointer correlations from the interaction and stability condition. § III.E(3.21)
8	1	16	34	Obtain Bohmian trajectories from the wave function and guiding equation. § IV.F(4.2)
9	17–18, 20	19	39	Obtain Heisenberg projectors from the history propagator. § IV.G(4.4)
10	17–18	20	38	Obtain the unitary propagator between history epochs. § IV.G(4.4),(4.11)

Grounding Conditions

Table 426: Schlosshauer 2005 Conditions for Transfers

Trans-fer Index	Conditions
1	Specified subsystem partition, initial state, total unitary evolution and partial trace over the environment.
2	Specified interaction and environmental states; the complex-conjugate orientation corresponds to the off-diagonal matrix element.
3	The averaging limit exists; factorization of the product average requires the corresponding nonresonance or phase-averaging conditions on the couplings.

Trans- fer Index	Conditions
4	Fix the coupling distribution, initial weights and approximation interval; the uniform approximation excludes recurrence epochs of the finite environment.
5	Use the temporal damping scale defined by the selected model.
6	Finite model spectrum and specified recurrence accuracy; the scale depends on the couplings and the accuracy.
7	Commutation in an interaction-dominated model, or the corresponding stability condition accounting for the total Hamiltonian.
8	Specified initial configuration and wave function, on the domain where the guiding velocity is defined.
9	Use the initial projectors, ordered times and the same unitary propagator.
10	Total-system Hamiltonian and evolution between the specified endpoints.

Results

Definition 8.565 (Schlosshauer 2005 Results). Let $\text{Idx}_R^{\text{audit,result}} = [30]$.

Table 427: Schlosshauer 2005 Results

q	Result	Source Location
1	Premeasurement correlations and conditions for explaining definite outcomes	§ II.A–C
2	Subsystem partitions, superposition and existence and uniqueness of decompositions	§ III.A–C;III.E
3	Local expectations and the distinction between proper and improper mixtures	§ III.B(3.1)–(3.4)
4	Temporal evolution of the reduced state under interaction	§ III.C–D
5	Local interference suppression through environmental overlap	§ III.D(3.8)–(3.9)
6	Coherence retained in the unitarily evolving total state	§ III.D
7	Time-dependent interference coefficient of the two-state model	§ III.D(3.10)–(3.16)
8	Long-time average of the interference coefficient	§ III.D(3.17)
9	Gaussian damping in a large-environment approximation	§ III.D(3.18)–(3.20)
10	Interference damping and the three time scales	§ III.D
11	Recurrence and practical irreversibility within the observation interval	§ III.D
12	Environmental records and persistent correlations	§ III.C–E;IV.C.3
13	Pointer selection appropriate to the interaction	§ III.E(3.21)–(3.22)
14	Predictability sieving over time	§ III.E.1

q	Result	Source Location
15	Instantaneous Schmidt bases and pointer stability	§ III.E.4(3.23)
16	Algebraic identities of envariance	§ III.F(3.24)–(3.27)
17	Argument for Born probabilities under explicit assumptions	§ III.F(A1)–(A4), (3.28)–(3.32)
18	Measurement and classicality conditions in the standard and Copenhagen interpretations	§ IV.B
19	Branches and records over time in relative-state interpretations	§ IV.C
20	Conditions on property assignment and trajectories in modal interpretations	§ IV.D
21	Time scales of stochastic collapse and spontaneous localization	§ IV.E
22	Joint action of decoherence and collapse models	§ IV.E.1–5
23	Bohmian wave function, configuration and guided trajectories	§ IV.F(4.2)
24	Conditions for the quasiclassical limit of Bohmian trajectories	§ IV.F.2
25	Construction of histories from ordered times and an initial state	§ IV.G(4.3)–(4.7)
26	History functional and probability additivity	§ IV.G(4.9)–(4.15)
27	Reduced history functional under conditions on environmental correlations	§ IV.G.4
28	Quasiclassical histories and approximate consistency of stable pointer projectors	§ IV.G.5–8
29	Openness of realistic quantum systems	§ V
30	Distinction between local classical appearance and a single outcome	§ IV.A;V

Table 428: Schlosshauer 2005 Conditions for Results

Result Index	Conditions and Domain
1	Unitary correlation formation in equation(2.1), distinguishing definite outcomes, the preferred basis and the eigenvalue–eigenstate link.
2	Bipartite Schmidt decomposition and tripartite decomposition satisfying existence, independence and noncollinearity; preserve basis freedom at degeneracies.
3	Observables acting only on the specified subsystem and expectations evaluated with the Born rule; joint correlations are determined by the full density matrix.
4	Specified subsystem partition, initial state, unitary interaction and partial trace.
5	Small environmental overlaps suppress the corresponding off-diagonal elements of the reduced state.

Result Index	Conditions and Domain
6	Unitary evolution of the total system; local suppression coexists with the full state including environmental correlations.
7	Source model with vanishing self-Hamiltonians, initial product environment and specified couplings; define the coefficient by the environmental-state overlap.
8	Existence of the averaging limit and the coupling/phase-averaging conditions; suppression of the average differs from vanishing at every instant.
9	Specified coupling distribution, initial weights and validity interval of the large-environment approximation.
10	Model-specific decoherence, thermal-relaxation and recurrence scales; the source comparisons retain the conditions of the models.
11	Finite model and observation conditions under which interference decays and recurrence lies beyond the observation interval.
12	Interaction with the record-bearing environment and stability of the correlations over the relevant interval.
13	Commutation is sufficient in the specified interaction-dominated model; when self-evolution matters, evaluate stability under the full dynamics.
14	Vary initial states to evaluate entropy or purity and select states that remain robust as time is varied.
15	Even small off-diagonal elements can yield eigenbases far from the stable pointer basis when eigenvalues are nearly degenerate.
16	Normalized composite state and paired system/environment transformations restoring that state.
17	A1–A3 on states, measurable properties and local transformations; subsequent-measurement correlation in A4; equal-amplitude symmetry, rational counting and continuous extension.
18	The source distinctions among the eigenvalue–eigenstate link, collapse rule, classical apparatus and environmental interaction.
19	Robustness enabling branch reidentification and the interpretive conditions on records and probabilities considered by the source.
20	Property-assignment rule, temporal evolution, degeneracy and dimensional conditions; instantaneous decomposition alone does not determine physically stable trajectories.
21	Specified stochastic law and rate per second; the source example combines 10^{23} particles at 10^{-16} per second per particle to give a mean waiting interval of 10^{-7} seconds.
22	Selected bases, the two time scales, compared evolution equations and observation conditions; finite-time localization retains model-dependent tails.
23	Unitary wave-function evolution, initial configuration, guiding equation and the quantum-equilibrium assumption.
24	Conditions on the model’s initial state and long-time behavior accompany environmental interaction; diagonality in position alone is not generally sufficient.

Result Index	Conditions and Domain
25	$t_0 < t_1 < \dots < t_n$, exhaustive orthogonal projectors at each epoch, and the specified Hamiltonian and propagator.
26	A union of histories is represented by the sum of their class operators; $\text{Re } D = 0$ between distinct histories gives additivity, while $D = 0$ is stronger.
27	The source condition that reduced evolution be independent of formed system–environment correlations; a time-local master equation is necessary but not always sufficient.
28	Projection intervals exceeding the decoherence time and stability in the selected model; consistency and probability sum rules are approximate according to the remaining off-diagonal terms.
29	Description based on the interaction between the selected system and its environment.
30	Distinction between full and reduced states and the interpretive conditions specifying realization or perception of one localized state.

Result Dependence

Definition 8.566 (Schlosshauer 2005 Result-Dependence Hyperedges). For each row of the table, let $\text{ResultEdge}_{R,q}$ ($q \in [30]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 429: Schlosshauer 2005 Result-Dependence Hyperedges

q	Time Premises i	Claim Support p
1	1–2	7–8
2	$\emptyset$	6
3	$\emptyset$	9
4	1, 3–4	10
5	3–5	11
6	1	18
7	3, 5	12
8	5–6	13
9	3, 5, 7	14
10	3, 5, 8–10	15–16
11	1, 8, 10, 21	17, 46
12	1, 3, 11	19
13	3, 11	20
14	1, 4, 11	21
15	4, 11–12	22–23
16	$\emptyset$	24
17	13	25–26
18	1–2	27

q	Time Premises i	Claim Support p
19	1, 11, 13	28
20	1, 4, 11–12	29
21	14–15	30–31
22	4, 8, 14–15, 21	32–33
23	1, 16	34
24	1, 11, 16	35
25	17–20	36–39
26	17–19	40–41
27	4, 17–20	42
28	8, 11, 18–19	43
29	3	44
30	1, 4, 11	45

Proposition 8.567 (Audit-Boundary Cardinalities).

$$\text{BoundaryCount}_R^{\text{audit}} = (97, 40).$$

Proof. The 46 claims, 21 time contents and 30 results give 97 nodes; the 10 transfers and 30 result dependencies give 40 edges. $\square$

Evidence Hierarchy The theoretical and interpretive discussion belongs to Schlosshauer’s review. Experimental examples are attributed to the studies cited by the source; this review reports zero newly acquired direct measurements. Results concerning premeasurement, interference suppression, pointer stability and the respective interpretations retain their model, time-domain and interpretive conditions.

Generation Judgment The complete LCTR generation count and source-evidence-conditioned local-reach count are both zero, leaving 21 primitive-time roots. Each transfer takes evolution time, interaction time, an interference history or ordered history epochs as input. Collapse rates and observation intervals also use stipulated temporal scales.

Result Dependence Of the 30 results, 27 depend on primitive-time roots. The other 3 are reached from source formal evidence concerning subsystem decomposition conditions, static partial traces and local expectations, or the algebra of enviance. There are 90 root–result dependence pairs, of which 40 are necessary.

Placement of the Grounding Obligation The dynamical models reviewed by Schlosshauer use quantum-evolution times, interaction durations, decay scales, record order, collapse processes and history epochs in their laws and predictions. These temporal uses carry the physical-time grounding obligation; the LCTR placement judgment is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The initial state, system–environment partition, interaction, approximation conditions and time domain of each model determine the domain of its results. Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 90 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope The repair targets are each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Definition 8.568 (Schlosshauer 2005 Finite Dependency Graph). There are 46 claims, 21 time contents, 10 transfers and 30 results. Counting claims, time contents and results as nodes, and transfers and result dependencies as edges,

$$\text{BoundaryCount}_R^{\text{audit}} = (97, 40).$$

Proposition 8.569 (Primitive Time Roots and Result Dependence in Schlosshauer 2005). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = (\emptyset, [21]; \\ \emptyset, \{2, 3, 16\}, \\ [30] \setminus \{2, 3, 16\}, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–5 remains those five nodes. Even with all 46 claims as source formal evidence, all 10 transfers require temporal inputs. The 3 results with empty temporal premises are reached from source formal evidence; adding all 21 time contents as seeds reaches all 30 results. $\square$

Proposition 8.570 (Minimal Primitive-Time-Root Support in Schlosshauer 2005). The table gives each result’s minimal-support index family, with 59 family members in total.

Proof. Start with singleton time-content seeds and empty supports for results with empty temporal premises, and apply Definition 7.46 to all transfers and result dependencies. By Proposition 7.47, the inclusion-minimal supports of the stable antichains are the displayed families. $\square$

Table 430: Minimal Primitive-Time-Root Support in Schlosshauer 2005

Result Indices	Minimal-Support Family Members
1, 18	$\{1, 2\}$
2–3, 16	$\emptyset$
4, 12	$\{1, 3\}$
5	$\{1, 3\}, \{3, 4\}$
6, 23	$\{1\}$
7, 9	$\{1, 3\}, \{3, 4\}, \{3, 5\}$
8	$\{1, 3\}, \{3, 4\}, \{5\}$
10	$\{1, 3, 9\}, \{3, 4, 9, 10\}, \{3, 5, 9, 10\}$

Result Indices	Minimal-Support Family Members
11	$\{1, 3, 21\}, \{1, 8, 21\}$
13, 29	$\{3\}$
14, 30	$\{1, 3\}, \{1, 4, 11\}$
15	$\{1, 3, 12\}, \{3, 4, 12\}, \{4, 11, 12\}$
17	$\{13\}$
19	$\{1, 3, 13\}, \{1, 11, 13\}$
20	$\{1, 3, 12\}, \{1, 4, 11, 12\}$
21	$\{14, 15\}$
22	$\{1, 3, 14, 15, 21\}, \{3, 4, 14, 15, 21\}, \{4, 8, 14, 15, 21\}$
24	$\{1, 3\}, \{1, 11\}$
25–26	$\{17, 18\}$
27	$\{1, 3, 17, 18\}, \{4, 17, 18\}$
28	$\{1, 3, 17, 18\}, \{3, 4, 17, 18\}, \{3, 5, 17, 18\}, \{3, 8, 17, 18\}, \{8, 11, 17, 18\}, \{1, 3, 18, 19\}, \{3, 4, 18, 19\}, \{3, 5, 18, 19\}, \{3, 8, 18, 19\}, \{8, 11, 18, 19\}$

Evidentiary Effect Premeasurement, local interference suppression, pointer stability and records use the respective interactions and time domains. Envariance identities rest on algebraic conditions, the Born-probability argument on A1–A4, and collapse and Bohmian trajectories on their dynamical assumptions. Consistent histories use an initial state, ordered times, propagators and projectors. Experimental results cited by Schlosshauer belong to the respective experimental studies.

Result-Preservation Domain Fix the models, initial states, interactions, approximation conditions and domains in Table 428. When temporal inputs can be realized by generated temporal representations preserving the same evolution, intervals, rates, record order and history epochs, time-dependent results are preserved within their domains. Results with empty temporal premises are determined by the nontemporal formal conditions stated in the tables.

Proposition 8.571 (Complete Audit of Time Use in Schlosshauer 2005). There are 21 time contents, 46 claims, 10 transfers and 30 results. The 95 claim/time-content use pairs have 147 role assignments. There are 107 source-specific obligation atoms. Complete generation and local reach are both zero, leaving 21 primitive-time roots and 27 root-dependent results.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual statements Statement 1

Original (English):

Environment-induced decoherence and superselection have been a subject of intensive research over the past two decades, yet their implications for the foundational problems of quantum mechanics, most notably the quantum measurement problem, have remained a matter of great controversy.

The source location is the abstract. Schlosshauer summarizes the research accumulated over the two decades preceding the review and the continuing controversy over its foundational implications, including the quantum measurement problem.

Statement 2

Original (English):

In short, decoherence brings about a local suppression of interference between preferred states selected by the interaction with the environment.

The source location is Section I. Schlosshauer describes quantum correlations formed through environmental interaction and identifies local interference suppression between preferred states as their local consequence.

Statement 3

Original (English):

Thus, without supplying an additional physical process (say, some collapse mechanism) or giving a suitable interpretation of such a superposition, it is not clear how to account, given the final composite state, for the definite pointer positions that are perceived as the result of an actual measurement—i.e., why do we seem to perceive the pointer to be in one position $|a_n\rangle$ but not in a superposition of positions?

The source location is the premeasurement discussion following equation (2.1) in Section II.A. Schlosshauer considers a physical process or interpretation for explaining a perceived definite pointer position from the correlated system–apparatus state after unitary evolution.

Statement 4

Original (English):

However, the crucial difficulty in the above reasoning is justifying the second step: How is one to interpret the local suppression of interference in spite of the fact that full coherence is retained in the total state that describes the system–environment combination? While the local destruction of interference allows one to infer the emergence of an (improper) ensemble of individually localized components of the wave function, one still needs to impose an interpretive framework that explains why only one of the localized states is realized and/or perceived.

The source location is Section IV.A. Schlosshauer examines the passage from preferred-basis selection and local interference suppression to definite outcomes, distinguishing coherence in the total state from the realization or perception of a single localized state.

Common evidence with specified domain of application The table gives the subjects, predicates and domains of the preceding four statements.

State- ment	Subject	Predicate	Explicit Domain
----------------	---------	-----------	-----------------

S46-Q001	Research on environmental decoherence and superselection and the quantum measurement problem	Schlosshauer describes two decades of intensive research and continuing controversy over foundational implications.	Research history up to this review, summarized in the abstract.
S46-Q002	Interference between preferred states selected by environmental interaction	Decoherence produces local interference suppression.	The system–environment partition, interaction and local description in Section I.
S46-Q003	The post-premeasurement composite state and a perceived definite pointer position	Without an additional physical process or suitable interpretation, it is unclear how the definite outcome is explained.	Premeasurement in equation (2.1) and the outcome problem in Section II.A.
S46-Q004	Coherence of the total state and realization or perception of one localized state	An interpretive framework is required to explain realization or perception of one localized state from the improper mixture produced by local interference suppression.	The full and reduced states and outcome interpretation in Section IV.A.

Source attribution

All four statements are attributed to Maximilian Schlosshauer.

Individual audit Research history and continuing controversy

Statement 1 is Schlosshauer’s summary of research history. It concerns the studies and interpretive disagreements reviewed in the source. Sections I and IV present the development of the research and the issues associated with the respective interpretations.

Local suppression between preferred states

In Statement 2, local suppression is the process by which environmental overlaps make the off-diagonal elements of the reduced density matrix small. Claims 10–12, Transfers 1–2 and Results 4–7 represent this process using the subsystem partition, interaction, total unitary evolution and partial trace. The full state includes environmental correlations and retains its coherence.

Conditions for explaining a definite pointer position

Statement 3 asks how a correlated superposition in source equation (2.1) accounts for the definite pointer position perceived in an actual measurement. Claims 7–8 and Result 1 represent premeasurement and the definite-outcome problem. The alternatives specified by Schlosshauer are an additional physical process or a suitable interpretation of the superposition.

The full state and a single localized state

Statement 4 distinguishes coherence of the total system–environment state from the improper mixture in the local reduced state. Claims 9–11, 18, 45 and Results 3–6, 30

represent this state distinction and the interpretive conditions explaining realization or perception of one localized state.

Common audit with specified domain of application The arguments concerning local interference suppression and definite outcomes use the specified unitary evolution, interaction and time domain. The generation and result-dependence judgments for the 21 time contents are given in Proposition 8.571.

8.19.47 Source audit

Audit Inputs

Source and Audit Scope

Source identifier 47.

Source name Kiefer 2005 Primitive-Time-Root Audit

Bibliographic identification Claus Kiefer, *Quantum Cosmology and the Arrow of Time*, *Brazilian Journal of Physics* 35, 296–299 (2005); arXiv:gr-qc/0502016v1, pp. 1–9.
 $R = \text{Ref}_{\text{Kie05QCAT}}$.

Audit-scope identifier 47-SC1.

Audit-scope content Abstract, Sections 1–5, Equations (1)–(6), references and notes.

Source-location identifier 47-L1.

Source-location content Tables identify sections, equations and printed pages.

Assumptions Kiefer adopts quantum universality and canonical quantum gravity for a closed universe. The continuous model uses a hyperbolic Wheeler–DeWitt equation; the loop model uses the specified difference equation and operator representation. Semiclassical time uses slowly evolving gravity and a Born–Oppenheimer-type approximation. The arrow construction uses perturbation potentials, a subsystem split and a small-scale product boundary. Recollapse and black-hole consequences use symmetric cosmological boundaries and the validity domains of the respective branches.

Nontemporal Physical Inputs The carriers, spatial configurations, nontemporal partitions and correlations, and state differences before ordering in Claims 1–5 form the generation-initial set.

Time Contents

Time-content identifier 47-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content specification The time contents of Definition 8.572 and their use roles in the claim table.

Source-location identifier set 47-L1.

Definition 8.572 (Kiefer 2005 Time Contents). Let $n_R = 22$. Each time content is scoped by the model, state and domain of the corresponding claims and results.

Table 432: Kiefer 2005 Time Contents

i	Time Content	Source Location
1	Time parameters and time reversal of fundamental laws	abstract;§ 1;§ 3, p. 1,5
2	Observed asymmetric direction from past to future	abstract;§ 3–4, p. 1,5,6
3	Order of unitary evolution of a composite quantum system	§ 1, p. 1
4	Master-equation evolution of local or reduced states	§ 1–2, p. 1,4
5	Time reparametrization in classical gravity	§ 2, p. 2
6	Continuous intrinsic time from scale factor or volume	§ 2(2);§ 4(4)–(5), p. 2,3,6,7
7	Discrete intrinsic-time label of loop quantum cosmology	§ 2(3), p. 3,4
8	Temporal order of cosmic expansion, condensation and recollapse	§ 2–5, p. 3,4,5,7,8
9	Collapse–reexpansion order in the null-dust-shell model	§ 2, p. 4
10	Slow evolution of gravitational degrees of freedom in the semiclassical approximation	§ 2, p. 4
11	Semiclassical time of the recovered functional Schrodinger equation	§ 2;§ 4, p. 4,7
12	Boundary of onset of classical evolution in the cited inflationary model	§ 2, p. 4
13	Retarded–advanced radiation relation and causal direction	§ 3–4, p. 5,6
14	Nondecreasing thermodynamic entropy order in closed systems	§ 3–4, p. 5,6
15	Record direction in quantum measurement and decoherence	§ 3–4, p. 5,6
16	Temporal boundary of the smooth early universe	§ 3, p. 5
17	Growth of entanglement and reduced entropy along the intrinsic scale	§ 4(4)–(6), p. 6,7
18	Time direction defined by increasing reduced entropy	§ 4, p. 6,7
19	Arrows on the two branches of a recollapsing model and their quantum turning boundary	§ 4–5, p. 7,8
20	Proper time of collapsing matter and an infalling observer	§ 5, p. 7,8
21	Schwarzschild time and cosmological turning time	§ 5, p. 7,8
22	Cross-time comparison and proper-time intervals in the collapse scenario	§ 5, p. 7,8

Definition 8.573 (Kiefer 2005 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(47)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([5], [47])$. Claims 1–5 form the generation-initial set; all 47 claims form the source formal-evidence environment.

Table 433: Kiefer 2005 Claims and Temporal Uses

p	Claim	i	Source Location
1	Physical carriers of the universe and three-dimensional spatial configurations.	$\emptyset$	§ 1–2, p. 1,2
2	Local systems, environments, matter degrees of freedom, and their nontemporal partitions and correlations.	$\emptyset$	§ 1–2;§ 4, p. 1,2,4,6
3	Apparatus, radiation, media carrying records and information, and state differences before temporal ordering.	$\emptyset$	§ 1;§ 3;§ 5, p. 1,5,8
4	Physical carriers of the null-dust-shell model.	$\emptyset$	§ 2, p. 4
5	Physical carriers specified by collapsing matter and black-hole models.	$\emptyset$	§ 5, p. 7,8
6	Kiefer adopts the universality of quantum theory and cites the absence of experimentally established violations of the superposition principle.	$\emptyset$	§ 1, p. 1
7	Under unitary Schrodinger evolution of the composite system, entanglement with the environment suppresses local interference, and the local system is described by an effective master equation.	$3^S, 4^{SP}$	§ 1, p. 1
8	Because an environment itself has a further environment, only the entire universe is a strictly closed quantum system, with no measuring agency outside it.	$\emptyset$	§ 1, p. 1
9	Kiefer lists candidates including superstring theory, quantum general relativity and causal sets, distinguishing path-integral and canonical quantum general relativity and, within the latter, quantum geometrodynamics and the loop representation; he selects the canonical framework for the conceptual discussion.	$\emptyset$	§ 1, p. 2
10	Kiefer contrasts time-reversal symmetry of fundamental laws with the asymmetry of observed arrows of time and develops an explanation through quantum-cosmological boundary conditions.	$1^S, 2^O$	ab- stract;§ 1;§ 3–4, p. 1,5,6
11	For a closed universe, the constraint operator containing gravity and other fields annihilates the state: $\hat{H}\Psi = 0$. Ψ depends on three-dimensional quantities and is invariant under three-dimensional coordinate transformations; the equation contains no external nondynamical time parameter.	$\emptyset$	§ 2(1);note7, p. 2,9

p	Claim	i	Source Location
12	Time-reparametrization invariance of the classical theory underlies the Hamiltonian constraint of the canonical quantum theory.	5 ^S	§ 2, p. 2
13	The hyperbolic principal part of quantum geometrodynamics selects an intrinsic time; in the closed Friedmann model with a homogeneous massive scalar field, the scale factor a , $\alpha = \ln a$, or spatial volume serves this role.	6 ^{KS}	§ 2(2), p. 2
14	The directly quantized equation is treated as an effective large-scale equation; modifications such as changes to the potential become an issue at the Planck scale.	$\emptyset$	§ 2, p. 3
15	Loop quantum cosmology uses a difference equation together with $\hat{p} n\rangle = (\beta\ell_{\text{P}}^2 n/6) n\rangle$ and $ p = a^2$; the integer n can be interpreted as playing the role of discrete intrinsic time.	7 ^{KS}	§ 2(3), p. 3
16	For large integers $n \gg 1$, the difference equation of loop quantum cosmology recovers the Wheeler–DeWitt differential equation.	7 ^S , 6 ^S	§ 2, p. 3
17	For a particular factor ordering, ψ_0 drops out of the difference equation, while a constraint among the remaining amplitudes survives and acts as a dynamical initial condition.	7 ^{BS}	§ 2, p. 3
18	In the loop model of source Section 2, the inverse-scale-factor operator is bounded and the state can be continued through $n = 0$ to negative n ; the sign of n distinguishes spatial orientations.	7 ^{BS}	§ 2, p. 3,4
19	Kiefer states that divergence at large scale factor appears unavoidable in a closed loop-universe model.	7 ^Q	§ 2, p. 4
20	Describing a packet along a recollapsing trajectory requires the returning packet to be included in the initial condition at small a ; the semiclassical approximation maintaining a narrow packet along the entire classical trajectory must break down somewhere.	6 ^B , 8 ^S	§ 2, p. 3
21	In a separate unitary null-dust-shell model, the shell collapses within the Schwarzschild radius and reexpands to infinity, avoiding a singularity through a superposition of black and white holes.	9 ^S	§ 2, p. 4

p	Claim	i	Source Location
22	A Born–Oppenheimer-type approximation defines a time parameter through slowly evolving gravitational degrees of freedom and yields a functional Schrodinger equation for nongravitational degrees of freedom.	$10^S, 11^S$	§ 2, p. 4
23	A decoherence description separating relevant and irrelevant degrees of freedom suppresses interference between macroscopic alternatives and yields reduced master dynamics in semiclassical time.	$11^S, 4^{SP}$	§ 2, p. 4
24	In the cited inflationary model, decoherence begins with inflation, and classical evolution emerges from the preceding timeless regime.	12^{BS}	§ 2, p. 4
25	If the forward semiclassical time of the master equation corresponds to increasing scale factor, that equation cannot remain valid across the turning point of recollapse.	$4^S, 6^S, 8^S, 11^S$	§ 2, p. 4
26	The source lists ordinary retarded radiation, nondecreasing thermodynamic entropy in closed systems, quantum measurement and decoherence, and cosmic expansion and condensation as four arrows of time.	$13^C, 14^{SQ}, 15^O, 8^S$	§ 3, p. 5
27	The view that cosmic expansion is the master arrow has a history extending back to Boltzmann; Kiefer takes up Zeh’s quantum-cosmological explanation.	$8^S, 2^O$	§ 3, p. 5
28	The source contrasts the low gravitational entropy of the homogeneous early universe with the maximum for matter condensed into one black hole, and asks for an explanation of initial smoothness.	$16^B, 8^S$	§ 3, p. 5
29	The potentials $V_i(\alpha, x_i)$ in the perturbed Wheeler–DeWitt equation are asymmetric in α and approach zero as $\alpha \rightarrow -\infty$.	6^{SB}	§ 4(4), p. 6
30	In that small-scale limit, Zeh’s boundary condition $\Psi \rightarrow \chi(\alpha) \prod_i \psi_i(x_i)$ can be imposed. This product condition sets initial entanglement between the specified degrees of freedom to zero.	6^B	§ 4(5), p. 6
31	Kiefer states that solutions of the equation with the product boundary condition (5) exhibit increasing entanglement as α increases.	$6^S, 17^S$	§ 4, p. 6
32	For a specified split into relevant and irrelevant degrees of freedom, tracing out the latter gives ρ and $S = -k_B \text{tr}(\rho \ln \rho)$. S depends on that split.	$\emptyset$	§ 4(6), p. 6

p	Claim	i	Source Location
33	Kiefer defines the direction of time by the reduced entropy increasing in the small-scale product-boundary construction.	$17^Q, 18^S$	§ 4, p. 6
34	Kiefer argues that, if the small-scale product-boundary construction applies, the four arrows of time would have a common root in this entropy increase.	$13^C, 14^S, 15^O, 8^S, 18^S$	§ 4, p. 6
35	Kiefer compares the emergence of time for subsystems from a timeless description of the whole with spontaneous symmetry breaking.	18^S	§ 4, p. 6
36	Kiefer identifies carrying out this entropy analysis in loop quantum cosmology as a subject for further investigation.	$\emptyset$	§ 4, p. 7
37	On a branch where the semiclassical approximation holds, the time parameter is a function of the scale factor, and the universe described in the direction of increasing entropy expands. In this sense, Kiefer calls cosmic expansion a tautology.	$6^S, 11^S, 18^S$	§ 4, p. 7
38	In a recollapsing model, the same product boundary condition applies at small a on both the big-bang and big-crunch sides, and entropy increases with increasing a on each side.	$6^B, 8^S, 18^S$	§ 4, p. 7
39	The arrow formally reverses at the turning point, but no classical observer persists through the fully quantum region, so no observer experiences the reversal.	$8^S, 18^S, 19^{BS}$	§ 4, p. 7
40	The recollapse argument also imposes vanishing of the wave function as $\alpha \rightarrow +\infty$, satisfied through destructive interference between quasiclassical components.	$6^B, 8^S$	§ 4, p. 7
41	In a recollapsing universe with symmetric cosmological boundary conditions, matter collapses along proper time τ , long before maximum expansion at Schwarzschild time t_{turn} .	$8^S, 20^S, 21^B, 22^X$	§ 5, p. 7,8
42	For $t > t_{\text{turn}}$, the object formally reexpands, but observers following the local arrow on either branch experience collapse.	$8^S, 19^S, 20^S, 21^S$	§ 5, p. 8
43	In the recollapsing black-hole scenario with symmetric cosmological boundaries, neither horizons nor singularities form, and no information-loss problem arises. Cosmic censorship is unnecessary because naked singularities do not arise either.	$8^S, 19^S, 20^S, 21^S$	§ 5, p. 8

p	Claim	i	Source Location
44	Large redshift prevents external testing of the source's black-hole consequences, whereas an infalling observer in this scenario would reach the quantum region in a short proper time. Kiefer calls for a full theory of quantum gravity and experimental tests to obtain a final answer.	$20^I, 21^X, 22^I$	§ 5, p. 8
45	Kiefer compares the Horowitz–Maldacena proposal imposing a final state at the singularity with this scenario, in which cosmological boundary conditions prevent singularity formation.	20^B	§ 5, p. 8
46	Quantum entanglement between matter and gravity produces mutual decoherence, which may mimic gravity-induced wave-function collapse.	4^{SP}	§ 2, p. 4
47	The source mentions eternal inflation as another attempt to explain initial smoothness.	$8^S, 16^B$	§ 3, p. 5

Definition 8.574 (Kiefer 2005 Formal Relations). Source Section 2, Equations (1)–(3), gives

$$\begin{aligned}\hat{H}\Psi &= 0, \quad a = e^\alpha, \\ \left(G\hbar^2\partial_\alpha^2 - \hbar^2\partial_\phi^2 + m^2\phi^2e^{6\alpha} - \frac{e^{4\alpha}}{G} \right) \psi(\alpha, \phi) &= 0, \\ \hat{p}|n\rangle &= \frac{\beta\ell_P^2 n}{6}|n\rangle, \quad \ell_P = \sqrt{8\pi G\hbar}, \quad |p| = a^2, \quad n \in \mathbb{Z}.\end{aligned}$$

The constants in the second equation follow the unit convention of source Note 8. The hyperbolic principal part defines intrinsic time in the continuous model. The loop model uses the integer-label/eigenvalue relation and recovery of the differential equation at large integer values. Source Section 4, Equations (4)–(6), uses

$$\begin{aligned}\left\{ \partial_\alpha^2 + \sum_i [-\partial_{x_i}^2 + V_i(\alpha, x_i)] \right\} \Psi &= 0, \quad V_i \longrightarrow 0 \quad (\alpha \rightarrow -\infty), \\ \Psi &\longrightarrow \chi(\alpha) \prod_i \psi_i(x_i) \quad (\alpha \rightarrow -\infty), \\ \rho &= \text{Tr}_{\text{irr}} |\Psi\rangle\langle\Psi|, \quad S(\alpha, \{y_i\}) = -k_B \text{tr}(\rho \ln \rho).\end{aligned}$$

The product state is imposed as a boundary condition, and the specified irrelevant degrees of freedom are traced out. Kiefer states that entropy under these conditions increases with scale factor and defines time direction through that increase. In the recollapsing model, vanishing of the wave function at large scale is a further boundary condition.

Definition 8.575 (Kiefer 2005 Finite Cardinalities). Superscripts in the claim table denote the use roles below. Auxiliary objects are specified within claims; the separate auxiliary-object index set is empty.

Table 434: Kiefer 2005 Time-Use Roles

Code	Role	Use Count
I	Interval	2
Q	Quantity	3
X	Cross-position correspondence	2
K	Clock reading	2
C	Causality	2
S	State evolution	54
P	Probabilistic evolution	3
O	Observation record	4
B	Reference boundary	13

Table 435: Kiefer 2005 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3–5, 7, 9–11, 20	member	instant, order_relation
2, 13–15, 17–18	member	order_relation
6	member	instant, order_relation, interval_boundary
8, 21	member	instant, order_relation, phase_boundary
12, 16	member	phase_boundary
19	member	order_relation, phase_boundary
22	member	interval_boundary, order_relation

Table 436: Kiefer 2005 Finite Cardinalities

Object	Cardinality
Time contents	22
Claims	47
Auxiliary objects	0
Transfers	9
Results	31
Claim/time-content uses	74
Role assignments	85
Obligation atoms	109
Audit-boundary nodes	100
Audit-boundary edges	40

Dependency Inputs

Definition 8.576 (Kiefer 2005 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [9]$. The table gives time inputs, time outputs, claim support and source locations.

Table 437: Kiefer 2005 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	3	4	7	From unitary composite evolution to local master evolution. § 1
2	$\emptyset$	6	13	From a hyperbolic constraint and scale variable to intrinsic time. § 2(2)
3	$\emptyset$	7	15	From triad eigenvalues and a difference equation to discrete intrinsic time. § 2(3)
4	7	6	16	From the large-integer limit of the discrete description to continuous intrinsic time. § 2
5	10	11	22	From slowly evolving gravity to semiclassical time. § 2
6	11	4	23	From semiclassical time to reduced master evolution. § 2
7	6	17	29–31	From the small-scale product boundary to entanglement growth along the intrinsic scale. § 4(4)–(5)
8	17	18	32–33	From increasing reduced entropy to time direction. § 4(6)
9	6, 8, 18	19	38–39	From the two small-scale boundaries of recollapse to formal reversal. § 4

Grounding Conditions

Table 438: Kiefer 2005 Conditions for Transfers

Transfer Index	Conditions
1	Specified system–environment split and effective decoherence description.
2	Use the scale coordinate associated with the hyperbolic principal part in the specified quantum-geometrodynamical minisuperspace model.
3	The source’s eigenvalue representation of $\hat{p}$ and interpretation of the integer label in the difference equation.
4	The $n \gg 1$ limit recovering the Wheeler–DeWitt differential equation.
5	A gravitational branch and nongravitational degrees of freedom for which the Born–Oppenheimer-type approximation holds.
6	Specified relevant–irrelevant split and decoherence approximation.
7	The growth asserted for the source’s perturbed equation, potentials and Zeh product boundary condition.

Trans-fer Index	Conditions
8	Reduced state for the specified split and definition of orientation through increasing entropy.
9	The same product boundary on both sides, entropy direction on each branch and the quantum turning region.

Results

Definition 8.577 (Kiefer 2005 Results). Let $\text{Idx}_R^{\text{audit,result}} = [31]$.

Table 439: Kiefer 2005 Results

q	Result	Source Location
1	Universality of quantum theory and closure of the universe	§ 1, p. 1,2
2	Contrast between fundamental laws and observed arrows	abstract;§ 1;§ 3, p. 1,5
3	Evolution descriptions of composite and local systems	§ 1, p. 1
4	Quantum constraint without external time	§ 2(1);note7, p. 2,9
5	Time-reparametrization invariance and the constraint	§ 2, p. 2
6	Continuous intrinsic time through a scale variable	§ 2(2), p. 2
7	Large-scale validity of direct quantization	§ 2, p. 3
8	Discrete intrinsic time and its large-integer limit	§ 2(3), p. 3
9	Dynamical initial condition and continuation through the singular label	§ 2, p. 3,4
10	Large-scale divergence issue in the closed loop model	§ 2, p. 4
11	Limit of semiclassical wave packets on a recollapsing trajectory	§ 2, p. 3
12	Singularity avoidance in a unitary null-dust-shell model	§ 2, p. 4
13	Semiclassical time and the functional Schrodinger equation	§ 2, p. 4
14	Reduced master dynamics and mutual decoherence	§ 2, p. 4
15	Onset of classical evolution in the inflationary model	§ 2, p. 4
16	Limit of the forward master equation in recollapse	§ 2, p. 4
17	Four arrows and the view of expansion as the master arrow	§ 3, p. 5
18	Problem of explaining initially low gravitational entropy	§ 3, p. 5
19	Perturbation potentials asymmetric in the intrinsic scale	§ 4(4), p. 6
20	Product boundary condition imposed at small scale	§ 4(5), p. 6
21	Entanglement growth along the intrinsic scale	§ 4(4)–(5), p. 6
22	Entropy from the reduced density matrix	§ 4(6), p. 6

q	Result	Source Location
23	Definition of time direction through entropy	§ 4, p. 6
24	Common root of four arrows and the symmetry-breaking analogy	§ 4, p. 6
25	Proposed extension of the entropy analysis to loop cosmology	§ 4, p. 7
26	Relation of scale factor, time and expansion on a semiclassical branch	§ 4, p. 7
27	Both recollapse boundaries and formal arrow reversal	§ 4, p. 7
28	Collapse and formal reexpansion in a recollapsing universe	§ 5, p. 7,8
29	Nonformation of horizons and singularities and its consequences	§ 5, p. 7,8
30	Restriction on external testing and conditions for a final answer	§ 5, p. 8
31	Comparison with a final-state condition at the singularity	§ 5, p. 8

Table 440: Kiefer 2005 Conditions for Results

Result Index	Conditions and Domain
1	Adopt quantum universality and use the containment relation of environments and the hierarchy of candidate theories.
2	Descriptions of time-reversal-symmetric laws and macroscopic asymmetry.
3	Specified partition and effective decoherence description.
4	Hamiltonian constraint of a closed universe and three-dimensional coordinate invariance.
5	Time-reparametrization invariance of the classical theory.
6	Hyperbolic quantum geometrodynamics and the specified minisuperspace model.
7	The equation as an effective theory and the issue of Planck-scale applicability.
8	Eigenvalue representation and difference equation of the specified loop model, and its $n \gg 1$ continuum limit.
9	Specified factor ordering and inverse-scale-factor operator, and the two triad orientations.
10	The difficulty reported for the source's closed-universe model.
11	Initial data at small a containing both incident and returning packets, and a classical recollapsing trajectory.
12	The unitary quantum shell model cited by the source.
13	Slowly evolving gravity and a Born–Oppenheimer-type approximation.
14	Relevant–irrelevant split and quantum entanglement of gravity and matter.
15	The inflation–decoherence setting of source Reference15.
16	The case in which forward semiclassical time corresponds to increasing a .

Result Index	Conditions and Domain
17	The four observed asymmetries and their cosmological explanation.
18	Comparison of the homogeneous early universe with gravitational condensation, and the alternative explanation mentioned by the source.
19	Specified perturbed equation and the small-scale limit $V_i \rightarrow 0$.
20	Zeh's boundary choice and the specified split of degrees of freedom.
21	The source's equation, potentials and product boundary condition.
22	Normalized state and specified relevant-irrelevant split.
23	Reduced-entropy growth in the preceding construction.
24	The quantum-cosmological construction discussed by Kiefer and the four physical arrows.
25	The analysis identified by the source for further investigation.
26	A branch admitting the semiclassical approximation and entropy-defined time orientation.
27	Small-scale product boundary, separately imposed large-scale vanishing condition, and the quantum turning region.
28	Symmetric cosmological boundary conditions and the specified collapse before maximum expansion.
29	The source's recollapsing black-hole scenario.
30	Redshift, the infalling observer's proper time, and a full theory with experimental tests.
31	The Horowitz-Maldacena final-state condition and the source's cosmological boundary condition.

Result Dependence

Definition 8.578 (Kiefer 2005 Result-Dependence Hyperedges). For each row, let $\text{ResultEdge}_{R,q}$ ($q \in [31]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 441: Kiefer 2005 Result-Dependence Hyperedges

q	Time Premises i	Claim Support p
1	$\emptyset$	6, 8–9
2	1–2	10
3	3–4	7
4	$\emptyset$	11
5	5	12
6	6	13
7	$\emptyset$	14
8	6–7	15–16
9	7	17–18
10	7	19
11	6, 8	20
12	9	21

q	Time Premises i	Claim Support p
13	10–11	22
14	4, 11	23, 46
15	12	24
16	4, 6, 8, 11	25
17	2, 8, 13–15	26–27
18	8, 16	28, 47
19	6	29
20	6	30
21	6, 17	29–31
22	$\emptyset$	32
23	17–18	33
24	8, 13–15, 18	34–35
25	$\emptyset$	36
26	6, 11, 18	37
27	6, 8, 18–19	38–40
28	8, 19–22	41–42
29	8, 19–22	41–43
30	20–22	44
31	20	45

$$\text{BoundaryCount}_R^{\text{audit}} = (100, 40).$$

The 47 claims, 22 time contents and 31 results give 100 nodes; the 9 transfers and 31 result dependencies give 40 edges.

Evidence Hierarchy Kiefer discusses quantum-cosmological constructions and model and approximation analyses from the cited literature. This source reports zero newly acquired direct measurements. The black-hole consequences in Section 5 depend on a recollapsing universe and symmetric boundary conditions. The source calls for a full theory of quantum gravity and experimental tests for a final answer.

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 4; remaining primitive-time roots 18. The locally reached contents are continuous intrinsic time, discrete intrinsic time, entanglement and reduced-entropy growth along the intrinsic scale, and the time direction defined by that growth.

Result Dependence Of the 31 results, 18 depend on primitive-time roots and 13 are reached from source formal evidence and local constructions. There are 39 root–result dependence pairs, of which 33 are necessary.

Placement of the Grounding Obligation The local constructions of intrinsic time and entropy direction use the specified constraints, operators and boundary conditions as source formal evidence. Quantum evolution, slowly evolving gravity, semiclassical evolution, the four observed arrows, proper time and Schwarzschild time enter laws and results. Complete generation from nontemporal physical inputs is zero; the placement of the physical-time grounding obligation is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure The source describes breakdown of the semiclassical wave-packet approximation on recollapsing trajectories and a continuation limit for a forward master equation corresponding to increasing scale factor. Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 39 root–result dependence pairs. Actual reach from failure of a specific physical-time generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns results dependent on it.

Audit Conclusions

Proposition 8.579 (Primitive Time Roots and Result Dependence in Kiefer 2005). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = (\emptyset, [22]; \\ \emptyset, \{1, 4, 6, 7, 8, 9, 10, 19, 20, 21, 22, 23, 25\}, \\ [31] \setminus \{1, 4, 6, 7, 8, 9, 10, 19, 20, 21, 22, 23, 25\}, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–5 remains those five nodes. Adding all 47 claims enables Transfers 2 and 3 to produce Time Contents 6 and 7, followed by Transfers 7 and 8 producing Time Contents 17 and 18. This closure reaches 13 results. Adding the remaining time contents as seeds reaches all 31 results. $\square$

Proposition 8.580 (Minimal Primitive-Time-Root Support in Kiefer 2005). The table gives each result’s minimal-support index family, with 34 family members in total.

Proof. Apply Definition 7.46 to all transfers and result dependencies, starting from singleton time-content seeds and empty supports for source formal evidence. By Proposition 7.47, the inclusion-minimal supports of the stable antichains give the displayed families. $\square$

Table 442: Minimal Primitive-Time-Root Support in Kiefer 2005

Result Indices	Minimal-Support Family Members
1, 4, 6–10, 19–23, 25	$\emptyset$
2	$\{1, 2\}$
3	$\{3\}$
5	$\{5\}$
11, 27	$\{8\}$
12	$\{9\}$
13	$\{10\}$
14, 26	$\{10\}, \{11\}$
15	$\{12\}$

Result Indices	Minimal-Support Family Members
16	{8, 10}, {8, 11}
17	{2, 8, 13, 14, 15}
18	{8, 16}
24	{8, 13, 14, 15}
28–29	{8, 20, 21, 22}
30	{20, 21, 22}
31	{20}

Result-Preservation Domain Fix the models, constraints, boundaries, subsystem splits and approximation domains in Table 440. When temporal inputs are realized by generated representations preserving the same intrinsic variables, evolution, entropy direction, proper time and cross-time comparisons, time-dependent results are preserved within their domains.

Proposition 8.581 (Complete Audit of Time Use in Kiefer 2005). There are 22 time contents, 47 claims, 9 transfers and 31 results. The 74 claim/time-content use pairs have 85 role assignments, with 109 source-specific obligation atoms. Complete generation is zero, local reach is 4, remaining primitive-time roots number 18, and 18 results depend on roots.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual statements

Object of statements The external time parameter at the most fundamental level, intrinsic time given by the scale factor, discrete intrinsic time given by an integer label, and the failure boundary of a semiclassical approximation along a classical trajectory.

Original (English) Quantum gravity does not contain an external time parameter at the most fundamental level.

intrinsic time, given in quantum cosmology by the scale factor

The number n can be interpreted as playing the role of ‘discrete intrinsic time’.

one cannot find a narrow wave packet all along the classical trajectory – the semiclassical approximation must breakdown somewhere

Source location The first and second quotations are in arXiv:gr-qc/0502016v1, p. 6, §4; the third and fourth are on p. 3, §2.

Minimal context The first and second quotations belong to the context in which an intrinsic-time role is assigned to the scale factor in quantum cosmology starting from quantum gravity without external time. The third belongs to the discussion of

the difference equation of loop quantum cosmology, the integer label and operator eigenvalues, and the large-integer limit. The fourth belongs to the discussion of the turning point and domain of validity of the semiclassical approximation when a wave packet represents a recollapsing classical trajectory.

Attribution All four quotations are attributed to the text by Claus Kiefer.

Content stated directly by the quotations The first quotation states the absence of an external time parameter at the most fundamental level, and the second states that the scale factor gives intrinsic time. The third states that the integer label can be interpreted as discrete intrinsic time; the fourth states that the semiclassical approximation cannot be maintained along the entire classical trajectory.

Explicit domain of application The first quotation is limited to quantum gravity at the most fundamental level; the second and third to the adopted quantum-cosmological models; and the fourth to the semiclassical wave-packet description of a recollapsing model.

Evidence provenance All four quotations belong to Kiefer, *Quantum Cosmology and the Arrow of Time*, arXiv:gr-qc/0502016v1, pp. 3,6.

Individual Audit Absence of External Time at the Most Fundamental Level

The first quotation rests on the closed-universe constraint $\hat{H}\Psi = 0$ and the theory's state dependence on three-dimensional quantities. It corresponds to Claim 11 and Result 4. Time-reparametrization invariance corresponds to Claim 12 and Time Content 5; the construction using a scale variable as intrinsic time corresponds to Claim 13.

Intrinsic Time through the Scale Factor

The second quotation rests on the hyperbolic principal part of quantum geometrodynamics and the scale variable of the specified minisuperspace model. With Claim 13 as evidence, Transfer 2 yields Time Content 6. Applications to recollapsing wave packets or semiclassical time additionally use the boundary, branch and approximation conditions of Results 11 and 26.

Discrete Intrinsic Time through an Integer Label

The third quotation rests on the eigenvalue representation of $\hat{p}$ in source Equation (3) and the specified loop-cosmological difference equation. With Claim 15 as evidence, Transfer 3 yields Time Content 7. Recovery of a continuous equation at large integer values, the factor-ordering-dependent dynamical initial condition, and continuation through the zero label follow the respective conditions of Claims 16–18.

Breakdown Boundary of the Semiclassical Approximation

The fourth quotation states that, in a recollapsing model whose small-scale initial data contain both incident and returning packets, a narrow-packet approximation cannot hold along the entire classical trajectory. It corresponds to Time Contents 6 and 8, Claim 20 and Result 11. Source Section 4 also discusses breakdown in the turning region, interference between quasiclassical components, vanishing of the wave function at large scale and the limit of classical observer continuity. These follow the conditions of Claims 38–40 and Result 27.

Common Audit with Specified Domain The results of the four statements concern the closed-universe constraint, constructions of continuous and discrete intrinsic time, and the semiclassical validity domain of the recollapsing model. Local reach of intrinsic time and complete generation from physical inputs are given in Proposition 8.579.

8.19.48 Source audit

Audit Inputs

Source and Audit Scope

Source identifier 48.

Source name Grøn–Elgarøy 2007 Primitive-Time-Root Audit

Bibliographic identification Øyvind Grøn and Øystein Elgarøy, *Is space expanding in the Friedmann universe models?*, *American Journal of Physics* 75, 151–157 (2007); arXiv:astro-ph/0603162v2, pp. 1–18. $R = \text{Ref}_{\text{GE07FE}}$.

Audit-scope identifier 48-SC1.

Audit-scope content Abstract, Sections I–V, Equations (1)–(52), Figures 1–2 and references.

Source-location identifier 48-L1.

Source-location content Tables identify sections, equations and printed pages.

Assumptions The analysis uses FLRW geometry and timelike radial geodesics. The power-law model assumes a flat single fluid with constant $w > -1$ and positive cosmic time. Tether release uses initial rest and a subluminal condition; the Milne model uses empty negatively curved space and a clock congruence with a common origin. Redshift calculations specify emission and reception events and a light path. The low-redshift potential comparison uses dust, no cosmological constant, and the stated velocity and approximation order.

Nontemporal Physical Inputs The carriers, spatial configurations, distance, density and wavelength records, and origin and endpoint markers in Claims 1–12 form the generation-initial set.

Time Contents

Time-content identifier 48-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content specification The time contents of Definition 8.582 and their use roles in the claim table.

Source-location identifier set 48-L1.

Definition 8.582 (Grøn–Elgarøy 2007 Time Contents). Let $n_R = 19$. Each time content is scoped by the model, state and domain of the corresponding claims and results.

Table 443: Grøn–Elgarøy 2007 Time Contents

i	Time Content	Source Location
1	FLRW cosmic time and reference time	§II–IV, p. 3,4,5,10,11
2	Temporal relations of scale factor, Hubble rate and deceleration parameter	§II;§IV–V, p. 3,4,5,6,7,10,11,12,14
3	Proper time and clock intervals on timelike worldlines	§II–III;Fig.2, p. 3,9,18
4	Coordinate trajectory of a free particle and its time derivatives	§II;Fig.1, p. 3,4,5,6,7,17
5	Peculiar velocity relative to a local comoving frame	§II;§IV(49)–(52), p. 4,5,13
6	Temporal change of proper distance on cosmic-time slices	§II–IV, p. 5,6,7,9,11,13
7	Event boundary specifying tether release or remote rest	§II(16)–(28);§IV(52), p. 5,6,7,13
8	Annual change in the cited bound-orbit model	end of §II, p. 8
9	Cosmic time of Milne reference particles	§III;Fig.2, p. 8,9,18
10	Minkowski inertial time and private simultaneity slices	§III;Fig.2, p. 8,9,18
11	Synchronization and public simultaneity of a clock congruence with a common origin	§III;§V;Fig.2, p. 9,14,18
12	Clock and velocity reference in a local inertial neighborhood	§III–IV, p. 9,10,13
13	Time correspondence between emission and reception events	§IV(34)–(52), p. 10,11,12,13
14	Light-propagation interval and time difference in the nearby-source expansion	§IV–V, p. 10,11,12,14
15	Wavelength ratio and cosmological shift along propagation	§IV–V, p. 10,11,12,13,14
16	Low-redshift Doppler term using the emission-time Hubble velocity	§IV(36)–(41),(49), p. 11,13
17	Frequency shift associated with a gravitational-potential difference	§IV(42)–(48), p. 11,12
18	Dynamical evolution of general-relativistic geometry	abstract;§I;§III;§V, p. 1,2,9,10,14
19	Doppler wavelength correspondence between the emitter and its local comoving frame	§IV(49)–(51), p. 13

Definition 8.583 (Grøn–Elgarøy 2007 Formal Relations). The cosmic-time radial geodesic and proper time satisfy

$$c^2 d\tau^2 = -ds^2, \quad \ddot{\chi} = \frac{a\dot{a}}{c^2} \dot{\chi}^3 - 2H\dot{\chi}, \quad \dot{\chi} = \frac{\pm c}{a\sqrt{1 + Aa^2}} \quad (A > 0).$$

The tether-release initial condition uses

$$a_0 = 1, \quad \dot{l}_0 = 0, \quad 0 < \chi_0 < c/H_0, \quad \ddot{l}_0 = H_0^2 \chi_0 \left[-q_0 - \left(\frac{H_0 \chi_0}{c} \right)^2 \right].$$

The Milne line element, including its angular part, is

$$\eta = \frac{\chi}{ct_0}, \quad T = t \cosh \eta, \quad R = ct \sinh \eta,$$

$$ds^2 = -c^2 dt^2 + \left(\frac{t}{t_0}\right)^2 \left[d\chi^2 + (ct_0)^2 \sinh^2\left(\frac{\chi}{ct_0}\right) d\Omega^2 \right].$$

With outward radial peculiar velocity positive relative to the local comoving frame,

$$1 + z = \frac{a_0}{a_e} \sqrt{\frac{1 + v_p/c}{1 - v_p/c}}, \quad |v_p| < c,$$

$$\dot{l}_e = 0 \quad \implies \quad v_p = -H_e l_e = -a_e H_e \chi_e.$$

The claim table specifies the temporal variables, events and model conditions for these relations.

Definition 8.584 (Grøn–Elgarøy 2007 Finite Cardinalities). Superscripts in the claim table denote the use roles below. Auxiliary objects are specified within claims; the separate auxiliary-object index set is empty.

Table 444: Grøn–Elgarøy 2007 Time-Use Roles

Code	Role	Use Count
I	Interval	9
Q	Quantity	61
X	Cross-position correspondence	28
K	Clock reading	14
C	Causality	1
S	State evolution	43
P	Probabilistic evolution	0
O	Observation record	0
B	Reference boundary	18

Table 445: Grøn–Elgarøy 2007 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3, 9	member	instant, order_relation, interval_boundary
2, 4, 6, 10, 12, 18	member	instant, order_relation
5, 15–17, 19	dependency	∅
7	member	instant, phase_boundary
8, 14	member	interval_boundary, order_relation
11	member	instant, order_relation, endpoint

Time Indices	Structural Type	Axes
13	member	endpoint, order_relation

Table 446: Grøn–Elgarøy 2007 Finite Cardinalities

Object	Cardinality
Time contents	19
Claims	51
Auxiliary objects	0
Transfers	18
Results	29
Claim/time-content uses	131
Role assignments	174
Obligation atoms	117
Audit-boundary nodes	99
Audit-boundary edges	47

Definition 8.585 (Grøn–Elgarøy 2007 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(51)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([12], [51])$. Claims 1–12 form the generation-initial set; all 51 claims form the source formal-evidence environment.

Table 447: Grøn–Elgarøy 2007 Claims and Temporal Uses

p	Claim	i	Source Location
1	Physical carriers of the reference particles.	$\emptyset$	§I, p. 2
2	Physical carriers of free and test particles.	$\emptyset$	§II, p. 3,5
3	Physical carriers of the matter fluid.	$\emptyset$	§II–III, p. 4,9
4	Physical carriers of dust matter.	$\emptyset$	§II;§IV, p. 5,12
5	Physical carriers of clock devices.	$\emptyset$	§III;Fig.2, p. 9,18
6	Physical carriers of electromagnetic signals.	$\emptyset$	§IV, p. 10,13
7	Spatial configurations of physical carriers.	$\emptyset$	§I–III, p. 2,3,9
8	Comparison data for spatial distances.	$\emptyset$	§II, p. 5,7
9	Matter-density data.	$\emptyset$	§II;§IV, p. 4,12
10	Media and values of wavelength records.	$\emptyset$	§IV, p. 10,13
11	Local physical neighborhoods.	$\emptyset$	§II–III, p. 8,9
12	Physical markers of reference origins and comparison endpoints.	$\emptyset$	§II;§IV, p. 5,13
13	Grøn and Elgaroy use general relativity for the global description of the universe and distinguish its domain from local Newtonian or special-relativistic explanations.	18^S	abstract;§I, p. 1,2

p	Claim	i	Source Location
14	The authors describe space through dynamical geometry and relations between reference particles, citing gravitational waves as an example of its physical character.	18 ^S	§I, p. 2
15	The FLRW line element is specified by cosmic time t , the scale factor $a(t)$, and the curvature function S_k .	1 ^K , 2 ^S	§II(1)–(2), p. 3
16	A timelike particle geodesic is parametrized by proper time τ ; conversion to coordinate time uses $\frac{d\chi}{dt} = \frac{d\chi/d\tau}{dt/d\tau}$. With τ in units of time, $c^2 d\tau^2 = -ds^2$.	1 ^X , 2 ^S , 3 ^{KQ} , 4 ^S	§II(3)–(8), p. 3
17	The radial coordinate velocity $u = d\chi/dt$ obeys $\dot{u} = (a\dot{a}/c^2)u^3 - 2Hu$; $u = 0$ is the comoving solution.	1 ^S , 2 ^{SQ} , 4 ^{SQ}	§II(9), p. 4
18	For a noncomoving massive particle, $u = \pm c/(a\sqrt{1 + Aa^2})$, $v_p = au$, and $A > 0$; $\chi(t)$ is obtained by integrating u over time.	2 ^S , 4 ^{SI} , 5 ^Q	§II(10)–(13), p. 4
19	On an expanding branch with $A > 0$ and a tending to infinity, both the coordinate velocity u and the local peculiar velocity v_p tend to zero.	2 ^S , 4 ^S , 5 ^Q	§II(12)–(13), p. 4
20	For a flat single-fluid model with constant $P = w\rho$, $w > -1$, $t, t_0 > 0$, and $a(t_0) = 1$, $a(t) = (t/t_0)^{2/[3(1+w)]}$ and $H_0 = 2/[3(1+w)t_0]$.	1 ^{KQB} , 2 ^{SQ}	§II(14)–(15), (18), p. 4,5
21	Defining $l = a\chi$ and $v_p = a d\chi/dt$ on the same cosmic-time slicing gives $\dot{l} = Hl + v_p$.	1 ^X , 2 ^Q , 4 ^{SQ} , 5 ^Q , 6 ^{SQ}	§II(16); §IV, p. 5,13
22	At the tether-release event with $a(t_0) = 1$, imposing $dl/dt = 0$ gives the initial peculiar velocity $v_{p0} = -H_0\chi_0$.	1 ^B , 2 ^Q , 5 ^Q , 6 ^Q , 7 ^B	§II(16)–(21), p. 5
23	For a massive particle with $H_0 > 0$ and $\chi_0 > 0$, this rest condition is compatible with $ v_{p0} < c$ precisely when $0 < \chi_0 < c/H_0$, giving $A = c^2/(H_0^2\chi_0^2) - 1 > 0$.	2 ^Q , 5 ^Q , 7 ^B	§II(21)–(22), p. 5
24	The proper-distance acceleration along a radial geodesic is $\ddot{l} = \ddot{a}\chi + (a^2\dot{a}/c^2)\dot{\chi}^3$; at tether release it becomes $\ddot{l}_0 = H_0^2\chi_0[-q_0 - (H_0\chi_0/c)^2]$.	2 ^{SQ} , 4 ^{SQ} , 6 ^{SQ} , 7 ^B	§II(23)–(27), p. 6,7
25	Within $0 < \chi_0 < c/H_0$, the initial acceleration is inward when $q_0 \geq 0$. For $-1 < q_0 < 0$, it is inward for $\sqrt{-q_0}c/H_0 < \chi_0 < c/H_0$, outward inside that threshold, and zero at the threshold.	2 ^Q , 6 ^Q , 7 ^B	§II(24)–(28), p. 6,7
26	The authors reject an analogy assigning particle motion to a separate pushing force of the cosmic flow and explain it using geodesics and cosmic-time distance changes.	2 ^S , 4 ^S , 6 ^S	§II; §V, p. 7,14

p	Claim	i	Source Location
27	At fixed proper distance l from the origin, the Hubble velocity is $H l$ and its time derivative is $l \dot{H}$. On the stated power-law branch with $w > -1$, $\dot{H} < 0$.	$2^{\text{SQ}}, 6^{\text{XQ}}$	§II;§V, p. 7,14
28	Application of a homogeneous FLRW model presupposes a homogeneous matter distribution; the source therefore distinguishes the model domain when discussing a local room or a bound system.	$1^{\text{S}}, 2^{\text{S}}$	end of §II, p. 8
29	In the cited model embedding a spherical mass in an Einstein–de Sitter background, the expansion of a circular orbit of radius 1 AU around a solar mass is estimated at about 10^{-23} m per year.	$2^{\text{S}}, 8^{\text{IQ}}$	end of §II;refs.11–13, p. 8
30	In the empty Milne model, $\eta = \chi/(ct_0)$, $T = t \cosh \eta$, and $R = ct \sinh \eta$ relate cosmic time to Minkowski coordinates, with $c^2 T^2 - R^2 = c^2 t^2$.	$9^{\text{KX}}, 10^{\text{KX}}$	§III(29)–(31), p. 8,9
31	Clocks following the Milne reference particles measure their proper times from the common origin. Events with the same reading t form a public simultaneity slice.	$3^{\text{KIB}}, 9^{\text{K}}, 11^{\text{KXB}}$	§III;Fig.2, p. 9,18
32	In the Milne region $T > 0$, $ R < cT$, a constant- T slice and a constant- t hyperboloid are different sets of events. The portion of the former within the region has a finite R range, whereas the latter has unbounded proper radial distance.	$9^{\text{BX}}, 10^{\text{BX}}, 11^{\text{X}}$	§III(31);Fig.2, p. 9,18
33	For a Milne particle at fixed χ , $dR/dT = c \tanh \eta$, while the distance $l = (t/t_0)\chi$ on public slices gives $dl/dt = \chi/t_0$. The two velocities use different pairs of time and distance variables.	$6^{\text{Q}}, 9^{\text{SQ}}, 10^{\text{SQ}}$	§III(32)–(33), p. 9
34	The empty $k = -1$ Milne spacetime is four-dimensionally flat. A spatially flat $w = -1/3$ fluid model with the same a proportional to t has four-dimensional curvature; a curved matter-filled spacetime has no global Minkowski chart.	$1^{\text{S}}, 2^{\text{S}}, 9^{\text{S}}, 18^{\text{S}}$	§II–III, p. 7,8,9,10
35	Special-relativistic clock and velocity descriptions in curved spacetime apply in local inertial neighborhoods and use proper-clock readings on worldlines.	$3^{\text{K}}, 12^{\text{KQ}}$	§III, p. 9,10
36	In discussing a curved universe containing matter, Grøn and Elgarøy state the view that public space is the only real space.	1^{X}	§III, p. 10

p	Claim	i	Source Location
37	For light connecting a comoving emitter and observer, the received-to-emitted wavelength ratio is a_0/a_e ; with $a_0 = 1$, $z = 1/a_e - 1$.	$2^S, 13^{XB}, 14^{IC}, 15^{QX}$	§IV(34), p. 10
38	For small $x = H_0(t_0 - t_e)$ and smooth a , with $a_0 = 1$, $z = x + (1 + q_0/2)x^2 + O(x^3)$.	$1^B, 2^Q, 13^X, 14^{IQ}, 15^Q$	§IV(35), p. 10,11
39	In the source comparison applying the special-relativistic Doppler formula to the selected emission-time Hubble velocity $v_e = \dot{a}_e \chi_e$, $z_D = x + (1 + q_0)x^2 + O(x^3)$.	$2^Q, 6^Q, 13^X, 14^{IQ}, 16^Q$	§IV(36)–(41), p. 11
40	In the source potential comparison for dust without a cosmological constant, $4\pi G\rho_0/3 = q_0 H_0^2$ is used to obtain $z_G = -q_0 x^2/2 + O(x^3)$ from the emitter-observer potential difference.	$2^Q, 13^X, 14^{IQ}, 17^{QX}$	§IV(42)–(47), p. 11,12
41	Under this velocity and potential comparison, $z = z_D + z_G + O(x^3)$ through second order, agreeing with the expansion of the cosmological wavelength ratio.	$15^Q, 16^Q, 17^Q$	§IV(48), p. 12
42	The source uses a Doppler interpretation through the global Minkowski description of Milne spacetime and treats propagation in a curved universe through cosmology in place of the same global special-relativistic description.	$9^X, 10^X, 13^X, 14^I, 15^Q$	§IV–V, p. 12,14,15
43	For an emitter with radial peculiar velocity v_p relative to its local comoving frame, positive outward, the local wavelength factor is $D = \sqrt{(1 + v_p/c)/(1 - v_p/c)}$ for $ v_p < c$.	$5^Q, 13^B, 19^{QX}$	§IV(49), p. 13
44	Including propagation to a comoving observer, the emitter local factor D and endpoint scale factors give $1 + z = (a_0/a_e)D$.	$2^Q, 13^{XB}, 15^{QX}, 19^{QX}$	§IV(50)–(51), p. 13
45	Instantaneous rest of the remote proper distance at emission, $dl/dt = 0$, requires $v_p = -H_e l_e = -a_e H_e \chi_e$, with $l_e = a_e \chi_e$.	$2^Q, 5^Q, 6^Q, 7^B, 13^B$	§II(16); §IV(51)–(52), p. 5,13
46	Combining this remote-rest condition with the local Doppler factor and cosmological propagation makes a zero instantaneous distance rate compatible with a nonzero observed shift.	$2^Q, 5^Q, 6^Q, 7^B, 13^X, 15^Q, 19^Q$	§IV(51)–(52), p. 13
47	Grøn and Elgarøy support the interpretation that galaxies follow expanding space on the basis of a global description using cosmic simultaneity, distance changes, and light propagation.	$1^X, 2^S, 6^S, 14^I, 15^Q, 18^S$	§V, p. 13,14,15

p	Claim	i	Source Location
48	Figure 2 depicts private simultaneity defined by T on clocks at rest in the selected inertial frame and public simultaneity defined by t on clocks following the Milne test particles.	$3^K, 9^{KX}, 10^{KX}, 11^{KX}$	Fig.2, p. 18
49	Figure 1 plots a particle after tether release against cosmic time in a dust universe, displaying its signed radial position as negative after passage through the origin. Dashed lines show comoving trajectories.	$1^S, 2^S, 4^S, 6^{SQ}, 7^B$	§II;Fig.1, p. 5,6,17
50	The massive-particle geodesic solution and relativistic momentum $p = m\gamma v_p$ give $a^2 p^2 / (m^2 c^2) = 1/A$. Thus the magnitude of p is proportional to $1/a$, and the relation between v_p and p includes the Lorentz factor gamma.	$2^S, 4^S, 5^Q$	§II(12);end of §II, p. 4,8
51	Using Milne cosmic time, the proper distance $l = (t/t_0)\chi$ to a comoving particle has rate of change χ/t_0 .	$9^{SQ}, 6^{SQ}$	§III(33), p. 9

Table 448: Grøn–Elgarøy 2007 Nontemporal Physical Inputs

p	Component
1	Physical carriers of the reference particles.
2	Physical carriers of free and test particles.
3	Physical carriers of the matter fluid.
4	Physical carriers of dust matter.
5	Physical carriers of clock devices.
6	Physical carriers of electromagnetic signals.
7	Spatial configurations of physical carriers.
8	Comparison data for spatial distances.
9	Matter-density data.
10	Media and values of wavelength records.
11	Local physical neighborhoods.
12	Physical markers of reference origins and comparison endpoints.

Dependency Inputs

Definition 8.586 (Grøn–Elgarøy 2007 Time-Content Transfers). Let $\text{ld}x_R^{\text{audit,transfer}} = [18]$. The table gives time inputs, time outputs, claim support and source locations.

Table 449: Grøn–Elgarøy 2007 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1–2	4	17	Radial geodesic solution in coordinate time. §II(9)
2	2	4–5	18	Coordinate and peculiar velocity along the scale factor. §II(10)–(13)
3	1	2	20	Scale-factor relation from power-law cosmic time. §II(14)–(15),(18)
4	1–2, 4	5–6	21	Distance rate and peculiar velocity from a coordinate trajectory. §II(16)
5	1–2, 6–7	5	22	Initial peculiar velocity from the tether-release condition. §II(16)–(21)
6	2, 4, 7	6	24	Proper-distance acceleration from a geodesic. §II(23)–(27)
7	9	10	30	Conversion from Milne cosmic time to inertial time. §III(29)
8	10	9	30	Conversion from inertial coordinates to Milne cosmic time. §III(31)
9	3	9, 11	31	Public synchronization through a common zero of proper clocks. §III;Fig.2
10	2, 13–14	15	37	Comoving redshift from the propagation path and scale-factor ratio. §IV(34)
11	2, 6, 13–14	16	39	Low-redshift Doppler term for the specified Hubble velocity. §IV(36)–(41)
12	2, 13–14	17	40	Gravitational-shift term from the dust potential. §IV(42)–(47)
13	16–17	15	41	Combination of second-order Doppler and gravitational terms. §IV(48)
14	2, 13, 19	15	44	Combination of the local Doppler factor and cosmological propagation. §IV(50)–(51)
15	5, 13	19	43	Emission-wavelength correspondence from local peculiar velocity. §IV(49)
16	2, 6–7, 13	5	45	Local peculiar velocity from the remote-rest event. §IV(52)
17	1–2, 4	3	16	Proper-time relation from a timelike metric and trajectory. §II(3)–(8)
18	9	6	51	Public distance rate from Milne cosmic time. §III(33)

Grounding Conditions

Table 450: Grøn–Elgarøy 2007 Conditions for Transfers

Trans-fer Index	Conditions
1	Initial data and the FLRW radial geodesic equation.
2	Massive noncomoving branch with $A > 0$ and initial values.
3	Flat single fluid with constant $w > -1$, $t, t_0 > 0$, $a(t_0) = 1$.
4	$l = a\chi$ and $v_p = a\dot{\chi}$ on the same cosmic-time slicing.
5	$a_0 = 1$, $\dot{l}_0 = 0$, $H_0\chi_0 < c$.
6	Tether-release initial event and a timelike radial geodesic.
7	$t > 0$, specified t_0 and radial position χ .
8	$T > 0$, $ R < cT$, $t = \sqrt{T^2 - R^2/c^2}$.
9	Specified Milne congruence and clock readings, with a common origin.
10	Light path connecting a comoving emitter and observer.
11	Small x ; expansion through second order using the source choice $v_e = \dot{a}_e\chi_e$.
12	Dust with no cosmological constant, weak-field comparison through second order.
13	Specified velocity and potential comparison at the same approximation order.
14	Local radial emission, a comoving observer, and the corresponding emission and reception events.
15	Local comoving frame, radial velocity, $ v_p < c$, positive outward.
16	$l_e = a_e\chi_e$, $\dot{l}_e = 0$, $ H_e l_e < c$.
17	Timelike worldline, $c^2 d\tau^2 = -ds^2$, and a chosen proper-time zero.
18	A comoving particle at fixed χ and $l = (t/t_0)\chi$.

Results

Definition 8.587 (Grøn–Elgarøy 2007 Results). Let $\text{ldx}_R^{\text{audit,result}} = [29]$.

Table 451: Grøn–Elgarøy 2007 Results

q	Result	Source Location
1	Authors' view of global description and dynamical geometry	abstract;§I; §I, p. 1,2
2	FLRW geometry and the proper-time–coordinate-time relation	§II(1)–(2); §II(3)–(8), p. 3
3	Radial geodesics, velocity decay and momentum	§II(9); §II(10)–(13); §II(12)–(13); §II(12);end of §II, p. 4,8
4	Power-law flat single-fluid universe	§II(14)–(15),(18), p. 4,5
5	Proper-distance rate and local peculiar velocity	§II(16);§IV, p. 5,13

q	Result	Source Location
6	Tether-release initial condition and admissible domain	§II(16)–(21); §II(21)–(22), p. 5
7	Conditions on acceleration at tether release	§II(23)–(27); §II(24)–(28), p. 6,7
8	Critique of treating cosmic flow as an independent force	§II;§V, p. 7,14
9	Hubble velocity compared at fixed distance	§II;§V, p. 7,14
10	Local domain of the homogeneous FLRW model	end of §II, p. 8
11	Cited annual change of a bound orbit	end of §II;refs.11–13, p. 8
12	Milne–Minkowski coordinate relation	§III(29)–(31), p. 8,9
13	Public synchronization of proper clocks with a common origin	§III;Fig.2, p. 9,18
14	Event sets of private and public slices	§III(31);Fig.2, p. 9,18
15	Inertial-frame velocity and cosmic-time distance rate	§III(32)–(33), p. 9
16	Distinction among Milne, a coasting fluid and a curved universe	§II–III, p. 7,8,9,10
17	Clock and velocity reference in a local inertial neighborhood	§III, p. 9,10
18	Authors’ view of the reality of public space	§III, p. 10
19	Cosmological wavelength ratio between comoving endpoints	§IV(34), p. 10
20	Second-order expansion of the cosmological shift	§IV(35), p. 10,11
21	Doppler comparison using the selected Hubble velocity	§IV(36)–(41), p. 11
22	Gravitational shift for the dust potential	§IV(42)–(47), p. 11,12
23	Sum of Doppler and gravitational terms through second order	§IV(48), p. 12
24	Propagation interpretations in Milne and curved universes	§IV–V, p. 12,14,15
25	Wavelength factors from radial peculiar velocity and cosmological propagation	§IV(49); §IV(50)–(51), p. 13
26	Compatibility of remote rest with a nonzero shift	§II(16);§IV(51)–(52); §IV(51)–(52), p. 5,13
27	Authors’ conclusion concerning cosmic expansion	§V, p. 13,14,15
28	The two clock congruences of Figure 2	Fig.2, p. 18
29	Cosmic time and signed trajectory in Figure 1	§II;Fig.1, p. 5,6,17

Table 452: Grøn–Elgarøy 2007 Conditions for Results

Result Index	Conditions and Domain
1	The domains of global general relativity and local Newtonian/SR descriptions compared by the authors.
2	FLRW line element, a timelike worldline, and proper-time units and zero.

Result Index	Conditions and Domain
3	The comoving solution and the noncomoving massive branch with $A > 0$; decay uses a tending to infinity.
4	Flat single fluid with constant $w > -1$, $t, t_0 > 0$ and $a(t_0) = 1$.
5	The same cosmic-time slicing, $l = a\chi$, and $v_p = a\dot{\chi}$.
6	$a_0 = 1$, $\dot{l}_0 = 0$, $H_0 > 0$ and $0 < \chi_0 < c/H_0$.
7	A timelike radial geodesic and admissible tethered initial data, separating $q_0 \geq 0$ from $-1 < q_0 < 0$.
8	The free-particle geodesic motion in an expanding universe discussed by the source.
9	Fixed proper distance l , the same origin and cosmic time; decrease requires $\dot{H} < 0$.
10	Applicability conditions of an FLRW model with a homogeneous matter distribution.
11	The cited spherical embedding model, Einstein–de Sitter background, solar mass and circular-orbit radius 1 AU.
12	Empty Milne model, $t > 0$, $t_0 > 0$ and the coordinate domain inside the future light cone.
13	Milne particle congruence with a common origin, proper clocks on the worldlines and a common zero.
14	A constant- T slice restricted to the future light cone and a constant- t hyperboloid.
15	The two coordinate and distance definitions for the same Milne comoving particle.
16	Empty $k = -1$ model, spatially flat $w = -1/3$ fluid model, and nonzero four-dimensional curvature.
17	Specified local inertial neighborhood and proper clock on a worldline.
18	The authors' view of public space in a curved universe containing matter.
19	Comoving emitter and observer, the connecting light path and endpoint scale factors.
20	Smooth scale factor, $a_0 = 1$, small $x = H_0(t_0 - t_e)$, through second order.
21	The velocity reference $v_e = \dot{a}_e\chi_e$ and small x , through second order.
22	Dust without a cosmological constant, a weak-potential comparison and small x .
23	Fixed velocity and gravitational models of Claims 39 and 40, compared through second order.
24	The global inertial frame of Milne and the curved-universe propagation description contrasted by the source.
25	Radial emission, local comoving frame, $ v_p < c$, comoving observer and emission/reception endpoints.
26	$\dot{l}_e = 0$ at emission, $l_e = a_e\chi_e$, $ H_e l_e < c$, and the connecting light path.
27	The authors' interpretation using cosmic simultaneity, a comoving congruence, distance changes and light propagation.
28	The Figure 2 clocks at rest in the selected inertial frame and clocks following Milne particles.
29	Dust universe, tether-release initial condition and signed radial position.

Result Dependence

Definition 8.588 (Grøn–Elgarøy 2007 Result-Dependence Hyperedges). For each row, let $\text{ResultEdge}_{R,q}$ ($q \in [29]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 453: Grøn–Elgarøy 2007 Result-Dependence Hyperedges

q	Time Premises i	Claim Support p
1	18	13–14
2	1–4	15–16
3	1–2, 4–5	17–19, 50
4	1–2	20
5	1–2, 4–6	21
6	1–2, 5–7	22–23
7	2, 4, 6–7	24–25
8	2, 4, 6	26
9	2, 6	27
10	1–2	28
11	2, 8	29
12	9–10	30
13	3, 9, 11	31
14	9–11	32
15	6, 9–10	33, 51
16	1–2, 9, 18	34
17	3, 12	35
18	1	36
19	2, 13–15	37
20	1–2, 13–15	38
21	2, 6, 13–14, 16	39
22	2, 13–14, 17	40
23	15–17	41
24	9–10, 13–15	42
25	2, 5, 13, 15, 19	43–44
26	2, 5–7, 13, 15, 19	45–46
27	1–2, 6, 14–15, 18	47
28	3, 9–11	48
29	1–2, 4, 6–7	49

$$\text{BoundaryCount}_R^{\text{audit}} = (99, 47).$$

The 51 claims, 19 time contents and 29 results give 99 nodes; the 18 transfers and 29 result dependencies give 47 edges.

Evidence Hierarchy Grøn and Elgarøy discuss FLRW geodesics, Milne coordinates and clocks, light propagation and the interpretation of cosmic expansion. Figures 1–2

illustrate theoretical models. The annual bound-orbit change is an estimate from a cited embedding model; this source reports zero newly acquired direct measurements.

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 0; remaining primitive-time roots 19. Every transfer has temporal input from cosmic time, proper time, synchronization, velocity references or propagation events.

Result Dependence Of the 29 results, 29 depend on primitive-time roots and 0 are reached from source formal evidence and local constructions. There are 108 root–result dependence pairs, of which 33 are necessary.

Placement of the Grounding Obligation Cosmic time and proper time support geodesics, initial conditions, velocities and distance rates; clock readings and a common zero support Milne synchronization. Emission and reception times, light-propagation intervals and velocity references enter redshift relations. Complete generation from nontemporal physical inputs is zero; the placement of the physical-time grounding obligation is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Initial rest at tether release requires $H_0\chi_0 < c$; the low-redshift additive formula follows the stated velocity and gravitational model and approximation order. Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 108 root–result dependence pairs. Actual reach from failure of a specific physical-time generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns results dependent on it.

Audit Conclusions

Proposition 8.589 (Primitive Time Roots and Result Dependence in Grøn–Elgarøy 2007). In the component order of Definition 8.115,

$$\text{AuditPartId}_{x_R} = (\emptyset, [19]; \\ \emptyset, \emptyset, \\ [29], \emptyset).$$

Proof. Closure from Initial Claims 1–12 remains those twelve nodes. Even with all 51 claims, no time content is newly reached because each transfer requires temporal input. The mutually inverse Milne/inertial-time conversions likewise require one of those temporal inputs. Adding all time contents as seeds reaches all 29 results. $\square$

Proposition 8.590 (Minimal Primitive-Time-Root Support in Grøn–Elgarøy 2007). The table gives each result’s minimal-support index family, with 80 family members in total.

Proof. Apply Definition 7.46 to all transfers and result dependencies, starting from singleton time-content seeds and empty supports for source formal evidence. By Proposition 7.47, the inclusion-minimal supports of the stable antichains give the displayed families. $\square$

Table 454: Minimal Primitive-Time-Root Support in
Grøn–Elgarøy 2007

Result Indices	Minimal-Support Family Members
1	{18}
2–5, 10, 18	{1}
6, 29	{1, 7}
7	{1, 7}, {2, 7}
8–9	{1}, {2, 3}, {2, 6}, {2, 7}, {2, 9}, {2, 10}
11	{1, 8}, {2, 8}
12, 15	{1}, {3}, {9}, {10}
13, 28	{1}, {3}
14	{1}, {3}, {9, 11}, {10, 11}
16	{1, 18}
17	{1, 12}, {3, 12}
19, 22	{1, 13, 14}, {2, 13, 14}
20	{1, 13, 14}
21	{1, 13, 14}, {2, 3, 13, 14}, {2, 6, 13, 14}, {2, 7, 13, 14}, {2, 9, 13, 14}, {2, 10, 13, 14}
23	{1, 13, 14}, {2, 3, 13, 14}, {2, 6, 13, 14}, {2, 7, 13, 14}, {2, 9, 13, 14}, {2, 10, 13, 14}, {2, 13, 14, 16}, {16, 17}
24	{1, 13, 14}, {2, 3, 13, 14}, {2, 9, 13, 14}, {2, 10, 13, 14}, {3, 13, 14, 15}, {9, 13, 14, 15}, {10, 13, 14, 15}, {3, 13, 14, 16, 17}, {9, 13, 14, 16, 17}, {10, 13, 14, 16, 17}
25	{1, 13}, {2, 13}
26	{1, 7, 13}, {2, 7, 13}
27	{1, 13, 14, 18}, {1, 14, 15, 18}, {1, 14, 16, 17, 18}

Result-Preservation Domain Fix the models, initial and endpoint conditions, clock synchronization, velocity references and approximation domains in Table 452. When temporal inputs are realized by generated representations preserving the same cosmic and proper times, trajectories, synchronization, propagation intervals and velocity references, time-dependent results are preserved within their domains.

Proposition 8.591 (Complete Audit of Time Use in Grøn–Elgarøy 2007). There are 19 time contents, 51 claims, 18 transfers and 29 results. The 131 claim/time-content use pairs have 174 role assignments, with 117 source-specific obligation atoms. Complete generation is zero, local reach is 0, remaining primitive-time roots number 19, and 29 results depend on roots.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual statements

Object of statements Attribution of reality to public space, the relation between expanding space and galaxies, simultaneity on following clocks, and cosmic time measured by following clocks.

Original (English) The only real space is the public space.

space expands and the galaxies follow the expanding space

simultaneity on the clocks following the reference particles

The cosmic time t is the time measured by clocks following all of the test particles.

Source location The first quotation is in Section III, p. 10; the second and third are in Section V, p. 14; the fourth is in the caption to Fig. 2, p. 18.

Minimal context Immediately before the first quotation, the authors move from the Milne comparison to the absence of a global Minkowski description in a curved universe containing matter. The second and third concern the interpretation of expansion using cosmic simultaneity and changing distances between comoving galaxies. Figure 2, containing the fourth quotation, identifies readings of proper clocks following Milne test particles from a common origin with cosmic time t , and compares them with private simultaneity defined by the selected inertial-frame time T .

Attribution All four statements are attributed to the text by Øyvind Grøn and Øystein Elgarøy.

Content stated directly by the quotations The first states that only public space is real space. The second states that space expands and galaxies follow the expanding space. The third states simultaneity on clocks following the reference particles. The fourth states that cosmic time is measured by clocks following the test particles.

Explicit domain of application The first quotation concerns the interpretation of public space in a curved universe containing matter; the second and third concern an FLRW description using a comoving congruence and cosmic time; the fourth concerns the Milne model with its specified common origin and following clocks.

Evidence provenance The quoted source is arXiv:astro-ph/0603162v2.

Individual Audit Attribution of Reality to Public Space

The first quotation states Grøn and Elgarøy's view that public space in a curved universe containing matter is the only real space. The source's limitation on global Minkowski charts corresponds to Claim 34; its attribution of reality to cosmic-time public space corresponds to Time Content 1, Claim 36 and Result 18. The preceding Milne coordinate and slice comparison corresponds to Claims 30–33.

Expanding Space and Following Galaxies

The second quotation is the authors' interpretation of the FLRW description in which proper distance $l = a\chi$ between galaxies at fixed comoving coordinates changes with the

scale factor. It corresponds to Time Contents 1, 2 and 6, Claims 21 and 47, and Results 5 and 27. Its domain is the comoving congruence and cosmic-time slicing specified by the source.

Simultaneity on Following Clocks

The third quotation defines public simultaneity by equal readings of clocks following the reference particles. Its concrete Milne construction uses each clock's proper time, a common origin and a common zero. It corresponds to Time Contents 3, 9 and 11, Claims 31 and 48, Transfer 9, and Results 13 and 28.

Cosmic Time Measured by Following Clocks

The fourth quotation identifies proper-time readings of clocks following the Milne test particles in Figure 2 with cosmic time t . Private time T is defined by a different clock congruence at rest in the selected inertial frame, and the two simultaneity slices contain different event sets. It corresponds to Time Contents 3 and 9–11, Claims 30–32 and 48, Transfers 7–9, and Results 12–14 and 28.

Common Audit with Specified Domain The references in the four statements are determined by cosmic time, proper-clock readings, congruence synchronization and the selected simultaneity slices. Their generation judgment and result dependence are given in Proposition 8.589.

8.19.49 Source audit

Audit input

Original source and audit scope

Source identifier 49.

Source title Primitive-time-root audit of Ryu–Takayanagi 2006

Bibliographic identification Let the original source Shinsei Ryu and Tadashi Takayanagi, *Holographic Derivation of Entanglement Entropy from AdS/CFT*, *Physical Review Letters* 96, 181602 (2006), DOI: 10.1103/PhysRevLett.96.181602, arXiv:hep-th/0603001v2 be $R = \text{Ref}_{\text{RT06HE}}$. The audit domain comprises the boundary spacetime, static bulk geometry, fixed-time spatial section, boundary-anchored minimal surface, Euclidean finite-temperature branch, Poincaré branch, and AdS black-hole branch.

Audit-scope identifier 49-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Source-location identifier 49-L1.

Source-location content The audit domain stated in the bibliographic identification and the source-location columns of the respective evidence tables.

Formal Objects

Object Identifier 49-F1.

Object Name Formal objects of the source

Object Type Formal object

Formal Content Definition 8.593 specifies nine time contents and sixteen auxiliary objects; Definition 8.594 specifies forty audit claims comprising six physical initial components and thirty-four source claims.

Primary-Source-Location Identifier Set 49-L1.

Assumptions

Assumption Identifier 49-A1.

Assumption Content The explicit assumptions in Definition 8.592.

Primary-Source-Location Identifier Set 49-L1.

Nontemporal Physical Inputs

Input Identifier 49-P1.

Input Name Physical carriers and spatial configurations

Input Type Nontemporal physical input

Input Content The six components in Table 459. Spatial carriers are distinguished from time coordinates, periods, temperature scales and static-slice conditions.

Primary-Source-Location Identifier Set 49-L1.

Time Contents

Time-Content Identifier 49-T1.

Time-Content Name Time coordinates, thermal scales and static-slice conditions in AdS/CFT

Time-Content Type Time content

Time-Content Description The nine contents in Table 455.

Primary-Source-Location Identifier Set 49-L1.

Formal Input Elements

Definition 8.592 (Formal Objects and Explicit Assumptions of Ryu–Takayanagi 2006). The source uses the AdS/CFT correspondence and dimension-dependent Newton constants to obtain entropy from the area of a static minimal surface anchored on ∂A . It uses $e^{\rho_0} \gg 1$ in the global branch, $\beta/L \ll 1$ in the finite-temperature CFT2 branch, and a large interval with a correlation-length IR cap in the massive branch. The higher-dimensional branches use the specified strip and disk regions, UV cutoffs and a gravitational description of the strongly coupled theory. The coefficient comparison with free Yang–Mills theory includes the estimate that a gauge field contributes as two real scalars. Complementary-entropy equality assumes a globally pure state; the horizon thermal contribution uses a large strip width and the finite-temperature metric.

Definition 8.593 (Time Contents and Auxiliary Objects of Ryu–Takayanagi 2006). Set $n_R = 9$ and define nine time contents and sixteen auxiliary objects by the following two tables.

Table 455: Temporal contents and source locations

Group	Time content i	Temporal content	Auxiliary index	Source location key
1	1	The Lorentzian time coordinate t identified between the boundary cylinder and global AdS	None	1,3,4
2	2	The time coordinate τ of the Euclidean BTZ metric	None	5,6
3	3	The time period $\Delta\tau = 2\pi R/r_+$ fixed by BTZ regularity	None	5
4	4	The boundary inverse temperature β and temperature scale $T = \beta^{-1}$	None	2,5,6
5	5	The Lorentzian time coordinate x_0 in the Poincaré branch	None	7,8,9,10,11
6	6	The Lorentzian time coordinate t in the AdS black-hole branch	None	12,13,14
7	7	The temperature scale T of the black-hole branch and its identification $u_0 = \pi T$ with the horizon parameter	None	12,13,14
8	8	The fixed-time slicing and time-independence conditions of the static prescription	None	1,3,4,5,6,7,8,9,10,11,12,13,14
9	9	The Euclidean time t_E and period $2\pi r_+/R$ after the BTZ-to-thermal-AdS coordinate exchange	None	6

In time content 9, t_E denotes t in the source’s exchanged coordinates. Time content 8 comprises the fixed-slice and time-independence conditions of the respective branches. Global AdS, Poincaré AdS and the AdS black hole hold t, x_0, t , respectively, fixed. Euclidean BTZ holds τ fixed; after the coordinate exchange $r_+\tau = R\theta$, $r_+\phi = Rt_E$, θ is fixed and the endpoints are separated along t_E . The period of time content 9 follows from $\phi \sim \phi + 2\pi$. The relations $T = \beta^{-1}$ and $u_0 = \pi T$ apply in their respective branches; the spatial role of u_0 also corresponds to auxiliary object 15.

Table 456: Auxiliary objects

a	Auxiliary object	Source location key
1	The algebra of states, reduced density matrices, partial traces and entropy	1,2
2	Spatial regions A, B , their boundaries, interval endpoints, and lengths l, L	1,2,3,4,7,9,13
3	The spatial metric, AdS radius R , boundary-anchored surfaces and area functional	1,3,5,7,9,12,13
4	Dimension-dependent Newton constants and area-entropy normalization	1,3,9,10,11,13
5	The central charge c	1,3,6,7,8
6	Spatial UV cutoffs a, ρ_0, z_{UV} and their identifications	1,3,6,7,8,9,13
7	The BTZ spatial coordinates, horizon radius and angular periodicity $\phi \sim \phi + 2\pi$	5,6
8	The normalization and thermal dictionary between the BTZ boundary and the CFT	5,6
9	Spatial-curve coordinates and integration parameters s, y	7,8,9
10	The correlation length ξ , massive deformation and spatial IR cap	1,8
11	Geodesic lengths between multiple-interval endpoints and their signed combination	4,6
12	Strip and disk regions, spatial dimension d , and area-expansion coefficients	9,10,11
13	Sphere reduction, N , string and brane constants, and gravitational-constant identifications	10,11
14	The free Yang–Mills multiplet and approximation to the gauge-field contribution	10
15	The spatial variables $u, u_*, u_0, h(u)$, and strip width in the black-hole branch	12,13
16	Horizon sections and their coverage by complementary regions	13,14

Definition 8.594 (Physical and Formal Claims of Ryu–Takayanagi 2006). Set $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(40)$. Claims 1–6 are physical initial components, and Claims 7–40 correspond to thirty-four source claims. The source-formal evidence environment comprises these forty claims. The dependency key d denotes a pair of time-content and auxiliary-object index sets; the source-location key ℓ denotes the locations below.

Table 457: Claim-dependency keys

d	Time indices	Auxiliary indices
1	$\emptyset$	$\{1\}$
2	$\emptyset$	$\{2\}$
3	$\emptyset$	$\{3, 16\}$
4	$\emptyset$	$\{13, 14\}$

d	Time indices	Auxiliary indices
5	$\emptyset$	$\{13\}$
6	$\emptyset$	$\{4, 16\}$
7	$\emptyset$	$\{3, 4, 5, 13\}$
8	$\emptyset$	$\{1, 2\}$
9	$\{1, 8\}$	$\{2, 5, 6\}$
10	$\{5, 8\}$	$\{1, 5, 6, 10\}$
11	$\{8\}$	$\llbracket 2, 4 \rrbracket$
12	$\emptyset$	$\llbracket 3, 5 \rrbracket$
13	$\{1\}$	$\{2, 3\}$
14	$\emptyset$	$\{2, 6\}$
15	$\{1, 8\}$	$\{2, 3, 6\}$
16	$\{1, 8\}$	$\llbracket 2, 6 \rrbracket$
17	$\{1, 8\}$	$\{2, 3, 4, 11\}$
18	$\{2\}$	$\{3, 7\}$
19	$\{3, 4\}$	$\{2, 7, 8\}$
20	$\{2\}$	$\{3, 7, 8\}$
21	$\{2, 4, 8, 9\}$	$\{2, 3, 4, 5, 6, 7, 8, 11\}$
22	$\{5, 8\}$	$\{2, 3, 6, 9\}$
23	$\{5, 8\}$	$\{2, 3, 4, 5, 6, 9\}$
24	$\{5, 8\}$	$\{3, 4, 5, 6, 9, 10\}$
25	$\{5\}$	$\{3\}$
26	$\{5, 8\}$	$\{2, 3, 6, 12\}$
27	$\{5, 8\}$	$\{2, 3, 6, 9, 12\}$
28	$\{5, 8\}$	$\{3, 4, 6, 12\}$
29	$\{5, 8\}$	$\{2, 3, 4, 6, 12, 13\}$
30	$\{5, 8\}$	$\{2, 12, 13, 14\}$
31	$\{6, 7\}$	$\{3, 15\}$
32	$\llbracket 6, 8 \rrbracket$	$\{2, 3, 6, 15\}$
33	$\llbracket 6, 8 \rrbracket$	$\{2, 3, 4, 13, 15, 16\}$
34	$\llbracket 6, 8 \rrbracket$	$\{2, 3, 4, 16\}$
35	$\llbracket 6, 8 \rrbracket$	$\{1, 2, 16\}$

The source-location keys ℓ refer to the following locations.

- 1** p. 1: Section I, Equations (1.1)–(1.5), definitions and static prescription
- 2** pp. 1–2: Section I, complementary equality for a globally pure state, subadditivity, and $T = \beta^{-1}$
- 3** p. 2: Section II, Equations (2.1)–(2.5), global metric, fixed-time geodesic and cutoff
- 4** pp. 2–3: Section II, Equations (2.6) and (2.7), signed multiple-interval geodesic combination and its finite-temperature extension
- 5** p. 2: Section II, Equation (2.8), Euclidean BTZ regularity, time period and boundary dictionary
- 6** p. 3: Section II, coordinate-circle exchange and finite-temperature Equation (2.9)

- 7** p. 3: Section II, Poincaré metric and Equations (2.10) and (2.11)
- 8** p. 3: Section II, massive deformation, correlation length and the one-boundary-point leading logarithm in Equation (2.12)
- 9** p. 3: Section III, Equations (3.1)–(3.3), strip and disk regions, and universal terms
- 10** pp. 4–5: Sections III and V, Equations (3.4) and (3.5), comparison of strongly coupled and free $\mathcal{N} = 4$ theory
- 11** p. 4: Section III, Equations (3.6)–(3.9), M2 and M5
- 12** p. 4: Section IV, Equation (4.1), h and $u_0 = \pi T$
- 13** p. 4: Section IV, Equation (4.2), strip width and the thermal contribution in Equation (4.3)
- 14** pp. 4–5: Sections IV and V, Figure 3, whole-horizon limit, finite-temperature complementary entropies and conclusion

Table 458: Physical and formal claims

p	Claim content	d	ℓ
1	Physical carriers of the quantum system	1	1
2	Spatial carriers of the boundary and its subregions	2	1,3
3	Spatial carriers of the bulk, surfaces, and horizon sections	3	1,3,5,7,9,12,14
4	Physical carriers of the Yang–Mills fields	4	10
5	Physical carriers of M2 branes	5	11
6	Physical carriers of M5 branes	5	11
7	Use Bekenstein–Hawking area entropy as an existing relation.	6	1
8	Use the AdS/CFT correspondence and dimensional and gravitational dictionaries.	7	1,3,10,11
9	Define the reduced state by a partial trace and its von Neumann entropy.	8	1
10	State that complementary entropies are equal for a globally pure state. Also state subadditivity of entropy.	8	2
11	Use the known CFT interval entropy for a periodic one-dimensional system.	9	1,3
12	Use the known massive-CFT entropy depending on correlation length and boundary-point count.	10	1,8
13	Propose an entropy prescription using the area of a boundary-anchored static minimal surface. The surface is likened to a holographic screen.	11	1,3
14	Use the Brown–Henneaux relation $c = 3R/(2G_N^{(3)})$.	12	3
15	Specify the global AdS metric and its boundary-cylinder time identification.	13	3

p	Claim content	d	ℓ
16	Relate the global radial and boundary UV cutoffs by $e^{\rho_0} \sim L/a$.	14	3
17	Calculate the fixed-time geodesic length between interval endpoints.	15	3
18	Recover the global single-interval entropy for $e^{\rho_0} \gg 1$.	16	3
19	Express the cited CFT result in the global-AdS branch as a signed combination of multiple-interval geodesic lengths.	17	4
20	Specify the Euclidean BTZ metric and spatial angular periodicity.	18	5
21	Fix $\Delta\tau = 2\pi R/r_+$ by regularity of the Euclidean BTZ metric.	18	5
22	Use the boundary dictionary $\beta/L = R/r_+$ and $T = \beta^{-1}$.	19	2,5
23	Exchange Euclidean-time and spatial circles using $r = r_+ \cosh \rho$, $r_+ \tau = R\theta$, and $r_+ \phi = Rt$.	20	6
24	Recover the finite-temperature single-interval entropy in Equation (2.9) for $\beta/L \ll 1$, and extend the calculation to multiple intervals with the form of Equation (2.7).	21	4,5,6
25	Give the fixed-time semicircular geodesic in the Poincaré metric.	22	7
26	Obtain the Poincaré single-interval entropy $(c/3) \log(l/a)$.	23	7
27	Obtain the leading logarithm $(c/6) \log(\xi/a)$ for one boundary point with the massive IR cap.	24	8
28	Use the higher-dimensional Poincaré metric.	25	9
29	Give the strip minimal-surface area in Equation (3.2).	26	9
30	Give the disk area and its even/odd-dimensional expansions in Equation (3.3).	27	9
31	Distinguish the UV area-law divergence from the universal finite term or logarithmic coefficient.	28	9
32	Give the AdS results corresponding to strongly coupled $\mathcal{N} = 4$ theory in Equations (3.4) and (3.5).	29	10
33	Compare functional forms with the free theory and, treating the gauge field as two real scalars, estimate the ratio of finite-term coefficients (free theory/gravitational calculation) as about $3/2$.	30	10
34	Give the M2/M5 area entropies and $N^{3/2}/N^3$ dependence in Equations (3.6)–(3.9).	29	11
35	Use the finite-temperature AdS black-hole metric with $h = 1 - u_0^4/u^4$ and $u_0 = \pi T$.	31	12
36	Use the strip-area integral and strip-width relation in Equation (4.2).	32	13
37	Obtain the near-horizon thermal contribution $\pi^2 N^2 T^3 L^2 l/2$ at large strip width.	33	13
38	State that the surface wraps the full horizon in the whole-system limit and reproduces its area entropy.	34	14
39	State that complementary-entropy equality does not hold generically at finite temperature.	35	2,14

p	Claim content	d	ℓ
40	State the difficulty, at the time of the source, of quantitative comparison with higher-dimensional strongly coupled theories.	4	10,14

Definition 8.595 (Semantic Components and Generation-Initial Set of Ryu–Takayanagi 2006). Set $\text{Node}_R^{\text{audit,semantic}} = \text{Node}_R^{\text{audit,claim}}$. Define the initial-set/evidence-environment pair by $\text{InitEnv}_R^{\text{audit}} = \text{InitEnv}_R^{\text{phys}} = \text{ClaimPrefix}_R(6)$.

Table 459: Generation-initial components

a	Physical carrier	Claim index
1	Physical carriers of the quantum system	$\{1\}$
2	Spatial carriers of the boundary and its subregions	$\{2\}$
3	Spatial carriers of the bulk, surfaces, and horizon sections	$\{3\}$
4	Physical carriers of the Yang–Mills fields	$\{4\}$
5	Physical carriers of M2 branes	$\{5\}$
6	Physical carriers of M5 branes	$\{6\}$

Dependent inputs

Index identifier S49-XI-DI1.

Existing-evidence reference set Definition 8.596.

Dependence and transfer elements

Definition 8.596 (Time-Content Transfers in Ryu–Takayanagi 2006). Set $\text{Idx}_R^{\text{audit,transfer}} = [3]$. Inputs and outputs are pairs of time-content and auxiliary-object index sets; formal support is a set of claim indices.

Table 460: Temporal-content transfers

j	Input (time; auxiliary)	Output (time; auxiliary)	Formal support	Transfer content
1	$(\{2\}; \{3, 7\})$	$(\{3\}; \emptyset)$	$\{21\}$	The time period from Euclidean BTZ regularity
2	$(\{3\}; \{2, 7, 8\})$	$(\{4\}; \emptyset)$	$\{22\}$	Boundary inverse temperature from the time period and boundary normalization
3	$(\{2\}; \{3, 7, 8\})$	$(\{9\}; \emptyset)$	$\{23\}$	Thermal-AdS Euclidean time from the BTZ coordinate exchange

Conditions of establishment Transfer 1 uses Euclidean time and regularity of the BTZ metric; Transfer 2 uses the time period and boundary normalization; Transfer 3 exchanges the circles of the Euclidean BTZ metric. The temporal role of t_E in Transfer 3 follows from the metric obtained by exchanging the existing Euclidean-time and spatial circles. Each transfer requires a time-content input, so none starts from nontemporal physical inputs alone or from the source-formal evidence environment alone. Both the complete and local-reach assessments therefore give zero.

Original-source support

Index identifier S49-XI-SS1.

Existing-evidence reference set Definition 8.596.

Results

Definition 8.597 (Results of Ryu–Takayanagi 2006). Set $\text{Idx}_R^{\text{audit,result}} = [16]$; the table specifies each result and its source locations.

Table 461: Results and source locations

j	Result	Source location key
1	Reduced state and von Neumann entropy	1
2	The area prescription for a boundary-anchored static minimal surface	1
3	Recovery of global-AdS3 single-interval entropy	3
4	The Brown–Henneaux central-charge relation	3
5	The radial-to-boundary UV-cutoff identification	3
6	The signed geodesic combination of the cited CFT result in the global-AdS branch	4
7	Area calculations for higher-dimensional strips and disks	9
8	Separation of UV divergences from universal finite terms and logarithmic coefficients	9
9	Comparison of functional forms and estimated coefficients for strongly coupled and free Yang–Mills theory	10
10	M2/M5 area entropies and scaling with degrees of freedom	11
11	The finite-temperature horizon thermal contribution and whole-system limit	12,13,14
12	Complementary-entropy equality for a globally pure state, and subadditivity	2
13	Finite-temperature single-interval entropy and the multiple-interval extension in the BTZ branch	4,5,6
14	Poincaré single-interval entropy	7
15	The leading logarithm for the massive deformation with one boundary point	8

j	Result	Source location key
16	Loss of generic complementary-entropy equality at finite temperature	2,14

Definition 8.598 (Result-Dependence Hyperedges in Ryu–Takayanagi 2006). The time-prerequisite, auxiliary-prerequisite and formal-support columns contain time-content, auxiliary-object and claim index sets, respectively. Each row specifies the corresponding premise sets in Definition 7.105.

Table 462: Result-dependency hyperedges

j	Time prerequisites	Auxiliary prerequisites	Formal support
1	$\emptyset$	$\{1, 2\}$	$\{1, 2, 9\}$
2	$\{8\}$	$\llbracket 2, 4 \rrbracket$	$\{2, 3, 7, 8, 13\}$
3	$\{1, 8\}$	$\llbracket 2, 6 \rrbracket$	$\{2, 3, 11, 14, 15, 16, 17, 18\}$
4	$\emptyset$	$\llbracket 3, 5 \rrbracket$	$\{14\}$
5	$\emptyset$	$\{2, 6\}$	$\{16\}$
6	$\{1, 8\}$	$\{2, 3, 4, 11\}$	$\{19\}$
7	$\{5, 8\}$	$\{2, 3, 4, 6, 9, 12\}$	$\{2, 3, 13, 28, 29, 30\}$
8	$\{5, 8\}$	$\{3, 4, 6, 12\}$	$\llbracket 29, 31 \rrbracket$
9	$\{5, 8\}$	$\{2, 3, 4, 6, 12, 13, 14\}$	$\{4, 32, 33, 40\}$
10	$\{5, 8\}$	$\{2, 3, 4, 6, 12, 13\}$	$\{5, 6, 34\}$
11	$\llbracket 6, 8 \rrbracket$	$\{2, 3, 4, 6, 13, 15, 16\}$	$\{3, 35, 36, 37, 38\}$
12	$\emptyset$	$\{1, 2\}$	$\{1, 10\}$
13	$\{2, 4, 8, 9\}$	$\{2, 3, 4, 5, 6, 7, 8, 11\}$	$\{3, 20, 21, 22, 23, 24\}$
14	$\{5, 8\}$	$\{2, 3, 4, 5, 6, 9\}$	$\{3, 14, 25, 26\}$
15	$\{5, 8\}$	$\{3, 4, 5, 6, 9, 10\}$	$\{3, 12, 14, 25, 27\}$
16	$\llbracket 6, 8 \rrbracket$	$\{1, 2, 16\}$	$\{3, 35, 38, 39\}$

There are 27 time-content prerequisite occurrences, 74 auxiliary-prerequisite occurrences and 61 supporting-claim occurrences.

Proposition 8.599 (Audit-Boundary Cardinalities for Ryu–Takayanagi 2006).

$$\text{BoundaryCount}_R^{\text{audit}} = (65, 19).$$

Proof. Forty claims, nine time contents and sixteen results give $40 + 9 + 16 = 65$ nodes; three transfers and sixteen result edges give $3 + 16 = 19$ edges. $\square$

Grounds of establishment Results 1, 4, 5 and 12 give the reduced state and entropy, central charge, cutoff dictionary, complementary equality for a globally pure state, and subadditivity. Results 2, 3, 6–11 and 13–16 use the static area prescription and the metric, boundary conditions, cutoff and temperature scale of the respective branch. The signed multiple-interval combination is Equation (2.7), which the source obtains from a cited CFT result. The free-theory coefficient ratio is an estimate with a gauge-field approximation; the thermal contribution is the near-horizon calculation at large strip width.

Evidence hierarchy The source supplies geometric calculations, agreement with known CFT results, scaling with degrees of freedom, and comparisons of strongly coupled and free theories. It has zero new direct-measurement objects and predicates. The higher-dimensional comparison concerns functional forms and coefficients supported by the specified gravitational description and free-theory estimate.

Generation assessment **Assessment:** complete 0 / local reach 0 / remaining roots 9.

All three transfers require a time-content input. Neither the six physical initial components nor the source-formal evidence environment alone reaches a time content. All nine contents remain primitive time roots.

Result dependence Results 1, 4, 5 and 12 are reachable from the source-formal evidence environment alone; Results 2, 3, 6–11 and 13–16 depend on primitive time roots.

Definition 8.600 (Evidential Scope and Boundaries of Ryu–Takayanagi 2006). The source evidence supports the algebra of reduced states and entropy, the static boundary-anchored area prescription, single and multiple intervals in AdS3/CFT2, higher-dimensional strips and disks, UV divergences and universal terms, scaling with field and brane degrees of freedom, and the finite-temperature horizon branch. The static prescription uses each branch’s fixed slice and time independence; the thermal calculations use a Euclidean period or black-hole temperature scale.

Result-preservation domain

Definition 8.601 (Result-Preservation Conditions for Ryu–Takayanagi 2006). Each result holds within the corresponding AdS/CFT dictionary, state conditions, boundary anchoring, metric, cutoff, dimensional and coupling conditions, and approximations. Complementary entropies are equal for a globally pure state; the equality does not hold generically at finite temperature. A result with temporal prerequisites is preserved as a conditional result when an LCTR generation sequence establishes the required coordinate, order, scale, period and slicing correspondences before use and its nontemporal conditions are also retained.

Allocation of the grounding obligation

Placement of grounding obligations The nine time contents audited in this source are not generated from nontemporal physical inputs before use but are accepted as already established time inputs. The placement judgment for their physical-time grounding obligations is “backgrounded by primitive time input.”

Evidence of actual failure

Evidence of actual failure The complete-generation judgment and remaining-root judgment concern physical-time qualification and are distinct from actual failure of a specific condition governing a result value or domain of application. Neither the formal evidence of the original source nor the LCTR audit evidence establishes actual failure of such a target condition. The actual-failure judgment is “indeterminate from the original-source evidence”; actual failure is not inferred from backgrounding alone.

Downstream reach

Downstream reach The extent of reach to primitive-time-root-dependent results is determined source by source by the primitive-time-root–result dependency relation and minimal primitive-time-root support reconstructed in this division. This is structural reach. Because actual failure of a specific condition has not been established, the actual-reach judgment from a failure location to a result is “indeterminate from the original-source evidence.” Structural reach does not substitute for actual reach.

Repair scope

Repair scope The repair targets given by structural reach are the corresponding time contents and the generation or establishment conditions required before their use. Recalculation is limited to results reached by dependence from a failure location after actual failure has been established independently. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If the condition is met, the target results are regenerated or recalculated from the repaired inputs, or their domain of establishment is restricted. Values after recalculation are determined as audit results.

Audit conclusion

Proposition 8.602 (Primitive Time Roots and Result Dependence in Ryu–Takayanagi 2006). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [9]; \emptyset, \{1, 4, 5, 12\}, \{2, 3, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16\}, \emptyset).$$

Proof. The temporal input sets of the three transfers are $\{2\}$, $\{3\}$, $\{2\}$, respectively. No transfer starts from a time-free initial set, so all nine contents remain roots. Results 1, 4, 5 and 12 have no temporal prerequisite and are reached using the corresponding source claims in the source-formal evidence environment. The other twelve results have temporal prerequisites and are reached from that environment with the nine roots added. The four-way result partition gives the displayed tuple. $\square$

Proposition 8.603 (Minimal Primitive-Root Supports for Ryu–Takayanagi 2006). There are 16 members across the minimal-support families, 25 root–result pairs and 25 necessary root–result pairs.

Table 463: Minimal Primitive-Root Supports

Result indices	Support statistics
1,4,5,12	(1; 0)
2	(1; 1)
3,6,7,8,9,10,13,14,15	(1; 2)
11,16	(1; 3)

Proof. Applying Definition 7.46 to the three transfers and sixteen result edges gives

$$\mathcal{A}_R(q) = \begin{cases} \{\emptyset\}, & q \in \{1, 4, 5, 12\}, \\ \{\{8\}\}, & q = 2, \\ \{\{1, 8\}\}, & q \in \{3, 6\}, \\ \{\{5, 8\}\}, & q \in \{7, 8, 9, 10, 14, 15\}, \\ \{\llbracket 6, 8 \rrbracket\}, & q \in \{11, 16\}, \\ \{\{2, 8\}\}, & q = 13. \end{cases}$$

□

Individual statements Statement 1

Original (English):

A holographic derivation of the entanglement entropy in quantum (conformal) field theories is proposed from AdS/CFT correspondence.

The source location is Shinsei Ryu and Tadashi Takayanagi, *Holographic Derivation of Entanglement Entropy from AdS/CFT*, arXiv:hep-th/0603001v2, p. 1, abstract. Its minimal context is the opening sentence of the abstract, which summarizes the AdS/CFT correspondence, the area of a minimal surface, and the analogy with the Bekenstein–Hawking formula.

The object of the statement is a proposal for a holographic derivation of entanglement entropy in quantum (conformal) field theories based on the AdS/CFT correspondence. Its predicate is that a prescription is proposed for deriving entanglement entropy from the area of a bulk minimal surface. Its explicit domain of application is the static AdS/CFT setting treated by the original source, the boundary subsystem and its boundary, the corresponding static minimal surface, and the cases compared within the original source.

Statement 2

Original (English):

Here, we present a simple proposal on this issue in the light of AdS/CFT duality.

The source location is p. 1, Section I of the same source. Its minimal context is the sentence positioning the source’s proposal immediately after the authors state that a comprehensive gravitational interpretation of entanglement entropy had been lacking.

The object of the statement is a research proposal based on AdS/CFT duality for the problem of a gravitational interpretation of entanglement entropy. Its predicate is that a simple proposal is presented on the issue. Its explicit domain of application is limited to the problem defined in Section I, the static prescription presented by the source, and the tests conducted within the source.

Statement 3

Original (English):

In this setup we propose the following ‘area law’

$$S_A = \frac{\text{Area of } \gamma_A}{4G_N^{(d+2)}}$$

where γ_A is the d dimensional static minimal surface in AdS_{d+2} whose boundary is given by ∂A , and $G_N^{(d+2)}$ is the $d + 2$ dimensional Newton constant.

The source location is p. 1, Section I, Eq. (1.5), and the immediately following definition in the same source. Its minimal context is the passage that defines subsystem A of the boundary CFT and its boundary ∂A , then gives the proposed area law and the meaning of the static minimal surface.

The object of the statement is the correspondence between the entanglement entropy of boundary subsystem A , its boundary ∂A , and the area of the static minimal surface γ_A anchored there. Its predicate is that $S_A = \text{Area}(\gamma_A)/(4G_N^{(d+2)})$ is proposed and that γ_A and the Newton constant are defined. Its explicit domain of application is the static AdS_{d+2} bulk, a subsystem of the boundary CFT, the stated boundary condition, and the AdS/CFT correspondence.

Statement 4

Original (English):

In higher dimensions ($d \geq 2$), a quantitative comparison is not easy because the gravity description corresponds to the strongly coupled gauge theory whose entanglement entropy is not known at present.

The source location is p. 5, Section V of the same source. Its minimal context is the sentence stating the limit of the higher-dimensional comparison after noting that, for $d = 1$, the formula in the original source agrees exactly with the entropy calculated directly from the CFT.

The object of the statement is the availability of a quantitative comparison for holographic entanglement entropy in strongly coupled gauge theories with $d \geq 2$. Its predicate is that such a quantitative comparison is not easy because the entanglement entropy of the corresponding strongly coupled gauge theory was unknown at the time of publication. Its explicit domain of application is the results known at publication for higher-dimensional strongly coupled gauge theories, distinguished from the quantitative comparison performed for $d = 1$.

Common evidence with specified domain of application Source attribution

All four statements are attributed to Shinsei Ryu and Tadashi Takayanagi.

Individual audit Proposal for a holographic derivation

Statement S49-Q001 proposes a holographic derivation of quantum-field-theory entanglement entropy through the AdS/CFT correspondence. Its independent grounds are the analogy with the Bekenstein–Hawking area formula, area calculations for a static minimal surface anchored on the boundary, and comparisons with known CFT results. The source treats static AdS/CFT settings: it demonstrates agreement with two-dimensional CFT formulas and compares higher-dimensional gravitational calculations with the functional forms and coefficients of the free theory. The corresponding formal evidence comprises Claims 7, 8, 13–18, 32, 33 and 40 in Table 458 and Results 2, 3, 9 and 13–15 in Table 461.

A simple proposal for the gravitational interpretation

Statement S49-Q002 concerns entanglement entropy, for which a comprehensive gravitational interpretation had been lacking. The proposal is a single prescription that uses AdS/CFT duality to normalize the area of the static bulk minimal surface corresponding to a boundary subsystem by the Newton constant. Its grounds within the source are the definitions of the reduced state and entropy, the boundary-anchoring condition, and calculated agreement with known CFT formulas. The specified dimensions, geometries

and states determine the scope of these comparisons. The corresponding formal evidence comprises Claims 8, 9 and 13 and Results 1–3 in Table 461.

Area law from a static minimal surface

Statement S49-Q003 proposes $S_A = \text{Area}(\gamma_A)/(4G_N^{(d+2)})$ for a d -dimensional static minimal surface in AdS_{d+2} satisfying $\partial\gamma_A = \partial A$. Its independent grounds are the area calculation for a surface anchored on the specified boundary and normalization by the Newton constant. The source’s AdS_3/CFT_2 calculations agree with known entropy formulas under the respective conditions of the global single-interval, finite-temperature Euclidean-BTZ and Poincaré branches. Each surface belongs to the fixed slice specified in its branch. The corresponding formal evidence comprises Claims 13–18 and 20–27, Results 2, 3 and 13–15 in Table 461, and the fixed-slice conditions in Definition 8.593.

Limits of quantitative comparison in higher dimensions

Statement S49-Q004 says that quantitative comparison at $d \geq 2$ is difficult because the entanglement entropy of the strongly coupled gauge theory used for comparison was then unknown. Its independent grounds are agreement with the known CFT formula at $d = 1$ and the comparisons between the strongly coupled gravitational calculation and free theory at $d \geq 2$. The source’s $\mathcal{N} = 4$ comparison gives agreement in functional form and an estimated finite-term coefficient ratio (free theory/gravitational calculation) of about $3/2$, assuming that the gauge field contributes as two real scalars. The corresponding formal evidence comprises Claims 32, 33 and 40 and Result 9 in Table 461.

Common audit with specified domain of application The evidential force of the four statements corresponds to the source’s static area prescription and its dimensional, state, geometric and comparative conditions. Fixed slices, Euclidean periods, temperature scales and the coordinate exchange supply time contents used by these calculations. Proposition 8.602 and Proposition 8.603 specify the reach relations of the nine time contents and the results.

8.19.50 Source audit

Audit Inputs

Source and Audit Scope

Source identifier 50.

Source name Francis–Barnes–James–Lewis 2007 Primitive-Time-Root Audit

Bibliographic identification Matthew J. Francis, Luke A. Barnes, J. Berian James and Geraint F. Lewis, *Expanding Space: the Root of all Evil?*, *Publications of the Astronomical Society of Australia* 24(2), 95–102 (2007). $R = \text{Ref}_{\text{FBJL07ES}}$.

Audit-scope identifier 50-SC1.

Audit-scope content Abstract, Sections 1–3, Equations (1)–(11) and references.

Source-location identifier 50-L1.

Source-location content Tables identify sections, equations and printed pages.

Assumptions Francis and colleagues note the absence of a single agreed meaning of expanding space and adopt the increase over time of separation between comoving observers as a working definition. They use the cosmological principle, FRW geometry, a

comoving congruence and a common zero for its clocks. Local bound systems, the empty Milne model, released tethered particles, supported bodies and free-particle sequences each have their own matter, initial and support conditions. Distance denotes proper distance on the specified cosmic-time slice; peculiar velocity is relative to the local comoving frame. Light propagation specifies emission and reception events and the endpoint observers.

Nontemporal Physical Inputs The carriers, spatial configurations, distance and density records and signal media, and origin and endpoint markers in Claims 1–14 form the generation-initial set.

Time Contents

Time-content identifier 50-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content specification The time contents of Definition 8.604 and their use roles in the claim table.

Source-location identifier set 50-L1.

Definition 8.604 (Francis–Barnes–James–Lewis 2007 Time Contents). Let $n_R = 23$. Each time content is scoped by the model, state and domain of the corresponding claims and results.

Table 464: Francis–Barnes–James–Lewis 2007 Time Contents

i	Time Content	Source Location
1	Cosmic time, reference times and their order	pp. 95–101
2	Scale factor, Hubble rate and temporal derivatives	pp. 96–101
3	Comoving congruence and cosmic-time slices	pp. 96–97, 101
4	Temporal change of cosmic-fluid density and pressure	pp. 96–98
5	Time dependence and evolution of electromagnetic fields	pp. 96
6	Proper time, readings and time zeros of comoving clocks	pp. 97
7	Synchronization relation of the comoving clock family	pp. 97
8	Temporal change of distance and volume on cosmic-time slices	pp. 97–98, 101
9	Coordinate trajectories and temporal evolution of radial particles	pp. 95, 98–100
10	Initial event and initial conditions of tether release	pp. 98–99
11	Peculiar velocity relative to the local comoving frame	pp. 98–100
12	Recession component and rate of change of proper distance	pp. 98–100
13	Particle-clock proper time and its relation to cosmic time	pp. 99

i	Time Content	Source Location
14	Emission–reception time correspondence along a null path	pp. 99–101
15	Correspondence between time representations under coordinate change	pp. 97, 99, 101
16	Center and endpoint trajectories and maintenance conditions of a supported object	pp. 97, 100
17	Departure times and departure interval of a particle sequence	pp. 100
18	Arrival times and arrival interval of a particle sequence	pp. 100
19	Emission–reception frequency comparison and redshift	pp. 101
20	Comparison order and frequency transfer along a comoving-observer sequence	pp. 101
21	Temporal evolution and stability of local geometry and bound systems	pp. 97–98, 101
22	Clock and velocity reference in a local inertial frame	pp. 97, 99–100
23	Length correspondence obtained from particle passage intervals	pp. 100–101

Definition 8.605 (Francis–Barnes–James–Lewis 2007 Formal Relations). In the physical equations, $R(t)$ denotes the scale factor. Proper distance on a cosmic-time slice and local peculiar velocity satisfy

$$r_p = R\chi, \quad \dot{r}_p = Hr_p + v_p, \quad v_p = R\dot{\chi}, \quad H = \dot{R}/R, \\ c^2 d\tau^2 = c^2 dt^2 - R^2 d\chi^2.$$

For an endpoint held at constant proper separation from its center,

$$\chi_f = \chi_c + \frac{L}{R}, \quad v_{p,f} = v_{p,c} - HL.$$

For a comoving center, the force in Equation (7) is

$$F = -mL \frac{\ddot{R}}{R} + O(L^3), \quad |HL|/c \ll 1.$$

For the free-particle sequence, on the positive radial branch,

$$v_p = \frac{1}{\sqrt{1 + CR^2}}, \quad \dot{\chi} = f(t, C) := \frac{1}{R(t)\sqrt{1 + CR(t)^2}}, \quad c = 1,$$

With a common integration constant for neighboring particles, to first order in their intervals,

$$f(t_0, C)\Delta t_0 = f(t_f, C)\Delta t_f, \quad l_0 = v_0\Delta t_0, \quad l_f = v_f\Delta t_f, \quad \frac{l_f}{l_0} = \frac{R_f}{R_0}.$$

For a family whose integration constant depends on launch time, with arrival time a function of launch time,

$$f(t_f, C) \frac{dt_f}{dt_0} - f(t_0, C) + \frac{dC}{dt_0} \int_{t_0}^{t_f} \frac{\partial f}{\partial C}(t, C) dt = 0.$$

Making the intervals small does not eliminate the last term. Redshift uses the comoving-endpoint relation

$$\frac{\nu_e}{\nu_r} = \frac{R_r}{R_e}.$$

Definition 8.606 (Francis–Barnes–James–Lewis 2007 Finite Cardinalities). Superscripts in the claim table denote the use roles below. Auxiliary objects are specified within claims; the separate auxiliary-object index set is empty.

Table 465: Francis–Barnes–James–Lewis 2007 Time-Use Roles

Code	Role	Use Count
I	Interval	14
Q	Quantity	31
X	Cross-position correspondence	18
K	Clock reading	20
C	Causality	0
S	State evolution	52
P	Probabilistic evolution	0
O	Observation record	3
B	Reference boundary	15

Table 466: Francis–Barnes–James–Lewis 2007 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 6, 13	member	instant, order_relation, interval_boundary
2–5, 8–9, 16, 21–22	member	instant, order_relation
7	member	instant, endpoint
10	member	instant, phase_boundary
11–12, 19	dependency	∅
14, 17–18	member	endpoint, order_relation, interval_boundary
15	member	instant, order_relation, endpoint
20	member	endpoint, order_relation
23	member	interval_boundary, order_relation

Table 467: Francis–Barnes–James–Lewis 2007 Finite Cardinalities

Object	Cardinality
Time contents	23
Claims	53
Auxiliary objects	0
Transfers	18
Results	24
Claim/time-content uses	119
Role assignments	153
Obligation atoms	118
Audit-boundary nodes	100
Audit-boundary edges	42

Definition 8.607 (Francis–Barnes–James–Lewis 2007 Physical and Formal Input Claims). Let the claim-node set be $\mathbf{ClaimNodes}_R(53)$, with claim boundary $\mathbf{ClaimBoundaryIdx}_R = ([14], [53])$. Claims 1–14 form the generation-initial set; all 53 claims form the source formal-evidence environment.

Table 468: Francis–Barnes–James–Lewis 2007 Claims and Temporal Uses

p	Claim	i	Source Location
1	Physical carriers of comoving observers.	$\emptyset$	§ 2.1, p. 97
2	Physical carriers of clock devices.	$\emptyset$	§ 2.1;§ 2.6.1, p. 97,99
3	Physical carriers of galaxies and test particles.	$\emptyset$	§ 2.5;§ 2.6.3, p. 98,99,101
4	Physical carriers of photons and electromagnetic signals.	$\emptyset$	§ 2.6.1;§ 2.6.4, p. 99,100,101
5	Physical carriers of the cosmic fluid.	$\emptyset$	§ 2.1, p. 96,97
6	Physical carriers of local bound systems.	$\emptyset$	§ 2.2;§ 2.6.2–2.6.3, p. 97,100, 101
7	Spatial configurations of objects.	$\emptyset$	§ 2.1–2.2;§ 2.5, p. 97,98
8	Physical interactions providing support forces.	$\emptyset$	§ 2.2;§ 2.6.2, p. 97,100
9	Physical interactions providing internal binding.	$\emptyset$	§ 2.2;§ 2.6.2, p. 97,100,101
10	Recorded spatial distances and lengths.	$\emptyset$	§ 2.1;§ 2.6.2, p. 97,100
11	Record media for emitted and received signals.	$\emptyset$	§ 2.6.1;§ 2.6.4, p. 99,100,101

p	Claim	i	Source Location
12	Local measurement apparatus.	$\emptyset$	§ 2;§ 2.6.1; § 2.6.4, p. 96,99, 101
13	Physical markers of origins, centers and endpoints.	$\emptyset$	§ 2.6.1–2.6.4, p. 99,100,101
14	Spatial distribution of matter density.	$\emptyset$	§ 2.1–2.3, p. 96,97,98
15	Francis and colleagues call the increase over time of the distance between comoving observers expanding space and present it as a pedagogical and conceptual aid to understanding the consequences of general relativity.	$1^S, 2^S, 8^{SQ}$	abstract;§ 1;§ 3, p. 95,96,101
16	The authors describe the dynamical relation between matter density and pressure and spacetime geometry using general relativity.	$1^S, 2^S, 4^S$	§ 2, p. 96
17	The authors compare the description of electromagnetic fields evolving according to Maxwell equations with the general-relativistic description using dynamical spacetime.	5^S	§ 2, p. 96
18	When the same physical events are reexpressed by a coordinate transformation, observational predictions agree if the metric and observers are represented under the same transformation.	$1^X, 15^X$	§ 2;§ 2.6.1;§ 3, p. 96,99,101
19	Homogeneity and isotropy under the cosmological principle specify FRW geometry and its description in terms of cosmic time, a scale factor and a comoving congruence.	$1^K, 2^S, 3^X$	§ 2.1, p. 96,97
20	Using the fluid continuity equation, the authors describe the generic change of cosmic-fluid density and pressure over cosmic time as a characteristic of expansion or contraction.	$1^S, 2^S, 4^S$	§ 2.1, p. 96
21	Proper time on a comoving worldline advances at the same rate as cosmic time, so each clock reading has the form $\tau = t + b$. The constant b specifies that clock's time zero.	$1^K, 3^X, 6^{KIB}$	§ 2.1, p. 97
22	Choosing a common time zero for the comoving clock family makes its readings agree on each cosmic-time slice.	$1^K, 3^X, 6^{KIB}, 7^{KX}$	§ 2.1, p. 97

<i>p</i>	Claim	<i>i</i>	Source Location
23	Between observers at fixed comoving coordinates, proper distance is proportional to $R(t)$, and the volume of a comoving cube is proportional to $R(t)^3$. Both increase on an interval where R increases.	$1^I, 2^{SQ}, 3^X, 8^{SQ}$	§ 2.1, p. 97
24	The authors describe the increase in proper distance between comoving observers by saying that space appears or wells up between them.	8^S	§ 2.1, p. 97
25	A Minkowski description in a local freely falling frame holds infinitesimally and approximates a finite small region to the extent that curvature effects can be neglected.	22^{KQ}	§ 2.1 and note2, p. 97
26	In the empty Milne model, FRW cosmic-time and Minkowski inertial-time representations can be transformed into one another within the coordinate domain inside the future light cone.	$1^{KX}, 15^{KX}$	§ 2.1 note2; § 2.6.1, p. 97,99
27	For a curved FRW spacetime containing matter, the domain of a local inertial frame differs from the Minkowski coordinate domain of the empty Milne model.	$2^S, 15^X, 22^K$	§ 2.1 note2; § 2.6.1, p. 97,99
28	In a bound room or galaxy, the inhomogeneous matter distribution, internal interactions and support conditions specify the local metric and the trajectories of its constituents.	$16^S, 21^S$	§ 2.2; § 2.6.2– 2.6.3, p. 97,100, 101
29	In the hypothetical room with homogeneous matter, whose wall particles experience no internal forces and are at rest relative to the cosmic fluid, the volume increases with the expansion.	$2^S, 3^X, 8^{SQ}, 9^S$	§ 2.2, p. 97
30	The authors relate the nonincrease of dark-energy density for $w \geq -1$ in an expanding universe to the stability of bound systems whose interiors are locally matter dominated.	$1^S, 2^S, 4^S, 21^S$	§ 2.3, p. 97,98
31	The balloon and raisin-bread analogies illustrate proportionality between velocity and distance in homogeneous expansion and the equivalent appearance from each point. The authors reject extending these analogies into a dynamical mechanism of friction or viscosity.	$2^S, 8^S, 12^Q$	§ 2.4, p. 98

p	Claim	i	Source Location
32	Setting the proper-distance rate to zero at tether release gives the initial local peculiar velocity $v_{p0} = -H_0 r_0$, cancelling the recession component.	$1^B, 2^Q, 10^B, 11^Q, 12^Q$	§ 2.5, p. 98,99
33	The particle trajectory after release follows its initial conditions and the evolution of the universe. Footnote 3 states that even in an accelerating universe a particle at a greater initial distance can initially approach the origin.	$2^S, 9^S, 10^B, 11^Q$	§ 2.5 note3, p. 98,99
34	The authors explain expansion through the past separation of galaxies, changing mean density and changing metric, treating expanding space as a description of that metric change.	$1^S, 2^S, 4^S, 8^S$	§ 2.5, p. 98
35	For $r_p = R\chi$, the proper-distance rate is $\dot{r}_p = \dot{R}\chi + R\dot{\chi} = Hr_p + v_p$, with $v_p = R\dot{\chi}$.	$1^K, 2^Q, 9^S, 11^Q, 12^Q$	§ 2.6.1(1), p. 99,100
36	The distant proper-distance rate on FRW slices and the speed measured in a local inertial frame use different pairs of time and distance variables. Even in the Milne model the former can exceed c , while the inertial speed of the same massive particle remains below c .	$12^Q, 15^X, 22^{KQ}$	§ 2.6.1, p. 99
37	The radial line element $c^2 d\tau^2 = c^2 dt^2 - R^2 d\chi^2$ gives $(d\tau/dt)^2 = 1 - (R\dot{\chi}/c)^2$. The clock of a timelike particle measures this proper time.	$1^K, 2^S, 9^S, 13^{KQ}$	§ 2.6.1(2)–(3), p. 99
38	The timelike branch of the line element has $ v_p < c$ and the null branch has $ v_p = c$. This bound applies to velocity relative to the local comoving frame.	$1^K, 11^Q, 13^Q, 14^S$	§ 2.6.1(2)–(3), p. 99
39	For light moving outward from the origin, $v_p = c$ and the distance rate on FRW slices is $\dot{r}_p = c + Hr_p$.	$1^K, 2^Q, 12^Q, 14^S$	§ 2.6.1, p. 100
40	Whether light reaches a distant observer is determined by integrating the null geodesic from the specified emission event.	$1^B, 2^S, 14^{SI}$	§ 2.6.1, p. 100
41	An endpoint maintained at proper separation L from the center trajectory $\chi_c(t)$ on cosmic-time slices satisfies $R\chi_f - R\chi_c = L$, hence $\chi_f = \chi_c + L/R$.	$1^K, 2^S, 9^S, 16^S$	§ 2.6.2(4)–(5), p. 100

p	Claim	i	Source Location
42	The endpoint four-force follows from the acceleration in Equation (6). Under the source condition $U = -HL$ the center is comoving; for small $ HL /c$, the radial force in a local orthonormal basis is $F = -mL\ddot{R}/R$ to first order in L .	$1^K, 2^{SQ}, 16^S$	§ 2.6.2(6)–(7), p. 100
43	In an object with sufficient internal interactions, those forces support the endpoint trajectories and maintain its dimensions.	16^S	§ 2.6.2, p. 100,101
44	The sequence without internal forces launches its first particle at t_0 and its last at $t_0 + \Delta t_0$, with arrivals at the comoving receiver specified by t_f and $t_f + \Delta t_f$.	$9^S, 17^{BKI}, 18^{BKI}$	§ 2.6.2, p. 100
45	On the positive radial geodesic branch with $c = 1$, $v_p = 1/\sqrt{1 + CR^2}$ and $\dot{\chi} = 1/[R\sqrt{1 + CR^2}]$. The constant C is fixed by each particle's initial conditions.	$2^S, 9^{SI}, 11^Q$	§ 2.6.2(8), p. 100
46	When neighboring particles have the same constant C , first-order expansion of their propagation integrals to the same comoving receiver gives $\frac{\Delta t_0}{R_0\sqrt{1+CR_0^2}} = \frac{\Delta t_f}{R_f\sqrt{1+CR_f^2}}$.	$2^S, 17^{IB}, 18^{IB}$	§ 2.6.2(8)–(10), p. 100
47	At the launch and receiving locations, the particle-sequence lengths are obtained as $l_0 = v_0\Delta t_0$ and $l_f = v_f\Delta t_f$, respectively.	$11^Q, 17^I, 18^I, 23^{QX}$	§ 2.6.2, p. 100
48	With common C and passage intervals treated to first order, the particle-sequence length ratio is $l_f/l_0 = R_f/R_0$.	$2^Q, 17^{IB}, 18^{IB}, 23^{QX}$	§ 2.6.2(11), p. 100,101
49	On galactic scales, concentrated matter specifies the local metric and the applicability conditions of a purely homogeneous and isotropic FRW metric are not met.	21^S	§ 2.6.3, p. 101
50	The authors reject a picture in which gravity already represented by the local metric and global FRW expansion are two independent forces competing over a galaxy.	$2^S, 21^S$	§ 2.6.3, p. 101
51	Frequencies compared in the local orthonormal frames of comoving emitters and receivers satisfy $\nu_e/\nu_r = R_r/R_e$ through the change in scale factor along null propagation.	$2^Q, 14^{BX}, 19^{QO}$	§ 2.6.4, p. 101
52	The authors explain cosmological redshift through the frequency comparison when a photon is received in an observer frame different from its emission frame.	$14^{BX}, 19^{QO}$	§ 2.6.4, p. 101

p	Claim	i	Source Location
53	A sequence of comoving observers along the light path describes the wavelength as changing with the scale factor. Composing the frequency ratios along this sequence gives the ratio between the emission and reception endpoints.	$2^Q, 14^{BX}, 19^{QO}, 20^{XI}$	§ 2.6.4, p. 101

Table 469: Francis–Barnes–James–Lewis 2007 Nontemporal Physical Inputs

p	Component
1	Physical carriers of comoving observers.
2	Physical carriers of clock devices.
3	Physical carriers of galaxies and test particles.
4	Physical carriers of photons and electromagnetic signals.
5	Physical carriers of the cosmic fluid.
6	Physical carriers of local bound systems.
7	Spatial configurations of objects.
8	Physical interactions providing support forces.
9	Physical interactions providing internal binding.
10	Recorded spatial distances and lengths.
11	Record media for emitted and received signals.
12	Local measurement apparatus.
13	Physical markers of origins, centers and endpoints.
14	Spatial distribution of matter density.

Dependency Inputs

Definition 8.608 (Francis–Barnes–James–Lewis 2007 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [18]$. The table gives time inputs, time outputs, claim support and source locations.

Table 470: Francis–Barnes–James–Lewis 2007 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1, 3	6	21	Comoving proper-time correspondence. § 2.1
2	1, 3, 6	7	22	Comoving-clock synchronization with a common zero. § 2.1
3	1–3	8	23	Temporal change of proper separation and volume for comoving observers. § 2.1

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
4	1–2, 10, 12	11	32	Cancellation by the tethered initial velocity. § 2.5
5	2, 10–11	9	33	Free-particle trajectory from initial conditions. § 2.5 note3
6	1–2, 9	11–12	35	Distance rate and peculiar velocity from the coordinate trajectory. § 2.6.1(1)
7	1–2, 9	13	37	Proper-time correspondence from the radial line element. § 2.6.1(2)–(3)
8	1–2, 14	12	39	Proper-distance rate of outward light. § 2.6.1
9	1–2	14	40	Endpoint-time correspondence of null propagation. § 2.6.1
10	1–2, 9	16	41	Supported endpoint trajectory from the center trajectory. § 2.6.2(4)–(5)
11	2	9, 11	45	Geodesic velocity along the scale factor. § 2.6.2(8)
12	2, 17	18	46	Departure–arrival interval correspondence for common C. § 2.6.2(8)–(10)
13	11, 17–18	23	47	Length correspondence from passage intervals and speeds. § 2.6.2
14	2, 17–18	23	48	Scale-factor length ratio for common C. § 2.6.2(11)
15	2, 14	19	51	Redshift between comoving endpoints. § 2.6.4
16	2, 14, 20	19	53	Composition of frequency ratios along an observer sequence. § 2.6.4
17	1	15	26	Milne cosmic time to inertial-time representation. § 2.1 note2;§ 2.6.1
18	15	1	26	Inertial-time representation to Milne cosmic time. § 2.1 note2;§ 2.6.1

Grounding Conditions

Table 471: Francis–Barnes–James–Lewis 2007 Conditions for Transfers

Trans-fer Index	Conditions
1	Comoving worldline and clock zero.
2	Specified congruence and common clock zero.
3	Fixed comoving configuration and FRW slicing.
4	Initially zero proper-distance rate and a timelike particle.
5	Radial geodesic, initial distance and velocity, and specified cosmology.
6	$r_p = R\chi$, with the same cosmic time and distance convention.

Trans-fer Index	Conditions
7	Timelike worldline, proper-time orientation and zero.
8	Outward null branch with $v_p = c$.
9	Integrate the null geodesic for the specified emission event and observer configuration.
10	Constant L and $\chi_f = \chi_c + L/R$.
11	Specified $C > 0$, $c = 1$ and positive radial branch.
12	Same comoving receiver, common C for neighboring particles, intervals to first order.
13	At each measurement location define length as local speed times passage interval.
14	Common C , the same comoving endpoints and first-order interval approximation.
15	Comoving emitter and receiver, local orthonormal frames and null propagation.
16	Ordered comoving observers along the same light path.
17	Empty Milne model in the specified future-light-cone coordinate domain.
18	Same Milne domain and inverse coordinate transformation.

Results

Definition 8.609 (Francis–Barnes–James–Lewis 2007 Results). Let $\text{ld}\mathbf{x}_R^{\text{audit,result}} = [24]$.

Table 472: Francis–Barnes–James–Lewis 2007 Results

q	Result	Source Location
1	Working definition and pedagogical role of expanding space	abstract;§ 1;§ 3, p. 95,96,101
2	Comparison of dynamical-spacetime and electromagnetic-field descriptions	§ 2, p. 96
3	Observational predictions independent of coordinate representation	§ 2;§ 2.6.1;§ 3, p. 96,99,101
4	FRW geometry and cosmic-fluid evolution	§ 2.1, p. 96,97
5	Comoving proper time and clock synchronization	§ 2.1, p. 97
6	Increase in proper separation and volume for comoving observers	§ 2.1, p. 97
7	Domain of the local Minkowski description	§ 2.1 and note2, p. 97
8	Coordinate domains of Milne and general FRW models	§ 2.1 note2;§ 2.6.1, p. 97,99
9	Bound room and hypothetical comoving walls	§ 2.2;§ 2.6.2–2.6.3; § 2.2, p. 97,100,101
10	Dark energy and local binding	§ 2.3, p. 97,98
11	Content of the homogeneous-expansion analogies	§ 2.4, p. 98
12	Velocity condition at tether release	§ 2.5, p. 98,99

q	Result	Source Location
13	Post-release trajectory and initial distance	§ 2.5 note3, p. 98,99
14	Account of galaxy separation and metric change	§ 2.5, p. 98
15	Decomposition into recession and local peculiar velocity	§ 2.6.1(1), p. 99,100
16	Distinction between distant distance rate and local speed	§ 2.6.1, p. 99
17	Proper-time relation and local speed bound	§ 2.6.1(2)–(3), p. 99
18	Light distance rate and reachability condition	§ 2.6.1, p. 100
19	Supported object of fixed length	§ 2.6.2(4)–(5); § 2.6.2(6)–(7); § 2.6.2, p. 100,101
20	Times and geodesics of the free-particle sequence	§ 2.6.2; § 2.6.2(8), p. 100
21	Departure and arrival intervals for common C	§ 2.6.2(8)–(10), p. 100
22	Passage-interval lengths and scale-factor ratio	§ 2.6.2; § 2.6.2(11), p. 100,101
23	Metric and expansion account on galactic scales	§ 2.6.3, p. 101
24	Redshift endpoint relation and observer sequence	§ 2.6.4, p. 101

Table 473: Francis–Barnes–James–Lewis 2007 Conditions
for Results

Result Index	Conditions and Domain
1	GR description of changing comoving-observer separation.
2	Field evolution and theory-specific laws in the authors' comparison.
3	Consistent coordinate transformation of the same events, metric and observers.
4	Cosmological principle and specified cosmic fluid.
5	Comoving congruence and common clock zero.
6	Fixed comoving configuration and increasing scale factor.
7	Specified local freely falling neighborhood.
8	Distinction between the empty Milne domain and curved FRW models.
9	Respective local matter/support conditions and homogeneous free-particle configuration.
10	Expanding branch, $w \geq -1$ and locally matter-dominated bound system.
11	Authors' account of the velocity–distance relation of homogeneous expansion.
12	Initially zero proper-distance rate and $ v_{p0} < c$.
13	Specified cosmology, timelike initial data and the distance conditions of footnote3.
14	Authors' general-relativistic explanatory context.
15	Same cosmic-time slicing with $r_p = R\chi$.
16	Specified FRW/Milne coordinates and local inertial frame.
17	Radial FRW line element, timelike and null branches.

Result Index	Conditions and Domain
18	Emission event, scale factor and null path.
19	Constant proper length and supported trajectory; Equation 7 uses $U = -HL$ and small $ HL /c$.
20	No internal forces, with specified initial data for each particle.
21	Common C, fixed comoving receiver and intervals to first order.
22	Length from local speed and passage interval; the scale-factor ratio uses common C.
23	Local metric of an inhomogeneous galaxy and the domain of global FRW.
24	Comoving emitter and receiver, a null path and an observer sequence along the same path.

Result Dependence

Definition 8.610 (Francis–Barnes–James–Lewis 2007 Result-Dependence Hyperedges). For each row, let $\text{ResultEdge}_{R,q}$ ($q \in [24]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 474: Francis–Barnes–James–Lewis 2007 Result-Dependence Hyperedges

q	Time Premises i	Claim Support p
1	1–2, 8	15
2	1–2, 4–5	16–17
3	1, 15	18
4	1–4	19–20
5	1, 3, 6–7	21–22
6	1–3, 8	23–24
7	22	25
8	1–2, 15, 22	26–27
9	2–3, 8–9, 16, 21	28–29
10	1–2, 4, 21	30
11	2, 8, 12	31
12	1–2, 10–12	32
13	2, 9–11	33
14	1–2, 4, 8	34
15	1–2, 9, 11–12	35
16	12, 15, 22	36
17	1–2, 9, 11, 13–14	37–38
18	1–2, 12, 14	39–40
19	1–2, 9, 16	41–43
20	2, 9, 11, 17–18	44–45
21	2, 17–18	46
22	2, 11, 17–18, 23	47–48
23	2, 21	49–50

q	Time Premises i	Claim Support p
24	2, 14, 19–20	51–53

$$\text{BoundaryCount}_R^{\text{audit}} = (100, 42).$$

The 53 claims, 23 time contents and 24 results give 100 nodes; the 18 transfers and 24 result dependencies give 42 edges.

Evidence Hierarchy Francis and colleagues discuss the general-relativistic account of expanding space and theoretical calculations involving comoving clocks, tethered particles, supported bodies, free-particle sequences and redshift. Their arguments, including cited earlier work, use the applicability conditions of each model. This source reports zero newly acquired direct measurements.

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 0; remaining primitive-time roots 23. Every transfer has temporal input from cosmic time, proper time, synchronization, velocity references or propagation events.

Result Dependence Of the 24 results, 24 depend on primitive-time roots and 0 are reached from source formal evidence and local constructions. There are 89 root–result dependence pairs, of which 41 are necessary.

Placement of the Grounding Obligation Cosmic time and proper time support geodesics, initial conditions, velocities and distance rates; clock readings and a common zero support comoving-clock synchronization. Emission and reception times, light-propagation intervals and velocity references enter redshift relations. Complete generation from nontemporal physical inputs is zero; the placement of the physical-time grounding obligation is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Initial rest at tether release requires a subluminal condition; the scale-factor length ratio requires a common integration constant and first-order interval approximation. Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 89 root–result dependence pairs. Actual reach from failure of a specific physical-time generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns results dependent on it.

Audit Conclusions

Proposition 8.611 (Primitive Time Roots and Result Dependence in Francis–Barnes–James–Lewis 2007). In the component order of Definition 8.115,

$$\text{AuditPartId}_{\times_R} = \text{AllRootDepIdx}(23, 24), \quad R \in \text{Ref}^{\text{audit,rootdep}}, \quad R \notin \text{Ref}^{\text{audit,rootunreach}}.$$

Proof. Closure from Initial Claims 1–14 remains those fourteen nodes. Even with all 53 claims, no time content is newly reached because each transfer requires temporal input. The mutually inverse Milne/inertial-time conversions likewise require one of those temporal inputs. Adding all time contents as seeds reaches all 24 results. $\square$

Proposition 8.612 (Minimal Primitive-Time-Root Support in Francis–Barnes–James–Lewis 2007).

The table gives each result’s minimal-support index family, with 53 family members in total.

Proof. Apply Definition 7.46 to all transfers and result dependencies, starting from singleton time-content seeds and empty supports for source formal evidence. By Proposition 7.47, the inclusion-minimal supports of the stable antichains give the displayed families. $\square$

Table 475: Minimal Primitive-Time-Root Support in Francis–Barnes–James–Lewis 2007

Result Indices	Minimal-Support Family Members
1	$\{1, 2, 3\}, \{1, 2, 8\}, \{2, 3, 15\}, \{2, 8, 15\}$
2	$\{1, 2, 4, 5\}, \{2, 4, 5, 15\}$
3	$\{1\}, \{15\}$
4	$\{1, 2, 3, 4\}, \{2, 3, 4, 15\}$
5	$\{1, 3\}, \{3, 15\}$
6	$\{1, 2, 3\}, \{2, 3, 15\}$
7	$\{22\}$
8	$\{1, 2, 22\}, \{2, 15, 22\}$
9	$\{1, 2, 3, 21\}, \{2, 3, 15, 21\}, \{2, 3, 8, 16, 21\}$
10	$\{1, 2, 4, 21\}, \{2, 4, 15, 21\}$
11	$\{1, 2, 3\}, \{1, 2, 8\}, \{2, 8, 12\}, \{2, 3, 15\}, \{2, 8, 15\}$
12	$\{1, 2, 10\}, \{2, 10, 15\}$
13	$\{2, 10\}$
14	$\{1, 2, 3, 4\}, \{1, 2, 4, 8\}, \{2, 3, 4, 15\}, \{2, 4, 8, 15\}$
15, 17–19	$\{1, 2\}, \{2, 15\}$
16	$\{1, 2, 22\}, \{1, 12, 22\}, \{2, 15, 22\}, \{12, 15, 22\}$
20–22	$\{2, 17\}$
23	$\{2, 21\}$
24	$\{1, 2, 20\}, \{2, 14, 20\}, \{2, 15, 20\}$

Result-Preservation Domain Fix the models, initial and endpoint conditions, common clock zero, distance and velocity conventions, and approximation domains in Table 473. When temporal inputs are realized by generated representations preserving the same times, trajectories, synchronization, propagation and passage intervals, and frequency comparisons, time-dependent results are preserved within their domains.

Proposition 8.613 (Complete Audit of Time Use in Francis–Barnes–James–Lewis 2007). There are 23 time contents, 53 claims, 18 transfers and 24 results. The 119 claim/time-content use pairs have 153 role assignments, with 118 source-specific obligation atoms. Complete generation is zero, local reach is 0, remaining primitive-time roots number 23, and 24 results depend on roots.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual statements

Statement objects Synchronization of the clocks of comoving observers, the expanding-space explanation, the departure-time interval of a sequence of freely falling particles, and the local application of expansion on galactic scales.

Original text (English) the watches of all privileged observers are synchronised

more space appeared, or ‘welled up’ between the two observers

the first particle leaving at time t_0 and the last at $t_0 + \Delta t_0$

There is no expansion for the galaxy to overcome

Original-source location The first and second quotations are in Section 2.1, p. 97; the third is in Section 2.6.2, p. 100; the fourth is in Section 2.6.3, p. 101.

Minimal context Section 2.1 introduces the comoving congruence and cosmic time in the FRW model, states the synchronization of comoving clocks, and explains increasing proper separation between observers in terms of expanding space. Section 2.6.2 calculates a length ratio from departure and arrival intervals of a free-particle sequence without internal forces. Section 2.6.3 states that the applicability conditions of the purely homogeneous and isotropic FRW metric are not met on galactic scales and rejects a picture of gravity and expansion as opposing forces.

Attribution The four statements are attributed to the main text by Matthew J. Francis, Luke A. Barnes, J. Berian James, and Geraint F. Lewis.

What the quotations state directly The first quotation states that the clocks of all privileged observers are synchronized. The second explains that space appears or wells up between observers. The third assigns different departure times to the first and last particles. The fourth denies expansion as something for a galaxy to overcome.

Explicit applicability domain The first and second quotations concern the comoving-observer congruence of the FRW model, the third concerns the specified finite sequence of particles, and the fourth concerns the local metric on galactic scales.

Individual Audit Comoving-Clock Synchronization

The clocks synchronized by Francis and colleagues in the first quotation belong to the specified comoving congruence of the FRW model. Along a comoving worldline, $d\tau/dt = 1$ gives $\tau = t + b$. Selecting a common zero for the clock family makes the readings agree on the same cosmic-time slice. The synchronization relation therefore depends on the comoving congruence, clock readings and zero. It corresponds to Time Contents 1,3,6,7, Claims 21–22, Transfers 1–2 and Result 5.

The Explanation That Space Appears

The object explained by the authors in the second quotation is the proper separation of observers with fixed comoving coordinates. An increasing scale factor increases the proper distance $r_p = R\chi$ and the proper volume of a comoving region. The authors describe this metric relation by saying that space appears. It corresponds to Time Contents 1–3,8, Claims 15,23–24, Transfer 3 and Results 1,6.

Departure Interval of a Free-Particle Sequence

The third quotation specifies the departure times of the first and last particles as t_0 and $t_0 + \Delta t_0$. Equations (8)–(11) use a common integration constant C to compare departure and arrival intervals, obtaining $l_f/l_0 = R_f/R_0$ to first order in the intervals. For a particle family whose C varies with launch time, differentiating the propagation integral adds a term involving dC/dt_0 ; the stated length ratio therefore requires the common- C condition. Departure and arrival times and the passage intervals are temporal inputs to the calculation. This corresponds to Time Contents 2,9,11,17–18,23, Claims 44–48, Transfers 11–14 and Results 20–22.

Local Metric on Galactic Scales

In the fourth quotation, the authors reject applying the pure FRW metric to a galaxy with concentrated matter and treating expansion and gravity as independent competing forces. Their domain argument assigns the motion to the local metric and binding conditions and distinguishes supported objects from free-particle sequences without internal forces. Temporal evolution of the local metric and bound state remains an input to this account. It corresponds to Time Contents 2,16,21, Claims 28,43,49–50 and Results 9,19,23.

8.19.51 Source audit

Audit Inputs

Source and Audit Scope

Source identifier 51.

Source name Cook–Burns 2009 Primitive-Time-Root Audit

Bibliographic identification Richard J. Cook and M. Shane Burns, *Interpretation of the cosmological metric*, *American Journal of Physics* 77, 59–66 (2009), doi:10.1119/1.2987790; arXiv:0803.2701v2, pp. 1–20. $R = \text{Ref}_{\text{CookBurns09Metric}}$.

Audit-scope identifier 51-SC1.

Audit-scope content Abstract, Sections I–V, Equations (1)–(31), Table I, references and notes.

Source-location identifier 51-L1.

Source-location content Tables identify sections, equations and pages of arXiv:0803.2701v2.

Assumptions The Robertson–Walker branch uses homogeneity, isotropy, a comoving congruence and a specified scale-factor history. Friedmann evolution uses specified matter and curvature conditions. Photon calculations specify an inward null path and its endpoints. The empty Milne branch uses $\rho = 0$, $\Lambda = 0$, $k = -1$, $a(t) = ct$ and $t > 0$; comparison with the rigid frame relates the same physical events and observers.

Nontemporal Physical Inputs The carriers, spatial configurations, distance records, record media and marks, and connections of light sources and receivers in Claims 1–12 form the generation-initial set. Temporal and frequency interpretations of records and the ordering of null propagation belong to the corresponding time contents and claims.

Time Contents

Time-content identifier 51-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content specification The time contents of Definition 8.614 and their use roles in the claim table.

Source-location identifier set 51-L1.

Definition 8.614 (Cook–Burns 2009 Time Contents). Let $n_R = 28$. Each time content is scoped by the model, state and domain of the corresponding claims and results.

Table 476: Cook–Burns 2009 Time Contents

i	Time Content	Source Location
1	Cosmic-time coordinate and reference times	pp. 3–4, 6, 8
2	Scale-factor history and Hubble rate	pp. 3–4, 6, 8–9
3	Comoving congruence and clock field	pp. 3, 5, 9
4	Proper-time readings of comoving clocks	pp. 3, 9, 14
5	Temporal evolution under the Friedmann equations	pp. 4, 8
6	Simultaneous slices defined by cosmic time	pp. 4, 7, 14, 16
7	Proper distance of comoving separations and its history	pp. 4–5, 9, 15
8	Proper-distance rate and Hubble recession velocity	pp. 4–5, 7, 9–10, 17
9	Hubble-sphere radius and its change over time	pp. 4, 6
10	Photon history from emission to reception	pp. 5–6, 11
11	Photon emission time	pp. 6, 11
12	Scale-factor ratio at emission and reception	pp. 6, 11
13	Cosmological redshift between comoving endpoints	pp. 6–7, 11
14	Curvature of a selected spatial slice and its time dependence	pp. 7, 12, 16
15	Relation between the present Hubble rate and model age	pp. 8
16	Coordinate relation between Milne cosmic time and rigid-frame time	pp. 8–9, 14–15
17	Proper time and synchronization of rigid-frame clocks	pp. 9, 14–15

i	Time Content	Source Location
18	Local-frame velocity comparison and composed rapidity	pp. 9–10
19	Rigid-frame Doppler redshift and endpoint comparison	pp. 11
20	Gravitational time-dilation component in curved cosmological models	pp. 13
21	Comparison of two clocks at the same event	pp. 13–15
22	Local relative velocity between the two observer congruences	pp. 14–15
23	Clock time dilation due to relative motion	pp. 14–15
24	Relation between the simultaneity conventions of the two frames	pp. 14–16
25	Time- and space-interval relation under a local Lorentz transformation	pp. 14–15
26	Predictions involving time for specified clock fields and foliations	pp. 13–17
27	Photon reception time	pp. 6, 11
28	Photon turnaround and Hubble-sphere crossing time	pp. 6

Definition 8.615 (Cook–Burns 2009 Formal Relations). Future-directed proper time on a comoving worldline and distance on a constant-cosmic-time slice satisfy

$$d\tau = dt, \quad \tau = t + b, \quad \ell = a(t)\chi, \quad H = \frac{\dot{a}}{a}, \quad v = H\ell, \quad \ell_H = \frac{c}{H} \quad (H > 0).$$

For inward radial light,

$$\dot{\ell}_\gamma = H\ell_\gamma - c.$$

In the linear model $a(t) = \alpha t$, with $\alpha > 0$ and $t_e > 0$,

$$\ell_\gamma(t) = \ell_G \frac{t}{t_e} - ct \log \frac{t}{t_e}, \quad t_r = t_e e^{\ell_G/(ct_e)}, \quad t_* = t_e e^{\ell_G/(ct_e)-1}.$$

Turnaround after emission occurs when $\ell_G > ct_e$. The redshift between comoving endpoints and curvature of a fixed-time slice are

$$z = \frac{a(t_r)}{a(t_e)} - 1, \quad {}^{(3)}\mathcal{R} = \frac{6k}{a(t)^2}.$$

The empty Milne branch $a = ct$, $t > 0$, and the rigid frame are related by

$$\bar{t} = t \cosh \chi, \quad \bar{r} = ct \sinh \chi, \quad t = \sqrt{\bar{t}^2 - \bar{r}^2/c^2}, \quad ct > |\bar{r}|, \quad \bar{t} > 0.$$

In this model,

$$\eta = \frac{\dot{a}}{c} \int_0^\chi d\chi' = \chi, \quad \bar{v} = c \tanh \chi, \quad z = \frac{t_r}{t_e} - 1 = \sqrt{\frac{c + \bar{v}}{c - \bar{v}}} - 1 = e^\chi - 1.$$

The argument of $\tanh$ on page 10 is printed as $a(t) \int d\chi$. The dimensionless argument consistent with the preceding $dv = \dot{a} d\chi$ is $(\dot{a}/c) \int d\chi$. For local intervals $dx^0 = c dt$ and $d\ell_\chi = ct d\chi$, the forward differential of equations (19)–(20) and its inverse are

$$\begin{pmatrix} d\bar{x}^0 \\ d\bar{\ell}_r \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} dx^0 \\ d\ell_\chi \end{pmatrix}, \quad \begin{pmatrix} dx^0 \\ d\ell_\chi \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} d\bar{x}^0 \\ d\bar{\ell}_r \end{pmatrix},$$

$$\beta = \bar{v}/c, \quad \gamma = (1 - \beta^2)^{-1/2}.$$

The minus signs in source equations (29a)–(29b) correspond to this inverse direction for the same positive relative velocity.

Definition 8.616 (Cook–Burns 2009 Finite Cardinalities). Superscripts in the claim table denote the use roles below. Auxiliary objects are specified within claims; the separate auxiliary-object index set is empty.

Table 477: Cook–Burns 2009 Time-Use Roles

Code	Role	Use Count
I	Interval	10
Q	Quantity	42
X	Cross-position correspondence	26
K	Clock reading	12
C	Causality	2
S	State evolution	17
P	Probabilistic evolution	0
O	Observation record	8
B	Reference boundary	18

Table 478: Cook–Burns 2009 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 4, 17	member	instant, order_relation, interval_boundary
2–3, 5, 7, 9, 14, 26	member	instant, order_relation
6, 24	member	instant, endpoint
8, 12–13, 18–19, 22	dependency	$\emptyset$
10	member	endpoint, order_relation
11, 27–28	member	instant, phase_boundary
15, 20, 23	member	instant, interval_boundary
16	member	instant, order_relation, endpoint
21	member	instant, endpoint, interval_boundary
25	member	interval_boundary, endpoint

Table 479: Cook–Burns 2009 Finite Cardinalities

Object	Cardinality
Time contents	28
Claims	46
Auxiliary objects	0
Transfers	23

Object	Cardinality
Results	22
Claim/time-content uses	110
Role assignments	135
Obligation atoms	119
Audit-boundary nodes	96
Audit-boundary edges	45

Definition 8.617 (Cook–Burns 2009 Physical and Formal Input Claims). Let the claim-node set be $\text{ClaimNodes}_R(46)$, with claim boundary $\text{ClaimBoundaryIdx}_R = ([12], [46])$. Claims 1–12 form the generation-initial set; all 46 claims form the source formal-evidence environment.

Table 480: Cook–Burns 2009 Claims and Temporal Uses

p	Claim	i	Source Location
1	Physical carriers of galaxies.	$\emptyset$	§II–V, p. 3,6,9
2	Physical carriers of clock devices.	$\emptyset$	§II–V, p. 3,9,14
3	Physical carriers of observers.	$\emptyset$	§II–V, p. 5,9,14
4	Physical carriers of photons and electromagnetic signals.	$\emptyset$	§II–V, p. 5,6,11
5	Physical carriers of rods and length standards.	$\emptyset$	§II–V, p. 14,15
6	Media and marks of emission and reception records.	$\emptyset$	§II–V, p. 6,11
7	Media and marks of wavelength and frequency records.	$\emptyset$	§II–V, p. 6,11
8	Spatial configurations of objects, observers and endpoints.	$\emptyset$	§II–V, p. 3,5,9,14
9	Physical carriers of cosmic fluid and matter distributions.	$\emptyset$	§II–V, p. 4,8,13
10	Local comparison and measurement apparatus.	$\emptyset$	§II–V, p. 9,14,15
11	Recorded spatial distances and lengths.	$\emptyset$	§II–V, p. 4,7,14,15,16
12	Physical carriers and connections of light sources, signals and receivers.	$\emptyset$	§II–V, p. 5,6,11
13	The homogeneous and isotropic Robertson–Walker metric is expressed using cosmic time, a scale factor and comoving spatial coordinates.	$1^K, 2^S, 3^X, 6^B$	§II(1)–(3), p. 3,4
14	On a future-directed comoving worldline, $d\tau = dt$; specifying the clock zero b gives $\tau = t + b$.	$1^K, 3^X, 4^{KI}$	§II(1), p. 3
15	The Friedmann equations with specified matter and curvature conditions determine the temporal evolution of the scale factor.	$1^S, 2^S, 5^S$	§II;§IV(15), p. 4,8

p	Claim	i	Source Location
16	On a slice of constant cosmic time, the proper distance across a comoving coordinate separation χ is $\ell = a(t)\chi$.	$2^Q, 6^B, 7^{QX}$	§III.A(4), p. 4
17	Differentiating the proper distance between comoving endpoints with respect to cosmic time gives $v = H\ell$; the Hubble distance on an expanding branch is $\ell_H = c/H$.	$1^I, 2^S, 3^X, 6^B, 7^Q, 8^Q, 9^Q$	§III.A(4)–(6), p. 4
18	The proper-distance rate between distant endpoints and relative velocity compared in one local inertial frame are different quantities; the former can exceed c consistently with the local light-speed bound.	$8^Q, 18^{QX}$	§III.B; §IV.A; §V, p. 5,9,10,17
19	The history of inward radial light is obtained by solving $\dot{\ell}_\gamma = H\ell_\gamma - c$ with specified emission time and initial position.	$1^S, 2^S, 10^{SC}, 11^B$	§III.C(7)–(9), p. 5,6
20	For the linear scale factor $a(t) = \alpha t$ with $t > 0$, $H = 1/t$ and $\ell_H = ct$.	$1^K, 2^S, 9^Q$	§III.C(9)–(10), p. 6
21	In the linear model, the photon history gives reception time $t_r = t_e \exp[\ell_G/(ct_e)]$ and turnaround time $t_* = t_e \exp[\ell_G/(ct_e) - 1]$.	$1^S, 2^S, 10^{SC}, 11^B, 27^B, 28^B$	§III.C(10), p. 6
22	The ratio $a(t_r)/a(t_e)$ is evaluated at the endpoints of the null path connecting the comoving emitter and receiver.	$2^Q, 11^B, 27^B, 12^Q$	§III.D(11), p. 6
23	The wavelength redshift compared at the comoving endpoints satisfies $z = a(t_r)/a(t_e) - 1$.	$12^Q, 13^{QO}$	§III.D(11), p. 6,7
24	Equation (12) compares a substitution of the Hubble recession velocity into the special-relativistic Doppler formula; it generally gives a value different from equation (11).	$8^Q, 13^Q, 19^Q$	§III.D(11)–(12), p. 6,7
25	The scalar curvature of a Robertson–Walker slice of constant cosmic time is ${}^{(3)}\mathcal{R} = 6k/a(t)^2$.	$2^Q, 6^B, 14^Q$	§III.F(13)–(14), p. 7
26	Choosing the expanding branch and time origin in the empty negative-curvature model gives $a(t) = ct$.	$1^B, 2^S, 5^S$	§IV(15)–(16), p. 8
27	Matching the present Hubble rate H_0 to $H(t) = 1/t$ in the linear model gives model age $t_0 = 1/H_0$.	$1^K, 2^Q, 15^{QI}$	§IV, p. 8
28	The $t > 0$ domain of the empty Milne model maps to a Minkowski representation inside the future light cone by $\bar{t} = t \cosh \chi$ and $\bar{r} = ct \sinh \chi$.	$1^K, 2^S, 16^X$	§IV(17)–(21), p. 8,9
29	Synchronizing clocks at rest at fixed spatial coordinates in the rigid frame permits their proper-time readings to be represented by $\bar{t}$.	$16^X, 17^{KIB}$	§IV(21), p. 9

p	Claim	i	Source Location
30	Composing collinear Lorentz transformations of comoving local frames along a spatial chain at fixed t gives $\eta = \chi$ and $\bar{v} = c \tanh \chi$ in the Milne model.	$2^Q, 6^B, 18^{QX}$	§IV.A(22), p. 9,10
31	The Doppler redshift of light from an emitter with velocity $\bar{v}$ in the rigid frame is $z = \sqrt{(c + \bar{v})/(c - \bar{v})} - 1$.	$18^Q, 19^{QO}$	§IV.D(25), p. 11
32	For the same emitter, receiver and light signal, the Milne cosmological redshift and rigid-frame Doppler redshift both equal $z = \exp \chi - 1$.	$13^{QO}, 19^{QO}$	§IV.D(23)–(25), p. 11
33	The same redshift is represented by the wavelength and scale-factor ratios in the comoving description and by the Doppler effect in the rigid description.	$2^Q, 11^B, 27^B,$ $12^Q, 13^{QO}, 19^{QO}$	§III.D;§IV.D, p. 6,7,11
34	The constant- t and constant- $\bar{t}$ foliations select different spatial slices. Negative curvature of Milne slices and flatness of rigid-frame slices coexist in the same four-dimensional spacetime.	$6^B, 14^Q, 16^X,$ $24^X, 26^X$	§V(30)–(31), p. 15,16
35	A comoving observer and a rigid-frame observer meeting at the same event compare standard clocks at rest in their respective local frames.	$4^K, 17^K, 21^{KX}$	§V, p. 13,14
36	At a meeting event specified in Milne coordinates, the local relative velocity between the congruences is $\bar{v} = c \tanh \chi$.	$16^X, 22^Q$	§V;§IV.A(22), p. 14,15
37	Differentiating coordinate equations (19)–(20) gives $d\bar{x}^0 = \gamma(dx^0 + \beta d\ell_\chi)$ and $d\bar{\ell}_r = \gamma(d\ell_\chi + \beta dx^0)$, where $dx^0 = c dt$ and $d\ell_\chi = ct d\chi$.	$16^X, 22^Q, 25^{IX}$	§IV(19)–(20); §V(27)–(29), p. 9,14,15
38	Each local observer obtains kinematic time dilation for the other observer's moving standard clock using the relative velocity and Lorentz relation.	$21^{KX}, 22^Q, 23^{IQ},$ 25^{IX}	§V, p. 14,15
39	The time component of the local Lorentz relation determines the time differences involved in relating one frame's simultaneous slice to the other frame's simultaneity convention.	$6^B, 24^X, 25^{IX}$	§V(27)–(29), p. 14,15
40	With simultaneity and the local Lorentz relation specified for each frame, the distance to a galaxy is $\ell = ct\chi$ in the expanding frame and $\bar{r} = c\bar{t} \tanh \chi$ in the rigid frame.	$7^Q, 16^X, 24^B,$ 25^{IX}	§V, p. 15

p	Claim	i	Source Location
41	The authors state that extending the discussion to curved cosmological models may permit a component of redshift to be attributed to gravitational time dilation, depending on the metric and interpretation.	20 ^{IQ}	§V, p. 13
42	Consistently transforming the same spacetime, physical events and observers preserves their physical predictions under a change of coordinate representation.	16 ^X , 26 ^X	§I;§V, p. 1,2,13,17
43	Cook and Burns reject interpreting recession velocities, redshift accounts and spatial curvatures relative to observer congruences as identical coordinate-invariant attributes.	8 ^Q , 13 ^{QO} , 14 ^Q , 19 ^{QO} , 26 ^X	§V;Table I, p. 13,16,17,20
44	Galaxies and light in the empty Milne model are test objects whose effects on geometry are neglected; cosmological models in which matter and radiation change the geometry use the corresponding matter conditions.	2 ^S , 5 ^S , 16 ^X	§IV–V, p. 8,13
45	Inside the future light cone, the positive branch of the inverse coordinate transformation, $t = \sqrt{\bar{t}^2 - \bar{r}^2/c^2}$, gives cosmic time.	1 ^K , 16 ^X	§IV(17)–(20), p. 8,9
46	Given a Hubble rate $H > 0$, the Hubble distance is $\ell_H = c/H$.	2 ^Q , 9 ^Q	§III.A(6), p. 4

Table 481: Cook–Burns 2009 Nontemporal Physical Inputs

p	Component
1	Physical carriers of galaxies.
2	Physical carriers of clock devices.
3	Physical carriers of observers.
4	Physical carriers of photons and electromagnetic signals.
5	Physical carriers of rods and length standards.
6	Media and marks of emission and reception records.
7	Media and marks of wavelength and frequency records.
8	Spatial configurations of objects, observers and endpoints.
9	Physical carriers of cosmic fluid and matter distributions.
10	Local comparison and measurement apparatus.
11	Recorded spatial distances and lengths.
12	Physical carriers and connections of light sources, signals and receivers.

Dependency Inputs

Definition 8.618 (Cook–Burns 2009 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [23]$. The table gives time inputs, time outputs, claim support and source locations.

Table 482: Cook–Burns 2009 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1, 3	4	14	Comoving proper-time relation. §II(1)
2	2, 6	7	16	Proper distance on a slice. §III.A(4)
3	1–3, 6–7	8–9	17	Hubble recession and Hubble distance. §III.A(4)–(6)
4	1–2, 11	10	19	Inward radial photon history. §III.C(7)–(9)
5	1–2, 10–11	27–28	21	Reception and turnaround times in the linear model. §III.C(10)
6	2, 11, 27	12	22	Endpoint scale-factor ratio. §III.D(11)
7	12	13	23	Cosmological redshift. §III.D(11)
8	1, 5	2	26	Scale factor of the empty model. §IV(15)–(16)
9	1–2	15	27	Model age from the Hubble rate. §IV
10	1–2	16	28	Rigid-frame representation of Milne time. §IV(17)–(21)
11	16	17	29	Rigid-frame clock readings. §IV(21)
12	2, 6	18	30	Lorentz composition along a spatial chain. §IV.A(22)
13	18	19	31	Rigid-frame Doppler redshift. §IV.D(25)
14	4, 17	21	35	Clock comparison at a meeting event. §V
15	16	22	36	Local relative velocity between congruences. §V; §IV.A(22)
16	16, 22	25	37	Forward local Lorentz relation. §IV(19)–(20); §V(27)–(29)
17	21–22, 25	23	38	Kinematic time dilation. §V
18	6, 25	24	39	Simultaneity relation. §V(27)–(29)
19	16, 24–25	7	40	Distance representations in the two frames. §V
20	6, 14, 16, 24	26	34	Relation between foliation-specific predictions. §V(30)–(31)
21	16	1	45	From rigid-frame representation to cosmic time. §IV(17)–(20)
22	2, 6	14	25	Spatial curvature of a fixed-time slice. §III.F(13)–(14)
23	2	9	46	Hubble distance from the Hubble rate. §III.A(6)

Grounding Conditions

Table 483: Cook–Burns 2009 Conditions for Transfers

Trans- fer Index	Conditions
1	Future-directed comoving worldline and clock zero.
2	Constant cosmic time and specified comoving separation.
3	Comoving endpoints, $H > 0$ and the same distance convention.
4	Emission time $t_e > 0$, initial distance and null path.
5	$a(t) = \alpha t$, $\alpha > 0$, $t_e > 0$; turnaround after emission requires $\ell_G > ct_e$.
6	Comoving endpoints of the same null path.
7	Comoving emitter and receiver and corresponding wavelength comparison.
8	$\rho = 0$, $\Lambda = 0$, $k = -1$, expanding branch and time origin.
9	$H(t) = 1/t$ and a specified H_0 .
10	$a(t) = ct$, $t > 0$ and the future-light-cone coordinate domain.
11	Fixed rigid-frame spatial coordinates and common clock zero.
12	Milne model, fixed t and collinear local frames.
13	Same emitter and receiver, with $ \bar{v} < c$.
14	Same event and standard clocks in the respective local rest frames.
15	Specified meeting event and the two Milne congruences.
16	$\beta = \bar{v}/c$, $\gamma = (1 - \beta^2)^{-1/2}$, and the direction of equations (19)–(20).
17	Local clock comparison, relative velocity and each observer’s simultaneity convention.
18	Transformation of the same local events and each frame’s simultaneity condition.
19	Same galaxy and the specified simultaneous slice in each frame.
20	Same Milne spacetime, with the respective foliations and observer congruences.
21	$c\bar{t} > \bar{r} $, $\bar{t} > 0$ and the positive square root.
22	Specified k and scale-factor value $a(t) > 0$.
23	Specified $H > 0$.

Results

Definition 8.619 (Cook–Burns 2009 Results). Let $\text{Idx}_R^{\text{audit,result}} = [22]$.

Table 484: Cook–Burns 2009 Results

q	Result	Source Location
1	Robertson–Walker geometry and Friedmann evolution	§II(1)–(3); §II;§IV(15), p. 3,4,8
2	Hubble relation from the proper-distance derivative	§III.A(4); §III.A(4)–(6), p. 4
3	Distant distance rate and local light-speed bound	§III.B;§IV.A;§V, p. 5,9,10,17
4	Radial photon history	§III.C(7)–(9), p. 5,6

q	Result	Source Location
5	Photon reception and turnaround in the linear model	§III.C(9)–(10); §III.C(10), p. 6
6	Endpoint scale-factor ratio and cosmological redshift	§III.D(11), p. 6,7
7	Spatial curvature of a fixed-time slice	§III.F(13)–(14), p. 7
8	Linear expansion and model age in the empty model	§IV(15)–(16); §IV, p. 8
9	Milne coordinate transformation and inverse	§IV(17)–(21); §IV(17)–(20), p. 8,9
10	Milne coordinate domain and Minkowski representation	§IV(17)–(21), p. 8,9
11	Comparison of comoving and rigid-frame clocks	§II(1); §IV(21); §V, p. 3,9,13,14
12	Bound on the composed endpoint relative velocity	§IV.A(22), p. 9,10
13	Spatial composition of neighboring local Lorentz transformations	§IV.A(22), p. 9,10
14	Redshift agreement between the two descriptions	§IV.D(23)–(25), p. 11
15	Rigid-frame Doppler description	§IV.D(25), p. 11
16	Relation between wavelength stretching and the Doppler description	§III.D(11)–(12); §III.D;§IV.D, p. 6,7,11
17	Spatial curvature of different simultaneous foliations	§V(30)–(31), p. 15,16
18	Lorentz relation of rods and clocks and kinematic time dilation	§V; §V;§IV.A(22); §IV(19)–(20); §V(27)–(29), p. 9,13, 14,15
19	Frame dependence of simultaneity and distance	§V(27)–(29); §V, p. 14,15
20	General covariance and correspondence of physical predictions	§I;§V, p. 1,2,13,17
21	Distinction between coordinate-dependent descriptions and invariants	§V;Table I, p. 13,16,17,20
22	Domains of the empty model and curved cosmological models	§V; §IV–V, p. 8,13

Table 485: Cook–Burns 2009 Conditions for Results

Result Index	Conditions and Domain
1	Homogeneity, isotropy and specified matter and curvature conditions.
2	Comoving endpoints, simultaneous slicing and expanding branch.
3	Proper-distance rate and velocity comparison in the same local frame.
4	Specified scale factor, emission time, initial position and inward null branch.
5	$a(t) = \alpha t$, $\alpha > 0$, $t_e > 0$; turnaround after emission requires $\ell_G > ct_e$.
6	Comoving emitter and receiver and the null path connecting them.
7	Slice t , k and $a(t) > 0$.

Result Index	Conditions and Domain
8	Empty $k = -1$ model; the age comparison uses the source's H_0 .
9	Positive-time branch inside the future light cone.
10	Corresponding domain $t > 0$ and $c\bar{t} > \bar{r} $.
11	Respective congruences, clock zeros and the same meeting event.
12	Finite χ and collinear comparison in the Milne model.
13	Spatial chain at fixed t and dimensionless rapidity.
14	Same emitter, receiver and signal in the Milne model.
15	Specified relative velocity and endpoint observers.
16	Distinguish the Hubble distance rate and local relative velocity according to each formula's definition.
17	Respective simultaneous slices of the same four-dimensional Milne spacetime.
18	Same event, local relative velocity and a transformation with specified direction.
19	Each frame's simultaneity condition and the same galaxy worldline.
20	Consistently transform the same spacetime, events and observers.
21	Specified observer congruences and the six comparisons in Table I.
22	Test-object approximation and matter conditions; attribution to gravitational time dilation depends on the curved model and interpretation.

Result Dependence

Definition 8.620 (Cook–Burns 2009 Result-Dependence Hyperedges). For each row, let $\text{ResultEdge}_{R,q}$ ($q \in [22]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 486: Cook–Burns 2009 Result-Dependence Hyperedges

q	Time Premises i	Claim Support p
1	1–3, 5–6	13, 15
2	1–3, 6–9	16–17, 46
3	8, 18	18
4	1–2, 10–11	19
5	1–2, 9–11, 27–28	20–21
6	2, 11–13, 27	22–23
7	2, 6, 14	25
8	1–2, 5, 15	26–27
9	1–2, 16	28, 45
10	1–2, 16	28
11	1, 3–4, 16–17, 21	14, 29, 35
12	2, 6, 18	30
13	2, 6, 18	30

q	Time Premises i	Claim Support p
14	13, 19	32
15	18–19	31
16	2, 8, 11–13, 19, 27	24, 33
17	6, 14, 16, 24, 26	34
18	4, 16–17, 21–23, 25	35–38
19	6–7, 16, 24–25	39–40
20	16, 26	42
21	8, 13–14, 19, 26	43
22	2, 5, 16, 20	41, 44

$$\text{BoundaryCount}_R^{\text{audit}} = (96, 45).$$

The 46 claims, 28 time contents and 22 results give 96 nodes; the 23 transfers and 22 result dependencies give 45 edges.

Evidence Hierarchy Cook and Burns describe the same empty cosmological model in comoving and rigid frames and compare distance rates, redshifts, clocks and rods, and the curvature of simultaneous slices. The source’s Hubble rate is an existing observational input; the paper reports zero newly acquired direct measurements.

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 0; remaining primitive-time roots 28. Every transfer has temporal input from cosmic time, proper time, synchronization, velocity references or propagation events.

Result Dependence Of the 22 results, 22 depend on primitive-time roots and 0 are reached from source formal evidence and local constructions. There are 137 root–result dependence pairs, of which 18 are necessary.

Placement of the Grounding Obligation Cosmic time and proper time support geodesics, initial conditions, velocities and distance rates; clock readings and a common zero support comoving-clock synchronization. Emission and reception times, light-propagation intervals and velocity references enter redshift relations. Complete generation from nontemporal physical inputs is zero; the placement of the physical-time grounding obligation is “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 137 root–result dependence pairs. Actual reach from failure of a specific physical-time generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns results dependent on it.

Audit Conclusions

Proposition 8.621 (Primitive Time Roots and Result Dependence in Cook–Burns 2009). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(28, 22), \quad R \in \text{Ref}^{\text{audit,rootdep}}, \quad R \notin \text{Ref}^{\text{audit,rootunreach}}.$$

Proof. Closure from Initial Claims 1–12 remains those twelve nodes. Even with all 46 claims, no time content is newly reached because each transfer requires temporal input. The mutually inverse Milne/inertial-time conversions likewise require one of those temporal inputs. Adding all time contents as seeds reaches all 22 results. $\square$

Proposition 8.622 (Physical-time admissibility in the Cook–Burns 2009 source). All 28 time contents in the source are primitive-time inputs and do not satisfy physical-time admissibility. No time content is reachable from the source-form evidence environment, and the local-reach determination reaches none of the 28 time contents. All 22 results depend on time contents that do not satisfy physical-time admissibility.

Proof. By Proposition 8.621, all 28 time contents are primitive-time roots and all 22 results are primitive-time-root-dependent results. The proof of the same proposition shows that no time content is reachable from the source-form evidence environment. The conclusion follows from Definition 7.10, Theorem 7.11, and Corollary 7.12. $\square$

Proposition 8.623 (Minimal Primitive-Time-Root Support in Cook–Burns 2009). The table gives the number of minimal supports and their cardinality range for each result, with 166 family members in total.

Proof. Apply Definition 7.46 to all transfers and result dependencies, starting from singleton time-content seeds and empty supports for source formal evidence. By Proposition 7.47, the inclusion-minimal supports of the stable antichains give the families with the tabulated statistics. $\square$

The support statistic $(m; a, b)$ gives the number of supports m , minimum cardinality a , and maximum cardinality b . When the bounds coincide, write $(m; a)$.

Table 487: Minimal Primitive-Time-Root Support in Cook–Burns 2009

Result Indices	Support Statistics
1	2; 4
2	4; 4
3	8; 2, 4
4–5	4; 3
6, 17	5; 3
7, 11–13, 19	3; 2, 3
8	2; 2

Result Indices	Support Statistics
9–10	4; 2
14	25; 2, 4
15	4; 1, 3
16	19; 5
18	6; 2, 3
20	8; 2, 3
21	45; 5, 7
22	2; 3

Result-Preservation Domain Fix the models, initial and endpoint conditions, common clock zero, distance and velocity conventions, and applicability domains in Table 485. When temporal inputs are realized by generated representations preserving the same times, trajectories, synchronization, null-propagation intervals, and frequency comparisons, time-dependent results are preserved within their domains.

Proposition 8.624 (Complete Audit of Time Use in Cook–Burns 2009). There are 28 time contents, 46 claims, 23 transfers and 22 results. The 110 claim/time-content use pairs have 135 role assignments, with 119 source-specific obligation atoms. Complete generation is zero, local reach is 0, remaining primitive-time roots number 28, and 22 results depend on roots.

Proof. This follows from the claim, transfer and result tables and the generation-reach and minimal-support judgments. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Individual statement Original text (English):

the time coordinate t is the proper time on clocks moving with the galaxies

The original-source location is Cook and Burns,
Interpretation of the Cosmological Metric,
arXiv:0803.2701v2, p. 3, §II, immediately after Eq. (1).

Statement object, minimal context, and attribution

The statement object is the correspondence between the time coordinate t of the Robertson–Walker metric and the proper time of clocks accompanying the galaxies. The quotation belongs to the context immediately after Eq. (1) in §II of the original source, which introduces the Robertson–Walker metric and explains the relation between comoving galaxies and the cosmic-time coordinate. The statement is attributed to R. J. Cook and M. S. Burns.

Predicate and explicit applicability domain

The quotation specifies clocks moving with the galaxies and identifies the proper time of those clocks with the time coordinate t . Its explicit applicability domain is the congruence of comoving galaxy observers in the Robertson–Walker model.

Individual Audit Comoving Clocks and Cosmic Time

The clocks specified by Cook and Burns in the quotation move with the galaxies of the Robertson–Walker model. Setting the spatial coordinates constant in equation (1) gives

$ds^2 = -c^2 dt^2 = -c^2 d\tau^2$. On a future-directed worldline, $d\tau = dt$, so $\tau = t + b$ for clock zero b . Clock readings agree with t when the common zero $b = 0$ is chosen.

Proper time for a noncomoving clock is obtained from the same metric along that clock's worldline. Comparing a comoving clock and a rigid-frame clock uses the same meeting event, the relative velocity of the two congruences and the local Lorentz relation.

The statement corresponds to Time Contents 1,3,4, Claims 13–14, Transfer 1 and Result 11. The proper-time relation takes cosmic time, the comoving congruence and clock readings as inputs; there are zero paths to these time contents from the generation-initial set in Proposition 8.621. Comparison with noncomoving clocks corresponds to Time Contents 16–17,21–25, Claims 35–39 and Results 18–19.

8.19.52 Source audit

Audit Inputs

Source and Audit Scope

Source identifier 52.

Source name Peacock 2008 Primitive-Time-Root Audit

Bibliographic identification J. A. Peacock, *A diatribe on expanding space*, arXiv:0809.4573v1 (2008), pp. 1–4. $R = \text{Ref}_{\text{Pea08ES}}$.

Audit-scope identifier 52-SC1.

Audit-scope content Introduction, Sections 1–3, Equations (1)–(16) and references.

Source-location identifier 52-L1.

Source-location content Tables give sections, equations and source pages.

Assumptions The RW representation uses homogeneity, isotropy, a positive scale factor and a comoving observer congruence. Closed-slice volume growth uses a fixed compact comoving slice and an expanding interval $H > 0$. Local Newtonian test-particle analysis neglects backreaction and assumes weak fields, low speeds, $a(t_0) = 1$, $r(t_0) = r_0$ and $\dot{r}(t_0) = 0$. The matter-only, empty-universe and pure de Sitter solutions retain their respective model conditions. The redshift path integral takes the continuous limit of infinitesimal intervals; the endpoint Doppler-plus-gravity expression holds through second order in the matter-only weak-field approximation.

Nontemporal Physical Inputs The physical carriers, spatial configurations, record media and marks, devices and physical connections in Claims 1–14 form the generation-initial set. Temporal and frequency interpretations of records, trajectories, propagation order and initial-time conditions belong to the corresponding time contents and claims.

Time Contents

Time-content identifier 52-T1.

Time-content name Time contents of the source.

Time-content type Time content.

Time-content specification Time contents in Definition 8.625 and their roles in the claim table.

Source-location identifier set 52-L1.

Definition 8.625 (Peacock 2008 Time Contents). Let $n_R = 22$. Each time content is scoped by the model, state and domain of the corresponding claims and results.

Table 488: Peacock 2008 Time Contents

i	Time Content	Source Location
1	Cosmic-time coordinate and basis for time differentiation	pp. 1–4
2	Scale-factor history	pp. 1–4
3	Proper-time intervals defined by the metric	pp. 1, 3
4	Relation between static and expanding de Sitter times	pp. 1
5	Volume history of closed spatial slices	pp. 1
6	Test-particle initial time and initial conditions	pp. 2–3
7	History of signed proper position	pp. 1–3
8	Time rate and coordinate acceleration of signed position	pp. 2–3
9	Hubble rate and scale-factor derivative	pp. 2–4
10	Einstein–de Sitter trajectory	pp. 3
11	Late-time asymptotic behavior of the Einstein–de Sitter trajectory	pp. 3
12	Stationary trajectory and linear scale factor in the empty universe	pp. 3
13	Initially stationary trajectory in the pure de Sitter model	pp. 3
14	Photon propagation time between neighboring observers	pp. 4
15	Emission and reception endpoints and their order	pp. 4
16	Scale-factor change along the light path	pp. 4
17	Local frequency change between neighboring observers	pp. 4
18	Integrated redshift from emission to reception	pp. 4
19	Peculiar-momentum history of a free particle	pp. 2, 4
20	Peculiar velocity relative to the Hubble flow	pp. 2–3
21	Gravitational frequency change between endpoints	pp. 4
22	Relative recession velocity of local endpoints	pp. 4

Definition 8.626 (Peacock 2008 Physical and Formal Input Claims). Let the claim-node set be $\mathbf{ClaimNodes}_R(57)$, with claim boundary $\mathbf{ClaimBoundaryIdx}_R = ([14], [57])$. Claims 1–14 form the generation-initial set; all 57 claims form the source formal-evidence environment.

Table 489: Peacock 2008 Claims and Temporal Uses

p	Claim	i	Source Location
1	Physical carriers of galaxies.	$\emptyset$	§ 1–3, p. 1,2

p	Claim	i	Source Location
2	Physical carriers of test particles.	$\emptyset$	§ 1–3, p. 2,3
3	Physical carriers of photons.	$\emptyset$	§ 1–3, p. 4
4	Physical carriers of observers.	$\emptyset$	§ 1–3, p. 1,4
5	Physical carriers of matter.	$\emptyset$	§ 1–3, p. 2,3
6	Physical carriers of fields.	$\emptyset$	§ 1–3, p. 2,3
7	Spatial configurations.	$\emptyset$	§ 1–3, p. 1,2,3,4
8	Physical connections for local interactions.	$\emptyset$	§ 1–3, p. 1,2,3,4
9	Media and marks of emission and reception records.	$\emptyset$	§ 1–3, p. 4
10	Media and marks of frequency and wavelength records.	$\emptyset$	§ 1–3, p. 4
11	Recorded distances and lengths.	$\emptyset$	§ 1–3, p. 1,2,3
12	Spatial configurations of cavity boundaries.	$\emptyset$	§ 1–3, p. 3
13	Physical carriers of local measurement devices.	$\emptyset$	§ 1–3, p. 2,4
14	Media and marks of endpoint and boundary-condition records.	$\emptyset$	§ 1–3, p. 2,3,4
15	Peacock develops his 2001 discussion.	$\emptyset$	Introduction, p. 1
16	Peacock examines the interpretation of cosmic expansion as a cause in local dynamics.	$2^S, 8^C$	§ 1, p. 1
17	Cosmic scale change and the dynamical terms in a local equation of motion represent different physical quantities.	$2^S, 8^C$	§ 1–2, p. 1,2,3
18	In comoving RW coordinates, fundamental observers have constant spatial coordinates.	1^X	§ 1, p. 1
19	The proper separation between observers at a fixed comoving coordinate separation is proportional to the scale factor.	$2^Q, 7^{QX}$	§ 1, p. 1
20	The Milne representation expresses the interior of the future light cone of Minkowski spacetime in open RW coordinates with a linear scale factor.	$1^I, 2^S, 3^I$	§ 1, p. 1
21	De Sitter spacetime admits expanding and static representations, with coordinate times and observer congruences determined by the respective metrics.	$1^I, 2^S, 3^I, 4^{IX}$	§ 1(1)–(2), p. 1
22	Different coordinate representations of the same geometry distinguish the representation of a scale factor from local physical effects.	$1^I, 2^S, 3^I, 4^X$	§ 1(1)–(2), p. 1
23	The finite volume of a fixed comoving slice is $V = V_0 a^3$ and increases on an expanding branch with $H > 0$.	$1^I, 2^S, 5^{QS}, 9^Q$	§ 1, p. 1

p	Claim	i	Source Location
24	Slice-volume growth is a global quantity; local dynamical terms are determined by matter, vacuum energy and initial conditions.	$5^Q, 6^B, 8^C$	§ 1–2, p. 1,2,3
25	Peacock applies Newtonian dynamics in a region much smaller than the horizon where weak-field and low-speed approximations hold.	$1^I, 7^Q, 8^{QC}$	§ 1; § 2(4)–(7), p. 1,2
26	The equivalence principle supplies a special-relativistic metric in a local inertial frame at a point; curvature determines tidal effects over a finite region.	$3^I, 8^C$	§ 1, p. 1
27	A nearby, low-redshift galaxy is treated as a test body with negligible backreaction and zero velocity relative to the origin at the initial time.	$6^B, 7^Q, 8^Q$	§ 2(4)–(8), p. 2
28	For a constant-mass particle, peculiar momentum and peculiar velocity satisfy $p = m\gamma v_{\text{pec}}$; in the nonrelativistic limit, $v_{\text{pec}} = p/m$.	$19^Q, 20^Q$	§ 2(3); § 3, p. 2,4
29	Peculiar-velocity decay and the sign of local coordinate acceleration depend on different conditions; in the empty model, decay coexists with zero coordinate acceleration.	$8^C, 12^S, 20^Q$	§ 2, p. 2,3
30	In the local Newtonian model of homogeneous matter, the coordinate acceleration of signed position is $\ddot{r} = -\Omega_m H_0^2 r / (2a^3)$.	$2^S, 7^Q, 8^C, 9^Q$	§ 2(4)–(6), p. 2
31	With matter and vacuum energy, the same model gives $\ddot{r} = -H_0^2 r (\Omega_m a^{-3} - 2\Omega_v) / 2$.	$2^S, 7^Q, 8^C, 9^Q$	§ 2(7), p. 2
32	The Friedmann relation and $v_{\text{pec}} = \dot{r} - Hr = -H_0 r_0 / a$ yield the equation of motion for the same matter-and-vacuum model.	$2^S, 6^B, 7^Q, 8^C, 9^Q, 20^Q$	§ 2(7)–(9), p. 2,3
33	In the Einstein–de Sitter model, the initially stationary solution is $r = r_0(2x^{1/3} - x^{2/3})$, $x = t/t_0$; its coordinate acceleration is inward before reaching the origin.	$1^I, 6^B, 8^C, 9^Q, 10^S$	§ 2(10)–(11), p. 3
34	This signed solution crosses the origin at $t = 8t_0$, and r/a tends to $-r_0$.	$10^S, 11^{SB}$	§ 2(11), p. 3
35	The approach to the Hubble flow on the opposite side belongs to the same Einstein–de Sitter trajectory that initially moves inward.	$10^S, 11^S$	§ 2(10)–(11), p. 3

p	Claim	i	Source Location
36	In the spherically symmetric Swiss-cheese model without vacuum energy, concentrating the cavity mass at the center gives a Schwarzschild interior metric.	$1^I, 3^I$	§ 2, Swiss-cheese discussion, p. 3
37	With neither matter nor vacuum energy, $a = t/t_0$ and an initially stationary particle follows $r = r_0$.	$1^I, 2^S, 6^B, 12^S$	§ 2, unnumbered empty-universe case, p. 3
38	For the initially stationary solution in the empty universe, $H = 1/t$ and $v_{\text{pec}} = -Hr_0$; $\ddot{r} = 0$ coexists with $1/a$ decay of peculiar velocity.	$9^Q, 12^S, 20^Q$	§ 2, unnumbered empty-universe case, p. 3
39	At the initial time with $a(t_0) = 1$ and $r_0 > 0$, the coordinate acceleration is outward when $\Omega_v > \Omega_m/2$.	$6^B, 7^Q, 8^C, 9^Q$	§ 2(7), vacuum case, p. 2,3
40	In the pure de Sitter model, $r(t_0) = r_0$ and $\dot{r}(t_0) = 0$ give $r = r_0 \cosh[H_0(t - t_0)]$.	$1^I, 6^B, 8^C, 9^Q, 13^S$	§ 2(12), p. 3
41	The outward coordinate acceleration of the initially stationary pure de Sitter solution arises from the vacuum-energy term.	$8^C, 13^S$	§ 2(7),(12), p. 2,3
42	The source's local Newtonian model determines test-particle motion through matter attraction and the repulsive vacuum-energy term.	$2^S, 7^Q, 8^C, 9^Q$	§ 2(4)–(12), p. 2,3
43	For sufficiently small relative recession velocity, the local frequency change is $d\nu/\nu = -dv/c$.	$17^{\text{QX}}, 22^{\text{QX}}$	§ 3(13), p. 4
44	The same null path is divided into infinitesimal propagation intervals between neighboring fundamental observers in future-directed order.	$14^I, 15^{\text{BX}}$	§ 3(13)–(14), p. 4
45	With neighboring-observer recession $dv = Hd$ and local light interval $d = c dt$, the frequency change is $d\nu/\nu = -H dt$.	$9^Q, 14^I, 17^{\text{QX}}$	§ 3(13)–(14), p. 4
46	Integrating $d\nu/\nu = -da/a$ along the light path gives $1 + z = a_{\text{rec}}/a_{\text{em}}$ between comoving endpoints.	$15^B, 16^S, 17^Q, 18^{\text{QX}}$	§ 3(14), p. 4
47	The peculiar momentum of a freely propagating particle obeys $p = p_0/a$, in both relativistic and nonrelativistic regimes.	$2^S, 19^{\text{QS}}$	§ 2 after (3); § 3 after (14), p. 2,4
48	Cosmological redshift is obtained by integrating the chain of infinitesimal Doppler changes along the same light path in the continuous limit.	$14^I, 15^{\text{BX}}, 17^Q, 18^{\text{QX}}$	§ 3(13)–(14), p. 4
49	For comoving endpoints in the RW model, the infinitesimal Doppler chain also applies at large redshift.	$15^{\text{BX}}, 17^Q, 18^{\text{QX}}$	§ 3 after (14), p. 4

p	Claim	i	Source Location
50	Local velocity comparisons along the light path are distinct from recession velocity between distant endpoints; an endpoint velocity requires a comparison convention.	$15^{\text{BX}}, 18^{\text{QX}}, 22^{\text{QX}}$	§ 3(15), p. 4
51	In the weak field of uniform nonrelativistic matter, the receiver-minus-emitter potential difference $\Delta\phi = -\Omega_{\text{m}}H_0^2r^2/4$ gives the gravitational frequency change.	$1^{\text{B}}, 9^{\text{Q}}, 21^{\text{QX}}$	§ 3 after (16), p. 4
52	In the source's nearby-endpoint representation, the product of the local Doppler factor and $1 + \Delta\phi/c^2$ gives the redshift through second order.	$18^{\text{QX}}, 21^{\text{QX}}, 22^{\text{QX}}$	§ 3(16), p. 4
53	This endpoint approximation is a second-order expression for a matter-only weak field, with a different domain from the full-redshift-range light-path integral.	$18^{\text{QX}}, 21^{\text{QX}}, 22^{\text{QX}}$	§ 3(16) and final paragraph, p. 4
54	The scale factor differentiated with respect to cosmic time defines $H = \dot{a}/a$.	$1^{\text{I}}, 2^{\text{S}}, 9^{\text{Q}}$	§ 2(8)–(9); § 3(14), p. 2,3,4
55	For signed position and its time rate, peculiar velocity relative to the Hubble flow is $v_{\text{pec}} = \dot{r} - Hr$.	$7^{\text{Q}}, 8^{\text{Q}}, 9^{\text{Q}}, 20^{\text{Q}}$	§ 2(8), p. 2
56	Evaluating the scale-factor history along the propagation times and emission/reception endpoints of the same light path determines da/a and its endpoint ratio.	$2^{\text{S}}, 14^{\text{I}}, 15^{\text{B}}, 16^{\text{S}}$	§ 3(13)–(14), p. 4
57	First and second differentiation of signed-position history with respect to cosmic time gives velocity and coordinate acceleration.	$1^{\text{I}}, 7^{\text{S}}, 8^{\text{Q}}$	§ 2(4)–(12), p. 2,3

Formal Input Elements

Definition 8.627 (Peacock 2008 Formal Inputs). For a fixed compact comoving slice Σ with reference spatial metric h_0 ,

$$h(t) = a(t)^2 h_0, \quad V_0 = \int_{\Sigma} \sqrt{\det h_0} d^3x, \quad V(t) = \int_{\Sigma} \sqrt{\det h(t)} d^3x = V_0 a(t)^3.$$

Let $\nu_{\text{em}} > 0$ and $\nu_{\text{rec}} > 0$ be the frequencies of the same photon measured by its comoving emitter and receiver, respectively. Use cosmic time t , with $a(t) > 0$ and $H = \dot{a}/a$. Let $r(t)$ be signed proper position along a line through the origin, and $|r(t)|$ the separation.

$$v_{\text{pec}} = \dot{r} - Hr, \quad \ddot{r} = -\frac{H_0^2 r}{2} (\Omega_{\text{m}} a^{-3} - 2\Omega_{\text{v}}).$$

Here $\ddot{r}$ is coordinate acceleration with respect to cosmic time. The proper acceleration measured by an accelerometer on a freely falling particle is zero. For initial conditions

$$r(t_0) = r_0 > 0 \text{ and } \dot{r}(t_0) = 0,$$

$$\begin{aligned} \text{Einstein-de Sitter: } & a = x^{2/3}, \quad r = r_0(2x^{1/3} - x^{2/3}), \quad x = t/t_0 > 0, \quad H_0 t_0 = 2/3, \\ \text{empty: } & a = t/t_0, \quad r = r_0, \\ \text{pure de Sitter: } & r = r_0 \cosh[H_0(t - t_0)], \quad H_0 > 0. \end{aligned}$$

The Einstein-de Sitter solution crosses the origin at $t = 8t_0$, with $r/a \rightarrow -r_0$. Each solution retains the domain of the local Newtonian model. Source Equation (13) prints $\delta\nu/\nu = \delta v/c$ after defining recession by $dv = Hd$. Combining that definition with the local Doppler relation gives

$$\frac{d\nu}{\nu} = -\frac{dv}{c} = -H dt = -\frac{da}{a}, \quad 1 + z = \frac{\nu_{\text{em}}}{\nu_{\text{rec}}} = \frac{a_{\text{rec}}}{a_{\text{em}}}.$$

A free particle has peculiar momentum $p = p_0/a$. For constant mass m , $p = m\gamma v_{\text{pec}}$, so the $1/a$ speed law holds in the nonrelativistic limit. For nearby endpoints in the matter-only weak-field approximation,

$$1 + z = \sqrt{\frac{1 + v/c}{1 - v/c}} \left(1 + \frac{\Delta\phi}{c^2} \right) + O((v/c)^3), \quad \Delta\phi = -\frac{\Omega_m H_0^2 r^2}{4}.$$

Here v is the local relative velocity in this endpoint comparison; the expression is valid through second order in v/c .

Definition 8.628 (Peacock 2008 Finite Cardinalities). Superscripts in the claim table denote the use roles below. Auxiliary objects belong to the corresponding conditional claims; the separate auxiliary-object index set is empty.

Table 490: Peacock 2008 Time-Use Roles

Code	Role	Use Count
I	Interval	21
Q	Quantity	54
X	Cross-position correspondence	23
K	Clock reading	0
C	Causality	14
S	State evolution	29
P	Probabilistic evolution	0
O	Observation record	0
B	Reference boundary	15

Table 491: Peacock 2008 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1	member	instant, order_relation, orientation
2, 4–5, 7, 12–13, 16, 19	member	instant, order_relation
3, 14	member	interval_boundary, endpoint
6	member	instant, phase_boundary
8–9, 17–18, 20–22	dependency	$\emptyset$
10	member	instant, order_relation, transition_position
11	member	order_relation, orientation
15	member	endpoint, order_relation

Table 492: Peacock 2008 Finite Cardinalities

Object	Cardinality
Time contents	22
Claims	57
Auxiliary objects	0
Transfers	18
Results	14
Claim/time-content uses	131
Role assignments	156
Obligation atoms	111
Audit-boundary nodes	93
Audit-boundary edges	32

Dependency Inputs

Definition 8.629 (Peacock 2008 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [18]$. The table gives time inputs, time outputs, claim support and source locations.

Table 493: Peacock 2008 Time-Content Transfers

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
1	1–2, 9	5	23	Volume history of closed slices. § 1
2	1, 7	8	57	Time derivatives of position. § 2(4)–(12)
3	1–2	9	54	Hubble rate. § 2(8)–(9); § 3(14)
4	1, 6, 8–9	10	33	Einstein–de Sitter initial-value solution. § 2(10)–(11)
5	10	11	34	Origin crossing and late-time asymptote. § 2(11)

s	Time In i	Time Out i	Claim Support p	Transfer Content and Source Location
6	1–2, 6	12	37	Initially stationary solution in the empty universe. § 2, unnumbered empty-universe case
7	1, 6, 8–9	13	40	Initially stationary pure de Sitter solution. § 2(12)
8	2, 14–15	16	56	Scale-factor change along the light path. § 3(13)–(14)
9	9, 14	17	45	Infinitesimal frequency change. § 3(13)–(14)
10	15–17	18	46	Integrated redshift. § 3(14)
11	2	19	47	Free-particle momentum evolution. § 2 after (3); § 3 after (14)
12	19	20	28	From momentum to peculiar velocity. § 2(3); § 3
13	7–9	20	55	Velocity comparison with the Hubble flow. § 2(8)
14	1, 9	21	51	Gravitational frequency change. § 3 after (16)
15	21–22	18	52	Endpoint redshift through second order. § 3(16)
16	22	17	43	Local Doppler comparison. § 3(13)
17	9, 12	20	38	Peculiar velocity in the empty universe. § 2, unnumbered empty-universe case
18	2	7	19	Proper separation across a comoving interval. § 1

Grounding Conditions

Table 494: Peacock 2008 Conditions for Transfers

Transfer Index	Conditions
1	Fixed compact comoving slice, $a > 0$ and an expanding interval $H > 0$.
2	Signed-position history twice differentiable in the same cosmic time.
3	Positive differentiable scale factor.
4	Matter only, spatial flatness, $x = t/t_0 > 0$, $H_0 t_0 = 2/3$, $r(t_0) = r_0$ and $\dot{r}(t_0) = 0$.
5	Continue the same signed solution through the origin to the opposite side.
6	Zero matter density and vacuum energy, $a = t/t_0$ and the specified initial-rest conditions.
7	$\Omega_m = 0$, $\Omega_v = 1$, $H_0 > 0$, local Newtonian model, $r(t_0) = r_0$ and $\dot{r}(t_0) = 0$.
8	Same null path, future-directed emission/reception order and positive scale factor.

Trans- fer Index	Conditions
9	Neighboring comoving observers, $d = c dt$, $dv = Hd$ and the infinitesimal-interval limit.
10	RW model, comoving endpoints of the same light ray and continuous integration of $d\nu/\nu = -da/a$.
11	Free propagation in the RW model and p_0 specified at a reference scale.
12	Constant mass and $p = m\gamma v_{\text{pec}}$; $v_{\text{pec}} = p/m$ in the nonrelativistic limit.
13	Evaluate $v_{\text{pec}} = \dot{r} - Hr$ in the same cosmic time and position convention.
14	Uniform nonrelativistic matter, weak field, nearby endpoints and receiver-minus-emitter potential difference.
15	The source's matter-only weak-field approximation, local velocity comparison and terms through second order in v/c .
16	Infinitesimal recession dv between neighboring endpoints and the corresponding infinitesimal frequency change.
17	$r = r_0$, $H = 1/t$ and the same initially stationary empty-universe solution.
18	Fixed comoving coordinate separation on a slice of the same cosmic time.

Results

Definition 8.630 (Peacock 2008 Results). Let $\text{ldx}_R^{\text{audit,result}} = [14]$.

Table 495: Peacock 2008 Results

q	Result	Source Location
1	Distinction between RW coordinate representation and local physical effects	§ 1; § 1–2; § 1(1)–(2), p. 1,2,3
2	Milne description of Minkowski spacetime	§ 1, p. 1
3	Expanding and static representations of de Sitter spacetime	§ 1(1)–(2), p. 1
4	Global growth of closed-slice volume	§ 1, p. 1
5	Distinction between global volume change and local dynamics	§ 1–2, p. 1,2,3
6	Domains of local Newtonian analysis and tangent frames	§ 1; § 2(4)–(7); § 1; § 2(4)–(6); § 2(7), p. 1,2
7	Relation of peculiar momentum and velocity decay to coordinate acceleration	§ 2(3); § 3; § 2; § 2(7)–(9); § 2 after (3); § 3 after (14), p. 2,3,4
8	Einstein–de Sitter trajectory, origin crossing and asymptote	§ 2(10)–(11); § 2(11), p. 3
9	Stationary particle and peculiar-velocity decay in the empty universe	§ 2, unnumbered empty-universe case, p. 3

q	Result	Source Location
10	Outward acceleration from vacuum energy and the pure de Sitter solution	§ 2(7), vacuum case; § 2(12); § 2(7),(12), p. 2,3
11	Interior metric of a Swiss-cheese cavity	§ 2, Swiss-cheese discussion, p. 3
12	Cosmological redshift from an infinitesimal Doppler chain	§ 3(13); § 3(13)–(14); § 3(14); § 3 after (14), p. 4
13	Distinction between a local chain and endpoint recession velocity	§ 3(13)–(14); § 3 after (14); § 3(15), p. 4
14	Endpoint redshift in a matter-only weak field	§ 3 after (16); § 3(16); § 3(16) and final paragraph, p. 4

Table 496: Peacock 2008 Conditions for Results

Result Index	Conditions and Domain
1	Geometric meaning of comoving coordinates and comparison of charts on the same geometry.
2	Expanding empty-universe branch in open RW coordinates inside the future light cone.
3	Same geometry, respective observer congruences and overlapping chart domains; the static patch has $r < c/H$.
4	Fixed compact comoving slice, $V = V_0 a^3$ and an interval with $H > 0$.
5	Definition of global volume and a local Newtonian equation with specified initial conditions.
6	Local weak-field, low-speed model with matter and vacuum terms; the equivalence-principle tangent frame is local to a point.
7	Free particles of constant mass, with $1/a$ velocity scaling restricted to low speeds; Equations (7)–(9) share one matter-and-vacuum model, and the zero-acceleration example is the empty universe.
8	$x = t/t_0 > 0$, initial rest and signed position; origin crossing at $x = 8$ and asymptote $r/a \rightarrow -r_0$.
9	Zero matter and vacuum energy, $a = t/t_0$, $r(t_0) = r_0$ and $\dot{r}(t_0) = 0$.
10	For $a_0 = 1$ and $r_0 > 0$, initial outward acceleration requires $\Omega_v > \Omega_m/2$; the cosh solution assumes pure de Sitter spacetime and initial rest.
11	Spherically symmetric model without vacuum energy, with mass and boundary matched to a Schwarzschild cavity.
12	Same null path in the RW model, comoving emitter and receiver and the continuous limit of infinitesimal changes.
13	Local velocity comparisons on the same light path and the convention defining velocity between distant endpoints.
14	Nonrelativistic matter, nearby endpoints, weak potential and Doppler and gravitational terms through second order in v/c .

Result Dependence

Definition 8.631 (Peacock 2008 Result-Dependence Hyperedges). For each row, let $\text{ResultEdge}_{R,q}$ ($q \in [14]$) be the hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 497: Peacock 2008 Result-Dependence Hyperedges

q	Time Premises i	Claim Support p
1	1–4, 7–8	16–19, 22
2	1–3	20
3	1–4	21
4	1–2, 5, 9	23
5	5–6, 8	24
6	1–3, 7–9	25–26, 30–31
7	2, 6–9, 12, 19–20	28–29, 32, 47
8	1, 6, 8–11	33–35
9	1–2, 6, 9, 12, 20	37–38
10	1, 6–9, 13	39–41
11	1, 3	36
12	9, 14–18, 22	43–46, 48–49
13	14–15, 17–18, 22	48–50
14	1, 9, 18, 21–22	51–53

Evidence Hierarchy Peacock analyzes coordinate representations, model-specific trajectories, local dynamics and redshift. The paper reports zero newly acquired direct measurements.

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 0; remaining primitive-time roots 22. Every transfer has temporal input from coordinate time, a temporal history, a time rate, a velocity reference or photon propagation.

Result Dependence All 14 results depend on primitive-time roots. There are 64 root–result dependence pairs, of which 36 are necessary.

Evidential Force The source addresses RW, Milne and de Sitter coordinate comparisons, closed-slice volume, model-specific local Newtonian motion, Swiss-cheese cavities and redshift. Each result holds under its own metric, model, initial values, local approximations and light-path conditions.

Placement of Grounding Obligations Cosmic time and its derivatives support scale factors, position rates and model-specific trajectories. Metric proper-time intervals and coordinate times support representation comparisons; photon propagation time and emission/reception order support frequency comparisons. Complete generation from nontemporal physical inputs is zero, and physical-time grounding obligations are “back-grounded by primitive-time inputs.”

Evidence of Actual Failure Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 64 root–result dependence pairs. Actual reach from failure of a specific physical-time generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Proposition 8.632 (Primitive-Time Roots and Result Dependence in Peacock 2008).

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(22, 14), \quad R \in \text{Ref}^{\text{audit, rootdep}}, \quad R \notin \text{Ref}^{\text{audit, rootunreach}}.$$

Proof. Closure from Initial Claims 1–14 remains those fourteen nodes. Even with all 57 claims, every transfer requires temporal input, so no time content or result is reached. Adding all 22 time contents as seeds reaches all 14 results. The common reconstruction in Definition 8.115 gives the displayed tuple. $\square$

Proposition 8.633 (Minimal Primitive-Time-Root Support in Peacock 2008). The table gives the number of minimal supports and their cardinality range for each result; there are 27 family members in total.

Proof. Apply Definition 7.46 to all transfers and result dependencies, starting from singleton time-content seeds and empty supports for source formal evidence. By Proposition 7.47, the inclusion-minimal supports of the stable antichains give the tabulated statistics. $\square$

The statistic $(m; a, b)$ gives the number of supports m , minimum cardinality a and maximum cardinality b ; write $(m; a)$ when the bounds coincide.

Table 498: Minimal Primitive-Time-Root Support in Peacock 2008

Result Indices	Support Statistics
1, 3	1; 4
2, 6, 9	1; 3
4, 11	1; 2
5, 8	3; 3, 4
7	2; 3, 5
10	2; 3, 4
12	3; 5
13	5; 4, 5
14	2; 3

Result-Preservation Domain Fix the models, initial conditions, coordinate domains, local approximations, cavity matching and light-path conditions in Table 496. When temporal inputs are realized by generated representations with the same temporal order, intervals, position and momentum histories, and frequency comparisons, all 14 results are preserved within their domains.

Proposition 8.634 (Audit of Time Use in Peacock 2008). There are 22 time contents, 57 claims, 18 transfers and 14 results. The 131 claim/time-content use pairs have 156 role assignments, with 111 obligation atoms. Complete generation and local reach are both zero; 22 primitive-time roots remain and all 14 results depend on roots.

Proof. This follows from the claim, time-content, transfer and result tables and the generation-reach and minimal-support judgments. Obligation atoms are the sum of claims, time contents, transfers and results. $\square$

Proposition 8.635 (Audit-Boundary Cardinalities in Peacock 2008).

$$\text{BoundaryCount}_R^{\text{audit}} = (93, 32).$$

Proof. The 57 claims, 22 time contents and 14 results give 93 nodes; the 18 transfers and 14 result dependencies give 32 edges. $\square$

Individual statements Statement evidence 1

Original text (English):

The idea of an expanding universe can easily lead to confusion, and this note tries to counter some of the more tenacious misconceptions. The worst of these is the ‘expanding space’ fallacy.

Original-source location: Peacock, *A Diatribe on Expanding Space*, arXiv:0809.4573v1, opening of §1. Minimal context: This is the introduction to §1, which enumerates misconceptions concerning cosmic expansion. Attribution: J. A. Peacock.

The statement object is the set of misconceptions accompanying explanations of an expanding universe, particularly the “expanding space” explanation. The predicate is that the idea of an expanding universe readily causes confusion, that the source opposes those misconceptions, and that it identifies “expanding space” as the worst fallacy among them. The explicit applicability domain comprises the Robertson–Walker, Milne, and de Sitter representations compared in the original source, together with its explanations of local dynamics and cosmological redshift.

Statement evidence 2

Original text (English):

This is most clear-cut in the case of closed universes, where the total volume is a well-defined quantity that increases with time, so undoubtedly space is expanding in that case.

Original-source location: Peacock, *A Diatribe on Expanding Space*, arXiv:0809.4573v1, §1, the paragraph discussing the total volume of a closed universe. Minimal context: Immediately after stating that the coordinate representation of the scale factor alone does not demonstrate expansion, the source presents a closed universe as a case that warrants a global expansion description. Attribution: J. A. Peacock.

The statement object is the total volume of spatial slices of a closed universe and its temporal development. The predicate is that, in a closed universe, total volume is well defined and increases with time, and that space therefore expands in that case. The explicit applicability domain is an expanding branch in which the volume of the adopted closed spatial slice increases.

Statement evidence 3

Original text (English):

It should be clear that Minkowski spacetime does not expand – indeed, the very idea that the motion of distant galaxies could affect local dynamics is profoundly anti-relativistic: the equivalence principle says that we can always find a tangent frame in which physics is locally special relativity.

Original-source location: Peacock, *A Datribe on Expanding Space*, arXiv:0809.4573v1, §1, the paragraph discussing local dynamics. Minimal context: This belongs to the examination of whether the global description of “expanding space” entails a local dynamical effect. Attribution: J. A. Peacock.

The statement object is whether Minkowski spacetime expands and the explanation that attributes an effect on local dynamics to the motion of distant galaxies. The predicate is that Minkowski spacetime does not expand and that the idea that the motion of distant galaxies affects local dynamics conflicts with the equivalence principle. The explicit applicability domain is Minkowski spacetime, the local tangent frame supplied by the equivalence principle, and local physics described as special relativity in that frame.

Individual Audits The expanding-space fallacy

Peacock criticizes the treatment of a description of cosmic expansion as an independent dynamical mechanism that pushes local bodies. The source recognizes global growth of closed-slice volume, while calculating local motion from matter and vacuum-energy terms and initial conditions, and explaining redshift through an infinitesimal Doppler chain along the light path.

Claims 16–24 and 42, and Results 1–7 and 12–13, express these distinctions among coordinates, volume, local motion and redshift.

Increase in the total volume of a closed universe

Peacock’s second statement treats increasing closed-slice volume as a global expansion quantity. For a fixed comoving slice, $V = V_0 a^3$ and $\dot{V} = 3HV$, so volume grows on intervals with $H > 0$. The family of slices and $a(t)$ used here are already temporally ordered.

Claims 23–24 and Results 4–5 give this volume change and its distinction from local dynamics.

Minkowski spacetime and local dynamics

In the third statement, Peacock uses the Milne representation of Minkowski spacetime and local tangent frames to criticize the treatment of distant-galaxy motion as an expansion force acting directly on local bodies. The special-relativistic tangent-frame metric is local to a point; tidal relative acceleration over a finite region depends on curvature.

Claims 20, 25–26 and 42, and Results 2 and 6, relate this geometry and approximation domain to the local equation of motion.

8.19.53 Source audit

Audit Inputs

Source and Audit Scope

Source identifier 53.

Source name Bunn–Hogg 2009 Primitive-Time-Root Audit

Bibliographic identification E. F. Bunn and D. W. Hogg, *The kinematic origin of the cosmological redshift*, *American Journal of Physics* 77, 688–694 (2009), doi:10.1119/1.3129103; arXiv:0808.1081v2. $R = \text{Ref}_{\text{Bun09Red}}$.

Audit-scope identifier 53-SC1.

Audit-scope content Abstract, Sections I–V, Equations (1)–(7), Figures 1–3 and notes 15 and 24.

Source-location identifier 53-L1.

Source-location content Sections, equations and figures in the tables, with page numbers from arXiv:0808.1081v2.

Assumptions The RW model uses a positive scale-factor history and a comoving observer family. The local Doppler approximation applies in a low-speed, weak-curvature neighborhood, with distinct curvature and expansion scales. The low-density approximation requires controlled curvature errors along the entire chosen light path. Distant velocity uses the endpoints of the same photon, normalized four-velocities, a selected null path and parallel transport. In general spacetime, the Doppler family is freely falling and the gravitational family satisfies neighboring rest at photon passage. Each family matches the emitter and absorber velocities at the endpoints.

Nontemporal Physical Inputs The physical carriers, spatial configurations, record media and marks, apparatus and physical connections in Claims 1–12 form the generation-initial set. Frequency and time interpretations of records, timelike worldlines and photon-passage order belong to the corresponding time contents and formal claims.

Formal Input Elements

Definition 8.636 (Bunn–Hogg 2009 Physical and Formal Input Claims). The semantic components of the 18 source-form descriptions $\{\text{FormDesc}_{R,p}\}_{p \in [18]}$ are represented by $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(72)$. Let $\text{ClaimBoundaryIdx}_R = ([12], [72])$: Claims 1–12 form the generation-initial set, and all 72 claims form the source formal-evidence environment. The claim table gives their contents and correspondence to source-form descriptions.

Table 499: Bunn–Hogg 2009 Source-Form Descriptions

p	Source-Form Description	Source Location
1	Observed cosmological redshift and the general-relativistic equations	Abstract;§I, pp. 1–2
2	Criticism of reifying the expanding-space description	§I,V, pp. 2, 8
3	Path dependence of distant vectors and relative velocity	§I,III, pp. 1–2, 6–7
4	Local Minkowski structure	§II;note15, pp. 3–4, 9

p	Source-Form Description	Source Location
5	Doppler interpretation of nearby redshift	§II(3), pp. 3–4
6	Milne model and low-density approximation	§I–II;note24, pp. 2–3, 9
7	Robertson–Walker metric and cosmic-time coordinate	§II(1), pp. 3
8	Riemann normal-coordinate approximation and error order	§II(2)–(3);Fig.1, pp. 3–4
9	Sequence of comoving observers and infinitesimal Doppler shifts	§III(4), pp. 5
10	Scale-factor redshift relation	§III(5), pp. 5
11	Distant relative velocity for events on the past light cone	§I;§III;Fig.2, pp. 2, 5–6
12	Parallel transport along a null geodesic	§III(6);Fig.3, pp. 6
13	Successive composition of special-relativistic velocities	§III(7), pp. 6
14	Local flatness in a thin world tube	§III,world-tube argument, pp. 6–7
15	Path-relative interpretation	§III final paragraph, pp. 7
16	Observer-family-specific decomposition in an arbitrary spacetime	§IV, pp. 7–8
17	Selection of the comoving observer family by symmetry and simplicity	§IV final paragraph, pp. 8
18	Criticism of interpretations that transfer a coordinate description to a spatial entity	§V, pp. 8

For cosmic time t , a positive scale factor $a(t)$ and fixed comoving separation χ ,

$$ds^2 = -c^2 dt^2 + a(t)^2 [d\chi^2 + S(\chi)^2 d\Omega^2], \quad H = \frac{\dot{a}}{a}, \quad D(t) = a(t)\chi, \quad v_{\text{coord}} = \frac{dD}{dt}.$$

$S(\chi) = \sin \chi, \chi, \sinh \chi$ corresponds to positive, zero and negative normalized spatial curvature, respectively; $d\Omega^2$ is the unit two-sphere metric. Let P_e, P_o be emission and reception of the same photon, with cosmic times t_e, t_o and endpoint frequencies $\nu_e, \nu_o > 0$. For neighboring comoving observers with local physical separation $\delta\ell = c \delta t$ and relative speed $\delta v = H \delta\ell$, the infinitesimal limit gives

$$\frac{d\nu}{\nu} = -H dt = -\frac{da}{a}, \quad 1 + z = \frac{\nu_e}{\nu_o} = \frac{a(t_o)}{a(t_e)}.$$

Let $U_{\text{e}\parallel}$ be the emitter four-velocity transported to reception along the chosen light path, and U_o the receiver four-velocity. With signature $(-+++)$ and $g(U, U) = -c^2$,

$$\gamma_{\text{rel}} = -\frac{g(U_{\text{e}\parallel}, U_o)}{c^2}.$$

For comoving RW endpoints, the signed longitudinal recession velocity satisfies

$$1 + z = \sqrt{\frac{c + v_{\text{rel}}}{c - v_{\text{rel}}}}, \quad v_2 = \frac{v_1 + u}{1 + v_1 u / c^2}, \quad |v_1|, |u| < c.$$

The locally inertial leading term $z = v/c$ has quadratic velocity corrections and curvature corrections. The curvature error in the metric is quadratic in the domain width, and expansion rate and curvature have independently specified conditions. In the Milne model, spacetime curvature vanishes while the scale factor is proportional to cosmic time. For $a(t_o)/a(t_e) = 5/2$, the example gives $z = 3/2$ and $v_{\text{rel}}/c = 21/29$.

Formal Input Elements

Definition 8.637 (Bunn–Hogg 2009 Time Contents). Let $n_R = 24$. Each time content retains the same objects, endpoints and paths, with the conditions specified by its claims and results.

Table 500: Bunn–Hogg 2009 Time Contents

i	Time Content	Source Location
1	Cosmic time and reference present time	pp. 2–5
2	History of the scale factor and expansion state	pp. 2–3, 5, 8
3	Observer proper time and four-velocity normalization	pp. 3, 6–7
4	Local inertial-time coordinate	pp. 3–4
5	Emission event and its spacetime coordinates	pp. 2, 5–7
6	Reception event and its spacetime coordinates	pp. 2, 5–7
7	Null propagation path from emission to reception	pp. 4–7
8	Timelike worldlines of the observer family and galaxy	pp. 3–8
9	Spacetime domain and temporal extent of the local approximation	pp. 2–4, 6–7, 9
10	Endpoint frequencies of the same photon	pp. 5, 7
11	Infinitesimal frequency shift between neighboring observers	pp. 5, 7
12	Photon-passage order and intervals between neighboring passages	pp. 5–7
13	Accumulation of frequency factors along the light path	pp. 5, 7
14	Total redshift between emission and reception	pp. 5–7
15	Scale-factor expansion rate	pp. 3, 5
16	Galaxy-distance history on simultaneous slices	pp. 2–3, 5
17	Coordinate velocity with respect to cosmic time	pp. 2, 5
18	Endpoint scale-factor ratio for the same light path	pp. 5–6
19	Relation between the present-time slice and the past light cone	pp. 2, 5
20	Relative velocity from transport along the selected light path	pp. 6–7
21	Successive velocity composition along the observer sequence	pp. 6
22	Doppler decomposition using freely falling observers	pp. 7–8
23	Gravitational decomposition using a family satisfying neighboring rest	pp. 7–8
24	Leading-order Doppler relation in a local neighborhood	pp. 3–4

Table 501: Bunn–Hogg 2009 Semantic Components and Temporal Uses

k	Claim	Temporal Use	Description and Location
1	Media and marks of redshift records	$\emptyset$	1; §I–IV, pp. 1–8
2	Media and marks of endpoint-frequency records	$\emptyset$	1, 5, 9–10; §I–IV, pp. 1–8
3	Physical carrier of the photon or electromagnetic signal	$\emptyset$	1, 9, 12, 14; §I–IV, pp. 1–8
4	Physical carrier of the emitting galaxy	$\emptyset$	1, 5, 11; §I–IV, pp. 1–8
5	Physical carrier of the receiving observer or detector	$\emptyset$	1, 5, 9, 11, 16–17; §I–IV, pp. 1–8
6	Record media and marks identifying the emitting apparatus	$\emptyset$	1, 11–12; §I–IV, pp. 1–8
7	Record media and marks identifying the receiving or absorbing apparatus	$\emptyset$	1, 11–12; §I–IV, pp. 1–8
8	Record of the spatial configuration of emitter and receiver	$\emptyset$	3, 11–12; §I–IV, pp. 1–8
9	Spatial configuration of the local observational apparatus	$\emptyset$	4–5, 9, 14, 16–17; §I–IV, pp. 1–8
10	Records of physical connections between signal paths and endpoints	$\emptyset$	3, 11–13; §I–IV, pp. 1–8
11	Record of spatial distance or separation	$\emptyset$	3–5, 14; §I–IV, pp. 1–8
12	Local frequency-comparison apparatus	$\emptyset$	5, 9, 14; §I–IV, pp. 1–8
13	Bunn and Hogg use the general-relativistic metric and equations of light propagation.	$1^I, 2^S, 3^I, 7^C$	1; §I–II(1), pp. 1–3
14	The ratio of endpoint frequencies gives the observed redshift.	$10^{QX}, 14^{QX}$	1; Abstract; §III(5), pp. 1, 5
15	RW coordinates describe the changing scale of spatial slices through the scale-factor history.	$1^I, 2^S$	2; §I–II(1); §V, pp. 2–3, 8
16	The expanding-rubber-sheet account treats scale change as the stretching of a material carrier.	2^S	2; §I, V, pp. 2, 8
17	Bunn and Hogg criticize the reification of cosmic expansion as the stretching of a material rubber sheet.	2^S	2; §I, V, pp. 2, 8
18	Velocity four-vectors at distant events are normalized by the proper time of each body.	$3^I, 5^B, 6^B, 8^{QX}$	3; §III before(6), pp. 6
19	Endpoint four-velocities compared along a specified path define a distant relative velocity.	$3^I, 5^B, 6^B, 7^X, 8^{QX}, 20^{QX}$	3; §III(6), pp. 6–7

k	Claim	Temporal Use	Description and Location
20	The distant comparison selects a null path from the emission event to the reception event.	$5^B, 6^B, 7^X$	3; §III; Fig.3, pp. 6
21	In curved spacetime, comparing endpoint four-velocities by parallel transport depends on the chosen path.	$3^I, 7^X, 8^{QX}, 20^{QX}$	3; §I, III, pp. 1, 6–7
22	Locally inertial coordinates give the Minkowski metric at a point.	$3^I, 4^I, 9^B$	4; §II(2); note15, pp. 3–4, 9
23	In a finite normal neighborhood, spacetime curvature and domain size determine the metric-approximation error.	$4^I, 9^B$	4; §II(2)–(3), pp. 3–4
24	The nearby galaxy and observer worldlines are compared in the same local approximation domain.	$8^X, 9^B$	5; §II; Fig.1, pp. 3–4
25	A nearby redshift is determined by the frequency ratio measured by the specified emitter and observer.	$10^{QX}, 14^{QX}$	5; §II(3), pp. 3–4
26	In a low-speed, weak-curvature neighborhood, the galaxy velocity compared in the local inertial frame gives the leading redshift.	$4^I, 8^{QX}, 9^B, 24^{QX}$	5; §II(2)–(3), pp. 3–4
27	The Milne model expresses the interior of the future light cone of Minkowski spacetime in open RW coordinates with a linear scale factor.	$1^I, 2^S, 3^I, 8^X$	6; §I–II; note24, pp. 2–3, 9
28	The weak-curvature approximation applies on a light path and neighborhood where curvature errors can be controlled uniformly.	$7^X, 9^B$	6; §I–II, pp. 2–4
29	Spacetime curvature vanishes in the Milne model; a low-density model admits the special-relativistic approximation where the weak-curvature condition holds along the entire chosen light path.	$1^I, 2^S, 3^I, 7^X, 8^{QX}, 9^B$	6; §I–II; note24, pp. 2–3, 9
30	The RW metric defines spacetime intervals using cosmic time and the scale-factor history.	$1^I, 2^S, 3^I$	7; §II(1), pp. 3
31	Cosmic time corresponds to proper time along comoving worldlines.	$1^I, 3^I, 8^X$	7; §II(1), pp. 3
32	Riemann normal coordinates centered on the observing event define a local inertial time T.	$4^I, 6^B, 9^B$	8; §II(2); Fig.1, pp. 3–4
33	Within that normal neighborhood, the metric is approximated by the Minkowski form.	$3^I, 4^I, 9^B$	8; §II(2), pp. 3–4

k	Claim	Temporal Use	Description and Location
34	The normal-neighborhood metric error is quadratic in the ratio of domain width to curvature radius, while the leading nearby-redshift term is set by local relative velocity.	$4^I, 8^{QX}, 9^B, 24^Q$	8; §II(2)–(3), pp. 3–4
35	Comoving observers are ordered by photon passage along the same light path.	$7^C, 8^X, 12^{IX}$	9; §III before(4), pp. 5
36	In the infinitesimal limit between neighboring comoving observers, $d\nu/\nu = -H dt$.	$11^{QX}, 12^I, 15^Q$	9; §III(4), pp. 5
37	The scale factor $a(t)$ expresses the cosmic scale in comoving coordinates as a function of cosmic time.	$1^I, 2^S$	10; §II(1);§III(5), pp. 3, 5
38	Evaluating the positive scale-factor history at the emission and reception times of the same photon gives the endpoint ratio $a(t_o)/a(t_e)$.	$2^S, 5^B, 6^B, 18^{QX}$	10; §III(5), pp. 5
39	Redshift between comoving RW endpoints satisfies $1 + z = a(t_o)/a(t_e)$.	$14^{QX}, 18^{QX}$	10; §III(5), pp. 5
40	An emission event on the receiving event's past light cone specifies the observed state of the galaxy.	$5^B, 6^B, 7^C, 19^{BX}$	11; §I;§III after Fig.2, pp. 2, 5
41	In the specified causal geometry, the receiving event's past light cone is distinguished from the present cosmic-time slice.	$1^B, 6^B, 19^{BX}$	11; §I;§III after Fig.2, pp. 2, 5
42	The distant-velocity comparison takes the galaxy at emission and the observer at reception as its endpoints.	$5^B, 6^B, 8^{QX}, 20^{QX}$	11; §III(6);Fig.3, pp. 5–6
43	Photon propagation is represented by a selected future-directed null geodesic joining emission and reception.	$5^B, 6^B, 7^C$	12; §III;Fig.3, pp. 5–6
44	Parallel transport by the connection carries the velocity four-vector along the selected light path to the comparison point.	$3^I, 7^X, 8^{QX}, 20^{QX}$	12; §III(6), pp. 6
45	The normalized emitter four-velocity is transported along the light path to reception, where its inner product with the receiver four-velocity determines the relative Lorentz factor and speed.	$3^I, 5^B, 6^B, 7^X, 8^{QX}, 20^{QX}$	12; §III(6), pp. 6
46	The collinear velocities of the galaxy and observers are compared in neighboring local Lorentz frames.	$8^{QX}, 12^{IX}$	13; §III(7), pp. 6
47	The local velocity sequence along the light path is composed successively using $v_2 = (v_1 + u)/(1 + v_1 u/c^2)$.	$8^{QX}, 12^{IX}, 21^{QX}$	13; §III(7), pp. 6

k	Claim	Temporal Use	Description and Location
48	Velocity composition follows the order in which the same photon passes the observers.	$7^C, 12^{IX}, 21^{QX}$	13; §III(7), pp. 6
49	A thin world tube is chosen around the selected null geodesic.	$7^X, 9^B$	14; §III, world-tube argument, pp. 6–7
50	The local world-tube approximation controls errors according to tube width and curvature.	$7^X, 9^B$	14; §III, world-tube argument, pp. 6–7
51	Bunn and Hogg explain the Doppler interpretation along the light path using locally inertial comparisons within a thin world tube.	$3^I, 7^X, 8^{QX}, 9^B, 20^{QX}$	14; §III, world-tube argument, pp. 6–7
52	Distant relative velocity uses the comparison of endpoint four-velocities along a specified comparison path.	$3^I, 7^X, 8^{QX}, 20^{QX}$	15; §III after(7), pp. 6–7
53	Bunn and Hogg recognize that light-path transport and comoving-coordinate velocity conventions yield different redshift decompositions.	$17^{QX}, 20^{QX}, 22^{QX}, 23^{QX}$	15; §III final paragraph, pp. 7
54	The general-spacetime argument uses a specified light path joining the endpoints and an observer family constructed around it.	$5^B, 6^B, 7^C, 8^X$	16; §IV, pp. 7
55	The smooth observer family is chosen to be at rest with the emitter at emission and with the absorber at reception.	$5^B, 6^B, 7^C, 8^X, 12^{IX}$	16; §IV, pp. 7
56	For a freely falling family satisfying the endpoint conditions, infinitesimal shifts between local inertial frames accumulate as Doppler shifts.	$5^B, 6^B, 7^C, 8^X, 12^{IX}, 22^{QX}$	16; §IV, Doppler family, pp. 7–8
57	The family used for the gravitational decomposition is an accelerated family whose neighboring pairs are momentarily at rest relative to each other at photon passage.	$5^B, 6^B, 7^C, 8^X, 12^{IX}, 23^{QX}$	16; §IV, gravitational family, pp. 7–8
58	For a smooth observer family matching the emitter and absorber at the endpoints and satisfying neighboring rest at photon passage, local shifts accumulate as gravitational shifts.	$5^B, 6^B, 7^C, 8^X, 12^{IX}, 23^{QX}$	16; §IV, gravitational family, pp. 7–8
59	The cosmological comoving family preserves the spatial symmetries of RW spacetime.	$1^I, 2^S, 8^X$	17; §IV final paragraph, pp. 8

k	Claim	Temporal Use	Description and Location
60	Bunn and Hogg cite the simplicity of the comoving description as a reason for choosing that family.	$1^I, 2^S, 8^X$	17; §IV final paragraph, pp. 8
61	Cosmic time and the scale factor in the RW metric define the freely falling comoving family with constant spatial coordinates.	$1^I, 2^S, 8^X$	17; §II(1); §IV final paragraph, pp. 3, 8
62	The same spacetime geometry is described in comoving and locally inertial coordinates.	$1^I, 2^S, 4^I, 9^B$	18; §II; Fig.1; §V, pp. 3–4, 8
63	Reifying the expanding rubber sheet attributes coordinate-dependent scale change to the properties of a spatial substance.	$1^I, 2^S$	18; §I, V, pp. 2, 8
64	Bunn and Hogg’s criticism targets the treatment of the rubber sheet as a material entity while retaining descriptions of scale change in terms of the metric, observers and coordinates.	$1^I, 2^S, 4^I, 8^X, 9^B$	18; §I, V, pp. 2, 8
65	Differentiating the positive scale-factor history with respect to cosmic time gives $H = \dot{a}/a$.	$1^I, 2^S, 15^Q$	7, 9–10; §II(1); §III(4)–(5), pp. 3, 5
66	For fixed comoving separation χ , the simultaneous proper-distance history is $D(t) = a(t)\chi$.	$1^I, 2^S, 16^{QS}$	3, 7; §I final paragraph; §II(1); Fig.2, pp. 2–3, 5
67	Differentiating the specified distance history with respect to the same cosmic time gives the coordinate velocity dD/dt .	$1^I, 16^{QS}, 17^Q$	3, 7; §I final paragraph; Fig.2, pp. 2, 5
68	Ordered infinitesimal frequency factors are accumulated in the continuous limit along the emission-to-reception interval of the same light path.	$5^B, 6^B, 7^C, 11^{QX}, 12^{IX}, 13^{QX}$	9–10; §III(4)–(5); §IV, pp. 5, 7
69	Positive endpoint frequencies of the same photon define $1 + z = \nu_e/\nu_o$.	$10^{QX}, 14^{QX}$	1, 10, 16; §III(5); §IV, pp. 5, 7
70	The inverse of the accumulated emission-to-reception frequency factor gives $1 + z$.	$13^{QX}, 14^{QX}$	9–10; §III(4)–(5), pp. 5
71	For the same fixed light path and endpoints, the continuous limit of local velocity composition agrees with comparison by parallel transport of normalized four-velocities.	$3^I, 5^B, 6^B, 7^X, 20^{QX}, 21^{QX}$	12–13; §III(6)–(7), pp. 6
72	The signed longitudinal recession velocity compared along the light path between comoving RW endpoints gives $1 + z = \sqrt{(c + v_{\text{rel}})/(c - v_{\text{rel}})}$.	$14^{QX}, 20^{QX}$	12, 14; §III(6), pp. 6

Superscripts on the temporal indices in the claim table denote the use roles below. Time differentiation and local velocity composition are constituent operations on the specified functions and observer sequences.

Table 502: Bunn–Hogg 2009 Time-Use Roles

Code	Role	Use Count
I	Interval	49
Q	Quantity	54
X	Cross-position correspondence	84
K	Clock reading	0
C	Causality	11
S	State evolution	19
P	Probabilistic evolution	0
O	Observation record	0
B	Reference boundary	49

Table 503: Bunn–Hogg 2009 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–2, 4, 16	member	instant, order_relation
3	member	interval_boundary, endpoint
5–6	member	endpoint, instant
7	member	order_relation, orientation, endpoint
8	member	order_relation, orientation
9	member	interval_boundary, instant
10–11, 13–15, 17–18, 20–24	dependency	$\emptyset$
12	member	order_relation, interval_boundary, endpoint
19	member	order_relation, endpoint, instant

Table 504: Bunn–Hogg 2009 Finite Cardinalities

Object	Cardinality
Time contents	24
Claims	72
Auxiliary objects	0
Transfers	18
Results	18
Claim/time-content uses	205
Role assignments	266
Obligation atoms	132

Object	Cardinality
Audit-boundary nodes	114
Audit-boundary edges	36

Dependency Inputs

Definition 8.638 (Bunn–Hogg 2009 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [18]$. The table gives time inputs, time outputs, claim support and source locations.

Table 505: Bunn–Hogg 2009 Time-Content Transfers

j	Time In i	Time Out i	Claim Support k	Transfer Content and Source Location
1	1–2	15	65	From scale factor to expansion rate. §II(1);§III(4)–(5)
2	1–2	16	66	From comoving separation to distance history. §I final paragraph;§II(1);Fig.2
3	1, 16	17	67	From distance history to coordinate velocity. §I final paragraph;Fig.2
4	4, 8–9	24	26	Nearby Doppler relation. §II(2)–(3)
5	12, 15	11	36	Infinitesimal shift between neighboring comoving observers. §III(4)
6	5–7, 11–12	13	68	Shift accumulation along the light path. §III(4)–(5);§IV
7	2, 5–6	18	38	Endpoint scale-factor ratio. §III(5)
8	18	14	39	From scale-factor ratio to total redshift. §III(5)
9	10	14	69	From endpoint frequencies to total redshift. §III(5);§IV
10	13	14	70	From accumulated frequency factor to total redshift. §III(4)–(5)
11	3, 5–8	20	45	Relative velocity by light-path transport. §III(6)
12	8, 12	21	47	Successive composition of local velocities. §III(7)
13	3, 5–7, 21	20	71	Local velocity composition and light-path transport. §III(6)–(7)
14	20	14	72	Doppler factor of the transported velocity. §III(6)
15	5–8, 12	22	56	Decomposition using freely falling observers. §IV,Doppler family
16	5–8, 12	23	58	Decomposition using a neighboring-rest family. §IV,gravitational family
17	1, 6	19	41	Distinction between the present slice and observed light cone. §I;§III after Fig.2

j	Time In i	Time Out i	Claim Support k	Transfer Content and Source Location
18	1–2	8	61	Specification of the comoving observer family. §II(1);§IV final paragraph

Grounding Conditions

Table 506: Bunn–Hogg 2009 Conditions for Transfers

Trans-fer Index	Conditions
1	Same cosmic time and positive differentiable $a(t)$.
2	Fixed comoving separation χ and simultaneous RW slices.
3	Differentiate the specified distance function using the same cosmic time.
4	Locally inertial frame, low speed and a weak-curvature neighborhood containing the ray; leading order with second-order corrections in speed divided by c and in domain width divided by curvature radius.
5	Local physical separation $\delta r = c \delta t$, $\delta v = H \delta r$ and the infinitesimal-interval limit.
6	Same photon and endpoints, ordered neighboring-observer sequence and a convergent continuous integral.
7	Positive $a(t)$ and emission t_e and reception t_o of the same photon.
8	Comoving RW emitter and receiver on the same null path.
9	Positive frequencies of the same photon measured by the corresponding endpoint observers.
10	Product of positive emission-to-reception frequency ratios, whose inverse is $1 + z$.
11	Metric and connection, normalized four-velocities at the same endpoints and selected light path; $\gamma_{\text{rel}} = -g(U_{e\parallel}, U_o)/c^2$ for signature $-+++$.
12	Neighboring local frames on the same light path, collinear speeds $ u $, $ v_1 < c$ and photon-passage order.
13	Same path and endpoints, normalized four-velocities and the continuous limit of the local velocity sequence.
14	Comoving RW endpoints, signed longitudinal recession along the chosen light path and $ v_{\text{rel}} < c$.
15	Smooth freely falling family at rest with the emitter and absorber at the endpoints, on infinitesimal segments of the same ray.
16	Endpoint matching, neighboring rest at photon passage and a smooth family on the same ray.
17	Specified spacetime metric and causal structure, reference t_0 and receiving event.
18	RW metric, worldlines of constant spatial coordinates and positive scale factor.

Results

Definition 8.639 (Bunn–Hogg 2009 Results). Let $\text{Idx}_R^{\text{audit,result}} = [18]$.

Table 507: Bunn–Hogg 2009 Results

q	Result	Source Location
1	Observed redshift and its general-relativistic description	§I–II(1); Abstract; §III(5), pp. 1–3, 5
2	Criticism of reifying the expanding rubber sheet	§I–II(1); §V; §I, V, pp. 2–3, 8
3	Path dependence of distant four-velocity comparison	§III before(6); §III(6); §III; Fig. 3; §I, III, pp. 1, 6–7
4	Local Minkowski structure and a finite neighborhood	§II(2); note 15; §II(2)–(3), pp. 3–4, 9
5	Doppler interpretation of a nearby redshift	§II; Fig. 1; §II(3); §II(2)–(3), pp. 3–4
6	Doppler interpretation in the Milne model	§I–II; note 24; §III(6), pp. 2–3, 6, 9
7	Weak-curvature approximation in a low-density model	§I–II; §I–II; note 24, pp. 2–4, 9
8	RW redshift and the scale-factor ratio	§II(1); §II(1); §III(5); §III(5), pp. 3, 5
9	Riemann normal-coordinate approximation and error order	§II(2); Fig. 1; §II(2); §II(2)–(3), pp. 3–4
10	Accumulation of infinitesimal Doppler shifts	§III before(4); §III(4); §III(4)–(5); §IV; §III(4)–(5), pp. 5, 7
11	Relative velocity of the galaxy at emission and observer at reception	§I; §III after Fig. 2; §III(6); Fig. 3; §III(6), pp. 2, 5–6
12	Distinction between the distant present and the observed event	§I; §III after Fig. 2, pp. 2, 5
13	Parallel transport of the velocity four-vector along the light path	§III; Fig. 3; §III(6), pp. 5–6
14	Transported velocity and the special-relativistic Doppler relation	§III, world-tube argument; §III(6), pp. 6–7
15	Successive local velocity composition and parallel transport	§III(7); §III(6)–(7), pp. 6
16	Doppler and gravitational decompositions by observer family	§IV; §IV, Doppler family; §IV, gravitational family, pp. 7–8
17	Symmetry and simplicity of the comoving family	§IV final paragraph; §II(1); §IV final paragraph, pp. 3, 8

q	Result	Source Location
18	Distinction between coordinate-dependent scale change and rubber-sheet reification	§II;Fig.1;§V; §I,V, pp. 2–4, 8

Table 508: Bunn–Hogg 2009 Conditions for Results

Result Index	Conditions and Domain
1	Specified metric and light path, and corresponding endpoint frequencies.
2	The criticism concerns an account that treats cosmic scale change as the stretching of a material carrier.
3	Normalized endpoint four-velocities and a specified comparison path in the same spacetime.
4	Inertial coordinates at a point and a finite normal neighborhood with specified curvature errors.
5	Same locally inertial frame, low-speed and weak-curvature domain, and specified endpoint frequencies.
6	Milne domain inside the future light cone, comoving endpoints and velocity comparison in Minkowski spacetime.
7	Model domain with uniformly controlled curvature errors along the entire selected light path and its neighborhood.
8	Positive scale factor and comoving emitter and receiver on the same light path.
9	Specified normal neighborhood, metric error quadratic in the ratio of domain width to curvature radius and leading order in local velocity.
10	Comoving observer sequence on the same ray, continuous limit of local shifts and both endpoints.
11	Selected null path and normalized four-velocities at emission and reception.
12	Specified causal geometry, receiving event’s past light cone and present-time slice.
13	Same endpoints, light path and connection, and proper-time-normalized four-velocities.
14	Comoving RW endpoints, specified path and longitudinal velocity; the world-tube explanation uses a local approximation with controlled curvature errors.
15	Collinear local velocity sequence on the same path, endpoints, normalized four-velocities and the continuous limit in photon-passage order.
16	Smooth families matching emitter and absorber at the endpoints; the Doppler family is freely falling, and the gravitational family satisfies neighboring rest at photon passage.
17	Freely falling comoving family of constant spatial coordinates in the RW metric.
18	Retain coordinate descriptions of the same geometry and the local approximation domain when assessing attribution of scale change to a material rubber sheet.

Result Dependence

Definition 8.640 (Bunn–Hogg 2009 Result-Dependence Hyperedges). Each row defines a hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 509: Bunn–Hogg 2009 Result-Dependence Hyperedges

q	Time Premises i	Claim Support k
1	1–3, 7, 10, 14	13–14
2	1–2	15–17
3	3, 5–8, 20	18–21
4	3–4, 9	22–23
5	4, 8–10, 14, 24	24–26
6	1–3, 8, 14, 20	27, 72
7	1–3, 7–9	28–29
8	1–3, 5–6, 14, 18	30, 37–39
9	3–4, 6, 8–9, 24	32–34
10	5–8, 11–15	35–36, 68, 70
11	3, 5–8, 19–20	40, 42, 45
12	1, 5–7, 19	40–41
13	3, 5–8, 20	43–45
14	3, 7–9, 14, 20	49–51, 72
15	3, 5–8, 12, 20–21	46–48, 71
16	5–8, 12, 22–23	54–58
17	1–2, 8	59–61
18	1–2, 4, 8–9	62–64

Proposition 8.641 (Bunn–Hogg 2009 Audit-Boundary Cardinalities).

$$\text{BoundaryCount}_R^{\text{audit}} = (114, 36).$$

Proof. The 72 claims, 24 time contents and 18 results give the node count; the 18 transfers and 18 result dependencies give the edge count. $\square$

Evidence Hierarchy Bunn and Hogg analyze redshift using metrics, coordinates, light paths, endpoint velocities and observer families. The paper reports zero newly acquired direct measurements.

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 0; remaining primitive-time roots 24.

Result Dependence All 18 results depend on primitive-time roots. There are 104 root–result dependence pairs, of which 68 are necessary.

Evidential Force Bunn and Hogg’s argument covers the nearby Doppler approximation, cosmological redshift, four-velocity comparison along a light path, observer-family decompositions and criticism of rubber-sheet reification. The result tables specify the conditions and source locations.

Placement of Grounding Obligations Cosmic time and the scale-factor history support expansion rate and coordinate velocity; proper time and four-velocities support velocity comparison; endpoints, the light path and passage order support frequency comparisons and shift accumulation. Complete generation from nontemporal physical inputs is zero, and physical-time grounding obligations are “backgrounded by primitive-time inputs.”

Evidence of Actual Failure Actual failure of a specific physical-time generation condition and the resulting change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by 104 root–result dependence pairs. Actual reach from failure of a specific physical-time generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns each temporal use and the physical-time generation conditions required before it. If actual failure of a specific condition is established, recalculation concerns the results dependent on it.

Audit Conclusions

Proposition 8.642 (Primitive-Time Roots and Result Dependence in Bunn–Hogg 2009).

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(24, 18), \quad R \in \text{Ref}^{\text{audit, rootdep}}, \quad R \notin \text{Ref}^{\text{audit, rootunreach}}.$$

Proof. Closure from Initial Claims 1–12 remains those twelve nodes. Even with all 72 claims, every transfer requires temporal input, so no time content or result is reached. Adding all 24 time contents as seeds reaches all 18 results. The common reconstruction in Definition 8.115 gives the displayed tuple. $\square$

Proposition 8.643 (Physical-Time Admissibility in Bunn–Hogg 2009). All 24 time contents are primitive-time inputs and lack physical-time admissibility. All 18 results depend on them.

Proof. Apply Theorem 7.11 and Corollary 7.12 to the generation-reach and root-dependence judgments. $\square$

Proposition 8.644 (Minimal Primitive-Time-Root Support in Bunn–Hogg 2009). There are 31 minimal supports in total; the table gives the statistics for each result.

Proof. Apply Definition 7.46 to all transfers and result dependencies, starting from singleton time-content seeds and empty supports for source formal evidence. By Proposition 7.47, the inclusion-minimal supports of the stable antichains give the tabulated statistics. $\square$

$(m; a, b)$ gives the number of supports and their minimum and maximum cardinalities; write $(m; a)$ when the bounds coincide.

Table 510: Minimal Primitive-Time-Root Support in
Bunn–Hogg 2009

Result Indices	Support Statistics
1, 7–8	1; 5
2, 17	1; 2
3, 9, 13, 16	2; 5, 6
4	1; 3
5	2; 4, 5
6	2; 4, 6
10	2; 6
11	3; 6
12, 18	1; 4
14	4; 5, 7
15	2; 6, 7

Result-Preservation Domain Fix the metrics, observer families, endpoints, light paths, parallel transport, local approximations and composition order specified in the result tables. When temporal inputs are realized by generated representations with the same temporal order, intervals, worldline and scale-factor histories, and frequency comparisons, all 18 results are preserved within their domains.

There are 24 time contents, 72 claims, 18 transfers and 18 results. The 205 claim/time-content use pairs have 266 role assignments, with 132 obligation atoms. Obligation atoms are the sum of claims, time contents, transfers and results.

Individual statements Excerpt from the original text (English):

the rate of change of the distance ... with respect to the cosmic time coordinate

The original-source location is E. F. Bunn and D. W. Hogg, *The kinematic origin of the cosmological redshift*,
arXiv:0808.1081v2, §I, final paragraph on p. 2.

Excerpt from the original text (English):

We argue that, contrary to this view, the most natural interpretation of the redshift is as a Doppler shift, or rather as the accumulation of many infinitesimal Doppler shifts.

The original-source location is the abstract of the same source.
Excerpt from the original text (English):

It therefore has no rightful place in the theory.

The original-source location is §I of the same source, in the paragraph beginning *The expanding rubber sheet*.

Statement Objects The distance rate defined by cosmic time, the interpretation of redshift as accumulated Doppler shifts, and accounts that reify an expanding rubber sheet.

Minimal Context “It” in the third quotation refers to the expanding rubber sheet in the preceding sentence. The source uses the RW metric and recognizes multiple decompositions based on observer families and distant-velocity conventions.

Attribution E. F. Bunn and D. W. Hogg.

Predicates and Explicit Domains The rate in the first quotation is evaluated on the chosen present cosmic-time slice. The second expresses an interpretive preference grounded in the symmetry and descriptive simplicity of the comoving family. The third criticizes an account that treats scale change as the stretching of a material rubber sheet.

Individual Audit First Quotation: Coordinate Velocity with Respect to Cosmic Time

The quantity defined by Bunn and Hogg in the first quotation is the derivative of a selected distance history with respect to cosmic time. The distance rate on the present cosmic-time slice and the velocity comparing the galaxy at emission with the observer at reception along the observed light path use different endpoints and comparison conventions.

This derivative corresponds to Time Contents 1 and 16–17 and Transfer 3; the distinction between the present slice and past light cone corresponds to Time Content 19 and Result 12.

Second Quotation: Grounds for Choosing the Doppler Interpretation

Bunn and Hogg regard the Doppler interpretation as the most natural because the comoving family preserves cosmological symmetry and simplifies the description. Source Equations (4)–(5) accumulate infinitesimal frequency ratios along the same light path, while Equations (6)–(7) give relative velocity through light-path transport and successive velocity composition. The gravitational decomposition in Section IV uses a different family satisfying endpoint matching and neighboring rest at photon passage.

Frequency accumulation, the scale-factor ratio and velocity comparison correspond to Transfers 5–16 and Results 8, 10–11 and 13–16; the grounds for choosing the comoving family correspond to Result 17.

Third Quotation: Criticism of Rubber-Sheet Reification

Bunn and Hogg criticize treating the expanding rubber sheet as a material entity. The RW metric used by the same source specifies geometry through cosmic time and the scale-factor history and supports comparisons in local coordinates and observer-family decompositions.

This criticism corresponds to Claims 15–17 and 62–64 and Results 2 and 18. Cosmic time, the scale-factor history, proper time, endpoints, the light path and frequency comparison belong to the generation-reach and root-dependence judgments for the 24 time contents.

Common Audit with an Applicability Scope The source’s redshift account is realized by 18 transfers and 18 results using prespecified temporal inputs. Generation reach from nontemporal physical inputs is zero, and all 18 results depend on primitive-time

roots. Each result's force is determined by the model, endpoints, observer family, light path and approximation domain in the result tables.

8.19.54 Source audit

Audit input

Original source and audit scope

Source identifier 54.

Source name Primitive-time-root audit of Kiefer 2009

Bibliographic identification Let $R = \text{Ref}_{\text{Kie09TQG}}$ denote Claus Kiefer, *Does Time Exist in Quantum Gravity?*, arXiv:0909.3767v1 [gr-qc] (2009). The audit domain comprises the original-source abstract, original-source §§1–4, original-source footnotes, and original-source equations.

Formal Objects Time parameters, temporal relations, temporal boundaries and references to absent time in the physical arguments and historical account of time concepts are represented by the 35 time contents in Table 511. Repeated references to the same model parameter or temporal relation have the same index. The associated descriptions and claim nodes specify the subsystems, branches, approximation domains and relations in use. Physical carriers, operations, boundary conditions and interpretive assessments are represented by 69 claim nodes; no separate auxiliary objects are indexed.

Assumptions The Wheeler–DeWitt representation uses quantum universality, the classical limit of general relativity and the adopted canonical formalism. The Mott and semiclassical constructions use the specified subsystem separation, phase and approximation. Interference suppression uses environmental degrees of freedom and selected model parameters; the time direction uses the small-scale product boundary and semiclassical approximation. The recollapsing-model conclusions use their large-scale boundary and interference condition near the classical turning point. The transfer and result tables specify where each condition is required.

Non-Temporal Physical Inputs Claims 1–10 in Table 513 place the non-temporal configurations and records of the spatial metric, non-gravitational fields, heavy and light subsystems, gravitational degrees of freedom, environment, subsystem partition, spatial scale, boundaries and physical correlations in the physical initial set. Claims 11–69 retain the source equations, model conditions, interpretations and audit distinctions as evidence-only components.

Time Contents External time, dynamical spacetime, model clocks, time direction, histories and temporal ends are distinguished. Mentioning time in a statement about its absence or in a historical account does not by itself make that time a necessary input to a physical operation.

Formal Input Elements

Definition 8.645 (Time contents in Kiefer 2009). Let $\text{ldx}_R^{\text{audit,time,src}} = [35]$. Table 511 defines the contents and their source pages. Page numbers refer to the printed pages of the bibliographically identified source.

Table 511: Time contents and source locations in Kiefer 2009

i	Time content	Source pages
1	Absolute external time parameter of standard quantum theory	1, 2, 5, 6
2	Dynamical spacetime of general relativity	1, 2, 5
3	External time excluded from the fundamental quantum-gravity description	1, 5, 7
4	Subsystem-relative semiclassical time parameter	1, 6, 7, 8
5	Periodic motion in the pre-Newtonian account	1
6	Countable ticks corresponding to periods	1
7	Newtonian absolute continuous time	1, 2
8	Minkowski spacetime as a fixed background	2
9	Quantum-state evolution with respect to external time	2, 4
10	Time-reversal relation of quantum evolution	2
11	Conservation of probability in time	2
12	Dynamical backreaction between matter, fields and spacetime	2
13	Irreversibility and the direction of time	3, 7, 8
14	End of time in singular regions of general relativity	3
15	Temporal meaning of the speed of light c	3, 6
16	Parameterized classical-particle trajectory	5
17	External time absent from Mott's total-system equation	5
18	Mott parameter defined from the heavy-subsystem phase	5
19	Proportionality between the Mott-parameter derivative and the directional derivative	5
20	Light-subsystem evolution with respect to the Mott parameter	5
21	Directional derivative defining semiclassical time	6, 7
22	Conditioned subsystem evolution with respect to semiclassical time	6, 7
23	Inverse-time meaning of the Hubble constant H_0	6
24	Validity boundary of semiclassical time	6, 7
25	Temporal tracking of a wave packet along a classical trajectory	7
26	Classical expansion, turning and recollapse history	7, 8
27	Order of approach to the big-brake singularity	7
28	End of classical evolution before the singularity is reached	7
29	Entropy growth inherited by semiclassical time	8
30	Order of cosmic expansion corresponding to the time direction	8
31	Big Bang end of the classical trajectory	8

i	Time content	Source pages
32	Big Crunch end of the classical trajectory	8
33	Observer-history condition under which recollapse is not observed	8
34	Region of the history corresponding to the classical turning point	8
35	Future end of classical evolution in the recollapsing model	8

The source-location keys in Table 512 are: 1, the abstract and source §1; 2, source §1; 3, source §2; 4, source §3; 5, source §4.

Definition 8.646 (Physical and formal input claims in Kiefer 2009). Table 512 defines the contents and locations of $\{\text{FormDesc}_{R,p}\}_{p \in [59]}$. Let $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(69)$, with components as specified in Table 513. The initial-set/evidence-environment pair is $\text{InitEnv}_R^{\text{phys}} = \text{ClaimPrefix}_R(10)$. Claims 1–10 are initial and evidence components; claims 11–69 are evidence-only components.

The corresponding time contents identify referents of the source-form descriptions. The necessary inputs of each transfer and result are specified in the respective dependency tables.

Table 512: Physical and formal input claims in Kiefer 2009

p	Physical or formal input claim	Corresponding time contents	Source-location key
1	Standard quantum theory uses an absolute external time parameter	$\{1\}$	1
2	General relativity treats spacetime as a dynamical variable	$\{2, 12\}$	1
3	Combining standard quantum theory with general relativity produces the tension known as the problem of time	$\{1, 2\}$	1
4	The assumptions stated by Kiefer 2009 are universality of the linearity of quantum theory and recovery of general relativity in the classical limit	$\{2\}$	1
5	Kiefer 2009 directly concludes fundamental timelessness, approximate semiclassical time, and a correlation between entropy and the size of the universe	$\{3, 4, 21\}$	1
6	Pre-Newtonian periodic and countable time is presented as historical background	$\{5, 6\}$	2
7	Newtonian absolute continuous time is required by the historical formulation of mechanics described by Kiefer 2009	$\{7\}$	2
8	In the contrast drawn by Kiefer 2009, Minkowski spacetime is a nondynamical background	$\{8\}$	2

p	Physical or formal input claim	Corresponding time contents	Source-location key
9	The source describes the Schrödinger equation as using external time t and the imaginary unit i , and as deterministic and time-reversal invariant with respect to t .	$\{1, 9, 10\}$	2
10	Kiefer 2009 connects conservation of probability with the existence of time and complex structure	$\{1, 9, 11\}$	2
11	The contrast between quantum theory and general relativity is reformulated in terms of background independence	$\{1, 2\}$	2
12	Universality of superposition is an empirically supported assumption open to future testing	$\emptyset$	3
13	Kiefer treats the argument from universal gravitational coupling to quantization as an argument of naturalness, not a logical inference.	$\{2\}$	3
14	Kiefer states that singularity theorems imply an end of classical time under their assumptions and presents this as a motivation for quantum gravity.	$\{14\}$	3
15	Kiefer rejects the argument that quantum-gravitational effects are restricted to the Planck scale.	$\emptyset$	3
16	Kiefer states that decoherence explains the classical appearance of most gravitational-field states.	$\{9, 22\}$	3
17	Kiefer describes the construction of a background-independent quantum theory of gravity as an outstanding task.	$\emptyset$	3
18	The direct-quantization and interaction-unification approaches are distinct classifications	$\emptyset$	3
19	Kiefer 2009 claims that an inference concerning temporal structure from tested physics and universality is valid	$\emptyset$	3
20	The Wheeler–DeWitt equation is obtained by analogy with the gravitational Hamilton–Jacobi structure	$\{2\}$	3
21	The state depends on the three-metric and nongravitational fields and obeys the spatial diffeomorphism constraint	$\emptyset$	3
22	The Wheeler–DeWitt equation may fail fundamentally at the Planck scale and is claimed to be a reliable approximation at large scales	$\{2\}$	3
23	The Wheeler–DeWitt wave functional does not depend on an external time parameter t .	$\{3\}$	3

p	Physical or formal input claim	Corresponding time contents	Source-location key
24	Kiefer 2009 compares the disappearance of classical spacetime through quantization with the disappearance of particle trajectories	$\{1, 2, 16\}$	3
25	The LCTR audit distinguishes the absence of classical trajectories or spacetime from a representation from the nonexistence of physical spacetime.	$\{2, 3, 16\}$	3
26	Kiefer directly asserts the fundamental timelessness of quantum gravity	$\{3\}$	3
27	Fundamental timelessness is argued for within the adopted canonical representation, using quantum universality and the large-scale limit of general relativity.	$\{2, 3\}$	3
28	The Mott example begins with a time-independent equation for the total system	$\{17\}$	4
29	The heavy-subsystem phase in Mott's construction defines a directional derivative proportional to the time derivative.	$\{18, 19\}$	4
30	The resulting time-dependent equation applies to the light subsystem	$\{18, 20\}$	4
31	The Mott analogy assigns the gravitational part to the clock subsystem. Source footnote 2 also allows some gravitational degrees of freedom to remain quantum and some non-gravitational variables to enter the clock definition.	$\{4, 21, 22\}$	4
32	Semiclassical time depends on subsystem separation, a phase branch, and approximation order	$\{4, 21, 22\}$	4
33	Kiefer 2009 acknowledges quantum-gravity correction terms	$\{4, 22\}$	4
34	Kiefer conjectures that Hilbert-space structure may itself be approximate	$\{1\}$	4
35	A real Wheeler–DeWitt solution decomposes into complex-conjugate components	$\{4\}$	4
36	Through decoherence, a selected complex component becomes dynamically independent within the reduced description	$\{4, 22\}$	4
37	The LCTR audit distinguishes dynamical independence of complex components from unique selection of one realized branch.	$\{4, 22\}$	4
38	Source Eq. (6) evaluates interference suppression in a simple cosmological model using the mass m , the constants c and $\hbar$, the Hubble constant H_0 , and selected numerical values. The essay does not list those selected values.	$\{4, 15, 23\}$	4

p	Physical or formal input claim	Corresponding time contents	Source-location key
39	For semiclassical time to be a viable approximation, wave packets must follow the classical trajectory without significant spreading.	{4, 25}	4
40	Breakdown of the semiclassical limit at the Planck scale is stated as an expectation; a recollapsing universe and the big brake are given as examples in other specified models.	{4, 24, 26}	4
41	In the specified big-brake model, the wave function of a packet approaching the classical singularity necessarily vanishes there, and classical evolution ends before the singularity is reached.	{4, 27, 28}	4
42	Disappearance of space in the big-brake model is presented as a speculative claim	{28}	4
43	Failure of applicability of the Copenhagen interpretation is argued from its presupposition of classical spacetime	{2, 3}	4
44	The LCTR audit distinguishes criticism of the Copenhagen interpretation's domain of applicability from selection of a particular alternative interpretation.	{2, 3}	4
45	The argument concerning the arrow of time uses the scale-factor asymmetry of the Wheeler–DeWitt equation	{13}	5
46	A pure product state with no entanglement between the global and local environment variables is imposed as a boundary condition in the small-scale-factor domain.	$\emptyset$	5
47	Zero reduced entropy is derived relative to a selected partition and a pure-product condition	$\emptyset$	5
48	In the specified model, an increasing scale factor increases entanglement and reduced entropy	$\emptyset$	5
49	Transfer of increasing entanglement and reduced entropy to semiclassical time requires the semiclassical approximation	{4, 29}	5
50	Temporal direction is identified with increasing entanglement	{13}	5
51	The numerical growth relation between scale factor and entropy is assigned a temporal direction where the semiclassical approximation is valid. In this sense, Kiefer describes cosmic expansion as tautological.	{4, 13, 29, 30}	5
52	A recollapsing branch imposes a large-scale-factor vanishing condition in addition to a small-scale product boundary	{26, 31, 32}	5

p	Physical or formal input claim	Corresponding time contents	Source-location key
53	Many quasiclassical components increase their entropy toward a larger universe	{26, 30, 31, 32, 33}	5
54	All components interfere destructively near the classical turning point to satisfy the large-scale boundary condition.	{26, 34}	5
55	Kiefer 2009 conditionally predicts the future termination of classical evolution within the recollapse-turning-region scenario	{26, 35}	5
56	Kiefer asserts that the stated consequences hold in every consistent quantum theory of gravity unless the assumed universality of quantum theory and its superposition principle breaks down.	{3, 4}	5
57	Each consequence depends on the canonical representation, subsystem separation, branches, decoherence, product boundary or recollapse boundary actually used in its argument.	{4, 13, 26}	5
58	The LCTR audit distinguishes local reach through model time constructions from complete generation from non-temporal physical inputs. The source supplies model-relative recovery and limits of time, but not complete generation.	{4, 24, 35}	5
59	Source Eq. (2) constructs the Planck length from the action constant $\hbar$, the gravitational constant G and the speed of light c , and gives its value as approximately 1.62×10^{-35} m.	{15}	3

Table 513: Semantic components and types in Kiefer 2009

k	Semantic component	Corresponding source-form description
1	Physical configuration of the three-metric and spatial geometry	Descriptions 20, 21
2	Physical configuration of nongravitational fields and local matter degrees of freedom	Description 21
3	Physical degrees of freedom of the heavy subsystem	Descriptions 28, 29, 30, 32, 34, 35, 39
4	Physical degrees of freedom of the light subsystem	Descriptions 28, 30, 32, 34, 39
5	Physical configuration of gravitational degrees of freedom	Descriptions 20, 21, 31, 32
6	Physical configuration of environmental degrees of freedom	Descriptions 16, 36, 38, 46, 47, 48, 49, 50

k	Semantic component	Corresponding source-form description
7	Record of the subsystem partition	Descriptions 28, 29, 30, 31, 32, 34, 35, 36, 37, 39, 46, 47, 48, 49, 50
8	Record of the spatial-scale configuration	Descriptions 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55
9	Record of boundary conditions	Descriptions 46, 52, 57
10	Record of physical correlations and entanglement	Descriptions 46, 47, 48, 49, 50, 53
11	Standard quantum theory uses an absolute external time parameter	Description 1
12	General relativity treats spacetime as a dynamical variable	Description 2
13	Combining standard quantum theory with general relativity produces the tension known as the problem of time	Description 3
14	The assumptions stated by Kiefer 2009 are universality of the linearity of quantum theory and recovery of general relativity in the classical limit	Description 4
15	Kiefer 2009 directly concludes fundamental timelessness, approximate semiclassical time, and a correlation between entropy and the size of the universe	Description 5
16	Pre-Newtonian periodic and countable time is presented as historical background	Description 6
17	Newtonian absolute continuous time is required by the historical formulation of mechanics described by Kiefer 2009	Description 7
18	In the contrast drawn by Kiefer 2009, Minkowski spacetime is a nondynamical background	Description 8
19	The source describes the Schrödinger equation as using external time t and the imaginary unit i , and as deterministic and time-reversal invariant with respect to t .	Description 9
20	Kiefer 2009 connects conservation of probability with the existence of time and complex structure	Description 10
21	The contrast between quantum theory and general relativity is reformulated in terms of background independence	Description 11
22	Universality of superposition is an empirically supported assumption open to future testing	Description 12
23	Kiefer treats the argument from universal gravitational coupling to quantization as an argument of naturalness, not a logical inference.	Description 13

k	Semantic component	Corresponding source-form description
24	Kiefer states that singularity theorems imply an end of classical time under their assumptions and presents this as a motivation for quantum gravity.	Description 14
25	Kiefer rejects the argument that quantum-gravitational effects are restricted to the Planck scale.	Description 15
26	Kiefer states that decoherence explains the classical appearance of most gravitational-field states.	Description 16
27	Kiefer describes the construction of a background-independent quantum theory of gravity as an outstanding task.	Description 17
28	The direct-quantization and interaction-unification approaches are distinct classifications	Description 18
29	Kiefer 2009 claims that an inference concerning temporal structure from tested physics and universality is valid	Description 19
30	The Wheeler–DeWitt equation is obtained by analogy with the gravitational Hamilton–Jacobi structure	Description 20
31	The state depends on the three-metric and nongravitational fields and obeys the spatial diffeomorphism constraint	Description 21
32	The Wheeler–DeWitt equation may fail fundamentally at the Planck scale and is claimed to be a reliable approximation at large scales	Description 22
33	The Wheeler–DeWitt wave functional does not depend on an external time parameter t .	Description 23
34	Kiefer 2009 compares the disappearance of classical spacetime through quantization with the disappearance of particle trajectories	Description 24
35	The LCTR audit distinguishes the absence of classical trajectories or spacetime from a representation from the nonexistence of physical spacetime.	Description 25
36	Kiefer directly asserts the fundamental timelessness of quantum gravity	Description 26
37	Fundamental timelessness is argued for within the adopted canonical representation, using quantum universality and the large-scale limit of general relativity.	Description 27
38	The Mott example begins with a time-independent equation for the total system	Description 28
39	The heavy-subsystem phase in Mott’s construction defines a directional derivative proportional to the time derivative.	Description 29
40	The resulting time-dependent equation applies to the light subsystem	Description 30

k	Semantic component	Corresponding source-form description
41	The Mott analogy assigns the gravitational part to the clock subsystem. Source footnote 2 also allows some gravitational degrees of freedom to remain quantum and some non-gravitational variables to enter the clock definition.	Description 31
42	Semiclassical time depends on subsystem separation, a phase branch, and approximation order	Description 32
43	Kiefer 2009 acknowledges quantum-gravity correction terms	Description 33
44	Kiefer conjectures that Hilbert-space structure may itself be approximate	Description 34
45	A real Wheeler–DeWitt solution decomposes into complex-conjugate components	Description 35
46	Through decoherence, a selected complex component becomes dynamically independent within the reduced description	Description 36
47	The LCTR audit distinguishes dynamical independence of complex components from unique selection of one realized branch.	Description 37
48	Source Eq. (6) evaluates interference suppression in a simple cosmological model using the mass m , the constants c and $\hbar$, the Hubble constant H_0 , and selected numerical values. The essay does not list those selected values.	Description 38
49	For semiclassical time to be a viable approximation, wave packets must follow the classical trajectory without significant spreading.	Description 39
50	Breakdown of the semiclassical limit at the Planck scale is stated as an expectation; a recollapsing universe and the big brake are given as examples in other specified models.	Description 40
51	In the specified big-brake model, the wave function of a packet approaching the classical singularity necessarily vanishes there, and classical evolution ends before the singularity is reached.	Description 41
52	Disappearance of space in the big-brake model is presented as a speculative claim	Description 42
53	Failure of applicability of the Copenhagen interpretation is argued from its presupposition of classical spacetime	Description 43
54	The LCTR audit distinguishes criticism of the Copenhagen interpretation’s domain of applicability from selection of a particular alternative interpretation.	Description 44

k	Semantic component	Corresponding source-form description
55	The argument concerning the arrow of time uses the scale-factor asymmetry of the Wheeler–DeWitt equation	Description 45
56	A pure product state with no entanglement between the global and local environment variables is imposed as a boundary condition in the small-scale-factor domain.	Description 46
57	Zero reduced entropy is derived relative to a selected partition and a pure-product condition	Description 47
58	In the specified model, an increasing scale factor increases entanglement and reduced entropy	Description 48
59	Transfer of increasing entanglement and reduced entropy to semiclassical time requires the semiclassical approximation	Description 49
60	Temporal direction is identified with increasing entanglement	Description 50
61	The numerical growth relation between scale factor and entropy is assigned a temporal direction where the semiclassical approximation is valid. In this sense, Kiefer describes cosmic expansion as tautological.	Description 51
62	A recollapsing branch imposes a large-scale-factor vanishing condition in addition to a small-scale product boundary	Description 52
63	Many quasiclassical components increase their entropy toward a larger universe	Description 53
64	All components interfere destructively near the classical turning point to satisfy the large-scale boundary condition.	Description 54
65	Kiefer 2009 conditionally predicts the future termination of classical evolution within the recollapse-turning-region scenario	Description 55
66	Kiefer asserts that the stated consequences hold in every consistent quantum theory of gravity unless the assumed universality of quantum theory and its superposition principle breaks down.	Description 56
67	Each consequence depends on the canonical representation, subsystem separation, branches, decoherence, product boundary or recollapse boundary actually used in its argument.	Description 57
68	The LCTR audit distinguishes local reach through model time constructions from complete generation from non-temporal physical inputs. The source supplies model-relative recovery and limits of time, but not complete generation.	Description 58

k	Semantic component	Corresponding source-form description
69	Source Eq. (2) constructs the Planck length from the action constant $\hbar$, the gravitational constant G and the speed of light c , and gives its value as approximately 1.62×10^{-35} m.	Description 59

Formal evidence for the Planck length and the model-dependent decoherence-suppression factor

Equation (2) in original-source §2 gives the Planck length as

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.62 \times 10^{-35} \text{ m.}$$

Equation (6) in original-source §3 evaluates the interference term between complex components in a simple cosmological model as

$$\exp\left(-\frac{\pi m c^2}{128 \hbar H_0}\right) \sim \exp(-10^{43})$$

where H_0 is the Hubble constant and m is the mass of the scalar field. The Planck-length formula and the interference-suppression estimate correspond, respectively, to source-form descriptions 59 and 38 in Table 512 and semantic components 69 and 48 in Table 513.

Dependency Inputs The Mott construction, the quantum-gravity semiclassical construction and the assignment of time direction are distinguished. Numerical ordering of scale or entropy is not itself temporal ordering. The transfer to a time direction uses semiclassical time within its valid approximation.

Dependency and Transfer Elements

Definition 8.647 (Time-content transfers in Kiefer 2009). Let $\text{ldx}_R^{\text{audit,transfer}} = [4]$. Table 514 specifies inputs and outputs; Table 515 specifies corresponding source-form descriptions and supporting claims; Table 516 maps the transfers to their support groups.

Table 514: Time-content transfers in Kiefer 2009

s	Time-content inputs i	Time-content outputs a	Auxiliary-support indices
1	$\emptyset$	$\{18, 19\}$	$\emptyset$
2	$\{18, 19\}$	$\{20\}$	$\emptyset$
3	$\emptyset$	$\{4, 21, 22\}$	$\emptyset$
4	$\{4, 21\}$	$\{13, 29, 30\}$	$\emptyset$

Table 515: Transfer support groups in Kiefer 2009

Group	Corresponding source-form descriptions	Supporting claim indices
1	{28, 29}	{3, 4, 7, 38, 39}
2	{28, 29, 30}	{3, 4, 7, 38, 39, 40}
3	{20, 21, 31, 32}	{1, 2, 5, 7, 30, 31, 41, 42}
4	{45, 46, 47, 48, 49, 50}	{6, 7, 8, 9, 10, 55, 56, 57, 58, 59, 60}

Table 516: Transfer-to-support-group correspondence in Kiefer 2009

Transfer	Support group
1	1
2	2
3	3
4	4

The indices specify the corresponding sets in Definition 7.105. Transfer 1 supplies the Mott parameter and its directional-derivative relation; transfer 2, light-subsystem evolution; transfer 3, semiclassical clocks and conditioned evolution; transfer 4, time direction and the temporal entropy-growth and expansion relations.

Applicability Conditions The Mott construction in source Eqs.(4)–(5) uses

$$\Psi(\mathbf{r}, \mathbf{R}) = \psi(\mathbf{r}, \mathbf{R})e^{i\mathbf{k} \cdot \mathbf{R}}, \quad i\frac{\partial}{\partial t} \propto i\mathbf{k} \cdot \nabla_{\mathbf{R}}.$$

Here $\mathbf{r}$ denotes the atom’s variables and $\mathbf{R}$ the alpha particle’s. The displayed relation alone does not specify the proportionality coefficient or normalization of physical duration. Transfers 1 and 3 have empty time-input sets but require the evidence claims expressing the source’s formal constructions. Neither fires from the physical initial set alone. Source-local reach of a model parameter is distinguished from complete LCTR generation from non-temporal physical inputs. Transfer 4 requires the small-scale product boundary, subsystem partition, scale-dependent reduced-entropy relation and its connection to semiclassical time. The boundary conditions specific to the later recollapsing model are not premises of this transfer.

Source Support Transfers 1–2 correspond to the Mott construction in source §3; transfer 3, to its quantum-gravity analogy and footnote 2; transfer 4, to the product-boundary and time-direction argument in source §4. Statements about absent time, interpretation criticism and conditional ends of evolution are not treated as constructions of new time contents.

Results

Definition 8.648 (Results in Kiefer 2009). Let $\text{ldx}_R^{\text{audit,result}} = [15]$. Table 517 defines the results and source locations. The descriptions distinguish the author’s interpretive claims, model constructions, conditional consequences and estimates using selected numerical values.

Table 517: Results and source locations in Kiefer 2009

j	Result	Source location
1	Contrast between quantum and relativistic time structures	§1
2	Reformulation as a problem of background independence	§1
3	Wheeler–DeWitt representation relative to the classical limit and canonical formalism	§2
4	Kiefer’s assertion of fundamental timelessness	§2
5	Mott’s model clock and light-subsystem equation	§3
6	Subsystem-relative semiclassical clock construction	§3
7	Quantum-gravitational correction terms in the approximation scheme	§3
8	Complex components and dynamical independence through decoherence	§3
9	Interference-suppression estimate in the specified cosmological model	§3
10	Wave-packet tracking condition for semiclassical time	§3
11	Conditional end of classical evolution in the big-brake model	§3
12	Interpretation-domain criticism relative to the classical-spacetime premise	§3
13	Time direction in the product-boundary model	§4
14	Observer history and future end in the recollapsing model	§4
15	Dimensional construction and displayed value of the Planck length	§2

Table 518: Result support in Kiefer 2009

q	Corresponding source-form descriptions	Supporting claim indices
1	{1, 2, 3}	{11, 12, 13}
2	{11}	{21}
3	{4, 20, 21, 22}	{1, 2, 5, 14, 30, 31, 32}
4	{4, 20, 21, 22, 23, 26}	{14, 30, 31, 32, 33, 36}
5	{28, 29, 30}	{3, 4, 7, 38, 39, 40}
6	{20, 21, 31, 32}	{1, 2, 5, 7, 30, 31, 41, 42}
7	{31, 32, 33}	{41, 42, 43}
8	{35, 36}	{6, 7, 45, 46}
9	{38}	{6, 48}
10	{39}	{49}
11	{39, 41}	{49, 51}
12	{23, 26, 43}	{33, 36, 53}

q	Corresponding source-form descriptions	Supporting claim indices
13	$\{45, 46, 47, 48, 49, 50\}$	$\{6, 7, 8, 9, 10, 55, 56, 57, 58, 59, 60\}$
14	$\{52, 53, 54, 55\}$	$\{8, 9, 10, 62, 63, 64, 65\}$
15	$\{59\}$	$\{69\}$

Result Dependencies

Definition 8.649 (Result-dependency hyperedges in Kiefer 2009). For each $q \in [15]$, the hyperedge $\text{ResultEdge}_{R,q}$ takes the time contents in Table 519 and the supporting claims in Table 518 as inputs and yields result q .

Table 519: Result dependencies in Kiefer 2009

q	Time-content input indices	Auxiliary-input indices
1	$\{1, 2\}$	$\emptyset$
2	$\{1, 2\}$	$\emptyset$
3	$\{2\}$	$\emptyset$
4	$\{2\}$	$\emptyset$
5	$\{18, 19, 20\}$	$\emptyset$
6	$\{4, 21, 22\}$	$\emptyset$
7	$\{4, 22\}$	$\emptyset$
8	$\{4, 22\}$	$\emptyset$
9	$\{15, 23\}$	$\emptyset$
10	$\{4, 25\}$	$\emptyset$
11	$\{4, 25\}$	$\emptyset$
12	$\{2\}$	$\emptyset$
13	$\{4, 13, 29, 30\}$	$\emptyset$
14	$\{4, 13, 26, 31, 32, 34\}$	$\emptyset$
15	$\{15\}$	$\emptyset$

The indices specify the corresponding sets in Definition 7.105.

Grounds Results 1–4 concern the contrast in time structures and the adopted quantum universality, classical GR limit and canonical representation. Result 4 is Kiefer’s assertion of fundamental timelessness within that framework. Results 5–8 concern the specified subsystems, phases, approximations and environmental interactions. Dynamical independence of complex components is distinguished from unique selection of one realized branch. Source Eq. (6), used in result 9, depends on $m, c, \hbar, H_0$ and selected numerical values. The quantities c , as the speed of light, and H_0 , as the Hubble constant, carry temporal meaning. The displayed finite suppression is nonzero and belongs to the stated model and numerical choices. Result 10 is the wave-packet tracking condition for the semiclassical approximation; result 11 is a conditional consequence of the specified big-brake model. The source anticipates breakdown at the Planck scale and speculates about the disappearance of space in the big-brake model (Descriptions 40 and 42). Result 12 concerns the source’s contrast between the classical-spacetime premise and its fundamental

quantum description. Result 13 concerns time direction from the small-scale product boundary. Result 14 concerns the recollapsing model's large-scale boundary, quasi-classical components and interference near the classical turning point. Result 15 constructs a length from the constants in source Eq. (2). Obtaining the dimension of length does not generate the temporal meaning of the speed used as an input.

Evidence Levels The sets of objects and predicates newly measured directly in this source are both empty. Its equations, subsystem approximations, branch analyses, boundary conditions and displayed numerical estimates supply formal evidence.

Generation Verdict **Verdict:** complete LCTR generation 0; source-evidence-conditioned local reach 9; residual primitive-time roots 26.

No time contents or results are reached from the physical initial set. Adding the source evidence environment reaches time contents $\{4, 13, 18, 19, 20, 21, 22, 29, 30\}$ and results $\{5, 6, 7, 8, 13\}$. This reach adds the evidence claims supplying the model constructions.

Result Dependencies The primitive-root-dependent results are $\{1, 2, 3, 4, 9, 10, 11, 12, 14, 15\}$; their minimal supports are given in Table 520.

Evidential Force Recognizing the source's time constructions leaves dependencies on quantum and relativistic time inputs and on the temporal meanings of the speed of light and the Hubble constant. The absence of external time from an equation does not prove the nonexistence of physical time. Construction of a model parameter is not by itself complete LCTR generation. Numerical scale and entropy ordering is treated as a relation prior to its assignment to time; the connection uses the semiclassical approximation. The future classical end in the recollapsing universe is relative to the specified model and boundaries.

Result-Preservation Domains Each result is preserved under the inputs in its dependency table and the source's conditions. The relevant results retain the Mott proportionality relation; the subsystems, branches and domains of the semiclassical approximation; the environment and partition for decoherence; the product boundary for time direction; and the large-scale boundary and interference conditions for recollapse. Verifying historical accounts or statements about absent time does not require generation of a physical clock. Physical uses of time contents require separate assessment of their generation, comparison and domain qualifications.

Placement of Grounding Obligations

Placement of grounding obligations For physical uses dependent on primitive-time inputs, the obligations are classified as backgrounded through primitive-time inputs. The source supplies construction procedures for model time parameters and directions, but not complete generation from non-temporal physical inputs alone. Mere mention of absent or historical time is not classified as a physical time input.

Evidence of Actual Failure

Evidence of actual failure The source makes wave-packet tracking of the classical trajectory without significant spreading a necessary condition for viable semiclassical time. If that condition fails, the semiclassical approximation is not viable. Results 11 and 14 state the conditional consequences of the specified big-brake and recollapsing models. Actual failure of LCTR’s physical-time grounding conditions in the universe is not determined by this source evidence. Failure of a model approximation is distinguished from such actual failure.

Downstream Reach

Downstream reach Table 520 specifies structural dependence of results on time inputs. A judgment that a particular failure has reached an actual result requires evidence about the failure location and that result’s applicability conditions.

Repair Scope

Repair scope If actual failure of a concrete grounding condition is established, repair the relevant time construction, comparison and domain and reassess the results dependent on that input. Specific recalculated values require a specified model and input values. Interpretation of absent time and historical accounts are distinguished from this physical recalculation.

Audit Conclusion

Proposition 8.650 (Primitive-time roots and result dependencies in Kiefer 2009).

$$\text{AuditPartIdx}_R = (\emptyset, [35]; \emptyset, \{5, 6, 7, 8, 13\}, \\ \{1, 2, 3, 4, 9, 10, 11, 12, 14, 15\}, \emptyset).$$

Proof. The boundary has 119 nodes: 69 claims, 35 time contents and 15 results. Its 19 edges consist of four transfers and 15 result edges. Every edge requires at least one claim from 11–69, so no time content or result is reached from the physical initial set. Under the evidence environment, transfers 1 and 3 fire, followed by transfers 2 and 4, reaching the nine time contents and five results listed above. Adding all time contents as seeds reaches all 15 results. Definition 8.115 gives the displayed six components. $\square$

Proposition 8.651 (Minimal primitive-root support in Kiefer 2009). Each result has the support family $\mathcal{A}_R(q) = \{S\}$, whose sole member is the set S in Table 520. When $S = \emptyset$, the family is $\{\emptyset\}$, not the empty family.

Table 520: Minimal primitive-root supports in Kiefer 2009

Result index	S
1	$\{1, 2\}$
2	$\{1, 2\}$
3	$\{2\}$

Result index	S
4	$\{2\}$
5	$\emptyset$
6	$\emptyset$
7	$\emptyset$
8	$\emptyset$
9	$\{15, 23\}$
10	$\{25\}$
11	$\{25\}$
12	$\{2\}$
13	$\emptyset$
14	$\{26, 31, 32, 34\}$
15	$\{15\}$

Proof. The nine locally reachable time contents have empty minimal support. Every other time content is absent from the outputs of the transfer system and has its own singleton as minimal support. Taking unions of the supports of each result's required time contents and minimizing by inclusion gives the table. Every displayed set is inclusion-minimal, and the total number of members across the minimal-support families is 15. $\square$

Individual statements

1. Original text (English)

Time is absolute in standard quantum theory and dynamical in general relativity.

Original-source location Abstract, p. 1. **Minimal context** This sentence opens the abstract and is followed immediately by the statement that combining the two theories produces the problem of time.

The statement object is the theoretical status of time in standard quantum theory and general relativity. Its predicate is that time is absolute in standard quantum theory and dynamical in general relativity. Its explicit applicability domain is standard quantum theory and general relativity as contrasted in the abstract, together with the problem of time that arises when the two theories are combined.

2. Original text (English)

It is generally argued that quantum-gravity effects can only be seen at a remote scale – the Planck scale, which originates from the combination of the three fundamental constants c (speed of light), G (gravitational constant), and $\hbar$ (quantum of action).

Original-source location p. 3, §2. **Minimal context** This sentence reports the common argument restricting quantum-gravity effects to the Planck scale. After the Planck-length formula, the same paragraph rejects that restriction as misleading.

The statement objects are the scale at which quantum-gravity effects are expected to be observable and the constants composing the Planck scale. Its predicate is that quantum-gravity effects are generally argued to be visible only at the Planck

scale and that this scale arises from the combination of c , G , and $\hbar$. Its explicit applicability domain is the scale-restriction argument reported and rejected in the same paragraph of §2, together with the dimensional construction of the Planck length given there. The author does not endorse restricting quantum-gravity effects to the Planck scale.

3. Original text (English)

Time emerges from the separation into two different subsystems: one subsystem (here: the gravitational part) defines the time with respect to which the other subsystem (here: the non-gravitational part) evolves.

Original-source location p. 6, §3. **Minimal context** Following the analogy with Mott’s heavy and light subsystems, this sentence states the general idea of semiclassical time.

The statement object is the emergence of semiclassical time through separation of the gravitational and nongravitational subsystems. Its predicate is that the gravitational part defines time and the nongravitational part evolves with respect to that time. Its explicit applicability domain is the Mott analogy in §3, the separation of gravitational and nongravitational subsystems, the selected semiclassical branch, and the approximation order.

4. Original text (English)

Time is thus only an approximate concept.

Original-source location p. 6, §3. **Minimal context** This sentence immediately follows the account of time emerging from two subsystems and precedes the discussion of quantum-gravitational correction terms.

The statement object is the precision and validity of the concept of time obtained from the two-subsystem separation. Its predicate is that this time is only an approximate concept. Its explicit applicability domain is the immediately preceding semiclassical subsystem separation, selected branch, approximation order, and the validity domain fixed by the subsequent quantum-gravitational correction terms.

5. Original text (English)

where H_0 is the Hubble constant and m the mass of a scalar field, and some standard numbers have been chosen.

Original-source location p. 6, §3, immediately after Eq. (6). **Minimal context** This sentence identifies the quantities and numerical choices used to evaluate the amount of interference suppression between complex components.

The statement objects are the Hubble constant, scalar-field mass, and selected numerical values used to evaluate the interference-suppression quantity in Eq. (6). Its predicate is that H_0 is the Hubble constant, m is the mass of a scalar field, and certain standard numerical values have been selected. Its explicit applicability domain is the simple cosmological model in the source, Eq. (6), and its numerical evaluation.

6. Original text (English)

In quantum gravity, the world is fundamentally timeless and does not contain classical parts. Classical appearance only emerges for subsystems through the process of decoherence – with limitations dictated by the solution of the full quantum equations.

Original-source location p. 7, §3. **Minimal context** This passage follows the statement that classical spacetime is absent at the most fundamental level and precedes §4 on the direction of time.

The statement objects are fundamental timelessness in quantum gravity, the absence of classical parts, and the emergence of classical appearance in subsystems. Its predicate is that the world is fundamentally timeless and contains no classical parts, while classical appearance emerges through decoherence under limitations imposed by solutions of the full quantum equations. Its explicit applicability domain is the quantum-gravity representation in §3, subsystem separation, the decoherence process, and the range permitted by solutions of the full quantum equations.

7. Original text (English)

The direction of time is thus defined by the direction of increasing entanglement. In this sense, the expansion of the universe would be a tautology.

Original-source location p. 8, §4. **Minimal context** This passage immediately follows the claim that entanglement and reduced entropy increase with the scale factor.

The statement objects are the correspondence between the direction of time and the direction of increasing entanglement, and the status of cosmic expansion under that correspondence. Its predicate is that the direction of time is defined by the direction of increasing entanglement and that cosmic expansion is, in this sense, a tautology. Its explicit applicability domain is the selected cosmological boundary condition in §4 and the relation among the scale factor, entanglement, and reduced entropy.

Common evidence with an applicability scope The seven statements are from Kiefer, *Does Time Exist in Quantum Gravity?*.

Source attribution

All seven statements are attributed to Claus Kiefer.

Individual audit Correspondence with formal evidence

Statement 1 corresponds to source-form descriptions 1–3 in Definition 8.646; statement 2 to source-form descriptions 15 and 59 and the Planck length in the original Eq. (2); statements 3–4 to source-form descriptions 31–33; statement 5 to source-form description 38 and the reproduced original Eq. (6); statement 6 to source-form descriptions 26–27 and 43–44; and statement 7 to source-form descriptions 45–51. The correspondences with time contents, transfers, results, and primitive-time roots are recorded respectively in Definition 8.645, Definition 8.647, Definition 8.648, and Proposition 8.650.

Available interpretations

The restriction to the Planck scale reported in the second statement is rejected by the author in the same paragraph. The dimensional construction of the Planck length is distinct from the claim that quantum-gravity effects are confined to that scale. The third and

fourth statements concern semiclassical time obtained through subsystem separation and its approximate character. The sixth states the author’s claims of fundamental timelessness and the emergence of classical appearance in subsystems through decoherence. Under the specified cosmological model and boundary conditions, the seventh defines the direction of time by the increase in entanglement accompanying an increase in the scale factor, in the domain where the semiclassical approximation holds. These statements belong to different equations, approximations, interpretations, and model conditions in the source. Obtaining a subsystem evolution parameter or a direction of time and establishing LCTR physical-time qualification from nontemporal physical inputs have different generation conditions.

Hubble constant, scalar-field mass, and selected numerical values

The fifth statement makes explicit that the Hubble constant H_0 , scalar-field mass m , speed of light c , action constant $\hbar$, and selected standard numerical values are taken as inputs to evaluate the interference-suppression quantity in the original Eq. (6). The independent basis is the numerical calculation in the original source that uses H_0 , m , c , $\hbar$, and the selected values in Eq. (6) to evaluate the amount of interference suppression between complex components. Its evidential force belongs to that model, Eq. (6), the selected values, and their numerical evaluation.

Formally, it corresponds to source-form description 38 in Definition 8.646, the reproduced original Eq. (6), the corresponding result in Definition 8.648, and the root determination in Proposition 8.650.

8.19.55 Source audit

Audit input

Original source and audit scope

Source identifier 55.

Source name Primitive-time-root audit of Rovelli 2011

Bibliographic identification Carlo Rovelli, *Forget Time, Foundations of Physics* 41, 1475–1490 (2011), DOI: 10.1007/s10701-011-9561-4, with arXiv:0903.3832v3 taken as the original source and denoted by $R = \text{Ref}_{\text{Rov11FT}}$. The audit domain comprises relational classical mechanics, quantum events, group averaging, approximate non-relativistic time, classical thermal time, quantum thermal flow, modular flow, and the correspondence to physical time.

Audit-scope identifier 55-SC1.

Audit-scope content The audit domain stated under bibliographic identification.

Original-source-location identifier 55-L1.

Original-source-location content The audit domain stated under bibliographic identification and the original-source-location columns of the respective evidence tables.

Formal object

Object identifier 55-F1.

Object name Formal objects in the original source.

Object type Formal object.

Formal content The individual formal objects are given in Definitions 8.655, 8.652, and 8.654.

Original-source-location identifier set 55-L1.

Assumption

Assumption identifier 55-A1.

Assumption content The individual assumptions of the source are given in Definition 8.654.

Original-source-location identifier set 55-L1.

Nontemporal physical input

Input identifier 55-P1.

Input name Nontemporal physical inputs in the original source.

Input type Nontemporal physical input.

Input content The individual nontemporal physical inputs are given in Definition 8.654.

Original-source-location identifier set 55-L1.

Time content

Time-content identifier 55-T1.

Time-content name Time contents in the original source.

Time-content type Time content.

Time-content description The individual time contents are given in Definitions 8.653 and 8.652.

Original-source-location identifier set 55-L1.

Formal input elements The audit distinguishes 23 time contents, 52 source claims, and 24 results. Source Claims 1–4 identify nontemporal physical carriers; Claims 5–52 give formal or operational descriptions in the source.

Definition 8.652 (Semantic objects in the Rovelli 2011 source). Time contents comprise times, intervals, clock comparisons, orders of evolution, and flows used as time. States, generators, algebras, and relational rules are included in the source claims in Table 522. The integration variable in group averaging and the arbitrary orbit parameter in Appendix A are formal parameters of the constrained system. The variable in the proper-time integral is an arbitrary curve parameter; the temporal structure determining proper time resides in the metric and worldline. Repeated occurrences of the same time content share an index. Newtonian true time belongs to independent Time Content 1; time identification by the special Hamiltonian form and its approximate recovery belong to Time Content 12; thermal time and its state-dependent physical interpretation belong to Time Content 16.

Definition 8.653 (Time contents in the Rovelli 2011 source). Table 521 gives the 23 time contents, their uses, and their source locations. Their use as clocks and the construction of mathematical flows are distinguished by the inputs and conditions of Table 523.

Table 521: Time contents and source locations in Rovelli 2011

ν	Time content	Source location
1	Pre-relativistic independent time and Newtonian true time	II, III.A; pp. 2–3
2	Coordinate-time label in general relativity	III; p. 2
3	Temporal and causal structure carried by a Lorentzian metric and timelike worldline	III; p. 2
4	Proper time along a worldline	III; p. 2
5	Pairs of terrestrial and satellite clock readings and their mutual prediction	III; p. 2
6	Earth-clock proper-time interval from emission to reflected reception	III; p. 2
7	Pulse-frequency modulation and progression of binary orbital cycles	III; p. 2
8	Relative progression of pulse count and orbital-cycle count	III; p. 2
9	Relative evolution of observables on an equal footing	III; pp. 2–3
10	Observable clock readings and changes in physical quantities relative to them	III.A, IV.A; pp. 3–4
11	Jointly read measurement values and the order of measurement sequences and subsequent measurements	IV.A–B, V; pp. 4, 6
12	Approximate nonrelativistic time singled out by the special Hamiltonian form	IV.C, VI; Eq. (3), pp. 5, 7
13	Clock/time arguments in the pendulum transition amplitude	V.A; p. 6
14	Experienced flow, irreversibility, and past–future asymmetry	VI–VII; pp. 7–9
15	Time of the Hamiltonian flow identified from an equilibrium distribution	VI; Eqs. (14)–(16), p. 7
16	Classical thermal time and its state-dependent physical-time interpretation	VI; Eqs. (17)–(18), pp. 7–8
17	Clock reading linearly related to the thermal flow	VI; p. 8
18	Friedmann time in the radiation-filled cosmological model	VI; p. 8
19	Evolution time in conventional quantum theory and the time argument of Gibbs–KMS correlations	VI; Eqs. (19)–(24), p. 8
20	Parameter of the general quantum thermal flow	VI; Eqs. (25)–(26), p. 9
21	Tomita modular flow and its thermal-time interpretation	VI; p. 9, note 11
22	Connes state-independent outer flow	VI; p. 9, note 13
23	Mutual comparison and isochronism of pendulum oscillations and pulse beats	III.A; p. 3

Definition 8.654 (Physical inputs and source formal evidence in Rovelli 2011). Let $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(52)$ and $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(4)$. Source-claim indices and semantic-component indices identify the same objects. The initial set

comprises spatial configurations and degrees of freedom of the carriers. Temporal and causal geometry, clock cycles, calibration, and ordering belong to time contents. Phase spaces, constraints, statistical states, operators, algebras, and construction rules belong to the source formal-evidence environment. The source formal-evidence environment consists of the 52 claims in Table 522.

Table 522: Physical inputs and source formal evidence in
Rovelli 2011

p	Source claim	Source location
1	Physical degrees of freedom of pendulums, clocks, measuring devices, and target systems	III.A, IV.A; pp. 3–4
2	Physical configurations of transmitters, receivers, signal carriers, and recording media	III; p. 2
3	Matter and spatial configurations of terrestrial, satellite, planetary, and binary systems	III; p. 2
4	Microscopic configurations and physical degrees of freedom of statistically described systems	VI; pp. 7–9
5	Provides a strategy for removing time from the fundamental description.	Abstract, I; p. 1
6	Provides the opposing view that time is irreducible.	I; p. 1
7	Provides pre-relativistic independent time.	II; p. 2
8	Provides Hamiltonian evolution with respect to time.	II; p. 2
9	Distinguishes coordinate time from proper time.	III; p. 2
10	Provides a proper-time relation between two clocks.	III; p. 2
11	Provides a distance from emission and reception proper times in the Solar System.	III; p. 2
12	Provides the counting relation for a binary pulsar.	III; p. 2
13	Provides relative evolution among physical quantities of the same standing.	III; pp. 2–3
14	Provides rejection of Newtonian universal time.	III.A; p. 3
15	Distinguishes a clock variable from the true-time assumption.	III.A; p. 3
16	Provides partial observables of a pendulum.	IV.A; p. 4
17	Provides event space.	IV.A; p. 4
18	Provides unparametrized physical motion.	IV.A; p. 4
19	Provides the space of motions and relativistic states.	IV.A; p. 4
20	Provides a correlation triple for a general classical system.	IV.B; p. 4
21	Provides predictions for subsequent measurements.	IV.B; p. 4
22	Provides a relativistic Hamiltonian constraint surface.	IV.C; p. 5
23	Provides a constrained variational principle.	IV.C; p. 5
24	Provides the special nonrelativistic constraint form.	IV.C; Eq. (3), p. 5
25	Provides a time-independent interpretation of classical mechanics.	IV.B–C; pp. 4–5
26	Provides a rigged kinematical Hilbert space.	V; p. 5
27	Provides partial-observable operators and quantum events.	V; p. 5

p	Source claim	Source location
28	Defines the group-averaging integral operator P .	V; Eq. (5), p. 5
29	Provides transition amplitudes and event probabilities.	V; Eqs. (6)–(7), (10), pp. 5–6
30	Provides a Wheeler–DeWitt physical state.	V; Eq. (8), p. 6
31	Provides the commutativity condition for complete observables.	V; Eq. (9), p. 6
32	Presents the quantum structure as a tentative proposal.	V; p. 6
33	Identifies the treatment of repeated measurements as unresolved.	V; p. 6
34	Provides recovery of approximate nonrelativistic time.	VI; p. 7
35	Takes flow, irreversibility, and past–future asymmetry as phenomena to explain.	VI; p. 7
36	Provides a classical statistical state.	VI; Eqs. (11)–(13), p. 7
37	Provides a Gibbs state and Hamiltonian reconstruction.	VI; Eqs. (14)–(16), p. 7
38	Provides a general thermal generator.	VI; Eq. (17), p. 7
39	Provides thermal flow and a thermal clock.	VI; Eq. (18), p. 8
40	Provides the thermal-time hypothesis.	VI; p. 8
41	Provides an interpretive relation between time and incomplete knowledge.	VI; p. 8
42	Provides an intra-model correspondence with Friedmann time.	VI; p. 8
43	Gives a quantum density matrix and the Gibbs–KMS relation.	VI; Eqs. (19)–(24), p. 8
44	Provides general quantum thermal flow.	VI; Eqs. (25)–(26), p. 9
45	Extends the thermal-time hypothesis to quantum field theory.	VI; p. 9
46	Uses Tomita’s theorem.	VI; p. 9, note 11
47	Uses the Connes outer automorphism class.	VI; p. 9, note 13
48	Identifies physical time with the Tomita flow of the statistical state for the chosen macroscopic description.	VI; pp. 8–9
49	Provides a thermodynamic explanation of temporal asymmetry.	VI–VII; pp. 7–9
50	Acknowledges disagreement with views treating time as irreducible and favors empirically effective theories.	VII; p. 9
51	Unparametrized orbits are obtained from the kernel of the presymplectic form on the constraint surface.	Appendix A; Eqs. (A1)–(A2), p. 10
52	Pendulum oscillations and pulse beats are compared using each as a clock for the other.	III.A; p. 3

Definition 8.655 (Finite audit domain for the Rovelli 2011 source). The finite audit domain consists of the 52 source claims in Table 522, 23 time contents in Table 521, 17 transfers in Table 523, and 24 results in Table 524. Table 525 gives result premises, and Table 526 gives minimal primitive-time-root supports.

Dependency inputs Definitions 8.654 and 8.653 specify the source claims and time premises, respectively.

Dependencies and transfers

Definition 8.656 (Time-content transfers in the Rovelli 2011 source). Table 523 gives the source claims, time inputs, and time outputs of the 17 transfers. Support indices refer to the source-claim table; input and output indices refer to the time-content table.

Table 523: Time-content transfers in Rovelli 2011

s	Transfer	Source claims	Time inputs	Time outputs
1	From metric and timelike worldline to proper time	{9}	{3}	{4}
2	From proper times along the worldlines to mutual clock prediction	{10}	{4}	{5}
3	From proper times of emission and reception to the ranging interval	{11}	{4}	{6}
4	From pulse and orbital-cycle counting to relative progression	{12}	{7}	{8}
5	From the two-clock relation to relative evolution	{10, 13}	{5}	{9}
6	From the distance-measurement sequence to relative evolution	{11, 13}	{6}	{9}
7	From the pulsar count relation to relative evolution	{12, 13}	{8}	{9}
8	Clock calibration using Newtonian true time	{15}	{1}	{10}
9	Identification of nonrelativistic time from the special Hamiltonian form	{24, 34}	$\emptyset$	{12}
10	Time arguments in the pendulum model with clock readings	{16, 24, 28, 29}	{10, 12}	{13}
11	From a Gibbs distribution to the Hamiltonian and flow parameter	{36, 37}	$\emptyset$	{15}
12	From a nonvanishing statistical state to the classical thermal flow	{36, 38}	$\emptyset$	{16}
13	From thermal flow and the linear-reading condition to a thermal clock	{39}	{16}	{17}
14	From thermal time and cosmological geometry to the Friedmann-time correspondence	{42}	{3, 16}	{18}
15	From the logarithm of a density operator to the quantum thermal flow	{43, 44}	$\emptyset$	{20}
16	From the algebra, state, and Tomita theorem to the modular flow	{45, 46}	$\emptyset$	{21}

s	Transfer	Source claims	Time inputs	Time outputs
17	From the modular flow modulo inner automorphisms to the outer flow	{47}	{21}	{22}

Conditions Transfer 1 integrates the line element determined by the metric and timelike worldline. The choice of coordinate time and the value of proper time are distinct objects. Transfers 2 and 3 are separate clock-comparison and ranging examples; Transfers 5–7 are three separate routes illustrating relative evolution. The ranging paragraph does not specify a concrete conversion formula.

Transfer 9 specifies a regime in which the constraint has the form of Eq. (3) for a partial observable and its conjugate momentum. Writing the canonical coordinates as q^0, p_0 , with $H = p_0 + H_0(q^i, p_i, q^0)$, gives $\{q^0, H\} = 1$ along the constraint-generated flow. This identifies q^0 as nonrelativistic time in that regime. For an approximate form, the identification has the same approximation domain. Transfer 10 uses the pendulum’s clock reading and special Hamiltonian form.

Transfer 11 obtains $-\beta^{-1} \ln \rho_0 = H_0 - \beta^{-1} \ln N$ from the normalized Gibbs distribution $\rho_0 = N e^{-\beta H_0}$. The additive constant leaves the Hamiltonian flow unchanged; a multiplicative constant fixes its unit. Transfer 12 constructs the Hamiltonian vector field of $H_\rho = -\ln \rho$ from a statistical phase space Γ with a symplectic structure and a positive smooth density on it. Where constraint reduction is required, the construction uses a regular region in which the reduced phase space has this structure. Its flow is defined on the domain of existence of its solutions; extension to all parameter values requires completeness of the vector field. The clock condition in Transfer 13 is that the device reading increase linearly along this flow. Transfer 14 belongs to the radiation-filled covariant cosmological model.

In Transfer 15, spectral calculus on the support space $(\ker \rho)^\perp$ of the positive density operator ρ defines the self-adjoint operator $H_\rho = -\ln \rho$ and the unitary group $e^{it_\rho H_\rho}$. In infinite dimension, H_ρ may be unbounded but is self-adjoint on its dense domain, and its exponential is unitary on the entire support space. Unitary conjugation acts on bounded operators on the support space; restricting observables to a subalgebra requires that this action preserve the subalgebra. In Transfer 16, the von Neumann algebra is the weak operator closure of the observable algebra in the chosen representation, equipped with a faithful normal state. The cyclic separating vector in the GNS representation determines the Tomita modular group. In Transfer 17, state independence belongs to the outer equivalence class of modular automorphisms modulo inner automorphisms. For a finite matrix algebra, automorphisms are inner and this outer flow is trivial. An admissible constant classical density gives a zero Hamiltonian vector field, while the maximally mixed state $\rho = I/d$ in dimension d gives $\alpha_{t_\rho}(A) = A$. Conditions for distinguishable change serving as a clock are therefore additional to conditions for the existence of a flow.

Results

Definition 8.657 (Results in the Rovelli 2011 source). Table 524 gives the 24 results and their source locations.

Table 524: Results and source locations in Rovelli 2011

q	Result	Source location
1	Distinction between pre-relativistic independent time and relativistic time concepts	II–III; pp. 2–3
2	Distinction between coordinate time and worldline proper time	III; p. 2
3	Relative prediction between two clocks	III; p. 2
4	Planetary ranging through the emission–reception proper-time interval	III; p. 2
5	Count relation for a binary pulsar	III; p. 2
6	Relative evolution of observables on an equal footing	III; pp. 2–3
7	Distinction between operational clocks and the Newtonian true-time assumption	III.A; p. 3
8	General formal structure of partial observables and event space	IV.A–B; p. 4
9	General unparametrized motion represented by a correlation curve	IV.A–B; Eqs. (1)–(2), p. 4
10	Prediction of subsequent measurements by the general triple (C, Γ, f)	IV.B; p. 4
11	Relativistic Hamiltonian constraint surface and variational principle	IV.C; p. 5
12	Orbit formulation using the presymplectic kernel	Appendix A; Eqs. (A1)–(A2), p. 10
13	Quantum events in a rigged Hilbert space	V; p. 5
14	Integral operator P associated with the constraint	V; Eq. (5), p. 5
15	Transition amplitudes, conditional event probabilities, and Wheeler–DeWitt states	V; Eqs. (6)–(8), (10), pp. 5–6
16	Commutativity condition for complete observables	V; Eq. (9), p. 6
17	Correspondence between the pendulum transition amplitude and the conventional propagator	V.A; p. 6
18	Recovery of approximate nonrelativistic time	VI; p. 7
19	Identification of the Hamiltonian and time flow from a Gibbs distribution	VI; Eqs. (14)–(16), p. 7
20	Classical thermal flow and a linear clock reading	VI; Eqs. (17)–(18), pp. 7–8
21	Thermal-time hypothesis that the statistical state selects physical time	VI; p. 8
22	Gibbs–KMS relation in conventional quantum theory	VI; Eqs. (19)–(24), p. 8
23	Tomita modular flow and state-independent outer flow	VI; p. 9, notes 11–13
24	Thermodynamic account of experienced temporal features	VI–VII; pp. 7–9

Definition 8.658 (Result-dependence hyperedges in Rovelli 2011). Table 525 gives the source claims and time premises of each result.

Table 525: Result-dependence hyperedges in Rovelli 2011

q	Supporting source claims	Time premises
1	{7, 9}	{1, 2, 4}
2	{9}	{2, 4}
3	{10}	{5}
4	{11}	{6}
5	{12}	{8}
6	{13}	{9}
7	{14, 15, 52}	{1, 10, 23}
8	{17, 20}	$\emptyset$
9	{18, 19, 20}	$\emptyset$
10	{20, 21}	{11}
11	{22, 23}	$\emptyset$
12	{51}	$\emptyset$
13	{26, 27}	$\emptyset$
14	{28}	$\emptyset$
15	{29, 30}	$\emptyset$
16	{31}	$\emptyset$
17	{24, 28, 29}	{13}
18	{24, 34}	{12}
19	{36, 37}	{15}
20	{38, 39}	{16, 17}
21	{40, 41, 48}	{16}
22	{43}	{19}
23	[[45, 47]]	{21, 22}
24	{35, 38, 39, 49}	{14, 16}

Proposition 8.659 (Audit-boundary counts for Rovelli 2011).

$$\text{BoundaryCount}_R^{\text{audit}} = (99, 41).$$

Proof. The 52 source claims, 23 time contents, and 24 results give 99 nodes; the 17 transfers and 24 result edges give 41 edges. $\square$

Grounds Results 8 and 9 use the general formalism of partial observables and correlations in Section IV.B. Results 11–16 are formal constructions of constraints, orbits, and operators. Result 10 applies the correlation formalism to subsequent measurements. In the pendulum example, β is the clock reading and α the elongation; the source writes the quantum-oscillator propagator with arguments t, t' .

The orbit construction in Appendix A uses the kernel of the presymplectic form on a regular constraint surface. Where $dH \neq 0$ and $a \neq 0$, replacing H by aH preserves the characteristic direction on the constraint surface. For H^2 , the differential vanishes on the same zero set, so equality of zero sets alone does not guarantee equality of characteristic directions. In the pendulum-state representation (A, ϕ) , all phases at $A = 0$ describe the same motion.

The range of the group-averaging integral and operator domains depend on the system. The source's $P : S \rightarrow S'$ includes distributional operators, and its discrete-

event probabilities use the normalization $\langle s|P|s\rangle = 1$. The quantum structure is presented as tentative, with repeated measurements listed as an unresolved issue.

The Gibbs–KMS branch uses the convention $\alpha_t(A) = e^{itH_0} A e^{-itH_0}$, for which Eq. (23) gives $W_{AB}(t) = W_{BA}(-t - i\beta)$. Mathematical existence of the Tomita flow, state independence of the outer flow, physical-time identification under the thermal-time hypothesis, and the thermodynamic account of experienced temporal features each have their own conditions.

Evidential hierarchy The source discusses relational mechanics and constructions recovering time. It reports no new quantitative measurements. Two clocks, planetary ranging, a binary pulsar, pendulum/pulse comparison, and clock readings serve as physical or operational examples. Time identification from a special Hamiltonian form or statistical state is a construction within the specified formal or approximation domain. The thermal-time hypothesis and thermodynamic origin of experienced temporal features are presented as the author’s physical interpretation. Agreement with Friedmann time is a correspondence in the radiation cosmological model cited by the source.

Generation judgment **Judgment:** LCTR complete generation: 0; source-evidence-conditioned local reach: 7; remaining primitive-time roots: 16.

Locally reachable time contents have indices $\{12, 15, 16, 17, 20, 21, 22\}$; root indices remaining after local reach are $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 18, 19, 23\}$.

Result dependence There are 30 primitive-time-root/result dependency pairs, of which eight are necessary. Root-dependent results have indices $\{1, 2, 3, 4, 5, 6, 7, 10, 17, 22, 24\}$; results reachable with the source formal-evidence environment alone have indices $\{8, 9, 11, 12, 13, 14, 15, 16, 18, 19, 20, 21, 23\}$.

Evidential force The special Hamiltonian form identifies a specified partial observable as an evolution parameter. Statistical states and algebraic conditions provide classical and quantum thermal flows, modular flow, and outer flow. These constructions support the seven local-reach judgments. Constituting physical carriers, retained readings, comparison operations, and their connection to laws from inputs that do not already use time is the subject of the LCTR complete-generation judgment. The source’s constructions hold under the phase-space, constraint, state, algebraic, and clock-calibration conditions supplied in the formal environment.

Result-preservation domain Formal correlation, constraint, and operator results hold under their respective regularity, spectral, domain, and normalization conditions. Clock comparison, relative evolution, subsequent measurements, and conventional quantum time correlations use the time premises in the result-dependence table. A thermal clock requires the linear-reading condition; physical-time identification uses the thermal-time hypothesis; the cosmological correspondence uses the specified model.

Placement of grounding obligations Construction conditions for the seven locally reachable contents are supplied by the source formal-evidence environment. The temporal/causal structure, clock readings and cycles, measurement order, experienced asymmetry,

and model-specific times among the 16 remaining roots are retained as temporal premises at their respective uses.

Evidence of actual failure Constant densities and maximally mixed states are mathematical boundary cases showing the need for additional conditions for nontrivial clock change. The source evidence does not establish an actual failure of a concrete physical condition or a resulting change in measured outcome values.

Downstream reach Structural reach is specified by the 30 root/result dependency pairs in the result-dependence and minimal-support tables. Actual propagation from a failed condition and recalculated values remain undetermined.

Repair scope Grounding obligations for physical-time use concern the readings, cycles, orderings, and geometry, together with the clock correspondence of locally constructed flows. If an actual failure of a concrete condition is established, recalculation concerns the results dependently reachable from that condition.

Audit conclusions

Proposition 8.660 (Primitive-time roots and result dependence in Rovelli 2011). The complete-generation judgment yields no time contents or results. Adding the source formal-evidence environment reaches seven time contents and thirteen results, leaving sixteen time contents and eleven results with temporal dependencies.

Proof. Every transfer or result edge requires at least one of Source Claims 5–52, so none applies from Initial Claims 1–4 alone. Adding the evidence environment enables Transfers 9, 11, 12, 15, and 16, followed by Transfers 13 and 17. Their outputs are the seven locally reachable time contents. The remaining transfers require existing time inputs. Comparing the time premises of each result with this reachable set yields thirteen reachable results and eleven with residual dependencies. $\square$

Proposition 8.661 (Minimal primitive-time-root supports in Rovelli 2011). Table 526 gives the minimal primitive-time-root supports of the 24 results. The support families have 39 members in total.

Proof. Start with singleton seeds for time contents and empty supports for transfers with no time inputs. Propagate unions of supports along transfer edges and remove strict supersets. Once time-content supports stabilize, combine the time premises of each result. Locally reachable time contents have empty support. $\square$

Table 526: Minimal primitive-time-root supports in Rovelli 2011

Result index	Minimal support indices
1	$\llbracket 1, 3 \rrbracket, \{1, 2, 4\}$
2	$\{2, 3\}, \{2, 4\}$
3	$\{3\}, \{4\}, \{5\}$
4	$\{3\}, \{4\}, \{6\}$

Result index	Minimal support indices
5	$\{7\}, \{8\}$
6	$\{3\}, \{4\}, \{5\}, \{6\}, \{7\}, \{8\}, \{9\}$
7	$\{1, 23\}$
8	$\emptyset$
9	$\emptyset$
10	$\{11\}$
11	$\emptyset$
12	$\emptyset$
13	$\emptyset$
14	$\emptyset$
15	$\emptyset$
16	$\emptyset$
17	$\{1\}, \{10\}, \{13\}$
18	$\emptyset$
19	$\emptyset$
20	$\emptyset$
21	$\emptyset$
22	$\{19\}$
23	$\emptyset$
24	$\{14\}$

Proposition 8.662 (Audit conclusion for Rovelli 2011). The source has 52 source claims, 23 time contents, 17 transfers, and 24 results. Complete generation is zero, local reach is seven, and sixteen roots remain; eleven results depend on primitive-time roots.

Proof. The result follows from the finite-audit tables and the reachability and minimal-support propositions above. $\square$

Individual statements Statement 1

Original text (English):

I summarize here this point of view, explaining why I think that in a fundamental description of nature we must “forget time”, and how this can be done in the classical and in the quantum theory.

The original-source location is Carlo Rovelli, *Forget Time*, arXiv:0903.3832v3, p. 1, abstract. The minimal context is a sentence in the abstract that states a strategy for constructing, for quantum gravity, a physical description in which time has no role and then summarizes a formalism that treats dependent and independent variables on the same footing.

The statement objects are the role of time in a fundamental description of nature and the construction policy for classical and quantum descriptions that do not privilege time. Its predicate is that a fundamental description must “forget time” and that such a description can be constructed in both classical and quantum theory. Its explicit applicability domain is the relational formalism supplied by the original source for relations among partial observables, unparametrized classical mechanics, constrained systems, and quantum events.

Statement 2

Original text (English):

The notion of time familiar to us may then be reconstructed in special physical situations, or within an approximation, as is the case for a number of familiar physical quantities that disappear when moving to a deeper level of description (for instance: the “surface of a liquid” disappears when going to the atomic level, or “temperature” is a notion that makes sense only in certain physical situations and when there are enough degrees of freedom.)

The original-source location is p. 1, Section I of the same source. The minimal context is a sentence that follows the proposal to construct a theory of quantum gravity in which the concept of time does not appear and states the conditions for reconstructing a concept of time by using the surface of a liquid and temperature as analogies.

The statement object is the reconstructibility of the familiar concept of time after it loses its privileged role in the fundamental description. Its predicate is that the concept of time may be reconstructed in special physical situations or within an approximation. Its explicit applicability domain comprises the approximate nonrelativistic time treated by the original source, selected statistical states, macroscopic descriptions with sufficiently many degrees of freedom, and the validity conditions of the respective recovery branches.

Statement 3

Original text (English):

Time is, that is to say, the expression of our ignorance of the full microstate.

The original-source location is p. 8, Section VI of the same source, in the paragraph stating the thermal-time hypothesis. The minimal context is the sentence immediately following the thermal-time hypothesis, which calls the thermal time of a statistical state selected by macroscopic variables “time.”

The statement object is the relation, under the thermal-time hypothesis, between time and a coarse-grained informational state relative to the complete microstate. Its predicate is that time is an expression of ignorance of the full microstate. Its explicit applicability domain is the thermal-time construction using selected macroscopic variables, an observable algebra, a statistical state, Gibbs and KMS flows, and Tomita modular flow.

Common evidence with an applicability scope Source attribution

All three statements are attributed to Carlo Rovelli.

Individual audits Forgetting time in the fundamental description

Statement 1 proposes constructing a fundamental classical and quantum description through relations among partial observables, without a privileged independent time. Its grounds are the constructions of event space, the correlation triple, the constraint surface, presymplectic orbits, quantum events, and transition amplitudes. General correlation, constraint, and operator constructions are reachable under the source formal-evidence environment; clock examples and applications to subsequent measurements use clock readings and measurement order. The corresponding formal evidence is Results 1–17.

Reconstruction in special situations or approximations

Statement 2 uses the surface of a liquid and temperature as analogies to state that the familiar concept of time may be reconstructed in special physical situations or within

an approximation. Its grounds are the identification of nonrelativistic time by a special Hamiltonian form, reconstruction of the Hamiltonian from a Gibbs distribution, and classical and quantum thermal flows obtained from statistical states. Time identification belongs to each construction’s formal or approximation domain; a thermal-clock reading additionally requires linear increase along the flow. The corresponding formal evidence is Results 18–23 and Transfers 9 and 11–17.

Time as an expression of ignorance of the complete microscopic state

In the context of the thermal-time hypothesis, Statement 3 identifies physical time with the thermal time of the statistical state fixed by selected macroscopic variables. Its grounds are the source’s constructions of a logarithmic generator and thermal flow from a statistical state, and of modular flow from a state and observable algebra. The source explicitly identifies this as a statement about macroscopic descriptions under incomplete microscopic knowledge, rather than about the fundamental mechanical structure of the world (Section VI, p. 8). The formal flow constructions and the hypothesis identifying their flow as physical time correspond, respectively, to Results 19, 20, 22, and 23 and to Result 21.

Common audit with scope The corresponding finite evidence consists of 23 time contents, 17 transfers, and 24 results. Complete generation is zero, local reach is seven, and sixteen roots remain; the tables specify conditions for the local constructions and dependencies on time roots.

8.19.56 Source audit

Audit input

Original source and audit scope

Source identifier 56.

Source name Complete audit of time use in Van Raamsdonk 2010

Bibliographic identification Write the audited source as $R := \text{Ref}_{\text{VR10BS}}^{\text{survey}}$. This denotes Mark Van Raamsdonk, *Building up Spacetime with Quantum Entanglement, General Relativity and Gravitation* 42, 2323–2329 (2010), DOI: 10.1007/s10714-010-1034-0. The audit covers every section, equations (1)–(3), Figures 1–5, and footnote 1.

Audit-scope identifier 56-SC1.

Audit-scope content The audited domain specified in the bibliographic identification.

Original-source-location identifier 56-L1.

Original-source-location content Abstract, pp. 1–6, equations (1)–(3), Figures 1–5, and footnote 1 of arXiv:1005.3035v1. Individual locations are given in the evidence tables.

Formal object

Object identifier 56-F1.

Object name Formal objects of the original source.

Object type Formal object.

Formal content The individual formal objects are given in Definitions 8.665 and 8.664.

Original-source-location identifier set 56-L1.

Assumptions

Assumption identifier 56-A1.

Assumption content The argument uses AdS/CFT, the identification of the TFD state with the eternal AdS black hole, a regulated spatial partition, the RT relation, and the heavy-operator geodesic approximation. Conditions for each result are given in Table 532.

Original-source-location identifier set 56-L1.

Nontemporal physical input

Input identifier 56-P1.

Input name Nontemporal physical inputs of the original source.

Input type Nontemporal physical input.

Input content The individual nontemporal physical inputs are given in Definitions 8.665 and 8.664.

Original-source-location identifier set 56-L1.

Time contents

Time-content identifier 56-T1.

Time-content name Time contents of the original source.

Time-content type Time content.

Time-content statement The individual time contents are given in Definition 8.666.

Original-source-location identifier set 56-L1.

Formal input components

Formal-component identifier 56-C1.

Formal-component name Formal-evidence environment of the original source.

Formal-component type Formal component.

Formal-component text The individual formal input components are given in Definitions 8.664 and 8.665.

Original-source-support identifier set 56-L1.

Definition 8.663 (Source and attribution components). The claims in the original source are attributed to Van Raamsdonk, while the proponents of the cited holographic relations are distinguished from the agent of the conditional inferences. The source itself reports no newly obtained direct measurement evidence.

Definition 8.664 (Physical and formal evidence in the original source). The 25 physical/formal claims comprise Nontemporal Carriers 1–7 in Table 529 and State/Geometry Descriptions 8–25 in Table 527. The 12 mathematical-object types are given in Table 528.

Table 527: Physical evidence in Van Raamsdonk 2010

Com- po- nent	Physical description	Original-source location
8	Correspondence between a quantum theory of gravity with asymptotically AdS boundary conditions and a CFT on a sphere	Introduction
9	Tensor product of two noninteracting CFT Hilbert spaces	Classical connectivity from quantum superposition
10	Correspondence between a product state and two disconnected bulk components	Product-state discussion in the same section
11	TFD state in the energy eigenbasis	Equation (1)
12	Correspondence between the TFD state and a connected eternal AdS black hole	Equation (1) and Figure 1
13	Partial trace over one CFT and the reduced thermal state	Partial trace following equation (1)
14	Partition of the boundary region and Hilbert-space factorization subject to a cutoff	A disentangling experiment
15	Entanglement entropy obtained from a reduced state	Reduced-state discussion in the same section
16	RT correspondence between entropy and the area of a minimal bulk surface	Figure 2 and the RT relation
17	Decrease in minimal-surface area accompanying a decrease in entanglement	Area discussion following Figure 2
18	Mutual information between two boundary regions	Immediately before equation (2)
19	Bound on connected correlations from mutual information	Equation (2)
20	Correspondence between a heavy-operator two-point correlation function and the length of the shortest bulk geodesic	Equation (3)
21	Conditional correspondence between the zero-mutual-information limit and divergence of proper distance	Equations (2)–(3) and Figure 3
22	Coupling of the area branch and the distance branch	Figure 4 and its discussion; the example in Figure 5
23	Separation between bulk regions accompanying a decrease in entanglement	A disentangling experiment
24	Expectation that a completely geometric description is likely to fail before entanglement is strictly zero	Footnote 1
25	Relation between entanglement and classical spatial connectivity	Abstract and conclusion

Table 528: Mathematical-object types in Van Raamsdonk 2010

m	Mathematical-object type
1	CFT Hilbert space
2	Hamiltonian
3	State family
4	Boundary region
5	Reduced state
6	Entanglement entropy
7	Bulk geometry
8	Anchored minimal surface
9	Mutual information
10	Bulk geodesic distance
11	Spatial connectivity
12	Temporal and causal structure of the boundary, asymptotic region, and eternal black hole

Definition 8.665 (Semantic components and generation-initial set for the Van Raamsdonk 2010 source). The 37 semantic components comprise 25 physical/formal claims and 12 mathematical-object types. Semantic components 1–25 correspond to physical/formal claims with the same indices; 26–37 correspond to Mathematical-Object Types 1–12. The generation-initial set consists of the nontemporal carriers in Claims 1–7. The initial set comprises the carriers’ spatial configuration and degrees of freedom; state/geometry relations belong to the source formal-evidence environment, and temporal and causal structures are supplied as time contents.

Table 529: Generation-initial semantic components of the Van Raamsdonk 2010 source

k	Semantic component	Formal evidence in the original source
1	Physical degrees of freedom of the two CFT subsystems	Description 9
2	Spatial boundary regions and subsystems	Description 14
3	Subsystem degrees of freedom represented by the reduced state	Description 15
4	Spatial bulk regions and anchored surfaces	Description 16
5	Degrees of freedom in the two regions whose correlations are compared	Description 18
6	Spatial operator locations and paths in the bulk	Description 20
7	Spatial bulk regions whose connectivity is compared	Description 23

Definition 8.666 (Time contents and time roles). The following table gives the three time contents used by the original source. There are no auxiliary objects. Table 531 identifies their uses and source locations.

Table 530: Time contents in the Van Raamsdonk 2010 source

i	Time content
1	Time in the boundary CFT
2	Temporal and causal structure of the asymptotically global AdS spacetime
3	Temporal and causal structure of the eternal black hole

Table 531: Time roles in the Van Raamsdonk 2010 source

Time content	Use in the original source	Source location
1	A CFT on a sphere times time is assigned a spacetime dual	p. 1; Figure 1 caption
2	Global AdS asymptotics specify the dual spacetime	pp. 1–2
3	The eternal AdS black hole has horizons that prevent communication between its two asymptotic regions	p. 2; Figures 1 and 5

Dependency input

Index identifier S56-XI-DI1.

Existing-evidence reference set Definition 8.667.

Dependency and transfer components

Definition 8.667 (Dependency and generation components). The dependency structure comprises 37 semantic components, 3 time contents, 0 time-content transfers, and 10 results. The transfer-index set is $\text{Idx}_R^{\text{audit,transfer}} = \emptyset$. Equation (1) gives a state in an energy eigenbasis, and Figure 5 compares spatial sections of states at different inverse temperatures. The time contents are input structures of the CFT and its dual spacetime.

Definition 8.668 (Audit-judgment components). All three time contents are used as inputs, and no generation edge outputs them. No time content therefore satisfies complete time generation, and none satisfies carrier-mediated qualified partial generation. The spatial connectivity constructed by the original source and the temporal and causal structures it uses as inputs are distinct objects.

Definition 8.669 (Result and preservation-condition components). Table 532 gives ten results and their conditions of validity. Results 1–8 are conditional state/geometry correspondences, Result 9 is spatial factorization, and Result 10 is a research question concerning the emergence mechanism. Preservation of a result with time-content premises requires its conditions to remain satisfied when those premises are replaced with generated time representations.

Table 532: Results and preservation conditions in Van Raamsdonk 2010

j	Result	Preservation conditions
1	Correspondence between quantum entanglement and spacetime connectivity	The specified AdS/CFT correspondence and its domain of boundary-state/bulk-geometry identification
2	Correspondence between a noninteracting CFT pair and a disconnected dual spacetime	Noninteraction, tensor factors, a product state, and a dual spacetime for each factor
3	Correspondence between the TFD state and an eternal AdS black hole	The TFD coefficients in equation (1), the reduced thermal state, and identification with the eternal AdS black hole
4	Relation between entanglement entropy and minimal area	A regulated boundary-region partition, a normalized reduced state, and the RT relation
5	Contraction of the geometric throat accompanying a decrease in entanglement	A fixed positive Newton constant, a common RT regime, and the compared state family
6	Relation between a correlation function and bulk geodesic length	Fixed operators, insertion points, positive mass, and normalization, the shortest bulk geodesic, and the heavy-operator approximation
7	Divergence of bulk geodesic distance in the zero-mutual-information limit	Correlator identification for a fixed bounded operator pair, the fixed conditions of Result 6, a fixed nonzero prefactor, and a geodesic approximation whose relative error vanishes in the zero limit
8	Construction of classical spatial connectivity by entanglement	Both area and distance relations in the same state family and the domain of the classical geometric description
9	Two-factor decomposition of Hilbert space induced by partitioning the boundary space into two regions	A local CFT, a boundary-region partition, and spatial factorization with a cutoff
10	Research question of spacetime emergence from quantum-information structure	The statement of incomplete understanding in the introduction and the proposed entanglement/geometry relation in the conclusion

Definition 8.670 (Evidential boundary). The formal evidence constructs spatial connectivity supported by quantum entanglement. Time in the boundary CFT, temporal and causal structure in the asymptotically global AdS spacetime, and temporal and causal structure in the eternal-black-hole example are used as inputs. When the connected correlation bounded by equation (2) is identified with the correlator in equation (3), and the fixed conditions and approximation domain in Table 532 remain valid through the

limit, vanishing mutual information implies a vanishing correlator and divergent geodesic distance. Footnote 1 states that a completely geometric description is likely to fail before entanglement is strictly zero.

Formation conditions

Index identifier S56-XI-FC1.

Existing-evidence reference set Definition 8.667.

Original-source support

Index identifier S56-XI-SS1.

Existing-evidence reference set Definition 8.667.

Results

Definition 8.671 (Typed finite audit input for Van Raamsdonk 2010). The audit input is a finite directed hypergraph comprising 37 semantic components, 3 time contents, 0 time-content transfers, and 10 results. The dependency structure follows Definition 8.667, and the results and preservation conditions follow Definition 8.669. Table 533 gives the source-formal-evidence indices and time-content premises for the ten results.

Table 533: Result inputs in Van Raamsdonk 2010

Re- sult	Content	Formal evidence	Time premises
1	Correspondence between quantum entanglement and spacetime connectivity	{25}	{1, 2}
2	Correspondence between a noninteracting CFT pair and a disconnected dual spacetime	{9, 10}	{1, 2}
3	Correspondence between the TFD state and an eternal AdS black hole	[[11, 13]]	[[1, 3]]
4	Relation between entanglement entropy and minimal area	[[14, 16]]	{1, 2}
5	Contraction of the geometric throat accompanying a decrease in entanglement	{16, 17}	{1, 2}
6	Relation between a correlation function and bulk geodesic length	{20}	{1, 2}
7	Divergence of bulk geodesic distance in the zero-mutual-information limit	[[18, 21]]	{1, 2}
8	Construction of classical spatial connectivity by entanglement	{16, 17, 21, 22, 23, 24, 25}	{1, 2}
9	Two-factor decomposition of Hilbert space induced by partitioning the boundary space into two regions	{14}	∅

Re- sult	Content	Formal evidence	Time premises
10	Research question of spacetime emergence from quantum-information structure	$\{8, 25\}$	$\emptyset$

Definition 8.672 (Canonical audit realization for Van Raamsdonk 2010). The generation-initial set and source formal-evidence environment are specified by the claim-boundary index pair of Definition 7.98, $\text{ClaimBoundaryIdx}_R := ([7], [25])$. The initial set corresponds to the nontemporal carriers in Table 529; the evidence environment contains the 25 physical/formal claims. Results 1–8 have time-content inputs, whereas Results 9 and 10 have empty time-content premises. None of the three time contents is reachable from nontemporal physical inputs; with the source formal-evidence environment added, Results 9 and 10 are reachable under their stated conditions.

Definition 8.673 (Semantic audit boundary for the Van Raamsdonk 2010 source). The source-specific semantic representation has 50 nodes: 37 semantic components, 3 time contents, and 10 results. The common audit boundary has 50 nodes: 25 physical/formal claims, 12 mathematical claims, 3 time contents, and 10 results. Its dependency edges are the ten result edges.

Definition 8.674 (Support-completion specification for Van Raamsdonk 2010). Table 533 specifies support within the 25 physical/formal claims and time-content premises for each of the ten results. The dependency structure consisting of the ten result edges is acyclic.

Result dependencies

Index identifier S56-XI-RD1.

Existing-evidence reference set Definition 8.671.

Grounds of validity

Index identifier S56-XI-RG1.

Existing-evidence reference set Definition 8.674.

Evidential hierarchy The source newly and directly measures neither an object nor a predicate. Conditioned on the domain of application of AdS/CFT, factorization of the two CFT Hilbert spaces, the product and TFD states, the RT relation, the heavy-operator geodesic approximation, and the distance limit, the formal evidence gives relations between quantum-entanglement structure and classical spatial connectivity and bulk geometry.

Generation judgment **Judgment values:** LCTR complete-generation judgment 0 / source-internal-evidence-conditioned local-reach judgment 0 / residual primitive-time roots 3.

Source-internal-evidence-conditioned local reach The source-internal-evidence-conditioned local-reach judgment is zero, and three primitive-time roots remain after that judgment.

Result dependencies There are 17 primitive-time-root–result dependency pairs, all of which are necessary dependencies.

Evidential force The domain of force of the formal evidence comprises the AdS/CFT correspondence, the product and TFD states, the RT relation, mutual information, the bulk-geodesic approximation, and the construction of spatial connectivity. The spatial connectivity constructed by the original source does not generate temporal and causal structure from nontemporal inputs. This source newly and directly measures zero objects; establishment of the source-specific results and the physical-time admissibility of the time contents are distinct determinations.

Result-preservation domain The ten results are preserved insofar as each time content occurring in them is generated through the LCTR generation sequence and the preservation conditions in Definition 8.669 and the source-specific approximation and correspondence domains are satisfied.

Placement of grounding obligations

Placement of grounding obligations The three time contents audited in this source are accepted as already formed temporal inputs rather than generated from nontemporal physical inputs before use. The placement determination for their physical-time grounding obligations is *backgrounded by primitive-time input*. This determination follows by comparing the use of each time content in a law operation or result derivation with the absence, before that use, of its corresponding generation or formation condition as an input, formation condition, evaluation condition, or failure condition.

Evidence of actual failure

Evidence of actual failure The complete determination and residual-root determination concern physical-time admissibility and are distinct from the actual failure of a concrete condition governing a result value or applicability domain. Neither the original-source formal evidence nor the LCTR audit evidence for this source establishes such an actual failure of a target condition. The actual-failure determination is *indeterminate from the original-source evidence*, and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream reach

Downstream reach The extent of reach to primitive-time-root-dependent results is determined source by source by the primitive-time-root–result dependence relation and minimal primitive-time-root support reconstructed in this division. This is structural reach. Because no actual failure of a concrete condition has been established, the determination of actual reach from a failure location to a result is *indeterminate from the original-source evidence*. Structural reach does not substitute for actual reach.

Repair scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation or formation conditions required before their use. The recalculation scope is limited to results reached from a failure location when actual failure has independently been established. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domains are restricted. Post-recalculation values are fixed by the audit result.

Audit conclusions

Proposition 8.675 (Complete audit of time use in Van Raamsdonk 2010). The source has 37 semantic components, 3 time contents, 0 time-content transfers, and 10 results. Both the complete-generation and local-reach counts for time contents are zero; Results 1–8 depend on primitive-time roots.

Proof. No edge outputs a time content, so adding the source formal-evidence environment to the nontemporal initial set reaches no time content. Results 1–8 have nonempty time premises; Results 9 and 10 have empty time premises. $\square$

Proposition 8.676 (Primitive-time roots and result dependencies in the Van Raamsdonk 2010 source). All three time contents are primitive-time roots. Results 1–8 depend on at least one root; Results 9 and 10 have empty primitive-time-root support.

Proof. With reachability conditioned on the evidence environment of Definition 8.672, Results 9 and 10 are reachable without primitive-time roots; adding all three roots makes all ten results reachable. $\square$

Proposition 8.677 (Minimal primitive-time-root support in the Van Raamsdonk 2010 source). Table 534 gives the minimal primitive-time-root support for the ten results. Each support family has one member, giving ten family members in total.

Table 534: Minimal primitive-time-root support in the Van Raamsdonk 2010 source

Result indices	Minimal-support indices
1, 2, 4–8	$\{1, 2\}$
3	$\llbracket 1, 3 \rrbracket$
9, 10	$\emptyset$

Proof. There are no transfer edges, so the time premises of each result edge form its unique minimal support. Their cardinalities sum to 17, and all 17 root–result dependency pairs are necessary. $\square$

Individual statements Statement evidence 1

Original text (English):

In this essay, we argue that the emergence of classically connected spacetimes is intimately related to the quantum entanglement of degrees of freedom in a non-perturbative description of quantum gravity.

The source location is Van Raamsdonk, *Building up Spacetime with Quantum Entanglement*, arXiv:1005.3035v1, abstract. The minimal context is the sentence summarizing the central claim of the paper as a whole. The agent to whom the statement is attributed is Mark Van Raamsdonk.

The statement objects are the emergence of classically connected spacetimes and its relation to the quantum entanglement of degrees of freedom in a nonperturbative description of quantum gravity. The predicate is that the two are argued to be intimately related. The explicit domain of application comprises the AdS/CFT correspondence, the state of two CFTs, boundary degrees of freedom, bulk geometry, and classical spatial connectivity employed by the original source.

Statement evidence 2

Original text (English):

Disentangling the degrees of freedom associated with two regions of spacetime results in these regions pulling apart and pinching off from each other in a way that can be quantified by standard measures of entanglement.

The source location is Van Raamsdonk, *Building up Spacetime with Quantum Entanglement*, arXiv:1005.3035v1, abstract. The minimal context is the sentence stating the relation between decreasing entanglement and the separation of spatial regions. The agent to whom the statement is attributed is Mark Van Raamsdonk.

The statement objects are the entanglement of degrees of freedom associated with two spacetime regions and the bulk-geometric connectivity between those regions. The predicate is that disentangling causes the regions to pull apart and pinch off, and that this change toward separation can be quantified by standard measures of entanglement. The explicit domain of application comprises the holographic state family, boundary-region partition, RT relation, mutual information, and heavy-operator geodesic approximation constructed in the original source.

Statement evidence 3

Original text (English):

However, despite a great deal of evidence for the validity of this correspondence, we do not have a deep understanding of why or how spacetime/gravity emerges from the degrees of freedom of the field theory.

The source location is Van Raamsdonk, *Building up Spacetime with Quantum Entanglement*, arXiv:1005.3035v1, p. 1, introduction. The minimal context is the passage distinguishing the validity of the gauge-theory/gravity correspondence from an understanding of the mechanism of emergence. The agent to whom the statement is attributed is Mark Van Raamsdonk.

The statement objects are the evidence supporting the validity of the gauge-theory/gravity correspondence and the degree of understanding of the mechanism by which spacetime and gravity emerge from field-theory degrees of freedom. The predicate is that, despite extensive evidence supporting the correspondence, no deep understanding of the reason for or manner of emergence has been obtained. The explicit domain of application comprises the agreements concerning the gauge-theory/gravity correspondence available at the time of publication and the research program pursued by the original source for investigating spacetime emergence from quantum-information structure.

Individual audits Quantum entanglement and classical spacetime connectivity

The first statement argues that quantum entanglement among the degrees of freedom in a nonperturbative description of quantum gravity is intimately related to the emergence of classically connected spacetime. The statement asserts a structural correspondence, within AdS/CFT, between quantum entanglement in the boundary state and the spatial connectivity of the corresponding classical bulk. Its independent grounds are the state–geometry comparisons in the original source that relate the product state of two noninteracting CFTs to a disconnected bulk, the TFD state to a connected eternal AdS black hole, and quantum entanglement to classical spatial connectivity. The evidential force belongs to the AdS/CFT correspondence, boundary states, bulk geometry, and the connectivity analyzed by the original source. In the two-CFT example, the correspondence between entanglement and classical connectivity is conditional on the identification of the TFD state with the eternal AdS black hole. The corresponding formal evidence comprises Physical/Formal Claims 8–13 and 25 in Definition 8.664, and Results 1–3, 8, and 10 in Definition 8.671.

Disentanglement and separation of regions

The second statement relates, within a state family in which the entanglement of degrees of freedom associated with two regions decreases, the geometric change by which the regions pull apart and pinch off to standard measures of entanglement. The statement relates a change in the bulk minimal area, correlation distance, and spatial connectivity to a family of states in which entanglement between boundary subsystems is varied. Its independent grounds are the three relations connecting reduced-state entanglement entropy to RT minimal area, mutual information to connected correlations, and heavy-operator two-point correlations to bulk geodesic distance. At fixed positive Newton constant, decreasing entropy implies decreasing minimal area. Divergent distance corresponds to the zero-mutual-information limit under the correlator identification and approximation conditions in Definition 8.670. The evidential force belongs to the holographic states, region partition, entanglement deformation, and bulk connectivity constructed by the original source. The deformation compares states with different entanglement. Figure 4 holds boundary geometry fixed, and Figure 5 compares spatial sections of eternal black holes at different inverse temperatures. The corresponding formal evidence comprises Physical/Formal Claims 14–24 in Definition 8.664, and Results 4–9 in Definition 8.671.

Evidence for the correspondence and incomplete understanding of the emergence mechanism

The third statement jointly states that extensive evidence supports the validity of the gauge-theory/gravity correspondence and that a deep understanding of why and how spacetime or gravity emerges from field-theory degrees of freedom has not been attained. The statement distinguishes particular agreements that support the correspondence from

theoretical understanding that explains the mechanism of spacetime and gravity emergence, and places the latter among unresolved research problems. Its independent grounds are that formal results support particular correspondences among states, geometry, and correlations, while the original source itself distinguishes them from a complete explanation of the emergence mechanism and identifies the investigation of spacetime emergence from quantum-information structure as a research program. The evidential force belongs to the existing agreements supporting the correspondence and the state of explanatory attainment at the time of publication. The supporting evidence is the statement of incomplete understanding in the introduction, Physical/Formal Claims 8 and 25 in Definition 8.664, and Result 10 in Definition 8.671.

Common audit with specified domain of application The formal evidence corresponding to the three statements comprises the 25 physical/formal claims and 12 mathematical-object types in Definition 8.664, the three time contents in Definition 8.666, and the ten result edges in Definition 8.671. Generation reach is 0/3 across the time contents. Results 1–8 give a primitive-time-root-dependence count of 8/10; Results 9 and 10 have empty time premises.

8.19.57 Source audit

Audit input

Original source and audit scope

Source identifier 57.

Source name Primitive-time-root audit of Greene’s *The Illusion of Time*

Bibliographic identification Let source *R* be the PBS/NOVA program *The Fabric of the Cosmos: The Illusion of Time* (2011), hosted and principally narrated by Brian Greene. The audited domain is the entire episode body in the official PBS subtitles (lines 23–420).

Audit-scope identifier 57-SC1.

Audit-scope content The audited domain specified in the bibliographic identification.

Original-source-location identifier 57-L1.

Original-source-location content The audited domain specified in the bibliographic identification and the original-source-location fields in the respective evidence tables.

Formal object

Object identifier 57-F1.

Object name Formal objects of the original source.

Object type Formal object.

Formal content The individual formal objects are given in Definitions 8.678 and 8.679.

Original-source-location identifier set 57-L1.

Assumptions The source assumptions are clock construction from periodic processes, realization of the second through the cesium transition, remote synchronization by signal exchange, special- and general-relativistic clock comparison, a four-dimensional spacetime representation, and a low-entropy initial condition for the universe.

Nontemporal physical input

Input identifier 57-P1.

Input name Nontemporal physical inputs of the original source.

Input type Nontemporal physical input.

Input content The individual nontemporal physical inputs are given in Definitions 8.678 and 8.679.

Original-source-location identifier set 57-L1.

Time contents Table 537 gives the uses and source locations of the 17 time contents.

Formal input components

Formal-component identifier 57-C1.

Formal-component name Formal-evidence environment of the original source.

Formal-component type Formal component.

Formal-component text The individual formal input components are given in Definitions 8.678 and 8.679.

Original-source-support identifier set 57-L1.

Definition 8.678 (Semantic components of the Greene source). The 21 semantic components comprise seven initial carriers and fourteen physical descriptions. Components 1–7 are the carriers in the following table; 8–21 are the physical descriptions in the source-claim table. The initial set comprises spatial configurations and degrees of freedom of the carriers. Process, law, statistical, and geometric relations belong to the source formal-evidence environment; cycles, elapsed quantities, synchronization, and earlier–later relations are supplied as time contents.

Table 535: Generation-initial semantic components of the Greene source

k	Semantic component	Corresponding claim indices
1	Physical degrees of freedom of oscillators, cesium atoms, and clocks	$\{8, 9, 12\}$
2	Signal carriers, transmitting and receiving devices, and recording media	$\{10\}$
3	Aircraft, ground stations, observers, and clock carriers	$\{11, 12, 15, 16\}$
4	Spatial configurations of gravitational sources, bodies, and observation sites	$\{13, 14, 15, 16, 21\}$
5	Physical degrees of freedom of microscopic systems and their surroundings	$\{17\}$
6	Microscopic configurations belonging to macroscopic classes	$\{17, 18\}$
7	Cosmic matter and spatial regions	$\{19, 20\}$

Definition 8.679 (Source claims in the Greene source). Source-claim indices 1–7 identify the initial carriers; 8–21 identify the physical descriptions in the source. Table 536 gives each object and its source location.

Table 536: Source claims in the Greene source

p	Claim	Original-source location
1	Physical degrees of freedom of oscillators, cesium atoms, and clocks	Official subtitles, lines 48–75
2	Signal carriers, transmitting and receiving devices, and recording media	Official subtitles, lines 98–102
3	Aircraft, ground stations, observers, and clock carriers	Official subtitles, lines 117–154
4	Spatial configurations of gravitational sources, bodies, and observation sites	Official subtitles, lines 170–203, 242–253
5	Physical degrees of freedom of microscopic systems and their surroundings	Official subtitles, lines 295–315
6	Microscopic configurations belonging to macroscopic classes	Official subtitles, lines 321–351
7	Cosmic matter and spatial regions	Official subtitles, lines 353–411
8	Clocks based on repeatable physical processes	Official subtitles, lines 48–54
9	Cycle counting of the cesium transition	Official subtitles, lines 65–74
10	Remote-clock synchronization by signal exchange	Official subtitles, lines 98–102
11	Clock-rate comparison between relatively moving observers	Official subtitles, lines 117–139
12	Account of the 1971 airborne atomic-clock comparison	Official subtitles, lines 149–155
13	Four-dimensional spacetime representation	Official subtitles, lines 159–166
14	Motion-direction-dependent simultaneity and interpretations of past, present, and future in spacetime	Official subtitles, lines 170–225, 416–417
15	Gravitational clock comparison between lower and upper floors of a building	Official subtitles, lines 242–247
16	Remote observation near a black hole and elapsed-time comparison upon return	Official subtitles, lines 248–263
17	Time-reversal symmetry of many equations and velocity reversal of all relevant degrees of freedom	Official subtitles, lines 295–315
18	Microscopic multiplicity corresponding to a macroscopic class and entropy	Official subtitles, lines 321–351

p	Claim	Original-source location
19	Account of the arrow of time using a low-entropy initial condition of the universe	Official subtitles, lines 353–382
20	Account of observed accelerated expansion and model-dependent scenarios for the distant future	Official subtitles, lines 166, 382–411
21	Past-directed travel and self-consistency conditional on a wormhole	Official subtitles, lines 264–282

Definition 8.680 (Time contents of the Greene source). Table 537 gives the 17 time contents. Newtonian common time is presented as a historical contrast with relativity (lines 104–115). Time Content 13 is the temporal placement and earlier–later relation of events, used in the spacetime interpretation of Description 14 and the hypothetical past-directed travel of Description 21.

Table 537: Time contents and original-source locations in the Greene source

i	Time content	Original-source location
1	Clock based on a repetitive process	Official subtitles, lines 48–54
2	Cesium frequency and the second	Official subtitles, lines 65–74
3	Clock readings and comparison of elapsed quantities	Official subtitles, lines 48–75, 149–154
4	Remote-clock synchronization	Official subtitles, lines 98–102
5	Observer-dependent rate of elapsed time	Official subtitles, lines 113–139, 242–253
6	Relation of relative motion, including its direction, to elapsed-time rates and simultaneity	Official subtitles, lines 117–139, 184–197
7	Difference in the tick rates of moving clocks	Official subtitles, lines 130–154
8	Difference between airborne atomic clocks and ground clocks	Official subtitles, lines 149–154
9	Event times and world-picture in spacetime	Official subtitles, lines 160–203
10	Relativity of simultaneity and present hypersurfaces	Official subtitles, lines 170–203
11	Difference in clock rates within a gravitational field	Official subtitles, lines 242–247
12	Comparison of elapsed quantities and signal arrival near and far from a black hole	Official subtitles, lines 248–263

i	Time content	Original-source location
13	Past, present, and future and earlier–later relations between events in spacetime	Official subtitles, lines 163–225, 264–282
14	Arrow of time	Official subtitles, lines 283–319, 353–382
15	State evolution along which entropy is compared	Official subtitles, lines 321–351
16	Low-entropy initial condition of the universe	Official subtitles, lines 353–382
17	Models of cosmic age, expansion, the distant future, and the end of time	Official subtitles, lines 166, 382–411

Dependency input

Index identifier S57-XI-DI1.

Existing-evidence reference set Definition 8.681.

Dependency and transfer components

Definition 8.681 (Time-content transfers in the Greene source). Table 538 gives the ten time-content transfers.

Table 538: Time-content transfers in the Greene source

Transfer	Dependency and transfer	Source claims	Time inputs	Time outputs
1	From cycle counting to clock-reading comparison	{8}	{1}	{3}
2	From the specified cesium cycle count to a reading in seconds	{9}	{2}	{3}
3	From clock readings and signal exchange to remote synchronization	{10}	{3}	{4}
4	From synchronization and the relative-motion relation to a moving-clock rate difference	{11}	{4, 6}	{7}
5	Use of the reported airborne/ground clock difference as an elapsed-time comparison	{12}	{8}	{3}

Transfer	Dependency and transfer	Source claims	Time inputs	Time outputs
6	From synchronization, motion direction, and spacetime coordinates to different present hypersurfaces	{14}	{4, 6, 9}	{10}
7	From spacetime coordinates and present hypersurfaces to temporal placement of events	{13, 14}	{9, 10}	{13}
8	From gravitational clock comparison to observer-dependent elapsed-time rates	{15}	{3, 11}	{5}
9	From near/far black-hole comparisons to observer-dependent elapsed-time rates	{16}	{12}	{5}
10	From state evolution with a low-entropy boundary to an account of the arrow of time	[[17, 19]]	{15, 16}	{14}

All ten transfers take at least one time content as input. Transfers 1 and 2 output clock readings; clock accuracy is an apparatus property stated separately in the source. Transfer 6 uses the simultaneity relation with motion direction, spatial separation, and the synchronization convention. Transfer 7 outputs temporal placement and earlier–later relations between events. Transfers 8 and 9 relate clock differences to observer-dependent elapsed-time rates under the specified worldlines and comparison conditions. Transfer 10 is a statistical account conditional on a low-entropy boundary. The source leaves the reason for the initial low entropy unresolved (lines 371–378). Source locations for the claims are given in Table 536.

Formation conditions

Index identifier S57-XI-FC1.

Existing-evidence reference set Definition 8.681.

Original-source support

Index identifier S57-XI-SS1.

Existing-evidence reference set Definition 8.681.

Results

Definition 8.682 (Results and preservation conditions in the Greene source). Table 539 gives the thirteen results, their source claims, time premises, and preservation conditions.

Table 539: Results, dependencies, and preservation conditions in the Greene source

Re- sult	Content	Source claims	Time premises	Preservation conditions
1	Operational account of periodic physical processes as clocks	{8}	{1, 3}	The repetitive process, cycle identification, counting, and clock carrier
2	Realization of the second through cesium oscillations	{9}	{2, 3}	The specified cesium transition and prescribed cycle count
3	Remote-clock synchronization	{10}	{3, 4}	Signal exchange, propagation conditions, and the synchronization procedure
4	Clock-rate differences due to relative motion	{11}	{4, 6, 7}	The compared observers, relative-motion relation, and synchronization convention
5	Account of the airborne atomic-clock comparison	{12}	{8}	The reported airborne/ground clock paths and reading comparison
6	Relativity of simultaneity and placement of events in spacetime	{13, 14}	{4, 6, 9, 10, 13}	Direction of relative motion, spatial separation, synchronization convention, and spacetime representation
7	Gravitational clock-rate differences	{15}	{3, 5, 11}	The specified positions and worldlines on the lower/upper floors and the clock comparison
8	Elapsed-time comparison near and far from a black hole	{16}	{5, 12}	The spacetime model, worldlines, signal paths, and compared events

Re- sult	Content	Source claims	Time premises	Preservation conditions
9	Distinction between microscopic reversibility and the macroscopic arrow of time	{17, 18}	{14, 15}	The time-reversal-symmetric equations considered in the source and relevant degrees of freedom including the surroundings
10	Connection between entropy and a low-entropy initial condition	{18, 19}	[[14, 16]]	Macroscopic classes, microscopic multiplicity, statistical reasoning, and a low-entropy boundary
11	Account of temporal direction from a low-entropy initial universe	{18, 19}	[[14, 16]]	The initial cosmic boundary and subsequent statistical state evolution
12	Account of cosmic age, expansion, and the distant future	{20}	{17}	Observational input, the accelerated-expansion model, and each far-future hypothesis
13	Discussion of past-directed travel conditional on a wormhole	{21}	{13}	A traversable wormhole connecting different times and self-consistency of the history

Result dependencies

Index identifier S57-XI-RD1.

Existing-evidence reference set Definition 8.682.

Grounds of validity

Index identifier S57-XI-RG1.

Existing-evidence reference set Definition 8.682.

Evidential hierarchy The source presents NIST time-standard operations, the 1971 airborne atomic-clock comparison, and astronomers' observations of accelerated expansion. It reports zero new quantitative measurements performed by the program itself. Results 1–4 and 6–11 are operational or theoretical explanations; Result 5 is a secondary account of an existing experiment. Result 12 comprises cosmic-age and accelerated-expansion explanations and model-dependent far-future scenarios; Result 13 is conditional on the existence of a wormhole. The equal reality of past, present, and future, discussed after

the geometric account of simultaneity, is an interpretation advanced in the program; its conclusion states that they “may all exist on equal footing” (lines 416–417). The low-entropy-boundary account of the arrow of time is presented as the source’s best understanding, while the origin of that boundary remains unresolved (lines 371–382). The far-future scenario is qualified by “one theory suggests” (line 405), and the feasibility of past-directed travel remains unresolved (lines 279–282).

Generation judgment Judgment values: LCTR complete-generation judgment 0 / source-internal-evidence-conditioned local-reach judgment 0 / residual primitive-time roots 17.

Source-internal-evidence-conditioned local reach The source-internal-evidence-conditioned local-reach judgment is zero, and 17 primitive-time roots remain after that judgment.

Result dependencies There are 43 primitive-time-root–result dependency pairs, of which 14 are necessary dependencies.

Evidential force The evidence concerns clock realization, remote synchronization, relativistic clock/worldline comparison, relativity of simultaneity, the statistical arrow of time, and conditional accounts of cosmic history and past-directed travel. The building example compares clock rates at the specified positions and worldlines. The black-hole example treats remotely received signals and elapsed quantities along a round-trip worldline under their respective comparison conditions. Entropy evolution uses temporal ordering and an initial boundary; cosmic age and far-future scenarios depend on observational input and the adopted cosmological model.

Result-preservation domain The 13 results are preserved insofar as each time-content premise is generated through the LCTR generation sequence and the preservation conditions in Table 539 and the source-specific realization, comparison, and cosmological-model conditions are satisfied.

Placement of grounding obligations

Placement of grounding obligations The 17 time contents audited in this source are accepted as already formed temporal inputs rather than generated from nontemporal physical inputs before use. The placement determination for their physical-time grounding obligations is *backgrounded by primitive-time input*. This determination follows by comparing the use of each time content in a law operation or result derivation with the absence, before that use, of its corresponding generation or formation condition as an input, formation condition, evaluation condition, or failure condition.

Evidence of actual failure

Evidence of actual failure The complete determination and residual-root determination concern physical-time admissibility and are distinct from the actual failure of a concrete condition governing a result value or applicability domain. Neither the

original-source formal evidence nor the LCTR audit evidence for this source establishes such an actual failure of a target condition. The actual-failure determination is *indeterminate from the original-source evidence*, and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream reach

Downstream reach The extent of reach to primitive-time-root-dependent results is determined source by source by the primitive-time-root–result dependence relation and minimal primitive-time-root support reconstructed in this division. This is structural reach. Because no actual failure of a concrete condition has been established, the determination of actual reach from a failure location to a result is *indeterminate from the original-source evidence*. Structural reach does not substitute for actual reach.

Repair scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation or formation conditions required before their use. The recalculation scope is limited to results reached from a failure location when actual failure has independently been established. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domains are restricted. Post-recalculation values are fixed by the audit result.

Audit conclusions

Definition 8.683 (Finite dependency graph of the Greene source). The finite dependency graph of source *R* comprises 21 semantic components, 17 time contents, 10 time-content transfers, and 13 results. It has 51 nodes and 23 transfer and result edges in total.

Proposition 8.684 (Complete audit of time use in the Greene source). The numbers of time contents generated by the complete determination and reached by the local-reach determination are both zero. All 17 time contents remain primitive-time roots, and each of the 13 results depends on at least one of them.

Proof. Each of the ten transfers in Table 538 and thirteen results in Table 539 takes at least one time content as input. Even after adding the source formal-evidence environment to the nontemporal initial set, no edge can output the first time content. Time-generation reach is therefore empty, and each result’s time premises yield its dependence on primitive-time roots. □

Proposition 8.685 (Minimal primitive-time-root support in the Greene source). Table 540 gives the minimal primitive-time-root support for the 13 results. There are 34 family members in total.

Table 540: Minimal primitive-time-root support in the
Greene source

Result	Family of minimal-support index sets
1	$\{1\}$
2	$\{2\}$
3	$\{1\}, \{2\}, \{3\}, \{8\}$
4	$\{1, 6\}, \{2, 6\}, \{3, 6\}, \{4, 6\}, \{6, 8\}$
5	$\{8\}$
6	$\{1, 6, 9\}, \{2, 6, 9\}, \{3, 6, 9\}, \{4, 6, 9\}, \{6, 8, 9\}$
7	$\{1, 11\}, \{2, 11\}, \{3, 11\}, \{8, 11\}$
8	$\{12\}$
9	$\{14, 15\}, \{15, 16\}$
10	$\{15, 16\}$
11	$\{15, 16\}$
12	$\{17\}$
13	$\{13\}, \{9, 10\}, \{1, 6, 9\}, \{2, 6, 9\}, \{3, 6, 9\}, \{4, 6, 9\}, \{6, 8, 9\}$

Proof. Updating the support antichain to a fixed point over the 10 transfer edges and 13 result edges yields the 34 family members in the table. $\square$

Individual statements Statement evidence 1

Original text (English):

It turns out time itself can speed up or slow down.

The source location is line 35 of the official PBS subtitles. The minimal context is the opening introduction of the concept of time. The agent to whom the statement is attributed is Brian Greene.

The object of the statement is the speeding up and slowing down of “time itself,” explained in the program through relativistic comparisons of clocks and physical processes. The predicate is that time itself can speed up or slow down. The explicit domain of application is the program’s opening general account introducing relativistic differences in clock rates.

Statement evidence 2

Original text (English):

That is, Einstein realized that if a stationary observer carefully analyzes what he sees, he’ll find that time itself elapses more slowly for me than it does for him.

The source location is line 137 of the official PBS subtitles. The minimal context is the account of relative motion and clock-rate comparison. The agent to whom the statement is attributed is Brian Greene.

The statement object is the passage of time for a stationary observer and for the relatively moving speaker as compared by the stationary observer. The predicate is that the observer finds time itself to elapse more slowly for the speaker than for the stationary observer. The explicit domain of application comprises the relative motion, inter-observer comparison, and clock-rate difference exemplified by the program.

Statement evidence 3

Original text (English):

It's as if gravity can pull on time, slowing its passage.

The source location is line 244 of the official PBS subtitles. The minimal context is the explicitly figurative account of gravity and clock-rate differences. The agent to whom the statement is attributed is Brian Greene.

The statement object is an explicit figure for the relation between gravity and the passage of time. The predicate is that it is as if gravity pulls on time and slows its passage. The explicit domain of application is the program's account of clock-rate differences induced by differences in gravitational field.

Statement evidence 4

Original text (English):

Compared to someone living on the top floor of a skyscraper, someone living on the bottom experiences time elapsing a little slower because gravity is just a tiny bit stronger closer to the ground.

The source location is line 247 of the official PBS subtitles. The minimal context is the application of the account of differences in gravitational field to people at different heights in a building. The agent to whom the statement is attributed is Brian Greene.

The statement object is a comparison of the passage of time for people living on the top and bottom floors of a skyscraper. The predicate is that the person on the bottom floor, where gravity is slightly stronger, experiences time as elapsing slightly more slowly. The explicit domain of application is the weak difference in gravitational field corresponding to a difference in height within a building and the comparison of people located there.

Statement evidence 5

Original text (English):

You near that black hole will appear to your friend far away to be moving slowly, talking slowly, biologically aging slowly.

The source location is line 251 of the official PBS subtitles. The minimal context is the extension of the comparison between locations near and far from a black hole to biological aging. The agent to whom the statement is attributed is Janna Levin.

The statement objects are the motion, speech, and biological aging of a person near a black hole as observed by a distant friend. The predicate is that the nearby person appears to the distant friend to move slowly, speak slowly, and age biologically slowly. The explicit domain of application comprises the two worldlines near and far from the black hole and remote observation between them.

Individual audits The speeding up and slowing down of time itself

The first statement previews the later account of differences in elapsed quantities for clocks and physical processes under relative motion and gravity. Its independent grounds are clock comparisons between relatively moving observers, the account of the 1971 airborne atomic-clock experiment, and worldline comparisons in gravity. Its evidential force concerns differences in elapsed-time rates under the respective worldline, synchronization, and signal-comparison conditions. The corresponding formal evidence comprises Time Contents 5–8, 11, and 12, Transfers 4, 5, 8, and 9, and Results 4, 5, 7, and 8.

Elapsed-time-rate comparison between relatively moving observers

The second statement is conditional on a stationary observer carefully analyzing what is seen; it states that time elapses more slowly for the relatively moving speaker. Its independent grounds are the program’s account of remote synchronization and moving-clock rate differences. Its evidential force concerns clock-rate comparison under the specified observers, synchronization convention, and motion conditions. The comparison concerns rates: agreement of rates after the motion ends and a difference in previously accumulated elapsed quantities are distinct quantities. The corresponding formal evidence comprises Time Contents 4, 6, and 7, Transfer 4, and Result 4.

The explicit metaphor of gravity pulling on time

The expression “as if” in the third statement explicitly marks the metaphor of gravity pulling on time and slowing its passage. Its independent grounds are the subsequent comparison of clocks on lower and upper floors. Its evidential force concerns elapsed-time-rate differences at the specified positions and worldlines. The corresponding formal evidence comprises Time Contents 3, 5, and 11, Transfer 8, and Result 7.

Time elapsed for people on lower and upper floors

In the fourth statement, “experiences” belongs to a relativistic comparison of elapsed quantities for clocks and bodily processes accompanying people on lower and upper floors. Its independent grounds are the relation between those worldlines and clock rates. Its evidential force concerns gravitational clock comparison at those positions. In a general gravitational clock comparison, spacetime geometry, worldlines, and signal paths specify the elapsed quantities being compared. The corresponding formal evidence comprises Time Contents 3, 5, and 11, Transfer 8, and Result 7.

Remote observation and aging near a black hole

The fifth statement says that motion, speech, and bodily processes appear slow to a distant friend observing signals received from a person near a black hole. Its independent grounds are the source’s comparison of worldlines and signal arrival in black-hole spacetime. If bodily processes follow proper time along each worldline and the source’s comparison conditions hold, the corresponding rate difference also applies to those processes. The source then considers a round trip, comparing the elapsed quantity near the black hole with that on Earth. This comparison requires the worldline through the outward journey, stay, and return, together with the reunion event. Its evidential force concerns theoretical consequences under the specified worldline and signal conditions. The corresponding formal evidence comprises Time Contents 5 and 12, Transfer 9, and Result 8.

Common audit with specified domain of application Time-content, transfer, and result indices refer respectively to Definitions 8.680, 8.681, and 8.682. The five statements describe relativistic comparisons of clocks and physical processes through a general explanation, a motion-conditioned comparison, a gravitational metaphor, a lower/upper-floor comparison, and remote observation near a black hole. Each comparison uses time contents of cycles, elapsed quantities, synchronization, and worldlines. Time-generation reach and result dependence on primitive-time roots are given in Proposition 8.684.

8.19.58 Source audit

Audit input

Original source and audit scope

Source identifier 58.

Source name Primitive-time-root audit of Wolf et al. 2011

Bibliographic identification Let source R be P. Wolf, L. Blanchet, C. J. Bordé, S. Reynaud, C. Salomon, and C. Cohen-Tannoudji, *Does an atom interferometer test the gravitational redshift at the Compton frequency?* (2011) [62]. The audited domain comprises the text, footnotes, figures, and Appendices A and B on pp. 1–26.

Audit-scope identifier 58-SC1.

Audit-scope content The audited domain specified in the bibliographic identification.

Original-source-location identifier 58-L1.

Original-source-location content The audited domain specified in the bibliographic identification and the original-source-location fields in the respective evidence tables.

Formal objects The audit uses 61 source claims, 70 time contents, 35 transfers, and 26 results. Claims 1–13 specify material and spatial initial components; Claims 14–61 specify phase laws, measurement rules, application conditions, arguments by Wolf et al., and prior work reported in their paper.

Assumptions The three-pulse action cancellation uses the same at-most-quadratic Lagrangian, its classical paths, position continuity and closure, and equal and opposite recoils. Vanishing of the whole free phase also requires the internal-state term to vanish, as it does for the symmetric geometry $T = T'$. Its intervals carry the proper-time interpretation in footnote 6 on p. 10, and each expression is used within its stated approximation. The two-pulse Ramsey geometry uses negligible recoil and wave-packet overlap within the coherence length.

Definition 8.686 (Semantic components and generation-initial set in Wolf et al. 2011). Let $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(61)$ and $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(13)$. Source-claim indices identify the same semantic components. The generation-initial set comprises material degrees of freedom and spatial configurations of the carriers.

Definition 8.687 (Physical and formal input claims in Wolf et al. 2011). The following table gives the 61 source claims and printed source pages.

Table 541: Physical and formal input claims in Wolf et al. 2011

p	Source claim	Source location
1	Material degrees of freedom and initial spatial positions of the two clock carriers.	p. 3
2	Material degrees of freedom and rest mass of the atomic carrier.	p. 4
3	Atomic wave-packet carriers and the spatial incidence structure of the two arms.	pp. 8, 9
4	Carriers of the two internal atomic states and their energy-level structure.	pp. 9, 11

p	Source claim	Source location
5	Raman-light carriers and the arrangement of their counter-propagating spatial wave vectors.	pp. 8, 10
6	Microwave or co-propagating Raman drive carriers and their spatial coupling arrangement.	p. 11
7	Material carriers of the platform and apparatus, and the spatial arrangement of the gravitational source.	pp. 3, 4, 12
8	Material coupling arrangement of the light and atomic carriers at the interaction positions.	pp. 8, 9
9	Material carrier and initial spatial position of the reference body.	pp. 3, 20
10	Material carriers and spatial configuration of the neutron beam.	pp. 24, 25
11	The three silicon crystal planes and their spatial lattice geometry.	pp. 24, 25
12	Material configuration of the gravitational source and radial spatial geometry.	pp. 25, 26
13	Atomic carriers in the optical dipole trap and the spatial configuration of the trapping potential.	p. 16
14	For two clocks with a common proper angular frequency, Eq. (1.1) relates their path proper-time difference to their phase difference.	p. 3
15	The Compton angular frequency in Eq. (1.2) is $\omega_C = mc^2/\hbar$.	p. 4
16	The closed atom-interferometer phase in Eq. (1.3) is the sum of matter-wave and light-interaction contributions.	p. 4
17	In the stated general-relativistic approximation and symmetric three-pulse geometry, the total phase in Eq. (1.4) is kgT^2 .	p. 4
18	The classical-action path-integral representations in Eqs. (2.5) and (2.9) apply to Lagrangians at most quadratic in position and velocity.	pp. 6, 7
19	In the single-Lagrangian calculation, the classical paths satisfy the Euler–Lagrange equations of the Lagrangian used for the phase.	pp. 7, 9, 17, 18, 19, 20
20	The three-pulse interferometer satisfies the position-continuity and initial/final closure conditions in Eqs. (2.12a)–(2.12d).	p. 9
21	The velocity boundary values at the middle pulse satisfy the equal and opposite recoil conditions in Eqs. (2.13a)–(2.13b).	p. 9
22	The internal-state action difference in Eq. (2.15) is $\hbar\omega_{gg'}(T - T')$.	p. 9
23	For classical paths of the same at-most-quadratic Lagrangian, the external action difference is a boundary term that cancels under Eqs. (2.12) and (2.13).	p. 10

p	Source claim	Source location
24	The free-propagation phase retains $\omega_{gg'}(T - T')$. This internal-state term also vanishes for the symmetric case $T = T'$. Its intervals carry the proper-time interpretation in footnote 6 on p. 10.	pp. 10, 12
25	The three-pulse light phase is the signed sum of $\phi(z, t) = kz - \omega t - \phi_0$ evaluated at the interaction points.	p. 10
26	The phase of a nonclosed interferometer includes the contributions associated with wave-packet separation at its endpoints.	p. 10
27	The two-pulse Ramsey geometry uses microwaves or co-propagating Raman beams, with negligible recoil and wave-packet overlap within the coherence length.	p. 11
28	The Ramsey phase is $\omega_{gg'}T + \Delta\varphi_\ell$; the leading light phase $-\omega T$ gives the detuning phase $(\omega_{gg'} - \omega)T$.	p. 11
29	The internal-transition clock uses a readable Ramsey resonance signal to lock the drive frequency to a stable internal transition.	pp. 6, 11
30	In the weak-field, nonrelativistic, linear-gravitational-potential approximation, the three-pulse geometry is symmetric, the free-propagation phase vanishes, and the light and total phases are kgT^2 .	p. 12
31	With known effective wave vector and pulse interval, the observed interferometer phase determines the atomic acceleration relative to the platform.	pp. 3, 12
32	The universality-of-free-fall test compares that atomic acceleration with the measured acceleration of a separately falling reference body.	pp. 3, 12, 17, 18, 19, 20
33	In the source's modified-Lagrangian model with anomalous position-dependent internal energy, the same Lagrangian determines trajectory and phase. Equations (3.17)–(3.22) use leading nonrelativistic terms and a linear potential, giving $\ddot{z} = -(1 + \beta_X^{(a)})g$, $Z = (1 + \alpha_X^{(a)})g\Delta z/c^2$, and $\beta_X^{(a)} = \alpha_X^{(a)}\overline{E}_X/(\overline{m}c^2)$.	pp. 17, 18, 19, 20
34	The multiple-Lagrangian model combines ordinary particle trajectories with an anomalous wave-phase law, giving $\Delta\varphi_S = \beta kgT^2$ and $\Delta\varphi = (1 + \beta)kgT^2$ in the stated leading-order approximation.	pp. 13, 14, 15, 16, 17
35	Wolf et al. argue that this multiple-Lagrangian assignment conflicts with the Feynman path-integral and least-action basis of the phase calculation and raises problems for energy conservation and Schiff's conjecture.	pp. 13, 14, 15, 16, 17, 22
36	The leading neutron-interferometer expression in Eq. (A1) is rewritten as $\Delta\varphi = \mathbf{k}_{\text{latt}} \cdot \mathbf{g} T^2$ through the de Broglie relation, $T = L/v$, and the Bragg condition.	pp. 24, 25

<i>p</i>	Source claim	Source location
37	Appendix B estimates the magnitude of the free phase and internal-state residual from higher gravity gradients. The source states that a correct general phase calculation requires revisiting the full path integral.	pp. 25, 26
38	The source's conclusion depends on the theoretical framework: the same-modified-Lagrangian model gives a universality-of-free-fall test, whereas the multiple-Lagrangian model yields the redshift interpretation criticized by the source.	p. 22
39	Müller, Peters, and Chu propose the Compton-clock redshift interpretation; Wolf et al. report and criticize that claim.	p. 1
40	Adding the same rest-mass energy to every level of the same Hamiltonian preserves energy differences and the Bloch frequency. Footnote 11 on p. 16 criticizes using different gravitational terms for states and motion.	p. 16
41	The redshift-isolation method uses independently measured clock trajectories and velocities to determine time-dilation and light-phase corrections, then compares the corrected term with the theoretical prediction.	pp. 3, 21
42	Light speed, the reduced Planck constant, gravitational acceleration, and velocity carry temporal-scale or rate calibration in their respective frequency, phase, and flight-time relations.	pp. 4, 7, 9, 11, 12, 25, 26
43	The physical-clock paths in Eq. (1.1) and the atom-interferometer arms in Eq. (1.3) are the respective carriers of their proper-time integrals.	pp. 3, 4
44	For clocks at two points at rest in a stationary gravitational field, Eq. (3.2) defines $Z = \Delta\tau_1/\Delta\tau_0 - 1$, with a nonzero proper-time interval in the denominator.	p. 14
45	Wolf et al. describe observations of planetary perihelion advance as a classical test of general relativity.	p. 2
46	Wolf et al. describe the calculation and measurement of the relativistic delay of radio waves passing near the Sun on a round trip to Mercury.	p. 2
47	Wolf et al. cite tests of the weak equivalence principle using torsion balances and lunar laser ranging, and contrast the sensitivity of macroscopic free-fall comparisons with atom-reference-body comparisons.	pp. 2, 3
48	Wolf et al. describe the Pound-Rebka experiment as a measurement of the frequency difference of gamma radiation emitted and absorbed at the top and bottom of a 22.5 m tower.	pp. 2, 21
49	Wolf et al. describe measurements of the gravitational redshift of spectral lines emitted at the solar surface.	p. 2

p	Source claim	Source location
50	Clock tests of gravitational redshift examine whether relative rates depend on the gravitational-potential difference independently of clock type and internal structure. The potential difference is determined from the trajectories of massive test bodies.	p. 2
51	Wolf et al. describe a redshift test at an accuracy of about 10^{-4} , comparing the frequency of a rocket-borne hydrogen-maser clock with that of a similar ground clock.	p. 2
52	Wolf et al. forecast a precision of about 10^{-6} for the gravitational-redshift test using ACES.	p. 2
53	Null-redshift tests compare the rates of clocks based on different physical processes or atomic species.	p. 2
54	Wolf et al. name “rotational acceleration” as the quantity measured through the Sagnac effect.	p. 3
55	Measuring the redshift between separated clocks by continuous electromagnetic signal exchange requires monitoring the clock trajectories and removing the Doppler shifts accompanying the exchange.	p. 3
56	The gravitational-redshift phase term is obtained by subtracting the time-dilation and light-phase contributions, determined using independent trajectory and velocity measurements, from the observed overall phase difference.	p. 21
57	In the idealization of clocks at rest at the top and bottom of the tower, position is constant and velocity is zero, so the first-order Doppler and special-relativistic time-dilation contributions from clock motion vanish.	p. 21
58	According to footnote 15 on p. 21, the actual Pound–Rebka experiment applies a first-order Doppler shift to compensate the gravitational redshift, and that shift provides the measured signal.	p. 21
59	The initial state in Eq. (2.8) is a plane wave with phase $(p_0 z_a - E_0 t_a)/\hbar$. Equation (2.9) gives the propagated wavefunction using this initial wavefunction, the time-dependent prefactor $G(t_b, t_a)$, and the phase factor from the classical action of the same Lagrangian.	pp. 6, 7
60	In the examples given by Wolf et al., whether anomalous coupling involves electromagnetic or hyperfine internal energy determines which is more sensitive: universality-of-free-fall tests or universality-of-clock-rates tests. The hyperfine-energy example uses GHz-scale transition frequencies.	pp. 19, 20

p	Source claim	Source location
61	Equation (2.3) defines S_{Γ} as the time integral of the Lagrangian along path Γ . Equation (2.1) gives the final state by multiplying the initial wavefunction by the propagator and integrating over initial position. For the same at-most-quadratic Lagrangian, Eq. (2.5) gives $K = F \exp(iS_{\text{cl}}/\hbar)$.	pp. 6, 7

Definition 8.688 (Time contents in Wolf et al. 2011). The following table gives the 70 time contents. The clock paths, three-pulse geometry, Ramsey geometry, and neutron geometry have their respective carriers and intervals.

Table 542: Time contents in Wolf et al. 2011

i	Time content	Source location
1	Atomic Compton angular frequency $\omega_C = mc^2/\hbar$.	pp. 4, 6
2	Atomic acceleration inferred from the same observed phase using the known effective wave vector and pulse interval.	pp. 3, 12, 17–20
3	Measured acceleration of the separately falling reference corner cube.	pp. 3, 12, 17–20
4	Matter-wave or free-propagation phase contribution $\Delta\varphi_S$ in the single-Lagrangian three-pulse geometry.	pp. 4, 9–10, 12
5	Orbital progression in the description of planetary perihelion advance.	p. 2
6	Relativistic propagation delay of radio waves passing near the Sun on a round trip to Mercury.	p. 2
7	Acceleration comparison in universality-of-free-fall tests using macroscopic bodies.	pp. 2, 3
8	Gamma-ray frequency difference associated with emission and absorption at the top and bottom of the tower.	pp. 2, 21
9	Gravitational redshift of spectral lines emitted at the solar surface.	p. 2
10	Relative rates of clocks at different gravitational potentials.	p. 2
11	Trajectories of massive test bodies used to determine gravitational-potential differences.	p. 2
12	Frequency of the rocket-borne hydrogen-maser clock.	p. 2
13	Frequency of the ground-based hydrogen-maser clock used for comparison.	p. 2
14	The planned clock-redshift comparison using ACES.	p. 2
15	Rate comparison between clocks based on different physical processes or atomic species.	p. 2
16	Internal-transition angular frequency $\omega_{gg'}$ in the three-pulse geometry.	pp. 9–10

<i>i</i>	Time content	Source location
17	Observed three-pulse atom-interferometer phase, shared by the GR and modified-single-Lagrangian interpretations of the experiment.	pp. 3, 8, 12, 17–20
18	Light-interaction phase contribution $\Delta\varphi_\ell$ in the three-pulse geometry.	pp. 4, 10, 12
19	The quantity termed “rotational acceleration” in the source’s account of Sagnac measurements.	p. 3
20	Phase difference $\Delta\varphi_{\text{clock}}$ between the two physical clocks.	p. 3
21	Independently measured clock trajectories and velocities used to isolate the redshift term.	pp. 3, 21
22	Doppler frequency shifts accompanying continuous electromagnetic signal exchange between two clocks.	p. 3
23	Proper angular frequency ω common to the two physical clocks.	p. 3
24	Proper-time integral along the path of physical clock I.	p. 3
25	Proper-time integral along the path of physical clock II.	p. 3
26	Proper-time difference between the two clock paths in Eq. (1.1).	p. 3
27	Spatial–temporal rate calibration carried by the speed of light c .	pp. 4, 12–20
28	Action/energy–time/frequency calibration carried by the reduced Planck constant $\hbar$.	pp. 4, 7, 9, 11, 25–26
29	Calculated total phase $\Delta\varphi$ in the single-Lagrangian three-pulse geometry.	pp. 4, 10, 12
30	Proper-time difference between the two atom-interferometer arms in Eq. (1.3).	pp. 4, 12
31	Local gravitational acceleration g .	pp. 3–4, 12–20, 25–26
32	First pulse interval T of the three-pulse geometry. The internal-state phase uses the proper-time interpretation in footnote 6 on p. 10 within the source’s approximation.	pp. 4, 8–10, 12
33	Atomic free-propagation phase in the modified single-Lagrangian model.	pp. 17–20
34	Quantum propagator $K(z_b, t_b; z_a, t_a)$ in Eqs. (2.1), (2.2), and (2.5).	pp. 6, 7
35	Coordinate time and integration intervals used for trajectory differentiation and action integration.	pp. 7, 9–10
36	Propagated wavefunction $\psi(z_b, t_b)$ in Eqs. (2.1) and (2.9).	pp. 6, 7
37	Initial wavefunction $\psi(z_a, t_a)$; Eq. (2.8) uses a plane wave.	pp. 6, 7
38	Action phase $S_\Gamma/\hbar$ along path Γ , evaluated on the classical path in Eqs. (2.5) and (2.9).	pp. 6, 7
39	Time-dependent prefactor $F(t_b, t_a)$ in the propagator in Eq. (2.5).	pp. 6, 7
40	Time-dependent propagation prefactor $G(t_b, t_a)$ in Eq. (2.9).	pp. 6, 7

i	Time content	Source location
41	Second pulse interval T' of the three-pulse geometry, with the same proper-time interpretation and approximation for the internal-state phase.	pp. 8–10
42	Velocity differences $\pm\hbar k/m$ induced by recoil at the middle pulse.	pp. 9–10, 25–26
43	Light angular frequency ω entering the interaction-phase field of the three-pulse geometry.	p. 10
44	Interrogation interval T in the two-pulse Ramsey geometry.	p. 11
45	Total Ramsey phase and its leading detuning term $(\omega_{gg'} - \omega)T$.	p. 11
46	Internal-transition angular frequency $\omega_{gg'}$ in the Ramsey geometry.	p. 11
47	Drive-field phase contribution $\Delta\varphi_\ell$ in the Ramsey geometry.	p. 11
48	Optical or microwave drive angular frequency ω in the Ramsey geometry.	p. 11
49	Clock-redshift quantity Z in the multiple-Lagrangian model.	pp. 13–17
50	Proper-time interval $\Delta\tau_1$ at position z_1 in the stationary-clock comparison of Eq. (3.2).	p. 14
51	Proper-time interval $\Delta\tau_0$ at position z_0 in the stationary-clock comparison of Eq. (3.2).	p. 14
52	Atomic free-propagation phase in the multiple-Lagrangian model.	pp. 13–17
53	Anomalous proper-time difference entering the wave-phase law of the multiple-Lagrangian model.	pp. 13–17
54	Total atom-interferometer phase in the multiple-Lagrangian model.	pp. 13–17
55	Bloch-oscillation frequency in the optical-dipole-trap gravimeter.	p. 16
56	Effective atomic free-fall acceleration $(1 + \beta_X^{(a)})g$ predicted by the modified single-Lagrangian model.	pp. 17–20
57	Clock-redshift quantity Z in the modified single-Lagrangian model.	pp. 17–20
58	GHz-scale transition frequencies used to illustrate hyperfine internal energies.	pp. 19, 20
59	Total atom-interferometer phase in the modified single-Lagrangian model.	pp. 17–20
60	Light-interaction phase in the modified single-Lagrangian model.	pp. 17–20
61	Gravitational-redshift phase term obtained by removing the time-dilation and light-phase contributions from the observed phase.	p. 21
62	Phase contribution from special-relativistic time dilation due to clock motion.	p. 21

i	Time content	Source location
63	Light-signal phase contribution in the clock comparison.	p. 21
64	Overall phase difference observed in a redshift measurement.	p. 21
65	First-order Doppler shift applied in the Pound–Rebka experiment to compensate the gravitational redshift.	p. 21
66	Gravitational phase $\Delta\varphi$ in the neutron interferometer.	pp. 24–25
67	Compton frequency associated with the neutron.	pp. 24, 25
68	Neutron velocity v .	p. 25
69	Neutron-interferometer flight time $T = L/v$.	p. 25
70	Estimated free-propagation phase from higher gravity gradients in Eq. (B2), including the internal-state residual in Eq. (B3).	pp. 25–26

Dependency inputs Source-claim and time-content indices refer to the tables above. Each input set specifies the inputs required jointly for that transfer.

Dependencies and transfers

Definition 8.689 (Time-content transfers in Wolf et al. 2011). The following table gives the 35 transfers and their conditions of application.

Table 543: Time-content transfers in Wolf et al. 2011

Transfer	Operation	Source claims	Time inputs	Time outputs
1	Subtract the two clock-path proper-time integrals to obtain the difference in Eq. (1.1). p. 3	{14}	{24, 25}	{26}
2	Multiply the path proper-time difference by the common proper angular frequency to obtain the clock phase difference. p. 3	{14}	{23, 26}	{20}
3	Multiply the proper-time difference of the closed atom-interferometer arms by ω_C to obtain the matter-wave term in Eq. (1.3). pp. 4, 6	{16, 18, 43}	{1, 30}	{4}
4	Add the matter-wave and light phases of the same three-pulse geometry. pp. 4, 10	{16}	{4, 18}	{29}

Transfer	Operation	Source claims	Time inputs	Time outputs
5	Use the same at-most-quadratic Lagrangian, classical paths, position conditions (2.12), and recoil conditions (2.13) to obtain the free phase $\omega_{gg'}(T - T')$. pp. 7, 9–10	[[18, 24]]	{16, 32, 35, 41, 42}	{4}
6	Add the light phases $kz - \omega t - \phi_0$ at the interaction points with the signs in Eq. (2.19). p. 10	{25}	{32, 41, 43}	{18}
7	Evaluate the light phase as kgT^2 for the symmetric geometry in the weak-field, nonrelativistic, linear-potential GR approximation. p. 12	{30, 42}	{31, 32}	{18}
8	Evaluate the leading drive-phase term $-\omega T$ in the two-pulse Ramsey geometry with negligible recoil. p. 11	{27, 28}	{44, 48}	{47}
9	Add the Ramsey internal-state phase $\omega_{gg'}T$ and drive-field phase to obtain the leading term $(\omega_{gg'} - \omega)T$. p. 11	{27, 28}	{44, 46, 47}	{45}
10	Evaluate the neutron phase $\mathbf{k}_{\text{latt}} \cdot \mathbf{g} T^2$ for the three-crystal Bragg geometry at the leading order of Eqs. (A1)–(A4). pp. 24–25	{36, 42}	{31, 69}	{66}
11	Estimate the free phase and internal-state residual from higher gravity gradients using the approximation in Appendix B. pp. 25–26	{37}	{16, 31, 32, 41, 42}	{70}
12	Multiply the anomalous proper-time difference of the multiple-Lagrangian model by ω_C to obtain its free-propagation phase. pp. 13–17	{34}	{1, 53}	{52}

Transfer	Operation	Source claims	Time inputs	Time outputs
13	Evaluate the free phase as βkgT^2 with that model's leading weak-field/nonrelativistic approximation, ordinary particle trajectories, and symmetric geometry. pp. 13–17	{34, 42}	{31, 32}	{52}
14	Add that model's free phase and the ordinary light phase to obtain $(1 + \beta)kgT^2$ in the same approximation. pp. 13–17	{34}	{18, 52}	{54}
15	Derive the leading free-fall acceleration from the modified single Lagrangian with anomalous position-dependent internal energy. pp. 17–20	{33}	{31}	{56}
16	Apply the external-action cancellation conditions and $T = T'$ to the modified single Lagrangian to obtain zero free-propagation phase. pp. 10, 17–20	{18, 19, 23, 33}	{16, 32, 35, 41, 42}	{33}
17	Evaluate the light phase from the effective acceleration and pulse interval in that model's symmetric three-pulse geometry. pp. 17–20	{33, 42}	{32, 56}	{60}
18	Add the free and light phases of that model. pp. 17–20	{33}	{33, 60}	{59}
19	Define $\omega_C = mc^2/\hbar$ from c , $\hbar$, and atomic rest mass. p. 4	{15, 42}	{27, 28}	{1}
20	Divide the internal energy splitting of the three-pulse geometry by $\hbar$ to obtain its transition angular frequency. pp. 9–10	{22, 42}	{28}	{16}
21	Divide the internal energy splitting of the Ramsey clock by $\hbar$ to obtain its transition angular frequency. p. 11	{29, 42}	{28}	{46}
22	Obtain the recoil velocity in Eq. (2.13) from recoil momentum $\hbar k$ and atomic mass. pp. 9, 25–26	{21, 42}	{28}	{42}

Transfer	Operation	Source claims	Time inputs	Time outputs
23	Obtain the neutron velocity from the de Broglie relation $Mv = 2\pi\hbar/\lambda$. p. 25	{36, 42}	{28}	{68}
24	Divide the neutron flight length L by velocity v to obtain $T = L/v$. p. 25	{36, 42}	{68}	{69}
25	Evaluate $Z = (1 + \beta)g\Delta z/c^2$ at leading order in the multiple-Lagrangian model. pp. 13–17	{34, 42}	{27, 31}	{49}
26	Evaluate $Z = (1 + \alpha_X^{(a)})g\Delta z/c^2$ at leading order in the modified single-Lagrangian model. pp. 17–20	{33, 42}	{27, 31}	{57}
27	Convert the observed phase in the standard geometry into atomic acceleration using the known effective wave vector and symmetric pulse interval. pp. 3, 8, 12	{31}	{17, 32}	{2}
28	Apply that measurement conversion within the modified single-Lagrangian model to infer atomic acceleration from the observed phase. pp. 3, 17–20	{33}	{17, 32}	{2}
29	Convert the level spacing of the same Hamiltonian to angular frequency using $\hbar$ to obtain the Bloch-oscillation frequency. p. 16	{40, 42}	{28}	{55}
30	Form the ratio of the two proper-time intervals in Eq. (3.2) and subtract one to obtain the clock redshift. p. 14	{44}	{50, 51}	{49}
31	Subtract the time-dilation and light-phase corrections determined for the same clock comparison from the observed overall phase difference. p. 21	{41, 56}	[[62, 64]]	{61}

Transfer	Operation	Source claims	Time inputs	Time outputs
32	Integrate the same Lagrangian along a path over the specified time interval and divide by $\hbar$ to obtain its action phase. The at-most-quadratic propagator representation uses its value on the classical path. pp. 6, 7	{61}	{28, 35}	{38}
33	Use the initial plane wave in Eq. (2.8), prefactor G , and classical action phase to obtain the final state in Eq. (2.9), with initial classical momentum equal to the plane-wave momentum p_0 . pp. 6, 7	{18, 19, 59}	{37, 38, 40}	{36}
34	For the same at-most-quadratic Lagrangian, combine prefactor F with the classical action phase to obtain the propagator in Eq. (2.5). pp. 6, 7	{18, 19, 61}	{38, 39}	{34}
35	Integrate the product of the propagator and initial wavefunction over initial position to obtain the final state in Eq. (2.1). pp. 6, 7	{18, 19, 61}	{34, 37}	{36}

Results

Definition 8.690 (Results in Wolf et al. 2011). The following table gives the 26 results.

Table 544: Results in Wolf et al. 2011

q	Result	Source location
1	For two clocks with a common proper angular frequency, their phase difference is determined by the path proper-time difference in Eq. (1.1).	p. 3
2	Equation (1.2), $\omega_C = mc^2/\hbar$, defines the Compton angular frequency using the calibrations carried by c and $\hbar$.	p. 4
3	The total phase of the closed atom interferometer is the sum of the matter-wave and light phases in Eq. (1.3).	pp. 4, 6

q	Result	Source location
4	For classical paths of the same at-most-quadratic Lagrangian, the position and recoil conditions in Eqs. (2.12) and (2.13) cancel the external action difference.	pp. 7, 9–10
5	Under those conditions, the free-propagation phase retains the internal-state term $\omega_{gg'}(T - T')$.	pp. 9–10
6	In the linear-gravitational-potential GR approximation and symmetric geometry $T = T'$, the internal-state term also vanishes.	pp. 10, 12
7	The calculated total phase in that approximation is kgT^2 . Known k, T convert the observed phase into atomic acceleration relative to the platform.	pp. 3–4, 12
8	In the two-pulse Ramsey geometry with negligible recoil, the resonance signal based on the leading detuning phase $(\omega_{gg'} - \omega)T$ enables drive-frequency locking.	p. 11
9	Under Wolf et al.'s operational clock criterion, the Ramsey internal transition supplies a readable resonance signal, whereas the whole-atom Compton phase under discussion supplies no readable signal at ω_C .	pp. 1, 6, 11
10	At leading order and in the symmetric geometry of the modified single-Lagrangian model, the free phase vanishes and the total phase is $(1 + \beta_X^{(a)})kgT^2$, with no Compton frequency in the expression.	pp. 17–20
11	The universality-of-free-fall test compares the atomic acceleration inferred from the observed phase with the measured acceleration of a separately falling corner cube.	pp. 3, 12, 20
12	The multiple-Lagrangian model combines ordinary particle trajectories and an anomalous wave-phase law to give, at leading order, free phase βkgT^2 and total phase $(1 + \beta)kgT^2$.	pp. 13–17
13	The source compares the free-fall interpretation in the modified single-Lagrangian model with the redshift interpretation in the multiple-Lagrangian model, criticizing the latter for conflicting with the principles underlying phase calculation.	pp. 1, 13–20, 22
14	The leading neutron phase takes the form $\mathbf{k}_{\text{latt}} \cdot \mathbf{g} T^2$ in Eq. (A4) through the de Broglie relation, flight time, and Bragg condition. At fixed lattice wave vector, gravity, flight time, and geometry, this expression is independent of mass and Compton frequency.	pp. 24–25
15	Appendix B estimates the free phase and internal-state residual from higher gradients. An exact general phase calculation requires revisiting the full path integral.	pp. 25–26
16	The clock redshift in the multiple-Lagrangian model is $Z = (1 + \beta)g\Delta z/c^2$ at leading order.	pp. 13–17
17	The atomic acceleration in the modified single-Lagrangian model satisfies $\ddot{z} = -(1 + \beta_X^{(a)})g$ in Eq. (3.18).	pp. 17–20

q	Result	Source location
18	That model gives the clock redshift $Z = (1 + \alpha_X^{(a)})g\Delta z/c^2$ in Eq. (3.19).	pp. 17–20
19	That model relates its free-fall and clock-redshift coefficients by $\beta_X^{(a)} = \alpha_X^{(a)}\overline{E}_X/(\overline{m}c^2)$ in Eq. (3.20).	pp. 17–20
20	The source’s redshift-isolation method supplies independently measured clock trajectories and velocities as inputs to the time-dilation and light-phase corrections.	pp. 3, 21
21	The difference between atomic and reference-body accelerations interpreted within the modified single-Lagrangian model constrains the difference in composition-dependent coefficients in Eq. (3.22).	pp. 3, 17–20
22	Adding a common rest-mass energy to the same Hamiltonian preserves its level spacings and Bloch-oscillation frequency.	p. 16
23	The stationary-clock redshift in Eq. (3.2) is defined by $Z = \Delta\tau_1/\Delta\tau_0 - 1$.	p. 14
24	The gravitational-redshift phase term is the observed overall phase difference minus the independently determined time-dilation and light-phase corrections.	p. 21
25	Under the same at-most-quadratic Lagrangian and initial-plane-wave conditions, Eq. (2.9) gives $\psi(z_b, t_b) = G(t_b, t_a)\psi(z_0, t_a)\exp(iS_{cl}/\hbar)$.	pp. 6, 7
26	The propagator and initial state determine the final state through Eq. (2.1); for an at-most-quadratic Lagrangian, the classical-action representation in Eq. (2.5) applies.	pp. 6, 7

Table 545: Result dependencies in Wolf et al. 2011

q	Source claims	Temporal premises
1	{14}	{23, 26}
2	{15, 42}	{27, 28}
3	{16, 18, 43}	{1, 18, 30}
4	{18, 19, 20, 21, 23}	{32, 35, 41, 42}
5	[[22, 24]]	{16, 32, 41}
6	{24, 30}	{16, 32, 41}
7	{17, 30, 31, 42}	{17, 31, 32}
8	[[27, 29]]	{44, 46, 48}
9	{29, 39}	{1, 46, 48}
10	{19, 33}	{33, 56, 60}
11	{31, 32}	{2, 3}
12	{34, 35}	{1, 18, 52, 53}
13	{33, 34, 35, 38, 39}	{49, 54, 57, 59}
14	{36, 42}	{28, 31, 68, 69}
15	{37}	{16, 31, 32, 41, 42}
16	{34, 42}	{27, 31}

q	Source claims	Temporal premises
17	$\{33, 42\}$	$\{31\}$
18	$\{33, 42\}$	$\{27, 31\}$
19	$\{33\}$	$\{27\}$
20	$\{41\}$	$\{21\}$
21	$\{32, 33\}$	$\{2, 3\}$
22	$\{40\}$	$\{55\}$
23	$\{44\}$	$\{50, 51\}$
24	$\{41, 56\}$	$\llbracket 62, 64 \rrbracket$
25	$\{18, 19, 59\}$	$\{37, 38, 40\}$
26	$\{18, 19, 61\}$	$\{34, 37\}$

Grounds for the results Following Definition 7.71, support for each result is calculated using the transfer set below. The physical conditions of each transfer are those specified in the transfer table and source claims.

Table 546: Result-specific audit branches in Wolf et al. 2011

Domain	Results	Transfer set
Two-clock path comparison	$\{1\}$	$\{1, 2\}$
Compton-frequency definition	$\{2\}$	$\{19\}$
Closed three-pulse geometry, single Lagrangian	$\llbracket 3, 5 \rrbracket$	$\{3, 4, 5, 6, 19, 20, 22\}$
Leading-order GR, linear gravity, symmetric geometry	$\{6, 7, 11\}$	$\{3, 4, 5, 6, 7, 19, 20, 22, 27\}$
Two-pulse Ramsey clock	$\{8\}$	$\{8, 9, 19, 21\}$
Multiple-Lagrangian model	$\{12, 16\}$	$\{6, 7, 12, 13, 14, 19, 25\}$
Modified single-Lagrangian model	$\{10, 17, 18, 19, 21\}$	$\{15, 16, 17, 18, 19, 20, 22, 26, 28\}$
Neutron Bragg geometry	$\{14\}$	$\{10, 23, 24\}$
Higher-gradient estimate	$\{15\}$	$\{11, 20, 22\}$
Operational-clock comparison	$\{9\}$	$\{8, 9, 19, 21\}$
Independent trajectory and velocity measurement	$\{20\}$	$\emptyset$
Common energy shift of the same Hamiltonian	$\{22\}$	$\{29\}$
Comparison of theoretical frameworks	$\{13\}$	$\{6, 7, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 25, 26, 28\}$
Proper-time ratio of stationary clocks	$\{23\}$	$\{30\}$
Isolation of the clock-redshift phase	$\{24\}$	$\{31\}$
Quadratic-Lagrangian propagation of an initial plane wave	$\{25\}$	$\{32, 33\}$
State evolution with the quadratic propagator	$\{26\}$	$\{32, 34, 35\}$

Generation judgment Judgment: 0 LCTR-complete generations; 0 locally reachable time contents under source-evidence conditions; 70 residual primitive time roots.

Result dependencies There are 119 primitive-time-root–result dependency pairs, including 40 necessary dependencies.

Evidential scope The domain consists of the physical configuration of each clock or interferometer, the Lagrangians used for trajectory and phase, the stated approximation, and the input conditions of each measurement or comparison.

Result-preservation domain The 26 results hold under their respective conditions on physical configuration, correspondence within the same carrier and measurement, time contents, calibration, and theoretical model. The domain of reuse under LCTR is the intersection of this evidential domain and the domain satisfying the generation and validity conditions in Definition 8.99.

Placement of grounding obligations The 70 time contents are used as already established temporal inputs without being generated from nontemporal physical inputs before use. The judgment on the placement of physical-time grounding obligations is “backgrounded through primitive temporal inputs.”

Evidence of actual failure Whether the input conditions specified here actually fail is undetermined by the source evidence.

Downstream reach The result-dependency table and minimal primitive-time-root support families specify structural reach. If an actual failure of a specific condition is identified, recalculation covers the results reached through dependencies from that failure site.

Scope of repair The conditions required to generate time contents before use are the objects of repair. If an actual failure of a specific condition is identified, the dependent results are recalculated from the repaired inputs or their domain of validity is restricted.

Definition 8.691 (Finite dependency graph in Wolf et al. 2011). The 61 semantic components, 70 time contents, 35 transfers, and 26 results form the finite dependency graph with the audit branches specified above.

Proposition 8.692 (Primitive time roots and result dependence in Wolf et al. 2011).

$$\text{AuditPartIdx}_R = \text{AllRootDepIdx}(70, 26).$$

Proof. The generation-initial set consists of Claims 1–13. Every one of the 35 transfers requires existing time contents, so no time content is reachable from that initial set. Adding the source formal-evidence environment supplies no transfer without temporal inputs. Complete generation and local reach under source-evidence conditions are therefore both zero, leaving 70 primitive time roots. All 26 results have temporal premises; supplying the time contents and source formal-evidence environment under the respective application conditions satisfies each result edge. $\square$

Proposition 8.693 (Minimal primitive-time-root support in Wolf et al. 2011). The minimal-support families below contain 115 members. There are 119 primitive-time-root–result dependency pairs, including 40 necessary dependencies.

Proof. Updating the support antichains to a fixed point over the transfer set and result edge specified for each result yields the following table. $\square$

Table 547: Minimal primitive-time-root support in Wolf et al. 2011

Result	Family of minimal-support index sets
1	$\{23, 26\}, \{23, 24, 25\}$
2	$\{27, 28\}$
3	$\{1, 18, 30\}, \{18, 27, 28, 30\}, \{1, 30, 32, 41, 43\}, \{27, 28, 30, 32, 41, 43\}$
4	$\{28, 32, 35, 41\}, \{32, 35, 41, 42\}$
5	$\{16, 32, 41\}, \{28, 32, 41\}$
6	$\{16, 32, 41\}, \{28, 32, 41\}$
7	$\{17, 31, 32\}$
8	$\{44, 46, 48\}, \{28, 44, 48\}$
9	$\{1, 46, 48\}, \{1, 28, 48\}, \{27, 28, 48\}$
10	$\{31, 32, 33\}, \{32, 33, 56\}, \{31, 33, 60\}, \{33, 56, 60\}, \{28, 31, 32, 35, 41\},$ $\{28, 32, 35, 41, 56\}, \{16, 31, 32, 35, 41, 42\}, \{16, 32, 35, 41, 42, 56\}$
11	$\{2, 3\}, \{3, 17, 32\}$
12	$\{1, 18, 53\}, \{1, 31, 32, 53\}, \{18, 27, 28, 53\}, \{1, 32, 41, 43, 53\}, \{27, 28, 31,$ $32, 53\}, \{27, 28, 32, 41, 43, 53\}$

Result	Family of minimal-support index sets
13	$\{27, 31, 32, 33\}$, $\{27, 31, 32, 59\}$, $\{27, 31, 54, 59\}$, $\{49, 54, 57, 59\}$, $\{31, 32, 33, 49, 57\}$, $\{31, 32, 49, 57, 59\}$, $\{18, 27, 31, 52, 59\}$, $\{18, 49, 52, 57, 59\}$, $\{27, 31, 33, 54, 60\}$, $\{33, 49, 54, 57, 60\}$, $\{1, 18, 27, 31, 53, 59\}$, $\{1, 18, 49, 53, 57, 59\}$, $\{27, 28, 31, 32, 35, 41\}$, $\{32, 33, 49, 54, 56, 57\}$, $\{18, 27, 28, 31, 53, 59\}$, $\{18, 27, 31, 33, 52, 60\}$, $\{18, 33, 49, 52, 57, 60\}$, $\{1, 18, 27, 31, 33, 53, 60\}$, $\{1, 18, 33, 49, 53, 57, 60\}$, $\{16, 27, 31, 32, 35, 41, 42\}$, $\{32, 41, 43, 49, 52, 57, 59\}$, $\{28, 31, 32, 35, 41, 49, 57\}$, $\{18, 32, 33, 49, 52, 56, 57\}$, $\{18, 27, 28, 31, 33, 53, 60\}$, $\{18, 27, 28, 49, 53, 57, 59\}$, $\{1, 32, 41, 43, 49, 53, 57, 59\}$, $\{1, 18, 32, 33, 49, 53, 56, 57\}$, $\{16, 31, 32, 35, 41, 42, 49, 57\}$, $\{32, 33, 41, 43, 49, 52, 56, 57\}$, $\{32, 33, 41, 43, 49, 52, 57, 60\}$, $\{28, 32, 35, 41, 49, 54, 56, 57\}$, $\{28, 32, 35, 41, 49, 54, 57, 60\}$, $\{18, 27, 28, 33, 49, 53, 57, 60\}$, $\{1, 32, 33, 41, 43, 49, 53, 56, 57\}$, $\{1, 32, 33, 41, 43, 49, 53, 57, 60\}$, $\{16, 32, 35, 41, 42, 49, 54, 56, 57\}$, $\{16, 32, 35, 41, 42, 49, 54, 57, 60\}$, $\{27, 28, 32, 41, 43, 49, 53, 57, 59\}$, $\{28, 32, 35, 41, 43, 49, 52, 56, 57\}$, $\{28, 32, 35, 41, 43, 49, 52, 57, 60\}$, $\{18, 28, 32, 35, 41, 49, 52, 56, 57\}$, $\{18, 28, 32, 35, 41, 49, 52, 57, 60\}$, $\{18, 27, 28, 32, 33, 49, 53, 56, 57\}$, $\{1, 28, 32, 35, 41, 43, 49, 53, 56, 57\}$, $\{1, 28, 32, 35, 41, 43, 49, 53, 57, 60\}$, $\{1, 18, 28, 32, 35, 41, 49, 53, 56, 57\}$, $\{1, 18, 28, 32, 35, 41, 49, 53, 57, 60\}$, $\{16, 32, 35, 41, 42, 43, 49, 52, 56, 57\}$, $\{16, 32, 35, 41, 42, 43, 49, 52, 57, 60\}$, $\{16, 18, 32, 35, 41, 42, 49, 52, 56, 57\}$, $\{16, 18, 32, 35, 41, 42, 49, 52, 57, 60\}$, $\{27, 28, 32, 35, 41, 43, 49, 53, 56, 57\}$, $\{27, 28, 32, 35, 41, 43, 49, 53, 57, 60\}$, $\{27, 28, 32, 33, 41, 43, 49, 53, 56, 57\}$, $\{27, 28, 32, 33, 41, 43, 49, 53, 57, 60\}$, $\{18, 27, 28, 32, 35, 41, 49, 53, 56, 57\}$, $\{18, 27, 28, 32, 35, 41, 49, 53, 57, 60\}$, $\{1, 16, 32, 35, 41, 42, 43, 49, 53, 56, 57\}$, $\{1, 16, 32, 35, 41, 42, 43, 49, 53, 57, 60\}$, $\{1, 16, 18, 32, 35, 41, 42, 49, 53, 56, 57\}$, $\{1, 16, 18, 32, 35, 41, 42, 49, 53, 57, 60\}$, $\{1, 16, 18, 32, 35, 41, 42, 49, 53, 56, 57\}$, $\{1, 16, 18, 32, 35, 41, 42, 49, 53, 57, 60\}$
14	$\{28, 31\}$
15	$\{28, 31, 32, 41\}$, $\{16, 31, 32, 41, 42\}$
16	$\{27, 31\}$
17	$\{31\}$
18	$\{27, 31\}$
19	$\{27\}$
20	$\{21\}$
21	$\{2, 3\}$, $\{3, 17, 32\}$
22	$\{28\}$, $\{55\}$
23	$\{50, 51\}$
24	$\{62, 63, 64\}$
25	$\{37, 38, 40\}$, $\{28, 35, 37, 40\}$
26	$\{34, 37\}$, $\{37, 38, 39\}$, $\{28, 35, 37, 39\}$

Individual statements Statement evidence 1

Original text (English):

clock operating at the Compton frequency

The source location is the abstract on p. 1 of Wolf et al., *Does an atom interferometer test the gravitational redshift at the Compton frequency?* The minimal context is the passage in which Wolf et al. report the claim by Müller, Peters, and Chu that an atom

is a clock operating at the Compton frequency associated with its rest mass. Wolf et al. are the authors of the quoted passage; Müller, Peters, and Chu are the proponents of the reported claim.

The statement object is the interpretation that treats an atom used in an atom interferometer as a clock operating at the Compton frequency. The predicate is that the atom is a clock operating at the Compton frequency. The explicit domain of application is the existing interpretation, introduced in the abstract, of the atom interferometer as a gravitational-redshift test.

Statement evidence 2

Original text (English):

not a real clock

The source location is p. 6 of the same source. The minimal context compares a clock whose resonant signal is read by tuning a drive to an internal atomic transition with the reinterpretation of the whole atom as a clock ticking at the Compton frequency. The agents to whom the statement is attributed are P. Wolf et al.

The statement object is the reinterpretation that treats the atom as a whole as a clock ticking at the Compton frequency. The predicate is that the whole atom supplies no physical signal at the Compton frequency and thus does not constitute a clock with a readable resonant signal in the sense just described. The explicit domain of application is the signal-based criterion for clock operation on p. 6.

Statement evidence 3

Original text (English):

does not deliver a physical signal

The source location is p. 6 of the same source. The minimal context is the statement that no readable physical signal is present at the Compton frequency. The agents to whom the statement is attributed are P. Wolf et al.

The statement object is a readable signal from the atom as a whole associated with the Compton frequency. The predicate is that the atom does not deliver a physical signal at the Compton frequency. The explicit domain of application is the readable-physical-signal output condition used to assess clock operation on p. 6.

Statement evidence 4

Original text (English):

constitutes an atomic clock

The source location is p. 11 of the same source. The minimal context is the passage in which Ramsey fringes from two-pulse interference driven by microwaves or co-propagating Raman beams are read to lock the drive frequency to the internal atomic transition frequency. The agents to whom the statement is attributed are P. Wolf et al.

The statement object is the internal-transition system that uses an internal atomic level difference and a driving signal. The predicate is that the system constitutes an atomic clock. The explicit domain of application is the clock construction on p. 11 of the original source that uses the internal-transition angular frequency and the driving frequency of a laser or microwave field.

Statement evidence 5

Original text (English):

independent of the Compton frequency

The source location is p. 24 of the same source. The minimal context is the account of the gravitational phase difference in a three-crystal-plane neutron interferometer and its reformulation in Appendix A, pp. 24–25. The agents to whom the statement is attributed are P. Wolf et al.

The statement object is the gravitationally induced phase difference in a neutron interferometer. The predicate is that the representation in Eq. (A4), obtained using the Bragg condition and $T = L/v$, contains no explicit mass or Compton frequency when the lattice wave vector, gravitational acceleration, flight time, and apparatus orientation are specified. The explicit domain comprises the three-crystal-plane geometry on pp. 24–25 and $\Delta\varphi = \mathbf{k}_{\text{latt}} \cdot \mathbf{g} T^2$, obtained from the leading expression in Eq. (A1) through the de Broglie relation and the Bragg condition.

Individual audit Clock criterion and phase relations

Müller, Peters, and Chu treat the whole atom as a clock operating at the Compton frequency. Wolf et al. use a readable resonant signal tuned to an internal transition as the criterion for clock operation and criticize applying the Compton-clock interpretation to a whole atom that supplies no signal at that frequency. The internal-transition clock on p. 11 uses Ramsey fringes from two-pulse interference to lock the drive frequency to the internal transition frequency. Interference between the two arms constitutes its clock operation.

For the three-pulse interferometer, classical paths obtained from the same at-most-quadratic Lagrangian, together with the position conditions in Eq. (2.12) and recoil conditions in Eq. (2.13), yield cancellation of the external action difference in Eq. (2.17). The free-propagation phase retains the internal-state term $\omega_{gg'}(T - T')$ in Eq. (2.18); the full free-propagation phase vanishes for the symmetric case $T = T'$. Footnote 6 on p. 10 specifies that the interval in this internal phase is, strictly, a proper-time interval. In the weak-field, nonrelativistic, linear-potential approximation to general relativity, the total phase is the light-interaction phase kgT^2 .

Comparison of the atomic acceleration inferred from the interferometer with the measured acceleration of a separately falling corner cube constitutes the universality-of-free-fall test. The source’s model using the same modified Lagrangian for trajectory and phase also makes this comparison. A model assigning different Lagrangians to the trajectory and wave phase yields a redshift-like phase; Wolf et al. argue that this assignment conflicts with the Feynman path-integral and least-action basis used to calculate the phase.

For the neutron interferometer, the leading expression in Eq. (A1) is transformed through the de Broglie relation, $T = L/v$, and the Bragg condition into the phase representation in Eq. (A4), using the lattice wave vector, gravitational acceleration, and flight time. The flight time carries temporal dependence through the neutron velocity. The source provides a theoretical analysis of these existing atom- and neutron-interferometer experiments.

Corresponding formal evidence

The clock interpretation introduced by Statement 1 corresponds to Source Claim 39 and Time Content 1. Source Claims 15 and 42, Transfer 19 and Result 2 give the definition of the Compton angular frequency. The readable-signal clock criterion in Statements 2 and 3 corresponds to Result 9; its comparison with an internal-transition clock uses Source Claim 29 and Time Contents 1, 46 and 48.

The Ramsey clock in Statement 4 corresponds to Source Claims 27–29, Time Contents 44–48, Transfers 8, 9 and 21, and Results 8 and 9. The neutron-phase representation in Statement 5 corresponds to Source Claims 36 and 42, Time Contents 28, 31 and 66–69, Transfers 10, 23 and 24, and Result 14. The preceding three-pulse phase calculations and comparison of the two Lagrangian frameworks correspond to Results 3–7 and 10–13.

The indices for these time contents, transfers and results refer respectively to Definitions 8.688, 8.689 and 8.690. Proposition 8.692 gives the reachability judgments for each time content and result.

Common audit with specified domain of application Boundary of evidential force

The source’s phase calculations and clock judgments depend on each interferometer’s geometry, the Lagrangians used for trajectory and phase, the approximation domain, and the readable signal. Temporal dependence reaches the results through proper-time intervals, pulse intervals, transition and drive frequencies, flight times, velocities, and accelerations.

8.19.59 Source audit

Audit Inputs

Source and Audit Scope

Source identifier 59.

Source name Anderson 2012 Primitive-Time-Root Audit

Bibliographic identification Edward Anderson, *Problem of Time in Quantum Gravity, Annalen der Physik* 524, 757–786 (2012), doi:10.1002/andp.201200147; arXiv:1206.2403v2. $R = \text{Ref}_{\text{And12PT}}$.

Audit-scope identifier 59-SC1.

Audit-scope content Abstract, Sections 1–14, Equations (16)–(18), (20)–(30), (32), (37) and (39)–(44), with the notes and domains of each strategy.

Source-location identifier 59-L1.

Source-location content Sections and equations in the tables, with page numbers from arXiv:1206.2403v2.

Assumptions Anderson examines facets of the Problem of Time in quantum gravity and several strategies. The ADM decomposition uses a specified spacelike foliation, and the JBB construction uses a real nonsingular integration domain. An internal-clock function, its local validity as a clock, and the solution for reduced evolution each have their corresponding conditions. The WKB construction uses a selected phase flow and approximation order. Conditional probabilities, records, histories and nonideal-clock evolution have their respective states, projectors, carriers and temporal variables.

Nontemporal Physical Inputs The nontemporal configuration data, metric and energy, wave function, clock and record carriers, projectors, state and Hamiltonian, spatial geometry and canonical variables in Claims 1–7 form the generation-initial set. Temporal order, evolution in time and the clock constructions belong to the corresponding time contents and formal claims.

Formal Input Elements

Definition 8.694 (Anderson 2012 Physical and Formal Input Claims). The semantic components of the 13 source-form descriptions $\{\text{FormDesc}_{R,p}\}_{p \in [13]}$ are represented by $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(48)$. Let $\text{ClaimBoundaryIdx}_R = ([7], [48])$: Claims 1–7 form the generation-initial set, and all 48 claims form the source formal-evidence environment. The claim table gives their contents and correspondence to source-form descriptions.

Table 548: Anderson 2012 Source-Form Descriptions

p	Source-Form Description	Source Location
1	Configurational change	§3,7, pp. 5–6, 15
2	Metric structure of configuration space	§3,7, pp. 5, 15
3	Energy difference	§3,7, pp. 5, 15
4	Heavy degrees of freedom	§6.2, pp. 10–11
5	Light degrees of freedom	§6.2, pp. 10–11
6	Wave function	§6.2–6.3, pp. 10–12
7	Clock subsystem	§6.3,14.2, pp. 12, 22
8	Record carrier	§6.3–6.4, pp. 12–13
9	History carrier	§6.4,14.1, pp. 13, 21
10	Quantum state	§6,14, pp. 8–13, 21–22
11	Hamiltonian	§6.4,14, pp. 13, 21–22
12	Spacetime geometry	§1–3,6.1, pp. 1–5, 8–9
13	Matter fields	§1–3,6, pp. 1–5, 8–13

Formal Objects For positive lapse α and slice coordinate time t , the ADM normal proper-time increment is $d\tau = \alpha dt$. With configurational line element ds along a solution of the Jacobi-type action and energy difference $E - V > 0$, JBB time, with fixed origin and clock unit, has the normalization of Eq. (18):

$$t^{\text{JBB}} = t_0 + \int \frac{ds}{\sqrt{E - V}}.$$

The group-corrected branch of Eq. (32) solves the action extremum; the direct reduced-space branch uses its metric and action. A sufficient totality of local relevant change (STLRC) considers contributions from all changes and retains locally significant contributions in the time calculation.

For heavy variable h and light variable l , in units with $\hbar = 1$,

$$\Psi(h, l) = \psi(h)\chi(h, l), \quad \psi(h) = e^{iS(h)}, \quad \frac{dh}{dt^{\text{WKB}}} = N^{hh} \frac{dS}{dh}.$$

Time is integrated locally along the nonzero flow of the selected real phase. With $N^{hh}M_{hh} = 1$, the cross derivative becomes a time derivative, giving $i\partial_{t^{\text{WKB}}}\chi = H_l\chi$ at the same BO/WKB order. The state χ depends nontrivially on h . Higher-order corrections have the validity domain associated with the light-sector contributions and omitted terms.

For a positive normalized state ρ and projectors P_A, P_B , Eq. (29) gives the conditional probability, provided $\text{tr}(P_A\rho) > 0$,

$$p(B | A) = \frac{\text{tr}(P_B P_A \rho P_A)}{\text{tr}(P_A \rho)}.$$

When the conditioning proposition is a clock reading, its projector is identified with the physical clock value. History class operators use time-ordered projector products; HPO uses tensor products over distinct time factors. With an adjoint on the conjugate branch,

$$D(a, b) = \text{tr}(C_a \rho C_b^\dagger), \quad D(a, a) \geq 0.$$

Class operators for avoiding a region use projected propagation, or a complex potential and scattering limit within a specified approximation regime.

Let T be the clock reading and t the auxiliary evolution parameter. For a clock distribution with $p(t | T) \geq 0$ and $\int_I p(t | T) dt = 1$ on a domain I ,

$$\rho(T) = \int_I p(t | T) U(t) \rho_0 U(t)^\dagger dt, \quad U(t) = e^{-iHt}.$$

This state preserves positivity and trace. At a common coarse-graining order for a narrow clock distribution with increasing width, and with the mean drift included in H_{eff} , the dissipative convention is

$$\frac{d\rho}{dT} = -i[H_{\text{eff}}, \rho] - \gamma(T)[H, [H, \rho]], \quad \gamma(T) \geq 0.$$

The coefficient corresponds to the width of the same clock distribution. Finite-window integration, averaging over the full parameter range and a shrinking-window limit have their respective domains and limiting conditions.

Formal Input Elements

Definition 8.695 (Anderson 2012 Time Contents). Let $n_R = 20$. Each time content retains the same objects, variables and branches, with the conditions specified by its claims and results.

Table 549: Anderson 2012 Time Contents

i	Time Content	Source Location
1	External time of ordinary quantum mechanics	pp. 1–2, 5, 12
2	Proper time in special relativity	pp. 1–2
3	Coordinate time in special relativity	pp. 1–2
4	Foliation time in general relativity	pp. 2–5, 18–19
5	Normal-time interval determined by the ADM lapse	pp. 3–4
6	JBB time	pp. 5–6, 15
7	Time candidate associated with the indefinite direction of Riem space	pp. 8, 10
8	Time candidates from scale factor and spatial volume	pp. 8, 19
9	General internal time selected by a canonical transformation	pp. 8–9, 19–20
10	York time defined from canonical geometric data	pp. 9, 15, 19
11	Time candidate from a scalar matter field	pp. 9
12	Reference-matter time labeling events	pp. 9
13	Four-volume time in unimodular theory	pp. 9, 18

i	Time Content	Source Location
14	WKB time constructed from the phase of the heavy degrees of freedom	pp. 10–11, 19, 21
15	Clock reading used in conditional probabilities	pp. 12, 22
16	The common instant of records and their represented temporal relations	pp. 11–13, 21
17	Label time and kinematical time ordering histories	pp. 13
18	Hamiltonian-evolution parameter in histories theory	pp. 13
19	Parameter used for the Heisenberg evolution of projectors	pp. 22
20	Density-state evolution with respect to a nonideal clock	pp. 22

Table 550: Anderson 2012 Semantic Components and Temporal Uses

k	Claim	Temporal Use	Description and Location
1	Nontemporal data for candidate configurations and differences between configurations.	$\emptyset$	1; §3;§7, pp. 5–6, 15
2	Candidate data for the configuration-space metric, energy and potential.	$\emptyset$	2–3; §3;§7, pp. 5, 15
3	Nontemporal candidate data for heavy and light degrees of freedom and the wave function.	$\emptyset$	4–6; §6.2, pp. 10–11
4	Physical configurations and states of clock subsystems and record carriers.	$\emptyset$	7–8; §6.3–6.4;§14.2, pp. 12–13, 22
5	Nontemporal data for projectors and carriers representing candidate histories.	$\emptyset$	9; §6.3–6.4, pp. 12–13
6	Algebraic data for quantum states and Hamiltonians.	$\emptyset$	10–11; §6.2–6.4;§14.2, pp. 10, 12–13, 22
7	Nontemporal candidate data for spatial geometry, matter fields and canonical variables.	$\emptyset$	12–13; §2;§6.1;§7, pp. 3–4, 8–9, 15
8	Anderson distinguishes properties of time and eight facets of the Problem of Time, and states that the facets interfere with one another.	$\emptyset$	$\emptyset$; §1;§4–5, pp. 1–2, 6–8
9	In ordinary quantum mechanics, the Schrödinger equation determines state evolution with respect to external time.	1 ^S	6, 10–11; §1.2;§3(17), pp. 2, 5
10	Special relativity uses proper time along worldlines and coordinate time relative to an inertial frame.	2 ^I , 3 ^B	12; §1.1–1.3, pp. 1–2

k	Claim	Temporal Use	Description and Location
11	A general-relativistic spacetime need not admit a timelike Killing vector and allows choices of spacelike foliation.	4^{BC}	12; §1.2;§2;§5;§9, pp. 2–3, 7, 18
12	The ADM decomposition determines the normal proper-time increment from the positive lapse and coordinate-time increment and, with the shift, describes evolution of the spatial metric.	$4^{\text{S}}, 5^{\text{IQ}}$	12; §2(2)–(4), pp. 3–4
13	Quantizing the Hamiltonian constraint yields the frozen formalism and its contrast with the ordinary external-time-dependent Schrödinger equation.	$1^{\text{S}}, 4^{\text{S}}$	10–12; §3(16)–(17), pp. 5–6
14	The configurational line element and energy difference along a solution of the Jacobi-type action define JBB time by the integral in Eq. (18).	6^{KQ}	1–3; §3(18), pp. 5
15	Equation (32) uses group-corrected change and an extremum of the action. RPMs built directly on reduced configuration space define the JBB integral without a separate group extremization.	6^{KQ}	1–3; §7(32), Strategies1–2, pp. 15
16	STLRC considers contributions from all changes and retains locally significant contributions in the time calculation. The clock readout and the system maintaining its calibration are distinct.	6^{KQ}	1, 7; §3 Notes2–4, pp. 6
17	Resolving the classical frozen-formalism problem through the JBB construction and unfreezing the quantum constraint are distinct tasks.	$\emptyset$	$\emptyset$; §3;§6.1;§7;§14, pp. 5, 8, 15, 21
18	For the proposal to choose the indefinite direction of Riem space as time, the gravitational potential fails to preserve the conformal Killing symmetry needed for a Klein–Gordon-type interpretation.	7^{B}	2–3, 12; §6.1(19);§6.2, pp. 8, 10
19	The scale factor a or spatial volume element $\sqrt{\det h}$ is selected as an internal-time candidate. The same scale clock cannot be continued through a region where recollapse destroys its monotonicity.	8^{KQB}	12; §6.1;§11 Example1.1, pp. 8, 19
20	Solving the constraint in a form linear in the clock momentum and specifying the canonical transformation, branch and quantum domain yields a formal Schrödinger equation in the selected internal time.	9^{S}	10–12; §6.1(20)–(22), pp. 8–9

k	Claim	Temporal Use	Description and Location
21	General hidden time requires a canonical transformation from the canonical variables to a clock variable and true degrees of freedom. Its domain of existence and interpretation as a spacetime coordinate have distinct conditions.	$9^{\text{KB}}, 4^{\text{B}}$	12; §6.1(22); §11; §13, pp. 9, 19–20
22	York time is defined, with a fixed normalization, as a function of canonical variables proportional to $h_{ij}\pi^{ij}/\sqrt{\det h}$.	10^{KQ}	12; §6.1, pp. 9
23	Reduced evolution in York time requires a CMC foliation and solutions of the momentum constraint and Lichnerowicz–York equation; its validity domain has global limitations.	$10^{\text{SB}}, 4^{\text{B}}$	12; §6.1; §7; §11, pp. 9, 15, 19
24	A one-component scalar matter field is selected as clock variable and its conjugate momentum is isolated from the constraint. The field choice, constraint branch and clock domain specify this construction.	11^{KQB}	13; §6.1, pp. 9
25	Reference matter couples to gravitational variables to label spacetime events. Compatibility between clock resolution and the physicality of matter constrains its use.	$12^{\text{KXB}}, 4^{\text{B}}$	12–13; §6.1, pp. 9
26	Unimodular time measures the four-volume between specified embeddings. Distinct embeddings can enclose the same four-volume, so its value alone does not determine a unique foliation.	$4^{\text{B}}, 13^{\text{IQB}}$	12–13; §6.1; §9, pp. 9, 18
27	The Born–Oppenheimer split and WKB phase of the heavy degrees of freedom yield a Hamilton–Jacobi equation at the specified approximation order.	14^{B}	4–6; §6.2(24), pp. 10
28	WKB time is defined along the nondegenerate flow of a selected real phase. The local branch, configuration metric and its inverse, and approximation regime are inputs to the construction.	14^{KQ}	2, 4–6; §6.2(24)–(26); §11, pp. 10–11, 19
29	Converting the heavy–light cross derivative to a derivative in the constructed WKB time yields the Schrödinger equation for the light degrees of freedom at the same approximation order. The light state depends nontrivially on the heavy variable.	14^{S}	4–6, 11; §6.2(25)–(26), Note3, pp. 10–11

k	Claim	Temporal Use	Description and Location
30	Higher-order time corrections involving the light degrees of freedom, and omissions of nonadiabatic, higher-derivative and averaged terms, change the approximation order and its validity domain.	14 ^{SB}	4–6, 11; §6.2 Notes4–6;§11, pp. 11, 19
31	The conditional probability in Eq. (29) is determined by the specified state and two projectors when its denominator is positive.	∅	6, 9–10; §6.3(29), pp. 12
32	Choosing a clock-subsystem value as the conditioning proposition makes Eq. (29) a conditional probability at that clock reading.	15 ^{KQ}	7, 9–10; §6.3(29), Note2, pp. 12
33	Equation (28) in ordinary quantum mechanics gives the conditional probability for sequential measurements at two specified times.	1 ^{SB}	9–10; §6.3(28), pp. 12
34	The records postulates use information-bearing subconfigurations localized within a common instant and the atemporal propositions associated with them.	16 ^{BO}	8–10; §6.3, Records Postulates1–2, pp. 12–13
35	Anderson requires record correlations to support a semblance of dynamics or history and adopts STLRC for the abstraction of time.	16 ^{SO}	8, 10; §6.3, Records Postulates3–4, pp. 12–13
36	Classical histories are configuration sequences or configuration-space curves ordered by specified label times, with corresponding momenta, brackets and constraints.	17 ^{SB}	9–11; §6.4, pp. 13
37	Ordinary class operators use time-ordered products of projectors, whereas HPO uses tensor products over distinct time factors. The latter gives projectors on the history Hilbert space.	17 ^{SB}	9–11; §6.4, items1Q–2Q, pp. 13
38	With a positive normalized state and the adjoint on the conjugate branch, the class-operator convention $D(a, b) = \text{tr}(C_a \rho C_b^\dagger)$ has nonnegative diagonal values.	17 ^P	9–11; §6.4(30), pp. 13
39	Histories theory distinguishes kinematical transformations of history labels from dynamical transformations generated by the Hamiltonian.	17 ^B , 18 ^S	9–11; §6.4 Note3, pp. 13

k	Claim	Temporal Use	Description and Location
40	Records that select a single time within a history use one projector, for which the distinction between ordinary and tensor products disappears.	$16^B, 17^B$	8–9; §6.4 Note4, pp. 13
41	History calculations use label time, an available emergent time, or internal or matter time in some approaches.	$\emptyset$	$\emptyset$; §6.4 Note1, pp. 13
42	Anderson describes decoherence of histories as a promising but not yet established mechanism for justifying the WKB approximation.	$\emptyset$	$\emptyset$; §14.1, pp. 21
43	Halliwell-type combinations use WKB flux and class operators to compute probabilities of histories entering configuration-space regions. This calculation follows a different route from deriving a time-dependent Schrödinger equation for the light state.	$14^B, 17^P$	4–6, 9–11; §8(37); §14.1, pp. 16, 21
44	The class operator for histories not entering a region is constructed with a specified complex potential, time interval, limit and operator domain. Softening discrete projections has approximation conditions.	17^{IPB}	9–11; §14.1(40)–(42), pp. 21
45	The projector P evolves as $P(t) = e^{iHt} P e^{-iHt}$ under the specified self-adjoint Hamiltonian and evolution parameter.	19^S	9–11; §14.2, after (43), pp. 22
46	Averaging the clock- and observable-projector conditional probability over the auxiliary parameter requires a specified integration domain, state and positive normalization denominator.	$15^{KQ}, 19^{IB}$	7, 9–11; §14.2(43), pp. 22
47	The approximate evolution of the density state conditioned on a nonideal clock contains a Hamiltonian commutator and a dissipative term depending on the clock-distribution width. Its coefficients and approximation order refer to the same coarse-graining.	$15^{KB}, 19^S, 20^{SP}$	7, 10–11; §14.2(44), pp. 22
48	Global use of internal clocks is limited by monotonicity, canonical-chart existence and CMC foliations; joining local domains requires compatible evolution.	$8^B, 9^B, 10^B, 11^B$	4–6, 12–13; §11–12, pp. 19–20

Superscripts on the temporal indices in the claim table denote the use roles below. The temporal uses of each claim correspond to its variables, order, intervals, clock readings

and evolution.

Table 551: Anderson 2012 Time-Use Roles

Code	Role	Use Count
I	Interval	5
Q	Quantity	11
X	Cross-position correspondence	1
K	Clock reading	12
C	Causality	1
S	State evolution	16
P	Probabilistic evolution	4
O	Observation record	2
B	Reference boundary	30

Table 552: Anderson 2012 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3–4, 6, 14, 16–19	member	instant, order_relation
2, 5	member	interval_boundary, endpoint
7–8, 10–11, 15	member	instant
9, 12	member	instant, endpoint
13	member	interval_boundary, orientation, endpoint
20	dependency	$\emptyset$

Table 553: Anderson 2012 Finite Cardinalities

Object	Cardinality
Time contents	20
Claims	48
Auxiliary objects	0
Transfers	9
Results	15
Claim/time-content uses	52
Role assignments	82
Obligation atoms	92
Audit-boundary nodes	83
Audit-boundary edges	24

Definition 8.696 (Semantic Components and Generation-Initial Set in Anderson 2012). The 48 semantic components in the claim table use the same claim nodes. With $\text{ClaimBoundaryIdx}_R = ([7], [48])$, the initial set comprises Claims 1–7 and the source formal-evidence environment comprises Claims 1–48.

Table 554: Generation-Initial Semantic Components in
Anderson 2012

k	Semantic Component	Source Description
1	Nontemporal data for candidate configurations and differences between configurations.	1
2	Candidate data for the configuration-space metric, energy and potential.	2–3
3	Nontemporal candidate data for heavy and light degrees of freedom and the wave function.	4–6
4	Physical configurations and states of clock subsystems and record carriers.	7–8
5	Nontemporal data for projectors and carriers representing candidate histories.	9
6	Algebraic data for quantum states and Hamiltonians.	10–11
7	Nontemporal candidate data for spatial geometry, matter fields and canonical variables.	12–13

Dependency Inputs

Definition 8.697 (Anderson 2012 Time-Content Transfers). Let $\text{Idx}_R^{\text{audit,transfer}} = [9]$. The table gives time inputs, time outputs, claim support and source locations.

Table 555: Anderson 2012 Time-Content Transfers

j	Time In i	Time Out i	Claim Support k	Transfer Content and Source Location
1	4	5	7, 12	Calculation of the ADM normal-time interval. §2(2)–(4)
2	$\emptyset$	6	1–2, 14	JBB integration from the configurational line element. §3(18)
3	$\emptyset$	6	1–2, 14–15	JBB construction by best matching or on reduced configuration space. §7(32), Strategies1–2
4	$\emptyset$	8	7, 19	Definition of clock coordinates from scale variables. §6.1
5	$\emptyset$	10	7, 22	Definition of York time from canonical data. §6.1
6	$\emptyset$	11	7, 24	Selection of a clock coordinate from a matter variable. §6.1
7	$\emptyset$	14	2–3, 27–28	Construction of WKB time from the heavy-sector phase flow. §6.2(24)–(26)
8	4	13	7, 26	Calculation of four-volume time between embeddings. §9

j	Time In i	Time Out i	Claim Support k	Transfer Content and Source Location
9	15, 19	20	4, 6, 45–47	Effective density evolution conditioned on a clock. §14.2(43)–(44)

Grounding Conditions

Table 556: Anderson 2012 Conditions for Transfers

Trans-fer Index	Conditions
1	Specified Lorentz metric, spacelike foliation, positive lapse and coordinate-time increment.
2	A solution of the specified Jacobi-type action, real line element, positive energy difference, integrability, orientation and origin; the group action is trivial or already removed.
3	The group-corrected branch solves the action extremum; the direct reduced-space branch uses its metric and action. A real nonsingular integration domain is specified for each branch.
4	A positive scale factor or positive-definite three-metric, with the choice of a or $\sqrt{\det h}$ fixed; monotonicity is separately required on the domain used for evolution.
5	Positive-definite three-metric, canonical momentum and fixed normalization.
6	Selected one-component scalar field and canonical structure; evolution after solving the constraint requires a specified real branch and valid clock domain.
7	BO/WKB approximation, selected real phase, invertible configuration metric, nonzero flow, and a local branch and origin.
8	Two specified embeddings in a Lorentz spacetime, orientation and a finite volume integral.
9	Distribution relating clock readings to the auxiliary parameter, projectors, positive normalization, specified coarse-graining and approximation order, and a dissipative convention preserving Hermiticity and positivity.

Results

Definition 8.698 (Anderson 2012 Results). Let $\text{ld}\mathbf{x}_R^{\text{audit,result}} = [15]$.

Table 557: Anderson 2012 Results

q	Result	Source Location
1	Classification of facets and strategies of the Problem of Time	§1,3–6,14, pp. 1–2, 5–8, 13, 21

q	Result	Source Location
2	ADM time and the frozen-formalism contrast	§1.2–3(16)–(17), pp. 2–5
3	Classical construction of JBB time and STLRC	§3(18);§7(32), pp. 5–6, 15
4	Internal-clock functions and reduced evolution	§6.1(20)–(22);§7,11–13, pp. 8–9, 15, 19–20
5	Validity domains of Riem, reference-matter and four-volume time	§6.1;§9, pp. 8–9, 18
6	Evolution of the light state in WKB time	§6.2(24)–(26);§11, pp. 10–11, 19
7	Conditional probability defined by projectors on one state	§6.3(29), pp. 12
8	Probability conditioned on a clock reading	§6.3(29), Note2, pp. 12
9	Records postulates and single-time records within histories	§6.3–6.4, pp. 12–13
10	History projectors, class operators and decoherence functionals	§6.4(30), pp. 13
11	Kinematical and dynamical time transformations	§6.4 Note3, pp. 13
12	Heisenberg time evolution of projectors	§14.2, after (43), pp. 22
13	Clock-conditioned probability averaged over the auxiliary parameter	§14.2(43), pp. 22
14	Effective density evolution with respect to a nonideal clock	§14.2(44), pp. 22
15	Combination of semiclassical, records and histories approaches	§8(37);§14.1(39)–(42), pp. 16, 21

Table 558: Anderson 2012 Conditions for Results

Result Index	Conditions and Domain
1	The objects and strategies distinguished by Anderson and the stated status of each account.
2	Specified foliation, lapse, constraints and quantization.
3	Integration domain and construction conditions of Transfer 2 or 3; STLRC retains locally significant changes.
4	Each branch specifies its clock function, invertible canonical transformation, constraint solution, applicable CMC conditions and local validity domain.
5	Potential symmetry, physicality of reference matter, and embedding and volume conditions, as appropriate to each proposal.
6	Selected BO/WKB branch, nondegenerate flow, a light state depending nontrivially on the heavy variable, and a common approximation order.
7	Positive normalized state, projectors and positive conditioning probability.

Result Index	Conditions and Domain
8	Conditions of Result 7 and the physical identification of the conditioning variable with a clock reading.
9	The records postulates use common-instant localization, information, correlations and logical structure. Single-time records within histories use a specified history label and one projector. Recovery of dynamics is a requirement of the postulates.
10	Label ordering, product type, state, conjugate-branch adjoint convention and operator domain.
11	Distinct generators for history-label transformations and Hamiltonian evolution.
12	Self-adjoint Hamiltonian, projector and specified auxiliary parameter; the source exponent uses units with $\hbar = 1$.
13	Specified integration domain and positive normalization denominator; any limit requires a specified direction and existence.
14	State and coefficients referring to the same clock distribution, coarse-graining and approximation order, with a specified dissipative convention.
15	Local semiclassical regime, valid operator and integration domains, and the unresolved parts identified by the source.

Result Dependence

Definition 8.699 (Anderson 2012 Result-Dependence Hyperedges). Each row defines a hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 559: Anderson 2012 Result-Dependence Hyperedges

q	Time Premises i	Claim Support k
1	$\emptyset$	8, 17, 41–42
2	1, 4–5	9, 11–13
3	6	14–17
4	4, 8–11	19–24, 48
5	4, 7, 12–13	18, 25–26
6	14	27–30
7	$\emptyset$	31
8	15	31–32
9	16–17	34–35, 40
10	17	36–38
11	17–18	39
12	19	45
13	15, 19	45–46
14	15, 19–20	47
15	6, 14, 16–17	17, 30, 42–44

Proposition 8.700 (Anderson 2012 Audit-Boundary Cardinalities).

$$\text{BoundaryCount}_R^{\text{audit}} = (83, 24).$$

Proof. The 48 claims, 20 time contents and 15 results give the node count; the 9 transfers and 15 result dependencies give the edge count. $\square$

Evidence Hierarchy Anderson examines classifications of the Problem of Time, conditional constructions, approximations and combined strategies. The paper reports zero newly acquired direct measurements.

Definition 8.701 (Audit Boundary in Anderson 2012). The audit boundary is the disjoint union of 48 claim nodes, 20 time contents and 15 results, with 83 nodes and 24 transfer or result-dependence edges.

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 5; remaining primitive-time roots 15. Local reach comprises Time Contents 6, 8, 10, 11 and 14, corresponding to the JBB, scale, York, scalar-matter and WKB constructions.

Result Dependence Results 1, 3, 6 and 7 are reachable from the source formal-evidence environment without added temporal seeds. The other 11 results depend on primitive-time roots. All 20 root–result dependence pairs are necessary.

Evidential Force Anderson’s classifications, clock functions, integral constructions, approximate evolution and combined strategies have the conditions given in the result tables. The JBB integral and WKB phase flow construct local times; scale, York and matter variables give specified clock functions. Reduced evolution with a clock function additionally requires the branch-specific conditions, including monotonicity, constraint solutions, canonical transformations or foliations. The source describes decoherence-based justification of WKB as an unestablished mechanism.

Placement of Grounding Obligations The construction conditions for the five locally reachable time contents occur in Transfers 2–7. Foliation, general internal clocks, the common instant of records, history ordering, clock readings and the auxiliary evolution parameter occur as temporal premises of the results that use them. Complete generation from nontemporal Initial Claims 1–7 is zero; all 20 time contents are primitive-time inputs in the physical-time-admissibility judgment.

Evidence of Actual Failure Actual failure of a specific physical-time generation condition and the consequent change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 20 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns the time contents used by each result and the physical-time generation conditions required before their use. If actual failure of a specific condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.702 (Primitive-Time Roots and Result Dependence in Anderson 2012). In the component order of Definition 8.115,

$$\begin{aligned}\text{AuditPartIdx}_R &= (\emptyset, [20]; \\ &\quad \emptyset, \{1, 3, 6, 7\}; \\ &\quad \{2, 4, 5\} \cup \llbracket 8, 15 \rrbracket, \emptyset).\end{aligned}$$

$$\text{TimeContent}_R^{\text{audit, evid}} = \{\text{TimeContent}_{R,i} \mid i \in \{6, 8, 10, 11, 14\}\}.$$

Thus $R \in \text{Ref}^{\text{audit, rootdep}}$ and $R \notin \text{Ref}^{\text{audit, rootunreach}}$.

Proof. Closure from Initial Claims 1–7 remains those seven nodes. With all 48 claims, Transfers 2–7, which require no temporal input, yield Time Contents 6, 8, 10, 11 and 14 and reach Results 1, 3, 6 and 7. The other transfers require temporal input. Adding all 20 time contents as seeds reaches all 15 results. $\square$

Proposition 8.703 (Minimal Primitive-Time-Root Support in Anderson 2012). Each result has one minimal support, giving 15 supports in total. Empty supports correspond to Results 1, 3, 6 and 7.

Proof. Apply Definition 7.46 to the 9 transfers and 15 result dependencies, starting from singleton temporal seeds and empty supports for source formal evidence. By Proposition 7.47, the inclusion-minimal supports of the stable antichains give the following table. $\square$

$(m; a, b)$ gives the number of supports and their minimum and maximum cardinalities; write $(m; a)$ when the bounds coincide.

Table 560: Minimal Primitive-Time-Root Support in Anderson 2012

Result Indices	Support Statistics
1, 3, 6–7	1; 0
2, 4, 9, 11, 13–15	1; 2
5	1; 3
8, 10, 12	1; 1

Result-Preservation Domain Fix the object classifications, actions, configuration metrics, constraints, clock functions, quantum states, projectors, approximation orders, operator domains and boundary conditions as specified in the result tables. With generated temporal representations realizing the same order, intervals, clock readings, flows and averaged evolution, all 15 results are preserved within their respective domains.

The finite counts are 20 time contents, 48 claims, 9 transfers and 15 results. The 52 claim/time-content use pairs have 82 role assignments, with 92 obligation atoms.

Individual statements Original text (English):

Heisenberg time evolutions of projectors P

The source location is Anderson, *Problem of Time in Quantum Gravity*, arXiv:1206.2403v2, p. 22, §14.2, immediately after Eq. (43). At that location, $P(t)$ denotes this time evolution.

Original text (English):

Simply calling an entity a time or a clock does not make it one.

The source location is p. 1, §1.1 of the same work.

Shared Evidence with Scope The first statement refers to the projectors $P(t)$ used in nonideal-clock conditional probabilities in §14.2. Their evolution parameter is the auxiliary parameter averaged over when conditioning on clock readings. The second statement belongs to the list of properties of time and clocks in §1.1 and distinguishes naming a candidate from the properties it satisfies. Both statements are made by Edward Anderson.

The JBB integral construction appears in Eq. (18) of §3 and Eq. (32) of §7; the WKB construction and approximation conditions appear in Eqs. (24)–(26) and Notes 3–6 of §6.2. The classical JBB construction, quantum-level WKB construction and clock-conditioned averaged evolution have their respective inputs and domains.

On pp. 10–11, Anderson identifies the risk that an insufficiently justified WKB assumption transfers the Problem of Time to a wavefront that strongly implies time.

Individual audit Dependencies of the respective objects

The projector evolution in the first statement uses the Heisenberg evolution parameter in Claim 45 and Result 12. Results 13–14, conditioned on clock readings, depend on both the clock variable and that auxiliary parameter. The second statement distinguishes naming from the properties of a clock. The JBB, scale, York, matter-variable and WKB constructions give local reach within the source under the conditions of Transfers 2–7. Proposition 8.702 gives the judgments for complete physical-time generation and local reach.

Corresponding formal evidence

Definitions 8.695, 8.696, 8.697 and 8.698 specify the correspondence among the 20 time contents, 48 claims, 9 transfers and 15 results.

Shared audit with scope Anderson’s classification of the Problem of Time, conditional constructions, approximations and combined strategies have the objects and domains given in the result tables. The JBB construction depends on the specified action and integration domain; the WKB construction depends on the specified phase flow and approximation order; histories and conditional averages depend on the specified projectors, states and temporal variables.

8.19.60 Source audit

Audit input

Source and audit scope

Source identifier 60.

Source title Primitive-time-root audit of Ellis 2012

Bibliographic identification G. F. R. Ellis, *Space time and the passage of time*, arXiv:1208.2611v4 [gr-qc] (2012). Let the source be $R = \text{Ref}_{\text{Ell12PT}}$.

Audit-scope identifier 60-SC1.

Audit-scope content The abstract, §§1–9, and the appendix by Goswami.

Source-location identifier 60-L1.

Source-location content Printed source pp. 1–25. Individual locations and relevant equations are given in the tables.

Formal objects The audit uses 47 source claims, 40 time contents, 17 transfers and 10 results, with 33 components of those results.

Assumptions Each relation uses its specified physical model, curves, metric, foliation and matter conditions. The EBU, CBU and reported quantum proposals retain their respective positions in the source. Conditions on uniqueness of eigendirections, regularity of time surfaces, clock approximation and reparameterization are given in the transfer table.

Nontemporal physical inputs The generation-initial set comprises the material degrees of freedom, spatial structures and state values in Claims 1–8. Time labels, histories, causal succession and proper time belong to the objects in the time-content table.

Definition 8.704 (Semantic components and generation-initial set in Ellis 2012). Let $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(47)$ and $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(8)$. Source-claim indices identify the same semantic components.

Definition 8.705 (Time contents in Ellis 2012). For $R = \text{Ref}_{\text{Ell12PT}}$, let $n_R = 40$. The time contents and their source locations are given below.

Table 561: Time contents and source locations in Ellis 2012

i	Time content	Source location
1	Time that Ellis argues physically exists and flows. Section 9 includes transition and continuity among the features of its passage.	pp. 1, 14, 23
2	The EBU temporal structure distinguishing the definite past, the present boundary and the indeterminate future.	pp. 1, 2, 3, 4, 23
3	The temporal reference that orders events. In §4.1, the event time t_E and comparison time t_1 are supplied on the same reference.	pp. 1, 2, 11
4	The boundaries between a particle’s determined path and possible future paths at two times satisfying $t_2 > t_1$.	p. 3

<i>i</i>	Time content	Source location
5	The parameter t used in equations of motion, Hamilton's equations, the specification of initial data and the time differentiation of conserved quantities.	pp. 5, 6
6	The time t_* at which the velocity in Eq. (13) vanishes and the resting state of Eq. (14) begins.	pp. 6, 7
7	The time parameter used for state evolution in Eqs. (15) and (44) and for specifying an initial state.	pp. 7, 18
8	The times of preparation, measurement, emission and collapse events, and the temporal positions of the states before and after them.	pp. 7, 8
9	The simultaneity relation between distant events, depending on the observer's motion and the adopted definition of simultaneity.	pp. 9, 23
10	The metric content determining the causal character of curves and proper-time intervals. In the FLRW model, it includes the scale-factor history $a(\tau)$.	pp. 9, 12, 15, 23, 24, 25
11	The causal ordering of events along timelike and null curves with a specified time orientation.	pp. 9, 23
12	The proper-time interval determined by Eq. (20) from the metric and integration endpoints on a specified timelike curve. Fixing an origin gives the proper time τ .	pp. 9, 11, 12, 13, 15, 20, 23, 25
13	The fundamental congruence along a timelike eigendirection satisfying Eq. (21). Geometrical uniqueness holds where the timelike eigenspace is one-dimensional and nondegenerate.	pp. 9, 10, 13, 25
14	The sets $S(\tau)$ of equal integrated proper time on a specified fundamental congruence with a common origin or initial surface.	pp. 9, 10, 24
15	The spacelike, null or timelike character of a regular equal-proper-time surface, determined by its induced metric.	pp. 10, 24
16	The classification of event E as future, present or past according to $t_1 < t_E$, $t_1 = t_E$ or $t_1 > t_E$ in §4.1.	p. 11
17	The relation between proper time along a timelike world line and an ideal clock reading, with real clocks treated under the stated approximation.	pp. 11, 12, 15
18	The coordinate time t labelling spacelike leaves Σ_t and describing the evolution of the metric and its conjugate quantities.	pp. 12, 13, 14, 24, 25
19	The normal projection N of a scaled flow relative to a fixed foliation. For normal flow it gives $d\tau = Ndt$.	pp. 13, 18, 25
20	The relation between the advances of proper time on two world lines once the correspondence between the compared events is specified.	p. 12
21	Equation (25), $d\tau/d\tau = 1$, obtained by setting $f(\tau) = \tau$ in Eq. (24).	pp. 12, 14

<i>i</i>	Time content	Source location
22	The representation of an existing time under an invertible correspondence $t' = t'(t)$, with $q(t(t'))$ as in Eq. (38).	p. 15
23	The proper time τ along the specified comoving curves in the normalized comoving coordinates of Eq. (40).	p. 15
24	The time η defined by the integral in Eq. (41), with a fixed origin on an interval where $a(\tau) > 0$.	p. 15
25	Effective macroscopic time discussed in the introduction to §5 as the target of proposals based on a timeless substratum. Ellis states its relation to the EBU conditional on its emergence.	p. 16
26	The time used to describe system evolution through the motion of a classically approximated environment in the Mott and Briggs–Rost construction discussed by Ellis.	p. 16
27	The time variable t_ρ and thermal clock associated with a statistical state ρ in the proposal discussed in §5.2.	p. 17
28	The correspondence $t \rightarrow -t$ combined with momentum reversal in Eq. (43).	pp. 17, 18
29	The direction from the already realized past towards the present that Ellis describes in §8.1.	pp. 7, 17, 18, 21, 22, 23
30	The t of the Schrödinger representation in §5.3 and the discussion of independence from that time associated with the Wheeler–DeWitt equation.	pp. 18, 19
31	The global time used for quantum-state evolution in the unimodular-gravity proposal introduced through Smolin’s account in §5.3.	p. 19
32	The τ_{exp} related in §6 to proper time and emotional and psychological context, together with the temporal width τ_{min} used in sensory averaging.	p. 20
33	The temporal positions of local transitions from quantum indeterminacy to definiteness that lead or lag the apparent present in the CBU proposal.	pp. 21, 23
34	The endpoint of fundamental curves and temporal extension at a spacetime boundary assumed in Ellis’s chronology-protection argument in §8.2.	p. 22
35	The rate $\sqrt{N^2 - h_{ab}N^aN^b}$ for a timelike coordinate flow $t^a = Nn^a + N^a$.	pp. 13, 25
36	The time histories of neuronal states $b_i(t)$ and the environmental state $E(t)$ supplied to the state function in Eq. (48).	p. 20
37	The history $\Phi(b_i(t), E(t))$ obtained by applying the same state function to those histories. Equation (49) concerns its time derivative when neither the brain nor the environment changes with time.	p. 20
38	The future and past Cauchy domains that Ellis discusses in §3 as regions determined by data on a time surface.	p. 10

<i>i</i>	Time content	Source location
39	The time horizon that Ellis suggests may occur as a null boundary where equal-proper-time surfaces change from spacelike to timelike. Coincidence with an event horizon is also presented as a possibility.	p. 10
40	The direction of time associated with coarse graining and entropy increase, together with the question of its agreement across the universe. Section 8.1 assigns the low-entropy initial state a role in how the arrow develops.	pp. 17, 18, 22

Definition 8.706 (Physical and formal input claims in Ellis 2012). Claims 1–8 specify the generation-initial components; Claims 9–47 describe equations, constructions, physical interpretations and proposals reported in the source.

Table 562: Physical and formal input claims in Ellis 2012

<i>p</i>	Source claim	Source location
1	Material degrees of freedom and spatial arrangements of bodies, samples, control apparatus and clock devices.	pp. 2, 3, 6, 7, 11, 12, 15
2	Masses, spatial positions, displacements and material coupling parameters of the classical systems.	pp. 5, 6, 11
3	State spaces, bases, state values and coefficients of the quantum systems.	pp. 7, 8, 18, 19
4	Points, surfaces and three-metrics on spatial leaves, and the spatial configurations of bodies.	pp. 9, 10, 11, 12, 13, 15, 24
5	The constitution of matter carriers and the state values of density, pressure and stress.	pp. 13, 14
6	The partition into environment, system and interaction, and their material degrees of freedom.	pp. 16, 17
7	The material constitution and state values of neurons, the brain and the environment.	p. 20
8	The material carriers of measurement records, outcome labels N and the corresponding state values.	pp. 7, 8
9	Ellis argues that time physically exists and flows, and includes transition and continuity among the features that a physical description should capture.	pp. 1, 14, 23
10	The standard Block Universe is presented as a fixed four-dimensional spacetime containing all events. Ellis argues that this representation removes a special present and the development of spacetime itself.	p. 1
11	In the EBU, the past that has occurred is definite, the future is indeterminate, and the present bounds the events being incorporated into the past. Spacetime grows as physical events occur.	pp. 1, 2, 3, 4, 23

p	Source claim	Source location
12	In the example of a macroscopic body controlled by radioactive-decay signals, the realized events determine its path and the resulting matter motion contributes to spacetime curvature. Figure 1 distinguishes the determined portion of the path from possible future paths at $t_2 > t_1$.	pp. 2, 3, 4, 8, 14
13	Equations (1)–(3) describe state evolution parametrized by t . The local unique evolution in Eq. (8) uses conditions ensuring existence and uniqueness of solutions to the equations of motion, together with compatible initial data.	pp. 5, 6
14	Constraints $C_m = 0$ imposed on Hamilton's equations (4) and (5) must be preserved by the evolution. Equations (9) and (10) express the time derivative of a state function through the chain rule and Hamilton's equations.	pp. 5, 6
15	For the Hamiltonian considered after Ellis excludes explicit time dependence, Eqs. (11) and (12) give the conservation relation. That relation is presented as a first integral of state evolution in time.	p. 6
16	The velocity $v_1(t) = v_0 - \mu g t$ in Eq. (13) vanishes at $t_* = v_0/(\mu g)$ when $v_0 > 0$ and $\mu g > 0$. The text immediately after that equation prints $t_* = \mu g/v_0$. Equation (14) describes the resting state and is used to illustrate loss of retrodictability when information in the dissipative degrees of freedom is coarse grained.	pp. 6, 7
17	Schrödinger evolution in Eqs. (15) and (44), together with the initial state in Eq. (45), describes the quantum state using a time argument.	pp. 7, 18
18	Ellis distinguishes unitary evolution from state preparation and measurement reduction. Equation (18) gives the probability of outcome N , whereas Eq. (19) expresses the reduced state using the realized outcome N . He states that the initial state does not determine the particular N or event time t_* .	pp. 7, 8
19	Distant simultaneity depends on the observer's motion and the adopted definition of simultaneity. Ellis treats it as secondary to physical interactions along world lines and signals.	pp. 9, 10, 23
20	The succession of physical interactions is described along time-oriented timelike and null curves. Their causal character is determined by the spacetime metric.	pp. 9, 23
21	Equation (20) integrates proper-time intervals using a specified timelike curve and metric. The integration endpoints and orientation are supplied, and an origin fixes the proper time τ .	p. 9

p	Source claim	Source location
22	Equation (21) uses a timelike eigendirection of the stress-energy or Ricci tensor for the fundamental curves. Where the timelike eigenspace is one-dimensional and nondegenerate, the required smoothness and orientation determine a local congruence along that direction.	pp. 9, 10, 25
23	A common origin or initial surface on the fundamental congruence defines the level sets $S(\tau)$ of accumulated proper time. Ellis interprets them as present surfaces. A congruence does not admit orthogonal hypersurfaces where its vorticity is nonzero, and equal-proper-time surfaces need not be spacelike in general.	pp. 9, 10
24	Ellis describes a possible time horizon at a null boundary where equal-proper-time surfaces change from spacelike to timelike, and also discusses possible coincidence with an event horizon. He argues that data on such surfaces determine their future and past Cauchy development, and identifies the associated initial-value problem as requiring investigation.	pp. 10, 24
25	Given the time t_E of event E and a comparison time t_1 on the same temporal reference, $t_1 < t_E$, $t_1 = t_E$ and $t_1 > t_E$ represent E as future, present and past, respectively.	p. 11
26	Using the solution in Eq. (2), the phase orbit is $q^2 + [p/(m\omega)]^2 = A^2$. Equations (22) and (23) print p/m instead. Ellis distinguishes the orbit obtained by eliminating time from the parametrized solution progressing along that orbit with t .	p. 11
27	An ideal clock reading is taken to measure proper time along its world line, with real clocks treated as approximations. Equation (24) expresses the rate of change of a physical quantity with respect to proper time τ .	pp. 11, 12
28	Ellis states that clocks on different world lines can advance at different rates according to the metric. A comparison of relative rates specifies the correspondence between the compared events on the two world lines.	p. 12
29	Setting $f(\tau) = \tau$ in Eq. (24) gives $d\tau/d\tau = 1$ in Eq. (25). Ellis interprets this identity as a rate of passage of one second per second.	pp. 12, 14
30	The constraints and evolution equations of the ADM decomposition describe the three-metric and conjugate quantities on time-labelled leaves. Matter equations of state and conservation equations supply the evolution of the matter terms. Ellis discusses cases in which dissipation or unpredictable matter responses prevent prediction of the future spacetime from initial data.	pp. 12, 13, 14

p	Source claim	Source location
31	Ellis proposes selecting matter flow along a Ricci eigendirection and taking coordinate time to measure proper time along that flow. In connection with Eq. (33), he claims that these conditions determine the shift and lapse uniquely.	pp. 13, 18
32	The appendix by Goswami uses a given spacelike foliation, its unit normal n^a and a timelike flow t^a with a specified scale, decomposing it as $N^a = h_b^a t^b$ and $N = -t^a n_a$. The proper-time rate along the coordinate flow $t^a = N n^a + N^a$ is $\sqrt{N^2 - h_{ab} N^a N^b}$, reducing to $d\tau = N dt$ when $N > 0$ and $N^a = 0$.	pp. 24, 25
33	Equation (38) re-expresses $q(t)$ as $q(t(t'))$ using an invertible $t' = t'(t)$. For $t' = \exp(Ht)$ with $H \neq 0$, the inverse is $t = \ln(t')/H$, differing from $\exp(-Ht')$ printed in Eq. (39). Ellis favours affine transformations of proper time as choices that display regular physical motion.	p. 15
34	The FLRW metric in Eq. (40) uses proper time τ along the specified comoving curves. Fixing an origin on an interval with $a(\tau) > 0$, the relation $\eta = \int d\tau/a(\tau)$ in Eq. (41) gives τ and η as orderings of the same leaves.	p. 15
35	Section 5 introduces proposals for obtaining effective time from a timeless substratum. Ellis argues that macroscopic effective time would give a macroscopic EBU, and identifies nonunitary state preparation and measurement as a problem common to the proposals.	p. 16
36	The proposals of Mott and Briggs–Rost are presented through a stationary Schrödinger equation partitioned into environment, system and interaction, with a classical approximation for the environment. In the quoted account, the motion of that classical environment introduces a time-dependent Hamiltonian and evolution equation for the system.	p. 16
37	Rovelli’s proposal is introduced as assigning a variable t_ρ to a statistical state ρ , measured by a thermal clock, with respect to which the system is in equilibrium. Ellis objects that the establishment of equilibrium through molecular collisions already uses time.	p. 17
38	For time-reversal-symmetric Hamiltonian dynamics, $t \rightarrow -t$ and $p \rightarrow -p$ in Eq. (43) give the reversal correspondence. Ellis discusses Loschmidt’s objection that a coarse-grained derivation of entropy increase can also be applied in the reverse direction, and the need to explain agreement of the arrow across the universe.	pp. 17, 18

<i>p</i>	Source claim	Source location
39	Ellis contrasts the time-dependent Schrödinger representation with the Wheeler–DeWitt equation (46), and examines the premise of a single unitary description of the universal wavefunction. Equation (47) concerns an assumed composition of that wavefunction from component wavefunctions.	pp. 18, 19
40	The quotation from ADM obtains a nonzero Hamiltonian through a coordinate condition identifying a canonical coordinate with τ . Separately, Smolin’s account introduces a proposal for quantum-state evolution in a global time within unimodular gravity.	pp. 18, 19
41	Equation (48) takes the mental state to be $\Phi = \Phi(b_i, E)$. With the same state function and the required differentiability, $db_i/dt = 0$ and $dE/dt = 0$ imply $d\Phi/dt = 0$ as in Eq. (49). Ellis uses changes in mental experience to criticize timeless interpretations that deny physical change in the brain.	p. 20
42	Experienced time τ_{exp} is related to proper time and emotional and psychological context, while τ_{min} is described as a temporal width of sensory averaging. Ellis states that the brain need not be a good clock and has finite temporal resolution.	p. 20
43	The CBU is proposed as an extension of the EBU allowing local transitions from quantum indeterminacy to classical definiteness to lead or lag. Ellis cites delayed-choice and quantum-eraser experiments, while leaving the relation to EPR experiments and entanglement for further investigation.	pp. 21, 23
44	Ellis argues that, in the EBU, the direction from the already realized past to the present supplies the direction of time. He distinguishes the existence of that direction from the role of a low-entropy initial state in how the thermodynamic arrow develops.	pp. 21, 22
45	Ellis argues that intersections of fundamental world lines produce a singularity under the stated matter and energy conditions, ending temporal extension there and preventing closed timelike curves in the EBU.	p. 22
46	Section 9 lists seven functions: event ordering, duration, simultaneity, direction of passage, transition, continuity and monotonicity. In relativity, it relates duration to world-line proper time and the metric, treats simultaneity as secondary, and connects monotonicity with the issue of closed timelike curves.	p. 23

p	Source claim	Source location
47	Ellis lists as open questions the preferred time surfaces in inhomogeneous cosmologies, ADM analysis when their causal character changes, the geometry of those surfaces at different averaging scales, and relations to EPR experiments and quantum gravity.	pp. 10, 21, 24

Dependency inputs Each input set gives the jointly required inputs. Source-claim and time-content indices refer to their respective tables.

Dependencies and transfers

Definition 8.707 (Time-content transfers in Ellis 2012). The following table gives the 17 transfers and their conditions of application.

Table 563: Time-content transfers in Ellis 2012

Transfer	Operation	Source claims	Time inputs	Time outputs
1	A time-oriented Lorentzian metric and timelike or null curves give causal succession along the curves. pp. 9, 23	{4, 20}	{10}	{11}
2	Integrating Eq. (20) on a specified regular timelike curve gives proper time with fixed endpoints, orientation and origin. p. 9	{4, 21}	{10}	{12}
3	Where the timelike eigenspace in Eq. (21) is one-dimensional and nondegenerate, smoothness, normalization and time orientation determine the fundamental congruence. pp. 9, 10, 25	{5, 22}	{10}	{13}
4	Proper time on a specified fundamental congruence, with a common origin, defines the level sets $S(\tau)$ where proper time is single-valued. These are regular surfaces wherever $d\tau \neq 0$. pp. 9, 10	{4, 23}	{12, 13}	{14}

Transfer	Operation	Source claims	Time inputs	Time outputs
5	Restricting the metric to the tangent spaces of a regular equal-proper-time surface determines its spacelike, null or timelike causal character. pp. 10, 24	{4, 24}	{10, 14}	{15}
6	In the friction model with $v_0 > 0$ and $\mu g > 0$, the zero of the velocity in Eq. (13) gives the stopping time $t_* = v_0/(\mu g)$ on the existing time t. pp. 6, 7	{1, 2, 16}	{5}	{6}
7	Comparing t_E and t_1 supplied on the same temporal reference classifies event E as future, present or past. p. 11	{25}	{3}	{16}
8	The ideal-clock approximation and the correspondence of units and origins relate world-line proper time to a clock reading. pp. 11, 12	{1, 27}	{12}	{17}
9	Setting $f(\tau) = \tau$ in Eq. (24) gives $d\tau/d\tau = 1$. pp. 12, 14	{29}	{12}	{21}
10	A specified invertible correspondence $t' = t'(t)$ re-expresses the existing-time representation as $q(t'(t))$, as in Eq. (38). p. 15	{33}	{5}	{22}
11	For a given spacelike foliation, a timelike flow with specified scale and orientation is decomposed by projection into its normal component N and spatial component N^a . pp. 13, 24, 25	{4, 22, 32}	{10, 13, 18}	{19}
12	FLRW symmetry and normalized comoving coordinates identify proper time along the selected comoving curves with the time coordinate τ of Eq. (40). p. 15	{4, 34}	{12, 13}	{23}

Transfer	Operation	Source claims	Time inputs	Time outputs
13	On an interval where $a(\tau) > 0$ and the integral exists, fixing an origin in $\eta = \int d\tau/a(\tau)$ of Eq. (41) gives conformal time. p. 15	{4, 34}	{10, 23}	{24}
14	The EBU relation between past and present supplies the direction from the already realized past towards the present. pp. 21, 22	{11, 44}	{2}	{29}
15	For time-reversal-symmetric Hamiltonian dynamics, combining $t \rightarrow -t$ and $p \rightarrow -p$ gives the reversal correspondence in Eq. (43). pp. 17, 18	{14, 38}	{5}	{28}
16	With metric signature $(-, +, +, +)$, a timelike $t^a = Nn^a + N^a$ and $n_a N^a = 0$ give the proper-time rate $\sqrt{N^2 - h_{ab}N^aN^b}$ along the coordinate flow. pp. 13, 25	{4, 32}	{10, 18, 19}	{35}
17	Applying the same state function Φ to the supplied brain and environmental histories gives the mental-state history $\Phi(b_i(t), E(t))$. The differential relation in Eq. (49) uses the required differentiability. p. 20	{7, 41}	{36}	{37}

Results

Definition 8.708 (Results in Ellis 2012). The ten results and their 33 components are given below. Each result comprises its corresponding components jointly.

Table 564: Results in Ellis 2012

q	Result	Source location
1	Ellis's claim that time physically exists and flows.	pp. 1, 14
2	The distinction between the fixed Block Universe, the growing EBU and the CBU allowing local lead and lag in becoming definite.	pp. 1, 2, 3, 4, 21, 23

q	Result	Source location
3	The examination of evolution, constraints and conservation in classical and Hamiltonian dynamics, loss of retrodictability through friction, and time-reversal symmetry.	pp. 5, 6, 7, 17, 18
4	The distinction between unitary Schrödinger evolution and state preparation, measurement and reduction at specified times.	pp. 7, 8
5	The relations among world-line proper time, distant simultaneity, causal succession, the fundamental congruence and equal-proper-time surfaces, including correspondence with clock readings.	pp. 9, 10, 11, 12
6	The ADM constraints and evolution equations, their dependence on matter terms, and lapse-shift relations for the selected flow and foliation.	pp. 12, 13, 14, 24, 25
7	The relations between time reparameterization, FLRW comoving proper time and conformal time describing the same leaves.	p. 15
8	Ellis's presentation of the Mott-type, thermal-time, Wheeler-DeWitt and unimodular-gravity proposals and his comparison of their uses of time.	pp. 16, 17, 18, 19
9	Ellis's arguments concerning brain, environmental and mental states and experienced time, local becoming-definite in the CBU, the EBU direction of time and chronology protection.	pp. 20, 21, 22, 23
10	The open questions identified in the source about the causal character and domains of determination of time surfaces, inhomogeneous cosmology and averaging, EPR, and quantum gravity.	pp. 10, 18, 19, 21, 23, 24, 25

Table 565: Result components in Ellis 2012

Item	Content	Temporal premises	Source location
1.1	thesis that time exists	{1}	pp. 1, 14
1.2	thesis that time flows	{1}	pp. 1, 14
2.1	Block Universe	{3}	pp. 1, 2
2.2	Evolving Block Universe	{2}	pp. 2, 3, 4, 23
2.3	Crystallizing Block Universe	{2, 33}	pp. 21, 23
3.1	reversible classical evolution equations	{5, 28}	pp. 5, 6, 17
3.2	Hamiltonian-constraint example	{5}	pp. 5, 6
3.3	first-integral/time-elimination example	{5}	pp. 6, 11
3.4	dissipative friction example	{5, 6}	pp. 6, 7
4.1	unitary Schrödinger evolution	{7}	p. 7
4.2	temporal order of measurements	{7, 8}	pp. 7, 8
4.3	collapse branch adopted by the source	{7, 8}	pp. 7, 8

Item	Content	Temporal premises	Source location
5.1	World-line proper-time integral and clock readings	{10, 12, 17}	pp. 9, 11, 12
5.2	distinction from distant simultaneity	{9}	pp. 9, 10
5.3	Proposed fundamental congruence and equal-proper-time surfaces	{10, 12, 13, 14}	pp. 9, 10
6.1	ADM constraint system	{18}	pp. 12, 13
6.2	ADM evolution system	{18}	pp. 13, 14
6.3	Lapse, shift and the selected flow	{10, 13, 18, 19, 35}	pp. 12, 13, 24, 25
6.4	dependence on matter equations and stress content	{18}	pp. 13, 14
7.1	FLRW proper time	{23}	p. 15
7.2	conformal time	{10, 23, 24}	p. 15
7.3	coordinate-time reparameterization	{22}	p. 15
8.1	Mott-type effective time from the environment	{26}	p. 16
8.2	Thermal time and the time-reversal problem	{27, 28, 40}	pp. 17, 18
8.3	Wheeler–DeWitt/unimodular comparison branch	{30, 31}	pp. 18, 19
9.1	Brain, environmental and mental states and experienced time	{32, 36, 37}	p. 20
9.2	delayed crystallization proposal	{2, 33}	pp. 21, 23
9.3	Direction of time and the thermodynamic arrow	{2, 29, 40}	pp. 21, 22
9.4	Chronology protection and a temporal endpoint	{11, 13, 34}	p. 22
10.1	Surface causal character, Cauchy domains and a proposed time horizon	{10, 14, 15, 38, 39}	pp. 9, 10, 24
10.2	EPR and nonlocality	{33}	pp. 21, 24
10.3	Quantum gravity	{30}	pp. 18, 19, 24
10.4	Time surfaces in inhomogeneous cosmology	{14, 15, 39}	pp. 9, 10, 24

Table 566: Result dependencies in Ellis 2012

q	Source claims	Temporal premises
1	{9}	{1}
2	{10, 11, 43}	{2, 3, 33}
3	{13, 14, 15, 16, 26, 38}	{5, 6, 28}
4	{17, 18}	{7, 8}
5	{19, 20, 21, 22, 23, 27}	{9, 10, 11, 12, 13, 14, 17}
6	[[30, 32]]	{10, 12, 13, 18, 19, 35}
7	{33, 34}	{10, 22, 23, 24}

q	Source claims	Temporal premises
8	$\llbracket 35, 40 \rrbracket$	$\{26, 27, 28, 30, 31, 40\}$
9	$\llbracket 41, 45 \rrbracket$	$\{2, 11, 13, 29, 32, 33, 34, 36, 37, 40\}$
10	$\{24, 39, 43, 47\}$	$\{10, 14, 15, 30, 33, 38, 39\}$

Grounds for the results Following Definition 7.71, each result uses the transfer set below under the conditions in the transfer table.

Table 567: Result-specific audit branches in Ellis 2012

Domain	Results	Transfer set
Claims about the reality and passage of time	$\{1\}$	$\emptyset$
Comparison of the three universe models	$\{2\}$	$\emptyset$
Classical and Hamiltonian dynamics and friction	$\{3\}$	$\{6, 15\}$
Quantum evolution and measurement	$\{4\}$	$\emptyset$
World lines, clocks and time surfaces	$\{5\}$	$\{1, 2, 3, 4, 8\}$
ADM and the selected foliation and flow	$\{6\}$	$\{2, 3, 11, 16\}$
FLRW with selected comoving curves and reparameterization	$\{7\}$	$\{2, 10, 12, 13\}$
Effective-time and quantum-gravity proposals	$\{8\}$	$\{15\}$
Mental states, CBU, time’s arrow and causality	$\{9\}$	$\{1, 3, 14, 17\}$
Open questions identified by the source	$\{10\}$	$\llbracket 2, 5 \rrbracket$

Generation judgment Judgment: 0 LCTR-complete generations; 0 locally reachable time contents under source-evidence conditions; 40 residual primitive time roots.

Result dependencies There are 37 primitive-time-root–result dependency pairs, including 28 necessary dependencies.

Evidential scope The domain covers the source’s claims about the reality and passage of time, its dynamical models, the relations among clocks, proper time and time surfaces, and the reported effective-time and quantum-gravity proposals.

Result-preservation domain Each relation uses its specified curves, metric, foliation, origin, clock approximation and matter conditions. Results concerning physical interpretation retain the content of the respective positions stated by Ellis.

Placement of grounding obligations The 40 time contents are used as already established inputs. The grounding obligation for physical time is classified as backgrounded by primitive-time inputs.

Evidence of actual invalidation Actual invalidation of the specified input conditions remains undetermined.

Downstream reach Structural reach from each primitive time root to the results is given by the dependency and minimal-support tables.

Repair scope A construction supplying the grounding obligation concerns generation of time contents before use. If failure of a specific input condition is identified, dependent results are recalculated from inputs satisfying that condition.

Definition 8.709 (Finite dependency graph in Ellis 2012). The 47 semantic components, 40 time contents, 17 transfers and 10 results form the finite dependency graph with the audit branches specified above.

Proposition 8.710 (Primitive time roots and result dependence in Ellis 2012).

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(40, 10).$$

Proof. The generation-initial set consists of Claims 1–8. Every one of the 17 transfers has temporal inputs, so no time content is reachable from that initial set. Adding the source formal-evidence environment supplies no transfer without temporal inputs. Complete generation and local reach under source-evidence conditions are therefore both zero, leaving 40 primitive time roots. All ten results have temporal premises; supplying time contents and the source formal-evidence environment under the respective application conditions satisfies each result edge. $\square$

Proposition 8.711 (Minimal primitive-time-root support in Ellis 2012). The minimal-support families below contain 15 members. There are 37 dependency pairs and 28 necessary dependencies.

Proof. Updating the support antichains to a fixed point over the transfer set and result edge of each result gives the following table. $\square$

Table 568: Minimal primitive-time-root support in Ellis 2012

Result	Family of minimal-support index sets
1	{1}
2	{2, 3, 33}
3	{5}
4	{7, 8}
5	{9, 10}
6	{10, 18}
7	{5, 10, 13}, {5, 10, 23}, {10, 13, 22}, {10, 22, 23}

Result	Family of minimal-support index sets
8	$\{5, 26, 27, 30, 31, 40\}, \{26, 27, 28, 30, 31, 40\}$
9	$\{2, 10, 32, 33, 34, 36, 40\}, \{2, 11, 13, 32, 33, 34, 36, 40\}$
10	$\{10, 30, 33, 38, 39\}$

Individual statements

Statement objects The source reports on the existence and passage of time; the priority of proper time; the common time parameter supplied by clocks; the relative flow of time between worldlines; the time parameter and proper time on fundamental flow lines; the rate of temporal passage stated as one second per second; the connection of time to physics, astronomy, geology, technology, and biology; the preferred time parameter in standard physics; and the emergence of effective time from a timeless substratum.

- Original text (English)**
1. I argue that (i) time both exists and flows;
 2. What really matters is proper time τ measured along timelines $x^i(v)$, determined from the metric tensor $g_{ij}(x^k)$ by the basic formula [49, 34]
 3. The whole edifice of physics is built on the assumption that we can build such clocks to a good approximation, giving a time parameter τ that appears equally in all dynamical equations of physics: Newton's equation of motion [36], Maxwell's equations [37], the Schrödinger equation [38], General Relativity expressed in a 1+3 covariant formalism [21], and so on;
 4. The relative flow of time along different world lines may be different: that is the phenomenon of time dilation, caused by the varying gravitational potentials represented by the metric tensor [78]. But this does not mean it is not well defined along each world line.
 5. We determine the lapse function $N(x^i)$ by the condition that the time parameter t measures proper time τ along the fundamental flow lines.
 6. Conclusion: time flows at the rate of one second per second, with the metric tensor determining what this rate is for clocks and every other physical system (the choice of units is of course arbitrary, but can be done in a way that makes sense).
 7. The usual choice as in (40), agreeing with the proposal made here in §(3), ties time τ in to all the rest of physics, astronomy, geology, technology, and biology.
 8. In conclusion: Standard physics is based in choice of a preferred time parameter τ along matter world lines. General Relativity both allows such a choice, and can itself be written in terms of that choice.
 9. There are a number of proposals for an effective time to emerge somehow in the context of a timeless substratum.

Original locations In Ellis, *Space time and the passage of time*, arXiv:1208.2611v4, the first quotation is in the Abstract, p. 1; the second immediately precedes Eq. (20) in

§3, p. 9; the third and fourth are in §4.3, p. 12; the fifth is in §4.3, p. 13; the sixth is in the Conclusion of §4.3, p. 14; the seventh is in §4.4, pp. 15–16; the eighth is in the conclusion of §4.4, p. 16; and the ninth is in §5, p. 16.

Minimal contexts The first quotation summarizes Ellis’s central claim that time exists and flows. The second belongs to the context in which proper time along a worldline is determined from the metric in place of distant simultaneity. The third moves from the assumption that ideal clocks can be constructed approximately to a common τ across multiple physical equations. The fourth belongs to the context in which differences in proper time between worldlines and differences in gravitational potential are called time dilation. The fifth belongs to the selection of fundamental flow lines and the alignment of their coordinate parameter with proper time. The sixth concludes a rate of temporal passage from correlations among the metric, clock readings, and particle motion. The seventh and eighth connect the selection of a proper-time coordinate to general relativity and temporal descriptions in other fields. The ninth introduces the ensuing examination of several proposals that posit a timeless substratum.

Agents of attribution Statements 1–8 are attributed to the position taken by G. F. R. Ellis in his 2012 paper. Statement 9 is Ellis’s introduction to existing proposals; the following sections address Mott-type, thermal-time and Wheeler–DeWitt-based proposals.

Predicates and explicit domains of application The first quotation attributes existence and passage to time. The second gives priority to worldline proper time determined from the metric. The third extends the common domain of application of the time parameter supplied by clocks to Newtonian mechanics, electromagnetism, quantum mechanics, and general relativity. The fourth describes differences between worldlines as a relative flow of time. The fifth states that t measures τ along the fundamental flow lines. The sixth takes clocks and every other physical system as the objects of the rate of temporal passage. The seventh explicitly takes the domain from physics through biology. The eighth explicitly takes standard physics and general relativity as its domain. The ninth reports effective time emerging from a timeless substratum as its object.

Individual audit

Interpretive possibilities Proper time in Quotation 2 is a quantity determined by the metric and a world line; in Quotation 3 it serves as a time parameter linking clock readings and several evolution equations. Quotations 5 and 8 concern selection of a fundamental congruence and coordinate time. Quotations 1 and 6 express Ellis’s position on the reality and passage of time, while Quotation 9 introduces proposals for effective time.

Independent grounds The corresponding grounds are the proper-time integral in Eq. (20), the fundamental congruence in Eq. (21), the clock-rate, ADM and matter equations (24)–(37), the reparameterization and FLRW time relations (38)–(41), and the comparison of proposals in §5.

Boundary of evidential force Quotation 4 concerns relative passage of time on different world lines; Quotation 7 connects a proper-time coordinate to physics, astronomy,

geology, technology and biology. The proper-time integral uses a metric and curve; present surfaces use a selected congruence and common origin; lapse–shift decomposition uses a foliation and a flow with a specified scale. These uses of time and Ellis’s claims about its reality and passage constitute the argument of his paper.

Corresponding formal evidence Quotation 1 corresponds to Source Claim 9, Time Content 1 and Result 1. The contrast in Quotation 2 between worldline proper time and distant simultaneity corresponds to Source Claims 19 and 21, Time Contents 9, 10 and 12, Transfer 2, and Result Components 5.1 and 5.2. The correspondence between clock readings and a common parameter in Quotation 3 is represented by Source Claim 27, Time Contents 12 and 17, Transfer 8 and Result Component 5.1. The classical, quantum and general-relativistic evolution descriptions cited in that quotation use Time Contents 5, 7 and 18 and correspond respectively to Result Components 3.1, 4.1 and 6.2. The relative rate between worldlines in Quotation 4 corresponds to Source Claim 28 and Time Content 20.

The choice of lapse and fundamental congruence in Quotation 5 corresponds to Source Claims 22, 31 and 32, Time Contents 10, 12, 13, 18, 19 and 35, Transfers 2, 3, 11 and 16, and Result Component 6.3. The rate of one second per second in Quotation 6 corresponds to Source Claim 29, Time Contents 12 and 21 and Transfer 9; its interpretation as temporal passage corresponds to Source Claim 9, Time Content 1 and Result 1. The correspondence of proper time with clock readings and comoving time in Quotations 7 and 8 is represented by Source Claims 27 and 34, Time Contents 12, 17 and 23, Transfers 8 and 12, and Result Components 5.1 and 7.1. The general-relativistic choice of flow in Quotation 8 also corresponds to Result Component 6.3, as in Quotation 5.

The introduction of proposals in Quotation 9 corresponds to Source Claim 35 and Time Content 25. The ensuing comparison of Mott-type, thermal-time, Wheeler–DeWitt and unimodular-gravity proposals corresponds to Source Claims 36–40, Time Contents 26–28, 30–31 and 40, Transfer 15, and Result Components 8.1–8.3.

These time contents, transfers and results are specified respectively in Definitions 8.705, 8.707 and 8.708. Temporal premises and minimal supports for each result are given in Propositions 8.710 and 8.711.

Common audit with specified domain of application The temporal dependence of results is assessed within the specified domains, according to the respective roles of the metric, world lines, congruence, foliation, clock readings and effective time.

8.19.61 Source audit

Audit input

Original source and audit scope

Source identifier 61.

Source name Primitive-time-root audit of Tong 2013

Bibliographic identification Let David Tong, *Dynamics and Relativity* (2013), be source *R*. The audit domain is §§7.1–7.4.1 of Chapter 7, *Special Relativity*.

Audit-scope identifier 61-SC1.

Audit-scope content The audit domain specified in the bibliographic identification.

Original-location identifier 61-L1.

Original-location content The audit domain specified in the bibliographic identification and the original-location columns of the respective evidence tables.

Formal objects The audit uses 19 time contents, 36 source claims, 10 transfers, and 26 results. Claims 1–12 specify material and spatial initial components; Claims 13–36 specify physical laws, formal operations, measurement rules, and historical accounts.

Assumptions The Lorentz transformation uses inertial frames with a common origin, linearity, the principle of relativity, and invariance of the vacuum speed of light. The temporal structures of the physical spacetime interval and Minkowski metric belong to Time Contents 17 and 18. Historical calculations under the ether hypothesis constitute the domain of Time Contents 13 and 19 and Results 22–24 and 26.

Nontemporal physical inputs The generation-initial set comprises the material degrees of freedom and initial spatial configurations in Claims 1–12. Temporal event order, clock-tick sequences, motion histories, reading intervals, and simultaneity belong to their respective time contents.

Definition 8.712 (Semantic components and generation-initial set in Tong 2013). Let $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(36)$ and $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(12)$. Source-claim and semantic-component indices identify the same objects. The source formal-evidence environment consists of the 36 claims in Table 569.

Definition 8.713 (Physical and formal input claims in Tong 2013). The following table gives the 36 source claims and their locations, using printed page numbers.

Table 569: Physical and formal input claims in Tong 2013

p	Source claim	Source location
1	Material degrees of freedom and initial spatial configuration of distributed recording devices	pp. 108, 117
2	Material carriers involved in events and their spatial positions and coincidence configuration	pp. 112, 115, 125–128
3	Material degrees of freedom and initial spatial positions of particles	pp. 112, 121–122, 124, 133–134
4	Emitters, receivers, signal carriers, and the initial spatial configuration of the optical path	pp. 110–113, 116, 119
5	Material states and initial spatial positions of clock devices	pp. 118–119
6	Material states and initial spatial positions of muon carriers	p. 120
7	Material states and initial spatial configuration of atomic clocks and aircraft	p. 120
8	Material carriers and initial spatial configuration of rod endpoints	pp. 122–124

p	Source claim	Source location
9	Material degrees of freedom and initial spatial configuration of Io, Jupiter, and Earth	p. 113
10	Material degrees of freedom of the rotating-mirror apparatus and spatial configuration of its optical path	p. 113
11	Material degrees of freedom of the interferometer and spatial configuration of its orthogonal optical paths	p. 114
12	Material carriers, internal degrees of freedom, and initial spatial configuration of general physical processes	pp. 119–120
13	Principle of relativity and invariance of the vacuum speed of light, with physical light speed using temporal scale 11.	pp. 107, 110–111, 115
14	Construction of a boost under common-origin, linearity, aligned-axis, and symmetry assumptions.	pp. 108–110; (7.2), (7.3), (7.4)
15	Lorentz transformation and its inverse: $t' = \gamma(t - vx/c^2)$, $x' = \gamma(x - vt)$, with $\gamma = (1 - v^2/c^2)^{-1/2}$.	pp. 110–111, 116, 119, 123–125; (7.5), (7.6)
16	Spacetime diagrams, coordinate axes, and worldline representations.	pp. 112–113, 115–117, 121–124
17	Classification of timelike, spacelike, and null intervals by the physical light cone.	pp. 117–118, 126–128; (7.11), (7.13)
18	Time-dilation law $T = \gamma T'$ for the same pair of tick events on the same clock.	p. 119
19	Application of the same argument to every physical process, including clocks, elementary particles, and human hearts.	pp. 119–120
20	Outward and return journeys and reassignment of simultaneity in the twin example; the turnaround gives the two routes different inertial-frame structures.	pp. 120–122; (7.5)
21	Length measurement by selecting simultaneous endpoints in each inertial frame, and Lorentz contraction.	pp. 122–124; (7.5), (7.6)
22	Collinear velocity addition, $u = (u' + v)/(1 + u'v/c^2)$.	pp. 124–125; (7.8), (7.9)
23	Invariant interval $\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$ for the same event pair and its Lorentz invariance.	pp. 125–127; (7.10), (7.11), (7.12)
24	Abstract matrix preservation condition $\Lambda^\top \eta \Lambda = \eta$ and group structure; the physical interpretation uses the metric in Time Content 18.	pp. 128–130; (7.14), (7.15), (7.16)
25	Classification of the Lorentz group by the sign of the determinant and by time orientation.	p. 130; (7.14)
26	Boost representation and collinear composition through rapidity ϕ , retaining the velocity sign through $\tanh \phi = v/c$ and $\sinh \phi = \gamma v/c$.	pp. 130–131; (7.17)
27	The $c = 1$ unit convention using light-seconds and the length–time conversion calibrated by Time Content 11.	pp. 131–132
28	Representation of a timelike worldline parametrized by proper time.	pp. 133–134; (7.18), (7.19)

p	Source claim	Source location
29	Calibration of metres and light-seconds through the second and physical c .	pp. 107, 131–132
30	Historical inference attributed to Rømer, explaining variations in the received periods of Io events through finite light propagation.	p. 113
31	Historical account of the rotating-mirror method of measuring the speed of light.	p. 113
32	Maxwell’s relation for light speed, $c = 1/\sqrt{\epsilon_0\mu_0}$.	pp. 113, 115; (7.7)
33	Directional light propagation under an ether rest frame and Newtonian velocity addition.	pp. 113–114
34	Interference comparison of directional round-trip times and the reported Michelson–Morley null result.	p. 114
35	The transformation $x' = \gamma(x - vt)$ of distances between matter proposed by Lorentz to explain the null directional round-trip time difference through contraction along the ether direction.	p. 114
36	The source’s alternative theoretical route from Maxwell’s equations holding in every inertial frame to invariance of the speed of light.	p. 115; (7.7)

Definition 8.714 (Time contents in Tong 2013). The audit uses the 19 time contents in the following table. Time Content 13 is a direction-indexed family of round-trip times whose members share the same ether hypothesis, path length, and velocity conditions. Time Contents 15 and 16 are rest-frame and moving-frame comparison families indexed by the same physical process.

Table 570: Time contents and source locations in Tong 2013

i	Time content	Source location
1	Coordinate time t in inertial frame S , using the coordinate times of distributed clocks.	pp. 108, 112, 116–117, 125, 133; (7.1), (7.5), (7.10), (7.18)
2	Coordinate time t' in inertial frame S' , related by a Lorentz transformation with a common origin, aligned spatial axes, and $ v < c$.	pp. 108, 110–111, 113, 116, 119; (7.5), (7.6)
3	Simultaneity as equality of coordinate time assigned by clocks distributed throughout the specified inertial frame. A single receiver receives a signal after the light-travel time from the event to that receiver. Disagreement in simultaneity between frames depends on the projection of the event separation along the boost direction.	pp. 115–117, 121–124; (7.5)

i	Time content	Source location
4	Causal temporal order defined by the future and past light cones with a fixed time orientation and transformations preserving that orientation.	pp. 117–118, 126–127, 130; (7.11), (7.14)
5	Interval T' between the same pair of tick events on one clock in its rest frame, $x' = 0$.	pp. 118–119
6	Interval $T = \gamma T'$ for the same pair of tick events compared in another inertial frame.	p. 119
7	One-way signal-propagation time from emission to arrival at the floor in Figure 51, with the same emission and arrival events as endpoints.	p. 119
8	Comparison of muon lifetime with atmospheric travel time. The source uses $v \simeq 0.99c$ and $\gamma \simeq 10$ as rough values; decay is stochastic.	p. 120
9	Comparison of elapsed readings of aircraft-borne and ground-based atomic clocks; both special and general relativity contribute to the difference.	p. 120
10	Proper time τ on a future-directed timelike worldline, using the local differential and path integral for a general trajectory.	pp. 133–134; (7.18), (7.19)
11	Calibration of the second and of length and time units: the common temporal scale used in physical c , v/c , u/c , γ , metre and light-second conversion, and the $c = 1$ representation.	pp. 107, 120, 131–132
12	Changes in the received periods of Io events and light-propagation delay in the historical account of Rømer’s inference.	p. 113
13	The family of directional round-trip light-travel times (T_x , T_y) under an ether rest frame, Newtonian velocity addition, and a common nominal path length L .	p. 114
14	The directional round-trip time-difference reading reported as null in the source’s account of the Michelson–Morley interference comparison.	p. 114
15	Rest-frame elapsed-interval and rate reference for a general physical process, with the same process, event cycle, or lifetime specified.	pp. 119–120
16	Moving-frame elapsed-interval and rate comparison for that same process, related by time dilation between inertial frames.	pp. 119–120
17	Temporal content of the physical invariant spacetime interval Δs^2 , determined by the coordinate differences of an arbitrary pair of events, and of its differential form ds^2 . This includes timelike, spacelike, and null intervals. Elapsed proper time along a future-directed timelike worldline is the path integral in Time Content 10 using that line element.	pp. 125–128, 133–134; (7.10), (7.11), (7.13), (7.18), (7.19)
18	Temporal direction and signature structure of the physical Minkowski metric with $X^0 = ct$ and signature $(+ - - -)$.	pp. 128–130; (7.13), (7.14)

i	Time content	Source location
19	Predicted time difference $\Delta T_{\text{pred}} = T_y - T_x$ calculated from the ether model's directional round-trip times.	p. 114

Dependency input Source-claim indices refer to Table 569; time-input and output indices refer to Table 570. Each input set lists jointly required inputs.

Dependencies and transfers

Definition 8.715 (Time-content transfers in Tong 2013). The following table gives the ten transfers and their inputs and outputs.

Table 571: Time-content transfers in Tong 2013

Transfer	Operation	Source claims	Time inputs	Time outputs
1	Transfer from t to t' by a Lorentz transformation between inertial frames with a common origin, aligned spatial axes, and $ v < c$. pp. 107–111; (7.2), (7.3), (7.4), (7.5), (7.6)	$\{1, 2, 4, 13, 14, 15\}$	$\{1, 11\}$	$\{2\}$
2	Obtain frame-relative simultaneity from equality of coordinate time in the specified inertial frame. pp. 115–117; (7.5)	$\{1, 2, 15\}$	$\{1, 2, 11\}$	$\{3\}$
3	Obtain the future–past causal order of the light cone from the physical interval and metric of the same event pair and a fixed time orientation. pp. 117–118, 126–128, 130; (7.11), (7.13), (7.14)	$\{2, 4, 17, 25\}$	$\{1, 17, 18\}$	$\{4\}$
4	Obtain T' by differencing the coordinate times of the same two tick events on the same clock at rest, $x' = 0$. pp. 118–119	$\{5\}$	$\{2\}$	$\{5\}$
5	Transfer from rest-frame T' to $T = \gamma T'$ in another inertial frame for the same pair of tick events. p. 119; (7.6)	$\{5, 15, 18\}$	$\{5, 11\}$	$\{6\}$

Transfer	Operation	Source claims	Time inputs	Time outputs
6	Obtain proper time along a general future-directed timelike worldline through $d\tau = dt\sqrt{1 - u^2/c^2}$ and $T = \int d\tau$; on a rest worldline, $\Delta\tau = \Delta t'$. pp. 133–134; (7.18), (7.19)	{3, 23, 28}	{1, 11, 17, 18}	{10}
7	Obtain the physical invariant interval from coordinate differences and the Lorentz transformation for the same event pair, including timelike, spacelike, and null cases. pp. 125–127; (7.10), (7.11)	{2, 15, 23}	{1, 2, 11}	{17}
8	Calculate directional round-trip times (T_x, T_y) from L, c, v under the historical ether-rest-frame and Newtonian-velocity-addition hypothesis. p. 114	{11, 33}	{11}	{13}
9	Calculate $\Delta T_{\text{pred}} = T_y - T_x$ from the predicted directional round-trip times. p. 114	{11}	{13}	{19}
10	Transfer the rest-frame elapsed interval and rate of the same physical process to its moving-frame elapsed and rate comparison under the inertial time-dilation law. pp. 119–120	{12, 18, 19}	{11, 15}	{16}

Conditions Operations directly using velocity ratios, γ , $c^2\Delta t^2$, or length–time conversion take Time Content 11 as input. Differences and comparisons of already calibrated time contents inherit calibration through those inputs.

Transfer 6 uses a future-directed timelike worldline and integrates $d\tau = dt\sqrt{1 - u^2/c^2}$ along the path. The rest-worldline identity $\Delta\tau = \Delta t'$ is its specialization.

Transfers 8 and 9 calculate round-trip times and their difference under the ether hypothesis. The interference-comparison reading is supplied as Time Content 14 and compared with predicted difference 19 in Result 23.

Results

Definition 8.716 (Results and preservation conditions in Tong 2013). The following table gives the 26 results and their conditions of application.

Table 572: Results and source locations in Tong 2013

q	Result	Source location
1	Lorentz transformation between inertial frames with a common origin and aligned spatial axes.	pp. 107–111; (7.2), (7.3), (7.4), (7.5), (7.6)
2	Relative simultaneity determined by equality of coordinate time in the specified inertial frame.	pp. 115–117; (7.5)
3	Causal temporal order of the light cone with a fixed time orientation.	pp. 117–118, 126–128, 130; (7.11), (7.13), (7.14)
4	The relation $T = \gamma T'$ for the same two tick events on the same clock.	pp. 118–119; (7.6)
5	The relation $t = \gamma t'$ for the same emission and floor-arrival events in Figure 51.	p. 119
6	Application of time dilation to inertial-frame comparisons of elapsed intervals and rates for every physical process, including clocks, elementary particles, and hearts.	pp. 119–120
7	Comparison of muon lifetime with atmospheric travel time, explained in the source using rough velocity and γ values.	p. 120
8	Comparison of aircraft-borne and ground-based atomic-clock readings in an experimental account involving both special- and general-relativistic contributions.	p. 120
9	Elapsed-age comparison and reassignment of simultaneity in the twin example, using the route with a turnaround and the route remaining at home.	pp. 120–122; (7.5)
10	Lorentz contraction obtained by selecting rod endpoints simultaneous in each inertial frame.	pp. 122–124; (7.5), (7.6)
11	Relativistic addition of constant collinear velocities.	pp. 124–125; (7.8), (7.9)
12	Physical invariant spacetime interval for the same event pair, covering timelike, spacelike, and null cases.	pp. 125–127; (7.10), (7.11), (7.12)
13	Correspondence between the physical Minkowski metric and the abstract Lorentz matrix preservation condition.	pp. 128–130; (7.13), (7.14), (7.15), (7.16)
14	Composition of collinear boosts through signed rapidity.	pp. 130–131; (7.17)
15	The $c = 1$ unit convention using length–time conversion through light-seconds.	pp. 131–132
16	The relations $d\tau = dt/\gamma$ and $T = \int d\tau$ on a general future-directed timelike worldline; on a rest trajectory, $\Delta\tau = \Delta t'$.	pp. 133–134; (7.18), (7.19)
17	Parametrization of a timelike worldline by proper time.	pp. 133–134; (7.18), (7.19)
18	Calibration of the metre through the second and the exact light-speed value, and conversion to light-seconds.	pp. 107, 131–132
19	Historical inference attributed to Rømer from variations in the received periods of Io events to finite light speed.	p. 113

q	Result	Source location
20	Historical account of light-speed measurement using rotating mirrors.	p. 113
21	Maxwell's relation $c = 1/\sqrt{\epsilon_0\mu_0}$.	pp. 113, 115; (7.7)
22	Prediction of directional round-trip times (T_x, T_y) under the ether hypothesis.	p. 114
23	Comparison of the ether model's predicted time difference with the reported null result of the interference comparison.	p. 114
24	Explanation of the null round-trip time difference by Lorentz's historical ether-direction contraction hypothesis, in which distances between matter transform as $x' = \gamma(x - vt)$. The factor $\gamma(v/c)$ uses temporal scale 11.	p. 114
25	The theoretical route from Maxwell's equations holding in every inertial frame to invariance of light speed.	p. 115; (7.7)
26	Predicted round-trip time difference $\Delta T_{\text{pred}} = T_y - T_x$ under the ether hypothesis.	p. 114

Result dependencies

Table 573: Result dependencies in Tong 2013

q	Source claims	Temporal premises
1	{1, 2, 4, 13, 14, 15}	{1, 11}
2	{1, 2, 15}	{1, 2, 11}
3	{2, 4, 17, 25}	{1, 17, 18}
4	{5, 15, 18}	{5, 11}
5	{4, 13, 18}	{7, 11}
6	{12, 18, 19}	{11, 15}
7	{6, 18, 19}	{8, 11, 15, 16}
8	{7, 19}	{9, 11}
9	{3, 20}	{3, 16}
10	{8, 15, 21}	{3, 11}
11	{3, 15, 22}	{1, 2, 11}
12	{2, 15, 23}	{1, 2, 11}
13	{2, 24, 25}	{11, 17, 18}
14	{3, 22, 26}	{1, 2, 11}
15	{27, 29}	{11}
16	{3, 23, 28}	{1, 11, 17, 18}
17	{3, 28}	{10}
18	{4, 29}	{11}
19	{4, 9, 30}	{11, 12}
20	{4, 10, 31}	{11}
21	{32}	{11}
22	{11, 33}	{11}
23	{11, 34}	{14, 19}
24	{11, 35}	{11, 13, 14, 19}

q	Source claims	Temporal premises
25	{13, 32, 36}	{11}
26	{11, 33}	{11, 13}

Evidence hierarchy Results 1–6 and 9–18 are derivations and generalizations under the specified inertial-frame, spacetime, and worldline conditions. Results 7 and 8 are accounts of muon and aircraft-clock experiments. Results 19–26 are historical accounts and theoretical relations concerning light speed, the ether, and Maxwell theory. The source names hearts within the generalization in Result 6.

Generation judgment Judgment: 0 LCTR-complete generations; 0 locally reachable time contents under source-evidence conditions; 19 residual primitive time roots.

Result dependencies There are 61 primitive-time-root–result dependency pairs, of which 40 are necessary dependencies.

Evidential scope The domain comprises Lorentz transformations, relative simultaneity, causal order, time dilation for clocks, signals, and general processes, worldline proper time, and the hypotheses and readings specified in each historical account.

Result-preservation domain The 26 results hold under their respective conditions on correspondence of the same event pair or process, inertial frames, time orientation, unit calibration, physical metric, and historical hypotheses.

Placement of grounding obligations The 19 time contents are used as already established inputs without prior generation from nontemporal physical inputs. The placement judgment is that the grounding obligation for physical time is backgrounded by primitive-time inputs.

Evidence of actual invalidation Whether the input conditions specified here actually fail cannot be determined from the source evidence.

Downstream reach Structural reach is given by the result-dependence table and minimal primitive-time-root support families. If an actual failure of a specific condition is located, the recalculation scope consists of results dependently reachable from that failure.

Repair scope Repair concerns the grounding conditions for generating time contents before use. If an actual failure of a specific condition is located, dependent results are recalculated from repaired inputs or restricted to the domain in which their conditions hold.

Definition 8.717 (Finite dependency graph of Tong 2013). The finite dependency graph comprises 36 semantic components, 19 time contents, 10 transfers, and 26 results.

Proposition 8.718 (Primitive time roots and result dependence in Tong 2013).

$$\text{AuditPartIdx}_R = \text{AllRootDepIdx}(19, 26).$$

Proof. The generation-initial set consists of Claims 1–12. All ten transfers take time contents as inputs, so no time content is reachable from that initial set. Adding the source formal-evidence environment provides no transfer without time inputs. Complete generation and local reach under source-evidence conditions are therefore both zero, and all 19 time contents remain primitive time roots. Every one of the 26 results has temporal premises, and supplying the time contents and source formal-evidence environment satisfies each result edge. $\square$

Proposition 8.719 (Minimal primitive-time-root support in Tong 2013). The minimal-support families in the following table contain 37 members in total.

Proof. Updating the support antichains to a fixed point over the ten transfer edges and 26 result edges yields the following table. $\square$

Table 574: Minimal primitive-time-root support in Tong 2013

Result	Family of minimal-support index sets
1	$\{1, 11\}$
2	$\{1, 11\}$
3	$\{1, 11, 18\}, \{1, 17, 18\}$
4	$\{1, 11\}, \{2, 11\}, \{5, 11\}$
5	$\{7, 11\}$
6	$\{11, 15\}$
7	$\{8, 11, 15\}$
8	$\{9, 11\}$
9	$\{3, 16\}, \{1, 11, 15\}, \{3, 11, 15\}, \{1, 11, 16\}$
10	$\{1, 11\}, \{3, 11\}$
11	$\{1, 11\}$
12	$\{1, 11\}$
13	$\{1, 11, 18\}, \{11, 17, 18\}$
14	$\{1, 11\}$
15	$\{11\}$
16	$\{1, 11, 18\}$
17	$\{10\}, \{1, 11, 18\}$
18	$\{11\}$
19	$\{11, 12\}$
20	$\{11\}$
21	$\{11\}$
22	$\{11\}$
23	$\{11, 14\}, \{13, 14\}, \{14, 19\}$
24	$\{11, 14\}$
25	$\{11\}$
26	$\{11\}$

Individual statements Statement evidence 1

Original text (English):

the same argument holds for any process, be it clocks, elementary particles or human hearts

The source location is Tong, *Dynamics and Relativity*, §7.2.3 *Time Dilation*, p. 119. The minimal context is the extension of the argument for the clock-tick interval ratio to physical processes other than clocks. The agent to whom the statement is attributed is David Tong.

The statement objects are the argument given for the clock-tick interval ratio and the physical processes to which it is applied. The predicate is that the same argument holds for any process, including clocks, elementary particles, and human hearts. The explicit domain of application comprises inter-frame time dilation in §7.2.3 and the clock, particle, and biological processes enumerated by the source.

Statement evidence 2

Original text (English):

The correct interpretation is that time itself runs more slowly in moving frames.

The source location is Tong, *Dynamics and Relativity*, §7.2.3 *Time Dilation*, p. 119. The minimal context is the interpretation stated after the derivation of the time-dilation ratio. The agent to whom the statement is attributed is David Tong.

The statement object is the rate of passage of time itself in a moving inertial frame. The predicate is that, on the correct interpretation, time itself runs more slowly. The explicit domain of application is the clock-tick interval ratio between moving inertial frames derived in §7.2.3.

Statement evidence 3

Original text (English):

it is time itself which is running slower.

The source location is Tong, *Dynamics and Relativity*, §7.2.3 *Time Dilation*, under the subheading *On Muons and Planes*, p. 120. The immediate context is the paragraph attributing the increased muon lifetime to time itself running more slowly, on the grounds that it cannot be explained by the slowing of an internal mechanical clock. The agent to whom the statement is attributed is David Tong.

The statement object is that to which slower running is attributed in the explanation of the increased muon lifetime. The predicate is that time itself is what runs more slowly. The explicit domain of application is the explanation of muon lifetimes in §7.2.3.

Individual audit Objects and claims of the statements

Statement 1 asserts that the same argument about clock-tick intervals holds for every physical process, explicitly including elementary particles and human hearts. Statement 2 presents time itself running more slowly in moving inertial frames as the “correct interpretation.” Statement 3 follows the explanation that elementary particles are not clocks with internal machinery and attributes the increased muon lifetime to time itself running more slowly.

For the same pair of tick events on the same clock, the interval T' in its rest frame and the interval T in another inertial frame are related by $T = \gamma T'$, with $\gamma = (1 - v^2/c^2)^{-1/2}$. Time Contents 5 and 6, Transfer 5, and Result 4 give this relation. The emission and floor-arrival events in Figure 51 correspond to Time Content 7 and Result 5; the geometry

of signal propagation yields the same factor γ . The generalization to every process corresponds to Transfer 10 and Result 6, relating the rest-frame elapsed content 15 to the moving-frame comparison content 16 for the same process. The physical velocities in these factors use the common temporal scale and unit calibration of Time Content 11.

Common audit with specified domain of application Theoretical generalization and experimental evidence

The source describes tests in particle accelerators such as the LHC and LEP, muon lifetimes, and comparisons of readings from aircraft-borne atomic clocks. The muon and aircraft-clock examples correspond to Time Contents 8 and 9 and Results 7 and 8. Both special and general relativity contribute to the aircraft-clock difference. Human hearts occur in the theoretical generalization as physical processes to which the same argument applies.

Relation to the time-content generation judgment

The coordinate times, tick intervals, signal-propagation time, elapsed-process contents, and temporal scale used in this argument are time contents in Table 570. Their relations are given by the transfers in Table 571. Every transfer requires prior time content, and the generation judgment in Proposition 8.718 is zero.

8.19.62 Source audit

Audit Input

Primary Source and Audit Scope

Source Identifier 62.

Source Name Primitive-Time-Root Audit of Lewis 2016

Bibliographic Identification Let Geraint F. Lewis, *On the Relativity of Redshifts: Does Space Really “Expand”?*, *Australian Physics*, Vol. 53, No. 3 (2016), pp. 95–100, arXiv:1605.08634v1 be denoted by

$$R = \text{Ref}_{\text{Lew16RR}}$$

The audit domain comprises pp. 95–100 of the primary source, its text and figures, and its bibliography.

Audit-Scope Identifier 62-SC1.

Audit-Scope Content The audit domain specified under Bibliographic Identification.

Primary-Source-Location Identifier 62-L1.

Primary-Source-Location Content The audit domain specified under Bibliographic Identification and the primary-source-location column of each evidence table.

Formal Objects The audit uses 14 time contents, 27 source claims, 3 transfers, and 12 results. Time Content 1 contains the temporal and causal structure of the physical metric, and Time Content 4 the observer’s unit time direction. Four-vector algebra, contraction, parallel transport, and coordinate transformation are formal operations using that structure.

Assumptions The path and emission and reception events of the same photon are fixed, with an observer specified at each measurement event. FRW comoving observers, the accelerating rocket, and general local measurements belong to their respective models and domains.

Nontemporal Physical Inputs The degrees of freedom of the radiation field, bodies, and devices, together with their spatial configuration in Source Claims 1–3, form the generation-initial set. Measurement-event order, motion histories, the temporal direction of the metric, and calibration by a clock unit belong to the corresponding time contents.

Definition 8.720 (Time contents in the Lewis 2016 source). Table 575 gives the time contents and their source locations. Time Content 8 is the family of readings at the emission, reception, and intermediate measurements specified in the source comparison, fixing the model, physical spacetime, light path, event, and observer of each member. A reading family in the same physical setting is represented uniformly by frequencies, using the same calibration for conversion to energy and wavelength. Time Contents 13 and 14 are the temporal contents of the energy-loss and wavelength-stretching pictures examined in the source.

Table 575: Time contents and source locations in Lewis 2016

i	Time content	Source location
1	Temporal and causal structure of the Lorentzian metric	pp. 96–99
2	Temporal order of emission and reception events	pp. 97–99
3	Motion history of an accelerating observer between emission and reception	pp. 96–97
4	Unit time direction of the observer tetrad at a measurement event	p. 98
5	Local measurement event at which the photon meets the observer	pp. 97–98
6	Temporal component of the photon four-vector at the initial event	p. 98
7	Temporal component of the photon four-vector transported along its path	p. 98
8	Family of local spectral readings relative to the specified observers	pp. 97–99
9	Ordered sequence of observer encounters and readings along the same light path	p. 99
10	Cosmic time along the comoving observers in FRW spacetime	pp. 98–99
11	FRW scale-factor history and its values at emission and reception	pp. 98–99
12	History of proper distance between fixed comoving observers on equal-cosmic-time slices	pp. 98–99
13	Continuous photon-energy-loss picture examined in the source	p. 96

i	Time content	Source location
14	Photon-wavelength-stretching picture used for FRW comoving observers	pp. 95, 99

Definition 8.721 (Physical and formal input claims in Lewis 2016). Table 576 gives the 27 source claims. Claims 1–3 identify physical carriers; Claims 4–27 give the physical and formal descriptions and attributions in the source.

Table 576: Physical and formal input claims in Lewis 2016

p	Source claim	Source location
1	Physical degrees of freedom of the radiation field carrying the photon	pp. 95–99
2	Physical degrees of freedom of emitters, detectors, laboratories and rockets	pp. 96–99
3	Spatial configuration of the devices and bodies being compared	pp. 96–99
4	A Lorentzian metric with the photon null condition is used.	p. 98
5	Emission and reception of the same photon are joined by a specified light path.	pp. 98–99
6	The photon four-vector is supplied at the specified initial event.	p. 98
7	The photon four-vector is parallel transported along the specified light path.	p. 98
8	An orthonormal observer tetrad and its time direction are supplied.	p. 98
9	Photon and observer four-vectors are compared at the same measurement event.	p. 98
10	Local energy is obtained by contracting the photon four-vector with the observer time direction.	p. 98
11	The local spectral readings of the emitter and receiver are compared.	pp. 98–99
12	Adjacent spectral-reading ratios of the same kind and orientation along one light path give the endpoint ratio by cancellation of shared intermediate readings.	p. 99
13	The equivalence principle is used for the local correspondence between gravity and acceleration.	pp. 96–97
14	The motions of the accelerating rocket and external inertial observers are specified.	pp. 96–97
15	FRW spacetime and observers at fixed comoving spatial coordinates are specified.	pp. 98–99
16	Redshift between FRW comoving endpoints is expressed by the scale-factor ratio at reception and emission.	p. 99

p	Source claim	Source location
17	Proper distance on an equal-cosmic-time slice is determined by the scale factor and fixed comoving distance.	pp. 98–99
18	Many comoving observers are evenly spread between FRW comoving endpoints A and B, and comparisons between adjacent observers are described as local Doppler effects.	p. 99
19	Different coordinate descriptions of the same physical situation give the same local readings.	p. 98
20	The measured wavelength is attributed to the photon–observer relation.	p. 98
21	The continuous energy-loss picture during propagation is examined in terms of local observations.	pp. 96–98
22	The expanding-space picture is useful for wavelength comparison by FRW comoving observers. Lewis points out that treating expanding space as a physical medium, like a river carrying objects, leads to incorrect conclusions about motion.	p. 99
23	The unified redshift description of Narlikar 1994 is reported.	pp. 98, 100
24	Doppler, gravitational and cosmological redshifts are compared.	pp. 95–99
25	The standard local correspondence between energy, frequency and wavelength is used.	pp. 98–99
26	Changes of coordinate components are distinguished from the geometry of the same spacetime.	p. 98
27	Statements by Peacock, Rees and Weinberg are quoted to introduce criticisms of the expanding-space picture.	p. 95

Definition 8.722 (Semantic components and generation-initial set in Lewis 2016). Let $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(27)$ and $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(3)$. Source-claim and semantic-component indices identify the same objects. The source formal-evidence environment consists of the 27 claims in Table 576.

Dependency Inputs Transfer-support indices refer to the source-claim table; time-input and time-output indices refer to the time-content table.

Dependencies and Transfers

Definition 8.723 (Time-content transfers in Lewis 2016). Table 577 gives the inputs, outputs, and source claims of the three transfers.

Table 577: Time-content transfers in Lewis 2016

Transfer	Transfer operation	Source claims	Time inputs	Time outputs
1	Transport of the initial photon four-vector along its light path	$\llbracket 4, 7 \rrbracket$	$\{1, 2, 6\}$	$\{7\}$
2	From the photon four-vector and observer time direction at the same event to local spectral readings	$\{8, 9, 10, 25\}$	$\{1, 4, 5, 7\}$	$\{8\}$
3	From the FRW scale-factor history to the proper-distance history between fixed comoving observers	$\{15, 17\}$	$\{10, 11\}$	$\{12\}$

Conditions Transfer 1 is parallel transport along a regular null geodesic segment of the fixed Lorentzian metric. The initial photon four-vector is supplied, with the same affine normalization and light path retained. The transported family in Time Content 7 agrees with the initial value at the initial point by identity transport and has the value transported to each subsequent encounter. Each value carries its own event index.

Transfer 2 contracts each member of this family with the observer tetrad time direction at that measurement event. The model, physical spacetime, and light path are fixed, and the photon and observer are matched at each event. With metric signature $(-+++)$, units $c = 1$, a future-directed unit timelike vector u , and a positive-energy null four-momentum p , the local energy is $E = -g(p, u)$. The standard spectral relations $E = h\nu$ and $\lambda\nu = 1$ determine frequency and wavelength. The temporal component belongs to a specified four-vector; contraction uses the full vector, including its spatial components, and the metric. Each reading retains the event, observer, and photon-data correspondence and the calibration of the observer's unit time direction.

Transfer 3 concerns the distance between fixed comoving observers on equal-cosmic-time slices of an FRW model. For a positive scale factor $a(t)$ and fixed comoving distance χ along a specified spatial path, the proper-distance history is $D(t) = a(t)\chi$.

Source Support The input–output relations above express the account of photon four-vector transport and projection onto the observer time direction on p. 98, and the description of FRW comoving observers on pp. 98–99.

Results

Definition 8.724 (Results in Lewis 2016). Table 578 gives the 12 results and their source locations.

Table 578: Results and source locations in Lewis 2016

q	Result	Source location
1	Local spectral measurement relative to an observer	pp. 97–98
2	Attribution of measured wavelength to the photon–observer relation	p. 98
3	Transport of the photon four-vector along the specified light path	p. 98
4	General redshift comparison between emission and reception endpoints	pp. 98–99
5	Equality of a finite product of ordered local-reading ratios and the endpoint ratio	p. 99
6	Correspondence between local observations in an accelerating rocket and a gravitational description	pp. 96–98
7	Redshift between FRW comoving endpoints and the scale-factor ratio	pp. 98–99
8	Proper-distance history between fixed comoving observers	pp. 98–99
9	Local Doppler description using a chain of FRW comoving observers	p. 99
10	Correspondence between coordinate descriptions and local observables for the same physical situation	p. 98
11	Use of the expanding-space picture for FRW comoving observers	p. 99
12	Distinction between the continuous energy-loss picture and observer-relative local measurement	pp. 96–98

Result Dependencies

Definition 8.725 (Result-dependence hyperedges in Lewis 2016). Table 579 gives the source claims and time premises of each result. All premises in a row jointly support that row’s result.

Table 579: Result-dependence hyperedges in Lewis 2016

q	Source claims	Time premises
1	{8, 9, 10, 25}	{8}
2	{8, 9, 20}	{4, 5, 8}
3	[[4, 7]]	{7}
4	{5, 11}	{2, 8}
5	{11, 12}	{2, 8, 9}
6	{13, 14, 19}	{1, 2, 3, 4, 5, 8}
7	{5, 11, 15, 16, 25}	{2, 8, 10, 11}
8	{15, 17}	{12}
9	{12, 15, 18}	{1, 4, 8, 9, 10}
10	{19, 26}	{1, 4, 5, 6, 7, 8}
11	{15, 22}	{8, 10, 11, 14}

q	Source claims	Time premises
12	$\{20, 21\}$	$\{4, 8, 13\}$

Grounds For positive frequencies ν_e, ν_r measured at the emission and reception endpoints of the same photon, the redshift is $1 + z = \nu_e/\nu_r$. For a positive finite sequence of frequency readings in the same physical model $\nu_0, \dots, \nu_n$ along the path, adjacent ratios sharing the same intermediate reading give

$$\prod_{i=0}^{n-1} \frac{\nu_i}{\nu_{i+1}} = \frac{\nu_0}{\nu_n}.$$

This is the finite-product equality in Result 5.

Result 7 specifies the emission and reception events and comoving observers on the same FRW light path, relating the endpoint-reading ratio $1 + z = \nu_e/\nu_r$ to the scale-factor ratio $a(t_r)/a(t_e)$. Result 9 describes comparisons between adjacent members of a large collection of comoving observers evenly spread between FRW comoving endpoints A and B as Doppler effects in local inertial frames. The source describes the endpoint redshift as a long series of Doppler shifts. Mathematical composition of this local description uses the specified FRW metric, comoving observer time directions, encounter order, and local readings, taking the limit as the adjacent intervals are refined.

In the accelerating rocket, observers whose motions agree at emission have different states of motion at reception and measure different local energies for the same photon. The local equivalence principle connects this account of acceleration to the corresponding gravitational description.

Under a coordinate transformation with the physical spacetime, photon, and observers fixed, the components of the four-vectors and metric change, while local measurements obtained by contraction, endpoint redshift, and spacetime curvature are preserved.

Evidential Hierarchy Lewis explains relativistic accounts of redshift using the Pound–Rebka experiment and photon exchange between rockets and observers as examples. The statement about a single mathematical description on p. 98 belongs to the account of Narlikar’s 1994 generalization of Synge’s argument. The usefulness of the expanding-space picture on p. 99 is tied to wavelength comparison by FRW comoving observers.

Generation Judgment Judgment: 0 LC^{TR}-complete generations; 0 locally reachable time contents under source-evidence conditions; 14 residual primitive time roots.

Result Dependencies The dependence relations between primitive time roots and results are determined by $\mathcal{A}_R(q)$ in Definition 7.108 and Definition 7.50.

Evidential Scope The domain comprises local spectral measurements by specified observers, transport along null light paths, emission–reception endpoint comparisons, the local equivalence principle, and redshift for FRW comoving observers.

Result-Preservation Domain The 12 results are placed in the generation-dependence order while preserving the photon and observer event indices, metric and light path, calibration of the unit time direction, and model-specific conditions. The endpoint-ratio equality, FRW scale-factor relation, and local Doppler description are retained with their respective conditions.

Placement of the Grounding Obligation The 14 time contents are used as already established temporal inputs without prior generation from nontemporal physical inputs. Photon transport and measurement readings are supported by transfers that presuppose the metric's temporal and causal structure, the observer time direction, and time contents already in use. The placement judgment is: the grounding obligation for physical time is backgrounded by primitive-time inputs.

Evidence of Actual Invalidation The source gives accounts of local measurement and redshift. Whether the input conditions specified here actually fail cannot be determined from the source evidence.

Downstream Reach Structural reach from time contents to the 12 results is given by the result-dependence table and minimal primitive-time-root support families. If actual failure of a specific condition is established, the recalculation scope consists of results dependently reachable from that failure.

Repair Scope Repair of the generation obligation concerns time contents before their use and their grounding conditions. If a failure of a specific condition is located, dependent results are recalculated from repaired inputs or restricted to the domain in which their conditions hold.

Audit Conclusion

Proposition 8.726 (Lewis 2016: primitive time roots and result dependence).

$$\text{AuditPartIdx}_R = \text{AllRootDepIdx}(14, 12).$$

Proof. The generation-initial set consists of Source Claims 1–3. All three transfers have time inputs, so the generation-initial set reaches no time content. Adding the source formal-evidence environment provides no transfer without time inputs. Both complete time generation and local time reach under source-evidence conditions are therefore empty, and all 14 time contents remain primitive time roots. All 12 result edges have time premises. Supplying the 14 primitive time roots and source formal-evidence environment satisfies the premises of the three transfers and every result edge. Proposition 7.49 and Definition 8.115 give the displayed index tuple. $\square$

Proposition 8.727 (Minimal primitive-time-root supports in Lewis 2016). Table 580 gives the minimal primitive-time-root support families $\mathcal{A}_R(q)$ and their statistics. The families contain 33 members in total.

Proof. Starting from the initial assignment in Definition 7.46, propagate the antichain products associated with the time premises of the three transfers to their outputs and minimize under inclusion. Combining the premise antichains for each result edge at the fixed point gives the displayed families. Proposition 7.47 identifies them as the minimal support families for the respective results. $\square$

Table 580: Minimal primitive-time-root supports in Lewis
2016

q	Support statistics	Members of the support family
1	(3; 1 – –5)	{8}, {1, 4, 5, 7}, {1, 2, 4, 5, 6}
2	(3; 3 – –5)	{4, 5, 8}, {1, 4, 5, 7}, {1, 2, 4, 5, 6}
3	(2; 1 – –3)	{7}, {1, 2, 6}
4	(3; 2 – –5)	{2, 8}, {1, 2, 4, 5, 6}, {1, 2, 4, 5, 7}
5	(3; 3 – –6)	{2, 8, 9}, {1, 2, 4, 5, 6, 9}, {1, 2, 4, 5, 7, 9}
6	(3; 6)	{1, 6}, {1, 2, 3, 4, 5, 7}, {1, 2, 3, 4, 5, 8}
7	(3; 4 – –7)	{2, 8, 10, 11}, {1, 2, 4, 5, 6, 10, 11}, {1, 2, 4, 5, 7, 10, 11}
8	(2; 1 – –2)	{12}, {10, 11}
9	(3; 5 – –7)	{1, 4, 8, 9, 10}, {1, 4, 5, 7, 9, 10}, {1, 2, 4, 5, 6, 9, 10}
10	(2; 5)	{1, 2, 4, 5, 6}, {1, 4, 5, 6, 7}
11	(3; 4 – –8)	{8, 10, 11, 14}, {1, 4, 5, 7, 10, 11, 14}, {1, 2, 4, 5, 6, 10, 11, 14}
12	(3; 3 – –6)	{4, 8, 13}, {1, 4, 5, 7, 13}, {1, 2, 4, 5, 6, 13}

Individual Statements **Statement evidence 1: a single mathematical description of redshift**

Original text (English):

The central thrust of this paper is that it is incorrect to think that there are three distinct mechanisms for redshifting photons in relativity, and that there is truly only a single underlying mathematical description for use in all occasions.

Primary-source location Lewis, *On the Relativity of Redshifts: Does Space Really “Expand”?*, *Australian Physics*, Vol. 53, No. 3, the *Universal Redshifting* section on p. 98.

Minimal context The paragraph introduces Narlikar’s 1994 generalization of Synge’s argument. “This paper” in the quotation refers to Narlikar’s paper.

Attribution Geraint F. Lewis’s account of the central claim of Narlikar’s 1994 paper.

The predicate is that relativistic redshift has a single underlying mathematical description applicable in all cases. The explicit domain of application is the relativistic description of Doppler, gravitational, and cosmological redshift compared in the primary source.

Statement evidence 2: relational attribution of wavelength

Original text (English):

Wavelength is not a property of the photon, but of the “photon+observer” system.

Primary-source location Lewis, *On the Relativity of Redshifts: Does Space Really “Expand”?*, *Australian Physics*, Vol. 53, No. 3, the account of observer-dependent wavelength on p. 98.

Minimal context The passage follows the account of the same physical situation in an external inertial description and in an accelerated or gravitational description: photon four-vector components differ, while local observations agree. Measured energy also depends on the observer's motion at the measurement event.

Attribution A statement in the main text by Geraint F. Lewis.

The predicate is that wavelength is not a property of the photon alone, but a relational property of the photon–observer system. The explicit domain of application is wavelength measurement at a specified measurement event by means of the photon four-quantity, observer four-velocity, and metric.

Statement evidence 3: limited utility of the expanding-space concept

Original text (English):

the concept of expanding space is useful in a particular scenario

Primary-source location Lewis, *On the Relativity of Redshifts: Does Space Really "Expand"?*, *Australian Physics*, Vol. 53, No. 3, the conclusion on p. 99.

Minimal context The passage follows the comparison of wavelength stretching and local Doppler effects for a photon passed between comoving observers in FRW spacetime.

Attribution A statement in the main text by Geraint F. Lewis.

The predicate is that the concept of expanding space is useful in a particular scenario. The explicit domain of application is the explanation of redshift measured between comoving observers in FRW spacetime.

Statement evidence 4: replacement of the expanding-space account by Doppler shift

Original text (English):

expanding space can be extinguished, replaced with the simple Doppler shift

Primary-source location Lewis, *On the Relativity of Redshifts: Does Space Really "Expand"?*, *Australian Physics*, Vol. 53, No. 3, the conclusion on p. 99.

Minimal context Following the local Doppler account along an FRW light path, the passage discusses the description of redshift under a change of coordinates.

Attribution A statement in the main text by Geraint F. Lewis.

The predicate is that a change of coordinates permits the expanding-space account to be replaced by a simple-Doppler-shift account. The explicit domain of application is the account of redshift with the same physical FRW spacetime, light path, and endpoint observers retained.

Individual Audit A single mathematical description of redshift

The Narlikar account introduced in the first statement parallel transports the same photon’s four-vector along its light path and compares local measurements obtained by contraction with the observer time direction at each endpoint. Transfers 1 and 2 in Table 577, together with Results 3 and 4, express this relation. Its inputs include emission–reception order, the metric’s temporal and causal structure, and the observer time directions.

Relational attribution of wavelength

The photon–observer relation in the second statement is expressed by contraction of the photon four-vector and observer time direction at the same measurement event. The energy and corresponding wavelength measured for the same photon depend on the observer’s state of motion. Under coordinate transformation with the specified observer fixed, this local reading is preserved. Time Contents 4–8 retain the inputs and readings of this comparison and connect them to Results 1, 2, and 10.

FRW comoving observers and the expanding-space picture

The particular scenario in the third statement is photon emission and reception by comoving observers in FRW spacetime. Their proper separation changes with the scale factor, and the ratio of wavelengths measured at the endpoints corresponds to the ratio of scale factors at reception and emission. Time Contents 2, 8, 10–12, and 14 represent emission–reception order, local readings, cosmic time, scale-factor history, proper-distance history, and the wavelength-stretching picture, supporting Results 7, 8, and 11. Lewis points out that treating space as a physical medium, like a river carrying objects, leads to incorrect conclusions about their motion.

Local Doppler description and coordinate representation

The source construction preceding the fourth statement compares adjacent members of a large collection of comoving observers evenly spread between FRW comoving endpoints A and B. The source describes the endpoint redshift as a long series of Doppler shifts. Cancellation of ratios for a finite reading sequence is Result 5; local comparisons between FRW comoving observers are Result 9. A change of coordinate representation for the same physical spacetime preserves redshift at the specified endpoints and spacetime curvature, as expressed by Result 10.

Common Audit The redshift accounts in the source depend on photon propagation, observer time directions, local readings, and the temporal histories of the models. Table 579 and Proposition 8.726 give these dependencies and the generation judgment.

8.19.63 Source Audit

Audit Inputs

Source and Audit Scope

Source identifier 63.

Source name Smolin 2018/2020 Primitive-Time-Root Audit

Bibliographic identification Lee Smolin, *Temporal Relationalism*, arXiv:1805.12468v1 (2018); *Beyond Spacetime*, Cambridge University Press (2020), pp. 143–175, doi:10.1017/9781108655705.010. $R = \text{Ref}_{\text{Smo18TR}}$.

Audit-scope identifier 63-SC1.

Audit-scope content Abstract, Sections 1–7, Equations (1)–(38), the conditions of each construction and the cited recovery mechanisms.

Source-location identifier 63-L1.

Source-location content Sections and equations in the tables, with page numbers from arXiv:1805.12468v1.

Assumptions Smolin’s temporal relationalism takes time, causation, energy and momentum as fundamental, while treating space, spacetime and locality as emergent. Event updating in causal sets, local geometric constructions from momentum space, coarse graining to macroscopic spacetime, stochastic hidden variables and quantum recovery from real ensembles each have distinct conditions.

Nontemporal Physical Inputs Nontemporal candidate data in Claims 1–7—event carriers, energy and momentum labels, relations, records, states, configurations and geometric coefficients—form the generation-initial set. Causal order, temporal updating and clock interpretations belong to the time contents of the claims that use them.

Formal Objects

Definition 8.728 (Formal Inputs in Smolin 2018/2020). Let $H_n = (V_n, E_n)$ be a finite causal-history state, where E_n is the parent-to-child relation and $\prec_n = E_n^+$ its transitive closure. Let $d_n^+(v)$ count children in the parent relation, and define the thick present by

$$\Pi_n = \{v \in V_n \mid d_n^+(v) < 2\}.$$

Choose distinct parents $a, b \in \Pi_n$ maximizing a defined distinctiveness $D_n(a, b)$, and add a novel event $e \notin V_n$. The update is

$$V_{n+1} = V_n \cup \{e\}, \quad E_{n+1} = E_n \cup \{(a, e), (b, e)\}.$$

When maximizing choices tie, this update is a relation $H_n \rightsquigarrow H_{n+1}$ between history-states. For the causal past $P_n(v) = \{u \mid u \prec_n v\}$,

$$P_{n+1}(e) = P_n(a) \cup P_n(b) \cup \{a, b\}.$$

The updated child counts determine Π_{n+1} . Although $|V_{n+1}| = |V_n| + 1$, parents that exhaust their capacity leave the thick present. Edges of the parent relation are distinguished from irreducible primary links of the causal order.

For the particle branch, set $c = 1$ and use a Lorentzian metric η , worldline parameter s and lapse N :

$$S_{\text{free}} = \int (-x^a \dot{p}_a - N(\eta^{ab} p_a p_b + m^2)) ds.$$

This convention gives $\dot{p}_a = 0$ and $\dot{x}^a = 2N\eta^{ab}p_b$. Adding $z^a(p_a^C - p_a^A - p_a^B)$ at a vertex makes the endpoints coincide at $-z^a$. With interaction coordinate $Z = -z$, the massless branch satisfies

$$\Delta Z^a = 2 \int N ds \eta^{ab} p_b, \quad \eta_{ab} \Delta Z^a \Delta Z^b = 0.$$

The relative-locality arrival-time difference uses simultaneity at emission and the receiving observer’s clock. Restoring c , with energy scale $E_P = \sqrt{\hbar c^5/G}$ and the first-order approximation of source Eq. (27),

$$\Delta t = \frac{d}{c} \frac{\Delta E}{E_P}.$$

This approximation specifies the non-metricity component and direction; d/c , simultaneous emission and clock readings are temporal premises of the calculation.

A view V_J of a general graph consists of nested rooted neighborhoods. On the domain where a first distinguishing order n_{JK} exists, define $D_{JK} = n_{JK}^{-p}$. An alternative branch specifies a distance between views modulo rooted isomorphism. For a finite nonempty set of objects, the variety is

$$\mathcal{V} = \frac{1}{M^2} \sum_{J \neq K} D_{JK}.$$

Here M is the number of objects. The causal-view branch replaces general neighborhoods with neighborhoods of the causal past. The claim and result tables specify the definitions of views, distinctiveness and variety and the conditions for temporal dynamics using them.

Definition 8.729 (Time Contents in Smolin). Let $n_R = 27$; the time-content table specifies the objects, variables and branches. Uses of the same content in expressions, inputs and outputs share an index when their objects and domains agree. Law domains and law evolution are the two specified components of Time Content 8, with the component used determined by each claim.

Table 581: Smolin 2018/2020 Time Contents

i	Time Content	Source Location
1	Fundamental, irreducible time in temporal relationalism	§1,3,6, pp. 1, 3, 15, 26
2	Fundamental causal order	§1,3,3.2,5.1,6, pp. 3, 16, 18, 25–26
3	Passage of the present	§1,3, pp. 3, 15–16
4	Objective distinction among past, present and future	§1,3, pp. 3, 15–16
5	Succession in the generation of novel events	§1,3,6, pp. 3, 16, 26
6	Thick present of events with remaining child capacity	§1,3, pp. 3, 16
7	Causal past of an event	§3(10)–(11);§5.1(36), pp. 16, 25
8	Temporal domains and cosmological evolution of physical laws	§3.1(12);§6, pp. 16–17, 27
9	Irreversibility of fundamental laws	§3.1,6, pp. 17, 26–27
10	Reversibility of effective dynamics	§3.1,6, pp. 17, 27
11	Time of the preferred foliation	§2.1,3.2, pp. 13, 17–18
12	Update stages in causal-set growth	§3,6, pp. 16, 26–27
13	Global temporal order	§3.2, pp. 18
14	Arrival-time difference in relative locality	§4.2(27), pp. 22

i	Time Content	Source Location
15	Initial disordered, irreversible phase	§6, pp. 27
16	Later ordered, quasi-reversible phase	§6, pp. 27
17	Arrow of time	§3.1,6, pp. 17, 27
18	Time in differential equations and canonical dynamics	§2.1(1),(3)–(8), pp. 10–12
19	Update order in general discrete dynamics	§2.1(2), pp. 10
20	Reparametrization order of a Jacobi-type action	§2.1(9), pp. 11
21	Parameter of relativistic worldlines	§4.1(16)–(25), pp. 20–21
22	Simultaneity relative to the observer at emission	§4.2(27), pp. 22
23	Propagation and reception intervals specified by observer clocks	§4.2(27), pp. 22–23
24	Temporal evolution of probability distributions	§2.1, pp. 14–15
25	Temporal evolution in real-ensemble quantum recovery	§5–6, pp. 24–26, 28
26	Lorentzian causal and temporal relations induced by the particle action	§4.1–4.2, pp. 20–22
27	Causal and temporal relations of macroscopic emergent spacetime	§3.2,6, pp. 18, 26–29

Definition 8.730 (Physical and Formal Input Claims in Smolin). Let $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(65)$. With $\text{ClaimBoundaryIdx}_R = ([7], [65])$, Claims 1–7 form the generation-initial set and all 65 claims form the source formal-evidence environment.

Table 582: Smolin 2018/2020 Claims and Temporal Uses

k	Claim	Temporal Use	Source Location
1	Carriers and identifiers of candidate events.	$\emptyset$	§3,4.1,5.1, pp. 16, 20, 25
2	Data for energy and momentum labels.	$\emptyset$	§4–4.2, pp. 19–20, 22
3	Candidate data for relations, neighborhoods and observation records.	$\emptyset$	§5, pp. 23–25
4	Carriers of candidate states and interaction records.	$\emptyset$	§2.1,3,4.1,6, pp. 14, 16, 20, 27
5	Carriers and configurations of physical systems, observers and signals.	$\emptyset$	§4.1–5, pp. 20, 22, 24
6	Algebraic candidate data for geometry and connection coefficients.	$\emptyset$	§4–4.2, pp. 19–22
7	Data for finite adjacency relations and vertex labels.	$\emptyset$	§3,5–5.1, pp. 16, 23, 25
8	Smolin sets a realist completion of quantum mechanics through relational hidden variables as a research goal.	$\emptyset$	§1, pp. 1–3
9	Relationalism describes relations among dynamical actors as the beables of a theory.	$\emptyset$	§1, pp. 3

k	Claim	Temporal Use	Source Location
10	Smolin takes fundamentality and irreducibility to mean that an element cannot be eliminated from a theory fully describing nature.	$\emptyset$	§1, pp. 3
11	Smolin's temporal relationalism takes the activity of time as the generation of novel events to be fundamental.	$1^B, 5^S$	§1,6, pp. 3, 26
12	Smolin contrasts his fundamental-time position with timeless relationalism, in which time is emergent.	$\emptyset$	§1, pp. 3
13	Smolin takes time and causation to be irreducible fundamental elements.	$1^B, 2^C$	§1,3,6, pp. 3, 15, 26
14	The branch taking energy and momentum as fundamental imposes their conservation laws from the outset.	$\emptyset$	§4, pp. 19
15	Temporal relationalism takes space to emerge from a dynamically evolving network of causal relations.	$2^C, 12^S$	§1,6, pp. 3, 26
16	Emergence of classical spacetime requires a specified approximation regime and correspondence between microscopic causal structure and macroscopic Lorentzian geometry.	$2^C, 27^{BC}$	§3.2,6, pp. 18, 26–27
17	In relative locality, localization of a remote event depends on the observer and the energy of the probe.	$21^S, 26^{CX}$	§4.1–4.2, pp. 21–23
18	Smolin asserts an objective distinction among past, present and future, the passage of the present and the openness of the future.	$3^S, 4^B$	§1,3, pp. 3, 15–16
19	At a specified update stage of the causal set, events with fewer than two children constitute the thick present.	$2^C, 6^B, 12^S$	§3, pp. 16
20	Adding a novel event updates a history-state containing causal pasts, child counts and parent selection.	$2^C, 5^S, 6^B, 7^C, 12^S$	§3(10)–(11), pp. 16
21	Smolin takes the future to be open in the sense that there are no present facts about future events.	4^B	§3, pp. 16
22	The principle of sufficient reason is methodological guidance to deepen the explanations given by a theory.	$\emptyset$	§2, pp. 5–6
23	The principle of background independence requires incorporating fixed external structures into interacting dynamical variables.	$\emptyset$	§2, pp. 6–8

k	Claim	Temporal Use	Source Location
24	Causal completeness requires chains of causes to trace back to prior events and properties of the same universe.	$2^C, 7^C$	§2, pp. 8
25	Reciprocity requires that when one object acts on another, the second acts back on the first.	$\emptyset$	§2, pp. 8
26	The identity of indiscernibles identifies two objects whose properties are all identical.	$\emptyset$	§2, pp. 8–9
27	Relational observables describe relations between two entities; spatial, temporal and motional properties are treated relationally.	$\emptyset$	§2–2.1, pp. 8–9, 12–14
28	The shape-dynamics branch uses a preferred CMC foliation; its correspondence with general relativity has the domain specified in the source.	11^B	§2.1, pp. 13
29	Smolin criticizes immutable deterministic laws for specifying a later state as a function of earlier states.	$8^B, 18^S$	§3.1(12), pp. 16–17
30	Smolin proposes limited spatiotemporal domains in which physical laws are meaningful and evolution of the laws themselves on cosmological scales.	8^{SB}	§3.1,6, pp. 17, 28
31	Irreversibility of the fundamental law producing novel events is distinguished from asymmetry of laws under time reversal.	$5^S, 9^B$	§3.1,6, pp. 17, 26–27
32	An effective reversible or quasi-reversible dynamics arises in the later ordered phase of the specified irreversible model.	$9^S, 10^S, 15^B, 16^S$	§3.1,6, pp. 17, 27
33	Energetic causal sets take events carrying energy and momentum and their causal relations as fundamental.	$2^C, 12^S$	§3,6, pp. 16, 26–27
34	The history of the universe is represented by a set of events with causal relations and additional properties.	$2^C, 12^S$	§3, pp. 16
35	A causal set carries a transitive strict causal partial order and permits incomparable pairs of events.	2^C	§5.1(34)–(35), pp. 25
36	Local finiteness requires finitely many events in the causal interval between two specified events.	2^{CB}	§5.1, pp. 25
37	Primary causal links are irreducible causal relations that generate the transitive causal order.	$2^C, 7^C$	§5.1(35)–(36), pp. 25
38	The causal past of an event consists of events that strictly precede it.	$2^C, 7^C$	§3(10)–(11); §5.1(36), pp. 16, 25

k	Claim	Temporal Use	Source Location
39	At each stage, the distinctiveness of eligible parent pairs is compared and a new event is added with a maximizing pair as its parents.	$5^S, 6^B, 7^C, 12^S$	§3(10)–(11), pp. 16
40	Each new event has two parents, and the total number of children of each parent is limited to two.	$5^S, 6^B, 12^S$	§3, pp. 16
41	The inverse problem of spacetime reconstruction is to select, among general causal sets, those corresponding to low-dimensional Lorentzian spacetimes.	$2^C, 27^B$	§3.2, pp. 18–19
42	Smolin conjectures an inverse Noether effect in which conserved fundamental energy and momentum yield an emergent spacetime with translation symmetries.	$21^S, 26^B$	§4–4.1, pp. 19–21
43	The momentum-space metric specifies the mass shell, while its connection specifies momentum composition.	$\emptyset$	§4.1–4.2, pp. 20–22
44	Conservation of incoming and outgoing momentum is imposed at each interaction vertex.	21^{SC}	§4.1(20)–(23), pp. 20–21
45	Endpoint variation of the worldline action makes the endpoint coordinates at the same interaction vertex coincide.	$21^S, 26^{CX}$	§4.1(16)–(23), pp. 20–21
46	For a flat momentum metric and a zero-mass constraint, the separation of interaction events is null with respect to the induced metric.	$21^{SQ}, 26^{CQ}$	§4.1(24)–(25), pp. 21
47	With a nontrivial momentum-space connection, the locality of a remote event depends on the observer position and probe.	$21^S, 26^{CX}$	§4.2, pp. 21–23
48	The displacement assigned by a distant observer depends on the specified non-metricity component, energy and distance.	$21^S, 26^{XQ}$	§4.2(26), pp. 22
49	For two photons emitted simultaneously at the source, the specified first-order approximation computes their arrival-time difference from distance, energy difference and non-metricity.	$14^{KQ}, 21^S, 22^X, 23^{KIQ}$	§4.2(27), pp. 22
50	A view on a general graph consists of rooted neighborhoods and their nested embeddings.	$\emptyset$	§5(28)–(29), pp. 23
51	The n th neighborhood contains vertices at shortest-path distance at most n from the root and the edges between them.	$\emptyset$	§5(28)–(29), pp. 23

k	Claim	Temporal Use	Source Location
52	Distinctiveness is defined by the first nonisomorphic rooted-neighborhood order, or by a specified distance between views.	$\emptyset$	§5(32), pp. 24–25
53	The variety of finitely many objects is the pairwise sum of defined distinctiveness values normalized by the squared number of objects.	$\emptyset$	§5(33), pp. 25
54	In the causal-view branch, causal-past neighborhoods define distinctiveness and variety, and dynamics extremizing that variety are proposed.	$2^C, 7^C, 25^{SP}$	§5.1(34)–(38), pp. 25–26
55	The real-ensemble branch uses ensembles of similar microscopic systems distributed through the universe and interactions comparing their views.	25^{SP}	§5,6, pp. 24–25, 28
56	Smolin describes a construction recovering Schrödinger dynamics under the conditions of the specified ensemble approximation and variety-based dynamics.	25^{SP}	§5–6, pp. 24–26, 28
57	Section 6 summarizes eight hypotheses concerning fundamental time, causation and energy, emergent spacetime and locality, and the irreversibility and evolution of laws.	$\emptyset$	§6, pp. 26–27
58	The differential-equation and Hamiltonian examples being compared use time derivatives and initial conditions.	18^{SQB}	§2.1(1),(3)–(9), pp. 10–12
59	The general discrete update $X(n+1) = F[X(n)]$ uses successive update stages.	19^S	§2.1(2), pp. 10
60	On a real nonsingular domain, a Jacobi-type action describes configuration curves invariant under orientation-preserving reparametrization.	$18^S, 20^B$	§2.1(8)–(9), pp. 11
61	The stochastic hidden-variable examples recover Schrödinger dynamics under the selected microscopic dynamics, large- N approximation and tuning for reversibility.	24^{SPB}	§2.1, pp. 14–15
62	Quantum recovery from causal views uses an ensemble of similar events, a nonrelativistic common frame and the variety term at the specified derivative-expansion order.	$2^C, 7^C, 25^{SP}$	§5.1,6; ref.[44],§4.3–5, pp. 25–26, 28

k	Claim	Temporal Use	Source Location
63	Recovery of the Einstein equations uses thermodynamic conditions based on fundamental causality, correspondence with classical spacetime, weak holography and the quantum equivalence principle.	$2^C, 27^{CQB}$	§6; ref.[27],§2–3, pp. 26, 28–29
64	The source explains the arrow of time through the mechanism of a specified model distinguishing an initial irreversible phase from a later effectively reversible phase.	$9^S, 10^S, 15^B, 16^B, 17^B$	§3.1,6, pp. 17, 27
65	Smolin takes a global temporal order as fundamental while distinguishing it from partial causal order and spacetime geometry.	$1^B, 2^C, 13^B, 27^B$	§3.2, pp. 18

Superscripts on the temporal indices denote the use roles in each claim.

Table 583: Smolin 2018/2020 Time-Use Roles

Code	Role	Use Count
I	Interval	1
Q	Quantity	7
X	Cross-position correspondence	5
K	Clock reading	2
C	Causality	31
S	State evolution	36
P	Probabilistic evolution	5
O	Observation record	0
B	Reference boundary	27

Table 584: Smolin 2018/2020 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3, 11–12, 18	member	instant, order_relation
2, 5, 7, 13, 19	member	order_relation
4	member	instant, orientation
6	member	instant, endpoint
8, 15–16	member	interval_boundary, order_relation
9–10	member	orientation, order_relation
14, 23	member	interval_boundary, endpoint
17	member	orientation
20	member	order_relation, orientation

Time Indices	Structural Type	Axes
21	member	instant, order_relation, endpoint
22	member	instant
24–25	dependency	$\emptyset$
26–27	member	order_relation, interval_boundary

Definition 8.731 (Semantic Components and Generation-Initial Set in Smolin). The 65 semantic components in the claim table use the same 65 claim nodes. The seven generation-initial components correspond to the nontemporal carriers and data in the following table.

Table 585: Smolin 2018/2020 Generation-Initial Components

k	Nontemporal Component
1	Carriers and identifiers of candidate events.
2	Data for energy and momentum labels.
3	Candidate data for relations, neighborhoods and observation records.
4	Carriers of candidate states and interaction records.
5	Carriers and configurations of physical systems, observers and signals.
6	Algebraic candidate data for geometry and connection coefficients.
7	Data for finite adjacency relations and vertex labels.

Definition 8.732 (Finite Audit Quantities for Smolin). The following table specifies the finite audit counts according to the types of claims, time contents, transfers and results.

Table 586: Smolin 2018/2020 Finite Cardinalities

Object	Cardinality
Time contents	27
Claims	65
Auxiliary objects	0
Transfers	7
Results	28
Claim/time-content uses	91
Role assignments	114
Obligation atoms	127
Audit-boundary nodes	120
Audit-boundary edges	35

Dependency Inputs

Definition 8.733 (Smolin 2018/2020 Time-Content Transfers). Let $\text{ldx}_R^{\text{audit,transfer}} = [7]$. The table specifies time inputs, time outputs, claim support and source locations.

Table 587: Smolin 2018/2020 Time-Content Transfers

j	Time In i	Time Out i	Claim Support k	Transfer Content and Source Location
1	2	7	35, 37–38	Define strict causal pasts from the causal partial order. §5.1(34)–(36)
2	2, 7, 12	5–6	19–20, 34, 39–40	Update novel events and the thick present through eligible-parent selection and child capacity. §3(10)–(11)
3	21	26	43–46	Construct local interactions and null separation on the specified classical branch from the worldline action, vertex conservation and mass shell. §4.1(16)–(25)
4	21–23	14	43–44, 47–49	Compute the relative-locality arrival-time difference under simultaneous-emission, clock and propagation conditions. §4.2(26)–(27)
5	9, 12, 15	16	31–33	Pass from the initial phase to the later ordered phase of the specified model. §6
6	9, 16	10	31–32	Obtain effective reversible dynamics in the specified late cyclic regime. §6
7	9–10, 15–16	17	31–32, 64	Characterize the arrow of time through the relation between irreversible global history and an effectively reversible phase. §3.1,6

Grounding Conditions

Table 588: Smolin 2018/2020 Conditions for Transfers

Transfer Index	Conditions
1	Use an irreflexive transitive causal order; the causal past of an event excludes that event itself.
2	Use a finite causal history, a parent relation and child capacities, a defined distinctiveness and at least two eligible parents. Tied maximizing pairs define a selection relation. Parent-relation edges are distinguished from irreducible links of the causal order.
3	Use a flat momentum metric and a consistent action, endpoint and lapse convention. Null separation applies to the massless branch.
4	Specify the first-order non-metricity approximation, the local emission frame, the receiving clock, propagation distance and a common energy scale.
5	Use the model and initial conditions described in the source. In the finite-state analogy, specify a basin of attraction leading to a cycle.
6	Restrict the dynamics to the specified cycle or effective regime in which it is reversible.

Trans-fer Index	Conditions
7	Use the specified initial and late phases and the relation between fundamental irreversibility and effective reversibility.

Primary-Source Support The index sets in each row realize the corresponding components of Definition 7.105. The seven transfers have no directed cycles or self-transfers.

Results

Definition 8.734 (Smolin 2018/2020 Results). Let $\text{Idx}_R^{\text{audit,result}} = [28]$.

Table 589: Smolin 2018/2020 Results

q	Result	Source Location
1	Distinction between fundamental-time and timeless relationalism	§1, pp. 3
2	Conceptual distinction between fundamentality and emergence	§1, pp. 3
3	Methodology of increasingly sufficient reason	§2, pp. 5–6
4	Background-independence principle replacing fixed external structures with dynamical objects	§2, pp. 6–8
5	Causal completeness tracing prior causes within one universe	§2, pp. 8
6	Reciprocity of interaction	§2, pp. 8
7	Principle of the identity of indiscernibles	§2, pp. 8–9
8	Temporal ontology of an objective present and open future	§1,3, pp. 3, 15–16
9	Thick present defined by child capacity	§3, pp. 16
10	History updating in energetic causal sets	§3,6, pp. 16, 27
11	Distinction between primary causal links and transitive causal order	§5.1(34)–(36), pp. 25
12	Local finiteness of causal intervals	§5.1, pp. 25
13	Parent selection by distinctiveness and the child-capacity constraint	§3(10)–(11), pp. 16
14	Critique of immutable laws and proposal of evolving laws	§3.1,6, pp. 16–17, 28
15	Relation between fundamental irreversibility and effective reversible regimes	§3.1,6, pp. 17, 27
16	Research construction for emergent Lorentzian spacetime	§3.2,6, pp. 18, 26–27
17	Inverse problem of selecting causal sets corresponding to low-dimensional spacetime	§3.2, pp. 18–19

q	Result	Source Location
18	Inverse Noether conjecture relating conservation laws to emergent spacetime	§4–4.1, pp. 19–21
19	Relative locality and conditional arrival-time difference	§4.1–4.2, pp. 21–23
20	Observer-dependent localization of remote events	§4.2(26), pp. 22–23
21	Views from rooted neighborhoods of a general graph	§5(28)–(29), pp. 23
22	Distinctiveness and variety on specified domains	§5(32)–(33), pp. 24–25
23	Conditional Schrödinger recovery from a real ensemble of causal views	§5–6; ref.[44],§4.3–5, pp. 24–26, 28
24	Einstein-equation recovery under thermodynamic conditions	§6; ref.[27],§2–3, pp. 26, 28–29
25	Quantum recovery in large-N stochastic hidden-variable approximations	§2.1, pp. 14–15
26	Time use in classical dynamics, discrete updates and reparametrization	§2.1(1)–(9), pp. 10–12
27	Explanation of the arrow of time in the specified model	§3.1,6, pp. 17, 27
28	Distinction between the preferred-foliation branch and global temporal order	§2.1,3.2, pp. 13, 17–18

Table 590: Smolin 2018/2020 Conditions for Results

Result Index	Conditions and Domain
1	Smolin’s research construction taking time as fundamental.
2	Smolin’s criterion of whether an element can be eliminated from the theory.
3	Methodological guidance to deepen the explanations of a theory.
4	The general principle of incorporating fixed structures into dynamical objects. Each concrete dynamical example has its own temporal premises.
5	Prior causal events within the same universe.
6	The principle of reciprocal action between two objects.
7	Identity of objects whose properties are all equal.
8	Smolin’s hypothesis of an objective present and an open future.
9	A specified causal-history state and a set of eligible events with remaining child capacity.
10	The parent-selection relation, child bound, event labels and causal history.
11	A strict partial order and the irreducible links generating its transitive relations.
12	Finiteness between any specified pair of causal endpoints.
13	Defined distinctiveness and eligible parents. With ties, selection remains a relation unless a selection rule is fixed.
14	Limited domains of application of laws and their evolution on cosmological scales.

Result Index	Conditions and Domain
15	The specified mechanism from an initial to a late phase and its domain of application.
16	The specified correspondence and coarse-graining regime connecting microscopic causal structure with macroscopic spacetime.
17	Correspondence with the target low-dimensional Lorentzian spacetime.
18	Smolin's general conjecture and the specified particle-action branch illustrating it.
19	The specified momentum geometry, locally simultaneous emission, propagation approximation and observer's clock.
20	The specified observer, probe and momentum-space connection.
21	A general graph, a root and nested embeddings.
22	A finite nonempty set of objects. The domain where a first distinguishing order exists, or a separately specified distance between views.
23	An ensemble of similar events, a nonrelativistic common frame, a continuum approximation and the specified derivative-expansion order.
24	Causality, classical correspondence, weak holography, quantum equivalence and the corresponding thermodynamic approximation regime.
25	The selected microscopic dynamics, order of the large- N approximation and tuning for reversibility.
26	The parameter, initial conditions, domain and normalization specific to each equation in the comparison.
27	The specified initial and late regimes and the distinction between fundamental irreversibility and effective reversibility.
28	The specified CMC/shape-dynamics domain and the separately posited hypothesis of global temporal order. Global temporal order and causal partial order are distinct assumptions.

Result Dependence

Definition 8.735 (Smolin 2018/2020 Result-Dependence Hyperedges). Each row defines a hyperedge from the union of its temporal premises and claim support to the corresponding result.

Table 591: Smolin 2018/2020 Result-Dependence Hyperedges

q	Time Premises i	Claim Support k
1	1	11–13
2	$\emptyset$	10
3	$\emptyset$	22
4	$\emptyset$	23
5	2, 7	24
6	$\emptyset$	25
7	$\emptyset$	26

q	Time Premises i	Claim Support k
8	3–4	18, 21
9	6	19–20
10	2, 5, 12	20, 33–34, 39–40
11	2, 7	35, 37–38
12	2	35–36
13	2, 7, 12	19–20, 39–40
14	8	29–30
15	9–10, 15–16	31–32
16	2, 27	16, 33, 65
17	2, 27	41
18	21, 26	42, 44–46
19	14, 26	43–44, 47–49
20	21, 26	47–48
21	$\emptyset$	50–51
22	$\emptyset$	52–53
23	2, 7, 25	54–56, 62
24	2, 27	63
25	24	61
26	18–20	58–60
27	17	31–32, 64
28	2, 11, 13, 27	28, 65

Proposition 8.736 (Smolin 2018/2020 Audit-Boundary Cardinalities).

$$\text{BoundaryCount}_R^{\text{audit}} = (120, 35).$$

Proof. The 65 claims, 27 time contents and 28 results give the node count; the 7 transfers and 28 result dependencies give the edge count. $\square$

Evidence Hierarchy Smolin discusses relational principles, constructions in specified models, conjectures and conditional recovery. The source reports zero newly acquired direct measurements.

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 0; remaining primitive-time roots 27.

Result Dependence Results 2–4, 6–7, 21–22 are reachable from the source formal-evidence environment without added temporal seeds. The remaining 21 results depend on primitive-time roots. There are 46 root–result dependence pairs, of which 31 are necessary.

Evidential Force Smolin’s temporal relationalism, general principles, causal-set updates, local geometric constructions from particle actions, relative locality and conditional recovery of dynamics have the conditions given in the result tables. Local particle geometry and correspondence with macroscopic spacetime are separate branches, as are the recovery conditions for stochastic hidden variables and causal real ensembles.

Placement of Grounding Obligations Causal-history updates use causal order, pasts and update stages; arrival-delay calculations use worldlines, local simultaneity and clock readings. Complete generation of the 27 time contents, including these contents, is not obtained from nontemporal Initial Claims 1–7. The inputs of each transfer and result specify the placement of its temporal premises.

Evidence of Actual Failure Actual failure of a specific physical-time generation condition and the consequent change in predicted values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 46 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns the time contents used by each result and the physical-time generation conditions required before their use. If actual failure of a specific condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.737 (Primitive-Time Roots and Result Dependence in Smolin 2018/2020). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = & (\emptyset, [27]; \\ & \emptyset, \{2, 3, 4, 6, 7, 21, 22\}; \\ & \{1, 5\} \cup \llbracket 8, 20 \rrbracket \cup \llbracket 23, 28 \rrbracket, \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–7 remains those seven nodes. Even with all 65 claims, each of the seven transfers requires nonempty temporal inputs. The seven results without temporal premises are reached. Adding all 27 time contents as seeds reaches all 28 results. $\square$

Proposition 8.738 (Local-Reach Judgment in Smolin 2018/2020). The set of time contents locally reachable from the source formal-evidence environment is empty, leaving 27 primitive-time roots.

Proof. Every transfer has a nonempty temporal input set, so no first transfer can fire from an initial closure containing no time content. $\square$

Proposition 8.739 (Minimal Primitive-Time-Root Support in Smolin 2018/2020). The minimal-support families contain 34 members in total, with the per-result statistics shown below.

Proof. Apply Definition 7.46 to the seven transfers and 28 result dependencies, starting from singleton temporal seeds and empty supports for source formal evidence. By Proposition 7.47, the stable antichains give the minimal supports of each result. $\square$

$(m; a, b)$ gives the number of supports and the minimum and maximum cardinalities; equal cardinalities are written $(m; a)$.

Table 592: Minimal Root Supports in Smolin

Result Indices	Support Statistics
1, 5, 11–12, 14, 18, 20, 25	1; 1
2–4, 6–7, 21–22	1; 0
8, 10, 13, 16–17, 23–24	1; 2
9	2; 1, 2
15	2; 3
19	3; 2, 3
26	1; 3
27	3; 1, 3
28	1; 4

Proposition 8.740 (Audit Conclusion for Smolin 2018/2020). There are 27 time contents, 65 claims, 7 transfers and 28 results, with 91 claim/time-content uses, 114 role assignments and 127 obligation atoms. Complete generation and local reach are both zero, and 21 results depend on residual primitive-time roots.

Proof. Combine the claim/time-use table, transfer table, result-dependence table and the reachability and minimal-support propositions in this division. $\square$

Individual Statements Statement 1

Original text (English):

I take the opposite view, that time is fundamental.

The source location is Lee Smolin, *Temporal Relationalism*, arXiv:1805.12468v1, p. 3, Section 1. The minimal context is the passage that states Smolin's position after setting out the opposition between the fundamentality and emergence of time in relational theories.

The statement object is Smolin's position on the fundamental status of time. The predicate is that time is fundamental. The explicit domain of application is the discussion in Section 1 that contrasts the fundamentality and emergence of time in relational theories.

Statement 2

Original text (English):

the aspect of time I assert is irreducible is its activity as the generator of novel events from present events

The source location is p. 3, Section 1 of the same source. The minimal context is the passage that identifies which aspect of time is taken to be fundamental and introduces the thick present and the addition of novel events.

The statement object is the event-generating activity treated as an irreducible aspect of time. The predicate is that the irreducible aspect of time is its activity of generating novel events from present events. The explicit domain of application is the construction in Section 1 that discusses the thick present and the addition of novel events.

Statement 3

Original text (English):

Instead, we propose that laws are only meaningful for limited ranges of space and time, and that on cosmological scales, laws evolve.

The source location is p. 17, Section 3.1 of the same source. The minimal context is the passage that rejects absolute, immutable, deterministic laws and states their ranges of application and cosmological evolution.

The statement object is the range within which physical laws are meaningful and their evolution on cosmological scales. The predicate is that Smolin proposes limited ranges of space and time in which laws are meaningful and evolution of the laws themselves on cosmological scales. The explicit domain of application is Section 3.1, which criticizes immutable laws and proposes limited domains of application and the evolution of laws.

Statement 4

Original text (English):

The state of the art concerning models in which classical spacetime emerges from an underlying dynamical quantum geometry is that those that succeed either have a fixed boundary, or assume a prior temporal or causal order.

The source location is p. 18, Section 3.2 of the same source. The minimal context is the passage that considers whether spacetime is fundamental or emergent and states the ordering between temporal or causal order and the emergence of spacetime.

The statement object is the condition under which models deriving classical spacetime from a dynamical quantum geometry have succeeded. The predicate is Smolin's assessment that the successful models considered in the source either have a fixed boundary or assume a prior temporal or causal order. The explicit domain of application is Section 3.2, which discusses the state of models for the emergence of classical spacetime.

Common Evidence with a Specified Domain of Application Source attribution

All four statements are attributed to Lee Smolin.

Individual Audit Fundamental time and the generation of novel events

In Statement 1, Smolin expressly adopts fundamental time; in Statement 2, he specifies its irreducible aspect as the activity of generating novel events from present events. These statements correspond to Claims 11 and 13 and to Claims 19–20 and 39–40 describing event updates. Updating uses causal order, causal pasts and update stages, which form premises of Transfer 2 and Results 9–10 and 13.

Domains of physical laws and evolution of the laws themselves

In Statement 3, Smolin proposes limited spatiotemporal domains in which laws are meaningful and evolution of the laws themselves on cosmological scales. These correspond to Claims 29–30 and Result 14, using the validity domains and ordering of law changes in Time Content 8. Specific research constructions for evolving laws belong respectively to cosmological natural selection, the principle of precedence and universality of metalaws described in Section 6.

Conditions for emergent-spacetime models

Statement 4 gives Smolin's assessment of the successful emergent-spacetime models considered in the source: they have a fixed boundary or assume a prior temporal or causal order. Fixed boundary and prior order are the two alternatives of the disjunction. The relation between temporal or causal order and macroscopic spacetime corresponds to Claims 16, 41 and 65 and Results 16–17 and 28.

Common Audit with a Specified Domain of Application The temporal premises of each statement and their connections to results are given by the finite representations in Definitions 8.729, 8.733 and 8.734. Smolin’s event generation extends causal histories; Proposition 8.737 gives the generation judgment for the time contents used by that construction.

8.19.64 Source Audit

Audit Inputs

Source and Audit Scope

Source identifier 64.

Source name BIPM 2019/2026 Primitive-Time-Root Audit

Bibliographic identification Bureau International des Poids et Mesures, *The International System of Units (SI)*, 9th edition, version 4.01, June 2026, doi:10.59161/AUEZ1291; Consultative Committee for Time and Frequency, *Mise en pratique for the definition of the second in the SI*, 20 May 2019. $R = \text{Ref}_{\text{BIPM19SI}}$.

Audit-scope identifier 64-SC1.

Audit-scope content Chapters 1–5, tables and Appendices 1 and 4 of the SI Brochure, and Sections 1–3 of the second mise en pratique.

Source-location identifier 64-L1.

Source-location content Sections, tables, resolutions and printed page numbers specified in each table. M denotes the second mise en pratique identified above, with its page numbers given separately.

Formal Objects The audit uses 34 time contents, 52 physical and formal claims, 12 transfers and 32 results.

Assumptions The BIPM defines the SI by fixing the numerical values of seven defining constants and realizes units through physical quantity relations and experiments. Exact defining values and realization uncertainty concern different objects. The second refers to the specified caesium-133 transition frequency and applies as a unit of proper time. Clock comparisons use signal propagation and delays, a synchronization convention and relativistic corrections appropriate to the domain. TAI computation uses a clock ensemble, long-term stability, calibration against standards, reference-surface corrections and frequency steering when necessary. UTC and legal time follow integral-second offsets and national time-offset conventions respectively.

Nontemporal Physical Inputs Candidate values, atomic and event carriers, count records, apparatus configurations, spatial paths and reference-surface shapes in Claims 1–8 form the generation-initial set. Periods, clock readings, synchronization, temporal comparisons and correction values belong to the time contents of the claims that use them.

Time Contents Uses of the same quantity or scale in expressions, inputs and outputs share an index when the object and domain agree. For grouped components, the expansion below and each claim's source location specify the component used.

Definition 8.741 (Time Contents in the BIPM Source). $n_R = 34$; the following table specifies the time contents and their source locations.

Table 593: BIPM 2019/2026 Time Contents

i	Time Content	Source Location
1	Time as a physical quantity, time variables and the time dimension	§§1.1,2.3.1,2.3.3; pp. 121, 126, 132
2	Unperturbed caesium-133 hyperfine transition frequency	§§2.2,2.3.1; App.1,4; pp. 123, 126, 164, 205
3	SI-second unit scale and its one-second duration	§2.3.1; pp. 126
4	Caesium-radiation period and duration corresponding to a number of periods	§2.3.1; pp. 126
5	Proper time in a local domain comoving with the atom	§§2.3.1,2.3.6; M§3.1; pp. 126, 137, M pp. 2
6	Atomic-clock time readings, signals and realized seconds	§2.3.1; App.1; M§§3.1,3.2; pp. 126, 166, M pp. 2–3
7	Transition frequencies and calibrated seconds in secondary representations	§2.3.1; M§3.1.2; pp. 126, M pp. 2
8	Frequency, period and angular frequency of periodic phenomena	§§2.3.4,2.3.5; pp. 133–134, 136–137
9	Stochastic decay activity of a radionuclide and the becquerel	Table 4(d); §2.3.4; pp. 134, 136
10	Temporal components of defining constants, derived quantities and realization equations	§§2.2,2.3; Tables 4–6,8; App.4; pp. 123–124, 127–129, 131, 133–137, 140, 201–202
11	Electromagnetic propagation times, equipment delays and time transfer	§§2.3.1,2.3.6; M§3.2; pp. 127, 137, M pp. 3
12	Relativistic frequency shifts and corrections for comparison positions and reference points	§§2.3.1,2.3.6; M§§3.1–3.3; pp. 126, 137, M pp. 2–4
13	Reading differences and time comparisons between separated clocks	M§§3.2,3.3; M pp. 3–5
14	Frequency comparison, calibration and scale-unit evaluation against standards	§§2.3.1,2.3.6; M§§3.1,3.3; pp. 126, 137, M pp. 2, 4
15	Local spacetime domains and shared spacetime coordinates	§2.3.6; M§3.1; pp. 137, M pp. 2
16	Conventions of simultaneity and synchronization	M§3.2; M pp. 3
17	Minute, hour and day unit scales expressed in seconds	Ch.4, Table 8; pp. 140
18	Historical mean solar day and its second	App.4; pp. 204
19	Fixed-epoch tropical year, ephemeris time and ephemeris second	App.1,4; pp. 158, 160, 204

<i>i</i>	Time Content	Source Location
20	Geocentric Coordinate Time, TCG	App.1, IAU 1991; pp. 167
21	Terrestrial Time, TT, as an ideal time scale	App.1, IAU 1991; M§3.3; pp. 167, M pp. 4
22	International Atomic Time, TAI, combining clock readings	§2.3.1; App.1; M§3.3; pp. 126, 166–167, M pp. 4
23	Constant TAI–TT offset and constant-rate TT–TCG relation	App.1, IAU 1991; pp. 167
24	Local atomic time scale $TA(k)$ within a laboratory	M§3.3; M pp. 4
25	Time scale from a weighted clock ensemble	M§3.3; M pp. 4
26	Intervals of long-term observation, deferred computation and monthly evaluation	M§§3.2,3.3; M pp. 3–5
27	Universal Time and mean solar time based on Earth rotation	App.1, CGPM 1975; M§3.3; pp. 168, M pp. 5
28	Coordinated Universal Time, UTC	App.1, CGPM 1975; M§3.3; pp. 168, M pp. 4–5
29	Local approximation $UTC(k)$ to UTC	M§3.3; M pp. 5
30	Integral-second UTC–TAI difference, leap seconds and local offsets	M§3.3; M pp. 4–5
31	Civil and legal time based on UTC	App.1, CGPM 1975; M§3.3; pp. 168, M pp. 5
32	Steering of TAI by frequency corrections	M§3.3; M pp. 4
33	Clock frequency stability and drift over specified evaluation intervals	M§§3.1.3,3.3; M pp. 2–4
34	GNSS system time, broadcast time signals and predictions of UTC	M§§3.2,3.3; M pp. 3, 5

Table 594: BIPM 2019/2026 Time-Content Components

Time Index	Components and Applications
1	Time quantity, variables and dimension. Claim 52 concerns time variation of the fine-structure constant.
8	Frequency, period, phase change and angular frequency. Claim 51 applies these to the frequency sensitivity of hearing.
10	Temporal components of light speed, action, charge, thermal energy and optical power. Claim 23 specifies the derived quantities of Tables 4–6; Claim 24 specifies base-unit defining equations; Claims 25–28 specify optical, Kibble-balance, blackbody, atom-counting and recoil realization equations; Claim 51 specifies photometric use.
12	Corrections for atomic motion, remote comparison, field nonuniformity within apparatus and a reference equipotential surface, under the comparison positions and accuracy conditions specified in each claim.
17	Minute, hour and day unit scales, with conversion factors 60, 3600 and 86400 respectively.

Time Index	Components and Applications
26	Averaging intervals for clock comparisons, long-term observations, computation delays, monthly evaluations and published values at five-day intervals. Each claim and source location specifies the interval used.
30	The integral-second UTC–TAI difference, leap seconds and local differences between UTC and UTC(k). The integral-second condition applies to UTC–TAI.

Definition 8.742 (Physical and Formal Input Claims in the BIPM Source). Let $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(52)$. With $\text{ClaimBoundaryIdx}_R = ([8], [52])$, Claims 1–8 form the generation-initial set and all 52 claims form the source formal-evidence environment.

Table 595: BIPM 2019/2026 Claims and Temporal Uses

k	Claim	Temporal Use	Source Location
1	Candidate nontemporal quantity values, numbers, unit symbols and formal algebraic data.	$\emptyset$	§§2.1,2.3.3; pp. 123, 132
2	Candidate data for caesium-133 atoms, level labels and external-field configurations.	$\emptyset$	§2.3.1; pp. 126
3	Carriers of identified events, integer counts and recording media.	$\emptyset$	§2.3.1; pp. 126, 130
4	Physical configurations of standards, atomic clocks and oscillators.	$\emptyset$	M§3.1; M pp. 1–2
5	Signal carriers, transmitting and receiving apparatus and spatial propagation paths.	$\emptyset$	§2.3.6; M§3.2; pp. 137, M pp. 3
6	Measurement and comparison apparatus, recording media and uncertainty-record formats.	$\emptyset$	§2.1; M§§3.1,3.2; pp. 123, M pp. 2–3
7	Spatial positions, geocentric geometry and spatial shapes of reference surfaces.	$\emptyset$	§2.3.6; App.1; M§3.3; pp. 137, 166, M pp. 4
8	Configurations of clock-ensemble, computing, communication and control apparatus.	$\emptyset$	M§3.3; M pp. 4–5
9	Express a quantity value as the product of a number and a unit, and associate uncertainty with a measurement result.	$\emptyset$	§2.1; pp. 123
10	Define the SI through the specified quantities and units of the seven defining constants.	$2^Q, 10^Q$	§§2.2,2.2.1; pp. 123–125
11	Numerical values fixed by definition are exact; unit realizations have uncertainty.	$\emptyset$	§2.2.1; M§1; pp. 124, M pp. 1

k	Claim	Temporal Use	Source Location
12	Separate a definition from a particular realization method and realize units through physical equations.	$\emptyset$	§§1.2,2.2.1,2.3.2; App.4; pp. 122, 124, 131, 200
13	Even under definitions permitting future realization methods, the second retains the specified caesium microwave transition as its reference.	$2^{\text{QB}}, 3^{\text{Q}}$	§2.3.2; App.4; pp. 131, 201
14	Treat time as a base quantity and use time variables and dimensions in physical quantity relations.	1^{Q}	§§1.1,2.3.1,2.3.3; pp. 121, 126, 132
15	Fix the numerical value of the caesium frequency at 9 192 631 770 Hz to define the second.	$2^{\text{Q}}, 3^{\text{Q}}$	§2.3.1; pp. 126
16	Take the duration of 9 192 631 770 periods of the specified transition radiation as one second.	$2^{\text{Q}}, 3^{\text{IQ}}, 4^{\text{IQ}}$	§2.3.1; M§2; pp. 126, M pp. 1
17	Specify the atomic species, levels and unperturbed conditions referred to by the second definition.	$\emptyset$	§2.3.1; M§3.1; pp. 126, M pp. 1
18	The second is a unit of proper time and is realized in a local comoving domain.	$3^{\text{Q}}, 5^{\text{IK}}, 15^{\text{B}}$	§2.3.1; M§3.1; pp. 126, M pp. 2
19	Primary standards correct frequency shifts due to atomic motion and environment and realize the proper second at a specified point with uncertainty.	$2^{\text{Q}}, 5^{\text{K}}, 6^{\text{K}}, 12^{\text{QB}}, 15^{\text{B}}$	M§§3.1,3.1.1; M pp. 2
20	Measure and calibrate transition frequencies for secondary representations against primary standards.	$2^{\text{Q}}, 6^{\text{K}}, 7^{\text{QK}}, 8^{\text{Q}}, 14^{\text{Q}}$	§2.3.1; M§3.1.2; pp. 126, M pp. 2
21	Distinguish period, frequency and angular frequency; one period corresponds to a phase change of 2π rad.	8^{IQ}	Table 4(b); §2.3.4; pp. 134, 136
22	Use Hz for periodic phenomena and Bq for stochastic activity of a radionuclide.	$8^{\text{Q}}, 9^{\text{QP}}$	Table 4(d); §2.3.4; pp. 134, 136
23	Dimensions and units of derived quantities follow physical quantity relations and include powers of the time dimension and the second.	$1^{\text{Q}}, 3^{\text{Q}}, 10^{\text{Q}}$	§§2.2,2.3.3,2.3.4; pp. 124, 132–136
24	Express the metre, kilogram, ampere, kelvin and candela through their specified constants and the caesium frequency.	$2^{\text{Q}}, 3^{\text{Q}}, 10^{\text{Q}}$	§2.3.1; pp. 127–129, 131
25	Realize an optical path length using light speed and propagation time, or the relation among light speed, frequency and wavelength.	$8^{\text{Q}}, 10^{\text{Q}}, 11^{\text{IQ}}$	§2.3.1; App.4; pp. 127, 201

k	Claim	Temporal Use	Source Location
26	A Kibble balance relates mechanical and electrical power, with measured frequencies included in its calibration constant.	$8^Q, 10^{QS}$	App.4; pp. 201
27	Realize thermodynamic temperature from total blackbody radiant exitance and the specified constants.	10^Q	App.4; pp. 201
28	Use silicon-sphere atom counting and atomic-recoil relations for mass realization.	10^Q	App.4; pp. 201–202
29	The mole definition fixes a number of specified entities; Celsius temperature t denotes a difference from a reference thermodynamic temperature.	$\emptyset$	§2.3.1; pp. 129–130
30	Remote frequency comparisons use electromagnetic signals and account for relativistic shifts.	$6^K, 11^{IX}, 12^{QX}, 14^{QX}$	§2.3.6; pp. 137
31	Curvature effects may be neglected in a small local spacetime domain; relativistic effects may be neglected between realizations sharing the same spacetime coordinates.	$12^Q, 15^{XB}$	§2.3.6; pp. 137
32	Clock synchronization uses a convention of simultaneity and synchronization.	$6^K, 16^{XB}$	M§3.2; M pp. 3
33	GNSS and two-way satellite clock comparisons use time signals, relativistic corrections and calibration of equipment delays.	$6^K, 11^{IX}, 12^{QX}, 13^{QX}, 16^X, 26^I, 34^{KX}$	M§3.2; M pp. 3
34	Define the minute, hour and day as units of 60, 3600 and 86400 seconds respectively.	$3^Q, 17^{IQ}$	Ch.4, Table 8; pp. 140
35	The historical reference for the second moved from the mean solar day to a fixed-epoch tropical year and then to the caesium transition.	$2^B, 18^{IB}, 19^{IB}$	App.1,4; pp. 158, 160, 162, 164, 204–205
36	In the 1970 definition, TAI is a time-reference coordinate established by the BIH from atomic-clock readings in different establishments.	$3^Q, 6^K, 22^{KX}$	App.1, CCDS 1970; pp. 166
37	The 1980 supplement specifies TAI in a geocentric reference frame, with the SI second realized on the rotating geoid as its scale unit.	$3^Q, 12^B, 22^{QB}$	App.1, CCDS 1980; pp. 166
38	Apart from a constant offset of 32.184 seconds, the ideal form of TAI is TT, which is related to TCG by a constant rate.	$20^Q, 21^Q, 22^Q, 23^{QB}$	App.1, IAU 1991; pp. 167

k	Claim	Temporal Use	Source Location
39	Combine data from independently operated clocks into a local scale $TA(k)$, based on comparisons within the laboratory.	$6^K, 13^X, 24^K, 33^Q$	M§3.3; M pp. 4
40	Calculate a weighted clock ensemble optimized for long-term stability from readings and comparisons over a long duration.	$6^K, 13^X, 25^K, 26^{IO}, 33^Q$	M§3.3; M pp. 4
41	Evaluate the TAI scale unit against primary and secondary standards, apply corrections to a reference equipotential surface, and steer its frequency when necessary.	$3^Q, 6^K, 7^Q, 12^{QB}, 14^Q, 22^{QK}, 25^K, 26^I, 32^Q, 33^Q$	M§3.3; M pp. 4
42	In the 2018 definition, TAI is a continuous scale produced by the BIPM from the best realizations of the SI second, and is a realization of TT.	$3^Q, 21^Q, 22^K$	M§3.3; M pp. 4
43	UTC has the same rate as TAI and is related to Earth-rotation time through an integral-second offset and leap seconds.	$22^Q, 27^B, 28^{QK}, 30^{QB}$	App.1, CGPM 1975; M§3.3; pp. 168, M pp. 4–5
44	Publish monthly the differences between each laboratory's $UTC(k)$ and UTC at five-day intervals.	$13^{QX}, 26^{IO}, 28^K, 29^K, 30^Q$	M§3.3; M pp. 5
45	Disseminate legal time based on UTC and its local realizations under the time-offset conventions of each country.	$28^K, 29^K, 30^Q, 31^K$	App.1, CGPM 1975; M§3.3; pp. 168, M pp. 5
46	GNSS broadcasts time signals and provides predictions of UTC.	$28^K, 34^{KX}$	M§§3.2,3.3; M pp. 3, 5
47	Frequency-standard stability, drift and calibration depend on specified observation intervals.	$6^K, 14^Q, 26^I, 33^{QS}$	M§§3.1.3,3.3; M pp. 2–4
48	Measurement traceability links values and their uncertainties to standards realizing defined units.	$\emptyset$	§§1.3,2.1; pp. 122–123
49	Establish the international adoption of the SI and rules for unit notation.	$\emptyset$	Chs.1,3,5; App.4; pp. 121, 138–139, 143, 202–204
50	Two-way satellite transfer compares clocks between laboratories using quasi-simultaneous signal exchange, relativistic corrections and equipment-delay calibration.	$6^K, 11^{IX}, 12^{QX}, 13^{QX}, 16^X, 26^I$	M§3.2; M pp. 3

k	Claim	Temporal Use	Source Location
51	Acoustics uses weighted quantities approximating the frequency sensitivity of hearing; photometry uses a standard observer's response at the specified frequency.	$8^Q, 10^Q$	§§2.2.1, 2.3.1, 2.3.5; pp. 125, 131, 137
52	The SI treats its defining constants as invariant. For theories allowing time variation of the fine-structure constant, the source excludes effects on foreseeable practical measurements on the basis of experimental upper limits.	$1^S, 10^Q$	§§2.2, 2.2.1; pp. 123–125

Superscripts on temporal indices denote the use roles in each claim.

Table 596: BIPM 2019/2026 Time-Use Roles

Code	Role	Use Count
I	Interval	17
Q	Quantity	71
X	Cross-position correspondence	19
K	Clock reading	29
C	Causality	0
S	State evolution	3
P	Probabilistic evolution	1
O	Observation record	2
B	Reference boundary	15

Table 597: BIPM 2019/2026 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1–2, 7–10, 12–14, 23, 30, 32–33	dependency	$\emptyset$
3, 17–18	member	interval_boundary
4	member	interval_boundary, phase_boundary
5, 15, 19, 27	member	instant, interval_boundary
6, 16, 20–22, 24–25, 28–29, 31, 34	member	instant, order_relation
11	member	endpoint, interval_boundary
26	member	interval_boundary, order_relation

Definition 8.743 (Semantic Components and Generation-Initial Set in the BIPM Source). The 52 semantic components in the claim table use the same 52 claim nodes. The eight

generation-initial components correspond to the nontemporal carriers and data in the following table.

Table 598: BIPM 2019/2026 Generation-Initial Components

k	Nontemporal Component
1	Candidate nontemporal quantity values, numbers, unit symbols and formal algebraic data.
2	Candidate data for caesium-133 atoms, level labels and external-field configurations.
3	Carriers of identified events, integer counts and recording media.
4	Physical configurations of standards, atomic clocks and oscillators.
5	Signal carriers, transmitting and receiving apparatus and spatial propagation paths.
6	Measurement and comparison apparatus, recording media and uncertainty-record formats.
7	Spatial positions, geocentric geometry and spatial shapes of reference surfaces.
8	Configurations of clock-ensemble, computing, communication and control apparatus.

Dependency Inputs

Index identifier S64-XI-DI1.

Existing-evidence reference set 8.744.

Dependency and Transfer Elements

Definition 8.744 (Time-Content Transfers in the BIPM Source). Let $\text{Idx}_R^{\text{audit,transfer}} = [12]$; the table specifies temporal inputs, outputs and claim support.

Table 599: BIPM 2019/2026 Time-Content Transfers

j	Time In i	Time Out i	Claim Support k	Transfer Content and Source Location
1	2	3–4	15–16	Determine the period and one-second scale from the caesium frequency. §2.3.1; pp. 126
2	2–3, 5, 12, 15	6	18–19	Realize the proper second with an atomic clock at a specified point. M§§3.1,3.1.1; M pp. 2
3	2, 6, 8, 14	7	20	Calibrate a secondary representation against primary standards. §2.3.1; M§3.1.2; pp. 126, M pp. 2

j	Time In i	Time Out i	Claim Support k	Transfer Content and Source Location
4	6, 11–12, 16, 26, 34	13	32–33	Determine differences between separated clock readings through GNSS time transfer. M§3.2; M pp. 3
5	6, 11–12	14	30	Compare remote frequencies using electromagnetic signals and relativistic corrections. §2.3.6; pp. 137
6	3	17	34	Convert the second to minute, hour and day unit scales. Ch.4, Table 8; pp. 140
7	6, 13, 33	24	39	Combine clock readings within a laboratory into a local atomic time scale. M§3.3; M pp. 4
8	6, 13, 26, 33	25	40	Combine compared clock readings into a weighted ensemble time scale. M§3.3; M pp. 4
9	3, 6–7, 12, 14, 25–26, 32–33	22	41	Obtain TAI by calibrating and correcting the ensemble scale against standards. M§3.3; M pp. 4
10	22, 27, 30	28	43	Determine UTC from the TAI rate and an integral-second offset. M§3.3; M pp. 4–5
11	28–30	31	45	Determine legal time from UTC and its local realizations. App.1, CGPM 1975; M§3.3; pp. 168, M pp. 5
12	6, 11–12, 16, 26	13	50	Determine differences between separated clock readings by two-way satellite transfer. M§3.2; M pp. 3

Establishment Conditions

Table 600: BIPM 2019/2026 Transfer Conditions

Transfer Index	Conditions and Domain
1	Use the frequency and period count of the specified unperturbed transition.
2	Correct atomic-motion and environmental effects; use general relativity and a reference point when field nonuniformity within the apparatus matters at the required accuracy.
3	Use a selected transition and retain the uncertainty of calibration against primary standards.
4	Use the observation interval, simultaneity convention, orbital and timing signals, relativistic corrections and equipment-delay calibration of the selected GNSS comparison method.
5	Use the signals and positions of the frequency standards and correct frequency shifts as required by the comparison accuracy.

Trans- fer Index	Conditions and Domain
6	Use the unit-conversion factors 60, 3600 and 86400.
7	Use comparisons and stability of independently operated clocks within the laboratory.
8	Use stability estimates and weights based on long-term observations, so the computed scale is obtained with a delay.
9	Use primary and secondary standards, a reference equipotential surface and scale-unit evaluation; reduce the difference by frequency steering when necessary.
10	Use the leap-second convention specified in the document to maintain the relation to Earth-rotation time.
11	Apply the adopted national conventions for time-zone offsets, daylight-saving time and related arrangements.
12	Use quasi-simultaneous two-way signals relayed through a geostationary satellite, with their delay and relativistic corrections.

The twelve transfers have 22 distinct temporal inputs, 12 distinct temporal outputs and 8 time contents common to both sets. There are no self-transfers or directed cycles.

Primary-Source Support The index sets in each row realize the corresponding components of Definition 7.105. GNSS and two-way satellite transfer belong to Transfers 4 and 12 respectively. TAI computation has the ensemble, calibration and correction conditions of Transfers 8–9.

Results

Definition 8.745 (Results in the BIPM Source). Let $\text{Idx}_R^{\text{audit,result}} = [32]$.

Table 601: BIPM 2019/2026 Results

q	Result	Source Location
1	Conceptual distinction among quantities, numbers and units	§2.1; pp. 123
2	Definition of the physical SI unit system by seven constants	§§2.2,2.2.1; pp. 123–125
3	Distinction between exact defining numerical values and realization uncertainty	§2.2.1; M§1; pp. 124, M pp. 1
4	Distinction between a definition and its realization methods	§§1.2,2.2.1,2.3.2; App.4; pp. 122, 124, 131, 200
5	Allowance for new realization methods and the caesium-reference constraint on the second	§2.3.2; App.4; pp. 131, 201
6	Definition of the second by the caesium frequency	§2.3.1; pp. 126
7	Specification of the atomic species, levels and unperturbed conditions	§2.3.1; M§3.1; pp. 126, M pp. 1

q	Result	Source Location
8	One-second duration corresponding to the period count	§2.3.1; M§2; pp. 126, M pp. 1
9	Application and local realization of the second as a proper-time unit	§2.3.1; M§3.1; pp. 126, M pp. 2
10	Realization and calibration of primary and secondary frequency standards	§2.3.1; M§3.1; pp. 126, M pp. 2
11	Frequency stability, drift and calibration over specified evaluation intervals	M§§3.1.3,3.3; M pp. 2–4
12	Remote comparison of clocks and frequencies	§2.3.6; M§3.2; pp. 137, M pp. 3
13	Relativistic conditions for local realization and comparison across positions	§2.3.6; M§3.1; pp. 137, M pp. 2
14	Rules for measurement traceability	§§1.3,2.1; pp. 122–123
15	History of second definitions using the mean solar day, ephemeris time and an atomic transition	App.1,4; pp. 158, 160, 162, 164, 204–205
16	Construction of TAI and its relation to TT and TCG	App.1, CCDS 1970/1980, IAU 1991; M§3.3; pp. 166–167, M pp. 4
17	Relations among UTC, its local realizations and civil time	App.1, CGPM 1975; M§3.3; pp. 168, M pp. 4–5
18	Conversion of minutes, hours and days to seconds	Ch.4, Table 8; pp. 140
19	Formation of local and ensemble scales and evaluation and steering of TAI	M§3.3; M pp. 4
20	International adoption of the SI and unit notation	Chs.1,3,5; App.4; pp. 121, 138–139, 143, 202–204
21	Quantity system including time and its dimension	§§1.1,2.3.1,2.3.3; pp. 121, 126, 132
22	Distinction among frequency, angular frequency and decay activity	§2.3.4; pp. 134, 136
23	Derived units containing the time dimension under physical quantity relations	§§2.2,2.3.3,2.3.4; pp. 124, 132–136
24	Defining equations for other SI base units involving the caesium frequency	§2.3.1; pp. 127–129, 131
25	Length realization by propagation-time and frequency–wavelength methods	§2.3.1; App.4; pp. 127, 201
26	Mass realization with a Kibble balance	App.4; pp. 201
27	Realization of thermodynamic temperature through blackbody radiation	App.4; pp. 201
28	Mass realization using silicon spheres and atomic recoil	App.4; pp. 201–202
29	Entity-count convention for the mole and reference difference for Celsius temperature	§2.3.1; pp. 129–130
30	Dissemination of UTC predictions through GNSS time signals	M§§3.2,3.3; M pp. 3, 5

q	Result	Source Location
31	Frequency-dependent weighted quantities for hearing and vision	§§2.2.1,2.3.1,2.3.5; pp. 125, 131, 137
32	Invariance of defining constants and its use in practical measurement	§§2.2,2.2.1; pp. 123–125

Table 602: BIPM 2019/2026 Result Conditions

Result Index	Conditions and Domain
1	Rules for expressing quantity values.
2	Use the physical quantities and units of the constants specified in the source.
3	A rule distinguishing fixed numerical values from measured values.
4	The relation between a definitional form and experimental methods.
5	Include the explicit exception made for the second.
6	The unperturbed ground-state hyperfine transition and the specified numerical value.
7	The specification of atomic configuration conditions.
8	Use the period of the specified radiation and the count of 9 192 631 770 periods.
9	A comoving domain and reference point, with atomic-motion, environmental and relativistic corrections as required by accuracy.
10	The calibration uncertainty of secondary representations depends on primary standards.
11	The observation intervals and apparatus conditions specified in the document.
12	Timing signals, a synchronization convention, delay calibration and relativistic corrections for the selected transfer method.
13	Treat the local proper second, nonuniformity within apparatus and remote-comparison corrections in their respective domains.
14	General rules relating measurement values and uncertainties to unit realizations.
15	Use the reference and epoch specified for each historical definition.
16	Distinguish realized TAI, ideal TT, constant-rate-related TCG and the specified offset.
17	Use integral-second offsets, local-scale comparisons and national time-offset conventions.
18	Equalities between the specified unit scales.
19	Long-term clock comparisons, stability-based weights, standard calibration, reference-surface corrections and steering when necessary.
20	Institutional and notational rules.
21	The base quantities and dimensions adopted in the SI.
22	Treat periodic phenomena, phase changes and stochastic decay as their respective kinds of quantity.
23	Retain each quantity definition and its powers of the time dimension.

Result Index	Conditions and Domain
24	Use the respective constants for the metre, kilogram, ampere, kelvin and candela.
25	Use the respective conditions of propagation-time and frequency–wavelength methods.
26	Mechanical and electrical power and calibration by measured frequencies.
27	The relation between total radiant exitance and the fourth power of temperature.
28	The specified constants, volume, lattice spacing and mass ratios, or the recoil relation.
29	Definitions concerning amount of substance and temperature respectively.
30	GNSS clocks and signals and prediction differences from UTC.
31	Use standard-observer responses and the respective physiological domains.
32	The experimental upper limits and domain of foreseeable practical measurement stated by the source.

Result Dependence The union of temporal premises and claim support in each row realizes the corresponding result dependence of Definition 7.105.

Table 603: BIPM 2019/2026 Result Dependencies

q	Time Premises i	Claim Support k
1	$\emptyset$	9
2	2, 10	10
3	$\emptyset$	11
4	$\emptyset$	12
5	2–3	13
6	2–3	15
7	$\emptyset$	17
8	2–4	16
9	2–3, 5–6, 12, 15	18–19
10	2, 5–8, 12, 14–15	19–20
11	6, 14, 26, 33	47
12	6, 11–14, 16, 26, 34	30, 32–33
13	2, 5–6, 11–12, 14–15	19, 30–31
14	$\emptyset$	48
15	2, 18–19	35
16	3, 6, 12, 20–23	36–38, 42
17	13, 22, 26–31	43–45
18	3, 17	34
19	3, 6–7, 12–14, 22, 24–26, 32–33	39–41
20	$\emptyset$	49
21	1	14
22	8–9	21–22
23	1, 3, 10	23
24	2–3, 10	24

q	Time Premises i	Claim Support k
25	8, 10–11	25
26	8, 10	26
27	10	27
28	10	28
29	$\emptyset$	29
30	28, 34	46
31	8, 10	51
32	1, 10	52

Grounds of Establishment The results have the unit conventions, quantity relations, realization methods, clock comparisons, relativistic corrections, historical definitions and time-scale operating conditions specified in the source. Each historical reference is associated with its specified object and epoch.

Evidence Hierarchy The BIPM and CCTF describe SI definitions and realization rules and the operation of clocks and time scales. Frequencies, stability, uncertainties and comparison values are objects of the measurements and calibrations cited in the source.

Table 604: BIPM 2019/2026 Finite Audit Counts

Object	Count
Time contents	34
Claims	52
Transfers	12
Results	32
Claim/time-content uses	120
Role assignments	157
Obligation atoms	130
Audit-boundary nodes	118
Audit-boundary edges	44

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 0; remaining primitive-time roots 34.

Result Dependence Results 1, 3–4, 7, 14, 20, 29 are reachable from the source formal-evidence environment without added temporal seeds. The remaining 25 results depend on primitive-time roots. There are 139 root–result dependence pairs, of which 60 are necessary.

Evidential Force The SI-second definition uses the specified caesium frequency; realization of the proper second uses a local domain and correction conditions; clock comparison uses propagation times and synchronization and comparison conditions. Relations among TAI, TT, TCG, UTC and legal time are specified by their respective scales, reference points, rates and offsets.

Result-Preservation Domain Each result has the claim support, temporal premises and establishment conditions specified in the result tables. Unit conventions, clock realizations, relativistic quantity relations and operational time scales have their respective source conditions.

Placement of Grounding Obligations Period counting uses radiation periodicity; clock comparisons use readings, signal delays and synchronization; formation of time scales uses a clock ensemble and scale calibration. Complete generation of the 34 time contents, including these contents, is not obtained from nontemporal Initial Claims 1–8. The inputs to each transfer and result specify the time contents required before their use.

Evidence of Actual Failure Actual failure of a specific physical-time generation condition and consequent changes in measured or calibrated values are undetermined by the source evidence.

Downstream Reach Structural reach is determined by the 139 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns the time contents used by each result and the physical-time generation conditions required before their use. If actual failure of a specific condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.746 (Primitive-Time Roots and Result Dependence in the BIPM Source). In the component order of Definition 8.115,

$$\begin{aligned} \text{AuditPartIdx}_R = (\emptyset, [34]; \\ \emptyset, \{1\} \cup [3, 4] \cup \{7\} \cup \{14\} \cup \{20\} \cup \{29\}; \\ \{2\} \cup [5, 6] \cup [8, 13] \cup [15, 19] \cup [21, 28] \cup [30, 32], \emptyset). \end{aligned}$$

Proof. Closure from Initial Claims 1–8 remains those eight nodes. Even with all 52 claims, each of the twelve transfers requires nonempty temporal inputs. The seven results without temporal premises are reached. Adding all 34 time contents as seeds reaches all 32 results, including the remaining 25. $\square$

Proposition 8.747 (Minimal Primitive-Time-Root Support in the BIPM Source). The minimal-support families of all results contain 123 members in total, with per-result statistics shown below.

Proof. Apply Definition 7.46 to the twelve transfers and 32 result dependencies, starting from singleton temporal seeds and empty supports for source formal evidence. By Proposition 7.47, the stable antichains give the minimal supports of each result. $\square$

$(m; a, b)$ gives the number of supports and their minimum and maximum cardinalities; equal cardinalities are written $(m; a)$.

Table 605: Minimal Root Supports in the BIPM Source

Result Indices	Support Statistics
1, 3–4, 7, 14, 20, 29	1; 0
2, 22, 24, 26, 31–32	1; 2
5–6, 8, 21, 27–28	1; 1
9	1; 4
10	2; 6
11	4; 4, 7
12	2; 6, 8
13	1; 5
15, 25	1; 3
16	28; 7, 13
17	18; 6, 13
18	2; 1
19	15; 9, 10
23	2; 3
30	27; 2, 13

Individual Statements Statement evidence 1

Original text (English edition):

The second, symbol s, is the SI unit of time.

Primary-source location BIPM, *The International System of Units (SI Brochure)*, 9th edition, version 4.01, p. 126, *The second*.

Minimal context The entry entitled “The second” in which the BIPM SI document, the *SI Brochure*, defines the second as an SI base unit.

Attribution The BIPM SI document cited above.

The statement object is the second, symbol s. The predicate is that the second is the SI unit of time. The explicit domain of application is the unit convention for the quantity time in the SI.

Statement evidence 2

Original text (English edition):

unit of proper time

Primary-source location The same document, p. 126, Section 2.3.1.

Minimal context The passage in Section 2.3.1 of the same SI document that applies the defined second to proper time.

Attribution The BIPM SI document cited above.

The statement object is proper time as used in general relativity and the second as its unit. The predicate is that the second is the unit of proper time. The explicit domain of application is the unit representation of the quantity proper time in general relativity.

Statement evidence 3

Original text (English edition):

TAI is a coordinate time scale

Primary-source location The same document, p. 166, the 1980 supplement in Appendix 1. The 1970 definition in that appendix defines TAI as a *time-reference coordinate* established by the Bureau International de l’Heure (BIH) on the basis of atomic-clock readings in various establishments.

Minimal context The account in the 1970 definition and 1980 supplement in Appendix 1 of the same SI document that constructs TAI from clock readings, corrections, and a reference coordinate.

Attribution The CCDS definition and supplement reproduced in the BIPM SI document cited above.

The statement object is TAI. The predicate is that TAI is a “coordinate time scale.” The explicit domain of application is International Atomic Time under the 1970 definition and 1980 supplement, with its geocentric reference frame and scale unit.

Individual Audit Unit convention for the second

Statement 1 gives the convention that the second is the SI unit of time. The immediately following definition fixes the numerical value of the unperturbed ground-state hyperfine transition frequency of caesium 133 at 9 192 631 770 Hz. The relation between this frequency and the duration of its periods corresponds to Claims 15–16, Transfer 1 and Results 6 and 8.

Proper-time application and local realization

Statement 2 applies the second as a unit of proper time in general relativity. The same section specifies relativistic frequency corrections when signals from clocks at different locations are combined. The second *mise en pratique* specifies a local comoving domain, shifts due to atomic motion and environment, and a reference point at accuracies where field nonuniformity within the apparatus matters. These correspond to Claims 18–19 and 30–31, Transfers 2 and 5 and Results 9 and 13.

Ensemble-based TAI and coordinate time scales

The 1980 supplement in Statement 3 gives TAI in a geocentric reference frame the SI second realized on the rotating geoid as its scale unit. The establishing body in the 1970 definition was the BIH; responsibility for TAI moved to the BIPM in 1988. TAI computation in the second *mise en pratique* uses clock comparisons, a weighted ensemble, calibration against standards, reference-surface corrections and steering when necessary. Claims 36–43, Transfers 8–10 and Results 16 and 19 express this operational scale and its relations to ideal TT, TCG and UTC under their respective conditions.

Common Audit with a Specified Domain of Application The temporal premises of each statement and their connections to results are specified in Definitions 8.741, 8.744 and 8.745. Proposition 8.746 gives the generation judgment for time contents used in unit definitions, clock realizations and time-scale formation.

8.19.65 Source Audit

Audit Input

Primary Source and Audit Scope

Source Identifier 65.

Source Name Primitive-Time-Root Audit of ESA 2019

Bibliographic Identification Let European Space Agency, *Clocks, gravity, and the limits of relativity*, 23 May 2019, be denoted by $R = \text{Ref}_{\text{ESA19CGR}}$. The audit uses the body of the official ESA article published on 23 May 2019 and takes as its domain the title, publication date, summary, body paragraphs, two quoted statements by Luigi Cacciapuoti, and the descriptions of integration, testing, and preparation status.

The correspondence between the source-location keys in the tables and primary-source locations is as follows.

- 1 Summary
- 2 Opening paragraph explaining relativity
- 3 Historical paragraph on light deflection and the Pound–Rebka experiment
- 4 First paragraph quoting Luigi Cacciapuoti
- 5 Paragraph on gravitational dilation and clocks at different altitudes
- 6 Paragraph on planned measurements at an altitude of 400 km
- 7 Paragraph on the atomic-clock network and ground–ISS comparison
- 8 Paragraph on comparison stability and two-way timing links
- 9 Paragraph on measuring geopotential differences using ground clocks
- 10 Paragraph on atmospheric propagation, time dissemination, and synchronization
- 11 Second paragraph quoting Luigi Cacciapuoti
- 12 Final paragraph on integration, testing, and launch preparation

Audit-Scope Identifier 65-SC1.

Audit-Scope Content The audit domain specified under Bibliographic Identification.

Primary-Source-Location Identifier 65-L1.

Primary-Source-Location Content The audit domain specified under Bibliographic Identification and the primary-source-location column of each evidence table.

Formal Objects Table 606 gives the time contents used in clock measurements, temporal comparisons, timing signals, frequencies, and synchronization. Table 607 gives apparatus and spatial configurations, qualitative gravitational conditions, numerical factors, and conditions of the propagation media.

Assumptions

Assumption Identifier 65-A1.

Assumption Content The explanation involving clocks at different altitudes uses a comparison with a surface clock. The ACES measurement plan uses the ISS at an altitude of 400 km, a ground atomic-clock network, two-way microwave and optical links, and their comparison stability. Future gravitational-wave observation is described as a candidate application of next-generation clocks and links; a possible implication for new physics is conditional on finding a tiny violation.

Primary-Source-Location Identifier Set 65-L1.

Nontemporal Physical Inputs

Input Identifier 65-P1.

Input Name Nontemporal physical inputs of the primary source.

Input Type Nontemporal physical input.

Input Content The individual nontemporal physical inputs are specified in Definitions 8.750 and 8.749.

Primary-Source-Location Identifier Set 65-L1.

Time Contents

Time-Content Identifier 65-T1.

Time-Content Name Time contents of the primary source.

Time-Content Type Time content.

Time-Content Record The individual time contents are specified in Definition 8.748.

Primary-Source-Location Identifier Set 65-L1.

Formal Input Elements

Definition 8.748 (Time Contents and Auxiliary Objects in the ESA 2019 Source). For source $R = \text{Ref}_{\text{ESA19CGR}}$, define the 24 time contents occurring in physical timing, comparison, and frequency operations and the nine auxiliary objects by Tables 606 and 607. The time contents form eight source-location groups; the order of entries in each row follows the ascending time-content indices.

Table 606: Time Contents and Primary-Source Locations
in the ESA 2019 Source

Location group	Time-content indices	Time contents (in index order)	Auxiliary indices	Source-location key
1	1–3	Error scale of one second; Comparison interval of 300 million years; Temporal quantity measured in the relativity test	None	1

Location group	Time-content indices	Time contents (in index order)	Auxiliary indices	Source-location key
2	4–6	Speed; Comparison of temporal rates under motion conditions; Comparison of temporal rates under gravitational conditions	1	2
3	7–9	Gravitational time-dilation relation; Comparison value for clocks at different altitudes; Relative rate of a higher-altitude clock with respect to a surface clock	1,2	5
4	10	Precision of the planned clock measurements at an altitude of 400 km	3	6
5	11–13	Timekeeping values of ground atomic clocks; Timekeeping values of clocks aboard the ISS; Timekeeping comparison between ground and ISS clocks	3,4	7
6	14–18	Time-comparison stability scale of hundreds of femtoseconds; Scale correspondence between seconds and femtoseconds; Timing information conveyed by the uplink signal; Timing information conveyed by the return signal; Correspondence and order of outgoing and returning timing events	5,6	8
7	19	Ground-clock comparison value used to measure geopotential differences	7	9
8	20–24	Laser-link frequency; Microwave-link frequency; Disseminated time value; Synchronization correspondence among clocks worldwide; Timing precision required by applications	4,8	10

Table 607: Auxiliary Objects in the ESA 2019 Source

a	Auxiliary object	Source- location key
1	Qualitative specification of gravitational conditions	2,5
2	Physical clock identities and altitude configuration	5
3	Configuration of the ground and the ISS at an altitude of 400 km	6,7
4	Ground-clock network and connection configuration	7,10
5	Numerical factor in the scale conversion	8
6	Outgoing and return paths of the microwave and optical links	8
7	Distinct ground-clock locations	9
8	Conditions of the tropospheric and ionospheric media	10
9	Proposed next-generation clocks and link techniques	11

Definition 8.749 (Physical-Input Claims in the ESA 2019 Source). Define the physical-input-claim node set by $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(23)$. Treat these 23 nodes as source-formal-evidence nodes. The nodes, dependency objects, and primary-source locations are given in Table 608.

The correspondence between the dependency-object keys and the time-content-index and auxiliary-object-index sets in Table 608 is as follows.

- 1 $(\emptyset, \{3\})$
- 2 $(\{1, 2\}, \emptyset)$
- 3 $(\llbracket 4, 6 \rrbracket, \{1\})$
- 4 $(\emptyset, \emptyset)$
- 5 $(\{7\}, \{1\})$
- 6 $(\{8\}, \{2\})$
- 7 $(\{9\}, \{1, 2\})$
- 8 $(\{10\}, \{3\})$
- 9 $(\emptyset, \{4\})$
- 10 $(\llbracket 11, 13 \rrbracket, \{3, 4\})$
- 11 $(\{14, 15\}, \{5\})$
- 12 $(\emptyset, \{6\})$
- 13 $(\llbracket 16, 18 \rrbracket, \{6\})$
- 14 $(\{19\}, \{7\})$
- 15 $(\{20, 21\}, \{8\})$

16 ($\{22, 23\}, \{4\}$)

17 ($\{24\}, \{4\}$)

18 ($\emptyset, \{9\}$)

19 ($\{3, 7, 13, 14\}, \{1, 3, 4\}$)

Table 608: Physical-Input Claims in the ESA 2019 Source

p	Physical-input claim	Dependency-object key	Primary-source-location key
1	ACES clock apparatus aboard the ISS	1	1,6
2	Performance statement of one second in 300 million years	2	1
3	Influence of gravity and speed on time	3	2
4	Account of light deflection observed during the eclipse of 29 May 1919	4	3
5	Account of the Pound–Rebka experiment forty years later and the search for theoretical limits a century later	4	3
6	Statement concerning inconsistency between general relativity and quantum mechanics	4	4
7	Research motivation for searches for equivalence-principle violations	4	4
8	Explanation of gravitational time dilation	5	5

p	Physical-input claim	Dependency-object key	Primary-source-location key
9	Comparison of clocks at different altitudes	6	5
10	Relative rate of a higher-altitude clock	7	5
11	Measurement plan at an altitude of 400 km	8	6
12	Worldwide network of atomic clocks	9	7
13	Timekeeping comparison between the ground and the ISS	10	7
14	Comparison stability of hundreds of femtoseconds	11	8
15	Microwave and optical links	12	8
16	Two-way exchange of timing signals	13	8
17	Measurement of geopotential differences using ground-clock comparison	14	9
18	Application of link frequencies to atmospheric propagation	15	10
19	Time dissemination and clock synchronization	16	10
20	Applications requiring precise timing	17	10
21	Candidate future application to gravitational-wave observation	18	11
22	Relativity tests and searches for new physics using ACES	19	1,11

p	Physical-input claim	Dependency-object key	Primary-source-location key
23	Completed clock integration, ongoing link tests, and planned launch readiness by 2020	4	12

Definition 8.750 (Semantic Components and Generation-Initial Set in the ESA 2019 Source). Let the semantic-component node set be $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(27)$. Components 1–4 are the physical objects and configurations in Table 609; its third column defines the corresponding claim-index sets $\text{ClaimIndex}_R^{\text{init}}(a)$. With $\text{ClaimNodeLift}_R(4; 23)$ and $\text{StdSemLift}_R(4; 23)$, set $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(4)$. Components 5–27 are evidence components retaining the 23 source descriptions.

Table 609: Generation-Initial Semantic Components in the ESA 2019 Source

k	Semantic component	Corresponding claim indices
1	Physical clock apparatus, displays, and their configurations	$\{1, 2, 9, 10, 12, 13, 14, 17, 19, 20, 23\}$
2	Spatial configuration of Earth, ground stations, and the ISS	$\{1, 9, 10, 11, 12, 13, 17\}$
3	Signal carriers, transmitting and receiving devices, and propagation paths	$\{13, 15, 16, 18, 19, 23\}$
4	Material media of the troposphere and ionosphere	$\{18\}$

The initial components as physical objects and configurations are related to the time values, speeds, frequencies, and comparison correspondences through the distinctions in Table 606.

Dependency Input

Index Identifier S65-XI-DI1.

Existing-Evidence Reference Set Definition 8.751.

Dependency and Transfer Elements

Definition 8.751 (Time-Content Transfer in the ESA 2019 Source). Define the transfer index set by $\text{Idx}_R^{\text{audit,transfer}} = [2]$. The time-content inputs, time-content outputs, auxiliary-object support, and physical-input-claim support of the transfer event are specified by the equalities in Table 610. The primary-source location of the transfer event is the union of the primary-source locations corresponding to its time-content inputs and outputs.

Table 610: Time-Content Transfer in the ESA 2019 Source

j	Time-content input indices	Time-content output indices	Auxiliary-object support indices	Supporting claim indices
1	{8}	{9}	{2}	{9, 10}
2	{11, 12, 16, 17, 18}	{13}	{3, 4, 6}	{13, 15, 16}

Establishment Conditions The union of transfer inputs is {8, 11, 12, 16, 17, 18}, and the union of outputs is {9, 13}. Their intersection and the set of self-loops are empty. The first transfer uses the comparison value for clocks at different altitudes with a surface clock as the reference. The second uses ground and ISS timekeeping values together with the outgoing and returning timing information and event correspondence. Both transfers have temporal prerequisites.

Primary-Source Support Take the supporting-claim-index, time-content-input-index, time-content-output-index, and auxiliary-object-support-index entries in each row of Table 610 as the corresponding index sets of Definition 7.105. The claim-support set, semantic-component-support set, time-content-input set, time-content-output set, and auxiliary-object-support set of each transfer are then determined by that definition.

The first transfer is supported by Claims 9 and 10 on clocks at different altitudes and their relative rates. The second is supported by Claims 13, 15, and 16 on clock comparison and two-way links. The auxiliary objects specify the clocks being compared, their altitude configuration, and the configurations of the ISS, ground-clock network, and outgoing and return links.

Results

Definition 8.752 (Results in the ESA 2019 Source). Define the result index set by $\text{idx}_R^{\text{audit,result}} = [9]$. The correspondence between the results and primary-source locations is given in Table 611.

The source-location keys refer to the paragraphs specified under Audit Input.

Table 611: Results and Primary-Source Locations in the ESA 2019 Source

j	Result	Primary-source-location key
1	Atomic-clock performance statement	1
2	Explanation of temporal rates under gravitational and motion conditions	2,5
3	Comparison and relative rates of clocks at different altitudes	5
4	Plan for precise comparison of ground and ISS clocks	6,7,8
5	Two-way timing links	8

j	Result	Primary-source-location key
6	Plan to measure geopotential differences using ground-clock comparison	9
7	Application of link frequencies to atmospheric propagation	10
8	Time dissemination, synchronization, and precision-timing applications	10
9	Relativity test using ACES clock comparison	1,2,5,6,7,8,11

Result Dependencies

Definition 8.753 (Result-Dependency Hyperedges in the ESA 2019 Source). For each $j \in [9]$, define the time-content inputs, auxiliary-object inputs, and physical-input-claim support of the result-dependency hyperedge $\text{ResultEdge}_{R,j}$ by the equalities in Table 612, and take its result output to be Result j .

Table 612: Result-Dependency Hyperedges in the ESA 2019 Source

j	Time-content input indices	Auxiliary-object input indices	Supporting claim indices
1	$\{1, 2\}$	$\emptyset$	$\{1, 2\}$
2	$\llbracket 4, 7 \rrbracket$	$\{1\}$	$\{3, 8\}$
3	$\{8, 9\}$	$\{1, 2\}$	$\{9, 10\}$
4	$\{10, 13, 14, 15\}$	$\llbracket 3, 5 \rrbracket$	$\llbracket 11, 14 \rrbracket$
5	$\llbracket 16, 18 \rrbracket$	$\{6\}$	$\{15, 16\}$
6	$\{19\}$	$\{7\}$	$\{17\}$
7	$\{20, 21\}$	$\{8\}$	$\{18\}$
8	$\llbracket 22, 24 \rrbracket$	$\{4\}$	$\{19, 20\}$
9	$\{3, 7, 13, 14\}$	$\{1, 3, 4\}$	$\{3, 8, 11, 13, 14, 22\}$

Take the supporting-claim-index, time-content-antecedent-index, and auxiliary-object-antecedent-index entries in each row of Table 612 as the corresponding index sets of Definition 7.105. The claim-support set, semantic-component-support set, time-content-antecedent set, and auxiliary-object-antecedent set of each result are then determined by that definition.

The nine results are supported by 17 distinct source claims. Their temporal prerequisites have 25 occurrences, and their auxiliary prerequisites have 13 occurrences. Each result requires the time contents shown in the table.

Grounds of Establishment

Grounds of establishment of the results Result 1 rests on the numerical clock-performance description; Results 2 and 3 on the accounts of relativity and clocks at different altitudes; Results 4 and 5 on the measurement plan for the clock network and two-way links; Results 6–8 on applications to geodesy, atmospheric propagation,

time dissemination, and synchronization; and Result 9 on the ACES plan to test relativity.

Evidence Hierarchy

Direct-measurement objects The article itself supplies zero newly measured objects.

Direct-measurement predicates The article itself supplies zero new direct-measurement predicates.

Direct-measurement conditions and domains of application The article itself supplies zero conditions or domains of application for a new direct measurement.

Planned-measurement objects The objects are the ground and ISS clocks and the two-way microwave and optical links.

Planned-measurement predicates The predicates concern clock timekeeping comparison, link-comparison stability, and deviations from relativistic predictions.

Planned-measurement conditions and domains of application The conditions are the ISS at an altitude of 400 km, the ground atomic-clock network, two-way links, and comparison stability of hundreds of femtoseconds.

Objects of formal generalization The applications lead from clock comparisons to geopotential differences, from link frequencies to information about atmospheric propagation, and from the clock network to time dissemination and synchronization.

Conditions and domains of application of formal generalization Each application uses the clock-comparison values, frequencies, time values, synchronization correspondences, and apparatus or medium conditions in Table 612.

Generation Judgment Judgment values: LCTR complete-generation judgment 0 / source-internal-evidence-conditioned local-reach judgment 0 / residual primitive-time roots 24.

Generation reach and primitive-time roots Generation reach from a starting set consisting solely of nontemporal physical inputs generates zero time contents. All 24 time contents in Definition 8.748 remain as primitive-time roots. Every transfer takes existing time contents as inputs.

Source-internal-evidence-conditioned local reach The source-internal-evidence-conditioned local-reach judgment reaches zero time contents, and 24 primitive-time roots remain after that judgment.

Result Dependencies For each $\text{Result}_{R,q} \in \text{Result}_R$, take $\mathcal{A}_R(q)$ from Definition 7.108 and apply Definition 7.50 to it. Aggregation over all results determines the primitive-time-root–result dependence relation and the necessary primitive-time-root–result dependence relation.

Result dependencies All nine results in Definition 8.752 are primitive-time-root-dependent results. There are zero generated results, evidence-only-reached results, and root-added-but-unreached results. The minimal primitive-time-root support of each result is given by Proposition 8.755.

Evidential Force

Domain of evidential force The source describes atomic-clock performance, clocks at different altitudes, ground–ISS comparison, two-way timing links, applications to geodesy, atmospheric propagation, time dissemination and synchronization, and planned relativity tests. Research motivation, the candidate future application to gravitational waves, and apparatus preparation status are included among the source statements in Table 608.

Boundary of evidential force Each measurement or application takes clock values, temporal comparisons, scales, frequencies, or synchronization correspondences as inputs. The article supplies no map or relation generating these time contents from nontemporal physical inputs.

Result-Preservation Domain

Result-preservation domain When an LCTR generation sequence supplies the same comparisons, scales, and correspondences for each time input, and the apparatus configurations and operations in Table 612 are retained, the dependency structures of the nine results are preserved.

Placement of Grounding Obligations

Placement of grounding obligations The 24 time contents audited in this source are accepted as already formed temporal inputs rather than generated from nontemporal physical inputs before use. The placement determination for their physical-time grounding obligations is *backgrounded by primitive-time input*. This determination follows by comparing the use of each time content in a law operation or result derivation with the absence, before that use, of its corresponding generation or formation condition as an input, formation condition, evaluation condition, or failure condition.

Evidence of Actual Failure

Evidence of actual failure The complete determination and residual-root determination concern physical-time admissibility and are distinct from the actual failure of a concrete condition governing a result value or applicability domain. Neither the original-source formal evidence nor the LCTR audit evidence for this source establishes such an actual failure of a target condition. The actual-failure determination is *indeterminate from the original-source evidence*, and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream reach The extent of reach to primitive-time-root-dependent results is determined source by source by the primitive-time-root–result dependence relation and minimal primitive-time-root support reconstructed in this division. This is structural reach. Because no actual failure of a concrete condition has been established, the determination of actual reach from a failure location to a result is *indeterminate from the original-source evidence*. Structural reach does not substitute for actual reach.

Repair Scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation or formation conditions required before their use. The recalculation scope is limited to results reached from a failure location when actual failure has independently been established. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domains are restricted. Post-recalculation values are fixed by the audit result.

Audit Conclusions

Proposition 8.754 (Primitive-Time Roots and Result Dependencies in the ESA 2019 Source). In the component order of Definition 8.115, $\text{AuditPartIdx}_R = \text{AllRootDepIdx}(24, 9)$ holds. Moreover, $R \in \text{Ref}^{\text{audit,rootdep}}$.

Proof. The two transfers require time contents $\{8\}$ and $\{11, 12, 16, 17, 18\}$, respectively. Neither nontemporal physical inputs nor the source-formal-evidence environment alone can initiate a transfer, so all 24 time contents remain primitive time roots.

Each of the nine result edges in Table 612 requires at least one time content. Supplying the source-formal-evidence environment and all 24 roots provides every support and temporal prerequisite and reaches all nine results. Proposition 7.49 and Definition 8.115 give the displayed index tuple. $\square$

Proposition 8.755 (Minimal Primitive-Time-Root Support in the ESA 2019 Source). For each $q \in \text{Idx}_R^{\text{audit,result}}$, the cardinality of $\mathcal{A}_R(q)$ from Definition 7.108, together with the minimum and maximum cardinalities of its elements, is given in Table 613.

Table 613: Minimal Primitive-Time-Root Support in the ESA 2019 Source

Result index	Support statistic	Result index	Support statistic
1	(1; 2)	2	(1; 4)
3	(1; 1)	4	(2; 4, 8)
5	(1; 3)	6	(1; 1)
7	(1; 2)	8	(1; 3)
9	(2; 4, 8)		

Proof. Applying Definition 7.46 to the two transfers and nine result edges leaves the temporal support families unchanged on the second scan. Proposition 7.47 gives the table. There are 11 minimal-support family members in total, 34 primitive-time-root/result dependency pairs, and 22 necessary primitive-time-root/result dependency pairs. $\square$

Individual Statements

Statement object The influence relation between gravity and speed and what the article calls time, together with the explanatory move from the results of comparisons between clocks at different altitudes to the passage of time.

- Original text (English)**
1. Albert Einstein's general theory of relativity predicted that gravity and speed influences time, the faster you travel the more time slows down, but also the more gravity pulling on you the more time slows down.
 2. Clocks at higher altitude show time passes faster with respect to a clock on the Earth surface as there is less gravity from Earth the farther you are from our planet.

Primary-source location In European Space Agency, *Clocks, gravity, and the limits of relativity*, 23 May 2019, the first quotation occurs in the opening paragraph explaining relativity, and the second occurs in the paragraph explaining gravitational time dilation.

Minimal context Before introducing the purpose of ACES, the first quotation presents the article's opening explanation connecting gravitational and kinematic conditions with a slowing of time. The second follows the statement that comparisons between clocks at different altitudes have been experimentally demonstrated and describes the relative rate of a higher-altitude clock as the rate at which time passes.

Attribution Both statements are attributed to the text of the official European Space Agency article.

Predicate and explicit domain of application The first quotation states that gravity and speed affect time and that time passes more slowly at greater speed and under stronger gravity. The second concerns clocks at different altitudes and states that a higher-altitude clock shows that time passes faster relative to a clock at Earth's surface.

Individual Audit

Interpretive possibilities First, the expressions can be read as explanatory shorthand describing gravitational and kinematic differences in clock readings or clock rates as an influence on time or a difference in its rate of passage. Second, they can be read as theoretical expressions describing the conditional dependence of metric and proper-time relations in general relativity as an influence on time or a difference in its rate of passage.

Boundary of evidential force The two quotations explain time dilation under gravitational and motion conditions and the relative rate of a higher-altitude clock with respect to a surface clock. The passage immediately preceding the second quotation describes confirmation of the effect by comparing clocks at different altitudes.

Independent grounds The grounds are the article's comparisons of clocks at different altitudes, its account of relative rates with a surface clock as the reference, and the ACES configuration of ground and ISS clocks and two-way links.

Corresponding formal evidence The quotations correspond to time contents 4–9, Claims 3 and 8–10, Results 2 and 3, and Proposition 8.754. The time contents, claims, and results are given in Definitions 8.748, 8.749, and 8.752, respectively.

Within-Reference Comparison The first quotation describes the influence of gravity and speed on time; the second describes the comparison of clocks at different altitudes in terms of the passage of time.

Common Audit with a Specified Domain of Application Under either the clock-comparison reading or the relativistic-time reading, comparison values, scales, and clock correspondences supply the temporal inputs to the explanation. Their generation dependencies are the subject of Proposition 8.754.

8.19.66 Source Audit

Audit Input

Primary Source and Audit Scope

Source Identifier 66.

Source Name Primitive-Time-Root Audit of NASA/JPL–Caltech 2020

Bibliographic Identification Let NASA Space Place / NASA/JPL–Caltech, *Is Time Travel Possible?*, 30 April 2020, be denoted by $R = \text{Ref}_{\text{NASA20TT}}$. The audit domain comprises the title, short answer, telescope, relativity, airborne-clock, GPS, and summary sections of the article.

Primary-Source-Location Identifier 66-L1.

Primary-Source-Location Content The title, short answer, telescope, contrast with fictional portrayals, relativity, airborne-clock, GPS, and summary sections of the article. The locations of the semantic objects and formal-evidence claim groups are given in Table 614 and Table 616.

Audit-Scope Identifier 66-SC1.

Audit-Scope Content The audit domain specified under Bibliographic Identification.

Formal Objects

Object Identifier 66-F1.

Object Name Formal object.

Object Type Formal object.

Formal Content Definitions, assumptions, and derivations involving temporal dependence in ordinary temporal experience, retrospective observation of distant celestial objects, relativistic clock comparison, airborne-clock comparison, differences and corrections in GPS clock rates, the upper bound imposed by the speed of light, and time-travel vocabulary.

Primary-Source-Location Identifier Set 66-L1.

Object Identifier 66-F2.

Object Name Objects that use time.

Object Type Formal object.

Formal Content Define 20 time contents and eight auxiliary objects used in clock comparison, light propagation, and GPS correction.

Primary-Source-Location Identifier Set 66-L1.

Assumptions

Assumption Identifier 66-A1.

Assumption Content The airborne-clock comparison uses a common initial reading, an eastward flight around the world, and comparison with the ground clock on return. The GPS kinematic and gravitational contributions are rate differences relative to the same ground reference, which is retained when adding them. Positioning uses corrected time.

Primary-Source-Location Identifier Set 66-L1.

Nontemporal Physical Inputs

Input Identifier 66-P1.

Input Name Nontemporal physical input.

Input Type Nontemporal physical input.

Input Content Of the 35 semantic components, five describing clock carriers, spatial projections of paths and the meeting location, and spatial lengths are generation-initial inputs.

Primary-Source-Location Identifier Set 66-L1.

Time Contents

Time-Content Identifier 66-T1.

Time-Content Name Time contents of the primary source.

Time-Content Type Time content.

Time-Content Record The individual time contents are specified in Definition 8.756.

Primary-Source-Location Identifier Set 66-L1.

Formal Input Elements

Formal-Component Identifier 66-C1.

Formal-Component Name Composition of the reference's formal-evidence environment.

Formal-Component Type Formal component.

Formal-Component Text All 35 semantic components are included in the source-formal-evidence environment.

Primary-Source-Support Identifier Set 66-L1.

Definition 8.756 (Time Contents and Auxiliary Objects in the NASA/JPL–Caltech 2020 Source). For source $R = \text{Ref}_{\text{NASA20TT}}$, define 20 time contents and eight auxiliary objects by Tables 614 and 615. Entries within each time-content row follow the ascending index order.

Table 614: Time Contents and Source Locations in the
NASA/JPL-Caltech 2020 Source

Location group	Time content i	Time contents (in index order)	Auxiliary indices	Source- location key
1	1–2	The elapsed year between birthdays; The one-second-per-second rate describing ordinary passage	None	1
2	3–4	Light-travel time from a celestial object to the observer; The temporal order of emission at the celestial object and reception by the observer	1	2
3	5	Comparison of passage at a rate other than one second per second	None	3,7
4	6–9	The temporal component linked to space; The speed of light in vacuum used as a speed limit; The travel speed entering the comparison; The comparison of clock time associated with travel speed	2	4
5	10–13	The common initial reading of the ground and airborne clocks; The ground-clock reading at reunion; The airborne-clock reading at reunion; The airborne-minus-ground clock-reading difference at reunion	3,7	5
6	14–17	The GPS satellite orbital speed of approximately 14000 km per hour; The kinematic contribution to the GPS-minus-ground clock-rate difference; The gravitational contribution to the GPS-minus-ground clock-rate difference; The net GPS-minus-ground clock-rate difference	4,5,6,7	6

Location group	Time content i	Time contents (in index order)	Auxiliary indices	Source-location key
7	18–20	The corrected GPS time used in positioning; The ground reference time used in GPS clock-rate comparison; The one-day interval over which uncorrected positioning errors accumulate	4,8	6

Table 615: Auxiliary Objects in the NASA/JPL–Caltech 2020 Source

a	Auxiliary object	Source-location key
1	The celestial object, telescope, observer, and spatial light path	2
2	Spatial lengths and configurations	4
3	The ground and airborne clock carriers, the spatial projection of the eastward flight path, and the reunion location	5
4	The GPS satellite and ground-clock carriers and their spatial configuration	6
5	The satellite altitude of approximately 20200 km	6
6	The qualitative gravitational conditions at the surface and satellite orbit	6
7	Addition and subtraction of signed numerical values	5,6
8	The spatial position to be determined and a positioning error of a few miles	6

Definition 8.757 (Formal-Evidence Claim Groups in the NASA/JPL–Caltech 2020 Source). Let $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(30)$. The source-formal-evidence environment consists of 35 semantic components corresponding to five physical initial components and 30 source descriptions. Table 616 specifies each claim, its dependencies, and its source location.

The dependency keys correspond to the following pairs of time-content and auxiliary-object index sets.

- 1 $(\emptyset; \emptyset)$
- 2 $(\{9, 17, 19\}; \{3, 4\})$
- 3 $(\{1\}; \emptyset)$
- 4 $(\{2\}; \emptyset)$
- 5 $(\{3\}; \{1\})$
- 6 $(\{4\}; \{1\})$

- 7** ($\{3, 4\}; \{1\}$)
- 8** ($\{5\}; \emptyset$)
- 9** ($\{6\}; \{2\}$)
- 10** ($\{7\}; \{2\}$)
- 11** ($\{8, 9\}; \emptyset$)
- 12** ($\{10\}; \{3\}$)
- 13** ($\emptyset; \{3\}$)
- 14** ($\{11, 12\}; \{3\}$)
- 15** ($\{13\}; \{3\}$)
- 16** ($\{18, 19\}; \{4, 8\}$)
- 17** ($\{14\}; \{4\}$)
- 18** ($\{14, 15\}; \{4\}$)
- 19** ($\emptyset; \{5\}$)
- 20** ($\{16\}; \llbracket 4, 6 \rrbracket$)
- 21** ($\{15, 16\}; \{7\}$)
- 22** ($\{17\}; \{7\}$)
- 23** ($\llbracket 17, 19 \rrbracket; \{4\}$)
- 24** ($\{18, 20\}; \{4, 8\}$)
- 25** ($\{17, 18\}; \emptyset$)
- 26** ($\{5, 9, 13, 17\}; \emptyset$)

The source-location keys correspond to the following locations in the article.

- 1** Short answer and birthday explanation
- 2** Telescopic observation of the celestial past
- 3** Contrast with fictional portrayals
- 4** Explanation of relativity
- 5** Airborne-clock comparison
- 6** GPS
- 7** Summary

Table 616: Formal-Evidence Claim Groups in the
NASA/JPL–Caltech 2020 Source

p	Content	d	ℓ
1	The opening short answer states that humans cannot hop into a time machine and go back in time.	1	1
2	Airborne and satellite clocks have reading rates different from those of ground clocks.	2	1
3	Associate one year between birthdays with ordinary time travel.	3	1
4	Express the ordinary experiential rate as one second per second.	4	1
5	Light from a distant celestial object has a positive propagation time ΔT_{light} .	5	2
6	Associate the received image with an earlier emission state of the celestial object.	6	2
7	Constitute retrospective observation by the emission, propagation, and reception of light.	7	2
8	Give the comparison called time travel at more than one second per second.	8	3
9	Explain the combination of space and time.	9	4
10	Describe the vacuum speed of light as an upper speed limit.	10	4
11	Associate greater speed with less experienced time.	11	4
12	Give two clocks a common initial reading.	12	5
13	Assign the two clocks to a ground path and a flight path.	13	5
14	Compare the two clock readings at the reunion event after travel around the world.	14	5
15	At reunion, the airborne clock reads slightly less than the ground clock.	15	5
16	Associate the negative difference in clock readings with slow time travel.	15	5
17	Take the eastward flight path and reunion comparison as the objects of description.	14	5
18	Use the relation aligning satellite- and ground-reference-clock readings in positioning calculations.	16	6
19	Take the satellite orbital speed to be approximately 14,000 km per hour.	17	6
20	Give $\varepsilon_{\text{kin}}^{\text{GPS}} < 0$.	18	6
21	Take the satellite orbital altitude to be approximately 20,200 km.	19	6
22	Give $\varepsilon_{\text{grav}}^{\text{GPS}} > 0$.	20	6
23	Take the absolute value of the gravitational component to exceed that of the kinematic component.	21	6
24	Give $\varepsilon_{\text{net}}^{\text{GPS}} > 0$.	22	6

p	Content	d	ℓ
25	The article describes the need to correct GPS clock differences mathematically.	23	6
26	GPS clock correction is used in positioning; without it, the article says, positioning errors would accumulate by a few miles each day.	24	6
27	Associate the clock-correction calculation with a time-travel calculation.	25	6
28	Give an explanation in terms of real time travel.	26	7
29	Associate that explanation with different rates of time passage.	26	7
30	Give an official educational explanation by NASA/JPL–Caltech.	1	1

Definition 8.758 (Semantic Components and Generation-Initial Set in the NASA/JPL–Caltech 2020 Source). Let $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(35)$. Components 1–5 are the physical objects and spatial configurations in Table 617; its third column defines $\text{ClaimIndex}_R^{\text{init}}(a)$. With $\text{ClaimNodeLift}_R(5;30)$ and $\text{StdSemLift}_R(5;30)$, set $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(5)$. Components 6–35 are evidence components retaining the 30 source descriptions.

Table 617: Generation-Initial Semantic Components in the NASA/JPL–Caltech 2020 Source

k	Semantic component	Corresponding claim indices
1	The ground-clock carrier and the spatial projection of its path	{13}
2	The airborne-clock carrier and the spatial projection of its eastward flight path	{13, 17}
3	The GPS satellite-clock carrier and the spatial projection of its orbit	{18, 19, 21}
4	The spatial configuration at the meeting location of the ground and airborne clock carriers	{14}
5	Spatial length between the ground reference location and the GPS satellite orbit	{21}

For the objects and spatial configurations supplied by the initial components, the initial and final clock readings, intervals, speeds, clock rates, comparison reference, and corrected time are used as the time contents in Table 614.

Dependent Inputs

Index Identifier S66-XI-DI1.

Existing-Evidence Reference Set Definition 8.759.

Dependency and Transfer Elements

Definition 8.759 (Time-Content Transfers in the NASA/JPL–Caltech 2020 Source). Let $\text{idx}_R^{\text{audit,transfer}} = [3]$. In the input and output columns of Table 618, the first component is a time-content index set and the second an auxiliary-object index set. Formal support is a claim-index set; each transfer expresses a relation among existing time contents.

Table 618: Time-Content Transfers in the NASA/JPL–Caltech 2020 Source

j	Input (time; auxiliary)	Output (time; auxiliary)	Formal support	Transfer content
1	$(\{3\}; \{1\})$	$(\{4\}; \emptyset)$	$\llbracket 5, 7 \rrbracket$	Emission-before-reception order from a positive light-travel time
2	$(\{11, 12\}; \{3, 7\})$	$(\{13\}; \emptyset)$	$\{12, 14, 15\}$	Difference of the two clock readings at reunion
3	$(\{15, 16\}; \{7\})$	$(\{17\}; \emptyset)$	$\{23, 24\}$	Addition of the kinematic and gravitational GPS clock-rate contributions

Establishment Conditions The first transfer obtains emission-before-reception order from a positive light-travel time between identified emission and reception events. The second subtracts the ground-clock reading from the airborne-clock reading at reunion. If C_0 is their common initial reading and C_{air} and C_{ground} are their readings at reunion, the difference of elapsed readings is

$$(C_{\text{air}} - C_0) - (C_{\text{ground}} - C_0) = C_{\text{air}} - C_{\text{ground}}.$$

The third transfer adds rate differences relative to the same ground reference:

$$\varepsilon_{\text{net}}^{\text{GPS}} = \varepsilon_{\text{kin}}^{\text{GPS}} + \varepsilon_{\text{grav}}^{\text{GPS}}.$$

The ground reference time is time content 19. Relative to itself, the ground-clock rate is 1, and the satellite-clock rate is $1 + \varepsilon_{\text{net}}^{\text{GPS}}$. Corrected GPS time 18 is a temporal input to positioning. All three transfers have nonempty time-content inputs.

Primary-Source Support

Index Identifier S66-XI-SS1.

Existing-Evidence Reference Set Definition 8.759.

Results

Definition 8.760 (Results in the NASA/JPL–Caltech 2020 Source). Let $\text{idx}_R^{\text{audit,result}} = [11]$. Table 619 relates each result to its source location.

Table 619: Results and Source Locations in the
NASA/JPL–Caltech 2020 Source

j	Result	Source-location key
1	The ordinary elapsed year between birthdays	1
2	Ordinary passage expressed as one second per second	1
3	Looking back at a distant celestial object by receiving light emitted earlier	2
4	The connection between space and time in relativity	4
5	Local speed comparison with the speed of light in vacuum as its upper bound	4
6	The comparison of clock time associated with travel speed	4
7	The negative airborne-minus-ground clock-reading difference at reunion after the eastward flight	5
8	The GPS satellite speed of approximately 14000 km per hour and the negative kinematic contribution	6
9	The positive gravitational contribution to the GPS clock-rate difference	6
10	The positive net GPS clock-rate difference resulting from the larger gravitational contribution	6
11	GPS time correction and the daily accumulation of uncorrected positioning error	6

Result Dependencies

Definition 8.761 (Result-Dependency Hyperedges in the NASA/JPL–Caltech 2020 Source). For each $j \in [11]$, define the hyperedge to Result j by the time-content prerequisites, auxiliary-object prerequisites, and formal-evidence support in Table 620. The indices uniquely identify the objects in Tables 614, 615, and 616.

Table 620: Result-Dependency Hyperedges in the
NASA/JPL–Caltech 2020 Source

j	Time-content prerequisite a	Auxiliary-object prerequisite indices	Formal-evidence support indices
1	$\{1\}$	$\emptyset$	$\{3\}$
2	$\{2\}$	$\emptyset$	$\{4\}$
3	$\{3, 4\}$	$\{1\}$	$\llbracket 5, 7 \rrbracket$
4	$\{6\}$	$\{2\}$	$\{9\}$
5	$\{7\}$	$\{2\}$	$\{10\}$
6	$\{8, 9\}$	$\emptyset$	$\{11\}$
7	$\{13\}$	$\{3\}$	$\{12, 14, 15, 16, 17\}$
8	$\{14, 15\}$	$\{4\}$	$\{19, 20\}$
9	$\{16\}$	$\llbracket 4, 6 \rrbracket$	$\{21, 22\}$
10	$\{17\}$	$\{7\}$	$\{20, 22, 23, 24\}$
11	$\llbracket 18, 20 \rrbracket$	$\{4, 8\}$	$\{18, 25, 26\}$

The 11 results have 16 time-content prerequisite occurrences, 11 auxiliary-object prerequisite occurrences, and 24 formal-evidence support occurrences. Every result has temporal prerequisites.

Grounds of Establishment

Grounds of establishment of the results Results 1 and 2 rest on descriptions of ordinary elapsed time and passage rate; Result 3 on light propagation; Results 4–6 on the explanation of relativity; Result 7 on airborne-clock comparison; and Results 8–11 on GPS clock rates and positioning correction.

Evidence Hierarchy

Direct-measurement objects The article itself supplies zero newly measured objects.

Direct-measurement predicates The article itself supplies zero new direct-measurement predicates.

Direct-measurement conditions and domains of application The article itself supplies zero conditions or domains of application for a new direct measurement.

Objects of measurements reported from other sources The objects are the airborne and ground clocks and the GPS satellite and ground reference clocks.

Predicates of measurements reported from other sources The predicates concern the reading difference at reunion, the kinematic, gravitational, and net GPS clock-rate contributions, and clock correction for positioning.

Conditions and domains of application of measurements reported from other sources The airborne-clock comparison uses a common initial reading, an eastward flight around the world, and comparison with the ground clock on return. The GPS kinematic and gravitational contributions are rate differences relative to the same ground reference, which is retained when adding them. Positioning uses corrected time.

Objects of formal generalization The explanation relates clock-reading and rate differences to experienced time, different rates of time passage, and real time travel.

Conditions and domains of application of formal generalization The explanation uses the eastward flight and reunion comparison of the airborne clock, together with GPS clock-rate comparison against a ground reference and time correction.

Generation Judgment Judgment values: LCTR complete-generation judgment 0 / source-internal-evidence-conditioned local-reach judgment 0 / residual primitive-time roots 20.

Time contents audit-reachable from nontemporal physical inputs The complete judgment generates zero time contents.

Residual primitive-time roots All 20 time contents remain under both the complete and local-reach judgments.

Source-internal-evidence-conditioned local reach Adding the source-formal-evidence environment activates no transfer without temporal inputs.

Result Dependencies

Dependence from primitive-time roots to results All 11 results are primitive-time-root-dependent. Neither the complete judgment nor the local judgment from the source-formal-evidence environment reaches a result without temporal roots.

Evidential Force

Domain of evidential force The evidential domain comprises light emission, propagation and reception, airborne-clock comparison, GPS clock-rate differences and correction, the speed-of-light limit, and the description of clock differences as time travel.

Boundary of evidential force Each comparison and correction uses clock readings, intervals, speeds, or clock rates. The article supplies no map or relation generating these time contents from nontemporal physical inputs.

Result-Preservation Domain

Conditions and domain of application under which source-specific results are preserved Retain the physical carriers of clocks and light, spatial paths, eastward flight, reunion comparison, satellite motion and altitude, comparison reference, correction operations, measurement procedures, and the validity domains of the approximations.

Results preserved after relocation into the LCTR generation order When an LCTR generation sequence supplies the same comparisons, scales, and correspondences for each temporal input, and the stated apparatus configurations and operations are retained, the dependency structures of the three transfers and 11 results are preserved.

Placement of Grounding Obligations

Placement of grounding obligations The 20 time contents audited in this source are accepted as already formed temporal inputs rather than generated from nontemporal physical inputs before use. The placement determination for their physical-time grounding obligations is *backgrounded by primitive-time input*. This determination follows by comparing the use of each time content in a law operation or result derivation with the absence, before that use, of its corresponding generation or formation condition as an input, formation condition, evaluation condition, or failure condition.

Evidence of Actual Failure

Evidence of actual failure The complete determination and residual-root determination concern physical-time admissibility and are distinct from the actual failure of a concrete condition governing a result value or applicability domain. Neither the original-source formal evidence nor the LCTR audit evidence for this source establishes such an actual failure of a target condition. The actual-failure determination is *indeterminate from the original-source evidence*, and actual failure is not inferred from the backgrounding of a grounding obligation alone.

Downstream Reach

Downstream reach The extent of reach to primitive-time-root-dependent results is determined source by source by the primitive-time-root–result dependence relation and minimal primitive-time-root support reconstructed in this division. This is structural reach. Because no actual failure of a concrete condition has been established, the determination of actual reach from a failure location to a result is *indeterminate from the original-source evidence*. Structural reach does not substitute for actual reach.

Repair Scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation or formation conditions required before their use. The recalculation scope is limited to results reached from a failure location when actual failure has independently been established. Because actual failure is indeterminate in this source, this repair and recalculation specification is conditional. If that condition is satisfied, the target results are regenerated or recalculated from the repaired inputs, or their validity domains are restricted. Post-recalculation values are fixed by the audit result.

Audit Conclusions

Proposition 8.762 (Primitive-Time Roots in the NASA/JPL–Caltech 2020 Source). The first two components in Definition 8.115 are

$$(J_R^{\text{gen}}, J_R^{\text{root}}) = (\emptyset, [20]).$$

Proof. The temporal input sets of the three transfers are $\{3\}$, $\{11, 12\}$, and $\{15, 16\}$, all nonempty. No transfer can start from nontemporal physical inputs; Proposition 7.43 therefore makes all 20 time contents primitive time roots. $\square$

Proposition 8.763 (Local-Reach Judgment in the NASA/JPL–Caltech 2020 Source). The source-formal-evidence environment reaches zero time contents; the residual primitive-time-root index set is $[20]$.

Proof. Adding each transfer support in the source-formal-evidence environment does not satisfy its nonempty temporal input set. $\square$

Proposition 8.764 (Result Dependencies in the NASA/JPL–Caltech 2020 Source).

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(20, 11).$$

Proof. Each of the 11 result edges has temporal prerequisites. The source-formal-evidence environment alone does not satisfy them; adding all 20 roots reaches every result. Proposition 7.49 and Definition 8.115 give the displayed formula. $\square$

Proposition 8.765 (Minimal Primitive-Time-Root Support in the NASA/JPL–Caltech 2020 Source). For $\mathcal{A}_R(q)$ in Definition 7.108, the table gives the number of family members and the minimum and maximum cardinalities of those members.

Table 621: Minimal Primitive-Time-Root Support in the
NASA/JPL–Caltech 2020 Source

Result index	Support statistic	Result index	Support statistic
1	(1; 1)	2	(1; 1)
3	(1; 1)	4	(1; 1)
5	(1; 1)	6	(1; 2)
7	(2; 1, 2)	8	(1; 2)
9	(1; 1)	10	(2; 1, 2)
11	(1; 3)		

Proof. Applying Definition 7.46 to the three transfers and 11 result edges leaves the temporal support antichains unchanged on the second scan. Proposition 7.47 gives the table. There are 13 minimal-support family members, 19 primitive-time-root/result dependency pairs, and 13 necessary primitive-time-root/result dependency pairs. $\square$

Individual Statements Statement 1: Travel Speed and Experienced Time
Original text (English):

the faster you travel, the slower you experience time

Primary-source location: the body paragraph beginning *What does this mean for time travel?* in NASA Space Place, *Is Time Travel Possible?*. Minimal context: the explanation for a general audience that connects ordinary temporal passage, relativistic clock differences, GPS corrections, and time-travel vocabulary.

Statement 2: Upper Limit Imposed by the Speed of Light
Original text (English):

nothing can travel faster than the speed of light

Primary-source location: NASA Space Place, *Is Time Travel Possible?*, in the paragraph explaining relativity. Minimal context: the explanation that states the speed of light as an upper limit on motion and introduces high-speed motion and the rate at which clocks advance.

Common Evidence with a Specified Domain of Application The two statements are attributed to the text of the NASA Space Place article. For each statement, the original text, primary-source location, and the minimal context required for interpretation are given.

Target state- ment	Statement object	Predicate	Explicit domain of application
--------------------------	------------------	-----------	-----------------------------------

S66-Q001	Travel speed and the time experienced by the traveler	States that the greater the travel speed, the more slowly time is experienced.	The domain of description in the NASA Space Place article’s explanation of relativity that connects high-speed motion, clock comparison, and experienced time.
S66-Q002	Moving objects or signals and the speed of light	States that nothing can travel faster than the speed of light.	The domain of description in the NASA Space Place article’s paragraph explaining relativity that introduces high-speed motion and the rate at which clocks advance by presenting the speed of light as an upper limit on motion.

Individual Audit Travel Speed and Experienced Time

Interpretive possibilities The first statement’s “experience time” corresponds to the airborne-clock reading difference at reunion and the GPS clock-rate differences explained subsequently in the article. For a traveler, it is expressed by proper elapsed time along the traveler’s worldline and readings of a comoving clock.

Independent grounds The article connects high-speed motion with the passage of time and gives the example of two clocks starting with the same reading: after an eastward flight, the airborne clock is slightly behind the ground clock.

Boundary of evidential force under each interpretive possibility The airborne-clock explanation is conditioned on eastward flight and comparison at reunion. The GPS comparison includes both kinematic and gravitational contributions; at the source’s stated altitude and speed, the gravitational contribution dominates.

Corresponding formal evidence The correspondence is to time contents 8–13, Claims 11–17, Results 6 and 7, and Proposition 8.762.

Within-source comparison The article’s “real time travel” is explanatory language bringing together the airborne-clock reading difference and GPS rate differences and correction. Each comparison uses clock readings, a comparison reference, and temporal scales.

Upper Limit Imposed by the Speed of Light

Interpretive possibilities The second statement explains the relativistic local upper limit on the propagation of matter, signals, and causal influence: the vacuum speed of light c , as measured in a local inertial frame.

Independent grounds The article states the speed-of-light limit in its explanation of relativity and then discusses travel speed and clock comparison.

Boundary of evidential force under each interpretive possibility The numerical representation of speed uses spatial and temporal differences in a local inertial frame, clocks, a length standard, and a measurement procedure. Emission-before-reception order rests on a positive propagation time between identified emission and reception events.

Corresponding formal evidence The vacuum speed of light corresponds to time content 7, Claim 10, and Result 5. Propagation time and emission–reception order correspond to time contents 3 and 4, Transfer 1, and Result 3.

Within-source comparison The telescope explanation concerns receiving light emitted in the past; the speed-of-light explanation concerns a local speed limit. Their temporal representations use intervals or speed scales in addition to the light carrier and spatial path.

8.19.67 Source Audit

Audit Inputs

Source and Audit Scope

Source identifier 67.

Source name Pössel 2021 Primitive-Time-Root Audit

Bibliographic identification Markus Pössel, *From Light Clocks to Time Dilation, Einstein Online* 13 (2021), 1004. $R = \text{Ref}_{\text{Poe21LC}}$.

Audit-scope identifier 67-SC1.

Audit-scope content The two-mirror light clock, inertial-frame comparison, paths, intervals and reading rates, generalization to all good clocks and physical processes, reciprocity, comparisons involving acceleration, counting and signal constraints, and the page’s glossary explanations of time and speed.

Source-location identifier 67-L1.

Source-location content Article passages or glossary entries identified in each table.

Formal Objects The audit uses 14 time contents, 33 physical and formal claims, 9 transfers and 19 results.

Assumptions Pössel treats two stations drifting freely far from large masses as inertial frames and applies the two postulates of special relativity. The mirror separation is constant in each clock’s rest frame and perpendicular to relative motion. Good clocks on each station have beats corresponding to its light clock. Comparisons use event times in the respective inertial frames.

Nontemporal Physical Inputs The physical carriers of the mirrors, pulse, detector, counter and stations, and the spatial arrangement and path in Claims 1–9 form the generation-initial set. Motion, arrival order, clock readings and signal delays belong to the time contents of the claims that use them.

Time Contents The same time content for the same object and domain retains its index across expression, input and output uses. The tables below specify its components and occurrences.

Definition 8.766 (Time Contents in the Pössel Source). $n_R = 14$; the table specifies the time contents and source locations.

Table 623: Pössel 2021 Time Contents

i	Time Content	Source Location
1	Event times, order and comparison simultaneity in an inertial frame	Introduction and inertial-frame comparison
2	Temporal component of the vacuum speed of light	Light-speed postulate and path calculation
3	Upper-arrival tick sequence and time readings	Light-clock construction
4	Round-trip propagation time and tick interval of the rest light clock	Rest light clock and numerical examples
5	Temporal component of the relative velocity between inertial frames	Frame comparison and moving light path
6	Moving-clock round-trip propagation time in the comparison frame	Moving-light-path comparison
7	Interval ratio and reading-rate ratio of rest and moving light clocks	Light-clock rate comparison
8	Rate correspondence among good clocks comoving with each station	Generalization to all clocks
9	Durations and comparative rates of comoving physical processes	Generalization to all processes
10	Reciprocal clock-rate judgment on exchanging comparison frames	Reciprocity
11	Motion histories with acceleration and clock comparison at reunion	Acceleration and twin comparison
12	Half-period and displayed rate under upper-and-lower arrival counting	Counting convention
13	Signal delay of lower-arrival information and arrival order at the upper mirror	Signal constraint
14	Second and nanosecond unit scales	Numerical examples and glossary second

Table 624: Pössel 2021 Time-Content Components

Time Index	Components and Applications
1	Frame-specific times, order and simultaneity. Local and distant timestamps in the entry time use the corresponding counts and simultaneity.

Time Index	Components and Applications
3	Upper-arrival events, their ordered tick sequence and counter readings. The physical counter carrier alone belongs to Claim 5.
4, 6	Durations of the same round trip in the rest and comparison frames. The indices refer to quantities in different inertial frames.
7–10	Light-clock rate ratio, comoving good-clock correspondence, comoving process rates and reciprocity under frame exchange. Each claim specifies its object and comparison frame.
12–14	Both-arrival counting, information-propagation order and delay, and second and nanosecond scales. The radiation period referred to in the entry second belongs to the explanation of this unit scale.

Definition 8.767 (Physical and Formal Input Claims in the Pössel Source). Let $\text{Node}_R^{\text{audit}, \text{claim}} = \text{ClaimNodes}_R(33)$. With $\text{ClaimBoundaryIdx}_R = ([9], [33])$, Claims 1–9 form the generation-initial set and all 33 claims form the source formal-evidence environment.

Table 625: Pössel 2021 Claims and Temporal Uses

k	Claim	Temporal Use	Source Location
1	Physical carrier of the upper mirror.	$\emptyset$	Introduction, clock construction and signal constraint
2	Physical carrier of the lower mirror.	$\emptyset$	Introduction, clock construction and signal constraint
3	Physical carrier of the light pulse.	$\emptyset$	Introduction, clock construction and signal constraint
4	Physical carrier of the upper detector.	$\emptyset$	Introduction, clock construction and signal constraint
5	Physical carrier of the upper counter.	$\emptyset$	Introduction, clock construction and signal constraint
6	Physical carrier of the first space station.	$\emptyset$	Introduction, clock construction and signal constraint
7	Physical carrier of the second space station.	$\emptyset$	Introduction, clock construction and signal constraint
8	Spatial arrangement of mirrors and stations.	$\emptyset$	Introduction, clock construction and signal constraint

k	Claim	Temporal Use	Source Location
9	Spatial path of the information signal.	$\emptyset$	Introduction, clock construction and signal constraint
10	Apply the two postulates of special relativity to a freely drifting inertial observer.	$1^B, 2^Q$	Introduction
11	State the vacuum light speed as 299,792,458 m/s and use 300,000 km/s in the subsequent calculations.	$2^Q, 14^Q$	Light-speed postulate
12	Light travels back and forth between two mirrors at constant separation, and successive upper arrivals are counted.	$1^B, 2^Q, 3^K$	Light-clock construction
13	Send arrival information from the upper detector to the upper counter and read the ticks.	3^K	Light-clock construction
14	Divide the rest-frame path length between upper arrivals by the light speed to obtain the tick interval.	$2^Q, 3^K, 4^I$	Rest-path calculation
15	Obtain a one-second round trip from the rounded light speed and a 150,000 km mirror separation.	$2^Q, 4^I, 14^Q$	One-second example
16	Obtain a one-nanosecond round trip from the rounded light speed and a 15 cm mirror separation.	$2^Q, 4^I, 14^Q$	One-nanosecond example
17	Treat the second as a time unit; its glossary explanation refers to the period of caesium radiation.	14^{QI}	Glossary second
18	The second space station also drifts freely and moves at constant velocity relative to the first.	$1^B, 5^Q$	Inertial-frame comparison
19	Relate the beats of good clocks at rest on each station to those of its light clock.	$4^I, 8^K$	Comoving-clock correspondence
20	Orient the mirror separation perpendicular to the relative-motion direction and take that transverse distance to be equal in both inertial frames.	$1^B, 5^Q$	Transverse arrangement
21	Obtain the zigzag path and round-trip time in the comparison frame from constant relative velocity and transverse geometry.	$1^B, 2^Q, 4^I, 5^Q, 6^I$	Moving-path calculation
22	Compare path times at the same light speed to obtain the larger interval and lower reading rate of the moving clock.	$4^I, 6^I, 7^Q$	Rate comparison

k	Claim	Temporal Use	Source Location
23	The article pairs an approximately 86.7% speed example with a doubled interval and describes two flashes versus three.	$4^I, 5^Q, 6^I, 7^Q$	Speed and flash example
24	Generalize the light-clock rate relation to all clocks using the relativity principle and good-clock correspondence within each frame.	$7^Q, 8^K$	Generalization to all clocks
25	Apply the same rate relation to all comoving processes, illustrated by egg cooking and musical performance.	$8^K, 9^{IQ}$	Generalization to all processes
26	On exchanging the comparison frame, each inertial observer judges the other clock to run more slowly.	$1^B, 5^Q, 7^Q, 8^K, 10^{KB}$	Reciprocity
27	A reunion comparison involving acceleration requires motion-history analysis beyond the constant-relative-velocity comparison of inertial frames.	$5^Q, 10^B, 11^{BI}$	Acceleration and twin comparison
28	Counting both upper and lower arrivals requires a count each half-period, compared with counting upper arrivals alone.	$3^K, 4^I, 12^{KI}$	Counting convention
29	Lower-arrival information travels at no more than light speed and therefore cannot reach the upper counter before the reflected pulse.	$1^B, 2^Q, 3^K, 13^{CI}$	Signal constraint
30	The article uses the name time dilation for the rate relation of clocks and comoving physical processes.	$\emptyset$	Title and generalization to all processes
31	The glossary entry time explains local timestamps from regular repetition and counting, and distant-event timestamps using simultaneity.	$1^{BX}, 3^K, 4^I$	Glossary time
32	The entries speed and speed of light give the distance-to-time ratio, the common light speed in inertial frames and the signal-speed bound.	$2^Q, 5^Q, 13^C$	Glossary speed and speed of light
33	The entry time dilation describes the same rate change for clocks and comoving processes in inertial frames, and its reciprocity.	$7^Q, 8^K, 9^Q, 10^B$	Glossary time dilation

Superscripts on time indices denote their roles in each claim.

Table 626: Pössel 2021 Time-Use Roles

Code	Role	Use Count
I	Interval	17
Q	Quantity	27
X	Cross-position correspondence	1
K	Clock reading	13
C	Causality	2
S	State evolution	0
P	Probabilistic evolution	0
O	Observation record	0
B	Reference boundary	12

Table 627: Pössel 2021 Structural Attributes of Time Contents

Time Indices	Structural Type	Axes
1, 3	member	instant, order_relation
2, 5, 7–10	dependency	$\emptyset$
4, 6, 14	member	interval_boundary
11	member	endpoint, interval_boundary
12	member	interval_boundary, order_relation
13	member	endpoint, order_relation

Definition 8.768 (Semantic Components and Generation-Initial Set in the Pössel Source). The 33 semantic components in the claim table use the same 33 claim nodes. The nine initial components are the nontemporal carriers, arrangement and path in Claims 1–9.

Formal Input Elements

Definition 8.769 (Light-Clock Comparison in the Pössel Source). Let $d > 0$ be the mirror separation, $c > 0$ the vacuum light speed and v the constant relative velocity, with $|v| < c$. Let $\Delta\tau$ and Δt be the round-trip durations in the rest and comparison frames. The transverse path gives

$$\Delta\tau = \frac{2d}{c}, \quad \left(\frac{c\Delta t}{2}\right)^2 = d^2 + \left(\frac{v\Delta t}{2}\right)^2, \quad \Delta t = \gamma\Delta\tau, \quad \gamma := \frac{1}{\sqrt{1 - v^2/c^2}}.$$

The reading-rate ratio is $1/\gamma$. Under the article’s explicit rounded convention $c = 300,000$ km/s, $d = 150,000$ km gives one second and $d = 15$ cm gives one nanosecond. Interpreting the article’s approximately 86.7% example as $v/c = 0.867$ gives $\gamma \simeq 2.006790$; exact $\gamma = 2$ corresponds to $v/c = \sqrt{3}/2 \simeq 0.8660254$. Two moving-clock flashes delimit one interval, while three rest-clock flashes delimit two intervals.

Dependency Inputs Time contents and claim support are specified in Definitions 8.766 and 8.767.

Dependency and Transfer Elements

Definition 8.770 (Time-Content Transfers in the Pössel Source). Let $\text{Idx}_R^{\text{audit,transfer}} = [9]$; the table specifies temporal inputs, outputs and claim support.

Table 628: Pössel 2021 Time-Content Transfers

j	Time In i	Time Out i	Claim Support k	Transfer Content and Source Location
1	1–2	3	12	Count upper arrivals to obtain a tick sequence. Light-clock construction
2	2–3	4	14	Obtain the rest-clock interval from its round-trip path and light speed. Rest-path calculation
3	1–2, 4–5	6	21	Obtain the comparison-frame round-trip time from the transverse light path. Moving-path calculation
4	4, 6	7	22	Compare the two clock intervals to obtain their reading-rate ratio. Rate comparison
5	4, 7	8	19, 24	Transfer the light-clock rate comparison to all good clocks. Generalization to all clocks
6	8	9	25	Transfer the rate relation to all comoving physical processes. Generalization to all processes
7	1, 5, 7–8	10	26	Exchange comparison frames to obtain the reciprocal rate judgment. Reciprocity
8	3–4	12	28	Obtain the half-period counting rate from the both-arrivals convention. Counting convention
9	1–3	13	29	Obtain information-arrival order at the upper mirror from finite signal speed. Signal constraint

Establishment Conditions

Table 629: Pössel 2021 Transfer Conditions

Transfer Index	Conditions and Domain
1	Use two mirrors at constant separation, a light pulse, arrival order and upper-arrival counting.
2	Use $2d/c$ between successive upper arrivals, with mirror separation d and light speed c .
3	Use constant relative velocity, transverse-distance invariance and the common light speed.

Transfer Index	Conditions and Domain
4	Count one round trip as one tick and take the reading-rate ratio as the reciprocal interval ratio.
5	Use the relativity principle and good-clock correspondence within each frame.
6	Describe the same comoving physical laws and processes in the time of the comparison inertial frame.
7	Use event times and simultaneity belonging to each inertial frame.
8	This is the rate specified by counting upper and lower arrival events; readout at one counter additionally has the signal condition of Transfer 9.
9	For rest mirror separation d , the delay from the lower mirror is at least d/c , no shorter than the reflected pulse's remaining travel time d/c .

The 9 transfers have 8 distinct temporal inputs, 9 distinct outputs and 5 contents common to both. There are no self-transfers or directed cycles.

Primary-Source Support The row index sets specify the corresponding components of Definition 7.105. Rest-path calculation, frame comparison, generalization to clocks and processes, counting conventions and signal arrival have the respective conditions specified in the tables.

Results

Definition 8.771 (Results in the Pössel Source). Let $\text{Id}x_R^{\text{audit,result}} = [19]$.

Table 630: Pössel 2021 Results

q	Result	Source Location
1	Inertial observers, two stations and the two postulates	Introduction and frame comparison
2	Common light speed and its rounded value for calculation	Light-speed postulate
3	Operation and counting of the two-mirror light clock	Light-clock construction
4	Tick readout at the upper detector and counter	Light-clock construction
5	Convention taking successive upper arrivals as ticks	Light-clock construction
6	150,000 km and one-second example	One-second example
7	15 cm and one-nanosecond example	One-nanosecond example
8	Zigzag path and propagation time of the transverse light clock	Moving-path calculation
9	Approximately 86.7% speed example, doubled interval and flash counts	Speed and flash example
10	Reading-rate ratio of moving and rest light clocks	Rate comparison

q	Result	Source Location
11	Generalization of the rate relation to all comoving good clocks	Generalization to all clocks
12	Generalization of the rate relation to all comoving physical processes	Generalization to all processes
13	Reciprocity of clock rates and comparisons involving acceleration	Reciprocity and acceleration
14	Both-arrival counting and the constraint on information at the upper mirror	Counting convention and signal constraint
15	Referent of the name time dilation	Title and generalization to all processes
16	Explanation of timestamp assignment using repetition, counting and simultaneity	Glossary time
17	Explanation of the temporal dependence of speed and the signal-speed bound	Glossary speed and speed of light
18	Clock and process rates and reciprocity in the glossary explanation	Glossary time dilation
19	Explanation of the second unit scale	Glossary second

Table 631: Pössel 2021 Result Conditions

Result Index	Conditions and Domain
1	Use free motion far from large masses and constant relative velocity.
2	Distinguish the exact SI value from the explicitly rounded value used in the article.
3	Use upper arrivals of light travelling between mirrors at constant separation.
4	Count arrival information received at the upper mirror.
5	Identify successive arrivals at the same upper location.
6	Holds under the article's rounded convention $c=300,000$ km/s.
7	Use the same rounding convention and a round-trip path.
8	Use transverse distance and the vacuum light speed in the two inertial frames.
9	Two flashes delimit one interval and three delimit two. Exact doubling occurs at $\sqrt{3}/2$ times the light speed.
10	Compare intervals for the same round trip in the comparison inertial frame.
11	Good clocks correspond to the light clock within each inertial frame.
12	Express the comoving duration of the same process in the time of the comparison inertial frame.
13	Distinguish constant-velocity inertial comparison from the motion history of a reunion comparison with acceleration.
14	Compare information arrival at the single upper counter with the pulse's arrival at the upper mirror.

Result Index	Conditions and Domain
15	Correspondence between the name and the clocks and processes described in the article.
16	Use regular repetition, event order and simultaneity relating distant events.
17	Use travelled distance, time intervals and the vacuum light speed.
18	Corresponds to the special-relativistic inertial-frame comparison in the entry.
19	The glossary explanation refers to the period of caesium radiation.

Result Dependence

Definition 8.772 (Result Dependence in the Pössel Source). The union of temporal premises and claim support in each row specifies the corresponding result dependence of Definition 7.105.

Table 632: Pössel 2021 Result Dependencies

q	Time Premises i	Claim Support k
1	1–2, 5	10, 18
2	2, 14	11
3	1–3	12
4	3	13
5	1–3	12–13
6	2, 4, 14	15
7	2, 4, 14	16
8	1–2, 4–6	20–21
9	4–7	23
10	4, 6–7	22
11	4, 7–8	19, 24
12	8–9	25
13	1, 5, 7–8, 10–11	26–27
14	1–4, 12–13	28–29
15	$\emptyset$	30
16	1, 3–4	31
17	2, 5, 13	32
18	7–10	33
19	14	17

Grounds of Establishment Pössel explains clock and process rate relations through the light-clock thought experiment, the two postulates of special relativity, transverse geometry, comoving-clock correspondence and the signal-speed bound. Numerical examples use the stated light-speed approximation and specified velocity.

Evidence Hierarchy The article’s light clock is a thought device, and the animations explain its motion and counting. The generalization to good clocks and comoving processes is a theoretical consequence for each object satisfying the same inertial-frame conditions.

Table 633: Pössel 2021 Finite Audit Counts

Object	Count
Time contents	14
Claims	33
Transfers	9
Results	19
Claim/time-content uses	65
Role assignments	72
Obligation atoms	75
Audit-boundary nodes	66
Audit-boundary edges	28

Generation Judgment Judgment: complete LCTR generation 0; source-evidence-conditioned local reach 0; residual primitive-time roots 14.

Result Dependence The naming correspondence in Result 15 is obtained from the source formal-evidence environment without added temporal seeds. The remaining 18 results depend on primitive-time roots. There are 80 root–result dependence pairs, of which 26 are necessary.

Evidential Force Pössel’s light-clock comparison uses inertial motion, light-speed invariance, transverse-distance correspondence and arrival counting. The same inter-frame rate relation holds for corresponding good clocks and comoving processes within each frame.

Result-Preservation Domain Each result holds under the claim support, temporal premises and establishment conditions in the result tables. Reunion comparisons involving acceleration use motion histories, and information about both arrivals at a single counter has the signal-arrival constraint.

Placement of Grounding Obligations Light-clock counting uses arrival order; propagation-time calculation uses the temporal component of light speed; rate comparison uses event times and intervals in each inertial frame. Timestamp assignment in the entry time also uses regular repetition, order and simultaneity. The fourteen time contents are not completely generated from nontemporal Initial Claims 1–9.

Evidence of Actual Failure Actual failure of a specific physical-time generation condition and consequent changes in clock or process comparisons are undetermined by the source evidence.

Downstream Reach Structural reach is given by the 80 root–result dependence pairs. Actual reach from failure of a specific generation condition to a result is undetermined by the source evidence.

Repair Scope Repair concerns each result’s time contents and the physical-time generation conditions required before use. If actual failure of a specific condition is established, recalculation concerns the results dependent on that condition.

Audit Conclusions

Proposition 8.773 (Primitive-Time Roots and Result Dependence in the Pössel Source). In the component order of Definition 8.115,

$$\text{AuditPartIdx}_R = (\emptyset, [14]; \emptyset, \{15\}; \llbracket 1, 14 \rrbracket \cup \llbracket 16, 19 \rrbracket, \emptyset).$$

Proof. Closure from Initial Claims 1–9 remains those nine nodes. Even with all 33 claims, all nine transfers require nonempty temporal inputs. Result 15, which has no temporal premise, is reached. Adding the fourteen time contents as seeds reaches all nineteen results, including the remaining eighteen. $\square$

Proposition 8.774 (Minimal Primitive-Time-Root Support in the Pössel Source). The minimal-support families contain 57 members in total, with per-result statistics below.

Proof. Apply the updates of Definition 7.46 to the nine transfers and nineteen result dependencies, starting from singleton temporal seeds and empty supports for source formal evidence. By Proposition 7.47, the stable antichains give the minimal result supports. $\square$

$(m; a, b)$ gives the number of supports and their minimum and maximum cardinalities; equal cardinalities are written $(m; a)$.

Table 634: Minimal Root Supports in the Pössel Source

Result Indices	Support Statistics
1, 8	1; 3
2–3, 5, 14	1; 2
4	2; 1, 2
6–7	3; 3
9	3; 3, 4
10	4; 2, 3
11	7; 2, 3
12	8; 1, 3
13	4; 4, 5
15	1; 0
16	2; 2, 3
17	2; 3
18	11; 3, 4
19	1; 1

Proposition 8.775 (Audit Conclusion for the Pössel Source). The audit has fourteen time contents, 33 claims, nine transfers and nineteen results. The 65 claim/time-content uses have 72 role assignments. Complete generation and local reach are zero, leaving fourteen primitive-time roots.

Proof. This follows from temporal uses in the claim table and the transfer and result-dependence tables. $\square$

Proposition 8.776 (Audit-Boundary Cardinalities of the Pössel Source).

$$\text{BoundaryCount}_R^{\text{audit}} = (66, 28).$$

Proof. The 33 claims, fourteen time contents and nineteen results give 66 nodes; nine transfers and nineteen result dependencies give 28 edges. The claims, time contents, transfers and results give 75 obligation atoms. $\square$

Individual Statements Reading Rate of a Moving Light Clock

Original text (English):

Apparently, from my point of view, the moving light clock moves much slower than my own identical light clock:

The primary-source location is the paragraph immediately following the animation comparing two light clocks in Pössel, *From Light Clocks to Time Dilation*. The article explains that the one interval delimited by two moving-clock flashes corresponds to the two intervals delimited by three rest-clock flashes. The statement is attributed to the text of Markus Pössel's article.

Generalization to All Clocks on the Moving Space Station

Original text (English):

Judged from my space station all clocks of the space station moved relative to me run slower than my own clocks.

The primary-source location is the paragraph following the comparison of the two light clocks in the conclusion of the same article. The minimal context is the explanation that treats light clocks as only one example of the more general special-relativistic time dilation and relates the beats of all clocks at rest on each space station to the beats of its light clock. The statement is attributed to the text of Markus Pössel's article.

Generalization to All Processes on the Moving Space Station

Original text (English):

Just as the moving clocks run slower, all processes on the other space station also run slower for me – five-minute eggs take longer to cook and still have the right consistency in the end, and the pianist on board the other station who plays the Minute Waltz takes considerably more time to do so than is normal performance practice.

The primary-source location is the same concluding paragraph that generalizes to all clocks. The minimal context is the explanation that, after the reading-rate relation for moving clocks, gives cooking a five-minute egg and performing the Minute Waltz as examples of processes on the other space station. The statement is attributed to the text of Markus Pössel's article.

Common Evidence with a Specified Domain of Application

Target state- ment	Statement object	Predicate	Explicit domain of application
S67- Q001	Two identically constructed light clocks in relative motion	States that, from the observer's point of view, the moving light clock runs much slower than the observer's own light clock.	The domain of description immediately following the animation comparing the two light clocks, where two flashes of the moving light clock are compared with three flashes of the rest light clock.
S67- Q002	All clocks on the space station moving relative to the observer	States that they run more slowly than the observer's own clocks.	The concluding passage that treats the light clock as one example of special-relativistic time dilation and relates the beats of all clocks at rest on each space station to those of the light clock.
S67- Q003	All processes on the space station moving relative to the observer	States that, like moving clocks, they run more slowly and that cooking and performance take longer.	The concluding passage that generalizes the moving-clock reading-rate relation to processes on the other space station, illustrated by cooking an egg and performing the Minute Waltz.

Individual Audit Reading Rate of a Moving Light Clock

Interpretive possibilities The first statement concerns the reading rates of two identically constructed light clocks compared using event times in the observer's inertial frame.

Independent grounds Transverse mirror separation, constant relative velocity and light-speed invariance give a longer round-trip interval and a lower reading rate for the moving clock. Two and three flashes delimit one and two intervals respectively.

Boundary of evidential force under each interpretive possibility The comparison concerns upper-arrival events of the two light clocks, using their intervals and reading rates in the inertial frame. The speed–interval relation is given in Definition 8.769.

Corresponding formal evidence Time contents 1–7 in Definition 8.766, Claims 20–23 in Definition 8.767, and Results 8–10 in Definition 8.771.

All Clocks on the Moving Space Station

Interpretive possibilities The second statement relates good clocks at rest on each station to its light clock and compares the reading rates of all clocks on the other station in the observer's inertial frame.

Independent grounds The grounds are the correspondence between each station’s clocks and its light clock, and the rate relation between the two frames’ light clocks.

Boundary of evidential force under each interpretive possibility Every comoving good clock satisfying the article’s correspondence conditions has the same comparative rate, irrespective of its mechanism.

Corresponding formal evidence Time contents 4, 7–8 in Definition 8.766, Claims 19, 22, 24 in Definition 8.767, and Results 10–11 in Definition 8.771.

All Processes on the Moving Space Station

Interpretive possibilities The third statement compares processes on the other station as described in their comoving frame and in the observer’s inertial frame.

Independent grounds Pössel generalizes the rate correspondence with clocks to all processes and gives egg cooking and piano performance as examples.

Boundary of evidential force under each interpretive possibility Expressing the duration of the same comoving process in the comparison frame’s time gives the same interval ratio as the light clock. The local physical laws and apparatus and material conditions are those of that process in its comoving frame.

Corresponding formal evidence Time contents 7–9 in Definition 8.766, Claims 24–25 in Definition 8.767, and Results 11–12 in Definition 8.771.

Common Audit with a Specified Domain of Application The three statements successively concern the rate relations of light clocks, comoving good clocks and comoving physical processes under the same inertial-frame comparison conditions. Light speed, event times, tick intervals and rates supply the temporal premises of the respective derivations.

8.19.68 Source Audit

Audit Input

Primary Source and Audit Scope

Source identifier 68.

Source title Primitive-time-root audit of the PDG 2023 source on resonances

Bibliographic identification Let D. M. Asner, C. Hanhart, and M. Mikhasenko, *50. Resonances* (2023) [69] be source R . The audit domain is p. 1 and Eq. (50.20) and the surrounding paragraphs on pp. 9–10.

Definition 8.777 (Semantic Components of the PDG 2023 Source on Resonances). The claim nodes are $\text{ClaimNodes}_R(15)$. Claims 1–4 are nontemporal initial components; Claim $4 + p$ has the content of source description $\text{FormDesc}_{R,p}$, for $p \in [11]$. The two mathematical-object types specify the types of Claims 9 and 11. The initial-set/evidence-environment pair is $\text{InitEnv}_R^{\text{phys}} = \text{ClaimPrefix}_R(4)$.

Audit-scope identifier 68-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Primary-source-location identifier 68-L1.

Primary-source-location content The audit domain stated in the bibliographic identification and the entries in the primary-source-location columns of the evidence tables.

Formal Objects

Object identifier 68-F1.

Object name Formal objects of the primary source.

Object type Formal object.

Formal content The individual formal objects are given in Definitions 8.777 and 8.778.

Set of primary-source-location identifiers 68-L1.

Assumptions

Assumption identifier 68-A1.

Assumption statement The condition for applying the width–lifetime relation is given in Description 6 of Table 637. The source presents its formulas for distinguishable scalar particles and predominantly uses two-body kinematics.

Set of primary-source-location identifiers 68-L1.

Nontemporal Physical Inputs

Input identifier 68-P1.

Input name Nontemporal physical inputs of the primary source.

Input type Nontemporal physical input.

Input content Claims 1–4 represent nontemporal configurations and numerical records of channel distinctions, couplings and pole residues, complex pole positions, and relative threshold positions. These initial components do not include the full S-matrix dynamics or the temporal interpretation of pole width as inverse lifetime.

Set of primary-source-location identifiers 68-L1.

Time Contents

Time-content identifier 68-T1.

Time-content name Time contents of the primary source.

Time-content type Time content.

Time-content description The individual time contents are given in Definition 8.779.

Set of primary-source-location identifiers 68-L1.

Formal Input Components

Formal-component identifier 68-C1.

Formal-component name Formal-evidence environment of the primary source.

Formal-component type Formal component.

Formal-component text The individual formal input components are given in Definitions 8.777 and 8.778.

Set of primary-source-support identifiers 68-L1.

Table 636: Nontemporal initial components of the PDG 2023 source on resonances

k	Semantic component	Corresponding source descriptions
1	Nontemporal configuration retaining the distinction between resonance and scattering channels	$\llbracket 1, 3 \rrbracket$
2	Coupling to each decay channel and pole residue	$\{2, 11\}$
3	Complex pole position	$\llbracket 3, 5 \rrbracket$
4	Relative energy positions of relevant thresholds and the resonance	$\{6, 7\}$

Definition 8.778 (Source Claims of the PDG 2023 Source on Resonances). The source descriptions $\{\text{FormDesc}_{R,p}\}_{p \in [11]}$ are given in the following table. Description p corresponds to Claim $4 + p$.

Table 637: Source descriptions of the PDG 2023 source on resonances

p	Source description	Primary-source location
1	Scattering structure that identifies resonances with poles of the S-matrix in the complex plane	p. 1
2	Resonance pole position and the pole residues of the couplings to the respective decay channels	p. 9, opening of Sec. 50.2
3	Pole position in the complex plane	p. 9, opening of Sec. 50.2
4	Pole mass and pole width	Eq. (50.20)
5	Pole-parameter equation	Eq. (50.20)
6	Condition that relevant thresholds lie well below the resonance position	Immediately after Eq. (50.20)
7	Resonance-lifetime equation under the same condition	Immediately after Eq. (50.20)
8	Conventional Breit–Wigner parameters	p. 10
9	Parameter differences due to finite-width effects, thresholds, and background terms	p. 10
10	Absence of a direct relation between pole position and total width or lifetime for broad resonances	p. 1
11	Pole residues quantifying the decay properties of broad resonances	p. 1

Table 638: Mathematical-object types of the PDG 2023 source on resonances

m	Mathematical-object type	Primary-source location
1	Complex pole-parameter equation	Eq. (50.20)
2	Threshold-conditioned width–lifetime relation	Immediately after Eq. (50.20)

Definition 8.779 (Time Contents of the PDG 2023 Source on Resonances). The two time contents are given in the following table. In source Eq. (50.20), Γ_R is an energy width. Its inverse-lifetime role in the following relation $\tau_R = 1/\Gamma_R$ is Time Content 1; the duration used as lifetime is Time Content 2. With time and energy dimensions restored, the relation is $\tau_R = \hbar/\Gamma_R$, and the inverse-lifetime quantity is $\Gamma_R/\hbar$. This notation restores dimensions to the source relation. The definition of the bare spectral parameter belongs to Claim 9 and is not itself a time content.

Table 639: Time contents of the PDG 2023 source on resonances

i	Time content	Primary-source location
1	Temporal content of pole width used as inverse lifetime in the width–lifetime relation	p. 1; pp. 9–10, immediately after Eq. (50.20)
2	Duration used as the resonance lifetime	p. 1; pp. 9–10, immediately after Eq. (50.20)

Dependency Inputs

Index identifier S68-XI-DI1.

Set of existing-evidence references Definition 8.780.

Dependency and Transfer Components

Definition 8.780 (Time-Content Transfer in the PDG 2023 Source on Resonances). The single time-content transfer is given in the following table.

Table 640: Time-content transfer in the PDG 2023 source
on resonances

Transfer	Dependency or transfer	Supporting claim indices	Time input	Time output	Primary-source location
1	From inverse lifetime to threshold-conditioned resonance lifetime	{4, 10, 11}	{1}	{2}	pp. 9–10, immediately after Eq. (50.20)

The single transfer inputs at least one existing time content before it outputs a time content. There are no transfers that output a time content solely from nontemporal initial components.

Formation Conditions

Index identifier S68-XI-FC1.

Set of existing-evidence references Definition 8.780.

Primary-Source Support

Index identifier S68-XI-SS1.

Set of existing-evidence references Definition 8.780.

Results

Definition 8.781 (Results and Preservation Conditions of the PDG 2023 Source on Resonances). The four source-specific results, source claims, time presuppositions, primary-source locations, and preservation conditions are given in the following table.

Table 641: Results, dependencies, and preservation conditions of the PDG 2023 source on resonances

Re-sult	Content	Supporting claim indices	Time inputs	Primary-source location	Preservation condition
1	Definition of pole mass and energy width from a complex pole position	{3, 9}	$\emptyset$	Eq. (50.20)	Preserve the complex pole position and its parameter representation.

Re- sult	Content	Supporting claim indices	Time inputs	Primary- source location	Preservation condition
2	Threshold- conditioned resonance lifetime	{4, 10, 11}	{1}	Immedi- ately after Eq. (50.20)	Preserve the condition that relevant thresholds lie well below the resonance position and the domain of pole width that yields a positive finite lifetime.
3	Difference between pole parameters and Breit–Wigner parameters	{9, 12, 13}	$\emptyset$	p. 10	Preserve finite-width effects, thresholds, and background terms.
4	Limit of applicability of a direct width–lifetime relation for broad resonances	{14, 15}	{2}	p. 1	Preserve the domain of broad resonances and the quantification by pole residues.

Result Dependencies

Index identifier S68-XI-RD1.

Set of existing-evidence references Definition 8.781.

Grounding

Index identifier S68-XI-RG1.

Set of existing-evidence references Definition 8.781.

Evidence Hierarchy This source is a theoretical review; the number of objects newly measured directly by the source itself within the audit domain is zero.

Generation Determination **Determined values:** 0 cases satisfying the LCTR complete-generation determination / 0 cases satisfying the source-internal-evidence-conditioned local-reach determination / 2 residual primitive time roots.

Source-internal-evidence-conditioned local reach There are no cases satisfying the source-internal-evidence-conditioned local-reach determination, and two primitive time roots remain after that determination.

Result Dependencies There are three root–result dependencies. The only necessary dependency is from Root 1 to Result 2. Result 4 is reachable from either Root 1 or Root 2 and does not require one particular root. The spectral definition and comparison in Results 1 and 3 do not presuppose either of these temporal roles.

Evidential Force The formal evidence supports the definition of energy width, conditioned lifetime, parameter differences, and the limit of applicability for broad resonances. Defining energy width and evaluating lifetime from an inverse-lifetime quantity are different operations. Successful evaluation in the latter operation does not imply physical generation of its temporal roles. Result 4 uses the lifetime meaning of Time Content 2 to state the limit of applicability of the relation.

Result-Preservation Domain Each result is preserved within its specific conditions in Table 641. For Results 2 and 4, the physical grounding of Time Contents 1 and 2, respectively, is additionally supplied by the LCTR generation sequence.

Placement of the Grounding Obligation

Placement of the grounding obligation The two time contents are not generated from nontemporal physical inputs before use. Transfer 1 gives the lifetime representation of Time Content 2 by presupposing the inverse-lifetime role of Time Content 1. The placement determination for the physical-time grounding obligation is “backgrounded by primitive time input.” This determination follows by comparing the use of each time content in a law operation or result derivation with the absence, before that use, of the corresponding generation condition or formation condition as an input, a formation condition, an evaluation condition, or a failure condition.

Evidence of Actual Failure

Evidence of actual failure The complete-generation and residual-root determinations concern physical-time admissibility and are distinct from the actual failure of a concrete condition governing a result’s value or domain of application. The source’s formal evidence and the LCTR audit evidence do not establish the actual failure of such a target condition. The actual-failure determination is “indeterminate from the primary-source evidence”; actual failure is not inferred from the backgrounding of the grounding obligation alone.

Downstream Reach

Downstream reach The reach to primitive-time-root-dependent results is determined source by source by the primitive-time-root–result dependency relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because the actual failure of a concrete condition has not been established, the actual-reach determination from such a failure location to a result is “indeterminate from the primary-source evidence.” Structural reach does not substitute for actual reach.

Repair Scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation conditions or formation conditions required before their use. The recalculation scope is limited to results to which an independently established actual failure reaches through dependency. Because actual failure is indeterminate in this source, this repair-and-recalculation specification is conditional. If that condition is met, the target results are regenerated or recalculated from the repaired inputs, or their domains of validity are restricted. The values after recalculation are determined as audit results.

Audit Conclusions

Definition 8.782 (Finite Dependency Graph of the PDG 2023 Source on Resonances). The finite dependency graph consists of 15 claims, two time contents, one transfer, and four results, with 21 nodes and five edges.

Proposition 8.783 (Complete Audit of Time Use in the PDG 2023 Source on Resonances). Both the number of generated time contents satisfying the complete determination and the number of reached time contents satisfying the local-reach determination are zero. Both time contents remain as primitive time roots. Results 1 and 3 are reachable under the source-internal evidence conditions. Result 2 requires Time Content 1, and Result 4 requires Time Content 2.

Proof. Every edge requires at least one of Claims 5–15, so no edge applies from initial Claims 1–4 alone. In the evidence environment, the inputs for Results 1 and 3 are available. Transfer 1 and Result 2 require Time Content 1, whereas Result 4 requires Time Content 2. Thus an empty time seed set reaches no time content and reaches only Results 1 and 3. Supplying Time Content 1 produces Time Content 2 through Transfer 1 and also reaches Results 2 and 4. Supplying only Time Content 2 reaches Result 4 but neither Time Content 1 nor Result 2. $\square$

Proposition 8.784 (Minimal Primitive-Time-Root Supports of the PDG 2023 Source on Resonances). The minimal-support index family is $\{\emptyset\}$ for Results 1 and 3, $\{\{1\}\}$ for Result 2, and $\{\{1\}, \{2\}\}$ for Result 4. The total number of family members is five.

Individual Statements Statement Evidence 1

Original text (English):

pole width Γ_R

The primary-source location is Asner, Hanhart, and Mikhasenko, 50. *Resonances*, p. 9, immediately before Eq. (50.20). The minimal context is the passage that represents the complex pole position by the pole mass and pole width. The statement is attributed to D. M. Asner, C. Hanhart, and M. Mikhasenko.

Statement Evidence 2

Original text (English):

the lifetime τ_R of the resonance

The primary-source location is pp. 9–10 of the same source, immediately after Eq. (50.20). The minimal context is the passage that states the lifetime equation for the condition in which relevant thresholds lie well below the resonance position. The statement is attributed to the authors of the same review.

Statement Evidence 3

Original text (English):

differ from the pole parameters

The primary-source location is p. 10 of the same source. The minimal context is the passage that compares conventional Breit–Wigner parameters with pole parameters. The statement is attributed to the authors of the same review.

Statement Evidence 4

Original text (English):

there is no direct relation

The primary-source location is p. 1 of the same source. The minimal context is the passage that limits the direct relation between pole position and total width or lifetime for broad resonances. The statement is attributed to the authors of the same review.

Common Evidence with a Specified Domain of Application

Target state-ment	Statement object	Predicate	Explicit domain of application
S68-Q001	Pole width Γ_R appearing in the complex pole position of a resonance	Treats it as the pole width of the pole parameters representing the complex pole position.	The domain of description immediately before Eq. (50.20) on p. 9, where the complex pole position is represented by the pole mass and pole width.
S68-Q002	Resonance lifetime τ_R	Gives a relation as the resonance lifetime under the condition that relevant thresholds lie well below the resonance position.	The domain of description immediately after Eq. (50.20) on pp. 9–10 for the state in which relevant thresholds lie well below the resonance position.
S68-Q003	Conventional Breit–Wigner parameters and pole parameters	States that the two differ.	The domain of description on p. 10 that compares the two parameters under finite-width effects, thresholds, and background terms.

S68-Q004	Pole position and total width or lifetime of a broad resonance	States that no direct relation exists.	The domain of description on p. 1 that limits the direct relation between pole position and total width or lifetime for broad resonances and states quantification by pole residues.
----------	--	--	--

Individual Audit

Independent grounds The review text and the description adjacent to Eq. (50.20) separately state the complex pole position, pole mass, pole width, threshold-conditioned resonance lifetime, difference from Breit–Wigner parameters, and absence of a direct relation between pole position and total width or lifetime for broad resonances.

Interpretive possibility 1 The source refers to the energy width that, together with pole mass, represents the complex pole position.

Boundary of evidential force 1 The quotation directly supports energy width as a pole parameter.

Interpretive possibility 2 The source gives the inverse relation between pole width and resonance lifetime for states whose relevant thresholds lie well below the resonance position.

Boundary of evidential force 2 The quotation and adjacent equation support this conditioned relation.

Interpretive possibility 3 The source states the difference between conventional Breit–Wigner parameters and pole parameters due to finite-width effects, thresholds, and background terms.

Boundary of evidential force 3 The quotation supports this parameter difference. It does not rule out agreement in a special approximation or limit.

Interpretive possibility 4 For broad resonances, the source states that there is no direct relation between pole position and total width or lifetime and that pole residues are needed to quantify decay properties.

Boundary of evidential force 4 The absence of a direct relation between width and lifetime concerns broad resonances; its domain differs from that of the threshold-conditioned relation in the second statement.

Corresponding formal evidence This audit corresponds to Definition 8.779, Definition 8.778, Results 1–4 of Definition 8.781, and Proposition 8.783.

Within-source collation The relation between pole width and resonance lifetime is conditioned on relevant thresholds lying well below the resonance position. The difference between Breit–Wigner parameters and pole parameters depends on finite-width effects, thresholds, and background terms. The statement that there is no direct relation concerns broad resonances; it does not exclude the conditioned relation within its stated domain.

Common Audit with a Specified Domain of Application The pole-width definition, threshold-conditioned lifetime evaluation, parameter difference, and broad-resonance applicability limit each retain their source context and conditions. The quantitative relation used to evaluate lifetime is distinct from physical generation of its temporal roles.

8.19.69 Source Audit

Audit Input

Primary Source and Audit Scope

Source identifier 69.

Source title Primitive-time-root audit of NIST 2025

Bibliographic identification Let the official article of the National Institute of Standards and Technology, *Putting Einstein to the Test*, published on April 3, 2025 and updated on April 17, 2025, be $R = \text{Ref}_{\text{NIST25E}}$. The audit domain comprises the article information and first two paragraphs; the accounts of relativity and clock tests, GPS clock corrections, the 2010 and 2024 clock comparisons, the ISS project, and the Mount Blue Sky project; and the image-caption and link layers.

Primary-source-location identifier 69-L1.

Primary-source-location content The article information; the accounts of relativity and clock tests, GPS clock corrections, the 2010 and 2024 clock comparisons, the ISS project, and the Mount Blue Sky project; and the image-caption and link layers. The locations of the semantic objects and formal-evidence claim groups are given in Tables 643 and 645.

Audit-scope identifier 69-SC1.

Audit-scope content The audit domain stated in the bibliographic identification.

Formal Objects

Object identifier 69-F1.

Object name Formal object.

Object type Formal object.

Formal content The temporal dependencies that occur in statements concerning clock-reading rates in response to relative motion and gravitational conditions; the motion-induced, gravity-induced, and net GPS contributions and corrections; clock comparisons across height differences; remote clock comparisons; timing-signal transfer; the vocabulary of time dilation; the spacetime metaphor; and the difference in age between twins.

Set of primary-source-location identifiers 69-L1.

Object identifier 69-F2.

Object name Objects employing time.

Object type Formal object.

Formal content Define 24 time contents and 14 auxiliary objects used in clock rates, proper elapsed time, clock comparisons, and timing signals.

Set of primary-source-location identifiers 69-L1.

Assumptions

Assumption Identifier 69-A1.

Assumption Content Clock comparisons use each clock’s motion and gravitational conditions and a common comparison reference. The three GPS contributions express clock differences in microseconds over the same one-day interval. The twin example compares ages at reunion when initial ages are equal and elapsed ages follow the proper times along the respective worldlines. The Jupiter illustration compares ages over 30 years relative to sea level on Earth, treating the respective body as the sole mass in each comparison[73]. The ISS and Mount Blue Sky experiments are plans as of the 2025 article; the latter requires simultaneous clock operation and signal comparison at a relay location.

Primary-Source-Location Identifier Set 69-L1.

Nontemporal Physical Inputs

Input identifier 69-P1.

Input name Nontemporal physical inputs.

Input type Nontemporal physical input.

Input content Of the 41 semantic components, the 14 that describe clock carriers, GPS satellites and ground stations, comparison configurations, optical fibers, lasers, the ISS configuration, and the Mount Blue Sky–Boulder configuration are taken as generation-initial inputs.

Set of primary-source-location identifiers 69-L1.

The generation-initial semantic-component set $\text{Init}_R^{\text{phys}}$ consists of clock carriers, GPS satellites and ground stations, comparison configurations, optical fibers, lasers, the ISS configuration, and the Mount Blue Sky–Boulder configuration. Clock readings, clock rates, the passage of time, time dilation, age differences, timing information, project dates, and source provenance belong to the time contents or to the source formal-evidence environment $\text{Env}_R^{\text{phys}}$.

Time Contents

Time-content identifier 69-T1.

Time-content name Time contents of the primary source.

Time-content type Time content.

Time-content description The individual time contents are given in Definition 8.785.

Set of primary-source-location identifiers 69-L1.

Formal Input Components

Formal-component identifier 69-C1.

Formal-component name Composition of the reference’s formal-evidence environment.

Formal-component type Formal component.

Formal-component text Include all 41 semantic components in the source formal-evidence environment.

Set of primary-source-support identifiers 69-L1.

Definition 8.785 (Time Contents and Auxiliary Objects in the NIST 2025 Source). For source $R = \text{Ref}_{\text{NIST25E}}$, define 24 time contents and 14 auxiliary objects by Tables 643 and 644. Entries within each time-content row follow the ascending index order.

Table 643: Time Contents and Source Locations in the NIST 2025 Source

Location group	Time content i	Time contents (in index order)	Auxiliary indices	Source-location key
1	1–3	The speed of light in vacuum used in comparing the spacecraft speed; The relative velocity of the compared clock or spacecraft; The clock rate associated with relative motion	1	1,3
2	4–5	The clock rate associated with gravitational conditions; The temporal component of spacetime	2	2,3
3	6–8	The proper elapsed time of the Earthbound twin; The proper elapsed time of the traveling twin; The twins' age difference at reunion	3,6	3
4	9–10	The comparison between a moving clock and its reference clock; The comparison between clocks under different gravitational conditions	1,2,4	4,5
5	11–13	The GPS kinematic contribution of approximately minus 7 microseconds per day; The GPS gravitational contribution of approximately plus 45 microseconds per day; The net GPS contribution of approximately plus 38 microseconds per day	5,6	6
6	14	The corrected GPS time used in positioning	5	7
7	15	The 2010 comparison of aluminum-ion clocks separated in height by about one foot	7	8

Location group	Time content i	Time contents (in index order)	Auxiliary indices	Source-location key
8	16	The submillimeter-scale gravitational effect described using the 2024 optical lattice clock	8	9
9	17	The rate difference between ISS and ground clocks	9,12	11
10	18	The clock-rate comparison between Mount Blue Sky and Boulder	10	12,13,15
11	19–21	The timing signal sent from the ground laboratory through optical fiber; The timing signal sent from the mountaintop clock by laser; Simultaneous operation of the remote clocks and correspondence of comparison events	10,11	13,14
12	22–23	The 30-year age-comparison interval in the Jupiter illustration; The Jupiter illustration's age difference of minus 18.4 seconds relative to sea level on Earth	13	16
13	24	The 10.23 MHz output frequency marked on the pictured rubidium clock for GPS	14	17

Table 644: Auxiliary Objects in the NIST 2025 Source

a	Auxiliary object	Source-location key
1	The compared clock carriers and spatial projections of their paths	1,5
2	The qualitative gravitational conditions and spatial configuration of the comparison locations	2,3,5
3	The twins as physical carriers, the spacecraft, and spatial projections of the paths	3
4	Portable atomic clocks and the apparatus of existing clock tests	4,5
5	The GPS satellite and ground-clock carriers and their spatial configuration	6,7
6	Addition and subtraction of signed numerical values	3,6
7	The aluminum-ion clock pair and its vertical separation of about one foot	8

a	Auxiliary object	Source- location key
8	The atomic ensemble of the optical lattice clock and the submillimeter spatial scale	9
9	The ISS at an altitude of 370–460 km and the two-clock apparatus configuration	11
10	The clock configuration on Mount Blue Sky at an elevation of 4348 m and in Boulder, their height difference of nearly 3 km, and their separation of 56 km	12,13,15
11	The spatial paths of the optical fiber, relay location, and mountaintop laser	14
12	The anticipated precision factor of approximately 50 relative to earlier tests	11
13	The gravitational configurations at sea level on Earth and at Jupiter, treating the respective body as the sole mass in each comparison	16
14	The physical carrier and output connector of the rubidium clock for GPS shown in the photograph	17

Definition 8.786 (Formal-Evidence Claim Groups in the NIST 2025 Source). Let $\text{Node}_R^{\text{audit,claim}} = \text{ClaimNodes}_R(27)$. The source-formal-evidence environment consists of 41 semantic components corresponding to 14 physical initial components and 27 source descriptions. Table 645 specifies each claim, its dependencies, and its source location.

The dependency keys correspond to the following pairs of time-content and auxiliary-object index sets.

- 1 $(\emptyset; \emptyset)$
- 2 $(\{2, 3\}; \{1\})$
- 3 $(\{4\}; \{2\})$
- 4 $(\{5\}; \{2\})$
- 5 $(\{4, 5\}; \{2\})$
- 6 $(\{1, 2, 6, 7, 8\}; \{2, 3\})$
- 7 $(\{9, 10\}; \{4\})$
- 8 $(\{9\}; \{1, 4\})$
- 9 $(\{10\}; \{2, 4\})$
- 10 $(\llbracket 11, 13 \rrbracket; \{5\})$
- 11 $(\{11\}; \{5\})$
- 12 $(\{12\}; \{5\})$
- 13 $(\{13\}; \{5, 6\})$

- 14** ($\{13, 14\}; \{5\}$)
- 15** ($\{15\}; \{7\}$)
- 16** ($\{16\}; \{8\}$)
- 17** ($\{9, 10, 15, 16\}; \{4, 7, 8\}$)
- 18** ($\{17\}; \{9\}$)
- 19** ($\{17\}; \{9, 12\}$)
- 20** ($\{18, 21\}; \{10\}$)
- 21** ($\llbracket 19, 21 \rrbracket; \{10, 11\}$)
- 22** ($\{17\}; \{2, 9\}$)
- 23** ($\{22, 23\}; \{13\}$)
- 24** ($\{24\}; \{14\}$)

The source-location keys correspond to the following locations in the article.

- 1** Paragraph 1: special relativity, motion, and time
- 2** Paragraph 2: general relativity, gravity, and time
- 3** Paragraph 3: spacetime, gravity, and the twin example
- 4** Paragraph 4: tests using portable atomic clocks
- 5** Paragraph 5: clocks on airplanes, satellites, and skyscrapers
- 6** Paragraph 6: the three GPS contributions
- 7** Paragraph 7: GPS correction
- 8** Paragraph 8: the 2010 clock comparison
- 9** Paragraph 9: the 2024 optical lattice clock
- 10** Paragraph 10: assessment of earlier tests and introduction of two plans
- 11** Paragraph 11: the ISS plan
- 12** Paragraph 12: clocks on Mount Blue Sky and in Boulder
- 13** Paragraph 13: the planned simultaneous operation in 2025
- 14** Paragraph 14: the relay location, optical fiber, and laser
- 15** Paragraph 15: the success condition and height difference
- 16** The embedded Jupiter illustration and the reference point and footnote in the linked age-comparison infographic

17 The photograph and caption of the rubidium clock for GPS, including the frequency and output-connector markings

Table 645: Formal-Evidence Claim Groups in the NIST 2025 Source

p	Content	d	ℓ
1	The article attributes special relativity in 1905 to Einstein and mentions $E = mc^2$ in that account.	1	1
2	The NIST article states that a clock in greater relative motion exhibits a lower reading rate.	2	1
3	The NIST article attributes the 1915 general theory of relativity to Einstein.	1	2
4	The NIST article states that gravitational conditions act on clock-reading rates.	3	2
5	The NIST article uses an account of space and time as an integrated fabric.	4	3
6	The NIST article states a close coupling between gravity and time.	5	3
7	The twin example describes an age difference at reunion along two worldlines, using either a near-light-speed spacecraft journey or a journey around a black hole.	6	3
8	The article states the history of testing relativity with portable atomic clocks.	7	4
9	The article states a comparison of reading rates between a moving clock and a reference clock.	8	5
10	The article states a comparison of reading rates between clocks under different gravitational conditions.	9	5
11	The article states a comparison system comprising GPS satellite atomic clocks and ground reference clocks.	10	6
12	The article gives the motion-induced GPS contribution as approximately minus seven microseconds per day.	11	6
13	The article gives the gravity-induced GPS contribution as approximately plus 45 microseconds per day.	12	6
14	The article gives the net GPS contribution as approximately plus 38 microseconds per day.	13	6
15	The article states that GPS uses relativistic clock corrections in the time information used for positioning.	14	7
16	The article summarizes the 2010 clock comparison over a height difference of approximately one foot.	15	8

p	Content	d	ℓ
17	The article describes the submillimeter-scale gravitational effect using an optical lattice clock in 2024.	16	9
18	The NIST article evaluates the empirical reach of tests of relativity.	17	10
19	The article describes a plan to install two high-performance atomic clocks on the ISS at an altitude of 370–460 km.	18	11
20	The article predicts approximately 50 times the precision of existing tests for the ISS experiment.	19	11
21	The article states a planned comparison between clocks on Mount Blue Sky and in Boulder.	20	12,13,15
22	The article states a planned remote clock comparison using optical-fiber and laser signals.	21	14
23	The article states that the ISS and Mount Blue Sky experiments were planned as of the 2025 article.	1	10,13
24	The NIST article was published on 3 April 2025 and updated on 17 April 2025.	1	1
25	The article explains that ISS clocks will run faster than ground clocks by invoking weaker gravity in space and an analogy with GPS.	22	11
26	The Jupiter illustration gives an age 18.4 seconds younger after a 30-year comparison relative to sea level on Earth, treating the respective body as the sole mass in each comparison.	23	16
27	The photograph of the rubidium clock for GPS shows 10.23 MHz in the frequency field and beside output connector J1.	24	17

Definition 8.787 (Semantic Components and Generation-Initial Set in the NIST 2025 Source). Let $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(41)$. Components 1–14 are the physical objects and spatial configurations in Table 646; its third column defines $\text{ClaimIndex}_R^{\text{init}}(a)$. With $\text{ClaimNodeLift}_R(14;27)$ and $\text{StdSemLift}_R(14;27)$, set $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(14)$. Components 15–41 are evidence components retaining the 27 source descriptions.

Table 646: Generation-Initial Semantic Components in the NIST 2025 Source

k	Semantic component	Corresponding claim indices
1	Physical carrier of a portable atomic clock	{8}
2	Physical carrier of a moving clock	{9}
3	Physical carrier of the reference clock compared with the moving clock	{9}

k	Semantic component	Corresponding claim indices
4	Physical carrier of the first clock under a distinct gravitational condition	$\{10\}$
5	Physical carrier of the second clock under a distinct gravitational condition	$\{10\}$
6	Physical carrier of a GPS satellite atomic clock	$\llbracket 11, 15 \rrbracket \cup \{27\}$
7	Physical carrier of a GPS ground reference clock	$\llbracket 11, 15 \rrbracket$
8	Spatial configuration and signal-carrier path between a GPS satellite and ground station	$\llbracket 11, 15 \rrbracket$
9	The 2010 clock pair and the vertical spatial configuration between the two clocks	$\{16\}$
10	The atomic ensemble and spatial configuration of the 2024 optical lattice clock	$\{17\}$
11	Physical carriers and spatial configuration of the planned two-clock ISS experiment	$\{19, 20, 23\}$
12	Physical carriers and spatial configuration of the Mount Blue Sky–Boulder clock pair	$\{21, 23\}$
13	Signal-carrier path through optical fiber for remote clock comparison	$\{22\}$
14	Laser signal carrier for remote clock comparison	$\{22\}$

Dependency Inputs

Index identifier S69-XI-DI1.

Set of existing-evidence references Definition 8.788.

Dependency and Transfer Components

Definition 8.788 (Time-Content Transfers in the NIST 2025 Source). Let $\text{Idx}_R^{\text{audit,transfer}} = [2]$. In the input and output columns of Table 647, the first component is a time-content index set and the second an auxiliary-object index set. Formal support is a claim-index set; each transfer expresses a relation among existing time contents.

Table 647: Time-Content Transfers in the NIST 2025 Source

j	Input (time; auxiliary)	Output (time; auxiliary)	Formal support	Transfer content
1	$(\{6, 7\}; \{3, 6\})$	$(\{8\}; \emptyset)$	$\{7\}$	Age difference at reunion from the proper elapsed times along the two paths

j	Input (time; auxiliary)	Output (time; auxiliary)	Formal support	Transfer content
2	({11, 12}; {6})	({13}; $\emptyset$)	$\llbracket 12, 14 \rrbracket$	Addition of the GPS kinematic and gravitational contributions over the same daily reference interval

Establishment Conditions In the first transfer, when initial ages are equal and elapsed ages follow the proper times along the respective worldlines, the Earthbound and traveling proper elapsed times τ_E, τ_T give the age difference at reunion:

$$\Delta a = \tau_E - \tau_T.$$

If $\tau_E > \tau_T$, the Earthbound twin is older.

The second transfer adds GPS clock differences expressed over the same ground-reference day. With the source’s rounded displayed values,

$$(-7 + 45) \mu\text{s}/\text{day} = 38 \mu\text{s}/\text{day}.$$

The day and microsecond are included in each contribution as its comparison interval and unit. Corrected GPS time 14 is a temporal input to positioning. The temporal input sets of the two transfers are {6, 7} and {11, 12}, both nonempty.

Primary-Source Support

Index identifier S69-XI-SS1.

Set of existing-evidence references Definition 8.788.

Results

Definition 8.789 (Results in the NIST 2025 Source). Let $\text{idx}_R^{\text{audit,result}} = [17]$. Table 648 relates each result to its source location.

Table 648: Results and Source Locations in the NIST 2025 Source

j	Result	Source-location key
1	Clock-rate change associated with relative motion	1
2	Clock-rate change associated with gravitational conditions	2
3	The account of spacetime and the relationship between gravity and time	3
4	The twins’ age difference at reunion determined by the proper elapsed times along the specified paths	3
5	Existing comparisons of moving and reference clocks	4,5
6	Existing comparisons of clocks under different gravitational conditions	4,5

j	Result	Source-location key
7	The negative daily kinematic contribution to GPS clock time	6
8	The positive daily gravitational contribution to GPS clock time	6
9	The positive net daily contribution to GPS clock time	6
10	Provision of GPS time with relativistic corrections	7
11	The 2010 height comparison using aluminum-ion clocks	8
12	The submillimeter-scale gravitational effect of the 2024 optical lattice clock	9
13	The planned relativity test using two ISS clocks and its anticipated precision improvement of approximately fiftyfold	11
14	The planned comparison through simultaneous operation of the Mount Blue Sky and Boulder clocks	12,13,15
15	The planned remote timing comparison using optical fiber, a relay location, and laser signals	14
16	The age difference relative to sea level on Earth shown in the Jupiter illustration for a 30-year comparison	16
17	The 10.23 MHz output of the rubidium clock for GPS shown in the photograph	17

Proposition 8.790 (Audit-Boundary Cardinalities in the NIST 2025 Source). The audit-boundary count for source R is $\text{BoundaryCount}_R^{\text{audit}} = (68, 19)$.

Proof. The audit-boundary nodes form the typed disjoint union of 27 claims, 24 time contents, and 17 results; the edges consist of two transfers and 17 result-dependency hyperedges. $\square$

Result Dependencies

Definition 8.791 (Result-Dependency Hyperedges in the NIST 2025 Source). For each $j \in [17]$, define the hyperedge to Result j by the time-content prerequisites, auxiliary-object prerequisites, and formal-evidence support in Table 649. The indices uniquely identify the objects in Tables 643, 644, and 645.

Table 649: Result-Dependency Hyperedges in the NIST 2025 Source

j	Time-content prerequisite a	Auxiliary-object prerequisite indices	Formal-evidence support indices
1	$\{2, 3\}$	$\{1\}$	$\{2\}$
2	$\{4\}$	$\{2\}$	$\{4\}$
3	$\{4, 5\}$	$\{2\}$	$\{5, 6\}$
4	$\{8\}$	$\{3\}$	$\{7\}$
5	$\{9\}$	$\{1, 4\}$	$\{8, 9\}$
6	$\{10\}$	$\{2, 4\}$	$\{8, 10\}$

j	Time-content prerequisite a	Auxiliary-object prerequisite indices	Formal-evidence support indices
7	{11}	{5}	{12}
8	{12}	{5}	{13}
9	{13}	{5, 6}	[[12, 14]]
10	{14}	{5}	{15}
11	{15}	{7}	{16}
12	{16}	{8}	{17}
13	{17}	{9, 12}	{19, 20}
14	{18, 21}	{10}	{21}
15	[[19, 21]]	{10, 11}	{22}
16	{22, 23}	{13}	{26}
17	{24}	{14}	{27}

The 17 results have 23 time-content prerequisite occurrences, 22 auxiliary-object prerequisite occurrences, and 23 formal-evidence support occurrences. Every result has temporal prerequisites.

Grounds of Establishment

Grounds of establishment of the results Results 1–4 rest on relativity and the twin example; Results 5 and 6 on earlier clock tests; Results 7–10 on GPS clock-rate differences and correction; Results 11 and 12 on the accounts of clocks in 2010 and 2024; and Results 13–15 on the ISS and Mount Blue Sky experiment plans. Result 16 rests on the Jupiter illustration’s age comparison, and Result 17 on the output frequency marked on the pictured rubidium clock for GPS.

Evidence Hierarchy

Direct-measurement objects The article itself supplies zero newly measured objects.

Direct-measurement predicates The article itself supplies zero new direct-measurement predicates.

Direct-measurement conditions and domains of application The article itself supplies zero conditions or domains of application for a new direct measurement.

Objects of measurements reported from other sources The objects are portable atomic clocks, the 2010 aluminum-ion clock pair, and the 2024 optical lattice clock.

Predicates of measurements reported from other sources The predicates concern motion- and gravity-dependent clock-rate differences and gravitational effects over height differences and submillimeter scales.

Conditions and domains of application of measurements reported from other sources The conditions are the compared clocks’ motion and gravitational conditions, the 2010 height difference of about one foot, and the atomic-ensemble configuration of the 2024 optical lattice clock.

Planned-measurement objects The planned objects are the two ISS clocks, and the Mount Blue Sky and Boulder clocks with optical-fiber and laser signals at a relay location.

Planned-measurement predicates The planned predicates concern ground–ISS clock-rate comparison and remote rate comparison between the mountain and ground laboratory clocks.

Planned-measurement conditions and domains of application The conditions include an ISS altitude of 370–460 km and an approximately fiftyfold precision target; the 4348 m summit and Boulder, separated by nearly 3 km in height and 56 km in distance; simultaneous clock operation; and a relay location compensating for the lack of a direct line of sight.

Objects of formal generalization The generalizations are the description of clock-rate differences as time dilation and the theoretical example relating proper elapsed-time differences along worldlines to the twins’ age difference at reunion. The Jupiter illustration is a theoretical example of the age difference over 30 years relative to sea level on Earth.

Conditions and domains of application of formal generalization Clock comparisons use each clock’s motion and gravitational conditions and a common comparison reference. The three GPS contributions express clock differences in microseconds over the same one-day interval. The twin example compares ages at reunion when initial ages are equal and elapsed ages follow the proper times along the respective worldlines. The Jupiter illustration compares ages over 30 years relative to sea level on Earth, treating the respective body as the sole mass in each comparison[73]. The ISS and Mount Blue Sky experiments are plans as of the 2025 article; the latter requires simultaneous clock operation and signal comparison at a relay location.

Generation Determination **Determined values:** LCTR complete-generation determination 0 / source-internal-evidence-conditioned local-reach determination 0 / residual primitive time roots 24.

Source-internal-evidence-conditioned local reach The source-formal-evidence environment alone reaches zero time contents, leaving all 24 time contents as primitive time roots.

Result Dependencies

Dependence from primitive-time roots to results All 17 results are primitive-time-root-dependent. Neither the complete determination nor the local determination from the source-formal-evidence environment reaches a result without temporal roots.

Evidential Force

Domain of evidential force The domain comprises clock-rate comparisons, GPS correction, height comparisons using optical clocks, planned remote timing-signal comparisons, the twin example using proper elapsed time, the Jupiter illustration’s

age comparison, and the output frequency of the pictured rubidium clock for GPS. The ISS and Mount Blue Sky measurement targets and precision goals are given as experiment plans in the 2025 article.

Boundary of evidential force Each comparison uses clock rates, proper elapsed time, time units, or timing signals. The article supplies no map or relation generating these time contents from nontemporal physical inputs.

Result-Preservation Domain

Conditions and domain of application under which source-specific results are preserved Retain the clock carriers, motion and gravitational conditions, comparison configurations, clock corrections, signal paths, simultaneous operation, conditions for the plans to succeed, the Jupiter illustration's comparison reference, interval and mass-source assumption, and the pictured output frequency.

Results preserved after relocation into the LCTR generation order When an LCTR generation sequence supplies the same comparisons, scales, and correspondences for each temporal input, and the stated configurations and operations are retained, the dependency structures of the two transfers and 17 results are preserved.

Placement of the Grounding Obligation

Placement of the grounding obligation The 24 time contents audited in this source are not generated from nontemporal physical inputs before use and are accepted as already-formed time inputs. The placement determination for the physical-time grounding obligation is "backgrounded by primitive time input." This determination follows by comparing the use of each time content in a law operation or result derivation with the absence, before that use, of the corresponding generation condition or formation condition as an input, a formation condition, an evaluation condition, or a failure condition.

Evidence of Actual Failure

Evidence of actual failure The complete-generation and residual-root determinations concern physical-time admissibility and are distinct from the actual failure of a concrete condition governing a result's value or domain of application. The source's formal evidence and the LCTR audit evidence do not establish the actual failure of such a target condition. The actual-failure determination is "indeterminate from the primary-source evidence"; actual failure is not inferred from the backgrounding of the grounding obligation alone.

Downstream Reach

Downstream reach The reach to primitive-time-root-dependent results is determined source by source by the primitive-time-root-result dependency relations and minimal primitive-time-root supports reconstructed in this division. This is structural reach. Because the actual failure of a concrete condition has not been established, the actual-reach determination from such a failure location to a result is "indeterminate from the primary-source evidence." Structural reach does not substitute for actual reach.

Repair Scope

Repair scope On the basis of structural reach, the repair targets are the corresponding time contents and the generation conditions or formation conditions required before their use. The recalculation scope is limited to results to which an independently established actual failure reaches through dependency. Because actual failure is indeterminate in this source, this repair-and-recalculation specification is conditional. If that condition is met, the target results are regenerated or recalculated from the repaired inputs, or their domains of validity are restricted. The values after recalculation are determined as audit results.

Audit Conclusions

Proposition 8.792 (Primitive Time Roots and Result Dependencies in the NIST 2025 Source).

$$\text{AuditPartIdx}_R = \text{AllRootDepIdx}(24, 17).$$

Both the complete determination and the local-reach determination from the source-formal-evidence environment reach zero time contents.

Proof. The temporal input sets of the two transfers are $\{6, 7\}$ and $\{11, 12\}$. Both are nonempty, so neither nontemporal physical inputs nor the source-formal-evidence environment alone can initiate a transfer. Proposition 7.43 therefore makes all 24 time contents primitive time roots. Each of the 17 result edges also has temporal prerequisites. Adding all 24 roots to that environment reaches every result; Proposition 7.49 and Definition 8.115 give the displayed formula. $\square$

Proposition 8.793 (Minimal Primitive-Time-Root Support in the NIST 2025 Source). For $\mathcal{A}_R(q)$ in Definition 7.108, the table gives the number of family members and the minimum and maximum cardinalities of those members.

Table 650: Minimal Primitive-Time-Root Support in the NIST 2025 Source

Result index	Support statistic	Result index	Support statistic
1	(1; 2)	2	(1; 1)
3	(1; 2)	4	(2; 1, 2)
5	(1; 1)	6	(1; 1)
7	(1; 1)	8	(1; 1)
9	(2; 1, 2)	10	(1; 1)
11	(1; 1)	12	(1; 1)
13	(1; 1)	14	(1; 2)
15	(1; 3)	16	(1; 2)
17	(1; 1)		

Proof. Applying Definition 7.46 to the two transfers and 17 result edges leaves the temporal support antichains unchanged on the second scan. Proposition 7.47 gives the table. There are 19 minimal-support family members, 27 primitive-time-root/result dependency pairs, and 21 necessary primitive-time-root/result dependency pairs. $\square$

Individual Statements Statement Evidence 1: Gravity and the Passage of Time

Original quotation According to the audacious new theory of general relativity he published that year, time is also affected by gravity. The more strongly you feel the force of gravity, Einstein argued, the slower time passes for you.

Primary-source location The second paragraph of the opening overview of relativity in National Institute of Standards and Technology, *Putting Einstein to the Test*.

Minimal context This paragraph follows the explanation of motion and the passage of time under the 1905 special theory of relativity and introduces the 1915 general theory of relativity and gravitational time dilation.

Attribution The official article text of the National Institute of Standards and Technology.

Statement Evidence 2: Difference in Age between Twins

Original quotation For example, if you had a twin who went on a long spaceship ride at close to the speed of light or who spent time near a black hole and then rejoined you on Earth, according to relativity, you would now be older than your own twin!

Primary-source location The third paragraph of the opening overview of relativity in the same source.

Minimal context This sentence follows immediately after the relation between space and time and that between gravity and time, and gives the example of a twin who travels in a high-speed spacecraft or remains near a black hole. The following paragraph turns to tests of relativity using portable atomic clocks.

Attribution The official article text of the National Institute of Standards and Technology.

Common Evidence with a Specified Domain of Application

Target state-ment	Statement object	Predicate	Explicit domain of application
S69-Q001	Gravitational conditions and the passage of time experienced by a person	States that time is affected by gravity and that the stronger the gravity a person feels, the more slowly time passes for that person.	The domain of description at the beginning of the official NIST article that introduces the 1915 general theory of relativity and gravitational time dilation after the account of the 1905 special theory of relativity.

S69-Q002	The ages, upon reunion, of twins one of whom travels in a high-speed spacecraft or remains near a black hole	States that the person who remains on Earth is older at reunion than the twin who travels in the spacecraft or remains near the black hole.	The domain of the twin example at the beginning of the official NIST article, placed immediately after the account of the relation between space and time and that between gravity and time, and immediately before the account of tests using portable atomic clocks.
----------	--	---	--

Individual Audit

Gravity and the Passage of Time

Interpretive possibilities In the context of the subsequent clock tests, GPS, and height comparisons, the first statement describes gravitationally conditioned comparisons of clock rates and proper time in everyday language.

Independent grounds Paragraph 5 describes clock comparisons under motion and gravitational conditions; Paragraph 6 gives the negative kinematic and positive gravitational GPS contributions; and Paragraphs 8 and 9 describe gravitational effects using optical clocks in 2010 and 2024.

Boundary of evidential force under each interpretive possibility The gravitational rate comparison of stationary clocks depends on the gravitational potential difference; comparisons involving moving clocks also depend on motion. For the ISS, the magnitude of the negative kinematic contribution exceeds the positive gravitational contribution relative to a ground clock, giving a negative net rate difference[74].

Corresponding formal evidence The correspondence is to time contents 4, 5, 10, and 12; Claims 4–6, 10, and 13; Results 2, 3, 6, and 8; and Proposition 8.792. The ISS rate comparison corresponds to time content 17, and its plan to Claims 19 and 20 and Result 13.

Within-source comparison Paragraph 6 distinguishes the kinematic, gravitational, and net GPS contributions. The explanation in Paragraph 11 that ISS clocks run faster than ground clocks is Claim 25; the gravitational contribution alone does not determine the sign of the net clock-rate difference. The two ISS clocks, their altitude of 370–460 km, and the approximately fiftyfold precision target constitute the experiment plan in Claims 19 and 20.

The Twins’ Age Difference at Reunion

Interpretive possibilities The second statement assumes a reunion after either a near-light-speed spacecraft journey or a stay near a black hole. When initial ages are equal and elapsed ages follow proper time, a larger Earthbound proper elapsed time makes the Earthbound twin older at reunion.

Independent grounds The original introduces the travel conditions with “if” and states the consequence “according to relativity.” The following paragraph on portable atomic clocks explains apparatus and measurements used to test relativistic clock predictions.

Boundary of evidential force under each interpretive possibility The age difference at reunion is determined by the difference of proper elapsed times along the two worldlines between the corresponding departure and reunion events. Determining that difference requires the travel speeds, gravitational conditions, and worldlines.

Corresponding formal evidence The correspondence is to time contents 1, 2, and 6–8; Claim 7; Transfer 1; Result 4; and Proposition 8.792.

Within-source comparison The twin example uses proper elapsed time along worldlines; the clock-test account uses clock-rate comparisons under motion and gravitational conditions. In each case, the temporal inputs accompany the specified objects and comparison conditions.

8.19.70 Source Audit

Audit Input

Primary Source and Audit Scope

Source identifier 70.

Source title Primitive-time-root audit of Feynman’s *Lectures on Gravitation*

Bibliographic identification Let Richard P. Feynman, Fernando B. Morinigo, and William G. Wagner, edited by Brian Hatfield, *Feynman Lectures on Gravitation*, Addison–Wesley, 1995, be source $R = \text{Ref}_{\text{Fey95LG}}$. The text records lectures delivered at Caltech in 1962–63.

Primary-source-location identifier 70-L1.

Primary-source-location content Lecture 5, §5.2, pp. 68–69 and §5.3, pp. 69–70; Lecture 7, §7.2, pp. 94–95. The derivations and application conditions include the adjoining pp. 67, 93 and 96. Page numbers refer to printed pages.

Audit-scope identifier 70-SC1.

Audit-scope content Formal and statement evidence concerning differences in gravitational potential, nuclear-transition frequencies, clock rates, all physical processes, human life and aging, the ages of objects, comparison of signals sent upward and downward, time scales, and the flow of time.

Formal Objects

Object identifier 70-F1.

Object name Comparison of frequencies, clock rates, and process rates in a gravitational field.

Object type Formal object.

Formal content The description that associates nuclear transitions, photons, clocks, persons, and objects at different gravitational potentials with frequency differences, clock-rate differences, rates of physical processes, rates of living and aging, time-scale differences, and differences in the flow of time.

Set of primary-source-location identifiers 70-L1.

Assumptions

Assumption identifier 70-A1.

Assumption content The nuclear-frequency route uses a weak field, nuclear transitions and energy conservation; the field-action route uses a locally constant weak field, stress-energy coupling and an action depending on fields and first derivatives; the signal-comparison route uses the equivalence principle, the emission–reception configuration and the low-velocity Doppler approximation.

Set of primary-source-location identifiers 70-L1.

Nontemporal Physical Inputs

Input identifier 70-P1.

Input name Nontemporal physical inputs of the primary source.

Input type Nontemporal physical input.

Input content The generation-initial set consists of the material states, spatial gradients and spatial configurations in Claims 1–3. Frequencies, process rates, time variables, comparison intervals and histories belong to the respective objects in the time-content table.

Set of primary-source-location identifiers 70-L1.

Definition 8.794 (Semantic components and generation-initial set in the Feynman gravitation lectures). Let $\text{Node}_R^{\text{audit,semantic}} = \text{SemanticNodes}_R(10)$ and $\text{InitEnv}_R^{\text{phys}} = \text{SemanticPrefix}_R(3)$. Source-claim indices identify the same semantic components.

Time Contents

Time-content identifiers 70-T1–70-T11.

Time-content name Eleven time contents of the primary source.

Time-content type Time content.

Time-content description Their names and primary-source locations are given in Table 652.

Set of primary-source-location identifiers 70-L1.

Formal Input Components

Formal-component identifiers 70-C1–70-C10.

Formal-component name Ten source claims.

Formal-component type Formal component.

Formal-component text The claims and primary-source locations are given in Table 653.

Set of primary-source-support identifiers 70-L1.

Definition 8.795 (Time Contents of the Feynman Gravitation Lectures). Define the eleven time contents as in the following table. Frequencies and periods, and a time variable with its derivative and integration element, are listed as components of the respective contents.

Table 652: Time contents and primary-source locations
in the Feynman gravitation lectures

i	Time content	Source location
1	Reference nuclear-transition and photon frequencies and their upper–lower comparison, including ω_0 , ω , $\Delta\omega/\omega$ and the corresponding periods.	pp. 68–69
2	Comparison of the reading rates of identical clocks at the upper and lower positions.	pp. 68–69
3	The rate of the same physical process relative to its reference rate in a locally constant potential.	pp. 67–69
4	Comparison of the rates of living of people at a mountaintop and at a lower elevation.	p. 69
5	Comparison of the rates of aging of people at a mountaintop and at a lower elevation.	p. 69
6	The accumulated age difference between objects at Earth’s center and surface.	p. 69
7	Emission and reception frequencies and propagation comparison for upper–lower signals, including f_{em} , f_{rec} , h/c , gh/c and the low-velocity Doppler relation.	pp. 93–94
8	The time-scale ratio ds/dt from clock and signal comparison, including the reference position and the correspondence of stationary clocks.	pp. 94–95
9	The difference in the flow of time described in the source through differing time scales.	p. 94
10	The time variable t and its transformation t' in the field action; the time derivative ∂_t and integration element dt are tied to this variable.	pp. 67–68
11	Initial ages, the endpoints of the comparison interval and rate histories along the objects’ paths used in an age comparison.	p. 69; pp. 95–96

Definition 8.796 (Source Claims of the Feynman Gravitation Lectures). The following table gives ten source claims. Claims 1–3 specify generation-initial components; Claims 4–10 describe theoretical relations, an account of an experiment and applications.

Table 653: Source claims of the Feynman gravitation
lectures

p	Source claim	Source location
1	Material states of nuclei, photons, crystals and clock devices, and the spatial arrangement of the upper and lower locations.	pp. 68–69, 93–94
2	State values and spatial gradients of matter fields, and the spatial configuration of bodies and gravitational sources.	pp. 67–68
3	The material constitution and spatial locations of people and of objects inside Earth and at its surface.	p. 69

p	Source claim	Source location
4	The lecture describes a gravitational frequency-shift experiment using sharp nuclear-transition frequencies and an artificial Doppler shift.	pp. 68–69
5	Excitation energy, the mass–energy relation, gravitational energy, $E = \hbar\omega$ and energy conservation give the weak-field frequency ratio relative to a reference frequency ω_0 .	p. 69
6	For a locally constant weak gravitational field, the argument uses a Lagrangian depending on matter fields and their first derivatives, with stress-energy coupling. Under $t' = t\sqrt{1 + \epsilon}$, the action takes its field-free form through first order in ϵ , changing the rate of the same process.	pp. 67–68, Eqs. (5.2.4)–(5.2.8)
7	The process-rate relation is applied to human living and aging at a mountaintop and at a lower elevation.	p. 69
8	The lecture discusses an age difference between objects at Earth’s center and surface. An accumulated difference depends on initial ages, a common comparison interval and the rate histories along the respective paths.	p. 69; pp. 95–96
9	An accelerating box under the equivalence principle and the low-velocity Doppler relation give $f_{\text{rec}}/f_{\text{em}} = 1 + gh/c^2$ for downward signals, with the opposite shift for upward signals. The calculation uses the light-travel time h/c and velocity increment gh/c .	pp. 93–94, Eq. (7.2.1)
10	Clock and signal comparisons are described as differences in time scale and flow, relating ds to dt referred to a reference position. The potential-only Eq. (7.2.3) gives the stationary-clock relation; elapsed proper time for a moving clock includes the velocity term in the subsequent Eq. (7.3.1).	pp. 94–96, Eqs. (7.2.2)–(7.3.1)

Dependency Inputs

Index identifier S70-XI-DI1.

Set of existing-evidence references Definition 8.797.

Dependency and Transfer Components

Definition 8.797 (Time-Content Transfers in the Feynman Gravitation Lectures). The following table gives the six dependencies and transfers. Source-claim and time-content indices refer to their respective tables; each input set lists jointly required inputs.

Table 654: Time-content transfers in the Feynman gravitation lectures

Transfer	Operation	Source claims	Time inputs	Time outputs
1	Translate a comparison of nuclear-transition frequencies into a comparison of the reading rates of identical nuclear clocks. pp. 68–69	{4, 5}	{1}	{2}
2	Transform the source’s matter-field action to t' to obtain the rate relation for the same physical process and clocks using it. pp. 67–68	{6}	{10}	{2, 3}
3	Apply the process-rate relation to the rates of human living and aging. p. 69	{7}	{3}	{4, 5}
4	Integrate the rate along each path over the specified comparison interval, with the specified initial ages, to obtain an age difference. p. 69; pp. 95–96	{8}	{3, 11}	{6}
5	Obtain the time-scale ratio of the corresponding stationary clocks from the signal-frequency comparison based on the equivalence principle. pp. 93–95	{9, 10}	{7}	{8}
6	Map the time-scale difference in the same clock and signal comparison to the source’s description of a difference in time flow. p. 94	{10}	{8}	{9}

Formation Conditions

Formation-condition identifier 70-K1.

Formation-condition content Each transfer uses the physical model, approximations and temporal inputs of its corresponding source claims. Accumulating an age difference uses initial ages, a comparison interval and the rate history along each path.

Set of primary-source-support identifiers 70-L1.

Primary-Source Support

Support identifier 70-SS1.

Set of supported-object identifiers 70-T1–70-T11, 70-C1–70-C10.

Set of primary-source-location identifiers 70-L1.

Support content Each time content, source claim, and transfer is supported by the corresponding equation, explanation, or statement within the audit scope.

Results

Definition 8.798 (Results and Preservation Conditions of the Feynman Gravitation Lectures). The seven source-specific results and their time presuppositions and preservation conditions are given in the following table.

Table 655: Results, dependencies and preservation conditions in the Feynman gravitation lectures

Re- sult	Content	Source claims	Temporal premises	Preservation conditions
1	The gravitational shift of nuclear-transition frequencies.	{4, 5}	{1}	A weak field, identical nuclear transitions, a reference frequency, the mass–energy relation and energy conservation.
2	The lower clock rate at the lower-potential position.	[[4, 6]]	{2}	Identical clocks and fixed comparison positions, with the conditions of either the nuclear-frequency route or the field-action route.
3	The process-rate relation derived from the field action.	{6}	{3, 10}	A locally constant weak field, the source’s stress-energy coupling, an action depending on fields and first derivatives, and first order in ϵ .
4	The theoretical application to rates of human living and aging.	{6, 7}	[[3, 5]]	Bodily processes follow the same matter-field description, with the compared processes and positions specified.
5	The application to an accumulated age difference between objects.	{6, 8}	{3, 6, 11}	Initial ages, a common comparison interval, and the histories of potential, motion and process rates along the respective paths.

Re- sult	Content	Source claims	Temporal premises	Preservation conditions
6	The upper–lower signal-frequency comparison from the equivalence principle.	{9}	{7}	A locally uniform gravitational field, the emission–reception configuration, light propagation and the low-velocity Doppler approximation.
7	The description of clock comparison in terms of differences in time scale and flow.	{9, 10}	[[7, 9]]	The same clock and signal comparison, reference position and correspondence of stationary clocks.

Result Dependencies

Result-dependency identifier 70-RD1.

Set of result identifiers 70-R1–70-R7.

Set of input identifiers 70-T1–70-T11, 70-C1–70-C10.

Dependency content All seven results presuppose at least one time content given in Table 655.

Grounds

Ground identifier 70-RG1.

Set of result identifiers 70-R1–70-R7.

Ground content The energy-conservation derivation for nuclear frequencies, the first-order rate relation from the matter-field action, the signal comparison from the equivalence principle, and their applications to clocks, people and objects.

Set of primary-source-support identifiers 70-L1.

Following Definition 7.71, each result uses the following transfer set under the conditions in the result table.

Table 656: Result-specific audit branches in the Feynman gravitation lectures

Domain	Results	Transfer set
The gravitational shift of nuclear-transition frequencies.	{1}	$\emptyset$
The lower clock rate at the lower-potential position.	{2}	{1, 2}
The process-rate relation derived from the field action.	{3}	{2}

Domain	Results	Transfer set
The theoretical application to rates of human living and aging.	{4}	{2, 3}
The application to an accumulated age difference between objects.	{5}	{2, 4}
The upper–lower signal-frequency comparison from the equivalence principle.	{6}	$\emptyset$
The description of clock comparison in terms of differences in time scale and flow.	{7}	{5, 6}

Evidence Hierarchy

Evidence-hierarchy identifier 70-MT1.

Hierarchy A secondary account of an existing experiment, theoretical derivations from the matter-field action and the equivalence principle, and their applications to people and objects.

Objects Nuclear-transition frequencies, clock rates, physical-process rates, human living and aging, object age, and the flow of time.

Predicates States comparative frequency and rate relations and the associated living, aging and accumulated age differences.

Conditions The physical model, approximation, comparison objects and history conditions of each result.

Domain of application Lecture 5, §5.2 and Lecture 7, §7.2, with boundary conditions from §5.3 and §7.3.

Set of primary-source-support identifiers 70-L1.

Generation Determination

Generation-determination identifier 70-GA1.

Set of object identifiers 70-T1–70-T11.

Complete-generation-determination identifier 70-GA-C.

Complete-generation determined value 0 cases.

Ground for complete-generation determination Every transfer within the source inputs one or more time contents, and the source has no time-generation path completed solely from nontemporal physical inputs.

Domain of the complete-generation determination Entire audit scope.

Local-reach-determination identifier 70-GA-L.

Local-reach determined value 0 cases.

Ground for local-reach determination Even under the local-reach determination, which does not require an intention to generate time, no transfer attains local reach without inputting a time content.

Domain of the local-reach determination Entire audit scope.

Residual-root-determination identifier 70-GA-R.

Residual-root determined value 11 cases.

Ground for residual-root determination All eleven time contents remain as primitive time roots.

Domain of the residual-root determination Entire audit scope.

Determined values: 0 cases satisfying the LCTR complete-generation determination / 0 cases satisfying the source-internal-evidence-conditioned local-reach determination / 11 residual primitive time roots.

Result Dependencies

Result-dependency identifier 70-RD2.

Set of result identifiers 70-R1–70-R7.

Set of input identifiers 70-T1–70-T11.

Dependency content The seven primitive-time-root-dependent results each depend on at least one residual primitive time root.

Evidential Force

Evidential-force identifier 70-EE1.

Force-domain identifier 70-EE-IN.

Boundary identifier 70-EE-OUT.

Set of evidence identifiers 70-L1, 70-SS1.

Set of supported-object identifiers 70-R1–70-R7.

Force-domain content Frequency, clock-rate and physical-process-rate relations under the source’s stated conditions, and their theoretical application to people and objects.

Within domain Weak-field nuclear-frequency shifts, action transformations in a locally constant field, signal comparisons from the equivalence principle, and rate and age comparisons under specified initial conditions and histories.

Outside domain Direct extrapolation of a relation obtained for small potential differences to large cosmological potential differences (§5.3).

Boundary content The nuclear-transition experiment supports the frequency comparison; the field action and equivalence principle yield theoretical relations under their respective conditions. Human living and aging are applications of the process-rate relation.

Result-Preservation Domain

Preservation-domain identifier 70-RP1.

Set of result identifiers 70-R1–70-R7.

Set of formation-condition identifiers 70-K1.

Preservation-domain content Each result holds under the conditions in Table 655. Preservation under LCTR requires generation of the time contents used there and preservation of their relations.

Placement of the Grounding Obligation

Obligation identifier 70-GO1.

Set of object identifiers 70-T1–70-T11.

Set of prior-requirement identifiers 70-P1, 70-K1.

Placement content The eleven time contents are used as already established inputs. The grounding obligation for physical time is classified as backgrounded by primitive-time inputs.

Evidence of Actual Failure

Actual-failure identifier 70-AF1.

Set of failure-condition identifiers None.

Set of evidence identifiers 70-L1, 70-SS1.

Failure content Actual failure of the input conditions specified here cannot be determined from the source evidence.

Downstream Reach

Reach identifier 70-DR1.

Set of origin identifiers 70-T1–70-T11.

Set of reached-result identifiers 70-R1–70-R7.

Reach content Structural reach from residual primitive time roots to results is given by the result-dependency and minimal-support tables. If actual failure of a specific condition is identified, results dependently reachable from that location require recalculation.

Repair Scope

Repair identifier 70-RS1.

Set of failed-item identifiers None.

Set of repair-item identifiers 70-T1–70-T11, 70-K1.

Regeneration order Generate time contents from nontemporal physical inputs before use, then connect them to comparison relations and results. If actual failure of a specific condition is identified, recalculate dependent results from repaired inputs.

Set of recalculated-result identifiers 70-R1–70-R7.

Audit Conclusions

Proposition 8.799 (Primitive time roots and result dependence in the Feynman gravitation lectures).

$$\text{AuditPartIdx}_R = \text{AllRootDepldx}(11, 7).$$

Proof. The generation-initial set consists of Claims 1–3. All six transfers have temporal inputs; adding the source formal-evidence environment provides no transfer without temporal inputs. Complete generation and local reach under source-evidence conditions are therefore both zero, leaving eleven primitive time roots. All seven results have temporal premises; supplying the time contents and source formal-evidence environment under the respective application conditions satisfies each result edge. $\square$

Proposition 8.800 (Minimal primitive-time-root support in the Feynman gravitation lectures). The minimal-support families below contain eleven members. There are twelve primitive-time-root–result dependency pairs, including five necessary dependencies.

Proof. Updating support antichains to a fixed point over each result’s transfer set and result edge gives the following table. $\square$

Table 657: Minimal primitive-time-root support in the Feynman gravitation lectures

Result	Family of minimal-support index sets
1	{1}
2	{1}, {2}, {10}
3	{10}
4	{3}, {10}
5	{3, 11}, {10, 11}
6	{7}
7	{7}

Individual Statements Statement Evidence 1: Generalization to All Physical Processes

Original quotation Our theory is then not inconsistent, and it suggests that the frequency shift required by energy conservation be considered as a general property of all physical processes, that they run more slowly in lower gravitational potentials.

Primary-source location Lecture 5, §5.2, printed p. 69, PDF p. 110.

Minimal context After deriving the process-rate relation from the matter-field action on pp. 67–68 and the nuclear-frequency shift from energy conservation on p. 69, the lecture states the process-rate relation common to both arguments.

Attribution The text of Richard P. Feynman’s lecture.

Statement Evidence 2: Application to Human Living and Aging

Original quotation The man on top of a mountain is living and aging at a faster rate than we.

Primary-source location Lecture 5, §5.2, printed p. 69, PDF p. 110.

Minimal context Immediately after the generalization to all physical processes, the predicate is transferred to human living and aging.

Attribution The text of Richard P. Feynman’s lecture.

Statement Evidence 3: Attribute of Temporal Flow

Original quotation One of the ways of describing this situation is to say that the time scale is faster at the top; time flows are different in different gravitational potentials, so that time flows are unequal in various parts of the world.

Primary-source location Lecture 7, §7.2, printed p. 94, PDF p. 135.

Minimal context Immediately after the comparison of frequencies between upper and lower clocks and light signals, this is presented as a way of describing the same situation.

Attribution The text of Richard P. Feynman’s lecture.

Common Evidence with a Specified Domain of Application

Target statement	Statement object	Predicate	Explicit domain of application
S70-Q001	Frequency shift and all physical processes	Generalizes that all physical processes run more slowly at lower gravitational potentials.	The first-order derivation from the matter-field action in a locally constant weak field, together with nuclear transitions and energy conservation.
S70-Q002	The living and aging of persons on a mountaintop and at a lower elevation	States that the person on the mountaintop lives and ages at a faster rate.	Application to humans of the process-rate generalization accompanying a gravitational-potential difference.
S70-Q003	The time scale and flow of time at different gravitational potentials	States that the time scale is faster at the upper position and that the flow of time differs by location.	Comparison of upper and lower clocks and signal frequencies by the equivalence principle.

Individual Audit Generalization to all physical processes

Interpretive possibility This is a theoretical generalization comparing the rate of the same physical process at different locally constant potentials.

Independent grounds The first-order derivation from the matter-field action, stress-energy coupling and transformation of the time variable on pp. 67–68. The nuclear-transition and energy-conservation frequency relation on p. 69 also supports the same rate comparison.

Boundary of evidential force The domain uses an action depending on fields and first derivatives, a locally constant weak gravitational field and the first-order approximation.

Corresponding formal evidence Time Contents 1–3 and 10, Transfers 1 and 2, and Results 1–3.

Application to human living and aging

Interpretive possibility The statement applies the matter-field process-rate relation to human bodily processes.

Independent grounds The preceding process-rate generalization and the gravitational-potential difference between the mountaintop and the lower elevation.

Boundary of evidential force This is a theoretical application to bodily processes following the same matter-field description, with the comparison positions specified. Accumulated age differences depend on initial ages, the comparison interval and rate histories along the paths.

Corresponding formal evidence Time Contents 3–6 and 11, Transfers 3 and 4, and Results 4 and 5.

Time scale and flow described through clock and signal comparison

Interpretive possibility The same comparison of upper and lower clocks and signal frequencies is described as a difference in time scale and flow.

Independent grounds The low-velocity Doppler relation based on the equivalence principle and the immediately following correspondence between ds and dt .

Boundary of evidential force The stationary-clock scale relation is given in Eqs. (7.2.2)–(7.2.3); elapsed proper time for a moving clock includes the velocity term in Eq. (7.3.1).

Corresponding formal evidence Time Contents 7–9, Transfers 5 and 6, and Results 6 and 7.

Common Audit with a Specified Domain of Application

Statement-audit-scope identifier S70-SA-SC1.

Target statements S70-Q001, S70-Q002, S70-Q003.

Audit scope Collates, by object and evidence type, frequency and clock-rate comparisons, the generalization to all physical processes, the application to human living and aging, and the attribution of time-scale and temporal-flow properties.

8.19.71 Source Audit

Audit Input

Primary Source and Audit Scope

Source identifier 71.

Source title Primitive-time-root audit of Narlikar's *An Introduction to Relativity*

Bibliographic identification Let Jayant V. Narlikar, *An Introduction to Relativity*, Cambridge University Press, 2010, be source $R = \text{Ref}_{\text{Nar10IR}}$.

Primary-source-location identifier 71-L1.

Primary-source-location content Chapter 13, §13.2, printed pp. 201–202 (PDF pp. 210–211). Equations (13.18) and (13.22)–(13.25) on printed pp. 199–200 identify the notation and model conditions.

Audit-scope identifier 71-SC1.

Audit-scope content Equations (13.26)–(13.30), Fig. 13.2, Example 13.2.2, and the explanation of time-flow rates on printed pp. 201–202. Proper time, external coordinate time, boundary evolution, signal propagation, initial and terminal conditions, and the statement are audited using the finite decomposition below.

Formal Objects

Object identifier 71-F1.

Object name Signal transfer from a collapsing boundary to an external observer.

Object type Formal object.

Formal content The correspondence between proper time t and external coordinate time T , boundary evolution, emission time T_1 , reception time T_2 , the propagation interval, horizon-crossing time t_s , collapse-completion time t_0 , the redshift explanation, and the statement about time-flow rates.

Set of primary-source-location identifiers 71-L1.

Assumptions

Assumption identifier 71-A1.

Assumption content Adopt the collapse branch of a spherically symmetric homogeneous dust model, a Schwarzschild exterior, $\alpha > 0$, $r_b > 0$, $0 < \alpha r_b^2 < 1$, $R_s = 2GM = \alpha r_b^3$, and $\gamma = \sqrt{1 - \alpha r_b^2} > 0$. Outgoing signals are considered for $R_2 > R_1 > R_s$, with the horizon limit taken from the exterior. A is at fixed radius R_2 , and B is on the contracting boundary.

Set of primary-source-location identifiers 71-L1.

Nontemporal Physical Inputs

Input identifier 71-P1.

Input name Nontemporal physical inputs of the primary source.

Input type Nontemporal physical input.

Input content The collapsing object, boundary radius, two observers, emission and reception events, light signals, wavelength-measurement configuration, and mass distribution.

Set of primary-source-location identifiers 71-L1.

Time Contents

Time-content identifiers 71-T1–71-T14.

Time-content name Fourteen time contents in the finite audit.

Time-content type Time content.

Time-content description Time Contents 1–14 are enumerated in Definition 8.801, with source correspondences and the decomposition granularity.

Set of primary-source-location identifiers 71-L1.

Formal Input Components

Formal-component identifiers 71-C1–71-C9.

Formal-component name Source claims concerning collapse, signals, redshift, and temporal flow.

Formal-component type Formal component.

Formal-component text Formal Components 1–9 are enumerated in Definition 8.802.

Set of primary-source-support identifiers 71-L1.

Definition 8.801 (Time Contents of the Narlikar Source). The fourteen time contents are (1) proper time t , (2) external coordinate time T , (3) the rate correspondence dT/dt , (4) boundary evolution $R(t), dR/dt$, (5) scale-factor evolution $S(t), dS/dt$, (6) emission coordinate time T_1 , (7) reception coordinate time T_2 , (8) the propagation interval $T_2 - T_1$, (9) proper time t_s of horizon crossing, (10) proper time t_0 of completed collapse, (11) the initial collapse condition $t = 0, S = 1, R = r_b$, (12) the external coordinate-time origin $T = 0$, (13) the correspondence between collapse parameter θ and proper time t , (14) the statement content of different rates of time flow for the two subjects. The unit of this finite decomposition is a temporal role; repeated occurrences of the same role belong to one content. A trajectory and its velocity belong to one evolutionary content, while clock scales, coordinate endpoints, and reference boundaries belong to distinct contents. The redshift account belongs to Formal Component 7. The two zeros in Fig. 13.2 are the respective origins of proper time t and external coordinate time T . Fourteen is the cardinality of the set defined by these decomposition rules.

Definition 8.802 (Source Claims of the Narlikar Source). The nine source-claim components are (1) the proper-time/external-coordinate-time correspondence, Eq. (13.26), (2) radial boundary motion, Eq. (13.27), (3) the boundary/scale-factor correspondence, Eq. (13.28), (4) model-parameter relations, Eq. (13.29), (5) outgoing null signals and the corrected form of Eq. (13.30), (6) the proper-time solution and initial and terminal conditions in Example 13.2.2, (7) the qualitative account of gravitational and Doppler redshift and dimming, (8) the statement of different time-flow rates, (9) the correspondence among the proper-time and coordinate-time origins, signals, and boundaries in Fig. 13.2.

Printed Equation and Grounds for Correction

The printed integrand in Eq. (13.30) is $(1 - 2GMR)^{-1}$. It is inconsistent with the metric in Eq. (13.22). The radial null condition for that metric gives the integrand $(1 - 2GM/R)^{-1}$, and the propagation interval for an outgoing signal is as follows.

$$\begin{aligned} T_2 - T_1 &= \int_{R_1}^{R_2} \frac{dR}{1 - R_s/R} \\ &= R_2 - R_1 + R_s \log \frac{R_2 - R_s}{R_1 - R_s}, \quad R_s = 2GM. \end{aligned}$$

For fixed $R_2 > R_s$, this interval diverges as $R_1 \rightarrow R_s^+$. By contrast, the proper-time solution with $S = \cos^2 \theta$ in Example 13.2.2 is

$$\begin{aligned} t(\theta) &= \frac{\theta + \sin \theta \cos \theta}{\sqrt{\alpha}}, \\ u &= \alpha r_b^2, \quad \theta_s = \arccos \sqrt{u}, \\ t_s &= \frac{\arccos \sqrt{u} + \sqrt{u(1-u)}}{\sqrt{\alpha}}, \\ t_0 &= \frac{\pi}{2\sqrt{\alpha}}, \quad 0 < t_s < t_0. \end{aligned}$$

For $0 < u < 1$, $0 < \theta_s < \pi/2$; the inequality follows from $dt/d\theta = 2 \cos^2 \theta / \sqrt{\alpha} > 0$. The finite horizon-arrival time t_s is a proper time on the collapsing boundary; the divergent $T_2 - T_1$ is the propagation interval of an outgoing signal in external coordinate time.

Source Coverage and Finite Decomposition

Table 659: Source coverage in the Narlikar source

Source unit	Time-content indices	Formal-component indices
13.26	1,2,3	1
13.27	1,3,4	2
13.28	1,4,5	3
13.29	—	4
13.30	6,7,8	5
After 13.30	1,2,3,5,6,7,8,9	1,3,4,5
Redshift and dimming paragraph	4	7
Time-flow paragraph	1,2,8,14	8
Fig. 13.2	1,2,6,7,9,10,11,12	9
Example 13.2.2	1,5,9,10,11,13	6

The algebraic parameter relation in Eq. (13.29) belongs to Formal Component 4. The example parameter θ mediates the scale factor; its proper-time correspondence belongs to Time Content 13. Table 659 maps the audited equations, figure, example, and statement to this finite decomposition.

Dependency Inputs

Index identifier S71-XI-DI1.

Set of existing-evidence references Definition 8.803.

Dependency and Transfer Components

Definition 8.803 (Time-Content Transfers in the Narlikar Source). The eight dependencies and transfers within the source are given in Table 660. Because every transfer inputs an existing time content, none generates a time content solely from nontemporal physical inputs.

Table 660: Time-content transfers in the Narlikar source

Transfer	Dependency or transfer	Source claims	Time input	Time output
1	From proper and external coordinate times to rate correspondence	$\{1\}$	$\{1, 2\}$	$\{3\}$
2	From rate correspondence, proper time, and initial condition to boundary evolution	$\{1, 2, 4, 6\}$	$\{1, 3, 11\}$	$\{4\}$

Transfer	Dependency or transfer	Source claims	Time input	Time output
3	From boundary evolution to scale-factor evolution	{3}	{1, 4}	{5}
4	From emission and reception coordinate times to propagation interval	{5}	{6, 7}	{8}
5	From evolution and initial condition to parameter/proper-time correspondence	{6}	{1, 5, 11}	{13}
6	From parameter/proper-time correspondence to horizon-crossing time	{6}	{13}	{9}
7	From parameter/proper-time correspondence to collapse-completion time	{6}	{13}	{10}
8	Explanation of time-flow rates through boundary motion and signal exchange	{1, 2, 5, 7, 8}	{1, 2, 4, 8}	{14}

Equation (13.26) relates proper time t on the boundary to external coordinate time T . The two axes in Fig. 13.2 likewise represent t and T , respectively. Transfer 8 is the explanatory dependence by which reception waiting time and redshift, arising from boundary motion and signal exchange, are described as different time-flow rates for the two subjects.

Formation Conditions

Formation-condition identifier 71-K1.

Formation-condition content Adopt the collapse branch of a spherically symmetric homogeneous dust model, a Schwarzschild exterior, $\alpha > 0$, $r_b > 0$, $0 < \alpha r_b^2 < 1$, $R_s = 2GM = \alpha r_b^3$, and $\gamma = \sqrt{1 - \alpha r_b^2} > 0$. Outgoing signals are considered for $R_2 > R_1 > R_s$, with the horizon limit taken from the exterior. A is at fixed radius R_2 , and B is on the contracting boundary. Equation (13.30) is evaluated in the explicitly corrected form.

Set of primary-source-support identifiers 71-L1.

Primary-Source Support

Support identifier 71-SS1.

Set of supported-object identifiers 71-T1–71-T14, 71-C1–71-C9.

Set of primary-source-location identifiers 71-L1.

Support content Equations (13.26)–(13.30), Fig. 13.2, Example 13.2.2, and the time-flow paragraph. Equations (13.18) and (13.22)–(13.25) identify origins, endpoints, and the metric.

Results

Definition 8.804 (Results of the Narlikar Source). Define the result index set by $\text{idx}_R^{\text{audit,result}} = [10]$. The correspondence between results and primary-source locations is given in Table 661.

Table 661: Results and primary-source locations in the Narlikar source

j	Result	Primary-source location
1	Rate correspondence between proper and external coordinate times	13.26
2	Collapse equations for boundary and scale factor	13.27–13.29
3	Outgoing-signal propagation interval in the corrected form	13.30
4	Divergence of external coordinate time and propagation interval in the horizon limit	13.26–13.30
5	Proper-time parametrization satisfying the initial condition	Example 13.2.2
6	Finite proper time of horizon crossing	Example 13.2.2
7	Finite proper time of completed collapse	Example 13.2.2; 13.18
8	Qualitative explanation of redshift and dimming	p. 201
9	Diagrammatic relation among the proper-time and coordinate-time origins, signals, and boundaries	Fig. 13.2
10	Explanation of signal exchange in terms of different time-flow rates for the two subjects	p. 201

Result Dependencies

Definition 8.805 (Result-Dependency Hyperedges of the Narlikar Source). For each $q \in [10]$, define the time-content inputs, formal-component support, and result output of the result-dependency hyperedge $\text{ResultEdge}_{R,q}$ by Table 662.

Table 662: Result-dependency hyperedges of the Narlikar source

q	Time-content input indices	Formal-component-support indices	Result-output indices
1	$\{1, 2, 3\}$	$\{1\}$	$\{1\}$
2	$\{1, 3, 4, 5\}$	$\{2, 3, 4\}$	$\{2\}$
3	$\{6, 7, 8\}$	$\{5\}$	$\{3\}$
4	$\{1, 2, 3, 5, 6, 7, 8, 9\}$	$\{1, 3, 4, 5\}$	$\{4\}$
5	$\{1, 5, 11, 13\}$	$\{6\}$	$\{5\}$

q	Time-content input indices	Formal-component-support indices	Result-output indices
6	{1, 9, 13}	{6}	{6}
7	{1, 10, 13}	{6}	{7}
8	{4}	{7}	{8}
9	{1, 2, 6, 7, 9, 10, 11, 12}	{9}	{9}
10	{1, 2, 4, 8, 14}	{1, 2, 5, 7, 8}	{10}

Grounds

Ground identifier 71-RG1.

Set of result identifiers 71-R1–71-R10.

Ground content Results 1–7 are supported by the stated model conditions and equations; Result 8 by the qualitative redshift and dimming account; Result 9 by Fig. 13.2; and Result 10 by the explanatory statement. Results 3–4 include the explicit correction of Eq. (13.30).

Set of primary-source-support identifiers 71-L1.

Evidence Hierarchy

Evidence-hierarchy identifier 71-MT1.

Hierarchy Theoretical derivation and explanatory statement within a textbook.

Objects Proper time and external coordinate time, the collapsing boundary, the signal interval, arrival times, redshift, and the time-flow statement.

Predicates Derives, depicts, and explains temporal relations within the model.

Conditions The adopted collapse model and observer configuration.

Domain of application Chapter 13, §13.2.

Set of primary-source-support identifiers 71-L1.

Generation Determination

Generation-determination identifier 71-GA1.

Set of object identifiers 71-T1–71-T14.

Complete-generation-determination identifier 71-GA-C.

Complete-generation determined value 0 cases.

Ground for complete-generation determination Every transfer inputs an existing time content.

Domain of the complete-generation determination Entire audit scope.

Local-reach-determination identifier 71-GA-L.

Local-reach determined value 0 cases.

Ground for local-reach determination Even under the local-reach determination, no time content is reached solely from nontemporal inputs.

Domain of the local-reach determination Entire audit scope.

Residual-root-determination identifier 71-GA-R.

Residual-root determined value 14 cases.

Ground for residual-root determination All fourteen time contents remain as primitive time roots.

Domain of the residual-root determination Entire audit scope.

Determined values: 0 cases satisfying the LCTR complete-generation determination / 0 cases satisfying the source-internal-evidence-conditioned local-reach determination / 14 residual primitive time roots.

Result Dependencies

Result-dependency identifier 71-RD2.

Set of result identifiers 71-R1–71-R10.

Set of input identifiers 71-T1–71-T14.

Dependency content Each of the ten primitive-time-root-dependent results depends on at least one residual primitive time root.

Evidential Force

Evidential-force identifier 71-EE1.

Force-domain identifier 71-EE-IN.

Boundary identifier 71-EE-OUT.

Set of evidence identifiers 71-L1, 71-SS1.

Set of supported-object identifiers 71-R1–71-R10.

Force-domain content Relations among proper time, external coordinate time, collapse, and signals within the model, together with their diagrammatic representation and qualitative explanation.

Within domain The collapsing-boundary and external-observer system satisfying Formation Condition 71-K1.

Outside domain Collapse outside the model conditions of spherically symmetric homogeneous dust and a Schwarzschild exterior.

Boundary content Results 1–7 concern equations and solutions within the model, Result 8 the qualitative account of redshift and dimming, Result 9 the depicted correspondence, and Result 10 the explanation in terms of time-flow rates.

Result-Preservation Domain

Preservation-domain identifier 71-RP1.

Set of result identifiers 71-R1–71-R10.

Set of formation-condition identifiers 71-K1.

Preservation-domain content Where the generated time contents realize the variables, initial conditions, and comparison relations of the source and Formation Condition 71-K1 holds, the equations and solutions in Results 1–7 and the model relations expressed by the explanations and diagram in Results 8–10 are preserved. Outgoing-signal relations apply for $R_2 > R_1 > R_s$; the proper-time solution applies to the collapse branch and its terminal limit.

Placement of the Grounding Obligation

Obligation identifier 71-GO1.

Set of object identifiers 71-T1–71-T14.

Set of prior-requirement identifiers 71-P1, 71-K1.

Placement content The placement determination is “backgrounded by primitive time input.” The ground is that the fourteen time contents are not generated from nontemporal physical inputs before use and are employed as already-formed inputs.

Evidence of Actual Failure

Actual-failure identifier 71-AF1.

Set of failure-condition identifiers None.

Set of evidence identifiers 71-L1, 71-SS1.

Failure content The actual-failure determination is “indeterminate from the primary-source evidence.”

Downstream Reach

Reach identifier 71-DR1.

Set of origin identifiers 71-T1–71-T14.

Set of reached-result identifiers 71-R1–71-R10.

Reach content Structural reach from the residual roots to the results is established. The actual-reach determination is “indeterminate from the primary-source evidence.”

Repair Scope

Repair identifier 71-RS1.

Set of failed-item identifiers None.

Set of repair-item identifiers 71-T1–71-T14, 71-K1.

Regeneration order Physical inputs, signal comparison, time representation, and results, in that order. If actual failure is established, the repair targets are the corresponding time contents and formation conditions; the recalculation scope comprises results reached through dependency from that failure location.

Set of recalculated-result identifiers 71-R1–71-R10.

Audit Conclusions

Proposition 8.806 (Time-Use Audit of the Explicit Finite Decomposition of the Narlikar Source). Both complete generation and local reach yield zero cases, and all fourteen time contents remain as primitive time roots. Each of the ten results depends on at least one of them.

Proof. No time content is supplied at the generation starting point. Each of the eight listed transfers inputs at least one time content; none has only nontemporal physical inputs. Thus the generation closure adds no time content, and all fourteen contents remain primitive time roots. Every one of the ten result-input sets is nonempty. This determination concerns the source mapping and finite decomposition above; the calculation of finite proper time depends on temporal variables in the model. $\square$

Individual Statements Statement Evidence 1: Different Rates of Temporal Flow for Two Subjects

Original quotation The time flows at different rates for the two protagonists.

Primary-source location Chapter 13, §13.2, printed p. 201, PDF p. 210.

Minimal context The paragraph explaining the signal from Observer B on the surface of the collapsing body to external Observer A, the reception waiting time, redshift, and dimming.

Attribution Jayant V. Narlikar.

Common Evidence with a Specified Domain of Application

Target statement	Statement object	Predicate	Explicit domain of application
S71-Q001	Time for the two subjects, one on the collapsing boundary and one outside	States that time flows at different rates for the two subjects.	The explanation of signal exchange, redshift, and difficulty of reception in the collapse model.

Individual Audit Transfer from Signal Relations to an Attribute of Temporal Flow

Interpretive possibility In the context of signal exchange between A and B, the original sentence makes time the subject that flows and its rate the attribute that differs.

Independent grounds The corrected form of Eq. (13.30), consistent with the radial null condition in Eq. (13.22); the correspondence between boundary proper time and external coordinate time in Eq. (13.26); finite proper time in Example 13.2.2; and the qualitative account of gravitational and Doppler redshift.

Boundary of evidential force The derivations and diagram concern relations among proper time t , coordinate time T , boundary motion, and light signals. The statement explains these relations by attributing different rates of flow to time for the two subjects.

Corresponding formal evidence This audit corresponds to Time Contents 1–14 in Definition 8.801, Definition 8.803, and Results 1–10.

Common Audit with a Specified Domain of Application

Statement-audit-scope identifier S71-SA-SC1.

Target statement S71-Q001.

Audit scope The objects and grounds of signal exchange, redshift, and reception waiting time in the collapse model, and of the time-flow-rate statement explaining them.

8.19.72 Source Audit

Audit Input

Primary Source and Audit Scope

Source identifier 72.

Source title Primitive-time-root audit of Heisenberg’s *Physics and Philosophy*

Bibliographic identification Let Werner Heisenberg, *Physics and Philosophy: The Revolution in Modern Science*, Harper & Brothers, 1958, be source $R = \text{Ref}_{\text{Hei58PP}}$.

Primary-source-location identifier 72-L1.

Primary-source-location content Chapter VII, “The Theory of Relativity,” printed pp. 124–125 (PDF pp. 143–144).

Audit-scope identifier 72-SC1.

Audit-scope content Augustine’s statements about the creation of the universe and time; Heisenberg’s presentation of them; the analysis that the word *created* presupposes a concept of time; and the connection to general-relativistic cosmological models.

Formal Objects

Object identifier 72-F1.

Object name The relation among cosmic geometry, mass distribution, and observations within general relativity.

Object type Formal object.

Formal content The source’s conditional relation between four-dimensional spacetime geometry and the distribution of masses in the universe, together with the investigation of cosmological models using galaxy observations. The presentation of Augustine’s argument and the analysis of the ordinary meaning of *created* are distinguished as statement evidence.

Set of primary-source-location identifiers 72-L1.

Assumptions

Assumption identifier 72-A1.

Assumption content Retain the source’s condition that general relativity correctly gives the relation between cosmic geometry and mass distribution. Preserve the attribution to Augustine in the historical statements and the ordinary meaning of *created* as something coming into being that did not exist before.

Set of primary-source-location identifiers 72-L1.

Nontemporal Physical Inputs

Input identifier 72-P1.

Input name Nontemporal physical inputs of the primary source.

Input type Nontemporal physical input.

Input content The nontemporal physical inputs are the spatial contents of the mass distribution in the universe and of astronomical observations of the distribution of galaxies. Observation times and correspondence with four-dimensional spacetime geometry are excluded from these nontemporal projections. Initial Components 1 and 2 denote the same claim nodes as Source Claims 1 and 2, respectively.

Set of primary-source-location identifiers 72-L1.

Time Contents

Time-content identifiers 72-T1–72-T15.

Time-content name Fifteen time contents of the primary source.

Time-content type Time content.

Time-content description Definition 8.807 and its table distinguish the time contents of historical and conceptual statements from the time content of the cosmological model.

Set of primary-source-location identifiers 72-L1.

Formal Input Components

Formal-component identifiers 72-C1–72-C5.

Formal-component name Source claims concerning the conditions and observational investigation of general-relativistic cosmological models.

Formal-component type Formal component.

Formal-component text The five source claims in Definition 8.808 are used. The attribution of historical statements and the ordinary-word analysis of created are not input claims for the physical derivation.

Set of primary-source-support identifiers 72-L1.

Definition 8.807 (Time Contents of the Heisenberg Source). The fifteen time contents are listed in Table 664. Repeated expressions referring to the same object have the same index. Time Content 1 concerns the question of the infinity of time; Time Contents 2–14 belong to historical or conceptual statements, and Time Content 15 belongs to the general-relativistic cosmological model. The comparison of a thousand years and one day is retained as a description concerning God, not as an equality between measured time quantities. There are no separate auxiliary objects.

Table 664: Time contents of the Heisenberg source

<i>i</i>	Time content	Source location
1	The infinity of time	pp. 124–125
2	Time before creation of the world: the question's presupposition and the object of the subsequent denial	pp. 124–125
3	Time passing for human beings	pp. 124–125
4	The expected future	pp. 124–125
5	The passing present	p. 125
6	The remembered past	p. 125
7	The description of God as not in time	p. 125
8	The temporal expression a thousand years	p. 125
9	The temporal expression one day	p. 125
10	The reciprocal comparison of a thousand years and one day for God	p. 125
11	Time created together with and belonging to the world	p. 125
12	The whole course of the universe being given at once for God	p. 125
13	The before–after relation in the ordinary meaning of created	p. 125
14	The concept of time presupposed by created	p. 125
15	Time in the four-dimensional spacetime geometry of general relativity	p. 125

Definition 8.808 (Source Claims of the Heisenberg Source). The five source claims are given in Table 665.

Table 665: Conditional physical inference in the Heisenberg source

j	Source claim	Source location
1	The spatial content of the distribution of masses in the universe	p. 125
2	The spatial content of astronomical observations of the distribution of galaxies	p. 125
3	The condition that general relativity correctly gives the relation between four-dimensional spacetime geometry and the distribution of masses in the universe	p. 125
4	Under that condition, astronomical observations of the distribution of galaxies provide information about the geometry of the universe as a whole	p. 125
5	The consequences of cosmological models can be compared with empirical facts	p. 125

Definition 8.809 (Claim Nodes and Initial Set of the Heisenberg Source). Take the five claims in Table 665 as the claim nodes, Claims 1 and 2 as the physical-input initial set, and all five claims as the source evidence environment.

$$\begin{aligned}\text{Node}_R^{\text{audit,claim}} &= \text{ClaimNodes}_R(5), \\ \text{InitEnv}_R^{\text{audit}} &= \text{ClaimPrefix}_R(2).\end{aligned}$$

Dependency Inputs

Index identifier S72-XI-DI1.

Set of existing-evidence references Definition 8.810.

Dependency and Transfer Components

Definition 8.810 (Time-Content Transfers in the Heisenberg Source). The audit scope supplies no physical construction or calculation that outputs a time content. Thus there are zero transfer events, and

$$\text{Idx}_R^{\text{audit,transfer}} = \emptyset.$$

The dependencies in historical statements and ordinary-word analysis shown in Table 669 are relations among statement contents, not transfers constructing time contents from nontemporal physical inputs.

Formation Conditions

Formation-condition identifier 72-K1.

Formation-condition content Historical attribution is preserved; Augustine’s statements are separated from Heisenberg’s own conceptual analysis; and, when a cosmological model is adopted, its geometrical, material, and observational conditions hold.

Set of primary-source-support identifiers 72-L1.

Primary-Source Support

Support identifier 72-SS1.

Set of supported-object identifiers 72-T1–72-T15, 72-C1–72-C5.

Set of primary-source-location identifiers 72-L1.

Support content Supported by the presentation of Augustine, conceptual analysis, and paragraph on general-relativistic cosmological models in Chapter VII.

Results

Definition 8.811 (Results of the Heisenberg Source). The single source-specific physical result is given in Table 666; its index set is $\text{Idx}_R^{\text{audit,result}} = [1]$. This conditional result comprises observational information about cosmic geometry and the empirical comparability of model consequences. The presentation of historical statements, Augustine’s argument for the co-creation of time and the world, and the ordinary-word analysis of created are given as statement evidence and in the statement audit.

Table 666: Results and primary-source locations in the Heisenberg source

j	Result	Primary-source location
1	Observational investigation of cosmic geometry and empirical comparability of cosmological-model consequences within general relativity	Chapter VII, p. 125

Result Dependencies

Definition 8.812 (Result-Dependency Hyperedges of the Heisenberg Source). The dependency hyperedge $\text{ResultEdge}_{R,1}$ of Result 1 is given in Table 667. It uses Time Content 15, the time in the four-dimensional spacetime geometry of general relativity. Source Claims 1 and 2 supply the spatial content of the physical inputs; Claims 3–5 supply the condition on the geometry–mass relation, the observational inference, and the comparison of model consequences.

Table 667: Result-dependency hyperedges of the Heisenberg source

q	Time-content input indices	Formal-component-support indices	Result-output indices
1	$\{15\}$	$\llbracket 1, 5 \rrbracket$	$\{1\}$

Grounds

Ground identifier 72-RG1.

Set of result identifiers 72-R1.

Ground content The paragraph on p. 125 giving an inference from galaxy observations conditional on the general-relativistic geometry–mass relation, and discussing empirical comparison of cosmological-model consequences.

Set of primary-source-support identifiers 72-L1.

Evidence Hierarchy

Evidence-hierarchy identifier 72-MT1.

Hierarchy Presentation of a historical statement, Heisenberg’s own conceptual analysis, and account of cosmological models.

Objects The existence of time, creation of the universe, the vocabulary of creation, and cosmological models.

Predicates States historical attribution, conceptual indefinability, and empirical comparability.

Conditions Separates the speakers of the statements and their evidence types.

Domain of application Audit scope in Chapter VII.

Set of primary-source-support identifiers 72-L1.

Generation Determination

Generation-determination identifier 72-GA1.

Set of object identifiers 72-T1–72-T15.

Complete-generation-determination identifier 72-GA-C.

Complete-generation determined value 0 cases.

Ground for complete-generation determination Initial Components 1 and 2 are nontemporal projections, and there are no transfer events that output time contents.

Domain of the complete-generation determination Entire audit scope.

Local-reach-determination identifier 72-GA-L.

Local-reach determined value 0 cases.

Ground for local-reach determination Providing Source Claims 1–5 supplies no edge that outputs a time content.

Domain of the local-reach determination Entire audit scope.

Residual-root-determination identifier 72-GA-R.

Residual-root determined value 15 cases.

Ground for residual-root determination All fifteen time contents remain primitive time roots. Physical Result 1 uses Time Content 15.

Domain of the residual-root determination Entire audit scope.

Determined values: 0 cases satisfying the LCTR complete-generation determination / 0 cases satisfying the source-internal-evidence-conditioned local-reach determination / 15 residual primitive time roots.

Result Dependencies

Result-dependency identifier 72-RD2.

Set of result identifiers 72-R1.

Set of input identifiers 72-T15.

Dependency content Result 1 is a primitive-time-root-dependent result whose prerequisite is Time Content 15 of the general-relativistic cosmological model.

Evidential Force

Evidential-force identifier 72-EE1.

Force-domain identifier 72-EE-IN.

Boundary identifier 72-EE-OUT.

Set of evidence identifiers 72-L1, 72-SS1.

Set of supported-object identifiers 72-R1, S72-Q001, S72-Q002.

Force-domain content Supports the presentation of the historical statement, conceptual analysis, and account of cosmological models.

Within domain Attribution, analysis, and explanation in the primary source.

Outside domain Identification of Augustine's own verbatim text, empirical establishment of time generation, and empirical determination of a particular cosmological model.

Boundary content Augustine's statement and Heisenberg's analysis are not combined into a statement by one speaker.

Result-Preservation Domain

Preservation-domain identifier 72-RP1.

Set of result identifiers 72-R1.

Set of formation-condition identifiers 72-K1.

Preservation-domain content Result 1 is preserved under the source’s condition on the geometry–mass relation and the conditions relating model consequences to observations. Qualified physical use of Time Content 15 requires its generation, comparison structure, and domain of application to be established. Preservation of historical attribution and ordinary-word analysis is determined by agreement with the original text and context.

Placement of the Grounding Obligation

Obligation identifier 72-GO1.

Set of object identifiers 72-T15.

Set of prior-requirement identifiers 72-P1, 72-K1.

Placement content For Time Content 15, the placement determination is “backgrounded by primitive time input.” The source uses spacetime geometry in a physical inference without first providing conditions that generate its time content from nontemporal physical inputs. Verification of the historical and conceptual descriptions in Time Contents 1–14 does not require physical time generation.

Evidence of Actual Failure

Actual-failure identifier 72-AF1.

Set of failure-condition identifiers None.

Set of evidence identifiers 72-L1, 72-SS1.

Failure content The actual-failure determination is “indeterminate from the primary-source evidence.” No actual failure of a concrete condition of a particular cosmological model is established, and none is inferred from the backgrounding of the obligation alone.

Downstream Reach

Reach identifier 72-DR1.

Set of origin identifiers 72-T15.

Set of reached-result identifiers 72-R1.

Reach content There is a structural dependency from Time Content 15 to Result 1. The primary-source evidence does not determine actual failure of the corresponding physical conditions or propagation of such failure to the result.

Repair Scope

Repair identifier 72-RS1.

Set of failed-item identifiers None.

Set of repair-item identifiers 72-T15, 72-K1.

Regeneration order If actual failure of the formation conditions for Time Content 15 is established, construct the relevant temporal representation and comparison structure, specify their domain of application, and then reassess the dependent cosmological model's comparison with observations. No numerical calculation for a particular model is supplied within this audit scope, so no recalculated value is determined. The historical presentation and ordinary-word analysis are outside this physical reassessment.

Set of recalculated-result identifiers 72-R1.

Audit Conclusions

Proposition 8.813 (Complete Audit of Time Use in the Heisenberg Source). Complete generation and local reach both yield zero cases, and all fifteen time contents remain primitive time roots. Physical Result 1 depends on Time Content 15.

Proof. There are no transfer edges that output time contents. The sole result edge takes Time Content 15 and Source Claims 1–5 as inputs. Thus neither the physical initial set nor the source evidence environment reaches a time content, and reaching Result 1 requires the addition of Time Content 15. $\square$

Individual Statements **Statement Evidence 1: Nonexistence of Time before the Universe**

Original quotation Time has been created together with the world, it belongs to the world, therefore time did not exist before the universe existed.

Primary-source location Chapter VII, printed p. 125, PDF p. 144.

Minimal context The paragraph in which Heisenberg presents Augustine's discussion of the creation of the universe and time in the *Confessions*.

Attribution Werner Heisenberg presenting Augustine's argument.

Statement Evidence 2: Temporal Presupposition in the Vocabulary of Creation

Original quotation Therefore, it is impossible to define in rational terms what could be meant by the phrase "time has been created."

Primary-source location Chapter VII, printed p. 125, PDF p. 144.

Minimal context The conclusion immediately following the observation that *created* means that something which did not exist before came into being, and that this meaning presupposes a concept of time.

Attribution Werner Heisenberg's own conceptual analysis.

Common Evidence with a Specified Domain of Application

Target statement	Statement object	Predicate	Explicit domain of application
S72-Q001	Time before the existence of the universe	States that time was created together with the world and did not exist before the universe.	Augustine's account of the creation of the universe and time as presented by Heisenberg.
S72-Q002	The phrase "time has been created"	States that a rational definition is impossible because <i>created</i> presupposes a concept of time.	Heisenberg's own immediately following conceptual analysis.

Individual Audit

Table 669: Dependencies among the statement contents of the Heisenberg source

Relation	Content	Time contents
1	Description of human temporal passage in terms of future, present, and past	4, 5, 6; 3
2	Presentation of Augustine's argument from the co-creation of time and the world to the denial of time before creation	11; 2
3	Analysis identifying the presupposition of time in the before–after relation contained in the ordinary meaning of created	13; 14

Nonexistence of Time before the Universe

Interpretive possibility The original text presents Augustine's argument that time belongs to the world and therefore did not exist before its creation. The absence of a defined time coordinate in a cosmological model and the nonexistence of time itself are distinct propositions.

Independent grounds Heisenberg's explicit presentation of Augustine's argument.

Boundary of evidential force The original text supports that Heisenberg presented the argument.

Corresponding Time Contents and Statement Dependencies Time Contents 11 and 2 and Relation 2 in Table 669. Here Time Contents 11 and 2 correspond to the presentation of Augustine's argument. Physical Result 1 concerns general-relativistic cosmological models.

Temporal Presupposition in the Vocabulary of Creation

Interpretive possibility This identifies a self-dependency in which the word asserting the creation of time already requires a temporal relation that represents before and after generation.

Independent grounds The immediately preceding sentence that explicitly gives the ordinary meaning of *created* as the coming into being of something that did not exist before.

Boundary of evidential force The original text identifies dependence on a concept of time through the before–after relation contained in the ordinary meaning of *created*.

Corresponding Time Contents and Statement Dependencies Time Contents 11, 13, and 14 and Relation 3 in Table 669.

Common Audit with a Specified Domain of Application

Statement-audit-scope identifier S72-SA-SC1.

Target statements S72-Q001, S72-Q002.

Audit scope The first statement is Augustine’s argument for the nonexistence of time before the universe, as presented by Heisenberg. The second is Heisenberg’s own conceptual analysis identifying the presupposition of time in the vocabulary of creation.

8.20 Cross-Source Audit

8.20.1 Cross-Source Primitive-Time-Root Projections

The source subsets determined by primitive-time-root and result decompositions, source-tagged primitive-time-root-dependent results and dependency relations, the cross-source decomposition of results outside audit reach, and the inclusion and result projection of cross-source primitive-time-root–result dependency relations are given, respectively, by Definition 7.52, Definition 7.59, Proposition 7.60, and Proposition 7.61.

Five audit outcomes across the 72-source corpus

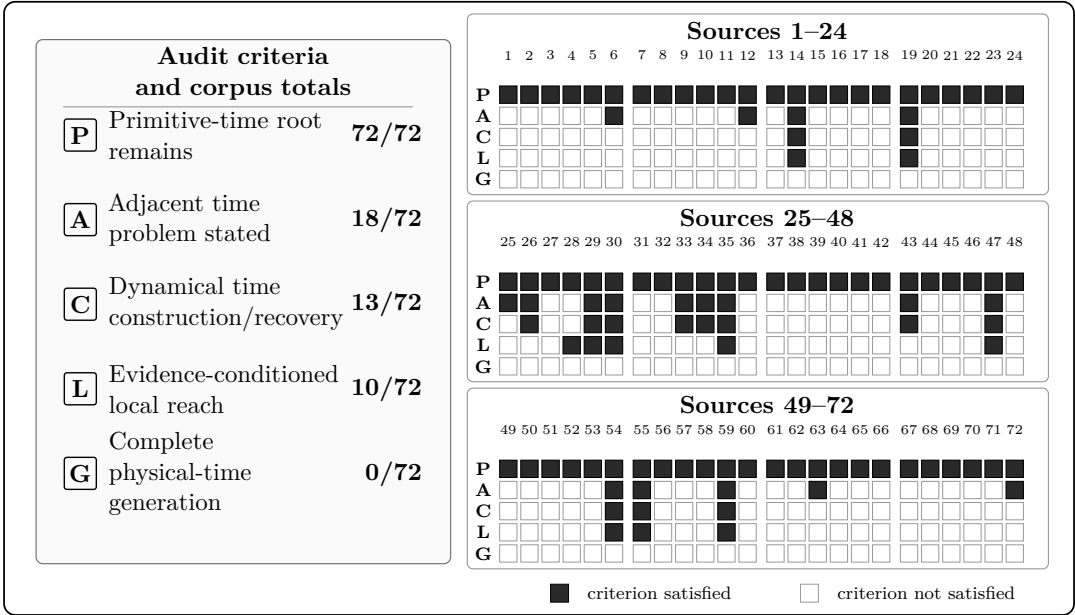


Figure 32: Source-by-source distribution of the five audit results across the 72 audited sources. Source numbers correspond to the sequence in Table 16.

Formal-Verification Assets

The supplementary archive `lctr-verification.zip` contains formal-verification assets for Sections 2–6. Its top-level directory after extraction is `lctr-verification/`. The assets include a complete inventory of the mathematical definitions, formulas, claims, and proof obligations in this scope for the same version as the published PDF, the Lean and Isabelle formalization sources, and a correspondence table linking locations in the Japanese and English PDFs to both formalizations. The table records types, quantification, premises, domains, all conclusions, and dependencies component by component, and links each formula occurrence to the mathematical object it expresses.

The mathematical proof obligations in this scope are formalized in both Lean and Isabelle, and their proofs are checked by the Lean kernel and the Isabelle/Pure inference kernel. Lean proofs are additionally rechecked by the separate implementations `nanoda` and `con-leche`. For each formalized object in Lean and Isabelle, the assets include verification results, the axioms used, and theorem dependencies.

For each proof obligation in the text, the assets include a comparison tracing the proof bodies, definitions used, actual theorem dependencies, and local assumptions in both formalizations. The comparison links the premises and conclusions in the text to the dependencies and dependent-type or carrier conditions in both formalizations.

The partial output map of a law-candidate relation has machine-readable representations in MMT, OpenMath, and Content MathML, together with the results of checking these representations. The assets also include SMT proofs and checks of their proof traces, symbolic computations using computer algebra systems, unit and dimension checks, and results from checking counterexamples and boundary cases, together with the conditions under which each check applies. Each result is linked to its object in the text and records the domain and conditions checked.

The release package contains pinned version information for the verifiers and their library dependencies, the inputs and outputs of every included check, reports for each object, and instructions for running each standard tool directly. To establish file identity, the accompanying file manifest associates the name of each published PDF, formalization source, check input, verification result, and rerun script with its SHA-256 hash. The `reproduce.sh` script runs the included checks and regenerates their reports in the pinned verification environment. The release metadata identifies the verification-assets version corresponding to this PDF, its scope, and the checks performed.

References

- [1] Isaac Newton. *Philosophiae Naturalis Principia Mathematica*. Royal Society, London, 1687.
- [2] Albert Einstein. Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 322(10):891–921, 1905.
- [3] Hermann Minkowski. Raum und Zeit. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 18:75–88, 1909. Lecture delivered in 1908.
- [4] Albert Einstein. Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik*, 354(7):769–822, 1916.

- [5] Aristotle. *Aristotle's Physics: Books III and IV*. Clarendon Aristotle Series. Oxford University Press, Oxford, 1983. Translated with introduction and notes by Edward Hussey; new impression with corrections and additions, 1993; Book IV, Chapters 11–14, Bekker 218b21–224a15.
- [6] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Feynman Lectures on Physics, Volume I*. Addison-Wesley, Reading, MA, 1963. Chapter 4, Section 4-4; Chapter 5; Chapters 15–17.
- [7] Percy W. Bridgman. *The Logic of Modern Physics*. Macmillan, New York, 1927.
- [8] Karel V. Kuchař. Time and interpretations of quantum gravity. In *Proceedings of the 4th Canadian Conference on General Relativity and Relativistic Astrophysics*, pages 211–314, 1992.
- [9] Chris J. Isham. Canonical quantum gravity and the problem of time. In L. A. Ibort and M. A. Rodríguez, editors, *Integrable Systems, Quantum Groups, and Quantum Field Theories*, pages 157–287. Kluwer Academic, Dordrecht, 1993.
- [10] Don N. Page and William K. Wootters. Evolution without evolution: Dynamics described by stationary observables. *Physical Review D*, 27(12):2885–2892, 1983.
- [11] Carlo Rovelli. Partial observables. *Physical Review D*, 65(12):124013, 2002.
- [12] Bureau International des Poids et Mesures. *The International System of Units (SI Brochure)*. BIPM, Sèvres, 9 edition, 2019. 9th edition, updated in 2026.
- [13] Edwin F. Taylor and John Archibald Wheeler. *Spacetime Physics: Introduction to Special Relativity*. W. H. Freeman and Company, New York, 2 edition, 1992.
- [14] Wolfgang Rindler. *Essential Relativity: Special, General, and Cosmological*. Springer-Verlag, New York, Heidelberg and Berlin, 2 edition, 1977.
- [15] Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler. *Gravitation*. W. H. Freeman and Company, San Francisco, 1973.
- [16] National Institute of Standards and Technology. Putting einstein to the test. <https://www.nist.gov/atomic-clocks/a-powerful-tool-for-science/putting-einstein-test>, April 2025. Created April 3, 2025; updated April 17, 2025; accessed July 29, 2026.
- [17] European Space Agency. Clocks, gravity, and the limits of relativity. https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/Research/Clocks_gravity_and_the_limits_of_relativity, May 2019. Published May 23, 2019; accessed July 29, 2026.
- [18] Markus Pössel. From light clocks to time dilation. *Einstein Online*, 13:1004, 2021. <https://www.einstein-online.info/en/spotlight/light-clocks-time-dilation/>; accessed July 29, 2026.
- [19] P. A. M. Dirac. The theory of gravitation in hamiltonian form. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 246(1246):333–343, 1958.

- [20] Richard Arnowitt, Stanley Deser, and Charles W. Misner. The dynamics of general relativity. In Louis Witten, editor, *Gravitation: An Introduction to Current Research*, pages 227–265. Wiley, New York, 1962.
- [21] Bryce S. DeWitt. Quantum theory of gravity. i. the canonical theory. *Physical Review*, 160(5):1113–1148, 1967.
- [22] James B. Hartle and Stephen W. Hawking. Wave function of the universe. *Physical Review D*, 28(12):2960–2975, 1983.
- [23] Roger Penrose. Singularities and time-asymmetry. In S. W. Hawking and W. Israel, editors, *General Relativity: An Einstein Centenary Survey*, pages 581–638. Cambridge University Press, Cambridge, 1979.
- [24] Jacob D. Bekenstein. Black holes and entropy. *Physical Review D*, 7(8):2333–2346, 1973.
- [25] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220, 1975.
- [26] S. W. Hawking. Breakdown of predictability in gravitational collapse. *Physical Review D*, 14(10):2460–2473, 1976.
- [27] Don N. Page. Information in black hole radiation. *Physical Review Letters*, 71(23):3743–3746, 1993.
- [28] John von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, Berlin, 1932.
- [29] Wojciech H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3):715–775, 2003.
- [30] Jürgen Ehlers, Felix A. E. Pirani, and Alfred Schild. The geometry of free fall and light propagation. In L. O’Raifeartaigh, editor, *General Relativity: Papers in Honour of J. L. Synge*, pages 63–84. Clarendon Press, Oxford, 1972.
- [31] Luca Bombelli, Joohan Lee, David Meyer, and Rafael D. Sorkin. Space-time as a causal set. *Physical Review Letters*, 59(5):521–524, 1987.
- [32] Juan M. Maldacena. The large n limit of superconformal field theories and supergravity. *Advances in Theoretical and Mathematical Physics*, 2:231–252, 1998.
- [33] Shinsei Ryu and Tadashi Takayanagi. Holographic derivation of entanglement entropy from the anti-de sitter space/conformal field theory correspondence. *Physical Review Letters*, 96(18):181602, 2006.
- [34] Steven Weinberg. The cosmological constant problem. *Reviews of Modern Physics*, 61(1):1–23, 1989.
- [35] Adam G. Riess, Alexei V. Filippenko, Peter Challis, Alejandro Clocchiatti, Alan Diercks, Peter M. Garnavich, Ron L. Gilliland, Craig J. Hogan, Saurabh Jha, Robert P. Kirshner, B. Leibundgut, M. M. Phillips, David Reiss, Brian P. Schmidt, Robert A. Schommer, R. Chris Smith, J. Spyromilio, Christopher Stubbs, Nicholas B. Suntzeff,

- and John Tonry. Observational evidence from supernovae for an accelerating universe and a cosmological constant. *The Astronomical Journal*, 116(3):1009–1038, 1998.
- [36] S. Perlmutter, G. Aldering, G. Goldhaber, R. A. Knop, P. Nugent, P. G. Castro, S. Deustua, S. Fabbro, A. Goobar, D. E. Groom, I. M. Hook, A. G. Kim, M. Y. Kim, J. C. Lee, N. J. Nunes, R. Pain, C. R. Pennypacker, R. Quimby, C. Lidman, R. S. Ellis, M. Irwin, R. G. McMahon, P. Ruiz-Lapuente, N. Walton, B. Schaefer, B. J. Boyle, A. V. Filippenko, T. Matheson, A. S. Fruchter, N. Panagia, H. J. M. Newberg, W. J. Couch, and The Supernova Cosmology Project. Measurements of Omega and Lambda from 42 high-redshift supernovae. *The Astrophysical Journal*, 517(2):565–586, 1999.
 - [37] J. Bailey, K. Borer, F. Combley, H. Drumm, F. Krienen, F. Lange, E. Picasso, W. von Ruden, F. J. M. Farley, J. H. Field, W. Flegel, and P. M. Hattersley. Measurements of relativistic time dilatation for positive and negative muons in a circular orbit. *Nature*, 268:301–305, 1977.
 - [38] Leon F. Curtiss. Measurements of radioactivity. Circular of the Bureau of Standards 476, National Bureau of Standards, October 1949.
 - [39] Robert V. Pound and Jr. Rebka, Glen A. Apparent weight of photons. *Physical Review Letters*, 4(7):337–341, 1960.
 - [40] J. C. Hafele and Richard E. Keating. Around-the-world atomic clocks: Observed relativistic time gains. *Science*, 177(4044):168–170, 1972.
 - [41] J. C. Hafele and Richard E. Keating. Around-the-world atomic clocks: Predicted relativistic time gains. *Science*, 177(4044):166–168, 1972.
 - [42] Joel L. Lebowitz. Boltzmann’s entropy and time’s arrow. *Physics Today*, 46(9):32–38, 1993.
 - [43] Julian B. Barbour. The timelessness of quantum gravity: I. the evidence from the classical theory. *Classical and Quantum Gravity*, 11(12):2853–2873, 1994.
 - [44] Julian B. Barbour. The timelessness of quantum gravity: II. the appearance of dynamics in static configurations. *Classical and Quantum Gravity*, 11(12):2875–2897, 1994.
 - [45] Alain Connes and Carlo Rovelli. Von neumann algebra automorphisms and time-thermodynamics relation in general covariant quantum theories. *Classical and Quantum Gravity*, 11(12):2899–2918, 1994.
 - [46] Clifford Headley, J. B. Clayton, and W. A. Reed. Methods of suppressing stimulated brillouin scattering in optical fibers by manipulation of the fiber properties. In Gordon W. Day, Douglas L. Franzen, and Paul A. Williams, editors, *Technical Digest: Symposium on Optical Fiber Measurements, 1996*, number 905 in NIST Special Publication, pages 105–110. National Institute of Standards and Technology, Boulder, CO, October 1996.
 - [47] Stephen L. Adler. Why decoherence has not solved the measurement problem: A response to p. w. anderson. *Studies in History and Philosophy of Modern Physics*, 34:135–142, 2003.

- [48] Philip W. Anderson. Science: A dappled world or a seamless web? *Studies in History and Philosophy of Modern Physics*, 32(3):487–494, 2001.
- [49] Charles W. Clark. Effects of configuration interaction on intensities and phase shifts. In David R. Lide, editor, *A Century of Excellence in Measurements, Standards, and Technology: A Chronicle of Selected NBS/NIST Publications, 1901–2000*, number 958 in NIST Special Publication, pages 116–119. U.S. Government Printing Office, Washington, DC, January 2001.
- [50] Tamara M. Davis and Charles H. Lineweaver. Expanding confusion: Common misconceptions of cosmological horizons and the superluminal expansion of the universe. *Publications of the Astronomical Society of Australia*, 21:97–109, 2004.
- [51] Maximilian Schlosshauer. Decoherence, the measurement problem, and interpretations of quantum mechanics. *Reviews of Modern Physics*, 76:1267–1305, 2005.
- [52] Claus Kiefer. Quantum cosmology and the arrow of time. *Brazilian Journal of Physics*, 35(2A):296–299, 2005.
- [53] Øyvind Grøn and Øystein Elgarøy. Is space expanding in the friedmann universe models? *American Journal of Physics*, 75:151–157, 2007.
- [54] Matthew J. Francis, Luke A. Barnes, J. Berian James, and Geraint F. Lewis. Expanding space: The root of all evil? *Publications of the Astronomical Society of Australia*, 24(2):95–102, 2007.
- [55] Richard J. Cook and M. Shane Burns. Interpretation of the cosmological metric. *American Journal of Physics*, 77:59–66, 2009.
- [56] J. A. Peacock. A diatribe on expanding space, 2008.
- [57] Emory F. Bunn and David W. Hogg. The kinematic origin of the cosmological redshift. *American Journal of Physics*, 77:688–694, 2009.
- [58] Claus Kiefer. Does time exist in quantum gravity?, 2009.
- [59] Carlo Rovelli. Forget time. *Foundations of Physics*, 41:1475–1490, 2011.
- [60] Mark Van Raamsdonk. Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42:2323–2329, 2010.
- [61] Brian Greene. NOVA: The fabric of the cosmos: The illusion of time. PBS/NOVA television program, 2011. Hosted by Brian Greene; produced, written, and directed by Randall MacLowry; official transcript.
- [62] Peter Wolf, Luc Blanchet, Christian J. Bordé, Serge Reynaud, Christophe Salomon, and Claude Cohen-Tannoudji. Does an atom interferometer test the gravitational redshift at the compton frequency? *Classical and Quantum Gravity*, 28(14):145017, 2011.
- [63] Edward Anderson. Problem of time in quantum gravity. *Annalen der Physik*, 524:757–786, 2012.
- [64] George F. R. Ellis. Space time and the passage of time, 2012.

- [65] David Tong. Dynamics and relativity. University of Cambridge lecture notes, 2013. Chapter 7, Special Relativity.
- [66] Geraint F. Lewis. On the relativity of redshifts: Does space really expand? *Australian Physics*, 53(3):95–100, 2016.
- [67] Lee Smolin. Temporal relationalism; beyond spacetime, 2018. Temporal Relationalism, arXiv:1805.12468v1 (2018); Beyond Spacetime, Cambridge University Press (2020), pp. 143–175, doi:10.1017/9781108655705.010.
- [68] NASA Space Place and NASA/JPL–Caltech. Is time travel possible? <https://spaceplace.nasa.gov/time-travel/en/>, April 2020. Published April 30, 2020.
- [69] D. M. Asner, Christoph Hanhart, and Mikhail Mikhasenko. 50. resonances. Review of Particle Physics, 2023 update, August 2023. Revised August 2023.
- [70] Richard P. Feynman, Fernando B. Morinigo, and William G. Wagner. *Feynman Lectures on Gravitation*. Addison–Wesley, 1995. Lectures delivered at Caltech during the 1962–63 academic year.
- [71] Jayant V. Narlikar. *An Introduction to Relativity*. Cambridge University Press, 2010.
- [72] Werner Heisenberg. *Physics and Philosophy: The Revolution in Modern Science*. Harper & Brothers, New York, 1958.
- [73] National Institute of Standards and Technology. Einstein’s general relativity and your age, 2022.
- [74] E. Savalle, C. Guerlin, P. Delva, F. Meynadier, C. le Poncin-Lafitte, and P. Wolf. Gravitational redshift test with the future ACES mission, 2019.